Revenue Management Developer Guide

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CONTENTS

CHAPTER 1 Get Started with Revenue Management Developer Resources

Get a single, unified system to automate your CRM processes. Use the developer sources of Revenue Management to automate the backend work to support the end-to-end revenue solution.

Revenue Cloud is now Revenue Management. You may see references to Revenue Cloud in our application and documentation.

Revenue Management provides extensible and API-first business components of the product-to-cash processes. Learn more about the developer resources that are available for these components.

EDITIONS

Available in: Lightning Experience

Available in: **Enterprise**, **Unlimited**, and **Developer** Editions

Product Catalog Management

Create and manage an entire product portfolio with components such as attributes, product classifications, simple and bundled products, and rules.

- Use standard objects and fields to manage products, rules, and catalogs.
- Use business APIs to serve catalog definitions to users or applications.
- Use metadata API types to access and manage the metadata types, such as product specification type and product specification record types.
- Use tooling API objects to retrieve and manage smaller pieces of metadata types through SOQL capabilities. Use REST or SOAP to access metadata.
- Use Product Discovery business APIs, which are composite APIs, to search products or to discover catalogs, products, and categories.

Salesforce Pricing

Create a reliable pricing solution for your users through customized price adjustment schedules. Get accurate pricing for your entire product portfolio.

- Use standard objects and fields to manage pricing processes such as product management, and the calculation and application of discounts.
- Use business APIs to get unified pricing experiences across product lines.
- Use invocable actions to invoke the pricing Connect API by providing the pricing, context, and price waterfall details.
- Use metadata API types to work with the metadata associated with Flows and Salesforce Pricing settings.

- Use tooling API objects to retrieve and manage smaller pieces of metadata types through SOQL capabilities such as pricing action parameters, pricing procedure output map, and pricing recipe details. Use REST or SOAP to access metadata.

Product Configurator

Customize the components and attributes of a product to meet the business requirement expectations.

- Use standard objects to manage product-related information.
- Use the business APIs to retrieve and update a product's configuration from a configurator or to access configurator capabilities by integrating with any front-end application.
- Use constraint modeling language on page 1078 to enforce business logic declaratively, without the need for extensive code in a general-purpose programming language.

Transaction Management

Manage subscription lifecycles from quotes and orders to contracts, assets, amendments, and renewals. Get insights into customer assets and see a consolidated list of all assets that belong to an account.

- Use standard objects and fields to manage transactions and details of a customer asset. Use the QuoteSaveEvent platform event to notify subscribers after saving of a quote is processed.
- Use business APIs to place, clone, or supplement a sales transaction. You can also initiate amendment, renewal, or cancellation of assets by using APIs.
- Use invocable actions to create and activate an order from a quote, or to initiate amendment, renewal, or cancellation of assets through invocable actions.
- Use metadata API types to work with the metadata associated with Flows.
- Use built-in Apex classes and interfaces grouped by namespace.

Usage Management

Ensure transparent, accurate, and efficient management of usage data and estimated usage amount.

- Use standard objects and fields to set up and manage consumption of usage-based products.
- Use metadata API types to work with the metadata associated with Usage Management.
- Use business APIs to get details of a usage-based product that's associated with an asset, an order item, or a quote line item.
- Use invocable actions to invoke usage summaries, process consumption overages, and refresh usage entitlements.

Rate Management

Quote and price products based on predefined rates for future use of the product or service.

- Use standard objects and fields to manage rates and discounts for a product's resource consumption.
- Use metadata API types to work with the metadata associated with Rate Management settings.
- Use business APIs to get details of a rate plan and persisted rating waterfall.

- Use invocable action to invoke the rating service to rate the usage records.

Dynamic Revenue Orchestrator

Get visibility into a product's fulfillment journey. Also, get a view of the entire fulfillment design processes.

- Use standard objects to manage details of a product's fulfillment.
- Use invocable actions to submit an order or a sales transaction to Dynamic Revenue Orchestrator for fulfillment.
- Use metadata API types to work with the metadata associated with Flows.
- Use callout step types to make HTTP calls to an external system.

Billing

Get an integrated and extensible subscription and usage-based billing solution. Automate processes such as payment processing, invoice generation, and usage-based billing.

- Use standard objects to manage billing and tax configurations, credit memos, and invoices.
- Use platform events types to know more about standard platform events.
- Use invocable actions to manage credit application, billing schedules, and invoices.
- Use business APIs to manage credit application and to handle billing scenarios.
- Use built-in Apex classes to access the same capabilities that are available in the Billing Business APIs.
- Use metadata API types to work with the metadata associated with Billing settings and Flows.

See the RevenueManagementSettings on page 4 metadata type to set up Revenue Management through configuration settings.

SEE ALSO:

Business Rules Engine

Context Service

Salesforce Contracts

CHAPTER 2 RevenueManagementSettings

Represents the configuration settings to set up Revenue Management.

Parent Type and Manifest Access

This type extends the Metadata metadata type and inherits its `fullName` field.

In the package manifest, all the settings metadata types for the org are accessed using the "Settings" name. See Settings for more details.

File Suffix and Directory Location

`RevenueManagementSettings` values are stored in the `revenuemanagement.settings` file in the `settings` folder. The `.settings` files are different from other named components, because there's only one settings file for each settings component.

Version

RevenueManagementSettings is available in API version 60.0 and later.

Special Access Rules

This metadata settings type is available with Revenue Management.

Fields

Field Name	Description
`enableAdvCreateOrdersFromQuote`	**Field Type** boolean **Description** Indicates whether to enable users to choose a method for generating multiple orders from a single quote (`true`) or not (`false`). This setting updates the functionality of the Create Order capability on quotes. Available in API version 65.0 and later.

Field Name	Description
`enableAdvancedDetailLinePricing`	**Field Type** boolean **Description** Indicates whether to enable advanced pricing for quote and order detail line items (`true`) or not (`false`). The default value is `false`. Available in API version 65.0 and later.
`enableAsIsRenewals`	**Field Type** boolean **Description** Indicates whether users can enable as-is renewals capability for existing assets (`true`) or not (`false`). The default value is `false`. Available in API version 64.0 and later.
`enableAutoAddDerivedAsset`	**Field Type** boolean **Description** Indicates whether to automatically add assets with derived pricing to a quote or an order when contributing products are added to it (`true`) or not (`false`). Available in API version 62.0 and later.
`enableCoreCPQ`	**Field Type** boolean **Description** Indicates whether to enable read and write access to Revenue Management features and objects (`true`) or not (`false`). See Enable Revenue Management Features in Your Scratch Org.
`enableDeltaPricing`	**Field Type** boolean **Description** Indicates whether to reprice faster by processing only the changes (delta) made to quotes and orders (`true`) or not (`false`). The default value is `false`. Available in API version 63.0 and later.
`enableGroupRamp`	**Field Type** boolean

Field Name	Description
	Description Reserved for internal use.
`enableGroupRampPref`	**Field Type** boolean **Description** Indicates whether to create ramp deals for multiple products by creating group ramp segments (`true`) or not (`false`). The default value is `false`. Configure group ramp segments with specific start dates and end dates, and different product quantities and prices. Available in API version 65.0 and later. Before you enable this setting, ensure the `groupsEnabled` and `enableTransactionCloning` fields are set to `true`.
`enableRampDeal`	**Field Type** boolean **Description** Indicates whether to create ramp deals for individual line items by creating segments with specific start dates, end dates, quantities, and prices (`true`) or not (`false`). Available in API version 62.0 and later.
`enableRevUnifiedSetup`	**Field Type** boolean **Description** Indicates whether to enable the usage of a procedure plan for price calculation (`true`) or not (`false`).
`enableTransactionCloning`	**Field Type** boolean **Description** Indicates whether to clone quotes or orders with their line items and groups, or clone the individual line items and groups of line items. Available in API version 64.0 and later.
`enableTransactionProcessor`	**Field Type** boolean

Field Name	Description
	Description Indicates whether to enable transaction types for quotes and orders (`true`) or not (`false`). Transaction types optimize the way transactions are processed, depending on their size, complexity, and the business needs of the org. You can't turn off transaction processing after it's turned on. Available in API version 63.0 and later. To set a default transaction type, create transaction types, and select the default. After you select a default transaction type, you can't be without a default.
`groupsEnabled`	**Field Type** boolean **Description** Indicates whether users can group line items in quotes and orders (`true`) or not (`false`). Available in API version 62.0 and later.
`hidePriceRefreshNtfcn`	**Field Type** boolean **Description** Indicates whether to hide the notification that appears when quote or order prices aren't updated (`true`) or not (`false`). Hiding this notification may affect saving quotes and creating orders because you could be using outdated prices. The default value is `false`. Available in API version 65.0 and later.
`relaxUniqueCipValidation`	**Field Type** boolean **Description** Indicates whether to enable fully customizable extensions to contract item prices by ignoring record validations (`true`) or not (`false`). Available in API version 64.0 and later.
`skipOrgSttPricing`	**Field Type** boolean **Description** Indicates whether to skip the default pricing procedure and any pricing procedure set for a sales transaction with quote or order type (`true`) or not (`false`). To

Field Name	Description
	enable this field, you must enable the `enableRevUnifiedSetup` field.

Declarative Metadata Sample Definition

Here's an example of a RevenueManagementSettings component.

```
<?xml version="1.0" encoding="UTF-8"?>
<RevenueManagementSettings
xmlns="http://soap.sforce.com/2006/04/metadata">
    <enableCoreCPQ>true</enableCoreCPQ>
    <enableDeltaPricing>true</enableDeltaPricing>
    <enableTransactionProcessor>true</enableTransactionProcessor>
    <enableAutoAddDerivedAsset>true</enableAutoAddDerivedAsset>
    <enableRampDeal>true</enableRampDeal>
    <enableRevUnifiedSetup>true</enableRevUnifiedSetup>
    <groupsEnabled>true</groupsEnabled>
    <enableTransactionCloning>true</enableTransactionCloning>
    <relaxUniqueCipValidation>true</relaxUniqueCipValidation>
    <skipOrgSttPricing>true</skipOrgSttPricing>
    <enableAsIsRenewals>true</enableAsIsRenewals>

<enableAdvCreateOrdersFromQuote>true</enableAdvCreateOrdersFromQuote>


<enableAdvancedDetailLinePricing>true</enableAdvancedDetailLinePricing>

    <enableGroupRampPref>true</enableGroupRampPref>
    <hidePriceRefreshNtfcn>true</hidePriceRefreshNtfcn>
</RevenueManagementSettings>
```

This example `package.xml` references the previous definition.

```
<?xml version="1.0" encoding="UTF-8"?>
<Package xmlns="http://soap.sforce.com/2006/04/metadata">
    <types>
        <members>RevenueManagement</members>
        <name>Settings</name>
    </types>
    <version>67.0</version>
</Package>
```

Wildcard Support in the Manifest File

The wildcard character `*` (asterisk) in the `package.xml` manifest file doesn't apply to metadata types for feature settings. The wildcard applies only when retrieving all settings, not for an individual setting. For details, see Settings. For information about using the manifest file, see Deploying and Retrieving Metadata with the Zip File.

CHAPTER 3 Revenue Management Deployment

In this chapter ...

This section provides a clear roadmap for accurately and efficiently deploying Revenue Management objects and metadata between development, staging, sandbox, and production orgs.

Developing and maintaining complex business applications on the Salesforce Platform requires a structured approach to managing changes and deployments. This is especially true for Revenue Management, which involves a complex data model and metadata components.

The interconnected nature of Revenue Management's architecture requires a meticulous and well-defined development operations (DevOps) process. Without it, organizations face significant risks, including data corruption, deployment failures, and disruption to critical business processes.

Who This Section Is For

This section is intended for advanced users who are familiar with Salesforce development, established DevOps concepts, and Salesforce administration.

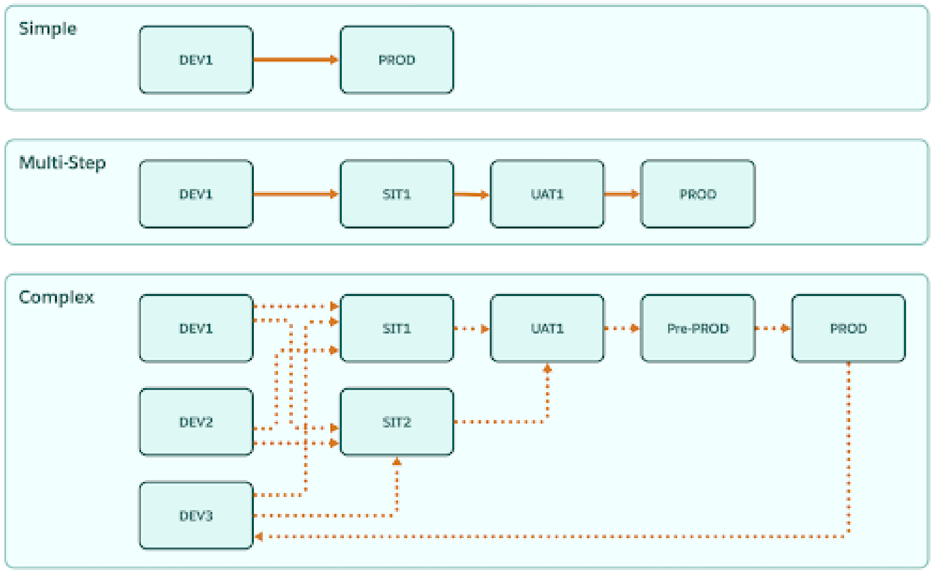
What's Not Included

- This section doesn't include information about data migration from external or legacy systems.
- There's no built-in, Revenue Management-specific deployment tool. Select the Salesforce or third-party deployment tools appropriate for your use case.
- Every Salesforce implementation is unique. This guide doesn't address how to deploy your custom objects, fields, Apex classes, metadata, Lightning Web Components (LWCs), and so on.

Deployment Workflows and Sequence

A typical dev ops cycle involves managing test orgs, sandboxes, and a production org. Your deployment workflow can be a simple deployment of all setup and data from a development org to production. Or, it can be much more complex, involving multiple staging and testing orgs before any changes are sent to production.

Here's a diagram illustrating several potential deployment plans.



Full vs. Incremental Deployment

Deployments take two primary forms: full and incremental.

A full deployment involves migrating the entire set of relevant metadata such as objects, fields, classes, triggers, and potentially a significant volume of data from a source org. This ensures that the target org is a complete, synchronized replica of the source at that point in time.

Full deployments are best suited for initial setup, major releases, or when deploying to a completely new, or non-synced environment, such as the first deployment to a sandbox or staging org.

Incremental deployments are used for ongoing DevOps activities. They focus only on the specific metadata components and associated data records that have been created, modified, or deleted since the last successful deployment.

This incremental approach is faster, reduces deployment risk by minimizing the scope of changes, and is essential for maintaining business continuity in a live production environment. Critical to this process is a version control system to track and compare differences between environments, ensuring only the necessary "delta" is packaged and deployed.

Planning for incremental deployments requires change tracking, dependency analysis, and careful sequencing of data and metadata components to ensure that all prerequisites are met in the target environment.

Deployment Sequence and Dependencies

The deployment process must navigate intricate dependencies involving both metadata and the data records themselves. For instance, before deploying the data records for an object, the necessary metadata must be in place first to ensure that the destination org is structurally prepared to receive the data.

Also, there are dependencies based on the record relationships. For example, if you're deploying a custom object with a lookup to the standard Account object, the Account records must be created first to satisfy the lookup relationship when the custom object's child records are subsequently inserted. You must also account for parent-child or master-detail relationships, requiring the parent record to be deployed before the child, to preserve referential integrity. Proper assessment and sequencing of these interdependent data and metadata components is vital to avoid validation errors and data corruption.

Sometimes, there are loops in the inter-relationships. For example, an object A depends on object B being deployed earlier, and object B depends on object A being deployed earlier. Let’s say object A goes first, then object B goes next. Because of circular dependencies, one has to redeploy object A a second time to ensure all the references are properly populated.

Here are a few deployment dependency examples.

- The Product Specification Record Type must be deployed before you can insert any Product records.
- The Attribute Based Adjustment Rule must be deployed before you can insert any Attribute Based Adjustment records.
- Decision Tables (metadata records) must be deployed before deploying Pricing Recipes (metadata records).

Note: For complete information on the required deployment sequence, see Object Deployment Sequence and Metadata Deployment Reference.

Deployment Considerations

In any deployment scenario, you must understand all dependencies and prerequisites related to your planned changes.

Here's a list of things to consider and plan for.

Key considerations include:

- What’s the sequence followed for object deployment? What dependencies exist on related objects?
- Do your objects have Draft, Active, or Inactive states?
- Does activation occur through an API or by simply setting a boolean field (active flag)?
- Are there special fields to consider, such as text fields with embedded IDs, JSON, or fields populated via triggers?
- Are there fields or objects that represent unique or sequential data that must be preserved in the target org?
- How would you handle system settings between the orgs? For example, making sure that a feature toggle is turned off in the target org so that automation doesn't start executing when you insert data from your source org.
- What's the impact of versioning on your deployment? For example, schema, app, or API versions.

Note: For detailed information about these deployment considerations, see Additional Deployment Information.

Global Unique ID Setup

The establishment of a Global Unique ID (GUID) column on all objects during day-one initialization is a recommended practice for Salesforce DevOps.

Salesforce uses its own internal ID for records, but this ID isn't portable. The same record in a testing, sandbox, or production org has different Salesforce IDs. This makes tracking a single piece of data across environments nearly impossible. By introducing your own custom, externally set GUID, you have a single, immutable, and universally consistent identifier.

Note: You can also choose your preferred method to maintain the mapping of records IDs across orgs.

Before performing any deployment work, establish your GUID across all database objects, and across all Salesforce orgs involved in your deployment plan.

Note: Add a GUID field to all objects used during your deployment. Since you can't add fields to metadata types (such as a Flow), use the API name for each metadata type as the unique identifier.

Create a GUID Field

Add a GUID field to all objects used during your deployment to ensure unique identification of records across environments.

1. From Setup, in the Quick Find box, find and select **Object Manager**.
2. Select an object.
3. Click **Fields & Relationships**.
4. Click **New**.
5. Select **Text** for the data type.
6. Enter a field label and field name.
7. Enter a length.

   We recommend 255 to avoid any errors related to ID length.
8. Select **Unique** and **External ID**.

   Important: Selecting these attributes ensures that every record gets a unique ID.
9. Click **Next**.
10. Select the appropriate profiles for field access, optionally add the field to page layouts, and then click **Save**.
11. Repeat this process for all Salesforce objects related to your deployment plan.

Note: Alternatively, you can create GUID fields by using the Metadata API. For more information, see Understanding Metadata API and the Custom Field metadata type.

GUID Design and Usage

The format and values of the Global Unique ID (GUID) are up to you. Here's some guidance on what makes a good versus poor GUID for deployment tracking.

The format and values of the GUID itself are up to you. Here's some guidance on what makes a good versus poor GUID.

A good GUID design includes these characteristics.

- Immutable
- Unique globally
- Non-translatable
- A single key
- Generated programmatically, and not manually

Poor GUID design includes these characteristics.

- Record Name field (mutable)
- Combination of mutable attributes such as Name + Version + Sequence
- Concatenated keys
- Conditional keys (Pricebook + Product OR Pricebook + Product + ISO Code)

Populate a GUID

Once you have created the GUID field on all the Salesforce objects related to your deployment plan, you're ready to populate the field with data.

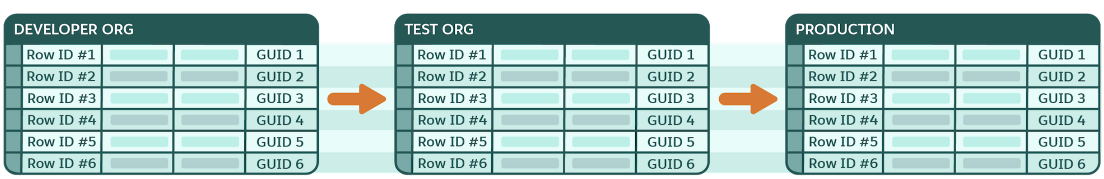
Add the GUID column to your import spreadsheets or import file definitions. Populate the field with the GUID format you've chosen. Now, you're ready to use the GUID in your deployment process.

Important: You must populate the GUID field not only at the beginning of your deployment process, but you must also populate it for any new records you generate along the way.

Usage of GUID in the Deployment Cycle

The GUID is the cornerstone of your deployment and data migration strategy. During the deployment process, leverage the GUID for accurate upserts and data mapping. When migrating data records from one org to another, your deployment method must use the GUID as the external ID field for lookup and matching. This practice ensures that updates are applied precisely to the correct record, preventing data duplication or corruption.

In the event of a deployment failure or a post-deployment data issue, administrators can use the GUID to quickly and confidently locate the problematic records in all orgs for precise auditing and troubleshooting.



Non-Extensible Objects

A few Revenue Management objects are protected, and therefore non-extensible. You can't add a GUID field to these objects. Instead, create an external reference table to store your GUID. Use this reference table to track the non-extensible object throughout your deployment process.

Managing Component States

Manage activation, versioning, and dependencies for components and objects as part of your deployment plan. A successful deployment makes sure that the system executes the intended, final, and active logic, preventing failures caused by stale or inactive component dependencies.

A component's state is its operational status, which includes the:

- Active, inactive, or draft state
- Component version

- Dependencies on other components and objects
- Component-specific attributes like rank on expression sets

Components like decision tables, expression sets, and context definitions must be deployed taking their state into account. Also, some objects use an Active checkbox to determine if their records must be used during run time operations or not.

You must also make sure that all related dependencies are moved correctly and in the proper sequence from the source org to the target org.

State Management Scenarios

Component deployments usually fall into one of these categories:

Deploy a New Component
: If the target org doesn't contain the component that you want to migrate, deploy it along with its versions from the source org to the target org.

Deploy Component Updates
: When the target org already contains the component you want to migrate, then state and version management take effect. For example, with expression sets, you must deactivate the version in the target org where you want to deploy changes. If you try to deploy to an active version, the deployment fails.

Deploy Components with Dependencies
: All related dependencies must be moved correctly and in the proper sequence from the source org to the target org. For example, expression sets reference elements such as decision matrices, decision tables, subexpressions, context definitions, field and object aliases, and explainability message templates. Migrate the dependent elements independently before you migrate the expression set version.

Helpful Links

Refer to these links for examples and component-specific information:

- Considerations for Migrating Decision Tables
- Considerations for Migrating Expression Sets
- Migration of Expression Sets with Dependencies
- Migrate Context Definitions

Note: Some Revenue Management components have unique activation requirements. For complete information about state management for these components, see Additional Deployment Information.

Post-Deployment Steps

Most deployments require that you take some actions after the deployment to the target org is complete to ensure proper functionality and data integrity.

Here are some examples of common post-deployment tasks.

- Refresh decision tables
- Sync pricing data
- Rebuild the product index
- Manually create elements that couldn't be deployed through the API
- Create or activate user accounts for testing and validation

Note: For information about the post-deployment steps necessary for each Revenue Cloud feature, see Additional Deployment Information.

Deployment Scenarios

Learn about specific deployment scenarios including new environment setup, refreshes, retiring records, deploying patches, and many more.

Product Catalog Management

Table 1: Product Definition: Full Deployment Scenarios

Scenario	Impact	Validation and Dependencies
Large Catalog Refresh Introduce 200+ new SKUs for accessories line	Metadata: See Additional Deployment Information. Data: bulk product load, product classes, selling models, price book entries, discount schedules, association to categories, catalogs	In the target org: Align baseline metadata. Validation: Run automated regression tests on bundles and pricing. Validate sample SKUs across categories.

Table 2: Product Definition: Incremental Deployment Scenarios

Scenario	Impact	Validation and Dependencies
New Product Launch Introduce a new product (simple) of type one time, subscription, and usage with standard pricing.	Metadata: product record type, updated page layouts, product catalog management, pricing, usage, and rating metadata Data: product records, product classifications, attributes, attribute picklists, price book entries, rate card entries, subscription term details	In the target org, a price book and rate card must exist, and billing rules are configured. Validation: Create a test quote with products, confirm price flows into quote or order, validate subscription billing cycle, validate rates.
Bundle Creation Launch a new product bundle that includes core products, add-ons, and attribute configurations with standard pricing.	Metadata: none Data: bundle definition, child products, option groups, option attribute configurations, bundle pricing	In the target org, attributes and picklists must be created or updated, child products must be created or updated, and a price book must exist. Validation: Configure a bundle in a quote, test configuration options, and ensure correct pricing roll up.
Update Product Classification Properties Add new attributes and update existing attribute properties to existing product classification.	Metadata: none Data: new attributes, new and updated product class attribute assignments.	In the target org, attributes and picklists must be created or updated first. Validation: Generate a quote with updated product, and validate that product attributes appear correctly.

Scenario	Impact	Validation and Dependencies
Update Product Classification Hierarchy Add a new sub-classification, configure attributes, and create a product by using the sub-class.	Metadata: none Data: new class, class relationship, and class attribute assignments	In the target org, attributes and picklists must be created or updated first. Validation: Generate a quote with a new product. Validate product attributes correctly appear.
Product Retirement Deactivate legacy products	Metadata: none Data: Set product status to Inactive. Remove the product from active price books.	Active quotes with retired products must be closed or replaced. Validation: Attempt to add a retired product to the quote and confirm it's blocked. Make sure that reporting reflects retirement.
Update Product Discovery Setup Change qualification procedure, product discovery flow, and turn on Indexed Product feature.	Metadata: See Appendix C Data: Index configuration (searchable, filterable, languages)	In target org, make sure that baseline metadata is created. Run full indexing in the target org before you enable the feature. Validation: Validate the product discovery setup changes and product indexing process status. Verify by discovering products with term-based and faceted search.

Table 3: Catalog, Categories & Related Products: Full Deployment Scenarios

Scenario	Impact	Validation and Dependencies
New Catalog Creation Introduce a new catalog for a business line. For example, Cloud Services.	Metadata: Catalog object configuration, and Salesforce Configure, Price, Quote (CPQ) page layouts Data: Create catalog records and linked products	Products must exist before adding to the catalog Validation: Validate that the catalog is visible in quotes and orders. Confirm that all linked products appear.
Large Catalog Expansion Add 500+ new SKUs across multiple categories.	Metadata: Update catalog metadata if new categories are required. Data: Bulk product-category relationships, and new catalog entries.	In the target org, make sure that baseline metadata is created. Validation: Confirm that sample SKUs appear in the correct categories. Perform a regression test of the catalog navigation.

Table 4: Catalog, Categories & Related Products: Incremental Deployment Scenarios

Scenario	Impact	Validation and Dependencies
Add New Category	Metadata: Update category hierarchy metadata	Parent catalog must exist in the target org.

Scenario	Impact	Validation and Dependencies
Add the Security Services category under the existing catalog.	Data: Create category record, and link products	Validation: Confirm that products appear under the new category in the catalog browser.
Category Reorganization Move products from the Hardware category to the Accessories category.	Metadata: Update category hierarchy and associations. Data: Update category-to-product mappings.	Validation: Validate that moved products appear only in the new category and that the old category is clean.
Product Retirement from Category Remove retired products from catalog and related category.	Metadata: Update product visibility settings. Data: Update product-category relationships.	Validation: Validate that the products aren't visible in the category browser or guided selling.
Seasonal Category Introduce a temporary Holiday Deals category.	Metadata: Configure the category (and time-based visibility if applicable). Data: Establish product-category relationships with seasonal tag.	Products must exist and be tagged for promotion. Validation: Confirm that the category is visible during the correct time period only. Verify that the correct products are included.

Table 5: Qualification Rules: Full Deployment Scenarios

Scenario	Impact	Validation and Dependencies
Large-Scale Decision Table Migration Introduce 100+ new eligibility conditions, such as new geographies and pricing models.	Metadata: Qualification Rule framework and Expression Set alignment Data: Insert decision tables in bulk.	Perform target org metadata baseline alignment and staging of decision tables for verification. Validation: Regression test eligibility across multiple conditions and customer profiles.
Composite Qualification Rule Combine multiple decision tables and expression sets into a new guided selling path.	Metadata: Create parent qualification rule, and update multiple expression sets. Data: Populate multiple decision tables with new mappings.	All child rules and data mappings must exist in the target org. Validation: Test end-to-end guided selling to confirm the composite flow works correctly.

Table 6: Qualification Rules: Incremental Deployment Scenarios

Scenario	Impact	Validation and Dependencies
New Qualification Rule with Decision Table Create a qualification rule to enforce eligibility for product bundles based on region and customer tier.	Metadata: New qualification rule metadata, and context definition setup	Product catalog and customer tiers must be available in the target org.

Scenario	Impact	Validation and Dependencies
	Data: Populate decision table entries with region and tier mappings	Validation: Create test quotes for each region and tier. Confirm that only eligible products appear.
Update Existing Decision Table Entries Add new conditions to an existing qualification rule such as adding a new country to eligibility.	Metadata: none Data: Update and add rows in decision tables.	Make sure that new values are aligned with the active product catalog. Validation: Confirm that the rule returns the correct product eligibility for the new condition.
Modify Expression Set Change logical criteria from OR to AND in the qualification logic. For example, the customer must be an enterprise and have a support contract.	Metadata: Update expression set definition Data: none	Make sure that the dependent context variable exists. Validation: Confirm both the positive and negative test cases (eligible vs. non-eligible customers).
Context Definition Expansion Add a new attribute to the context (for example, Industry) for use in qualification rules.	Metadata: Add field reference in context definition Data: None (until data is mapped into a decision table)	Ensure that the Industry field exists and is populated. Validation: Test that qualification rules can read Industry context during quote creation.
Retirement of Qualification Rule Deactivate old eligibility rule no longer needed such as a legacy discount program.	Metadata: Update rule status to inactive, and remove references from related expression set. Data: Remove or disable related decision table entries.	Make sure that the rule isn't referenced by other bundles or flows. Validation: Confirm that the retired rule no longer impacts guided selling or product selection.

Salesforce Pricing

Table 7: Pricing: Incremental Deployment Scenarios

Scenario	Impact	Validation and Dependencies
Pricing Update Update base license cost by +5% for existing products.	Metadata: None (if there are no new rules) Data: Update price book entries and adjust discount tiers.	Make sure that no dependent contracts reference the old pricing. Validation: Generate a quote with the updated product. Confirm that pricing reflects the new value. Confirm no impact to active contracts.
Promotional Discount Introduce a three-month promotional discount for enterprise customers.	Metadata: New time-based pricing rule and approval process update.	Approval workflow for discount must be active

Scenario	Impact	Validation and Dependencies
	Data: Create a discount schedule and promo product entry.	Validation: Confirm that the discount applies only during the promotion period and verify that approvals trigger correctly.

Object Deployment Reference

Get to know the object deployment sequence and associated properties.

Product Catalog Management Objects

This table provides the deployment sequence, object types, API names, lookup fields, and data translation requirements for Product Catalog Management objects in Revenue Cloud.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Note: Internal objects aren't accessible.

Object Use Type	Object Name	Object API	Deployment Sequence	Lookup Fields (Foreign Keys)
Metadata	Product Specification Type	ProductSpecificationType	1	User
Metadata	Product Specification Record Type	ProductSpecification Record Type	2	Product Specification Type
Configuration	Attribute Picklist	AttributePickist	3	User, User Group, Unit of Measure
Configuration	Attribute Picklist Value Translation table: Attribute Picklist Value Data Translation	AttributePicklistValue Translation table: AttributePicklist ValueDataTranslation	4 46	User, Attribute Picklist (Master-Detail)
Configuration	Unit of Measure Class	UnitOfMeasureClass	5	User, Unit of Measure
Configuration	Unit of Measure	UnitOfMeasure	6	User Group, Unit of Measure Class
Configuration	Attribute Definition Translation table: Attribute Definition Data Translation	AttributeDefinition Translation table: AttributeDefinitionDataTranslation	7 47	User, Attribute Picklist, Unit of Measure
Configuration	Attribute Category Translation table: Attribute Category Data Translation	AttributeCategory Translation table: AttributeCategoryDataTranslation	8 48	User Group
Configuration	Attribute Category Attribute	AttributeCategoryAttribute	9	User Group, Attribute Category, Attribute Definition
Configuration	Product Classification Translation table: Product Classification Data Translation	ProductClassification Translation table: ProductClassificationData Translation	10 49	User, User Group, Product Classification
Configuration	Product Classification Attribute Translation table: Product Classification Attribute Data Translation	ProductClassificationAttr Translation table: ProductClassificationAttrData Translation	11 50	Attribute Definition, Attribute Category, Product Classification Attribute, Product Classification Attribute, User, User Group, Unit of Measure
Configuration	Tax Policy	TaxPolicy	12	User, Tax Treatment
Configuration	Product Translation table: Product2 Data Translation	Product2 Translation table: Product2DataTranslation	13 51	Product Classification, Billing Policy, User, External Data Source, Tax Policy, Unit of Measure

Object Use Type	Object Name	Object API	Deployment Sequence	Lookup Fields (Foreign Keys)
Configuration	Tax Engine	TaxEngine	14	Named Credential, Tax Engine Provider
Configuration	Tax Treatment	TaxTreatment	15	Legal Entity, Product, Tax Policy, Tax Engine
Configuration	Product Attribute Definition	ProductAttributeDefinition	16	Attribute Definition, Attribute Category, User, Product Attribute Definition, Product Classification Attribute, Unit of Measure
Configuration	Attribute Picklist Excluded Value	AttrPicklistExcludedValue	17	Product Classification Attribute, Product Attribute Definition (polymorphic), Attribute Picklist Value
Configuration	Product Attribute Scope	ProdtAttrScope	18	User, User Group
Configuration	Product Attribute Mapped Scope	ProdtAttrMappedScope	19	Product Classification Attribute, Product Attribute Definition (polymorphic), Product Attribute Mapped Scope, Product Attribute Scope
Configuration	Product Selling Model	ProductSellingModel	20	User
Configuration	Product Selling Model Option	ProductSellingModelOption	21	User, Product Selling Model, Proration Policy
Configuration	Product Ramp Segment	ProductRampSegment	22	User, Product Selling Model, Product
Configuration	Product Relationship Type	ProductRelationshipType	23	User
Configuration	Product Component Group	ProductComponentGroup	24	User, User Group, Product Component Group
Configuration	Product Related Component	ProductRelatedComponent	25	User, Product, Product Classification, Product Selling Model, Product Component Group, Product Relationship Type, Unit of Measure
Configuration	Product Related Group Override	ProductComponentGrpOverride	26	User, User Group, Product, Product Component Group
Configuration	Product Related Component Override	ProductRelComponentOverride	27	UserGroup, Product, Product Related Component, Unit of Measure
Configuration	Catalog	ProductCatalog	28	User, User Group

Object Use Type	Object Name	Object API	Deployment Sequence	Lookup Fields (Foreign Keys)
	Translation table: Product Catalog Data Translation	Translation table: ProductCatalogDataTranslation	52	
Configuration	Category Translation table: Product Category Data Translation	ProductCategory Translation table: ProductCategoryDataTranslation	29 53	User, Catalog (Master-Detail)
Configuration	Product Category Product	ProductCategoryProduct	30	Product, Category (Master-Detail)
Configuration	Product Qualification	ProductQualification	31	User, User Group, Product
Configuration	Product Disqualification	ProductDisqualification	32	User, User Group, Product
Configuration	Product Category Qualification	ProductCategoryQualification	33	User, User Group, Category
Configuration	Product Category Disqualification	ProductCategoryDisqual	34	User, User Group, Category
Configuration	Runtime Catalog Index Settings (Internal)	RuntimeCatalogIndexSetting	35	
Configuration	WebStore Search Attr Settings (Internal)	WebStoreSearchAttrSettings	36	
Configuration	Assessment Question	AssessmentQuestion	37	Assessment Question Version, Assessment Question, User, User Group
Configuration	Assessment Question Version	AssessmentQuestionVersion	38	Assessment Question (Master-Detail)
Configuration	Assessment	Assessment	39	Account, Contact, User, User Group, Omni Process, Assessment
Configuration	Assessment Question Response	AssessmentQuestionResponse	40	Assessment Question Version, Assessment (Master-Detail), User, User Group
Configuration	Omni Process	OmniProcess	41	User, User Group
Configuration	Omni Process Element	OmniProcessElement	42	Omni Process (Master-Detail), Omni Process Element
Configuration	OmniProcess Assessment Question Version	OmniProcessAsmtQuestionVer	43	Assessment Question Version, Omni Process, Omni Process Element, User Group
Configuration	Assessment Question Set	AssessmentQuestionSet	44	User, User Group

Object Use Type	Object Name	Object API	Deployment Sequence	Lookup Fields (Foreign Keys)
Configuration	Assessment Question Assignment	AssessmentQuestionAssignment	45	Assessment Question Set, User, User Group

SEE ALSO:

Revenue Cloud Developer Guide: Product Catalog Management Standard Objects

Explore the Revenue Cloud Data Model

Salesforce Pricing Objects

This table provides the deployment sequence, object types, API names, lookup fields, and data translation requirements for Salesforce Pricing objects in Revenue Management.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Note: Internal objects aren't accessible.

Object Use Type	Object Name	Object API	Deployment Sequence	Lookup Fields (Foreign Keys)
Configuration	Product Selling Model Translation table: Product Selling Model Data Translation	ProductSellingModel Translation table: ProductSelling ModelDataTranslation	1	
Configuration	Product Selling Model Option Translation table: Product Selling Model Option Data Translation	ProductSellingModelOption Translation table: ProductSelling ModelOptionDataTranslation	2	ProductSellingModel (Master-Detail), Product2 (Foreign key), ProrationPolicy (Foreign key)
Configuration	Price Book	Pricebook2	3	Pricebook2 (Foreign Key)
Configuration	Cost Book	CostBook	4	
Configuration	Price Book Entry	PriceBookEntry	5	Pricebook2 (Foreign Key), Product2 (Foreign key), ProductSellingModel (Foreign Key)
Configuration	Cost Book Entry	CostBookEntry	6	CostBook (Master-Detail), Product (Foreign Key)
Configuration	Price Adjustment Schedule	PriceAdjustmentSchedule	7	Pricebook2 (Foreign Key), Contract (Foreign Key)

Object Use Type	Object Name	Object API	Deployment Sequence	Lookup Fields (Foreign Keys)
Configuration	Price Adjustment Tier	Price Adjustment Tier	8	PriceAdjustmentSchedule (Master-Detail), ProductSellingModel (Foreign Key), Product2 (Foreign key)
Configuration	Price Book Entry Derived Price	PriceBookEntryDerivedPrice	9	Product2 (Foreign Key), PricebookEntry (Foreign Key), Pricebook2 (Foreign Key), ProductSellingModel (Foreign Key)
Configuration	Bundle Based Adjustment	BundleBasedAdjustment	10	PriceAdjustmentSchedule (Master-Detail), Product2 (Foreign key), ProductSellingModel (Foreign Key)
Configuration	Attribute Based Adjustment Rule	AttributeBasedAdjRule	11	
Configuration	Attribute Adjustment Condition	AttributeAdjustmentCondition	12	AttributeBasedAdjRule (Master-Detail), AttributeDefinition (Foreign Key), Product2 (Foreign Key)
Configuration	Attribute Based Adjustment	AttributeBasedAdjustment	13	PriceAdjustmentSchedule (Master-Detail), ProductSellingModel (Foreign Key), AttributeBasedAdjRule (Foreign Key), Product2 (Foreign key)
Configuration	Index rate (extended from Financial Services Cloud)	IndexRate	30	
Metadata	Price Book Price Guidance	PriceBookPriceGuidance	35	PricebookEntry (Foreign Key), Pricebook (Foreign Key), Product (Foreign Key), ProductSellingModel (Foreign Key)
Metadata	Pricing Procedure Resolution	PricingProcedureResolution	40	ExpressionSet (Foreign Key)
Configuration	Pricing Procedure Output Map	PricingProcedureOutputMap	40	PricingRecipeTableMapping (Foreign Key), OutputFieldName (Foreign Key)
Metadata	Pricing Recipe	PricingRecipe	50	ExpressionSetDefinition (Foreign Key)
Configuration	Proration Policy	ProrationPolicy	50	
Configuration	Product Price Range	ProductPriceRange	90	Pricebook2 (Foreign Key)

Object Use Type	Object Name	Object API	Deployment Sequence	Lookup Fields (Foreign Keys)
Configuration	Price Revision Policy	PriceRevisionPolicy		
Metadata	Pricing Recipe Table Mapping (Internal)	PricingRecipeTableMapping		PricingRecipe (Master-Detail), LookupTable (Foreign Key)
Configuration	Product Price History Log (Internal)	ProductPriceHistoryLog		ProductPriceRange (Master-Detail)
Configuration	Pricing Adjustment Batch Job (Internal)	PricingAdjBatchJob		
Configuration	Pricing Adjustment Batch Job Log (Internal)	PricingAdjBatchJobLog		PricingAdjBatchJob (Master-Detail)
Configuration	Procedure Plan Definition (Internal)	ProcedurePlanDefinition		
Configuration	Procedure Plan Definition Version (Internal)	ProcedurePlanDefinitionVersion		ProcedurePlanDefinition (Master-Detail)
Configuration	Procedure Plan Criterion (Internal)	ProcedurePlanCriterion		ProcedurePlanOption (Master-Detail), DecisionTableParameter (Foreign Key)
Configuration	Procedure Plan Option (Internal)	ProcedurePlanOption		ProcedurePlanSection (Master-Detail), ExpressionSetDefinition (Foreign Key), DecisionTable (Foreign Key), DecisionTableParameter (Foreign Key), DecisionTableParameter (Foreign Key), DecisionTableParameter (Foreign Key), ApexClass (Foreign Key)
Configuration	Procedure Plan Variable (Internal)	ProcedurePlanVariable		ProcedurePlanDefinitionVersion (Master-Detail)
Configuration	Procedure Plan Section (Internal)	ProcedurePlanSection		ProcedurePlanDefinitionVersion (Master-Detail)

SEE ALSO:

Revenue Cloud Developer Guide: Salesforce Pricing Standard Objects

Explore the Revenue Cloud Data Model

Product Configurator Objects

This table provides the object deployment sequence and properties for Product Configurator in Revenue Management, including object types, API names, deployment sequences, and lookup fields.

Object Use Type	Object Name	Object API	Deployment Sequence	Lookup Fields (Foreign Keys)
Configuration	Product Configuration Rule	ProductConfigurationRule	1	User
Configuration	Product Configuration Flow	ProductConfigurationFlow	1	UserFlowIdentifier
Configuration	ExpressionSetConstraintObj	ExpressionSetConstraintObj	1	ExpressionSetId, ReferenceObjectId (Polymorphic)
Configuration	Product Configuration Flow Assignment	ProductConfigFlowAssignment	2	User, ProductId, ProductClassificationId, ProductConfigurationFlow

Transaction Management Objects

This table provides the deployment sequence, object types, and API names for Transaction Management objects in Revenue Management.

Object Use Type	Object Name	Object API	Deployment Sequence	Lookup Fields (Foreign Keys)
Metadata	App Usage Assignment	AppUsageAssignment	1	Order, Quote, Contract, Asset
Metadata	Sales Transaction Type	SalesTransactionType	1	PricingProcedure
Metadata	Quote Template Rich Text Data	QuoteTemplateRichTextData	1	None
Metadata	Transaction Processing Type	TransactionProcessingType	1	None

SEE ALSO:

Revenue Cloud Developer Guide: Transaction Management Standard Objects

Explore the Revenue Cloud Data Model

Dynamic Revenue Orchestrator Objects

This table provides the deployment sequence, object types, API names, and lookup fields for Dynamic Revenue Orchestrator objects in Revenue Management.

Object Use Type	Object Name	Object API	Deployment Sequence	Lookup Fields (Foreign Keys)
Configuration	Fulfillment Step Definition Group	FulfillmentStepDefinitionGroup	1	None
Configuration	Fulfillment Step Definition	FulfillmentStepDefinition	2	Ruleset, ExpressionSet, FulfillmentStepDefinitionGroup, IntegrationProviderDef, User, Queue

Object Use Type	Object Name	Object API	Deployment Sequence	Lookup Fields (Foreign Keys)
Configuration	Fulfillment Step Dependency Definition	FulfillmentStepDependencyDef	3	FulfillmentStepDefinition
Configuration	Product Fulfillment Scenario	ProductFulfillmentScenario	4	FulfillmentStepDefinitionGroup, Ruleset, Product2, ProductClassification, FlowDefinition, StageDefinition, FlowRecord, FlowOrchestration
Configuration	Fulfillment Workspace	FulfillmentWorkspace	5	None
Configuration	Fulfillment Workspace Item	FulfillmentWorkspaceItem	6	FulfillmentWorkspace, FulfillmentStepDefinitionGroup
Configuration	Fulfillment Fallout Rule	FulfillmentFalloutRule	7	IntegrationProviderDef, Group
Configuration	Fulfillment Step Jeopardy Rule	FulfillmentStepJeopardyRule	8	IntegrationProviderDef
Configuration	Fulfillment Task Assignment Rule	FulfillmentTaskAssignmentRule	9	Ruleset, ExpressionSet, User, Queue
Configuration	Product Fulfillment Decomposition Rule	ProductFulfillmentDecompRule	1	Ruleset, Product2, ProductClassification
Configuration	Value Transformation Group	ValTfrmGrp	2	None
Configuration	Value Transformation	ValTfrm	3	ValTfrmGrp, AttributePicklistValue
Configuration	Product Decomposition Enrichment Rule	ProductDecompEnrichmentRule	4	ProductFulfillmentDecompRule, ExpressionSet, AttributeDefinition, ValTfrmGrp, DecisionMatrixDefinition
Configuration	Product Decomposition Enrichment Variable Mapping	ProdtDecompEnrchVarMap	5	ProductDecompEnrichmentRule, AttributeDefinition

SEE ALSO:

Revenue Cloud Developer Guide: Dynamic Revenue Orchestrator Standard Objects

Explore the Revenue Cloud Data Model

Usage Management Objects

This table provides the deployment sequence, object types, API names, and lookup fields for Usage Management objects in Revenue Management.

Object Use Type	Object Name	Object API	Deployment Sequence	Lookup Fields (Foreign Keys)
Configuration	Usage Aggregation Policy	UsageResourceBillingPolicy	1	None
Configuration	Usage Grant Rollover Policy	UsageGrantRolloverPolicy	2	None
Configuration	Usage Grant Renewal Policy	UsageGrantRenewalPolicy	3	None
Configuration	Usage Overage Policy	UsageOveragePolicy	4	None
Configuration	Usage Commitment Policy	UsageCommitmentPolicy	5	None
Configuration	Rate Card	RateCard	6	None
Configuration	Usage Resource	UsageResource	7	UnitOfMeasure, UnitOfMeasureClass, Product2, UsageResourceBillingPolicy
Configuration	Price Book Rate Card	PriceBookRateCard	8	PriceBook2, RateCard
Configuration	Rating Frequency Policy	RatingFrequencyPolicy	9	Product2, UsageResource
Configuration	Product Usage Resource	ProductUsageResource	10	Product2, UsageResource
Configuration	Rate Card Entry	RateCardEntry	11	RateCard, UnitOfMeasure, UnitOfMeasureClass, UsageResource, Product2, ProductSellingModel
Configuration	Usage Resource Policy	UsageResourcePolicy	12	UsageResource, UsageOveragePolicy, RatingFrequencyPolicy, UsageResourceBillingPolicy, UsageCommitmentPolicy
Configuration	Product Usage Grant	ProductUsageGrant	13	ProductUsageResource, UnitOfMeasure, UnitOfMeasureClass, UsageGrantRolloverPolicy, UsageGrantRenewalPolicy, Product2, ProductSellingModel
Configuration	Product Usage Resource Policy	ProductUsageResourcePolicy	14	ProductUsageResource, UsageOveragePolicy, RatingFrequencyPolicy, UsageResourceBillingPolicy, UsageCommitmentPolicy, ProductSellingModel
Configuration	Rate Adjustment By Tier	RateAdjustmentByTier	15	RateCardEntry

Object Use Type	Object Name	Object API	Deployment Sequence	Lookup Fields (Foreign Keys)
Configuration	Rate Adjustment By Attribute	RateAdjustmentByAttribute	16	RateCardEntry, AttributeBasedAdjRule

SEE ALSO:

Revenue Cloud Developer Guide: Usage Management Standard Objects

Explore the Revenue Cloud Data Model

Billing Objects

This table provides the deployment sequence, object types, API names, and lookup fields for Billing objects in Revenue Management.

Object Use Type	Object Name	Object API	Deployment Sequence	Lookup Fields (Foreign Keys)
Configuration	Profile	Profile	1	None
Configuration	User	User	2	None
Configuration	User Role	UserRole	3	None
Configuration	Legal Entity	LegalEntity	4	None
Configuration	Billing Policy	BillingPolicy	5	None
Configuration	Billing Treatment	BillingTreatment	6	LegalEntity, BillingPolicy
Configuration	Billing Treatment Item	BillingTreatmentItem	7	BillingTreatment
Configuration	Tax Engine Provider	TaxEngineProvider	8	ApexAdapter
Configuration	Tax Engine	TaxEngine	9	NamedCredential, TaxEngineProvider
Configuration	Tax Policy	TaxPolicy	10	None
Configuration	Tax Treatment	TaxTreatment	11	TaxEngine, TaxPolicy
Configuration	Tax Treatment Item	TaxTreatmentItem	12	Product2
Configuration	Payment Term	PaymentTerm	13	None
Configuration	Payment Term Item	PaymentTermItem	14	PaymentTerm
Configuration	Accounting Period	AccountingPeriod	15	None
Configuration	Legal Entity Accounting Period	LegalEntityAccountingPeriod	16	LegalEntity, AccountingPeriod
Configuration	General Ledger Account	GeneralLedgerAccount	17	LegalEntity
Configuration	General Ledger Account Assignment Rules	GeneralLedgerAcctAsgntRule	18	LegalEntity, GeneralLedgerAccount

Object Use Type	Object Name	Object API	Deployment Sequence	Lookup Fields (Foreign Keys)
Configuration	Payment Schedule Policy	PaymentSchedulePolicy	19	None
Configuration	Payment Schedule Treatment	PaymentScheduleTreatment	20	PaymentSchedulePolicy
Configuration	Payment Schedule Treatment Detail	PaymentScheduleTreatmentDtl	21	PaymentScheduleTreatment, PymtSchdDistributionMethod
Configuration	Payment Schedule Distribution Method	PymtSchdDistributionMethod	22	None
Configuration	Billing Milestone Plan	BillingMilestonePlan	23	BillingTreatment
Configuration	Billing Milestone Plan Item	BillingMilestonePlanItem	24	BillingMilestonePlan
Configuration	General Ledger Journal Entry Rule	GeneralLedgerJrnlEntryRule	25	GeneralLedgerAccount, GeneralLedgerAcctAsgntRule
Configuration	Payment Retry Rule	PaymentRetryRule	26	PaymentGateway
Configuration	Payment Retry Rule Set	PaymentRetryRuleSet	27	None
Configuration	Billing Arrangement	BillingArrangement	28	None
Configuration	Billing Arrangement Line	BillingArrangementLine	29	Account, BillingAccount

SEE ALSO:

Revenue Cloud Developer Guide: Billing Standard Objects

Explore the Revenue Cloud Data Model

Salesforce Contracts Objects

This table provides the deployment sequence, object types, API names, and lookup fields for Salesforce Contracts in Revenue Management.

Object Use Type	Object Name	Object API	Deployment Sequence	Lookup Fields (Foreign Keys)
Metadata	Clause Category Configuration	ClauseCatgConfiguration	1	None
Configuration	Document Clause Set	DocumentClauseSet	2	ClauseCatgConfiguration
Configuration	Document Clause	DocumentClause	3	DocumentClauseSet, ContentDocument

Metadata Deployment Reference

Get to know the metadata deployment sequence and associated details.

Product Catalog Management Metadata

This table provides the metadata deployment reference for Product Catalog Management in Revenue Management, including setup paths and configuration details.

Type	Label	Setup Path	Details
Setup	Omnistudio Settings	Setup > Apps > Packaging > Installed Packages	Prerequisite: Omnistudio Managed package must be installed.
Flag	Managed Package Runtime	Setup > Feature Settings > Omni interaction > Omnistudio Settings	Disable
Flag	Define Custom Lightning Web Components in Standard Runtime	Setup > Feature Settings > Omni interaction > Omnistudio Settings	Enable

Type	Label	Setup Path	Details
Setup	Discovery Framework	Setup > Discovery Framework	Prerequisite: Enable guided product selection through this path. Setup> Product Discovery > Product Discovery Settings- > Guided Product Selection
Flag	Discovery Framework	Setup > Discovery Framework > General Settings	Enable
Flag	Import & Export	Setup > Discovery Framework > General Settings	Enable
Flag	Sample Templates	Setup > Discovery Framework > General Settings	Enable. Also, refer to these steps. • In the **Sample Templates** section, click the Discovery Framework Sample Template page link. The Discovery Framework Sample Templates page appears. • Click **Deploy** if not already deployed.
Setup	Procedure Plan Definition	Procedure Plan Setup > Procedure Plan Definitions	
Setup	Product Discovery Settings	Setup> Product Discovery > Product Discovery Settings	
Field	Select Context Definition	Setup> Product Discovery > Product Discovery Settings	
Field + Flag	Select Pricing Procedure	Setup> Product Discovery > Product Discovery Settings	
Field + Flag	Select Qualification Procedure	Setup> Product Discovery > Product Discovery Settings	
Field	Select a Custom Flow for Browsing and Adding Products in Revenue Cloud	Setup> Product Discovery > Product Discovery Settings	
Field	Select Default Catalog	Setup> Product Discovery > Product Discovery Settings	
Flag	Product Field Search	Setup> Product Discovery > Product Discovery Settings	
Flag	Use Indexed Data for Product Listing and Search	Setup> Product Discovery > Product Discovery Settings	

Type	Label	Setup Path	Details
Flag	Dynamic Product Facets	Setup> Product Discovery > Product Discovery Settings	
Flag	Semantic Search	Setup> Product Discovery > Product Discovery Settings	
Flag	Guided Product Selection	Setup> Product Discovery > Product Discovery Settings	
Flag	Generate Product Descriptions with Einstein AI	Setup> Product Discovery > Product Discovery Settings	
Flag	Product Catalog Management Cache	Setup> Product Discovery > Product Discovery Settings	
Permission Sets	Product Catalog Management Customer Community User	Setup > Users > Permission Sets	
Permission Sets	Product Catalog Management Partner Community User	Setup > Users > Permission Sets	
Permission Sets	Product Catalog Management Cache	Setup > Users > Permission Sets	
Permission Sets	Product Discovery Admin	Setup > Users > Permission Sets	
Permission Sets	Product Discovery User	Setup > Users > Permission Sets	
Permission Sets	ProductImport API	Setup > Users > Permission Sets	
Permission Sets	DecimalQuantityDesigntime	Setup > Users > Permission Sets	
Permission Sets	DecimalQuantityRuntime	Setup > Users > Permission Sets	
Permission Sets	Advanced CSV Data Import	Setup > Users > Permission Sets	Data Processing Engine (DPE) is included. Data Cloud is licensed and deployed separately.
Permission Sets	Basic CSV Data Import	Setup > Users > Permission Sets	
Permission Sets	Context Service Admin	Setup > Users > Permission Sets	
Permission Sets	Context Service Runtime	Setup > Users > Permission Sets	
Permission Sets	Fulfillment Designer	Setup > Users > Permission Sets	
Permission Sets	Omnistudio Admin	Setup > Users > Permission Sets	
Permission Sets	Omnistudio User	Setup > Users > Permission Sets	
Flow	Discover Products	Setup > Flow	Supports Flow version, activation, and deactivation

Type	Label	Setup Path	Details
Omniscript	ProductGuidedSelectionIntegration	All Apps > Omnistudio	Supports Omniscript version, activation, and deactivation
Decision Table Definition	ProductQualificationDT	Setup > Decision Table	
Decision Table Definition	ProductDisqualificationQualificationDT	Setup > Decision Table	
Decision Table Definition	ProductCategoryQualificationDT	Setup > Decision Table	
Decision Table Definition	ProductCategory DisqualificationQualificationDT	Setup > Decision Table	

Salesforce Pricing Metadata

This table provides the metadata deployment reference for Salesforce Pricing in Revenue Cloud, including setup paths and configuration details.

Type	Label	Setup Path	Details
Setup	Context Service	Context Service > Context Service Settings	Enable. This is a prerequisite to Salesforce Pricing settings.
Setup	Salesforce Pricing Settings	Setup > Salesforce Pricing > Salesforce Pricing Settings	Enable
Setup	Pricing Recipes	Setup > Salesforce Pricing > Pricing Recipes	
Setup	Salesforce Pricing Setup	Setup > Salesforce Pricing > Salesforce Pricing Setup	
Field	Select a Pricing Recipe	Setup > Salesforce Pricing > Salesforce Pricing Setup	
Field	Select a Pricing Procedure	Setup > Salesforce Pricing > Salesforce Pricing Setup	
Flag	Sync Pricing Data	Setup > Salesforce Pricing > Salesforce Pricing Setup	Decision tables are refreshed.
Flag	Activate Price Waterfall for API Responses	Setup > Salesforce Pricing > Salesforce Pricing Setup	
Flag	Turn On Price Waterfall Persistence	Setup > Salesforce Pricing > Salesforce Pricing Setup	
Field	Price Tracking History	Setup > Salesforce Pricing > Salesforce Pricing Setup	Includes two fields of type flag. 1. Enable Maximum Price 2. Enable Minimum Price

Type	Label	Setup Path	Details
Field	Proration Settings	Setup > Salesforce Pricing > Salesforce Pricing Setup	Includes two fields of type field. 1. Evergreen 2. One Time
Flag	Turn On Price Logs Capture	Setup > Salesforce Pricing > Salesforce Pricing Setup	
Flag	Turn on Parallel Execution	Setup > Salesforce Pricing > Salesforce Pricing Setup	
Setup	Procedure Plan Definition	Procedure Plan Setup > Procedure Plan Definitions	For Procedure Plan definitions, if Apex is selected, the Apex class must be migrated. Packaging isn't supported in Winter' 26. See Customize Your Procedure Plans With Apex Hooks.
Permission Sets	Salesforce Pricing Admin	Setup > Users > Permission Sets	
Permission Sets	Salesforce Pricing Design Time User	Setup > Users > Permission Sets	
Permission Sets	Salesforce Pricing Manager	Setup > Users > Permission Sets	
Permission Sets	Salesforce Pricing Run Time User	Setup > Users > Permission Sets	
Decision Table Definition	Asset Action Source Entries	Setup > Decision Table	
Decision Table Definition	Asset Action Source Entries V2	Setup > Decision Table	
Decision Table Definition	Attribute Discount Entries	Setup > Decision Table	
Decision Table Definition	Bundle Based Adjustment Entries	Setup > Decision Table	
Decision Table Definition	Contract Pricing Adjustment Tiers	Setup > Decision Table	
Decision Table Definition	Contract Pricing Entries	Setup > Decision Table	
Decision Table Definition	Contract Pricing Volume Tiers	Setup > Decision Table	
Decision Table Definition	Contract Pricing Volume Tiers V2	Setup > Decision Table	

Type	Label	Setup Path	Details
Decision Table Definition	Derived Pricing Entries	Setup > Decision Table	
Decision Table Definition	Index Rate	Setup > Decision Table	
Decision Table Definition	Price Book Entries	Setup > Decision Table	
Decision Table Definition	Price Book Entries V2	Setup > Decision Table	
Decision Table Definition	Pricebook Rate Card Entries	Setup > Decision Table	
Decision Table Definition	Product Price Range Entries	Setup > Decision Table	
Decision Table Definition	Product Price Range Entries V2	Setup > Decision Table	
Decision Table Definition	Tiered Adjustment Entries	Setup > Decision Table	
Decision Table Definition	Volume Discount Entries	Setup > Decision Table	

Product Configurator Metadata

This table provides the metadata deployment reference for Product Configurator in Revenue Management, including setup paths and configuration details.

Type	Label	Setup Path	Comments
Setup	Context Service	Context Service > Context Service Settings	Enable (prerequisite to Product Configurator).
Setup	Configure Products at Runtime	Setup > Feature Settings > Revenue Cloud > Revenue Settings	
Setup	Set Up Configuration Rules with Business Rules Engine	Setup > Feature Settings > Revenue Cloud > Revenue Settings	Standard Configurator Prerequisite: Create a Rule Library Version (Set Up Configurator With Business Rules Engine)
Setup	Set Up Configuration Rules and Constraints with Constraints Engine	Setup > Feature Settings > Revenue Cloud > Revenue Settings	
Setup	Transaction processing for quotes and orders	Setup > Feature Settings > Revenue Cloud > Revenue Settings	If both Business Rules Engine and Constraint Builder are enabled in the org,

Type	Label	Setup Path	Comments
			ConstraintBuilder is used. Exception is Transaction Processing Type on Quotes and Orders override.
Setup	Set Up Asset Context for Product Configurator	Setup > Feature Settings > Revenue Cloud > Revenue Settings	
Custom Field	ConstraintEngineNodeStatus (Text Area (Long), length 5000)	Setup > Object Manager > Quote Line Item	Prerequisite for Constraint Builder.
Custom Field	ConstraintEngineNodeStatus (Text Area (Long), length 5000)	Setup > Object Manager > Order Product	Prerequisite for Constraint Builder.
Custom Field	ConstraintEngineNodeStatus (Text Area (Long), length 5000)	Setup > Object Manager > Asset Action Source	Prerequisite for Constraint Builder.
Context Mapping	ConstraintEngineNodeStatus	Setup > Feature Settings > Context Definitions > [Context]	Map the three custom fields, which are mentioned in the previous rows, with the ConstraintEngineNodeStatus tag.
Permission Set	Product Configurator	Setup > Users > Permission Sets	
Permission Set	Product Configuration Rules Designer	Setup > Users > Permission Sets	Prerequisite for Business Rules Engine Configurator.
Permission Set	Product Configuration Constraints Designer	Setup > Users > Permission Sets	Prerequisite for Constraint Builder.
Flow	Default Product Configurator Flow	Setup > Flow	Supports Flow version, and activation or deactivation. Other Flows can be created.

Transaction Management Metadata

This table provides the metadata deployment reference for Transaction Management in Revenue Management, including setup paths and configuration details.

Label	Setup Path
App Usage Type	
Transaction Processing Type	
RevenueManagement	Setup > Revenue Cloud > Revenue Settings
IncludeEstTaxInQuoteCPQ	Setup > Revenue Cloud > Revenue Settings
Quote	Setup > Feature Settings > Sales > Quotes > Quote Settings
Order	Setup > Feature Settings > Sales > Order Settings

Label	Setup Path
LargeQuotesandOrdersForRlm	Setup > Revenue Cloud > Revenue Settings
User UI Preference	

Dynamic Revenue Orchestrator Metadata

This table provides the metadata deployment reference for Dynamic Revenue Orchestrator (DRO) in Revenue Management, including setup paths and configuration details.

Type	Label	Setup Path	Details
Setup	Dynamic Revenue Orchestrator	Setup > Feature Settings > Dynamic Revenue Orchestrator	
Flag	Dynamic Revenue Orchestrator	Setup > Feature Settings > Dynamic Revenue Orchestrator > Dynamic Revenue Orchestrator Settings	
Flag	In-flight Amendments	Setup > Feature Settings > Dynamic Revenue Orchestrator > Dynamic Revenue Orchestrator Settings	
Flag	Future Dated Steps	Setup > Feature Settings > Dynamic Revenue Orchestrator > Dynamic Revenue Orchestrator Settings	
Flag	Link Task to Step Source	Setup > Feature Settings > Dynamic Revenue Orchestrator > Dynamic Revenue Orchestrator Settings	
Field	Fulfillment User	Setup > Feature Settings > Dynamic Revenue Orchestrator > Dynamic Revenue Orchestrator Settings	
Field	Context Definition	Setup > Feature Settings > Dynamic Revenue Orchestrator > Context Definition Settings (Sales Transaction Context Definition)	
Field	Context Node for Sales Transaction Header	Setup > Feature Settings > Dynamic Revenue Orchestrator > Context Definition Settings (Sales Transaction Context Definition)	
Field	Context field for Orchestration Group Key (Optional)	Setup > Feature Settings > Dynamic Revenue Orchestrator > Context Definition Settings (Sales Transaction Context Definition)	
Field	Context Node for Sales Transaction Item	Setup > Feature Settings > Dynamic Revenue Orchestrator > Context Definition Settings (Sales Transaction Context Definition)	
Field	Context Node for Sales Transaction Item Relationship	Setup > Feature Settings > Dynamic Revenue Orchestrator > Context Definition Settings (Sales Transaction Context Definition)	
Field	Context Node for Sales Transaction Item Attribute	Setup > Feature Settings > Dynamic Revenue Orchestrator > Context Definition Settings (Sales Transaction Context Definition)	
Field	Context Node for Fulfillment Transaction	Setup > Feature Settings > Dynamic Revenue Orchestrator > Context Definition Settings (Sales Transaction Context Definition)	
Field	Context Node for Fulfillment Transaction Item	Setup > Feature Settings > Dynamic Revenue Orchestrator > Context Definition Settings (Sales Transaction Context Definition)	

Type	Label	Setup Path	Details
Field	Context Node for Fulfillment Transaction Item Relationship	Setup > Feature Settings > Dynamic Revenue Orchestrator > Context Definition Settings (Sales Transaction Context Definition)	
Field	Context Node for Fulfillment Transaction Item Attribute	Setup > Feature Settings > Dynamic Revenue Orchestrator > Context Definition Settings (Sales Transaction Context Definition)	
Field	Context Node for Fulfillment Transaction Item Source Relationship	Setup > Feature Settings > Dynamic Revenue Orchestrator > Context Definition Settings (Sales Transaction Context Definition)	
Field	Context Definition	Setup > Feature Settings > Dynamic Revenue Orchestrator > Context Definition Settings (Fulfillment Asset Context Definition)	
Field	Context Node for Fulfillment Asset	Setup > Feature Settings > Dynamic Revenue Orchestrator > Context Definition Settings (Fulfillment Asset Context Definition)	
Field	Context Node for Fulfillment Asset Attribute	Setup > Feature Settings > Dynamic Revenue Orchestrator > Context Definition Settings (Fulfillment Asset Context Definition)	
Flag	Fallout	Setup > Feature Settings > Dynamic Revenue Orchestrator > Fallout and SLA Settings	
Flag	Service Level Agreement	Setup > Feature Settings > Dynamic Revenue Orchestrator > Fallout and SLA Settings	
Permission Sets	DRO Admin User	Setup > Users > Permission Sets	
Permission Sets	Submit Transactions and Fulfillment User	Setup > Users > Permission Sets	
Permission Sets	Fulfillment Designer	Setup > Users > Permission Sets	
Permission Sets	Fulfillment Manager/Operator	Setup > Users > Permission Sets	
Setup	Procedure Plan Definition	Procedure Plan Setup > Procedure Plan Definitions	The value of `UsageType` field is `Dfo`. This is used to define an alternate context mapping for sales transaction context in an order submission flow. See Submit Orders for Decomposition and Order Fulfillment
User Permission	Submit Transactions and Orchestrate User		Enables user to submit and orchestrate

Type	Label	Setup Path	Details
			transactions for any object by using Dynamic Revenue Orchestrator.

SEE ALSO:

Revenue Cloud Developer Guide: DynamicFulfillmentOrchestratorSettings

Usage Management Metadata

This table provides the metadata deployment reference for Usage Management in Revenue Cloud, including setup paths and configuration details.

Type	Label	Setup Path	Details
Setup	Context Service	Context Service > Context Service Settings	Enable
Setup	Rate Management	Setup > Usage Management > Usage Management Settings	Enable Rate Management.
Setup	Rating Setup	Setup > Usage Management > Rate Management Setup	Rating Waterfall -> Enable / Disable based on preference. Rating Waterfall Persistence -> Enable / Disable based on preference.
Permission Sets	Rate Management: Admin	Setup > Users > Permission Sets	
Permission Sets	Rate Management: Design Time User	Setup > Users > Permission Sets	
Permission Sets	Rate Management: Manager	Setup > Users > Permission Sets	
Permission Sets	Rate Management Run Time User	Setup > Users > Permission Sets	
Decision Table Definition	Binding Object Rate Adjustment Resolution Entries	Setup > Decision Tables	
Decision Table Definition	Binding Object Rate Card Entry Resolution Entries	Setup > Decision Tables	
Decision Table Definition	Rate Card Entry Resolution Entries 2	Setup > Decision Tables	

Type	Label	Setup Path	Details
Decision Table Definition	Rate Adjustment by Attribute Resolution Entries	Setup > Decision Tables	
Decision Table Definition	Rate Adjustment by Tier Resolution Entries	Setup > Decision Tables	
Decision Table Definition	Pricebook Rate Card Entries	Setup > Decision Tables	
Decision Table Definition	Binding Object Volume-based Rate Adjustment	Setup > Decision Tables	
Decision Table Definition	Binding Object Rate	Setup > Decision Tables	
Decision Table Definition	Binding Object Tier-based Rate Adjustment	Setup > Decision Tables	
Decision Table Definition	Binding Object Rate Card Entry	Setup > Decision Tables	
Decision Table Definition	Asset Volume-based Rate Adjustment	Setup > Decision Tables	
Decision Table Definition	Asset Rate	Setup > Decision Tables	
Decision Table Definition	Asset Rate Card Entry	Setup > Decision Tables	
Decision Table Definition	Asset Tier-based Rate Adjustment	Setup > Decision Tables	
Decision Table Definition	Attribute-based Rate Adjustment by Rate Card Entry ID	Setup > Decision Tables	
Decision Table Definition	Volume-based Rate Adjustment by Rate Card Entry ID	Setup > Decision Tables	
Decision Table Definition	Tier-based Rate Adjustment by Rate Card Entry ID	Setup > Decision Tables	
Decision Table Definition	Rate Adjustment by Attribute Entries 2	Setup > Decision Tables	
Decision Table Definition	Rate Adjustment by Tier Entries 2	Setup > Decision Tables	
Decision Table Definition	Rate Adjustment by Volume Entries 2	Setup > Decision Tables	
Decision Table Definition	Rate Card Entries 2	Setup > Decision Tables	
Flow	Call Rating Service	Setup > Flows	

Type	Label	Setup Path	Details
Flow	Call Entitlement Refresh Service	Setup > Flows	
Flow	Create Summary	Setup > Flows	
Flow	Generate Liable Summary	Setup > Flows	
Flow	Generate Usage Rateable Summary	Setup > Flows	
Flow	Generate Usage Summary	Setup > Flows	
Flow	Orchestrate Usage Management	Setup > Flows	
Data Processing Engine	Create Liable Summary Template	Setup > Data Processing Engine	
Data Processing Engine	Create Usage Summary Template	Setup > Data Processing Engine	

Billing Metadata

This table provides the metadata deployment reference for Billing in Revenue Management, including setup paths and configuration details.

Type	Label	Setup Path	Details
Setup	Context Service	Context Service > Context Service Settings	
Setup	Data Pipeline	Setup > Feature Settings > Analytics > Data Pipeline	
Permission Set Assignment	Billing Admin	Setup > Users > Permission Sets	
Setup	Enable Billing	Setup > Feature Settings > Billing > Billing Settings	
Setup	Document Generation		
Setup	Create Transaction Journals for Transactions	Setup > Feature Settings > Billing > Billing Settings	
Setup	Apply Credits to Posted Invoices	Setup > Feature Settings > Billing > Billing Settings	
Setup	Enable Foreign Exchange Transaction Journal		
Setup	Apply Credits to Posted Invoices	Setup > Feature Settings > Billing > Billing Settings	
Setup	Configure Email Delivery Settings	Setup > Feature Settings > Billing > Billing Settings	
Setup	Configure Gapless Sequential Numbering for Billing	Setup > Feature Settings > Billing > Billing Settings	

Type	Label	Setup Path	Details
Setup	Store Transaction Amounts in Corporate Currency	Setup > Feature Settings > Billing > Billing Settings	
Setup	Create Payment Schedules and Payment Schedule Items	Setup > Feature Settings > Billing > Billing Settings	
Setup	Share Payment Accounts	Setup > Feature Settings > Billing > Billing Settings	
Setup	Convert Negative Invoice Lines to Credit Memo Lines	Setup > Feature Settings > Billing > Billing Settings	
Setup	Credit and Payment Application Level	Setup > Feature Settings > Billing > Billing Settings	
Setup	Select Default Credit Memo Flow	Setup > Feature Settings > Billing > Billing Settings	
Permission Sets	Data Pipeline User	Setup > Users > Permission Sets	
Permission Sets	Billing Admin	Setup > Users > Permission Sets	
Permission Sets	Context Service Admin	Setup > Users > Permission Sets	
Permission Sets	Tableau Next Admin	Setup > Users > Permission Sets	For Analytics Dashboard Enabling
Permission Sets	Data Cloud Admin	Setup > Users > Permission Sets	For Analytics Dashboard Enabling
Permission Sets	BillingAdvanced PaymentAdministrator	Setup > Users > Permission Sets	
Permission Sets	RevenueLifecycle ManagementBilling CustomerService	Setup > Users > Permission Sets	
Permission Sets	BillingAdvancedPaymentOperations	Setup > Users > Permission Sets	
Permission Sets	RevenueLifecycle ManagementBillingCreditMemo Operations	Setup > Users > Permission Sets	
Setup	Default Invoice Template		
Setup	FX Gain/Loss Account Lookups		
Setup	Context Definition		
Setup	Context Mapping		
Setup	Intra Context Custom Mapping		

Type	Label	Setup Path	Details
Setup	Default DPE Definition to Close Legal Entity Accounting Periods		
Setup	Default Invoice Preview Template		
Setup	Default Invoice Email Template		
Setup	Default Credit Memo Flow		

Industries Common Component Metadata

This table provides the metadata deployment reference for Industries common components in Revenue Management, including setup paths and configuration details.

Type	Label	Setup Path	Details
Setup	Context Service	Context Service > Context Service Settings	Enable
Permission Sets	Permission Sets	Setup > Users > Permission Sets	Data Pipeline User
Permission Sets	Permission Sets	Setup > Users > Permission Sets	Data Cloud Admin
Data Processing Engine (DPE) Definition	datasources -> sourceName	v{}/tooling/sobjects/BatchCalcJobDefinition	Read permission to the user who is creating the Data Processing Engine (DPE) with the data sources and fields.
Data Processing Engine (DPE) Definition	datasources -> sourceName	v{}/tooling/sobjects/BatchCalcJobDefinition	Read permission to the Analytics Integration User with the data sources and fields.
Data Processing Engine (DPE) Definition	writebacks -> targetObjectName	v{}/tooling/sobjects/BatchCalcJobDefinition	Create, update, or delete permission on the targetObjectName object based on the operationType value defined in the writebacks.
Data Processing Engine (DPE) Definition	writebacks -> writebackUser	v{}/tooling/sobjects/BatchCalcJobDefinition	Delete this value if it exists.

Salesforce Contracts Metadata

This table provides the metadata deployment reference for Salesforce Contracts in Revenue Management, including setup paths and configuration details.

Type	Label	Setup Path
Flag	Document Templates Export	Setup > Document Generation > General Settings
Setup	Context Service	Context Service > Context Service Settings > Context Definitions

Additional Deployment Information

Get to know additional deployment information for each Revenue Management feature domain, ensuring successful deployments and migrations.

Product Catalog Management Additional Information

Get to know additional deployment information for Product Catalog Management in Revenue Cloud, including active or inactive states, object information, and migration considerations.

Object-Specific Information

Object Name	Object API	Notes
Product Qualification	ProductQualification	• Considerations for Migrating Decision Tables • Considerations for Migrating Expression Sets • Migrate Context Definitions
Product Disqualification	ProductDisqualification	• Considerations for Migrating Decision Tables • Considerations for Migrating Expression Sets • Migrate Context Definitions
Product Category Qualification	ProductCategoryQualification	• Considerations for Migrating Decision Tables • Considerations for Migrating Expression Sets • Migrate Context Definitions
Product Category Disqualification	ProductCategoryDisqual	• Considerations for Migrating Decision Tables • Considerations for Migrating Expression Sets • Migrate Context Definitions

Other Information

- All product data (product, class, product attributes, attributes, picklist, catalog, categories, and so on) definitions must be active to appear in the sales channels.
  - These definitions need not be deactivated in the target to propagate updates as part of an org-to-org deployment. Exceptions are qualification rules that use decision tables, expression sets, and context definitions. The definitions in the target org must be deactivated to promote changes.
- Use the Full Index Rebuild to rebuild the entire search index, when enabling the feature for the first time, or changing the index settings. Use the Partial Index Rebuild to update recent changes to products and categories assignment. For more information about indexing products, see Manage Your Product Index.

  - If indexing is enabled in source but not enabled in the target, then while deploying the indexing flag (enable indexed product=true), the Full indexing must be run on the target org before the feature flag can be enabled.
- Migrating qualification rules records requires refreshing Qualification Rules Decision Table definitions (Product and Category Qualification and Disqualification Decision Tables).
- Migrating price book Entries requires Decision Table refresh for Product Discovery to show list price.
- Core to Near Core Sync Processes—Configure Product Catalog Management Cache in the new org to populate the product details cache, essential for large-scale implementations.
- Cross-Module Synchronization—Synchronization of catalog date to Constraint Modeling Language (CML) to support constraint rules.
- Incremental Changes and cohesive Sync (Recommended)—Depending on the deployment scenario, for example, new or changed product, new or changed price, new price book, new or changed rules, and so on.

Salesforce Pricing Additional Information

Get to know additional deployment information for Salesforce Pricing in Revenue Management, including active or inactive states, object information, and migration considerations.

Object-Specific Information

Object Name	Object API	Notes
Attribute Based Adjustment	AttributeBasedAdjustment	• Considerations for Migrating Decision Tables • Considerations for Migrating Expression Sets • Migrate Context Definitions
Price Adjustment Tier	PriceAdjustmentTier	• Considerations for Migrating Decision Tables • Considerations for Migrating Expression Sets • Migrate Context Definitions
Bundle Based Adjustment	BundleBasedAdjustment	• Considerations for Migrating Decision Tables • Considerations for Migrating Expression Sets • Migrate Context Definitions
Product Price Range	ProductPriceRange	• Considerations for Migrating Decision Tables • Considerations for Migrating Expression Sets • Migrate Context Definitions

Other Information

- You need active context definitions for pricing recipes. A pricing recipe is marked as default in an org. A System Admin can access the PricingRecipe object whereas a Pricing Admin can't access this object internally. You can't create or update fields of a PricingRecipe object.
- You must activate decision tables and pricing procedures.
- You need a refresh of the decision table for any new or modified records. For example, any modification in price book entries requires a refresh of the Price Book Entries decision table.
- You must migrate the Apex class if Apex is selected for Procedure Plan definitions. See Customize Your Procedure Plans with Apex Hooks.
- You can't delete a standard price book. A price book can be deleted after all references are removed. Instead, set the price book to Inactive status.
- If you delete a price adjustment schedule, all associated data is deleted without any warning or error messages. The adjustment method and status of a price adjustment schedule aren't considered for resolution. If you edit or delete an active price adjustment schedule, an error is thrown.
- The EffectiveFrom and EffectiveTo field values of a price adjustment tier, attribute-based adjustment, and bundle-based adjustment must be within the EffectiveFrom and EffectiveTo field values.
- Product and product selling model must be a part of the product selling model option before you create values in a price book entry.
- You can't delete an inactive or active product selling model. You can delete a draft product selling model by deleting all references. You can't change status from Active or Inactive to Draft.
- You can't edit a product price range after you've created it. Enable price tracking to edit a product price range.
- Salesforce Pricing doesn't consider the Active status of a price book when fetching the values. Also, it doesn't consider the validity date ranges of a price book.
- The pricing term and pricing term unit of a price adjustment tier aren't considered for resolution.
- The IsActive field of a price book entry and price adjustment schedule aren't considered for resolution.

Salesforce Pricing Migration Scenarios

Review these considerations to understand the Salesforce Pricing data migration process along with migration order and prerequisites.

Here are the possible migration scenarios.

- One sandbox org to another sandbox org
- Sandbox org to production org

To migrate pricing procedures, see Export and Import Procedure Plans.

To import and export pricing data, see Considerations for Importing and Exporting Pricing Data. Keep these considerations in mind for first-time migration.

First-Time Migration

- You must migrate metadata in this specific sequence to ensure a successful deployment.
  - You must migrate context definitions before any pricing procedures. If multiple pricing procedures are linked to various context definitions, include all associated definitions in the migration.

- You must deploy all decision tables (DTs) before you migrate the pricing procedures that reference them.
- Pricing recipe migration also migrates the associated default pricing procedure.

- You must manually refresh decision tables after migration within the target environment to make sure that the data is updated and active. If a decision table migration encounters missing metadata, such as a custom field or element, the deployment fails.
- You must manually update any pricing procedure constants that contain reference sandbox org IDs to the corresponding production org IDs. This step is required before migration to avoid logic failures in the production environment.
- You must manually update the price adjustment schedule IDs in the pricing procedure constants of the production org before or after migration.

Subsequent Production Migrations

- You must migrate context definitions before you migrate pricing procedures. If multiple pricing procedures are linked to various context definitions, then you must migrate all associated metadata to ensure system integrity.
- You must migrate decision tables along with any relevant metadata changes. If a decision table migration fails, the system must roll back to the previous snapshot, making sure that the existing decision tables are usable.
- You must migrate pricing procedures and expression set versions after the successful deployment of context definitions, decision tables, and pricing recipes in a sequential order.

Product Configurator Additional Information

Get to know additional deployment information for Product Configurator in Revenue Cloud, including active or inactive states, object information, and migration considerations.

Object-Specific Information

Object Name	Object API	Notes
Product Configuration Rule	ProductConfigurationRule	ProductConfigurationRule has a Binary Large Object (BLOB) for the rule content. ConfigurationRuleDefinition contains a JSON with the rules details. The BLOB references product IDs, which can't be migrated as is. You must use the npm migration utility for the migration. ProductConfigurationRule has a status field.
Product Configuration Flow	ProductConfigurationFlow	ProductConfigurationFlow has a status field. You don't require an API to set the status.
ExpressionSetConstraintObj	ExpressionSetConstraintObj	The ExpressionSet for Constraints has a status field. The ReferenceObjectId lookup field is polymorphic and references object IDs.

Other Information

- You can't update rules in active status.
- Product catalog management data must be migrated before the associations (ExpressionSetConstraintObj) are migrated and before the Business Rules Engine rules (ProductConfigurationRule) are migrated. When updating an Expression Set (Constraint Model), the target must be deactivated.
- Cross-Module Synchronization—Constraint Modeling Language (CML) and Business Rules Engine rules both reference product catalog management data. This data must be migrated before the business rules or the constraint rules.
- These components have dependencies on Industries common features.
  - Business Rules Product Configuration Rules—Business Rules Engine and Context Service
  - Constraint Builder Product Configuration Rules—Expression Set and Context Service
  - Product Configurator—Flow

In a future version of this guide, we plan to add information for moving Constraint Modeling Language (CML) code and related expression sets.

Dynamic Revenue Orchestrator Additional Information

Get to know additional deployment information for Dynamic Revenue Orchestrator (DRO) in Revenue Management, including active or inactive states, object information, and migration considerations.

RuleSet API Naming Pattern

In API version 67.0 and later, RuleSet API naming is decoupled from the parent record ID and uses a global key (UUID). This naming pattern removes the pre-commit dependency on record IDs and enables condition JSON processing during INSERT operation for records.

Existing RuleSet records that use the legacy record ID-based naming patterns keep their original API naming pattern and aren't automatically renamed. Legacy and current naming patterns can coexist in the same org. No mandatory migration is required solely to normalize RuleSet API names.

Special Fields

- These objects use a large text field to contain a JSON representation of Dynamic Revenue Orchestrator (DRO) condition data, backed up by internal Business Rules Engine (BRE) RuleSet objects.
  - FulfillmentStepDefinition
    - ExecuteOnConditionData. This is related to ExecuteOnRule field (RuleSet reference).
    - ResumeOnConditionData. This is related to ResumeOnRule field (RuleSet reference).
  - ProductFulfillmentScenario
    - ConditionData. This is related to ScenarioRule field (RuleSet reference).
  - ProductFulfillmentDecompRule
    - ConditionData. This is related to ExecuteOnRule field (RuleSet reference).
  - FulfillmentTaskAssignmentRule
    - ConditionData. This is related to Condition field (RuleSet reference).

- Condition JSON fields support full lifecycle behavior for global key-based naming pattern records. You can provide condition JSON at insert, modify it through updates, and remove it to delete associated RuleSet artifacts where applicable. Here are the details for each operation type.
  - You can create the record and provide condition JSON in the same transaction, including bulk operations.
  - When you change condition JSON, the related RuleSet reference and runtime evaluation logic are updated.
  - When you remove condition JSON, the related RuleSet artifacts are deleted where applicable, including resume on rule scenarios.

Object-Specific Information

Object Name	Object API	Notes
Product Fulfillment Decomposition Rule	ProductFulfillmentDecompRule	
Fulfillment Step Definition	FulfillmentStepDefinition	
Product Fulfillment Scenario	ProductFulfillmentScenario	
Fulfillment Task Assignment Rule	FulfillmentTaskAssignmentRule	
Fulfillment Fallout Rule	FulfillmentFalloutRule	After migration, refresh the related Decision Tables.
Orchestration Plan Context Mapping	OrchestrationPlanCtxMapping	This setup object is always active after creation. Here are some considerations. • The object must have an active context definition to create the mapping. • The context definition must remain active for the mapping to be valid. • Context nodes or referenced mappings must exist in the active context definition version.

Other Information

- Migration Prerequisites
  - When DRO is enabled in a new org, the system provides an built-in library named as DRORuleLibrary. See Dynamic Fulfillment Orchestrator Settings.
  - Context rules library is Active with `Usage Type=Dfo` and is linked with Sales Context Definition.
  - The active rule library version must point to the context definition currently configured in the DRO Admin settings. If you change the context definition in the settings, you must create a new rule library version that links to the new definition and activate it.

    When creating a version, you must clone from the latest rule library version. If you clone from an older version instead of the latest, any rule sets created in the latest version is lost in any other newly created version. This leads to evaluation errors where rules evaluate to `false` as the engine is unable to find the rule set in the active library.

    When moving to a new version, you must first deactivate the older version and then activate the cloned version.
  - Context definitions must have the necessary nodes, mappings, and context tags used in rule sets.

  - DRO rules include Product Fulfillment Decomposition Rule, Fulfillment Step Definition, Product Fulfillment Scenario, and Fulfillment Task Assignment Rule objects. To migrate DRO rules or scenarios, make sure the AttributeDefinition and AttributePicklistValue records are referenced in condition data fields by attribute code and picklist value name in the org before migration. The code value is defined in product management records.
  - The SourceAttributeIdentifier and DestinationAttributeIdentifier fields of a ProductDecompEnrichmentRule record can include attribute IDs from the source org, leading to invalid references in the target org. Refresh the identifier fields by saving the ProductDecompEnrichmentRule records again. You can also set the identifier fields to null during migration to make sure that they auto-populate correctly.
  - Before you can migrate the Product Fulfillment Decomposition Rule, Fulfillment Step Definition, Product Fulfillment Scenario, and Fulfillment Task Assignment Rule objects with their related child records, several other key records must exist in the target Salesforce org. Migrating these records first is crucial to maintain data integrity and to make sure that the relationships are correctly established in rules.

    Here are the required records in the target org as per the required migration sequence.

    - Products (Product2)—The top-level ProductFulfillmentDecompRule record is directly linked to a specific product via the ProductId field. You can't create the rule without associating it with an existing product record.

      Make sure that the Product2 records in the source and target orgs can be uniquely identified. For example, use a custom external ID field such as GlobalKey__c or any other ExternalId field, or use the standard ProductCode field.

    - Attribute Definitions (AttributeDefinition)—The ProductDecompEnrichmentRule records, which are children of the decomposition rule, use attribute definitions to specify which product attributes to use for enrichment or transformation. These rules reference AttributeDefinition records in fields, such as SourceAttributeDefinitionId and TargetAttributeDefinitionId. DRO Condition Data fields and rule sets use AttributeCode values in Condition Expressions.

      Attribute Definitions must be migrated before the enrichment rules to make sure that the lookup relationships are resolved successfully. Attribute Code values must be defined as these values are used in DRO Conditions as unique identifiers for attribute-type expressions.

    - Attribute Definition Picklist Values (AttributePicklistValue)—The ValTfrm (Value Transform) records, which are used for mapping and transforming data, often rely on picklist values defined in the AttributePicklistValue object. These records define the specific values used in the transformation logic. DRO Condition Data fields and rule sets use AttributeCode values in Condition Expressions.

      If your transformation rules involve picklist fields, you must migrate the corresponding AttributePicklistValue records first.

- DRO Rules and Scenario Migration
  - Internal rule set tables may contain attribute ID values, which are the Salesforce IDs of the relevant AttributeDefinition records of the Product or Product Classification.
  - DRO stores rule set-related data in string representation with AttributeDefinition. During migration, the JSON representation of a rule set uses Attribute Code as the natural key to resolve AttributeDefinition records and set as key in internal rule set tables. As a prerequisite, make sure that attribute code values are populated and consistent in both the source and target orgs before you migrate DRO rules.
  - Rule sets also contain attribute picklist values in condition expressions. The AttributePicklistValue records of the associated AttributeDefinition records must exist in target org before you migrate DRO rules.
  - In API version 67.0 and later, you can insert DRO rule and scenario records with condition JSON in one transaction, including bulk inserts, when prerequisites are satisfied.
  - Deleting a DRO parent record also deletes related RuleSet and rule artifacts to prevent orphan records. If condition JSON is removed from a record, related RuleSet artifacts are also removed where applicable.
- After Migration:

- Refresh the Decision Tables (DT) used by DRO Fallout Management—Migrating the Fulfillment Fallout Rules records requires a refreshed Fallout Rules DT definition.
- Refresh DTs used by DRO Jeopardy Management—Migrating the Fulfillment Step Jeopardy Rule records requires a refreshed Fulfillment Step Jeopardy Rule DT definition.
- Activate technical products in your target org as part of your deployment.

- These components have dependencies on specific features.
  - Enrichment Rule—Expression Set
  - Decomposition Rule with condition—Business Rules Engine Context Rule and Context Service
  - Callout step—Integration Definition
  - Autotask step—Flow
  - Manual Task step—Queue
  - Product Fulfillment Scenario with condition—Business Rules Engine Context Rule and Context Service

SEE ALSO:

Bulk API 2.0 and Bulk API Developer Guide: Introduction to Bulk API 2.0 and Bulk API

Usage Management Additional Information

Get to know additional deployment information for Usage Management in Revenue Management, including active or inactive states, object information, and migration considerations.

Object-Specific Information

Object Name	Object API	Notes
Rate Card	RateCard	• Considerations for Migrating Decision Tables • Considerations for Migrating Expression Sets • Migrate Context Definitions
Price Book Rate Card	PriceBookRateCard	• Considerations for Migrating Decision Tables • Considerations for Migrating Expression Sets • Migrate Context Definitions
Product Usage Resource	ProductUsageResource	Effective end date can be extended in Active state.
Rate Card Entry	RateCardEntry	• Considerations for Migrating Decision Tables

Object Name	Object API	Notes
		• Considerations for Migrating Expression Sets • Migrate Context Definitions • No updates are allowed in Active or Inactive state.
Product Usage Grant	ProductUsageGrant	The Effective End Date can be extended in Active state.
Rate Adjustment By Tier	RateAdjustmentByTier	• Considerations for Migrating Decision Tables • Considerations for Migrating Expression Sets • Migrate Context Definitions
Rate Adjustment By Attribute	RateAdjustmentByAttribute	• Considerations for Migrating Decision Tables • Considerations for Migrating Expression Sets • Migrate Context Definitions
Usage Commitment Asset Related Object	UsageCmtAssetRelatedObj	The RelatedObjectId lookup field is polymorphic.

Other Information

- For Selling (Revenue Cloud Advanced), perform these actions:
  - Confirm that Rates, Grants, and Policies for usage products are set up correctly.
  - Extend and sync the SalesTransaction context definition.
  - Confirm that the pricing procedure is active and set up correctly in Revenue Settings.
  - Refresh Decision Tables (DT) being referred in pricing procedures.
  - Set up and activate the rating discovery procedure.
  - Sync the RatingDiscovery context definition.
  - Refresh all Decision Tables (DTs) being used in the rating discovery procedure.
- For Consumption (Revenue Cloud Billing), perform these actions:
  - Clone and set up the Orchestration Flow.
  - Configure the Data Processing Engine (DPE) jobs.
  - Set up and activate the rating procedure.
  - Sync the Rate Management context definition.
  - Refresh all Decision Tables (DTs) being used in the rating procedure.
- These components have dependencies on Industries common Features:

  - Rating or Discovery Procedures: Expression Set
  - Rating, Discovery, or Selling Journey: Business Rules Engine and Context Service
  - Rating or summary creation: Batch and Data Processing Engine

- After migration, selling and consumption-related object records such as UnitOfMeasureClass, UsageResource, RateCardEntry, ProductUsageResource, ProductUsageGrant must be active before you use it.

For ProductUsageResource and ProductUsageGrant objects, review these considerations.

- You can delete records in Draft or Inactive states. You can't delete a record in Active state.
- After you activate a record, you can extend end date only.
- The applicable status transitions are Draft, Active, and Inactive.

You can edit a RateCardEntry object in Draft status only. You can't edit this object after you've activated it. The applicable status transitions are:

- Draft, Active, and Inactive
- Inactive and Active

Billing Additional Information

Get to know additional deployment information for Billing in Revenue Management, including active or inactive states, object information, and migration considerations.

Object-Specific Information

Object Name	Object API	Notes
Legal Entity	LegalEntity	This object contains a polymorphic field for address.
Billing Policy	BillingPolicy	Activate after Billing Treatment is activated.
Billing Treatment	BillingTreatment	Activate after Billing Treatment Item is activated.
Billing Treatment Item	BillingTreatmentItem	Activate first when activating Billing Policy.
Tax Policy	TaxPolicy	Activate after Tax Treatment is activated.
Tax Treatment	TaxTreament	Activate first when activating Tax Policy.
Tax Treatment	TaxTreatment	Review these considerations for the TaxTreatment object. • Create a tax treatment in Draft or Active status. • While the tax treatment is Active, you can't edit LegalEntity, TaxEngine, TaxCode, ProductCode, IsTaxable, or ShouldUseTaxTreatmentItems. • Status can move among Draft, Active, and Inactive states.

Object Name	Object API	Notes
Payment Term	PaymentTerm	Activate after Payment Term Item is activated.
Payment Term Item	PaymentTermItem	Activate first when activating Payment Term.
Payment Schedule Policy	PaymentSchedulePolicy	Activate after Payment Schedule Treatment is activated
Payment Schedule Treatment	PaymentScheduleTreatment	Activate first when activating Payment Schedule Policy.
Payment Retry Rule	PaymentRetryRule	If `RetryIntervalType` field value is specified, you must also specify values for `IntervalUnit` and `IntervalValue` fields.
Payment Retry Rule Set	PaymentRetryRuleSet	You can't delete or modify the active org-default rule set after the feature is enabled. The org can't have multiple active org-default rule sets.
Tax Treatment Item	TaxTreatmentItem	You can't delete a tax treatment item when the parent tax treatment is active.
Accounting Period	AccountingPeriod	Review these considerations for the AccountingPeriod object. • The StartDate, EndDate, and FinancialYear fields can't be edited if child records exist. • The Name field is auto-populated. • The Status field is mutable. • The TotalAssetsAmount, TotalLiabilitiesAmount, TotalEquitiesAmount, TotalRevenueAmount, and TotalExpensesAmount are derived or read-only fields. • The creation of a record can happen only in Open status. • For AccountingPeriod as a parent record and LegalEntyAccountingPeriod as a child record, the status of the parent record must be Open when the child is created. The parent record can't close if any non-closed child record exists. The parent fields, such as StartDate, EndDate, or FinancialYear, become immutable if any child record exists.

Object Name	Object API	Notes
Billing Arrangement	BillingArrangement	Review these considerations for the BillingArrangement object. • You can create a billing arrangement only in Draft status. • Before you activate it, add at least one billing arrangement line. • You can't change ShouldBillRemainderToAccount field after the billing arrangement is linked to a billing schedule group. • Status can move among Draft, Active, and Inactive states.
Billing Arrangement Line	BillingArrangementLine	Review these considerations for the BillingArrangementLine object. • You can't insert or update a line while the parent billing arrangement is in Inactive status. • You can delete a line only when the parent billing arrangement is in Draft status.
Billing Milestone Plan	BillingMilestonePlan	Review these considerations for the BillingMilestonePlan object. • Create the plan in Draft status. • Activate only after you add at least one billing milestone plan item. • Updates apply only to lines that use the latest version number. • You can't delete a line that has ShouldBillRemainder field set to `true`. • The related billing treatment must be active before you activate the plan. • At least one billing milestone plan item must exist before activation. • While the plan is active, you can't edit ReferenceItem, ExternalReference, ReferenceItemAmount, BillingTreatment, Name, or Description fields. • You can't change ReferenceItem or ExternalReference fields after the plan is linked to a billing schedule.

Object Name	Object API	Notes
		• Status can move among Draft, Active, and Inactive states. • The plan must be in Draft status before you can delete it.
Billing Milestone Plan Item	BillingMilestonePlanItem	Review these considerations for the BillingMilestonePlanItem object. • Create items while the record is in Draft status. • The parent billing milestone plan must stay in Draft status while you create, update, or delete items.
General Ledger Account	GeneralLedgerAccount	Review these considerations for the GeneralLedgerAccount object. • You can't edit LegalEntity after the account is tied to a general ledger account assignment rule. • This object doesn't use status transitions. • A general ledger account assignment rule can be active only when both a credit and a debit general ledger account are set.
General Ledger Account Assignment Rules	GeneralLedgerAcctAsgntRule	Review these considerations for the GeneralLedgerAcctAsgntRule object. • While the rule is active, you can't edit the debit account, credit account, or currency fields. • Status can move from Inactive to Active states. • The rule can be active only when both a credit and a debit general ledger account are present.
General Ledger Journal Entry Rule	GeneralLedgerJrnlEntryRule	Review these considerations for the GeneralLedgerJrnlEntryRule object. • You can't edit the journal entry rule while its general ledger account assignment rule is active. • This object doesn't use status transitions.

Object Name	Object API	Notes
		• The credit account, debit account, and assignment rule must belong to the same legal entity.
Legal Entity Accounting Period	LegalEntityAccountingPeriod	Review these considerations for the LegalEntityAccountingPeriod object. • After creation, you can't change the AccountingPeriod or LegalEntity fields. • The Name and CurrencyIsoCode fields are auto-populated. • You can change status according to the product's transition rules. For example, Open to Pending Closure, Pending Closure to Closed or Error, Closed to Pending Reopen, and related paths for Reopened and Error. • The TotalAssetsAmount, TotalLiabilitiesAmount, TotalEquitiesAmount, TotalRevenueAmount, and TotalExpensesAmount fields are derived or read-only. • The ClosureStage field is read-only. • Create a record only when the status is Open. • When `AccountingPeriod` record is the parent and `LegalEntityAccountingPeriod` is the child record, the parent record must be Open when the child record is created. The parent record can't close if any child record is still open. The StartDate, EndDate, and FinancialYear fields on the parent record can't be edited if any child record exists. • When `LegalEntity` record is the parent record, it must be active when the child record is created. It can't move to Inactive status when any child accounting period is open. • For `LegalEntityAccountingPeriod` as parent record and `GeneralLdgrAcctPrdSummary` as child record, the parent can't be

Object Name	Object API	Notes
		deleted if children exist. Children can be created or updated only when the parent is Open or Reopened. Child summary records can't be deleted.
Payment Schedule Treatment Detail	PaymentScheduleTreatmentDtl	Review these considerations for the PaymentScheduleTreatmentDtl object. • While the parent payment schedule treatment is Active, Inactive, or Canceled, you can't change the InstallmentPaymentType, Percentage, ProcessingDateReference, DateOffset, PaymentMethodSelectionType, PymtSchdDistributionMethod, or PaymentRunMatchingValue fields. • Create a detail record only when the parent payment schedule treatment is in Draft status. • Delete a detail record only when the parent payment schedule treatment is in Draft status.

Other Information

- Data pipeline must be enabled before enabling Billing.
- Order to Billing Schedule Flow must be copied from the template and activated.

Business Rules Engine and Decision Tables Additional Information

Get to know additional Revenue Management deployment information for Industries common features such as Business Rules Engine, Expression Sets, and Decision Tables.

Object-Specific Information

Object Name	Object API	Notes
Expression Set Definition	ExpressionSetDefinition	Only call the setup object via Metadata API to get child setup objects.
Expression Set Definition Version	ExpressionSetDefinitionVersion	Only call the setup object via Metadata API to get child setup objects.
Rule Library	RuleLibrary	Not supported by Metadata API.
Rule Library Version	RuleLibraryVersion	Not supported by Metadata API.

Helpful Links

- Considerations for Migrating Expression Sets
- Migration of Expression Sets with Dependencies

Deployment Considerations

- Initial Deployment—Expression Sets Decision Tables can be deployed in any state during their first deployment to a target org.
- Subsequent Updates—To update an expression set that exists in the target org, migrate a new expression set version with the updates. You must not update the existing active version.
- When deploying an active decision table, it goes through activation again in the target org and may fail if the required custom object isn't present in the target org.
- Two expression set versions of the same parent expression set can't have the same rank, regardless of the `Draft` or `Active` states. In this case:
  - Create an expression set version in the source org with the updates.
  - Activate the new version in the source org.
  - Then, migrate the new version or the entire expression set to the target org.
  - The migrated expression set version is invoked in the target org as per the specified rank.
  - Migration fails if the target org contains an expression set version with the specified rank.
- Activate and refresh of decision tables after deployment.
- Activate sub expressions after deployment.
- Dynamic rules are stored in platform objects, and not in setup objects. There's no platform support for deploying dynamic rules.

Dependencies

- Industries common features can have dependencies among themselves. For example, expression sets can have sub expressions and reference decision tables or context definitions. Decision Tables can have a dependency on a custom object, and so on.
- For a deployment to be successful, all dependent components must be migrated correctly before the expression set version itself.
- You must migrate dependencies before the expression sets that reference them. If there are nested dependencies, you must migrate the lowest-level components first. Here's a general deployment order.
  - Standard and custom objects
  - Context Definitions, Decision Tables, Decision Matrices, Object Aliases, and Explainability Message Templates
  - Sub Expression Sets
  - Parent Expression Set Version

Other Information

- Both Expression Set and Decision Matrix have versioned objects.
- Decision Tables don't have versions.

Context Service Additional Information

Get to know additional deployment information for Context Service in Revenue Cloud.

Helpful Links

- Migrate Context Definitions

Deployment Considerations

- Context tags and individual sObject mappings must be unique for every definition.
- Deployment can lead to conflict if there are multiple tags with the same name, or if there are multiple mappings to the same sObject and sObject fields.
- Standard definitions are versioned and new changes must have a higher version than the version, which is available in the org.
- You can deploy updates that introduce new custom nodes, attributes, and their corresponding mappings to existing context definitions.
- Deployments that modify (update or delete) existing custom mappings for context definitions in an activated or a deactivated state are supported.

Unsupported Deployment Scenarios

- Deploying context definitions from the current release to an older release is not supported.
- Modifying custom nodes and attributes for context definitions that are in a deactivated state aren't supported. To make changes, replicate your desired modifications manually on the target context definition. You can also delete the context definition on the target org and deploy again.
- Modifying standard nodes and attributes for context definitions that are in an activated state isn't supported. To make changes, deactivate the definition on the target org first, make your desired modifications, and then activate the context definition again.
- You can't activate or deactivate an active context definition as an action within a deployment package. Activation or deactivation must be performed as a separate, manual step outside the deployment process.
- You can't make a context mapping default or non-default as an action within a deployment. Changing the default status of context mappings must be done as a separate, manual step.

Data Processing Engine Additional Information

Get to know additional deployment information for Data Processing Engine in Revenue Cloud.

Deployment Considerations

- Data Processing Engine objects have `Draft` and `Active` states.
- The objects must be created in `Draft` state and activated later. The activation is done through API.
- Configuration can't be changed after an object is updated to `Active` state.
- Set the state of the object to `Inactive` for any modifications, and then set the state to `Active`.

Other Information

- Data Processing Engine has dependencies on these components.
  - CRM Analytics or Data Cloud
  - Bulk API

- You can deploy Data Processing Engine definitions from one organization to another. Both organizations must be on the same Salesforce release version.

Salesforce Contracts Additional Information

Get to know additional deployment information for Salesforce Contracts in Revenue Cloud, including active or inactive states, and migration considerations.

Document clauses can be in different states, such as Draft, In Approval, Active, and Archived. Keep these considerations in mind regarding the clause statuses.

- A clause is created in Draft status.
- During the import process, the clause must be in Draft status. You can update the status after you complete the import process.
- An archived clause can't be updated.
- An active clause can't be updated, but it can change to Draft status only if it's not used.
- An active clause must be moved to Draft status before you update it to Active status.

Clause Migration Considerations

Clause migration is a prerequisite for Microsoft 365 template migration. Review these considerations to understand how clause structure, versions, and relationships must exist in the target org before migrating document templates.

Clause Validations and Migration Constraints

Validation rules, data model constraints, and known migration behaviors that affect clause records during migration. Review this reference to understand sequencing, status handling, and dependency requirements that can impact migration success.

Clause Migration Considerations

Clause migration is a prerequisite for Microsoft 365 template migration. Review these considerations to understand how clause structure, versions, and relationships must exist in the target org before migrating document templates.

Clause migration prepares the target org with all required clause data so that template migration can resolve clause references correctly. Clause records don't retain source org record IDs during migration. Instead, template migration resolves references based on clause attributes and relationships that already exist in the target org.

Clause migration requires careful sequencing and coordination because clause records enforce strict validation rules around status, hierarchy, and versioning.

Microsoft 365 template migration requires all clause records, clause categories, and related clause sets to exist in the target org.

During template migration, clause references resolve in the target org by matching this combination of attributes. Make sure that the target org contains clauses with the exact attribute values referenced by the Microsoft 365 template.

- ClauseSetName
- ClauseName
- ClauseVersion
- ClauseLanguage

If a referenced clause is missing in the target org, template migration stops to prevent broken references. This alignment helps maintain template integrity and reduces the need for manual corrections after migration.

Migrate clause-related objects separately by using a data migration tool, such as Data Loader.

Migrate the three document clause objects in the required order.

- Clause Category Configuration
- Document Clause Set
- Document Clause

Important: Avoid modifying clause data in the target org after migration to prevent mismatches between source and target.

Avoid modifying clause data in the source org after exporting clause data and before Microsoft 365 template migration to prevent reference mismatches.

Clause Category Configuration

Clause Category Configuration is a setup object that you migrate by using metadata deployment tools, such as Package Manager, and Salesforce CLI. Migrate all clause categories referenced by your clause sets.

Clause Set

Clause sets must be migrated before you can migrate clauses. This needs to be either created explicitly on the target org or migrated from the source org by using data migration utilities. Resolve these clause category configuration references to IDs on the target org in your export set before you migrate.

- `CategoryReferenceId`—Replace with the mapped ID obtained from clause category migration
- `Category`—This is a 15-character ID of the clause category. Update CategoryReferenceId by removing the last three characters.

For iterative migrations, use upsert operation to update existing clause sets and insert newly created clause sets since the previous migration.

Clause

Migrate clause category configuration and clause sets before initiating clause migration. Resolve the clause set references (DocumentClauseSet) to the IDs on the target in your export set before migrating.

Important: All clauses are inserted only in Draft status, regardless of their source status. Run a subsequent migration that includes status mapping to update the clauses to the correct status. See Clause Status Mapping During Migration on page 66.

First-Time Migration

Use this approach when migrating clause data for the first time from the source org to the destination org for Microsoft 365 template migration. Clauses enforce validation rules that affect insert and update operations during migration.

- Insert main clauses before alternate clauses.
- Clauses are inserted in Draft status. Run a subsequent migration to update the clause status. See Clause Status Mapping During Migration on page 66 for details on how clause statuses are applied and updated.

Iterative Migration

Use iterative migration when clause data exists in the target org and you must migrate updates or newly created clauses from the source org. Track the timestamp of the last successful export and migrate only records that were created or updated after that timestamp. This approach avoids validation failures when attempting to update unchanged clauses, especially clauses in Archived status.

- Migrate clause records that were modified after the last export timestamp and created on or before the last export timestamp. This step migrates only updated records and preserves the correct status for existing clauses before introducing new draft versions.

- Migrate main clause records that were created after the last export timestamp. Clauses are inserted in Draft status. Run a subsequent migration to update the clause status. See Clause Status Mapping During Migration on page 66 for details on how clause statuses are applied and updated.
- Migrate alternate clause records that were created after the last export timestamp. Clauses are inserted in Draft status. Run a subsequent migration to update the status.

Clause Validations and Migration Constraints

Validation rules, data model constraints, and known migration behaviors that affect clause records during migration. Review this reference to understand sequencing, status handling, and dependency requirements that can impact migration success.

Clause Migration Validations

These validations apply during insert, update, and delete operations on clause records. Operations fail when a validation is violated.

Operation	Validation	Description
Insert	Draft-only creation	Allows only Draft clauses during initial insert operations.
Insert	Version conflict prevention	Prevents inserting a clause when a Draft version exists for the same clause.
Insert	Clause set requirement	Requires each clause to reference a valid DocumentClauseSet that exists in the target org.
Insert	Main and alternate clause hierarchy	• Requires an existing main clause before inserting alternate clauses and allows only one main clause per clause group. • Allows only one main clause per language within a clause set, and allows multiple alternate clauses for the same language.
Update	Restricted fields	Prevents updating the DocumentClauseSet, Language, IsAlternateClause, and Version fields after clause creation.
Update	Status transition rules	Allows changing an Active clause only to Draft (without modifying other fields) or to Archived.
Update	Archived clause immutability	Prevents updates to Archived clauses.
Update	Content requirement for activation	Requires clause content before changing status to Active.

Operation	Validation	Description
Delete	Status-based deletion	Allows deletion of Draft and Archived clauses only.
Delete	Main clause protection	Prevents deleting a main clause while alternate clauses still exist.

Clause Status Mapping During Migration

Clauses insert only in Draft status, regardless of their source status. Run a subsequent migration with status mapping to update the clause statuses. For example, a clause in the source org has three versions: Version 1 is Archived, Version 2 is Active, and Version 3 is Draft. When the clause is inserted, all three versions are created in Draft status in the target org. Perform a follow-up upsert migration with status mapping to align each clause version with its source org status.

Source Org Status (Pre-Migration)	Guidance
Archived	Archived clauses insert as Draft. Run a subsequent migration with status mapping to restore the Archived status. After a clause is archived, you can't change its status.
Active	Active clauses insert as Draft. Run a subsequent migration with status mapping to restore the Active status.
Draft	Draft clauses insert as Draft, so no status change is required.
In Approval	You must not migrate In Approval clauses because their workflows don't migrate. However, if In Approval clauses are migrated, they're inserted in Draft status, and the approval process must be reinitiated from the UI.

Clause Migration Constraints

These constraints describe behavioral rules, sequencing requirements, and tooling limitations that affect clause migration, especially during iterative and incremental migrations.

Category	Constraint	Description
Tooling	No automated migration	Requires that you use supported data migration tools, such as Data Loader, because no automated clause migration utility exists.
Tooling	Authentication requirements for Data Loader	Requires OAuth-based authentication. Enforcing OAuth, Proof Key for Code Exchange (PKCE), or IP restrictions requires completing additional connected app and access configuration.

Category	Constraint	Description
Data Model	Clause category limitations	Prevents adding custom fields to Clause Category Configuration because it's a setup object.
Data Model	Manual ID creation	Creation of an external ID field in the target org to store source org IDs for upsert operations on clause sets and clause records.
Data Model	Manual ID mapping	Requires manually mapping clause category IDs between source and target orgs.
Data Dependency	Clause set dependency	Requires clause sets and clause category configuration to exist in the target org before importing clauses.
Iterative Migration	Migration order dependency	Requires that you update existing clauses before you insert newly created clauses. Also, insert main clauses before alternate clauses.
Template Dependency	Missing clause references	Prevents template migration when referenced clauses, versions, or languages don't exist in the target org.
Template Dependency	Duplicate clauses	Can cause clause reference mismatches during template migration when duplicate clauses exist in the target org.
Template Dependency	Office Open XML failures	Can cause template migration to fail when Office Open XML updates fail during document import.

CHAPTER 4 Product Catalog Management

In this chapter ...

Create and manage a product catalog with components, such as attributes, product classifications, simple and bundled products, and rules.

Product Catalog Management

Manage an entire product portfolio with components such as attributes, product classifications, simple and bundled products, and rules.

EDITIONS

Available in: Lightning Experience

Available in: **Enterprise**, **Unlimited**, and **Developer** Editions of Revenue Management

Product Catalog Management Standard Objects

The Product Catalog Management data model provides objects and fields to manage products, rules, and catalogs.

Product Catalog Management Fields on Standard Objects

Product Catalog Management adds standard and custom fields to some standard Salesforce objects. These fields are available only in orgs where Product Catalog Management is enabled. This object is available in API version 60.0 and later.

Product Catalog Management Business APIs

Use primitive APIs of Product Catalog Management that serve catalog definitions to users or applications.

Product Catalog Management Metadata API Types

Metadata API enables you to access some types and feature settings that you can customize in the user interface.

Product Catalog Management Tooling API Objects

Tooling API exposes metadata used in developer tooling that you can access through REST or SOAP. Tooling API's SOQL capabilities for many metadata types allow you to retrieve smaller pieces of metadata.

SEE ALSO:

Salesforce Help: Product Catalog Management Permission Set Licenses

Salesforce Help: User Permissions for Product Catalog Management

Product Catalog Management Standard Objects

The Product Catalog Management data model provides objects and fields to manage products, rules, and catalogs.

AttributeCategory

Represents a logical grouping of attributes that can be reused while defining products. Attribute Categories are used for searching and managing product attributes. For example, the "Mobile Handset Properties" category has color, storage and make model, and size attributes. This object is available in API version 60.0 and later.

AttributeCategoryAttribute

Represents a relationship between an attribute category and the attribute definition. This object is available in API version 60.0 and later.

AttrPicklistExcludedValue

Represents the excluded picklist values for a product classification attribute or a product attribute definition. This object is available in API version 61.0 and later.

ProductAttributeDefinition

Represents the relationship between a product and its attributes. This object is available in API version 60.0 and later.

ProductCategoryDisqual

Represents disqualification rules for product categories. The rules determine when the product category doesn't qualify to be displayed to users. This object is available in API version 60.0 and later.

ProductCategoryQualification

Represents qualification rules for product categories. The rules determine when the product category qualifies to be displayed to users. This object is available in API version 60.0 and later.

ProductClassification

Represents a template that holds a collection of dynamic attributes. Product classification is used to quickly define and create multiple products that are similar yet different. This object is available in API version 60.0 and later.

ProductClassificationAttr

Represents the relationship between a product classification and its attributes. This is the default configuration for products based on the product classification. This object is available in API version 60.0 and later.

ProductComponentGrpOverride

Represents override information for a Product Component Group. The cardinality of the product component group can be overridden in the context of a product bundle. This object is available in API version 60.0 and later.

ProductDisqualification

Represents disqualification rules for products. The rules determine when the product doesn't qualify to be displayed to users. The rules are based on user context. This object is available in API version 60.0 and later.

ProductQualification

Represents qualification rules for products. The rules determine when the product qualifies to be displayed to users. The rules are based on user context. This object is available in API version 60.0 and later.

ProductRampSegment

Represents the ramp period within a ramp deal where terms, volumes, and other commitments change over time. This object is available in API version 62.0 and later.

ProductRelComponentOverride

Represents the cardinality overrides for product components in a bundle. This object is available in API version 60.0 and later.

ProductSpecificationRecType

Represents the relationship between industry-specific product specifications and the product record type. This object is available in API version 60.0 and later.

ProductSpecificationType

Represents the type of product specification provided by the user to make the product terminology unique to an industry. A product specification type is associated with a product specification record type. This object is available in API version 60.0 and later.

SEE ALSO:

Object Reference for the Salesforce Platform: Overview of Salesforce Objects and Fields

SOAP API Developer Guide: Introduction to SOAP API

AttributeCategory

Represents a logical grouping of attributes that can be reused while defining products. Attribute Categories are used for searching and managing product attributes. For example, the "Mobile Handset Properties" category has color, storage and make model, and size attributes. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

Product Catalog Management must be enabled to access this object.

Fields

Field	Details
`Code`	**Type** string **Properties** Create, Filter, Group, idLookup, Nillable, Sort, Update **Description** The unique code for the attribute category. The maximum size is 80 alphanumeric characters. The code can include the following special characters: @ ! - < > * ? + = % # ( ) / \ & ‘ £ € $ ”.
`Description`	**Type** textarea **Properties** Create, Nillable, Update **Description** The description of the attribute category that's used only during design time.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the attribute category was last referenced.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the attribute category was last viewed.
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The unique name of the attribute category. The maximum length is 80 characters (of any type).

Field	Details
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The owner of the attribute category. This field is a polymorphic relationship field. **Relationship Name** Owner **Relationship Type** Lookup **Refers To** Group, User

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

AttributeCategoryFeed on page 2866
Feed tracking is available for the object.

AttributeCategoryHistory on page 2873
History is available for tracked fields of the object.

AttributeCategoryShare on page 2877
Sharing is available for the object.

AttributeCategoryAttribute

Represents a relationship between an attribute category and the attribute definition. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

Product Catalog Management must be enabled to access this object.

Fields

Field	Details
AttributeCategoryId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The ID of the attribute category that the attribute is associated with. The ID is unique within the organization. This field is a relationship field. **Relationship Name** AttributeCategory **Relationship Type** Lookup **Refers To** AttributeCategory
AttributeDefinitionId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The ID of the attribute definition associated with the attribute category. The ID is unique within the organization. This field is a relationship field. **Relationship Name** AttributeDefinition **Relationship Type** Lookup **Refers To** AttributeDefinition
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the attribute category attribute was last referenced.
LastViewedDate	**Type** dateTime

Field	Details
	Properties Filter, Nillable, Sort **Description** The date the attribute category attribute was last viewed.
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** A unique name for the attribute. The maximum length is 80 characters (of any type).
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The owner of the attribute category attribute. This field is a polymorphic relationship field. **Relationship Name** Owner **Relationship Type** Lookup **Refers To** Group, User

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

AttributeCategoryAttributeFeed on page 2866
Feed tracking is available for the object.

AttributeCategoryAttributeHistory on page 2873
History is available for tracked fields of the object.

AttributeCategoryAttributeShare on page 2877
Sharing is available for the object.

AttrPicklistExcludedValue

Represents the excluded picklist values for a product classification attribute or a product attribute definition. This object is available in API version 61.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

Product Catalog Management must be enabled to access this object.

Fields

Field	Details
`AttributeId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The ID of the product classification attribute or the product attribute definition of the picklist data type. This field is a polymorphic relationship field. **Relationship Name** Attribute **Relationship Type** Lookup **Refers To** ProductAttributeDefinition, ProductClassificationAttr
`AttributePicklistValueId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The ID of the attribute picklist value that's excluded in the product classification attribute or product attribute definition. This field is a relationship field. **Relationship Name** AttributePicklistValue **Relationship Type** Lookup **Refers To** AttributePicklistValue

Field	Details
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the excluded attribute picklist value was last referenced.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the excluded attribute picklist value was last viewed.
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The name of the excluded attribute picklist value.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The owner of the excluded attribute picklist value. This field is a polymorphic relationship field. **Relationship Name** Owner **Relationship Type** Lookup **Refers To** Group, User

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

AttrPicklistExcludedValueFeed on page 2866
Feed tracking is available for the object.

AttrPicklistExcludedValueHistory on page 2873
History is available for tracked fields of the object.

AttrPicklistExcludedValueShare on page 2877
Sharing is available for the object.

ProductAttributeDefinition

Represents the relationship between a product and its attributes. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

Product Catalog Management must be enabled to access this object.

Fields

Field	Details
`AttributeCategoryId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort **Description** The ID of the attribute category assigned to the parent object. This field is a relationship field. **Relationship Name** AttributeCategory **Relationship Type** Lookup **Refers To** AttributeCategory
`AttributeDefinitionId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The ID of the attribute associated with the product. This field is a relationship field.

Field	Details
	Relationship Name AttributeDefinition **Relationship Type** Lookup **Refers To** AttributeDefinition
`AttributeNameOverride`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The name to display for the attribute when shown for this object. This display name overrides the name on the attribute. For example, the attribute "Color" is overridden to display as "Laptop Color." The maximum size is 255 alphanumeric characters. The name can include these special characters: @ ! - < > * ? + = % # ( ) / \ & ' £ € $ ".
`DefaultValue`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The default value for the product attribute. The attribute value can be changed. This default overrides the default value set for a picklist for the attribute.
`Description`	**Type** textarea **Properties** Create, Nillable, Update **Description** A description of the attribute definition. The maximum size is 32,000 alphanumeric characters. The description can include these special characters: @ ! - < > * ? + = % # ( ) / \ & ' £ € $ ".
`DisplayType`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The type to display data for the selected data type. Possible values are: • `CheckBox`—Checkbox

Field	Details
	• `ComboBox`—Combobox • `Date` • `Datetime`—Date Time • `Number` • `RadioButton`—Radio Button • `Slider`—Available in API version 61.0 and later • `Text` • `Toggle`
`HelpText`	**Type** textarea **Properties** Create, Nillable, Update **Description** The help text to display when end users are configuring this attribute. This field overrides the help text defined for the attribute itself.
`IsHidden`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates if this product attribute is hidden from end users in the run time (`true`) or not (`false`). The default value is `false`.
`IsPriceImpacting`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether this attribute dictates the price of a product (`true`) or not (`false`). The default value is `false`.
`IsReadOnly`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the attribute is read-only for users in the run time (`true`) or not (`false`). The default value is `false`.

Field	Details
`IsRequired`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether this attribute requires a value when assigned to a parent object (`true`) or not (`false`). The default value is `false`.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the product attribute was last referenced.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the product attribute was last viewed.
`MaximumCharacterCount`	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The maximum number of characters that can be entered for an attribute value.
`MaximumValue`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The maximum value that can be entered as an attribute value.
`MinimumCharacterCount`	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update

Field	Details
	Description The minimum number of characters that can be entered for an attribute value.
`MinimumValue`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The minimum value that can be entered as an attribute value.
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The name of the product attribute.
`OverriddenProductAttributeDefinitionId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort **Description** The ID associated with the overridden product attribute definition. This field is a relationship field. **Relationship Name** OverriddenProductAttributeDefinition **Relationship Type** Lookup **Refers To** ProductAttributeDefinition
`OverrideContextId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort **Description** The ID associated with the root product record in a bundle. This field is a polymorphic relationship field. **Relationship Name** OverrideContext

Field	Details
	Relationship Type Lookup **Refers To** Product2
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The owner of the product attribute definition. This field is a polymorphic relationship field. **Relationship Name** Owner **Relationship Type** Lookup **Refers To** Group, User
Product2Id	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The product ID associated with the product attribute definition. This field is a relationship field. **Relationship Name** Product2 **Relationship Type** Lookup **Refers To** Product2
ProductClassificationAttributeId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The ID of the attribute assigned to the product classification. This field is a relationship field.

Field	Details
	Relationship Name ProductClassificationAttribute **Relationship Type** Lookup **Refers To** ProductClassificationAttr
`Sequence`	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The display sequence of the attribute when configuring the product during run time.
`Status`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The lifecycle state of the product attribute definition. Valid values are: • `Active` • `Draft` • `Inactive` The default value is `Draft`.
`StepValue`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The increment or decrement by which a slider's value changes as the user adjusts the product attribute value. Available in API version 61.0 and later.
`ValueDescription`	**Type** textarea **Properties** Create, Nillable, Update **Description** The description of the value assigned to this attribute. This field takes on the value description from the attribute definition.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

ProductAttributeDefinitionFeed on page 2866
: Feed tracking is available for the object.

ProductAttributeDefinitionHistory on page 2873
: History is available for tracked fields of the object.

ProductAttributeDefinitionShare on page 2877
: Sharing is available for the object.

ProductCategoryDisqual

Represents disqualification rules for product categories. The rules determine when the product category doesn't qualify to be displayed to users. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

Product Catalog Management must be enabled to access this object.

Fields

Field	Details
`CategoryId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The product category associated with the product disqualification record. This field is a relationship field. **Relationship Name** Category **Relationship Type** Lookup **Refers To** ProductCategory
`EffectiveFromDate`	**Type** date

Field	Details
	Properties Create, Filter, Group, Sort, Update **Description** The date from which the disqualification rule for the product category comes into effect.
`EffectiveToDate`	**Type** date **Properties** Create, Filter, Group, Sort, Update **Description** The date to which the disqualification rule for the product category ceases to be in effect.
`IsDisqualified`	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Indicates whether the product category is disqualified (`true`) or not (`false`) based on the disqualification rules. The default value is `false`.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the product category disqualification record was last referenced.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the product category disqualification record was last viewed.
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The name of the product category disqualification record.

Field	Details
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the owner of the product category disqualification record. This field is a polymorphic relationship field. **Relationship Name** Owner **Relationship Type** Lookup **Refers To** Group, User
Reason	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The reason to disqualify the product category.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

ProductCategoryDisqualFeed on page 2866
: Feed tracking is available for the object.

ProductCategoryDisqualHistory on page 2873
: History is available for tracked fields of the object.

ProductCategoryDisqualShare on page 2877
: Sharing is available for the object.

ProductCategoryQualification

Represents qualification rules for product categories. The rules determine when the product category qualifies to be displayed to users. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

Product Catalog Management must be enabled to access this object.

Fields

Field	Details
CategoryId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The product category associated with the category qualification record. This field is a relationship field. **Relationship Name** Category **Relationship Type** Lookup **Refers To** ProductCategory
EffectiveFromDate	**Type** date **Properties** Create, Filter, Group, Sort, Update **Description** The date from which the qualification rule for the product category comes into effect.
EffectiveToDate	**Type** date **Properties** Create, Filter, Group, Sort, Update **Description** The date to which the qualification rule for the product category ceases to be in effect.
IsQualified	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Indicates whether the product category is qualified (`true`) or not (`false`) based on the qualification rules. The default value is `false`.

Field	Details
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the product category qualification record was last referenced.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the product category qualification record was last viewed.
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The name of the product category qualification record.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the owner of the product category qualification record. This field is a polymorphic relationship field. **Relationship Name** Owner **Relationship Type** Lookup **Refers To** Group, User

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

ProductCategoryQualificationFeed on page 2866
Feed tracking is available for the object.

ProductCategoryQualificationHistory on page 2873
History is available for tracked fields of the object.

ProductCategoryQualificationShare on page 2877
Sharing is available for the object.

ProductClassification

Represents a template that holds a collection of dynamic attributes. Product classification is used to quickly define and create multiple products that are similar yet different. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

Product Catalog Management must be enabled to access this object.

Fields

Field	Details
`Code`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** A unique code for the product classification. The maximum size is 80 alphanumeric characters. The code can include the following special characters: @ ! - < > * ? + = % # ( ) / \ & ‘ £ € $ ”.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the product classification record was last referenced.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the product classification record was last viewed.

Field	Details
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The name of the product classification. The maximum length is 80 characters (of any type).
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The owner of the product classification. This field is a polymorphic relationship field. **Relationship Name** Owner **Relationship Type** Lookup **Refers To** Group, User
Status	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The lifecycle status of the product classification. Possible values are: • `Active` • `Draft` • `Inactive` The default value is `Draft`.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

ProductClassificationFeed on page 2866
Feed tracking is available for the object.

ProductClassificationHistory on page 2873
History is available for tracked fields of the object.

Sharing rules are available for the object.

ProductClassificationShare on page 2877
Sharing is available for the object.

ProductClassificationAttr

Represents the relationship between a product classification and its attributes. This is the default configuration for products based on the product classification. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

Product Catalog Management must be enabled to access this object.

Fields

Field	Details
`AttributeCategoryId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort **Description** The ID of the attribute category assigned to the parent object. This field is a relationship field. **Relationship Name** AttributeCategory **Relationship Type** Lookup **Refers To** AttributeCategory
`AttributeDefinitionId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The ID of the attribute assigned to the parent object.

Field	Details
	This field is a relationship field. **Relationship Name** AttributeDefinition **Relationship Type** Lookup **Refers To** AttributeDefinition
`AttributeNameOverride`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The overridden attribute name to display for the attribute when shown for this object. For example, "Color" overridden to "Laptop Color."
`DefaultValue`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The default value of the attribute for a product based on the product classification. This value can be changed.
`Description`	**Type** textarea **Properties** Create, Nillable, Update **Description** The description of this product classification attribute definition.
`DisplayType`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The type to display data for the selected data type. Valid values are: • `CheckBox`—Checkbox • `ComboBox`—Combobox • `Date`

Field	Details
	• `Datetime`—Date Time • `Number` • `RadioButton`—Radio Button • `Slider`—Available in API version 61.0 and later • `Text` • `Toggle`
`ExcludedPicklistValues`	**Type** textarea **Properties** Create, Nillable, Update **Description** The picklist values excluded from the attribute picklist. This field ensures that the product classification attribute only has valid values.
`HelpText`	**Type** textarea **Properties** Create, Nillable, Update **Description** The help text to display when end users are configuring this attribute. This field overrides the help text defined for the attribute itself.
`IsHidden`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates if this product attribute is hidden from end users in the run time (`true`) or not (`false`). The default value is `false`.
`IsPriceImpacting`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether this attribute dictates the price of a product (`true`) or not (`false`). The default value is `false`.

Field	Details
`IsReadOnly`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the attribute is read-only for users in the run time (`true`) or not (`false`). The default value is `false`.
`IsRequired`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether this attribute requires a value when assigned to a parent object (`true`) or not (`false`). The default value is `false`.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the product classification attribute was last referenced.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the product classification attribute was last viewed.
`MaximumCharacterCount`	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The maximum number of characters that can be entered for an attribute value.
`MaximumValue`	**Type** string

Field	Details
	Properties Create, Filter, Group, Nillable, Sort, Update **Description** The maximum value that can be entered as an attribute value.
MinimumCharacterCount	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The minimum number of characters that can be entered for an attribute value.
MinimumValue	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The minimum value that can be entered as an attribute value.
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The name of the product classification attribute.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the product classification attribute owner. This field is a polymorphic relationship field. **Relationship Name** Owner **Relationship Type** Lookup **Refers To** Group, User

Field	Details
`ProductClassificationId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The ID of the product classification that the attribute is associated with. This field is unique within your organization. This field is a relationship field. **Relationship Name** ProductClassification **Relationship Type** Lookup **Refers To** ProductClassification
`Sequence`	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The display sequence of the attribute when configuring the product during run time.
`Status`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The lifecycle status of the product classification attribute. Valid values are: • `Active` • `Draft` • `Inactive` The default value is `Draft`.
`StepValue`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update

Field	Details
	Description The increment or decrement by which a slider's value changes as the user adjusts the product classification attribute value. Available in API version 61.0 and later.
`ValueDescription`	**Type** textarea **Properties** Create, Nillable, Update **Description** The description of the value assigned to this attribute. This field takes on the value description from the attribute definition.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

ProductClassificationAttrFeed on page 2866
Feed tracking is available for the object.

ProductClassificationAttrHistory on page 2873
History is available for tracked fields of the object.

Sharing rules are available for the object.

ProductClassificationAttrShare on page 2877
Sharing is available for the object.

ProductComponentGrpOverride

Represents override information for a Product Component Group. The cardinality of the product component group can be overridden in the context of a product bundle. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

Product Catalog Management must be enabled to access this object.

Fields

Field	Details
`IsExcluded`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the product component group is excluded from the product bundle in the runtime. Excluding a group automatically excludes all child components of the group. The default value is `false`.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the product component override record was last referenced.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the product component override record was last viewed.
`MaxBundleComponents`	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The maximum number of components that can be added to a group.
`MinBundleComponents`	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The minimum number of components that must be added to a group.
`Name`	**Type** string

Field	Details
	Properties Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The name of the overridden product component group.
`OverrideContextId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The root bundle product in whose context the group cardinality is overridden. This field is a polymorphic relationship field. **Relationship Name** OverrideContext **Relationship Type** Lookup **Refers To** Product2
`OwnerId`	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The owner of the product component group override record. This field is a polymorphic relationship field. **Relationship Name** Owner **Relationship Type** Lookup **Refers To** Group, User
`ProductComponentGroupId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The ID of the product component group record. This field is a relationship field.

Field	Details
	Relationship Name ProductComponentGroup **Relationship Type** Lookup **Refers To** ProductComponentGroup

ProductDisqualification

Represents disqualification rules for products. The rules determine when the product doesn't qualify to be displayed to users. The rules are based on user context. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

Product Catalog Management must be enabled to access this object.

Fields

Field	Details
`EffectiveFromDate`	**Type** date **Properties** Create, Filter, Group, Sort, Update **Description** The date from which the disqualification rule for the product comes into effect.
`EffectiveToDate`	**Type** date **Properties** Create, Filter, Group, Sort, Update **Description** The date to which the disqualification rule for the product ceases to be in effect.
`IsDisqualified`	**Type** boolean

Field	Details
	Properties Defaulted on create, Filter, Group, Sort **Description** Indicates whether the product is disqualified based on the disqualification rules (`true`) or not (`false`). The default value is `false`.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the product disqualification record was last referenced.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the product disqualification record was last viewed.
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The name of the product disqualification record.
`OwnerId`	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The owner of the product disqualification record. This field is a polymorphic relationship field. **Relationship Name** Owner **Relationship Type** Lookup **Refers To** Group, User

Field	Details
ParentProductId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the immediate parent product in the product bundle hierarchy. This field is a relationship field. **Relationship Name** ParentProduct **Relationship Type** Lookup **Refers To** Product2
ProductId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The product for which the disqualification rule is defined. This field is a relationship field. **Relationship Name** Product **Relationship Type** Lookup **Refers To** Product2
Reason	**Type** textarea **Properties** Create, Nillable, Update **Description** The reason to disqualify the product.
RootProductId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the root product in the product bundle hierarchy.

Field	Details
	This field is a relationship field. **Relationship Name** RootProduct **Relationship Type** Lookup **Refers To** Product2

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

ProductDisqualificationFeed on page 2866
Feed tracking is available for the object.

ProductDisqualificationHistory on page 2873
History is available for tracked fields of the object.

ProductDisqualificationShare on page 2877
Sharing is available for the object.

ProductQualification

Represents qualification rules for products. The rules determine when the product qualifies to be displayed to users. The rules are based on user context. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

Product Catalog Management must be enabled to access this object.

Fields

Field	Details
`EffectiveFromDate`	**Type** date **Properties** Create, Filter, Group, Sort, Update **Description** The date from which the qualification rule for the product comes into effect.

Field	Details
EffectiveToDate	**Type** date **Properties** Create, Filter, Group, Sort, Update **Description** The date to which the qualification rule for the product ceases to be in effect.
IsQualified	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Indicates whether the product is qualified based on the qualification rules (`true`) or not (`false`). For a product to qualify, this field should be true. The default value is `false`.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the product qualification record was last referenced.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the product qualification record was last viewed.
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The name of the product qualification record.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update

Field	Details
	Description The owner of the product qualification record. This field is a polymorphic relationship field. **Relationship Name** Owner **Relationship Type** Lookup **Refers To** Group, User
ParentProductId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the immediate parent product in the product bundle hierarchy. This field is a relationship field. **Relationship Name** ParentProduct **Relationship Type** Lookup **Refers To** Product2
ProductId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The product for which the qualification rule is defined. This field is a relationship field. **Relationship Name** Product **Relationship Type** Lookup **Refers To** Product2
RootProductId	**Type** reference

Field	Details
	Properties Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the root product in the product bundle hierarchy. This field is a relationship field. **Relationship Name** RootProduct **Relationship Type** Lookup **Refers To** Product2

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

ProductQualificationFeed on page 2866
: Feed tracking is available for the object.

ProductQualificationHistory on page 2873
: History is available for tracked fields of the object.

ProductQualificationShare on page 2877
: Sharing is available for the object.

ProductRampSegment

Represents the ramp period within a ramp deal where terms, volumes, and other commitments change over time. This object is available in API version 62.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

Product Catalog Management must be enabled to access this object.

Fields

Field	Details
`DurationType`	**Type** picklist

Field	Details
	Properties Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The unit of time within which users can try the product for free. Valid values are: • `Days` • `Months`
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the product ramp segment was last referenced.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the product ramp segment was last viewed.
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The name of the product ramp segment.
`OwnerId`	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the owner of the product ramp segment. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User

Field	Details
ProductId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The ID of the product associated with the product ramp segment. This field is a relationship field. **Relationship Name** Product **Refers To** Product2
ProductSellingModelId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The ID of the product selling model associated with the product ramp segment. This field is a relationship field. **Relationship Name** ProductSellingModel **Refers To** ProductSellingModel
SegmentType	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** The time duration within the ramp deal where specific terms, volumes, and commitments are applied to the subscription product. Valid values are: • `Custom` • `FreeTrial` • `Yearly` The default value is `Yearly`.
TrialDuration	**Type** int

Field	Details
	Properties Create, Filter, Group, Nillable, Sort, Update **Description** The duration within which users can try the product for free.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

ProductRampSegmentFeed on page 2866
Feed tracking is available for the object.

ProductRampSegmentHistory on page 2873
History is available for tracked fields of the object.

ProductRampSegmentShare on page 2877
Sharing is available for the object.

ProductRelComponentOverride

Represents the cardinality overrides for product components in a bundle. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

Product Catalog Management must be enabled to access this object.

Fields

Field	Details
`DoesBundlePriceIncludeChild`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the bundle price includes the associated child component's price (`true`) or not (`false`). The default value is `false`.

Field	Details
`IsComponentRequired`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the component is a required component in the product bundle. The default value is `false`.
`IsDefaultComponent`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether this component is included in the product component group by default. The default value is `false`.
`IsExcluded`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the bundle excludes the component (`true`) or not (`false`). The default value is `false`.
`IsQuantityEditable`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the product component quantity can be edited (`true`) or not (`false`). The default value is `false`.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the product related component override record was last referenced.
`LastViewedDate`	**Type** dateTime

Field	Details
	Properties Filter, Nillable, Sort **Description** The date the product related component override record was last viewed.
`MaxQuantity`	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The maximum quantity for the product component in the product bundle.
`MinQuantity`	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The minimum quantity for the product component in the product bundle
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The name of the product related component override. The maximum length is 255 characters (of any type).
`OverrideContextId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The ID associated with the root product in a bundle. This field is a polymorphic relationship field. **Relationship Name** OverrideContext **Relationship Type** Lookup **Refers To** Product2

Field	Details
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The owner ID of the product related component override record. This field is a polymorphic relationship field. **Relationship Name** Owner **Relationship Type** Lookup **Refers To** Group, User
ProductRelatedComponentId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The ID associated with the product related component record. This field is a relationship field. **Relationship Name** ProductRelatedComponent **Relationship Type** Lookup **Refers To** ProductRelatedComponent
Quantity	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The default number of child product related components.
QuantityScaleMethod	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update

Field	Details
	Description The scaling method that determines how the child product quantity changes as the quantity of the parent product changes in the runtime cart. Possible values are: • `Constant` • `Proportional` The default value is `Proportional`.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

ProductRelComponentOverrideFeed **on page 2866**
Feed tracking is available for the object.

ProductRelComponentOverrideHistory **on page 2873**
History is available for tracked fields of the object.

ProductRelComponentOverrideShare **on page 2877**
Sharing is available for the object.

ProductSpecificationRecType

Represents the relationship between industry-specific product specifications and the product record type. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeSObjects()`, `query()`, `retrieve()`, `update()`, `upsert()`

Special Access Rules

Product Catalog Management must be enabled to access this object.

Fields

Field	Details
`DeveloperName`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** The unique name of the ProductSpecificationRecType object in the API. The name:

Field	Details
	• must be 40 characters or fewer. • can contain only underscores and alphanumeric characters. • must begin with a letter. • can contain only underscores and alphanumeric characters. • can't include spaces • can't end with an underscore • can't contain 2 consecutive underscores
`IsCommercial`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the product is sold commercially (`true`) or not (`false`)
`Language`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The combined language and locale ISO code, which controls the language of the Product Specification Record Type. Possible values are: • `da`—Danish • `de`—German • `en_US`—English • `es`—Spanish • `es_MX`—Spanish (Mexico) • `fi`—Finnish • `fr`—French • `it`—Italian • `ja`—Japanese • `ko`—Korean • `nl_NL`—Dutch • `no`—Norwegian • `pt_BR`—Portuguese (Brazil) • `ru`—Russian • `sv`—Swedish • `th`—Thai

Field	Details
	• `zh_CN`—Chinese (Simplified) • `zh_TW`—Chinese (Traditional)
`MasterLabel`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** Label for this Product Specification Record Type value. This display value is the internal label that doesn't get translated.
`NamespacePrefix`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The namespace prefix that is associated with this object. Each Developer Edition org that creates a managed package has a unique namespace prefix. Limit: 15 characters. You can refer to a component in a managed package by using the ***`namespacePrefix__componentName`*** The namespace prefix can have one of the following values. • In Developer Edition orgs, `NamespacePrefix` is set to the namespace prefix of the org for all objects that support it, unless an object is in an installed managed package. In that case, the object has the namespace prefix of the installed managed package. This field's value is the namespace prefix of the Developer Edition org of the package developer. • In orgs that aren't Developer Edition orgs, `NamespacePrefix` is set only for objects that are part of an installed managed package. All other objects have no namespace prefix.
`ProductSpecificationType`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The product specification type that's associated with the record type.
`RecordTypeId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update

Field	Details
	Description The ID of the record type that's associated with the product specification type. This field is a relationship field. **Relationship Name** RecordType **Relationship Type** Lookup **Refers To** RecordType

ProductSpecificationType

Represents the type of product specification provided by the user to make the product terminology unique to an industry. A product specification type is associated with a product specification record type. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeSObjects()`, `query()`, `retrieve()`, `update()`, `upsert()`

Special Access Rules

Product Catalog Management must be enabled to access this object.

Fields

Field	Details
`Description`	**Type** textarea **Properties** Create, Nillable, Update **Description** The description of the Product Specification Type.
`DeveloperName`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** The unique name of the Product Specification Type object in the API. The name: • must be 40 characters or fewer. • can contain only underscores and alphanumeric characters.

Field	Details
	• must begin with a letter. • can contain only underscores and alphanumeric characters. • can't include spaces • can't end with an underscore • can't contain 2 consecutive underscores In managed packages, this field prevents naming conflicts on package installations. With this field, a developer can change the object's name in a managed package and the changes are reflected in a subscriber's organization.
`Language`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The combined language and locale ISO code, which controls the language of the Product Specification Type. Possible values are: • `da`—Danish • `de`—German • `en_US`—English • `es`—Spanish • `es_MX`—Spanish (Mexico) • `fi`—Finnish • `fr`—French • `it`—Italian • `ja`—Japanese • `ko`—Korean • `nl_NL`—Dutch • `no`—Norwegian • `pt_BR`—Portuguese (Brazil) • `ru`—Russian • `sv`—Swedish • `th`—Thai • `zh_CN`—Chinese (Simplified) • `zh_TW`—Chinese (Traditional)
`MasterLabel`	**Type** string **Properties** Create, Filter, Group, Sort, Update

Field	Details
	Description Label for this Product Specification Type value. This display value is the internal label that doesn't get translated.
`NamespacePrefix`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The namespace prefix that is associated with this object. Each Developer Edition org that creates a managed package has a unique namespace prefix. Limit: 15 characters. You can refer to a component in a managed package by using the ***`namespacePrefix_componentName`*** The namespace prefix can have one of the following values. • In Developer Edition orgs, `NamespacePrefix` is set to the namespace prefix of the org for all objects that support it, unless an object is in an installed managed package. In that case, the object has the namespace prefix of the installed managed package. This field's value is the namespace prefix of the Developer Edition org of the package developer. • In orgs that aren't Developer Edition orgs, `NamespacePrefix` is set only for objects that are part of an installed managed package. All other objects have no namespace prefix.

Product Catalog Management Fields on Standard Objects

Product Catalog Management adds standard and custom fields to some standard Salesforce objects. These fields are available only in orgs where Product Catalog Management is enabled. This object is available in API version 60.0 and later.

Product Catalog Management Fields on Attribute Definition

Standard and custom fields extend the standard Attribute Definition object for use in Product Catalog Management.

Product Catalog Management Fields on Product2

Standard and custom fields extend the standard Product2 object for use in Product Catalog Management to represent information about products.

Product Catalog Management Fields on Product Catalog

Standard and custom fields extend the standard Product Catalog object for use in Product Catalog Management. This object is available in API version 60.0 and later.

Product Catalog Management Fields on Product Category

Standard and custom fields extend the standard Product Category object for use in Product Catalog Management.

Product Catalog Management Fields on Product Component Group

Standard and custom fields extend the standard Product Component Group object for use in Product Catalog Management.

Product Catalog Management Fields on Product Related Component

Standard and custom fields extend the standard Product Related Component object for use in Product Catalog Management.

Product Catalog Management Fields on Product Relationship Type

Standard and custom fields extend the standard Product Relationship Type object for use in Product Catalog Management.

Product Catalog Management Fields on Product Selling Model Option

Standard and custom fields extend the standard Product Selling Model Option object for use in Product Catalog Management. This object is available in API version 60.0 and later.

Product Catalog Management Fields on Attribute Definition

Standard and custom fields extend the standard Attribute Definition object for use in Product Catalog Management.

Fields

Field	Details
`DataType`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort **Description** The data type of the attribute definition. Valid values are: • `Checkbox` • `Currency`—Available in API version 61.0 and later • `Date` • `Datetime` • `Number` • `Percent`—Available in API version 61.0 and later • `Picklist` • `Text`

SEE ALSO:

Attribute Definition

Product Catalog Management Fields on Product2

Standard and custom fields extend the standard Product2 object for use in Product Catalog Management to represent information about products.

Fields

Field	Details
`Based On`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the product classification from which this product inherits. This field is a relationship field. **Relationship Name** BasedOn **Relationship Type** Lookup **Refers To** ProductClassification
`Help Text`	**Type** textarea **Properties** Create, Nillable, Update **Description** The help text that appears at runtime for the product. The maximum size is 32,000 alphanumeric characters. The help text can include these special characters: @ ! - < > * ? + = % # ( ) / \ & ' £ € $ ”.
`Availability Date`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date when the product is available.
`CanRamp`	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Indicates whether the product's terms, volumes, and other commitments can be ramped (true) at run time or not (false) The default value is `false`.

Field	Details
`Discontinued Date`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date when the product is discontinued.
`End Of Life Date`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date and time after which a product isn't supported, ordered, or maintained.
`Specification Type`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The type of product specification that's being created.
`DecompositionScope`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The number of fulfillment order line items that must be generated. Available in API version 61.0 and later. Valid values are: • `Account` • `Bundle` • `Order` • `OrderLineItem`
`FulfillmentQtyCalcMethod`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Determines whether the quantity of fulfillment order line items must always be one or must be aggregated from the source line items. Available in API version 61.0 and later.

Field	Details
	Valid values are: • `Aggregate` • `AlwaysOne`
`UsageModelType`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The type of usage model for a product or service. Anchor is the main subscription product or service. Pack is the add-on product or service that grants additional usage resources for consumption. Available in API version 62.0 and later. Valid values are: • `Anchor` • `Pack`

SEE ALSO:

Product2

Product Catalog Management Fields on Product Catalog

Standard and custom fields extend the standard Product Catalog object for use in Product Catalog Management. This object is available in API version 60.0 and later.

Fields

Field	Details
`Code`	**Type** string **Properties** Create, Filter, Group, idLookup, Nillable, Sort, Update **Description** A unique ID associated with the catalog. The maximum size is 80 alphanumeric characters.
`Description`	**Type** textarea **Properties** Create, Nillable, Update

Field	Details
	Description The description of the catalog that's used during design time. The maximum size is 255 alphanumeric characters.
`EffectiveEndDate`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date after which the product catalog is unavailable to end users.
`EffectiveStartDate`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date on which the product catalog is available to end users.
`CatalogType`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The category of an entry in the catalog. Possible values are: • `Sales` • `ServiceProcess`—Service Process The default value is `Sales`.

SEE ALSO:

Product Catalog

Product Catalog Management Fields on Product Category

Standard and custom fields extend the standard Product Category object for use in Product Catalog Management.

Fields

Field	Details
`Code`	**Type** string **Properties** Create, Filter, Group, idLookup, Nillable, Sort, Update **Description** A unique ID associated with the catalog. The maximum size is 80 alphanumeric characters.
`IsNavigational`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the category or subcategory is shown in the menu as a navigational breadcrumb (`true`) or not (`false`). Available in API version 62.0 and later. The default value is `false`.

SEE ALSO:

Product Category

Product Catalog Management Fields on Product Component Group

Standard and custom fields extend the standard Product Component Group object for use in Product Catalog Management.

Fields

Field	Details
`ParentGroupId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The parent product component group in a nested group hierarchy for the same parent product. Available in API version 62.0 and later. This field is a relationship field. **Relationship Name** ParentGroup

Field	Details
	Refers To ProductComponentGroup

SEE ALSO:

Product Component Group

Product Catalog Management Fields on Product Related Component

Standard and custom fields extend the standard Product Related Component object for use in Product Catalog Management.

Fields

Field	Details
`ChildProductClassificationId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort **Description** The child product classification that's associated with a product. This field is a relationship field. **Relationship Name** ChildProductClassification **Refers To** ProductClassification
`QuoteVisibility`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies the product-related component to display as a quote line item in the Transaction Line Editor and the quote document. The default value is Always. Possible values are: • Always • Transaction Line Editor Only • Quote Document Only

Field	Details
	• Never

SEE ALSO:

Product Related Component

Product Catalog Management Fields on Product Relationship Type

Standard and custom fields extend the standard Product Relationship Type object for use in Product Catalog Management.

Fields

Field	Details
AssociatedProductRoleCat	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort **Description** The role that the associated component plays in the relationship. Valid values are: • BundleComponent— The associated product is part of a bundle. • ClassificationComponent— The associated component is a product classification. Available in API version 61.0 and later

SEE ALSO:

Product Relationship Type

Product Catalog Management Fields on Product Selling Model Option

Standard and custom fields extend the standard Product Selling Model Option object for use in Product Catalog Management. This object is available in API version 60.0 and later.

Fields

Field	Details
IsDefault	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update

Field	Details
	Description Indicates the default product selling model for a product. Setting a product selling model as default is optional. A product can only have one default product selling model. The default value is `false`.

SEE ALSO:

Product Selling Model Option

Product Catalog Management Business APIs

Use primitive APIs of Product Catalog Management that serve catalog definitions to users or applications.

This table lists the available Product Catalog Management resources.

Resource	Description
`/connect/pcm/catalogs` (POST)	Retrieve, search, filter, or sort catalog records.
`/connect/pcm/catalogs/`***`catalogId`*** (GET)	Retrieve details of catalog records based on a catalog ID.
`/connect/pcm/catalogs/`***`catalogId`***`/categories` (GET)	Retrieve the root-level categories of a catalog based on a catalog ID, or subcategories based on a parent category. You can also search, filter, or sort the categories.
`/connect/pcm/categories/`***`categoryId`*** (GET)	Retrieve details of individual category records based on a category ID.
`/connect/pcm/products` (POST)	Retrieve products. You can also search, filter, or sort the products.
`/connect/pcm/products/`***`productId`*** (GET)	Retrieve details of individual product records or a bundle based on a product ID.
`/revenue/product-catalog-management/product-classifications/details` (POST)	Retrieve the details for a list of product classification records.
`/revenue/product-discovery/products/recommendations` (POST)	Get a list of recommended products based on your underlying business rules.

Resource	Description
`/revenue/product-configurator/rules/actions/execute` (POST)	This API is used in Product Catalog Management to disable rules, get product recommendations, and get message rules.
`/connect/pcm/products/bulk` (POST)	Retrieve details for multiple products.
`/connect/pcm/products/variants` (POST)	Retrieve the variation product associated with one or more parent variant products.
`/connect/pcm/index/configurations` (GET, PUT)	Retrieve the saved index configurations. Additionally, you can persist the index configuration.
`/connect/pcm/relatedRecords/`***`entityName`*** (POST)	Retrieve related ProductRampSegment or ProductUsageGrant records for Product2 object.
`/connect/pcm/index/snapshots` (GET)	Retrieve the created snapshots and snapshot indexes.
`/connect/pcm/index/deploy` (POST)	Create indexes for a snapshot. Indexes improve search results and make it easier to find products at run time through search terms.
`/connect/pcm/index/setting` (GET, PATCH)	Fetch and update settings related to indexing and search.
`/connect/pcm/index/error` (GET)	Get the count and details of the errors that occurred during the indexing process.
`/connect/pcm/deep-clone` (POST)	Copy related records of an object along with the main product record.
`/connect/pcm/unit-of-measure/info` (GET)	Get details about the unit of measure for a specific set of records.
`/connect/pcm/unit-of-measure/rounded-data` (POST)	Round off and scale decimal data for a specific set of fields.

Resources

Learn more about the available Product Catalog Management API resources.

Request Bodies

Learn more about the available Product Catalog Management API request bodies.

Response Bodies

Learn more about the available Product Catalog Management API response bodies.

SEE ALSO:

Connect REST API Developer Guide: Introduction

Resources

Learn more about the available Product Catalog Management API resources.

Bulk Product Details (POST)

Retrieve details for multiple products.

Catalog List (POST)

Retrieve, search, filter, or sort catalog records.

Catalog By ID (GET)

Retrieve details of catalog records based on a catalog ID.

Categories List (GET)

Retrieve the root-level categories of a catalog based on a catalog ID, or subcategories based on a parent category. You can also search, filter, or sort the categories.

Category By ID (GET)

Retrieve details of individual category records based on a category ID.

Deep Clone (POST)

Copy related records of an object along with the main product record.

Index Configuration Collection (GET, PUT)

Retrieve the saved index configurations. Additionally, you can persist the index configuration.

Index Setting (GET, PATCH)

Fetch and update settings related to indexing and search.

Product Classification Details (POST)

Retrieve the details for a list of product classification records.

Product Classification List (POST)

Retrieve a list of product classification records. You can also search, filter, or sort the product classifications.

Product Details (GET)

Retrieve details of individual product records or a bundle based on a product ID.

Product Related Records List (POST)

Retrieve related ProductRampSegment or ProductUsageGrant records for Product2 object.

Products List (POST)

Retrieve products. You can also search, filter, or sort the products.

Product Variants (POST)
Retrieve the variation product associated with one or more parent variant products.

Snapshot Collection (GET)
Retrieve the created snapshots and snapshot indexes.

Snapshot Deployment (POST)
Create indexes for a snapshot. Indexes improve search results and make it easier to find products at run time through search terms.

Snapshot Index Error (GET)
Get the count and details of the errors that occurred during the indexing process.

Unit of Measure Info (GET)
Get details about the unit of measure for a specific set of records.

Unit of Measure Rounded Data (POST)
Round off and scale decimal data for a specific set of fields.

Bulk Product Details (POST)

Retrieve details for multiple products.

Resource

```
/connect/pcm/products/bulk
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/pcm/products/bulk
```

Available version
61.0

HTTP methods
POST

Request body for POST

JSON example

```
{
  "correlationId": "cbfabffb-093f-45b3-8c2d-88b4acbd4867",
  "productIds": [
    "01tT1000000F0afIAC",
    "01tT1000000F0afIAC"
  ],
  "uptoLevel": 1,
  "language": "french",
  "additionalFields": {
    "Product2": {
      "fields": [
        "code__c"
      ]
    },
    "ProductAttributeDefinition": {
      "fields": [
        "scope"
      ]
```

```
    }
  }
}
```

This example shows a sample request to fetch data from Enterprise Product Catalog.

```
{
  "productIds": [
    "01tLT000009vGXfYAM"
  ],
  "catalogSystems": [
    "epc"
  ],
  "additionalFields": {
    "AttributeDefinition": {
      "fields": [
        "OptOutAssetization",
        "OptOutDecompositionAction",
        "OptOutSupplementalAction"
      ]
    }
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`additional Fields`	Map<String, Additional Fields Input>	Map of object and list of additional standard or custom fields to be included in the response. The supported objects are: • Product2 • ProductAttributeDefinition—If the fields defined for the ProductAttributeDefinition object aren't available for the ProductClassificationAttr object, then the API request fails. If the Dynamic Revenue Orchestrator permission is enabled, then this property supports AttributeDefinition as key with these supported values. • `OptOutAssetization` • `OptOutDecompositionAction` • `OptOutSupplementalAction`	Optional	61.0

Name	Type	Description	Required or Optional	Available Version
`catalogSystems`	String[]	Name of the catalog system. Valid values are: • `epc`—Enterprise Product Catalog • `pcm`—Product Catalog Management Although this property accepts a list, you can pass only one value. If you don't specify a value, the default behavior is to fetch data from the `pcm` catalog.	Optional	66.0
`correlationId`	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Optional	61.0
`language`	String	Custom language that you can specify to get translated data for the fields of an object that's enabled for translation. See Translate Product and Product Category Data.	Optional	64.0
`productIds`	String[]	List of product IDs that details must be returned for. If any product ID is blank, invalid, or not found, then the request is processed with valid and available product IDs.	Required	61.0
`uptoLevel`	Integer	Hierarchy level to follow to return the product details. For a bundle, this property determines the number of levels of child components to be returned. You can specify up to a hierarchy level of 1. If unspecified, the default level is the full bundle hierarchy.	Optional	61.0

Response body for POST

Products Output

Catalog List (POST)

Retrieve, search, filter, or sort catalog records.

Resource

```
/connect/pcm/catalogs
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/pcm/catalogs
```

Available version

60.0

Requires Chatter

No

HTTP methods

POST

Note: The POST method is used to retrieve the catalog records instead of the GET method as a request payload is sent to filter the records.

Request body for POST

JSON example

This example shows how to retrieve catalogs that contain `apple` in the catalog name.

```
{
"pageSize": 100,
"offset": 0,
"language": "french",
"filter": {
"criteria": [
{
"property": "name",
"operator": "contains",
"value": "apple"
}
]
}
}
```

This example shows how to retrieve catalogs with `ServiceProcess` as the catalog type.

```
{
"pageSize": 100,
"offset": 0,
"sort": {
  "orders": [
   {
   "property": "name",
   "direction": "desc"
   }
  ]
 },
"filter": {
"criteria": [
{
"property": "catalogType",
"operator": "eq",
"value": "ServiceProcess"
}
]
```

```
}
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
correlationId	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Optional	60.0
filter	Filter	Criteria to filter the records. Filters are applicable to the fields of the ProductCatalog object. The supported operators are: • `eq` • `in` • `contains` The supported properties are `name` and `catalogType`.	Optional	60.0
language	String	Custom language that you can specify to get translated data for the fields of an object that's enabled for translation. See Translate Product and Product Category Data.	Optional	64.0
offset	Integer	Number of records to skip. The default value is 0.	Optional	60.0
pageSize	Integer	Number of records per page. Valid values are from 1 through 100. If unspecified, defaults to 100.	Optional	60.0
sort	Sort	Sort order of the catalog records. The supported operators are: • `asc` • `desc`	Optional	60.0

Response body for POST

Catalogs Output

Catalog By ID (GET)

Retrieve details of catalog records based on a catalog ID.

Resource

```
/connect/pcm/catalogs/catalogId
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/pcm/catalogs/0ZST100000000kUOAQ
```

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/pcm/catalogs/0ZST100000000kUOAQ?language=spanish
```

Available version

60.0

Requires Chatter

No

HTTP methods

GET

Request parameters for GET

Parameter Name	Type	Description	Required or Optional	Available Version
correlationId	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Optional	60.0
fields	String[]	For internal use only.	Optional	60.0
language	String	Custom language that you can specify to get translated data for the fields of an object that's enabled for translation. See Translate Product and Product Category Data.	Optional	64.0

Response body for GET

Catalogs Output

Categories List (GET)

Retrieve the root-level categories of a catalog based on a catalog ID, or subcategories based on a parent category. You can also search, filter, or sort the categories.

Resource

```
/connect/pcm/catalogs/catalogId/categories
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/pcm/catalogs/0ZST100000000kUOAQ/categories
```

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/pcm/catalogs/0ZST100000000kUOAQ/categories?language=spanish
```

Available version

60.0

Requires Chatter

No

HTTP methods

GET

Request parameters for GET

Parameter Name	Type	Description	Required or Optional	Available Version
`correlationId`	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Optional	60.0
`depth`	Integer	Number of levels in the category hierarchy to return. The default value is 1.	Optional	60.0
`fields`	String[]	For internal use only.	Optional	60.0
`language`	String	Custom language that you can specify to get translated data for the fields of an object that's enabled for translation. See Translate Product and Product Category Data.	Optional	64.0
`parent CategoryId`	String	ID of the category to fetch the associated hierarchy of subcategories. If unspecified, then the root-level categories are returned.	Optional	60.0

Response body for GET

Categories Output

Category By ID (GET)

Retrieve details of individual category records based on a category ID.

Resource

```
/connect/pcm/categories/categoryId
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/pcm/categories/0ZGT100000000qqOAA
```

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/pcm/categories/0ZGT100000000qqOAA?language=spanish
```

Available version

60.0

Requires Chatter
No

HTTP methods
GET

Request parameters for GET

Parameter Name	Type	Description	Required or Optional	Available Version
`correlationId`	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Optional	60.0
`fields`	String[]	For internal use only.	Optional	60.0
`language`	String	Custom language that you can specify to get translated data for the fields of an object that's enabled for translation. See Translate Product and Product Category Data.	Optional	64.0

Response body for GET
Categories Output

Deep Clone (POST)

Copy related records of an object along with the main product record.

Resource

```
/connect/pcm/deep-clone
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/pcm/deep-clone
```

Available version
63.0

HTTP methods
POST

Request body for POST

JSON example

```
{
  "mainRecordId": "01tSG0000028kcSYAQ",
  "mainObjectApiName": "Product2",
  "mainRecordFieldValues": {
    "Name": "New Cloud Storage"
```

```
    }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
mainObjectApiName	String	API name of the object. The supported object is Product2.	Required	63.0
mainRecordFieldValues	Map<String, String>	Mapping of the API name of the field to its value. The values passed through this map are set for the created record. You can pass the Name field only through this map.	Optional	63.0
mainRecordId	String	ID of the record.	Required	63.0

Response body for POST

Deep Clone Response

Index Configuration Collection (GET, PUT)

Retrieve the saved index configurations. Additionally, you can persist the index configuration.

Resource

```
/connect/pcm/index/configurations
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/pcm/index/configurations
```

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/pcm/index/configurations?includeMetadata=false&fieldTypes=Standard,Custom
```

Available version

62.0

HTTP methods

GET, PUT

Request parameters for GET

Parameter Name	Type	Description	Required or Optional	Available Version
correlationId	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Optional	62.0

Parameter Name	Type	Description	Required or Optional	Available Version
`fieldTypes`	String[]	Filters and returns only the persisted index configurations, based on the index configuration type specified in the query parameters. The supported types of filters are: • `STANDARD` • `CUSTOM` • `ProductDynamicAttribute` • `ProductAttributeDefinitionStandard` • `ProductAttributeDefinitionCustom`	Optional	62.0
`include Metadata`	Boolean	Indicates whether to include metadata (`true`) or not (`false`).	Optional	62.0

Response body for GET

Index Configuration Collection

Request body for PUT

JSON example


```
{
  "correlationId": "8545b5aa-f3e6-429a-8f21-9cc4ce50b1d7",
  "indexConfigurations": [
    {
      "attributeDefinitionId": "0tjT1000000002bIAA",
      "name": "Color",
      "type": "ProductDynamicAttribute",
      "isSearchable": true
    },
    {
      "attributeFieldId": "00Nxx000001FwnABII",
      "name": "Message__c",
      "type": "Custom",
      "isSearchable": true
    },
    {
      "name": "Code",
      "type": "Standard",
      "isSearchable": true
    },
    {
      "facetDisplayRank": 1,
      "isFacetable": false,
      "isSearchable": true,
      "name": "Family",
      "type": "Standard"
    }
```

```
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`correlationId`	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Optional	62.0
`index Configurations`	Index Configuration Input[]	List of index configurations.	Required	62.0

Response body for PUT

Index Configurations Update

Index Setting (GET, PATCH)

Fetch and update settings related to indexing and search.

Resource

```
/connect/pcm/index/setting
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/pcm/index/setting
```

Available version

63.0

HTTP methods

GET, PATCH

Response body for GET

Index Setting Results

Request body for PATCH

JSON example

```
{
    "setting" : {
        "supportedLanguages" : ["en_US","ja","es","nl_NL"],
        "defaultLanguage" : "en_US",
        "productsGrouping": "GROUPING_VARIATION"
    }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`setting`	Setting Input[]	Object containing the setting-related details.	Required	63.0

Request parameters for PATCH

Parameter Name	Type	Description	Required or Optional	Available Version
`settingId`	String	ID of the setting to update the details for.	Required	63.0

Response body for PATCH

Index Setting Update

Product Classification Details (POST)

Retrieve the details for a list of product classification records.

This API fetches metadata, attributes, and attribute categories associated with product classifications across supported catalog systems.

Resource

```
/revenue/product-catalog-management/product-classifications/details
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/revenue/product-catalog-management/product-classifications/details
```

Available version

66.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "productClassificationIds": [
    "01txx0000006iFMAAY",
    "01txx0000006iGxAAY"
  ],
  "catalogSystems": [
    "epc"
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`catalogSystems`	String[]	Name of the catalog system. Valid value is: • `epc`—Enterprise Product Catalog	Optional	66.0
`productClassificationIds`	String[]	List of product classification IDs for which you want to retrieve product details. In the `epc` catalog system, these values are the `Product2` record IDs.	Required	66.0

Response body for POST

Product Classification Details Collection on page 214

Product Classification List (POST)

Retrieve a list of product classification records. You can also search, filter, or sort the product classifications.

This API retrieves a paginated list of product classification records from the Product Catalog Management (PCM) or Enterprise Product Catalog (EPC) catalog system. You can search for product classifications by name, apply filters, and sort the results.

Resource

```
/revenue/product-catalog-management/product-classifications/list
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/revenue/product-catalog-management/product-classifications/list
```

Available version

67.0

HTTP methods

POST

Note: POST methods typically create an item, but for this resource POST is used to retrieve information.

Request body for POST

JSON example

```
{
  "catalogSystem": "pcm",
  "searchTerm": "Mobile",
  "filter": {
    "criteria": [
      {
        "property": "name",
        "operator": "contains",
        "value": "Mobile"
      }
```

```
    ]
  },
  "sort": {
    "orders": [
      {
        "property": "name",
        "direction": "asc"
      }
    ]
  },
  "pageSize": 25,
  "offset": 0
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`catalogSystem`	String	Name of the catalog system. Valid values are: • `pcm`—Product Catalog Management • `epc`—Enterprise Product Catalog If unspecified, the default catalog system is `pcm`.	Optional	67.0
`filter`	Criteria Input	Criteria to filter the product classification records. The supported property is `name`. The supported operators are: • `eq` • `in` • `contains` If multiple criteria are specified, they're combined by using the `and` operator. Each criterion supports only the `property`, `operator`, and `value` fields.	Optional	67.0
`offset`	Integer	Number of records to skip. The default value is 0.	Optional	67.0
`pageSize`	Integer	Specifies the number of records per page. Valid values are 5, 10, 25, 50, and 100. If unspecified, the default value is 100.	Optional	67.0
`searchTerm`	String	String used to search for product classifications with the product classification name containing the search term.	Optional	67.0

Name	Type	Description	Required or Optional	Available Version
`sort`	Order Input	Sort order for the product classifications. If unspecified, the default sort order is by name in ascending order.	Optional	67.0

Response body for POST

Product Classification List Collection on page 216

Product Details (GET)

Retrieve details of individual product records or a bundle based on a product ID.

Resource

```
/connect/pcm/products/productId
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/pcm/products/01tT1000000F0afIAC
```

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/pcm/products/01tT1000000F0afIAC?language=spanish
```

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/pcm/products/01tT1000000F0afIAC/catalogSystems=epc
```

Available version

60.0

Requires Chatter

No

HTTP methods

GET

Note: You must invoke this API request by using GET method only. If the request is invoked by using POST method, the request is considered as a Products List API request.

Request parameters for GET

Parameter Name	Type	Description	Required or Optional	Available Version
`catalogSystems`	String[]	Name of the catalog system. Valid values are: • `epc`—Enterprise Product Catalog • `pcm`—Product Catalog Management Although this parameter accepts a list, you can pass only one value. If you don't specify a value, the default behavior is to fetch data from the `pcm` catalog.	Optional	66.0

Parameter Name	Type	Description	Required or Optional	Available Version
`correlationId`	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Optional	60.0
`fields`	String[]	For internal use only.	Optional	60.0
`language`	String	Custom language that you can specify to get translated data for the fields of an object that's enabled for translation. See Translate Product and Product Category Data.	Optional	64.0

Response body for GET

Products

Product Related Records List (POST)

Retrieve related ProductRampSegment or ProductUsageGrant records for Product2 object.

Resource

```
/connect/pcm/relatedRecords/entityName
```

The supported entity or object is Product2.

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/pcm/relatedRecords/product2
```

Available version

62.0

HTTP methods

POST

Note: POST methods typically create an item, but for this resource POST is used to retrieve information.

Request body for POST

JSON example

```
{
  "recordIds": [
    "01txx0000006i44AAA",
    "01txx0000006i5gAAA"
  ],
  "relatedObjectNodes": [
    {
      "relatedObjectAPIName": "ProductRampSegment",
      "pageSize": 20,
      "offSet": 0
```

```
    },
    {
      "relatedObjectAPIName": "ProductUsageGrant",
      "pageSize": 10,
      "offSet": 0,
      "filter": {
        "criteria": [
          {
            "property": "status",
            "operator": "eq",
            "value": "active"
          },
          {
            "property": "effectivestartdate",
            "operator": "lte",
            "value": "2024-06-25"
          },
          {
            "criteriaType": "CustomWhereCondition",
            "value": "(effectiveenddate = null OR effectiveenddate >= 2024-06-25)"
          }
        ]
      }
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`correlationId`	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Optional	62.0
`recordIds`	String[]	List of record IDs to return the relatedObjects records for. The maximum number of record IDs supported is 20.	Required	62.0
`relatedObjectNodes`	Related Object Node Input[]	List of nodes for the related objects. The maximum number of related object nodes supported is two.	Required	62.0

Response body for POST

Related Records List

Products List (POST)

Retrieve products. You can also search, filter, or sort the products.

Resource

```
/connect/pcm/products
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/pcm/products?productClassificationId=11BT10000004C9SMAU
```

Available version

60.0

Requires Chatter

No

HTTP methods

POST

Note: POST methods typically create an item, but for this resource POST is used to retrieve information.

Request body for POST

JSON example

This example is a search request for products that contain `Bundle Product` in the product name.

```
{
  "catalogIds": [
    "0ZST10000004D03OAE"
  ],
  "language": "spanish",
  "filter": {
    "criteria": [
      {
        "property": "name",
        "operator": "contains",
        "value": "Bundle Product"
      }
    ]
  },
  "relatedObjectFilters": [
    {
      "objectName": "ProductSpecificationRecType",
      "criteria": [
        {
          "property": "IsCommercial",
          "operator": "eq",
          "value": false
        }
      ],
      "additionalFields": {
        "Product2": {
          "fields": [
            "code__c"
          ]
        }
      }
    }
```

```
  ]
}
```

This example is a search request for products that contain the specified search term in the product name and are part of a catalog.

```
  {
  "catalogIds": [
    "0ZSDU0000002Og54AE"
  ],
  "searchTerm": "Slack"
}
```

Query parameters

Name	Type	Description	Required or Optional	Available Version
`productClassificationId`	String	ID of the product classification template that contains a collection of dynamic attributes and can be reused to create multiple products. Products that are based on a product classification inherit all the attributes of the product classification. Specify either the product classification ID, list of category IDs, or list of catalog IDs to retrieve products.	Optional	60.0

Properties

Name	Type	Description	Required or Optional	Available Version
`additionalFields`	Map<String, Additional Fields Input>	Map of the object and the list of standard and custom fields to be queried along with the standard response. The supported object is Product2.	Optional	61.0
`catalogIds`	List<String>	List of comma-separated catalog IDs. The API returns the list of product records that are associated with the specified catalog IDs. Specify either the product classification ID, list of category IDs, or list of catalog IDs to retrieve products.	Optional	60.0
`categoryIds`	List<String>	List of comma-separated category IDs. The API returns the list of product records	Optional	60.0

Name	Type	Description	Required or Optional	Available Version
		that are associated with the specified category IDs. If unspecified, then returns all products that are added to at least one category. Specify either the product classification ID, list of category IDs, or list of catalog IDs to retrieve products.		
`correlationId`	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Optional	60.0
`filter`	Criteria Input	Criteria to filter the records. Filters are applicable to the fields of Product2 object. The supported operators are: • `eq` • `in` • `contains`—This value isn't applicable if the **Use Indexed Data For Product Listing and Search** toggle from the Product Discovery Settings page from Setup is enabled. • `gt`—Specifies a greater than criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only. • `lt`—Specifies a less than criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only. • `gte`—Specifies a greater than or equal to criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only. • `lte`—Specifies a less than or equal to criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only. If multiple criteria are specified, then the criteria are combined by using the `and` operator.	Optional	60.0

Name	Type	Description	Required or Optional	Available Version
		The supported properties are `name`, `description`, and `isActive`. If the **Use Indexed Data For Product Listing and Search** toggle from the Product Discovery Settings page from Setup is enabled, then the supported property is `name` only.		
`language`	String	Custom language that you can specify to get translated data for the fields of an object that's enabled for translation. See Translate Product and Product Category Data.	Optional	64.0
`offset`	Integer	Number of records to skip. The default value is 0.	Optional	60.0
`pageSize`	Integer	Specifies the number of records per page. Valid values are from 1 through 200. If the value is unspecified, it defaults to 100.	Optional	60.0
`relatedObjectFilters`	Related Object Filter[]	Criteria for the related objects to filter the records. The supported operator is `eq`. The supported object is `ProductSpecificationRecType`. The supported values are `true` and `false`. The supported property is `IsCommercial`.	Optional	60.0
`searchTerm`	String	String used to get products with the product name containing the search term. See Search Considerations When Using Indexed Data.	Optional	62.0
`sort`	Sort	Sort order for the products. If the **Use Indexed Data For Product Listing and Search** toggle from the Product Discovery Settings page from Setup is enabled, then you can sort products by using name only.	Optional	60.0

Response body for POST

Products Output

Product Variants (POST)

Retrieve the variation product associated with one or more parent variant products.

A parent variant product is a non-purchasable product that groups related variations. Use this API to retrieve the mapping between parent variant product IDs and their associated variation product IDs.

Resource

```
/connect/pcm/products/variants
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/pcm/products/variants
```

Available version

67.0

Requires Chatter

No

HTTP methods

POST

Note: POST methods typically create an item, but for this resource POST is used to retrieve information.

Request body for POST

JSON example

```
{
  "correlationId": "9b6bc520-3c82-4d6c-a458-47590370681a",
  "parentVariantsIds": [
    "01tT1000000F0afIAC",
    "01tT1000000F0agIAC"
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
correlationId	String	Unique ID to track and associate related events or transactions. If unspecified, a Universally Unique Identifier (UUID) is generated.	Optional	67.0
parentVariantsIds	String[]	List of product IDs for parent variant products whose variations you want to retrieve. If any product ID is blank, invalid, or not found, the API skips that ID and processes the remaining valid IDs. The skipped IDs appear in the `inValidProductIds` response property. If a product ID is valid but isn't a parent variant product, the API	Required	67.0

Name	Type	Description	Required or Optional	Available Version
		includes it in the `nonVariantParentIds` response property.		

Response body for POST

Product Variants

Snapshot Collection (GET)

Retrieve the created snapshots and snapshot indexes.

Resource

```
/connect/pcm/index/snapshots
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/pcm/index/snapshots
```

Available version

62.0

HTTP methods

GET

Query Parameter for GET

Parameter Name	Type	Description	Required or Optional	Available Version
`numberOfIndexLogs`	Integer	Number of index logs to include in the response. Specify a number from 0 through 100. The default value is 25.	Optional	63.0

Response body for GET

Snapshot Collection

Snapshot Deployment (POST)

Create indexes for a snapshot. Indexes improve search results and make it easier to find products at run time through search terms.

Resource

```
/connect/pcm/index/deploy
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/pcm/index/deploy
```

Available version

62.0

HTTP methods

POST

Request body for POST

JSON example

This example shows a sample request to build a new snapshot with immediate activation.

```
{
  "snapshot": {
    "activationType": "IMMEDIATE"
  },
  "buildType": "FULL"
}
```

This example shows a sample request to rebuild a snapshot in the `active` status.

```
{
  "snapshot": {
    "activationType": "IMMEDIATE",
    "id": "1Avxx0000005DFe1AM"
  },
  "buildType": "FULL"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`buildType`	String	Build type of the snapshot index. Valid value is: • `FULL`—Specifies a full index build. • `INCREMENTAL`—Specifies an incremental index build. Available from API version 63.0 and later.	Required	62.0
`snapshot`	Run-time Catalog Snapshot Input[]	Snapshot to deploy.	Required	62.0

Response body for POST

Snapshot Deployment

Snapshot Index Error (GET)

Get the count and details of the errors that occurred during the indexing process.

Resource

```
/connect/pcm/index/error
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/pcm/index/error
```

Available version

63.0

HTTP methods

GET

Request parameters for GET

Parameter Name	Type	Description	Required or Optional	Available Version
indexId	String	ID of the index.	Required	63.0
snapshot IndexId	String	ID of the snapshot index.	Required	63.0

Response body for GET

Snapshot Index Error

Unit of Measure Info (GET)

Get details about the unit of measure for a specific set of records.

Resource

```
/connect/pcm/unit-of-measure/info
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/pcm/unit-of-measure/info
```

Available version

63.0

HTTP methods

GET

Query parameters for GET

Parameter Name	Type	Description	Required or Optional	Available Version
correlationId	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Optional	63.0
ids	String	IDs of the unit of measure records.	Optional	63.0

Response body for GET

Bulk Unit Of Measure Info

Unit of Measure Rounded Data (POST)

Round off and scale decimal data for a specific set of fields.

Resource

```
/connect/pcm/unit-of-measure/rounded-data
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/pcm/unit-of-measure/rounded-data
```

Available version

63.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "dataRowInputs": [
    {
      "key": "PRC1",
      "fieldDataInputs": [
        {
          "fieldApiName": "MaxQuantity",
          "originalValue": 1234.5678,
          "unitOfMeasureId": "0hExx0000000001EAA"
        },
        {
          "fieldApiName": "MinQuantity",
          "originalValue": "987462848934739347.32232590183756545",
          "unitOfMeasureId": "0hExx000000001dEAA"
        }
      ]
    },
    {
      "key": "PRC2",
      "fieldDataInputs": [
        {
          "fieldApiName": "MaxQuantity",
          "originalValue": 1234.5678,
          "unitOfMeasureId": "uomId1"
        },
        {
          "fieldApiName": "MinQuantity",
          "originalValue": 987.4628,
          "unitOfMeasureId": "Kgs Id"
        }
      ]
    },
    {
      "key": "PRC3",
      "fieldDataInputs": [
        {
```

```
          "fieldApiName": "MaxQuantity",
          "originalValue": 0.437584,
          "unitOfMeasureId": "uomId2"
        },
        {
          "fieldApiName": "MinQuantity",
          "originalValue": 7364.58923,
          "unitOfMeasureId": "uomId2"
        }
      ]
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
correlationId	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Optional	63.0
dataRowInputs	Data Row Input[]	List of row inputs for rounding the data.	Required	63.0

Response body for POST

Data Rounding

Request Bodies

Learn more about the available Product Catalog Management API request bodies.

Additional Fields Input

Input representation of the additional standard or custom fields to be included in the response.

Bulk Product Details Input

Input representation of the request to retrieve the details of multiple products.

Catalog Input

Input representation of the request to retrieve catalog records.

Criteria Input

Input representation of the filter criteria item request.

Data Rounding Input

Input representation of the details of the data rounding input.

Data Row Input

Input representation of the details of the input for a data rounding request.

Deep Clone Input

Input representation of the details of the object and associated record to be cloned.

Field Data Input

Input representation of the details of the field data input.

Filter Input

Input representation of the filter request.

Include Object Input

Input representation of the object to include in the response.

Index Configuration Collection Input

Input representation of the collection of index configurations.

Index Configuration Input

Input representation of the request to persist the index configuration.

Index Setting Input

Input representation of the index setting.

Options Input

Reserved for internal use.

Order Input

Input representation of the sort order item request.

Product Classification Details Input

Input representation of the request to fetch details of product classification records, including their attributes and attribute categories.

Product Classification List Input

Input representation of the request to retrieve a list of product classification records.

Product Input

Input representation of a product in the catalog.

Product Variants Input

Input representation of the request to retrieve the variation products associated with parent variant products.

Related Object Filters Input

Input representation of the request to filter related objects.

Related Object Node Input

Input representation of the details of a related object node.

Related Records Input

Input representation of the request to retrieve related ProductRampSegment or ProductUsageGrant records for Product2 object.

Run-time Catalog Snapshot Input

Input representation of the details of a run-time catalog snapshot for deployment.

Setting Input

Input representation of the details of the index setting.

Snapshot Deployment Input

Input representation of the request to deploy a run-time catalog snapshot.

Sort Input

Input representation of the sort request.

Additional Fields Input

Input representation of the additional standard or custom fields to be included in the response.

JSON example

```
"additionalFields": {
    "Product2": {
      "fields": [
        "code__c"
      ]
    }
```

This example shows a sample request with attribute definition field details.

```
"additionalFields":{
  "OptOutAssetization":true
  "OptOutDecompositionAction":false
  "OptOutSupplementalAction":false
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`fields`	String[]	List of additional standard or custom fields to be included in the response.	Required	61.0

Bulk Product Details Input

Input representation of the request to retrieve the details of multiple products.

JSON example

```
{
  "correlationId": "cbfabffb-093f-45b3-8c2d-88b4acbd4867",
  "productIds": [
    "01tT1000000F0afIAC",
    "01tT1000000F0afIAC"
  ],
  "uptoLevel": 1,
  "language": "french",
  "additionalFields": {
    "Product2": {
      "fields": [
        "code__c"
      ]
    },
    "ProductAttributeDefinition": {
      "fields": [
        "scope"
      ]
    }
  }
}
```

This example shows a sample request to fetch data from Enterprise Product Catalog.

```
{
  "productIds": [
    "01tLT000009vGXfYAM"
  ],
  "catalogSystems": [
    "epc"
  ],
  "additionalFields": {
    "AttributeDefinition": {
      "fields": [
        "OptOutAssetization",
        "OptOutDecompositionAction",
        "OptOutSupplementalAction"
      ]
    }
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`additional Fields`	Map<String, Additional Fields Input>	Map of object and list of additional standard or custom fields to be included in the response. The supported objects are: • Product2 • ProductAttributeDefinition—If the fields defined for the ProductAttributeDefinition object aren't available for the ProductClassificationAttr object, then the API request fails. If the Dynamic Revenue Orchestrator permission is enabled, then this property supports AttributeDefinition as key with these supported values. • `OptOutAssetization` • `OptOutDecompositionAction` • `OptOutSupplementalAction`	Optional	61.0
`catalogSystems`	String[]	Name of the catalog system. Valid values are: • `epc`—Enterprise Product Catalog • `pcm`—Product Catalog Management	Optional	66.0

Name	Type	Description	Required or Optional	Available Version
		Although this property accepts a list, you can pass only one value. If you don't specify a value, the default behavior is to fetch data from the `pcm` catalog.		
`correlationId`	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Optional	61.0
`language`	String	Custom language that you can specify to get translated data for the fields of an object that's enabled for translation. See Translate Product and Product Category Data.	Optional	64.0
`productIds`	String[]	List of product IDs that details must be returned for. If any product ID is blank, invalid, or not found, then the request is processed with valid and available product IDs.	Required	61.0
`uptoLevel`	Integer	Hierarchy level to follow to return the product details. For a bundle, this property determines the number of levels of child components to be returned. You can specify up to a hierarchy level of 1. If unspecified, the default level is the full bundle hierarchy.	Optional	61.0

Catalog Input

Input representation of the request to retrieve catalog records.

JSON example

This example shows how to retrieve catalogs that contain `apple` in the catalog name.

```
{
"pageSize": 100,
"offset": 0,
"language": "french",
"filter": {
"criteria": [
{
"property": "name",
"operator": "contains",
```

```
"value": "apple"
}
]
}
}
```

This example shows how to retrieve catalogs with `ServiceProcess` as the catalog type.

```
{
"pageSize": 100,
"offset": 0,
"sort": {
  "orders": [
   {
   "property": "name",
   "direction": "desc"
   }
  ]
 },
"filter": {
"criteria": [
{
"property": "catalogType",
"operator": "eq",
"value": "ServiceProcess"
}
]
}
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`correlationId`	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Optional	60.0
`filter`	Filter	Criteria to filter the records. Filters are applicable to the fields of the ProductCatalog object. The supported operators are: • `eq` • `in` • `contains` The supported properties are `name` and `catalogType`.	Optional	60.0

Name	Type	Description	Required or Optional	Available Version
`language`	String	Custom language that you can specify to get translated data for the fields of an object that's enabled for translation. See Translate Product and Product Category Data.	Optional	64.0
`offset`	Integer	Number of records to skip. The default value is 0.	Optional	60.0
`pageSize`	Integer	Number of records per page. Valid values are from 1 through 100. If unspecified, defaults to 100.	Optional	60.0
`sort`	Sort	Sort order of the catalog records. The supported operators are: • `asc` • `desc`	Optional	60.0

Criteria Input

Input representation of the filter criteria item request.

JSON example

```
"criteria":
[{
   "attributeType": "ProductStandard",
   "property": "name",
   "operator": "eq",
   "value": "iPhone"
},
{
   "criteriaType": "CustomWhereCondition",
   "value": "(effectiveenddate = null OR effectiveenddate >= 2024-06-25)"
}
]
```

Properties

Name	Type	Description	Required or Optional	Available Version
`attributeType`	String	Search attribute type of the facet for a faceted search. Valid values are: • `ProductStandard` • `ProductCustom` • `ProductDynamicAttribute`	Optional	63.0

Name	Type	Description	Required or Optional	Available Version
		• `ProductAttributeStandard` • `ProductAttributeCustom` This property is applicable if the **Use Indexed Data For Product Listing and Search** toggle from the Product Discovery Settings page from Setup is enabled.		
`criteriaType`	String	Type of criteria for the filter. Valid value is: • `CustomWhereCondition`	Required	60.0
`operator`	String	Operator used for the filter criteria. The supported operators are: • `eq` • `in` • `contains`—This value isn't applicable if the **Use Indexed Data For Product Listing and Search** toggle from the Product Discovery Settings page from Setup is enabled. • `gt`—Specifies a greater than criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only. • `lt`—Specifies a less than criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only. • `gte`—Specifies a greater than or equal to criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only. • `lte`—Specifies a less than or equal to criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only.	Required	60.0
`property`	String	Property name to use in the filter, which must be the same as the object field.	Required	60.0
`value`	Object	Value for the filter criteria.	Required	60.0

Data Rounding Input

Input representation of the details of the data rounding input.

JSON example

```
{
  "dataRowInputs": [
    {
      "key": "PRC1",
      "fieldDataInputs": [
        {
          "fieldApiName": "MaxQuantity",
          "originalValue": 1234.5678,
          "unitOfMeasureId": "0hExx0000000001EAA"
        },
        {
          "fieldApiName": "MinQuantity",
          "originalValue": "987462848934739347.32232590183756545",
          "unitOfMeasureId": "0hExx000000001dEAA"
        }
      ]
    },
    {
      "key": "PRC2",
      "fieldDataInputs": [
        {
          "fieldApiName": "MaxQuantity",
          "originalValue": 1234.5678,
          "unitOfMeasureId": "uomId1"
        },
        {
          "fieldApiName": "MinQuantity",
          "originalValue": 987.4628,
          "unitOfMeasureId": "Kgs Id"
        }
      ]
    },
    {
      "key": "PRC3",
      "fieldDataInputs": [
        {
          "fieldApiName": "MaxQuantity",
          "originalValue": 0.437584,
          "unitOfMeasureId": "uomId2"
        },
        {
          "fieldApiName": "MinQuantity",
          "originalValue": 7364.58923,
          "unitOfMeasureId": "uomId2"
        }
      ]
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`correlationId`	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Optional	63.0
`dataRowInputs`	Data Row Input[]	List of row inputs for rounding the data.	Required	63.0

Data Row Input

Input representation of the details of the input for a data rounding request.

JSON example

```
JSON example
{
  "dataRowInputs": [
    {
      "key": "PRC1",
      "fieldDataInputs": [
        {
          "fieldApiName": "MaxQuantity",
          "originalValue": 1234.5678,
          "unitOfMeasureId": "0hExx0000000001EAA"
        },
        {
          "fieldApiName": "MinQuantity",
          "originalValue": "987462848934739347.32232590183756545",
          "unitOfMeasureId": "0hExx000000001dEAA"
        }
      ]
    },
    {
      "key": "PRC2",
      "fieldDataInputs": [
        {
          "fieldApiName": "MaxQuantity",
          "originalValue": 1234.5678,
          "unitOfMeasureId": "uomId1"
        },
        {
          "fieldApiName": "MinQuantity",
          "originalValue": 987.4628,
          "unitOfMeasureId": "Kgs Id"
        }
      ]
    },
    {
      "key": "PRC3",
```

```
      "fieldDataInputs": [
        {
          "fieldApiName": "MaxQuantity",
          "originalValue": 0.437584,
          "unitOfMeasureId": "uomId2"
        },
        {
          "fieldApiName": "MinQuantity",
          "originalValue": 7364.58923,
          "unitOfMeasureId": "uomId2"
        }
      ]
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`fieldDataInputs`	Field Data Input[]	List of field-level data inputs.	Required	63.0
`key`	String	Key that identifies a unique data row.	Required	63.0

Deep Clone Input

Input representation of the details of the object and associated record to be cloned.

JSON example

```
{
  "mainRecordId": "01tSG0000028kcSYAQ",
  "mainObjectApiName": "Product2",
  "mainRecordFieldValues": {
    "Name": "New Cloud Storage"
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`mainObjectApiName`	String	API name of the object. The supported object is Product2.	Required	63.0
`mainRecordFieldValues`	Map<String, String>	Mapping of the API name of the field to its value. The values passed through this map are set for the created record. You can pass the Name field only through this map.	Optional	63.0
`mainRecordId`	String	ID of the record.	Required	63.0

Field Data Input

Input representation of the details of the field data input.

JSON example

```
"fieldDataInputs": [
  {
    "fieldApiName": "MaxQuantity",
    "originalValue": 0.437584,
    "unitOfMeasureId": "uomId2"
  },
  {
    "fieldApiName": "MinQuantity",
    "originalValue": 7364.58923,
    "unitOfMeasureId": "uomId2"
  }
]
```

Properties

Name	Type	Description	Required or Optional	Available Version
`fieldApiName`	String	Unique API name of the field.	Required	63.0
`originalValue`	String	Original value of the fields.	Required	63.0
`unitOfMeasureId`	String	ID of the unit of measure record that's associated to the field.	Required	63.0

Filter Input

Input representation of the filter request.

JSON example

```
"filter":
{
"criteria": [ {
"property": "name",
"operator": "eq",
"value": "iPhone"
},
{
"criteriaType": "CustomWhereCondition",
"value": "(effectiveenddate = null OR effectiveenddate >= 2024-06-25)"
}
]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`criteria`	Criteria[]	Details of the filter criteria.	Required if the `filter` property is specified.	60.0

Include Object Input

Input representation of the object to include in the response.

JSON example

```
"includeObjects":
[{
"objectName": "ProductCategory"
}]
```

Properties

Name	Type	Description	Required or Optional	Available Version
`objectName`	String	Name of the object to include in the response. The supported object is `ProductCategory`.	Required if the `options` property is specified.	60.0

Index Configuration Collection Input

Input representation of the collection of index configurations.

JSON example

```
{
  "correlationId": "8545b5aa-f3e6-429a-8f21-9cc4ce50b1d7",
  "indexConfigurations": [
    {
      "attributeDefinitionId": "0tjT1000000002bIAA",
      "name": "Color",
      "type": "ProductDynamicAttribute",
      "isSearchable": true
    },
    {
      "attributeFieldId": "00Nxx000001FwnABII",
      "name": "Message__c",
      "type": "Custom",
      "isSearchable": true
    },
    {
      "name": "Code",
      "type": "Standard",
```

```
      "isSearchable": true
    },
    {
      "facetDisplayRank": 1,
      "isFacetable": false,
      "isSearchable": true,
      "name": "Family",
      "type": "Standard"
    }
  ]
}
```


Properties

Name	Type	Description	Required or Optional	Available Version
`correlationId`	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Optional	62.0
`index Configurations`	Index Configuration Input[]	List of index configurations.	Required	62.0

Index Configuration Input

Input representation of the request to persist the index configuration.

JSON example


```
  "indexConfigurations": [
    {
      "attributeDefinitionId": "0tjT1000000002bIAA",
      "name": "Color",
      "type": "ProductDynamicAttribute",
      "isSearchable": true
    },
    {
      "attributeFieldId": "00Nxx000001FwnABII",
      "name": "Message__c",
      "type": "Custom",
      "isSearchable": true
    },
    {
      "name": "Code",
      "type": "Standard",
      "isSearchable": true
    },
    {
      "facetDisplayRank": 1,
      "isFacetable": false,
      "isSearchable": true,
```

```
      "name": "Family",
      "type": "Standard"
    }
  ]
```

Properties

Name	Type	Description	Required or Optional	Available Version
`attribute DefinitionId`	String	ID of the attribute definition.	Required if the `attribute FieldId`property isn't specified.	62.0
`attribute FieldId`	String	ID of the attribute field.	Required if the `attribute DefinitionId`property isn't specified.	62.0
`facet DisplayRank`	Integer	Sort order for displaying the facets at run time.	Optional	63.0
`isFacetable`	Boolean	Indicates whether the field is facetable (`true`) or not (`false`).	Optional	63.0
`isSearchable`	Boolean	Indicates whether the index-configured field is searchable (`true`) or not (`false`).	Optional	62.0
`name`	String	Name of the index-configured field.	Required	62.0
`type`	String	Type of the index-configured field.	Required	62.0

Index Setting Input

Input representation of the index setting.

JSON example

```
{
    "setting" : {
        "supportedLanguages" : ["en_US","ja","es","nl_NL"],
        "defaultLanguage" : "en_US",
        "productsGrouping": "GROUPING_VARIATION"
   }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`setting`	Setting Input[]	Object containing the setting-related details.	Required	63.0

Options Input

Reserved for internal use.

Order Input

Input representation of the sort order item request.

JSON example

```
"sort":{
"orders":
[{
   "property": "name",
   "direction": "asc"
}]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`direction`	String	Direction to sort the list items, such as in ascending order or descending order.	Required	60.0
`property`	String	Property to use for the sorting of the list items.	Required	60.0

Product Classification Details Input

Input representation of the request to fetch details of product classification records, including their attributes and attribute categories.

JSON example

```
{
  "productClassificationIds": [
    "01txx0000006iFMAAY",
    "01txx0000006iGxAAY"
  ],
  "catalogSystems": [
    "epc"
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`catalogSystems`	String[]	Name of the catalog system. Valid value is: • `epc`—Enterprise Product Catalog	Optional	66.0
`product ClassificationIds`	String[]	List of product classification IDs for which you want to retrieve product details. In the	Required	66.0

Name	Type	Description	Required or Optional	Available Version
		epc catalog system, these values are the Product2 record IDs.		

Product Classification List Input

Input representation of the request to retrieve a list of product classification records.

JSON example

```
{
  "catalogSystem": "pcm",
  "searchTerm": "Mobile",
  "filter": {
    "criteria": [
      {
        "property": "name",
        "operator": "contains",
        "value": "Mobile"
      }
    ]
  },
  "sort": {
    "orders": [
      {
        "property": "name",
        "direction": "asc"
      }
    ]
  },
  "pageSize": 25,
  "offset": 0
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
catalogSystem	String	Name of the catalog system. Valid values are: • pcm—Product Catalog Management • epc—Enterprise Product Catalog If unspecified, the default catalog system is pcm.	Optional	67.0
filter	Criteria Input	Criteria to filter the product classification records. The supported property is name. The supported operators are: • eq	Optional	67.0

Name	Type	Description	Required or Optional	Available Version
		• `in` • `contains` If multiple criteria are specified, they're combined by using the `and` operator. Each criterion supports only the `property`, `operator`, and `value` fields.		
`offset`	Integer	Number of records to skip. The default value is 0.	Optional	67.0
`pageSize`	Integer	Specifies the number of records per page. Valid values are 5, 10, 25, 50, and 100. If unspecified, the default value is 100.	Optional	67.0
`searchTerm`	String	String used to search for product classifications with the product classification name containing the search term.	Optional	67.0
`sort`	Order Input	Sort order for the product classifications. If unspecified, the default sort order is by name in ascending order.	Optional	67.0

Product Input

Input representation of a product in the catalog.

JSON example

This example is a search request for products that contain `Bundle Product` in the product name.

```
{
  "catalogIds": [
    "0ZST10000004D03OAE"
  ],
  "language": "spanish",
  "filter": {
    "criteria": [
      {
        "property": "name",
        "operator": "contains",
        "value": "Bundle Product"
      }
    ]
  },
  "relatedObjectFilters": [
    {
      "objectName": "ProductSpecificationRecType",
```

```
      "criteria": [
        {
          "property": "IsCommercial",
          "operator": "eq",
          "value": false
        }
      ],
      "additionalFields": {
        "Product2": {
          "fields": [
            "code__c"
          ]
        }
      }
    }
  ]
}
```

This example is a search request for products that contain the specified search term in the product name and are part of a catalog.

```
  {
  "catalogIds": [
    "0ZSDU00000020g54AE"
  ],
  "searchTerm": "Slack"
}
```

Query parameters

Name	Type	Description	Required or Optional	Available Version
`productClassificationId`	String	ID of the product classification template that contains a collection of dynamic attributes and can be reused to create multiple products. Products that are based on a product classification inherit all the attributes of the product classification. Specify either the product classification ID, list of category IDs, or list of catalog IDs to retrieve products.	Optional	60.0

Properties

Name	Type	Description	Required or Optional	Available Version
`additionalFields`	Map<String, Additional Fields Input>	Map of the object and the list of standard and custom fields to be queried along with the standard response. The supported object is Product2.	Optional	61.0

Name	Type	Description	Required or Optional	Available Version
`catalogIds`	List<String>	List of comma-separated catalog IDs. The API returns the list of product records that are associated with the specified catalog IDs. Specify either the product classification ID, list of category IDs, or list of catalog IDs to retrieve products.	Optional	60.0
`categoryIds`	List<String>	List of comma-separated category IDs. The API returns the list of product records that are associated with the specified category IDs. If unspecified, then returns all products that are added to at least one category. Specify either the product classification ID, list of category IDs, or list of catalog IDs to retrieve products.	Optional	60.0
`correlationId`	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Optional	60.0
`filter`	Criteria Input	Criteria to filter the records. Filters are applicable to the fields of Product2 object. The supported operators are: • `eq` • `in` • `contains`—This value isn't applicable if the **Use Indexed Data For Product Listing and Search** toggle from the Product Discovery Settings page from Setup is enabled. • `gt`—Specifies a greater than criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only. • `lt`—Specifies a less than criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only. • `gte`—Specifies a greater than or equal to criteria. Available from API	Optional	60.0

Name	Type	Description	Required or Optional	Available Version
		version 63.0 and later for Number, Date, and Datetime data types only. • `lte`—Specifies a less than or equal to criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only. If multiple criteria are specified, then the criteria are combined by using the `and` operator. The supported properties are `name`, `description`, and `isActive`. If the **Use Indexed Data For Product Listing and Search** toggle from the Product Discovery Settings page from Setup is enabled, then the supported property is `name` only.		
`language`	String	Custom language that you can specify to get translated data for the fields of an object that's enabled for translation. See Translate Product and Product Category Data.	Optional	64.0
`offset`	Integer	Number of records to skip. The default value is 0.	Optional	60.0
`pageSize`	Integer	Specifies the number of records per page. Valid values are from 1 through 200. If the value is unspecified, it defaults to 100.	Optional	60.0
`relatedObjectFilters`	Related Object Filter[]	Criteria for the related objects to filter the records. The supported operator is `eq`. The supported object is `ProductSpecificationRecType`. The supported values are `true` and `false`. The supported property is `IsCommercial`.	Optional	60.0
`searchTerm`	String	String used to get products with the product name containing the search term. See Search Considerations When Using Indexed Data.	Optional	62.0
`sort`	Sort	Sort order for the products. If the **Use Indexed Data For Product Listing and Search** toggle from the	Optional	60.0

Name	Type	Description	Required or Optional	Available Version
		Product Discovery Settings page from Setup is enabled, then you can sort products by using name only.		

Product Variants Input

Input representation of the request to retrieve the variation products associated with parent variant products.

JSON example

```
{
  "correlationId": "9b6bc520-3c82-4d6c-a458-47590370681a",
  "parentVariantsIds": [
    "01tT1000000F0afIAC",
    "01tT1000000F0agIAC"
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`correlationId`	String	Unique ID to track and associate related events or transactions. If unspecified, a Universally Unique Identifier (UUID) is generated.	Optional	67.0
`parent VariantsIds`	String[]	List of product IDs for parent variant products whose variations you want to retrieve. If any product ID is blank, invalid, or not found, the API skips that ID and processes the remaining valid IDs. The skipped IDs appear in the `inValidProductIds` response property. If a product ID is valid but isn't a parent variant product, the API includes it in the `nonVariantParentIds` response property.	Required	67.0

Related Object Filters Input

Input representation of the request to filter related objects.

JSON example

```
"relatedObjectFilters":
[{
   "criteria": [
    {
    "property": "IsCommercial",
    "operator": "eq",
    "value": true
    }
  ],
   "objectName": "ProductSpecificationRecType"
 }]
```

Properties

Name	Type	Description	Required or Optional	Available Version
`criteria`	Criteria[]	Criteria to filter the related objects.	Required if the `relatedObjectFilters` property is specified.	60.0
`objectName`	String	API name of the object that's related to the main object.	Required if the `relatedObjectFilters` property is specified.	60.0

Related Object Node Input

Input representation of the details of a related object node.

JSON example

```
 "relatedObjectNodes": [
    {
      "relatedObjectAPIName": "ProductRampSegment",
      "pageSize": 20,
      "offSet": 0
    },
    {
      "relatedObjectAPIName": "ProductUsageGrant",
      "pageSize": 10,
      "offSet": 0,
      "filter": {
        "criteria": [
          {
            "property": "status",
            "operator": "eq",
            "value": "active"
          },
          {
            "property": "effectivestartdate",
            "operator": "lte",
            "value": "2024-06-25"
```

```
        },
        {
          "criteriaType": "CustomWhereCondition",
          "value": "(effectiveenddate = null OR effectiveenddate >= 2024-06-25)"
        }
      ]
    }
  }
]
```

Properties

Name	Type	Description	Required or Optional	Available Version
`filter`	Criteria[]	Criteria to filter records. The supported properties are: • `StartDate` • `EndDate` • `Status` The supported operators are: • `eq` • `gte` • `lte` The supported related object is ProductUsageGrant. If multiple criteria are specified, then the resultant criteria are combined by using the `and` operator.	Optional	62.0
`offSet`	Integer	Number of records to skip. The default value is 0.	Optional	62.0
`pageSize`	Integer	Number of records per page. Valid values are from 1 through 100. If unspecified, the default value is 100.	Optional	62.0
`relatedObject APIName`	String	API name of the related object to return the records for. The supported related objects are ProductRampSegment and ProductUsageGrant.	Required	62.0

Related Records Input

Input representation of the request to retrieve related ProductRampSegment or ProductUsageGrant records for Product2 object.

JSON example

```
{
  "recordIds": [
    "01txx0000006i44AAA",
    "01txx0000006i5gAAA"
  ],
  "relatedObjectNodes": [
    {
      "relatedObjectAPIName": "ProductRampSegment",
      "pageSize": 20,
      "offSet": 0
    },
    {
      "relatedObjectAPIName": "ProductUsageGrant",
      "pageSize": 10,
      "offSet": 0,
      "filter": {
        "criteria": [
          {
            "property": "status",
            "operator": "eq",
            "value": "active"
          },
          {
            "property": "effectivestartdate",
            "operator": "lte",
            "value": "2024-06-25"
          },
          {
            "criteriaType": "CustomWhereCondition",
            "value": "(effectiveenddate = null OR effectiveenddate >= 2024-06-25)"
          }
        ]
      }
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`correlationId`	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Optional	62.0
`recordIds`	String[]	List of record IDs to return the relatedObjects records for. The maximum number of record IDs supported is 20.	Required	62.0

Name	Type	Description	Required or Optional	Available Version
`relatedObjectNodes`	Related Object Node Input[]	List of nodes for the related objects. The maximum number of related object nodes supported is two.	Required	62.0

Run-time Catalog Snapshot Input

Input representation of the details of a run-time catalog snapshot for deployment.

JSON example

```
"snapshot": {
  "activationType": "IMMEDIATE",
  "activationDate": "2024-05-06T05:12:59.000Z",
  "id": "1Avxx0000005DFe1AM"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`activationDate`	String	Activation date of the snapshot.	Optional	62.0
`activationType`	String	Activation type of the snapshot. Valid value is: • `IMMEDIATE`—Snapshot is activated immediately after a successful build.	Required	62.0
`id`	String	ID of the snapshot.	Required	62.0

Setting Input

Input representation of the details of the index setting.

JSON example

```
  "setting" : {
      "supportedLanguages" : ["en_US","ja","es","nl_NL"],
      "defaultLanguage" : "en_US"
 }
```

Properties

Name	Type	Description	Required or Optional	Available Version
`defaultLanguage`	String	Default language for the API.	Required	63.0
`supportedLanguages`	String[]	List of supported language locales for indexing.	Required	63.0

Snapshot Deployment Input

Input representation of the request to deploy a run-time catalog snapshot.

JSON example

This example shows a sample request to build a new snapshot with immediate activation.

```
{
  "snapshot": {
    "activationType": "IMMEDIATE"
  },
  "buildType": "FULL"
}
```

This example shows a sample request to rebuild a snapshot in the `active` status.

```
{
  "snapshot": {
    "activationType": "IMMEDIATE",
    "id": "1Avxx0000005DFe1AM"
  },
  "buildType": "FULL"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`buildType`	String	Build type of the snapshot index. Valid value is: • `FULL`—Specifies a full index build. • `INCREMENTAL`—Specifies an incremental index build. Available from API version 63.0 and later.	Required	62.0
`snapshot`	Run-time Catalog Snapshot Input[]	Snapshot to deploy.	Required	62.0

Sort Input

Input representation of the sort request.

JSON example

```
"sort":
{
"orders":
[{
"property": "name",
"direction": "asc"
}]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`orders`	Order[]	Details of the sort order.	Required if the `sort` property is specified.	60.0

Response Bodies

Learn more about the available Product Catalog Management API response bodies.

Attribute Category
Output representation of the attribute category.

Attribute Definition
Output representation of the attribute definition.

Attribute Picklist
Output representation of the attribute picklist.

Attribute Picklist Value
Output representation of the attribute picklist value.

Bulk Unit Of Measure Info
Output representation of the details of the unit of measure records along with error details.

Data Rounding
Output representation of the data rounding response.

Data Row
Output representation of the details of a data row.

Catalog Output
Output representation of the catalog definition.

Catalogs Output
Output representation of the retrieved catalog result.

Categories Output
Output representation of the retrieved categories result.

Category Output
Output representation of the category definition.

Deep Clone Error
Output representation of the error details related to the deep clone request.

Deep Clone Record Response
Output representation of the details of the cloned related records.

Deep Clone Response
Output representation of the details of the cloned record.

Error Output

Output representation of the error details.

Facet Value

Output representation of the facet values found in the search result.

Field Data

Output representation of the field data.

Fields Info

Output representation of the metadata fields in an object.

Index Configuration Collection

Output representation of the collection of index configuration details.

Index Configuration Field

Output representation of the details of the index-configured field.

Index Configurations Update

Output representation of the updated index configuration.

Index Error

Output representation of the error details related to an index.

Index Setting

Output representation of the retrieved index settings.

Index Setting Update

Output representation of the details of the updated index setting.

Invalid Related Object Node

Output representation of the invalid related object node with details of errors.

Metadata

Output representation of the metadata details for objects.

Object Info

Output representation of the object details along with its fields.

Product Catalog Management Error

Output representation that contains error details, including error codes and messages.

Product Classification

Output representation of the product classification details.

Product Classification Details

Output representation that contains the details of a single product classification, including its attributes and categories.

Product Classification Details Collection

Output representation that contains a collection of product classification details along with any processing errors.

Product Classification List Collection

Output representation that contains a collection of product classification records along with any processing errors.

Product Component Group

Output representation of the product component group.

Product

Output representation of the product definition.

Product Related Component

Output representation of the product-related component.

Product Selling Model

Output representation of the definition of the product selling model.

Product Selling Model Option

Output representation of the definition of the product selling model option.

Product Specification Type

Output representation of the product specification type.

Products

Output representation of the list of retrieved products.

Product Variants

Output representation of the variation products associated with the specified parent variant products.

Related Object Records

Output representation of the related records for a specified record ID and related object API name.

Related Records

Output representation of the list of relatedObject records for a specified record ID.

Related Records List

Output representation of the list of related records.

Search Facet

Output representation of the details of the faceted search.

Setting

Output representation of the setting that's used in indexing.

Setting Metadata

Output representation of the metadata associated with a setting.

Snapshot

Output representation of the list of active snapshots.

Snapshot Collection

Output representation of the retrieved snapshot collection.

Snapshot Deployment

Output representation of the snapshot deployment.

Snapshot Index

Output representation of the snapshot index of a run-time catalog.

Snapshot Index Error

Output representation of the error details related to a snapshot index.

Snapshot Index Info

Output representation of the details of a snapshot index.

Snapshot Index Log

Output representation of a snapshot index log.

Status

Output representation of the status of the request.

Unit of Measure Error

Output representation of the details of errors encountered during the processing of the Unit of Measure API request.

Unit of Measure Info

Output representation of the details of a unit of measure record.

Unit of Measure Status

Output representation of the status of the Unit of Measure API request.

Attribute Category

Output representation of the attribute category.

JSON example


```
"attributeCategory": [
        {
          "attributes": [
            {
              "additionalFields": {
                "scope": "Order"
              },
              "attributeNameOverride": "AD Text",
              "code": "AD02",
              "dataType": "Text",
              "defaultValue": "AD Text DV",
              "description": "AD Text Desc",
              "displayType": "Text",
              "helpText": "AD Text DHT",
              "id": "0tjT1000000002bIAA",
              "isHidden": false,
              "isPriceImpacting": true,
              "isReadOnly": true,
              "isRequired": true,
              "label": "AD Text Label",
              "maximumCharacterCount": "20",
              "maximumValue": "100",
              "minimumCharacterCount": "1",
              "minimumValue": "50",
              "name": "AD Text",
              "sequence": 1,
              "status": "Active",
              "valueDescription": "AD Text VD"
            }
          ],
          "code": "AC001",
          "id": "0v3T1000000000BIAQ",
          "name": "build and make"
    }
]
```

Property Name	Type	Description	Filter Group and Version	Available Version
attributes	Attribute Definition[]	List of categorized attributes associated with the product.	Small, 60.0	60.0
code	String	Code of the attribute category.	Small, 60.0	60.0
id	String	ID associated with the attribute category.	Small, 60.0	60.0
name	String	Name of the attribute category. If data translation is set up and specified in the org, the translated description is available.	Small, 60.0	60.0

Attribute Definition

Output representation of the attribute definition.

JSON example


```
"attributes": [
     {
      "additionalFields": {
      "scope": "Order"
       },
      "attributeNameOverride": "AD Text",
      "code": "AD02",
      "dataType": "Text",
      "defaultValue": "AD Text DV",
      "description": "AD Text Desc",
      "displayType": "Text",
      "helpText": "AD Text DHT",
      "id": "0tjT1000000002bIAA",
      "isHidden": false,
      "isPriceImpacting": true,
      "isReadOnly": true,
      "isRequired": true,
      "label": "AD Text Label",
      "maximumCharacterCount": 20,
      "maximumValue": "100",
      "minimumCharacterCount": 1,
      "minimumValue": "50",
      "name": "AD Text",
      "sequence": 1,
      "status": "Active",
      "valueDescription": "AD Text VD"
     }
   ],
    "code": "AC001",
    "id": "0v3T1000000000BIAQ",
    "name": "build and make"
}]
```

Property Name	Type	Description	Filter Group and Version	Available Version
`additional Fields`	Map<String, Additional Fields Input>	Key-value pair of additional standard or custom fields to include in the response.	Small, 61.0	61.0
`attributeName Override`	String	Name to display for the attribute, which overrides the name on the attribute. For example, the Color attribute is overridden to display as Laptop Color.	Small, 60.0	60.0
`code`	String	Unique code of the attribute definition.	Small, 60.0	60.0
`dataType`	String	Data type of the attribute definition value.	Small, 60.0	60.0
`defaultValue`	String	Default value of the attribute.	Small, 60.0	60.0
`description`	String	Description of the attribute.	Small, 60.0	60.0
`displayType`	String	Display types of the attribute. Valid values are: • `Radio Button` • `Checkbox` • `Toggle` • `Input Date` • `DateTime` • `Currency Symbol` • `Currency Code` • `Currency Name` • `Percentage` • `Text` • `Combobox` • `Radio Button` • `MultiSelect` • `MultiSelectCheckboxes`	Small, 60.0	60.0
`helpText`	String	Help text that appears at run time for the attribute. If data translation is set up and specified in the org, the translated description is available.	Small, 60.0	60.0
`id`	String	ID of the attribute definition.	Small, 60.0	60.0
`is Configurable`	Boolean	Reserved for future use.	Small, 60.0	60.0

Property Name	Type	Description	Filter Group and Version	Available Version
`isHidden`	Boolean	Indicates whether to hide the attribute from the users in the order capture interface (`true`) or not (`false`).	Small, 60.0	60.0
`isPrice Impacting`	Boolean	Indicates whether the attribute impacts the product price (`true`) or not (`false`).	Small, 60.0	60.0
`isReadOnly`	Boolean	Indicates whether the product attribute is read-only for the end users in the order capture page (`true`) or not (`false`).	Small, 60.0	60.0
`isRequired`	Boolean	Indicates whether a value for the attribute is required for the assigned parent object (`true`) or not (`false`).	Small, 60.0	60.0
`isValue Cloneable`	Boolean	Reserved for future use.	Small, 60.0	60.0
`label`	String	Label of the attribute. If data translation is set up and specified in the org, the translated description is available.	Small, 60.0	60.0
`maximum Character Count`	Integer	Maximum number of alphanumeric characters that can be entered for attributes of type number and text in run time.	Small, 60.0	60.0
`maximumValue`	String	Maximum value that can be entered for attributes of type number, currency, and percent in run time.	Small, 60.0	60.0
`minimum Character Count`	Integer	Minimum number of alphanumeric characters that can be entered for attributes of type number and text in run time. The minimum character count must be less than or equal to the maximum character count.	Small, 60.0	60.0
`minimumValue`	String	Minimum value that can be entered for attributes of type number, currency, and percent in run time. The minimum value must be less than or equal to the maximum value.	Small, 60.0	60.0
`name`	String	Name of the attribute.	Small, 60.0	60.0
`picklist`	Attribute Picklist	ID of the attribute picklist that provides the valid values for the attribute.	Small, 60.0	60.0
`sequence`	Integer	Order in which the attribute values appear in the attribute definition when the product is configured at run time.	Small, 60.0	60.0

Property Name	Type	Description	Filter Group and Version	Available Version
`status`	String	Lifecycle state of the attribute picklist. Valid values are: • `Active` • `Draft` • `Inactive`	Small, 60.0	60.0
`stepValue`	String	Reserved for future use.	Small, 60.0	60.0
`valueDecoder`	String	Reserved for future use.	Small, 60.0	60.0
`value Description`	String	Description of the value assigned to the attribute.	Small, 60.0	60.0

Attribute Picklist

Output representation of the attribute picklist.

JSON example

```
"picklist": {
     "dataType": "Text",
     "description": "Fabric Module options",
     "id": "0v51Q000000TNDkQAO",
     "name": "Fabric Module options",
     "values": [
      {
        "abbreviation": "IFM1"
        "code": "PV0051",
        "displayValue": "25G Intelligent Fabric Module with 8x 25G SFP28 ports",
        "id": "0v61Q0000008OMYQA2",
        "name": "25G Intelligent Fabric Module with 8x 25G SFP28 ports",
        "sequence": "1",
        "value": "25G Intelligent Fabric Module with 8x 25G SFP28 ports",
        "status" : "Active"
      },
      {
        "abbreviation": "IFM2"
        "code": "PV0052",
        "displayValue": "100G Intelligent Fabric Module with 8x 100G QSFP28 ports",
        "id": "0v61Q0000008OMZQA2",
        "name": "100G Intelligent Fabric Module with 8x 100G QSFP28 ports",
        "sequence": "2",
        "value": "100G Intelligent Fabric Module with 8x 100G QSFP28 ports",
        "status" : "Active"
      }
     ]
  }
```

Property Name	Type	Description	Filter Group and Version	Available Version
`dataType`	String	Data type of the values in the picklist. Valid values are: • `Boolean` • `Date` • `Datetime` • `Number` • `Text` • `Currency` • `Percent`	Small, 60.0	60.0
`description`	String	Description of the picklist, such as the picklist purpose or the associated product.	Small, 60.0	60.0
`id`	String	ID associated with the attribute picklist record.	Small, 60.0	60.0
`name`	String	Name of the picklist value.	Small, 60.0	60.0
`values`	Attribute Picklist Value[]	List of values associated with the picklist.	Small, 60.0	60.0

Attribute Picklist Value

Output representation of the attribute picklist value.

JSON example

```
"values": [
      {
        "abbreviation": "IFM1"
        "code": "PV0051",
        "displayValue": "25G Intelligent Fabric Module with 8x 25G SFP28 ports",
        "id": "0v61Q0000008OMYQA2",
        "name": "25G Intelligent Fabric Module with 8x 25G SFP28 ports",
        "sequence": "1",
        "value": "25G Intelligent Fabric Module with 8x 25G SFP28 ports",
        "status" : "Active"
      }
     ]
```

Property Name	Type	Description	Filter Group and Version	Available Version
`abbreviation`	String	Name of the picklist value that appears at run time.	Small, 60.0	60.0
`code`	String	Unique code of the picklist value within the picklist.	Small, 60.0	60.0

Property Name	Type	Description	Filter Group and Version	Available Version
`displayValue`	String	Picklist value that appears at run time in the order capture page. If data translation is set up and specified in the org, the translated description is available.	Small, 60.0	60.0
`id`	String	ID associated with the attribute picklist value.	Small, 60.0	60.0
`name`	String	Name of the picklist value.	Small, 60.0	60.0
`sequence`	String	Order in which the picklist value appears in the picklist.	Small, 60.0	60.0
`status`	String	Status of the attribute picklist value.	Small, 62.0	62.0
`value`	String	Value of the picklist item. Value must be unique within the picklist.	Small, 60.0	60.0

Bulk Unit Of Measure Info

Output representation of the details of the unit of measure records along with error details.

JSON example


```
{
  "correlationId": "928ea35f-8a2f-4932-9f7e-ec6cdbcabdbe",
  "errorCodeToErrorMap": {
    "UNIT_OF_MEASURE_INFO_INVALID_UOM_IDS": {
      "errorCode": "UOM_INFO_API_003",
      "messageDetail": "Invalid uomId is passed. Please specify a valid uomId.",
      "messageTitle": "Invalid uomId is passed.",
      "recordIds": [
        "sample"
      ],
      "source": "Unit_Of_Measure_Info_Api"
    }
  },
  "status": {
    "errors": [],
    "httpStatusCode": "200",
    "message": " Successfully fetched UnitOfMeasure Info. "
  },
  "uomIdToUnitOfMeasureInfo": {
    "0hEU200000003M5MAI": {
      "id": "0hEU200000003M5MAI",
      "name": "Pounds",
      "roundingMethod": "Nearest",
      "scale": 1,
      "unitCode": "Pounds"
    },
    "0hEU200000003KTMAY": {
      "id": "0hEU200000003KTMAY",
```

```
        "name": "Grams",
        "roundingMethod": "Down",
        "scale": 5,
        "unitCode": "Grams"
      }
    }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
correlationId	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Small, 63.0	63.0
errorCode ToErrorMap	Map<String, Unit Of Measure Error>	Error codes mapped to their details.	Small, 63.0	63.0
status	Unit Of Measure Status[]	Status of the API request.	Small, 63.0	63.0
uomIdToUnit OfMeasureInfo	Map<String, Unit Of Measure Info>	Unit of measure record IDs mapped to their details.	Small, 63.0	63.0

Data Rounding

Output representation of the data rounding response.

JSON example

```
{
  "keyToUomDataRowOutput": {
    "PRC1": {
      "key": "PRC1",
      "fieldApiNameToFieldDataOutput": {
        "MaxQuantity": {
          "fieldApiName": "MaxQuantity",
          "originalValue": 1234.5678,
          "isRoundingApplicable": true,
          "roundedValue": 1234.56,
          "unitOfMeasureId": "uomId1",
          "errorCodeToErrorMap" : []
        },
        "MinQuantity": {
          "fieldApiName": "MinQuantity",
          "originalValue": 643.1,
          "isRoundingApplicable": true,
          "roundedValue": 643.1,
          "unitOfMeasureId": "uomId1"
        }
      },
      "errorCodeToErrorMap" : []
```

```
    },
    "PRC2": {
      "key": "PRC2",
      "fieldApiNameToFieldDataOutput": {
        "MaxQuantity": {
          "fieldApiName": "MaxQuantity",
          "originalValue": 1234.5678,
          "isRoundingApplicable": true,
          "roundedValue": 1234.56,
          "unitOfMeasureId": "uomId1"
        },
        "MinQuantity": {
          "fieldApiName": "MinQuantity",
          "originalValue": 987.4628,
          "isRoundingApplicable": true,
          "errorCodeToErrorMap": {
            "message": "arithrmetic operation"
          },
          "unitOfMeasureId": "uomId1"
        }
      },
      "errorCodeToErrorMap": []
    },
    "PRC3": {
      "key": "PRC3",
      "fieldApiNameToFieldDataOutput": {
        "MaxQuantity": {
          "fieldApiName": "MaxQuantity",
          "originalValue": 1234.5678,
          "isRoundingApplicable": false,
          "unitOfMeasureId": "uomId2"
        },
        "MinQuantity": {
          "fieldApiName": "MinQuantity",
          "originalValue": 987.4628,
          "isRoundingApplicable": false,
          "unitOfMeasureId": "uomId2"
        }
      },
      "errorCodeToErrorMap": []
    }
  }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`correlationId`	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Small, 63.0	63.0

Property Name	Type	Description	Filter Group and Version	Available Version
`errorCodeToErrorMap`	Map<String, Unit Of Measure Error>	Error codes mapped to their details.	Small, 63.0	63.0
`keyToDataRowOutput`	Map<String, Data Row>	Data row key mapped to the associated data row.	Small, 63.0	63.0
`status`	Unit Of Measure Status[]	Status of the API request.	Small, 63.0	63.0

Data Row

Output representation of the details of a data row.

JSON example

```
{
  "keyToUomDataRowOutput": {
    "PRC1": {
      "key": "PRC1",
      "fieldApiNameToFieldDataOutput": {
        "MaxQuantity": {
          "fieldApiName": "MaxQuantity",
          "originalValue": 1234.5678,
          "isRoundingApplicable": true,
          "roundedValue": 1234.56,
          "unitOfMeasureId": "uomId1",
          "errorCodeToErrorMap" : []
        },
        "MinQuantity": {
          "fieldApiName": "MinQuantity",
          "originalValue": 643.1,
          "isRoundingApplicable": true,
          "roundedValue": 643.1,
          "unitOfMeasureId": "uomId1"
        }
      },
      "errorCodeToErrorMap" : []
    },
    "PRC2": {
      "key": "PRC2",
      "fieldApiNameToFieldDataOutput": {
        "MaxQuantity": {
          "fieldApiName": "MaxQuantity",
          "originalValue": 1234.5678,
          "isRoundingApplicable": true,
          "roundedValue": 1234.56,
          "unitOfMeasureId": "uomId1"
        },
        "MinQuantity": {
          "fieldApiName": "MinQuantity",
          "originalValue": 987.4628,
```

```
          "isRoundingApplicable": true,
          "errorCodeToErrorMap": {
            "message": "arithrmetic operation"
          },
          "unitOfMeasureId": "uomId1"
        }
      },
      "errorCodeToErrorMap": []
    },
    "PRC3": {
      "key": "PRC3",
      "fieldApiNameToFieldDataOutput": {
        "MaxQuantity": {
          "fieldApiName": "MaxQuantity",
          "originalValue": 1234.5678,
          "isRoundingApplicable": false,
          "unitOfMeasureId": "uomId2"
        },
        "MinQuantity": {
          "fieldApiName": "MinQuantity",
          "originalValue": 987.4628,
          "isRoundingApplicable": false,
          "unitOfMeasureId": "uomId2"
        }
      },
      "errorCodeToErrorMap": []
    }
  }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`errorCodeToErrorMap`	Map<String, Unit Of Measure Error>	Map of error codes to their details.	Small, 63.0	63.0
`fieldApiNameToFieldDataOutput`	Map<String, Field Data>	Map of field API name to associated field data.	Small, 63.0	63.0
`key`	String	Unique key of the data row.	Small, 63.0	63.0

Catalog Output

Output representation of the catalog definition.

JSON example

```
catalogs": [
    {
      "catalogType": "Sales",
      "code": "CAT009",
      "description": "SmartBytes B2B Catalog",
      "effectiveEndDate": "31-07-2023",
```

```
        "effectiveStartDate": "24-07-2023",
        "id": "0ZS1Q000000XbZAWA0",
        "name": "SmartBytes B2B Catalog",
        "numberOfCategories": 8
    }
]
```

Property Name	Type	Description	Filter Group and Version	Available Version
catalogType	String	The category of an entry in the catalog, which is customizable. For example, catalog types, such as sellable products, services, parts, technical services, or technical resources.	Small, 60.0	60.0
code	String	Unique ID associated with the catalog.	Small, 60.0	60.0
description	String	Description of the catalog. If data translation is set up and specified in the org, the translated description is available.	Small, 60.0	60.0
effective EndDate	String	Date and time from when the catalog isn't available to the end users.	Small, 60.0	60.0
effective StartDate	String	Date and time from when the catalog is available to the end users.	Small, 60.0	60.0
id	String	ID of the catalog.	Small, 60.0	60.0
name	String	Name of the catalog. If data translation is set up and specified in the org, the translated description is available.	Small, 60.0	60.0
numberOf Categories	Integer	Number of categories in the catalog.	Small, 60.0	60.0

Catalogs Output

Output representation of the retrieved catalog result.

JSON example

```
{
  "catalogs": [
    {
      "catalogType": "Sales",
      "code": "CAT009",
      "id": "0ZS1Q000000XbZAWA0",
      "name": "SmartBytes B2B Catalog",
      "numberOfCategories": 8
    }
  ],
  "correlationId": "0b7b6a30-895c-407a-91b3-e67482d339a3",
```

```
  "count": 1,
  "status": {
    "code": "200",
    "errors": [],
    "message": "Successfully fetched the catalog records."
  }
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`catalogs`	Catalog Output[]	List of the catalogs.	Small, 60.0	60.0
`correlationId`	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Small, 60.0	60.0
`count`	Integer	Total number of the catalog records retrieved after the query execution, wherein the `pageSize` property determines the number of records returned in every page.	Small, 60.0	60.0
`status`	Status	Status of the request.	Small, 60.0	60.0

Categories Output

Output representation of the retrieved categories result.

JSON example


```
{
  "categories": [
    {
      "catalogId": "0ZS1Q000000XbZAWA0",
      "code": "B2B Category",
      "description": "Products Category",
      "hasSubCategories": true,
      "id": "0ZG1Q000000XbVGWA0",
      "name": "Unified Computing",
      "numberOfProducts": 2,
      "parentCategoryId": "0ZGT100000000ql0AA",
      "sortOrder": 2,
      "subCategories": [],
      "isNavigational: false
    }
  ],
  "correlationId": "30230973-0a09-405e-b148-f085bb6dd66e",
  "status": {
    "code": "200",
    "errors": [],
    "message": "Successfully fetched the category records."
```

```
  }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`categories`	Category Output[]	List of the retrieved categories.	Small, 60.0	60.0
`correlationId`	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Small, 60.0	60.0
`status`	Status	Status of the request.	Small, 60.0	60.0

Category Output

Output representation of the category definition.

JSON example

```
"categories": [
    {
      "catalogId": "0ZS1Q000000XbZAWA0",
      "code": "B2B Category",
      "description": "Products Category",
      "hasSubCategories": true,
      "id": "0ZG1Q000000XbVGWA0",
      "name": "Unified Computing",
      "numberOfProducts": 2,
      "parentCategoryId": "0ZGT100000000ql0AA",
      "sortOrder": 2,
      "subCategories": [],
      "isNavigational: false
    }
  ]
```

Property Name	Type	Description	Filter Group and Version	Available Version
`catalogId`	String	ID of the catalog that the category is associated with.	Small, 60.0	60.0
`code`	String	Unique code of the product category.	Small, 60.0	60.0
`description`	String	Description of the category. If data translation is set up and specified in the org, the translated description is available.	Small, 60.0	60.0
`hasSub Categories`	Boolean	Indicates whether the subcategories are available (true) or not (false).	Small, 60.0	60.0
`id`	String	ID of the category.	Small, 60.0	60.0

Property Name	Type	Description	Filter Group and Version	Available Version
isNavigational	Boolean	Indicates whether the category node is navigational (`true`) or not (`false`).	Small, 62.0	62.0
name	String	Name of the category. If data translation is set up and specified in the org, the translated name is available.	Small, 60.0	60.0
numberOf Products	Integer	Number of products associated with the category.	Small, 60.0	60.0
parent CategoryId	String	ID of the parent category.	Small, 60.0	60.0
sortOrder	Integer	Display order of the product category relative to the siblings with the same parent category.	Small, 60.0	60.0
subCategories	Category Output[]	List of subcategories, if available. This property is returned with the Categories List (GET) API response.	Small, 60.0	60.0

Deep Clone Error

Output representation of the error details related to the deep clone request.

Property Name	Type	Description	Filter Group and Version	Available Version
errorCode	String	Code of the error message related to the deep clone request.	Small, 63.0	63.0
errorMessage	String	Details of the error message related to the deep clone request.	Small, 63.0	63.0

Deep Clone Record Response

Output representation of the details of the cloned related records.

JSON example


```
"createdRecordList": [
  {
    "createdRecordId": "01tSG0000030Yb3YAE",
    "entityApiName": "Product2",
    "entityLabel": "Product"
  },
  {
    "createdRecordId": "0iOSG0000002rMn2AI",
    "entityApiName": "ProductSellingModelOption",
    "entityLabel": "Product Selling Model Option"
```

```
    },
    {
      "createdRecordId": "0v7SG0000001ktdYAA",
      "entityApiName": "ProductAttributeDefinition",
      "entityLabel": "Product Attribute Definition"
    }
  ]
```

Property Name	Type	Description	Filter Group and Version	Available Version
`createdRecordId`	String	ID of the created related record.	Small, 63.0	63.0
`entityApiName`	String	API name of the created object.	Small, 63.0	63.0
`entityLabel`	String	Label of the created object.	Small, 63.0	63.0

Deep Clone Response

Output representation of the details of the cloned record.

JSON example

```
{
  "createdRecordList": [
    {
      "createdRecordId": "01tSG0000030Yb3YAE",
      "entityApiName": "Product2",
      "entityLabel": "Product"
    },
    {
      "createdRecordId": "0iOSG0000002rMn2AI",
      "entityApiName": "ProductSellingModelOption",
      "entityLabel": "Product Selling Model Option"
    },
    {
      "createdRecordId": "0v7SG0000001ktdYAA",
      "entityApiName": "ProductAttributeDefinition",
      "entityLabel": "Product Attribute Definition"
    }
  ],
  "createdRootRecordId": "01tSG0000030Yb3YAE",
  "errorList": [],
  "isSuccessful": true
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`createdRecordList`	Deep Clone Record Response[]	List of cloned related records of the main record.	Small, 63.0	63.0

Property Name	Type	Description	Filter Group and Version	Available Version
`createdRootRecordId`	String	ID of the created root record.	Small, 63.0	63.0
`errorList`	Deep Clone Error[]	Details of errors, if any.	Small, 63.0	63.0
`errorMessage`	String	Error message if the API request fails.	Small, 63.0	63.0
`isSuccessful`	Boolean	Indicates whether the API request is successful (`true`) or not (`false`).	Small, 63.0	63.0

Error Output

Output representation of the error details.

Property Name	Type	Description	Filter Group and Version	Available Version
`errorCode`	String	Code of the error message.	Small, 60.0	60.0
`messageDetail`	String	Details of the error message.	Small, 60.0	60.0
`messageTitle`	String	Title of the error message.	Small, 60.0	60.0
`nodeProductId`	String	ID of the product node.	Small, 61.0	61.0
`recordId`	String	ID of the record.	Small, 60.0	60.0
`recordName`	String	Name of the record.	Small, 60.0	60.0
`relatedObjectNodes`	Invalid Related Object Node[]	List of related object nodes with errors.	Small, 62.0	62.0
`source`	String	Name of the API that's the source of the error.	Small, 60.0	60.0

Facet Value

Output representation of the facet values found in the search result.

JSON example

```
"values": [
  {
    "displayName": "Simple",
    "nameOrId": "Simple",
    "productCount": 9
  }
]
```

Property Name	Type	Description	Filter Group and Version	Available Version
`displayName`	String	Display name of the facet value.	Small, 63.0	63.0
`nameOrId`	String	Facet value name or ID. Reserved for internal use.	Small, 63.0	63.0
`productCount`		Number of products in the search result that match the facet value.	Small, 63.0	63.0

Field Data

Output representation of the field data.

Property Name	Type	Description	Filter Group and Version	Available Version
`errorCodeToErrorMap`	Map<String, Unit Of Measure Error>	Error codes mapped to their details.	Small, 63.0	63.0
`fieldApiName`	String	Unique API Name of the field.	Small, 63.0	63.0
`isRoundingApplicable`	Boolean	Indicates whether data rounding is applicable to the decimal (`true`) or not (`false`).	Small, 63.0	63.0
`originalValue`	String	Original value of the field.	Small, 63.0	63.0
`roundedValue`	String	Rounded field value that corresponds to the original value, if data rounding is applicable.	Small, 63.0	63.0
`unitOfMeasureId`	String	ID of the unit of measure record that's associated to the field.	Small, 63.0	63.0

Fields Info

Output representation of the metadata fields in an object.

JSON example


```
        "fields": [
          {
            "dataType": "text",
            "isFacetableConfigurable": true,
            "isSearchableConfigurable": false,
            "label": "Product Name",
            "name": "Name",
            "type": "Standard"
          },
          {
            "dataType": "multilinetext",
            "isFacetableConfigurable": false,
            "isSearchableConfigurable": true,
```

```
            "label": "Product Description",
            "name": "Description",
            "type": "Standard"
          }
        ]
```

Property Name	Type	Description	Filter Group and Version	Available Version
customFieldId	String	ID of the custom field.	Small, 62.0	62.0
dataType	String	Type of data.	Small, 62.0	62.0
is Configurable	Boolean	Reserved for internal use.	Small, 62.0	62.0
isFacetable Configurable	Boolean	Indicates whether the field is facetable (`true`) or not (`false`).	Small, 63.0	Small, 63.0
isSearchable Configurable	Boolean	Indicates whether the field is searchable (`true`) or not (`false`).	Small, 63.0	Small, 63.0
label	String	Label of the object field.	Small, 62.0	62.0
name	String	Name of the object field.	Small, 62.0	62.0
type	String	Type of the object field.	Small, 62.0	62.0

Index Configuration Collection

Output representation of the collection of index configuration details.

JSON example

```
{
  "correlationId": "ad960cb6-392d-4d11-bac3-3824baedf67e",
  "errors": [],
  "indexConfigurations": [
    {
      "isSearchable": true,
      "name": "Name",
      "type": "Standard"
    }
  ],
  "metadata": {
    "objectInfos": [
      {
        "fields": [
          {
            "dataType": "text",
            "isFacetableConfigurable": true,
            "isSearchableConfigurable": false,
            "label": "Product Name",
            "name": "Name",
            "type": "Standard"
```

```
          },
          {
            "dataType": "multilinetext",
            "isFacetableConfigurable": false,
            "isSearchableConfigurable": true,
            "label": "Product Description",
            "name": "Description",
            "type": "Standard"
          }
        ],
        "name": "Product2"
      },
      {
        "fields": [
          {
            "dataType": "stringplusclob",
            "label": "Description",
            "name": "Description",
            "type": "ProductAttributeDefinitionStandard"
          },
          {
            "dataType": "text",
            "label": "Name",
            "name": "Name",
            "type": "ProductAttributeDefinitionStandard"
          },
        ],
        "name": "ProductAttributeDefinition"
      }
    ]
  },
  "statusCode": "200"
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`correlationId`	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Small, 62.0	62.0
`errors`	Error Output[]	List of errors, if any.	Small, 62.0	62.0
`index Configurations`	Index Configuration Field[]	Details of the index-configured fields.	Small, 62.0	62.0
`metadata`	Metadata[]	Details of the metadata for objects.	Small, 62.0	62.0
`statusCode`	String	Code that indicates the status of the request.	Small, 62.0	62.0

Index Configuration Field

Output representation of the details of the index-configured field.

JSON example


```
"indexConfigurations": [
  {
    "attributeDefinitionId": "0tjT1000000002bIAA",
    "name": "Color",
    "type": "ProductDynamicAttribute",
    "isSearchable": true
  },
  {
    "attributeFieldId": "00Nxx000001FwnABII",
    "name": "Message__c",
    "type": "Custom",
    "isSearchable": true
  },
  {
    "name": "Code",
    "type": "Standard",
    "isSearchable": true
  },
  {
    "facetDisplayRank": 1,
    "isFacetable": false,
    "isSearchable": true,
    "name": "Family",
    "type": "Standard"
  }
]
```


Property Name	Type	Description	Filter Group and Version	Available Version
`attributeDefinitionId`	String	ID of the attribute definition.	Small, 62.0	62.0
`attributeFieldId`	String	ID of the attribute field.	Small, 62.0	62.0
`facetDisplayRank`	Integer	Sort order for displaying the facets at run time.	Small, 63.0	63.0
`isFacetable`	Boolean	Indicates whether the field is facetable (`true`) or not (`false`).	Small, 63.0	63.0
`isSearchable`	Boolean	Indicates whether the index-configured field is searchable (`true`) or not (`false`).	Small, 62.0	62.0
`name`	String	Name of the index-configured field.	Small, 62.0	62.0
`type`	String	Type of the index-configured field.	Small, 62.0	62.0

Index Configurations Update

Output representation of the updated index configuration.

JSON example

```
{
  "correlationId": "8545b5aa-f3e6-429a-8f21-9cc4ce50b1d7",
  "errors": [],
  "indexConfigurations": [
    {
      "attributeDefinitionId": "0tjT1000000002bIAA",
      "name": "Color",
      "type": "ProductDynamicAttribute",
      "isSearchable": true
    },
    {
      "attributeFieldId": "00Nxx000001FwnABII",
      "name": "Message__c",
      "type": "Custom",
      "isSearchable": true
    },
    {
      "name": "Code",
      "type": "Standard",
      "isSearchable": true
    },
    {
      "facetDisplayRank": 1,
      "isFacetable": false,
      "isSearchable": true,
      "name": "Family",
      "type": "Standard"
    }
  ],
  "statusCode": "200"
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`correlationId`	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Small, 62.0	62.0
`errors`	Error Output[]	List of errors, if any.	Small, 62.0	62.0
`index Configurations`	Index Configuration Field[]	Details of the index-configured fields.	Small, 62.0	62.0
`statusCode`	String	Code that indicates the status of the request.	Small, 62.0	62.0

Index Error

Output representation of the error details related to an index.

JSON example

```
  "indexErrorDetails": {
    "errorFileId": "069xx0000004C92AAE",
    "indexCreatedDate": "2024-10-03T05:24:18.000Z",
    "indexErrorsCount": 1,
    "indexLastUpdatedDate": "2024-10-03T05:27:00.000Z",
    "itemLevelErrorsCount": 1
  }
```

Property Name	Type	Description	Filter Group and Version	Available Version
errorFileId	String	ID of the exported error file that contains the index errors.	Small, 63.0	63.0
indexCreated Date	String	Date on which the index was created.	Small, 63.0	63.0
index ErrorsCount	Integer	Number of index-level errors.	Small, 63.0	63.0
indexLast UpdatedDate	String	Date on which the index was last updated.	Small, 63.0	63.0
itemLevel ErrorsCount	Integer	Number of item-level errors.	Small, 63.0	63.0

Index Setting

Output representation of the retrieved index settings.

JSON example

```
{
  "errors": [],
  "metadata": {
    "activeLanguages": ["en_US","ja","es","nl_NL"]
  },
  "setting": {
    "defaultLanguage": "en_US",
    "id": "1JySG0000000GUb0AM",
    "supportedLanguages": ["en_US","ja","es","nl_NL"]
  },
  "statusCode": "200"
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
errors	Error Output[]	List of errors, if any.	Small, 63.0	63.0

Property Name	Type	Description	Filter Group and Version	Available Version
`metadata`	Setting Metadata[]	Metadata associated with the setting.	Small, 63.0	63.0
`setting`	Setting[]	Setting that's used in indexing and maintained for an org.	Small, 63.0	63.0
`statusCode`	String	Code that indicates the status of the request.	Small, 63.0	63.0

Index Setting Update

Output representation of the details of the updated index setting.

JSON example

```
{
   "setting" : {
        "supportedLanguages" : ["en_US","ja","es","nl_NL"],
        "id": "1JySG0000000GUb0AM",
        "defaultLanguage" : "en_US"
   },
   "errors" : [],
   "statusCode" : "200"
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`errors`	Error Output[]	List of errors, if any.	Small, 63.0	63.0
`setting`	Setting[]	Setting that's used in indexing and maintained for an org.	Small, 63.0	63.0
`statusCode`	String	Code that indicates the status of the API request.	Small, 63.0	63.0

Invalid Related Object Node

Output representation of the invalid related object node with details of errors.

JSON example

```
To add
```

Property Name	Type	Description	Filter Group and Version	Available Version
`errorMessages`	String[]	List of error messages.	Small, 62.0	62.0
`relatedObject APIName`	String	API name of the related object.	Small, 62.0	62.0

Metadata

Output representation of the metadata details for objects.

JSON example

```
"metadata": {
  "objectInfos": [
    {
      "fields": [
        {
          "dataType": "text",
          "isFacetableConfigurable": true,
          "isSearchableConfigurable": false,
          "label": "Product Name",
          "name": "Name",
          "type": "Standard"
        },
        {
          "dataType": "multilinetext",
          "isFacetableConfigurable": false,
          "isSearchableConfigurable": true,
          "label": "Product Description",
          "name": "Description",
          "type": "Standard"
        }
      ],
      "name": "Product2"
    },
    {
      "fields": [
        {
          "dataType": "stringplusclob",
          "label": "Description",
          "name": "Description",
          "type": "ProductAttributeDefinitionStandard"
        },
        {
          "dataType": "text",
          "label": "Name",
          "name": "Name",
          "type": "ProductAttributeDefinitionStandard"
        },
      ],
      "name": "ProductAttributeDefinition"
    }
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`objectInfos`	Object Info[]	Metadata details for objects.	Small, 62.0	62.0

Object Info

Output representation of the object details along with its fields.

JSON example


```
"objectInfos": [
  {
    "fields": [
      {
        "dataType": "text",
        "isFacetableConfigurable": true,
        "isSearchableConfigurable": false,
        "label": "Product Name",
        "name": "Name",
        "type": "Standard"
      },
      {
        "dataType": "multilinetext",
        "isFacetableConfigurable": false,
        "isSearchableConfigurable": true,
        "label": "Product Description",
        "name": "Description",
        "type": "Standard"
      }
    ],
    "name": "Product2"
  },
  {
    "fields": [
      {
        "dataType": "stringplusclob",
        "label": "Description",
        "name": "Description",
        "type": "ProductAttributeDefinitionStandard"
      },
      {
        "dataType": "text",
        "label": "Name",
        "name": "Name",
        "type": "ProductAttributeDefinitionStandard"
      },
    ],
    "name": "ProductAttributeDefinition"
  }
]
```


Property Name	Type	Description	Filter Group and Version	Available Version
`fields`	Fields Info[]	Fields of the object.	Small, 62.0	62.0
`name`	String	Name of the object.	Small, 62.0	62.0

Product Catalog Management Error

Output representation that contains error details, including error codes and messages.

JSON example

```
{
  "errorCode": "INSUFFICIENT_ACCESS",
  "message": "Insufficient access rights on cross-reference ID"
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`errorCode`	String	Error code that identifies the type of error.	Small, 66.0	66.0
`message`	String	Message that explains the reason for the error.	Small, 66.0	66.0

Product Classification

Output representation of the product classification details.

JSON example

```
{
  "productClassification": {
    "id": "11BT10000004C9SMAU",
    "name": "class",
    "code": "code",
    "parentProductClassificationId": "11BDU0000004JXq2AM",
    "status": "Active"
  }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`code`	String	Code of the product classification record.	Small, 61.0	61.0
`id`	String	ID of the product classification record.	Small, 60.0	60.0
`name`	String	Name of the product classification record. If data translation is set up and specified in the org, the translated description is available.	Small, 61.0	61.0
`parentProductClassificationId`	String	ID of the parent product classification.	Small, 65.0	65.0
`status`	String	Status of the product classification record.	Small, 61.0	61.0

Product Classification Details

Output representation that contains the details of a single product classification, including its attributes and categories.

JSON example

```
{
  "id": "dummyId",
  "name": "Dummy Product Classification",
  "code": "DUMMY_CODE",
  "attributeCategories": [{
          "attributes": [
            {
              "attributeNameOverride": "Dummy_Attribute__c",
              "code": "ATTR_CODE_1",
              "dataType": "String",
              "defaultValue": "default",
              "description": "A dummy attribute for demonstration.",
              "developerName": "Dummy_Attribute",
              "displayType": "Text",
              "helpText": "Help text for dummy attribute",
              "id": "attrId1",
              "isConfigurable": true,
              "isHidden": false,
              "isPriceImpacting": false,
              "isReadOnly": false,
              "isRequired": false,
              "isValueCloneable": true,
              "label": "Dummy Attribute Label",
              "maximumCharacterCount": 100,
              "maximumValue": "100",
              "minimumCharacterCount": 1,
              "minimumValue": "1",
              "name": "Dummy Attribute",
              "sequence": 1,
              "status": "Active",
              "stepValue": "1"
            }
          ],
          "code": "GENERAL",
          "id": "catId1",
          "name": "General"
        }],
  "attributes": [{
          "attributeNameOverride": "Dummy_Attribute__c",
          "code": "ATTR_CODE_1",
          "dataType": "String",
          "defaultValue": "default",
          "description": "A dummy attribute for demonstration.",
          "developerName": "Dummy_Attribute",
          "displayType": "Text",
          "helpText": "Help text for dummy attribute",
          "id": "attrId1",
          "isConfigurable": true,
          "isHidden": false,
          "isPriceImpacting": false,
```

```
            "isReadOnly": false,
            "isRequired": false,
            "isValueCloneable": true,
            "label": "Dummy Attribute Label",
            "maximumCharacterCount": 100,
            "maximumValue": "100",
            "minimumCharacterCount": 1,
            "minimumValue": "1",
            "name": "Dummy Attribute",
            "sequence": 1,
            "status": "Active",
            "stepValue": "1"
          }]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
attributeCategories	Product Classification Attribute Category[]	List of attribute categories applicable to the product classification.	Small, 66.0	66.0
attributes	Product Classification Attribute Definition[]	List of uncategorized attributes applicable to the product classification.	Small, 66.0	66.0
code	String	Code of the product classification.	Small, 66.0	66.0
id	String	ID of the product classification.	Small, 66.0	66.0
name	String	Name of the product classification.	Small, 66.0	66.0

Product Classification Details Collection

Output representation that contains a collection of product classification details along with any processing errors.

JSON example

```
{
  "success": false,
  "errors": [
    {
      "errorCode": "INSUFFICIENT_ACCESS",
      "message": "Insufficient access rights on cross-reference ID"
    }
  ],
  "productClassifications": [
    {
      "id": "dummyId",
      "name": "Dummy Product Classification",
      "code": "DUMMY_CODE",
      "attributeCategories": [
        {
          "attributes": [
```

```
          {
            "attributeNameOverride": "Dummy_Attribute__c",
            "code": "ATTR_CODE_1",
            "dataType": "String",
            "defaultValue": "default",
            "description": "A dummy attribute for demonstration.",
            "developerName": "Dummy_Attribute",
            "displayType": "Text",
            "helpText": "Help text for dummy attribute",
            "id": "attrId1",
            "isConfigurable": true,
            "isHidden": false,
            "isPriceImpacting": false,
            "isReadOnly": false,
            "isRequired": false,
            "isValueCloneable": true,
            "label": "Dummy Attribute Label",
            "maximumCharacterCount": 100,
            "maximumValue": "100",
            "minimumCharacterCount": 1,
            "minimumValue": "1",
            "name": "Dummy Attribute",
            "sequence": 1,
            "status": "Active",
            "stepValue": "1"
          }
        ],
        "code": "GENERAL",
        "id": "catId1",
        "name": "General"
      }
    ],
    "attributes": [
      {
        "attributeNameOverride": "Dummy_Attribute__c",
        "code": "ATTR_CODE_1",
        "dataType": "String",
        "defaultValue": "default",
        "description": "A dummy attribute for demonstration.",
        "developerName": "Dummy_Attribute",
        "displayType": "Text",
        "helpText": "Help text for dummy attribute",
        "id": "attrId1",
        "isConfigurable": true,
        "isHidden": false,
        "isPriceImpacting": false,
        "isReadOnly": false,
        "isRequired": false,
        "isValueCloneable": true,
        "label": "Dummy Attribute Label",
        "maximumCharacterCount": 100,
        "maximumValue": "100",
        "minimumCharacterCount": 1,
        "minimumValue": "1",
```

```
            "name": "Dummy Attribute",
            "sequence": 1,
            "status": "Active",
            "stepValue": "1"
          }
        ]
      }
    ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`errors`	Product Catalog Management Error[]	List of errors encountered during the processing of the API request.	Small, 66.0	66.0
`productClassifications`	Product Classification Details on page 213[]	List of product classification detail records that match the requested product classification IDs.	Small, 66.0	66.0
`success`	Boolean	Indicates whether the API request is successful (`true`) or has failed (`false`).	Small, 66.0	66.0

Product Classification List Collection

Output representation that contains a collection of product classification records along with any processing errors.

JSON example

```
{
  "success": true,
  "errors": [],
  "productClassifications": [
    {
      "id": "11BT10000004C9SMAU",
      "name": "Mobile Devices",
      "code": "MOB_DEV",
      "parentProductClassificationId": "11BDU0000004JXq2AM",
      "status": "Active"
    },
    {
      "id": "11BT10000004C9TMAU",
      "name": "Mobile Accessories",
      "code": "MOB_ACC",
      "status": "Active"
    }
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
errors	Product Catalog Management Error[]	List of errors encountered during the processing of the API request.	Small, 67.0	67.0
productClassifications	Product Classification[]	List of product classification records that match the request query.	Small, 67.0	67.0
success	Boolean	Indicates whether the API request is successful (true) or has failed (false).	Small, 67.0	67.0

Product Component Group

Output representation of the product component group.

JSON example


```
"productComponentGroups":[
  {
    "childGroups":[
     {
      "childGroups":[],
    "    components":[
          {
            "additionalFields":{},
            "attributeCategory":[],
            "attributes":[],
            "catalogs":[],
            "categories":[],
            "childProducts":[],
            "id":"01txx0000006i2aAAA",
            "isActive":true,
            "isAssetizable":true,
            "isSoldOnlyWithOtherProds":true,
            "name":"GenWatt Diesel 1000kW",
            "nodeType":"simpleProduct",
            "productCode":"GC1060",
            "productComponentGroups":[],
            "productRelatedComponent":
             {
              "childProductId":"01txx0000006i2aAAA",
              "doesBundlePriceIncludeChild":true,
              "id":"0dSxx00000001P7EAI",
              "isComponentRequired":false,
              "isDefaultComponent":true,
              "isExcluded":false,
              "isQuantityEditable":false,
              "parentProductId":"01txx0000006iC8AAI",
              "productRelationshipTypeId":"0yoxx000000001dAAA",
              "quantity":1,
              "quantityScaleMethod":"Proportional"
              },
              "productSellingModelOptions":[]
```

```
            },
            {
            "additionalFields":{},
            "attributeCategory":[],
            "attributes":[],
            "catalogs":[],
            "categories":[],
            "childProducts":[],
            "id":"01txx0000006i2TAAQ",
            "isActive":true,
            "isAssetizable":true,
            "isSoldOnlyWithOtherProds":true,
            "name":"GenWatt Diesel 10kW",
            "nodeType":"simpleProduct",
            "productCode":"GC1020",
            "productComponentGroups":[],
            "productRelatedComponent":{
                "childProductId":"01txx0000006i2TAAQ",
                "doesBundlePriceIncludeChild":true,
                "id":"0dSxx00000001P8EAI",
                "isComponentRequired":false,
                "isDefaultComponent":true,
                "isExcluded":false,
                "isQuantityEditable":false,
                "parentProductId":"01txx0000006iC8AAI",
                "productRelationshipTypeId":"0yoxx000000001dAAA",
                "quantity":1,"quantityScaleMethod":"Proportional"
             },
            "productSellingModelOptions":[]
             },
             {
              "additionalFields":{},
              "attributeCategory":[],
              "attributes":[],
              "catalogs":[],
              "categories":[],
              "childProducts":[],
              "id":"01txx0000006i2SAAQ",
              "isActive":true,
              "isAssetizable":true,
              "isSoldOnlyWithOtherProds":true,
              "name":"GenWatt Diesel 200kW",
              "nodeType":"simpleProduct",
              "productCode":"GC1040",
              "productComponentGroups":[],
              "productRelatedComponent":
                {
                  "childProductId":"01txx0000006i2SAAQ",
                  "doesBundlePriceIncludeChild":true,
                  "id":"0dSxx00000001P9EAI",
                  "isComponentRequired":false,
                  "isDefaultComponent":true,
                  "isExcluded":false,
                  "isQuantityEditable":false,
```

```
                    "parentProductId":"01txx0000006iC8AAI",
                    "productRelationshipTypeId":"0yoxx000000001dAAA",
                    "quantity":1,
                    "quantityScaleMethod":"Proportional"
                   },
                  "productSellingModelOptions":[]
                    }
                  ],
                  "id":"0y7xx0000000151AAA",
                  "isExcluded":false,
                  "name":"G1.1",
                  "parentGroupId":"0y7xx0000000149AAA",
                  "parentProductId":"01txx0000006iC8AAI"
                   }
                  ],
                "components":[],
                "id":"0y7xx0000000149AAA",
                "isExcluded":false,
                "name":"G1",
                "parentProductId":"01txx0000006iC8AAI"
                }
          ]
```

Property Name	Type	Description	Filter Group and Version	Available Version
`code`	String	Unique code of the product component group, which is used only during design time.	Small, 60.0	60.0
`components`	Product[]	List of the product details.	Small, 60.0	60.0
`childGroups`	Product Component Group[]	List of child product components groups.	Small, 62.0	62.0
`description`	String	Description of the product component group.	Small, 60.0	60.0
`id`	String	ID of the record.	Small, 60.0	60.0
`isExcluded`	Boolean	Indicates whether the product component group is excluded from the product bundle for selection in the run time (`true`) or not (`false`).	Small, 60.0	60.0
`maxBundleComponents`	Integer	Maximum number of product components that can be added to a group.	Small, 60.0	60.0
`minBundleComponents`	Integer	Minimum number of product components that can be added to a group.	Small, 60.0	60.0
`name`	String	Name of the record. If data translation is set up and specified in the org, the translated description is available.	Small, 60.0	60.0

Property Name	Type	Description	Filter Group and Version	Available Version
parentProductId	String	ID associated with the parent product record.	Small, 60.0	60.0
parentGroupId	String	ID of the parent group record.	Small, 62.0	62.0
sequence	Integer	Order in which the groups are listed in the bundle.	Small, 60.0	60.0

Product

Output representation of the product definition.

JSON example

This example shows a sample of the Product List (POST) API response.

```
{
  "products": [
      {
      "additionalFields": {
        "code__c": "SWX445"
      },
       "attributeCategory": [
        {
          "attributes": [
            {
              "additionalFields": {
                "scope": "Order"
              },
              "attributeNameOverride": "AD Text",
              "code": "AD02",
              "dataType": "Text",
              "displayType": "Text",
              "MinimumCharacterCount": "1",
              "MaximumCharacterCount": "20",
              "defaultValue": "AD Text DV",
              "description": "AD Text Desc",
              "helpText": "AD Text DHT",
              "id": "0tjT1000000002bIAA",
              "isHidden": false,
              "isPriceImpacting": true,
              "isReadOnly": true,
              "isRequired": true,
              "label": "AD Text Label",
              "name": "AD Text",
              "sequence": 1,
              "status": "Active",
              "valueDescription": "AD Text VD"
            }
          ],
          "code": "AC001",
          "id": "0v3T1000000000BIAQ",
```

```
          "name": "build and make"
        }
      ],
      "attributes": [
        {
          "additionalFields": {
            "scope": "SWX445"
          },
          "attributeNameOverride": "AD Picklist",
          "code": "AD001",
          "dataType": "Picklist",
          "defaultValue": "Red",
          "description": "AD Picklist Description",
          "helpText": "AD Picklist DHT",
          "id": "0tjT1000000002WIAQ",
          "isHidden": false,
          "isPriceImpacting": false,
          "isReadOnly": false,
          "isRequired": true,
          "label": "AD Picklist Label",
          "name": "AD Picklist",
          "picklist": {
            "dataType": "Text",
            "description": "APV Description",
            "id": "0v5T10000000001IAA",
            "name": "Color",
            "values": [
              {
                "abbreviation": "Blue Abb",
                "code": "APV03",
                "displayValue": "Blue DV",
                "id": "0v6T10000000006IAA",
                "name": "Blue",
                "sequence": "3",
                "value": "Blue b",
                "status": "Active"
              },
              {
                "abbreviation": "Red Abb",
                "code": "APV04",
                "displayValue": "Red",
                "id": "0v6T10000000001IAA",
                "name": "Red",
                "sequence": "4",
                "value": "Red",
                "status": "Active"
              },
              {
                "abbreviation": "One Abb",
                "code": "APV02",
                "displayValue": "One DV",
                "id": "0v6T1000000000uIAA",
                "name": "One",
                "sequence": "2",
```

```
                "value": "One 1",
                "status": "Active"
              },
              {
                "abbreviation": "Red Abbreviation",
                "code": "APV01",
                "displayValue": "Red Display Value",
                "id": "0v6T10000000001OIAQ",
                "name": "Red",
                "sequence": "1",
                "value": "red12",
                "status": "Active"
              }
            ]
          },
          "sequence": 1,
          "status": "Active",
          "valueDescription": "AD Picklist VD"
        }
      ],
      "categories": [],
      "childProducts": [
        {
          "attributeCategory": [],
          "attributes": [],
          "categories": [],
          "childProducts": [],
          "configureDuringSale": "NotAllowed",
          "id": "01tZ7000000AJkaIAG",
          "isActive": false,
          "isAssetizable": true,
          "isSoldOnlyWithOtherProds": false,
          "name": "Earphones",
          "nodeType": "bundleProduct",
          "productComponentGroups": [],
          "productRelatedComponent": {
            "childProductId": "01tZ7000000AJkaIAG",
            "doesBundlePriceIncludeChild": true,
            "id": "0dSZ700000000cdMAA",
            "isComponentRequired": false,
            "isDefaultComponent": true,
            "isExcluded": false,
            "isQuantityEditable": false,
            "parentProductId": "01tZ7000000AJXOIA4",
            "productRelationshipTypeId": "0yoZ700000000kPIAQ",
            "quantity": 1,
            "quantityScaleMethod": "Proportional"
          },
          "productSellingModelOptions": []
        }
      ],
      "description": "Keep your organization connected with seamless collaboration across
 distributed teams. No matter where employees are located, organizations are seeking
stronger employee engagement and customer experiences to enable more productivity and
```

```
greater business agility. More effective collaboration helps organizations work smarter.",

      "displayUrl":
"https://dispatch.m.io/wp-content/uploads/2023/01/History-of-Webex.png",
      "id": "01t1Q000008CD2eQAG",
      "isActive": true,
      "isAssetizable": true,
      "isSoldOnlyWithOtherProds": false,
      "name": "SmartBytes Collaboration Suite",
      "nodeType": "bundleProduct",
      "productClassification": {
        "id": "11B1Q0000008OMGUA2"
      },
      "productCode": "P0143",
      "productComponentGroups": [
        {
          "components": [
            {
              "attributeCategory": [],
              "attributes": [],
              "categories": [],
              "childProducts": [],
              "id": "01tZ7000000AJXTIA4",
              "isActive": false,
              "isAssetizable": true,
              "isSoldOnlyWithOtherProds": false,
              "name": "Charger",
              "nodeType": "bundleProduct",
              "productComponentGroups": [],
              "productRelatedComponent": {
                "childProductId": "01tZ7000000AJXTIA4",
                "doesBundlePriceIncludeChild": true,
                "id": "0dSZ700000000YLMAY",
                "isComponentRequired": false,
                "isDefaultComponent": true,
                "isExcluded": false,
                "isQuantityEditable": false,
                "parentProductId": "01tZ7000000AJXOIA4",
                "productRelationshipTypeId": "0yoZ700000000kPIAQ",
                "quantity": 1,
                "quantityScaleMethod": "Proportional"
              },
              "productSellingModelOptions": []
            }
          ],
          "id": "0y7Z700000000TtIAI",
          "isExcluded": false,
          "name": "Box",
          "parentProductId": "01tZ7000000AJXOIA4"
        }
      ],
      "productSellingModelOptions": [
        {
          "id": "0iO1Q0000008OkeUAE",
```

```
          "productId": "01t1Q000008CD2eQAG",
          "productSellingModel": {
            "id": "0jP1Q000000CaVFUA0",
            "isDefault": true,
            "name": "One Time",
            "sellingModelType": "OneTime",
            "status": "Active"
          }
        }
      ],
      "productSpecificationType": {
        "name": "None"
      }
    }
],
  "status": {
    "code": "200",
    "correlationId": "fd158d80-d73c-4a1f-a009-9225db804d70",
    "errors": [],
    "message": "Successfully fetched product records."
  }
}
```

This example shows a sample of the Bulk Product Details (POST) API response.

```
{
  "products": [
      {
      "additionalFields": {
        "code__c": "SWX445"
      },
       "attributeCategory": [
        {
          "attributes": [
            {
              "additionalFields": {
                "scope": "Order"
              },
              "attributeNameOverride": "AD Text",
              "code": "AD02",
              "dataType": "Text",
              "displayType": "Text",
              "MinimumCharacterCount": "1",
              "MaximumCharacterCount": "20",
              "defaultValue": "AD Text DV",
              "description": "AD Text Desc",
              "helpText": "AD Text DHT",
              "id": "0tjT1000000002bIAA",
              "isHidden": false,
              "isPriceImpacting": true,
              "isReadOnly": true,
              "isRequired": true,
              "label": "AD Text Label",
              "name": "AD Text",
              "sequence": 1,
```

```
            "status": "Active",
            "valueDescription": "AD Text VD"
          }
        ],
        "code": "AC001",
        "id": "0v3T1000000000BIAQ",
        "name": "build and make"
      }
    ],
    "attributes": [
      {
        "additionalFields": {
          "scope": "SWX445"
        },
        "attributeNameOverride": "AD Picklist",
        "code": "AD001",
        "dataType": "Picklist",
        "defaultValue": "Red",
        "description": "AD Picklist Description",
        "helpText": "AD Picklist DHT",
        "id": "0tjT1000000002WIAQ",
        "isHidden": false,
        "isPriceImpacting": false,
        "isReadOnly": false,
        "isRequired": true,
        "label": "AD Picklist Label",
        "name": "AD Picklist",
        "picklist": {
          "dataType": "Text",
          "description": "APV Description",
          "id": "0v5T10000000001IAA",
          "name": "Color",
          "values": [
            {
              "abbreviation": "Blue Abb",
              "code": "APV03",
              "displayValue": "Blue DV",
              "id": "0v6T10000000006IAA",
              "name": "Blue",
              "sequence": "3",
              "value": "Blue b",
              "status": "Active"
            },
            {
              "abbreviation": "Red Abb",
              "code": "APV04",
              "displayValue": "Red",
              "id": "0v6T10000000001IAA",
              "name": "Red",
              "sequence": "4",
              "value": "Red",
              "status": "Active"
            },
            {
```

```
          "abbreviation": "One Abb",
          "code": "APV02",
          "displayValue": "One DV",
          "id": "0v6T1000000000uIAA",
          "name": "One",
          "sequence": "2",
          "value": "One 1",
          "status": "Active"
        },
        {
          "abbreviation": "Red Abbreviation",
          "code": "APV01",
          "displayValue": "Red Display Value",
          "id": "0v6T10000000001OIAQ",
          "name": "Red",
          "sequence": "1",
          "value": "red12",
          "status": "Active"
        }
      ]
    },
    "sequence": 1,
    "status": "Active",
    "valueDescription": "AD Picklist VD"
  }
],
"categories": [],
"childProducts": [
  {
    "attributeCategory": [],
    "attributes": [],
    "categories": [],
    "childProducts": [],
    "configureDuringSale": "NotAllowed",
    "id": "01tZ7000000AJkaIAG",
    "isActive": false,
    "isAssetizable": true,
    "isSoldOnlyWithOtherProds": false,
    "name": "Earphones",
    "nodeType": "bundleProduct",
    "productComponentGroups": [],
    "productRelatedComponent": {
      "childProductId": "01tZ7000000AJkaIAG",
      "doesBundlePriceIncludeChild": true,
      "id": "0dSZ700000000cdMAA",
      "isComponentRequired": false,
      "isDefaultComponent": true,
      "isExcluded": false,
      "isQuantityEditable": false,
      "parentProductId": "01tZ7000000AJXOIA4",
      "productRelationshipTypeId": "0yoZ700000000kPIAQ",
      "quantity": 1,
      "quantityScaleMethod": "Proportional"
    },
```

```
          "productSellingModelOptions": []
        }
      ],
     "description": "Keep your organization connected with seamless collaboration across
 distributed teams. No matter where employees are located, organizations are seeking
stronger employee engagement and customer experiences to enable more productivity and
greater business agility. More effective collaboration helps organizations work smarter.",

      "displayUrl":
"https://dispatch.m.io/wp-content/uploads/2023/01/History-of-Webex.png",
      "id": "01t1Q000008CD2eQAG",
      "isActive": true,
      "isAssetizable": true,
      "isSoldOnlyWithOtherProds": false,
      "name": "SmartBytes Collaboration Suite",
      "nodeType": "bundleProduct",
      "productClassification": {
        "id": "11B1Q0000008OMGUA2",
        "name" : "class",
        "code" : "code",
        "status" : "Active"
      },
      "productCode": "P0143",
      "productComponentGroups": [
        {
          "components": [
            {
              "attributeCategory": [],
              "attributes": [],
              "categories": [],
              "childProducts": [],
              "id": "01tZ7000000AJXTIA4",
              "isActive": false,
              "isAssetizable": true,
              "isSoldOnlyWithOtherProds": false,
              "name": "Charger",
              "nodeType": "bundleProduct",
              "productComponentGroups": [],
              "productRelatedComponent": {
                "childProductId": "01tZ7000000AJXTIA4",
                "doesBundlePriceIncludeChild": true,
                "id": "0dSZ700000000YLMAY",
                "isComponentRequired": false,
                "isDefaultComponent": true,
                "isExcluded": false,
                "isQuantityEditable": false,
                "parentProductId": "01tZ7000000AJXOIA4",
                "productRelationshipTypeId": "0yoZ700000000kPIAQ",
                "quantity": 1,
                "quantityScaleMethod": "Proportional"
              },
              "productSellingModelOptions": []
            }
          ],
```

```
          "id": "0y7Z700000000TtIAI",
          "isExcluded": false,
          "name": "Box",
          "parentProductId": "01tZ7000000AJXOIA4"
        }
      ],
      "productSellingModelOptions": [
        {
          "id": "0iO1Q0000008OkeUAE",
          "productId": "01t1Q000008CD2eQAG",
          "productSellingModel": {
            "id": "0jP1Q000000CaVFUA0",
            "isDefault": true,
            "name": "One Time",
            "sellingModelType": "OneTime",
            "status": "Active"
          }
        }
      ],
      "productSpecificationType": {
        "name": "None"
      }
    }
],
  "status": {
    "code": "200",
    "correlationId": "fd158d80-d73c-4a1f-a009-9225db804d70",
    "errors": [],
    "message": "Successfully fetched product records."
  }
}
```


This example shows a sample of the Product By ID (GET) API response.


```
{
  "products": [
    {
      "attributeCategory": [
        {
          "attributes": [
            {
              "attributeNameOverride": "AD Text",
              "code": "AD02",
              "dataType": "Text",
              "displayType": "Text",
              "MinimumCharacterCount": "1",
              "MaximumCharacterCount": "20",
              "defaultValue": "AD Text DV",
              "description": "AD Text Desc",
              "helpText": "AD Text DHT",
              "id": "0tjT1000000002bIAA",
              "isHidden": false,
              "isPriceImpacting": true,
              "isReadOnly": true,
              "isRequired": true,
```

```
          "label": "AD Text Label",
          "name": "AD Text",
          "sequence": 1,
          "status": "Active",
          "valueDescription": "AD Text VD"
        }
      ],
      "code": "AC001",
      "id": "0v3T1000000000BIAQ",
      "name": "build and make"
    }
  ],
  "attributes": [
    {
      "attributeNameOverride": "AD Picklist",
      "code": "AD001",
      "dataType": "Picklist",
      "defaultValue": "Red",
      "description": "AD Picklist Description",
      "helpText": "AD Picklist DHT",
      "id": "0tjT1000000002WIAQ",
      "isHidden": false,
      "isPriceImpacting": false,
      "isReadOnly": false,
      "isRequired": true,
      "label": "AD Picklist Label",
      "name": "AD Picklist",
      "picklist": {
        "dataType": "Text",
        "description": "APV Description",
        "id": "0v5T10000000001IAA",
        "name": "Color",
        "values": [
          {
            "abbreviation": "Blue Abb",
            "code": "APV03",
            "displayValue": "Blue DV",
            "id": "0v6T10000000006IAA",
            "name": "Blue",
            "sequence": "3",
            "value": "Blue b",
            "status": "Active"
          },
          {
            "abbreviation": "Red Abb",
            "code": "APV04",
            "displayValue": "Red",
            "id": "0v6T10000000001IAA",
            "name": "Red",
            "sequence": "4",
            "value": "Red",
            "status": "Active"
          },
          {
```

```
            "abbreviation": "One Abb",
            "code": "APV02",
            "displayValue": "One DV",
            "id": "0v6T1000000000uIAA",
            "name": "One",
            "sequence": "2",
            "value": "One 1",
            "status": "Active"
          },
          {
            "abbreviation": "Red Abbreviation",
            "code": "APV01",
            "displayValue": "Red Display Value",
            "id": "0v6T10000000001OIAQ",
            "name": "Red",
            "sequence": "1",
            "value": "red12",
            "status": "Active"
          }
        ]
      },
      "sequence": 1,
      "status": "Active",
      "valueDescription": "AD Picklist VD"
    }
  ],
  "availabilityDate": "2023-07-12T19:00:00.000Z",
  "categories": [],
  "childProducts": [],
  "configureDuringSale": "Allowed",
  "description": "Bundle Product Description",
  "discontinuedDate": "2023-07-27T19:00:00.000Z",
  "displayUrl": "www.google.com",
  "endOfLifeDate": "2023-07-31T19:00:00.000Z",
  "id": "01tT1000000F0afIAC",
  "isActive": true,
  "isAssetizable": true,
  "isSoldOnlyWithOtherProds": true,
  "name": "Bundle Product",
  "nodeType": "bundleProduct",
  "productClassification": {
    "id": "11BT10000004C9SMAU",
    "name" : "class",
    "code" : "code",
    "status" : "Active"
  },
  "productCode": "P001",
  "productComponentGroups": [
    {
      "code": "PCG002",
      "components": [
        {
          "attributeCategory": [],
          "attributes": [],
```

```
        "categories": [],
        "childProducts": [],
        "nodeType": "productClass",
        "productClassification": {
          "id": "11BT10000004C9SMAU",
          "name" : "class",
          "code" : "code",
          "status" : "Active"
        },
        "productComponentGroups": [],
        "productRelatedComponent": {
          "childSellingModelId": "0jPT10000004CAfMAM",
          "doesBundlePriceIncludeChild": true,
          "id": "0dST100000000rlMAA",
          "isComponentRequired": false,
          "isDefaultComponent": false,
          "isQuantityEditable": true,
          "maxQuantity": 5,
          "minQuantity": 1,
          "parentProductId": "01tT1000000F0afIAC",
          "productClassificationId": "11BT10000004C9SMAU",
          "productRelationshipTypeId": "0yoT10000004CBEIA2",
          "quantity": 1,
          "quantityScaleMethod": "Proportional",
          "sequence": 2,
          "isExcluded": false
        },
        "productSellingModelOptions": []
      }
    ],
    "description": "PCG002 desc",
    "id": "0y7T10000004C9IIAU",
    "maxBundleComponents": 5,
    "minBundleComponents": 1,
    "name": "PCG002",
    "parentProductId": "01tT1000000F0afIAC",
    "sequence": 2,
    "isExcluded": false
  },
  {
    "code": "PCG001",
    "components": [
      {
        "attributeCategory": [],
        "attributes": [
          {
            "attributeNameOverride": "AD Picklist",
            "code": "AD001",
            "dataType": "Picklist",
            "defaultValue": "Red",
            "description": "AD Picklist Description",
            "helpText": "AD Picklist DHT",
            "id": "0tjT1000000002WIAQ",
            "isHidden": false,
```

```
"isPriceImpacting": false,
"isReadOnly": false,
"isRequired": true,
"label": "AD Picklist Label",
"name": "AD Picklist",
"picklist": {
  "dataType": "Text",
  "description": "APV Description",
  "id": "0v5T10000000001IAA",
  "name": "Color",
  "values": [
    {
      "abbreviation": "Blue Abb",
      "code": "APV03",
      "displayValue": "Blue DV",
      "id": "0v6T10000000006IAA",
      "name": "Blue",
      "sequence": "3",
      "value": "Blue b",
      "status": "Active"
    },
    {
      "abbreviation": "Red Abb",
      "code": "APV04",
      "displayValue": "Red",
      "id": "0v6T10000000001IAA",
      "name": "Red",
      "sequence": "4",
      "value": "Red",
      "status": "Active"
    },
    {
      "abbreviation": "One Abb",
      "code": "APV02",
      "displayValue": "One DV",
      "id": "0v6T1000000000uIAA",
      "name": "One",
      "sequence": "2",
      "value": "One 1",
      "status": "Active"
    },
    {
      "abbreviation": "Red Abbreviation",
      "code": "APV01",
      "displayValue": "Red Display Value",
      "id": "0v6T1000000001OIAQ",
      "name": "Red",
      "sequence": "1",
      "value": "red12",
      "status": "Active"
    }
  ]
},
"sequence": 1,
```

```
        "status": "Active",
        "valueDescription": "AD Picklist VD"
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        "displayType": "Text",
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        "MaximumCharacterCount": "20",
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        "helpText": "AD Text DHT",
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        "isReadOnly": false,
        "isRequired": true,
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        "valueDescription": "AD Text VD"
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    "endOfLifeDate": "2023-07-28T19:00:00.000Z",
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    "isAssetizable": true,
    "isSoldOnlyWithOtherProds": false,
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      "name" : "class",
      "code" : "code",
      "status" : "Active"
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        "code": "PCG2",
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      "childProducts": [],
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        "isDefaultComponent": false,
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        "productRelationshipTypeId": "0yoT1000000002WIAQ",
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            "productSpecificationRecordType": null
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            "isExcluded": false
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            "productSpecificationRecordType": null
          }
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      "parentProductId": "01tT1000000F0YyIAK",
      "isExcluded": false
    }
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```

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            "isDefaultComponent": false,
            "isQuantityEditable": true,
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            "productRelationshipTypeId": "0yoT1000000002WIAQ",
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                "status": "Active"
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            "productSpecificationRecordType": null
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      "minBundleComponents": 1,
      "name": "PCG001",
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```

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      "productSpecificationRecordType": null
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            "name": "AD Text",
            "sequence": 1,
            "status": "Active",
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        "helpText": "AD Picklist DHT",
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      "values": [
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          "code": "APV03",
          "displayValue": "Blue DV",
          "id": "0v6T10000000006IAA",
          "name": "Blue",
          "sequence": "3",
          "value": "Blue b",
          "status": "Active"
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        {
          "abbreviation": "Red Abb",
          "code": "APV04",
          "displayValue": "Red",
          "id": "0v6T10000000001IAA",
          "name": "Red",
          "sequence": "4",
          "value": "Red",
          "status": "Active"
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          "code": "APV02",
          "displayValue": "One DV",
          "id": "0v6T1000000000uIAA",
          "name": "One",
          "sequence": "2",
          "value": "One 1",
          "status": "Active"
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          "code": "APV01",
          "displayValue": "Red Display Value",
          "id": "0v6T1000000001OIAQ",
          "name": "Red",
          "sequence": "1",
          "value": "red12",
          "status": "Active"
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    "sequence": 1,
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    "valueDescription": "AD Picklist VD"
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```

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    "description": "Bundle Product Description",
    "discontinuedDate": "2023-07-27T19:00:00.000Z",
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    "endOfLifeDate": "2023-07-31T19:00:00.000Z",
    "id": "01tT1000000F0afIAC",
    "isActive": true,
    "isAssetizable": true,
    "isSoldOnlyWithOtherProds": true,
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      "name" : "class",
      "code" : "code",
      "status" : "Active"
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            "attributes": [],
            "categories": [],
            "childProducts": [],
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            "productClassification": {
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              "name" : "class",
              "code" : "code",
              "status" : "Active"
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              "id": "0dST100000000rlMAA",
              "isComponentRequired": false,
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              "maxQuantity": 5,
              "minQuantity": 1,
              "parentProductId": "01tT1000000F0afIAC",
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    "parentProductId": "01tT1000000F0afIAC",
    "sequence": 2,
    "isExcluded": false
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            "code": "AD001",
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            "defaultValue": "Red",
            "description": "AD Picklist Description",
            "helpText": "AD Picklist DHT",
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            "isReadOnly": false,
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                  "code": "APV03",
                  "displayValue": "Blue DV",
                  "id": "0v6T10000000006IAA",
                  "name": "Blue",
                  "sequence": "3",
                  "value": "Blue b",
                  "status": "Active"
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                {
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            "status": "Active"
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            "code": "APV02",
            "displayValue": "One DV",
            "id": "0v6T1000000000uIAA",
            "name": "One",
            "sequence": "2",
            "value": "One 1",
            "status": "Active"
          },
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            "code": "APV01",
            "displayValue": "Red Display Value",
            "id": "0v6T1000000001OIAQ",
            "name": "Red",
            "sequence": "1",
            "value": "red12",
            "status": "Active"
          }
        ]
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      "sequence": 1,
      "status": "Active",
      "valueDescription": "AD Picklist VD"
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      "displayType": "Text",
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      "MaximumCharacterCount": "20",
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      "description": "AD Text Desc",
      "helpText": "AD Text DHT",
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      "isHidden": false,
      "isPriceImpacting": false,
      "isReadOnly": false,
      "isRequired": true,
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      "name": "AD Text",
      "status": "Active",
      "valueDescription": "AD Text VD"
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"displayUrl": "www.google.com",
"endOfLifeDate": "2023-07-28T19:00:00.000Z",
"id": "01tT1000000F0YyIAK",
"isActive": false,
"isAssetizable": true,
"isSoldOnlyWithOtherProds": false,
"name": "Child1 - Bundle with PCG",
"nodeType": "bundleProduct",
"productClassification": {
  "id": "11BT10000004C9SMAU",
  "name" : "class",
  "code" : "code",
  "status" : "Active"
},
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"productComponentGroups": [
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    "code": "PCG2",
    "components": [
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        "attributes": [],
        "categories": [],
        "childProducts": [],
        "id": "01tT1000000F0Z8IAK",
        "isActive": false,
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        "isSoldOnlyWithOtherProds": false,
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          "isDefaultComponent": false,
          "isQuantityEditable": false,
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          "sequence": 2,
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          "productSpecificationRecordType": null
```

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    }
  },
  {
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    "categories": [],
    "childProducts": [],
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    "displayUrl": "Test",
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    "id": "01tT1000000F0YzIAK",
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      "productSpecificationRecordType": null
    }
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  {
    "attributeCategory": [],
    "attributes": [],
    "categories": [],
    "childProducts": [],
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    "nodeType": "bundleProduct",
    "productCode": "PC003",
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              "productRelationshipTypeId": "0yoT1000000002WIAQ",
              "quantity": 1,
              "quantityScaleMethod": "Proportional",
              "isExcluded": false
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            "productSellingModelOptions": [],
            "productSpecificationType": {
              "name": "NonCommercialSpecType",
              "productSpecificationRecordType": null
            }
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        "minBundleComponents": 1,
        "name": "PCG2",
        "parentProductId": "01tT1000000F0YyIAK",
        "isExcluded": false
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      "childSellingModelId": "0jPT10000004CAfMAM",
      "doesBundlePriceIncludeChild": true,
      "id": "0dST100000000rgMAA",
      "isComponentRequired": false,
      "isDefaultComponent": false,
      "isQuantityEditable": true,
      "maxQuantity": 3,
      "minQuantity": 1,
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      "productRelationshipTypeId": "0yoT1000000002WIAQ",
      "quantity": 1,
      "quantityScaleMethod": "Proportional",
      "sequence": 1,
      "isExcluded": false
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        "productId": "01tT1000000F0YyIAK",
        "isDefault": false,
        "productSellingModel": {
          "id": "0jPT10000004CAfMAM",
```

```
                    "name": "OneTimePSM",
                    "sellingModelType": "OneTime",
                    "status": "Active"
                  }
                }
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                "name": "NonCommercialSpecType",
                "productSpecificationRecordType": null
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            }
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          "id": "0y7T10000004C9DIAU",
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            "name": "OneTimePSM",
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            "status": "Active"
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      "productSpecificationType": {
        "name": "NonCommercialSpecType",
        "productSpecificationRecordType": null
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    }
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    "correlationId": "fd158d80-d73c-4a1f-a009-9225db804d70",
    "errors": [],
    "message": "Successfully fetched Product records."
  }
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`additional Fields`	Map<String, Additional Fields Input>	Key-value pair of additional standard or custom fields with their values.	Small, 61.0	61.0

Property Name	Type	Description	Filter Group and Version	Available Version
`attribute Category`	Attribute Category[]	List of categorized attributes related to the product.	Small, 60.0	60.0
`attributes`	Attribute Definition[]	List of uncategorized attributes related to the product.	Small, 60.0	60.0
`availability Date`	String	Date when the part is used in the product or is made available for sale.	Small, 60.0	60.0
`catalogs`	Catalog[]	List of the associated catalogs returned with the Product List API (POST) response. The Product By ID API (GET) returns an empty catalog list in the response. Returns the `name` and `id` values only.	Small, 61.0	61.0
`categories`	Category[]	List of the associated categories returned with the Product List API (POST) response. The Product By ID API (GET) returns an empty category list in the response. Returns the `name` and `id` values only.	Small, 60.0	60.0
`childProducts`	Product[]	Hierarchy of the child products.	Small, 60.0	60.0
`configure DuringSale`	String	Determines whether to allow or prevent configuration when a bundle is sold.	Small, 60.0	60.0
`description`	String	Description of the product. If data translation is set up and specified in the org, the translated description is available.	Small, 60.0	60.0
`discontinued Date`	String	Date from when the part can't be used in the product or sold.	Small, 60.0	60.0
`displayUrl`	String	Display image URL of the product.	Small, 60.0	60.0
`endOfLifeDate`	String	Date after which a product isn't supported, ordered, or maintained.	Small, 60.0	60.0
`id`	String	ID of the product.	Small, 60.0	60.0
`isActive`	Boolean	Indicates if the product is active (`true`) or not (`false`).	Small, 60.0	60.0
`isAssetizable`	Boolean	Indicates if the product instance remains a customer asset after it's purchased (`true`) or not (`false`).	Small, 60.0	60.0
`isSoldOnly WithOther Prods`	Boolean	Indicates whether the product can't be sold separately (`true`) or not (`false`).	Small, 60.0	60.0

Property Name	Type	Description	Filter Group and Version	Available Version
name	String	Name of the product. If data translation is set up and specified in the org, the translated name is available.	Small, 60.0	60.0
nodeType	String	Type of the node, such as a product or bundled product.	Small, 60.0	60.0
productClassification	Product Classification	Details of the product classification that the product is based on.	Small, 60.0	60.0
productCode	String	Universal product code that's used to track the part that's used in the product.	Small, 60.0	60.0
productComponentGroups	Product Component Group[]	Logical grouping of the component products in a bundle and the group cardinality for ordering the product components.	Small, 60.0	60.0
productRelatedComponent	Product Related Component	Details of the related components of a product.	Small, 60.0	60.0
productSellingModelOptions	Product Selling Model Option[]	Details of the product selling model options.	Small, 60.0	60.0
productSpecificationType	Product Specification Type	Details of the product specification type.	Small, 60.0	60.0
quantityScaleMethod	String	Method to scale the quantity of the child product in relation to the quantity of the parent.	Small, 60.0	60.0

Product Related Component

Output representation of the product-related component.

JSON example

```
"productRelatedComponent": {
      "childProductId": "01tT1000000F0YyIAK",
      "childSellingModelId": "0jPT10000004CAfMAM",
      "doesBundlePriceIncludeChild": true,
      "id": "0dST100000000rgMAA",
      "isComponentRequired": false,
      "isDefaultComponent": false,
      "isExcluded": false,
      "isQuantityEditable": true,
      "maxQuantity": 3,
      "minQuantity": 1,
```

```
        "parentProductId": "01tT1000000F0afIAC",
        "parentSellingModelId": "0jPT10000004CAfMAM",
        "productClassificationId": "11BRO00000000222AA",
        "productRelationshipTypeId": "0yoT1000000002WIAQ",
        "quantity": 1,
        "quantityScaleMethod": "Proportional",
        "quoteVisibility": "Quote Document Only",
        "sequence": 1
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`childProductId`	String	Lookup to the child product in the bundle.	Small, 60.0	60.0
`childSellingModelId`	String	ID of the child product selling model record.	Small, 60.0	60.0
`doesBundlePriceIncludeChild`	Boolean	Indicates whether the price of the bundle includes the child product (`true`) or not (`false`).	Small, 60.0	60.0
`id`	String	ID of the record.	Small, 60.0	60.0
`isComponentRequired`	Boolean	Indicates whether the component is required in the bundle (`true`) or not (`false`).	Small, 60.0	60.0
`isDefaultComponent`	Boolean	Indicates whether to select the component in the bundle group by default (`true`) or not (`false`).	Small, 60.0	60.0
`isExcluded`	Boolean	Indicates whether the component is excluded in the bundle group (`true`) or not (`false`).	Small, 60.0	60.0
`isQuantityEditable`	Boolean	Indicates whether to allow changes to the quantity of the component in the bundle (`true`) or not (`false`).	Small, 60.0	60.0
`maxQuantity`	Double	Maximum quantity of the product in the opportunity, quote, or order line item.	Small, 60.0	60.0
`minQuantity`	Double	Minimum quantity of the product in the opportunity, quote, or order line item.	Small, 60.0	60.0
`parentProductId`	String	Lookup to the parent product.	Small, 60.0	60.0
`parentSellingModelId`	String	ID of the product selling model record.	Small, 60.0	60.0

Property Name	Type	Description	Filter Group and Version	Available Version
`productClassificationId`	String	ID of the product classification record.	Small, 60.0	60.0
`productInstanceReuse`	String	Reserved for future use.	Small, 62.0	62.0
`productRelationshipTypeId`	String	ID of the product relationship type record.	Small, 60.0	60.0
`quantity`	Double	Quantity of the child products.	Small, 60.0	60.0
`quantityScaleMethod`	String	Method to scale the quantity of the child product in relation to the quantity of the parent. Valid values are: • `Constant` • `Proportional`	Small, 60.0	60.0
`quoteVisibility`	String	Specifies whether a quote line item must be shown on the transaction line editor or quote document. Valid values are: • `Always` • `Transaction Line Editor Only`—Specifies whether to show a quote line item on quote editor only. • `Quote Document Only`—Specifies whether to show a quote line item on quote proposal only. • `Never` The API returns this property only if the CoreCPQ permission set is available.	Small, 64.0	64.0
`sequence`	Integer	Order in which the child products are displayed.	Small, 60.0	60.0

Product Selling Model

Output representation of the definition of the product selling model.

JSON example


```
"productSellingModel":
{
  "id": "0jPT10000004CAfMAM",
  "name": "OneTimePSM",
```

```
  "pricingTerm": 1,
  "pricingTermUnit": "Months",
  "sellingModelType": "TermDefined",
  "status": "Active"
}
}]
```

Property Name	Type	Description	Filter Group and Version	Available Version
id	String	ID of the record.	Small, 60.0	60.0
name	String	Name of the record.	Small, 60.0	60.0
pricingTerm	Integer	Duration of the selling model.	Small, 60.0	60.0
pricingTermUnit	String	Units of the pricing term.	Small, 60.0	60.0
sellingModelType	String	Different models of selling the product. Valid values are: • OneTime • TermDefined • Evergreen	Small, 60.0	60.0
status	String	Status of the selling model. For example, whether the selling model is active and can be used in transactions.	Small, 60.0	60.0

Product Selling Model Option

Output representation of the definition of the product selling model option.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

JSON example

```
"productSellingModelOptions": [
   {
    "id": "0iOT10000004CMrMAM",
    "isDefault": false,
    "productId": "01tT1000000F0YyIAK",
    "productSellingModel": {
    "id": "0jPT10000004CAfMAM",
    "name": "OneTimePSM",
    "pricingTerm": 1,
    "pricingTermUnit": "Months",
    "sellingModelType": "TermDefined",
    "status": "Active"
  }
}]
```

Property Name	Type	Description	Filter Group and Version	Available Version
id	String	ID of the record.	Small, 60.0	60.0
isDefault	Boolean	Indicates whether this model option is the default product selling model option (true) or not (false).	Small, 60.0	60.0
productId	String	ID of the product.	Small, 60.0	60.0
product SellingModel	Product Selling Model	Master-detail field to the product selling model.	Small, 60.0	60.0

Product Specification Type

Output representation of the product specification type.

JSON example

```
"productSpecificationType":
{
"name": "NonCommercialSpecType",
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
name	String	Name of the product specification type.	Small, 60.0	60.0

Products

Output representation of the list of retrieved products.

JSON example

```
{
  "correlationId": "9b6bc520-3c82-4d6c-a458-47590370681a",
  "facets":[
    {
      "attributeType": "ProductStandard",
      "displayName": "Product Type",
      "displayRank": 1,
      "displayType": "MultiSelect",
      "nameOrId": "Type",
      "values": [
        {
          "displayName": "Simple",
          "nameOrId": "Simple",
          "productCount": 9
        }
      ]
    },
    {
```

```
      "attributeType": "ProductStandard",
      "displayName": "Active",
      "displayRank": 2,
      "displayType": "MultiSelect",
      "nameOrId": "IsActive",
      "values": [
        {
          "displayName": "true",
          "nameOrId": "true",
          "productCount": 47
        }
      ]
    }
  ],
  "count": 3,
  "products": [
    {
      "attributeCategory": [],
      "attributes": [],
      "categories": [],
      "childProducts": [],
     "description": "Keep your organization connected with seamless collaboration across
 distributed teams. No matter where employees are located, organizations are seeking
stronger employee engagement and customer experiences to enable more productivity and
greater business agility. More effective collaboration helps organizations work smarter.",

      "displayUrl":
"https://dispatch.m.io/wp-content/uploads/2023/01/History-of-Webex.png",
      "id": "01t1Q000008CD2eQAG",
      "isActive": true,
      "isAssetizable": true,
      "isSoldOnlyWithOtherProds": false,
      "name": "SmartBytes Collaboration Suite",
      "nodeType": "bundleProduct",
      "productClassification": {
        "id": "11B1Q0000008OMGUA2"
      },
      "productCode": "P0143",
      "productComponentGroups": [],
      "productSellingModelOptions": [
        {
          "id": "0iO1Q00000008OkeUAE",
          "productId": "01t1Q000008CD2eQAG",
          "productSellingModel": {
            "id": "0jP1Q000000CaVFUA0",
            "name": "One Time",
            "sellingModelType": "OneTime",
            "status": "Active"
          }
        }
      ],
      "productSpecificationType": {
        "name": "None"
      }
```

```
    },
    {
      "attributeCategory": [],
      "attributes": [],
      "categories": [],
      "childProducts": [],
      "configureDuringSale": "Allowed",
      "displayUrl":
"https://www.cisco.com/c/dam/en/us/products/collateral/wireless/small-business-300-series-wireless-access-points/datasheet-c78-732143.doc/_jcr_content/renditions/datasheet-c78-732143_1.jpg",

      "id": "01t1Q000008CD36QAG",
      "isActive": true,
      "isAssetizable": true,
      "isSoldOnlyWithOtherProds": false,
      "name": "SmartBytes Repeater",
      "nodeType": "simpleProduct",
      "productClassification": {
        "id": "11B1Q0000008OMMUA2"
      },
      "productCode": "BS_R001",
      "productComponentGroups": [],
      "productSellingModelOptions": [
        {
          "id": "0iO1Q0000008OlsUAE",
          "productId": "01t1Q000008CD36QAG",
          "productSellingModel": {
            "id": "0jP1Q000000CaVFUA0",
            "name": "One Time",
            "sellingModelType": "OneTime",
            "status": "Active"
          }
        }
      ],
      "productSpecificationType": {
        "name": "COM_Offer"
      }
    },
    {
      "attributeCategory": [],
      "attributes": [],
      "categories": [],
      "childProducts": [],
      "configureDuringSale": "Allowed",
      "description": "A converged data center architecture that integrates computing,
networking and storage resources to increase efficiency and enable centralized management.
 UCS products are designed and configured to work together effectively.",
      "displayUrl":
"https://www.cisco.com/c/dam/en/us/products/collateral/servers-unified-computing/ucs-x-series-modular-system/sap-hana-scale-appliance-with-x-series-wp.docx/_jcr_content/renditions/sap-hana-scale-appliance-with-x-series-wp_3.jpg",

      "id": "01t1Q000008CD2sQAG",
      "isActive": true,
      "isAssetizable": true,
      "isSoldOnlyWithOtherProds": false,
      "name": "SmartBytes Unified Computing System",
```

```
      "nodeType": "bundleProduct",
      "productCode": "P0157",
      "productComponentGroups": [],
      "productSellingModelOptions": [
        {
          "id": "0iO1Q0000008OkzUAE",
          "productId": "01t1Q000008CD2sQAG",
          "productSellingModel": {
            "id": "0jP1Q000000CaVFUA0",
            "name": "One Time",
            "sellingModelType": "OneTime",
            "status": "Active"
          }
        },
        {
          "id": "0iO1Q0000008OlWUAU",
          "productId": "01t1Q000008CD2sQAG",
          "productSellingModel": {
            "id": "0jP1Q000000CaVHUA0",
            "name": "Evergreen",
            "pricingTerm": 1,
            "pricingTermUnit": "Months",
            "sellingModelType": "Evergreen",
            "status": "Active"
          }
        }
      ],
      "productSpecificationType": {
        "name": "None"
      }
    }
  ],
  "status": {
    "code": "200",
    "errors": [],
    "message": "Successfully fetched the product records."
  }
}
```


This example shows a sample response with data from Enterprise Product Catalog.


```
{
  "correlationId": "4cf1997d-5d86-4997-9c6b-55842442f325",
  "facets": [],
  "products": [
    {
      "additionalFields": {},
      "attributeCategory": [
        {
          "attributes": [
            {
              "additionalFields": {
                "OptOutAssetization": true,
                "OptOutDecompositionAction": false,
                "OptOutSupplementalAction": false
```

```
          },
          "attributeNameOverride": "ATTRIBUTE-ABP-INT",
          "code": "ABP-INT",
          "dataType": "Number",
          "developerName": "ABP-INT",
          "id": "ABP-INT",
          "isHidden": false,
          "isPriceImpacting": false,
          "isReadOnly": false,
          "isRequired": false,
          "name": "ATTRIBUTE-ABP-INT",
          "sequence": 3,
          "status": "Active"
        },
        {
          "additionalFields": {
            "OptOutAssetization": false,
            "OptOutDecompositionAction": false,
            "OptOutSupplementalAction": false
          },
          "attributeNameOverride": "ATTRIBUTE-ABP-CHECKBOX",
          "code": "ABP-CHECKBOX",
          "dataType": "Checkbox",
          "developerName": "ABP-CHECKBOX",
          "id": "ABP-CHECKBOX",
          "isHidden": false,
          "isPriceImpacting": false,
          "isReadOnly": false,
          "isRequired": false,
          "name": "ATTRIBUTE-ABP-CHECKBOX",
          "sequence": 2,
          "status": "Active"
        }
      ],
      "code": "CATEGORY-ABP",
      "id": "a0Zxx000000c3tNEAQ",
      "name": "ATTRIBUTE_CATEGORY-ABP"
    }
  ],
  "attributes": [],
  "catalogs": [],
  "categories": [],
  "childProducts": [],
  "id": "01txx0000006kFYAAY",
  "isActive": true,
  "isAssetizable": true,
  "name": "ARProduct",
  "nodeType": "simpleProduct",
  "productCode": "ARProduct",
  "productComponentGroups": [],
  "productSellingModelOptions": [],
  "productUnitOfMeasures": []
}
],
```

```
  "status": {
    "code": "200",
    "errors": [],
    "message": "Successfully fetched Product records."
  }
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
correlationId	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Small, 60.0	60.0
count	Integer	Total count of the products matching the request query.	Small, 60.0	60.0
facets	Search Facet	Details of the faceted search. This property is applicable if the **Use Indexed Data For Product Listing and Search** toggle from the Product Discovery Settings page from Setup is enabled.	Small, 63.0	63.0
products	Product[]	List of products matching the request query.	Small, 60.0	60.0
status	Status	Status of the request.	Small, 60.0	60.0

Product Variants

Output representation of the variation products associated with the specified parent variant products.

JSON example


```
{
  "correlationId": "9b6bc520-3c82-4d6c-a458-47590370681a",
  "count": 2,
  "details": {
    "01tT1000000F0afIAC": [
      "01tT1000000F0ahIAC",
      "01tT1000000F0aiIAC"
    ],
    "01tT1000000F0agIAC": [
      "01tT1000000F0ajIAC"
    ]
  },
  "inValidProductIds": [],
  "nonVariantParentIds": [],
  "status": {
    "code": "200",
    "errors": [],
    "message": "Successfully fetched the product variants."
```

```
  }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
correlationId	String	Unique ID to track and associate related events or transactions.	Small, 67.0	67.0
count	Integer	Number of parent variant products in the response that have at least one associated variation.	Small, 67.0	67.0
details	Map<String, String[]>	Specifies a map where each key is a parent variant product ID and the value is a list of its associated variation product IDs. Only parent variants that have at least one variation are included in this map.	Small, 67.0	67.0
inValid ProductIds	String[]	List of product IDs from the request that are blank, invalid, or not found in the org. The API excludes these IDs from processing but doesn't fail the request.	Small, 67.0	67.0
nonVariant ParentIds	String[]	List of product IDs from the request that are valid products but don't have the VariationParent product class. For example, simple products or bundle products passed in the request appear in this list.	Small, 67.0	67.0
status	Status	Status of the request.	Small, 67.0	67.0

Related Object Records

Output representation of the related records for a specified record ID and related object API name.

JSON example

```
      "relatedObjectRecords": [
        {
          "count": 2,
          "records": [
            {
              "SegmentType": "Yearly",
              "DurationType": "Months",
              "TrialDuration": null,
              "ProductSellingModelId": "0jPxx000000001dEAA",
              "ProductId": "01txx0000006i44AAA",
              "Id": "1FTxx0000004CDtGAM",
              "Name": "PPRS-000000005"
            },
            {
```

```
            "SegmentType": "FreeTrial",
            "DurationType": "Days",
            "TrialDuration": null,
            "ProductSellingModelId": "0jPxx000000001dEAA",
            "ProductId": "01txx0000006i44AAA",
            "Id": "1FTxx0000004CFUGA2",
            "Name": "PPRS-000000006"
          }
        ],
        "relatedObjectAPIName": "ProductRampSegment"
      },
      {
        "count": 2,
        "records": [
          {
            "UsageMetricId": "1BRxx0000004CAeGAM",
            "UsageMetricName": "Test Usage Metric 2",
            "UsageDefinitionProductId": null,
            "Label": "PUG-103",
            "Quantity": 100,
            "Id": "1BXxx0000004CCGGA2"
          },
          {
            "UsageMetricId": "1BRxx0000004CCGGA2",
            "UsageMetricName": "Test Usage Metric 3",
            "UsageDefinitionProductId": "01txx0000006i2eAAA",
            "Label": "PUG-105",
            "Quantity": 500,
            "Id": "1BXxx0000004CFUGA2"
          }
        ],
        "relatedObjectAPIName": "ProductUsageGrant"
      }
    ]
```


Property Name	Type	Description	Filter Group and Version	Available Version
`count`	Integer	Total count of the related records that are returned.	Small, 62.0	62.0
`records`	Map<String, Object>	List of related object records.	Small, 62.0	62.0
`relatedObjectAPIName`	String	API name of the related object to return the records for.	Small, 62.0	62.0

Related Records

Output representation of the list of relatedObject records for a specified record ID.

JSON example


```
"relatedRecords": [
  {
    "recordId": "01txx0000006i44AAA",
    "relatedObjectRecords": [
      {
        "count": 2,
        "records": [
          {
            "SegmentType": "Yearly",
            "DurationType": "Months",
            "TrialDuration": null,
            "ProductSellingModelId": "0jPxx000000001dEAA",
            "ProductId": "01txx0000006i44AAA",
            "Id": "1FTxx0000004CDtGAM",
            "Name": "PPRS-000000005"
          },
          {
            "SegmentType": "FreeTrial",
            "DurationType": "Days",
            "TrialDuration": null,
            "ProductSellingModelId": "0jPxx000000001dEAA",
            "ProductId": "01txx0000006i44AAA",
            "Id": "1FTxx0000004CFUGA2",
            "Name": "PPRS-000000006"
          }
        ],
        "relatedObjectAPIName": "ProductRampSegment"
      },
      {
        "count": 2,
        "records": [
          {
            "UsageMetricId": "1BRxx0000004CAeGAM",
            "UsageMetricName": "Test Usage Metric 2",
            "UsageDefinitionProductId": null,
            "Label": "PUG-103",
            "Quantity": 100,
            "Id": "1BXxx0000004CCGGA2"
          },
          {
            "UsageMetricId": "1BRxx0000004CCGGA2",
            "UsageMetricName": "Test Usage Metric 3",
            "UsageDefinitionProductId": "01txx0000006i2eAAA",
            "Label": "PUG-105",
            "Quantity": 500,
            "Id": "1BXxx0000004CFUGA2"
          }
        ],
        "relatedObjectAPIName": "ProductUsageGrant"
      }
    ]
  },
```

```
  {
    "recordId": "01txx0000006i5gAAA",
    "relatedObjectRecords": [
      {
        "count": 4,
        "records": [
          {
            "SegmentType": "Yearly",
            "DurationType": "Months",
            "TrialDuration": null,
            "ProductSellingModelId": "0jPxx0000000001EAA",
            "ProductId": "01txx0000006i5gAAA",
            "Id": "1FTxx0000004C92GAE",
            "Name": "PPRS-000000001"
          },
          {
            "SegmentType": "Custom",
            "DurationType": "Days",
            "TrialDuration": null,
            "ProductSellingModelId": "0jPxx0000000001EAA",
            "ProductId": "01txx0000006i5gAAA",
            "Id": "1FTxx0000004CAeGAM",
            "Name": "PPRS-000000002"
          },
          {
            "SegmentType": "FreeTrial",
            "DurationType": "Months",
            "TrialDuration": 6,
            "ProductSellingModelId": "0jPxx0000000001EAA",
            "ProductId": "01txx0000006i5gAAA",
            "Id": "1FTxx0000004CCGGA2",
            "Name": "PPRS-000000003"
          },
          {
            "SegmentType": "Custom",
            "DurationType": null,
            "TrialDuration": null,
            "ProductSellingModelId": "0jPxx0000000001EAA",
            "ProductId": "01txx0000006i5gAAA",
            "Id": "1FTxx0000004CDsGAM",
            "Name": "PPRS-000000004"
          }
        ],
        "relatedObjectAPIName": "ProductRampSegment"
      },
      {
        "count": 0,
        "records": [],
        "relatedObjectAPIName": "ProductUsageGrant"
      }
    ]
  }
]
```

Property Name	Type	Description	Filter Group and Version	Available Version
`recordId`	String	ID of the record.	Small, 62.0	62.0
`relatedObjectRecords`	Related Object Records[]	List of related object records.	Small, 62.0	62.0

Related Records List

Output representation of the list of related records.

JSON example


```
{
  "correlationId": "f4d6b42a-d9b7-49c9-8fa8-1c7bb6fe99aa",
  "relatedRecords": [
    {
      "recordId": "01txx0000006i44AAA",
      "relatedObjectRecords": [
        {
          "count": 2,
          "records": [
            {
              "SegmentType": "Yearly",
              "DurationType": "Months",
              "TrialDuration": null,
              "ProductSellingModelId": "0jPxx000000001dEAA",
              "ProductId": "01txx0000006i44AAA",
              "Id": "1FTxx0000004CDtGAM",
              "Name": "PPRS-000000005"
            },
            {
              "SegmentType": "FreeTrial",
              "DurationType": "Days",
              "TrialDuration": null,
              "ProductSellingModelId": "0jPxx000000001dEAA",
              "ProductId": "01txx0000006i44AAA",
              "Id": "1FTxx0000004CFUGA2",
              "Name": "PPRS-000000006"
            }
          ],
          "relatedObjectAPIName": "ProductRampSegment"
        },
        {
          "count": 2,
          "records": [
            {
              "UsageMetricId": "1BRxx0000004CAeGAM",
              "UsageMetricName": "Test Usage Metric 2",
              "UsageDefinitionProductId": null,
              "Label": "PUG-103",
              "Quantity": 100,
              "Id": "1BXxx0000004CCGGA2"
```

```
          },
          {
            "UsageMetricId": "1BRxx0000004CCGGA2",
            "UsageMetricName": "Test Usage Metric 3",
            "UsageDefinitionProductId": "01txx0000006i2eAAA",
            "Label": "PUG-105",
            "Quantity": 500,
            "Id": "1BXxx0000004CFUGA2"
          }
        ],
        "relatedObjectAPIName": "ProductUsageGrant"
      }
    ]
  },
  {
    "recordId": "01txx0000006i5gAAA",
    "relatedObjectRecords": [
      {
        "count": 4,
        "records": [
          {
            "SegmentType": "Yearly",
            "DurationType": "Months",
            "TrialDuration": null,
            "ProductSellingModelId": "0jPxx0000000001EAA",
            "ProductId": "01txx0000006i5gAAA",
            "Id": "1FTxx0000004C92GAE",
            "Name": "PPRS-000000001"
          },
          {
            "SegmentType": "Custom",
            "DurationType": "Days",
            "TrialDuration": null,
            "ProductSellingModelId": "0jPxx0000000001EAA",
            "ProductId": "01txx0000006i5gAAA",
            "Id": "1FTxx0000004CAeGAM",
            "Name": "PPRS-000000002"
          },
          {
            "SegmentType": "FreeTrial",
            "DurationType": "Months",
            "TrialDuration": 6,
            "ProductSellingModelId": "0jPxx0000000001EAA",
            "ProductId": "01txx0000006i5gAAA",
            "Id": "1FTxx0000004CCGGA2",
            "Name": "PPRS-000000003"
          },
          {
            "SegmentType": "Custom",
            "DurationType": null,
            "TrialDuration": null,
            "ProductSellingModelId": "0jPxx0000000001EAA",
            "ProductId": "01txx0000006i5gAAA",
            "Id": "1FTxx0000004CDsGAM",
```

```
              "Name": "PPRS-000000004"
            }
          ],
          "relatedObjectAPIName": "ProductRampSegment"
        },
        {
          "count": 0,
          "records": [],
          "relatedObjectAPIName": "ProductUsageGrant"
        }
      ]
    }
  ],
  "status": {
    "code": "200",
    "errors": [
    ],
    "message": ""
  }
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
correlationId	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Small, 62.0	62.0
related Records	Related Records[]	List of related records.	Small, 62.0	62.0
status	Status[]	Status of the API request.	Small, 62.0	62.0

Search Facet

Output representation of the details of the faceted search.

JSON example


```
  "facets":[
    {
      "attributeType": "ProductStandard",
      "displayName": "Product Type",
      "displayRank": 1,
      "displayType": "MultiSelect",
      "nameOrId": "Type",
      "values": [
        {
          "displayName": "Simple",
          "nameOrId": "Simple",
          "productCount": 9
```

```
        }
      ]
    },
    {
      "attributeType": "ProductStandard",
      "displayName": "Active",
      "displayRank": 2,
      "displayType": "MultiSelect",
      "nameOrId": "IsActive",
      "values": [
        {
          "displayName": "true",
          "nameOrId": "true",
          "productCount": 47
        }
      ]
    }
  ]
```

Property Name	Type	Description	Filter Group and Version	Available Version
attributeType	String	Search attribute type of the facet.	Small, 63.0	63.0
displayName	String	Display name of the facet.	Small, 63.0	63.0
displayRank	Integer	Display rank of the facet.	Small, 63.0	63.0
displayType	String	Display type of the face.	Small, 63.0	63.0
nameOrId	String	Facet name or ID. Reserved for internal use.	Small, 63.0	63.0
values	Facet Value[]	Values of the facet found in the search result. Sorted by display name in alphabetical order.	Small, 63.0	63.0

Setting

Output representation of the setting that's used in indexing.

JSON example

```
"setting": {
  "defaultLanguage": "en_US",
  "id": "1JySG0000000GUb0AM",
  "supportedLanguages": ["en_US","ja","es","nl_NL"]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
default Language	String	Default language for the API.	Small, 63.0	63.0

Property Name	Type	Description	Filter Group and Version	Available Version
id	String	ID of the setting.	Small, 63.0	63.0
supported Languages	String[]	List of supported language locales for indexing.	Small, 63.0	63.0

Setting Metadata

Output representation of the metadata associated with a setting.

JSON example

```
"metadata": {
  "activeLanguages": [
    "en_US"
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
active Languages	String[]	List of active languages in an org.	Small, 63.0	63.0

Snapshot

Output representation of the list of active snapshots.

JSON example

```
"snapshots": [
  {
    "activationDate": "2024-11-06T06:58:02.000Z",
    "activationStatus": "ACTIVE",
    "activationType": "IMMEDIATE",
    "id": "1Avxx0000004C92CAE",
    "snapshotIndexes": [
      {
        "createdDate": "2024-11-06T06:56:49.000Z",
        "id": "1D6xx0000004C92CAE",
        "indexBuildType": "FULL",
        "indexInfos": [
          {
            "buildType": "FULL",
            "id": "0axxx00000000T3AAI",
            "isIncrementable": true,
            "usageType": "LIVE"
          }
        ],
        "indexLogs": [
          {
```

```
            "catalogSnapshotTime": "2024-11-06T16:14:30.000Z",
            "completionTime": "2024-11-06T16:16:02.000Z",
            "createdById": "005xx000001X7x7AAC",
            "indexBuildStatus": "COMPLETED",
            "indexBuildType": "FULL",
            "indexId": "0axxx00000000T3AAI",
            "numberOfChanges": 7
          },
          {
            "catalogSnapshotTime": "2024-11-06T15:03:32.000Z",
            "completionTime": "2024-11-06T15:05:02.000Z",
            "createdById": "005xx000001X7x7AAC",
            "indexBuildStatus": "COMPLETED_WITH_ERRORS",
            "indexBuildType": "INCREMENTAL",
            "indexId": "0axxx00000000RRAAY",
            "message": "Warning: Product errors found.",
            "numberOfChanges": 3
          },
          {
            "catalogSnapshotTime": "2024-11-06T12:35:34.000Z",
            "completionTime": "2024-11-06T12:35:34.000Z",
            "createdById": "005xx000001X7x7AAC",
            "indexBuildStatus": "COMPLETED",
            "indexBuildType": "INCREMENTAL",
            "indexId": "0axxx00000000RRAAY",
            "message": "There are no changes for the partial update.",
            "numberOfChanges": 0
          },
          {
            "catalogSnapshotTime": "2024-11-06T12:07:32.000Z",
            "completionTime": "2024-11-06T12:09:02.000Z",
            "createdById": "005xx000001X7x7AAC",
            "indexBuildStatus": "COMPLETED_WITH_ERRORS",
            "indexBuildType": "FULL",
            "message": "Warning: Product errors found.",
            "numberOfChanges": 1
          }
        ],
        "indexType": "PRODUCT",
        "lastBuildStatus": "IN_PROGRESS",
        "venueId": "1D6xx0000004C92CAE"
      }
    ]
  }
]
```


Property Name	Type	Description	Filter Group and Version	Available Version
`activation Date`	String	Activation date of the snapshot.	Small, 62.0	62.0

Property Name	Type	Description	Filter Group and Version	Available Version
activation Status	String	Activation status of the snapshot. Valid values are: • NONE • ACTIVE—Snapshot is in active status. • EXPIRED—Snapshot is in expired status.	Small, 62.0	62.0
activation Type	String	Activation type of the snapshot. Valid values are: • IMMEDIATE—Snapshot is activated immediately after a successful build.	Small, 62.0	62.0
id	String	ID of the snapshot.	Small, 61.0	61.0
snapshot Indexes	Snapshot Index[]	List of indexes created in the snapshot.	Small, 62.0	62.0

Snapshot Collection

Output representation of the retrieved snapshot collection.

JSON example


```
{
  "errors": [],
  "snapshots": [
    {
      "activationDate": "2024-11-06T06:58:02.000Z",
      "activationStatus": "ACTIVE",
      "activationType": "IMMEDIATE",
      "id": "1Avxx0000004C92CAE",
      "snapshotIndexes": [
        {
          "createdDate": "2024-11-06T06:56:49.000Z",
          "id": "1D6xx0000004C92CAE",
          "indexBuildType": "FULL",
          "indexInfos": [
            {
              "buildType": "FULL",
              "id": "0axxx00000000T3AAI",
              "isIncrementable": true,
              "usageType": "LIVE"
            }
          ],
          "indexLogs": [
            {
              "catalogSnapshotTime": "2024-11-06T16:14:30.000Z",
              "completionTime": "2024-11-06T16:16:02.000Z",
              "createdById": "005xx000001X7x7AAC",
```

```
                "indexBuildStatus": "COMPLETED",
                "indexBuildType": "FULL",
                "indexId": "0axxx00000000T3AAI",
                "numberOfChanges": 7
              },
              {
                "catalogSnapshotTime": "2024-11-06T15:03:32.000Z",
                "completionTime": "2024-11-06T15:05:02.000Z",
                "createdById": "005xx000001X7x7AAC",
                "indexBuildStatus": "COMPLETED_WITH_ERRORS",
                "indexBuildType": "INCREMENTAL",
                "indexId": "0axxx00000000RRAAY",
                "message": "Warning: Product errors found.",
                "numberOfChanges": 3
              },
              {
                "catalogSnapshotTime": "2024-11-06T12:35:34.000Z",
                "completionTime": "2024-11-06T12:35:34.000Z",
                "createdById": "005xx000001X7x7AAC",
                "indexBuildStatus": "COMPLETED",
                "indexBuildType": "INCREMENTAL",
                "indexId": "0axxx00000000RRAAY",
                "message": "There are no changes for the partial update.",
                "numberOfChanges": 0
              },
              {
                "catalogSnapshotTime": "2024-11-06T12:07:32.000Z",
                "completionTime": "2024-11-06T12:09:02.000Z",
                "createdById": "005xx000001X7x7AAC",
                "indexBuildStatus": "COMPLETED_WITH_ERRORS",
                "indexBuildType": "FULL",
                "message": "Warning: Product errors found.",
                "numberOfChanges": 1
              }
            ],
            "indexType": "PRODUCT",
            "lastBuildStatus": "IN_PROGRESS",
            "venueId": "1D6xx0000004C92CAE"
          }
        ]
      }
    ],
    "statusCode": "200"
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`errors`	Error Output[]	List of errors, if any.	Small, 62.0	62.0
`snapshots`	Snapshot[]	List of active snapshots.	Small, 62.0	62.0
`statusCode`	String	Code indicating the status of the request.	Small, 62.0	62.0

Snapshot Deployment

Output representation of the snapshot deployment.

JSON example

This example shows a sample response to the request to build a new snapshot with immediate activation.

```
{
  "errors": [],
  "snapshot": {
    "activationStatus": "NONE",
    "activationType": "IMMEDIATE",
    "id": "1Avxx0000004CFU",
    "snapshotIndexes": [
      {
        "createdDate": "2024-07-24T21:10:48.000Z",
        "id": "1D6xx0000004CFU",
        "indexBuildType": "FULL",
        "indexType": "PRODUCT",
        "lastBuildStatus": "IN_PROGRESS"
      }
    ]
  },
  "statusCode": "200"
}
```

This example shows a sample response of the request to rebuild a snapshot in the `active` status.

```
{
  "errors": [],
  "snapshot": {
    "activationStatus": "NONE",
    "activationType": "IMMEDIATE",
    "id": "1Avxx0000004CH6",
    "snapshotIndexes": [
      {
        "createdDate": "2024-07-24T21:13:05.000Z",
        "id": "1D6xx0000004CH6",
        "indexBuildType": "FULL",
        "indexType": "PRODUCT",
        "lastBuildStatus": "IN_PROGRESS"
      }
    ]
  },
  "statusCode": "200"
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`errors`	Error Output[]	List of errors, if any.	Small, 62.0	62.0
`snapshot`	Snapshot[]	Run-time catalog snapshot associated with the created index.	Small, 62.0	62.0
`statusCode`	String	Code indicating the status of the request.	Small, 62.0	62.0

Snapshot Index

Output representation of the snapshot index of a run-time catalog.

JSON example


```
"snapshotIndexes": [
  {
    "createdDate": "2024-11-06T06:56:49.000Z",
    "id": "1D6xx0000004C92CAE",
    "indexBuildType": "FULL",
    "indexInfos": [
      {
        "buildType": "FULL",
        "id": "0axxx00000000T3AAI",
        "isIncrementable": true,
        "usageType": "LIVE"
      }
    ],
    "indexLogs": [
      {
        "catalogSnapshotTime": "2024-11-06T16:14:30.000Z",
        "completionTime": "2024-11-06T16:16:02.000Z",
        "createdById": "005xx000001X7x7AAC",
        "indexBuildStatus": "COMPLETED",
        "indexBuildType": "FULL",
        "indexId": "0axxx00000000T3AAI",
        "numberOfChanges": 7
      },
      {
        "catalogSnapshotTime": "2024-11-06T15:03:32.000Z",
        "completionTime": "2024-11-06T15:05:02.000Z",
        "createdById": "005xx000001X7x7AAC",
        "indexBuildStatus": "COMPLETED_WITH_ERRORS",
        "indexBuildType": "INCREMENTAL",
        "indexId": "0axxx00000000RRAAY",
        "message": "Warning: Product errors found.",
        "numberOfChanges": 3
      },
      {
        "catalogSnapshotTime": "2024-11-06T12:35:34.000Z",
        "completionTime": "2024-11-06T12:35:34.000Z",
        "createdById": "005xx000001X7x7AAC",
        "indexBuildStatus": "COMPLETED",
        "indexBuildType": "INCREMENTAL",
        "indexId": "0axxx00000000RRAAY",
        "message": "There are no changes for the partial update.",
        "numberOfChanges": 0
      },
      {
        "catalogSnapshotTime": "2024-11-06T12:07:32.000Z",
        "completionTime": "2024-11-06T12:09:02.000Z",
        "createdById": "005xx000001X7x7AAC",
        "indexBuildStatus": "COMPLETED_WITH_ERRORS",
        "indexBuildType": "FULL",
        "message": "Warning: Product errors found.",
```

```
                "numberOfChanges": 1
              }
            ],
            "indexType": "PRODUCT",
            "lastBuildStatus": "IN_PROGRESS",
            "venueId": "1D6xx0000004C92CAE"
          }
        ]
```

Property Name	Type	Description	Filter Group and Version	Available Version
completedDate	String	Completed date of the snapshot index.	Small, 62.0	62.0
createdDate	String	Created date of the snapshot index.	Small, 62.0	62.0
id	String	ID of the snapshot index.	Small, 62.0	62.0
indexBuild Type	String	Build type of the snapshot index. Valid value is: • FULL—Specifies a full index build. • INCREMENTAL—Specifies an incremental index build. Available in API version 63.0 and later.	Small, 62.0	62.0
indexInfos	Index Info	Index information records associated with the snapshot index.	Small, 63.0	63.0
indexLogs	Index Logs[]	Index logs associated with the snapshot index.	Small, 63.0	63.0
indexType	String	Index type of the snapshot index. Valid value is: • PRODUCT—Snapshot index is of product type.	Small, 62.0	62.0
lastBuild Status	String	Last build status of the snapshot index. Valid values are: • IN_PROGRESS—Snapshot index build is in progress. • FAILED—Snapshot index build failed. • COMPLETED—Snapshot index build completed successfully.	Small, 62.0	62.0
numberOf Records	Integer	Number of indexed records.	Small, 62.0	62.0
venueId	String	Venue ID of the snapshot index.	Small, 63.0	63.0

Snapshot Index Error

Output representation of the error details related to a snapshot index.

JSON example

```
{
  "errors": [],
  "indexErrorDetails": {
    "errorFileId": "069xx0000004C92AAE",
    "indexCreatedDate": "2024-10-03T05:24:18.000Z",
    "indexErrorsCount": 1,
    "indexLastUpdatedDate": "2024-10-03T05:27:00.000Z",
    "itemLevelErrorsCount": 1
  },
  "statusCode": "200"
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`errors`	Error Output[]	List of errors encountered during the processing of the API request.	Small, 63.0	63.0
`indexErrorDetails`	Index Error[]	Count and details of errors that occurred during indexing.	Small, 63.0	63.0
`statusCode`	String	Code that indicates the status of the API request.	Small, 63.0	63.0

Snapshot Index Info

Output representation of the details of a snapshot index.

JSON example

```
 "indexInfos": [
            {
              "buildType": "FULL",
              "id": "0axxx00000000T3AAI",
              "isIncrementable": true,
              "usageType": "LIVE"
            }
          ]
```

Property Name	Type	Description	Filter Group and Version	Available Version
`buildType`	String	Build type of the index. Valid values are: • `FULL`—Specifies a full index build. • `INCREMENTAL`—Specifies an incremental index build. Available in API version 63.0 and later.	Small, 63.0	63.0

Property Name	Type	Description	Filter Group and Version	Available Version
`id`	String	ID of the index information record.	Small, 63.0	63.0
`isIncrementable`	Boolean	Indicates whether a partial build is enabled (`true`) or disabled (`false`).	Small, 63.0	63.0
`usageType`	String	Usage type of the index. Valid values are: • `LIVE`—Index usage type is live. • `OUT_OF_USE`—Index usage type is out of use.	Small, 63.0	63.0

Snapshot Index Log

Output representation of a snapshot index log.

JSON example


```
          "indexLogs": [
            {
              "catalogSnapshotTime": "2024-11-06T16:14:30.000Z",
              "completionTime": "2024-11-06T16:16:02.000Z",
              "createdById": "005xx000001X7x7AAC",
              "indexBuildStatus": "COMPLETED",
              "indexBuildType": "FULL",
              "indexId": "0axxx00000000T3AAI",
              "numberOfChanges": 7
            },
            {
              "catalogSnapshotTime": "2024-11-06T15:03:32.000Z",
              "completionTime": "2024-11-06T15:05:02.000Z",
              "createdById": "005xx000001X7x7AAC",
              "indexBuildStatus": "COMPLETED_WITH_ERRORS",
              "indexBuildType": "INCREMENTAL",
              "indexId": "0axxx00000000RRAAY",
              "message": "Warning: Product errors found.",
              "numberOfChanges": 3
            },
            {
              "catalogSnapshotTime": "2024-11-06T12:35:34.000Z",
              "completionTime": "2024-11-06T12:35:34.000Z",
              "createdById": "005xx000001X7x7AAC",
              "indexBuildStatus": "COMPLETED",
              "indexBuildType": "INCREMENTAL",
              "indexId": "0axxx00000000RRAAY",
              "message": "There are no changes for the partial update.",
              "numberOfChanges": 0
            },
            {
              "catalogSnapshotTime": "2024-11-06T12:07:32.000Z",
              "completionTime": "2024-11-06T12:09:02.000Z",
              "createdById": "005xx000001X7x7AAC",
```

```
            "indexBuildStatus": "COMPLETED_WITH_ERRORS",
            "indexBuildType": "FULL",
            "message": "Warning: Product errors found.",
            "numberOfChanges": 1
          }
        ]
```


Property Name	Type	Description	Filter Group and Version	Available Version
`catalog SnapshotTime`	String	Catalog snapshot time of the index.	Small, 63.0	63.0
`completion Time`	String	Completion time of the index.	Small, 63.0	63.0
`createdById`	String	ID of the user that initiated the index build.	Small, 63.0	63.0
`indexBuild Status`	String	Status of the index build.	Small, 63.0	63.0
`indexBuild Type`	String	Type of the index build. Valid values are: • `FULL`—Specifies a full index build. • `INCREMENTAL`—Specifies an incremental index build. Available in API version 63.0 and later.	Small, 63.0	63.0
`indexId`	String	ID of the index.	Small, 63.0	63.0
`message`	String	Message for the index status.	Small, 63.0	63.0
`numberOf Changes`	Integer	Number of new or changed products included in the index.	Small, 63.0	63.0

Status

Output representation of the status of the request.

JSON example


```
"status": {
    "code": "200",
    "errors": [],
    "message": "Successfully fetched the catalog records."
  }
```


Property Name	Type	Description	Filter Group and Version	Available Version
`code`	String	Code of the error message.	Small, 60.0	60.0
`errors`	Error[]	Details of the error.	Small, 60.0	60.0

Property Name	Type	Description	Filter Group and Version	Available Version
`message`	String	Error message.	Small, 60.0	60.0

Unit of Measure Error

Output representation of the details of errors encountered during the processing of the Unit of Measure API request.

JSON example

```
"errorCodeToErrorMap": {
  "UNIT_OF_MEASURE_INFO_INVALID_UOM_IDS": {
    "errorCode": "UOM_INFO_API_003",
    "messageDetail": "Invalid uomId is passed. Please specify a valid uomId.",
    "messageTitle": "Invalid uomId is passed.",
    "recordIds": [
      "sample"
    ],
    "source": "Unit_Of_Measure_Info_Api"
  }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`errorCode`	String	Localized error code.	Small, 63.0	63.0
`messageDetail`	String	Localized details of the error message.	Small, 63.0	63.0
`messageTitle`	String	Localized title of the error message.	Small, 63.0	63.0
`recordIds`	String[]	List of erroneous record IDs.	Small, 63.0	63.0
`source`	String	Localized source of the error.	Small, 63.0	63.0

Unit of Measure Info

Output representation of the details of a unit of measure record.

JSON example

```
"uomIdToUnitOfMeasureInfo": {
  "0hEU200000003M5MAI": {
    "id": "0hEU200000003M5MAI",
    "name": "Pounds",
    "roundingMethod": "Nearest",
    "scale": 1,
    "unitCode": "Pounds"
  },
  "0hEU200000003KTMAY": {
    "id": "0hEU200000003KTMAY",
    "name": "Grams",
    "roundingMethod": "Down",
    "scale": 5,
```

```
      "unitCode": "Grams"
    }
  }
```

Property Name	Type	Description	Filter Group and Version	Available Version
`id`	String	ID of the unit of measure record.	Small, 63.0	63.0
`name`	String	Name of the unit of measure record.	Small, 63.0	63.0
`roundingMethod`	String	Data rounding method of the unit of measure record.	Small, 63.0	63.0
`scale`	Integer	Scale of the unit of measure record.	Small, 63.0	63.0
`unitCode`	String	Code of the unit of measure record.	Small, 63.0	63.0

Unit of Measure Status

Output representation of the status of the Unit of Measure API request.

JSON example

```
  "status": {
    "errors": [],
    "httpStatusCode": "200",
    "message": " Successfully fetched UnitOfMeasure Info. "
  }
```

Property Name	Type	Description	Filter Group and Version	Available Version
`errors`	Unit Of Measure Error[]	Errors encountered during the processing of the API request.	Small, 63.0	63.0
`httpStatusCode`	String	HTTP status code of the API request.	Small, 63.0	63.0
`message`	String	Localized response message.	Small, 63.0	63.0

Product Catalog Management Metadata API Types

Metadata API enables you to access some types and feature settings that you can customize in the user interface.

ProductCatalogManagementSettings
Represents the settings for Product Catalog Management.

ProductSpecificationType
Represents the specification types in your org that define products with unique terminology specific to the industry.

ProductSpecificationRecType

Represents the association of a product specification type with record types defined on the Product object. The product specification record type also determines if the product specification is sold commercially or not.

SEE ALSO:

Metadata API Developer Guide: Understanding Metadata API

ProductCatalogManagementSettings

Represents the settings for Product Catalog Management.

Parent Type and Manifest Access

This type extends the Metadata metadata type and inherits its `fullName` field.

In the package manifest, all the settings metadata types for the org are accessed using the "Settings" name. See Settings for more details.

File Suffix and Directory Location

`ProductCatalogManagementSettings` values are stored in the `ProductCatalogManagementSettings.settings` file in the `settings` folder. The `.settings` files are different from other named components, because there is only one settings file for each settings component.

Version

ProductCatalogManagementSettings components are available in API version 64.0 and later.

Special Access Rules

These settings are available when Product Catalog Management is enabled.

Fields

Field Name	Description
`productDeepCloneContextDefOrgValue`	**Field Type** string **Description** Name of the context definition that you want to use to deep clone the product.
`productDeepCloneExpressionSetOrgValue`	**Field Type** string **Description** Expression set that contains the rules that you want to apply to deep clone the product.

Declarative Metadata Sample Definition

The following is an example of a ProductCatalogManagementSettings component.

```
<ProductCatalogManagementSettings xmlns="http://soap.sforce.com/2006/04/metadata">
<productDeepCloneContextDefOrgValue>ProductDeepCloneContextDefinition</productDeepCloneContextDefOrgValue>
<productDeepCloneExpressionSetOrgValue>ProductDeepCloneExpressionSet</productDeepCloneExpressionSetOrgValue>
</ProductCatalogManagementSettings>
```

The following is an example `package.xml` that references the previous definition.

```
<?xml version="1.0" encoding="UTF-8"?>
<Package xmlns="http://soap.sforce.com/2006/04/metadata">
    <types>
        <members>ProductCatalogManagement</members>
        <name>Settings</name>
    </types>
    <version>67.0</version>
</Package>
```

Wildcard Support in the Manifest File

The wildcard character `*` (asterisk) in the `package.xml` manifest file doesn't apply to metadata types for feature settings. The wildcard applies only when retrieving all settings, not for an individual setting. For details, see Settings. For information about using the manifest file, see Deploying and Retrieving Metadata with the Zip File.

ProductSpecificationType

Represents the specification types in your org that define products with unique terminology specific to the industry.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Parent Type

This type extends the Metadata metadata type and inherits its `fullName` field.

File Suffix and Directory Location

ProductSpecificationType components have the suffix `.productSpecificationType` and are stored in the `productSpecificationTypes` folder.

Version

ProductSpecificationType components are available in API version 60.0 and later.

Special Access Rules

Ensure Product Catalog Management is enabled to access this metadata type.

Fields

Field Name	Description
description	**Field Type** string **Description** Required. Description of the product specification type.
masterLabel	**Field Type** string **Description** Required. A user-friendly name for the product specification record type, which is defined when the metadata component is created.

Declarative Metadata Sample Definition

The following is an example of a ProductSpecificationType component.

```
<ProductSpecificationType xmlns="http://soap.sforce.com/2006/04/metadata">
    <masterLabel>sample</masterLabel>
    <description>Sample Description</description>
</ProductSpecificationType>
```

The following is an example package.xml that references the previous definition.

```
<Package xmlns="http://soap.sforce.com/2006/04/metadata">
    <types>
        <members>*</members>
        <name>ProductSpecificationType</name>
    </types>
    <version>67.0</version>
</Package>
```

Wildcard Support in the Manifest File

This metadata type supports the wildcard character * (asterisk) in the package.xml manifest file. For information about using the manifest file, see Deploying and Retrieving Metadata with the Zip File.

ProductSpecificationRecType

Represents the association of a product specification type with record types defined on the Product object. The product specification record type also determines if the product specification is sold commercially or not.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Parent Type

This type extends the Metadata metadata type and inherits its `fullName` field.

File Suffix and Directory Location

ProductSpecificationRecType components have the suffix `.productSpecificationRecType` and are stored in the `productSpecificationRecTypes` folder.

Version

ProductSpecificationRecType components are available in API version 60.0 and later.

Special Access Rules

Ensure Product Catalog Management is enabled to access this metadata type.

Fields

Field Name	Description
`isCommercial`	**Field Type** boolean **Description** Required. Indicates whether the product is sold commercially (`true`) or not (`false`). The default value is `true`.
`masterLabel`	**Field Type** string **Description** Required. A user-friendly name for the product specification record type, which is defined when the metadata component is created.
`productSpecificationType`	**Field Type** string **Description** Required. Product specification type that's associated with the record type. This field is unique within your organization.

Field Name	Description
recordType	**Field Type** string **Description** Required. Custom record type of Product2 object.

Declarative Metadata Sample Definition

The following is an example of a ProductSpecificationRecType component.

```
<ProductSpecificationRecType xmlns="http://soap.sforce.com/2006/04/metadata">
    <masterLabel>sample</masterLabel>
   <recordType>Product2.Offer</recordType>
   <productSpecificationType>Placeholder</productSpecificationType>
   <isCommercial>true</isCommercial>
</ProductSpecificationRecType>
```

The following is an example `package.xml` that references the previous definition.

```
<Package xmlns="http://soap.sforce.com/2006/04/metadata">
    <types>
        <members>*</members>
        <name>ProductSpecificationRecType</name>
    </types>
    <types>
        <members>*</members>
        <name>ProductSpecificationType</name>
    </types>
    <types>
        <members>Product2.Offer</members>
        <name>RecordType</name>
    </types>
    <version>67.0</version>
</Package>
```

Wildcard Support in the Manifest File

This metadata type supports the wildcard character `*` (asterisk) in the `package.xml` manifest file. For information about using the manifest file, see Deploying and Retrieving Metadata with the Zip File.

Product Catalog Management Tooling API Objects

Tooling API exposes metadata used in developer tooling that you can access through REST or SOAP. Tooling API's SOQL capabilities for many metadata types allow you to retrieve smaller pieces of metadata.

ProductSpecificationType

Represents the specification types in your org that define products with unique terminology specific to the industry. This object is available in API version 60.0 and later.

ProductSpecificationRecType

Represents the association of a product specification type with record types defined on the Product object. The product specification record type also determines if the product specification is sold commercially or not. This object is available in API version 60.0 and later.

SEE ALSO:

Tooling API Developer Guide: Introducing Tooling API

ProductSpecificationType

Represents the specification types in your org that define products with unique terminology specific to the industry. This object is available in API version 60.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported SOAP API Calls

`create()`, `delete()`, `describeSObjects()`, `query()`, `retrieve()`, `update()`, `upsert()`

Supported REST API Methods

`DELETE, GET, HEAD, PATCH, POST, Query`

Special Access Rules

Ensure Product Catalog Management is enabled to access this object.

Fields

Field	Details
`Description`	**Type** textarea **Properties** Create, Nillable, Update **Description** Description of the product specification type.
`DeveloperName`	**Type** string **Properties** Create, Filter, Group, Sort, Update

Field	Details
	Description The unique name of the object in the API. This name can contain only underscores and alphanumeric characters, and must be unique in your org. It must begin with a letter, not include spaces, not end with an underscore, and not contain two consecutive underscores. In managed packages, this field prevents naming conflicts on package installations. With this field, a developer can change the object's name in a managed package and the changes are reflected in a subscriber's organization. Label is **Record Type Name**.
`Language`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Language of the product specification type instance. Possible values are: • `da`—Danish • `de`—German • `en_US`—English • `es`—Spanish • `es_MX`—Spanish (Mexico) • `fi`—Finnish • `fr`—French • `it`—Italian • `ja`—Japanese • `ko`—Korean • `nl_NL`—Dutch • `no`—Norwegian • `pt_BR`—Portuguese (Brazil) • `ru`—Russian • `sv`—Swedish • `th`—Thai • `zh_CN`—Chinese (Simplified) • `zh_TW`—Chinese (Traditional)
`ManageableState`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort

Field	Details
	Description **Type** ManageableState enumerated list **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** Indicates the manageable state of the specified component that is contained in a package: • `beta` • `deleted` • `deprecated` • `deprecatedEditable` • `installed` • `installedEditable` • `released` • `unmanaged`
`MasterLabel`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** Label assigned to the ProductSpecificationType object.
`NamespacePrefix`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The namespace prefix associated with this object. Each Developer Edition organization that creates a managed package has a unique namespace prefix. Limit: 15 characters. You can refer to a component in a managed package by using the ***`namespacePrefix__componentName`*** notation. The namespace prefix can have one of the following values: • In Developer Edition organizations, the namespace prefix is set to the namespace prefix of the organization for all objects that support it. There is an exception if an object is in an installed managed package. In that case, the object has the namespace prefix of the installed managed package. This field's value is the namespace prefix of the Developer Edition organization of the package developer. • In organizations that are not Developer Edition organizations, `NamespacePrefix` is only set for objects that are part of an installed managed package. There is no namespace prefix for all other objects.

ProductSpecificationRecType

Represents the association of a product specification type with record types defined on the Product object. The product specification record type also determines if the product specification is sold commercially or not. This object is available in API version 60.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported SOAP API Calls

`create()`, `delete()`, `describeSObjects()`, `query()`, `retrieve()`, `update()`, `upsert()`

Supported REST API Methods

`DELETE, GET, HEAD, PATCH, POST, Query`

Special Access Rules

Ensure Product Catalog Management is enabled to access this object.

Fields

Field	Details
`DeveloperName`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** The unique name of the object in the API. This name can contain only underscores and alphanumeric characters, and must be unique in your org. It must begin with a letter, not include spaces, not end with an underscore, and not contain two consecutive underscores. In managed packages, this field prevents naming conflicts on package installations. With this field, a developer can change the object's name in a managed package and the changes are reflected in a subscriber's organization. Label is **Record Type Name**.
`IsCommercial`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the product is sold commercially (`true`) or not (`false`). The default value is `true`.
`Language`	**Type** picklist

Field	Details
	Properties Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Language of the product specification record type instance. Possible values are: • `da`—Danish • `de`—German • `en_US`—English • `es`—Spanish • `es_MX`—Spanish (Mexico) • `fi`—Finnish • `fr`—French • `it`—Italian • `ja`—Japanese • `ko`—Korean • `nl_NL`—Dutch • `no`—Norwegian • `pt_BR`—Portuguese (Brazil) • `ru`—Russian • `sv`—Swedish • `th`—Thai • `zh_CN`—Chinese (Simplified) • `zh_TW`—Chinese (Traditional)
`ManageableState`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** **Type** ManageableState enumerated list **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** Indicates the manageable state of the specified component that is contained in a package: • `beta` • `deleted` • `deprecated` • `deprecatedEditable`

Field	Details
	• `installed` • `installedEditable` • `released` • `unmanaged`
`MasterLabel`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** Label assigned to the ProductSpecificationRecType object.
`NamespacePrefix`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The namespace prefix associated with this object. Each Developer Edition organization that creates a managed package has a unique namespace prefix. Limit: 15 characters. You can refer to a component in a managed package by using the ***`namespacePrefix__componentName`*** notation. The namespace prefix can have one of the following values: • In Developer Edition organizations, the namespace prefix is set to the namespace prefix of the organization for all objects that support it. There is an exception if an object is in an installed managed package. In that case, the object has the namespace prefix of the installed managed package. This field's value is the namespace prefix of the Developer Edition organization of the package developer. • In organizations that are not Developer Edition organizations, `NamespacePrefix` is only set for objects that are part of an installed managed package. There is no namespace prefix for all other objects.
`ProductSpecificationType`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. Product specification type that's associated with the record type. This field is unique within your organization. The picklist values that are available to you depend on your industry solution and permission sets.

Field	Details
RecordType	**Type** sObject **Properties** Nillable **Description** Custom record type of Product2 object.
RecordTypeId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** Required. ID of the product record type associated with the product specification type. This field is unique within your organization. This field is a relationship field. **Relationship Name** RecordType **Relationship Type** Lookup **Refers To** RecordType

Product Discovery

Product Discovery provides a hierarchical catalog browsing experience to identify suitable products with text-based and faceted search.

EDITIONS

Available in: Lightning Experience

Available in: **Enterprise**, **Unlimited**, and **Developer** Editions of Revenue Management

Product Discovery Business APIs

Use the Product Discovery Business APIs, which are composite APIs, to search products or to discover catalogs, products, and categories during the product browsing experience.

Product Discovery Standard Invocable Actions

Use the standard invocable actions available with Product Discovery to find and retrieve product, category, and catalog details. Additionally, execute a qualification procedure, and search products with guided selection.

Product Discovery Apex Reference

Use built-in Apex classes and interfaces grouped by namespace.

Product Discovery Metadata API Types
Metadata API enables you to access some types and feature settings that you can customize in the user interface.

SEE ALSO:

Salesforce Help: Configure Product Discovery Settings

Salesforce Help: Invoke Qualification Procedures During Product Discovery

Product Discovery Business APIs

Use the Product Discovery Business APIs, which are composite APIs, to search products or to discover catalogs, products, and categories during the product browsing experience.

This table lists the available Product Discovery resources.

Resource	Description
`/connect/cpq/catalogs/`***`catalogId`*** (POST)	Get catalog details for a specified catalog ID. This API is a composite API for Product Discovery.
`/connect/cpq/catalogs` (POST)	Get a paginated list of catalogs. This API is a composite API for Product Discovery.
`/connect/cpq/categories` (POST)	Get a list of categories and subcategories of a specified catalog. This API is a composite API for Product Discovery.
`/connect/cpq/categories/`***`categoryId`*** (POST)	Get details of a category for a specified category ID. This API is a composite API for Product Discovery.
`/connect/cpq/qualification` (POST)	Run the qualification procedure on a list of product IDs. This API is a composite API for Product Discovery.
`/connect/cpq/products/search` (POST)	Retrieves a list of products based on a search query or search term. This API is a composite API for Product Discovery.
`/connect/cpq/products/`***`productId`*** (POST)	Get product details, such as attributes, hierarchy, or cardinality, for a specified product ID. This API is a

Resource	Description
	composite API for Product Discovery.
`/connect/cpq/products` (POST)	Get a list of products for a specified catalog, category, or subcategory. This API is a composite API for Product Discovery.
`/connect/cpq/products/bulk` (POST)	Retrieve details for multiple products. This API is a composite API for Product Discovery.
`/connect/cpq/products/guided-selection` (POST)	Retrieve a list of products based on the response identifier or search terms of a guided selection. Guided selection captures user requirements to show suitable products.
`/revenue/product-discovery/products/recommendations` (POST)	Get a list of recommended products based on your underlying business rules.

Resources
Learn more about the available Product Discovery API resources.

Request Bodies
Learn more about the available Product Discovery API request bodies.

Response Bodies
Learn more about the available Product Discovery API response bodies.

Resources

Learn more about the available Product Discovery API resources.

Catalog Details (POST)
Get catalog details for a specified catalog ID. This API is a composite API for Product Discovery.

Catalog List (POST)
Get a paginated list of catalogs. This API is a composite API for Product Discovery.

Categories List (POST)
Get a list of categories and subcategories of a specified catalog. This API is a composite API for Product Discovery.

Category Details (POST)
Get details of a category for a specified category ID. This API is a composite API for Product Discovery.

[Bulk Product Details (POST)](#)
Retrieve details for multiple products. This API is a composite API for Product Discovery.

[Global Search (POST)](#)
Retrieves a list of products based on a search query or search term. This API is a composite API for Product Discovery.

[Guided Selection (POST)](#)
Retrieve a list of products based on the response identifier or search terms of a guided selection. Guided selection captures user requirements to show suitable products.

[Product Details (POST)](#)
Get product details, such as attributes, hierarchy, or cardinality, for a specified product ID. This API is a composite API for Product Discovery.

[Product Recommendations (POST)](#)
Get a list of recommended products based on your underlying business rules.

[Products List (POST)](#)
Get a list of products for a specified catalog, category, or subcategory. This API is a composite API for Product Discovery.

[Qualification (POST)](#)
Run the qualification procedure on a list of product IDs. This API is a composite API for Product Discovery.

Catalog Details (POST)

Get catalog details for a specified catalog ID. This API is a composite API for Product Discovery.

Resource

```
/connect/cpq/catalogs/catalogId
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/cpq/catalogs/0ZSxx000000009hGAA
```

Available version

60.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "correlationId": "9cbb9650-48c5-11ed-96d1-0afcf185843b",
  "userContext": {
     "accountId": "001xx0000000001AAA",
     "contactId": "003xx00000000D7AAI"
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`correlationId`	String	Unique identifier value that's attached to the requests and messages, and accepts references to a particular transaction or event chain.	Optional	60.0
`userContext`	User Context Input	User context details. For example, account ID or contact ID.	Optional	60.0

Response body for POST

CPQ Base Details

Catalog List (POST)

Get a paginated list of catalogs. This API is a composite API for Product Discovery.

Resource

```
/connect/cpq/catalogs
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/cpq/catalogs
```

Available version

60.0

HTTP methods

POST

Request body for POST

JSON example

```
{
"correlationId": "9cbb9650-48c5-11ed-96d1-0afcf185843b",
"limit": 10,
"offset": 0,
"orderBy": [
    "name:asc",
    "id:desc"
],
"userContext": {
    "accountId": "001xx0000000001AAA",
    "contactId": "003xx00000000D7AAI"
}
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`correlationId`	String	Unique identifier value that's attached to the requests and messages, and accepts references to a particular transaction or event chain.	Optional	60.0
`limit`	Integer	Number of items to include in the response.	Optional	60.0
`offset`	Integer	Offset size from which to get the catalog count.	Optional	60.0
`orderBy`	String[]	Sort order for the catalogs.	Optional	60.0
`userContext`	User Context Input	User context details. For example, account ID or contact ID.	Optional	60.0

Response body for POST

CPQ Base List

Categories List (POST)

Get a list of categories and subcategories of a specified catalog. This API is a composite API for Product Discovery.

Resource

```
/connect/cpq/categories
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/cpq/categories
```

Available version

60.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "correlationId": "9cbb9650-48c5-11ed-96d1-0afcf185843b",
  "catalogId": "0ZSxx000000009hGAA",
  "userContext": {
      "accountId": "001xx0000000001AAA",
      "contactId": "003xx00000000D7AAI"
    }
}
```

This example shows a sample request to get a list of categories with eligible promotions.

```
{
  "correlationId": "9cbb9650-48c5-11ed-96d1-0afcf185843b",
  "catalogId": "0ZSxx000000009hGAA",
  "userContext": {
    "accountId": "001xx0000000001AAA",
    "contactId": "003xx00000000D7AAI"
  },
  "usePromotions": true
}
```

Response body for POST

CPQ Base List

Category Details (POST)

Get details of a category for a specified category ID. This API is a composite API for Product Discovery.

Resource

```
/connect/cpq/categories/categoryId
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/cpq/categories/0ZGxx000000001dGAA
```

Available version

60.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "correlationId": "9cbb9650-48c5-11ed-96d1-0afcf185843b",
  "userContext": {
      "accountId": "001xx0000000001AAA",
      "contactId": "003xx00000000D7AAI"
  }
}
```

This example shows a sample request to get category details with eligible promotions.

```
{
  "correlationId": "9cbb9650-48c5-11ed-96d1-0afcf185843b",
  "userContext": {
    "accountId": "001xx0000000001AAA",
    "contactId": "003xx00000000D7AAI"
  },
  "usePromotions": true
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`additional ContextData`	Context Data Input	Additional nodes that are added to the custom or default context definition. The maximum number of supported nodes is 10.	Optional	60.0
`catalogId`	String	ID of the catalog. If the catalog ID is specified, then the API returns the list of offers from the catalog with the pricing details related to the catalog.	Optional	60.0
`context Definition`	String	API name of the custom context definition that's sent for context creation. If this property isn't specified, then the default context definition is used.	Optional	60.0
`context Mapping`	String	Default context mapping of the context definition. If a context mapping is specified, then the API checks whether the mapping belongs to the specified context definition to process the details for hydration.	Optional	60.0
`customFields`	String[]	List of category fields to retrieve in the response.	Optional	60.0
`correlation Id`	String	Unique identifier value that's attached to the requests and messages, and accepts references to a particular transaction or event chain.	Optional	60.0
`enable Qualification`	Boolean	Indicates whether to enable qualification rules for the categories (`true`) or not (`false`). The default value is `true`. The **Qualification Procedure** toggle from the Product Discovery Settings page from Setup overrides this property. For example, if the **Qualification Procedure** toggle is disabled, then setting the `enableQualification` property to `true` has no effect and the `qualificationContext` property in the API response isn't returned.	Optional	60.0
`filter`	Filter Input	Filters records based on supported criteria. The supported property is `name`.	Optional	60.0

Name	Type	Description	Required or Optional	Available Version
		The supported operators are: • `eq` • `in` • `contains` • `gt`—Specifies a greater than criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only. • `lt`—Specifies a less than criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only. • `gte`—Specifies a greater than or equal to criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only. • `lte`—Specifies a less than or equal to criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only. If multiple criteria are specified, then the resultant criteria are combined by using the `and` operator.		
`qualificationProcedure`	String	API name of the custom qualification procedure that's used for the qualification process. If this property isn't specified, then the default qualification procedure is executed.	Optional	60.0
`userContext`	User Context Input	User context details. For example, account ID or contact ID.	Optional	60.0
`usePromotions`	Boolean	Indicates whether to fetch eligible promotions from Global Promotion Management (GPM) for the requested category and its products (`true`) or not (`false`). If Promotion feature is enabled in the org and this property isn't specified, then the default value is `true`. If the Promotion feature isn't enabled, the default value is `false`.	Optional	66.0

Response body for POST

CPQ Base Details

Bulk Product Details (POST)

Retrieve details for multiple products. This API is a composite API for Product Discovery.

Resource

```
/connect/cpq/products/bulk
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/cpq/products/bulk
```

Available version

61.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "productData": [
    {
      "productId": "01txx0000006ivJAAQ",
      "productSellingModelId": "0jPxx000000009hEAA"
    },
    {
      "productId": "01txx0000006ivLAAQ",
      "productSellingModelId": "0jPxx000000009iEAABB"
    }
  ],
  "correlationId": "de9a674c-1807-438c-ac78-2c96f4655325",
  "priceBookId" : "01sxx0000005qxxAAA",
  "userContext": {
      "accountId": "001xx0000000001AAA",
      "contactId": "003xx00000000D7AAI"
  }
}
```

This example shows a sample request with proration policy details requested through the `additionalFields` property.

```
{
  "additionalFields": {
    "ProductSellingModelOption": {
      "additionalFields": {
        "ProrationPolicy": {
          "fields": [
            "ArePartialPeriodsAllowed",
            "ProrationPolicyType"
          ]
        }
      }
```

```
      }
    },
    "productData": [
      {
        "productId": "01txx0000006ivJAAQ",
        "productSellingModelId": "0jPxx000000009hEAA"
      },
      {
        "productId": "01txx0000006ivLAAQ",
        "productSellingModelId": "0jPxx000000009iEAABB"
      }
    ],
    "correlationId": "de9a674c-1807-438c-ac78-2c96f4655325",
    "priceBookId": "01sxx0000005qxxAAA",
    "userContext": {
      "accountId": "001xx0000000001AAA",
      "contactId": "003xx00000000D7AAI"
    }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
additional ContextData	Context Data Input[]	Additional nodes to add to the custom or default context definition. This data is appended to the context input and sent for hydration and qualification. The maximum limit of supported nodes is 10.	Optional	61.0
additional Fields	Map<String, Additional Fields Input>	Additional standard or custom fields of the Product2 object to include in the response. The field values are returned in the response for each of the products. In API version 66.0 and later, you can request proration policy details in the response for each product selling model option through this property.	Optional	61.0
context Definition	String	Name of the custom context definition that's sent for the context creation. If unspecified, the default context definition is used.	Optional	61.0
context Mapping	String	Context mapping details from the context definition. If specified, the API validates if the context mapping belongs to the specified context definition and considers the mapping for hydration.	Optional	61.0

Name	Type	Description	Required or Optional	Available Version
		If unspecified, the default context mapping of the context definition is used.		
`correlationId`	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Optional	61.0
`currencyCode`	String	Currency code to consider for pricing and filtering.	Optional	61.0
`enablePricing`	Boolean	Indicates whether to enable pricing for the products (`true`) or not (`false`). The default value is `true`. The **Pricing Procedure** toggle from the Product Discovery Settings page from Setup overrides this property. For example, if the **Pricing Procedure** toggle is disabled, then setting the `enablePricing` property to `true` has no effect and the `prices` property in the API response is returned empty.	Optional	61.0
`enableQualification`	Boolean	Indicates whether to enable qualification rules for the products (`true`) or not (`false`). The default value is `true`. The **Qualification Procedure** toggle from the Product Discovery Settings page from Setup overrides this property. For example, if the **Qualification Procedure** toggle is disabled, then setting the `enableQualification` property to `true` has no effect and the `qualificationContext` property in the API response isn't returned.	Optional	61.0
`priceBookId`	String	ID of the price book to fetch the prices from.	Optional	61.0
`pricingProcedure`	String	Name of the custom pricing procedure to send for processing. If unspecified, the default pricing procedure is executed.	Optional	61.0
`productData`	Product Data Input[]	List of maps that contain product IDs and product selling model IDs.	Required	61.0

Name	Type	Description	Required or Optional	Available Version
`qualificationProcedure`	String	Name of the custom qualification procedure to send for processing. If unspecified, the default qualification procedure is executed.	Optional	61.0
`userContext`	User Context Input[]	User context details. For example, account ID or contact ID.	Optional	61.0

Response body for POST

Bulk Product Details

Global Search (POST)

Retrieves a list of products based on a search query or search term. This API is a composite API for Product Discovery.

Resource

```
/connect/cpq/products/search
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/cpq/products/search
```

Available version

60.0

HTTP methods

POST

Request body for POST

JSON example

This example shows a sample request to search products by using a query.

```
{
  "query": {
    "textQuery": {
      "searchPhrase": "firstproduct"
    }
  },
  "catalogId": "0ZSxx0000000001GAA",
  "categoryId": "0ZGT100000000q1OAA",
  "correlationId": "9cbb9650-48c5-11ed-96d1-0afcf185843b",
  "limit": 10,
  "cursor": "MTAwMDAwMDAwNg==",
  "orderBy": [
    "name:asc",
    "id:desc"
  ],
  "userContext": {
    "accountId": "001xx0000000001AAA",
    "contactId": "003xx00000000D7AAI"
```

```
  },
  "additionalFields": {
    "Product2": {
      "fields": [
        "CustomField1__c",
        "CustomField2__c",
        "StandardField1"
      ]
    }
  }
}
```

This example shows a sample request to search products by using the `searchTerm` property.

```
{
  "searchTerm": "Laptop",
  "catalogId": "0ZSDU0000002Og64AE",
  "categoryId": "0ZGDU0000002P0A4AU",
  "correlationId": "d9d8f898-19f5-464a-ba2b-6a070783f6c4",
  "limit": 10,
  "cursor": "MTAwMDAwMDAwNw==",
  "orderBy": [
    "name:asc",
    "id:desc"
  ],
  "userContext": {
    "accountId": "001DU000001o2V0YAI"
  }
}
```

If a parent category ID is specified in the request body, then the API returns all products associated to all child categories.

This example shows a sample request to search products with eligible promotions. To fetch eligible promotions, specify a value for the `query` or `searchTerm` property, and set the `usePromotions` property to `true`.

```
{
  "query": {
    "textQuery": {
      "searchPhrase": "laptop"
    }
  },
  "catalogId": "0ZSxx0000000001GAA",
  "categoryId": "0ZGT100000000qlOAA",
  "correlationId": "9cbb9650-48c5-11ed-96d1-0afcf185843b",
  "limit": 10,
  "cursor": "MTAwMDAwMDAwNg==",
  "orderBy": [
    "name:asc",
    "id:desc"
  ],
  "userContext": {
    "accountId": "001xx0000000001AAA",
    "contactId": "003xx00000000D7AAI"
  },
  "additionalFields": {
    "Product2": {
```

```
        "fields": [
          "CustomField1__c",
          "CustomField2__c",
          "StandardField1"
        ]
      }
    },
    "usePromotions": true
}
```

This example shows a sample request to run visibility rules.

```
{
  "enableQualification": true,
  "enablePricing": true,
  "includeCatalogDetails": true,
  "searchTerm": "128GB LRDIMM",
  "catalogId": "0ZSVW000000AhdC4AS",
  "limit": 12,
  "userContext": {
    "accountId": null
  },
  "priceBookId": "01sVW0000024PZlYAM",
  "currencyCode": "USD",
  "transactionId": "0Q0VW000001190f0AA",
  "filter": {
    "criteria": [
      {
        "property": "isActive",
        "operator": "eq",
        "value": true
      },
      {
        "property": "UsedFor",
        "operator": "eq",
        "value": ""
      }
    ]
  },
  "orderBy": [
    "name:asc"
  ],
  "executeConfigurationRules": true
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`additionalContextData`	Context Data Input[]	Additional nodes that are added to the custom or default context definition. The maximum number of supported nodes is 10.	Optional	60.0

Name	Type	Description	Required or Optional	Available Version
additional Fields	Map<String, Additional Fields Input>	Additional standard or custom fields of the Product2 object to include in the response. If the requested fields are invalid or access to fields isn't available, then the API throws an error.	Optional	61.0
catalogId	String	ID of the catalog. If the catalog ID is specified, then the API returns the list of offers from the catalog with the pricing details related to the catalog.	Optional	60.0
categoryId	String	ID of the category. If the category ID isn't specified, then the API returns the matching query offers from the catalog.	Optional	60.0
context Definition	String	API name of the custom context definition that's sent for context creation. If this property isn't specified, then the default context definition is used.	Optional	60.0
context Mapping	String	Default context mapping of the context definition. If a context mapping is specified, then the API checks whether the mapping belongs to the specified context definition to process the details for hydration.	Optional	60.0
correlation Id	String	Unique identifier of the request.	Optional	60.0
currencyCode	String	Currency code that's considered for pricing and filtering request.	Optional	60.0
cursor	String	Unique ID to represent the position of each product in the dataset.	Optional	60.0
enable Pricing	Boolean	Indicates whether to enable pricing for the products (`true`) or not (`false`). The default value is `true`. The **Pricing Procedure** toggle from the Product Discovery Settings page from Setup overrides this property. For example, if the **Pricing Procedure** toggle is disabled, then setting the `enablePricing` property to `true` has no effect and the `prices` property in the API response is returned empty.	Optional	60.0

Name	Type	Description	Required or Optional	Available Version
`enableQualification`	Boolean	Indicates whether to enable qualification rules for the products (`true`) or not (`false`). The default value is `true`. The **Qualification Procedure** toggle from the Product Discovery Settings page from Setup overrides this property. For example, if the **Qualification Procedure** toggle is disabled, then setting the `enableQualification` property to `true` has no effect and the `qualificationContext` property in the API response isn't returned.	Optional	60.0
`executeConfigurationRules`	Boolean	Indicates whether to execute configuration rules (`true`) or not (`false`).	Optional	67.0
`filter`	Filter Input	Filters records based on supported criteria. The supported property is `name`. The supported operators are: • `eq` • `in` • `contains` • `gt`—Specifies a greater than criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only. • `lt`—Specifies a less than criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only. • `gte`—Specifies a greater than or equal to criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only. • `lte`—Specifies a less than or equal to criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only. If multiple criteria are specified, then the resultant criteria are combined by using the `and` operator.	Optional	60.0

Name	Type	Description	Required or Optional	Available Version
include CatalogDetails	Boolean	Indicates whether to include catalog details in the response (`true`) or not (`false`).	Optional	61.0
limit	Integer	Number of items to include in the response. The default value is 10.	Optional	60.0
offset	Integer	Reserved for internal use.	Optional	60.0
orderBy	String[]	Sort order of the results, which is either ascending or descending order. The default sort order is ascending order. The default value is `asc`.	Optional	60.0
priceBookId	String	ID of the price book to get the prices from. If this property isn't specified, then prices from the standard price book are fetched.	Optional	60.0
pricingProcedure	String	API name of the custom pricing procedure that's used for the pricing process. If this property isn't specified, then the default pricing procedure is executed.	Optional	60.0
product ClassificationId	String	ID of the product classification.	Optional	60.0
qualification Procedure	String	API name of the custom qualification procedure that's used for the product qualification process. If this property isn't specified, then the default qualification procedure is executed.	Optional	60.0
query	Map<String, Object>	Query to search the products.	Required	60.0
related ObjectFilter	Related Object Filter Input[]	Filter records based on supported criteria for related objects. The supported object is `ProductSpecificationRecType`. The supported property is `IsCommerical`. The supported operator is `eq`. The supported values are `true` and `false`.	Optional	60.0

Name	Type	Description	Required or Optional	Available Version
`searchTerm`	String	String used to get products with the product name containing the search term. See Search Considerations When Using Indexed Data.	Optional	62.0
`transactionContextId`	String	ID of the transaction context.	Optional	66.0
`transactionId`	String	ID of the transaction.	Optional	66.0
`usePromotions`	Boolean	Indicates whether to retrieve eligible promotions from Global Promotion Management (GPM) for each product in the search results (`true`) or not (`false`). If Promotion feature is enabled in the org and this property isn't specified, then the default value is `true`. If the Promotion feature isn't enabled, the default value is `false`.	Optional	66.0
`userContext`	User Context Input	User context details. For example, account ID or contact ID.	Optional	60.0

Response body for POST

CPQ Base List

Guided Selection (POST)

Retrieve a list of products based on the response identifier or search terms of a guided selection. Guided selection captures user requirements to show suitable products.

Resource

```
/connect/cpq/products/guided-selection
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/cpq/products/guided-selection
```

Available version

62.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "correlationId": "corrId",
  "catalogId": "0ZSxx0000000001GAA",
```

```
  "priceBookId": "pricebookId",
  "limit": 10,
  "cursor": "MTAwMDAwMDAwNg==",
  "userContext": {
    "accountId": "accId"
  },
  "guidedSelectionResponseId": "ABCxx0000000001GAA",
  "searchTerms": [
    {
      "term": "IPhone",
      "tags": [
        "deviceType",
        "mobile"
      ]
    },
    {
      "term": "4GB",
      "tags": [
        "RAM"
      ]
    },
    {
      "term": "64GB",
      "tags": [
        "Storage"
      ]
    }
  ],
  "enableQualification": true,
  "enablePricing": true,
  "includeCatalogDetails": false
}
```

This example shows a sample request to fetch eligible promotions.

```
{
  "correlationId": "corrId",
  "catalogId": "0ZSxx0000000001GAA",
  "priceBookId": "pricebookId",
  "limit": 10,
  "cursor": "MTAwMDAwMDAwNg==",
  "userContext": {
    "accountId": "accId"
  },
  "guidedSelectionResponseId": "ABCxx0000000001GAA",
  "searchTerms": [
    {
      "term": "IPhone",
      "tags": [
        "deviceType",
        "mobile"
      ]
    },
    {
      "term": "4GB",
```

```
      "tags": [
        "RAM"
      ]
    },
    {
      "term": "64GB",
      "tags": [
        "Storage"
      ]
    }
  ],
  "enableQualification": true,
  "enablePricing": true,
  "includeCatalogDetails": false,
  "transactionContextId": "context123",
  "transactionId": "trans456",
  "usePromotions": true
}
```

This example shows a sample request to run visibility rules.

```
{
  "catalogId": "0ZSVW000000AhdC4AS",
  "currencyCode": "USD",
  "enablePricing": true,
  "enableQualification": true,
  "executeConfigurationRules": true,
  "filter": {
    "criteria": [
      {
        "property": "isActive",
        "operator": "eq",
        "value": true
      },
      {
        "property": "UsedFor",
        "operator": "eq",
        "value": ""
      }
    ]
  },
  "guidedSelectionResponseId": "0U3VW0000000a1t0AA",
  "includeCatalogDetails": true,
  "limit": 12,
  "offset": 0,
  "orderBy": [
    "name:asc"
  ],
  "priceBookId": "01sVW0000024PZlYAM",
  "transactionId": "0Q0VW000001190f0AA",
  "userContext": {
    "accountId": null
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
additionalContextData	Context Data Input[]	Additional nodes that are added to the custom or default context definition. The maximum number of supported nodes is 10.	Optional	62.0
additionalFields	Map<String, Additional Fields Input>	Additional standard or custom fields of the Product2 object to include in the response. If the requested fields are invalid or access to fields isn't available, then the API throws an error.	Optional	62.0
catalogId	String	ID of the catalog.	Required	62.0
categoryId	String	ID of the category.	Optional	62.0
contextDefinition	String	API name of the custom context definition that's sent for context creation. If this property isn't specified, then the default context definition is used.	Optional	62.0
contextMapping	String	Default context mapping of the context definition. If a context mapping is specified, then the API checks whether the mapping belongs to the specified context definition to process the details for hydration.	Optional	62.0
correlationId	String	Unique identifier value that's attached to the requests and messages, and accepts references to a particular transaction or event chain.	Optional	62.0
currencyCode	String	Currency code that's considered for pricing and filtering request. If multiple currencies are enabled for the org, then the `currencyCode` property is required.	Optional	62.0
cursor	String	Unique ID to represent the position of each product in the dataset.	Optional	62.0
enablePricing	Boolean	Indicates whether to enable pricing for the products (`true`) or not (`false`). The default value is `true`.	Optional	62.0

Name	Type	Description	Required or Optional	Available Version
enable Qualification	Boolean	Indicates whether to enable qualification rules for the products (`true`) or not (`false`). The default value is `true`.	Optional	62.0
execute ConfigurationRules	Boolean	Indicates whether to execute configuration rules (`true`) or not (`false`).	Optional	67.0
filter	Filter Input	Filters records based on supported criteria. The supported property is `name`. The supported operators are: • `eq` • `in` • `contains`—This value isn't applicable if the **Use Indexed Data For Product Listing and Search** toggle from the Product Discovery Settings page from Setup is enabled. If multiple criteria are specified, then the resultant criteria are combined by using the `and` operator.	Optional	62.0
guided Selection ResponseId	String	Response identifier of the guided selection.	Required if the `searchTerms` property isn't specified.	62.0
include Catalog Details	Boolean	Indicates whether to include catalog details in the response (`true`) or not (`false`).	Optional	62.0
limit	Integer	Number of items to include in the response. The default value is 10.	Optional	62.0
orderBy	String[]	Sort order of the results, which is either ascending (`asc`) or descending order (`desc`). The default sort order is ascending order. The default value is `asc`. If the **Use Indexed Data For Product Listing and Search** toggle from the Product Discovery Settings page from Setup is enabled, then you can sort products by using name only.	Optional	62.0

Name	Type	Description	Required or Optional	Available Version
priceBookId	String	ID of the price book to get the prices from. If this property isn't specified, then prices from the standard price book are fetched.	Required	62.0
pricing Procedure	String	API name of the custom pricing procedure that's used for the pricing process. If this property isn't specified, then the default pricing procedure is executed.	Optional	62.0
product Classification Id	String	ID of the product classification.	Optional	62.0
qualification Procedure	String	API name of the custom qualification procedure that's used for the product qualification process. If this property isn't specified, then the default qualification procedure is executed.	Optional	62.0
searchTerms	Guided Selection Search Term Input[]	Search terms of the guided selection.	Required if the guided Selection ResponseId property isn't specified.	62.0
transaction ContextId	String	ID of the transaction context.	Optional	67.0
transactionId	String	ID of the transaction.	Optional	67.0
usePromotions	Boolean	Indicates whether to fetch applicable promotions from Global Promotion Management (GPM) for the guided selection (true) or not (false). If Promotion feature is enabled in the org and this property isn't specified, then the default value is true. If the Promotion feature isn't enabled, the default value is false.	Optional	66.0
userContext	User Context Input	User context details. For example, account ID or contact ID.	Optional	62.0

If both the guidedSelectionResponseId and searchTerms properties are specified, then the searchTerms property is considered in the input request.

Response body for POST

Guided Selection

Product Details (POST)

Get product details, such as attributes, hierarchy, or cardinality, for a specified product ID. This API is a composite API for Product Discovery.

Resource

```
/connect/cpq/products/productId
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/cpq/products/01txx0000006j08AAA
```

Available version

60.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "correlationId": "9cbb9650-48c5-11ed-96d1-0afcf185843b",
  "catalogId": "0ZSxx0000000001GAA",
  "priceBookId": "01s26000002ZT71AAG",
  "productSellingModelId": "0jP1Q000000CaVFUA0",
  "userContext": {
    "accountId": "001xx0000000001AAA",
    "contactId": "003xx00000000D7AAI"
  },
  "enablePricing": true,
  "enableQualification": true,
  "qualificationProcedure": "QualificationProcedure",
  "pricingProcedure": "Preview",
  "contextDefinition": "TestDefinition",
  "contextMapping": "TestDefinitionNode",
  "additionalFields": {
    "ProductSellingModelOption": {
      "additionalFields": {
        "ProrationPolicy": {
          "fields": [
            "ArePartialPeriodsAllowed",
            "ProrationPolicyType"
          ]
        }
      }
    },
    "Product2": {
      "fields": [
        "field1",
        "field2"
      ]
    },
```

```
    "ProductAttributeDefinition": {
      "fields": [
        "field3",
        "field4"
      ]
    }
  },
  "additionalContextData": [
    {
      "nodeName": "Contract",
      "nodeData": {
        "id": "xxxxx231",
        "name": "Contract1"
      }
    },
    {
      "nodeName": "Lead",
      "nodeData": {
        "id": "lllllll31",
        "name": "Lead1"
      }
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`additionalContextData`	Context Data Input[]	Additional nodes that are added to the custom or default context definition. The maximum number of supported nodes is 10.	Optional	60.0
`additionalFields`	Map<String, Additional Fields Input>	Additional standard or custom fields to include in the response. The supported objects are: • Product2 • ProductAttributeDefinition—If the fields defined for the ProductAttributeDefinition object aren't available for the ProductClassificationAttr object, then the API request fails. In API version 66.0 and later, you can request proration policy details in the response for each product selling model option through this property.	Optional	61.0
`catalogId`	String	ID of the catalog. If the catalog ID is specified, then the API returns the list of	Optional	60.0

Name	Type	Description	Required or Optional	Available Version
		offers from the catalog with the pricing details related to the catalog.		
`contextDefinition`	String	API name of the custom context definition that's sent for context creation. If this property isn't specified, then the default context definition is used.	Optional	60.0
`contextMapping`	String	Default context mapping of the context definition. If a context mapping is specified, then the API checks whether the mapping belongs to the specified context definition to process the details for hydration.	Optional	60.0
`correlationId`	String	Unique identifier value that's attached to the requests and messages, and accepts references to a particular transaction or event chain.	Optional	60.0
`currencyCode`	String	Currency code that's considered for pricing and filtering request. If multiple currencies are enabled for the org, then the `currencyCode` property is required.	Optional	60.0
`enablePricing`	Boolean	Indicates whether to enable pricing for the products (`true`) or not (`false`). The default value is `true`. The **Pricing Procedure** toggle from the Product Discovery Settings page from Setup overrides this property. For example, if the **Pricing Procedure** toggle is disabled, then setting the `enablePricing` property to `true` has no effect and the `prices` property in the API response is returned empty.	Optional	60.0
`enableQualification`	Boolean	Indicates whether to enable qualification rules for the products (`true`) or not (`false`). The default value is `true`. The **Qualification Procedure** toggle from the Product Discovery Settings page from Setup overrides this property. For example, if the **Qualification Procedure** toggle is disabled, then setting the `enableQualification` property to `true` has no effect and the	Optional	60.0

Name	Type	Description	Required or Optional	Available Version
		`qualificationContext` property in the API response isn't returned.		
`priceBookId`	String	ID of the price book to fetch the prices from. If this property isn't specified, then the prices from the standard price book are fetched.	Required	60.0
`pricingProcedure`	String	API name of the custom pricing procedure that's used for the pricing process. If this property isn't specified, then the default pricing procedure is executed.	Optional	60.0
`productSellingModelId`	String	ID of the product selling model.	Optional	60.0
`qualificationProcedure`	String	API name of the custom qualification procedure that's used for the product qualification process. If this property isn't specified, then the default qualification procedure is executed.	Optional	60.0
`userContext`	User Context Input	User context details. For example, account ID or contact ID.	Optional	60.0

Response body for POST

CPQ Base Details

Product Recommendations (POST)

Get a list of recommended products based on your underlying business rules.

This API returns a list of products based on the current quote or order and applicable constraint rules, along with the recommendation reasons and compatibility indicators. This enables sales reps to define accurate and complete, quoting process, minimizing both manual effort and errors.

Resource

```
/revenue/product-discovery/products/recommendations
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/revenue/product-discovery/products/recommendations
```

Available version

67.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "currencyCode": "USD",
  "enablePricing": true,
  "enableQualification": true,
  "filter": {
    "criteria": [
      {
        "property": "isActive",
        "operator": "eq",
        "value": true
      },
      {
        "property": "isQualified",
        "operator": "eq",
        "value": true
      }
    ]
  },
  "limit": 12,
  "priceBookId": "01sSG00000DQCjhYAH",
  "transactionId": "0Q0SG0000014Ui50AE"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
additionalContextData	Context Data Input[]	Additional nodes that are added to the custom or default context definition. The maximum number of supported nodes is 10.	Optional	67.0
additionalFields	Map<String, AdditionalFieldsInputRepresentation>	Additional standard or custom fields of the Product2 object to include in the response. If the requested fields are invalid or access to fields isn't available, then the API throws an error.	Optional	67.0
catalogId	String	ID of the catalog to fetch the recommended products from.	Optional	67.0
contextDefinition	String	API name of the custom context definition that's sent for context creation. If you don't specify this property, then the default context definition is used.	Optional	67.0
contextMapping	String	Default context mapping of the context definition. If you specify a context	Optional	67.0

Name	Type	Description	Required or Optional	Available Version
		mapping, then the API checks whether the mapping belongs to the specified context definition to process the details for hydration.		
`currencyCode`	String	Currency code that's considered for pricing and filtering request. If multiple currencies are enabled in the org, then the `currencyCode` property is required. If you don't specify a currency code, then the value is fetched from the account.	Optional	67.0
`cursor`	String	Unique ID to represent the position of each product in the data set.	Optional	67.0
`enablePricing`	Boolean	Indicates whether to enable pricing for products in orgs where Salesforce Pricing is enabled (`true`) or not (`false`). Set the value to `false` to skip the execution of Salesforce Pricing. In orgs where Salesforce Pricing is disabled, you can't override this value to `true`. The default value is `true`.	Optional	67.0
`enableQualification`	Boolean	Indicates whether to enable qualification rules for products in orgs where Qualification Procedure is enabled (`true`) or not (`false`). Set the value to `false` to skip the execution of Business Rules Engine qualification rules. In orgs where Qualification Procedure is disabled, you can't override this value to `true`. The default value is `true`.	Optional	67.0
`filter`	Filter Input[]	Filters records based on supported criteria. The supported property is `name`. The supported operators are: • `eq` • `in`	Optional	67.0

Name	Type	Description	Required or Optional	Available Version
		• `contains`—This value isn't applicable if the **Use Indexed Data For Product Listing and Search** toggle from the Product Discovery Settings page from Setup is enabled. If you specify multiple criteria, then the resultant criteria are combined by using the `and` operator.		
`limit`	Integer	Number of recommended products to include in the response. The default value is `10`.	Optional	67.0
`priceBookId`	String	ID of the price book to get prices from. If you don't specify a price book ID, then prices from the standard price book are fetched.	Optional	67.0
`pricingProcedure`	String	API name of the custom pricing procedure that's used for the pricing process. If you don't specify this property, then the default pricing procedure is executed.	Optional	67.0
`qualificationProcedure`	String	API name of the custom qualification procedure that's used for the product qualification process. If you don't specify this property, then the default qualification procedure is executed.	Optional	67.0
`transactionContextId`	String	ID of the sales transaction context instance.	Optional	67.0
`transactionId`	String	ID of the quote or order.	Optional	67.0
`usePromotions`	Boolean	Indicates whether to fetch applicable promotions from Global Promotion Management (GPM) for the guided selection (`true`) or not (`false`). The default value is `false`.	Optional	67.0
`userContext`	User Context Input[]	User context details. For example, account ID or contact ID.	Optional	67.0

Response body for POST

Product Recommendations

Products List (POST)

Get a list of products for a specified catalog, category, or subcategory. This API is a composite API for Product Discovery.

Resource

```
/connect/cpq/products
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/cpq/products
```

Available version

60.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "correlationId": "eeaa1db2-f371-4227-a886-c77e2f66ce1d",
  "limit": 60,
  "cursor": "MTAwMDAwMDAwNg==",
  "orderBy": [
    "name:asc"
  ],
  "catalogId": "0ZSDU0000002Og74AE",
  "categoryId": "0ZGDU0000002P0H4AU",
  "priceBookId": "01sDU000000JVsVYAW",
  "productClassificationId": "11BDU0000002TCC2A2",
  "currencyCode": "USD",
  "userContext": {
    "accountId": "001DU000001o2UzYAI"
  },
  "includeCatalogDetails": true,
  "enableQualification": true,
  "enablePricing": true,
  "qualificationProcedure": "ProductQualification",
  "pricingProcedure": "pricingProcedure",
  "contextDefinition": "BrowseContextDefinitionExt",
  "contextMapping": "ProductDiscoveryMapping",
  "filter": {
    "criteria": [
      {
        "property": "name",
        "operator": "eq",
        "value": "Laptop Pro Bundle"
      }
    ]
  },
  "relatedObjectFilters": [
    {
      "objectName": "ProductSpecificationRecType",
      "criteria": [
        {
```

```
          "property": "IsCommercial",
          "operator": "eq",
          "value": true
        }
      ]
    }
  ],
  "additionalContextData": [
    {
      "nodeName": "Account",
      "nodeData": {
        "id": "001DU000001o2UzYAI",
        "name": "Cloud Kicks"
      }
    }
  ],
  "additionalFields": {
    "Product2": {
      "fields": [
        "CanRamp",
        "DecompositionScope",
        "ProductCode"
      ]
    }
  }
}
```

If a parent category ID is specified in the request body, then the API returns all products associated to all child categories.

This example shows a sample request to retrieve a product list with promotions.

```
{
  "correlationId": "eeaa1db2-f371-4227-a886-c77e2f66ce1d",
  "limit": 60,
  "cursor": "MTAwMDAwMDAwNg==",
  "orderBy": [
    "name:asc"
  ],
  "catalogId": "0ZSDU0000002Og74AE",
  "categoryId": "0ZGDU0000002P0H4AU",
  "priceBookId": "01sDU000000JVsVYAW",
  "productClassificationId": "11BDU0000002TCC2A2",
  "currencyCode": "USD",
  "userContext": {
    "accountId": "001DU000001o2UzYAI"
  },
  "includeCatalogDetails": true,
  "enableQualification": true,
  "enablePricing": true,
  "qualificationProcedure": "ProductQualification",
  "pricingProcedure": "pricingProcedure",
  "contextDefinition": "BrowseContextDefinitionExt",
  "contextMapping": "ProductDiscoveryMapping",
  "filter": {
    "criteria": [
```

```
      {
        "property": "name",
        "operator": "eq",
        "value": "Laptop Pro Bundle"
      }
    ]
  },
  "relatedObjectFilters": [
    {
      "objectName": "ProductSpecificationRecType",
      "criteria": [
        {
          "property": "IsCommercial",
          "operator": "eq",
          "value": true
        }
      ]
    }
  ],
  "additionalContextData": [
    {
      "nodeName": "Account",
      "nodeData": {
        "id": "001DU000001o2UzYAI",
        "name": "Cloud Kicks"
      }
    }
  ],
  "additionalFields": {
    "Product2": {
      "fields": [
        "CanRamp",
        "DecompositionScope",
        "ProductCode"
      ]
    }
  },
  "executeConfigurationRules": false,
  "transactionContextId": "a1b2c3d4e5f6",
  "transactionId": "trans789",
  "usePromotions": true
}
```

This example shows a sample request to run visibility rules.

```
{
  "enableQualification": true,
  "enablePricing": true,
  "includeCatalogDetails": true,
  "catalogId": "0ZSVW000000AhdC4AS",
  "limit": 12,
  "offset": 0,
  "userContext": {
    "accountId": null
  },
```

```
  "priceBookId": "01sVW0000024PZlYAM",
  "currencyCode": "USD",
  "transactionId": "0Q0VW000001190f0AA",
  "filter": {
    "criteria": [
      {
        "property": "isActive",
        "operator": "eq",
        "value": true
      },
      {
        "property": "UsedFor",
        "operator": "eq",
        "value": ""
      }
    ]
  },
  "orderBy": [
    "name:asc"
  ],
  "executeConfigurationRules": true
}
```


Properties

Name	Type	Description	Required or Optional	Available Version
additional ContextData	Context Data Input[]	Additional nodes that are added to the custom or default context definition. The maximum number of supported nodes is 10.	Optional	60.0
additional Fields	Map<String, Additional Fields Input>	Additional standard or custom fields of the Product2 object to include in the response. If the requested fields are invalid or access to fields isn't available, then the API throws an error.	Optional	61.0
catalogId	String	ID of the catalog. If the catalog ID is specified, then the API returns the list of offers from the catalog with the pricing details related to the catalog.	Optional	60.0
categoryId	String	ID of the category. If the category ID isn't specified, then the API returns the list of offers from the catalog.	Optional	60.0
context Definition	String	API name of the custom context definition that's sent for context creation. If this property isn't specified, then the default context definition is used.	Optional	60.0

Name	Type	Description	Required or Optional	Available Version
context Mapping	String	Default context mapping of the context definition. If a context mapping is specified, then the API checks whether the mapping belongs to the specified context definition to process the details for hydration.	Optional	60.0
correlation Id	String	Unique identifier of the request.	Optional	60.0
currencyCode	String	Currency code that's considered for pricing and filtering request. If multiple currencies are enabled for the org, then the `currencyCode` property is required.	Optional	60.0
cursor	String	Unique ID to represent the position of each product in the dataset.	Optional	60.0
enable Pricing	Boolean	Indicates whether to enable pricing for the products (`true`) or not (`false`). The default value is `true`. The **Pricing Procedure** toggle from the Product Discovery Settings page from Setup overrides this property. For example, if the **Pricing Procedure** toggle is disabled, then setting the `enablePricing` property to `true` has no effect and the `prices` property in the API response is returned empty.	Optional	60.0
enable Qualification	Boolean	Indicates whether to enable qualification rules for the products (`true`) or not (`false`). The default value is `true`. The **Qualification Procedure** toggle from the Product Discovery Settings page from Setup overrides this property. For example, if the **Qualification Procedure** toggle is disabled, then setting the `enableQualification` property to `true` has no effect and the `qualificationContext` property in the API response isn't returned.	Optional	60.0
execute ConfigurationRules	Boolean	Indicates whether to execute configuration rules (`true`) or not (`false`).	Optional	67.0

Name	Type	Description	Required or Optional	Available Version
`filter`	Filter Input	Filters records based on supported criteria. The supported property is `name`. The supported operators are: • `eq` • `in` • `contains`—This value isn't applicable if the **Use Indexed Data For Product Listing and Search** toggle from the Product Discovery Settings page from Setup is enabled. If multiple criteria are specified, then the resultant criteria are combined by using the `and` operator.	Optional	60.0
`include Catalog Details`	Boolean	Indicates whether to include catalog details in the response (`true`) or not (`false`).	Optional	61.0
`limit`	Integer	Number of items to include in the response. The default value is 10.	Optional	60.0
`offset`	Integer	Reserved for internal use.	Optional	60.0
`orderBy`	String[]	Sort order of the results, which is either ascending (`asc`) or descending order (`desc`). The default sort order is ascending order. The default value is `asc`. If the **Use Indexed Data For Product Listing and Search** toggle from the Product Discovery Settings page from Setup is enabled, then you can sort products by using name only.	Optional	60.0
`priceBookId`	String	ID of the price book to get the prices from. If this property isn't specified, then prices from the standard price book are fetched.	Required	60.0
`pricing Procedure`	String	API name of the custom pricing procedure that's used for the pricing process. If this property isn't specified, then the default pricing procedure is executed.	Optional	60.0

Name	Type	Description	Required or Optional	Available Version
`productClassificationId`	String	ID of the product classification.	Optional	60.0
`qualificationProcedure`	String	API name of the custom qualification procedure that's used for the product qualification process. If this property isn't specified, then the default qualification procedure is executed.	Optional	60.0
`relatedObjectFilters`	Related Object Filter Input[]	Filter records based on supported criteria for related objects. The supported object is `ProductSpecificationRecType`. The supported property is `IsCommerical`. The supported operator is `eq`. The supported values are `true` and `false`.	Optional	60.0
`transactionContextId`	String	ID of the transaction context.	Optional	67.0
`transactionId`	String	ID of the transaction.	Optional	67.0
`usePromotions`	Boolean	Indicates whether to fetch applicable promotions from Global Promotion Management (GPM) for each product in the list (`true`) or not (`false`). If Promotion feature is enabled in the org and this property isn't specified, then the default value is `true`. If the Promotion feature isn't enabled, the default value is `false`.	Optional	66.0
`userContext`	User Context Input	User context details. For example, account ID or contact ID.	Optional	60.0

Response body for POST

CPQ Base List

Qualification (POST)

Run the qualification procedure on a list of product IDs. This API is a composite API for Product Discovery.

Resource

```
/connect/cpq/qualification
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/cpq/qualification
```

Available version

60.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "productIds": [
    "01txx0000006i7PAAQ",
    "01txx0000006i7QAAQ",
    "01txx0000006i7IAAQ"
  ],
  "userContext": {
    "accountId": "001xx000003GZHgAAO"
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
additional ContextData	Context Data Input[]	Additional nodes that are added to the custom or default context definition. The maximum number of supported nodes is 10.	Optional	60.0
catalogId	String	ID of the catalog.	Optional	60.0
categoryId	String	ID of the category.	Optional	60.0
context Definition	String	API name of the custom context definition that's sent for context creation. If this property isn't specified, the default context definition is used.	Optional	60.0
context Mapping	String	Default context mapping of the context definition. If a context mapping is specified, then the API checks whether the mapping belongs to the specified context definition to process the details for hydration.	Optional	60.0
correlation Id	String	Unique identifier of the request.	Optional	60.0

Name	Type	Description	Required or Optional	Available Version
productIds	String[]	List of product IDs for qualification check.	Required	60.0
qualificationProcedure	String	API name of the custom qualification procedure that's sent for processing. If this property isn't specified, the default qualification procedure is executed.	Optional	60.0
userContext	User Context Input	User context details. For example, account ID or contact ID.	Optional	60.0

Response body for POST

CPQ Base List

Request Bodies

Learn more about the available Product Discovery API request bodies.

Additional Fields Input

Input representation of the additional standard or custom fields to include in the response.

Bulk Product Details Input

Input representation of the request to retrieve details of multiple products.

Catalog Details Input

Input representation of the request to get the catalog details.

Catalog List Input

Input representation of the request to get a list of catalogs.

Category Details Input

Input representation of the request to get category details.

Category List Input

Input representation of the request to get a list of categories.

Context Data Input

Input representation of the context data.

Guided Selection Input

Input representation of the guided selection details.

Guided Selection Search Term Input

Input representation of the search terms of a guided selection.

Filter Criteria Input

Input representation of the criteria to filter records based on supported properties.

Filter Input

Input representation of the request to filter records.

Product Details Input

Input representation of the request to get product details.

Product Data Input

Input representation of the product details such as the product ID and product selling model ID.

Product List Input

Input representation of the request to retrieve a list of products.

Product Recommendation Input

Input representation of the details of a request to get product recommendations.

Products Search Input

Input representation of the request to search products.

QOC Qualification

Input representation of the qualification request.

Related Object Filter Input

Input representation of the request to filter records of a related object.

User Context Input

Input representation of the details with the user context.

Additional Fields Input

Input representation of the additional standard or custom fields to include in the response.

JSON example

```
  "additionalFields" : {
    "Product2" : {
    "fields" : ["CustomField1__c","StandardField1"]
}
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`fields`	String[]	List of additional standard or custom fields to include in the response.	Optional	61.0

Bulk Product Details Input

Input representation of the request to retrieve details of multiple products.

JSON example

```
{
  "productData": [
    {
      "productId": "01txx0000006ivJAAQ",
      "productSellingModelId": "0jPxx000000009hEAA"
    },
    {
      "productId": "01txx0000006ivLAAQ",
      "productSellingModelId": "0jPxx000000009iEAABB"
```

```
    }
  ],
  "correlationId": "de9a674c-1807-438c-ac78-2c96f4655325",
  "priceBookId" : "01sxx0000005qxxAAA",
  "userContext": {
      "accountId": "001xx0000000001AAA",
      "contactId": "003xx00000000D7AAI"
  }
}
```

This example shows a sample request with proration policy details requested through the `additionalFields` property.

```
{
  "additionalFields": {
    "ProductSellingModelOption": {
      "additionalFields": {
        "ProrationPolicy": {
          "fields": [
            "ArePartialPeriodsAllowed",
            "ProrationPolicyType"
          ]
        }
      }
    }
  },
  "productData": [
    {
      "productId": "01txx0000006ivJAAQ",
      "productSellingModelId": "0jPxx000000009hEAA"
    },
    {
      "productId": "01txx0000006ivLAAQ",
      "productSellingModelId": "0jPxx000000009iEAABB"
    }
  ],
  "correlationId": "de9a674c-1807-438c-ac78-2c96f4655325",
  "priceBookId": "01sxx0000005qxxAAA",
  "userContext": {
    "accountId": "001xx0000000001AAA",
    "contactId": "003xx00000000D7AAI"
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`additional ContextData`	Context Data Input[]	Additional nodes to add to the custom or default context definition. This data is appended to the context input and sent for hydration and qualification. The maximum limit of supported nodes is 10.	Optional	61.0

Name	Type	Description	Required or Optional	Available Version
additional Fields	Map<String, Additional Fields Input>	Additional standard or custom fields of the Product2 object to include in the response. The field values are returned in the response for each of the products. In API version 66.0 and later, you can request proration policy details in the response for each product selling model option through this property.	Optional	61.0
context Definition	String	Name of the custom context definition that's sent for the context creation. If unspecified, the default context definition is used.	Optional	61.0
context Mapping	String	Context mapping details from the context definition. If specified, the API validates if the context mapping belongs to the specified context definition and considers the mapping for hydration. If unspecified, the default context mapping of the context definition is used.	Optional	61.0
correlationId	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Optional	61.0
currencyCode	String	Currency code to consider for pricing and filtering.	Optional	61.0
enablePricing	Boolean	Indicates whether to enable pricing for the products (`true`) or not (`false`). The default value is `true`. The **Pricing Procedure** toggle from the Product Discovery Settings page from Setup overrides this property. For example, if the **Pricing Procedure** toggle is disabled, then setting the `enablePricing` property to `true` has no effect and the `prices` property in the API response is returned empty.	Optional	61.0

Name	Type	Description	Required or Optional	Available Version
enable Qualification	Boolean	Indicates whether to enable qualification rules for the products (`true`) or not (`false`). The default value is `true`. The **Qualification Procedure** toggle from the Product Discovery Settings page from Setup overrides this property. For example, if the **Qualification Procedure** toggle is disabled, then setting the `enableQualification` property to `true` has no effect and the `qualificationContext` property in the API response isn't returned.	Optional	61.0
priceBookId	String	ID of the price book to fetch the prices from.	Optional	61.0
pricing Procedure	String	Name of the custom pricing procedure to send for processing. If unspecified, the default pricing procedure is executed.	Optional	61.0
productData	Product Data Input[]	List of maps that contain product IDs and product selling model IDs.	Required	61.0
qualification Procedure	String	Name of the custom qualification procedure to send for processing. If unspecified, the default qualification procedure is executed.	Optional	61.0
userContext	User Context Input[]	User context details. For example, account ID or contact ID.	Optional	61.0

Catalog Details Input

Input representation of the request to get the catalog details.

JSON example

```
{
  "correlationId": "9cbb9650-48c5-11ed-96d1-0afcf185843b",
  "userContext": {
     "accountId": "001xx0000000001AAA",
     "contactId": "003xx00000000D7AAI"
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
correlationId	String	Unique identifier value that's attached to the requests and messages, and accepts references to a particular transaction or event chain.	Optional	60.0
userContext	User Context Input	User context details. For example, account ID or contact ID.	Optional	60.0

Catalog List Input

Input representation of the request to get a list of catalogs.

JSON example

```
{
"correlationId": "9cbb9650-48c5-11ed-96d1-0afcf185843b",
"limit": 10,
"offset": 0,
"orderBy": [
    "name:asc",
    "id:desc"
],
"userContext": {
    "accountId": "001xx0000000001AAA",
    "contactId": "003xx00000000D7AAI"
}
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
correlationId	String	Unique identifier value that's attached to the requests and messages, and accepts references to a particular transaction or event chain.	Optional	60.0
limit	Integer	Number of items to include in the response.	Optional	60.0
offset	Integer	Offset size from which to get the catalog count.	Optional	60.0
orderBy	String[]	Sort order for the catalogs.	Optional	60.0
userContext	User Context Input	User context details. For example, account ID or contact ID.	Optional	60.0

Category Details Input

Input representation of the request to get category details.

JSON example

```
{
  "correlationId": "9cbb9650-48c5-11ed-96d1-0afcf185843b",
  "userContext": {
      "accountId": "001xx0000000001AAA",
      "contactId": "003xx00000000D7AAI"
  }
}
```

This example shows a sample request to get category details with eligible promotions.

```
{
  "correlationId": "9cbb9650-48c5-11ed-96d1-0afcf185843b",
  "userContext": {
    "accountId": "001xx0000000001AAA",
    "contactId": "003xx00000000D7AAI"
  },
  "usePromotions": true
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
additional ContextData	Context Data Input	Additional nodes that are added to the custom or default context definition. The maximum number of supported nodes is 10.	Optional	60.0
catalogId	String	ID of the catalog. If the catalog ID is specified, then the API returns the list of offers from the catalog with the pricing details related to the catalog.	Optional	60.0
context Definition	String	API name of the custom context definition that's sent for context creation. If this property isn't specified, then the default context definition is used.	Optional	60.0
context Mapping	String	Default context mapping of the context definition. If a context mapping is specified, then the API checks whether the mapping belongs to the specified context definition to process the details for hydration.	Optional	60.0
customFields	String[]	List of category fields to retrieve in the response.	Optional	60.0

Name	Type	Description	Required or Optional	Available Version
`correlationId`	String	Unique identifier value that's attached to the requests and messages, and accepts references to a particular transaction or event chain.	Optional	60.0
`enable Qualification`	Boolean	Indicates whether to enable qualification rules for the categories (`true`) or not (`false`). The default value is `true`. The **Qualification Procedure** toggle from the Product Discovery Settings page from Setup overrides this property. For example, if the **Qualification Procedure** toggle is disabled, then setting the `enableQualification` property to `true` has no effect and the `qualificationContext` property in the API response isn't returned.	Optional	60.0
`filter`	Filter Input	Filters records based on supported criteria. The supported property is `name`. The supported operators are: • `eq` • `in` • `contains` • `gt`—Specifies a greater than criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only. • `lt`—Specifies a less than criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only. • `gte`—Specifies a greater than or equal to criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only. • `lte`—Specifies a less than or equal to criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only. If multiple criteria are specified, then the resultant criteria are combined by using the `and` operator.	Optional	60.0

Name	Type	Description	Required or Optional	Available Version
`qualificationProcedure`	String	API name of the custom qualification procedure that's used for the qualification process. If this property isn't specified, then the default qualification procedure is executed.	Optional	60.0
`userContext`	User Context Input	User context details. For example, account ID or contact ID.	Optional	60.0
`usePromotions`	Boolean	Indicates whether to fetch eligible promotions from Global Promotion Management (GPM) for the requested category and its products (`true`) or not (`false`). If Promotion feature is enabled in the org and this property isn't specified, then the default value is `true`. If the Promotion feature isn't enabled, the default value is `false`.	Optional	66.0

Category List Input

Input representation of the request to get a list of categories.

JSON example

```
{
  "correlationId": "9cbb9650-48c5-11ed-96d1-0afcf185843b",
  "catalogId": "0ZSxx000000009hGAA",
  "userContext": {
      "accountId": "001xx0000000001AAA",
      "contactId": "003xx00000000D7AAI"
    }
}
```

This example shows a sample request to get a list of categories with eligible promotions.

```
{
  "correlationId": "9cbb9650-48c5-11ed-96d1-0afcf185843b",
  "catalogId": "0ZSxx000000009hGAA",
  "userContext": {
    "accountId": "001xx0000000001AAA",
    "contactId": "003xx00000000D7AAI"
  },
  "usePromotions": true
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`additionalContextData`	Context Data Input[]	Additional nodes that are added to the custom or default context definition. The maximum number of supported nodes is 10.	Optional	60.0
`catalogId`	String	ID of the catalog.	Required	60.0
`contextDefinition`	String	API name of the custom context definition that's sent for context creation. If this property isn't specified, then the default context definition is used.	Optional	60.0
`contextMapping`	String	Default context mapping of the context definition. If a context mapping is specified, then the API checks whether the mapping belongs to the specified context definition to process the details for hydration.	Optional	60.0
`correlationId`	String	Unique identifier value that's attached to the requests and messages, and accepts references to a particular transaction or event chain.	Optional	60.0
`customFields`	String[]	List of additional custom fields to include in the response.	Optional	60.0
`depth`	Integer	Specifies the levels of subcategories to retrieve beneath the parent category. This parameter only applies when `parentCategoryId` is provided.	Optional	61.0
`enableQualification`	Boolean	Indicates whether to enable qualification rules for the products (`true`) or not (`false`). The default value is `true`.	Optional	60.0
`filter`	Filter Input	Filters records based on supported criteria. The supported property is `isQualified`. The supported operators are: • `eq` • `in` • `contains`—This value isn't applicable if the **Use Indexed Data For Product Listing and Search**	Optional	62.0

Name	Type	Description	Required or Optional	Available Version
		toggle from the Product Discovery Settings page from Setup is enabled.		
parent CategoryId	String	ID of the parent category whose subcategories you want to retrieve. If not specified, only root-level categories from the catalog are returned.	Optional	61.0
qualification Procedure	String	API name of the custom qualification procedure that's used for the product qualification process. If this property isn't specified, then the default qualification procedure is executed.	Optional	60.0
usePromotions	Boolean	Indicates whether to fetch eligible promotions from Global Promotion Management (GPM) for the requested category and its products (`true`) or not (`false`). If Promotion feature is enabled in the org and this property isn't specified, then the default value is `true`. If the Promotion feature isn't enabled, the default value is `false`.	Optional	66.0
userContext	User Context Input	User context details. For example, account ID or contact ID.	Optional	60.0

Context Data Input

Input representation of the context data.

JSON example

```
"additionalContextData":[
  {
  "nodeName": "Contract",
  "nodeData": {
    "id": "xxxxx231",
    "name": "Contract1"

  }
},
{
  "nodeName": "Lead",
  "nodeData": {
    "id": "11111131",
    "name": "Lead1"
```

```
  }
}]
```

Properties

Name	Type	Description	Required or Optional	Available Version
`nodeData`	Map<String, Object>	Details of the node.	Optional	60.0
`nodeName`	String	Name of the node.	Optional	60.0

Guided Selection Input

Input representation of the guided selection details.

JSON example

```
{
  "correlationId": "corrId",
  "catalogId": "0ZSxx0000000001GAA",
  "priceBookId": "pricebookId",
  "limit": 10,
  "cursor": "MTAwMDAwMDAwNg==",
  "userContext": {
    "accountId": "accId"
  },
  "guidedSelectionResponseId": "ABCxx0000000001GAA",
  "searchTerms": [
    {
      "term": "IPhone",
      "tags": [
        "deviceType",
        "mobile"
      ]
    },
    {
      "term": "4GB",
      "tags": [
        "RAM"
      ]
    },
    {
      "term": "64GB",
      "tags": [
        "Storage"
      ]
    }
  ],
  "enableQualification": true,
  "enablePricing": true,
  "includeCatalogDetails": false
}
```

This example shows a sample request to fetch eligible promotions.


```
{
  "correlationId": "corrId",
  "catalogId": "0ZSxx0000000001GAA",
  "priceBookId": "pricebookId",
  "limit": 10,
  "cursor": "MTAwMDAwMDAwNg==",
  "userContext": {
    "accountId": "accId"
  },
  "guidedSelectionResponseId": "ABCxx0000000001GAA",
  "searchTerms": [
    {
      "term": "IPhone",
      "tags": [
        "deviceType",
        "mobile"
      ]
    },
    {
      "term": "4GB",
      "tags": [
        "RAM"
      ]
    },
    {
      "term": "64GB",
      "tags": [
        "Storage"
      ]
    }
  ],
  "enableQualification": true,
  "enablePricing": true,
  "includeCatalogDetails": false,
  "transactionContextId": "context123",
  "transactionId": "trans456",
  "usePromotions": true
}
```


This example shows a sample request to run visibility rules.


```
{
  "catalogId": "0ZSVW000000AhdC4AS",
  "currencyCode": "USD",
  "enablePricing": true,
  "enableQualification": true,
  "executeConfigurationRules": true,
  "filter": {
    "criteria": [
      {
        "property": "isActive",
        "operator": "eq",
        "value": true
```

```
      },
      {
        "property": "UsedFor",
        "operator": "eq",
        "value": ""
      }
    ]
  },
  "guidedSelectionResponseId": "0U3VW0000000a1t0AA",
  "includeCatalogDetails": true,
  "limit": 12,
  "offset": 0,
  "orderBy": [
    "name:asc"
  ],
  "priceBookId": "01sVW0000024PZlYAM",
  "transactionId": "0Q0VW000001190f0AA",
  "userContext": {
    "accountId": null
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`additional ContextData`	Context Data Input[]	Additional nodes that are added to the custom or default context definition. The maximum number of supported nodes is 10.	Optional	62.0
`additional Fields`	Map<String, Additional Fields Input>	Additional standard or custom fields of the Product2 object to include in the response. If the requested fields are invalid or access to fields isn't available, then the API throws an error.	Optional	62.0
`catalogId`	String	ID of the catalog.	Required	62.0
`categoryId`	String	ID of the category.	Optional	62.0
`context Definition`	String	API name of the custom context definition that's sent for context creation. If this property isn't specified, then the default context definition is used.	Optional	62.0
`context Mapping`	String	Default context mapping of the context definition. If a context mapping is specified, then the API checks whether the mapping belongs to the specified context definition to process the details for hydration.	Optional	62.0

Name	Type	Description	Required or Optional	Available Version
correlationId	String	Unique identifier value that's attached to the requests and messages, and accepts references to a particular transaction or event chain.	Optional	62.0
currencyCode	String	Currency code that's considered for pricing and filtering request. If multiple currencies are enabled for the org, then the currencyCode property is required.	Optional	62.0
cursor	String	Unique ID to represent the position of each product in the dataset.	Optional	62.0
enablePricing	Boolean	Indicates whether to enable pricing for the products (true) or not (false). The default value is true.	Optional	62.0
enable Qualification	Boolean	Indicates whether to enable qualification rules for the products (true) or not (false). The default value is true.	Optional	62.0
execute ConfigurationRules	Boolean	Indicates whether to execute configuration rules (true) or not (false).	Optional	67.0
filter	Filter Input	Filters records based on supported criteria. The supported property is name. The supported operators are: • eq • in • contains—This value isn't applicable if the **Use Indexed Data For Product Listing and Search** toggle from the Product Discovery Settings page from Setup is enabled. If multiple criteria are specified, then the resultant criteria are combined by using the and operator.	Optional	62.0
guided Selection ResponseId	String	Response identifier of the guided selection.	Required if the searchTerms property isn't specified.	62.0
include Catalog Details	Boolean	Indicates whether to include catalog details in the response (true) or not (false).	Optional	62.0

Name	Type	Description	Required or Optional	Available Version
limit	Integer	Number of items to include in the response. The default value is 10.	Optional	62.0
orderBy	String[]	Sort order of the results, which is either ascending (asc) or descending order (desc). The default sort order is ascending order. The default value is asc. If the **Use Indexed Data For Product Listing and Search** toggle from the Product Discovery Settings page from Setup is enabled, then you can sort products by using name only.	Optional	62.0
priceBookId	String	ID of the price book to get the prices from. If this property isn't specified, then prices from the standard price book are fetched.	Required	62.0
pricingProcedure	String	API name of the custom pricing procedure that's used for the pricing process. If this property isn't specified, then the default pricing procedure is executed.	Optional	62.0
productClassificationId	String	ID of the product classification.	Optional	62.0
qualificationProcedure	String	API name of the custom qualification procedure that's used for the product qualification process. If this property isn't specified, then the default qualification procedure is executed.	Optional	62.0
searchTerms	Guided Selection Search Term Input[]	Search terms of the guided selection.	Required if the guidedSelectionResponseId property isn't specified.	62.0
transactionContextId	String	ID of the transaction context.	Optional	67.0
transactionId	String	ID of the transaction.	Optional	67.0
usePromotions	Boolean	Indicates whether to fetch applicable promotions from Global Promotion Management (GPM) for the guided selection (true) or not (false). If Promotion feature is enabled in the org	Optional	66.0

Name	Type	Description	Required or Optional	Available Version
		and this property isn't specified, then the default value is `true`. If the Promotion feature isn't enabled, the default value is `false`.		
`userContext`	User Context Input	User context details. For example, account ID or contact ID.	Optional	62.0

If both the `guidedSelectionResponseId` and `searchTerms` properties are specified, then the `searchTerms` property is considered in the input request.

Guided Selection Search Term Input

Input representation of the search terms of a guided selection.

JSON example

```
"searchTerms": [
  {
    "term": "IPhone",
    "tags": [
      "deviceType",
      "mobile"
    ]
  },
  {
    "term": "4GB",
    "tags": [
      "RAM"
    ]
  },
  {
    "term": "64GB",
    "tags": [
      "Storage"
    ]
  }
]
```

Properties

Name	Type	Description	Required or Optional	Available Version
`tags`	String[]	Search term tags for the guided selection.	Optional	62.0
`term`	String	Search term for the guided selection.	Required	62.0

Filter Criteria Input

Input representation of the criteria to filter records based on supported properties.

JSON example

```
"criteria":
 [{
   "attributeType": "ProductStandard",
   "property": "name",
   "operator": "eq",
   "value": "iPhone"
 }]
```

Properties

Name	Type	Description	Required or Optional	Available Version
`attributeType`	String	Search attribute type of the facet for a faceted search. Valid values are: • `ProductStandard` • `ProductCustom` • `ProductDynamicAttribute` • `ProductAttributeStandard` • `ProductAttributeCustom`	Optional	63.0
`operator`	String	Operator used for the filter criteria. The supported operators are: • `eq` • `in` • `contains` • `gt`—Specifies a greater than criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only. • `lt`—Specifies a less than criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only. • `gte`—Specifies a greater than or equal to criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only. • `lte`—Specifies a less than or equal to criteria. Available from API version	Required	60.0

Name	Type	Description	Required or Optional	Available Version
		63.0 and later for Number, Date, and Datetime data types only.		
`property`	String	Property name to use in the filter, which must be the same as the object field. The supported property is `name`.	Required	60.0
`value`	Object	Value for the filter criteria.	Required	60.0

Filter Input

Input representation of the request to filter records.

JSON example

```
"filter":
 {"criteria":
 [ {
   "attributeType": "ProductStandard",
   "property": "name",
   "operator": "eq",
   "value": "iPhone"
 }]}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`criteria`	Filter Criteria Input[]	Filter criteria to filter the records.	Optional	60.0

Product Details Input

Input representation of the request to get product details.

JSON example

```
{
  "correlationId": "9cbb9650-48c5-11ed-96d1-0afcf185843b",
  "catalogId": "0ZSxx0000000001GAA",
  "priceBookId": "01s26000002ZT71AAG",
  "productSellingModelId": "0jP1Q000000CaVFUA0",
  "userContext": {
    "accountId": "001xx0000000001AAA",
    "contactId": "003xx00000000D7AAI"
  },
  "enablePricing": true,
  "enableQualification": true,
  "qualificationProcedure": "QualificationProcedure",
  "pricingProcedure": "Preview",
```

```
  "contextDefinition": "TestDefinition",
  "contextMapping": "TestDefinitionNode",
  "additionalFields": {
    "ProductSellingModelOption": {
      "additionalFields": {
        "ProrationPolicy": {
          "fields": [
            "ArePartialPeriodsAllowed",
            "ProrationPolicyType"
          ]
        }
      }
    },
    "Product2": {
      "fields": [
        "field1",
        "field2"
      ]
    },
    "ProductAttributeDefinition": {
      "fields": [
        "field3",
        "field4"
      ]
    }
  },
  "additionalContextData": [
    {
      "nodeName": "Contract",
      "nodeData": {
        "id": "xxxxx231",
        "name": "Contract1"
      }
    },
    {
      "nodeName": "Lead",
      "nodeData": {
        "id": "lllllll31",
        "name": "Lead1"
      }
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
additional ContextData	Context Data Input[]	Additional nodes that are added to the custom or default context definition. The maximum number of supported nodes is 10.	Optional	60.0

Name	Type	Description	Required or Optional	Available Version
additionalFields	Map<String, Additional Fields Input>	Additional standard or custom fields to include in the response. The supported objects are: • Product2 • ProductAttributeDefinition—If the fields defined for the ProductAttributeDefinition object aren't available for the ProductClassificationAttr object, then the API request fails. In API version 66.0 and later, you can request proration policy details in the response for each product selling model option through this property.	Optional	61.0
catalogId	String	ID of the catalog. If the catalog ID is specified, then the API returns the list of offers from the catalog with the pricing details related to the catalog.	Optional	60.0
contextDefinition	String	API name of the custom context definition that's sent for context creation. If this property isn't specified, then the default context definition is used.	Optional	60.0
contextMapping	String	Default context mapping of the context definition. If a context mapping is specified, then the API checks whether the mapping belongs to the specified context definition to process the details for hydration.	Optional	60.0
correlationId	String	Unique identifier value that's attached to the requests and messages, and accepts references to a particular transaction or event chain.	Optional	60.0
currencyCode	String	Currency code that's considered for pricing and filtering request. If multiple currencies are enabled for the org, then the `currencyCode` property is required.	Optional	60.0
enablePricing	Boolean	Indicates whether to enable pricing for the products (`true`) or not (`false`). The default value is `true`. The **Pricing Procedure** toggle from the Product Discovery Settings page from	Optional	60.0

Name	Type	Description	Required or Optional	Available Version
		Setup overrides this property. For example, if the **Pricing Procedure** toggle is disabled, then setting the `enablePricing` property to `true` has no effect and the `prices` property in the API response is returned empty.		
`enableQualification`	Boolean	Indicates whether to enable qualification rules for the products (`true`) or not (`false`). The default value is `true`. The **Qualification Procedure** toggle from the Product Discovery Settings page from Setup overrides this property. For example, if the **Qualification Procedure** toggle is disabled, then setting the `enableQualification` property to `true` has no effect and the `qualificationContext` property in the API response isn't returned.	Optional	60.0
`priceBookId`	String	ID of the price book to fetch the prices from. If this property isn't specified, then the prices from the standard price book are fetched.	Required	60.0
`pricingProcedure`	String	API name of the custom pricing procedure that's used for the pricing process. If this property isn't specified, then the default pricing procedure is executed.	Optional	60.0
`productSellingModelId`	String	ID of the product selling model.	Optional	60.0
`qualificationProcedure`	String	API name of the custom qualification procedure that's used for the product qualification process. If this property isn't specified, then the default qualification procedure is executed.	Optional	60.0
`userContext`	User Context Input	User context details. For example, account ID or contact ID.	Optional	60.0

Product Data Input

Input representation of the product details such as the product ID and product selling model ID.

JSON example

```
"productData": [
  {
    "productId": "01txx0000006ivJAAQ",
    "productSellingModelId": "0jPxx000000009hEAA"
  },
  {
    "productId": "01txx0000006ivLAAQ",
    "productSellingModelId": "0jPxx000000009iEAABB"
  }
]
```

Properties

Name	Type	Description	Required or Optional	Available Version
productId	String	ID of the product.	Required	61.0
product Selling ModelId	String	ID of the product selling model.	Optional	61.0

Product List Input

Input representation of the request to retrieve a list of products.

JSON example

```
{
  "correlationId": "eeaa1db2-f371-4227-a886-c77e2f66ce1d",
  "limit": 60,
  "cursor": "MTAwMDAwMDAwNg==",
  "orderBy": [
    "name:asc"
  ],
  "catalogId": "0ZSDU0000002Og74AE",
  "categoryId": "0ZGDU0000002P0H4AU",
  "priceBookId": "01sDU000000JVsVYAW",
  "productClassificationId": "11BDU0000002TCC2A2",
  "currencyCode": "USD",
  "userContext": {
    "accountId": "001DU000001o2UzYAI"
  },
  "includeCatalogDetails": true,
  "enableQualification": true,
  "enablePricing": true,
  "qualificationProcedure": "ProductQualification",
  "pricingProcedure": "pricingProcedure",
  "contextDefinition": "BrowseContextDefinitionExt",
  "contextMapping": "ProductDiscoveryMapping",
  "filter": {
    "criteria": [
```

```
      {
        "property": "name",
        "operator": "eq",
        "value": "Laptop Pro Bundle"
      }
    ]
  },
  "relatedObjectFilters": [
    {
      "objectName": "ProductSpecificationRecType",
      "criteria": [
        {
          "property": "IsCommercial",
          "operator": "eq",
          "value": true
        }
      ]
    }
  ],
  "additionalContextData": [
    {
      "nodeName": "Account",
      "nodeData": {
        "id": "001DU000001o2UzYAI",
        "name": "Cloud Kicks"
      }
    }
  ],
  "additionalFields": {
    "Product2": {
      "fields": [
        "CanRamp",
        "DecompositionScope",
        "ProductCode"
      ]
    }
  }
}
```

If a parent category ID is specified in the request body, then the API returns all products associated to all child categories.

This example shows a sample request to retrieve a product list with promotions.

```
{
  "correlationId": "eeaa1db2-f371-4227-a886-c77e2f66ce1d",
  "limit": 60,
  "cursor": "MTAwMDAwMDAwNg==",
  "orderBy": [
    "name:asc"
  ],
  "catalogId": "0ZSDU0000002Og74AE",
  "categoryId": "0ZGDU0000002P0H4AU",
  "priceBookId": "01sDU000000JVsVYAW",
  "productClassificationId": "11BDU0000002TCC2A2",
  "currencyCode": "USD",
```

```
  "userContext": {
    "accountId": "001DU000001o2UzYAI"
  },
  "includeCatalogDetails": true,
  "enableQualification": true,
  "enablePricing": true,
  "qualificationProcedure": "ProductQualification",
  "pricingProcedure": "pricingProcedure",
  "contextDefinition": "BrowseContextDefinitionExt",
  "contextMapping": "ProductDiscoveryMapping",
  "filter": {
    "criteria": [
      {
        "property": "name",
        "operator": "eq",
        "value": "Laptop Pro Bundle"
      }
    ]
  },
  "relatedObjectFilters": [
    {
      "objectName": "ProductSpecificationRecType",
      "criteria": [
        {
          "property": "IsCommercial",
          "operator": "eq",
          "value": true
        }
      ]
    }
  ],
  "additionalContextData": [
    {
      "nodeName": "Account",
      "nodeData": {
        "id": "001DU000001o2UzYAI",
        "name": "Cloud Kicks"
      }
    }
  ],
  "additionalFields": {
    "Product2": {
      "fields": [
        "CanRamp",
        "DecompositionScope",
        "ProductCode"
      ]
    }
  },
  "executeConfigurationRules": false,
  "transactionContextId": "a1b2c3d4e5f6",
  "transactionId": "trans789",
  "usePromotions": true
}
```

This example shows a sample request to run visibility rules.

```
{
  "enableQualification": true,
  "enablePricing": true,
  "includeCatalogDetails": true,
  "catalogId": "0ZSVW000000AhdC4AS",
  "limit": 12,
  "offset": 0,
  "userContext": {
    "accountId": null
  },
  "priceBookId": "01sVW0000024PZlYAM",
  "currencyCode": "USD",
  "transactionId": "0Q0VW000001190f0AA",
  "filter": {
    "criteria": [
      {
        "property": "isActive",
        "operator": "eq",
        "value": true
      },
      {
        "property": "UsedFor",
        "operator": "eq",
        "value": ""
      }
    ]
  },
  "orderBy": [
    "name:asc"
  ],
  "executeConfigurationRules": true
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`additional ContextData`	Context Data Input[]	Additional nodes that are added to the custom or default context definition. The maximum number of supported nodes is 10.	Optional	60.0
`additional Fields`	Map<String, Additional Fields Input>	Additional standard or custom fields of the Product2 object to include in the response. If the requested fields are invalid or access to fields isn't available, then the API throws an error.	Optional	61.0
`catalogId`	String	ID of the catalog. If the catalog ID is specified, then the API returns the list of	Optional	60.0

Name	Type	Description	Required or Optional	Available Version
		offers from the catalog with the pricing details related to the catalog.		
`categoryId`	String	ID of the category. If the category ID isn't specified, then the API returns the list of offers from the catalog.	Optional	60.0
`context Definition`	String	API name of the custom context definition that's sent for context creation. If this property isn't specified, then the default context definition is used.	Optional	60.0
`context Mapping`	String	Default context mapping of the context definition. If a context mapping is specified, then the API checks whether the mapping belongs to the specified context definition to process the details for hydration.	Optional	60.0
`correlationId`	String	Unique identifier of the request.	Optional	60.0
`currencyCode`	String	Currency code that's considered for pricing and filtering request. If multiple currencies are enabled for the org, then the `currencyCode` property is required.	Optional	60.0
`cursor`	String	Unique ID to represent the position of each product in the dataset.	Optional	60.0
`enablePricing`	Boolean	Indicates whether to enable pricing for the products (`true`) or not (`false`). The default value is `true`. The **Pricing Procedure** toggle from the Product Discovery Settings page from Setup overrides this property. For example, if the **Pricing Procedure** toggle is disabled, then setting the `enablePricing` property to `true` has no effect and the `prices` property in the API response is returned empty.	Optional	60.0
`enable Qualification`	Boolean	Indicates whether to enable qualification rules for the products (`true`) or not (`false`). The default value is `true`. The **Qualification Procedure** toggle from the Product Discovery Settings page from Setup overrides this property. For example, if the **Qualification Procedure** toggle is disabled, then setting the	Optional	60.0

Name	Type	Description	Required or Optional	Available Version
		`enableQualification` property to `true` has no effect and the `qualificationContext` property in the API response isn't returned.		
`executeConfigurationRules`	Boolean	Indicates whether to execute configuration rules (`true`) or not (`false`).	Optional	67.0
`filter`	Filter Input	Filters records based on supported criteria. The supported property is `name`. The supported operators are: • `eq` • `in` • `contains`—This value isn't applicable if the **Use Indexed Data For Product Listing and Search** toggle from the Product Discovery Settings page from Setup is enabled. If multiple criteria are specified, then the resultant criteria are combined by using the `and` operator.	Optional	60.0
`includeCatalogDetails`	Boolean	Indicates whether to include catalog details in the response (`true`) or not (`false`).	Optional	61.0
`limit`	Integer	Number of items to include in the response. The default value is 10.	Optional	60.0
`offset`	Integer	Reserved for internal use.	Optional	60.0
`orderBy`	String[]	Sort order of the results, which is either ascending (`asc`) or descending order (`desc`). The default sort order is ascending order. The default value is `asc`. If the **Use Indexed Data For Product Listing and Search** toggle from the Product Discovery Settings page from Setup is enabled, then you can sort products by using name only.	Optional	60.0
`priceBookId`	String	ID of the price book to get the prices from. If this property isn't specified, then prices from the standard price book are fetched.	Required	60.0

Name	Type	Description	Required or Optional	Available Version
pricing Procedure	String	API name of the custom pricing procedure that's used for the pricing process. If this property isn't specified, then the default pricing procedure is executed.	Optional	60.0
product Classification Id	String	ID of the product classification.	Optional	60.0
qualification Procedure	String	API name of the custom qualification procedure that's used for the product qualification process. If this property isn't specified, then the default qualification procedure is executed.	Optional	60.0
relatedObject Filters	Related Object Filter Input[]	Filter records based on supported criteria for related objects. The supported object is `ProductSpecificationRecType`. The supported property is `IsCommerical`. The supported operator is `eq`. The supported values are `true` and `false`.	Optional	60.0
transaction ContextId	String	ID of the transaction context.	Optional	67.0
transactionId	String	ID of the transaction.	Optional	67.0
usePromotions	Boolean	Indicates whether to fetch applicable promotions from Global Promotion Management (GPM) for each product in the list (`true`) or not (`false`). If Promotion feature is enabled in the org and this property isn't specified, then the default value is `true`. If the Promotion feature isn't enabled, the default value is `false`.	Optional	66.0
userContext	User Context Input	User context details. For example, account ID or contact ID.	Optional	60.0

Product Recommendation Input

Input representation of the details of a request to get product recommendations.

JSON example

```
{
  "currencyCode": "USD",
  "enablePricing": true,
  "enableQualification": true,
  "filter": {
    "criteria": [
      {
        "property": "isActive",
        "operator": "eq",
        "value": true
      },
      {
        "property": "isQualified",
        "operator": "eq",
        "value": true
      }
    ]
  },
  "limit": 12,
  "priceBookId": "01sSG00000DQCjhYAH",
  "transactionId": "0Q0SG0000014Ui50AE"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
additionalContextData	Context Data Input[]	Additional nodes that are added to the custom or default context definition. The maximum number of supported nodes is 10.	Optional	67.0
additionalFields	Map<String, AdditionalFieldInputRepresentation>	Additional standard or custom fields of the Product2 object to include in the response. If the requested fields are invalid or access to fields isn't available, then the API throws an error.	Optional	67.0
catalogId	String	ID of the catalog to fetch the recommended products from.	Optional	67.0
contextDefinition	String	API name of the custom context definition that's sent for context creation. If you don't specify this property, then the default context definition is used.	Optional	67.0
contextMapping	String	Default context mapping of the context definition. If you specify a context mapping, then the API checks whether the mapping belongs to the specified context	Optional	67.0

Name	Type	Description	Required or Optional	Available Version
		definition to process the details for hydration.		
`currencyCode`	String	Currency code that's considered for pricing and filtering request. If multiple currencies are enabled in the org, then the `currencyCode` property is required. If you don't specify a currency code, then the value is fetched from the account.	Optional	67.0
`cursor`	String	Unique ID to represent the position of each product in the data set.	Optional	67.0
`enablePricing`	Boolean	Indicates whether to enable pricing for products in orgs where Salesforce Pricing is enabled (`true`) or not (`false`). Set the value to `false` to skip the execution of Salesforce Pricing. In orgs where Salesforce Pricing is disabled, you can't override this value to `true`. The default value is `true`.	Optional	67.0
`enableQualification`	Boolean	Indicates whether to enable qualification rules for products in orgs where Qualification Procedure is enabled (`true`) or not (`false`). Set the value to `false` to skip the execution of Business Rules Engine qualification rules. In orgs where Qualification Procedure is disabled, you can't override this value to `true`. The default value is `true`.	Optional	67.0
`filter`	Filter Input[]	Filters records based on supported criteria. The supported property is `name`. The supported operators are: • `eq` • `in` • `contains`—This value isn't applicable if the **Use Indexed Data For Product Listing and Search** toggle from the Product Discovery Settings page from Setup is enabled.	Optional	67.0

Name	Type	Description	Required or Optional	Available Version
		If you specify multiple criteria, then the resultant criteria are combined by using the `and` operator.		
`limit`	Integer	Number of recommended products to include in the response. The default value is `10`.	Optional	67.0
`priceBookId`	String	ID of the price book to get prices from. If you don't specify a price book ID, then prices from the standard price book are fetched.	Optional	67.0
`pricingProcedure`	String	API name of the custom pricing procedure that's used for the pricing process. If you don't specify this property, then the default pricing procedure is executed.	Optional	67.0
`qualificationProcedure`	String	API name of the custom qualification procedure that's used for the product qualification process. If you don't specify this property, then the default qualification procedure is executed.	Optional	67.0
`transactionContextId`	String	ID of the sales transaction context instance.	Optional	67.0
`transactionId`	String	ID of the quote or order.	Optional	67.0
`usePromotions`	Boolean	Indicates whether to fetch applicable promotions from Global Promotion Management (GPM) for the guided selection (`true`) or not (`false`). The default value is `false`.	Optional	67.0
`userContext`	User Context Input[]	User context details. For example, account ID or contact ID.	Optional	67.0

Products Search Input

Input representation of the request to search products.

JSON example

This example shows a sample request to search products by using a query.

```
{
  "query": {
    "textQuery": {
      "searchPhrase": "firstproduct"
    }
  },
  "catalogId": "0ZSxx0000000001GAA",
```

```
  "categoryId": "0ZGT100000000qlOAA",
  "correlationId": "9cbb9650-48c5-11ed-96d1-0afcf185843b",
  "limit": 10,
  "cursor": "MTAwMDAwMDAwNg==",
  "orderBy": [
    "name:asc",
    "id:desc"
  ],
  "userContext": {
    "accountId": "001xx0000000001AAA",
    "contactId": "003xx00000000D7AAI"
  },
  "additionalFields": {
    "Product2": {
      "fields": [
        "CustomField1__c",
        "CustomField2__c",
        "StandardField1"
      ]
    }
  }
}
```

This example shows a sample request to search products by using the `searchTerm` property.

```
{
  "searchTerm": "Laptop",
  "catalogId": "0ZSDU0000002Og64AE",
  "categoryId": "0ZGDU0000002P0A4AU",
  "correlationId": "d9d8f898-19f5-464a-ba2b-6a070783f6c4",
  "limit": 10,
  "cursor": "MTAwMDAwMDAwNw==",
  "orderBy": [
    "name:asc",
    "id:desc"
  ],
  "userContext": {
    "accountId": "001DU000001o2V0YAI"
  }
}
```

If a parent category ID is specified in the request body, then the API returns all products associated to all child categories.

This example shows a sample request to search products with eligible promotions. To fetch eligible promotions, specify a value for the `query` or `searchTerm` property, and set the `usePromotions` property to `true`.

```
{
  "query": {
    "textQuery": {
      "searchPhrase": "laptop"
    }
  },
  "catalogId": "0ZSxx0000000001GAA",
  "categoryId": "0ZGT100000000qlOAA",
  "correlationId": "9cbb9650-48c5-11ed-96d1-0afcf185843b",
  "limit": 10,
```

```
  "cursor": "MTAwMDAwMDAwNg==",
  "orderBy": [
    "name:asc",
    "id:desc"
  ],
  "userContext": {
    "accountId": "001xx0000000001AAA",
    "contactId": "003xx00000000D7AAI"
  },
  "additionalFields": {
    "Product2": {
      "fields": [
        "CustomField1__c",
        "CustomField2__c",
        "StandardField1"
      ]
    }
  },
  "usePromotions": true
}
```

This example shows a sample request to run visibility rules.

```
{
  "enableQualification": true,
  "enablePricing": true,
  "includeCatalogDetails": true,
  "searchTerm": "128GB LRDIMM",
  "catalogId": "0ZSVW000000AhdC4AS",
  "limit": 12,
  "userContext": {
    "accountId": null
  },
  "priceBookId": "01sVW0000024PZlYAM",
  "currencyCode": "USD",
  "transactionId": "0Q0VW000001190f0AA",
  "filter": {
    "criteria": [
      {
        "property": "isActive",
        "operator": "eq",
        "value": true
      },
      {
        "property": "UsedFor",
        "operator": "eq",
        "value": ""
      }
    ]
  },
  "orderBy": [
    "name:asc"
  ],
  "executeConfigurationRules": true
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
additional ContextData	Context Data Input[]	Additional nodes that are added to the custom or default context definition. The maximum number of supported nodes is 10.	Optional	60.0
additional Fields	Map<String, Additional Fields Input>	Additional standard or custom fields of the Product2 object to include in the response. If the requested fields are invalid or access to fields isn't available, then the API throws an error.	Optional	61.0
catalogId	String	ID of the catalog. If the catalog ID is specified, then the API returns the list of offers from the catalog with the pricing details related to the catalog.	Optional	60.0
categoryId	String	ID of the category. If the category ID isn't specified, then the API returns the matching query offers from the catalog.	Optional	60.0
context Definition	String	API name of the custom context definition that's sent for context creation. If this property isn't specified, then the default context definition is used.	Optional	60.0
context Mapping	String	Default context mapping of the context definition. If a context mapping is specified, then the API checks whether the mapping belongs to the specified context definition to process the details for hydration.	Optional	60.0
correlationId	String	Unique identifier of the request.	Optional	60.0
currencyCode	String	Currency code that's considered for pricing and filtering request.	Optional	60.0
cursor	String	Unique ID to represent the position of each product in the dataset.	Optional	60.0
enablePricing	Boolean	Indicates whether to enable pricing for the products (`true`) or not (`false`). The default value is `true`. The **Pricing Procedure** toggle from the Product Discovery Settings page from Setup overrides this property. For example, if the **Pricing Procedure** toggle is	Optional	60.0

Name	Type	Description	Required or Optional	Available Version
		disabled, then setting the `enablePricing` property to `true` has no effect and the `prices` property in the API response is returned empty.		
`enableQualification`	Boolean	Indicates whether to enable qualification rules for the products (`true`) or not (`false`). The default value is `true`. The **Qualification Procedure** toggle from the Product Discovery Settings page from Setup overrides this property. For example, if the **Qualification Procedure** toggle is disabled, then setting the `enableQualification` property to `true` has no effect and the `qualificationContext` property in the API response isn't returned.	Optional	60.0
`executeConfigurationRules`	Boolean	Indicates whether to execute configuration rules (`true`) or not (`false`).	Optional	67.0
`filter`	Filter Input	Filters records based on supported criteria. The supported property is `name`. The supported operators are: • `eq` • `in` • `contains` • `gt`—Specifies a greater than criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only. • `lt`—Specifies a less than criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only. • `gte`—Specifies a greater than or equal to criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only. • `lte`—Specifies a less than or equal to criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only.	Optional	60.0

Name	Type	Description	Required or Optional	Available Version
		If multiple criteria are specified, then the resultant criteria are combined by using the `and` operator.		
`includeCatalogDetails`	Boolean	Indicates whether to include catalog details in the response (`true`) or not (`false`).	Optional	61.0
`limit`	Integer	Number of items to include in the response. The default value is 10.	Optional	60.0
`offset`	Integer	Reserved for internal use.	Optional	60.0
`orderBy`	String[]	Sort order of the results, which is either ascending or descending order. The default sort order is ascending order. The default value is `asc`.	Optional	60.0
`priceBookId`	String	ID of the price book to get the prices from. If this property isn't specified, then prices from the standard price book are fetched.	Optional	60.0
`pricingProcedure`	String	API name of the custom pricing procedure that's used for the pricing process. If this property isn't specified, then the default pricing procedure is executed.	Optional	60.0
`productClassificationId`	String	ID of the product classification.	Optional	60.0
`qualificationProcedure`	String	API name of the custom qualification procedure that's used for the product qualification process. If this property isn't specified, then the default qualification procedure is executed.	Optional	60.0
`query`	Map<String, Object>	Query to search the products.	Required	60.0
`relatedObjectFilter`	Related Object Filter Input[]	Filter records based on supported criteria for related objects. The supported object is `ProductSpecificationRecType`. The supported property is `IsCommerical`. The supported operator is `eq`. The supported values are `true` and `false`.	Optional	60.0

Name	Type	Description	Required or Optional	Available Version
searchTerm	String	String used to get products with the product name containing the search term. See Search Considerations When Using Indexed Data.	Optional	62.0
transaction ContextId	String	ID of the transaction context.	Optional	66.0
transactionId	String	ID of the transaction.	Optional	66.0
usePromotions	Boolean	Indicates whether to retrieve eligible promotions from Global Promotion Management (GPM) for each product in the search results (`true`) or not (`false`). If Promotion feature is enabled in the org and this property isn't specified, then the default value is `true`. If the Promotion feature isn't enabled, the default value is `false`.	Optional	66.0
userContext	User Context Input	User context details. For example, account ID or contact ID.	Optional	60.0

QOC Qualification

Input representation of the qualification request.

JSON example

```
{
  "productIds": [
    "01txx0000006i7PAAQ",
    "01txx0000006i7QAAQ",
    "01txx0000006i7IAAQ"
  ],
  "userContext": {
    "accountId": "001xx000003GZHgAAO"
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
additional ContextData	Context Data Input[]	Additional nodes that are added to the custom or default context definition. The maximum number of supported nodes is 10.	Optional	60.0
catalogId	String	ID of the catalog.	Optional	60.0

Name	Type	Description	Required or Optional	Available Version
categoryId	String	ID of the category.	Optional	60.0
context Definition	String	API name of the custom context definition that's sent for context creation. If this property isn't specified, the default context definition is used.	Optional	60.0
context Mapping	String	Default context mapping of the context definition. If a context mapping is specified, then the API checks whether the mapping belongs to the specified context definition to process the details for hydration.	Optional	60.0
correlationId	String	Unique identifier of the request.	Optional	60.0
productIds	String[]	List of product IDs for qualification check.	Required	60.0
qualification Procedure	String	API name of the custom qualification procedure that's sent for processing. If this property isn't specified, the default qualification procedure is executed.	Optional	60.0
userContext	User Context Input	User context details. For example, account ID or contact ID.	Optional	60.0

Related Object Filter Input

Input representation of the request to filter records of a related object.

JSON example

```
"relatedObjectFilters":
 [
    {
      "objectName": "ProductSpecificationRecType",
      "criteria":
 [{
      "property": "IsCommercial",
      "operator": "eq",
      "value": true
   }]
  }
  ]
```

Properties

Name	Type	Description	Required or Optional	Available Version
`criteria`	Filter Criteria Input[]	Criteria to filter the related objects.	Required if the `relatedObjectFilters` property is specified.	60.0
`objectName`	String	Name of the object that's related to the main object.	Required if the `relatedObjectFilters` property is specified.	60.0

User Context Input

Input representation of the details with the user context.

JSON example

```
"userContext": {
    "accountId": "001xx0000000001AAA",
    "contactId": "003xx00000000D7AAI",
    "contextId": "e055bb18-d4e8-41c3-881e-0132b9561708"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`accountId`	String	ID of the account in a user context.	Optional	60.0
`contactId`	String	ID of the contact in a user context.	Optional	60.0
`contextId`	String	ID of the context that represents the created session.	Optional	60.0

Response Bodies

Learn more about the available Product Discovery API response bodies.

API Status
Output representation of the API status.

Catalog Details
Output representation of the details of a catalog definition.

Category Details
Output representation of the details of a category.

Bulk Product Details
Output representation of the response that contains the details of multiple products.

CPQ Base Details
Output representation of the catalog, category, or product details based on the request.

CPQ Base List
Output representation of the list of catalogs, categories, or products based on the request.

CPQ Message
Output representation of the API messages.

Facet Value
Output representation of the facet values found in the search result.

Guided Selection
Output representation of the details of a guided selection.

Guided Selection Search Term
Output representation of the search term details for a guided selection.

Pricing Model
Output representation of the details of the pricing model.

Product Classification
Output representation of the details of a product classification.

Product Configuration Rule Details
Output representation of the product configuration rule details.

Product Configuration Rules
Output representation of the details of the product configuration rules.

Product Prices
Output representation of the details of the product prices.

Product Quantity
Output representation of the details of the quantity constraints and current quantity for a product in the product discovery context.

Product Recommendation Errors
Output representation of the details of the errors encountered during the processing of the Product Recommendations API request.

Product Recommendations
Output representation of the fetched product recommendations.

Product Selling Model Option
Output representation of the product selling model option component.

Product Selling Model
Product Selling Model Component output representation

Product Specification Record Type
Output representation of the details of the product specification record type.

Product Specification Type
Output representation of the details of the product specification type.

Promotion Details
Output representation of the details of applicable promotions.

Proration Policy
Output representation of the details of the proration policy component.

[Qualification Context](#)
Output representation of the details about the product qualification.

[Recommended Products](#)
Output representation of the list of recommended products.

[Search Products Facet](#)
Output representation of the details of the faceted search.

[User Context](#)
Output representation of the user context details.

API Status

Output representation of the API status.

Property Name	Type	Description	Filter Group and Version	Available Version
`messages`	CPQ Message[]	Status messages of the API execution.	Small, 60.0	60.0
`statusCode`	String	Status code of the API execution.	Small, 60.0	60.0
`statusMessage`	String	Display label for the API status.	Small, 60.0	60.0

Catalog Details

Output representation of the details of a catalog definition.

JSON example

```
{
  "catalogs": [
    {
      "customFields": {},
      "id": "0ZSSG0000018750 4AQ",
      "name": "Hardware Catalog",
      "numberOfCategories": 4
    }
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`catalogCode`	String	Unique ID associated with the catalog.	Small, 67.0	67.0
`catalogType`	String	Category of an entry in the catalog, which is customizable. For example, catalog types, such as sellable products, services, parts, technical services, or technical resources.	Small, 67.0	67.0
`customFields`	Map<String, Object>	Details of the custom fields associated with a catalog.	Small, 67.0	67.0

Property Name	Type	Description	Filter Group and Version	Available Version
description	String	Description of the catalog.	Small, 67.0	67.0
effectiveEndDate	String	Date and time from when the catalog isn't available to the end users.	Small, 67.0	67.0
effectiveStartDate	String	Date and time from when the catalog is available to the end users.	Small, 67.0	67.0
id	String	ID of the catalog.	Small, 67.0	67.0
name	String	Name of the catalog.	Small, 67.0	67.0
numberOfCategories	Integer	Number of categories in the catalog.	Small, 67.0	67.0
status	String	Status of the catalog.	Small, 67.0	67.0

Category Details

Output representation of the details of a category.

JSON example


```
{
  "categories": [
    {
      "catalogId": "0ZSSG00000187504AQ",
      "childCategories": [],
      "customFields": {},
      "eligiblePromotions": [],
      "id": "0ZGSG000001DJtv4AG",
      "name": "Accessories",
      "qualificationContext": {
        "isQualified": true
      }
    }
  ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
catalogId	String	ID of the catalog the requested category belong to.	Small, 67.0	67.0
customFields	Map<String, Object>	Details of the custom fields associated with a catalog.	Small, 67.0	67.0
description	String	Description of the category.	Small, 67.0	67.0
eligiblePromotions	Promotion Output[]	List of eligible promotions for the product.	Small, 67.0	67.0
hasSubCategories	Boolean	Indicates whether the subcategories are available (true) or not (false).	Small, 67.0	67.0

Property Name	Type	Description	Filter Group and Version	Available Version
`id`	String	ID of the category.	Small, 67.0	67.0
`isNavigational`	Boolean	Indicates whether the category node is navigational (`true`) or not (`false`).	Small, 67.0	67.0
`name`	String	Name of the category.	Small, 67.0	67.0
`parentCategoryId`	String	ID of the parent category.	Small, 67.0	67.0
`qualificationContext`	Qualification Context Output[]	Context details of a user, which are used for qualification rules.	Small, 67.0	67.0
`sortOrder`	Integer	Display order of the product category relative to the siblings with the same parent category.	Small, 67.0	67.0

Bulk Product Details

Output representation of the response that contains the details of multiple products.

JSON example


```
{
  "apiStatus": {
    "messages": [],
    "statusCode": "FetchedDetailsSuccessfully"
  },
  "contextId": "c68c7c7e85f3ea5b0e7bcfefc0f2dba9bbd24bfe2f4240ca589af50e473e2242",
  "correlationId": "de9a674c-1807-438c-ac78-2c96f4655325",
  "result": [
    {
      "additionalFields": [],
      "attributeCategories": [],
      "attributes": [],
      "catalogs": [],
      "childProducts": [],
      "id": "01txx0000006ivJAAQ",
      "isActive": true,
      "isAssetizable": true,
      "isSoldOnlyWithOtherProds": false,
      "name": "iPhone12",
      "nodeType": "simpleProduct",
      "prices": [
        {
          "currencyIsoCode": "USD",
          "isDefault": false,
          "isSelected": true,
          "price": 100,
          "priceBookEntryId": "01uxx0000008zUkAAI",
          "priceBookId": "01sxx0000005qxxAAA",
          "pricingModel": {
            "id": "0jPxx000000009hEAA",
```

```
        "name": "OneTime",
        "pricingModelType": "OneTime"
      }
    },
    {
      "currencyIsoCode": "USD",
      "isDefault": true,
      "isSelected": false,
      "price": 15,
      "priceBookEntryId": "01uxx0000008zUmAAI",
      "priceBookId": "01sxx0000005qxxAAA",
      "pricingModel": {
        "frequency": "Months",
        "id": "0jPxx000000009iEAA",
        "name": "Monthly",
        "occurrence": 1,
        "pricingModelType": "TermDefined"
      }
    }
  ],
  "productClassification": {},
  "productCode": "iPhone12",
  "productComponentGroups": [],
  "productSellingModelOptions": [
    {
      "id": "0iOxx00000000EfEAI",
      "productId": "01txx0000006ivJAAQ",
      "productSellingModel": {
        "id": "0jPxx000000009iEAA",
        "name": "Monthly",
        "pricingTerm": 1,
        "pricingTermUnit": "Months",
        "sellingModelType": "TermDefined",
        "status": "Active"
      },
      "productSellingModelId": "0jPxx000000009iEAA"
    },
    {
      "id": "0iOxx00000000EgEAI",
      "productId": "01txx0000006ivJAAQ",
      "productSellingModel": {
        "id": "0jPxx000000009hEAA",
        "name": "OneTime",
        "sellingModelType": "OneTime",
        "status": "Active"
      },
      "productSellingModelId": "0jPxx000000009hEAA"
    }
  ],
  "productSpecificationType": {
    "name": "ProdSpecRecType1",
    "productSpecificationRecordType": {}
  },
  "qualificationContext": {
```

```
      "isQualified": true
    }
  },
  {
    "additionalFields": [],
    "attributeCategories": [],
    "attributes": [],
    "catalogs": [],
    "childProducts": [],
    "id": "01txx0000006ivLAAQ",
    "isActive": true,
    "isAssetizable": true,
    "isSoldOnlyWithOtherProds": false,
    "name": "iPhone13",
    "nodeType": "simpleProduct",
    "prices": [
      {
        "currencyIsoCode": "USD",
        "isDefault": true,
        "isSelected": false,
        "price": 1520,
        "priceBookEntryId": "01uxx0000008zUpAAI",
        "priceBookId": "01sxx0000005qxxAAA",
        "pricingModel": {
          "id": "0jPxx000000009hEAA",
          "name": "OneTime",
          "pricingModelType": "OneTime"
        }
      },
      {
        "currencyIsoCode": "USD",
        "isDefault": false,
        "isSelected": false,
        "price": 152,
        "priceBookEntryId": "01uxx0000008zUqAAI",
        "priceBookId": "01sxx0000005qxxAAA",
        "pricingModel": {
          "frequency": "Months",
          "id": "0jPxx000000009iEAA",
          "name": "Monthly",
          "occurrence": 1,
          "pricingModelType": "TermDefined"
        }
      }
    ],
    "productClassification": {},
    "productCode": "iPhone13",
    "productComponentGroups": [],
    "productSellingModelOptions": [
      {
        "id": "0iOxx00000000EbEAI",
        "productId": "01txx0000006ivLAAQ",
        "productSellingModel": {
          "id": "0jPxx000000009iEAA",
```

```
            "name": "Monthly",
            "pricingTerm": 1,
            "pricingTermUnit": "Months",
            "sellingModelType": "TermDefined",
            "status": "Active"
          },
          "productSellingModelId": "0jPxx000000009iEAA"
        },
        {
          "id": "0iOxx00000000EeEAI",
          "productId": "01txx0000006ivLAAQ",
          "productSellingModel": {
            "id": "0jPxx000000009hEAA",
            "name": "OneTime",
            "sellingModelType": "OneTime",
            "status": "Active"
          },
          "productSellingModelId": "0jPxx000000009hEAA"
        }
      ],
      "productSpecificationType": {
        "name": "ProdSpecRecType1",
        "productSpecificationRecordType": {}
      },
      "qualificationContext": {
        "isQualified": true
      }
    }
  ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`apiStatus`	API Status[]	Status of the API request.	Small, 61.0	61.0
`contextId`	String	ID of the context.	Small, 61.0	61.0
`correlationId`	String	Unique identifier of the request.	Small, 61.0	61.0
`result`	Any response body	Result that contains the details of products.	Small, 61.0	61.0
`userContext`	User Context[]	User context details. For example, account ID or contact ID.	Small, 61.0	61.0

CPQ Base Details

Output representation of the catalog, category, or product details based on the request.

JSON example

This example shows the sample catalog details.


```
{
  "apiStatus": {
    "messages": [
```

```
    ],
    "statusCode": "FetchedDetailsSuccessfully"
  },
  "correlationId": "32595ed6-1922-41f7-9c2c-373c677a7d62",
  "result": {
    "catalogCode": "Mobiles",
    "catalogType": "Sales",
    "description": "Catalog for mobile phones",
    "effectiveEndDate": "2028-04-01T19:00Z",
    "effectiveStartDate": "2024-04-01T19:00Z",
    "id": "0ZSxx0000000001GAA",
    "name": "Mobiles",
    "numberOfCategories": 3
  }
}
```

This example shows the sample category details.

```
{
  "apiStatus": {
    "messages": [

    ],
    "statusCode": "FetchedDetailsSuccessfully"
  },
  "correlationId": "f9fb90de-36aa-44a1-9961-a9ef5fc0cad8",
  "result": {
    "catalogId": "0ZSxx0000000001GAA",
    "childCategories": [

    ],
    "description": "Category for Samsung phones",
    "id": "0ZGxx000000004rGAA",
    "name": "Samsung",
    "isNavigational": true,
    "sortOrder": 2
  }
}
```

This example shows the sample product details.

```
{
  "apiStatus": {
    "messages": [

    ],
    "statusCode": "FetchedDetailsSuccessfully"
  },
  "correlationId": "822a8941-7412-4883-bb80-e488f908471e",
  "result": {
    "additionalFields": {
        "CustomField1__c": "TextValue",
        "CustomField2__c": "10",
        "StandardField1": "false"
```

```
    },
    "attributeCategories": [

    ],
    "attributes": [

    ],
    "catalogs": [

    ],
    "childProducts": [

    ],
    "id": "01txx0000006i2WAAQ",
    "isActive": true,
    "isAssetizable": true,
    "isSoldOnlyWithOtherProds": false,
    "name": "iPhone15",
    "nodeType": "simpleProduct",
    "prices": [

    ],
    "productClassification": {

    },
    "productCode": "iPhone15",
    "productComponentGroups": [

    ],
    "productSellingModelOptions": [
      {
        "id": "0iOxx0000000003EAA",
        "productId": "01txx0000006i2WAAQ",
        "productSellingModel": {
          "id": "0jPxx0000000001EAA",
          "name": "OneTime",
          "sellingModelType": "OneTime",
          "status": "Active"
        },
        "productSellingModelId": "0jPxx0000000001EAA"
      },
      {
        "id": "0iOxx0000000004EAA",
        "productId": "01txx0000006i2WAAQ",
        "productSellingModel": {
          "id": "0jPxx0000000002EAA",
          "name": "Monthly",
          "pricingTerm": 1,
          "pricingTermUnit": "Months",
          "sellingModelType": "TermDefined",
          "status": "Active"
        },
        "productSellingModelId": "0jPxx0000000002EAA"
      }
```

```
    ],
    "productSpecificationType": {
      "name": "ProdSpecRecType1",
      "productSpecificationRecordType": {

      }
    }
  }
}
```

This example shows a sample of category details with eligible promotions.

```
{
  "apiStatus": {
    "messages": [],
    "statusCode": "FetchedDetailsSuccessfully"
  },
  "correlationId": "f9fb90de-36aa-44a1-9961-a9ef5fc0cad8",
  "result": {
    "catalogId": "0ZSxx0000000001GAA",
    "childCategories": [],
    "description": "Category for Samsung phones",
    "id": "0ZGxx000000004rGAA",
    "name": "Samsung",
    "isNavigational": true,
    "sortOrder": 2,
    "eligiblePromotions": [
      {
        "id": "0ZPxx0000000001AAA",
        "name": "Summer Sale 2025",
        "displayName": "Summer Electronics Sale",
        "description": "Get 20% off on all Samsung devices",
        "priority": 100,
        "startDateTime": "2025-06-01T00:00:00Z",
        "endDateTime": "2025-08-31T23:59:59Z",
        "isAutomatic": true,
        "isCategoryPromo": true,
        "isProductPromo": false,
        "couponCode": null,
        "termsAndConditions": "Valid on all Samsung products. Cannot be combined with
other offers."
      },
      {
        "id": "0ZPxx0000000002AAA",
        "name": "Electronics Bundle Deal",
        "displayName": "Bundle & Save",
        "description": "Save 15% when you buy 2 or more items",
        "priority": 90,
        "startDateTime": "2025-05-15T00:00:00Z",
        "endDateTime": "2025-12-31T23:59:59Z",
        "isAutomatic": false,
        "isCategoryPromo": false,
        "isProductPromo": true,
        "couponCode": "BUNDLE15",
        "termsAndConditions": "Minimum 2 items required."
```

```
      }
    ]
  }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`apiStatus`	API Status	Status of the API request.	Small, 60.0	60.0
`contextId`	String	ID of the context.	Small, 60.0	60.0
`correlationId`	String	Unique identifier of the request.	Small, 60.0	60.0
`result`	Any response body	Result that contains the details of catalogs, categories, or products as per the requested resource.	Small, 60.0	60.0
`userContext`	User Context	User context details. For example, account ID or contact ID.	Small, 60.0	60.0

CPQ Base List

Output representation of the list of catalogs, categories, or products based on the request.

JSON example

This example shows a sample catalog list.

```
{
  "apiStatus": {
    "messages": [

    ],
    "statusCode": "FetchedDetailsSuccessfully"
  },
  "correlationId": "9f417514-9587-4063-9e48-18a2cf2477c0",
  "limit": 10,
  "offSet": 0,
  "query": {

  },
  "result": [
    {
      "catalogCode": "Mobiles",
      "catalogType": "Sales",
      "description": "Catalog for mobile phones",
      "effectiveEndDate": "2028-04-01T19:00Z",
      "effectiveStartDate": "2024-04-01T19:00Z",
      "id": "0ZSxx0000000001GAA",
      "name": "Mobiles",
      "numberOfCategories": 3
    }
  ],
```

```
  "total": 1
}
```

This example shows a sample category list.

```
{
  "apiStatus": {
    "messages": [

    ],
    "statusCode": "FetchedDetailsSuccessfully"
  },
  "correlationId": "3f2a8f45-e7d2-42ec-bc4c-b981d750e912",
  "query": {

  },
  "result": [
    {
      "catalogId": "0ZSxx0000000001GAA",
      "childCategories": [

      ],
      "description": "Category for Apple phones and iPads",
      "id": "0ZGxx0000000001GAA",
      "name": "Apple",
      "sortOrder": 1,
      "isNavigational": true
    },
    {
      "catalogId": "0ZSxx0000000001GAA",
      "childCategories": [

      ],
      "description": "Category for Samsung phones",
      "id": "0ZGxx000000004rGAA",
      "name": "Samsung",
      "sortOrder": 2,
      "isNavigational": true
    },
    {
      "catalogId": "0ZSxx0000000001GAA",
      "childCategories": [

      ],
      "description": "Category for Android phones",
      "id": "0ZGxx000000006TGAQ",
      "name": "Android",
      "isNavigational": true
    }
  ],
  "total": 3
}
```

This example shows a sample product list.

```
{
  "apiStatus": {
    "messages": [],
    "statusCode": "FetchedDetailsSuccessfully"
  },
  "contextId": "f36f8e73f1fc338cc4e93c61613cba07a6a0129941d97e5dd6e52a2885776ce4",
  "correlationId": "eeaa1db2-f371-4227-a886-c77e2f66ce1d",
  "cursor": "MTAwMDAwMDAwNg==",
  "query": {},
  "result": [
    {
      "additionalFields": {
        "DecompositionScope": "OrderLineItem",
        "ProductCode": "LPB001",
        "CanRamp": false
      },
      "attributeCategories": [],
      "catalogs": [
        {
          "customFields": {},
          "id": "0ZSDU0000002Og74AE",
          "name": "Service Catalog",
          "numberOfCategories": 5
        }
      ],
      "categories": [
        {
          "catalogId": "0ZSDU0000002Og74AE",
          "childCategories": [],
          "customFields": {},
          "hasSubCategories": false,
          "id": "0ZGDU0000002P0H4AU",
          "name": "Cloud Services",
          "qualificationContext": {
            "isQualified": true
          }
        }
      ],
      "childProducts": [],
      "configureDuringSale": "Allowed",
      "description": "The laptop pro bundle includes a Laptop, mouse, warranty for 2
years, premium support and printer bundle",
      "displayUrl":
"https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcSCrGjPR1fvJqg4yP3RMyqjI0H9eL6tk1fvzw&amp;usqp=CAU",

      "id": "01tDU000000ExkZYAS",
      "isActive": true,
      "isAssetizable": true,
      "isSoldOnlyWithOtherProds": false,
      "name": "Laptop Pro Bundle",
      "nodeType": "bundleProduct",
      "prices": [],
      "productClassification": {},
```

```
      "productCode": "LPB001",
      "productComponentGroups": [],
      "productSellingModelOptions": [
        {
          "id": "0iODU0000002TBN2A2",
          "productId": "01tDU000000ExkZYAS",
          "productSellingModel": {
            "id": "0jPDU0000002OTv2AM",
            "name": "One Time",
            "sellingModelType": "OneTime",
            "status": "Active"
          },
          "productSellingModelId": "0jPDU0000002OTv2AM"
        }
      ],
      "productSpecificationType": {
        "name": "Commercial",
        "productSpecificationRecordType": {}
      },
      "productType": "Bundle",
      "qualificationContext": {
        "isQualified": true
      }
    }
  ]
}
```


This example shows a sample of the list of products retrieved based on the `Laptop` search term.


```
{
  "apiStatus": {
    "messages": [],
    "statusCode": "FetchedDetailsSuccessfully"
  },
  "correlationId": "d9d8f898-19f5-464a-ba2b-6a070783f6c4",
  "cursor": "MTAwMDAwMDAwMw==",
  "facets": [
    {
      "attributeType": "ProductStandard",
      "displayName": "Product Type",
      "displayRank": 2,
      "nameOrId": "Type",
      "values": [
        {
          "displayName": "Bundle",
          "nameOrId": "Bundle"
        }
      ]
    },
    {
      "attributeType": "ProductDynamicAttribute",
      "displayName": "Display",
      "displayRank": 3,
      "nameOrId": "0tjDU0000003K5BYAU",
      "values": [
```

```
        {
          "displayName": "1080p Built-in Display",
          "nameOrId": "1080p Built-in Display"
        },
        {
          "displayName": "2k Built-in Display",
          "nameOrId": "2k Built-in Display"
        },
        {
          "displayName": "4k Built-in Display",
          "nameOrId": "4k Built-in Display"
        }
      ]
    }
  ],
  "limit": 10,
  "query": {},
  "result": [
    {
      "additionalFields": {},
      "attributeCategories": [],
      "catalogs": [],
      "categories": [
        {
          "catalogId": "0ZSDU0000002Og64AE",
          "childCategories": [],
          "customFields": {},
          "hasSubCategories": false,
          "id": "0ZGDU0000002P0A4AU",
          "name": "Laptops",
          "qualificationContext": {
            "isQualified": true
          }
        }
      ],
      "configureDuringSale": "Allowed",
      "description": "Battery- or AC-powered personal computer (PC) smaller than a
briefcase",
      "displayUrl":
"https://media.istockphoto.com/id/1023428598/photo/3d-illustration-laptop-isolated-on-white-background-laptop-with-empty-space-screen-laptop-at.jpg?s=612x612&w=0&k=20&c=sK6e51fGSjGoyS08rbb1CxgfQl2xU8=",

      "id": "01tDU000000ExkWYAS",
      "isActive": true,
      "isAssetizable": true,
      "isSoldOnlyWithOtherProds": false,
      "name": "Laptop",
      "nodeType": "simpleProduct",
      "prices": [],
      "productClassification": {
        "id": "11BDU0000002TCD2A2"
      },
      "productCode": "LP001",
      "productComponentGroups": [],
      "productSellingModelOptions": [
```

```
        {
          "id": "0iODU0000002TBF2A2",
          "productId": "01tDU000000ExkWYAS",
          "productSellingModel": {
            "id": "0jPDU0000002OTv2AM",
            "name": "One Time",
            "sellingModelType": "OneTime",
            "status": "Active"
          },
          "productSellingModelId": "0jPDU0000002OTv2AM"
        }
      ],
      "productSpecificationType": {
        "name": "Commercial",
        "productSpecificationRecordType": {}
      },
      "qualificationContext": {
        "isQualified": true
      }
    },
    {
      "additionalFields": {},
      "attributeCategories": [],
      "catalogs": [],
      "categories": [
        {
          "catalogId": "0ZSDU0000002Og64AE",
          "childCategories": [],
          "customFields": {},
          "hasSubCategories": false,
          "id": "0ZGDU0000002P0A4AU",
          "name": "Laptops",
          "qualificationContext": {
            "isQualified": true
          }
        }
      ],
      "configureDuringSale": "Allowed",
      "description": "The laptop basic bundle includes a Laptop, mouse, and warranty
for 1 year.",
      "displayUrl":
"https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcTbf49JG4zZogCmZMJuXU38qOkR9X36MN4bSw&amp;usqp=CAU",

      "id": "01tDU000000ExkXYAS",
      "isActive": true,
      "isAssetizable": true,
      "isSoldOnlyWithOtherProds": false,
      "name": "Laptop Basic Bundle",
      "nodeType": "bundleProduct",
      "prices": [],
      "productClassification": {},
      "productCode": "LB001",
      "productComponentGroups": [],
      "productSellingModelOptions": [
```

```
        {
          "id": "0iODU0000002TBD2A2",
          "productId": "01tDU000000ExkXYAS",
          "productSellingModel": {
            "id": "0jPDU0000002OTv2AM",
            "name": "One Time",
            "sellingModelType": "OneTime",
            "status": "Active"
          },
          "productSellingModelId": "0jPDU0000002OTv2AM"
        }
      ],
      "productSpecificationType": {
        "name": "Commercial",
        "productSpecificationRecordType": {}
      },
      "productType": "Bundle",
      "qualificationContext": {
        "isQualified": true
      }
    },
    {
      "additionalFields": {},
      "attributeCategories": [],
      "catalogs": [],
      "categories": [
        {
          "catalogId": "0ZSDU0000002Og64AE",
          "childCategories": [],
          "customFields": {},
          "hasSubCategories": false,
          "id": "0ZGDU0000002P0A4AU",
          "name": "Laptops",
          "qualificationContext": {
            "isQualified": true
          }
        }
      ],
      "configureDuringSale": "Allowed",
      "description": "The laptop pro bundle includes a Laptop, mouse, warranty for 2
years, premium support and printer bundle",
      "displayUrl":
"https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcSCrGjPR1fvJqg4yP3RMyqjI0H9eL6tk1fvzw&amp;usqp=CAU",

      "id": "01tDU000000ExkZYAS",
      "isActive": true,
      "isAssetizable": true,
      "isSoldOnlyWithOtherProds": false,
      "name": "Laptop Pro Bundle",
      "nodeType": "bundleProduct",
      "prices": [],
      "productClassification": {},
      "productCode": "LPB001",
      "productComponentGroups": [],
```

```
      "productSellingModelOptions": [
        {
          "id": "0iODU0000002TBN2A2",
          "productId": "01tDU000000ExkZYAS",
          "productSellingModel": {
            "id": "0jPDU0000002OTv2AM",
            "name": "One Time",
            "sellingModelType": "OneTime",
            "status": "Active"
          },
          "productSellingModelId": "0jPDU0000002OTv2AM"
        }
      ],
      "productSpecificationType": {
        "name": "Commercial",
        "productSpecificationRecordType": {}
      },
      "productType": "Bundle",
      "qualificationContext": {
        "isQualified": true
      }
    },
    {
      "additionalFields": {},
      "attributeCategories": [],
      "catalogs": [],
      "categories": [
        {
          "catalogId": "0ZSDU0000002Og64AE",
          "childCategories": [],
          "customFields": {},
          "hasSubCategories": false,
          "id": "0ZGDU0000002P0A4AU",
          "name": "Laptops",
          "qualificationContext": {
            "isQualified": true
          }
        }
      ],
      "configureDuringSale": "Allowed",
      "description": "Laptop, Laptop Bag, Laptop stand, Mouse, Keyboard, USB-C
Hub,External Hard Drive, Noise Cancelling Headphones, office 365",
      "displayUrl": "https://m.media-amazon.com/images/I/613Fno-NLYL._AC_SL1000_.jpg",
      "id": "01tDU000000ExlAYAS",
      "isActive": true,
      "isAssetizable": true,
      "isSoldOnlyWithOtherProds": false,
      "name": "Laptop Productivity Bundle",
      "nodeType": "bundleProduct",
      "prices": [],
      "productClassification": {},
      "productCode": "LPB001",
      "productComponentGroups": [],
      "productSellingModelOptions": [
```

```
        {
          "id": "0iODU0000002TBq2AM",
          "productId": "01tDU000000ExlAYAS",
          "productSellingModel": {
            "id": "0jPDU0000002OTv2AM",
            "name": "One Time",
            "sellingModelType": "OneTime",
            "status": "Active"
          },
          "productSellingModelId": "0jPDU0000002OTv2AM"
        }
      ],
      "productSpecificationType": {
        "name": "Commercial",
        "productSpecificationRecordType": {}
      },
      "productType": "Bundle",
      "qualificationContext": {
        "isQualified": true
      }
    }
  ],
  "total": 4
}
```


This example shows a sample of the results of a qualification procedure that's executed on a list of product IDs.


```
{
  "apiStatus": {
    "messages": [

    ],
    "statusCode": "FetchedDetailsSuccessfully"
  },
  "contextId": "e055bb18-d4e8-41c3-881e-0132b9561708",
  "correlationId": "c280c1b0-fd3f-4eac-9b08-075bdf1cbefc",
  "query": {

  },
  "result": [
    {
      "productId": "01txx0000006i7PAAQ",
      "qualificationContext": {
        "isQualified": true
      }
    },
    {
      "productId": "01txx0000006i7QAAQ",
      "qualificationContext": {
        "isQualified": true
      }
    },
    {
      "productId": "01txx0000006i7IAAQ",
      "qualificationContext": {
```

```
        "isQualified": true
      }
    }
  ]
}
```

This example shows a list of products with eligible promotions.

```
{
  "apiStatus": {
    "messages": [],
    "statusCode": "FetchedDetailsSuccessfully"
  },
  "correlationId": "f9fb90de-36aa-44a1-9961-a9ef5fc0cad8",
  "result": {
    "catalogId": "0ZSxx0000000001GAA",
    "childCategories": [],
    "description": "Category for Samsung phones",
    "id": "0ZGxx000000004rGAA",
    "name": "Samsung",
    "isNavigational": true,
    "sortOrder": 2,
    "eligiblePromotions": [
      {
        "id": "0ZPxx0000000001AAA",
        "name": "Summer_Sale_2025",
        "displayName": "Summer Electronics Sale",
        "description": "Get 20% off on all Samsung devices",
        "priority": 100,
        "startDateTime": "2025-06-01T00:00:00Z",
        "endDateTime": "2025-08-31T23:59:59Z",
        "isAutomatic": true,
        "isCategoryPromo": true,
        "isProductPromo": false,
        "couponCode": null,
        "termsAndConditions": "Valid on all Samsung products. Cannot be combined with
other offers."
      },
      {
        "id": "0ZPxx0000000002AAA",
        "name": "Electronics_Bundle",
        "displayName": "Bundle & Save",
        "description": "Save 15% when you buy 2 or more items",
        "priority": 90,
        "startDateTime": "2025-05-15T00:00:00Z",
        "endDateTime": "2025-12-31T23:59:59Z",
        "isAutomatic": false,
        "isCategoryPromo": false,
        "isProductPromo": true,
        "couponCode": "BUNDLE15",
        "termsAndConditions": "Minimum 2 items required."
      }
    ]
  }
}
```

This example shows a sample response with visibility rule details.


```
{
  "apiStatus": {
    "messages": [],
    "statusCode": "FetchedDetailsSuccessfully"
  },
  "contextId": "0000000r25tp21g0025177564847265 23db13b6a0ce245dc98062eb32a583a69",
  "correlationId": "81e072e2-e1ae-4321-9b73-d591f75854ec",
  "cursor": "MTAwMDAwMDAwNg==",
  "facets": [],
  "limit": 12,
  "query": {},
  "result": [
    {
      "additionalFields": {},
      "attributeCategories": [],
      "catalogs": [
        {
          "customFields": {},
          "id": "0ZSVW000000AhdC4AS",
          "name": "QuantumBit Hardware",
          "numberOfCategories": 11
        }
      ],
      "categories": [
        {
          "catalogId": "0ZSVW000000AhdC4AS",
          "childCategories": [],
          "customFields": {},
          "eligiblePromotions": [],
          "id": "0ZGVW000000IUEC4A4",
          "name": "Memory"
        }
      ],
      "childVariationIds": [],
      "configurationRules": [
        {
          "details": [
            {
              "message": "32GB RDIMM disables 128GB LRDIMM"
            }
          ],
          "type": "disable"
        }
      ],
      "configureDuringSale": "Allowed",
      "description": "128GB RDIMM",
      "displayUrl": "/resource/ram_32GB_RDIMM",
      "eligiblePromotions": [],
      "id": "01tVW000003l7v5YAA",
      "isActive": true,
      "isAssetizable": true,
      "isSoldOnlyWithOtherProds": false,
      "name": "128GB LRDIMM",
```

```
        "nodeType": "simpleProduct",
        "prices": [
          {
            "currencyIsoCode": "USD",
            "isDefault": true,
            "isDerived": false,
            "isSelected": false,
            "price": 0,
            "priceBookEntryId": "01uVW000000jzfJYAQ",
            "priceBookId": "01sVW0000024PZlYAM",
            "pricingModel": {
              "id": "0jPVW0000001fh42AA",
              "name": "One-Time",
              "pricingModelType": "OneTime"
            }
          }
        ],
        "productClass": "Simple",
        "productClassification": {},
        "productCode": "QB-MEM 128GB",
        "productComponentGroups": [],
        "productSellingModelOptions": [
          {
            "id": "0iOVW00000049xS2AQ",
            "isDefault": true,
            "productId": "01tVW000003l7v5YAA",
            "productSellingModel": {
              "doesAutoRenewByDefault": false,
              "id": "0jPVW0000001fh42AA",
              "name": "One-Time",
              "sellingModelType": "OneTime",
              "status": "Active"
            },
            "productSellingModelId": "0jPVW0000001fh42AA"
          }
        ],
        "productUnitOfMeasures": []
      }
    ],
    "total": 1
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
apiStatus	API Status	Status of the API request.	Small, 60.0	60.0
contextId	String	ID of the context.	Small, 60.0	60.0
correlationId	String	Unique ID of the request.	Small, 60.0	60.0
cursor	String	Unique ID to represent the position of each product in the dataset.	Small, 60.0	60.0

Property Name	Type	Description	Filter Group and Version	Available Version
facets	Search Products Facet	Details of the faceted search.	Small, 63.0	63.0
limit	Integer	Number of items fetched in the response.	Small, 60.0	60.0
offSet	Integer	Offset size from which the item count is fetched.	Small, 60.0	60.0
query	Map<String, Object>	Query that was used for the search request.	Small, 60.0	60.0
result	Any response body	Result that contains the list of catalogs, categories, or products as per the requested resource.	Small, 60.0	60.0
total	Integer	Number of fetched records.	Small, 60.0	60.0
userContext	User Context	User context details. For example, account ID or contact ID.	Small, 60.0	60.0

CPQ Message

Output representation of the API messages.

Property Name	Type	Description	Filter Group and Version	Available Version
code	String	Code specifying the type of message. Valid value is `CartValidationError`.	Small, 60.0	60.0
detail	String	Required details other than the message text.	Small, 60.0	60.0
message	String	Text of the API message.	Small, 60.0	60.0
severity	String	Severity of the API message. Valid values are: • `Error` • `Info` • `Warning`	Small, 60.0	60.0

Facet Value

Output representation of the facet values found in the search result.

JSON example

```
"values": [
  {
```

```
          "displayName": "1080p Built-in Display",
          "nameOrId": "1080p Built-in Display"
        },
        {
          "displayName": "2k Built-in Display",
          "nameOrId": "2k Built-in Display"
        },
        {
          "displayName": "4k Built-in Display",
          "nameOrId": "4k Built-in Display"
        }
      ]
```

Property Name	Type	Description	Filter Group and Version	Available Version
displayName	String	Display name of the facet value.	Small, 63.0	63.0
nameOrId	String	ID or the internal name of the facet value.	Small, 63.0	63.0

Guided Selection

Output representation of the details of a guided selection.

JSON example

```
{
  "apiStatus": {
    "messages": [],
    "statusCode": "FETCHED_DETAILS_SUCCESSFULLY"
  },
  "correlationId": "corrId",
  "cursor": "MTAwMDAwMDAwNg==",
  "searchTerms": [
    {
      "term": "IPhone"
    },
    {
      "term": "4GB"
    },
    {
      "term": "64GB"
    }
  ],
  "result": [
    {
      "additionalFields": {
        "CustomField1__c": "TextValue",
        "CustomField2__c": "10",
        "StandardField1": "false"
      },
      "description": "IPhone-13",
      "id": "01txx0000006kYwAAI",
      "name": "Sample product 1",
```

```
      "prices": [
        {
          "price": 150,
          "priceBookEntryId": "12Axx0000004DF7EAM",
          "priceBookId": "01sxx0000005puLAAQ",
          "pricingModel": {
            "frequency": "Monthly",
            "id": "12Bxx000000CiCDEA0",
            "name": "IPhone-13",
            "occurrence": 6,
            "pricingModelType": "Recurring"
          }
        },
        {
          "price": 400,
          "priceBookEntryId": "12Axx0000004DGjEAM",
          "priceBookId": "01sxx0000005puLAAQ",
          "pricingModel": {
            "id": "12Bxx000000CiCCEA0",
            "name": "IPhone-13",
            "pricingModelType": "OneTime"
          }
        }
      ],
      "qualificationContext": {
        "isQualified": true
      }
    }
  ]
}
```


This example shows a sample response with details of eligible promotions.


```
{
  "apiStatus": {
    "messages": [],
    "statusCode": "FETCHED_DETAILS_SUCCESSFULLY"
  },
  "correlationId": "corrId",
  "cursor": "MTAwMDAwMDAwNg==",
  "searchTerms": [
    {
      "term": "IPhone",
      "tags": [
        "deviceType",
        "mobile"
      ]
    },
    {
      "term": "4GB",
      "tags": [
        "RAM"
      ]
    },
    {
```

```
      "term": "64GB",
      "tags": [
        "Storage"
      ]
    }
  ],
  "result": [
    {
      "additionalFields": {
        "CustomField1__c": "TextValue",
        "CustomField2__c": "10",
        "StandardField1": "false"
      },
      "description": "IPhone-13",
      "id": "01txx0000006kYwAAI",
      "name": "Sample product 1",
      "prices": [
        {
          "price": 150,
          "priceBookEntryId": "12Axx0000004DF7EAM",
          "priceBookId": "01sxx0000005puLAAQ",
          "pricingModel": {
            "frequency": "Monthly",
            "id": "12Bxx000000CiCDEA0",
            "name": "IPhone-13",
            "occurrence": 6,
            "pricingModelType": "Recurring"
          }
        },
        {
          "price": 400,
          "priceBookEntryId": "12Axx0000004DGjEAM",
          "priceBookId": "01sxx0000005puLAAQ",
          "pricingModel": {
            "id": "12Bxx000000CiCCEA0",
            "name": "IPhone-13",
            "pricingModelType": "OneTime"
          }
        }
      ],
      "qualificationContext": {
        "isQualified": true
      },
      "eligiblePromotions": [
        {
          "id": "0ZPxx0000000001AAA",
          "name": "IPhone_Promotion_2025",
          "displayName": "iPhone Special Offer",
          "description": "Get 15% off on iPhone 13",
          "priority": 100,
          "startDateTime": "2025-03-01T00:00:00Z",
          "endDateTime": "2025-03-31T23:59:59Z",
          "isAutomatic": true,
          "isCategoryPromo": false,
```

```
          "isProductPromo": true,
          "couponCode": null,
          "termsAndConditions": "Valid on iPhone 13 models only."
        },
        {
          "id": "0ZPxx0000000002AAA",
          "name": "Memory_Upgrade_Deal",
          "displayName": "Free Memory Upgrade",
          "description": "Upgrade to 128GB for the price of 64GB",
          "priority": 90,
          "startDateTime": "2025-02-01T00:00:00Z",
          "endDateTime": "2025-12-31T23:59:59Z",
          "isAutomatic": false,
          "isCategoryPromo": false,
          "isProductPromo": true,
          "couponCode": "MEMORY128",
          "termsAndConditions": "Applies to select iPhone models."
        }
      ]
    }
  ]
}
```

This example shows a sample request to run visibility rules.

```
{
  "apiStatus": {
    "messages": [],
    "statusCode": "FetchedDetailsSuccessfully"
  },
  "correlationId": "ef4f324d-b6d9-47ca-9e78-c6ec60ca5b23",
  "cursor": "MTAwMDAwMDAwNg==",
  "offSet": 1,
  "result": [
    {
      "additionalFields": {},
      "attributeCategories": [],
      "catalogs": [
        {
          "customFields": {},
          "id": "0ZSVW000000AhdC4AS",
          "name": "QuantumBit Hardware",
          "numberOfCategories": 11
        }
      ],
      "categories": [
        {
          "catalogId": "0ZSVW000000AhdC4AS",
          "childCategories": [],
          "customFields": {},
          "eligiblePromotions": [],
          "id": "0ZGVW000000IUEC4A4",
          "name": "Memory"
        }
      ],
```

```
"childProducts": [],
"childVariationIds": [],
"configurationRules": [
  {
    "details": [
      {
        "message": "32GB RDIMM disables 128GB LRDIMM"
      }
    ],
    "type": "disable"
  }
],
"configureDuringSale": "Allowed",
"description": "128GB RDIMM",
"displayUrl": "/resource/ram_32GB_RDIMM",
"eligiblePromotions": [],
"id": "01tVW000003l7v5YAA",
"isActive": true,
"isAssetizable": true,
"isSoldOnlyWithOtherProds": false,
"name": "128GB LRDIMM",
"nodeType": "simpleProduct",
"prices": [
  {
    "currencyIsoCode": "USD",
    "isDefault": true,
    "isDerived": false,
    "isSelected": false,
    "price": 0,
    "priceBookEntryId": "01uVW000000jzfJYAQ",
    "priceBookId": "01sVW0000024PZlYAM",
    "pricingModel": {
      "id": "0jPVW0000001fh42AA",
      "name": "One-Time",
      "pricingModelType": "OneTime"
    }
  }
],
"productClass": "Simple",
"productClassification": {},
"productCode": "QB-MEM 128GB",
"productComponentGroups": [],
"productSellingModelOptions": [
  {
    "id": "0iOVW00000049xS2AQ",
    "isDefault": true,
    "productId": "01tVW000003l7v5YAA",
    "productSellingModel": {
      "doesAutoRenewByDefault": false,
      "id": "0jPVW0000001fh42AA",
      "name": "One-Time",
      "sellingModelType": "OneTime",
      "status": "Active"
    },
```

```
          "productSellingModelId": "0jPVW0000001fh42AA"
        }
      ],
      "productUnitOfMeasures": []
    }
  ],
  "searchTerms": [
    {
      "tags": [],
      "term": "128GB LRDIMM"
    }
  ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
apiStatus	API Status	Status of the API request.	Small, 62.0	62.0
correlationId	String	Unique ID of the request.	Small, 62.0	62.0
cursor	String	Unique ID to represent the position of each product in the dataset.	Small, 62.0	62.0
result	Any response body	Result that contains the list of products as per the requested resource.	Small, 62.0	62.0
searchTerms	Guided Selection Search Term[]	Search terms for the guided selection.	Small, 62.0	62.0

Guided Selection Search Term

Output representation of the search term details for a guided selection.

JSON example


```
{
  "searchTerms": [
    {
      "term": "IPhone",
      "tags": [
        "deviceType",
        "mobile"
      ]
    },
    {
      "term": "4GB",
      "tags": [
        "RAM"
      ]
    },
    {
      "term": "64GB",
      "tags": [
```

```
          "Storage"
        ]
      }
    ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`tags`	String[]	Search term tags for the guided selection.	Small, 62.0	62.0
`term`	String	Search term value for the guided selection.	Small, 62.0	62.0

Pricing Model

Output representation of the details of the pricing model.

JSON example

```
{
  "pricingModel": {
    "id": "0jPSG000000Avcv2AC",
    "name": "One Time",
    "pricingModelType": "OneTime"
  }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`frequency`	String	Details about the frequency of recurrence of the pricing model.	Small, 67.0	67.0
`id`	String	ID of the pricing model.	Small, 67.0	67.0
`name`	String	Name of the pricing model.	Small, 67.0	67.0
`occurrence`	Integer	Details about the number of occurrences of the pricing model.	Small, 67.0	67.0
`pricingModelType`	String	Type of the pricing model.	Small, 67.0	67.0
`unitOfMeasure`	String	Unit of measure for the pricing model.	Small, 67.0	67.0

Product Classification

Output representation of the details of a product classification.

JSON example

```
{
  "productClassification": {
    "id": "11BSG00000UNijx2AD"
```

```
  }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
code	String	Code of the product classification.	Small, 67.0	67.0
id	String	ID of the product classification.	Small, 67.0	67.0
name	String	Name of the product classification.	Small, 67.0	67.0
status	String	Status of the product classification.	Small, 67.0	67.0

Product Configuration Rule Details

Output representation of the product configuration rule details.

JSON example

```
{
  "configurationRules": [
    {
      "details": [
        {
          "message": "recommend Mouse from monitor"
        }
      ],
      "type": "recommend"
    }
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
message	String	Message associated with the product configuration rule.	Small, 67.0	67.0

Product Configuration Rules

Output representation of the details of the product configuration rules.

JSON example

```
{
  "configurationRules": [
    {
      "details": [
        {
          "message": "recommend Mouse from monitor"
        }
      ],
      "type": "recommend"
```

```
      }
    ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`details`	Product Configuration Rule Details[]	Details of the product configuration rule.	Small, 67.0	67.0
`type`	String	Type of the product configuration rule.	Small, 67.0	67.0

Product Prices

Output representation of the details of the product prices.

JSON example


```
{
  "prices": [
    {
      "currencyIsoCode": "USD",
      "isDefault": true,
      "isDerived": false,
      "isSelected": false,
      "price": 7.99,
      "priceBookEntryId": "01uSG000004wTsEYAU",
      "priceBookId": "01sSG00000DQCjhYAH",
      "pricingModel": {
        "id": "0jPSG000000Avcv2AC",
        "name": "One Time",
        "pricingModelType": "OneTime"
      }
    }
  ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`currencyIsoCode`	String	Currency ISO code of the given price.	Small, 67.0	67.0
`effectiveFrom`	String	Date from when the given price is effective.	Small, 67.0	67.0
`effectiveTo`	String	Date until when the given price is effective.	Small, 67.0	67.0
`isDefault`	Boolean	Indicates whether the given price is default (`true`) or not (`false`).	Small, 67.0	67.0
`isDerived`	Boolean	Indicates whether the given price is derived (`true`) or not (`false`).	Small, 67.0	67.0

Property Name	Type	Description	Filter Group and Version	Available Version
`isSelected`	Boolean	Indicates whether the given price is selected (`true`) or not (`false`).	Small, 67.0	67.0
`price`	Double	Price of the product.	Small, 67.0	67.0
`priceBookEntryId`	String	Price book entry ID of the given price.	Small, 67.0	67.0
`priceBookId`	String	Price book ID of the given price.	Small, 67.0	67.0
`pricingModel`	Pricing Model[]	Pricing model of the given price.	Small, 67.0	67.0

Product Quantity

Output representation of the details of the quantity constraints and current quantity for a product in the product discovery context.

Property Name	Type	Description	Filter Group and Version	Available Version
`maxQuantity`	Double	Maximum quantity allowed for a product.	Small, 67.0	67.0
`minQuantity`	Double	Minimum quantity allowed for a product.	Small, 67.0	67.0
`quantity`	Double	Default or current quantity of the product.	Small, 67.0	67.0

Product Recommendation Errors

Output representation of the details of the errors encountered during the processing of the Product Recommendations API request.

Property Name	Type	Description	Filter Group and Version	Available Version
`errorCode`	String	Code that indicates the type of error.	Small, 66.0	66.0
`message`	String	Message stating the reason for error, if any.	Small, 66.0	66.0

Product Recommendations

Output representation of the fetched product recommendations.

JSON example

```
{
  "cursor": "MTAwMDAwMDAwNQ==",
  "errors": [],
  "products": [
    {
      "additionalFields": {},
      "catalogs": [
        {
          "customFields": {},
          "id": "0ZSSG00000187504AQ",
```

```
          "name": "Hardware Catalog",
          "numberOfCategories": 4
        }
      ],
      "categories": [
        {
          "catalogId": "0ZSSG000001875O4AQ",
          "childCategories": [],
          "customFields": {},
          "eligiblePromotions": [],
          "id": "0ZGSG000001DJtv4AG",
          "name": "Accessories",
          "qualificationContext": {
            "isQualified": true
          }
        }
      ],
      "configurationRules": [
        {
          "details": [
            {
              "message": "recommend Mouse from monitor"
            }
          ],
          "type": "recommend"
        }
      ],
      "configureDuringSale": "Allowed",
      "description": "HeyMicky Mouse compatible comes with choice of USB or Wireless
connectivity.",
      "displayUrl": "https://5.imimg.com/data5/IS/JP/MY-58678879/computer-mouse.jpg",
      "eligiblePromotions": [],
      "id": "01tSG00000BiywkYAB",
      "isActive": true,
      "isAssetizable": true,
      "isSoldOnlyWithOtherProds": false,
      "name": "Mouse",
      "nodeType": "simpleProduct",
      "prices": [
        {
          "currencyIsoCode": "USD",
          "isDefault": true,
          "isDerived": false,
          "isSelected": false,
          "price": 7.99,
          "priceBookEntryId": "01uSG000004wTsEYAU",
          "priceBookId": "01sSG00000DQCjhYAH",
          "pricingModel": {
            "id": "0jPSG000000Avcv2AC",
            "name": "One Time",
            "pricingModelType": "OneTime"
          }
        }
      ],
```

```
        "productClassification": {
          "id": "11BSG00000UNijx2AD"
        },
        "productCode": "MOU001",
        "productSellingModelOptions": [
          {
            "id": "0iOSG000000J64x2AC",
            "isDefault": true,
            "productId": "01tSG00000BiywkYAB",
            "productSellingModel": {
              "doesAutoRenewByDefault": false,
              "id": "0jPSG000000Avcv2AC",
              "name": "One Time",
              "sellingModelType": "OneTime",
              "status": "Active"
            },
            "productSellingModelId": "0jPSG000000Avcv2AC"
          }
        ],
        "productSpecificationType": {
          "name": "Commercial",
          "productSpecificationRecordType": {}
        },
        "qualificationContext": {
          "isQualified": true
        }
      }
    ],
    "success": true,
    "transactionContextId":
  "0000000p18dq18g002917732061529307c5e047346664b2f81ff3d4845acf2d0"
  }
```


Property Name	Type	Description	Filter Group and Version	Available Version
`cursor`	String	Unique ID to represent the position of each product in the data set.	Small, 67.0	67.0
`errors`	Product Recommendation Errors[]	List of errors encountered during the processing of the API request.	Small, 67.0	67.0
`products`	Recommended Products[]	List of recommended products.	Small, 67.0	67.0
`success`	Boolean	Indicates whether the API request is successful (`true`) or not (`false`).	Small, 67.0	67.0
`transactionContextId`	String	ID of the sales transaction context.	Small, 67.0	67.0

Product Selling Model Option

Output representation of the product selling model option component.

JSON example


```
{
  "productSellingModelOptions": [
    {
      "id": "0iOSG000000J64x2AC",
      "isDefault": true,
      "productId": "01tSG00000BiywkYAB",
      "productSellingModel": {
        "doesAutoRenewByDefault": false,
        "id": "0jPSG000000Avcv2AC",
        "name": "One Time",
        "sellingModelType": "OneTime",
        "status": "Active"
      },
      "productSellingModelId": "0jPSG000000Avcv2AC"
    }
  ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`id`	String	ID of the product selling model option.	Small, 67.0	67.0
`isDefault`	Boolean	Indicates whether this product selling model option is default (`true`) or not (`false`).	Small, 67.0	67.0
`productId`	String	ID of the product.	Small, 67.0	67.0
`productSellingModel`	Product Selling Model[]	Details of the product selling model.	Small, 67.0	67.0
`productSellingModelId`	String	ID of the product selling model.	Small, 67.0	67.0
`prorationPolicy`	Proration Policy[]	Details of the proration policy.	Small, 67.0	67.0

Product Selling Model

Product Selling Model Component output representation

JSON example


```
{
  "productSellingModel": {
    "doesAutoRenewByDefault": false,
    "id": "0jPSG000000Avcv2AC",
    "name": "One Time",
    "sellingModelType": "OneTime",
    "status": "Active"
  }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
doesAutoRenewByDefault	Boolean	Indicates whether the product is automatically renewed by default (true) or not (false).	Small, 67.0	67.0
id	String	ID of the product selling model.	Small, 67.0	67.0
name	String	Name of the product selling model.	Small, 67.0	67.0
pricingTerm	Integer	Pricing term of the product selling model.	Small, 67.0	67.0
pricingTermUnit	String	Pricing term unit of the product selling model.	Small, 67.0	67.0
sellingModelType	String	Selling model type associated with the product selling model.	Small, 67.0	67.0
status	String	Status of the product selling model.	Small, 67.0	67.0

Product Specification Record Type

Output representation of the details of the product specification record type.

Property Name	Type	Description	Filter Group and Version	Available Version
isCommercial	Boolean	Indicates whether this product specification record type is commercial (true) or not (false).	Small, 67.0	67.0

Product Specification Type

Output representation of the details of the product specification type.

Property Name	Type	Description	Filter Group and Version	Available Version
name	String	Name of the product specification type.	Small, 67.0	67.0
productSpecificationRecordType	Product Specification Record Type[]	Details of the product specification record type.	Small, 67.0	67.0

Promotion Details

Output representation of the details of applicable promotions.

Property Name	Type	Description	Filter Group and Version	Available Version
couponCode	String	Coupon code for the promotion.	Small, 67.0	67.0

Property Name	Type	Description	Filter Group and Version	Available Version
description	String	Description of the promotion.	Small, 67.0	67.0
displayName	String	Display name of the promotion.	Small, 67.0	67.0
endDateTime	String	End date and time of the promotion.	Small, 67.0	67.0
id	String	ID of the promotion.	Small, 67.0	67.0
isAutomatic	Boolean	Indicates whether the promotion is automatic (true) or not (false).	Small, 67.0	67.0
isCategoryPromo	Boolean	Indicates whether the promotion is a category-level promotion (true) or not (false).	Small, 67.0	67.0
isProductPromo	Boolean	Indicates whether the promotion is a product-level promotion (true) or not (false).	Small, 67.0	67.0
name	String	Name of the promotion.	Small, 67.0	67.0
priority	Integer	Priority of the promotion.	Small, 67.0	67.0
startDateTime	String	Start date and time of the promotion.	Small, 67.0	67.0
termsAndConditions	String	Terms and conditions of the promotion.	Small, 67.0	67.0

Proration Policy

Output representation of the details of the proration policy component.

Property Name	Type	Description	Filter Group and Version	Available Version
additionalFields	Map<String, Object>	Fields associated with the proration policy.	Small, 67.0	67.0
id	String	ID of the proration policy.	Small, 67.0	67.0

Qualification Context

Output representation of the details about the product qualification.

JSON example

```
{
  "qualificationContext": {
    "isQualified": true
  }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`isQualified`	Boolean	Indicates whether the product is qualified (`true`) or not (`false`).	Small, 67.0	67.0
`reason`	String	Specifies the reason for product qualification or disqualification.	Small, 67.0	67.0

Recommended Products

Output representation of the list of recommended products.

JSON example


```
{
  "products": [
    {
      "additionalFields": {},
      "catalogs": [
        {
          "customFields": {},
          "id": "0ZSSG000001875O4AQ",
          "name": "Hardware Catalog",
          "numberOfCategories": 4
        }
      ],
      "categories": [
        {
          "catalogId": "0ZSSG000001875O4AQ",
          "childCategories": [],
          "customFields": {},
          "eligiblePromotions": [],
          "id": "0ZGSG000001DJtv4AG",
          "name": "Accessories",
          "qualificationContext": {
            "isQualified": true
          }
        }
      ],
      "configurationRules": [
        {
          "details": [
            {
              "message": "recommend Mouse from monitor"
            }
          ],
          "type": "recommend"
        }
      ],
      "configureDuringSale": "Allowed",
      "description": "HeyMicky Mouse compatible comes with choice of USB or Wireless
connectivity.",
      "displayUrl": "https://5.imimg.com/data5/IS/JP/MY-58678879/computer-mouse.jpg",
```

```
      "eligiblePromotions": [],
      "id": "01tSG00000BiywkYAB",
      "isActive": true,
      "isAssetizable": true,
      "isSoldOnlyWithOtherProds": false,
      "name": "Mouse",
      "nodeType": "simpleProduct",
      "prices": [
        {
          "currencyIsoCode": "USD",
          "isDefault": true,
          "isDerived": false,
          "isSelected": false,
          "price": 7.99,
          "priceBookEntryId": "01uSG000004wTsEYAU",
          "priceBookId": "01sSG00000DQCjhYAH",
          "pricingModel": {
            "id": "0jPSG000000Avcv2AC",
            "name": "One Time",
            "pricingModelType": "OneTime"
          }
        }
      ],
      "productClassification": {
        "id": "11BSG00000UNijx2AD"
      },
      "productCode": "MOU001",
      "productSellingModelOptions": [
        {
          "id": "0iOSG000000J64x2AC",
          "isDefault": true,
          "productId": "01tSG00000BiywkYAB",
          "productSellingModel": {
            "doesAutoRenewByDefault": false,
            "id": "0jPSG000000Avcv2AC",
            "name": "One Time",
            "sellingModelType": "OneTime",
            "status": "Active"
          },
          "productSellingModelId": "0jPSG000000Avcv2AC"
        }
      ],
      "productSpecificationType": {
        "name": "Commercial",
        "productSpecificationRecordType": {}
      },
      "qualificationContext": {
        "isQualified": true
      }
    }
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
additionalFields	Map<String, Object>	Additional fields of the product.	Small, 67.0	67.0
availabilityDate	String	Date when the part is used in the product or is made available for sale.	Small, 67.0	67.0
catalogs	Catalog[]	List of the associated catalogs. This property returns the name and id values only.	Small, 67.0	67.0
categories	Category[]	List of the associated categories. This property returns the name and id values only.	Small, 67.0	67.0
configurationRules	Product Configuration Rules[]	List of configuration rules of the product.	Small, 67.0	67.0
configureDuringSale	String	Specifies whether to allow or prevent configuration when a bundle is sold.	Small, 67.0	67.0
description	String	Description of the product.	Small, 67.0	67.0
discontinuedDate	String	Date from when the part can't be used in the product or sold.	Small, 67.0	67.0
displayUrl	String	Display image URL of the product.	Small, 67.0	67.0
eligiblePromotions	Promotion Details[]	List of applicable promotions of the product.	Small, 67.0	67.0
endOfLifeDate	String	Date after which a product isn't supported, ordered, or maintained.	Small, 67.0	67.0
id	String	ID of the product.	Small, 67.0	67.0
isActive	Boolean	Indicates whether the product is active (true) or not (false).	Small, 67.0	67.0
isAssetizable	Boolean	Indicates whether a product instance remains a customer asset after it's purchased (true) or not (false). The default value is true.	Small, 67.0	67.0
isComponentRequired	Boolean	Indicates whether a product component is required (true) or not (false).	Small, 67.0	67.0
isDefaultComponent	Boolean	Indicates whether a product component is a default component (true) or not (false).	Small, 67.0	67.0
isQuantityEditable	Boolean	Indicates whether the quantity of a product is editable (true) or not (false).	Small, 67.0	67.0
isSoldOnlyWithOtherProds	Boolean	Indicates whether a product can be sold only with other products (true) or not (false).	Small, 67.0	67.0

Property Name	Type	Description	Filter Group and Version	Available Version
`name`	String	Name of the product.	Small, 67.0	67.0
`nodeType`	String	Specifies whether a node is a product node or a product classification node.	Small, 67.0	67.0
`prices`	Product Prices[]	Pricing details of a product.	Small, 67.0	67.0
`productClassification`	Product Classification[]	Details of the product classification that the product is based on.	Small, 67.0	67.0
`productCode`	String	Universal product code that's used to track the part that's used in the product.	Small, 67.0	67.0
`productQuantity`	Product Quantity[]	Quantity details of a product.	Small, 67.0	67.0
`productSellingModelOptions`	Product Selling Model Option[]	Details of the product selling model options.	Small, 67.0	67.0
`productSpecificationType`	Product Specification Type[]	Details of the product specification type.	Small, 67.0	67.0
`productType`	String	Details about the product type.	Small, 67.0	67.0
`qualificationContext`	Qualification Context[]	Context details of a user, which are used for qualification rules.	Small, 67.0	67.0
`status`	String	Status of the product, such as active or inactive.	Small, 67.0	67.0

Search Products Facet

Output representation of the details of the faceted search.

JSON example

```
"facets": [
  {
    "attributeType": "ProductStandard",
    "displayName": "Product Type",
    "displayRank": 2,
    "nameOrId": "Type",
    "values": [
      {
        "displayName": "Bundle",
        "nameOrId": "Bundle"
      }
    ]
  },
  {
    "attributeType": "ProductDynamicAttribute",
    "displayName": "Display",
    "displayRank": 3,
    "nameOrId": "0tjDU0000003K5BYAU",
```

```
      "values": [
        {
          "displayName": "1080p Built-in Display",
          "nameOrId": "1080p Built-in Display"
        },
        {
          "displayName": "2k Built-in Display",
          "nameOrId": "2k Built-in Display"
        },
        {
          "displayName": "4k Built-in Display",
          "nameOrId": "4k Built-in Display"
        }
      ]
    }
  ]
```

Property Name	Type	Description	Filter Group and Version	Available Version
attributeType	String	Search attribute type of the facet.	Small, 63.0	63.0
displayName	String	Display name of the facet.	Small, 63.0	63.0
displayRank	Integer	Display rank for the facet.	Small, 63.0	63.0
nameOrId	String	ID or the internal name of the facet.	Small, 63.0	63.0
values	Facet Value[]	Values of the facet found in the search result. Sorted by display name in alphabetical order.	Medium, 63.0	63.0

User Context

Output representation of the user context details.

Property Name	Type	Description	Filter Group and Version	Available Version
contextId	String	ID of the context.	Small, 60.0	60.0

Product Discovery Standard Invocable Actions

Use the standard invocable actions available with Product Discovery to find and retrieve product, category, and catalog details. Additionally, execute a qualification procedure, and search products with guided selection.

Find Products Action

Search for the products from a catalog, category, or subcategory by using the specified search term.

Execute Qualification Procedure Action

Execute a qualification procedure, which returns the qualification status for the specified products.

Get Catalogs Action
Get a list of catalog records.

Get Catalog Details Action
Get details of a catalog record.

Get Categories Action
Get the list of categories associated with a catalog record.

Get Category Details Action
Get details of a category record.

Get Multiple Product Details Action
Get product details for a list of products.

Get Products Action
Get products from the specified catalog, category, or subcategory, including product qualification and pricing details.

Get Product Details Action
Get details such as attributes, hierarchy, and cardinality for the specified product.

Search Product with Guided Selection Action
Use guided product selection to search for products.

Get Product Recommendations Action
Retrieve a list of recommended products for a quote or order by using the Constraint Rule Engine.

Find Products Action

Search for the products from a catalog, category, or subcategory by using the specified search term.

This action is available in API version 62.0 and later.

Special Access Rules

The Find Products action is available in Enterprise, Unlimited, and Developer Editions where Product Discovery is enabled.

You can invoke this action via Apex and Flows only.

Inputs

Input	Details
`additionalContextData`	**Type** Apex-defined **Description** An array of Apex `runtime_industries_cpq.AdditionalContextData` records that contain the additional nodes that are used along with the context definition nodes for data hydration. The maximum number of supported nodes is 10.

Input	Details
`additionalFields`	**Type** Apex-defined **Description** An Apex `runtime_industries_cpq.AdditionalFields` record that contains an array of additional standard or custom fields to include in the response. The supported objects are: • `Product2` • `ProductAttributeDefinition`—If the fields defined for the `ProductAttributeDefinition` object aren't available for the `ProductClassificationAttr` object, then the API request fails.
`catalogId`	**Type** string **Description** Catalog ID that's used to find and retrieve the products.
`categoryId`	**Type** string **Description** ID of the category or subcategory to get the products for.
`contextDefinition`	**Type** string **Description** API name of the context definition used for context creation. If you don't specify a value, the context selected on the Product Discovery Settings page from Setup is used.
`contextMapping`	**Type** string **Description** API name of the context mapping that's used for data hydration. The value of this parameter is used only if it belongs to the specified context definition.
`correlationId`	**Type** string **Description** Currency code that's used to calculate and show prices. Only the products with the currency code matching the specified currency code are fetched.

Input	Details
`currencyCode`	**Type** string **Description** Currency code that's used to calculate and show prices.
`cursor`	**Type** string **Description** A unique identifier that represents the position of the product from which the next set of results are retrieved.
`executeConfigurationRules`	**Type** boolean **Description** Indicates whether configuration rules must run (`true`) or not (`false`). Available in API version 67.0 and later.
`enablePricing`	**Type** boolean **Description** Indicates whether the pricing procedure must run (`true`) or not (`false`). The default value is `true`. To use this parameter, you must enable the Pricing Procedure setting from Setup.
`enableQualification`	**Type** boolean **Description** Indicates whether the qualification procedure must run (`true`) or not (`false`). The default value is `true`. To use this parameter, you must enable the Qualification Procedure setting from Setup.
`filter`	**Type** Apex-defined **Description** A collection of Apex `runtime_industries_cpq.FilterInputRepresentation` records where each record contains a related object and the filter criteria that's applied on the object. The `filter` parameter supports only the `name` property.

Input	Details
	The supported operators are: • `eq` • `in` • `contains` If this parameter contains multiple criteria, all the criteria are applied.
`includeCatalogDetails`	**Type** boolean **Description** Indicates whether catalog details must be included in the response (`true`) or not (`false`).
`limit`	**Type** integer **Description** Maximum number of results to be returned in the response. Enter a value from 1 through 100. The default value is 10.
`orderBy`	**Type** string **Description** Comma-separated string of key-value pairs that specify how results are sorted. Each string must contain a field name and its sort order. For example, `["name:asc",”custom_field:asc”]`.
`priceBookId`	**Type** string **Description** ID of the pricebook from which you wan to retrieve the pricing details.
`productClassificationId`	**Type** string **Description** ID of the product classification that’s used to filter products.
`pricingProcedure`	**Type** string **Description** API name of the pricing procedure to calculate product prices. If you don’t specify a value, the pricing procedure selected on the Product Discovery Settings page from Setup is used.

Input	Details
`qualificationProcedure`	**Type** string **Description** API name of the qualification procedure to evaluate product eligibility. If you don't specify a value, the qualification procedure selected on the Product Discovery Settings page from Setup is used.
`transactionContextId`	**Type** string **Description** Context ID of the quote or order. Available in API version 67.0 and later.
`transactionId`	**Type** string **Description** ID of the quote or order. Available in API version 67.0 and later.
`relatedObjectFilters`	**Type** Apex-defined **Description** A collection of Apex `runtime_industries_cpq.RelatedObjectFilterInputRepresentation` records, where each record contains a related object and the filter criteria that's applied on the object.
`searchTerm`	**Type** string **Description** Required. Search term to find and retrieve products.
`userContext`	**Type** Apex-defined **Description** An Apex `ConnectApi.UserContextInputRepresentation` record that contains the user details to evaluate product eligibility and calculate prices.

Outputs

Output	Details
`apiStatus`	**Type** Apex-defined **Description** An Apex `runtime_industries_cpq.ApiStatusRepresentation` record that contains a status code and message.
`contextId`	**Type** string **Description** ID of the context that's created by using the specified context definition.
`correlationId`	**Type** string **Description** ID to reference a series of related actions.
`cursor`	**Type** string **Description** Unique identifier that represents the position of the next product in the dataset. It's used as an input to retrieve the next set of products.
`facets`	**Type** Apex-defined **Description** Collection of Apex ProductFacetsRepresentation records that contain details of the facet that's retrieved.
`results`	**Type** Apex-defined **Description** An Apex `runtime_industries_cpq.SearchProductsRepresentation` record that contains the products that match the query.
`userContext`	**Type** Apex-defined

Output	Details
	Description An Apex `ConnectApi.UserContextRepresentation` record that includes the user details.

Example

Here's a sample input to call this invocable action from Apex code.

```
Invocable.Action action = Invocable.Action.createStandardAction('findProducts');
action.setInvocationParameter('searchTerm', 'Additional API');
action.setInvocationParameter('executeConfigurationRules', true);
action.setInvocationParameter('catalogId', '0ZSVW000000AhdB4AS');
action.setInvocationParameter('transactionId', '0Q0VW0000018J6X0AU');
List<String> orderBy = new List<String>();
orderBy.add('name:asc');
action.setInvocationParameter('orderBy', orderBy);
List<Invocable.Action.Result> results = action.invoke();

for (Invocable.Action.Result res : results) {
    if (res.isSuccess()) {
        Map<String, Object> outMap = res.getOutputParameters();
        String jsonOutput = JSON.serialize(outMap);
        System.debug(jsonOutput);
    }
}
```

Here’s a sample response for this action.


```
{
  "facets": [],
  "apiStatus": {
    "statusMessage": null,
    "statusCode": "FetchedDetailsSuccessfully",
    "messages": []
  },
  "results": [
    {
      "unitOfMeasure": null,
      "status": null,
      "qualificationContext": {
        "reason": null,
        "isQualified": true
      },
      "productType": null,
      "productSpecificationType": null,
      "productSellingModelOptions": [
        {
          "productSellingModelId": "0jPVW0000001fh32AA",
          "productSellingModel": {
            "status": "Active",
            "sellingModelType": "TermDefined",
```

```
        "pricingTermUnit": "Annual",
        "pricingTerm": 1,
        "name": "Term Annual",
        "id": "0jPVW0000001fh32AA"
      },
      "productId": "01tVW000003l7txYAA",
      "id": "0iOVW00000049w22AA"
    },
    {
      "productSellingModelId": "0jPVW0000001fh52AA",
      "productSellingModel": {
        "status": "Active",
        "sellingModelType": "Evergreen",
        "pricingTermUnit": "Months",
        "pricingTerm": 1,
        "name": "Evergreen Monthly",
        "id": "0jPVW0000001fh52AA"
      },
      "productId": "01tVW000003l7txYAA",
      "id": "0iOVW00000049w12AA"
    },
    {
      "productSellingModelId": "0jPVW0000001fh62AA",
      "productSellingModel": {
        "status": "Active",
        "sellingModelType": "TermDefined",
        "pricingTermUnit": "Months",
        "pricingTerm": 1,
        "name": "Term Monthly",
        "id": "0jPVW0000001fh62AA"
      },
      "productId": "01tVW000003l7txYAA",
      "id": "0iOVW00000049w32AA"
    }
  ],
  "productRelatedComponent": null,
  "productQuantity": null,
  "productPricingInformation": null,
  "productInformation": null,
  "productComponentGroups": [],
  "productCode": "QB-API",
  "productClassification": {
    "id": "11BVW000004ljiZ2AQ"
  },
  "prices": [
    {
      "pricingModel": {
        "unitOfMeasure": null,
        "pricingModelType": "TermDefined",
        "occurrence": 1,
        "name": "Term Annual",
        "id": "0jPVW0000001fh32AA",
        "frequency": "Annual"
      },
```

```
            "priceBookId": "01sVW0000024PZlYAM",
            "priceBookEntryId": "01uVW000000jzfGYAQ",
            "price": 15000,
            "isSelected": false,
            "isDerived": false,
            "isDefault": true,
            "effectiveTo": null,
            "effectiveFrom": null,
            "currencyIsoCode": "USD"
          },
          {
            "pricingModel": {
              "unitOfMeasure": null,
              "pricingModelType": "Evergreen",
              "occurrence": 1,
              "name": "Evergreen Monthly",
              "id": "0jPVW0000001fh52AA",
              "frequency": "Months"
            },
            "priceBookId": "01sVW0000024PZlYAM",
            "priceBookEntryId": "01uVW000000jzg8YAA",
            "price": 2000,
            "isSelected": false,
            "isDerived": false,
            "isDefault": false,
            "effectiveTo": null,
            "effectiveFrom": null,
            "currencyIsoCode": "USD"
          },
          {
            "pricingModel": {
              "unitOfMeasure": null,
              "pricingModelType": "TermDefined",
              "occurrence": 1,
              "name": "Term Monthly",
              "id": "0jPVW0000001fh62AA",
              "frequency": "Months"
            },
            "priceBookId": "01sVW0000024PZlYAM",
            "priceBookEntryId": "01uVW000000jzg9YAA",
            "price": 1500,
            "isSelected": false,
            "isDerived": false,
            "isDefault": false,
            "effectiveTo": null,
            "effectiveFrom": null,
            "currencyIsoCode": "USD"
          }
        ],
        "nodeType": "simpleProduct",
        "name": "Additional API",
        "isSoldOnlyWithOtherProds": false,
        "isQuantityEditable": null,
        "isDefaultComponent": null,
```

```
      "isComponentRequired": null,
      "isAssetizable": true,
      "isActive": true,
      "id": "01tVW000003l7txYAA",
      "endOfLifeDate": null,
      "displayUrl": "/resource/add_api",
      "discontinuedDate": null,
      "description": null,
      "configureDuringSale": "Allowed",
      "configurationRules": [
        {
          "type": "disable",
          "details": [
            {
              "message": "This is disabled"
            }
          ]
        }
      ],
      "catalogs": [],
      "availabilityDate": null,
      "attributeCategories": [],
      "additionalFields": []
    },
    {
      "unitOfMeasure": null,
      "status": null,
      "qualificationContext": {
        "reason": null,
        "isQualified": true
      },
      "productType": null,
      "productSpecificationType": null,
      "productSellingModelOptions": [
        {
          "productSellingModelId": "0jPVW0000001fh32AA",
          "productSellingModel": {
            "status": "Active",
            "sellingModelType": "TermDefined",
            "pricingTermUnit": "Annual",
            "pricingTerm": 1,
            "name": "Term Annual",
            "id": "0jPVW0000001fh32AA"
          },
          "productId": "01tVW000003l7tyYAA",
          "id": "0iOVW00000049w42AA"
        }
      ],
      "productRelatedComponent": null,
      "productQuantity": null,
      "productPricingInformation": null,
      "productInformation": null,
      "productComponentGroups": [],
      "productCode": "QB-API-FLEX",
```

```
      "productClassification": {
        "id": null
      },
      "prices": [
        {
          "pricingModel": {
            "unitOfMeasure": null,
            "pricingModelType": "TermDefined",
            "occurrence": 1,
            "name": "Term Annual",
            "id": "0jPVW0000001fh32AA",
            "frequency": "Annual"
          },
          "priceBookId": "01sVW0000024PZlYAM",
          "priceBookEntryId": "01uVW000000jzfFYAQ",
          "price": 450,
          "isSelected": false,
          "isDerived": false,
          "isDefault": true,
          "effectiveTo": null,
          "effectiveFrom": null,
          "currencyIsoCode": "USD"
        }
      ],
      "nodeType": "simpleProduct",
      "name": "Additional API Flex (100M)",
      "isSoldOnlyWithOtherProds": false,
      "isQuantityEditable": null,
      "isDefaultComponent": null,
      "isComponentRequired": null,
      "isAssetizable": true,
      "isActive": true,
      "id": "01tVW000003l7tyYAA",
      "endOfLifeDate": null,
      "displayUrl": "/resource/api_flex",
      "discontinuedDate": null,
      "description": "API instances remain under management until they are deleted.
Instances of API Manager are aggregated using a Max Concurrent model. The usage for a
month is the highest number of APIs Managed in a single given hour during a month.",
      "configureDuringSale": null,
      "configurationRules": [],
      "catalogs": [],
      "availabilityDate": null,
      "attributeCategories": [],
      "additionalFields": []
    },
    {
      "unitOfMeasure": null,
      "status": null,
      "qualificationContext": {
        "reason": null,
        "isQualified": true
      },
      "productType": null,
```

```
"productSpecificationType": null,
"productSellingModelOptions": [
  {
    "productSellingModelId": "0jPVW0000001fh32AA",
    "productSellingModel": {
      "status": "Active",
      "sellingModelType": "TermDefined",
      "pricingTermUnit": "Annual",
      "pricingTerm": 1,
      "name": "Term Annual",
      "id": "0jPVW0000001fh32AA"
    },
    "productId": "01tVW000003l7tzYAA",
    "id": "0iOVW00000049w52AA"
  }
],
"productRelatedComponent": null,
"productQuantity": null,
"productPricingInformation": null,
"productInformation": null,
"productComponentGroups": [],
"productCode": "QB-API-GOVT",
"productClassification": {
  "id": null
},
"prices": [
  {
    "pricingModel": {
      "unitOfMeasure": null,
      "pricingModelType": "TermDefined",
      "occurrence": 1,
      "name": "Term Annual",
      "id": "0jPVW0000001fh32AA",
      "frequency": "Annual"
    },
    "priceBookId": "01sVW0000024PZlYAM",
    "priceBookEntryId": "01uVW000000jzfHYAQ",
    "price": 1450,
    "isSelected": false,
    "isDerived": false,
    "isDefault": true,
    "effectiveTo": null,
    "effectiveFrom": null,
    "currencyIsoCode": "USD"
  }
],
"nodeType": "simpleProduct",
"name": "Additional API Gov",
"isSoldOnlyWithOtherProds": false,
"isQuantityEditable": null,
"isDefaultComponent": null,
"isComponentRequired": null,
"isAssetizable": true,
"isActive": true,
```

```
      "id": "01tVW000003l7tzYAA",
      "endOfLifeDate": null,
      "displayUrl": "/resource/api_govt",
      "discontinuedDate": null,
      "description": "API instances remain under management until they are deleted.
Instances of API Manager are aggregated using a Max Concurrent model. The usage for a
month is the highest number of APIs Managed in a single given hour during a month.",
      "configureDuringSale": null,
      "configurationRules": [],
      "catalogs": [],
      "availabilityDate": null,
      "attributeCategories": [],
      "additionalFields": []
    },
    {
      "unitOfMeasure": null,
      "status": null,
      "qualificationContext": {
        "reason": null,
        "isQualified": true
      },
      "productType": null,
      "productSpecificationType": null,
      "productSellingModelOptions": [
        {
          "productSellingModelId": "0jPVW0000001fh32AA",
          "productSellingModel": {
            "status": "Active",
            "sellingModelType": "TermDefined",
            "pricingTermUnit": "Annual",
            "pricingTerm": 1,
            "name": "Term Annual",
            "id": "0jPVW0000001fh32AA"
          },
          "productId": "01tVW000003l7u0YAA",
          "id": "0iOVW00000049w92AA"
        }
      ],
      "productRelatedComponent": null,
      "productQuantity": null,
      "productPricingInformation": null,
      "productInformation": null,
      "productComponentGroups": [],
      "productCode": "QB-API-PREP",
      "productClassification": {
        "id": null
      },
      "prices": [
        {
          "pricingModel": {
            "unitOfMeasure": null,
            "pricingModelType": "TermDefined",
            "occurrence": 1,
            "name": "Term Annual",
```

```
          "id": "0jPVW0000001fh32AA",
          "frequency": "Annual"
        },
        "priceBookId": "01sVW0000024PZlYAM",
        "priceBookEntryId": "01uVW000000jzfSYAQ",
        "price": 3240,
        "isSelected": false,
        "isDerived": false,
        "isDefault": true,
        "effectiveTo": null,
        "effectiveFrom": null,
        "currencyIsoCode": "USD"
      }
    ],
    "nodeType": "simpleProduct",
    "name": "Additional API Pre-Prod",
    "isSoldOnlyWithOtherProds": false,
    "isQuantityEditable": null,
    "isDefaultComponent": null,
    "isComponentRequired": null,
    "isAssetizable": true,
    "isActive": true,
    "id": "01tVW000003l7u0YAA",
    "endOfLifeDate": null,
    "displayUrl": "/resource/api_preprod",
    "discontinuedDate": null,
    "description": "API instances remain under management until they are deleted.
Instances of API Manager are aggregated using a Max Concurrent model. The usage for a
month is the highest number of APIs Managed in a single given hour during a month.",
    "configureDuringSale": null,
    "configurationRules": [],
    "catalogs": [],
    "availabilityDate": null,
    "attributeCategories": [],
    "additionalFields": []
  },
  {
    "unitOfMeasure": null,
    "status": null,
    "qualificationContext": {
      "reason": null,
      "isQualified": true
    },
    "productType": null,
    "productSpecificationType": null,
    "productSellingModelOptions": [
      {
        "productSellingModelId": "0jPVW0000001fh32AA",
        "productSellingModel": {
          "status": "Active",
          "sellingModelType": "TermDefined",
          "pricingTermUnit": "Annual",
          "pricingTerm": 1,
          "name": "Term Annual",
```

```
          "id": "0jPVW0000001fh32AA"
        },
        "productId": "01tVW000003l7u1YAA",
        "id": "0iOVW00000049wA2AQ"
      }
    ],
    "productRelatedComponent": null,
    "productQuantity": null,
    "productPricingInformation": null,
    "productInformation": null,
    "productComponentGroups": [],
    "productCode": "QB-API-PROD",
    "productClassification": {
      "id": null
    },
    "prices": [
      {
        "pricingModel": {
          "unitOfMeasure": null,
          "pricingModelType": "TermDefined",
          "occurrence": 1,
          "name": "Term Annual",
          "id": "0jPVW0000001fh32AA",
          "frequency": "Annual"
        },
        "priceBookId": "01sVW0000024PZlYAM",
        "priceBookEntryId": "01uVW000000jzfdYAA",
        "price": 3240,
        "isSelected": false,
        "isDerived": false,
        "isDefault": true,
        "effectiveTo": null,
        "effectiveFrom": null,
        "currencyIsoCode": "USD"
      }
    ],
    "nodeType": "simpleProduct",
    "name": "Additional API Prod",
    "isSoldOnlyWithOtherProds": false,
    "isQuantityEditable": null,
    "isDefaultComponent": null,
    "isComponentRequired": null,
    "isAssetizable": true,
    "isActive": true,
    "id": "01tVW000003l7u1YAA",
    "endOfLifeDate": null,
    "displayUrl": "/resource/api_prod",
    "discontinuedDate": null,
    "description": "API instances remain under management until they are deleted.
Instances of API Manager are aggregated using a Max Concurrent model. The usage for a
month is the highest number of APIs Managed in a single given hour during a month.",
    "configureDuringSale": null,
    "configurationRules": [],
    "catalogs": [],
```

```
      "availabilityDate": null,
      "attributeCategories": [],
      "additionalFields": []
    }
  ],
  "contextId": "0000000r25tp21g002517775323419202e0743b1f275423d8a489c12ca99da72",
  "correlationId": "4ac34e38-13d6-4720-97c4-4db7b3d41678"
}
```

Usage of an Apex-Defined Data Type in a Flow

To use an Apex-defined input parameter in a flow, follow these guidelines.

Create an Apex Class

Create an Apex class defining the input and output parameters. In the flow, include the Apex-defined input parameters for which you want to add the details. In this example, we've created a class named ProductServiceAction that takes an object's API name and record ID as input, and returns the additional context data.

```
public class ProductServiceAction {
    // Define input parameters
    public class FlowInput{
        @InvocableVariable(required=false)
        public String objectApiName;

        @InvocableVariable(required=false)
        public String recordId;
    }

    // Define output parameters
    public class FlowOutput{
        @invocableVariable
        public runtime_industries_cpq.AdditionalContextData
additionalContextDataFinalOutput = new runtime_industries_cpq.AdditionalContextData();
    }

    // This method is invoked from a Flow
    @InvocableMethod(label='Process Input' description='Creates the Array of
ContextDataInput for additional Context Data')
    public static List<FlowOutput> processContextData(List<FlowInput> inputs){
        String apiName;
        String recId;
        FlowOutput output = new FlowOutput();

    // Capture input from the flow
        for(FlowInput input: inputs){
            apiName = input.objectApiName;
            recId = input.recordId;
            }
    // Populate the ContextDataInput list to store additional context data
        List<runtime_industries_cpq.ContextDataInput> listContextData = new
List<runtime_industries_cpq.ContextDataInput>();
        runtime_industries_cpq.ContextDataInput cd1 = new
runtime_industries_cpq.ContextDataInput();
        cd1.nodeName = apiName;
```

```
        cd1.nodeData = new Map<String, Object>();
        cd1.nodeData.put('id',recId);
        listContextData.add(cd1);

       output.additionalContextDataFinalOutput.additionalContextData = listContextData;

        List<FlowOutput> flowOutputs = new List<FlowOutput>();
        flowOutputs.add(output);

        List<runtime_industries_cpq.RelatedObjectFilter> relatedObjectFilterList = new
 List<runtime_industries_cpq.RelatedObjectFilter>();

        runtime_industries_cpq.RelatedObjectFilter relatedObjectFilter = new
runtime_industries_cpq.RelatedObjectFilter();

        relatedObjectFilter.objectName = 'ProductSpecificationRecType';
        List<runtime_industries_cpq.FilterCriteriaInputRepresentation> criteriaList =
new  List<runtime_industries_cpq.FilterCriteriaInputRepresentation>();
        runtime_industries_cpq.FilterCriteriaInputRepresentation criteria = new
runtime_industries_cpq.FilterCriteriaInputRepresentation();
        criteria.property = 'IsCommercial';
        criteria.operator = 'eq';
        criteria.value = 'true';
        criteriaList.add(criteria);
        relatedObjectFilter.criteria = criteriaList;

        relatedObjectFilterList.add(relatedObjectFilter);
        output.relatedObjectFilter.relatedObjectFilter  = relatedObjectFilterList;

  output.userContext.accountId = '001DU000001nx9BYAQ';

        return flowOutputs;

    }

}
```

Create a Flow with the Necessary Variables and Components

Create a flow that enables users to add a search term to find products. Add the ProductService action that you've created above by using Apex. When a flow is invoked from a record, the flow sends the record's objectApiName and recordId to the Apex class, which then generates the flow output. The flow passes the objectApiName and recordId of the record that the flow is invoked from to the Apex class to generate the flow output. See Example of How to Create a Flow for Product Discovery.

Configure the Action

Configure the action (for example, Find Products action) to add values for the Apex-defined input parameters. Use the output of the created Apex class as the input of the Apex-defined parameter in the Find Products action, which users can use to find products.

Execute Qualification Procedure Action

Execute a qualification procedure, which returns the qualification status for the specified products.

This action is available in API version 64.0 and later.

You can invoke this action via Apex and Flows only.

Special Access Rules

The Execute Qualification Procedure action is available in Enterprise, Unlimited, and Developer Editions where Product Discovery is enabled.

Inputs

Input	Details
`additionalContextData`	**Type** Apex-defined **Description** Collection of Apex AdditionalContextData records that contain additional context data for nodes of the custom context definition, if applicable. You can add details for up to 10 nodes. See AdditionalContextData.
`contextDefinitionName`	**Type** string **Description** Name of the custom context definition that's used to create context data for categories. If null, the default context definition is used.
`contextMappingName`	**Type** string **Description** Name of the context mapping. By default, the default context mapping associated with the context definition is used.
`correlationId`	**Type** string **Description** Unique identifier attached to requests and messages, allowing reference to a specific transaction or event chain.
`productIds`	**Type** string **Description** Required. Collection of IDs of products that are to be checked for qualification.
`qualificationProcedureName`	**Type** string

Input	Details
	Description Name of the custom qualification procedure that's executed to determine the category list. If null, the default qualification procedure is executed.
`userContextInputRepresentation`	**Type** Apex-defined **Description** An Apex UserContextInputRepresentation record that contains user details, such as account ID, geographical location, language preferences, and more.

Outputs

Output	Details
`apiStatusOutputRepresentation`	**Type** Apex-defined **Description** An Apex ApiStatusOutputRepresentation record that contains the status of the request, including the status code and message.
`contextId`	**Type** string **Description** ID of the context that's created by using the specified context definition.
`correlationId`	**Type** string **Description** Unique identifier attached to requests and messages, allowing reference to a specific transaction or event chain.
`qualificationResultRepresentations`	**Type** Apex-defined **Description** Collection of Apex QualificationResultRepresentation records that contain details about the qualified product.

Example

Here's a sample input to call this invocable action from Apex code.

```
Invocable.Action action =
Invocable.Action.createStandardAction('executeQualificationProcedure');

ConnectApi.UserContextInputRepresentation userContext = new
ConnectApi.UserContextInputRepresentation();
userContext.accountId = '001xx000003GYiEAAW';

runtime_industries_cpq.ContextDataInput data = new
runtime_industries_cpq.ContextDataInput();
String nodeNameVal = 'Quote__c';
Map<String,Object> nodeDataVal = new Map<String,Object>();
nodeDataVal.put('id',(Object)'0Q0xx0000004CDsCAM');
nodeDataVal.put('businessObjectType',(Object)'Quote');
data.nodeName = nodeNameVal;
data.nodeData = nodeDataVal;
List<runtime_industries_cpq.ContextDataInput> contextData= new
List<runtime_industries_cpq.ContextDataInput>();
contextData.add(data);
runtime_industries_cpq.AdditionalContextData additionalContextDataOut = new
runtime_industries_cpq.AdditionalContextData();
additionalContextDataOut.additionalContextData = new
List<runtime_industries_cpq.ContextDataInput>();
additionalContextDataOut.additionalContextData.add(data);

List<String> productIds = new List<String>();
prodIds.add('01txx0000006i2ZAAQ');
prodIds.add('01txx0000006i35AAA');
action2.setInvocationParameter('productIds', productIds);
action.setInvocationParameter('correlationId', '9cbb9650-48c5-11ed-96d1-0afcf185843b');
action.setInvocationParameter('userContextInputRepresentation', userContext);
action.setInvocationParameter('qualificationProcedureName', 'CatQual02');
action.setInvocationParameter('contextDefinitionName', 'CategoryCD');
action.setInvocationParameter('contextMappingName', 'ProductDiscoveryMapping');
action.setInvocationParameter('additionalContextData', additionalContextDataOut);

List<Invocable.Action.Result> results = action.invoke();
System.debug('Execute Qualification Procedure Action + '+results);
```

Here's a sample response when you call this action.

```
[
  {
    "actionName": "executeQualificationProcedure",
    "errors": null,
    "invocationId": null,
    "isSuccess": true,
    "outcome": null,
    "outputValues": {
      "apiStatusOutputRepresentation": {
        "statusMessage": null,
        "statusCode": "FetchedDetailsSuccessfully",
        "messages": []
```

```
      },
      "qualificationResultRepresentations": [
        {
          "qualificationContext": {
            "reason": null,
            "isQualified": true
          },
          "productId": "01tSG000007uDL8YAM"
        },
        {
          "qualificationContext": {
            "reason": "Product is not qualified because one or more field(s) do not
match the qualification criteria. Fields:- ProductId - 01tSG000007uDLBYA2
Max_Number_of_Employees - 50 Min_Number_of_Employees - 50 RootProductId - ",
            "isQualified": false
          },
          "productId": "01tSG000007uDLBYA2"
        }
      ],
      "correlationId": "9cbb9650-48c5-11ed-96d1-0afcf185843b",
      "contextId": "0000000b28op21g0025174722465473 9cade16577eac4c8fb5d94a20d952fdab"
    },
    "sortOrder": -1,
    "version": 1
  }
]
```

Get Catalogs Action

Get a list of catalog records.

This action is available in API version 64.0 and later.

You can invoke this action via Apex and Flows only.

Special Access Rules

Inputs

Input	Details
`recordLimit`	**Type** integer **Description** Number of catalog records to get. The minimum is 1, the maximum is 100, and the default is 100.
`recordOffset`	**Type** integer

Input	Details
	Description Number of catalog records to skip in the request. The default is 0.
`correlationId`	**Type** string **Description** Unique identifier for tracking requests and messages, allowing reference to a specific transaction or event chain.
`orderBy`	**Type** string **Description** Sort records in ascending or descending order.

Outputs

Output	Details
`resultCatalogList`	**Type** Apex-defined **Description** List of filtered catalog records. See CatalogOutputRepresentation.
`apiStatusOutputRepresentation`	**Type** Apex-defined **Description** An Apex `runtime_industries_cpq.ApiStatusRepresentation` record that contains the status of the request, including the status code and message.
`recordLimit`	**Type** integer **Description** Number of catalog records to show per page.
`resultListCount`	**Type** integer **Description** Number of catalog records in the result.

Output	Details
correlationId	**Type** string **Description** Unique identifier for tracking requests and messages, allowing reference to a specific transaction or event chain.
recordOffset	**Type** integer **Description** Number of catalog records to skip in the request. The default is 0.

Example

POST

Here's a sample input to call this invocable action from Apex code.

```
Invocable.Action action = Invocable.Action.createStandardAction('getCatalogs');

action.setInvocationParameter('correlationId', '9cbb9650-48c5-11ed-96d1-0afcf185843b');
action.setInvocationParameter('recordLimit', 1);
action.setInvocationParameter('recordOffset', 1);
String[] sortOrder = new String[]{'Name:ASC'};
action.setInvocationParameter('orderBy', sortOrder);

List<Invocable.Action.Result> results = action.invoke();
System.debug('Catalog List Action + '+results);
```

Here's a sample response when you call this action.


```
[
  {
    "actionName": "getCatalogs",
    "errors": null,
    "invocationId": null,
    "isSuccess": true,
    "outcome": null,
    "outputValues": {
      "resultListCount": 2,
      "apiStatusOutputRepresentation": {
        "statusMessage": null,
        "statusCode": "FetchedDetailsSuccessfully",
        "messages": []
      },
      "resultCatalogList": [
        {
          "status": null,
          "numberOfCategories": 2,
          "name": "Hardware Catalog",
```

```
          "id": "0ZSZ6000000CtXYOA0",
          "effectiveStartDate": null,
          "effectiveEndDate": null,
          "description": "Hardware Catalog Desc",
          "customFields": [],
          "catalogType": "Sales",
          "catalogCode": "HC"
        }
      ],
      "correlationId": "9cbb9650-48c5-11ed-96d1-0afcf185843b",
      "recordOffset": 0,
      "recordLimit": 1
    },
    "sortOrder": -1,
    "version": 1
  }
]
```


Get Catalog Details Action

Get details of a catalog record.

This action is available in API version 64.0 and later.

You can invoke this action via Apex and Flows only.

Special Access Rules

The Get Catalog Details action is available in Enterprise, Unlimited, and Developer Editions where Product Discovery is enabled.

Inputs

Input	Details
`catalogId`	**Type** string **Description** Required. ID of the catalog record.
`correlationId`	**Type** string **Description** Unique identifier attached to requests and messages, enabling reference to a specific transaction or event chain.

Outputs

Output	Details
apiStatusOutputRepresentation	**Type** Apex-defined **Description** An Apex `runtime_industries_cpq.ApiStatusRepresentation` record that contains the status of the request, including the status code and message.
`catalogDetailsResult`	**Type** Apex-defined **Description** Details of the catalog record.
`correlationId`	**Type** string **Description** Unique identifier attached to requests and messages, allowing reference to a specific transaction or event chain.

Example

Here's a sample input to call this invocable action from Apex code.

```
Invocable.Action action = Invocable.Action.createStandardAction('getCatalogDetails');

action.setInvocationParameter('correlationId', '9cbb9650-48c5-11ed-96d1-0afcf185843b');
action.setInvocationParameter('catalogId', '0ZSZ6000000CtXYOA0');

List<Invocable.Action.Result> results = action.invoke();
System.debug('Catalog Details Action + '+results);
```

Here's a sample response when you call this action.


```
[
  {
    "actionName": "getCatalogDetails",
    "errors": null,
    "invocationId": null,
    "isSuccess": true,
    "outcome": null,
    "outputValues": {
      "apiStatusOutputRepresentation": {
        "statusMessage": null,
        "statusCode": "FetchedDetailsSuccessfully",
        "messages": []
      },
      "correlationId": "9cbb9650-48c5-11ed-96d1-0afcf185843b",
```

```
      "catalogDetailsResult": {
        "status": null,
        "numberOfCategories": 2,
        "name": "Hardware Catalog",
        "id": "0ZSZ6000000CtXYOA0",
        "effectiveStartDate": null,
        "effectiveEndDate": null,
        "description": "Hardware Catalog Desc",
        "customFields": null,
        "catalogType": "Sales",
        "catalogCode": "HC"
      }
    },
    "sortOrder": -1,
    "version": 1
  }
]
```


Get Categories Action

Get the list of categories associated with a catalog record.

This action is available in API version 64.0 and later.

You can invoke this action via Apex and Flows only.

Special Access Rules

The Get Categories action is available in Enterprise, Unlimited, and Developer Editions where Product Discovery is enabled.

Inputs

Input	Details
`additionalContextData`	**Type** Apex-defined **Description** Collection of Apex `runtime_industries_cpq.AdditionalContextData` records that contain additional context data for nodes of the custom context definition, if applicable. You can add details for up to 10 nodes.
`catalogId`	**Type** string **Description** Required. ID of the catalog record.

Input	Details
`categoryNestLevel`	**Type** integer **Description** Level of nesting within the category hierarchy to include in the request.
`contextDefinitionName`	**Type** string **Description** Name of the custom context definition that's used to create context data for categories. If null, the default context definition is used.
`correlationId`	**Type** string **Description** Unique identifier attached to requests and messages, allowing reference to a specific transaction or event chain.
`contextMappingName`	**Type** string **Description** Name of the context mapping. By default, the default context mapping associated with the context definition is used.
`enableQualificationProcedure`	**Type** boolean **Description** Indicates whether qualification rules are applied to categories (`true`) or not (`false`).
`filterInputRepresentation`	**Type** Apex-defined **Description** Collection of Apex `runtime_industries_cpq.FilterInputRepresentation` records that contain the filter criteria applied to the category records.
`parentCategoryId`	**Type** string **Description** ID of the parent category record.

Input	Details
qualificationProcedureName	**Type** string **Description** Name of the custom qualification procedure that's executed to determine the category list. If null, the default qualification procedure is executed.
userContextInputRepresentation	**Type** Apex-defined **Description** An Apex UserContextInputRepresentation record that contains user details, such as account ID, geographical location, language preferences, and more.

Outputs

Output	Details
apiStatusOutputRepresentation	**Type** Apex-defined **Description** An Apex `runtime_industries_cpq.ApiStatusRepresentation` record that contains the status of the request, including the status code and message.
correlationId	**Type** string **Description** Unique identifier attached to requests and messages, allowing reference to a specific transaction or event chain.
resultCategoryList	**Type** Apex-defined **Description** List of filtered category records. See CategoryOutputRepresentation.
resultListCount	**Type** integer **Description** Number of category records in the result.

Example

Here's a sample input to call this invocable action from Apex code.

```
Invocable.Action action = Invocable.Action.createStandardAction('getCategories');

ConnectApi.UserContextInputRepresentation userContext = new
ConnectApi.UserContextInputRepresentation();
userContext.accountId = '001xx000003GYiEAAW';

runtime_industries_cpq.ContextDataInput data = new
runtime_industries_cpq.ContextDataInput();
String nodeNameVal = 'Quote__c';
Map<String,Object> nodeDataVal = new Map<String,Object>();
nodeDataVal.put('id',(Object)'0Q0xx0000004CDsCAM');
nodeDataVal.put('businessObjectType',(Object)'Quote');
data.nodeName = nodeNameVal;
data.nodeData = nodeDataVal;
List<runtime_industries_cpq.ContextDataInput> contextData= new
List<runtime_industries_cpq.ContextDataInput>();
contextData.add(data);
runtime_industries_cpq.AdditionalContextData additionalContextDataOut = new
runtime_industries_cpq.AdditionalContextData();
additionalContextDataOut.additionalContextData = new
List<runtime_industries_cpq.ContextDataInput>();
additionalContextDataOut.additionalContextData.add(data);


action.setInvocationParameter('catalogId', '0ZSxx0000000002GAA');
action.setInvocationParameter('correlationId', '9cbb9650-48c5-11ed-96d1-0afcf185843b');
action.setInvocationParameter('userContextInputRepresentation', userContext);
action.setInvocationParameter('enableQualificationProcedure', True);
action.setInvocationParameter('qualificationProcedureName', 'CatQual02');
action.setInvocationParameter('contextDefinitionName', 'CategoryCD');
action.setInvocationParameter('contextMappingName', 'ProductDiscoveryMapping');
action.setInvocationParameter('additionalContextData', additionalContextDataOut);
//action.setInvocationParameter('categoryDepth', 4);
//action.setInvocationParameter('parentCategoryId', '0ZGxx0000000001GAA');

List<Invocable.Action.Result> results = action.invoke();
System.debug('Search Action + '+results);
```

Here's a sample response when you call this action.


```
[
  {
    "actionName": "getCategories",
    "errors": null,
    "invocationId": null,
    "isSuccess": true,
    "outcome": null,
    "outputValues": {
      "resultCategoryList": [
        {
          "sortOrder": null,
          "qualificationContext": {
```

```
            "reason": "Category is not qualified because one or more field(s) do not
match the qualification criteria. Fields:- CategoryId - 0ZGxx0000000001GAA quotename -
 null Max_Number_of_Employees - 1500 Min_Number_of_Employees - 1500",
            "isQualified": false
          },
          "parentCategoryId": null,
          "name": "Laptops",
          "isNavigational": true,
          "id": "0ZGxx0000000001GAA",
          "hasSubCategories": true,
          "description": null,
          "childCategories": [],
          "catalogId": "0ZSxx0000000002GAA"
        },
        {
          "sortOrder": null,
          "qualificationContext": {
            "reason": null,
            "isQualified": true
          },
          "parentCategoryId": null,
          "name": "Desktops",
          "isNavigational": true,
          "id": "0ZGxx0000000002GAA",
          "hasSubCategories": false,
          "description": null,
          "childCategories": [],
          "catalogId": "0ZSxx0000000002GAA"
        },
        {
          "sortOrder": null,
          "qualificationContext": {
            "reason": null,
            "isQualified": true
          },
          "parentCategoryId": null,
          "name": "Accessories",
          "isNavigational": true,
          "id": "0ZGxx0000000003GAA",
          "hasSubCategories": false,
          "description": null,
          "childCategories": [],
          "catalogId": "0ZSxx0000000002GAA"
        }
      ],
      "resultListCount": 3,
      "apiStatusOutputRepresentation": {
        "statusMessage": null,
        "statusCode": "FetchedDetailsSuccessfully",
        "messages": []
      },
      "correlationId": "9cbb9650-48c5-11ed-96d1-0afcf185843b"
    },
    "sortOrder": -1,
```

```
    "version": 1
  }
]
```

Get Category Details Action

Get details of a category record.

This action is available in API version 64.0 and later.

You can invoke this action via Apex and Flows only.

Special Access Rules

The Get Category Details action is available in Enterprise, Unlimited, and Developer Editions where Product Discovery is enabled.

Inputs

Input	Details
additionalContextData	**Type** Apex-defined **Description** Collection of Apex runtime_industries_cpq.AdditionalContextData records that contain additional context data for nodes of the custom context definition, if applicable. You can add details for up to 10 nodes.
catalogId	**Type** string **Description** ID of the catalog record that's used to search for products within a category.
categoryId	**Type** string **Description** Required. ID of the category or subcategory that's used to search for products.
contextDefinitionName	**Type** string **Description** Name of the custom context definition that's used to create context data for categories. If null, the default context definition is used.

Input	Details
contextMappingName	**Type** string **Description** Name of the context mapping. By default, the default context mapping associated with the context definition is used.
correlationId	**Type** string **Description** Unique identifier attached to requests and messages, allowing reference to a specific transaction or event chain.
enableQualificationProcedure	**Type** boolean **Description** Indicates whether qualification rules are applied to categories (`true`) or not (`false`).
filterInputRepresentation	**Type** Apex-defined **Description** Collection of Apex `runtime_industries_cpq.FilterInputRepresentation` records that contain the filter criteria applied to the category records.
qualificationProcedureName	**Type** string **Description** Name of the custom qualification procedure that's executed to determine the category list. If null, the default qualification procedure is executed.
userContextInputRepresentation	**Type** Apex-defined **Description** An Apex UserContextInputRepresentation record that contains user details, such as account ID, geographical location, language preferences, and more.

Outputs

Output	Details
apiStatusOutputRepresentation	**Type** Apex-defined **Description** An Apex `runtime_industries_cpq.ApiStatusRepresentation` record that contains the status of the request, including the status code and message.
categoryDetailsRepresentations	**Type** Apex-defined **Description** Collection of Apex CategoryDetailsRepresentation records that contain details about the retrieved category.
correlationId	**Type** string **Description** Unique identifier attached to requests and messages, allowing reference to a specific transaction or event chain.

Example

Here's a sample input to call this invocable action from Apex code.

```
Invocable.Action action = Invocable.Action.createStandardAction('getCategoryDetails');

ConnectApi.UserContextInputRepresentation userContext = new
ConnectApi.UserContextInputRepresentation();
userContext.accountId = '001xx000003GYiEAAW';

runtime_industries_cpq.ContextDataInput data = new
runtime_industries_cpq.ContextDataInput();
String nodeNameVal = 'Quote__c';
Map<String,Object> nodeDataVal = new Map<String,Object>();
nodeDataVal.put('id',(Object)'0Q0xx0000004CDsCAM');
nodeDataVal.put('businessObjectType',(Object)'Quote');
data.nodeName = nodeNameVal;
data.nodeData = nodeDataVal;
List<runtime_industries_cpq.ContextDataInput> contextData= new
List<runtime_industries_cpq.ContextDataInput>();
contextData.add(data);
runtime_industries_cpq.AdditionalContextData additionalContextDataOut = new
runtime_industries_cpq.AdditionalContextData();
additionalContextDataOut.additionalContextData = new
List<runtime_industries_cpq.ContextDataInput>();
additionalContextDataOut.additionalContextData.add(data);
```

```
action.setInvocationParameter('categoryId', '0ZGxx0000000001GAA');
action.setInvocationParameter('correlationId', '9cbb9650-48c5-11ed-96d1-0afcf185843b');
action.setInvocationParameter('userContext', userContext);
action.setInvocationParameter('enableQualification', True);
action.setInvocationParameter('qualificationProcedure', 'CatQual02');
action.setInvocationParameter('contextDefinition', 'CategoryCD');
action.setInvocationParameter('contextMapping', 'ProductDiscoveryMapping');
action.setInvocationParameter('additionalContextData', additionalContextDataOut);

List<Invocable.Action.Result> results = action.invoke();
System.debug('Search Action + '+results);
```

Here's a sample response when you call this action.


```
{
  "root": [
    {
      "actionName": "getCategoryDetails",
      "errors": null,
      "invocationId": null,
      "isSuccess": true,
      "outcome": null,
      "outputValues": {
        "apiStatusOutputRepresentation": {
          "statusMessage": null,
          "statusCode": "FetchedDetailsSuccessfully",
          "messages": []
        },
        "correlationId": "9cbb9650-48c5-11ed-96d1-0afcf185843b",
        "categoryDetailsRepresentations": {
          "sortOrder": null,
          "qualificationContext": {
            "reason": "Category is not qualified because one or more field(s) do not
match the qualification criteria. Fields:- CategoryId - 0ZGxx0000000001GAA quotename -
 null Max_Number_of_Employees - 1500 Min_Number_of_Employees - 1500",
            "isQualified": false
          },
          "parentCategoryId": null,
          "name": "Laptops",
          "isNavigational": true,
          "id": "0ZGxx0000000001GAA",
          "hasSubCategories": true,
          "description": null,
          "childCategories": [
            {
              "sortOrder": null,
              "qualificationContext": {
                "reason": null,
                "isQualified": true
              },
              "parentCategoryId": "0ZGxx0000000001GAA",
              "name": "level1",
              "isNavigational": true,
```

```
                "id": "0ZGxx000000004rGAA",
                "hasSubCategories": true,
                "description": null,
                "childCategories": [],
                "catalogId": null
              }
            ],
            "catalogId": "0ZSxx0000000002GAA"
          }
        },
        "sortOrder": -1,
        "version": 1
      }
    ]
}
```


Get Multiple Product Details Action

Get product details for a list of products.

This action is available in API version 64.0 and later.

You can invoke this action via Apex and Flows only.

Special Access Rules

The Get Multiple Product Details action is available in Enterprise, Unlimited, and Developer Editions where Product Discovery is enabled.

Inputs

Input	Details
`additionalContextData`	**Type** Apex-defined **Description** Collection of Apex `runtime_industries_cpq.AdditionalContextData` records that contain additional context data for nodes of the custom context definition, if applicable. You can add details for up to 10 nodes.
`additionalFields`	**Type** Apex-defined **Description** An Apex AdditionalFields record that contains the additional fields that are passed for the Product2 object.
`catalogId`	**Type** string

Input	Details
	Description ID of the catalog record.
`contextDefinitionName`	**Type** string **Description** Name of the custom context definition that’s used to create context data for categories. If null, the default context definition is used.
`contextMappingName`	**Type** string **Description** Name of the context mapping. By default, the default context mapping associated with the context definition is used.
`correlationId`	**Type** string **Description** Unique identifier attached to requests and messages, allowing reference to a specific transaction or event chain.
`currencyCode`	**Type** string **Description** Currency code that’s used to calculate and show prices.
`enablePricing`	**Type** boolean **Description** Indicates whether pricing procedure must run (`true`) or not (`false`). The default value is `true`. To use this parameter, the **Pricing Procedure** setting must be enabled.
`enableQualificationProcedure`	**Type** boolean **Description** Indicates whether the qualification procedure is applied to categories (`true`) or not (`false`).
`priceBookId`	**Type** string

Input	Details
	Description ID of the pricebook that the pricing information is retrieved from.
`pricingProcedureName`	**Type** string **Description** Name of the pricing procedure to calculate product prices. If you don't enter a value, the pricing procedure selected on the **Product Discovery Settings** page is used.
`productDataInputs`	**Type** Apex-defined **Description** Required. Collection of Apex BulkProductDetailsInputBody records that contain details about the products that are to be retrieved.
`qualificationProcedureName`	**Type** string **Description** Name of the custom qualification procedure that's executed to determine the category list. If null, the default qualification procedure is executed.
`userContextInputRepresentation`	**Type** Apex-defined **Description** An Apex UserContextInputRepresentation record that contains user details, such as account ID, geographical location, language preferences, and more.

Outputs

Output	Details
`apiStatusOutputRepresentation`	**Type** Apex-defined **Description** An Apex `runtime_industries_cpq.ApiStatusRepresentation` record that contains the status of the request, including the status code and message.
`contextId`	**Type** string

Output	Details
	Description ID of the context that's created using the specified context definition.
`correlationId`	**Type** string **Description** Unique identifier attached to requests and messages, allowing reference to a specific transaction or event chain.
`productDetailsOutputRepresentation`	**Type** Apex-defined **Description** Collection of Apex BulkProductDetailsRepresentation records that contain details of available products.
`userContext`	**Type** Apex-defined **Description** An Apex ConnectApi.UserContextRepresentation record containing user information.

Example

Here's a sample input to call this invocable action from Apex code.

```
Invocable.Action action =
Invocable.Action.createStandardAction('getMultipleProductDetails');

List<runtime_industries_cpq.BulkProductDetailsInputBody> productDataList = new
List<runtime_industries_cpq.BulkProductDetailsInputBody>();
runtime_industries_cpq.BulkProductDetailsInputBody productData = new
runtime_industries_cpq.BulkProductDetailsInputBody();
productData.productId = '01tIY000000nCxhYAE';

productDataList.add(productData);

runtime_industries_cpq.BulkProductDetailsInputBodyList productList = new
runtime_industries_cpq.BulkProductDetailsInputBodyList();
productList.productData = productDataList;

action.setInvocationParameter('productDataInputs', productList);

List<Invocable.Action.Result> results = action.invoke();
System.debug('Result === ' + results);
```

Here's a sample response when you call this action.


```
[
  {
    "actionName": "getMultipleProductDetails",
    "errors": null,
    "invocationId": null,
    "isSuccess": true,
    "outcome": null,
    "outputValues": {
      "apiStatusOutputRepresentation": {
        "statusMessage": null,
        "statusCode": "FetchedDetailsSuccessfully",
        "messages": []
      },
      "correlationId": "9cbb9650-48c5-11ed-96d1-0afcf185843b",
      "contextId": null,
      "productDetailsOutputRepresentation": [
        {
          "unitOfMeasure": {
            "unitCode": null,
            "scale": null,
            "roundingMethod": null,
            "name": null,
            "id": null
          },
          "status": null,
          "qualificationContext": null,
          "productType": null,
          "productSpecificationType": null,
          "productSellingModelOptions": [
            {
              "productSellingModelId": "0jPxx0000000002EAA",
              "productSellingModel": {
                "status": "Active",
                "sellingModelType": "TermDefined",
                "pricingTermUnit": "Annual",
                "pricingTerm": 1,
                "name": "Term Based - Yearly",
                "id": "0jPxx0000000002EAA"
              },
              "productId": "01txx0000006i2oAAA",
              "id": "0iOxx000000000JEAQ"
            }
          ],
          "productRelatedComponent": {
            "unitOfMeasure": null,
            "sequence": null,
            "quoteVisibility": null,
            "quantityScaleMethod": null,
            "quantity": null,
            "productRelationshipTypeId": null,
            "productComponentGroupId": null,
            "productClassificationId": null,
            "parentSellingModelId": null,
```

```
      "parentProductId": null,
      "minQuantity": null,
      "maxQuantity": null,
      "isQuantityEditable": null,
      "isExcluded": null,
      "isDefaultComponent": null,
      "isComponentRequired": null,
      "id": null,
      "doesBundlePriceIncludeChild": null,
      "childSellingModelId": null,
      "childProductId": null
    },
    "productQuantity": {
      "quantity": null,
      "minQuantity": null,
      "maxQuantity": null
    },
    "productPricingInformation": null,
    "productInformation": null,
    "productComponentGroups": [],
    "productCode": "ASTCERT001",
    "productClassification": {
      "id": null
    },
    "prices": [
      {
        "pricingModel": {
          "unitOfMeasure": null,
          "pricingModelType": "TermDefined",
          "occurrence": 1,
          "name": "Term Based - Yearly",
          "id": "0jPxx0000000002EAA",
          "frequency": "Annual"
        },
        "priceBookId": "01sxx0000005ptpAAA",
        "priceBookEntryId": "01uxx0000008yXaAAI",
        "price": 2000,
        "isSelected": true,
        "isDerived": false,
        "isDefault": true,
        "effectiveTo": null,
        "effectiveFrom": null,
        "currencyIsoCode": "USD"
      }
    ],
    "nodeType": "simpleProduct",
    "name": "AI Specialist Certification",
    "isSoldOnlyWithOtherProds": false,
    "isQuantityEditable": null,
    "isDefaultComponent": null,
    "isComponentRequired": null,
    "isAssetizable": true,
    "isActive": true,
    "id": "01txx0000006i2oAAA",
```

```
          "endOfLifeDate": null,
          "displayUrl":
"https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcQsp4GoUP_nGiCXJ-wzYkPx-RlA_UCUX0zIDv3slxnJACqXaRd1vHJUZe6yQklUiXvEDgo&usqp=CAU",

          "discontinuedDate": null,
         "description": "The Certified Artificial Intelligence Expert certification is
 typically designed for individuals who have an in-depth understanding of artificial
intelligence (AI) concepts, algorithms, and applications and want to validate their
expertise in the field.",
          "configureDuringSale": "NotAllowed",
          "childProducts": [],
          "catalogs": [],
          "availabilityDate": null,
          "attributes": [],
          "attributeCategories": [],
          "additionalFields": []
        },
        {
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            "unitCode": null,
            "scale": null,
            "roundingMethod": null,
            "name": null,
            "id": null
          },
          "status": null,
          "qualificationContext": null,
          "productType": null,
          "productSpecificationType": null,
          "productSellingModelOptions": [
            {
              "productSellingModelId": "0jPxx0000000002EAA",
              "productSellingModel": {
                "status": "Active",
                "sellingModelType": "TermDefined",
                "pricingTermUnit": "Annual",
                "pricingTerm": 1,
                "name": "Term Based - Yearly",
                "id": "0jPxx0000000002EAA"
              },
              "productId": "01txx0000006i2rAAA",
              "id": "0iOxx000000000QEAQ"
            }
          ],
          "productRelatedComponent": {
            "unitOfMeasure": null,
            "sequence": null,
            "quoteVisibility": null,
            "quantityScaleMethod": null,
            "quantity": null,
            "productRelationshipTypeId": null,
            "productComponentGroupId": null,
            "productClassificationId": null,
            "parentSellingModelId": null,
```

```
      "parentProductId": null,
      "minQuantity": null,
      "maxQuantity": null,
      "isQuantityEditable": null,
      "isExcluded": null,
      "isDefaultComponent": null,
      "isComponentRequired": null,
      "id": null,
      "doesBundlePriceIncludeChild": null,
      "childSellingModelId": null,
      "childProductId": null
    },
    "productQuantity": {
      "quantity": null,
      "minQuantity": null,
      "maxQuantity": null
    },
    "productPricingInformation": null,
    "productInformation": null,
    "productComponentGroups": [],
    "productCode": "ACERT001",
    "productClassification": {
      "id": null
    },
    "prices": [
      {
        "pricingModel": {
          "unitOfMeasure": null,
          "pricingModelType": "TermDefined",
          "occurrence": 1,
          "name": "Term Based - Yearly",
          "id": "0jPxx0000000002EAA",
          "frequency": "Annual"
        },
        "priceBookId": "01sxx0000005ptpAAA",
        "priceBookEntryId": "01uxx0000008yXdAAI",
        "price": 2000,
        "isSelected": true,
        "isDerived": false,
        "isDefault": true,
        "effectiveTo": null,
        "effectiveFrom": null,
        "currencyIsoCode": "USD"
      }
    ],
    "nodeType": "simpleProduct",
    "name": "Admin Certification",
    "isSoldOnlyWithOtherProds": false,
    "isQuantityEditable": null,
    "isDefaultComponent": null,
    "isComponentRequired": null,
    "isAssetizable": true,
    "isActive": true,
    "id": "01txx0000006i2rAAA",
```

```
          "endOfLifeDate": null,
          "displayUrl": null,
          "discontinuedDate": null,
          "description": null,
          "configureDuringSale": "Allowed",
          "childProducts": [],
          "catalogs": [],
          "availabilityDate": null,
          "attributes": [],
          "attributeCategories": [],
          "additionalFields": []
        }
      ]
    },
    "sortOrder": -1,
    "version": 1
  }
]
```


Get Products Action

Get products from the specified catalog, category, or subcategory, including product qualification and pricing details.

If a catalog, category, or subcategory isn't specified, then this action retrieves all the products from all catalogs. This action is available in API version 62.0 and later.

You can invoke this action via Apex and Flows only.

Special Access Rules

The Get Products action is available in Enterprise, Unlimited, and Developer Editions where Product Discovery is enabled.

Inputs

Input	Details
additionalContextData	**Type** Apex-defined **Description** An array of Apex `runtime_industries_cpq.AdditionalContextData` records that contain the additional nodes that are used along with the context definition nodes for data hydration. The maximum number of supported nodes is 10.
additionalFields	**Type** Apex-defined

Input	Details
	Description An Apex `runtime_industries_cpq.AdditionalFields` record that contains an array of additional standard or custom fields to include in the response. The supported objects are: • `Product2` • `ProductAttributeDefinition`—If the fields defined for the `ProductAttributeDefinition` object aren't available for the `ProductClassificationAttr` object, then the API request fails.
`catalogId`	**Type** string **Description** Catalog ID that's used to find and retrieve the products.
`categoryId`	**Type** string **Description** ID of the category or subcategory to get the products for.
`contextDefinition`	**Type** string **Description** API name of the context definition that's used for context creation. If you don't specify a value, the context selected on the Product Discovery Settings page from Setup is used.
`contextMapping`	**Type** string **Description** API name of the context mapping that's used for data hydration. The value of this parameter is used only if it belongs to the specified context definition.
`correlationId`	**Type** string **Description** Unique ID that's used to reference a series of related actions.

Input	Details
`currencyCode`	**Type** string **Description** Currency code that's used to calculate and show prices. Only the products with the currency code matching the specified currency code are fetched.
`cursor`	**Type** string **Description** Unique identifier that represents the position of the product from which the next set of results are retrieved.
`enablePricing`	**Type** boolean **Description** Indicates whether the pricing procedure must run (`true`) or not (`false`). The default value is `true`. To use this parameter, you must enable the Pricing Procedure setting from Setup.
`enableQualification`	**Type** boolean **Description** Indicates whether the qualification procedure must run (`true`) or not (`false`). The default value is `true`. To use this parameter, you must enable the Qualification Procedure setting from Setup.
`executeConfigurationRules`	**Type** boolean **Description** Indicates whether configuration rules must run (`true`) or not (`false`). Available in API version 67.0 and later.
`filter`	**Type** Apex-defined **Description** A collection of Apex `runtime_industries_cpq.FilterInputRepresentation` records where each record contains a related object and the filter criteria that's applied on the object. The `filter` parameter supports only the `name` property.

Input	Details
	The supported operators are: • `eq` • `in` • `contains` If this parameter contains multiple criteria, all the criteria are applied.
`includeCatalogDetails`	**Type** boolean **Description** Indicates whether catalog details must be included in the response (`true`) or not (`false`).
`limit`	**Type** integer **Description** Maximum number of results to be returned in the response. Specify a value from 1 through 100. The default value is 10.
`orderBy`	**Type** string **Description** Comma-separated string of key-value pairs that specify how the results are sorted. Each string must contain a field name and its sort order. For example, `["name:asc",”custom_field:asc”]`.
`priceBookId`	**Type** string **Description** ID of the pricebook that the pricing details are retrieved from.
`pricingProcedure`	**Type** string **Description** API name of the pricing procedure to calculate product prices. If you don’t specify a value, the pricing procedure selected on the Product Discovery Settings page from Setup is used.
`productClassificationId`	**Type** string **Description** ID of the product classification that’s used to filter the products.

Input	Details
`qualificationProcedure`	**Type** string **Description** API name of the qualification procedure to evaluate product eligibility. If you don't specify a value, the qualification procedure selected on the Product Discovery Settings page from Setup is used.
`transactionContextId`	**Type** string **Description** Context ID of the quote or order. Available in API version 67.0 and later.
`transactionId`	**Type** string **Description** ID of the quote or order. Available in API version 67.0 and later.
`relatedObjectFilters`	**Type** Apex-defined **Description** A collection of Apex `runtime_industries_cpq.RelatedObjectFilterInputRepresentation` records, where each record contains a related object and the filter criteria that's applied on the object.
`userContext`	**Type** Apex-defined **Description** An Apex `ConnectApi.UserContextInputRepresentation` record that contains the user details to evaluate product eligibility and calculate prices.

Outputs

Output	Details
`apiStatus`	**Type** Apex-defined

Output	Details
	Description An Apex `runtime_industries_cpq.ApiStatusRepresentation` record that contains a status code and message.
`contextId`	**Type** string **Description** ID of the context that's created by using the specified context definition.
`correlationId`	**Type** string **Description** ID to reference a series of related actions.
`cursor`	**Type** string **Description** Unique identifier that represents the position of the product from which the next set of results are retrieved.
`results`	**Type** Apex-defined **Description** An Apex `runtime_industries_cpq.ProductListRepresentation` record that contains the retrieved products.
`userContext`	**Type** Apex-defined **Description** An Apex `ConnectApi.UserContextRepresentation` record that includes the user details.

Example

Here's a sample input to call this invocable action from Apex code.

```
Invocable.Action action = Invocable.Action.createStandardAction('getProducts');
action.setInvocationParameter('executeConfigurationRules', true);
action.setInvocationParameter('catalogId', '0ZSVW000000AhdB4AS');
action.setInvocationParameter('transactionId', '0Q0VW0000018J6X0AU');
```

```
List<String> orderBy = new List<String>();
orderBy.add('name:asc');
action.setInvocationParameter('orderBy', orderBy);
List<Invocable.Action.Result> results = action.invoke();

for (Invocable.Action.Result res : results) {
    if (res.isSuccess()) {
        Map<String, Object> outMap = res.getOutputParameters();
        String jsonOutput = JSON.serialize(outMap);
        System.debug(jsonOutput);
    }
}
```

Here's a sample response for this action.


```
{
  "facets": [],
  "apiStatus": {
    "statusMessage": null,
    "statusCode": "FetchedDetailsSuccessfully",
    "messages": []
  },
  "results": [
    {
      "unitOfMeasure": null,
      "status": null,
      "qualificationContext": {
        "reason": null,
        "isQualified": true
      },
      "productType": null,
      "productSpecificationType": null,
      "productSellingModelOptions": [
        {
          "productSellingModelId": "0jPVW0000001fh32AA",
          "productSellingModel": {
            "status": "Active",
            "sellingModelType": "TermDefined",
            "pricingTermUnit": "Annual",
            "pricingTerm": 1,
            "name": "Term Annual",
            "id": "0jPVW0000001fh32AA"
          },
          "productId": "01tVW000003l7txYAA",
          "id": "0iOVW00000049w22AA"
        },
        {
          "productSellingModelId": "0jPVW0000001fh52AA",
          "productSellingModel": {
            "status": "Active",
            "sellingModelType": "Evergreen",
            "pricingTermUnit": "Months",
            "pricingTerm": 1,
            "name": "Evergreen Monthly",
            "id": "0jPVW0000001fh52AA"
```

```
        },
        "productId": "01tVW000003l7txYAA",
        "id": "0iOVW00000049w12AA"
      },
      {
        "productSellingModelId": "0jPVW0000001fh62AA",
        "productSellingModel": {
          "status": "Active",
          "sellingModelType": "TermDefined",
          "pricingTermUnit": "Months",
          "pricingTerm": 1,
          "name": "Term Monthly",
          "id": "0jPVW0000001fh62AA"
        },
        "productId": "01tVW000003l7txYAA",
        "id": "0iOVW00000049w32AA"
      }
    ],
    "productRelatedComponent": null,
    "productQuantity": null,
    "productPricingInformation": null,
    "productInformation": null,
    "productComponentGroups": [],
    "productCode": "QB-API",
    "productClassification": {
      "id": "11BVW000004ljiZ2AQ"
    },
    "prices": [
      {
        "pricingModel": {
          "unitOfMeasure": null,
          "pricingModelType": "TermDefined",
          "occurrence": 1,
          "name": "Term Annual",
          "id": "0jPVW0000001fh32AA",
          "frequency": "Annual"
        },
        "priceBookId": "01sVW0000024PZlYAM",
        "priceBookEntryId": "01uVW000000jzfGYAQ",
        "price": 15000,
        "isSelected": false,
        "isDerived": false,
        "isDefault": true,
        "effectiveTo": null,
        "effectiveFrom": null,
        "currencyIsoCode": "USD"
      },
      {
        "pricingModel": {
          "unitOfMeasure": null,
          "pricingModelType": "Evergreen",
          "occurrence": 1,
          "name": "Evergreen Monthly",
          "id": "0jPVW0000001fh52AA",
```

```
          "frequency": "Months"
        },
        "priceBookId": "01sVW0000024PZlYAM",
        "priceBookEntryId": "01uVW000000jzg8YAA",
        "price": 2000,
        "isSelected": false,
        "isDerived": false,
        "isDefault": false,
        "effectiveTo": null,
        "effectiveFrom": null,
        "currencyIsoCode": "USD"
      },
      {
        "pricingModel": {
          "unitOfMeasure": null,
          "pricingModelType": "TermDefined",
          "occurrence": 1,
          "name": "Term Monthly",
          "id": "0jPVW0000001fh62AA",
          "frequency": "Months"
        },
        "priceBookId": "01sVW0000024PZlYAM",
        "priceBookEntryId": "01uVW000000jzg9YAA",
        "price": 1500,
        "isSelected": false,
        "isDerived": false,
        "isDefault": false,
        "effectiveTo": null,
        "effectiveFrom": null,
        "currencyIsoCode": "USD"
      }
    ],
    "nodeType": "simpleProduct",
    "name": "Additional API",
    "isSoldOnlyWithOtherProds": false,
    "isQuantityEditable": null,
    "isDefaultComponent": null,
    "isComponentRequired": null,
    "isAssetizable": true,
    "isActive": true,
    "id": "01tVW000003l7txYAA",
    "endOfLifeDate": null,
    "displayUrl": "/resource/add_api",
    "discontinuedDate": null,
    "description": null,
    "configureDuringSale": "Allowed",
    "configurationRules": [
      {
        "type": "disable",
        "details": [
          {
            "message": "This is disabled"
          }
        ]
```

```
      }
    ],
    "childProducts": [],
    "categories": [
      {
        "sortOrder": null,
        "qualificationContext": {
          "reason": null,
          "isQualified": true
        },
        "parentCategoryId": null,
        "name": "API",
        "isNavigational": null,
        "id": "0ZGVW000000IUEG4A4",
        "hasSubCategories": null,
        "description": null,
        "childCategories": null,
        "catalogId": "0ZSVW000000AhdB4AS"
      }
    ],
    "catalogs": [],
    "availabilityDate": null,
    "attributeCategories": [],
    "additionalFields": []
  },
  {
    "unitOfMeasure": null,
    "status": null,
    "qualificationContext": {
      "reason": null,
      "isQualified": true
    },
    "productType": null,
    "productSpecificationType": null,
    "productSellingModelOptions": [
      {
        "productSellingModelId": "0jPVW0000001fh32AA",
        "productSellingModel": {
          "status": "Active",
          "sellingModelType": "TermDefined",
          "pricingTermUnit": "Annual",
          "pricingTerm": 1,
          "name": "Term Annual",
          "id": "0jPVW0000001fh32AA"
        },
        "productId": "01tVW000003l7tyYAA",
        "id": "0iOVW00000049w42AA"
      }
    ],
    "productRelatedComponent": null,
    "productQuantity": null,
    "productPricingInformation": null,
    "productInformation": null,
    "productComponentGroups": [],
```

```
      "productCode": "QB-API-FLEX",
      "productClassification": {
        "id": null
      },
      "prices": [
        {
          "pricingModel": {
            "unitOfMeasure": null,
            "pricingModelType": "TermDefined",
            "occurrence": 1,
            "name": "Term Annual",
            "id": "0jPVW0000001fh32AA",
            "frequency": "Annual"
          },
          "priceBookId": "01sVW0000024PZlYAM",
          "priceBookEntryId": "01uVW000000jzfFYAQ",
          "price": 450,
          "isSelected": false,
          "isDerived": false,
          "isDefault": true,
          "effectiveTo": null,
          "effectiveFrom": null,
          "currencyIsoCode": "USD"
        }
      ],
      "nodeType": "simpleProduct",
      "name": "Additional API Flex (100M)",
      "isSoldOnlyWithOtherProds": false,
      "isQuantityEditable": null,
      "isDefaultComponent": null,
      "isComponentRequired": null,
      "isAssetizable": true,
      "isActive": true,
      "id": "01tVW000003l7tyYAA",
      "endOfLifeDate": null,
      "displayUrl": "/resource/api_flex",
      "discontinuedDate": null,
      "description": "API instances remain under management until they are deleted.
Instances of API Manager are aggregated using a Max Concurrent model. The usage for a
month is the highest number of APIs Managed in a single given hour during a month.",
      "configureDuringSale": null,
      "configurationRules": [],
      "childProducts": [],
      "categories": [
        {
          "sortOrder": null,
          "qualificationContext": {
            "reason": null,
            "isQualified": true
          },
          "parentCategoryId": null,
          "name": "API",
          "isNavigational": null,
          "id": "0ZGVW000000IUEG4A4",
```

```
        "hasSubCategories": null,
        "description": null,
        "childCategories": null,
        "catalogId": "0ZSVW000000AhdB4AS"
      }
    ],
    "catalogs": [],
    "availabilityDate": null,
    "attributeCategories": [],
    "additionalFields": []
  },
  {
    "unitOfMeasure": null,
    "status": null,
    "qualificationContext": {
      "reason": null,
      "isQualified": true
    },
    "productType": null,
    "productSpecificationType": null,
    "productSellingModelOptions": [
      {
        "productSellingModelId": "0jPVW0000001fh32AA",
        "productSellingModel": {
          "status": "Active",
          "sellingModelType": "TermDefined",
          "pricingTermUnit": "Annual",
          "pricingTerm": 1,
          "name": "Term Annual",
          "id": "0jPVW0000001fh32AA"
        },
        "productId": "01tVW000003l7tzYAA",
        "id": "0iOVW00000049w52AA"
      }
    ],
    "productRelatedComponent": null,
    "productQuantity": null,
    "productPricingInformation": null,
    "productInformation": null,
    "productComponentGroups": [],
    "productCode": "QB-API-GOVT",
    "productClassification": {
      "id": null
    },
    "prices": [
      {
        "pricingModel": {
          "unitOfMeasure": null,
          "pricingModelType": "TermDefined",
          "occurrence": 1,
          "name": "Term Annual",
          "id": "0jPVW0000001fh32AA",
          "frequency": "Annual"
        },
```

```
          "priceBookId": "01sVW0000024PZlYAM",
          "priceBookEntryId": "01uVW000000jzfHYAQ",
          "price": 1450,
          "isSelected": false,
          "isDerived": false,
          "isDefault": true,
          "effectiveTo": null,
          "effectiveFrom": null,
          "currencyIsoCode": "USD"
        }
      ],
      "nodeType": "simpleProduct",
      "name": "Additional API Gov",
      "isSoldOnlyWithOtherProds": false,
      "isQuantityEditable": null,
      "isDefaultComponent": null,
      "isComponentRequired": null,
      "isAssetizable": true,
      "isActive": true,
      "id": "01tVW000003l7tzYAA",
      "endOfLifeDate": null,
      "displayUrl": "/resource/api_govt",
      "discontinuedDate": null,
      "description": "API instances remain under management until they are deleted.
Instances of API Manager are aggregated using a Max Concurrent model. The usage for a
month is the highest number of APIs Managed in a single given hour during a month.",
      "configureDuringSale": null,
      "configurationRules": [],
      "childProducts": [],
      "categories": [
        {
          "sortOrder": null,
          "qualificationContext": {
            "reason": null,
            "isQualified": true
          },
          "parentCategoryId": null,
          "name": "API",
          "isNavigational": null,
          "id": "0ZGVW000000IUEG4A4",
          "hasSubCategories": null,
          "description": null,
          "childCategories": null,
          "catalogId": "0ZSVW000000AhdB4AS"
        }
      ],
      "catalogs": [],
      "availabilityDate": null,
      "attributeCategories": [],
      "additionalFields": []
    },
    {
      "unitOfMeasure": null,
      "status": null,
```

```
"qualificationContext": {
  "reason": null,
  "isQualified": true
},
"productType": null,
"productSpecificationType": null,
"productSellingModelOptions": [
  {
    "productSellingModelId": "0jPVW0000001fh32AA",
    "productSellingModel": {
      "status": "Active",
      "sellingModelType": "TermDefined",
      "pricingTermUnit": "Annual",
      "pricingTerm": 1,
      "name": "Term Annual",
      "id": "0jPVW0000001fh32AA"
    },
    "productId": "01tVW000003l7u0YAA",
    "id": "0iOVW00000049w92AA"
  }
],
"productRelatedComponent": null,
"productQuantity": null,
"productPricingInformation": null,
"productInformation": null,
"productComponentGroups": [],
"productCode": "QB-API-PREP",
"productClassification": {
  "id": null
},
"prices": [
  {
    "pricingModel": {
      "unitOfMeasure": null,
      "pricingModelType": "TermDefined",
      "occurrence": 1,
      "name": "Term Annual",
      "id": "0jPVW0000001fh32AA",
      "frequency": "Annual"
    },
    "priceBookId": "01sVW0000024PZlYAM",
    "priceBookEntryId": "01uVW000000jzfSYAQ",
    "price": 3240,
    "isSelected": false,
    "isDerived": false,
    "isDefault": true,
    "effectiveTo": null,
    "effectiveFrom": null,
    "currencyIsoCode": "USD"
  }
],
"nodeType": "simpleProduct",
"name": "Additional API Pre-Prod",
"isSoldOnlyWithOtherProds": false,
```

```
      "isQuantityEditable": null,
      "isDefaultComponent": null,
      "isComponentRequired": null,
      "isAssetizable": true,
      "isActive": true,
      "id": "01tVW000003l7u0YAA",
      "endOfLifeDate": null,
      "displayUrl": "/resource/api_preprod",
      "discontinuedDate": null,
      "description": "API instances remain under management until they are deleted.
Instances of API Manager are aggregated using a Max Concurrent model. The usage for a
month is the highest number of APIs Managed in a single given hour during a month.",
      "configureDuringSale": null,
      "configurationRules": [],
      "childProducts": [],
      "categories": [
        {
          "sortOrder": null,
          "qualificationContext": {
            "reason": null,
            "isQualified": true
          },
          "parentCategoryId": null,
          "name": "API",
          "isNavigational": null,
          "id": "0ZGVW000000IUEG4A4",
          "hasSubCategories": null,
          "description": null,
          "childCategories": null,
          "catalogId": "0ZSVW000000AhdB4AS"
        }
      ],
      "catalogs": [],
      "availabilityDate": null,
      "attributeCategories": [],
      "additionalFields": []
    },
    {
      "unitOfMeasure": null,
      "status": null,
      "qualificationContext": {
        "reason": null,
        "isQualified": true
      },
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Instances of API Manager are aggregated using a Max Concurrent model. The usage for a
month is the highest number of APIs Managed in a single given hour during a month.",
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            "frequency": "Months"
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      "id": "01tVW000003l7u3YAA",
      "endOfLifeDate": null,
      "displayUrl": "/resource/add_auto_credit",
      "discontinuedDate": null,
      "description": "Your QuantumBit Automation subscription plan includes a quota of
Automation QB Credits that you can use with QB RPA, QB Orchestrator, or a combination
of both products. As you use processes to create automations, your total credit quota
is depleted. 1 QB Automation Credit = 2 RPA Bot Minutes, 1 QB Automation Credit = 100
RPA API Calls, 1 QB Automation Credit = 50 Orchestrator Tasks",
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```

```
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```

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      "endOfLifeDate": null,
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      "discontinuedDate": null,
     "description": "With QuantumBit DataGraph, you can reuse multiple APIs in a single
 request. Enterprise architects can easily unify APIs into one data service — all without
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      "isActive": true,
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      "displayUrl": "/resource/api_access",
      "discontinuedDate": null,
      "description": "QuantumBit API Experience Hub enables you to create vibrant
ecosystems and grow engagement for your API products",
      "configureDuringSale": null,
      "configurationRules": [],
      "childProducts": [],
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          "description": null,
          "childCategories": null,
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          "id": "0iOVW00000049x22AA"
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      "productInformation": null,
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            "id": "0jPVW0000001fh32AA",
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      "id": "01tVW000003l7uhYAA",
      "endOfLifeDate": null,
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      "discontinuedDate": null,
      "description": "System maintenance, administration, and configuration performed
by a certified technician. Also includes 24x7 phone service and monthly on-site check-ups.
 20% of hardware.",
      "configureDuringSale": null,
      "configurationRules": [],
      "childProducts": [],
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          "description": null,
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```

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        "id": "0jPVW0000001fh32AA"
      },
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      "id": "0iOVW0000005X1l2AE"
    },
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  "productPricingInformation": null,
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        "occurrence": 1,
        "name": "Term Annual",
        "id": "0jPVW0000001fh32AA",
        "frequency": "Annual"
      },
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```

```
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      "effectiveFrom": null,
      "currencyIsoCode": "USD"
    }
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  "isComponentRequired": null,
  "isAssetizable": true,
  "isActive": true,
  "id": "01tVW000003olLNYAY",
  "endOfLifeDate": null,
  "displayUrl": null,
  "discontinuedDate": null,
  "description": null,
  "configureDuringSale": null,
  "configurationRules": [],
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  "categories": [
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        "reason": null,
        "isQualified": true
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      "id": "0ZGVW000000ItXV4A0",
      "hasSubCategories": null,
      "description": null,
      "childCategories": null,
      "catalogId": "0ZSVW000000AhdB4AS"
    }
  ],
```

```
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  ],
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  "correlationId": "ee883b85-2c99-48fb-88e6-95cb35e8751e",
  "cursor": "MTAwMDAwMDAxMw=="
}
```


Usage of an Apex-Defined Data Type in a Flow

To use an Apex-defined input parameter in a Flow, follow these guidelines.

Create an Apex class

Create an Apex class defining the input and output parameters. In the flow, include the Apex-defined input parameters for which you want to add the details. In this example, we have created a class named ProductServiceAction that takes an object's API name and record ID as input, and returns the additional context data.

```
public class ProductServiceAction {
    // Define input parameters
    public class FlowInput{
        @InvocableVariable(required=false)
        public String objectApiName;

        @InvocableVariable(required=false)
        public String recordId;
    }

    // Define output parameters
    public class FlowOutput{
        @invocableVariable
        public runtime_industries_cpq.AdditionalContextData
additionalContextDataFinalOutput = new runtime_industries_cpq.AdditionalContextData();

        @invocableVariable
        public runtime_industries_cpq.RelatedObjectFilterInputRepresentation
relatedObjectFilter = new runtime_industries_cpq.RelatedObjectFilterInputRepresentation();


        @invocableVariable
        public ConnectApi.UserContextInputRepresentation userContext = new
ConnectApi.UserContextInputRepresentation();

        }

    // This method is invoked from a Flow
    @InvocableMethod(label='Process Input' description='Creates the Array of
ContextDataInput for additional Context Data')
    public static List<FlowOutput> processContextData(List<FlowInput> inputs){
        String apiName;
        String recId;
        FlowOutput output = new FlowOutput();
```

```
    // Capture input from the flow
        for(FlowInput input: inputs){
            apiName = input.objectApiName;
            recId = input.recordId;
            }
    // Populate the ContextDataInput list to store additional context data
        List<runtime_industries_cpq.ContextDataInput> listContextData = new
List<runtime_industries_cpq.ContextDataInput>();
        runtime_industries_cpq.ContextDataInput cd1 = new
runtime_industries_cpq.ContextDataInput();
        cd1.nodeName = apiName;
        cd1.nodeData = new Map<String, Object>();
        cd1.nodeData.put('id',recId);
        listContextData.add(cd1);

       output.additionalContextDataFinalOutput.additionalContextData = listContextData;

        List<FlowOutput> flowOutputs = new List<FlowOutput>();
        flowOutputs.add(output);

        List<runtime_industries_cpq.RelatedObjectFilter> relatedObjectFilterList = new
 List<runtime_industries_cpq.RelatedObjectFilter>();

        runtime_industries_cpq.RelatedObjectFilter relatedObjectFilter = new
runtime_industries_cpq.RelatedObjectFilter();

        relatedObjectFilter.objectName = 'ProductSpecificationRecType';
        List<runtime_industries_cpq.FilterCriteriaInputRepresentation> criteriaList =
new  List<runtime_industries_cpq.FilterCriteriaInputRepresentation>();
        runtime_industries_cpq.FilterCriteriaInputRepresentation criteria = new
runtime_industries_cpq.FilterCriteriaInputRepresentation();
        criteria.property = 'IsCommercial';
        criteria.operator = 'eq';
        criteria.value = 'true';
        criteriaList.add(criteria);
        relatedObjectFilter.criteria = criteriaList;

       relatedObjectFilterList.add(relatedObjectFilter);

        output.relatedObjectFilter.relatedObjectFilter  = relatedObjectFilterList;

  output.userContext.accountId = '001Hs00003r17HNIAY';

        return flowOutputs;

    }

}
```

Create a Flow with the necessary variables and components

Create a flow that enables users to add a search term to find products. Add the ProductService action that you've created above by using Apex. When a flow is invoked from a record, the flow sends the record's objectApiName and recordId to the Apex class, which then generates the flow output. The flow passes the objectApiName and recordId of the record that the flow is invoked from to the Apex class to generate the flow output. See Example of How to Create a Flow for Product Discovery.

Configure the action

Configure the action (for example, Get Products action) to add values for the Apex-defined input parameters. Use the output of the created Apex class as the input of the Apex-defined parameter in the Get Products action, which users can use to get products.

Get Product Details Action

Get details such as attributes, hierarchy, and cardinality for the specified product.

This action is available in API version 62.0 and later.

You can invoke this action via Apex and Flows only.

Special Access Rules

The Get Product Details action is available in Enterprise, Unlimited, and Developer Editions where Product Discovery is enabled.

Inputs

Input	Details
`additionalContextData`	**Type** Apex-defined **Description** An array of Apex `runtime_industries_cpq.AdditionalContextData` records that contain the additional nodes that are used along with the context definition nodes for data hydration. The maximum number of supported nodes is 10.
`additionalFields`	**Type** Apex-defined **Description** An Apex `runtime_industries_cpq.AdditionalFields` record that contains an array of additional standard or custom fields to include in the response. The supported objects are: • `Product2` • `ProductAttributeDefinition`—If the fields defined for the `ProductAttributeDefinition` object aren't available for the `ProductClassificationAttr` object, then the API request fails.
`catalogId`	**Type** string **Description** Catalog ID that's used to find and retrieve the products.

Input	Details
`contextDefinition`	**Type** string **Description** API name of the context definition for context creation. If you don't provide a value, the context selected on the Product Discovery Settings page from Setup is used.
`contextMapping`	**Type** string **Description** API name of the context mapping for data hydration. The value of this parameter is used only if it belongs to the specified context definition.
`correlationId`	**Type** string **Description** Currency code that's used to calculate and show prices. Only the products with the currency code matching the specified currency code are fetched.
`currencyCode`	**Type** string **Description** Currency code that's used to calculate and show prices.
`enablePricing`	**Type** boolean **Description** Indicates whether the pricing procedure must run (`true`) or not (`false`). The default value is `true`. To use this parameter, you must enable the Pricing Procedure setting from Setup.
`enableQualification`	**Type** boolean **Description** Indicates whether the qualification procedure must run (`true`) or not (`false`). The default value is `true`. To use this parameter, you must enable the Qualification Procedure setting from Setup.

Input	Details
`priceBookId`	**Type** string **Description** ID of the pricebook from which you want to retrieve the pricing details.
`pricingProcedure`	**Type** string **Description** API name of the pricing procedure to calculate product prices. If you don't specify a value, the pricing procedure selected on the Product Discovery Settings page from Setup is used.
`productId`	**Type** string **Description** Required. ID of the product to get the details for.
`productSellingModelId`	**Type** string **Description** ID of the product selling model.
`qualificationProcedure`	**Type** string **Description** API name of the qualification procedure to evaluate product eligibility. If you don't specify a value, the qualification procedure selected on the Product Discovery Settings page from Setup is used.
`userContext`	**Type** Apex-defined **Description** An Apex `ConnectApi.UserContextInputRepresentation` record that contains the user details to evaluate product eligibility and calculate prices.

Outputs

Output	Details
`apiStatus`	**Type** Apex-defined **Description** An Apex `runtime_industries_cpq.ApiStatusRepresentation` record that contains a status code and message.
`contextId`	**Type** string **Description** ID of the context that's created by using the specified context definition.
`correlationId`	**Type** string **Description** ID to reference a series of related actions.
`results`	**Type** Apex-defined **Description** An Apex `runtime_industries_cpq.ProductDetailsRepresentation` record that contains the product details.
`userContext`	**Type** Apex-defined **Description** An Apex `ConnectApi.UserContextRepresentation` record that contains the user details.

Example

Usage of an Apex-Defined Data Type in a Flow

To use an Apex-defined input parameter in a flow, follow these guidelines.

Create an Apex Class

Create an Apex class defining the input and output parameters. In the flow, include the Apex-defined input parameters for which you want to add the details. In this example, we have created a class named ProductServiceAction that takes an object's API name and record ID as input, and returns the additional context data.

```
public class ProductServiceAction {
    // Define input parameters
    public class FlowInput{
        @InvocableVariable(required=false)
        public String objectApiName;

        @InvocableVariable(required=false)
        public String recordId;
    }

    // Define output parameters
    public class FlowOutput{
        @invocableVariable
        public runtime_industries_cpq.AdditionalContextData
additionalContextDataFinalOutput = new runtime_industries_cpq.AdditionalContextData();

    }

    // This method is invoked from a Flow
    @InvocableMethod(label='Process Input' description='Creates the Array of
ContextDataInput for additional Context Data')
    public static List<FlowOutput> processContextData(List<FlowInput> inputs){
        String apiName;
        String recId;
        FlowOutput output = new FlowOutput();

    // Capture input from the flow
        for(FlowInput input: inputs){
            apiName = input.objectApiName;
            recId = input.recordId;
            }
    // Populate the ContextDataInput list to store additional context data
        List<runtime_industries_cpq.ContextDataInput> listContextData = new
List<runtime_industries_cpq.ContextDataInput>();
        runtime_industries_cpq.ContextDataInput cd1 = new
runtime_industries_cpq.ContextDataInput();
        cd1.nodeName = apiName;
        cd1.nodeData = new Map<String, Object>();
        cd1.nodeData.put('id',recId);
        listContextData.add(cd1);

       output.additionalContextDataFinalOutput.additionalContextData = listContextData;

        List<FlowOutput> flowOutputs = new List<FlowOutput>();
        flowOutputs.add(output);

        List<runtime_industries_cpq.RelatedObjectFilter> relatedObjectFilterList = new
 List<runtime_industries_cpq.RelatedObjectFilter>();
```

```
        runtime_industries_cpq.RelatedObjectFilter relatedObjectFilter = new
runtime_industries_cpq.RelatedObjectFilter();

        relatedObjectFilter.objectName = 'ProductSpecificationRecType';
        List<runtime_industries_cpq.FilterCriteriaInputRepresentation> criteriaList =
new  List<runtime_industries_cpq.FilterCriteriaInputRepresentation>();
        runtime_industries_cpq.FilterCriteriaInputRepresentation criteria = new
runtime_industries_cpq.FilterCriteriaInputRepresentation();
        criteria.property = 'IsCommercial';
        criteria.operator = 'eq';
        criteria.value = 'true';
        criteriaList.add(criteria);
        relatedObjectFilter.criteria = criteriaList;

        relatedObjectFilterList.add(relatedObjectFilter);
        output.relatedObjectFilter.relatedObjectFilter  = relatedObjectFilterList;

  output.userContext.accountId = '001DU000001nx9BYAQ';

        return flowOutputs;

    }

}
```

Create a Flow with the Necessary Variables and Components

Create a flow that enables users to add a search term to find products. Add the ProductService action that you've created above by using Apex. When a flow is invoked from a record, the flow sends the record's objectApiName and recordId to the Apex class, which then generates the flow output. The flow passes the objectApiName and recordId of the record that the flow is invoked from to the Apex class to generate the flow output. See Example of How to Create a Flow for Product Discovery.

Configure the action

Configure the action (for example, Get Product Details action) to add values for the Apex-defined input parameters. Use the output of the created Apex class as the input of the Apex-defined parameter in the Get Product Details action, which users can use to get the product details.

Search Product with Guided Selection Action

Use guided product selection to search for products.

This action is available in API version 64.0 and later.

You can invoke this action via Apex and Flows only.

Special Access Rules

The Search Product with Guided Selection action is available in Enterprise, Unlimited, and Developer Editions where Product Discovery is enabled.

Inputs

Input	Details
`additionalContextData`	**Type** Apex-defined **Description** Collection of Apex `runtime_industries_cpq.AdditionalContextData` records that contain the nodes used in addition to context definition nodes for data hydration. This parameter can contain up to 10 nodes.
`additionalFields`	**Type** Apex-defined **Description** An Apex `runtime_industries_cpq.AdditionalFields` record that contains the collection of additional fields to be included in the response. This parameter supports only the fields from the Product2 and ProductAttributeDefinition objects. The fields defined for the ProductAttributeDefinition object must also be available on the ProductClassificationAttr object.
`catalogId`	**Type** string **Description** ID of the catalog to find and retrieve products.
`categoryId`	**Type** string **Description** ID of the category or subcategory to find and retrieve products.
`contextDefinition`	**Type** string **Description** API name of the context definition that's used for context creation. If you don't specify a value, the context selected on the **Product Discovery Settings** page is used.
`contextMapping`	**Type** string **Description** API name of the context mapping that's used for data hydration. The value of this parameter is used only if it belongs to the specified context definition.
`correlationId`	**Type** string

Input	Details
	Description Unique ID that's used to reference a series of related actions.
`currencyCode`	**Type** string **Description** Currency code that's used to calculate and show prices. Only the products with the currency code matching the entered currency code are fetched.
`executeConfigurationRules`	**Type** boolean **Description** Indicates whether configuration rules must run (`true`) or not (`false`). Available in API version 67.0 and later.
`enablePricing`	**Type** boolean **Description** Indicates whether pricing procedure must run (`true`) or not (`false`). The default value is `true`. To use this parameter, the **Pricing Procedure** setting must be enabled.
`enableQualification`	**Type** boolean **Description** Indicates whether qualification procedure must run (`true`) or not (`false`). The default value is `true`. To use this parameter, the **Qualification Procedure** setting must be enabled.
`filterInputRepresentation`	**Type** Apex-defined **Description** An Apex `runtime_industries_cpq.FilterInputRepresentation` record that contains the filter criteria. This parameter supports only the `name` property and the `eq`, `in`, or `contains` operators. If it contains multiple criteria, all the criteria are applied.
`guidedSelectionResponseId`	**Type** string **Description** Response identifier that stores user responses specified in the guided product selection window.
`includeCatalogDetails`	**Type** boolean

Input	Details
	Description Indicates whether catalog details must be included in the response (`true`) or not (`false`).
`orderBy`	**Type** string **Description** Comma-delimited string of key-value pairs that specify how results are sorted. Each string must contain a field name and its sort order. For example, ["name:asc",”custom_field:asc”].
`priceBookId`	**Type** string **Description** ID of the pricebook that the pricing information is retrieved from.
`pricingProcedure`	**Type** string **Description** API name of the pricing procedure to calculate product prices. If you don’t specify a value, the pricing procedure selected on the **Product Discovery Settings** page is used.
`productClassificationId`	**Type** string **Description** ID of the product classification that's used to filter products.
`productCursor`	**Type** string **Description** Unique identifier that represents the position of the product from which the next set of results are retrieved.
`qualificationProcedure`	**Type** string **Description** API name of the qualification procedure to evaluate product eligibility. If you don’t specify a value, the qualification procedure selected on the **Product Discovery Settings** page is used.
`recordLimit`	**Type** integer

Input	Details
	Description Maximum number of results to be returned in the response. Specify a value from 1 through 100. Default value is 10.
`relatedObjectFilters`	**Type** Apex-defined **Description** Collection of Apex `runtime_industries_cpq.RelatedObjectFilterInputRepresentation` records, each containing a related object and the filter criteria that's applied on the object.
`searchTerms`	**Type** Apex-defined **Description** Collection of terms that are used to search products. See GuidedSelectionSearchTerm.
`transactionContextId`	**Type** string **Description** Context ID of the quote or order. Available in API version 67.0 and later.
`transactionId`	**Type** string **Description** ID of the quote or order. Available in API version 67.0 and later.
`userContext`	**Type** Apex-defined **Description** An Apex ConnectApi.UserContextInputRepresentation record containing user information to evaluate product eligibility and calculate pricing.

Outputs

Output	Details
`apiStatusOutputRepresentation`	**Type** Apex-defined

Output	Details
	Description An Apex `runtime_industries_cpq.ApiStatusRepresentation` record that contains the status of the request, including the status code and message.
`contextId`	**Type** string **Description** ID of the context that's created by using the specified context definition.
`correlationId`	**Type** string **Description** Unique identifier attached to requests and messages, allowing reference to a specific transaction or event chain.
`productCursor`	**Type** string **Description** Unique identifier that represents the position of the product from which the next set of results are retrieved.
`productListOutputRepresentations`	**Type** Apex-defined **Description** Collection of Apex ProductListOutputRepresentation records that contain details about the product shown by the Guided Product Selection.
`recordOffset`	**Type** integer **Description** Number of catalog records to skip in the request. The default is 0.
`searchTerms`	**Type** Apex-defined **Description** Collection of terms that are used to search products. See GuidedSelectionSearchTerm.
`userContext`	**Type** Apex-defined

Output	Details
	Description An Apex ConnectApi.UserContextRepresentation record containing user information.

Example

Here's a sample input to call this invocable action from Apex code.

```
Invocable.Action action =
Invocable.Action.createStandardAction('searchPrdctWithGuidedSelection');

action.setInvocationParameter('correlationId', '77f9dc6a-8ecc-44a3-8d89-4050179cc846');
//action.setInvocationParameter('catalogId', '0ZSxx000000004sGAA');
action.setInvocationParameter('guidedSelectionResponseId', '0U3xx0000004CPACA2');
//action.setInvocationParameter('priceBookId', '01sxx0000005pyfAAA');
//action.setInvocationParameter('categoryId', '0ZGxx000000004rGAA');
action.setInvocationParameter('enableQualification', true);
action.setInvocationParameter('enablePricing', true);
//action.setInvocationParameter('contextDefinition', 'PDACDCtx');
//action.setInvocationParameter('contextMapping', 'ProductDiscoveryMapping');
//action.setInvocationParameter('qualificationProcedure', 'PDQualProceWithQuote');
//action.setInvocationParameter('pricingProcedure', 'IconpricingProcedure');
action.setInvocationParameter('includeCatalogDetails', true);
action.setInvocationParameter('currencyCode', 'USD');
action.setInvocationParameter('executeConfigurationRules', true);
List<String> orderByInputs = new List<String>();
orderByInputs.add('name:asc');
orderByInputs.add('id:desc');
action.setInvocationParameter('orderBy', orderByInputs);
List<runtime_industries_cpq.GuidedSelectionSearchTerm> searchTerms = new
List<runtime_industries_cpq.GuidedSelectionSearchTerm>();
runtime_industries_cpq.GuidedSelectionSearchTerm searchTerm = new
runtime_industries_cpq.GuidedSelectionSearchTerm();
searchTerm.term = 'Laptop Basic Bundle';
List<String> tags = new List<String>();
tags.add('Laptop');
tags.add('Desktop');
searchTerm.tags = tags;
searchTerms.add(searchTerm);

runtime_industries_cpq.GuidedSelectionSearchTermList searchTermList = new
runtime_industries_cpq.GuidedSelectionSearchTermList();
searchTermList.searchTerms = searchTerms;
action.setInvocationParameter('searchTerms', searchTermList);
List<Invocable.Action.Result> results = action.invoke();
System.debug('Guided Selection result = ' + results);
```

Here's a sample response when you call this action.


```
{
  "recordOffset": 5,
  "contextId": null,
```

```
"correlationId": "854c487d-183e-40b6-ba42-f079a6e695a3",
"apiStatusOutputRepresentation": {
  "statusMessage": null,
  "statusCode": "FetchedDetailsSuccessfully",
  "messages": []
},
"searchTerms": [
  {
    "term": "Additional API",
    "tags": []
  }
],
"productListOutputRepresentations": [
  {
    "unitOfMeasure": null,
    "status": null,
    "qualificationContext": {
      "reason": null,
      "isQualified": true
    },
    "productType": null,
    "productSpecificationType": null,
    "productSellingModelOptions": [
      {
        "productSellingModelId": "0jPVW0000001fh32AA",
        "productSellingModel": {
          "status": "Active",
          "sellingModelType": "TermDefined",
          "pricingTermUnit": "Annual",
          "pricingTerm": 1,
          "name": "Term Annual",
          "id": "0jPVW0000001fh32AA"
        },
        "productId": "01tVW000003l7txYAA",
        "id": "0iOVW00000049w22AA"
      },
      {
        "productSellingModelId": "0jPVW0000001fh52AA",
        "productSellingModel": {
          "status": "Active",
          "sellingModelType": "Evergreen",
          "pricingTermUnit": "Months",
          "pricingTerm": 1,
          "name": "Evergreen Monthly",
          "id": "0jPVW0000001fh52AA"
        },
        "productId": "01tVW000003l7txYAA",
        "id": "0iOVW00000049w12AA"
      },
      {
        "productSellingModelId": "0jPVW0000001fh62AA",
        "productSellingModel": {
          "status": "Active",
          "sellingModelType": "TermDefined",
```

```
      "pricingTermUnit": "Months",
      "pricingTerm": 1,
      "name": "Term Monthly",
      "id": "0jPVW0000001fh62AA"
    },
    "productId": "01tVW000003l7txYAA",
    "id": "0iOVW00000049w32AA"
  }
],
"productRelatedComponent": null,
"productQuantity": null,
"productPricingInformation": null,
"productInformation": null,
"productComponentGroups": [],
"productCode": "QB-API",
"productClassification": {
  "id": "11BVW000004ljiZ2AQ"
},
"prices": [
  {
    "pricingModel": {
      "unitOfMeasure": null,
      "pricingModelType": "TermDefined",
      "occurrence": 1,
      "name": "Term Annual",
      "id": "0jPVW0000001fh32AA",
      "frequency": "Annual"
    },
    "priceBookId": "01sVW0000024PZlYAM",
    "priceBookEntryId": "01uVW000000jzfGYAQ",
    "price": 15000,
    "isSelected": false,
    "isDerived": false,
    "isDefault": true,
    "effectiveTo": null,
    "effectiveFrom": null,
    "currencyIsoCode": "USD"
  },
  {
    "pricingModel": {
      "unitOfMeasure": null,
      "pricingModelType": "Evergreen",
      "occurrence": 1,
      "name": "Evergreen Monthly",
      "id": "0jPVW0000001fh52AA",
      "frequency": "Months"
    },
    "priceBookId": "01sVW0000024PZlYAM",
    "priceBookEntryId": "01uVW000000jzg8YAA",
    "price": 2000,
    "isSelected": false,
    "isDerived": false,
    "isDefault": false,
    "effectiveTo": null,
```

```
        "effectiveFrom": null,
        "currencyIsoCode": "USD"
      },
      {
        "pricingModel": {
          "unitOfMeasure": null,
          "pricingModelType": "TermDefined",
          "occurrence": 1,
          "name": "Term Monthly",
          "id": "0jPVW0000001fh62AA",
          "frequency": "Months"
        },
        "priceBookId": "01sVW0000024PZlYAM",
        "priceBookEntryId": "01uVW000000jzg9YAA",
        "price": 1500,
        "isSelected": false,
        "isDerived": false,
        "isDefault": false,
        "effectiveTo": null,
        "effectiveFrom": null,
        "currencyIsoCode": "USD"
      }
    ],
    "nodeType": "simpleProduct",
    "name": "Additional API",
    "isSoldOnlyWithOtherProds": false,
    "isQuantityEditable": null,
    "isDefaultComponent": null,
    "isComponentRequired": null,
    "isAssetizable": true,
    "isActive": true,
    "id": "01tVW000003l7txYAA",
    "endOfLifeDate": null,
    "displayUrl": "/resource/add_api",
    "discontinuedDate": null,
    "description": null,
    "configureDuringSale": "Allowed",
    "configurationRules": [
      {
        "type": "disable",
        "details": [
          {
            "message": "This is disabled"
          }
        ]
      }
    ],
    "childProducts": [],
    "categories": [
      {
        "sortOrder": null,
        "qualificationContext": {
          "reason": null,
          "isQualified": true
```

```
        },
        "parentCategoryId": null,
        "name": "API",
        "isNavigational": null,
        "id": "0ZGVW000000IUEG4A4",
        "hasSubCategories": null,
        "description": null,
        "childCategories": null,
        "catalogId": "0ZSVW000000AhdB4AS"
      }
    ],
    "catalogs": [],
    "availabilityDate": null,
    "attributeCategories": [],
    "additionalFields": []
  },
  {
    "unitOfMeasure": null,
    "status": null,
    "qualificationContext": {
      "reason": null,
      "isQualified": true
    },
    "productType": null,
    "productSpecificationType": null,
    "productSellingModelOptions": [
      {
        "productSellingModelId": "0jPVW0000001fh32AA",
        "productSellingModel": {
          "status": "Active",
          "sellingModelType": "TermDefined",
          "pricingTermUnit": "Annual",
          "pricingTerm": 1,
          "name": "Term Annual",
          "id": "0jPVW0000001fh32AA"
        },
        "productId": "01tVW000003l7tyYAA",
        "id": "0iOVW00000049w42AA"
      }
    ],
    "productRelatedComponent": null,
    "productQuantity": null,
    "productPricingInformation": null,
    "productInformation": null,
    "productComponentGroups": [],
    "productCode": "QB-API-FLEX",
    "productClassification": {
      "id": null
    },
    "prices": [
      {
        "pricingModel": {
          "unitOfMeasure": null,
          "pricingModelType": "TermDefined",
```

```
            "occurrence": 1,
            "name": "Term Annual",
            "id": "0jPVW0000001fh32AA",
            "frequency": "Annual"
          },
          "priceBookId": "01sVW0000024PZlYAM",
          "priceBookEntryId": "01uVW000000jzfFYAQ",
          "price": 450,
          "isSelected": false,
          "isDerived": false,
          "isDefault": true,
          "effectiveTo": null,
          "effectiveFrom": null,
          "currencyIsoCode": "USD"
        }
      ],
      "nodeType": "simpleProduct",
      "name": "Additional API Flex (100M)",
      "isSoldOnlyWithOtherProds": false,
      "isQuantityEditable": null,
      "isDefaultComponent": null,
      "isComponentRequired": null,
      "isAssetizable": true,
      "isActive": true,
      "id": "01tVW000003l7tyYAA",
      "endOfLifeDate": null,
      "displayUrl": "/resource/api_flex",
      "discontinuedDate": null,
      "description": "API instances remain under management until they are deleted.
Instances of API Manager are aggregated using a Max Concurrent model. The usage for a
month is the highest number of APIs Managed in a single given hour during a month.",
      "configureDuringSale": null,
      "configurationRules": [],
      "childProducts": [],
      "categories": [
        {
          "sortOrder": null,
          "qualificationContext": {
            "reason": null,
            "isQualified": true
          },
          "parentCategoryId": null,
          "name": "API",
          "isNavigational": null,
          "id": "0ZGVW000000IUEG4A4",
          "hasSubCategories": null,
          "description": null,
          "childCategories": null,
          "catalogId": "0ZSVW000000AhdB4AS"
        }
      ],
      "catalogs": [],
      "availabilityDate": null,
      "attributeCategories": [],
```

```
    "additionalFields": []
  },
  {
    "unitOfMeasure": null,
    "status": null,
    "qualificationContext": {
      "reason": null,
      "isQualified": true
    },
    "productType": null,
    "productSpecificationType": null,
    "productSellingModelOptions": [
      {
        "productSellingModelId": "0jPVW0000001fh32AA",
        "productSellingModel": {
          "status": "Active",
          "sellingModelType": "TermDefined",
          "pricingTermUnit": "Annual",
          "pricingTerm": 1,
          "name": "Term Annual",
          "id": "0jPVW0000001fh32AA"
        },
        "productId": "01tVW000003l7tzYAA",
        "id": "0iOVW00000049w52AA"
      }
    ],
    "productRelatedComponent": null,
    "productQuantity": null,
    "productPricingInformation": null,
    "productInformation": null,
    "productComponentGroups": [],
    "productCode": "QB-API-GOVT",
    "productClassification": {
      "id": null
    },
    "prices": [
      {
        "pricingModel": {
          "unitOfMeasure": null,
          "pricingModelType": "TermDefined",
          "occurrence": 1,
          "name": "Term Annual",
          "id": "0jPVW0000001fh32AA",
          "frequency": "Annual"
        },
        "priceBookId": "01sVW0000024PZlYAM",
        "priceBookEntryId": "01uVW000000jzfHYAQ",
        "price": 1450,
        "isSelected": false,
        "isDerived": false,
        "isDefault": true,
        "effectiveTo": null,
        "effectiveFrom": null,
        "currencyIsoCode": "USD"
```

```
        }
      ],
      "nodeType": "simpleProduct",
      "name": "Additional API Gov",
      "isSoldOnlyWithOtherProds": false,
      "isQuantityEditable": null,
      "isDefaultComponent": null,
      "isComponentRequired": null,
      "isAssetizable": true,
      "isActive": true,
      "id": "01tVW00000317tzYAA",
      "endOfLifeDate": null,
      "displayUrl": "/resource/api_govt",
      "discontinuedDate": null,
      "description": "API instances remain under management until they are deleted.
Instances of API Manager are aggregated using a Max Concurrent model. The usage for a
month is the highest number of APIs Managed in a single given hour during a month.",
      "configureDuringSale": null,
      "configurationRules": [],
      "childProducts": [],
      "categories": [
        {
          "sortOrder": null,
          "qualificationContext": {
            "reason": null,
            "isQualified": true
          },
          "parentCategoryId": null,
          "name": "API",
          "isNavigational": null,
          "id": "0ZGVW000000IUEG4A4",
          "hasSubCategories": null,
          "description": null,
          "childCategories": null,
          "catalogId": "0ZSVW000000AhdB4AS"
        }
      ],
      "catalogs": [],
      "availabilityDate": null,
      "attributeCategories": [],
      "additionalFields": []
    },
    {
      "unitOfMeasure": null,
      "status": null,
      "qualificationContext": {
        "reason": null,
        "isQualified": true
      },
      "productType": null,
      "productSpecificationType": null,
      "productSellingModelOptions": [
        {
          "productSellingModelId": "0jPVW0000001fh32AA",
```

```
    "productSellingModel": {
      "status": "Active",
      "sellingModelType": "TermDefined",
      "pricingTermUnit": "Annual",
      "pricingTerm": 1,
      "name": "Term Annual",
      "id": "0jPVW0000001fh32AA"
    },
    "productId": "01tVW000003l7u0YAA",
    "id": "0iOVW00000049w92AA"
  }
],
"productRelatedComponent": null,
"productQuantity": null,
"productPricingInformation": null,
"productInformation": null,
"productComponentGroups": [],
"productCode": "QB-API-PREP",
"productClassification": {
  "id": null
},
"prices": [
  {
    "pricingModel": {
      "unitOfMeasure": null,
      "pricingModelType": "TermDefined",
      "occurrence": 1,
      "name": "Term Annual",
      "id": "0jPVW0000001fh32AA",
      "frequency": "Annual"
    },
    "priceBookId": "01sVW0000024PZlYAM",
    "priceBookEntryId": "01uVW000000jzfSYAQ",
    "price": 3240,
    "isSelected": false,
    "isDerived": false,
    "isDefault": true,
    "effectiveTo": null,
    "effectiveFrom": null,
    "currencyIsoCode": "USD"
  }
],
"nodeType": "simpleProduct",
"name": "Additional API Pre-Prod",
"isSoldOnlyWithOtherProds": false,
"isQuantityEditable": null,
"isDefaultComponent": null,
"isComponentRequired": null,
"isAssetizable": true,
"isActive": true,
"id": "01tVW000003l7u0YAA",
"endOfLifeDate": null,
"displayUrl": "/resource/api_preprod",
"discontinuedDate": null,
```

```
      "description": "API instances remain under management until they are deleted.
Instances of API Manager are aggregated using a Max Concurrent model. The usage for a
month is the highest number of APIs Managed in a single given hour during a month.",
      "configureDuringSale": null,
      "configurationRules": [],
      "childProducts": [],
      "categories": [
        {
          "sortOrder": null,
          "qualificationContext": {
            "reason": null,
            "isQualified": true
          },
          "parentCategoryId": null,
          "name": "API",
          "isNavigational": null,
          "id": "0ZGVW000000IUEG4A4",
          "hasSubCategories": null,
          "description": null,
          "childCategories": null,
          "catalogId": "0ZSVW000000AhdB4AS"
        }
      ],
      "catalogs": [],
      "availabilityDate": null,
      "attributeCategories": [],
      "additionalFields": []
    },
    {
      "unitOfMeasure": null,
      "status": null,
      "qualificationContext": {
        "reason": null,
        "isQualified": true
      },
      "productType": null,
      "productSpecificationType": null,
      "productSellingModelOptions": [
        {
          "productSellingModelId": "0jPVW0000001fh32AA",
          "productSellingModel": {
            "status": "Active",
            "sellingModelType": "TermDefined",
            "pricingTermUnit": "Annual",
            "pricingTerm": 1,
            "name": "Term Annual",
            "id": "0jPVW0000001fh32AA"
          },
          "productId": "01tVW000003l7u1YAA",
          "id": "0iOVW00000049wA2AQ"
        }
      ],
      "productRelatedComponent": null,
      "productQuantity": null,
```

```
      "productPricingInformation": null,
      "productInformation": null,
      "productComponentGroups": [],
      "productCode": "QB-API-PROD",
      "productClassification": {
        "id": null
      },
      "prices": [
        {
          "pricingModel": {
            "unitOfMeasure": null,
            "pricingModelType": "TermDefined",
            "occurrence": 1,
            "name": "Term Annual",
            "id": "0jPVW0000001fh32AA",
            "frequency": "Annual"
          },
          "priceBookId": "01sVW0000024PZlYAM",
          "priceBookEntryId": "01uVW000000jzfdYAA",
          "price": 3240,
          "isSelected": false,
          "isDerived": false,
          "isDefault": true,
          "effectiveTo": null,
          "effectiveFrom": null,
          "currencyIsoCode": "USD"
        }
      ],
      "nodeType": "simpleProduct",
      "name": "Additional API Prod",
      "isSoldOnlyWithOtherProds": false,
      "isQuantityEditable": null,
      "isDefaultComponent": null,
      "isComponentRequired": null,
      "isAssetizable": true,
      "isActive": true,
      "id": "01tVW000003l7u1YAA",
      "endOfLifeDate": null,
      "displayUrl": "/resource/api_prod",
      "discontinuedDate": null,
      "description": "API instances remain under management until they are deleted.
Instances of API Manager are aggregated using a Max Concurrent model. The usage for a
month is the highest number of APIs Managed in a single given hour during a month.",
      "configureDuringSale": null,
      "configurationRules": [],
      "childProducts": [],
      "categories": [
        {
          "sortOrder": null,
          "qualificationContext": {
            "reason": null,
            "isQualified": true
          },
          "parentCategoryId": null,
```

```
          "name": "API",
          "isNavigational": null,
          "id": "0ZGVW000000IUEG4A4",
          "hasSubCategories": null,
          "description": null,
          "childCategories": null,
          "catalogId": "0ZSVW000000AhdB4AS"
        }
      ],
      "catalogs": [],
      "availabilityDate": null,
      "attributeCategories": [],
      "additionalFields": []
    }
  ]
}
```


Get Product Recommendations Action

Retrieve a list of recommended products for a quote or order by using the Constraint Rule Engine.

This action is available in API version 67.0 and later.

Special Access Rules

The Get Product Details action is available in Enterprise, Unlimited, and Developer Editions where Product Discovery is enabled.

You can invoke this action via Apex and Flows only.

Inputs

Input	Details
additionalContextData	**Type** Apex-defined **Description** An array of Apex `runtime_industries_cpq.AdditionalContextData` records that contain the additional nodes used along with the context definition nodes for data hydration. The maximum number of supported nodes is 10.
additionalFields	**Type** Apex-defined **Description** An Apex `runtime_industries_cpq.AdditionalFields` record that contains an array of additional standard or custom fields from the `Product2` object to include in the response.

Input	Details
`catalogId`	**Type** string **Description** ID of the catalog from which to retrieve recommended products.
`contextDefinition`	**Type** string **Description** API name of the context definition used for context creation. If you don't specify a value, the context definition selected on the Product Discovery Settings page from Setup is used.
`contextMapping`	**Type** string **Description** API name of the context mapping used for data hydration. The value of this parameter is used only if it belongs to the specified context definition. If you don't specify a value, the default context mapping of the context definition is used.
`currencyCode`	**Type** string **Description** Currency code used for pricing calculations and product filtering.
`enablePricing`	**Type** boolean **Description** Indicates whether the pricing procedure must run (`true`) or not (`false`). For orgs where pricing is enabled, you can override to `false` to skip Salesforce Pricing execution. For orgs where pricing is disabled, you can't override to `true`.
`enableQualification`	**Type** boolean **Description** Indicates whether the qualification procedure must run (`true`) or not (`false`). For orgs where qualification is enabled, you can override to `false` to skip BRE qualification rules. For orgs where qualification is disabled, you can't override to `true`.

Input	Details
`filterCriteria`	**Type** Apex-defined **Description** A collection of Apex `runtime_industries_cpq.FilterInputRepresentation` records used to filter products. The `filterCriteria` parameter supports only the `name` property. The supported operators are: • `eq` • `in` • `contains` If this parameter contains multiple criteria, all criteria are applied using the `and` operator.
`nextCursor`	**Type** string **Description** A unique identifier that represents the position of the product from which the next set of results are retrieved.
`pageSize`	**Type** integer **Description** Maximum number of recommended products to return in the response.
`priceBookId`	**Type** string **Description** ID of the pricebook from which to retrieve pricing details.
`pricingProcedure`	**Type** string **Description** API name of the pricing procedure used to calculate product prices. If you don't specify a value, the pricing procedure selected on the Product Discovery Settings page from Setup is used.
`qualificationProcedure`	**Type** string

Input	Details
	Description API name of the qualification procedure used to evaluate product eligibility. If you don't specify a value, the qualification procedure selected on the Product Discovery Settings page from Setup is used.
`transactionContextId`	**Type** string **Description** Context ID of the transaction.
`transactionId`	**Type** string **Description** ID of the transaction.
`usePromotions`	**Type** boolean **Description** Indicates whether to fetch promotions for recommended products (`true`) or not (`false`). Applies only when promotions are enabled for the org.
`userContext`	**Type** Apex-defined **Description** An Apex `ConnectApi.UserContextInputRepresentation` record that contains the user details used to evaluate product eligibility and calculate prices.

Outputs

Output	Details
`errorMessages`	**Type** Apex-defined **Description** A list of Apex `runtime_industries_cpq.ProductRecommendationErrorsOutputRepresentation` records that contain error codes and messages returned by the action.

Output	Details
`isSuccessful`	**Type** boolean **Description** Indicates whether the action call succeeded (`true`) or not (`false`).
`nextCursor`	**Type** string **Description** Unique identifier that represents the position of the next product in the dataset. Use this value as an input to retrieve the next page of results.
`recommendedProducts`	**Type** Apex-defined **Description** A list of Apex `runtime_industries_cpq.RecommendedProductsOutputRepresentation` records that contain the recommended products returned by the Constraint Rule Engine.
`transactionContextId`	**Type** string **Description** Context ID of the transaction.

Example

Here's a sample input to call this invocable action from Apex code.

```
Invocable.Action action = Invocable.Action.createStandardAction('getRecommendedProducts');
action.setInvocationParameter('catalogId', '0ZSVW000000AhdB4AS');
action.setInvocationParameter('transactionId', '0Q0VW0000018J6X0AU');
List<Invocable.Action.Result> results = action.invoke();

for (Invocable.Action.Result res : results) {
    if (res.isSuccess()) {
        Map<String, Object> outMap = res.getOutputParameters();
        String jsonOutput = JSON.serialize(outMap);
        System.debug(jsonOutput);
    }
}
```

Here's a sample response for the Get Product Recommendations action.


```
{
  "prodRecomErrorOutRepresentation": [],
```

```
  "isSuccessful": true,
  "productCursor": "MTAwMDAwMDAwNg==",
  "transactionContextId":
"0000000r25tq18g00291777531846434b162e02f878f425680e0d95dc5f650fe",
  "recomProdOutputRepresentations": [
    {
      "status": null,
      "qualificationContext": {
        "reason": null,
        "isQualified": true
      },
      "productType": null,
      "productSpecificationType": null,
      "productSellingModelOptions": [
        {
          "productSellingModelId": "0jPVW0000001fh32AA",
          "productSellingModel": {
            "status": "Active",
            "sellingModelType": "TermDefined",
            "pricingTermUnit": "Annual",
            "pricingTerm": 1,
            "name": "Term Annual",
            "id": "0jPVW0000001fh32AA"
          },
          "productId": "01tVW000003l7txYAA",
          "id": "0iOVW00000049w22AA"
        },
        {
          "productSellingModelId": "0jPVW0000001fh52AA",
          "productSellingModel": {
            "status": "Active",
            "sellingModelType": "Evergreen",
            "pricingTermUnit": "Months",
            "pricingTerm": 1,
            "name": "Evergreen Monthly",
            "id": "0jPVW0000001fh52AA"
          },
          "productId": "01tVW000003l7txYAA",
          "id": "0iOVW00000049w12AA"
        },
        {
          "productSellingModelId": "0jPVW0000001fh62AA",
          "productSellingModel": {
            "status": "Active",
            "sellingModelType": "TermDefined",
            "pricingTermUnit": "Months",
            "pricingTerm": 1,
            "name": "Term Monthly",
            "id": "0jPVW0000001fh62AA"
          },
          "productId": "01tVW000003l7txYAA",
          "id": "0iOVW00000049w32AA"
        }
      ],
```

```
    "productQuantity": null,
    "productCode": "QB-API",
    "productClassification": {
      "id": "11BVW000004ljiZ2AQ"
    },
    "prices": [
      {
        "pricingModel": {
          "unitOfMeasure": null,
          "pricingModelType": "TermDefined",
          "occurrence": 1,
          "name": "Term Annual",
          "id": "0jPVW0000001fh32AA",
          "frequency": "Annual"
        },
        "priceBookId": "01sVW0000024PZlYAM",
        "priceBookEntryId": "01uVW000000jzfGYAQ",
        "price": 15000,
        "isSelected": false,
        "isDerived": false,
        "isDefault": true,
        "effectiveTo": null,
        "effectiveFrom": null,
        "currencyIsoCode": "USD"
      },
      {
        "pricingModel": {
          "unitOfMeasure": null,
          "pricingModelType": "Evergreen",
          "occurrence": 1,
          "name": "Evergreen Monthly",
          "id": "0jPVW0000001fh52AA",
          "frequency": "Months"
        },
        "priceBookId": "01sVW0000024PZlYAM",
        "priceBookEntryId": "01uVW000000jzg8YAA",
        "price": 2000,
        "isSelected": false,
        "isDerived": false,
        "isDefault": false,
        "effectiveTo": null,
        "effectiveFrom": null,
        "currencyIsoCode": "USD"
      },
      {
        "pricingModel": {
          "unitOfMeasure": null,
          "pricingModelType": "TermDefined",
          "occurrence": 1,
          "name": "Term Monthly",
          "id": "0jPVW0000001fh62AA",
          "frequency": "Months"
        },
        "priceBookId": "01sVW0000024PZlYAM",
```

```
        "priceBookEntryId": "01uVW000000jzg9YAA",
        "price": 1500,
        "isSelected": false,
        "isDerived": false,
        "isDefault": false,
        "effectiveTo": null,
        "effectiveFrom": null,
        "currencyIsoCode": "USD"
      }
    ],
    "nodeType": "simpleProduct",
    "name": "Additional API",
    "isSoldOnlyWithOtherProds": false,
    "isQuantityEditable": null,
    "isDefaultComponent": null,
    "isComponentRequired": null,
    "isAssetizable": true,
    "isActive": true,
    "id": "01tVW000003l7txYAA",
    "endOfLifeDate": null,
    "displayUrl": "/resource/add_api",
    "discontinuedDate": null,
    "description": null,
    "configureDuringSale": "Allowed",
    "configurationRules": [
      {
        "type": "recommend",
        "details": [
          {
            "message": "APIAccessRequests recommends AdditionalAPI"
          }
        ]
      },
      {
        "type": "disable",
        "details": [
          {
            "message": "This is disabled"
          }
        ]
      }
    ],
    "categories": [
      {
        "sortOrder": null,
        "qualificationContext": {
          "reason": null,
          "isQualified": true
        },
        "parentCategoryId": null,
        "name": "API",
        "isNavigational": null,
        "id": "0ZGVW000000IUEG4A4",
        "hasSubCategories": null,
```

```
          "description": null,
          "childCategories": null,
          "catalogId": "0ZSVW000000AhdB4AS"
        }
      ],
      "catalogs": [
        {
          "status": null,
          "numberOfCategories": 8,
          "name": "QuantumBit Software",
          "id": "0ZSVW000000AhdB4AS",
          "effectiveStartDate": null,
          "effectiveEndDate": null,
          "description": null,
          "customFields": [],
          "catalogType": null,
          "catalogCode": null
        }
      ],
      "availabilityDate": null,
      "additionalFields": []
    }
  ]
}
```


Product Discovery Apex Reference

Use built-in Apex classes and interfaces grouped by namespace.

runtime_industries_cpq Namespace

The runtime_industries_cpq namespace provides classes and methods to search products or to manage products, catalogs, and categories.

The `runtime_industries_cpq` namespace includes these classes.

Usage

You can access the `runtime_industries_cpq` classes if your org has the Product Discovery feature.

AdditionalFieldsInput Class

Contains properties to include the additional standard or custom fields in the request.

ApiStatusRepresentation Class

Stores details of the API request such as execution messages, status code, and status message.

AttributeCategoryOutputRepresentation Class

Stores details of an attribute such as code, description, usage type, and so on.

AttributePickListOutputRepresentation Class

Stores details of an attribute picklist.

AttributePickListValueOutputRepresentation Class

Stores details of an attribute picklist value.

BulkProductDetailsInputBody Class

Contains the details of the request to retrieve product details by using product ID and product selling model ID.

BulkProductDetailsInputBodyList Class

Contains details of the request to retrieve a list of products.

BulkProductDetailsRepresentation Class

Get the details of multiple product definitions in a single request. This class is used for bulk product retrieval operations in Product Discovery.

CatalogOutputRepresentation Class

Contains properties to store details of a catalog definition.

ConfigRuleResult Class

Contains the results of configuration rule evaluation, including message rules, product recommendation rules, and visibility rules that are applied during product configuration.

CategoryOutputRepresentation Class

Contains properties to store details of a category.

ContextDataInput Class

Get details of a context.

FacetValueRepresentation Class

Get details of the facet values that are found in the search result.

Filter Class

Contains the criteria property to store the details of a filter criteria, which is used to filter records.

FilterCriteriaInputRepresentation Class

Contains properties to store criteria details to filter records based on supported properties.

FilterInputRepresentation Class

Contains the filter property to filters records based on supported criteria.

GuidedSelectionRepresentation Class

Contains properties to represent a product in a guided selection flow. This class is used in Product Discovery to provide structured product information during guided product selection processes.

GuidedSelectionSearchTerm Class

Represents a search term used in guided product selection. Contains the search term text and associated tags for filtering and searching products in Product Discovery.

GuidedSelectionSearchTermList Class

Contains a list of search terms used in guided product selection. This class is used to pass multiple search terms for filtering and searching products in Product Discovery.

MessageRule Class

Represents a message rule that is evaluated during product configuration. Message rules display informational, warning, or error messages to users based on configuration conditions.

PricingModelOutputRepresentation Class

Contains details of a pricing model in a product configuration.

ProductAttributeOutputRepresentation Class

Contains details about the attribute in a product configuration.

ProductClassificationOutputRepresentation Class

Get details of the product classification in a product configuration.

ProductComponentGroupOutputRepresentation Class

Get details of the product component group in a product classification.

ProductComponentGroupRepresentation Class

Represents a product component group used in bulk product operations. This class is similar to ProductComponentGroupOutputRepresentation but is used specifically for bulk product detail representations where components are represented as BulkProductDetailsRepresentation objects.

ProductDetailsRepresentation Class

Get the details of a product definition.

ProductListRepresentation Class

Get the list of retrieved products.

ProductOutputRepresentation Class

Get the details of a product definition.

ProductPricesOutputRepresentation Class

Get the price details of a product.

ProductQuantityOutputRepresentation Class

Represents the quantity constraints and current quantity for a product in the product discovery context.

ProductRecommendationRule Class

Represents a product recommendation rule that is evaluated during product configuration. Product recommendation rules suggest additional products to users based on configuration conditions.

ProductRelatedComponentOutputRepresentation Class

Represents a related component product in a bundle or product relationship, including component configuration details such as quantity constraints, required status, and relationship metadata.

ProductSellingModelOptionOutputRepresentation Class

Represents a selling model option available for a product, which defines how the product can be sold (such as subscription, one-time, or usage-based).

ProductSellingModelOutputRepresentation Class

Represents a product selling model that defines how a product is sold, including the selling model type, pricing term, and status.

ProductSpecificationRecordTypeOutputRepresentation Class

Represents the record type information for a product specification, which defines the type of record used to store product specification data.

ProductSpecificationTypeOutputRepresentation Class

Represents a product specification type that defines the structure and attributes available for configuring a product.

QocQualificationOutputRepresentation Class

Represents a quote, order, or contract qualification that determines whether a product can be sold based on specific business rules and conditions.

QualificationContextOutputRepresentation Class

Represents the context information used for product qualification, including account, opportunity, and other relevant context data for determining product eligibility.

RelatedObjectFilter Class

Represents a filter for related objects used in product search and discovery, allowing you to filter products based on related object criteria.

RelatedObjectFilterInputRepresentation Class

Represents input criteria for filtering products based on related object information, such as account, opportunity, or contract data.

SearchProductsFacetRepresentation Class

Represents a search facet that provides filtering and categorization options for product search results, such as categories, attributes, or other product characteristics.

SearchProductsRepresentation Class

Represents the results of a product search operation, including the list of products, search facets, pagination information, and total count of matching products.

UnitOfMeasureOutputRepresentation Class

Represents the unit of measure for a product. This class contains information about how product quantities are measured, including the unit code, name, scale, and rounding method.

VisibilityRule Class

Represents a visibility rule that is evaluated during product configuration. Visibility rules control the visibility of products, attributes, or other UI elements based on configuration conditions.

AdditionalContextData Class

Contains properties to include a list of additional context data nodes. These nodes are used along with the context definition nodes for data hydration.

Namespace

runtime_industries_cpq on page 512

AdditionalContextData Properties

Set the AdditionalContextData class property to specify a list of additional nodes for data hydration.

AdditionalContextData Properties

Set the AdditionalContextData class property to specify a list of additional nodes for data hydration.

The `AdditionalContextData` class includes this property.

additionalContextData

Include a list of additional nodes that are used along with the context definition nodes for data hydration. The maximum number of supported nodes is 10.

additionalContextData

Include a list of additional nodes that are used along with the context definition nodes for data hydration. The maximum number of supported nodes is 10.

Signature

```
public List<runtime_industries_cpq.ContextDataInput> additionalContextData {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.ContextDataInput on page 548>

AdditionalFields Class

Contains properties to include a map where the key is a string and the value is an instance of the AdditionalFieldsInput class.

Namespace

runtime_industries_cpq on page 512

AdditionalFields Properties

Set the AdditionalFields class property to include a map where the key is a string and the value is an instance of the AdditionalFieldsInput class.

AdditionalFields Properties

Set the AdditionalFields class property to include a map where the key is a string and the value is an instance of the AdditionalFieldsInput class.

The `AdditionalFields` class includes this property.

additionalFields

Includes a map where the key is a string and the value is an instance of the AdditionalFieldsInput class.

additionalFields

Includes a map where the key is a string and the value is an instance of the AdditionalFieldsInput class.

Signature

```
public Map<String,runtime_industries_cpq.AdditionalFieldsInput> additionalFields {get;
set;}
```

Property Value

Type: Map<String,runtime_industries_cpq.AdditionalFieldsInput on page 517>

AdditionalFieldsInput Class

Contains properties to include the additional standard or custom fields in the request.

Namespace

runtime_industries_cpq on page 512

AdditionalFieldsInput Properties
Set the AdditionalFieldsInput class property to include the additional standard or custom fields in the response.

AdditionalFieldsInput Properties

Set the AdditionalFieldsInput class property to include the additional standard or custom fields in the response.

The `AdditionalFieldsInput` class includes this property.

fields
Include the additional standard or custom fields in the response.

`fields`

Include the additional standard or custom fields in the response.

Signature

```
public List<String> fields {get; set;}
```

Property Value

Type: List<String>

ApiStatusRepresentation Class

Stores details of the API request such as execution messages, status code, and status message.

Namespace

runtime_industries_cpq on page 512

ApiStatusRepresentation Properties
Contains properties to include the API response details.

ApiStatusRepresentation Properties

Contains properties to include the API response details.

The `ApiStatusRepresentation` class includes these properties.

[messages](#messages)
Get the status messages of the API execution.

[statusCode](#statuscode)
Get the status code of the API execution.

[statusMessage](#statusmessage)
Get the display label for the API status.

messages

Get the status messages of the API execution.

Signature

```
public List<ConnectApi.CpqMessageOutputRepresentation> messages {get; set;}
```

Property Value

Type: List<ConnectApi.CpqMessageOutputRepresentation>

statusCode

Get the status code of the API execution.

Signature

```
public String statusCode {get; set;}
```

Property Value

Type: String

statusMessage

Get the display label for the API status.

Signature

```
public String statusMessage {get; set;}
```

Property Value

Type: String

AttributeCategoryOutputRepresentation Class

Stores details of an attribute such as code, description, usage type, and so on.

Namespace

runtime_industries_cpq on page 512

AttributeCategoryOutputRepresentation Properties
Contains properties to include details of an attribute.

AttributeCategoryOutputRepresentation Properties

Contains properties to include details of an attribute.

The `AttributeCategoryOutputRepresentation` class includes these properties.

code
Get the code of the attribute category.

description
Get the description of the attribute category.

id
Get the ID of the attribute category.

name
Get the name of the attribute category.

records
Get the attributes of the attribute category.

status
Get the status of the attribute category.

totalSize
Get the total size of the attribute category.

usageType
Get the usage type of the attribute category.

`code`

Get the code of the attribute category.

Signature

```
public String code {get; set;}
```

Property Value

Type: String

`description`

Get the description of the attribute category.

Signature

```
public String description {get; set;}
```

Property Value

Type: String

id

Get the ID of the attribute category.

Signature

```
public String id {get; set;}
```

Property Value

Type: String

name

Get the name of the attribute category.

Signature

```
public String name {get; set;}
```

Property Value

Type: String

records

Get the attributes of the attribute category.

Signature

```
public List<runtime_industries_cpq.ProductAttributeOutputRepresentation> records {get;
set;}
```

Property Value

Type: List<runtime_industries_cpq.ProductAttributeOutputRepresentation on page 571>

status

Get the status of the attribute category.

Signature

```
public String status {get; set;}
```

Property Value

Type: String

`totalSize`

Get the total size of the attribute category.

Signature

`public Integer totalSize {get; set;}`

Property Value

Type: Integer

`usageType`

Get the usage type of the attribute category.

Signature

`public String usageType {get; set;}`

Property Value

Type: String

AttributePickListOutputRepresentation Class

Stores details of an attribute picklist.

Namespace

runtime_industries_cpq on page 512

AttributePickListOutputRepresentation Properties
Contains properties to include details of an attribute picklist.

AttributePickListOutputRepresentation Properties

Contains properties to include details of an attribute picklist.

The `AttributePickListOutputRepresentation` class includes these properties.

description
Get the description of the attribute picklist.

id
Get the ID of the attribute picklist.

name
Get the name of the attribute picklist.

status
Get the status of the attribute picklist.

values
Get the values of the attribute picklist.

dataType
Get the datatype value.

description

Get the description of the attribute picklist.

Signature

```
public String description {get; set;}
```

Property Value

Type: String

id

Get the ID of the attribute picklist.

Signature

```
public String id {get; set;}
```

Property Value

Type: String

name

Get the name of the attribute picklist.

Signature

```
public String name {get; set;}
```

Property Value

Type: String

status

Get the status of the attribute picklist.

Signature

```
public String status {get; set;}
```

Property Value

Type: String

values

Get the values of the attribute picklist.

Signature

```
public List<runtime_industries_cpq.AttributePickListValueOutputRepresentation> values
{get; set;}
```

Property Value

Type: List<runtime_industries_cpq.AttributePickListValueOutputRepresentation on page 523 >

dataType

Get the datatype value.

Signature

```
public String dataType {get; set;}
```

Property Value

Type: String

AttributePickListValueOutputRepresentation Class

Stores details of an attribute picklist value.

Namespace

runtime_industries_cpq on page 512

AttributePickListValueOutputRepresentation Properties
Contains properties to include details of an attribute picklist value.

AttributePickListValueOutputRepresentation Properties

Contains properties to include details of an attribute picklist value.

The `AttributePickListValueOutputRepresentation` class includes these properties.

code
Get the code of the attribute picklist value.

description
Get the description of attribute picklist value.

displayValue
Get the display value of the attribute value.

id
Get the ID of the attribute picklist value.

isBooleanValue
Indicates whether this attribute value is a boolean value (true) or not (false).

label
Get the label of the attribute picklist value.

name
Get the name of the attribute picklist value.

sequence
Get the sequence of the attribute picklist value.

status
Get the status of the attribute picklist value.

textValue
Get the text value of the attribute picklist value.

code

Get the code of the attribute picklist value.

Signature

```
public String code {get; set;}
```

Property Value

Type: String

description

Get the description of attribute picklist value.

Signature

```
public String description {get; set;}
```

Property Value

Type: String

`displayValue`

Get the display value of the attribute value.

Signature

`public String displayValue {get; set;}`

Property Value

Type: String

`id`

Get the ID of the attribute picklist value.

Signature

`public String id {get; set;}`

Property Value

Type: String

`isBooleanValue`

Indicates whether this attribute value is a boolean value (true) or not (false).

Signature

`public Boolean isBooleanValue {get; set;}`

Property Value

Type: Boolean

`label`

Get the label of the attribute picklist value.

Signature

`public String label {get; set;}`

Property Value

Type: String

`name`

Get the name of the attribute picklist value.

Signature

```
public String name {get; set;}
```

Property Value

Type: String

sequence

Get the sequence of the attribute picklist value.

Signature

```
public Integer sequence {get; set;}
```

Property Value

Type: Integer

status

Get the status of the attribute picklist value.

Signature

```
public String status {get; set;}
```

Property Value

Type: String

textValue

Get the text value of the attribute picklist value.

Signature

```
public String textValue {get; set;}
```

Property Value

Type: String

BulkProductDetailsInputBody Class

Contains the details of the request to retrieve product details by using product ID and product selling model ID.

Namespace

runtime_industries_cpq on page 512

[BulkProductDetailsInputBody Properties](#)
Contains properties to retrieve details of products.

BulkProductDetailsInputBody Properties

Contains properties to retrieve details of products.

The `BulkProductDetailsInputBody` class includes these properties.

[productId](#)
Set the ID of the product to return the details for.

[productSellingModelId](#)
Set the ID of the product selling model to return the details for.

`productId`

Set the ID of the product to return the details for.

Signature

```
public String productId {get; set;}
```

Property Value

Type: String

`productSellingModelId`

Set the ID of the product selling model to return the details for.

Signature

```
public String productSellingModelId {get; set;}
```

Property Value

Type: String

BulkProductDetailsInputBodyList Class

Contains details of the request to retrieve a list of products.

Namespace

runtime_industries_cpq on page 512

[BulkProductDetailsInputBodyList Properties](#)
Contains properties to retrieve details of a list of products.

BulkProductDetailsInputBodyList Properties

Contains properties to retrieve details of a list of products.

The `BulkProductDetailsInputBodyList` class includes these properties.

productData
: Set the list of maps that contain product IDs and product selling model IDs.

`productData`

Set the list of maps that contain product IDs and product selling model IDs.

Signature

```
public List<runtime_industries_cpq.BulkProductDetailsInputBody> productData {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.BulkProductDetailsInputBody on page 526 >

BulkProductDetailsRepresentation Class

Get the details of multiple product definitions in a single request. This class is used for bulk product retrieval operations in Product Discovery.

Namespace

runtime_industries_cpq on page 512

BulkProductDetailsRepresentation Constructor
: Learn more about the constructors that are available with the BulkProductDetailsRepresentation class.

BulkProductDetailsRepresentation Properties
: Contains properties to include details of product definitions retrieved in bulk operations.

BulkProductDetailsRepresentation Constructor

Learn more about the constructors that are available with the BulkProductDetailsRepresentation class.

The `BulkProductDetailsRepresentation` class includes these constructors.

BulkProductDetailsRepresentation(apexObj)
: Constructor to create a BulkProductDetailsRepresentation instance from a ConnectApi CPQProductDetailsOutputRepresentation object.

BulkProductDetailsRepresentation()
: Default constructor to create an empty BulkProductDetailsRepresentation instance.

`BulkProductDetailsRepresentation(apexObj)`

Constructor to create a BulkProductDetailsRepresentation instance from a ConnectApi CPQProductDetailsOutputRepresentation object.

Signature

`public BulkProductDetailsRepresentation(ConnectApi.CPQProductDetailsOutputRepresentation apexObj)`

Parameters

apexObj
Type: ConnectApi.CPQProductDetailsOutputRepresentation

The ConnectApi product details representation object to convert to BulkProductDetailsRepresentation.

BulkProductDetailsRepresentation()

Default constructor to create an empty BulkProductDetailsRepresentation instance.

Signature

`public BulkProductDetailsRepresentation()`

BulkProductDetailsRepresentation Properties

Contains properties to include details of product definitions retrieved in bulk operations.

The `BulkProductDetailsRepresentation` class includes these properties.

additionalFields
Get the list of additionalfield.

attributeCategories
Get the list of attributecategorie.

attributes
Get the list of attribute.

availabilityDate
Get the availability date.

catalogs
Get the list of catalog.

childProducts
Get the list of childproduct.

configureDuringSale
Get the configureduringsale value.

description
Get the description of the bulkproductdetails.

discontinuedDate
Get the discontinued date.

displayUrl
Get the displayurl value.

endOfLifeDate

Get the endoflife date.

isActive

Indicates whether the item is active.

isAssetizable

Indicates whether assetizable is true or false.

isComponentRequired

Indicates whether componentrequired is true or false.

id

Get the ID of the product.

isDefaultComponent

Indicates whether defaultcomponent is true or false.

isQuantityEditable

Indicates whether quantityeditable is true or false.

isSoldOnlyWithOtherProds

Indicates whether soldonlywithotherprods is true or false.

name

Get the name of the bulkproductdetails.

nodeType

Get the nodetype value.

prices

Get the list of price.

productClassification

Get the productclassification value.

productCode

Get the productcode value.

productComponentGroups

Get the list of productcomponentgroup.

productInformation

Get the productinformation value.

productPricingInformation

Get the productpricinginformation value.

productQuantity

Get the productquantity value.

productRelatedComponent

Get the productrelatedcomponent value.

productSellingModelOptions

Get the list of productsellingmodeloption.

productSpecificationType

Get the productspecificationtype value.

productType

Get the producttype value.

qualificationContext

Get the qualificationcontext value.

status

Get the status of the bulkproductdetails.

unitOfMeasure

Get the unitofmeasure value.

additionalFields

Get the list of additionalfield.

Signature

```
public List<runtime_industries_cpq.AdditionalFieldsWrapper> additionalFields {get;
set;}
```

Property Value

Type: List<runtime_industries_cpq.AdditionalFieldsWrapper>

attributeCategories

Get the list of attributecategorie.

Signature

```
public List<runtime_industries_cpq.AttributeCategoryOutputRepresentation>
attributeCategories {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.AttributeCategoryOutputRepresentation on page 518>

attributes

Get the list of attribute.

Signature

```
public List<runtime_industries_cpq.ProductAttributeOutputRepresentation> attributes
{get; set;}
```

Property Value

Type: List<runtime_industries_cpq.ProductAttributeOutputRepresentation on page 571>

`availabilityDate`

Get the availability date.

Signature

`public Datetime availabilityDate {get; set;}`

Property Value

Type: Datetime

`catalogs`

Get the list of catalog.

Signature

`public List<runtime_industries_cpq.CatalogOutputRepresentation> catalogs {get; set;}`

Property Value

Type: List<runtime_industries_cpq.CatalogOutputRepresentation on page 539 >

`childProducts`

Get the list of childproduct.

Signature

`public List<runtime_industries_cpq.BulkProductDetailsRepresentation> childProducts {get; set;}`

Property Value

Type: List<runtime_industries_cpq.BulkProductDetailsRepresentation on page 528 >

`configureDuringSale`

Get the configureduringsale value.

Signature

`public String configureDuringSale {get; set;}`

Property Value

Type: String

`description`

Get the description of the bulkproductdetails.

Signature

```
public String description {get; set;}
```

Property Value

Type: String

discontinuedDate

Get the discontinued date.

Signature

```
public Datetime discontinuedDate {get; set;}
```

Property Value

Type: Datetime

displayUrl

Get the displayurl value.

Signature

```
public String displayUrl {get; set;}
```

Property Value

Type: String

endOfLifeDate

Get the endoflife date.

Signature

```
public Datetime endOfLifeDate {get; set;}
```

Property Value

Type: Datetime

isActive

Indicates whether the item is active.

Signature

```
public Boolean isActive {get; set;}
```

Property Value

Type: Boolean

isAssetizable

Indicates whether assetizable is true or false.

Signature

```
public Boolean isAssetizable {get; set;}
```

Property Value

Type: Boolean

isComponentRequired

Indicates whether componentrequired is true or false.

Signature

```
public Boolean isComponentRequired {get; set;}
```

Property Value

Type: Boolean

id

Get the ID of the product.

Signature

```
public String id {get; set;}
```

Property Value

Type: String

isDefaultComponent

Indicates whether defaultcomponent is true or false.

Signature

```
public Boolean isDefaultComponent {get; set;}
```

Property Value

Type: Boolean

`isQuantityEditable`

Indicates whether quantityeditable is true or false.

Signature

`public Boolean isQuantityEditable {get; set;}`

Property Value

Type: Boolean

`isSoldOnlyWithOtherProds`

Indicates whether soldonlywithotherprods is true or false.

Signature

`public Boolean isSoldOnlyWithOtherProds {get; set;}`

Property Value

Type: Boolean

`name`

Get the name of the bulkproductdetails.

Signature

`public String name {get; set;}`

Property Value

Type: String

`nodeType`

Get the nodetype value.

Signature

`public String nodeType {get; set;}`

Property Value

Type: String

`prices`

Get the list of price.

Signature

```
public List<runtime_industries_cpq.ProductPricesOutputRepresentation> prices {get;
set;}
```

Property Value

Type: List<runtime_industries_cpq.ProductPricesOutputRepresentation on page 617>

productClassification

Get the productclassification value.

Signature

```
public runtime_industries_cpq.ProductClassificationOutputRepresentation
productClassification {get; set;}
```

Property Value

Type: runtime_industries_cpq.ProductClassificationOutputRepresentation on page 579

productCode

Get the productcode value.

Signature

```
public String productCode {get; set;}
```

Property Value

Type: String

productComponentGroups

Get the list of productcomponentgroup.

Signature

```
public List<runtime_industries_cpq.ProductComponentGroupRepresentation>
productComponentGroups {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.ProductComponentGroupRepresentation on page 583>

productInformation

Get the productinformation value.

Signature

```
public String productInformation {get; set;}
```

Property Value

Type: String

productPricingInformation

Get the productpricinginformation value.

Signature

```
public String productPricingInformation {get; set;}
```

Property Value

Type: String

productQuantity

Get the productquantity value.

Signature

```
public runtime_industries_cpq.ProductQuantityOutputRepresentation productQuantity {get;
set;}
```

Property Value

Type: runtime_industries_cpq.ProductQuantityOutputRepresentation on page 620

productRelatedComponent

Get the productrelatedcomponent value.

Signature

```
public runtime_industries_cpq.ProductRelatedComponentOutputRepresentation
productRelatedComponent {get; set;}
```

Property Value

Type: runtime_industries_cpq.ProductRelatedComponentOutputRepresentation on page 624

productSellingModelOptions

Get the list of productsellingmodeloption.

Signature

```
public List<runtime_industries_cpq.ProductSellingModelOptionOutputRepresentation>
productSellingModelOptions {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.ProductSellingModelOptionOutputRepresentation on page 630>

`productSpecificationType`

Get the productspecificationtype value.

Signature

```
public runtime_industries_cpq.ProductSpecificationTypeOutputRepresentation
productSpecificationType {get; set;}
```

Property Value

Type: runtime_industries_cpq.ProductSpecificationTypeOutputRepresentation on page 635

`productType`

Get the producttype value.

Signature

```
public String productType {get; set;}
```

Property Value

Type: String

`qualificationContext`

Get the qualificationcontext value.

Signature

```
public runtime_industries_cpq.QualificationContextOutputRepresentation
qualificationContext {get; set;}
```

Property Value

Type: runtime_industries_cpq.QualificationContextOutputRepresentation on page 637

`status`

Get the status of the bulkproductdetails.

Signature

```
public String status {get; set;}
```

Property Value

Type: String

unitOfMeasure

Get the unitofmeasure value.

Signature

```
public runtime_industries_cpq.UnitOfMeasureOutputRepresentation unitOfMeasure {get;
set;}
```

Property Value

Type: runtime_industries_cpq.UnitOfMeasureOutputRepresentation on page 651

CatalogOutputRepresentation Class

Contains properties to store details of a catalog definition.

Namespace

runtime_industries_cpq on page 512

CatalogOutputRepresentation Properties
Contains properties to include details of a catalog definition.

CatalogOutputRepresentation Properties

Contains properties to include details of a catalog definition.

The `CatalogOutputRepresentation` class includes these properties.

catalogCode
Get the unique ID associated with the catalog.

catalogType
Get the category of an entry in the catalog, which is customizable. For example, catalog types, such as sellable products, services, parts, technical services, or technical resources.

customFields
Get details of the custom fields associated with a catalog.

description
Get the description of the catalog.

effectiveEndDate
Get the date and time from when the catalog isn't available to the end users.

effectiveStartDate

Get the date and time from when the catalog is available to the end users.

id

Get the ID of the catalog.

name

Get the name of the catalog.

numberOfCategories

Get the number of categories in the catalog.

status

Get the status of the catalog.

catalogCode

Get the unique ID associated with the catalog.

Signature

```
public String catalogCode {get; set;}
```

Property Value

Type: String

catalogType

Get the category of an entry in the catalog, which is customizable. For example, catalog types, such as sellable products, services, parts, technical services, or technical resources.

Signature

```
public String catalogType {get; set;}
```

Property Value

Type: String

customFields

Get details of the custom fields associated with a catalog.

Signature

```
public List<runtime_industries_cpq.AdditionalFieldsWrapper> customFields {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.AdditionalFieldsWrapper>

description

Get the description of the catalog.

Signature

```
public String description {get; set;}
```

Property Value

Type: String

effectiveEndDate

Get the date and time from when the catalog isn't available to the end users.

Signature

```
public String effectiveEndDate {get; set;}
```

Property Value

Type: String

effectiveStartDate

Get the date and time from when the catalog is available to the end users.

Signature

```
public String effectiveStartDate {get; set;}
```

Property Value

Type: String

id

Get the ID of the catalog.

Signature

```
public String id {get; set;}
```

Property Value

Type: String

name

Get the name of the catalog.

Signature

```
public String name {get; set;}
```

Property Value

Type: String

`numberOfCategories`

Get the number of categories in the catalog.

Signature

```
public Integer numberOfCategories {get; set;}
```

Property Value

Type: Integer

`status`

Get the status of the catalog.

Signature

```
public String status {get; set;}
```

Property Value

Type: String

ConfigRuleResult Class

Contains the results of configuration rule evaluation, including message rules, product recommendation rules, and visibility rules that are applied during product configuration.

Namespace

runtime_industries_cpq on page 512

ConfigRuleResult Constructor

Learn more about the constructor that's available with the ConfigRuleResult class.

The `ConfigRuleResult` class includes this constructor.

ConfigRuleResult(transactionContextId, messageRules, productRecommendationRules, visibilityRules, errors)
Constructor to create a ConfigRuleResult instance with configuration rule evaluation results.

`ConfigRuleResult(transactionContextId, messageRules, productRecommendationRules, visibilityRules, errors)`

Constructor to create a ConfigRuleResult instance with configuration rule evaluation results.

Signature

```
public ConfigRuleResult(String transactionContextId,
List<runtime_industries_cpq.MessageRule> messageRules,
List<runtime_industries_cpq.ProductRecommendationRule> productRecommendationRules,
List<runtime_industries_cpq.VisibilityRule> visibilityRules, List<String> errors)
```

Parameters

`transactionContextId`
Type: String

The ID of the transaction context for this configuration rule evaluation.

`messageRules`
Type: List<runtime_industries_cpq.MessageRule>

List of message rules that were evaluated during configuration.

`productRecommendationRules`
Type: List<runtime_industries_cpq.ProductRecommendationRule>

List of product recommendation rules that were evaluated during configuration.

`visibilityRules`
Type: List<runtime_industries_cpq.VisibilityRule>

List of visibility rules that were evaluated during configuration.

`errors`
Type: List<String>

List of error messages from the configuration rule evaluation, if any.

ConfigRuleResult Properties

Contains properties to store configuration rule evaluation results.

The `ConfigRuleResult` class includes these properties.

errors
Get the list of error messages.

messageRules
Get the list of messagerule.

productRecommendationRules
Get the list of productrecommendationrule.

transactionContextId
Get the ID of the transactioncontext.

visibilityRules
Get the list of visibilityrule.

`errors`

Get the list of error messages.

Signature

```
public List<String> errors {get; set;}
```

Property Value

Type: List<String>

`messageRules`

Get the list of messagerule.

Signature

```
public List<runtime_industries_cpq.MessageRule> messageRules {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.MessageRule on page 567>

`productRecommendationRules`

Get the list of productrecommendationrule.

Signature

```
public List<runtime_industries_cpq.ProductRecommendationRule> productRecommendationRules
{get; set;}
```

Property Value

Type: List<runtime_industries_cpq.ProductRecommendationRule on page 621>

`transactionContextId`

Get the ID of the transactioncontext.

Signature

```
public String transactionContextId {get; set;}
```

Property Value

Type: String

`visibilityRules`

Get the list of visibilityrule.

Signature

`public List<runtime_industries_cpq.VisibilityRule> visibilityRules {get; set;}`

Property Value

Type: List<runtime_industries_cpq.VisibilityRule on page 653>

CategoryOutputRepresentation Class

Contains properties to store details of a category.

Namespace

runtime_industries_cpq on page 512

CategoryOutputRepresentation Properties
Learn more about the properties available with the CategoryOutputRepresentation class.

CategoryOutputRepresentation Properties

Learn more about the properties available with the CategoryOutputRepresentation class.

The `CategoryOutputRepresentation` class includes these properties.

catalogId
Get the ID of the catalog that the category is associated with.

description
Get the description of a catalog.

hasSubCategories
Indicates whether the subcategories are available (true) or not (false).

id
Get the ID of the category.

isNavigational
Indicates whether the category node is navigational (true) or not (false).

name
Get the name of the category. If data translation is set up and specified in the org, the translated name is available.

parentCategoryId
Get the ID of the parent category.

qualificationContext
Get the context details of a user, which are used for qualification rules.

sortOrder
Get the display order of the product category relative to the siblings with the same parent category.

childCategories
Get the list of childcategorie.

catalogId

Get the ID of the catalog that the category is associated with.

Signature

```
public String catalogId {get; set;}
```

Property Value

Type: String

description

Get the description of a catalog.

Signature

```
public String description {get; set;}
```

Property Value

Type: String

hasSubCategories

Indicates whether the subcategories are available (true) or not (false).

Signature

```
public Boolean hasSubCategories {get; set;}
```

Property Value

Type: Boolean

id

Get the ID of the category.

Signature

```
public String id {get; set;}
```

Property Value

Type: String

isNavigational

Indicates whether the category node is navigational (true) or not (false).

Signature

```
public Boolean isNavigational {get; set;}
```

Property Value

Type: Boolean

name

Get the name of the category. If data translation is set up and specified in the org, the translated name is available.

Signature

```
public String name {get; set;}
```

Property Value

Type: String

parentCategoryId

Get the ID of the parent category.

Signature

```
public String parentCategoryId {get; set;}
```

Property Value

Type: String

qualificationContext

Get the context details of a user, which are used for qualification rules.

Signature

```
public runtime_industries_cpq.QualificationContextOutputRepresentation
qualificationContext {get; set;}
```

Property Value

Type: runtime_industries_cpq.QualificationContextOutputRepresentation on page 637

sortOrder

Get the display order of the product category relative to the siblings with the same parent category.

Signature

```
public Integer sortOrder {get; set;}
```

Property Value

Type: Integer

childCategories

Get the list of childcategorie.

Signature

```
public List<runtime_industries_cpq.CategoryOutputRepresentation> childCategories {get;
set;}
```

Property Value

Type: List<runtime_industries_cpq.CategoryOutputRepresentation on page 545>

ContextDataInput Class

Get details of a context.

Namespace

runtime_industries_cpq on page 512

ContextDataInput Properties
Learn more about the properties available with the ContextDataInput class.

ContextDataInput Properties

Learn more about the properties available with the ContextDataInput class.

The `ContextDataInput` class includes these properties.

nodeData
Get details of the node.

nodeName
Get the name of the node.

nodeData

Get details of the node.

Signature

```
public Map<String,ANY> nodeData {get; set;}
```

Property Value

Type: Map<String,ANY>

nodeName

Get the name of the node.

Signature

```
public String nodeName {get; set;}
```

Property Value

Type: String

FacetValueRepresentation Class

Get details of the facet values that are found in the search result.

Namespace

runtime_industries_cpq on page 512

FacetValueRepresentation Properties
Learn more about the properties available with the FacetValueRepresentation class.

FacetValueRepresentation Properties

Learn more about the properties available with the FacetValueRepresentation class.

The `FacetValueRepresentation` class includes these properties.

displayName
Get the display name of the facet value.

nameOrId
Get the name or ID of the facet. Reserved for internal use.

displayName

Get the display name of the facet value.

Signature

```
public String displayName {get; set;}
```

Property Value

Type: String

`nameOrId`

Get the name or ID of the facet. Reserved for internal use.

Signature

```
public String nameOrId {get; set;}
```

Property Value

Type: String

Filter Class

Contains the criteria property to store the details of a filter criteria, which is used to filter records.

Namespace

runtime_industries_cpq on page 512

Filter Properties
Learn more about the properties available with the Filter class.

Filter Properties

Learn more about the properties available with the Filter class.

The `Filter` class includes these properties.

criteria
Get the filter criteria to filter the records.

`criteria`

Get the filter criteria to filter the records.

Signature

```
public List<runtime_industries_cpq.FilterCriteriaInputRepresentation> criteria {get;
set;}
```

Property Value

Type: List<runtime_industries_cpq.FilterCriteriaInputRepresentation on page 551>

FilterCriteriaInputRepresentation Class

Contains properties to store criteria details to filter records based on supported properties.

Namespace

runtime_industries_cpq on page 512

FilterCriteriaInputRepresentation Properties

Learn more about the properties available with the FilterCriteriaInputRepresentation class.

FilterCriteriaInputRepresentation Properties

Learn more about the properties available with the FilterCriteriaInputRepresentation class.

The `FilterCriteriaInputRepresentation` class includes these properties.

attributeType

Get details of the search attribute type of the facet for a faceted search.

operator

Get the operator that's used for the filter criteria.

property

Get the property name to use in the filter, which must be the same as the object field. The supported property is name.

value

Get the value for the filter criteria.

`attributeType`

Get details of the search attribute type of the facet for a faceted search.

Signature

```
public String attributeType {get; set;}
```

Property Value

Type: String

Usage

Valid values are:

- `ProductStandard`
- `ProductCustom`
- `ProductDynamicAttribute`
- `ProductAttributeStandard`
- `ProductAttributeCustom`

`operator`

Get the operator that's used for the filter criteria.

Signature

`public String operator {get; set;}`

Property Value

Type: String

Usage

The supported operators are:

- `eq`
- `in`
- `contains`
- `gt`—Specifies a greater than criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only.
- `lt`—Specifies a less than criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only.
- `gte`—Specifies a greater than or equal to criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only.
- `lte`—Specifies a less than or equal to criteria. Available from API version 63.0 and later for Number, Date, and Datetime data types only.

`property`

Get the property name to use in the filter, which must be the same as the object field. The supported property is name.

Signature

`public String property {get; set;}`

Property Value

Type: String

`value`

Get the value for the filter criteria.

Signature

`public String value {get; set;}`

Property Value

Type: String

FilterInputRepresentation Class

Contains the filter property to filters records based on supported criteria.

Namespace

runtime_industries_cpq on page 512

FilterInputRepresentation Properties
Learn more about the available properties with the FilterInputRepresentation class.

FilterInputRepresentation Properties

Learn more about the available properties with the FilterInputRepresentation class.

The `FilterInputRepresentation` class includes these properties.

filter
Filters records based on supported criteria. The supported property is name.

`filter`

Filters records based on supported criteria. The supported property is name.

Signature

```
public runtime_industries_cpq.Filter filter {get; set;}
```

Property Value

Type: runtime_industries_cpq.Filter on page 550

Usage

The supported operators are:

- `eq`
- `in`
- `contains`—This value isn't applicable if the **Use Indexed Data For Product Listing and Search** toggle from the Product Discovery Settings page from Setup is enabled.

If multiple criteria are specified, then the resultant criteria are combined by using the `and` operator.

GuidedSelectionRepresentation Class

Contains properties to represent a product in a guided selection flow. This class is used in Product Discovery to provide structured product information during guided product selection processes.

Namespace

runtime_industries_cpq on page 512

GuidedSelectionRepresentation Properties

Contains properties to include details of a product in a guided selection flow.

GuidedSelectionRepresentation Properties

Contains properties to include details of a product in a guided selection flow.

The `GuidedSelectionRepresentation` class includes these properties.

additionalFields

Get the list of additionalfield.

attributeCategories

Get the list of attributecategorie.

availabilityDate

Get the availability date.

catalogs

Get the list of catalog.

categories

Get the list of categorie.

childProducts

Get the list of childproduct.

configureDuringSale

Get the configureduringsale value.

description

Get the description of the guidedselection.

discontinuedDate

Get the discontinued date.

displayUrl

Get the displayurl value.

endOfLifeDate

Get the endoflife date.

id

Get the ID of the guidedselection.

isActive

Indicates whether the item is active.

isAssetizable

Indicates whether assetizable is true or false.

isComponentRequired

Indicates whether componentrequired is true or false.

isDefaultComponent

Indicates whether defaultcomponent is true or false.

isQuantityEditable

Indicates whether quantityeditable is true or false.

isSoldOnlyWithOtherProds

Indicates whether soldonlywithotherprods is true or false.

name

Get the name of the guidedselection.

nodeType

Get the nodetype value.

prices

Get the list of price.

productClassification

Get the productclassification value.

productCode

Get the productcode value.

productComponentGroups

Get the list of productcomponentgroup.

productInformation

Get the productinformation value.

productPricingInformation

Get the productpricinginformation value.

productQuantity

Get the productquantity value.

productRelatedComponent

Get the productrelatedcomponent value.

productSellingModelOptions

Get the list of productsellingmodeloption.

productSpecificationType

Get the productspecificationtype value.

productType

Get the producttype value.

qualificationContext

Get the qualificationcontext value.

status

Get the status of the guidedselection.

unitOfMeasure

Get the unitofmeasure value.

`additionalFields`

Get the list of additionalfield.

Signature

```
public List<runtime_industries_cpq.AdditionalFieldsWrapper> additionalFields {get;
set;}
```

Property Value

Type: List<runtime_industries_cpq.AdditionalFieldsWrapper>

attributeCategories

Get the list of attributecategorie.

Signature

```
public List<runtime_industries_cpq.AttributeCategoryOutputRepresentation>
attributeCategories {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.AttributeCategoryOutputRepresentation on page 518>

availabilityDate

Get the availability date.

Signature

```
public Datetime availabilityDate {get; set;}
```

Property Value

Type: Datetime

catalogs

Get the list of catalog.

Signature

```
public List<runtime_industries_cpq.CatalogOutputRepresentation> catalogs {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.CatalogOutputRepresentation on page 539>

categories

Get the list of categorie.

Signature

```
public List<runtime_industries_cpq.CategoryOutputRepresentation> categories {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.CategoryOutputRepresentation on page 545>

childProducts

Get the list of childproduct.

Signature

```
public List<runtime_industries_cpq.ProductListRepresentation> childProducts {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.ProductListRepresentation on page 598>

configureDuringSale

Get the configureduringsale value.

Signature

```
public String configureDuringSale {get; set;}
```

Property Value

Type: String

description

Get the description of the guidedselection.

Signature

```
public String description {get; set;}
```

Property Value

Type: String

discontinuedDate

Get the discontinued date.

Signature

```
public Datetime discontinuedDate {get; set;}
```

Property Value

Type: Datetime

`displayUrl`

Get the displayurl value.

Signature

```
public String displayUrl {get; set;}
```

Property Value

Type: String

`endOfLifeDate`

Get the endoflife date.

Signature

```
public Datetime endOfLifeDate {get; set;}
```

Property Value

Type: Datetime

`id`

Get the ID of the guidedselection.

Signature

```
public String id {get; set;}
```

Property Value

Type: String

`isActive`

Indicates whether the item is active.

Signature

```
public Boolean isActive {get; set;}
```

Property Value

Type: Boolean

`isAssetizable`

Indicates whether assetizable is true or false.

Signature

`public Boolean isAssetizable {get; set;}`

Property Value

Type: Boolean

`isComponentRequired`

Indicates whether componentrequired is true or false.

Signature

`public Boolean isComponentRequired {get; set;}`

Property Value

Type: Boolean

`isDefaultComponent`

Indicates whether defaultcomponent is true or false.

Signature

`public Boolean isDefaultComponent {get; set;}`

Property Value

Type: Boolean

`isQuantityEditable`

Indicates whether quantityeditable is true or false.

Signature

`public Boolean isQuantityEditable {get; set;}`

Property Value

Type: Boolean

`isSoldOnlyWithOtherProds`

Indicates whether soldonlywithotherprods is true or false.

Signature

```
public Boolean isSoldOnlyWithOtherProds {get; set;}
```

Property Value

Type: Boolean

`name`

Get the name of the guidedselection.

Signature

```
public String name {get; set;}
```

Property Value

Type: String

`nodeType`

Get the nodetype value.

Signature

```
public String nodeType {get; set;}
```

Property Value

Type: String

`prices`

Get the list of price.

Signature

```
public List<runtime_industries_cpq.ProductPricesOutputRepresentation> prices {get;
set;}
```

Property Value

Type: List<runtime_industries_cpq.ProductPricesOutputRepresentation on page 617 >

`productClassification`

Get the productclassification value.

Signature

```
public runtime_industries_cpq.ProductClassificationOutputRepresentation
productClassification {get; set;}
```

Property Value

Type: runtime_industries_cpq.ProductClassificationOutputRepresentation on page 579

productCode

Get the productcode value.

Signature

```
public String productCode {get; set;}
```

Property Value

Type: String

productComponentGroups

Get the list of productcomponentgroup.

Signature

```
public List<runtime_industries_cpq.ProductComponentGroupOutputRepresentation>
productComponentGroups {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.ProductComponentGroupOutputRepresentation on page 580 >

productInformation

Get the productinformation value.

Signature

```
public String productInformation {get; set;}
```

Property Value

Type: String

productPricingInformation

Get the productpricinginformation value.

Signature

```
public String productPricingInformation {get; set;}
```

Property Value

Type: String

`productQuantity`

Get the productquantity value.

Signature

```
public runtime_industries_cpq.ProductQuantityOutputRepresentation productQuantity {get;
set;}
```

Property Value

Type: runtime_industries_cpq.ProductQuantityOutputRepresentation on page 620

`productRelatedComponent`

Get the productrelatedcomponent value.

Signature

```
public ConnectApi.CPQProductRelatedComponentOutputRepresentation productRelatedComponent
{get; set;}
```

Property Value

Type: ConnectApi.CPQProductRelatedComponentOutputRepresentation

`productSellingModelOptions`

Get the list of productsellingmodeloption.

Signature

```
public List<runtime_industries_cpq.ProductSellingModelOptionOutputRepresentation>
productSellingModelOptions {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.ProductSellingModelOptionOutputRepresentation on page 630 >

`productSpecificationType`

Get the productspecificationtype value.

Signature

```
public runtime_industries_cpq.ProductSpecificationTypeOutputRepresentation
productSpecificationType {get; set;}
```

Property Value

Type: runtime_industries_cpq.ProductSpecificationTypeOutputRepresentation on page 635

productType

Get the producttype value.

Signature

```
public String productType {get; set;}
```

Property Value

Type: String

qualificationContext

Get the qualificationcontext value.

Signature

```
public runtime_industries_cpq.QualificationContextOutputRepresentation
qualificationContext {get; set;}
```

Property Value

Type: runtime_industries_cpq.QualificationContextOutputRepresentation on page 637

status

Get the status of the guidedselection.

Signature

```
public String status {get; set;}
```

Property Value

Type: String

unitOfMeasure

Get the unitofmeasure value.

Signature

```
public runtime_industries_cpq.UnitOfMeasureOutputRepresentation unitOfMeasure {get;
set;}
```

Property Value

Type: runtime_industries_cpq.UnitOfMeasureOutputRepresentation on page 651

GuidedSelectionSearchTerm Class

Represents a search term used in guided product selection. Contains the search term text and associated tags for filtering and searching products in Product Discovery.

Namespace

runtime_industries_cpq on page 512

GuidedSelectionSearchTerm Constructor
Learn more about the constructors that are available with the GuidedSelectionSearchTerm class.

GuidedSelectionSearchTerm Properties
Contains properties to include search term details for guided product selection.

GuidedSelectionSearchTerm Constructor

Learn more about the constructors that are available with the GuidedSelectionSearchTerm class.

The `GuidedSelectionSearchTerm` class includes these constructors.

GuidedSelectionSearchTerm(apexObj)
Constructor to create a GuidedSelectionSearchTerm instance from a ConnectApi CPQGuidedSelectionSearchTermOutputRepresentation object.

GuidedSelectionSearchTerm()
Default constructor to create an empty GuidedSelectionSearchTerm instance.

GuidedSelectionSearchTerm(apexObj)

Constructor to create a GuidedSelectionSearchTerm instance from a ConnectApi CPQGuidedSelectionSearchTermOutputRepresentation object.

Signature

```
public
GuidedSelectionSearchTerm(ConnectApi.CPQGuidedSelectionSearchTermOutputRepresentation
apexObj)
```

Parameters

apexObj

Type: ConnectApi.CPQGuidedSelectionSearchTermOutputRepresentation

The ConnectApi guided selection search term representation object to convert to GuidedSelectionSearchTerm.

`GuidedSelectionSearchTerm()`

Default constructor to create an empty GuidedSelectionSearchTerm instance.

Signature

`public GuidedSelectionSearchTerm()`

GuidedSelectionSearchTerm Properties

Contains properties to include search term details for guided product selection.

The `GuidedSelectionSearchTerm` class includes these properties.

tags
Get the list of tags.

term
Get the term value.

`tags`

Get the list of tags.

Signature

`public List<String> tags {get; set;}`

Property Value

Type: List<String>

`term`

Get the term value.

Signature

`public String term {get; set;}`

Property Value

Type: String

GuidedSelectionSearchTermList Class

Contains a list of search terms used in guided product selection. This class is used to pass multiple search terms for filtering and searching products in Product Discovery.

Namespace

runtime_industries_cpq on page 512

GuidedSelectionSearchTermList Constructor
Learn more about the constructor that's available with the GuidedSelectionSearchTermList class.

GuidedSelectionSearchTermList Properties
Contains properties to include a list of search terms for guided product selection.

GuidedSelectionSearchTermList Constructor

Learn more about the constructor that's available with the GuidedSelectionSearchTermList class.

The `GuidedSelectionSearchTermList` class includes this constructor.

GuidedSelectionSearchTermList()
Default constructor to create an empty GuidedSelectionSearchTermList instance.

`GuidedSelectionSearchTermList()`

Default constructor to create an empty GuidedSelectionSearchTermList instance.

Signature

```
public GuidedSelectionSearchTermList()
```

GuidedSelectionSearchTermList Properties

Contains properties to include a list of search terms for guided product selection.

The `GuidedSelectionSearchTermList` class includes these properties.

searchTerms
Get the list of searchterm.

`searchTerms`

Get the list of searchterm.

Signature

```
public List<runtime_industries_cpq.GuidedSelectionSearchTerm> searchTerms {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.GuidedSelectionSearchTerm on page 564>

MessageRule Class

Represents a message rule that is evaluated during product configuration. Message rules display informational, warning, or error messages to users based on configuration conditions.

Namespace

runtime_industries_cpq on page 512

MessageRule Constructor

Learn more about the constructor that's available with the MessageRule class.

The `MessageRule` class includes this constructor.

`MessageRule(stiId, severity, messages)`

Constructor to create a MessageRule instance with the specified STI ID, severity, and messages.

Signature

```
public MessageRule(String stiId, String severity, List<String> messages)
```

Parameters

stiId
Type: String

The ID of the Sales Transaction Item (STI) associated with this message rule.

severity
Type: String

The severity level of the message (for example, Error, Warning, or Info).

messages
Type: List<String>

List of message strings to display to the user.

MessageRule Properties

Contains properties to store message rule details from configuration rule evaluation.

The `MessageRule` class includes these properties.

[messages](#)
Get the list of messages.

[severity](#)
Get the severity value.

[stiId](#)
Get the ID of the sti.

`messages`

Get the list of messages.

Signature

```
public List<String> messages {get; set;}
```

Property Value

Type: List<String>

`severity`

Get the severity value.

Signature

```
public String severity {get; set;}
```

Property Value

Type: String

`stiId`

Get the ID of the sti.

Signature

```
public String stiId {get; set;}
```

Property Value

Type: String

PricingModelOutputRepresentation Class

Contains details of a pricing model in a product configuration.

Namespace

runtime_industries_cpq on page 512

PricingModelOutputRepresentation Properties
Learn more about the properties available with the PricingModelOutputRepresentation class.

PricingModelOutputRepresentation Properties

Learn more about the properties available with the PricingModelOutputRepresentation class.

The `PricingModelOutputRepresentation` class includes these properties.

frequency
Contains frequency details of the pricing model.

id
Contains the ID of the pricing model.

name
Contains the name of the pricing model.

occurrence
Contains details about the occurrence of the pricing model.

pricingModelType
Contains the type of the pricing model.

unitOfMeasure
Contains details about the unit of measure for the pricing model.

`frequency`

Contains frequency details of the pricing model.

Signature

```
public String frequency {get; set;}
```

Property Value

Type: String

`id`

Contains the ID of the pricing model.

Signature

```
public String id {get; set;}
```

Property Value

Type: String

name

Contains the name of the pricing model.

Signature

```
public String name {get; set;}
```

Property Value

Type: String

occurrence

Contains details about the occurrence of the pricing model.

Signature

```
public Integer occurrence {get; set;}
```

Property Value

Type: Integer

pricingModelType

Contains the type of the pricing model.

Signature

```
public String pricingModelType {get; set;}
```

Property Value

Type: String

unitOfMeasure

Contains details about the unit of measure for the pricing model.

Signature

```
public String unitOfMeasure {get; set;}
```

Property Value

Type: String

ProductAttributeOutputRepresentation Class

Contains details about the attribute in a product configuration.

Namespace

runtime_industries_cpq on page 512

ProductAttributeOutputRepresentation Properties

Learn more about the properties available with the ProductAttributeOutputRepresentation class.

ProductAttributeOutputRepresentation Properties

Learn more about the properties available with the ProductAttributeOutputRepresentation class.

The `ProductAttributeOutputRepresentation` class includes these properties.

additionalFields

Get the key-value pairs of additional standard or custom fields and their values for the product attribute.

attributeCategoryId

Get the ID of the attribute category.

attributeNameOverride

Get the name override value of the attribute.

attributePickList

Get the picklist values of the attribute.

code

Get the code of the attribute.

dataType

Get the data type of the attribute.

defaultHelpText

Get the default help text value of the attribute.

defaultValue

Get the default value of the attribute.

description

Get the description of the attribute.

developerName

Get the developer name of the product attribute.

displayTypeOverride

Get the override value of the display type of the attribute.

hidden

Indicates whether the attribute is hidden (true) or not (false).

id
Get the ID of the attribute.

isCloneable
Indicates whether the attribute is cloneable (true) or not (false).

isConfigurable
Indicates whether the attribute is configurable (true) or not (false).

isEncrypted
Indicates whether the attribute is encrypted (true) or not (false).

isPriceImpacting
Indicates whether this is a price impacting attribute (true) or not (false).

isReadOnly
Indicates whether the attribute is read-only (true) or not (false).

isRequired
Indicates whether the attribute is required (true) or not (false).

label
Get the label of the attribute.

maximumValue
Get the maximum value for the product attribute.

minimumValue
Get the minimum value for the product attribute.

name
Get the name of the attribute.

sequence
Get the sequence values of the attribute.

status
Get the status of the attribute.

stepValue
Get the step value for the attribute.

unitOfMeasure
Get details about the unit of measure associated with an attribute.

userValue
Get the user value of the attribute.

valueDescription
Get the value description of the attribute.

`additionalFields`

Get the key-value pairs of additional standard or custom fields and their values for the product attribute.

Signature

```
public List<runtime_industries_cpq.AdditionalFieldsWrapper> additionalFields {get;
set;}
```

Property Value

Type: List<runtime_industries_cpq.AdditionalFieldsWrapper>

attributeCategoryId

Get the ID of the attribute category.

Signature

```
public String attributeCategoryId {get; set;}
```

Property Value

Type: String

attributeNameOverride

Get the name override value of the attribute.

Signature

```
public String attributeNameOverride {get; set;}
```

Property Value

Type: String

attributePickList

Get the picklist values of the attribute.

Signature

```
public runtime_industries_cpq.AttributePickListOutputRepresentation attributePickList
{get; set;}
```

Property Value

Type: runtime_industries_cpq.AttributePickListOutputRepresentation on page 521

code

Get the code of the attribute.

Signature

`public String code {get; set;}`

Property Value

Type: String

`dataType`

Get the data type of the attribute.

Signature

`public String dataType {get; set;}`

Property Value

Type: String

`defaultHelpText`

Get the default help text value of the attribute.

Signature

`public String defaultHelpText {get; set;}`

Property Value

Type: String

`defaultValue`

Get the default value of the attribute.

Signature

`public String defaultValue {get; set;}`

Property Value

Type: String

`description`

Get the description of the attribute.

Signature

`public String description {get; set;}`

Property Value

Type: String

`developerName`

Get the developer name of the product attribute.

Signature

`public String developerName {get; set;}`

Property Value

Type: String

`displayTypeOverride`

Get the override value of the display type of the attribute.

Signature

`public String displayTypeOverride {get; set;}`

Property Value

Type: String

`hidden`

Indicates whether the attribute is hidden (true) or not (false).

Signature

`public Boolean hidden {get; set;}`

Property Value

Type: Boolean

`id`

Get the ID of the attribute.

Signature

`public String id {get; set;}`

Property Value

Type: String

isCloneable

Indicates whether the attribute is cloneable (true) or not (false).

Signature

```
public Boolean isCloneable {get; set;}
```

Property Value

Type: Boolean

isConfigurable

Indicates whether the attribute is configurable (true) or not (false).

Signature

```
public Boolean isConfigurable {get; set;}
```

Property Value

Type: Boolean

isEncrypted

Indicates whether the attribute is encrypted (true) or not (false).

Signature

```
public Boolean isEncrypted {get; set;}
```

Property Value

Type: Boolean

isPriceImpacting

Indicates whether this is a price impacting attribute (true) or not (false).

Signature

```
public Boolean isPriceImpacting {get; set;}
```

Property Value

Type: Boolean

isReadOnly

Indicates whether the attribute is read-only (true) or not (false).

Signature

```
public Boolean isReadOnly {get; set;}
```

Property Value

Type: Boolean

`isRequired`

Indicates whether the attribute is required (true) or not (false).

Signature

```
public Boolean isRequired {get; set;}
```

Property Value

Type: Boolean

`label`

Get the label of the attribute.

Signature

```
public String label {get; set;}
```

Property Value

Type: String

`maximumValue`

Get the maximum value for the product attribute.

Signature

```
public String maximumValue {get; set;}
```

Property Value

Type: String

`minimumValue`

Get the minimum value for the product attribute.

Signature

```
public String minimumValue {get; set;}
```

Property Value

Type: String

name

Get the name of the attribute.

Signature

```
public String name {get; set;}
```

Property Value

Type: String

sequence

Get the sequence values of the attribute.

Signature

```
public Integer sequence {get; set;}
```

Property Value

Type: Integer

status

Get the status of the attribute.

Signature

```
public String status {get; set;}
```

Property Value

Type: String

stepValue

Get the step value for the attribute.

Signature

```
public String stepValue {get; set;}
```

Property Value

Type: String

unitOfMeasure

Get details about the unit of measure associated with an attribute.

Signature

```
public ConnectApi.UnitOfMeasureOutputRepresentation unitOfMeasure {get; set;}
```

Property Value

Type: ConnectApi.UnitOfMeasureOutputRepresentation

userValue

Get the user value of the attribute.

Signature

```
public String userValue {get; set;}
```

Property Value

Type: String

valueDescription

Get the value description of the attribute.

Signature

```
public String valueDescription {get; set;}
```

Property Value

Type: String

ProductClassificationOutputRepresentation Class

Get details of the product classification in a product configuration.

Namespace

runtime_industries_cpq on page 512

ProductClassificationOutputRepresentation Properties
Learn more about the properties available with the ProductClassificationOutputRepresentation class.

ProductClassificationOutputRepresentation Properties

Learn more about the properties available with the ProductClassificationOutputRepresentation class.

The `ProductClassificationOutputRepresentation` class includes these properties.

[id](#)
Get the ID of the product classification.

id

Get the ID of the product classification.

Signature

```
public String id {get; set;}
```

Property Value

Type: String

ProductComponentGroupOutputRepresentation Class

Get details of the product component group in a product classification.

Namespace

runtime_industries_cpq on page 512

[ProductComponentGroupOutputRepresentation Properties](#)
Learn more about the properties available with the ProductComponentGroupOutputRepresentation class.

ProductComponentGroupOutputRepresentation Properties

Learn more about the properties available with the ProductComponentGroupOutputRepresentation class.

The `ProductComponentGroupOutputRepresentation` class includes these properties.

[childGroups](#)
Get the child groups associated with a product component group.

[classifications](#)
Get the list of classifications for the product component group.

[code](#)
Get the code of the product component group.

[components](#)
Get the details of components within the product component group.

[description](#)
Get the description of the product component group.

[id](#)
Get the ID of the product component group.

[name](#)
Get the name of the product component group.

parentGroupId
Get the ID of the parent group.

parentProductId
Get the parent Product2 ID of the product component group.

sequence
Get the sequence of the product component group.

childGroups

Get the child groups associated with a product component group.

Signature

```
public List<runtime_industries_cpq.ProductComponentGroupOutputRepresentation> childGroups
{get; set;}
```

Property Value

Type: List<runtime_industries_cpq.ProductComponentGroupOutputRepresentation on page 580>

classifications

Get the list of classifications for the product component group.

Signature

```
public List<runtime_industries_cpq.ProductClassificationOutputRepresentation>
classifications {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.ProductClassificationOutputRepresentation on page 579>

code

Get the code of the product component group.

Signature

```
public String code {get; set;}
```

Property Value

Type: String

components

Get the details of components within the product component group.

Signature

```
public List<runtime_industries_cpq.ProductDetailsRepresentation> components {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.ProductDetailsRepresentation on page 587>

description

Get the description of the product component group.

Signature

```
public String description {get; set;}
```

Property Value

Type: String

id

Get the ID of the product component group.

Signature

```
public String id {get; set;}
```

Property Value

Type: String

name

Get the name of the product component group.

Signature

```
public String name {get; set;}
```

Property Value

Type: String

parentGroupId

Get the ID of the parent group.

Signature

```
public String parentGroupId {get; set;}
```

Property Value

Type: String

`parentProductId`

Get the parent Product2 ID of the product component group.

Signature

`public String parentProductId {get; set;}`

Property Value

Type: String

`sequence`

Get the sequence of the product component group.

Signature

`public Integer sequence {get; set;}`

Property Value

Type: Integer

ProductComponentGroupRepresentation Class

Represents a product component group used in bulk product operations. This class is similar to ProductComponentGroupOutputRepresentation but is used specifically for bulk product detail representations where components are represented as BulkProductDetailsRepresentation objects.

Namespace

runtime_industries_cpq on page 512

ProductComponentGroupRepresentation Constructor

Learn more about the constructor that's available with the ProductComponentGroupRepresentation class.

The `ProductComponentGroupRepresentation` class includes this constructor.

ProductComponentGroupRepresentation(apexObj)

Constructor to create a ProductComponentGroupRepresentation instance from a ConnectApi CPQProductComponentGroupOutputRepresentation object.

`ProductComponentGroupRepresentation(apexObj)`

Constructor to create a ProductComponentGroupRepresentation instance from a ConnectApi CPQProductComponentGroupOutputRepresentation object.

Signature

```
public
ProductComponentGroupRepresentation(ConnectApi.CPQProductComponentGroupOutputRepresentation
apexObj)
```

Parameters

`apexObj`

Type: ConnectApi.CPQProductComponentGroupOutputRepresentation

The ConnectApi product component group representation object to convert to ProductComponentGroupRepresentation.

ProductComponentGroupRepresentation Properties

Contains properties to include details of a product component group used in bulk operations.

The `ProductComponentGroupRepresentation` class includes these properties.

childGroups

Get the list of childgroup.

classifications

Get the list of product classifications.

code

Get the code of the productcomponentgroup.

components

Get the list of component.

description

Get the description of the productcomponentgroup.

id

Get the ID of the productcomponentgroup.

name

Get the name of the productcomponentgroup.

parentGroupId

Get the ID of the parentgroup.

parentProductId

Get the ID of the parentproduct.

sequence
Get the sequence value.

childGroups

Get the list of childgroup.

Signature

```
public List<runtime_industries_cpq.ProductComponentGroupRepresentation> childGroups
{get; set;}
```

Property Value

Type: List<runtime_industries_cpq.ProductComponentGroupRepresentation on page 583>

classifications

Get the list of product classifications.

Signature

```
public List<runtime_industries_cpq.ProductClassificationOutputRepresentation>
classifications {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.ProductClassificationOutputRepresentation on page 579>

code

Get the code of the productcomponentgroup.

Signature

```
public String code {get; set;}
```

Property Value

Type: String

components

Get the list of component.

Signature

```
public List<runtime_industries_cpq.BulkProductDetailsRepresentation> components {get;
set;}
```

Property Value

Type: List<runtime_industries_cpq.BulkProductDetailsRepresentation on page 528>

description

Get the description of the productcomponentgroup.

Signature

`public String description {get; set;}`

Property Value

Type: String

id

Get the ID of the productcomponentgroup.

Signature

`public String id {get; set;}`

Property Value

Type: String

name

Get the name of the productcomponentgroup.

Signature

`public String name {get; set;}`

Property Value

Type: String

parentGroupId

Get the ID of the parentgroup.

Signature

`public String parentGroupId {get; set;}`

Property Value

Type: String

`parentProductId`

Get the ID of the parentproduct.

Signature

```
public String parentProductId {get; set;}
```

Property Value

Type: String

`sequence`

Get the sequence value.

Signature

```
public Integer sequence {get; set;}
```

Property Value

Type: Integer

ProductDetailsRepresentation Class

Get the details of a product definition.

Namespace

runtime_industries_cpq on page 512

ProductDetailsRepresentation Constructor
Learn more about the constructor that's available with the ProductDetailsRepresentation class.

ProductDetailsRepresentation Properties
Learn more about the properties available with the ProductDetailsRepresentation class.

ProductDetailsRepresentation Constructor

Learn more about the constructor that's available with the ProductDetailsRepresentation class.

The `ProductDetailsRepresentation` class includes this constructor.

ProductDetailsRepresentation()
Constructor to get the product details for a product definition.

`ProductDetailsRepresentation()`

Constructor to get the product details for a product definition.

Signature

`public ProductDetailsRepresentation()`

ProductDetailsRepresentation Properties

Learn more about the properties available with the ProductDetailsRepresentation class.

The `ProductDetailsRepresentation` includes these properties.

additionalFields
Get the key-value pair of additional standard or custom fields with their values.

attributeCategories
Get the list of categorized attributes related to the product.

attributes
Get the list of uncategorized attributes related to the product.

availabilityDate
Get the date when the part is used in the product or is made available for sale.

catalogs
Get the list of the associated catalogs returned with the Product List API (POST) response. The Product By ID API (GET) returns an empty catalog list in the response. Returns the name and id values only.

childProducts
Get the hierarchy of the child products.

configureDuringSale
Determines whether to allow or prevent configuration when a bundle is sold.

description
Get the description of the product.

discontinuedDate
Get the date from when the part can’t be used in the product or sold.

displayUrl
Get the display image URL of the product.

endOfLifeDate
Get the date after which a product isn’t supported, ordered, or maintained.

id
Get the ID of the product.

isActive
Indicates whether the product is active (true) or not (false).

isAssetizable
Indicates whether the product instance remains a customer asset after it's purchased (true) or not (false).

isComponentRequired
Indicates whether the product component is required (true) or not (false).

isDefaultComponent
Indicates whether the product component is the default component (true) or not (false).

isQuantityEditable

Indicates whether the product quantity is editable (true) or not (false).

isSoldOnlyWithOtherProds

Indicates whether the product can't be sold separately (true) or not (false).

name

Get the name of the product.

nodeType

Get the type of the node, such as a product or bundled product.

prices

Get the price details associated with the products.

productClassification

Get the details of the product classification that the product is based on.

productCode

Get the universal product code that's used to track the part that's used in the product.

productComponentGroups

Get the logical grouping of the component products in a bundle and the group cardinality for ordering the product components.

productInformation

Get the details of a product.

productPricingInformation

Get the pricing details of a product.

productQuantity

Get the quantity of a product.

productRelatedComponent

Get the details of the related components of a product.

productSellingModelOptions

Get the details of the product selling model options.

productSpecificationType

Get the details of the product specification type.

productType

Get the product type.

qualificationContext

Get the context details of a user, which are used for qualification rules.

status

Get or set the status of the product, such as Active or Inactive.

unitOfMeasure

Get details about the unit of measure for a specific set of records.

`additionalFields`

Get the key-value pair of additional standard or custom fields with their values.

Signature

```
public List<runtime_industries_cpq.AdditionalFieldsWrapper> additionalFields {get;
set;}
```

Property Value

Type: List<runtime_industries_cpq.AdditionalFieldsWrapper>

attributeCategories

Get the list of categorized attributes related to the product.

Signature

```
public List<runtime_industries_cpq.AttributeCategoryOutputRepresentation>
attributeCategories {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.AttributeCategoryOutputRepresentation on page 518>

attributes

Get the list of uncategorized attributes related to the product.

Signature

```
public List<runtime_industries_cpq.ProductAttributeOutputRepresentation> attributes
{get; set;}
```

Property Value

Type: List<runtime_industries_cpq.ProductAttributeOutputRepresentation on page 571>

availabilityDate

Get the date when the part is used in the product or is made available for sale.

Signature

```
public Datetime availabilityDate {get; set;}
```

Property Value

Type: Datetime

catalogs

Get the list of the associated catalogs returned with the Product List API (POST) response. The Product By ID API (GET) returns an empty catalog list in the response. Returns the name and id values only.

Signature

```
public List<runtime_industries_cpq.CatalogOutputRepresentation> catalogs {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.CatalogOutputRepresentation on page 539>

childProducts

Get the hierarchy of the child products.

Signature

```
public List<runtime_industries_cpq.ProductDetailsRepresentation> childProducts {get;
set;}
```

Property Value

Type: List<runtime_industries_cpq.ProductDetailsRepresentation on page 587>

configureDuringSale

Determines whether to allow or prevent configuration when a bundle is sold.

Signature

```
public String configureDuringSale {get; set;}
```

Property Value

Type: String

description

Get the description of the product.

If data translation is set up and specified in the org, the translated description is available.

Signature

```
public String description {get; set;}
```

Property Value

Type: String

discontinuedDate

Get the date from when the part can't be used in the product or sold.

Signature

```
public Datetime discontinuedDate {get; set;}
```

Property Value

Type: Datetime

displayUrl

Get the display image URL of the product.

Signature

```
public String displayUrl {get; set;}
```

Property Value

Type: String

endOfLifeDate

Get the date after which a product isn't supported, ordered, or maintained.

Signature

```
public Datetime endOfLifeDate {get; set;}
```

Property Value

Type: Datetime

id

Get the ID of the product.

Signature

```
public String id {get; set;}
```

Property Value

Type: String

isActive

Indicates whether the product is active (true) or not (false).

Signature

```
public Boolean isActive {get; set;}
```

Property Value

Type: Boolean

isAssetizable

Indicates whether the product instance remains a customer asset after it's purchased (true) or not (false).

Signature

```
public Boolean isAssetizable {get; set;}
```

Property Value

Type: Boolean

isComponentRequired

Indicates whether the product component is required (true) or not (false).

Signature

```
public Boolean isComponentRequired {get; set;}
```

Property Value

Type: Boolean

isDefaultComponent

Indicates whether the product component is the default component (true) or not (false).

Signature

```
public Boolean isDefaultComponent {get; set;}
```

Property Value

Type: Boolean

isQuantityEditable

Indicates whether the product quantity is editable (true) or not (false).

Signature

```
public Boolean isQuantityEditable {get; set;}
```

Property Value

Type: Boolean

`isSoldOnlyWithOtherProds`

Indicates whether the product can't be sold separately (true) or not (false).

Signature

```
public Boolean isSoldOnlyWithOtherProds {get; set;}
```

Property Value

Type: Boolean

`name`

Get the name of the product.

If data translation is set up and specified in the org, the translated name is available.

Signature

```
public String name {get; set;}
```

Property Value

Type: String

`nodeType`

Get the type of the node, such as a product or bundled product.

Signature

```
public String nodeType {get; set;}
```

Property Value

Type: String

`prices`

Get the price details associated with the products.

Signature

```
public List<runtime_industries_cpq.ProductPricesOutputRepresentation> prices {get;
set;}
```

Property Value

Type: List<runtime_industries_cpq.ProductPricesOutputRepresentation on page 617 >

productClassification

Get the details of the product classification that the product is based on.

Signature

```
public runtime_industries_cpq.ProductClassificationOutputRepresentation
productClassification {get; set;}
```

Property Value

Type: runtime_industries_cpq.ProductClassificationOutputRepresentation on page 579

productCode

Get the universal product code that's used to track the part that’s used in the product.

Signature

```
public String productCode {get; set;}
```

Property Value

Type: String

productComponentGroups

Get the logical grouping of the component products in a bundle and the group cardinality for ordering the product components.

Signature

```
public List<runtime_industries_cpq.ProductComponentGroupOutputRepresentation>
productComponentGroups {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.ProductComponentGroupOutputRepresentation on page 580 >

productInformation

Get the details of a product.

Signature

```
public String productInformation {get; set;}
```

Property Value

Type: String

`productPricingInformation`

Get the pricing details of a product.

Signature

```
public String productPricingInformation {get; set;}
```

Property Value

Type: String

`productQuantity`

Get the quantity of a product.

Signature

```
public runtime_industries_cpq.ProductQuantityOutputRepresentation productQuantity {get;
set;}
```

Property Value

Type: runtime_industries_cpq.ProductQuantityOutputRepresentation on page 620

`productRelatedComponent`

Get the details of the related components of a product.

Signature

```
public ConnectApi.CPQProductRelatedComponentOutputRepresentation productRelatedComponent
{get; set;}
```

Property Value

Type: ConnectApi.CPQProductRelatedComponentOutputRepresentation

`productSellingModelOptions`

Get the details of the product selling model options.

Signature

```
public List<runtime_industries_cpq.ProductSellingModelOptionOutputRepresentation>
productSellingModelOptions {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.ProductSellingModelOptionOutputRepresentation on page 630 >

productSpecificationType

Get the details of the product specification type.

Signature

```
public runtime_industries_cpq.ProductSpecificationTypeOutputRepresentation
productSpecificationType {get; set;}
```

Property Value

Type: runtime_industries_cpq.ProductSpecificationTypeOutputRepresentation on page 635

productType

Get the product type.

Signature

```
public String productType {get; set;}
```

Property Value

Type: String

qualificationContext

Get the context details of a user, which are used for qualification rules.

Signature

```
public runtime_industries_cpq.QualificationContextOutputRepresentation
qualificationContext {get; set;}
```

Property Value

Type: runtime_industries_cpq.QualificationContextOutputRepresentation on page 637

status

Get or set the status of the product, such as Active or Inactive.

Signature

```
public String status {get; set;}
```

Property Value

Type: String

unitOfMeasure

Get details about the unit of measure for a specific set of records.

Signature

```
public ConnectApi.UnitOfMeasureOutputRepresentation unitOfMeasure {get; set;}
```

Property Value

Type: ConnectApi.UnitOfMeasureOutputRepresentation

ProductListRepresentation Class

Get the list of retrieved products.

Namespace

runtime_industries_cpq on page 512

ProductListRepresentation Properties
Learn more about the properties available with the ProductListRepresentation class.

ProductListRepresentation Properties

Learn more about the properties available with the ProductListRepresentation class.

The following are properties for `ProductListRepresentation`.

additionalFields
Get the key-value pair of additional standard or custom fields with their values.

attributeCategories
Get the list of categorized attributes related to the product.

availabilityDate
Get the date when the part is used in the product or is made available for sale.

catalogs
Get the list of associated catalogs. Returns the name and id values only.

categories
Get the list of associated categories. Returns the name and id values only.

childProducts
Get the hierarchy of the child products.

configureDuringSale
Determines whether to allow or prevent configuration when a bundle is sold.

description
Get the description of the product. If data translation is set up and specified in the org, the translated description is available.

discontinuedDate
Get the date from when the part can't be used in the product or sold.

displayUrl

Get the display image URL of the product.

endOfLifeDate

Get the date after which a product isn't supported, ordered, or maintained.

id

Get the ID of the product.

isActive

Indicates whether the product is active (true) or not (false).

isAssetizable

Indicates whether the product instance remains a customer asset after it's purchased (true) or not (false).

isComponentRequired

Indicates whether the product component is required (true) or not (false).

isDefaultComponent

Indicates whether the product component is the default component (true) or not (false).

isQuantityEditable

Indicates whether the product quantity is editable (true) or not (false).

isSoldOnlyWithOtherProds

Indicates whether the product can't be sold separately (true) or not (false).

name

Get the name of the product. If data translation is set up and specified in the org, the translated name is available.

nodeType

Get the type of the node, such as a product or bundled product.

prices

Get the price details associated with the products.

productClassification

Get the details of the product classification that the product is based on.

productCode

Get the universal product code that's used to track the part that's used in the product.

productComponentGroups

Get the logical grouping of the component products in a bundle and group cardinality for ordering the product components.

productInformation

Get the details of a product.

productPricingInformation

Get the pricing details of a product.

productQuantity

Get the quantity of a product.

productRelatedComponent

Get the details of the related components of a product.

productSellingModelOptions

Get the details of the product selling model options.

productSpecificationType
Get the details of the product specification type.

productType
Get the product type.

qualificationContext
Get the context details of a user, which are used for qualification rules.

status
Get or set the status of the product, such as Active or Inactive.

unitOfMeasure
Get details about the unit of measure for a specific set of records.

additionalFields

Get the key-value pair of additional standard or custom fields with their values.

Signature

```
public List<runtime_industries_cpq.AdditionalFieldsWrapper> additionalFields {get;
set;}
```

Property Value

Type: List<runtime_industries_cpq.AdditionalFieldsWrapper>

attributeCategories

Get the list of categorized attributes related to the product.

Signature

```
public List<runtime_industries_cpq.AttributeCategoryOutputRepresentation>
attributeCategories {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.AttributeCategoryOutputRepresentation on page 518 >

availabilityDate

Get the date when the part is used in the product or is made available for sale.

Signature

```
public Datetime availabilityDate {get; set;}
```

Property Value

Type: Datetime

`catalogs`

Get the list of associated catalogs. Returns the name and id values only.

Signature

```
public List<runtime_industries_cpq.CatalogOutputRepresentation> catalogs {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.CatalogOutputRepresentation on page 539>

`categories`

Get the list of associated categories. Returns the name and id values only.

Signature

```
public List<runtime_industries_cpq.CategoryOutputRepresentation> categories {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.CategoryOutputRepresentation on page 545>

`childProducts`

Get the hierarchy of the child products.

Signature

```
public List<runtime_industries_cpq.ProductListRepresentation> childProducts {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.ProductListRepresentation on page 598>

`configureDuringSale`

Determines whether to allow or prevent configuration when a bundle is sold.

Signature

```
public String configureDuringSale {get; set;}
```

Property Value

Type: String

`description`

Get the description of the product. If data translation is set up and specified in the org, the translated description is available.

Signature

```
public String description {get; set;}
```

Property Value

Type: String

discontinuedDate

Get the date from when the part can't be used in the product or sold.

Signature

```
public Datetime discontinuedDate {get; set;}
```

Property Value

Type: Datetime

displayUrl

Get the display image URL of the product.

Signature

```
public String displayUrl {get; set;}
```

Property Value

Type: String

endOfLifeDate

Get the date after which a product isn't supported, ordered, or maintained.

Signature

```
public Datetime endOfLifeDate {get; set;}
```

Property Value

Type: Datetime

id

Get the ID of the product.

Signature

```
public String id {get; set;}
```

Property Value

Type: String

isActive

Indicates whether the product is active (true) or not (false).

Signature

```
public Boolean isActive {get; set;}
```

Property Value

Type: Boolean

isAssetizable

Indicates whether the product instance remains a customer asset after it's purchased (true) or not (false).

Signature

```
public Boolean isAssetizable {get; set;}
```

Property Value

Type: Boolean

isComponentRequired

Indicates whether the product component is required (true) or not (false).

Signature

```
public Boolean isComponentRequired {get; set;}
```

Property Value

Type: Boolean

isDefaultComponent

Indicates whether the product component is the default component (true) or not (false).

Signature

```
public Boolean isDefaultComponent {get; set;}
```

Property Value

Type: Boolean

`isQuantityEditable`

Indicates whether the product quantity is editable (true) or not (false).

Signature

`public Boolean isQuantityEditable {get; set;}`

Property Value

Type: Boolean

`isSoldOnlyWithOtherProds`

Indicates whether the product can't be sold separately (true) or not (false).

Signature

`public Boolean isSoldOnlyWithOtherProds {get; set;}`

Property Value

Type: Boolean

`name`

Get the name of the product. If data translation is set up and specified in the org, the translated name is available.

Signature

`public String name {get; set;}`

Property Value

Type: String

`nodeType`

Get the type of the node, such as a product or bundled product.

Signature

`public String nodeType {get; set;}`

Property Value

Type: String

`prices`

Get the price details associated with the products.

Signature

```
public List<runtime_industries_cpq.ProductPricesOutputRepresentation> prices {get;
set;}
```

Property Value

Type: List<runtime_industries_cpq.ProductPricesOutputRepresentation on page 617>

productClassification

Get the details of the product classification that the product is based on.

Signature

```
public runtime_industries_cpq.ProductClassificationOutputRepresentation
productClassification {get; set;}
```

Property Value

Type: runtime_industries_cpq.ProductClassificationOutputRepresentation on page 579

productCode

Get the universal product code that's used to track the part that's used in the product.

Signature

```
public String productCode {get; set;}
```

Property Value

Type: String

productComponentGroups

Get the logical grouping of the component products in a bundle and group cardinality for ordering the product components.

Signature

```
public List<runtime_industries_cpq.ProductComponentGroupOutputRepresentation>
productComponentGroups {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.ProductComponentGroupOutputRepresentation on page 580>

productInformation

Get the details of a product.

Signature

```
public String productInformation {get; set;}
```

Property Value

Type: String

productPricingInformation

Get the pricing details of a product.

Signature

```
public String productPricingInformation {get; set;}
```

Property Value

Type: String

productQuantity

Get the quantity of a product.

Signature

```
public runtime_industries_cpq.ProductQuantityOutputRepresentation productQuantity {get;
set;}
```

Property Value

Type: runtime_industries_cpq.ProductQuantityOutputRepresentation on page 620

productRelatedComponent

Get the details of the related components of a product.

Signature

```
public ConnectApi.CPQProductRelatedComponentOutputRepresentation productRelatedComponent
{get; set;}
```

Property Value

Type: ConnectApi.CPQProductRelatedComponentOutputRepresentation

productSellingModelOptions

Get the details of the product selling model options.

Signature

```
public List<runtime_industries_cpq.ProductSellingModelOptionOutputRepresentation>
productSellingModelOptions {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.ProductSellingModelOptionOutputRepresentation on page 630>

`productSpecificationType`

Get the details of the product specification type.

Signature

```
public runtime_industries_cpq.ProductSpecificationTypeOutputRepresentation
productSpecificationType {get; set;}
```

Property Value

Type: runtime_industries_cpq.ProductSpecificationTypeOutputRepresentation on page 635

`productType`

Get the product type.

Signature

```
public String productType {get; set;}
```

Property Value

Type: String

`qualificationContext`

Get the context details of a user, which are used for qualification rules.

Signature

```
public runtime_industries_cpq.QualificationContextOutputRepresentation
qualificationContext {get; set;}
```

Property Value

Type: runtime_industries_cpq.QualificationContextOutputRepresentation on page 637

`status`

Get or set the status of the product, such as Active or Inactive.

Signature

```
public String status {get; set;}
```

Property Value

Type: String

unitOfMeasure

Get details about the unit of measure for a specific set of records.

Signature

```
public ConnectApi.UnitOfMeasureOutputRepresentation unitOfMeasure {get; set;}
```

Property Value

Type: ConnectApi.UnitOfMeasureOutputRepresentation

ProductOutputRepresentation Class

Get the details of a product definition.

Namespace

runtime_industries_cpq on page 512

ProductOutputRepresentation Properties
Learn more about the properties available with the ProductOutputRepresentation class.

ProductOutputRepresentation Properties

Learn more about the properties available with the ProductOutputRepresentation class.

The `ProductOutputRepresentation` class includes these properties.

additionalFields
Get the key-value pair of additional standard or custom fields with their values.

attributeCategories
Get the list of categorized attributes related to the product.

availabilityDate
Get the date when the part is used in the product or is made available for sale.

catalogs
Get the list of associated catalogs. Returns the name and id values only.

configureDuringSale
Determines whether to allow or prevent configuration when a bundle is sold.

description

Get the description of the product. If data translation is set up and specified in the org, the translated description is available.

discontinuedDate

Get the date from when the part can’t be used in the product or sold.

displayUrl

Get the display image URL of the product.

endOfLifeDate

Get the date after which a product isn’t supported, ordered, or maintained.

id

Get the ID of the product.

isActive

Indicates whether the product is active (true) or not (false).

isAssetizable

Indicates whether the product instance remains a customer asset after it's purchased (true) or not (false).

isComponentRequired

Indicates whether the product component is required (true) or not (false).

isDefaultComponent

Indicates whether the product component is the default component (true) or not (false).

isQuantityEditable

Indicates whether the product quantity is editable (true) or not (false).

isSoldOnlyWithOtherProds

Indicates whether the product can't be sold separately (true) or not (false).

name

Get the name of the product. If data translation is set up and specified in the org, the translated name is available.

nodeType

Get the type of the node, such as a product or bundled product.

productClassification

Get the details of the product classification that the product is based on.

productCode

Get the universal product code that's used to track the part that’s used in the product.

productComponentGroups

Get the logical grouping of the component products in a bundle and group cardinality for ordering the product components.

productInformation

Get the details of a product.

productPricingInformation

Get the pricing details of a product.

productQuantity

Get the quantity of a product.

productRelatedComponent

Get the details of the related components of a product.

productSellingModelOptions
Get the details of the product selling model options.

productSpecificationType
Get the details of the product specification type.

productType
Get the product type.

qualificationContext
Get the context details of a user, which are used for qualification rules.

status
Get or set the status of the product, such as Active or Inactive.

additionalFields

Get the key-value pair of additional standard or custom fields with their values.

Signature

```
public List<runtime_industries_cpq.AdditionalFieldsWrapper> additionalFields {get;
set;}
```

Property Value

Type: List<runtime_industries_cpq.AdditionalFieldsWrapper>

attributeCategories

Get the list of categorized attributes related to the product.

Signature

```
public List<runtime_industries_cpq.AttributeCategoryOutputRepresentation>
attributeCategories {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.AttributeCategoryOutputRepresentation on page 518 >

availabilityDate

Get the date when the part is used in the product or is made available for sale.

Signature

```
public Datetime availabilityDate {get; set;}
```

Property Value

Type: Datetime

catalogs

Get the list of associated catalogs. Returns the name and id values only.

Signature

```
public List<runtime_industries_cpq.CatalogOutputRepresentation> catalogs {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.CatalogOutputRepresentation on page 539>

configureDuringSale

Determines whether to allow or prevent configuration when a bundle is sold.

Signature

```
public String configureDuringSale {get; set;}
```

Property Value

Type: String

description

Get the description of the product. If data translation is set up and specified in the org, the translated description is available.

Signature

```
public String description {get; set;}
```

Property Value

Type: String

discontinuedDate

Get the date from when the part can't be used in the product or sold.

Signature

```
public Datetime discontinuedDate {get; set;}
```

Property Value

Type: Datetime

displayUrl

Get the display image URL of the product.

Signature

```
public String displayUrl {get; set;}
```

Property Value

Type: String

endOfLifeDate

Get the date after which a product isn’t supported, ordered, or maintained.

Signature

```
public Datetime endOfLifeDate {get; set;}
```

Property Value

Type: Datetime

id

Get the ID of the product.

Signature

```
public String id {get; set;}
```

Property Value

Type: String

isActive

Indicates whether the product is active (true) or not (false).

Signature

```
public Boolean isActive {get; set;}
```

Property Value

Type: Boolean

isAssetizable

Indicates whether the product instance remains a customer asset after it's purchased (true) or not (false).

Signature

```
public Boolean isAssetizable {get; set;}
```

Property Value

Type: Boolean

`isComponentRequired`

Indicates whether the product component is required (true) or not (false).

Signature

```
public Boolean isComponentRequired {get; set;}
```

Property Value

Type: Boolean

`isDefaultComponent`

Indicates whether the product component is the default component (true) or not (false).

Signature

```
public Boolean isDefaultComponent {get; set;}
```

Property Value

Type: Boolean

`isQuantityEditable`

Indicates whether the product quantity is editable (true) or not (false).

Signature

```
public Boolean isQuantityEditable {get; set;}
```

Property Value

Type: Boolean

`isSoldOnlyWithOtherProds`

Indicates whether the product can't be sold separately (true) or not (false).

Signature

```
public Boolean isSoldOnlyWithOtherProds {get; set;}
```

Property Value

Type: Boolean

name

Get the name of the product. If data translation is set up and specified in the org, the translated name is available.

Signature

```
public String name {get; set;}
```

Property Value

Type: String

nodeType

Get the type of the node, such as a product or bundled product.

Signature

```
public String nodeType {get; set;}
```

Property Value

Type: String

productClassification

Get the details of the product classification that the product is based on.

Signature

```
public runtime_industries_cpq.ProductClassificationOutputRepresentation
productClassification {get; set;}
```

Property Value

Type: runtime_industries_cpq.ProductClassificationOutputRepresentation on page 579

productCode

Get the universal product code that's used to track the part that's used in the product.

Signature

```
public String productCode {get; set;}
```

Property Value

Type: String

productComponentGroups

Get the logical grouping of the component products in a bundle and group cardinality for ordering the product components.

Signature

```
public List<runtime_industries_cpq.ProductComponentGroupOutputRepresentation>
productComponentGroups {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.ProductComponentGroupOutputRepresentation on page 580>

productInformation

Get the details of a product.

Signature

```
public String productInformation {get; set;}
```

Property Value

Type: String

productPricingInformation

Get the pricing details of a product.

Signature

```
public String productPricingInformation {get; set;}
```

Property Value

Type: String

productQuantity

Get the quantity of a product.

Signature

```
public runtime_industries_cpq.ProductQuantityOutputRepresentation productQuantity {get;
set;}
```

Property Value

Type: runtime_industries_cpq.ProductQuantityOutputRepresentation on page 620

productRelatedComponent

Get the details of the related components of a product.

Signature

```
public ConnectApi.CPQProductRelatedComponentOutputRepresentation productRelatedComponent
{get; set;}
```

Property Value

Type: ConnectApi.CPQProductRelatedComponentOutputRepresentation

`productSellingModelOptions`

Get the details of the product selling model options.

Signature

```
public List<runtime_industries_cpq.ProductSellingModelOptionOutputRepresentation>
productSellingModelOptions {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.ProductSellingModelOptionOutputRepresentation on page 630 >

`productSpecificationType`

Get the details of the product specification type.

Signature

```
public runtime_industries_cpq.ProductSpecificationTypeOutputRepresentation
productSpecificationType {get; set;}
```

Property Value

Type: runtime_industries_cpq.ProductSpecificationTypeOutputRepresentation on page 635

`productType`

Get the product type.

Signature

```
public String productType {get; set;}
```

Property Value

Type: String

`qualificationContext`

Get the context details of a user, which are used for qualification rules.

Signature

```
public runtime_industries_cpq.QualificationContextOutputRepresentation
qualificationContext {get; set;}
```

Property Value

Type: runtime_industries_cpq.QualificationContextOutputRepresentation on page 637

status

Get or set the status of the product, such as Active or Inactive.

Signature

```
public String status {get; set;}
```

Property Value

Type: String

ProductPricesOutputRepresentation Class

Get the price details of a product.

Namespace

runtime_industries_cpq on page 512

ProductPricesOutputRepresentation Properties
Learn more about the properties available with the ProductPricesOutputRepresentation class.

ProductPricesOutputRepresentation Properties

Learn more about the properties available with the ProductPricesOutputRepresentation class.

The following are properties for `ProductPricesOutputRepresentation`.

currencyIsoCode
Get or set the ISO currency code for the price.

effectiveFrom
Get or set the date and time when this price becomes effective.

effectiveTo
Get or set the date and time when this price expires.

isDefault
Indicates whether this is the default price for the product.

isDerived
Indicates whether this price is derived from another price.

- isSelected
  Indicates whether this price is currently selected.
- price
  Get or set the price value for the product.
- priceBookEntryId
  Get or set the ID of the price book entry.
- priceBookId
  Get or set the ID of the price book containing this price.
- pricingModel
  Get or set the pricing model associated with this price.

currencyIsoCode

Get or set the ISO currency code for the price.

Signature

```
public String currencyIsoCode {get; set;}
```

Property Value

Type: String

effectiveFrom

Get or set the date and time when this price becomes effective.

Signature

```
public String effectiveFrom {get; set;}
```

Property Value

Type: String

effectiveTo

Get or set the date and time when this price expires.

Signature

```
public String effectiveTo {get; set;}
```

Property Value

Type: String

`isDefault`

Indicates whether this is the default price for the product.

Signature

```
public Boolean isDefault {get; set;}
```

Property Value

Type: Boolean

`isDerived`

Indicates whether this price is derived from another price.

Signature

```
public Boolean isDerived {get; set;}
```

Property Value

Type: Boolean

`isSelected`

Indicates whether this price is currently selected.

Signature

```
public Boolean isSelected {get; set;}
```

Property Value

Type: Boolean

`price`

Get or set the price value for the product.

Signature

```
public Double price {get; set;}
```

Property Value

Type: Double

`priceBookEntryId`

Get or set the ID of the price book entry.

Signature

```
public String priceBookEntryId {get; set;}
```

Property Value

Type: String

priceBookId

Get or set the ID of the price book containing this price.

Signature

```
public String priceBookId {get; set;}
```

Property Value

Type: String

pricingModel

Get or set the pricing model associated with this price.

Signature

```
public runtime_industries_cpq.PricingModelOutputRepresentation pricingModel {get; set;}
```

Property Value

Type: runtime_industries_cpq.PricingModelOutputRepresentation on page 569

ProductQuantityOutputRepresentation Class

Represents the quantity constraints and current quantity for a product in the product discovery context.

Namespace

runtime_industries_cpq on page 512

ProductQuantityOutputRepresentation Properties

The following are properties for `ProductQuantityOutputRepresentation`.

maxQuantity
Get or set the maximum quantity allowed for the product.

minQuantity
Get or set the minimum quantity allowed for the product.

[quantity](#)
Get or set the current quantity of the product.

`maxQuantity`

Get or set the maximum quantity allowed for the product.

Signature

```
public Double maxQuantity {get; set;}
```

Property Value

Type: Double

`minQuantity`

Get or set the minimum quantity allowed for the product.

Signature

```
public Double minQuantity {get; set;}
```

Property Value

Type: Double

`quantity`

Get or set the current quantity of the product.

Signature

```
public Double quantity {get; set;}
```

Property Value

Type: Double

ProductRecommendationRule Class

Represents a product recommendation rule that is evaluated during product configuration. Product recommendation rules suggest additional products to users based on configuration conditions.

Namespace

runtime_industries_cpq on page 512

ProductRecommendationRule Constructor
Learn more about the constructor that's available with the ProductRecommendationRule class.

[ProductRecommendationRule Properties](#)
Contains properties to store product recommendation rule details from configuration rule evaluation.

ProductRecommendationRule Constructor

Learn more about the constructor that's available with the ProductRecommendationRule class.

The `ProductRecommendationRule` class includes this constructor.

[ProductRecommendationRule(referenceId, productIds, message, recordType, target, scope)](#)
Constructor to create a ProductRecommendationRule instance with the specified recommendation details.

`ProductRecommendationRule(referenceId, productIds, message, recordType, target, scope)`

Constructor to create a ProductRecommendationRule instance with the specified recommendation details.

Signature

```
public ProductRecommendationRule(String referenceId, List<String> productIds, String
message, String recordType, String target, String scope)
```

Parameters

referenceId
Type: String

The reference ID of the product recommendation rule.

productIds
Type: List<String>

List of product IDs that are recommended.

message
Type: String

The message to display with the product recommendation.

recordType
Type: String

The record type associated with the recommendation.

target
Type: String

The target of the recommendation rule.

scope
Type: String

The scope of the recommendation rule.

ProductRecommendationRule Properties

Contains properties to store product recommendation rule details from configuration rule evaluation.

The `ProductRecommendationRule` class includes these properties.

- message
  Get the message value.
- productIds
  Get the list of productid.
- recordType
  Get the recordtype value.
- referenceId
  Get the ID of the reference.
- scope
  Get the scope value.
- target
  Get the target value.

`message`

Get the message value.

Signature

```
public String message {get; set;}
```

Property Value

Type: String

`productIds`

Get the list of productid.

Signature

```
public List<String> productIds {get; set;}
```

Property Value

Type: List<String>

`recordType`

Get the recordtype value.

Signature

```
public String recordType {get; set;}
```

Property Value

Type: String

referenceId

Get the ID of the reference.

Signature

```
public String referenceId {get; set;}
```

Property Value

Type: String

scope

Get the scope value.

Signature

```
public String scope {get; set;}
```

Property Value

Type: String

target

Get the target value.

Signature

```
public String target {get; set;}
```

Property Value

Type: String

ProductRelatedComponentOutputRepresentation Class

Represents a related component product in a bundle or product relationship, including component configuration details such as quantity constraints, required status, and relationship metadata.

Namespace

runtime_industries_cpq on page 512

ProductRelatedComponentOutputRepresentation Properties

The following are properties for `ProductRelatedComponentOutputRepresentation`.

childProductId

Get or set the identifier of the child product in the relationship.

childSellingModelId

Get or set the identifier of the child product's selling model in the relationship.

doesBundlePriceIncludeChild

Get or set whether the bundle price includes the price of this child component.

id

Get or set the unique identifier of the product related component relationship.

isComponentRequired

Get or set whether this component is required in the bundle or product relationship.

isDefaultComponent

Get or set whether this component is selected by default in the bundle or product relationship.

isQuantityEditable

Get or set whether the quantity of this component can be edited.

maxQuantity

Get or set the maximum quantity allowed for this component.

minQuantity

Get or set the minimum quantity required for this component.

parentProductId

Get or set the identifier of the parent product in the relationship.

parentSellingModelId

Get or set the identifier of the parent product's selling model in the relationship.

productClassificationId

Get or set the identifier of the product classification for this component.

productComponentGroupId

Get or set the identifier of the product component group that this component belongs to.

productRelationshipTypeId

Get or set the identifier of the product relationship type that defines this relationship.

quantity

Get or set the current quantity of this component.

quantityScaleMethod

Get or set the method used to scale the quantity of this component, such as "PerUnit" or "PerBundle".

sequence

Get or set the display sequence order of this component within the bundle or product relationship.

unitOfMeasure

Get or set the unit of measure for the quantity of this component.

isExcluded
Indicates whether excluded is true or false.

quoteVisibility
Get the quotevisibility value.

childProductId

Get or set the identifier of the child product in the relationship.

Signature

```
public String childProductId {get; set;}
```

Property Value

Type: String

childSellingModelId

Get or set the identifier of the child product's selling model in the relationship.

Signature

```
public String childSellingModelId {get; set;}
```

Property Value

Type: String

doesBundlePriceIncludeChild

Get or set whether the bundle price includes the price of this child component.

Signature

```
public Boolean doesBundlePriceIncludeChild {get; set;}
```

Property Value

Type: Boolean

id

Get or set the unique identifier of the product related component relationship.

Signature

```
public String id {get; set;}
```

Property Value

Type: String

isComponentRequired

Get or set whether this component is required in the bundle or product relationship.

Signature

```
public Boolean isComponentRequired {get; set;}
```

Property Value

Type: Boolean

isDefaultComponent

Get or set whether this component is selected by default in the bundle or product relationship.

Signature

```
public Boolean isDefaultComponent {get; set;}
```

Property Value

Type: Boolean

isQuantityEditable

Get or set whether the quantity of this component can be edited.

Signature

```
public Boolean isQuantityEditable {get; set;}
```

Property Value

Type: Boolean

maxQuantity

Get or set the maximum quantity allowed for this component.

Signature

```
public Double maxQuantity {get; set;}
```

Property Value

Type: Double

minQuantity

Get or set the minimum quantity required for this component.

Signature

```
public Double minQuantity {get; set;}
```

Property Value

Type: Double

parentProductId

Get or set the identifier of the parent product in the relationship.

Signature

```
public String parentProductId {get; set;}
```

Property Value

Type: String

parentSellingModelId

Get or set the identifier of the parent product's selling model in the relationship.

Signature

```
public String parentSellingModelId {get; set;}
```

Property Value

Type: String

productClassificationId

Get or set the identifier of the product classification for this component.

Signature

```
public String productClassificationId {get; set;}
```

Property Value

Type: String

productComponentGroupId

Get or set the identifier of the product component group that this component belongs to.

Signature

```
public String productComponentGroupId {get; set;}
```

Property Value

Type: String

productRelationshipTypeId

Get or set the identifier of the product relationship type that defines this relationship.

Signature

```
public String productRelationshipTypeId {get; set;}
```

Property Value

Type: String

quantity

Get or set the current quantity of this component.

Signature

```
public Double quantity {get; set;}
```

Property Value

Type: Double

quantityScaleMethod

Get or set the method used to scale the quantity of this component, such as "PerUnit" or "PerBundle".

Signature

```
public String quantityScaleMethod {get; set;}
```

Property Value

Type: String

sequence

Get or set the display sequence order of this component within the bundle or product relationship.

Signature

```
public Integer sequence {get; set;}
```

Property Value

Type: Integer

`unitOfMeasure`

Get or set the unit of measure for the quantity of this component.

Signature

`public ConnectApi.UnitOfMeasureOutputRepresentation unitOfMeasure {get; set;}`

Property Value

Type: ConnectApi.UnitOfMeasureOutputRepresentation

`isExcluded`

Indicates whether excluded is true or false.

Signature

`public Boolean isExcluded {get; set;}`

Property Value

Type: Boolean

`quoteVisibility`

Get the quotevisibility value.

Signature

`public String quoteVisibility {get; set;}`

Property Value

Type: String

ProductSellingModelOptionOutputRepresentation Class

Represents a selling model option available for a product, which defines how the product can be sold (such as subscription, one-time, or usage-based).

Namespace

runtime_industries_cpq on page 512

ProductSellingModelOptionOutputRepresentation Properties
Learn more about the properties available with the ProductSellingModelOptionOutputRepresentation class.

ProductSellingModelOptionOutputRepresentation Properties

Learn more about the properties available with the ProductSellingModelOptionOutputRepresentation class.

The `ProductSellingModelOptionOutputRepresentation` class includes these properties.

- id
  Get or set the unique identifier of the product selling model option.
- productId
  Get or set the identifier of the product that this selling model option applies to.
- productSellingModel
  Get or set the product selling model details for this option.
- productSellingModelId
  Get or set the identifier of the product selling model for this option.

`id`

Get or set the unique identifier of the product selling model option.

Signature

```
public String id {get; set;}
```

Property Value

Type: String

`productId`

Get or set the identifier of the product that this selling model option applies to.

Signature

```
public String productId {get; set;}
```

Property Value

Type: String

`productSellingModel`

Get or set the product selling model details for this option.

Signature

```
public runtime_industries_cpq.ProductSellingModelOutputRepresentation productSellingModel
{get; set;}
```

Property Value

Type: runtime_industries_cpq.ProductSellingModelOutputRepresentation

`productSellingModelId`

Get or set the identifier of the product selling model for this option.

Signature

```
public String productSellingModelId {get; set;}
```

Property Value

Type: String

ProductSellingModelOutputRepresentation Class

Represents a product selling model that defines how a product is sold, including the selling model type, pricing term, and status.

Namespace

runtime_industries_cpq on page 512

ProductSellingModelOutputRepresentation Properties
Learn more about the properties available with the ProductSellingModelOutputRepresentation class.

ProductSellingModelOutputRepresentation Properties

Learn more about the properties available with the ProductSellingModelOutputRepresentation class.

The `ProductSellingModelOutputRepresentation` class includes these properties.

id
Get or set the unique identifier of the product selling model.

name
Get or set the name of the product selling model.

pricingTerm
Get or set the pricing term value, which represents the duration or quantity for the pricing term unit.

pricingTermUnit
Get or set the unit for the pricing term, such as "Month", "Year", or "Day".

sellingModelType
Get or set the type of selling model, such as "Subscription", "OneTime", or "Usage".

status
Get or set the status of the product selling model, such as "Active" or "Inactive".

id

Get or set the unique identifier of the product selling model.

Signature

```
public String id {get; set;}
```

Property Value

Type: String

name

Get or set the name of the product selling model.

Signature

```
public String name {get; set;}
```

Property Value

Type: String

pricingTerm

Get or set the pricing term value, which represents the duration or quantity for the pricing term unit.

Signature

```
public Integer pricingTerm {get; set;}
```

Property Value

Type: Integer

pricingTermUnit

Get or set the unit for the pricing term, such as "Month", "Year", or "Day".

Signature

```
public String pricingTermUnit {get; set;}
```

Property Value

Type: String

sellingModelType

Get or set the type of selling model, such as "Subscription", "OneTime", or "Usage".

Signature

```
public String sellingModelType {get; set;}
```

Property Value

Type: String

status

Get or set the status of the product selling model, such as "Active" or "Inactive".

Signature

```
public String status {get; set;}
```

Property Value

Type: String

ProductSpecificationRecordTypeOutputRepresentation Class

Represents the record type information for a product specification, which defines the type of record used to store product specification data.

Namespace

runtime_industries_cpq on page 512

ProductSpecificationRecordTypeOutputRepresentation Properties

Learn more about the properties available with the ProductSpecificationRecordTypeOutputRepresentation class.

The `ProductSpecificationRecordTypeOutputRepresentation` class includes these properties.

isCommercial

Get or set whether this product specification record type is commercial.

Signature

```
public Boolean isCommercial {get; set;}
```

Property Value

Type: Boolean

ProductSpecificationTypeOutputRepresentation Class

Represents a product specification type that defines the structure and attributes available for configuring a product.

Namespace

runtime_industries_cpq on page 512

ProductSpecificationTypeOutputRepresentation Properties
Learn more about the properties available with the ProductSpecificationTypeOutputRepresentation class.

ProductSpecificationTypeOutputRepresentation Properties

Learn more about the properties available with the ProductSpecificationTypeOutputRepresentation class.

The `ProductSpecificationTypeOutputRepresentation` class includes these properties.

name
Get or set the name of the product specification type.

productSpecificationRecordType
Get or set the product specification record type associated with this specification type.

`name`

Get or set the name of the product specification type.

Signature

```
public String name {get; set;}
```

Property Value

Type: String

`productSpecificationRecordType`

Get or set the product specification record type associated with this specification type.

Signature

```
public runtime_industries_cpq.ProductSpecificationRecordTypeOutputRepresentation
productSpecificationRecordType {get; set;}
```

Property Value

Type: runtime_industries_cpq.ProductSpecificationRecordTypeOutputRepresentation

QocQualificationOutputRepresentation Class

Represents a quote, order, or contract qualification that determines whether a product can be sold based on specific business rules and conditions.

Namespace

runtime_industries_cpq on page 512

QocQualificationOutputRepresentation Constructors

The following are constructors for `QocQualificationOutputRepresentation`.

`QocQualificationOutputRepresentation()`

Constructs an empty QocQualificationOutputRepresentation instance.

Signature

```
public QocQualificationOutputRepresentation()
```

QocQualificationOutputRepresentation Properties

Learn more about the properties available with the QocQualificationOutputRepresentation class.

The `QocQualificationOutputRepresentation` class includes these properties.

`productId`

Get or set the identifier of the product being qualified.

Signature

```
public String productId {get; set;}
```

Property Value

Type: String

`qualificationContext`

Get or set the qualification context that contains the qualification result and reason.

Signature

```
public runtime_industries_cpq.QualificationContextOutputRepresentation
qualificationContext {get; set;}
```

Property Value

Type: runtime_industries_cpq.QualificationContextOutputRepresentation

QualificationContextOutputRepresentation Class

Represents the context information used for product qualification, including account, opportunity, and other relevant context data for determining product eligibility.

Namespace

runtime_industries_cpq on page 512

QualificationContextOutputRepresentation Properties
Learn more about the properties available with the QualificationContextOutputRepresentation class.

QualificationContextOutputRepresentation Properties

Learn more about the properties available with the QualificationContextOutputRepresentation class.

The `QualificationContextOutputRepresentation` class includes these properties.

isQualified
Get or set whether the product is qualified based on the qualification rules.

reason
Get or set the reason for the qualification result, explaining why the product is qualified or not qualified.

`isQualified`

Get or set whether the product is qualified based on the qualification rules.

Signature

```
public Boolean isQualified {get; set;}
```

Property Value

Type: Boolean

`reason`

Get or set the reason for the qualification result, explaining why the product is qualified or not qualified.

Signature

```
public String reason {get; set;}
```

Property Value

Type: String

RelatedObjectFilter Class

Represents a filter for related objects used in product search and discovery, allowing you to filter products based on related object criteria.

Namespace

runtime_industries_cpq on page 512

RelatedObjectFilter Properties
Learn more about the properties available with the RelatedObjectFilter class.

RelatedObjectFilter Properties

Learn more about the properties available with the RelatedObjectFilter class.

The `RelatedObjectFilter` class includes these properties.

criteria
Get or set the list of filter criteria to apply to the related object.

objectName
Get or set the name of the related object to filter by, such as "Account" or "Opportunity".

`criteria`

Get or set the list of filter criteria to apply to the related object.

Signature

```
public List<runtime_industries_cpq.FilterCriteriaInputRepresentation> criteria {get;
set;}
```

Property Value

Type: List<runtime_industries_cpq.FilterCriteriaInputRepresentation>

`objectName`

Get or set the name of the related object to filter by, such as "Account" or "Opportunity".

Signature

```
public String objectName {get; set;}
```

Property Value

Type: String

RelatedObjectFilterInputRepresentation Class

Represents input criteria for filtering products based on related object information, such as account, opportunity, or contract data.

Namespace

runtime_industries_cpq on page 512

RelatedObjectFilterInputRepresentation Properties
Learn more about the properties available with the RelatedObjectFilterInputRepresentation

RelatedObjectFilterInputRepresentation Properties

Learn more about the properties available with the RelatedObjectFilterInputRepresentation

The `RelatedObjectFilterInputRepresentation` class includes these properties.

relatedObjectFilter
Get or set the list of related object filters to apply when searching for products.

`relatedObjectFilter`

Get or set the list of related object filters to apply when searching for products.

Signature

```
public List<runtime_industries_cpq.RelatedObjectFilter> relatedObjectFilter {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.RelatedObjectFilter>

SearchProductsFacetRepresentation Class

Represents a search facet that provides filtering and categorization options for product search results, such as categories, attributes, or other product characteristics.

Namespace

runtime_industries_cpq on page 512

SearchProductsFacetRepresentation Properties
Learn more about the properties available with the SearchProductsFacetRepresentation class.

SearchProductsFacetRepresentation Properties

Learn more about the properties available with the SearchProductsFacetRepresentation class.

The `SearchProductsFacetRepresentation` class includes these properties.

attributeType
Get or set the type of attribute this facet represents, such as "Picklist" or "Text".

displayName
Get or set the display name of the search facet.

displayRank
Get or set the display rank that determines the order in which this facet appears in search results.

nameOrId
Get or set the name or identifier of the attribute or category that this facet represents.

values
Get or set the list of facet values available for filtering, each representing a distinct option within this facet.

`attributeType`

Get or set the type of attribute this facet represents, such as "Picklist" or "Text".

Signature

```
public String attributeType {get; set;}
```

Property Value

Type: String

`displayName`

Get or set the display name of the search facet.

Signature

```
public String displayName {get; set;}
```

Property Value

Type: String

`displayRank`

Get or set the display rank that determines the order in which this facet appears in search results.

Signature

`public Integer displayRank {get; set;}`

Property Value

Type: Integer

`nameOrId`

Get or set the name or identifier of the attribute or category that this facet represents.

Signature

`public String nameOrId {get; set;}`

Property Value

Type: String

`values`

Get or set the list of facet values available for filtering, each representing a distinct option within this facet.

Signature

`public List<runtime_industries_cpq.FacetValueRepresentation> values {get; set;}`

Property Value

Type: List<runtime_industries_cpq.FacetValueRepresentation>

SearchProductsRepresentation Class

Represents the results of a product search operation, including the list of products, search facets, pagination information, and total count of matching products.

Namespace

runtime_industries_cpq on page 512

SearchProductsRepresentation Properties
Learn more about the properties available with the SearchProductsRepresentation class.

SearchProductsRepresentation Properties

Learn more about the properties available with the SearchProductsRepresentation class.

The `SearchProductsRepresentation` class includes these properties.

additionalFields
Get or set additional custom fields for the product in the search results.

attributeCategories
Get or set the list of attribute categories that contain product attributes for configuration.

availabilityDate
Get or set the date when the product becomes available for sale.

catalogs
Get or set the list of catalogs that this product belongs to.

configureDuringSale
Get or set whether the product can be configured during the sales process.

description
Get or set the description of the product in the search results. If data translation is set up and specified in the org, the translated description is available.

discontinuedDate
Get or set the date when the product was discontinued.

displayUrl
Get or set the URL to display additional information about the product.

endOfLifeDate
Get or set the end of life date for the product.

id
Get or set the ID of the product in the search results.

isActive
Get or set whether the product is active (true) or not (false) in the search results.

isAssetizable
Get or set whether the product can be converted to an asset after sale.

isComponentRequired
Get or set whether this component is required in a bundle or product relationship.

isDefaultComponent
Get or set whether this component is selected by default in a bundle or product relationship.

isQuantityEditable
Get or set whether the quantity of this product can be edited.

isSoldOnlyWithOtherProds
Get or set whether this product can only be sold with other products.

name
Get or set the name of the product in the search results. If data translation is set up and specified in the org, the translated name is available.

nodeType
Get or set the node type of the product, such as "Product" or "Bundle".

prices
Get or set the list of prices available for this product.

productClassification
Get or set the product classification information for this product.

productCode
Get or set the universal product code that's used to track the part that's used in the product.

productComponentGroups
Get or set the list of product component groups that organize related components.

productInformation
Get or set additional product information details.

productPricingInformation
Get or set the pricing information for this product, including price lists and pricing models.

productQuantity
Get or set the quantity constraints and current quantity for this product.

productRelatedComponent
Get or set the related component information for this product in bundle or product relationships.

productSellingModelOptions
Get or set the list of selling model options available for this product.

productSpecificationType
Get or set the product specification type that defines the structure and attributes for configuring this product.

productType
Get or set the type of product, such as "Product", "Service", or "Bundle".

qualificationContext
Get or set the qualification context that contains the qualification result and reason for this product.

status
Get or set the status of the product, such as "Active" or "Inactive".

unitOfMeasure
Get or set the unit of measure for the quantity of this product.

additionalFields

Get or set additional custom fields for the product in the search results.

Signature

```
public List<runtime_industries_cpq.AdditionalFieldsWrapper> additionalFields {get;
set;}
```

Property Value

Type: List<runtime_industries_cpq.AdditionalFieldsWrapper>

attributeCategories

Get or set the list of attribute categories that contain product attributes for configuration.

Signature

```
public List<runtime_industries_cpq.AttributeCategoryOutputRepresentation>
attributeCategories {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.AttributeCategoryOutputRepresentation>

availabilityDate

Get or set the date when the product becomes available for sale.

Signature

```
public Datetime availabilityDate {get; set;}
```

Property Value

Type: Datetime

catalogs

Get or set the list of catalogs that this product belongs to.

Signature

```
public List<runtime_industries_cpq.CatalogOutputRepresentation> catalogs {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.CatalogOutputRepresentation>

configureDuringSale

Get or set whether the product can be configured during the sales process.

Signature

```
public String configureDuringSale {get; set;}
```

Property Value

Type: String

`description`

Get or set the description of the product in the search results. If data translation is set up and specified in the org, the translated description is available.

Signature

```
public String description {get; set;}
```

Property Value

Type: String

`discontinuedDate`

Get or set the date when the product was discontinued.

Signature

```
public Datetime discontinuedDate {get; set;}
```

Property Value

Type: Datetime

`displayUrl`

Get or set the URL to display additional information about the product.

Signature

```
public String displayUrl {get; set;}
```

Property Value

Type: String

`endOfLifeDate`

Get or set the end of life date for the product.

Signature

```
public Datetime endOfLifeDate {get; set;}
```

Property Value

Type: Datetime

`id`

Get or set the ID of the product in the search results.

Signature

```
public String id {get; set;}
```

Property Value

Type: String

isActive

Get or set whether the product is active (true) or not (false) in the search results.

Signature

```
public Boolean isActive {get; set;}
```

Property Value

Type: Boolean

isAssetizable

Get or set whether the product can be converted to an asset after sale.

Signature

```
public Boolean isAssetizable {get; set;}
```

Property Value

Type: Boolean

isComponentRequired

Get or set whether this component is required in a bundle or product relationship.

Signature

```
public Boolean isComponentRequired {get; set;}
```

Property Value

Type: Boolean

isDefaultComponent

Get or set whether this component is selected by default in a bundle or product relationship.

Signature

```
public Boolean isDefaultComponent {get; set;}
```

Property Value

Type: Boolean

isQuantityEditable

Get or set whether the quantity of this product can be edited.

Signature

```
public Boolean isQuantityEditable {get; set;}
```

Property Value

Type: Boolean

isSoldOnlyWithOtherProds

Get or set whether this product can only be sold with other products.

Signature

```
public Boolean isSoldOnlyWithOtherProds {get; set;}
```

Property Value

Type: Boolean

name

Get or set the name of the product in the search results. If data translation is set up and specified in the org, the translated name is available.

Signature

```
public String name {get; set;}
```

Property Value

Type: String

nodeType

Get or set the node type of the product, such as "Product" or "Bundle".

Signature

```
public String nodeType {get; set;}
```

Property Value

Type: String

prices

Get or set the list of prices available for this product.

Signature

```
public List<runtime_industries_cpq.ProductPricesOutputRepresentation> prices {get;
set;}
```

Property Value

Type: List<runtime_industries_cpq.ProductPricesOutputRepresentation>

productClassification

Get or set the product classification information for this product.

Signature

```
public runtime_industries_cpq.ProductClassificationOutputRepresentation
productClassification {get; set;}
```

Property Value

Type: runtime_industries_cpq.ProductClassificationOutputRepresentation

productCode

Get or set the universal product code that's used to track the part that's used in the product.

Signature

```
public String productCode {get; set;}
```

Property Value

Type: String

productComponentGroups

Get or set the list of product component groups that organize related components.

Signature

```
public List<runtime_industries_cpq.ProductComponentGroupOutputRepresentation>
productComponentGroups {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.ProductComponentGroupOutputRepresentation>

productInformation

Get or set additional product information details.

Signature

```
public String productInformation {get; set;}
```

Property Value

Type: String

productPricingInformation

Get or set the pricing information for this product, including price lists and pricing models.

Signature

```
public String productPricingInformation {get; set;}
```

Property Value

Type: String

productQuantity

Get or set the quantity constraints and current quantity for this product.

Signature

```
public runtime_industries_cpq.ProductQuantityOutputRepresentation productQuantity {get;
set;}
```

Property Value

Type: runtime_industries_cpq.ProductQuantityOutputRepresentation

productRelatedComponent

Get or set the related component information for this product in bundle or product relationships.

Signature

```
public ConnectApi.CPQProductRelatedComponentOutputRepresentation productRelatedComponent
{get; set;}
```

Property Value

Type: ConnectApi.CPQProductRelatedComponentOutputRepresentation

`productSellingModelOptions`

Get or set the list of selling model options available for this product.

Signature

```
public List<runtime_industries_cpq.ProductSellingModelOptionOutputRepresentation>
productSellingModelOptions {get; set;}
```

Property Value

Type: List<runtime_industries_cpq.ProductSellingModelOptionOutputRepresentation>

`productSpecificationType`

Get or set the product specification type that defines the structure and attributes for configuring this product.

Signature

```
public runtime_industries_cpq.ProductSpecificationTypeOutputRepresentation
productSpecificationType {get; set;}
```

Property Value

Type: runtime_industries_cpq.ProductSpecificationTypeOutputRepresentation

`productType`

Get or set the type of product, such as "Product", "Service", or "Bundle".

Signature

```
public String productType {get; set;}
```

Property Value

Type: String

`qualificationContext`

Get or set the qualification context that contains the qualification result and reason for this product.

Signature

```
public runtime_industries_cpq.QualificationContextOutputRepresentation
qualificationContext {get; set;}
```

Property Value

Type: runtime_industries_cpq.QualificationContextOutputRepresentation

`status`

Get or set the status of the product, such as "Active" or "Inactive".

Signature

```
public String status {get; set;}
```

Property Value

Type: String

`unitOfMeasure`

Get or set the unit of measure for the quantity of this product.

Signature

```
public ConnectApi.UnitOfMeasureOutputRepresentation unitOfMeasure {get; set;}
```

Property Value

Type: ConnectApi.UnitOfMeasureOutputRepresentation

UnitOfMeasureOutputRepresentation Class

Represents the unit of measure for a product. This class contains information about how product quantities are measured, including the unit code, name, scale, and rounding method.

Namespace

runtime_industries_cpq on page 512

UnitOfMeasureOutputRepresentation Constructor
Learn more about the constructor that's available with the UnitOfMeasureOutputRepresentation class.

UnitOfMeasureOutputRepresentation Properties
Contains properties to include unit of measure details for products.

UnitOfMeasureOutputRepresentation Constructor

Learn more about the constructor that's available with the UnitOfMeasureOutputRepresentation class.

The `UnitOfMeasureOutputRepresentation` class includes this constructor.

UnitOfMeasureOutputRepresentation(apexObj)
Constructor to create a UnitOfMeasureOutputRepresentation instance from a ConnectApi UnitOfMeasureOutputRepresentation object.

`UnitOfMeasureOutputRepresentation(apexObj)`

Constructor to create a UnitOfMeasureOutputRepresentation instance from a ConnectApi UnitOfMeasureOutputRepresentation object.

Signature

```
public UnitOfMeasureOutputRepresentation(ConnectApi.UnitOfMeasureOutputRepresentation
apexObj)
```

Parameters

apexObj

Type: ConnectApi.UnitOfMeasureOutputRepresentation

The ConnectApi unit of measure representation object to convert to UnitOfMeasureOutputRepresentation.

UnitOfMeasureOutputRepresentation Properties

Contains properties to include unit of measure details for products.

The `UnitOfMeasureOutputRepresentation` class includes these properties.

id

Get the ID of the unitofmeasure.

name

Get the name of the unitofmeasure.

roundingMethod

Get the roundingmethod value.

scale

Get the scale value.

unitCode

Get the unitcode value.

`id`

Get the ID of the unitofmeasure.

Signature

```
public String id {get; set;}
```

Property Value

Type: String

`name`

Get the name of the unitofmeasure.

Signature

```
public String name {get; set;}
```

Property Value

Type: String

`roundingMethod`

Get the roundingmethod value.

Signature

```
public String roundingMethod {get; set;}
```

Property Value

Type: String

`scale`

Get the scale value.

Signature

```
public Integer scale {get; set;}
```

Property Value

Type: Integer

`unitCode`

Get the unitcode value.

Signature

```
public String unitCode {get; set;}
```

Property Value

Type: String

VisibilityRule Class

Represents a visibility rule that is evaluated during product configuration. Visibility rules control the visibility of products, attributes, or other UI elements based on configuration conditions.

Namespace

runtime_industries_cpq on page 512

[VisibilityRule Constructor](#)
Learn more about the constructor that's available with the VisibilityRule class.

[VisibilityRule Properties](#)
Contains properties to store visibility rule details from configuration rule evaluation.

VisibilityRule Constructor

Learn more about the constructor that's available with the VisibilityRule class.

The `VisibilityRule` class includes this constructor.

[VisibilityRule(stiId, prcId, attributeId, attributePicklistValueId, target, scope, type, productIds, message)](#)
Constructor to create a VisibilityRule instance with the specified visibility rule details.

VisibilityRule(stiId, prcId, attributeId, attributePicklistValueId, target, scope, type, productIds, message)

Constructor to create a VisibilityRule instance with the specified visibility rule details.

Signature

```
public VisibilityRule(String stiId, String prcId, String attributeId, String
attributePicklistValueId, String target, String scope, String type, List<String>
productIds, String message)
```

Parameters

stiId
Type: String

The ID of the Sales Transaction Item (STI) associated with this visibility rule.

prcId
Type: String

The ID of the Product Relationship Configuration (PRC) associated with this visibility rule.

attributeId
Type: String

The ID of the attribute associated with this visibility rule.

attributePicklistValueId
Type: String

The ID of the attribute picklist value associated with this visibility rule.

target
Type: String

The target of the visibility rule (for example, product, attribute, or component).

scope
Type: String

The scope of the visibility rule.

type
Type: String

The type of visibility rule (for example, Show or Hide).

productIds
Type: List<String>

List of product IDs affected by this visibility rule.

message
Type: String

The message to display when the visibility rule is applied.

VisibilityRule Properties

Contains properties to store visibility rule details from configuration rule evaluation.

The `VisibilityRule` class includes these properties.

attributeId
Get the ID of the attribute.

attributePicklistValueId
Get the ID of the attributepicklistvalue.

message
Get the message value.

prcId
Get the ID of the prc.

productId
Get the ID of the product.

scope
Get the scope value.

stiId
Get the ID of the sti.

target
Get the target value.

type
Get the type value.

attributeId

Get the ID of the attribute.

Signature

```
public String attributeId {get; set;}
```

Property Value

Type: String

attributePicklistValueId

Get the ID of the attributepicklistvalue.

Signature

```
public String attributePicklistValueId {get; set;}
```

Property Value

Type: String

message

Get the message value.

Signature

```
public String message {get; set;}
```

Property Value

Type: String

prcId

Get the ID of the prc.

Signature

```
public String prcId {get; set;}
```

Property Value

Type: String

productId

Get the ID of the product.

Signature

```
public List<String> productId {get; set;}
```

Property Value

Type: List<String>

`scope`

Get the scope value.

Signature

`public String scope {get; set;}`

Property Value

Type: String

`stiId`

Get the ID of the sti.

Signature

`public String stiId {get; set;}`

Property Value

Type: String

`target`

Get the target value.

Signature

`public String target {get; set;}`

Property Value

Type: String

`type`

Get the type value.

Signature

`public String type {get; set;}`

Property Value

Type: String

Product Discovery Metadata API Types

Metadata API enables you to access some types and feature settings that you can customize in the user interface.

Flow for Product Discovery

Represents the metadata associated with a flow. With Flow, you can create an application that takes users through a series of pages to query and update the records in the database. You can also run logic and provide branching capability based on user input to build dynamic applications.

ProductDiscoverySettings

Represents the settings for Product Discovery.

Flow for Product Discovery

Represents the metadata associated with a flow. With Flow, you can create an application that takes users through a series of pages to query and update the records in the database. You can also run logic and provide branching capability based on user input to build dynamic applications.

FlowActionCall

Product Discovery exposes additional actionType values for the FlowActionCall metadata type.

Field Name	Field Type	Description
`actionType`	InvocableActionType (enumeration of type string)	Required. The action type. Additional valid values for Product Discovery are: • `findProducts`—Search for the products from a catalog, category, or subcategory by using the specified search term. • `GetProducts`—Get products from the specified catalog, category, or subcategory, including product qualification and pricing details. • `GetProductDetails`—Get details such as attributes, hierarchy, and cardinality for the specified product. • `executeQualificationProcedure`—Execute a qualification procedure, which returns the qualification status for the specified products. • `getCatalogDetails`—Get details of a catalog record. • `getCatalogs`—Get a list of catalog records. • `getCategories`—Get the list of categories associated with a catalog record. • `getCategoryDetails`—Get details of a category record. • `getMultipleProductDetails`—Get product details for a list of products. • `searchPrdctWithGuidedSelection`—Use guided product selection to search for products. • `getRecommendedProducts`—Retrieve a list of recommended products for a quote or order by using the Constraint Rule Engine.

ProductDiscoverySettings

Represents the settings for Product Discovery.

Parent Type and Manifest Access

This type extends the Metadata metadata type and inherits its `fullName` field.

In the package manifest, all the settings metadata types for the org are accessed using the "Settings" name. See Settings for more details.

File Suffix and Directory Location

`ProductDiscoverySettings` values are stored in the `ProductDiscoverySettings.settings` file in the `settings` folder. The `.settings` files are different from other named components, because there is only one settings file for each settings component.

Version

ProductDiscoverySettings components are available in API version 62.0 and later.

Fields

Field Name	Description
`enableGuidedSelling`	**Field Type** boolean **Description** Indicates whether guided product selection is enabled (`true`) or not (`false`). Using guided product selection, you can manage dynamic forms that assess user requirements and show relevant products. The default value is `false`. Available in API version 63.0 and later.
`enableProdDiscConstraintRules`	**Field Type** boolean **Description** Indicates whether to enable the Constraint Rules Engine within Product Discovery (`true`) or not (`false`). The default value is `false`. If set to `true`, this setting activates constraint-based configuration rules so that the Constraint Rules Engine evaluates product selections during configuration. This step determines valid combinations based on defined constraints. Available in API version 67.0 and later.
`discoverProductsFlowNameOrgValue`	**Field Type** string **Description** Name of the custom flow for browsing and adding products. If this field isn't specified, the Discover Products flow is used for browsing and adding products. Available in API version 63.0 and later.

Field Name	Description
prodDiscBrowseContextDefOrgValue	**Field Type** string **Description** Context definition that gets updated based on the user-selected options and provides summary data.
prodDiscDefaultCatalogOrgValue	**Field Type** string **Description** Default catalog that determines the products to be displayed on the product list page. Available in API version 64.0 and later.
prodDiscPricingEnabledOrgValue	**Field Type** string **Description** Indicates whether pricing is enabled (`true`) or not (`false`). The default value is `false`.
prodDiscProcedureOrgValue	**Field Type** string **Description** Pricing procedure that calculates the list price.
prodDiscQualEnabledOrgValue	**Field Type** string **Description** Indicates whether product qualification is enabled (`true`) or not (`false`). The default value is `false`.
prodDiscQualificationOrgValue	**Field Type** string **Description** Qualification procedure that determines product eligibility.

Declarative Metadata Sample Definition

The following is an example of a ProductDiscoverySettings component.

```
<ProductDiscoverySettings xmlns="http://soap.sforce.com/2006/04/metadata">
    <enableGuidedSelling>true</enableGuidedSelling>

<discoverProductsFlowNameOrgValue>revenue_products__DiscoverProducts</discoverProductsFlowNameOrgValue>

    <prodDiscPricingEnabledOrgValue>true</prodDiscPricingEnabledOrgValue>
```

```
    <prodDiscQualEnabledOrgValue>true</prodDiscQualEnabledOrgValue>
    <prodDiscProcedureOrgValue>PricingProcedure</prodDiscProcedureOrgValue>
    <prodDiscQualificationOrgValue>QualificationProcedure</prodDiscQualificationOrgValue>


<prodDiscBrowseContextDefOrgValue>BrowseContextDefinition</prodDiscBrowseContextDefOrgValue>
</ProductDiscoverySettings>
```

The following is an example `package.xml` that references the previous definition.

```
<?xml version="1.0" encoding="UTF-8"?>
<Package xmlns="http://soap.sforce.com/2006/04/metadata">
    <types>
        <members>ProductDiscovery</members>
        <name>Settings</name>
    </types>
    <version>67.0</version>
</Package>
```

Wildcard Support in the Manifest File

The wildcard character `*` (asterisk) in the `package.xml` manifest file doesn't apply to metadata types for feature settings. The wildcard applies only when retrieving all settings, not for an individual setting. For details, see Settings. For information about using the manifest file, see Deploying and Retrieving Metadata with the Zip File.

CHAPTER 5 Salesforce Pricing

In this chapter ...

Create customized price adjustment methods and pricing procedures. Determine the discounts to apply to your products and services by using price adjustment methods. Get visibility into the pricing calculation process by using pricing procedures.

SEE ALSO:

Salesforce Help: Permissions for Salesforce Pricing

EDITIONS

Available in: Lightning Experience

Available in: **Enterprise**, **Unlimited**, and **Developer** Editions of Revenue Management where Salesforce Pricing is enabled

Salesforce Pricing Standard Objects

The Salesforce Pricing data model provides objects and fields to manage pricing processes, such as product management and calculation and application of discounts.

AttributeAdjustmentCondition

Represents the condition applied to an attribute that determines the price of a product or service being sold. This object is available in API version 60.0 and later.

AttributeBasedAdjRule

Represents the attribute conditions in a rule associated with the attribute based adjustment made for a product or service being sold. This object is available in API version 60.0 and later.

AttributeBasedAdjustment

Represents the association between the product selling model and the price adjustment for product or service being sold based on its attributes. This object stores information about the attributes that define the price of the product or service, the discounts applied, along with its value for a given date range. This object is available in API version 60.0 and later.

AttributeDefinition

Represents a product, asset, or object attribute, for example, a hardware specification or software detail. This object is available in API version 60.0 and later.

BundleBasedAdjustment

Represents the association between the product selling model and the price adjustment for a product or service being sold as a bundle. This object stores information of the product or service's price, the discounts applied, along with its value for a given date range. This object is available in API version 60.0 and later.

CostBook

Represents the cost book that contains multiple cost book entries. This object is available in API version 61.0 and later.

ContractItemPrice

Represents the price of a product on the contract. This object is available in API version 61.0 and later.

CostBookEntry

Represents the total cost of a product or service that’s determined based on various factors that affect a product's price. For example, when a product is manufactured, the weight of the raw material can be a cost factor based on the amount of material required and its shipping cost. This object is available in API version 61.0 and later.

PriceAdjustmentSchedule

Represents a series of tiered discounts based on the number of items purchased. This object is available in API version 60.0 and later.

PriceAdjustmentTier

Represents a discount tier in a price adjustment schedule. This object is available in API version 60.0 and later.

PriceBook2

Represents a price book that contains the list of products that your org sells. This object is available in API version 60.0 and later.

PriceBookEntry

Represents a product entry (an association between a Pricebook2 and Product2) in a price book. This object is available in API version 60.0 and later.

PriceBookEntryDerivedPrice

Represents the price of a product that’s derived from another source such as a product or an asset. This object is available in API version 61.0 and later.

PriceRevisionPolicy

Represents the guidelines and methods used to modify product or service prices, often incorporating formulas based on price revision entries and various adjustments. For example, a policy might dictate that prices are revised based on a formula that considers the regional Consumer Price Index (CPI) with a specific adjustment percentage, effective from a defined date, and categorized as either a flat adjustment or one directly based on the price revision entry data. This object is available in API version 65.0 and later.

PricingAdjBatchJob

Represents the collective update of multiple records on their prices and other adjustments. This object is available in API version 62.0 and later.

PricingAdjBatchJobLog

Represents the report that contains a list of failed adjustment requests along with an error message that describes the reason for failure. This object is available in API version 62.0 and later.

PricingAPIExecution

Represents the pricing resolution for an pricing element determined using strategy name and formula. This object is available in API version 63.0 and later.

PricingProcedureResolution

Represents a selection for a pricing procedure to execute a pricing process from a list of pricing procedures available. This object is available in API version 60.0 and later.

PricingProcessExecution

Represents a record generated during the execution of a discovery or pricing procedure. Multiple procedures may be performed within a single API call, with each recorded in a Pricing API Execution record. This object is available in API version 63.0 and later.

ProductPriceHistoryLog

Stores historical pricing data based on the product's price range. This object is available in API version 62.0 and later.

ProductPriceRange

Represents the price range of a product determined by using a product selling model that's stored in the relevant price book. This object is available in API version 62.0 and later.

ProductSellingModel

Defines one method by which a product can be sold; for example, as a one-time sale, an evergreen subscription, or a termed subscription. If the product is sold on subscription, this object defines the subscription's term. A product can have multiple product selling models. This object is available in API version 60.0 and later.

ProductSellingModelDataTranslation

Represents the translated values of the data stored within the ProductSellingModel record's fields. This object is available in API version 61.0 and later.

ProductSellingModelOption

A junction object between Product Selling Model and Product2. This object is available in API version 60.0 and later.

ProcedurePlanCriterion

The procedure plan option associated with the procedure plan criterion record. This object is available in API version 67.0 and later.

ProrationPolicy

Represents the proration policy associated with a Product Selling Model Option that determines how a product's price is calculated based on subscription duration or billing periods. This object is available in API version 67.0 and later.

SEE ALSO:

Object Reference for the Salesforce Platform: Overview of Salesforce Objects and Fields

SOAP API Developer Guide: Introduction to SOAP API

AttributeAdjustmentCondition

Represents the condition applied to an attribute that determines the price of a product or service being sold. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `update()`, `upsert()`

Fields

Field	Details
`AttributeBasedAdjRuleId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** Specifies the attribute adjustment rule record for which the condition is to be applied. This field is a relationship field. **Relationship Name** AttributeBasedAdjRule **Relationship Type** Lookup **Refers To** AttributeBasedAdjRule
`AttributeDefinitionId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** Specifies the attribute definition record for which the condition is to be applied. This field is a relationship field. **Relationship Name** AttributeDefinition **Relationship Type** Lookup **Refers To** AttributeDefinition
`BooleanValue`	**Type** picklist

Field	Details
	Properties Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies the boolean value of the operator. Possible values are: • `False` • `True`
`DateTimeValue`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date time value of the attribute.
`DateValue`	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The date value of the attribute.
`DoubleValue`	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The double value of the attribute.
`IntegerValue`	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The integer value of the attribute.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort

Field	Details
	Description The timestamp for when the current user last referred to this record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed a record related to this record.
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Name of the attribute adjustment condition.
`Operator`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** Operator used by the attribute. Possible values are: • `doesnotexistin`—Does Not Exist In • `equals`—Equals • `existsin`—Exists In • `greaterorequal`—Greater Or Equal • `greaterthan`—Greater Than • `lessorequal`—Less Or Equal • `lessthan`—Less Than • `matches`—Matches • `notequals`—Not Equals The default value is `equals`.
`ProductId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update

Field	Details
	Description The ID of the product associated with the attribute adjustment condition. This field is a relationship field. **Relationship Name** Product **Relationship Type** Lookup **Refers To** Product2
`StringValue`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The string value of the attribute.
`UsageType`	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** The type of record where the attribute adjustment condition is used. Possible values are: • `Pricing` • `Rating`

Associated Objects

This object has these associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

AttributeAdjustmentConditionFeed on page 2866
Feed tracking is available for the object.

AttributeAdjustmentConditionHistory on page 2873
History is available for tracked fields of the object.

AttributeBasedAdjRule

Represents the attribute conditions in a rule associated with the attribute based adjustment made for a product or service being sold. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`AttributeCount`	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The number of attributes.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the attribute based adjustment rule was last referenced.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed a record related to this record.
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** Name of the attribute based adjustment rule.
`OwnerId`	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** ID of the owner of the attribute based adjustment rule. This field is a polymorphic relationship field.

Field	Details
	Relationship Name Owner **Relationship Type** Lookup **Refers To** Group, User
`UsageType`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The type of record where the attribute-based adjustment rule is used. Possible values are: • `Pricing` • `Rating`

Associated Objects

This object has these associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

AttributeBasedAdjRuleFeed on page 2866
Feed tracking is available for the object.

AttributeBasedAdjRuleHistory on page 2873
History is available for tracked fields of the object.

AttributeBasedAdjRuleShare on page 2877
Sharing is available for the object.

AttributeBasedAdjustment

Represents the association between the product selling model and the price adjustment for product or service being sold based on its attributes. This object stores information about the attributes that define the price of the product or service, the discounts applied, along with its value for a given date range. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`AdjustmentType`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The type of pricing adjustment being made. Possible values are: • `Amount` • `Override` • `Percentage`
`AdjustmentValue`	**Type** double **Properties** Create, Filter, Sort, Update **Description** The value of the adjustment being made based on the adjustment type.
`AttributeBasedAdjRuleId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The attribute based adjustment rule associated with this attribute based adjustment record. This field is a relationship field. **Relationship Name** AttributeBasedAdjRule **Relationship Type** Lookup **Refers To** AttributeBasedAdjRule
`AttributeCount`	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The number of attributes.

Field	Details
`CurrencyIsoCode`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** ISO code of the currency. Must be one of the valid alphabetic, three-letter currency ISO codes defined by the ISO 4217 standard, such as USD, GBP, or JPY. Must be unique within your organization. Possible values are: • `BHD`—Bahraini Dinar • `JPY`—Japanese Yen • `USD`—U.S. Dollar The default value is `USD`.
`EffectiveFrom`	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** The date and time when the price list entry comes into effect.
`EffectiveTo`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date and time till when the price list entry remains effective.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the attribute based adjustment was last referenced.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed a record related to this record.

Field	Details
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Name of the attribute based adjustment.
PriceAdjustmentScheduleId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The price adjustment schedule associated with the attribute based adjustment record. This field is a relationship field. **Relationship Name** PriceAdjustmentSchedule **Relationship Type** Lookup **Refers To** PriceAdjustmentSchedule
PricingTerm	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The number of pricing term units in the pricing term. Used with PricingTermUnit to define the length of the pricing term. For example, if PricingTermUnit is Months and this field is 1, the subscription is priced monthly.If the selling model is one-time, this field must be null.
PricingTermUnit	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The unit of time used to define the pricing term. Used with PricingTerm to define the length of the pricing term. For example, if this field is Months and PricingTerm is 1, the subscription is priced monthly. If the selling model is one-time, this field must be null. Possible values are: • `Annual`—Years • `Months`

Field	Details
	• `Quarterly` • `Semi-Annual`
`ProductId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The ID of the product associated with the product attribute set. This field is a relationship field. **Relationship Name** Product **Relationship Type** Lookup **Refers To** Product2
`ProductSellingModelId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort **Description** The ID of the ProductSellingModel record associated with this attribute based adjustment record. This field is a relationship field. **Relationship Name** ProductSellingModel **Relationship Type** Lookup **Refers To** ProductSellingModel
`SellingModelType`	**Type** picklist **Properties** Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort **Description** Indicates whether the product is sold as a one-time sale, an evergreen subscription, or a subscription with a defined term. Possible values are: • `Evergreen`

Field	Details
	• `OneTime`—One Time • `TermDefined`—Term-Defined The default value is `OneTime`.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

AttributeBasedAdjustmentFeed on page 2866
Feed tracking is available for the object.

AttributeBasedAdjustmentHistory on page 2873
History is available for tracked fields of the object.

AttributeDefinition

Represents a product, asset, or object attribute, for example, a hardware specification or software detail. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`Code`	**Type** string **Properties** Create, Filter, Group, idLookup, Nillable, Sort, Update **Description** Code for the attribute definition. This field is unique within your organization.
`CurrencyIsoCode`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Available only if the multicurrency feature is enabled. Contains the ISO code for any currency allowed by the organization. Possible values are:

Field	Details
	• `BHD`—Bahraini Dinar • `JPY`—Japanese Yen • `USD`—U.S. Dollar The default value is `USD`.
`DataType`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort **Description** The data type of the attribute definition. Possible values are: • `Checkbox` • `Currency` • `Date` • `Datetime` • `Multipicklist` • `Number` • `Percent` • `Picklist` • `Text`
`DefaultHelpText`	**Type** textarea **Properties** Create, Nillable, Update **Description** The default help text for this attribute.
`DefaultValue`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The default value for this attribute.
`Description`	**Type** textarea **Properties** Create, Nillable, Update

Field	Details
	Description Description of this attribute.
`DeveloperName`	**Type** string **Properties** Create, Filter, Group, Sort **Description** The unique name of the attribute definition record. This name must begin with a letter and use only alphanumeric characters and underscores. It can't include spaces, end with an underscore, or have two consecutive underscores.
`IsActive`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates that the attribute definition is active. Active attributes definitions can be selected for products. The default value is `false`.
`IsRequired`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the attribute definition is required for a product. The default value is `false`.
`Label`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** The label for the attribute.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort

Field	Details
	Description The date the attribute definition was last referenced.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the attribute definition was last viewed.
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The name of the attribute.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The owner of the attribute definition. This field is a polymorphic relationship field. **Relationship Name** Owner **Relationship Type** Lookup **Refers To** Group, User
PicklistId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort **Description** The ID of the attribute picklist with the valid values for this attribute. This field is a relationship field. **Relationship Name** Picklist

Field	Details
	Relationship Type Lookup **Refers To** AttributePicklist
`SourceSystemIdentifier`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The identifier of the attribute definition in an external system.
`ValueDescription`	**Type** textarea **Properties** Create, Nillable, Update **Description** The default value description for this attribute.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

AttributeDefinitionFeed on page 2866
Feed tracking is available for the object.

AttributeDefinitionHistory on page 2873
History is available for tracked fields of the object.

AttributeDefinitionShare on page 2877
Sharing is available for the object.

BundleBasedAdjustment

Represents the association between the product selling model and the price adjustment for a product or service being sold as a bundle. This object stores information of the product or service's price, the discounts applied, along with its value for a given date range. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
AdjustmentType	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The type of pricing adjustment being made. Possible values are: • Amount • Override • Percentage
AdjustmentValue	**Type** double **Properties** Create, Filter, Sort, Update **Description** The type of pricing adjustment being made.
CurrencyIsoCode	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** ISO code of the currency. Must be one of the valid alphabetic, three-letter currency ISO codes defined by the ISO 4217 standard, such as USD, GBP, or JPY. Must be unique within your organization. Possible values are: • BHD—Bahraini Dinar • JPY—Japanese Yen • USD—U.S. Dollar The default value is USD.
EffectiveFrom	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** The date and time when the bundle based adjustment comes into effect.

Field	Details
EffectiveTo	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date and time till when the bundle based adjustment remains effective.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the bundle based adjustment was last referenced.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the bundle based adjustment was last viewed.
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The name of the bundle based adjustment record.
ParentProductId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** Represents the immediate parent in the product bundle that the price is configured for. This field is a relationship field. **Relationship Name** ParentProduct **Relationship Type** Lookup **Refers To** Product2

Field	Details
ParentProductSellingModelId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort **Description** The immediate parent node of a product selling model when nodes are represented in a hierarchial structure. This field is a relationship field. **Relationship Name** ParentProductSellingModel **Relationship Type** Lookup **Refers To** ProductSellingModel
PriceAdjustmentScheduleId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The price adjustment schedule associated with the bundle based adjustment record. This field is a relationship field. **Relationship Name** PriceAdjustmentSchedule **Relationship Type** Lookup **Refers To** PriceAdjustmentSchedule
PricingTerm	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The number of pricing term units in the pricing term. Used with PricingTermUnit to define the length of the pricing term. For example, if PricingTermUnit is Months and this field is 1, the subscription is priced monthly.If the selling model is one-time, this field must be null.
PricingTermUnit	**Type** picklist

Field	Details
	Properties Filter, Group, Nillable, Restricted picklist, Sort **Description** The unit of time used to define the pricing term. Used with PricingTerm to define the length of the pricing term. For example, if this field is Months and PricingTerm is 1, the subscription is priced monthly. If the selling model is one-time, this field must be null. Possible values are: • `Annual`—Years • `Months` • `Quarterly` • `Semi-Annual`
`ProductId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The ID of the product associated with the product bundle set. This field is a relationship field. **Relationship Name** Product **Relationship Type** Lookup **Refers To** Product2
`ProductSellingModelId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort **Description** The ID of the ProductSellingModel record associated with this bundle based adjustment record. This field is a relationship field. **Relationship Name** ProductSellingModel **Relationship Type** Lookup **Refers To** ProductSellingModel

Field	Details
`RootBundleId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** Represents the structural root product in a product's bundle that the price is configured for. This field is a relationship field. **Relationship Name** RootBundle **Relationship Type** Lookup **Refers To** Product2
`RootProductSellingModelId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort **Description** The root product selling model. The primary or base product selling method. This field is a relationship field. **Relationship Name** RootProductSellingModel **Relationship Type** Lookup **Refers To** ProductSellingModel
`SellingModelType`	**Type** picklist **Properties** Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort **Description** Indicates whether the product is sold as a one-time sale, an evergreen subscription, or a subscription with a defined term. Possible values are: • `Evergreen` • `OneTime`—One Time • `TermDefined`—Term-Defined The default value is `OneTime`.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

BundleBasedAdjustmentFeed on page 2866
: Feed tracking is available for the object.

BundleBasedAdjustmentHistory on page 2873
: History is available for tracked fields of the object.

CostBook

Represents the cost book that contains multiple cost book entries. This object is available in API version 61.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`EffectiveFrom`	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** Date and time when the cost book comes into effect.
`EffectiveTo`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** Date and time till when the cost book is no longer in effect.
`IsDefault`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the cost book is default (`true`) or not (`false`). The default value is `true`.

Field	Details
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Timestamp for when the current user last referred to this record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Timestamp for when the current user last viewed a record related to this record.
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** Name of the cost book.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** ID of the user who created the record. This field is a polymorphic relationship field. **Relationship Name** Owner **Relationship Type** Lookup **Refers To** Group, User

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

CostBookFeed on page 2866
Feed tracking is available for the object.

CostBookHistory on page 2873
History is available for tracked fields of the object.

CostBookShare on page 2877
Sharing is available for the object.

ContractItemPrice

Represents the price of a product on the contract. This object is available in API version 61.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`ContractId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The contract record associated with this contract item price record. This field is a relationship field. **Relationship Name** Contract **Relationship Type** Master-detail **Refers To** Contract (the master object)
`CurrencyIsoCode`	**Type** picklist **Properties** Defaulted on create, Filter, Group, Restricted picklist, Sort **Description** Contains the ISO code for any currency allowed by the organization. Available only if the multicurrency feature is enabled. Valid value is: • `USD`—U.S. Dollar

Field	Details
	The default value is `USD`.
`DiscountType`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The unit of the discount. Valid values are: • `AdjustmentAmount` • `AdjustmentPercentage`
`DiscountValue`	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The value of the discount.
`EndDate`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date and time till when the contract item price is no longer in effect.
`ItemId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The ID of the line item record associated with this contract item price record. This field is a polymorphic relationship field. **Relationship Name** Item **Relationship Type** Lookup **Refers To** Product2, ProductCategory

Field	Details
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last referred to this record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed a record related to this record.
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The name of the contract item price record.
Price	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The unit price of the product that's being sold as part of the contract.
ProductSellingModelId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the product selling model record associated with this contract item price record. This field is a relationship field. **Relationship Name** ProductSellingModel **Relationship Type** Lookup **Refers To** ProductSellingModel

Field	Details
SellingModelType	**Type** picklist **Properties** Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort **Description** Specifies whether the line item is sold as a one-time sale, an evergreen subscription, or a subscription with a defined term. Valid values are: • `Evergreen` • `OneTime` • `TermDefined` The default value is `OneTime`.
StartDate	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** The date and time when the contract item price comes into effect.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

ContractItemPriceFeed on page 2866
Feed tracking is available for the object.

ContractItemPriceHistory on page 2873
History is available for tracked fields of the object.

CostBookEntry

Represents the total cost of a product or service that's determined based on various factors that affect a product's price. For example, when a product is manufactured, the weight of the raw material can be a cost factor based on the amount of material required and its shipping cost. This object is available in API version 61.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`Cost`	**Type** currency **Properties** Create, Filter, Sort, Update **Description** Total cost of the product.
`CostBookId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** ID of the Cost Book record with which this record is associated. This field is a relationship field. **Relationship Name** CostBook **Relationship Type** Master-detail **Refers To** CostBook (the master object)
`CurrencyIsoCode`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** ISO code for any currency allowed by the organization. Available only for organizations with the multicurrency feature enabled. Valid values are: • `BHD`—Bahraini Dinar • `JPY`—Japanese Yen • `USD`—U.S. Dollar The default value is `USD`.
`Description`	**Type** textarea **Properties** Create, Filter, Group, Nillable, Sort, Update

Field	Details
	Description Description of this cost book entry record.
EffectiveFrom	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** Date and time when the cost book entry comes into effect.
EffectiveTo	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** Date and time till when the cost book entry is no longer in effect.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Timestamp for when the current user last referred to this record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Timestamp for when the current user last viewed a record related to this record.
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Name of the cost book entry.
ProductId	**Type** reference **Properties** Create, Filter, Group, Sort

Field	Details
	Description Required. ID of the Product2 record with which this record is associated. This field must be specified when creating Product2 records. This field can't be changed in an update. This field is a relationship field. **Relationship Name** Product **Relationship Type** Lookup **Refers To** Product2

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

CostBookEntryFeed on page 2866
Feed tracking is available for the object.

CostBookEntryHistory on page 2873
History is available for tracked fields of the object.

PriceAdjustmentSchedule

Represents a series of tiered discounts based on the number of items purchased. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`AdjustmentMethod`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** The method for applying tiered pricing. Possible values are: • Range—All items receive the discount of the highest tier the quantity falls in. • Slab—Items receive the discount defined for the tier they fall in.

Field	Details
	Possible values are: • `Range` • `Slab` The default value is `Range`.
`CurrencyIsoCode`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Available only if the multicurrency feature is enabled. Contains the ISO code for any currency allowed by the organization. Possible values are: • `BHD`—Bahraini Dinar • `JPY`—Japanese Yen • `USD`—U.S. Dollar The default value is `USD`.
`Description`	**Type** textarea **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Text description of the price adjustment schedule.
`EffectiveFrom`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date and time when the price adjustment schedule comes into effect.
`EffectiveTo`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date and time when the price adjustment schedule remains effective.
`IsActive`	**Type** boolean

Field	Details
	Properties Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the price adjustment schedule is active (true) or not (false). You can change this field's value as often as necessary. Label is **Active**. Default value is `False`.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Indicates whether the price adjustment schedule has been archived (true) or not (false). This field is read-only.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed this record. If this value is null, it's possible that this record was referenced (LastReferencedDate) and not viewed.
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** Required. The name of the price adjustment schedule. This field is read-only. Label is Price Adjustment Schedule Name.
`OwnerId`	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The Salesforce ID of the sales representative who owns the price adjustment schedule. This field is a polymorphic relationship field. **Relationship Name** Owner **Relationship Type** Lookup

Field	Details
	Refers To Group, User
`Pricebook2Id`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The price book associated with this price adjustment schedule record. This field is a relationship field. **Relationship Name** Pricebook2 **Relationship Type** Lookup **Refers To** Pricebook2
`ScheduleType`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** The type of price adjustment schedule. Possible values are: • `Attribute` • `Bundle` • `Custom` • `Term` • `Volume` The default value is `Volume`.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

PriceAdjustmentScheduleShare on page 2877
Sharing is available for the object.

PriceAdjustmentTier

Represents a discount tier in a price adjustment schedule. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`AdjustmentType`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The type of price adjustment.
`CurrencyIsoCode`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Available only if the multicurrency feature is enabled. Contains the ISO code for any currency allowed by the organization. Possible values are: • `BHD`—Bahraini Dinar • `JPY`—Japanese Yen • `USD`—U.S. Dollar The default value is `USD`.
`EffectiveFrom`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date and time when the price adjustment tier comes into effect.
`EffectiveTo`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update

Field	Details
	Description The date and time when the price adjustment tier remains effective.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Indicates when the user last referred to this record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed this record. If this value is null, it’s possible that this record was referenced (LastReferencedDate) and not viewed.
LowerBound	**Type** double **Properties** Create, Filter, Sort, Update **Description** The minimum quantity the discount can be applied to. It must be a positive integer and less than or equal to the upper bound of the tier.
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The name of the price adjustment tier.
PriceAdjustmentScheduleId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The ID of the price adjustment schedule that the discount is applied to. This field is a relationship field.

Field	Details
	Relationship Name PriceAdjustmentSchedule **Relationship Type** Lookup **Refers To** PriceAdjustmentSchedule
PricingTerm	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The number of pricing term units in the pricing term. Used with PricingTermUnit to define the length of the pricing term. For example, if PricingTermUnit is Months and this field is 1, the subscription is priced monthly.If the selling model is one-time, this field must be null.
PricingTermUnit	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The unit of time used to define the pricing term. Used with PricingTerm to define the length of the pricing term. For example, if this field is Months and PricingTerm is 1, the subscription is priced monthly. If the selling model is one-time, this field must be null. Possible values are: • `Annual`—Years • `Months`
Product2Id	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the product associated with the price adjustment tier. This field is a relationship field. **Relationship Name** Product2 **Relationship Type** Lookup **Refers To** Product2

Field	Details
`ProductSellingModelId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the ProductSellingModel record associated with this price adjustment tier record. This field is a relationship field. **Relationship Name** ProductSellingModel **Relationship Type** Lookup **Refers To** ProductSellingModel
`SellingModelType`	**Type** picklist **Properties** Defaulted on create, Filter, Group, Restricted picklist, Sort **Description** Indicates whether the product is sold as a one-time sale, an evergreen subscription, or a subscription with a defined term. Possible values are: • `Evergreen` • `OneTime`—One Time • `TermDefined`—Term-Defined The default value is `OneTime`.
`TierType`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The unit of the discount. Possible values are: • AdjustmentAmount—An amount discounted from an item's list price • AdjustmentPercentage—A percentage discounted from an item's list price Possible values are: • `AdjustmentAmount`—Amount • `AdjustmentPercentage`—Percentage • `OverrideAmount`—Override

Field	Details
TierValue	**Type** double **Properties** Create, Filter, Sort, Update **Description** The value of the discount.
UpperBound	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The maximum quantity the discount can be applied to. Must be a positive integer. Not inclusive. Set this value one digit higher than the quantity you want the tier to include. For example, if a tier's upper bound is 99, set the value of UpperBound to 100. For the last tier, the value is optional.

Usage

To use PriceAdjustmentTiers, associate them with a PriceAdjustmentSchedule.

Tiers can't overlap, and no gaps are allowed between tiers.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

PriceAdjustmentTierFeed on page 2866
Feed tracking is available for the object.

PriceAdjustmentTierHistory on page 2873
History is available for tracked fields of the object.

PriceBook2

Represents a price book that contains the list of products that your org sells. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`CurrencyIsoCode`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Available only if the multicurrency feature is enabled. Contains the ISO code for any currency allowed by the organization. Possible values are: • `BHD`—Bahraini Dinar • `JPY`—Japanese Yen • `USD`—U.S. Dollar The default value is `USD`.
`Description`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Text description of the price book.
`IsActive`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the price book is active (true) or not (false). Inactive price books are hidden in many areas in the user interface. You can change this field's value as often as necessary. Label is **Active**. The default value is `false`.
`IsArchived`	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Indicates whether the price book has been archived (true) or not (false). This field is read only. The default value is `false`.

Field	Details
IsStandard	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Indicates whether the price book is the standard price book for the org (true) or not (false). Every org has one standard price book—all other price books are custom price books. The default value is `false`.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed a record related to this record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed this record. If this value is null, it's possible that this record was referenced (LastReferencedDate) and not viewed.
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** Required. Name of this object. This field is read-only for the standard price book.
ValidFrom	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date and time when a price book is initially valid. If this field is null, the price book is valid immediately when active.
ValidTo	**Type** dateTime

Field	Details
	Properties Create, Filter, Nillable, Sort, Update **Description** The date and time when a price book is valid to. If this field is null, the price book is valid until it's deactivated.

Usage

A price book is a list of products that your org sells.

- Each org has one standard price book that defines the standard or generic list price for each product or service that it sells.
- An org can have multiple custom price books to use for specialized purposes, such as for discounts, different channels or markets, or select accounts or opportunities. While your client application can create, delete, and update custom price books, your client application can only update the standard price book.
- For some orgs, the standard price book is the only price needed. If you set up other price books, you can reference the standard price book when setting up list prices in custom price books.

Use this object to query standard and custom price books that have been configured for your org. A common use of this object is to allow your client application to obtain valid Pricebook2 object IDs for use when configuring PricebookEntry records via the API.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

PriceBook2ChangeEvent **on page 2864**
Change events are available for the object.

PriceBook2History **on page 2873**
History is available for tracked fields of the object.

PriceBookEntry

Represents a product entry (an association between a Pricebook2 and Product2) in a price book. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `update()`, `upsert()`

Fields

Note: Salesforce Object Search Language (SOSL) allows you to search records across standard and custom objects. When filtering records in the PriceBookEntry object using SOSL, you can only sort by fields related to Product2.

Field	Details
CurrencyIsoCode	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort **Description** Available only for organizations with the multicurrency feature enabled. Contains the ISO code for any currency allowed by the organization. Possible values are: • BHD—Bahraini Dinar • JPY—Japanese Yen • USD—U.S. Dollar The default value is USD.
IsActive	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether this price book entry is active (true) or not (false). Although you can never delete PricebookEntry records, your client application can set this flag to false. Inactive PricebookEntry records are hidden in many areas in the user interface. You can change this flag on a PricebookEntry record as often as necessary. The default value is false.
IsArchived	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Indicates whether the PricebookEntry has been archived (true) or not (false). This field is set to true when the Product2 record it's associated with is archived, or when the Pricebook2 record is archived. This field is read only. The default value is false.
Name	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** Name of this price book entry record. This read-only field references the value in the Name field of the Product2 record.

Field	Details
Pricebook2Id	**Type** reference **Properties** Create, Filter, Group, Sort **Description** Required. ID of the Pricebook2 record with which this record is associated. This field must be specified when creating Pricebook2 records. It can’t be changed in an update. This field is a relationship field. **Relationship Name** Pricebook2 **Relationship Type** Lookup **Refers To** Pricebook2
Product2Id	**Type** reference **Properties** Create, Filter, Group, Sort **Description** Required. ID of the Product2 record with which this record is associated. This field must be specified when creating Product2 records. It can’t be changed in an update. This field is a relationship field. **Relationship Name** Product2 **Relationship Type** Lookup **Refers To** Product2
ProductCode	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** Product code for this record. This read-only field references the value in the ProductCode field of the associated Product2 record.
ProductSellingModelId	**Type** reference

Field	Details
	Properties Create, Filter, Group, Nillable, Sort **Description** The ID of the related product selling model. This field is a relationship field. **Relationship Name** ProductSellingModel **Relationship Type** Lookup **Refers To** ProductSellingModel
`UnitPrice`	**Type** currency **Properties** Create, Filter, Sort, Update **Description** Required. Unit price for this price book entry. You can specify a value only if UseStandardPrice is set to false.
`UseStandardPrice`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether this price book entry uses the standard price defined in the standard Pricebook2 record (true) or not (false). If set to true, then the UnitPrice field is read-only, and the value is the same as the UnitPrice value in the corresponding PricebookEntry in the standard price book (that is, the PricebookEntry record whose Pricebook2Id refers to the standard price book and whose Product2Id and CurrencyIsoCode are the same as this record). For PricebookEntry records associated with the standard Pricebook2 record, this field must be set to true. The default value is `false`.

Usage

Use this object to define the association between your organization’s products (Product2) and your organization’s standard price book or to custom price books (Pricebook2). Create one PricebookEntry record for each standard or custom price and currency combination for a product in a Pricebook2.

When creating these records, you must specify the IDs of the associated Pricebook2 record and Product2 record. Once these records are created, your client application can’t update these IDs.

This object is defined only for those organizations that have products enabled as a feature. If the organization doesn't have the products feature enabled, then the PricebookEntry object doesn't appear in the describeGlobal call, and you can't access it.

If you delete a PriceBookEntry that is referenced by a line item, the line item is unaffected, but the PriceBookEntry is archived and unavailable from the API. Deleted PriceBookEntry records can't be recovered.

You must load the standard price for a product before you're permitted to load its custom prices.

Create PriceBookEntry records by using this sObject API.

```
https://yourInstance.salesforce.com/services/data/v67.0/sobjects/PricebookEntry
```

This example shows a sample request that specifies the details of a price book entry.

```
{
  "ProductSellingModelId": "0jPxx0000000005EAA",
  "Product2Id": "01tLT00000A0YTlYAN",
  "IsActive": true,
  "Pricebook2Id": "01s1W000000SYXNQA4",
  "UnitPrice": "100.00"
}
```

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

PriceBookEntryChangeEvent on page 2864
Change events are available for the object.

PriceBookEntryHistory on page 2873
History is available for tracked fields of the object.

PriceBookEntryDerivedPrice

Represents the price of a product that's derived from another source such as a product or an asset. This object is available in API version 61.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
ContributingProductId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort

Field	Details
	Description Source product from which the derived price is calculated. The source product is associated with the derived price product. This field is a relationship field. **Relationship Name** ContributingProduct **Relationship Type** Lookup **Refers To** Product2
`CurrencyIsoCode`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** ISO code for any currency allowed by the organization. Available only if the multicurrency feature is enabled. Valid values are: • `BHD`—Bahraini Dinar • `JPY`—Japanese Yen • `USD`—U.S. Dollar The default value is `USD`.
`DerivedPricingScope`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Scope of the product based on which the derived price is calculated. Valid values are: • `Both` • `NonTransactional` • `Transactional`
`EffectiveFrom`	**Type** dateTime **Properties** Create, Filter, Sort, Update

Field	Details
	Description Date and time when the derived pricing comes into effect.
`EffectiveTo`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** Date and time when the derived pricing is no longer in effect.
`Formula`	**Type** string **Properties** Create, Filter, Sort, Update **Description** Coded format of the formula used to calculate the derived price of a product from another product or asset.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Timestamp for when the current user last referred to this record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Timestamp for when the current user last viewed this record. If this value is null, it’s possible that this record was referenced (`LastReferencedDate`) and not viewed.
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Name of the derived price record.
`OwnerId`	**Type** reference

Field	Details
	Properties Create, Defaulted on create, Filter, Group, Sort, Update **Description** ID of the user who created the record. This field is a polymorphic relationship field. **Relationship Name** Owner **Relationship Type** Lookup **Refers To** Group, User
`PricebookEntryId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** Price book entry associated with the source product. This field is a relationship field. **Relationship Name** PricebookEntry **Relationship Type** Lookup **Refers To** PricebookEntry
`PricebookId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort **Description** Price book associated with the source product. This field is a relationship field. **Relationship Name** Pricebook **Relationship Type** Lookup **Refers To** Pricebook2

Field	Details
PricingSource	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort **Description** Pricing type used to calculate the derived price of the product. Valid values are: • Header • Product
ProductId	**Type** reference **Properties** Filter, Group, Sort **Description** Product associated with the derived pricing. This field is a relationship field. **Relationship Name** Product **Relationship Type** Lookup **Refers To** Product2
ProductSellingModelId	**Type** reference **Properties** Filter, Group, Sort **Description** Product selling model associated with this record. This field is a relationship field. **Relationship Name** ProductSellingModel **Relationship Type** Lookup **Refers To** ProductSellingModel

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

PriceBookEntryDerivedPriceFeed on page 2866
Feed tracking is available for the object.

PriceBookEntryDerivedPriceHistory on page 2873
History is available for tracked fields of the object.

PriceBookEntryDerivedPriceShare on page 2877
Sharing is available for the object.

PriceRevisionPolicy

Represents the guidelines and methods used to modify product or service prices, often incorporating formulas based on price revision entries and various adjustments. For example, a policy might dictate that prices are revised based on a formula that considers the regional Consumer Price Index (CPI) with a specific adjustment percentage, effective from a defined date, and categorized as either a flat adjustment or one directly based on the price revision entry data. This object is available in API version 65.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`EffectiveFrom`	**Type** date **Properties** Create, Filter, Group, Sort, Update **Description** The date and time when the price revision policy comes into effect.
`EffectiveTo`	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The date and time when the price revision policy is no longer in effect.
`Formula`	**Type** string **Properties** Create, Filter, Group, Sort, Update

Field	Details
	Description The coded format of the formula used to calculate the revised price of a product from a quote and order, or contract.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed a record related to this record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed this record. If this value is null, it's possible that this record was referenced (LastReferencedDate) and not viewed.
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The name of the price revision policy.
`OwnerId`	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** ID of the user who created the record. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
`PolicyType`	**Type** picklist

Field	Details
	Properties Create, Filter, Group, Restricted picklist, Sort, Update **Description** The type of price revision policy. Valid values are: • `Flat`—You can't define a price index-based formula for revision. • `PriceIndex`—Price Index
`Region`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The region where the price revision policy is valid.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

PriceRevisionPolicyFeed on page 2866
Feed tracking is available for the object.

PriceRevisionPolicyHistory on page 2873
History is available for tracked fields of the object.

PriceRevisionPolicyShare on page 2877
Sharing is available for the object.

PricingAdjBatchJob

Represents the collective update of multiple records on their prices and other adjustments. This object is available in API version 62.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`EffectiveFrom`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date and time when the updated value can be considered for a pricing adjustment batch job.
`EffectiveTo`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date and time until when the updated value is effective and can be considered for a pricing adjustment batch job.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Indicates whether the pricing adjustment batch job has been archived (`true`) or not (`false`). This field is read-only.
`LastTriggeredDate`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date and time when the pricing adjustment batch job was last triggered.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed this record. If this value is null, it's possible that this record was referenced (`LastReferencedDate`) and not viewed.

Field	Details
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The name of the pricing adjustment batch job.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The Salesforce ID of the sales representative who owns the pricing procedure resolution. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
ProcessedRecordsCount	**Type** long **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The total number of records that were successfully updated.
RecordCount	**Type** long **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The total number of records that have been processed.
RecordList	**Type** textarea **Properties** Create **Description** The list of record IDs eligible for a pricing adjustment batch job.

Field	Details
`ShouldSkipBulkRetry`	**Type** boolean **Properties** Create, Filter, Group, Nillable, Sort **Description** For internal use only.
`Status`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The processing status of the pricing adjustment batch job. Valid values are: • `Completed` • `Failed` • `InProgress` • `New` • `PartiallyCompleted` • `Rerun`
`TargetObject`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort **Description** The target object of the pricing adjustment batch job. Valid values are: • `AttributeBasedAdjustment` • `BundleBasedAdjustment` • `PriceAdjustmentTier` • `PricebookEntry`
`UpdateType`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The type of update being made by the pricing adjustment batch job. Valid values are:

Field	Details
	• `Amount` • `Override` • `Percentage`
`UpdateValue`	**Type** double **Properties** Create, Filter, Sort, Update **Description** The numerical value of the update.

Usage

To execute a pricing adjustment batch job through an API request, make a POST request to the `/services/data/v`***`67.0`***`/sobjects/PricingAdjBatchJob` resource. Here's a sample request payload.

```
{
  "TargetObject": "PriceAdjustmentTier",
  "UpdateType": "Amount",
  "UpdateValue": "10",
  "RecordList": "84YDU00000010ig2AA,84YDU00000010ig2AB,84YDU00000010ig2AC",
  "EffectiveFrom": "2024-08-01T10:07:09.000+0000",
  "EffectiveTo": "2024-08-05T10:07:09.000+0000"
}
```

You can specify a comma-separated list of record IDs that are eligible for a pricing adjustment batch job.

To rerun a pricing adjustment batch job, make a PATCH request to the `/services/data/v`***`67.0`***`/sobjects/PricingAdjBatchJob/`***`pricingAdjBatchJobID`*** resource. Here's a sample request payload.

```
{
  "Status": "Rerun"
}
```

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

PricingAdjBatchJobFeed on page 2866
Feed tracking is available for the object.

PricingAdjBatchJobHistory on page 2873
History is available for tracked fields of the object.

PricingAdjBatchJobShare on page 2877
Sharing is available for the object.

PricingAdjBatchJobLog

Represents the report that contains a list of failed adjustment requests along with an error message that describes the reason for failure. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`AdjustedValue`	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The adjusted value of a record. The stored value is used even if another pricing adjustment batch job is triggered again.
`EffectiveFrom`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date and time when the failed versioned record is generated. This is only applicable for Price Adjustment records.
`EffectiveTo`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date and time until when the failed versioned record is available. This is applicable only for Price Adjustment records.
`ErrorCode`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update

Field	Details
	Description The error code for the failure during the record update process.
`ErrorMessage`	**Type** textarea **Properties** Create, Update **Description** The error message that's generated for the failure during the record update process.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Indicates whether the pricing adjustment batch job has been archived (`true`) or not (`false`). This field is read-only.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed this record. If this value is null, it's possible that this record was referenced (`LastReferencedDate`) and not viewed.
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. The name of the pricing adjustment batch job.
`PricingAdjBatchJobId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The pricing adjustment batch job associated with the pricing adjustment batch job log. This field is a relationship field.

Field	Details
	Relationship Name PricingAdjBatchJob **Relationship Type** Master-detail **Refers To** PricingAdjBatchJob (the master object)
`TargetRecord`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The ID of the record for which a pricing adjustment error was generated.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

PricingAdjBatchJobLogFeed
Feed tracking is available for the object.

PricingAdjBatchJobLogHistory **on page 2873**
History is available for tracked fields of the object.

PricingAPIExecution

Represents the pricing resolution for an pricing element determined using strategy name and formula. This object is available in API version 63.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`ApiEndpoint`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update

Field	Details
	Description The unique API endpoint that is called.
`ApiType`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** Specifies the API type of the pricing API execution. Possible values are: • `NGP` The default value is `NGP`.
`CurrencyIsoCode`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Available only if the multicurrency feature is enabled. Contains the ISO code for any currency allowed by the organization. Possible values are: • `BHD`—Bahraini Dinar • `JPY`—Japanese Yen • `USD`—U.S. Dollar The default value is `USD`.
`ExecutionKey`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** The unique execution ID generated each time a pricing API runs.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed a record related to this record.

Field	Details
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed this record. If this value is null, it's possible that this record was referenced (LastReferencedDate) and not viewed.
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The name of the record.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The Salesforce ID of the sales representative who owns the pricing procedure resolution. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
ReferenceKey	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Optional. The reference ID that a consuming workstream must pass in the API to search for specific logs in the Pricing Operations Console.
Status	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** The status of the API response.

Field	Details
	Possible values are: • `Failure` • `Partial_Success`—Partial Success • `Success` The default value is `Success`.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

PricingAPIExecutionFeed on page 2866
Feed tracking is available for the object.

PricingAPIExecutionHistory on page 2873
History is available for tracked fields of the object.

PricingAPIExecutionShare on page 2877
Sharing is available for the object.

PricingProcedureResolution

Represents a selection for a pricing procedure to execute a pricing process from a list of pricing procedures available. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`CurrencyIsoCode`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Available only if the multicurrency feature is enabled. Contains the ISO code for any currency allowed by the organization. Possible values are: • `BHD`—Bahraini Dinar • `JPY`—Japanese Yen

Field	Details
	• `USD`—U.S. Dollar The default value is `USD`.
`EffectiveFrom`	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** The date and time when the pricing procedure resolution comes into effect.
`EffectiveTo`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date and time till when the pricing procedure resolution remains effective.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Indicates whether the pricing procedure resolution has been archived (true) or not (false). This field is read-only.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed this record. If this value is null, it's possible that this record was referenced (LastReferencedDate) and not viewed.
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. The name of the pricing procedure resolution.
`OwnerId`	**Type** reference

Field	Details
	Properties Create, Defaulted on create, Filter, Group, Sort, Update **Description** The Salesforce ID of the sales representative who owns the pricing procedure resolution. This field is a polymorphic relationship field. **Relationship Name** Owner **Relationship Type** Lookup **Refers To** Group, User
PricingProcedureId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The pricing procedure record associated with this pricing procedure resolution. This field is a relationship field. **Relationship Name** PricingProcedure **Relationship Type** Lookup **Refers To** ExpressionSet
ProcedureType	**Type** picklist **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The pricing data store associated with this pricing recipe field mappings.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

PricingProcedureResolutionFeed on page 2866
Feed tracking is available for the object.

PricingProcedureResolutionHistory on page 2873
History is available for tracked fields of the object.

PricingProcedureResolutionShare on page 2877
Sharing is available for the object.

PricingProcessExecution

Represents a record generated during the execution of a discovery or pricing procedure. Multiple procedures may be performed within a single API call, with each recorded in a Pricing API Execution record. This object is available in API version 63.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`CurrencyIsoCode`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Available only if the multicurrency feature is enabled. Contains the ISO code for any currency allowed by the organization. Possible values are: • `BHD`—Bahraini Dinar • `JPY`—Japanese Yen • `USD`—U.S. Dollar The default value is `USD`.
`ExecutionKey`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The unique execution ID generated each time a pricing API runs.
`ExecutionType`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update

Field	Details
	Description The type of execution defined internally within the pricing API. Possible values are: • `Api_Execution`—Api Execution • `Discovery` • `Discovery_Line`—Discovery Line • `Pricing` • `Pricing_Line`—Pricing Line The default value is `Pricing`.
`ExecutionTypeKey`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** The unique execution type ID generated internally for procedure executions, such as pricing or discovery procedures.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed a record related to this record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed this record. If this value is null, it's possible that this record was referenced (LastReferencedDate) and not viewed.
`Message`	**Type** textarea **Properties** Create, Nillable, Update **Description** The message generated upon running a pricing process.

Field	Details
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The name of the record.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The Salesforce ID of the sales representative who owns the pricing procedure resolution. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
Status	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** Specifies the status of the execution type. Possible values are: • `Failure` • `Partial_Success`—Partial Success • `Success` The default value is `Success`.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

PricingProcessExecutionFeed on page 2866
Feed tracking is available for the object.

PricingProcessExecutionHistory on page 2873
History is available for tracked fields of the object.

PricingProcessExecutionShare on page 2877
Sharing is available for the object.

ProductPriceHistoryLog

Stores historical pricing data based on the product's price range. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`CurrencyIsoCode`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** ISO code for any currency allowed by the organization. Available only for organizations with the multicurrency feature enabled. Valid values are: • `BHD`—Bahraini Dinar • `JPY`—Japanese Yen • `USD`—U.S. Dollar The default value is `USD`.
`Date`	**Type** date **Properties** Create, Filter, Group, Sort, Update **Description** The date when the product price history log record is created.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort

Field	Details
	Description Indicates whether the product price history log has been archived (`true`) or not (`false`). This field is read-only.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed this record. If this value is null, it's possible that this record was referenced (`LastReferencedDate`) and not viewed.
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. The name of the product price history log.
`ProductPriceRangeId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The product price range associated with this product price history log record. This field is a relationship field. **Relationship Name** ProductPriceRange **Relationship Type** Master-detail **Refers To** ProductPriceRange (the master object)
`TrackedPrice`	**Type** currency **Properties** Create, Filter, Sort, Update **Description** The price for a product recorded for a particular date.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

ProductPriceHistoryLogFeed on page 2866
Feed tracking is available for the object.

ProductPriceHistoryLogHistory on page 2873
History is available for tracked fields of the object.

ProductPriceRange

Represents the price range of a product determined by using a product selling model that's stored in the relevant price book. This object is available in API version 62.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`CurrencyIsoCode`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** ISO code for any currency allowed by the organization. Available only for organizations with the multicurrency feature enabled. Valid values are: • `BHD`—Bahraini Dinar • `JPY`—Japanese Yen • `USD`—U.S. Dollar The default value is `USD`.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Indicates whether the product price range has been archived (`true`) or not (`false`). This field is read-only.

Field	Details
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed this record. If this value is null, it's possible that this record was referenced (`LastReferencedDate`) and not viewed.
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. The name of the product price range.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The Salesforce ID of the sales representative who owns the product price range. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
ProductId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The product for which the price range is being determined. This field is a relationship field. **Relationship Name** Product **Refers To** Product2
ProductSellingModelId	**Type** reference

Field	Details
	Properties Filter, Group, Nillable, Sort **Description** The product selling model used to determine the price range of the product. This field is a relationship field. **Relationship Name** ProductSellingModel **Refers To** ProductSellingModel
`RecordedPrice`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The selected price of the product over a range of prices.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

ProductPriceRangeFeed on page 2866
Feed tracking is available for the object.

ProductPriceRangeHistory on page 2873
History is available for tracked fields of the object.

ProductPriceRangeShare on page 2877
Sharing is available for the object.

ProductSellingModel

Defines one method by which a product can be sold; for example, as a one-time sale, an evergreen subscription, or a termed subscription. If the product is sold on subscription, this object defines the subscription's term. A product can have multiple product selling models. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Indicates when the user last referred to this record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Indicates when the user last viewed this record.
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The name given to the product selling model.
`PricingTerm`	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The number of pricing term units in the pricing term. Used with PricingTermUnit to define the length of the pricing term. For example, if PricingTermUnit is Months and this field is 1, the subscription is priced monthly.If the selling model is one-time, this field must be null.
`PricingTermUnit`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The unit of time used to define the pricing term. Used with PricingTerm to define the length of the pricing term. For example, if this field is Months and PricingTerm is 1, the subscription is priced monthly. If the selling model is one-time, this field must be null. Possible values are: • `Months`

Field	Details
	• `Quarterly` • `Semi-Annual` • `Annual`
`SellingModelType`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** Indicates whether the product is sold as a one-time sale, an evergreen subscription, or a subscription with a defined term. Possible values are: • `Evergreen` • `OneTime` • `TermDefined` The default value is `OneTime`.
`Status`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** The status of the product selling model. Possible values are: • `Active` • `Draft` • `Inactive` The default value is `Draft`.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

ProductSellingModelFeed on page 2866
Feed tracking is available for the object.

ProductSellingModelHistory on page 2873
History is available for tracked fields of the object.

ProductSellingModelDataTranslation

Represents the translated values of the data stored within the ProductSellingModel record's fields. This object is available in API version 61.0 and later.

Supported Calls

`create()`, `delete()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

- Your organization must be using Enterprise, Unlimited, or Developer edition.
- Translation Workbench and data translation must be enabled in your org.
- To view this object, you must have the "View Setup and Configuration" permission.

Fields

Field	Details
`IsOutOfDate`	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Indicates whether the translation is out-of-date (`true`) or current (`false`). A translation is out-of-date if the parent Product2 record is updated after the last translation was filed.
`Language`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort **Description** The language for these translated values.
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The translated value for the ProductSellingModel record name. This field is required to translate the text in other fields.
`ParentId`	**Type** reference

Field	Details
	Properties Create, Filter, Group, Sort, Update **Description** The ID of the ProductSellingModel record associated with the data that's being translated. This field is a relationship field. **Relationship Name** Parent **Relationship Type** Lookup **Refers To** ProductSellingModel

Usage

Use this object to translate the data stored in a ProductSellingModel record into the different languages supported by Salesforce. If data translation is enabled for custom fields on the ProductSellingModel object, additional ProductSellingModelDataTranslation fields exist for translating the data contained within those fields.

You can't use a custom external id field in an upsert call for a ProductSellingModelDataTranslation object.

ProductSellingModelOption

A junction object between Product Selling Model and Product2. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`Description`	**Type** textarea **Properties** Create, Nillable, Update **Description** The description of the product selling model option.
`DisplayName`	**Type** string

Field	Details
	Properties Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The name of the product selling model option to display to customers.
`Increment`	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The number of pricing term units that can be used to increase a subscription term.
`IsDefault`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indcates the default product selling model for a product. Setting a default is optional. A product can only have one default product selling model. The default value is `false`. This field requires Industries EPC.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed this record, a record related to this record, or a list view.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed this record or list view. If this value is null, the user might have only accessed this record or list view but not viewed it.
`Maximum`	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update

Field	Details
	Description The maximum number of pricing term units for a subscription term.
`Minimum`	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The minimum number of pricing term units for a subscription term.
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The name of the product selling model option.
`Product2Id`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The ID of the Product2 record associated with this ProductSellingModelOption record. This field is a relationship field. **Relationship Name** Product2 **Relationship Type** Lookup **Refers To** Product2
`ProductSellingModelId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The ID of the ProductSellingModel record associated with this ProductSellingModelOption record. This field is a relationship field. **Relationship Name** ProductSellingModel

Field	Details
	Relationship Type Lookup **Refers To** ProductSellingModel
`ProrationPolicyId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort **Description** The ID of the ProrationPolicy record associated with this ProductSellingModelOption record. This field is a relationship field. **Relationship Name** ProrationPolicy **Relationship Type** Lookup **Refers To** ProrationPolicy

ProcedurePlanCriterion

The procedure plan option associated with the procedure plan criterion record. This object is available in API version 67.0 and later.

Supported Calls

`create()`, `delete()`, `describeSObjects()`, `query()`, `retrieve()`, `update()`, `upsert()`

Fields

Field	Details
`ActualValue`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The user-defined value that is compared to the value of the SObject field value.
`DataType`	**Type** picklist

Field	Details
	Properties Create, Filter, Group, Restricted picklist, Sort, Update **Description** The data type of the field from the selected object.
`FieldPath`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** The path of the field used in a procedure in relation to the object that the field belongs to.
`ObjectField`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** The object field value used to resolve the procedure plan option.
`Operator`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Specifies the operator used by the procedure plan criterion. Possible values are: • Equals • GreaterThan • GreaterThanOrEquals • In • IsNotNull • IsNull • LessThan • LessThanOrEquals • NotEquals • NotIn
`ProcedurePlanOptionId`	**Type** reference

Field	Details
	Properties Create, Filter, Group, Sort **Description** This field is a relationship field. **Relationship Name** ProcedurePlanOption **Relationship Type** Master-detail **Refers To** ProcedurePlanOption (the master object)
`Sequence`	**Type** int **Properties** Create, Filter, Group, Sort **Description** The sequence in which the conditions defined in the procedure plan criteria are processed.

ProrationPolicy

Represents the proration policy associated with a Product Selling Model Option that determines how a product's price is calculated based on subscription duration or billing periods. This object is available in API version 67.0 and later.

Supported Calls

`create()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`

Fields

Field	Details
`ArePartialPeriodsAllowed`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort **Description** Indicates whether partial periods should be allowed for standard time periods. The default value is false.

Field	Details
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the proration policy was last referenced.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the proration policy was last viewed.
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort **Description** The name of the proration policy.
ProrationPolicyType	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort **Description** The type of proration policy to be used. Possible values are: • `StandardTimePeriods`—Standard Time Periods
RemainderStrategy	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort **Description** The type of remainder strategy to be used.

Salesforce Pricing Fields on Standard Objects

Salesforce Pricing adds standard fields to some standard Salesforce objects of other features to represent information specific to pricing. These fields are available only in orgs where Salesforce Pricing is enabled.

IndexRate

Standard fields extend the IndexRate object for use in Salesforce Pricing to represent information for a given rate. This object is available in API version 65.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`Region`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Specified the region associated with the given rate.
`UsageType`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies the usage type associated with the given rate. Possible values are: • `Pricing`

Salesforce Pricing Business APIs

Perform pricing request, create context instance, sync pricing data, and manage pricing recipes and pricing waterfall details by using Salesforce Pricing Business APIs.

This table lists the available Salesforce Pricing resources.

Resource	Description
`/connect/core-pricing/price-contexts/`***`contextid`*** (POST)	Perform a pricing request by using the instance ID of a context.
`/connect/core-pricing/pricing` (POST)	Create and hydrate context instance in a single request. Provide a comprehensive response that contains final pricing details per line items and related errors, if any.
`/connect/core-pricing/sync/`***`pricingSyncOrigin`*** (GET)	Sync pricing data to ensure that the lookup tables contain the latest pricing data.
`/connect/core-pricing/recipe` (GET)	Get the mapping details of pricing recipes to the associated pricing recipe table.
`/connect/core-pricing/recipe/mapping` (POST)	Create a mapping between the pricing recipe and the Decision Tables. Post recipes with lookup tables or procedures.
`/connect/core-pricing/versioned-revise-details` (POST)	Create revisions of a pricing request with versions for adjustment entities.
`/connect/core-pricing/waterfall/`***`lineItemId`***`/`***`executionId`*** (GET)	Get the persisted price waterfall that stores the process logs. Price waterfall provides insights into every step of the pricing process.
`/connect/core-pricing/waterfall` (POST)	Create a log of price waterfall. Price waterfall provides insights into every step of the pricing process.
`/connect/core-pricing/pbeDerivedPricingSourceProduct` (POST)	Get the source product for the Price Book Entry (PBE) derived pricing.

Resource	Description
/connect/core-pricing/apiexecutionlogs/*executionId* (GET)	Get the log details of a pricing API execution record by using the execution ID.
/connect/core-pricing/pricing-process-execution/*executionId* (GET)	Get the execution details of a pricing process by using the execution ID.
/connect/core-pricing/pricing-process-execution/lineitems/*executionId*/*executionType* (GET)	Get the pricing execution details for the line items of a pricing process by using the execution ID and execution type.
/connect/core-pricing/simulationInputVariablesWithData (GET)	Get details of the pricing simulation input variables along with associated data.

This section lists the available Procedure Plan Definition-related resources. Use procedure plan definitions to define criteria for all pricing process-related requirements in one central location, and to set up the procedures based on these requirements.

Resource	Description
`/connect/procedure-plan-definitions` (GET, POST)	Get the records of procedure plan definitions. Additionally, create a record of a procedure plan definition.
`/connect/procedure-plan-definitions/`***`procedurePlanDefinitionId`***(GET, PATCH, DELETE)	Get, update, or delete a procedure plan definition record by using the record ID.
`/connect/procedure-plan-definitions/evaluate` (POST)	Evaluate a procedure plan definition based on a primary object to check for prerequisites such as usage type and context mapping details.
`/connect/procedure-plan-definitions/evaluate/`***`procedurePlanDefinitionName`*** (POST)	Evaluate a procedure plan definition based on the name of a definition to check for prerequisites such as usage type and context mapping details.
`/connect/procedure-plan-definitions/`***`procedurePlanDefinitionId`***`/version` (POST)	Create records of a procedure plan version with details.
`/connect/procedure-plan-definitions/versions/`***`procedurePlanVersionId`*** (GET, PATCH, DELETE)	Get, update, or delete a procedure plan definition version record by using the record ID.

Resources

Learn more about the available Salesforce Pricing resources.

Request Bodies

Learn more about the available Salesforce Pricing API request bodies.

Response Bodies

Learn more about the available Salesforce Pricing API response bodies.

SEE ALSO:

Connect REST API Developer Guide: Introduction

Resources

Learn more about the available Salesforce Pricing resources.

PBE Derived Pricing (POST)

Get the source product for the Price Book Entry (PBE) derived pricing.

Price Context (POST)

Perform a pricing request by using the instance ID of a context.

Pricing (POST)

Create and hydrate context instance in a single request. Provide a comprehensive response that contains final pricing details per line items and related errors, if any.

API Execution Logs (GET)

Get the log details of a pricing API execution record by using the execution ID.

Pricing Process Execution (GET)

Get the execution details of a pricing process by using the execution ID.

Pricing Process Execution for Line Items (GET)

Get the pricing execution details for the line items of a pricing process by using the execution ID and execution type.

Pricing Data Sync (GET)

Sync pricing data to ensure that the lookup tables contain the latest pricing data.

Pricing Recipe (GET)

Get the mapping details of pricing recipes to the associated pricing recipe table.

Pricing Recipe Mapping (POST)

Create a mapping between the pricing recipe and the Decision Tables. Post recipes with lookup tables or procedures.

Pricing Versioned Revision Details (POST)

Create revisions of a pricing request with versions for adjustment entities.

Pricing Waterfall (GET)

Get the persisted price waterfall that stores the process logs. Price waterfall provides insights into every step of the pricing process.

Pricing Waterfall (POST)

Create a log of price waterfall. Price waterfall provides insights into every step of the pricing process.

Procedure Plan Definitions (GET, POST)

Get the records of procedure plan definitions. Additionally, create a record of a procedure plan definition.

Procedure Plan Definition By ID (GET, PATCH, DELETE)

Get, update, or delete a procedure plan definition record by using the record ID.

Procedure Plan Evaluation By Object (POST)

Evaluate a procedure plan definition based on a primary object to check for prerequisites such as usage type and context mapping details.

Procedure Plan Evaluation By Definition Name (POST)

Evaluate a procedure plan definition based on the name of a definition to check for prerequisites such as usage type and context mapping details.

Procedure Plan Version (POST)

Create records of a procedure plan version with details.

Procedure Plan Version Details (GET, PATCH, DELETE)

Get, update, or delete a procedure plan definition version record by using the record ID.

Pricing Simulation Input Variables With Data (GET)

Get details of the pricing simulation input variables along with associated data.

PBE Derived Pricing (POST)

Get the source product for the Price Book Entry (PBE) derived pricing.

Resource

```
/connect/core-pricing/pbeDerivedPricingSourceProduct
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/core-pricing/pbeDerivedPricingSourceProduct
```

Available version

61.0

HTTP methods

POST

Request body for POST

JSON example

```
{
"productId":"01txx0000006i2SAAQ",
"pricebookEntryId":"01uxx0000008yYcAAI",
"effectiveFrom":"2020-01-01T22:53:20.000Z",
"effectiveTo":"2021-01-01T22:53:20.000Z"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
effectiveFrom	String	Date from when the price book entry is effective.	Required	61.0

Name	Type	Description	Required or Optional	Available Version
`effectiveTo`	String	Date until when the price book entry is effective.	Required	61.0
`pricebookEntryId`	String	ID of the price book entry.	Required	61.0
`productId`	String	ID of the price book.	Required	61.0

Response body for POST

PBE Derived Pricing

Price Context (POST)

Perform a pricing request by using the instance ID of a context.

If price waterfall is disabled from Salesforce Pricing Setup in your org, this API doesn't return the waterfall details. You can use the Price Waterfall API to retrieve the waterfall details if price waterfall persistence is enabled in Salesforce Pricing Setup.

Resource

```
/connect/core-pricing/price-contexts/contextid
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/core-pricing/price-contexts/0U3RM00000000SR0AY
```

Available version

60.0

HTTP methods

POST

Request body for POST

JSON example

```
{
    "configurationOverrides": {
       "skipWaterfall": true,
       "useSessionScopedContext": true,
       "persistContext": true,
       "taggedData": false
    }
    "procedureName": "ES1"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
configuration Overrides	Configuration Override Input	Parameters to override pricing configuration.	Optional	60.0
procedure Name	String	Name of the pricing procedure.	Optional	60.0

Response body for POST

Pricing Response

Pricing (POST)

Create and hydrate context instance in a single request. Provide a comprehensive response that contains final pricing details per line items and related errors, if any.

If price waterfall is disabled from Salesforce Pricing Setup in your org, this API doesn't return the waterfall details. You can use the Price Waterfall API to retrieve the waterfall details if price waterfall persistence is enabled in Salesforce Pricing Setup.

Resource

```
/connect/core-pricing/pricing
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/core-pricing/pricing
```

Available version

60.0

Requires Chatter

No

HTTP methods

POST

Request body for POST

JSON example

```
{
  "contextDefinitionId": "11Oxx0000006PdxEAE",
  "contextMappingId": "11jxx0000004LDDAA2",
  "jsonDataString": {
    "Cart": [
      {
        "id": "cart_1001",
        "cart_id": "cart_1001",
        "PriceBookId": "PriceBookId_1001",
        "businessObjectType": "Cart",
        "CartItem": [
          {
            "id": "lineItem_1001",
```

```
      "line_item_id": "lineItem_1001",
      "Quantity": 7,
      "PriceType": "OneTime",
      "Frequency": "",
      "UOM": "",
      "businessObjectType": "CartItem",
      "product_id": "01txx0000006i44AAA",
      "UnitPrice": 6.8,
      "NetUnitPrice": 0,
      "Attribute": [
        {
          "name": "Color",
          "code": "RED",
          "isPriceImpacting": true,
          "businessObjectType": "Attribute",
          "id": "Attribute_1001",
          "attribute_id": "Attribute_1001"
        },
        {
          "name": "Size",
          "code": "10INCH",
          "isPriceImpacting": true,
          "businessObjectType": "Attribute",
          "id": "Attribute_1002",
          "attribute_id": "Attribute_1002"
        }
      ]
    },
    {
      "id": "lineItem_1002",
      "line_item_id": "lineItem_1002",
      "quantity": 3,
      "PriceType": "OneTime",
      "Frequency": "",
      "UOM": "",
      "businessObjectType": "CartItem",
      "product_id": "01txx0000006i2SAAQ",
      "unitprice": 6,
      "NetUnitPrice": 0,
      "Attribute": [
        {
          "name": "Color",
          "code": "BLUE",
          "isPriceImpacting": true,
          "businessObjectType": "Attribute",
          "id": "Attribute_1003",
          "attribute_id": "Attribute_1003"
        },
        {
          "name": "Size",
          "code": "6INCH",
          "isPriceImpacting": true,
          "businessObjectType": "Attribute",
          "id": "Attribute_1004",
```

```
                "attribute_id": "Attribute_1004"
              }
            ]
          }
        ]
      }
    ]
  },
  "pricingProcedureId": "9QMxx0000004CKKGA2",
  "configurationOverrides": {
    "skipWaterfall": true,
    "useSessionScopedContext": true,
    "persistContext": true,
    "referenceKey": "referenceKey-12345",
    "displayContext" : false,
    "taggedData": false,
    "isHighVolumeLineItems": false
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`configurationOverrides`	Configuration Override Input	Parameters to override the pricing configuration.	Optional	60.0
`contextDefinitionId`	String	ID of the context definition that defines the structure of the input data.	Required	60.0
`contextMappingId`	String	ID of the context mapping that maps the input data to the context instance.	Required	60.0
`jsonDataString`	String	Data to hydrate the context, which must be in JSON format and passed as String. Pass the JSON data as String by using the stringify() method to convert the object to string. The keys in the `jsonDataString` property must be in accordance to the `contextMappingId` property sent in the request. Make sure that the `businessObjectType` value within this property node is set to the sObject used in the context mappings.	Required	60.0

Name	Type	Description	Required or Optional	Available Version
pricing ProcedureId	String	ID or API name of the pricing procedure used for calculating the prices. A pricing procedure is represented as an Expression Set Definition in the system. If you're an Experience Cloud user, specify the name of the pricing procedure.	Optional	60.0

Response body for POST

Pricing Output

API Execution Logs (GET)

Get the log details of a pricing API execution record by using the execution ID.

Resource

```
/connect/core-pricing/apiexecutionlogs/executionId
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/core-pricing/apiexecutionlogs/29646938297972
```

Available version

63.0

HTTP methods

GET

Path parameter for GET

Parameter Name	Type	Description	Required or Optional	Available Version
executionId	String	ID of the pricing process execution record.	Required	63.0

Response body for GET

Pricing Execution Waterfall Response

Pricing Process Execution (GET)

Get the execution details of a pricing process by using the execution ID.

Resource

```
/connect/core-pricing/pricing-process-execution/executionId
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/core-pricing/pricing-process-execution/29646938297972
```

Available version

63.0

HTTP methods

GET

Path parameter for GET

Parameter Name	Type	Description	Required or Optional	Available Version
`executionId`	String	ID of the pricing process execution record. The ID is generated each time a pricing process is executed.	Required	63.0

Query parameter for GET

Parameter Name	Type	Description	Required or Optional	Available Version
`executionType`	String	Type of execution that's defined internally within the pricing API. Valid values are: • `API_Execution` • `Discovery`—Discovery procedure • `Discovery_Line`—Discovery procedure for the line items. • `Pricing`—Pricing procedure • `Pricing_Line`—Pricing procedure for the line items. If the `executionType` parameter isn't specified, the API retrieves records for all the execution types that are associated with the specified execution ID.	Optional	63.0

Response body for GET

Pricing Process Execution Response

Pricing Process Execution for Line Items (GET)

Get the pricing execution details for the line items of a pricing process by using the execution ID and execution type.

Resource

```
/connect/core-pricing/pricing-process-execution/lineitems/executionId/executionType
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/core-pricing/pricing-process-execution/lineitems/29646938297972/Pricing_Line
```

Available version

63.0

HTTP methods

GET

Path parameters for GET

Parameter Name	Type	Description	Required or Optional	Available Version
executionId	String	ID of the pricing process execution record.	Required	63.0
executionType	String	Type of the execution that's defined internally within the pricing API. Valid values are: • Pricing_Line • Discovery_Line	Required	63.0

Response body for GET

Pricing Process Execution Details for Line Items

Pricing Data Sync (GET)

Sync pricing data to ensure that the lookup tables contain the latest pricing data.

To partially synchronize pricing data, use the Decision Table Refresh Action in a Flow. See Decision Table Refresh Action.

Resource

```
/connect/core-pricing/sync/pricingSyncOrigin
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/core-pricing/sync/syncData
```

This example shows a sample resource to filter by pricing recipe.

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/core-pricing/sync/syncData?pricingRecipeId=12Gxx0000005IzhEAE
```

Available version

60.0

HTTP methods

GET

Request parameters for GET

Parameter Name	Type	Description	Required or Optional	Available Version
`pricingRecipeId`	String	ID of the pricing recipe whose decision tables you want to sync. If not specified, the default pricing recipe is used.	Optional	67.0

Response body for GET

Pricing Generic Response

Pricing Recipe (GET)

Get the mapping details of pricing recipes to the associated pricing recipe table.

Resource

```
/connect/core-pricing/recipe
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/core-pricing/recipe
```

Available version

60.0

HTTP methods

GET

Response body for GET

Pricing Recipe Response

Pricing Recipe Mapping (POST)

Create a mapping between the pricing recipe and the Decision Tables. Post recipes with lookup tables or procedures.

Resource

```
/connect/core-pricing/recipe/mapping
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/core-pricing/recipe/mapping
```

Available version

60.0

HTTP methods

POST

Request body for POST

JSON example

```
{
```

```
    "recipeId" : "12Gxx0000005J9MEAU",
    "pricingRecipeLookUpTableInputRepresentations": [
    {
      lookupId: “12Gxx0000005J9MEAU”,
      pricingComponentType: “CustomDiscount”
  },
    {
      lookupId: “12Gxx0000005J9MEAU”,
      pricingComponentType: “CustomDiscount”
  }
 ],
    "pricingRecipeProcedureInputRepresentation" : {
    "procedureId" : "9QLxx0000004C92GAE"
}
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`pricingRecipeLookUpTableInputRepresentations`	Pricing Recipe LookUp Table Input[]	Input representation of the recipe mapping.	Required	60.0
`pricingRecipeProcedureInputRepresentation`	Pricing Recipe Procedure Input	Input representation of the procedure that's used in the pricing recipe.	Required	60.0
`recipeId`	String	ID of the pricing recipe.	Required	60.0

Response body for POST

Pricing Recipe Post

Pricing Versioned Revision Details (POST)

Create revisions of a pricing request with versions for adjustment entities.

Resource

```
/connect/core-pricing/versioned-revise-details
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/core-pricing/versioned-revise-details
```

Available version

60.0

HTTP methods

POST

Request body for POST

JSON example

This example shows the input for versioned revision details for attribute-based adjustment.

```
{
        "entityName":"AttributeBasedAdjustment",
        "id":"entityId",
        "priceAdjustmentId":"priceAdjustmentScheduleId",
        "productId":"ProductId",
        "productSellingModelId":"PsmId",
        "adjustmentType":"AdjustmentType",
        "adjustmentValue":"AdjustmentValue(Numeric)"",
        "effectiveFrom":"EffectiveFrom date",
        "effectiveTo":"EffectiveTo Date",
        "additionalFieldsToValueMap":{
        "attributeBasedAdjRuleId":"AttributeBasedAdjRuleId"
}
}
```

This example shows the input for versioned revision details for bundle-based adjustment.

```
 {
        "entityName": "BundleBasedAdjustment",
        "id": "entityId",
        "priceAdjustmentScheduleId": "priceAdjustmentScheduleId",
        "productId": "ProductId",
        "productSellingModelId": "PsmId",
        "adjustmentType": "AdjustmentType",
        "adjustmentValue": "AdjustmentValue(Numeric)",
        "effectiveFrom":"EffectiveFrom date",
        "effectiveTo":"EffectiveTo Date",
        "additionalFieldsToValueMap": {
          "rootBundleId": "RootBundleId",
          "parentProductId": "ParentProductId"
        }
 }
```

Properties

Name	Type	Description	Required or Optional	Available Version
additional Fields ToValueMap	Map<String, String>	Map containing the additional fields specific to the entity.	Optional	60.0
adjustment Type	String	Adjustment type such as, percentage, amount, or override.	Required	60.0
adjustment Value	String	Value for the adjustment.	Required	60.0

Name	Type	Description	Required or Optional	Available Version
`effectiveFrom`	String	Date from when the adjustment is effective.	Required	60.0
`effectiveTo`	String	Date until when the adjustment is effective.	Optional	60.0
`entityName`	String	Name of the entity such as AttributeBasedAdjustment entity or BundleBasedAdjustment entity.	Required	60.0
`id`	String	ID of the record.	Required	60.0
`priceAdjustmentScheduleId`	String	ID of the price adjustment schedule record.	Required	60.0
`productId`	String	Product ID of the record.	Required	60.0
`productSellingModelId`	String	Product selling model ID associated to the record.	Optional	60.0

Response body for POST

Pricing Versioned Revision Details

Pricing Waterfall (GET)

Get the persisted price waterfall that stores the process logs. Price waterfall provides insights into every step of the pricing process.

If price waterfall persistence is disabled from Salesforce Pricing Setup in your org, this API doesn't return the waterfall details. You can view the waterfall details in the Pricing API or Price Context API response if price waterfall is enabled in Salesforce Pricing Setup.

Advanced Price Logs

You can set up advanced price logs to capture exception details for complex pricing elements. The API response captures input and output values to trace any exceptions. Refer to the diagnostic data available in the price waterfall details to identify and fix performance issues. See Advanced Price Logs.

Resource

```
/connect/core-pricing/waterfall/lineItemId/executionId
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/core-pricing/waterfall/Gold/2yHdNNEFOZr9jAe4gHS7?tagsToFilter=UnitPrice
```

Available version

60.0

HTTP methods

GET

Query parameters

Name	Type	Description	Required or Optional	Available Version
`tagsToFilter`	String	Comma-separated tags to filter.	Optional	61.0
`usageType`	String	Usage type of the waterfall log record. Valid values are: • `Pricing` • `Discovery` • `Rating`—Specifies that the record type is `Rating`. If this value is specified, the API creates a log of rating waterfall. See Rating Waterfall. The default value is `Pricing`.	Optional	62.0

Response body for GET

Line Item Waterfall Response

Pricing Waterfall (POST)

Create a log of price waterfall. Price waterfall provides insights into every step of the pricing process.

Resource

```
/connect/core-pricing/waterfall
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/core-pricing/waterfall
```

Available version

60.0

HTTP methods

POST

Request body for POST

JSON example

```
{
      "currencyCode": "USD",
      "executionEndTimestamp": "2023-07-31T20:11:29.625Z",
      "executionId": "executionId1",
      "executionStartTimestamp": null,
      "lineItemId": "item1",
      "output": {
          "Subtotal": 38.25,
          "ListPrice": 10,
```

```
        "NetUnitPrice": 7.65
},
    "waterfall": [{
      "fieldToTagNameMapping": {
          "Product2Id": "ItemProduct",
          "Subtotal": "Subtotal",
          "Pricebook2Id": "Pricebook",
          "Quantity": "ItemQuantity",
          "LineItemId": "SalesTransactionSource",
          "ListPrice": "ItemListPrice"
},
    "inputParameters": {
        "Product2Id": "01txx0000006i44AAA",
        "Pricebook2Id": "01sxx0000005q9xAAA",
        "Quantity": 5,
        "LineItemId": "item1"
  },
    "outputParameters": {
        "Subtotal": 50,
        "ListPrice": 10
    },
    "pricingElement": {
        "adjustments": [{
            "AdjustmentValue": "95.00",
            "AdjustmentType": "Amount"
         }],
        "description": null,
        "elementType": "ListPrice",
        "name": "List Price"
    },
        "sequence": 1
  },
  {
      "fieldToTagNameMapping": {
          "PriceAdjustmentScheduleId": "ItemDescription",
          "NetUnitPrice": "ItemNetUnitPrice",
          "Product2Id": "ItemProduct",
          "LowerBound": "ItemQuantity",
          "UpperBound": "ItemQuantity",
          "Subtotal": "Subtotal",
          "Quantity": "ItemQuantity",
          "LineItemId": "SalesTransactionSource",
          "InputUnitPrice": "ItemListPrice"
    },
      "inputParameters": {
          "PriceAdjustmentScheduleId": "84Xxx0000004CGSEA2",
          "Product2Id": "01txx0000006i44AAA",
          "LowerBound": 5,
          "UpperBound": 5,
          "Quantity": 5,
          "LineItemId": "item1",
          "InputUnitPrice": 10
     },
      "outputParameters": {
```

```
                "NetUnitPrice": 8.5,
                "Subtotal": 42.5
  },
          "pricingElement": {
                "adjustments": [{
                "AdjustmentValue": "15.00",
                "AdjustmentType": "Percentage"
  }],
        "description": null,
        "elementType": "VolumeDiscount",
        "name": "Volume Discount"
    },
        "sequence": 2
        }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`contextDefinitionVersionId`	String	Context definition version ID of the pricing procedure.	Optional	60.0
`contextMappingId`	String	Context mapping ID of the pricing procedure.	Optional	60.0
`currencyCode`	String	Currency code such as, USD or INR.	Optional	60.0
`executionEndTimestamp`	String	End timestamp of procedure execution.	Optional	60.0
`executionId`	String	Execution ID for a particular execution of a pricing procedure.	Required	60.0
`executionStartTimestamp`	String	Start timestamp of procedure execution.	Optional	60.0
`lineItemId`	String	Line item ID for which the price is being calculated.	Required	60.0
`output`	Map<String, Object>	Output of the pricing procedure.	Optional	60.0
`waterfall`	Pricing Waterfall Input[]	Details of the pricing waterfall.	Required	60.0

Response body for POST

Pricing Generic Response

Procedure Plan Definitions (GET, POST)

Get the records of procedure plan definitions. Additionally, create a record of a procedure plan definition.

Resource

```
/connect/procedure-plan-definitions
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/
procedure-plan-definitions?isTemplate=true
```

Available version

62.0

HTTP methods

GET, POST

Request parameters for GET

Parameter Name	Type	Description	Required or Optional	Available Version
`isTemplate`	Boolean	Indicates whether to return a list of file-based definitions (`true`) or not (`false`). This API request returns a list of database-based definitions, by default.	Optional	62.0

Response body for GET

Procedure Plan Definitions

Request body for POST

JSON example

This example shows a sample request to create a procedure plan definition record by using the Procedure Plan Definitions (POST) API.

```
  {
  "description": "Definition for Quote",
  "developerName": "Quote_Definition_Sample",
  "name": "Quote_Definition_Sample",
  "processType": "Default",
  "primaryObject": "BusinessHours",
  "procedurePlanDefinitionVersions": [
    {
      "active": false,
      "contextDefinition": "SalesTransactionContext__stdctx",
      "readContextMapping": "QuoteEntitiesMapping",
      "saveContextMapping": "QuoteEntitiesMapping",
      "effectiveFrom": "2024-07-15T10:15:30.000Z",
      "developerName": "Quote_Definition_V1",
      "rank": 1
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`description`	String	Description of the procedure plan definition.	Optional	62.0
`developerName`	String	Developer name of the procedure plan definition.	Required if you're invoking the Procedure Plan Definitions API (POST).	62.0
`name`	String	Name of the procedure plan definition.	Optional	62.0
`primaryObject`	String	Source object that's used to create a procedure with rule-based criteria. This property value must be a valid object name and must be unique in the ProcedurePlanDefinition object.	Required if you're invoking the Procedure Plan Definitions API (POST) and if you're creating a procedure with rule-based criteria.	62.0
`procedurePlanDefinitionVersions`	Procedure Plan Definition Version Input[]	List of versions of a procedure plan definition.	Required	62.0
`processType`	String	Specifies the business processes that need a procedure plan for each sObject and definition. Valid values are: • `Billing` • `DRO` • `DeepClone` • `ProductDiscovery` • `Revenue Cloud` These values can be used based on the available license. If unspecified, the value is set to `Default`.	Required	63.0
`recordId`	String	ID of the procedure plan definition record.	Required if you're invoking the Procedure Plan Definition By ID API (PATCH).	62.0

Response body for POST

Procedure Plan Generic

Procedure Plan Definition By ID (GET, PATCH, DELETE)

Get, update, or delete a procedure plan definition record by using the record ID.

Resource

```
/connect/procedure-plan-definitions/procedurePlanDefinitionId
```

The `procedurePlanDefinitionId` property value is the ID or name of the procedure plan definition record to perform the request for.

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/procedure-plan-definitions/1FNxx0000004EsOGAU
```

Available version

62.0

HTTP methods

DELETE, GET, PATCH

You can delete a procedure plan definition only if it doesn't include any active procedure plan version.

Response body for GET

Procedure Plan Definition

Request body for PATCH

JSON example

This example shows a sample request to update a procedure plan definition by using the Procedure Plan Definition By ID (PATCH) API.

Note: The properties that aren't specified in the input are deleted when updating the record.

```
{
  "description": "Default definition patch update",
  "developerName": "Quote_Definition",
  "name": "Quote_Definition",
  "primaryObject": "Quote",
  "recordId": "1FNxx0000004EsOGAU"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
description	String	Description of the procedure plan definition.	Optional	62.0
developerName	String	Developer name of the procedure plan definition.	Required if you're invoking the Procedure Plan Definitions API (POST).	62.0
name	String	Name of the procedure plan definition.	Optional	62.0

Name	Type	Description	Required or Optional	Available Version
primary Object	String	Source object that's used to create a procedure with rule-based criteria. This property value must be a valid object name and must be unique in the ProcedurePlanDefinition object.	Required if you're invoking the Procedure Plan Definitions API (POST) and if you're creating a procedure with rule-based criteria.	62.0
procedurePlan Definition Versions	Procedure Plan Definition Version Input[]	List of versions of a procedure plan definition.	Required	62.0
processType	String	Specifies the business processes that need a procedure plan for each sObject and definition. Valid values are: • `Billing` • `DRO` • `DeepClone` • `ProductDiscovery` • `Revenue Cloud` These values can be used based on the available license. If unspecified, the value is set to `Default`.	Required	63.0
recordId	String	ID of the procedure plan definition record.	Required if you're invoking the Procedure Plan Definition By ID API (PATCH).	62.0

Response body for PATCH

Procedure Plan Definition

Procedure Plan Evaluation By Object (POST)

Evaluate a procedure plan definition based on a primary object to check for prerequisites such as usage type and context mapping details.

Resource

```
/connect/procedure-plan-definitions/evaluate
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/
procedure-plan-definitions/evaluate
```

Available version

62.0

HTTP methods

POST

Request body for POST

JSON example

This example shows a sample request to evaluate a procedure plan definition by using a primary object.

```
{
  "idList": ["a01DU000000BylcYAC"],
  "evaluationDate": "2024-07-08T10:15:30.000Z",
  "processType": "Default",
  "sectionType": ["PricingProcedure"],
  "subSectionType": ["Revenue"]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`evaluationDate`	String	Date when the evaluation is applicable. This property value must be within the date range when the procedure plan definition is effective.	Required	62.0
`idList`	String[]	List of record IDs of the procedure plan definitions to be evaluated.	Required only if you're invoking the Procedure Plan Evaluation By Object (POST) API.	62.0
`processType`	String	Specifies the business processes that need a procedure plan for each sObject and definition. Valid values based on the available are: • `Billing` • `DRO` • `DeepClone` • `ProductDiscovery` • `Revenue Cloud` These values can be used based on the available license. If unspecified, the value is set to `Default`.	Optional	63.0

Name	Type	Description	Required or Optional	Available Version
		If a procedure plan definition exist in the org with `processType` value as `null`, modify the value to `Default`.		
`sectionType`	String[]	Name of section to be evaluated. Valid values are: • `PricingProcedure` • `ProductDiscoveryProcedure` • `ProductQualificationProcedure` • `PricingDiscoveryProcedure` • `DiscountSpreadServiceProcedure` • `RatingProcedure` • `Custom` • `RatingDiscoveryProcedure`	Optional	62.0
`subSectionType`	String[]	Name of subsection to be evaluated.	Optional	62.0

The combination of the `sectionType` and `subSectionType` property values must be unique for every procedure plan version.

Response body for POST

Procedure Plan Evaluation Response

Procedure Plan Evaluation By Definition Name (POST)

Evaluate a procedure plan definition based on the name of a definition to check for prerequisites such as usage type and context mapping details.

Resource

```
/connect/procedure-plan-definitions/evaluate/procedurePlanDefinitionName
```

Resource example

```
https://yourInstance.salesforce.com/services/data
/v67.0/connect/procedure-plan-definitions/evaluate/Sample_Definition
```

Available version

62.0

HTTP methods

POST

Request body for POST

JSON example

This example shows a sample request to evaluate a procedure plan definition by using a definition name.

```
{
  "evaluationDate": "2024-07-08T10:15:30.000Z",
  "processType": "Default",
  "sectionType": ["PricingProcedure"],
  "subSectionType": ["Revenue"]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`evaluationDate`	String	Date when the evaluation is applicable. This property value must be within the date range when the procedure plan definition is effective.	Required	62.0
`idList`	String[]	List of record IDs of the procedure plan definitions to be evaluated.	Required only if you're invoking the Procedure Plan Evaluation By Object (POST) API.	62.0
`processType`	String	Specifies the business processes that need a procedure plan for each sObject and definition. Valid values based on the available are: • `Billing` • `DRO` • `DeepClone` • `ProductDiscovery` • `Revenue Cloud` These values can be used based on the available license. If unspecified, the value is set to `Default`. If a procedure plan definition exist in the org with `processType` value as `null`, modify the value to `Default`.	Optional	63.0
`sectionType`	String[]	Name of section to be evaluated. Valid values are: • `PricingProcedure` • `ProductDiscoveryProcedure` • `ProductQualificationProcedure`	Optional	62.0

Name	Type	Description	Required or Optional	Available Version
		• `PricingDiscoveryProcedure` • `DiscountSpreadServiceProcedure` • `RatingProcedure` • `Custom` • `RatingDiscoveryProcedure`		
`subSectionType`	String[]	Name of subsection to be evaluated.	Optional	62.0

The combination of the `sectionType` and `subSectionType` property values must be unique for every procedure plan version.

Response body for POST

Procedure Plan Evaluation Response

Procedure Plan Version (POST)

Create records of a procedure plan version with details.

Resource

```
/connect/procedure-plan-definitions/procedurePlanDefinitionId/version
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/
procedure-plan-definitions/1FNxx0000004EsOGAU/version
```

Available version

62.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "active": false,
  "developerName": "sample_version_input",
  "effectiveFrom": "2024-07-09T00:00:00.000Z",
  "contextDefinition": "SalesTransactionContext__stdctx",
  "procedurePlanSections": [
    {
      "isInherited": false,
      "procedurePlanOptions": [
        {
          "saveContextMapping": "AssetToSalesTransactionMapping",
          "expressionSetDefinition": "9QAZ60000004ECOOA2",
          "expressionSetLabel": "Revenue_Default_Pricing_Procedure",
          "expressionSetApiName": "Revenue Default Pricing Procedure",
```

```
          "logic": "1 AND 2 AND 3",
          "priority": 1,
          "procedurePlanCriterion": [
            {
              "conditionSequence": 1,
              "fieldObject": "BillingCountry",
              "fieldPath": "BillingCountry",
              "literalValue": "test",
              "operator": "Equals",
              "dataType": "Text"
            },
            {
              "conditionSequence": 2,
              "fieldObject": "BillingPostalCode",
              "fieldPath": "BillingPostalCode",
              "literalValue": "sample",
              "operator": "Equals",
              "dataType": "Text"
            },
            {
              "conditionSequence": 3,
              "fieldObject": "LastActivityDate",
              "fieldPath": "LastActivityDate",
              "literalValue": "2024-07-14",
              "operator": "LessThan",
              "dataType": "Date"
            }
          ]
        }
      ],
      "resolutionType": "RuleBased",
      "sectionType": "PricingProcedure",
      "sequence": 1,
      "subSectionType": "PricingProcedure",
      "recordId": "1FRZ60000008OIAOA2"
    }
  ],
  "rank": 1,
  "readContextMapping": "ProductDiscoveryContextMapping",
  "saveContextMapping": "OrderEntitiesMapping"
}
```

Note: The properties that aren’t specified in the input are deleted when updating the record.

Properties

Name	Type	Description	Required or Optional	Available Version
`active`	Boolean	Indicates whether this procedure plan definition version is active (`true`) or not (`false`). You can’t edit or delete a procedure plan version that’s in the active state.	Required	62.0

Name	Type	Description	Required or Optional	Available Version
`contextDefinition`	String	Context definition that's associated with the procedure plan definition version record.	Required	62.0
`developerName`	String	Unique developer name of the procedure plan definition version.	Required	62.0
`effectiveFrom`	String	Date and time from when the procedure plan definition version comes into effect.	Required	62.0
`effectiveTo`	String	Date and time from when the procedure plan definition version is no longer in effect.	Required	62.0
`inheritedFrom`	String	Template this procedure plan definition version is created from.	This property is read-only.	62.0
`procedurePlanSections`	Procedure Plan Section Input[]	Procedure setup sections for a procedure plan definition. Each section enables the setup of a procedure type by using a rule-based criteria. Keep these considerations in mind when you modify this property. • You can edit or delete a procedure plan section if it isn't associated with an active procedure plan version. • You can create a procedure plan section with rule-based resolution type if the primary object isn't empty in the definition.	Required	62.0
`rank`	Integer	Current rank of the procedure plan definition version that's used to decide the sequence of execution of a procedure plan definition version.	Required	62.0
`readContextMapping`	String	Mapping that's used to read data from the mapped object and populate the context definition. This property value must be associated with a context definition.	Optional	62.0
`recordId`	String	ID of the procedure plan definition version record.	Required	62.0

Name	Type	Description	Required or Optional	Available Version
`saveContextMapping`	String	Mapping that's used to save data from the context definition and populate the mapped object. This property value must be associated with a context definition.	Optional	62.0
`status`	String	Status of the procedure plan definition version record.	Optional	62.0

Response body for POST

Procedure Plan Generic

Procedure Plan Version Details (GET, PATCH, DELETE)

Get, update, or delete a procedure plan definition version record by using the record ID.

Resource

```
/connect/procedure-plan-definitions/versions/procedurePlanVersionId
```

The `procedurePlanVersionId` property value is the ID or name of the procedure plan version record to perform the request for.

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/
procedure-plan-definitions/versions/1Cvxx0000004E1ACAU
```

Available version

62.0

HTTP methods

DELETE, GET, PATCH

You can't delete a procedure plan version if it's the only procedure plan version in a procedure plan definition.

Response body for GET

Procedure Plan Definition Version

Request body for PATCH

JSON example

```
{
  "active": false,
  "developerName": "sample_version_input",
  "effectiveFrom": "2024-07-09T00:00:00.000Z",
  "contextDefinition": "SalesTransactionContext__stdctx",
  "procedurePlanSections": [
    {
      "isInherited": false,
      "procedurePlanOptions": [
```

```
        {
          "saveContextMapping": "AssetToSalesTransactionMapping",
          "expressionSetDefinition": "9QAZ60000004ECOOA2",
          "expressionSetLabel": "Revenue_Default_Pricing_Procedure",
          "expressionSetApiName": "Revenue Default Pricing Procedure",
          "logic": "1 AND 2 AND 3",
          "priority": 1,
          "procedurePlanCriterion": [
            {
              "conditionSequence": 1,
              "fieldObject": "BillingCountry",
              "fieldPath": "BillingCountry",
              "literalValue": "test",
              "operator": "Equals",
              "dataType": "Text"
            },
            {
              "conditionSequence": 2,
              "fieldObject": "BillingPostalCode",
              "fieldPath": "BillingPostalCode",
              "literalValue": "sample",
              "operator": "Equals",
              "dataType": "Text"
            },
            {
              "conditionSequence": 3,
              "fieldObject": "LastActivityDate",
              "fieldPath": "LastActivityDate",
              "literalValue": "2024-07-14",
              "operator": "LessThan",
              "dataType": "Date"
            }
          ]
        }
      ],
      "resolutionType": "RuleBased",
      "sectionType": "PricingProcedure",
      "sequence": 1,
      "subSectionType": "PricingProcedure",
      "recordId": "1FRZ60000008OIAOA2"
    }
  ],
  "rank": 1,
  "readContextMapping": "ProductDiscoveryContextMapping",
  "saveContextMapping": "OrderEntitiesMapping"
}
```


Note: The properties that aren't specified in the input are deleted when updating the record.

Properties

Name	Type	Description	Required or Optional	Available Version
`active`	Boolean	Indicates whether this procedure plan definition version is active (`true`) or not (`false`). You can't edit or delete a procedure plan version that's in the active state.	Required	62.0
`contextDefinition`	String	Context definition that's associated with the procedure plan definition version record.	Required	62.0
`developerName`	String	Unique developer name of the procedure plan definition version.	Required	62.0
`effectiveFrom`	String	Date and time from when the procedure plan definition version comes into effect.	Required	62.0
`effectiveTo`	String	Date and time from when the procedure plan definition version is no longer in effect.	Required	62.0
`inheritedFrom`	String	Template this procedure plan definition version is created from.	This property is read-only.	62.0
`procedurePlanSections`	Procedure Plan Section Input[]	Procedure setup sections for a procedure plan definition. Each section enables the setup of a procedure type by using a rule-based criteria. Keep these considerations in mind when you modify this property. • You can edit or delete a procedure plan section if it isn't associated with an active procedure plan version. • You can create a procedure plan section with rule-based resolution type if the primary object isn't empty in the definition.	Required	62.0
`rank`	Integer	Current rank of the procedure plan definition version that's used to decide the sequence of execution of a procedure plan definition version.	Required	62.0
`readContextMapping`	String	Mapping that's used to read data from the mapped object and populate the context definition.	Optional	62.0

Name	Type	Description	Required or Optional	Available Version
		This property value must be associated with a context definition.		
`recordId`	String	ID of the procedure plan definition version record.	Required	62.0
`saveContext Mapping`	String	Mapping that's used to save data from the context definition and populate the mapped object. This property value must be associated with a context definition.	Optional	62.0
`status`	String	Status of the procedure plan definition version record.	Optional	62.0

Response body for PATCH

Procedure Plan Generic

Pricing Simulation Input Variables With Data (GET)

Get details of the pricing simulation input variables along with associated data.

Resource

```
/connect/core-pricing/simulationInputVariablesWithData
```

Resource example

```
[illegible]
```

Available version

64.0

HTTP methods

GET

Request parameters for GET

Parameter Name	Type	Description	Required or Optional	Available Version
`context DefinitionId`	String	ID or developer name of the context definition.	Required	64.0
`contextMapping Id`	String	ID or name of the context mapping that's used.	Required	64.0
`entityId`	String	ID of a quote or an order.	Required	64.0

Parameter Name	Type	Description	Required or Optional	Available Version
expressionSetVersionId	String	ID of the expression set that starts with 9QM.	Required	64.0

Response body for GET

Pricing Simulation Input Variables With Data

Request Bodies

Learn more about the available Salesforce Pricing API request bodies.

Adjustment Details Input

Input representation of the adjustment details.

Configuration Override Input

Input representation of the details to override for a Pricing API configuration.

PBE Derived Pricing Input

Input representation of the request to get the source product for the Price Book Entry (PBE) derived pricing.

Pricing Input

Input representation of the details of a Pricing API request.

Pricing Recipe Input

Input representation to set up a pricing recipe page.

Pricing Recipe LookUp Table Input

Input representation of the lookup tables for the setup page recipe.

Pricing Recipe Procedure Input

Input representation of the procedure for the setup page recipe.

Pricing Request Input

Input representation of a pricing request.

Pricing Versioned Revision Details Input

Input representation of the versioned revision details.

Pricing Waterfall Input

Input representation of the pricing waterfall details.

Pricing Waterfall Log Input

Input representation of the request to create an explainability action log.

Procedure Plan Criterion Input

Input representation of the details of a procedure plan criterion.

Procedure Plan Definition Input

Input representation of the details of a procedure plan definition.

Procedure Plan Definition Version Input

Input representation of the details of a procedure plan definition version.

[Procedure Plan Evaluation Input](#)

Input representation of the details used to evaluate a procedure plan definition.

[Procedure Plan Section Input](#)

Input representation of the details of a procedure plan section.

[Procedure Plan Option Input](#)

Input representation of the details of a procedure plan option.

Adjustment Details Input

Input representation of the adjustment details.

JSON example

```
  "pricingElement": {
    "adjustments": [{
    "AdjustmentValue": "15.00",
    "AdjustmentType": "Percentage"
 }],
  "description": null,
  "elementType": "VolumeDiscount",
  "name": "Volume Discount"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`adjustments`	Map<String, Object>[]	Details of the pricing element.	Optional	60.0
`description`	String	Description of the pricing element.	Optional	60.0
`elementType`	String	Type of the pricing element.	Optional	60.0
`name`	String	Name of the pricing element.	Optional	60.0

Configuration Override Input

Input representation of the details to override for a Pricing API configuration.

JSON example

```
   "configurationOverrides": {
        "skipWaterfall": true,
        "useSessionScopedContext": true,
        "persistContext": true,
        "referenceKey": "referenceKey-12345",
        "displayContext" : false,
        "taggedData": false,
```

```
        "isHighVolumeLineItems": false
    }
```

Properties

Name	Type	Description	Required or Optional	Available Version
discovery Procedure	String	Name of the discovery procedure to use to fetch the details of assets.	Optional	61.0
display Context	Boolean	Indicates whether the context structure for pricing must be displayed (`true`) or not (`false`).	Optional	61.0
isHighVolume LineItems	Boolean	Indicates whether the pricing API returns pricing details for more than 100 line items (`true`) or not (`false`).	Optional	63.0
persist Context	Boolean	Indicates whether the context must be persisted as per the mapping (`true`) or not (`false`). If set to `true`, the user must have edit access to all sObject fields used in the context mapping.	Optional	60.0
referenceKey	String	Reference ID that a consuming workstream provides in the API to search for specific logs in the Pricing Operations Console.	Optional	63.0
skipDiscovery	Boolean	Indicates whether the discovery procedure must be skipped (`true`) or not (`false`).	Optional	61.0
skipWaterfall	Boolean	Indicates whether the price waterfall must be skipped in the output response (`true`) or not (`false`).	Optional	60.0
taggedData	Boolean	Indicates whether the JSON data string can specify tags in the input instead of attributes (`true`) or not (`false`).	Optional	60.0
useSession ScopedContext	Boolean	Indicates whether a session scoped context must be created (`true`) or request scoped context (`false`). The default is `false`.	Optional	60.0

PBE Derived Pricing Input

Input representation of the request to get the source product for the Price Book Entry (PBE) derived pricing.

JSON example

```
{
"productId":"01txx0000006i2SAAQ",
"pricebookEntryId":"01uxx0000008yYcAAI",
```

```
"effectiveFrom":"2020-01-01T22:53:20.000Z",
"effectiveTo":"2021-01-01T22:53:20.000Z"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`effectiveFrom`	String	Date from when the price book entry is effective.	Required	61.0
`effectiveTo`	String	Date until when the price book entry is effective.	Required	61.0
`pricebook EntryId`	String	ID of the price book entry.	Required	61.0
`productId`	String	ID of the price book.	Required	61.0

Pricing Input

Input representation of the details of a Pricing API request.

JSON example

```
{
  "contextDefinitionId": "11Oxx0000006PdxEAE",
  "contextMappingId": "11jxx0000004LDDAA2",
  "jsonDataString": {
    "Cart": [
      {
        "id": "cart_1001",
        "cart_id": "cart_1001",
        "PriceBookId": "PriceBookId_1001",
        "businessObjectType": "Cart",
        "CartItem": [
          {
            "id": "lineItem_1001",
            "line_item_id": "lineItem_1001",
            "Quantity": 7,
            "PriceType": "OneTime",
            "Frequency": "",
            "UOM": "",
            "businessObjectType": "CartItem",
            "product_id": "01txx0000006i44AAA",
            "UnitPrice": 6.8,
            "NetUnitPrice": 0,
            "Attribute": [
              {
                "name": "Color",
                "code": "RED",
                "isPriceImpacting": true,
                "businessObjectType": "Attribute",
```

```
                "id": "Attribute_1001",
                "attribute_id": "Attribute_1001"
              },
              {
                "name": "Size",
                "code": "10INCH",
                "isPriceImpacting": true,
                "businessObjectType": "Attribute",
                "id": "Attribute_1002",
                "attribute_id": "Attribute_1002"
              }
            ]
          },
          {
            "id": "lineItem_1002",
            "line_item_id": "lineItem_1002",
            "quantity": 3,
            "PriceType": "OneTime",
            "Frequency": "",
            "UOM": "",
            "businessObjectType": "CartItem",
            "product_id": "01txx0000006i2SAAQ",
            "unitprice": 6,
            "NetUnitPrice": 0,
            "Attribute": [
              {
                "name": "Color",
                "code": "BLUE",
                "isPriceImpacting": true,
                "businessObjectType": "Attribute",
                "id": "Attribute_1003",
                "attribute_id": "Attribute_1003"
              },
              {
                "name": "Size",
                "code": "6INCH",
                "isPriceImpacting": true,
                "businessObjectType": "Attribute",
                "id": "Attribute_1004",
                "attribute_id": "Attribute_1004"
              }
            ]
          }
        ]
      }
    ]
  },
  "pricingProcedureId": "9QMxx0000004CKKGA2",
  "configurationOverrides": {
    "skipWaterfall": true,
    "useSessionScopedContext": true,
    "persistContext": true,
    "referenceKey": "referenceKey-12345",
    "displayContext" : false,
```

```
    "taggedData": false,
    "isHighVolumeLineItems": false
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`configurationOverrides`	Configuration Override Input	Parameters to override the pricing configuration.	Optional	60.0
`contextDefinitionId`	String	ID of the context definition that defines the structure of the input data.	Required	60.0
`contextMappingId`	String	ID of the context mapping that maps the input data to the context instance.	Required	60.0
`jsonDataString`	String	Data to hydrate the context, which must be in JSON format and passed as String. Pass the JSON data as String by using the stringify() method to convert the object to string. The keys in the `jsonDataString` property must be in accordance to the `contextMappingId` property sent in the request. Make sure that the `businessObjectType` value within this property node is set to the sObject used in the context mappings.	Required	60.0
`pricingProcedureId`	String	ID or API name of the pricing procedure used for calculating the prices. A pricing procedure is represented as an Expression Set Definition in the system. If you're an Experience Cloud user, specify the name of the pricing procedure.	Optional	60.0

Pricing Recipe Input

Input representation to set up a pricing recipe page.

JSON example

```
{
    "recipeId" : "12Gxx0000005J9MEAU",
    "pricingRecipeLookUpTableInputRepresentations": [
    {
      lookupId: "12Gxx0000005J9MEAU",
      pricingComponentType: "CustomDiscount"
  },
    {
      lookupId: "12Gxx0000005J9MEAU",
      pricingComponentType: "CustomDiscount"
  }
 ],
    "pricingRecipeProcedureInputRepresentation" : {
    "procedureId" : "9QLxx0000004C92GAE"
}
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`pricingRecipeLookUpTableInputRepresentations`	Pricing Recipe LookUp Table Input[]	Input representation of the recipe mapping.	Required	60.0
`pricingRecipeProcedureInputRepresentation`	Pricing Recipe Procedure Input	Input representation of the procedure that's used in the pricing recipe.	Required	60.0
`recipeId`	String	ID of the pricing recipe.	Required	60.0

Pricing Recipe LookUp Table Input

Input representation of the lookup tables for the setup page recipe.

JSON example

```
    "pricingRecipeLookUpTableInputRepresentations": [
    {
      lookupId: "12Gxx0000005J9MEAU",
      pricingComponentType: "CustomDiscount"
  },
    {
      lookupId: "12Gxx0000005J9MEAU",
```

```
        pricingComponentType: “CustomDiscount”
    }
  ]
```

Properties

Name	Type	Description	Required or Optional	Available Version
`lookupTableId`	String	ID of the decision table.	Optional	60.0
`pricingComponentType`	String	Pricing component types such as volume discount, custom discount, attribute-based discount, and bundle-based discount.	Optional	60.0

Pricing Recipe Procedure Input

Input representation of the procedure for the setup page recipe.

JSON example

```
    "pricingRecipeProcedureInputRepresentation" : {
    "procedureId" : "9QLxx0000004C92GAE"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`procedureId`	String	ID of the expression set.	Required	60.0

Pricing Request Input

Input representation of a pricing request.

JSON example

```
{
    "configurationOverrides": {
       "skipWaterfall": true,
       "useSessionScopedContext": true,
       "persistContext": true,
       "taggedData": false
    }
    "procedureName": "ES1"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`configurationOverrides`	Configuration Override Input	Parameters to override pricing configuration.	Optional	60.0
`procedureName`	String	Name of the pricing procedure.	Optional	60.0

Pricing Versioned Revision Details Input

Input representation of the versioned revision details.

JSON example

This example shows the input for versioned revision details for attribute-based adjustment.

```
{
        "entityName":"AttributeBasedAdjustment",
        "id":"entityId",
        "priceAdjustmentId":"priceAdjustmentScheduleId",
        "productId":"ProductId",
        "productSellingModelId":"PsmId",
        "adjustmentType":"AdjustmentType",
        "adjustmentValue":"AdjustmentValue(Numeric)"",
        "effectiveFrom":"EffectiveFrom date",
        "effectiveTo":"EffectiveTo Date",
        "additionalFieldsToValueMap":{
        "attributeBasedAdjRuleId":"AttributeBasedAdjRuleId"
}
}
```

This example shows the input for versioned revision details for bundle-based adjustment.

```
 {
        "entityName": "BundleBasedAdjustment",
        "id": "entityId",
        "priceAdjustmentScheduleId": "priceAdjustmentScheduleId",
        "productId": "ProductId",
        "productSellingModelId": "PsmId",
        "adjustmentType": "AdjustmentType",
        "adjustmentValue": "AdjustmentValue(Numeric)",
        "effectiveFrom":"EffectiveFrom date",
        "effectiveTo":"EffectiveTo Date",
        "additionalFieldsToValueMap": {
          "rootBundleId": "RootBundleId",
          "parentProductId": "ParentProductId"
        }
 }
```

Properties

Name	Type	Description	Required or Optional	Available Version
additionalFieldsToValueMap	Map<String, String>	Map containing the additional fields specific to the entity.	Optional	60.0
adjustmentType	String	Adjustment type such as, percentage, amount, or override.	Required	60.0
adjustmentValue	String	Value for the adjustment.	Required	60.0
effectiveFrom	String	Date from when the adjustment is effective.	Required	60.0
effectiveTo	String	Date until when the adjustment is effective.	Optional	60.0
entityName	String	Name of the entity such as AttributeBasedAdjustment entity or BundleBasedAdjustment entity.	Required	60.0
id	String	ID of the record.	Required	60.0
priceAdjustmentScheduleId	String	ID of the price adjustment schedule record.	Required	60.0
productId	String	Product ID of the record.	Required	60.0
productSellingModelId	String	Product selling model ID associated to the record.	Optional	60.0

Pricing Waterfall Input

Input representation of the pricing waterfall details.

JSON example

```
      "waterfall": [{
        "fieldToTagNameMapping": {
            "Product2Id": "ItemProduct",
            "Subtotal": "Subtotal",
            "Pricebook2Id": "Pricebook",
            "Quantity": "ItemQuantity",
            "LineItemId": "SalesTransactionSource",
            "ListPrice": "ItemListPrice"
  },
      "inputParameters": {
```

```
            "Product2Id": "01txx0000006i44AAA",
            "Pricebook2Id": "01sxx0000005q9xAAA",
            "Quantity": 5,
            "LineItemId": "item1"
     },
       "outputParameters": {
            "Subtotal": 50,
            "ListPrice": 10
       },
       "pricingElement": {
            "adjustments": [{
                "AdjustmentValue": "95.00",
                "AdjustmentType": "Amount"
             }],
            "description": null,
            "elementType": "ListPrice",
            "name": "List Price"
       },
            "sequence": 1
     },
     {
         "fieldToTagNameMapping": {
              "PriceAdjustmentScheduleId": "ItemDescription",
              "NetUnitPrice": "ItemNetUnitPrice",
              "Product2Id": "ItemProduct",
              "LowerBound": "ItemQuantity",
              "UpperBound": "ItemQuantity",
              "Subtotal": "Subtotal",
              "Quantity": "ItemQuantity",
              "LineItemId": "SalesTransactionSource",
              "InputUnitPrice": "ItemListPrice"
       },
         "inputParameters": {
              "PriceAdjustmentScheduleId": "84Xxx0000004CGSEA2",
              "Product2Id": "01txx0000006i44AAA",
              "LowerBound": 5,
              "UpperBound": 5,
              "Quantity": 5,
              "LineItemId": "item1",
              "InputUnitPrice": 10
        },
         "outputParameters": {
                "NetUnitPrice": 8.5,
                "Subtotal": 42.5
   },
         "pricingElement": {
                "adjustments": [{
                "AdjustmentValue": "15.00",
                "AdjustmentType": "Percentage"
   }],
       "description": null,
       "elementType": "VolumeDiscount",
       "name": "Volume Discount"
     },
```

```
        "sequence": 2
        }
  ]
```

Properties

Name	Type	Description	Required or Optional	Available Version
`fieldToTagNameMapping`	Map<String, String>	Mappings of field to tag names.	Optional	60.0
`inputParameters`	Map<String, Object>	Input parameters of the pricing element.	Optional	60.0
`outputParameters`	Map<String, Object>	Output parameters of the pricing element.	Optional	60.0
`pricingElement`	Adjustment Details Input	Details of the pricing element.	Optional	60.0
`sequence`	Integer	Sequence of the pricing element execution.	Optional	60.0

Pricing Waterfall Log Input

Input representation of the request to create an explainability action log.

JSON example


```
{
      "currencyCode": "USD",
      "executionEndTimestamp": "2023-07-31T20:11:29.625Z",
      "executionId": "executionId1",
      "executionStartTimestamp": null,
      "lineItemId": "item1",
      "output": {
          "Subtotal": 38.25,
          "ListPrice": 10,
          "NetUnitPrice": 7.65
  },
      "waterfall": [{
        "fieldToTagNameMapping": {
            "Product2Id": "ItemProduct",
            "Subtotal": "Subtotal",
            "Pricebook2Id": "Pricebook",
            "Quantity": "ItemQuantity",
            "LineItemId": "SalesTransactionSource",
            "ListPrice": "ItemListPrice"
  },
      "inputParameters": {
          "Product2Id": "01txx0000006i44AAA",
```

```
          "Pricebook2Id": "01sxx0000005q9xAAA",
          "Quantity": 5,
          "LineItemId": "item1"
    },
      "outputParameters": {
          "Subtotal": 50,
          "ListPrice": 10
      },
      "pricingElement": {
          "adjustments": [{
              "AdjustmentValue": "95.00",
              "AdjustmentType": "Amount"
           }],
          "description": null,
          "elementType": "ListPrice",
          "name": "List Price"
      },
          "sequence": 1
    },
    {
        "fieldToTagNameMapping": {
            "PriceAdjustmentScheduleId": "ItemDescription",
            "NetUnitPrice": "ItemNetUnitPrice",
            "Product2Id": "ItemProduct",
            "LowerBound": "ItemQuantity",
            "UpperBound": "ItemQuantity",
            "Subtotal": "Subtotal",
            "Quantity": "ItemQuantity",
            "LineItemId": "SalesTransactionSource",
            "InputUnitPrice": "ItemListPrice"
      },
        "inputParameters": {
            "PriceAdjustmentScheduleId": "84Xxx0000004CGSEA2",
            "Product2Id": "01txx0000006i44AAA",
            "LowerBound": 5,
            "UpperBound": 5,
            "Quantity": 5,
            "LineItemId": "item1",
            "InputUnitPrice": 10
       },
        "outputParameters": {
              "NetUnitPrice": 8.5,
              "Subtotal": 42.5
  },
        "pricingElement": {
              "adjustments": [{
              "AdjustmentValue": "15.00",
              "AdjustmentType": "Percentage"
  }],
      "description": null,
      "elementType": "VolumeDiscount",
      "name": "Volume Discount"
    },
      "sequence": 2
```

```
      }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
contextDefinitionVersionId	String	Context definition version ID of the pricing procedure.	Optional	60.0
contextMappingId	String	Context mapping ID of the pricing procedure.	Optional	60.0
currencyCode	String	Currency code such as, USD or INR.	Optional	60.0
executionEndTimestamp	String	End timestamp of procedure execution.	Optional	60.0
executionId	String	Execution ID for a particular execution of a pricing procedure.	Required	60.0
executionStartTimestamp	String	Start timestamp of procedure execution.	Optional	60.0
lineItemId	String	Line item ID for which the price is being calculated.	Required	60.0
output	Map<String, Object>	Output of the pricing procedure.	Optional	60.0
waterfall	Pricing Waterfall Input[]	Details of the pricing waterfall.	Required	60.0

Procedure Plan Criterion Input

Input representation of the details of a procedure plan criterion.

JSON example

```
          "procedurePlanCriterion": [
            {
              "conditionSequence": 1,
              "fieldObject": "BillingCountry",
              "fieldPath": "BillingCountry",
              "literalValue": "test",
              "operator": "Equals",
              "dataType": "Text"
            },
            {
              "conditionSequence": 2,
```

```
        "fieldObject": "BillingPostalCode",
        "fieldPath": "BillingPostalCode",
        "literalValue": "sample",
        "operator": "Equals",
        "dataType": "Text"
      },
      {
        "conditionSequence": 3,
        "fieldObject": "LastActivityDate",
        "fieldPath": "LastActivityDate",
        "literalValue": "2024-07-14",
        "operator": "LessThan",
        "dataType": "Date"
      }
    ]
```

Properties

Name	Type	Description	Required or Optional	Available Version
`conditionSequence`	Integer	Sequence to be followed to process the conditions defined in the procedure plan option. This property value must be unique within a procedure plan option.	Required	62.0
`dataType`	String	Data type of the field from the selected object.	Required	62.0
`fieldObject`	String	Value of the object field that's used to resolve the procedure plan option. This property value must belong to the primary object that's associated with the procedure plan definition, at a maximum two levels up in the hierarchy.	Required	62.0
`fieldPath`	String	Path of the field that's used in a procedure in relation to the object that the field belongs to. The field path must end with the object field that's associated with the procedure plan criterion.	Required	62.0
`literalValue`	String	User-defined value that's compared to the value of the sObject field value.	Optional	62.0
`operator`	String	Operator that's used by the procedure plan criterion.	Required	62.0
`recordId`	String	ID of the procedure plan criterion record.	Required	62.0

Procedure Plan Definition Input

Input representation of the details of a procedure plan definition.

JSON example

This example shows a sample request to create a procedure plan definition record by using the Procedure Plan Definitions (POST) API.

```
  {
  "description": "Definition for Quote",
  "developerName": "Quote_Definition_Sample",
  "name": "Quote_Definition_Sample",
  "processType": "Default",
  "primaryObject": "BusinessHours",
  "procedurePlanDefinitionVersions": [
    {
      "active": false,
      "contextDefinition": "SalesTransactionContext__stdctx",
      "readContextMapping": "QuoteEntitiesMapping",
      "saveContextMapping": "QuoteEntitiesMapping",
      "effectiveFrom": "2024-07-15T10:15:30.000Z",
      "developerName": "Quote_Definition_V1",
      "rank": 1
    }
  ]
}
```

This example shows a sample request to update a procedure plan definition by using the Procedure Plan Definition By ID (PATCH) API.

Note: The properties that aren't specified in the input are deleted when updating the record.

```
{
  "description": "Default definition patch update",
  "developerName": "Quote_Definition",
  "name": "Quote_Definition",
  "primaryObject": "Quote",
  "recordId": "1FNxx0000004EsOGAU"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
description	String	Description of the procedure plan definition.	Optional	62.0
developerName	String	Developer name of the procedure plan definition.	Required if you're invoking the Procedure Plan Definitions API (POST).	62.0
name	String	Name of the procedure plan definition.	Optional	62.0

Name	Type	Description	Required or Optional	Available Version
`primaryObject`	String	Source object that's used to create a procedure with rule-based criteria. This property value must be a valid object name and must be unique in the ProcedurePlanDefinition object.	Required if you're invoking the Procedure Plan Definitions API (POST) and if you're creating a procedure with rule-based criteria.	62.0
`procedurePlan Definition Versions`	Procedure Plan Definition Version Input[]	List of versions of a procedure plan definition.	Required	62.0
`processType`	String	Specifies the business processes that need a procedure plan for each sObject and definition. Valid values are: • `Billing` • `DRO` • `DeepClone` • `ProductDiscovery` • `Revenue Cloud` These values can be used based on the available license. If unspecified, the value is set to `Default`.	Required	63.0
`recordId`	String	ID of the procedure plan definition record.	Required if you're invoking the Procedure Plan Definition By ID API (PATCH).	62.0

Procedure Plan Definition Version Input

Input representation of the details of a procedure plan definition version.

JSON example

```
{
  "active": false,
  "developerName": "sample_version_input",
  "effectiveFrom": "2024-07-09T00:00:00.000Z",
  "contextDefinition": "SalesTransactionContext__stdctx",
  "procedurePlanSections": [
    {
      "isInherited": false,
      "procedurePlanOptions": [
```

```
        {
          "saveContextMapping": "AssetToSalesTransactionMapping",
          "expressionSetDefinition": "9QAZ60000004ECOOA2",
          "expressionSetLabel": "Revenue_Default_Pricing_Procedure",
          "expressionSetApiName": "Revenue Default Pricing Procedure",
          "logic": "1 AND 2 AND 3",
          "priority": 1,
          "procedurePlanCriterion": [
            {
              "conditionSequence": 1,
              "fieldObject": "BillingCountry",
              "fieldPath": "BillingCountry",
              "literalValue": "test",
              "operator": "Equals",
              "dataType": "Text"
            },
            {
              "conditionSequence": 2,
              "fieldObject": "BillingPostalCode",
              "fieldPath": "BillingPostalCode",
              "literalValue": "sample",
              "operator": "Equals",
              "dataType": "Text"
            },
            {
              "conditionSequence": 3,
              "fieldObject": "LastActivityDate",
              "fieldPath": "LastActivityDate",
              "literalValue": "2024-07-14",
              "operator": "LessThan",
              "dataType": "Date"
            }
          ]
        }
      ],
      "resolutionType": "RuleBased",
      "sectionType": "PricingProcedure",
      "sequence": 1,
      "subSectionType": "PricingProcedure",
      "recordId": "1FRZ60000008OIAOA2"
    }
  ],
  "rank": 1,
  "readContextMapping": "ProductDiscoveryContextMapping",
  "saveContextMapping": "OrderEntitiesMapping"
}
```

Note: The properties that aren't specified in the input are deleted when updating the record.

Properties

Name	Type	Description	Required or Optional	Available Version
`active`	Boolean	Indicates whether this procedure plan definition version is active (`true`) or not (`false`). You can't edit or delete a procedure plan version that's in the active state.	Required	62.0
`contextDefinition`	String	Context definition that's associated with the procedure plan definition version record.	Required	62.0
`developerName`	String	Unique developer name of the procedure plan definition version.	Required	62.0
`effectiveFrom`	String	Date and time from when the procedure plan definition version comes into effect.	Required	62.0
`effectiveTo`	String	Date and time from when the procedure plan definition version is no longer in effect.	Required	62.0
`inheritedFrom`	String	Template this procedure plan definition version is created from.	This property is read-only.	62.0
`procedurePlanSections`	Procedure Plan Section Input[]	Procedure setup sections for a procedure plan definition. Each section enables the setup of a procedure type by using a rule-based criteria. Keep these considerations in mind when you modify this property. • You can edit or delete a procedure plan section if it isn't associated with an active procedure plan version. • You can create a procedure plan section with rule-based resolution type if the primary object isn't empty in the definition.	Required	62.0
`rank`	Integer	Current rank of the procedure plan definition version that's used to decide the sequence of execution of a procedure plan definition version.	Required	62.0
`readContextMapping`	String	Mapping that's used to read data from the mapped object and populate the context definition.	Optional	62.0

Name	Type	Description	Required or Optional	Available Version
		This property value must be associated with a context definition.		
`recordId`	String	ID of the procedure plan definition version record.	Required	62.0
`saveContextMapping`	String	Mapping that's used to save data from the context definition and populate the mapped object. This property value must be associated with a context definition.	Optional	62.0
`status`	String	Status of the procedure plan definition version record.	Optional	62.0

Procedure Plan Evaluation Input

Input representation of the details used to evaluate a procedure plan definition.

JSON example

This example shows a sample request to evaluate a procedure plan definition by using a primary object.

```
{
  "idList": ["a01DU000000BylcYAC"],
  "evaluationDate": "2024-07-08T10:15:30.000Z",
  "processType": "Default",
  "sectionType": ["PricingProcedure"],
  "subSectionType": ["Revenue"]
}
```

This example shows a sample request to evaluate a procedure plan definition by using a definition name.

```
{
  "evaluationDate": "2024-07-08T10:15:30.000Z",
  "processType": "Default",
  "sectionType": ["PricingProcedure"],
  "subSectionType": ["Revenue"]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`evaluationDate`	String	Date when the evaluation is applicable. This property value must be within the date range when the procedure plan definition is effective.	Required	62.0

Name	Type	Description	Required or Optional	Available Version
`idList`	String[]	List of record IDs of the procedure plan definitions to be evaluated.	Required only if you're invoking the Procedure Plan Evaluation By Object (POST) API.	62.0
`processType`	String	Specifies the business processes that need a procedure plan for each sObject and definition. Valid values based on the available are: • `Billing` • `DRO` • `DeepClone` • `ProductDiscovery` • `Revenue Cloud` These values can be used based on the available license. If unspecified, the value is set to `Default`. If a procedure plan definition exist in the org with `processType` value as `null`, modify the value to `Default`.	Optional	63.0
`sectionType`	String[]	Name of section to be evaluated. Valid values are: • `PricingProcedure` • `ProductDiscoveryProcedure` • `ProductQualificationProcedure` • `PricingDiscoveryProcedure` • `DiscountSpreadServiceProcedure` • `RatingProcedure` • `Custom` • `RatingDiscoveryProcedure`	Optional	62.0
`subSectionType`	String[]	Name of subsection to be evaluated.	Optional	62.0

The combination of the `sectionType` and `subSectionType` property values must be unique for every procedure plan version.

Procedure Plan Section Input

Input representation of the details of a procedure plan section.

JSON example

```
"procedurePlanSections": [
  {
    "isInherited": false,
    "procedurePlanOptions": [
      {
        "saveContextMapping": "AssetToSalesTransactionMapping",
        "expressionSetDefinition": "9QAZ60000004ECOOA2",
        "expressionSetLabel": "Revenute_Default_Pricing_Procedure",
        "expressionSetApiName": "Revenue Default Pricing Procedure",
        "logic": "1 AND 2 AND 3",
        "priority": 1,
        "procedurePlanCriterion": [
          {
            "conditionSequence": 1,
            "fieldObject": "BillingCountry",
            "fieldPath": "BillingCountry",
            "literalValue": "test",
            "operator": "Equals",
            "dataType": "Text"
          },
          {
            "conditionSequence": 2,
            "fieldObject": "BillingPostalCode",
            "fieldPath": "BillingPostalCode",
            "literalValue": "sample",
            "operator": "Equals",
            "dataType": "Text"
          },
          {
            "conditionSequence": 3,
            "fieldObject": "LastActivityDate",
            "fieldPath": "LastActivityDate",
            "literalValue": "2024-07-14",
            "operator": "LessThan",
            "dataType": "Date"
          }
        ]
      }
    ],
    "resolutionType": "RuleBased",
    "sectionType": "PricingProcedure",
    "sequence": 1,
    "subSectionType": "PricingProcedure",
    "recordId": "1FRZ60000008OIAOA2"
  }
]
```

Properties

Name	Type	Description	Required or Optional	Available Version
`isInherited`	Boolean	Indicates whether the procedure plan section is inherited from a template (`true`) or not (`false`).	This property is read-only.	62.0
`procedurePlanOptions`	Procedure Plan Option Input[]	List of procedure plan options that defines a group of criteria. You can edit or delete a procedure plan option only if it isn't associated with an active procedure plan version.	Required	62.0
`recordId`	String	ID of the procedure plan section record.	Required	62.0
`resolutionType`	String	Type of resolution used to filter the procedure. You can't edit this property value if the procedure plan section includes a procedure plan option record.	Required	62.0
`sectionType`	String	Type of section. Valid values are: • `PricingProcedure` • `ProductDiscoveryProcedure` • `ProductQualificationProcedure` • `PricingDiscoveryProcedure` • `DiscountSpreadServiceProcedure` • `RatingProcedure` • `Custom` • `RatingDiscoveryProcedure`	Required	62.0
`sequence`	Integer	Sequence to be followed for the processing of the procedures. This property value must be greater than 0 and must be unique for a procedure plan section associated with a procedure plan version.	Required	62.0
`subSectionType`	String	Procedure subsection added to the procedure plan definition.	Required	62.0

Procedure Plan Option Input

Input representation of the details of a procedure plan option.

JSON example

```
      "procedurePlanOptions": [
        {
          "saveContextMapping": "AssetToSalesTransactionMapping",
          "expressionSetDefinition": "9QAZ60000004ECOOA2",
          "expressionSetLabel": "Revenute_Default_Pricing_Procedure",
          "expressionSetApiName": "Revenue Default Pricing Procedure",
          "logic": "1 AND 2 AND 3",
          "priority": 1,
          "procedurePlanCriterion": [
            {
              "conditionSequence": 1,
              "fieldObject": "BillingCountry",
              "fieldPath": "BillingCountry",
              "literalValue": "test",
              "operator": "Equals",
              "dataType": "Text"
            },
            {
              "conditionSequence": 2,
              "fieldObject": "BillingPostalCode",
              "fieldPath": "BillingPostalCode",
              "literalValue": "sample",
              "operator": "Equals",
              "dataType": "Text"
            },
            {
              "conditionSequence": 3,
              "fieldObject": "LastActivityDate",
              "fieldPath": "LastActivityDate",
              "literalValue": "2024-07-14",
              "operator": "LessThan",
              "dataType": "Date"
            }
          ]
        }
      ]
```

Properties

Name	Type	Description	Required or Optional	Available Version
`expressionSetApiName`	String	API name of the expression set.	Optional	62.0
`expressionSetDefinition`	String	Expression set definition that's associated with this procedure plan option record.	Required	62.0
`expressionSetLabel`	String	Label of the expression set that's associated with this procedure plan option record.	Optional	62.0

Name	Type	Description	Required or Optional	Available Version
`logic`	String	Computation logic for the conditions applied to a procedure plan option. This property value must be blank if the resolution type is default.	Optional	62.0
`priority`	Integer	Priority for the specified criteria. This property value must be greater than 0 and must be unique within a procedure plan section.	Required	62.0
`procedurePlanCriterion`	Procedure Plan Criterion Input[]	Details of the rule-based criteria for the procedure. You can edit or delete a procedure plan criterion only if it isn't associated with an active procedure plan version.	Optional	62.0
`readContextMapping`	String	Mapping that's used to read from the mapped object and populate the context definition. This property value must be associated with a context definition.	Optional	62.0
`recordId`	String	ID of the procedure plan option record.	Required	62.0
`saveContextMapping`	String	Mapping that's used to save data from the context definition and populate the mapped object. This property value must be associated with a context definition.	Optional	62.0

Response Bodies

Learn more about the available Salesforce Pricing API response bodies.

Adjustment Details
Output representation of a pricing adjustment request.

API Execution Log Response
Output representation of the execution log of a pricing waterfall request.

Line Item Waterfall Response
Output representation of the line item waterfall response.

PBE Derived Pricing
Output representation of the response that includes the source product for the Price Book Entry (PBE) derived pricing.

[Pricing Error Response]
Output representation of the pricing error response.

[Pricing Execution Waterfall Response]
Output representation of the execution process that's associated with a pricing waterfall.

[Pricing Generic Response]
Output representation of a pricing data sync request.

[Pricing Output]
Output representation of a Salesforce pricing request.

[Pricing Recipe LookUp Table Response]
Output representation of a pricing recipe lookup table.

[Pricing Recipe]
Output representation of the pricing recipe information table.

[Pricing Recipe Post]
Output representation of the pricing recipe after the API request.

[Pricing Recipe Response]
Output representation of the pricing recipe.

[Pricing Response]
Output representation of the pricing request.

[Pricing Result]
Output representation of the pricing result.

[Pricing Result Error]
Output representation of the pricing result error.

[Pricing Versioned Revision Details]
Output representation of the versioned revision details.

[Pricing Waterfall Response]
Output representation of a pricing waterfall request.

[Procedure Plan Criterion]
Output representation of the details of a procedure plan criterion.

[Procedure Plan Definition]
Output representation of the details of a single procedure plan definition.

[Procedure Plan Definition Version]
Output representation of the version details of a procedure plan definition.

[Procedure Plan Definitions]
Output representation of the details of procedure plan definitions.

[Procedure Plan Generic]
Output representation of the details of the created procedure plan definition record.

[Procedure Plan Generic Error]
Output representation of the error details related to the procedure plan definitions.

[Procedure Plan Option]
Output representation of the details of a procedure plan option.

Procedure Plan Section

Output representation of the details of a procedure plan section.

Procedure Plan Section Evaluation Runtime

Output representation of the results from the procedure plan evaluation.

Procedure Plan Evaluation

Output representation of the evaluation details of a procedure plan definition.

Procedure Plan Evaluation Response

Output representation of the evaluation details of a procedure plan definition.

Procedure Plan Evaluation Result

Output representation of the evaluation result of a procedure plan definition.

Pricing Process Execution Details for Line Items

Output representation of the pricing process execution details for the line items along with the error details and response generation status.

Line Item Details Response

Output representation of the pricing process execution details for the line items.

Pricing Process Execution Response

Output representation of the details of a pricing process execution.

Pricing Process Execution List

Output representation of the execution details for different types of the pricing processes.

Pricing Simulation Input Variables With Data

Output representation of the pricing simulation variables with data.

Adjustment Details

Output representation of a pricing adjustment request.

JSON example

```
"pricingElement": {
      "adjustments": [{
      "adjustmentType": null,
      "adjustmentValue": null
    }],
      "name": "List Price",
      "elementType": "ListPrice"
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`adjustments`	Map<String, Object>[]	Details of the pricing element.	Small, 60.0	60.0
`description`	String	Description of the pricing element.	Small, 60.0	60.0
`elementType`	String	Type of the pricing element.	Small, 60.0	60.0

Property Name	Type	Description	Filter Group and Version	Available Version
name	String	Name of the pricing element.	Small, 60.0	60.0

API Execution Log Response

Output representation of the execution log of a pricing waterfall request.

JSON example

```
{
  "message": {The Pricing API execution was successful.},
  "pricingElement": {
    "adjustments": [
      {
        "adjustmentType": null,
        "adjustmentValue": null
      }
    ],
    "name": "List Price",
    "elementType": "ListPrice"
  }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
message	String []	Message of the API execution.	Small, 63.0	63.0
pricingElement	Adjustment Details	Details of the price adjustment of a pricing element.	Small, 63.0	63.0

Line Item Waterfall Response

Output representation of the line item waterfall response.

JSON example

```
{
       "currencyCode": "USD",
       "error": null,
       "executionEndTimestamp": "2023-07-31T20:11:29.625Z",
       "executionId": "gdLVwn2x1uats2xWMAjV",
       "executionStartTimestamp": null,
       "lineItemId": "item1",
       "success": true,
       "usageType":"Pricing",
       "output": {
          "quantity": "10",
          "netUnitPrice": "10",
          "subtotal": "100"
```

```
        },
        "waterfall": []
}
```

This sample response includes diagnostic data if you've enabled Advanced Logging settings under Salesforce Pricing from Setup.

```
{
  "apiExecutionId": "395526033950891",
  "contextDefinitionVersionId": "11OSG000000Eiu12AC",
  "contextMappingId": "11jSG00001YqfHVYAZ",
  "currencyCode": "USD",
  "executionEndTimestamp": "2026-02-17T10:43:46.197Z",
  "executionId": "395527509949481",
  "executionStartTimestamp": "2026-02-17T10:43:44.063Z",
  "id": "0QLSG000001OXv84AG",
  "lineItemId": "0QLSG000001OXv84AG",
  "output": {
    "NetUnitPrice": 20,
    "TotalSubscriptionPrice": 40,
    "Subtotal": 40,
    "diagnosticData": {
      "lineItemId": "0QLSG000001OXv84AG",
      "exceptionDetails": {},
      "inputParams": {
        "ContributingNetUnitPrice": [
          100
        ],
        "ContributingSubTotal": [
          100
        ],
        "DerivedFormula": [
          "PERCENTAGE(ListPrice,10)",
          "PERCENTAGE(ListPrice,10)"
        ],
        "PRODUCT_REFERENCE_IDS": null,
        "Subtotal": 40,
        "Quantity": 2,
        "TransactionalListPrice": 0,
        "ContractId": null,
        "CurrencyIsoCode": "USD",
        "ContributingSource": [
          "Product",
          "Product"
        ],
        "isDerivedProcessed": true,
        "IsDerived": true,
        "NetUnitPrice": 20,
        "ContributingId": [
          "02iSG000001fenKYAQ",
          "0QLSG000001OXv74AG"
        ],
        "HeaderTotal": 100,
        "ContributingScope": [
          "NonTransactional",
          "Transactional"
```

```
      ],
      "Non-TransactionalListPrice": [
        100
      ],
      "ContributingProduct": [
        "01tSG00000CC6NhYAL",
        "01tSG00000CC6NhYAL"
      ],
      "LineItemId": "0QLSG000001OXv84AG",
      "hasError": false,
      "lineItemDetailId": "0QLSG000001OXv84AG",
      "lineItemDetailIndex": 1
    },
    "contributorCount": 2,
    "contributingLines": [
      {
        "ContributingNetUnitPrice": 100,
        "DerivedFormula": "PERCENTAGE(ListPrice,10)",
        "ContributingSubTotal": 100,
        "ContributingId": "0QLSG000001OXv84AG",
        "ContributingScope": "NonTransactional",
        "HeaderTotal": 100,
        "isSkipped": false,
        "Non-TransactionalListPrice": 100,
        "ContributingProduct": "01tSG00000CC6NhYAL",
        "TransactionalListPrice": 0,
        "hasError": false,
        "ContributingSource": "Product"
      },
      {
        "ContributingNetUnitPrice": null,
        "DerivedFormula": "PERCENTAGE(ListPrice,10)",
        "ContributingSubTotal": null,
        "ContributingId": "0QLSG000001OXv84AG",
        "ContributingScope": "Transactional",
        "HeaderTotal": 100,
        "isSkipped": false,
        "Non-TransactionalListPrice": null,
        "ContributingProduct": "01tSG00000CC6NhYAL",
        "TransactionalListPrice": 0,
        "hasError": false,
        "ContributingSource": "Product"
      }
    ]
  },
  "isParallelExecution": true,
  "SubscriptionNetUnitPrice": 20,
  "section-1-output": 40,
  "section-0-output": 40
},
"success": true,
"usageType": "Pricing",
"waterfall": [
  {
```

```
"fieldToTagNameMapping": {
  "ContributingNetUnitPrice": "ContributorUnitPrice",
  "ContributingSubTotal": "ContributorTotalPrice",
  "DerivedFormula": "ContributorFormulaInput",
  "Subtotal": "ItemNetTotalPrice",
  "Quantity": "LineItemQuantity",
  "TransactionalListPrice": "ListPrice",
  "ContractId": "ItemContract",
  "CurrencyIsoCode": "CurrencyIsoCode",
  "PriceWaterfall": "price_water_fall",
  "ContributingSource": "ContributorSource",
  "IsDerived": "DerivedPricingAttribute",
  "NetUnitPrice": "NetUnitPrice",
  "ContributingId": "Contributor",
  "ContributingScope": "ContributorScope",
  "HeaderTotal": "TotalAmount",
  "Non-TransactionalListPrice": "ContributorListPrice",
  "ContributingProduct": "ContributorProduct",
  "LineItemId": "LineItem"
},
"hideWaterfall": false,
"inputParameters": {
  "ContributingNetUnitPrice": [
    100
  ],
  "ContributingSubTotal": [
    100
  ],
  "DerivedFormula": [
    "PERCENTAGE(ListPrice,10)",
    "PERCENTAGE(ListPrice,10)"
  ],
  "Quantity": 2,
  "TransactionalListPrice": 0,
  "ContractId": null,
  "CurrencyIsoCode": "USD",
  "ContributingSource": [
    "Product",
    "Product"
  ],
  "IsDerived": true,
  "ContributingId": [
    "02iSG000001fenKYAQ",
    "0QLSG000001OXv74AG"
  ],
  "HeaderTotal": 100,
  "ContributingScope": [
    "NonTransactional",
    "Transactional"
  ],
  "Non-TransactionalListPrice": [
    100
  ],
  "diagnosticData": {
```

```
      "lineItemId": "0QLSG000001OXv84AG",
      "exceptionDetails": {},
      "inputParams": {
        "ContributingNetUnitPrice": [
          100
        ],
        "ContributingSubTotal": [
          100
        ],
        "DerivedFormula": [
          "PERCENTAGE(ListPrice,10)",
          "PERCENTAGE(ListPrice,10)"
        ],
        "PRODUCT_REFERENCE_IDS": null,
        "Subtotal": 40,
        "Quantity": 2,
        "TransactionalListPrice": 0,
        "ContractId": null,
        "CurrencyIsoCode": "USD",
        "ContributingSource": [
          "Product",
          "Product"
        ],
        "isDerivedProcessed": true,
        "IsDerived": true,
        "NetUnitPrice": 20,
        "ContributingId": [
          "02iSG000001fenKYAQ",
          "0QLSG000001OXv74AG"
        ],
        "HeaderTotal": 100,
        "ContributingScope": [
          "NonTransactional",
          "Transactional"
        ],
        "Non-TransactionalListPrice": [
          100
        ],
        "ContributingProduct": [
          "01tSG00000CC6NhYAL",
          "01tSG00000CC6NhYAL"
        ],
        "LineItemId": "0QLSG000001OXv84AG",
        "hasError": false,
        "lineItemDetailId": "0QLSG000001OXv84AG",
        "lineItemDetailIndex": 1
      },
      "contributorCount": 2,
      "contributingLines": [
        {
          "ContributingNetUnitPrice": 100,
          "DerivedFormula": "PERCENTAGE(ListPrice,10)",
          "ContributingSubTotal": 100,
          "ContributingId": "0QLSG000001OXv84AG",
```

```
        "ContributingScope": "NonTransactional",
        "HeaderTotal": 100,
        "isSkipped": false,
        "Non-TransactionalListPrice": 100,
        "ContributingProduct": "01tSG00000CC6NhYAL",
        "TransactionalListPrice": 0,
        "hasError": false,
        "ContributingSource": "Product"
      },
      {
        "ContributingNetUnitPrice": null,
        "DerivedFormula": "PERCENTAGE(ListPrice,10)",
        "ContributingSubTotal": null,
        "ContributingId": "0QLSG000001OXv84AG",
        "ContributingScope": "Transactional",
        "HeaderTotal": 100,
        "isSkipped": false,
        "Non-TransactionalListPrice": null,
        "ContributingProduct": "01tSG00000CC6NhYAL",
        "TransactionalListPrice": 0,
        "hasError": false,
        "ContributingSource": "Product"
      }
    ]
  },
  "ContributingProduct": [
    "01tSG00000CC6NhYAL",
    "01tSG00000CC6NhYAL"
  ],
  "LineItemId": "0QLSG000001OXv84AG"
},
"outputParameters": {
  "Subtotal": 40,
  "diagnosticData": {
    "lineItemId": "0QLSG000001OXv84AG",
    "exceptionDetails": {},
    "inputParams": {
      "ContributingNetUnitPrice": [
        100
      ],
      "ContributingSubTotal": [
        100
      ],
      "DerivedFormula": [
        "PERCENTAGE(ListPrice,10)",
        "PERCENTAGE(ListPrice,10)"
      ],
      "PRODUCT_REFERENCE_IDS": null,
      "Subtotal": 40,
      "Quantity": 2,
      "TransactionalListPrice": 0,
      "ContractId": null,
      "CurrencyIsoCode": "USD",
      "ContributingSource": [
```

```
          "Product",
          "Product"
        ],
        "isDerivedProcessed": true,
        "IsDerived": true,
        "NetUnitPrice": 20,
        "ContributingId": [
          "02iSG000001fenKYAQ",
          "0QLSG000001OXv74AG"
        ],
        "HeaderTotal": 100,
        "ContributingScope": [
          "NonTransactional",
          "Transactional"
        ],
        "Non-TransactionalListPrice": [
          100
        ],
        "ContributingProduct": [
          "01tSG00000CC6NhYAL",
          "01tSG00000CC6NhYAL"
        ],
        "LineItemId": "0QLSG000001OXv84AG",
        "hasError": false,
        "lineItemDetailId": "0QLSG000001OXv84AG",
        "lineItemDetailIndex": 1
      },
      "contributorCount": 2,
      "contributingLines": [
        {
          "ContributingNetUnitPrice": 100,
          "DerivedFormula": "PERCENTAGE(ListPrice,10)",
          "ContributingSubTotal": 100,
          "ContributingId": "0QLSG000001OXv84AG",
          "ContributingScope": "NonTransactional",
          "HeaderTotal": 100,
          "isSkipped": false,
          "Non-TransactionalListPrice": 100,
          "ContributingProduct": "01tSG00000CC6NhYAL",
          "TransactionalListPrice": 0,
          "hasError": false,
          "ContributingSource": "Product"
        },
        {
          "ContributingNetUnitPrice": null,
          "DerivedFormula": "PERCENTAGE(ListPrice,10)",
          "ContributingSubTotal": null,
          "ContributingId": "0QLSG000001OXv84AG",
          "ContributingScope": "Transactional",
          "HeaderTotal": 100,
          "isSkipped": false,
          "Non-TransactionalListPrice": null,
          "ContributingProduct": "01tSG00000CC6NhYAL",
          "TransactionalListPrice": 0,
```

```
                "hasError": false,
                "ContributingSource": "Product"
              }
            ]
          },
          "NetUnitPrice": 20
        },
        "pricingElement": {
          "adjustments": [
            {
              "NetUnitPrice": 10,
              "ContributingId": "02iSG000001fenKYAQ",
              "ContributingScope": "NonTransactional",
              "ContributingProduct": "01tSG00000CC6NhYAL",
              "ContributingSource": "Product"
            },
            {
              "NetUnitPrice": 10,
              "ContributingId": "0QLSG000001OXv74AG",
              "ContributingScope": "Transactional",
              "ContributingProduct": "01tSG00000CC6NhYAL",
              "ContributingSource": "Product"
            }
          ],
          "elementType": "DerivedPricing",
          "name": "Derived Price"
        },
        "profileAccess": [],
        "sequence": 3,
        "tasksInfo": [
          {
            "executionEndTimestamp": "2026-02-17T10:43:46.077280883Z",
            "executionStartTimestamp": "2026-02-17T10:43:46.037719198Z",
            "taskName": "DerivedPrice-DerivedCalculate"
          },
          {
            "executionEndTimestamp": "2026-02-17T10:43:45.657762503Z",
            "executionStartTimestamp": "2026-02-17T10:43:45.657283237Z",
            "taskName": "DerivedPrice-UpdateCalculationPayload"
          }
        ]
      }
    ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`contextDefinitionVersionId`	String	Context definition version ID of the pricing procedure.	Small, 60.0	60.0
`contextMappingId`	String	Context mapping ID of the record.	Small, 60.0	60.0

Property Name	Type	Description	Filter Group and Version	Available Version
currencyCode	String	Currency code. For example, USD or INR.	Small, 60.0	60.0
error	Pricing Error Response	Details of any errors.	Small, 60.0	60.0
executionEnd Timestamp	String	End timestamp of procedure execution.	Small, 60.0	60.0
executionId	String	Execution ID of a particular execution of a pricing procedure.	Small, 60.0	60.0
execution Start Timestamp	String	Start timestamp of procedure execution.	Small, 60.0	60.0
lineItemId	String	Line item ID for which the price is being calculated.	Small, 60.0	60.0
output	Map<String, Object>	Output of the pricing procedure.	Small, 60.0	60.0
success	Boolean	Indicates whether the API request is successful (`true`) or not (`false`).	Small, 60.0	60.0
usageType	String	Usage type of the waterfall log record.	Small, 62.0	62.0
waterfall	Pricing Waterfall Response[]	Details of the price waterfall.	Small, 60.0	60.0

PBE Derived Pricing

Output representation of the response that includes the source product for the Price Book Entry (PBE) derived pricing.

JSON example

```
{
"productId":"01txx0000006i2SAAQ",
"pricebookEntryId":"01uxx0000008yYcAAI",
"effectiveFrom":"2020-01-01T22:53:20.000Z",
"effectiveTo":"2021-01-01T22:53:20.000Z"
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
error	Pricing Error Response[]	Displays the error while processing the request.	Small, 61.0	61.0
isSuccess	Boolean	Indicates whether the request is successful (`true`) or not (`false`).	Small, 61.0	61.0

Property Name	Type	Description	Filter Group and Version	Available Version
`source ProceductId`	String	ID of the source product.	Small, 61.0	61.0

Pricing Error Response

Output representation of the pricing error response.

Property Name	Type	Description	Filter Group and Version	Available Version
`errorCode`	String	Indicates the error code.	Small, 60.0	60.0
`message`	String	Specifies the message stating the reason for the error, if any.	Small, 60.0	60.0

Pricing Execution Waterfall Response

Output representation of the execution process that's associated with a pricing waterfall.

JSON example

```
{
  "apiEndpoint": "/connect/core-pricing/pricing",
  "apiExecutionId": "263369316770986",
  "apiExecutionLogRepresentationList": [
    {
      "message": [
        "The Pricing API couldn't be run. Try again, and if the issue persists, ask your
 admin for help."
      ]
    }
  ],
  "currencyCode": "USD",
  "executionId": "263369316895959",
  "id": "263369316895960",
  "lineItemId": null,
  "referenceKey": "referenceKey-ABCD",
  "success": false,
  "usageType": "Api_Execution"
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`apiEndpoint`	String	API endpoint that ran during the pricing request.	Small, 63.0	Small, 63.0
`apiExecutionId`	String	Unique execution ID that's generated each time a pricing API is executed.	Small, 63.0	Small, 63.0

Property Name	Type	Description	Filter Group and Version	Available Version
`apiExecutionLogRepresentationList`	API Execution Log Response[]	List of API execution logs.	Small, 63.0	63.0
`currencyCode`	String	Currency code that's stored in each API log record for the pricing execution.	Small, 67.0	Small, 67.0
`error`	Pricing Error Response	Error details of the pricing execution process.	Small, 63.0	Small, 63.0
`executionId`	String	Unique ID that's generated each time a pricing process is executed.	Small, 63.0	Small, 63.0
`id`	String	Unique record ID of the waterfall response.	Small, 65.0	Small, 65.0
`lineItemId`	String	Unique ID of the line item that's associated with this pricing execution.	Small, 59.0	Small, 59.0
`referenceKey`	String	The reference ID that a consuming workstream provides in the API to search for the specific logs in the Pricing Operations Console.	Small, 63.0	Small, 63.0
`success`	Boolean	Indicates whether the API execution is successful (`true`) or not (`false`).	Small, 63.0	Small, 63.0
`usageType`	String	Usage type of the API execution.	Small, 63.0	Small, 63.0

Pricing Generic Response

Output representation of a pricing data sync request.

JSON example

```
{
  "success": true
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`error`	Pricing Error Response	Details from the pricing error response.	Small, 60.0	60.0
`success`	Boolean	Indicates whether the request is successful (`true`) or not (`false`).	Small, 60.0	60.0

Pricing Output

Output representation of a Salesforce pricing request.

JSON example


```
{
"apiExecutionId": "612228038743152",
"pricingExecutionId": "612229738898095",
"pricingResult": {
"subtotal": [
{
"dataPath": [
"cart_1001",
"lineItem_1002"
],
"value": 300.0,
"errors": [],
"isSuccess": true
},
{
"dataPath": [
"cart_1001",
"lineItem_1001"
],
"value":400.0,
"errors": [],
"isSuccess": true
}
],
"netunitprice": [
{
"dataPath": [
"cart_1001",
"lineItem_1002"
],
"value": xx,
"errors": [],
"isSuccess": true
},
{
"dataPath": [
"cart_1001",
"lineItem_1001"
],
"value": xx,
"errors": [],
"isSuccess": true
}
]
},
"pricingResultErrors": [],
"status": "Completed",
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
apiExecutionId	String	Unique ID that's generated each time a pricing API is executed.	Small, 63.0	63.0
error	Pricing Error Response	Displays the error encountered when the request is processed. For example, a pricing procedure isn't found.	Small, 60.0	60.0
pricingExecutionId	String	Unique ID that's generated each time a pricing process is executed.	Small, 63.0	63.0
pricingResult	Pricing Result	Represents the outcomes associated with the output tags defined in the contextual definition for which the pricing engine establishes values. The initial attribute name is substituted for the output tag's designation. For instance, if the original attribute name specified in the Context Definition is "Subtotal," but during contextual setup, the output tag is denoted as "Total Price," the API output exhibits the initial attribute name "Subtotal" in the response.	Small, 60.0	60.0
pricingResult Errors	Pricing Result Error[]	Errors from the pricing request, if any.	Small, 60.0	60.0
status	String	Status of the pricing request. Valid values are: • Completed — Pricing is completed for all the line items. • Partially Completed — Pricing is completed for some line items. • Failed — Pricing isn't completed for the line items.	Small, 60.0	60.0

Pricing Recipe LookUp Table Response

Output representation of a pricing recipe lookup table.

JSON example

```
"decisionTables": [
  {
    "id": "0lDxx00000000T3EAI",
    "isInternal": true,
    "pricingComponentType": "ListPrice"
  },
```

```
        {
          "id": "0lDxx00000000T4EAI",
          "isInternal": true,
          "pricingComponentType": "VolumeDiscount"
        },
        {
          "id": "0lDxx00000000HlEAI",
          "isInternal": false,
          "pricingComponentType": "CustomDiscount"
        }
      ]
```

Property Name	Type	Description	Filter Group and Version	Available Version
`id`	String	ID of the pricing recipe table mapping.	Small, 60.0	60.0
`isInternal`	Boolean	Indicates if the decision table is available (`true`) or not (`false`).	Small, 60.0	60.0
`pricing ComponentType`	String	Price component types such as, custom discount, volume discount, attribute-based discount, bundle-based discount, and list price.	Small, 60.0	60.0

Pricing Recipe

Output representation of the pricing recipe information table.

JSON example

```
  "recipes": [
    {
      "active": false,
      "createdBy": "autoproc@00dxx0000006gmjea2",
      "createdOn": "2023-07-15T13:12:38.000Z",
      "decisionTables": [
        {
          "id": "0lDxx00000000T3EAI",
          "isInternal": true,
          "pricingComponentType": "ListPrice"
        },
        {
          "id": "0lDxx00000000T4EAI",
          "isInternal": true,
          "pricingComponentType": "VolumeDiscount"
        },
        {
          "id": "0lDxx00000000HlEAI",
          "isInternal": false,
          "pricingComponentType": "CustomDiscount"
        }
      ],
```

```
        "developerName": "NGPDefaultRecipe",
        "id": "12Gxx0000005Ka4EAE",
        "name": "NGPDefaultRecipe",
        "procedureCreatedBy": "",
        "procedureCreatedOn": "2023-09-19T11:39:18.983Z",
        "procedureId": "",
        "procedureName": ""
    }]
```

Property Name	Type	Description	Filter Group and Version	Available Version
`active`	Boolean	Indicates whether the recipe is active (`true`) or not (`false`).	Small, 60.0	60.0
`createdBy`	String	Details on who created the recipe.	Small, 60.0	60.0
`createdOn`	String	Date when the recipe was created.	Small, 60.0	60.0
`decision Tables`	Pricing Recipe LookUp Table Response []	Decision tables linked to the recipe.	Small, 60.0	60.0
`developerName`	String	API name of the recipe.	Small, 60.0	60.0
`id`	String	ID of the recipe.	Small, 60.0	60.0
`name`	String	Name of the recipe.	Small, 60.0	60.0
`procedure CreatedBy`	String	Details on who created the procedure.	Small, 60.0	60.0
`procedure CreatedOn`	String	Date when the procedure was created.	Small, 60.0	60.0
`procedureId`	String	ID of the procedure.	Small, 60.0	60.0
`procedureName`	String	Name of the procedure.	Small, 60.0	60.0

Pricing Recipe Post

Output representation of the pricing recipe after the API request.

JSON example

```
{
isSuccess : true
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`error`	Pricing Error Response	Details from the pricing error response.	Small, 60.0	60.0

Property Name	Type	Description	Filter Group and Version	Available Version
`isSuccess`	Boolean	Indicates whether the response was calculated successfully (`true`) or not (`false`).	Small, 60.0	60.0

Pricing Recipe Response

Output representation of the pricing recipe.

JSON example


```
{
  "recipes": [
    {
      "active": false,
      "createdBy": "autoproc@00dxx0000006gmjea2",
      "createdOn": "2023-07-15T13:12:38.000Z",
      "decisionTables": [
        {
          "id": "0lDxx00000000T3EAI",
          "isInternal": true,
          "pricingComponentType": "ListPrice"
        },
        {
          "id": "0lDxx00000000T4EAI",
          "isInternal": true,
          "pricingComponentType": "VolumeDiscount"
        },
        {
          "id": "0lDxx00000000HlEAI",
          "isInternal": false,
          "pricingComponentType": "CustomDiscount"
        }
      ],
      "developerName": "NGPDefaultRecipe",
      "id": "12Gxx0000005Ka4EAE",
      "name": "NGPDefaultRecipe",
      "procedureCreatedBy": "",
      "procedureCreatedOn": "2023-09-19T11:39:18.983Z",
      "procedureId": "",
      "procedureName": ""
    }],
  "success": true
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`recipes`	Pricing Recipe Output Representation[]	Representation of the pricing recipe.	Small, 60.0	60.0

Property Name	Type	Description	Filter Group and Version	Available Version
`success`	Boolean	Indicates if the request is successful (`true`) or not (`false`).	Small, 60.0	60.0

Pricing Response

Output representation of the pricing request.

JSON example

```
{
    "success": true,
    "executionId": "zu81o5hBCrFzyd5LWZk"
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`error`	Pricing Error Response	Errors while processing the request, if any.	Small, 60.0	60.0
`executionId`	String	Auto-generated alphanumeric string for correlation to extract async waterfall and context persistence status.	Small, 60.0	60.0
`success`	Boolean	Indicates if the request is successful (`true`) or not (`false`).	Small, 60.0	60.0

Pricing Result

Output representation of the pricing result.

JSON example

```
"pricingResult": {
"subtotal": [
{
"dataPath": [
"cart_1001",
"lineItem_1002"
],
"value": 300.0,
"errors": [],
"isSuccess": true
},
{
"dataPath": [
"cart_1001",
"lineItem_1001"
],
```

```
"value":400.0,
"errors": [],
"isSuccess": true
}
],
"netunitprice": [
{
"dataPath": [
"cart_1001",
"lineItem_1002"
],
"value": xx,
"errors": [],
"isSuccess": true
},
{
"dataPath": [
"cart_1001",
"lineItem_1001"
],
"value": xx,
"errors": [],
"isSuccess": true
}
]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`dataPath`	String	Includes the entire data route for the specific element starting from the root node. The request must include the ID to construct the accurate data route. For example, if a `jsonDataString` property comprises a Cart [Id = Cart1] and its associated Cart Item [Id = CartItem1], then the data route for CartItem appears as [Cart1, CartItem1].	Small, 60.0	60.0
`errors`	Pricing Error Response[]	Displays processing errors related to the element as recognized by the data path.	Small, 60.0	60.0
`isSuccess`	Boolean	Displays if processing of the element for the specified data path is successful or not.	Small, 60.0	60.0
`value`	Object	Displays the value of the element into consideration. Element is uniquely identify by the data path.	Small, 60.0	60.0

Pricing Result Error

Output representation of the pricing result error.

JSON example

```
"pricingResultErrors": {
"Aggregateprice": [
{
"dataPath": [
"cart_1001",
],


"errors": [
{
"errorCode": "Dummy"
"message":
 }
        ]
   }
 ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
dataPath	String[]	Includes the entire data route for the specific element starting from the root node. The request must include the ID to construct the accurate data route. For example, if a `jsonDataString` property comprises a Cart [Id = Cart1] and its associated Cart Item [Id = CartItem1], then the data route for CartItem appears as [Cart1, CartItem1].	Small, 60.0	60.0
errors	Pricing Error Response	Displays processing errors related to the element as recognized by the data path.	Small, 60.0	60.0

Pricing Versioned Revision Details

Output representation of the versioned revision details.

Property Name	Type	Description	Filter Group and Version	Available Version
error	Pricing Error Response	Details from the pricing error response.	Small, 60.0	60.0
success	Boolean	Indicates whether the request is successful (`true`) or not (`false`).	Small, 60.0	60.0

Pricing Waterfall Response

Output representation of a pricing waterfall request.

JSON example

```
{
  "inputParameters": {
    "productId": "01txx0000006i2SAAQ",
    "pricebookId": "01sxx0000005ptpAAA",
    "pricingModelType": "OneTime"
  },
  "fieldToTagNameMapping": {
    "Product2Id": "ItemProduct",
    "Subtotal": "Subtotal",
    "Pricebook2Id": "Pricebook",
    "Quantity": "ItemQuantity",
    "LineItemId": "SalesTransactionSource",
    "ListPrice": "ItemListPrice"
  },
  "sequence": 0,
  "outputParameters": {
    "listPrice": "10"
  },
  "pricingElement": {
    "adjustments": [
      {
        "adjustmentType": null,
        "adjustmentValue": null
      }
    ],
    "name": "List Price"
  }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`fieldToTagNameMapping`	Map<String, String>	Mappings of field to tag names.	Small, 60.0	60.0
`inputParameters`	Map<String, Object>	Parameters of pricing element input.	Small, 60.0	60.0
`outputParameters`	Map<String, Object>	Parameters of pricing element output.	Small, 60.0	60.0
`pricingElement`	Adjustment Details	Details of the price adjustment of a pricing element.	Small, 60.0	60.0
`sequence`	Integer	Sequence of pricing element execution.	Small, 60.0	60.0

Procedure Plan Criterion

Output representation of the details of a procedure plan criterion.

JSON example


```
"procedurePlanCriterion": [
        {
          "conditionSequence": 1,
          "dataType": "Text",
          "fieldObject": "BillingCountry",
          "fieldPath": "BillingCountry",
          "isSuccess": true,
          "literalValue": "test",
          "operator": "Equals",
          "recordId": "1FiZ60000004C9cKAE"
         },
         {
          "conditionSequence": 2,
          "dataType": "Text",
          "fieldObject": "BillingPostalCode",
          "fieldPath": "BillingPostalCode",
          "isSuccess": true,
          "literalValue": "pramit",
          "operator": "Equals",
          "recordId": "1FiZ60000004C9dKAE"
          },
          {
            "conditionSequence": 3,
            "dataType": "Date",
            "fieldObject": "LastActivityDate",
            "fieldPath": "LastActivityDate",
            "isSuccess": true,
            "literalValue": "2024-07-14",
            "operator": "LessThan",
            "recordId": "1FiZ60000004C9eKAE"
           }
         ]
```


Property Name	Type	Description	Filter Group and Version	Available Version
`conditionSequence`	Integer	Sequence to be followed to process the conditions defined in the procedure plan option.	Small, 62.0	62.0
`dataType`	String	Data type of the field from the selected object.	Small, 62.0	62.0
`error`	Procedure Plan Generic Error[]	Details of the error encountered during the processing of the API request.	Small, 62.0	62.0
`fieldObject`	String	Value of the object field that's used to resolve the procedure plan option.	Small, 62.0	62.0

Property Name	Type	Description	Filter Group and Version	Available Version
`fieldPath`	String	Path of the field that's used in a procedure in relation to the object that the field belongs to.	Small, 62.0	62.0
`isSuccess`	Boolean	Indicates whether the API request is successful (`true`) or not (`false`).	Small, 62.0	62.0
`literalValue`	String	User-defined value that's compared to the value of the sObject field value.	Small, 62.0	62.0
`operator`	String	Operator that's used by the procedure plan criterion.	Small, 62.0	62.0
`recordId`	String	ID of the procedure plan criterion record.	Small, 62.0	62.0

Procedure Plan Definition

Output representation of the details of a single procedure plan definition.

JSON example

This example shows a sample response for the Procedure Plan Definition By ID (GET) request.

```
       {
     "description": "Default Definition",
     "developerName": "Quote_Definition",
     "name": "Quote_Definition",
     "primaryObject": "Quote",
     "procedurePlanDefinitionVersions": [
     {
      "active": false,
      "contextDefinition": "11Oxx0000006PZ7EAM",
      "effectiveFrom": "2024-02-03T10:15:30.000Z",
      "effectiveTo": "2024-02-03T10:15:30.000Z",
      "readContextMapping": "MedicalHistoryMapping",
      "recordId": "1Cvxx0000004E1ACAU",
      "saveContextMapping": "MedicalHistoryMapping",
      "success": true,
      "processType": "Default"
    }
  ],
  "recordId": "1FNxx0000004GkWGAU",
  "processType": "Default",
  "success": true
}
```

This example shows a sample response for the Procedure Plan Definition By ID (PATCH) request.

```
  {
  "procedurePlanDefinitionVersions":[],
  "recordId":"1FNDU00000000EX4AY",
  "processType": "Default",
```

```
"success":true
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`description`	String	Description for the procedure plan definition.	Small, 62.0	62.0
`developerName`	String	Developer name of the procedure plan definition.	Small, 62.0	62.0
`error`	Procedure Plan Generic Error[]	Details of the error encountered during the processing of the API request.	Small, 62.0	62.0
`name`	String	Name of the procedure plan definition.	Small, 62.0	62.0
`primaryObject`	String	Object that's associated with the procedure plan definition.	Small, 62.0	62.0
`procedurePlanDefinitionVersions`	Procedure Plan Definition Version[]	Details of the versions of a procedure plan definition.	Small, 62.0	62.0
`processType`	String	Business processes that's specified that requires a procedure plan for each sObject and definition.	Small, 63.0	63.0
`recordId`	String	ID of the procedure plan definition record.	Small, 62.0	62.0
`success`	Boolean	Indicates whether the API request is successful (`true`) or not (`false`).	Small, 62.0	62.0

Procedure Plan Definition Version

Output representation of the version details of a procedure plan definition.

JSON example

```
"procedurePlanDefinitionVersions": [
 {
   "active": false,
   "developerName": "sample_test",
   "effectiveFrom": "2024-07-09T00:00:00.000Z",
   "contextDefinition": "SalesTransactionContext__stdctx",
   "procedurePlanSections": [],
   "rank": 1,
   "readContextMapping": "ProductDiscoveryContextMapping",
   "recordId": "1CvZ60000008OIaKAM",
   "success": true,
   "processType": "Default"
 }
]
```

Property Name	Type	Description	Filter Group and Version	Available Version
active	Boolean	Indicates whether the procedure plan definition version is active (true) or not (false).	Small, 62.0	62.0
context Definition	String	Context definition that's associated with the procedure plan definition version.	Small, 62.0	62.0
developerName	String	Developer name of the procedure plan definition version.	Small, 62.0	62.0
effectiveFrom	String	Date and time from when the procedure plan definition version is effective.	Small, 62.0	62.0
effectiveTo	String	Date and time until when the procedure plan definition version is effective.	Small, 62.0	62.0
error	Procedure Plan Generic Error[]	Details of the error encountered during the processing of the API request.	Small, 62.0	62.0
inheritedFrom	String	Name of the template the procedure plan definition is extended from.	Small, 62.0	62.0
procedure PlanSections	Procedure Plan Section[]	List of sections of the procedure plan definition that you can organize in any order. Each section must include a procedure or a set of procedures to be executed for a specific criteria.	Small, 62.0	62.0
processType	String	Business processes that's specified that requires a procedure plan for each sObject and definition.	Small, 64.0	64.0
rank	Integer	Rank or the order of sequence to follow for the processing of the procedure plan definition version.	Small, 62.0	62.0
readContext Mapping	String	Mapping that's used to read data from the mapped object and populate the context definition.	Small, 62.0	62.0
recordId	String	ID of the procedure plan definition version record.	Small, 62.0	62.0
saveContext Mapping	String	Mapping that's used to save data from the context definition and populate the mapped object.	Small, 62.0	62.0
success	Boolean	Indicates whether the API request is successful (true) or not (false).	Small, 62.0	62.0

Procedure Plan Definitions

Output representation of the details of procedure plan definitions.

JSON example


```
{
             "isSuccess": true,
             "procedurePlanDefinitions": [
                 {
                     "description": "test description",
                     "developerName": "sample_test",
                     "name": "sample_test",
                     "primaryObject": "Account",
                     "procedurePlanDefinitionVersions": [
                         {
                             "active": false,
                             "developerName": "sample_test",
                             "effectiveFrom": "2024-07-09T00:00:00.000Z",
                             "contextDefinition": "SalesTransactionContext__stdctx",
                             "procedurePlanSections": [],
                             "rank": 1,
                             "readContextMapping": "ProductDiscoveryContextMapping",
                             "recordId": "1CvZ60000008OIaKAM",
                             "success": true
                         }
                     ],
                     "recordId": "1FNZ60000004CAHOA2",
                     "success": true
                 },
                 {
                     "developerName": "PriceAdjustmentSchedule",
                     "name": "PriceAdjustmentSchedule",
                     "primaryObject": "PriceAdjustmentSchedule",
                     "procedurePlanDefinitionVersions": [
                         {
                             "active": false,
                             "developerName": "PriceAdjustmentSchedule",
                             "effectiveFrom": "2024-07-10T00:00:00.000Z",
                             "contextDefinition": "SalesTransactionContext__stdctx",
                             "procedurePlanSections": [],
                             "rank": 1,
                             "recordId": "1CvZ6000000CaRbKAK",
                             "success": true
                         }
                     ],
                     "recordId": "1FNZ6000000CaSAOA0",
                     "success": true
                 }
                 ]
             }
```

Property Name	Type	Description	Filter Group and Version	Available Version
`error`	Procedure Plan Generic Error[]	Details of the error encountered during the processing of the API request.	Small, 62.0	62.0
`isSuccess`	Boolean	Indicates whether the API request is successful (`true`) or not (`false`).	Small, 62.0	62.0
`procedurePlanDefinitions`	Procedure Plan Definition[]	Details of a single procedure plan definition.	Small, 62.0	62.0

Procedure Plan Generic

Output representation of the details of the created procedure plan definition record.

JSON example

This example shows a sample response of the details of a procedure plan definition record, created by using the Procedure Plan Definitions (POST) API.

```
{
 "isSuccess":true,
 "recordId":"1FNDU00000000EX4AY"
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`error`	Procedure Plan Generic Error[]	Details of the error encountered during the processing of the API request.	Small, 62.0	62.0
`isSuccess`	Boolean	Indicates whether the API request is successful (`true`) or not (`false`).	Small, 62.0	62.0
`recordId`	String	ID of the created procedure plan definition record.	Small, 62.0	62.0

Procedure Plan Generic Error

Output representation of the error details related to the procedure plan definitions.

Property Name	Type	Description	Filter Group and Version	Available Version
`errorCode`	String	Code indicating the type of error.	Small, 62.0	62.0
`message`	String	Message stating the reason for the error, if any.	Small, 62.0	62.0

Procedure Plan Option

Output representation of the details of a procedure plan option.

JSON example

```
                "procedurePlanOptions": [
                  {
                    "expressionSetApiName": "Revenue_Mgmt_Default_Pricing_Procedure",
                    "expressionSetDefinition": "9QAZ60000004ECOOA2",
                 "expressionSetLabel": "Revenue Management Default Pricing Procedure",

                    "isSuccess": true,
                    "logic": "1 AND 2 AND 3",
                    "primaryObject": "Account",
                    "priority": 1,
                    "procedurePlanCriterion": [
                      {
                        "conditionSequence": 1,
                        "dataType": "Text",
                        "fieldObject": "BillingCountry",
                        "fieldPath": "BillingCountry",
                        "isSuccess": true,
                        "literalValue": "test",
                        "operator": "Equals",
                        "recordId": "1FiZ60000004C9cKAE"
                       },
                       {
                        "conditionSequence": 2,
                        "dataType": "Text",
                        "fieldObject": "BillingPostalCode",
                        "fieldPath": "BillingPostalCode",
                        "isSuccess": true,
                        "literalValue": "pramit",
                        "operator": "Equals",
                        "recordId": "1FiZ60000004C9dKAE"
                        },
                        {
                          "conditionSequence": 3,
                          "dataType": "Date",
                          "fieldObject": "LastActivityDate",
                          "fieldPath": "LastActivityDate",
                          "isSuccess": true,
                          "literalValue": "2024-07-14",
                          "operator": "LessThan",
                          "recordId": "1FiZ60000004C9eKAE"
                         }
                       ],
                      "recordId": "1FYZ6000000000fOAA",
                      "saveContextMapping": "AssetToSalesTransactionMapping"
                       }
                     ]
```

Property Name	Type	Description	Filter Group and Version	Available Version
`error`	Procedure Plan Generic Error[]	Details of the error encountered during the processing of the API request.	Small, 62.0	62.0

Property Name	Type	Description	Filter Group and Version	Available Version
expression SetApiName	String	API name of the expression set.	Small, 62.0	62.0
expressionSet Definition	String	Expression set definition that's associated with this procedure plan option record.	Small, 62.0	62.0
expression SetLabel	String	Label of the expression set that's associated with this procedure plan option record.	Small, 62.0	62.0
isSuccess	Boolean	Indicates whether the API request is successful (`true`) or not (`false`).	Small, 62.0	62.0
logic	String	Computation logic for the conditions applied to a procedure plan option.	Small, 62.0	62.0
primaryObject	String	Source object that's used to create a procedure with rule-based criteria.	Small, 62.0	62.0
priority	Integer	Priority for the specified criteria.	Small, 62.0	62.0
procedure PlanCriterion	Procedure Plan Criterion[]	Details of the rule-based criteria for the procedure.	Small, 62.0	62.0
readContext Mapping	String	Mapping that's used to read from the mapped object and populate the context definition.	Small, 62.0	62.0
recordId	String	ID of the procedure plan option record.	Small, 62.0	62.0
saveContext Mapping	String	Mapping that's used to save data from the context definition and populate the mapped object.	Small, 62.0	62.0

Procedure Plan Section

Output representation of the details of a procedure plan section.

JSON example


```
                "procedurePlanSections": [
              {
                 "isInherited": false,
                 "isSuccess": true,
                 "procedurePlanOptions": [
                   {
                     "expressionSetApiName": "Revenue_Mgmt_Default_Pricing_Procedure",
                     "expressionSetDefinition": "9QAZ60000004ECOOA2",
                  "expressionSetLabel": "Revenue Management Default Pricing Procedure",

                     "isSuccess": true,
                     "logic": "1 AND 2 AND 3",
                     "primaryObject": "Account",
```

```
                    "priority": 1,
                    "procedurePlanCriterion": [
                      {
                        "conditionSequence": 1,
                        "dataType": "Text",
                        "fieldObject": "BillingCountry",
                        "fieldPath": "BillingCountry",
                        "isSuccess": true,
                        "literalValue": "test",
                        "operator": "Equals",
                        "recordId": "1FiZ60000004C9cKAE"
                       },
                       {
                        "conditionSequence": 2,
                        "dataType": "Text",
                        "fieldObject": "BillingPostalCode",
                        "fieldPath": "BillingPostalCode",
                        "isSuccess": true,
                        "literalValue": "pramit",
                        "operator": "Equals",
                        "recordId": "1FiZ60000004C9dKAE"
                        },
                        {
                          "conditionSequence": 3,
                          "dataType": "Date",
                          "fieldObject": "LastActivityDate",
                          "fieldPath": "LastActivityDate",
                          "isSuccess": true,
                          "literalValue": "2024-07-14",
                          "operator": "LessThan",
                          "recordId": "1FiZ60000004C9eKAE"
                         }
                       ],
                      "recordId": "1FYZ6000000000fOAA",
                      "saveContextMapping": "AssetToSalesTransactionMapping"
                       }
                       ],
                     "recordId": "1FRZ60000008OIAOA2",
                     "resolutionType": "RuleBased",
                     "sectionType": "PricingProcedure",
                     "sequence": 1,
                     "subSectionType": "PricingProcedure"
                   }
               ]
```


Property Name	Type	Description	Filter Group and Version	Available Version
`error`	Procedure Plan Generic Error[]	Details of the error encountered during the processing of the API request.	Small, 62.0	62.0
`isInherited`	Boolean	Indicates whether the procedure plan section is inherited from a template (`true`) or not (`false`).	Small, 62.0	62.0

Property Name	Type	Description	Filter Group and Version	Available Version
isSuccess	Boolean	Indicates whether the API request is successful (true) or not (false).	Small, 62.0	62.0
procedure PlanOptions	Procedure Plan Option[]	List of procedure plan options.	Small, 62.0	62.0
recordId	String	ID of the procedure plan option record.	Small, 62.0	62.0
resolution Type	String	Type of resolution that's used to filter the procedure.	Small, 62.0	62.0
sectionType	String	Type of section. Valid values are: • PricingProcedure • ProductDiscoveryProcedure • ProductQualificationProcedure • PricingDiscoveryProcedure • DiscountSpreadServiceProcedure • RatingProcedure • Custom • RatingDiscoveryProcedure	Small, 62.0	62.0
sequence	Integer	Sequence that's followed for the processing of the procedures.	Small, 62.0	62.0
subSection Type	String	Subsection that's added to the procedure plan definition.	Small, 62.0	62.0

Procedure Plan Section Evaluation Runtime

Output representation of the results from the procedure plan evaluation.

JSON example

```
                "procedurePlanSections": [
                    {
                        "expressionSetApiName": "pricingProcedure_usageType_3",
                        "expressionSetDefinitionId": "9QAZ60000004Ef6OAE",
                        "expressionSetLabel": "pricingProcedure_usageType_3",
                        "sectionType": "PricingProcedure",
                        "sequence": 1,
                        "subSectionType": "Section1",
                        "usageType": "DefaultPricing"
                    },
                    {
                        "expressionSetApiName": "productQualification_usageType_3",
                        "expressionSetDefinitionId": "9QAZ60000004EfFOAU",
                        "expressionSetLabel": "productQualification_usageType_3",
                        "sectionType": "ProductQualificationProcedure",
```

```
                "sequence": 3,
                "subSectionType": "Section2",
                "usageType": "ProductQualification"
            },
            {
                "expressionSetApiName": "rating_usageType_2",
                "expressionSetDefinitionId": "9QAZ60000004EfHOAU",
                "expressionSetLabel": "rating_usageType_2",
                "sectionType": "RatingProcedure",
                "sequence": 2,
                "subSectionType": "Section3",
                "usageType": "DefaultRating"
            }
        ]
```


Property Name	Type	Description	Filter Group and Version	Available Version
`expression SetApiName`	String	API name of the expression set.	Small, 62.0	62.0
`expressionSet DefinitionId`	String	ID of the expression set definition.	Small, 62.0	62.0
`expression SetLabel`	String	Label of the expression set.	Small, 62.0	62.0
`readContext Mapping`	String	Mapping that's used to read data from the mapped object and populate the context definition.	Small, 62.0	62.0
`saveContext Mapping`	String	Mapping that's used to save data from the context definition and populate the mapped object.	Small, 62.0	62.0
`sectionType`	String	Name of the evaluated section. Valid values are: • `PricingProcedure` • `ProductDiscoveryProcedure` • `ProductQualificationProcedure` • `PricingDiscoveryProcedure` • `DiscountSpreadServiceProcedure` • `RatingProcedure` • `Custom` • `RatingDiscoveryProcedure`	Small, 62.0	62.0
`sequence`	Integer	Sequence that's followed for the processing of the procedures.	Small, 62.0	62.0
`subSection Type`	String	Name of the evaluated subsection.	Small, 62.0	62.0

Property Name	Type	Description	Filter Group and Version	Available Version
`usageType`	String	Usage type of the procedure.	Small, 62.0	62.0

Procedure Plan Evaluation

Output representation of the evaluation details of a procedure plan definition.

JSON example

```
"procedurePlanEvaluations":[
{
  "errorMessage":"",
  "id":"a01DU000000BylcYAC",
  "isSuccess":true,
  "primaryObject":"SignallingCustomEvaluation__c",
  "result":{
  "contextDefinition":"11ODU00000008Sw2AI",
  "procedurePlanSections":[]
}
}
]
```

Property Name	Type	Description	Filter Group and Version	Available Version
`errorMessage`	String	Message indicating the error details, if any.	Small, 62.0	62.0
`id`	String	ID of the object used for evaluation.	Small, 62.0	62.0
`isSuccess`	Boolean	Indicates whether the API request is successful (`true`) or not (`false`).	Small, 62.0	62.0
`primaryObject`	String	Name of the object used for evaluation.	Small, 62.0	62.0
`result`	Procedure Plan Evaluation Result[]	Results from the procedure plan evaluation.	Small, 62.0	62.0

Procedure Plan Evaluation Response

Output representation of the evaluation details of a procedure plan definition.

JSON example

```
{
"isSuccess":true,
"procedurePlanEvaluations":[
{
  "errorMessage":"",
  "id":"a01DU000000BylcYAC",
  "isSuccess":true,
  "primaryObject":"SignallingCustomEvaluation__c",
```

```
    "result":{
    "contextDefinition":"11ODU00000008Sw2AI",
    "procedurePlanSections":[]
  }
  }
  ]
  }
```


Property Name	Type	Description	Filter Group and Version	Available Version
`errorMessage`	String	Message indicating the error details, if any.	Small, 62.0	62.0
`isSuccess`	Boolean	Indicates whether the API request is successful (`true`) or not (`false`).	Small, 62.0	62.0
`procedurePlanDefinitionName`	String	Name of the procedure plan definition.	Small, 62.0	62.0
`procedurePlanEvaluations`	Procedure Plan Evaluation[]	Evaluation details of the procedure plan.	Small, 62.0	62.0

Procedure Plan Evaluation Result

Output representation of the evaluation result of a procedure plan definition.

JSON example


```
    "result":{
    "contextDefinition":"11ODU00000008Sw2AI",
    "procedurePlanSections":[]
  }
```


Property Name	Type	Description	Filter Group and Version	Available Version
`contextDefinition`	String	Context definition that's associated with the procedure plan evaluation.	Small, 62.0	62.0
`procedurePlanSections`	Procedure Plan Section Evaluation Runtime[]	Results from the procedure plan evaluation.	Small, 62.0	62.0

Pricing Process Execution Details for Line Items

Output representation of the pricing process execution details for the line items along with the error details and response generation status.

JSON example


```
 {
```

```
  "error": {},
  "isSuccess": true,
  "lineItemDetailsList": [
    {
      "lineItemId": "LineItem1",
      "status": "Success"
    },
    {
      "lineItemId": "LineItem2",
      "status": "Success"
    },
    {
      "lineItemId": "LineItem3",
      "status": "Failure"
    }
  ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`error`	Pricing Error Response	Error encountered during the processing of the API request.	Small, 63.0	63.0
`isSuccess`	Boolean	Indicates whether the response was generated successfully (`true`) or not (`false`).	Small, 63.0	63.0
`lineItemDetailsList`	Line Item Details Response []	List of the line items for which the pricing process is executed.	Small, 63.0	63.0

Line Item Details Response

Output representation of the pricing process execution details for the line items.

JSON example


```
 {
  "lineItemDetailsList": [
    {
      "lineItemId": "LineItem1",
      "status": "Success"
    },
    {
      "lineItemId": "LineItem2",
      "status": "Success"
    },
    {
      "lineItemId": "LineItem3",
      "status": "Failure"
    }
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`lineItemId`	String	ID of the line item that the pricing process is executed for.	Small, 63.0	63.0
`status`	String	Specifies whether the pricing process execution for the line item is successful or has failed. Valid values are: • `Success` • `Failure`	Small, 63.0	63.0

Pricing Process Execution Response

Output representation of the details of a pricing process execution.

JSON example


```
{
 "error": {},
 "isSuccess": true,
 "pricingProcessExecutionList": [
   {
     "executionId": "12345",
     "executionType": "Pricing_Line",
     "executionTypeId": "111_LineItem1",
     "message": "The Pricing API execution was successful.",
     "status": "Success"
   },
   {
     "executionId": "12345",
     "executionType": "Api_Execution",
     "executionTypeId": "333",
     "status": "Partial_Success"
   },
   {
     "executionId": "12345",
     "executionType": "Discovery",
     "executionTypeId": "222",
     "status": "Success"
   },
   {
     "executionId": "12345",
     "executionType": "Pricing",
     "executionTypeId": "111",
     "status": "Failure"
   },
   {
     "executionId": "12345",
     "executionType": "Discovery_Line",
     "executionTypeId": "222_LineItem1",
     "status": "Partial_Success"
```

```
    },
    {
      "executionId": "12345",
      "executionType": "Pricing_Line",
      "executionTypeId": "111_LineItem2",
      "status": "Failure"
    }
  ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`error`	Pricing Error Response	Error encountered during the processing of the API request.	Small, 63.0	63.0
`isSuccess`	Boolean	Indicates whether the response was generated successfully (`true`) or not (`false`).	Small, 63.0	63.0
`pricingProcessExecutionList`	Pricing Process Execution List []	List of the execution details of the pricing process.	Small, 63.0	63.0

Pricing Process Execution List

Output representation of the execution details for different types of the pricing processes.

JSON example


```
{
 "pricingProcessExecutionList": [
    {
      "executionId": "12345",
      "executionType": "Pricing_Line",
      "executionTypeId": "111_LineItem1",
      "message": "The Pricing API execution was successful.",
      "status": "Success"
    },
    {
      "executionId": "12345",
      "executionType": "Api_Execution",
      "executionTypeId": "333",
      "status": "Success"
    },
    {
      "executionId": "12345",
      "executionType": "Discovery",
      "executionTypeId": "222",
      "status": "Success"
    },
    {
      "executionId": "12345",
      "executionType": "Pricing",
```

```
      "executionTypeId": "111",
      "status": "Failure"
    },
    {
      "executionId": "12345",
      "executionType": "Discovery_Line",
      "executionTypeId": "222_LineItem1",
      "status": "Success"
    },
    {
      "executionId": "12345",
      "executionType": "Pricing_Line",
      "executionTypeId": "111_LineItem2",
      "status": "Failure"
    }
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`executionId`	String	Unique ID that's generated each time a pricing process is executed.	Small, 63.0	63.0
`executionType`	String	Type of the execution that's defined internally within the pricing API.	Small, 63.0	63.0
`executionTypeId`	String	Unique execution type ID that's generated internally for process executions, such as pricing or discovery procedures.	Small, 63.0	63.0
`message`	String	Message that's generated when a pricing process is executed.	Small, 63.0	63.0
`status`	String	Execution process status for a line item. Valid values are: • `Failure` • `Partial_Success`—Applies to `Pricing` and `Discovery` procedures when execution for some line items fails. • `Success`	Small, 63.0	63.0

Pricing Simulation Input Variables With Data

Output representation of the pricing simulation variables with data.

JSON example

```
{
  "error": "",
  "simulationInputJsonWithData": "{\"SalesTransaction\": [{\"PriceBooks\":
```

```
\"01sxx0000005ptpAAA\",\"SalesTransactionItem\": [{\"LineItemQuantity\":
4,\"ProductSellingModel\": null,\"Product\": \"01txx0000006i2SAAQ\",\"LineItem\":
\"0QLxx0000004C92GAE\"},{\"LineItemQuantity\": 3,\"ProductSellingModel\":
null,\"Product\": \"01txx0000006i2TAAQ\",\"LineItem\": \"0QLxx0000004C93GAE\"}]}]}",
  "success": true
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`error`	String	Returns the cause of error, if any. For a successful request, this API returns an empty string.	Small, 64.0	64.0
`simulationInputJsonWithData`	String	Resultant simulation input variables with quote or order data such as ID, which was specified in the query parameters.	Small, 64.0	64.0
`success`	Boolean	Indicates whether the request was successful (`true`) or not (`false`).	Small, 64.0	64.0

Salesforce Pricing Apex Reference

Use built-in Apex classes and interfaces grouped by namespace.

RevSignaling Namespace
The RevSignaling Namespace includes properties and methods to extend the standard procedure plan implementation through custom logic. Using this extension support, you can tailor implementations to your unique requirements.

SEE ALSO:

Apex Developer Guide: Getting Started with Apex

RevSignaling Namespace

The RevSignaling Namespace includes properties and methods to extend the standard procedure plan implementation through custom logic. Using this extension support, you can tailor implementations to your unique requirements.

Usage

To use this namespace, enable the **Procedure Plan Orchestration for Pricing** toggle from the Revenue Settings page from Setup.

The `RevSignaling` namespace includes these classes.

ProcedurePlan Class
Represents the instance of the current pricing procedure plan that you're working on.

SignalingApexProcessor Interface
Defines the context-driven orchestration logic based on procedure plan instance and contextual instance.

TransactionRequest Class

Represents the transaction request to the signaling Apex processor.

TransactionResponse Class

Represents the transaction response from the signaling Apex processor.

TransactionStatus Enum

Specifies the status of the transaction request.

ProcedurePlan Class

Represents the instance of the current pricing procedure plan that you're working on.

Namespace

RevSignaling on page 843

ProcedurePlan Properties

Learn more about the properties that are available with the ProcedurePlan class.

ProcedurePlan Properties

Learn more about the properties that are available with the ProcedurePlan class.

The `ProcedurePlan` class includes these properties.

prevStepOutput

Output of the previous step that's executed and passed to the next step.

`prevStepOutput`

Output of the previous step that's executed and passed to the next step.

Signature

`public Map<String,ANY> prevStepOutput {get; set;}`

```
RevSignaling.ProcedurePlan, prevStepOutput
```

Property Value

Type: Map<String,Object>

SignalingApexProcessor Interface

Defines the context-driven orchestration logic based on procedure plan instance and contextual instance.

Namespace

RevSignaling on page 843

Usage

Here's a sample implementation of the SignalingApexProcessor interface.

```
public class SignalingApexProcessorImpl implements RevSignaling.SignalingApexProcessor {

   public RevSignaling.TransactionResponse execute(RevSignaling.TransactionRequest request)
 {
        System.debug('Executing SampleValidClass...');
        System.debug('Procedure Plan: ' + request.procedurePlanInstance);
        System.debug('Context Instance: ' + request.ctxInstanceId);

        // Add your logic here

        // Return the response
       RevSignaling.TransactionResponse response = new RevSignaling.TransactionResponse();

        response.status = RevSignaling.TransactionStatus.SUCCESS;
        response.message = 'Apex method was successfully executed!';
        return response;
    }
}
```

Refer to Customize Your Pricing Procedures With Apex Hooks for additional samples that cover unique pricing scenarios by implementing this interface.

SignalingApexProcessor Methods

The SignalingApexProcessor method executes the specified transaction request, which returns the corresponding response.

SignalingApexProcessor Example Implementation

Refer to the example implementation of the SignalingApexProcessor interface to define a context-driven orchestration logic.

SignalingApexProcessor Methods

The SignalingApexProcessor method executes the specified transaction request, which returns the corresponding response.

The `SignalingApexProcessor` class includes these methods.

execute(var1)

Executes the parameter that's specified in the instance of a transaction request.

execute(var1)

Executes the parameter that's specified in the instance of a transaction request.

Signature

`public RevSignaling.TransactionResponse execute(RevSignaling.TransactionRequest var1)`

```
RevSignaling.SignalingApexProcessor, execute, [RevSignaling.TransactionRequest],
RevSignaling.TransactionResponse
```

Parameters

var1

Type: RevSignaling.TransactionRequest on page 846

Instance of the TransactionRequest class containing the execution parameter.

Return Value

Type: RevSignaling.TransactionResponse on page 848

Response from the orchestration.

SignalingApexProcessor Example Implementation

Refer to the example implementation of the SignalingApexProcessor interface to define a context-driven orchestration logic.

This is an example implementation of the `RevSignaling.SignalingApexProcessor` interface.

```
public class SignalingApexProcessorImpl implements RevSignaling.SignalingApexProcessor {

    public RevSignaling.TransactionResponse execute(RevSignaling.TransactionRequest request)
 {
        System.debug('Executing SampleValidClass...');
        System.debug('Procedure Plan: ' + request.procedurePlanInstance);
        System.debug('Context Instance: ' + request.ctxInstanceId);

        // Add your logic here

        // Return the response
       RevSignaling.TransactionResponse response = new RevSignaling.TransactionResponse();

        response.status = RevSignaling.TransactionStatus.SUCCESS;
        response.message = 'Apex method was successfully executed!';
        return response;
    }
}
```

TransactionRequest Class

Represents the transaction request to the signaling Apex processor.

Namespace

RevSignaling on page 843

TransactionRequest Constructors

Learn more about the available constructors with the TransactionRequest class.

TransactionRequest Properties

Learn more about the properties that are available with the TransactionRequest class.

TransactionRequest Constructors

Learn more about the available constructors with the TransactionRequest class.

The `TransactionRequest` class includes these constructors.

TransactionRequest(procedurePlanInstance, ctxInstanceId)
Creates an instance of the TransactionRequest class to specify the procedure plan and context instance ID.

TransactionRequest(procedurePlanInstance, ctxInstanceId)

Creates an instance of the TransactionRequest class to specify the procedure plan and context instance ID.

Signature

`public TransactionRequest(RevSignaling.ProcedurePlan procedurePlan, String ctxInstanceId)`

```
RevSignaling.TransactionRequest, newinstance, [RevSignaling.ProcedurePlan, String],
RevSignaling.TransactionRequest
```

Parameters

procedurePlan
Type: RevSignaling.ProcedurePlan on page 844

Instance of the procedure plan.

ctxInstanceId
Type: String

ID of the context.

TransactionRequest Properties

Learn more about the properties that are available with the TransactionRequest class.

The `TransactionRequest` class includes these properties.

ctxInstanceId
Set the context ID.

procedurePlanInstance
Set the instance of the procedure plan.

ctxInstanceId

Set the context ID.

Signature

`public String ctxInstanceId {get; set;}`

```
RevSignaling.TransactionRequest, ctxInstanceId
```

Property Value

Type: String

`procedurePlanInstance`

Set the instance of the procedure plan.

Signature

`public RevSignaling.ProcedurePlan procedurePlanInstance {get; set;}`

```
RevSignaling.TransactionRequest, procedurePlanInstance
```

Property Value

Type: RevSignaling.ProcedurePlan on page 844

TransactionResponse Class

Represents the transaction response from the signaling Apex processor.

Namespace

RevSignaling on page 843

TransactionResponse Properties
Learn more about the properties that are available with the TransactionResponse class.

TransactionResponse Properties

Learn more about the properties that are available with the TransactionResponse class.

The `TransactionResponse` class includes these properties.

message
Get the message from the transaction response.

status
Get the status of the request from the transaction response.

`message`

Get the message from the transaction response.

Signature

`public String message {get; set;}`

```
RevSignaling.TransactionResponse, message
```

Property Value

Type: String

`status`

Get the status of the request from the transaction response.

Signature

`public RevSignaling.TransactionStatus status {get; set;}`

```
RevSignaling.TransactionResponse, status
```

Property Value

Type: RevSignaling.TransactionStatus on page 849

TransactionStatus Enum

Specifies the status of the transaction request.

Enum Values

The `RevSignaling.TransactionStatus` enum includes these values.

Value	Description
`FAILED` `RevSignaling.TransactionStatus, FAILED`	The transaction request has failed.
`SUCCESS` `RevSignaling.TransactionStatus, SUCCESS`	The transaction request was successful.

Salesforce Pricing Standard Invocable Actions

Learn more about the standard invocable actions available with Salesforce Pricing.

Run Salesforce Headless Pricing Action
Invoke the Pricing Connect API by providing the pricing data and details of a context, pricing procedure, and price waterfall.

Run Salesforce Pricing Action

Invoke the Pricing Connect API by providing the context, pricing procedure, and price waterfall details.

SEE ALSO:

Actions Developer Guide: Overview

REST API Developer Guide: Invocable Actions Standard

Run Salesforce Headless Pricing Action

Invoke the Pricing Connect API by providing the pricing data and details of a context, pricing procedure, and price waterfall.

This action is available in API version 60.0 and later.

Special Access Rules

The Run Salesforce Headless Pricing action is available in Enterprise, Unlimited, and Developer Editions where Salesforce Pricing is enabled.

Supported REST HTTP Methods

URI

`/services/data/v67.0/actions/standard/runSalesforceHeadlessPricing`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer token`

Inputs

Input	Details
`contextDefinitionId`	**Type** string **Description** Required. ID of the context definition record that's used to build the pricing data.
`contextMappingId`	**Type** string **Description** Required.

Input	Details
	ID of the context-mapping record that identifies which Salesforce object and mappings to use to build the pricing data.
`discoveryProcedure`	**Type** string **Description** Name of the discovery procedure that's used to execute the discovery process.
`displayContext`	**Type** boolean **Description** Indicates whether to display the context structure for pricing (`true`) or not (`false`).
`effectiveDate`	**Type** string **Description** Date when the pricing rules, as specified in the pricing procedure, are applied. The `effectiveDate` parameter determines which pricing procedure to execute when multiple active versions of pricing procedures are available with different date ranges.
`isHighVolumeLineItems`	**Type** boolean **Description** Indicates whether the pricing API returns pricing details for more than 100 line items (`true`) or not (`false`).
`isSkipWaterfall`	**Type** boolean **Description** Indicates whether to generate the price waterfall data (`true`) or not (`false`).
`persistContext`	**Type** boolean **Description** Indicates whether to store the context (`true`) or not (`false`). If set to `true`, the user must have edit access to all sObject fields used in the context mapping.
`pricingData`	**Type** string

Input	Details
	Description Required. JSON data that's used to build the pricing data. The JSON must be escaped. If you're using this data within a Flow, then the JSON data must not be escaped. Also, make sure that the `businessObjectType` value within the `pricingData` property node is set to the sObject used in the context mappings.
`pricingProcedureId`	**Type** string **Description** Required. ID or API name of the pricing procedure used for calculating the prices. A pricing procedure is represented as an Expression Set Definition in the system. If you're an Experience Cloud user, specify the name of the pricing procedure.
`skipDiscovery`	**Type** boolean **Description** Indicates whether to skip executing the discovery procedure (`true`) or not (`false`).
`taggedData`	**Type** boolean **Description** Indicates whether the associated context node contains a key (`true`) or not (`false`).
`useSessionScopedContext`	**Type** boolean **Description** Indicates whether to store the context in a session (`true`) or a request (`false`).

Outputs

Output	Details
`contextDetails`	**Type** string **Description** Context structure for pricing that's generated when the `displayContext` parameter is set to `true`.

Output	Details
`executionId`	**Type** string **Description** ID of the executed pricing data.
`pricingProcessErrors`	**Type** string **Description** Errors that are generated due to context tags in a pricing process.
`pricingProcessStatus`	**Type** string **Description** Status of the pricing process, which is executed as an API call.
`pricingResult`	**Type** string **Description** Outcome of the executed pricing process that's based on the output tags defined in the associated context definition.

Example

POST

This sample request is for the Run Salesforce Headless Pricing action.

```
{
  "inputs": [
    {
      "contextMappingId": "11jSB000002Bn13YAC",
      "contextDefinitionId": "11OSB0000000WSv2AM",
      "pricingProcedureId": "9QLSB0000001lT74AI",
      "discoveryProcedure": "ES1",
      "displayContext": true,
      "effectiveDate": "2023-11-16T12:20:00.000Z",
      "isHighVolumeLineItems": true,
      "skipDiscovery": true,
      "taggedData": false,
      "pricingData":
[illegible]

      "isSkipWaterfall": false,
      "persistContext": true,
      "useSessionScopedContext": false
    }
```

```
  ]
}
```

This sample response is for the Run Salesforce Headless Pricing action.


```
{
  "actionName": "runSalesforceHeadlessPricing",
  "errors": null,
  "isSuccess": true,
  "outputValues": {
    "contextDetails": {
      "SalesTransaction": [
        {
          "PriceBooks": "01sxx0000005q3VAAQ",
          "Subtotal": 587.2095,
          "PriceAdjustmentSchedule": "84Xxx0000004CMxEAM",
          "EffectiveDate": "2024-02-12T00:00:00.000Z",
          "SalesTransactionItem": [
            {
              "LineItemQuantity": 5,
              "ProductSellingModel": "0jPxx000000009hEAA",
              "DerivedPricingAttribute": false,
              "Product": "01txx0000006iLwAAI",
              "LineItem": "LineItem1",
              "NetUnitPrice": 117.4419,
              "SalesTrxnAdjustmentGroup": [
                {
                  "AdjustmentType": "Percentage",
                  "AdjustmentValue": 10
                }
              ]
            }
          ]
        }
      ]
    },
    "executionId": "1708488641379821",
    "pricingProcessStatus": "Completed",
    "pricingResult": {
      "StartDate": [
        {
          "value": 1700137200000,
          "dataPath": [
            "CTX_ece1667e-7a09-40ab-9718-23bdc179a0a5",
            "CTX_d3b9ffd5-2be7-4366-92d8-c2bcf03b69ed"
          ],
          "errors": [],
          "isSuccess": true
        }
      ],
      "NetUnitPrice": [
        {
          "value": 115,
          "dataPath": [
            "CTX_ece1667e-7a09-40ab-9718-23bdc179a0a5",
```

```
        "CTX_d3b9ffd5-2be7-4366-92d8-c2bcf03b69ed"
      ],
      "errors": [],
      "isSuccess": true
    }
  ],
  "ProductSellingModel": [
    {
      "value": "0jPDE000000042K2AQ",
      "dataPath": [
        "CTX_ece1667e-7a09-40ab-9718-23bdc179a0a5",
        "CTX_d3b9ffd5-2be7-4366-92d8-c2bcf03b69ed"
      ],
      "errors": [],
      "isSuccess": true
    }
  ],
  "Pricebook": [
    {
      "value": "01sDE0000000LoeYAE",
      "dataPath": [
        "CTX_ece1667e-7a09-40ab-9718-23bdc179a0a5"
      ],
      "errors": [],
      "isSuccess": true
    }
  ],
  "NetTotalPrice": [
    {
      "value": 1150,
      "dataPath": [
        "CTX_ece1667e-7a09-40ab-9718-23bdc179a0a5",
        "CTX_d3b9ffd5-2be7-4366-92d8-c2bcf03b69ed"
      ],
      "errors": [],
      "isSuccess": true
    }
  ],
  "Subtotal": [
    {
      "value": 1150,
      "dataPath": [
        "CTX_ece1667e-7a09-40ab-9718-23bdc179a0a5"
      ],
      "errors": [],
      "isSuccess": true
    }
  ],
  "Quantity": [
    {
      "value": 10,
      "dataPath": [
        "CTX_ece1667e-7a09-40ab-9718-23bdc179a0a5",
        "CTX_d3b9ffd5-2be7-4366-92d8-c2bcf03b69ed"
```

```
          ],
          "errors": [],
          "isSuccess": true
        }
      ],
      "Product": [
        {
          "value": "01tDE000000FU99YAG",
          "dataPath": [
            "CTX_ece1667e-7a09-40ab-9718-23bdc179a0a5",
            "CTX_d3b9ffd5-2be7-4366-92d8-c2bcf03b69ed"
          ],
          "errors": [],
          "isSuccess": true
        }
      ],
      "CurrencyIsoCode": [
        {
          "value": "USD",
          "dataPath": [
            "CTX_ece1667e-7a09-40ab-9718-23bdc179a0a5"
          ],
          "errors": [],
          "isSuccess": true
        }
      ]
    }
  },
  "version": 1
}
```

SEE ALSO:

Salesforce Help: Invoke Salesforce Headless Pricing in a Flow

Run Salesforce Pricing Action

Invoke the Pricing Connect API by providing the context, pricing procedure, and price waterfall details.

This action is available in API version 60.0 and later. You can use this action with Flows only. To use this action with an API tool such as Postman, see Run Salesforce Headless Pricing Action.

Special Access Rules

The Run Salesforce Pricing action is available in Developer, Enterprise, and Unlimited Editions where Salesforce Pricing is enabled.

Supported REST HTTP Methods

URI

/services/data/v***67.0***/actions/standard/runSalesforcePricing

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer` ***`token`***

Inputs

Input	Details
`contextInstanceId`	**Type** string **Description** Required. ID of the context data that's used to build the pricing procedure. Get the context instance ID by invoking the Context Service API. See Context Service (POST).
`discoveryProcedure`	**Type** string **Description** Name of the discovery procedure that's used to execute the discovery process of the pricing data.
`effectiveDate`	**Type** string **Description** Date when the pricing rules, as specified in the pricing procedure, are applied. The `effectiveDate` parameter determines which pricing procedure to execute when multiple active versions of pricing procedures are available with different date ranges.
`isDeveloperName`	**Type** boolean **Description** Indicates whether the input value in a procedure must use the API name of the pricing (`true`) or the field name (`false`).
`isSkipWaterfall`	**Type** boolean **Description** Indicates whether the price waterfall data must be generated (`true`) or not (`false`).
`pricingProcedureName`	**Type** string

Input	Details
	Description Required. Name of the pricing procedure record that's used to execute the pricing process.
`skipDiscovery`	**Type** boolean **Description** Indicates whether to skip executing the discovery procedure (`true`) or not (`false`).

Outputs

Output	Details
`executionId`	**Type** string **Description** ID of the executed pricing data.

Example

POST

This sample request is for the Run Salesforce Pricing action.

```
{
  "inputs": [
    {
      "contextInstanceId": "32f2c894-ba5e-41c0-91e4-2ab5826f579b",
      "pricingProcedureName": "PricingAction",
      "isSkipWaterfall": false,
      "skipDiscovery": false,
      "isDeveloperName": true,
      "effectiveDate": "2023-11-16T12:20:00.000Z",
      "discoveryProcedure": "ES1"
    }
  ]
}
```

This sample response is for the Run Salesforce Pricing action.

```
{
  "actionName": "runSalesforcePricing",
  "errors": null,
  "isSuccess": true,
  "outputValues": {
    "executionId": "2QTurzG2NRQ5bgrjvvqyh"
```

```
  },
  "version": 1
}
```

SEE ALSO:

Salesforce Help: Invoke Salesforce Pricing in a Flow

Salesforce Pricing Metadata API Types

Metadata API enables you to access some types and feature settings that you can customize in the user interface.

Flow for Salesforce Pricing

Represents the metadata associated with a flow. With Flow, you can create an application that navigates users through a series of screens to query and update records in the database. You can also execute logic and provide branching capability based on user input to build dynamic applications.

IndustriesPricingSettings

Represents the settings for Salesforce Pricing.

PricingActionParameters

Represents the pricing action that's associated with a context definition and pricing procedure.

PricingRecipe

Represents the data models or sets of objects of a particular cloud that the pricing data store consumes during design time and run time.

ProcedureOutputResolution

Represents the pricing resolution for a pricing element determined by using strategy name and formula.

SEE ALSO:

Metadata API Developer Guide: Understanding Metadata API

Flow for Salesforce Pricing

Represents the metadata associated with a flow. With Flow, you can create an application that navigates users through a series of screens to query and update records in the database. You can also execute logic and provide branching capability based on user input to build dynamic applications.

FlowActionCall

Salesforce Pricing exposes additional actionType values for the FlowActionCall Metadata type.

Field Name	Field Type	Description
`actionType`	InvocableActionType (enumeration of type string)	Required. The action type. Additional valid values only for Salesforce Pricing include:

Field Name	Field Type	Description
		• `runSalesforcePricing`—Invoke the Pricing Connect API by providing the context, pricing procedure, and price waterfall details. • `runSalesforceHeadlessPricing`—Invoke the Pricing Connect API by providing the pricing data and details of a context, pricing procedure, and price waterfall.

IndustriesPricingSettings

Represents the settings for Salesforce Pricing.

Parent Type and Manifest Access

This type extends the Metadata metadata type and inherits its `fullName` field.

In the package manifest, all the settings metadata types for the org are accessed using the "Settings" name. See Settings for more details.

File Suffix and Directory Location

`IndustriesPricingSettings` values are stored in the `IndustriesPricingSettings.settings` file in the `settings` folder. The `.settings` files are different from other named components, because there's only one settings file for each settings component.

Version

IndustriesPricingSettings components are available in API version 60.0 and later.

Special Access Rules

This metadata type is available with Salesforce Pricing.

Fields

Field Name	Description
`enableDebugPriceLogs`	**Field Type** boolean **Description** Indicates whether to use price logs to diagnose and resolve pricing issues (`true`) or not (`false`). The default value is `false`. Available in API version 63.0 and later.
`enableHighAvailability`	**Field Type** boolean

Field Name	Description
	Description Reserved for internal use.
enableHighestPriceCompliance	**Field Type** boolean **Description** Indicates whether to track the maximum price of a product over a period of 30 days (`true`) or not (`false`). The default value is `false`. Available in API version 64.0 and later.
enableLowestPriceCompliance	**Field Type** boolean **Description** Indicates whether to track the minimum price of a product over a period of 30 days (`true`) or not (`false`). The default value is `false`. Available in API version 62.0 and later.
enablePricingProcParallelization	**Field Type** boolean **Description** Indicates whether to run pricing elements in parallel within a pricing procedure to optimize the performance of the pricing execution process (`true`) or not (`false`). The default value is `false`. Available in API version 64.0 and later.
enablePricingWaterfall	**Field Type** boolean **Description** Indicates whether to enable Price Waterfall (`true`) or not (`false`). The default value is `false`. Price Waterfall provides insights that include price breakups and reasons for every step of the pricing process.
enablePricingWaterfallPersistence	**Field Type** boolean **Description** Indicates whether to enable Price Waterfall Persistence (`true`) or not (`false`). The default value is `false`. Price Waterfall Persistence stores the process logs that provide insights into the internal pricing processes.
enableSalesforcePricing	**Field Type** boolean **Description** Indicates whether to enable Salesforce Pricing (`true`) or not (`false`). The default value is `false`.

Declarative Metadata Sample Definition

This example shows a sample IndustriesPricingSettings component.

```
<IndustriesPricingSettings xmlns="http://soap.sforce.com/2006/04/metadata">
     <enableDebugPriceLogs>true</enableDebugPriceLogs>
     <enableHighAvailability>true</enableHighAvailability>
     <enableHighestPriceCompliance>true</enableHighestPriceCompliance>
     <enableLowestPriceCompliance>true</enableLowestPriceCompliance>
     <enablePricingProcParallelization>true</enablePricingProcParallelization>
     <enablePricingWaterfall>true</enablePricingWaterfall>
     <enablePricingWaterfallPersistence>true</enablePricingWaterfallPersistence>
     <enableSalesforcePricing>true</enableSalesforcePricing>
</IndustriesPricingSettings>
```

This example shows a sample `package.xml` that references the previous definition.

```
<?xml version="1.0" encoding="UTF-8"?>
<Package xmlns="http://soap.sforce.com/2006/04/metadata">
    <types>
        <members>IndustriesPricing</members>
        <name>Settings</name>
    </types>
    <version>67.0</version>
</Package>
```

Wildcard Support in the Manifest File

The wildcard character `*` (asterisk) in the `package.xml` manifest file doesn't apply to metadata types for feature settings. The wildcard applies only when retrieving all settings, not for an individual setting. For details, see Settings. For information about using the manifest file, see Deploying and Retrieving Metadata with the Zip File.

PricingActionParameters

Represents the pricing action that's associated with a context definition and pricing procedure.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Parent Type

This type extends the Metadata metadata type and inherits its `fullName` field.

File Suffix and Directory Location

PricingActionParameters components have the suffix `.pricingActionParameters` and are stored in the `pricingActionParameters` folder.

Version

PricingActionParameters components are available in API version 60.0 and later.

Special Access Rules

This metadata type is available with Salesforce Pricing.

Fields

Field Name	Description
`contextDefinition`	**Field Type** string **Description** Required. Context definition record that's associated with the pricing action.
`contextMapping`	**Field Type** string **Description** Required. Context mapping record that's associated with the pricing action.
`developerName`	**Field Type** string **Description** Required. Unique name of the pricing action parameter record. The name must begin with a letter and use only alphanumeric characters and underscores. The name must not include spaces, end with an underscore, or have two consecutive underscores.
`effectiveFrom`	**Field Type** dateTime **Description** Required. Date and time from when the pricing action becomes effective.
`effectiveTo`	**Field Type** dateTime **Description** Date and time till when the pricing action is in effect.
`masterLabel`	**Field Type** string

Field Name	Description
	Description Required. Master label of the pricing action parameter.
objectName	**Field Type** string **Description** Name of the object that's associated with the pricing action. Valid values are: • Case • Contract • Opportunity • Order • Quote • SalesAgreement • WorkOrder
pricingProcedure	**Field Type** string **Description** Pricing procedure record that's associated with this pricing action.

Declarative Metadata Sample Definition

The following is an example of a PricingActionParameters component.

```
<PricingActionParameters xmlns="http://soap.sforce.com/2006/04/metadata">
    <developerName>CMEDefaultActionParameters</developerName>
    <objectName>ORDER</objectName>
    <pricingProcedure>PP</pricingProcedure>
    <effectiveFrom>2024-04-08T07:32:00.000Z</effectiveFrom>
    <effectiveTo>2024-04-11T07:32:00.000Z</effectiveTo>
    <contextDefinition>SalesTransactionContext__stdctx</contextDefinition>
    <contextMapping>SalesAgreementEntitiesMapping</contextMapping>
    <masterLabel>PAP_test</masterLabel>
</PricingActionParameters>
```

The following is an example `package.xml` that references the previous definition.

```
<?xml version="1.0" encoding="UTF-8"?>
<Package xmlns="http://soap.sforce.com/2006/04/metadata">
    <types>
        <members>*</members>
        <name>PricingActionParameters</name>
    </types>
```

```
    <version>67.0</version>
</Package>
```

Wildcard Support in the Manifest File

This metadata type supports the wildcard character `*` (asterisk) in the `package.xml` manifest file. For information about using the manifest file, see Deploying and Retrieving Metadata with the Zip File.

PricingRecipe

Represents the data models or sets of objects of a particular cloud that the pricing data store consumes during design time and run time.

Parent Type

This type extends the Metadata metadata type and inherits its `fullName` field.

File Suffix and Directory Location

PricingRecipe components have the suffix `.pricingRecipe` and are stored in the `pricingRecipe` folder.

Version

PricingRecipe components are available in API version 60.0 and later.

Special Access Rules

This metadata type is available with Salesforce Pricing.

Fields

Field Name	Description
`defaultPricingProcedure`	**Field Type** ExpressionSetDefinition **Description** Expression set definition that's associated with this pricing recipe setting.
`defaultPricingProcedureDeveloperName`	**Field Type** string **Description** For internal use only.
`defaultPricingProcedureId`	**Field Type** string

Field Name	Description
	Description ID of the pricing procedure of the pricing recipe.
`developerName`	**Field Type** string **Description** Required. API name of the pricing recipe.
`isActive`	**Field Type** boolean **Description** Indicates whether the pricing recipe is active (`true`) or not (`false`). The default value is `false`
`isInternal`	**Field Type** boolean **Description** Indicates whether the price recipe record is created internally by the Salesforce platform (`true`) or not (`false`). The default value is `false`
`masterLabel`	**Field Type** string **Description** Required. Name for pricing recipe that's defined when the pricing recipe is created.
`pricingRecipeTableMapping`	**Field Type** PricingRecipeTableMapping[] **Description** Mapping of the pricing components of a lookup table with the chosen pricing recipe.

PricingRecipeTableMapping

Represents the mapping of the lookup table with the chosen pricing recipe.

Field Name	Description
`isInternal`	**Field Type** boolean

Field Name	Description
	Description Indicates whether the price recipe field mapping record is created internally by the Salesforce platform (`true`) or not (`false`). The default value is `false`.
`lookupTable`	**Field Type** DecisionTable DecisionMatrixDefinition **Description** Lookup table that's associated with either a decision matrix or decision table.
`lookupTableDeveloperName`	**Field Type** string **Description** For internal use only.
`pricingComponentType`	**Field Type** string **Description** Pricing component field data that the decision table is built on. Valid values are: • `AttributeDiscount` • `BundleDiscount` • `DerivedPricing` • `ListPrice` • `PriceAdjustmentMatrix` • `PromotionsDiscount` • `VolumeDiscount` • `VolumeTierDiscount` • `DiscountDistributionService`. This value is available in API version 60.0 and later. • `MinimumPrice`. Available in API version 62.0 and later.
`pricingProcedureOutputMapList`	**Field Type** PricingProcedureOutputMap[] **Description** List of the mappings of the outputs of the pricing procedures to the associated lookup tables. Available in API version 60.0 and later.
`pricingRecipe`	**Field Type** string

Field Name	Description
	Description Required. Pricing data store that's associated with this pricing recipe field mapping.

PricingProcedureOutputMap

Represents the mapping of the outputs of the pricing procedures to the associated lookup tables. Each record specifies the output mapping of the associated lookup table based on the pricing component type specified in the PricingRecipeTableMapping object.

Field Name	Description
`fieldName`	**Field Type** string **Description** For internal use only.
`isPricingRecipeActive`	**Field Type** boolean **Description** Indicates whether the associated pricing recipe is active (`true`) or not (`false`). The default value is `false`.
`outputFieldName`	**Field Type** string **Description** Field name that contains the output type that's generated from the pricing element.
`outputFieldNameString`	**Field Type** string **Description** Derived field that references a specific column in a decision table or decision matrix.
`outputType`	**Field Type** string **Description** Output type that's generated from a pricing element. Valid values are: • `AdjustmentType` • `AdjustmentValue` • `CustomOutput` • `HashOutput`

Field Name	Description
	• `UnitPrice`
`pricingElementType`	**Field Type** PricingElementType (enumeration of type string) **Description** Type of pricing element, which is a derived field from `PricingRecipeTableMapping.PricingComponentType`. Valid values are: • `AssetDiscovery` • `AttributeDiscount` • `BundleDiscount` • `DerivedPricing` • `DiscountDistributionService` • `ListPrice` • `MinimumPrice` • `PriceAdjustmentMatrix` • `PriceRevision` • `PromotionsDiscount` • `RuleFetch` • `VolumeDiscount` • `VolumeTierDiscount`

Declarative Metadata Sample Definition

The following is an example of a PricingRecipe component.

```
<PricingRecipe xmlns="http://soap.sforce.com/2006/04/metadata">
    <defaultPricingProcedureId> </defaultPricingProcedureId>
    <developerName>CMEDefaultRecipe</developerName>
    <isActive>false</isActive>
    <isInternal>false</isInternal>
    <masterLabel>CMEDefaultRecipe</masterLabel>
    <pricingRecipeTableMapping>
        <isInternal>false</isInternal>

<lookupTableDeveloperName>Bundle_Based_Adjustment_Decision_Table</lookupTableDeveloperName>

        <pricingComponentType>CUSTOMDISCOUNT</pricingComponentType>
        <fileBasedDecisionTableName>Bundle Based Adjustment
Entries</fileBasedDecisionTableName>
        <pricingProcedureOutputMapList>
            <fieldName>AdjustmentValue</fieldName>
            <isPricingRecipeActive>false</isPricingRecipeActive>
            <outputFieldName>0lPxx000000000f</outputFieldName>
```

```
            <outputFieldNameString>false</outputFieldNameString>
            <outputType>AdjustmentValue</outputType>
      <pricingElementType>BundleDiscount</pricingElementType>
        </pricingProcedureOutputMapList>
        <pricingProcedureOutputMapList>
            <fieldName>AdjustmentType</fieldName>
            <isPricingRecipeActive>false</isPricingRecipeActive>
            <outputFieldName>01Pxx000000000m</outputFieldName>
            <outputFieldNameString>false</outputFieldNameString>
            <outputType>AdjustmentType</outputType>
      <pricingElementType>BundleDiscount</pricingElementType>
        </pricingProcedureOutputMapList>
        <pricingRecipe>CMEDefaultRecipe</pricingRecipe>
    </pricingRecipeTableMapping>
</PricingRecipe>
```

The following is an example `package.xml` that references the previous definition.

```
<?xml version="1.0" encoding="UTF-8"?>
<Package xmlns="http://soap.sforce.com/2006/04/metadata">
    <types>
        <members>*</members>
        <name>PricingRecipe</name>
    </types>
    <version>67.0</version>
</Package>
```

Wildcard Support in the Manifest File

This metadata type supports the wildcard character `*` (asterisk) in the `package.xml` manifest file. For information about using the manifest file, see Deploying and Retrieving Metadata with the Zip File.

ProcedureOutputResolution

Represents the pricing resolution for a pricing element determined by using strategy name and formula.

Parent Type

This type extends the Metadata metadata type and inherits its `fullName` field.

File Suffix and Directory Location

ProcedureOutputResolution components have the suffix `.procedureOutputResolution` and are stored in the `procedureOutputResolution` folder.

Version

ProcedureOutputResolution components are available in API version 63.0 and later.

Special Access Rules

This metadata type is available with Salesforce Pricing.

Fields

Field Name	Description
`developerName`	**Field Type** string **Description** Required. API name of the procedure output resolution.
`formula`	**Field Type** string **Description** Required. Stores the encoded formula as text.
`isActive`	**Field Type** boolean **Description** Required. Indicates whether the strategy is active (`true`) or not (`false`).
`isInternal`	**Field Type** boolean **Description** Reserved for internal use.
`masterLabel`	**Field Type** string **Description** Required. A user-friendly name for the procedure output resolution, which is defined when the ProcedureOutputResolution record is created.
`pricingElement`	**Field Type** string **Description** Required.

Field Name	Description
	Pricing element on which the procedure output resolution is defined.

Declarative Metadata Sample Definition

Here's an example of a ProcedureOutputResolution component.

```
<?xml version="1.0" encoding="UTF-8"?>
<ProcedureOutputResolution xmlns="http://soap.sforce.com/2006/04/metadata">
    <developerName>ProcedureOutputResolution</developerName>
    <isActive>false</isActive>
    <isInternal>false</isInternal>
    <masterLabel>Procedure Output Resolution</masterLabel>
    <pricingElement>ListPrice</pricingElement>
    <formula>MAX(ListPrice)</formula>
</ProcedureOutputResolution>
```

Here's an example `package.xml` that references the previous definition.

```
<?xml version="1.0" encoding="UTF-8"?>
<Package xmlns="http://soap.sforce.com/2006/04/metadata">
    <types>
        <members>*</members>
        <name>ProcedureOutputResolution</name>
    </types>
    <version>67.0</version>
</Package>
```

Wildcard Support in the Manifest File

This metadata type supports the wildcard character `*` (asterisk) in the `package.xml` manifest file. For information about using the manifest file, see Deploying and Retrieving Metadata with the Zip File.

Salesforce Pricing Tooling API Objects

Tooling API exposes metadata used in developer tooling that you can access through REST or SOAP. Tooling API's SOQL capabilities for many metadata types allow you to retrieve smaller pieces of metadata.

PricingActionParameters

Represents a pricing action associated to a context definition and a pricing procedure. This object is available in API version 60.0 and later.

PricingProcedureOutputMap

Represents the mapping of the outputs of the pricing procedures to the associated lookup tables. Each record specifies the output mapping of the associated lookup table based on the pricing component type specified in the Pricing Recipe Table Mapping object. This object is available in API version 60.0 and later.

PricingRecipe

Represents one out of various data models or sets of entities of a particular cloud that'll be consumed by the pricing data store during design and run time. This object is available in API version 60.0 and later.

PricingRecipeTableMapping

Represents the mapping of pricing components of a lookup table with the chosen pricing recipe. This object is available in API version 60.0 and later.

ProcedureOutputResolution

Represents the pricing resolution for an pricing element determined using strategy name and formula. This object is available in API version 63.0 and later.

ProcedurePlanCriterion

Represents a criterion within a procedure plan option record. This object is available in API version 62.0 and later.

ProcedurePlanDefinition

Represents the setup of a unified procedure from a list of multiple procedures that can be sequenced in any order based on business needs. Each procedure plan definition contains sections and subsections where procedures can be configured by using a lookup table or rule-based criteria. This object is available in API version 62.0 and later.

ProcedurePlanDefinitionVersion

Represents the versions for a procedure plan definition. Multiple versions under a procedure plan definition must be active at a time, which can be resolved at run time using the rank field. This object is available in API version 62.0 and later.

ProcedurePlanOption

Represents the selection criteria of how a procedure can be configured for a selected procedure plan section record. This object is available in API version 62.0 and later.

ProcedurePlanSection

Represents various procedure setup sections for a procedure plan definition. Each section enables the setup of a procedure of a type that can be further determined by using a rule-based criteria or it can be set based on a selected lookup table. This object is available in API version 62.0 and later.

ProcedurePlanVariable

Represents the setup for any adhoc user-defined variable that can be linked to a procedure plan definition record. This object is available in API version 62.0 and later.

SEE ALSO:

Tooling API Developer Guide: Introducing Tooling API

PricingActionParameters

Represents a pricing action associated to a context definition and a pricing procedure. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeSObjects()`, `query()`, `retrieve()`, `update()`, `upsert()`

Fields

Field	Details
ContextDefinition	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** The context definition record associated with the pricing action.
ContextMapping	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** The context mapping record that's associated with the pricing action.
DeveloperName	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** The unique name of the pricing action parameter record. This name must begin with a letter and use only alphanumeric characters and underscores. It can't include spaces, end with an underscore, or have two consecutive underscores.
EffectiveFrom	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** The date and time when the pricing action comes into effect.
EffectiveTo	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date and time till when the pricing action is in effect.
Language	**Type** picklist

Field	Details
	Properties Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The language of the pricing action parameter. Possible values are: • `da`—Danish • `de`—German • `en_US`—English • `es`—Spanish • `es_MX`—Spanish (Mexico) • `fi`—Finnish • `fr`—French • `it`—Italian • `ja`—Japanese • `ko`—Korean • `nl_NL`—Dutch • `no`—Norwegian • `pt_BR`—Portuguese (Brazil) • `ru`—Russian • `sv`—Swedish • `th`—Thai • `zh_CN`—Chinese (Simplified) • `zh_TW`—Chinese (Traditional)
`MasterLabel`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** The master label of the pricing action parameter.
`NamespacePrefix`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The namespace prefix that is associated with this object. Each Developer Edition org that creates a managed package has a unique namespace prefix. Limit: 15 characters. You can refer to a component in a managed package by using the namespacePrefix__componentName notation.

Field	Details
	The namespace prefix can have one of the following values. • In Developer Edition organizations, the namespace prefix is set to the namespace prefix of the organization for all objects that support it. There's an exception if an object is in an installed managed package. In that case, the object has the namespace prefix of the installed managed package. This field's value is the namespace prefix of the Developer Edition organization of the package developer. • In organizations that are Developer Edition organizations, NamespacePrefix is only set for objects that are part of an installed managed package. There's no namespace prefix for all other objects.
`ObjectName`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Specifies the object associated to the pricing action. Possible values are: • `Case` • `Contract` • `Opportunity` • `Order` • `Quote` • `SalesAgreement` • `WorkOrder`
`PricingProcedure`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The pricing procedure record associated with this pricing action.

PricingProcedureOutputMap

Represents the mapping of the outputs of the pricing procedures to the associated lookup tables. Each record specifies the output mapping of the associated lookup table based on the pricing component type specified in the Pricing Recipe Table Mapping object. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeSObjects()`, `query()`, `retrieve()`, `update()`, `upsert()`

Fields

Field	Details
`IsPricingRecipeActive`	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Indicates if the pricing recipe is active (true) or not (false). The default value is `false`.
`LookupField`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** Definition of the fields that are used for this lookup.
`OutputFieldNameId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The field name containing the output type generated from the pricing element. This field is a polymorphic relationship field. **Relationship Name** OutputFieldName **Relationship Type** Lookup **Refers To** CalculationMatrixColumn, DecisionTableParameter
`OutputFieldNameString`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** This is a derived field that references a specific column in a decision table or decision matrix.
`OutputType`	**Type** picklist

Field	Details
	Properties Create, Filter, Group, Restricted picklist, Sort, Update **Description** The output type generated from a pricing element. Possible values are: • `AdjustmentType`—Adjustment Type • `AdjustmentValue`—Adjustment Value • `CustomOutput`—Custom Output • `HashOutput`—Hash Output • `PriceIndex`—Price Index • `UnitPrice`—Unit Price • `UpperBound`—Unit Price • `LowerBound`—Unit Price • `TierValue`—Unit Price • `TierType`—Unit Price
`PricingComponentType`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The pricing component field data on which the decision table is built. Possible values are: • `AttributeDiscount`—Attribute Based Discount • `BundleDiscount`—Bundle Based Discount • `DerivedPricing` • `DiscountDistributionService`—Discount Distribution Service • `ListPrice`—List Price • `PriceAdjustmentMatrix` • `PromotionsDiscount` • `VolumeDiscount`—Volume Based Discount • `VolumeTierDiscount`—Tier Discount • `RuleFetch` • `AssetDiscovery`
`PricingRecipeTableMappingId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update

Field	Details
	Description The mapping of pricing components of a lookup table with the chosen pricing recipe. This field is a relationship field. **Relationship Name** PricingRecipeTableMapping **Relationship Type** Lookup **Refers To** PricingRecipeTableMapping

PricingRecipe

Represents one out of various data models or sets of entities of a particular cloud that'll be consumed by the pricing data store during design and run time. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeSObjects()`, `query()`, `retrieve()`, `update()`, `upsert()`

Special Access Rules

Fields

Field	Details
`DefaultPricingProcedureId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The expression set definition or Salesforce flow definition associated with this pricing recipe settings. This field is a relationship field. **Relationship Name** DefaultPricingProcedure **Relationship Type** Lookup **Refers To** ExpressionSetDefinition

Field	Details
DeveloperName	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** The developer name of the pricing recipe.
IsActive	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the pricing recipe is active (true) or not (false). The default value is `false`.
IsInternal	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Indicates if the price recipe record is created internally by the Salesforce platform (true) or not (false). The default value is `false`.
Language	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The languages in which pricing recipe is supported. Possible values are: • `da`—Danish • `de`—German • `en_US`—English • `es`—Spanish • `es_MX`—Spanish (Mexico) • `fi`—Finnish • `fr`—French • `it`—Italian • `ja`—Japanese

Field	Details
	• `ko`—Korean • `nl_NL`—Dutch • `no`—Norwegian • `pt_BR`—Portuguese (Brazil) • `ru`—Russian • `sv`—Swedish • `th`—Thai • `zh_CN`—Chinese (Simplified) • `zh_TW`—Chinese (Traditional)
MasterLabel	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** A user-friendly name for pricing recipe, which is defined when the pricing recipe is created.
NamespacePrefix	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The namespace prefix that is associated with this object. Each Developer Edition org that creates a managed package has a unique namespace prefix. Limit: 15 characters. You can refer to a component in a managed package by using the namespacePrefix__componentName notation. The namespace prefix can have one of the following values. • In Developer Edition organizations, the namespace prefix is set to the namespace prefix of the organization for all objects that support it. There's an exception if an object is in an installed managed package. In that case, the object has the namespace prefix of the installed managed package. This field's value is the namespace prefix of the Developer Edition organization of the package developer. • In organizations that are Developer Edition organizations, NamespacePrefix is only set for objects that are part of an installed managed package. There's no namespace prefix for all other objects.

PricingRecipeTableMapping

Represents the mapping of pricing components of a lookup table with the chosen pricing recipe. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeSObjects()`, `query()`, `retrieve()`, `update()`, `upsert()`

Fields

Field	Details
`FileBasedDecisionTableName`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Name of the file-based decision table.
`IsInternal`	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Indicates if the price recipe field mapping record is created internally by the Salesforce platform (true) or not (false). The default value is `false`.
`LookupTableId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The lookup table associated with the mapped fields. This field is a polymorphic relationship field. **Relationship Name** LookupTable **Relationship Type** Lookup **Refers To** DecisionMatrixDefinition, DecisionTable
`PricingComponentType`	**Type** Pricing Element Type enumerated list **Properties** Create, Filter, Group, Restricted picklist, Sort, Update

Field	Details
	Description The pricing component field data on which the decision table is built. Possible values are: • `AttributeDiscount`—Attribute Based Discount • `BundleDiscount`—Bundle Based Discount • `DerivedPricing` • `ListPrice`—List Price • `PriceAdjustmentMatrix` • `PromotionsDiscount` • `VolumeDiscount`—Volume Based Discount • `VolumeTierDiscount`—Tier Discount • `DiscountDistributionService`. This value is available in API version 60.0 and later. • `MinimumPrice`. This value is available in API version 62.0 and later. • `RuleFetch`. This value is available in API version 64.0 and later. • `AssetDiscovery`. This value is available in API version 64.0 and later.
`PricingRecipeId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The pricing data store associated with this pricing recipe field mappings. This field is a relationship field. **Relationship Name** PricingRecipe **Relationship Type** Lookup **Refers To** PricingRecipe

ProcedureOutputResolution

Represents the pricing resolution for an pricing element determined using strategy name and formula. This object is available in API version 63.0 and later.

Supported Calls

`create()`, `delete()`, `describeSObjects()`, `query()`, `retrieve()`, `update()`, `upsert()`

Fields

Field	Details
BusinessVertical	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The business vertical associated with the procedure output resolution record. Possible values are: • RLM
DeveloperName	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** The name of the procedure output resolution.
Formula	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** The formula function used to determine the minimum or maximum price of a product. The supported operations are MIN and MAX.
IsActive	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the procedure output resolution is active (true) or not (false). Only active procedure output resolutions can be applied to a procedure. The default value is `false`.
IsInternal	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort

Field	Details
	Description Indicates if the procedure output resolution record is created internally by the Salesforce platform (true) or not (false). The default value is `false`.
`Language`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The languages in which pricing recipe is supported. Possible values are: • `da`—Danish • `de`—German • `en_US`—English • `es`—Spanish • `es_MX`—Spanish (Mexico) • `fi`—Finnish • `fr`—French • `it`—Italian • `ja`—Japanese • `ko`—Korean • `nl_NL`—Dutch • `no`—Norwegian • `pt_BR`—Portuguese (Brazil) • `ru`—Russian • `sv`—Swedish • `th`—Thai • `zh_CN`—Chinese (Simplified) • `zh_TW`—Chinese (Traditional)
`MasterLabel`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** A user-friendly name for procedure output resolution, which is defined when the procedure output resolution record is created.

Field	Details
NamespacePrefix	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The namespace prefix that is associated with this object. Each Developer Edition org that creates a managed package has a unique namespace prefix. Limit: 15 characters. You can refer to a component in a managed package by using the namespacePrefix__componentName notation. The namespace prefix can have one of the following values. • In Developer Edition organizations, the namespace prefix is set to the namespace prefix of the organization for all objects that support it. There's an exception if an object is in an installed managed package. In that case, the object has the namespace prefix of the installed managed package. This field's value is the namespace prefix of the Developer Edition organization of the package developer. • In organizations that are Developer Edition organizations, NamespacePrefix is only set for objects that are part of an installed managed package. There's no namespace prefix for all other objects.
PricingElement	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Specifies the pricing element on which the procedure output resolution is defined. Possible values are: • `ListPrice`—List Price • `MinimumPrice`—Price Tracking • `PriceAdjustmentMatrix`—Price Adjustment Matrix

ProcedurePlanCriterion

Represents a criterion within a procedure plan option record. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported SOAP API Calls

`create()`, `delete()`, `describeSObjects()`, `query()`, `retrieve()`, `update()`, `upsert()`

Supported REST API Methods

DELETE, GET, HEAD, PATCH, POST, Query

Fields

Field	Details
ActualValue	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The user-defined value that's compared to the value of the sObject field value.
DataType	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The data type of the field from the selected object.
FieldPath	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** The path of the field used in a procedure in relation to the object that the field belongs to.
ObjectField	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** The Salesforce object field value used to resolve the procedure plan option.
Operator	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The operator used by the procedure plan criterion. Valid values are:

Field	Details
	• `Equals` • `GreaterThan` • `GreaterThanOrEquals` • `In` • `IsNotNull` • `IsNull` • `LessThan` • `LessThanOrEquals` • `NotEquals` • `NotIn`
`ProcedurePlanOptionId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The procedure plan option associated with the procedure plan criterion record. This field is a relationship field. **Relationship Name** ProcedurePlanOption **Relationship Type** Master-detail **Refers To** ProcedurePlanOption (the master object)
`Sequence`	**Type** int **Properties** Create, Filter, Group, Sort **Description** The sequence in which the conditions defined in the procedure plan criteria are processed.

ProcedurePlanDefinition

Represents the setup of a unified procedure from a list of multiple procedures that can be sequenced in any order based on business needs. Each procedure plan definition contains sections and subsections where procedures can be configured by using a lookup table or rule-based criteria. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported SOAP API Calls

`create()`, `delete()`, `describeSObjects()`, `query()`, `retrieve()`, `update()`, `upsert()`

Supported REST API Methods

`DELETE, GET, HEAD, PATCH, POST, Query`

Fields

Field	Details
`Description`	**Type** textarea **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The description of the procedure plan definition.
`DeveloperName`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** The unique name of the procedure plan definition record. This name must begin with a letter and use only alphanumeric characters and underscores. It can't include spaces, end with an underscore, or have two consecutive underscores.
`Language`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The language of the procedure plan definition. Valid values are: • `da`—Danish • `de`—German • `en_US`—English • `es`—Spanish • `es_MX`—Spanish (Mexico) • `fi`—Finnish • `fr`—French • `it`—Italian

Field	Details
	• `ja`—Japanese • `ko`—Korean • `nl_NL`—Dutch • `no`—Norwegian • `pt_BR`—Portuguese (Brazil) • `ru`—Russian • `sv`—Swedish • `th`—Thai • `zh_CN`—Chinese (Simplified) • `zh_TW`—Chinese (Traditional)
`MasterLabel`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** The master label of the procedure plan definition.
`NamespacePrefix`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The namespace prefix that is associated with this object. Each Developer Edition org that creates a managed package has a unique namespace prefix. Limit: 15 characters. You can refer to a component in a managed package by using the namespacePrefix__componentName notation. The namespace prefix can have one of the following values. • In Developer Edition organizations, the namespace prefix is set to the namespace prefix of the organization for all objects that support it. There's an exception if an object is in an installed managed package. In that case, the object has the namespace prefix of the installed managed package. This field's value is the namespace prefix of the Developer Edition organization of the package developer. • In organizations that are Developer Edition organizations, NamespacePrefix is only set for objects that are part of an installed managed package. There's no namespace prefix for all other objects.
`PrimaryObject`	**Type** string **Properties** Create, Filter, Group, idLookup, Nillable, Sort

Field	Details
	Description The object associated to the procedure plan definition. The fields in the object are used as variables in the procedure plan criterion.
ProcessType	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Identify the business processes that need a procedure plan for each SObject and definition. Possible values are: • `Billing` • `DRO` • `DeepClone`—Deep Clone • `Default` • `ProductDiscovery`—Product Discovery • `RLM` The default value is `Default`.

ProcedurePlanDefinitionVersion

Represents the versions for a procedure plan definition. Multiple versions under a procedure plan definition must be active at a time, which can be resolved at run time using the rank field. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported SOAP API Calls

`create()`, `delete()`, `describeSObjects()`, `query()`, `retrieve()`, `update()`, `upsert()`

Supported REST API Methods

`DELETE, GET, HEAD, PATCH, POST, Query`

Fields

Field	Details
`ContextDefinition`	**Type** picklist

Field	Details
	Properties Create, Filter, Group, Restricted picklist, Sort, Update **Description** The context definition associated with the procedure plan definition record. Valid values are: • `11ODU00000007Zx2AI` • `11ODU000000084F2AQ` • `CollectionPlanEvent__stdctx` • `CommerceCartContextDefinition__stdctx` • `SalesTransactionContext__stdctx` • `TestContextService__stdctx` • `TestDynamicAttribute__stdctx` • `TestExtendedDefinition__stdctx`
`DefaultReadContextMapping`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The default read context mapping used to read from the mapped object and populate the context definition.
`DefaultSaveContextMapping`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The save context mapping used to save from the context definition and populate the mapped object.
`DeveloperName`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** The unique name of the procedure plan definition version record. This name must begin with a letter and use only alphanumeric characters and underscores. It can't include spaces, end with an underscore, or have two consecutive underscores.
`EffectiveFrom`	**Type** dateTime

Field	Details
	Properties Create, Filter, Sort, Update **Description** The date and time when the procedure plan definition comes into effect.
EffectiveTo	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date and time when the procedure plan definition is no longer in effect.
InheritedFrom	**Type** string **Properties** Create, Filter, Group, idLookup, Nillable, Sort **Description** The template from which this procedure plan definition is created.
IsActive	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates if this procedure plan definition version is active (`true`) or not (`false`). The default value is `false`.
ProcedurePlanDefinitionId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The procedure plan definition associated with this procedure plan definition version record. This field is a relationship field. **Relationship Name** ProcedurePlanDefinition **Relationship Type** Master-detail **Refers To** ProcedurePlanDefinition (the master object)

Field	Details
`Rank`	**Type** int **Properties** Create, Filter, Group, Sort, Update **Description** The current rank of the procedure plan definition version that's used to determine which procedure plan definition version is executed .

ProcedurePlanOption

Represents the selection criteria of how a procedure can be configured for a selected procedure plan section record. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported SOAP API Calls

`create()`, `delete()`, `describeSObjects()`, `query()`, `retrieve()`, `update()`, `upsert()`

Supported REST API Methods

`DELETE, GET, HEAD, PATCH, POST, Query`

Fields

Field	Details
`ApexClassId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The Apex class associated with the procedure plan option record. This field is a relationship field. **Relationship Name** ApexClass **Refers To** ApexClass
`ApexClassName`	**Type** string

Field	Details
	Properties Filter, Group, Nillable, Sort **Description** The name of the Apex class associated with the procedure plan option record.
CriteriaLogic	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The computation logic for the various conditions applied to an option.
CtxDefinitionOutputFieldId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The context definition field that's associated with the decision table. This field is a relationship field. **Relationship Name** CtxDefinitionOutputField **Refers To** DecisionTableParameter
CtxMappingOutputFieldId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The context mapping field that's associated with the decision table. This field is a relationship field. **Relationship Name** CtxMappingOutputField **Refers To** DecisionTableParameter
DecisionTableId	**Type** reference **Properties** Filter, Group, Nillable, Sort

Field	Details
	Description The decision table associated with the pricing procedure. This field is a relationship field. **Relationship Name** DecisionTable **Refers To** DecisionTable
`ExpressionSetApiName`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The API name of the expression set.
`ExpressionSetDefinitionId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The expression set definition associated with the procedure plan option record. This field is a relationship field. **Relationship Name** ExpressionSetDefinition **Refers To** ExpressionSetDefinition
`ExpressionSetLabel`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The label of the expression set definition.
`ExpressionSetOutputFieldId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The expression set output field that’s associated with the decision table. This field is a relationship field.

Field	Details
	Relationship Name ExpressionSetOutputField **Refers To** DecisionTableParameter
`PrimaryObject`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The procedure plan definition associated with the procedure plan option record.
`Priority`	**Type** int **Properties** Create, Filter, Group, Sort **Description** The order in which the options are executed.
`ProcedurePlanSectionId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The procedure plan section associated with the procedure plan option record. This field is a relationship field. **Relationship Name** ProcedurePlanSection **Relationship Type** Master-detail **Refers To** ProcedurePlanSection (the master object)
`ReadContextMapping`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The read context mapping used to read from the mapped object and populate the context definition.

Field	Details
SaveContextMapping	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The save context mapping used to save from the context definition and populate the mapped object.

ProcedurePlanSection

Represents various procedure setup sections for a procedure plan definition. Each section enables the setup of a procedure of a type that can be further determined by using a rule-based criteria or it can be set based on a selected lookup table. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported SOAP API Calls

`create()`, `delete()`, `describeSObjects()`, `query()`, `retrieve()`, `update()`, `upsert()`

Supported REST API Methods

`DELETE, GET, HEAD, PATCH, POST, Query`

Fields

Field	Details
Description	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The description of the procedure plan section.
IsInherited	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort **Description** Indicates if the section is inherited from a template (`true`) ot not (`false`).

Field	Details
	The default value is `false`.
`Phase`	**Type** string **Properties** Create, Filter, Nillable, Sort, Update **Description** The phase associated with the procedure plan section record.
`ProcedurePlanVersionId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The procedure plan version associated with this procedure plan section record. This field is a relationship field. **Relationship Name** ProcedurePlanVersion **Relationship Type** Master-detail **Refers To** ProcedurePlanDefinitionVersion (the master object)
`ResolutionType`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The type of resolution used to filter the procedure. Valid values are: • `Default` • `RuleBased`
`SectionType`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort **Description** The type of procedure section. Valid values are:

Field	Details
	• `Custom` • `PricingDiscoveryProcedure` • `PricingProcedure` • `ProductDiscoveryProcedure`
`Sequence`	**Type** int **Properties** Create, Filter, Group, Sort, Update **Description** The sequence in which the procedures are processed.
`SubSectionType`	**Type** string **Properties** Create, Filter, Group, Sort **Description** The procedure subsection added to the procedure plan definition.

ProcedurePlanVariable

Represents the setup for any adhoc user-defined variable that can be linked to a procedure plan definition record. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported SOAP API Calls

`create()`, `delete()`, `describeSObjects()`, `query()`, `retrieve()`, `update()`, `upsert()`

Supported REST API Methods

`DELETE, GET, HEAD, PATCH, POST, Query`

Fields

Field	Details
`DataType`	**Type** picklist

Field	Details
	Properties Create, Filter, Group, Restricted picklist, Sort, Update **Description** The data type of the input procedure plan variable.
`DefaultValue`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The default value for the user-defined procedure plan variable.
`DeveloperName`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** The unique name of the procedure plan variable record. This name must begin with a letter and use only alphanumeric characters and underscores. It can't include spaces, end with an underscore, or have two consecutive underscores.
`Label`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** The label of the procedure plan variable.
`ProcedurePlanVersionId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The procedure plan version associated with the procedure plan variable record. This field is a relationship field. **Relationship Name** ProcedurePlanVersion **Relationship Type** Master-detail **Refers To** ProcedurePlanDefinitionVersion (the master object)

CHAPTER 6 Rate Management

In this chapter ...

Quote and price products based on predefined rates for future use of the product or service.

SEE ALSO:

Salesforce Help: Rate Management Permission Set Licenses

EDITIONS

Available in: Lightning Experience

Available in: **Enterprise**, **Unlimited**, and **Developer** Editions of Revenue Management where Rate Management is enabled

Rate Management Standard Objects

The Rate Management data model provides objects and fields to manage rates and discounts for a product's resource consumption.

BindingObjectCustomExt

Represents the external or custom target object that's bound to the entitlements granted with the sellable product. This object is available in API version 64.0 and later.

BindingObjectRateAdjustment

Represents the rate adjustments of the usage resource associated with the binding object that's used to charge over consumption. This object is available in API version 64.0 and later.

BindingObjectRateCardEntry

Represents the rate card entry details of the usage resource associated with the binding object that's used to charge over consumption. This object is available in API version 64.0 and later.

PriceBookRateCard

Represents a junction between price book and rate card objects. This object is available in API version 62.0 and later.

RateAdjustmentByAttribute

Represents the adjustments that determine the rate of a resource based on its rate-impacting attributes. These attributes are linked to the usage product record. Rates are then influenced by conditions specified in the Attribute Based Adjustment Condition object. Finally, the charge rate is determined by using the Attribute Based Adjustment Rule object. This object is available in API version 62.0 and later.

RateAdjustmentByTier

Represents the adjustments for the rate of a resource that’s determined based on the specified tiers. This object stores information about the type of adjustment used, the value of the adjustment type, and any applicable boundaries. This object is available in API version 62.0 and later.

RateCard

Represents the rules used to rate the consumption of a group of resources within a product. Usage of a resource is billed at a specified rate if the user consumes more than their allowance for a time period. This object is available in API version 62.0 and later.

RateCardEntry

Represents a rule that determines the charge rate for using a product's resource. Each entry is linked to one rate card exclusively, and its activation or deactivation can be controlled by assigning effective dates. This object is available in API version 62.0 and later.

RatingFrequencyPolicy

Represents the policy that defines the frequency at which rating is triggered for the ratable summary records. This object is available in API version 62.0 and later.

RatingRequest

Represents the common run-time parameters, such as context definition and rating procedure for a set of records in the rateable summary table. This object is available in API version 62.0 and later.

RatingRequestBatchJob

Represents a junction between the rating request and batch job objects. This object is available in API version 62.0 and later.

SEE ALSO:

Object Reference for the Salesforce Platform: Overview of Salesforce Objects and Fields

SOAP API Developer Guide: Introduction to SOAP API

BindingObjectCustomExt

Represents the external or custom target object that's bound to the entitlements granted with the sellable product. This object is available in API version 64.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

- This object is available in Revenue Management when Rate Management is enabled.
- Users with any Rate Management permission set (Admin, Manager, Designtime, Runtime) can view records. Only Admins can create, edit, and delete records.

Fields

Field	Details
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when this record was last referenced.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed a record related to this record.
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The name of the binding custom object record.
`OwnerId`	**Type** reference

Field	Details
	Properties Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the owner of the product usage grant. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User

BindingObjectRateAdjustment

Represents the rate adjustments of the usage resource associated with the binding object that's used to charge over consumption. This object is available in API version 64.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`

Special Access Rules

- This object is available in Revenue Management when Rate Management is enabled.
- Users with any Rate Management permission set (Admin, Manager, Designtime, Runtime) can view records.

Fields

Field	Details
`AdjustmentType`	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** The type of rate adjustment related to the associated rate card entry. Valid values are: • `Amount` • `Override` • `Percentage`

Field	Details
AdjustmentValue	**Type** double **Properties** Filter, Sort **Description** The value of the rate adjustment based on the adjustment type.
BindingObjectRateCardEntryId	**Type** reference **Properties** Filter, Group, Sort **Description** The binding object rate card entry associated with the binding object rate adjustment. This field is a relationship field. **Relationship Name** BindingObjectRateCardEntry **Relationship Type** Master-detail **Refers To** BindingObjectRateCardEntry (the master object)
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Timestamp for when the current user last referred to this record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Timestamp for when the current user last viewed a record related to this record.
LowerBound	**Type** double **Properties** Filter, Sort

Field	Details
	Description The minimum quantity of the rate adjustment value that can be applied to the binding object rate adjustment record. This must be a positive number that's less than or equal to the upper bound of the tier.
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The autogenerated identifier for the binding object rate adjustment record.
`UpperBound`	**Type** double **Properties** Filter, Nillable, Sort **Description** The maximum quantity of the rate adjustment value that can be applied to the binding object rate adjustment record. This must be a positive number. Set this value one digit higher than the quantity you want the tier to include.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

BindingObjectRateAdjustmentFeed
Feed tracking is available for the object.

BindingObjectRateAdjustmentHistory
History is available for tracked fields of the object.

BindingObjectRateAdjustmentShare
Sharing is available for the object.

BindingObjectRateCardEntry

Represents the rate card entry details of the usage resource associated with the binding object that's used to charge over consumption. This object is available in API version 64.0 and later.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`

Special Access Rules

- This object is available in Revenue Management when Rate Management is enabled.
- Users with any Rate Management permission set (Admin, Manager, Designtime, Runtime) can view records.

Fields

Field	Details
`BindingObjectId`	**Type** reference **Properties** Filter, Group, Sort **Description** The object associated with the entitlements that are granted with the sellable product. This field is a polymorphic relationship field. **Relationship Name** BindingObject **Refers To** Account, BindingObjectCustomExt, Contract
BindingObjectRateOrder	**Type** double **Properties** Filter, Nillable, Sort **Description** The order that determines the applicable binding object rate when multiple rates are defined for an Anchor target within an effective period. Available in API version 65.0 and later.
`BindingObjectUsageResourceKey`	**Type** string **Properties** Filter, Group, idLookup, Nillable, Sort **Description** Specifies the autogenerated composite key combining the BindingObjectId and UsageResourceId fields. The format used is `BindingObjectId_UsageResourceId` and must be within 18 characters. The key is generated on saving the BindingObjectRateCardEntry record. For example,`001xx000003Gc4FAAS_1BRxx0000004C95GAE`. Available in API version 66.0 and later.
`EffectiveFrom`	**Type** dateTime

Field	Details
	Properties Filter, Sort **Description** The date and time when the associated rate card entry comes into effect.
`EffectiveTo`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date and time until which the associated rate card entry remains effective.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Timestamp for when the current user last referred to this record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Timestamp for when the current user last viewed a record related to this record.
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The autogenerated identifier for the binding object rate card entry record. For example, `BORCE-00004` or `BORCE-4567`.
`NegotiatedRate`	**Type** double **Properties** Filter, Nillable, Sort **Description** The rate negotiated for the associated binding object.

Field	Details
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the user who created the record. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
RateCardEntryId	**Type** reference **Properties** Filter, Group, Sort **Description** The rate card entry associated with the binding object. This field is a relationship field. **Relationship Name** RateCardEntry **Refers To** RateCardEntry
RateCardId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The rate card associated to the binding object. This field is a relationship field. **Relationship Name** RateCard **Refers To** RateCard
RateCardType	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort

Field	Details
	Description The type of the rate card associated with the binding object rate card entry. Valid values are: • `Attribute` • `Base` • `Tier`
`RateUnitOfMeasureId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The standard unit of measure associated with the rate defined for the binding object. This field is a relationship field. **Relationship Name** RateUnitOfMeasure **Refers To** UnitOfMeasure
`SourceAssetId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The asset that's used to create the binding object rate card entry. This field is a relationship field. **Relationship Name** SourceAsset **Refers To** Asset
`UsageResourceId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The usage resource associated with the binding object rate card entry. This field is a relationship field. **Relationship Name** UsageResource

Field	Details
	Refers To UsageResource

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

BindingObjectRateAdjustmentFeed
Feed tracking is available for the object.

BindingObjectRateAdjustmentHistory
History is available for tracked fields of the object.

BindingObjectRateAdjustmentShare
Sharing is available for the object.

PriceBookRateCard

Represents a junction between price book and rate card objects. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Timestamp for when the current user last referred to this record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort

Field	Details
	Description Timestamp for when the current user last viewed a record related to this record.
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Auto-generated identifier for the price book rate card record.
PriceBookId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** Price book ID that's associated with the rate cards IDs. For Quote, Order, and Contracts, the price book IDs identify the associated rate cards. This field is a relationship field. **Relationship Name** PriceBook **Relationship Type** Master-detail **Refers To** Pricebook2 (the master object)
RateCardId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** Rate card ID that's associated with the price book. This field is a relationship field. **Relationship Name** RateCard **Relationship Type** Master-detail **Refers To** RateCard (the detail object)
RateCardType	**Type** picklist

Field	Details
	Properties Filter, Group, Restricted picklist, Sort **Description** Type of rate card associated with the price book. Valid values are: • `Attribute` • `Base` • `Tier`

Associated Objects

This object has these associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

PriceBookRateCardFeed
Feed tracking is available for the object.

PriceBookRateCardHistory
History is available for tracked fields of the object.

RateAdjustmentByAttribute

Represents the adjustments that determine the rate of a resource based on its rate-impacting attributes. These attributes are linked to the usage product record. Rates are then influenced by conditions specified in the Attribute Based Adjustment Condition object. Finally, the charge rate is determined by using the Attribute Based Adjustment Rule object. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`AdjustmentType`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Type of rate adjustment.

Field	Details
	Valid values are: • `Amount` • `Override` • `Percentage`
`AdjustmentValue`	**Type** double **Properties** Create, Filter, Sort, Update **Description** Value of the rate adjustment based on the selected adjustment type.
`AttributeBasedAdjRuleId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** ID of the attribute based adjustment rule associated with this rate adjustment by attribute record. This field is a relationship field. **Relationship Name** AttributeBasedAdjRule **Refers To** AttributeBasedAdjRule
`EffectiveFrom`	**Type** dateTime **Properties** Filter, Sort **Description** Date and time when the associated rate card entry comes into effect.
`EffectiveTo`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Date and time until when the associated rate card entry remains effective.
`LastReferencedDate`	**Type** dateTime

Field	Details
	Properties Filter, Nillable, Sort **Description** Timestamp for when the current user last referred to this record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Timestamp for when the current user last viewed a record related to this record.
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Auto-generated identifier for the rate adjustment by attribute record.
ProductId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** ID of the product whose resource is being used as the associated rate card entry. This field is a relationship field. **Relationship Name** Product **Refers To** Product2
ProductSellingModelId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** ID of the product selling model for the associated rate card entry. This field is a relationship field. **Relationship Name** ProductSellingModel

Field	Details
	Refers To ProductSellingModel
RateCardEntryId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** ID of the rate card entry associated with this rate adjustment by attribute record. This field is a relationship field. **Relationship Name** RateCardEntry **Relationship Type** Master-detail **Refers To** RateCardEntry (the master object)
RateCardEntryStatus	**Type** picklist **Properties** Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort **Description** Status of the rate card entry associated with this rate adjustment by attribute. Valid values are: • `Active` • `Draft` • `Inactive` The default value is `Draft`. Available in API version 63.0 and later.
RateCardId	**Type** reference **Properties** Filter, Group, Sort **Description** ID of the rate card of the associated rate card entry. This field is a relationship field. **Relationship Name** RateCard **Refers To** RateCard

Field	Details
RateUnitOfMeasureId	**Type** reference **Properties** Filter, Group, Sort **Description** ID of the standard unit of measure record of the associated rate card entry. This field is a relationship field. **Relationship Name** RateUnitOfMeasure **Refers To** UnitOfMeasure
RateUnitOfMeasureName	**Type** string **Properties** Filter, Group, Sort **Description** Name of the standard unit of measure record of the associated rate card entry.
UsageResourceId	**Type** reference **Properties** Filter, Group, Sort **Description** ID of the resource selected for the associated rate card entry. This field is a relationship field. **Relationship Name** UsageResource **Refers To** UsageResource

Associated Objects

This object has these associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

RateAdjustmentByAttributeFeed
Feed tracking is available for the object.

RateAdjustmentByAttributeHistory
History is available for tracked fields of the object.

RateAdjustmentByTier

Represents the adjustments for the rate of a resource that's determined based on the specified tiers. This object stores information about the type of adjustment used, the value of the adjustment type, and any applicable boundaries. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`AdjustmentType`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Type of rate adjustment. Valid values are: • `Amount` • `Override` • `Percentage`
`AdjustmentValue`	**Type** double **Properties** Create, Filter, Sort, Update **Description** Value of the rate adjustment based on the selected adjustment type.
`EffectiveFrom`	**Type** dateTime **Properties** Filter, Sort **Description** Date and time when the associated rate card entry comes into effect.
`EffectiveTo`	**Type** dateTime

Field	Details
	Properties Filter, Nillable, Sort **Description** Date and time until when the associated rate card entry remains effective.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Timestamp for when the current user last referred to this record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Timestamp for when the current user last viewed a record related to this record.
LowerBound	**Type** double **Properties** Create, Filter, Sort, Update **Description** Minimum quantity of the rate adjustment value that can be applied to the record. This value must be a positive integer and must be less than the upper bound of the tier.
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Auto-generated identifier for the rate adjustment by tier record.
ProductId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** ID of the product whose resource is being used as the associated rate card entry. This field is a relationship field.

Field	Details
	Relationship Name Product **Refers To** Product2
`ProductSellingModelId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** ID of the product selling model for the associated rate card entry record. This field is a relationship field. **Relationship Name** ProductSellingModel **Refers To** ProductSellingModel
`RateCardEntryId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** ID of the rate card entry associated with this rate adjustment by tier record. This field is a relationship field. **Relationship Name** RateCardEntry **Relationship Type** Master-detail **Refers To** RateCardEntry (the master object)
`RateCardEntryStatus`	**Type** picklist **Properties** Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort **Description** Status of the rate card entry associated with this rate adjustment by tier. Valid values are: • `Active` • `Draft`

Field	Details
	• `Inactive` The default value is `Draft`. Available in API version 63.0 and later.
`RateCardId`	**Type** reference **Properties** Filter, Group, Sort **Description** ID of the rate card of the associated rate card entry. This field is a relationship field. **Relationship Name** RateCard **Refers To** RateCard
`RateUnitOfMeasureId`	**Type** reference **Properties** Filter, Group, Sort **Description** ID of the standard unit of measure record of the associated rate card entry. This field is a relationship field. **Relationship Name** RateUnitOfMeasure **Refers To** UnitOfMeasure
`RateUnitOfMeasureName`	**Type** string **Properties** Filter, Group, Sort **Description** Name of the standard unit of measure record of the associated rate card entry.
`UpperBound`	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** Maximum quantity of the rate adjustment value that can be applied to the record. This value must be a positive double and must be greater than the lower bound of the tier.

Field	Details
`UsageResourceId`	**Type** reference **Properties** Filter, Group, Sort **Description** ID of the resource selected for the associated rate card entry. This field is a relationship field. **Relationship Name** UsageResource **Refers To** UsageResource

Associated Objects

This object has these associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

RateAdjustmentByTierFeed
Feed tracking is available for the object.

RateAdjustmentByTierHistory
History is available for tracked fields of the object.

RateCard

Represents the rules used to rate the consumption of a group of resources within a product. Usage of a resource is billed at a specified rate if the user consumes more than their allowance for a time period. This object is available in API version 62.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`Description`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Description about the rate card.

Field	Details
EffectiveFrom	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** Date and time when the rate card becomes effective.
EffectiveTo	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** Date and time until when the rate card remains effective.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Timestamp for when the current user last referred to this record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Timestamp for when the current user last viewed a record related to this record.
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** Name of the rate card.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** ID of the user who created the record.

Field	Details
	This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
`Type`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Type of rate card. Valid values are: • `Attribute` • `Base` • `Tier`

Associated Objects

This object has these associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

RateCardFeed
Feed tracking is available for the object.

RateCardHistory
History is available for tracked fields of the object.

RateCardShare
Sharing is available for the object.

RateCardEntry

Represents a rule that determines the charge rate for using a product's resource. Each entry is linked to one rate card exclusively, and its activation or deactivation can be controlled by assigning effective dates. This object is available in API version 62.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
DefaultUnitOfMeasureClassId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** ID of the default unit of measure classification record associated with this rate card entry. This field is a relationship field. **Relationship Name** DefaultUnitOfMeasureClass **Refers To** UnitOfMeasureClass
DefaultUnitOfMeasureId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** ID of the default unit of measure record associated with this rate card entry. This field is a relationship field. **Relationship Name** DefaultUnitOfMeasure **Refers To** UnitOfMeasure
EffectiveFrom	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** Date and time when the rate card entry comes into effect.
EffectiveTo	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** Date and time until when the rate card entry remains effective.

Field	Details
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Timestamp for when the current user last referred to this record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Timestamp for when the current user last viewed a record related to this record.
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Auto-generated identifier for the rate card entry record.
ProductId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** ID of the product whose resource is being used as a rate card entry. This field is a relationship field. **Relationship Name** Product **Refers To** Product2
ProductSellingModelId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** ID of the product selling model associated with this rate card entry. This field is a relationship field.

Field	Details
	Relationship Name ProductSellingModel **Refers To** ProductSellingModel
`Rate`	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** Value of the rate card entry.
`RateCardId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** ID of the rate card associated with this rate card entry. This field is a relationship field. **Relationship Name** RateCard **Relationship Type** Master-detail **Refers To** RateCard (the master object)
`RateCardType`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** Type of rate card associated with this rate card entry. Valid values are: • `Attribute` • `Base` • `Tier` Available in API version 63.0 and later.
`RateNegotiation`	**Type** picklist

Field	Details
	Properties Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** Type of rate negotiation applicable to the rate card entry. Valid values are: • `Negotiable` • `NonNegotiable` The default value is `Negotiable`. Available in API version 63.0 and later.
`RateUnitOfMeasureClassId`	**Type** reference **Properties** Filter, Group, Sort **Description** ID of the unit of measure classification record associated with this rate card entry. This field is a relationship field. **Relationship Name** RateUnitOfMeasureClass **Refers To** UnitOfMeasureClass
`RateUnitOfMeasureId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** ID of the standard unit of measure record associated with this rate card entry. This field is a relationship field. **Relationship Name** RateUnitOfMeasure **Refers To** UnitOfMeasure
`RateUnitOfMeasureName`	**Type** string **Properties** Filter, Group, Sort **Description** Name of the standard unit of measure record of the associated rate card entry.

Field	Details
Status	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** Status of the rate card entry. • Active • Draft • Inactive The default value is Draft. Available in API version 63.0 and later.
UsageProductId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** ID of the product associated with the resource for which the rate is specified. This field is a relationship field. **Relationship Name** UsageProduct **Refers To** Product2
UsageResourceId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** ID of the resource associated with this rate card entry. This field is a relationship field. **Relationship Name** UsageResource **Refers To** UsageResource

Associated Objects

This object has these associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

RateCardEntryFeed
Feed tracking is available for the object.

RateCardEntryHistory
History is available for tracked fields of the object.

RatingFrequencyPolicy

Represents the policy that defines the frequency at which rating is triggered for the ratable summary records. This object is available in API version 62.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Timestamp for when the current user last referred to this record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Timestamp for when the current user last viewed a record related to this record.
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** Name of the rating frequency policy. This is a mandatory field.
`OwnerId`	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update

Field	Details
	Description ID of the user who created the record. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
ProductId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** ID of the product for which the rating policy is defined. This field is a relationship field. This field is deprecated and will be retired in a future version. **Relationship Name** Product **Refers To** Product2
RatingDelayDuration	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Duration of delay—in hours—post the billing period after which the rating is to be triggered.
RatingDelayDurationUnit	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies the unit for the specified rating delay duration. Available in API version 65.0 and later. Valid values are: • Days • Hours
RatingPeriod	**Type** picklist

Field	Details
	Properties Create, Filter, Group, Restricted picklist, Sort, Update **Description** Period for which the usage of a product and usage resource combination is to be rated. Valid values are: • `Daily` • `Monthly`
`UsageResourceId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** ID of the usage resource for which the rating policy is defined. This field is a relationship field. This field is deprecated and will be retired in a future version. **Relationship Name** UsageResource **Refers To** UsageResource

Associated Objects

This object has these associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

RatingFrequencyPolicyFeed
Feed tracking is available for the object.

RatingFrequencyPolicyHistory
History is available for tracked fields of the object.

RatingFrequencyPolicyOwnerSharingRule
Sharing rules are available for the object.

RatingFrequencyPolicyShare
Sharing is available for the object.

RatingRequest

Represents the common run-time parameters, such as context definition and rating procedure for a set of records in the rateable summary table. This object is available in API version 62.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`ContextDefinition`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** Context definition that's used for context instance creation, which encapsulates all aggregated records that are stamped for the rating request.
`ContextMapping`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Context mapping that's used for context instance creation, which encapsulates all aggregated records that are stamped for the rating request. If no ID is provided, default context mapping is used.
`DoesExcludeWaterfall`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the waterfall isn't generated for the rating request (`true`) or is generated (`false`). The default value is `false`. Available in API version 64.0 and later.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Timestamp for when the current user last referred to this record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort

Field	Details
	Description Timestamp for when the current user last viewed a record related to this record.
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Auto-generated identifier for the rating request record.
`OwnerId`	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** ID of the user who created the record. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
`RatingProcedureName`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** Procedure name that's used to rate the aggregated records that are stamped for rating request.
`Status`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Status of the rating request. Valid values are: • `Failed` • `Pending` • `RatingComplete` • `RatingInProgress`

Field	Details
	• `ReadyForRating`

Associated Objects

This object has these associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

RatingRequestFeed
Feed tracking is available for the object.

RatingRequestHistory
History is available for tracked fields of the object.

RatingRequestShare
Sharing is available for the object.

RatingRequestBatchJob

Represents a junction between the rating request and batch job objects. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`BatchJobId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** ID of the batch job that triggered the rating request on the aggregated records. This field is a relationship field. **Relationship Name** BatchJob **Refers To** BatchJob

Field	Details
ErrorCode	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Error code that defines the batch job failure. Valid values are: • BadRequest • InternalError
ErrorMessage	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Error message that describes the cause of the batch job failure.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Timestamp for when the current user last referred to this record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Timestamp for when the current user last viewed a record related to this record.
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Name of the rating request batch job record.
RatingRequestId	**Type** reference

Field	Details
	Properties Create, Filter, Group, Sort **Description** ID of the rating request record associated with the batch job. This field is a relationship field. **Relationship Name** RatingRequest **Relationship Type** Master-detail **Refers To** RatingRequest (the master object)

Associated Objects

This object has these associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

RatingRequestBatchJobFeed
Feed tracking is available for the object.

RatingRequestBatchJobHistory
History is available for tracked fields of the object.

Rate Management Metadata API Types

Metadata API enables you to access some types and feature settings that you can customize in the user interface.

IndustriesRatingSettings
Represents the settings for Rate Management.

Flow for Rate Management
Represents the metadata associated with a flow. With Flow, you can create an application that takes users through a series of pages to query and update the records in the database. You can also run logic and provide branching capability based on user input to build dynamic applications.

IndustriesRatingSettings

Represents the settings for Rate Management.

Parent Type and Manifest Access

This type extends the Metadata metadata type and inherits its `fullName` field.

In the package manifest, all the settings metadata types for the org are accessed using the "Settings" name. See Settings for more details.

File Suffix and Directory Location

The `IndustriesRatingSettings` values are stored in the `IndustriesRating.settings` file in the `settings` folder. The `.settings` files are different from other named components, because there's only one settings file for each settings component.

Version

IndustriesRatingSettings components are available in API version 62.0 and later.

Special Access Rules

This metadata type is available with Rate Management.

Fields

Field Name	Description
`enableRating`	**Field Type** boolean **Description** Indicates whether to enable Rate Management (`true`) or not (`false`). The default value is `false`.
`enableRatingWaterfall`	**Field Type** boolean **Description** Indicates whether to enable Rating Waterfall (`true`) or not (`false`). The default value is `false`. Rating Waterfall provides insights into the rating data, which you can synchronize with your rating lookup tables.
`enableRatingWaterfallPersistence`	**Field Type** boolean **Description** Indicates whether to enable Rating Waterfall Persistence (`true`) or not (`false`). The default value is `false`. Rating Waterfall Persistence stores rating data, which you can use to enhance the internal processes and increase efficiency.

Declarative Metadata Sample Definition

The following is an example of an IndustriesRatingSettings component.

```
<IndustriesRatingSettings xmlns="http://soap.sforce.com/2006/04/metadata">
     <enableRating>true</enableRating>
     <enableRatingWaterfall>true</enableRatingWaterfall>
     <enableRatingWaterfallPersistence>true</enableRatingWaterfallPersistence>
</IndustriesRatingSettings>
```

The following is an example `package.xml` that references the previous definition.

```
<?xml version="1.0" encoding="UTF-8"?>
<Package xmlns="http://soap.sforce.com/2006/04/metadata">
    <types>
        <members>IndustriesRating</members>
        <name>Settings</name>
    </types>
    <version>67.0</version>
</Package>
```

Wildcard Support in the Manifest File

The wildcard character `*` (asterisk) in the `package.xml` manifest file doesn't apply to metadata types for feature settings. The wildcard applies only when retrieving all settings, not for an individual setting. For details, see Settings. For information about using the manifest file, see Deploying and Retrieving Metadata with the Zip File.

Flow for Rate Management

Represents the metadata associated with a flow. With Flow, you can create an application that takes users through a series of pages to query and update the records in the database. You can also run logic and provide branching capability based on user input to build dynamic applications.

FlowActionCall

Rate Management exposes additional actionType values for the FlowActionCall metadata type.

Field Name	Field Type	Description
`actionType`	InvocableActionType (enumeration of type string)	Required. The action type. Additional valid values for Rate Management include: • `invokeRatingService`—Invoke the rating service to rate the usage records.

Rate Management Business APIs

Use the Rate Management Business APIs to get rate plan and persisted rating waterfall details.

Resources
Learn more about the available Rate Management API resources.

Response Bodies
Learn more about the available response bodies of Rate Management APIs.

SEE ALSO:

Connect REST API Developer Guide: Introduction

Resources

Learn more about the available Rate Management API resources.

Rate Plan (GET)
: Get a rate plan for a specified set of context input. Use this API to retrieve rate cards, rate card entries, and related adjustments based on the filter criteria for the context input.

Rating Waterfall (GET)
: Get the persisted rating waterfall that stores the process logs. Rating waterfall provides insights into the internal rating process.

Rate Plan (GET)

Get a rate plan for a specified set of context input. Use this API to retrieve rate cards, rate card entries, and related adjustments based on the filter criteria for the context input.

Keep these considerations in mind when you use this API.

- This API request supports one pricebook and one sellable product.
- The ID of the product is required to invoke this API.
- Invoke this API even if a hydrated context is available to fetch the rate card, rate card entry, and adjustment details.

Special Access Rules

The org must have the Rate Management: Run Time User permission set to use this API. Additionally, the org must also have a default usage rating discovery procedure defined in the Revenue Settings from Setup.

Resource

```
/connect/core-rating/rate-plan
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/core-rating/rate-plan
?contextId=858a3ad3e5a0e5c319652a6ab92f6fdb2b4fa8be72b390506d014596c6da62c9&procedure
ApiName=SampleProcedure
```

Available version

62.0

HTTP methods

GET

Request parameters for GET

Parameter Name	Type	Description	Required or Optional	Available Version
`contextId`	String	ID of the context to specify as an input to the procedure.	Required	62.0
`procedureApiName`	String	API name of the procedure to be executed.	Required	62.0

Response body for GET

[Rate Plan Response]

Rating Waterfall (GET)

Get the persisted rating waterfall that stores the process logs. Rating waterfall provides insights into the internal rating process.

Resource

```
/connect/core-pricing/waterfall/lineItemId/executionId
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/core-pricing/waterfall/Gold/2yHbNNEFOZr9jAe4gH57?tagsToFilter=UnitPrice&usageType=Rating
```

Available version

62.0

HTTP methods

GET

Query parameters

Name	Type	Description	Required or Optional	Available Version
`tagsToFilter`	String	Comma-separated tags to filter.	Optional	62.0
`usageType`	String	Usage type of the waterfall log record. Valid values are: • `Rating` • `Pricing`—Specifies that the record type is `Pricing`. If this value is specified, the API creates a log of pricing waterfall. See Pricing Waterfall. The default value is `Pricing`.	Optional	62.0

Response body for GET

[Line Item Waterfall Response]

Response Bodies

Learn more about the available response bodies of Rate Management APIs.

[Adjustment Details]

Output representation of a rate adjustment request.

[Line Item Waterfall Response]

Output representation of the line item waterfall response.

Rate Plan Response

Output representation of the details of a rate plan.

Rating Error Response

Output representation of the error details related to the API request.

Rating Waterfall Response

Output representation of a rating waterfall request.

Adjustment Details

Output representation of a rate adjustment request.

JSON example

```
 "pricingElement": {
      "adjustments": [{
      "adjustmentType": null,
      "adjustmentValue": null
    }],
      "name": "List Price",
      "elementType": "ListPrice"
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
adjustments	Map<String, Object>[]	Details of the rate element.	Small, 60.0	60.0
description	String	Description of the rate element.	Small, 60.0	60.0
elementType	String	Type of the rate element.	Small, 60.0	60.0
name	String	Name of the rate element.	Small, 60.0	60.0

Line Item Waterfall Response

Output representation of the line item waterfall response.

JSON example

```
{
       "currencyCode": "USD",
       "error": null,
       "executionEndTimestamp": "2023-07-31T20:11:29.625Z",
       "executionId": "gdLVwn2x1uats2xWMAjV",
       "executionStartTimestamp": null,
       "lineItemId": "item1",
       "success": true,
       "usageType":"Rating",
       "output": {
          "quantity": "10",
          "netUnitPrice": "10",
```

```
            "subtotal": "100"
        },
        "waterfall": []
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
contextDefinitionVersionId	String	Context definition version ID of the rating procedure.	Small, 62.0	62.0
contextMappingId	String	Context mapping ID of the record.	Small, 62.0	62.0
currencyCode	String	Currency code. For example, USD or INR.	Small, 62.0	62.0
error	Rating Error Response	Details of any errors.	Small, 62.0	62.0
executionEndTimestamp	String	End timestamp of procedure execution.	Small, 62.0	62.0
executionId	String	Execution ID of a particular execution of a rating procedure.	Small, 62.0	62.0
executionStartTimestamp	String	Start timestamp of procedure execution.	Small, 62.0	62.0
lineItemId	String	Line item ID for which the price is being calculated.	Small, 62.0	62.0
output	Map<String, Object>	Output of the rating procedure.	Small, 62.0	62.0
success	Boolean	Indicates whether the API request is successful (`true`) or not (`false`).	Small, 62.0	62.0
usageType	String	Usage type of the waterfall log record.	Small, 62.0	62.0
waterfall	Rating Waterfall Response[]	Details of the rating waterfall.	Small, 62.0	62.0

Rate Plan Response

Output representation of the details of a rate plan.

JSON example

```
{
  "success": true,
  "executionId" : "a521d592-71c3-4db3-8048-r64504df1605",
  "error": {}
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`error`	Rating Error Response[]	Error response for the API request, if any.	Small, 62.0	62.0
`executionId`	String	ID of the procedure execution record.	Small, 62.0	62.0
`success`	Boolean	Indicates whether the request is successful (`true`) or not (`false`)	Small, 62.0	62.0

Rating Error Response

Output representation of the error details related to the API request.

Property Name	Type	Description	Filter Group and Version	Available Version
`errorCode`	String	Code indicating the type of error.	Small, 62.0	62.0
`message`	String	Message stating the reason for error, if any.	Small, 62.0	62.0

Rating Waterfall Response

Output representation of a rating waterfall request.

JSON example


```
{
  "inputParameters": {
    "productId": "01txx0000006i2SAAQ",
    "pricebookId": "01sxx0000005ptpAAA",
    "pricingModelType": "OneTime"
  },
  "fieldToTagNameMapping": {
    "Product2Id": "ItemProduct",
    "Subtotal": "Subtotal",
    "Pricebook2Id": "Pricebook",
    "Quantity": "ItemQuantity",
    "LineItemId": "SalesTransactionSource",
    "ListPrice": "ItemListPrice"
  },
  "sequence": 0,
  "outputParameters": {
    "listPrice": "10"
  },
  "pricingElement": {
    "adjustments": [
      {
        "adjustmentType": null,
        "adjustmentValue": null
      }
```

```
    ],
    "name": "List Price"
  }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`fieldToTagNameMapping`	Map<String, String>	Mappings of field to tag names.	Small, 62.0	62.0
`inputParameters`	Map<String, Object>	Parameters of rating element input.	Small, 62.0	62.0
`outputParameters`	Map<String, Object>	Parameters of rating element output.	Small, 62.0	62.0
`pricingElement`	Adjustment Details	Details of the rate adjustment of a rating element.	Small, 62.0	62.0
`sequence`	Integer	Sequence of rating element execution.	Small, 62.0	62.0

Rate Management Standard Invocable Actions

Learn more about the standard invocable actions available with Rate Management.

Invoke Rating Service Action
Invoke the rating service to rate the usage records.

SEE ALSO:

Actions Developer Guide: Overview

REST API Developer Guide: Invocable Actions Standard

Metadata API Developer Guide: Understanding Metadata API

Invoke Rating Service Action

Invoke the rating service to rate the usage records.

The Invoke Rating Service action acts as a connector between batch management and the rating service. This action is available in API version 62.0 and later.

Special Access Rules

The Invoke Rating Service action is available in Enterprise, Unlimited, and Developer Editions where Rate Management is enabled.

Supported REST HTTP Methods

URI

`/services/data/v67.0/actions/standard/invokeRatingService`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer token`

Inputs

Input	Details
`attributeRateCardID`	**Type** string **Description** ID of the rate card that's used to define adjustments based on the attributes that impact the rate.
`baseRateCardID`	**Type** string **Description** ID of the rate card that includes the base rate for the resource to be rated, based on its consumption.
`contextDefinitionId`	**Type** string **Description** Required. ID of the context definition that's used to create the context instance.
`contextMappingID`	**Type** string **Description** ID of the context mapping that maps a standard object, context definition object, or any other input data source to the node that's defined in the context definition.
`isSkipWaterfall`	**Type** boolean **Description** Indicates whether to skip the generation of price waterfall data (`true`) or not (`false`).

Input	Details
`procedureName`	**Type** string **Description** Name of the rating procedure that's used to calculate the rates.
`recordID`	**Type** reference **Description** Required. ID of the usage ratable summary record to be rated.
`tierRateCardID`	**Type** string **Description** ID of the rate card that's used to define adjustments for different tiers of a resource.

Outputs

None

Example

POST

This example shows a sample request for the Invoke Rating Service action.

```
{
  "inputs": [
    {
      "recordIDs": "56jSB000002Bn12CXC",
      "contextMappingId": "11jSB000002Bn13YAC",
      "contextDefinitionId": "11OSB0000000WSv2AM",
      "procedureName": "Invoke Rate",
      "isSkipWaterfall": false,
      "baseRateCardID": "11jSB000002Bn13YAC",
      "tierRateCardID": "fgjjSB0sdf987dsf",
      "attributeRateCardID": "asdfgh563034lk"
    }
  ]
}
```

This example shows a sample response for the Invoke Rating Service action.

```
[
    {
        "actionName": "invokeRatingService",
        "errors": null,
```

```
        "isSuccess": true,
        "outputValues": {}
    }
]
```

CHAPTER 7 Product Configurator

In this chapter ...

Customize the components and attributes of a product to align with specific business requirements.

EDITIONS

Available in: Lightning Experience

Available in: **Enterprise**, **Unlimited**, and **Developer** Editions of Revenue Management where Product Configurator is enabled

Product Configurator Standard Objects

The Product Configurator data model provides objects and fields to manage the product configurator flow.

ExpressionSetConstraintObj
Represents the association between a Product object and the constraint model tags defined in a given constraint model. This object is available in API version 63.0 and later.

ProductConfigurationFlow
Specifies the many-to-many relationship between Product Classification, Product, and Flow Definition objects. The flow definition is used to configure standalone and bundled products of a specific product classification along with the product attributes, quantities, and product selling models. This object is available in API version 60.0 and later.

ProductConfigFlowAssignment
A junction object that represents the many-to-many relationship between Product Configuration Flow, Product, and Product Classification. This object is available in API version 60.0 and later.

ProductConfigurationRule
Represents the validation, inclusion, and exclusion rules for products in the context of the selling process. The selling process can be quoting, configuration, or ordering. This object is available in API version 61.0 and later.

SEE ALSO:
Object Reference for the Salesforce Platform: Overview of Salesforce Objects and Fields
SOAP API Developer Guide: Introduction to SOAP API

ExpressionSetConstraintObj

Represents the association between a Product object and the constraint model tags defined in a given constraint model. This object is available in API version 63.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

This object is available in orgs where Revenue Cloud is enabled.

Fields

Field	Details
`ConstraintModelTag`	**Type** string **Properties** Create, Filter, Group, Sort, Update

Field	Details
	Description The product tag that is defined in the constraint model, for example, `Laptop`.
`ConstraintModelTagType`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** The type of the product tag that is defined in the constraint model. Possible values are: • `Port` • `Type` The default value is `Type`.
`ExpressionSetId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The expression set associated with the expression set constraint object. This field is a relationship field. **Relationship Name** ExpressionSet **Refers To** ExpressionSet
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed this record, a record related to this record, or a list view.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed this record or list view. If this value is null, the user accessed this record or list view (LastReferencedDate) but didn't view it.

Field	Details
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The name of the expression set constraint.
`OwnerId`	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** For internal use only. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
`ReferenceObjectId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The object associated with the expression set constraint object. This field is a polymorphic relationship field. **Relationship Name** ReferenceObject **Refers To** Product2, ProductClassification, ProductRelatedComponent

ProductConfigurationFlow

Specifies the many-to-many relationship between Product Classification, Product, and Flow Definition objects. The flow definition is used to configure standalone and bundled products of a specific product classification along with the product attributes, quantities, and product selling models. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`FlowIdentifier`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** Stores the flow API name.
`IsDefault`	**Type** Boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates the default configurator flow. The default value is `false`.
`Status`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Indicates the status of the product configuration flow. Possible values include Draft, Active, and Inactive Possible values are: • `Active` • `Draft` • `Inactive` The default value is `Draft`.

ProductConfigFlowAssignment

A junction object that represents the many-to-many relationship between Product Configuration Flow, Product, and Product Classification. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`AssignmentType`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies whether the Product Configuration Flow is assigned to the primary product or classification, or to dynamic components added within a bundle. Valid values are: • `DynamicAdditionFlow` • `PrimaryConfiguratorFlow` The default value is `PrimaryConfiguratorFlow`. Available in API version 65.0 and later.
`ProductClassificationId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The product classification associated with the Product Configuration Flow. This field is a relationship field. **Relationship Name** ProductClassification **Relationship Type** Lookup **Refers To** ProductClassification
`ProductConfigurationFlowId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The Product Configuration Flow associated with the Product Classification or Product. This field is a relationship field. **Relationship Name** ProductConfigurationFlow **Relationship Type** Lookup

Field	Details
	Refers To ProductConfigurationFlow
`ProductId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The product associated with the Product Configuration Flow. This field is a relationship field. **Relationship Name** Product **Relationship Type** Lookup **Refers To** Product2

ProductConfigurationRule

Represents the validation, inclusion, and exclusion rules for products in the context of the selling process. The selling process can be quoting, configuration, or ordering. This object is available in API version 61.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`ApiName`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description**
`ConfigurationRuleDefinition`	**Type** textarea **Properties** Create, Nillable, Update

Field	Details
	Description The configuration rule criteria and actions.
Description	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The description of the configuration rule.
EffectiveFromDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date and time from which the configuration rules comes into effect.
EffectiveToDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date and time to which the configuration rules ceases to be in effect.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the configuration rule record was last referenced.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the configuration rule record was last viewed.
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update

Field	Details
	Description The name of the configuration rule.
`OwnerId`	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the configuraion rule owner. This field is a polymorphic relationship field. **Relationship Name** Owner **Relationship Type** Lookup **Refers To** Group, User
`ProcessScope`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** The scope of the configuration rule. Possible values are: • `Bundle` • `Product` The default value is `Product`.
`RuleSubType`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** The segregation of products into subsets such that the configuration rules only apply to the products that fall under the ambit of the selected rule subtype. Possible values are: • `BundleProduct` • `BundleProductClassification` • `Product` • `ProductClassification`

Field	Details
	The default value is `Product`.
`RuleType`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** Indicates the industry vertical or the feature of the industry vertical that's using the configuration rule. Possible values are: • `Configurator` • `Promotions` The default value is `Configurator`.
`Sequence`	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** Indicates the order for executing the configuration rule. Rules with lower numbers run first when multiple rules are triggered at once.
`Status`	**Type** picklist **Properties** Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort **Description** The lifecycle status of the configuration rule. Possible values are: • `Active` • `Draft` • `Inactive` The default value is `Draft`.

Product Configurator Business APIs

Use the Product Configurator Business APIs to customize a product or a service according to your business-specific requirements.

Perform product configuration-related operations by using the Product Configurator Business APIs. Integrate these APIs with any front-end application to access the configurator capabilities.

These APIs are stateless and don't recall prior user actions between calls unless state is explicitly persisted and reloaded.

For example, if a user deselects a child product and later reselects it, the engine treats the request as adding a new item that doesn't currently exist in the model. It can't infer that this request is a re-selection of a previously removed item. As a result, the original line item isn't restored and a new line item is created.

This table lists the available Product Configurator resources.

Resource	Description
`/connect/cpq/configurator/actions/configure` (POST)	Retrieve and update a product's configuration from a configurator. Execute configuration rules and notify users of any violations for changes to product bundle, attributes, or product quantity within a bundle. Additionally, get pricing details for the configured bundle.
`/connect/cpq/configurator/actions/load-instance` (POST)	Create a session for the product configuration instance using the transaction ID. Get the session ID that includes the results of actions, such as configuration rules, qualification rules, and pricing management.
`/connect/cpq/configurator/actions/set-instance` (POST)	Set a product configuration instance. This API is used in scenarios where the configuration instance is available in a different database than Salesforce and the product catalog management data is in Salesforce.
`/connect/cpq/configurator/actions/get-instance` (POST)	Fetch the JSON representation of a product configuration. Use the response to display the details of the product configuration instance on the Salesforce user interface, or save the product configuration instance to an external system.
`/connect/cpq/configurator/actions/save-instance` (POST)	Save a configuration instance after a successful product configuration.
`/connect/cpq/configurator/actions/set-product-quantity` (POST)	Set the quantity of a product through the runtime system.

Resource	Description
`/connect/cpq/configurator/actions/add-nodes` (POST)	Add a node to the context through the runtime system without using the Salesforce user interface.
`/connect/cpq/configurator/actions/update-nodes` (POST)	Update nodes in a product configuration.
`/connect/cpq/configurator/actions/delete-nodes` (POST)	Delete nodes from a product configuration.
`/revenue/product-configurator/rules/actions/execute` (POST)	Run rules for a specific quote or order based on a context ID or transaction ID.

Resources

Learn more about the available Product Configurator API resources.

Request Bodies

Learn more about the available Product Configurator API request bodies.

Response Bodies

Learn more about the available Product Configurator API response bodies.

SEE ALSO:

Connect REST API Developer Guide: Introduction

Salesforce Help: Product Configurator Permissions

Resources

Learn more about the available Product Configurator API resources.

Configuration (POST)

Retrieve and update a product's configuration from a configurator. Execute configuration rules and notify users of any violations for changes to product bundle, attributes, or product quantity within a bundle. Additionally, get pricing details for the configured bundle.

Saved Configuration (GET, POST)

Save and reuse a record's configurations, and get a list of the saved configurations for a record.

Saved Configuration (DELETE, PUT)

Update or delete a record's saved configuration by using the configuration ID.

Configuration Get Instance (POST)

Fetch the JSON representation of a product configuration. Use the response to display the details of the product configuration instance on the Salesforce user interface, or save the product configuration instance to an external system.

Configuration Load Instance (POST)

Create a session for the product configuration instance using the transaction ID. Get the session ID that includes the results of actions, such as configuration rules, qualification rules, and pricing management.

Configuration Save Instance (POST)
: Save a configuration instance after a successful product configuration.

Configuration Set Instance (POST)
: Set a product configuration instance. This API is used in scenarios where the configuration instance is available in a different database than Salesforce and the product catalog management data is in Salesforce.

Configurator Add Nodes (POST)
: Add a node to the context through the runtime system without using the Salesforce user interface.

Configurator Delete Nodes (POST)
: Delete nodes from a product configuration.

Configurator Update Nodes (POST)
: Update nodes in a product configuration.

Product Set Quantity (POST)
: Set the quantity of a product through the runtime system.

Config Rules (POST)
: Run rules for a specific quote or order based on a context ID or transaction ID.

Configuration (POST)

Retrieve and update a product's configuration from a configurator. Execute configuration rules and notify users of any violations for changes to product bundle, attributes, or product quantity within a bundle. Additionally, get pricing details for the configured bundle.

Resource

```
/connect/cpq/configurator/actions/configure
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/cpq/configurator/actions/configure
```

Available version

60.0

HTTP methods

POST

Request body for POST

JSON example

This example shows a sample to initiate a context based on a transaction ID.

```
{
    "transactionLineId": "0QLDE000000IBXw4AO",
    "transactionId": "0Q0xx0000000001GAA",
    "correlationId": "c95246d4-102c-4ecd-a263-f74ac525d1e5",
    "configuratorOptions": {
        "executePricing": true,
        "returnProductCatalogData": true
    },
    "qualificationContext": {
        "accountId": "001xx0000000001AAA",
        "contactId": "003xx00000000D7AAI"
```

```
    }
}
```

This example shows a sample to add, update, or delete a node in an existing context.

```
{
    "transactionLineId": "0QLDE000000IBXw4AO",
    "transactionId": "0Q0DE000000ISHJs81",
    "correlationId": "c95246d4-102c-4ecd-a263-f74ac525d1e5",
    "configuratorOptions": {
        "executePricing": true,
        "returnProductCatalogData": true,
        "qualifyAllProductsInTransaction": true,
        "validateProductCatalog": true,
        "validateAmendRenewCancel": true,
        "executeConfigurationRules": true,
        "addDefaultConfiguration": true
    },
    "contextResponseType": "Full",
    "qualificationContext": {
        "accountId": "001xx0000000001AAA",
        "contactId": "003xx00000000D7AAI"
    },
    "transactionContextId": "008d27d7-e004-4906-a949-ee7d7c323c77",
    "addedNodes": [
        {
            "path": ["0Q0DE000000ISHJs81", "sti2_id"],
            "addedObject": {
                "id": "ref_sti2_id",
                "SalesTransactionSource": "sti2_id",
                "PricebookEntry": "01uxx0000000001AAA",
                "ProductSellingModel": "0jPxx0000000001AAA",
                "businessObjectType": "QuoteLineItem",
                "Quantity": 10,
                "UnitPrice": 2.0,
                "Product": "01txx0000000001AAA"
            }
        },
        {
            "path": ["0Q0DE000000ISHJs81", "ref_sti2_id","ref_stir1_id"],
            "addedObject": {
                "id": "ref_stir1_id",
                "businessObjectType": "QuoteLineItemRelationship",
                "MainItem": "0QLDE000000IBXw4AO",
                "AssociatedItem": "ref_sti2_id",
                "ProductRelatedComponent": "0dSxx0000000001AAA",
                "ProductRelationshipType": "0yoxx0000000001AAA",
                "AssociatedItemPricing": "IncludedInBundlePrice"

            }
        }
    ],
    "updatedNodes": [
        {
            "path": ["0Q0DE000000ISHJs81", "0QLDE000000IBXw4AO"],
```

```
            "updatedAttributes": {
                "Quantity": 5
            }
        }
    ],
    "deletedNodes": [
        {
            "path": ["0Q0DE000000ISHJs81", "0QLDE000000IBXw4AO"]
        }
    ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`addedNodes`	Configurator Added Node Input[]	List of added context nodes that's passed to the product configurator.	Optional	60.0
`configuratorOptions`	Configurator Options Input[]	Options to pass to the configurator.	Optional	60.0
`contextResponseType`	String	Specifies the type of transaction context response. Valid values are: • `Delta`—Returns the sales transaction items that are added or updated. • `Full`—Returns all sales transaction items in a transaction. • `None`—Returns empty transaction context response. • `Product`—Returns the sales transaction items related to the product that's being configured.	Required for large sales transactions with more than 1000 line items and less than 15K line items.	65.0
`correlationId`	String	ID that's specified for traceability of logs.	Optional	60.0
`deletedNodes`	Configurator Deleted Node Input[]	List of deleted context nodes that's passed to the product configurator.	Optional	60.0
`qualificationContext`	User Context Input[]	Details such as account ID, contact ID, and context ID that are used for executing qualification rules.	Optional	60.0
`transactionContextId`	String	ID of the transaction context.	Optional	60.0

Name	Type	Description	Required or Optional	Available Version
`transactionId`	String	ID of the sales transaction that's being configured such as a quote or an order.	Required	60.0
`transactionLineId`	String	ID of the top-level line item that's being configured.	Optional	60.0
`updatedNodes`	Configurator Updated Node Input[]	List of updated context nodes that's passed to the product configurator.	Optional	60.0

Response body for POST

Configuration Details

Saved Configuration (GET, POST)

Save and reuse a record's configurations, and get a list of the saved configurations for a record.

Resource

```
/connect/cpq/configurator/saved-configuration
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/cpq/configurator/saved-configuration
```

Available version

63.0

HTTP methods

GET, POST

Request parameters for GET

Parameter Name	Type	Description	Required or Optional	Available Version
`referenceRecordId`	String	ID of the record whose saved configurations must be retrieved.	Required	63.0

Response body for GET

Configuration List

Request body for POST

JSON example

```
{
  "data":
[illegible]
 Built-in
[illegible]
```

```
[redacted]
  "description": "This configuration is saved for reuse.",
  "name": "Favorite Configuration",
  "referenceRecordId": "01txx0000006iCFAAY"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`data`	String	JSON object that contains the details of the sales transaction, formatted as a string.	Optional	63.0
`description`	String	Description of the saved configuration.	Optional	63.0
`name`	String	Name of the saved configuration.	Optional	63.0
`referenceRecordId`	String	ID of the record for which the configuration must be saved.	Required	63.0

Response body for POST

Configuration Record Save

SEE ALSO:

Salesforce Help: Considerations to Save and Reuse Configurations

Saved Configuration (DELETE, PUT)

Update or delete a record's saved configuration by using the configuration ID.

Resource

```
/connect/cpq/configurator/saved-configuration/id
```

The `id` parameter is the ID of the configuration that you want to update or delete.

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/cpq/configurator/saved-configuration/5KPxx0025063GSmSAX
```

Available version

63.0

HTTP methods

DELETE, PUT

Request body for PUT

JSON example

```
{
  "data":
[illegible]
 Built-in
[illegible]

[illegible]

  "description": "This configuration is updated.",
  "name": "Updated Configuration"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`data`	String	JSON object that contains the details of the sales transaction, formatted as a string.	Required	63.0
`description`	String	Description of the configuration.	Required	63.0
`name`	String	Name of the configuration.	Required	63.0

Response body for PUT

Configuration Update

SEE ALSO:

Salesforce Help: Considerations to Save and Reuse Configurations

Configuration Get Instance (POST)

Fetch the JSON representation of a product configuration. Use the response to display the details of the product configuration instance on the Salesforce user interface, or save the product configuration instance to an external system.

Resource

```
/connect/cpq/configurator/actions/get-instance
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/cpq/configurator/actions/get-instance
```

Available version

60.0

HTTP methods

POST

Request body for POST

JSON example

```
{
"contextId": "008d27d7-e004-4906-a949-ee7d7c323c77"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`contextId`	String	Transaction context ID of the product configuration instance that's to be fetched.	Required	60.0

Response body for POST

Configuration Get Instance

Configuration Load Instance (POST)

Create a session for the product configuration instance using the transaction ID. Get the session ID that includes the results of actions, such as configuration rules, qualification rules, and pricing management.

Resource

```
/connect/cpq/configurator/actions/load-instance
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/cpq/configurator/actions/load-instance
```

Available version

60.0

HTTP methods

POST

Request body for POST

JSON example

```
{
    "configuratorOptions":
      {
        "addDefaultConfiguration": true,
        "executeConfigurationRules": true,
        "executePricing": true,
        "qualifyAllProductsInTransaction": true,
        "validateAmendRenewCancel": true,
        "validateProductCatalog": true
    },
    "qualificationContext": {
```

```
        "accountId": "001DU000001nHUGYA2"
    },
    "transactionId": "0Q0DU0000000XoN0AU"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
configurator Options	Configurator Options Input	List of the configurator options to execute.	Optional	60.0
context MappingId	String	ID of the context mapping record.	Optional	60.0
qualification Context	User Context Input	Context details that are used for the qualification rules.	Optional	60.0
transaction Id	String	Transaction ID of the header entity that's used to create a session. For example, a Quote or an Order.	Required	60.0

Response body for POST

Configuration Load Instance

Configuration Save Instance (POST)

Save a configuration instance after a successful product configuration.

Use the Configuration Save Instance API to save the changes to the source after a successful configuration. For example, save changes to the quote line item of a product, which is the source used to load the configuration.

Resource

```
/connect/cpq/configurator/actions/save-instance
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/cpq/configurator/actions/save-instance
```

Available version

60.0

HTTP methods

POST

Request body for POST

JSON example

```
{
"contextId": "008d27d7-e004-4906-a949-ee7d7c323c77"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`contextId`	String	Transaction context ID of the product configuration instance that's to be saved.	Required	60.0

Response body for POST

Configuration Save Instance

Configuration Set Instance (POST)

Set a product configuration instance. This API is used in scenarios where the configuration instance is available in a different database than Salesforce and the product catalog management data is in Salesforce.

Resource

```
/connect/cpq/configurator/actions/set-instance
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/cpq/configurator/actions/set-instance
```

Available version

60.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "configuratorOptions": {
    "addDefaultConfiguration": true,
    "executeConfigurationRules": true,
    "executePricing": false,
    "qualifyAllProductsInTransaction": false,
    "validateAmendRenewCancel": false,
    "validateProductCatalog": false
  },
  "contextMappingId": "11jEk000017YdyUIAS",
  "qualificationContext": {
    "accountId": "001DU000001nHUGYA2"
  },
  "transaction":
[illegible]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`configuratorOptions`	Configurator Options Input	List of the configurator options to execute.	Optional	60.0
`contextMappingId`	String	ID of the context mapping record.	Required	60.0
`qualificationContext`	User Context Input	Context details that are used for the qualification rules.	Optional	60.0
`transaction`	String	Transaction JSON payload representing an object in an external system that's used to create a session.	Required	60.0

Response body for POST

Configuration Set Instance

Configurator Add Nodes (POST)

Add a node to the context through the runtime system without using the Salesforce user interface.

Resource

```
/connect/cpq/configurator/actions/add-nodes
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/cpq/configurator/actions/add-nodes
```

Available version

60.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "configuratorOptions": {
    "executePricing": true,
    "returnProductCatalogData": true,
    "qualifyAllProductsInTransaction": true,
    "validateProductCatalog": true,
    "validateAmendRenewCancel": true,
    "executeConfigurationRules": true,
    "addDefaultConfiguration": true
  },
  "qualificationContext": {
    "accountId": "001xx0000000001AAA",
```

```
    "contactId": "003xx00000000D7AAI"
  },
  "contextId": "008d27d7-e004-4906-a949-ee7d7c323c77",
  "addedNodes": [
    {
      "path": [
        "0Q0xx0000004EvcCAE",
        "ref_d3a3f8d2_e031_4517_ae28_69ce16cb6589"
      ],
      "addedObject": {
        "id": "ref_d3a3f8d2_e031_4517_ae28_69ce16cb6589",
        "SalesTransactionItemSource": "ref_d3a3f8d2_e031_4517_ae28_69ce16cb6589",
        "SalesTransactionItemParent": "0Q0xx0000004EvcCAE",
        "PricebookEntry": "01uxx00000090VuAAI",
        "ProductSellingModel": "0jPxx00000001KHEAY",
        "UnitPrice": 15.26,
        "Quantity": 1,
        "Product": "01txx0000006lfHAAQ",
        "businessObjectType": "QuoteLineItem"
      }
    },
    {
      "path": [
        "0Q0xx0000004EvcCAE",
        "ref_d3a3f8d2_e031_4517_ae28_69ce16cb6589",
        "ref_d85b036d_d305_4bb6_aba8_a1dff645a664"
      ],
      "addedObject": {
        "id": "ref_d85b036d_d305_4bb6_aba8_a1dff645a664",
        "MainItem": "0QLxx0000004QdRGAU",
        "AssociatedItem": "ref_d3a3f8d2_e031_4517_ae28_69ce16cb6589",
        "ProductRelatedComponent": "0dSxx00000001p6EAA",
        "ProductRelationshipType": null,
        "AssociatedItemPricing": "NotIncludedInBundlePrice",
        "AssociatedQuantScaleMethod": "Proportional",
        "businessObjectType": "QuoteLineRelationship"
      }
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`addedNodes`	Configurator Added Node Input[]	List of the nodes to be added.	Required	60.0
`configurator Options`	Configurator Options Input	List of the configuration options to execute.	Optional	60.0
`contextId`	String	ID of the context object that's being considered.	Required	60.0

Name	Type	Description	Required or Optional	Available Version
qualification Context	User Context Input	Context details that are used for the qualification rules.	Optional	60.0

Response body for POST

Configurator Add Nodes

Configurator Delete Nodes (POST)

Delete nodes from a product configuration.

Resource

```
/connect/cpq/configurator/actions/delete-nodes
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/cpq/configurator/actions/delete-nodes
```

Available version

60.0

HTTP methods

POST

Request body for POST

JSON example

```
{
    "configuratorOptions": {
        "executePricing": true,
        "returnProductCatalogData": true,
        "qualifyAllProductsInTransaction": true,
        "validateProductCatalog": true,
        "validateAmendRenewCancel": true,
        "executeConfigurationRules": true,
        "addDefaultConfiguration": true
    },
    "qualificationContext": {
        "accountId": "001xx0000000001AAA",
        "contactId": "003xx00000000D7AAI"
    },
    "contextId": "008d27d7-e004-4906-a949-ee7d7c323c77",
    "deletedNodes": [
        {
            "path": ["0Q0DE000000ISHJs81", "0QLDE000000IBXw4AO"]
        }
    ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
configuratorOptions	Configurator Options Input	List of the configuration options to execute.	Optional	60.0
contextId	String	ID of the context object that's being considered.	Required	60.0
deletedNodes	Configurator Deleted Node Input[]	List of the nodes to be deleted.	Required	60.0
qualificationContext	User Context Input	Context details that are used for the qualification rules.	Optional	60.0

Response body for POST

Configurator Delete Nodes

Configurator Update Nodes (POST)

Update nodes in a product configuration.

Resource

```
/connect/cpq/configurator/actions/update-nodes
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/cpq/configurator/actions/update-nodes
```

Available version

60.0

HTTP methods

POST

Request body for POST

JSON example

```
{
    "configuratorOptions": {
        "executePricing": true,
        "returnProductCatalogData": true,
        "qualifyAllProductsInTransaction": true,
        "validateProductCatalog": true,
        "validateAmendRenewCancel": true,
        "executeConfigurationRules": true,
        "addDefaultConfiguration": true
    },
    "qualificationContext": {
        "accountId": "001xx0000000001AAA",
```

```
        "contactId": "003xx00000000D7AAI"
    },
    "contextId": "008d27d7-e004-4906-a949-ee7d7c323c77",
    "updatedNodes": [
        {
            "path": ["0Q0DE000000ISHJs81", "0QLDE000000IBXw4AO"],
            "updatedAttributes": {
                "Quantity": 5
            }
        }
    ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`configuratorOptions`	Configurator Options Input	List of the configuration options to execute.	Optional	60.0
`contextId`	String	ID of the context object that's being considered.	Required	60.0
`qualificationContext`	User Context Input	Context details that are used for the qualification rules.	Optional	60.0
`updatedNodes`	Configurator Updated Node Input[]	List of the nodes to be updated.	Required	60.0

Response body for POST

Configurator Update Nodes

Product Set Quantity (POST)

Set the quantity of a product through the runtime system.

Resource

```
/connect/cpq/configurator/actions/set-product-quantity
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/cpq/configurator/actions/set-product-quantity
```

Available version

60.0

HTTP methods

POST

Request body for POST

JSON example

```
{
    "configuratorOptions": {
        "executePricing": true,
        "returnProductCatalogData": true,
        "qualifyAllProductsInTransaction": true,
        "validateProductCatalog": true,
        "validateAmendRenewCancel": true,
        "executeConfigurationRules": true,
        "addDefaultConfiguration": true
    },
    "qualificationContext": {
        "accountId": "001xx0000000001AAA",
        "contactId": "003xx00000000D7AAI"
    },
    "contextId": "008d27d7-e004-4906-a949-ee7d7c323c77",
    "quantity": 20,
    "transactionLinePath": "Quote.QuoteLineItem.Quantity"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`configuratorOptions`	Configurator Options Input	List of the configuration options to execute.	Optional	60.0
`contextId`	String	ID of the context object that's being considered.	Required	60.0
`qualificationContext`	User Context Input	Context details that are used for the qualification rules.	Optional	60.0
`quantity`	Integer	Value of the product quantity.	Required	60.0
`transactionLinePath`	String[]	Path to the line item where the update to the quantity is applied. For example, Quote.QuoteLineItem.Quantity.	Required	60.0

Response body for POST

Product Quantity Set Configurator

Config Rules (POST)

Run rules for a specific quote or order based on a context ID or transaction ID.

Resource

```
/revenue/product-configurator/rules/actions/execute
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/revenue/product-configurator/rules/actions/execute
```

Available version

67.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "transactionContextId": "008d27d7-e004-4906-a949-ee7d7c323c77",
  "transactionId": "0Q0DU0000005tJh0AI",
  "ruleOptions": {
    "isUpdateContextRequired": false
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`ruleOptions`	Config Rule Options Input[]	Details of the options to run specific steps in rules.	Optional	67.0
`transactionContextId`	String	ID of the sales transaction context instance.	Required if the `transactionId` property isn't specified.	67.0
`transactionId`	String	ID of the quote or order.	Required if the `transactionContextId` property isn't specified.	67.0

Response body for POST

Configuration Rule Response

Request Bodies

Learn more about the available Product Configurator API request bodies.

Configuration Get Instance Input

Input representation of the request to get a product configuration instance.

Configuration Load Instance Input

Input representation of the request to load a product configuration instance.

Configuration Save Input

Input representation of the details to save a configuration.

Configuration Update Input

Input representation of the details to update a configuration.

Configuration Save Instance Input

Input representation of the request to save a product configuration instance.

Configuration Set Instance Input

Input representation of the request to set a product configuration instance.

Configurator Add Nodes Input

Input representation of the request to add nodes within a root node.

Configurator Added Node Input

Input representation of the nodes to be added to a product configuration.

Configurator Delete Nodes Input

Input representation of the request to delete nodes from a product configuration.

Configurator Deleted Node Input

Input representation of the nodes to be deleted from a product configuration.

Configurator Input

Input representation of the request to modify the product configuration.

Configurator Options Input

Input representation of the request to get the product configuration options that's passed to the configurator.

Configuration Rule Input

Input representation of the details of a configuration rule.

Configurator Update Nodes Input

Input representation of the request to update the nodes in a product configuration.

Configurator Updated Node Input

Input representation of the nodes to be updated in a product configuration.

Product Quantity Set Configurator Input

Input representation of the request to set the quantity of a product.

User Context Input

Input representation of the request to get the context details of a user, which are used for qualification rules.

Configuration Get Instance Input

Input representation of the request to get a product configuration instance.

JSON example

```
{
"contextId": "008d27d7-e004-4906-a949-ee7d7c323c77"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
contextId	String	Transaction context ID of the product configuration instance that's to be fetched.	Required	60.0

Configuration Load Instance Input

Input representation of the request to load a product configuration instance.

JSON example


```
{
    "configuratorOptions":
      {
        "addDefaultConfiguration": true,
        "executeConfigurationRules": true,
        "executePricing": true,
        "qualifyAllProductsInTransaction": true,
        "validateAmendRenewCancel": true,
        "validateProductCatalog": true
    },
    "qualificationContext": {
        "accountId": "001DU000001nHUGYA2"
    },
    "transactionId": "0Q0DU0000000XoN0AU"
}
```


Properties

Name	Type	Description	Required or Optional	Available Version
configurator Options	Configurator Options Input	List of the configurator options to execute.	Optional	60.0
context MappingId	String	ID of the context mapping record.	Optional	60.0
qualification Context	User Context Input	Context details that are used for the qualification rules.	Optional	60.0
transactionId	String	Transaction ID of the header entity that's used to create a session. For example, a Quote or an Order.	Required	60.0

Configuration Save Input

Input representation of the details to save a configuration.

JSON example

```
{
  "data":
[illegible]
 Built-in
[illegible]

[illegible]

  "description": "This configuration is saved for reuse.",
  "name": "Favorite Configuration",
  "referenceRecordId": "01txx0000006iCFAAY"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
data	String	JSON object that contains the details of the sales transaction, formatted as a string.	Optional	63.0
description	String	Description of the saved configuration.	Optional	63.0
name	String	Name of the saved configuration.	Optional	63.0
referenceRecordId	String	ID of the record for which the configuration must be saved.	Required	63.0

Configuration Update Input

Input representation of the details to update a configuration.

JSON example

```
{
  "data":
[illegible]
 Built-in
[illegible]

[illegible]

  "description": "This configuration is updated.",
  "name": "Updated Configuration"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
data	String	JSON object that contains the details of the sales transaction, formatted as a string.	Required	63.0

Name	Type	Description	Required or Optional	Available Version
`description`	String	Description of the configuration.	Required	63.0
`name`	String	Name of the configuration.	Required	63.0

Configuration Save Instance Input

Input representation of the request to save a product configuration instance.

JSON example

```
{
"contextId": "008d27d7-e004-4906-a949-ee7d7c323c77"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`contextId`	String	Transaction context ID of the product configuration instance that's to be saved.	Required	60.0

Configuration Set Instance Input

Input representation of the request to set a product configuration instance.

JSON example

```
{
  "configuratorOptions": {
    "addDefaultConfiguration": true,
    "executeConfigurationRules": true,
    "executePricing": false,
    "qualifyAllProductsInTransaction": false,
    "validateAmendRenewCancel": false,
    "validateProductCatalog": false
  },
  "contextMappingId": "11jEk000017YdyUIAS",
  "qualificationContext": {
    "accountId": "001DU000001nHUGYA2"
  },
  "transaction":
[illegible]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`configuratorOptions`	Configurator Options Input	List of the configurator options to execute.	Optional	60.0
`contextMappingId`	String	ID of the context mapping record.	Required	60.0
`qualificationContext`	User Context Input	Context details that are used for the qualification rules.	Optional	60.0
`transaction`	String	Transaction JSON payload representing an object in an external system that's used to create a session.	Required	60.0

Configurator Add Nodes Input

Input representation of the request to add nodes within a root node.

JSON example


```
{
  "configuratorOptions": {
    "executePricing": true,
    "returnProductCatalogData": true,
    "qualifyAllProductsInTransaction": true,
    "validateProductCatalog": true,
    "validateAmendRenewCancel": true,
    "executeConfigurationRules": true,
    "addDefaultConfiguration": true
  },
  "qualificationContext": {
    "accountId": "001xx0000000001AAA",
    "contactId": "003xx00000000D7AAI"
  },
  "contextId": "008d27d7-e004-4906-a949-ee7d7c323c77",
  "addedNodes": [
    {
      "path": [
        "0Q0xx0000004EvcCAE",
        "ref_d3a3f8d2_e031_4517_ae28_69ce16cb6589"
      ],
      "addedObject": {
        "id": "ref_d3a3f8d2_e031_4517_ae28_69ce16cb6589",
        "SalesTransactionItemSource": "ref_d3a3f8d2_e031_4517_ae28_69ce16cb6589",
        "SalesTransactionItemParent": "0Q0xx0000004EvcCAE",
        "PricebookEntry": "01uxx00000090VuAAI",
        "ProductSellingModel": "0jPxx00000001KHEAY",
        "UnitPrice": 15.26,
        "Quantity": 1,
        "Product": "01txx0000006lfHAAQ",
```

```
          "businessObjectType": "QuoteLineItem"
        }
      },
      {
        "path": [
          "0Q0xx0000004EvcCAE",
          "ref_d3a3f8d2_e031_4517_ae28_69ce16cb6589",
          "ref_d85b036d_d305_4bb6_aba8_a1dff645a664"
        ],
        "addedObject": {
          "id": "ref_d85b036d_d305_4bb6_aba8_a1dff645a664",
          "MainItem": "0QLxx0000004QdRGAU",
          "AssociatedItem": "ref_d3a3f8d2_e031_4517_ae28_69ce16cb6589",
          "ProductRelatedComponent": "0dSxx00000001p6EAA",
          "ProductRelationshipType": null,
          "AssociatedItemPricing": "NotIncludedInBundlePrice",
          "AssociatedQuantScaleMethod": "Proportional",
          "businessObjectType": "QuoteLineRelationship"
        }
      }
    ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
addedNodes	Configurator Added Node Input[]	List of the nodes to be added.	Required	60.0
configurator Options	Configurator Options Input	List of the configuration options to execute.	Optional	60.0
contextId	String	ID of the context object that's being considered.	Required	60.0
qualification Context	User Context Input	Context details that are used for the qualification rules.	Optional	60.0

Configurator Added Node Input

Input representation of the nodes to be added to a product configuration.

JSON example

```
{
  "addedNodes": [
    {
      "path": [
        "0Q0xx0000004EvcCAE",
        "ref_d3a3f8d2_e031_4517_ae28_69ce16cb6589"
      ],
      "addedObject": {
```

```
        "id": "ref_d3a3f8d2_e031_4517_ae28_69ce16cb6589",
        "SalesTransactionItemSource": "ref_d3a3f8d2_e031_4517_ae28_69ce16cb6589",
        "SalesTransactionItemParent": "0Q0xx0000004EvcCAE",
        "PricebookEntry": "01uxx00000090VuAAI",
        "ProductSellingModel": "0jPxx00000001KHEAY",
        "UnitPrice": 15.26,
        "Quantity": 1,
        "Product": "01txx0000006lfHAAQ",
        "businessObjectType": "QuoteLineItem"
      }
    },
    {
      "path": [
        "0Q0xx0000004EvcCAE",
        "ref_d3a3f8d2_e031_4517_ae28_69ce16cb6589",
        "ref_d85b036d_d305_4bb6_aba8_a1dff645a664"
      ],
      "addedObject": {
        "id": "ref_d85b036d_d305_4bb6_aba8_a1dff645a664",
        "MainItem": "0QLxx0000004QdRGAU",
        "AssociatedItem": "ref_d3a3f8d2_e031_4517_ae28_69ce16cb6589",
        "ProductRelatedComponent": "0dSxx00000001p6EAA",
        "ProductRelationshipType": null,
        "AssociatedItemPricing": "NotIncludedInBundlePrice",
        "AssociatedQuantScaleMethod": "Proportional",
        "businessObjectType": "QuoteLineRelationship"
      }
    }
  ]
}
```

This example shows a sample request for orders.

```
{
  "addedNodes": [
    {
      "path": [
        "0Q0xx0000004EvcCAE",
        "ref_d3a3f8d2_e031_4517_ae28_69ce16cb6589"
      ],
      "addedObject": {
        "id": "ref_d3a3f8d2_e031_4517_ae28_69ce16cb6589",
        "SalesTransactionItemSource": "ref_d3a3f8d2_e031_4517_ae28_69ce16cb6589",
        "SalesTransactionItemParent": "0Q0xx0000004EvcCAE",
        "PricebookEntry": "01uxx00000090VuAAI",
        "ProductSellingModel": "0jPxx00000001KHEAY",
        "UnitPrice": 15.26,
        "Quantity": 1,
        "Product": "01txx0000006lfHAAQ",
        "businessObjectType": "OrderItem"
      }
    },
    {
      "path": [
        "0Q0xx0000004EvcCAE",
```

```
          "ref_d3a3f8d2_e031_4517_ae28_69ce16cb6589",
          "ref_d85b036d_d305_4bb6_aba8_a1dff645a664"
        ],
        "addedObject": {
          "id": "ref_d85b036d_d305_4bb6_aba8_a1dff645a664",
          "MainItem": "0QLxx0000004QdRGAU",
          "AssociatedItem": "ref_d3a3f8d2_e031_4517_ae28_69ce16cb6589",
          "ProductRelatedComponent": "0dSxx00000001p6EAA",
          "ProductRelationshipType": null,
          "AssociatedItemPricing": "NotIncludedInBundlePrice",
          "AssociatedQuantScaleMethod": "Proportional",
          "businessObjectType": "OrderItemRelationship"
        }
      }
    ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`addedObject`	Map<String, Object>	Details of the object that's being added. This property supports fields of objects from the Sales Transaction context definition, including custom objects and fields in your extended context definition.	Required	60.0
`path`	String[]	Path to the node that's being added. The path includes the unique ID of the context node in the data structure. This ID must match the ID of the sales transaction item source such as a quote line or an order line item. Keep these considerations in mind when setting the `path` value. • If the `businessObjectType` property value is `QuoteLineItem`, the path must contain 2 IDs. The first ID is the quote ID, and the second ID is the quote line item ID. • If the `businessObjectType` property value is `QuoteLineItem`, the path must contain `SalesTransactionItemSource` and `SalesTransactionItemParent`. • If the `businessObjectType` property value is `QuoteLineItemRelationship`,	Required	60.0

Name	Type	Description	Required or Optional	Available Version
		the path must contain 3 IDs. The first ID is the quote ID. The second ID is the quote line item ID. The third ID is the quote line item relationship ID.		

Configurator Delete Nodes Input

Input representation of the request to delete nodes from a product configuration.

JSON example

```
{
    "configuratorOptions": {
        "executePricing": true,
        "returnProductCatalogData": true,
        "qualifyAllProductsInTransaction": true,
        "validateProductCatalog": true,
        "validateAmendRenewCancel": true,
        "executeConfigurationRules": true,
        "addDefaultConfiguration": true
    },
    "qualificationContext": {
        "accountId": "001xx0000000001AAA",
        "contactId": "003xx00000000D7AAI"
    },
    "contextId": "008d27d7-e004-4906-a949-ee7d7c323c77",
    "deletedNodes": [
        {
            "path": ["0Q0DE000000ISHJs81", "0QLDE000000IBXw4AO"]
        }
    ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`configuratorOptions`	Configurator Options Input	List of the configuration options to execute.	Optional	60.0
`contextId`	String	ID of the context object that's being considered.	Required	60.0
`deletedNodes`	Configurator Deleted Node Input[]	List of the nodes to be deleted.	Required	60.0
`qualificationContext`	User Context Input	Context details that are used for the qualification rules.	Optional	60.0

Configurator Deleted Node Input

Input representation of the nodes to be deleted from a product configuration.

JSON example

```
"deletedNodes": [
    {
        "path": ["0Q0DE000000ISHJs81", "0QLDE000000IBXw4AO"]
    }
]
```

Properties

Name	Type	Description	Required or Optional	Available Version
`path`	String[]	Path to the node that's being deleted.	Required	60.0

Configurator Input

Input representation of the request to modify the product configuration.

JSON example

This example shows a sample to initiate a context based on a transaction ID.

```
{
    "transactionLineId": "0QLDE000000IBXw4AO",
    "transactionId": "0Q0xx0000000001GAA",
    "correlationId": "c95246d4-102c-4ecd-a263-f74ac525d1e5",
    "configuratorOptions": {
        "executePricing": true,
        "returnProductCatalogData": true
    },
    "qualificationContext": {
        "accountId": "001xx0000000001AAA",
        "contactId": "003xx00000000D7AAI"
    }
}
```

This example shows a sample to add, update, or delete a node in an existing context.

```
{
    "transactionLineId": "0QLDE000000IBXw4AO",
    "transactionId": "0Q0DE000000ISHJs81",
    "correlationId": "c95246d4-102c-4ecd-a263-f74ac525d1e5",
    "configuratorOptions": {
        "executePricing": true,
        "returnProductCatalogData": true,
        "qualifyAllProductsInTransaction": true,
        "validateProductCatalog": true,
        "validateAmendRenewCancel": true,
        "executeConfigurationRules": true,
        "addDefaultConfiguration": true
    },
```

```
    "contextResponseType": "Full",
    "qualificationContext": {
        "accountId": "001xx0000000001AAA",
        "contactId": "003xx00000000D7AAI"
    },
    "transactionContextId": "008d27d7-e004-4906-a949-ee7d7c323c77",
    "addedNodes": [
        {
            "path": ["0Q0DE000000ISHJs81", "sti2_id"],
            "addedObject": {
                "id": "ref_sti2_id",
                "SalesTransactionSource": "sti2_id",
                "PricebookEntry": "01uxx0000000001AAA",
                "ProductSellingModel": "0jPxx0000000001AAA",
                "businessObjectType": "QuoteLineItem",
                "Quantity": 10,
                "UnitPrice": 2.0,
                "Product": "01txx0000000001AAA"
            }
        },
        {
            "path": ["0Q0DE000000ISHJs81", "ref_sti2_id","ref_stir1_id"],
            "addedObject": {
                "id": "ref_stir1_id",
                "businessObjectType": "QuoteLineItemRelationship",
                "MainItem": "0QLDE000000IBXw4AO",
                "AssociatedItem": "ref_sti2_id",
                "ProductRelatedComponent": "0dSxx0000000001AAA",
                "ProductRelationshipType": "0yoxx0000000001AAA",
                "AssociatedItemPricing": "IncludedInBundlePrice"

            }
        }
    ],
    "updatedNodes": [
        {
            "path": ["0Q0DE000000ISHJs81", "0QLDE000000IBXw4AO"],
            "updatedAttributes": {
                "Quantity": 5
            }
        }
    ],
    "deletedNodes": [
        {
            "path": ["0Q0DE000000ISHJs81", "0QLDE000000IBXw4AO"]
        }
    ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
addedNodes	Configurator Added Node Input[]	List of added context nodes that's passed to the product configurator.	Optional	60.0
configurator Options	Configurator Options Input[]	Options to pass to the configurator.	Optional	60.0
context ResponseType	String	Specifies the type of transaction context response. Valid values are: • Delta—Returns the sales transaction items that are added or updated. • Full—Returns all sales transaction items in a transaction. • None—Returns empty transaction context response. • Product—Returns the sales transaction items related to the product that's being configured.	Required for large sales transactions with more than 1000 line items and less than 15K line items.	65.0
correlationId	String	ID that's specified for traceability of logs.	Optional	60.0
deletedNodes	Configurator Deleted Node Input[]	List of deleted context nodes that's passed to the product configurator.	Optional	60.0
qualification Context	User Context Input[]	Details such as account ID, contact ID, and context ID that are used for executing qualification rules.	Optional	60.0
transaction ContextId	String	ID of the transaction context.	Optional	60.0
transactionId	String	ID of the sales transaction that's being configured such as a quote or an order.	Required	60.0
transaction LineId	String	ID of the top-level line item that's being configured.	Optional	60.0
updatedNodes	Configurator Updated Node Input[]	List of updated context nodes that's passed to the product configurator.	Optional	60.0

SEE ALSO:

Salesforce Help: Context Service

Developer Documentation: Context Service

Configurator Options Input

Input representation of the request to get the product configuration options that's passed to the configurator.

JSON example

```
"configuratorOptions":
  {
    "addDefaultConfiguration": true,
    "executeConfigurationRules": true,
    "executePricing": true,
    "qualifyAllProductsInTransaction": true,
    "validateAmendRenewCancel": true,
    "validateProductCatalog": true
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`addDefault Configuration`	Boolean	Indicates whether to add the default configurations (`true`) or not (`false`).	Optional	60.0
`execute Configuration Rules`	Boolean	Indicates whether to execute the configuration rules (`true`) or not (`false`).	Optional	60.0
`execute Pricing`	Boolean	Indicates whether to execute pricing (`true`) or not (`false`).	Optional	60.0
`explainability Enabled`	Boolean	Indicates whether additional metadata about how the solver achieved the execution request solution must be collected and made available to the caller (`true`) or not (`false`). If you set this property to `true`, you can get explainability action logs by using the Action Logs API. See Troubleshoot Product Configurations to set up configuration logs.	Optional	66.0
`pricing Procedure`	String	Name of the pricing procedure to use during the API calls to Salesforce Pricing Management.	Optional	60.0
`qualifyAll ProductsIn Transaction`	Boolean	Indicates whether to run the qualification rules on all the products in the context (`true`) or not (`false`).	Optional	60.0
`returnProduct CatalogData`	Boolean	Indicates whether to return the product catalog data (`true`) or not (`false`).	Optional	60.0

Name	Type	Description	Required or Optional	Available Version
		Exclude this property or specify the property value as `false` if you're using the API without the Product Configurator UI.		
`validateAmendRenewCancel`	Boolean	Indicates whether to run the amend, renew, cancel-related validations (`true`) or not (`false`).	Optional	60.0
`validateProductCatalog`	Boolean	Indicates whether to run the validations against the product catalog (`true`) or not (`false`).	Optional	60.0

Configuration Rule Input

Input representation of the details of a configuration rule.

JSON example

```
{
  "transactionContextId": "008d27d7-e004-4906-a949-ee7d7c323c77",
  "transactionId": "0Q0DU0000005tJh0AI",
  "ruleOptions": {
    "isUpdateContextRequired": false
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`ruleOptions`	Config Rule Options Input[]	Details of the options to run specific steps in rules.	Optional	67.0
`transactionContextId`	String	ID of the sales transaction context instance.	Required if the `transactionId` property isn't specified.	67.0
`transactionId`	String	ID of the quote or order.	Required if the `transactionContextId` property isn't specified.	67.0

Configuration Rule Options Input

Input representation of the details of the configuration rule options.

Configuration Rule Options Input

Input representation of the details of the configuration rule options.

JSON example

```
{
  "ruleOptions": {
    "isUpdateContextRequired": false
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
isUpdateContextRequired	Boolean	Indicates whether context update is required with the capability to automatically add or delete a product and its components (`true`) or not (`false`). If Place Sales Transaction API is invoked with configuration enabled, set this property to `false` to avoid any redundant execution of context logic.	Optional	67.0

Configurator Update Nodes Input

Input representation of the request to update the nodes in a product configuration.

JSON example

```
{
    "configuratorOptions": {
        "executePricing": true,
        "returnProductCatalogData": true,
        "qualifyAllProductsInTransaction": true,
        "validateProductCatalog": true,
        "validateAmendRenewCancel": true,
        "executeConfigurationRules": true,
        "addDefaultConfiguration": true
    },
    "qualificationContext": {
        "accountId": "001xx0000000001AAA",
        "contactId": "003xx00000000D7AAI"
    },
    "contextId": "008d27d7-e004-4906-a949-ee7d7c323c77",
    "updatedNodes": [
        {
            "path": ["0Q0DE000000ISHJs81", "0QLDE000000IBXw4AO"],
            "updatedAttributes": {
                "Quantity": 5
            }
```

```
        }
    ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
configurator Options	Configurator Options Input	List of the configuration options to execute.	Optional	60.0
contextId	String	ID of the context object that's being considered.	Required	60.0
qualification Context	User Context Input	Context details that are used for the qualification rules.	Optional	60.0
updatedNodes	Configurator Updated Node Input[]	List of the nodes to be updated.	Required	60.0

Configurator Updated Node Input

Input representation of the nodes to be updated in a product configuration.

JSON example

```
    "updatedNodes": [
        {
            "path": ["0Q0DE000000ISHJs81", "0QLDE000000IBXw4AO"],
            "updatedAttributes": {
                "Quantity": 5
            }
        }
    ]
```

Properties

Name	Type	Description	Required or Optional	Available Version
path	String[]	Path to the node that's being updated.	Required	60.0
updated Attributes	Map<String, Object>	Details of the object that's being updated. This property supports fields of objects from the Sales Transaction context definition, including custom objects and fields in your extended context definition.	Required	60.0

Product Quantity Set Configurator Input

Input representation of the request to set the quantity of a product.

JSON example


```
{
    "configuratorOptions": {
        "executePricing": true,
        "returnProductCatalogData": true,
        "qualifyAllProductsInTransaction": true,
        "validateProductCatalog": true,
        "validateAmendRenewCancel": true,
        "executeConfigurationRules": true,
        "addDefaultConfiguration": true
    },
    "qualificationContext": {
        "accountId": "001xx0000000001AAA",
        "contactId": "003xx00000000D7AAI"
    },
    "contextId": "008d27d7-e004-4906-a949-ee7d7c323c77",
    "quantity": 20,
    "transactionLinePath": "Quote.QuoteLineItem.Quantity"
}
```


Properties

Name	Type	Description	Required or Optional	Available Version
`configuratorOptions`	Configurator Options Input	List of the configuration options to execute.	Optional	60.0
`contextId`	String	ID of the context object that's being considered.	Required	60.0
`qualificationContext`	User Context Input	Context details that are used for the qualification rules.	Optional	60.0
`quantity`	Integer	Value of the product quantity.	Required	60.0
`transactionLinePath`	String[]	Path to the line item where the update to the quantity is applied. For example, Quote.QuoteLineItem.Quantity.	Required	60.0

User Context Input

Input representation of the request to get the context details of a user, which are used for qualification rules.

JSON example


```
    "qualificationContext": {
        "accountId": "001DU000001nHUGYA2",
        "contactId": "003xx00000000D7AAI"
    }
```

Properties

Name	Type	Description	Required or Optional	Available Version
`accountId`	String	ID of the account in a user context.	Optional	60.0
`contactId`	String	ID of the contact in a user context.	Optional	60.0
`contextId`	String	ID of the context that represents the created session.	Optional	60.0

Response Bodies

Learn more about the available Product Configurator API response bodies.

Configuration Details
Output representation of the product configuration details.

Configuration Get Instance
Output representation of the request to retrieve the configuration instance.

Configuration Load Instance
Output representation of the details of the context or session that are returned with a load configuration request.

Configuration Save Details
Output representation of the details of a saved configuration.

Configuration Record Save
Output representation of the details of a saved configuration.

Configuration List
Output representation of the details of the saved configuration.

Configuration Save Instance
Output representation of the response that's returned with a save configuration request.

Configuration Set Instance
Output representation of the details of the context or session that are returned with a set configuration request.

Configurator Add Nodes
Output representation of the configuration request details to add nodes.

Configurator Additional Fields
Output representation of the additional fields of a product configuration.

Configurator Attribute
Output representation of the attribute in a product configuration.

Configurator Attribute Category
Output representation of the attribute category in a product configuration.

Configurator Attribute Picklist
Output representation of the attribute picklist in a product configuration.

Configurator Attribute Picklist Value

Output representation of the values of an attribute picklist in a product configuration.

Configurator Delete Nodes

Output representation of the details of the configuration request to delete nodes.

Configurator Message

Output representation of the messages of a product configurator.

Configurator Price

Output representation of the pricing details in a product configuration.

Configurator Pricing Model

Output representation of the details of a pricing model in a product configuration.

Configurator Product Catalog

Output representation of the product catalog.

Configurator Product Classification

Output representation of the product classification in a product configuration.

Configurator Product Component Group

Output representation of the product component group in a product classification.

Configurator Product Recommendations

Output representation of the details of the product recommendations.

Configurator Product Related Component

Output representation of the product related component in a product configuration.

Configurator Product Selling Model

Output representation of the product selling model in a product configuration.

Configurator Product Selling Model Option

Output representation of the product selling model option in a product configuration.

Configurator Qualification Context

Output representation of the qualification context in a product configuration.

Configurator UI Treatment

Output representation of the details of the UI treatments of a product configurator. The details include the product configuration rule actions to override the disable or hide behavior in the UI for product options, product attributes, and attribute picklist values.

Configurator Unit Of Measure

Output representation of the details of the unit of measure record.

Configurator Update Nodes

Output representation of the configuration request details to update nodes.

Configuration Update

Output representation of the details of the updated configuration.

Error Response

Output representation of the details of the error.

Product Quantity Set Configurator

Output representation of the request details to set product quantity.

Configuration Rule Response

Output representation of the details of the configuration rule response.

Configuration Details

Output representation of the product configuration details.

JSON example


```
{
  "catalogProducts": {
    "additionalFields": [],
    "attributeCategories": [
      {
        "attributes": [
          {
            "attributeCategoryId": "0v3xx0000000001AAA",
            "attributeNameOverride": "Load Capacity",
            "code": "CAP",
            "dataType": "NUMBER",
            "defaultValue": "1500",
           "description": "Server racks are designed to support a specific load capacity,
 commonly measured in kilograms (kg) or pounds (lbs). Typical load capacities range from
 500 kg (1102 lbs) to 1500 kg (3307 lbs) depending on the model.",
            "id": "0tjxx00000001DpAAI",
            "isCloneable": false,
            "isConfigurable": true,
            "isHidden": false,
            "isPriceImpacting": false,
            "isReadOnly": false,
            "isRequired": false,
            "label": "Load Capacity",
            "name": "Load Capacity"
          },
          {
            "attributeCategoryId": "0v3xx0000000001AAA",
            "attributeNameOverride": "Expansion Slots",
            "code": "SLOTCAP",
            "dataType": "NUMBER",
            "defaultValue": "12",
            "id": "0tjxx00000001H3AAI",
            "isCloneable": false,
            "isConfigurable": true,
            "isHidden": false,
            "isPriceImpacting": false,
            "isReadOnly": false,
            "isRequired": true,
            "label": "Expansion Slots",
            "name": "Expansion Slots"
          },
          {
            "attributeCategoryId": "0v3xx0000000001AAA",
            "attributeNameOverride": "Memory",
            "attributePicklist": {
              "id": "0v5xx0000000001AAA",
              "values": [
                {
                  "code": "MEM",
```

```
                  "displayValue": "25",
                  "id": "0v6xx0000000001AAA",
                  "isBooleanValue": false,
                  "name": "25Mem",
                  "sequence": 0,
                  "textValue": "25"
                },
                {
                  "code": "50MEM",
                  "displayValue": "50",
                  "id": "0v6xx000000001eAAA",
                  "isBooleanValue": false,
                  "name": "50Mem",
                  "sequence": 1,
                  "textValue": "50"
                },
                {
                  "code": "100MEM",
                  "displayValue": "100",
                  "id": "0v6xx000000003FAAQ",
                  "isBooleanValue": false,
                  "name": "100Mem",
                  "sequence": 2,
                  "textValue": "100"
                }
              ]
            },
            "dataType": "MULTIPICKLIST",
            "defaultValue": "25",
            "id": "0tjxx00000001IfAAI",
            "isCloneable": false,
            "isConfigurable": true,
            "isHidden": false,
            "isPriceImpacting": false,
            "isReadOnly": false,
            "isRequired": true,
            "label": "Memory",
            "name": "Memory"
          }
        ],
        "code": "SPEC",
        "name": "Server Rack Specifications"
      }
    ],
    "description": "Introducing the Cisco Server Rack, a sleek and robust solution
designed to streamline your data center infrastructure. With its scalable design and
advanced cable management features, it ensures optimal performance, efficiency, and easy
 maintenance for your critical network equipment.",
    "displayUrl":
"https://www.cisco.com/content/dam/en/us/products/servers-unified-computing/ucs-c240-m4-rack-server/product-large.jpg",

    "id": "01txx0000006jkuAAA",
    "isActive": true,
    "isAssetizable": true,
```

```
    "isConfigurable": true,
    "isSoldOnlyWithOtherProds": false,
    "name": "Cisco Server Rack NX44",
    "nodeType": "bundleProduct",
    "prices": [],
    "productClassification": {
      "id": "11Bxx000002CC02EAG"
    },
    "productCode": "RACK",
    "productComponentGroups": [
      {
        "classifications": [],
        "code": "SERVICE",
        "components": [
          {
            "additionalFields": [],
            "attributeCategories": [],
            "description": "Introducing the Cisco Rack Server NX44 Service, a
comprehensive protection plan designed to safeguard your valuable data infrastructure.
 With extended coverage and rapid response times, this warranty ensures peace of mind
and uninterrupted performance for your critical business operations.",
            "id": "01txx0000006jmWAAQ",
            "isActive": true,
            "isAssetizable": true,
            "isComponentRequired": false,
            "isConfigurable": false,
            "isDefaultComponent": false,
            "isQuantityEditable": false,
            "isSoldOnlyWithOtherProds": false,
            "name": "Cisco Rack Server Service - 1 Year",
            "nodeType": "simpleProduct",
            "prices": [],
            "productClassification": {},
            "productCode": "SERVICE",
            "productComponentGroups": [],
            "productRelatedComponent": {
              "childProductId": "01txx0000006jmWAAQ",
              "childSellingModelId": "0jPxx000000004rEAA",
              "doesBundlePriceIncludeChild": true,
              "id": "0dSxx000000001dEAA",
              "isComponentRequired": false,
              "isDefaultComponent": false,
              "isQuantityEditable": false,
              "parentProductId": "01txx0000006jkuAAA",
              "parentSellingModelId": "0jPxx000000004rEAA",
              "productComponentGroupId": "0y7xx000000001dAAA",
              "productRelationshipTypeId": "0yoxx00000001IfAAI",
              "quantity": 1,
              "quantityScaleMethod": "Proportional"
            },
            "productSellingModelOptions": [
              {
                "id": "0iOxx000000009hEAA",
                "productId": "01txx0000006jmWAAQ",
```

```
                "productSellingModel": {
                  "id": "0jPxx000000004rEAA",
                  "name": "Termed Annually",
                  "pricingTerm": 1,
                  "pricingTermUnit": "Annual",
                  "sellingModelType": "TermDefined",
                  "status": "Active"
                },
                "productSellingModelId": "0jPxx000000004rEAA"
              },
              {
                "id": "0iOxx00000000PpEAI",
                "productId": "01txx0000006jmWAAQ",
                "productSellingModel": {
                  "id": "0jPxx0000000085EAA",
                  "name": "Evergreen Annually",
                  "pricingTerm": 1,
                  "pricingTermUnit": "Annual",
                  "sellingModelType": "Evergreen",
                  "status": "Active"
                },
                "productSellingModelId": "0jPxx0000000085EAA"
              }
            ]
          }
        ],
        "description": "The services available for the Cisco Server Rack NX44 product
provide comprehensive coverage and support for optimal performance and reliability,
ensuring peace of mind for your data center infrastructure.",
        "id": "0y7xx000000001dAAA",
        "maxBundleComponents": 1,
        "minBundleComponents": 0,
        "name": "Services",
        "parentProductId": "01txx0000006jkuAAA",
        "sequence": 1
      },
      {
        "classifications": [],
        "code": "WARRANTY",
        "components": [
          {
            "additionalFields": [],
            "attributeCategories": [],
            "description": "Introducing the Cisco Rack Server NX44 Warranty, a
comprehensive protection plan designed to safeguard your valuable data infrastructure.
 With extended coverage and rapid response times, this warranty ensures peace of mind
and uninterrupted performance for your critical business operations.",
            "id": "01txx0000006jjIAAQ",
            "isActive": true,
            "isAssetizable": true,
            "isComponentRequired": false,
            "isConfigurable": false,
            "isDefaultComponent": true,
            "isQuantityEditable": true,
```

```
"isSoldOnlyWithOtherProds": false,
"name": "Cisco Rack Server Warranty - 1 Year",
"nodeType": "simpleProduct",
"prices": [],
"productClassification": {},
"productCode": "WARRANTY",
"productComponentGroups": [],
"productRelatedComponent": {
  "childProductId": "01txx0000006jjIAAQ",
  "childSellingModelId": "0jPxx000000001dEAA",
  "doesBundlePriceIncludeChild": false,
  "id": "0dSxx0000000001EAA",
  "isComponentRequired": false,
  "isDefaultComponent": true,
  "isQuantityEditable": true,
  "maxQuantity": 1,
  "parentProductId": "01txx0000006jkuAAA",
  "parentSellingModelId": "0jPxx000000001dEAA",
  "productComponentGroupId": "0y7xx0000000001AAA",
  "productRelationshipTypeId": "0yoxx00000001IfAAI",
  "quantity": 1,
  "quantityScaleMethod": "Proportional",
  "sequence": 0
},
"productSellingModelOptions": [
  {
    "id": "0iOxx000000001dEAA",
    "productId": "01txx0000006jjIAAQ",
    "productSellingModel": {
      "id": "0jPxx000000001dEAA",
      "name": "One Time",
      "sellingModelType": "OneTime",
      "status": "Active"
    },
    "productSellingModelId": "0jPxx000000001dEAA"
  },
  {
    "id": "0iOxx00000000HlEAI",
    "productId": "01txx0000006jjIAAQ",
    "productSellingModel": {
      "id": "0jPxx000000003FEAQ",
      "name": "Termed Monthly",
      "pricingTerm": 1,
      "pricingTermUnit": "Months",
      "sellingModelType": "TermDefined",
      "status": "Active"
    },
    "productSellingModelId": "0jPxx000000003FEAQ"
  },
  {
    "id": "0iOxx00000000JNEAY",
    "productId": "01txx0000006jjIAAQ",
    "productSellingModel": {
      "id": "0jPxx000000004rEAA",
```

```
                "name": "Termed Annually",
                "pricingTerm": 1,
                "pricingTermUnit": "Annual",
                "sellingModelType": "TermDefined",
                "status": "Active"
              },
              "productSellingModelId": "0jPxx000000004rEAA"
            },
            {
              "id": "0iOxx00000000KzEAI",
              "productId": "01txx0000006jjIAAQ",
              "productSellingModel": {
                "id": "0jPxx0000000085EAA",
                "name": "Evergreen Annually",
                "pricingTerm": 1,
                "pricingTermUnit": "Annual",
                "sellingModelType": "Evergreen",
                "status": "Active"
              },
              "productSellingModelId": "0jPxx0000000085EAA"
            },
            {
              "id": "0iOxx00000000MbEAI",
              "productId": "01txx0000006jjIAAQ",
              "productSellingModel": {
                "id": "0jPxx000000006TEAQ",
                "name": "Evergreen Monthly",
                "pricingTerm": 1,
                "pricingTermUnit": "Months",
                "sellingModelType": "Evergreen",
                "status": "Active"
              },
              "productSellingModelId": "0jPxx000000006TEAQ"
            }
          ]
        }
      ],
      "description": "The warranties available for the Cisco Server Rack NX44 product
 provide comprehensive coverage and support for optimal performance and reliability,
ensuring peace of mind for your data center infrastructure.",
      "id": "0y7xx0000000001AAA",
      "maxBundleComponents": 1,
      "minBundleComponents": 0,
      "name": "Warranties",
      "parentProductId": "01txx0000006jkuAAA",
      "sequence": 0
    }
  ],
  "productSellingModelOptions": [
    {
      "id": "0iOxx000000003FEAQ",
      "productId": "01txx0000006jkuAAA",
      "productSellingModel": {
        "id": "0jPxx000000001dEAA",
```

```
          "name": "One Time",
          "sellingModelType": "OneTime",
          "status": "Active"
        },
        "productSellingModelId": "0jPxx000000001dEAA"
      },
      {
        "id": "0iOxx000000004rEAA",
        "productId": "01txx0000006jkuAAA",
        "productSellingModel": {
          "id": "0jPxx000000003FEAQ",
          "name": "Termed Monthly",
          "pricingTerm": 1,
          "pricingTermUnit": "Months",
          "sellingModelType": "TermDefined",
          "status": "Active"
        },
        "productSellingModelId": "0jPxx000000003FEAQ"
      },
      {
        "id": "0iOxx000000006TEAQ",
        "productId": "01txx0000006jkuAAA",
        "productSellingModel": {
          "id": "0jPxx000000004rEAA",
          "name": "Termed Annually",
          "pricingTerm": 1,
          "pricingTermUnit": "Annual",
          "sellingModelType": "TermDefined",
          "status": "Active"
        },
        "productSellingModelId": "0jPxx000000004rEAA"
      }
    ],
    "productType": "Bundle"
  },
  "uiTreatments": [
    {
      "details": {
        "attributeId": "0tjxx0000000007AAA",
        "prcId": "0dSxx0000000007EAA",
        "stiId": "0QLxx0000004CU0GAM",
        "attributePicklistValueId": "0v6xx0000000005AAA"
      },
      "uiTreatmentScope": "Bundle",
      "uiTreatmentTarget": "Attribute_Picklist_Value",
      "uiTreatmentType": "Hide"
    },
    {
      "details": {
        "stiId": "ref_f0f2da7b_c431_482d_bf4b_599052f3a2e1"
      },
      "uiTreatmentScope": "Product",
      "uiTreatmentTarget": "Component",
      "uiTreatmentType": "Disable"
```

```
    }
  ],
  "errors": [],
  "success": true,
  "productRecommendations": [
    {
      "referenceId": "CORE_BUNDLE_001",
      "productIds": [
        "01t000000001234",
        "01t000000005678"
      ]
    }
  ],
  "transactionContext": {
    "SalesTransaction": [
      {
        "Status": "Draft",
        "Account": "001xx000003GeIxAAK",
        "BillingCity": "San Francisco",
        "Subtotal": 152500,
        "LastPricedDate": "2023-08-22T05:55:39Z",
        "businessObjectType": "Quote",
        "TotalAmount": 152500,
        "ShippingStreet": "415 Mission St",
        "SalesTransactionItem": [
          {
            "ProrationPolicy": null,
            "Discount": null,
            "ProductSellingModel": "0jPxx000000001dEAA",
            "Product": "01txx0000006jkuAAA",
            "businessObjectType": "QuoteLineItem",
            "BasisTransactionItem": null,
            "PartnerUnitPrice": null,
            "StartingUnitPriceSource": "System",
            "ListPrice": 150000,
            "ItemTotalAdjustmentAmount": 0,
            "SalesTransactionItemSource": "0QLxx0000004CQmGAM",
            "SubscriptionTerm": null,
            "StartDate": null,
            "NetTotalPrice": 150000,
            "TotalLineAmount": 150000,
            "PeriodBoundaryStartMonth": null,
            "ListPriceTotal": 150000,
            "PartnerDiscountPercent": null,
            "id": "0QLxx0000004CQmGAM",
            "PriceWaterFall": {
              "currencyCode": "USD",
              "executionEndTimestamp": "2023-09-18T20:11:15.016Z",
              "executionId": "ruepwmHn2ZFvnQo5bjot",
              "executionStartTimestamp": "2023-09-18T20:11:14.906Z",
              "lineItemId": "0QLxx0000004CQmGAM",
              "output": {
                "NetUnitPrice": 150000,
                "Subtotal": 0
```

```
          },
          "success": true,
          "waterfall": [
            {
              "fieldToTagNameMapping": {
                "NetUnitPrice": "ItemUnitPrice",
                "AdjustmentValue": "ItemAdjustmentValue",
                "Subtotal": "ItemTotalAdjustmentAmount",
                "Quantity": "ItemQuantity",
                "LineItemId": "SalesTransactionItemSource",
                "InputUnitPrice": "ItemUnitPrice"
              },
              "inputParameters": {
                "Quantity": 1,
                "LineItemId": "0QLxx0000004CQmGAM",
                "InputUnitPrice": 150000,
                "AdjustmentType": "Amount"
              },
              "outputParameters": {
                "NetUnitPrice": 150000,
                "Subtotal": 0
              },
              "pricingElement": {
                "adjustments": [
                  {
                    "AdjustmentType": "Amount"
                  }
                ],
                "elementType": "MANUALDISCOUNT",
                "name": "ManualDiscount"
              },
              "sequence": 1
            }
          ]
        },
        "BillingFrequency": null,
        "SalesTransactionItemParent": "0Q0xx0000004CAeCAM",
        "StartingPriceTotal": 150000,
        "Quantity": 1,
        "PeriodBoundary": null,
        "SalesTransactionItemAttribute": [
          {
            "AttributeKey": "0tjxx00000001H3AAI",
            "AttributeValue": "30.0",
            "AttributePicklistValue": null,
            "IsPriceImpacting": true,
            "businessObjectType": "QuoteLineItemAttribute",
            "AttributeName": "Expansion Slots",
            "AttributeDefinitionCode": "SLOTCAP",
            "id": "0zuxx0000000001AAA",
            "SalesTransactionItemAttrParent": "0QLxx0000004CQmGAM"
          }
        ],
        "EndDate": null,
```

```
    "DiscountAmount": null,
    "PricebookEntry": "01uxx0000009154AAA",
    "PricingTermCount": 1,
    "NetUnitPrice": 150000,
    "UnitPrice": 150000,
    "StartingUnitPrice": 150000,
    "SalesTrxnItemRelationship": [
      {
        "ProductRelationshipType": "0yoxx00000001IfAAI",
        "MainItemRole": "Bundle",
        "AssociatedItem": "0QLxx0000004CQnGAM",
        "ProductRelatedComponent": "0dSxx0000000001EAA",
        "MainItem": "0QLxx0000004CQmGAM",
        "AssociatedQuantScaleMethod": "Proportional",
        "businessObjectType": "QuoteLineRelationship",
        "AssociatedItemRole": "BundleComponent",
        "SalesTrnItemRelationshipParent": "0Q0xx0000004CAeCAM",
        "id": "0yQxx000000001dEAA",
        "AssociatedItemPricing": "NotIncludedInBundlePrice"
      }
    ],
    "TotalPrice": 150000,
    "PeriodBoundaryDay": null
  },
  {
    "ProrationPolicy": null,
    "Discount": null,
    "ProductSellingModel": "0jPxx000000001dEAA",
    "Product": "01txx0000006jjIAAQ",
    "businessObjectType": "QuoteLineItem",
    "BasisTransactionItem": null,
    "PartnerUnitPrice": null,
    "StartingUnitPriceSource": "System",
    "ListPrice": 2000,
    "ItemTotalAdjustmentAmount": 0,
    "SalesTransactionItemSource": "0QLxx0000004CQnGAM",
    "SubscriptionTerm": null,
    "StartDate": null,
    "NetTotalPrice": 2000,
    "TotalLineAmount": 2000,
    "PeriodBoundaryStartMonth": null,
    "ListPriceTotal": 2000,
    "PartnerDiscountPercent": null,
    "id": "0QLxx0000004CQnGAM",
    "PriceWaterFall": {
      "currencyCode": "USD",
      "executionEndTimestamp": "023-09-18T20:11:15.016Z",
      "executionId": "ruepwmHn2ZFvnQo5bjot",
      "executionStartTimestamp": "2023-09-18T20:11:14.906Z",
      "lineItemId": "0QLxx0000004CQnGAM",
      "output": {
        "NetUnitPrice": 2000,
        "Subtotal": 0
      },
```

```
        "success": true,
        "waterfall": [
          {
            "fieldToTagNameMapping": {
              "NetUnitPrice": "ItemUnitPrice",
              "AdjustmentValue": "ItemAdjustmentValue",
              "Subtotal": "ItemTotalAdjustmentAmount",
              "Quantity": "ItemQuantity",
              "LineItemId": "SalesTransactionItemSource",
              "InputUnitPrice": "ItemUnitPrice"
            },
            "inputParameters": {
              "Quantity": 1,
              "LineItemId": "0QLxx0000004CQnGAM",
              "InputUnitPrice": 2000,
              "AdjustmentType": "Amount"
            },
            "outputParameters": {
              "NetUnitPrice": 2000,
              "Subtotal": 0
            },
            "pricingElement": {
              "adjustments": [
                {}
              ],
              "elementType": "MANUALDISCOUNT",
              "name": "ManualDiscount"
            },
            "sequence": 1
          }
        ]
      },
      "BillingFrequency": null,
      "SalesTransactionItemParent": "0Q0xx0000004CAeCAM",
      "StartingPriceTotal": 2000,
      "Quantity": 1,
      "PeriodBoundary": null,
      "EndDate": null,
      "DiscountAmount": null,
      "PricebookEntry": "01uxx000000913SAAQ",
      "PricingTermCount": 1,
      "NetUnitPrice": 2000,
      "UnitPrice": 2000,
      "StartingUnitPrice": 2000,
      "TotalPrice": 2000,
      "PeriodBoundaryDay": null
    }
  ],
  "BillingCountry": "US",
  "BillingStreet": "415 Mission St",
  "Pricebook": "01sxx0000005uDZAAY",
  "ShippingPostalCode": "94105",
  "SalesTransactionSource": "0Q0xx0000004CAeCAM",
  "ShippingCountry": "US",
```

```
        "ShippingCity": "San Francisco",
        "ShippingState": "CA",
        "BillingPostalCode": "94105",
        "id": "0Q0xx0000004CAeCAM",
        "BillToContact": null,
        "Contract": null,
        "BillingState": "CA"
      }
    ]
  },
  "transactionContextId": "cda87acd-45ed-4913-903e-9dd33cec85a6",
  "transactionContextMappingId": "11jxx0000004LwOAAU"
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
catalog Products	Configurator Product Catalog[]	Structure that contains the product catalog data.	Small, 61.0	61.0
errors	ConnectApiErrorResponse	List of errors that contains a message and an error code.	Small, 60.0	60.0
messages	Map<String, Configurator Message>>	Messages from the validation, Business Rules Engine (BRE), or Salesforce Pricing calls.	Small, 60.0	60.0
product Recommendations	Configurator Product Recommendations	List of product recommendations.	Small, 65.0	65.0
success	Boolean	Indicates whether the request is successful (`true`) or not (`false`).	Small, 60.0	60.0
transaction Context	Map<String, Object>	Serialized JSON representation of the transaction context.	Small, 60.0	60.0
transaction ContextId	String	ID of the transaction context.	Small, 60.0	60.0
transactionContext MappingId	String	ID of the context mapping.	Small, 60.0	60.0
transaction Qualification	Map<String, Configurator Qualification Context>	Map of the product IDs to the qualification context.	Small, 60.0	60.0
uiTreatments	Configurator UI Treatment[]	Details of the UI treatments that specify the product configuration rule actions to override the disable or hide behavior in the UI for product options, product attributes, and attribute picklist values.	Small, 62.0	62.0

Configuration Get Instance

Output representation of the request to retrieve the configuration instance.

Property Name	Type	Description	Filter Group and Version	Available Version
`errors`	Error Response	List of errors, which contains an error code and a message.	Small, 60.0	60.0
`success`	Boolean	Indicates whether the call was successful (`true`) not (`false`).	Small, 60.0	60.0
`transaction`	Map<String, Object>	Transaction JSON payload representing an object in an external system that's used to create a session.	Small, 60.0	60.0

Configuration Load Instance

Output representation of the details of the context or session that are returned with a load configuration request.

JSON Example

```
{
  "configuratorMessages": {},
  "configuratorUITreatments": [
    {
      "details": {
        "attributeId": "0tjxx0000000007AAA",
        "prcId": "0dSxx0000000007EAA",
        "stiId": "0QLxx0000004CU0GAM",
        "attributePicklistValueId": "0v6xx0000000005AAA"
      },
      "uiTreatmentScope": "Bundle",
      "uiTreatmentTarget": "Attribute_Picklist_Value",
      "uiTreatmentType": "Hide"
    },
    {
      "details": {
        "stiId": "ref_f0f2da7b_c431_482d_bf4b_599052f3a2e1"
      },
      "uiTreatmentScope": "Product",
      "uiTreatmentTarget": "Component",
      "uiTreatmentType": "Disable"
    }
  ],
  "contextId": "831f07b01cf0cbd2d046adf5350420f85f0611b4b1e22e183921a063857a1377",
  "errors": [],
  "productQualifications": {
    "01tDU000000EOTCYA4": {
      "isQualified": true
    }
  },
```

```
  "success": true
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`configurator Messages`	Map<String, <Configurator Message>>	Map of the product IDs to the list of configurator messages. Configurator messages are results from any validations, Business Rules Engines (BRE) calls, or Salesforce Pricing calls.	Small, 60.0	60.0
`configurator UITreatments`	Configurator UI Treatment[]	Details of the UI treatments that specify the product configuration rule actions to override the disable or hide behavior in the UI for product options, product attributes, and attribute picklist values.	Small, 62.0	62.0
`contextId`	String	ID of the transaction context.	Small, 60.0	60.0
`errors`	Error Response	List of errors, which contains an error code and a message.	Small, 60.0	60.0
`product Qualifications`	Map<String, Configurator Qualification Context>	Map of the product IDs to the execution results from qualification rules.	Small, 60.0	60.0
`success`	Boolean	Indicates whether the call was successful (`true`) not (`false`).	Small, 60.0	60.0

Configuration Save Details

Output representation of the details of a saved configuration.

JSON example

```
{
  "savedConfigurations": [
    {
      "data":
[illegible]
 Built-in
[illegible]

[illegible]

      "description": "This configuration is for cline XYZ",
      "id": "1Nyxx0000004CFUCA2",
      "name": "Client XYZ Favorite",
      "referenceRecordId": "01txx0000006iCFAAY"
    },
    {
```

```
      "data":
[illegible]
 Built-in
[illegible]

[illegible]

      "description": "This configuration is for cline XYZ",
      "id": "1Nyxx0000004CH6CAM",
      "name": "Client XYZ Favorite",
      "referenceRecordId": "01txx0000006iCFAAY"
    }
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
data	String	JSON object that contains the details of the sales transaction, formatted as a string.	Small, 63.0	63.0
description	String	Description of the saved configuration.	Small, 63.0	63.0
id	String	ID of the saved configuration.	Small, 63.0	63.0
name	String	Name of the saved configuration.	Small, 63.0	63.0
referenceRecordId	String	ID of the record that the saved configuration belongs to.	Small, 63.0	63.0

Configuration Record Save

Output representation of the details of a saved configuration.

JSON example

This example shows a sample when the save operation is successful.

```
{
  "errors": [],
  "id": "1Nyxx0000004CNYCA2"
}
```

This example shows a sample when the save operation has errors.

```
{
  "errors": [{
  "code": "INTERNAL_SERVER_ERROR",
  "message": "INVALID_REFERENCEOBJECTID"
}]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`errors`	Error Response	List of errors that contains a message and an error code.	Small, 63.0	63.0
`id`	String	ID of the configuration that's saved. This property isn't shown if the operation has errors.	Small, 63.0	63.0

Configuration List

Output representation of the details of the saved configuration.

JSON example

```
{
  "errors": [],
  "savedConfigurations": [
    {
      "data":
[illegible]
 Built-in
[illegible]


[illegible]


      "description": "This configuration is saved for reuse.",
      "id": "1Nyxx0000004CFUCA2",
      "name": "Favorite Configuration",
      "referenceRecordId": "01txx0000006iCFAAY"
    },
    {
      "data":
[illegible]
 Built-in
[illegible]


[illegible]


      "description": "This configuration is saved for reuse.",
      "id": "1Nyxx0000004CH6CAM",
      "name": "Configuration 2",
      "referenceRecordId": "01txx0000006iCFAAY"
    }
  ],
  "success": true
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`errors`	Error Response	List of errors that contains a message and an error code.	Small, 63.0	63.0
`saved Configurations`	Configuration Save Details	Configuration details associated with the `referenceRecordID` request parameter.	Small, 63.0	63.0
`success`	Boolean	Indicates whether the operation is successful (`true`) or not (`false`).	Small, 63.0	63.0

Configuration Save Instance

Output representation of the response that's returned with a save configuration request.

Property Name	Type	Description	Filter Group and Version	Available Version
`errors`	Error Response	List of errors, which contains an error code and a message.	Small, 60.0	60.0
`success`	Boolean	Indicates whether the call was successful (`true`) not (`false`).	Small, 60.0	60.0

Configuration Set Instance

Output representation of the details of the context or session that are returned with a set configuration request.

JSON Example


```
{
  "configuratorMessages": {},
  "configuratorUITreatments": [
    {
      "details": {
        "attributeId": "0tjxx0000000007AAA",
        "prcId": "0dSxx0000000007EAA",
        "stiId": "0QLxx0000004CU0GAM",
        "attributePicklistValueId": "0v6xx0000000005AAA"
      },
      "uiTreatmentScope": "Bundle",
      "uiTreatmentTarget": "Attribute_Picklist_Value",
      "uiTreatmentType": "Hide"
    },
    {
      "details": {
        "stiId": "ref_f0f2da7b_c431_482d_bf4b_599052f3a2e1"
      },
      "uiTreatmentScope": "Product",
      "uiTreatmentTarget": "Component",
      "uiTreatmentType": "Disable"
```

```
    }
  ],
  "contextId": "831f07b01cf0cbd2d046adf5350420f85f0611b4b1e22e183921a063857a1377",
  "errors": [],
  "productQualifications": {
    "01tDU000000EOTCYA4": {
      "isQualified": true
    }
  },
  "success": true
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
configurator Messages	Map<String, <Configurator Message>>	Map of the product IDs to the list of configurator messages. Configurator messages are results from any validations, Business Rules Engines (BRE) calls, or Salesforce Pricing calls.	Small, 60.0	60.0
configurator UITreatments	Configurator UI Treatment[]	Details of the UI treatments that specify the product configuration rule actions to override the disable or hide behavior in the UI for product options, product attributes, and attribute picklist values.	Small, 62.0	62.0
contextId	String	ID of the transaction context.	Small, 60.0	60.0
errors	Error Response	List of errors, which contains an error code and a message.	Small, 60.0	60.0
product Qualifications	Map<String, Configurator Qualification Context>	Map of the product IDs to the execution results from qualification rules.	Small, 60.0	60.0
success	Boolean	Indicates whether the call was successful (`true`) not (`false`).	Small, 60.0	60.0

Configurator Add Nodes

Output representation of the configuration request details to add nodes.

JSON Example

```
{
  "configuratorMessages": {},
  "configuratorUITreatments": [
    {
      "details": {
        "attributeId": "0tjxx0000000007AAA",
        "prcId": "0dSxx0000000007EAA",
```

```
          "stiId": "0QLxx0000004CU0GAM",
          "attributePicklistValueId": "0v6xx0000000005AAA"
        },
        "uiTreatmentScope": "Bundle",
        "uiTreatmentTarget": "Attribute_Picklist_Value",
        "uiTreatmentType": "Hide"
      },
      {
        "details": {
          "stiId": "ref_f0f2da7b_c431_482d_bf4b_599052f3a2e1"
        },
        "uiTreatmentScope": "Product",
        "uiTreatmentTarget": "Component",
        "uiTreatmentType": "Disable"
      }
    ],
    "errors": [],
    "productQualifications": {
      "01tDU000000EOTCYA4": {
        "isQualified": true
      }
    },
    "success": true
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
configurator Messages	Map<String, Configurator Message>	Map of the product IDs to the list of configurator messages. Configurator messages are results from any validations, Business Rules Engines (BRE) calls, or Salesforce Pricing calls.	Small, 60.0	60.0
configurator UITreatments	Configurator UI Treatment[]	Details of the UI treatments that specify the product configuration rule actions to override the disable or hide behavior in the UI for product options, product attributes, and attribute picklist values.	Small, 62.0	62.0
errors	Error Response	List of errors, which contains an error code and a message.	Small, 60.0	60.0
product Qualifications	Map<String, Configurator Qualification Context>	Map of the product IDs to the qualification rule execution results.	Small, 60.0	60.0
success	Boolean	Indicates whether the call was successful (`true`) not (`false`).	Small, 60.0	60.0

Configurator Additional Fields

Output representation of the additional fields of a product configuration.

Property Name	Type	Description	Filter Group and Version	Available Version
`fieldApiName`	String	API name of the field.	Small, 60.0	60.0
`fieldName`	String	Name of the field.	Small, 60.0	60.0
`value`	String	Value of the field.	Small, 60.0	60.0

Configurator Attribute

Output representation of the attribute in a product configuration.

JSON example


```
    "attributeCategories": [
      {
        "attributes": [
          {
            "attributeCategoryId": "0v3xx0000000001AAA",
            "attributeNameOverride": "Load Capacity",
            "code": "CAP",
            "dataType": "NUMBER",
            "defaultValue": "1500",
           "description": "Server racks are designed to support a specific load capacity,
commonly measured in kilograms (kg) or pounds (lbs). Typical load capacities range from
500 kg (1102 lbs) to 1500 kg (3307 lbs) depending on the model.",
            "id": "0tjxx00000001DpAAI",
            "isCloneable": false,
            "isConfigurable": true,
            "isHidden": false,
            "isPriceImpacting": false,
            "isReadOnly": false,
            "isRequired": false,
            "label": "Load Capacity",
            "name": "Load Capacity"
          },
          {
            "attributeCategoryId": "0v3xx0000000001AAA",
            "attributeNameOverride": "Expansion Slots",
            "code": "SLOTCAP",
            "dataType": "NUMBER",
            "defaultValue": "12",
            "id": "0tjxx00000001H3AAI",
            "isCloneable": false,
            "isConfigurable": true,
            "isHidden": false,
            "isPriceImpacting": false,
            "isReadOnly": false,
            "isRequired": true,
```

```
        "label": "Expansion Slots",
        "name": "Expansion Slots"
      },
      {
        "attributeCategoryId": "0v3xx0000000001AAA",
        "attributeNameOverride": "Memory",
        "attributePicklist": {
          "id": "0v5xx0000000001AAA",
          "values": [
            {
              "code": "MEM",
              "displayValue": "25",
              "id": "0v6xx0000000001AAA",
              "isBooleanValue": false,
              "name": "25Mem",
              "sequence": 0,
              "textValue": "25"
            },
            {
              "code": "50MEM",
              "displayValue": "50",
              "id": "0v6xx000000001eAAA",
              "isBooleanValue": false,
              "name": "50Mem",
              "sequence": 1,
              "textValue": "50"
            },
            {
              "code": "100MEM",
              "displayValue": "100",
              "id": "0v6xx000000003FAAQ",
              "isBooleanValue": false,
              "name": "100Mem",
              "sequence": 2,
              "textValue": "100"
            }
          ]
        },
        "dataType": "MULTIPICKLIST",
        "defaultValue": "25",
        "id": "0tjxx00000001IfAAI",
        "isCloneable": false,
        "isConfigurable": true,
        "isHidden": false,
        "isPriceImpacting": false,
        "isReadOnly": false,
        "isRequired": true,
        "label": "Memory",
        "name": "Memory"
      }
    ],
    "code": "SPEC",
    "name": "Server Rack Specifications"
  }
```

```
    ],
    "description": "Introducing the Cisco Server Rack, a sleek and robust solution
designed to streamline your data center infrastructure. With its scalable design and
advanced cable management features, it ensures optimal performance, efficiency, and easy
 maintenance for your critical network equipment.",
    "displayUrl":
"https://www.cisco.com/content/dam/en/us/products/servers-unified-computing/ucs-c240-m4-rack-server/product-large.jpg",

    "id": "01txx0000006jkuAAA",
    "isActive": true,
    "isAssetizable": true,
    "isConfigurable": true,
    "isSoldOnlyWithOtherProds": false,
    "name": "Cisco Server Rack NX44",
    "nodeType": "bundleProduct",
    "prices": [],
    "productClassification": {
      "id": "11Bxx000002CC02EAG"
    },
    "productCode": "RACK",
    "productComponentGroups": [
      {
        "classifications": [],
        "code": "SERVICE",
        "components": [
          {
            "additionalFields": [],
            "attributeCategories": [],
            "description": "Introducing the Cisco Rack Server NX44 Service, a
comprehensive protection plan designed to safeguard your valuable data infrastructure.
 With extended coverage and rapid response times, this warranty ensures peace of mind
and uninterrupted performance for your critical business operations.",
            "id": "01txx0000006jmWAAQ",
            "isActive": true,
            "isAssetizable": true,
            "isComponentRequired": false,
            "isConfigurable": false,
            "isDefaultComponent": false,
            "isQuantityEditable": false,
            "isSoldOnlyWithOtherProds": false,
            "name": "Cisco Rack Server Service - 1 Year",
            "nodeType": "simpleProduct",
            "prices": [],
            "productClassification": {},
            "productCode": "SERVICE",
            "productComponentGroups": [],
            "productRelatedComponent": {
              "childProductId": "01txx0000006jmWAAQ",
              "childSellingModelId": "0jPxx000000004rEAA",
              "doesBundlePriceIncludeChild": true,
              "id": "0dSxx000000001dEAA",
              "isComponentRequired": false,
              "isDefaultComponent": false,
              "isQuantityEditable": false,
```

```
              "parentProductId": "01txx0000006jkuAAA",
              "parentSellingModelId": "0jPxx000000004rEAA",
              "productComponentGroupId": "0y7xx000000001dAAA",
              "productRelationshipTypeId": "0yoxx00000001IfAAI",
              "quantity": 1,
              "quantityScaleMethod": "Proportional"
            },
            "productSellingModelOptions": [
              {
                "id": "0iOxx000000009hEAA",
                "productId": "01txx0000006jmWAAQ",
                "productSellingModel": {
                  "id": "0jPxx000000004rEAA",
                  "name": "Termed Annually",
                  "pricingTerm": 1,
                  "pricingTermUnit": "Annual",
                  "sellingModelType": "TermDefined",
                  "status": "Active"
                },
                "productSellingModelId": "0jPxx000000004rEAA"
              },
              {
                "id": "0iOxx00000000PpEAI",
                "productId": "01txx0000006jmWAAQ",
                "productSellingModel": {
                  "id": "0jPxx0000000085EAA",
                  "name": "Evergreen Annually",
                  "pricingTerm": 1,
                  "pricingTermUnit": "Annual",
                  "sellingModelType": "Evergreen",
                  "status": "Active"
                },
                "productSellingModelId": "0jPxx0000000085EAA"
              }
            ]
          }
        ],
        "description": "The services available for the Cisco Server Rack NX44 product
provide comprehensive coverage and support for optimal performance and reliability,
ensuring peace of mind for your data center infrastructure.",
        "id": "0y7xx000000001dAAA",
        "maxBundleComponents": 1,
        "minBundleComponents": 0,
        "name": "Services",
        "parentProductId": "01txx0000006jkuAAA",
        "sequence": 1
      },
      {
        "classifications": [],
        "code": "WARRANTY",
        "components": [
          {
            "additionalFields": [],
            "attributeCategories": [],
```

```
            "description": "Introducing the Cisco Rack Server NX44 Warranty, a
comprehensive protection plan designed to safeguard your valuable data infrastructure.
 With extended coverage and rapid response times, this warranty ensures peace of mind
and uninterrupted performance for your critical business operations.",
            "id": "01txx0000006jjIAAQ",
            "isActive": true,
            "isAssetizable": true,
            "isComponentRequired": false,
            "isConfigurable": false,
            "isDefaultComponent": true,
            "isQuantityEditable": true,
            "isSoldOnlyWithOtherProds": false,
            "name": "Cisco Rack Server Warranty - 1 Year",
            "nodeType": "simpleProduct",
            "prices": [],
            "productClassification": {},
            "productCode": "WARRANTY",
            "productComponentGroups": [],
            "productRelatedComponent": {
              "childProductId": "01txx0000006jjIAAQ",
              "childSellingModelId": "0jPxx000000001dEAA",
              "doesBundlePriceIncludeChild": false,
              "id": "0dSxx0000000001EAA",
              "isComponentRequired": false,
              "isDefaultComponent": true,
              "isQuantityEditable": true,
              "maxQuantity": 1,
              "parentProductId": "01txx0000006jkuAAA",
              "parentSellingModelId": "0jPxx000000001dEAA",
              "productComponentGroupId": "0y7xx0000000001AAA",
              "productRelationshipTypeId": "0yoxx00000001IfAAI",
              "quantity": 1,
              "quantityScaleMethod": "Proportional",
              "sequence": 0
            },
            "productSellingModelOptions": [
              {
                "id": "0iOxx000000001dEAA",
                "productId": "01txx0000006jjIAAQ",
                "productSellingModel": {
                  "id": "0jPxx000000001dEAA",
                  "name": "One Time",
                  "sellingModelType": "OneTime",
                  "status": "Active"
                },
                "productSellingModelId": "0jPxx000000001dEAA"
              },
              {
                "id": "0iOxx00000000HlEAI",
                "productId": "01txx0000006jjIAAQ",
                "productSellingModel": {
                  "id": "0jPxx000000003FEAQ",
                  "name": "Termed Monthly",
                  "pricingTerm": 1,
```

```
                "pricingTermUnit": "Months",
                "sellingModelType": "TermDefined",
                "status": "Active"
              },
              "productSellingModelId": "0jPxx000000003FEAQ"
            },
            {
              "id": "0iOxx00000000JNEAY",
              "productId": "01txx0000006jjIAAQ",
              "productSellingModel": {
                "id": "0jPxx000000004rEAA",
                "name": "Termed Annually",
                "pricingTerm": 1,
                "pricingTermUnit": "Annual",
                "sellingModelType": "TermDefined",
                "status": "Active"
              },
              "productSellingModelId": "0jPxx000000004rEAA"
            },
            {
              "id": "0iOxx00000000KzEAI",
              "productId": "01txx0000006jjIAAQ",
              "productSellingModel": {
                "id": "0jPxx0000000085EAA",
                "name": "Evergreen Annually",
                "pricingTerm": 1,
                "pricingTermUnit": "Annual",
                "sellingModelType": "Evergreen",
                "status": "Active"
              },
              "productSellingModelId": "0jPxx0000000085EAA"
            },
            {
              "id": "0iOxx00000000MbEAI",
              "productId": "01txx0000006jjIAAQ",
              "productSellingModel": {
                "id": "0jPxx000000006TEAQ",
                "name": "Evergreen Monthly",
                "pricingTerm": 1,
                "pricingTermUnit": "Months",
                "sellingModelType": "Evergreen",
                "status": "Active"
              },
              "productSellingModelId": "0jPxx000000006TEAQ"
            }
          ]
        }
      ],
      "description": "The warranties available for the Cisco Server Rack NX44 product
 provide comprehensive coverage and support for optimal performance and reliability,
ensuring peace of mind for your data center infrastructure.",
      "id": "0y7xx0000000001AAA",
      "maxBundleComponents": 1,
      "minBundleComponents": 0,
```

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        "name": "Warranties",
        "parentProductId": "01txx0000006jkuAAA",
        "sequence": 0
      }
    ],
    "productSellingModelOptions": [
      {
        "id": "0iOxx000000003FEAQ",
        "productId": "01txx0000006jkuAAA",
        "productSellingModel": {
          "id": "0jPxx000000001dEAA",
          "name": "One Time",
          "sellingModelType": "OneTime",
          "status": "Active"
        },
        "productSellingModelId": "0jPxx000000001dEAA"
      },
      {
        "id": "0iOxx000000004rEAA",
        "productId": "01txx0000006jkuAAA",
        "productSellingModel": {
          "id": "0jPxx000000003FEAQ",
          "name": "Termed Monthly",
          "pricingTerm": 1,
          "pricingTermUnit": "Months",
          "sellingModelType": "TermDefined",
          "status": "Active"
        },
        "productSellingModelId": "0jPxx000000003FEAQ"
      },
      {
        "id": "0iOxx000000006TEAQ",
        "productId": "01txx0000006jkuAAA",
        "productSellingModel": {
          "id": "0jPxx000000004rEAA",
          "name": "Termed Annually",
          "pricingTerm": 1,
          "pricingTermUnit": "Annual",
          "sellingModelType": "TermDefined",
          "status": "Active"
        },
        "productSellingModelId": "0jPxx000000004rEAA"
      }
    ],
    "productType": "Bundle"
  },
  "errors": [],
  "success": true,
  "transactionContext": {
    "SalesTransaction": [
      {
        "Status": "Draft",
        "Account": "001xx000003GeIxAAK",
        "BillingCity": "San Francisco",
```

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        "Subtotal": 152500,
        "LastPricedDate": "2023-08-22T05:55:39Z",
        "businessObjectType": "Quote",
        "TotalAmount": 152500,
        "ShippingStreet": "415 Mission St",
        "SalesTransactionItem": [
          {
            "ProrationPolicy": null,
            "Discount": null,
            "ProductSellingModel": "0jPxx000000001dEAA",
            "Product": "01txx0000006jkuAAA",
            "businessObjectType": "QuoteLineItem",
            "BasisTransactionItem": null,
            "PartnerUnitPrice": null,
            "StartingUnitPriceSource": "System",
            "ListPrice": 150000,
            "ItemTotalAdjustmentAmount": 0,
            "SalesTransactionItemSource": "0QLxx0000004CQmGAM",
            "SubscriptionTerm": null,
            "StartDate": null,
            "NetTotalPrice": 150000,
            "TotalLineAmount": 150000,
            "PeriodBoundaryStartMonth": null,
            "ListPriceTotal": 150000,
            "PartnerDiscountPercent": null,
            "id": "0QLxx0000004CQmGAM",
            "PriceWaterFall": "{
                "currencyCode":"USD",
                "executionEndTimestamp":"2023-09-18T20:11:15.016Z",
                "executionId":"ruepwmHn2ZFvnQo5bjot",
                "executionStartTimestamp":"2023-09-18T20:11:14.906Z",
                "lineItemId":"0QLxx0000004CQmGAM",
                "output":{
                    "NetUnitPrice":150000.0,
                    "Subtotal":0.0
                 },
                "success":true,
                "waterfall":[
                {
                "fieldToTagNameMapping":{
                "NetUnitPrice":"ItemUnitPrice",
                "AdjustmentValue":"ItemAdjustmentValue",
                "Subtotal":"ItemTotalAdjustmentAmount",
                "Quantity":"ItemQuantity",
                "LineItemId":"SalesTransactionItemSource",
                "InputUnitPrice":"ItemUnitPrice"
                },
                "inputParameters":{
                    "Quantity":1.0,
                    "LineItemId":"0QLxx0000004CQmGAM",
                    "InputUnitPrice":150000.0,
                    "AdjustmentType":"Amount"
                },
                "outputParameters":{
```

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        "NetUnitPrice":150000.0,
        "Subtotal":0.0
    },
  "pricingElement":{
    "adjustments":[
      {
        "AdjustmentType":"Amount"
      }
  ],
  "elementType":"MANUALDISCOUNT",
  "name":"ManualDiscount"
  },
  "sequence":1
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  ]
  }",
  "BillingFrequency": null,
  "SalesTransactionItemParent": "0Q0xx0000004CAeCAM",
  "StartingPriceTotal": 150000,
  "Quantity": 1,
  "PeriodBoundary": null,
  "SalesTransactionItemAttribute": [
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      "AttributeKey": "0tjxx00000001H3AAI",
      "AttributeValue": "30.0",
      "AttributePicklistValue": null,
      "IsPriceImpacting": true,
      "businessObjectType": "QuoteLineItemAttribute",
      "AttributeName": "Expansion Slots",
      "AttributeDefinitionCode": "SLOTCAP",
      "id": "0zuxx0000000001AAA",
      "SalesTransactionItemAttrParent": "0QLxx0000004CQmGAM"
    }
  ],
  "EndDate": null,
  "DiscountAmount": null,
  "PricebookEntry": "01uxx0000009154AAA",
  "PricingTermCount": 1,
  "NetUnitPrice": 150000,
  "UnitPrice": 150000,
  "StartingUnitPrice": 150000,
  "SalesTrxnItemRelationship": [
    {
      "ProductRelationshipType": "0yoxx00000001IfAAI",
      "MainItemRole": "Bundle",
      "AssociatedItem": "0QLxx0000004CQnGAM",
      "ProductRelatedComponent": "0dSxx0000000001EAA",
      "MainItem": "0QLxx0000004CQmGAM",
      "AssociatedQuantScaleMethod": "Proportional",
      "businessObjectType": "QuoteLineRelationship",
      "AssociatedItemRole": "BundleComponent",
      "SalesTrnItemRelationshipParent": "0Q0xx0000004CAeCAM",
      "id": "0yQxx000000001dEAA",
      "AssociatedItemPricing": "NotIncludedInBundlePrice"
```

```
        }
      ],
      "TotalPrice": 150000,
      "PeriodBoundaryDay": null
    },
    {
      "ProrationPolicy": null,
      "Discount": null,
      "ProductSellingModel": "0jPxx000000001dEAA",
      "Product": "01txx0000006jjIAAQ",
      "businessObjectType": "QuoteLineItem",
      "BasisTransactionItem": null,
      "PartnerUnitPrice": null,
      "StartingUnitPriceSource": "System",
      "ListPrice": 2000,
      "ItemTotalAdjustmentAmount": 0,
      "SalesTransactionItemSource": "0QLxx0000004CQnGAM",
      "SubscriptionTerm": null,
      "StartDate": null,
      "NetTotalPrice": 2000,
      "TotalLineAmount": 2000,
      "PeriodBoundaryStartMonth": null,
      "ListPriceTotal": 2000,
      "PartnerDiscountPercent": null,
      "id": "0QLxx0000004CQnGAM",
      "PriceWaterFall": "{
          "currencyCode":"USD",
          "executionEndTimestamp":"023-09-18T20:11:15.016Z",
          "executionId":"ruepwmHn2ZFvnQo5bjot",
          "executionStartTimestamp":"2023-09-18T20:11:14.906Z",
          "lineItemId":"0QLxx0000004CQnGAM",
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              "Subtotal":0.0
           },
          "success":true,
          "waterfall":[
          {
          "fieldToTagNameMapping":{
          "NetUnitPrice":"ItemUnitPrice",
          "AdjustmentValue":"ItemAdjustmentValue",
          "Subtotal":"ItemTotalAdjustmentAmount",
          "Quantity":"ItemQuantity",
          "LineItemId":"SalesTransactionItemSource",
          "InputUnitPrice":"ItemUnitPrice"
          },
          "inputParameters":{
              "Quantity":1.0,
              "LineItemId":"0QLxx0000004CQnGAM",
              "InputUnitPrice":2000.0,
              "AdjustmentType":"Amount"
          },
          "outputParameters":{
              "NetUnitPrice":2000.0,
```

```
                    "Subtotal":0.0
                },
            "pricingElement":{
                "adjustments":[
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              ],
            "elementType":"MANUALDISCOUNT",
            "name":"ManualDiscount"
            },
            "sequence":1
            }
            ]
            }",
            "BillingFrequency": null,
            "SalesTransactionItemParent": "0Q0xx0000004CAeCAM",
            "StartingPriceTotal": 2000,
            "Quantity": 1,
            "PeriodBoundary": null,
            "EndDate": null,
            "DiscountAmount": null,
            "PricebookEntry": "01uxx000000913SAAQ",
            "PricingTermCount": 1,
            "NetUnitPrice": 2000,
            "UnitPrice": 2000,
            "StartingUnitPrice": 2000,
            "TotalPrice": 2000,
            "PeriodBoundaryDay": null
          }
        ],
        "BillingCountry": "US",
        "BillingStreet": "415 Mission St",
        "Pricebook": "01sxx0000005uDZAAY",
        "ShippingPostalCode": "94105",
        "SalesTransactionSource": "0Q0xx0000004CAeCAM",
        "ShippingCountry": "US",
        "ShippingCity": "San Francisco",
        "ShippingState": "CA",
        "BillingPostalCode": "94105",
        "id": "0Q0xx0000004CAeCAM",
        "BillToContact": null,
        "Contract": null,
        "BillingState": "CA"
      }
    ]
```


Property Name	Type	Description	Filter Group and Version	Available Version
`attributeCategoryId`	String	ID of the attribute category.	Small, 60.0	60.0
`attributeNameOverride`	String	Name override value of the attribute.	Small, 60.0	60.0

Property Name	Type	Description	Filter Group and Version	Available Version
attribute Picklist	Configurator Attribute Picklist[]	Picklist values of the attribute.	Small, 60.0	60.0
code	String	Code of the attribute.	Small, 60.0	60.0
dataType	String	Data type of the attribute.	Small, 60.0	60.0
default HelpText	String	Default help text value of the attribute.	Small, 60.0	60.0
defaultValue	String	Default value of the attribute.	Small, 60.0	60.0
description	String	Description of the attribute.	Small, 60.0	60.0
displayType	String	Display type of the attribute.	Small, 60.0	60.0
id	String	ID of the attribute.	Small, 60.0	60.0
isCloneable	Boolean	Indicates if the attribute is cloneable (`true`) or not (`false`).	Small, 60.0	60.0
is Configurable	Boolean	Indicates if the attribute is configurable (`true`) or not (`false`).	Small, 60.0	60.0
isEncrypted	Boolean	Indicates if the attribute is encrypted (`true`) or not (`false`).	Small, 60.0	60.0
isHidden	Boolean	Indicates if the attribute is hidden (`true`) or not (`false`).	Small, 60.0	60.0
isPrice Impacting	Boolean	Indicates if this is a price impacting attribute (`true`) or not (`false`).	Small, 60.0	60.0
isReadOnly	Boolean	Indicates if the attribute is read-only (`true`) or not (`false`).	Small, 60.0	60.0
isRequired	Boolean	Indicates if the attribute is required (`true`) or not (`false`).	Small, 60.0	60.0
label	String	Label of the attribute.	Small, 60.0	60.0
maximumValue	String	Maximum value for the product attribute.	Small, 60.0	60.0
minimumValue	String	Minimum value for the product attribute.	Small, 60.0	60.0
name	String	Name of the attribute.	Small, 60.0	60.0
sequence	Integer	Sequence values of the attribute.	Small, 60.0	60.0
status	String	Status of the attribute.	Small, 60.0	60.0
stepValue	String	Step value for the attribute.	Small, 60.0	60.0
unitOfMeasure	Configurator Unit Of Measure[]	Details about the unit of measure associated with an attribute.	Small, 63.0	63.0

Property Name	Type	Description	Filter Group and Version	Available Version
userValue	String	User value of the attribute.	Small, 60.0	60.0
valueDecoder	String	Value decoder for the attribute.	Small, 60.0	60.0
value Description	String	Value description of the attribute.	Small, 60.0	60.0

Configurator Attribute Category

Output representation of the attribute category in a product configuration.

JSON example


```
   "attributeCategories": [
     {
       "attributes": [
         {
           "attributeCategoryId": "0v3xx0000000001AAA",
           "attributeNameOverride": "Load Capacity",
           "code": "CAP",
           "dataType": "NUMBER",
           "defaultValue": "1500",
          "description": "Server racks are designed to support a specific load capacity,
commonly measured in kilograms (kg) or pounds (lbs). Typical load capacities range from
500 kg (1102 lbs) to 1500 kg (3307 lbs) depending on the model.",
           "id": "0tjxx00000001DpAAI",
           "isCloneable": false,
           "isConfigurable": true,
           "isHidden": false,
           "isPriceImpacting": false,
           "isReadOnly": false,
           "isRequired": false,
           "label": "Load Capacity",
           "name": "Load Capacity"
         },
         {
           "attributeCategoryId": "0v3xx0000000001AAA",
           "attributeNameOverride": "Expansion Slots",
           "code": "SLOTCAP",
           "dataType": "NUMBER",
           "defaultValue": "12",
           "id": "0tjxx00000001H3AAI",
           "isCloneable": false,
           "isConfigurable": true,
           "isHidden": false,
           "isPriceImpacting": false,
           "isReadOnly": false,
           "isRequired": true,
           "label": "Expansion Slots",
           "name": "Expansion Slots"
```

```
        },
        {
          "attributeCategoryId": "0v3xx0000000001AAA",
          "attributeNameOverride": "Memory",
          "attributePicklist": {
            "id": "0v5xx0000000001AAA",
            "values": [
              {
                "code": "MEM",
                "displayValue": "25",
                "id": "0v6xx0000000001AAA",
                "isBooleanValue": false,
                "name": "25Mem",
                "sequence": 0,
                "textValue": "25"
              },
              {
                "code": "50MEM",
                "displayValue": "50",
                "id": "0v6xx000000001eAAA",
                "isBooleanValue": false,
                "name": "50Mem",
                "sequence": 1,
                "textValue": "50"
              },
              {
                "code": "100MEM",
                "displayValue": "100",
                "id": "0v6xx000000003FAAQ",
                "isBooleanValue": false,
                "name": "100Mem",
                "sequence": 2,
                "textValue": "100"
              }
            ]
          },
          "dataType": "MULTIPICKLIST",
          "defaultValue": "25",
          "id": "0tjxx00000001IfAAI",
          "isCloneable": false,
          "isConfigurable": true,
          "isHidden": false,
          "isPriceImpacting": false,
          "isReadOnly": false,
          "isRequired": true,
          "label": "Memory",
          "name": "Memory"
        }
      ],
      "code": "SPEC",
      "name": "Server Rack Specifications"
    }
  ],
  "description": "Introducing the Cisco Server Rack, a sleek and robust solution
```

```
designed to streamline your data center infrastructure. With its scalable design and
advanced cable management features, it ensures optimal performance, efficiency, and easy
 maintenance for your critical network equipment.",
    "displayUrl":
"https://www.cisco.com/content/dam/en/us/products/servers-unified-computing/ucs-c240-m4-rack-server/product-large.jpg",

    "id": "01txx0000006jkuAAA",
    "isActive": true,
    "isAssetizable": true,
    "isConfigurable": true,
    "isSoldOnlyWithOtherProds": false,
    "name": "Cisco Server Rack NX44",
    "nodeType": "bundleProduct",
    "prices": [],
    "productClassification": {
      "id": "11Bxx000002CC02EAG"
    },
    "productCode": "RACK",
    "productComponentGroups": [
      {
        "classifications": [],
        "code": "SERVICE",
        "components": [
          {
            "additionalFields": [],
            "attributeCategories": [],
            "description": "Introducing the Cisco Rack Server NX44 Service, a
comprehensive protection plan designed to safeguard your valuable data infrastructure.
 With extended coverage and rapid response times, this warranty ensures peace of mind
and uninterrupted performance for your critical business operations.",
            "id": "01txx0000006jmWAAQ",
            "isActive": true,
            "isAssetizable": true,
            "isComponentRequired": false,
            "isConfigurable": false,
            "isDefaultComponent": false,
            "isQuantityEditable": false,
            "isSoldOnlyWithOtherProds": false,
            "name": "Cisco Rack Server Service - 1 Year",
            "nodeType": "simpleProduct",
            "prices": [],
            "productClassification": {},
            "productCode": "SERVICE",
            "productComponentGroups": [],
            "productRelatedComponent": {
              "childProductId": "01txx0000006jmWAAQ",
              "childSellingModelId": "0jPxx000000004rEAA",
              "doesBundlePriceIncludeChild": true,
              "id": "0dSxx000000001dEAA",
              "isComponentRequired": false,
              "isDefaultComponent": false,
              "isQuantityEditable": false,
              "parentProductId": "01txx0000006jkuAAA",
              "parentSellingModelId": "0jPxx000000004rEAA",
```

```
              "productComponentGroupId": "0y7xx000000001dAAA",
              "productRelationshipTypeId": "0yoxx00000001IfAAI",
              "quantity": 1,
              "quantityScaleMethod": "Proportional"
            },
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                "id": "0iOxx000000009hEAA",
                "productId": "01txx0000006jmWAAQ",
                "productSellingModel": {
                  "id": "0jPxx000000004rEAA",
                  "name": "Termed Annually",
                  "pricingTerm": 1,
                  "pricingTermUnit": "Annual",
                  "sellingModelType": "TermDefined",
                  "status": "Active"
                },
                "productSellingModelId": "0jPxx000000004rEAA"
              },
              {
                "id": "0iOxx00000000PpEAI",
                "productId": "01txx0000006jmWAAQ",
                "productSellingModel": {
                  "id": "0jPxx0000000085EAA",
                  "name": "Evergreen Annually",
                  "pricingTerm": 1,
                  "pricingTermUnit": "Annual",
                  "sellingModelType": "Evergreen",
                  "status": "Active"
                },
                "productSellingModelId": "0jPxx0000000085EAA"
              }
            ]
          }
        ],
        "description": "The services available for the Cisco Server Rack NX44 product
provide comprehensive coverage and support for optimal performance and reliability,
ensuring peace of mind for your data center infrastructure.",
        "id": "0y7xx000000001dAAA",
        "maxBundleComponents": 1,
        "minBundleComponents": 0,
        "name": "Services",
        "parentProductId": "01txx0000006jkuAAA",
        "sequence": 1
      },
      {
        "classifications": [],
        "code": "WARRANTY",
        "components": [
          {
            "additionalFields": [],
            "attributeCategories": [],
            "description": "Introducing the Cisco Rack Server NX44 Warranty, a
comprehensive protection plan designed to safeguard your valuable data infrastructure.
```

```
 With extended coverage and rapid response times, this warranty ensures peace of mind
and uninterrupted performance for your critical business operations.",
            "id": "01txx0000006jjIAAQ",
            "isActive": true,
            "isAssetizable": true,
            "isComponentRequired": false,
            "isConfigurable": false,
            "isDefaultComponent": true,
            "isQuantityEditable": true,
            "isSoldOnlyWithOtherProds": false,
            "name": "Cisco Rack Server Warranty - 1 Year",
            "nodeType": "simpleProduct",
            "prices": [],
            "productClassification": {},
            "productCode": "WARRANTY",
            "productComponentGroups": [],
            "productRelatedComponent": {
              "childProductId": "01txx0000006jjIAAQ",
              "childSellingModelId": "0jPxx000000001dEAA",
              "doesBundlePriceIncludeChild": false,
              "id": "0dSxx0000000001EAA",
              "isComponentRequired": false,
              "isDefaultComponent": true,
              "isQuantityEditable": true,
              "maxQuantity": 1,
              "parentProductId": "01txx0000006jkuAAA",
              "parentSellingModelId": "0jPxx000000001dEAA",
              "productComponentGroupId": "0y7xx0000000001AAA",
              "productRelationshipTypeId": "0yoxx00000001IfAAI",
              "quantity": 1,
              "quantityScaleMethod": "Proportional",
              "sequence": 0
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            "productSellingModelOptions": [
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                "id": "0iOxx000000001dEAA",
                "productId": "01txx0000006jjIAAQ",
                "productSellingModel": {
                  "id": "0jPxx000000001dEAA",
                  "name": "One Time",
                  "sellingModelType": "OneTime",
                  "status": "Active"
                },
                "productSellingModelId": "0jPxx000000001dEAA"
              },
              {
                "id": "0iOxx00000000HlEAI",
                "productId": "01txx0000006jjIAAQ",
                "productSellingModel": {
                  "id": "0jPxx000000003FEAQ",
                  "name": "Termed Monthly",
                  "pricingTerm": 1,
                  "pricingTermUnit": "Months",
                  "sellingModelType": "TermDefined",
```

```
                  "status": "Active"
                },
                "productSellingModelId": "0jPxx000000003FEAQ"
              },
              {
                "id": "0iOxx00000000JNEAY",
                "productId": "01txx0000006jjIAAQ",
                "productSellingModel": {
                  "id": "0jPxx000000004rEAA",
                  "name": "Termed Annually",
                  "pricingTerm": 1,
                  "pricingTermUnit": "Annual",
                  "sellingModelType": "TermDefined",
                  "status": "Active"
                },
                "productSellingModelId": "0jPxx000000004rEAA"
              },
              {
                "id": "0iOxx00000000KzEAI",
                "productId": "01txx0000006jjIAAQ",
                "productSellingModel": {
                  "id": "0jPxx0000000085EAA",
                  "name": "Evergreen Annually",
                  "pricingTerm": 1,
                  "pricingTermUnit": "Annual",
                  "sellingModelType": "Evergreen",
                  "status": "Active"
                },
                "productSellingModelId": "0jPxx0000000085EAA"
              },
              {
                "id": "0iOxx00000000MbEAI",
                "productId": "01txx0000006jjIAAQ",
                "productSellingModel": {
                  "id": "0jPxx000000006TEAQ",
                  "name": "Evergreen Monthly",
                  "pricingTerm": 1,
                  "pricingTermUnit": "Months",
                  "sellingModelType": "Evergreen",
                  "status": "Active"
                },
                "productSellingModelId": "0jPxx000000006TEAQ"
              }
            ]
          }
        ],
        "description": "The warranties available for the Cisco Server Rack NX44 product
 provide comprehensive coverage and support for optimal performance and reliability,
ensuring peace of mind for your data center infrastructure.",
        "id": "0y7xx0000000001AAA",
        "maxBundleComponents": 1,
        "minBundleComponents": 0,
        "name": "Warranties",
        "parentProductId": "01txx0000006jkuAAA",
```

```
        "sequence": 0
      }
    ],
    "productSellingModelOptions": [
      {
        "id": "0iOxx000000003FEAQ",
        "productId": "01txx0000006jkuAAA",
        "productSellingModel": {
          "id": "0jPxx000000001dEAA",
          "name": "One Time",
          "sellingModelType": "OneTime",
          "status": "Active"
        },
        "productSellingModelId": "0jPxx000000001dEAA"
      },
      {
        "id": "0iOxx000000004rEAA",
        "productId": "01txx0000006jkuAAA",
        "productSellingModel": {
          "id": "0jPxx000000003FEAQ",
          "name": "Termed Monthly",
          "pricingTerm": 1,
          "pricingTermUnit": "Months",
          "sellingModelType": "TermDefined",
          "status": "Active"
        },
        "productSellingModelId": "0jPxx000000003FEAQ"
      },
      {
        "id": "0iOxx000000006TEAQ",
        "productId": "01txx0000006jkuAAA",
        "productSellingModel": {
          "id": "0jPxx000000004rEAA",
          "name": "Termed Annually",
          "pricingTerm": 1,
          "pricingTermUnit": "Annual",
          "sellingModelType": "TermDefined",
          "status": "Active"
        },
        "productSellingModelId": "0jPxx000000004rEAA"
      }
    ],
    "productType": "Bundle"
  },
  "errors": [],
  "success": true,
  "transactionContext": {
    "SalesTransaction": [
      {
        "Status": "Draft",
        "Account": "001xx000003GeIxAAK",
        "BillingCity": "San Francisco",
        "Subtotal": 152500,
        "LastPricedDate": "2023-08-22T05:55:39Z",
```

```
        "businessObjectType": "Quote",
        "TotalAmount": 152500,
        "ShippingStreet": "415 Mission St",
        "SalesTransactionItem": [
          {
            "ProrationPolicy": null,
            "Discount": null,
            "ProductSellingModel": "0jPxx000000001dEAA",
            "Product": "01txx0000006jkuAAA",
            "businessObjectType": "QuoteLineItem",
            "BasisTransactionItem": null,
            "PartnerUnitPrice": null,
            "StartingUnitPriceSource": "System",
            "ListPrice": 150000,
            "ItemTotalAdjustmentAmount": 0,
            "SalesTransactionItemSource": "0QLxx0000004CQmGAM",
            "SubscriptionTerm": null,
            "StartDate": null,
            "NetTotalPrice": 150000,
            "TotalLineAmount": 150000,
            "PeriodBoundaryStartMonth": null,
            "ListPriceTotal": 150000,
            "PartnerDiscountPercent": null,
            "id": "0QLxx0000004CQmGAM",
            "PriceWaterFall": "{
                "currencyCode":"USD",
                "executionEndTimestamp":"2023-09-18T20:11:15.016Z",
                "executionId":"ruepwmHn2ZFvnQo5bjot",
                "executionStartTimestamp":"2023-09-18T20:11:14.906Z",
                "lineItemId":"0QLxx0000004CQmGAM",
                "output":{
                    "NetUnitPrice":150000.0,
                    "Subtotal":0.0
                 },
                "success":true,
                "waterfall":[
                {
                "fieldToTagNameMapping":{
                "NetUnitPrice":"ItemUnitPrice",
                "AdjustmentValue":"ItemAdjustmentValue",
                "Subtotal":"ItemTotalAdjustmentAmount",
                "Quantity":"ItemQuantity",
                "LineItemId":"SalesTransactionItemSource",
                "InputUnitPrice":"ItemUnitPrice"
                },
                "inputParameters":{
                    "Quantity":1.0,
                    "LineItemId":"0QLxx0000004CQmGAM",
                    "InputUnitPrice":150000.0,
                    "AdjustmentType":"Amount"
                },
                "outputParameters":{
                    "NetUnitPrice":150000.0,
                    "Subtotal":0.0
```

```
    },
"pricingElement":{
    "adjustments":[
      {
        "AdjustmentType":"Amount"
      }
  ],
"elementType":"MANUALDISCOUNT",
"name":"ManualDiscount"
},
"sequence":1
}
]
}",
"BillingFrequency": null,
"SalesTransactionItemParent": "0Q0xx0000004CAeCAM",
"StartingPriceTotal": 150000,
"Quantity": 1,
"PeriodBoundary": null,
"SalesTransactionItemAttribute": [
  {
    "AttributeKey": "0tjxx00000001H3AAI",
    "AttributeValue": "30.0",
    "AttributePicklistValue": null,
    "IsPriceImpacting": true,
    "businessObjectType": "QuoteLineItemAttribute",
    "AttributeName": "Expansion Slots",
    "AttributeDefinitionCode": "SLOTCAP",
    "id": "0zuxx0000000001AAA",
    "SalesTransactionItemAttrParent": "0QLxx0000004CQmGAM"
  }
],
"EndDate": null,
"DiscountAmount": null,
"PricebookEntry": "01uxx0000009154AAA",
"PricingTermCount": 1,
"NetUnitPrice": 150000,
"UnitPrice": 150000,
"StartingUnitPrice": 150000,
"SalesTrxnItemRelationship": [
  {
    "ProductRelationshipType": "0yoxx00000001IfAAI",
    "MainItemRole": "Bundle",
    "AssociatedItem": "0QLxx0000004CQnGAM",
    "ProductRelatedComponent": "0dSxx0000000001EAA",
    "MainItem": "0QLxx0000004CQmGAM",
    "AssociatedQuantScaleMethod": "Proportional",
    "businessObjectType": "QuoteLineRelationship",
    "AssociatedItemRole": "BundleComponent",
    "SalesTrnItemRelationshipParent": "0Q0xx0000004CAeCAM",
    "id": "0yQxx000000001dEAA",
    "AssociatedItemPricing": "NotIncludedInBundlePrice"
  }
],
```

```
        "TotalPrice": 150000,
        "PeriodBoundaryDay": null
      },
      {
        "ProrationPolicy": null,
        "Discount": null,
        "ProductSellingModel": "0jPxx000000001dEAA",
        "Product": "01txx0000006jjIAAQ",
        "businessObjectType": "QuoteLineItem",
        "BasisTransactionItem": null,
        "PartnerUnitPrice": null,
        "StartingUnitPriceSource": "System",
        "ListPrice": 2000,
        "ItemTotalAdjustmentAmount": 0,
        "SalesTransactionItemSource": "0QLxx0000004CQnGAM",
        "SubscriptionTerm": null,
        "StartDate": null,
        "NetTotalPrice": 2000,
        "TotalLineAmount": 2000,
        "PeriodBoundaryStartMonth": null,
        "ListPriceTotal": 2000,
        "PartnerDiscountPercent": null,
        "id": "0QLxx0000004CQnGAM",
        "PriceWaterFall": "{
            "currencyCode":"USD",
            "executionEndTimestamp":"023-09-18T20:11:15.016Z",
            "executionId":"ruepwmHn2ZFvnQo5bjot",
            "executionStartTimestamp":"2023-09-18T20:11:14.906Z",
            "lineItemId":"0QLxx0000004CQnGAM",
            "output":{
                "NetUnitPrice":2000.0,
                "Subtotal":0.0
             },
            "success":true,
            "waterfall":[
            {
            "fieldToTagNameMapping":{
            "NetUnitPrice":"ItemUnitPrice",
            "AdjustmentValue":"ItemAdjustmentValue",
            "Subtotal":"ItemTotalAdjustmentAmount",
            "Quantity":"ItemQuantity",
            "LineItemId":"SalesTransactionItemSource",
            "InputUnitPrice":"ItemUnitPrice"
            },
            "inputParameters":{
                "Quantity":1.0,
                "LineItemId":"0QLxx0000004CQnGAM",
                "InputUnitPrice":2000.0,
                "AdjustmentType":"Amount"
            },
            "outputParameters":{
                "NetUnitPrice":2000.0,
                "Subtotal":0.0
            },
```

```
            "pricingElement":{
                "adjustments":[
                  {}
              ],
            "elementType":"MANUALDISCOUNT",
            "name":"ManualDiscount"
            },
            "sequence":1
            }
            ]
            }",
            "BillingFrequency": null,
            "SalesTransactionItemParent": "0Q0xx0000004CAeCAM",
            "StartingPriceTotal": 2000,
            "Quantity": 1,
            "PeriodBoundary": null,
            "EndDate": null,
            "DiscountAmount": null,
            "PricebookEntry": "01uxx000000913SAAQ",
            "PricingTermCount": 1,
            "NetUnitPrice": 2000,
            "UnitPrice": 2000,
            "StartingUnitPrice": 2000,
            "TotalPrice": 2000,
            "PeriodBoundaryDay": null
          }
        ],
        "BillingCountry": "US",
        "BillingStreet": "415 Mission St",
        "Pricebook": "01sxx0000005uDZAAY",
        "ShippingPostalCode": "94105",
        "SalesTransactionSource": "0Q0xx0000004CAeCAM",
        "ShippingCountry": "US",
        "ShippingCity": "San Francisco",
        "ShippingState": "CA",
        "BillingPostalCode": "94105",
        "id": "0Q0xx0000004CAeCAM",
        "BillToContact": null,
        "Contract": null,
        "BillingState": "CA"
    }
]
```


Property Name	Type	Description	Filter Group and Version	Available Version
`attributes`	Configurator Attribute[]	Attributes of the attribute category.	Small, 60.0	60.0
`code`	String	Code of the attribute category.	Small, 60.0	60.0
`description`	String	Description of the attribute category.	Small, 60.0	60.0
`id`	String	ID of the attribute category.	Small, 60.0	60.0

Property Name	Type	Description	Filter Group and Version	Available Version
name	String	Name of the attribute category.	Small, 60.0	60.0
status	String	Status of the attribute category.	Small, 60.0	60.0
totalSize	Integer	Total size of the attribute category.	Small, 60.0	60.0
usageType	String	Usage type of the attribute category.	Small, 60.0	60.0

Configurator Attribute Picklist

Output representation of the attribute picklist in a product configuration.

Property Name	Type	Description	Filter Group and Version	Available Version
description	String	Description of the attribute picklist.	Small, 60.0	60.0
id	String	ID of the attribute picklist.	Small, 60.0	60.0
name	String	Name of the attribute picklist.	Small, 60.0	60.0
status	String	Status of the attribute picklist.	Small, 60.0	60.0
values	Configurator Attribute Picklist Value[]	Values of the attribute picklist.	Small, 60.0	60.0

Configurator Attribute Picklist Value

Output representation of the values of an attribute picklist in a product configuration.

Property Name	Type	Description	Filter Group and Version	Available Version
code	String	Code of the attribute value.	Small, 60.0	60.0
description	String	Description of the attribute value.	Small, 60.0	60.0
displayValue	String	Display value of the attribute value.	Small, 60.0	60.0
id	String	ID of the attribute value.	Small, 60.0	60.0
isBooleanValue	Boolean	Indicates if this attribute value is a boolean value (true) or not (false).	Small, 60.0	60.0
label	String	Label of the attribute value.	Small, 60.0	60.0
name	String	Name of the attribute value.	Small, 60.0	60.0
sequence	Integer	Sequence of the attribute value.	Small, 60.0	60.0
status	String	Status of the attribute value.	Small, 60.0	60.0

Property Name	Type	Description	Filter Group and Version	Available Version
textValue	String	Text value of the attribute value.	Small, 60.0	60.0

Configurator Delete Nodes

Output representation of the details of the configuration request to delete nodes.

JSON Example


```
{
  "configuratorMessages": {},
  "configuratorUITreatments": [
    {
      "details": {
        "attributeId": "0tjxx0000000007AAA",
        "prcId": "0dSxx0000000007EAA",
        "stiId": "0QLxx0000004CU0GAM",
        "attributePicklistValueId": "0v6xx0000000005AAA"
      },
      "uiTreatmentScope": "Bundle",
      "uiTreatmentTarget": "Attribute_Picklist_Value",
      "uiTreatmentType": "Hide"
    },
    {
      "details": {
        "stiId": "ref_f0f2da7b_c431_482d_bf4b_599052f3a2e1"
      },
      "uiTreatmentScope": "Product",
      "uiTreatmentTarget": "Component",
      "uiTreatmentType": "Disable"
    }
  ],
  "errors": [],
  "productQualifications": {
    "01tDU000000EOTCYA4": {
      "isQualified": true
    }
  },
  "success": true
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
configurator Messages	Map<String, Configurator Message>	Map of the product IDs to the list of configurator messages. Configurator messages are results from any validations, Business Rules Engine (BRE) calls, or Salesforce Pricing calls.	Small, 60.0	60.0

Property Name	Type	Description	Filter Group and Version	Available Version
configurator UITreatments	Configurator UI Treatment[]	Details of the UI treatments that specify the product configuration rule actions to override the disable or hide behavior in the UI for product options, product attributes, and attribute picklist values.	Small, 62.0	62.0
errors	Error Response	List of errors, which contains an error code and a message.	Small, 60.0	60.0
product Qualifications	Map<String, Configurator Qualification Context>	Map of the product IDs to the qualification rule execution results.	Small, 60.0	60.0
success	Boolean	Indicates whether the call was successful (`true`) not (`false`).	Small, 60.0	60.0

Configurator Message

Output representation of the messages of a product configurator.

JSON example

```
{
  "messages": {
    "0Q0xx0000004CDsCAM": [
      {
        "message": "This is a quote with warranty",
        "messageType": "Info",
        "category": "ConfigurationRules",
        "primaryRecordId": "0Q0xx0000004CDsCAM"
      },
      {
        "message": "It is a group 1C9xx0000004CCGCA2",
        "messageType": "Warning",
        "category": "ConfigurationRules",
        "primaryRecordId": "0Q0xx0000004CDsCAM",
        "groupByValue": "1C9xx0000004CCGCA2"
      },
      {
        "message": "We're in group virtual bundle",
        "messageType": "Warning",
        "category": "ConfigurationRules",
        "primaryRecordId": "0Q0xx0000004CDsCAM",
        "groupByValue": "1C9xx0000004CCGCA2"
      },
      {
        "message": "It is a group 1C9xx0000004CCOCA2",
        "messageType": "Warning",
        "category": "ConfigurationRules",
        "primaryRecordId": "0Q0xx0000004CDsCAM",
```

```
        "groupByValue": "1C9xx0000004CCOCA2"
      },
      {
        "message": "We're in group virtual bundle",
        "messageType": "Warning",
        "category": "ConfigurationRules",
        "primaryRecordId": "0Q0xx0000004CDsCAM",
        "groupByValue": "1C9xx0000004CCOCA2"
      }
    ],
    "stiId-Laptop1": [
      {
        "message": "Only laptop",
        "messageType": "Info",
        "category": "ConfigurationRules",
        "primaryRecordId": "0Q0xx0000004CDsCAM",
        "relatedRecordId": "stiId-Laptop1"
      }
    ]
  }
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`category`	String	Category or type of the error message. Valid values are: • `ArcResolutionService` • `ArcValidationService` • `BundleValidation` • `ConfigurationRules` • `Pricing`	Small, 60.0	60.0
`groupByValue`	String	Specifies the value from Constraint Modeling Language rule action details.	Small, 67.0	67.0
`message`	String	Message that contains the error details.	Small, 60.0	60.0
`messageType`	String	Type of error message. Valid values are: • `Error` • `Info` • `Warning`	Small, 60.0	60.0
`primary RecordId`	String	Primary record ID that contains the error.	Small, 60.0	60.0
`related RecordId`	String	Related record ID for the error, if any.	Small, 60.0	60.0

Configurator Price

Output representation of the pricing details in a product configuration.

Property Name	Type	Description	Filter Group and Version	Available Version
currency IsoCode	String	Currency ISO code of the price book entry.	Small, 60.0	60.0
effectiveFrom	String	Date from when the price book entry is effective.	Small, 60.0	60.0
effectiveTo	String	Date until when the price book entry is effective.	Small, 60.0	60.0
isDefault	Boolean	Indicates if this price book entry is the default pricing model (true) or not (false).	Small, 60.0	60.0
isSelected	Boolean	Indicates if this price book entry is selected (true) or not (false).	Small, 60.0	60.0
pricebook EntryId	String	ID of the price book entry.	Small, 60.0	60.0
pricebookId	String	Pricebook2 ID of the price book entry.	Small, 60.0	60.0
pricingModel	Configurator Pricing Model[]	Pricing model details of the price book entry.	Small, 60.0	60.0
unitPrice	Double	Unit price of the price book entry.	Small, 60.0	60.0

Configurator Pricing Model

Output representation of the details of a pricing model in a product configuration.

Property Name	Type	Description	Filter Group and Version	Available Version
frequency	String	Frequency of the pricing model.	Small, 60.0	60.0
id	String	ID of the pricing model.	Small, 60.0	60.0
name	String	Name of the pricing model.	Small, 60.0	60.0
occurrence	Integer	Occurrence of the pricing model.	Small, 60.0	60.0
pricing ModelType	String	Type of the pricing model.	Small, 60.0	60.0
unitOfMeasure	String	Unit of measure for the pricing model.	Small, 60.0	60.0

Configurator Product Catalog

Output representation of the product catalog.

JSON example


```
 "catalogProducts": {
   "additionalFields": [],
   "attributeCategories": [
     {
       "attributes": [
         {
           "attributeCategoryId": "0v3xx0000000001AAA",
           "attributeNameOverride": "Load Capacity",
           "code": "CAP",
           "dataType": "NUMBER",
           "defaultValue": "1500",
          "description": "Server racks are designed to support a specific load capacity,
commonly measured in kilograms (kg) or pounds (lbs). Typical load capacities range from
500 kg (1102 lbs) to 1500 kg (3307 lbs) depending on the model.",
           "id": "0tjxx00000001DpAAI",
           "isCloneable": false,
           "isConfigurable": true,
           "isHidden": false,
           "isPriceImpacting": false,
           "isReadOnly": false,
           "isRequired": false,
           "label": "Load Capacity",
           "name": "Load Capacity"
         },
         {
           "attributeCategoryId": "0v3xx0000000001AAA",
           "attributeNameOverride": "Expansion Slots",
           "code": "SLOTCAP",
           "dataType": "NUMBER",
           "defaultValue": "12",
           "id": "0tjxx00000001H3AAI",
           "isCloneable": false,
           "isConfigurable": true,
           "isHidden": false,
           "isPriceImpacting": false,
           "isReadOnly": false,
           "isRequired": true,
           "label": "Expansion Slots",
           "name": "Expansion Slots"
         },
         {
           "attributeCategoryId": "0v3xx0000000001AAA",
           "attributeNameOverride": "Memory",
           "attributePicklist": {
             "id": "0v5xx0000000001AAA",
             "values": [
               {
                 "code": "MEM",
```

```
                  "displayValue": "25",
                  "id": "0v6xx0000000001AAA",
                  "isBooleanValue": false,
                  "name": "25Mem",
                  "sequence": 0,
                  "textValue": "25"
                },
                {
                  "code": "50MEM",
                  "displayValue": "50",
                  "id": "0v6xx000000001eAAA",
                  "isBooleanValue": false,
                  "name": "50Mem",
                  "sequence": 1,
                  "textValue": "50"
                },
                {
                  "code": "100MEM",
                  "displayValue": "100",
                  "id": "0v6xx000000003FAAQ",
                  "isBooleanValue": false,
                  "name": "100Mem",
                  "sequence": 2,
                  "textValue": "100"
                }
              ]
            },
            "dataType": "MULTIPICKLIST",
            "defaultValue": "25",
            "id": "0tjxx00000001IfAAI",
            "isCloneable": false,
            "isConfigurable": true,
            "isHidden": false,
            "isPriceImpacting": false,
            "isReadOnly": false,
            "isRequired": true,
            "label": "Memory",
            "name": "Memory"
          }
        ],
        "code": "SPEC",
        "name": "Server Rack Specifications"
      }
    ],
    "description": "Introducing the Cisco Server Rack, a sleek and robust solution
designed to streamline your data center infrastructure. With its scalable design and
advanced cable management features, it ensures optimal performance, efficiency, and easy
 maintenance for your critical network equipment.",
    "displayUrl":
"https://www.cisco.com/content/dam/en/us/products/servers-unified-computing/ucs-c240-m4-rack-server/product-large.jpg",

    "id": "01txx0000006jkuAAA",
    "isActive": true,
    "isAssetizable": true,
```

```
    "isConfigurable": true,
    "isSoldOnlyWithOtherProds": false,
    "name": "Cisco Server Rack NX44",
    "nodeType": "bundleProduct",
    "prices": [],
    "productClassification": {
      "id": "11Bxx000002CC02EAG"
    },
    "productCode": "RACK",
    "productComponentGroups": [
      {
        "classifications": [],
        "code": "SERVICE",
        "components": [
          {
            "additionalFields": [],
            "attributeCategories": [],
            "description": "Introducing the Cisco Rack Server NX44 Service, a
comprehensive protection plan designed to safeguard your valuable data infrastructure.
 With extended coverage and rapid response times, this warranty ensures peace of mind
and uninterrupted performance for your critical business operations.",
            "id": "01txx0000006jmWAAQ",
            "isActive": true,
            "isAssetizable": true,
            "isComponentRequired": false,
            "isConfigurable": false,
            "isDefaultComponent": false,
            "isQuantityEditable": false,
            "isSoldOnlyWithOtherProds": false,
            "name": "Cisco Rack Server Service - 1 Year",
            "nodeType": "simpleProduct",
            "prices": [],
            "productClassification": {},
            "productCode": "SERVICE",
            "productComponentGroups": [],
            "productRelatedComponent": {
              "childProductId": "01txx0000006jmWAAQ",
              "childSellingModelId": "0jPxx000000004rEAA",
              "doesBundlePriceIncludeChild": true,
              "id": "0dSxx000000001dEAA",
              "isComponentRequired": false,
              "isDefaultComponent": false,
              "isQuantityEditable": false,
              "parentProductId": "01txx0000006jkuAAA",
              "parentSellingModelId": "0jPxx000000004rEAA",
              "productComponentGroupId": "0y7xx000000001dAAA",
              "productRelationshipTypeId": "0yoxx00000001IfAAI",
              "quantity": 1,
              "quantityScaleMethod": "Proportional"
            },
            "productSellingModelOptions": [
              {
                "id": "0iOxx000000009hEAA",
                "productId": "01txx0000006jmWAAQ",
```

```
        "productSellingModel": {
          "id": "0jPxx000000004rEAA",
          "name": "Termed Annually",
          "pricingTerm": 1,
          "pricingTermUnit": "Annual",
          "sellingModelType": "TermDefined",
          "status": "Active"
        },
        "productSellingModelId": "0jPxx000000004rEAA"
      },
      {
        "id": "0iOxx00000000PpEAI",
        "productId": "01txx0000006jmWAAQ",
        "productSellingModel": {
          "id": "0jPxx0000000085EAA",
          "name": "Evergreen Annually",
          "pricingTerm": 1,
          "pricingTermUnit": "Annual",
          "sellingModelType": "Evergreen",
          "status": "Active"
        },
        "productSellingModelId": "0jPxx0000000085EAA"
      }
    ]
  }
],
"description": "The services available for the Cisco Server Rack NX44 product
provide comprehensive coverage and support for optimal performance and reliability,
ensuring peace of mind for your data center infrastructure.",
"id": "0y7xx000000001dAAA",
"maxBundleComponents": 1,
"minBundleComponents": 0,
"name": "Services",
"parentProductId": "01txx0000006jkuAAA",
"sequence": 1
},
{
"classifications": [],
"code": "WARRANTY",
"components": [
  {
    "additionalFields": [],
    "attributeCategories": [],
    "description": "Introducing the Cisco Rack Server NX44 Warranty, a
comprehensive protection plan designed to safeguard your valuable data infrastructure.
 With extended coverage and rapid response times, this warranty ensures peace of mind
and uninterrupted performance for your critical business operations.",
    "id": "01txx0000006jjIAAQ",
    "isActive": true,
    "isAssetizable": true,
    "isComponentRequired": false,
    "isConfigurable": false,
    "isDefaultComponent": true,
    "isQuantityEditable": true,
```

```
"isSoldOnlyWithOtherProds": false,
"name": "Cisco Rack Server Warranty - 1 Year",
"nodeType": "simpleProduct",
"prices": [],
"productClassification": {},
"productCode": "WARRANTY",
"productComponentGroups": [],
"productRelatedComponent": {
  "childProductId": "01txx0000006jjIAAQ",
  "childSellingModelId": "0jPxx000000001dEAA",
  "doesBundlePriceIncludeChild": false,
  "id": "0dSxx0000000001EAA",
  "isComponentRequired": false,
  "isDefaultComponent": true,
  "isQuantityEditable": true,
  "maxQuantity": 1,
  "parentProductId": "01txx0000006jkuAAA",
  "parentSellingModelId": "0jPxx000000001dEAA",
  "productComponentGroupId": "0y7xx0000000001AAA",
  "productRelationshipTypeId": "0yoxx00000001IfAAI",
  "quantity": 1,
  "quantityScaleMethod": "Proportional",
  "sequence": 0
},
"productSellingModelOptions": [
  {
    "id": "0iOxx000000001dEAA",
    "productId": "01txx0000006jjIAAQ",
    "productSellingModel": {
      "id": "0jPxx000000001dEAA",
      "name": "One Time",
      "sellingModelType": "OneTime",
      "status": "Active"
    },
    "productSellingModelId": "0jPxx000000001dEAA"
  },
  {
    "id": "0iOxx00000000HlEAI",
    "productId": "01txx0000006jjIAAQ",
    "productSellingModel": {
      "id": "0jPxx000000003FEAQ",
      "name": "Termed Monthly",
      "pricingTerm": 1,
      "pricingTermUnit": "Months",
      "sellingModelType": "TermDefined",
      "status": "Active"
    },
    "productSellingModelId": "0jPxx000000003FEAQ"
  },
  {
    "id": "0iOxx00000000JNEAY",
    "productId": "01txx0000006jjIAAQ",
    "productSellingModel": {
      "id": "0jPxx000000004rEAA",
```

```
                  "name": "Termed Annually",
                  "pricingTerm": 1,
                  "pricingTermUnit": "Annual",
                  "sellingModelType": "TermDefined",
                  "status": "Active"
                },
                "productSellingModelId": "0jPxx000000004rEAA"
              },
              {
                "id": "0iOxx00000000KzEAI",
                "productId": "01txx0000006jjIAAQ",
                "productSellingModel": {
                  "id": "0jPxx0000000085EAA",
                  "name": "Evergreen Annually",
                  "pricingTerm": 1,
                  "pricingTermUnit": "Annual",
                  "sellingModelType": "Evergreen",
                  "status": "Active"
                },
                "productSellingModelId": "0jPxx0000000085EAA"
              },
              {
                "id": "0iOxx00000000MbEAI",
                "productId": "01txx0000006jjIAAQ",
                "productSellingModel": {
                  "id": "0jPxx000000006TEAQ",
                  "name": "Evergreen Monthly",
                  "pricingTerm": 1,
                  "pricingTermUnit": "Months",
                  "sellingModelType": "Evergreen",
                  "status": "Active"
                },
                "productSellingModelId": "0jPxx000000006TEAQ"
              }
            ]
          }
        ],
        "description": "The warranties available for the Cisco Server Rack NX44 product
 provide comprehensive coverage and support for optimal performance and reliability,
ensuring peace of mind for your data center infrastructure.",
        "id": "0y7xx0000000001AAA",
        "maxBundleComponents": 1,
        "minBundleComponents": 0,
        "name": "Warranties",
        "parentProductId": "01txx0000006jkuAAA",
        "sequence": 0
      }
    ],
    "productSellingModelOptions": [
      {
        "id": "0iOxx000000003FEAQ",
        "productId": "01txx0000006jkuAAA",
        "productSellingModel": {
          "id": "0jPxx000000001dEAA",
```

```
          "name": "One Time",
          "sellingModelType": "OneTime",
          "status": "Active"
        },
        "productSellingModelId": "0jPxx000000001dEAA"
      },
      {
        "id": "0iOxx000000004rEAA",
        "productId": "01txx0000006jkuAAA",
        "productSellingModel": {
          "id": "0jPxx000000003FEAQ",
          "name": "Termed Monthly",
          "pricingTerm": 1,
          "pricingTermUnit": "Months",
          "sellingModelType": "TermDefined",
          "status": "Active"
        },
        "productSellingModelId": "0jPxx000000003FEAQ"
      },
      {
        "id": "0iOxx000000006TEAQ",
        "productId": "01txx0000006jkuAAA",
        "productSellingModel": {
          "id": "0jPxx000000004rEAA",
          "name": "Termed Annually",
          "pricingTerm": 1,
          "pricingTermUnit": "Annual",
          "sellingModelType": "TermDefined",
          "status": "Active"
        },
        "productSellingModelId": "0jPxx000000004rEAA"
      }
    ],
    "productType": "Bundle"
  }
```


Property Name	Type	Description	Filter Group and Version	Available Version
additional Fields	Configurator Additional Fields[]	Additional fields for this product.	Small, 60.0	60.0
attribute Categories	Configurator Attribute Category[]	List of attribute categories. The categories that aren't categorized are specified in the uncategorized entry in this list.	Small, 60.0	60.0
availability Date	String	Availability date of this product.	Small, 60.0	60.0
description	String	Description of this product.	Small, 60.0	60.0
discontinued Date	String	Date when this product is discontinued.	Small, 60.0	60.0
displayUrl	String	Display URL of this product.	Small, 60.0	60.0

Property Name	Type	Description	Filter Group and Version	Available Version
endOfLifeDate	String	End of life date for this product.	Small, 60.0	60.0
id	String	ID of the product.	Small, 60.0	60.0
isActive	Boolean	Indicates whether this product is active (true) or not (false).	Small, 60.0	60.0
isAssetizable	Boolean	Indicates whether this product is assetizable (true) or not (false).	Small, 60.0	60.0
isComponent Required	Boolean	Indicates whether the component is required (true) or not (false).	Small, 60.0	60.0
is Configurable	Boolean	Indicates whether the component is configurable (true) or not (false).	Small, 60.0	60.0
isDefault Component	Boolean	Indicates whether the component is a default component (true) or not (false).	Small, 60.0	60.0
isQuantity Editable	Boolean	Indicates whether the quantity of the component is editable (true) or not (false).	Small, 60.0	60.0
isSoldOnly WithOtherProds	Boolean	Indicates whether this product is sold only with other products (true) or not (false).	Small, 60.0	60.0
name	String	Name of the product.	Small, 60.0	60.0
nodeType	String	Node type of the product.	Small, 60.0	60.0
prices	Configurator Price[]	List of prices from the product catalog.	Small, 60.0	60.0
product Classification	Configurator Product Classification[]	Classification details of the product.	Small, 60.0	60.0
productCode	String	Code of the product.	Small, 60.0	60.0
product ComponentGroups	Configurator Product Component Group[]	List of product component groups for this product.	Small, 60.0	60.0
product RelatedComponent	Configurator Product Related Component[]	Details of the product related component of this product.	Small, 60.0	60.0
productSelling ModelOptions	Configurator Product Selling Model Option[]	List of product selling model options for this product.	Small, 60.0	60.0
productType	String	Type of product.	Small, 60.0	60.0
qualification Context	Configurator Qualification Context[]	Details of the qualification context of the product.	Small, 60.0	60.0

Property Name	Type	Description	Filter Group and Version	Available Version
`status`	String	Status of the product.	Small, 60.0	60.0
`unitOfMeasure`	Configurator Unit Of Measure[]	Details about the unit of measure associated with a product.	Small, 63.0	63.0

Configurator Product Classification

Output representation of the product classification in a product configuration.

JSON example

```
"productClassification": {
  "id": "11Bxx000002CC02EAG"
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`id`	String	ID of the product classification.	Small, 60.0	60.0

Configurator Product Component Group

Output representation of the product component group in a product classification.

JSON example

```
"productComponentGroups": [
  {
    "classifications": [],
    "code": "SERVICE",
    "components": [
      {
        "additionalFields": [],
        "attributeCategories": [],
        "description": "Introducing the Cisco Rack Server NX44 Service, a
comprehensive protection plan designed to safeguard your valuable data infrastructure.
 With extended coverage and rapid response times, this warranty ensures peace of mind
and uninterrupted performance for your critical business operations.",
        "id": "01txx0000006jmWAAQ",
        "isActive": true,
        "isAssetizable": true,
        "isComponentRequired": false,
        "isConfigurable": false,
        "isDefaultComponent": false,
        "isQuantityEditable": false,
        "isSoldOnlyWithOtherProds": false,
        "name": "Cisco Rack Server Service - 1 Year",
        "nodeType": "simpleProduct",
```

```
            "prices": [],
            "productClassification": {},
            "productCode": "SERVICE",
            "productComponentGroups": [],
            "productRelatedComponent": {
              "childProductId": "01txx0000006jmWAAQ",
              "childSellingModelId": "0jPxx000000004rEAA",
              "doesBundlePriceIncludeChild": true,
              "id": "0dSxx000000001dEAA",
              "isComponentRequired": false,
              "isDefaultComponent": false,
              "isQuantityEditable": false,
              "parentProductId": "01txx0000006jkuAAA",
              "parentSellingModelId": "0jPxx000000004rEAA",
              "productComponentGroupId": "0y7xx000000001dAAA",
              "productRelationshipTypeId": "0yoxx00000001IfAAI",
              "quantity": 1,
              "quantityScaleMethod": "Proportional"
            },
            "productSellingModelOptions": [
              {
                "id": "0iOxx000000009hEAA",
                "productId": "01txx0000006jmWAAQ",
                "productSellingModel": {
                  "id": "0jPxx000000004rEAA",
                  "name": "Termed Annually",
                  "pricingTerm": 1,
                  "pricingTermUnit": "Annual",
                  "sellingModelType": "TermDefined",
                  "status": "Active"
                },
                "productSellingModelId": "0jPxx000000004rEAA"
              },
              {
                "id": "0iOxx00000000PpEAI",
                "productId": "01txx0000006jmWAAQ",
                "productSellingModel": {
                  "id": "0jPxx0000000085EAA",
                  "name": "Evergreen Annually",
                  "pricingTerm": 1,
                  "pricingTermUnit": "Annual",
                  "sellingModelType": "Evergreen",
                  "status": "Active"
                },
                "productSellingModelId": "0jPxx0000000085EAA"
              }
            ]
          }
        ],
        "description": "The services available for the Cisco Server Rack NX44 product
provide comprehensive coverage and support for optimal performance and reliability,
ensuring peace of mind for your data center infrastructure.",
        "id": "0y7xx000000001dAAA",
        "maxBundleComponents": 1,
```

```
        "minBundleComponents": 0,
        "name": "Services",
        "parentProductId": "01txx0000006jkuAAA",
        "sequence": 1
      },
      {
        "classifications": [],
        "code": "WARRANTY",
        "components": [
          {
            "additionalFields": [],
            "attributeCategories": [],
            "description": "Introducing the Cisco Rack Server NX44 Warranty, a
comprehensive protection plan designed to safeguard your valuable data infrastructure.
 With extended coverage and rapid response times, this warranty ensures peace of mind
and uninterrupted performance for your critical business operations.",
            "id": "01txx0000006jjIAAQ",
            "isActive": true,
            "isAssetizable": true,
            "isComponentRequired": false,
            "isConfigurable": false,
            "isDefaultComponent": true,
            "isQuantityEditable": true,
            "isSoldOnlyWithOtherProds": false,
            "name": "Cisco Rack Server Warranty - 1 Year",
            "nodeType": "simpleProduct",
            "prices": [],
            "productClassification": {},
            "productCode": "WARRANTY",
            "productComponentGroups": [],
            "productRelatedComponent": {
              "childProductId": "01txx0000006jjIAAQ",
              "childSellingModelId": "0jPxx000000001dEAA",
              "doesBundlePriceIncludeChild": false,
              "id": "0dSxx0000000001EAA",
              "isComponentRequired": false,
              "isDefaultComponent": true,
              "isQuantityEditable": true,
              "maxQuantity": 1,
              "parentProductId": "01txx0000006jkuAAA",
              "parentSellingModelId": "0jPxx000000001dEAA",
              "productComponentGroupId": "0y7xx0000000001AAA",
              "productRelationshipTypeId": "0yoxx00000001IfAAI",
              "quantity": 1,
              "quantityScaleMethod": "Proportional",
              "sequence": 0
            },
            "productSellingModelOptions": [
              {
                "id": "0iOxx000000001dEAA",
                "productId": "01txx0000006jjIAAQ",
                "productSellingModel": {
                  "id": "0jPxx000000001dEAA",
                  "name": "One Time",
```

```
        "sellingModelType": "OneTime",
        "status": "Active"
      },
      "productSellingModelId": "0jPxx000000001dEAA"
    },
    {
      "id": "0iOxx00000000HlEAI",
      "productId": "01txx0000006jjIAAQ",
      "productSellingModel": {
        "id": "0jPxx000000003FEAQ",
        "name": "Termed Monthly",
        "pricingTerm": 1,
        "pricingTermUnit": "Months",
        "sellingModelType": "TermDefined",
        "status": "Active"
      },
      "productSellingModelId": "0jPxx000000003FEAQ"
    },
    {
      "id": "0iOxx00000000JNEAY",
      "productId": "01txx0000006jjIAAQ",
      "productSellingModel": {
        "id": "0jPxx000000004rEAA",
        "name": "Termed Annually",
        "pricingTerm": 1,
        "pricingTermUnit": "Annual",
        "sellingModelType": "TermDefined",
        "status": "Active"
      },
      "productSellingModelId": "0jPxx000000004rEAA"
    },
    {
      "id": "0iOxx00000000KzEAI",
      "productId": "01txx0000006jjIAAQ",
      "productSellingModel": {
        "id": "0jPxx0000000085EAA",
        "name": "Evergreen Annually",
        "pricingTerm": 1,
        "pricingTermUnit": "Annual",
        "sellingModelType": "Evergreen",
        "status": "Active"
      },
      "productSellingModelId": "0jPxx0000000085EAA"
    },
    {
      "id": "0iOxx00000000MbEAI",
      "productId": "01txx0000006jjIAAQ",
      "productSellingModel": {
        "id": "0jPxx000000006TEAQ",
        "name": "Evergreen Monthly",
        "pricingTerm": 1,
        "pricingTermUnit": "Months",
        "sellingModelType": "Evergreen",
        "status": "Active"
```

```
                },
                "productSellingModelId": "0jPxx000000006TEAQ"
              }
            ]
          }
        ],
        "description": "The warranties available for the Cisco Server Rack NX44 product
 provide comprehensive coverage and support for optimal performance and reliability,
ensuring peace of mind for your data center infrastructure.",
        "id": "0y7xx0000000001AAA",
        "maxBundleComponents": 1,
        "minBundleComponents": 0,
        "name": "Warranties",
        "parentProductId": "01txx0000006jkuAAA",
        "sequence": 0
      }
    ]
```


Property Name	Type	Description	Filter Group and Version	Available Version
classifications	Configurator Product Classification[]	List of classifications for this product component group.	Small, 60.0	60.0
code	String	Code of the product component group.	Small, 60.0	60.0
components	Configurator Product Catalog[]	Components within the product component group.	Small, 60.0	60.0
description	String	Description of the product component group.	Small, 60.0	60.0
id	String	ID of the product component group.	Small, 60.0	60.0
maxBundle Components	Integer	Maximum number of bundle components within the product component group.	Small, 60.0	60.0
minBundle Components	Integer	Minimum number of bundle components within the product component group.	Small, 60.0	60.0
name	String	Name of the product component group.	Small, 60.0	60.0
parent ProductId	String	Parent Product2 ID of the product component group.	Small, 60.0	60.0
sequence	Integer	Sequence of the product component group.	Small, 60.0	60.0

Configurator Product Recommendations

Output representation of the details of the product recommendations.

JSON Example


```
{
  "productRecommendations": [
```

```
    {
      "referenceId": "CORE_BUNDLE_001",
      "productIds": [
        "01t000000001234",
        "01t000000005678"
      ]
    }
  ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
productIds	String[]	List of recommended product IDs.	Small, 65.0	65.0
referenceId	String	Reference ID for the recommendation.	Small, 65.0	65.0

Configurator Product Related Component

Output representation of the product related component in a product configuration.

JSON example


```
        "productRelatedComponent": {
          "childProductId": "01txx0000006jmWAAQ",
          "childSellingModelId": "0jPxx000000004rEAA",
          "doesBundlePriceIncludeChild": true,
          "id": "0dSxx000000001dEAA",
          "isComponentRequired": false,
          "isDefaultComponent": false,
          "isQuantityEditable": false,
          "parentProductId": "01txx0000006jkuAAA",
          "parentSellingModelId": "0jPxx000000004rEAA",
          "productComponentGroupId": "0y7xx000000001dAAA",
          "productRelationshipTypeId": "0yoxx00000001IfAAI",
          "quantity": 1,
          "quantityScaleMethod": "Proportional",
          "quoteVisibility": "Quote Document Only"
        }
```


Property Name	Type	Description	Filter Group and Version	Available Version
childProductId	String	ID of the child product in the bundle.	Small, 60.0	60.0
childSellingModelId	String	ID of the child product selling model record.	Small, 60.0	60.0
doesBundlePriceIncludeChild	Boolean	Indicates whether the price of the bundle includes the child product (`true`) or not (`false`).	Small, 60.0	60.0

Property Name	Type	Description	Filter Group and Version	Available Version
id	String	ID of the product related component.	Small, 60.0	60.0
isComponentRequired	Boolean	Indicates whether the component is required in the bundle (true) or not (false).	Small, 60.0	60.0
isDefaultComponent	Boolean	Indicates whether to select the component in the bundle group by default (true) or not (false).	Small, 60.0	60.0
isQuantityEditable	Boolean	Indicates whether to allow changes to the quantity of the component in the bundle (true) or not (false).	Small, 60.0	60.0
maxQuantity	Double	Maximum quantity of the product in the opportunity, quote, or order line item.	Small, 60.0	60.0
minQuantity	Double	Minimum quantity of the product in the opportunity, quote, or order line item.	Small, 60.0	60.0
parentProductId	String	ID of the parent product.	Small, 60.0	60.0
parentSellingModelId	String	ID of the parent product selling model record.	Small, 60.0	60.0
productClassificationId	String	ID of the product classification record.	Small, 60.0	60.0
productComponentGroupId	String	ID of the product component group.	Small, 60.0	60.0
productRelationshipTypeId	String	ID of the product relationship type record.	Small, 60.0	60.0
quantity	Double	Quantity of the child products.	Small, 60.0	60.0
quantityScaleMethod	String	Method to scale the quantity of the child product in relation to the quantity of the parent. Valid values are: • Constant • Proportional	Small, 60.0	60.0
quoteVisibility	String	Specifies whether a quote line item must be shown on the transaction line editor or quote document. Valid values are: • Always	Small, 64.0	64.0

Property Name	Type	Description	Filter Group and Version	Available Version
		• `Transaction Line Editor Only`—Specifies whether to show a quote line item on quote editor only. • `Quote Document Only`—Specifies whether to show a quote line item on quote proposal only. • `Never` The API returns this property only if the CoreCPQ permission set is available.		
`sequence`	Integer	Order in which the child products are displayed.	Small, 60.0	60.0

Configurator Product Selling Model

Output representation of the product selling model in a product configuration.

JSON example

```
    "productSellingModel": {
        "id": "0jPxx0000000085EAA",
        "name": "Evergreen Annually",
        "pricingTerm": 1,
        "pricingTermUnit": "Annual",
        "sellingModelType": "Evergreen",
        "status": "Active"
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`id`	String	ID of the product selling model.	Small, 60.0	60.0
`name`	String	Name of the product selling model.	Small, 60.0	60.0
`pricingTerm`	Integer	Pricing term of the product selling model.	Small, 60.0	60.0
`pricing TermUnit`	String	Pricing term unit of the product selling model.	Small, 60.0	60.0
`selling ModelType`	String	Selling model type of the product selling model.	Small, 60.0	60.0
`status`	String	Status of the product selling model.	Small, 60.0	60.0

Configurator Product Selling Model Option

Output representation of the product selling model option in a product configuration.

JSON example

```
        "productSellingModelOptions": [
          {
            "id": "0iOxx000000009hEAA",
            "productId": "01txx0000006jmWAAQ",
            "productSellingModel": {
              "id": "0jPxx000000004rEAA",
           "name": "Termed Annually",
              "pricingTerm": 1,
              "pricingTermUnit": "Annual",
              "sellingModelType": "TermDefined",
              "status": "Active"
            },
            "productSellingModelId": "0jPxx000000004rEAA"
          },
          {
            "id": "0iOxx00000000PpEAI",
            "productId": "01txx0000006jmWAAQ",
            "productSellingModel": {
              "id": "0jPxx0000000085EAA",
              "name": "Evergreen Annually",
              "pricingTerm": 1,
              "pricingTermUnit": "Annual",
              "sellingModelType": "Evergreen",
              "status": "Active"
            },
            "productSellingModelId": "0jPxx0000000085EAA"
          }
        ]
```

Property Name	Type	Description	Filter Group and Version	Available Version
id	String	ID of the product selling model option.	Small, 60.0	60.0
productId	String	ID of the product that's associated with the product selling model option.	Small, 60.0	60.0
product SellingModel	Configurator Product Selling Model[]	Product selling model that's associated with the product selling model option.	Small, 60.0	60.0
product Selling ModelId	String	ID of the product selling model that's associated with the product selling model option.	Small, 60.0	60.0

Configurator Qualification Context

Output representation of the qualification context in a product configuration.

Property Name	Type	Description	Filter Group and Version	Available Version
isQualified	Boolean	Indicates whether the product is qualified (true) or not (false).	Small, 60.0	60.0
reason	String	Reason for the qualification of this product.	Small, 60.0	60.0

Configurator UI Treatment

Output representation of the details of the UI treatments of a product configurator. The details include the product configuration rule actions to override the disable or hide behavior in the UI for product options, product attributes, and attribute picklist values.

JSON Example


```
[
  {
    "details": {
      "attributeId": "0tjxx0000000007AAA",
      "prcId": "0dSxx0000000007EAA",
      "stiId": "0QLxx0000004CU0GAM",
      "attributePicklistValueId": "0v6xx0000000005AAA"
    },
    "uiTreatmentScope": "Bundle",
    "uiTreatmentTarget": "Attribute_Picklist_Value",
    "uiTreatmentType": "Hide"
  },
  {
    "details": {
      "stiId": "ref_f0f2da7b_c431_482d_bf4b_599052f3a2e1"
    },
    "uiTreatmentScope": "Product",
    "uiTreatmentTarget": "Component",
    "uiTreatmentType": "Disable"
  }
]
```


Property Name	Type	Description	Filter Group and Version	Available Version
details	Map<String, Object>	Key-value pair that specifies the items to apply the rules on, which includes these details. • ID of the sales transaction item • ID of the product-related component • ID of the attribute • ID of the attribute picklist value	Small, 62.0	62.0
uiTreatment Scope	String	Type of the UI treatment to be performed. Valid values are:	Small, 62.0	62.0

Property Name	Type	Description	Filter Group and Version	Available Version
		• `Product`—UI treatment is applicable to a certain product only. • `Bundle`—UI treatment is applicable to the whole bundle.		
`uiTreatment Target`	String	Target of the UI treatment. Valid values are: • `Component`—Represents a product option or bundle component. • `Quantity`—Represents a quantity field. • `Attribute`—Represents a certain attribute of the product. • `Attribute_Picklist_Value`—Represents one of the picklist values of a product attribute.	Small, 62.0	62.0
`uiTreatment Type`	String	Type of UI treatment to be performed. Valid values are: • `Hide`—Hide the associated target. • `Disable`—Disable the associated target.	Small, 62.0	62.0

Configurator Unit Of Measure

Output representation of the details of the unit of measure record.

JSON Example


```
{
  "unitOfMeasure": {
    "id": "0hEXR00000000BJ2AY",
    "name": "Litres",
    "roundingMethod": "Down",
    "scale": 2,
    "unitCode": "Ltrs"
  }
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`id`	String	ID of the unit of measure record.	Small, 63.0	63.0
`name`	String	Name of the unit of measure record.	Small, 63.0	63.0

Property Name	Type	Description	Filter Group and Version	Available Version
roundingMethod	String	Rounding method associated with the unit of measure record.	Small, 63.0	63.0
scale	Integer	Scale associated with the unit of measure record.	Small, 63.0	63.0
unitCode	String	Unit code associated with the unit of measure record.	Small, 63.0	63.0

Configurator Update Nodes

Output representation of the configuration request details to update nodes.

JSON Example


```
{
  "configuratorMessages": {},
  "configuratorUITreatments": [
    {
      "details": {
        "attributeId": "0tjxx0000000007AAA",
        "prcId": "0dSxx0000000007EAA",
        "stiId": "0QLxx0000004CU0GAM",
        "attributePicklistValueId": "0v6xx0000000005AAA"
      },
      "uiTreatmentScope": "Bundle",
      "uiTreatmentTarget": "Attribute_Picklist_Value",
      "uiTreatmentType": "Hide"
    },
    {
      "details": {
        "stiId": "ref_f0f2da7b_c431_482d_bf4b_599052f3a2e1"
      },
      "uiTreatmentScope": "Product",
      "uiTreatmentTarget": "Component",
      "uiTreatmentType": "Disable"
    }
  ],
  "errors": [],
  "productQualifications": {
    "01tDU000000EOTCYA4": {
      "isQualified": true
    }
  },
  "success": true
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`configuratorMessages`	Map<String, Configurator Message>	Map of the product IDs to the list of configurator messages. Configurator messages are results from any validations, Business Rules Engines (BRE) calls, or Salesforce Pricing calls.	Small, 60.0	60.0
`configuratorUITreatments`	Configurator UI Treatment[]	Details of the UI treatments that specify the product configuration rule actions to override the disable or hide behavior in the UI for product options, product attributes, and attribute picklist values.	Small, 62.0	62.0
`errors`	Error Response	List of errors, which contains an error code and a message.	Small, 60.0	60.0
`productQualifications`	Map<String, Configurator Qualification Context>	Map of the product IDs to the qualification rule execution results.	Small, 60.0	60.0
`success`	Boolean	Indicates whether the call was successful (`true`) not (`false`).	Small, 60.0	60.0

Configuration Update

Output representation of the details of the updated configuration.

JSON example

This example shows a sample when the update operation is successful.

```
{
  "errors": [],
  "success": true
}
```

This example shows a sample when the update operation has errors.

```
{
  "errors": [
    {
      "code": "INTERNAL_SERVER_ERROR",
      "message": "INVALID_REFERENCEOBJECTID"
    }
  ],
  "success": false
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`errors`	Error Response	List of errors that contains a message and an error code.	Small, 63.0	63.0
`success`	Boolean	Indicates whether the update operation is successful (`true`) or not (`false`)	Small, 63.0	63.0

Error Response

Output representation of the details of the error.

JSON example

```
{
  "errors": [
    {
      "code": "BAD_REQUEST",
      "message": "MISSING_REFERENCEOBJECTID"
    }
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`code`	String	Code of the error.	Small, 63.0	63.0
`message`	String	Description of the error.	Small, 63.0	63.0

Product Quantity Set Configurator

Output representation of the request details to set product quantity.

JSON Example

```
{
  "configuratorMessages": {},
  "configuratorUITreatments": [
    {
      "details": {
        "attributeId": "0tjxx0000000007AAA",
        "prcId": "0dSxx0000000007EAA",
        "stiId": "0QLxx0000004CU0GAM",
        "attributePicklistValueId": "0v6xx0000000005AAA"
      },
      "uiTreatmentScope": "Bundle",
      "uiTreatmentTarget": "Attribute_Picklist_Value",
      "uiTreatmentType": "Hide"
    },
    {
      "details": {
```

```
        "stiId": "ref_f0f2da7b_c431_482d_bf4b_599052f3a2e1"
      },
      "uiTreatmentScope": "Product",
      "uiTreatmentTarget": "Component",
      "uiTreatmentType": "Disable"
    }
  ],
  "errors": [],
  "productQualifications": {
    "01tDU000000EOTCYA4": {
      "isQualified": true
    }
  },
  "success": true
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`configurator Messages`	Map<String, Configurator Message>	Map of the product IDs to the list of configurator messages. Configurator messages are results from any validations, Business Rules Engines (BRE) calls, or Salesforce Pricing calls.	Small, 60.0	60.0
`configurator UITreatments`	Configurator UI Treatment[]	Details of the UI treatments that specify the product configuration rule actions to override the disable or hide behavior in the UI for product options, product attributes, and attribute picklist values.	Small, 62.0	62.0
`errors`	Error Response	List of errors, which contains an error code and a message.	Small, 60.0	60.0
`product Qualifications`	Map<String, Configurator Qualification Context>	Map of the product IDs to the qualification rule execution results.	Small, 60.0	60.0
`success`	Boolean	Indicates whether the call was successful (`true`) not (`false`).	Small, 60.0	60.0

Configuration Rule Response

Output representation of the details of the configuration rule response.

JSON example


```
{
  "errors": [],
  "messageRules": [
    {
      "message": "You have a 128GB LRDIMM QLI",
```

```
      "messageType": "error",
      "primaryRecordId": "0Q0VW000000z8yN0AQ"
    },
    {
      "message": "You have a 16GB RDIMM QLI",
      "messageType": "warning",
      "primaryRecordId": "0Q0VW000000z8yN0AQ"
    },
    {
      "message": "You have a 32GB RDIMM QLI",
      "messageType": "info",
      "primaryRecordId": "0Q0VW000000z8yN0AQ"
    },
    {
      "message": "128GB LRDIMM disables 16GB RDIMM",
      "messageType": "info",
      "primaryRecordId": "0Q0VW000000z8yN0AQ"
    },
    {
      "message": "Additional API disables Additional API Gov",
      "messageType": "info",
      "primaryRecordId": "0Q0VW000000z8yN0AQ"
    },
    {
      "message": "Disable All other API Products",
      "messageType": "info",
      "primaryRecordId": "0Q0VW000000z8yN0AQ"
    },
    {
      "message": "32GB RDIMM disables 128GB LRDIMM",
      "messageType": "info",
      "primaryRecordId": "0Q0VW000000z8yN0AQ"
    },
    {
      "message": "API Access Requests AEH disables Additional API Prod",
      "messageType": "info",
      "primaryRecordId": "0Q0VW000000z8yN0AQ"
    }
  ],
  "productRecommendationRules": [
    {
      "message": "32GB RDIMM recommends 16GB RDIMM",
      "productIds": [
        "01tVW000003l7uaYAA"
      ],
      "recordType": "Type",
      "referenceId": "0Q0VW000000z8yN0AQ"
    },
    {
      "message": "32GB RDIMM recommends 64GB RDIMM",
      "productIds": [
        "01tVW000003l7v6YAA"
      ],
      "recordType": "Type",
```

```
      "referenceId": "0Q0VW000000z8yN0AQ"
    }
  ],
  "success": true,
  "transactionContextId":
"0000000r25tq18g00291775730228818e689c3c5756e409fb3f886f68937ab13",
  "visibilityRules": [
    {
      "message": "128GB LRDIMM disables 16GB RDIMM",
      "productIds": [
        "01tVW000003l7uaYAA"
      ],
      "scope": "virtual",
      "target": "product",
      "type": "disable"
    },
    {
      "message": "Additional API disables Additional API Gov",
      "productIds": [
        "01tVW000003l7tzYAA"
      ],
      "scope": "virtual",
      "target": "product",
      "type": "disable"
    },
    {
      "message": "Disable All other API Products",
      "productIds": [
        "01tVW000003l7u0YAA",
        "01tVW000003l7u1YAA"
      ],
      "scope": "virtual",
      "target": "product",
      "type": "disable"
    },
    {
      "message": "32GB RDIMM disables 128GB LRDIMM",
      "productIds": [
        "01tVW000003l7v5YAA"
      ],
      "scope": "virtual",
      "target": "product",
      "type": "disable"
    },
    {
      "message": "API Access Requests AEH disables Additional API Prod",
      "productIds": [
        "01tVW000003l7u1YAA"
      ],
      "scope": "virtual",
      "target": "product",
      "type": "disable"
    }
```

```
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
errors	Config Rule Errors[]	List of error messages from the configuration rule evaluation, if any.	Small, 67.0	67.0
messageRules	Message Rules[]	List of message rules that were evaluated during configuration.	Small, 67.0	67.0
product RecommendationRules	Product Recommendation Rules[]	List of product recommendation rules that were evaluated during configuration.	Small, 67.0	67.0
success	Boolean	Indicates whether the API request is successful (true) or not (false).	Small, 67.0	67.0
transaction ContextId	String	ID of the transaction context for this configuration rule evaluation.	Small, 67.0	67.0
visibility Rules	Visibility Rules[]	List of visibility rules that were evaluated during configuration.	Small, 67.0	67.0

Configuration Rule Errors
Output representation of the error details, if applicable, when you run config rules.

Message Rules
Output representation of the details of the message rules.

Product Recommendation Rules
Output representation of the details of the product recommendation rules.

Visibility Rules
Output representation of the details of the visibility rules.

Configuration Rule Errors

Output representation of the error details, if applicable, when you run config rules.

Property Name	Type	Description	Filter Group and Version	Available Version
errorCode	String	Code indicating the type of error.	Small, 67.0	67.0
message	String	Message stating the reason for error, if any.	Small, 67.0	67.0

Message Rules

Output representation of the details of the message rules.

JSON example

```
{
  "message": "Constraints is running for laptop",
  "messageType": "Info",
  "primaryRecordId": "0QLxx0000004CU0GAM",
  "relatedRecordId": "0QLxx0000004CU1GAM"
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`message`	String	List of message strings to display to the user.	Small, 67.0	67.0
`messageType`	String	Severity level of the message. Valid values are: • `INFO` • `WARNING` • `ERROR`	Small, 67.0	67.0
`primaryRecordId`	String	ID of the primary sales transaction item record.	Small, 67.0	67.0
`relatedRecordId`	String	ID of the related record.	Small, 67.0	67.0

Product Recommendation Rules

Output representation of the details of the product recommendation rules.

JSON example

```
{
  "productRecommendationRules": [
    {
      "message": "32GB RDIMM recommends 16GB RDIMM",
      "productIds": [
        "01tVW00000317uaYAA"
      ],
      "recordType": "Type",
      "referenceId": "0Q0VW000000z8yN0AQ"
    },
    {
      "message": "32GB RDIMM recommends 64GB RDIMM",
      "productIds": [
        "01tVW00000317v6YAA"
      ],
      "recordType": "Type",
      "referenceId": "0Q0VW000000z8yN0AQ"
    }
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
message	String	Message to display with the product recommendation.	Small, 67.0	67.0
productIds	String[]	List of recommended Product IDs.	Small, 67.0	67.0
recordType	String	Record type associated with the recommendation.	Small, 67.0	67.0
referenceId	String	Reference ID of the product recommendation rule.	Small, 67.0	67.0

Visibility Rules

Output representation of the details of the visibility rules.

JSON example


```
{
  "visibilityRules": [
    {
      "message": "128GB LRDIMM disables 16GB RDIMM",
      "productIds": [
        "01tVW00000317uaYAA"
      ],
      "scope": "virtual",
      "target": "product",
      "type": "disable"
    },
    {
      "message": "Additional API disables Additional API Gov",
      "productIds": [
        "01tVW00000317tzYAA"
      ],
      "scope": "virtual",
      "target": "product",
      "type": "disable"
    },
    {
      "message": "Disable All other API Products",
      "productIds": [
        "01tVW00000317u0YAA",
        "01tVW00000317u1YAA"
      ],
      "scope": "virtual",
      "target": "product",
      "type": "disable"
    },
    {
      "message": "32GB RDIMM disables 128GB LRDIMM",
      "productIds": [
        "01tVW00000317v5YAA"
      ],
```

```
      "scope": "virtual",
      "target": "product",
      "type": "disable"
    },
    {
      "message": "API Access Requests AEH disables Additional API Prod",
      "productIds": [
        "01tVW000003l7u1YAA"
      ],
      "scope": "virtual",
      "target": "product",
      "type": "disable"
    }
  ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`attributeId`	String	ID of the attribute associated with this visibility rule.	Small, 67.0	67.0
`attribute PicklistValueId`	String	ID of the attribute picklist value associated with this visibility rule.	Small, 67.0	67.0
`message`	String	Message to display when the visibility rule is applied.	Small, 67.0	67.0
`prcId`	String	ID of the Product Relationship Configuration (PRC) associated with this visibility rule.	Small, 67.0	67.0
`productIds`	String[]	List of product IDs affected by this visibility rule.	Small, 67.0	67.0
`scope`	String	Scope of the visibility rule. Valid values are: • `PRODUCT` • `BUNDLE` • `VIRTUAL`	Small, 67.0	67.0
`stiId`	String	ID of the sales transaction item associated with this visibility rule.	Small, 67.0	67.0
`target`	String	Target of the visibility rule. Valid values are: • `COMPONENT` • `QUANTITY` • `ATTRIBUTE` • `ATTRIBUTE_PICKLIST_VALUE` • `PRODUCT`	Small, 67.0	67.0

Property Name	Type	Description	Filter Group and Version	Available Version
`type`	String	Type of visibility rule. Valid values are: • `HIDE` • `DISABLE`	Small, 67.0	67.0

Product Configurator Standard Invocable Actions

Learn more about the standard invocable actions available with Product Configurator.

Run Config Rules Action
Run rules for a specific quote or order based on a context ID or transaction ID, and process other steps that are part of the configuration directly within a Flow. This action decouples rule execution from configurations to enable independent execution of rules and for easier retrieval of responses.

SEE ALSO:

Actions Developer Guide: Overview

REST API Developer Guide: Invocable Actions Standard

Run Config Rules Action

Run rules for a specific quote or order based on a context ID or transaction ID, and process other steps that are part of the configuration directly within a Flow. This action decouples rule execution from configurations to enable independent execution of rules and for easier retrieval of responses.

This action is available in API version 65.0 and later.

Supported REST HTTP Methods

URI

`/services/data/v`***`67.0`***`/actions/standard/runConfigRules`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer`***`token`***

Inputs

Input	Details
`transactionContextId`	**Type** string **Description** Unique identifier for the transaction context.
`transactionId`	**Type** string **Description** Required. Unique identifier for the transaction.

Outputs

Output	Details
`configRuleResult`	**Type** Apex-defined **Description** An `runtime_industries_cpq.ConfigRuleResult` record that contains the configuration rule execution results including validation messages, product recommendations, visibility rules, and errors from rule processing.
`transactionContextId`	**Type** string **Description** Unique identifier for the transaction context.

Example

POST

This example shows a sample request to run rules for the specified quote based on the transaction context ID and transaction ID.

```
{
  "inputs": [
    {
      "transactionContextId": "008d27d7-e004-4906-a949-ee7d7c323c77",
      "transactionId": "0Q0DU0000005tJh0AI"
    }
```

```
  ]
}
```

This example shows a sample successful response.


```
[
  {
    "actionName": "runConfigRules",
    "errors": null,
    "invocationId": null,
    "isSuccess": true,
    "outcome": null,
    "outputValues": {
      "transactionContextId":
"0000000p18dq18g0029175793402786243c3d5ea94c241f09c11388ac1b865f9",
      "configRuleResult": {
        "visibilityRules": [
          {
            "stiId": "0QLxx0000004CU0GAM",
            "prcId": "PRC1",
            "attributeId": "Color",
            "attributePicklistValueId": "Red",
            "target": "Attribute",
            "scope": "Product",
            "type": "Hide"
          },
          {
            "stiId": "0QLxx0000004CU0GAM",
            "prcId": "PRC2",
            "attributeId": "Size",
            "attributePicklistValueId": "Large",
            "target": "Attribute",
            "scope": "Bundle",
            "type": "Disable"
          }
        ],
        "transactionContextId":
"0000000p18dq18g0029175793402786243c3d5ea94c241f09c11388ac1b865f9",
        "productRecommendationRules": [
          {
            "referenceId": "CORE_BUNDLE_001",
            "productIds": [
              "01t000000001234",
              "01t000000005678"
            ]
          }
        ],
        "messageRules": [
          {
            "stiId": "0QLxx0000004CU0GAM",
            "severity": "INFO",
            "messages": [
              "Product configuration validated successfully"
            ]
          },
```

```
            {
              "stiId": "0QLxx0000004CU0GAM",
              "severity": "INFO",
              "messages": [
                "All required attributes are configured"
              ]
            },
            {
              "stiId": "0QLxx0000004CU0GAM",
              "severity": "INFO",
              "messages": [
                "Bundle compatibility check passed"
              ]
            }
          ],
          "errors": []
        }
      },
      "sortOrder": -1,
      "version": 1
    }
]
```

Here's a sample input to call this invocable action by using `transactionContextId` property from Apex code.

```
// Create the invocable action with namespace
Invocable.Action action = Invocable.Action.createStandardAction('runConfigRules');

// Set input parameters using setInvocationParameter
String contextId = '008d27d7-e004-4906-a949-ee7d7c323c77';
System.debug('Setting transactionContextId parameter with value: ' + contextId);

// Use the exact parameter name format from the debug output
action.setInvocationParameter('transactionContextId', contextId);

// Debug the action parameters
System.debug('Action parameters: ' + action);

// Execute the action
System.debug('Invoking action...');
List<Invocable.Action.Result> results = action.invoke();

System.debug('Number of results: ' + results.size());
```

Product Configurator Metadata API Types

Metadata API enables you to access some types and feature settings that you can customize in the user interface.

Flow for Product Configurator

Represents the metadata associated with a flow. With Flow, you can create an application that navigates users through a series of screens to query and update records in the database. You can also execute logic and provide branching capability based on user input to build dynamic applications.

ProductConfiguratorSettings
Represents the settings for Product Configurator.

SEE ALSO:

Metadata API Developer Guide: Understanding Metadata API

Flow for Product Configurator

Represents the metadata associated with a flow. With Flow, you can create an application that navigates users through a series of screens to query and update records in the database. You can also execute logic and provide branching capability based on user input to build dynamic applications.

FlowActionCall

Product Configurator exposes additional actionType values for the FlowActionCall Metadata type.

Field Name	Field Type	Description
`actionType`	InvocableActionType (enumeration of type string)	Required. The action type. Additional valid values only for Product Configurator include: • `runConfigRules`—Run rules for a specific quote or order based on a context ID or transaction ID, and process other steps that are part of the configuration directly within a Flow. This action decouples rule execution from configurations to enable independent execution of rules and for easier retrieval of responses.

ProductConfiguratorSettings

Represents the settings for Product Configurator.

Parent Type and Manifest Access

This type extends the Metadata metadata type and inherits its `fullName` field.

In the package manifest, all the settings metadata types for the org are accessed using the "Settings" name. See Settings for more details.

File Suffix and Directory Location

`ProductConfiguratorSettings` values are stored in the `ProductConfiguratorSettings.settings` file in the `settings` folder. The `.settings` files are different from other named components, because there is only one settings file for each settings component.

Version

ProductConfiguratorSettings components are available in API version 60.0 and later.

Fields

Field Name	Description
enableProductConfigurator	**Field Type** boolean **Description** Indicates whether to enable Product Configurator at run time to customize product components and attributes during the sales process (`true`) or not (`false`). The default value is `false`.

Declarative Metadata Sample Definition

Here's an example of a ProductConfiguratorSettings component.

```
<?xml version="1.0" encoding="UTF-8"?>
<ProductConfiguratorSettings xmlns="http://soap.sforce.com/2006/04/metadata">
    <enableProductConfigurator>true</enableProductConfigurator>
</ProductConfiguratorSettings>
```

Here's an example `package.xml` that references the previous definition.

```
<?xml version="1.0" encoding="UTF-8"?>
<Package xmlns="http://soap.sforce.com/2006/04/metadata">
    <types>
        <members>ProductConfigurator</members>
        <name>Settings</name>
    </types>
    <version>67.0</version>
</Package>
```

Wildcard Support in the Manifest File

The wildcard character `*` (asterisk) in the `package.xml` manifest file doesn't apply to metadata types for feature settings. The wildcard applies only when retrieving all settings, not for an individual setting. For details, see Settings. For information about using the manifest file, see Deploying and Retrieving Metadata with the Zip File.

Constraint Modeling Language

Constraint Modeling Language (CML) is a domain-specific language that defines models for complex systems. For product configuration, constraint models describe real-world entities and their relationships to each other.

Constraint models enforce business logic declaratively, without the need for extensive code in a general-purpose programming language. In Product Configurator, a product designers or admin create a constraint model in CML, then Constraint Rules Engine compiles the CML code. The constraint model enforces requirements in a product configuration to comply with the specified constraints.

To build a constraint model in CML, use this basic workflow.

- Create global properties and settings that are header-level declarations in CML that define the foundational, fixed values for the entire constraint model. They're crucial for setting up the core configuration environment and ensuring reusability across the model.

- Create variables to define the properties or characteristics of a type. Variables can hold different kinds of data, such as strings, numbers, or lists, and can be calculated from other variables and values. In Revenue Management, variables represent product fields, product attributes, and sometimes context tags. See Create a Context Definition in Salesforce Help.
- Define types, which represent entities or objects in the model. Types are the building blocks of CML. They're similar to classes in object-oriented programming. In Revenue Management, types represent standalone products, bundles, product components, and product classes.
- Define relationships that describe how different types are associated with each other. In Revenue Management, relationships represent the product structure in a bundle. For example, the root product has a relationship with its components.
- Apply constraints to define logical restrictions, and enforce rules and conditions on types, variables, and relationships.

In Product Configurator, use the Visual Builder and CML Editor in Constraint Builder to create constraint models with CML.

- Use Constraint Builder With Constraint Rules Engine
- Define Constraints and Rules with the Visual Builder
- Use Code to Define Constraints and Rules in the CML Editor

See the linked topics for information on working with CML, including detailed code examples.

Modeling a Generator Set

The Constraint Model for a Generator Set examples use CML to define a technical power configuration, illustrating concepts such as calculated variables, enforcement of external standards, and component selection based on requirements.

Core Concepts

Constraint Modeling Language (CML) includes components that cover high-level global configurations to specific data types and constraints.

Core Concept Examples

These examples illustrate core Constraint Modeling Language (CML) concepts including type, relationships, constraints, and so on.

Annotation Examples

Constraint Modeling Language (CML) annotations are labels that you add to parts of a model, such as types, variables, relationships, and constraints. Annotations control how these elements are shown and how they behave in the configurator. Annotations help fine-tune the configurator and the constraint engine without changing the actual structure of the model.

Constraint Modeling Language (CML) Best Practices

To prevent performance degradation or unexpected behaviors when the constraint engine executes CML code, follow these practices when writing code.

Business-Centric Constraint Modeling Language (CML) Guidelines

Constraint Modeling Language (CML) must accurately calculate the total sum or aggregate of specific attributes like quantity or userCount across child components, especially in complex configurations requiring group-level aggregation

Debugging Constraint Modeling Language (CML)

To debug constraint models and troubleshoot performance issues, enable debug logging in Apex and set the debug log level to FINE.

Model Structure

The tables on the following pages show the structure for the constraint model in Core Concept Examples.

Modeling a Generator Set

The Constraint Model for a Generator Set examples use CML to define a technical power configuration, illustrating concepts such as calculated variables, enforcement of external standards, and component selection based on requirements.

The examples on page 1135 correspond to the CML core concepts on page 1080 linked here. See the Generator Set examples for code samples that use the core concepts.

- Global Properties and Settings—`VOLTAGE_REGEX` is a global constant that defines a fixed regular expression pattern used for validation or parsing throughout the model.
- Types—`GeneratorSet` is the root type that represents the main entity. `GeneralModel` represents a related component type.
- Variables—The `GeneratorSet` type defines variables like `requiredKW` (the user's power requirement), `Voltage`, and calculated variables like `surgeLoadKW` and `Voltage3` (derived from parsing the `Voltage` string).
- Relationships—The `GeneralModels` relation connects the `GeneratorSet` type to its possible configurations (`GeneralModel`).
- Constraints—Constraints enforce critical business rules and safety standards, such as ensuring the selected generator model's power meets the required threshold, or restricting configuration options (such as `Voltage`) based on the specified compliance standards (`Listing-UL 2200`).

Core Concepts

Constraint Modeling Language (CML) includes components that cover high-level global configurations to specific data types and constraints.

See these topics for information on each core concept and the ways they work together.

Note: CML supports single-line code comments with // and block comments with /* */.

Global Properties and Settings
Header-level declarations define the global properties and settings for a model, including constants, properties, and external values that set up the foundation of the CML code.

Variables
Variables are the properties or characteristics defined within a type. Variables can hold different types of data and can be calculated from other values.

Types
In Constraint Modeling Language (CML), you define types to represent entities or objects in the model. Types are the foundational building blocks of CML. A type encapsulates the property, relationships, constraint, and rules for the entity.

Relationships
Relationships in Constraint Modeling Language (CML) define how different product types are associated with each other, forming the structural hierarchy of a product bundle. Relationships are also referred to as ports.

Constraints
Constraints enforce rules and conditions on types, variables, and relationships. Use constraints to define logical restrictions and ensure consistency within the model.

Global Properties and Settings

Header-level declarations define the global properties and settings for a model, including constants, properties, and external values that set up the foundation of the CML code.

Use these declarations to create reusable components and configuration settings that you can reference throughout the model.

Global Constants

Use global constants to define values that remain fixed throughout the model. These constants can be numeric values, strings, lists, or other supported data types. Use constants to create standardized settings or options that you can reference multiple times. See Example 1: Use Regex Global Variable on page 1080.

In the example, MAX_COUNT is a global constant that is hard-coded to 100: define MAX_COUNT 100.

Regex (regular expressions) can be used to define global constants. The generalized abstract syntax structure for regex expressions is define <CONSTANT_NAME> "^<REGEX_PATTERN_STRING>$".

Regex Pattern Components

This table lists regex components and their details.

Regex Component	Description	Generalization
^ and $	Anchors that ensure the pattern matches the entire string, from the beginning (^) to the end ($).	Ensures strict adherence to the required format.
()	Capturing Groups used to isolate portions of the matched string. You can reference the captured parts later by using $1, $2, and so on, in functions such as regexpreplace. See String Variable Functions and Operators on page 1084.	Allows extraction of specific data fields from a string.
+	Character Class and Quantifier that matches one or more (+) digits or characters.	Defines the permitted characters and minimum occurrences for the data fields.
/	Literal Character matching the forward slash separator present in the input data.	Matches fixed delimiters in the input string.

In the example, VOLTAGE_REGEX is a global constant that defines a fixed regular expression pattern used for validation or parsing throughout the model define VOLTAGE_REGEX "^([0-9]+)/([0-9]+)$".

For more on the usage of global properties, see External Variables.

Variables

Variables are the properties or characteristics defined within a type. Variables can hold different types of data and can be calculated from other values.

Variable Domains and Domain Restrictions
A variable can have a fixed domain that defines a set of permitted values. You can specify the domain as a list of discrete values, a continuous range, or a combination.

Variable Data Types
Variables support multiple data types including boolean, date, decimal, and so on. Variables without a domain definition may remain unbound, leading to errors.

Variable Domains and Domain Restrictions

A variable can have a fixed domain that defines a set of permitted values. You can specify the domain as a list of discrete values, a continuous range, or a combination.

Variable Domains and Domain Restrictions

A variable can have a fixed domain that defines a set of permitted values. You can specify the domain as:

- A list of discrete values
- A continuous range
- A combination of ranges and discrete values

For more information, see domainComputation in Variable Annotations on page 1086.

Example

This example defines a `SwitchgearBay` type with three variables, each having a fixed domain.

- `BayType` can be one of the specified string values.
- `Bay_Number` can be any integer from 1 through 9 (inclusive).
- `Total_Power_Required_kW` can be any integer from 1 through 100000 (inclusive).

Note: For string (picklist) variables, if you don't specify a `defaultValue` annotation and don't define a default in the Product Attribute Definition in Product Catalog Management (PCM), the configurator uses the first value in the domain as the initial value. In the example, `BayType` defaults to `"load"`. To set a different default, use the `defaultValue` annotation on the variable. To avoid inconsistent behavior at run time, use the same default value in CML as the default value you define in PCM.

```
type SwitchgearBay {
  string BayType = ["load", "lv", "mv"];
  int Bay_Number = [1..9];
  int Total_Power_Required_kW = [1..100000];
}
```

Variable Data Types

Variables support multiple data types including boolean, date, decimal, and so on. Variables without a domain definition may remain unbound, leading to errors.

Data Type	Description	Defaulting Example
boolean	Only true, false, or null can be assigned as a value.	`@(defaultValue="true") boolean isActive;`
date	A value that indicates a particular day, the same as local date in Java.	`date shipDate = ["2023-01-01", "2023-12-31"];` If there's no `defaultValue` specified, the variable defaults to the first value in the domain. See the Constraints using Format Specifiers (%s, %d) and Dates example on page 1135.
double(n)	A 64-bit number that includes a decimal point, the same as double in Java.	`double(2) percentage = [0.00..100.00];` If there's no defaultValue specified, the variable defaults to the first value in the domain.
decimal(n)	A fixed-point numeric value with n decimal places. Note: The decimal variable data type isn't supported for the quantity field in quote line items. Use the integer data type for the quantity field.	`decimal(2) TaxRate = 0.08;`
int	Integer. A 32-bit number that doesn't include a decimal point, the same as int in Java.	`@(defaultValue = "5") int defaultQty = [1..10];` If there's no `defaultValue` specified, the variable defaults to the first value in the domain.
string	Any set of characters surrounded by double quotes (")	`@(defaultValue = "Red") string color = ["Red", "Green", "Blue"];` If there's no `defaultValue` specified, the attribute defaults to the first value in the domain.
string[]	Used in multi-select picklists for the user to select more than one item from multiple options. For example, if a user selects "Red", "Green", and "Blue" values in a color picker, this variable holds those selected values.	`@(defaultValue = '["Red", "Green"]') string[] selectedColors;` If there's no `defaultValue` specified, the attribute picklist defaults to the first value in the domain.

Mathematical Functions (Numerical Derivation)

Mathematical functions and operators are used to calculate derived values based on arithmetic relationships between variables.

Function/Operator	Purpose	CML Keyword/Operator [Source]
Arithmetic Operators	Perform standard arithmetic: addition (+), subtraction (-), multiplication (*), division (/), modulo (% or mod), and power (^).	`+, -, *, /, %, ^`
ceil()	Returns the smallest integer greater than or equal to the argument (rounds up).	`ceil`

Usage Example

```
surgeLoadKW == requiredKW * 1.25);
ceil(totalItems / itemsPerCrate)
```

See Arithmetic Calculations and Functions on page 1135 and examples here on page 1197.

String Variable Functions and Operators

Constraint Modeling Language (CML) provides string manipulation and conversion functions, and string comparison and validation operators.

The functions can be used to modify strings, extract information, or convert strings to numeric types. The operators can be used to validate string values against sets or regular expressions.

Function	Purpose	Syntax, Examples, and Additional Details
strlen()	Returns the length of the string as an integer.	`strlen(inputString)` `strlen("Hello"` returns 5.
substr()	Returns a substring from the input string.	`substr(inputString, startIndex)` or `substr(inputString, startIndex, endIndex)` The index starts with 0. The character at `startIndex` is included, and the character at `endIndex` is excluded.
strconcat()	Concatenates multiple strings using a specified separator.	`strconcat(separator, stringArgsToConcatenate)` `strconcat("-", "a", "b")` results in "a-b".
join()	An aggregate function that concatenates string values across related components, typically using a separator. The separator is a comma by default.	`names = join(name)`

Function	Purpose	Syntax, Examples, and Additional Details
trim()	Returns the string after removing any leading or trailing spaces.	`trim(strToTrim)`
strsplit()	Splits a string into an array of strings around a given separator.	`strsplit(stringToSplit, separator)`
strcontain()	Returns a boolean value (true or false) to specify whether the input string contains the specified search string.	`strcontain(inputString, searchString)`
strshare()	Splits two strings using a delimiter (comma by default) and returns true if the resulting lists have any elements in common.	`strshare(string1, string2, delimiter)`
strformat()	Returns a formatted string based on parameters and format specifiers.	`strformat("%d person", quantity)` Specifiers include `%d` for integer, `%s` for string, `%.2f` for decimal or float, and `%b` for boolean.
strtoint()	Converts a string to an integer.	`strtoint(inputString, defaultValue)` If the string can't be parsed as an integer (for example, it contains text or a decimal point), the provided defaultValue is returned.
strtofloat()	Converts a string to a decimal (floating point) value.	`strtofloat(inputString, defaultValue)` If the conversion fails, the provided `defaultValue` is returned. The resulting decimal attribute must be declared with the intended precision. For example, `decimal(3)`.
regexpreplace()	Take an input string and a defined Regular Expression (regex). Replace the entire input string with a specific captured group from the regex match.	`regexpreplace(InputString, RegexPattern, ReplacementGroup)` This function is commonly used to derive numeric data from complex, formatted strings before they are converted into integers or decimals.
get(index, array)	Used to retrieve an element at a specified index (starting at 0) from an array or list generated by a function like strsplit.	`string splitStr1 = get(0, strsplit("hello constraint engine", " ")); // returns hello`

Function	Purpose	Syntax, Examples, and Additional Details
		Using Get with Table Constraint—If your table constraint output for three columns is stored in a string[] variable named picklistValues, you would access the first row's attributes like this (assuming the columns are stored in a predictable sequence): `// Retrieve the first element (Row 0, Column 0) string value0_0 = get(0, picklistValues); // Retrieve the second element (Row 0, Column 1)`

String Comparison and Validation Operators

The in operator checks if the value of a string attribute is present within a defined set or array of strings.

Example

```
color in ["red", "blue"]
```

Variable Annotations

You can annotate variables with properties such as configurable, defaultValue, domainComputation, and relatedAttributes.

In this example, the `gc_runningKw` variable is annotated to indicate that it's not configurable and has a default value of `0.00`.

Note: This example requires an enclosing type.

```
@(configurable = false, defaultValue = 0.00)
decimal(2) gc_runningKw;
```

Property	Values	Description
allowOverride	true, false	Allows the engine to recalculate a value even if a value was already received from core. This annotation helps save on performance by allowing early calculation.
configurable	true, false	Indicates whether the variable is configurable (`true`) or not (`false`). If the configurable annotation isn't explicitly specified, the engine sets it implicitly as `true` for the variable.

Property	Values	Description
		If the configurable annotation is explicitly specified as `true`, the variable is indicated as configurable. The engine can set the value to the variable according to the defined logic, and users can modify it. If the configurable annotation is explicitly specified as `false`, the engine cannot set a value to the variable, and users can't update it. The variable value in this case is set through the PCM default. See examples here on page 1141.
defaultValue	literal	Indicates the default value for the variable. The configurator uses the default value defined in PCM (in Product Attribute Definition). If no PCM default is available, the configurator uses the first value in the variable domain as the initial value. If no default value is defined in PCM and a defaultValue is specified in CML, the configurator uses the value defined in CML as the initial value of the variable. See examples here on page 1141.
domainComputation	true, false	domainComputation is a CML annotation that specifies how the domain of a model element is determined, either by using a fixed domain or by computing the domain dynamically during configuration. If domainComputation is not explicitly specified, the engine sets it implicitly as false for the variable. If the domainComputation is specified as true, the variable domain is dynamically determined based on the configuration and constraint logic. If the domainComputation is specified as false, the variable domain is fixed. See examples here on page 1141.
nullAssignable	true, false	Sets an initial value for the calculated variable if the expression value can't be calculated.

Property	Values	Description
Peelable	true, false	Indicates whether the constraint engine can override the variable's value (whether set by default or user selection) to resolve a conflict (`true`) or not (`false`). If set to `true`, the engine treats the value as a "soft selection." When a configuration conflict occurs, the engine attempts to "peel" (unbind) this variable and retry the solution. If a valid configuration is found, the engine automatically changes the value to satisfy constraints, rather than displaying an error message to the user. If not explicitly set, or set to false, the engine treats the value as a "hard selection." If the value causes a conflict with a constraint, the engine will not attempt to change it automatically. Instead, it will stop and display a conflict error message to the user, requiring manual intervention to resolve the issue. See examples here on page 1141.
productGroup	integer	Used to represent the group cardinality for relationships under a type, either for specified relationships, or for all relationships (using`*`). We recommend using `maxInstanceQty` and `minInstanceQty` type annotations instead of productGroup. See Type Annotations on page 1093.
relatedAttributes	string value	relatedAttributesannotation is required to reset the domain to the original domain for domain computation. If relatedAttributes annotation is not specified, the engine updates the variable domain according to domainComputation and constraint logic. If the relatedAttributes annotation is specified with one or multiple values (separated by comma), the variable domain is reset to the original domain. See examples here on page 1141.

Property	Values	Description
sequence	integer	Indicates the sequence in which variables are configured. If a sequence value is not explicitly defined, the configurator implicitly determines the order based on the variable declaration order in the CML model. If a sequence value is explicitly defined, the configurator uses the sequence number to control the order in which variables are configured. Variables with lower sequence values are assigned first. See examples here on page 1141.
setDefault	true, false	Sets the variable status to default.
source	string value	Data source defined in the model.
sourceAttribute	Variable name in string	Sets the domain of the current variable to be the domain of the source variable.
strategy	descending, ascending, string	Defines the strategy to configure the variable. • If the strategy is ascending, the engine tries values from low to high. • If the strategy is descending, the engine tries values from high to low. See example here on page 1141.
tagName	string value	To create an external variable linked to a Sales Transaction header, use the tagName annotation with the contextPath annotation to reference context tags on SalesTransactionItem within a type. See contextPath in External Variable Annotations on page 1089.

External Variables

External variables are global Constraint Modeling Language (CML) variables defined within a virtual CML type.

See virtual in Type Annotations on page 1093. The values for external variables are usually set by the environment that launches the constraint solver engine. Use external variables to import runtime data from the context header (such as Quote or Sales Transaction) into the configuration model, with the contextPath annotation to denote header fields, or with tagName annotation to denote lineItem fields. See External Variable Annotations section.

If the external variable isn't mapped to any external data source, it must have a default value. Otherwise, it may remain unbound and cause errors.

Here's a basic declaration syntax.

```
extern int MAX_VALUE = 9999;
extern decimal(2) threshold = 1.5;
```

Example: Using External Variables with Context Path Annotation

In this example, the constraint engine needs access to the quote header (Sales Transaction) field, which defines the shipping location to enforce region-specific compliance requirements. The `contextPath` annotation is used to map the field (`SalesTransaction.ShippingCountry`) to an external CML variable (`ShippingCountry`).

Note: The CML variable name can be different from the context path value.

```
// External variable declaration with context path annotation
@(contextPath = "SalesTransaction.ShippingCountry")
extern string ShippingCountry; // ShippingCountry Value is pulled from the Quote/Order
header
```

See the full example in Using ContextPath and tagName annotations on page 1135.

External Variable Annotations

Here are the details of external variable annotations.

Annotation	Possible Value	Description
contextPath	"SalesTransaction.<ST_FIELD>", where the sales transaction field is pulled directly from the context definition.	References sales transaction values directly from their context definition, such as account name, sales transaction total, or address. The contextPath annotation can only be used for header fields. To create a variable linked to a SalesTransactionItem, use the tagName annotation to reference context tags on SalesTransactionItem within a type. See tagName in Variable Annotations on page 1086.

Types

In Constraint Modeling Language (CML), you define types to represent entities or objects in the model. Types are the foundational building blocks of CML. A type encapsulates the property, relationships, constraint, and rules for the entity.

A type is similar to a class in object-oriented programming. You can define relationships that represent associations between different types. See Relationships on page 1102.

Note: To define a constraint for a child product in a bundle, you must include the entire bundle in the constraint model. For example, if you define a constraint for a laptop, and the laptop is a child product in the Laptop Pro Bundle, you must include the Laptop Pro Bundle in the constraint model for the constraint on the laptop to run.

Note: Product variants aren't supported in constraint models.

Generic Structure of a Type

This table explains the generic structure of types with examples.

Element	Purpose	Example
Declaration	Defines the entity name, optionally preceded by annotations and optionally followed by inheritance, if applicable.	`type Product {}` `Or optionally: @annotation_name("annotation parameters") type Product: BaseProduct {}`
Variables (Attributes)	Defines the properties or characteristics of the entity, including data type and domain.	`int requiredKW = [101..10000]; string color = ["Red", "Blue"];`
Relations	Defines one-to-many associations with other types, specifying cardinality (the quantity range) and optionally, the configuration order.	`relation items : LineItem[1..10] {}`
Constraints and Rules	Enforces business logic and restrictions that must be satisfied by the entity's variables and relationships.	`constraint(condition); require(condition, items [type]);`

Example: Basic Type Declaration with Variables

This example shows the declaration of the main `GeneratorSet` type. It defines several core attributes (variables) that characterize the product.

```
type GeneratorSet{
// Declaration only (inherits LineItem properties)
// Attributes with explicit domains
int requiredKW = [101..10000];
string Voltage = ["220/380", "240/416", "255/440", "277/480", "347/600", "2400/4160",
"7200/12470", "7621/13200", "7976/13800];
string DutyRating = ["Prime Power (PRP)", "Continuous Power (COP)", "Data Center Continuous
 (DCC)", "Emergency Standby Power (ESP)"];
}
```

Type Hierarchies

Constraint Modeling Language (CML) supports inheritance and overriding, which allow you to create hierarchies between types. By establishing these hierarchies, constraint models become more modular and efficient.

Type Annotations

You can annotate types to add information. Type annotations are metadata applied to a type declaration to provide instructions to the constraint engine regarding how instances of that type should be handled, instantiated, or used in the configuration structure.

Group Type

In Constraint Modeling Language (CML), a Group Type is used to logically containerize related components within a bundle configuration, primarily when product component groups are imported from Product Catalog Management (PCM).

Type Hierarchies

Constraint Modeling Language (CML) supports inheritance and overriding, which allow you to create hierarchies between types. By establishing these hierarchies, constraint models become more modular and efficient.

How Hierarchies Function

- Inheritance: This mechanism enables a specialized child type to automatically share common variables (attributes) and relationships defined in its parent type. By inheriting from a base type, child types don't need to redefine shared properties, which ensures consistency across the model.
- Overriding: While child types inherit the structure of the parent, they can override or specialize those properties. For example, a parent type might define a variable with a broad range of possible values, while a child type overrides that variable with a fixed value specific to that individual product.

Practical Examples of Hierarchy

- Simple Product Extension: A `BaseProduct` might define an `id` and `name`. A `PhysicalProduct` can then inherit from `BaseProduct` to gain those fields while adding its own unique characteristics, such as `weight` or `color`.
- Multi-Level Nesting: Hierarchies can extend through multiple layers. For instance, a `Room` type can be the parent to a `Bedroom` type, and a `MasterBedroom` can further inherit from the `Bedroom` type, carrying all properties down the chain.
- Abstract Base Models: In complex configurations such as a `GeneratorSet`, a parent type such as `GeneralModel` defines the necessary attributes (such as `powerKW` and `dB`), while specific child types such as `GeneralModel900` or `GeneralModel1200` inherit those attributes and override them with their specific ratings.

Core Benefits

By establishing these hierarchies, constraint models become more modular and efficient. You can create header-level declarations and base types to serve as a foundation for the entire model, allowing you to reference reusable components multiple times rather than writing redundant code for every product variation. This structural organization allows the constraint engine to effectively enforce business logic and validate configurations across all related types.

Example 1: Simple Product Extension

In this pattern, a specialized type inherits from a broader base type. In the provided generator set model, the `GeneratorSet` type inherits from `LineItem`, gaining any properties defined at the line-item level while adding its own specific configuration fields like `requiredKW` and `Voltage`.

```
// The ultimate base type in the system
type LineItem;
// Child type extending LineItem with generator-specific attributes
type GeneratorSet : LineItem {
int requiredKW = [101..10000];
string Voltage = ["220/380", "240/416", "255/440"];
string DutyRating;
}
```

Example 2: Multi-Level Nesting

CML supports hierarchies with multiple layers of depth. In this example, properties flow from the top-level `LineItem` down to the `GeneratorSet`, and finally to a highly specialized `EmergencyGenerator`. Each level inherits all attributes from the levels above it.

```
type LineItem;
// Level 2: Adds basic generator capacity attributes
type GeneratorSet : LineItem {
int requiredKW = [101..10000];
decimal(2) surgeLoadKW = requiredKW * 1.25; // Calculation shared down the chain
}
// Level 3: Specialized version inheriting everything from GeneratorSet and LineItem
type EmergencyGenerator : GeneratorSet {
// Automatically inherits requiredKW and surgeLoadKW
string DutyRating = "Emergency Standby Power (ESP)"; // Specific fixed rating
}
```

Example 3: Abstract Base Models (Polymorphism & Overriding)

This pattern uses an abstract type as a structural blueprint. Specific components (the "General Models") inherit from this blueprint and override its generic attributes with fixed, real-world ratings.

```
// Base model acting as a blueprint for all engine options
type GeneralModel{
int powerKW = [100..2000]; // Broad domain
int dB = [60..100];
}
// Specific product type that overrides parent domains with fixed values
type GeneralModel900 : GeneralModel {
int powerKW = 900; // Overrides broad range with exact value
int dB = 78;
}
// Another specialized model with different fixed properties
type GeneralModel1500 : GeneralModel {
int powerKW = 1500;
int dB = 83;
}
```

Type Annotations

You can annotate types to add information. Type annotations are metadata applied to a type declaration to provide instructions to the constraint engine regarding how instances of that type should be handled, instantiated, or used in the configuration structure.

Annotation	Possible Values	Description
virtual	true, false	If `true`, specifies whether the indicated type refers to the transaction header (such as Quote or Order) or to a logical container (sub group of the Quote or Order). If the value is `false`, then it's the default behavior for types and doesn't need to be explicitly specified.

Annotation	Possible Values	Description
groupBy	Variable name	Used with `virtual = true`, the `groupBy` annotation organizes child products—the individual instances populating a relationship—into virtual containers based on a shared attribute value. See Relationships on page 1102 and the Grouping Generators by Voltage example on page 1135 .
maxInstanceQty	Integer	Specifies the maximum cardinality for a component in a group. See Group Type on page 1096.
minInstanceQty	Integer	Specifies the minimum cardinality for a component in a group. See Group Type on page 1096.
source	String	Specifies the data source defined in the model.
split	true, false, none	Specifies whether the type should be split or not. • If `split=true`, there can be multiple instances of the type, and the quantity of each instance is always 1. • If `split=false`, there is only one instance in the relationship. If the user adds more instances, the engine adds more quantity to the existing instance. • If `split=none` (the default), there are multiple instances of the same type in the relationship, with different quantities. The `split=true` annotation isn't supported for child products within a dynamic bundle. See examples here on page 1141.
sharingcount	Integer	Specifies the maximum number of times a single instance of a specific type can be shared or reused across different relationships within the configuration model.

Annotation	Possible Values	Description
		This annotation is used in conjunction with the @(`split=true`) annotation. When a type is marked for splitting, the constraint engine can process multiple instances in parallel to improve performance. The `sharingCount` tells the engine exactly how many times it can "split" or reuse that instance to satisfy the configuration requirements without generating entirely new, unique instances. It's a critical tool for managing large-scale configurations (for example, models with over 1,000 components). By setting a sharing limit, you reduce the number of variables the engine must instantiate, which helps prevent performance degradation and system timeouts.The `sharingCount` annotation works with the @(`sharing=true`) annotation applied to Relations. The relation annotation enables the general capability to share components across instances, while the `sharingCount` on the child type sets the numerical limit for that behavior. See Relationship Annotations on page 1107 and the Sharing Accessories in a Generator Set example on page 1135.

Creating a Virtual Container (@virtual = true)

In this example, the `@virtual = true` annotation is applied to a logical container type, `System`, which is primarily used to define relationships. These relationships aggregate data across line items in the quote that forms a sub-group called `system`. See Relationships on page 1102.

```
@(virtual = true)
type System {
// This relation gathers all GeneratorSet line items on the sales transaction
@(sourceContextNode = "SalesTransaction.SalesTransactionItem")
relation generators : GeneratorSet[0..10];
// This variable aggregates the surge load (calculated inside GeneratorSet) from all
collected generators
int totalQuotedLoad = generators.sum(surgeLoadKW);
}
type GeneratorSet {
// The attribute calculated here is aggregated in the virtual 'System' type above
@(configurable = false)
int requiredKW = [101..10000];
```

```
string DutyRating = ["Prime Power (PRP)", "Continuous Power (COP)", "Data Center Continuous
 (DCC)", "Emergency Standby Power (ESP)"];
decimal(2) surgeLoadKW = requiredKW * 1.25;
}
```

Group Type

In Constraint Modeling Language (CML), a Group Type is used to logically containerize related components within a bundle configuration, primarily when product component groups are imported from Product Catalog Management (PCM).

The Group Type uses specific annotations to control the total quantity of components selected from that group, regardless of the individual component type.

Bundles and Group Types (also known as Product Component Groups or PCGs) represent different levels of a product hierarchy and serve distinct functional roles in configuration logic.

Conceptual Hierarchy

- Bundles are high-level parent products that contain multiple child products sold together as a package. In CML, they are defined as "root types" that encapsulate the properties, relationships, and constraints for the entire entity.
- Group Types are structural containers within a bundle. They act as intermediate folders that organize related components imported from Product Catalog Management (PCM). Instances of these Group Types are declared as variables inside the root Bundle type.

Role in Cardinality and Selection

- Bundles establish the primary relationship between a root product and its broad categories of components.
- Group Types are specifically designed to enforce cardinality rules for a collection of products. They use the `@(minInstanceQty)` and `@(maxInstanceQty)` annotations to control exactly how many instances can be selected from a specific set of options (for example, "select at least 1 but no more than 2 accessories"). While the selected component can have a high quantity, the Group Type restricts the number of unique instances chosen from that group.

Syntactic Implementation

- Accessing Components: For standard bundles, you reference components directly via their relation name. For components within a Group Type, you must use dot notation starting with the group variable name defined in the root type (for example, `accessoryGroup.mouse.Wireless == true`).
- Constraint Limitations: You cannot write a constraint directly on a Group Type's attribute to apply it to all components within that group; constraints must reference the specific child components.
- Identification: Group Types are automatically identified by a Group suffix and the presence of instance cardinality annotations during the import from PCM.

Note: The following examples are partial. See the complete code in the final CML code sample.

Example 1: Defining Generator Set Group Types

We can define two group types for a Generator Set configuration: `CoreModelGroup` (mandatory selection of a primary generator model) and `EnclosureGroup` (optional selection of a specific enclosure type).

For the `CoreModelGroup`, setting `minInstanceQty` to 1 and `maxInstanceQty` to 1 means that at least 1 is required, and a maximum of 1 is permitted.

```
@(minInstanceQty=1, maxInstanceQty=1)
type CoreModelGroup {
// Relation holds the actual Generator Model products
relation generalModel : GeneralModel[0..9999];
}
type GeneralModel : LineItem {
int powerKW;
}
```

For the `EnclosureGroup`, setting `minInstanceQty` to 0 and `maxInstanceQty` to 1 means that selecting an enclosure is optional, but if selected, the user can add at most one instance of an enclosure component (such as WeatherproofEnclosure).

```
@(minInstanceQty=0, maxInstanceQty=1)
type EnclosureGroup {
relation enclosure : Enclosure;
relation cabinetHeater : ControlCabinetHeater;
}
type Enclosure : LineItem;
type ControlCabinetHeater : LineItem;
```

Example 2: Referencing Group Types in the Root Bundle

Once defined, instances of these group types are used as variables within the root type, which represents the entire bundle. In this example, the `coreModelGroup` and `enclosureGroup` are instances of the respective group types, defined within the root type `GeneratorSetBundle`.

```
type GeneratorSetBundle {
CoreModelGroup coreModelGroup;
EnclosureGroup enclosureGroup;
}
```

Example 3: Writing Constraints on Group Components

To write constraints or rules that reference components inside a group, use dot notation starting with the group variable name defined in the root type.

This example shows how a constraint might be defined within the `GeneratorSetBundle` type to enforce business logic on components within the groups.

```
// If the selected generator model has a powerKW greater than 1500,
// then a Control Cabinet Heater must be included in the Enclosure Group.
constraint(coreModelGroup.generalModel.powerKW > 1500 ->
enclosureGroup.cabinetHeater[ControlCabinetHeater] == 1);
// Require an Enclosure component if the set requires seismic certification.
require(seismicCertification == "IBC Seismic Certification",
enclosureGroup.enclosure[Enclosure],
"Enclosure required for seismic certification");
```

Final CML Code Sample with Group Types

This structure defines the complete model, showing the component hierarchy and the relationship constraints.

Parent Type	Relation Name	Child Type	Cardinality	Key Attribute Usage
`GeneratorSetBundle`	`coreModelGroup` (Group Instance)	`CoreModelGroup`	Group Cardinality: 1	The group type defined by `@minInstanceQty=1, maxInstanceQty=1`. This enforces that exactly one component choice must be made from the internal generalModel relation.
`GeneratorSetBundle`	`enclosureGroup` (Group Instance)	`EnclosureGroup`	Group Cardinality: [0..1]	The group type defined by `@minInstanceQty=0, maxInstanceQty=1`. This enforces that selecting components from this group is optional (min 0) and at most one total component instance can be selected (max 1).

```
/**
* The Root Bundle: GeneratorSetBundle
* This type holds instances of the defined Group Types and the definition of seismic
certification */
type GeneratorSetBundle {
string seismicCertification = ["IBC Seismic Certification", "OSHPD Seismic Certification"];
CoreModelGroup coreModelGroup;
EnclosureGroup enclosureGroup;
// Constraint 1: If the selected GeneralModel has a powerKW greater than 1500,
// then a Control Cabinet Heater must be included in the Enclosure Group.
constraint(coreModelGroup.generalModel.powerKW > 1500 ->
enclosureGroup.cabinetHeater[ControlCabinetHeater] == 1);
// Constraint 2: Require an Enclosure component if the set requires seismic certification.
require(seismicCertification == "IBC Seismic Certification",
enclosureGroup.enclosure[Enclosure],
"Enclosure required for seismic certification");
// Action Rule Example: Disable a specific enclosure type if the core model is low power
(e.g., 900kW).
rule(coreModelGroup.generalModel.powerKW == 900, "disable", "relation",
"enclosureGroup.enclosure", "type", "ReinforcedEnclosure");
}
/**
* Group Type 1: Core Model Group (Mandatory, Single Select)
* Min/Max Instance Quantity controls the selection rules for components within this group.
*/
@(minInstanceQty=1, maxInstanceQty=1)
type CoreModelGroup {
```

```
// Relation holds the actual Generator Model products
relation generalModel : GeneralModel[0..9999];
}
/**
* Group Type 2: Enclosure and Accessories Group (Optional)
* minInstanceQty=0 means selection is optional. maxInstanceQty=1 limits the selection to
a single enclosure or accessory component instance.
*/
@(minInstanceQty=0, maxInstanceQty=1)
type EnclosureGroup {
relation enclosure : Enclosure;
relation cabinetHeater : ControlCabinetHeater;
}
/**
* Component Types (Children)
*/
type GeneralModel {
int powerKW;
}
type Enclosure {
@(defaultValue = "false")
boolean Weatherproof;
}
type ReinforcedEnclosure : Enclosure; // Subtype of Enclosure
type ControlCabinetHeater;
```

Key Considerations

When reviewing Group Types in the context of the Generator Set model, keep these architectural points in mind.

- Group Cardinality Enforcement: The annotations `@(minInstanceQty)` (minimum instance quantity) and `@(maxInstanceQty)` (maximum instance quantity) are defined on the Group Type itself (`ElectricalSafetyGroup`). These annotations control the overall cardinality for all the components contained within that group, regardless of the individual relation cardinality defined (for example `[0..1]`, `[1..2]`).
- Root Reference: The `GeneratorSetBundle` includes the groups as variables (`coreModelGroup` and `enclosureGroup`).
- Constraint Syntax for Group Components: Constraints access attributes or components inside the groups using dot notation starting with the group variable name (for example, `coreModelGroup.generalModel.powerKW`).
- Limitation on Group Attributes: You cannot write a constraint directly on a group's attribute and expect it to apply to all components within that group (for example, `constraint(enclosureGroup.color == "Black")`) is not a valid constraint).

Define Constraints for Quote Groups, Ramps, and Ramp Segments
Apply rules to quote groups, ramps, and ramp segments by using the SalesTransactionItemGroup context tag. Assign a groupby value to messages defined in Constraint Rules Engine to include the messages in custom grouping strategies.

Define Constraints for Quote Groups, Ramps, and Ramp Segments

Apply rules to quote groups, ramps, and ramp segments by using the SalesTransactionItemGroup context tag. Assign a groupby value to messages defined in Constraint Rules Engine to include the messages in custom grouping strategies.

[Define a Constraint for a Quote Group](#)

To define a constraint for a quote group, use the require rule to assign the SalesTransactionItemGroup attribute that's contained on a type to the value of the QuoteGroup container.

[Define a Constraint for a Ramp Group](#)

To define a constraint that applies to a ramp group, use the IsLineGroupRamped__std attribute in the require rule to specify that the group is a ramp group.

[Define a Constraint for a Ramp Segment](#)

To define a rule that applies to a ramp segment for defined conditions, use the ItemSegmentType attribute.

Define a Constraint for a Quote Group

To define a constraint for a quote group, use the require rule to assign the SalesTransactionItemGroup attribute that's contained on a type to the value of the QuoteGroup container.

In this example for a GeneratorSet and its Enclosure, all the relations defined in the groupBy container include the `SalesTransactionItemGroup` attribute. Constraint Rules Engine creates a virtual container for each unique value of the `SalesTransactionItemGroup` attribute defined in the `groupBy` container. The engine creates one virtual container with its own self-contained rule execution for each group that contains one or more of the products defined in the `QuoteGroup` container.

Note: Groups don't support rules such as Hide, Disable, or Recommend, that perform an action when specified conditions are true.

Example:

```
type LineItem {
    @(tagName = "SalesTransactionItemGroup")
    string SalesTransactionItemGroup;
}

type GeneratorSet : LineItem;
type Enclosure : LineItem;

// Transaction scoped container
@(virtual = true)
type Quote {
    relation quotegroup : QuoteGroup;
   @(sourceContextNode = "SalesTransaction.SalesTransactionItem") relation generators
 : GeneratorSet;
   @(sourceContextNode = "SalesTransaction.SalesTransactionItem") relation enclosures
 : Enclosure;
}

// Group scoped container
@(virtual = true, groupBy = SalesTransactionItemGroup)
type QuoteGroup {
    string SalesTransactionItemGroup; // Needed for the require rule to save

   @(sourceContextNode = "SalesTransaction.SalesTransactionItem") relation generators
 : GeneratorSet;
   @(sourceContextNode = "SalesTransaction.SalesTransactionItem") relation enclosures
 : Enclosure;

   require(generators[GeneratorSet], enclosures[Enclosure] {SalesTransactionItemGroup
```

```
 = SalesTransactionItemGroup});
}
```

Define a Constraint for a Ramp Group

To define a constraint that applies to a ramp group, use the IsLineGroupRamped__std attribute in the require rule to specify that the group is a ramp group.

In this example, the Enclosure is added to the group only if the GeneratorSet is in a ramp group. In the last lines of the example, the `IsLineGroupRamped__std` attribute in the require rule specifies that the GeneratorSet is in a ramp group.

Note: Groups don't support rules such as Hide, Disable, or Recommend, that perform an action when specified conditions are true.

Note: To add a type to an unramped group only and exclude the type from ramp groups, set the `IsLineGroupRamped__std` attribute to NOT (!) in the require rule. For example, `require(!(generators[GeneratorSet].IsLineGroupRamped__std),`.

Example:

```
type LineItem {
    @(tagName = "SalesTransactionItemGroup")
    string SalesTransactionItemGroup;
    @(tagName = "IsLineGroupRamped__std")
    boolean IsLineGroupRamped__std;
}

type GeneratorSet : LineItem;
type Enclosure : LineItem;

// Transaction scoped container
@(virtual = true) type Quote;

// Group scoped container
@(virtual = true, groupBy = SalesTransactionItemGroup)
type QuoteGroup {
    string SalesTransactionItemGroup; // Needed for the require rule to save

   @(sourceContextNode = "SalesTransaction.SalesTransactionItem") relation generators
 : GeneratorSet;
   @(sourceContextNode = "SalesTransaction.SalesTransactionItem") relation enclosures
 : Enclosure;

    require(generators[GeneratorSet].IsLineGroupRamped__std, enclosures[Enclosure]
{SalesTransactionItemGroup = SalesTransactionItemGroup});
}
```

Define a Constraint for a Ramp Segment

To define a rule that applies to a ramp segment for defined conditions, use the ItemSegmentType attribute.

In this example, Enclosure is included in the group with GeneratorSet only if `ItemSegmentType` for GeneratorSet isn't Trial. To set a value for `ItemSegmentType`, the type must be in a ramp group and be ramped.

Note: Groups don't support rules such as Hide, Disable, or Recommend, that perform an action when specified conditions are true.

Example:

```
type LineItem {
@(tagName = "SalesTransactionItemGroup")
string SalesTransactionItemGroup;
@(tagName = "ItemSegmentType")
string ItemSegmentType;
}

type GeneratorSet : LineItem;
type Enclosure : LineItem;

// Transaction scoped container
@(virtual = true) type Quote;

// Group scoped container
@(virtual = true, groupBy = SalesTransactionItemGroup)
type QuoteGroup {
string SalesTransactionItemGroup; // Needed for the require rule to save

@(sourceContextNode = "SalesTransaction.SalesTransactionItem") relation generators :
GeneratorSet;
@(sourceContextNode = "SalesTransaction.SalesTransactionItem") relation enclosures :
Enclosure;

require(generators[GeneratorSet].ItemSegmentType != "Trial", enclosures[Enclosure]
{SalesTransactionItemGroup = SalesTransactionItemGroup});
}
```

Relationships

Relationships in Constraint Modeling Language (CML) define how different product types are associated with each other, forming the structural hierarchy of a product bundle. Relationships are also referred to as ports.

Here's a comprehensive overview of relationships, their syntax, purpose, and key features, particularly using examples relevant to the Generator Set model.

Definition and Syntax of Relationships

Relationships define the one-to-many connections between a parent type (such as a bundle) and its component types (children).

- Keyword: The keyword used is `relation`.
- Syntax: A basic relationship declaration includes the relation name, the target type, and cardinality bounds.

```
relation <relation name> : <Target Type>[min..max] { /* Optional content */ }
```

- Purpose: Relationships represent the product structure in a bundle. For example, the root product (GeneratorSet) has relationships with its components (`MainAlternators`, `TemperatureSensors`).

Note: Specifying the smallest required cardinality (quantity range) is a best practice to avoid unnecessary testing of value combinations, which improves performance.

Set Maximum Relationship Size

The `maxRelationSize` property is the maximum cardinality for a single relationship in the constraint model. It controls how many instances of a child component can exist under one parent through the relationship. The maximum value for `maxRelationSize` is 1073741824. The default value is 9999.

Use `maxRelationSize` to set a limit based on the number of instances you need for your business purposes. For example, if the most a customer can order of a child component is 200,000, set `maxRelationSize` to 200,000. An overly high cardinality limit increases the risk of unbound variables that cause the constraint engine to backtrack through the entire range to resolve a conflict. Use `maxRelationSize` with the `property`.

```
property maxRelationSize = 100000;

 type Quote {


   relation items : LineItem[0..99999999]; // cardinality uses this max

 }
```

The `maxRelationSize` property doesn't limit these values:

- The number of relationship declarations in a type
- The number of types in the constraint model
- The total component count across all relationships

Omit Unnecessary Relationships

When using the visual builder or the CML editor to create a CML code for a bundle, the system by default imports all the relationships for the selected bundle from the structure defined in Product Catalog Management (PCM). In large and complex CML code, some of these relationships may not be relevant to any constraint and can be potentially omitted.

To enable import of a subset of bundle components, add this property at the top of the constraint model CML file.

```
property allowMissingRelations = "true";
```

If your PCM bundle contains many different relations but your CML code defines only one, the engine will validate the model but this often results in a configuration run-time failure. By setting `allowMissingRelations = "true"`, you do not have to define every relation found in the PCM (such as GeneralModels in the Relationship Ordering example) if they do not require specific configuration logic in your CML file.

Here's an example with allowMissingRelations property.

```
// 1. Enable skipping of unneeded relations from the Product Catalog (PCM)
property allowMissingRelations = "true";

type ConciseGeneratorBundle  {
    // Define only the specific accessory needed for this logic
    relation enclosures : Enclosure;

    // A simple variable to trigger the logic
    int requiredKW = [100..5000];

    // Logic: High power requirements force a specific enclosure type
    // This omits other accessories like filters, batteries, and heaters [2, 3].
    constraint(requiredKW > 2000 -> enclosures[ReinforcedEnclosure] == 1,
```

```
                "Power levels above 2000kW require a Reinforced Enclosure.");
}

// 2. Define the accessory and its specific subtype
type Enclosure ;
type ReinforcedEnclosure : Enclosure;
```

For more information, see Constraints on page 1110.

To ensure run-time stability without the allowMissingRelations property, you must manually define every single relation and type present in the PCM bundle, even if you don't intend to write logic for them. This creates large CML files with a high number of variables and components, which lead to performance degradation, and even timeout issues.

Note: This code isn't recommended.

```
// EXPENSIVE FIX: Mirroring everything
type GeneratorSet {
    relation engineModels : EngineModel; // Unused in CML logic
    relation alternators : Alternator;    // Unused in CML logic
    relation fuelFilters : FuelFilter;    // Unused in CML logic
    relation starterMotors : StarterMotor; // Unused in CML logic
    relation enclosures : Enclosure;      // The only one we need
    // ... potentially 15+ more relations ...

    constraint(requiredKW > 2000 -> enclosures[ReinforcedEnclosure] == 1);
}
type Enclosure ;
type ReinforcedEnclosure : Enclosure;
```

Order Keyword

The `order()` keyword is used within a `relation` declaration to define the specific sequence in which the constraint engine evaluates and attempts to instantiate the child subtypes available in that relationship. This controls the prioritization of component selection.

Example: Relationship Ordering

```
// --- Component Subtypes (Specific Generator Models) ---
// Define a base type for generator models with a power attribute
type GeneralModel {
int powerKW = [0..3000]; // Explicit domain
}
// Specific subtypes that inherit from GeneralModel
type GeneralModel2500 : GeneralModel {
int powerKW = 2500;
}
type GeneralModel1750 : GeneralModel {
int powerKW = 1750;
}
type GeneralModel900 : GeneralModel {
int powerKW = 900;
}
// --- Parent Type (GeneratorSet) ---
```

```
type GeneratorSet {
// Required power defined by the parent (non-configurable)
@(configurable = false)
int requiredKW = [100..3000];
// Relation Declaration using order()
// Cardinality requires exactly one model to be selected.
// order() sets the selection priority (2500 KW model is checked before 1750 KW model).
relation GeneralModels : GeneralModel  order(GeneralModel2500, GeneralModel1750,
GeneralModel900);
}
```


Relationship Variable Functions

CML variable functions are fundamental tools used to perform both aggregation (summarizing data from related components) and complex mathematical calculations on attribute values (variables) within a configuration model. These functions are crucial for enforcing dimensional validity and calculating derived attributes.

You can use functions such as `count()`, `min()`, `max()`, `sum()` and `total()` to calculate values from all variables with the same name in the descendants of the current type. Aggregate functions are used primarily as relationship attributes within a relation to calculate values across multiple instances of a component type (descendants).

Function	Purpose	CML Keyword [Source]
`count()`	Counts the number of component instances within a relationship that match a specific logical condition.	`count`
`max() / min()`	Returns the maximum or minimum value of an attribute found across all instances in a relationship.	`max, min`
`sum()`	Calculates the sum of a specific numeric variable across all instances in a relationship. Note: executes multiplication of a variable value by a product quantity.	`sum`
`total()`	Calculates the sum of a specific numeric variable across all instances in a relationship. Note: Unlike the sum() function, does not execute multiplication of a variable value by line item quantity.	`total`

Example: Using Aggregate Functions

```
type LineItem;
type GeneratorSet : LineItem {
relation modelclassification : ModelClassification {
sumPowerKW = sum(powerKW);
maxPowerKW = max(powerKW);
highPowerModelsAmount = count(powerKW > 500);
}
relation warranties : Warranty  {
totalCoverageDays = total(coverageDays);
}
constraint(modelclassification.sumPowerKW > 2500 && modelclassification.sumPowerKW < 3500);
message(true, "The maximum `powerKW` from selected GeneralModels is: {}",
modelclassification.maxPowerKW);
constraint(warranties[Warranty] == modelclassification.highPowerModelsAmount);
message(true, "Effective Warranty coverage days: {}", warranties.totalCoverageDays);
}
type ModelClassification : LineItem {
int powerKW = [900, 1750, 2500];
}
type GeneralModel1750 : ModelClassification {
int powerKW = 1750;
}
type GeneralModel2500 : ModelClassification {
int powerKW = 2500;
}
type GeneralModel900 : ModelClassification {
int powerKW = 900;
}
type Warranty {
int coverageDays;
}
type Warranty_PRP : Warranty {
int coverageDays = 50;
}
type Warranty_DCC : Warranty {
int coverageDays = 100;
}
type Warranty_ESP : Warranty {
int coverageDays = 200;
}
```

sum()

In the example, the `sum()` function aggregates the `powerKW` variable values of the selected products in the `modelClassification` relationship. The model includes a constraint that enforces the selection rule based on the calculated total: the `sum` of `powerKW` for the selected products must be between 2500 and 3500.

```
constraint(modelclassification.sumPowerKW > 2500 && modelclassification.sumPowerKW < 3500);
```

As a result, the engine selects two products from the `modelClassification` relation (`GeneralModel1750` and `GeneralModel900`), since their calculated sum of `powerKW` values (2650) satisfies the constraint.

max()

In the example, the `max()` function returns the maximum `powerKW` variable value among the selected products in the `modelClassification` relationship. The model defines a message that displays the maximum `powerKW` value in the Product Configurator.

```
message(true, "The maximum `powerKW` from selected GeneralModels is: {}",
modelclassification.maxPowerKW);
```

As a result, because the previously described `sum()` function and constraint selected two products (`GeneralModel1750` and `GeneralModel900`), the maximum returned value is 1750.

count()

In the example, the `count()` function calculates the amount of selected `modelClassification` relationship products that contain the `powerKW` variable value greater than 500. The model includes a constraint that enforces the selection of `warranties` product for each `GeneralModel` counted by the function.

```
constraint(warranties[Warranty] == modelclassification.highPowerModelsAmount);
```

As a result, based on the previous `sum()` description, the engine selects two products (`GeneralModel1750` and `GeneralModel900`). Since both products meet the count() condition (`powerKW` > 500), the engine adds one of the warranties product with quantity 2.

total()

In the example, the `total()` function aggregates the values of the `coverageDays` variable for the selected products of the `warranties` relationship. The model defines a message that displays the calculated total in the Product Configuration.

```
message(true, "Effective Warranty coverage days: {}", warranties.totalCoverageDays);
```

As a result, based on previously described `count()` example, the system selects one of the `warranties` product (e.g., `Warranty_DCC`) with a quantity of 2 and calculates the total value of its `coverageDays` variable (100).

A key difference between `total()` and `sum()` functions is that for `total()` the engine ignores the `warranties` product quantity. The system does not multiply the quantity of the selected warranty by the `coverageDays` variable value when computing the overall total.

See also Arithmetic Calculations and Functions on page 1135 and examples here on page 1197.

Relationship Annotations

You can annotate relationships by using annotations, such as configurable, allowNewInstance, closeRelation, sourceContextNode, and so on.

You can annotate relationships, as in this example, with `configurable=true`.

```
// 1. Define the target type
type Component;
// 2. Define the parent type to hold the relation
type System {
// 3. Add the relation with cardinality and proper nesting
@(configurable = true)
relation components : Component[0..10];
}
```

Here are the details of relationship annotations.

Annotation	Values	Description
allowNewInstance	true, false	Enables require rule constraints to work on relationships that are otherwise closed. Must be set to true to enable require rule constraints on closed relationships.
closeRelation	true, false	If the value is true, prevents the addition of new line items to the relationship. false is the default value and allows the addition of new line items to the relationship. See examples here on page 1141.
compNumberVar	true, false	Avoids creating a component number variable if it is set to false.
disableCardinalityConstraint	true, false	Disable cardinality constraint in the relationship to optimize the performance.
domainComputation	true, false	`domainComputation` is a Constraint Modeling Language (CML) annotation that specifies how the domain of a model element is determined, either by using a fixed domain or by computing the domain dynamically during configuration. If `domainComputation` is not explicitly specified, the engine sets it implicitly as `true` for the relationship. If the domainComputation is specified as `true`, the relationship domain is dynamically determined based on the configuration and constraint logic. If the `domainComputation` is specified as `false`, the relationship domain is fixed. See examples here on page 1141.
generic	true, false	Indicates if generic instance is allowed in the relationship or not. Generic instance is used to prompt the user that they need to select a product in the relationship.
noneLeafCardVar	true, false	Avoids creating cardinality variables for none leaf type (a node with no children) in the relationship to optimize the performance.
propagateUp	true, false	Aggregates values from child elements to parent elements.

Annotation	Values	Description
		If propagateUp is not specified, the engine sets it implicitly as false for the relationship. If the propagateUp annotation is specified as true, the engine aggregates values from children to parent elements (upward propagation). The engine cannot modify this value from the parent level (e.g. via constraint), so the children relation domain will not be affected. If the propagateUp is specified as false, both upward and downward propagations are applicable. The engine aggregates values from children to parent elements (upward propagation). Meanwhile the engine can modify this value (e.g. via constraint) from the parent level. The value is propagated downward and might affect the relation domain (downward propagation). See examples here on page 1141.
readOnly	true, false	Sets a relationship and all child relationships to read-only, to prevent the engine or user from modifying.
relatedAttributes	string value	`relatedAttributes` is a CML annotation that resets the domain to the original one for `domainComputation`. If `domainComputation` is not explicitly specified, the engine sets it implicitly as `true` for the relationship. If the `domainComputation` is specified as `true`, the relationship domain is dynamically determined based on the configuration and constraint logic. If the `domainComputation` is specified as `false`, the relationship domain is fixed. See examples here on page 1141.
relatedRelationships	string value	Related relationships whose cardinality variables must be unbound for domain computation.
sequence	integer	Indicates the sequence in which the relationship is configured and executed.

Annotation	Values	Description
sharing	true, false	Indicates if the relationship is shared or not. If the relationship is shared, the engine connects the instance from another relationship to this relationship instead of instantiating the instance in the relationship itself.
singleton	true, false	Indicates if all types in the relationship must be singleton or none.
source	string	Data source defined in the model.
sourceAttribute	Variable name in string	Sets the domain of the current relationship to the domain of the source attribute.
sourceContextNode	string	For cases that use a virtual container, specifies the path in the context service for the instances in the relationship.

Constraints

Constraints enforce rules and conditions on types, variables, and relationships. Use constraints to define logical restrictions and ensure consistency within the model.

For more information about the supported constraints, see:

- Logical Constraints on page 1113
- Table Constraints on page 1118
- Using Proxy Variables with Constraints on Types and Relationships on page 1120
- Group Type on page 1096
- Message Rule on page 1127
- Preference Rule on page 1128
- Require Rule on page 1128
- Require Rule vs Constraint on page 1129
- SetDefault Rule on page 1129
- Exclude Rule on page 1130
- Action Rule on page 1130
- Hide/Disable Rule on page 1131
- Recommendation Rule on page 1133
- Set Product Selling Model in a Constraint on page 1134

Supported Logic Operators

These logic operators are supported in CML.

Arithmetic Operators

- Multiplication (*)
- Division (/)
- Remainder (%)
- Addition (+)
- Subtraction (-)

Relational Operators

- Greater than (>)
- Greater than or equal to (>=)
- Less than (<)
- Less than or equal to (<=)

Equality Operators

- Equal (==)
- Not equal (!=)

Logic Operators

- Not (!)
- And (&&)
- XOR/Exclusive or (^)
- Or (||)
- Bi-conditional (<->)
- Conditional (?:)
- Implication (->)

Operator Precedence

In resolving equations, operator precedence determines the order in which operations are performed. Operators in CML have precedence in this order:

- Arithmetic operators have the first precedence.
- Relational operators have the second precedence.
- Equality operators have the third precedence.
- Logic operators have a lower precedence than equality operators, in decreasing order as listed, with Implication having the lowest precedence.

Constraint Annotation

Here are the details of `abort`, a constraint annotation.

Annotation	Possible Values	Description
abort	true, false	Specifies that, if this constraint fails, abort search and return `false` for configuration.

Logical Constraints

A logical constraint defines a statement that must hold true logically. The constraint can be any logical expression by using a logical operator.

Table Constraints

The table constraint in Constraint Modeling Language (CML) is used to define a set of valid combinations of values for two or more attributes. These combinations are specified in rows within the constraint definition.

Proxy Variables with Constraints on Types and Relationships

Use proxy variables to reference the variables of related types, including parent, root, and sibling types.

Message Rule

The message rule displays a message to users based on specified conditions.

Preference Rule

The preference rule encourages the constraint solver to satisfy the condition, but doesn't enforce it if the condition can't be met.

Require Rule

The require rule requires certain components to be included in a relationship when specified conditions are met.

Require Rule vs Constraint

In Constraint Modeling Language (CML), constraint() and require() can both enforce behavior, but they operate differently: constraint focuses on logical consistency, require focuses on physical presence of products.

Setdefault Rule

The setdefault rule allows component selection with attribute values and quantity, similar to the require rule.

Exclude Rule

The exclude rule is used to automatically remove a specific type in a relationship if a certain condition is met.

Action Rule

The CML Action Rule is defined using the rule() keyword. Its primary purpose is to execute a designated action, specified as a string literal, when a condition is met.

Hide or Disable Rule

The Hide or Disable Rule uses the rule() keyword to conditionally remove an element from the selection menu (hide) or preserve it in the menu while preventing user selection (disable).

Recommendation Rule

The recommend keyword is used within a Constraint Modeling Language (CML) rule to display suggestions for related products in the Product Configurator. The rule defines the condition under which a specific product type or relation should be suggested to the user.

Set Product Selling Model in a Constraint

Use the productSellingModel tagname to write a constraint that sets the Product Selling Model (PSM) for a type. You can define a PSM as one time, time-deferred (subscription with end date), or evergreen (recurring subscription with no preset end date). The PSM is updated for new line items at runtime, based on the constraint.

Logical Constraints

A logical constraint defines a statement that must hold true logically. The constraint can be any logical expression by using a logical operator.

For example, the statement `c0 ? c1 : c2` means that if `c0` is true, then `c1` needs to be true, otherwise `c2` needs to be true.

Constraint Syntax Patterns

Here are the details of the constraint syntax patterns.

- `constraint(logic expression);`: The simplest form, enforcing a logic statement.
- `constraint(logic expression, string literal | string variable);`: Includes an optional failure explanation that is displayed if the constraint is violated.
- `constraint(logic expression, string literal | string variable, arg, …, arg);`: Includes a failure explanation with additional arguments to be interpolated into the string of the failure explanation.

If a constraint is violated, it takes an optional string variable or string literal as the failure explanation. The failure explanation could be in a string format with additional arguments. CML supports two string formats. One is the standard string format and another is a string with {} placeholder, to be replaced with the actual argument value.

Key Components of the Constraint Syntax

Here are the details of the key components of the constraint syntax.

Component	Description	Code Sample
Logic Expression	This can be a basic mathematical expression (using operators like +, -, * , /) or a relational expression (using ==, !=, >, <, and so on). It can also include logical operators such as AND (&&), OR (||), NOT (!), the implication operator (->), or the bi-directional operator (< - >).	Basic `constraint(generator.sum(quantity) > 20);`
Failure Explanation	This optional string literal or variable provides a human-readable reason for the violation. CML supports two formatting styles for these strings: • Java string format (for example, using `%d` or `%s`). • Placeholder format using `{}`	With Explanation `constraint(x + y <= 100, "Total must not exceed 100");`
Arguments (arg)	Arguments are used to replace the placeholders in the failure explanation string with actual values at runtime.	With Explanation and Arguments `constraint(requiredKW <= 2500, "The required capacity of {} kW exceeds the maximum supported limit of 2500 kW.",requiredKW);`

See the complete example in Constraints using Format Specifiers (%s, %d) and Dates on page 1135.

Constraint Example

In this example, the first constraint specifies that, if the `DutyRating` value isn't `Prime Power (PRP)`, then the quantity of `Warranty_PRP (Warranties subtype)` must be 0. Similarly in the second constraint, if `DutyRating` value isn't `Data Center Continuous (DCC)` then the quantity of `Warranty_DCC` must be 0. Lastly, in the third constraint, if `DutyRating` value isn't `Emergency Standby Power (ESP)`, then the quantity of `Warranty_ESP` must be 0.

```
type Warranty;
type Warranty_PRP : Warranty;
type Warranty_DCC : Warranty;
type Warranty_ESP : Warranty;
type GeneratorSet  {
string DutyRating = ["Prime Power (PRP)", "Continuous Power (COP)", "Data Center Continuous
 (DCC)", "Emergency Standby Power (ESP)"];
relation Warranties : Warranty[3];
constraint(DutyRating != "Prime Power (PRP)" -> Warranties[Warranty_PRP] == 0);
constraint(DutyRating != "Data Center Continuous (DCC)" -> Warranties[Warranty_DCC] == 0);
constraint(DutyRating != "Emergency Standby Power (ESP)" -> Warranties[Warranty_ESP] ==
0);
}
```

Example: Constraints Using String and Logical Operators

This example demonstrates how a GeneratorSet uses string functions to extract a numeric voltage value, and then uses logical operators (`->`, `&&`, `||`) to enforce safety and certification requirements.

Explanation of Operators Used in the Example

Here are the details of the operators used in the example.

Operator/Function	Category	Usage in Example
`strtoint()`	String Function	Converts the extracted voltage string to the integer `Voltage3`.
`regexpreplace()`	String Function	Replaces the full Voltage string with only the second matched group (the high voltage number) using the defined `VOLTAGE_REGEX`.
`strcontain()`	String Function	Checks if the `DutyRating` string contains the substring "Power".
`->`	Logical Implication	Defines a conditional rule: If the condition on the left is true, the condition on the right must be true.
`&&`	Logical AND	Requires both conditions (high requiredKW AND duty rating contains "Power") to be true.

Operator/Function	Category	Usage in Example
`||`	Logical OR	Requires `standardsAndCompliance` to be "`Listing-UL 2200`" or "`Certification-CSA`".
`!=`	Comparison/Relational	Used to check if the string value is "Not Equal" to a specific string literal.

```
// --- Constants (Required for String Manipulation) ---
define VOLTAGE_REGEX "^(+)/(+)$"

// --- Base Type and Configuration Attributes ---
type Warranty;
type Warranty_ESP : Warranty;

type GeneratorSet {
    // String input attribute for user selection
    string Voltage = ["277/480", "2400/4160", "7976/13800"];

    // String input attribute for operational mode
    string DutyRating = ["Prime Power (PRP)", "Continuous Power (COP)", "Emergency Standby
 Power (ESP)"];

    // String input attribute for compliance
    string standardsAndCompliance = ["Certification-CSA", "Listing-UL 2200"];
    // Required KW (Int attribute)
    int requiredKW = [101..10000];
// warranties for generator set
relation Warranties : Warranty[3];

    // 1. STRING OPERATORS/FUNCTIONS: Deriving Numeric Data
    // We use strtoint() combined with regexpreplace() to extract the second number (the
high voltage)
    // from the Voltage string (e.g., extracting 480 from "277/480").
    int Voltage3 = strtoint(regexpreplace(Voltage, VOLTAGE_REGEX, "$2"), 0);
  // Prime Power or Continuous Power using strcontain
    boolean NonstandbyPower = !strcontain(DutyRating, "ESP");

    // 2. LOGICAL OPERATOR (Implication ->): Certification Validation
    // Constraint: If the standard is UL 2200 (precondition), THEN Voltage3 must be <= 600
 (postcondition).
    constraint(
        standardsAndCompliance == "Listing-UL 2200" -> Voltage3 <= 600,
        "The UL 2200 standard covers stationary engine generator assemblies rated at 600
volts or less."
    );


    // 3. LOGICAL OPERATORS (AND &&, OR ||) and STRING FUNCTION (strcontain)
    // Constraint: If the required power is high AND the unit is used for Prime Power OR
Continuous Power,
    // THEN a specific standard must be selected.
```

```
    constraint(
        (requiredKW >= 5000 && NonstandbyPower)
        ->
        (standardsAndCompliance == "Listing-UL 2200" || standardsAndCompliance ==
"Certification-CSA"),  "High power and continuous use requires a major compliance
standard.");


    // 4. LOGICAL OPERATOR (Implication ->) and String Comparison (!=)
    // Constraint: If the Duty Rating is NOT Emergency Standby Power, THEN a specific
warranty (Warranties[Warranty_ESP]) must be excluded/zero.
    constraint(
        DutyRating != "Emergency Standby Power (ESP)" -> Warranties[Warranty_ESP] == 0,
       "The DutyRating when not equal to Emergency Standby Power, implies that the Warranty
 must be 0."
    );
}
```

How User Input Order Affects Constraint Engine Behavior

The constraint engine is architecturally designed to never override or modify a user-selected value when evaluating constraints, except in the specific case of an `exclude` rule. If a constraint violation occurs due to user input, the engine generates an error message rather than attempting to fix the value itself.

Example: Constraint Evaluation Based on Input Order (Generator Set)

The order in which a user configures attributes for a product (such as a `GeneratorSet`) determines whether the constraint engine performs an automatic update or enforces a validation error.

Consider a constraint within the `GeneratorSet` type that links the operational requirement (`DutyRating`) to the required certification (`standardsAndCompliance`).

```
type GeneratorSet {
// Attribute 1 (Precondition): User selects operational mode
string DutyRating = ["Prime Power (PRP)", "Continuous Power (COP)"];
// Attribute 2 (Dependent): Certification standard
string standardsAndCompliance = ["Certification-CSA", "Listing-UL 2200"];
// Constraint: If the unit is configured for Prime Power, it must have UL 2200 Listing.
constraint(
DutyRating == "Prime Power (PRP)" -> standardsAndCompliance == "Listing-UL 2200",
"Prime Power Duty Rating requires UL 2200 Listing"
);
}
```

The outcomes depend on the user's input sequence.

Scenario	User Input Sequence	Constraint Engine Result
Scenario 1: Successful Update (Precondition first)	The user first sets `DutyRating` to "`Prime Power (PRP)`".	The constraint engine recognizes the precondition is met and updates the dependent attribute, setting `standardsAndCompliance` to "`Listing-UL 2200`". The constraint is validated.

Scenario	User Input Sequence	Constraint Engine Result
Scenario 2: Constraint Violation (Conflicting Value first)	The user first sets `standardsAndCompliance` to "`Certification-CSA`", and then sets `DutyRating` to "`Prime Power (PRP)`".	The existing user-selected value ("`Certification-CSA`") violates the constraint. The constraint engine will not override the user's prior selection to change it to "`Listing-UL 2200`". Instead, the engine displays an error.

To resolve the error in Scenario 2, the user must manually adjust one of their selections: either change the `DutyRating` (precondition) or manually update the `standardsAndCompliance` (dependent value) to "`Listing-UL 2200`".

Left-Hand Side and Right-Hand Side Behavior in Constraint Resolution

Operator precedence and constraint engine resolution process determines whether the left-hand side (LHS) or right-hand side (RHS) of a constraint changes or is constrained to maintain logical validity. Variable origin, which means whether variables are user-selected, calculated, or restricted by system limitations, can also affect the outcome for the LHS or RHS in an expression.

These examples show how different user inputs lead to different outcomes for the LHS and RHS in the same expression.

Example 1. Implication Operator (->): Directional Enforcement

```
type Order {
int quantity = [1..1000];
boolean requiresApproval;
constraint bulkOrderApproval (
quantity >= 100 -> requiresApproval == true
);
}
```

The implication constraint `A -> B` (if A, then B) is the primary method for defining a mandatory outcome. The engine evaluates the LHS (A) first. If the `quantity` is 150 (LHS TRUE), then the engine forces `requiresApproval` (RHS) to be `TRUE`.

Scenario	Outcome
Scenario A: LHS is True, RHS Changes	The user sets a quantity greater than or equal to 100 on the LHS, which makes the condition true. The engine sets or changes the value of requiresApproval, the RHS, to true.
Scenario B: RHS is False, LHS is Constrained to be False	The user manually sets requiresApproval, the RHS variable, to false. Since the implication true -> false is forbidden, the LHS condition must be false to satisfy the constraint. The engine constrains quantity on the LHS to be less than 100 to make the LHS false.

Example 2. Bi-conditional (<->): Symmetrical Equivalence

```
type BulkOrderSystem {
int quantity = [1..1000];
boolean requiresApproval;
boolean isBulkOrder;
```

```
constraint bulkOrderStatus (
isBulkOrder <-> (quantity >= 100 && requiresApproval),
"Bulk order status requires 100+ quantity AND management approval."
);
}
```

The bi-conditional constraint A `<->` B (A if and only if B) requires that the LHS and RHS must share the same Boolean truth value. Either side can act as the driver, forcing the other side to change. In the example above:

- A is the boolean variable `isBulkOrder`.
- B is the complex condition `(quantity >= 100 && requiresApproval)`.

Scenario	Outcome
Scenario A: LHS Drives RHS	If the user manually sets the attribute isBulkOrder to true (making the LHS true), the engine immediately forces the RHS (quantity >= 100 && requiresApproval) to also be true, ensuring that both quantity is at least 100 and requiresApproval is set to true.
Scenario B: RHS Drives LHS	If the user selects a configuration where the quantity is high (for example, 500) and then sets requiresApproval to false (making the RHS condition false), the engine immediately forces the LHS attribute isBulkOrder to false to maintain equivalence.

Exception: The exclude Rule

The only scenario where the constraint engine intentionally overrides user input is when processing the `exclude` rule. If a user selects a component or sets an attribute value that violates an `exclude` rule, the engine will automatically override that user input to satisfy the exclusion constraint. In all other constraint types (like implication constraints shown previously), the engine relies on the user to fix the error. For more information, see Exclude Rule on page 1130.

Table Constraints

The table constraint in Constraint Modeling Language (CML) is used to define a set of valid combinations of values for two or more attributes. These combinations are specified in rows within the constraint definition.

The table constraint has this syntax:

```
table(variable, …, variable, {value, .. value}, …, {value, …, value});
```

Each row inside {} defines a valid combination of values.

Example: Table Constraint

In this example:

- Variables: The attributes `Voltage` and `DutyRating` are listed as the columns for the table.
- Table Rows: Each row defined within the curly braces ({}) specifies a valid combination. For instance, `{"7976/13800", "Continuous Power (COP)"}` is a valid pairing.

- Enforcement: If a user attempts to select a high voltage (`"7976/13800"`) while choosing a Prime Power rating (`"Prime Power (PRP)"`), the table constraint is violated, and the engine displays the error message: "Selected Voltage is not compatible with the required Duty Rating."

```
// --- Component Types ---
type GeneratorSet {
// 1. Attributes whose values must align according to the table
string Voltage = ["220/380", "277/480", "7976/13800"];
string DutyRating = ["Prime Power (PRP)", "Continuous Power (COP)", "Emergency Standby
Power (ESP)"];
// 2. Table Constraint
// Defines valid combinations where Voltage and DutyRating are mutually dependent.
constraint validOperationalModes(
table(
Voltage,
DutyRating,
{"220/380", "Prime Power (PRP)"},
{"220/380", "Continuous Power (COP)"},
{"220/380", "Emergency Standby Power (ESP)"},
{"277/480", "Prime Power (PRP)"},
{"277/480", "Emergency Standby Power (ESP)"},
{"7976/13800", "Continuous Power (COP)"}
),
"Selected Voltage is not compatible with the required Duty Rating."
);
}
```

Import Data from a Salesforce Object to Populate a Table Constraint

Import data from a standard or custom Salesforce object to use in a table constraint in a constraint model. The imported data populates the columns and rows in the table constraint in CML, and saves you the step of manually entering the data.

To import data from a Salesforce object, first assign Read, Create, Edit, and Delete permissions for the object to the Constraint Rules Engine Licenseless permission set. See Import Data from Salesforce Objects to Use in Constraint Models in Salesforce Help.

In CML, use the SalesforceTable keyword and the syntax shown here to import data from a Salesforce object. This example uses the `GeneratorSet` type to constrain the calculated running capacity (`gc_runningKw`) based on a user's selection of the nominal output (`Nominal_Power_Output`), referencing an external Salesforce custom object named PowerCst__c.

Example: Imported Table Constraint

```
type GeneratorSet {
// 1. Attribute storing the user-selected power output (String)
string Nominal_Power_Output = ["100 kW", "300 kW", "500 kW", "700 kW"];
// 2. Attribute storing the resulting Running kW (Decimal, calculated)
@(configurable = false, defaultValue = "0")
decimal(2) gc_runningKw;
// Constraint ensures the pairing of Nominal_Power_Output and gc_runningKw is found in the
 imported Salesforce table.
constraint(
table(
Nominal_Power_Output, gc_runningKw,
SalesforceTable("PowerCst__c","Nominal__c,Running__c")
)
```

```
);
}
```

Explanation of the Imported Table

The table constraint ensures that the selected values for `Nominal_Power_Output` and `gc_runningKw` must form one of the valid combinations defined in the external source.

- Table (`Nominal_Power_Output`, `gc_runningKw`, ...): These are the CML attributes whose values must correlate. They define the columns of the required combination table.
- Salesforce Table ("`PowerCst__c", "Nominal__c,Running__c`"): This function keyword directs the constraint engine to import data from the Salesforce custom object PowerCst__c.
- Field Mapping: The fields specified ("`Nominal__c,Running__c`") define the columns in the Salesforce object that correspond to the CML attributes listed in the table function. Nominal__c maps to `Nominal_Power_Output`, and Running__c maps to `gc_runningKw`.

Proxy Variables with Constraints on Types and Relationships

Use proxy variables to reference the variables of related types, including parent, root, and sibling types.

The supported proxy variables include:

- Cardinality
- Parent
- this.quantity

Cardinality

Cardinality is a fundamental concept in Constraint Modeling Language (CML), controlling the quantity of product instances allowed in a defined relationship. It is crucial for ensuring product structure validity and optimizing performance.

The `cardinality` proxy variable refers to the cardinality of a relationship, that is, the quantity of instances of the same type in a relationship. The first parameter is the type name. The second, optional parameter is the port name. This variable differs from the this.quantity proxy variable, which refers to the quantity of the current instance.

Use this format:

```
cardinality(<type name>, <relation name>) or cardinality(<type name>)
```

Each parameter value can be a string or a string variable. If the relation name isn't specified, the engine searches all ports to find the type.

The cardinality bounds are defined within the relation declaration using square brackets, formatted as `[minimum..maximum]`.

Cardinality Examples

These examples illustrate how cardinality is defined and managed in the Generator Set model, focusing on standard definition, conditional enforcement, and reading the quantity.

Example 1: Using Cardinality Full Syntax

```
type Heater ;
type GeneratorSet  {
```

```
//Define the relation and specify the smallest cardinality required for performance
relation Heaters : Heater[0..10];
//Declare a variable to hold the count
int totalHeaterCount = [0..10];
// This counts instances of type "Heater" specifically within the "Heaters" relation.
int InstancesofHeater = cardinality(Heater, Heaters);
constraint(totalHeaterCount == InstancesofHeater);
// Optional message to display the count to the user
message(totalHeaterCount > 0, "Current configuration includes {} heaters.", totalHeaterCount,
 "Info");
}
```

See Message Rule on page 1127.

Example 2: Using Cardinality Partial Syntax

```
type Heater ;
type OutputTerminal ;
// Main Generator Set type
type GeneratorSet {
// Define multiple relations that might contain specific types, each with a specified
cardinality
relation Heaters : Heater[2];
relation OutputTerminals : OutputTerminal[0..99];
// Define derived attributes with an explicit domain
int totalHeatersFound = [0..100];
int totalTerminalsFound = [0..100];
// Partial cardinality SYNTAX
// The engine searches all relations to find the specified type
// Counts all instances of "Heater" across the entire GeneratorSet
int AllHeaterInstances = cardinality(Heater);
constraint(totalHeatersFound == AllHeaterInstances);
// Counts all instances of "OutputTerminal" across all relations
int AllOutputTerminals = cardinality(OutputTerminal);
constraint(totalTerminalsFound == AllOutputTerminals);
// Optional message for user visibility
message(totalHeatersFound > 0, "System found {} heaters in this configuration.",
totalHeatersFound, "Info");
}
```

Example 3: Defining Cardinality in Relations

This example shows how the Generator Set uses different cardinality ranges to define fixed requirements, optional components, and minimum quantities for necessary components.

```
type GeneratorSet {
// 1. Fixed Cardinality: Exactly one General Model (component) is required.
// means min=1 and max=1.
relation GeneralModels : GeneralModel [1..1];
// 2. Bounded Optional Cardinality: Between 0 and 5 Temperature Sensors are allowed.
// [0..5] means the component is optional but limited.
relation TemperatureSensors : TemperatureSensor[0..5];
// 3. Minimum Required Cardinality: At least 1 Test record is required, up to 99.
// [1..99] enforces inclusion.
```

```
relation Testing : Test[1..99];
}
type GeneralModel;
type TemperatureSensor;
type Test ;
```

Key Concepts:

- Syntax: Cardinality is specified immediately after the component type in the relation declaration.
- Best Practice: Specifying the smallest required range, such as [1..1] or [0..5], ensures the constraint engine tests fewer combinations, which prevents performance degradation.

Example 4: Enforcing Conditional Cardinality via require()

Cardinality can be dynamically enforced based on attributes of the parent product using the `require()` constraint keyword. For more information, see Require Rule on page 1128.

```
type GeneratorSet {
int requiredKW = [101..10000];
string standardsAndCompliance = ["Certification-CSA", "Listing-UL 2200"];
// Relation for mandatory installation accessories
relation Accessories : Accessory[1..99];
relation Testing : Test[1..99];
// Conditional Requirement based on power level:
// If the required power is over 5000 kW, exactly 5 specific Accessory instances are
required.
require(
requiredKW > 5000,
Accessories [Accessory] == 5,
"High capacity generators require exactly 5 specialized accessory kits."
);
// Conditional Requirement based on compliance standard:
// If UL 2200 is selected, exactly 2 Test records must be included.
require(
standardsAndCompliance == "Listing-UL 2200",
Testing [Test] == 2,
"UL 2200 listing mandates exactly two certification tests."
);
}
type Accessory;
type Test;
```

This pattern (`require(condition, relation[type] == N, ...)`) allows the CML model to enforce a precise fixed number of child components when a quantity threshold or attribute condition is met on the parent.

Cardinality vs. Count

The `cardinality()` keyword is a proxy variable used to refer to the size of a relationship. The `count()` keyword is an aggregate function that counts matching components or attribute conditions within a relation. Both can be used to read quantities for validation or aggregation rules.

Example for Similarity

Both `cardinality()` and `count()` can be used to determine the total quantity of items in a relationship. In this scenario, they return the same value because the `count()` condition covers all possible active instances.

```
type Accessory{
boolean isPresent = true;
}
type Bundle {
relation items : Accessory[0..10];
int totalByCount = [0..10];
// Similarity: Both can read the headcount of the relation
// Use the proxy variable to set an attribute
int totalByCardinality = cardinality(Accessory, items);
// Use count() aggregate function within a constraint
constraint(totalByCount == items.count(isPresent == true));
}
```

Example for Difference

The core difference is that cardinality is a proxy variable representing the headcount or size of the relation, whereas count is a function used to evaluate specific attribute conditions or filters within those instances.

```
type Accessory {
int weight = [1..100];
boolean isEssential;
}
type Bundle {
relation items : Accessory[1..20];
// 1. Declare variables with explicit domains
int totalHeadcount = cardinality(Accessory,items);
int heavyItemsCount = [0..20];
int essentialItemsCount = [0..20];
// DIFFERENCE 1: Cardinality (Proxy Variable)
// Cardinality refers to the relationship size (how many instances are present).
// It does not look at the values of the attributes within the instances.
// DIFFERENCE 2: Count (Aggregate Function)
// Count MUST use a logical expression to evaluate conditions within the relation.
// It filters the instances based on attribute logic before returning a total.
// Example: Counting only accessories that weigh more than 50kg
constraint(heavyItemsCount == items.count(weight > 50));
// Example: Counting only accessories marked as 'essential'
constraint(essentialItemsCount == items.count(isEssential == true));
// Business Logic using both:
message(
heavyItemsCount > (totalHeadcount / 2),
"Warning: More than half of your accessories ({}) are heavy items.",
heavyItemsCount
);
}
```

For more information, see Message Rule on page 1127.

Parent

The `parent()` proxy variable is used to enable components at any level of the product hierarchy (child, grandchild, etc.) to access the attributes (variables) defined by their ancestor types. This mechanism facilitates the flow of configuration data and calculated values from the top of the bundle down to its components.

The `parent()` keyword functions as a proxy variable used to reference attributes residing in the immediate parent or any ancestor type in the configuration model.

Variable Name	Syntax	Purpose
`parent()`	`parent(<ancestor variable name>, <level>)`	Retrieves the value of an attribute from a parent or ancestor type.

Key Characteristics

- Targeting Ancestors: The first parameter is the name of the attribute in the ancestor type that you wish to reference.
- Level Parameter: The second parameter (`<level>`) is optional and specifies how many levels up the hierarchy the engine should search. If omitted, it typically references the immediate parent. The level index for the `parent()` function effectively starts from 0 (implicit) for the immediate parent.
- Level 0 (Default): When you use `parent(attributeName)`, the engine references the attribute in the immediate parent.
- Level n: When you specify `parent(attributeName, 1)`, the engine reaches one level beyond the immediate parent, to the grandparent.
- Data Flow: `parent()` ensures that the data flows unidirectionally, where the child reads attributes calculated or defined by the parent, thereby helping to prevent complex circular dependencies.
- Single Inheritance: CML follows a single inheritance model, where a type can only extend one other type at a time.
- Same Variable Name: In CML, reusing the same variable name between a parent bundle and its child accessories is a standard architectural pattern for cascading values down a product hierarchy. This allows a child component to automatically inherit or synchronize its properties with the parent type, ensuring configuration consistency.

Example: Standards and Compliance Synchronization

In this example, the `standardsAndCompliance` selection made at the bundle level is automatically passed down to every accessory within that bundle.

```
type AccessoryBundle {
// 1. Define the attribute at the Bundle level
string standardsAndCompliance = ["Certification-CSA", "Listing-UL 2200"];
// Relation to accessories
relation accessories : Accessory[1..10];
}
type Accessory {
// 2. Reuse the same variable name
// The parent() function targets the attribute in the immediate parent
string standardsAndCompliance = parent(standardsAndCompliance);
// Business Logic: If the inherited standard is UL 2200, price is affected
decimal(2) testingFee = [0.00..500.00];
constraint(standardsAndCompliance == "Listing-UL 2200" -> testingFee == 250.00);
}
```

The `parent()` proxy variable is essential for cascading configuration data and constraints down the product hierarchy, allowing child components to reference attributes defined by their ancestors.

Example: Parent Attribute Reference

This example utilizes attributes found on the `GeneratorSet` type (the parent) and demonstrates how related components (like TemperatureSensor) access those values using the parent() proxy variable. Here is the hierarchy context.

Parent Type	Relation Name	Child Type	Cardinality	Key Attribute Usage
Root `GeneratorSet`	`TemperatureSensors`	`TemperatureSensor`	`[0..5]`	Defines primary inputs (`requiredKW`) and a derived attribute (`ULComplianceEnforced`) based on the `standardsAndCompliance` selection.
Child `TemperatureSensor` (Instance within `GeneratorSet` relation)	Not Applicable	Not Applicable	Not Applicable	Uses the `parent()` proxy variable to reference the `requiredKW` attribute from its ancestor (`GeneratorSet`). A constraint ensures the component's capacity (`maxOperatingKW`) meets the requirement inherited from the parent.
Grandchild `StatorTemperatureSensor` (Subtype of `TemperatureSensor`)	Not Applicable	Not Applicable	Not Applicable	Inherits functionality from `TemperatureSensor`. Uses parent() to retrieve the calculated `ULComplianceEnforced` boolean from the `GeneratorSet`. Enforces a conditional installation rule using the Implication Operator (->).

```
// --- Parent Type (GeneratorSet) ---
type GeneratorSet {
// Attributes defined by the parent
int requiredKW = [101..10000]; // Required power rating
string standardsAndCompliance = ["Certification-CSA", "Listing-UL 2200"]; // Compliance
choice
// Derived status: True if UL 2200 is selected (precondition for compliance checks)
boolean ULComplianceEnforced = standardsAndCompliance == "Listing-UL 2200";
// Relation to child components
```

```
relation TemperatureSensors : TemperatureSensor[0..5];
}
// --- Child Component Type (TemperatureSensor) ---
type TemperatureSensor {
// Temperature sensors may have a model that defines max operating KW
int maxOperatingKW = [1000..10000];
// Retrieve the required KW value from the immediate parent (GeneratorSet)
int parentRequiredKW = parent(requiredKW);
// Constraint ensures the sensor can handle the power defined by the parent
constraint(maxOperatingKW >= parentRequiredKW, "Sensor capacity must meet the required KW
 set by the generator set.");
}
// --- Specific Sensor Type (Grandchild) ---
type StatorTemperatureSensor : TemperatureSensor {
// Retrieve the boolean compliance flag from the immediate parent (GeneratorSet)
boolean enforceULCompliance = parent(ULComplianceEnforced);
// Constraint ensures that if UL Compliance is enforced by the parent,
// this sensor must meet a specific installation requirement (e.g., must be shielded)
string installationMethod = ["Standard", "Shielded"];
constraint(enforceULCompliance -> installationMethod == "Shielded",
"UL Compliance requires a shielded temperature sensor installation."
);
}
// Note: To reference a value further up the hierarchy, the optional level parameter can
be used:
// int grandParentValue = parent(requiredKW, 1); // Accesses attribute 1 level up
```

Explanation of Parent Reference

- Direct Reference (Immediate Parent): In the `TemperatureSensor` type, `int parentRequiredKW = parent(requiredKW);` accesses the `requiredKW` attribute defined in the immediately superior type, which is `GeneratorSet`.
- Unidirectional Data Flow: The `parent()` proxy variable is critical for enabling unidirectional data flow, where calculated or defined values in the parent are read by children to enforce constraints. The engine ensures that parent aggregation and calculation are complete before the `parent()` function executes in the child component.

this.quantity

`this.quantity` is a proxy variable used specifically to access the quantity of the current instance (the specific product line item) within a configuration.

Key Characteristics and Usage

- Scope: It refers only to the quantity of the specific instance in which it is used. It is used at the component level to determine the quantity of that item chosen by the user or set by the constraint engine.
- Calculation Rule: The `this.quantity` proxy variable can be used only within a calculation rule.
- Distinction from `cardinality()`: `this.quantity` differs from the `cardinality()` proxy variable, which refers to the total number of instances of a specific type within a relationship and includes the quantity of other instances of the same type.
- Read-Only/Validation: When accessing quantity in CML, attributes like `this.quantity`, or external equivalents like `lineItemQuantity` or `ItemEndQuantity`, are treated as read-only and should be used only in calculation or evaluation rules. It is not recommended to use it to drive component creation, which should be done via cardinality.

Example: Using this.quantity for Calculation

In the Generator Set configuration, a component like the `NaturalGasReformer` may use `this.quantity` to define a local attribute representing how many units were selected, which is then used for internal calculations (like total capacity or total weight contribution).

```
type LineItem;
type NaturalGasReformer : LineItem {
// 1. Definition: The LineItemQuantity attribute captures the quantity of this specific
instance.
// The CML syntax dictates defining an attribute (LineItemQuantity) that equals
this.quantity.
int LineItemQuantity = this.quantity;
// Placeholder: Attribute representing unit capacity per reformer unit
int unitCapacityKW = 100;
// 2. Calculation: Used in a calculation rule to determine total capacity based on the
quantity selected for this line item instance.
int totalReformerCapacity = LineItemQuantity * unitCapacityKW;
}
```

Message Rule

The message rule displays a message to users based on specified conditions.

Message rules have this syntax:

```
message(logical expression, string literal | string variable, argument, .., argument,
severity);
message(logical expression, string literal | string variable, severity);
message(logical expression, string literal | string variable);
```

A message rule can take optional arguments to generate the message and indicate the severity of the message as the last argument. Message severity can be `Info`, `Warning`, or `Error`. Without an explicit message severity argument, the message will be treated as `Info`.

Understand the behaviour of each message severity type at runtime.

- The `Info` message type doesn't require the user to take any action in order to continue with the current task. Info messages display a gray banner.
- The `Warning` message type allows the user to continue working on the current task, but blocks them from taking the next step until they take action to address the issue described in the message. Warning messages display a yellow banner.
- The `Error` message type blocks the user from continuing with the current task until they fix the error described in the message. Error messages display a red banner. Note: An Error message doesn't block a user working in the Transaction Line Editor (Transaction Line Table, or TLT). In that component, the user can still make changes and save the quote, even when the quote contains conditions that trigger an Error message.

Message format can be a Java string, or a string with {} as a placeholder. The constraint solver replaces each {} with arguments specified after the string, in the order they are written.

Example

In this example, if `requiredKW` is greater than `2500`, a message is displayed that the specified required kW is larger than the supported options and must be changed.

```
type GeneratorSet {
int requiredKW = [101..10000];
message(requiredKW > 2500, "The required kW is above what the current options can support.
 Please adjust to 2500 kW or select a new generator set that meets your requirements.");
}
```

Preference Rule

The preference rule encourages the constraint solver to satisfy the condition, but doesn't enforce it if the condition can't be met.

The system tries to satisfy the condition in a preference rule, but if for some reason it can't, the system delivers a failure message to the user with `Info` severity.

The preference rule has this syntax.

```
preference(logic expression, string literal | string variable, argument, .., argument);
preference(logic expression, string literal | string variable);
preference(logic expression);
```

A preference rule can include an optional explanation message for failure. The message is of `Info` severity, meaning it does not block the user from continuing with the action.

In this example, the preference rule encourages the user to mention the `dBMax` value as `90` and the `requiredKW` value as `500`.

```
type GeneratorSet {
   int requiredKW = [101..10000];
   int dBMax = [0..140];
   preference(dBMax == 90, "90 preferred for dbMax");
   preference(requiredKW == 500,"50 preferred for requiredKW");
}
```

Require Rule

The require rule requires certain components to be included in a relationship when specified conditions are met.

Required components can have attributes and quantity specified. The require rule can include an optional explanation message, for the rule failure explanation.

In certain scenarios, you can independently add a type at the header level. This means you can include a specific type even if it isn't explicitly defined as part of any of the relationships you've configured. This capability offers flexibility in managing and including necessary types that might not always fall under a specific relationship structure.

Note: When you assign a require rule to a virtual bundle (a bundle related to the sales transaction, where the parent product has no associated price), set one Product Selling Model Option on the required product to Default. For more information on Product Selling Model Options, see Manage Product Selling Model in Revenue Cloud.

The require rule has this syntax:

```
require(logic expression, relationship[type]{var=value,…,var=value}==integer value,
"Explanation message");
```

In this example, the require rule specifies that if the number of engineers is more than 0, installation is required. The installation will be automatically added upon adding an engineer.

```
type GeneratorSet {
    relation engineers : engineer[0..99];
relation installation : install[0..5];
   require(engineers[engineer] > 0, installation[install], "Installation is required if
engineers are present");
}
type engineer{}
type install{}
```

Require Rule vs Constraint

In Constraint Modeling Language (CML), constraint() and require() can both enforce behavior, but they operate differently: constraint focuses on logical consistency, require focuses on physical presence of products.

Here's a comparison between `constraint()` and `require()`.

Feature	constraint()	require()
Primary goal	Validates if a condition is met (LHS) and operates on the result (RHS).	Forces a product to be present.
Engine action	Resolves the constraint or displays an error if there are no options to resolve.	Adds the required product to the quote.

Setdefault Rule

The setdefault rule allows component selection with attribute values and quantity, similar to the require rule.

Unlike the require rule, the setdefault rule applies a default configuration status and uses a triggering mechanism to control when the solver attempts to satisfy the rule. The setdefault rule can include an optional explanation message.

The setdefault has this syntax, similar to the require rule.

```
setdefault(condition, expression, message);
```

The setdefault rule evaluates the condition. If the condition is false, the solver ignores the expression and doesn't display a message, regardless of whether any part of the condition is changed. If the condition is true, the solver performs one of these actions.

- If any part of the condition is changed or the parent component is new, the solver attempts to satisfy the expression. If the solver can't satisfy the expression, an explanation message is displayed (if included).
- If no part of the condition is changed, the solver evaluates the expression without attempting to satisfy it. If the expression evaluates to false, an explanation message is displayed (if included).

The key difference between the setdefault rule and the require rule is that the setdefault rule attempts to satisfy the expression only when a condition is changed. If no condition is changed, the setdefault rule performs a passive evaluation. The require rule always attempts to satisfy the expression when the condition is true.

In this scenario, we use the requiredKW attribute (the user's power requirement) as the condition and the Accessories relation as the target for the recommended cardinality.

```
type Accessory;
type GeneratorSet  {
int requiredKW = [101..10000];
```

```
relation Accessories : Accessory[1..99];
/**
* @Title High Power Accessory Recommendation
* The setdefault constraint specifies that 2 accessory units are
* recommended when the required power capacity is greater than 2000 kW.
*/
setdefault(
requiredKW > 2000,
Accessories[Accessory] == 2,
"2 specialized accessory kits are recommended for power levels above 2000 kW"
);
}
```

Exclude Rule

The exclude rule is used to automatically remove a specific type in a relationship if a certain condition is met.

The exclude rule has this syntax.

```
exclude(logic expression, relationship[type],"Explanation message");
```

The type must be leaf type, a node without children.

In the exclude rule, if a user sets attribute values in Product Catalog Management (PCM) that violate the rule requirements, the constraint engine overrides the user input in order to validate the constraint. This behavior is different than other constraints, in which the constraint engine doesn't override user input, but displays an error if user input violates the constraint. See How User Input Order Affects Constraint Engine Behavior section in Logical Constraints on page 1113.

In this example, the exclude rule automatically removes the `Heater_120` heater from the type `GeneratorSet` if the `Voltage3` is greater than or equal to `4160`.

```
type GeneratorSet {
int Voltage3 = [120..13800];
relation Heaters : Heater_120 [1..3];
exclude(Voltage3 >= 4160, Heaters[Heater_120]);
}
type Heater_120 {}
```

Action Rule

The CML Action Rule is defined using the rule() keyword. Its primary purpose is to execute a designated action, specified as a string literal, when a condition is met.

This action is typically handled by external systems, such as the Product Configurator API or custom code, to manage business processes, workflows, or complex constraints that fall outside the constraint engine's primary scope.

Action rules have this syntax.

```
rule(condition, action, arg, ..., arg)
rule(<condition>, <action>, "attribute", <attribute>);
rule(<condition>, <action>, "attribute", <attribute>, "value", [<attribute values>]);
rule(<condition>, <action>, "relation", <relation>, "type", <type>);
```

`condition` is any logic expression such as a constraint in CML.

`action` is a string literal that specifies an action that can be interpreted by the Product Configurator API. The Product Configurator API supports these actions.

- Hide: hide attribute, attribute value, product option
- Disable: disable attribute, attribute value, product option
- args are a list of arguments needed to execute the action. An argument is a pair including a string literal and an identifier, a literal, or a domain, enclosed in brackets [ ] to specify multiple values. The string literal specifies what kind of argument follows. The identifier attribute can be defined in the type. The engine retrieves the argument value and passes it to the caller to execute the action.

Hide or Disable Rule

The Hide or Disable Rule uses the rule() keyword to conditionally remove an element from the selection menu (hide) or preserve it in the menu while preventing user selection (disable).

This functionality can be applied.

- On a bundle, hide a component to remove it from the selection menu, or disable a component to preserve it in the menu but prevent users from selecting options for it.
- On an individual product, hide an attribute to remove it from the selection menu, or disable an attribute to preserve it in the menu but prevent users from selecting options for it.
- On an attribute, hide or disable an attribute value to preserve it in the menu but prevent users from selecting options for it. For attribute values, the hide and disable rules have the same behavior.

Note: In Visual Builder in Salesforce, for attribute values, only the hide rule is enabled. When you apply the hide rule to an attribute value in Visual Builder, the value appears in the menu but selections are disabled.

The hide and disable rules use this syntax, where `action` is replaced by either `hide` or `disable`.

```
rule(logic expression, action, actionScope, actionTarget)
rule(logic expression, action, actionScope, actionTarget, actionClassification,
actionValueTarget)
```

Example: Hiding and Disabling Features

In the example in this section, rules rely on specific variables to define the scope and target of the action.

Variable in Rule	Purpose	Example from Generator Set	Description
`logic expression` (Declaration)	Condition upon which the rule occurs.	`requiredKW <= 500`	The logical test that triggers the action.
`action`	Designates whether the rule is hide or disable.	`"hide", "disable"`	Determines if the element is removed from view or made unselectable.
`actionScope`	Designates whether the rule acts on an attribute or relation scope.	`"attribute", "relation"`	Specifies if the target is a variable property or a component relationship.
`actionTarget`	Designates the specific variable or relation that the rule acts on.	`"Voltage", "StarterMotors"`	The Constraint Modeling Language (CML) name of the attribute or relation.
`actionClassification`	Designates whether the rule acts on a type or a value.	`"type", "value"`	Used when targeting components within a relation

Variable in Rule	Purpose	Example from Generator Set	Description
			(type) or specific options within an attribute domain (value).
`actionValueTarget`	Designates the specific type or value that the rule acts on.	`"7976/13800"`, `"StarterMotor_Advanced"`	The specific string value or type name to hide or disable.

```
// --- Component Definitions (For relations referenced below) ---
type LineItem;
type EngineModel : LineItem;
type StarterMotor : LineItem;
type OutputTerminal : LineItem;
type OutputTerminals2HoleLugNEMA : OutputTerminal; // Specific terminal type
type GeneratorSet : LineItem {
// Attributes subject to hide/disable rules
int requiredKW = [101..10000];
string Voltage = ["220/380", "240/416", "4160/7200", "7976/13800"]; // Voltage is the
attribute being hidden/disabled
string specialApplication = ["None - Standard", "Motor Starting"]; // specialApplication
attribute contains a value to hide
// Relations subject to hide/disable rules
relation StarterMotors : StarterMotor; // Relation target (component) to hide/disable
relation OutputTerminals : OutputTerminal[0..99]; // Relation being acted upon
// 1. Disable a Component (Type) in a Relation
// If requiredKW is too low (<= 500 kW), the advanced starter motor component is disabled
 (visible but unselectable).
rule(
requiredKW <= 500,
"disable",
"relation",
"StarterMotors",
"type",
"StarterMotor_Advanced"     );
// 2. Hide an Attribute
// If the Generator is configured for a special purpose (Motor Starting), hide the Voltage
 attribute entirely.
rule(
specialApplication == "Motor Starting",
"hide",
"attribute",
"Voltage"
);
// 3. Hide a Specific Attribute Value
// If the power requirement is low (< 2000 kW), hide the high voltage option (7976/13800)
 from the Voltage attribute domain.
rule(
requiredKW < 2000,
"hide",
"attribute",
"Voltage",
"value",
```

```
"7976/13800"
);
// 4. Disable a Component Type (Alternate Syntax using the component name)
// Disable the OutputTerminals2HoleLugNEMA component if the required KW is high.
rule(
requiredKW >= 5000,
"disable",
"relation",
"OutputTerminals",
"type",
"OutputTerminals2HoleLugNEMA"
);
}
type StarterMotor_Advanced : StarterMotor; // Subtype used in the rule
```

Recommendation Rule

The recommend keyword is used within a Constraint Modeling Language (CML) rule to display suggestions for related products in the Product Configurator. The rule defines the condition under which a specific product type or relation should be suggested to the user.

You can recommend a type, a relation, or both in the same rule. The recommendation rule can be added inside a standalone product, a product bundle, or a virtual container.

Unlike action rules that are interpreted directly by the Product Configurator API, when the condition is met, the engine forces a UI change (hiding or disabling a product option, attribute, or value). Recommendations are not automatically applied to the UI by the configuration engine alone. To suggest types or relations (products/bundles), typically for up-selling or cross-selling based on their current selections at runtime, use the Run Config Rules action within a Salesforce Flow. See Run Config Rules Action in the Revenue Cloud Developer Guide.

- Use an action rule when a selection makes another option invalid or irrelevant. For example, if a user selects a basic warranty, you should "hide" or "disable" the premium support options to prevent a conflicting or impossible setup.
- Use a recommendation rule when you want to nudge the user toward a beneficial add-on. For example, if a user buys a high-end generator, you "recommend" a maintenance service contract. This does not block the user if they choose not to add it.

Here are 3 examples demonstrating how to implement product recommendation rules.

Example 1: Recommending a Type (Based on Attribute Selection)

This rule is placed within the `GeneratorSet` type. If a user selects an extremely high voltage, the configuration engine recommends a specialized engineer component required for installation or commissioning.

```
type GeneratorSet {
// Attribute input
string Voltage = ["277/480", "7976/13800"];
// Relation to the component type being recommended
relation engineers : engineer[0..99];
// Recommend Engineer Specialist (type) for High Voltage
rule(
Voltage == "7976/13800", "recommend", "type", "EngineerSpecialist"
);
}
// Recommended type
type EngineerSpecialist ;
type engineer;
```

Example 2: Recommending a Relation (Based on Component Quantity)

This rule recommends adding items to an existing relation (`Accessories`) if the user selects a high-end component (such as the most robust enclosure, `Enclosure_SA3`).

```
type GeneratorSet  {
// Relations defined in the GeneratorSet
relation Enclosures : Enclosure;
relation Accessories : Accessory[1..99]; // The relation being recommended
// Recommend Accessories when maximum sound dB is chosen
rule(
Enclosures[Enclosure_SA3] == 1,
"recommend",
"relation",
"Accessories");
}
type Enclosure ;
type Enclosure_SA3 : Enclosure;
type Accessory ;
```

Example 3: Recommending a Type (From a Virtual/System Container)

This rule is applied at the Quote or System level (using a `virtual` type) and recommends a system integration product if multiple generators are being configured, reflecting the requirement that large projects often need central control components.

```
@(virtual = true)
type Quote {
// Relation referencing all GeneratorSet instances on the quote
@(sourceContextNode = "SalesTransaction.SalesTransactionItem")
relation lineItems : GeneratorSet[0..10];
// Recommend Switchgear for multi-unit orders
rule(
lineItems[GeneratorSet] > 1,
"recommend",
"type",
"Switchgear"
);
}
type GeneratorSet;
type Switchgear;
```

Set Product Selling Model in a Constraint

Use the productSellingModel tagname to write a constraint that sets the Product Selling Model (PSM) for a type. You can define a PSM as one time, time-deferred (subscription with end date), or evergreen (recurring subscription with no preset end date). The PSM is updated for new line items at runtime, based on the constraint.

You can't use a Constraint Modeling Language (CML) constraint to update the PSM for an existing quote line. For more information on product selling models, see Manage Product Selling Model in Revenue Cloud in Salesforce Help.

Constraint Example

Using the `GeneratorSet` model, a constraint can be defined that sets the PSM based on a specific operational attribute chosen by the user, such as the `DutyRating`. This assumes that different duty ratings correspond to different billing models (for example, permanent installation versus temporary rental).

This example sets the PSM to a specific ID (for example, `PSM_EVERGREEN_ID`) if the user selects the "Continuous Power (COP)" duty rating.

```
//Global variable PSM ID
define PSM_EVERGREEN_ID "a00Tx000000Qz1g"
type GeneratorSet {
// Use the productSellingModel tag from the context definition
@(tagName = "ProductSellingModel")
string productSellingModel;
string DutyRating = ["Prime Power (PRP)", "Continuous Power (COP)", "Emergency Standby
Power (ESP)"];
// Set PSM based on Duty Rating
constraint(
DutyRating == 'Continuous Power (COP)' ->    productSellingModel == PSM_EVERGREEN_ID
);
}
```

Core Concept Examples

These examples illustrate core Constraint Modeling Language (CML) concepts including type, relationships, constraints, and so on.

For an explanation of the constraint model structure in these examples, see Model Structure on page 1202.

Example 1: Use Regex Global Variable

```
// Global Constant: Regex used to parse voltage strings
define VOLTAGE_REGEX "^([11-19]+)/([11-19]+)$"

type LineItem;
type GeneratorSet : LineItem {
   // Core attributes
   @(configurable = false)
   int requiredKW = [101..10000]; // Required power capacity
   string Voltage = ["220/380", "240/416", "277/480", "7976/13800"];
   string standardsAndCompliance = ["Certification-CSA", "Listing-UL 2200"];

   // Calculated attributes
   // Surge load is 1.25 times the required KW
   decimal(2) surgeLoadKW = requiredKW * 1.25;
   // Voltage3 extracts the secondary voltage (e.g., 380 or 416) for checks
   int Voltage3 = strtoint(regexpreplace(Voltage, VOLTAGE_REGEX, "$2"), 0);

   // Relation to model components
   relation GeneralModels : GeneralModel[11];

   // Constraint 1: Ensure model capacity meets the required KW
   constraint(GeneralModels[GeneralModel].powerKW >= requiredKW,
      "Selected Generator Model is insufficient for the required power.");
```

```
   // Constraint 2: UL 2200 standard restricts voltage to 600V or less (Implication rule)

   constraint(standardsAndCompliance == "Listing-UL 2200" -> Voltage3 <= 600,
      "The UL 2200 standard covers stationary engine generator assemblies rated at 600 
volts or less.");
}

// Component type definition (GeneralModel is a product component)
type GeneralModel : LineItem {
   int powerKW = [900, 1200, 1500, 1750, 2500];
   int dB = [20..23];
}
```

Key Technical Details

- The `GeneratorSet` type is related to the `GeneralModel` type through a single relation.

Parent Type	Relation Name	Child Type	Cardinality	Key Attribute Usage
`GeneratorSet`	`GeneralModels`	`GeneralModel`	Not Applicable	Constrained by `requiredKW` on the parent type.

- Parent Type (`GeneratorSet`): Defined as a `LineItem` component, the `GeneratorSet` holds configuration parameters such as the user's power requirement (`requiredKW`), the specified voltage (`Voltage`), and compliance standards (`standardsAndCompliance`). It also includes calculated attributes like `surgeLoadKW` and `Voltage3`.
- Child Type (`GeneralModel`): Defined as a `LineItem` component, the `GeneralModel` specifies the technical attributes of the selected generator configuration, including its power output (`powerKW`) and noise rating (`dB`).
- Cardinality: The relation `GeneralModels : GeneralModel` specifies the quantity bounds for the component, indicating that exactly 11 instances of the `GeneralModel` type must be included in this configuration.

Example 2: Use Groupby Annotation to Create Virtual Group

Using Groupby annotation for types, this model creates a virtual `VoltageGroup` for every unique voltage (for example, "220/380", "480/277") found in the transaction.

```
// 1. The physical product type
type GeneratorSet {
   @(configurable = false)
   int requiredKW = [101..10000];

   string Voltage = ["220/380", "240/416", "255/440", "277/480"];

   // Calculation with explicit domain for best practice
   decimal(2) surgeLoadKW = [126.25..12500.00];
   constraint(surgeLoadKW == requiredKW * 1.25);
}

// 2. The virtual container grouped by the "Voltage" attribute
```

```
@(split=true, virtual=true, groupBy=Voltage)
type VoltageGroup {
   string Voltage; // The attribute used for grouping
   decimal(2) groupTotalSurgeKW = [0.00..99999.99];

   // Map instances to this group from the transaction line items
   @(sourceContextNode="SalesTransaction.SalesTransactionItem")
   relation generators : GeneratorSet[0..50];

   // Aggregation: Sum only the surge load of generators in THIS voltage group
   constraint(groupTotalSurgeKW == generators.sum(surgeLoadKW));

   // Business Rule: Limit total surge load per voltage branch
   message(groupTotalSurgeKW > 10000, "Warning: Surge load for Voltage {} exceeds branch
capacity!", Voltage);
}

// 3. The top-level system managing the groups
@(virtual = true)
type GeneratorSystem {
   relation generators : GeneratorSet[0..100];
   relation voltageGroups : VoltageGroup[1..10];

   decimal(2) systemTotalKW = voltageGroups.sum(groupTotalSurgeKW);
}
```

Key Technical Details

- `groupBy=Voltage`: The engine scans all `GeneratorSet` instances and creates one virtual `VoltageGroup` for each unique voltage value detected (for example, one group for all "220/380" units, another for all "277/480" units).
- `sourceContextNode`: This tells the virtual group to pull its "children" (the generators) from the specific path in the Salesforce context where quote line items are stored.
- Bottom-Up Aggregation: Each `VoltageGroup` independently calculates its `groupTotalSurgeKW` based strictly on the generators assigned to it by the groupBy logic.
- Explicit Domains: Following best practices, the `surgeLoadKW` and `groupTotalSurgeKW` use explicit domains (for example, [0.00..99999.99]) to prevent "NullPointer" or initialization errors during solving.

Example 3: Use Sharingcount Annotation to Reuse Accessory Instances

In this example, we apply sharingcount annotation to the `Accessory` type to allow the engine to reuse accessory instances across the configuration up to a specified limit.

```
// 1. Define the component type with split and sharingCount
// split=true enables parallel solving by splitting quantity into multiple instances
// sharingCount=5 allows a single Accessory instance to be reused 5 times in the model
@(split = true, sharingCount = 5)
type Accessory {
   string category;
   decimal(2) weight;
}
```

```
type GeneratorSet {
   @(configurable = false)
   int requiredKW = [101..10000];

   // 2. Define the relationship with the sharing annotation
   // @(sharing = true) allows the engine to satisfy this relation by
   // "pointing" to existing instances instead of creating new ones
   @(sharing = true)
   relation Accessories : Accessory[1..99];
}
```

Key Technical Details

- Parallel Solving: By setting `split = true`, the engine can process the 99 possible accessories in parallel rather than sequentially, which is critical for large-scale generator configurations.
- Resource Management: The `sharingCount = 5` tells the solver that it doesn't need to instantiate 99 unique `Accessory` objects; it can reuse the same object definition up to five times across different parts of the configuration graph.
- Relationship Requirement: For `sharingCount` to take effect, the Relation (the port) must also be annotated with `@(sharing = true)` to grant the engine permission to reuse instances.

Example 4: Use contextPath and tagName Annotations

```
// 1. GLOBAL EXTERN DECLARATIONS
// Unifying contextPath (header field) and tagName (context identifier)
@(contextPath = "SalesTransaction.ShippingCountry", tagName = "Region_Identifier")
extern string ShippingCountry = "International";

@(contextPath = "SalesTransaction.ProjectUrgency", tagName = "Priority_Level")
extern string ProjectUrgency = "Standard";

// 2. PHYSICAL PRODUCT TYPE
type GeneratorSet {
   // Core technical attributes
   @(configurable = false)
   int requiredKW = [101..10000];

   string DutyRating = ["Prime Power (PRP)", "Emergency Standby Power (ESP)"];

   // Technical calculation with explicit domain (Best Practice)
   decimal(2) surgeLoadKW = [126.25..12500.00];
   constraint(surgeLoadKW == requiredKW * 1.25);

   // 3. LOGIC INTEGRATING EXTERNAL DATA
   // Using contextPath/tagName data to drive technical warnings
   message(ShippingCountry == "US",
           "Regional Notice: Generator must comply with US-specific UL 2200 standards.");


   message(ProjectUrgency == "Critical" && requiredKW > 5000,
           "High Priority Alert: Large scale power requirement in a Critical project
requires site inspection.",
```

```
            requiredKW, "Warning");
}
```

Key Technical Details

- External Variable (`extern`): These are declared outside of any type to hold values supplied by the environment (Salesforce).
- `contextPath` Annotation: This maps the variable directly to a header-level field in the Sales Transaction.
- `tagName` Annotation: This links the variable to a specific Context Tag identifier within the Salesforce Context Definition.

Note: Rules depending on these external variables (such as the `ShippingCountry` message) only re-evaluate when a Line Item change occurs (For example, updating `requiredKW`), not solely when the header field changes.

Example 5: Use Format Specifiers (%s, %d) and Dates in Constraints

In this example, we define a generator configuration that validates the required power (`requiredKW`) against the duty rating and voltage, providing specific feedback when a mismatch occurs.

```
type GeneratorSet  {
   // 1. Core Technical Attributes with explicit domains
   @(configurable = true)
   int requiredKW = [101..10000];

   string DutyRating = ["Prime Power (PRP)", "Continuous Power (COP)",
                        "Data Center Continuous (DCC)", "Emergency Standby Power (ESP)"];


   string Voltage = ["220/380", "277/480", "347/600", "2400/4160"];

   string standardsAndCompliance = ["Certification-CSA", "Listing-UL 2200"];

   // Date Attribute Definition
   // Date variables represent a specific day.
   // You can define a fixed domain for a date as a range between two dates.
   date requestedDeliveryDate = ["2024-01-01", "2025-12-31"];

   // Message Rule with Multiple Arguments using placeholders
   message(requiredKW > 5000 && DutyRating == "Emergency Standby Power (ESP)",
           "High Capacity Alert: The %s rating for %d kW requires an additional cooling
system.",
           DutyRating, requiredKW, "Warning");

   // Constraint with Multiple Arguments using placeholders
   constraint(Voltage == "2400/4160" && standardsAndCompliance == "Listing-UL 2200",
              "Configuration Error: Voltage %s cannot be used with %s standards due to
safety limits.",
              Voltage, standardsAndCompliance);

   // Dates can be used in logical expressions with comparison operators like < or >=.
   message(requestedDeliveryDate < "2024-09-01" && requiredKW > 7000,
           "Schedule Warning: Delivery for %d kW units on %s may require expedited
manufacturing.",
```

```
            requiredKW, requestedDeliveryDate, "Info");
}
```

Key Technical Details

- Multiple Arguments: You can pass several variables after the string literal. The engine maps them to the placeholders in the exact order they appear.
- Format Specifiers
- `%s` is used for string variables (for example, `DutyRating` or `Voltage`).
- `%d` is used for integer variables (for example, `requiredKW`).
- Placeholder Usage: Alternatively, you can use the {} syntax, which the constraint engine automatically replaces with the actual argument values during runtime evaluation.
- Comparison Logic with Dates: You can treat dates as comparable values in `constraint` or `message` rules to enforce scheduling logic (for example, ensuring a delivery date is not too early for a complex build).

Example 6: Use Arithmetic Calculations and Functions

This example demonstrates using aggregate functions (a virtual type for transaction-level aggregation) and mathematical functions within the `GeneratorSet` type for precise quantity calculation.

```
// --- Component Definition —
type GeneratorSet  {
    int requiredKW = [101..10000];
    int quantity = [1..10];
    // BEST PRACTICE: Define derived attributes with an explicit domain
    decimal(2) surgeLoadKW = [0.00..12500.00];
    // BEST PRACTICE: Put calculations inside of constraints
    constraint(surgeLoadKW == requiredKW * 1.25);
}
// --- Accessory Definition ---
type Accessory  {
    int weight_kg = [1..100];
}
// --- Virtual System Type (Aggregation Context) ---
@(virtual = true)
type System {
    @(sourceContextNode = "SalesTransaction.SalesTransactionItem")
    relation generators : GeneratorSet[0..10];
 relation shipmentbatch : ShipmentBatch [0..10];
    // BEST PRACTICE: Attributes must have explicit domains
    decimal(2) totalQuotedLoad = [0.00..125000.00];
    int highPowerCount = [0..10];
    // Aggregate calculations in separate constraints
    constraint(totalQuotedLoad == generators.sum(surgeLoadKW));
    // VARIATION: Condition-based count
    // count() requires a logical expression
    constraint(highPowerCount == generators.count(requiredKW > 5000));
    // Ensuring Shipment batch crates are similar to the number of generators
    message(
        shipmentbatch.requiredCrates != generators[GeneratorSet],
        "The Shipment items ({}) must match the number of Generators ({})",
```

```
        shipmentbatch.requiredCrates, generators[GeneratorSet],
        "Error"
    );
}
type ShipmentBatch {
    int totalItems = [1..100];
    int itemsPerCrate = 10;
    // BEST PRACTICE: Define derived attributes for the relation in the parent type
    int totalWeight = [0..1000];
    int accessoryCount = [0..10];
    // Aggregates within the relation block
    relation accessories : Accessory[0..10] {
        accessoryWeight = sum(weight_kg);
        accessoryCount = count(weight_kg > 0);
    }
    // Mapping relation attributes to parent variables via constraints
    constraint(totalWeight == accessories.accessoryWeight);
    constraint(accessoryCount == accessories.accessoryCount);
    int requiredCrates = [0..100];
    // Mathematical Function Example: CEIL
    constraint calculateCeil(requiredCrates == ceil(totalItems / itemsPerCrate));
}
```

Key Considerations for Calculations and Aggregations

When implementing these functions in CML, follow these architectural best practices for robustness and performance.

- Explicit domains are required: All the derived attributes that result from a calculation or aggregation must have an explicit variable domain definition. This practice ensures accurate aggregation and helps prevent runtime errors.
- Separate calculation from declaration: Define the aggregation or calculation in a separate constraint rather than using an inline derived attribute declaration (for example, `int total = items.sumPrice;`). This separation helps avoid issues where domains may not initialize correctly.
- Correct Pattern: Define the relation aggregate (`totalQty = sum(quantity);`) and then enforce the result via a constraint (`constraint(totalItemCount == items.totalQty);`).

Annotation Examples

Constraint Modeling Language (CML) annotations are labels that you add to parts of a model, such as types, variables, relationships, and constraints. Annotations control how these elements are shown and how they behave in the configurator. Annotations help fine-tune the configurator and the constraint engine without changing the actual structure of the model.

The examples explain what each annotation does, where it can be used in the model, what kinds of values it supports, and how it behaves when the configurator runs and evaluates constraints. CML code samples show how the annotation works in practice.

domainComputation Annotation

`domainComputation` is a CML annotation that specifies how the domain of a model element is determined, either by using a fixed domain or by computing the domain dynamically during configuration.

peelable Annotation

The `peelable` annotation is used to create soft selection values and allow the engine to modify these selections to satisfy a constraint.

productField Annotation

productField is a CML annotation that defines the Product2 field on a variable. productField loads the value from Product Catalog Management (PCM) during constraint model activation.

propagateUp Annotation

`propagateUp` is a Constraint Modeling Language (CML) annotation that controls aggregation propagation between children and parent elements.

relatedAttributes Annotation

`relatedAttributes` is a Constraint Modeling Language (CML) annotation that resets the domain to the original one for domainComputation.

sequence Annotation

The `sequence` annotation defines the execution and configuration order of elements in a Constraint Modeling Language (CML) model.

split Annotation

`split` is a Constraint Modeling Language (CML) annotation that specifies whether the instances of the type should be split or not.

closeRelation Annotation

closeRelation is a CML annotation that controls addition of new line items to the relationship by the engine.

Annotation	`closeRelation`
Applicable to	Relationship
Value Type/Values	true, false
Description	If `closeRelation` is not specified, the engine sets it implicitly as false for the relationship. If `closeRelation` is specified as true, the engine prevents the addition of new line items to the relationship. If `closeRelation` is specified as false, the engine allows the addition of new line items to the relationship.

Example 1

In this example, the `closeRelation` annotation isn't specified for the relationship (`mainalternatorclassification`). The model contains the constraint.

```
type LineItem;
type GeneratorSet : LineItem {
    @(defaultValue = "Emergency Standby Power (ESP)")
```

```
    string dutyRating = ["Emergency Standby Power (ESP)", "Prime Power (PRP)", "Continuous
 Power (COP)", "Data Center Continuous (DCC)"];
    relation mainalternatorclassification : MainAlternatorClassification;
    constraint(dutyRating == "Continuous Power (COP)" ->
mainalternatorclassification[Alternator_240] >= 1);
}
type MainAlternatorClassification : LineItem;
type Alternator_220 : MainAlternatorClassification;
type Alternator_440 : MainAlternatorClassification;
type Alternator_240 : MainAlternatorClassification;
```

Example 2

In this example, the `closeRelation`annotation is defined as `false` for the relationship (`mainalternatorclassification`). The model contains the constraint.

```
type LineItem;
type GeneratorSet : LineItem {
    @(defaultValue = "Emergency Standby Power (ESP)")
    string dutyRating = ["Emergency Standby Power (ESP)", "Prime Power (PRP)", "Continuous
 Power (COP)", "Data Center Continuous (DCC)"];
    @(closeRelation = false)
    relation mainalternatorclassification : MainAlternatorClassification;
    constraint(dutyRating == "Continuous Power (COP)" ->
mainalternatorclassification[Alternator_240] >= 1);
}
type MainAlternatorClassification : LineItem;
type Alternator_220 : MainAlternatorClassification;
type Alternator_440 : MainAlternatorClassification;
type Alternator_240 : MainAlternatorClassification;
```

Example Description and Configurator Result

In example 1, the `mainalternatorclassification` relation is not specified with the `closeRelation` annotation, but the system considers it as `false` by default. In example 2, the relation is explicitly defined with `false`. As a result, if the user updates the `dutyRating` variable to `"Continuous Power (COP)"`, the system adds the `Alternator_240` product according to annotation and constraint logic and allows addition of other products from the `mainalternatorclassification` relation (`Alternator_220,Alternator_440`). The system does not allow to unselect the `Alternator_240` product specified in the constraint while the `"Continuous Power (COP)"` is selected. If the user updates the `dutyRating` variable to another value (For example, `Emergency Standby Power (ESP)"`), the engine removes the `Alternator_240` product, but the user can add/select/unselect all products from the relationship manually.

Example 3

In this example, the `closeRelation` annotation is specified as `true` for the relationship (`mainalternatorclassification`). The model contains the constraint

```
type LineItem;
type GeneratorSet : LineItem {
    @(defaultValue = "Emergency Standby Power (ESP)")
    string dutyRating = ["Emergency Standby Power (ESP)", "Prime Power (PRP)", "Continuous
 Power (COP)", "Data Center Continuous (DCC)"];
```

```
    @(closeRelation = true)
    relation mainalternatorclassification : MainAlternatorClassification;
    constraint(dutyRating == "Continuous Power (COP)" ->
mainalternatorclassification[Alternator_240] >= 1);
}
type MainAlternatorClassification : LineItem;
type Alternator_220 : MainAlternatorClassification;
type Alternator_440 : MainAlternatorClassification;
type Alternator_240 : MainAlternatorClassification;
```

In example 3, the `mainalternatorclassification` relation is specified with the `closeRelation` annotation as `true`. As a result, if a user updates the dutyRating variable to `"Continuous Power (COP)"`, the engine resets it back to `"Emergency Standby Power (ESP)"` to prevent adding the `Alternator_240` product according to the annotation. The user can add all `mainalternatorclassification` products manually (including the `Alternator_240` product specified in the constraint).

Table 8: closeRelation Configuration Settings

Associated Example	Product Group Structure	Applicable to	"closeRelation" annotation	User Action	Engine Action	UI Behavior
N/A	Individual product	Relationship	not specified	Change variable value specified in constraint condition	Add the product according to the constraint. Allow addition of other products to the relationship	Display product specified in constraint as pre-selected: if a User unselects it, the product returns back to selected state. User can select other products from the relationship
N/A	Individual product	Relationship	TRUE	Change variable value specified in constraint condition	Reset the variable value back to the default value to avoid addition the relation product specified in the constraint	Products from the relation are not added or selected (including the product specified in constraint). User can manually select any products from the relationship
N/A	Individual product	Relationship	FALSE	Change variable value specified in constraint condition	Add the product according to the constraint. Allow addition of other	Display product specified in constraint as pre-selected: if a

Associated Example	Product Group Structure	Applicable to	"closeRelation" annotation	User Action	Engine Action	UI Behavior
					products to the relationship	User unselects it, the product returns back to selected state. User can select other products from the relationship
Example 1	Product Classification	Relationship	not specified	Change variable value specified in constraint condition	Add the product according to the constraint. Allow addition of other products to the relationship from browse products pop up	Display product specified in constraint as pre-selected: if a User unselects it, the product returns back to selected state. User can add other products from the relationship in browse products pop up
Example 2	Product Classification	Relationship	FALSE	Change variable value specified in constraint condition	Add the product according to the constraint. Allow addition of other products to the relationship from browse products pop up	Display product specified in constraint as pre-selected: if a User unselects it, the product returns back to selected state. User can add other products from the relationship in browse products pop up
Example 3	Product Classification	Relationship	TRUE	Change variable value specified in constraint condition	Reset the variable value back to the default value to avoid addition the relation	Products from the relation are not added or selected (including the product specified

Associated Example	Product Group Structure	Applicable to	"closeRelation" annotation	User Action	Engine Action	UI Behavior
					product specified in the constraint	in constraint). User can manually browse and add any products from the relationship

configurable Annotation

`configurable` is a CML annotation that controls whether a model element can be configured.

`configurable` annotation specification: Variable

Annotation	`configurable`
Applicable to	Variable
Value Type/Values	true, false
Description	If the `configurable` annotation is not explicitly specified, the engine sets it implicitly as true for the variable. If the `configurable` annotation is explicitly specified as true, the variable is indicated as configurable. The engine sets the value to the variable. If the `configurable` annotation is explicitly specified as false, the engine doesn't set a value to the variable.

Example 1

In this example, the `configurable` annotation is not specified for the variable (Cable Entry) of the VoltageConnection child products. The `defaultValue` is not defined.

```
type GeneratorSet {
    relation voltageConnections : VoltageConnection[1..999999];
}
type VoltageConnection {
    string cableEntry = ["Top Entry", "Bottom Entry", "Side Entry"];
}
```

Example Description and Configurator Result

In this example, the `configurable` annotation is not explicitly specified for the variable `cableEntry`, but the system considers it as `configurable = true` by default. As a result, the system sets the `"Top Entry"` (the first value in the `["Top Entry", "Bottom Entry", "Side Entry"]` CML domain) as the initial value for the configurable `Cable Entry` variable.

Example 2

In this example, the `configurable` annotation is specified as `true` for the variable (Cable Entry) of the VoltageConnection child products. The `defaultValue` is not specified.

```
type GeneratorSet {
    relation voltageConnections : VoltageConnection[1..999999];
}
type VoltageConnection {
    @(configurable = true)
    string cableEntry = ["Top Entry", "Bottom Entry", "Side Entry"];
}
```

Example Description and Configurator Result

In this example, the `configurable` annotation is `true` for the variable. As a result, the system specifies the variable as configurable and sets the `"Top Entry"` (the first value in the `["Top Entry", "Bottom Entry", "Side Entry"]` CML domain) as the initial value for the `Cable Entry`.

Example 3

In this example, the `configurable` annotation is specified as `false` for the variable (Cable Entry) of the VoltageConnection child products. The `defaultValue` is not specified

```
type GeneratorSet {
    relation voltageConnections : VoltageConnection[1..999999];
}
type VoltageConnection {
    @(configurable = false)
    string cableEntry = ["Top Entry", "Bottom Entry", "Side Entry"];
}
```

Example Description and Configurator Result

In this example, the `configurable` annotation is `false` for the variable. As a result, the engine doesn't configure the variable with an initial value, and it is displayed empty on the Product Configuration UI.

Example 4

In this example, the `configurable` is `false` and `defaultValue` annotation is `true` for the variable (Cable Entry) of the VoltageConnection child products.

```
type GeneratorSet {
    relation voltageConnections : VoltageConnection[1..999999];
}
type VoltageConnection {
    @(configurable = true, defaultValue = "Side Entry")
    string cableEntry = ["Top Entry", "Bottom Entry", "Side Entry"];
}
```

Example Description and Configurator Result

In this example, the system indicates the variable as `configurable` and sets the `"Side Entry"` as the initial value for the `Cable Entry` defined in the `defaultValue` annotation.

Example 5

In this example, the `configurable` and `defaultValue` annotations are specified for the variable (Cable Entry) of the VoltageConnection child products.

```
type GeneratorSet {
    relation voltageConnections : VoltageConnection[1..999999];
}
type VoltageConnection {
    @(configurable = false, defaultValue = "Side Entry")
    string cableEntry = ["Top Entry", "Bottom Entry", "Side Entry"];
}
```

Example Description and Configurator Result

In this example, the system sets the `"Side Entry"` as the initial value for the `Cable Entry` according to the `defaultValue` annotation.

Table 9: configurable Configuration Settings

Associated Example	Product Group Structure	Applicable to	"CONFIGURABLE" annotation	User Action	Engine Action	UI Behavior
Example 1	Individual product	Variable	not specified	Configure the product containing variable	Set the first value from the variable domain as initial value. If the User changes the variable value manually, reset it back to the first value from the variable domain. If the default value is defined in PCM, reset it back to the first value from the variable domain	Display defined by engine variable value as the default value
Example 2 Example 4	Individual product	Variable	TRUE	Configure the product containing variable	Set the first value from the variable domain as initial value	Display defined by engine variable value as the default value

Associated Example	Product Group Structure	Applicable to	"CONFIGURABLE" annotation	User Action	Engine Action	UI Behavior
					If the User changes the variable value manually, reset it back to the first value from the variable domain. If the default value is defined in PCM, reset it back to the first value from the variable domain If the defaultValue annotation is specified for the variable, set it as initial value	
Example 3 Example 5	Individual product	Variable	FALSE	Configure the product containing variable	Set emptiness for the variable as initial value (If the default value is specified in PCM for the variable, set it as initial value instead of emptiness). If the defaultValue annotation is specified for the variable, set it as initial value (despite of the PCM default value)	Display defined by engine or PCM variable value as the default value

defaultValue Annotation

The `defaultValue` annotation is used on a variable to define the value it should start with when configuration begins.

Annotation	`defaultValue`
Applicable to	Variable
Value Type/Values	Literal
Description	The configurator uses the default value defined in PCM (Product Attribute Definition). If no PCM default is available, the configurator uses the first value in the variable domain as the initial value. If no default value is defined in PCM and a defaultValue is specified in CML, the configurator uses the value defined in CML as the initial value of the variable.

Example 1

In this example, neither PCM nor CML defines a default value for the Cable Entry variable.

```
type GeneratorSet {
    relation voltageConnections : VoltageConnection[1..999999];
}
type VoltageConnection {
    string cableEntry = ["Top Entry", "Bottom Entry", "Side Entry"];
}
```

Example Description and Configurator Result

The configurator sets `"Top Entry"` as the initial value for `Cable Entry`, because it is the first value in the variable domain `["Top Entry", "Bottom Entry", "Side Entry"`.

Example 2

In this example, PCM (Product Attribute Definition) defines `"Bottom Entry"` as the default value for `Cable Entry`, and no `defaultValue` annotation is defined for this variable in CML.

```
type GeneratorSet {
    relation voltageConnections : VoltageConnection[1..999999];
}
// This annotation is added automatically if PCM has a default before the product is added
 to CML. If the product already exists in CML, the annotation is added or updated only
after running Sync.
@(defaultValue = "Bottom Entry")
type VoltageConnection {
    string cableEntry = ["Top Entry", "Bottom Entry", "Side Entry"];
}
```

Example Description and Configurator Result

The configurator sets the `"Bottom Entry"` as the initial value for the `Cable Entry` of the `VoltageConnection` child products.

As a best practice, define the default value in PCM and use the Sync function in CML to keep the model aligned whenever PCM settings are updated.

Example 3

In this example, a `defaultValue` annotation with `"Side Entry"` is defined in CML, and no default value is defined in PCM for the `Cable Entry` variable.

```
type GeneratorSet {
    relation voltageConnections : VoltageConnection[1..999999];
}

type VoltageConnection {
    @(defaultValue = "Side Entry")
    string cableEntry = ["Top Entry", "Bottom Entry", "Side Entry"];
}
```

Example Description and Configurator Result

The configurator sets the `"Side Entry"` annotation value as the initial value for Cable Entry.

Note: We recommend that you define default values in PCM (Product Attribute Definition) whenever possible, as this approach promotes consistency across products and simplifies maintenance.

The defaultValue annotation in CML should be used only when a default value is not suitable to be defined in PCM due to specific modeling requirements.

When a default value is defined in PCM but the CML model has not been synced, the configurator still applies the PCM default. However, the value displayed in the CML model may become inconsistent with PCM.

To avoid such inconsistencies, it is recommended to run the Sync action in CML after updating default values in PCM, so that the CML model remains aligned with the latest PCM settings.

defaultValue Configuration Settings

Table 10: defaultValue Configuration Settings

Associated Example	Configurable Specified	Configurable Value	defaultValue Specified	Initial Value Behavior	Example UI Result
Example 1	No	default = TRUE	No	Uses the first value in the domain	Automatically set to "Top Entry"
Example 2	Yes	TRUE	No	Uses the first value in the domain	Automatically set to "Bottom Entry"
Example 3	Yes	FALSE	No	No initial value is set	Field remains empty
N/A	Yes	TRUE	Yes	Uses the value defined in defaultValue	Automatically set to "Side Entry"
N/A	Yes	FALSE	Yes	Still uses the value defined in defaultValue	Automatically set to "Side Entry"

domainComputation Annotation

`domainComputation` is a CML annotation that specifies how the domain of a model element is determined, either by using a fixed domain or by computing the domain dynamically during configuration.

`domainComputation` annotation specification: Variable and Relationship

Annotation	`domainComputation`
Applicable to	Variable and Relationship
Value Type/Values	true, false
Description	`If domainComputation annotation is not explicitly specified, the engine sets it implicitly as false for the variable, and true for a relationship` If the `domainComputation` annotation is specified as `true`, the variable or relationship domain is dynamically determined based on the configuration and constraint logic. If the `domainComputation` annotation is specified as `false`, the variable domain is fixed.

Example 1

In this example, the `domainComputation` annotation is not specified for the variable (voltage) of the `GeneratorSet` type. The model contains the constraints.

```
type GeneratorSet {
    @(defaultValue = "Emergency Standby Power (ESP)")
    string dutyRating = ["Emergency Standby Power (ESP)", "Prime Power (PRP)", "Continuous
 Power (COP)", "Data Center Continuous (DCC)"];
    string voltage = ["220/380", "240/416", "347/600", "255/440", "277/480", "2400/4160",
 "7200/12470", "7621/13200", "7976/13800"]
    constraint(dutyRating == "Continuous Power (COP)" -> voltage != "220/380");
    constraint(dutyRating == "Data Center Continuous (DCC)" -> voltage == "255/440");
}
```

Example 2

In this example, the `domainComputation` annotation is explicitly specified as false for the variable (voltage) of the `GeneratorSet` type. The model contains the constraints.

```
type GeneratorSet {
    @(defaultValue = "Emergency Standby Power (ESP)")
    string dutyRating = ["Emergency Standby Power (ESP)", "Prime Power (PRP)", "Continuous
 Power (COP)", "Data Center Continuous (DCC)"];
    @(domainComputation = false)
    string voltage = ["220/380", "240/416", "347/600", "255/440", "277/480", "2400/4160",
 "7200/12470", "7621/13200", "7976/13800"];
    constraint(dutyRating == "Continuous Power (COP)" -> voltage != "220/380");
```

```
    constraint(dutyRating == "Data Center Continuous (DCC)" -> voltage == "255/440");
}
```

Example Description and Configurator Result

In example 1, the `domainComputation` annotation is not explicitly specified for the `voltage` variable, but the system considers it as `false` by default. In example 2, the `domainComputation` annotation is explicitly defined as `false`. As a result, the system remains the variable domain unchanged `(["220/380", "240/416", "347/600", "255/440", "277/480", "2400/4160", "7200/12470", "7621/13200", "7976/13800"])` and executes the constraints:

If the `dutyRating` is selected with the `"Continuous Power (COP)"` value, the `voltage` is pre-populated with the next possible value from the variable domain `("240/416")` to satisfy the constraint.

If the `dutyRating` is selected with the `"Data Center Continuous (DCC)"` value, the `voltage` is pre-populated with the `"255/440"` value to satisfy the constraint.

Example 3

In this example, the `domainComputation` annotation is explicitly specified as `true` for the variable (`voltage`) of the `GeneratorSet` type. The model contains the constraints.

```
type GeneratorSet {
    @(defaultValue = "Emergency Standby Power (ESP)")
    string dutyRating = ["Emergency Standby Power (ESP)", "Prime Power (PRP)", "Continuous
 Power (COP)", "Data Center Continuous (DCC)"];
    @(domainComputation = true)
    string voltage = ["220/380", "240/416", "347/600", "255/440", "277/480", "2400/4160",
 "7200/12470", "7621/13200", "7976/13800"];
    constraint(dutyRating == "Continuous Power (COP)" -> voltage != "220/380");
    constraint(dutyRating == "Data Center Continuous (DCC)" -> voltage == "255/440");
}
```

Example Description and Configurator Result

In example 3, the `domainComputation` annotation is explicitly specified as `true` for the `voltage` variable. As a result, the system updates the variable domain based on the constraint logic.

If the `dutyRating` is selected with the `"Continuous Power (COP)"` value, the voltage is pre-populated with the next possible value from the variable domain `("240/416")` to satisfy the constraint. The variable domain is updated to exclude the `"220/380"` value.

If the `dutyRating` is selected with the `"Data Center Continuous (DCC)"` value, the `voltage` is pre-populated with the `"255/440"` value to satisfy the constraint. The variable domain is updated to contain only the `"255/440"` value.

Example 4

In this example, the `domainComputation` annotation is not specified for the relationship (temperatureSensors). The model contains the constraints.

```
type GeneratorSet {
    @(defaultValue = "Emergency Standby Power (ESP)")
    string dutyRating = ["Emergency Standby Power (ESP)", "Prime Power (PRP)", "Continuous
 Power (COP)", "Data Center Continuous (DCC)"];
```

```
    relation temperatureSensors : TemperatureSensor[0..1];
    constraint(dutyRating == "Continuous Power (COP)" ->
temperatureSensors[BearingTemperatureSensor] == 0);
    constraint(dutyRating == "Data Center Continuous (DCC)" ->
temperatureSensors[StatorTemperatureSensor] == 0);
}
type TemperatureSensor;
type BearingTemperatureSensor : TemperatureSensor;
type StatorTemperatureSensor : TemperatureSensor;
```

Example 5

In this example, the `domainComputation` annotation is explicitly specified as `true` for the relationship (temperatureSensors). The model contains the constraints.

```
type GeneratorSet {
    @(defaultValue = "Emergency Standby Power (ESP)")
    string dutyRating = ["Emergency Standby Power (ESP)", "Prime Power (PRP)", "Continuous
 Power (COP)", "Data Center Continuous (DCC)"];
    @(domainComputation = true)
    relation temperatureSensors : TemperatureSensor[0..1];
    constraint(dutyRating == "Continuous Power (COP)" ->
temperatureSensors[BearingTemperatureSensor] == 0);
    constraint(dutyRating == "Data Center Continuous (DCC)" ->
temperatureSensors[StatorTemperatureSensor] == 0);
}
type TemperatureSensor;
type BearingTemperatureSensor : TemperatureSensor;
type StatorTemperatureSensor : TemperatureSensor;
```

Example Description and Configurator Result

In example 4, the `domainComputation` annotation is not explicitly specified for the `temperatureSensors` relationship, but the system considers it as `true` by default. In example 5, the `domainComputation` annotation is explicitly specified as `true` for the relationship. As a result, the system updates the relationship domain based on the constraint logic.

- If the `dutyRating` is selected with the `"Continuous Power (COP)"` value, the system re-computes the relationship domain and excludes the `BearingTemperatureSensor` product to satisfy the constraint. The `StatorTemperatureSensor` product remains and can be selected in scope of the `GeneratorSet` bundle product;
- If the `dutyRating` is selected with the `"Data Center Continuous (DCC)"` value: the system re-computes the relationship domain and excludes the `StatorTemperatureSensor` product to satisfy the constraint. The `BearingTemperatureSensor` product remains and can be selected in scope of the `GeneratorSet` bundle product.

Example 6

In this example, the `domainComputation` annotation is explicitly specified as `false` for the relationship (`temperatureSensors`). The model contains the constraints.

```
type GeneratorSet {
    @(defaultValue = "Emergency Standby Power (ESP)")
    string dutyRating = ["Emergency Standby Power (ESP)", "Prime Power (PRP)", "Continuous
 Power (COP)", "Data Center Continuous (DCC)"];
```

```
    @(domainComputation = false)
    relation temperatureSensors : TemperatureSensor[0..1];
    constraint(dutyRating == "Continuous Power (COP)" ->
temperatureSensors[BearingTemperatureSensor] == 0);
    constraint(dutyRating == "Data Center Continuous (DCC)" ->
temperatureSensors[StatorTemperatureSensor] == 0);
}
type TemperatureSensor;
type BearingTemperatureSensor : TemperatureSensor;
type StatorTemperatureSensor : TemperatureSensor;
```

Example Description and Configurator Result

In example 6, the `domainComputation` annotation is explicitly specified as `false` for the relationship. As a result, the system remains the relationship domain unchanged and executes the constraints:

If the `dutyRating` is selected with the `"Continuous Power (COP)"` value, the system displays both `BearingTemperatureSensor` and `StatorTemperatureSensor` products in scope of the `GeneratorSet` bundle product. The `BearingTemperatureSensor` product cannot be selected based on the constraint.

If the `dutyRating` is selected with the `"Data Center Continuous (DCC)"` value, the system displays both `BearingTemperatureSensor` and `StatorTemperatureSensor` products in scope of the GeneratorSet bundle product. The `StatorTemperatureSensor` product cannot be selected based on the constraint.

domainComputation Configuration Settings

Table 11: defaultValue Configuration Settings

Applicable to	"domainComputation" annotation	User Action	Actual UI Behavior	Actual Engine Behavior	Constraint Model
string voltage	not specified	Case 1: Duty Rating = any except specified in constraints (e.g. "Prime Power (PRP)", "Emergency Standby Power (ESP)"): • Initial value for the voltage = the value specified in the @defaultValue annotation ("220/380") • The user can select a different voltage value (e.g. "240/416", "255/440" etc.). The system remains User selected voltage	Always show the full voltage domain regardless of the Duty Rating selection.	Remain the full voltage domain. Case 1: Duty Rating = any except specified in constraints (e.g. "Prime Power (PRP)", "Emergency Standby Power (ESP)"): • If the configuration is saved with initial (default) voltage value ("220/380"): attributeName:"voltage", configStatus:"Default" • If the configuration is saved with User selected voltage value (e.g.	`type GeneratorSet { @(defaultValue = "Emergency Standby Power (ESP)") string dutyRating = ["Emergency Standby Power (ESP)", "Prime Power (PRP)", "Continuous Power (COP)",`

Applicable to	"domainComputation" annotation	User Action	Actual UI Behavior	Actual Engine Behavior	Constraint Model

		• The user can save the configuration with both values: initial voltage or user-selected voltage
		
		Case 2: Duty Rating = "Continuous Power (COP)"
		
		• Initial value for the voltage = next possible value from the domain to satisfy the constraint ("240/416")
		
		• Warning: If the user selects a different voltage (e.g. "220/380", "255/440", "2400/4160" etc.), the system reverts it to the initial value ("240/416")
		
		• The user can save the configuration with the initial voltage value (the configuration cannot be saved with other voltage due to reverting)
		
		Case 3: Duty Rating = "Data Center Continuous (DCC)"
		
		• Initial value for the voltage = specified value in constraint that comes first in the domain ("220/380")
		
		• The user can select a different vaild voltage specified in constraint ("255/440"). The system remains User selected voltage
				"255/440"): attributeName":"voltage","configStatus":"Default"
				
				Case 2: Duty Rating = "Continuous Power (COP)"
				
				• If the configuration is saved with initial/reverted to initial voltage value ("240/416"): "attributeName":"voltage","configStatus":"Engine"
				
				Case 3: Duty Rating = "Data Center Continuous (DCC)"
				
				• If the configuration is saved with initial ("220/380")/other constraint specified("255/440")/reverted to constraint specified voltage values ("220/380", "255/440"): "attributeName":"voltage","configStatus":"Default"
					`"Data Center Continuous (DCC)"];`
					
					`@(defaultValue = "220/380")`
					`string voltage = ["220/380", "240/416", "347/600", "255/440", "277/480", "2400/4160",`
					`"7200/12470",`
					`"7621/13200",`
					`"7976/13800"];`
					
					`constraint(dutyRating == "Continuous Power (COP)" -> voltage != "220/380");`
					
					`constraint(dutyRating == "Data Center Continuous (DCC)" -> voltage == "255/440" || voltage == "220/380");`
					`}`

Applicable to	"domainComputation" annotation	User Action	Actual UI Behavior	Actual Engine Behavior	Constraint Model
		• If the user selects a voltage that not vaild in the constraint (e.g. "240/416", "2400/4160" etc.), the system will revert it to one of the values defined in constraint ("220/380" or "255/440") • The user can save the configuration with the specified in constraint voltage value (the configuration cannot be saved with other voltage due to reverting)			

string voltage	TRUE	Case 1: Duty Rating = any except specified in constraints (e.g. "Prime Power (PRP)", "Emergency Standby Power (ESP)"):
		• Display the full voltage domain
		• Initial value for the voltage = the value specified in the @defaultValue annotation ("220/380")
		• The user can select a different voltage value (e.g. "240/416", "255/440" etc.). The system remains User selected voltage
		• The user can save the configuration with both values: initial
			Show updated voltage domain based on Duty Rating selection and constraint logic.	Update voltage domain based on Duty Rating selection and constraint logic.
				Case 1: Duty Rating = any except specified in constraints (e.g. "Prime Power (PRP)", "Emergency Standby Power (ESP)"):
				• Remain the full voltage domain
				• If the configuration is saved with initial (default) voltage value ("220/380"): attributeName:"Voltage", configStatus:"Default"
				• If the configuration is saved with User selected voltage value (e.g.
					`type GeneratorSet {`
					`@(defaultValue = "Emergency Standby Power (ESP)")`
					`string dutyRating = ["Emergency Standby Power (ESP)", "Prime Power (PRP)", "Continuous Power (COP)", "Data Center Continuous (DCC)"];`
					`@(defaultValue`

Applicable to	"domainComputation" annotation	User Action	Actual UI Behavior	Actual Engine Behavior	Constraint Model
		voltage or User selected voltage Case 2: Duty Rating = "Continuous Power (COP)" • Display updated voltage domain according to the constraint (the "220/380" is not shown) • Initial value for the voltage = next possible value from the domain to satisfy the constraint ("240/416") • Warning: If the user selects a different voltage (e.g. "255/440", "2400/4160" etc.), the system reverts it to the initial value ("240/416") • The user can save the configuration with the initial ("240/416") voltage value (the configuration cannot be saved with other voltage due to reverting) Case 3: Duty Rating = "Data Center Continuous (DCC)" • Display updated voltage domain according to the constraint (only "220/380" and "255/440" are shown, other values are excluded)		"255/440"): attributeName":"voltage","configStatus":"Default" Case 2: Duty Rating = "Continuous Power (COP)" • Update voltage domain according to the constraint (exclude the "220/380" value) • If the configuration is saved with initial/reverted voltage value ("240/416"): "attributeName":"voltage","configStatus":"Engine" Case 3: Duty Rating = "Data Center Continuous (DCC)" • If the configuration is saved with initial ("220/380")/reverted to "220/380" constraint specified voltage value: "attributeName":"voltage","configStatus":"Engine"	`= "220/380", domainComputation = true) string voltage = ["220/380", "240/416", "347/600", "255/440", "277/480", "2400/4160", "7200/12470", "7621/13200", "7976/13800"]; constraint(dutyRating == "Continuous Power (COP)" -> voltage != "220/380"); constraint(dutyRating == "Data Center Continuous (DCC)" -> voltage == "255/440" || voltage == "220/380"); }`

Applicable to	"domainComputation" annotation	User Action	Actual UI Behavior	Actual Engine Behavior	Constraint Model
		• Initial value for the voltage = specified value in constraint that comes first in the domain ("220/380") • Warning: If the user selects a different voltage specified in constraint ("255/440"), the system reverts it to the initial value ("220/380") • The user can save the configuration with the initial voltage value (the configuration cannot be saved with the "255/440" voltage due to reverting)			

string voltage	FALSE	Case 1: Duty Rating = any except specified in constraints (e.g. "Prime Power (PRP)", "Emergency Standby Power (ESP)"):
		• Initial value for the voltage = the value specified in the @defaultValue annotation ("220/380")
		• The user can select a different voltage value (e.g. "240/416", "255/440" etc.). The system remains User selected voltage
		• The user can save the configuration with both values: initial voltage or User selected voltage
			Same as not specified
			Always show the full voltage domain regardless of the Duty Rating selection.
				Same as not specified
				Remain the full voltage domain.
				Case 1: Duty Rating = any except specified in constraints (e.g. "Prime Power (PRP)", "Emergency Standby Power (ESP)"):
				• If the configuration is saved with initial (default) voltage value ("220/380"): attributeName":"voltage","flagStatus":"Default"
				• If the configuration is saved with User selected voltage value (e.g. "255/440"): attributeName":"voltage","flagStatus":"Default"
					`type GeneratorSet {`
					`@(defaultValue = "Emergency Standby Power (ESP)") string dutyRating = ["Emergency Standby Power (ESP)", "Prime Power (PRP)", "Continuous Power (COP)", "Data Center Continuous (DCC)"];`
					`@(defaultValue`

Applicable to	"domainComputation" annotation	User Action	Actual UI Behavior	Actual Engine Behavior	Constraint Model

		Case 2: Duty Rating = "Continuous Power (COP)"
		• Initial value for the voltage = next possible value from the domain to satisfy the constraint ("240/416")
		• Warning: If the user selects a different voltage (e.g. "220/380", "255/440", "2400/4160" etc.), the system reverts it to the initial value ("240/416")
		• The user can save the configuration with the initial voltage value (the configuration cannot be saved with other voltage due to reverting)
		Case 3: Duty Rating = "Data Center Continuous (DCC)"
		• Initialvalue for the voltage = specified value in constraint that comes first in the domain ("220/380")
		• The user can select a different voltage specified in constraint ("255/440"). The system remains User selected voltage
		• If the user selects a voltage that not specified in the constraint (e.g. "240/416",
				Case 2: Duty Rating = "Continuous Power (COP)"
				• If the configuration is saved with initial/reverted to initial voltage value ("240/416"): "attributeName":"voltage","configStatus":"Engine"
				Case 3: Duty Rating = "Data Center Continuous (DCC)"
				• If the configuration is saved with initial ("220/380")/other constraint specified("255/440")/reverted to constraint specified voltage values ("220/380", "255/440"): "attributeName":"voltage","configStatus":"Default"
					`= "220/380", domainComputation = false) string voltage = ["220/380", "240/416", "347/600", "255/440", "277/480", "2400/4160", "7200/12470", "7621/13200", "7976/13800"];`
					
					`constraint(dutyRating == "Continuous Power (COP)" -> voltage != "220/380");`
					
					`constraint(dutyRating == "Data Center Continuous (DCC)" -> voltage == "255/440" || voltage == "220/380"); }`

Applicable to	"domainComputation" annotation	User Action	Actual UI Behavior	Actual Engine Behavior	Constraint Model
		"2400/4160" etc.), the system will revert it to one of the values defined in constraint ("220/380" or "255/440") • The user can save the configuration with the specified in constraint voltage value (the configuration cannot be saved with other voltage due to reverting)			

peelable Annotation

The `peelable` annotation is used to create soft selection values and allow the engine to modify these selections to satisfy a constraint.

Annotation	`peelable`
Applicable to	Variable
Value Type/Values	true, false
Description	Indicates whether the constraint engine can override the variable's value (whether set by default or user selection) to resolve a conflict. • If `peelable` annotation is set to `true`, the engine treats the value as a "soft selection." When a configuration conflict occurs, the engine attempts to "peel" (unbind) this variable and retry the solution. If a valid configuration is found, the engine automatically changes the value to satisfy constraints rather than displaying an error message. • If `peelable` annotation is set to `false`, the engine treats the value as a "hard selection." If the value causes a conflict with a constraint, the engine will not attempt to change it automatically. Instead, it will stop and display a conflict error message to the user, requiring manual intervention to resolve the issue.

Example 1: peelable = true (Soft Selection)

In this scenario, the `ServiceLevel` defaults to `"Standard"`. Because it is marked `peelable = true`, the Constraint Engine treats this value as a "soft pick." It is authorized to override this value automatically if a conflict arises with another selection.

```
type CloudSubscription {

    // User adds 5TB of storage
    int storageTB = [1..3];

    /*
     * peelable=true: The engine acts as a "negotiator."
     * If a constraint needs to change this value, the engine is allowed
     * to "peel" (remove) the user's selection and apply the required value.
     */
    @(defaultValue = "Standard", peelable = true)
    string ServiceLevel = ["Standard", "Premium"];

    // Constraint: 5TB or more requires Premium Service
    constraint(storageTB >= 5 -> ServiceLevel == "Premium", "High storage requires Premium
 service");
}
```

Example 2: peelable = false (Hard Selection)

In this scenario, `peelable` is omitted (defaulting to `false`) or explicitly set to `false`. The engine treats the value `"Standard"` as a "hard constraint" or a firm user commitment. It is not authorized to change it automatically.

```
type CloudSubscription {

    // User adds 5TB of storage
    int storageTB = [1..3];

    /*
     * peelable=false (Default): The engine acts as a "validator."
     * It respects the current value as a hard fact.
     * It cannot change "Standard" to "Premium" on its own.
     */
    @(defaultValue = "Standard", peelable = false)
    string ServiceLevel = ["Standard", "Premium"];

    // Constraint: 5TB or more requires Premium Service
    constraint(storageTB >= 5 -> ServiceLevel == "Premium", "High storage requires Premium
 service");
}
```

Example Description and Configurator Result

In example 1, the `ServiceLevel` attribute is specified with the `peelable` annotation set to `true`. In example 2, the annotation is not specified, so the system considers it as false by default. As a result, if the user updates the `storageTB` variable to 5, the system automatically changes the `ServiceLevel` to `"Premium"` in example 1 according to constraint logic, effectively "peeling" the default `"Standard"` value to resolve the conflict. The system allows this override without displaying an error message to the user. In example 2, the system does not allow the engine to override the default `"Standard"` value automatically. Consequently, the system displays a conflict error message, and the user must manually update the `ServiceLevel` to `"Premium"` to satisfy the constraint.

Example 3: System-Driven Soft Selection (configurable = false, peelable = true)

In this example, the `Voltage` is set to a high voltage (" `2400/4160`") by default. The annotation `configurable = false` prevents the user from manually changing this value in the UI. However, `peelable = true` is added to allow the constraint engine to override this default if a specific compliance standard is selected.

```
type GeneratorSet {

    string standardsAndCompliance = ["None", "Listing-UL 2200"];

    /*
     * configurable=false: The user cannot change this directly.
     * peelable=true: The engine is authorized to change the default
     * to resolve conflicts with the standardsAndCompliance selection.
     */
    @(defaultValue = "2400/4160", configurable = false, peelable = true)
    string Voltage = ["220/380", "277/480", "2400/4160"];

    // Constraint: UL 2200 listing requires a specific low voltage (277/480)
    constraint(standardsAndCompliance == "Listing-UL 2200" -> Voltage == "277/480");
}
```

Example Description and Configurator Result

In example 3, the `Voltage` is specified with configurable set to `false`, preventing user interaction, but peelable is set to `true`. Initially, the system sets the voltage to "`2400/4160`". If the user updates the `standardsAndCompliance` variable to "`Listing-UL 2200`", the system detects a conflict with the default voltage. Because the variable is peelable, the system automatically "peels" the default "`2400/4160`" value and changes it to "`277/480`" to satisfy the constraint. The transition happens silently without a conflict error, even though the user cannot manually edit the field.

Example 4: System-Driven Hard Constraint (configurable = false, peelable = false)

In this example, the `EmissionsTier` is set to "`Tier 3`" by default. It is marked `configurable = false` (system-controlled) and `peelable` is omitted (defaults to false), treating the default value as a hard constraint.

```
type GeneratorSet {

    string Location = ["US", "International"];

    /*
     * configurable=false: The user cannot change this.
     * peelable=false (Default): The engine treats "Tier 3" as a hard fact.
     */
    @(defaultValue = "Tier 3", configurable = false)
    string EmissionsTier = ["Tier 3", "Tier 4 Final"];

    // Constraint: US Location requires Tier 4 Final
    constraint(Location == "US" -> EmissionsTier == "Tier 4 Final", "US requires Tier 4
Final emissions");
}
```

Example Description and Configurator Result

In example 4, the `EmissionsTier` is specified with configurable set to `false`, and `peelable` is not specified (defaulting to `false`). As a result, the system considers the default value "`Tier 3`" as a hard constraint that cannot be overridden. If the user updates the Location variable to "`US`", the constraint engine attempts to enforce "`Tier 4 Final`" but finds it cannot overwrite the fixed "`Tier 3`" value. Consequently, the system displays a conflict error message stating "`US requires Tier 4 Final emissions`," and the configuration enters an invalid state because the user cannot manually change the EmissionsTier to resolve it.

Example 5: Auto-Correcting User Input (`configurable = true`, `peelable = true`)

In this example, the `EnclosureType` is user-selectable (`configurable = true`) and defaults to "Standard". It is marked with `peelable = true`. This allows the engine to override even an explicit user selection if a subsequent choice makes the enclosure invalid.

```
type GeneratorAccessories {
    /*
     * configurable=true: User explicitly selects 'Standard'.
     * peelable=true: Grants permission to override the User's selection
     * to resolve conflicts with the Environment.
     */
    @(defaultValue = "Standard", configurable = true, peelable = true)
    string EnclosureType = ["Standard", "Heated", "Reinforced"];

    string Environment = ["Indoor", "Arctic"];

    // Constraint: Arctic requires Heated Enclosure
    constraint(Environment == "Arctic" -> EnclosureType == "Heated");
}
```

Example Description and Configurator Result

In example 5, the `EnclosureType` is specified with `configurable` set to `true` and `peelable` set to `true`. Initially, the system allows the user to confirm the selection of "`Standard`". If the user subsequently updates the `Environment` variable to "`Arctic`", the system detects that "`Standard`" is invalid. Typically, the engine protects user selections and would throw an error. However, because `EnclosureType` is peelable, the system treats the user's choice as a "soft pick." It automatically "peels" the "`Standard`" selection and changes it to "`Heated`" to satisfy the logic. The user sees their previous selection update automatically to match the new environment requirement, preventing a "dead end" configuration state.

Example 6: Upstream Correction (`sequence`, `peelable = true`)

In this example, the `Voltage` is configured early in the process (Sequence 1) with a default value of "`Low`". It is marked with `peelable = true` to allow downstream selections to override this initial setting.

```
type GeneratorSet {
    /*
     * sequence=1: Evaluated first. Defaults to 'Low'.
     * peelable=true: Allows 'Application' (seq=2) to force a change to this value.
     */
    @(defaultValue = "Low", sequence = 1, peelable = true)
    string Voltage = ["Low", "Medium", "High"];

    /*
     * sequence=2: Evaluated after Voltage is set.
```

```
     * The user picks 'DataCenter', which requires High Voltage.
     */
    @(sequence = 2)
    string Application = ["Residential", "DataCenter"];

    // Constraint: DataCenter application forces High Voltage
    constraint(Application == "DataCenter" -> Voltage == "High");
}
```

Example Description and Configurator Result

In example 6, the `Voltage` attribute is specified with `sequence` set to `1` and `peelable` set to `true`. As a result, the system initially sets the value to "`Low`" before evaluating the `Application` attribute. If the user updates the `Application` variable to "`DataCenter`" (which runs at sequence 2), the system detects a conflict between the established "`Low`" voltage and the constraint requirement for "`High`" voltage. Because `Voltage` is peelable, the system automatically "peels" the default "`Low`" value and changes it to "`High`" to satisfy the constraint. The transition happens silently without a conflict error, effectively allowing a later user choice to correct an earlier default assumption.

Example 7: Guided Fallback ('strategy`, `peelable = true`)

In this example, the `ServiceTier` defaults to "`Bronze`" (the lowest value). It is marked with `peelable = true` and `strategy = "descending"`. If a constraint forces a change from the default, the strategy directs the engine to try the highest possible valid values first.

```
type ServicePlan {
    /*
     * defaultValue="Bronze": Start cheap.
     * peelable=true: Allow upgrade if needed.
     * strategy="descending": If peeled, try 'Platinum' next (Max value),
     * instead of creeping up to 'Silver'.
     */
    @(defaultValue = "Bronze", strategy = "descending", peelable = true)
    string ServiceTier = ["Bronze", "Silver", "Gold", "Platinum"];

    int UserCount = [1..1000];

    // Constraint: > 100 users requires Premium tiers (Gold or Platinum)
    constraint(UserCount > 100 -> ServiceTier in ["Gold", "Platinum"]);
}
```

Example Description and Configurator Result

In example 7, the `ServiceTier` is specified with `strategy` set to "`descending`" and `peelable` set to `true`. Initially, the system sets the value to "`Bronze`". If the user updates the `UserCount` variable to 150, the constraint requires `ServiceTier` to be either "`Gold`" or "`Platinum`". Because the variable is peelable, the system removes the "`Bronze`" selection. Instead of testing the next closest value ("`Silver`"), the "`descending`" strategy instructs the engine to attempt resolution starting from the top of the domain. Consequently, the system automatically selects "`Platinum`" (the highest valid option) rather than "`Gold`". The user sees the plan upgrade immediately to the highest tier without an error message.

productField Annotation

productField is a CML annotation that defines the Product2 field on a variable. productField loads the value from Product Catalog Management (PCM) during constraint model activation.

Annotation	`productfield`
Applicable to	Variable
Value Type/Values	Literal (case sensitive)
Description	Used to load the value from the corresponding Product2 field defined in Product Catalog Management(PCM). Defined under either a type or supertype. If defined under a supertype, the types, which inherit from the supertype, load the Product2 field value for the corresponding product. Supports a maximum of 50 Product2 fields. Loads product field values for a maximum of 20,000 products. Read-only. Doesn't support a null value.

Example 1

In this example, "TestNumber__c" is a custom field defined on the Product2 object. The constraint rules engine loads the value of "TestNumber__c" for the Laptop product during constraint model activation.

```
type Laptop {
    @(productField = "TestNumber__c")
    int TestNumber;
}
```

Example 2

In this example, the product field variable "productName" is defined under a supertype "LineItem". Any type that inherits the supertype loads the value of "Name" for the corresponding product during constraint model activation.

```
type LineItem {
    @(productField = "Name")
    string productName;
}

type Laptop : LineItem;
type Mouse : LineItem;
}
```

Example 3

In this example, the product field variable "productName" is defined under the parent type "LaptopProBundle". "Laptop" is a child of "LaptopProBundle", and it can access the "productCode" variable from its parent by using the parent() function.

```
type LaptopProBundle {
    @(productField = "ProductCode")
    string productCode;

    relation laptop : Laptop;
}

type Laptop {
    string parentProductCode = parent(productCode);
}
```

propagateUp Annotation

`propagateUp` is a Constraint Modeling Language (CML) annotation that controls aggregation propagation between children and parent elements.

Annotation	`propagateUp`
Applicable to	Relationship
Value Type/Values	true, false
Description	If propagateUp is not specified, the engine sets it implicitly as false for the relationship. • If the propagateUp annotation is specified as true, the engine aggregates values from children to parent elements (upward propagation). The engine cannot modify this value from the parent level (e.g. via constraint), so the children relation domain will not be affected. • If the propagateUp is specified as false, both upward and downward propagations are applicable. The engine aggregates values from children to parent elements (upward propagation). Meanwhile the engine can modify this value (e.g. via constraint) from the parent level. The value is propagated downward and might affect the relation domain (downward propagation).

Example 1

In this example, the `propagateUp` annotation is not specified for the relationship (`warranties`). The model contains technical variable (`coverageDays`) and constraint.

```
type GeneratorSet {

        relation warranties : Warranty {
        totalDays = sum(coverageDays);
    }
```

```
    constraint(warranties.totalDays <= 299);
}

type Warranty {
    int coverageDays = 100;
}

type Warranty_PRP : Warranty;

type Warranty_DCC : Warranty;

type Warranty_ESP : Warranty;
```

Example 2

In this example, the `propagateUp` annotation is defined as `false` for the relationship (`warranties`). The model contains technical variable (coverageDays) and constraint.

```
type GeneratorSet {

        @(propagateUp = false)
        relation warranties : Warranty {
        totalDays = sum(coverageDays);
    }
    constraint(warranties.totalDays <= 299);
}

type Warranty {
    int coverageDays = 100;
}

type Warranty_PRP : Warranty;

type Warranty_DCC : Warranty;

type Warranty_ESP : Warranty;
```

Example Description and Configurator Result

In example 1, the `warranties` relation is not specified with the propagateUp annotation, and the system considers it as `false` by default. In example 2, the relation is explicitly defined with a `false` annotation value. Each product of the `warranties` relation is assigned with the technical `coverageDays` variable equalled to 100. The `totalDays` calculates the `coverageDays` sum of selected `warranties` products. As a result, the system validates the constraint and relation domain in the following way:

- The user selects one `warranties` product (e.g. `Warranty_PRP`). The system calculates the `totalDays` aggregated value for the relation (equal to 100). The system validates the constraint on the parent based on the calculated aggregation variable. The constraint is satisfied (100<= 299). The system also verifies the possibility to add extra products to the relation. The capacity of the constraint allows to add one more product to the relation, so the system will not reduce the relation domain.
- The user selects the second `warranties` product (e.g. `Warranty_DCC`). The system calculates the `totalDays` aggregated value for the relation (equal to 200). The system validates the constraint on the parent based on the calculated aggregation variable. The constraint is satisfied (200<= 299). The system verifies the possibility to add extra products to the relation. The capacity of the constraint (300<= 299) does not allow to add one more product to the relation, so the system will reduce the relation domain.

- The user unselects one of the selected `warranties` products (e.g. `Warranty_PRP`). The system calculates the `totalDays` aggregated value for the relation (equal to 100). The system validates the constraint on the parent based on the calculated aggregation variable. The constraint is satisfied (100<= 299). The system verifies the possibility to add extra products to the relation. The capacity of the constraint allows to add one more product to the relation, so the system will not reduce the relation domain (both `Warranty_DCC` and `Warranty_ESP` products are available for selection).

Example 3

In this example, the `propagateUp` annotation is specified for the relationship (`warranties`) as `true`. The model contains technical variable (`coverageDays`) and constraint.

```
type GeneratorSet {

        @(propagateUp = true)
        relation warranties : Warranty {
        totalDays = sum(coverageDays);
    }
    constraint(warranties.totalDays <= 299);
}

type Warranty {
    int coverageDays = 100;
}

type Warranty_PRP : Warranty;

type Warranty_DCC : Warranty;

type Warranty_ESP : Warranty;
```

Example Description and Configurator Result

In example 3, the relation is specified with `propagateUp` annotation as `true` value. Each product of the `warranties` relation is assigned with the technical `coverageDays` variable equalled to 100. The `totalDays` calculates the `coverageDays` sum of selected `warranties` products. As a result, the system validates the constraint and relation domain in the following way:

The user selects one `warranties` product (e.g. `Warranty_PRP`). The system calculates the `totalDays` aggregated value for the relation (equal to 100). The system validates the constraint on the parent based on the calculated aggregation variable. The constraint is satisfied (100<=299);

The user selects the second `warranties` product (e.g. `Warranty_DCC`). The system calculates the `totalDays` aggregated value for the relation (equal to 200). The system validates the constraint on the parent based on the calculated aggregation variable. The constraint is satisfied (200<=299);

The user selects the third `warranties` product (e.g. `Warranty_ESP`). The system calculates the `totalDays` aggregated value for the relation (equal to 300). The system validates the constraint on the parent based on the calculated aggregation variable. The constraint is not satisfied (300<=299). The system rejects the selection of one of the warranties products to satisfy the constraint.

As a best practice, use the `propagateUp = true` for models with few relationships and rare constraint violations, since it minimizes propagation work and avoids extra domain updates. Meanwhile, set the `propagateUp = false` for large product structures or frequent violations, since bidirectional propagation prevents invalid states early and reduces expensive backtracking.

propagateUp Configuration Settings

Table 12: propagateUp Configuration Settings

Associated Example	Product Group Structure	Applicable to	"propagateUp" annotation	User Action	Engine Action	UI Behavior
Example 1	Individual product	Relationship	not specified	Selects or deselects the products	Calculates the aggregation variable value on the relation, validation of the constraint, limitation of the relation domain when the constraint capacity is used up	Displays the selected products. When constraint capacity is used up, in addition to the selected products the limited list of the available to be added products will be displayed(if any)
Example 2	Individual product	Relationship	FALSE	Selects or deselects the products	Calculates the aggregation variable value on the relation, validation of the constraint, limitation of the relation domain when the constraint capacity is used up	Displays the selected products. When constraint capacity is used up, in addition to the selected products the limited list of the available to be added products will be displayed(if any)
Example 3	Individual product	Relationship	TRUE	Selects or deselects the products	Calculates the aggregation variable value on the relation, validation of the constraint, rejection of the user selection when the constraint validation is failed	Displays the selected products. No limitations for the list of available products
N/A	Product Classification	Relationship	not specified	Selects or deselects the products	Calculates the aggregation variable value on the relation, validation of the	Displays the selected products. No limitations for the list of available

Associated Example	Product Group Structure	Applicable to	"propagateUp" annotation	User Action	Engine Action	UI Behavior
					constraint. The engine might replace one of the previously selected products with a newly selected	products in the browse popup for the classification
N/A	Product Classification	Relationship	FALSE	Selects or deselects the products	Calculates the aggregation variable value on the relation, validation of the constraint. The engine might replace one of the previously selected products with a newly selected	Displays the selected products. No limitations for the list of available products in the browse popup for the classification
N/A	Product Classification	Relationship	TRUE	Selects or deselects the products	Calculates the aggregation variable value on the relation, validation of the constraint, rejection of the user selection when the constraint validation is failed	Displays the selected products. No limitations for the list of available products in the browse popup for the classification

relatedAttributes Annotation

`relatedAttributes` is a Constraint Modeling Language (CML) annotation that resets the domain to the original one for domainComputation.

`relatedAttributes` annotation specification: Variable

Annotation	`relatedAttributes`
Applicable to	Variable and Relationship
Value Type/Values	String
Description	For Variables

Annotation	relatedAttributes
	If `relatedAttributes` annotation is not specified, the engine updates the variable domain according to domainComputation and constraint logic. If the `relatedAttributes` annotation is specified with one or multiple values (separated by comma), the variable domain is reset to the original domain. For relationships If `domainComputation` is not explicitly specified, the engine sets it implicitly as true for the relationship. If the `domainComputation` is specified as `true`, the relationship domain is dynamically determined based on the configuration and constraint logic. If the `domainComputation` is specified as `false`, the relationship domain is fixed.

Example 1

In this example, the `relatedAttributes` annotation is not specified for the variables (`controlLanguage`, `controlPlacement`, `commissioningScope`) of the `Control` type. The `domainComputation` annotation is defined for the variables. The model contains the constraints.

```
type GeneratorSet {
    relation controls : Control;
}

type Control {
    @(domainComputation = true, defaultValue = "English")
    string controlLanguage = ["English", "Danish", "French"];

    @(domainComputation = true, defaultValue = "Left")
    string controlPlacement = ["Left", "Right", "Top"];

    @(domainComputation = true, defaultValue = "None")
    string commissioningScope = ["None", "Remote Support", "On-site Commissioning"];

    constraint(controlPlacement == "Right" -> commissioningScope != "Remote Support");

   constraint(controlLanguage == "Danish" -> commissioningScope != "On-site Commissioning");
```

Example Description and Configurator Result

In example 1, the variables of the `Control` type are not specified with the `relatedAttributes` annotation, but defined with the `domainComputation` annotation. As a result, the system updates the variable domain based on the constraint logic.

If the `controlPlacement` is selected with the "`Right`" value, the `commissioningScope` variable domain is updated to exclude the "`Remote Support`" value;

If the `controlLanguage` is selected with the "`Danish`" value, the `commissioningScope` variable domain is updated to exclude the "`On-site Commissioning`" value.

Example 2

In this example, the `relatedAttributes` annotation is specified for the variable with one value. The `domainComputation` annotation is defined for the variables. The model contains the constraints.

```
type GeneratorSet {
    relation controls : Control;
}

type Control {
    @(domainComputation = true, defaultValue = "English")
    string controlLanguage = ["English", "Danish", "French"];

    @(domainComputation = true, defaultValue = "Left")
    string controlPlacement = ["Left", "Right", "Top"];

     @(domainComputation = true, defaultValue = "None", relatedAttributes =
"controlPlacement")
    string commissioningScope = ["None", "Remote Support", "On-site Commissioning"];

    constraint(controlPlacement == "Right" -> commissioningScope != "Remote Support");

   constraint(controlLanguage == "Danish" -> commissioningScope != "On-site Commissioning");
}
```

Example 3

In this example, the `relatedAttributes` annotation is specified for the variable with one value. The `domainComputation` annotation is defined for the variables. The model contains the constraints.

```
type GeneratorSet {
    relation controls : Control;
}

type Control {
    @(domainComputation = true, defaultValue = "English")
    string controlLanguage = ["English", "Danish", "French"];

    @(domainComputation = true, defaultValue = "Left")
    string controlPlacement = ["Left", "Right", "Top"];

   @(domainComputation = true, defaultValue = "None", relatedAttributes = "controlLanguage")

    string commissioningScope = ["None", "Remote Support", "On-site Commissioning"];

    constraint(controlPlacement == "Right" -> commissioningScope != "Remote Support");

   constraint(controlLanguage == "Danish" -> commissioningScope != "On-site Commissioning");
}
```

Example Description and Configurator Result

In Example 2, the `relatedAttributes` annotation is defined with the "`controlPlacement`" value for the `commissioningScope` variable. As a result, the system controls the domain for the `commissioningScope` differently depending on `controlPlacement` or `controlLanguage` variable selection.

If the `controlPlacement` is selected with the "`Right`" value, the `commissioningScope` variable domain remains unchanged, because the `controlPlacement` is listed in the `relatedAttributes` annotation.

If the controlLanguage is selected with the “`Danish`" value, the `commissioningScope` variable domain is updated to exclude the "`On-site Commissioning`" value according to the constraint, because the `controlLanguage` is not listed in the `relatedAttributes` annotation.

In example 3, the `relatedAttributes` annotation is defined with the "`controlLanguage`" value for the `commissioningScope` variable. As a result, the system controls the domain for the `commissioningScope` differently depending on `controlPlacement` or `controlLanguage` variable selection.

If the `controlPlacement` is selected with the "`Right`" value, the `commissioningScope` variable domain is updated to exclude the "`Remote` Support" value according to the constraint, because the `controlPlacement` is not listed in the `relatedAttributes` annotation.

If the `controlLanguage` is selected with the "`Danish`" value, the `commissioningScope` variable domain remains unchanged, because the `controlLanguage` is listed in the `relatedAttributes` annotation.

Example 4

In this example, the `relatedAttributes` annotation is specified for the variable with several values (separated by comma). The domainComputation annotation is defined for the variables. The model contains the constraints.

```
type GeneratorSet {
    relation controls : Control;
}

type Control {
    @(domainComputation = true, defaultValue = "English")
    string controlLanguage = ["English", "Danish", "French"];

    @(domainComputation = true, defaultValue = "Left")
    string controlPlacement = ["Left", "Right", "Top"];

   @(domainComputation = true, defaultValue = "None", relatedAttributes = "controlPlacement,
 controlLanguage")
    string commissioningScope = ["None", "Remote Support", "On-site Commissioning"];

    constraint(controlPlacement == "Right" -> commissioningScope != "Remote Support");

   constraint(controlLanguage == "Danish" -> commissioningScope != "On-site Commissioning");
}
```

Example Description and Configurator Result

In example 4, the `relatedAttributes` annotation is defined with the "`controlPlacement`" and "`controlLanguage`" values for the commissioningScope variable. As a result, the system remains the `commissioningScope` variable domain unchanged.

If the `controlPlacement` is selected with the "`Right`" value, the `commissioningScope` variable domain remains unchanged, because the `controlPlacement` is listed in the `relatedAttributes` annotation.

If the controlLanguage is selected with the "Danish" value, the commissioningScope variable domain remains unchanged, because the controlLanguage is listed in the relatedAttributes annotation.

Example 5

In this example, the relatedAttributes annotation is not specified for the relationship (temperatureSensors). The domainComputation annotation is defined for the relationship. The model contains the constraints.

```
type GeneratorSet {
    @(defaultValue = "Emergency Standby Power (ESP)")
    string dutyRating = ["Emergency Standby Power (ESP)", "Prime Power (PRP)", "Continuous
 Power (COP)", "Data Center Continuous (DCC)"];

    @(domainComputation = true)
    relation temperatureSensors : TemperatureSensor[0..1];

    constraint(dutyRating == "Continuous Power (COP)" ->
temperatureSensors[BearingTemperatureSensor] == 0);

    constraint(dutyRating == "Data Center Continuous (DCC)" ->
temperatureSensors[StatorTemperatureSensor] == 0);
}

type TemperatureSensor;

type BearingTemperatureSensor : TemperatureSensor;

type StatorTemperatureSensor : TemperatureSensor;
```

Example Description and Configurator Result

In example 5, the temperatureSensors relation is not specified with the relatedAttributes annotation, but defined with the domainComputation as true. As a result, the system updates the relationship domain based on the constraint logic.

If the dutyRating is selected with the "Continuous Power (COP)" value, the system re-computes the relationship domain and excludes the BearingTemperatureSensor product to satisfy the constraint. The StatorTemperatureSensor product remains and can be selected in scope of the GeneratorSet bundle product.

If the dutyRating is selected with the "Data Center Continuous (DCC)" value, the system re-computes the relationship domain and excludes the StatorTemperatureSensor product to satisfy the constraint. The BearingTemperatureSensor product remains and can be selected in scope of the GeneratorSet bundle product.

Example 6

In this example, the relatedAttributes annotation is specified for the relationship (temperatureSensors). The domainComputation annotation is defined for the relationship. The model contains the constraints.

```
type GeneratorSet {
    @(defaultValue = "Emergency Standby Power (ESP)")
    string dutyRating = ["Emergency Standby Power (ESP)", "Prime Power (PRP)", "Continuous
 Power (COP)", "Data Center Continuous (DCC)"];

    @(domainComputation = true, relatedAttributes = 'dutyRating')
    relation temperatureSensors : TemperatureSensor[0..1];
```

```
    constraint(dutyRating == "Continuous Power (COP)" ->
temperatureSensors[BearingTemperatureSensor] == 0);

    constraint(dutyRating == "Data Center Continuous (DCC)" ->
temperatureSensors[StatorTemperatureSensor] == 0);
}

type TemperatureSensor;

type BearingTemperatureSensor : TemperatureSensor;

type StatorTemperatureSensor : TemperatureSensor;
```

Example Description and Configurator Result

In example 6, the `temperatureSensors` relation is specified with the `relatedAttributes` annotation containing the `dutyRating` value, the `domainComputation` annotation is defined as `true`. As a result, the system remains the relationship domain unchanged and executes the constraints.

If the `dutyRating` is selected with the "`Continuous Power (COP)`" value, the system displays both `BearingTemperatureSensor` and `StatorTemperatureSensor` products in scope of the `GeneratorSet` bundle product. The `BearingTemperatureSensor` product cannot be selected based on the constraint.

If the `dutyRating` is selected with the "`Data Center Continuous (DCC)`" value, the system displays both `BearingTemperatureSensor` and `StatorTemperatureSensor` products in scope of the `GeneratorSet` bundle product. The `StatorTemperatureSensor` product cannot be selected based on the constraint.

relatedAttributes Configuration Settings

Table 13: relatedAttributes Configuration Settings

Associated Example	Product Group Structure	Applicable to	"relatedAttributes" annotation	"domainComputation" annotation	User Action	Engine Action	UI Behavior
Example 1	Individual product	Variable	not specified	TRUE	Change variable value specified in constraint condition	Update the variable (specified with domainComputation) domain based on domainComputation and constraint logic	Display updated variable domain
Example 2 Example 3	Individual product	Variable	specified (with one variable)	TRUE	Change variable value specified in constraint condition and relatedAttributes	Reset variable (specified in relatedAttributes) domain to the original domain	Display original variable domain

Associated Example	Product Group Structure	Applicable to	"relatedAttributes" annotation	"domainComputation" annotation	User Action	Engine Action	UI Behavior
Example 4	Individual product	Variable	specified (with several variables)	TRUE	Change variable values specified in constraint conditions and relatedAttributes	Reset variable (specified in relatedAttributes) domain to the original domain	Display original variable domain
Example 5	Individual product	Relationship	not specified	TRUE	Change variable value specified in constraint condition	Update the relationship (specified with domainComputation) domain based on domainComputation and constraint logic	Display updated relationship domain
Example 6	Individual product	Relationship	specified	TRUE	Change variable value specified in constraint condition and relatedAttributes	Reset relationship domain to the original domain	Display original relationship domain (products). If the product is allowed according to constraint: the product can be selected If the product is not allowed according to constraint: selected product returns back to unselected state
N/A	Product Classification	Relationship	not specified	TRUE	• Change variable value specified in constraint condition > Add products on browse	Update the relationship (specified with domainComputation) domain based on domainComputation	Display updated relationship domain. If the variable is changed again by the User, the value is

Associated Example	Product Group Structure	Applicable to	"relatedAttributes" annotation	"domainComputation" annotation	User Action	Engine Action	UI Behavior
					popup from product classification • Add products on browse popup from product classification > Change variable value specified in constraint condition	and constraint logic	reverted to the initial value
N/A	Product Classification	Relationship	specified	TRUE	• Change variable value specified in constraint condition > Add products on browse popup from product classification •Add products on browse popup from product classification > Change variable value specified in constraint condition	• If the variable of constraint condition is changed first: update the relationship (specified with domainComputation) domain based on constraint logic • if products are added first from browse popup: remain the relationship domain unchanged	• If the variable of constraint condition is changed first: display updated relationship domain. If the variable is changed again by the user, the value is reverted to the initial value • if products are added first from browse popup: display added products as selected. If constraint variable is changed again by the user, the value is reverted to the initial value

sequence Annotation

The `sequence` annotation defines the execution and configuration order of elements in a Constraint Modeling Language (CML) model.

`sequence` annotation specification: Variable and Relationship

Annotation	`sequence`
Applicable to	Variable and Relationship
Value Type/Values	integer Example: 1, 2, 10
Description	If a sequence value is not explicitly defined, the configurator implicitly determines the order based on the variable or relationship declaration order in the CML model. If a sequence value is explicitly defined, the configurator uses the sequence number to control the order in which variables or relationships are configured. Variables or Relationships with lower sequence values are assigned first.

Example 1

In this example, the `sequence` annotation is not explicitly specified. The `defaultValue` is defined for 3 variables (`controlPlacement`, `commissioningScope`, and `controlLanguage`) of the `Control` products. The model contains a constraint.

```
type GeneratorSet {
    relation controls : Control[1..999999];
}

type Control {
 @(defaultValue = "Left")
 string controlPlacement = ["Left", "Right", "Top"];

 @(defaultValue = "None")
 string commissioningScope = ["None", "Remote Support", "On-site Commissioning"];

 @(defaultValue = "English")
 string controlLanguage = ["English", "Danish", "French"];

 constraint(controlPlacement == "Left" && controlLanguage == "English" -> commissioningScope
 == "Remote Support");
}
```

Example Description and Configurator Result

The model defines 3 variables with `defaultValue` annotation.

```
controlPlacement = "Left"
commissioningScope = "None"
controlLanguage = "English"
```

Even when sequence is not explicitly specified for the variables, the engine assigns it implicitly based on the variable declaration order in the CML type. For this example, the sequence for the variables is:

```
controlPlacement: sequence = 1 (highest priority)
commissioningScope: sequence = 2
controlLanguage: sequence = 3 (lowest priority)
```

The constraint specifies that a type with `controlPlacement == "Left"` and `controlLanguage == "English"` will have a `commissioningScope == "Remote Support":`.

```
constraint(controlPlacement == "Left" && controlLanguage == "English" -> commissioningScope
 == "Remote Support");
```

For any constraint, the engine enforces logical equivalence between the left-hand side (before ->) and the right-hand side (after ->): if the left side evaluates to true, the right side must also be true (and vice versa).

In this example, the engine resolves the true -> false constraint (to be false -> false): it modifies the controlLanguage with the highest sequence.

As a result, the Product Configurator will have next values for the variables:

```
controlPlacement = "Left"
commissioningScope = "None"
controlLanguage = "Danish"
```

Example 2

In this example, the `sequence` annotation is explicitly specified and the defaultValue is defined for 3 variables (`controlPlacement`, `controlLanguage`, `commissioningScope`) of the `Control` products. The model contains a constraint.

```
type GeneratorSet {
    relation controls : Control[1..999999];
}

type Control{

@(defaultValue = "Left", sequence = 1)
string controlPlacement = ["Left", "Right", "Top"];

@(defaultValue = "English", sequence = 3)
string controlLanguage = ["English", "Danish", "French"];

@(defaultValue = "None", sequence = 2)
string commissioningScope = ["None", "Remote Support", "On-site Commissioning"];

constraint(controlPlacement == "Left" && controlLanguage == "English" -> commissioningScope
 == "Remote Support");
}
```

Example Description and Configurator Result

This example is similar to Example 1, but the `sequence` annotation is specified explicitly for the variables to control solver priority:

```
controlPlacement: sequence = 1 (highest priority)
controlLanguage: sequence = 3 (lowest priority)
commissioningScope: sequence = 2
```

The constraint specifies that a type with `controlPlacement == "Left"` and `controlLanguage == "English"` will have a `commissioningScope == "Remote Support":`.

```
constraint(controlPlacement == "Left" && controlLanguage == "English" -> commissioningScope
 == "Remote Support");
```

- Following the same logical equivalence for constraints described in Example 1 and considering the default values specified in the Example 2, the left side of the constraint is true (`controlPlacement = "Left"` and `controlLanguage = "English"`), while the right side is false (`commissioningScope = "None"`).
- Because the `commissioningScope` has higher priority than the `controlLanguage`, the engine avoids changing the `commissioningScope` and instead resolves the constraint by making the left side false.
- The `controlPlacement` has the highest priority sequence than the `controlLanguage`, so the engine modifies the controlLanguage as the most efficient way to satisfy the constraint.

Available values for the `controlLanguage: ["English", "Danish", "French"]`. To make the left side of the constraint false, the engine selects the first non-matching value (specified in CML domain): `"Danish"`.

As a result, the Product Configurator will have next values for the variables.

```
controlPlacement = "Left"
controlLanguage = "Danish"
commissioningScope = "None"
```

Example 3

In this example, the `sequence` annotation is explicitly specified and the defaultValue is defined for 3 variables (`controlPlacement`, `controlLanguage`, and `commissioningScope`) of the `Control` products. The model contains a constraint.

```
type GeneratorSet {
    relation controls : Control[1..999999];
}

type Control{

@(defaultValue = "Left", sequence = 1)
string controlPlacement = ["Left", "Right", "Top"];

@(defaultValue = "English", sequence = 2)
string controlLanguage = ["English", "Danish", "French"];

@(defaultValue = "None", sequence = 3)
string commissioningScope = ["None", "Remote Support", "On-site Commissioning"];

constraint(controlPlacement == "Left" && controlLanguage == "English" -> commissioningScope
 == "Remote Support");
}
```

Example Description and Configurator Result

This example is similar to Example 2, but the lowest priority sequence is specified for the `commissioningScope`. This is a trigger for the engine to modify the `commissioningScope` first to resolve the true -> false constraint to be true -> true).

As a result, the Product Configurator will have next values for the variables:

```
controlPlacement = "Left"
controlLanguage = "English"
commissioningScope = "Remote Support"
```

Example 4

In this example, the sequence annotation is not explicitly specified for 3 relations (`voltageConnections`, `controls`, and `alternators`). The model contains a constraint.

```
type GeneratorSet {
    relation voltageConnections : VoltageConnection[0..999999];
    relation controls : Control[1..999999];
    relation alternators : Alternator[1..999999];

    int alternatorsQty = controls[Control] + voltageConnections[VoltageConnection];
    constraint(alternators[Alternator] > 0 -> alternators[Alternator] == alternatorsQty,
"alternators[Alternator] = controls[Control] + voltageConnections[VoltageConnection]");
}

type Alternator;

type VoltageConnection;

type Control;
```

Example Description and Configurator Result

Even when sequence is not explicitly specified for the relations, the engine sets it implicitly based on the relation declaration order in the CML type. For this example, the sequence for the relations is:

```
relation voltageConnections: sequence = 1 (highest priority)
relation controls: sequence = 2
relation alternators: sequence = 3 (lowest priority)
```

Note: Use mindful sequencing in CML for variables and relations to avoid backtracking issues. Define them in the order you expect the engine to modify them during constraint resolution.

In this example, the CML model works without conflicts because the relations are implicitly defined with the correct sequence. The `voltageConnections` and controls relations are evaluated first, then the `alternatorsQty` is calculated from their quantities. Finally, the `alternators` relation is adjusted based on that result. Because the `alternators` is processed last, the engine avoids backtracking and errors.

Example 5

In this example, the `sequence` annotation is explicitly specified for 3 relations (`voltageConnections`, `controls`, and `alternators`). The model contains a constraint.

```
type GeneratorSet {
 @(sequence=3)
    relation alternators : Alternator[1..999999];
    @(sequence=1)
    relation voltageConnections : VoltageConnection[0..999999];
```

```
 @(sequence=2)
    relation controls : Control[1..999999];

    int alternatorsQty = controls[Control] + voltageConnections[VoltageConnection];
    constraint(alternators[Alternator] > 0 -> alternators[Alternator] == alternatorsQty,
"alternators[Alternator] = controls[Control] + voltageConnections[VoltageConnection]");
}
type Alternator;

type VoltageConnection;

type Control;
```

Example Description and Configurator Result

This example is similar to Example 4, but the `sequence` is explicitly specified for the relations to control the processing priority and ensure the engine evaluates and modifies them in the correct order during configuration.

sequence Configuration Settings

Table 14: sequence Configuration Settings

Associated Example	Product Group Structure	Applicable to	"sequence" annotation	User Action	Engine Action	UI Behavior
Example 1	N/A	Variable	not specified	Configure the product containing variables	Assign the sequence for the variables based on the variable declaration order in the type. Enforce logical equivalence between left-hand side and right-hand side for the constraint by modifying the variable with the highest sequence	Display appropriate variable values based on sequence and constraint logic
Example 2 Example 3	N/A	Variable	specified	Configure the product containing variables	Assign the sequence for the variables based on defined sequence annotation. Enforce logical equivalence between	Display appropriate variable values based on sequence and constraint logic

Associated Example	Product Group Structure	Applicable to	"sequence" annotation	User Action	Engine Action	UI Behavior
					left-hand side and right-hand side for the constraint by modifying the variable with the highest sequence	
Example 4	Individual product	Relationship	not specified	Configure the bundle product containing relationship products	Assign the sequence for the relationships based on the relation declaration order in the type. Execute the constraint (according to assigned sequence)	Display adjusted relationships based on the relationship sequence and constraint logic
Example 5	Individual Product	Relationship	specified	Configure the bundle product containing relationship products	Assign the sequence for the relationships based on defined sequence annotation. Execute the constraint (according to specified sequence)	Display adjusted relationships based on the relationship sequence and constraint logic
N/A	Product Classification	Relationship	not specified	Configure the bundle product containing relationship products	Assign the sequence for the relationships based on the relation declaration order in the type. Execute the constraint (according to assigned sequence)	Display adjusted relationships based on the relationship sequence and constraint logic
N/A	Product Classification	Relationship	specified	Configure the bundle product	Assign the sequence for the	Display adjusted relationships

Associated Example	Product Group Structure	Applicable to	"sequence" annotation	User Action	Engine Action	UI Behavior
				containing relationship products	relationships based on defined sequence annotation. Execute the constraint (according to specified sequence)	based on the relationship sequence and constraint logic

split Annotation

`split` is a Constraint Modeling Language (CML) annotation that specifies whether the instances of the type should be split or not.

Annotation	`split`
Applicable to	Type
Value Type/Values	true, false
Description	If split isn't specified, there are multiple instances of the same type in the relationship with different quantities. If the split is specified as `true`, there can be multiple instances of the type, and the quantity of each instance is always 1. If the split is specified as `false`, there is only one instance in the relationship. If the user adds more instances, the engine adds more quantity to the existing instance.

Example 1

In this example, the split annotation isn't specified for the type (Model).

```
type FESBAGeneratorSet {
    relation models : Model;
}
type Model;
```

Example Description and Configurator Result

In this example, the system allows the multiple instances of the Model type with different quantities defined by the user.

Example 2

In this example, the split annotation is specified as `true` for the type (Model).

```
type FESBAGeneratorSet {
    relation models : Model;
}
@(split = true)
type Model;
```

Example Description and Configurator Result

In this example, the annotation `split = true` specifies that the multiple Model instances are split into individual items with fixed quantity 1.

Example 3

In this example, the split annotation is specified as `false` for the type (Model).

```
type FESBAGeneratorSet {
    relation models : Model;
}
@(split = false)
type Model;
```

Use `@(split=false)` when the line item record must persist across configuration changes and record continuity matters for your business process. The `@(split=false)` annotation prevents the configurator from deleting and recreating existing line item records when the quantity or selection changes. When you apply `@(split=false)`, the configurator updates the quantity on the existing record instead of replacing it with a new one. This preserves the record ID and any data associated with it.

Example Description and Configurator Result

In this example, the system allows one instance of the Model type with different quantity defined by the user.

Note: If there are multiple products of type Model (for example, `FESBA 900kW` and `FESBA 1500kW`), the user can add both models to the configuration and adjust the quantity for each one. However, the user can't add additional separate instances of the same product (for example, to configure unique instances with different variable values).

split Configuration Settings

Table 15: split Configuration Settings

Product Group Structure	"split" annotation	Engine Action	user Action	UI Behavior
Product Classification	TRUE	Add multiple products	N/A	N/A
Product Classification	TRUE	N/A	Selecting or adding the product multiple times manually	Multiple instances created (qty = 1 each). User can add more instances, but can't change quantity per instance.

Product Group Structure	"split" annotation	Engine Action	user Action	UI Behavior
Product Classification	FALSE	Add multiple products	N/A	One instance created with qty > 1. User can change the quantity of this instance, but can't add more instances.
Product Classification	FALSE	N/A	Selecting or adding the product multiple times manually	One instance created with qty = 1. User can change the quantity of this instance, but can't add more instances.
Product Classification	(empty)	Add multiple products	N/A	One instance created with qty > 1. User can change quantity and can add more instances.
Product Classification	(empty)	N/A	Selecting or adding the product multiple times manually	One instance created with qty = 1. User can change quantity and can add more instances.
Individual Product	TRUE	Add multiple products	N/A	Multiple line items created, but Configurator displays only one instance. Users cannot modify quantity for that instance.
Individual Product	TRUE	N/A	Selecting or adding the product manually	One instance created with qty = 1. Users cannot change quantity.
Individual Product	FALSE	Add multiple products		One instance created with qty > 1. Users can't modify quantity for that instance.
Individual Product	FALSE	N/A	Selecting or adding the product manually	One instance created with qty = 1. Users can modify quantity for that instance.
Individual Product	(empty)	Add multiple products		One instance created with qty > 1. Users can't modify quantity for that instance.
Individual Product	(empty)	N/A	Selecting or adding the product manually	One instance created with qty = 1. Users can

Product Group Structure	"split" annotation	Engine Action	user Action	UI Behavior
				modify quantity for that instance.

Constraint Modeling Language (CML) Best Practices

To prevent performance degradation or unexpected behaviors when the constraint engine executes CML code, follow these practices when writing code.

For tips on troubleshooting, see Debugging Constraint Modeling Language (CML) on page 1200.

Relationship Cardinality: Specify the Smallest Range Required

In a relationship, cardinality is the quantity of instances of the same type. Specify the smallest required cardinality for a variable, to avoid testing unneeded combinations of values. If you specify a higher cardinality than required, or don't specify cardinality, the constraint engine tests more combinations, which impacts performance.

This example doesn't specify cardinality. The constraint engine tries to set a quantity with 1, 2, 3, all the way up to 9,999:

```
relation engine : Engine;
```

This example specifies minimum and maximum cardinality as 0 and 1, so the constraint engine sets the quantity to 1. The engine tests fewer combinations to find a solution.

```
relation engine : Engine[0..1];
```

Decimals and Doubles: Consider the Impact of Scale on Performance

In a decimal or double, scale is the number of digits that follow the decimal point. Using decimals and doubles in expressions can cause performance problems due to the number of permutations.

In this example, myNumber is a double with a scale of 2. The value can be 0.00, 0.01, 0.02, all the way up to 2.99, which can impact constraint engine performance:

```
double(2) myNumber = [0..3];
```

In this example, myNumber is an integer. The value can only be 0, 1, 2 or 3, which has less impact on constraint engine performance:

```
int myNumber = [0..3];
```

Variable Domains: Keep Domains as Small as Possible

A variable domain is the set of all possible values that the variable can take. In this example, the variable color has a domain with three values:

```
string color = ["Red", "Yellow", "Green"];
```

The larger the domain, the more possible values for the variable, which means more combinations for the engine to test. A large domain can impact performance and lead to slower searches, errors, or unexpected behaviors.

Calculating Values: Put Calculations Inside of Constraints

To calculate a value, put the calculation inside of a constraint, instead of in an inline expression.

For example, to calculate area, use this constraint:

```
constraint(area == length * width)
```

Avoid this example, which calculates the area with an inline expression, and can impact performance:

```
area = length * width.
```

Relationships: Combine Relationships to Reduce Performance Impact

Creating multiple relationships on a type can impact performance. When possible, combine relationships to improve performance.

When possible, avoid this example, which includes separate relationships for Mouse and Keyboard, two accessories in a product bundle:

```
relation mouse : Mouse;
relation keyboard : Keyboard;
```

Follow this example, which uses one relationship for Accessories, which can include Mouse, Keyboard, and other accessories.

```
relation accessories : Accessories;
```

Sequence: Use the Sequence Variable Annotation to Specify the Order of Execution

If a constraint model includes multiple attributes and relationships that should follow a certain order of execution, use the sequence variable annotation to specify the order. The constraint engine follows sequence designations in satisfying constraint requirements and resolving constraint violations.

In this example, for the Desktop type, the sequence annotation directs the constraint engine to set the default values for attributes in this order:

- Display: sequence=1
- Windows_Processor: sequence=2
- Display_Size: sequence=3

```
type Desktop {
    @(defaultValue = "1080p Built-in Display", sequence=1)
    string Display = ["1080p Built-in Display", "4k Built-in Display", "2k Built-in
Display"];

    @(defaultValue = "15 Inch", sequence=3)
    string Display_Size = ["15 Inch", "24 Inch", "13 Inch", "27 Inch"];

    @(defaultValue = "i5-CPU 4.4GHz", sequence=2)
   string Windows_Processor = ["i5-CPU 4.4GHz", "i7-CPU 4.7GHz", "Intel Core i9 5.2 GHz"];


    constraint(Display == "1080p Built-in Display" && Display_Size == "15 Inch" ->
Windows_Processor == "i7-CPU 4.7GHz");
}
```

For Desktop, Display is set to 1080p, Windows_Processor to i5-CPU, and Display_Size to 15 Inch.

The constraint specifies that a type with Display of 1080p and Display_Size of 15 Inch must have a Windows_Processor of i7-CPU.

```
constraint(Display == "1080p Built-in Display" && Display_Size == "15 Inch" ->
Windows_Processor == "i7-CPU 4.7GHz");
```

The Windows_Processor default value of i5-CPU for Desktop violates the constraint. In order to satisfy the constraint and resolve the violation, the constraint engine uses a different Display_Size for Desktop, such as 24 Inch.

If the user manually updates Display_Size for Desktop to 15 Inch in the Product Configurator, the constraint engine updates Windows_Processor to i7-CPU to satisfy the constraint.

Sequence and Configurable

Using the `configurable` property with the `sequence` variable affects how the solver handles attributes. When configurable is set to `true`, the user can set attribute values, and the solver doesn't override user input unless no other solution exists. When `configurable` is set to `false`, the solver sets attribute values.

In this example for type A, `attribute1` is set to `configurable = true`. The user can set the value, and the solver doesn't override the user input unless no other solution exists. When the left-hand side of the rule is true, the constraint doesn't change `attribute1` to "Enum3".

```
type A : System {
    @(defaultValue = "Enum1", domainComputation = "true", configurable = true, sequence =
 48)
    string attribute1 = ["Enum1", "Enum2", "Enum3"];

  @(configurable = false, defaultValue = "0", sequence = 30)
    decimal(2) attribute2 = f(x, y, z);
    constraint((attribute2 > 3 && attribute2 <= 5) -> attribute1 == "Enum3", "message");
}
```

In this example for type B, `attribute1` is set to `configurable = false`. The solver propagates values to satisfy constraints. When the left-hand side of the rule is true, the constraint automatically sets `attribute1` to "Enum3".

```
type B : System {
    @(defaultValue = "Enum1", domainComputation = "true", configurable = false, sequence
= 48)
    string attribute1 = ["Enum1", "Enum2", "Enum3"];

  @(configurable = false, defaultValue = "0", sequence = 30)
    decimal(2) attribute2 = f(x, y, z);
    constraint((attribute2 > 3 && attribute2 <= 5) -> attribute1 == "Enum3", "message");
```

Use mindful sequencing in CML to avoid backtracking by the solver when looking for a solution, as in this example.

```
type LaptopProBundle {

    //relation mouse : Mouse[1..20];
    //The relation mouse has the highest sequence
    relation warranty : Warranty[0..10];
    relation software : Software;
    relation printerBundle : PrinterBundle;
    relation laptop : Laptop[1..10];
    relation mouse : Mouse[1..20];
    //Put highest sequence last to avoid backtracking
```

```
    int mouseQty = laptop[Laptop] + warranty[Warranty];


    constraint(mouse[Mouse] > 0 -> mouse[Mouse] == mouseQty,
               "mouse[Mouse] = laptop[Laptop] + warranty[Warranty]");
```

Automatically Add a Product: Define as a Separate Constraint

If you need to automatically add a product, and also set attributes on the product, define these procedures as separate constraints, as in this example.

```
constraint(laptop[Laptop] > 0, warranty[Warranty] > 0);
constraint(warranty[Warranty] > 0, warranty[Warranty].type == “Premium”);
```

Avoid this example, which automatically adds a product and sets attributes on the product, in the same constraint.

```
constraint(laptop[Laptop] > 0, warranty[Warranty] > 0 && warranty[Warranty].type ==
"Premium");
```

Access Quantity in CML: Cardinality and Attribute Constraints

There are two ways to access quantity in CML:

A cardinality constraint creates or validates the presence of components. The constraint engine adds or removes instances to satisfy the conditions of the constraint.

An attribute constraint, such as lineItemQuantity or ItemEndQuantity, only reads or validates. The constraint engine validates the expression, but doesn't configure to satisfy the conditions of the constraint

For best performance, follow these guidelines:

- Use a cardinality constraint whenever possible. Use `lineItemQuantity` or `ItemEndQuantity` only when a cardinality constraint can't meet the business need.
- Treat `lineItemQuantity` and `ItemEndQuantity` as read-only. Use only in calculation or evaluation rules.
- Keep scope in mind. When a product can have multiple instances, read `lineItemQuantity` or `ItemEndQuantity` per instance. Avoid reading the attribute at a parent or aggregate scope where it can be unbound or ambiguous.
- Don't use `lineItemQuantity` or `ItemEndQuantity` to create components. Drive component creation by cardinality, not by attribute references.

Use constraint patterns similar to these examples. Do this.

```
constraint(mouse[Mouse] == warranty[Warranty])
```

Avoid this.

```
constraint(mouse[Mouse].lineItemQuantity == warranty[Warranty].lineItemQuantity)
```

Do this.

```
constraint(mouse[Mouse] == 3)
```

Avoid this.

```
constraint(mouse[Mouse].lineItemQuantity == 3)
```

Pricing Fields Not Supported in CML

Pricing fields, such as ListPrice, NetUnitPrice, and others, are not supported in CML and should not be used in constraint models. CML is designed to enforce configuration logic for products, not to perform pricing calculations. Attempting to reference or manipulate pricing fields in CML code leads to errors and unexpected behaviors in the constraint engine. Use dedicated pricing or calculation mechanisms outside of the CML constraint model for such functionality.

Configure Child or Grandchild Products Based on Parent Product

Use the `parent` keyword to configure child and grandchild products dynamically based on the parent product. Control visibility or availability for the child or grandchild using the keyword.

Note: The `parent` keyword isn't supported with Group Type.

In this example, grandchild products are excluded based on which parent product they belong to.

```
type LineItem;

/*
 * Parent 1: Industrial Bundle
 * Sets the context 'ApplicationType' to "Industrial"
 */
type IndustrialGeneratorBundle : LineItem {
    relation enclosurePackage : EnclosurePackage[1..999];

    relation enclosurePackage1 : EnclosurePackage[1..999];

    relation enclosurePackage2 : EnclosurePackage;

    string ApplicationType = "Industrial";
}

/*
 * Parent 2: Residential Bundle
 * Sets the context 'ApplicationType' to "Residential"
 */
type ResidentialGeneratorBundle : LineItem {
    relation enclosurePackage3 : EnclosurePackage[1..999];

    relation enclosurePackage4 : EnclosurePackage;

    string ApplicationType = "Residential";
}

/*
 * Shared Component: Enclosure Package
 * Uses parent() to read the ApplicationType and excludes items accordingly.
 */
type EnclosurePackage : LineItem {
    relation soundInsulation : SoundInsulation[0..999];
    relation weatherProofing : WeatherProofing[0..999];
    relation heater : Heater[0..999];
    relation lighting : InternalLighting[0..999];
```

```
// Retrieve the context tag from the parent bundle
    string parentApp = parent(ApplicationType);

    message(true, parentApp);

    // Logic 1: Exclude Lighting and Heater for BOTH bundles
    exclude(parentApp == "Industrial" || parentApp == "Residential",
lighting[InternalLighting]);
    exclude(parentApp == "Industrial" || parentApp == "Residential", heater[Heater]);

    // Logic 2: Residential (Basic) excludes Sound Insulation
    exclude(parentApp == "Residential", soundInsulation[SoundInsulation]);

    // Logic 3: Industrial (Pro) excludes Weather Proofing
    exclude(parentApp == "Industrial", weatherProofing[WeatherProofing]);
}

// --- Sub-components ---

type SoundInsulation : LineItem {
    @(defaultValue = "Foam")
    string Material = ["Foam", "Fiberglass"];
}

type WeatherProofing : LineItem {
    @(defaultValue = "Standard")
    string Grade = ["Standard", "Marine"];
}

type Heater : LineItem;

type InternalLighting : LineItem;
```

PCG Group Relations: Follow Order in Constraints

Order the Product Component Group (PCG) group relations in the way they appear in the constraints. Specify the groups used in the LHS of the constraints first, followed by those in the RHS.

In this example, the order of PCG groups is incorrect as `Test1Group` is part of the LHS and should be declared first.

```
type TestParent : LineItem {
    Test2Group test2group;

    Test1Group test1group;

    constraint(test1group.testchild1[TestChild1] > 0 -> test2group.testchild2[TestChild2]
 == test1group.testchild1[TestChild1]);

}
```

This example shows the correct order of PCG groups, where `Test1Group` is declared first because it is part of the LHS.

```
type Test20260204Parent : LineItem {

    Test1Group test1group;
```

```
    Test2Group test2group;

    constraint(test1group.testchild1[TestChild1] > 0 -> test2group.testchild2[TestChild2]
 == test1group.testchild1[TestChild1]);

}
```

Relations Without PCG: Follow Order in Constraints

Order of relations matter for constraint evaluation. Specify the relations used in the LHS of the constraints first, followed by those in the RHS.

Relation Aggregates: Stabilize Preferences Using Staged Variables

To improve solver stability and prevent backtracking, avoid referencing relation aggregates directly within preferences when the preferences also influence the same relation. Instead, stage these aggregates through local variables and state anchors.

In this example, the preferences and the calculated quantity depend directly on live relation aggregates.

```
relation ml : ModelTraining {
   totalNumSubscriptions = total(NumSubscriptions);
   subCount = count(Sublicense == true);
}
preference(ml[ModelTraining] > 0 && ml.subCount == 0 -> ml[ModelTraining] == parentQty);
preference(ml[ModelTraining] > 0 && ml.subCount > 0 -> ml[ModelTraining] ==
calculatedQuantity);
constraint(calculatedQuantity == ((ml.totalNumSubscriptions + 99) / 100) * 100);
```

When the solver evaluates preferences that depend directly on a relation's own aggregates, it creates a tight feedback loop:

Relation Quantity Relation Aggregates Preference Guards/Targets Relation Quantity (loop repeats)

The feedback loop can cause the solver to oscillate or backtrack repeatedly before finding a stable solution.

This example shows the stabilized pattern. It introduces staged variables that act as anchors to separate the relation aggregates from the preference logic.

```
relation ml : ModelTraining {
     totalNumSubscriptions = total(NumSubscriptions);
     subCount = count(Sublicense == true);
}

// Stage 1: Convert relation conditions into stable booleans
boolean hasSub;
constraint(hasSub <-> (ml.subCount > 0));

// Stage 2: Copy relation aggregates into local variables
int localTotalNumSubscriptions;
constraint(localTotalNumSubscriptions == ml.totalNumSubscriptions);

// Stage 3: Perform calculations by using the local variables
int calculatedQuantity;
constraint(calculatedQuantity == ((localTotalNumSubscriptions + 99) / 100) * 100);

// Stage 4: Use the staged variables in the preferences
```

```
preference(ml[ModelTraining] > 0 && hasSub == false -> ml[ModelTraining] == parentQty);
preference(ml[ModelTraining] > 0 && hasSub == true -> ml[ModelTraining] ==
calculatedQuantity);
```

With this pattern, instead of solving everything simultaneously, the solver logic processes a pipeline:

Relation Aggregates Anchored State/Local Variables Derived Calculations Preferences

This process reduces oscillation and makes the solving path more deterministic.

Dependent Logic: Use Configurable Variable Annotation to Prevent Premature Default Assignment

To ensure preference rules correctly set target attributes based on calculated values, use the `@(configurable=false)` annotation on the target attribute. This prevents a race condition where the engine assigns a default value during initialization before your calculation and preference rules execute.

When an inline expression is used, CML evaluates the model top-down. If the target attribute is declared before the inline calculation, the engine immediately assigns a default value to the attribute. As a result, any subsequent `preference()` rules are ignored because the engine treats the existing default value as an established selection and avoids overwriting it with the calculated value. Annotating the target attribute with `@(configurable=false)`) forces the engine to leave the variable as `null` until the calculation and preference rules are executed.

In this example, the engine processes `ShippingMethod` first and immediately assigns the default value "Standard" to it. Later, when the calculation for `totalWeight` returns a value high enough to require the “Freight” shipping method, the engine ignores this preference rule because the `ShippingMethod` variable is already bound to "Standard".

```
type Order : LineItem {
    int quantity = [1..100];
    int weightPerUnit = [10..50];

    // Problem: Engine assigns "Standard" immediately during initialization
    @(defaultValue = "Standard")
    string ShippingMethod = ["Standard", "Freight"];

    // Calculated Attribute via Inline Expression

    // The calculation happens after the default is already set
    int totalWeight = quantity * weightPerUnit;

    // This rule is ignored because ShippingMethod is already bound to "Standard"
    preference(totalWeight > 1000 -> ShippingMethod == "Freight");
}
```

In this example, using the `@(configurable=false)` annotation ensures the `ShippingMethod` attribute remains as `null` until the calculation completes and the rules execute.

Note: Provide a second preference rule for the fallback condition (else) to ensure a value is always set.

```
ype Order : LineItem {
    int quantity = [1..100];
    int weightPerUnit = [10..50];

    // Solution: Prevent premature default assignment
    @(configurable=false)
```

```
    string ShippingMethod = ["Standard", "Freight"];


    // Calculated Attribute via Inline Expression
    int totalWeight = quantity * weightPerUnit;

    // Rule 1: Handle the specific condition (high weight)
    preference(totalWeight > 1000 -> ShippingMethod == "Freight");

    // Rule 2: Handle the fallback (low weight)
    preference(totalWeight <= 1000 -> ShippingMethod == "Standard");
}
```

Consolidate Multiple Default Configurations

Apply dynamic, overridable default quantities to a single component based on multiple possible states of a driving attribute. This process avoids rule conflicts where only the last rule evaluates.

When multiple `setdefault` rules target the same assignment in the RHS, evaluations can conflict. For example, a conflict can result when the rule targets the same relation and type, such as `accessories[Accessory] == ...`. The condition side can use different expressions, but if the rules resolve to the same target on the assignment side, behavior can become order-dependent and lead to inconsistent outcomes.

For example, if you want to set a default for the quantity of `Accessories` based on the selected `DutyRating` of a Generator Set, writing three separate `setdefault` rules causes inconsistent behavior depending on the quantity changes of the accessories.

To prevent the conflict, consolidate the conditional logic into a single derived variable using nested ternary expressions, such as `? :`. Then, apply a single `setdefault` rule using that newly derived variable to drive the relation's quantity.

This example shows the recommended process, consolidating the logic into a single variable and rule.

```
type GeneratorSet : LineItem {
    string DutyRating = ["Prime Power (PRP)", "Continuous Power (COP)", "Emergency Standby
 Power (ESP)"];

    relation accessories : Accessory[1..99] {
        default Accessory(1);
    }

    // 1. Consolidate the conditional logic into a single integer variable
    int defaultAccessoryQty = (DutyRating == "Prime Power (PRP)") ? 2 : ((DutyRating ==
"Continuous Power (COP)") ? 4 : 6);

    // 2. Create a dynamic informational message
    string infoMsg = DutyRating + " accessory default overridden";

    // 3. Apply a single setdefault rule using the derived variable
    setdefault(
        DutyRating in ["Prime Power (PRP)", "Continuous Power (COP)", "Emergency Standby
Power (ESP)"],
        accessories[Accessory] == defaultAccessoryQty,
        infoMsg
    );
}
```

Avoid the process shown in this example, which includes multiple conflicting `setdefault` rules.

```
type GeneratorSet : LineItem {
    string DutyRating = ["Prime Power (PRP)", "Continuous Power (COP)", "Emergency Standby
 Power (ESP)"];

    relation accessories : Accessory[1..99] {
        default Accessory(1);
    }

    // X AVOID: Multiple setdefault rules targeting the same relation.
    // Only the last rule will reliably evaluate.
   setdefault(DutyRating == "Prime Power (PRP)", accessories[Accessory] == 2, "PRP error");

    setdefault(DutyRating == "Continuous Power (COP)", accessories[Accessory] == 4, "COP
error");
    setdefault(DutyRating == "Emergency Standby Power (ESP)", accessories[Accessory] ==
6, "ESP error");
}
```

Business-Centric Constraint Modeling Language (CML) Guidelines

Constraint Modeling Language (CML) must accurately calculate the total sum or aggregate of specific attributes like quantity or userCount across child components, especially in complex configurations requiring group-level aggregation

The main modeling obstacles when performing aggregation in CML involve:

- Initialization Errors: Preventing runtime errors, such as `NullPointerException`, which can occur if derived aggregate attributes lack explicit domains.
- Circular Dependencies: Avoiding calculation loops where the parent and children mutually rely on aggregated totals, often involving the `total()` function. If these loops are not broken, the aggregated variable becomes "not bound", which causes the solution to fail.

User Workflow

As a sales representative, when I am configuring a bundle product in the Configurator window, I modify the quantities or specific attributes of the individual child components. I expect the constraint engine to immediately and accurately calculate the overall aggregated totals for the parent product, such as the totalItemCount or sumOfUsers.

Business-Centric CML Examples

These Constraint Modeling Language (CML) structures implement quantity aggregation and resolve calculation dependencies.

Example 1: Derived Aggregates (Total Quantity or Sum)

This CML example shows how to model bundle rollups by separating aggregation from validation, ensuring correct calculation order, solver stability, and business-rule consistency

```
type LineItem;
type GeneratorSet : LineItem {
  int totalItemPowerKW = modelclassification.totalPowerKW;
  // Aggregate `powerKW`
  relation modelclassification : ModelClassification[0..10] {
    totalPowerKW = sum(powerKW);
  }
  message(true, "GeneratorSet totalItemPowerKW = {} kW (calculated roll-up). Sum of selected
 ModelClassification powerKW values = {} kW.", totalItemPowerKW,
modelclassification.totalPowerKW);
}
type ModelClassification : LineItem {
  int powerKW = [900, 1750, 2500];
}
type GeneralModel1750 : ModelClassification {
  int powerKW = 1750;
}
type GeneralModel2500 : ModelClassification {
  int powerKW = 2500;
}
type GeneralModel900 : ModelClassification {
  int powerKW = 900;
}
```

Example Description and Configurator Result

In example 1, the `GeneratorSet` type contains the `totalItemPowerKW` variable that is assigned with the `totalPowerKW`. The model specifies a relationship based on the product classification (`modelClassification`). This relationship includes aggregation using the sum() function that calculates the `totalPowerKW` from the products selected in the `modelClassification` during bundle configuration. The model contains a message for displaying the `totalItemPowerKW` with the `modelClassification.totalPowerKW` on the Product Configuration.

```
message(true, "GeneratorSet totalItemPowerKW = {} kW (calculated roll-up). Sum of selected
 ModelClassification powerKW values = {} kW.", totalItemPowerKW,
modelclassification.totalPowerKW);
```

As a result, the engine calculates the `totalPowerKW` of user selected products in the `modelClassification` and assigns this value to the `totalItemPowerKW`. The system displays both values in the information message on the Product Configuration.

Example 2: Resolving Circular Dependencies

This pattern breaks unsolvable loops by enforcing unidirectional data flow (Parent dictates Child quantity based on Child aggregates).

```
type LineItem;
type GeneratorSet : LineItem {
  // RELATION: THE "READ" PHASE
  // Data flows UP from children to the parent via the port attribute
  relation voltageclassification : VoltageClassification[1..10] {
    sumOfChildPower = total(power);
```

```
  }
  // Explicit domain ensures the solver initializes correctly for calculations
  decimal(2) totalDistributionValue = [0.00..1000.00];
  // CONSTRAINT 1: THE "CALCULATION" PHASE
  // @(sequence = 1) ensures the engine reads child data and calculates first
  @(sequence = 1)
  constraint calcTotalDistribution(totalDistributionValue ==
voltageclassification.sumOfChildPower / 100);
  // CONSTRAINT 2: THE "WRITE/ENFORCE" PHASE
  // @(sequence = 2) ensures this happens last to push values DOWN to children
  @(sequence = 2)
  constraint setChildQuantity(voltageclassification[VoltageClassification] ==
totalDistributionValue * 2);
}
type VoltageClassification : LineItem {
  int power = [1..100];
}
type VoltageConnection_220_380 : VoltageClassification;
type VoltageConnection_240_416 : VoltageClassification;
type VoltageConnection_255_440 : VoltageClassification;
```

Example Description and Configurator Result

In example 2, the relationship includes aggregation using the `total()` function that calculates the `sumOfChildPower` from the selected products in the `voltageclassification`. The `GeneratorSet` type contains the `totalDistributionValue` with the explicit domain ([0.00..1000.00]). The model contains two constraints with corresponding `@sequence` annotation to enforce the logical order for calculation of the `calcTotalDistribution` (using child products data) and defining the quantity for the `voltageclassification` child items.

Example 3: Grouped Aggregation (Sum of Users Across Regions)

This advanced pattern uses the `@(groupBy=attribute)` annotation to create virtual components (`RegionGroup`) for aggregation, referencing the source data relation in the parent type (`SubscriptionOrder.licenses`) using the `sourceContextNode` annotation.

```
// 1. Base Product Type (e.g., Regional License)
type RegionalLicense {
  int regionId = [0..100]; // Attribute for grouping
  int userCount = [0..100];
}
// 2. Virtual Group Type (Performs grouping and user sum)
@(split=true, virtual=true, groupBy=regionId)
type RegionGroup {
  int regionId;
  int groupTotalUsers; // Sum of userCount for this region
  // Relation to licenses matching this region group
  @(sourceContextNode="SubscriptionOrder.licenses")
  relation licenses: RegionalLicense[0..100];
  // Constraint: Calculate the aggregate user count within this group
  constraint(groupTotalUsers == licenses.sum(userCount));
}
// 3. Parent Order Management (Aggregates all region totals)
@(virtual=true)
```

```
type SubscriptionOrder {
  // Overall total users is calculated by referencing the group totals
  int totalUsers = regionGroups.groupTotalUsers;
  relation licenses: RegionalLicense[0..500];
  relation regionGroups: RegionGroup[1..10];
}
```

Example Description and Configurator Result

In example 3, the `licenses` relationship that is brought as an external source by `@(sourceContextNode="SubscriptionOrder.licenses")`, includes an aggregation using the `sum()` function that calculates the `groupTotalUsers` from the selected products in the `RegionalLicense`. The `RegionGroup` type contains the `groupTotalUsers` variable that serves the aggregation. The `SubscriptionOrder` type contains the aggregate metric `totalUsers` which is calculated by summing all `groupTotalUsers`. The model contains one core constraint on `RegionGroup` to enforce that `groupTotalUsers` equals the sum of all `userCount` from its member licenses.

Debugging Constraint Modeling Language (CML)

To debug constraint models and troubleshoot performance issues, enable debug logging in Apex and set the debug log level to FINE.

For more information on debug logging in Salesforce, see these topics in Salesforce Help:

- Set Up Debug Logging
- Debug Log
- Debug Log Levels

Use the Apex log to get information about configurator engine performance when running a constraint model, including performance degradation or unexpected behavior. To improve performance, modify the constraint model based on information in the log.

For tips on writing trouble-free CML, see Constraint Modeling Language (CML) Best Practices on page 1188.

About the Apex Debugging Log File

The Apex debugging log file contains three sections: RLM_CONFIGURATOR_BEGIN, RLM_CONFIGURATOR_STATS, and RLM_CONFIGURATOR_END.

RLM_CONFIGURATOR_BEGIN

JSON representation of the request payload to ExecuteConstraintsRESTService.

```
"contextProperties" : { },
"rootLineItems" : [ {
  "attributes" : { },
  "properties" : {   },
```

```
  "ruleActions" : null,
  "attributeDomains" : { },
  "portDomainsToHide" : { },
  "lineItems" : [ {} ]
} ],
"orgId" : "00Dxx0000006H2F"
}
```


RLM_CONFIGURATOR_STATS

Key statistics of the request execution by the constraint engine, as in this example.

```
"rootId" : "0QLxx0000004D1uGAE",  //Root ID that is being configured
"Product" : "SFDC License",  //Root product name
"Total Execution Time" : "2ms",  //Total solver time
"Constraints Execution Stats" : "Distinct: 18 Total: 70",  //Number of distinct and total
 constraint satisfaction attempts
"Solving goal AndGoal([ConfigureComponentGoal(RootProduct RootProduct_0)]) took " : "2ms",
  //Total solver time for the goal
"Configurator Stats" : "Total Time 2ms",  //Total time
"Number of Component" : "6",  //Number of components instantiated
"Number of Variables" : "42",  //Number of variables instantiated
"Number of Constraints" : "13",  //Number of constraints instantiated
"Number of Backtracks" : "0",  //Number of backtracks solver did for the last choice point
"Constraints Violation Stats" : "Distinct: 0 Total: 0",  //Distinct and total number of
constraint violations followed by a list of top 10
"ChoicePoint Backtracking Stats" : "Distinct: 0 Total: 0"  // Distinct and total number
of backtracked choice points followed by a list of top 10
}
```

RLM_CONFIGURATOR_END

JSON representation of the response payload from ExecuteConstraintsRESTService.


```
"id" : "0QLxx0000004D1uGAE",
"rootId" : null,
"parentId" : null,
"cfgStatus" : "User",
"name" : "RootProduct",
"relation" : null,
"source" : "SalesTransaction.SalesTransactionItem",
"qty" : 1,
"actionCode" : null,
"modelName" : "Support_instance_variable_in_CML",
"productId" : "01txx0000006iP2AAI",
"productRelatedComponentId" : null,
"attributes" : {},  "properties" : {},
"ruleActions" : [ {} ],   "attributeDomains" : {},  "portDomainsToHide" : {},
"lineItems" : [ {} ]
}
```

Use the Apex Debugging Log File

To find possible reasons for the performance problems and identify solutions, look at the RLM_CONFIGURATOR_STATS section of the log file.

See the values for Total Execution Time, Constraints Violation Stats, and ChoicePoint Backtracking Stats.

For example, consider how the constraint engine performs with this sample constraint model. In the constraint model, the value of the `volts` variable is greater than 110/10000 (`volts = power/amps * 9999;`). The constraint engine must backtrack the `power` variable to find a value that satisfies the constraint, starting with 0.01, 0.02, and so on until it reaches a valid value.

```
relation laptops : Laptop[1..9999];
@(sequence = 1)
decimal(2) power = [0..500];
@(sequence = 1)
int amps = [1..5];
decimal(2)  volts = (power / amps) * laptops[Laptop];
constraint(volts > 110);
```

In the log file for this constraint model, see the execution statistics for Total Execution Time, Constraints Violation Stats, and ChoicePoint Backtracking Stats:

```
"rootId" : "ref_a67c6632_fa1f_40b4_8093_226a9ab8a4d0",
"Product" : "Laptop",
"Total Execution Time" : "676ms",
"Constraints Execution Stats" : "Distinct: 2 Total: 132006",
"Solving goal AndGoal([ConfigureComponentGoal(Laptop Laptop_0)]) took " : "677ms",
"Configurator Stats" : "Total Time 677ms",
"Number of Component" : "1",
"Number of Variables" : "4",
"Number of Constraints" : "1",
"Number of Backtracks" : "49500",
"Constraints Violation Stats" : "Distinct: 1 Total: 41250",
"IntComparison(GT,[DecimalVar(volts)])" : "41250",
"ChoicePoint Backtracking Stats" : "Distinct: 2 Total: 98999",
"VariableChoicePoint(DecimalVar(power))" : "49500",
"VariableChoicePoint(IntVar(amps))" : "49499"
```

Optimally, execution time for a constraint model is less than 100 milliseconds, with fewer than 1,000 backtracks and no violations. Values for the constraint model example are significantly higher, indicating that the constraint engine is performing inefficiently. To improve performance in this example, reduce the domain of the power variable without reducing the solution space. For example, define the domain as `[110..500]` instead of `[0..500]`. This change reduces the number of backtracks the constraint engine performs to find a solution.

Model Structure

The tables on the following pages show the structure for the constraint model in Core Concept Examples.

See Core Concept Examples on page 1135.

Level	Product Group	Product	Product Type Name	Product Attribute	API Name OR CML Variable	Configurable or Static	Type	Comments
0	Bundle	Generator Set	GeneratorSet					
				Required KW	requiredKW	CONFIGURABLE	int	
				Surge Load KW	surgeLoadKW	EXPRESSION	decimal	requiredKW * 1.25
				Reserve Capacity KW	reserveCapacityKW	EXPRESSION	decimal	surgeLoadKW - requiredKW
				Voltage	voltage	CONFIGURABLE	Picklist Voltage [illegible]	
				Duty Rating	dutyRating	CONFIGURABLE	Picklist DutyRating ["Prime Power", "Continuous Power", "Data Center Continuous", "Emergency Standby Power"]	
				Standards and Compliance	standardsAndCompliance		Picklist standardsAndCompliance ["Certification-CSA", "Listing-UL 2200"]	
				max dB level	dBMax	CONFIGURABLE	int	
				Nominal Power Output	nominalPowerOutput		Picklist Power Output Picklist ["100 kW", "300 kW", "500 kW", "700 kW"]	
1	General							
2	General - Model		GeneralModel					

Level	Product Group	Product	Product Type Name	Product Attribute	API Name OR CML Variable	Configurable or Static	Type	Comments
				Power KW	powerKW	STATIC	Picklist powerKW [900, 1750, 2500]	powerKW >= Needs.requiredKW
				dB	dB	STATIC	Picklist dB [78, 90, 94]	
		General Model 900	GeneralModel900					
		General Model 1750	GeneralModel1750					
		General Model 2500	GeneralModel2500					
2	General - Voltage Connection		VoltageConnection					
				Voltage	voltage	STATIC	Picklist voltage ["220/380", "240/416", "255/440"]	voltage == Needs.voltage
				Cable Entry	cableEntry	CONFIGURABLE	Picklist cableEntry ["Top Entry", "Bottom Entry", "Side Entry"]	
		220/380,3 Phase,Wye,4 Wire	VoltageConnection_220_380					
		240/416,3 Phase,Wye,4 Wire	VoltageConnection_240_416					
		255/440,3 Phase,Wye,4 Wire	VoltageConnection_255_440					
1	Alternator							
2	Alternator - Main Alternator		MainAlternator					

Level	Product Group	Product	Product Type Name	Product Attribute	API Name OR CML Variable	Configurable or Static	Type	Comments
					voltage	STATIC	Picklist Voltage	Needs.voltage == voltage
					PRP	STATIC	boolean	contains(Needs.dutyRating == "Prime Power"->PRP == true)
					COP	STATIC	boolean	contains(Needs.dutyRating == "Continuous Power"->COP == true)
					DCC	STATIC	boolean	contains(Needs.dutyRating == "Data Center Continuous"->DCC == true)
					ESP	STATIC	boolean	contains(Needs.dutyRating == "Emergency Standby Power"->ESP == true)
		[illegible] amb	FESBA_B595_2					
		Alternator 60Hz, Wye, 240/416 Volt, 105/80C - Stby/Prm	FESBA_B715_2					
		[illegible] amb	FESBA_B691_2					
2	Alternator - Heater							
2	Alternator - Temperature Sensors		TemperatureSensor					
				Max Operating KW	maxOperatingKW	STATIC		
				Parent Required KW	parentRequiredKW	EXPRESSION		
		Temp Sens-Stator, 2 RTD/Ph	StatorTemperatureSensor					
		Bearing,1 RTD NDE	BearingTemperatureSensor					
2	Alternator - Output Terminals		OutputTerminal					

Level	Product Group	Product	Product Type Name	Product Attribute	API Name OR CML Variable	Configurable or Static	Type	Comments
		Output Terminals-2-Hole Lug, NEMA	OutputTerminals2HoleLugNEMA					
1	Engine							
2	Engine - Engine Model		EngineModel					
		Engine - QSK60-G6	FESBA_2940					
2	Engine - Starter Motor		StarterMotor					
		Electric Starter Motor - 24V DC	FESBA_A334-2					
2	Engine - Fuel Filter		FuelFilter					
		Fuel Filters-Engine, Standard	FESBA_3090					
		Fuel Filters-Engine, Duplex	FESBA_C278-2					
1	Control							
2	Control - Control Main		Control					
				Control Placement	controlPlacement	CONFIGURABLE	Picklist ["Left", "Right", "Top"]	
				Commissioning Scope	commissioningScope	CONFIGURABLE	Picklist ["None", "Remote Support", "On-site Commissioning"]	
				Control Language	controlLanguage	CONFIGURABLE	Picklist ["English",	

Level	Product Group	Product	Product Type Name	Product Attribute	API Name OR CML Variable	Configurable or Static	Type	Comments
							"Danish", "French"]	
		PowerCommand 3.3	FESBA_H704_2					
		PowerCommand 3.3 with MLD	FESBA_KX21_2					
2	Control - Control Cabinet Heater		ControlCabinetHeater					
		120/240VAC compatible	FESBA_A460_2					
1	Services							
2	Services - Warranty		Warranty					
		Warranty PRP	Warranty_PRP					
		Warranty DCC	Warranty_DCC					
		Warranty ESP	Warranty_ESP					
2	Services - Maintenance		Maintenance					
				Maintenance Duration	maintenanceDuration	CONFIGURABLE	int [12..60]	
				Coverage Level	coverageLevel	CONFIGURABLE	Picklist ["Standard", "Premium"]	
		Standard Maintenance Kit	StandardMaintenanceKit					
2	Services - Testing		Test					
		Test - Standard Factory	StandardFactoryTest					
		Test Record-Certified	TestRecord					

Level	Product Group	Product	Product Type Name	Product Attribute	API Name OR CML Variable	Configurable or Static	Type	Comments
		Test-Independent Laboratory	IndependentLaboratoryTest					
		Test - Witness	WitnessTest					
		Test-Extended, Standby Load, 2	WitnessTestService					
1	Accessories							
2	Accessories - Main		Accessory					
				Category	category	CONFIGURABLE	Picklist	
				Weight	weight			
2	Accessories - Enclosure		Enclosure					
				dB Reduction	dBReduction	STATIC	Picklist dBReduction [0, 1, 3, 6, 9]	
		Enclosure None	Enclosure_None					
		Enclosure Weather	Enclosure_Weather					
		Enclosure SA1	Enclosure_SA1					
		Enclosure SA2	Enclosure_SA2					
		Enclosure SA3	Enclosure_SA3					
1	Install & Misc							
2	Install & Misc - Installation		install					

CHAPTER 8 Transaction Management

In this chapter ...

Configure, price, and sell products with Transaction Management in Revenue Management. Transaction Management supports subscription lifecycles and ensures end-to-end integrity of your quotes and orders.

SEE ALSO:

Salesforce Help: Permissions for Quote and Order Capture

EDITIONS

Available in: Lightning Experience

Available in: **Enterprise**, **Unlimited**, and **Developer** Editions of Revenue Cloud where Transaction Management is enabled

Transaction Management Standard Objects

The Transaction Management data model provides objects and fields to manage transactions.

Asset

Represents an item of commercial value, such as a product sold by your company or a competitor, that a customer has purchased.

AssetAction

Represents a change made to a lifecycle-managed asset. The fields can't be edited. This object is available in API version 50.0 and later.

AssetActionSource

Represents an optional way to record what transactions caused changes to lifecycle-managed assets. Use it to trace financial and other information about asset actions. This object supports Salesforce order products and work order line items, and transaction IDs from other systems. The fields can't be edited. This object is available in API version 50.0 and later.

AssetActionSrcPriceAdjustment

Each row represents a junction between an asset and the calculated price adjustment that's applied to an asset. This object is available in API version 66.0 and later.

AssetContractRelationship

Represents a relationship between an asset and a contract. This object is available in API version 60.0 and later.

AssetDowntimePeriod

Represents a period during which an asset is not able to perform as expected. Downtime periods include planned activities, such as maintenance, and unplanned events, such as mechanical breakdown. This object is available in API version 49.0 and later.

AssetOwnerSharingRule

Represents the rules for sharing an Asset with users other than the owner. This object is available in API version 33.0 and later.

AssetRateAdjustment

Stores the tier rate adjustments for the asset rate card entries.This object is available in API version 62.0 and later.

AssetRateCardEntry

Stores the negotiated rate card entries that are associated with an asset in Revenue Cloud. This object is available in API version 62.0 and later.

AssetRelationship

Represents a non-hierarchical relationship between assets due to an asset modification; for example, a replacement, upgrade, or other circumstance. In Revenue Lifecycle Management, this object represents an asset or assets grouped in a bundle or set. This object is available in API version 41.0 and later.

AssetShare

Represents a sharing entry on an Asset. This object is available in API version 33.0 and later.

AssetStatePeriod

Represents a time span when an asset has the same quantity, amount, and monthly recurring revenue (MRR). An asset has as many asset state periods as there are changes to it (asset actions) during its lifecycle. The dashboard and related pages show the current asset state period. The fields can't be edited. This object is available in API version 50.0 and later.

AssetStatePeriodAttribute

Represents a virtual object that holds the key-value pair of the asset attribute in a specified asset state period. This object is a child object of AssetStatePeriod. This object is available in API version 60.0 and later.

AssetTag

Associates a word or short phrase with an Asset.

AssetTokenEvent

The documentation has moved to AssetTokenEvent in the *Platform Events Developer Guide*.

AssetWarranty

Defines the warranty terms applicable to an asset along with any exclusions and extensions. This object is available in API version 50.0 and later.

BindingObjUsageRsrcPlcy

Represents the policies that are used for the usage resource that's associated with an asset or a binding object. This object is available in API version 65.0 and later.

ContractItemPrice

Represents an object that's used to capture a price for a product on a contract. This object is available in API version 61.0 and later.

ContractItemPriceAdjTier

Represents the tiers of a price adjustment to a product on a contract. This object is available in API version 63.0 and later.

ContractItemPriceHistory

Represents the history of changes to the values in the fields of a ContractItemPrice object. This object is available in API version 61.0 and later.

OrderDeliveryMethod

Shows the customizations and options that a buyer selected for their delivery method. This object is available in API version 48.0 and later.

OrderItemAttribute

Represents a virtual object that stores an attribute specified for an order item.This object is available in API version 60.0 and later.

OrderItemDetail

Represents the breakdown details of an order product. Revenue Management generates these records to capture pricing and quantity changes, such as negative quantity reductions, early renewals, derived pricing or repricing during an amendment, and bundle or product attribute reconfigurations. This object is available in API version 60.0 and later.

OrderItemRateAdjustment

Represents the negotiated rate adjustment for an order product. This object is available in API version 62.0 and later.

OrderItemRateCardEntry

Represents the catalog and negotiated rates of a usage metric associated with an order item that's used to charge overage consumption. This object is available in API version 62.0 and later.

OrderItemUsageRsrcGrant

Represents the negotiated grants for the usage resource that's associated with the usage product added in the order item. This object is available in API version 65.0 and later.

OrderItemUsageRsrcPlcy

Represents the policies that are used for the usage resource that's associated with the usage product added in the order item. This object is available in API version 65.0 and later.

SalesTransactionType

Represents the type of the sales transaction. This object is available in API version 61.0 and later.

QuoteAction

Indicates the type of sales transaction that's being quoted; for example, a renewal sale. This object is available in API version 59.0 and later.

QuoteItemTaxItem

The tax that is applied to a quote line item. This object is available in API version 55.0 and later.

QuoteLineDetail

Represents the breakdown details of a quote line item. Revenue Management generates these records to capture pricing and quantity changes, such as negative quantity reductions, early renewals, derived pricing or repricing during an amendment, and bundle or product attribute reconfigurations. This object is available in API version 60.0 and later.

QuoteLineGroup

Stores the group information for line items in a quote. It also stores the aggregated line field information (subtotal). It contains a parent-child relationship to quote. This object is available in API version 61.0 and later.

QuoteLineItemAttribute

Represents a virtual object that stores an attribute specified for a quote line item. This object is available in API version 59.0 and later.

QuotLineItmUseRsrcGrant

Represents the negotiated grants for the usage resource that's associated with the usage product added in the quote line item. This object is available in API version 65.0 and later.

QuotLineItmUsageRsrcPlcy

Represents the policies that are used for the usage resource that's associated with the usage product added in the quote line item. This object is available in API version 65.0 and later.

QuoteLineRateAdjustment

Represents the negotiated rate adjustment for a quote line item. This object is available in API version 62.0 and later.

QuoteLineRateCardEntry

Represents the catalog and negotiated rates of a usage resource associated with a quote line item that's used to charge overage consumption. This object is available in API version 62.0 and later.

SEE ALSO:

Object Reference for the Salesforce Platform: Overview of Salesforce Objects and Fields

SOAP API Developer Guide: Introduction to SOAP API

Asset

Represents an item of commercial value, such as a product sold by your company or a competitor, that a customer has purchased.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`AccountId`	**Type** reference

Field	Details
	Properties Create, Filter, Group, Nillable, Sort, Update **Description** (Required) ID of the Account associated with this asset. Must be a valid account ID. Required if `ContactId` isn't specified. This field is a relationship field. **Relationship Name** Account **Relationship Type** Lookup **Refers To** Account
`Address`	**Type** address **Properties** Filter, Nillable **Description** Represents the physical address or geolocation of the asset.
`AssetLevel`	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The asset's position in an asset hierarchy. If the asset has no parent or child assets, its level is 1. Assets that belong to a hierarchy have a level of 1 for the root asset, 2 for the child assets of the root asset, 3 for their children, and so forth. On assets created before the introduction of this field, the asset level defaults to –1. After the asset record is updated, the asset level is calculated and automatically updated.
`AssetProvidedById`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The account that provided the asset, typically a manufacturer. This field is a relationship field. **Relationship Name** AssetProvidedBy

Field	Details
	Relationship Type Lookup **Refers To** Account
AssetServicedById	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The account in charge of servicing the asset. This field is a relationship field. **Relationship Name** AssetServicedBy **Relationship Type** Lookup **Refers To** Account
AssetTypeId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The asset type associated with the asset. This field is a relationship field. This field is available in API version 62.0 and later for users with the Health Cloud Appointment Management permission set. **Relationship Name** AssetType **Relationship Type** Lookup **Refers To** AssetType
Availability	**Type** percent **Properties** Filter, Nillable, Sort **Description** The percentage of expected uptime where the asset was available for use.

Field	Details
AveragetimetoRepair	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** Represents the number of hours it typically takes to repair an asset after a failure.
AveragetimeBetweenFailure	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** Represents the number of hours that typically elapses before the asset is likely to fail again.
AverageUptimePerDay	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The average number of hours per day the asset is expected to be available for use.
City	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The city detail for the address.
ConsequenceOfFailure	**Type** picklist **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The business impact associated with the asset's failure. Using this field, you can address the asset's health and take action using Flows. To enable this field, use Object Manager to update the field availability. Make sure that the field is visible for field-level security and for page layout. To learn more, see What Determines Field Access. The picklist values aren't predefined in orgs created before Winter '22 that aren't Field Service enabled. This field is available in API version 53.0 and later. Possible values are: • `Insignificant`

Field	Details
	• `Minor` • `Moderate` • `Major` • `Critical`
`ContactId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Required if `AccountId` isn't specified. ID of the Contact associated with this asset. Must be a valid contact ID that has an account parent (but doesn't need to match the asset's `AccountId`). This field is a relationship field. **Relationship Name** Contact **Relationship Type** Lookup **Refers To** Contact
`Country`	**Type** String **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The country detail for the address.
`CurrencyIsoCode`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Three-letter ISO 4217 currency code associated with the invoice. The default value is USD. This field is available in API version 55.0 and later. This field is available when CPQ Plus, Salesforce Billing, or Revenue Cloud is enabled.
`CurrentAmount`	**Type** currency **Properties** Filter, Nillable, Sort

Field	Details
	Description Reserved for future use. This field is available in API version 50.0 and later. This field is available when CPQ Plus, Salesforce Billing, or Revenue Cloud is enabled.
`CurrentLifecycleEndDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Represents the end of the period shown as current. System-populated field inherited from the end date of the current asset state period. If that field is empty, as with an evergreen subscription, the Current Lifecycle End Date field is also empty. This field is available in API version 50.0 and later. This field is available when CPQ Plus, Salesforce Billing, or Revenue Cloud is enabled.
`CurrentMrr`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The asset's monthly recurring revenue during the current asset state period. System-populated field inherited from the monthly recurring revenue on the current asset state period. If no asset state period is current, the value is `0`. Label is Current Monthly Recurring Revenue. This field is available in API version 50.0 and later. This field is available when CPQ Plus, Salesforce Billing, or Revenue Cloud is enabled.
`CurrentQuantity`	**Type** double **Properties** Filter, Nillable, Sort **Description** The asset's quantity during the current asset state period. System-populated field inherited from the quantity on the current asset state period. If no asset state period is current, the value is `0`. This field is available in API version 50.0 and later. This field is available when CPQ Plus, Salesforce Billing, or Revenue Cloud is enabled.
`Description`	**Type** textarea **Properties** Create, Nillable, Update

Field	Details
	Description Description of the asset.
`DigitalAssetStatus`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Status of digital tracking of the asset. The default picklist includes the following values: • `On` • `Off` • `Warning` • `Error`
`ExternalIdentifier`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the matching record in an external system. This field is available in API version 49.0 and later.
`GeocodeAccuracy`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Accuracy level of the geocode for the address.
`HasLifecycleManagement`	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** True if this asset is a lifecycle-managed asset, otherwise false. You can't switch an asset to a lifecycle-managed asset or the reverse. This field is system populated. The default value is *`false`*. This field is available in API version 50.0 and later. This field is available when CPQ Plus, Salesforce Billing, or Revenue Cloud is enabled.

Field	Details
InstallDate	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Date when the asset was installed.
IsCompetitorProduct	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether this Asset represents a product sold by a competitor (`true`) or not (`false`). The default value is `false`. Its UI label is Competitor Asset.
IsInternal	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates that the asset is produced or used internally (`true`) or not (`false`). The default value is `false`. Its UI label is Internal Asset.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date and time that the asset was last modified. Its UI label is Last Modified Date.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date and time that the asset was last viewed.
Latitude	**Type** double **Properties** Create, Filter, Group, Nillable, Sort, Update

Field	Details
	Description Used with Longitude to specify the precise geolocation of the address.
LifecycleEndDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Represents the end of the asset's lifecycle. System-populated field inherited from the end date of the final asset state period. If that field is empty, as with an evergreen subscription, the lifecycle has no end date. This field is available in API version 50.0 and later. This field is available when CPQ Plus, Salesforce Billing, or Revenue Cloud is enabled.
LifecycleStartDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Represents the beginning of the asset's lifecycle. System-populated field inherited from the start date of the earliest asset state period. This field can't be edited. When a new asset action affects the start date of an asset state period, the period is deleted and a new one is generated. This field is available in API version 50.0 and later. This field is available when CPQ Plus, Salesforce Billing, or Revenue Cloud is enabled.
LocationId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The asset's location. Typically, this location is the place where the asset is stored, such as a warehouse or van. If you have access to the location entity, it doesn't necessarily mean you can access the location id field. To access the location, you must have `userHasLocation` user access.
Longitude	**Type** double **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Used with Latitude to specify the precise geolocation of the address.
ManufactureDate	**Type** date

Field	Details
	Properties Create, Filter, Group, Nillable, Sort, Update **Description** The date when the asset was manufactured. This field is available from API version 49.0 and later.
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** (Required) Name of the asset. Label is Asset Name.
`OwnerId`	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The asset’s owner. By default, the asset owner is the user who created the asset record. Its UI label is Asset Owner. This field is a relationship field. **Relationship Name** Owner **Relationship Type** Lookup **Refers To** User
`ParentId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The asset’s parent asset. Its UI label is Parent Asset. This field is a relationship field. **Relationship Name** Parent **Relationship Type** Lookup **Refers To** Asset

Field	Details
PostalCode	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The postal code for the address.
Price	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** Price paid for this asset.
PricingSource	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Pricing source to use when amending or renewing an asset. Valid values are: • LastTransaction—Last Transaction • PriceBookListPrice—Price Book or List Price Available in API version 60.0 and later.
Product2Id	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** (Optional) ID of the Product2 associated with this asset. Must be a valid Product2 ID. Its UI label is Product. This field is a relationship field. **Relationship Name** Product2 **Relationship Type** Lookup **Refers To** Product2

Field	Details
ProductCode	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The product code of the related product.
ProductDescription	**Type** string **Properties** Filter, Sort, Nillable **Description** The product description of the related product.
ProductFamily	**Type** picklist **Properties** Filter, Group, Sort, Nillable **Description** The product family of the related product.
PurchaseDate	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Date on which this asset was purchased.
Quantity	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** Quantity purchased or installed. The Quantity field value isn't set by Customer Asset Lifecycle Management. Instead, you can populate the field as you need.
QuantityIncreasePricingType	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update

Field	Details
	Description Specify which pricing type to use when the quantity of this asset is increased. Its UI label is Pricing Type for Quantity Increase. This field is available in API version 56.0 and later. This field is available when Revenue Cloud is enabled. Possible values are: • `LastNegotiatedPrice`—Available in API version 58.0 and later. • `ListPrice`
`RecordTypeId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The unique identifier for the asset. This field is a relationship field. **Relationship Name** RecordType **Relationship Type** Lookup **Refers To** RecordType
`Reliability`	**Type** percent **Properties** Filter, Nillable, Sort **Description** The percentage of expected uptime where the asset wasn't subject to unplanned downtime.
`RenewalPricingType`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The price used when renewing a subscription. Its UI label is Pricing Type for Renewal. This field is available in API version 55.0 and later. This field is available when Revenue Cloud is enabled. Possible values are: • `LastNegotiatedPrice` • `ListPrice`

Field	Details
RenewalTerm	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** With Renewal Term Unit, defines the default subscription term for renewal quotes. This field is available in API version 55.0 and later. This field is available when Revenue Cloud is enabled.
RenewalTermUnit	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The unit of time for a subscription term. This field is available in API version 55.0 and later. This field is available when Revenue Cloud is enabled. Possible values are: • `Annual`—Available in API version 58.0 and later. —UI label is `Years`. • `Months`
SalesStoreId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** ID of the RetailStore or WebStore associated with this Asset. This field is a polymorphic relationship field. To access this field, your org must have a Salesforce Order Management license or a B2B Commerce License. This field is available in API v60.0 and later. **Relationship Name** SalesStore **Relationship Type** Lookup **Refers To** RetailStore, WebStore
SerialNumber	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update

Field	Details
	Description Serial number for this asset.
State	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The state detail for the address.
Status	**Type** picklist **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Customizable picklist of values. The default picklist includes the following values: • Purchased • Shipped • Installed • Registered • Obsolete
StatusReason	**Type** picklist **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The explanation of the device status. This field is available from API version 49.0 and later. Possible values are: • Not Ready • Off • Offline • Online • Paused • Standby
StockKeepingUnit	**Type** string **Properties** Filter, Group, Nillable, Sort

Field	Details
	Description The SKU assigned to the related product.
Street	**Type** textarea **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The street detail for the address.
SumDowntime	**Type** double **Properties** Filter, Nillable, Sort **Description** Accumulated downtime (planned and unplanned), determined as follows: • When only UptimeRecordStart is set, the sum of all downtime from UptimeRecordStart • When UptimeRecordStart and UptimeRecordEnd are set, the sum of all downtime from UptimeRecordStart to UptimeRecordEnd Otherwise, downtime isn't accumulated.
SumUnplannedDowntime	**Type** double **Properties** Filter, Nillable, Sort **Description** Accumulated unplanned downtime, determined as follows: • When only UptimeRecordStart is set, the sum of all unplanned downtime from UptimeRecordStart • When UptimeRecordStart and UptimeRecordEnd are set, the sum of all unplanned downtime from UptimeRecordStart to UptimeRecordEnd Otherwise, unplanned downtime isn't accumulated.
TotalLifecycleAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The total amount of revenue for the asset, including revenue from each stage in the asset lifecycle. This field is available when CPQ Plus, Salesforce Billing, or Revenue Cloud is enabled.

Field	Details
UptimeRecordEnd	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date until which SumDowntime and SumUnplannedDowntime are accumulated.
UptimeRecordStart	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date from which SumDowntime and SumUnplannedDowntime are accumulated.
UsageEndDate	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Date when usage for this asset ends or expires.
Uuid	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The unique ID for the asset. This field is available in API version 49.0 and later.

Usage

Use this object to track products sold to customers. With asset tracking, a client application can quickly determine which products were previously sold or are currently installed at a specific account. You can also create hierarchies of up to 10,000 assets.

For example, suppose that your company wants to renew and upsell opportunities on products sold in the past. Similarly, your company can track competitive products in a customer environment where products can be replaced or swapped out.

Asset tracking is also useful for product support, providing detailed information to assist with product-specific support issues. For example, the PurchaseDate or SerialNumber can indicate whether a given product has certain maintenance requirements, including product recalls. Similarly, the UsageEndDate can indicate when the asset was removed from service or when a license or warranty expires.

If an application creates an Asset record, it must specify a Name and either an AccountId, ContactId, or both.

With REST API, use the `getRelatedListInfo` function to get information about related lists on the asset. Note that when requesting information about *`PrimaryAssets`*, the response is labeled `Related Assets`, and the response for *`RelatedAssets`* is labeled `Primary Assets`.

Associated Objects

This object has the following associated objects. If the API version isn't specified, those objects are available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

AssetChangeEvent (API version 44.0)
Change events are available for the object.

AssetFeed
Feed tracking is available for the object.

AssetHistory
History is available for tracked fields of the object.

AssetOwnerSharingRule
Sharing rules are available for the object.

AssetShare
Sharing is available for the object.

AssetAction

Represents a change made to a lifecycle-managed asset. The fields can't be edited. This object is available in API version 50.0 and later.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`

Special Access Rules

To use Customer Asset Lifecycle Management APIs, you must have the Access Customer Asset Lifecycle Management APIs permission and Read access to the Asset, Asset Action, Asset Action Source, and Asset State Period objects.

Fields

Field	Details
`ActionDate`	**Type** dateTime **Properties** Filter, Sort **Description** The date when an asset action change is recorded. This date can differ from the start date of the related asset state period. For example, suppose that a customer cancels a subscription in June, and the subscription expires in October. The date the customer cancels the

Field	Details
	subscription (June) is the action date of the asset action. The cancellation's effective date (October) is the start date of the asset state period.
ActualTaxChange	**Type** currency **Properties** Filter, Nillable, Sort **Description** The rollup of actual tax from all asset action sources. This field is populated by the system. Label is **Change in Actual Tax**. This field is a calculated field.
AdjustmentAmountChange	**Type** currency **Properties** Filter, Nillable, Sort **Description** The rollup of adjustment amount from all asset action sources. This field is populated by the system. Label is **Change in Adjustment Amount**. This field is a calculated field.
Amount	**Type** currency **Properties** Filter, Sort **Description** The delta in the total asset amount resulting from an asset action.
AssetActionNumber	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The ID of the asset action. Label is **Name**.
AssetId	**Type** reference **Properties** Filter, Group, Sort **Description** The ID of the related lifecycle-managed asset. Label is **Asset**.

Field	Details
	This field is a relationship field. **Relationship Name** Asset **Relationship Type** Lookup **Refers To** Asset
`CanRollBack`	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Indicates whether the last asset action can be rolled back (`true`). If this property is set to `false`, the asset and the last asset action can't be rolled back. The default value is `false`. This field is available in API version 65.0 and later.
`CategoryEnum`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The business category of the asset action, for use in reporting. Asset action totals are broken out by the picklist values on this required field, and those totals are in turn reflected on assets. These categories are available and aren't customizable. Label is **Business Category**. Possible values are: • `Cancellations` • `Cross-Sells` • `Downgrades` Indicates a transition to a lower-level version or tier of an asset. • `Downsells` Indicates a negative quantity amendment or a decreased Line Item total price with no change in quantity. • `Initial Sale` Indicates that the asset is initially purchased by an account. • `Other` • `Renewals` • `Swaps` Indicates the exchange of one asset for another. Applies to both swapped-out and swapped-in actions. • `Terms And Conditions Changes` • `Transfers` Indicates that an asset is transferred from one account to another. • `Upgrades` Indicates a transition to a higher-level version or tier of an asset. • `Upsells` Indicates a positive quantity amendment or an increased Line Item total price with no change in quantity.

Field	Details
`EstimatedTaxChange`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The rollup of estimated tax from all asset action sources. This field is populated by the system. Label is **Change in Estimated Tax**. This field is a calculated field.
`MrrChange`	**Type** currency **Properties** Filter, Sort **Description** The delta in the asset's monthly recurring revenue resulting from an asset action. For example, suppose that the MRR during an asset state period is $200 and the next asset action adds $100. Then this field's value is $100. Label is **Change in Monthly Recurring Revenue**.
`ProductAmountChange`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The rollup of product amount from all asset action sources. This field is populated by the system. Label is **Change in Product Amount**. This field is a calculated field.
`QuantityChange`	**Type** double **Properties** Filter, Sort **Description** The delta in the asset quantity resulting from an asset action. For example, suppose that the asset quantity during an asset state period is 20 and the next asset action adds 10. Then this field's value is 10. Label is **Change in Quantity**.
RolledbackAssetAction	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update

Field	Details
	Description The last asset action rolled back in the current rollback transaction. This field is available in API version 65.0 and later.
`SubtotalChange`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The rollup of subtotal from all asset action sources. This field is populated by the system. Label is **Change in Subtotal**. This field is a calculated field.
`Subtype`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The subtype of the action on the asset. Valid values are: • `DowngradeFrom`—Available in API version 66.0 and later. • `DowngradeTo`—Available in API version 66.0 and later. • `FieldAmendment` • `Rollback` • `StartDateAdjustment` • `SwapIn`—Available in API version 66.0 and later. • `SwapOut`—Available in API version 66.0 and later. • `TransferFrom` • `TransferTo` • `UpgradeFrom`—Available in API version 66.0 and later. • `UpgradeTo`—Available in API version 66.0 and later. This field is available in API version 65.0 and later.
`TotalAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of the current and previous asset action amount. This field is populated by the system.

Field	Details
	This field is a calculated field.
`TotalCancellationsAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of current and previous asset action amounts categorized as `Cancellations`. This field is populated by the system.
`TotalCrossSellsAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of current and previous asset action amounts categorized as `Cross-Sells`. This field is populated by the system.
`TotalDowngradesAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of current and previous asset action amounts categorized as `Downgrades`. This field is populated by the system and is available in API version 66.0 and later.
`TotalDownsellsAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of current and previous asset action amounts categorized as `Downsells`. This field is populated by the system.
`TotalInitialSaleAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of current and previous asset action amounts categorized as `Initial Sale`. This field is populated by the system.

Field	Details
TotalMrr	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of the monthly recurring revenue for the current and previous asset action. This field is populated by the system. Label is **Total Monthly Recurring Revenue**.
TotalOtherAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of current and previous asset action amounts categorized as `Other`. This field is populated by the system.
TotalQuantity	**Type** double **Properties** Filter, Nillable, Sort **Description** The sum of the changes in quantity for the current and previous asset action. This field is populated by the system.
TotalRenewalsAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of current and previous asset action amounts categorized as `Renewals`. This field is populated by the system.
TotalSwapsAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of current and previous asset action amounts categorized as `Swaps`. This field is populated by the system and is available in API version 66.0 and later.
TotalTermsAndConditionsAmount	**Type** currency

Field	Details
	Properties Filter, Nillable, Sort **Description** The sum of current and previous asset action amounts categorized as `Terms and Conditions Changes`. This field is populated by the system. Label is **Total Terms and Conditions Changes Amount**.
`TotalTransfersAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of current and previous asset action amounts categorized as `Transfers`. This field is populated by the system.
`TotalUpgradesAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of current and previous asset action amounts categorized as `Upgrades`. This field is populated by the system and is available in API version 66.0 and later.
`TotalUpsellsAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of current and previous asset action amounts categorized as `Upsells`. This field is populated by the system.
`Type`	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** The REST API used to generate the asset action. This field is populated by the system. Valid values are: • `Cancel` • `Change` • `Convert`

Field	Details
	• `Generate`

AssetActionSource

Represents an optional way to record what transactions caused changes to lifecycle-managed assets. Use it to trace financial and other information about asset actions. This object supports Salesforce order products and work order line items, and transaction IDs from other systems. The fields can't be edited. This object is available in API version 50.0 and later.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`.

Special Access Rules

To use Customer Asset Lifecycle Management APIs, you must have the Access Customer Asset Lifecycle Management APIs permission and Read access to the Asset, Asset Action, Asset Action Source, and Asset State Period objects.

Fields

Field	Details
`ActualTax`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The region-specific tax amount determined at time of the order. This field is not used for price and tax calculations.
`AdjustmentAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** An adjustment to the product amount, such as a discount.
`AssetActionId`	**Type** reference **Properties** Filter, Group, Sort **Description** The related asset action, that is, the change caused by an asset action source transaction.

Field	Details
	This field is a relationship field. **Relationship Name** AssetAction **Relationship Type** Lookup **Refers To** AssetAction
AssetActionSourceNumber	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The ID of the asset action source. Label is **Name**.
BillingReference	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The ID of the OrderItem or OrderItemDetail record that this AssetActionSource record is created for.
Discount	**Type** percent **Properties** Filter, Nillable, Sort **Description** The discount, expressed as a percentage, that's applied to the asset. This field is available in API version 62.0 and later.
DiscountAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The discount, expressed as currency, that's applied to the asset. This field is available in API version 62.0 and later.
EffectiveGrantDate	**Type** dateTime

Field	Details
	Properties Filter, Nillable, Sort **Description** The date when the resources associated with the asset were granted. This field is available in orgs that have Revenue Cloud when Rate Management is enabled. This field is available in API version 62.0 and later.
`EndDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The end date of the service or change.
`EstimatedTax`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The estimate of the region-specific tax amount made at time of the transaction.
`ExternalReference`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The ID of an asset action source transaction originating in a system outside of Salesforce.
`ExternalReferenceDataSource`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** A system outside of Salesforce that contains asset action source transactions.
`LegalEntityId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the legal entity record associated with the asset action source transaction.

Field	Details
	This field is a relationship field. This field is available in API version 62.0 and later. **Relationship Name** LegalEntity **Relationship Type** Lookup **Refers To** LegalEntity
ListPrice	**Type** currency **Properties** Filter, Nillable, Sort **Description** List price for the order product. Value is inherited from the associated PriceBookEntry upon order product creation.
NetUnitPrice	**Type** currency **Properties** Filter, Nillable, Sort **Description** The final adjusted unit price, inclusive of all adjustments, but exclusive of tax. The unit price after all price adjustments are applied.
ObligatedAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** When a line amount is prorated, this amount shows the service amount that's been consumed.
OriginalLineNumber	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The number of the original order item detail line. Salesforce uses this information to create a record to amend, renew, or cancel an order. This field is available in API version 64.0 and later. **Relationship Name** OrderItemDetail

Field	Details
	Relationship Type Lookup **Refers To** LineNumber
PeriodBoundary	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** Boundary delimiters for periods. It determines when a period starts and/or ends. Valid values are: • AlignToCalendar • Anniversary • DayOfPeriod • LastDayOfPeriod
PeriodBoundaryDay	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The number specifying the day number when Period Boundary is a specific day in a week/month/year. It only applies when PeriodBoundary is set to "day of period."
PeriodBoundaryStartMonth	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** Field is populated based on input in the StartDate, PeriodBoundary, and PeriodBoundaryDay when BillingFrequency2 is Annual or by manual user entry. Possible values are: 1-January 2-February 3-March 4-April 5-May 6-June 7-July 8-August

Field	Details
	9-September 10-October 11-November 12-December
PricebookEntryId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** PricebookEntry is used as a lookup for price information in order to pre-populate OrderItem's ListPrice and UnitPrice.
PricingTermCount	**Type** double **Properties** Filter, Nillable, Sort **Description** Number of pricing terms is this subscription product.
ProductAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The product amount after the asset action source transaction.
ProductSellingModelId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** Specifies the product selling model type. Foreignkey to ProductSellingModel entity.
ProrationPolicyId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the ProrationPolicy used for pricing.

Field	Details
`Quantity`	**Type** double **Properties** Filter, Nillable, Sort **Description** The product quantity or the change in product quantity after the asset action source transaction.
`ReferenceEntityItemId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of an asset action source transaction originating in Salesforce. The transaction can be an order product or a work order line item. This field is a polymorphic relationship field. **Relationship Name** ReferenceEntityItem **Relationship Type** Lookup **Refers To** OrderItem, WorkOrderLineItem
`SegmentIdentifier`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The ID of the ramp segment associated with the asset action source transaction. The maximum supported length is 255 characters from API version 67.0 and later. This field is available in API version 62.0 and later.
`StartDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The start date of the service or change.
`Subtotal`	**Type** currency

Field	Details
	Properties Filter, Nillable, Sort **Description** The sum of the product amount and the adjustment amount. This field is a calculated field.
TaxTreatmentId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** Lookup to Tax Treatment entity. It's used to calculate tax.
TotalLineAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The price of the line before any price adjustments were applied. SalesTransactionItem: ProratedStartingTotal / StartingPriceTotal. Note: TotalPrice is computed using the UnitPrice, which includes discounts (price adjustments), while TotalLineAmount doesn't include price adjustments.
TotalPrice	**Type** currency **Properties** Filter, Nillable, Sort **Description** Calculated by the pricing engine for ARC. Summation of TotalAdjustmentAmount plus TotalLineAmount for this item.
TransactionDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date of a source transaction, such as an order date.
UnitPrice	**Type** currency **Properties** Filter, Nillable, Sort

Field	Details
	Description The unit price of the item before any discounts or tax calculation.

AssetActionSrcPriceAdjustment

Each row represents a junction between an asset and the calculated price adjustment that's applied to an asset. This object is available in API version 66.0 and later.

Supported Calls

`create()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`

Fields

Field	Details
`AdjustmentValue`	**Type** double **Properties** Create, Filter, Nillable, Sort **Description** The value of the price adjustment that was applied to the asset.
`AppliedPriceAdjustmentDate`	**Type** dateTime **Properties** Create, Filter, Sort **Description** The date and time on which the price adjustment was applied to the asset.
`AssetActionSourceId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The ID of the asset action source related to the asset for which the price adjustment was applied. This field is a relationship field. **Relationship Name** AssetActionSource

Field	Details
	Relationship Type Master-detail **Refers To** AssetActionSource (the master object)
AssetId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the asset that this price adjustment applies to. This field is a relationship field. **Relationship Name** Asset **Refers To** Asset
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The name of the asset price adjustment record.
PriceAdjustmentCauseId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The ID of the record that caused the price adjustment. This field is a polymorphic relationship field. **Relationship Name** PriceAdjustmentCause **Refers To** Promotion
PriceAdjustmentSource	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort

Field	Details
	Description Specifies the source of price adjustment. Possible values are: • `Discretionary`—Reserved for future use. • `Promotion` • `Rule`—Reserved for future use. • `System`—Reserved for future use.
`PrioritySequence`	**Type** int **Properties** Create, Filter, Group, Nillable, Sort **Description** The priority sequence of the applied price adjustment.

AssetContractRelationship

Represents a relationship between an asset and a contract. This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `update()`, `upsert()`

Special Access Rules

This object is available in Enterprise, Unlimited, and Developer Editions of Revenue Cloud.

Fields

Field	Details
`AssetId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The ID of the asset related to the contract. This field is a relationship field. **Relationship Name** Asset

Field	Details
	Relationship Type Lookup **Refers To** Asset
`ContractId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The ID of the contract related to the asset. This field is a relationship field. **Relationship Name** Contract **Relationship Type** Lookup **Refers To** Contract
`EndDate`	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** The end date and time of the relationship between contract and asset.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed this record, a record related to this record, or a list view. The associated UI label is **Last Modified Date**.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed this record or list view. If this value is null, the user accessed this record or list view (`LastReferencedDate`) but didn't view it.

Field	Details
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The auto-generated number assigned to AssetContractRelationship. (Read Only)
`StartDate`	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** The start date and time of the relationship between contract and asset.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

AssetContractRelationshipFeed
Feed tracking is available for the object.

AssetContractRelationshipHistory
History is available for tracked fields of the object.

AssetDowntimePeriod

Represents a period during which an asset is not able to perform as expected. Downtime periods include planned activities, such as maintenance, and unplanned events, such as mechanical breakdown. This object is available in API version 49.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`AssetDowntimePeriodNumber`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort

Field	Details
	Description The unique number of this asset downtime period record.
`AssetId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The ID of the asset this asset downtime period record is for.
`Description`	**Type** textarea **Properties** Create, Nillable, Update **Description** The description of this asset downtime period.
`DowntimeType`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The type of this asset downtime period. Possible values are: • `Planned` • `Unplanned`
`EndTime`	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** The time this asset downtime period ended.
`IsExcluded`	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Whether this asset downtime period is excluded from the calculation of accumulated downtime and accumulated unplanned downtime, and therefore not included in availability and reliability calculations.

Field	Details
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed a record related to this record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed this record. If this value is null, this record might only have been referenced (LastReferencedDate) and not viewed.
StartTime	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** The time this asset downtime period started.

AssetOwnerSharingRule

Represents the rules for sharing an Asset with users other than the owner. This object is available in API version 33.0 and later.

Note: To enable access to this object for your org, contact Salesforce customer support. However, we recommend that you instead use Metadata API to programmatically update owner sharing rules because it triggers automatic sharing rule recalculation. The SharingRules Metadata API type is enabled for all orgs.

Supported Calls

create(), delete(), describeSObjects(), getDeleted(), getUpdated(), query(), retrieve(), update(), upsert()

Special Access Rules

Customer Portal users can’t access this object.

Fields

Field	Details
AssetAccessLevel	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** A value that represents the type of sharing being allowed. The possible values are: • Read • Edit
Description	**Type** textarea **Properties** Create, Filter, Nillable, Sort, Update **Description** A description of the sharing rule. Maximum size is 1000 characters.
DeveloperName	**Type** string **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The unique name of the object in the API. This name can contain only underscores and alphanumeric characters, and must be unique in your org. It must begin with a letter, not include spaces, not end with an underscore, and not contain two consecutive underscores. In managed packages, this field prevents naming conflicts on package installations. With this field, a developer can change the object's name in a managed package and the changes are reflected in a subscriber's organization. Corresponds to **Rule Name** in the user interface. Note: When creating large sets of data, always specify a unique DeveloperName for each record. If no DeveloperName is specified, performance may slow while Salesforce generates one for each record.
GroupId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The ID representing the source group. Cases owned by users in the source group trigger the rule to give access.

Field	Details
Name	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** Label of the sharing rule as it appears in the user interface. Limited to 80 characters. Corresponds to **Label** on the user interface.
UserOrGroupId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The ID representing the target user or group. Target users or groups are given access.

Usage

Use this object to manage the sharing rules for assets. General sharing uses this object.

AssetRateAdjustment

Stores the tier rate adjustments for the asset rate card entries.This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`

Special Access Rules

This object is available in orgs where Revenue Cloud is enabled.

Fields

Field	Details
AdjustmentType	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update

Field	Details
	Description The type of rate adjustment. Valid values are: • `Amount`—Adjusts rate by using a specific amount. • `Override`—Adjusts rate by using the override rate. • `Percentage`—Adjusts rate by using a percentage.
`AdjustmentValue`	**Type** double **Properties** Create, Filter, Sort, Update **Description** The value of the adjustment.
`AssetRateCardEntryId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The ID of the parent asset rate card entry record associated with the asset rate adjustment. This field is a relationship field. **Relationship Name** AssetRateCardEntry **Relationship Type** Master-detail **Refers To** AssetRateCardEntry (the master object)
`LowerBound`	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The minimum quantity for the adjustment to be applicable.
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The name of the asset rate adjustment.

Field	Details
UpperBound	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The maximum quantity for the adjustment to be applicable.

AssetRateCardEntry

Stores the negotiated rate card entries that are associated with an asset in Revenue Cloud. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`

Special Access Rules

This object is available in orgs where Revenue Cloud is enabled.

Fields

Field	Details
AssetId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The ID of the asset rate card entry record. This field is a relationship field. **Relationship Name** Asset **Relationship Type** Master-detail **Refers To** Asset (the master object)

Field	Details
BindingObjectFormula	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The formula that returns the ID of the associated binding object, if specified. If binding object isn't added, the formula returns the asset ID of the asset related to this asset rate card entry. This field is read-only. Available in API version 65.0 and later.
BindingObjectId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the binding object associated with the asset rate card entry. Available in API version 65.0 and later. This field is a relationship field. **Relationship Name** BindingObject **Refers To** Asset
BindingObjectRateOrder	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The order that determines the applicable binding object rate when multiple rates are defined for an Anchor binding object within a effective period. Available in API version 65.0 and later.
CurrencyIsoCode	**Type** picklist **Properties** Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The ID of the binding object associated with the asset rate card entry. Possible values are: • AED - UAE Dirham • AUD - Australian Dollar • BRL - Brazilian Real • CAD - Canadian Dollar

Field	Details
	• EUR - Euro • GBP - British Pound • INR - Indian Rupee • JPY - Japanese Yen • SEK - Swedish Krona • USD - U.S. Dollar The default value is USD. Available in API version 65.0 and later.
`EndDate`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date when the rate card's time period becomes inactive. The rate card becomes inactive at 11:59:00 PM on the end date.
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** An auto-generated number assigned to the asset rate card entry. Read-only.
`NegotiatedRate`	**Type** double **Properties** Create, Filter, Sort, Update **Description** The base negotiated rate used to charge overage consumption.
`RateCardEntryId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the rate card entry record containing the catalog rates that's associated with the asset rate card entry. This field is a relationship field. **Relationship Name** RateCardEntry

Field	Details
	Refers To RateCardEntry
`RateCardId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the rate card record that's associated with the asset rate card entry. This field is a relationship field. **Relationship Name** RateCard **Refers To** RateCard
`RateUnitOfMeasureId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the unit of measure record that's associated with the asset rate card entry. This field is a relationship field. **Relationship Name** RateUnitOfMeasure **Refers To** UnitOfMeasure
`StartDate`	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** The date when the rate card's time period becomes active. The rate card becomes active at 12:00:00 AM on the start date.
`UsageResourceId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the usage resource record that's associated with the asset rate card entry.

Field	Details
	This field is a relationship field. **Relationship Name** UsageResource **Refers To** UsageResource

AssetRelationship

Represents a non-hierarchical relationship between assets due to an asset modification; for example, a replacement, upgrade, or other circumstance. In Revenue Lifecycle Management, this object represents an asset or assets grouped in a bundle or set. This object is available in API version 41.0 and later.

Asset relationships appear in the Primary Assets and Related Assets related lists on asset records in the UI.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

Some fields are available only in Revenue Cloud. Field availability is noted in the field detail column.

Fields

Field Name	Details
`AssetId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The unique identifier of the new asset, which is the asset that is taking the place of the existing asset. This field is a relationship field. **Relationship Name** Asset **Relationship Type** Lookup **Refers To** Asset

Field Name	Details
AssetRelationshipNumber	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** An auto-generated number identifying the asset relationship.
AssetRole	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** Describes the position of the main asset relative to the other assets in the relationship. This field is available in API version 58.0 and later. This field is available in orgs with Revenue Cloud. Possible values are: • `Add-on`—The main asset is an add-on. • `Bundle`—The main asset is the bundle parent. • `Set`—The asset is the main asset in the set.
CurrencyIsoCode	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Three-letter ISO 4217 currency code associated with the asset. The default value is USD.
FromDate	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date when the new asset was installed.
GroupingKey	**Type** string **Properties** Filter, Group, idLookup, Nillable, Sort

Field Name	Details
	Description Read-only field used to indicate the bundle that an asset belongs to. For example, if two assets have the same GroupingKey value, then it means that the assets are bundled together. This field is available in API v.60.0 and later. This field is available in orgs with Revenue Cloud.
`ProductRelationshipTypeId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The unique identifier of the record that describes the relationship between the main and associated assets. This field is available in API version 58.0 and later. This field is available in orgs with Revenue Cloud. This field is a relationship field. **Relationship Name** ProductRelationshipType **Relationship Type** Lookup **Refers To** ProductRelationshipType
`ProductRelatedComponent`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The product related component that's associated with the asset relationship. This field is a relationship field. This field is available in API 60.0 and later in Revenue Cloud. **Relationship Name** ProductRelatedComponent **Relationship Type** Lookup **Refers To** ProductRelatedComponent
`RelatedAssetId`	**Type** reference

Field Name	Details
	Properties Create, Filter, Group, Sort, Update **Description** The existing asset that is being modified. This field is a relationship field. **Relationship Name** RelatedAsset **Relationship Type** Lookup **Refers To** Asset
`RelatedAssetPricing`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** Specifies whether the price of the related asset is included in the bundle price. Valid values are: • `IncludedInBundlePrice` • `NotIncludedInBundlePrice` This field is available in API version 59.0 and later in Revenue Cloud.
`RelatedAssetQtyScaleMethod`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** Specifies how the quantity of the related asset changes relative to the quantity of the parent asset. Valid values are: • `Constant` • `Proportional` This field is available in API version 59.0 and later in Revenue Cloud.
`RelatedAssetRole`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort

Field Name	Details
	Description Describes the position of the associated asset relative to other assets in the relationship. This field is available in API version 58.0 and later. This field is available in orgs with Revenue Cloud. Valid values are: • `Add-on`—The main asset is an add-on. • `Bundle`—The main asset is the bundle parent. • `Set`—The asset is the main asset in the set. • `Simple`—The asset is purchased individually and isn't associated with variations. • `Variation Parent`——The main asset is the variation parent.
`RelationshipType`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Sort, Update **Description** The type of relationship between the existing asset and the new asset. This field comes with three values—Replacement, Upgrade, and Crossgrade—, but you can create more values in Setup. Possible values are: • `Crossgrade`—The new asset is a crossgrade of an existing asset. For example, changing a subscription to a plan with the same service, but that runs for a longer amount of time. • `Replacement`—The new asset is replacing an existing asset. For example, a customer's faulty widget that was under warranty is being replaced with a new one. • `Upgrade`—The new asset is an upgrade of an existing asset. For example, upgrading a customer's existing subscription plan to a new plan with more services. The default value is `Replacement`.
`ToDate`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date when the modified asset is uninstalled.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

AssetRelationshipChangeEvent (API version 62.0)
Change events are available for the object.

AssetRelationshipFeed
Feed tracking is available for the object.

AssetRelationshipHistory
History is available for tracked fields of the object.

AssetRelationshipOwnerSharingRule (API version 58.0)
Sharing rules are available for the object.

AssetRelationshipShare (API version 58.0)
Sharing is available for the object.

AssetShare

Represents a sharing entry on an Asset. This object is available in API version 33.0 and later.

You can only create, edit, and delete sharing entries for standard objects whose `RowCause` field is set to `Manual`. Sharing entries for standard objects with different `RowCause` values are created as a result of your Salesforce org's sharing configuration and are read-only. For some sharing mechanisms, such as sharing sets, sharing entries aren't stored at all.

Note: While Salesforce currently maintains read-only sharing entries for multiple sharing mechanisms, it's possible that we'll stop storing certain share records to improve performance. As a best practice, don't create customizations that rely on the availability of these sharing entries. Any changes to sharing behavior will be communicated before they occur.

Supported Calls

`describeSObjects()`, `query()`, `retrieve()`

Special Access Rules

Customer Portal users can't access this object.

Fields

The properties available for some fields depend on the default organization-wide sharing settings. The properties listed are true for the default settings of such fields.

Field	Details
`AssetAccessLevel`	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort

Field	Details
	Description Level of access that the User or Group has to the Asset. The possible values are: • `Read` • `Edit` • `All` This value is not valid for creating or deleting records. This field must be set to an access level that is higher than the organization's default access level for cases.
`AssetId`	**Type** reference **Properties** Filter, Group, Sort **Description** ID of the Asset associated with this sharing entry. This field can't be updated. This is a relationship field. **Relationship Name** Asset **Relationship Type** Lookup **Refers To** Asset
`IsDeleted`	**Type** boolean **Properties** Defaulted on create, Filter **Description** Indicates whether the object has been moved to the Recycle Bin (`true`) or not (`false`). Label is **Deleted**.
`RowCause`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** Reason that this sharing entry exists. If you're creating a sharing entry, the only permitted value is `Manual`. If no value is specified, the field defaults to `Manual`. All other `RowCause` values are read-only. After the sharing entry is created, this field can't be edited. Valid values include: • `Manual`—The User or Group has access because a user with "All" access manually shared the Asset with them.

Field	Details
	• `Owner`—The User is the owner of the Asset. • `Rule`—The User or Group has access via an Asset sharing rule. • `GuestRule`—The User or Group has access via an Asset guest user sharing rule. • `LpuImplicit`—The User has access to records owned by high-volume Experience Cloud site users via a share group.
`UserOrGroupId`	**Type** reference **Properties** Filter, Group, Sort **Description** ID of the User or Group that has been given access to the Asset. This field can't be updated. This is a polymorphic relationship field. **Relationship Name** UserOrGroup **Relationship Type** Lookup **Refers To** Group, User

Usage

This object allows you to determine which users and groups can view and edit Asset records owned by other users.

If you attempt to create a new record that matches an existing record, request updates any modified fields and returns the existing record.

AssetStatePeriod

Represents a time span when an asset has the same quantity, amount, and monthly recurring revenue (MRR). An asset has as many asset state periods as there are changes to it (asset actions) during its lifecycle. The dashboard and related pages show the current asset state period. The fields can't be edited. This object is available in API version 50.0 and later.

Supported Calls

`createable()`, `deletable()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `updateable()`.

Special Access Rules

To use Customer Asset Lifecycle Management APIs, you must have the Access Customer Asset Lifecycle Management APIs permission and Read access to the Asset, Asset Action, Asset Action Source, and Asset State Period objects.

Fields

Field	Details
`Amount`	**Type** currency **Properties** Createable, Filter, Sort, Updateable **Description** An asset's total amount during an asset state period. Revenue Cloud doesn't set or use this field's value currently.
`AssetId`	**Type** reference **Properties** Createable, Filter, Group, Sort **Description** The asset related to an asset state period. Label is **Asset**. This field is a relationship field. **Relationship Name** Asset **Relationship Type** Lookup **Refers To** Asset
`AssetStatePeriodNumber`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The ID of the asset state period. Label is **Name**.
`BillingFrequency`	**Type** picklist **Properties** Createable, Filter, Group, Nillable, Restricted picklist, Sort, Updateable **Description** The time period that indicates how often the line item is billed. Possible values are: • `Annual` • `Monthly` • `Quarterly`

Field	Details
	• Semi-Annual Available in API version 65.0 and later.
BindingInstanceTargetId	**Type** reference **Properties** Createable, Filter, Group, Nillable, Sort, Updateable **Description** The ID of a custom product target for a usage-based quote line item, order Item, or asset allocation. This field is a polymorphic relationship field. **Relationship Name** BindingInstanceTarget **Refers To** Account, Asset, BindingObjectCustomExt, Contract
Discount	**Type** percent **Properties** Createable, Filter, Nillable, Sort, Updateable **Description** Editable number from 0 to 100. Available in API version 65.0 and later.
DiscountAmount	**Type** currency **Properties** Createable, Filter, Nillable, Sort, Updateable **Description** The fixed amount discount to apply to the line item. Available in API version 65.0 and later.
EndDate	**Type** dateTime **Properties** Createable, Filter, Nillable, Sort, Updateable **Description** The end date and time of an asset state period. On an asset that is an evergreen subscription, the last asset state period has no end date.
LegalEntityId	**Type** reference

Field	Details
	Properties Createable, Filter, Group, Nillable, Sort, Updateable **Description** The ID of the related legal entity. This field is a relationship field. **Relationship Name** LegalEntity **Refers To** LegalEntity
`Mrr`	**Type** currency **Properties** Createable, Filter, Sort, Updateable **Description** An asset's monthly recurring revenue during an asset state period.
`PriceRevisionPolicy`	**Type** reference **Properties** Createable, Filter, Group, Sort, Updateable **Description** Specifies the price uplift policy associated with this asset state period. This field is a relationship field. This field is available in API version 65.0 and later. **Relationship Name** Price Revision Policy **Relationship Type** Lookup **Refers To** PriceRevisionPolicy
`Quantity`	**Type** double **Properties** Createable, Filter, Sort, Updateable **Description** The total quantity of an asset during an asset state period.
`RampIdentifier`	**Type** string

Field	Details
	Properties Filter, Group, Nillable, Sort **Description** The ramp record used to group order item segments for this asset state period. This field is available in orgs that have Revenue Cloud when the Ramp Deals setting is enabled. The maximum supported length is 255 characters from API version 67.0 and later. This field is available in API version 62.0 and later.
`SegmentIdentifier`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The order item segment for this asset state period. This field is available in orgs that have Revenue Cloud when the Ramp Deals setting is enabled. The maximum supported length is 255 characters from API version 67.0 and later. This field is available in API version 62.0 and later.
`SegmentName`	**Type** string **Properties** Createable, Filter, Group, Nillable, Sort, Updateable **Description** The name of the order item segment for this asset state period. This field is available in orgs that have Revenue Cloud when the Ramp Deals setting is enabled. The maximum supported length is 255 characters from API version 67.0 and later. This field is available in API version 62.0 and later.
`SegmentType`	**Type** picklist **Properties** Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Updateable **Description** The period for the order item segment for this asset state period. Valid values are: • `Custom` • `Free Trial` • `Yearly` The default value is `Yearly`. This field is available in orgs that have Revenue Cloud when the Ramp Deals setting is enabled.

Field	Details
	This field is available in API version 62.0 and later.
StartDate	**Type** dateTime **Properties** Createable, Filter, Sort, Updateable **Description** The start date and time of an asset state period.
UnitPrice	**Type** currency **Properties** Createable, Filter, Nillable, Sort, Updateable **Description** The price per unit for the line item. Available in API version 65.0 and later. Revenue Cloud won't populate this field in API version 66.0 and later.
UnitPriceUplift	**Type** percent **Properties** Createable, Filter, Nillable, Sort, Updateable **Description** Indicates the percentage increase of a line item's unit price. Available in API version 65.0 and later.

AssetStatePeriodAttribute

Represents a virtual object that holds the key-value pair of the asset attribute in a specified asset state period. This object is a child object of AssetStatePeriod. This object is available in API version 60.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`describeLayout()`, `describeSObjects()`, `query()`, `retrieve()`

Special Access Rules

This object is available in Enterprise, Unlimited, and Developer Editions of Revenue Cloud with the Access Lifecycle-Managed Assets user permission. This object is editable only through API and not the UI.

Fields

Field	Details
AssetStatePeriodId	**Type** reference **Properties** Filter, Group, Sort **Description** The asset state period that's associated with the asset attribute. This field is a relationship field. **Relationship Name** AssetStatePeriod **Relationship Type** Master-detail **Refers To** AssetStatePeriod (the master object)
AttributeDefinitionId	**Type** reference **Properties** Filter, Group, Sort **Description** The attribute definition that's associated with the asset state period attribute. This field is a relationship field. **Relationship Name** AttributeDefinition **Relationship Type** Lookup **Refers To** AttributeDefinition
AttributeName	**Type** string **Properties** Filter, Group, idLookup, Nillable, Sort **Description** The name of the asset attribute.
AttributePicklistValueId	**Type** reference **Properties** Filter, Group, Nillable, Sort

Field	Details
	Description The value specified in the picklist type field that corresponds to the attribute in the AttributePicklistValue object. This field is a relationship field. **Relationship Name** AttributePicklistValue **Relationship Type** Lookup **Refers To** AttributePicklistValue
`AttributeValue`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The value of the asset state period attribute. For example, a shirt can have the value of `blue`, which indicates the shirt's color, or it can have the value of `small`, which indicates the shirt's size. You can use this field to filter records only if the DataType value in the related `AttributeDefinitionId` record is `Text`. If the DataType value is `Picklist`, use the value in the `AttributePicklistValueId` field for filtering. You can't use this field to filter records if the DataType value is `Checkbox`, `Currency`, `Date`, `Datetime`, `Multipicklist`, `Number`, or `Percent`.

Usage

This object doesn't support custom fields, validation rules, or triggers. In SOQL queries, you can filter records by using `Id` and `AttributeDefinition`. You can't use `AttributeValue` in the `WHERE` clause.

AssetTag

Associates a word or short phrase with an Asset.

Supported Calls

`create()`, `delete()`, `describeSObjects()`, `query()`, `retrieve()`

Fields

Field Name	Details
`ItemId`	**Type** reference **Properties** Create, Filter **Description** ID of the tagged item.
`Name`	**Type** string **Properties** Create, Filter **Description** Name of the tag. If this value does not already exist, a new TagDefinition is created and becomes the parent of this Tag object. Otherwise, a TagDefinition with the same name becomes the parent of this Tag object. Parent relationships are created automatically.
`TagDefinitionId`	**Type** reference **Properties** Filter **Description** ID of the parent TagDefinition object that owns the tag.
`Type`	**Type** picklist **Properties** Create, Filter, Restricted picklist **Description** Defines the visibility of a tag. Valid values: • `Public`—The tag can be viewed and manipulated by all users in an organization. • `Personal`—The tag can be viewed or manipulated only by a user with a matching `OwnerId`.

Usage

AssetTag stores the relationship between its parent TagDefinition and the Asset being tagged. Tag objects act as metadata, allowing users to describe and organize their data.

When a tag is deleted, its parent TagDefinition will also be deleted if the name is not being used; otherwise, the parent remains. Deleting a TagDefinition sends it to the Recycle Bin, along with any associated tag entries.

AssetTokenEvent

The documentation has moved to AssetTokenEvent in the *Platform Events Developer Guide*.

AssetWarranty

Defines the warranty terms applicable to an asset along with any exclusions and extensions. This object is available in API version 50.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`AssetId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The ID of the asset this warranty term applies to.
`AssetWarrantyNumber`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The identifier of the asset warranty record.
`EndDate`	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The date on which this warranty term expires.
`ExchangeType`	**Type** picklist

Field	Details
	Properties Create, Filter, Group, Nillable, Sort, Update **Description** The type of exchange offered by this warranty term.
Exclusions	**Type** textarea **Properties** Create, Nillable, Update **Description** Description of any exclusions.
ExpensesCovered	**Type** percent **Properties** Create, Filter, Nillable, Sort, Update **Description** The percentage of expenses covered.
ExpensesCoveredEndDate	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The date on which cover for expenses ends.
IsTransferable	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Defines whether the warranty term can be transferred to a new owner.
LaborCovered	**Type** percent **Properties** Create, Filter, Nillable, Sort, Update **Description** The percentage of labor covered.
LaborCoveredEndDate	**Type** date

Field	Details
	Properties Create, Filter, Group, Nillable, Sort, Update **Description** The date on which cover for labor ends.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when the asset warranty term was last modified. Its label in the user interface is `Last Modified Date`.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when the asset warranty term was last viewed.
`PartsCovered`	**Type** percent **Properties** Create, Filter, Nillable, Sort, Update **Description** The percentage of parts covered.
`PartsCoveredEndDate`	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The date on which cover for parts ends.
`Pricebook2Id`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the price book item associated with this asset warranty term.

Field	Details
`StartDate`	**Type** date **Properties** Create, Filter, Group, Sort, Update **Description** The date on which cover under this warranty term starts.
`WarrantyTermId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the warranty term this asset warranty term extends.
`WarrantyType`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The type of the warranty.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

AssetWarrantyChangeEvent
Change events are available for the object.

BindingObjUsageRsrcPlcy

Represents the policies that are used for the usage resource that's associated with an asset or a binding object. This object is available in API version 65.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
BindingObjectId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The object that's bounded with the quote line policy or order policy. This field is a polymorphic relationship field. **Relationship Name** BindingObject **Refers To** Account, Asset, BindingObjectCustomExt, Contract
DrawdownOrder	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies the order that's used to debit consumption of entitlements related to the usage resource from the usage entitlement bucket. Valid values are: • ExpiringFirst • GrantedFirst • GrantedLast
EffectiveEndDate	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date and time until when the policy remains effective.
EffectiveStartDate	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** The date and time when the policy becomes effective.

Field	Details
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when this record was last referenced.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed a record related to this record.
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The auto-generated identifier for the quote line item usage resource policy record. For example, BOURP-000004.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the owner of the binding object usage resource policy. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
RatingFrequencyPolicyId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The rating frequency policy associated with the usage resource. This field is a relationship field.

Field	Details
	Relationship Name RatingFrequencyPolicy **Refers To** RatingFrequencyPolicy
UsageAggregationPolicyId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The usage aggregation policy associated with the usage resource. This field is a relationship field. **Relationship Name** UsageAggregationPolicy **Refers To** UsageResourceBillingPolicy
UsageCommitmentPolicyId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The usage commitment policy associated with the usage resource. This field is a relationship field. **Relationship Name** UsageCommitmentPolicy **Refers To** UsageCommitmentPolicy
UsageOveragePolicyId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The usage overage policy associated with the usage resource. This field is a relationship field. **Relationship Name** UsageOveragePolicy **Refers To** UsageOveragePolicy

Field	Details
`UsageResourceId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The usage resource associated with the usage product. This field is a relationship field. **Relationship Name** UsageResource **Refers To** UsageResource

ContractItemPrice

Represents an object that's used to capture a price for a product on a contract. This object is available in API version 61.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

This object is available in Enterprise, Unlimited, and Developer Editions of Revenue Cloud.

Fields

Field	Details
`AdjustmentMethod`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Method used to apply discount. Valid values are: • `Range`—Apply the discount to all items after you reach the discount tier. For example, suppose that you give a 10% discount for 50 or more items. If a customer orders 50 products, and the type is `range`, apply the 10% discount to all 50 items.

Field	Details
	• `Slab`—Apply discounts in tiers. For example, suppose that you order 30 products, and the type is `slab`, you can apply a 10% discount to units 1–9, a 20% discount to units 10–19, and a 30% discount to units 20–30.
`ContractId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** ID of the contract. This field is a relationship field. **Relationship Name** Contract **Relationship Type** Master-detail **Refers To** Contract (the master object)
`DiscountType`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The type of discount applied, a percentage of the price or an amount. Valid values are: • `AdjustmentAmount` • `AdjustmentPercentage`
`DiscountValue`	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The value of the discount applied, based on the discount type.
`EndDate`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** End date and time of the relationship between the contract and contract item price.

Field	Details
`ItemId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** ID of the product or product category related to a price in a contract. This field is a polymorphic relationship field. **Relationship Name** Item **Relationship Type** Lookup **Refers To** Product2, ProductCategory
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Timestamp when the current user last accessed this record, a record related to this record, or a list view.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Timestamp when the current user last viewed this record or list view. If this value is null, the user accessed this record or list view (`LastReferencedDate`) but didn't view it.
ListPrice	**Type** currency **Properties** Filter, Nillable **Description** The list price of the product. This value is read-only and inherited from the price book related to the contract when the contract item price is created. Use the list price to compare the advertised price to prices that customers receive during contract negotiations. This field is available in API version 66.0 and later.
`Name`	**Type** string

Field	Details
	Properties Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Auto-generated number assigned to the contract item price. (Read only)
`Price`	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** Unit price for the product sold as part of the contract.
`ProductSellingModelId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Product selling model for the product associated with the price. This field is a relationship field. **Relationship Name** ProductSellingModel **Relationship Type** Lookup **Refers To** ProductSellingModel
`SellingModelType`	**Type** picklist **Properties** Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort **Description** Selling mode type to specify whether the product sold is a one-time sale, an evergreen subscription, or a subscription with a defined term. Valid values are: • `Evergreen` • `OneTime` • `TermDefined` The value derived from the product selling model.
`StartDate`	**Type** dateTime

Field	Details
	Properties Create, Filter, Sort, Update **Description** Start date and time of the relationship between the contract and contract item price.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

ContractItemPriceHistory on page 1288
History is available for tracked fields of the object.

ContractItemPriceAdjTier

Represents the tiers of a price adjustment to a product on a contract. This object is available in API version 63.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

This object is available with Revenue Management.

Fields

Field	Details
`ContractItemPriceId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The contract item price ID associated with the contract item price adjustment tier. This field is a relationship field. **Relationship Name** ContractItemPrice

Field	Details
	Relationship Type Master-detail **Refers To** ContractItemPrice (the master object)
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed this record, a record related to this record, or a list view.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Timestamp when the current user last viewed this record or list view. If this value is null, the user accessed this record or list view (`LastReferencedDate`) but didn't view it.
`LowerBound`	**Type** double **Properties** Create, Filter, Sort, Update **Description** The minimum quantity for the adjustment to be applicable.
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The name of the record.
`TierType`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The type of adjustment tier.

Field	Details
	Valid values are: • `AdjustmentAmount` • `AdjustmentPercentage` • `OverrideAmount`
`TierValue`	**Type** double **Properties** Create, Filter, Sort, Update **Description** The price adjustment value.
`UpperBound`	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The maximum quantity for the adjustment to be applicable.

Associated Objects

This object has associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

ContractItemPriceAdjTierFeed
Feed tracking is available for the object.

ContractItemPriceAdjTierHistory
History is available for tracked fields of the object.

ContractItemPriceHistory

Represents the history of changes to the values in the fields of a ContractItemPrice object. This object is available in API version 61.0 and later.

Supported Calls

`describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`

Special Access Rules

This object is available in Enterprise, Unlimited, and Developer Editions of Revenue Cloud.

Fields

Field	Details
ContractItemPriceId	**Type** reference **Properties** Filter, Group, Sort **Description** ID of the ContractItemPrice record. This field is a relationship field. **Relationship Name** ContractItemPrice **Relationship Type** Lookup **Refers To** ContractItemPrice
DataType	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** Data type of the field that was changed. Valid values are: • Address • AnyType • AutoNumber • Base64 • BitVector • Boolean • Content • Currency • DataCategoryGroupReference • DateOnly • DateTime • Division • Double • DynamicEnum • Email • EncryptedBase64

Field	Details
	• `EncryptedText` • `EntityId` • `EnumOrId` • `ExternalId` • `Fax` • `File` • `HtmlMultiLineText` • `HtmlStringPlusClob` • `InetAddress` • `Json` • `Location` • `MultiEnum` • `MultiLineText` • `Namespace` • `Percent` • `PersonName` • `Phone` • `Raw` • `RecordType` • `SfdcEncryptedText` • `SimpleNamespace` • `StringPlusClob` • `Switchable_PersonName` • `Text` • `TimeOnly` • `Url` • `YearQuarter`
`Field`	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** Name of the field that was changed. Possible values are: • `AdjustmentMethod` • `Contract` • `DiscountType`

Field	Details
	• `DiscountValue` • `EndDate` • `Item` • `Name` • `Price` • `ProductSellingModel` • `StartDate` • `created` • `customPersonMerged` • `feedEvent` • `individualMerged` • `locked`—Record locked. • `ownerAccepted`—Owner (Accepted) • `ownerAssignment`—Owner (Assignment) • `unlocked`—Record unlocked.
`NewValue`	**Type** anyType **Properties** Nillable, Sort **Description** New value of the field that was changed.
`OldValue`	**Type** anyType **Properties** Nillable, Sort **Description** Latest value of the field before it was changed.

OrderDeliveryMethod

Shows the customizations and options that a buyer selected for their delivery method. This object is available in API version 48.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

To access Commerce Orders entities, your org must have a Salesforce Order Management license. Commerce Orders entities are available only in Lightning Experience.

Fields

Field	Details
`Carrier`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The carrier that the buyer chose for their delivery method. Developers must add values to this field.
`ClassOfService`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The carrier class of service that the buyer chose for their delivery method. Developers must add values to this field.
`Description`	**Type** textarea **Properties** Create, Nillable, Update **Description** Description of the delivery method.
`IsActive`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Assign new delivery groups to active delivery methods.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort

Field	Details
	Description The timestamp for when the current user last viewed a record related to this record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed this record. If this value is null, this record might only have been referenced (LastReferencedDate) and not viewed.
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** Required. Default name of this record.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The user who owns an order delivery method record.
ProductId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Optional. This product represents a delivery charge order product for a delivery using this delivery method. For example, you could create a product that represents an overnight express charge and assign it to an overnight express delivery method.
ReferenceNumber	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Reference number for an external delivery method.

Field	Details
ShippingCarrierMethod	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Optional. A specific shipping service provided by a shipping carrier, such as Ground, 2Day, and NextDay. Depends on the range of transit times available for each carrier.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

OrderDeliveryMethodChangeEvent (API version 62.0)
Change events are available for the object.

OrderItemAttribute

Represents a virtual object that stores an attribute specified for an order item.This object is available in API version 60.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `query()`, `retrieve()`, `update()`, `upsert()`

Special Access Rules

This object is available in Enterprise, Unlimited, and Developer Editions of Revenue Cloud.

Fields

Field	Details
AttributeDefinitionId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The ID of the attribute definition for this order item attribute. This field is a relationship field. **Relationship Name** AttributeDefinition **Relationship Type** Lookup

Field	Details
	Refers To AttributeDefinition
AttributeName	**Type** string **Properties** Filter, Group, idLookup, Nillable, Sort **Description** The name given to order item attribute.
AttributePicklistValueId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the attribute picklist value if the attribute is a picklist type. This field is a relationship field. **Relationship Name** AttributePicklistValue **Relationship Type** Lookup **Refers To** AttributePicklistValue
AttributeValue	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Stores the value of the order item attribute. For example 5-TB storage. You can use this field to filter records only if the DataType value in the related AttributeDefinitionId record is Text. If the DataType value is Picklist, use the value in the AttributePicklistValueId field for filtering. You can't use this field to filter records if the DataType value is Checkbox, Currency, Date, Datetime, Multipicklist, Number, or Percent.
ExternalId	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update

Field	Details
	Description An auto-generated ID of the attribute record saved in an external system (for example an HBase database).
`IsPriceImpacting`	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** The pricing impacting the status of the attribute. The default value is `false`.
`OrderItemId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The parent order item associated with the order item attribute. This field is a relationship field. **Relationship Name** OrderItem **Relationship Type** Lookup **Refers To** OrderItem

OrderItemDetail

Represents the breakdown details of an order product. Revenue Management generates these records to capture pricing and quantity changes, such as negative quantity reductions, early renewals, derived pricing or repricing during an amendment, and bundle or product attribute reconfigurations. This object is available in API version 60.0 and later.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`

Special Access Rules

This object is available in Enterprise, Unlimited, and Developer Editions of Revenue Cloud.

Fields

Field	Details
BillingReference	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** Reference to the original order item for which this amend or cancel record is created.
EffectiveFrom	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when the transaction becomes effective.
EffectiveTo	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when the transaction ends.
LineNumber	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The line number of the detail record.
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The name of the record.
NetUnitPrice	**Type** currency **Properties** Filter, Sort

Field	Details
	Description The unit price of the item that's calculated after applying any discounts and before tax calculation.
`OrderItemId`	**Type** reference **Properties** Filter, Group, Sort **Description** The ID of the related order item. This field is a relationship field. **Relationship Name** OrderItem **Relationship Type** Lookup **Refers To** OrderItem
`PriceWaterfallIdentifier`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The price waterfall identifier generated by Salesforce Pricing that's associated with the pricing of the detail record.
`Quantity`	**Type** double **Properties** Filter, Sort **Description** The quantity specified for the item in the detail record.
`ReferenceDate`	**Type** date **Properties** Filter, Group, Nillable, Sort **Description** The reference date of the item. For example, the start date of the subscription that's associated with the detail record.

Field	Details
ReferenceNumber	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The reference number of the item. For example, the order number that's associated with the detail record.
TotalLineAmount	**Type** currency **Properties** Filter, Sort **Description** The net total price of the order product, before price adjustments, inclusive of quantity and subscription term.
TotalPrice	**Type** currency **Properties** Filter, Sort **Description** The total price of the item in the detail record that's calculated using the quantity, net unit price, and after applying the pricing terms.
UnitPrice	**Type** currency **Properties** Filter, Sort **Description** The unit price of the item before any discounts or tax calculation.

OrderItemRateAdjustment

Represents the negotiated rate adjustment for an order product. This object is available in API version 62.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `update()`, `upsert()`

Special Access Rules

This object is available with Revenue Cloud.

Fields

Field	Details
AdjustmentType	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Specifies the type of rate adjustment. Possible values are: • Amount • Override • Percentage
AdjustmentValue	**Type** double **Properties** Create, Filter, Sort, Update **Description** The value of the adjustment.
LowerBound	**Type** double **Properties** Create, Filter, Sort, Update **Description** The minimum quantity for the adjustment to be applicable.
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The name of the order item rate adjustment.
OrderItemRateCardEntryId	**Type** reference

Field	Details
	Properties Create, Filter, Group, Sort **Description** The parent order item rate card entry associated with the order item rate adjustment. This field is a relationship field. **Relationship Name** OrderItemRateCardEntry **Relationship Type** Master-detail **Refers To** OrderItemRateCardEntry (the master object)
`UpperBound`	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The quantity below which the adjustment must be applicable. For example, if you want the adjustment to be applicable when the quantity is 99 or less, set this value to 100.

OrderItemRateCardEntry

Represents the catalog and negotiated rates of a usage metric associated with an order item that's used to charge overage consumption. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `update()`, `upsert()`

Special Access Rules

This object is available with Revenue Cloud.

Fields

Field	Details
IsChosenRate	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether this rate is the chosen rate for the associated binding target and usage resource (`true`) or not (`false`). The default value is `false`. Available in API version 64.0 and later.
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** An auto-generated number assigned to the order item rate card entry record.
`NegotiatedRate`	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The base negotiated rate used to charge overage consumption.
`OrderItemId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The parent order item associated with the order item rate card entry. This field is a relationship field. **Relationship Name** OrderItem **Relationship Type** Master-detail **Refers To** OrderItem (the master object)
`RateCardEntryId`	**Type** reference

Field	Details
	Properties Create, Filter, Group, Sort **Description** The rate card entry containing catalog rates that's associated with the order item rate card entry. This field is a relationship field. **Relationship Name** RateCardEntry **Refers To** RateCardEntry
RateCardId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The rate card associated with the order item rate card entry. This field is a relationship field. **Relationship Name** RateCard **Refers To** RateCard
RateUnitOfMeasureId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The standard unit of measure containing the unit for the negotiated rate that's associated with the order item rate card entry. This field is a relationship field. **Relationship Name** RateUnitOfMeasure **Refers To** UnitOfMeasure
UsageResourceId	**Type** reference **Properties** Filter, Group, Nillable, Sort

Field	Details
	Description The usage resource associated with the order item rate card entry. This field is a relationship field. **Relationship Name** UsageResource **Refers To** UsageResource

OrderItemUsageRsrcGrant

Represents the negotiated grants for the usage resource that's associated with the usage product added in the order item. This object is available in API version 65.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `update()`, `upsert()`

Fields

Field	Details
`GrantQuantity`	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The granted or negotiated quantity of a usage resource associated with the usage product.
`GrantType`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** Specifies the type of model that defines how the usage resource is consumed. Valid values are: • `Commit` • `Grant`

Field	Details
	The default value is `Grant`.
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The auto-generated identifier for the order item usage resource grant record. For example, OIURG-00004 or OIURG-4567.
`OrderItemId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The order item associated with the usage product. This field is a relationship field. **Relationship Name** OrderItem **Relationship Type** Master-detail **Refers To** OrderItem (the master object)
`ProductUsageGrantId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The product usage grant associated with the order item. This field is a relationship field. **Relationship Name** ProductUsageGrant **Refers To** ProductUsageGrant
`TokenResourceId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update

Field	Details
	Description The usage resource of category Token associated with the usage resource related to the usage product added in the order item. This field is a relationship field. **Relationship Name** TokenResource **Refers To** UsageResource
`UsageGrantRefreshPolicyId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The usage grant refresh policy associated with the usage resource related to the usage product added in the order item. This field is a relationship field. **Relationship Name** UsageGrantRefreshPolicy **Refers To** UsageGrantRenewalPolicy
`UsageGrantRolloverPolicyId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The usage grant rollover policy associated with the usage resource related to the usage product added in the order item. This field is a relationship field. **Relationship Name** UsageGrantRolloverPolicy **Refers To** UsageGrantRolloverPolicy
`UsageResourceId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The usage resource associated with the usage product that's added in the order item.

Field	Details
	This field is a relationship field. **Relationship Name** UsageResource **Refers To** UsageResource
`ValidityPeriodTerm`	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The duration for which the usage resource grant is valid, when used with the validity period units.
`ValidityPeriodUnit`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The length of a validity period for the usage resource grant, when used with the validity period term. Valid values are: • `Month` • `None` • `Year`

OrderItemUsageRsrcPlcy

Represents the policies that are used for the usage resource that's associated with the usage product added in the order item. This object is available in API version 65.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `update()`, `upsert()`

Fields

Field	Details
DrawdownOrder	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies the order that's used to debit consumption of entitlements related to the usage resource from the usage entitlement bucket. Valid values are: • ExpiringFirst • GrantedFirst • GrantedLast
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** An auto-generated number assigned to the order item for usage product grant record. For example, OIURG-4567
OrderItemId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The order item associated with the usage product. This field is a relationship field. **Relationship Name** OrderItem **Relationship Type** Master-detail **Refers To** OrderItem (the master object)
ProductUsageResourcePolicyId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update

Field	Details
	Description The product usage resource policy associated with the usage resource related to the usage product added in the order item. This field is a relationship field. **Relationship Name** ProductUsageResourcePolicy **Refers To** ProductUsageResourcePolicy
RatingFrequencyPolicyId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The rating frequency policy associated with the usage resource related to the usage product added in the order item. This field is a relationship field. **Relationship Name** RatingFrequencyPolicy **Refers To** RatingFrequencyPolicy
UsageAggregationPolicyId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The usage aggregation policy associated with the usage resource related to the usage product added in the order item. This field is a relationship field. **Relationship Name** UsageAggregationPolicy **Refers To** UsageResourceBillingPolicy
UsageCommitmentPolicyId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update

Field	Details
	Description The usage commitment policy associated with the usage resource related to the usage product added in the order item. This field is a relationship field. **Relationship Name** UsageCommitmentPolicy **Refers To** UsageCommitmentPolicy
`UsageOveragePolicyId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The usage overage policy associated with the usage resource related to the usage product added in the order item. This field is a relationship field. **Relationship Name** UsageOveragePolicy **Refers To** UsageOveragePolicy
`UsageResourceId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The usage resource associated with the usage product that's added in the order item. This field is a relationship field. **Relationship Name** UsageResource **Refers To** UsageResource

SalesTransactionType

Represents the type of the sales transaction. This object is available in API version 61.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

This object is available in Enterprise, Unlimited, and Developer Editions of Revenue Cloud.

Fields

Field	Details
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed this record, a record related to this record, or a list view.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed this record or list view. If this value is null, the user might have only accessed this record or list view but not viewed it directly.
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The name of the sales transaction type.

QuoteAction

Indicates the type of sales transaction that's being quoted; for example, a renewal sale. This object is available in API version 59.0 and later.

If a quote doesn't have a quote action, Salesforce treats it as a quote of the `Add` type. When such a quote is used to create an order, Salesforce automatically creates an order action of the `Add` type.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

This object is available in orgs with Revenue Cloud. It's also available in Industries Automotive and Industries Field Service.

Fields

Field	Details
`CurrencyIsoCode`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** ISO code of the currency. Use only one of the valid alphabetic, three-letter currency ISO codes defined by the ISO 4217 standard, such as `USD`, `GBP`, or `JPY`. Must be unique within your organization. Label is **Currency ISO Code**. The default value is `USD`. See Supported Currencies (ICU) for a list of currency codes Salesforce supports. This field is available in Revenue Cloud in API version 66.0 and later.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed this record indirectly, for example, through a list view or related record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed this record or list view. If this value is null, the user accessed this record or list view indirectly, but didn't view it.
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort

Field	Details
	Description The name given to the quote action.
`QuoteId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The quote related to this quote action. This field is a relationship field. **Relationship Name** Quote **Relationship Type** Lookup **Refers To** Quote
`SourceAssetId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The asset changed by this sales transaction. For example, if the quote action is a quantity amendment, this field contains the ID of the asset that's amended. This field is a relationship field. **Relationship Name** SourceAsset **Relationship Type** Lookup **Refers To** Asset
`Subtype`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The subtype of the action on the quote line item. Valid values are: • `DowngradeFrom`—Available in API version 66.0 and later.

Field	Details
	• `DowngradeTo`—Available in API version 66.0 and later. • `FieldAmendment` • `Rollback` • `StartDateAdjustment` • `SwapIn`—Available in API version 66.0 and later. • `SwapOut`—Available in API version 66.0 and later. • `TransferFrom` • `TransferTo` • `UpgradeFrom`—Available in API version 66.0 and later. • `UpgradeTo`—Available in API version 66.0 and later. This field is available with Revenue Cloud in API version 64.0 and later.
`Type`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort **Description** The type of sales transaction that the related quote is for. Valid values are: • `Add` • `Amend` • `Association`—Available in API version 66.0 and later. • `Cancel` • `No Change` • `Renew` • `Transfer`—Available with Revenue Cloud in API version 65.0 and later.

QuoteItemTaxItem

The tax that is applied to a quote line item. This object is available in API version 55.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

This object is available if Subscription Management is enabled in your org. This object is also available in Enterprise, Unlimited, and Developer Editions of Revenue Cloud.

Fields

Field	Details
`Amount`	**Type** currency **Properties** Create, Filter, Sort, Update **Description** The tax amount for the quote line item.
`CurrencyIsoCode`	**Type** picklist **Properties** Defaulted on create, Filter, Group, Restricted picklist, Sort **Description** Available only for orgs with the multicurrency feature enabled. Contains the ISO code for any currency allowed by the org. Possible values are: • `BHD`—Bahraini Dinar • `EUR`—Euro • `JPY`—Japanese Yen • `USD`—U.S. Dollar The default value is 'USD'.
`Description`	**Type** textarea **Properties** Create, Nillable, Update **Description** User-defined description of the tax. For example, state sales tax or value-added tax (VAT).
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** Name of the tax.
`ProductSurchargeId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update

Field	Details
	Description The product surcharge associated with the quote item tax item. This field is a relationship field. This field is available with Digital Insurance. **Relationship Name** ProductSurcharge **Refers To** ProductSurcharge
ProratedSurchargeAmount	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The prorated portion of the tax amount based on the quote line item's effective dates. This field is available with Digital Insurance.
QuoteLineItemId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** ID of the related quote line item. This is a relationship field. **Relationship Name** QuoteLineItem **Relationship Type** Lookup **Refers To** QuoteLineItem
Rate	**Type** percent **Properties** Create, Filter, Nillable, Sort, Update **Description** If the tax is a percentage tax, then this field contains the percent value. If the tax is a fixed amount, then this field is null.
TaxEffectiveDate	**Type** date

Field	Details
	Properties Create, Filter, Group, Sort, Update **Description** The date used to calculate the tax rate.
`Type`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Whether the tax is estimated or calculated by the tax provider. Possible values are: • `Actual` • `Estimated`

QuoteLineDetail

Represents the breakdown details of a quote line item. Revenue Management generates these records to capture pricing and quantity changes, such as negative quantity reductions, early renewals, derived pricing or repricing during an amendment, and bundle or product attribute reconfigurations. This object is available in API version 60.0 and later.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`

Special Access Rules

This object is available in Enterprise, Unlimited, and Developer Editions of Revenue Cloud.

Fields

Field	Details
`BillingReference`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** Reference to the original order item for which this amend or cancel record is created.

Field	Details
EffectiveFrom	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when the transaction becomes effective.
EffectiveTo	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when the transaction ends.
LineNumber	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The line number of the detail record.
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The name of the record.
NetUnitPrice	**Type** currency **Properties** Filter, Sort **Description** The unit price of the item that's calculated after applying any discounts and before tax calculation.
PriceWaterfallIdentifier	**Type** string **Properties** Filter, Group, Nillable, Sort

Field	Details
	Description The price waterfall identifier generated by Salesforce Pricing that's associated with the pricing of the detail record.
`Quantity`	**Type** double **Properties** Filter, Sort **Description** The quantity specified for the item in the detail record.
`QuoteLineItemId`	**Type** reference **Properties** Filter, Group, Sort **Description** The ID of the related quote line. This field is a relationship field. **Relationship Name** QuoteLineItem **Relationship Type** Lookup **Refers To** QuoteLineItem
`ReferenceDate`	**Type** date **Properties** Filter, Group, Nillable, Sort **Description** The reference date of the item. For example, the start date of the subscription that's associated with the detail record.
`ReferenceNumber`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The reference number of the item. For example, the order number that's associated with the detail record.

Field	Details
TotalLineAmount	**Type** currency **Properties** Filter, Sort **Description** The net total price of the order product, before price adjustments, inclusive of quantity and the subscription term.
TotalPrice	**Type** currency **Properties** Filter, Sort **Description** The total price of the item in the detail record that's calculated using the quantity, net unit price, and after applying the pricing terms.
UnitPrice	**Type** currency **Properties** Filter, Sort **Description** The unit price of the item before any discounts or tax calculation.

QuoteLineGroup

Stores the group information for line items in a quote. It also stores the aggregated line field information (subtotal). It contains a parent-child relationship to quote. This object is available in API version 61.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `update()`, `upsert()`

Fields

Field	Details
Description	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update

Field	Details
	Description The description of the group.
Discount	**Type** percent **Properties** Create, Filter, Nillable, Sort, Update **Description** The optional discount percentage, specified by the sales representative at the group level.
DiscountAmount	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The optional discount amount, specified by the sales representative at the group level.
EndDate	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The end date of the group ramp segment.
IsRamped	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Indicates whether the group is a group ramp segment, which is a period in a group ramp deal with specific prices and volume. The default value is `false`.
Margin	**Type** percent **Properties** Create, Filter, Nillable, Sort, Update **Description** The optional margin percentage, specified by the sales representative at the group level.
MarginAmount	**Type** currency

Field	Details
	Properties Create, Filter, Nillable, Sort, Update **Description** The optional margin amount, specified by the sales representative at the group level. This amount can also be considered as the summary margin amount calculated by subtracting the total cost from the summary subtotal.
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The name of the group.
`ParentQuoteLineGroupId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the parent group for a nested quote line group. This field is a relationship field. **Relationship Name** ParentQuoteLineGroup **Refers To** QuoteLineGroup
`QuoteId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The ID of the related quote. This field is a relationship field. **Relationship Name** Quote **Relationship Type** Master-detail **Refers To** Quote (the master object)

Field	Details
SegmentType	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The duration type of the segment. Possible values are: • Custom • Yearly
SortOrder	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description**
StartDate	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The start date of the group ramp segment.
SummarySubtotal	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The aggregated subtotal amount of nested group lines.
SummaryTotalAmount	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The aggregated total amount of nested group lines before any discounts are applied.
TotalAdjustment	**Type** percent **Properties** Filter, Nillable, Sort

Field	Details
	Description The total discount percentage applied at the group level. This percentage is calculated by using the formula: (Summary Total Amount - Summary Subtotal) / Summary Total Amount.
TotalAdjustmentAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The total discount amount at the group level. This amount is calculated by subtracting the summary subtotal from the summary total amount.
TotalCost	**Type** currency **Properties** Filter, Nillable, Sort **Description** The aggregated total cost of nested group lines.
TotalMargin	**Type** percent **Properties** Filter, Nillable, Sort **Description** The summary margin percentage at the line item level. This percentage is calculated by using the formula: (Summary Subtotal - Total Cost) / Summary Subtotal.
TotalMarginAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The summary margin amount calculated at the group level.
Type	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The type of quote line group. Possible values are:

Field	Details
	• `AssetDowngrade` • `AssetSwap` • `AssetUpgrade` • `CPQQuoteGroup`—CPQ Line Grouping • `RampScheduleGroup` The default value is `CPQQuoteGroup`.

QuoteLineItemAttribute

Represents a virtual object that stores an attribute specified for a quote line item. This object is available in API version 59.0 and later.

Note: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `query()`, `retrieve()`, `update()`, `upsert()`

Special Access Rules

This object is available in Enterprise, Unlimited, and Developer Editions of Revenue Management.

Fields

Field	Details
`AttributeDefinitionId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The ID of the attribute definition for this quote line item attribute. This field is a relationship field. **Relationship Name** AttributeDefinition **Refers To** AttributeDefinition
`AttributeName`	**Type** string **Properties** Filter, Group, idLookup, Nillable, Sort

Field	Details
	Description The name of the quote line item attribute.
`AttributePicklistValueId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the attribute picklist value if the attribute is a picklist type. This field is a relationship field. **Relationship Name** AttributePicklistValue **Refers To** AttributePicklistValue
`AttributeValue`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The value of the quote line item attribute. For example 5-TB storage. You can use this field to filter records only if the DataType value in the related `AttributeDefinitionId` record is `Text`. If the DataType value is `Picklist`, use the value in the `AttributePicklistValueId` field for filtering. You can't use this field to filter records if the DataType value is `Checkbox`, `Currency`, `Date`, `Datetime`, `Multipicklist`, `Number`, or `Percent`.
`ExternalId`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** An auto-generated ID of the attribute record saved in an external system, such as an HBase database.
`IsPriceImpacting`	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** The pricing impacting the status of the attribute.

Field	Details
	The default value is `false`.
`QuoteLineItemId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The associated parent quote line item. This field is a relationship field. **Relationship Name** QuoteLineItem **Relationship Type** Master-detail **Refers To** QuoteLineItem (the master object)

Usage

This object doesn't support custom fields, validation rules, or triggers. In SOQL queries, you can filter records by using `Id` and `AttributeDefinition`. You can't use `AttributeValue` in the `WHERE` clause.

QuotLineItmUseRsrcGrant

Represents the negotiated grants for the usage resource that's associated with the usage product added in the quote line item. This object is available in API version 65.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `update()`, `upsert()`

Fields

Field	Details
`GrantQuantity`	**Type** double **Properties** Create, Filter, Nillable, Sort, Update

Field	Details
	Description The granted or negotiated quantity of a usage resource associated with the usage product.
`GrantType`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** Specifies the type of model that defines how the usage resource is consumed. Valid values are: • `Commit` • `Grant` The default value is `Grant`.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when this record was last referenced.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed a record related to this record.
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The auto-generated identifier for the quote line item usage resource grant record. For example, QLIURG-00004 or QLIURG-4567.
`ProductUsageGrantId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update

Field	Details
	Description The product usage grant associated with the quote line item. This field is a relationship field. **Relationship Name** ProductUsageGrant **Refers To** ProductUsageGrant
`QuoteLineItemId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The quote line item associated with the usage product. This field is a relationship field. **Relationship Name** QuoteLineItem **Relationship Type** Master-detail **Refers To** QuoteLineItem (the master object)
`TokenResourceId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The usage resource of category Token associated with the usage resource related to the usage product added in the order item. This field is a relationship field. **Relationship Name** TokenResource **Refers To** UsageResource
`UsageGrantRefreshPolicyId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update

Field	Details
	Description The usage grant refresh policy associated with the usage resource related to the usage product added in the quote line item. This field is a relationship field. **Relationship Name** UsageGrantRefreshPolicy **Refers To** UsageGrantRenewalPolicy
`UsageGrantRolloverPolicyId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The usage grant rollover policy associated with the usage resource related to the usage product added in the quote line item. This field is a relationship field. **Relationship Name** UsageGrantRolloverPolicy **Refers To** UsageGrantRolloverPolicy
`UsageResourceId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The usage resource associated with the usage product that's added as the quote line item. This field is a relationship field. **Relationship Name** UsageResource **Refers To** UsageResource
`ValidityPeriodTerm`	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The duration for which the usage resource grant is valid, when used with the validity period units.

Field	Details
ValidityPeriodUnit	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The length of a validity period for the usage resource grant, when used with the validity period term. Valid values are: • Month • None • Year

QuotLineItmUsageRsrcPlcy

Represents the policies that are used for the usage resource that's associated with the usage product added in the quote line item. This object is available in API version 65.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

create(), delete(), describeLayout(), describeSObjects(), getDeleted(), getUpdated(), query(), retrieve(), search(), update(), upsert()

Fields

Field	Details
DrawdownOrder	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies the order that's used to debit consumption of entitlements related to the usage resource from the usage entitlement bucket. Valid values are: • ExpiringFirst • GrantedFirst • GrantedLast

Field	Details
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when this record was last referenced.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed a record related to this record.
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The auto-generated identifier for the quote line item usage resource policy record. For example, QLIURP-00004 or QLIURP-4567.
ProductUsageResourcePolicyId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The product usage resource policy associated with the quote line item. This field is a relationship field. **Relationship Name** ProductUsageResourcePolicy **Refers To** ProductUsageResourcePolicy
QuoteLineItemId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The quote line item associated with the usage product. This field is a relationship field.

Field	Details
	Relationship Name QuoteLineItem **Relationship Type** Master-detail **Refers To** QuoteLineItem (the master object)
`RatingFrequencyPolicyId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The rating frequency policy associated with the usage resource related to the usage product added in the quote line item. This field is a relationship field. **Relationship Name** RatingFrequencyPolicy **Refers To** RatingFrequencyPolicy
`UsageAggregationPolicyId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The usage aggregation policy associated with the usage resource related to the usage product added in the quote line item. This field is a relationship field. **Relationship Name** UsageAggregationPolicy **Refers To** UsageResourceBillingPolicy
`UsageCommitmentPolicyId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The usage commitment policy associated with the usage resource related to the usage product added in the quote line item. This field is a relationship field.

Field	Details
	Relationship Name UsageCommitmentPolicy **Refers To** UsageCommitmentPolicy
UsageOveragePolicyId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The usage overage policy associated with the quote line item. This field is a relationship field. **Relationship Name** UsageOveragePolicy **Refers To** UsageOveragePolicy
UsageResourceId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The usage resource associated with the usage product that's added in the quote line item. This field is a relationship field. **Relationship Name** UsageResource **Refers To** UsageResource

QuoteLineRateAdjustment

Represents the negotiated rate adjustment for a quote line item. This object is available in API version 62.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `update()`, `upsert()`

Special Access Rules

This object is available with Revenue Cloud.

Fields

Field	Details
AdjustmentType	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Specifies the type of rate adjustment. Possible values are: • Amount • Override • Percentage
AdjustmentValue	**Type** double **Properties** Create, Filter, Sort, Update **Description** The value of the adjustment.
LowerBound	**Type** double **Properties** Create, Filter, Sort, Update **Description** The minimum quantity for the adjustment to be applicable.
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The name of the quote line rate adjustment record.
QuoteLineRateCardEntryId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The parent quote line rate card entry associated with the quote line rate adjustment. This field is a relationship field.

Field	Details
	Relationship Name QuoteLineRateCardEntry **Relationship Type** Master-detail **Refers To** QuoteLineRateCardEntry (the master object)
`UpperBound`	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The quantity below which the adjustment must be applicable. For example, if you want the adjustment to be applicable when the quantity is 99 or less, set this value to 100.

QuoteLineRateCardEntry

Represents the catalog and negotiated rates of a usage resource associated with a quote line item that's used to charge overage consumption. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `update()`, `upsert()`

Fields

Field	Details
IsChosenRate	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether this rate is the chosen rate for the associated binding target and usage resource (`true`) or not (`false`). The default value is `false`. Available in API version 64.0 and later.
`Name`	**Type** string

Field	Details
	Properties Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** An auto-generated number assigned to the quote line rate card entry record.
NegotiatedRate	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The base negotiated rate used to charge overage consumption.
QuoteLineItemId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The parent quote line item associated with the quote line rate card entry. This field is a relationship field. **Relationship Name** QuoteLineItem **Relationship Type** Master-detail **Refers To** QuoteLineItem (the master object)
RateCardEntryId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The rate card entry containing catalog rates that's associated with the quote line rate card entry. This field is a relationship field. **Relationship Name** RateCardEntry **Refers To** RateCardEntry
RateCardId	**Type** reference

Field	Details
	Properties Filter, Group, Nillable, Sort **Description** The rate card associated with the quote line rate card entry. This field is a relationship field. **Relationship Name** RateCard **Refers To** RateCard
`RateUnitOfMeasureId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The standard unit of measure containing the unit for the negotiated rate that's associated with the quote line rate card entry. This field is a relationship field. **Relationship Name** RateUnitOfMeasure **Refers To** UnitOfMeasure
`UsageResourceId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The usage resource associated with the quote line rate card entry. This field is a relationship field. **Relationship Name** UsageResource **Refers To** UsageResource

Transaction Management Fields on Standard Objects

Transaction Management adds standard and custom fields to some standard Salesforce objects. These fields are available only in orgs where Transaction Management is enabled.

Transaction Management Fields on Object State Definition

Standard and custom fields extend the standard Object State Definition object for use in Transaction Management to represent the object state model for a particular status field for an entity. This object is available in API version 60.0 and later.

Fields

Field	Details
AppUsageType	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** This field indicates under which AppUsageType the transition applies to. For example, ObjectStateDefinition associated with "Revenue Lifecycle Management" AppUsageType will apply to quotes, assets, or orders associated with "Revenue Lifecycle Management".

Transaction Management Fields on Object State Transition

Standard and custom fields extend the standard Object State Transition object for use in Transaction Management to define the valid transition between two statuses. This object is available in API version 60.0 and later.

Fields

Field	Details
CustomPermissionId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the associated custom permission. This field is a relationship field. **Relationship Name** CustomPermission **Relationship Type** Lookup **Refers To** CustomPermission

Transaction Management Fields on Object State Value

Standard and custom fields extend the standard Object State Transition object for use in Transaction Management This object is available in API version 60.0 and later.

Fields

Field	Details
CustomPermissionId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the associated custom permission. This field is a relationship field. **Relationship Name** CustomPermission **Relationship Type** Lookup **Refers To** CustomPermission

Transaction Management Fields on Order

Standard and custom fields extend the standard Order object for use in Transaction Management.

Special Access Rules

To view these fields, you must have the Revenue Cloud Advanced license. See Order for fields on the Salesforce platform object.

Fields

Field	Details
AdjustmentDistributionLogic	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies how the overall discount amount distributes among all the order line items that have prices associated with them. The amount distributed is either the value specified in the `AppliedDiscountAmount` field or the difference between the values in the calculated `TotalAmount` and the user-specified `TotalAmountOverride` fields. Valid values are: • `Equal`—Distributes the discount amount equally among all the order items.

Field	Details
	• `Proportionate`—Distributes the discount amount in proportion to the `ListPriceTotal` values of the order items. Available in API version 65.0 and later.
`AppliedDiscount`	**Type** percent **Properties** Create, Filter, Nillable, Sort, Update **Description** The discount amount that's distributed among all the order items that have prices associated with them. This amount is distributed based on the logic specified in the `AdjustmentDistributionLogic` field. Available in API version 65.0 and later.
`AppliedDiscountAmount`	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The percent discount applied to each order item. Available in API version 65.0 and later.
`CalculationStatus`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The status of the price and tax calculations for the order. Valid values are: • `CompletedWithPricing`—Indicates that pricing is complete and tax will now be calculated. • `CompletedWithTax`—Indicates that pricing and tax calculation are complete. • `CompletedWithoutPricing`—Indicates that pricing and tax calculation were skipped. For a sales rep, this value appears as `Unknown` on the order page. • `ConfigurationFailed`—Indicates that configuration failed. Available in API version 62.0 • `ConfigurationInProgress`—Indicates that the configuration is in progress. Available in API version 62.0 • `GroupRampConfigurationFailed`—Indicates that the checks for group ramps have failed. Available in API version 65.0 and later. • `OrderRequestFailed`—Indicates that the requested order changes weren't saved. Available in API version 62.0 • `OrderRequestPartiallySaved`—Indicates that the requested order changes were partially saved. Available in API version 62.0

Field	Details
	• PriceCalculationFailed—Indicates that pricing failed. • ReconciliationFailed—Indicates that the arrangement of order data failed. Available in API version 62.0 • ReconciliationInProgress—Indicates that the arrangement of data is in progress. For a sales rep, this value appears as Saving on the order page. Available in API version 62.0 • SaveFailedOrIncomplete—Indicates that the recent changes to the order weren't saved. For a sales rep, this value appears as Some Records Weren't Saved on the order page. • TaxCalculationFailed—Indicates that pricing is complete but tax calculation failed. • TaxCalculationWaiting—Indicates that pricing is complete and a request sent to the tax engine, but tax calculation isn't complete. This read-only field is available in API version 61.0 and later.
DiscountPercent	**Type** percent **Properties** Filter, Nillable, Sort **Description** The percentage of discount applied to the order. The discount percent calculation is ((TotalRoundedLineAmount - TotalAmount) / TotalRoundedLineAmount) * 100. Available in API version 60.0 and later.
FulfillmentPlanId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The unique ID of the fulfillment plan associated with the order. This field is a relationship field. This field is available only in orgs where Dynamic Revenue Orchestrator is enabled. Available in API version 60.0 and later. **Relationship Name** FulfillmentPlan **Relationship Type** Lookup **Refers To** FulfillmentPlan
LastPricedDate	**Type** dateTime

Field	Details
	Properties Filter, Nillable, Sort **Description** The date when the order price was last calculated. Available in API version 60.0 and later.
`OrchestrationSbmsStatus`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The status of order submission for orchestration. This field is available only in orgs where Dynamic Revenue Orchestrator is enabled. Valid values are: • `Completed` • `Rejected` • `Submitted` This read-only field is available in API version 61.0 and later.
`OriginalActionType`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** Specifies the action that created the order. Valid values are: • `Amend`—Indicates that the order was created to amend assets. • `Cancel`—Indicates that the order was created to cancel assets. • `Renew`—Indicates that the order was created to renew assets. • `Transfer`—Indicates that the order was created to transfer assets. Available in API version 61.0 and later.
`SalesTransactionTypeId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The foreign key to the Sales Transaction Type object. Available in API version 61.0 and later. This field is a relationship field.

Field	Details
	Relationship Name SalesTransactionType **Relationship Type** Lookup **Refers To** SalesTransactionType
`TotalAmountOverride`	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The value that the `TotalAmount` field must be set to by applying overall discounts. Transaction Management calculates the overall discount amount by finding the difference between the value in the calculated `TotalAmount` field and the value in this field. It then uses the logic specified in the `AdjustmentDistributionLogic` field to distribute the discount amount among all the order items that have prices associated with them. Available in API version 65.0 and later.
`TotalRoundedLineAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The total amount of all line items in an order without pricing adjustments, such as discounts or tax calculations. Available in API version 60.0 and later.
`TransactionType`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies the type of order being processed. The valid value is: • `AdvancedConfigurator`—Indicates that the order must be processed by using the configuration rules and constraints set up in Constraint Rules Engine. Available in API version 61.0 and later.
`ValidationResult`	**Type** picklist

Field	Details
	Properties Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies whether the order was configured and priced. Orders can be activated only after they're configured and priced. Valid values are: • `MissingContributor`—Indicates that the order contains a derived product but not its pricing source. • `TransactionIncomplete`—Indicates that the order wasn't configured and priced. Available in API version 61.0 and later.

Transaction Management Fields on Order Item

Standard and custom fields extend the standard Order Item object for use in Transaction Management.

Special Access Rules

To view these fields, you must have the Revenue Cloud Advanced license. See Order Item for fields on the Salesforce platform object.

Fields

Field	Details
`CustomProductName`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The custom name of a product that's used to override the product name. The maximum supported length is 80 characters. Available in API version 61.0 and later.
`Discount`	**Type** percent **Properties** Create, Filter, Nillable, Sort, Update **Description** The manual discount percentage for the order item. Available in API version 60.0 and later.
`DiscountAmount`	**Type** currency

Field	Details
	Properties Create, Filter, Nillable, Sort, Update **Description** The manual discount amount for the order item. Available in API version 60.0 and later.
`EffectiveGrantDate`	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The date on which the resources associated with the order item are granted.
`EndDateTime`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The end date and time of the order item, which is calculated by using the values in the `EndDate`, `EndTime`, and `ServiceEndTimeZone` fields. If the `EndTime` field doesn't have a value, 23:59:59 is used for the calculation. If the `ServiceEndTimeZone` field doesn't have a value, GMT is used for the calculation. Available in API version 65.0 and later.
`EndQuantity`	**Type** double **Properties** Filter, Nillable, Sort **Description** The revised quantity of the item after adjusting changes. The field is read-only. It's calculated by adding the Start Quantity and the Quantity fields. Available in API version 60.0 and later.
`EndTime`	**Type** time **Properties** Create, Filter, Nillable, Sort, Update **Description** The end time of the order item. Available in API version 65.0 and later.
`Margin`	**Type** percent

Field	Details
	Properties Create, Filter, Nillable, Sort, Update **Description** The optional margin percentage, specified by the sales representative at the line item level. Available in API version 65.0 and later.
`MarginAmount`	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The optional margin amount, specified by the sales representative at the line item level. Available in API version 65.0 and later.
`NetTotalPrice`	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The total price after applying all price adjustments. Available in API version 60.0 and later.
`OrderItemGroupId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The order item group associated with the order item. This field is a relationship field. **Relationship Name** OrderItemGroup **Refers To** OrderItemGroup
`PartnerDiscountPercent`	**Type** percent **Properties** Create, Filter, Nillable, Sort, Update **Description** The discount percentage given to the partner for the order item. Available in API version 60.0 and later.

Field	Details
PartnerUnitPrice	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The unit price after applying the discount given to the partner for the order item. Available in API version 60.0 and later.
PriceWaterfallIdentifier	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The price waterfall identifier generated by Salesforce Pricing that's associated with the pricing of this order item record. Available in API version 60.0 and later.
SegmentType	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The duration type of the segment. You can use this field from Revenue Cloud only when the **Ramp Deals for Groups in Quotes and Orders** setting is turned on. Valid values are: • `Custom`—Available in API version 65.0 and later. • `Prorated`—Available in API version 67.0 and later. • `Trial`—Available in API version 67.0 and later. • `Yearly`—Available in API version 65.0 and later. The default value is `Yearly`.
ServiceDateTime	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The start date and time of the order item, which is calculated by using the values in the `ServiceDate`, `ServiceTime`, and `ServiceEndTimeZone` fields. If the `ServiceTime` field doesn't have a value, 00:00:00 is used for the calculation. If the `ServiceEndTimeZone` field doesn't have a value, GMT is used for the calculation. Available in API version 65.0 and later.

Field	Details
ServiceEndTimeZone	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The time zone for the order item's start and end dates, times, and datetimes. Available in API version 65.0 and later.
ServiceTime	**Type** time **Properties** Create, Filter, Nillable, Sort, Update **Description** The start time of the order item. Available in API version 65.0 and later.
StartQuantity	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The quantity available on the order item start date. Available in API version 60.0 and later.
Status	**Type** picklist **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The status of the order item. Available in API version 60.0 and later.
SubscriptionTerm	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The number of terms in the subscription for the item. The unit of the subscription term is stored in the PricingTermUnit field of this order item's related product selling model record (OrderItem.ProductSellingModel.PricingTermUnit). Available in API version 61.0 and later.
TotalAdjustment	**Type** percent

Field	Details
	Properties Filter, Nillable, Sort **Description** The total discount percentage applied at the line item level. This percentage is calculated by using the formula: (Total Line Amount - Net Total Price) / Total Line Amount. Available in API version 65.0 and later.
`TotalCost`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The total cost of all products sold in the order, calculated by multiplying the quantity by the unit cost. Available in API version 65.0 and later.
`TotalMargin`	**Type** percent **Properties** Filter, Nillable, Sort **Description** The effective margin percentage at the line item level. This percentage is calculated by using the formula: (Net Total Price - Total Cost) / Net Total Price. Available in API version 65.0 and later.
`TotalMarginAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The effective margin amount at the line item level. This amount is calculated by subtracting total cost from net total price. Available in API version 65.0 and later.
`UnitCost`	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The unit cost of a product being sold as part of the order. Available in API version 65.0 and later.
`ValidationResult`	**Type** picklist

Field	Details
	Properties Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies whether the order item was configured and priced. An order can be activated only after all its order items are configured and priced. Valid value is: • `Warning`—Indicates that the order item isn't configured and priced. Available in API version 60.0 and later.

Transaction Management Fields on Order Item Group

Standard and custom fields extend the standard Order Item Group object for use in Transaction Management.

Special Access Rules

To view these fields, you must have the Revenue Cloud Advanced license. See Order Item Group for fields on the Salesforce platform object.

Fields

Field	Details
`Discount`	**Type** percent **Properties** Create, Filter, Nillable, Sort, Update **Description** The optional discount percentage, specified by the sales representative at the group level. Available in API version 65.0 and later.
`DiscountAmount`	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The optional discount amount, specified by the sales representative at the group level. Available in API version 65.0 and later.
`EndDate`	**Type** date

Field	Details
	Properties Create, Filter, Group, Nillable, Sort, Update **Description** The end date of the group. If the `IsRamped` field is set to `true`, Transaction Management sets this date as the end date of all the line items in the group that have the Term-Defined product selling model. Available in API version 65.0 and later.
`IsRamped`	**Type** boolean **Properties** Create, Defaulted on Create, Filter, Group, Sort, Update **Description** Indicates whether the group is a group ramp segment, which is a period in a group ramp schedule with specific products, quantities, and discounts. You can use this field from Revenue Management only when the **Ramp Deals for Groups in Quotes and Orders** setting is turned on. The default value is `false`. Available in API version 65.0 and later.
`Margin`	**Type** percent **Properties** Create, Filter, Nillable, Sort, Update **Description** The optional margin percentage, specified by the sales representative at the group level. Available in API version 65.0 and later.
`MarginAmount`	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The optional margin amount, specified by the sales representative at the group level. This amount can also be considered as the summary margin amount calculated by subtracting the total cost from the summary subtotal. Available in API version 65.0 and later.
`ParentOrderItemGroupId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update

Field	Details
	Description The ID of the parent group for a nested quote line group. Available in API version 65.0 and later. This field is a relationship field. **Relationship Name** ParentOrderItemGroup **Refers To** OrderItemGroup
`SegmentType`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The duration type of the segment. You can use this field from Revenue Management only when the **Ramp Deals for Groups in Quotes and Orders** setting is turned on. Valid values are: • `Custom`—Available in API version 65.0 and later. • `Prorated`—Available in API version 67.0 and later. • `Trial`—Available in API version 67.0 and later. • `Yearly`—Available in API version 65.0 and later.
`StartDate`	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The start date of the group. If the `IsRamped` field is set to `true`, Transaction Management sets this date as the start date of all the line items in the group. Available in API version 65.0 and later.
`SummaryTotalAmount`	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The aggregated total amount of nested group lines before any discounts are applied. Available in API version 65.0 and later.

Field	Details
`TotalAdjustment`	**Type** percent **Properties** Filter, Nillable, Sort **Description** The total discount percentage applied at the group level. This percentage is calculated by using the formula: (Summary Total Amount - Summary Subtotal) / Summary Total Amount. Available in API version 65.0 and later.
`TotalAdjustmentAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The total discount amount at the group level. This amount is calculated by subtracting the summary subtotal from the summary total amount. Available in API version 65.0 and later.
`TotalCost`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The aggregated total cost of nested group lines. Available in API version 65.0 and later.
`TotalMargin`	**Type** percent **Properties** Filter, Nillable, Sort **Description** The summary margin percentage at the line item level. This percentage is calculated by using the formula: (Summary Subtotal Total Cost) / Summary Subtotal. Available in API version 65.0 and later.
`TotalMarginAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The summary margin amount calculated at the group level. Available in API version 65.0 and later.

Transaction Management Fields on Order Action

Standard and custom fields extend the standard Order Action object for use in Transaction Management. This object is available in API version 55.0 and later.

Special Access Rules

To view these fields, you must have the Revenue Cloud Advanced license. See Order Action for fields on the Salesforce platform object.

Fields

Field	Details
`Type`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The business action that created the order product. Valid values are: • `Add` • `Amend` • `Cancel` • `No Change`—A child product was added to the bundle, but the top-level product in the bundle was otherwise unchanged. • `Renew`

Transaction Management Fields on Order Item Relationship

Standard and custom fields extend the standard Order Item Relationship object for use in Transaction Management. This object is available in API version 58.0 and later.

Special Access Rules

To view these fields, you must have the Revenue Cloud Advanced license. See Order Item Relationship for fields on the Salesforce platform object.

Fields

Field	Details
`ProductRelatedComponentId`	**Type** reference

Field	Details
	Properties Create, Filter, Group, Nillable, Sort **Description** The ID of the product that is included in a product bundle, a set, or a product and an add-on. This field is a relationship field. **Relationship Name** ProductRelatedComponent **Relationship Type** Lookup **Refers To** ProductRelatedComponent

Transaction Management Fields on Quote

Standard and custom fields extend the standard Quote object for use in Transaction Management to represent information about quotes. This object is available in API version 60.0 and later.

Special Access Rules

To view these fields, you must have the Revenue Cloud Advanced license. See Quote for fields on the Salesforce platform object.

Fields

Field	Details
`AdjustmentDistributionLogic`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies how the overall discount amount is distributed among all the quote line items that have prices associated with them. The amount distributed is either the value specified in the `AppliedDiscountAmount` field or the difference between the values in the calculated `TotalPrice` and the user-specified `TotalPriceOverride` fields. Valid values are: • `Equal`—Distributes the discount amount equally among all the quote line items. • `Proportionate`—Distributes the discount amount in proportion to the `ListPriceTotal` values of the quote line items. Available in API version 65.0 and later.

Field	Details
`AppliedDiscount`	**Type** percent **Properties** Create, Filter, Nillable, Sort, Update **Description** The percent discount applied to each quote line item. Available in API version 65.0 and later.
`AppliedDiscountAmount`	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The discount amount that's distributed among all the quote line items that have prices associated with them. This amount is distributed based on the logic specified in the `AdjustmentDistributionLogic` field. Available in API version 65.0 and later.
`LastPricedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when the quote is last priced.
`OriginalActionType`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** Specifies the action that created the quote. Valid values are: • `Amend`—Indicates that the quote was created to amend assets. • `Cancel`—Indicates that the quote was created to cancel assets. • `Renew`—Indicates that the quote was created to renew assets. • `Transfer`—Indicates that the quote was created to transfer assets. Available in API version 61.0 and later.
`PartnerAccountId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update

Field	Details
	Description ID of the related partner account. This field is a relationship field. **Relationship Name** PartnerAccount **Relationship Type** Lookup **Refers To** Account
`TotalPriceOverride`	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The value that the `TotalPrice` field must be set to by applying overall discounts. Transaction Management calculates the overall discount amount by finding the difference between the value in the calculated `TotalPrice` field and the value in this field. It then uses the logic specified in the `AdjustmentDistributionLogic` field to distribute the discount amount among all the quote line items that have prices associated with them. Available in API version 65.0 and later.
`TotalPriceWithTax`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of `TotalPrice` and `TotalTaxAmount`. This field is available only when you turn on Add Estimated Tax to Quotes and Orders settings and enable Revenue Cloud in your org. Available in API version 64.0 and later.
`TotalTaxAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The total amount of all taxes. This field is available only when you turn on Add Estimated Tax to Quotes and Orders settings and enable Revenue Cloud in your org. Available in API version 64.0 and later.

Field	Details
	This field is a calculated field.
`TransactionType`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies the type of quote being processed. Valid value is: • `AdvancedConfigurator`—Indicates that the transaction must be processed by using the configuration rules and constraints set up in Constraint Rules Engine. Available in API version 62.0 and later.
`ValidationResult`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies whether the quote was configured and priced. This field serves as a staleness indicator. When this value is blank, pricing is up-to-date. When this value is set, it signals that the quote needs attention before its pricing can be trusted. Valid values are: • `Blank/null`—Specifies that the quote pricing is current and valid. • `MissingContributor`—Specifies that the quote contains a derived product but not its pricing source. This occurs when the quote contains derived products without its necessary pricing sources. • `TransactionIncomplete`—Specifies that the quote wasn't configured and priced. This typically occurs when a line item or action is modified outside the standard pricing flow, such as saving changes through a method other than the Place Quote or Place Sales Transaction API. Available in API version 61.0 and later.

Transaction Management Fields on Quote Line Group

Standard and custom fields extend the standard Quote Line Group object for use in Transaction Management.

Special Access Rules

To view these fields, you must have the Revenue Cloud Advanced license. See Quote Line Group for fields on the Salesforce platform object.

Fields

Field	Details
`Discount`	**Type** percent **Properties** Create, Filter, Nillable, Sort, Update **Description** The optional discount percentage, specified by the sales representative at the group level. Available in API version 65.0 and later.
`DiscountAmount`	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The optional discount amount, specified by the sales representative at the group level. Available in API version 65.0 and later.
`EndDate`	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The end date of the group. If the `IsRamped` field is set to `true`, Transaction Management sets this date as the end date of all the line items in the group that have the Term-Defined product selling model. Available in API version 65.0 and later.
`IsRamped`	**Type** boolean **Properties** Create, Defaulted on Create, Filter, Group, Sort, Update **Description** Indicates whether the group is a group ramp segment, which is a period in a group ramp schedule with specific products, quantities, and discounts. You can use this field from Revenue Management only when the **Ramp Deals for Groups in Quotes and Orders** setting is turned on. The default value is `false`. Available in API version 65.0 and later.

Field	Details
Margin	**Type** percent **Properties** Create, Filter, Nillable, Sort, Update **Description** The optional margin percentage, specified by the sales representative at the group level. Available in API version 65.0 and later.
MarginAmount	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The optional margin amount, specified by the sales representative at the group level. This amount can also be considered as the summary margin amount calculated by subtracting the total cost from the summary subtotal. Available in API version 65.0 and later.
ParentQuoteLineGroupId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the parent group for a nested quote line group. Available in API version 65.0 and later. This field is a relationship field. **Relationship Name** ParentQuoteLineGroup **Refers To** QuoteLineGroup
SegmentType	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The duration type of the segment. You can use this field from Revenue Management only when the **Ramp Deals for Groups in Quotes and Orders** setting is turned on. Valid values are: • `Custom`—Available in API version 65.0 and later. • `Prorated`—Available in API version 67.0 and later. • `Trial`—Available in API version 67.0 and later.

Field	Details
	• `Yearly`—Available in API version 65.0 and later.
`StartDate`	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The start date of the group. If the `IsRamped` field is set to `true`, Transaction Management sets this date as the start date of all the line items in the group. Available in API version 65.0 and later.
`SummaryTotalAmount`	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The aggregated total amount of nested group lines before any discounts are applied. Available in API version 65.0 and later.
`TotalAdjustment`	**Type** percent **Properties** Filter, Nillable, Sort **Description** The total discount percentage applied at the group level. This percentage is calculated by using the formula: (Summary Total Amount - Summary Subtotal) / Summary Total Amount. Available in API version 65.0 and later.
`TotalAdjustmentAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The total discount amount at the group level. This amount is calculated by subtracting the summary subtotal from the summary total amount. Available in API version 65.0 and later.
`TotalCost`	**Type** currency **Properties** Filter, Nillable, Sort

Field	Details
	Description The aggregated total cost of nested group lines. Available in API version 65.0 and later.
TotalMargin	**Type** percent **Properties** Filter, Nillable, Sort **Description** The summary margin percentage at the line item level. This percentage is calculated by using the formula: (Summary Subtotal Total Cost) / Summary Subtotal. Available in API version 65.0 and later.
TotalMarginAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The summary margin amount calculated at the group level. Available in API version 65.0 and later.

Transaction Management Fields on Quote Line Item

Standard and custom fields extend the standard Quote Line Item object for use in Transaction Management to represent information about line items in a quote. This object is available in API version 60.0 and later.

Special Access Rules

To view these fields, you must have the Revenue Cloud Advanced license. See Quote Line Item for fields on the Salesforce platform object.

Fields

Field	Details
DiscountAmount	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** Specifies the fixed amount discount to apply to the quote line item.

Field	Details
`EffectiveGrantDate`	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The date on which the resources associated with the quote line item are granted.
`EndDateTime`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The end date and time of the quote line item, which is calculated by using the values in the `EndDate`, `EndTime`, and `StartEndTimeZone` fields. If the `EndTime` field doesn't have a value, 23:59:59 is used for the calculation. If the `StartEndTimeZone` field doesn't have a value, GMT is used for the calculation. Available in API version 65.0 and later.
`EndQuantity`	**Type** double **Properties** Filter, Nillable, Sort **Description** The quantity available on the quote line item end date. The field is read-only. It is calculated by adding the Start Quantity and the existing Quantity fields.
`EndTime`	**Type** time **Properties** Create, Filter, Nillable, Sort, Update **Description** The end time of the quote line item. Available in API version 65.0 and later.
`Margin`	**Type** percent **Properties** Create, Filter, Nillable, Sort, Update **Description** The optional margin percentage, specified by the sales representative at the line item level. Available in API version 65.0 and later.

Field	Details
MarginAmount	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The optional margin amount, specified by the sales representative at the line item level. Available in API version 65.0 and later.
PartnerDiscountPercent	**Type** percent **Properties** Create, Filter, Nillable, Sort, Update **Description** The discount percentage given to the partner for the quote line.
PartnerUnitPrice	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The unit price after discount given to the partner for the quote line.
PriceWaterfallIdentifier	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The price waterfall identifier generated by Salesforce Pricing that's associated with the pricing of the detail record.
StartDateTime	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The start date and time of the quote line item, which is calculated by using the values in the StartDate, StartTime, and StartEndTimeZone fields. If the StartTime field doesn't have a value, 00:00:00 is used for the calculation. If the StartEndTimeZone field doesn't have a value, GMT is used for the calculation. Available in API version 65.0 and later.

Field	Details
StartEndTimeZone	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The time zone for the quote line item's start and end dates, times, and datetimes. Available in API version 65.0 and later.
StartTime	**Type** time **Properties** Create, Filter, Nillable, Sort, Update **Description** The start time of the quote line item. Available in API version 65.0 and later.
StartQuantity	**Type** double **Properties** Create, Filter, Nillable, Sort **Description** The quantity available on the quote line item start date.
TotalAdjustment	**Type** percent **Properties** Filter, Nillable, Sort **Description** The total discount percentage applied at the line item level. This percentage is calculated by using the formula: (Total Line Amount - Net Total Price) / Total Line Amount. Available in API version 65.0 and later.
TotalCost	**Type** currency **Properties** Filter, Nillable, Sort **Description** The total cost of all products sold in the order, calculated by multiplying the quantity by the unit cost. Available in API version 65.0 and later.
TotalMargin	**Type** percent

Field	Details
	Properties Filter, Nillable, Sort **Description** The effective margin percentage at the line item level. This percentage is calculated by using the formula: (Net Total Price - Total Cost) / Net Total Price. Available in API version 65.0 and later.
TotalMarginAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The effective margin amount at the line item level. This amount is calculated by subtracting total cost from net total price. Available in API version 65.0 and later.
UnitCost	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The unit cost of a product being sold as part of the order. Available in API version 65.0 and later.
ValidationResult	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies whether the quote line item is configured and priced. A quote can be activated only after all its quote line items are configured and priced. Valid value is: • `Warning`—Indicates that the quote line item isn’t configured and priced. Available in API version 60.0 and later.

Transaction Management Fields on Quote Document

Standard and custom fields extend the standard Quote Document object for use in Transaction Management to represent information about quote documents. This object is available in API version 61.0 and later.

Special Access Rules

To view these fields, you must have the Revenue Cloud Advanced license. See Quote Document for fields on the Salesforce platform object.

Fields

Field	Details
`Document Template`	**Type** String **Properties** Create, Filter, Group, Nillable, Sort **Description** The template ID used for generating the quote document.
`Status`	**Type** Picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The status of the quote document. Possible values are: • `Completed` • `Failed` • `Generating` • `In Progress` • `None` • `Queued` The default value is `None`.

Transaction Management Tooling API Objects

Tooling API exposes metadata used in developer tooling that you can access through REST or SOAP. Tooling API's SOQL capabilities for many metadata types allow you to retrieve smaller pieces of metadata.

TransactionProcessingType
Represents the settings to configure the processing constraints for a request.. This object is available in API version 63.0 and later.

TransactionProcessingType

Represents the settings to configure the processing constraints for a request.. This object is available in API version 63.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Refer to the Usage section to learn more about creating Transaction Processing Type records based on your requirements. See the setup details to specify the default rule engine on the Revenue Settings page.

Supported SOAP API Calls

`create()`, `describeSObjects()`, `query()`, `retrieve()`

Supported REST API Methods

`GET, HEAD, POST, Query`

Fields

Field	Details
`Description`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The description of the transaction processing configuration to help Salesforce admins with configuration in their orgs.
`DeveloperName`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** Required. The unique name of the object in the API. This name can contain only underscores and alphanumeric characters, and must be unique in your org. It must begin with a letter, not include spaces, not end with an underscore, and not contain two consecutive underscores. In managed packages, this field prevents naming conflicts on package installations. With this field, a developer can change the object's name in a managed package and the changes are reflected in a subscriber's organization. Label is **Record Type Name**.
`Language`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Nillable, Update **Description** The language of the TransactionProcessingType object. Valid values are:

Field	Details
	• `da`—Danish • `de`—German • `en_US`—English • `es`—Spanish • `es_MX`—Spanish (Mexico) • `fi`—Finnish • `fr`—French • `it`—Italian • `ja`—Japanese • `ko`—Korean • `nl_NL`—Dutch • `no`—Norwegian • `pt_BR`—Portuguese (Brazil) • `ru`—Russian • `sv`—Swedish • `th`—Thai • `zh_CN`—Chinese (Simplified) • `zh_TW`—Chinese (Traditional)
`MasterLabel`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** The label for the TransactionProcessingType object.
`PricingPreference`	**Type** string **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies whether to execute the price calculation step for each sales transaction record. Valid values are: • `Force`—Reprices all lines. • `System`—Performs a delta pricing request on the unprocessed lines when Delta Pricing is enabled in the org. • `Skip`—Skips the pricing request on all lines. Available in API version 65.0 and later.

Field	Details
RatingPreference	**Type** string **Properties** **Description** Specifies whether catalog rates are fetched and saved during quote creation. Valid value is `Fetch`. Use this value to retrieve and save catalog rates for usage resources associated with each sales transaction record. If this value isn't specified, catalog rates aren't saved by default when a quote line item is added to a quote. Available in API version 66.0 and later if Rate Management is enabled.
RuleEngine	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort **Description** The rule engine to be used for processing rules. Valid values are: • `AdvancedConfigurator` • `StandardConfigurator`
SaveType	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Specifies how the transaction results are processed when saved for Salesforce administrators to adjust the user experience as desired. Valid values are: • `Standard` • `Large`—Reserved for future use.
TaxPreference	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies whether to execute or skip the tax calculation step for each sales transaction record. Valid value is `Skip`. If this value isn't specified, then tax calculation request is performed by default. Available in API version 65.0 and later.

Usage

Create transaction type records by calling this resource through a POST method.

```
/services/data/v67.0/tooling/sobjects/TransactionProcessingType
```

Here's a sample payload that specifies the rule engine to use and steps to skip for each sales transaction record.

```
{
  "SaveType": "Standard",
  "Description": "Setup for Transaction Processing Type",
  "DeveloperName": "SkipPricingAndTaxStep",
  "MasterLabel": "SkipPricingAndTaxStep",
  "RuleEngine": "StandardConfigurator",
  "PricingPreference": "Skip",
  "TaxPreference": "Skip"
}
```

Here's a sample payload that specifies a value for rating preference and the steps to skip for each sales transaction record.

```
{
  "SaveType": "Standard",
  "Description": "Setup for Transaction Processing Type",
  "DeveloperName": "SkipPricingAndTaxStep",
  "MasterLabel": "SkipPricingAndTaxStep",
  "RuleEngine": "StandardConfigurator",
  "PricingPreference": "Skip",
  "TaxPreference": "Skip",
  "RatingPreference": "Fetch"
}
```

Transaction Management Platform Event

Use the QuoteSaveEvent event to notify subscribers after saving of a quote is processed.

EDITIONS

Available in: Lightning Experience

Available in all Salesforce orgs where the admin settings for products related to the use case are enabled. The Salesforce org must have a Revenue Management or Subscription Management license.

CreateAssetOrderEvent
: Notifies subscribers that the process started by the `/actions/standard/createOrUpdateAssetFromOrder` or `/actions/standard/createOrUpdateAssetFromOrderItem` request is complete. If the process is successful, use this event to learn about the new assets. If the request isn't successful, use this event to learn about the errors and how to fix them. This object is available in API version 55.0 and later.

PlaceOrderCompletedEvent
: Notifies subscribers of an order being created or updated by invoking the Place Order API or the Place Sales Transaction API. This object is available in API version 63.0 and later.

QuoteSaveEvent
: Notifies subscribers that the process started by the Place Quote or Place Sales Transaction API request is complete. If the process is successful, use this event to learn about the updated quote. If the request isn't successful, use this event to learn about the errors and how to fix them. This object is available in API version 60.0 and later.

QuoteToOrderCompletedEvent

Notifies subscribers when the `/actions/standard/createOrderFromQuote` REST request is complete. If the request is successful, use this event to learn about the Order record. If the request isn't successful, use this event to learn about the errors associated with the request. This object is available in API version 56.0 and later.

CreateAssetOrderEvent

Notifies subscribers that the process started by the `/actions/standard/createOrUpdateAssetFromOrder` or `/actions/standard/createOrUpdateAssetFromOrderItem` request is complete. If the process is successful, use this event to learn about the new assets. If the request isn't successful, use this event to learn about the errors and how to fix them. This object is available in API version 55.0 and later.

Supported Calls

`describeSObjects()`

Supported Subscribers

Subscriber	Supported?
Apex Triggers	✓
Flows	✓
Processes	✓
Pub/Sub API	✓
Streaming API (CometD)	✓

Subscription Channel

`/event/CreateAssetOrderEvent`

Event Delivery Allocation Enforced

No

Special Access Rules

This object is available if Subscription Management or Revenue Cloud is enabled in your org. Users must have Read access on this event to receive or view event notifications.

Fields

Field	Details
`AssetDetails`	**Type** CreateAssetOrderDtlEvent on page 1377 **Properties** Nillable **Description** A list of AssetDetail records created as a result of a successful request. Each AssetDetail contains an order item ID, asset ID, and IsSuccess flag. If the request failed, the AssetDetail also contains an error code and error message.
`CorrelationIdentifier`	**Type** string **Properties** Nillable **Description** Reserved for future use.
`EventUuid`	**Type** string **Properties** Nillable **Description** A universally unique identifier (UUID) that identifies a platform event message.
`IsLastEvent`	**Type** boolean **Properties** Defaulted on create **Description** Indicates whether this event is the final event in the request (`true`) or not (`false`). If `true`, then there are no more events associated with the request. This field is populated only in the final event in the request. The default value is `false`. This field is available in API version 62.0 and later.
`OrderIdentifier`	**Type** string **Properties** Nillable

Field	Details
	Description The ID of the order associated with this event. Available with Revenue Cloud in API version 64.0 and later.
`ReplayId`	**Type** string **Properties** Nillable **Description** Represents an ID value that is populated by the system and refers to the position of the event in the event stream. Replay ID values aren't guaranteed to be contiguous for consecutive events. A subscriber can store a replay ID value and use it on resubscription to retrieve missed events that are within the retention window.
`RequestIdentifier`	**Type** string **Properties** Nillable **Description** Id of the request that triggered the event.

Example: A user successfully runs a `createOrUpdateAssetFromOrder` request on an order with two order items. The published createAssetOrderEvent contains this information.

- RequestId: 0001
- AssetDetail
  - OrderItemId: 802XX0000000001
  - AssetId: 02iXX000000001
  - IsSuccess: True
- AssetDetail
  - OrderItemId: 802XX0000000001
  - AssetId: 02iXX000000002
  - IsSuccess: True

Example: A user runs a `createOrUpdateAssetFromOrder` request on an order with two order items, but doesn't have Create access on assets. The request fails, and the published createAssetOrderEvent contains this information.

- RequestId: 0002
- AssetDetail
  - OrderItemId: 802XX0000000001
  - IsSuccess: False

    - ErrorCode: `INSUFFICIENT_ACCESS`
    - ErrorMessage: User doesn’t have Create Access to asset.
- AssetDetail
    - OrderItemId: 802XX0000000001
    - IsSuccess: False
    - ErrorCode: `INSUFFICIENT_ACCESS`
    - ErrorMessage: User doesn’t have Create Access to asset.

CreateAssetOrderDtlEvent
Contains information about an attempt to create or update an asset as a result of `/actions/standard/createOrUpdateAssetFromOrder`. If the request was successful, the event shows information about the asset. If the request failed, the event shows error information. This object is included in an `CreateAssetOrderEvent` message. You can't subscribe to `CreateAssetOrderDtlEvent` directly. This object is available in API version 55.0 and later.

CreateAssetOrderDtlEvent

Contains information about an attempt to create or update an asset as a result of `/actions/standard/createOrUpdateAssetFromOrder`. If the request was successful, the event shows information about the asset. If the request failed, the event shows error information. This object is included in an `CreateAssetOrderEvent` message. You can't subscribe to `CreateAssetOrderDtlEvent` directly. This object is available in API version 55.0 and later.

Supported Calls

`describeSObjects()`

Supported Subscribers

Subscriber	Supported?
Apex Triggers	✓
Flows	✓
Processes	✓
Pub/Sub API	✓
Streaming API (CometD)	✓

Subscription Channel

`/event/CreateAssetOrderDtlEvent`

Special Access Rules

This object is available if Revenue Cloud is installed in your org. Users must have Read access on this event to receive or view event notifications.

Fields

Field	Details
`AssetId`	**Type** reference **Properties** Nillable **Description** The ID of the asset that was created or updated. This field is a relationship field. **Relationship Name** Asset **Relationship Type** Lookup **Refers To** Asset
`ErrorCode`	**Type** string **Properties** Nillable **Description** Reference code for the type of error that occurred.
`ErrorMessage`	**Type** string **Properties** Nillable **Description** Information about the error that occurred after the request was made.
`EventUuid`	**Type** string **Properties** Nillable **Description** A universally unique identifier (UUID) that identifies a platform event message.
`IsSuccess`	**Type** boolean **Properties** Defaulted on create

Field	Details
	Description Indicates whether the request to create the asset for the order item was successful (true) or not (false). The default value is `false`. Available in API version 61.0 and later.
`OrderItemId`	**Type** reference **Properties** **Description** The ID of the order item used in the request. Available in API version 61.0 and later. This field is a relationship field. **Relationship Name** OrderItem **Relationship Type** Lookup **Refers To** OrderItem
`ReplayId`	**Type** string **Properties** Nillable **Description** Represents an ID value that is populated by the system and refers to the position of the event in the event stream. Replay ID values aren't guaranteed to be contiguous for consecutive events. A subscriber can store a replay ID value and use it on resubscription to retrieve missed events that are within the retention window.

PlaceOrderCompletedEvent

Notifies subscribers of an order being created or updated by invoking the Place Order API or the Place Sales Transaction API. This object is available in API version 63.0 and later.

Supported Calls

`describeSObjects()`

Supported Subscribers

Subscriber	Supported?
Apex Triggers	✓
Flows	✓
Processes	✓
Pub/Sub API	✓
Streaming API (CometD)	✓

Subscription Channel

`/event/PlaceOrderCompletedEvent`

Event Delivery Allocation Enforced

Yes

Fields

Field	Details
`AppUsageTypes`	**Type** string **Properties** Nillable **Description** Tag that represents the application that's using the order and determines how an order is processed. For example, the `AppUsageTypes` field value for Revenue Cloud orders is `RevenueLifecycleManagement`.
`CorrelationIdentifier`	**Type** string **Properties** Nillable **Description** Reserved for future use.
`EventUuid`	**Type** string **Properties** Nillable

Field	Details
	Description A universally unique identifier (UUID) that identifies a platform event message.
`HasErrors`	**Type** boolean **Properties** Defaulted on create **Description** Indicates whether errors occurred when creating or updating the order (`true`) or not (`false`). The default value is `false`.
`OrderId`	**Type** reference **Properties** Nillable **Description** ID of the order record. This field is a relationship field. **Relationship Name** Order **Refers To** Order
`ReplayId`	**Type** string **Properties** Nillable **Description** Represents an ID value that is populated by the system and refers to the position of the event in the event stream. Replay ID values aren't guaranteed to be contiguous for consecutive events. A subscriber can store a replay ID value and use it on resubscription to retrieve missed events that are within the retention window.
`RequestIdentifier`	**Type** string **Properties** Nillable **Description** ID of the request that triggered the event.

QuoteSaveEvent

Notifies subscribers that the process started by the Place Quote or Place Sales Transaction API request is complete. If the process is successful, use this event to learn about the updated quote. If the request isn't successful, use this event to learn about the errors and how to fix them. This object is available in API version 60.0 and later.

Supported Calls

`describeSObjects()`

Supported Subscribers

Subscriber	Supported?
Apex Triggers	✔
Flows	✔
Processes	✔
Streaming API (CometD)	✔

Streaming API Subscription Channel

`/event/QuoteSaveEvent`

Special Access Rules

This object is available in orgs with Subscription Management or Revenue Management.

Fields

Field	Details
`CorrelationIdentifier`	**Type** string **Properties** Nillable **Description** Reserved for future use.
`EventUuid`	**Type** string **Properties** Nillable

Field	Details
	Description A universally unique identifier (UUID) that identifies a platform event message.
HasErrors	**Type** boolean **Properties** Defaulted on create **Description** The default value is false. Possible values are: • false • true
QuoteId	**Type** reference **Properties** Nillable **Description** The ID of the quote associated with this event. This field is a relationship field.
ReplayId	**Type** string **Properties** Nillable **Description** Represents an ID value that is populated by the system and refers to the position of the event in the event stream. Replay ID values aren't guaranteed to be contiguous for consecutive events. A subscriber can store a replay ID value and use it on resubscription to retrieve missed events that are within the retention window.
RequestIdentifier	**Type** string **Properties** Nillable **Description** ID of the request that triggered the event.

QuoteToOrderCompletedEvent

Notifies subscribers when the `/actions/standard/createOrderFromQuote` REST request is complete. If the request is successful, use this event to learn about the Order record. If the request isn't successful, use this event to learn about the errors associated with the request. This object is available in API version 56.0 and later.

Supported Calls

`describeSObjects()`

Supported Subscribers

Subscriber	Supported?
Apex Triggers	✓
Flows	✓
Processes	✓
Pub/Sub API	✓
Streaming API (CometD)	✓

Subscription Channel

`/event/QuoteToOrderCompletedEvent`

Event Delivery Allocation Enforced

No

Special Access Rules

This object is available with Revenue Cloud.

Field	Details
`CorrelationIdentifier`	**Type** string **Properties** Nillable **Description** Reserved for future use.
`EventUuid`	**Type** string

Field	Details
	Properties Nillable **Description** A universally unique identifier (UUID) that identifies a platform event message.
`HasErrors`	**Type** boolean **Properties** Defaulted on create **Description** Contains `true` if errors occurred during the process; otherwise `false`. The default value is `false`.
`OrderId`	**Type** string **Properties** Nillable **Description** The ID of the order created from the quote. If the process failed, this field is null.
`OrderNumber`	**Type** string **Properties** Nillable **Description** The user-friendly, unique number assigned to the order created from the quote.
`QuoteToOrderErrorDetailEvents`	**Type** QuoteToOrderErrDtlEvent[] **Properties** Nillable **Description** Contains a list of error messages and error codes if the request failed.
`ReplayId`	**Type** string **Properties** Nillable **Description** Represents an ID value that is populated by the system and refers to the position of the event in the event stream. Replay ID values aren't guaranteed to be contiguous for consecutive

Field	Details
	events. A subscriber can store a replay ID value and use it on resubscription to retrieve missed events that are within the retention window.
`RequestIdentifier`	**Type** string **Properties** Nillable **Description** The unique ID returned in the `actions/standard/createOrderFromQuote` response. Use this ID to identify the event for a specific request.

Transaction Management Business APIs

Use the Transaction Management Business APIs to fetch instant pricing data on a quote or an order, to create a quote, or to create an order.

This table lists the available Transaction Management resources.

Resource	Description
`/connect/revenue-management/assets/actions/amend` (POST)	Initiate and execute the amendment of a quote or an order.
`/connect/revenue-management/assets/actions/cancel` (POST)	Initiate and execute the cancellation of an asset.
`/connect/revenue-management/assets/actions/renew` (POST)	Initiate and execute the renewal of an asset.
`/industries/cpq/quotes/actions/get-instant-price` (POST)	Fetch instant pricing data on the quote or order line data grid and associated summary component. This API offers capabilities to either create a context or update the existing one based on the provided context ID.
`/commerce/sales-orders/actions/place` (POST)	Place orders with integrated pricing, configuration, and validation, and manage them throughout their entire lifecycle. Additionally, update an order or insert order items.

Resource	Description
/commerce/quotes/actions/place (POST)	Create a quote to discover and price products and services. Additionally, insert, update, or delete a quote line item.
/connect/revenue-management/sales-transaction-contexts/**resourceId**/actions/ramp-deal-create (POST)	Create a ramp deal for a customer on a product. Sales reps can use ramp deals to provide yearly deals to a customer, resulting in long-term revenue and customer relationship. A customer can create, update, or view multiple segments of periods for their subscription term with different attributes for each segment.
/connect/revenue-management/sales-transaction-contexts/**resourceId**/actions/ramp-deal-update (POST)	Modify a ramp deal in scenarios where a segment has updates such as quantity, discount, or date change.
/connect/revenue-management/sales-transaction-contexts/**resourceId**/actions/ramp-deal-view (GET)	View a ramp deal related to a quote line item or an order item.
/connect/revenue-management/sales-transaction-contexts/**resourceId**/actions/ramp-deal-delete (POST)	Delete a ramp deal to convert a ramped product to include a single quote line item or order item.
/connect/rev/sales-transaction/actions/place (POST)	Create a sales transaction, such as an order or a quote, with integrated pricing and configuration. Also, update an order or a quote, and insert and delete order or quote line items to calculate the estimated tax.
/connect/rev/sales-transaction/actions/clone (POST)	Create a clone of a sales transaction, such as a quote or an order. You can also clone a quote line item or an order item record with its related records and configurations.
/connect/rev/sales-transaction/actions/place-supplemental-transaction (POST)	Create a supplemental order or change orders after they are submitted for processing, such as during the fulfillment process.

Resource	Description
/connect/revenue/transaction-management/sales-transactions/actions/read (POST)	Retrieve sales transaction data efficiently from an initialized or a hydrated context.
/global-promotions-management/promotions (GET, POST, PUT)	Get rewards based on a product selling model template.
/revenue/transaction-management/sales-transactions/actions/get-eligible-promotions (POST)	Get eligible promotions for line items within a quote or an order.
/revenue/transaction-management/assets/actions/swap (POST)	Exchange one product for another of equivalent or different value. The change is tracked as a swap request with linked asset actions and a net-zero order total where applicable. The API creates an amendment quote and order with order actions and quote action subtypes.
/revenue/transaction-management/assets/actions/upgrade (POST)	Move a lower-tier product to a higher-tier product. The change is tracked as an upgrade request with linked asset actions and quote or order line linkage for reporting and auditing. This API creates an amendment quote and order with order actions and quote action subtypes.
/revenue/transaction-management/assets/actions/downgrade (POST)	Move to a lower-tier or lower-value product. The change is tracked as a downgrade request with linked asset actions and quote or order line linkage for reporting and auditing. This API creates an amendment quote and order with downgrade-specific order actions and quote action subtypes.

Resources
Learn more about the available Quote and Order Capture resources.

Request Bodies
Learn more about the available API request bodies.

Response Bodies

Learn more about the available response bodies.

SEE ALSO:

Connect REST API Developer Guide: Introduction

Resources

Learn more about the available Quote and Order Capture resources.

Asset Amendment (POST)

Initiate and execute the amendment of a quote or an order.

Asset Cancellation (POST)

Initiate and execute the cancellation of an asset.

Asset Renewal (POST)

Initiate and execute the renewal of an asset.

Clone Sales Transaction (POST)

Create a clone of a sales transaction, such as a quote or an order. You can also clone a quote line item or an order item record with its related records and configurations.

Create Promotions (GET, POST, PUT)

Get rewards based on a product selling model template.

Get Eligible Promotions (POST)

Get eligible promotions for line items within a quote or an order.

Initiate Downgrade (POST)

Move to a lower-tier or lower-value product. The change is tracked as a downgrade request with linked asset actions and quote or order line linkage for reporting and auditing. This API creates an amendment quote and order with downgrade-specific order actions and quote action subtypes.

Initiate Swap (POST)

Exchange one product for another of equivalent or different value. The change is tracked as a swap request with linked asset actions and a net-zero order total where applicable. The API creates an amendment quote and order with order actions and quote action subtypes.

Initiate Upgrade (POST)

Move a lower-tier product to a higher-tier product. The change is tracked as an upgrade request with linked asset actions and quote or order line linkage for reporting and auditing. This API creates an amendment quote and order with order actions and quote action subtypes.

Instant Pricing (POST)

Fetch instant pricing data on the quote or order line data grid and associated summary component. This API offers capabilities to either create a context or update the existing one based on the provided context ID.

Place Order (POST)

Place orders with integrated pricing, configuration, and validation, and manage them throughout their entire lifecycle. Additionally, update an order or insert order items.

Place Quote (POST)

Create a quote to discover and price products and services. Additionally, insert, update, or delete a quote line item.

Place Sales Transaction (POST)

Create a sales transaction, such as an order or a quote, with integrated pricing and configuration. Also, update an order or a quote, and insert and delete order or quote line items to calculate the estimated tax.

Place Supplemental Transaction (POST)

Create a supplemental order or change orders after they are submitted for processing, such as during the fulfillment process.

Read Sales Transaction (POST)

Retrieve sales transaction data efficiently from an initialized or a hydrated context.

Retrieve Sales Transaction API Errors (GET)

Retrieve any asynchronous error details associated with a sales transaction request.

Create Ramp Deal (POST)

Create a ramp deal for a customer on a product. Sales reps can use ramp deals to provide yearly deals to a customer, resulting in long-term revenue and customer relationship. A customer can create, update, or view multiple segments of periods for their subscription term with different attributes for each segment.

Delete Ramp Deal (POST)

Delete a ramp deal to convert a ramped product to include a single quote line item or order item.

Update Ramp Deal (POST)

Modify a ramp deal in scenarios where a segment has updates such as quantity, discount, or date change.

View Ramp Deal (GET)

View a ramp deal related to a quote line item or an order item.

Asset Amendment (POST)

Initiate and execute the amendment of a quote or an order.

Considerations

For usage products, creating an order directly by setting `outputRecordType` to `Order` is currently not supported. The direct-to-order flow can create Order Products without required related Rate Card Entry records, which can cause order activation to fail.

When you amend usage assets, use a two-step flow.

- Create an amendment quote by calling `initiateAmendment` with `outputRecordType` set to `Quote`.
- Create an order from that amendment quote.

Special Access Rules

To use this API, you need the InitiateAmend API permission set.

Resource

```
/connect/revenue-management/assets/actions/amend
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/revenue-management/assets/actions/amend
```

Available version

62.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "assetIds": [
    "02iSG0000003NMhYAM",
    "02iSG0000006DvSYAU"
  ],
  "amendmentStartDate": "2023-10-04T00:00:00",
  "contractId": "800SG00000CFpepYAD",
  "opportunityId": "006SG000004W5tVYAS",
  "outputRecordId": "801SG00000DX1jWYAT",
  "outputRecordType": "Quote",
  "quantityChange": 5
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
assetIds	String[]	IDs of the assets that you want to add to the amendment record.	Required	62.0
amendmentStartDate	String	Start date of the amendment.	Required	62.0
contractId	String	ID of the Contract record that you want to sync with the amendment quote.	Optional	62.0
opportunityId	String	ID of the Opportunity record that you want to sync with the amendment quote.	Optional	62.0
outputRecordId	String	ID of the quote or order record that you want to add the assets to.	Optional	62.0
outputRecordType	String	Type of amendment record that you want to create. For usage products, set this value to Quote. The usage-related records (QuoteLineItemUseResourceGrant, QuoteLineItemUsageResourcePolicy, and QuoteLineRateCardEntry) are copied in the quote action flow, not in the amend order flow.	Required	62.0
quantityChange	Double	Quantity to add to or reduce from the asset's existing quantity.	Required	62.0

Response body for POST

Amendment

Asset Cancellation (POST)

Initiate and execute the cancellation of an asset.

Special Access Rules

To use this API, you need the InitiateCancellation API permission set.

Resource

```
/connect/revenue-management/assets/actions/cancel
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/revenue-management/assets/actions/cancel
```

Available version

62.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "assetIds": [
    "02iSG0000003NMhYAM",
    "02iSG0000006DvSYAU"
  ],
  "cancellationDate": "2023-10-04T00:00:00",
  "contractId": "800SG00000CFpepYAD",
  "opportunityId": "006SG000004W5tVYAS",
  "outputRecordId": "801SG00000DX1jWYAT",
  "outputRecordType": "Quote"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
assetIds	String[]	IDs of the assets that you want to cancel. All assets in a request must belong to the same price book.	Required	62.0
cancellationDate	String	Effective date of the cancellation.	Required	62.0
contractId	String	ID of the Contract record that you want to sync with the cancellation quote.	Optional	62.0
opportunityId	String	ID of the Opportunity record that you want to sync with the cancellation quote.	Optional	62.0
outputRecordId	String	ID of the quote or order that you want to cancel.	Optional	62.0

Name	Type	Description	Required or Optional	Available Version
`outputRecordType`	String	Type of cancellation record that you want to create.	Required	62.0

Response body for POST

Cancellation

Asset Renewal (POST)

Initiate and execute the renewal of an asset.

Special Access Rules

To use this API, you need the InitiateRenewal API permission set.

Resource

```
/connect/revenue-management/assets/actions/renew
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/revenue-management/assets/actions/renew
```

Available version

62.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "assetIds": [
    "02iSG0000003NMhYAM",
    "02iSG0000006DvSYAU"
  ],
  "contractId": "800SG00000CFpepYAD",
  "opportunityId": "006SG000004W5tVYAS",
  "outputRecordId": "801SG00000DX1jWYAT",
  "outputRecordType": "Quote",
  "renewalEndDate": "2024-10-03T23:59:59",
  "renewalStartDate": "2023-10-04T00:00:00"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`assetIds`	String[]	IDs of the assets that you want to renew.	Required	62.0

Name	Type	Description	Required or Optional	Available Version
contractId	String	ID of the Contract record that you want to sync with the renewal of the Quote or Order record.	Optional	62.0
opportunityId	String	ID of the Opportunity record that you want to sync with the renewal quote.	Optional	62.0
outputRecordId	String	ID of the Quote or Order record that you want to renew.	Optional	62.0
outputRecordType	String	Type of renewal record that you want to create.	Required	62.0
renewalEndDate	String	End date of the renewal process for the assets.	Optional	62.0
renewalStartDate	String	Start date of the renewal process for the assets. Required for early asset renewals and renewing expired assets, using today's date or a future date.	Optional	62.0

Response body for POST

Renewal

Clone Sales Transaction (POST)

Create a clone of a sales transaction, such as a quote or an order. You can also clone a quote line item or an order item record with its related records and configurations.

This API supports the cloning of records for these objects.

- Quote
- QuoteLineItem
- OrderItem
- QuoteLineGroup
- Order
- OrderItemGroup

You can clone all items in a quote line group or order item group when the record to clone is a quote line group or an order item group record.

Resource

```
/connect/rev/sales-transaction/actions/clone
```

Resource example

```
https://<varname>yourInstance</varname>.salesforce.com/services/data/v64.0/connect/rev/sales-transaction/actions/clone
```

Available version

64.0

HTTP methods

POST

Request body for POST

JSON example

This is a sample request to clone a record within a sales transaction.

```
{
  "recordIds": ["0QLxx0000004CBYGA2"],
  "salesTransactionId": "0Q0xx0000004CE0CAM"
}
```

This is a sample request to clone all line items in a ramped group within a sales transaction.

```
{
  "recordIds": ["0QLxx0000004CBYGA2"],
  "salesTransactionId": "0Q0xx0000004CE0CAM",
  "options": {
    "lineScope": "AllLines"
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`recordIds`	String[]	ID of the record to be cloned. You can specify a single record ID only.	Required	64.0
`salesTransactionId`	String	ID of the sales transaction related to the record IDs to clone.	Required	64.0
`options`	Clone Options Input	Specifies options to clone a ramp segment within a sales transaction. You can clone only the last ramp segment.	Optional	65.0

Response body for POST

Clone Sales Transaction

Create Promotions (GET, POST, PUT)

Get rewards based on a product selling model template.

Resource

```
/global-promotions-management/promotions
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/global-promotions-management/promotions
```

Available version

66.0

HTTP methods

GET, POST, PUT

Request and response body

This example shows a sample request to create a promotion.

```
{
  "promotionDetails": {
    "additionalFieldValues": {
      "attributes": {}
    },
    "displayName": "10% off on Cisco Router",
    "isAutomatic": true,
    "isEmailActivated": false,
    "name": "10% off on Cisco Router",
    "promotionEligibility": {
      "eligibleCustomerEvents": {},
      "eligibleEnrollmentPeriod": {
        "isEnrollmentRequired": false
      },
      "eligibleProducts": [
        {
          "id": "01txx0000006igmAAA",
          "name": "Cisco Router",
          "productType": "SimpleProduct"
        }
      ]
    },
    "promotionLimits": {},
    "ruleLibrary": {
      "id": "9Qsxx0000004H76CAE",
      "name": "RLMSales"
    },
    "startDateTime": "2025-11-01T08:43:00.000Z"
  },
  "rules": [
    {
      "eventConfiguration": [],
      "journalSubType": null,
      "journalSubTypeName": null,
      "journalType": "Customer Purchase",
      "priority": 10,
      "rewardConfiguration": [
        {
          "scope": "SimpleProduct",
          "scopeDetails": [
            {
              "name": "Cisco Router",
              "id": "01txx0000006igmAAA"
            }
          ],
          "doNotDefineRewards": false,
          "rewardDetailsList": [
            {
              "productSellingModel": {
```

```
                "name": "Monthly",
                "id": "0jPxx0000000001EAA"
              },
              "discountType": "PercentageOff",
              "discountValue": 10,
              "termBasedRewards": {
                "psmTenure": {
                  "tenure": "SpecificTerm",
                  "operator": "Equals",
                  "value": 12
                },
                "rewardDuration": {
                  "tenure": "SpecificTerm",
                  "value": 3
                }
              }
            }
          ],
          "childProducts": [],
          "type": "PSMDiscount",
          "isPrimaryReward": false
        }
      ],
      "ruleName": "rule",
      "templateName": "GetRewardsBasedOnSellingModel"
    }
  ]
}
```


See Promotions Creation API reference to get additional details of the request and response properties.

Get Eligible Promotions (POST)

Get eligible promotions for line items within a quote or an order.

This API accepts line item IDs and sales transaction ID as the input and then initializes the context by filtering on the specific line items.

Resource

```
/revenue/transaction-management/sales-transactions/actions/get-eligible-promotions
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/revenue/transaction-management/sales-transactions/actions/get-eligible-promotions
```

Available version

66.0

HTTP methods

POST

Request body for POST

JSON example


```
{
  "salesTransactionId": "0Q0xx0000004EOECA2",
  "lineItemIds": [
```

```
    "0QLxx0000004E7eGAE",
    "0QLxx0000004GCeGAM",
    "0QLxx0000004E7gGAE"
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`lineItemIds`	String[]	List of line item IDs to evaluate for promotions. The object type is auto-determined from the sales transaction ID.	Required	66.0
`salesTransactionId`	String	The sales transaction ID, such as an order ID or a quote ID, for the promotion evaluation.	Required	66.0

Response body for POST

Get Eligible Promotions

Initiate Downgrade (POST)

Move to a lower-tier or lower-value product. The change is tracked as a downgrade request with linked asset actions and quote or order line linkage for reporting and auditing. This API creates an amendment quote and order with downgrade-specific order actions and quote action subtypes.

After assetization, the original asset receives an asset action with business category as Downgrade (or equivalent). This step indicates that the downgrade-from product and the new asset is created with an asset action (downgraded to), with relationships between the two. This step also enables sales reps to process downgrades and makes sure that downgrades are auditable and reportable separately from cancellations and new sales.

Resource

```
/revenue/transaction-management/assets/actions/downgrade
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/revenue/transaction-management/assets/actions/downgrade
```

Available version

66.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "swapStartDate": "2025-12-01T00:00:00Z",
  "outputRecordType": "Quote",
```

```
  "swapGroups": {
    "groups": [
      {
        "referenceId": "DOWNGRADE-001",
        "outGroup": {
          "swapAssets": [
            {
              "assetId": "02ixx0000004HOAAA2",
              "quantity": 1
            }
          ]
        },
        "inGroup": {
          "graphId": "downgradeRequest",
          "records": [
            {
              "referenceId": "refQuoteLine0",
              "record": {
                "attributes": {
                  "type": "QuoteLineItem",
                  "method": "POST"
                },
                "Product2Id": "01txx0000006iVlAAI",
                "PricebookEntryId": "01uxx0000008ym4AAA",
                "UnitPrice": 1049,
                "Quantity": "1",
                "StartDate": "2022-09-22"
              }
            }
          ]
        }
      }
    ]
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
contractId	String	ID of the contract record to downgrade.	Optional	66.0
opportunityId	String	ID of the opportunity record to downgrade.	Optional	66.0
outputRecordType	String	Record type of the output for the downgrade.	Required	66.0
swapGroups	Swap Group[]	Groups that contain the asset details for the downgrade.	Required	66.0
swapStartDate	String	Amendment start date for the downgrade action.	Required	66.0

Response body for POST

Initiate Downgrade Response

SEE ALSO:

Salesforce Help: Swap, Upgrade, or Downgrade Assets

Initiate Swap (POST)

Exchange one product for another of equivalent or different value. The change is tracked as a swap request with linked asset actions and a net-zero order total where applicable. The API creates an amendment quote and order with order actions and quote action subtypes.

When the order is assetized, the source asset gets an asset action with business category as `Swap` (reduced quantity). The new asset is created with an asset action that identifies it as swapped in, with relationships that link the swapped-from and swapped-to assets. The swaps are auditable and reportable separately from cancellations and new sales. This API also supports use cases such as trading unused licenses for credits or moving spend between products while preserving contract intent.

Resource

```
/revenue/transaction-management/assets/actions/swap
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/revenue/transaction-management/assets/actions/swap
```

Available version

66.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "swapStartDate": "2025-12-01T00:00:00Z",
  "outputRecordType": "Quote",
  "swapGroups": {
    "groups": [
      {
        "referenceId": "SWAP-001",
        "outGroup": {
          "swapAssets": [
            {
              "assetId": "02ixx0000004HOAAA2",
              "quantity": 1
            }
          ]
        },
        "inGroup": {
          "graphId": "swapRequest",
          "records": [
            {
              "referenceId": "refQuoteLine0",
```

```
                "record": {
                  "attributes": {
                    "type": "QuoteLineItem",
                    "method": "POST"
                  },
                  "Product2Id": "01txx0000006iVlAAI",
                  "PricebookEntryId": "01uxx0000008ym4AAA",
                  "UnitPrice": 1049,
                  "Quantity": "1",
                  "StartDate": "2022-09-22"
                }
              }
            ]
          }
        }
      ]
    }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`contractId`	String	ID of the contract record to swap.	Optional	66.0
`opportunityId`	String	ID of the opportunity record to swap.	Optional	66.0
`outputRecordType`	String	Record type of the output for the swap.	Required	66.0
`swapGroups`	Swap Group[]	Groups that contain the asset details for the swap.	Required	66.0
`swapStartDate`	String	Amendment start date for the swap action.	Required	66.0

Response body for POST

Initiate Swap Response

SEE ALSO:

Salesforce Help: Swap, Upgrade, or Downgrade Assets

Initiate Upgrade (POST)

Move a lower-tier product to a higher-tier product. The change is tracked as an upgrade request with linked asset actions and quote or order line linkage for reporting and auditing. This API creates an amendment quote and order with order actions and quote action subtypes.

The original asset receives an asset action with an `Upgrade` (or equivalent) business category. This step indicates that the upgrade-from product and the new asset is created with an asset action (upgraded to), with relationships between the two. This step also enables sales reps to process upgrades and makes sure that upgrades are distinguishable in reporting and analytics from cancellations plus new sales.

Resource

```
/revenue/transaction-management/assets/actions/upgrade
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/revenue/transaction-management/assets/actions/upgrade
```

Available version

66.0

HTTP methods

POST

Request body for POST

JSON example


```
{
  "swapStartDate": "2025-12-01T00:00:00Z",
  "outputRecordType": "Quote",
  "swapGroups": {
    "groups": [
      {
        "referenceId": "UPGRADE-001",
        "outGroup": {
          "swapAssets": [
            {
              "assetId": "02ixx0000004HOAAA2",
              "quantity": 1
            }
          ]
        },
        "inGroup": {
          "graphId": "upgradeRequest",
          "records": [
            {
              "referenceId": "refQuoteLine0",
              "record": {
                "attributes": {
                  "type": "QuoteLineItem",
                  "method": "POST"
                },
                "Product2Id": "01txx0000006iVlAAI",
                "PricebookEntryId": "01uxx0000008ym4AAA",
                "UnitPrice": 1049,
                "Quantity": "1",
                "StartDate": "2026-03-22"
              }
            }
          ]
        }
      }
    ]
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`contractId`	String	ID of the contract record to upgrade.	Optional	66.0
`opportunityId`	String	ID of the opportunity record to upgrade.	Optional	66.0
`outputRecordType`	String	Record type of the output for the upgrade.	Required	66.0
`swapGroups`	Swap Group[]	Groups that contain the asset details for the upgrade.	Required	66.0
`swapStartDate`	String	Amendment start date for the upgrade action.	Required	66.0

Response body for POST

Initiate Upgrade Response

SEE ALSO:

Salesforce Help: Swap, Upgrade, or Downgrade Assets

Instant Pricing (POST)

Fetch instant pricing data on the quote or order line data grid and associated summary component. This API offers capabilities to either create a context or update the existing one based on the provided context ID.

You can also group quote line items or order items based on location, work types, or departments, if groups are enabled for your org. Groups provide a visualization of the products to view large quotes.

Resource

```
/industries/cpq/quotes/actions/get-instant-price
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/industries/cpq/quotes/actions/get-instant-price
```

Available version

60.0

Requires Chatter

No

HTTP methods

POST

Request body for POST

JSON example

```
{
  "correlationId": "7847127122596",
```

```
  "contextId": "0000000b28op21g00281751271041790cc1e5686282849a387dd3c59b310418b",
  "records": [
    {
      "referenceId": "0Q0xx0000004DOSCA2",
      "record": {
        "attributes": {
          "type": "Quote",
          "method": "POST"
        },
        "Name": "Test Quote Proration Pricing",
        "OpportunityId": "006xx000001a4ISAAY",
        "Pricebook2Id": "01sxx0000005ptpAAA"
      }
    },
    {
      "referenceId": "refQuoteLine",
      "record": {
        "attributes": {
          "type": "QuoteLineItem",
          "method": "POST"
        },
        "QuoteId": "0Q0xx0000004DOSCA2",
        "PricebookEntryId": "01uxx0000008zHmAAI",
        "Product2Id": "01txx0000006jmWAAQ",
        "Quantity": 2,
        "UnitPrice": 25,
        "StartDate": "2022-09-28",
        "EndDate": "2028-09-27",
        "PeriodBoundary": "ANNIVERSARY",
        "BillingFrequency": "ANNUAL"
      }
    }
  ]
}
```

This example shows a sample request to specify grouping of lines based on criteria.

```
{
    "contextId": "0000000b28op21g00281751271041790cc1e5686282849a387dd3c59b310418b",

    "correlationId": "7847127122596",
    "records": [
        {
            "referenceId": "0Q0xx0000004DOSCA2",
            "record": {
                "attributes": {
                    "type": "Quote",
                    "method": "PUT"
                }
            }
        },
        {
            "referenceId": "0QLxx0000004F3gGAE",
            "record": {
                "attributes": {
```

```
                    "type": "QuoteLineItem",
                    "method": "PUT"
                },
                "Quantity": 5
            }
        }, {
            "referenceId": "0QLxx0000004F3hGAE",
            "record": {
                "attributes": {
                    "type": "QuoteLineItem",
                    "method": "PUT"
                },
                "Quantity": 2
            }
        },
        {
            "referenceId": "GroupId1",
            "record": {
                "attributes": {
                    "type": "QuoteLineGroup",
                    "method": "POST",
                    "action": "GroupBy",
                    "criteria": {
                        "Quantity": 5
                    }
                },
                "Name": "record"
            }
        }, {
            "referenceId": "GroupId2",
            "record": {
                "attributes": {
                    "type": "QuoteLineGroup",
                    "method": "POST",
                    "action": "GroupBy",
                    "criteria": {
                        "Quantity": 2
                    }
                },
                "Name": "record1",
            }
        }
    ]
}
```

This example shows a sample request for the initial grouping of the quote with the quote lines assigned to the first group.

```
{
    "contextId": "0000000b28op21g00281751271041790cc1e5686282849a387dd3c59b310418b",

    "correlationId": "7847127122596",
    "records": [
        {
            "referenceId": "0Q0xx0000004CAgCAM",
            "record": {
```

```
                "attributes": {
                    "type": "Quote",
                    "method": "PUT"
                }
            }
        },
        {
            "referenceId": "GroupId1",
            "record": {
                "attributes": {
                    "type": "QuoteLineGroup",
                    "method": "POST",
                    "action": "GroupAll"
                },
                "Name": "sample"
            }
        }
    ]
}
```

This example shows a sample request to ungroup a quote but retain the quote lines.

```
{
    "contextId": "0000000b28op21g00281751271041790cc1e5686282849a387dd3c59b310418b",

    "correlationId": "7847127122596",
    "records": [
        {
            "referenceId": "0Q0xx0000004CAgCAM",
            "record": {
                "attributes": {
                    "type": "Quote",
                    "method": "PUT"
                }
            }
        },
         {
            "referenceId": "0QLxx0000004CBYGA2",
            "record": {
                "attributes": {
                    "type": "QuoteLineItem",
                    "method": "PUT"
                },
                "Quantity": 2,
                "QuoteLineGroupId": null
            }
        },
        {
            "referenceId": "GroupId1",
            "record": {
                "attributes": {
                    "type": "QuoteLineGroup",
                    "method": "DELETE",
                    "action": "Ungroup"
                }
```

```
            }
        }
    ]
}
```


This example shows a sample request to create a new group.


```
{
    "contextId": "0000000b28op21g00281751271041790cc1e5686282849a387dd3c59b310418b",

    "correlationId": "7847127122596",
    "records": [
        {
            "referenceId": "0Q0xx0000004CAgCAM",
            "record": {
                "attributes": {
                    "type": "Quote",
                    "method": "PUT"
                }
            }
        },
        {
            "referenceId": "GroupId1",
            "record": {
                "attributes": {
                    "type": "QuoteLineGroup",
                    "method": "POST"
                },
                "Name": "sample"
            }
        }
    ]
}
```


This example shows a sample request to delete a group.


```
{
    "contextId": "0000000b28op21g00281751271041790cc1e5686282849a387dd3c59b310418b",

    "correlationId": "7847127122596",
    "records": [
        {
            "referenceId": "0Q0xx0000004CAgCAM",
            "record": {
                "attributes": {
                    "type": "Quote",
                    "method": "PUT"
                }
            }
        },
        {
            "referenceId": "GroupId1",
            "record": {
                "attributes": {
                    "type": "QuoteLineGroup",
```

```
                    "method": "DELETE",
                    "action": "DeleteGroup"
                },
                "Name": "sample"
            }
        }
    ]
}
```

This example shows a sample request to move a group.

```
{
    "contextId": "0000000b28op21g00281751271041790cc1e5686282849a387dd3c59b310418b",

    "correlationId": "7847127122596",
    "records": [
        {
            "referenceId": "0Q0xx0000004CAgCAM",
            "record": {
                "attributes": {
                    "type": "Quote",
                    "method": "PUT"
                }
            }
        },
        {
            "referenceId": "0QLxx0000004CBYGA2",
            "record": {
                "attributes": {
                    "type": "QuoteLineItem",
                    "method": "PUT"
                },
                "Quantity": 2
                "QuoteLineGroupId": "{@GroupId2}"
            }
        },
    ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`contextId`	String	ID generated by the context service. If you don't specify a context ID, a new context is created.	Optional	59.0
`correlationId`	String	Client-generated ID for tracking multiple related API requests.	Optional	59.0
`records`	Object with Reference Input[]	List of pricing data to be fetched.	Required	59.0

Response body for POST

Instant Pricing

Place Order (POST)

Place orders with integrated pricing, configuration, and validation, and manage them throughout their entire lifecycle. Additionally, update an order or insert order items.

You can also group order items based on location, work types, or departments, if groups are enabled for your org. Groups provide a visualization of the products to view large quotes.

This API supports a maximum of 300 transaction line items.

Note: This API has been deprecated as of API version 63.0. In API version 63.0 and later, use the new Place Sales Transaction API.

Special Access Rules

You need the PlaceOrder API permission set to use this API.

Resource

```
/commerce/sales-orders/actions/place
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/sales-orders/actions/place
```

Available version

60.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "pricingPref": "System",
  "configurationInput": "RunAndAllowErrors",
  "configurationOptions": {
    "validateProductCatalog": true,
    "validateAmendRenewCancel": true,
    "executeConfigurationRules": true,
    "addDefaultConfiguration": true
  },
  "graph": {
    "graphId": "graphId",
    "records": [
      {
        "referenceId": "refOrder",
        "record": {
          "attributes": {
            "type": "Order",
            "method": "POST",
            "Id": "POST"
          }
        }
      },
```

```
        {
           "referenceId":"refOrderItem",
           "record":{
           "attributes":{
               "type":"OrderItem",
               "method":"POST"
              },
              "OrderId":"@{refOrder.id}",
              "OrderActionId":"@{refOrderAction.id}",
              "ListPrice":"144.99",
              "Quantity":3,
              "PricebookEntryId":"01uxx0000008yXPAAY",
              "Product2Id":"01txx0000006i2UAAQ",
              "UnitPrice":"199.49"
         }
        }
      ]
    }
}
```

This example shows a sample request to define grouping of order items.

```
{
    "pricingPref": "system",
    "graph": {
        "graphId": "placeOrder",
        "records": [
            {
                "referenceId": "refOrder",
                "record": {
                    "attributes": {
                        "type": "Order",
                        "method": "PATCH",
                        "id": "801xx000003GZ9bAAG"
                    }
                }
            },
            {
                "referenceId": "refOlg1",
                "record": {
                    "attributes": {
                        "type": "OrderItemGroup",
                        "method": "POST"
                    },
                    "Name": "New Group",
                    "OrderId": "@{refOrder.id}"
                }
            }
        ]
    }
}
```

This example shows a sample request to ungroup order items.

```
{
    "pricingPref": "system",
    "graph": {
        "graphId": "placeOrder",
        "records": [
            {
                "referenceId": "refOrder",
                "record": {
                    "attributes": {
                        "type": "Order",
                        "method": "PATCH",
                        "id": "refOrder"
                    }
                }
            },
            {
                "referenceId": "refOlg1",
                "record": {
                    "attributes": {
                        "type": "OrderItemGroup",
                        "method": "DELETE",
                        "id": "refOlg1",
                        "action": "Ungroup"
                    }
                }
            }
        ]
    }
}
```

This example shows a sample request to create a new group.

```
{
    "pricingPref": "system",
    "graph": {
        "graphId": "placeOrder",
        "records": [
            {
                "referenceId": "refOrder",
                "record": {
                    "attributes": {
                        "type": "Order",
                        "method": "PATCH",
                        "id": "refOrder"
                    }
                }
            },
            {
                "referenceId": "refOlg",
                "record": {
                    "attributes": {
                        "type": "OrderItemGroup",
                        "method": "POST"
```

```
                    },
                    "Name": "New Group",
                    "OrderId": "@{refOrder.id}"
                }
            }
        ]
    }
}
```

This example shows a sample request to delete a group.

```
{
    "contextId": "",
    "correlationId": "",
    "records": [
        {
            "referenceId": "refOrder",
            "record": {
                "attributes": {
                    "type": "Order",
                    "method": "PATCH",
                    "id": "refOrder"
                }
            }
        },
        {
            "referenceId": "refOlg",
            "record": {
                "attributes": {
                    "type": "OrderItemGroup",
                    "method": "DELETE",
                    "id": "refOlg",
                    "action": "DeleteGroup"
                }
            }
        }
    ]
}
```

This example shows a sample request to group order items based on criteria.

```
{
    "pricingPref": "system",
    "graph": {
        "graphId": "placeOrder",
        "records": [
            {
                "referenceId": "refOrder",
                "record": {
                    "attributes": {
                        "type": "Order",
                        "method": "PATCH",
                        "id": "refOrder"
                    }
                }
```

```
            },
            {
                "referenceId": "g0",
                "record": {
                    "attributes": {
                        "type": "OrderItemGroup",
                        "method": "POST",
                        "action": "GroupBy",
                        "criteria": {
                            "BillingFrequency2": null
                        }
                    },
                    "Name": "Billing Frequency: ",
                    "OrderId": "@{refOrder.id}"
                }
            },
            {
                "referenceId": "g1",
                "record": {
                    "attributes": {
                        "type": "OrderItemGroup",
                        "method": "POST",
                        "action": "GroupBy",
                        "criteria": {
                            "BillingFrequency2": "Monthly"
                        }
                    },
                    "Name": "Billing Frequency: Monthly",
                    "OrderId": "@{refOrder.id}"
                }
            }
        ]
    }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`catalog RatesPref`	String	Rate card entries defined in the catalog that must be fetched for order items with usage-based pricing during the order creation process. Valid values are: • `Fetch`—Retrieves the rate card entries defined in the catalog for order items during the order creation process. • `Skip`—Skips the retrieval of rate card entries for order items during the order creation process. The default value is `Skip`.	Optional	62.0

Name	Type	Description	Required or Optional	Available Version
		This property is available when the Usage-Based Selling feature is enabled.		
configurationInput	String	Configuration input for the place order process. Valid values are: • `RunAndAllowErrors`—Specifies to run the configuration and to proceed order ingestion upon encountering any configuration errors. • `RunAndBlockErrors`—Specifies to run configuration and to block order ingestion upon encountering any configuration errors. • `Skip`—Specifies to skip configuration. The default value is `RunAndBlockErrors`.	Optional	60.0
configurationOptions	Configuration Options Input[]	Configuration options during the ingestion process.	Optional	60.0
graph	Object Graph Input	The sObject graph of the order payload to be ingested. You can perform create, update, or delete operations on objects from the Sales Transaction context definition by using this property. Additionally, perform create, update, or delete operations on custom objects and fields in your extended context definition.	Required	60.0
pricingPref	String	Pricing preference during the create order process. Valid values are: • `Force`—Specifies to force pricing during the order ingestion process. • `Skip`—Specifies to skip pricing during the order ingestion process. • `System`—Specifies the system to determine whether a pricing calculation is required. The default value is `System`.	Optional	60.0

Response body for POST

Place Order

Place Quote (POST)

Create a quote to discover and price products and services. Additionally, insert, update, or delete a quote line item.

You can also group quote line items based on location, work types, or departments, if groups are enabled for your org. Groups provide a visualization of the products to view large quotes.

This API supports a maximum of 300 transaction line items.

Note: This API has been deprecated as of API version 63.0. In API version 63.0 and later, use the new Place Sales Transaction API.

Special Access Rules

You need the Create on Quotes user permission to create quotes.

Resource

```
/commerce/quotes/actions/place
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/quotes/actions/place
```

Available version

60.0

HTTP methods

POST

Request body for POST

JSON example

This example shows a sample request to create a quote.

```
{
  "pricingPref": "System",
  "configurationInput": "RunAndAllowErrors",
  "configurationOptions": {
    "validateProductCatalog": true,
    "validateAmendRenewCancel": true,
    "executeConfigurationRules": true,
    "addDefaultConfiguration": true
  },  "graph": {
    "graphId": "createQuote",
    "records": [{
        "referenceId": "refQuote",
        "record": {
          "attributes": {
            "type": "Quote",
            "method": "POST"
          },
          "opportunityId": "---",
          "quoteProp1": "value1",
          "quoteProp2": "value2"
        }
      },
      {
        "referenceId": "refQuoteLineItem1",
        "record": {
          "attributes": {
```

```
              "type": "QuoteLineItem",
              "method": "POST"
            },
            "QuoteLineItemProp1": "value1",
            "QuoteLineItemProp2": "value2"
          }
        },
        {
        "referenceId": "refQuoteLineItemAttribute",
        "record": {
          "attributes": {
                "type": "QuoteLineItemAttribute",
                "method": "POST"
         },
         "QuoteLineItemId": "@{refQuoteLineItem1.id}",
         "AttributeDefinitionId": "0tjxx0000000001AAA"
        }
}
    ]
  }
}
```

This example shows a sample request to insert, update, or delete a quote line item.

```
{
  "pricingPref": "System",
  "configurationInput": "skip",
  "graph": {
    "graphId": "updateQuote",
    "records": [
      {
        "referenceId": "refQuote",
        "record": {
          "attributes": {
            "type": "Quote",
            "method": "PATCH",
            "id": "0Q0xx0000004E2mCAE"
          },
          "Name": "Quote_Acme"
        }
      },
      {
        "referenceId": "refQuoteLineItemToCreate1",
        "record": {
          "attributes": {
            "type": "QuoteLineItem",
            "method": "POST"
          },
          "QuoteId": "0Q0xx0000004E2mCAE",
          "PricebookEntryId": "01uxx0000008yXPAAY",
          "Product2Id": "01txx0000006i2UAAQ",
          "Quantity": 2.0,
          "UnitPrice": 800.0,
          "PeriodBoundary": "Anniversary",
          "BillingFrequency": "Monthly",
```

```
          "StartDate": "2024-03-11"
        }
      },
      {
        "referenceId": "refQuoteLineItemToPatch2",
        "record": {
          "attributes": {
            "type": "QuoteLineItem",
            "method": "PATCH",
            "id": "0Q0xx0000004E2mCAE"
          },
          "Quantity": 2.0,
          "UnitPrice": 600.0
        }
      },
      {
        "referenceId": "refQuoteLineItemToDelete3",
        "record": {
          "attributes": {
            "type": "QuoteLineItem",
            "method": "DELETE",
            "id": "0Q0xx0000004E2mYLK"
          }
        }
      }
    ]
  }
}
```

This example shows a sample request to define grouping of quote line items.

```
{
   "pricingPref": "Force",
   "configurationInput": "skip",
   "graph": {
      "graphId": "groupLines",
      "records": [
         {
            "referenceId": "refQuote",
            "record": {
               "attributes": {
                  "type": "Quote",
                  "method": "PATCH",
                  "id":"0Q0xx0000004C99CAE"
               },
               "Name": "From Place Quote API"
            }
         },
         {
            "referenceId": "refQlg1",
            "record": {
               "attributes": {
                  "type": "QuoteLineGroup",
                  "method": "POST",
                  "action": "GroupBy",
```

```
                "criteria": {
                  "Quantity": 1
                }
              },
              "Name": "From Place Quote API Group",
              "QuoteId": "@{refQuote.id}"
            }
          },
          {
            "referenceId": "refQuoteItem1",
            "record": {
              "attributes": {
                "type": "QuoteLineItem",
                "method": "PATCH",
                "id":"0QLxx0000004DJcGAM"
              },
              "QuoteLineGroupId": "@{refQlg1.id}",
              "Quantity": 1
            }
          }
        ]
    }
}
```

This example shows a sample request for the initial grouping of the quote with the quote lines assigned to the first group.

```
{
  "pricingPref": "Force",
  "graph": {
    "graphId": "test",
    "records": [
      {
        "referenceId": "refQuote",
        "record": {
          "attributes": {
            "type": "Quote",
            "method": "PATCH",
            "id": "0Q0xx0000004CAmCAM"
          }
        }
      },
      {
        "referenceId": "refQlg1",
        "record": {
          "attributes": {
            "type": "QuoteLineGroup",
            "method": "POST",
            "action": "GroupAll"
          },
          "Name": "From PQ API Group",
          "QuoteId": "@{refQuote.id}"
        }
      }
    ]
```

```
  }
}
```

This example shows a sample request to ungroup a quote but retain the quote lines.

```
{
   "pricingPref": "Force",
   "configurationInput": "skip",
   "graph": {
      "graphId": "test",
      "records": [
         {
            "referenceId": "refQuote",
            "record": {
               "attributes": {
                  "type": "Quote",
                  "method": "PATCH",
                  "id":"0Q0xx0000004C99CAE"
               },
               "Name": "From Place Quote API"
            }
         },
         {
            "referenceId": "refQlg1",
            "record": {
               "attributes": {
                  "type": "QuoteLineGroup",
                  "method": "DELETE",
                  "id": "{GroupId}",
                  "action": "Ungroup"
               }
            }
         }
      ]
   }
}
```

This example shows a sample request to create a new group.

```
{
   "pricingPref": "Force",
   "configurationInput": "skip",
   "graph": {
      "graphId": "test",
      "records": [
         {
            "referenceId": "refQuote",
            "record": {
               "attributes": {
                  "type": "Quote",
                  "method": "PATCH",
                  "id":"0Q0xx0000004C99CAE"
               },
               "Name": "From Place Quote API"
            }
```

```
         },
         {
            "referenceId": "refQlg1",
            "record": {
               "attributes": {
                  "type": "QuoteLineGroup",
                  "method": "POST"
               },
               "Name": "From PQ API Group",
               "QuoteId": "@{refQuote.id}"
            }
         }
      ]
   }
}
```

This example shows a sample request to delete a group.

```
{
   "pricingPref": "Force",
   "configurationInput": "skip",
   "graph": {
      "graphId": "test",
      "records": [
         {
            "referenceId": "refQuote",
            "record": {
               "attributes": {
                  "type": "Quote",
                  "method": "PATCH",
                  "id":"0Q0xx0000004C99CAE"
               },
               "Name": "From Place Quote API"
            }
         },
         {
            "referenceId": "refQlg1",
            "record": {
               "attributes": {
                  "type": "QuoteLineGroup",
                  "method": "DELETE",
                  "id": "{GroupId}",
                  "action": "DeleteGroup"
               }
            }
         }
      ]
   }
}
```

This example shows a sample request to move a group.

```
{
   "pricingPref": "Force",
   "configurationInput": "skip",
```

```
    "graph": {
       "graphId": "test",
       "records": [
          {
             "referenceId": "refQuote",
             "record": {
                "attributes": {
                   "type": "Quote",
                   "method": "PATCH",
                   "id":"0Q0xx0000004C99CAE"
                },
                "Name": "From PlaceQuote Api"
             }
          },
         {
             "referenceId": "0QLxx0000004CBYGA2",
             "record": {
                 "attributes": {
                     "type": "QuoteLineItem",
                     "method": "PATCH"
                     "id": "0QLxx0000004CBYGA2"
                 },
                 "Quantity": 2,
                 "QuoteLineGroupId": "@{GroupId2}"
             }
       ]
    }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`catalog RatesPref`	String	Rate card entries defined in the catalog that must be fetched for quote line items with usage-based pricing during the quote creation process. Valid values are: • `Fetch`—Retrieves the rate card entries defined in the catalog for quote line items during the quote creation process. • `Skip`—Skips the retrieval of rate card entries for quote line items during the quote creation process. The default value is `Skip`. This property is available when the Usage-Based Selling feature is enabled.	Optional	62.0

Name	Type	Description	Required or Optional	Available Version
configuration Input	String	Configuration input for the place quote process. Valid values are: • RunAndAllowErrors • RunAndBlockErrors • Skip The default value is RunAndBlockErrors.	Optional	60.0
configuration Options	Configuration Options Input	Configuration options during the ingestion process.	Optional	60.0
graph	Object Graph Input	The sObject graph representing the quote structure. You can perform create, update, or delete operations on objects from the Sales Transaction context definition by using this property. Additionally, perform create, update, or delete operations on custom objects and fields in your extended context definition.	Required	60.0
pricingPref	String	Pricing preference during the quote process. Valid values are: • Force • Skip • System The default value is System.	Optional	60.0

Response body for POST

Place Quote

Place Sales Transaction (POST)

Create a sales transaction, such as an order or a quote, with integrated pricing and configuration. Also, update an order or a quote, and insert and delete order or quote line items to calculate the estimated tax.

You can also group order or quote line items based on location, work types, or departments, if groups are enabled for your org. Groups provide a visualization of the products to view large quotes.

Keep these considerations in mind when you use this API.

- You can add up to 1000 quote line items for a quote, and 1000 order products for an order. For complex flows that involve a large volume of records, make sure that the number of line items that are sent to this API are within this limit.
- A quote can have up to 3000 quote line item attributes, and an order can have up to 3000 order line item attributes.
- This API doesn't support creation of amendment, renewal, or cancellation quote or order. Use the amendment, renewal, or cancellation APIs or invocable actions.

- Create or update quote lines with an associated group ID if you plan to use the sales transaction line editor after the API request.
- Bundle quantity handling depends on the AssociatedQuantScaleMethod field in the payload. If AssociatedQuantScaleMethod field value is `Proportional`, child quantity is `parent × child`. For example, 10 × 2 = 20. Without this value, the API can enforce quantity or cardinality restrictions and return validation errors. Include it explicitly for predictable results. See OrderItemRelationship for details of AssociatedQuantScaleMethod field.
- For custom objects at level 2 or deeper in the sales transaction hierarchy, configure context mapping so that `parentReference` maps to the custom object's immediate parent field. For custom object `child_one` under QuoteLineItem object, map `parentReference` to quote_line_item_c in QuoteEntitiesMapping object. Without this mapping, this API fails as delta processing can't reconcile all input records.
- This API saves and commits the quote header first, then processes configuration, pricing, and persistence for components, such as line items and groups. If a later step fails, the header isn't rolled back.

Resource

```
/connect/rev/sales-transaction/actions/place
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/rev/sales-transaction/actions/place
```

Available version

63.0

HTTP methods

POST

Request body for POST

JSON example

This is a sample request to create a sales transaction for a quote. This example also skips tax calculation by specifying a value for the optional `taxPref` property.


```
{
  "pricingPref": "System",
  "catalogRatesPref": "Skip",
  "configurationPref": {
    "configurationMethod": "Skip",
    "configurationOptions": {
      "validateProductCatalog": true,
      "validateAmendRenewCancel": true,
      "executeConfigurationRules": true,
      "addDefaultConfiguration": true
    }
  },
  "taxPref": "Skip",
  "contextDetails": {
    "contextId": "e055bb18-d4e8-41c3-881e-0132b9561708"
  },
  "graph": {
    "graphId": "createQuote",
    "records": [
      {
        "referenceId": "refQuote",
        "record": {
          "attributes": {
```

```
        "method": "POST",
        "type": "Quote"
      },
      "Name": "Quote_Acme",
      "Pricebook2Id": "01sDU000000JvhbYAC"
    }
  },
  {
    "referenceId": "refQuoteLine0",
    "record": {
      "attributes": {
        "type": "QuoteLineItem",
        "method": "POST"
      },
      "QuoteId": "@{refQuote.id}",
      "Product2Id": "01tDU000000F7b8YAC",
      "PricebookEntryId": "01uDU000000fxt2YAA",
      "UnitPrice": 100,
      "Quantity": "1",
      "StartDate": "2024-10-29",
      "EndDate": "2025-03-01",
      "PeriodBoundary": "Anniversary"
    }
  },
  {
    "referenceId": "refQuoteLine1",
    "record": {
      "attributes": {
        "type": "QuoteLineItem",
        "method": "POST"
      },
      "QuoteId": "@{refQuote.id}",
      "Product2Id": "01tLT00000AA2M9YAL",
      "PricebookEntryId": "01uLT000007PTafYAG",
      "Quantity": "1"
    }
  },
  {
    "referenceId": "refQuoteLineItemAttribute1",
    "record": {
      "attributes": {
        "type": "QuoteLineItemAttribute",
        "method": "POST"
      },
      "QuoteLineItemId": "@{refQuoteLine0.id}",
      "AttributeDefinitionId": "0tjRT0000000XbtYAE",
      "AttributeValue": "True"
    }
  },
  {
    "referenceId": "refQuoteLineRelationship1",
    "record": {
      "attributes": {
        "type": "QuoteLineRelationship",
```

```
            "method": "POST"
          },
          "ProductRelationshipTypeId": "0yoLT000000wHq6YAE",
          "ProductRelatedComponentId": "0dSLT00000076IP2AY",
          "MainQuoteLineId": "@{refQuoteLine0.id}",
          "AssociatedQuoteLineId": "@{refQuoteLine1.id}",
          "AssociatedQuoteLinePricing": "NotIncludedInBundlePrice"
        }
      }
    ]
  }
}
```

This sample request assigns a TransactionProcessingType record to a quote without any additional preferences. In this example, the TransactionType value for a record is set to a TransactionProcessingType record. See TransactionProcessingType on page 1369 tooling object for more details.

```
{
  "pricingPref": "Force",
  "contextDetails": {
   "contextId": "f1c9e3e1c335f7959a88de09d3a867cc2b95e08709b99de8e2edeb8f5039e8ed",

    "scope": "Session"
  },
  "graph": {
    "graphId": "updateQuote",
    "records": [
      {
        "referenceId": "refQuote",
        "record": {
          "attributes": {
            "type": "Quote",
            "method": "POST"
          },
          "OpportunityId": "006xx000001a2oWAAQ",
          "PriceBook2Id": "01sxx0000005ptpAAA",
          "TransactionType": "SkipPricingAndRunTax",
          "Name": "Quote_No_Tax_System"
        }
      },
      {
        "referenceId": "refQLI1",
        "record": {
          "attributes": {
            "type": "QuoteLineItem",
            "method": "POST"
          },
          "QuoteId": "@{refQuote.id}",
          "UnitPrice": 49.99,
          "Product2Id": "01txx0000006i2aAAA",
          "PricebookEntryId": "01uxx0000008yX0AAI",
          "Quantity": 10
        }
      }
    ]
```

```
  }
}
```

This is a sample request to insert, update, or delete a quote line item.

```
{
  {
  "pricingPref": "System",
  "catalogRatesPref": "Skip",
  "configurationPref": {
    "configurationMethod": "Skip",
    "configurationOptions": {
      "validateProductCatalog": true,
      "validateAmendRenewCancel": true,
      "executeConfigurationRules": true,
      "addDefaultConfiguration": true
    }
  },
  "contextDetails": {
    "contextId": "e055bb18-d4e8-41c3-881e-0132b9561708"
  },
  "graph": {
    "graphId": "updateQuote",
    "records": [
      {
        "referenceId": "refQuote",
        "record": {
          "attributes": {
            "method": "PATCH",
            "type": "Quote",
            "id": "801xx000003GZ9bAAG"
          }
        }
      },
      {
        "referenceId": "refQuoteLine0",
        "record": {
          "attributes": {
            "type": "QuoteLineItem",
            "method": "PATCH",
            "id": "402xx000003KY5vJGH"
          },
          "Quantity": "5"
        }
      }
    ]
  }
}
```

This is a sample request to define grouping of quote line items.

```
{
  "pricingPref": "Force",
  "catalogRatesPref": "Skip",
  "configurationPref": {
```

```
    "configurationMethod": "Skip",
    "configurationOptions": {
      "validateProductCatalog": true,
      "validateAmendRenewCancel": true,
      "executeConfigurationRules": true,
      "addDefaultConfiguration": true
    }
  },
  "contextDetails": {
    "contextId": "e055bb18-d4e8-41c3-881e-0132b9561708"
  },
  "graph": {
    "graphId": "groupQuoteLines",
    "records": [
      {
        "referenceId": "refQuote",
        "record": {
          "attributes": {
            "method": "PATCH",
            "type": "Quote",
            "id": "801xx000003GZ9bAAG"
          }
        }
      },
      {
        "referenceId": "refQuoteLine0",
        "record": {
          "attributes": {
            "type": "QuoteLineItem",
            "method": "PATCH",
            "id": "402xx000003KY5vJGH"
          },
          "QuoteLineGroupId": "@{refQuote.id}"
        }
      }
    ]
  }
}
```

This is s a sample request for the initial grouping of the quote with all the quote lines assigned to the first group.

```
{
  "pricingPref": "Force",
  "catalogRatesPref": "Skip",
  "graph": {
    "graphId": "groupQuote",
    "records": [
      {
        "referenceId": "refQuote",
        "record": {
          "attributes": {
            "type": "Quote",
            "method": "PATCH",
            "id": "0Q0xx0000004CAmCAM"
          }
```

```
        }
      },
      {
        "referenceId": "refQlg1",
        "record": {
          "attributes": {
            "type": "QuoteLineGroup",
            "method": "POST",
            "action": "GroupAll"
          },
          "Name": "From PST API Group",
          "QuoteId": "@{refQuote.id}"
        }
      }
    ]
  }
}
```

This is a sample request to ungroup a quote but retain the quote lines.

```
{
   "catalogRatesPref": "Skip",
   "pricingPref": "Force",
   "graph": {
      "graphId": "ungroupQuote",
      "records": [
         {
            "referenceId": "refQuote",
            "record": {
               "attributes": {
                  "type": "Quote",
                  "method": "PATCH",
                  "id":"0Q0xx0000004C99CAE"
               },
               "Name": "Grouped Quote with PST API"
            }
         },
         {
            "referenceId": "refQlg1",
            "record": {
               "attributes": {
                  "type": "QuoteLineGroup",
                  "method": "DELETE",
                  "id": "402xx000003KY5vJGH",
                  "action": "Ungroup"
               }
            }
         }
      ]
   }
}
```

This is a sample request to create a new group.


```
{
   "catalogRatesPref": "Skip",
   "pricingPref": "Force",
   "graph": {
      "graphId": "createGroup",
      "records": [
         {
            "referenceId": "refQuote",
            "record": {
               "attributes": {
                  "type": "Quote",
                  "method": "PATCH",
                  "id":"0Q0xx0000004C99CAE"
               },
               "Name": "Grouped Quote with PST API"
            }
         },
         {
            "referenceId": "refQlg1",
            "record": {
               "attributes": {
                  "type": "QuoteLineGroup",
                  "method": "POST"
               },
               "Name": "From PQ API Group",
               "QuoteId": "@{refQuote.id}"
            }
         }
      ]
   }
}
```


This example shows a sample request to delete a group.


```
{
   "catalogRatesPref": "Skip",
   "pricingPref": "Force",
   "graph": {
      "graphId": "deleteGroup",
      "records": [
         {
            "referenceId": "refQuote",
            "record": {
               "attributes": {
                  "type": "Quote",
                  "method": "PATCH",
                  "id":"0Q0xx0000004C99CAE"
               },
               "Name": "Grouped Quote with PST API"
            }
         },
         {
            "referenceId": "refQlg1",
```

```
            "record": {
               "attributes": {
                  "type": "QuoteLineGroup",
                  "method": "DELETE",
                  "id": "402xx000003KY5vJGH",
                  "action": "DeleteGroup"
               }
            }
         }
      ]
   }
}
```

This is a sample request to group order items based on criteria.

```
{
    "catalogRatesPref": "Skip",
    "pricingPref": "Force",
    "graph": {
        "graphId": "groupOrderItems",
        "records": [
            {
                "referenceId": "refOrder",
                "record": {
                    "attributes": {
                        "type": "Order",
                        "method": "PATCH",
                        "id": "0Q0xx0000004C99CAE"
                    }
                }
            },
            {
                "referenceId": "refOlg1",
                "record": {
                    "attributes": {
                        "type": "OrderItemGroup",
                        "method": "POST",
                        "action": "GroupBy",
                        "criteria": {
                            "BillingFrequency2": null
                        }
                    },
                    "Name": "Billing Frequency: ",
                    "OrderId": "@{refOrder.id}"
                }
            },
            {
                "referenceId": "g1",
                "record": {
                    "attributes": {
                        "type": "OrderItemGroup",
                        "method": "POST",
                        "action": "GroupBy",
                        "criteria": {
                            "BillingFrequency2": "Monthly"
```

```
                          }
                      },
                      "Name": "Billing Frequency: Monthly",
                      "OrderId": "@{refOrder.id}"
                  }
              }
          ]
      }
}
```

This is a sample request to save changes to a ramp deal by using context ID. The context ID is returned by the Ramp Deal APIs. See Create Ramp Deal (POST).

```
{
  "pricingPref": "Force",
  "contextDetails": {
    "contextId": "f1c9e3e1c335f7959a88de09d3a867cc2b95e08709b99de8e2edeb8f5039e8ed",

    "scope": "Session"
  },
  "graph": {
    "graphId": "updateQuote",
    "records": [
      {
        "referenceId": "refQuote",
        "record": {
          "attributes": {
            "type": "Quote",
            "method": "PATCH",
            "id": "0Q0xx0000004DQ4CAM"
          }
        }
      }
    ]
  }
}
```

To see examples that specify actions to create ramp deals for groups, see Group Ramp Action Input on page 1491.

To see examples that apply to usage-based products, see Usage-Based Product Input on page 1502.

Properties

Name	Type	Description	Required or Optional	Available Version
catalogRatesPref	String	Rate card entries defined in the catalog that must be fetched for sales items with usage-based pricing during the creation of the sales transaction. Valid values are: • `Fetch`—Retrieves the rate card entries defined in the catalog for sales items during the creation of the sales transaction.	Optional	63.0

Name	Type	Description	Required or Optional	Available Version
		• `Skip`—Skips the retrieval of rate card entries for sales items during the creation of the sales transaction. The default value is `Skip`. This property is available when the Usage-Based Selling feature is enabled.		
`configurationPref`	Configurator Preference Input	Configuration preference during the quote process. These preferences ensure that quotes are defined as per the requirement.	Optional	63.0
`contextDetails`	Context Input	Context details that are created for a sales transaction.	Required if the `graph` property isn't specified.	63.0
`graph`	Object Graph Input	The sObject graph of the sales transaction to be ingested. You can perform create, update, or delete operations on objects from the Sales Transaction context definition by using this property. Additionally, perform create, update, or delete operations on custom objects and fields in your extended context definition. To create custom objects that are at the grandchildren level from a line item, you must create the hierarchy of objects until the grandchild object in the same request.	Required if the `contextDetails` property isn't specified.	63.0
`groupRampAction`	String	Specifies the action that you want to perform on group ramp segments. You can also convert a non-ramped group into a ramped group. Valid values are: • `AddProducts`—Adds rampable products to group ramp segments. • `DeleteProducts`—Deletes ramped products. • `EditGroup`—Converts a non-ramped group into a group ramp segment, or edit group ramp segment attributes such as name and description, except the start and end dates.	Optional	65.0

Name	Type	Description	Required or Optional	Available Version
		• `EditRampSchedule`—Edits details of the group ramp segments, including start and end dates. • `DeleteSegment`—Deletes the first or last segment in a group ramp schedule. • `ConvertToNonRampedGroup`—Converts the first or last group ramp segment into a non-ramped group. To add or delete ramped line items from multiple group ramp segments, specify the applicable values in the `graph` property. See Group Ramp Action Input on page 1491 to refer to examples.		
`pricingPref`	String	Pricing preference during the creation of a sales transaction. Valid values are: • `Force`—Enforces pricing during the creation of sales transactions. • `Skip`—Skips pricing during the creation of sales transactions. • `System`—Determines whether a pricing calculation is required. The default value is `System`.	Optional	63.0
`taxPref`	String	Specifies whether to execute or skip the tax calculation step for each sales transaction record. Valid value is `Skip`. If you don't specify this value, then tax calculation request is performed by default.	Optional	65.0

Response body for POST

Place Sales Transaction

Place Supplemental Transaction (POST)

Create a supplemental order or change orders after they are submitted for processing, such as during the fulfillment process.

Keep these considerations in mind when you use this API.

- The original order must not be assetized.
- If Billing is enabled and configured for the order, verify that the original order hasn't been billed.
- If Dynamic Revenue Orchestration (DRO) is enabled and configured for the order, ensure the original order hasn't reached the point of no return milestone. If point of no return milestone hasn't been reached, the fulfillment plan is frozen.

Resource

```
/connect/rev/sales-transaction/actions/place-supplemental-transaction
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/rev/sales-transaction/actions/place-supplemental-transaction
```

Available version

64.0

HTTP methods

POST

Request body for POST

JSON example

This sample creates a supplemental order, which is a clone of the original order. The supplemental order is related to the original order.

```
{
  "relatedSalesTransactionId": "801S70000001VKgIAM"
}
```

This sample overrides a field value of an order line item to supplement the order item with ID value as `802SG000003vZ15YAE`.

```
{
    "relatedSalesTransactionId": "801S70000001VKgIAM",
    "pricingPref": "System",
    "supplementalGraph": {
        "graphId": "1",
        "records": [
            {
                "referenceId": "refOrder",
                "record": {
                    "attributes": {
                        "type": "Order",
                        "method": "PATCH",
                        "id": "801S70000001VKgIAM"
                    },
                    "EffectiveDate": "2025-03-01",
                    "QuoteId": "0Q0xx0000004DQ4CAM"
                }
            },
            {
                "referenceId": "refOrderItem",
                "record": {
                    "attributes": {
                        "type": "OrderItem",
                        "method": "PATCH",
                        "id": "802SG000003vZ15YAE"
                    },
                    "QuoteLineItemId": "0Q0xx0000004E2mYLK"
                }
            }
        ]
```

```
    }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`pricingPref`	String	Pricing preference for this supplemental transaction or order ingestion process. Valid values are: • `Force`—Enforces pricing during the creation of sales transactions. • `Skip`—Skips pricing during the creation of sales transactions. • `System`—Determines whether a pricing calculation is required. If `pricingPref` value is defined as either `Force` or `System`, the supplemental order can have a different pricing from the original order.	Optional	64.0
`relatedSalesTransactionId`	String	Related or the original sales transaction upon which a supplemental transaction is created.	Required	64.0
`supplementalGraph`	Object Graph Input	The sObject graph that represents a payload with the additional changes to be ingested. The attribute's HTTP method must be PATCH. The attribute ID must be the ID of the original order or order item that you want to supplement.	Optional	64.0

Response body for POST

Supplemental Transaction

Read Sales Transaction (POST)

Retrieve sales transaction data efficiently from an initialized or a hydrated context.

Resource

```
/connect/revenue/transaction-management/sales-transactions/actions/read
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/revenue/transaction-management/sales-transactions/actions/read
```

Available version

65.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "contextId": "008d27d7-e004-4906-a949-ee7d7c323c77",
  "queryTags": [
    "Quote",
    "QuoteLineItem",
    "Product"
  ],
  "sobjectFieldMap": {
    "Quote": [],
    "QuoteLineItem": [
      "Quantity",
      "Product2Id"
    ]
  },
  "filters": [
    {
      "sObjectName": "Quote",
      "fieldName": "Status",
      "operator": "Equals",
      "operands": [
        {
          "value": "Draft",
          "type": "STRING"
        }
      ]
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`contextId`	String	ID of the context to retrieve the data records.	Required	65.0
`queryTags`	List<String>	List of objects that must be retrieved from the context.	Optional	65.0
`sobjectFieldMap`	Map<String, List<String>>	Mapping of an sObject name to a list. The list includes the sObject field names on the object or can be an empty list. An empty list specifies that all fields on the object must be queried.	Optional	67.0

Name	Type	Description	Required or Optional	Available Version
`filters`	List<Sales Transaction Filter Condition>	Filter conditions to query the context data.	Optional	67.0

Response body for POST

Read Sales Transaction

Retrieve Sales Transaction API Errors (GET)

Retrieve any asynchronous error details associated with a sales transaction request.

This API returns detailed error status and a retryable payload from Place Sales Transaction API that runs asynchronously. Also, view any blocking errors that prevent a subrequest from persisting. This request doesn't return any non-blocking warnings, such as configuration or tax warnings.

You can view the list of `rollbackedReferenceIds`, which shows synthetic or reference IDs that roll back when the batch fails.

Resource

```
connect/revenue/transaction-management/sales-transactions/actions/place/trackerId/errors
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/revenue/transaction-management/sales-transactions/actions/place/16PRM000000004DBq/errors
```

Available version

66.0

HTTP methods

GET

Request parameter for GET

Parameter Name	Type	Description	Required or Optional	Available Version
`includeRetryablePayload`	Boolean	Indicates whether to return a subset of the original Place Sales Transaction API payload errors (`true`) or not (`false`). The default value is `false`.	Optional	66.0

Response body for GET

Sales Transaction Async Error

Create Ramp Deal (POST)

Create a ramp deal for a customer on a product. Sales reps can use ramp deals to provide yearly deals to a customer, resulting in long-term revenue and customer relationship. A customer can create, update, or view multiple segments of periods for their subscription term with different attributes for each segment.

This API request creates segments based on the specified input properties such as term, segment type, and trial details. The API response includes the context ID and the updated context object for the sales transaction. You must call the Place Sales Transaction API by specifying this context ID to apply the ramp deal updates. See Place Sales Transaction API.

Note: This API is applicable when you're working with line ramps. To work with ramp deals for groups, you must use the Place Sales Transaction API and specify the `groupRampActions` property.

Resource

```
/connect/revenue-management/sales-transaction-contexts/resourceId/actions/ramp-deal-create
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/revenue-management/sales-transaction-contexts/0QLxx0000004CfIGAU/actions/ramp-deal-create
```

Available version

62.0

HTTP methods

POST

Path parameter for POST

Name	Type	Description	Required or Optional	Available Version
`resourceId`	String	ID of the quote line item, order item, or context.	Required	62.0

Request body for POST

JSON example

```
{
  "transactionId": "0Q0xx0000004C92CAE",
  "transactionLineId": "0QLxx0000004C9VGAU",
  "subscriptionTerm": 14,
  "subscriptionTermUnit": "MONTHS",
  "trialTerm": 45,
  "trialTermUnit": "DAYS",
  "segmentType": "YEARLY",
  "executionSettings": {
       "executePricing": true,
       "executeConfigRules": false
   }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`execution Settings`	Execution Settings Input[]	Settings to run the pricing or configuration rules.	Optional	62.0

Name	Type	Description	Required or Optional	Available Version
`segmentType`	String	Type of segment that the user wants to create. Valid values are: • `FREE_TRIAL` • `CUSTOM` • `YEARLY`	Required	62.0
`subscriptionTerm`	Integer	Subscription length of the term-defined product.	Required	62.0
`subscriptionTermUnit`	String	Unit of time for the subscription length. Valid value is: • `MONTHS`	Required	62.0
`transactionId`	String	ID of the sales transaction that's configured, such as quote or order.	Required	62.0
`transactionLineId`	String	Quote line item ID or order item ID that the price ramp is created for.	Required	62.0
`trialTerm`	Integer	Length of the trial period, if any.	Optional	62.0
`trialTermUnit`	String	Unit of time for the trial period. Valid value is: • `DAYS`	Optional. Required if `trialTerm` property is specified.	62.0

Response body for POST

Ramp Deal Service

Delete Ramp Deal (POST)

Delete a ramp deal to convert a ramped product to include a single quote line item or order item.

This API request deletes the segments related to the product. The API response includes the updated context with the context ID. You must call the Place Sales Transaction (POST) API by specifying this context ID to apply the ramp deal updates. See Place Sales Transaction (POST) API.

Note: This API is applicable when you're working with line ramps. To work with ramp deals for groups, you must use the Place Sales Transaction API and specify the `groupRampActions` property.

Resource

```
/connect/revenue-management/sales-transaction-contexts/resourceId/actions/ramp-deal-delete
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/revenue-management/sales-transaction-contexts/0QLxx0000004CfIGAU/actions/ramp-deal-delete
```

Available version

62.0

HTTP methods

POST

Path parameter for POST

Name	Type	Description	Required or Optional	Available Version
`resourceId`	String	ID of the context.	Required	62.0

Request body for POST

JSON example

```
{
  "rampDealIds": [
    "0Q0xx0000004CDxCAM",
    "0QLxx0000004CSOGA2"
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`rampDealIds`	String[]	Ramp identifier on the quote line item or order item.	Required	62.0

Response body for POST

Ramp Deal Service

Update Ramp Deal (POST)

Modify a ramp deal in scenarios where a segment has updates such as quantity, discount, or date change.

Update a ramp deal in these scenarios.

- A segment has quantity or discount changes.
- A trial segment or custom segment has a date change. A custom segment is an added or deleted segment. In this scenario, you can update a ramp deal during the initial sale before assetization.

This API request returns the updated context with the context ID. You must call the Place Sales Transaction (POST) API by specifying this context ID to apply the ramp deal updates. See Place Sales Transaction (POST) API.

Note: This API is applicable when you're working with line ramps. To work with ramp deals for groups, you must use the Place Sales Transaction API and specify the `groupRampActions` property.

Resource

```
/connect/revenue-management/sales-transaction-contexts/resourceId/actions/ramp-deal-update
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/revenue-management/sales-transaction-contexts/4f23961a5c98806f89305e064c67b397e93f1bb8a2a7a3a80db506f1d4110ee9/actions/ramp-deal-update
```

Available version

62.0

HTTP methods

POST

Path parameter for POST

Name	Type	Description	Required or Optional	Available Version
`resourceId`	String	ID of the context data that's used to build the pricing procedure. Get the context instance ID by invoking the Context Service API. See Context Service (POST).	Required	62.0

Request body for POST

JSON example

```
{
  "executionSettings": {
       "executePricing": true,
       "executeConfigRules": false
  },
  "addedNodes": [
    {
      "contextNodePath": [
       "4f23961a5c98806f89305e064c67b397e93f1bb8a2a7a3a80db506f1d4110ee9", // Context
 ID
        "0Q0xx0000004CPACA2", //Quote or Order ID
        "RandomUUID" // random UUID for Quote Line Item or Order Item ID
      ],
      "contextNode": {
         "Discount": 10,
         "Quantity": 5,
         "ItemSegmentName": "Year 5",
         "StartDate":"2024-09-07T00:00:00.000Z",
         "EndDate":"2024-09-07T00:00:00.000Z"
      }
    }
  ],
  "updatedNodes": [
     {
      "contextNodePath": [
       "4f23961a5c98806f89305e064c67b397e93f1bb8a2a7a3a80db506f1d4110ee9", // Context
```

```
ID
        "0Q0xx0000004CPACA2", //Quote or Order ID
        "0QLxx0000004CfIGAU" // Quote Line ID or Order Line ID to update
      ],
      "contextNode": {
          "Discount": 10,
          "Quantity": 5
      }
    }
  ],
  "deletedNodes": [
    {
      "contextNodePath": [
        "4f23961a5c98806f89305e064c67b397e93f1bb8a2a7a3a80db506f1d4110ee9",
        "0Q0xx0000004CPACA2",
        "0QLxx0000004CfIGAU" // Quote Line Item ID to delete
      ]
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
addedNodes	Context Node Input[]	Details of the nodes to be added.	Required	62.0
deletedNodes	Context Node Input[]	Details of the nodes to be deleted.	Required	62.0
executionSettings	Execution Settings Input[]	Settings to run the pricing or configuration rules.	Optional	62.0
updatedNodes	Context Node Input[]	Details of the nodes to be updated.	Required	62.0

Response body for POST

Ramp Deal Service

View Ramp Deal (GET)

View a ramp deal related to a quote line item or an order item.

This API request retrieves the segments if the ramp deal already exists.

Note: This API is applicable when you're working with line ramps. To work with ramp deals for groups, you must use the Place Sales Transaction API and specify the `groupRampActions` property.

Resource

```
/connect/revenue-management/sales-transaction-contexts/resourceId/actions/ramp-deal-view
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/revenue-management/sales-transaction-contexts/0Qxx0000004S0A2/actions/ramp-deal-view?transactionId=0Q0xx0000004DXA&transactionLineId=0QLxx0000004S0A2
```

Available version

62.0

HTTP methods

GET

Path parameter for GET

Name	Type	Description	Required or Optional	Available Version
`resourceId`	String	ID of the quote line item, order item, or context.	Required	62.0

Request parameters for GET

Parameter Name	Type	Description	Required or Optional	Available Version
`transactionId`	String	ID of the quote or order required to hydrate the context and retrieve the quote lines.	Required	62.0
`transactionLineId`	String	ID of the quote or order line required to retrieve the segmented details.	Required	62.0

Response body for GET

Ramp Deal Service

Request Bodies

Learn more about the available API request bodies.

Amendment Input

Input representation of the details of the request to create an amendment record.

Cancellation Input

Input representation of the details of the request to cancel a quote or an order.

Create Ramp Deal Input

Input representation of the request to create a ramp deal.

Clone Options Input

Input representation of the options to clone a sales transaction.

Clone Sales Transaction Input

Input representation of the request to clone records within a sales transaction.

Configuration Options Input

Input representation for the configuration options.

Configurator Preference Input

Input representation of the configuration preference for the place sales transaction request.

Context Input

Input representation of the context that's associated with a sales transaction for a quote or an order.

Context Node Input

Input representation of the details of the context nodes for ramp segments.

Delete Ramp Deal Input

Input representation of the request to delete a ramp deal.

Eligible Promotions Input

Input representation of the request to get eligible promotions for line items. This representation includes the details to accept line item IDs and a sales transaction ID.

Execution Settings Input

Input representation of the execution settings for a ramp deal.

Initiate Downgrade Input

Input representation of the details of the request to initiate a downgrade action.

Initiate Swap Input

Input representation of the details of the request to initiate a swap action.

Initiate Upgrade Input

Input representation of the details of the request to initiate an upgrade action. The response includes the ID of the sales transaction that the upgrade action creates.

Instant Pricing Input

Input representation to fetch the instant pricing details.

Object Graph Input

Input representation of an sObject with a graph ID.

Object Input Map

Input representation of an sObject record in a key-value map format.

Object with Reference Input

Input representation of a list of records to be inserted or updated. To update a record, specify the record ID.

Place Order Input

Input representation of the request to create or update an order.

Place Quote Input

Input representation of the request to create or update a quote.

Read Sales Transaction Input

Input representation of the filter criteria details to read a sales transaction.

Renewal Input

Input representation of the details of the request to initiate the renewal of an asset.

Sales Transaction Filter Condition Input

Input representation of filter conditions used to query sales transaction context data.

[Sales Transaction Input](#)

Input representation of the details of the request to place a sales transaction, such as a quote or an order.

[Sales Transaction Operand Input](#)

Input representation of a filter operand for reading sales transaction context data.

[Supplemental Transaction Input](#)

Input representation of the details of the request to create a supplemental order.

[Swap Group Input](#)

Input representation of the details of the swap groupings for swap operations.

[Update Ramp Deal Input](#)

Input representation of the request to update a ramp deal.

[Usage-Based Product Input](#)

Understand the sample request structure to specify and manage usage-based products within a sales transaction.

Amendment Input

Input representation of the details of the request to create an amendment record.

JSON example

```
{
  "assetIds": [
    "02iSG0000003NMhYAM",
    "02iSG0000006DvSYAU"
  ],
  "amendmentStartDate": "2023-10-04T00:00:00",
  "contractId": "800SG00000CFpepYAD",
  "opportunityId": "006SG000004W5tVYAS",
  "outputRecordId": "801SG00000DX1jWYAT",
  "outputRecordType": "Quote",
  "quantityChange": 5
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`assetIds`	String[]	IDs of the assets that you want to add to the amendment record.	Required	62.0
`amendmentStartDate`	String	Start date of the amendment.	Required	62.0
`contractId`	String	ID of the Contract record that you want to sync with the amendment quote.	Optional	62.0
`opportunityId`	String	ID of the Opportunity record that you want to sync with the amendment quote.	Optional	62.0
`outputRecordId`	String	ID of the quote or order record that you want to add the assets to.	Optional	62.0

Name	Type	Description	Required or Optional	Available Version
outputRecordType	String	Type of amendment record that you want to create. For usage products, set this value to `Quote`. The usage-related records (`QuoteLineItemUseResourceGrant`, `QuoteLineItemUsageResourcePolicy`, and `QuoteLineRateCardEntry`) are copied in the quote action flow, not in the amend order flow.	Required	62.0
quantityChange	Double	Quantity to add to or reduce from the asset's existing quantity.	Required	62.0

Cancellation Input

Input representation of the details of the request to cancel a quote or an order.

JSON example

```
{
  "assetIds": [
    "02iSG0000003NMhYAM",
    "02iSG0000006DvSYAU"
  ],
  "cancellationDate": "2023-10-04T00:00:00",
  "contractId": "800SG00000CFpepYAD",
  "opportunityId": "006SG000004W5tVYAS",
  "outputRecordId": "801SG00000DX1jWYAT",
  "outputRecordType": "Quote"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
assetIds	String[]	IDs of the assets that you want to cancel. All assets in a request must belong to the same price book.	Required	62.0
cancellationDate	String	Effective date of the cancellation.	Required	62.0
contractId	String	ID of the Contract record that you want to sync with the cancellation quote.	Optional	62.0
opportunityId	String	ID of the Opportunity record that you want to sync with the cancellation quote.	Optional	62.0

Name	Type	Description	Required or Optional	Available Version
output RecordId	String	ID of the quote or order that you want to cancel.	Optional	62.0
output RecordType	String	Type of cancellation record that you want to create.	Required	62.0

Create Ramp Deal Input

Input representation of the request to create a ramp deal.

JSON example

```
{
  "transactionId": "0Q0xx0000004C92CAE",
  "transactionLineId": "0QLxx0000004C9VGAU",
  "subscriptionTerm": 14,
  "subscriptionTermUnit": "MONTHS",
  "trialTerm": 45,
  "trialTermUnit": "DAYS",
  "segmentType": "YEARLY",
  "executionSettings": {
       "executePricing": true,
       "executeConfigRules": false
   }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
execution Settings	Execution Settings Input[]	Settings to run the pricing or configuration rules.	Optional	62.0
segmentType	String	Type of segment that the user wants to create. Valid values are: • FREE_TRIAL • CUSTOM • YEARLY	Required	62.0
subscription Term	Integer	Subscription length of the term-defined product.	Required	62.0
subscription TermUnit	String	Unit of time for the subscription length. Valid value is: • MONTHS	Required	62.0

Name	Type	Description	Required or Optional	Available Version
`transactionId`	String	ID of the sales transaction that's configured, such as quote or order.	Required	62.0
`transaction LineId`	String	Quote line item ID or order item ID that the price ramp is created for.	Required	62.0
`trialTerm`	Integer	Length of the trial period, if any.	Optional	62.0
`trialTermUnit`	String	Unit of time for the trial period. Valid value is: • `DAYS`	Optional. Required if `trialTerm` property is specified.	62.0

Clone Options Input

Input representation of the options to clone a sales transaction.

JSON example

This is a sample request to clone all line items in a ramped group within a sales transaction.

```
{
  "recordIds": ["0QLxx0000004CBYGA2"],
  "salesTransactionId": "0Q0xx0000004CE0CAM",
  "options": {
    "lineScope": "AllLines"
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`recordTypeId`	String	ID of the record type related to the record to clone.	Optional	65.0
`lineScope`	String	Specifies the scope for cloning a ramp segment. You can clone only the last ramp segment. This property determines which line items must be cloned and added to the cloned segment. Valid values are: • `AllLines`—Specifies whether all line items in a ramped group must be cloned. • `RampedLinesOnly`—Specifies whether only the ramped line items must be cloned. A segment identifier is created for the newly cloned line items, ensuring date	Optional	65.0

Name	Type	Description	Required or Optional	Available Version
		continuity between the existing and cloned segment.		

Clone Sales Transaction Input

Input representation of the request to clone records within a sales transaction.

JSON example

This is a sample request to clone a record within a sales transaction.

```
{
  "recordIds": ["0QLxx0000004CBYGA2"],
  "salesTransactionId": "0Q0xx0000004CE0CAM"
}
```

This is a sample request to clone all line items in a ramped group within a sales transaction.

```
{
  "recordIds": ["0QLxx0000004CBYGA2"],
  "salesTransactionId": "0Q0xx0000004CE0CAM",
  "options": {
    "lineScope": "AllLines"
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
recordIds	String[]	ID of the record to be cloned. You can specify a single record ID only.	Required	64.0
salesTransactionId	String	ID of the sales transaction related to the record IDs to clone.	Required	64.0
options	Clone Options Input	Specifies options to clone a ramp segment within a sales transaction. You can clone only the last ramp segment.	Optional	65.0

Configuration Options Input

Input representation for the configuration options.

JSON example

```
{
  "configurationOptions": {
    "validateProductCatalog": true,
    "validateAmendRenewCancel": true,
    "executeConfigurationRules": true,
```

```
    "addDefaultConfiguration": true
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
addDefault Configuration	Boolean	Indicates whether to automatically add default configurations to the order (true) or not (false).	Optional	60.0
execute Configuration Rules	Boolean	Indicates whether the order must adhere to configuration rules during processing (true) or bypass them (false).	Optional	60.0
validateAmend RenewCancel	Boolean	Indicates whether to run validations related to amend, renew, or cancel processes (true) or not (false).	Optional	60.0
validate Product Catalog	Boolean	Indicates whether the order must be validated against the product catalog (true) or not (false).	Optional	60.0

Configurator Preference Input

Input representation of the configuration preference for the place sales transaction request.

JSON example


```
{
  "configurationPref": {
    "configurationMethod": "Skip",
    "configurationOptions": {
      "validateProductCatalog": true,
      "validateAmendRenewCancel": true,
      "executeConfigurationRules": true,
      "addDefaultConfiguration": true
    }
  }
}
```


Properties

Name	Type	Description	Required or Optional	Available Version
configuration Method	String	Configuration method for the place sales transaction request. Valid values are: • Force—Specifies to enforce the predefined configuration process during the sales transaction process.	Optional	63.0

Name	Type	Description	Required or Optional	Available Version
		• `Skip`—Specifies to skip the configuration process during the quote creation process. • `System`—Specifies the system to determine whether the configuration process is required. The default value is `Skip`.		
`configurationOptions`	Configuration Options Input	Configuration options during the ingestion process.	Optional	63.0

Context Input

Input representation of the context that's associated with a sales transaction for a quote or an order.

JSON example

```
{
  "contextDetails": {
    "contextId": "e055bb18-d4e8-41c3-881e-0132b9561708"
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`contextId`	String	ID of the context that represents the created session for the sales transaction. This property is supported only for a PATCH request. If the `contextId` property isn't specified, the Place Sales Transaction API generates the context ID for the sales transaction.	Optional	63.0

Context Node Input

Input representation of the details of the context nodes for ramp segments.

JSON example

```
  "updatedNodes": [
     {
      "contextNodePath": [
       "4f23961a5c98806f89305e064c67b397e93f1bb8a2a7a3a80db506f1d4110ee9", // ContextId
```

```
        "0Q0xx0000004CPACA2", //Quote or OrderId
        "0QLxx0000004CfIGAU" // Quote Line ID or Order Line ID to update
      ],
      "contextNode": {
          "Discount": 10,
          "Quantity": 5
      }
    },
    {
      "contextNodePath": [
        "4f23961a5c98806f89305e064c67b397e93f1bb8a2a7a3a80db506f1d4110ee9",
        "0Q0xx0000004CPACA2",
        "2b6401d144904e10aa"
      ],
      "contextNode": {
          "Discount": 20,
          "Quantity": 15
      }
    }
  ]
```

Properties

Name	Type	Description	Required or Optional	Available Version
contextNode	Map<String, Object>	Details of the context node to be added, updated, or deleted.	Required	62.0
contextNode Path	String[]	Path to the context node to be added, updated, or deleted.	Required	62.0

Delete Ramp Deal Input

Input representation of the request to delete a ramp deal.

JSON example

```
{
  "rampDealIds": [
    "0Q0xx0000004CDxCAM",
    "0QLxx0000004CSOGA2"
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
rampDealIds	String[]	Ramp identifier on the quote line item or order item.	Required	62.0

Eligible Promotions Input

Input representation of the request to get eligible promotions for line items. This representation includes the details to accept line item IDs and a sales transaction ID.

JSON example

```
{
  "salesTransactionId": "0Q0xx0000004EOECA2",
  "lineItemIds": [
    "0QLxx0000004E7eGAE",
    "0QLxx0000004GCeGAM",
    "0QLxx0000004E7gGAE"
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`lineItemIds`	String[]	List of line item IDs to evaluate for promotions. The object type is auto-determined from the sales transaction ID.	Required	66.0
`salesTransactionId`	String	The sales transaction ID, such as an order ID or a quote ID, for the promotion evaluation.	Required	66.0

Execution Settings Input

Input representation of the execution settings for a ramp deal.

JSON example

```
  "executionSettings": {
       "executePricing": true,
       "executeConfigRules": false
   }
```

Properties

Name	Type	Description	Required or Optional	Available Version
`executeConfigRules`	Boolean	Indicates whether to run configuration rules (`true`) or not (`false`). The default value is `true`.	Optional	62.0
`executePricing`	Boolean	Indicates whether to run pricing request (`true`) or not (`false`). The default value is `true`.	Optional	62.0

Initiate Downgrade Input

Input representation of the details of the request to initiate a downgrade action.

JSON example


```
{
  "swapStartDate": "2025-12-01T00:00:00Z",
  "outputRecordType": "Quote",
  "swapGroups": {
    "groups": [
      {
        "referenceId": "DOWNGRADE-001",
        "outGroup": {
          "swapAssets": [
            {
              "assetId": "02ixx0000004HOAAA2",
              "quantity": 1
            }
          ]
        },
        "inGroup": {
          "graphId": "downgradeRequest",
          "records": [
            {
              "referenceId": "refQuoteLine0",
              "record": {
                "attributes": {
                  "type": "QuoteLineItem",
                  "method": "POST"
                },
                "Product2Id": "01txx0000006iVlAAI",
                "PricebookEntryId": "01uxx0000008ym4AAA",
                "UnitPrice": 1049,
                "Quantity": "1",
                "StartDate": "2022-09-22"
              }
            }
          ]
        }
      }
    ]
  }
}
```


Properties

Name	Type	Description	Required or Optional	Available Version
`contractId`	String	ID of the contract record to downgrade.	Optional	66.0
`opportunityId`	String	ID of the opportunity record to downgrade.	Optional	66.0

Name	Type	Description	Required or Optional	Available Version
outputRecordType	String	Record type of the output for the downgrade.	Required	66.0
swapGroups	Swap Group[]	Groups that contain the asset details for the downgrade.	Required	66.0
swapStartDate	String	Amendment start date for the downgrade action.	Required	66.0

Initiate Swap Input

Input representation of the details of the request to initiate a swap action.

JSON example

```
{
  "swapStartDate": "2025-12-01T00:00:00Z",
  "outputRecordType": "Quote",
  "swapGroups": {
    "groups": [
      {
        "referenceId": "SWAP-001",
        "outGroup": {
          "swapAssets": [
            {
              "assetId": "02ixx0000004HOAAA2",
              "quantity": 1
            }
          ]
        },
        "inGroup": {
          "graphId": "swapRequest",
          "records": [
            {
              "referenceId": "refQuoteLine0",
              "record": {
                "attributes": {
                  "type": "QuoteLineItem",
                  "method": "POST"
                },
                "Product2Id": "01txx0000006iVlAAI",
                "PricebookEntryId": "01uxx0000008ym4AAA",
                "UnitPrice": 1049,
                "Quantity": "1",
                "StartDate": "2022-09-22"
              }
            }
          ]
        }
      }
    ]
```

```
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`contractId`	String	ID of the contract record to swap.	Optional	66.0
`opportunityId`	String	ID of the opportunity record to swap.	Optional	66.0
`outputRecordType`	String	Record type of the output for the swap.	Required	66.0
`swapGroups`	Swap Group[]	Groups that contain the asset details for the swap.	Required	66.0
`swapStartDate`	String	Amendment start date for the swap action.	Required	66.0

Initiate Upgrade Input

Input representation of the details of the request to initiate an upgrade action. The response includes the ID of the sales transaction that the upgrade action creates.

JSON example

```
{
  "swapStartDate": "2025-12-01T00:00:00Z",
  "outputRecordType": "Quote",
  "swapGroups": {
    "groups": [
      {
        "referenceId": "UPGRADE-001",
        "outGroup": {
          "swapAssets": [
            {
              "assetId": "02ixx0000004HOAAA2",
              "quantity": 1
            }
          ]
        },
        "inGroup": {
          "graphId": "upgradeRequest",
          "records": [
            {
              "referenceId": "refQuoteLine0",
              "record": {
                "attributes": {
                  "type": "QuoteLineItem",
                  "method": "POST"
                },
                "Product2Id": "01txx0000006iVlAAI",
                "PricebookEntryId": "01uxx0000008ym4AAA",
                "UnitPrice": 1049,
                "Quantity": "1",
```

```
                "StartDate": "2026-03-22"
              }
            }
          ]
        }
      }
    ]
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`contractId`	String	ID of the contract record to upgrade.	Optional	66.0
`opportunityId`	String	ID of the opportunity record to upgrade.	Optional	66.0
`outputRecordType`	String	Record type of the output for the upgrade.	Required	66.0
`swapGroups`	Swap Group[]	Groups that contain the asset details for the upgrade.	Required	66.0
`swapStartDate`	String	Amendment start date for the upgrade action.	Required	66.0

Instant Pricing Input

Input representation to fetch the instant pricing details.

JSON example


```
{
  "correlationId": "7847127122596",
  "contextId": "0000000b28op21g00281751271041790cc1e5686282849a387dd3c59b310418b",
  "records": [
    {
      "referenceId": "0Q0xx0000004DOSCA2",
      "record": {
        "attributes": {
          "type": "Quote",
          "method": "POST"
        },
        "Name": "Test Quote Proration Pricing",
        "OpportunityId": "006xx000001a4ISAAY",
        "Pricebook2Id": "01sxx0000005ptpAAA"
      }
    },
    {
      "referenceId": "refQuoteLine",
      "record": {
        "attributes": {
          "type": "QuoteLineItem",
          "method": "POST"
```

```
        },
        "QuoteId": "0Q0xx0000004DOSCA2",
        "PricebookEntryId": "01uxx0000008zHmAAI",
        "Product2Id": "01txx0000006jmWAAQ",
        "Quantity": 2,
        "UnitPrice": 25,
        "StartDate": "2022-09-28",
        "EndDate": "2028-09-27",
        "PeriodBoundary": "ANNIVERSARY",
        "BillingFrequency": "ANNUAL"
      }
    }
  ]
}
```

This example shows a sample request to specify grouping of lines based on criteria.

```
{
    "contextId": "0000000b28op21g00281751271041790cc1e5686282849a387dd3c59b310418b",
    "correlationId": "7847127122596",
    "records": [
        {
            "referenceId": "0Q0xx0000004DOSCA2",
            "record": {
                "attributes": {
                    "type": "Quote",
                    "method": "PUT"
                }
            }
        },
        {
            "referenceId": "0QLxx0000004F3gGAE",
            "record": {
                "attributes": {
                    "type": "QuoteLineItem",
                    "method": "PUT"
                },
                "Quantity": 5
            }
        }, {
            "referenceId": "0QLxx0000004F3hGAE",
            "record": {
                "attributes": {
                    "type": "QuoteLineItem",
                    "method": "PUT"
                },
                "Quantity": 2
            }
        },
        {
            "referenceId": "GroupId1",
            "record": {
                "attributes": {
                    "type": "QuoteLineGroup",
                    "method": "POST",
```

```
                    "action": "GroupBy",
                    "criteria": {
                        "Quantity": 5
                    }
                },
                "Name": "record"
            }
        }, {
            "referenceId": "GroupId2",
            "record": {
                "attributes": {
                    "type": "QuoteLineGroup",
                    "method": "POST",
                    "action": "GroupBy",
                    "criteria": {
                        "Quantity": 2
                    }
                },
                "Name": "record1",
            }
        }
    ]
}
```

This example shows a sample request for the initial grouping of the quote with the quote lines assigned to the first group.

```
{
    "contextId": "0000000b28op21g00281751271041790cc1e5686282849a387dd3c59b310418b",
    "correlationId": "7847127122596",
    "records": [
        {
            "referenceId": "0Q0xx0000004CAgCAM",
            "record": {
                "attributes": {
                    "type": "Quote",
                    "method": "PUT"
                }
            }
        },
        {
            "referenceId": "GroupId1",
            "record": {
                "attributes": {
                    "type": "QuoteLineGroup",
                    "method": "POST",
                    "action": "GroupAll"
                },
                "Name": "sample"
            }
        }
    ]
}
```

This example shows a sample request to ungroup a quote but retain the quote lines.


```
{
    "contextId": "0000000b28op21g00281751271041790cc1e5686282849a387dd3c59b310418b",
    "correlationId": "7847127122596",
    "records": [
        {
            "referenceId": "0Q0xx0000004CAgCAM",
            "record": {
                "attributes": {
                    "type": "Quote",
                    "method": "PUT"
                }
            }
        },
         {
            "referenceId": "0QLxx0000004CBYGA2",
            "record": {
                "attributes": {
                    "type": "QuoteLineItem",
                    "method": "PUT"
                },
                "Quantity": 2,
                "QuoteLineGroupId": null
            }
        },
        {
            "referenceId": "GroupId1",
            "record": {
                "attributes": {
                    "type": "QuoteLineGroup",
                    "method": "DELETE",
                    "action": "Ungroup"
                }
            }
        }
    ]
}
```


This example shows a sample request to create a new group.


```
{
    "contextId": "0000000b28op21g00281751271041790cc1e5686282849a387dd3c59b310418b",
    "correlationId": "7847127122596",
    "records": [
        {
            "referenceId": "0Q0xx0000004CAgCAM",
            "record": {
                "attributes": {
                    "type": "Quote",
                    "method": "PUT"
                }
            }
        },
        {
```

```
            "referenceId": "GroupId1",
            "record": {
                "attributes": {
                    "type": "QuoteLineGroup",
                    "method": "POST"
                },
                "Name": "sample"
            }
        }
    ]
}
```

This example shows a sample request to delete a group.

```
{
    "contextId": "0000000b28op21g0028175127104179 0cc1e5686282849a387dd3c59b310418b",
    "correlationId": "7847127122596",
    "records": [
        {
            "referenceId": "0Q0xx0000004CAgCAM",
            "record": {
                "attributes": {
                    "type": "Quote",
                    "method": "PUT"
                }
            }
        },
        {
            "referenceId": "GroupId1",
            "record": {
                "attributes": {
                    "type": "QuoteLineGroup",
                    "method": "DELETE",
                    "action": "DeleteGroup"
                },
                "Name": "sample"
            }
        }
    ]
}
```

This example shows a sample request to move a group.

```
{
    "contextId": "0000000b28op21g00281751271041790cc1e5686282849a387dd3c59b310418b",
    "correlationId": "7847127122596",
    "records": [
        {
            "referenceId": "0Q0xx0000004CAgCAM",
            "record": {
                "attributes": {
                    "type": "Quote",
                    "method": "PUT"
                }
            }
```

```
        },
        {
            "referenceId": "0QLxx0000004CBYGA2",
            "record": {
                "attributes": {
                    "type": "QuoteLineItem",
                    "method": "PUT"
                },
                "Quantity": 2
                "QuoteLineGroupId": "{@GroupId2}"
            }
        },
    ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
contextId	String	ID generated by the context service. If you don't specify a context ID, a new context is created.	Optional	59.0
correlationId	String	Client-generated ID for tracking multiple related API requests.	Optional	59.0
records	Object with Reference Input[]	List of pricing data to be fetched.	Required	59.0

Object Graph Input

Input representation of an sObject with a graph ID.

JSON example

```
{
  "graph": {
    "graphId": "1",
    "records": [
      {
        "referenceId": "refOrder",
        "record": {
          "attributes": {
            "type": "Order",
            "method": "POST"
          }
        }
      }
    ]
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
graphId	String	The ID of the graph.	Required	60.0
records	Object with Reference Input on page 1464[]	List of the records to be ingested.	Required	60.0

Object Input Map

Input representation of an sObject record in a key-value map format.

JSON example

```
{
  "records": [
    {
      "referenceId": "refOrder",
      "record": {
        "attributes": {
          "type": "Order",
          "method": "PATCH",
          "id": "402xx000003KY5vJGH"
        },
        "Quantity": 5
      }
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
attributes	Map <String, String>	Configuration input for the record process. Valid values are: • type—Type of sales transaction such as Quote or Order. • method—HTTP methods such as POST, PATCH, and DELETE. • id—Unique identifier for the record. Required for PATCH and DELETE operations.	Required	60.0

Name	Type	Description	Required or Optional	Available Version
		• `criteria`—Criteria to group order or quote line items. For example, group order or quote line items based on a monthly billing frequency. • `action`—Action to group order or quote line items. Valid values are: – `GroupBy` – `Group` – `Ungroup` – `GroupAll` – `DeleteGroup`		

Object with Reference Input

Input representation of a list of records to be inserted or updated. To update a record, specify the record ID.

This is a sample request to create a sales transaction for an order line item.

JSON example

```
{
  "referenceId": "refOrderItem0",
  "record": {
    "attributes": {
      "type": "OrderItem",
      "method": "POST"
    },
    "OrderId": "@{refOrder.id}",
    "OrderActionId": "@{refOrderAction.id}",
    "PricebookEntryId": "01uRM000000igZG",
    "Quantity": 2
  }
}
```

This is a sample request to update an order line item.

JSON example

```
{
  "referenceId": "refOrderItem0",
  "record": {
    "attributes": {
      "type": "OrderItem",
      "method": "PATCH",
      "id": "402xx000003KY5vJGH"
```

```
    },
    "OrderId": "@{refOrder.id}",
    "OrderActionId": "@{refOrderAction.id}",
    "PricebookEntryId": "01uRM000000igZG",
    "Quantity": 2,
    "UnitPrice": 800
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`referenceId`	String	Reference ID that maps to the response and can be used as a reference in later subrecords. This property value starts with a letter or number only and can contain letters, numbers, and underscores. It's also case-sensitive when used for referencing.	Required	60.0
`records`	Object Input Map on page 1463	Details of a record to be ingested.	Required	60.0

Place Order Input

Input representation of the request to create or update an order.

JSON example

```
{
  "pricingPref": "System",
  "configurationInput": "RunAndAllowErrors",
  "configurationOptions": {
    "validateProductCatalog": true,
    "validateAmendRenewCancel": true,
    "executeConfigurationRules": true,
    "addDefaultConfiguration": true
  },
  "graph": {
    "graphId": "graphId",
    "records": [
      {
        "referenceId": "refOrder",
        "record": {
          "attributes": {
            "type": "Order",
            "method": "POST",
            "Id": "POST"
          }
        }
      },
      {
         "referenceId":"refOrderItem",
```

```
          "record":{
          "attributes":{
              "type":"OrderItem",
              "method":"POST"
             },
             "OrderId":"@{refOrder.id}",
             "OrderActionId":"@{refOrderAction.id}",
             "ListPrice":"144.99",
             "Quantity":3,
             "PricebookEntryId":"01uxx0000008yXPAAY",
             "Product2Id":"01txx0000006i2UAAQ",
             "UnitPrice":"199.49"
        }
       }
     ]
   }
}
```

This example shows a sample request to define grouping of order items.

```
{
    "pricingPref": "system",
    "graph": {
        "graphId": "placeOrder",
        "records": [
            {
                "referenceId": "refOrder",
                "record": {
                    "attributes": {
                        "type": "Order",
                        "method": "PATCH",
                        "id": "801xx000003GZ9bAAG"
                    }
                }
            },
            {
                "referenceId": "refOlg1",
                "record": {
                    "attributes": {
                        "type": "OrderItemGroup",
                        "method": "POST"
                    },
                    "Name": "New Group",
                    "OrderId": "@{refOrder.id}"
                }
            }
        ]
    }
}
```

This example shows a sample request to ungroup order items.

```
{
    "pricingPref": "system",
    "graph": {
```

```
        "graphId": "placeOrder",
        "records": [
            {
                "referenceId": "refOrder",
                "record": {
                    "attributes": {
                        "type": "Order",
                        "method": "PATCH",
                        "id": "refOrder"
                    }
                }
            },
            {
                "referenceId": "refOlg1",
                "record": {
                    "attributes": {
                        "type": "OrderItemGroup",
                        "method": "DELETE",
                        "id": "refOlg1",
                        "action": "Ungroup"
                    }
                }
            }
        ]
    }
}
```

This example shows a sample request to create a new group.

```
{
    "pricingPref": "system",
    "graph": {
        "graphId": "placeOrder",
        "records": [
            {
                "referenceId": "refOrder",
                "record": {
                    "attributes": {
                        "type": "Order",
                        "method": "PATCH",
                        "id": "refOrder"
                    }
                }
            },
            {
                "referenceId": "refOlg",
                "record": {
                    "attributes": {
                        "type": "OrderItemGroup",
                        "method": "POST"
                    },
                    "Name": "New Group",
                    "OrderId": "@{refOrder.id}"
                }
            }
```

```
        ]
    }
}
```

This example shows a sample request to delete a group.

```
{
    "contextId": "",
    "correlationId": "",
    "records": [
        {
            "referenceId": "refOrder",
            "record": {
                "attributes": {
                    "type": "Order",
                    "method": "PATCH",
                    "id": "refOrder"
                }
            }
        },
        {
            "referenceId": "refOlg",
            "record": {
                "attributes": {
                    "type": "OrderItemGroup",
                    "method": "DELETE",
                    "id": "refOlg",
                    "action": "DeleteGroup"
                }
            }
        }
    ]
}
```

This example shows a sample request to group order items based on criteria.

```
{
    "pricingPref": "system",
    "graph": {
        "graphId": "placeOrder",
        "records": [
            {
                "referenceId": "refOrder",
                "record": {
                    "attributes": {
                        "type": "Order",
                        "method": "PATCH",
                        "id": "refOrder"
                    }
                }
            },
            {
                "referenceId": "g0",
                "record": {
                    "attributes": {
```

```
                    "type": "OrderItemGroup",
                    "method": "POST",
                    "action": "GroupBy",
                    "criteria": {
                        "BillingFrequency2": null
                    }
                },
                "Name": "Billing Frequency: ",
                "OrderId": "@{refOrder.id}"
            }
        },
        {
            "referenceId": "g1",
            "record": {
                "attributes": {
                    "type": "OrderItemGroup",
                    "method": "POST",
                    "action": "GroupBy",
                    "criteria": {
                        "BillingFrequency2": "Monthly"
                    }
                },
                "Name": "Billing Frequency: Monthly",
                "OrderId": "@{refOrder.id}"
            }
        }
    ]
  }
}
```


Properties

Name	Type	Description	Required or Optional	Available Version
catalog RatesPref	String	Rate card entries defined in the catalog that must be fetched for order items with usage-based pricing during the order creation process. Valid values are: • Fetch—Retrieves the rate card entries defined in the catalog for order items during the order creation process. • Skip—Skips the retrieval of rate card entries for order items during the order creation process. The default value is Skip. This property is available when the Usage-Based Selling feature is enabled.	Optional	62.0

Name	Type	Description	Required or Optional	Available Version
configurationInput	String	Configuration input for the place order process. Valid values are: • `RunAndAllowErrors`—Specifies to run the configuration and to proceed order ingestion upon encountering any configuration errors. • `RunAndBlockErrors`—Specifies to run configuration and to block order ingestion upon encountering any configuration errors. • `Skip`—Specifies to skip configuration. The default value is `RunAndBlockErrors`.	Optional	60.0
configurationOptions	Configuration Options Input[]	Configuration options during the ingestion process.	Optional	60.0
graph	Object Graph Input	The sObject graph of the order payload to be ingested. You can perform create, update, or delete operations on objects from the Sales Transaction context definition by using this property. Additionally, perform create, update, or delete operations on custom objects and fields in your extended context definition.	Required	60.0
pricingPref	String	Pricing preference during the create order process. Valid values are: • `Force`—Specifies to force pricing during the order ingestion process. • `Skip`—Specifies to skip pricing during the order ingestion process. • `System`—Specifies the system to determine whether a pricing calculation is required. The default value is `System`.	Optional	60.0

Place Quote Input

Input representation of the request to create or update a quote.

JSON example

This example shows a sample request to create a quote.

```
{
  "pricingPref": "System",
  "configurationInput": "RunAndAllowErrors",
  "configurationOptions": {
    "validateProductCatalog": true,
    "validateAmendRenewCancel": true,
    "executeConfigurationRules": true,
    "addDefaultConfiguration": true
  },  "graph": {
    "graphId": "createQuote",
    "records": [{
        "referenceId": "refQuote",
        "record": {
          "attributes": {
            "type": "Quote",
            "method": "POST"
          },
          "opportunityId": "---",
          "quoteProp1": "value1",
          "quoteProp2": "value2"
        }
      },
      {
        "referenceId": "refQuoteLineItem1",
        "record": {
          "attributes": {
            "type": "QuoteLineItem",
            "method": "POST"
          },
          "QuoteLineItemProp1": "value1",
          "QuoteLineItemProp2": "value2"
        }
      },
      {
      "referenceId": "refQuoteLineItemAttribute",
      "record": {
        "attributes": {
              "type": "QuoteLineItemAttribute",
              "method": "POST"
       },
       "QuoteLineItemId": "@{refQuoteLineItem1.id}",
       "AttributeDefinitionId": "0tjxx0000000001AAA"
      }
}
    ]
  }
}
```

This example shows a sample request to insert, update, or delete a quote line item.

```
{
  "pricingPref": "System",
```

```
"configurationInput": "skip",
"graph": {
  "graphId": "updateQuote",
  "records": [
    {
      "referenceId": "refQuote",
      "record": {
        "attributes": {
          "type": "Quote",
          "method": "PATCH",
          "id": "0Q0xx0000004E2mCAE"
        },
        "Name": "Quote_Acme"
      }
    },
    {
      "referenceId": "refQuoteLineItemToCreate1",
      "record": {
        "attributes": {
          "type": "QuoteLineItem",
          "method": "POST"
        },
        "QuoteId": "0Q0xx0000004E2mCAE",
        "PricebookEntryId": "01uxx0000008yXPAAY",
        "Product2Id": "01txx0000006i2UAAQ",
        "Quantity": 2.0,
        "UnitPrice": 800.0,
        "PeriodBoundary": "Anniversary",
        "BillingFrequency": "Monthly",
        "StartDate": "2024-03-11"
      }
    },
    {
      "referenceId": "refQuoteLineItemToPatch2",
      "record": {
        "attributes": {
          "type": "QuoteLineItem",
          "method": "PATCH",
          "id": "0Q0xx0000004E2mCAE"
        },
        "Quantity": 2.0,
        "UnitPrice": 600.0
      }
    },
    {
      "referenceId": "refQuoteLineItemToDelete3",
      "record": {
        "attributes": {
          "type": "QuoteLineItem",
          "method": "DELETE",
          "id": "0Q0xx0000004E2mYLK"
        }
      }
    }
```

```
    ]
  }
}
```

This example shows a sample request to define grouping of quote line items.

```
{
   "pricingPref": "Force",
   "configurationInput": "skip",
   "graph": {
      "graphId": "groupLines",
      "records": [
         {
            "referenceId": "refQuote",
            "record": {
               "attributes": {
                  "type": "Quote",
                  "method": "PATCH",
                  "id":"0Q0xx0000004C99CAE"
               },
               "Name": "From Place Quote API"
            }
         },
         {
            "referenceId": "refQlg1",
            "record": {
               "attributes": {
                  "type": "QuoteLineGroup",
                  "method": "POST",
                  "action": "GroupBy",
                  "criteria": {
                    "Quantity": 1
                  }
               },
               "Name": "From Place Quote API Group",
               "QuoteId": "@{refQuote.id}"
            }
         },
         {
            "referenceId": "refQuoteItem1",
            "record": {
               "attributes": {
                  "type": "QuoteLineItem",
                  "method": "PATCH",
                  "id":"0QLxx0000004DJcGAM"
               },
               "QuoteLineGroupId": "@{refQlg1.id}",
               "Quantity": 1
            }
         }
      ]
   }
}
```

This example shows a sample request for the initial grouping of the quote with the quote lines assigned to the first group.

```
{
  "pricingPref": "Force",
  "graph": {
    "graphId": "test",
    "records": [
      {
        "referenceId": "refQuote",
        "record": {
          "attributes": {
            "type": "Quote",
            "method": "PATCH",
            "id": "0Q0xx0000004CAmCAM"
          }
        }
      },
      {
        "referenceId": "refQlg1",
        "record": {
          "attributes": {
            "type": "QuoteLineGroup",
            "method": "POST",
            "action": "GroupAll"
          },
          "Name": "From PQ API Group",
          "QuoteId": "@{refQuote.id}"
        }
      }
    ]
  }
}
```

This example shows a sample request to ungroup a quote but retain the quote lines.

```
{
   "pricingPref": "Force",
   "configurationInput": "skip",
   "graph": {
      "graphId": "test",
      "records": [
         {
            "referenceId": "refQuote",
            "record": {
               "attributes": {
                  "type": "Quote",
                  "method": "PATCH",
                  "id":"0Q0xx0000004C99CAE"
               },
               "Name": "From Place Quote API"
            }
         },
         {
            "referenceId": "refQlg1",
            "record": {
```

```
                "attributes": {
                    "type": "QuoteLineGroup",
                    "method": "DELETE",
                    "id": "{GroupId}",
                    "action": "Ungroup"
                }
            }
        }
    ]
  }
}
```

This example shows a sample request to create a new group.

```
{
   "pricingPref": "Force",
   "configurationInput": "skip",
   "graph": {
      "graphId": "test",
      "records": [
         {
            "referenceId": "refQuote",
            "record": {
               "attributes": {
                  "type": "Quote",
                  "method": "PATCH",
                  "id":"0Q0xx0000004C99CAE"
               },
               "Name": "From Place Quote API"
            }
         },
         {
            "referenceId": "refQlg1",
            "record": {
               "attributes": {
                  "type": "QuoteLineGroup",
                  "method": "POST"
               },
               "Name": "From PQ API Group",
               "QuoteId": "@{refQuote.id}"
            }
         }
      ]
   }
}
```

This example shows a sample request to delete a group.

```
{
   "pricingPref": "Force",
   "configurationInput": "skip",
   "graph": {
      "graphId": "test",
      "records": [
         {
```

```
            "referenceId": "refQuote",
            "record": {
               "attributes": {
                  "type": "Quote",
                  "method": "PATCH",
                  "id":"0Q0xx0000004C99CAE"
               },
               "Name": "From Place Quote API"
            }
         },
         {
            "referenceId": "refQlg1",
            "record": {
               "attributes": {
                  "type": "QuoteLineGroup",
                  "method": "DELETE",
                  "id": "{GroupId}",
                  "action": "DeleteGroup"
               }
            }
         }
      ]
   }
}
```

This example shows a sample request to move a group.

```
{
   "pricingPref": "Force",
   "configurationInput": "skip",
   "graph": {
      "graphId": "test",
      "records": [
         {
            "referenceId": "refQuote",
            "record": {
               "attributes": {
                  "type": "Quote",
                  "method": "PATCH",
                  "id":"0Q0xx0000004C99CAE"
               },
               "Name": "From PlaceQuote Api"
            }
         },
        {
            "referenceId": "0QLxx0000004CBYGA2",
            "record": {
                "attributes": {
                    "type": "QuoteLineItem",
                    "method": "PATCH"
                    "id": "0QLxx0000004CBYGA2"
                },
                "Quantity": 2,
                "QuoteLineGroupId": "@{GroupId2}"
            }
```

```
        ]
    }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`catalogRatesPref`	String	Rate card entries defined in the catalog that must be fetched for quote line items with usage-based pricing during the quote creation process. Valid values are: • `Fetch`—Retrieves the rate card entries defined in the catalog for quote line items during the quote creation process. • `Skip`—Skips the retrieval of rate card entries for quote line items during the quote creation process. The default value is `Skip`. This property is available when the Usage-Based Selling feature is enabled.	Optional	62.0
`configurationInput`	String	Configuration input for the place quote process. Valid values are: • `RunAndAllowErrors` • `RunAndBlockErrors` • `Skip` The default value is `RunAndBlockErrors`.	Optional	60.0
`configurationOptions`	Configuration Options Input	Configuration options during the ingestion process.	Optional	60.0
`graph`	Object Graph Input	The sObject graph representing the quote structure. You can perform create, update, or delete operations on objects from the Sales Transaction context definition by using this property. Additionally, perform create, update, or delete operations on custom objects and fields in your extended context definition.	Required	60.0
`pricingPref`	String	Pricing preference during the quote process. Valid values are: • `Force` • `Skip`	Optional	60.0

Name	Type	Description	Required or Optional	Available Version
		• `System` The default value is `System`.		

Read Sales Transaction Input

Input representation of the filter criteria details to read a sales transaction.

JSON example

```
{
  "contextId": "008d27d7-e004-4906-a949-ee7d7c323c77",
  "queryTags": [
    "Quote",
    "QuoteLineItem",
    "Product"
  ],
  "sobjectFieldMap": {
    "Quote": [],
    "QuoteLineItem": [
      "Quantity",
      "Product2Id"
    ]
  },
  "filters": [
    {
      "sObjectName": "Quote",
      "fieldName": "Status",
      "operator": "Equals",
      "operands": [
        {
          "value": "Draft",
          "type": "STRING"
        }
      ]
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`contextId`	String	ID of the context to retrieve the data records.	Required	65.0
`queryTags`	List<String>	List of objects that must be retrieved from the context.	Optional	65.0

Name	Type	Description	Required or Optional	Available Version
sobjectFieldMap	Map<String, List<String>>	Mapping of an sObject name to a list. The list includes the sObject field names on the object or can be an empty list. An empty list specifies that all fields on the object must be queried.	Optional	67.0
filters	List<Sales Transaction Filter Condition>	Filter conditions to query the context data.	Optional	67.0

Renewal Input

Input representation of the details of the request to initiate the renewal of an asset.

JSON example

```
{
  "assetIds": [
    "02iSG0000003NMhYAM",
    "02iSG0000006DvSYAU"
  ],
  "contractId": "800SG00000CFpepYAD",
  "opportunityId": "006SG000004W5tVYAS",
  "outputRecordId": "801SG00000DX1jWYAT",
  "outputRecordType": "Quote",
  "renewalEndDate": "2024-10-03T23:59:59",
  "renewalStartDate": "2023-10-04T00:00:00"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
assetIds	String[]	IDs of the assets that you want to renew.	Required	62.0
contractId	String	ID of the Contract record that you want to sync with the renewal of the Quote or Order record.	Optional	62.0
opportunityId	String	ID of the Opportunity record that you want to sync with the renewal quote.	Optional	62.0
outputRecordId	String	ID of the Quote or Order record that you want to renew.	Optional	62.0
outputRecordType	String	Type of renewal record that you want to create.	Required	62.0
renewalEndDate	String	End date of the renewal process for the assets.	Optional	62.0

Name	Type	Description	Required or Optional	Available Version
`renewal StartDate`	String	Start date of the renewal process for the assets. Required for early asset renewals and renewing expired assets, using today's date or a future date.	Optional	62.0

Sales Transaction Filter Condition Input

Input representation of filter conditions used to query sales transaction context data.

JSON example

```
{
  "filters": [
    {
      "sObjectName": "Quote",
      "fieldName": "Status",
      "operator": "Equals",
      "operands": [
        {
          "value": "Draft",
          "type": "STRING"
        }
      ]
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`sObjectName`	String	Name of the sObject that contains the field to filter on.	Required if the `fieldName` property value isn't specified. Or, specify the `nodeName` and `attributeName` property values.	67.0
`fieldName`	String	Name of the field on the specified sObject to filter on.	Required if the `sObjectName` property value isn't specified. Or, specify the `nodeName` and `attributeName` property values.	67.0

Name	Type	Description	Required or Optional	Available Version
`operator`	String	Operator to use to filter the results. Valid values are: • `Equals` • `NotEquals` • `Gt`—Greater Than • `GtEq`—Greater Than or Equals • `Lt`—Less Than • `LtEq`—Less Than or Equals • `In`	Required	67.0
`nodeName`	String	Name of the context node to apply filtering on.	Optional	67.0
`attributeName`	String	Name of the node attribute to apply filtering on.	Optional	67.0
`operands`	Sales Transaction Operand[]	Operand values with explicit value and type metadata.	Required	67.0
`underlyingFilters`	Sales Transaction Filter Condition[]	Nested filter conditions used to compose a composite filter.	Optional	67.0
`operatorLogic`	String	Logical expression that defines how underlying filters are combined. Valid values are: • `AND` • `OR`	Optional	67.0

Sales Transaction Input

Input representation of the details of the request to place a sales transaction, such as a quote or an order.

JSON example

This is a sample request to create a sales transaction for a quote. This example also skips tax calculation by specifying a value for the optional `taxPref` property.

```
{
  "pricingPref": "System",
  "catalogRatesPref": "Skip",
  "configurationPref": {
    "configurationMethod": "Skip",
    "configurationOptions": {
      "validateProductCatalog": true,
      "validateAmendRenewCancel": true,
      "executeConfigurationRules": true,
      "addDefaultConfiguration": true
```

```
    }
  },
  "taxPref": "Skip",
  "contextDetails": {
    "contextId": "e055bb18-d4e8-41c3-881e-0132b9561708"
  },
  "graph": {
    "graphId": "createQuote",
    "records": [
      {
        "referenceId": "refQuote",
        "record": {
          "attributes": {
            "method": "POST",
            "type": "Quote"
          },
          "Name": "Quote_Acme",
          "Pricebook2Id": "01sDU000000JvhbYAC"
        }
      },
      {
        "referenceId": "refQuoteLine0",
        "record": {
          "attributes": {
            "type": "QuoteLineItem",
            "method": "POST"
          },
          "QuoteId": "@{refQuote.id}",
          "Product2Id": "01tDU000000F7b8YAC",
          "PricebookEntryId": "01uDU000000fxt2YAA",
          "UnitPrice": 100,
          "Quantity": "1",
          "StartDate": "2024-10-29",
          "EndDate": "2025-03-01",
          "PeriodBoundary": "Anniversary"
        }
      },
      {
        "referenceId": "refQuoteLine1",
        "record": {
          "attributes": {
            "type": "QuoteLineItem",
            "method": "POST"
          },
          "QuoteId": "@{refQuote.id}",
          "Product2Id": "01tLT00000AA2M9YAL",
          "PricebookEntryId": "01uLT000007PTafYAG",
          "Quantity": "1"
        }
      },
      {
        "referenceId": "refQuoteLineItemAttribute1",
        "record": {
          "attributes": {
```

```
            "type": "QuoteLineItemAttribute",
            "method": "POST"
          },
          "QuoteLineItemId": "@{refQuoteLine0.id}",
          "AttributeDefinitionId": "0tjRT0000000XbtYAE",
          "AttributeValue": "True"
        }
      },
      {
        "referenceId": "refQuoteLineRelationship1",
        "record": {
          "attributes": {
            "type": "QuoteLineRelationship",
            "method": "POST"
          },
          "ProductRelationshipTypeId": "0yoLT000000wHq6YAE",
          "ProductRelatedComponentId": "0dSLT00000076IP2AY",
          "MainQuoteLineId": "@{refQuoteLine0.id}",
          "AssociatedQuoteLineId": "@{refQuoteLine1.id}",
          "AssociatedQuoteLinePricing": "NotIncludedInBundlePrice"
        }
      }
    ]
  }
}
```

This sample request assigns a TransactionProcessingType record to a quote without any additional preferences. In this example, the TransactionType value for a record is set to a TransactionProcessingType record. See TransactionProcessingType on page 1369 tooling object for more details.

```
{
  "pricingPref": "Force",
  "contextDetails": {
    "contextId": "f1c9e3e1c335f7959a88de09d3a867cc2b95e08709b99de8e2edeb8f5039e8ed",
    "scope": "Session"
  },
  "graph": {
    "graphId": "updateQuote",
    "records": [
      {
        "referenceId": "refQuote",
        "record": {
          "attributes": {
            "type": "Quote",
            "method": "POST"
          },
          "OpportunityId": "006xx000001a2oWAAQ",
          "PriceBook2Id": "01sxx0000005ptpAAA",
          "TransactionType": "SkipPricingAndRunTax",
          "Name": "Quote_No_Tax_System"
        }
      },
      {
        "referenceId": "refQLI1",
        "record": {
```

```
          "attributes": {
            "type": "QuoteLineItem",
            "method": "POST"
          },
          "QuoteId": "@{refQuote.id}",
          "UnitPrice": 49.99,
          "Product2Id": "01txx0000006i2aAAA",
          "PricebookEntryId": "01uxx0000008yX0AAI",
          "Quantity": 10
        }
      }
    ]
  }
}
```

This is a sample request to insert, update, or delete a quote line item.

```
{
  {
  "pricingPref": "System",
  "catalogRatesPref": "Skip",
  "configurationPref": {
    "configurationMethod": "Skip",
    "configurationOptions": {
      "validateProductCatalog": true,
      "validateAmendRenewCancel": true,
      "executeConfigurationRules": true,
      "addDefaultConfiguration": true
    }
  },
  "contextDetails": {
    "contextId": "e055bb18-d4e8-41c3-881e-0132b9561708"
  },
  "graph": {
    "graphId": "updateQuote",
    "records": [
      {
        "referenceId": "refQuote",
        "record": {
          "attributes": {
            "method": "PATCH",
            "type": "Quote",
            "id": "801xx000003GZ9bAAG"
          }
        }
      },
      {
        "referenceId": "refQuoteLine0",
        "record": {
          "attributes": {
            "type": "QuoteLineItem",
            "method": "PATCH",
            "id": "402xx000003KY5vJGH"
          },
          "Quantity": "5"
```

```
        }
      }
    ]
  }
}
```

This is a sample request to define grouping of quote line items.

```
{
  "pricingPref": "Force",
  "catalogRatesPref": "Skip",
  "configurationPref": {
    "configurationMethod": "Skip",
    "configurationOptions": {
      "validateProductCatalog": true,
      "validateAmendRenewCancel": true,
      "executeConfigurationRules": true,
      "addDefaultConfiguration": true
    }
  },
  "contextDetails": {
    "contextId": "e055bb18-d4e8-41c3-881e-0132b9561708"
  },
  "graph": {
    "graphId": "groupQuoteLines",
    "records": [
      {
        "referenceId": "refQuote",
        "record": {
          "attributes": {
            "method": "PATCH",
            "type": "Quote",
            "id": "801xx000003GZ9bAAG"
          }
        }
      },
      {
        "referenceId": "refQuoteLine0",
        "record": {
          "attributes": {
            "type": "QuoteLineItem",
            "method": "PATCH",
            "id": "402xx000003KY5vJGH"
          },
          "QuoteLineGroupId": "@{refQuote.id}"
        }
      }
    ]
  }
}
```

This is s a sample request for the initial grouping of the quote with all the quote lines assigned to the first group.

```
{
  "pricingPref": "Force",
```

```
  "catalogRatesPref": "Skip",
  "graph": {
    "graphId": "groupQuote",
    "records": [
      {
        "referenceId": "refQuote",
        "record": {
          "attributes": {
            "type": "Quote",
            "method": "PATCH",
            "id": "0Q0xx0000004CAmCAM"
          }
        }
      },
      {
        "referenceId": "refQlg1",
        "record": {
          "attributes": {
            "type": "QuoteLineGroup",
            "method": "POST",
            "action": "GroupAll"
          },
          "Name": "From PST API Group",
          "QuoteId": "@{refQuote.id}"
        }
      }
    ]
  }
}
```

This is a sample request to ungroup a quote but retain the quote lines.

```
{
   "catalogRatesPref": "Skip",
   "pricingPref": "Force",
   "graph": {
      "graphId": "ungroupQuote",
      "records": [
         {
            "referenceId": "refQuote",
            "record": {
               "attributes": {
                  "type": "Quote",
                  "method": "PATCH",
                  "id":"0Q0xx0000004C99CAE"
               },
               "Name": "Grouped Quote with PST API"
            }
         },
         {
            "referenceId": "refQlg1",
            "record": {
               "attributes": {
                  "type": "QuoteLineGroup",
                  "method": "DELETE",
```

```
                    "id": "402xx000003KY5vJGH",
                    "action": "Ungroup"
                }
            }
        }
    ]
  }
}
```

This is a sample request to create a new group.

```
{
   "catalogRatesPref": "Skip",
   "pricingPref": "Force",
   "graph": {
      "graphId": "createGroup",
      "records": [
         {
            "referenceId": "refQuote",
            "record": {
               "attributes": {
                  "type": "Quote",
                  "method": "PATCH",
                  "id":"0Q0xx0000004C99CAE"
               },
               "Name": "Grouped Quote with PST API"
            }
         },
         {
            "referenceId": "refQlg1",
            "record": {
               "attributes": {
                  "type": "QuoteLineGroup",
                  "method": "POST"
               },
               "Name": "From PQ API Group",
               "QuoteId": "@{refQuote.id}"
            }
         }
      ]
   }
}
```

This example shows a sample request to delete a group.

```
{
   "catalogRatesPref": "Skip",
   "pricingPref": "Force",
   "graph": {
      "graphId": "deleteGroup",
      "records": [
         {
            "referenceId": "refQuote",
            "record": {
               "attributes": {
```

```
                    "type": "Quote",
                    "method": "PATCH",
                    "id":"0Q0xx0000004C99CAE"
                 },
                 "Name": "Grouped Quote with PST API"
              }
           },
           {
              "referenceId": "refQlg1",
              "record": {
                 "attributes": {
                    "type": "QuoteLineGroup",
                    "method": "DELETE",
                    "id": "402xx000003KY5vJGH",
                    "action": "DeleteGroup"
                 }
              }
           }
        ]
     }
}
```

This is a sample request to group order items based on criteria.

```
{
    "catalogRatesPref": "Skip",
    "pricingPref": "Force",
    "graph": {
        "graphId": "groupOrderItems",
        "records": [
            {
                "referenceId": "refOrder",
                "record": {
                    "attributes": {
                        "type": "Order",
                        "method": "PATCH",
                        "id": "0Q0xx0000004C99CAE"
                    }
                }
            },
            {
                "referenceId": "refOlg1",
                "record": {
                    "attributes": {
                        "type": "OrderItemGroup",
                        "method": "POST",
                        "action": "GroupBy",
                        "criteria": {
                            "BillingFrequency2": null
                        }
                    },
                    "Name": "Billing Frequency: ",
                    "OrderId": "@{refOrder.id}"
                }
            },
```

```
            {
                "referenceId": "g1",
                "record": {
                    "attributes": {
                        "type": "OrderItemGroup",
                        "method": "POST",
                        "action": "GroupBy",
                        "criteria": {
                            "BillingFrequency2": "Monthly"
                        }
                    },
                    "Name": "Billing Frequency: Monthly",
                    "OrderId": "@{refOrder.id}"
                }
            }
        ]
    }
}
```

This is a sample request to save changes to a ramp deal by using context ID. The context ID is returned by the Ramp Deal APIs. See Create Ramp Deal (POST).

```
{
  "pricingPref": "Force",
  "contextDetails": {
    "contextId": "f1c9e3e1c335f7959a88de09d3a867cc2b95e08709b99de8e2edeb8f5039e8ed",
    "scope": "Session"
  },
  "graph": {
    "graphId": "updateQuote",
    "records": [
      {
        "referenceId": "refQuote",
        "record": {
          "attributes": {
            "type": "Quote",
            "method": "PATCH",
            "id": "0Q0xx0000004DQ4CAM"
          }
        }
      }
    ]
  }
}
```

To see examples that specify actions to create ramp deals for groups, see Group Ramp Action Input on page 1491.

To see examples that apply to usage-based products, see Usage-Based Product Input on page 1502.

Properties

Name	Type	Description	Required or Optional	Available Version
catalogRates Pref	String	Rate card entries defined in the catalog that must be fetched for sales items with	Optional	63.0

Name	Type	Description	Required or Optional	Available Version
		usage-based pricing during the creation of the sales transaction. Valid values are: • `Fetch`—Retrieves the rate card entries defined in the catalog for sales items during the creation of the sales transaction. • `Skip`—Skips the retrieval of rate card entries for sales items during the creation of the sales transaction. The default value is `Skip`. This property is available when the Usage-Based Selling feature is enabled.		
`configurationPref`	Configurator Preference Input	Configuration preference during the quote process. These preferences ensure that quotes are defined as per the requirement.	Optional	63.0
`contextDetails`	Context Input	Context details that are created for a sales transaction.	Required if the `graph` property isn't specified.	63.0
`graph`	Object Graph Input	The sObject graph of the sales transaction to be ingested. You can perform create, update, or delete operations on objects from the Sales Transaction context definition by using this property. Additionally, perform create, update, or delete operations on custom objects and fields in your extended context definition. To create custom objects that are at the grandchildren level from a line item, you must create the hierarchy of objects until the grandchild object in the same request.	Required if the `contextDetails` property isn't specified.	63.0
`groupRampAction`	String	Specifies the action that you want to perform on group ramp segments. You can also convert a non-ramped group into a ramped group. Valid values are: • `AddProducts`—Adds rampable products to group ramp segments. • `DeleteProducts`—Deletes ramped products. • `EditGroup`—Converts a non-ramped group into a group ramp	Optional	65.0

Name	Type	Description	Required or Optional	Available Version
		segment, or edit group ramp segment attributes such as name and description, except the start and end dates. • `EditRampSchedule`—Edits details of the group ramp segments, including start and end dates. • `DeleteSegment`—Deletes the first or last segment in a group ramp schedule. • `ConvertToNonRampedGroup`—Converts the first or last group ramp segment into a non-ramped group. To add or delete ramped line items from multiple group ramp segments, specify the applicable values in the `graph` property. See Group Ramp Action Input on page 1491 to refer to examples.		
`pricingPref`	String	Pricing preference during the creation of a sales transaction. Valid values are: • `Force`—Enforces pricing during the creation of sales transactions. • `Skip`—Skips pricing during the creation of sales transactions. • `System`—Determines whether a pricing calculation is required. The default value is `System`.	Optional	63.0
`taxPref`	String	Specifies whether to execute or skip the tax calculation step for each sales transaction record. Valid value is `Skip`. If you don't specify this value, then tax calculation request is performed by default.	Optional	65.0

Group Ramp Action Input

Understand the sample request to specify group ramp actions during initial sale.

Group Ramp Action Input

Understand the sample request to specify group ramp actions during initial sale.

Keep these considerations in mind when you specify ramp actions.

- Use the Clone Sales Transaction API to clone a ramp segment, and specify the clone option.

- Use the Place Sales Transaction API to specify a group ramp action by using the `groupRampAction` property. You can refer to the sections in this topic for examples.

JSON example to edit a group

This is a sample request that creates the first ramp segment. This request accepts IDs of a quote and quote line group. Additionally, the request accepts attributes of quote line group such as `IsRamped`, `SegmentType`, `StartDate`, and `EndDate`. A ramp segment is created with a ramp identifier and segment identifier added to all the quote line items available in the ramp segment.

This process converts a group into a segment, which becomes the first segment in the ramp schedule. A quote can contain a single ramp schedule only. To create another segment in the ramp schedule, use the Clone Sales Transaction API.

```
{
  "groupRampAction": "EditGroup",
  "pricingPref": "System",
  "graph": {
    "graphId": "updateQuote",
    "records": [
      {
        "referenceId": "0Q0xx0000004CYqCAM",
        "record": {
          "attributes": {
            "type": "Quote",
            "method": "PATCH",
            "id": "0Q0xx0000004CYqCAM"
          }
        }
      },
      {
        "referenceId": "1C9xx0000004CVcCAM",
        "record": {
          "attributes": {
            "type": "QuoteLineGroup",
            "method": "PATCH",
            "id": "1C9xx0000004CVcCAM"
          },
          "StartDate": "2025-05-01",
          "EndDate": "2025-06-30",
          "SortOrder": 1,
          "IsRamped": true,
          "SegmentType": "Custom"
        }
      }
    ]
  }
}
```

JSON example to edit a ramp segment

This is a sample request to edit multiple ramp segments simultaneously, maintaining date continuity among ramp segments.

```
{
  "groupRampAction": "EditRampSchedule",
  "pricingPref": "System",
  "graph": {
    "graphId": "updateQuote",
    "records": [
```

```
      {
        "referenceId": "0Q0xx0000004CYqCAM",
        "record": {
          "attributes": {
            "type": "Quote",
            "method": "PATCH",
            "id": "0Q0xx0000004CYqCAM"
          }
        }
      },
      {
        "referenceId": "1C9xx0000004CVcCAM",
        "record": {
          "attributes": {
            "type": "QuoteLineGroup",
            "method": "PATCH",
            "id": "1C9xx0000004CVcCAM"
          },
          "StartDate": "2025-05-01",
          "EndDate": "2025-06-30"
        }
      },
      {
        "referenceId": "1C9xx0000004CVcAAM",
        "record": {
          "attributes": {
            "type": "QuoteLineGroup",
            "method": "PATCH",
            "id": "1C9xx0000004CVcAAM"
          },
          "StartDate": "2025-07-01",
          "EndDate": "2025-08-30"
        }
      },
      {
        "referenceId": "1C9xx0000004CVcBAM",
        "record": {
          "attributes": {
            "type": "QuoteLineGroup",
            "method": "PATCH",
            "id": "1C9xx0000004CVcBAM"
          },
          "StartDate": "2025-09-01",
          "EndDate": "2025-10-30"
        }
      }
    ]
  }
}
```

JSON example to add a product

This is a sample request to add a product to the current and subsequent segments. A ramp identifier and segment identifier are added to the quote line items.

```
{
  "groupRampAction": "AddProducts",
  "pricingPref": "System",
  "graph": {
    "graphId": "updateQuote",
    "records": [
      {
        "referenceId": "0Q0xx0000004CKKCA2",
        "record": {
          "attributes": {
            "type": "Quote",
            "method": "PATCH",
            "id": "0Q0xx0000004CKKCA2"
          }
        }
      },
      {
        "referenceId": "1C9xx0000004CCGCA2",
        "record": {
          "attributes": {
            "type": "QuoteLineGroup",
            "method": "PATCH",
            "id": "1C9xx0000004CCGCA2"
          }
        }
      },
      {
        "referenceId": "ref_01txx0000006iCXAAY_0",
        "record": {
          "attributes": {
            "type": "QuoteLineItem",
            "method": "POST",
            "id": "ref_01txx0000006iCXAAY_0"
          },
          "QuoteId": "@{0Q0xx0000004CKKCA2.id}",
          "Id": "ref_01txx0000006iCXAAY_0",
          "UnitPrice": 2000,
          "Product2Id": "01txx0000006iCXAAY",
          "PricebookEntryId": "01uxx0000008yciAAA",
          "Quantity": 1,
          "StartDate": "2025-05-28T00:00:00.000Z",
          "BillingFrequency": null,
          "PeriodBoundary": null,
          "QuoteLineGroupId": "1C9xx0000004CCGCA2"
        }
      },
      {
        "referenceId": "1C9xx0000004CCGCAB",
        "record": {
          "attributes": {
```

```
            "type": "QuoteLineGroup",
            "method": "PATCH",
            "id": "1C9xx0000004CCGCAB"
          }
        }
      },
      {
        "referenceId": "ref_01txx0000006iCXABY_0",
        "record": {
          "attributes": {
            "type": "QuoteLineItem",
            "method": "POST",
            "id": "ref_01txx0000006iCXABY_0"
          },
          "QuoteId": "@{0Q0xx0000004CKKCA2.id}",
          "Id": "ref_01txx0000006iCXAAY_0",
          "UnitPrice": 2000,
          "Product2Id": "01txx0000006iCXAAY",
          "PricebookEntryId": "01uxx0000008yciAAA",
          "Quantity": 1,
          "StartDate": "2025-05-28T00:00:00.000Z",
          "BillingFrequency": null,
          "PeriodBoundary": null,
          "QuoteLineGroupId": "1C9xx0000004CCGCAB"
        }
      }
    ]
  }
}
```

JSON example to delete a product

This is a sample request to delete a product from the current and subsequent ramp segments.

```
{
  "pricingPref": "System",
  "groupRampAction": "DeleteProducts",
  "graph": {
    "graphId": "updateQuote",
    "records": [
      {
        "referenceId": "0Q0SG000000L5r70AC",
        "record": {
          "attributes": {
            "type": "Quote",
            "method": "PATCH",
            "id": "0Q0SG000000L5r70AC"
          }
        }
      },
      {
        "referenceId": "0QLSG000000WuTh4AK",
        "record": {
          "attributes": {
            "type": "QuoteLineItem",
            "method": "DELETE",
```

```
            "id": "0QLSG000000WuTh4AK"
          }
        }
      },
      {
        "referenceId": "0QLSG000000WuTh4BK",
        "record": {
          "attributes": {
            "type": "QuoteLineItem",
            "method": "DELETE",
            "id": "0QLSG000000WuTh4AK"
          }
        }
      }
    ]
  }
}
```

JSON example to delete a segment

This is a sample request to delete the first and last segment in a ramp schedule. The API throws an error if the specified segment isn't the first and last segment, ensuring there are no gaps between quote line items in different ramp segments.

```
{
  "pricingPref": "System",
  "groupRampAction": "DeleteSegment",
  "graph": {
    "graphId": "updateQuote",
    "records": [
      {
        "referenceId": "0Q0xx0000004CfICAU",
        "record": {
          "attributes": {
            "type": "Quote",
            "method": "PATCH",
            "id": "0Q0xx0000004CfICAU"
          }
        }
      },
      {
        "referenceId": "1C9xx0000004FjcCAE",
        "record": {
          "attributes": {
            "type": "QuoteLineGroup",
            "method": "DELETE",
            "id": "1C9xx0000004FjcCAE",
            "action": "DeleteGroup"
          }
        }
      }
    ]
  }
}
```

JSON example to remove a segment from a ramp schedule

This is a sample request to remove the first or last ramp segment in a ramp schedule. This request removes the ramp-specific fields from a quote line group such as `IsRamped` and `SegmentType`. Additionally, this request removes the `RampIdentifier` and `SegmentIdentifier` fields from a quote line item.

```
{
  "pricingPref": "System",
  "groupRampAction": "ConvertToNonRampedGroup",
  "graph": {
    "graphId": "updateQuote",
    "records": [
      {
        "referenceId": "0Q0xx0000004CfICAU",
        "record": {
          "attributes": {
            "type": "Quote",
            "method": "PATCH",
            "id": "0Q0xx0000004CfICAU"
          }
        }
      },
      {
        "referenceId": "1C9xx0000004FjcCAE",
        "record": {
          "attributes": {
            "type": "QuoteLineGroup",
            "method": "PATCH",
            "id": "1C9xx0000004FjcCAE"
          }
        }
      }
    ]
  }
}
```

Sales Transaction Operand Input

Input representation of a filter operand for reading sales transaction context data.

JSON example

```
{
  "operands": [
    {
      "value": "Draft",
      "type": "STRING"
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`value`	String	Operand value used by the filter condition.	Required	67.0
`type`	String	Data type of the operand value. Valid values are: • `STRING` • `NUMBER` • `BOOLEAN` • `DATETIME` • `DATE`	Required	67.0

Supplemental Transaction Input

Input representation of the details of the request to create a supplemental order.

JSON example

This sample creates a supplemental order, which is a clone of the original order. The supplemental order is related to the original order.

```
{
  "relatedSalesTransactionId": "801S70000001VKgIAM"
}
```

This sample overrides a field value of an order line item to supplement the order item with ID value as `802SG000003vZ15YAE`.

```
{
    "relatedSalesTransactionId": "801S70000001VKgIAM",
    "pricingPref": "System",
    "supplementalGraph": {
        "graphId": "1",
        "records": [
            {
                "referenceId": "refOrder",
                "record": {
                    "attributes": {
                        "type": "Order",
                        "method": "PATCH",
                        "id": "801S70000001VKgIAM"
                    },
                    "EffectiveDate": "2025-03-01",
                    "QuoteId": "0Q0xx0000004DQ4CAM"
                }
            },
            {
                "referenceId": "refOrderItem",
                "record": {
                    "attributes": {
```

```
                    "type": "OrderItem",
                    "method": "PATCH",
                    "id": "802SG000003vZ15YAE"
                },
                "QuoteLineItemId": "0Q0xx0000004E2mYLK"
            }
        }
    ]
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`pricingPref`	String	Pricing preference for this supplemental transaction or order ingestion process. Valid values are: • `Force`—Enforces pricing during the creation of sales transactions. • `Skip`—Skips pricing during the creation of sales transactions. • `System`—Determines whether a pricing calculation is required. If `pricingPref` value is defined as either `Force` or `System`, the supplemental order can have a different pricing from the original order.	Optional	64.0
`relatedSalesTransactionId`	String	Related or the original sales transaction upon which a supplemental transaction is created.	Required	64.0
`supplementalGraph`	Object Graph Input	The sObject graph that represents a payload with the additional changes to be ingested. The attribute's HTTP method must be PATCH. The attribute ID must be the ID of the original order or order item that you want to supplement.	Optional	64.0

Swap Group Input

Input representation of the details of the swap groupings for swap operations.

JSON example

```
{
  "swapGroups": {
    "groups": [
      {
        "referenceId": "SWAP-001",
        "outGroup": {
          "swapAssets": [
            {
              "assetId": "02ixx0000004HOAAA2",
              "quantity": 1
            }
          ]
        },
        "inGroup": {
          "graphId": "swapRequest",
          "records": [
            {
              "referenceId": "refQuoteLine0",
              "record": {
                "attributes": {
                  "type": "QuoteLineItem",
                  "method": "POST"
                },
                "Product2Id": "01txx0000006iVlAAI",
                "PricebookEntryId": "01uxx0000008ym4AAA",
                "UnitPrice": 1049,
                "Quantity": "1",
                "StartDate": "2022-09-22"
              }
            }
          ]
        }
      }
    ]
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`inGroup`	Object	Products to include in the swap. These objects are supported. • QuoteLineItem • OrderItem	Required	66.0
`outGroup`	Swap Group[]	Assets to exclude from the swap. You can't use string values for properties, such as quantity and price.	Required	66.0

Name	Type	Description	Required or Optional	Available Version
`referenceId`	String	Reference ID of the operation.	Required	66.0

Update Ramp Deal Input

Input representation of the request to update a ramp deal.

JSON example

```
{
  "executionSettings": {
       "executePricing": true,
       "executeConfigRules": false
  },
  "addedNodes": [
    {
      "contextNodePath": [
        "4f23961a5c98806f89305e064c67b397e93f1bb8a2a7a3a80db506f1d4110ee9", // Context
 ID
        "0Q0xx0000004CPACA2", //Quote or Order ID
        "RandomUUID" // random UUID for Quote Line Item or Order Item ID
      ],
      "contextNode": {
         "Discount": 10,
         "Quantity": 5,
         "ItemSegmentName": "Year 5",
         "StartDate":"2024-09-07T00:00:00.000Z",
         "EndDate":"2024-09-07T00:00:00.000Z"
      }
    }
  ],
  "updatedNodes": [
     {
      "contextNodePath": [
        "4f23961a5c98806f89305e064c67b397e93f1bb8a2a7a3a80db506f1d4110ee9", // Context
 ID
        "0Q0xx0000004CPACA2", //Quote or Order ID
        "0QLxx0000004CfIGAU" // Quote Line ID or Order Line ID to update
      ],
      "contextNode": {
          "Discount": 10,
          "Quantity": 5
      }
    }
  ],
  "deletedNodes": [
    {
      "contextNodePath": [
        "4f23961a5c98806f89305e064c67b397e93f1bb8a2a7a3a80db506f1d4110ee9",
        "0Q0xx0000004CPACA2",
        "0QLxx0000004CfIGAU" // Quote Line Item ID to delete
      ]
```

```
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`addedNodes`	Context Node Input[]	Details of the nodes to be added.	Required	62.0
`deletedNodes`	Context Node Input[]	Details of the nodes to be deleted.	Required	62.0
`execution Settings`	Execution Settings Input[]	Settings to run the pricing or configuration rules.	Optional	62.0
`updatedNodes`	Context Node Input[]	Details of the nodes to be updated.	Required	62.0

Usage-Based Product Input

Understand the sample request structure to specify and manage usage-based products within a sales transaction.

JSON example to create a quote record with default rates and default grants

The example shows a sample request to creates these records with these updates.

- `Quote`—Creates the main quote record with default rates and default grants.
- `Quote Line Item`—Adds a product to the quote.
- `Rate Card Entry`—Links default pricing from rate card. When `NegotiatedRate` property value is null, the rate card price is used.
- `Usage Grant`—Allocates all default grants.

```
{
  "pricingPref": "system",
  "configurationPref": {
    "configurationMethod": "skip",
    "configurationOptions": {
      "validateProductCatalog": false,
      "validateAmendRenewCancel": false,
      "executeConfigurationRules": false,
      "addDefaultConfiguration": false
    }
  },
  "graph": {
    "graphId": "graphId",
    "records": [
      {
        "referenceId": "refQuote",
        "record": {
          "attributes": {
            "type": "Quote",
```

```
            "method": "POST"
          },
          "Name": "Q-2024-001",
          "QuoteAccountId": "001xx000003DHP0AAO",
          "OpportunityId": "006xx000004TmiGAAS",
          "Status": "Draft",
          "Pricebook2Id": "01sxx0000001SGEAA2"
        }
      },
      {
        "referenceId": "refQuoteLineItem",
        "record": {
          "attributes": {
            "type": "QuoteLineItem",
            "method": "POST"
          },
          "QuoteId": "@{refQuote.id}",
          "Product2Id": "01txx0000001SGEAA2",
          "StartDate": "2024-10-29",
          "PeriodBoundary": "Anniversary",
          "EndDate": "2027-03-01",
          "PricebookEntryId": "01uxx0000001SGEAA2",
          "UnitPrice": 100,
          "Quantity": 1
        }
      },
      {
        "referenceId": "refRateCardEntry",
        "record": {
          "attributes": {
            "type": "QuoteLineRateCardEntry",
            "method": "POST"
          },
          "RateCardEntryId": "1CJxx0000004CXEGA2",
          "NegotiatedRate": null,
          "QuoteLineItemId": "@{refQuoteLineItem.id}"
        }
      }
    ]
  }
}
```

JSON example to specify negotiated grants

This example shows a sample response to create a QuotLineItmUseRsrcGrant record that links a quote line item to a product usage grant for usage-based products. This request performs these updates.

- Associates a usage grant with a quote line item.
- Allocates a quantity of usage resources.
- Applies a usage policy that defines how the resources can be consumed.
- Links to a base product usage grant that defines the grant structure.

```
{
  "pricingPref": "system",
  "configurationPref": {
```

```
    "configurationMethod": "skip",
    "configurationOptions": {
      "validateProductCatalog": false,
      "validateAmendRenewCancel": false,
      "executeConfigurationRules": false,
      "addDefaultConfiguration": false
    }
  },
  "graph": {
    "graphId": "graphId",
    "records": [
      {
        "referenceId": "ref1CJxx0000004CXGGA2",
        "record": {
          "attributes": {
            "type": "QuotLineItmUseRsrcGrant",
            "method": "POST"
          },
          "GrantQuantity": 4,
          "ProductUsageResourcePolicyId": "7Suxx0000004C92CAE",
          "QuoteLineItemId": "0QLxx0000004CnMGAU",
          "ProductUsageGrantId": "1BXxx0000004C92GAE"
        }
      }
    ]
  }
}
```


This example shows a sample response to update the previously created QuotLineItmUseRsrcGrant record with additional units, which is specified by using the `GrantQuantity` property. The quote line item now has 97 units of usage resource allocation.


```
{
  "pricingPref": "system",
  "configurationPref": {
    "configurationMethod": "skip",
    "configurationOptions": {
      "validateProductCatalog": false,
      "validateAmendRenewCancel": false,
      "executeConfigurationRules": false,
      "addDefaultConfiguration": false
    }
  },
  "graph": {
    "graphId": "graphId",
    "records": [
      {
        "referenceId": "ref1X6xx000000003GCAQ",
        "record": {
          "attributes": {
            "method": "PATCH",
            "type": "QuotLineItmUseRsrcGrant",
            "id": "1X6xx000000003GCAQ"
          },
          "GrantQuantity": 97
        }
```

```
      }
    ]
  }
}
```

JSON example to specify tiered pricing and usage grants

This example shows a sample request to update an existing quote with tiered pricing and usage grants. This request sets up a multi-tier rate structure and updates a usage grant quantity along with these additional updates.

- Creates a base rate card entry with a negotiated rate of 10. Sets the base price for the quote line item.
- Creates a second rate card entry for tiered pricing without any negotiated rate. The default rate card is used. This rate card entry is the base for tier adjustments.
- Creates a tier adjustment for quantities 0 through 50 and adds $5 adjustment to the base rate.
- Creates a tier adjustment for quantities 50 through 100 and adds an adjustment of $10.
- Updates an existing usage resource grant record to set grant quantity to 150 units.

For example, if an order is placed for 75 units, the base rate is $10. As tier 2 is applicable for this order, an adjustment of $10 is applicable. The final price per unit is $20 with a total of $1500 for 75 units.

```
{
  "pricingPref": "system",
  "configurationPref": {
    "configurationMethod": "skip",
    "configurationOptions": {
      "validateProductCatalog": false,
      "validateAmendRenewCancel": false,
      "executeConfigurationRules": false,
      "addDefaultConfiguration": false
    }
  },
  "graph": {
    "graphId": "graphId",
    "records": [
      {
        "referenceId": "refLineItemParent",
        "record": {
          "attributes": {
            "type": "Quote",
            "method": "PATCH",
            "id": "0Q0xx0000004CXEGA2"
          }
        }
      },
      {
        "referenceId": "ref1",
        "record": {
          "attributes": {
            "type": "QuoteLineRateCardEntry",
            "method": "POST"
          },
          "RateCardEntryId": "1CJxx0000004CXEGA2",
          "NegotiatedRate": 10,
          "QuoteLineItemId": "0QLxx0000004Eh6GAE"
        }
```

```
    },
    {
      "referenceId": "refTierRateCardEntry0",
      "record": {
        "attributes": {
          "type": "QuoteLineRateCardEntry",
          "method": "POST"
        },
        "RateCardEntryId": "1CJxx0000004CXFGA2",
        "QuoteLineItemId": "0QLxx0000004Eh6GAE"
      }
    },
    {
      "referenceId": "refTier0",
      "record": {
        "attributes": {
          "type": "QuoteLineRateAdjustment",
          "method": "POST"
        },
        "Name": "Tier 1",
        "AdjustmentType": "Amount",
        "AdjustmentValue": 5,
        "LowerBound": 0,
        "UpperBound": 50,
        "QuoteLineRateCardEntryId": "@{refTierRateCardEntry0.id}"
      }
    },
    {
      "referenceId": "refTier1",
      "record": {
        "attributes": {
          "type": "QuoteLineRateAdjustment",
          "method": "POST"
        },
        "Name": "Tier 2",
        "AdjustmentType": "Amount",
        "AdjustmentValue": 10,
        "LowerBound": 50,
        "UpperBound": 100,
        "QuoteLineRateCardEntryId": "@{refTierRateCardEntry0.id}"
      }
    },
    {
      "referenceId": "refTier2",
      "record": {
        "attributes": {
          "type": "QuoteLineRateAdjustment",
          "method": "POST"
        },
        "Name": "Tier 3",
        "AdjustmentType": "Amount",
        "AdjustmentValue": 15,
        "LowerBound": 100,
        "UpperBound": 150,
```

```
            "QuoteLineRateCardEntryId": "@{refTierRateCardEntry0.id}"
          }
        },
        {
          "referenceId": "refGt2xzdUP",
          "record": {
            "attributes": {
              "type": "QuotLineItmUseRsrcGrant",
              "method": "PATCH",
              "id": "1X6xx000000003GCAQ"
            },
            "GrantQuantity": 150
          }
        }
      ]
    }
}
```

Response Bodies

Learn more about the available response bodies.

Amendment

Output representation of the details of an amendment record.

ARC Base Error

Output representation of the error response related to the amendment, renewal, or cancellation of assets.

Cancellation

Output representation of the details of a cancellation record.

Clone Sales Transaction

Output representation for the result of cloning records within a sales transaction.

Clone Sales Transaction Error Response

Output representation of the errors that occur during the clone sales transaction operation.

Eligible Promotions Response

Output representation of the details of the eligible promotions.

Initiate Downgrade Response

Output representation of the request to initiate a downgrade action.

Initiate Swap Response

Output representation of the request to initiate a swap action.

Initiate Upgrade Response

Output representation of the request to initiate an upgrade action.

Instant Pricing

Output representation containing the results of the instant pricing request.

Object Reference

Output representation of an sObject with a reference ID along with any potential error.

Place Order Error Response

Output representation of the error response for the place order request.

Place Order Response

Output representation of the request to create or update an order.

Place Quote Error Response

Output representation of the error responses of a place quote request.

Place Quote

Output representation of the request to create or update a quote.

Sales Transaction Async Error

Output representation of the details of errors encountered during the async processing of the Place Sales Transaction API request.

Promotion Coupon

Output representation of the details of a coupon that's eligible for the recommended promotion.

Promotion Coupon Availability

Output representation of the details of the coupon that's eligible for the promotion. Additionally, this representation specifies the reason for any coupon ineligibility.

Promotion Details Response

Output representation of the eligible promotion and its details. This representation includes details of any coupons, eligibility rules, and terms and conditions.

Promotion Limit

Output representation of the details of the promotion limit of an eligible promotion.

Promotion Reward Details

Output representation of the details of the rewards of an eligible promotion rule.

Promotion Rules List

Output representation of the details of the rules of an eligible promotion.

Ramp Deal Service Error Response

Output representation of the details of errors encountered during the processing of the API request.

Ramp Deal Service

Output representation of the details of a created, updated, or deleted ramp deal.

Read Sales Transaction

Output representation of the request to read a sales transaction.

Read Sales Transaction Records

Output representation of the details of a map of keys and associated values. The keys are record type names, such as a Quote or QuoteLineItem, and values are lists of records of that type.

Renewal

Output representation of the details of a renewal record.

Sales Transaction

Output representation of the request to create a sales transaction.

Sales Transaction Context

Output representation of the context details that are associated with a sales transaction.

Sales Transaction Error Response

Output representation of the error details associated with the API request.

Sales Transaction Record

Generic output representation for any sales transaction record type.

Supplemental Transaction Error Response

Output representation of the error details associated with the Place Supplemental Transaction API.

Supplemental Transaction

Output representation of the details of the created supplemental order.

Amendment

Output representation of the details of an amendment record.

JSON example

```
{
  "amendmenRecordId": "0Q0xx0000004NsSCAU",
  "errors": [
    {
      "errorCode": "REQUIRED_FIELD_MISSING",
      "errorMessage": "Specify a value for quantityChange, and try again."
    }
  ],
  "requestId": "16Pxx0000004NIy",
  "success": true
}
```

Properties

Property Name	Type	Description	Filter Group and Version	Available Version
amendment RecordId	String	ID of the amendment record that's created for a quote or an order.	Small, 62.0	62.0
errors	ARC Base Error on page 1509[]	Error responses if the creation of an amendment record fails.	Small, 62.0	62.0
requestId	String	Request ID that's used to track an async request.	Small, 62.0	62.0
success	Boolean	Indicates whether the API request is successful (`true`) or not (`false`).	Small, 62.0	62.0

ARC Base Error

Output representation of the error response related to the amendment, renewal, or cancellation of assets.

```
  "errors": [
      {
          "errorCode": "REQUIRED_FIELD_MISSING",
          "errorMessage": "Specify a value for quantityChange, and try again."
      }
  ]
```

Property Name	Type	Description	Filter Group and Version	Available Version
errorCode	String	Code for the resultant error.	Big, 62.0	62.0
errorMessage	String	Error message for the resultant error.	Big, 62.0	62.0

Cancellation

Output representation of the details of a cancellation record.

JSON example

```
{
  "cancellationRecordId": "0Q0xx0000004NsSCAU",
  "errors": [
    {
      "errorCode": "REQUIRED_FIELD_MISSING",
      "errorMessage": "Specify a value for quantityChange, and try again."
    }
  ],
  "requestId": "16Pxx0000004NIy",
  "success": true
}
```

Properties

Property Name	Type	Description	Filter Group and Version	Available Version
cancellationRecordId	String	ID of the cancellation record that's created for a quote or an order.	Small, 62.0	62.0
errors	ARC Base Error on page 1509[]	Error responses if the creation of a cancellation record fails.	Small, 62.0	62.0
requestId	String	Request ID that's used to track the async request.	Small, 62.0	62.0
success	Boolean	Indicates whether the API request is successful (true) or not (false).	Small, 62.0	62.0

Clone Sales Transaction

Output representation for the result of cloning records within a sales transaction.

JSON example

This example shows a sample of a successful response.

```
{
    "requestId": "9356bcbf04f06e22360a09807c13e1d4e395",
    "salesTransactionId": "0Q0SG000000ACxf0AG",
    "errors": [],
```

```
    "success": true
}
```

This example shows a sample error response.

```
{
    "requestId": "9356bcbf04f06e22360a09807c13e1d4e395",
    "salesTransactionId": "0Q0SG000000ACxf0AG",
    "errors": [
        {
            "errorCode": "INVALID_API_INPUT",
            "message": "Specify only one record",
            "referenceId": "0QLxx0000004CBYGA2"
        }
    ],
    "success": false
}
```

Properties

Name	Type	Description	Available Version
`requestId`	String	Request ID of the process that can be used to query the async status.	64.0
`salesTransactionId`	String	ID of the quote line item, order item, quote line group, or order item group record.	64.0
`success`	Boolean	Indicates whether the synchronous part of the processing is successful (`true`) or not (`false`).	64.0
`errors`	Clone Sales Transaction Error Response[]	List of errors encountered during synchronous processing.	64.0

Clone Sales Transaction Error Response

Output representation of the errors that occur during the clone sales transaction operation.

JSON example

```
{
  "errors": [
    {
      "errorCode": "INVALID_API_INPUT",
      "message": "Specify only one record",
      "referenceId": "0QLxx0000004CBYGA2"
    }
  ]
}
```

Properties

Name	Type	Description	Available Version
`errorCode`	String	Code associated with the error.	64.0

Name	Type	Description	Available Version
message	String	Message associated with the error.	64.0
referenceId	String	Reference ID associated with the error.	64.0

Eligible Promotions Response

Output representation of the details of the eligible promotions.

JSON example

```
{
  "eligiblePromotions": [
    {
      "additionalPromotionFields": {},
      "couponDetails": {
        "coupon": {
          "couponCode": "COUPON_002",
          "endDateTime": null,
          "startDateTime": "2025-10-08T19:00:00.000Z",
          "status": "Active"
        },
        "couponAvailabilityMessage": null
      },
      "currencyIsoCode": "USD",
      "description": "15% on Accessories",
      "displayName": "Comms Manual Promotion 15% on Accessories",
      "endDateTime": null,
      "isAutomatic": false,
      "priorityNumber": null,
      "promotionCode": null,
      "promotionEligibleRules": [
        {
          "ruleName": "CommsAutoPromotionRule",
          "rulePriority": 1,
          "ruleRewards": [
            {
              "rewardDetails": {
                "discountLevel": "Line",
                "discountType": "PercentageOff",
                "discountValue": "10.0",
                "discountAppliedAt": "LineItem",
                "discountCategory": "Accessories"
              },
              "rewardType": "ProvideDiscount"
            }
          ]
        }
      ],
      "promotionId": "0c8DX00000001UeYAI",
      "promotionLimits": [],
      "startDateTime": "2025-07-01T17:02:00.000Z",
      "termsAndCondition": "&lt;p&gt;test&lt;/p&gt;"
```

```
    },
    {
      "additionalPromotionFields": {},
      "couponDetails": {
        "coupon": {
          "couponCode": "COUPON_001",
          "endDateTime": null,
          "startDateTime": null,
          "status": "Active"
        },
        "couponAvailabilityMessage": null
      },
      "currencyIsoCode": "USD",
      "description": "Promo_Printer_Child",
      "displayName": "Promo_Printer_Child",
      "endDateTime": null,
      "isAutomatic": false,
      "priorityNumber": null,
      "promotionCode": null,
      "promotionEligibleRules": [
        {
          "ruleName": "Rule001",
          "rulePriority": 1,
          "ruleRewards": [
            {
              "rewardDetails": {
                "discountLevel": "Line",
                "discountProduct": "Printer Paper",
                "discountType": "AmountOff",
                "discountValue": "2.0",
                "discountAppliedAt": "LineItem"
              },
              "rewardType": "ProvideDiscount"
            }
          ]
        }
      ],
      "promotionId": "0c8DX00000001h5YAA",
      "promotionLimits": [],
      "startDateTime": "2025-11-03T12:44:00.000Z",
      "termsAndCondition": "&lt;p&gt;Promo_Printer_Child&lt;/p&gt;"
    }
  ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
eligiblePromotions	Promotion Details Response	List of eligible promotions for the specified line items.	Big, 66.0	66.0

Initiate Downgrade Response

Output representation of the request to initiate a downgrade action.

JSON example

```
{
  "salesTransactionId": "0Q09Q000005IkEPSA0"
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
salesTransactionId	String	ID of the sales transaction that's created by the downgrade action.	Small, 66.0	66.0

Initiate Swap Response

Output representation of the request to initiate a swap action.

JSON example

```
{
  "salesTransactionId": "0Q09Q000005IkEPSA0"
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
salesTransactionId	String	ID of the sales transaction that's created by the swap action.	Small, 66.0	66.0

Initiate Upgrade Response

Output representation of the request to initiate an upgrade action.

JSON example

```
{
  "salesTransactionId": "0Q09Q000005IkEPSA0"
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
salesTransactionId	String	ID of the sales transaction that's created by the upgrade action.	Small, 66.0	66.0

Instant Pricing

Output representation containing the results of the instant pricing request.

Sample Response


```
{
  "contextId": "0000000p18dq18g0025177552281982954d1f35410374495bc923a2319a1247c",
  "correlationId": "7847127122596",
  "records": [
    {
      "error": {},
      "record": {
        "NetUnitPrice": 900.8972677417262,
        "Discount": null,
        "EndQuantity": 5,
        "NetTotalPrice": 3603.5890709669047,
        "AggregatedQuantity": 4,
        "TotalLineAmount": 3603.5890709669047,
        "Quantity": 4,
        "attributes": {
          "type": "QuoteLineItem"
        },
        "ListPrice": 900.8972677417262
      },
      "referenceId": "0QLLT0000018DY14AM"
    },
    {
      "error": {},
      "record": {
        "NetUnitPrice": 720.3738964743471,
        "Discount": null,
        "EndQuantity": 5,
        "NetTotalPrice": 2881.4955858973885,
        "AggregatedQuantity": 4,
        "TotalLineAmount": 2881.4955858973885,
        "Quantity": 4,
        "attributes": {
          "type": "QuoteLineItem"
        },
        "ListPrice": 720.3738964743471
      },
      "referenceId": "0QLLT0000018DYr4AM"
    },
    {
      "error": {},
      "record": {
        "NetUnitPrice": 218.36199390885784,
        "Discount": null,
        "EndQuantity": 5,
        "NetTotalPrice": 873.4479756354314,
        "AggregatedQuantity": 4,
        "TotalLineAmount": 873.4479756354314,
        "Quantity": 4,
        "attributes": {
          "type": "QuoteLineItem"
        },
        "ListPrice": 218.36199390885784
```

```
      },
      "referenceId": "0QLLT0000018DYG4A2"
    },
    {
      "error": {},
      "record": {
        "AccountId": "001LT000011SA6bYAG",
        "TotalPrice": 406014.84549259685,
        "attributes": {
          "type": "Quote"
        },
        "Discount": 0,
        "psmSummaryFields": {
          "evergreen##months##1": {
            "aggregatedTotal": 140349.76846114534,
            "pricingTerm": 1,
            "pricingTermUnit": "Months",
            "sellingModelType": "Evergreen"
          },
          "onetime": {
            "aggregatedTotal": 128927.94222255558,
            "sellingModelType": "OneTime"
          },
          "termdefined##annual##1": {
            "aggregatedTotal": 136737.13480889596,
            "pricingTerm": 1,
            "pricingTermUnit": "Annual",
            "sellingModelType": "TermDefined"
          }
        },
        "Subtotal": 406014.84549259685
      },
      "referenceId": "0Q0LT000000Q1LR0A0"
    }
  ],
  "success": true
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`contextId`	String	Context ID returned by the context service.	Small, 59.0	59.0
`correlationId`	String	Client-generated ID for tracking multiple related API calls.	Small, 59.0	59.0
`records`	Object Reference[]	List of records related to pricing results.	Small, 59.0	59.0
`success`	Boolean	Indicates whether the fetching of instant pricing is successful (`true`) or not (`false`).	Small, 59.0	59.0

Object Reference

Output representation of an sObject with a reference ID along with any potential error.

Sample Response

```
{
  "referenceid": "refQuote",
  "record": {
    "attributes": {
      "type": "Quote",
      "method": "POST"
    },
    "quantity": "2"
  },
  "error": {
    "errorCode": "INVALID_API_INPUT",
    "message": "Reference Id format is irrelevant."
  }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`referenceId`	String	ID that identifies the specific Salesforce object that's returned in the API response.	Small, 59.0	59.0
`record`	Map<String, Object>	The sObject record data represented as a map of attribute names to their values.	Small, 59.0	59.0
`error`	[illegible]	Detailed information about any error associated with the sObject in the response.	Small, 59.0	59.0

Place Order Error Response

Output representation of the error response for the place order request.

Property Name	Type	Description	Filter Group and Version	Available Version
`errorCode`	String	Code representing the type of error encountered during the place order create request.	Small, 60.0	60.0
`message`	String	Message stating the reason for the error, if any.	Small, 60.0	60.0
`referenceId`	String	Reference ID associated with the specific error instance for tracking and reference purposes.	Small, 60.0	60.0

Place Order Response

Output representation of the request to create or update an order.

JSON example


```
{
  "requestId": "16PRM0000004DBq",
  "orderId": "801S70000001VKgIAM",
  "success": true,
  "errors": [],
  "statusURL": "/services/data/vXX.X/sobjects/AsyncOperationTracker/16PRM0000004DBq"
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`errors`	Place OrderError Response on page 1517[]	List of errors encountered during the synchronous processing.	Small, 60.0	60.0
`orderId`	String	ID of the order created after a successful request.	Small, 60.0	60.0
`requestId`	String	Request ID of the process to query asynchronous status of the place order API.	Small, 60.0	60.0
`statusURL`	String	Asynchronous status URL of the request, if available.	Small, 60.0	60.0
`success`	Boolean	Indicates whether the synchronous part of the processing is successful (`true`) or not (`false`).	Small, 60.0	60.0

Place Quote Error Response

Output representation of the error responses of a place quote request.

JSON Example


```
{
 "errorCode": "INVALID_API_INPUT",
 "message": "Include record type and method in the request and try again.",
 "referenceId": "refQuoteItem2"
 }
```


Property Name	Type	Description	Filter Group and Version	Available Version
`errorCode`	String	Error code representing the type of error encountered in the create place quote request.	Small, 60.0	60.0
`message`	String	Message stating the reason for the error, if any.	Small, 60.0	60.0

Property Name	Type	Description	Filter Group and Version	Available Version
`referenceId`	String	Reference ID associated with the specific error instance for tracking and reference purposes.	Small, 60.0	60.0

Place Quote

Output representation of the request to create or update a quote.

JSON Example

This example shows a sample response of the place quote request.

```
{
    "quoteId": "0Q0xx0000004E2mCAE",
    "requestIdentifier": "95Txx0000004Cx2",
    "responseError": [],
   "statusURL": "/services/data/v60.0/sobjects/RevenueAsyncOperation/95Txx0000004Cx2EAE",

    "success": true
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`quoteId`	String	ID of the quote created after a successful request.	Small, 60.0	60.0
`request Identifier`	String	Unique request identifier that can be used to poll the async request.	Big, 60.0	60.0
`responseError`	Place Quote Error Response []	List of errors encountered during the synchronous processing.	Small, 60.0	60.0
`statusURL`	String	Asynchronous status URL to track the operation, if available.	Big, 60.0	60.0
`success`	Boolean	Indicates whether the synchronous part of the processing is successful (`true`) or not (`false`).	Small, 60.0	60.0

Sales Transaction Async Error

Output representation of the details of errors encountered during the async processing of the Place Sales Transaction API request.

JSON example

```
{
  "errors": [
    {
      "errorCode": "FIELD_CUSTOM_VALIDATION_EXCEPTION",
      "message": "Quantity cannot exceed 10",
```

```
      "referenceId": "refQuoteLineItem2"
    }
  ],
  "jobStatus": "Completed",
  "retryablePayload": {
    "pricingPref": "System",
    "configurationPref": {
      "configurationMethod": "System",
      "configurationOptions": {
        "validateProductCatalog": true,
        "validateAmendRenewCancel": true,
        "executeConfigurationRules": true,
        "addDefaultConfiguration": true
      }
    },
    "graph": {
      "graphId": "1",
      "records": [
        {
          "referenceId": "refQuoteLineItem2",
          "record": {
            "attributes": {
              "type": "QuoteLineItem",
              "method": "POST"
            },
            "QuoteId": "000xx0000004CNYCA2",
            "PricebookEntryId": "01uxx0000009CL2",
            "Product2Id": "01txx0000006uu2",
            "Quantity": 15,
            "UnitPrice": 100
          }
        },
        {
          "referenceId": "refQuoteLineItem3",
          "record": {
            "attributes": {
              "type": "QuoteLineItem",
              "method": "POST"
            },
            "QuoteId": "000xx0000004CNYCA2",
            "PricebookEntryId": "01uxx0000009CL3",
            "Product2Id": "01txx0000006uu3",
            "Quantity": 20,
            "UnitPrice": 100
          }
        },
        {
          "referenceId": "refQuoteLinePriceAdjustment1",
          "record": {
            "attributes": {
              "type": "QuoteLinePriceAdjustment",
              "method": "POST"
            },
            "QuoteLineItemId": "@{refQuoteLineItem3.id}",
```

```
                "Name": "Some Promo",
                "Amount": -25
              }
            }
          ]
        }
      },
      "rolledBackReferenceIds": [
        "refQuoteLineItem3",
        "refQuoteLinePriceAdjustment1"
      ]
    }
```

Property Name	Type	Description	Filter Group and Version	Available Version
`errors`	Place Sales Transaction Error[]	List of async errors from Place Sales Transaction API. The details include a reference ID for the failed subrequest, an error code, and a detailed message.	Small, 66.0	66.0
`jobStatus`	String	Status of the async tracker ID. Error results are returned after the job is in a `Completed` status.	Small, 66.0	66.0
`retryablePayload`	Sales Transaction Input	Subset of the original Place Sales Transaction API payload errors that can be used for retry attempts. Reference IDs that are saved are replaced by their corresponding record IDs.This property is included in the response only when the `includeRetryablePayload` query parameter is set to `true`.	Small, 66.0	66.0
`rolledBack ReferenceIds`	String[]	List of rolled-back reference IDs for subrequests due to failure of a subrequest in the same transaction.	Small, 66.0	66.0

Promotion Coupon

Output representation of the details of a coupon that's eligible for the recommended promotion.

JSON example

```
{
  "coupon": {
    "couponCode": "COUPON_002",
    "endDateTime": null,
    "startDateTime": "2025-10-08T19:00:00.000Z",
    "status": "Active"
  }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
couponCode	String	Unique code of the coupon.	Big, 66.0	66.0
endDateTime	String	End date and time of the coupon.	Big, 66.0	66.0
startDateTime	String	Start date and time of the coupon.	Big, 66.0	66.0
status	String	Status of the coupon.	Big, 66.0	66.0

Promotion Coupon Availability

Output representation of the details of the coupon that's eligible for the promotion. Additionally, this representation specifies the reason for any coupon ineligibility.

JSON example

```
{
  "couponDetails": {
    "coupon": {
      "couponCode": "COUPON_002",
      "endDateTime": null,
      "startDateTime": "2025-10-08T19:00:00.000Z",
      "status": "Active"
    },
    "couponAvailabilityMessage": null
  }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
coupon	Promotion Coupon[]	Coupon that's eligible for the customers' cart.	Big, 66.0	66.0
couponAvailability Message	String	Specifies the reason for coupon ineligibility for the promotion. Valid values are: • MultipleActiveCoupons • NoActiveCoupons • SingleActiveCouponLimitReached	Big, 66.0	66.0

Promotion Details Response

Output representation of the eligible promotion and its details. This representation includes details of any coupons, eligibility rules, and terms and conditions.

JSON example

```
{
  "eligiblePromotions": [
    {
```

```
      "additionalPromotionFields": {},
      "couponDetails": {
        "coupon": {
          "couponCode": "COUPON_002",
          "endDateTime": null,
          "startDateTime": "2025-10-08T19:00:00.000Z",
          "status": "Active"
        },
        "couponAvailabilityMessage": null
      },
      "currencyIsoCode": "USD",
      "description": "15% on Accessories",
      "displayName": "Comms Manual Promotion 15% on Accessories",
      "endDateTime": null,
      "isAutomatic": false,
      "priorityNumber": null,
      "promotionCode": null,
      "promotionEligibleRules": [
        {
          "ruleName": "CommsAutoPromotionRule",
          "rulePriority": 1,
          "ruleRewards": [
            {
              "rewardDetails": {
                "discountLevel": "Line",
                "discountType": "PercentageOff",
                "discountValue": "10.0",
                "discountAppliedAt": "LineItem",
                "discountCategory": "Accessories"
              },
              "rewardType": "ProvideDiscount"
            }
          ]
        }
      ],
      "promotionId": "0c8DX00000001UeYAI",
      "promotionLimits": [],
      "startDateTime": "2025-07-01T17:02:00.000Z",
      "termsAndCondition": "&lt;p&gt;test&lt;/p&gt;"
    },
    {
      "additionalPromotionFields": {},
      "couponDetails": {
        "coupon": {
          "couponCode": "COUPON_001",
          "endDateTime": null,
          "startDateTime": null,
          "status": "Active"
        },
        "couponAvailabilityMessage": null
      },
      "currencyIsoCode": "USD",
      "description": "Promo_Printer_Child",
      "displayName": "Promo_Printer_Child",
```

```
        "endDateTime": null,
        "isAutomatic": false,
        "priorityNumber": null,
        "promotionCode": null,
        "promotionEligibleRules": [
          {
            "ruleName": "Rule001",
            "rulePriority": 1,
            "ruleRewards": [
              {
                "rewardDetails": {
                  "discountLevel": "Line",
                  "discountProduct": "Printer Paper",
                  "discountType": "AmountOff",
                  "discountValue": "2.0",
                  "discountAppliedAt": "LineItem"
                },
                "rewardType": "ProvideDiscount"
              }
            ]
          }
        ],
        "promotionId": "0c8DX00000001h5YAA",
        "promotionLimits": [],
        "startDateTime": "2025-11-03T12:44:00.000Z",
        "termsAndCondition": "&lt;p&gt;Promo_Printer_Child&lt;/p&gt;"
      }
    ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
additional PromotionFields	Map<String, Object>	Values of the additional promotion fields that are mapped in the context definition.	Big, 66.0	66.0
couponDetails	Promotion Coupon Availability[]	Specifies the details of the coupon eligible for the promotion or specifies the reason why there isn't an eligible coupon for the promotion.	Big, 66.0	66.0
currencyIso Code	String	Three-letter ISO 4217 code of the monetary currency associated with the promotion.	Big, 66.0	66.0
description	String	Description of the promotion.	Big, 66.0	66.0
displayName	String	Display name of the eligible promotion.	Big, 66.0	66.0
endDateTime	String	End date and time of the promotion in the ISO 8601 date format.	Big, 60.0	66.0
isAutomatic	Boolean	Indicates whether the promotion is automatically applied for eligible carts (`true`) or not (`false`).	Big, 66.0	66.0

Property Name	Type	Description	Filter Group and Version	Available Version
`priorityNumber`	Integer	Priority number of the eligible promotion for promotion ordering.	Big, 66.0	66.0
`promotionCode`	String	Code of the eligible promotion.	Big, 66.0	66.0
`promotionEligibleRules`	Promotion Rules List[]	List of promotion rules that are eligible for the customer's cart.	Big, 66.0	66.0
`promotionId`	String	ID of the eligible promotion.	Big, 66.0	66.0
`promotionLimits`	Promotion Limit[]	List of the promotion's cart level and cart item level limits.	Big, 66.0	66.0
`startDateTime`	String	Start date and time of the promotion in the ISO 8601 date format.	Big, 66.0	66.0
`termsAndCondition`	String	Terms and conditions of the promotion.	Big, 66.0	66.0

Promotion Limit

Output representation of the details of the promotion limit of an eligible promotion.

Property Name	Type	Description	Filter Group and Version	Available Version
`limitCurrentAttainment`	Double	Current attainment of the promotion's limit value.	Big, 66.0	66.0
`limitTarget`	Double	Value beyond which the promotion can't be applied to customer carts.	Big, 66.0	66.0
`limitType`	String	Limit type for the promotion. Valid values are: • `CartItemCount`—Specifes the limit type that represents the maximum number of cart items the promotion can applied for. • `OrderCount`—Specifies the limit type that represents the maximum number of carts the promotion can applied for.	Big, 66.0	66.0

Promotion Reward Details

Output representation of the details of the rewards of an eligible promotion rule.

JSON example

```
{
  "ruleRewards": [
    {
      "rewardDetails": {
        "discountLevel": "Line",
        "discountProduct": "Printer Paper",
        "discountType": "AmountOff",
        "discountValue": "2.0",
        "discountAppliedAt": "LineItem"
      },
      "rewardType": "ProvideDiscount"
    }
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`rewardDetails`	Map<String, Object>	Details of the reward offered by the promotion rule.	Big, 66.0	66.0
`rewardType`	String	Reward type of the promotion. Valid values are: • `AssignBadge` • `AssignGame` • `CreditFixedPoints` • `CreditMultiplierPoints` • `GiveFreeProduct` • `IssueVoucher` • `ProvideDiscount`	Big, 66.0	66.0

Promotion Rules List

Output representation of the details of the rules of an eligible promotion.

JSON example

```
{
  "promotionEligibleRules": [
    {
      "ruleName": "Rule001",
      "rulePriority": 1,
      "ruleRewards": [
        {
          "rewardDetails": {
            "discountLevel": "Line",
            "discountProduct": "Printer Paper",
            "discountType": "AmountOff",
```

```
            "discountValue": "2.0",
            "discountAppliedAt": "LineItem"
          },
          "rewardType": "ProvideDiscount"
        }
      ]
    }
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`ruleName`	String	Rule name in the promotion which is eligible for rewards.	Big, 66.0	66.0
`rulePriority`	Integer	Rule priority in the promotion which is eligible for rewards.	Big, 66.0	66.0
`ruleRewards`	Promotion Reward Details[]	Rewards associated with the eligible rule.	Big, 66.0	66.0

Ramp Deal Service Error Response

Output representation of the details of errors encountered during the processing of the API request.

Property Name	Type	Description	Filter Group and Version	Available Version
`errorCode`	String	Error code from the API request. For example, `INVALID_INPUT_ERROR`.	Small, 62.0	62.0
`message`	String	Error message from the API request.	Small, 62.0	62.0

Ramp Deal Service

Output representation of the details of a created, updated, or deleted ramp deal.

JSON Example

This example shows the sample response for the create, update, or view ramp deal requests.

```
{
  "correlationId": "0QLDU0000002t0Z4AQ",
  "errors": [],
  "salesTransactionContext": {
    "SalesTransaction": [
      {
        "LegalEntity": null,
        "Account": null,
        "HeaderDistributionType": null,
        "BillingCity": null,
        "AccountBusinessType": null,
```

```
"HeaderDiscountType": null,
"businessObjectType": "Quote",
"QuoteAccount": null,
"EmployeeCount": null,
"SalesTransactionName": "WarrantyPriceRampAR",
"StartDate": null,
"HeaderDiscountValue": null,
"SalesTransactionType": null,
"Pricebook": "01sDU000000JGf4YAG",
"Opportunity": null,
"ShippingCountry": null,
"ShippingCity": null,
"BillingPostalCode": null,
"id": "0Q0DU0000002f3d0AA",
"BillToContact": null,
"CalculationStatus": "CompletedWithTax",
"Status": "Draft",
"LastPricedDate": null,
"Subtotal": 108.47145205479453,
"OriginalActionType": null,
"TotalAmount": 99.98,
"CurrencyIsoCode": null,
"ShippingStreet": null,
"SalesTransactionItem": [
  {
    "LegalEntity": null,
    "ProductName": "Warranty",
    "businessObjectType": "QuoteLineItem",
    "Product": "01tDU000000EsWSYA0",
    "ItemIsPrimarySegment": true,
    "ListPrice": 49.99,
    "ValidationResult": null,
    "StartDate": "2024-08-23T00:00:00.000Z",
    "ContractVolumePasId": null,
    "BillingTreatment": null,
    "PeriodBoundaryStartMonth": null,
    "SalesTransactionSourceAsset": null,
    "id": "0QLDU0000002t0Z4AQ",
    "PartnerDiscountPercent": null,
    "PriceWaterFall": "0QLDU0000002t0Z4AQ:548201414593252",
    "ItemProductRecipient": null,
    "BillingFrequency": "Annual",
    "ProductCode": "W001",
    "DerivedPricingAttribute": null,
    "TaxTreatment": "1ttDU0000001oGKYAY",
    "Subtotal": 8.491452054794523,
    "ItemRampIdentifier": "RDI5b5ce52b2db4484",
    "ItemSegmentName": "Trial",
    "PricebookEntry": "01uDU000000f4LSYAY",
    "DiscountAmount": null,
    "PricingTermCount": 0.0849315068493151,
    "NetUnitPrice": 0,
    "ItemEffectiveGrantDate": null,
    "ProductCategory": null,
```

```
    "SalesTransactionAction": null,
    "SalesTransactionActionType": null,
    "SalesTransactionItemGroup": null,
    "PeriodBoundaryDay": null,
    "LineItemDistributionType": null,
    "ProrationPolicy": "0muDU00000029ryYAA",
    "ContractDiscountType": null,
    "TransactionType": null,
    "ParentReference": null,
    "Discount": 100,
    "ProductSellingModel": "0jPDU00000029zb2AA",
    "PricingTermUnit": "Annual",
    "PricingSource": null,
    "StockKeepingUnit": null,
    "PartnerUnitPrice": null,
    "ItemTotalAdjustmentAmount": -8.491452054794523,
    "SalesTransactionItemSource": "0QLDU0000002t0Z4AQ",
    "ContractAttributePasId": null,
    "SubscriptionTerm": 1,
    "SellingModelType": "TermDefined",
    "EndQuantity": 2,
    "NetTotalPrice": 0,
    "TotalLineAmount": 8.491452054794523,
    "ItemSegmentType": "FreeTrial",
    "ProductBasedOn": null,
    "Deleted": null,
    "BillingReference": null,
    "ArePartialPeriodsAllowed": true,
    "ItemRecordedPrice": null,
    "CustomProductName": "Warranty",
    "ItemSegmentIdentifier": "SEG4380006a1c2b416",
    "SalesTransactionItemParent": "0Q0DU0000002f3d0AA",
    "Quantity": 2,
    "PeriodBoundary": "Anniversary",
    "ContractDiscountValue": null,
    "LineItemDiscountValue": null,
    "ContractId": null,
    "EndDate": "2024-09-22T00:00:00.000Z",
    "ItemGroupSummarySubtotal": null,
    "IsContracted": null,
    "UnitPrice": 49.99,
    "StartQuantity": 0,
    "ContractPrice": null,
    "TotalPrice": 0,
    "ItemPath": null,
    "LineItemDiscountType": null
  },
  {
    "LegalEntity": null,
    "ProductName": "Warranty",
    "businessObjectType": "QuoteLineItem",
    "Product": "01tDU000000EsWSYA0",
    "ItemIsPrimarySegment": false,
    "ListPrice": 49.99,
```

```
"ValidationResult": null,
"StartDate": "2024-09-23T00:00:00.000Z",
"ContractVolumePasId": null,
"BillingTreatment": null,
"PeriodBoundaryStartMonth": null,
"SalesTransactionSourceAsset": null,
"id": "0QLDU0000003CZ94AM",
"PartnerDiscountPercent": null,
"PriceWaterFall": "0QLDU0000003CZ94AM:548201414593252",
"ItemProductRecipient": null,
"BillingFrequency": "Annual",
"ProductCode": "W001",
"DerivedPricingAttribute": null,
"TaxTreatment": "1ttDU0000001oGKYAY",
"Subtotal": 99.98,
"ItemRampIdentifier": "RDI5b5ce52b2db4484",
"ItemSegmentName": "Year-1",
"PricebookEntry": "01uDU000000f4LSYAY",
"DiscountAmount": null,
"PricingTermCount": 1,
"NetUnitPrice": 49.99,
"ItemEffectiveGrantDate": null,
"ProductCategory": null,
"SalesTransactionAction": null,
"SalesTransactionActionType": null,
"SalesTransactionItemGroup": null,
"PeriodBoundaryDay": null,
"LineItemDistributionType": null,
"ProrationPolicy": "0muDU00000029ryYAA",
"ContractDiscountType": null,
"TransactionType": null,
"ParentReference": null,
"Discount": null,
"ProductSellingModel": "0jPDU00000029zb2AA",
"PricingTermUnit": "Annual",
"PricingSource": null,
"StockKeepingUnit": null,
"PartnerUnitPrice": null,
"ItemTotalAdjustmentAmount": 0,
"SalesTransactionItemSource": "0QLDU0000003CZ94AM",
"ContractAttributePasId": null,
"SubscriptionTerm": 1,
"SellingModelType": "TermDefined",
"EndQuantity": 2,
"NetTotalPrice": 99.98,
"TotalLineAmount": 99.98,
"ItemSegmentType": "Yearly",
"ProductBasedOn": null,
"Deleted": null,
"BillingReference": null,
"ArePartialPeriodsAllowed": true,
"ItemRecordedPrice": null,
"CustomProductName": "Warranty",
"ItemSegmentIdentifier": "SEG73ad7378e1ed4c5",
```

```
            "SalesTransactionItemParent": "0Q0DU0000002f3d0AA",
            "Quantity": 2,
            "PeriodBoundary": "Anniversary",
            "ContractDiscountValue": null,
            "LineItemDiscountValue": null,
            "ContractId": null,
            "EndDate": "2025-08-22T00:00:00.000Z",
            "ItemGroupSummarySubtotal": null,
            "IsContracted": null,
            "UnitPrice": 49.99,
            "StartQuantity": 0,
            "ContractPrice": null,
            "TotalPrice": 99.98,
            "ItemPath": null,
            "LineItemDiscountType": null
          }
        ],
        "AppUsageAssignment": [
          {
            "businessObjectType": "AppUsageAssignment",
            "ParentReference": null,
            "Record": "0Q0DU0000002f3d0AA",
            "id": "0j8DU0000002VKiYAM",
            "AppUsageType": "RevenueLifecycleManagement"
          }
        ],
        "BillingCountry": null,
        "BillingStreet": null,
        "ShippingPostalCode": null,
        "SalesTransactionSource": "0Q0DU0000002f3d0AA",
        "PrimaryIndustry": null,
        "ShippingState": null,
        "HeaderDistributionLogic": null,
        "Contract": null,
        "BillingState": null,
        "AnnualRevenue": null,
        "EffectiveDate": null
      }
    ]
  },
  "success": true,
  "transactionContextId":
"d3fd83b007418ce4980340313b40fd45665b194973486ebac3674c2b8002336f"
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`correlationId`	String	Resource ID to correlate the API request with the response.	Small, 62.0	62.0
`errors`	Ramp Deal Service Error Response[]	List of errors encountered during the processing of the API request.	Small, 62.0	62.0

Property Name	Type	Description	Filter Group and Version	Available Version
sales Transaction Context	Map<String, Object>	Context object for the sales transaction with updated segment details.	Small, 62.0	62.0
success	Boolean	Indicates whether the API request is successful (`true`) or not (`false`).	Small, 62.0	62.0
transaction ContextId	String	ID of the sales transaction context record instance.	Small, 62.0	62.0

Read Sales Transaction

Output representation of the request to read a sales transaction.

JSON example

```
{
  "response": {
    "records": {
      "Quote": [
        {
          "data": {
            "Id": "0Q05g000000AJK954",
            "Name": "Sample Quote",
            "Status": "Draft",
            "TotalPrice": 1500
          }
        }
      ],
      "QuoteLineItem": [
        {
          "data": {
            "Id": "0QL5g000000DEF456",
            "Product2Id": "01t5g000000GUE752",
            "Quantity": 2,
            "UnitPrice": 750,
            "TotalPrice": 1500
          }
        }
      ]
    }
  },
  "isSuccess": true,
  "errorResponse": []
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`errors`	Place Sales Transaction Error Response[]	List of errors encountered during the processing of the API request.	Small, 65.0	65.0
`isSuccess`	Boolean	Indicates whether the API request is successful (`true`) or not (`false`).	Small, 65.0	65.0
`response`	Read Sales Transaction Records[]	Contains a map of keys and associated values. The keys are record type names, such as a Quote or QuoteLineItem, and values are lists of records of that type.	Small, 65.0	65.0

Read Sales Transaction Records

Output representation of the details of a map of keys and associated values. The keys are record type names, such as a Quote or QuoteLineItem, and values are lists of records of that type.

JSON example


```
{
  "response": {
    "records": {
      "Quote": [
        {
          "data": {
            "Id": "0Q05g000000AJK954",
            "Name": "Sample Quote",
            "Status": "Draft",
            "TotalPrice": 1500
          }
        }
      ],
      "QuoteLineItem": [
        {
          "data": {
            "Id": "0QL5g000000DEF456",
            "Product2Id": "01t5g000000GUE752",
            "Quantity": 2,
            "UnitPrice": 750,
            "TotalPrice": 1500
          }
        }
      ]
    }
  },
  "isSuccess": true,
  "errorResponse": []
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`records`	Map<String, Sales Transaction Record>>	Map of record type names to the list of records.	Small, 65.0	65.0

Renewal

Output representation of the details of a renewal record.

JSON example


```
{
  "renewalRecordId": "0Q0xx0000004NsSCAU",
  "errors": [
    {
      "errorCode": "REQUIRED_FIELD_MISSING",
      "errorMessage": "Specify a value for quantityChange, and try again."
    }
  ],
  "requestId": "16Pxx0000004NIy",
  "success": true
}
```


Properties

Property Name	Type	Description	Filter Group and Version	Available Version
`errors`	ARC Base Error on page 1509[]	Error responses if the creation of a renewal record fails.	Small, 62.0	62.0
`renewalRecordId`	String	ID of the renewal record that's created for the Quote or Order record.	Small, 62.0	62.0
`requestId`	String	Request ID that's used to track the async request.	Small, 62.0	62.0
`success`	Boolean	Indicates whether the API request is successful (`true`) or not (`false`).	Small, 62.0	62.0

Sales Transaction

Output representation of the request to create a sales transaction.

JSON example


```
{
  "contextDetails": {
    "contextId": "e055bb18-d4e8-41c3-881e-0132b9561708",
    "isBuiltInTransaction": true
  },
  "errorResponse": {
```

```
    "errorCode": "INVALID_API_INPUT",
    "message": "Include record type and method in the request and try again.",
    "referenceId": "refQuoteItem2"
  },
  "isSuccess": true,
  "salesTransactionId": "0Q0xx0000004CNYCA2",
  "statusUrl": null,
  "trackerId": null
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`contextDetails`	Sales Transaction Context	Details of the context that's created for the sales transaction.	Small, 63.0	63.0
`errorResponse`	Sales Transaction Error Response[]	Details of the error if the operation fails.	Small, 63.0	63.0
`isSuccess`	Boolean	Indicates if the operation is successful (`true`) or not (`false`).	Small, 63.0	63.0
`salesTransactionId`	String	ID of the sales transaction, such as a quote or an order.	Small, 63.0	63.0
`statusUrl`	String	URL to check the status of the operation.	Small, 63.0	63.0
`trackerId`	String	Unique identifier assigned to a specific operation or request that's used for tracking and referencing the operation.	Small, 63.0	63.0

The **Calculation Status** field for a quote or an order shows an intermediate status as `Saving` during the creation of a sales transaction. If the pricing calculation fails, then the **Calculation Status** field shows the `Pricing Calculation Failed` status. See Quote standard object for a list of applicable calculation status values.

Sales Transaction Context

Output representation of the context details that are associated with a sales transaction.

JSON example

```
{
  "contextDetails": {
    "contextId": "e055bb18-d4e8-41c3-881e-0132b9561708",
    "isBuiltInTransaction": true
  }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`contextId`	String	ID of the context that's created for a session of the sales transaction.	Small, 63.0	63.0

Property Name	Type	Description	Filter Group and Version	Available Version
`isBuiltIn Transaction`	Boolean	Indicates whether a new context ID is created for the sales transaction (`true`) or not (`false`). If the `contextId` property isn't specified, the Place Sales Transaction API generates it.	Small, 63.0	63.0

Sales Transaction Error Response

Output representation of the error details associated with the API request.

JSON example

```
{
  "errorResponse": {
    "errorCode": "INVALID_API_INPUT",
    "message": "Include record type and method in the request and try again.",
    "referenceId": "refQuoteItem2"
  }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`errorCode`	String	Code for the resultant error.	Small, 63.0	63.0
`message`	String	Error message for the resultant error.	Small, 63.0	63.0
`referenceId`	String	Unique ID that's associated with the specific error for tracking and reference purposes.	Small, 63.0	63.0

Sales Transaction Record

Generic output representation for any sales transaction record type.

JSON example

```
{
  "response": {
    "records": {
      "Quote": [
        {
          "data": {
            "Id": "0Q05g000000AJK954",
            "Name": "Sample Quote",
            "Status": "Draft",
            "TotalPrice": 1500
          }
        }
      ],
```

```
      "QuoteLineItem": [
        {
          "data": {
            "Id": "0QL5g000000DEF456",
            "Product2Id": "01t5g000000GUE752",
            "Quantity": 2,
            "UnitPrice": 750,
            "TotalPrice": 1500
          }
        }
      ]
    }
  },
  "isSuccess": true,
  "errorResponse": []
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
data	Map<String, Object>	Represents the data map for any sales transaction record.	Small, 65.0	65.0

Supplemental Transaction Error Response

Output representation of the error details associated with the Place Supplemental Transaction API.

Property Name	Type	Description	Filter Group and Version	Available Version
errorCode	String	Code for the resultant error.	Small, 64.0	64.0
message	String	Message stating the reason for error, if any.	Small, 64.0	64.0
referenceId	String	Unique ID that's associated with the specific error for tracking and reference purposes.	Small, 64.0	64.0

Supplemental Transaction

Output representation of the details of the created supplemental order.

JSON example

```
{
  "requestId": "16PRM0000004DBq",
  "statusURL": "/services/data/vXX.X/sobjects/AsyncOperationTracker/16PRM0000004DBq",
  "orderId": "801S70000001VKgIAM",
  "success": true,
  "errors": []
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`errors`	Supplemental Transaction Error Response[]	List of errors encountered during synchronous processing.	Small, 64.0	64.0
`requestId`	String	Request ID of the process that can be used to query the async status.	Small, 64.0	64.0
`statusURL`	String	URL to check the status of the async operation, if available.	Small, 64.0	64.0
`success`	Boolean	Indicates whether the synchronous part of the processing is successful (`true`) or not (`false`).	Small, 64.0	64.0
`supplementalTransactionId`	String	ID of the created supplemental transaction.	Small, 64.0	64.0

Transaction Management Apex Reference

Use built-in Apex classes and interfaces grouped by namespace.

CommerceOrders Namespace
The `CommerceOrders` namespace provides classes and methods to place orders with integrated pricing, configuration, and validation.

ConnectApi Namespace
The `ConnectApi` namespace (also called Connect in Apex) provides an Apex class to specify details for asset transfer request from one account to another.

CommerceTax Namespace
Manage the communication between Salesforce and an external tax engine.

PlaceQuote Namespace
The `PlaceQuote` namespace provides classes and methods to create or update quotes with pricing preferences and configuration options.

renew_assets_summary Namespace
Retrieve details about renewable assets to create renewal opportunities.

RevSalesTrxn Namespace
Create a sales transaction, such as a quote or an order, with integrated pricing and configuration. Additionally, update an order or a quote, and insert and delete order or quote line items to calculate the estimated tax.

CommerceOrders Namespace

The `CommerceOrders` namespace provides classes and methods to place orders with integrated pricing, configuration, and validation.

Note: This namespace has been deprecated as of API version 63.0. In API version 63.0 and later, use the new RevSalesTrxn namespace.

The `CommerceOrders` namespace includes these classes.

CatalogRatesPreferenceEnum Enum
Specifies the rate card entries defined in the catalog that must be fetched for order items, with usage-based selling during the order creation process.

ConfigurationInputEnum Enum
Specifies the configuration input for the request to place an order.

ConfigurationOptionsInput Class
Contains methods and properties to set the configuration options for the input to the product configurator.

GraphRequest Class
Contains constructors and properties to set the graph ID and a list of records to be ingested. The list of records is specified in a key-value map format that contains the field values of an order.

PlaceOrderExecutor Class
Contains methods to place an order with details of the graph request, pricing preferences, and configuration options.

PlaceOrderResult Class
Contains properties to hold the response to the place order request.

PricingPreferenceEnum Enum
Specifies the pricing preference during the create order process.

RecordResource Class
Contains constructors and properties to create a record object from field values of an order.

RecordWithReferenceRequest Class
Contains constructors and properties to associate a record object with a reference identifier.

CatalogRatesPreferenceEnum Enum

Specifies the rate card entries defined in the catalog that must be fetched for order items, with usage-based selling during the order creation process.

Usage

This enum is available when the Usage-Based Selling feature is enabled.

Enum Values

The `commerceorders.CatalogRatesPreferenceEnum` enum includes these values.

Value	Description
`Fetch`	Retrieves the rate card entries defined in the catalog for order items during the order creation process.
`Skip`	Skips the retrieval of rate card entries for order items during the order creation process. The default value is `Skip`.

ConfigurationInputEnum Enum

Specifies the configuration input for the request to place an order.

Usage

Use these enum values for the `configurationInputEnum` property in the PlaceOrderExecutor Class

Enum Values

The `commerceorders.ConfigurationInputEnum` enum has these values.

Value	Description
`RunAndAllowErrors`	Run the configuration and proceed with order ingestion upon encountering any configuration errors.
`RunAndBlockErrors`	Run the configuration and block order ingestion upon encountering any configuration errors.
`Skip`	Skip the configuration execution.

ConfigurationOptionsInput Class

Contains methods and properties to set the configuration options for the input to the product configurator.

Namespace

CommerceOrders

Usage

This class holds the required details of the product configuration input. Set the class properties to enable default configuration, execution of configuration rules, and validation of the product catalog. Use these class properties as an input to the PlaceOrderExecutor Class method.

Example

```
CommerceOrders.ConfigurationOptionsInput configurationInput = new
CommerceOrders.ConfigurationOptionsInput();
configurationInput.validateProductCatalog = true;
configurationInput.validateAmendRenewCancel = true;
configurationInput.executeConfigurationRules = true;
configurationInput.addDefaultConfiguration = true;

CommerceOrders.GraphRequest graph = new CommerceOrders.GraphRequest('testGraph',
recordNodes);
CommerceOrders.PlaceOrderResult result = CommerceOrders.PlaceOrderExecutor.execute(graph,
 pricingPreference, configurationPreference, configurationInput);
```

ConfigurationOptionsInput Properties

Set the `ConfigurationOptionsInput` class properties to add default configuration, execute configuration rules, and validate the product catalog.

ConfigurationOptionsInput Methods

Learn more about the methods available with the `ConfigurationOptionsInput` class.

ConfigurationOptionsInput Properties

Set the `ConfigurationOptionsInput` class properties to add default configuration, execute configuration rules, and validate the product catalog.

The `ConfigurationOptionsInput` class includes these properties.

addDefaultConfiguration

Sets the default product configuration, such as bundle and product attributes, for an order request.

executeConfigurationRules

Sets the requirement for an order to adhere to the configuration rules.

validateAmendRenewCancel

Sets the requirement to run validations related to amend, renew, or cancel processes.

validateProductCatalog

Sets the requirement to validate an order against the product catalog.

addDefaultConfiguration

Sets the default product configuration, such as bundle and product attributes, for an order request.

Signature

```
public Boolean addDefaultConfiguration {get; set;}
```

Property Value

Type: Boolean

Indicates whether to automatically add default configuration to the order (`true`) or not (`false`).

executeConfigurationRules

Sets the requirement for an order to adhere to the configuration rules.

Signature

```
public Boolean executeConfigurationRules {get; set;}
```

Property Value

Type: Boolean

Indicates whether the order must adhere to configuration rules during processing (`true`) or bypass them (`false`).

`validateAmendRenewCancel`

Sets the requirement to run validations related to amend, renew, or cancel processes.

Signature

`public Boolean validateAmendRenewCancel {get; set;}`

Property Value

Type: Boolean

Indicates whether to run validations related to amend, renew, or cancel processes (`true`) or not (`false`).

`validateProductCatalog`

Sets the requirement to validate an order against the product catalog.

Signature

`public Boolean validateProductCatalog {get; set;}`

Property Value

Type: Boolean

Indicates whether the order must be validated against the product catalog (`true`) or not (`false`).

ConfigurationOptionsInput Methods

Learn more about the methods available with the `ConfigurationOptionsInput` class.

The `ConfigurationOptionsInput` class includes these methods.

- equals(obj)
  Determines the equality of external objects in a list. This method is dynamic and is based on the `equals()` method in Java.
- hashCode()
  Determines the uniqueness of the external object records in a list.
- toString()
  Converts a value to a string.

`equals(obj)`

Determines the equality of external objects in a list. This method is dynamic and is based on the `equals()` method in Java.

Signature

`public Boolean equals(Object obj)`

Parameters

obj

Type: Object

Reference object that's used to compare with the class object.

Return Value

Type: Boolean

Indicates if the class object is same as the reference object (`true`) or not (`false`).

hashCode()

Determines the uniqueness of the external object records in a list.

Signature

`public Integer hashCode()`

Return Value

Type: Integer

Integer hash code that represents the value of the object. Equal objects as per the `equals()` method must return the same hash code.

toString()

Converts a value to a string.

Signature

`public String toString()`

Return Value

Type: String

GraphRequest Class

Contains constructors and properties to set the graph ID and a list of records to be ingested. The list of records is specified in a key-value map format that contains the field values of an order.

Namespace

CommerceOrders

Example

Create the list of records to be ingested by using these steps.

- Create the list of records by constructing the `Map<String, Object>` map of field values of an order.

```
List<CommerceOrders.RecordWithReferenceRequest> recordNodes = new
List<CommerceOrders.RecordWithReferenceRequest>();

// Prepare for the Order
Map<String,Object> orderFieldValues = new Map<String,Object>();
orderFieldValues.put('Pricebook2Id', '01sDU0000001EIYAY');
orderFieldValues.put('AccountId', '001DU000001nIPKYA2');
orderFieldValues.put('EffectiveDate', '2024-01-01');
```

- To create a record object from the field values, create an instance of the `RecordResource` class.

```
CommerceOrders.RecordResource orderRecord = new
CommerceOrders.RecordResource(Order.getSobjectType(), 'POST');
orderRecord.fieldValues = orderFieldValues;
```

- To associate the Record object with a reference identifier, create an instance of the `RecordWithReferenceRequest` class.

```
CommerceOrders.RecordWithReferenceRequest orderRecordNode = new
CommerceOrders.RecordWithReferenceRequest('refOrder', orderRecord);
recordNodes.add(orderRecordNode);

// Prepare for the App Usage Assignment
Map<String,Object> auaFieldValues = new Map<String,Object>();
auaFieldValues.put('AppUsageType', 'RevenueLifecycleManagement');
auaFieldValues.put('RecordId', '@{refOrder.id}');

CommerceOrders.RecordResource auaRecord = new
CommerceOrders.RecordResource(AppUsageAssignment.getSobjectType(), 'POST');
auaRecord.fieldValues = auaFieldValues;

CommerceOrders.RecordWithReferenceRequest auaRecordNode = new
CommerceOrders.RecordWithReferenceRequest('refAppTag', auaRecord);
recordNodes.add(auaRecordNode);

// Prepare for the Order Item
Map<String,Object> oiFieldValues = new Map<String,Object>();
oiFieldValues.put('OrderId', '@{refOrder.id}');
oiFieldValues.put('PricebookEntryId', '01uDU000000YPkIYAW');
oiFieldValues.put('Product2Id', '01tDU000000ESCSYA4');
oiFieldValues.put('Quantity', 2);
oiFieldValues.put('UnitPrice', 800);

CommerceOrders.RecordResource oiRecord = new
CommerceOrders.RecordResource(OrderItem.getSobjectType(), 'POST');
oiRecord.fieldValues = oiFieldValues;

CommerceOrders.RecordWithReferenceRequest oiRecordNode = new
CommerceOrders.RecordWithReferenceRequest('refOrderItem', oiRecord);
recordNodes.add(oiRecordNode);
```

- Invoke the Place Order Apex API.

Note: The `CatalogRatesPreferenceEnum` enum is available when the Usage-Based Selling feature is enabled.

```
// Invoke the Place Order Apex API
CommerceOrders.PricingPreferenceEnum pricingPreference =
CommerceOrders.PricingPreferenceEnum.System;
CommerceOrders.CatalogRatesPreferenceEnum catalogRatesPreference =
CommerceOrders.CatalogRatesPreferenceEnum.Fetch;
CommerceOrders.ConfigurationInputEnum configurationPreference =
CommerceOrders.ConfigurationInputEnum.RunAndAllowErrors;
CommerceOrders.ConfigurationOptionsInput configurationInput = new
CommerceOrders.ConfigurationOptionsInput();
configurationInput.validateProductCatalog = true;
configurationInput.validateAmendRenewCancel = true;
configurationInput.executeConfigurationRules = true;
configurationInput.addDefaultConfiguration = true;
```

- To contain all record objects, create an instance of the `GraphRequest` class.

```
CommerceOrders.GraphRequest graph = new CommerceOrders.GraphRequest('testGraph',
recordNodes);
CommerceOrders.PlaceOrderResult result = CommerceOrders.PlaceOrderExecutor.execute(graph,
 pricingPreference, catalogRatesPreference, configurationPreference, configurationInput);

// Process any error, if exists
if (!result.success) {
  List<ConnectApi.PlaceOrderErrorResponse> errors = result.responseError;
  for (ConnectApi.PlaceOrderErrorResponse error : errors) {
    System.debug(error.errorCode + ': ' + error.message);
  }
}
```

GraphRequest Constructors

Learn more about the available constructors with the `GraphRequest` class.

GraphRequest Properties

Learn more about the available properties with the `GraphRequest` class.

GraphRequest Constructors

Learn more about the available constructors with the `GraphRequest` class.

The `GraphRequest` class includes these constructors.

GraphRequest(graphId, records)

Creates an instance of the `GraphRequest` class to assign the graph ID and a list of records to be ingested.

`GraphRequest(graphId, records)`

Creates an instance of the `GraphRequest` class to assign the graph ID and a list of records to be ingested.

Signature

```
public GraphRequest(String graphId, List<commerceorders.RecordWithReferenceRequest>
records)
```

Parameters

graphId
Type: String

ID of the graph.

records
Type: List<commerceorders.RecordWithReferenceRequest>

List of records to be ingested.

GraphRequest Properties

Learn more about the available properties with the `GraphRequest` class.

The `GraphRequest` class includes these properties.

graphId
Set the `graphId` property to assign the ID value of the graph.

graphId

Set the `graphId` property to assign the ID value of the graph.

Signature

```
public String graphId {get; set;}
```

Property Value

Type: String

PlaceOrderExecutor Class

Contains methods to place an order with details of the graph request, pricing preferences, and configuration options.

Namespace

CommerceOrders

Example

```
CommerceOrders.PlaceOrderResult resp =
CommerceOrders.PlaceOrderExecutor.execute(graph,internalEnum,cEnum,cInput,catalogRatesPreference);
```

PlaceOrderExecutor Methods

Learn more about the methods available with the `PlaceOrderExecutor` class.

PlaceOrderExecutor Example Implementation

Place orders with integrated pricing, configuration, and validation, and manage them throughout their entire lifecycle. To place an order from Apex, refer to the example implementation of the `PlaceOrderExecutor` class.

PlaceOrderExecutor Methods

Learn more about the methods available with the `PlaceOrderExecutor` class.

The `PlaceOrderExecutor` class includes these methods.

execute(graphRequest, pricingPreferenceEnum, configurationInputEnum, configurationOptionsInput)

Use the method in the `PlaceOrderExecutor` class to execute the Place Order Apex API request by assigning the properties for graph request, pricing reference, and configuration options.

execute(graphRequest, pricingPreferenceEnum, catalogRatesPreference, configurationInputEnum, configurationOptionsInput)

Use the method in the `PlaceOrderExecutor` class to execute the Place Order Apex API request by assigning the properties for graph request, pricing reference, and configuration options. This method also includes the property to define fetching of rate card entries.

execute(graphRequest)

Use the method in the `PlaceOrderExecutor` class to execute the Place Order Apex API request by assigning the properties for graph request.

`execute(graphRequest, pricingPreferenceEnum, configurationInputEnum, configurationOptionsInput)`

Use the method in the `PlaceOrderExecutor` class to execute the Place Order Apex API request by assigning the properties for graph request, pricing reference, and configuration options.

Signature

```
public static commerceorders.PlaceOrderResult execute(commerceorders.GraphRequest
graphRequest, commerceorders.PricingPreferenceEnum pricingPreferenceEnum,
commerceorders.ConfigurationInputEnum configurationInputEnum,
commerceorders.ConfigurationOptionsInput configurationOptionsInput)
```

Parameters

graphRequest

Type: commerceorders.GraphRequest

The sObject graph values of the order payload to be ingested.

pricingPreferenceEnum

Type: commerceorders.PricingPreferenceEnum

Pricing preference during the order process.

configurationInputEnum

Type: commerceorders.ConfigurationInputEnum

Configuration input for the place order process.

configurationOptionsInput

Type: commerceorders.ConfigurationOptionsInput on page 1540

Configuration options during the ingestion process.

Return Value

Type: commerceorders.PlaceOrderResult

execute(graphRequest, pricingPreferenceEnum, catalogRatesPreference, configurationInputEnum, configurationOptionsInput)

Use the method in the `PlaceOrderExecutor` class to execute the Place Order Apex API request by assigning the properties for graph request, pricing reference, and configuration options. This method also includes the property to define fetching of rate card entries.

Signature

```
public static commerceorders.PlaceOrderResult execute(commerceorders.GraphRequest
graphRequest, commerceorders.PricingPreferenceEnum pricingPreferenceEnum,
commerceorders.CatalogRatesPreferenceEnum catalogRatesPreferenceEnum,
commerceorders.ConfigurationInputEnum configurationInputEnum,
commerceorders.ConfigurationOptionsInput configurationOptionsInput)
```

Parameters

graphRequest

Type: commerceorders.GraphRequest

The sObject graph values of the order payload to be ingested.

pricingPreferenceEnum

Type: commerceorders.PricingPreferenceEnum

Pricing preference during the order process.

catalogRatesPreference

Type: commerceorders.CatalogRatesPreferenceEnum

The rate card entries defined in the catalog that must be fetched for order items, with usage-based pricing during the order creation process. The `CatalogRatesPreferenceEnum` enum is available when the Usage-Based Selling feature is enabled.

configurationInputEnum

Type: commerceorders.ConfigurationInputEnum

Configuration input for the place order process.

configurationOptionsInput

Type: commerceorders.ConfigurationOptionsInput on page 1540

Configuration options during the ingestion process.

Return Value

Type: commerceorders.PlaceOrderResult

`execute(graphRequest)`

Use the method in the `PlaceOrderExecutor` class to execute the Place Order Apex API request by assigning the properties for graph request.

Signature

`public static commerceorders.PlaceOrderResult execute(commerceorders.GraphRequest graphRequest)`

Parameters

graphRequest
Type: commerceorders.GraphRequest

The sObject graph values of the order payload to be ingested.

Return Value

Type: commerceorders.PlaceOrderResult

PlaceOrderExecutor Example Implementation

Place orders with integrated pricing, configuration, and validation, and manage them throughout their entire lifecycle. To place an order from Apex, refer to the example implementation of the `PlaceOrderExecutor` class.

Namespace

commerceorders

Usage

Customize this example to suit your requirements. Create the list of records to be ingested by using these steps. Replace the respective IDs with the values that are present in your org. For example, replace the value of `${Pricebook2Id}` field with the price book ID that's present in the org.

Example:

- Create the list of records by constructing the `Map<String, Object>` map of field values of an order.

```
List<CommerceOrders.RecordWithReferenceRequest> recordNodes = new
List<CommerceOrders.RecordWithReferenceRequest>();

// Prepare for the Order
Map<String,Object> orderFieldValues = new Map<String,Object>();
orderFieldValues.put('Pricebook2Id', '01sDU0000001EIYAY');
orderFieldValues.put('AccountId', '001DU000001nIPKYA2');
orderFieldValues.put('EffectiveDate', '2024-01-01');
```

- To create a record object from the field values, create an instance of the `RecordResource` class.

```
CommerceOrders.RecordResource orderRecord = new
```

```
CommerceOrders.RecordResource(Order.getSobjectType(), 'POST');
orderRecord.fieldValues = orderFieldValues;
```

- To associate the Record object with a reference identifier, create an instance of the `RecordWithReferenceRequest` class.

```
CommerceOrders.RecordWithReferenceRequest orderRecordNode = new
CommerceOrders.RecordWithReferenceRequest('refOrder', orderRecord);
recordNodes.add(orderRecordNode);

// Prepare for the App Usage Assignment
Map<String,Object> auaFieldValues = new Map<String,Object>();
auaFieldValues.put('AppUsageType', 'RevenueLifecycleManagement');
auaFieldValues.put('RecordId', '@{refOrder.id}');

CommerceOrders.RecordResource auaRecord = new
CommerceOrders.RecordResource(AppUsageAssignment.getSobjectType(), 'POST');
auaRecord.fieldValues = auaFieldValues;

CommerceOrders.RecordWithReferenceRequest auaRecordNode = new
CommerceOrders.RecordWithReferenceRequest('refAppTag', auaRecord);
recordNodes.add(auaRecordNode);

// Prepare for the Order Item
Map<String,Object> oiFieldValues = new Map<String,Object>();
oiFieldValues.put('OrderId', '@{refOrder.id}');
oiFieldValues.put('PricebookEntryId', '01uDU000000YPkIYAW');
oiFieldValues.put('Product2Id', '01tDU000000ESCSYA4');
oiFieldValues.put('Quantity', 2);
oiFieldValues.put('UnitPrice', 800);

CommerceOrders.RecordResource oiRecord = new
CommerceOrders.RecordResource(OrderItem.getSobjectType(), 'POST');
oiRecord.fieldValues = oiFieldValues;

CommerceOrders.RecordWithReferenceRequest oiRecordNode = new
CommerceOrders.RecordWithReferenceRequest('refOrderItem', oiRecord);
recordNodes.add(oiRecordNode);
```

- Invoke the Place Order Apex API.

Note: The `CatalogRatesPreferenceEnum` enum is available when the Usage-Based Selling feature is enabled.

```
// Invoke the Place Order Apex API
CommerceOrders.PricingPreferenceEnum pricingPreference =
CommerceOrders.PricingPreferenceEnum.System;
CommerceOrders.CatalogRatesPreferenceEnum catalogRatesPreference =
CommerceOrders.CatalogRatesPreferenceEnum.Fetch;
CommerceOrders.ConfigurationInputEnum configurationPreference =
CommerceOrders.ConfigurationInputEnum.RunAndAllowErrors;
CommerceOrders.ConfigurationOptionsInput configurationInput = new
```

```
CommerceOrders.ConfigurationOptionsInput();
configurationInput.validateProductCatalog = true;
configurationInput.validateAmendRenewCancel = true;
configurationInput.executeConfigurationRules = true;
configurationInput.addDefaultConfiguration = true;
```

- To contain all record objects, create an instance of the `GraphRequest` class.

```
CommerceOrders.GraphRequest graph = new CommerceOrders.GraphRequest('testGraph',
recordNodes);
CommerceOrders.PlaceOrderResult result =
CommerceOrders.PlaceOrderExecutor.execute(graph, pricingPreference,
catalogRatesPreference, configurationPreference, configurationInput);


// Process any error, if exists
if (!result.success) {
  List<ConnectApi.PlaceOrderErrorResponse> errors = result.responseError;
  for (ConnectApi.PlaceOrderErrorResponse error : errors) {
    System.debug(error.errorCode + ': ' + error.message);
  }
}
```

PlaceOrderResult Class

Contains properties to hold the response to the place order request.

Namespace

CommerceOrders

Example

```
CommerceOrders.PlaceOrderResult resp =
CommerceOrders.PlaceOrderExecutor.execute(graph,internalEnum,cEnum,cInput,catalogRatesPreference);
```

PlaceOrderResult Properties

Learn more about the available properties with the `PlaceOrderResult` class.

PlaceOrderResult Properties

Learn more about the available properties with the `PlaceOrderResult` class.

The `PlaceOrderResult` class includes these properties.

orderId

Get the ID of the order that's created after a successful request.

requestIdentifier
Get the request ID of the process to query the asynchronous status of the Place Order Apex API.

responseError
Get the list of errors encountered during the synchronous processing of the API request.

statusURL
Get the asynchronous status URL of the request, if available.

success
Get the request status of the synchronous part of the processing.

orderId

Get the ID of the order that's created after a successful request.

Signature

`public String orderId {get; set;}`

Property Value

Type: String

requestIdentifier

Get the request ID of the process to query the asynchronous status of the Place Order Apex API.

Signature

`public String requestIdentifier {get; set;}`

Property Value

Type: String

responseError

Get the list of errors encountered during the synchronous processing of the API request.

Signature

`public List<commerceorders.PlaceOrderErrorResponse> responseError {get; set;}`

Property Value

Type: List<ConnectApi.PlaceOrderErrorResponse>

statusURL

Get the asynchronous status URL of the request, if available.

Signature

```
public String statusURL {get; set;}
```

Property Value

Type: String

success

Get the request status of the synchronous part of the processing.

Signature

```
public Boolean success {get; set;}
```

Property Value

Type: Boolean

Indicates whether the synchronous part of the processing is successful (`true`) or not (`false`).

PricingPreferenceEnum Enum

Specifies the pricing preference during the create order process.

Usage

Used by the PlaceOrderExecutor class.

Enum Values

The `commerceorders.PricingPreferenceEnum` enum includes these values.

Value	Description
`Force`	Enforce pricing during the order ingestion process.
`Skip`	Skip pricing during the order ingestion process.
`System`	Determine whether a pricing calculation is required.

RecordResource Class

Contains constructors and properties to create a record object from field values of an order.

Namespace

CommerceOrders

Example

```
CommerceOrders.RecordResource orderRecord = new
CommerceOrders.RecordResource(Order.getSobjectType(), 'POST');
orderRecord.fieldValues = orderFieldValues;
```

RecordResource Constructors

Learn more about the available constructors with the `RecordResource` class.

RecordResource Properties

Learn more about the available properties with the `RecordResource` class.

RecordResource Constructors

Learn more about the available constructors with the `RecordResource` class.

The `RecordResource` class includes these constructors.

RecordResource(type, method, id)

Creates an instance of the `RecordResource` class to assign values to the fields of an order item by using the sObject type, API method, and order ID properties.

RecordResource(type, method)

Creates an instance of the `RecordResource` class to assign the values to the fields of an order item by using the sObject type and API method properties.

RecordResource(type, method, id)

Creates an instance of the `RecordResource` class to assign values to the fields of an order item by using the sObject type, API method, and order ID properties.

Signature

`public RecordResource(Schema.SObjectType type, String method, Id id)`

Parameters

type

Type: Schema.SObjectType

Object that's returned from the field describe result using the `getReferenceTo()` method or from the sObject describe result using the `getSObjectType()` method.

method

Type: String

Method for the API request, such as POST or PATCH.

id

Type: Id

ID of the order.

`RecordResource(type, method)`

Creates an instance of the `RecordResource` class to assign the values to the fields of an order item by using the sObject type and API method properties.

Signature

`public RecordResource(Schema.SObjectType type, String method)`

Parameters

`type`
Type: Schema.SObjectType

Object that's returned from the field describe result using the `getReferenceTo()` method or from the sObject describe result using the `getSObjectType()` method.

`method`
Type: String

Method for the API request, such as POST or PATCH.

RecordResource Properties

Learn more about the available properties with the `RecordResource` class.

The `RecordResource` class includes these properties.

fieldValues
Set the `fieldValues` property to assign values to the fields to update the order record.

id
Set the `id` property to assign the ID of the order record.

method
Set the `method` property to specify the API request method, such as POST or PATCH.

type
Set the `type` property to assign the object type that's returned from the field describe result using the `getReferenceTo()` method or from the sObject describe result using the `getSObjectType()` method.

`fieldValues`

Set the `fieldValues` property to assign values to the fields to update the order record.

Signature

`public Map<String,ANY> fieldValues {get; set;}`

Property Value

Type: List<Map<String,ANY>>

`id`

Set the `id` property to assign the ID of the order record.

Signature

`public String id {get; set;}`

Property Value

Type: String

`method`

Set the `method` property to specify the API request method, such as POST or PATCH.

Signature

`public String method {get; set;}`

Property Value

Type: String

`type`

Set the `type` property to assign the object type that's returned from the field describe result using the `getReferenceTo()` method or from the sObject describe result using the `getSObjectType()` method.

Signature

`public Schema.SObjectType type {get; set;}`

Property Value

Type: Schema.SObjectType

RecordWithReferenceRequest Class

Contains constructors and properties to associate a record object with a reference identifier.

Namespace

CommerceOrders

Example

```
CommerceOrders.RecordWithReferenceRequest orderRecordNode = new
CommerceOrders.RecordWithReferenceRequest('refOrder', orderRecord);
```

RecordWithReferenceRequest Constructors
Learn more about the available constructors with the `RecordWithReferenceRequest` class.

RecordWithReferenceRequest Properties
Learn more about the available properties with the `RecordWithReferenceRequest` class.

RecordWithReferenceRequest Constructors

Learn more about the available constructors with the `RecordWithReferenceRequest` class.

The `RecordWithReferenceRequest` class includes these constructors.

RecordWithReferenceRequest(referenceId, record)
Creates an instance of the `RecordWithReferenceRequest` class to associate a record object with a reference identifier by using the `referenceId` and `record` object properties.

`RecordWithReferenceRequest(referenceId, record)`

Creates an instance of the `RecordWithReferenceRequest` class to associate a record object with a reference identifier by using the `referenceId` and `record` object properties.

Signature

`public RecordWithReferenceRequest(String referenceId, commerceorders.RecordResource record)`

Parameters

`referenceId`
Type: String

Reference ID that maps to the subrequest response and can be used to reference the response in subsequent subrequests. You can reference the `referenceId` in either the body or URL of a subrequest. Use this syntax to include a reference: `@{referenceId.FieldName}`. See referenceId property of a composite subrequest.

`record`
Type: commerceorders.RecordResource

Record object that's defined using the `RecordResource` class.

RecordWithReferenceRequest Properties

Learn more about the available properties with the `RecordWithReferenceRequest` class.

The `RecordWithReferenceRequest` class includes these properties.

record
Set the `record` property to specify the record object that's defined by using the `RecordResource` class.

referenceId
Set the `referenceId` property to specify the reference ID that maps to the subrequest response. This reference ID can be used to reference the response in subsequent subrequests.

`record`

Set the `record` property to specify the record object that's defined by using the `RecordResource` class.

Signature

```
public commerceorders.RecordResource record {get; set;}
```

Property Value

Type: commerceorders.RecordResource

`referenceId`

Set the `referenceId` property to specify the reference ID that maps to the subrequest response. This reference ID can be used to reference the response in subsequent subrequests.

Signature

```
public String referenceId {get; set;}
```

Property Value

Type: String

`ConnectApi` Namespace

The `ConnectApi` namespace (also called Connect in Apex) provides an Apex class to specify details for asset transfer request from one account to another.

These classes are available in the `ConnectApi` namespace.

ConnectApi Input Classes
Transaction Management includes this Apex input class.

ConnectApi Input Classes

Transaction Management includes this Apex input class.

ConnectApi.RampOptionInputRepresentation
Input representation of the ramp option details during an asset renewal.

ConnectApi.TransferRecordInputRepresentation
Input representation of the details of the assets to be transferred.

ConnectApi.RampOptionInputRepresentation

Input representation of the ramp option details during an asset renewal.

This Apex class is used by the `rampOptionDetails` Apex-defined input variable. See Initiate Renewal Action.

Property	Type	Description	Required or Optional	Available Version
durationInMonths	Double	Maximum duration of all ramp segments.	Required	67.0
segmentCount	Double	Number of created ramp segments based on the specified duration.	Required if you specify Custom value for the segmentType property.	67.0
segmentType	ConnectApi.RampSegmentType	Type of ramp schedule. Valid values are: • Custom—Specifies the number of segments. The total duration is divided equally across segments. • Yearly—Creates 12-month segments. If the total duration isn't a multiple of 12, the last segment covers the remaining months. If the duration divided by the number of months has a remainder, that segment is listed first as a prorated segment.	Required	67.0

ConnectApi.TransferRecordInputRepresentation

Input representation of the details of the assets to be transferred.

This Apex class is used by the `transferRecords` apex-defined input variable. See Initiate Transfer Action on page 1758.

Property	Type	Description	Required or Optional	Available Version
assetId	String	ID of the asset to transfer.	Required	65.0
transferQuantity	Double	Transfer quantity for the request.	Required	65.0

CommerceTax Namespace

Manage the communication between Salesforce and an external tax engine.

The `CommerceTax` namespace includes these classes.

AbstractTransactionResponse Class
Abstract class that contains methods for setting tax fields based on the external tax provider's response. Response classes that extend `AbstractTransactionResponse` inherit these methods.

AddressesResponse Class
Sets the tax address fields based on a response from the external tax engine. Contains setter methods for the Ship From, Ship To, and Sold To addresses.

AddressResponse Class

Contains a location code sent from the external tax engine.

AmountDetailsResponse Class

Sets tax amount fields based on a response from the external tax engine.

CalculateTaxRequest Class

Represents a request to an external tax engine to calculate tax. Extends the TaxTransactionRequest class and is the top-level request class.

CalculateTaxResponse Class

Sets the values of the tax transaction following a response from the external tax engine. Extends the AbstractTransactionResponse class and is the top-level response class.

CalculateTaxType Enum

Shows whether a tax calculation request is for estimated or actual tax.

CustomTaxAttributesResponse Class

Sets additional data or custom attributes in the tax response.

ErrorResponse Class

Use to respond with an error after receiving errors from the PaymentGatewayAdapter methods of the CommercePayments namespace, such as request-forbidden responses, custom validation errors, or expired API tokens.

HeaderTaxAddressesRequest Class

Captures the address values that are applicable for the quote or order transaction.

ImpositionResponse Class

Stores details of tax impositions from the external tax engine.

JurisdictionResponse Class

Stores details from the external tax engine about the tax jurisdiction used in the tax calculation process. A tax jurisdiction represents a government entity that collects tax.

LineItemResponse Class

Response class that stores details of a list of one or more line items on which the tax engine has calculated tax.

LineTaxAddressesRequest Class

Stores details of the addresses applied per line item in a tax calculation request.

RequestType Enum

Shows the type of tax request made to the tax engine.

ResultCode Enum

Code that represents the results of a tax request made to the tax engine.

RuleDetailsResponse Class

Contains details about the tax rules used for tax calculation.

TaxAddressesRequest Class

Contains methods to get and set tax address values.

TaxAddressRequest Class

Contains address details used for tax calculation.

TaxApiException Class

Contains details about any exceptions during the tax calculation process. Extends the `ApexBaseException` class.

TaxCustomerDetailsRequest Class

Contains customer details used in tax calculation.

TaxDetailsResponse Class

Stores details of the tax values that an external tax engine calculates in response to a tax calculation request.

TaxEngineAdapter Interface

Retrieves information from the tax engine and evaluates the information to define tax details.

TaxEngineContext Class

Wrapper class that stores details about the type of a tax calculation request.

TaxLineItemRequest Class

Contains line item details of a tax request.

TaxSellerDetailsRequest Class

Contains tax code details used in the tax calculation request.

TaxTransactionRequest Class

Abstract class for storing customer details used in tax calculation and estimation requests.

TaxTransactionStatus Enum

Shows whether the tax transaction has been committed or uncommitted.

TaxTransactionType Enum

Shows whether the tax transaction is for a credit or debit transaction.

AbstractTransactionResponse Class

Abstract class that contains methods for setting tax fields based on the external tax provider's response. Response classes that extend `AbstractTransactionResponse` inherit these methods.

Namespace

CommerceTax

AbstractTransactionResponse Methods

Learn more about the methods for AbstractTransactionResponse class.

AbstractTransactionResponse Methods

Learn more about the methods for AbstractTransactionResponse class.

The `AbstractTransactionResponse` class includes these methods.

setAddresses(addresses)

Uses an instance of `AddressesResponse` to set the values of tax address fields.

setAmountDetails(amountDetails)

Uses an instance of `AmountDetailsResponse` to set tax amount fields such as exemption amount and tax amount.

setCurrencyIsoCode(currencyIsoCode)

Sets the currencyIsoCode field.

setCustomTaxAttributes(customTaxAttributes)
: Uses an instance of `CustomTaxAttributesResponse` class to include additional attributes in the tax response at the header level.

setDescription(dscptn)
: Sets the Description field.

setDocumentCode(documentCode)
: Sets the DocumentCode field. Document codes are often used to reference tax documents that the external tax engine uses in the tax calculation process. Document code acts as a unique link to chain-related transactions, such as amendment or refunds.

setEffectiveDate(effectiveDate)
: Sets the EffectiveDate field. Effective Date fields are optional fields that store the date that a transaction takes effect. We provide these fields only for recordkeeping purposes – for example, if you must report an effective date to an external general ledger system. Salesforce doesn't use them to calculate any tax or payment values.

setLineItems(lineItems)
: Uses an instance of the `LineItemResponse` class to set a list of line items. Each line item represents an item sent to an external tax engine for tax calculation.

setReferenceDocumentCode(referenceDocumentCode)
: Sets the ReferenceDocumentCode field. Use this field to store the code of an additional document used in the tax calculation process. For example, use this field in case of a refund for a previously taxed purchase.

setReferenceEntityId(referenceEntityId)
: Sets the ID of a reference entity. In Commerce Tax, a reference entity represents a record related to the items sent to the external tax engine for tax calculation. For example, if you sent order items for tax calculation, you could define the parent order as the reference entity.

setTaxTransactionId(taxTrxnId)
: Sets the TaxTransactionId field using the ID of a tax transaction record. In Commerce Tax, a tax transaction record stores information about a specific tax calculation process.

setTransactionDate(transactionDate)
: Sets the TransactionDate field.

`setAddresses(addresses)`

Uses an instance of `AddressesResponse` to set the values of tax address fields.

Signature

`global void setAddresses(commercetax.AddressesResponse addresses)`

Parameters

`addresses`
: Type: AddressesResponse

  Class that contains methods to set the Ship To, Ship From, and Sold To address information.

Return Value

Type: void

setAmountDetails(amountDetails)

Uses an instance of `AmountDetailsResponse` to set tax amount fields such as exemption amount and tax amount.

Signature

`global void setAmountDetails(commercetax.AmountDetailsResponse amountDetails)`

Parameters

amountDetails
Type: AmountDetailsResponse

Class that contains methods to set the tax exemption amount, tax amount, total amount, and total amount with tax.

Return Value

Type: void

setCurrencyIsoCode(currencyIsoCode)

Sets the currencyIsoCode field.

Signature

`global void setCurrencyIsoCode(String currencyIsoCode)`

Parameters

currencyIsoCode
Type: String

Three-letter ISO 4217 currency code associated with a tax object.

Return Value

Type: void

setCustomTaxAttributes(customTaxAttributes)

Uses an instance of `CustomTaxAttributesResponse` class to include additional attributes in the tax response at the header level.

Signature

`global void setCustomTaxAttributes(commercetax.CustomTaxAttributesResponse customTaxAttributes)`

Parameters

customTaxAttributes
Type: CustomTaxAttributesResponse

Additional data or custom attributes to include in the tax response.

Return Value

Type: void

setDescription(dscptn)

Sets the Description field.

Signature

`global void setDescription(String dscptn)`

Parameters

dscptn
Type: String

Optional field for providing additional information about a record.

Return Value

Type: void

setDocumentCode(documentCode)

Sets the DocumentCode field. Document codes are often used to reference tax documents that the external tax engine uses in the tax calculation process. Document code acts as a unique link to chain-related transactions, such as amendment or refunds.

Signature

`global void setDocumentCode(String documentCode)`

Parameters

documentCode
Type: String

Code for a tax document used in the tax calculation process.

Return Value

Type: void

setEffectiveDate(effectiveDate)

Sets the EffectiveDate field. Effective Date fields are optional fields that store the date that a transaction takes effect. We provide these fields only for recordkeeping purposes – for example, if you must report an effective date to an external general ledger system. Salesforce doesn't use them to calculate any tax or payment values.

Signature

```
global void setEffectiveDate(Datetime effectiveDate)
```

Parameters

effectiveDate
Type: Datetime

Optional field that stores the date that a transaction takes effect.

Return Value

Type: void

`setLineItems(lineItems)`

Uses an instance of the `LineItemResponse` class to set a list of line items. Each line item represents an item sent to an external tax engine for tax calculation.

Signature

```
global void setLineItems(List<commercetax.LineItemResponse> lineItems)
```

Parameters

lineItems
Type: List<LineItemResponse>

A list of line items sent to an external tax engine for tax calculation.

Return Value

Type: void

`setReferenceDocumentCode(referenceDocumentCode)`

Sets the ReferenceDocumentCode field. Use this field to store the code of an additional document used in the tax calculation process. For example, use this field in case of a refund for a previously taxed purchase.

Signature

```
global void setReferenceDocumentCode(String referenceDocumentCode)
```

Parameters

referenceDocumentCode
Type: String

The code for a document used in the tax calculation process.

Return Value

Type: void

`setReferenceEntityId(referenceEntityId)`

Sets the ID of a reference entity. In Commerce Tax, a reference entity represents a record related to the items sent to the external tax engine for tax calculation. For example, if you sent order items for tax calculation, you could define the parent order as the reference entity.

Signature

```
global void setReferenceEntityId(String referenceEntityId)
```

Parameters

referenceEntityId
Type: String

ID of a record related to the items sent for tax calculation.

Return Value

Type: void

`setTaxTransactionId(taxTrxnId)`

Sets the TaxTransactionId field using the ID of a tax transaction record. In Commerce Tax, a tax transaction record stores information about a specific tax calculation process.

Signature

```
global void setTaxTransactionId(String taxTrxnId)
```

Parameters

taxTrxnId
Type: String

The ID of a tax transaction record in Commerce Tax.

Return Value

Type: void

`setTransactionDate(transactionDate)`

Sets the TransactionDate field.

Signature

```
global void setTransactionDate(Datetime transactionDate)
```

Parameters

transactionDate
Type: Datetime

Date that a tax transaction occurred.

Return Value

Type: void

AddressesResponse Class

Sets the tax address fields based on a response from the external tax engine. Contains setter methods for the Ship From, Ship To, and Sold To addresses.

Namespace

CommerceTax

Usage

Because `AddressesResponse` contains multiple addresses, we recommend using multiple instances of `AddressResponse` to set unique values for each address.

Example

This code sample represents a portion of the code used in a mock tax adapter. In this example, you create three `AddressResponse` classes, set their location codes, and pass them to the `Ship To`, `Ship From`, and `Sold To` setter methods in `AddressesResponse`. In an actual implementation, your `AddressResponse` classes already have a location code based on the response from the external tax engine.

```
commercetax.AddressesResponse addressesRes = new commercetax.AddressesResponse();

//AddressResponse containing ShipTo information
commercetax.AddressResponse shipToAddress = new commercetax.AddressResponse();
shipToAddress.setLocationCode('1234567');

//AddressResponse containing ShipFrom information
commercetax.AddressResponse shipFromAddress = new commercetax.AddressResponse();
shipFromAddress.setLocationCode('84720385');

//AddressResponse containing Sold To information
commercetax.AddressResponse soldToAddress = new commercetax.AddressResponse();
soldToAddress.setLocationCode('92381749');

//set values of addressesRes
addressesRes.setShipFrom(shipFromAddress);
addressesRes.setShipTo(shipToAddress);
addressesRes.setSoldTo(soldToAddress);
```

AddressesResponse Methods

Learn more about the methods for AddressesResponse class.

AddressesResponse Methods

Learn more about the methods for AddressesResponse class.

The `AddressesResponse` class includes these methods.

setShipFrom(shipFrom)

Sets the value of a ShipFrom address field.

setShipTo(shipTo)

Sets the value of a ShipTo address field.

setSoldTo(soldTo)

Sets the value of a SoldTo address field.

setShipFrom(shipFrom)

Sets the value of a ShipFrom address field.

Signature

```
global void setShipFrom(commercetax.AddressResponse shipFrom)
```

Parameters

shipFrom

Type: AddressResponse

A single address. Use this generic address parameter to store any type of address, such as Ship From, Ship To, and Sold To details. Users set the specific address in an `AddressResponse` instance and then pass that instance to the `AddressesResponse`'s `setShipTo()`, `setShipFrom()`, and `setSoldTo()` methods as needed.

Return Value

Type: void

setShipTo(shipTo)

Sets the value of a ShipTo address field.

Signature

```
global void setShipTo(commercetax.AddressResponse shipTo)
```

Parameters

shipTo

Type: AddressResponse

Stores a single address. This is a generic address parameter and can be used to store any type of address, such as Ship From, Ship To, and Sold To details. Users set the specific address in an `AddressResponse` instance and then pass that instance to the `AddressesResponse`'s `setShipTo()`, `setShipFrom()`, and `setSoldTo()` methods as needed.

Return Value

Type: void

`setSoldTo(soldTo)`

Sets the value of a SoldTo address field.

Signature

`global void setSoldTo(commercetax.AddressResponse soldTo)`

Parameters

`soldTo`
Type: AddressResponse

Stores a single address. This is a generic address parameter and can be used to store any type of address, such as Ship From, Ship To, Sold To details. Users set the specific address in an AddressResponse instance and then pass that instance to the `AddressesResponse`'s `setShipTo()`, `setShipFrom()`, and `setSoldTo()` methods as needed.

Return Value

Type: void

AddressResponse Class

Contains a location code sent from the external tax engine.

Namespace

CommerceTax

Usage

Use the `AddressResponse` class to set unique values for each address.

```
commercetax.AddressesResponse addressesRes = new commercetax.AddressesResponse();

//AddressResponse containing ShipTo information
commercetax.AddressResponse shipToAddress = new commercetax.AddressResponse();
shipToAddress.setLocationCode('1234567');

//AddressResponse containing ShipFrom information
commercetax.AddressResponse shipFromAddress = new commercetax.AddressResponse();
shipFromAddress.setLocationCode('84720385');

//AddressResponse containing Sold To information
```

```
commercetax.AddressResponse soldToAddress = new commercetax.AddressResponse();
soldToAddress.setLocationCode('92381749');

//set values of addressesRes
addressesRes.setShipFrom(shipFromAddress);
addressesRes.setShipTo(shipToAddress);
addressesRes.setSoldTo(soldToAddress);
```


AddressResponse Methods

Learn more about the available methods with the `AddressResponse` class.

The `AddressResponse` class includes these methods.

`setLocationCode(locationCode)`

Sets the value of a LocationCode field.

Signature

`global void setLocationCode(String locationCode)`

Parameters

`locationCode`
Type: String

A code that contains address information. This value can be passed to a method that sets the value of an address field.

Return Value

Type: void

AmountDetailsResponse Class

Sets tax amount fields based on a response from the external tax engine.

Namespace

CommerceTax

Example

In this example, an instance of `AmountDetailsResponse` class in a mock adapter calculates several tax amount fields. The `totalTax` and `totalAmount` parameters were defined in an instance of `LineItemResponse` class. The adapter then assigns the instance to `lineItemResponse`.

```
commercetax.AmountDetailsResponse amountResponse = new commercetax.AmountDetailsResponse();
amountResponse.setTotalAmountWithTax(totalTax+totalAmount);
amountResponse.setExemptAmount(0);
amountResponse.setTotalAmount(totalAmount);
amountResponse.setTaxAmount(totalTax);
lineItemResponse.setAmountDetails(amountResponse);
```

AmountDetailsResponse Methods
: Learn more about the methods available from the `AmountDetailsResponse` class.

AmountDetailsResponse Methods

Learn more about the methods available from the `AmountDetailsResponse` class.

The following are methods for `AmountDetailsResponse`.

setExemptAmount(exemptAmount)
: Sets the value of the ExemptAmount field.

setTaxAmount(taxAmount)
: Sets the value of the TaxAmount field.

setTotalAmount(totalAmount)
: Sets the value of the TotalAmount field.

setTotalAmountWithTax(totalAmtWithTax)
: Sets the value of the TotalAmountWithTax field.

setExemptAmount(exemptAmount)

Sets the value of the ExemptAmount field.

Signature

`global void setExemptAmount(Double exemptAmount)`

Parameters

exemptAmount
: Type: Double
: The amount of a line item's total amount that's exempt from tax calculation.

Return Value

Type: void

setTaxAmount(taxAmount)

Sets the value of the TaxAmount field.

Signature

```
global void setTaxAmount(Double taxAmount)
```

Parameters

taxAmount
Type: Double

The calculated amount of tax for a line item.

Return Value

Type: void

setTotalAmount(totalAmount)

Sets the value of the TotalAmount field.

Signature

```
global void setTotalAmount(Double totalAmount)
```

Parameters

totalAmount
Type: Double

The total amount of a line item, excluding tax.

Return Value

Type: void

setTotalAmountWithTax(totalAmtWithTax)

Sets the value of the TotalAmountWithTax field.

Signature

```
global void setTotalAmountWithTax(Double totalAmtWithTax)
```

Parameters

totalAmtWithTax
Type: Double

The total amount of a line item combined with the calculated tax for that line item.

Return Value

Type: void

CalculateTaxRequest Class

Represents a request to an external tax engine to calculate tax. Extends the TaxTransactionRequest class and is the top-level request class.

Namespace

CommerceTax

Usage

Keep these considerations in mind when you use this class.

- If the `shouldVoidTax` property value is set to `true`, then the operation returns a response with `documentCode` property value updated to `referenceDocumentCode` property value that was originally sent in the request payload. The response also includes the `taxTransactionType` property value as `Void`. This indicates that the document specified in the `referenceDocumentCode` property value is voided.
- If document is locked or you can't void the tax transaction for any reason, then you can use the Tax Calculation request to perform another transaction such as a Credit Tax request. In this scenario, the response includes the `documentCode` property value that was sent in the request payload.
- If the document that's mentioned in the `referenceDocumentCode` property value isn't available in the tax engine, then an error response occurs with ResultCode on page 1607 value as `ReferenceDocumentCodeMissing`.

Example

See TaxEngineAdapter Example Implementation for more details on how to access information from the `CalculateTaxRequest` class.

CalculateTaxRequest Constructors

Learn more about the constructors that are available with the `CalculateTaxRequest` class. This constructor is intended for test usage and throws an exception if used outside of the Apex test context.

CalculateTaxRequest Properties

Learn more about the available properties with the `CalculateTaxRequest` class.

CalculateTaxRequest Methods

Learn more about the available methods with the `CalculateTaxRequest` class.

CalculateTaxRequest Constructors

Learn more about the constructors that are available with the `CalculateTaxRequest` class. This constructor is intended for test usage and throws an exception if used outside of the Apex test context.

The `CalculateTaxRequest` class includes these constructors.

CalculateTaxRequest(taxType)

This constructor is intended for test usage only and throws an exception if used outside of the Apex test context.

`CalculateTaxRequest(taxType)`

This constructor is intended for test usage only and throws an exception if used outside of the Apex test context.

Signature

```
global CalculateTaxRequest(commercetax.CalculateTaxType taxType)
```

Parameters

`taxType`

Type: CalculateTaxType

Indicates whether the tax calculation is for estimated tax or actual tax.

CalculateTaxRequest Properties

Learn more about the available properties with the `CalculateTaxRequest` class.

The `CalculateTaxRequest` class includes these properties.

isCommit

Indicates whether the tax calculation has to be committed or reported to government authorities.

isHeaderTaxRequested

Indicates whether header tax is enabled in the tax engine (`true`) or not (`false`).

shouldVoidTax

Indicates whether to void the tax transaction associated with a document that's mentioned in the `referenceDocumentCode` property value with `taxType` property value set to `Actual` and `isCommit` property value set to `true`.

taxTransactionType

Shows whether the tax transaction is for a credit or debit transaction.

taxType

Shows whether the tax calculation is for estimated or actual tax wherein only actual tax can be submitted.

`isCommit`

Indicates whether the tax calculation has to be committed or reported to government authorities.

Signature

```
global Boolean isCommit {get; set;}
```

Property Value

Type: Boolean

`isHeaderTaxRequested`

Indicates whether header tax is enabled in the tax engine (`true`) or not (`false`).

Signature

```
global Boolean isHeaderTaxRequested {get; set;}
```

Property Value

Type: Boolean

shouldVoidTax

Indicates whether to void the tax transaction associated with a document that's mentioned in the `referenceDocumentCode` property value with `taxType` property value set to `Actual` and `isCommit` property value set to `true`.

Signature

```
global commercetax.CalculateTaxType shouldVoidTax {get; set;}
```

Property Value

Type: Boolean

taxTransactionType

Shows whether the tax transaction is for a credit or debit transaction.

Signature

```
global commercetax.TaxTransactionType taxTransactionType {get; set;}
```

Property Value

Type: TaxTransactionType

taxType

Shows whether the tax calculation is for estimated or actual tax wherein only actual tax can be submitted.

Signature

```
global commercetax.CalculateTaxType taxType {get; set;}
```

Property Value

Type: CalculateTaxType

CalculateTaxRequest Methods

Learn more about the available methods with the `CalculateTaxRequest` class.

The `CalculateTaxRequest` class includes these methods.

equals(obj)
: Maintains the integrity of lists of type `CalculateTaxRequest` by determining the equality of external objects in a list. This method is dynamic and is based on the `equals()` method in Java.

hashCode()
: Maintains the integrity of lists of type `CalculateTaxRequest` by determining the uniqueness of the external object records in a list.

toString()
: Converts a value to a string.

equals(obj)

Maintains the integrity of lists of type `CalculateTaxRequest` by determining the equality of external objects in a list. This method is dynamic and is based on the `equals()` method in Java.

Signature

`global Boolean equals(Object obj)`

Parameters

obj
: Type: Object

  External object whose key is to be validated.

Return Value

Type: Boolean

hashCode()

Maintains the integrity of lists of type `CalculateTaxRequest` by determining the uniqueness of the external object records in a list.

Signature

`global Integer hashCode()`

Return Value

Type: Integer

toString()

Converts a value to a string.

Signature

`global String toString()`

Return Value

Type: String

CalculateTaxResponse Class

Sets the values of the tax transaction following a response from the external tax engine. Extends the AbstractTransactionResponse class and is the top-level response class.

Namespace

CommerceTax

Example

```
if(requestType == commercetax.RequestType.CalculateTax){
             commercetax.calculatetaxtype type = request.taxtype;
             String docCode='';
             if(request.DocumentCode == 'simulateEmptyDocumentCode')
                 docCode = '';
             else if(request.DocumentCode != null)
                 docCode =request.DocumentCode;
           else if(request.ReferenceEntityId != null) docCode = request.ReferenceEntityId;

             else docCode =  String.valueOf(getRandomInteger(0,2147483647));
             commercetax.CalculateTaxResponse response = new
commercetax.CalculateTaxResponse();
             if(request.isCommit == true) {
                 response.setStatus(commercetax.TaxTransactionStatus.Committed);
             } else {
                 response.setStatus(commercetax.TaxTransactionStatus.Uncommitted);
             }
             response.setDocumentCode(docCode);
             response.setReferenceDocumentCode(request.referenceDocumentCode);
             response.setTaxType(type);
             response.setStatusDescription('statusDescription');
             if(request.sellerDetails.code == 'testSellerCode') {
                 response.setDescription('SellerCode fetched from TaxEngine entity');
             }
             else {
                 response.setDescription('description');
             }
            response.setEffectiveDate(system.now());
            if(request.transactionDate == null) {
               response.setTransactionDate(system.now());
            } else {
               response.setTransactionDate(request.transactionDate);
            }
            if(request.taxTransactionType == null) {
                response.setTaxTransactionType(commercetax.TaxTransactionType.Debit);
            } else {
                response.setTaxTransactionType(request.taxTransactionType);
            }
```

```
            if(request.currencyIsoCode == null || request.currencyIsoCode == '') {
               response.setCurrencyIsoCode('USD');
            } else {
               response.setCurrencyIsoCode(request.currencyIsoCode);
            }
            response.setReferenceEntityId(request.ReferenceEntityId);
}
```

CalculateTaxResponse Methods
Learn more about the available methods with the `CalculateTaxResponse` class.

CalculateTaxResponse Methods

Learn more about the available methods with the `CalculateTaxResponse` class.

The `CalculateTaxResponse` class includes these methods.

setAddresses(addresses)
Sets the value of the Addresses field using the addresses contained in an instance of the AddressesResponse class.

setAmountDetails(amountDetails)
Sets the value of the AmountDetails field using an instance of `AmountDetailsResponse`.

setCurrencyIsoCode(currencyIsoCode)
Sets the value of the CurrencyIsoCode field of the `CalculateTaxResponse` object.

setDescription(dscptn)
Sets the value of the Description field of the `CalculateTaxResponse` object.

setDocumentCode(documentCode)
Sets the value of the DocumentCode field of the `CalculateTaxResponse` object.

setEffectiveDate(effectiveDate)
Sets the value of the EffectiveDate field of the `CalculateTaxResponse` object.

setLineItems(lineItems)
Sets the value of the LineItems field of the `CalculateTaxResponse` object.

setReferenceDocumentCode(referenceDocumentCode)
Sets the value of the ReferenceDocumentCode field of the `CalculateTaxResponse` object.

setReferenceEntityId(referenceEntityId)
Sets the value of the ReferenceEntityId field of the `CalculateTaxResponse` object.

setStatus(status)
Sets the value of the Status field of the `CalculateTaxResponse` object.

setStatusDescription(statusDescription)
Sets the value of the StatusDescription field of the `CalculateTaxResponse` object.

setTaxTransactionId(taxTrxnId)
Sets the value of the TaxTransactionId field of the `CalculateTaxResponse` object.

setTaxTransactionType(taxTransactionType)
Sets the value of the TaxTransactionType field of the `CalculateTaxResponse` object.

setTaxType(taxType)
Sets the value of the TaxType field of the `CalculateTaxResponse` object.

setTransactionDate(transactionDate)
Sets the value of the TransactionDate field of the `CalculateTaxResponse` object.

setAddresses(addresses)

Sets the value of the Addresses field using the addresses contained in an instance of the AddressesResponse class.

Signature

```
global void setAddresses(commercetax.AddressesResponse addresses)
```

Parameters

addresses
Type: AddressesResponse

Contains Ship To, Ship From, and Sold To addresses.

Return Value

Type: void

setAmountDetails(amountDetails)

Sets the value of the AmountDetails field using an instance of `AmountDetailsResponse`.

Signature

```
global void setAmountDetails(commercetax.AmountDetailsResponse amountDetails)
```

Parameters

amountDetails
Type: AmountDetailsResponse

The tax amount details for a line item on which tax was calculated.

Return Value

Type: void

setCurrencyIsoCode(currencyIsoCode)

Sets the value of the CurrencyIsoCode field of the `CalculateTaxResponse` object.

Signature

```
global void setCurrencyIsoCode(String currencyIsoCode)
```

Parameters

currencyIsoCode
Type: String

Three-letter ISO 4217 currency code associated with a tax object.

Return Value

Type: void

setDescription(dscptn)

Sets the value of the Description field of the `CalculateTaxResponse` object.

Signature

```
global void setDescription(String dscptn)
```

Parameters

dscptn
Type: String

Optional description for providing more information about the calculate tax response.

Return Value

Type: void

setDocumentCode(documentCode)

Sets the value of the DocumentCode field of the `CalculateTaxResponse` object.

Signature

```
global void setDocumentCode(String documentCode)
```

Parameters

documentCode
Type: String

Code for a tax document that's created by the tax engine for the calculation process.

Return Value

Type: void

setEffectiveDate(effectiveDate)

Sets the value of the EffectiveDate field of the `CalculateTaxResponse` object.

Signature

```
global void setEffectiveDate(Datetime effectiveDate)
```

Parameters

effectiveDate
Type: Datetime

The date a tax calculation action takes effect. This parameter is optional and is provided only for recordkeeping purpose. Additionally, this parameter is used to determine the tax rates or rules and overrides the transaction date. For example, if the tax calculation request is placed on January 3 and the transaction date is January 1, you can set the effective date as January 1.

Return Value

Type: void

setLineItems(lineItems)

Sets the value of the LineItems field of the `CalculateTaxResponse` object.

Signature

```
global void setLineItems(List<commercetax.LineItemResponse> lineItems)
```

Parameters

lineItems
Type: List<LineItemResponse>

Response object that the tax adapter populates from the response of the external tax engine.

Return Value

Type: void

setReferenceDocumentCode(referenceDocumentCode)

Sets the value of the ReferenceDocumentCode field of the `CalculateTaxResponse` object.

Signature

```
global void setReferenceDocumentCode(String referenceDocumentCode)
```

Parameters

referenceDocumentCode
Type: String

Code for a reference document used in the tax calculation process.

Return Value

Type: void

setReferenceEntityId(referenceEntityId)

Sets the value of the ReferenceEntityId field of the `CalculateTaxResponse` object.

Signature

```
global void setReferenceEntityId(String referenceEntityId)
```

Parameters

referenceEntityId
Type: String

ID of an entity related to the line items submitted for tax calculation. For example, if order items were sent for tax calculation, you could use the ID of their parent order.

Return Value

Type: void

setStatus(status)

Sets the value of the Status field of the `CalculateTaxResponse` object.

Signature

```
global void setStatus(commercetax.TaxTransactionStatus status)
```

Parameters

status
Type: TaxTransactionStatus

Indicates whether a tax transaction has been committed.

Return Value

Type: void

setStatusDescription(statusDescription)

Sets the value of the StatusDescription field of the `CalculateTaxResponse` object.

Signature

```
global void setStatusDescription(String statusDescription)
```

Parameters

statusDescription
Type: String

Optional value for providing more information about a tax transaction's status.

Return Value

Type: void

setTaxTransactionId(taxTrxnId)

Sets the value of the TaxTransactionId field of the `CalculateTaxResponse` object.

Signature

```
public void setTaxTransactionId(String taxTrxnId)
```

Parameters

taxTrxnId
Type: String

The ID of the Salesforce tax transaction entity that stores information about the tax calculation transaction.

Return Value

Type: void

setTaxTransactionType(taxTransactionType)

Sets the value of the TaxTransactionType field of the `CalculateTaxResponse` object.

Signature

```
global void setTaxTransactionType(commercetax.TaxTransactionType taxTransactionType)
```

Parameters

taxTransactionType
Type: TaxTransactionType

Whether the tax transaction was for a credit, debit, or voided transaction.

Return Value

Type: void

setTaxType(taxType)

Sets the value of the TaxType field of the `CalculateTaxResponse` object.

Signature

```
global void setTaxType(commercetax.CalculateTaxType taxType)
```

Parameters

taxType
Type: CalculateTaxType

Indicates whether a tax calculation request is for estimated or actual tax.

Return Value

Type: void

setTransactionDate(transactionDate)

Sets the value of the TransactionDate field of the `CalculateTaxResponse` object.

Signature

```
global void setTransactionDate(Datetime transactionDate)
```

Parameters

transactionDate
Type: Datetime

The date that the tax transaction occurred.

Return Value

Type: void

CalculateTaxType Enum

Shows whether a tax calculation request is for estimated or actual tax.

Usage

Used by the CalculateTaxRequest and CalculateTaxResponse class methods.

Enum Values

The `commercetax.CalculateTaxType` enum includes these values.

Value	Description
`Actual`	Specifies that the tax calculation service should calculate the finalized (actual) tax for the requested line items.
`Estimated`	Specifies that the tax calculation service should estimate the tax for the requested line items.

CustomTaxAttributesResponse Class

Sets additional data or custom attributes in the tax response.

Namespace

CommerceTax

CustomTaxAttributesResponse Constructors

Learn more about the available constructors with the `CustomTaxAttributesResponse` class.

CustomTaxAttributesResponse Methods

Learn more about the available methods with the `CustomTaxAttributesResponse` class.

CustomTaxAttributesResponse Constructors

Learn more about the available constructors with the `CustomTaxAttributesResponse` class.

The `CustomTaxAttributesResponse` class includes these constructors.

CustomTaxAttributesResponse()

Constructor to set additional data or custom attributes in the tax response.

`CustomTaxAttributesResponse()`

Constructor to set additional data or custom attributes in the tax response.

Signature

```
global CustomTaxAttributesResponse()
```

CustomTaxAttributesResponse Methods

Learn more about the available methods with the `CustomTaxAttributesResponse` class.

The `CustomTaxAttributesResponse` class includes these methods.

setData(data)

Sets additional data or custom attributes in the tax response.

`setData(data)`

Sets additional data or custom attributes in the tax response.

Signature

```
global void setData(Map<String, Object> data)
```

Parameters

data

Type: Map<String, Object>

Additional data or custom attributes to be included in the tax response.

Return Value

Type: void

ErrorResponse Class

Use to respond with an error after receiving errors from the PaymentGatewayAdapter methods of the CommercePayments namespace, such as request-forbidden responses, custom validation errors, or expired API tokens.

Namespace

CommerceTax

Example

This example snippet of a mock tax adapter shows a hypothetical scenario to demo an error response. The adapter receives request information from `TaxEngineContext` and stores it in an instance of `CalculateTaxRequest`. If the request's `documentCode` property is null or indicates an error, the adapter returns an error response with information about the error.

```
global virtual class MockAdapter implements commercetax.TaxEngineAdapter {

     global commercetax.TaxEngineResponse processRequest(commercetax.TaxEngineContext
taxEngineContext) {
          commercetax.RequestType requestType = taxEngineContext.getRequestType();
          commercetax.CalculateTaxRequest request =
(commercetax.CalculateTaxRequest)taxEngineContext.getRequest();

if(request.documentCode == null) {
              return new commercetax.ErrorResponse(commercetax.resultcode.TaxEngineError,
'404', 'documentCode  is mandatory');
          }
          if(request.documentCode == 'TaxEngineError') {
              return new commercetax.ErrorResponse(commercetax.resultcode.TaxEngineError,
'504', 'documentCode  - not supported');
          }
          if(request.documentCode == 'simulateValidationFailureInAdapter') {
              return new commercetax.ErrorResponse(commercetax.resultcode.TaxEngineError,
'400', 'validations for documentCode failed in adapter');
          }
          if(request.documentCode == 'simulateMalformedErrorInAdapter') {
                      return new
commercetax.ErrorResponse(commercetax.resultcode.TaxEngineError, null, 'malformed adapter
 error response');
          }
          if(request.documentCode == 'simulateTaxEngineProcessFailure') {
              return new commercetax.ErrorResponse(commercetax.resultcode.TaxEngineError,
```

```
'500', 'Tax Engine couldnt process your request');
         }
```

ErrorResponse Constructors
Learn more about the available constructors with the `ErrorResponse` class.

ErrorResponse Constructors

Learn more about the available constructors with the `ErrorResponse` class.

The `ErrorResponse` class includes these constructors.

ErrorResponse(resultCode, errorCode, errorMessage)
Constructor to initialize an `ErrorResponse` object from the result code, error code, and error message sent from the tax engine.

`ErrorResponse(resultCode, errorCode, errorMessage)`

Constructor to initialize an `ErrorResponse` object from the result code, error code, and error message sent from the tax engine.

Signature

```
global ErrorResponse(commercetax.ResultCode resultCode, String errorCode, String
errorMessage)
```

Parameters

resultCode
Type: ResultCode

Code for the type of result sent by the tax engine.

errorCode
Type: String

Code for the type of error sent by the tax engine.

Codes must match the HTTP status codes to be returned to the user. Here are a few examples:

- If the status code is for a bad request, set `errorCode` to `400`.
- If the status code is for a forbidden request, set `errorCode` to `403`.
- If `errorCode` isn't a valid HTTP status code, a 500 internal server error is returned.

errorMessage
Type: String

The error message sent by the tax engine.

HeaderTaxAddressesRequest Class

Captures the address values that are applicable for the quote or order transaction.

Namespace

CommerceTax

HeaderTaxAddressesRequest Constructors

Learn more about the constructors available with the `HeaderTaxAddressesRequest` class.

HeaderTaxAddressesRequest Properties

Learn more about the available properties with the `HeaderTaxAddressesRequest` class.

HeaderTaxAddressesRequest Methods

Learn more about the available methods with the `HeaderTaxAddressesRequest` class.

HeaderTaxAddressesRequest Constructors

Learn more about the constructors available with the `HeaderTaxAddressesRequest` class.

The `HeaderTaxAddressesRequest` class includes these constructors.

HeaderTaxAddressesRequest(shipFrom, shipTo, soldTo, billTo, taxEngineAddress)

Constructor for initializing the required addresses of the tax addresses request such as the ship from, ship to, sold to, and bill to addresses. This constructor is intended for test usage and throws an exception if used outside of the Apex test context.

HeaderTaxAddressesRequest(shipFrom, shipTo, soldTo, billTo, taxEngineAddress)

Constructor for initializing the required addresses of the tax addresses request such as the ship from, ship to, sold to, and bill to addresses. This constructor is intended for test usage and throws an exception if used outside of the Apex test context.

Signature

```
global HeaderTaxAddressesRequest(commercetax.TaxAddressRequest shipFrom,
commercetax.TaxAddressRequest shipTo, commercetax.TaxAddressRequest soldTo,
commercetax.TaxAddressRequest billTo, commercetax.TaxAddressRequest taxEngineAddress)
```

Parameters

shipFrom

Type: TaxAddressRequest

Address where a line item was shipped from.

shipTo

Type: TaxAddressRequest

Address where a line item was shipped to.

soldTo

Type: TaxAddressRequest

Address of the line item's buyer.

billTo

Type: TaxAddressRequest

Person or group who was billed for the line item.

taxEngineAddress

Type: TaxAddressRequest

Address that the tax engine uses to calculate tax.

HeaderTaxAddressesRequest Properties

Learn more about the available properties with the `HeaderTaxAddressesRequest` class.

The `HeaderTaxAddressesRequest` class includes these properties.

billTo
Specifies the billTo address for a line item on which tax was calculated.

shipFrom
Specifies the shipFrom address for a line item on which tax was calculated.

shipTo
Specifies the shipTo address for a line item on which tax was calculated.

soldTo
Specifies the soldTo address for a line item on which tax was calculated.

taxEngineAddress
Address used by the tax engine when calculating tax for a line item.

billTo

Specifies the billTo address for a line item on which tax was calculated.

Signature

```
global commercetax.TaxAddressRequest billTo {get; set;}
```

Property Value

Type: TaxAddressRequest

shipFrom

Specifies the shipFrom address for a line item on which tax was calculated.

Signature

```
global commercetax.TaxAddressRequest shipFrom {get; set;}
```

Property Value

Type: TaxAddressRequest

shipTo

Specifies the shipTo address for a line item on which tax was calculated.

Signature

```
global commercetax.TaxAddressRequest shipTo {get; set;}
```

Property Value

Type: TaxAddressRequest

soldTo

Specifies the soldTo address for a line item on which tax was calculated.

Signature

```
global commercetax.TaxAddressRequest soldTo {get; set;}
```

Property Value

Type: TaxAddressRequest

taxEngineAddress

Address used by the tax engine when calculating tax for a line item.

Signature

```
global commercetax.TaxAddressRequest taxEngineAddress {get; set;}
```

Property Value

Type: TaxAddressRequest

HeaderTaxAddressesRequest Methods

Learn more about the available methods with the `HeaderTaxAddressesRequest` class.

The `HeaderTaxAddressesRequest` class includes these methods.

equals(obj)
: Maintains the integrity of lists of type `HeaderTaxAddressesRequest` by determining the equality of external objects in a list. This method is dynamic and is based on the `equals()` method in Java.

hashCode()
: Maintains the integrity of lists of type `TaxAddressesRequest` by determining the uniqueness of the external objects in a list.

toString()
: Converts a value to a string.

equals(obj)

Maintains the integrity of lists of type `HeaderTaxAddressesRequest` by determining the equality of external objects in a list. This method is dynamic and is based on the `equals()` method in Java.

Signature

```
global Boolean equals(Object obj)
```

Parameters

obj
Type: Object

External object whose key is to be validated.

Return Value

Type: Boolean

hashCode()

Maintains the integrity of lists of type `TaxAddressesRequest` by determining the uniqueness of the external objects in a list.

Signature

```
global Integer hashCode()
```

Return Value

Type: Integer

toString()

Converts a value to a string.

Signature

```
global String toString()
```

Return Value

Type: String

ImpositionResponse Class

Stores details of tax impositions from the external tax engine.

Namespace

CommerceTax

Example

In this mock adapter example, the adapter sets the `TaxDetailsResponse.setImposition()` method parameter to null if the request's document code indicates that the tax calculation didn't require any exceptions. Otherwise, it creates an instance of `ImpositionResponse` and sets its SubType and Type values, and then assigns it to `TaxDetailsResponse`.

```
if(request.DocumentCode == 'SetsNullForResponseWithoutException'){
  taxDetailsResponse.setImposition(null);
                     }else{
                        commercetax.ImpositionResponse imposition = new
commercetax.ImpositionResponse();
                        imposition.setSubType('subtype');
                        imposition.setType('type');
                        taxDetailsResponse.setImposition(imposition);
                   }
```

ImpositionResponse Methods
Learn more about the available methods with the `ImpositionResponse` class.

ImpositionResponse Methods

Learn more about the available methods with the `ImpositionResponse` class.

The `ImpositionResponse` class includes these methods.

setId(id)
Sets the ID field of the `ImpositionResponse` class.

setName(name)
Sets the Name field of the `ImpositionResponse` class.

setSubType(subType)
Sets the SubType field of the `ImpositionResponse` class.

setType(type)
Sets the Type field of the `ImpositionResponse` class.

`setId(id)`

Sets the ID field of the `ImpositionResponse` class.

Signature

`global void setId(String id)`

Parameters

`id`
Type: String
User-defined ID value used for referencing the tax imposition.

Return Value

Type: void

`setName(name)`

Sets the Name field of the `ImpositionResponse` class.

Signature

```
global void setName(String name)
```

Parameters

`name`
Type: String

Optional user-defined name for the tax imposition response.

Return Value

Type: void

`setSubType(subType)`

Sets the SubType field of the `ImpositionResponse` class.

Signature

```
global void setSubType(String subType)
```

Parameters

`subType`
Type: String

Many tax calculation organizations use types and subtypes to categorize their tax imposition procedures. If the tax engine you use follows this process, set the subtype with this parameter.

Return Value

Type: void

`setType(type)`

Sets the Type field of the `ImpositionResponse` class.

Signature

```
public void setType(String type)
```

Parameters

type

Type: String

Many tax calculation organizations use types and subtypes to categorize their tax imposition procedures. If the tax engine you use follows this process, set the type with this parameter.

Return Value

Type: void

JurisdictionResponse Class

Stores details from the external tax engine about the tax jurisdiction used in the tax calculation process. A tax jurisdiction represents a government entity that collects tax.

Namespace

CommerceTax

Example

In this mock adapter example, the adapter sets the `TaxDetailsResponse.setJurisdiction()` method parameter to null if the request's document code indicates that the tax calculation didn't require any exceptions. Otherwise, it creates an instance of `JurisdictionResponse` and sets its address values. Because this code represents a mock adapter, the example defines the address parameters directly. In a standard implementation, the jurisdiction's setters receive values passed from the eternal tax engine.

```
if(request.DocumentCode == 'SetsNullForResponseWithoutException'){
                        taxDetailsResponse.setJurisdiction(null);
                    }else{
                       commercetax.JurisdictionResponse jurisdiction = new
commercetax.JurisdictionResponse();
                       jurisdiction.setCountry('country');
                       jurisdiction.setRegion('region');
                       jurisdiction.setName('name');
                       jurisdiction.setStateAssignedNumber('stateAssignedNo');
                       jurisdiction.setId('id');
                       jurisdiction.setLevel('level');
                       taxDetailsResponse.setJurisdiction(jurisdiction);

}
```

JurisdictionResponse Methods

Learn more about the available methods with the `JurisdictionResponse` class.

JurisdictionResponse Methods

Learn more about the available methods with the `JurisdictionResponse` class.

The `JurisdictionResponse` class includes these methods.

[setCountry(country)](#)
Sets the Country field of the `JurisdictionResponse` class.

[setId(id)](#)
Sets the ID field of the `JurisdictionResponse` class.

[setLevel(level)](#)
Sets the Level field of the `JurisdictionResponse` class.

[setName(name)](#)
Sets the Name field of the `JurisdictionResponse` class.

[setRegion(region)](#)
Sets the Region value of the `JurisdictionResponse` class.

[setStateAssignedNumber(stateAssignedNo)](#)
Sets the StateAssignedNumber field of the `JurisdictionResponse` class.

setCountry(country)

Sets the Country field of the `JurisdictionResponse` class.

Signature

```
global void setCountry(String country)
```

Parameters

country
Type: String

The country of the tax jurisdiction entity's address.

Return Value

Type: void

setId(id)

Sets the ID field of the `JurisdictionResponse` class.

Signature

```
global void setId(String id)
```

Parameters

id
Type: String

User-defined Id value used to reference the jurisdiction response.

Return Value

Type: void

setLevel(level)

Sets the Level field of the `JurisdictionResponse` class.

Signature

```
global void setLevel(String level)
```

Parameters

level
Type: String

Level value used in the jurisdiction entity's address.

Return Value

Type: void

setName(name)

Sets the Name field of the `JurisdictionResponse` class.

Signature

```
global void setName(String name)
```

Parameters

name
Type: String

Optional user-defined name field for referencing the jurisdiction response.

Return Value

Type: void

setRegion(region)

Sets the Region value of the `JurisdictionResponse` class.

Signature

```
global void setRegion(String region)
```

Parameters

region
Type: String

Region value used in the tax jurisdiction entity's address.

Return Value

Type: void

setStateAssignedNumber(stateAssignedNo)

Sets the StateAssignedNumber field of the `JurisdictionResponse` class.

Signature

`global void setStateAssignedNumber(String stateAssignedNo)`

Parameters

stateAssignedNo
Type: String

State assigned number value of the tax jurisdiction entity's address.

Return Value

Type: void

LineItemResponse Class

Response class that stores details of a list of one or more line items on which the tax engine has calculated tax.

Namespace

CommerceTax

Example

This example uses a `LineItemResponse` list to store information about each line item that was processed as part of the request. For simplicity, the sample code uses a static value of 1 for the tax rate. However, most integrations typically have a more complex process for determining a tax rate. Most integrations also build a `TaxDetailsResponse` list to store the actual tax value information that they assign to each line item in the `LineItemResponse` list.

```
Double totalTax = 0.0;
             Double totalAmount = 0.0;
             List<commercetax.LineItemResponse> lineItemResponses = new
List<commercetax.LineItemResponse>();
             for(Commercetax.TaxLineItemRequest lineItem : request.lineItems){
                 commercetax.AddressesResponse addressesRes = new
commercetax.AddressesResponse();
                 if(request.DocumentCode == 'SetsNullForResponseWithoutException'){
```

```
                    addressesRes.setShipFrom(null);
                    addressesRes.setShipTO(null);
                    addressesRes.setSoldTo(null);
                }else{
                 commercetax.AddressResponse addRes = new commercetax.AddressResponse();

                    addRes.setLocationCode('locationCode');
                    addressesRes.setShipFrom(addRes);
                    addressesRes.setShipTO(addRes);
                    addressesRes.setSoldTo(addRes);
                }
                commercetax.LineItemResponse lineItemResponse = new
commercetax.LineItemResponse();
                Double totalLineTax = 0;
                List<commercetax.TaxDetailsResponse> taxDetailsResponses = new
List<commercetax.TaxDetailsResponse>();
                for(integer i =0;i<1;i++){
                    Integer rate  = 1;
                    Double taxableAmount = lineItem.amount;
                    commercetax.TaxDetailsResponse taxDetailsResponse = new
commercetax.TaxDetailsResponse();
                    taxDetailsResponse.setRate(Double.valueOf(rate));
                    taxDetailsResponse.setTaxableAmount(taxableAmount);
                    Double tax = taxableAmount*rate;
                    totalLineTax+=tax;
                    taxDetailsResponse.setTax(taxableAmount*rate);
                    taxDetailsResponse.setExemptAmount(0);
                    taxDetailsResponse.setExemptReason('exemptReason');
                    taxDetailsResponse.setTaxRegionId('taxRegionId');

taxDetailsResponse.setTaxId(String.valueOf(getRandomInteger(0,2323233)));
                    taxDetailsResponse.setSerCode('serCode');
                    taxDetailsResponse.setTaxAuthorityTypeId('taxAuthorityTypeId');
                    if(request.DocumentCode == 'SetsNullForResponseWithoutException'){
                        taxDetailsResponse.setImposition(null);
                    }else{
                        commercetax.ImpositionResponse imposition = new
commercetax.ImpositionResponse();
                        imposition.setSubType('subtype');
                        imposition.setType('type');
                        taxDetailsResponse.setImposition(imposition);
                    }

                    if(request.DocumentCode == 'SetsNullForResponseWithoutException'){
                        taxDetailsResponse.setJurisdiction(null);
                    }else{
                        commercetax.JurisdictionResponse jurisdiction = new
commercetax.JurisdictionResponse();
                        jurisdiction.setCountry('country');
                        jurisdiction.setRegion('region');
                        jurisdiction.setName('name');
                        jurisdiction.setStateAssignedNumber('stateAssignedNo');
                        jurisdiction.setId('id');
                        jurisdiction.setLevel('level');
```

```
                taxDetailsResponse.setJurisdiction(jurisdiction);
            }

            taxDetailsResponses.add(taxDetailsResponse);
        }
        lineItemResponse.setTaxes(taxDetailsResponses);
        totalTax +=totalLineTax;
        totalAmount+=lineItem.amount;
```

LineItemResponse Methods

Learn more about the available methods with the `LineItemResponse` class.

LineItemResponse Methods

Learn more about the available methods with the `LineItemResponse` class.

The `LineItemResponse` class includes these methods.

setAddresses(addresses)

Sets the Addresses field on the `LineItemResponse` using an instance of `AddressesResponse` class.

setAmountDetails(amountDetails)

Sets the Amount Details field on the `LineItemResponse` using an instance of `AmountDetails`.

setCustomTaxAttributes(customTaxAttributes)

Uses an instance of `CustomTaxAttributesResponse` class to include additional attributes in the tax response at line item level.

setEffectiveDate(effectiveDate)

Sets the EffectiveDate field on the `LineItemResponse` class. Effective Date fields are optional fields that store the date that a transaction takes effect. We provide these fields only for recordkeeping purposes – for example, if you must report an effective date to an external general ledger system. Salesforce doesn't use them to calculate any tax or payment values.

setIsTaxable(isTaxable)

Sets the IsTaxable field on the `LineItemResponse` class.

setLineNumber(lineNumber)

Sets the LineNumber field on the `LineItemResponse` class.

setProductCode(productCode)

Sets the ProductCode field on the `LineItemResponse` class.

setQuantity(quantity)

Sets the Quantity field on the `LineItemResponse` class.

setTaxCode(taxCode)

Sets the TaxCode field on the `LineItemResponse`.

setTaxes(taxes)

Sets the Taxes field on a `LineItemResponse`.

`setAddresses(addresses)`

Sets the Addresses field on the `LineItemResponse` using an instance of `AddressesResponse` class.

Signature

```
global void setAddresses(commercetax.AddressesResponse addresses)
```

Parameters

addresses
Type: AddressesResponse

Class that contains methods to set the Ship To, Ship From, and Sold To address information.

Return Value

Type: void

setAmountDetails(amountDetails)

Sets the Amount Details field on the `LineItemResponse` using an instance of `AmountDetails`.

Signature

```
global void setAmountDetails(commercetax.AmountDetailsResponse amountDetails)
```

Parameters

amountDetails
Type: AmountDetailsResponse

Class that contains methods to set the tax amount, total amount with tax, total amount, and exempt amount.

Return Value

Type: void

setCustomTaxAttributes(customTaxAttributes)

Uses an instance of `CustomTaxAttributesResponse` class to include additional attributes in the tax response at line item level.

Signature

```
global void setCustomTaxAttributes(commercetax.CustomTaxAttributesResponse
customTaxAttributes)
```

Parameters

customTaxAttributes
Type: CustomTaxAttributesResponse

Additional data or custom attributes to include in the tax response.

Return Value

Type: void

setEffectiveDate(effectiveDate)

Sets the EffectiveDate field on the `LineItemResponse` class. Effective Date fields are optional fields that store the date that a transaction takes effect. We provide these fields only for recordkeeping purposes – for example, if you must report an effective date to an external general ledger system. Salesforce doesn't use them to calculate any tax or payment values.

Signature

```
global void setEffectiveDate(Datetime effectiveDate)
```

Parameters

effectiveDate
: Type: Datetime

  Optional field that stores the date that a transaction takes effect.

Return Value

Type: void

setIsTaxable(isTaxable)

Sets the IsTaxable field on the `LineItemResponse` class.

Signature

```
global void setIsTaxable(Boolean isTaxable)
```

Parameters

isTaxable
: Type: Boolean

  Whether line items were taxed as part of the tax calculation request.

Return Value

Type: void

setLineNumber(lineNumber)

Sets the LineNumber field on the `LineItemResponse` class.

Signature

```
global void setLineNumber(String lineNumber)
```

Parameters

lineNumber
: Type: String

User-defined number used to identify a line item.

Return Value

Type: void

setProductCode(productCode)

Sets the ProductCode field on the `LineItemResponse` class.

Signature

`global void setProductCode(String productCode)`

Parameters

productCode
Type: String

Code for the product that a line item represents.

Return Value

Type: void

setQuantity(quantity)

Sets the Quantity field on the `LineItemResponse` class.

Signature

`global void setQuantity(Double quantity)`

Parameters

quantity
Type: Double

Quantity of a line item.

Return Value

Type: void

setTaxCode(taxCode)

Sets the TaxCode field on the `LineItemResponse`.

Signature

`global void setTaxCode(String taxCode)`

Parameters

taxCode

Type: String

Federal code that an individual or business uses to pay their taxes to a federal or state government. The tax engine uses this code during the tax calculation process.

Return Value

Type: void

setTaxes(taxes)

Sets the Taxes field on a `LineItemResponse`.

Signature

```
global void setTaxes(List<commercetax.TaxDetailsResponse> taxes)
```

Parameters

taxes

Type: List<TaxDetailsResponse>

Tax values applied to a line item in the `LineItemResponse` list. This information is stored in a list of `TaxDetailsResponses`, which contains values such as tax, taxable amount, and tax rate.

Return Value

Type: void

LineTaxAddressesRequest Class

Stores details of the addresses applied per line item in a tax calculation request.

Namespace

CommerceTax

LineTaxAddressesRequest Constructors
Learn more about the constructors available with the `LineTaxAddressesRequest` class.

LineTaxAddressesRequest Properties
Learn more about the available properties with the `LineTaxAddressesRequest` class.

LineTaxAddressesRequest Methods
Learn more about the available methods with the `LineTaxAddressesRequest` class.

LineTaxAddressesRequest Constructors

Learn more about the constructors available with the `LineTaxAddressesRequest` class.

The `LineTaxAddressesRequest` class includes these constructors.

LineTaxAddressesRequest(shipFrom, shipTo, soldTo, billTo, taxEngineAddress)

Constructor for initializing the required addresses for a line item of the tax addresses request such as the ship to, ship from, and bill to addresses. This constructor is intended for test usage and throws an exception if used outside of the Apex test context.

LineTaxAddressesRequest(shipFrom, shipTo, soldTo, billTo, taxEngineAddress)

Constructor for initializing the required addresses for a line item of the tax addresses request such as the ship to, ship from, and bill to addresses. This constructor is intended for test usage and throws an exception if used outside of the Apex test context.

Signature

```
global LineTaxAddressesRequest(commercetax.TaxAddressRequest shipFrom,
commercetax.TaxAddressRequest shipTo, commercetax.TaxAddressRequest soldTo,
commercetax.TaxAddressRequest billTo, commercetax.TaxAddressRequest taxEngineAddress)
```

Parameters

shipFrom
TaxAddressRequest

Address where a line item was shipped from.

shipTo
TaxAddressRequest

Address where a line item is shipped to.

soldTo
TaxAddressRequest

Address of the line item's buyer.

billTo
TaxAddressRequest

Person or group who was billed for the line item.

taxEngineAddress
TaxAddressRequest

Address that the tax engine uses to calculate tax.

LineTaxAddressesRequest Properties

Learn more about the available properties with the `LineTaxAddressesRequest` class.

The `LineTaxAddressesRequest` class includes these properties.

billTo

The Bill To address for a line item.

shipFrom

The Ship From address for a line item.

shipTo

The Ship To address for a line item.

soldTo

The Sold To address for a line item.

billTo

The Bill To address for a line item.

Signature

```
global commercetax.TaxAddressRequest billTo {get; set;}
```

Property Value

Type: TaxAddressRequest

shipFrom

The Ship From address for a line item.

Signature

```
global commercetax.TaxAddressRequest shipFrom {get; set;}
```

Property Value

Type: TaxAddressRequest

shipTo

The Ship To address for a line item.

Signature

```
global commercetax.TaxAddressRequest shipTo {get; set;}
```

Property Value

Type: TaxAddressRequest

soldTo

The Sold To address for a line item.

Signature

```
global commercetax.TaxAddressRequest soldTo {get; set;}
```

Property Value

Type: TaxAddressRequest

LineTaxAddressesRequest Methods

Learn more about the available methods with the `LineTaxAddressesRequest` class.

The `LineTaxAddressesRequest` class includes these methods.

equals(obj)
Maintains the integrity of lists of type `LineTaxAddressesRequest` by determining the equality of external objects in a list. This method is dynamic and is based on the `equals()` method in Java.

hashCode()
Maintains the integrity of lists of type `LineTaxAddressesRequest` by determining the uniquness of the external object records in a list.

toString()
Converts a value to a string.

`equals(obj)`

Maintains the integrity of lists of type `LineTaxAddressesRequest` by determining the equality of external objects in a list. This method is dynamic and is based on the `equals()` method in Java.

Signature

```
global Boolean equals(Object obj)
```

Parameters

`obj`
Type: Object

External object whose key is to be validated.

Return Value

Type: Boolean

`hashCode()`

Maintains the integrity of lists of type `LineTaxAddressesRequest` by determining the uniquness of the external object records in a list.

Signature

```
global Integer hashCode()
```

Return Value

Type: Integer

toString()

Converts a value to a string.

Signature

```
global String toString()
```

Return Value

Type: String

RequestType Enum

Shows the type of tax request made to the tax engine.

Usage

Used by the TaxEngineContext class method.

Enum Values

The `commercetax.RequestType` enum includes these values.

Value	Description
`CalculateTax`	Represents a request to calculate tax on a list of taxable line items.

ResultCode Enum

Code that represents the results of a tax request made to the tax engine.

Usage

Used by the ErrorResponse class method.

Enum Values

The `commercetax.ResultCode` enum includes these values.

Value	Description
`TaxEngineError`	Represents an error that occurred during the tax request process.
`ReferenceDocumentCodeMissing`	Specifies if the document mentioned as a `referenceDocumentCode` value isn't available in the tax engine.

RuleDetailsResponse Class

Contains details about the tax rules used for tax calculation.

Namespace

CommerceTax

RuleDetailsResponse Methods
Learn more about the available methods with the `RuleDetailsResponse` class.

RuleDetailsResponse Methods

Learn more about the available methods with the `RuleDetailsResponse` class.

The `RuleDetailsResponse` includes these methods.

RuleDetailsResponse()
Contains information about the tax rules used when calculating tax for line items.

setNonTaxableRuleId(nonTaxableRuleId)
Sets the NonTaxableRuleId field of the `RuleDetailsResponse`.

setNonTaxableType(nonTaxableType)
Sets the NonTaxableType field of the `RuleDetailsResponse`.

setRateRuleId(rateRuleId)
Sets the RateRuleId field of the `RuleDetailsResponse`.

setRateSourceId(rateSourceId)
Sets the RateSourceId field on the `RuleDetailsResponse`.

RuleDetailsResponse()

Contains information about the tax rules used when calculating tax for line items.

Signature

```
global void RuleDetailsResponse()
```

Return Value

Type: void

setNonTaxableRuleId(nonTaxableRuleId)

Sets the NonTaxableRuleId field of the `RuleDetailsResponse`.

Signature

```
global void setNonTaxableRuleId(String nonTaxableRuleId)
```

Parameters

nonTaxableRuleId
Type: String

ID of the tax rule applied to non-taxable line items.

Return Value

Type: void

`setNonTaxableType(nonTaxableType)`

Sets the NonTaxableType field of the `RuleDetailsResponse`.

Signature

```
global void setNonTaxableType(String nonTaxableType)
```

Parameters

`nonTaxableType`
Type: String

Reason (from several possible types) that a line item is non-taxable.

Return Value

Type: void

`setRateRuleId(rateRuleId)`

Sets the RateRuleId field of the `RuleDetailsResponse`.

Signature

```
global void setRateRuleId(String rateRuleId)
```

Parameters

`rateRuleId`
Type: String

ID of the tax rule used to determine a tax rate.

Return Value

Type: void

`setRateSourceId(rateSourceId)`

Sets the RateSourceId field on the `RuleDetailsResponse`.

Signature

```
global void setRateSourceId(String rateSourceId)
```

Parameters

rateSourceId
Type: String

ID of the source object used for calculating tax rate.

Return Value

Type: void

TaxAddressesRequest Class

Contains methods to get and set tax address values.

Namespace

CommerceTax

TaxAddressesRequest Constructors
Learn more about the available constructors with the `TaxAddressesRequest` class.

TaxAddressesRequest Properties
Learn more about the available properties with the `TaxAddressesRequest` class.

TaxAddressesRequest Methods
Learn more about the available methods with the `TaxAddressesRequest` class.

TaxAddressesRequest Constructors

Learn more about the available constructors with the `TaxAddressesRequest` class.

The `TaxAddressesRequest` class includes these constructors.

TaxAddressesRequest(shipFrom, shipTo, soldTo, billTo, taxEngineAddress)
Constructor for defining addresses for the tax addresses request. This constructor is intended for test usage and throws an exception if used outside of the Apex test context.

TaxAddressesRequest(shipFrom, shipTo, soldTo, billTo, taxEngineAddress)

Constructor for defining addresses for the tax addresses request. This constructor is intended for test usage and throws an exception if used outside of the Apex test context.

Signature

```
global TaxAddressesRequest(commercetax.TaxAddressRequest shipFrom,
commercetax.TaxAddressRequest shipTo, commercetax.TaxAddressRequest soldTo,
commercetax.TaxAddressRequest billTo, commercetax.TaxAddressRequest taxEngineAddress)
```

Parameters

shipFrom

TaxAddressRequest

The address where a line item was shipped from.

shipTo

TaxAddressRequest

The address where a line item is shipped to.

soldTo

TaxAddressRequest

The address of the line item's buyer.

billTo

TaxAddressRequest

The person or group who was billed for the line item.

taxEngineAddress

TaxAddressRequest

The address that the tax engine uses to calculate tax.

TaxAddressesRequest Properties

Learn more about the available properties with the `TaxAddressesRequest` class.

The `TaxAddressesRequest` class includes these properties.

billTo

The Bill To address for a line item.

shipFrom

The Ship From address for a line item.

shipTo

The Ship To address for a line item.

soldTo

The Sold To address for a line item.

taxEngineAddress

The Tax Engine Address for a line item.

billTo

The Bill To address for a line item.

Signature

```
global commercetax.TaxAddressRequest billTo {get; set;}
```

Property Value

TaxAddressRequest

shipFrom

The Ship From address for a line item.

Signature

```
global commercetax.TaxAddressRequest shipFrom {get; set;}
```

Property Value

TaxAddressRequest

shipTo

The Ship To address for a line item.

Signature

```
public commercetax.TaxAddressRequest shipTo {get; set;}
```

Property Value

TaxAddressRequest

soldTo

The Sold To address for a line item.

Signature

```
global commercetax.TaxAddressRequest soldTo {get; set;}
```

Property Value

TaxAddressRequest

taxEngineAddress

The Tax Engine Address for a line item.

Signature

```
global commercetax.TaxAddressRequest taxEngineAddress {get; set;}
```

Property Value

TaxAddressRequest

TaxAddressesRequest Methods

Learn more about the available methods with the `TaxAddressesRequest` class.

The `TaxAddressesRequest` class includes these methods.

[equals(obj)]
Maintains the integrity of lists of type `TaxAddressesRequest` by determining the equality of external objects in a list. This method is dynamic and is based on the `equals()` method in Java.

[hashCode()]
Maintains the integrity of lists of type `TaxAddressesRequest` by determining the uniqueness of the external object records in a list.

[toString()]
Converts a value to a string.

equals(obj)

Maintains the integrity of lists of type `TaxAddressesRequest` by determining the equality of external objects in a list. This method is dynamic and is based on the `equals()` method in Java.

Signature

```
global Boolean equals(Object obj)
```

Parameters

obj
Type: Object

External object whose key is to be validated.

Return Value

Type: Boolean

hashCode()

Maintains the integrity of lists of type `TaxAddressesRequest` by determining the uniqueness of the external object records in a list.

Signature

```
global Integer hashCode()
```

Return Value

Type: Integer

toString()

Converts a value to a string.

Signature

```
global String toString()
```

Return Value

Type: String

TaxAddressRequest Class

Contains address details used for tax calculation.

Namespace

CommerceTax

TaxAddressRequest Constructors
Learn more about the available constructors with the `TaxAddressRequest` class.

TaxAddressRequest Properties
Learn more about the available properties with the `TaxAddressRequest` class.

TaxAddressRequest Methods
Learn more about the available methods with the `TaxAddressRequest` class.

TaxAddressRequest Constructors

Learn more about the available constructors with the `TaxAddressRequest` class.

The `TaxAddressRequest` class includes these constructors.

TaxAddressRequest(city, country, latitude, longitude, postalCode, state, street, locationCode)
Initializes the `TaxAddressRequest` object using address details. This constructor is intended for test usage and throws an exception if used outside of the Apex test context.

`TaxAddressRequest(city, country, latitude, longitude, postalCode, state, street, locationCode)`

Initializes the `TaxAddressRequest` object using address details. This constructor is intended for test usage and throws an exception if used outside of the Apex test context.

Signature

```
global TaxAddressRequest(String city, String country, Double latitude, Double longitude,
String postalCode, String state, String street, String locationCode)
```

Parameters

city
Type: String

City used in an address, which is required for tax calculation.

country

Type: String

Country used in an address, which is required for tax calculation.

latitude

Type: Double

Latitude used in an address, which is required for tax calculation.

longitude

Type: Double

Longitude used in an address, which is required for tax calculation.

postalCode

Type: String

Postal code used in an address, which is required for tax calculation.

state

Type: String

State used in an address, which is required for tax calculation.

street

Type: String

Street used in an address, which is required for tax calculation.

locationCode

Type: String

Location code used in an address, which is required for tax calculation.

TaxAddressRequest Properties

Learn more about the available properties with the `TaxAddressRequest` class.

The `TaxAddressRequest` class includes these properties.

city

City used in an address, which is required for tax calculation.

country

Country used in an address, which is required for tax calculation.

countryCode

Country code used in an address, which is required for tax calculation.

latitude

Latitude used in an address, which is required for tax calculation.

locationCode

Location code used in an address, which is required for tax calculation.

longitude

Longitude used in an address, which is required for tax calculation.

postalCode

Postal code used in an address, which is required for tax calculation.

state
State used in an address, which is required for tax calculation.

stateCode
State code used in an address, which is required for tax calculation.

street
Street used in an address, which is required for tax calculation.

city

City used in an address, which is required for tax calculation.

Signature

```
global String city {get; set;}
```

Property Value

Type: String

country

Country used in an address, which is required for tax calculation.

Signature

```
global String country {get; set;}
```

Property Value

Type: String

countryCode

Country code used in an address, which is required for tax calculation.

Signature

```
global String countryCode {get; set;}
```

Property Value

Type: String

latitude

Latitude used in an address, which is required for tax calculation.

Signature

```
global Double latitude {get; set;}
```

Property Value

Type: Double

locationCode

Location code used in an address, which is required for tax calculation.

Signature

```
global String locationCode {get; set;}
```

Property Value

Type: String

longitude

Longitude used in an address, which is required for tax calculation.

Signature

```
global Double longitude {get; set;}
```

Property Value

Type: Double

postalCode

Postal code used in an address, which is required for tax calculation.

Signature

```
global String postalCode {get; set;}
```

Property Value

Type: String

state

State used in an address, which is required for tax calculation.

Signature

```
global String state {get; set;}
```

Property Value

Type: String

`stateCode`

State code used in an address, which is required for tax calculation.

Signature

```
global String stateCode {get; set;}
```

Property Value

Type: String

`street`

Street used in an address, which is required for tax calculation.

Signature

```
global String street {get; set;}
```

Property Value

Type: String

TaxAddressRequest Methods

Learn more about the available methods with the `TaxAddressRequest` class.

The `TaxAddressRequest` class includes these methods.

- equals(obj)
  Maintains the integrity of lists of type TaxAddressRequest by determining the equality of external objects in a list. This method is dynamic and based on the `equals()` method in Java.
- hashCode()
  Maintains the integrity of lists of type `TaxAddressRequest` by determining the uniqueness of the external object in a list.
- toString()
  Converts a date to a string.

`equals(obj)`

Maintains the integrity of lists of type TaxAddressRequest by determining the equality of external objects in a list. This method is dynamic and based on the `equals()` method in Java.

Signature

```
global Boolean equals(Object obj)
```

Parameters

obj
Type: Object

External object whose key is to be validated.

Return Value

Type: Boolean

hashCode()

Maintains the integrity of lists of type `TaxAddressRequest` by determining the uniqueness of the external object in a list.

Signature

```
global Integer hashCode()
```

Return Value

Type: Integer

toString()

Converts a date to a string.

Signature

```
global String toString()
```

Return Value

Type: String

TaxApiException Class

Contains details about any exceptions during the tax calculation process. Extends the `ApexBaseException` class.

Namespace

CommerceTax

TaxApiException Constructors
Learn more about the available constructors with the `TaxApiException` class.

TaxApiException Constructors

Learn more about the available constructors with the `TaxApiException` class.

The `TaxApiException` class includes these constructors.

TaxApiException(var1, var2)

Initializes the `TaxApiException` class using an `Exception` and a string to provide more details about the exception. This constructor is intended for test usage and throws an exception if used outside of the Apex test context.

TaxApiException(var1)

Initializes the `TaxApiException` class using an `Exception`. This constructor is intended for test usage and throws an exception if used outside of the Apex test context.

TaxApiException()

Initializes the `TaxApiException` class without any initialized parameters. This constructor is intended for test usage and throws an exception if used outside of the Apex test context.

TaxApiException(var1, var2)

Initializes the `TaxApiException` class using an `Exception` and a string to provide more details about the exception. This constructor is intended for test usage and throws an exception if used outside of the Apex test context.

Signature

```
global TaxApiException(String var1, Exception var2)
```

Parameters

var1

Type: String

Text that provides more information about the returned exception.

var2

Type: Exception

An exception denotes an error that disrupts the normal flow of code execution. You can use Apex built-in exceptions or create custom exceptions. All exceptions have common methods.

TaxApiException(var1)

Initializes the `TaxApiException` class using an `Exception`. This constructor is intended for test usage and throws an exception if used outside of the Apex test context.

Signature

```
global TaxApiException(Exception var1)
```

Parameters

var1

Type: Exception

An exception denotes an error that disrupts the normal flow of code execution. You can use Apex built-in exceptions or create custom exceptions. All exceptions have common methods.

TaxApiException()

Initializes the `TaxApiException` class without any initialized parameters. This constructor is intended for test usage and throws an exception if used outside of the Apex test context.

Signature

```
global TaxApiException()
```

TaxCustomerDetailsRequest Class

Contains customer details used in tax calculation.

Namespace

CommerceTax

TaxCustomerDetailsRequest Constructors
Learn more about the available constructors with the `TaxCustomerDetailsRequest` class.

TaxCustomerDetailsRequest Properties
Learn more about the available properties with the `TaxCustomerDetailsRequest` class.

TaxCustomerDetailsRequest Methods
Learn more about the available methods with the `TaxCustomerDetailsRequest` class.

TaxCustomerDetailsRequest Constructors

Learn more about the available constructors with the `TaxCustomerDetailsRequest` class.

The `TaxCustomerDetailsRequest` class includes these constructors.

TaxCustomerDetailsRequest(accountId, code, exemptionNo, exemptionReason)
Initializes the `TaxCustomerDetailsRequest` object. This constructor is intended for test usage and throws an exception if used outside of the Apex test context.

TaxCustomerDetailsRequest(accountId, code, exemptionNo, exemptionReason)

Initializes the `TaxCustomerDetailsRequest` object. This constructor is intended for test usage and throws an exception if used outside of the Apex test context.

Signature

```
global TaxCustomerDetailsRequest(String accountId, String code, String exemptionNo,
String exemptionReason)
```

Parameters

accountId
Type: String

The customer account ID for the line items sent for tax calculation.

code

Type: String

The tax code used during tax calculation.

exemptionNo

Type: String

The exemption number applied to any tax exempt line items.

exemptionReason

Type: String

The reason that certain line items are tax exempt.

TaxCustomerDetailsRequest Properties

Learn more about the available properties with the `TaxCustomerDetailsRequest` class.

The `TaxCustomerDetailsRequest` class includes these properties.

accountId
Customer account that contains the line items sent for tax calculation.

code
Tax code used during tax calculation.

exemptionNo
Number used to qualify a line item for tax exemption.

exemptionReason
Reason why a line item qualifies for tax exemption.

taxCertificateId
ID of a tax certificate used for tax calculation.

accountId

Customer account that contains the line items sent for tax calculation.

Signature

```
global String accountId {get; set;}
```

Property Value

Type: String

code

Tax code used during tax calculation.

Signature

```
global String code {get; set;}
```

Property Value

Type: String

exemptionNo

Number used to qualify a line item for tax exemption.

Signature

```
global String exemptionNo {get; set;}
```

Property Value

Type: String

exemptionReason

Reason why a line item qualifies for tax exemption.

Signature

```
global String exemptionReason {get; set;}
```

Property Value

Type: String

taxCertificateId

ID of a tax certificate used for tax calculation.

Signature

```
global String taxCertificateId {get; set;}
```

Property Value

Type: String

TaxCustomerDetailsRequest Methods

Learn more about the available methods with the `TaxCustomerDetailsRequest` class.

The `TaxCustomerDetailsRequest` class includes these methods.

- equals(obj)
  Maintains the integrity of lists of type `TaxCustomerDetailsRequest` by determining the equality of external objects in a list. This method is dynamic and based on the `equals()` method in Java.

hashCode()

Maintains the integrity of lists of type `TaxCustomerDetailsRequest` by determining the uniqueness of the external objects in a list.

toString()

Converts a value to a string.

equals(obj)

Maintains the integrity of lists of type `TaxCustomerDetailsRequest` by determining the equality of external objects in a list. This method is dynamic and based on the `equals()` method in Java.

Signature

`global Boolean equals(Object obj)`

Parameters

obj

Type: Object

External object whose key is to be validated.

Return Value

Type: Boolean

hashCode()

Maintains the integrity of lists of type `TaxCustomerDetailsRequest` by determining the uniqueness of the external objects in a list.

Signature

`global Integer hashCode()`

Return Value

Type: Integer

toString()

Converts a value to a string.

Signature

`global String toString()`

Return Value

Type: String

TaxDetailsResponse Class

Stores details of the tax values that an external tax engine calculates in response to a tax calculation request.

Namespace

CommerceTax

Usage

If your tax calculation request contains multiple line items, we recommend building your adapter using a list of `TaxDetailsResponse` instances. Each instance represents the tax details calculated for a given line item.

Example

```
List<commercetax.TaxDetailsResponse> taxDetailsResponses = new
List<commercetax.TaxDetailsResponse>();
                 for(integer i =0;i<1;i++){
                     Integer rate  = 1;
                     Double taxableAmount = lineItem.amount;
                     commercetax.TaxDetailsResponse taxDetailsResponse = new
commercetax.TaxDetailsResponse();
                     taxDetailsResponse.setRate(Double.valueOf(rate));
                     taxDetailsResponse.setTaxableAmount(taxableAmount);
                     Double tax = taxableAmount*rate;
                     totalLineTax+=tax;
                     taxDetailsResponse.setTax(taxableAmount*rate);
                     taxDetailsResponse.setExemptAmount(0);
                     taxDetailsResponse.setExemptReason('exemptReason');
                     taxDetailsResponse.setTaxRegionId('taxRegionId');

taxDetailsResponse.setTaxId(String.valueOf(getRandomInteger(0,2323233)));
                     taxDetailsResponse.setSerCode('serCode');
                     taxDetailsResponse.setTaxAuthorityTypeId('taxAuthorityTypeId');
                     if(request.DocumentCode == 'SetsNullForResponseWithoutException'){
                         taxDetailsResponse.setImposition(null);
                     }else{
                         commercetax.ImpositionResponse imposition = new
commercetax.ImpositionResponse();
                         imposition.setSubType('subtype');
                         imposition.setType('type');
                         taxDetailsResponse.setImposition(imposition);
                     }

                     if(request.DocumentCode == 'SetsNullForResponseWithoutException'){
                         taxDetailsResponse.setJurisdiction(null);
                     }else{
                         commercetax.JurisdictionResponse jurisdiction = new
commercetax.JurisdictionResponse();
                         jurisdiction.setCountry('country');
                         jurisdiction.setRegion('region');
                         jurisdiction.setName('name');
                         jurisdiction.setStateAssignedNumber('stateAssignedNo');
```

```
                jurisdiction.setId('id');
                jurisdiction.setLevel('level');
                taxDetailsResponse.setJurisdiction(jurisdiction);
            }

            taxDetailsResponses.add(taxDetailsResponse);
        }
        lineItemResponse.setTaxes(taxDetailsResponses);
        totalTax +=totalLineTax;
        totalAmount+=lineItem.amount;
```

TaxDetailsResponse Methods
Learn more about the available methods with the `TaxDetailsResponse` class.

TaxDetailsResponse Methods

Learn more about the available methods with the `TaxDetailsResponse` class.

The `TaxDetailsResponse` class includes these methods.

[setTaxRuleDetails(taxRuleDetails)](#)
Sets the TaxRuleDetails field of the `TaxDetailsResponse` class.

[setTaxableAmount(taxableAmount)](#)
Sets the TaxableAmount field of the `TaxDetailsResponse class`.

setCustomTaxAttributes(customTaxAttributes)

Uses an instance of `CustomTaxAttributesResponse` class to include additional attributes in the tax response at the tax line item level.

Signature

```
global void setCustomTaxAttributes(commercetax.CustomTaxAttributesResponse
customTaxAttributes)
```

Parameters

customTaxAttributes
Type: CustomTaxAttributesResponse

Additional data or custom attributes to include in the tax response.

Return Value

Type: void

setExemptAmount(exemptAmount)

Sets the ExemptAmount field of the `TaxDetailsResponse` class.

Signature

```
global void setExemptAmount(Double exemptAmount)
```

Parameters

exemptAmount
Type: Double

Amount of tax on a line item that is exempt from tax calculation.

Return Value

Type: void

setExemptReason(reason)

Sets the ExemptReason field of the `TaxDetailsResponse` class.

Signature

```
global void setExemptReason(String reason)
```

Parameters

reason
Type: String

Optional user-defined information on why a tax exemption applies to a line item.

Return Value

Type: void

setImposition(imposition)

Sets the Imposition field of the `TaxDetailsResponse` class using an instance of the `ImpositionResponse` class.

Signature

```
global void setImposition(commercetax.ImpositionResponse imposition)
```

Parameters

imposition
Type: ImpositionResponse

Contains information about why tax was imposed on a line item.

Return Value

Type: void

setJurisdiction(jurisdiction)

Sets the Jurisdiction field of the `TaxDetailsResponse` using an instance of the `JurisdictionResponse` class.

Signature

```
global void setJurisdiction(commercetax.JurisdictionResponse jurisdiction)
```

Parameters

jurisdiction
Type: JurisdictionResponse

Contains address information about the tax jurisdiction used in the tax calculation process.

Return Value

Type: void

setRate(rate)

Sets the Rate field of the `TaxDetailsResponse` class.

Signature

```
global void setRate(Double rate)
```

Parameters

rate
: Type: Double

  Tax used during tax calculation. This value is often a decimal amount, such as 0.1 or 0.06, based on the applied tax percentage.

Return Value

Type: void

setSerCode(serCode)

Sets the Service Code field of the `TaxDetailsResponse` class.

Signature

```
global void setSerCode(String serCode)
```

Parameters

serCode
: Type: String

  Service code used in tax calculation.

Return Value

Type: void

setTax(tax)

Sets the Tax field of the `TaxDetailsResponse` class.

Signature

```
global void setTax(Double tax)
```

Parameters

tax
: Type: Double

  Amount of tax for a line item.

Return Value

Type: void

`setTaxAuthorityTypeId(taxAuthorityTypeId)`

Sets the TaxAuthorityTypeId field of the `TaxDetailsResponse` class.

Signature

`global void setTaxAuthorityTypeId(String taxAuthorityTypeId)`

Parameters

`taxAuthorityTypeId`
Type: String

ID of the organization that oversees tax collection.

Return Value

Type: void

`setTaxId(taxId)`

Sets the TaxId field of the `TaxDetailsResponse` class.

Signature

`global void setTaxId(String taxId)`

Parameters

`taxId`
Type: String

ID value used to determine the tax for an individual or business.

Return Value

Type: void

`setTaxRegionId(taxRegionId)`

Sets the TaxRegionId field on the `TaxDetailsResponse` class.

Signature

`global void setTaxRegionId(String taxRegionId)`

Parameters

`taxRegionId`
Type: String

ID of the tax region used in tax calculation. A tax region represents a geographical area where tax is applied.

Return Value

Type: void

`setTaxRuleDetails(taxRuleDetails)`

Sets the TaxRuleDetails field of the `TaxDetailsResponse` class.

Signature

```
global void setTaxRuleDetails(commercetax.RuleDetailsResponse taxRuleDetails)
```

Parameters

`taxRuleDetails`
Type: RuleDetailsResponse

Information about the Salesforce tax rules used during tax calculation.

Return Value

Type: void

`setTaxableAmount(taxableAmount)`

Sets the TaxableAmount field of the `TaxDetailsResponse class`.

Signature

```
global void setTaxableAmount(Double taxableAmount)
```

Parameters

`taxableAmount`
Type: Double

Amount that can be taxed on a line item.

Return Value

Type: void

TaxEngineAdapter Interface

Retrieves information from the tax engine and evaluates the information to define tax details.

Namespace

CommerceTax

TaxEngineAdapter Methods
Learn more about the available methods with the `TaxEngineAdapter` class.

TaxEngineAdapter Example Implementation

Refer to the example implementation of the `TaxEngineAdapter` interface to accept information from a tax engine and evaluate the information to define tax details.

Tax Mappings for Quotes and Orders

You can extend and customize the tax interface for quotes and orders by using custom metadata types and tax mappings. These customizations help you with unique business requirements such as the inclusion of specific data for accurate calculations and audits.

TaxEngineAdapter Methods

Learn more about the available methods with the `TaxEngineAdapter` class.

The `TaxEngineAdapter` class includes these methods.

processRequest(requestType)

The `processRequest` method takes an instance of `TaxEngineContext` class and returns a response with the calculated tax details through the `TaxDetailsResponse` class or an error response through the `ErrorResponse` class.

processRequest(requestType)

The `processRequest` method takes an instance of `TaxEngineContext` class and returns a response with the calculated tax details through the `TaxDetailsResponse` class or an error response through the `ErrorResponse` class.

Signature

```
global commercetax.TaxEngineResponse processRequest(commercetax.TaxEngineContext var1)
```

Parameters

var1

Type: TaxEngineContext

Wrapper class that stores information about the type of a tax calculation request.

Return Value

Type: TaxEngineResponse

Generic interface representing a response from a tax engine.

TaxEngineAdapter Example Implementation

Refer to the example implementation of the `TaxEngineAdapter` interface to accept information from a tax engine and evaluate the information to define tax details.

Namespace

commercetax

Usage

The `TaxEngineAdapter` interface accepts information from the tax engine through the `TaxEngineContext` class. The interface evaluates the information to define tax in the response with details, such as tax amount and addresses. The response is used to update and create entities in the Salesforce org.

Use these steps to build a sample tax adapter implementation. Each tax adapter implementation varies based on your implementation requirements. Customize this example to suit your business requirements.

Example:

- The custom adapter class implements the `TaxEngineAdapter` interface. The `processRequest` method takes an instance of `TaxEngineContext` class and returns a response with the calculated tax details through the `TaxDetailsResponse` class or an error response through the `ErrorResponse` class.

```
global virtual class AvalaraAdapter implements commercetax.TaxEngineAdapter {
    global commercetax.TaxEngineResponse processRequest(commercetax.TaxEngineContext
 taxEngineContext) {
            commercetax.RequestType requestType = taxEngineContext.getRequestType();
            if(requestType == commercetax.RequestType.CalculateTax){
                return CalculateTaxService.getTax(taxEngineContext);
            }
            else
                return null;
     }
}
```

- This example shows the `CalculateTaxService` class.

```
global class CalculateTaxService {
    // ============================================================================

     // CONSTANT
    // ============================================================================

     private static final String AVALARA_ENDPOINT_URL_SANDBOX =
'https://sandbox-rest.avatax.com/api/v2';
     // Avalara Endpoint URL Production
     private static final String AVALARA_ENDPOINT_URL_PRODUCTION =
'https://rest.avatax.com/api/v2';
    private static final String TEST_REQUEST_BODY = '{  "id": -1,  "code": "00000131",
  "companyId": -1,  "date": "2017-02-03T00:00:00",  "taxDate": "2017-02-03T00:00:00",
  "status": "Temporary",  "type": "SalesOrder",  "reconciled": false,  "totalAmount":
 4000,  "totalExempt": 0,  "totalTax": 290,  "totalTaxable": 4000,
"totalTaxCalculated": 290,  "adjustmentReason": "NotAdjusted",  "locked": false,
"version": 1,  "modifiedDate": "2017-02-03T12:18:18.7347388Z",  "modifiedUserId":
53894,  "lines": [    {      "id": -1,      "transactionId": -1,      "lineNumber":
 "80241000000jNDCAA2",      "discountAmount": 0,      "exemptAmount": 0,
"exemptCertId": 0,      "isItemTaxable": true,      "lineAmount": 1000,
"reportingDate": "2017-02-03T00:00:00",      "tax": 72.5,      "taxableAmount":
1000,      "taxCalculated": 72.5,      "taxCode": "P0000000",      "taxDate":
"2017-02-03T00:00:00",      "taxIncluded": false,      "details": [        {
    "id": -1,          "transactionLineId": -1,          "transactionId": -1,
     "country": "US",          "region": "CA",          "exemptAmount": 0,
  "jurisCode": "06",          "jurisName": "CALIFORNIA",          "stateAssignedNo":
```

```
 "",          "jurisType": "STA",          "nonTaxableAmount": 0,          "rate":
 0.06,          "tax": 60,          "taxableAmount": 1000,          "taxType":
"Sales",          "taxName": "CA STATE TAX",          "taxAuthorityTypeId": 45,
       "taxCalculated": 60,          "rateType": "General"        },        {
     "id": -1,          "transactionLineId": -1,          "transactionId": -1,
      "country": "US",          "region": "CA",          "exemptAmount": 0,
   "jurisCode": "075",          "jurisName": "SAN FRANCISCO",
"stateAssignedNo": "",          "jurisType": "CTY",          "nonTaxableAmount": 0,
          "rate": 0.0025,          "tax": 2.5,          "taxableAmount": 1000,
      "taxType": "Sales",          "taxName": "CA COUNTY TAX",
"taxAuthorityTypeId": 45,          "taxCalculated": 2.5,          "rateType":
"General"        },        {          "id": -1,          "transactionLineId": -1,
         "transactionId": -1,          "country": "US",          "region": "CA",
        "exemptAmount": 0,          "jurisCode": "EMTV0",          "jurisName": "SAN
 FRANCISCO CO LOCAL TAX SL",          "stateAssignedNo": "38",          "jurisType":
 "STJ",          "nonTaxableAmount": 0,          "rate": 0.01,          "tax": 10,
          "taxableAmount": 1000,          "taxType": "Sales",          "taxName":
"CA SPECIAL TAX",          "taxAuthorityTypeId": 45,          "taxCalculated": 10,
          "rateType": "General"        }      ]    }  ]}';

    private static String getTestResponseString(){

     List<String> jsonResponse = new List<String> {
                                   '"id": 0',
                                   '"code": "testDocCode1231245984"',
                                   '"companyId": 468039',
                                   '"date": "2020-07-15"',
                                   '"paymentDate": "2020-07-15"',
                                   '"status": "Temporary"',
                                   '"type": "SalesOrder"',
                                   '"customerVendorCode": "testDocCode1234"',
                                   '"customerCode": "testDocCode1234"',
                                   '"reconciled": false',
                                   '"totalAmount": 232',
                                   '"totalExempt": 0',
                                   '"totalDiscount": 0',
                                   '"totalTax": 23.43',
                                   '"totalTaxable": 232',
                                   '"totalTaxCalculated": 23.43',
                                   '"adjustmentReason": "NotAdjusted"',
                                   '"locked": false',
                                   '"version": 1',
                                   '"exchangeRateEffectiveDate": "2020-07-15"',
                                   '"exchangeRate": 1',
                                 '"modifiedDate": "2020-08-13T11:19:20.4836636Z"',

                                   '"modifiedUserId": 53894',
                                   '"taxDate": "2020-07-15T00:00:00"',
                                   '"lines": [{"id": 0,"transactionId":
0,"lineNumber": "1","discountAmount": 0,"exemptAmount": 0,"exemptCertId":
0,"isItemTaxable": true,"itemCode": "","lineAmount": 232,"quantity":
1,"reportingDate": "2020-07-15","tax": 23.43,"taxableAmount": 232,"taxCalculated":
 23.43,"taxCode": "P0000000","taxCodeId": 8087,"taxDate":
"2020-07-15","taxOverrideType": "None","taxOverrideAmount": 0,"taxIncluded":
```

```
false,"details": [{"id": 0,"transactionLineId": 0,"transactionId": 0,"country":
"US","region": "WA","exemptAmount": 0,"jurisCode": "53","jurisName":
"WASHINGTON","stateAssignedNo": "","jurisType": "STA","jurisdictionType":
"State","nonTaxableAmount": 0,"rate": 0.065,"tax": 15.08,"taxableAmount":
232,"taxType": "Sales","taxSubTypeId": "S","taxName": "WA STATE
TAX","taxAuthorityTypeId": 45,"taxCalculated": 15.08,"rateType":
"General","rateTypeCode": "G","unitOfBasis": "PerCurrencyUnit","isNonPassThru":
false,"isFee": false},{"id": 0,"transactionLineId": 0,"transactionId": 0,"country":
 "US","region": "WA","exemptAmount": 0,"jurisCode": "033","jurisName":
"KING","stateAssignedNo": "1700","jurisType": "CTY","jurisdictionType":
"County","nonTaxableAmount": 0,"rate": 0,"tax": 0,"taxableAmount": 232,"taxType":
"Sales","taxSubTypeId": "S","taxName": "WA COUNTY TAX","taxAuthorityTypeId":
45,"taxCalculated": 0,"rateType": "General","rateTypeCode": "G","unitOfBasis":
"PerCurrencyUnit","isNonPassThru": false,"isFee": false}],"nonPassthroughDetails":
 [],"hsCode": "","costInsuranceFreight": 0,"vatCode": "","vatNumberTypeId": 0}]',
                                    '"addresses": [{"id": 0,"transactionId":
0,"boundaryLevel": "Address","line1": "255 S. King Street","line2": "","line3":
"","city": "Seattle","region": "WA","postalCode": "98104","country":
"US","taxRegionId": 2109700,"latitude": "47.59821","longitude": "-122.33108"}]',
                                    '"summary": [{"country": "US","region":
"WA","jurisType": "State","jurisCode": "53","jurisName":
"WASHINGTON","taxAuthorityType": 45,"stateAssignedNo": "","taxType":
"Sales","taxSubType": "S","taxName": "WA STATE TAX","rateType": "General","taxable":
 232,"rate": 0.065,"tax": 15.08,"taxCalculated": 15.08,"nonTaxable": 0,"exemption":
 0},{"country": "US","region": "WA","jurisType": "County","jurisCode":
"033","jurisName": "KING","taxAuthorityType": 45,"stateAssignedNo": "1700","taxType":
 "Sales","taxSubType": "S","taxName": "WA COUNTY TAX","rateType": "General","taxable":
 232,"rate": 0,"tax": 0,"taxCalculated": 0,"nonTaxable": 0,"exemption": 0}]'
                                };
            return '{' + String.join(jsonResponse, ',') + '}';
        }

    public static commercetax.TaxEngineResponse getTax(commercetax.TaxEngineContext
 taxEngineContext)
    {
        commercetax.CalculateTaxRequest request =
(commercetax.CalculateTaxRequest)taxEngineContext.getRequest();
        commercetax.calculatetaxtype requestType = request.taxtype;
        string referenceEntity = request.ReferenceEntityId;
        try{
            List<commercetax.TaxLineItemRequest> listOfLines = request.lineItems;
            if(!listOfLines.isEmpty()){
                HttpService sendHttpRequest = new HttpService();
                sendHttpRequest.addHeader('Content-type', 'application/json');
                String requestBody =
AvalaraJSONBuilder.getInstance().frameJsonForGetTaxOrderItem(request);
                sendHttpRequest.post('/transactions/create',requestBody);
                //system.debug('Request '+requestBody);
                String responseString = '';
                if(Test.isRunningTest()){
                    responseString = getTestResponseString();
                } else{
                    responseString = sendHttpRequest.getResponse().getBody();
                }
```

```
                //system.debug(sendHttpRequest.getResponse());
                //system.debug('response'+responseString);
                //responseString = TEST_REQUEST_BODY;
                system.debug('Heap size used ' +Limits.getHeapSize());

                if(!responseString.contains('error'))
                {
                    commercetax.CalculateTaxResponse response = new
commercetax.CalculateTaxResponse();
                    JsonSuccessParser jsonSuccessParserClass =
JsonSuccessParser.parse(responseString);
                    response.setTaxTransactionType(request.taxTransactionType);
                    response.setDocumentCode(jsonSuccessParserClass.code);

response.setReferenceDocumentCode(jsonSuccessParserClass.referenceCode);
                    if(jsonSuccessParserClass.status == 'Temporary')  {

response.setStatus(commercetax.TaxTransactionStatus.Uncommitted);
                    }
                    if(jsonSuccessParserClass.status == 'Committed') {

response.setStatus(commercetax.TaxTransactionStatus.Committed);
                    }
                    response.setTaxType(requestType);
                    commercetax.AmountDetailsResponse headerAmountResponse = new
commercetax.AmountDetailsResponse();

headerAmountResponse.setTotalAmountWithTax(jsonSuccessParserClass.totalAmount +
jsonSuccessParserClass.totaltax);

headerAmountResponse.setExemptAmount(jsonSuccessParserClass.totalExempt);

headerAmountResponse.setTotalAmount(jsonSuccessParserClass.totalAmount);

headerAmountResponse.setTaxAmount(jsonSuccessParserClass.totalTax);
                    response.setAmountDetails(headerAmountResponse);

response.setStatusDescription(jsonSuccessParserClass.adjustmentReason);

response.setEffectiveDate(date.valueof(jsonSuccessParserClass.taxDate));

response.setTransactionDate(date.valueof(jsonSuccessParserClass.transactionDate));
                    response.setReferenceEntityId(referenceEntity);
                    response.setTaxTransactionId(jsonSuccessParserClass.id);
                    response.setCurrencyIsoCode(request.currencyIsoCode);
                    List<commercetax.LineItemResponse> lineItemResponses = new
List<commercetax.LineItemResponse>();
                    for(JsonSuccessParser.Lines linesToProcess:
jsonSuccessParserClass.lines)
                    {
                        commercetax.LineItemResponse lineItemResponse = new
commercetax.LineItemResponse();
                        Double rateCalculated = 0.0;
                        List<commercetax.TaxDetailsResponse> taxDetailsResponses =
```

```
 new List<commercetax.TaxDetailsResponse>();
                        for(JsonSuccessParser.details linesDetails :
linesToProcess.details)
                        {
                            commercetax.TaxDetailsResponse taxDetailsResponse = new
 commercetax.TaxDetailsResponse();
                            if(linesDetails.exemptAmount != 0){

taxDetailsResponse.setExemptAmount(linesDetails.exemptAmount);
                                taxDetailsResponse.setExemptReason('Some reason we
 dont know');
                            }
                                commercetax.ImpositionResponse imposition = new
commercetax.ImpositionResponse();
                                    imposition.setSubType(linesDetails.taxName);
                                    imposition.setType(linesDetails.ratetype);
                                    imposition.setSubType(linesDetails.taxName);
                                    taxDetailsResponse.setImposition(imposition);
                                commercetax.JurisdictionResponse jurisdiction = new
 commercetax.JurisdictionResponse();
                                    jurisdiction.setCountry(linesDetails.country);
                                    jurisdiction.setRegion(linesDetails.region);
                                    jurisdiction.setName(linesDetails.jurisName);

jurisdiction.setStateAssignedNumber(linesDetails.stateAssignedNo);
                                    jurisdiction.setId(linesDetails.jurisCode);
                                    jurisdiction.setLevel(linesDetails.jurisType);
                                 taxDetailsResponse.setJurisdiction(jurisdiction);

                                    rateCalculated += linesDetails.rate;
                                taxDetailsResponse.setRate(rateCalculated);
                            taxDetailsResponse.setTax(linesDetails.taxCalculated);


taxDetailsResponse.setTaxableAmount(linesDetails.taxableAmount);

taxDetailsResponse.setTaxAuthorityTypeId(String.valueOf(linesDetails.taxAuthorityTypeId));

                                taxDetailsResponse.setTaxId(linesDetails.id);

taxDetailsResponse.setTaxRegionId(linesDetails.region);
                                taxDetailsResponses.add(taxDetailsResponse);

                        }
                            lineItemResponse.setTaxes(taxDetailsResponses);

lineItemResponse.setEffectiveDate(date.valueof(linesToProcess.taxDate));
                            lineItemResponse.setIsTaxable(true);
                                commercetax.AmountDetailsResponse amountResponse =
 new commercetax.AmountDetailsResponse();

amountResponse.setTaxAmount(linesToProcess.taxCalculated);

amountResponse.setTotalAmount(linesToProcess.lineAmount);
```

```
amountResponse.setTotalAmountWithTax(linesToProcess.lineAmount+linesToProcess.taxCalculated);


amountResponse.setExemptAmount(linesToProcess.exemptAmount);
                                lineItemResponse.setAmountDetails(amountResponse);

lineItemResponse.setIsTaxable(linesToProcess.isItemTaxable);
                         lineItemResponse.setProductCode(linesToProcess.itemCode);

                            lineItemResponse.setTaxCode(linesToProcess.taxCode);
                        lineItemResponse.setLineNumber(linesToProcess.lineNumber);

                            lineItemResponse.setQuantity(linesToProcess.quantity);
                            lineItemResponses.add(lineItemResponse);
                    }
                    response.setLineItems(lineItemResponses);
                    return response;
                }
                else
                {
                    JsonErrorParser jsonErrorParserClass =
JsonErrorParser.parse(responseString);
                    String message = null;
                    if(String.isNotBlank(jsonErrorParserClass.error.message))
                    {
                       message=jsonErrorParserClass.error.message;
                    }else{
                           String errorMessage = '';
                            for(JsonErrorParser.cls_details messageString :
jsonErrorParserClass.error.details)
                            {
                                if(String.isNotBlank(messageString.message) )
                                {
                                    errorMessage = messageString.message;
                                }
                            }
                            message = errorMessage;
                        }
                     return new
commercetax.ErrorResponse(commercetax.resultcode.TaxEngineError, '501', message);

                }
            }else return null;
        }
        catch (Exception e)
        {
            throw e;
        }
    }
}
```

- In the `HttpService` class, replace the `test` value in the endpoint variable with the name of the `TaxTypedNamedCredential` record. This class contains the credentials that are required to access your Avalara account through Salesforce.

```
public with sharing class HttpService
{
    // Attribute to implement singleton pattern for Order Product Service class
    private static HttpService httpServiceInstance;

    // VARIABLES

    private HttpResponse httpResponse;
    private Map<String,String> mapOfHeaderParameter = new Map<String,String>();
    private enum Method {GET, POST}

    /**
    * @name getInstance
    * @description get an Instance of Service class
    * @params NA
    * @return Http Service Class Instance
    */
    public static HttpService getInstance()
    {
        if (NULL == httpServiceInstance)
        {
            httpServiceInstance =  new HttpService();
        }
        return httpServiceInstance;
    }

    /**
    * @name get
    * @description Get Method to get a HTTP request
    */
    public void get(String endPoint)
    {
        send(newRequest(Method.GET, endPoint));
    }

    /**
    * @name post
    * @description Post Method to Post a HTTP request
    */
    public void post(String path, String requestBody)
    {
       String endPoint = 'callout:commerce.tax.TaxTypedNamedCredential:test'+path;
        send(newRequest(Method.POST, endPoint, requestBody));
    }

    /**
    * @name addHeader
    * @description addHeader Methods to add all the defualt Header's required fo
rthe request
```

```
*/
public void addHeader(String name, String value)
{
    mapOfHeaderParameter.put(name, value);
}

/**
* @name setHeader
* @description setHeader Methods to set setHeader for the request
*/
private void setHeader(HttpRequest request)
{
    for(String headerValue : mapOfHeaderParameter.keySet())
    {
        request.setHeader(headerValue, mapOfHeaderParameter.get(headerValue));
    }
}
/**
* @name newRequest
* @description newRequest Methods to make a new request
*/
private HttpRequest newRequest(Method method, String endPoint)
{
    return newRequest(method, endPoint, NULL);
}

/**
* @name newRequest
* @description newRequest Methods to make a new request
*/
private HttpRequest newRequest(Method method, String endPoint, String requestBody)

{
    HttpRequest request = new HttpRequest();
    request.setMethod(Method.name());
    setHeader(request);
    request.setEndpoint(endPoint);
    if (String.isNotBlank(requestBody))
    {
        request.setBody(requestBody);
    }
    request.setTimeout(120000);
    return request;
}

/**
* @name send
* @description send Methods to send a request
*/
private void send(HttpRequest request)
{
    try
    {
        Http http = new Http();
```

```
            httpResponse = http.send(request);
        }
        catch(System.CalloutException e)
        {
            system.debug('callout exception happened' + e.getMessage());
        }
        catch(Exception e)
        {
            system.debug('callout did not happen' + e.getMessage());
        }
    }

    /**
    * @name getResponse
    * @description getResponse Method to get the Response
    */
    public HttpResponse getResponse()
    {
        return httpResponse;
    }

    /**
    * @name getResponseToString
    * @description getResponse Method to get the Responses
    */
    public String getResponseToString()
    {
        return getResponse().toString();
    }
}
```

- Parse the `JsonSuccessParser` response object by using the `AvalaraJSONBuilder` class to build the response for your adapter.

  This example shows the `JsonSuccessParser` class.

```
global with sharing class JsonSuccessParser
{
  public static void consumeObject(JSONParser parser)
  {
    Integer depth = 0;
    do {
      JSONToken curr = parser.getCurrentToken();
      if (curr == JSONToken.START_OBJECT ||
        curr == JSONToken.START_ARRAY) {
        depth++;
      } else if (curr == JSONToken.END_OBJECT ||
        curr == JSONToken.END_ARRAY) {
        depth--;
      }
    } while (depth > 0 && parser.nextToken() != null);
  }

    public class Addresses {
        public String id {get;set;}
```

```
        public String transactionId {get;set;}
        public String boundaryLevel {get;set;}
        public String line1 {get;set;}
        public String city {get;set;}
        public String region {get;set;}
        public String postalCode {get;set;}
        public String country {get;set;}
        public Integer taxRegionId {get;set;}

        public Addresses(JSONParser parser) {
            while (parser.nextToken() != JSONToken.END_OBJECT) {
                if (parser.getCurrentToken() == JSONToken.FIELD_NAME) {
                    String text = parser.getText();
                    if (parser.nextToken() != JSONToken.VALUE_NULL) {
                        if (text == 'id') {
                            id = parser.getText();
                        } else if (text == 'transactionId') {
                            transactionId = parser.getText();
                        } else if (text == 'boundaryLevel') {
                            boundaryLevel = parser.getText();
                        } else if (text == 'line1') {
                            line1 = parser.getText();
                        } else if (text == 'city') {
                            city = parser.getText();
                        } else if (text == 'region') {
                            region = parser.getText();
                        } else if (text == 'postalCode') {
                            postalCode = parser.getText();
                        } else if (text == 'country') {
                            country = parser.getText();
                        } else if (text == 'taxRegionId') {
                            taxRegionId = parser.getIntegerValue();
                        } else {
                            consumeObject(parser);
                        }
                    }
                }
            }
        }
    }

    public class Details {
        public String id {get;set;}
        public String transactionLineId {get;set;}
        public String transactionId {get;set;}
        public String country {get;set;}
        public String region {get;set;}
        public Integer exemptAmount {get;set;}
        public String jurisCode {get;set;}
        public String jurisName {get;set;}
        public String stateAssignedNo {get;set;}
        public String jurisType {get;set;}
        public Integer nonTaxableAmount {get;set;}
        public Double rate {get;set;}
```

```
        public Double tax {get;set;}
        public Integer taxableAmount {get;set;}
        public String taxType {get;set;}
        public String taxName {get;set;}
        public Integer taxAuthorityTypeId {get;set;}
        public Double taxCalculated {get;set;}
        public String rateType {get;set;}

        public Details(JSONParser parser) {
            while (parser.nextToken() != JSONToken.END_OBJECT) {
                if (parser.getCurrentToken() == JSONToken.FIELD_NAME) {
                    String text = parser.getText();
                    if (parser.nextToken() != JSONToken.VALUE_NULL) {
                        if (text == 'id') {
                            id = parser.getText();
                        } else if (text == 'transactionLineId') {
                            transactionLineId = parser.getText();
                        } else if (text == 'transactionId') {
                            transactionId = parser.getText();
                        } else if (text == 'country') {
                            country = parser.getText();
                        } else if (text == 'region') {
                            region = parser.getText();
                        } else if (text == 'exemptAmount') {
                            exemptAmount = parser.getIntegerValue();
                        } else if (text == 'jurisCode') {
                            jurisCode = parser.getText();
                        } else if (text == 'jurisName') {
                            jurisName = parser.getText();
                        } else if (text == 'stateAssignedNo') {
                            stateAssignedNo = parser.getText();
                        } else if (text == 'jurisType') {
                            jurisType = parser.getText();
                        } else if (text == 'nonTaxableAmount') {
                            nonTaxableAmount = parser.getIntegerValue();
                        } else if (text == 'rate') {
                            rate = parser.getDoubleValue();
                        } else if (text == 'tax') {
                            tax = parser.getDoubleValue();
                        } else if (text == 'taxableAmount') {
                            taxableAmount = parser.getIntegerValue();
                        } else if (text == 'taxType') {
                            taxType = parser.getText();
                        } else if (text == 'taxName') {
                            taxName = parser.getText();
                        } else if (text == 'taxAuthorityTypeId') {
                            taxAuthorityTypeId = parser.getIntegerValue();
                        } else if (text == 'taxCalculated') {
                            taxCalculated = parser.getDoubleValue();
                        } else if (text == 'rateType') {
                            rateType = parser.getText();
                        } else {
                            consumeObject(parser);
                        }
```

```
                    }
                }
            }
        }
    }

    public class Messages {
        public String summary {get;set;}
        public String details {get;set;}
        public String refersTo {get;set;}
        public String severity {get;set;}
        public String source {get;set;}

        public Messages(JSONParser parser) {
            while (parser.nextToken() != JSONToken.END_OBJECT) {
                if (parser.getCurrentToken() == JSONToken.FIELD_NAME) {
                    String text = parser.getText();
                    if (parser.nextToken() != JSONToken.VALUE_NULL) {
                        if (text == 'summary') {
                            summary = parser.getText();
                        } else if (text == 'details') {
                            details = parser.getText();
                        } else if (text == 'refersTo') {
                            refersTo = parser.getText();
                        } else if (text == 'severity') {
                            severity = parser.getText();
                        } else if (text == 'source') {
                            source = parser.getText();
                        } else {
                            consumeObject(parser);
                        }
                    }
                }
            }
        }
    }

    public String id {get;set;}
    public String code {get;set;}
    public String referenceCode {get;set;}
    public Integer companyId {get;set;}
    public String taxDate {get;set;}
    public String transactionDate {get;set;}
    public String status {get;set;}
    public String type_Z {get;set;} // in json: type
    public Boolean reconciled {get;set;}
    public Integer totalAmount {get;set;}
    public Integer totalExempt {get;set;}
    public Double totalTax {get;set;}
    public Integer totalTaxable {get;set;}
    public Double totalTaxCalculated {get;set;}
    public String adjustmentReason {get;set;}
    public Boolean locked {get;set;}
    public Integer version {get;set;}
```

```
public String modifiedDate {get;set;}
public Integer modifiedUserId {get;set;}
public List<Lines> lines {get;set;}
public List<Addresses> addresses {get;set;}
public List<Summary> summary {get;set;}
public List<Messages> messages {get;set;}

public JsonSuccessParser(JSONParser parser) {
    while (parser.nextToken() != JSONToken.END_OBJECT) {
        if (parser.getCurrentToken() == JSONToken.FIELD_NAME) {
            String text = parser.getText();
            if (parser.nextToken() != JSONToken.VALUE_NULL) {
                if (text == 'id') {
                    id = parser.getText();
                } else if (text == 'code') {
                    code = parser.getText();
                } else if (text == 'referenceCode'){
                    referenceCode = parser.getText();
                } else if (text == 'companyId') {
                    companyId = parser.getIntegerValue();
                } else if (text == 'taxDate') {
                    taxDate = parser.getText();
                } else if (text == 'date') {
                    transactionDate = parser.getText();
                } else if (text == 'status') {
                    status = parser.getText();
                } else if (text == 'type') {
                    type_Z = parser.getText();
                } else if (text == 'reconciled') {
                    reconciled = parser.getBooleanValue();
                } else if (text == 'totalAmount') {
                    totalAmount = parser.getIntegerValue();
                } else if (text == 'totalExempt') {
                    totalExempt = parser.getIntegerValue();
                } else if (text == 'totalTax') {
                    totalTax = parser.getDoubleValue();
                } else if (text == 'totalTaxable') {
                    totalTaxable = parser.getIntegerValue();
                } else if (text == 'totalTaxCalculated') {
                    totalTaxCalculated = parser.getDoubleValue();
                } else if (text == 'adjustmentReason') {
                    adjustmentReason = parser.getText();
                } else if (text == 'locked') {
                    locked = parser.getBooleanValue();
                } else if (text == 'version') {
                    version = parser.getIntegerValue();
                } else if (text == 'modifiedDate') {
                    modifiedDate = parser.getText();
                } else if (text == 'modifiedUserId') {
                    modifiedUserId = parser.getIntegerValue();
                } else if (text == 'lines') {
                    lines = new List<Lines>();
                    while (parser.nextToken() != JSONToken.END_ARRAY) {
                        lines.add(new Lines(parser));
```

```
                    }
                } else if (text == 'addresses') {
                    addresses = new List<Addresses>();
                    while (parser.nextToken() != JSONToken.END_ARRAY) {
                        addresses.add(new Addresses(parser));
                    }
                } else if (text == 'summary') {
                    summary = new List<Summary>();
                    while (parser.nextToken() != JSONToken.END_ARRAY) {
                        summary.add(new Summary(parser));
                    }
                } else if (text == 'messages') {
                    messages = new List<Messages>();
                    while (parser.nextToken() != JSONToken.END_ARRAY) {
                        messages.add(new Messages(parser));
                    }
                } else {
                    consumeObject(parser);
                }
            }
        }
    }
}

public class Summary {
    public String country {get;set;}
    public String region {get;set;}
    public String jurisType {get;set;}
    public String jurisCode {get;set;}
    public String jurisName {get;set;}
    public Integer taxAuthorityType {get;set;}
    public String stateAssignedNo {get;set;}
    public String taxType {get;set;}
    public String taxName {get;set;}
    public String taxGroup {get;set;}
    public String rateType {get;set;}
    public Integer taxable {get;set;}
    public Double rate {get;set;}
    public Double tax {get;set;}
    public Double taxCalculated {get;set;}
    public Integer nonTaxable {get;set;}
    public Integer exemption {get;set;}

    public Summary(JSONParser parser) {
        while (parser.nextToken() != JSONToken.END_OBJECT) {
            if (parser.getCurrentToken() == JSONToken.FIELD_NAME) {
                String text = parser.getText();
                if (parser.nextToken() != JSONToken.VALUE_NULL) {
                    if (text == 'country') {
                        country = parser.getText();
                    } else if (text == 'region') {
                        region = parser.getText();
                    } else if (text == 'jurisType') {
                        jurisType = parser.getText();
```

```
                    } else if (text == 'jurisCode') {
                        jurisCode = parser.getText();
                    } else if (text == 'jurisName') {
                        jurisName = parser.getText();
                    } else if (text == 'taxAuthorityType') {
                        taxAuthorityType = parser.getIntegerValue();
                    } else if (text == 'stateAssignedNo') {
                        stateAssignedNo = parser.getText();
                    } else if (text == 'taxType') {
                        taxType = parser.getText();
                    } else if (text == 'taxName') {
                        taxName = parser.getText();
                    } else if (text == 'taxGroup') {
                        taxGroup = parser.getText();
                    } else if (text == 'rateType') {
                        rateType = parser.getText();
                    } else if (text == 'taxable') {
                        taxable = parser.getIntegerValue();
                    } else if (text == 'rate') {
                        rate = parser.getDoubleValue();
                    } else if (text == 'tax') {
                        tax = parser.getDoubleValue();
                    } else if (text == 'taxCalculated') {
                        taxCalculated = parser.getDoubleValue();
                    } else if (text == 'nonTaxable') {
                        nonTaxable = parser.getIntegerValue();
                    } else if (text == 'exemption') {
                        exemption = parser.getIntegerValue();
                    } else {
                        consumeObject(parser);
                    }
                }
            }
        }
    }
}

public class Lines {
    public String id {get;set;}
    public String transactionId {get;set;}
    public String lineNumber {get;set;}
    public Integer discountAmount {get;set;}
    public Integer exemptAmount {get;set;}
    public Integer exemptCertId {get;set;}
    public Boolean isItemTaxable {get;set;}
    public Integer lineAmount {get;set;}
    public Double quantity {get;set;}
    public String reportingDate {get;set;}
    public Double tax {get;set;}
    public Integer taxableAmount {get;set;}
    public Double taxCalculated {get;set;}
    public String taxCode {get;set;}
    public String taxDate {get;set;}
    public Boolean taxIncluded {get;set;}
```

```
        public List<Details> details {get;set;}
        public String itemCode {get;set;}
        public Lines(JSONParser parser) {
            while (parser.nextToken() != JSONToken.END_OBJECT) {
                if (parser.getCurrentToken() == JSONToken.FIELD_NAME) {
                    String text = parser.getText();
                    if (parser.nextToken() != JSONToken.VALUE_NULL) {
                        if (text == 'id') {
                            id = parser.getText();
                        } else if (text == 'transactionId') {
                            transactionId = parser.getText();
                        }else if (text == 'itemCode') {
                            itemCode = parser.getText();
                        }else if (text == 'lineNumber') {
                            lineNumber = parser.getText();
                        } else if (text == 'discountAmount') {
                            discountAmount = parser.getIntegerValue();
                        } else if (text == 'exemptAmount') {
                            exemptAmount = parser.getIntegerValue();
                        } else if (text == 'exemptCertId') {
                            exemptCertId = parser.getIntegerValue();
                        } else if (text == 'isItemTaxable') {
                            isItemTaxable = parser.getBooleanValue();
                        } else if (text == 'lineAmount') {
                            lineAmount = parser.getIntegerValue();
                        } else if (text == 'quantity') {
                            quantity = parser.getDoubleValue();
                        } else if (text == 'reportingDate') {
                            reportingDate = parser.getText();
                        } else if (text == 'tax') {
                            tax = parser.getDoubleValue();
                        } else if (text == 'taxableAmount') {
                            taxableAmount = parser.getIntegerValue();
                        } else if (text == 'taxCalculated') {
                            taxCalculated = parser.getDoubleValue();
                        } else if (text == 'taxCode') {
                            taxCode = parser.getText();
                        } else if (text == 'taxDate') {
                            taxDate = parser.getText();
                        } else if (text == 'taxIncluded') {
                            taxIncluded = parser.getBooleanValue();
                        } else if (text == 'details') {
                            details = new List<Details>();
                            while (parser.nextToken() != JSONToken.END_ARRAY) {
                                details.add(new Details(parser));
                            }
                        } else {
                            consumeObject(parser);
                        }
                    }
                }
            }
        }
    }
```

```
    public static JsonSuccessParser parse(String json)
    {
        return new JsonSuccessParser(System.JSON.createParser(json));
    }
}
```

Prepare your JSON request to call the Avalara endpoint by using the `AvalaraJSONBuilder` class.

```
public with sharing class AvalaraJSONBuilder
{
    private static AvalaraJSONBuilder avalaraJSONBuilderInstance;

    public static AvalaraJSONBuilder getInstance()
    {
        if (NULL == avalaraJSONBuilderInstance)
        {
            avalaraJSONBuilderInstance = new AvalaraJSONBuilder();
        }
        return avalaraJSONBuilderInstance;
    }

    public String frameJsonForGetTaxOrderItem(commercetax.CalculateTaxRequest
calculateTaxRequest)
    {
        try
        {
            Id accountid  = null;
            if(calculateTaxRequest.CustomerDetails.AccountId != null &&
calculateTaxRequest.CustomerDetails.AccountId != '')
             accountid = Id.valueof(calculateTaxRequest.CustomerDetails.AccountId);

            JSONGenerator jsonGeneratorInstance = JSON.createGenerator(true);
            jsonGeneratorInstance.writeStartObject();
            String type = null;
            if(calculateTaxRequest.taxtype == commercetax.CalculateTaxType.Actual)
                type ='SalesInvoice';
                else type = 'SalesOrder';
            jsonGeneratorInstance.writeStringField('type', type);
            if(calculateTaxRequest.SellerDetails != null)
                jsonGeneratorInstance.writeStringField('companyCode',
calculateTaxRequest.SellerDetails.code);
            else
                jsonGeneratorInstance.writeStringField('companyCode', 'billing2');
            if(calculateTaxRequest.isCommit != null) {
                jsonGeneratorInstance.writeBooleanField('commit',
calculateTaxRequest.isCommit);
            }
            if(calculateTaxRequest.documentcode != null){
                jsonGeneratorInstance.writeStringField('code',
calculateTaxRequest.documentcode);
            }else if(calculateTaxRequest.referenceEntityId != null) {
                jsonGeneratorInstance.writeStringField('code',
calculateTaxRequest.referenceEntityId);
```

```
            }
           if(calculateTaxRequest.CustomerDetails.code == null && accountid !=null)
 {
                Account acc = [select id, name from account where id=:accountid];
                jsonGeneratorInstance.writeStringField('customerCode', acc.name);
            } else {
                jsonGeneratorInstance.writeStringField('customerCode',
calculateTaxRequest.CustomerDetails.code);
            }
            if(calculateTaxRequest.EffectiveDate == null)
                jsonGeneratorInstance.writeDateField('date', system.today());
            else
                jsonGeneratorInstance.writeDateTimeField('date',
calculateTaxRequest.EffectiveDate);

            jsonGeneratorInstance.writeFieldName('lines');
            jsonGeneratorInstance.writeStartArray();
            for(integer i=0;i<1;i++){
                for(Commercetax.TaxLineItemRequest lineItem :
calculateTaxRequest.LineItems)
                {
                    jsonGeneratorInstance.writeStartObject();
                    if(lineItem.linenumber != null){
                        jsonGeneratorInstance.writeStringField('number',
lineItem.linenumber);
                    }
                    jsonGeneratorInstance.writeNumberField('quantity',
lineItem.Quantity);
                    jsonGeneratorInstance.writeNumberField('amount',
(lineItem.Amount));

jsonGeneratorInstance.writeStringField('taxCode',lineItem.taxCode);

                    jsonGeneratorInstance.writeFieldName('addresses');
                    jsonGeneratorInstance.writeStartObject();
                    jsonGeneratorInstance.writeFieldName('ShipFrom');
                    jsonGeneratorInstance.writeStartObject();
                    jsonGeneratorInstance.writeStringField('line1',
lineItem.addresses.shipfrom.street);
                    jsonGeneratorInstance.writeStringField('line2',
lineItem.addresses.shipfrom.street);
                    jsonGeneratorInstance.writeStringField('city',
lineItem.addresses.shipfrom.city);
                    jsonGeneratorInstance.writeStringField('region',
lineItem.addresses.shipfrom.state);
                    jsonGeneratorInstance.writeStringField('country',
lineItem.addresses.shipfrom.country);

jsonGeneratorInstance.writeStringField('postalCode',lineItem.addresses.shipfrom.postalcode);

                    jsonGeneratorInstance.writeEndObject();

                    jsonGeneratorInstance.writeFieldName('ShipTo');
                    jsonGeneratorInstance.writeStartObject();
```

```
                    jsonGeneratorInstance.writeStringField('line1',
lineItem.addresses.shipto.street);
                    jsonGeneratorInstance.writeStringField('line2',
lineItem.addresses.shipto.street);
                    jsonGeneratorInstance.writeStringField('city',
lineItem.addresses.shipto.city);
                    jsonGeneratorInstance.writeStringField('region',
lineItem.addresses.shipto.state);
                    jsonGeneratorInstance.writeStringField('country',
lineItem.addresses.shipto.country);

jsonGeneratorInstance.writeStringField('postalCode',lineItem.addresses.shipto.postalcode);

                    jsonGeneratorInstance.writeEndObject();

                    jsonGeneratorInstance.writeFieldName('pointOfOrderOrigin');
                    jsonGeneratorInstance.writeStartObject();
                    jsonGeneratorInstance.writeStringField('line1',
lineItem.addresses.soldto.street);
                    jsonGeneratorInstance.writeStringField('line2',
lineItem.addresses.soldto.street);
                    jsonGeneratorInstance.writeStringField('city',
lineItem.addresses.soldto.city);
                    jsonGeneratorInstance.writeStringField('region',
lineItem.addresses.soldto.state);
                    jsonGeneratorInstance.writeStringField('country',
lineItem.addresses.soldto.country);

jsonGeneratorInstance.writeStringField('postalCode',lineItem.addresses.soldto.postalcode);

                    jsonGeneratorInstance.writeEndObject();


                    if(lineItem.effectiveDate != null)
                    {
                        jsonGeneratorInstance.writeFieldName('taxOverride');
                        jsonGeneratorInstance.writeStartObject();
                        jsonGeneratorInstance.writeDateTimeField('taxDate',
lineItem.effectiveDate);
                        jsonGeneratorInstance.writeEndObject();
                    }
                    jsonGeneratorInstance.writeEndObject();
                    jsonGeneratorInstance.writeEndObject();
                }
            }
                jsonGeneratorInstance.writeEndArray();
            jsonGeneratorInstance.writeEndObject();
            return jsonGeneratorInstance.getAsString();
        }
        catch (Exception e)
        {
             throw e;
        }
```

```
    }
}
```

- Use the `JsonErrorParser` class to extract the error details, if any.

```
global with sharing class JsonErrorParser
{
    public cls_error error;

    public class cls_error
    {
        public String code;
        public String message;
        public String target;
        public cls_details[] details;
    }

    public class cls_details
    {
        public String code;
        public String message;
        public String description;
        public String faultCode;
        public String helpLink;
        public String severity;
    }
    public static JsonErrorParser parse(String json)
    {
       return (JsonErrorParser) System.JSON.deserialize(json, JsonErrorParser.class);

    }
}
```

Tax Mappings for Quotes and Orders

You can extend and customize the tax interface for quotes and orders by using custom metadata types and tax mappings. These customizations help you with unique business requirements such as the inclusion of specific data for accurate calculations and audits.

Tax callout extensions are supported for the Quote, QuoteLineItem, Order, and OrderItem objects to include additional fields to tax requests. You must manually write back tax response extensions to the objects. See custom metadata types to specify all your tax mapping definitions.

Request Mappings for Header Attributes

This table defines the request mappings between the header attributes of a tax callout and fields of applicable quote and order objects.

Header Attributes	Quote Mapping	Order Mapping
currencyIsoCode	If multi-currency is enabled, then this value is `Quote.CurrencyISOCode`. Otherwise, this value is NULL.	If multi-currency is enabled, then this value is `Order.CurrencyISOCode`. Otherwise, this value is NULL.
isCommit	`False`	`False`

Header Attributes	Quote Mapping	Order Mapping
referenceEntityId	Quote.ID	Order.ID
taxEngineId	TaxTreatment.TaxEngine.ID	TaxTreatment.TaxEngine.ID
transactionDate	Current System Date	System Date
sellerDetails	NULL	
code		TaxEngine.SellerCode
customerDetails		
accountId	Quote.AccountId	Order.AccountId
code	NULL	NULL
exemptionNo	NULL	NULL
exemptionReason	NULL	NULL
taxType	`Estimated`	`Estimated`
taxTransactionType	NULL	NULL
effectiveDate	NULL	NULL
addresses		
billTo	NULL	NULL
shipTo	NULL	NULL
shipFrom	NULL	NULL
soldTo	NULL	NULL
taxEngineAddress	TaxEngine.Address	TaxEngine.Address
referenceDocumentCode	NULL	NULL
description	NULL	NULL
documentCode	`Quote.ID-TaxEngineId`	`Order.ID-TaxEngineId`
shouldVoid	`FALSE`	`FALSE`
lineItems	Refer to the next line attributes section.	Refer to the next line attributes section.

Request Mappings for Line Attributes

This table defines the request mappings between the line attributes of a tax callout and fields of applicable quote line items and order products.

Line Attributes	Quote Line Item Mapping	Order Product Mapping
taxCode	TaxTreatment.TaxCode	TaxTreatment.TaxCode
productCode	TaxTreatment.ProductCode	TaxTreatment.ProductCode

Line Attributes	Quote Line Item Mapping	Order Product Mapping
productId	QuoteLineItem.Product2.Id	OrderItem.Product2.Id
amount	QuoteLineItem.TotalPrice	OrderItem.TotalPrice
effectiveDate	Current System Date	Current System Date
lineNumber	QuoteLineItem.Id	OrderItem.Id
description	NULL	NULL
quantity	QuoteLineItem.Quantity	OrderItem.Quantity
addresses		
billTo	Quote.BillingAddress. If Quote.BillingAddress is null, then this value is Quote.Account.BillingAddress.	Order.BillingAddress
shipTo	Quote.ShippingAddress. If Quote.ShippingAddress is null, then this value is Quote.Account.ShippingAddress.	Order.ShippingAddress
shipFrom	NULL	NULL
soldTo	NULL	NULL
productsku	QuoteLineItem.Product2.ProductCode	OrderItem.Product2.ProductCode
referenceDocumentCode	NULL	NULL

Response Mappings for Header Attributes

This table defines the response mappings between the header attributes of a tax callout and fields of applicable objects. Most response data is used for tax calculation and isn't persisted on quote or order records.

Header Attributes	Quote Mapping	Order Mapping
currencyIsoCode	Quote.CurrencyISOCode	Order.CurrencyISOCode
isCommit	Not returned.	Not returned.
referenceEntityId	Quote.ID	Order.ID
taxEngineId	TaxTreatment.TaxEngine.ID	TaxTreatment.TaxEngine.ID
transactionDate	System Date	System Date
sellerDetails	Not returned.	Not returned.
code	Not returned.	Not returned.
customerDetails	Not returned.	Not returned.
accountId	Not returned.	Not returned.
code	Not returned.	Not returned.

Header Attributes	Quote Mapping	Order Mapping
exemptionNo	Not returned.	Not returned.
exemptionReason	Not returned.	Not returned.
taxType	`Estimated`	`Estimated`
taxTransactionType	Not returned.	Not returned.
effectiveDate	System Date	System Date
addresses		
billTo	Not returned.	Not returned.
shipTo	locationCode -> locationCode	locationCode -> locationCode
shipFrom	Not returned.	Not returned.
soldTo	Not returned.	Not returned.
taxEngineAddress	Not returned.	Not returned.
referenceDocumentCode	Not returned.	Not returned.
description	Not returned.	Not returned.
documentCode	`Quote.ID-TaxEngineId`	`Order.ID-TaxEngineId`
status	`Uncommitted`	`Uncommitted`
taxEngineLogs	Not returned.	Not returned.
resultCode	Not returned.	Not returned.
transactionDate	System Date	System Date
amountDetails		
exemptAmount	Actual exemptAmount from response.	Actual exemptAmount from response.
taxAmount	Actual taxAmount from response.	Actual taxAmount from response.
totalAmount	Quote.Subtotal	Order.Subtotal
totalAmountWithTax	TaxAmount + TotalAmount	TaxAmount + TotalAmount
lineItems	Refer to the next line attributes section.	Refer to the next line attributes section.

Response Mappings for Line Attributes

This table defines the response mappings between the line attributes of a tax callout and fields of applicable objects.

Line Attributes	Quote Line Item Mapping	Order Product Mapping
taxCode	TaxTreatment.TaxCode	TaxTreatment.TaxCode
productCode	TaxTreatment.ProductCode	TaxTreatment.ProductCode

Line Attributes	Quote Line Item Mapping	Order Product Mapping
productId	Not returned.	Not returned.
amountDetails		
exemptAmount	Actual exemptAmount from response	Actual exemptAmount from response
taxAmount	Actual taxAmount from response	Actual taxAmount from response
totalAmount	QuoteLineItem.Subtotal	OrderItem.Subtotal
totalAmountWithTax	TaxAmount + TotalAmount	TaxAmount + TotalAmount
effectiveDate	System Date	System Date
lineNumber	QuoteLineItem.Id	OrderItem.Id
description	Not returned.	Not returned.
quantity	Not returned.	Not returned.
addresses		
billTo	Not persisted.	Not persisted.
shipTo	locationCode -> locationCode	locationCode -> locationCode
shipFrom	Not returned.	Not returned.
soldTo	Not returned.	Not returned.
productsku	Not returned.	Not returned.
referenceDocumentCode	Not returned.	Not returned.
taxes	Refer to the next tax attributes section.	Refer to the next tax attributes section.

Response Mappings for Tax Attributes

This table defines the response mappings between the tax attributes of a tax callout and fields of applicable objects.

Tax Attributes	Quote Mapping	Order Mapping
exemptAmount	Not returned.	Not returned.
exemptReason	Not returned.	Not returned.
imposition		
type	Not returned.	Not returned.
Name	Not returned.	Not returned.
jurisdiction		
country	Not returned.	Not returned.
id	Not returned.	Not returned.

Tax Attributes	Quote Mapping	Order Mapping
level	Not returned.	Not returned.
name	Not returned.	Not returned.
region	Not returned.	Not returned.
stateAssignedNo	Not returned.	Not returned.
rate	QuoteItemTaxItem.Rate	OrderItemTaxItem.Rate
tax	QuoteItemTaxItem.amount	OrderItemTaxItem.amount
taxId	Not returned.	Not returned.
taxableAmount	Not returned.	Not returned.

TaxEngineContext Class

Wrapper class that stores details about the type of a tax calculation request.

Namespace

CommerceTax

Example

At the beginning of a tax adapter, use `TaxEngineContext` class to pass the value of a request type to an instance of `RequestType`.

```
global virtual class MockAdapter implements commercetax.TaxEngineAdapter {

     global commercetax.TaxEngineResponse processRequest(commercetax.TaxEngineContext
taxEngineContext) {
         commercetax.RequestType requestType = taxEngineContext.getRequestType();
         commercetax.CalculateTaxRequest request =
(commercetax.CalculateTaxRequest)taxEngineContext.getRequest();
```

Build the rest of your adapter based on the type of request that you got from `TaxEngineContext` class.

```
if(requestType == commercetax.RequestType.CalculateTax){
             commercetax.calculatetaxtype type = request.taxtype;
             String docCode='';
             if(request.DocumentCode == 'simulateEmptyDocumentCode')
                 docCode = '';
             else if(request.DocumentCode != null)
                 docCode =request.DocumentCode;
           else if(request.ReferenceEntityId != null) docCode = request.ReferenceEntityId;

             else docCode =  String.valueOf(getRandomInteger(0,2147483647));
             commercetax.CalculateTaxResponse response = new
commercetax.CalculateTaxResponse();
             if(request.isCommit == true) {
                 response.setStatus(commercetax.TaxTransactionStatus.Committed);
             } else {
```

```
                response.setStatus(commercetax.TaxTransactionStatus.Uncommitted);
            }
}
```

TaxEngineContext Constructors

Learn more about the available constructors with the `TaxEngineContext` class.

TaxEngineContext Methods

Learn more about the available methods with the `TaxEngineContext` class.

TaxEngineContext Constructors

Learn more about the available constructors with the `TaxEngineContext` class.

The `TaxEngineContext` class includes these constructors.

TaxEngineContext(request, requestType, namedUri)

Initializes the `TaxEngineContext` object. This constructor is intended for test usage and throws an exception if used outside of the Apex test context.

`TaxEngineContext(request, requestType, namedUri)`

Initializes the `TaxEngineContext` object. This constructor is intended for test usage and throws an exception if used outside of the Apex test context.

Signature

`TaxEngineContext(commercetax.TaxEngineRequest request, commercetax.RequestType requestType, String namedUri)`

Parameters

request
Type: TaxEngineRequest

Information about the request.

requestType
Type: RequestType

Whether the tax request is to calculate or estimate tax.

namedUri
Type: String

URI that was called as part of the tax calculation request.

TaxEngineContext Methods

Learn more about the available methods with the `TaxEngineContext` class.

The `TaxEngineContext` class includes these methods.

getNamedUri()
Retrieves the value of the NamedUri field of the `TaxEngineContext` class.

getRequest()
Gets the value of the `TaxEngineContext`'s Request field.

getRequestType()
Gets the value of the RequestType field of the `TaxEngineContext` class.

getNamedUri()

Retrieves the value of the NamedUri field of the `TaxEngineContext` class.

Signature

```
global String getNamedUri()
```

Return Value

Type: String

getRequest()

Gets the value of the `TaxEngineContext`'s Request field.

Signature

```
global commercetax.TaxEngineRequest getRequest()
```

Return Value

Type: TaxEngineRequest

An implemented instance of an external tax engine's interface for processing requests. We've provided the `TaxEngineRequest` interface for you to test within mock adapters with classes that implement it, such as CalculateTaxRequest. However, don't use it outside of a testing context.

getRequestType()

Gets the value of the RequestType field of the `TaxEngineContext` class.

Signature

```
global commercetax.RequestType getRequestType()
```

Return Value

Type: RequestType

Indicates whether the calculation request was for actual or calculated tax.

TaxLineItemRequest Class

Contains line item details of a tax request.

Namespace

CommerceTax

TaxLineItemRequest Constructors
Learn more about the constructors available with the `TaxLineItemRequest` class.

TaxLineItemRequest Properties
Learn more about the available properties with the `TaxLineItemRequest` class.

TaxLineItemRequest Methods
Learn more about the available methods with the `TaxLineItemRequest` class.

TaxLineItemRequest Constructors

Learn more about the constructors available with the `TaxLineItemRequest` class.

The `TaxLineItemRequest` class includes these constructors.

TaxLineItemRequest(addresses, amount, description, productCode, quantity, lineNumber, taxCode, effectiveDate)
Initializes the request for the tax line item. This constructor is intended for test usage and throws an exception if used outside of the Apex test context.

`TaxLineItemRequest(addresses, amount, description, productCode, quantity, lineNumber, taxCode, effectiveDate)`

Initializes the request for the tax line item. This constructor is intended for test usage and throws an exception if used outside of the Apex test context.

Signature

```
global TaxLineItemRequest(commercetax.LineTaxAddressesRequest addresses, Double amount,
String description, String productCode, Double quantity, String lineNumber, String
taxCode, Datetime effectiveDate)
```

```
commercetax.TaxLineItemRequest, newinstance, [commercetax.LineTaxAddressesRequest, Double,
 String, String, Double, String, String, Datetime], commercetax.TaxLineItemRequest
```

Parameters

addresses
Type: LineTaxAddressesRequest

Information about the addresses applied to each line item in a tax calculation request.

amount
Type: Double

Total amount (in a given currency) represented by a line item sent for tax calculation.

description

Type: String

User-defined description for a tax line item.

productCode

Type: String

Catalog code for the product represented by the tax line item.

quantity

Type: Double

Number of units of a given product that the tax line item represents.

lineNumber

Type: String

Unique number used to identify a tax line item.

taxCode

Type: String

Code used to identify how tax is calculated for a tax line item.

effectiveDate

Type: Datetime

This is a user-defined date used for reporting only. For negative invoice lines, this parameter represents the invoice date from the original invoice. In other cases, it represents the date when the tax transaction takes effect on the line item. The previous tax transaction type is always `Debit` for negative invoice lines.

TaxLineItemRequest Properties

Learn more about the available properties with the `TaxLineItemRequest` class.

The `TaxLineItemRequest` class includes these properties.

addresses

Contains the list of addresses of a line item.

amount

Total amount (in a given currency) represented by a line item sent for tax calculation.

customTaxAttributes

Customised tax contract to include additional attributes at the line item level.

description

User-defined description for a tax line item.

effectiveDate

The date that a tax transaction takes effect on a line item. This is a user-defined date used for reporting only.

lineNumber

Unique number used to identify a tax line item.

productCode

Catalog code for the product represented by the tax line item.

productSKU
Unique identifier of a product that can be used to identify products that are exempted from tax.

quantity
Number of units of a given product that the tax line item represents.

referenceDocumentCode
Identifier that combines the original invoice ID, previous tax transaction type, and tax engine ID, used in tax calculations for negative invoice lines.

taxCode
Code used to identify how tax is calculated for a tax line item.

addresses

Contains the list of addresses of a line item.

Signature

```
public commercetax.LineTaxAddressesRequest addresses {get; set;}
```

Property Value

Type: commercetax.LineTaxAddressesRequest

amount

Total amount (in a given currency) represented by a line item sent for tax calculation.

Signature

```
global Double amount {get; set;}
```

Property Value

Type: Double

customTaxAttributes

Customised tax contract to include additional attributes at the line item level.

Signature

```
global commercetax.TaxLineItemRequest customTaxAttributes {get; set;}
```

Property Value

Type: Map<String, Object>

description

User-defined description for a tax line item.

Signature

```
global String description {get; set;}
```

Property Value

Type: String

effectiveDate

The date that a tax transaction takes effect on a line item. This is a user-defined date used for reporting only.

Signature

```
global Datetime effectiveDate {get; set;}
```

Property Value

Type: Datetime

lineNumber

Unique number used to identify a tax line item.

Signature

```
global String lineNumber {get; set;}
```

Property Value

Type: String

productCode

Catalog code for the product represented by the tax line item.

Signature

```
global String productCode {get; set;}
```

Property Value

Type: String

productSKU

Unique identifier of a product that can be used to identify products that are exempted from tax.

Signature

```
global String productSKU {get; set;}
```

Property Value

Type: String

quantity

Number of units of a given product that the tax line item represents.

Signature

```
global Double quantity {get; set;}
```

Property Value

Type: Double

referenceDocumentCode

Identifier that combines the original invoice ID, previous tax transaction type, and tax engine ID, used in tax calculations for negative invoice lines.

For example, a referenceDocumentCode parameter value `3ttxx00000004Bh_Debit-4wAxx0000000001EAA` indicates `3ttxx00000004Bh` is the original invoice ID and `4wAxx0000000001EAA` is the tax engine ID. The previous tax transaction type is always `Debit` for negative invoice lines.

Signature

```
global String referenceDocumentCode {get; set;}
```

Property Value

Type: String

taxCode

Code used to identify how tax is calculated for a tax line item.

Signature

```
global String taxCode {get; set;}
```

Property Value

Type: String

TaxLineItemRequest Methods

Learn more about the available methods with the `TaxLineItemRequest` class.

The `TaxLineItemRequest` class includes these methods.

equals(obj)
: Maintains the integrity of lists of type `TaxLineItemRequest` by determining the equality of external objects in a list. This method is dynamic and is based on the `equals()` method in Java.

hashCode()
: Maintains the integrity of lists of type `TaxLineItemRequest` by determining the uniqueness of the external object records in a list.

toString()
: Converts a value to a string.

`equals(obj)`

Maintains the integrity of lists of type `TaxLineItemRequest` by determining the equality of external objects in a list. This method is dynamic and is based on the `equals()` method in Java.

Signature

```
global Boolean equals(Object obj)
```

Parameters

obj
: Type: Object
: External object whose key is to be validated.

Return Value

Type: Boolean

`hashCode()`

Maintains the integrity of lists of type `TaxLineItemRequest` by determining the uniqueness of the external object records in a list.

Signature

```
global Integer hashCode()
```

Return Value

Type: Integer

`toString()`

Converts a value to a string.

Signature

```
global String toString()
```

Return Value

Type: String

TaxSellerDetailsRequest Class

Contains tax code details used in the tax calculation request.

Namespace

CommerceTax

TaxSellerDetailsRequest Constructors
Learn more about the available constructors with the `TaxSellerDetailsRequest` class.

TaxSellerDetailsRequest Properties
Learn more about the available properties with the `TaxSellerDetailsRequest` class.

TaxSellerDetailsRequest Methods
Learn more about the available methods with the `TaxSellerDetailsRequest` class.

TaxSellerDetailsRequest Constructors

Learn more about the available constructors with the `TaxSellerDetailsRequest` class.

The `TaxSellerDetailsRequest` class includes these constructors.

TaxSellerDetailsRequest(code)
Initializes the request for the tax seller details. This constructor is intended for test usage and throws an exception if used outside of the Apex test context

`TaxSellerDetailsRequest(code)`

Initializes the request for the tax seller details. This constructor is intended for test usage and throws an exception if used outside of the Apex test context

Signature

```
global TaxSellerDetailsRequest(String code)
```

Parameters

code
Type: String

Tax code used for tax calculation.

TaxSellerDetailsRequest Properties

Learn more about the available properties with the `TaxSellerDetailsRequest` class.

The `TaxSellerDetailsRequest` class includes these properties.

code
: Tax code used for tax calculation.

code

Tax code used for tax calculation.

Signature

```
global String code {get; set;}
```

Property Value

Type: String

TaxSellerDetailsRequest Methods

Learn more about the available methods with the `TaxSellerDetailsRequest` class.

The `TaxSellerDetailsRequest` class includes these methods.

equals(obj)
: Maintains the integrity of lists of type `TaxSellerDetailsRequest` by determining the equality of the external objects in a list. This method is dynamic and based on the `equals()` method in Java.

hashCode()
: Maintains the integrity of lists of type `TaxSellerDetailsRequest` by determining the uniqueness of the external objects in a list.

toString()
: Converts a value to a string.

equals(obj)

Maintains the integrity of lists of type `TaxSellerDetailsRequest` by determining the equality of the external objects in a list. This method is dynamic and based on the `equals()` method in Java.

Signature

```
global Boolean equals(Object obj)
```

Parameters

obj
: Type: Object

  External object whose key is to be validated.

Return Value

Type: Boolean

`hashCode()`

Maintains the integrity of lists of type `TaxSellerDetailsRequest` by determining the uniqueness of the external objects in a list.

Signature

`global Integer hashCode()`

Return Value

Type: Integer

`toString()`

Converts a value to a string.

Signature

`global String toString()`

Return Value

Type: String

TaxTransactionRequest Class

Abstract class for storing customer details used in tax calculation and estimation requests.

Namespace

CommerceTax

Usage

Specify the `CommerceTax` namespace when creating an instance of this class. The constructor of this class takes no arguments. For example, let's say you create an instance of `CalculateTaxRequest` class, which extends the `TaxTransactionRequest` class.

TaxTransactionRequest Constructors
Learn more about the available constructors with the `TaxTransactionRequest` class.

TaxTransactionRequest Properties
Learn more about the available properties with the `TaxTransactionRequest` class.

TaxTransactionRequest Methods

TaxTransactionRequest Constructors

Learn more about the available constructors with the `TaxTransactionRequest` class.

The `TaxTransactionRequest` class includes these constructors.

TaxTransactionRequest(addresses, currencyIsoCode, customerDetails, description, documentCode, referenceDocumentCode, transactionDate, effectiveDate, lineItems, referenceEntityId, sellerDetails, customTaxAttributes)

Initializes the request for the tax transaction. This constructor is intended for test usage and throws an exception if used outside of the Apex test context.

`TaxTransactionRequest(addresses, currencyIsoCode, customerDetails, description, documentCode, referenceDocumentCode, transactionDate, effectiveDate, lineItems, referenceEntityId, sellerDetails, customTaxAttributes)`

Initializes the request for the tax transaction. This constructor is intended for test usage and throws an exception if used outside of the Apex test context.

Signature

```
global TaxTransactionRequest(commercetax.HeaderTaxAddressesRequest addresses, String
currencyIsoCode, commercetax.TaxCustomerDetailsRequest customerDetails, String
description, String documentCode, String referenceDocumentCode, Datetime transactionDate,
Datetime effectiveDate, List<commercetax.TaxLineItemRequest> lineItems, String
referenceEntityId, commercetax.TaxSellerDetailsRequest sellerDetails,Map<String,Object>
customTaxAttributes)
```

Parameters

addresses

Type: HeaderTaxAddressesRequest

Tax addresses, such as Ship To and Bill From.

currencyIsoCode

Type: String

Three-letter ISO 4217 currency code associated with the `TaxTransactionRequest`.

customerDetails

Type: TaxCustomerDetailsRequest

Customer information used in tax calculation.

description

Type: String

Optional user-defined description for providing more information about the tax transaction request.

documentCode

Type: String

Code for documents that are used to provide more information in the tax calculation process.

referenceDocumentCode

Type: String

Identifier that combines the original invoice ID, previous tax transaction type, and tax engine ID, used in tax calculations for negative invoice lines. For example, a referenceDocumentCode parameter value `3ttxx00000004Bh_Debit-4wAxx0000000001EAA` indicates `3ttxx00000004Bh` is the original invoice ID and `4wAxx0000000001EAA` is the tax engine ID.

transactionDate

Type: Datetime

The date that the tax transaction occurred.

`effectiveDate`

Type: Datetime

The date that the tax transaction takes effect. User-defined and used only for reporting purposes.

`lineItems`

Type: List<TaxLineItemRequest>

A list of line items on which tax is calculated.

`referenceEntityId`

Type: String

ID of an object related to the line items sent for tax calculation.

`sellerDetails`

Type: TaxSellerDetailsRequest

Contains tax code information used in a tax calculation request.

`customTaxAttributes`

Type: Map<String, Object>

Customised tax contract to include additional attributes at the header level.

TaxTransactionRequest Properties

Learn more about the available properties with the `TaxTransactionRequest` class.

The `TaxTransactionRequest` class includes these properties.

addresses

A list of addresses (such as Ship To and Sold To) used as part of the tax transaction.

currencyIsoCode

Three-letter ISO 4217 currency code associated with the `TaxTransactionRequest`.

customerDetails

Customer information used in tax calculation.

customTaxAttributes

Customised tax contract to include additional attributes at the header level.

description

Optional user-defined description for providing more information about the tax transaction request.

documentCode

Code for documents used to provide more information in the tax calculation process.

effectiveDate

The date that the tax transaction takes effect. User-defined and used only for reporting purposes.

lineItems

A list of line items on which tax will be calculated.

referenceDocumentCode

Identifier that combines the original invoice ID, previous tax transaction type, and tax engine ID, used in tax calculations for negative invoice lines.

referenceEntityId

ID of an object related to the line items sent for tax calculation.

sellerDetails

Contains tax code information used in a tax calculation request.

transactionDate

The date that the tax transaction occurred.

addresses

A list of addresses (such as Ship To and Sold To) used as part of the tax transaction.

Signature

```
global commercetax.HeaderTaxAddressesRequest addresses {get; set;}
```

Property Value

Type: HeaderTaxAddressesRequest

currencyIsoCode

Three-letter ISO 4217 currency code associated with the `TaxTransactionRequest`.

Signature

```
global String currencyIsoCode {get; set;}
```

Property Value

Type: String

customerDetails

Customer information used in tax calculation.

Signature

```
global CommerceTax.TaxCustomerDetailsRequest customerDetails {get; set;}
```

Property Value

Type: TaxCustomerDetailsRequest

customTaxAttributes

Customised tax contract to include additional attributes at the header level.

Signature

```
global commercetax.TaxTransactionRequest customTaxAttributes {get; set;}
```

Property Value

Type: Map<String, Object>

description

Optional user-defined description for providing more information about the tax transaction request.

Signature

```
global String description {get; set;}
```

Property Value

Type: String

documentCode

Code for documents used to provide more information in the tax calculation process.

Signature

```
global String documentCode {get; set;}
```

Property Value

Type: String

effectiveDate

The date that the tax transaction takes effect. User-defined and used only for reporting purposes.

Signature

```
global Datetime effectiveDate {get; set;}
```

Property Value

Type: Datetime

lineItems

A list of line items on which tax will be calculated.

Signature

```
global List<CommerceTax.TaxLineItemRequest> lineItems {get; set;}
```

Property Value

Type: List<TaxLineItemRequest>

referenceDocumentCode

Identifier that combines the original invoice ID, previous tax transaction type, and tax engine ID, used in tax calculations for negative invoice lines.

For example, a referenceDocumentCode parameter value `3ttxx00000004Bh_Debit-4wAxx0000000001EAA` indicates `3ttxx00000004Bh` is the original invoice ID and `4wAxx0000000001EAA` is the tax engine ID.

Signature

```
global String referenceDocumentCode {get; set;}
```

Property Value

Type: String

referenceEntityId

ID of an object related to the line items sent for tax calculation.

Signature

```
global String referenceEntityId {get; set;}
```

Property Value

Type: String

sellerDetails

Contains tax code information used in a tax calculation request.

Signature

```
global commercetax.TaxSellerDetailsRequest sellerDetails {get; set;}
```

Property Value

Type: TaxSellerDetailsRequest

transactionDate

The date that the tax transaction occurred.

Signature

```
global Datetime transactionDate {get; set;}
```

Property Value

Type: Datetime

TaxTransactionRequest Methods

The following are methods for `TaxTransactionRequest`.

- equals(obj)
  Maintains the integrity of lists of type `TaxTransactionRequest` by determining the equality of external objects in a list. This method is dynamic and based on the `equals()` method in Java.
- hashCode()
  Maintains the integrity of lists of type `TaxTransactionRequest` by determining the uniqueness of the external object records in a list.
- toString()
  Converts a value to a string.

`equals(obj)`

Maintains the integrity of lists of type `TaxTransactionRequest` by determining the equality of external objects in a list. This method is dynamic and based on the `equals()` method in Java.

Signature

```
global Boolean equals(Object obj)
```

Parameters

obj
Type: Object

Return Value

Type: Boolean

`hashCode()`

Maintains the integrity of lists of type `TaxTransactionRequest` by determining the uniqueness of the external object records in a list.

Signature

```
global Integer hashCode()
```

Return Value

Type: Integer

`toString()`

Converts a value to a string.

Signature

```
global String toString()
```

Return Value

Type: String

TaxTransactionStatus Enum

Shows whether the tax transaction has been committed or uncommitted.

Usage

Used by the CalculateTaxResponse class method.

Enum Values

The `commercetax.TaxTransactionStatus` enum includes these values.

Value	Description
`Committed`	Tax has been calculated and committed.
`Uncommitted`	Tax has been calculated but hasn't been committed.

TaxTransactionType Enum

Shows whether the tax transaction is for a credit or debit transaction.

Usage

Used by the CalculateTaxResponse and CalculateTaxRequest class methods.

Enum Values

The `commercetax.TaxTransactionType` enum includes these values.

Value	Description
`Credit`	Represents a credit transaction.
`Debit`	Represents a debit transaction.
`Void`	Specifies that the tax engine has voided the document that's mentioned in the `referenceDocumentCode` property value.

PlaceQuote Namespace

The `PlaceQuote` namespace provides classes and methods to create or update quotes with pricing preferences and configuration options.

Note: This namespace has been deprecated as of API version 63.0. In API version 63.0 and later, use the new RevSalesTrxn namespace.

The `PlaceQuote` namespace includes these classes.

CatalogRatesPreferenceEnum Enum
Specifies the rate card entries defined in the catalog that must be fetched for quote line items, with usage-based selling during the quote creation process.

ConfigurationInputEnum Enum
Specifies the configuration input for the request to place a quote.

ConfigurationOptionsInput Class
Contains methods and properties to set the configuration options for the input to the product configurator.

GraphRequest Class
Contains constructors and properties to set the graph ID and a list of records to be ingested. The list of records is specified in a key-value map format that contains the field values of an order.

PlaceQuoteException Class
Contains methods to hold the exception details for the place quote request.

PlaceQuoteResponse Class
Contains properties to hold the response to the place quote request.

PlaceQuoteRLMApexProcessor Class
Contains methods to place a quote with details of the graph request, pricing preferences, and configuration options.

PricingPreferenceEnum Enum
Specifies the pricing preference during the create quote process.

RecordResource Class
Contains constructors and properties to create a record object from the field values of a quote.

RecordWithReferenceRequest Class
Contains constructors and properties to associate a record object with a reference identifier.

CatalogRatesPreferenceEnum Enum

Specifies the rate card entries defined in the catalog that must be fetched for quote line items, with usage-based selling during the quote creation process.

Usage

This enum is available when the Usage-Based Selling feature is enabled.

Enum Values

The `placequote.CatalogRatesPreferenceEnum` enum includes these values.

Value	Description
`Fetch`	Retrieves the rate card entries defined in the catalog for quote line items during the quote creation process.

Value	Description
`Skip`	Skips the retrieval of rate card entries for quote line items during the quote creation process. The default value is `Skip`.

ConfigurationInputEnum Enum

Specifies the configuration input for the request to place a quote.

Usage

Use these enum values for the `configurationInputEnum` property in the PlaceQuoteRLMApexProcessor class.

Enum Values

The `placequote.ConfigurationInputEnum` enum has these values.

Value	Description
`RunAndAllowErrors`	Run the configuration and proceed with order ingestion upon encountering any configuration errors.
`RunAndBlockErrors`	Run the configuration and block order ingestion upon encountering any configuration errors.
`Skip`	Skip the configuration execution.

ConfigurationOptionsInput Class

Contains methods and properties to set the configuration options for the input to the product configurator.

Namespace

PlaceQuote

Usage

This class holds the required details of the product configuration input. Set the class properties to enable default configuration, execution of configuration rules, and validation of the product catalog. Use these class properties as an input to the PlaceQuoteRLMApexProcessor class method.

Example

```
PlaceQuote.GraphRequest graph = new PlaceQuote.GraphRequest('test',listOfRecords);
    PlaceQuote.PricingPreferenceEnum pricingPreference =
PlaceQuote.PricingPreferenceEnum.System;
    PlaceQuote.ConfigurationInputEnum configurationPreference =
PlaceQuote.ConfigurationInputEnum.RunAndAllowErrors;
   PlaceQuote.ConfigurationOptionsInput cInput = new PlaceQuote.ConfigurationOptionsInput();
```

```
    cInput.addDefaultConfiguration = true;
    cInput.executeConfigurationRules = true;
    cInput.validateAmendRenewCancel = true;
    cInput.validateProductCatalog = true;
        //Place Quote Call
     PlaceQuote.PlaceQuoteResponse resp =
PlaceQuote.PlaceQuoteRLMApexProcessor.execute(pricingPreference,graph,configurationPreference,cInput);
```

ConfigurationOptionsInput Properties

Set the `ConfigurationOptionsInput` class properties to add default configuration, execute configuration rules, and validate the product catalog.

ConfigurationOptionsInput Methods

Learn more about the methods available with the `ConfigurationOptionsInput` class.

ConfigurationOptionsInput Properties

Set the `ConfigurationOptionsInput` class properties to add default configuration, execute configuration rules, and validate the product catalog.

The `ConfigurationOptionsInput` class includes these properties.

addDefaultConfiguration

Sets the default product configuration, such as bundle and product attributes, for a quote request.

executeConfigurationRules

Sets the requirement for a quote to adhere to the configuration rules.

validateAmendRenewCancel

Sets the requirement to run validations related to amend, renew, or cancel processes.

validateProductCatalog

Sets the requirement to validate a quote against the product catalog.

`addDefaultConfiguration`

Sets the default product configuration, such as bundle and product attributes, for a quote request.

Signature

`public Boolean addDefaultConfiguration {get; set;}`

Property Value

Type: Boolean

Indicates whether to automatically add default configuration to the order (`true`) or not (`false`).

`executeConfigurationRules`

Sets the requirement for a quote to adhere to the configuration rules.

Signature

```
public Boolean executeConfigurationRules {get; set;}
```

Property Value

Type: Boolean

Indicates whether the order must adhere to configuration rules during processing (`true`) or bypass them (`false`).

`validateAmendRenewCancel`

Sets the requirement to run validations related to amend, renew, or cancel processes.

Signature

```
public Boolean validateAmendRenewCancel {get; set;}
```

Property Value

Type: Boolean

Indicates whether to run validations related to amend, renew, or cancel processes (`true`) or not (`false`).

`validateProductCatalog`

Sets the requirement to validate a quote against the product catalog.

Signature

```
public Boolean validateProductCatalog {get; set;}
```

Property Value

Type: Boolean

Indicates whether the quote must be validated against the product catalog (`true`) or not (`false`).

ConfigurationOptionsInput Methods

Learn more about the methods available with the `ConfigurationOptionsInput` class.

The `ConfigurationOptionsInput` class includes these methods.

- equals(obj)
  Determines the equality of external objects in a list. This method is dynamic and is based on the `equals()` method in Java.
- hashCode()
  Determines the uniqueness of the external object records in a list.
- toString()
  Converts a value to a string.

equals(obj)

Determines the equality of external objects in a list. This method is dynamic and is based on the `equals()` method in Java.

Signature

`public Boolean equals(Object obj)`

Parameters

obj
Type: Object

Reference object that's used to compare with the class object.

Return Value

Type: Boolean

Indicates if the class object is same as the reference object (`true`) or not (`false`).

hashCode()

Determines the uniqueness of the external object records in a list.

Signature

`public Integer hashCode()`

Return Value

Type: Integer

Integer hash code that represents the value of the object. Equal objects as per the `equals()` method must return the same hash code.

toString()

Converts a value to a string.

Signature

`public String toString()`

Return Value

Type: String

GraphRequest Class

Contains constructors and properties to set the graph ID and a list of records to be ingested. The list of records is specified in a key-value map format that contains the field values of an order.

Namespace

PlaceQuote

Example

Invoke the Place Quote Apex API by using these steps.

- Set up a quote and quote line item. To associate the Record object with a reference identifier, create an instance of the `RecordWithReferenceRequest` class.

```
         //Quote Setup
    PlaceQuote.RecordResource quoteRecord = new
PlaceQuote.RecordResource(Quote.getSobjectType(),'POST');
    Map<String,Object> quoteFieldValues = new Map<String,Object>();
    quoteFieldValues.put('Name','q-ap12');
    quoteFieldValues.put('OpportunityId','006xx000001aBFcAAM');
    quoteFieldValues.put('Pricebook2Id','01sxx0000005pvRAAQ');
    quoteRecord.fieldValues = quoteFieldValues;

           //Quote Line Item Setup
    PlaceQuote.RecordResource quoteLineItemRecord1 = new
PlaceQuote.RecordResource(QuoteLineItem.getSobjectType(),'POST');
    Map<String,Object> quoteLineItemFieldValues = new Map<String,Object>();
    quoteLineItemFieldValues.put('Product2Id','01txx0000006ibwAAA');
    quoteLineItemFieldValues.put('PricebookEntryId','01uxx0000008zPqAAI');
    quoteLineItemFieldValues.put('Quantity','2.0');
    quoteLineItemFieldValues.put('UnitPrice','5.0');
    quoteLineItemFieldValues.put('StartDate','2023-03-15');
    quoteLineItemFieldValues.put('QuoteId','@{refQuote.id}');
    quoteLineItemRecord1.fieldValues = quoteLineItemFieldValues;
    PlaceQuote.RecordWithReferenceRequest quoteItemRecords = new
PlaceQuote.RecordWithReferenceRequest('refQuote',quoteRecord);
    PlaceQuote.RecordWithReferenceRequest quoteLineItemRecords1 = new
PlaceQuote.RecordWithReferenceRequest('refQuoteItem1',quoteLineItemRecord1);
```

- To create a quote line relationship, create an instance of the `RecordResource` class.

```
PlaceQuote.RecordResource quoteLineRelationship1 = new
PlaceQuote.RecordResource(QuoteLineRelationship.getSobjectType(),'POST');
    Map<String,Object> quoteLineRelationshipValues = new Map<String,Object>();
    quoteLineRelationshipValues.put('ProductRelationshipTypeId','0yoxx00000000JNAAY');
    quoteLineRelationshipValues.put('MainQuoteLineId','@{refQuoteItem2.id}');
    quoteLineRelationshipValues.put('AssociatedQuoteLineId','@{refQuoteItem1.id}');
   quoteLineRelationshipValues.put('AssociatedQuoteLinePricing','IncludedInBundlePrice');

    quoteLineRelationship1.fieldValues = quoteLineRelationshipValues;
```

- Create the list of records to be ingested.

```
    PlaceQuote.RecordWithReferenceRequest quoteLineRelationship = new
PlaceQuote.RecordWithReferenceRequest('QuoteLineRelationship',quoteLineRelationship1);
    List<PlaceQuote.RecordWithReferenceRequest> listOfRecords = new
```

```
List<PlaceQuote.RecordWithReferenceRequest>();
    listOfRecords.add(quoteItemRecords);
    listOfRecords.add(quoteLineItemRecords1);
    listOfRecords.add(quoteLineItemRecords2);
    listOfRecords.add(quoteLineRelationship);
```

- To contain all record objects, create an instance of the `GraphRequest` class.

  Note: The `CatalogRatesPreferenceEnum` enum is available when the Usage-Based Selling feature is enabled.

```
    PlaceQuote.GraphRequest graph = new PlaceQuote.GraphRequest('test',listOfRecords);
    PlaceQuote.ConfigurationOptionsInput cInput = new
PlaceQuote.ConfigurationOptionsInput();
    PlaceQuote.CatalogRatesPreferenceEnum catalogRatesPreference =
PlaceQuote.CatalogRatesPreferenceEnum.Fetch;
....PlaceQuote.ConfigurationInputEnum configurationPreference =
PlaceQuote.ConfigurationInputEnum.RunAndAllowErrors;
    PlaceQuote.PricingPreferenceEnum pricingPreference =
PlaceQuote.PricingPreferenceEnum.System;
    //System.debug(graph);

    //Place Quote Call
    PlaceQuote.PlaceQuoteResponse resp =
PlaceQuote.PlaceQuoteRLMApexProcessor.execute(pricingPreference, catalogRatesPreference,
 graph, configurationPreference, cInput);
    System.debug(resp);
```

GraphRequest Constructors
Learn more about the available constructors with the `GraphRequest` class.

GraphRequest Properties

GraphRequest Constructors

Learn more about the available constructors with the `GraphRequest` class.

The `GraphRequest` class includes these constructors.

GraphRequest(graphId, records)
Creates an instance of the `GraphRequest` class to assign the graph ID and a list of records to be ingested.

`GraphRequest(graphId, records)`

Creates an instance of the `GraphRequest` class to assign the graph ID and a list of records to be ingested.

Signature

`public GraphRequest(String graphId, List<placequote.RecordWithReferenceRequest> records)`

Parameters

graphId
Type: String

ID of the graph.

records
Type: List<placequote.RecordWithReferenceRequest on page 1693>

List of records to be ingested.

GraphRequest Properties

The following are properties for `GraphRequest`.

graphId
Set the `graphId` property to assign the ID value of the graph.

`graphId`

Set the `graphId` property to assign the ID value of the graph.

Signature

`public String graphId {get; set;}`

Property Value

Type: String

PlaceQuoteException Class

Contains methods to hold the exception details for the place quote request.

Namespace

PlaceQuote

PlaceQuoteException Methods
Learn more about the methods available with the `PlaceQuoteException` class.

PlaceQuoteException Methods

Learn more about the methods available with the `PlaceQuoteException` class.

The `PlaceQuoteException` class includes these methods.

getErrorCode()
Gets the error code that's associated to the place quote request.

getErrorCode()

Gets the error code that's associated to the place quote request.

Signature

`public String getErrorCode()`

Return Value

Type: String

PlaceQuoteResponse Class

Contains properties to hold the response to the place quote request.

Namespace

PlaceQuote

Example

```
PlaceQuote.PlaceQuoteResponse resp =
PlaceQuote.PlaceQuoteExecutor.execute(internalEnum,graph);
```

PlaceQuoteResponse Properties
Learn more about the available properties with the `PlaceQuoteResponse` class.

PlaceQuoteResponse Properties

Learn more about the available properties with the `PlaceQuoteResponse` class.

The `PlaceQuoteResponse` class includes these properties.

quoteId
Get the ID of the quote that's created after a successful request.

requestIdentifier
Get the request ID of the process to query the asynchronous status of the Place Quote Apex API.

responseError
Get the list of errors encountered during the synchronous processing of the API request.

statusURL
Get the asynchronous status URL of the request, if available.

success
Get the request status of the synchronous part of the processing.

quoteId

Get the ID of the quote that's created after a successful request.

Signature

```
public String quoteId {get; set;}
```

Property Value

Type: String

`requestIdentifier`

Get the request ID of the process to query the asynchronous status of the Place Quote Apex API.

Signature

```
public String requestIdentifier {get; set;}
```

Property Value

Type: String

`responseError`

Get the list of errors encountered during the synchronous processing of the API request.

Signature

```
public List<ConnectApi.PlaceQuoteErrorResponse> responseError {get; set;}
```

Property Value

Type: List<ConnectApi.PlaceQuoteErrorResponse>

`statusURL`

Get the asynchronous status URL of the request, if available.

Signature

```
public String statusURL {get; set;}
```

Property Value

Type: String

`success`

Get the request status of the synchronous part of the processing.

Signature

```
public Boolean success {get; set;}
```

Property Value

Type: Boolean

Indicates whether the synchronous part of the processing is successful (`true`) or not (`false`).

PlaceQuoteRLMApexProcessor Class

Contains methods to place a quote with details of the graph request, pricing preferences, and configuration options.

Namespace

PlaceQuote

PlaceQuoteRLMApexProcessor Methods
Learn more about the methods available with the `PlaceQuoteRLMApexProcessor` class.

PlaceQuoteRLMApexProcessor Example Implementation
To place a quote from Apex, refer to the example implementation of the `PlaceQuoteRLMApexProcessor` class.

PlaceQuoteRLMApexProcessor Methods

Learn more about the methods available with the `PlaceQuoteRLMApexProcessor` class.

The `PlaceQuoteRLMApexProcessor` class includes these methods.

execute(pricingPreferenceEnum, graphRequest, configurationInputEnum, configurationOptionsInput)
Use the method in the `PlaceQuoteExecutor` class to execute the Place Quote Apex API request by assigning the properties for graph request, pricing references, and configuration options.

execute(pricingPreferenceEnum, catalogRatesPreference, graphRequest, configurationInputEnum, configurationOptionsInput)
Use the method in the `PlaceQuoteExecutor` class to execute the Place Quote Apex API request by assigning the properties for graph request, pricing references, and configuration options. This method also includes the property to define fetching of rate card entries.

execute(pricingPreferenceEnum, graphRequest, configurationInputEnum, configurationOptionsInput)

Use the method in the `PlaceQuoteExecutor` class to execute the Place Quote Apex API request by assigning the properties for graph request, pricing references, and configuration options.

Signature

```
public static placequote.PlaceQuoteResponse execute(placequote.PricingPreferenceEnum
pricingPreferenceEnum, placequote.GraphRequest graphRequest,
placequote.ConfigurationInputEnum configurationInputEnum,
placequote.ConfigurationOptionsInput configurationOptionsInput)
```

Parameters

pricingPreferenceEnum
Type: placequote.PricingPreferenceEnum

Pricing preference during the quote process.

graphRequest
Type: placequote.GraphRequest

The sObject graph values of the quote payload to be ingested.

configurationInputEnum
Type: placequote.ConfigurationInputEnum

Configuration input for the quote process.

configurationOptionsInput
Type: placequote.ConfigurationOptionsInput

Configuration options during the ingestion process.

Return Value

Type: placequote.PlaceQuoteResponse

execute(pricingPreferenceEnum, catalogRatesPreference, graphRequest, configurationInputEnum, configurationOptionsInput)

Use the method in the `PlaceQuoteExecutor` class to execute the Place Quote Apex API request by assigning the properties for graph request, pricing references, and configuration options. This method also includes the property to define fetching of rate card entries.

Signature

```
public static placequote.PlaceQuoteResponse execute(placequote.PricingPreferenceEnum
pricingPreferenceEnum, placequote.CatalogRatesPreferenceEnum catalogRatesPreference,
placequote.GraphRequest graphRequest, placequote.ConfigurationInputEnum
configurationInputEnum, placequote.ConfigurationOptionsInput configurationOptionsInput)
```

Parameters

pricingPreferenceEnum
Type: placequote.PricingPreferenceEnum

Pricing preference during the quote process.

catalogRatesPreference
Type: placequote.CatalogRatesPreferenceEnum

The rate card entries defined in the catalog that must be fetched for quote line items, with usage-based pricing during the quote creation process. The `CatalogRatesPreferenceEnum` enum is available when the Usage-Based Selling feature is enabled.

graphRequest
Type: placequote.GraphRequest

The sObject graph values of the quote payload to be ingested.

configurationInputEnum
Type: placequote.ConfigurationInputEnum

Configuration input for the quote process.

configurationOptionsInput
Type: placequote.ConfigurationOptionsInput

Configuration options during the ingestion process.

Return Value

Type: placequote.PlaceQuoteResponse

PlaceQuoteRLMApexProcessor Example Implementation

To place a quote from Apex, refer to the example implementation of the `PlaceQuoteRLMApexProcessor` class.

Namespace

PlaceQuote

Usage

Customize this example to suit your requirements. Create the list of records to be ingested by using these steps. Replace the respective IDs with the values that are present in your org. For example, replace the value of `${Pricebook2Id}` field with the price book ID that's present in the org.

Example:

- Set up a quote and quote line item. To associate the Record object with a reference identifier, create an instance of the `RecordWithReferenceRequest` class.

```
        //Quote Setup
    PlaceQuote.RecordResource quoteRecord = new
PlaceQuote.RecordResource(Quote.getSobjectType(),'POST');
    Map<String,Object> quoteFieldValues = new Map<String,Object>();
    quoteFieldValues.put('Name','q-ap12');
    quoteFieldValues.put('OpportunityId','006xx000001aBFcAAM');
    quoteFieldValues.put('Pricebook2Id','01sxx0000005pvRAAQ');
    quoteRecord.fieldValues = quoteFieldValues;

          //Quote Line Item Setup
    PlaceQuote.RecordResource quoteLineItemRecord1 = new
PlaceQuote.RecordResource(QuoteLineItem.getSobjectType(),'POST');
    Map<String,Object> quoteLineItemFieldValues = new Map<String,Object>();
    quoteLineItemFieldValues.put('Product2Id','01txx0000006ibwAAA');
    quoteLineItemFieldValues.put('PricebookEntryId','01uxx0000008zPqAAI');
    quoteLineItemFieldValues.put('Quantity','2.0');
    quoteLineItemFieldValues.put('UnitPrice','5.0');
    quoteLineItemFieldValues.put('StartDate','2023-03-15');
    quoteLineItemFieldValues.put('QuoteId','@{refQuote.id}');
    quoteLineItemRecord1.fieldValues = quoteLineItemFieldValues;
    PlaceQuote.RecordWithReferenceRequest quoteItemRecords = new
```

```
PlaceQuote.RecordWithReferenceRequest('refQuote',quoteRecord);
    PlaceQuote.RecordWithReferenceRequest quoteLineItemRecords1 = new
PlaceQuote.RecordWithReferenceRequest('refQuoteItem1',quoteLineItemRecord1);
```

- To create a quote line relationship, create an instance of the `RecordResource` class.

```
PlaceQuote.RecordResource quoteLineRelationship1 = new
PlaceQuote.RecordResource(QuoteLineRelationship.getSobjectType(),'POST');
    Map<String,Object> quoteLineRelationshipValues = new Map<String,Object>();
   quoteLineRelationshipValues.put('ProductRelationshipTypeId','0yoxx00000000JNAAY');

    quoteLineRelationshipValues.put('MainQuoteLineId','@{refQuoteItem2.id}');
   quoteLineRelationshipValues.put('AssociatedQuoteLineId','@{refQuoteItem1.id}');


quoteLineRelationshipValues.put('AssociatedQuoteLinePricing','IncludedInBundlePrice');

    quoteLineRelationship1.fieldValues = quoteLineRelationshipValues;
```

- Create the list of records to be ingested.

```

    PlaceQuote.RecordWithReferenceRequest quoteLineRelationship = new
PlaceQuote.RecordWithReferenceRequest('QuoteLineRelationship',quoteLineRelationship1);

    List<PlaceQuote.RecordWithReferenceRequest> listOfRecords = new
List<PlaceQuote.RecordWithReferenceRequest>();
    listOfRecords.add(quoteItemRecords);
    listOfRecords.add(quoteLineItemRecords1);
    listOfRecords.add(quoteLineItemRecords2);
    listOfRecords.add(quoteLineRelationship);
```

- To contain all record objects, create an instance of the `GraphRequest` class.

  Note: The `CatalogRatesPreferenceEnum` enum is available when the Usage-Based Selling feature is enabled.

```

   PlaceQuote.GraphRequest graph = new PlaceQuote.GraphRequest('test',listOfRecords);

    PlaceQuote.ConfigurationOptionsInput cInput = new
PlaceQuote.ConfigurationOptionsInput();
    PlaceQuote.CatalogRatesPreferenceEnum catalogRatesPreference =
PlaceQuote.CatalogRatesPreferenceEnum.Fetch;
....PlaceQuote.ConfigurationInputEnum configurationPreference =
PlaceQuote.ConfigurationInputEnum.RunAndAllowErrors;
    PlaceQuote.PricingPreferenceEnum pricingPreference =
PlaceQuote.PricingPreferenceEnum.System;
    //System.debug(graph);

    //Place Quote Call
    PlaceQuote.PlaceQuoteResponse resp =
PlaceQuote.PlaceQuoteRLMApexProcessor.execute(pricingPreference,
catalogRatesPreference, graph, configurationPreference, cInput);
    System.debug(resp);
```

PricingPreferenceEnum Enum

Specifies the pricing preference during the create quote process.

Usage

Used by the PlaceQuoteRLMApexProcessor class.

Enum Values

The `placequote.PricingPreferenceEnum` enum class includes these values.

Value	Description
`Force`	Enforce pricing during the quote ingestion process.
`Skip`	Skip pricing during the quote ingestion process.
`System`	Determine whether a pricing calculation is required.

RecordResource Class

Contains constructors and properties to create a record object from the field values of a quote.

Namespace

PlaceQuote

Example

```
PlaceQuote.RecordResource quoteLineRelationship1 = new
PlaceQuote.RecordResource(QuoteLineRelationship.getSobjectType(),'POST');
```

See PlaceQuoteRLMApexProcessor to refer to an example implementation.

RecordResource Constructors
Learn more about the available constructors with the `RecordResource` class.

RecordResource Properties
Learn more about the available properties with the `RecordResource` class.

RecordResource Constructors

Learn more about the available constructors with the `RecordResource` class.

The `RecordResource` class includes these constructors.

RecordResource(type, method, id)
Creates an instance of the `RecordResource` class to assign values to the fields of a quote item by using the sObject type, API method, and quote ID properties.

RecordResource(type, method)

Creates an instance of the `RecordResource` class to assign the values to the fields of a quote item by using the sObject type and API method properties.

`RecordResource(type, method, id)`

Creates an instance of the `RecordResource` class to assign values to the fields of a quote item by using the sObject type, API method, and quote ID properties.

Signature

`public RecordResource(Schema.SObjectType type, String method, Id id)`

Parameters

type
Type: Schema.SObjectType

Object that's returned from the field describe result using the `getReferenceTo()` method or from the sObject describe result using the `getSObjectType()` method.

method
Type: String

Method for the API request, such as POST or PATCH.

id
Type: Id

ID of the quote.

`RecordResource(type, method)`

Creates an instance of the `RecordResource` class to assign the values to the fields of a quote item by using the sObject type and API method properties.

Signature

`public RecordResource(Schema.SObjectType type, String method)`

Parameters

type
Type: Schema.SObjectType

Object that's returned from the field describe result using the `getReferenceTo()` method or from the sObject describe result using the `getSObjectType()` method.

method
Type: String

Method for the API request, such as POST or PATCH.

RecordResource Properties

Learn more about the available properties with the `RecordResource` class.

The `RecordResource` class includes these properties.

fieldValues
Set the `fieldValues` property to assign values to the fields to update the quote record.

id
Set the `id` property to assign the ID of the quote record.

method
Set the `method` property to specify the API request method, such as POST or PATCH.

type
Set the type property to assign the object type that's returned from the field describe result by using the `getReferenceTo()` method or from the sObject describe result by using the `getSObjectType()` method.

`fieldValues`

Set the `fieldValues` property to assign values to the fields to update the quote record.

Signature

```
public Map<String,ANY> fieldValues {get; set;}
```

Property Value

Type: List <Map<String,ANY>>

`id`

Set the `id` property to assign the ID of the quote record.

Signature

```
public String id {get; set;}
```

Property Value

Type: String

`method`

Set the `method` property to specify the API request method, such as POST or PATCH.

Signature

```
public String method {get; set;}
```

Property Value

Type: String

type

Set the type property to assign the object type that's returned from the field describe result by using the `getReferenceTo()` method or from the sObject describe result by using the `getSObjectType()` method.

Signature

`public Schema.SObjectType type {get; set;}`

Property Value

Type: Schema.SObjectType

RecordWithReferenceRequest Class

Contains constructors and properties to associate a record object with a reference identifier.

Namespace

PlaceQuote

Example

```
PlaceQuote.RecordWithReferenceRequest quoteLineRelationship = new
PlaceQuote.RecordWithReferenceRequest('QuoteLineRelationship',quoteLineRelationship1);
```

See PlaceQuoteRLMApexProcessor to refer to an example implementation.

RecordWithReferenceRequest Constructors

Learn more about the available constructors with the `RecordWithReferenceRequest` class.

RecordWithReferenceRequest Properties

Learn more about the available properties with the `RecordWithReferenceRequest` class.

RecordWithReferenceRequest Constructors

Learn more about the available constructors with the `RecordWithReferenceRequest` class.

The `RecordWithReferenceRequest` class includes these constructors.

RecordWithReferenceRequest(referenceId, record)

Creates an instance of the `RecordWithReferenceRequest` class to associate a record object with a reference identifier by using the `referenceId` and `record` object properties.

`RecordWithReferenceRequest(referenceId, record)`

Creates an instance of the `RecordWithReferenceRequest` class to associate a record object with a reference identifier by using the `referenceId` and `record` object properties.

Signature

`public RecordWithReferenceRequest(String referenceId, placequote.RecordResource record)`

Parameters

`referenceId`
Type: String

Reference ID that maps to the subrequest response and can be used to reference the response in subsequent subrequests. You can reference the referenceId in either the body or URL of a subrequest. Use this syntax to include a reference: @{referenceId.FieldName}. See referenceId property of a composite subrequest.

`record`
Type: placequote.RecordResource

Record object that's defined using the `RecordResource` class.

RecordWithReferenceRequest Properties

Learn more about the available properties with the `RecordWithReferenceRequest` class.

The `RecordWithReferenceRequest` class includes these properties.

record
Set the `record` property to specify the record object that's defined by using the `RecordResource` class.

referenceId
Set the `referenceId` property to specify the reference ID that maps to the subrequest response. This reference ID can be used to reference the response in subsequent subrequests.

`record`

Set the `record` property to specify the record object that's defined by using the `RecordResource` class.

Signature

`public placequote.RecordResource record {get; set;}`

Property Value

Type: placequote.RecordResource

`referenceId`

Set the `referenceId` property to specify the reference ID that maps to the subrequest response. This reference ID can be used to reference the response in subsequent subrequests.

Signature

```
public String referenceId {get; set;}
```

Property Value

Type: String

renew_assets_summary Namespace

Retrieve details about renewable assets to create renewal opportunities.

The `renew_assets_summary` namespace includes these classes.

RenewalOpptyDetail Class

Contains asset details and renewal pricing information for creating renewal opportunities.

Namespace

renew_assets_summary on page 1695

RenewalOpptyDetail Constructors

Learn more about the constructors available with the RenewalOpptyDetail class.

The `RenewalOpptyDetail` class includes these constructors.

RenewalOpptyDetail(assetId, startDate, endDate, productId, account, priceBookId, priceBookEntryId, lastAssetActionSubtype, renewalPriceDetails, opportunityProductId, orderItem, rootAssetOpportunity, lastAssetAction, totalLineAmount)

Initializes the RenewalOpptyDetail class that contains asset details and renewal pricing information.

Signature

```
public RenewalOpptyDetail(String assetId, Date startDate, Date endDate, String productId,
String account, String priceBookId, String priceBookEntryId, String
lastAssetActionSubtype, List<renew_assets_summary.RenewalPriceDetail>
renewalPriceDetails, String opportunityProductId, String orderItem, String
rootAssetOpportunity, String lastAssetAction, Decimal totalLineAmount)
```

```
renew_assets_summary.RenewalOpptyDetail, newinstance, [String, Date, Date, String, String,
 String, String, String, List<renew_assets_summary.RenewalPriceDetail>, String, String,
String, String, Decimal], renew_assets_summary.RenewalOpptyDetail
```

Parameters

assetId

Type: String

The ID of the asset.

startDate

Type: Date

The start date of the asset.

endDate

Type: Date

The end date of the asset.

productId

Type: String

The ID of the product associated with the asset.

account

Type: String

The ID of the account.

priceBookId

Type: String

The ID of the price book.

priceBookEntryId

Type: String

The ID of the price book entry.

lastAssetActionSubtype

Type: String

The subtype of the last action performed on the asset. Valid value is `Start Date Adjustment`.

renewalPriceDetails

Type: List<renew_assets_summary.RenewalPriceDetail on page 1702 >

List of renewal price details for the asset.

opportunityProductId

Type: String

The ID of the associated opportunity product.

orderItem

Type: String

The ID of the associated order item.

rootAssetOpportunity

Type: String

The ID of the root asset opportunity.

lastAssetAction

Type: String

The last action performed on the asset.

- `Initial Sale`
- `Upsell`
- `Downsell`
- `Renewal`
- `Cancellation`

totalLineAmount

Type: Decimal

The total line amount for the asset.

RenewalOpptyDetail Properties

Learn more about the properties available with the RenewalOpptyDetail class.

The `RenewalOpptyDetail` class includes these properties.

account

Get or set the ID of the account.

assetId

Get or set the ID of the asset.

endDate

Get or set the end date of the asset.

lastAssetAction

Get or set the last action performed on the asset.

lastAssetActionSubtype

Get or set the subtype of the last action performed on the asset.

opportunityProductId

Get or set the ID of the associated opportunity product.

orderItem

Get or set the ID of the associated order item.

priceBookEntryId

Get or set the ID of the price book entry.

priceBookId

Get or set the ID of the price book.

productId
Get or set the ID of the product associated with the asset.

renewalPriceDetails
Get or set the list of renewal price details for the asset.

rootAssetOpportunity
Get or set the ID of the root asset opportunity.

startDate
Get or set the start date of the asset.

totalLineAmount
Get or set the total line amount for the asset.

account

Get or set the ID of the account.

Signature

`public String account {get; set;}`

```
renew_assets_summary.RenewalOpptyDetail, account
```

Property Value

Type: String

assetId

Get or set the ID of the asset.

Signature

`public String assetId {get; set;}`

```
renew_assets_summary.RenewalOpptyDetail, assetId
```

Property Value

Type: String

endDate

Get or set the end date of the asset.

Signature

`public Date endDate {get; set;}`

```
renew_assets_summary.RenewalOpptyDetail, endDate
```

Property Value

Type: Date

lastAssetAction

Get or set the last action performed on the asset.

Signature

`public String lastAssetAction {get; set;}`

```
renew_assets_summary.RenewalOpptyDetail, lastAssetAction
```

Property Value

Type: String

Valid values are:

- `Initial Sale`
- `Upsell`
- `Downsell`
- `Renewal`
- `Cancellation`

lastAssetActionSubtype

Get or set the subtype of the last action performed on the asset.

Signature

`public String lastAssetActionSubtype {get; set;}`

```
renew_assets_summary.RenewalOpptyDetail, lastAssetActionSubtype
```

Property Value

Type: String

Valid value is `Start Date Adjustment`.

opportunityProductId

Get or set the ID of the associated opportunity product.

Signature

`public String opportunityProductId {get; set;}`

```
renew_assets_summary.RenewalOpptyDetail, opportunityProductId
```

Property Value

Type: String

orderItem

Get or set the ID of the associated order item.

Signature

`public String orderItem {get; set;}`

```
renew_assets_summary.RenewalOpptyDetail, orderItem
```

Property Value

Type: String

priceBookEntryId

Get or set the ID of the price book entry.

Signature

`public String priceBookEntryId {get; set;}`

```
renew_assets_summary.RenewalOpptyDetail, priceBookEntryId
```

Property Value

Type: String

priceBookId

Get or set the ID of the price book.

Signature

`public String priceBookId {get; set;}`

```
renew_assets_summary.RenewalOpptyDetail, priceBookId
```

Property Value

Type: String

productId

Get or set the ID of the product associated with the asset.

Signature

`public String productId {get; set;}`

```
renew_assets_summary.RenewalOpptyDetail, productId
```

Property Value

Type: String

renewalPriceDetails

Get or set the list of renewal price details for the asset.

Signature

`public List<renew_assets_summary.RenewalPriceDetail> renewalPriceDetails {get; set;}`

```
renew_assets_summary.RenewalOpptyDetail, renewalPriceDetails
```

Property Value

Type: List<renew_assets_summary.RenewalPriceDetail on page 1702 >

rootAssetOpportunity

Get or set the ID of the root asset opportunity.

Signature

`public String rootAssetOpportunity {get; set;}`

```
renew_assets_summary.RenewalOpptyDetail, rootAssetOpportunity
```

Property Value

Type: String

startDate

Get or set the start date of the asset.

Signature

`public Date startDate {get; set;}`

```
renew_assets_summary.RenewalOpptyDetail, startDate
```

Property Value

Type: Date

totalLineAmount

Get or set the total line amount for the asset.

Signature

`public Decimal totalLineAmount {get; set;}`

```
renew_assets_summary.RenewalOpptyDetail, totalLineAmount
```

Property Value

Type: Decimal

RenewalPriceDetail Class

Contains net unit price and quantity details for an asset to set as sales price for renewal opportunities.

Namespace

renew_assets_summary on page 1695

RenewalPriceDetail Constructors

Learn more about the constructors available with the RenewalPriceDetail class.

The `RenewalPriceDetail` class includes these constructors.

RenewalPriceDetail(netUnitPrice, quantity)

Initializes the RenewalPriceDetail class that contains net unit price and quantity information.

Signature

`public RenewalPriceDetail(Decimal netUnitPrice, Decimal quantity)`

```
renew_assets_summary.RenewalPriceDetail, newinstance, [Decimal, Decimal],
renew_assets_summary.RenewalPriceDetail
```

Parameters

netUnitPrice
Type: Decimal

The net unit price of the asset.

quantity
Type: Decimal

The quantity of the asset.

RenewalPriceDetail Properties

Learn more about the properties available with the RenewalPriceDetail class.

The `RenewalPriceDetail` class includes these properties.

[netUnitPrice](#)
Get or set the net unit price of the asset.

[quantity](#)
Get or set the quantity of the asset.

netUnitPrice

Get or set the net unit price of the asset.

Signature

`public Decimal netUnitPrice {get; set;}`

```
renew_assets_summary.RenewalPriceDetail, netUnitPrice
```

Property Value

Type: Decimal

quantity

Get or set the quantity of the asset.

Signature

`public Decimal quantity {get; set;}`

```
renew_assets_summary.RenewalPriceDetail, quantity
```

Property Value

Type: Decimal

RevSalesTrxn Namespace

Create a sales transaction, such as a quote or an order, with integrated pricing and configuration. Additionally, update an order or a quote, and insert and delete order or quote line items to calculate the estimated tax.

The `RevSalesTrxn` namespace includes these classes.

CatalogRatesPreferenceEnum Enum

Specifies the rate card entries defined in the catalog that must be fetched for quote line items, with usage-based selling during the place sales transaction process.

Usage

This enum is available when Usage Selling is enabled.

Enum Values

The `RevSalesTrxn.CatalogRatesPreferenceEnum` enum includes these values.

Value	Description
`Fetch`	Retrieves the rate card entries defined in the catalog for quote line items during the quote creation process.
`Skip`	Skips the retrieval of rate card entries for quote line items during the quote creation process. The default value is `Skip`.

ConfigurationExecutionEnum Enum

Specifies the configuration method for the place sales transaction request.

Usage

Use these enum values for the `configurationExecutionEnum` property in the PlaceSalesTransactionExecutor class.

Enum Values

The `RevSalesTrxn.ConfigurationExecutionEnum` enum has these values.

Value	Description
`Force`	Specifies to enforce the predefined configuration process during the sales transaction process.
`Skip`	Specifies to skip the configuration process during the quote creation process. The default value is `Skip`.
`System`	Specifies the system to determine whether the configuration process is required.

ConfigurationOptionsInput Class

Contains methods and properties to set the configuration options for the input to the product configurator.

Namespace

RevSalesTrxn

Usage

This class holds the required details of the product configuration input. Set the class properties to enable default configuration, execution of configuration rules, and validation of the product catalog. Use these class properties as an input to the PlaceSalesTransactionExecutor class method.

Example

```
        RevSalesTrxn.GraphRequest graph = new RevSalesTrxn.GraphRequest('test',
listOfRecords);
    RevSalesTrxn.PricingPreferenceEnum pricingPrefEnum =
RevSalesTrxn.PricingPreferenceEnum.SYSTEM;
    RevSalesTrxn.ConfigurationExecutionEnum configurationExecutionEnum =
RevSalesTrxn.ConfigurationExecutionEnum.SYSTEM;
    RevSalesTrxn.ConfigurationOptionsInput cInput = new
RevSalesTrxn.ConfigurationOptionsInput();
    cInput.addDefaultConfiguration = true;
    cInput.executeConfigurationRules = true;
    cInput.validateAmendRenewCancel = true;
    cInput.validateProductCatalog = true;
        //Place Sales Transaction API Call
     RevSalesTrxn.PlaceSalesTransactionResponse resp =
PlaceQuote.PlaceSalesTransactionExecutor.execute(graph,pricingPrefEnum,configurationExecutionEnum,cInput,null);
```

ConfigurationOptionsInput Properties

Set the ConfigurationOptionsInput class properties to add default configuration, execute configuration rules, and validate the product catalog.

ConfigurationOptionsInput Methods

Learn more about the methods available with the ConfigurationOptionsInput class.

ConfigurationOptionsInput Properties

Set the ConfigurationOptionsInput class properties to add default configuration, execute configuration rules, and validate the product catalog.

The `ConfigurationOptionsInput` class includes these properties.

addDefaultConfiguration

Sets the default product configuration, such as bundle and product attributes, for a quote request.

executeConfigurationRules

Sets the requirement for a quote to adhere to the configuration rules.

validateAmendRenewCancel

Sets the requirement to run validations related to amend, renew, or cancel processes.

validateProductCatalog

Sets the requirement to validate a quote against the product catalog.

addDefaultConfiguration

Sets the default product configuration, such as bundle and product attributes, for a quote request.

Signature

`public Boolean addDefaultConfiguration {get; set;}`

Property Value

Type: Boolean

Indicates whether to automatically add default configuration to the order (`true`) or not (`false`).

`executeConfigurationRules`

Sets the requirement for a quote to adhere to the configuration rules.

Signature

```
public Boolean executeConfigurationRules {get; set;}
```

Property Value

Type: Boolean

Indicates whether the order must adhere to configuration rules during processing (`true`) or bypass them (`false`).

`validateAmendRenewCancel`

Sets the requirement to run validations related to amend, renew, or cancel processes.

Signature

```
public Boolean validateAmendRenewCancel {get; set;}
```

Property Value

Type: Boolean

Indicates whether to run validations related to amend, renew, or cancel processes (`true`) or not (`false`).

`validateProductCatalog`

Sets the requirement to validate a quote against the product catalog.

Signature

```
public Boolean validateProductCatalog {get; set;}
```

Property Value

Type: Boolean

Indicates whether the quote must be validated against the product catalog (`true`) or not (`false`).

ConfigurationOptionsInput Methods

Learn more about the methods available with the ConfigurationOptionsInput class.

The `ConfigurationOptionsInput` class includes these methods.

[equals(obj)](#)
Determines the equality of external objects in a list. This method is dynamic and is based on the equals() method in Java.

[hashCode()](#)
Determines the uniqueness of the external object records in a list.

[toString()](#)
Converts a value to a string.

`equals(obj)`

Determines the equality of external objects in a list. This method is dynamic and is based on the equals() method in Java.

Signature

`public Boolean equals(Object obj)`

Parameters

obj
Type: Object

Reference object that's used to compare with the class object.

Return Value

Type: Boolean

Indicates if the class object is same as the reference object (`true`) or not (`false`).

`hashCode()`

Determines the uniqueness of the external object records in a list.

Signature

`public Integer hashCode()`

Return Value

Type: Integer

Integer hash code that represents the value of the object. Equal objects as per the `equals()` method must return the same hash code.

`toString()`

Converts a value to a string.

Signature

`public String toString()`

Return Value

Type: String

GraphRequest Class

Contains constructors and properties to set the graph ID and a list of records to be ingested. The list of records is specified in a key-value map format that contains field values.

Namespace

RevSalesTrxn

GraphRequest Constructors
Learn more about the available constructors with the GraphRequest class.

GraphRequest Properties
Learn more about the available properties with the GraphRequest class.

GraphRequest Constructors

Learn more about the available constructors with the GraphRequest class.

The `GraphRequest` class includes these constructors.

GraphRequest(graphId, records)
Creates an instance of the GraphRequest class to assign the graph ID and a list of records to be ingested.

`GraphRequest(graphId, records)`

Creates an instance of the GraphRequest class to assign the graph ID and a list of records to be ingested.

Signature

```
public GraphRequest(String graphId, List<RevSalesTrxn.RecordWithReferenceRequest>
records)
```

Parameters

`graphId`
Type: String

ID of the graph.

`records`
Type: List<revsalestrxn.RecordWithReferenceRequest on page 1731 >

List of records to be ingested.

GraphRequest Properties

Learn more about the available properties with the GraphRequest class.

The `GraphRequest` class includes these properties.

graphId
Set the `graphId` property to assign the ID value of the graph.

graphId

Set the `graphId` property to assign the ID value of the graph.

Signature

```
public String graphId {get; set;}
```

Property Value

Type: String

GroupRampActionEnum Enum

Specifies the action that you want to perform on group ramp segments. Additionally, you can also convert a non-ramped group into a ramped group.

Enum Values

The `RevSalesTrxn.GroupRampActionEnum` enum includes these values.

Value	Description
`AddProducts`	Specifies to add rampable products to group ramp segments.
`DeleteProducts`	Specifies to delete ramped products.
`EditGroup`	Specifies to convert a non-ramped group into a group ramp segment, or edit group ramp segment attributes such as name and description, except the start and end dates.
`EditRampSchedule`	Specifies to edit details of the group ramp segments, including start and end dates.
`DeleteSegment`	Specifies to delete the first or last segment in a group ramp schedule.
`ConvertToNonRampedGroup`	Specifies to convert the first or last group ramp segment into a non-ramped group.

To add or delete ramped line items from multiple group ramp segments, pass all the applicable values in the `graph` request. To refer to Connect API examples that specify actions to create ramp deals for groups, see Group Ramp Action Input on page 1491.

PersistPreferenceEnum Enum

Specifies whether to persist pricing changes for each sales transaction record. Available in API version 65.0 and later.

Enum Values

The `RevSalesTrxn.PersistPreferenceEnum` enum includes this value.

Value	Description
`Skip`	Skips the persistence of pricing changes for each sales transaction record. To persist pricing changes, specify `null` as the value in the method signature. If this value isn't specified, then request to persist pricing changes is performed by default.

PlaceSalesTransactionException Class

Contains methods to hold the exception details for the place sales transaction request.

Namespace

RevSalesTrxn

PlaceSalesTransactionException Methods
Learn more about the methods available with the PlaceSalesTransactionException class.

PlaceSalesTransactionException Methods

Learn more about the methods available with the PlaceSalesTransactionException class.

The `PlaceSalesTransactionException` class includes these methods.

getErrorCode()
Gets the error code that's associated to the place sales transaction request.

`getErrorCode()`

Gets the error code that's associated to the place sales transaction request.

Signature

`public String getErrorCode()`

Return Value

Type: String

PlaceSalesTransactionExecutor Class

Contains methods to place a sales transaction with details of the graph request, pricing preferences, and configuration options.

Namespace

RevSalesTrxn

[PlaceSalesTransactionExecutor Methods](#)

Learn more about the methods available with the `PlaceSalesTransactionExecutor` class.

[PlaceSalesTransactionExecutor Example Implementation](#)

To place a sales transaction from Apex, refer to the example implementation of the `PlaceSalesTransactionExecutor` class.

PlaceSalesTransactionExecutor Methods

Learn more about the methods available with the `PlaceSalesTransactionExecutor` class.

The `PlaceSalesTransactionExecutor` class includes these methods.

[execute(graphRequest, pricingPreferenceEnum, configurationExecutionEnum, configuratorOptions, contextId)](#)

Use the method in the `PlaceSalesTransactionExecutor` class to execute the Place Sales Transaction Apex API request by assigning the properties for graph request, pricing references, and configurator options.

[execute(graphRequest, pricingPreferenceEnum, configurationExecutionEnum, configuratorOptions, contextId, catalogRatesPreferenceEnum)](#)

Use the method in the `PlaceSalesTransactionExecutor` class to execute the Place Sales Transaction Apex API request by assigning the properties for graph request, pricing references, configurator options, and catalog rates.

[execute(graphRequest, pricingPreferenceEnum, configurationExecutionEnum, configuratorOptions, contextId, catalogRatesPreferenceEnum, taxPreferenceEnum, persistPreferenceEnum)](#)

Use the method in the `PlaceSalesTransactionExecutor` class to execute the Place Sales Transaction Apex API request by assigning the properties for graph request, pricing and tax preferences, configurator options, and catalog rates.

[execute(graphRequest, pricingPreferenceEnum, configurationExecutionEnum, configuratorOptions, contextId, taxPreferenceEnum, persistPreferenceEnum, groupRampActionEnum)](#)

Use the method in the `PlaceSalesTransactionExecutor` class to execute the Place Sales Transaction Apex API request by assigning the properties for graph request, pricing and tax preferences, configurator options, persist preference, and group ramp action.

[execute(graphRequest, pricingPreferenceEnum, configurationExecutionEnum, configuratorOptions, contextId, catalogRatesPreferenceEnum, taxPreferenceEnum, persistPreferenceEnum, groupRampActionEnum)](#)

Use the method in the `PlaceSalesTransactionExecutor` class to execute the Place Sales Transaction Apex API request by assigning the properties for graph request, pricing and tax preferences, configurator options, catalog rates, persist preference, and group ramp action.

`execute(graphRequest, pricingPreferenceEnum, configurationExecutionEnum, configuratorOptions, contextId)`

Use the method in the `PlaceSalesTransactionExecutor` class to execute the Place Sales Transaction Apex API request by assigning the properties for graph request, pricing references, and configurator options.

Signature

```
public static revsalestrxn.PlaceSalesTransactionResponse
execute(revsalestrxn.GraphRequest graphRequest, revsalestrxn.PricingPreferenceEnum
pricingPreferenceEnum, revsalestrxn.ConfigurationExecutionEnum
configurationExecutionEnum, revsalestrxn.ConfiguratorOptions configuratorOptions,
revsalestrxn.ContextId contextId)
```

Parameters

graphRequest
Type: revsalestrxn.GraphRequest

The sObject graph values of the order payload to be ingested.

pricingPreferenceEnum
Type: revsalestrxn.PricingPreferenceEnum

Pricing preference during the sales transaction process.

configurationExecutionEnum
Type: revsalestrxn.ConfigurationExecutionEnum

Configuration method for the sales transaction request.

configuratorOptions
Type: revsalestrxn.ConfigurationOptionsInput

Configuration options during the creation of the sales transaction.

contextId
Type: String

Context ID to assign to the sales transaction.

Return Value

Type: revsalestrxn.PlaceSalesTransactionResponse

execute(graphRequest, pricingPreferenceEnum, configurationExecutionEnum, configuratorOptions, contextId, catalogRatesPreferenceEnum)

Use the method in the `PlaceSalesTransactionExecutor` class to execute the Place Sales Transaction Apex API request by assigning the properties for graph request, pricing references, configurator options, and catalog rates.

Signature

Usage Selling must be enabled to use this signature.

```
public static revsalestrxn.PlaceSalesTransactionResponse
execute(revsalestrxn.GraphRequest graphRequest, revsalestrxn.PricingPreferenceEnum
pricingPreferenceEnum, revsalestrxn.ConfigurationExecutionEnum
configurationExecutionEnum, revsalestrxn.ConfiguratorOptions configuratorOptions,
revsalestrxn.ContextId contextId, revsalestrxn.CatalogRatesPreferenceEnum
catalogRatesPreferenceEnum)
```

Parameters

graphRequest
Type: revsalestrxn.GraphRequest

The sObject graph values of the order payload to be ingested.

pricingPreferenceEnum
Type: revsalestrxn.PricingPreferenceEnum

Pricing preference during the sales transaction process.

configurationExecutionEnum
Type: revsalestrxn.ConfigurationExecutionEnum

Configuration method for the sales transaction request.

configuratorOptions
Type: revsalestrxn.ConfigurationOptionsInput

Configuration options during the creation of the sales transaction.

contextId
Type: String

Context ID to assign to the sales transaction.

catalogRatesPreferenceEnum
Type: revsalestrxn.CatalogRatesPreferenceEnum

Rate card entries defined in the catalog that must be fetched for quote line items, with usage-based selling during the place sales transaction process.

Return Value

Type: revsalestrxn.PlaceSalesTransactionResponse

execute(graphRequest, pricingPreferenceEnum, configurationExecutionEnum, configuratorOptions, contextId, catalogRatesPreferenceEnum, taxPreferenceEnum, persistPreferenceEnum)

Use the method in the `PlaceSalesTransactionExecutor` class to execute the Place Sales Transaction Apex API request by assigning the properties for graph request, pricing and tax preferences, configurator options, and catalog rates.

Signature

Usage Selling must be enabled to use this signature.

```
public static revsalestrxn.PlaceSalesTransactionResponse
execute(revsalestrxn.GraphRequest graphRequest, revsalestrxn.PricingPreferenceEnum
pricingPreferenceEnum, revsalestrxn.ConfigurationExecutionEnum
configurationExecutionEnum, revsalestrxn.ConfiguratorOptions configuratorOptions,
revsalestrxn.ContextId contextId, revsalestrxn.CatalogRatesPreferenceEnum
catalogRatesPreferenceEnum, revsalestrxn.TaxPreferenceEnum taxPreferenceEnum,
revsalestrxn.PersistPreferenceEnum persistPreferenceEnum)
```

Parameters

graphRequest
Type: revsalestrxn.GraphRequest

The sObject graph values of the order payload to be ingested.

pricingPreferenceEnum
Type: revsalestrxn.PricingPreferenceEnum

Pricing preference during the sales transaction process.

configurationExecutionEnum
Type: revsalestrxn.ConfigurationExecutionEnum

Configuration method for the sales transaction request.

configuratorOptions
Type: revsalestrxn.ConfigurationOptionsInput

Configuration options during the creation of the sales transaction.

contextId
Type: String

Context ID to assign to the sales transaction.

catalogRatesPreferenceEnum
Type: revsalestrxn.CatalogRatesPreferenceEnum

Rate card entries defined in the catalog that must be fetched for quote line items, with usage-based selling during the place sales transaction process.

taxPreferenceEnum
Type: revsalestrxn.taxPreferenceEnum

Tax preference during the sales transaction process. Available in API version 65.0 and later.

persistPreferenceEnum
Type: revsalestrxn.persistPreferenceEnum

Persist preference during the sales transaction process. Available in API version 65.0 and later.

Return Value

Type: revsalestrxn.PlaceSalesTransactionResponse

execute(graphRequest, pricingPreferenceEnum, configurationExecutionEnum, configuratorOptions, contextId, taxPreferenceEnum, persistPreferenceEnum, groupRampActionEnum)

Use the method in the `PlaceSalesTransactionExecutor` class to execute the Place Sales Transaction Apex API request by assigning the properties for graph request, pricing and tax preferences, configurator options, persist preference, and group ramp action.

Signature

```
public static revsalestrxn.PlaceSalesTransactionResponse
execute(revsalestrxn.GraphRequest graphRequest, revsalestrxn.PricingPreferenceEnum
pricingPreferenceEnum, revsalestrxn.ConfigurationExecutionEnum
configurationExecutionEnum, revsalestrxn.ConfiguratorOptions configuratorOptions,
revsalestrxn.ContextId contextId, revsalestrxn.TaxPreferenceEnum taxPreferenceEnum,
revsalestrxn.PersistPreferenceEnum persistPreferenceEnum,
revsalestrxn.GroupRampActionEnum groupRampActionEnum)
```

Parameters

graphRequest
Type: revsalestrxn.GraphRequest

The sObject graph values of the order payload to be ingested.

pricingPreferenceEnum

Type: revsalestrxn.PricingPreferenceEnum

Pricing preference during the sales transaction process.

configurationExecutionEnum

Type: revsalestrxn.ConfigurationExecutionEnum

Configuration method for the sales transaction request.

configuratorOptions

Type: revsalestrxn.ConfigurationOptionsInput

Configuration options during the creation of the sales transaction.

contextId

Type: String

Context ID to assign to the sales transaction.

taxPreferenceEnum

Type: revsalestrxn.taxPreferenceEnum

Tax preference during the sales transaction process. Available in API version 65.0 and later.

persistPreferenceEnum

Type: revsalestrxn.persistPreferenceEnum

Persist preference during the sales transaction process. Available in API version 65.0 and later.

groupRampActionEnum

Type: revsalestrxn.GroupRampActionEnum

Specifies the action that you want to perform on group ramp segments.

Return Value

Type: revsalestrxn.PlaceSalesTransactionResponse

execute(graphRequest, pricingPreferenceEnum, configurationExecutionEnum, configuratorOptions, contextId, catalogRatesPreferenceEnum, taxPreferenceEnum, persistPreferenceEnum, groupRampActionEnum)

Use the method in the `PlaceSalesTransactionExecutor` class to execute the Place Sales Transaction Apex API request by assigning the properties for graph request, pricing and tax preferences, configurator options, catalog rates, persist preference, and group ramp action.

Signature

Usage Selling must be enabled to use this signature.

```
public static revsalestrxn.PlaceSalesTransactionResponse
execute(revsalestrxn.GraphRequest graphRequest, revsalestrxn.PricingPreferenceEnum
pricingPreferenceEnum, revsalestrxn.ConfigurationExecutionEnum
configurationExecutionEnum, revsalestrxn.ConfiguratorOptions configuratorOptions,
revsalestrxn.ContextId contextId, revsalestrxn.CatalogRatesPreferenceEnum
catalogRatesPreferenceEnum, revsalestrxn.TaxPreferenceEnum taxPreferenceEnum,
```

```
revsalestrxn.PersistPreferenceEnum persistPreferenceEnum,
revsalestrxn.GroupRampActionEnum groupRampActionEnum)
```

Parameters

graphRequest
Type: revsalestrxn.GraphRequest

The sObject graph values of the order payload to be ingested.

pricingPreferenceEnum
Type: revsalestrxn.PricingPreferenceEnum

Pricing preference during the sales transaction process.

configurationExecutionEnum
Type: revsalestrxn.ConfigurationExecutionEnum

Configuration method for the sales transaction request.

configuratorOptions
Type: revsalestrxn.ConfigurationOptionsInput

Configuration options during the creation of the sales transaction.

contextId
Type: String

Context ID to assign to the sales transaction.

catalogRatesPreferenceEnum
Type: revsalestrxn.CatalogRatesPreferenceEnum

Rate card entries defined in the catalog that must be fetched for quote line items, with usage-based selling during the place sales transaction process.

taxPreferenceEnum
Type: revsalestrxn.taxPreferenceEnum

Tax preference during the sales transaction process. Available in API version 65.0 and later.

persistPreferenceEnum
Type: revsalestrxn.persistPreferenceEnum

Persist preference during the sales transaction process. Available in API version 65.0 and later.

groupRampActionEnum
Type: revsalestrxn.GroupRampActionEnum

Specifies the action that you want to perform on group ramp segments.

Return Value

Type: revsalestrxn.PlaceSalesTransactionResponse

PlaceSalesTransactionExecutor Example Implementation

To place a sales transaction from Apex, refer to the example implementation of the `PlaceSalesTransactionExecutor` class.

Namespace

RevSalesTrxn

Usage

Customize this example to suit your requirements. Create the list of records to be ingested by using these steps. Replace the respective IDs with the values that are present in your org. For example, replace the value of `${Pricebook2Id}` field with the price book ID that's present in the org.

Example: This example shows a sample request to create a sales transaction, update a quote with a quote line item, delete a quote line item, or skip or persist pricing changes. It includes these steps.

- Set up a quote and quote line item.
- To create a quote line relationship, create an instance of the `RecordResource` class. To create an object with its fields, specify the object that you want to create by using POST method. Create a map of the fields and its values by using put(key, value) operation. Assign this field map to the record object.
- To update an object's specific fields, specify the object you want to update by using the PATCH method along with the ID of the object. Create a map of the fields and its values by using put(key, value) operation. Only certain fields on an object can be updated. Add the fields that you want to update to the map. Assign this field map to the record object.
- To associate the Record object with a reference identifier, create an instance of the `RecordWithReferenceRequest` class.
- To contain all record objects, create an instance of the `GraphRequest` class.

```
public class PlaceSalesTransactionTest {
    public static void callPSTAPI_Post() {
        RevSalesTrxn.PricingPreferenceEnum pricingPrefEnum =
RevSalesTrxn.PricingPreferenceEnum.SYSTEM;
        RevSalesTrxn.ConfigurationExecutionEnum configurationExecutionEnum =
RevSalesTrxn.ConfigurationExecutionEnum.SYSTEM;

        //Quote setup
        RevSalesTrxn.RecordResource quoteRecord = new
RevSalesTrxn.RecordResource(Quote.getSobjectType(),'POST');
        Map<String,Object> quoteFieldValues = new Map<String,Object>();
        quoteFieldValues.put('Name','q-ap12');
        quoteFieldValues.put('OpportunityId','006xx000001a3e8AAA');
        quoteFieldValues.put('Pricebook2Id','01sDU000000JRX8YAO');
        quoteRecord.fieldValues = quoteFieldValues;

        //Quote line item setup

        //1st quote line item
        RevSalesTrxn.RecordResource quoteLineItemRecord1 = new
RevSalesTrxn.RecordResource(QuoteLineItem.getSobjectType(),'POST');
        Map<String,Object> quoteLineItemFieldValues = new Map<String,Object>();
        quoteLineItemFieldValues.put('Product2Id','01txx0000006i7JAAQ');
        quoteLineItemFieldValues.put('PricebookEntryId','01uxx0000008yc6AAA');
        quoteLineItemFieldValues.put('Quantity','2.0');
        quoteLineItemFieldValues.put('UnitPrice','1000');
        quoteLineItemFieldValues.put('StartDate','2025-03-15');
        quoteLineItemFieldValues.put('QuoteId','@{refQuote.id}');
        quoteLineItemRecord1.fieldValues = quoteLineItemFieldValues;
```

```
        RevSalesTrxn.RecordWithReferenceRequest quoteItemRecords = new
RevSalesTrxn.RecordWithReferenceRequest('refQuote',quoteRecord);
        RevSalesTrxn.RecordWithReferenceRequest quoteLineItemRecords1 = new
RevSalesTrxn.RecordWithReferenceRequest('refQuoteItem1',quoteLineItemRecord1);

        //2nd quote line item
        RevSalesTrxn.RecordResource quoteLineItemRecord2 = new
RevSalesTrxn.RecordResource(QuoteLineItem.getSobjectType(),'POST');
        Map<String,Object> quoteLineItemFieldValues2 = new Map<String,Object>();
        quoteLineItemFieldValues2.put('Product2Id','01txx0000006i7RAAQ');
        quoteLineItemFieldValues2.put('PricebookEntryId','01uxx0000008ybvAAA');
        quoteLineItemFieldValues2.put('Quantity','2.0');
        quoteLineItemFieldValues2.put('UnitPrice','7.0');
        quoteLineItemFieldValues2.put('StartDate','2025-03-15');
        quoteLineItemFieldValues2.put('QuoteId','@{refQuote.id}');
        quoteLineItemRecord2.fieldValues = quoteLineItemFieldValues2;
        RevSalesTrxn.RecordWithReferenceRequest quoteLineItemRecords2 = new
RevSalesTrxn.RecordWithReferenceRequest('refQuoteItem2',quoteLineItemRecord2);

        List<RevSalesTrxn.RecordWithReferenceRequest> listOfRecords = new
List<RevSalesTrxn.RecordWithReferenceRequest>();
        listOfRecords.add(quoteItemRecords);
        listOfRecords.add(quoteLineItemRecords1);
        listOfRecords.add(quoteLineItemRecords2);


        RevSalesTrxn.GraphRequest graph = new
RevSalesTrxn.GraphRequest('test',listOfRecords);
        System.debug(graph);

        //Place sales transaction API call - There are different executor signatures
detailed in the documentation
        RevSalesTrxn.PlaceSalesTransactionResponse resp =
RevSalesTrxn.PlaceSalesTransactionExecutor.execute(graph, pricingPrefEnum,
configurationExecutionEnum, new RevSalesTrxn.ConfigurationOptionsInput(), null, null,
 null, null);
        //If you do not have Usage Based Selling Feature, you can use the signatures
without catalog rates enum
        //RevSalesTrxn.PlaceSalesTransactionResponse resp =
RevSalesTrxn.PlaceSalesTransactionExecutor.execute(graph, pricingPrefEnum,
configurationExecutionEnum, new RevSalesTrxn.ConfigurationOptionsInput(), null);
        System.debug(resp);
    }

    public static void callPSTAPI_Post_Skip_Tax_Skip_Persist() {
        RevSalesTrxn.PricingPreferenceEnum pricingPrefEnum =
RevSalesTrxn.PricingPreferenceEnum.SYSTEM;
        RevSalesTrxn.ConfigurationExecutionEnum configurationExecutionEnum =
RevSalesTrxn.ConfigurationExecutionEnum.SYSTEM;
       RevSalesTrxn.TaxPreferenceEnum taxPrefEnum = RevSalesTrxn.TaxPreferenceEnum.SKIP;

        RevSalesTrxn.PersistPreferenceEnum persistPrefEnum =
RevSalesTrxn.PersistPreferenceEnum.SKIP;
```

```
        //Quote setup
        RevSalesTrxn.RecordResource quoteRecord = new
RevSalesTrxn.RecordResource(Quote.getSobjectType(),'POST');
        Map<String,Object> quoteFieldValues = new Map<String,Object>();
        quoteFieldValues.put('Name','q-ap12');
        quoteFieldValues.put('OpportunityId','006xx000001a3e8AAA');
        quoteFieldValues.put('Pricebook2Id','01sDU000000JRX8YAO');
        quoteRecord.fieldValues = quoteFieldValues;

        //Quote line item setup

        //1st quote line item
        RevSalesTrxn.RecordResource quoteLineItemRecord1 = new
RevSalesTrxn.RecordResource(QuoteLineItem.getSobjectType(),'POST');
        Map<String,Object> quoteLineItemFieldValues = new Map<String,Object>();
        quoteLineItemFieldValues.put('Product2Id','01txx0000006i7JAAQ');
        quoteLineItemFieldValues.put('PricebookEntryId','01uxx0000008yc6AAA');
        quoteLineItemFieldValues.put('Quantity','2.0');
        quoteLineItemFieldValues.put('UnitPrice','1000');
        quoteLineItemFieldValues.put('StartDate','2025-03-15');
        quoteLineItemFieldValues.put('QuoteId','@{refQuote.id}');
        quoteLineItemRecord1.fieldValues = quoteLineItemFieldValues;
        RevSalesTrxn.RecordWithReferenceRequest quoteItemRecords = new
RevSalesTrxn.RecordWithReferenceRequest('refQuote',quoteRecord);
        RevSalesTrxn.RecordWithReferenceRequest quoteLineItemRecords1 = new
RevSalesTrxn.RecordWithReferenceRequest('refQuoteItem1',quoteLineItemRecord1);

        //2nd quote line item
        RevSalesTrxn.RecordResource quoteLineItemRecord2 = new
RevSalesTrxn.RecordResource(QuoteLineItem.getSobjectType(),'POST');
        Map<String,Object> quoteLineItemFieldValues2 = new Map<String,Object>();
        quoteLineItemFieldValues2.put('Product2Id','01txx0000006i7RAAQ');
        quoteLineItemFieldValues2.put('PricebookEntryId','01uxx0000008ybvAAA');
        quoteLineItemFieldValues2.put('Quantity','2.0');
        quoteLineItemFieldValues2.put('UnitPrice','7.0');
        quoteLineItemFieldValues2.put('StartDate','2025-03-15');
        quoteLineItemFieldValues2.put('QuoteId','@{refQuote.id}');
        quoteLineItemRecord2.fieldValues = quoteLineItemFieldValues2;
        RevSalesTrxn.RecordWithReferenceRequest quoteLineItemRecords2 = new
RevSalesTrxn.RecordWithReferenceRequest('refQuoteItem2',quoteLineItemRecord2);

        List<RevSalesTrxn.RecordWithReferenceRequest> listOfRecords = new
List<RevSalesTrxn.RecordWithReferenceRequest>();
        listOfRecords.add(quoteItemRecords);
        listOfRecords.add(quoteLineItemRecords1);
        listOfRecords.add(quoteLineItemRecords2);


        RevSalesTrxn.GraphRequest graph = new
RevSalesTrxn.GraphRequest('test',listOfRecords);
        System.debug(graph);

        //Place sales transaction API call
        RevSalesTrxn.PlaceSalesTransactionResponse resp =
```

```
RevSalesTrxn.PlaceSalesTransactionExecutor.execute(graph, pricingPrefEnum,
configurationExecutionEnum, new RevSalesTrxn.ConfigurationOptionsInput(), null, null,
 taxPrefEnum, persistPrefEnum);
        System.debug(resp);
    }

    public static void callPSTAPI_Patch() {

        // Apex test to update a quote with a quote line item
        RevSalesTrxn.PricingPreferenceEnum pricingPrefEnum =
RevSalesTrxn.PricingPreferenceEnum.SYSTEM;
        RevSalesTrxn.ConfigurationExecutionEnum configurationExecutionEnum =
RevSalesTrxn.ConfigurationExecutionEnum.SYSTEM;

        //Quote setup
        RevSalesTrxn.RecordResource quoteRecord = new
RevSalesTrxn.RecordResource(Quote.getSobjectType(),'PATCH','0Q0xx0000004CYsCAM');
        RevSalesTrxn.RecordWithReferenceRequest quoteItemRecords = new
RevSalesTrxn.RecordWithReferenceRequest('refQuote',quoteRecord);

        //Quote line item setup
        //New quote line item
        RevSalesTrxn.RecordResource quoteLineItemRecord = new
RevSalesTrxn.RecordResource(QuoteLineItem.getSobjectType(),'POST');
        Map<String,Object> quoteLineItemFieldValues = new Map<String,Object>();
        quoteLineItemFieldValues.put('Product2Id','01txx0000006i7KAAQ');
        quoteLineItemFieldValues.put('PricebookEntryId','01uxx0000008ycFAAQ');
        quoteLineItemFieldValues.put('Quantity','2.0');
        quoteLineItemFieldValues.put('UnitPrice','7.0');
        quoteLineItemFieldValues.put('StartDate','2025-03-15');
        quoteLineItemFieldValues.put('QuoteId','@{refQuote.id}');
        quoteLineItemRecord.fieldValues = quoteLineItemFieldValues;
        RevSalesTrxn.RecordWithReferenceRequest quoteLineItemRecords = new
RevSalesTrxn.RecordWithReferenceRequest('refQuoteItem',quoteLineItemRecord);
        List<RevSalesTrxn.RecordWithReferenceRequest> listOfRecords = new
List<RevSalesTrxn.RecordWithReferenceRequest>();
        listOfRecords.add(quoteItemRecords);
        listOfRecords.add(quoteLineItemRecords);
        RevSalesTrxn.GraphRequest graph = new RevSalesTrxn.GraphRequest('test',
listOfRecords);

        //Place sales transaction API call
        RevSalesTrxn.PlaceSalesTransactionResponse resp =
RevSalesTrxn.PlaceSalesTransactionExecutor.execute(graph, pricingPrefEnum,
configurationExecutionEnum, new RevSalesTrxn.ConfigurationOptionsInput(), null, null,
 null, null);
        System.debug(resp);
    }

    public static void callPSTAPI_Patch_Skip_Config_Skip_Pricing() {

        // Apex test to update a quote with a quote line item
        RevSalesTrxn.PricingPreferenceEnum pricingPrefEnum =
RevSalesTrxn.PricingPreferenceEnum.SKIP;
```

```
        RevSalesTrxn.ConfigurationExecutionEnum configurationExecutionEnum =
RevSalesTrxn.ConfigurationExecutionEnum.SKIP;

        //Quote setup
        RevSalesTrxn.RecordResource quoteRecord = new
RevSalesTrxn.RecordResource(Quote.getSobjectType(),'PATCH','0Q0xx0000004CYsCAM');
        RevSalesTrxn.RecordWithReferenceRequest quoteItemRecords = new
RevSalesTrxn.RecordWithReferenceRequest('refQuote',quoteRecord);

        //Quote line item setup
        //New quote line item
        RevSalesTrxn.RecordResource quoteLineItemRecord = new
RevSalesTrxn.RecordResource(QuoteLineItem.getSobjectType(),'POST');
        Map<String,Object> quoteLineItemFieldValues = new Map<String,Object>();
        quoteLineItemFieldValues.put('Product2Id','01txx0000006i7KAAQ');
        quoteLineItemFieldValues.put('PricebookEntryId','01uxx0000008ycFAAQ');
        quoteLineItemFieldValues.put('Quantity','2.0');
        quoteLineItemFieldValues.put('UnitPrice','7.0');
        quoteLineItemFieldValues.put('StartDate','2025-03-15');
        quoteLineItemFieldValues.put('QuoteId','@{refQuote.id}');
        quoteLineItemRecord.fieldValues = quoteLineItemFieldValues;
        RevSalesTrxn.RecordWithReferenceRequest quoteLineItemRecords = new
RevSalesTrxn.RecordWithReferenceRequest('refQuoteItem',quoteLineItemRecord);
        List<RevSalesTrxn.RecordWithReferenceRequest> listOfRecords = new
List<RevSalesTrxn.RecordWithReferenceRequest>();
        listOfRecords.add(quoteItemRecords);
        listOfRecords.add(quoteLineItemRecords);
        RevSalesTrxn.GraphRequest graph = new RevSalesTrxn.GraphRequest('test',
listOfRecords);

        //Place sales transaction API call
        RevSalesTrxn.PlaceSalesTransactionResponse resp =
RevSalesTrxn.PlaceSalesTransactionExecutor.execute(graph, pricingPrefEnum,
configurationExecutionEnum, new RevSalesTrxn.ConfigurationOptionsInput(), null, null,
 null, null);
        System.debug(resp);
    }

    public static void callPSTAPI_Patch_With_ContextId() {

          // Apex test to update a quote with a quote line item
        RevSalesTrxn.PricingPreferenceEnum pricingPrefEnum =
RevSalesTrxn.PricingPreferenceEnum.SYSTEM;
        RevSalesTrxn.ConfigurationExecutionEnum configurationExecutionEnum =
RevSalesTrxn.ConfigurationExecutionEnum.SYSTEM;

        //Quote setup
       String contextId = 'MY_CONTEXT_ID';
        RevSalesTrxn.RecordResource quoteRecord = new
RevSalesTrxn.RecordResource(Quote.getSobjectType(),'PATCH','0Q0xx0000004CYsCAM');
        RevSalesTrxn.RecordWithReferenceRequest quoteItemRecords = new
RevSalesTrxn.RecordWithReferenceRequest('refQuote',quoteRecord);

        //Quote line item setup
```

```
        //New quote line item
        RevSalesTrxn.RecordResource quoteLineItemRecord = new
RevSalesTrxn.RecordResource(QuoteLineItem.getSobjectType(),'POST');
        Map<String,Object> quoteLineItemFieldValues = new Map<String,Object>();
        quoteLineItemFieldValues.put('Product2Id','01txx0000006i7KAAQ');
        quoteLineItemFieldValues.put('PricebookEntryId','01uxx0000008ycFAAQ');
        quoteLineItemFieldValues.put('Quantity','2.0');
        quoteLineItemFieldValues.put('UnitPrice','7.0');
        quoteLineItemFieldValues.put('StartDate','2025-03-15');
        quoteLineItemFieldValues.put('QuoteId','@{refQuote.id}');
        quoteLineItemRecord.fieldValues = quoteLineItemFieldValues;
        RevSalesTrxn.RecordWithReferenceRequest quoteLineItemRecords = new
RevSalesTrxn.RecordWithReferenceRequest('refQuoteItem',quoteLineItemRecord);
        List<RevSalesTrxn.RecordWithReferenceRequest> listOfRecords = new
List<RevSalesTrxn.RecordWithReferenceRequest>();
        listOfRecords.add(quoteItemRecords);
        listOfRecords.add(quoteLineItemRecords);
        RevSalesTrxn.GraphRequest graph = new RevSalesTrxn.GraphRequest('test',
listOfRecords);

        //Place sales transaction API call
        RevSalesTrxn.PlaceSalesTransactionResponse resp =
RevSalesTrxn.PlaceSalesTransactionExecutor.execute(graph, pricingPrefEnum,
configurationExecutionEnum, new RevSalesTrxn.ConfigurationOptionsInput(), contextId,
null, null, null);
        System.debug(resp);
    }

    public static void callPSTAPI_Delete() {

        // Apex test to update a quote with a quote line item
        RevSalesTrxn.PricingPreferenceEnum pricingPrefEnum =
RevSalesTrxn.PricingPreferenceEnum.SYSTEM;
        RevSalesTrxn.ConfigurationExecutionEnum configurationExecutionEnum =
RevSalesTrxn.ConfigurationExecutionEnum.SYSTEM;

        //Quote setup
        RevSalesTrxn.RecordResource quoteRecord = new
RevSalesTrxn.RecordResource(Quote.getSobjectType(),'PATCH','0Q0xx0000004CYsCAM');
        RevSalesTrxn.RecordWithReferenceRequest quoteItemRecords = new
RevSalesTrxn.RecordWithReferenceRequest('refQuote',quoteRecord);

        //Quote line item setup
        //Delete a quote line item
        RevSalesTrxn.RecordResource quoteLineItemRecord = new
RevSalesTrxn.RecordResource(QuoteLineItem.getSobjectType(),'DELETE','0QLxx0000004CYsCAM');

        RevSalesTrxn.RecordWithReferenceRequest quoteLineItemRecords = new
RevSalesTrxn.RecordWithReferenceRequest('refQuoteItem',quoteLineItemRecord);
        List<RevSalesTrxn.RecordWithReferenceRequest> listOfRecords = new
List<RevSalesTrxn.RecordWithReferenceRequest>();
        listOfRecords.add(quoteItemRecords);
        listOfRecords.add(quoteLineItemRecords);
        RevSalesTrxn.GraphRequest graph = new
```

```
RevSalesTrxn.GraphRequest('test',listOfRecords);

        //Place sales transaction API call
        RevSalesTrxn.PlaceSalesTransactionResponse resp =
           RevSalesTrxn.PlaceSalesTransactionExecutor.execute(graph, pricingPrefEnum,
 configurationExecutionEnum, new RevSalesTrxn.ConfigurationOptionsInput(), null,null,
 null, null);
        System.debug(resp);
    }
}
```

PlaceSalesTransactionResponse Class

Contains properties to hold the response to the place sales transaction request.

Namespace

RevSalesTrxn

Example

```
RevSalesTrxn.PlaceSalesTransactionResponse resp =
RevSalesTrxn.PlaceSalesTransactionExecutor.execute(graph, pricingPrefEnum,
configurationExecutionEnum, new RevSalesTrxn.ConfigurationOptionsInput(), null);
```

PlaceSalesTransactionResponse Properties
Learn more about the available properties with the PlaceSalesTransactionResponse class.

PlaceSalesTransactionResponse Properties

Learn more about the available properties with the PlaceSalesTransactionResponse class.

The `PlaceSalesTransactionResponse` class includes these properties.

contextDetails
Get the details of the context that’s created for the sales transaction.

errorResponse
Get the list of errors encountered during the synchronous processing of the API request.

isSuccess
Get the request status of the synchronous part of the processing.

salesTransactionId
Get the ID of the sales transaction, such as a quote or an order.

statusUrl
Get the asynchronous status URL of the request, if available.

trackerId
Get the unique identifier assigned to a specific operation or request that's used for tracking and referencing the operation.

contextDetails

Get the details of the context that's created for the sales transaction.

Signature

```
public ConnectApi.ContextDetails contextDetails {get; set;}
```

Property Value

Type: ConnectApi.ContextDetails

errorResponse

Get the list of errors encountered during the synchronous processing of the API request.

Signature

```
public List<ConnectApi.PlaceSalesTransactionErrorResponse> errorResponse {get; set;}
```

Property Value

Type: List<ConnectApi.PlaceSalesTransactionErrorResponse>

isSuccess

Get the request status of the synchronous part of the processing.

Signature

```
public Boolean isSuccess {get; set;}
```

Property Value

Type: Boolean

Indicates whether the synchronous part of the processing is successful (`true`) or not (`false`).

salesTransactionId

Get the ID of the sales transaction, such as a quote or an order.

Signature

```
public String salesTransactionId {get; set;}
```

Property Value

Type: String

statusUrl

Get the asynchronous status URL of the request, if available.

Signature

`public String statusUrl {get; set;}`

Property Value

Type: String

trackerId

Get the unique identifier assigned to a specific operation or request that's used for tracking and referencing the operation.

Signature

`public String trackerId {get; set;}`

Property Value

Type: String

PricingPreferenceEnum Enum

Specifies the pricing preference during the creation of a sales transaction.

Usage

Used by the PlaceSalesTransactionExecutor class.

Enum Values

The `RevSalesTrxn.PricingPreferenceEnum` enum includes these values.

Value	Description
`Force`	Specifies to enforce pricing during the creation of sales transactions.
`Skip`	Specifies to skip pricing during the creation of sales transactions.
`System`	Specifies the system to determine whether a pricing calculation is required. The default value is `System`.

RecordResource Class

Contains constructors and properties to create a record object from the field values of a sales transaction.

Namespace

RevSalesTrxn

Example

```
RevSalesTrxn.RecordResource quoteRecord = new
RevSalesTrxn.RecordResource(Quote.getSobjectType(),'POST');
```

RecordResource Constructors

Learn more about the available constructors with the RecordResource class.

RecordResource Properties

Learn more about the available properties with the RecordResource class.

RecordResource Constructors

Learn more about the available constructors with the RecordResource class.

The `RecordResource` class has these constructors.

RecordResource(type, method, groupAction, criteria)

Creates an instance of the RecordResource class to assign values to the fields of a sales transaction by using the sObject type, API method, and sales transaction ID properties. Additionally, you can group order or quote line items based on a criteria by using the groupAction and criteria properties.

RecordResource(type, method, id)

Creates an instance of the RecordResource class to assign values to the fields of a sales transaction by using the sObject type, API method, and sales transaction ID properties.

RecordResource(type, method)

Creates an instance of the RecordResource class to assign the values to the fields of a sales transaction by using the sObject type and API method properties.

`RecordResource(type, method, groupAction, criteria)`

Creates an instance of the RecordResource class to assign values to the fields of a sales transaction by using the sObject type, API method, and sales transaction ID properties. Additionally, you can group order or quote line items based on a criteria by using the groupAction and criteria properties.

Signature

`public RecordResource(Schema.SObjectType type, String method, String groupAction, Map<String,ANY> criteria)`

Parameters

`type`

Type: Schema.SObjectType

Object that's returned from the field describe result using the `getReferenceTo()` method or from the sObject describe result using the `getSObjectType()` method.

method

Type: String

Method for the API request, such as POST, PATCH, or DELETE.

groupAction

Type: String

Action to group order or quote line items. Valid values are:

- `GroupBy`
- `Group`
- `Ungroup`
- `GroupAll`
- `DeleteGroup`

criteria

Type: Map<String,ANY>

Criteria to group order or quote line items. For example, group order or quote line items based on a monthly billing frequency.

`RecordResource(type, method, id)`

Creates an instance of the RecordResource class to assign values to the fields of a sales transaction by using the sObject type, API method, and sales transaction ID properties.

Signature

`public RecordResource(Schema.SObjectType type, String method, Id id)`

Parameters

type

Type: Schema.SObjectType

Object that's returned from the field describe result using the `getReferenceTo()` method or from the sObject describe result using the `getSObjectType()` method.

method

Type: String

Method for the API request, such as POST, PATCH, or DELETE.

id

Type: Id

ID of the sales transaction, such as a quote or an order.

`RecordResource(type, method)`

Creates an instance of the RecordResource class to assign the values to the fields of a sales transaction by using the sObject type and API method properties.

Signature

`public RecordResource(Schema.SObjectType type, String method)`

Parameters

type

Type: Schema.SObjectType

Object that's returned from the field describe result using the `getReferenceTo()` method or from the sObject describe result using the `getSObjectType()` method.

method

Type: String

Method for the API request, such as POST, PATCH, or DELETE.

RecordResource Properties

Learn more about the available properties with the RecordResource class.

The `RecordResource` class includes these properties.

criteria

Set the criteria property to group order or quote line items. For example, group order or quote line items based on a monthly billing frequency.

fieldValues

Set the fieldValues property to assign values to the fields to update the sales transaction.

groupAction

Set the groupAction property to group order or quote line items.

id

Set the id property to assign the ID of the sales transaction record.

method

Set the method property to specify the API request method, such as POST, PATCH, or DELETE.

type

Set the type property to assign the object type that's returned from the field describe result by using the getReferenceTo() method or from the sObject describe result by using the getSObjectType() method.

`criteria`

Set the criteria property to group order or quote line items. For example, group order or quote line items based on a monthly billing frequency.

Signature

```
public Map<String,ANY> criteria {get; set;}
```

Property Value

Type: Map<String,ANY>

`fieldValues`

Set the fieldValues property to assign values to the fields to update the sales transaction.

Signature

```
public Map<String,ANY> fieldValues {get; set;}
```

Property Value

Type: ListMap<String,ANY>

groupAction

Set the groupAction property to group order or quote line items.

You can group order or quote line items based on location, work types, or departments, if groups are enabled for your org. Groups provide a visualization of the products to view large quotes.

Signature

```
public String groupAction {get; set;}
```

Property Value

Type: String

Valid values are:

- `GroupBy`
- `Group`
- `Ungroup`
- `GroupAll`
- `DeleteGroup`

id

Set the id property to assign the ID of the sales transaction record.

Signature

```
public String id {get; set;}
```

Property Value

Type: String

method

Set the method property to specify the API request method, such as POST, PATCH, or DELETE.

Signature

```
public String method {get; set;}
```

Property Value

Type: String

type

Set the type property to assign the object type that's returned from the field describe result by using the getReferenceTo() method or from the sObject describe result by using the getSObjectType() method.

Signature

```
public Schema.SObjectType type {get; set;}
```

Property Value

Type: Schema.SObjectType

RecordWithReferenceRequest Class

Contains constructors and properties to associate a record object with a reference identifier.

Namespace

RevSalesTrxn

Example

```
RevSalesTrxn.RecordWithReferenceRequest quoteLineItemRecords = new
RevSalesTrxn.RecordWithReferenceRequest('refQuoteItem',quoteLineItemRecord);
```

RecordWithReferenceRequest Constructors
Learn more about the available constructors with the RecordWithReferenceRequest class.

RecordWithReferenceRequest Properties
Learn more about the available properties with the RecordWithReferenceRequest class.

RecordWithReferenceRequest Constructors

Learn more about the available constructors with the RecordWithReferenceRequest class.

The `RecordWithReferenceRequest` class includes these constructors.

RecordWithReferenceRequest(referenceId, record)
Creates an instance of the RecordWithReferenceRequest class to associate a record object with a reference identifier by using the referenceId and record object properties.

RecordWithReferenceRequest(referenceId, record)

Creates an instance of the RecordWithReferenceRequest class to associate a record object with a reference identifier by using the referenceId and record object properties.

Signature

```
public RecordWithReferenceRequest(String referenceId, RevSalesTrxn.RecordResource
record)
```

Parameters

referenceId
Type: String

Reference ID that maps to the subrequest response and can be used to reference the response in subsequent subrequests. You can reference the referenceId in either the body or URL of a subrequest. Use this syntax to include a reference: @{referenceId.FieldName}. See referenceId property of a composite subrequest.

record
Type: RevSalesTrxn.RecordResource

Record object that's defined using the `RecordResource` class.

RecordWithReferenceRequest Properties

Learn more about the available properties with the RecordWithReferenceRequest class.

The `RecordWithReferenceRequest` class has these properties.

record
Set the record property to specify the record object that's defined by using the RecordResource class.

referenceId
Set the referenceId property to specify the reference ID that maps to the subrequest response. This reference ID can be used to reference the response in subsequent subrequests.

record

Set the record property to specify the record object that's defined by using the RecordResource class.

Signature

```
public RevSalesTrxn.RecordResource record {get; set;}
```

Property Value

Type: RevSalesTrxn.RecordResource

referenceId

Set the referenceId property to specify the reference ID that maps to the subrequest response. This reference ID can be used to reference the response in subsequent subrequests.

Signature

```
public String referenceId {get; set;}
```

Property Value

Type: String

TaxPreferenceEnum Enum

Specifies whether to execute or skip the tax calculation step for each sales transaction record. Available in API version 65.0 and later.

Enum Values

The `RevSalesTrxn.TaxPreferenceEnum` enum includes this value.

Value	Description
`Skip`	Specifies to skip tax calculation request for each sales transaction record. If this value isn't specified, then tax calculation request is performed by default.

Transaction Management Standard Invocable Actions

Learn more about the standard invocable actions available with Transaction Management.

Create Contract Action

Create a contract from a specific quote record.

Create or Update Asset From Order Action

Create an asset for each order item in the specified order. New assets are created for a new order. Modify existing assets for change order requests, such as a renewal or a cancellation.

Create or Update Asset From Order Item Action

Create assets from individual order items within an order. Track assets after the individual line items of an order reach a certain stage in their lifecycle, such as submitted, fulfilled, or provisioned. If the order item is part of a renewal, an amendment, or a cancellation, existing assets are changed.

Create Order From Quote Action

Create an order from a quote record.

Create Orders From Quote Action

Create multiple orders from a single quote instead of a single order, ensuring easier order management and fulfillment operations.

Deep Clone Sales Transaction

Clone a quote or an order, including full object graph with related objects, selected lines, or selected groups.

Get Renewable Assets Summary Action

Retrieve details about renewable assets in a given order. You can use this information to create renewal opportunities.

Initiate Amendment Action

Initiate and execute the amendment of an asset.

Initiate Cancellation Action

Initiate and execute the cancellation of an asset.

Initiate Renewal Action

Initiate and execute the renewal of an asset.

[Initiate Rollback on Last Action](#)

Initiate the reversal of the last action on an asset to rectify any transactional errors or to meet changing business requirements.

[Initiate Transfer Action](#)

Transfer an asset or multiple assets from one account to another.

SEE ALSO:

Actions Developer Guide: Overview

REST API Developer Guide: Invocable Actions Standard

Create Contract Action

Create a contract from a specific quote record.

This action is available in API version 60.0 and later.

Special Access Rules

The Create Contract action is available in Developer, Enterprise, and Unlimited Editions of Revenue Management.

Supported REST HTTP Methods

URI

`/services/data/v67.0/actions/standard/createContract`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer token`

Inputs

Input	Details
`contractPriceOption`	**Type** string **Description** Optional. Determines how the contract price is set for quote line items based on the selected value. Valid values are: • `CONTRACT_HEADER_ONLY`—Creates a contract with only the header information, without using net prices or discounts. • `NET_UNIT_PRICE_ONLY`—Creates a contract specifically for quote line items with a net unit price, saving all net unit prices of the quote as contract prices.

Input	Details
	• `DISCOUNT_ONLY`—Creates contract prices specifically for quote line items with discounts, saving all discounts of the quote as contract prices. The default value is `CONTRACT_HEADER_ONLY`.
`sourceId`	**Type** string **Description** Required. ID of the quote or order that you want to create a contract from.

Outputs

Output	Details
`contractId`	**Type** string **Description** ID of the contract created for the specified order or quote.

Example

POST

This sample request is for the Create Contract action.

```
{
  "inputs": [
    {
      "sourceId": "0Q0RO0000003LyU",
      "contractPriceOption": "NET_UNIT_PRICE_ONLY"
    }
  ]
}
```

This sample response is for the Create Contract action.

```
[
  {
    "actionName": "createContract",
    "errors": null,
    "isSuccess": true,
    "outputValues": {
      "contractId": "800XXX123456789"
    }
```

```
    }
]
```

SEE ALSO:

Salesforce Help: Use a Custom Flow to Create Contracts

Create or Update Asset From Order Action

Create an asset for each order item in the specified order. New assets are created for a new order. Modify existing assets for change order requests, such as a renewal or a cancellation.

When the custom product name for an order line item has a value, the asset name is set to custom product name.

This action is available in API version 60.0 and later.

Special Access Rules

You need the Assetize Order permission set to use this invocable action.

Supported REST HTTP Methods

URI

`/services/data/v`***`67.0`***`/actions/standard/createOrUpdateAssetFromOrder`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer` ***`token`***

Inputs

Input	Details
`orderId`	**Type** string **Description** Required. ID of the order.

Outputs

Output	Details
`requestId`	**Type** string **Description** ID of the request to create an asset.

Example

POST

This sample request is for the Create or Update Asset From Order action.

```
{
  "inputs": [
    {
      "orderId": "801DE000000oJfAYAU"
    }
  ]
}
```

This sample response is for the Create or Update Asset From Order action.

```
[
    {
        "actionName": "createOrUpdateAssetFromOrder",
        "errors": null,
        "isSuccess": true,
        "outputValues": {
            "requestId": "3b89392d-6987-40d9-9190-d18fdb5cb849"
        }
    }
]
```

Create or Update Asset From Order Item Action

Create assets from individual order items within an order. Track assets after the individual line items of an order reach a certain stage in their lifecycle, such as submitted, fulfilled, or provisioned. If the order item is part of a renewal, an amendment, or a cancellation, existing assets are changed.

This action is available in API version 60.0 and later.

Special Access Rules

You need the Assetize Order permission set to use this invocable action.

Supported REST HTTP Methods

URI

/services/data/v*67.0*/actions/standard/createOrUpdateAssetFromOrderItem

Formats

JSON, XML

HTTP Methods

POST

Authentication

Authorization: Bearer ***token***

Inputs

Input	Details
orderId	**Type** string **Description** Required. ID of the order.

Outputs

Output	Details
requestId	**Type** string **Description** ID of the request to create an asset.

Example

POST

This sample request is for the Create or Update Asset From Order Item action.

```
{
    "inputs": [
        {
            "orderItemIds": ["802SG000002HixxYAC"]
        }
    ]
}
```

This sample response is for the Create or Update Asset From Order Item action.


```
{
  "actionName": "createOrUpdateAssetFromOrderItem",
  "errors": null,
  "isSuccess": true,
  "outputValues": {
    "requestId": "b2a2b4b9-845b-4078-980a-759308389604"
  },
  "version": 1
}
```


Create Order From Quote Action

Create an order from a quote record.

This action is available in API version 60.0 and later.

Special Access Rules

The Create Order From Quote action is available in Enterprise, Unlimited, and Developer Editions of Revenue Management.

Supported REST HTTP Methods

URI
`/services/data/v`***`67.0`***`/actions/standard/createOrderFromQuote`

Formats
JSON, XML

HTTP Methods
POST

Authentication
`Authorization: Bearer` ***`token`***

Inputs

Input	Details
`quoteRecordId`	**Type** datetime **Description** Required. ID of the quote record.

Outputs

Output	Details
`requestId`	**Type** string **Description** ID of the request.

Example

POST

This sample request is for the Create Order From Quote action.

```
{
    "inputs": [
        {
        "quoteRecordId": "0Q0D200000000DhKAI"
        }
    ]
}
```

This sample response is for the Create Order From Quote action.

```
[
  {
    "actionName": "createOrderFromQuote",
    "errors": null,
    "invocationId": null,
    "isSuccess": true,
    "outcome": null,
    "outputValues": {
      "orderNumber": "00000122",
      "orderId": "801oB000000DCrNQAW"
    },
    "sortOrder": -1,
    "version": 1
  }
]
```

Create Orders From Quote Action

Create multiple orders from a single quote instead of a single order, ensuring easier order management and fulfillment operations.

You can split a quote to create orders in these ways.

- You can create an order by selecting a subset of quote line items.
- You can split a quote into multiple orders based on quote line group.
- You can split a quote into multiple orders based on a quote line item field, such as Product2.

See Considerations for Creating Multiple Orders from a Quote. This action is available in API version 65.0 and later.

Special Access Rules

Ensure the **Advanced Order Creation From Quote** toggle from Revenue Settings from Setup is enabled to access this action.

Supported REST HTTP Methods

URI

`/services/data/v67.0/actions/standard/createOrdersFromQuote`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer token`

Inputs

Input	Details
`orderCreationMethod`	**Type** string **Description** If specified, represents the method used to create orders. Valid values are: • `CreateSingleOrder`—Creates a single order from a quote. This method creates the order header and order line items synchronously. • `CreateOrderByGroup`—Create multiple orders based on a quote line group. • `CreateOrderByField`—Create multiple orders based on a specified quote line item field. If you're creating multiple orders from a quote, the order header records are created synchronously and order line items are created asynchronously. If unspecified, the default value is `CreateSingleOrder`.
`orderCreationParameters`	**Type** Apex-defined **Description** An Apex `ConnectApi.OrderCreationParametersInputRepresentation` record that contains the additional parameters required by the specified order creation method.
`quoteId`	**Type** id **Description** Required. The ID of the source quote tied to the orders created.

Input	Details
`quoteLineItemIds`	**Type** id **Description** List of quote line item IDs to create the orders from. If specified, the orders are created for the list of provided quote line items. If unspecified, orders are created for all quote line items.

Outputs

Output	Details
`orderIds`	**Type** id **Description** List of the created order IDs.
`requestId`	**Type** string **Description** The request ID used to query the async status.
`statusUrl`	**Type** string **Description** The status URL that's used to track the async status.

Examples

Create a partial order from a quote

This example shows a sample request to create a partial order from a quote.

```
{
  "inputs": [
    {
      "quoteId": "0Q0DU0000005tJc0AI",
      "quoteLineItemIds": [
        "0QLDU000000ay2G4AQ",
        "0QLDU000000ay2n4AA"
      ],
      "orderCreationMethod": "CreateSingleOrder"
    }
  ]
}
```

This example shows a sample successful response.

```
{
    "actionName": "createOrdersFromQuote",
    "errors": null,
    "invocationId": null,
    "isSuccess": true,
    "outcome": null,
    "outputValues": {
      "requestId": null,
      "statusUrl": null,
      "orderIds": [
        "801DU000000EYySYAW"
      ]
    },
    "sortOrder": -1,
    "version": 1
  }
```

Create orders from a quote based on a quote line group

This example shows a sample request to create orders from a quote based on a quote line group.

```
{
  "inputs": [
    {
      "quoteId": "0Q0DU0000005tJk0AI",
      "orderCreationMethod": "CreateOrderByGroup"
    }
  ]
}
```

This example shows a sample successful response.

```
{
  "actionName": "createOrdersFromQuote",
  "errors": null,
  "invocationId": null,
  "isSuccess": true,
  "outcome": null,
  "outputValues": {
    "requestId": "16PRM0000004DBq",
    "orderIds": [
      "801S70000001VKgIAM"
    ],
    "success": true,
    "errors": [],
    "statusURL": "/services/data/v67.0/sobjects/AsyncOperationTracker/16PRM0000004DBq"
  },
  "sortOrder": -1,
  "version": 1
}
```

Create orders from a quote based on a quote line item field

This example shows a sample request to create orders from a quote based on a quote line item field


```
{
  "inputs": [
    {
      "quoteId": "0Q0DU0000005tJW0AY",
      "orderCreationMethod": "CreateOrderByField",
      "orderCreationParameters": {
        "splitFieldName": "Product2Id"
      }
    }
  ]
}
```


This example shows a sample successful response.


```
{
  "actionName": "createOrdersFromQuote",
  "errors": null,
  "invocationId": null,
  "isSuccess": true,
  "outcome": null,
  "outputValues": {
    "requestId": "16PRM0000004DBq",
    "orderIds": [
      "801S70000001VKgIAM",
      "801S70000001WKgIBN"
    ],
    "success": true,
    "errors": [],
    "statusURL": "/services/data/v67.0/sobjects/AsyncOperationTracker/16PRM0000004DBq"
  },
  "sortOrder": -1,
  "version": 1
}
```


Deep Clone Sales Transaction

Clone a quote or an order, including full object graph with related objects, selected lines, or selected groups.

This action is available in API version 67.0 and later.

Supported REST HTTP Methods

URI

`/services/data/v67.0/actions/standard/deepCloneSalesTransaction`

Formats

JSON, XML

HTTP Methods

POST

Authentication

```
Authorization: Bearer token
```

Inputs

Input	Details
options	**Type** Apex-defined **Description** Specifies options to clone a ramp segment within a sales transaction. You can clone only the last ramp segment. Valid values are: • `recordTypeId`—ID of the record type related to the record to clone. Optional. • `lineScope`—Specifies the scope for cloning a ramp segment. This property determines which line items must be cloned and added to the cloned segment. You can specify these options. – `AllLines`—Specifies whether all line items in a ramped group must be cloned. – `RampedLinesOnly`—Specifies whether only the ramped line items must be cloned. A segment identifier is created for the newly cloned line items, ensuring date continuity between the existing and cloned segment.
recordIds	**Type** List<String> **Description** Required. ID of the record to be cloned. You can specify a single record ID only.
salesTransactionId	**Type** String **Description** Required. ID of the sales transaction related to the record IDs to clone.

Outputs

Output	Details
newRecordId	**Type** String

Output	Details
	Description ID of the cloned sales transaction.

Example

POST

Here's a sample request for the Deep Clone Sales Transaction action.

```
{
  "inputs": [
    {
      "recordIds": ["0QLxx0000004CBYGA2"],
      "salesTransactionId": "0Q0xx0000004CE0CAM",
      "options": {
        "lineScope": "AllLines"
      }
    }
  ]
}
```

Here's a sample response for the Deep Clone Sales Transaction action.

```
{
  "actionName": "deepCloneSalesTransaction",
  "errors": null,
  "isSuccess": true,
  "outputValues": {
    "newRecordId": "0Q0SG000000ACxf0AG"
  }
}
```

Get Renewable Assets Summary Action

Retrieve details about renewable assets in a given order. You can use this information to create renewal opportunities.

This action gets pricing data from the OrderEntitiesMapping context mapping within the SalesTransactionContext context definition. Before you use this action, edit the context mapping to map the objects and fields used in your pricing procedure to the nodes and attributes in the context definition.

This action doesn't support providing a summary with procedure plans. As a result, renewal line items may return a price of zero.

Note: We recommend adding the same fields and mappings for the Quote and Order objects, and for the Quote Line Item and Order Product objects.

This action is available in API version 64.0 and later.

Supported REST HTTP Methods

URI

/services/data/v***67.0***/actions/standard/getRenewableAssetsSummary

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer` ***`token`***

Inputs

Input	Details
`orderId`	**Type** id **Description** Required. ID of the order related to the assets to check for renewal opportunities.

Outputs

Output	Details
`renewableAssetsSummary`	**Type** Apex-defined **Description** Summary of the assets associated with the order, including details about renewal opportunities such as renewal pricing information. See renew_assets_summary Apex Namespace.

Example

POST

Here's a sample request for the Get Renewable Assets Summary action.

```
{
  "inputs": [
    {
      "orderId": "801xx000003GZ39AAG"
    }
  ]
}
```

Here's a sample response for the Get Renewable Assets Summary action.

```
[
  {
```

```
    "actionName": "getRenewableAssetsSummary",
    "errors": null,
    "invocationId": null,
    "isSuccess": true,
    "outcome": null,
    "outputValues": {
      "renewableAssetsSummary": [
        {
          "startDate": "2025-07-22",
          "rootAssetOpportunity": null,
          "renewalPriceDetails": [
            {
              "quantity": 1,
              "netUnitPrice": 0
            }
          ],
          "productId": "01txx0000006i3DAAQ",
          "priceBookId": "01sxx0000005ptpAAA",
          "priceBookEntryId": "01uxx0000008yXCAAY",
          "orderItem": "802xx000001nb1LAAQ",
          "opportunityProductId": null,
          "lastAssetActionSubtype": null,
          "lastAssetAction": "Initial Sale",
          "endDate": "2025-08-21",
          "assetId": "02ixx0000004HKwAAM",
          "account": "001xx000003GZ1XAAW"
        }
      ]
    },
    "sortOrder": -1,
    "version": 1
  }
]
```


Initiate Amendment Action

Initiate and execute the amendment of an asset.

Specify the IDs of assets to amend, the amendment start date, and any quantity changes. You can choose whether the action creates an amendment quote or order.

Considerations

For usage products, creating an order directly by setting `amendOutputType` to `Order` is currently not supported. The direct-to-order flow can create Order Products without required related Rate Card Entry records, which can cause order activation to fail.

When you amend usage assets, use a two-step flow.

1. Create an amendment quote by calling `initiateAmendment` with `amendOutputType` set to `Quote`.
2. Create an order from that amendment quote.

Supported REST HTTP Methods

URI

`/services/data/v67.0/actions/standard/initiateAmendment`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer token`

Inputs

Input	Details
amendAssetIds	**Type** string **Description** Required. The IDs of the assets that you want to amend.
amendContractId	**Type** string **Description** The ID of the contract record to be synced with the amendment.
amendStartDate	**Type** datetime **Description** Required. Effective start date of the amendment.
amendOpportunityId	**Type** string **Description** The ID of the Opportunity record to be synced with the amendment quote.
amendOutputType	**Type** string **Description** Required. Type of amendment record to create such as a quote or an order.

Input	Details
	For usage products, set this value to `Quote`. The usage-related records (`QuoteLineItemUseResourceGrant`, `QuoteLineItemUsageResourcePolicy`, and `QuoteLineRateCardEntry`) are copied in the quote action flow, not in the amend order flow.
`quantityChange`	**Type** double **Description** Required. Quantity to add to or reduce from the asset's existing quantity.
`skipPricing`	**Type** boolean **Description** Indicates whether the pricing procedure must be skipped (`true`) or performed (`false`). Available in API version 64.0 and later.

Outputs

Output	Details
`amendRecordId`	**Type** string **Description** The ID of the amendment record that's created.
`requestIdentifier`	**Type** string **Description** Request ID that's used to track the async request.

Example

POST

This sample request is for the Initiate Amendment action.

```
{
    "inputs": [
        {
            "amendAssetIds": ["02iI8000000HPzXIAW"],
            "amendStartDate": "2023-10-21T00:00:00.000Z",
            "quantityChange": 5,
```

```
            "amendOutputType": "Quote",
            "amendContractId": "800DU0000000lZlYAI",
            "amendOpportunityId": "006DU0000025AanYAE",
            "skipPricing": false
        }
    ]
}
```

This sample response is for the Initiate Amendment action.

```
[
    {
        "actionName": "initiateAmendment",
        "errors": null,
        "isSuccess": true,
        "outputValues": {
            "record_id": "0Q0xx0000004NsSCAU",
            "requestIdentifier": "16Pxx0000004NIy"
        },
        "version": 1
    }
]
```

Initiate Cancellation Action

Initiate and execute the cancellation of an asset.

Specify the IDs of the assets that you want to add to cancel by specifying a start date. You can also specify the type of cancellation record that you want to create, such as a quote or an order.

This action is available in API version 60.0 and later.

Supported REST HTTP Methods

URI

`/services/data/v`***`67.0`***`/actions/standard/initiateCancellation`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer` ***`token`***

Inputs

Input	Details
`cancelAssetIds`	**Type** string

Input	Details
	Description Required. The IDs of the assets that you want to cancel. All assets in a request must belong to the same price book.
`cancelContractId`	**Type** string **Description** The ID of the contract record to sync with the cancellation.
`cancelOpportunityId`	**Type** string **Description** The ID of the Opportunity record to sync with the cancellation quote.
`cancelOutputType`	**Type** string **Description** Required. Type of cancellation record to create such as a quote or an order.
`cancelStartDate`	**Type** datetime **Description** Required. Effective date of the cancellation.
`skipPricing`	**Type** boolean **Description** Indicates whether the pricing procedure must be skipped (`true`) or performed (`false`). Available in API version 64.0 and later.

Outputs

Output	Details
`cancelRecordId`	**Type** string

Output	Details
	Description The ID of the cancellation record that's created.
`requestIdentifier`	**Type** string **Description** Request ID that's used to track the async request.

Example

POST

This sample request is for the Initiate Cancellation action.

```
{
    "inputs": [
        {
            "cancelAssetIds": [
                "02iI8000000Lc5fIAC"
            ],
            "cancelStartDate": "2023-11-09T00:00:00",
            "cancelOutputType": "Quote",
            "cancelContractId": "800DU000000lZlYAI",
            "cancelOpportunityId": "006DU0000025AanYAE",
            "skipPricing": false
        }
    ]
}
```

This sample response is for the Initiate Cancellation action.

```
[
    {
        "actionName": "initiateCancellation",
        "errors": null,
        "isSuccess": true,
        "outputValues": {
            "record_id": "0Q0xx0000004P32CAE",
            "requestIdentifier": "16Pxx0000004OTY"
        },
        "version": 1
    }
]
```

Initiate Renewal Action

Initiate and execute the renewal of an asset.

Specify the IDs of the assets that you want to add to renew by specifying a start date. You can also specify the type of renewal record that you want to create, such as a quote or an order.

This action is available in API version 60.0 and later.

Supported REST HTTP Methods

URI

`/services/data/v67.0/actions/standard/initiateRenewal`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer token`

Inputs

Input	Details
`rampOptionsDetails`	**Type** Apex-defined **Description** An Apex RampOptionInputRepresentation record that contains the details of ramp options. The details include segment type, duration, and segment count. Specify this property if you want to include assets in a group ramp with a ramp schedule. See RampOptionInputRepresentation on page 1558 to refer to the associated properties for ramp options. Available in API version 67.0 and later.
`renewAssetIds`	**Type** string **Description** Required. The IDs of the assets that you want to renew.
`renewContractId`	**Type** string **Description** ID of the contract record to sync with the renewal quote.
`renewEndDate`	**Type** datetime **Description** Effective end date of the renewal. Available in API version 62.0 and later.

Input	Details
renewOpportunityId	**Type** string **Description** ID of the Opportunity record to sync with the renewal quote.
renewOutputType	**Type** string **Description** Required. Type of renewal record to create such as a quote or an order.
renewStartDate	**Type** datetime **Description** Optional Effective start date of the renewal. Required for early asset renewals and renewing expired assets, by using today's date or a future date. Available in API version 62.0 and later.
skipPricing	**Type** boolean **Description** Indicates whether the pricing procedure must be skipped (`true`) or performed (`false`). Available in API version 64.0 and later.

Outputs

Output	Details
renewRecordId	**Type** string **Description** The ID of the amendment record that's created.
requestIdentifier	**Type** string **Description** Request ID that's used to track the async request.

Example

POST

This sample request is for the Initiate Renewal action.

```
{
  "inputs": [
    {
      "rampOptionsDetails": {
        "segmentType": "Custom",
        "duration": 40,
        "numberOfSegments": 10
      },
      "renewAssetIds": [
        "02ixx0000004LMwAAM"
      ],
      "renewOutputType": "Quote",
      "renewContractId": "800DU00000001ZlYAI",
      "renewOpportunityId": "006DU0000025AanYAE",
      "renewStartDate": "2023-10-21T00:00:00.000Z",
      "renewEndDate": "2024-10-21T00:00:00.000Z",
      "skipPricing": false
    }
  ]
}
```

This sample response is for the Initiate Renewal action.

```
[
    {
        "actionName": "initiateRenewal",
        "errors": null,
        "isSuccess": true,
        "outputValues": {
            "renewRecordId": "0Q0xx0000004P32CAE",
            "requestIdentifier": "16Pxx0000004OTY"
        },
        "version": 1
    }
]
```

Initiate Rollback on Last Action

Initiate the reversal of the last action on an asset to rectify any transactional errors or to meet changing business requirements.

Use this action to revert the last amendment or renewal on a particular asset, restoring the asset to its previous state. This action creates a quote or an order based on the specified output type. Use the created reversal quote to verify the reversal and convert the quote into an order.

Keep these considerations in mind when you use this action.

- You can roll back only future dated transactions.
- Rollback action isn't supported for legacy assets.
- The rollback operation is supported for amendment and renewal only.

This action is available in API version 65.0 and later.

Supported REST HTTP Methods

URI

`/services/data/v67.0/actions/standard/initiateRollBackLastAction`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearertoken`

Inputs

Input	Details
`assetIds`	**Type** string **Description** Required. List of asset IDs to include in the last action rollback.
`outputType`	**Type** string **Description** Required. The type of record to create for reversal. Valid values are: • `Quote`—Creates a reversal quote. • `Order`—Creates a reversal order.

Outputs

Output	Details
`recordId`	**Type** id **Description** The ID of the created quote or order.

Example

POST

This example shows a sample request that initiates the rollback action on an asset and converts it into a quote.

```
{
  "inputs": [
    {
      "assetIds": [
        "02iDU0000006UisYAE"
      ],
      "outputType": "Quote"
    }
  ]
}
```

This example shows a sample successful response.

```
{
  "actionName": "initiateRollBackLastAction",
  "errors": null,
  "isSuccess": true,
  "outputValues": {
    "recordId": "0Q0xx0000004NsSCAU"
  },
  "version": 1
}
```

Initiate Transfer Action

Transfer an asset or multiple assets from one account to another.

Keep these considerations in mind when you use this invocable action.

- This action generates 2 quotes or 2 orders.
- One transaction is considered the source, which is used to calculate the reduction amount of the transfer. The other transaction is the target, which is used to calculate the new amount on the target account.
- The quantity on the source is negative, which reduces the existing asset's quantity. The quantity on the target is positive, which creates a new asset on the target account.
- You can change the quantity on the source or the target, but the quantities must be equal and opposite. For example, the source quote line can have a quantity value as `-5`, and the target quote line can have a quantity value `5`. If you change the source value to `-10`, you must update the target value to `10`. You must update quotes or orders manually.
- The transfer is complete after the orders are assetized.

This action is available in API version 65.0 and later.

Supported REST HTTP Methods

URI

`/services/data/v67.0/actions/standard/initiateTransfer`

Formats

JSON, XML

HTTP Methods
POST

Authentication
`Authorization: Bearer`***token***

Inputs

Input	Details
`outputRecordType`	**Type** string **Description** Required. Specifies either a quote or order record type that's being transferred.
`shouldSkipPricing`	**Type** boolean **Description** Indicates whether to skip pricing at the time of asset transfer (`true`) or not (`false`).
`targetAccountId`	**Type** string **Description** Required. The ID of the target account where the asset is transferred.
`targetContractId`	**Type** string **Description** The ID of the target contract that's associated with the asset that's being transferred.
`transferDate`	**Type** datetime **Description** Required. The date of the asset transfer.
`transferRecords`	**Type** Apex-defined

Input	Details
	Description Required. A collection of Apex `connectapi__TransferRecordInputRepresentation` records that contain details about the assets to be transferred. See ConnectApi.TransferRecordInputRepresentation.

Outputs

Output	Details
`assetTransferSourceId`	**Type** id **Description** The ID of the quote or order that's related to the source account used to start the transfer.
`assetTransferTargetId`	**Type** id **Description** The ID of the quote or order that's related to the target account used to start the transfer.
`requestIdentifier`	**Type** id **Description** The request ID can be used to track the async request.

Example

POST

Here's a sample input to call this invocable action.

```
{
  "inputs": [
    {
      "transferRecords": [
        {
          "assetId": "02ixx0000004HZbAAM",
          "transferQuantity": 1
        }
      ],
      "transferDate": "2025-10-21T00:00:00.000Z",
      "targetAccountId": "001xx000003GbeXAAS",
      "targetContractId": "800DU0000000lZlYAI",
```

```
      "outputRecordType": "Quote"
    }
  ]
}
```

Here's a sample input to call this invocable action from Apex code.

```
ConnectApi.TransferRecordInputRepresentation transferRecord = new
ConnectApi.TransferRecordInputRepresentation();
transferRecord.assetId = '02ixx0000004HHjAAM';
transferRecord.transferQuantity = 1;
List<ConnectApi.TransferRecordInputRepresentation> transferRecords = new
List<ConnectApi.TransferRecordInputRepresentation>();
transferRecords.add(transferRecord);

Invocable.Action action = Invocable.Action.createStandardAction('initiateTransfer');
action.setInvocationParameter('transferRecords', transferRecords);
action.setInvocationParameter('targetAccountId', '001xx000003GYiHAAW');
action.setInvocationParameter('transferDate', '2025-03-01T00:00:00.000Z');
action.setInvocationParameter('outputRecordType', 'Quote');
List<Invocable.Action.Result> results = action.invoke();
System.assertEquals(true, results[0].isSuccess());
Assert.isNotNull(results[0].getOutputParameters().get('assetTransferSourceId'));
```

Here's a sample response when you call this action.

```
[
  {
    "actionName": "initiateTransfer",
    "errors": null,
    "invocationId": null,
    "isSuccess": true,
    "outcome": null,
    "outputValues": {
      "assetTransferSourceId": "0Q0DU0000006SYF0A2",
      "assetTransferTargetId": "0Q0DU0000006SYE0A2"
    },
    "sortOrder": -1,
    "version": 1
  }
]
```

Transaction Management Metadata API Types

Metadata API enables you to access some types and feature settings that you can customize in the user interface.

[Flow for Transaction Management](#)

The flow for Transaction Management represents the metadata associated with a flow. With Flow, you can create an application that takes users through a series of pages to query and update the records in the database. You can also run logic and provide branching capability based on user input to build dynamic applications.

SEE ALSO:

Metadata API Developer Guide: Understanding Metadata API

Flow for Transaction Management

The flow for Transaction Management represents the metadata associated with a flow. With Flow, you can create an application that takes users through a series of pages to query and update the records in the database. You can also run logic and provide branching capability based on user input to build dynamic applications.

FlowActionCall

Transaction Management exposes additional actionType values for the FlowActionCall metadata type.

Field Name	Field Type	Description
`actionType`	InvocableActionType (enumeration of type string)	Required. The action type. Additional valid values for Transaction Management are: • `createContract`—Create a contract from a specific quote record. • `createOrderFromQuote`—Create an order from a quote record. • `createServiceDocument`—Create service documents from work orders, work order line items, or service appointments. • `deepCloneSalesTransaction`—Clone a quote or an order, including full object graph with related objects, selected lines, or selected groups. Added in API version 67.0 and later. • `getRenewableAssetsSummary`—Retrieve details about renewable assets in a given order. You can use this information to create renewal opportunities. Added in API version 64.0 and later. • `createOrUpdateAssetFromOrder`—Create an asset for each order item in the specified order. New assets are created for a new order. Modify existing assets for change order requests, such as a renewal or a cancellation. • `createOrUpdateAssetFromOrderItem`—Create assets from individual order items within an order. Track assets after the individual line items of an order reach a certain stage in their lifecycle, such as submitted, fulfilled, or provisioned. If the order item is part of a renewal, an amendment, or a cancellation, existing assets are changed.

Field Name	Field Type	Description
		• `initiateAmendment`—Initiate and execute the amendment of an asset. • `initiateRenewal`—Initiate and execute the renewal of an asset. • `initiateCancellation`—Initiate and execute the cancellation of an asset. • `initiateRollBackLastAction`—Initiate the reversal of the last action on an asset to rectify any transactional errors or to meet changing business requirements. • `createOrdersFromQuote`—Create multiple orders from a single quote instead of a single order, ensuring easier order management and fulfillment operations. • `initiateTransfer`—Transfer an asset or multiple assets from one account to another.

/

CHAPTER 9 Advanced Approvals

In this chapter ...

Advanced Approvals feature provides invocable actions to streamline complex business processes. These actions automate and manage intricate parallel or sequential approval chains with detailed audit trails.

SEE ALSO:

Salesforce Help: Advanced Approvals

EDITIONS

Available in: Lightning Experience

Available in: **Enterprise**, **Unlimited**, and **Developer** Editions of Revenue Management where Advanced Approvals is enabled

Advanced Approvals Fields on Standard Objects

Advanced Approvals adds standard and custom fields to some standard Salesforce objects. These fields are available only in Revenue Management orgs where Advanced Approvals is enabled.

ApprovalSubmission
Represents the instance of an approval request that's submitted for a record of the related object. This object is available in API version 62.0 and later.

Advanced Approvals Fields on Approval Work Item
Standard and custom fields extend the standard Approval Work Item object for use in Advanced Approvals.

ApprovalSubmission

Represents the instance of an approval request that's submitted for a record of the related object. This object is available in API version 62.0 and later.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`

Special Access Rules

This object is available in Enterprise, Unlimited, and Developer Editions of Revenue Management.

Fields

Field	Details
`Comments`	**Type** textarea **Properties** Nillable **Description** The comments about the request that's submitted for approval.
`DoesSendApprovalEmail`	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Required. Indicates whether approval request emails are sent to approvers and delegates (`true`) or not (`false`). The default value is `false`.

Field	Details
FlowOrchestrationInstanceId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the flow orchestration instance record that's associated with the approval. This field is a relationship field. **Relationship Name** FlowOrchestrationInstance **Refers To** FlowOrchestrationInstance
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The auto-generated name for the approval submission.
OwnerId	**Type** reference **Properties** Filter, Group, Sort **Description** The ID of the user or the group that owns the approval submission record. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
RelatedRecordId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** Required. The ID of the related record that's submitted for approval. This field is a polymorphic relationship field. **Relationship Name** RelatedRecord

Field	Details
	Refers To Account, AdAvailabilityViewConfig, Address, AnalyticsUserAttrFuncTkn, ApprovalSubmission, ApprovalSubmissionDetail, ApprovalWorkItem, Asset, AssetAction, AssetActionSource, AssetContractRelationship, AssetRateAdjustment, AssetRateCardEntry, AssetRelationship, AssetStatePeriod, AssociatedLocation, AsyncOperationTracker, AttrPicklistExcludedValue, AttributeAdjustmentCondition, AttributeBasedAdjRule, AttributeBasedAdjustment, AttributeCategory, AttributeCategoryAttribute, AttributeDefinition, AttributePicklist, AttributePicklistValue, AuthorizationForm, AuthorizationFormConsent, AuthorizationFormDataUse, AuthorizationFormText, BatchJob, BatchJobPart, BatchJobPartFailedRecord, BundleBasedAdjustment, BusinessBrand, Case, CaseComment, ChannelProgram, ChannelProgramLevel, ChannelProgramMember, CollaborationGroup, CommSubscription, CommSubscriptionChannelType, CommSubscriptionConsent, CommSubscriptionTiming, Contact, ContactPointAddress, ContactPointConsent, ContactPointEmail, ContactPointPhone, ContactPointTypeConsent, ContactRequest, ContextDefinitionSync, Contract, ContractItemPrice, ContractItemPriceAdjTier, CostBook, CostBookEntry, Customer, DTRecordsetReplica, DataUseLegalBasis, DataUsePurpose, DecisionTblFileImportData, DelegatedAccount, DocGenerationQueryResult, DocTemplateSectionCondition, DocumentEnvelope, DocumentGenerationProcess, DocumentRecipient, DocumentTemplate, DocumentTemplateContentDoc, DocumentTemplateSection, DocumentTemplateToken, DuplicateRecordItem, DuplicateRecordSet, EmailMessage, EngagementChannelType, ExpressionSetConstraintObj, ExternalEventMapping, FlowOrchestrationInstance, FulfillmentOrder, FulfillmentOrderItemAdjustment, FulfillmentOrderItemTax, FulfillmentOrderLineItem, GeneratedDocument, GeneratedDocumentSection, Idea, Image, Individual, IntegrationProviderDcsnRqmt, IntegrationProviderExecution, Lead, Location, LocationTrustMeasure, ManagedContentVariant, ObjectStateDefinition, ObjectStateTransition, ObjectStateValue, Obligation, Opportunity, OpportunityRelatedDeleteLog, Order, OrderAction, OrderAdjustmentGroup, OrderDeliveryGroup, OrderDeliveryMethod, OrderItem, OrderItemAdjustmentLineItem, OrderItemDetail, OrderItemRateAdjustment, OrderItemRateCardEntry, OrderItemRecipient, OrderItemRelationship, OrderItemTaxLineItem, OrgMetricScanResult, OrgMetricScanSummary, Organization, PartnerFundAllocation, PartnerFundClaim, PartnerFundRequest, PartnerMarketingBudget, PartyConsent, PriceBookEntryDerivedPrice, PriceBookRateCard, PricingAdjBatchJob, PricingAdjBatchJobLog, PricingApiExecution, PricingProcessExecution, ProcessException, Product2, ProductAttributeDefinition, ProductCatalog, ProductCategory, ProductCategoryDisqual, ProductCategoryProduct, ProductCategoryQualification, ProductClassification, ProductClassificationAttr, ProductComponentGroup, ProductComponentGrpOverride, ProductConfigFlowAssignment, ProductConfigurationFlow, ProductConfigurationRule, ProductDisqualification, ProductPriceHistoryLog, ProductPriceRange, ProductQualification, ProductRampSegment, ProductRelComponentOverride, ProductUsageGrant, ProfileSkill, ProfileSkillEndorsement, ProfileSkillUser, PromptAction, PromptError, QuickText, QuickTextUsage, Quote, QuoteLineDetail, QuoteLineItem, QuoteLineItemRecipient, QuoteLineRateAdjustment, QuoteLineRateCardEntry, RateAdjustmentByAttribute, RateAdjustmentByTier, RateCard, RateCardEntry, RatingFrequencyPolicy, SalesTransactionType, Seller, Shipment, ShipmentItem, Site, SocialPersona, SocialPost, Solution, StreamingChannel, TableauHostMapping, Topic, UnitOfMeasure, UnitOfMeasureClass, UsageGrantRenewalPolicy, UsageGrantRolloverPolicy, UsageResource, UsageResourceBillingPolicy, User,

Field	Details
	UserEsignVendorIdentifier, UserLicense, UserLocalWebServerIdentity, UserProvisioningRequest, WorkBadge, WorkBadgeDefinition, WorkOrder, WorkOrderLineItem, WorkThanks
RelatedRecordObjectName	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** Required. The type of record that was submitted for approval.
Status	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** Required. The status of the approval. Valid values are: • Approved • Canceled • Errored • InProgress • Recalled • Rejected • Suspended
SubmittedById	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** Required. The ID of the user who submitted the record for approval. This field is a relationship field. **Relationship Name** SubmittedBy **Refers To** User

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

ApprovalSubmissionOwnerSharingRule on page 2875
Sharing rules are available for the object.

ApprovalSubmissionShare on page 2877
Sharing is available for the object.

Advanced Approvals Fields on Approval Work Item

Standard and custom fields extend the standard Approval Work Item object for use in Advanced Approvals.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `update()`

Special Access Rules

This object is available in Enterprise, Performance, Unlimited, and Developer Editions for users with access to the Approval Submission object.

Fields

Field	Details
`ApprovalChainName`	**Type** string **Properties** Filter, Group, Nillable, Sort, Update **Description** The name of the related approval chain. This field is populated when there are multiple approval chains that are run in parallel. This field is only available with Advanced Approvals enabled.
`CurrencyIsoCode`	**Type** picklist **Properties** Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Available only for organizations with the multicurrency feature enabled. Contains the ISO code for any currency allowed by the organization. Valid values are: • `AED`—UAE Dirham • `AUD`—Australian Dollar

Field	Details
	• BRL—Brazilian Real • CAD—Canadian Dollar • EUR—Euro • GBP—British Pound • INR—Indian Rupee • JPY—Japanese Yen • SEK—Swedish Krona • USD—U.S. Dollar The default value is USD.
IsAutoReviewed	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the approval work item is eligible for smart approval (true) or not (false). The default value is false.
IsEligibleForSmartApproval	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the approval work item is eligible for smart approval (true) or not (false). The default value is false.
SmartApprovalBasisWorkItemId	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The previous approval work item used as a reference for the auto-approval evaluation. This field is a relationship field. **Relationship Name** SmartApprovalBasisWorkItem **Refers To** ApprovalWorkItem

Advanced Approvals Standard Invocable Actions

Learn more about the standard invocable actions available with Advanced Approvals.

Cancel Approval Submission Action
: Cancels an approval submission and all child approval work items that haven't been completed. You can also add comments about why the approval admin made the cancellation.

Get Previous Related Record Details Action
: Get the related record details submitted for approval before the current approval submission. The details are required for approval steps that use custom logic for auto-approvals.

Override Approval Work Item Action
: Update an approval work item status with the approval admin decision and any comments that the approval admin added.

Reassign Approval Work Item Action
: Reassign an approval work item that hasn't been completed. You can also add comments about why the approval admin reassigned the approval work item.

Recall Approval Submission Action
: Recall an approval submission that isn't completed. You can also add comments that the submitter or approval admin made the recall.

Review Approval Work Item Action
: Update an approval work item status with the assignee or reviewer's decision and any comments that the assignee or reviewer added.

SEE ALSO:

Actions Developer Guide: Overview

REST API Developer Guide: Invocable Actions Standard

Cancel Approval Submission Action

Cancels an approval submission and all child approval work items that haven't been completed. You can also add comments about why the approval admin made the cancellation.

This action also validates if a user has required permissions to cancel an approval submission. Keep these considerations in mind when you use this invocable action.

- The user must have the Approval Admin user permission.
- The status of the approval submission must be in `In progress` or `Suspended` status.

This action is available in API version 62.0 and later.

Supported REST HTTP Methods

URI

`/services/data/v67.0/actions/standard/cancelApprovalSubmission`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer`***`token`***

Inputs

Input	Details
`approvalSubmissionId`	**Type** string **Description** Required. ID of the Approval Submission to be canceled.
`comments`	**Type** string **Description** Comments entered by the approval admin about why they canceled the Approval Submission.

Outputs

None.

Example

POST

Here's a sample request for the Cancel Approval Submission action.

```
{
  "inputs": [
    {
      "approvalSubmissionId": "9iPxx00000001lhEBA",
      "comments": "Cancellation comments."
    }
  ]
}
```

Here's a sample response for the Cancel Approval Submission action.

```
{
  "actionName": "cancelApprovalSubmission",
  "errors": null,
  "invocationId": null,
  "isSuccess": true,
  "outcome": null,
```

```
  "outputValues": null,
  "sortOrder": -1,
  "version": 1
}
```


Get Previous Related Record Details Action

Get the related record details submitted for approval before the current approval submission. The details are required for approval steps that use custom logic for auto-approvals.

This Apex action is available in API version 66.0 and later.

Supported REST HTTP Methods

URI

`/services/data/v67.0/actions/standard/getPreviousRelaRecDetails`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearertoken`

Inputs

Input	Details
flowOrchestrationInstanceId	**Type** string **Description** Required. The ID of the flow orchestration instance associated with the approval submission.
stepApiNamesList	**Type** string **Description** Required. The comma-delimited list of step API names to retrieve previous related record details for.

Outputs

Output	Details
previousRelatedRecordDetails	**Type** sObject **Description** The collection of previous related record details for approval steps that use custom logic for auto-approvals.

Usage

To use this Apex action in approval workflows, see Define Rules and Conditions for Auto-Approval Resubmissions.

Example

POST

Here's a sample request for the Get Previous Related Record Details action.

```
{
  "flowOrchestrationInstanceId": "0jEDU0000001zeN",
  "stepApiNameList": [
    "ManagerApprovalStep",
    "FinancialApprovalStep"
  ]
}
```

Override Approval Work Item Action

Update an approval work item status with the approval admin decision and any comments that the approval admin added.

This action also validates if a user has required permissions to override an approval work item and update its status. Keep these considerations in mind when you use this invocable action.

- The user must have the Approval Admin user permission.
- This action enables approval admins to interject the approval decision for any assignee.
- The status of the approval work item must be in `Assigned` status.

This action is available in API version 62.0 and later.

Supported REST HTTP Methods

URI

/services/data/v***67.0***/actions/standard/overrideApprovalWorkItem

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer`***`token`***

Inputs

Input	Details
`approvalDecision`	**Type** string **Description** Required. Action that the overriding approval admin made for the unreviewed approval work item. Valid values are: • `approve` • `reject`
`approvalWorkItemId`	**Type** string **Description** Required. ID of the unreviewed approval work item to be reviewed by an approval admin.
`channelType`	**Type** picklist **Description** Type of channel where the request to review the approval work item originated. Valid values are: • `InvocableAction` • `Slack` • `ApprovalRecord` The default value is `InvocableAction`. Available in API version 65.0 and later.
`comments`	**Type** string **Description** Comments entered by the approval admin about their decision to approve or reject an unreviewed approval work item.

Outputs

None.

Example

POST

Here's a sample request for the Override Approval Work Item action.

```
{
  "inputs": [
    {
      "approvalWorkItemId": "9jRxx00000001lhEAA",
      "approvalDecision": "Reject",
      "channelType": "InvocableAction",
      "comments": "Needs to be reviewed."
    }
  ]
}
```

Here's a sample response for the Override Approval Work Item action.

```
{
  "actionName": "overrideApprovalWorkItem",
  "errors": null,
  "invocationId": null,
  "isSuccess": true,
  "outcome": null,
  "outputValues": null,
  "sortOrder": -1,
  "version": 1
}
```

Reassign Approval Work Item Action

Reassign an approval work item that hasn't been completed. You can also add comments about why the approval admin reassigned the approval work item.

This action also validates if a user has required permissions to reassign an approval work item and update the assignee. Keep these considerations in mind when you use this invocable action.

- The user must have the Approval Admin user permission.
- The status of the approval work item must be in `Assigned` status.

This action is available in API version 62.0 and later.

Supported REST HTTP Methods

URI

`/services/data/v67.0/actions/standard/reassignApprovalWorkItem`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer`***`token`***

Inputs

Input	Details
`approvalWorkItemId`	**Type** string **Description** Required. ID of the approval work item to be reassigned.
`assigneeId`	**Type** string **Description** Required. ID of the user, group, or queue to reassign the approval work item to.
`comments`	**Type** string **Description** Comments entered by the approval admin about why they reassigned the approval work item.

Outputs

None.

Example

POST

Here's a sample request for the Reassign Approval Work Item action.

```
{
  "inputs": [
    {
      "approvalWorkItemId": "9jRDU00000015C22AI",
      "assigneeId": "005DU000000I3zHYAS",
      "comments": "Needs to be reviewed."
    }
```

```
  ]
}
```

Here's a sample response for the Reassign Approval Work Item action.

```
{
  "actionName": "reassignApprovalWorkItem",
  "errors": null,
  "invocationId": null,
  "isSuccess": true,
  "outcome": null,
  "outputValues": null,
  "sortOrder": -1,
  "version": 1
}
```

Recall Approval Submission Action

Recall an approval submission that isn't completed. You can also add comments that the submitter or approval admin made the recall.

This action also validates if a user has required permissions to recall an approval submission. Keep these considerations in mind when you use this invocable action.

- The user must have the Approval Admin user permission.
- The user must also be a submitter of this approval submission.
- The status of the approval submission must be in `In progress` or `Suspended` status.

This action is available in API version 62.0 and later.

Supported REST HTTP Methods

URI
`/services/data/v67.0/actions/standard/recallApprovalSubmission`

Formats
JSON, XML

HTTP Methods
POST

Authentication
`Authorization: Bearer token`

Inputs

Input	Details
`approvalSubmissionId`	**Type** string **Description** Required.

Input	Details
	ID of the approval submission to be recalled.
comments	**Type** string **Description** Comments entered by the approval admin or approval submitter about why they recalled the approval submission.

Outputs

None.

Example

POST

Here's a sample request for the Recall Approval Submission action.

```
{
  "inputs": [
    {
      "approvalSubmissionId": "9iPxx0000000llhEBA",
      "comments": "Recall comments."
    }
  ]
}
```

Here's a sample response for the Recall Approval Submission action.

```
{
  "actionName": "recallApprovalSubmission",
  "errors": null,
  "invocationId": null,
  "isSuccess": true,
  "outcome": null,
  "outputValues": null,
  "sortOrder": -1,
  "version": 1
}
```

Review Approval Work Item Action

Update an approval work item status with the assignee or reviewer's decision and any comments that the assignee or reviewer added.

Keep these considerations in mind when you use this invocable action.

- The user must be an assignee of the approval work item or be a member of a group or queue if the approval work item is assigned to the group or queue. Additionally, a user has access to Approval Submissions if they're a delegate of the assignee or has a role higher than the assignee.

- The status of the approval work item must be in `Assigned` status.
- The user can also use this action if inherited access to group or queue membership, nested group membership, roles hierarchy, or delegates is available. See Public Group Considerations.

This action is available in API version 62.0 and later.

Supported REST HTTP Methods

URI

`/services/data/v67.0/actions/standard/reviewApprovalWorkItem`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer token`

Inputs

Input	Details
`approvalDecision`	**Type** string **Description** Required. Action that the assigned approver made for the approval work item. Valid values are: • `approve` • `reject`
`approvalWorkItemId`	**Type** string **Description** Required. ID of the approval work item to be reviewed by the assigned approver.
`channelType`	**Type** picklist **Description** Type of channel where the request to review the approval work item originated. Valid values are: • `InvocableAction` • `Slack`

Input	Details
	• `ApprovalRecord` The default value is `InvocableAction`. Available in API version 65.0 and later.
`comments`	**Type** string **Description** Approval comments for the decision.

Outputs

None.

Example

POST

Here's a sample request for the Review Approval Work Item action.

```
{
  "inputs": [
    {
      "approvalWorkItemId": "9jRxx00000001lhEAA",
      "approvalDecision": "Approve",
      "channelType": "InvocableAction",
      "comments": "Looks good."
    }
  ]
}
```

Here's a sample response for the Review Approval Work Item action.

```
{
  "actionName": "reviewApprovalWorkItem",
  "errors": null,
  "invocationId": null,
  "isSuccess": true,
  "outcome": null,
  "outputValues": null,
  "sortOrder": -1,
  "version": 1
}
```

Advanced Approvals Business APIs

Use the Advanced Approvals Business APIs to preview approval levels and related details before you submit a record for approval.

This table lists the available Advanced Approvals resources.

Resource	Description
/connect/advanced-approvals/approval-submission/preview (POST)	Preview the approval levels of a record and associated level details, approval chains, approvers, and conditions before you submit the record for an approval.

Resources
Learn more about the available Advanced Approvals resources.

Request Bodies
Learn more about the available request bodies.

Response Bodies
Learn more about the available response bodies.

Resources

Learn more about the available Advanced Approvals resources.

Preview Approval (POST)
Preview the approval levels of a record and associated level details, approval chains, approvers, and conditions before you submit the record for an approval.

Preview Approval (POST)

Preview the approval levels of a record and associated level details, approval chains, approvers, and conditions before you submit the record for an approval.

For example, a sales rep working on a quote can preview the approval levels for a quote before submitting the quote for approval.

Resource

```
/connect/advanced-approvals/approval-submission/preview
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/advanced-approvals/approval-submission/preview
```

Available version

65.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "flowApiName": "QuoteApprovals",
  "objectApiName": "Quote",
```

```
  "recordId": "0Q0DU0000005HZC0A2",
  "inputParameters": {
    "approverComments": "Submitted for approval",
    "requestType": "Standard"
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`flowApiName`	String	API name of the auto-launched flow.	Required	65.0
`objectApiName`	String	API name of the object to preview the approvals for.	Required	65.0
`recordId`	String	ID of the record to preview the approvals for.	Required	65.0
`inputParameters`	Map<String, Object>	List of input parameters to preview.	Optional	67.0

Response body for POST

Preview Approval

Request Bodies

Learn more about the available request bodies.

Preview Approval Input

Input representation of the details of the request to preview an approval.

Preview Approval Input

Input representation of the details of the request to preview an approval.

JSON example

```
{
  "flowApiName": "QuoteApprovals",
  "objectApiName": "Quote",
  "recordId": "0Q0DU0000005HZC0A2",
  "inputParameters": {
    "approverComments": "Submitted for approval",
    "requestType": "Standard"
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`flowApiName`	String	API name of the auto-launched flow.	Required	65.0
`objectApiName`	String	API name of the object to preview the approvals for.	Required	65.0
`recordId`	String	ID of the record to preview the approvals for.	Required	65.0
`inputParameters`	Map<String, Object>	List of input parameters to preview.	Optional	67.0

Response Bodies

Learn more about the available response bodies.

Preview Approval
Output representation of the details of a preview approval request.

Preview Approval Chain Item
Output representation of the details of an approval chain item for a specific group.

Preview Approval Error
Output representation of the error details associated with the Preview Approval API.

Preview Approval Item
Output representation of the details of a specific approval item with an approval chain.

Preview Approval

Output representation of the details of a preview approval request.

JSON example


```
{
  "approvalChainItems": [
    {
      "approvalChainName": "FINANCE",
      "approvalItems": [
        {
          "stepApiName": "CPIFC",
          "approvalConditionName": "Payment Term Greater Than 90 Days",
          "assignedTo": "005XX000001ABCQYA4",
          "assigneeType": "User",
          "assigneeDetails": {
            "id": "005XX000001ABCQYA4",
            "name": "Sam Smith",
            "type": "User",
            "apiName": "",
```

```
          "email": "sam.smith@example.com"
        },
        "objectId": "0Q0bsOgSBEhlexWC3Q",
        "objectApiName": "Quote",
        "status": "Approved",
        "stage": {
          "apiName": "stage101",
          "label": "Stage 101"
        },
        "dependencies": [],
        "parents": [],
        "level": 1,
        "additionalFields": {
          "reviewedBy": "005CVxhW4MBIUbPYUX",
          "isEligibleForSmartApproval": true,
          "smartApprovalBasisWI": "000000000000000",
          "isAutoReviewed": false
        }
      },
      {
        "stepApiName": "CPIFC2",
        "approvalConditionName": "Payment Term Greater Than 90 Days",
        "assignedTo": "00GXX000001XYZA2A4",
        "assigneeType": "Group",
        "assigneeDetails": {
          "id": "00GXX000001XYZA2A4",
          "name": "Admin Group",
          "type": "Group",
          "apiName": "Admin_Group",
          "email": "admin.group@example.com"
        },
        "objectId": "0Q0bsOgSBEhlexWC3Q",
        "objectApiName": "Quote",
        "status": "In Progress",
        "stage": {
          "apiName": "stage101",
          "label": "Stage 101"
        },
        "dependencies": [
          "CPIFC"
        ],
        "parents": [
          "CPIFC"
        ],
        "level": 2,
        "additionalFields": {
          "reviewedBy": "00GvUUtUVGmTDUHU34",
          "isEligibleForSmartApproval": true,
          "smartApprovalBasisWI": "000000000000000",
          "isAutoReviewed": false
        }
      }
    ]
  }
```

```
  ],
  "flowOrchestrationDefinitionVersionId": "0jEDU0000001nZm",
  "status": "Success"
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
approvalChainItems	Preview Approval Chain Item[]	Details of the approval items of an approval chain.	Small, 65.0	65.0
correlationId	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Small, 65.0	65.0
error	Preview Approval Error[]	Details of the error encountered during the processing of the API request.	Small, 65.0	65.0
flowOrchestrationDefinitionVersionId	String	ID of the flow orchestration definition version.	Small, 65.0	65.0
status	String	Status of the API request.	Small, 65.0	65.0

Preview Approval Chain Item

Output representation of the details of an approval chain item for a specific group.

JSON example

```
{
  "approvalChainItems": [
    {
      "approvalChainName": "FINANCE",
      "approvalItems": [
        {
          "stepApiName": "CPIFC",
          "approvalConditionName": "Payment Term Greater Than 90 Days",
          "assignedTo": "005XX000001ABCQYA4",
          "assigneeType": "User",
          "assigneeDetails": {
            "id": "005XX000001ABCQYA4",
            "name": "Sam Smith",
            "type": "User",
            "apiName": "",
            "email": "sam.smith@example.com"
          },
          "objectId": "0Q0bsOgSBEhlexWC3Q",
          "objectApiName": "Quote",
          "status": "Approved",
          "stage": {
            "apiName": "stage101",
```

```
            "label": "Stage 101"
          },
          "dependencies": [],
          "parents": [],
          "level": 1,
          "additionalFields": {
            "reviewedBy": "005CVxhW4MBIUbPYUX",
            "isEligibleForSmartApproval": true,
            "smartApprovalBasisWI": "000000000000000",
            "isAutoReviewed": false
          }
        },
        {
          "stepApiName": "CPIFC2",
          "approvalConditionName": "Payment Term Greater Than 90 Days",
          "assignedTo": "00GXX000001XYZA2A4",
          "assigneeType": "Group",
          "assigneeDetails": {
            "id": "00GXX000001XYZA2A4",
            "name": "Admin Group",
            "type": "Group",
            "apiName": "Admin_Group",
            "email": "admin.group@example.com"
          },
          "objectId": "0Q0bsOgSBEhlexWC3Q",
          "objectApiName": "Quote",
          "status": "In Progress",
          "stage": {
            "apiName": "stage101",
            "label": "Stage 101"
          },
          "dependencies": [
            "CPIFC"
          ],
          "parents": [
            "CPIFC"
          ],
          "level": 2,
          "additionalFields": {
            "reviewedBy": "00GvUUtUVGmTDUHU34",
            "isEligibleForSmartApproval": true,
            "smartApprovalBasisWI": "000000000000000",
            "isAutoReviewed": false
          }
        }
      ]
    }
  ],
  "flowOrchestrationDefinitionVersionId": "0jEDU0000001nZm",
  "status": "Success"
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
approvalChainName	String	Name of the approval chain.	Small, 65.0	65.0
approvalItems	Preview Approval Item[]	Details of the approval items.	Small, 65.0	65.0

Preview Approval Error

Output representation of the error details associated with the Preview Approval API.

JSON example

This example shows a sample error scenario.

```
{
  "approvalChainItems": [],
  "error": {
    "correlationId": "0Q0DU0000005HZC0A2",
    "errorCode": "[xmlrpc=-1, statusCode=INVALID_API_INPUT, exceptionCode=null,
scope=PublicApi, http=400]",
    "errorMessage": "Looks like the flow associated with this approval workflow for the
 current record isn't active. Activate the flow and try again.",
    "source": "PreviewApprovalDataProcessingException"
  },
  "status": "Failure"
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
correlationId	String	Unique token to track and associate related events or transactions across different components of the application. If unspecified, a Universally Unique Identifier (UUID) is generated.	Small, 65.0	65.0
errorCode	String	Code for the resultant error.	Small, 65.0	65.0
errorMessage	String	Error message for the resultant error.	Small, 65.0	65.0
source	String	Details about the source of the error.	Small, 65.0	65.0

Preview Approval Item

Output representation of the details of a specific approval item with an approval chain.

JSON example

```
{
  "approvalChainItems": [
    {
```

```
    "approvalChainName": "FINANCE",
    "approvalItems": [
      {
        "stepApiName": "CPIFC",
        "approvalConditionName": "Payment Term Greater Than 90 Days",
        "assignedTo": "005XX000001ABCQYA4",
        "assigneeType": "User",
        "assigneeDetails": {
          "id": "005XX000001ABCQYA4",
          "name": "Sam Smith",
          "type": "User",
          "apiName": "",
          "email": "sam.smith@example.com"
        },
        "objectId": "0Q0bsOgSBEhlexWC3Q",
        "objectApiName": "Quote",
        "status": "Approved",
        "stage": {
          "apiName": "stage101",
          "label": "Stage 101"
        },
        "dependencies": [],
        "parents": [],
        "level": 1,
        "additionalFields": {
          "reviewedBy": "005CVxhW4MBIUbPYUX",
          "isEligibleForSmartApproval": true,
          "smartApprovalBasisWI": "000000000000000",
          "isAutoReviewed": false
        }
      },
      {
        "stepApiName": "CPIFC2",
        "approvalConditionName": "Payment Term Greater Than 90 Days",
        "assignedTo": "00GXX000001XYZA2A4",
        "assigneeType": "Group",
        "assigneeDetails": {
          "id": "00GXX000001XYZA2A4",
          "name": "Admin Group",
          "type": "Group",
          "apiName": "Admin_Group",
          "email": "admin.group@example.com"
        },
        "objectId": "0Q0bsOgSBEhlexWC3Q",
        "objectApiName": "Quote",
        "status": "In Progress",
        "stage": {
          "apiName": "stage101",
          "label": "Stage 101"
        },
        "dependencies": [
          "CPIFC"
        ],
        "parents": [
```

```
              "CPIFC"
            ],
            "level": 2,
            "additionalFields": {
              "reviewedBy": "00GvUUtUVGmTDUHU34",
              "isEligibleForSmartApproval": true,
              "smartApprovalBasisWI": "000000000000000",
              "isAutoReviewed": false
            }
          }
        ]
      }
    ],
    "flowOrchestrationDefinitionVersionId": "0jEDU0000001nZm",
    "status": "Success"
  }
```


Property Name	Type	Description	Filter Group and Version	Available Version
additional Fields	Map<String, String>	Details of any additional fields in the approval workflow.	Small, 65.0	65.0
approval ConditionName	String	Details of the configured conditions in the approval workflow.	Small, 65.0	65.0
assignedTo	String	Name of the assignee that the approval request is assigned to.	Small, 65.0	65.0
assigneeType	String	Type of assignee.	Small, 65.0	65.0
assigneeDetails	Map<String, Object>	Includes the details of an assignee, such as ID, name, type, API name, and email.	Small, 65.0	65.0
dependencies	String[]	Specifies dependencies among approval steps, which are entry conditions based on the status of a previous step within the same stage.	Small, 65.0	65.0
level	Integer	Hierarchy level of the approval item.	Small, 65.0	65.0
objectApiName	String	API name of the object to preview the approval for.	Small, 65.0	65.0
objectId	String	ID of the object to preview the approval for.	Small, 65.0	65.0
parents	String[]	Details of the parent step.	Small, 65.0	65.0
stage	Map<String, String>	Includes the API name and label.	Small, 65.0	65.0
status	String	Status of the approval request.	Small, 65.0	65.0
stepApiName	String	API name of the step.	Small, 65.0	65.0

Advanced Approvals Metadata API Types

Metadata API enables you to access some types and feature settings that you can customize in the user interface.

Flow for Advanced Approvals
The flow for Advanced Approvals represents the metadata associated with a flow. With Flow, you can create an application that takes users through a series of pages to query and update the records in the database. You can also run logic and provide branching capability based on user input to build dynamic applications.

SEE ALSO:

Metadata API Developer Guide: Understanding Metadata API

Flow for Advanced Approvals

The flow for Advanced Approvals represents the metadata associated with a flow. With Flow, you can create an application that takes users through a series of pages to query and update the records in the database. You can also run logic and provide branching capability based on user input to build dynamic applications.

FlowActionCall

Advanced Approvals exposes additional actionType values for the FlowActionCall metadata type.

Field Name	Field Type	Description
`actionType`	InvocableActionType (enumeration of type string)	Required. The action type. Additional valid values for Advanced Approvals are: • `cancelApprovalSubmission`—Cancels an approval submission and all child approval work items that haven't been completed. You can also add comments about why the approval admin made the cancellation. Added in API version 62.0 and later. • `overrideApprovalWorkItem`—Update an approval work item status with the approval admin decision and any comments that the approval admin added. Added in API version 62.0 and later. • `reassignApprovalWorkItem`—Reassign an approval work item that hasn't been completed. You can also add comments about why the approval admin reassigned the approval work item. Added in API version 62.0 and later. • `recallApprovalSubmission`—Recall an approval submission that isn't completed. You can also add comments that the submitter or approval admin made the recall. Added in API version 62.0 and later. • `reviewApprovalWorkItem`—Update an approval work item status with the assignee or reviewer's decision and any comments that the assignee or reviewer added. Added in API version 62.0 and later.

Field Name	Field Type	Description
		• `getPreviousRelaRecDetails`—Get the related record details submitted for approval before the current approval submission. The details are required for approval steps that use custom logic for auto-approvals. Added in API version 66.0 and later.

/

CHAPTER 10 Dynamic Revenue Orchestrator

In this chapter ...

Get visibility into a product's fulfillment journey. Also, get a complete view of the entire fulfillment design that includes processes such as order decomposition, fulfillment plans, and jeopardy management.

EDITIONS

Available in: **Enterprise**, **Unlimited**, and **Developer** Editions

Dynamic Revenue Orchestrator Standard Objects

The Dynamic Revenue Orchestrator data model provides objects and fields to manage details of a product's fulfillment.

AssetFulfillmentDecomp

Represents the relationship between an ordered asset and its corresponding fulfillment asset. This object is available in API version 62.0 and later.

FulfillmentAsset

Represents an instance of a technical product used to provide a customer asset. This object is available in API version 61.0 and later.

FulfillmentAssetAttribute

Represents an attribute of a fulfillment asset. This object is available in API version 61.0 and later.

FulfillmentAssetRelationship

Represents a relationship between two fulfillment assets. This object is available in API version 61.0 and later.

FulfillmentAssetStatePeriod

Represents the period during which the fulfillment asset configuration is applicable. This object is available in API version 67.0 and later.

FulfmtAssetStatePeriodAttr

Represents the key-value pair of a fulfillment asset attribute applicable during a specific asset state period. This object is available in API version 67.0 and later.

FulfillmentFalloutRule

Represents the fulfillment fallout handling rule. This object is available in API version 61.0 and later.

FulfillmentLineAttribute

Represents an attribute of a fulfillment order line. This object is available in API version 61.0 and later.

FulfillmentLineRel

Represents a relationship between two fulfillment order lines. This object is available in API version 61.0 and later.

FulfillmentLineSourceRel

Represents the relationship between a fulfillment order line and its decomposition source. This object is available in API version 61.0 and later.

FulfillmentPlan

Represents a set of steps to be created to fulfill the order. This object is available in API version 61.0 and later.

FulfillmentStep

Represents a task that's required to perform a certain action as part of order fulfillment. This task can be manual or automated. This object is available in API version 61.0 and later.

FulfillmentStepDefinition

Represents a definition of a step that must be executed during fulfillment orchestration. This object is available in API version 61.0 and later.

FulfillmentStepDefinitionGroup

Represents a set of fulfillment step definitions. This object is available in API version 61.0 and later.

FulfillmentStepDependency

Represents a dependency between tasks by defining the order between a task and one that depends on it. This object is available in API version 61.0 and later.

FulfillmentStepDependencyDef

Represents a dependency that must be created between two fulfillment step records. This object is available in API version 62.0 and later.

FulfillmentStepJeopardyRule

Represents the duration and tolerance for the step in the fulfillment process to allow the overall tracking of rules and risks. This object is available in API version 61.0 and later.

FulfillmentStepSource

Represents a link between a fulfillment step and the corresponding order lines. This object is available in API version 61.0 and later.

FulfillmentTaskAssignmentRule

Represents a set of actions that assign a task to a user or queue. This object is available in API version 63.0 and later.

FulfillmentWorkspace

Represents a visual designer for fulfillment plans that can have multiple step groups and their dependencies. This object is available in API version 61.0 and later.

FulfillmentWorkspaceItem

Represents information about the attributes that are used in the definition for a fulfillment step group. This object is available in API version 61.0 and later.

ProductDecompEnrichmentRule

Represents mappings between fields and attributes. Enrichment rules are part of a decomposition rule, and are used to propagate data to fulfillment order lines. This object is available in API version 61.0 and later.

ProdtDecompEnrchVarMap

Represents the mapping of a field context tag or an attribute to a variable within an expression set. This object is available in API version 64.0 and later.

ProductFulfillmentDecompRule

Represents a rule that determines how an order is broken into sub-orders with specific technical details that help in order fulfillment. It can be applied to a commercial or a technical product. This object is available in API version 61.0 and later.

ProductFulfillmentScenario

Represents a link between a product and the corresponding group of fulfillment steps that's necessary to fulfill that product. This object is available in API version 61.0 and later.

SalesTrxnDeleteEvent

Represents the platform event that triggers the deletion of sales transaction fulfillment request records when the corresponding reference records are deleted. This object is available in API version 64.0 and later.

SalesTransactionFulfillReq

Represents the statuses of all the sub-orders that belong to the selected commercial or technical product. This object is available in API version 62.0 and later.

ValTfrm

Represents mappings between fields and attributes. Enrichment rules are part of a decomposition rule, and are used to propagate data to fulfillment order lines. This object is available in API version 61.0 and later.

ValTfrmGrp
Represents a rule that determines how an order is broken into sub-orders with specific technical details that help in order fulfillment. The rule can be applied to a commercial or a technical product. This object is available in API version 61.0 and later.

SEE ALSO:

Object Reference for the Salesforce Platform: Overview of Salesforce Objects and Fields

SOAP API Developer Guide: Introduction to SOAP API

AssetFulfillmentDecomp

Represents the relationship between an ordered asset and its corresponding fulfillment asset. This object is available in API version 62.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`FulfillmentSourceAssetId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The identifier of the relationship between an asset and a fulfillment asset. This field is a polymorphic relationship field. **Relationship Name** FulfillmentSourceAsset **Refers To** Asset, FulfillmentAsset
`FulfillmentTargetAssetId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The identifier of the target asset that's being fulfilled. This field is a relationship field. **Relationship Name** FulfillmentTargetAsset

Field	Details
	Refers To FulfillmentAsset
IsUsedForFulfmtAssetActivation	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates if internal jobs are allowed to activate the related fulfillment asset. The default value is true. This field is available in API version 6.0 and later.
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The name of the asset that's being fulfilled.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the user who created the request record. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
RelationshipType	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The type of relationship between an asset and a fulfillment asset. Valid values are: • SourceBundleRoot • SourceLineItem

Field	Details
SegmentIdentifier	**Type** string **Properties** Create, Group, Nillable, Sort, **Description** The ID of the ramp segment associated with the asset state period. This field is available in API version 63.0 and later in orgs that have Revenue Management when the Ramp Deals setting is enabled.
Status	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The status of the fulfillment asset. Valid values are: • Active • Inactive

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

AssetFulfillmentDecompHistory on page 2873
History is available for tracked fields of the object starting API version 65.0.

AssetFulfillmentDecompShare on page 2877
Sharing is available for the object.

FulfillmentAsset

Represents an instance of a technical product used to provide a customer asset. This object is available in API version 61.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
AccountId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The identifier of the account. This field is a relationship field. **Relationship Name** Account **Relationship Type** Lookup **Refers To** Account
IsTimeAware	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates if the fulfillment asset's configuration data is tracked in a time-aware or time-agnostic manner. Available in API version 67.0 and later. The default value is `false`.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **For internal use only.**
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **For internal use only.**
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update

Field	Details
	Description The name of the fulfillment asset.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** For internal use only. This field is a polymorphic relationship field. **Relationship Name** Owner **Relationship Type** Lookup **Refers To** Group, User
ProductId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The identifier of the corresponding product. This field is a relationship field. **Relationship Name** Product **Relationship Type** Lookup **Refers To** Product2
Quantity	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** For internal use only.
ScopeIdentifierText	**Type** string

Field	Details
	Properties Create, Filter, Group, idLookup, Nillable, Sort **Description** The scope in which this fulfillment asset record is created. This field is available in API version 65.0 and later.
`StateEndTime`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** Represents the end time of the current fulfillment asset state period. Available in API version 67.0 and later.
`StateStartTime`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** Represents the start time of the current fulfillment asset state period. Available in API version 67.0 and later.
`Status`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The status of the fulfillment asset. Valid values are: • `Active` • `InActive`
`UnitOfMeasureId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort **Description** The unit of measure for the asset quantity such as, unit, gallon, ton, or case. This field is available in API version 63.0 and later. This field is a relationship field.

Field	Details
	Relationship Name UnitOfMeasure **Refers To** UnitOfMeasure

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

FulfillmentAssetHistory on page 2873
History is available for tracked fields of the object starting API version 65.0.

FulfillmentAssetShare on page 2877
Sharing is available for the object.

FulfillmentAssetAttribute

Represents an attribute of a fulfillment asset. This object is available in API version 61.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `query()`, `update()`, `upsert()`

Fields

Field	Details
`AttributeDefinitionId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The attribute definition in the catalog. This field is a relationship field. **Relationship Name** AttributeDefinition **Relationship Type** Lookup **Refers To** AttributeDefinition

Field	Details
AttributeName	**Type** string **Properties** Filter, Group, idLookup, Nillable, Sort **Description** The name of the attribute.
AttributePicklistValueId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** For internal use only. This field is a relationship field. **Relationship Name** AttributePicklistValue **Relationship Type** Lookup **Refers To** AttributePicklistValue
AttributeValue	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The value of the attribute.
ExternalId	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** For internal use only.
FulfillmentAssetId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The identifier of the fulfillment asset.

Field	Details
	This field is a relationship field. **Relationship Name** FulfillmentAsset **Relationship Type** Master-detail **Refers To** FulfillmentAsset (the master object)

FulfillmentAssetRelationship

Represents a relationship between two fulfillment assets. This object is available in API version 61.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
AssociatedFulfillAssetRole	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The role of the associated fulfillment asset. Valid values are: • `BundleComponent` • `ClassificationComponent`—Product Classification Component
AssociatedFulfillmentAssetId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The name of the associated fulfillment asset. This field is a relationship field.

Field	Details
	Relationship Name AssociatedFulfillmentAsset **Refers To** FulfillmentAsset
EndTime	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The time until when this asset relationship is valid.
MainFulfillmentAssetId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The name of the primary fulfillment asset. This field is a relationship field. **Relationship Name** MainFulfillmentAsset **Relationship Type** Master-detail **Refers To** FulfillmentAsset (the master object)
MainFulfillmentAssetRole	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The role of the primary fulfillment asset. Valid value is `Bundle Parent`.
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The name of the fulfillment asset relationship.

Field	Details
ProductRelationshipTypeId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The type of relationship between two assets. This field is a relationship field. **Relationship Name** ProductRelationshipType **Refers To** ProductRelationshipType
StartTime	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The time from when this asset relationship is valid.

FulfillmentAssetStatePeriod

Represents the period during which the fulfillment asset configuration is applicable. This object is available in API version 67.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

Fields

Field	Details
CurrencyIsoCode	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Possible values are:

Field	Details
	• `EUR`—Euro • `INR`—Indian Rupee • `USD`—U.S. Dollar The default value is `USD`.
`EndTime`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** Represents the end time of the state period. If null, the configuration is effective from Start Time. If Start Time is also null, the configuration is effective at all times.
`FulfillmentAssetId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** Represents the fulfillment asset to which this fulfillment asset state period applies. This field is a relationship field. **Relationship Name** FulfillmentAsset **Relationship Type** Master-detail **Refers To** FulfillmentAsset (the master object)
`IsSuperseded`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates if the state period is superseded and the record is replaced by a newer version due to updates by the fulfillment process. The default value is `false`.
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort

Field	Details
	Description The name of the fulfillment asset state period.
`Quantity`	**Type** double **Properties** Create, Filter, Sort, Update **Description** Represents the fulfillment product quantity applicable for the state period.
`StartTime`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** Represents the start time of the state period. If null, the configuration is effective until End Time. If End Time is also null, the configuration is effective at all times.

FulfmtAssetStatePeriodAttr

Represents the key-value pair of a fulfillment asset attribute applicable during a specific asset state period. This object is available in API version 67.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `query()`, `retrieve()`, `update()`, `upsert()`

Special Access Rules

Fields

Field	Details
`AttributeDefinitionId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** Represents the definition of the fulfillment asset state period attribute. This field is a relationship field.

Field	Details
	Relationship Name AttributeDefinition **Refers To** AttributeDefinition
AttributeName	**Type** string **Properties** Filter, Group, idLookup, Nillable, Sort **Description** Represents the name of the fulfillment asset state period attribute.
AttributePicklistValueId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Represents the reference identifier of the selected picklist value. This field is a relationship field. **Relationship Name** AttributePicklistValue **Refers To** AttributePicklistValue
AttributeValue	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Represents the value of the fulfillment asset state period attribute.
ExternalId	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Represents the external ID of the fulfillment asset state period attribute.
FulfillmentAssetStatePeriodId	**Type** reference

Field	Details
	Properties Create, Filter, Group, Sort **Description** Represents the fulfillment asset period applicable to the object. This field is a relationship field. **Relationship Name** FulfillmentAssetStatePeriod **Relationship Type** Master-detail **Refers To** FulfillmentAssetStatePeriod (the master object)

FulfillmentFalloutRule

Represents the fulfillment fallout handling rule. This object is available in API version 61.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`ErrorCode`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The failure error code of the fulfillment step that's associated with the rule.
`FalloutQueueId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The fallout queue that's associated with the fallout task. This field is available in API version 62.0 and later. This field is a relationship field.

Field	Details
	Relationship Name FalloutQueue **Refers To** Group
FlowDefinitionName	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The name of the flow definition that's associated with the `AutoTask` type of fulfillment step.
IntegrationDefinitionId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The integration definition that's associated with the `Callout` type of fulfillment step. This field is a relationship field. **Relationship Name** IntegrationDefinition **Relationship Type** Lookup **Refers To** IntegrationProviderDef
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The most recent date when a user referenced this record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The most recent date when a user viewed this record.

Field	Details
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The name of the flow definition that's associated with the `AutoTask` type of fulfillment step.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the user who created the fulfillment record. This field is a polymorphic relationship field. **Relationship Name** Owner **Relationship Type** Lookup **Refers To** Group, User
RetriesAllowed	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The maximum number of times a retry policy is run before the fulfillment step is considered failed.
RetryIntervals	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The interval after which the selected retry policy is run when the fulfillment step fails. This field is available in API version 62.0 and later.
RetryPolicy	**Type** picklist

Field	Details
	Properties Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies the retry policy used when the fulfillment step fails. Valid value is: • `Immediate` • `Monotonous` • `Staggered`
`StepType`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies the type of fulfillment step associated with the fallout rule. Valid values are: • `AutoTask` • `Callout` • `ManualTask` • `Milestone` • `Pause`

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

FulfillmentFalloutRuleHistory on page 2873
History is available for tracked fields of the object.

FulfillmentFalloutRuleShare on page 2877
Sharing is available for the object.

FulfillmentLineAttribute

Represents an attribute of a fulfillment order line. This object is available in API version 61.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `query()`, `update()`, `upsert()`

Fields

Field	Details
AttributeDefinitionId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** A unique identifier for the attribute definition in the catalog. This field is a relationship field. **Relationship Name** AttributeDefinition **Refers To** AttributeDefinition
AttributeName	**Type** string **Properties** Filter, Group, idLookup, Nillable, Sort **Description** The name of the attribute.
AttributePicklistValueId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** For internal use only. This field is a relationship field. **Relationship Name** AttributePicklistValue **Refers To** AttributePicklistValue
AttributeValue	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The value of the attribute.

Field	Details
`ExternalId`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID that uniquely identifies the relationship in an external data source.
`FulfillmentOrderLineItemId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** A unique identifier for the fulfillment order line item. This field is a relationship field. **Relationship Name** FulfillmentOrderLineItem **Relationship Type** Master-detail **Refers To** FulfillmentOrderLineItem (the master object)

FulfillmentLineRel

Represents a relationship between two fulfillment order lines. This object is available in API version 61.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`AssociatedFoItemInventory`	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort

Field	Details
	Description Inventory level for the associated fulfillment order item. Valid values are: • `Included in Main Inventory` • `Not Included in Main Inventory` This field is available in API version 63.0 and later.
`AssociatedFulfillOrderItemId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The identifier of the associated fulfillment order line item. This field is a relationship field. **Relationship Name** AssociatedFulfillOrderItem **Refers To** FulfillmentOrderLineItem
`AssociatedLineRole`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The role of the associated fulfillment order line item. Valid values are: • `BundleComponent` • `ClassificationComponent`—Product Classification Component
`AssociatedQuanScaleMethod`	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** Method used to scale the quantity of the associated order item summary relative to the main fulfillment order item. Valid values are: • `Constant` • `Proportional`

Field	Details
	The default value is `Proportional`. This field is available in API version 63.0 and later.
`FulfillmentOrderId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The ID of the parent fulfillment order. This field is a relationship field. **Relationship Name** FulfillmentOrder **Refers To** FulfillmentOrder
`MainFulfillOrderItemRole`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The role of the primary fulfillment order line item. Valid value is `Bundle`.
`MainFulfillmentOrderItemId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The ID of the associated fulfillment order line item. This field is a relationship field. **Relationship Name** MainFulfillmentOrderItem **Relationship Type** Master-detail **Refers To** FulfillmentOrderLineItem (the master object)
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort

Field	Details
	Description The name of the fulfillment order line relationship.
`ProductRelationshipTypeId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The type of relationship between two assets. This field is a relationship field. **Relationship Name** ProductRelationshipType **Refers To** ProductRelationshipType

FulfillmentLineSourceRel

Represents the relationship between a fulfillment order line and its decomposition source. This object is available in API version 61.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`Action`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort **Description** Specifies the action to be performed on the asset fulfillment decomposition record. This field is available in API version 66.0 and later. Possible values are: • `Add` • `Cancel` • `NoChange`—No Change

Field	Details
FulfilmentOrderLineId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The ID of the fulfillment order line. This field is a relationship field. **Relationship Name** FulfilmentOrderLine **Refers To** FulfillmentOrderLineItem
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The relation between the two order line sources.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the user who created the relationship record. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
SourceItemIdentifier	**Type** string **Properties** Create, Filter, Group, idLookup, Nillable, Sort, Update **Description** The ID of the decomposition source that's related to the fulfilment order line. This field is available in API version 64.0 and later.
SourceLineItemId	**Type** reference

Field	Details
	Properties Create, Filter, Group, Sort, Update **Description** The ID of the source line item. This field is a polymorphic relationship field. **Relationship Name** SourceLineItem **Refers To** FulfillmentOrderLineItem, OrderItem
`SourceType`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The type of source for the line item. Valid values are: • `SourceBundleRoot` • `SourceLineItem`
`SupplementalAction`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort **Description** The supplemental action that's applied to the line item based on the run-time changes made to the original fulfillment request. This field is available in API version 62.0 and later. Valid values are: • `Add` • `Amend` • `Cancel` • `NoChange`
`UniqueIdentifier`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort **Description** For internal use only.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

FulfillmentLineSourceRelShare on page 2877
Sharing is available for the object.

FulfillmentPlan

Represents a set of steps to be created to fulfill the order. This object is available in API version 61.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
ExecutionUserId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** For internal use only. **Relationship Name** ExecutionUser **Refers To** User
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The most recent date when a user referenced this record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The most recent date when a user viewed this record.

Field	Details
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The name of the fulfillment plan.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the user who created the record. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
Priority	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The priority of the fulfillment plan execution. This field is available in API version 63.0 and later. Valid values are: • Default • High • Bulk
SourceIdentifier	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update, idLookup (Available in API version 64.0 and later) **Description** For internal use only.
SourceType	**Type** string

Field	Details
	Properties Create, Filter, Group, Nillable, Sort **Description** The type of source for the fulfillment plan. This field is available in API version 62.0 and later.
State	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The status of the fulfillment plan. Valid values are: • `Completed` • `InProgress` • `NotStarted`
UsageType	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** The business that uses Fulfillment Orchestration. Valid values are: • `IntegrationOrchestrator` • `Generic` • `StageManagement`

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

FulfillmentPlanChangeEvent on page 2864
Change events are available for the object.

FulfillmentPlanHistory on page 2873
History is available for tracked fields of the object starting API version 65.0.

FulfillmentPlanShare on page 2877
Sharing is available for the object.

FulfillmentStep

Represents a task that's required to perform a certain action as part of order fulfillment. This task can be manual or automated. This object is available in API version 61.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`ActualCompletionDate`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date and time when the fulfillment step state changed to `Completed` or `Skipped`.
`ActualStartDate`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date and time when the fulfillment step state changed to `Ready`.
`AssignedToId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The user or queue associated with the fulfillment step. This field is a polymorphic relationship field. **Relationship Name** AssignedTo **Relationship Type** Lookup **Refers To** Queue, User
`CustomConfigParameter`	**Type** string

Field	Details
	Properties Create, Filter, Group, Nillable, Sort, Update **Description** Represents the custom configuration context set by the designer in the Fulfillment Step Definition record. This context is passed to fulfillment steps during execution to enable flow reusability.
`CompensatedStepId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort **Description** The alternative step that's executed when a particular step in the fulfillment plan is amended or canceled. This field is a relationship field. This field is available in API version 62.0 and later. **Relationship Name** CompensatedStep **Refers To** FulfillmentStep
`DelayOf`	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The value of the delay. This field is available in API version 63.0 and later.
`DelayUnit`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The unit for the delay of the fulfillment step. This field is available in API version 63.0 and later. Valid values are: • `Days` • `Hours` • `Minutes`
`ExecuteOn`	**Type** picklist

Field	Details
	Properties Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies when to execute the fulfillment step. This field is available in API version 63.0 and later. Valid values are: • `PreviousStepExecutionDate` • `SourceLineStartDate`
`ExecuteOnRuleId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The name of the expression set. The fulfillment step is executed only when its corresponding expression set returns the value `true`. This field is a polymorphic relationship field. **Relationship Name** ExecuteOnRule **Relationship Type** Lookup **Refers To** ExpressionSet
`ExecutionMessage`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** For internal use only.
`FalloutQueueId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The fallout queue that's associated with the fallout task. This field is available in API version 62.0 and later. This field is a relationship field. **Relationship Name** FalloutQueue

Field	Details
	Refers To Group
FlowDefinitionName	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The name of the flow definition that's associated with the `AutoTask` type of fulfillment step. This field is available in API version 62.0 and later.
FlowInterviewId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The flow interview associated with the fulfillment step. This field is a relationship field. **Relationship Name** FlowInterview **Relationship Type** Lookup **Refers To** FlowInterview
ForcePlanFreezeDuringExecution	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies whether to freeze the plan while the step is in progress. If enabled, specifies how to complete the step before resuming the plan. This field is available in API version 63.0 and later. Valid values are: • `Never` • `YesButWaitForStepCompletion` The default value is `Never`.
FulfillmentPlanId	**Type** reference

Field	Details
	Properties Create, Filter, Group, Sort, Update **Description** The fulfillment plan associated with the fulfillment step. This field is a relationship field. **Relationship Name** FulfillmentPlan **Relationship Type** Lookup **Refers To** FulfillmentPlan
`FulfillmentStepDefinitionId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The fulfillment step definition associated with the fulfillment step. This field is a relationship field. **Relationship Name** FulfillmentStepDefinition **Relationship Type** Lookup **Refers To** FulfillmentStepDefinition
`IntegrationDefinitionNameId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** For internal use only. This field is a relationship field. **Relationship Name** IntegrationDefinitionName **Relationship Type** Lookup **Refers To** IntegrationProviderDef

Field	Details
IsSkipBranch	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the remaining steps in the fulfillment step group are skipped from execution when the Execute On Rule condition is set (`true`) or not (`false`). This field is available in API version 62.0 and later. The default value is `false`.
JeopardyStatus	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The jeopardy status of the fulfillment step. This field is a calculated field.
JeopardyThreshold	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The number of days, hours, minutes, or seconds, counting back from the expected duration, before a fulfillment step is in jeopardy.
JeopardyThresholdUnit	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The time unit of the jeopardy threshold. Valid values are: • `Days` • `Hours` • `Minutes` • `Seconds` The default value is `Minutes`.
LastReferencedDate	**Type** dateTime

Field	Details
	Properties Filter, Nillable, Sort **Description** The most recent date when a user referenced this record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The most recent date when a user viewed this record.
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The name of the fulfillment step.
NextEarliestRunTime	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The next available time for the process execution. This field is available in API version 62.0 and later.
OmniscriptName	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** For inyternal use only.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the user who created the record. This field is a polymorphic relationship field.

Field	Details
	Relationship Name Owner **Relationship Type** Lookup **Refers To** Group, User
`PlannedCompletionDate`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date and time Dynamic Revenue Orchestrator estimates that the fulfillment step will complete.
`PlannedStartDate`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date and time when the Dynamic Revenue Orchestrator calculates the fulfillment step state change to `Ready`.
`PointOfNoReturn`	**Type** multipicklist **Properties** Create, Filter, Nillable **Description** The type of source change applied to the line item. This field is available in API version 62.0 and later. Valid value is: • `Changes Denied`
`RequestedCompletionDate`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The planned completion date and time of the fulfillment step.

Field	Details
`RequestedStartDate`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The planned start date and time of the fulfillment step.
`ResumeOnRuleId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The rule set or expression set for the fulfillment step. The step is completed only when the corresponding expression set returns the `isExecuteStep` output as `true`. This field is a polymorphic relationship field. **Relationship Name** ResumeOnRule **Relationship Type** Lookup **Refers To** ExpressionSet
`RetryAttempts`	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The number of attempts allowed for retry as set up in the Fallout Qualification Rule table.
`RunAsUserId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Overrides user context for an automated step. The default user context is AutomatedProc user. This field is a relationship field. **Relationship Name** RunAsUser **Relationship Type** Lookup

Field	Details
	Refers To User
`State`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The state of the fulfillment step. For example, `In Progress` or `Completed`. Valid values are: • `Completed` • `Failed` • `FatallyFailed` • `InProgress` • `Pending` • `Ready` • `Scheduled` • `Skipped`
`StepType`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The fulfillment step type associated with the fulfillment step. Valid values are: • `AutoTask` • `Callout` • `ManualTask` • `Milestone` • `Pause` • `StagedAssetize`—Available in API version 63.0 and later.
`TaskAllocationType`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The method of assigning the manual step. This field is available in API version 63.0 and later.

Field	Details
	Valid values are: • `ContextBased` • `LeastLoaded` • `RoundRobin`
`TaskId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the task assigned to a user or queue. This field is available in API version 63.0 and later. This field is a relationship field. **Relationship Name** Task **Refers To** Task
`UsageType`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The details about the business that uses Fulfillment Orchestration. Some examples of UsageBy include Financial Services Cloud and CPQ. Valid values are: • `Fulfillment` • `InsuranceRuleAction` • `IntegrationOrchestrator` • `OrderFulfillment`

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

FulfillmentStepChangeEvent on page 2864
Change events are available for the object.

FulfillmentStepHistory on page 2873
History is available for tracked fields of the object.

FulfillmentStepShare on page 2877
Sharing is available for the object.

FulfillmentStepDefinition

Represents a definition of a step that must be executed during fulfillment orchestration. This object is available in API version 61.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`AmendGroupId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The fulfillment step group that's added to the fulfillment plan when the step is amended. This field is available in API version 62.0 and later. This field is a relationship field. **Relationship Name** AmendGroup **Refers To** FulfillmentStepDefinitionGroup
`AssignedToId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The user or queue associated with the fulfillment step definition. This field is a polymorphic relationship field. **Relationship Name** AssignedTo **Relationship Type** Lookup

Field	Details
	Refers To Queue, User
CancelledGroupId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The fulfillment step group that's added to the fulfillment plan when the step is canceled. This field is available in API version 62.0 and later. This field is a relationship field. **Relationship Name** CancelledGroup **Refers To** FulfillmentStepDefinitionGroup
CustomBaseExecutionDate	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The context tag containing a custom date that's used to calculate the execution time for a future-dated step. This field is available in API version 65.0 and later.
CustomConfigParameter	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Represents the custom configuration context given by the designer. This context is passed to fulfillment steps during execution to support flow reusability.
CustomFulfillmentScope	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The custom scope to use during order fulfillment. This field is available in API version 65.0 and later.
DelayOf	**Type** int

Field	Details
	Properties Create, Filter, Group, Nillable, Sort, Update **Description** The value of the delay. This field is available in API version 63.0 and later.
`DelayUnit`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The unit for the delay of the fulfillment step. This field is available in API version 63.0 and later. Valid values are: • `Days` • `Hours` • `Minutes`
`ExecuteOn`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies when to execute the fulfillment step. This field is available in API version 63.0 and later. Valid values are: • `PreviousStepsStartDate` • `SourceLineStartDate`
`ExecuteOnConditionData`	**Type** textarea **Properties** Create, Nillable, Update **Description** The condition for executing the fulfillment step. The condition is defined as a rule or a set of rules in JSON format. This field is available in API version 66.0 and later.
`ExecuteOnRuleId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update

Field	Details
	Description The expression set for the fulfillment step. The step is executed only when the corresponding expression set is `true`. This field is a polymorphic relationship field. **Relationship Name** ExecuteOnRule **Relationship Type** Lookup **Refers To** ExpressionSet
`FlowDefinitionName`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The name of the associated flow.
`ForcePlanFreezeDuringExecution`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies whether to freeze the plan while the step is in progress. If enabled, specifies how to complete the step before resuming the plan. This field is available in API version 63.0 and later. Valid values are: • `Never` • `YesButForcefullyCompleteStep` The default value is `Never`.
`IntegrationDefinitionNameId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The name of the integration definition that's used to set up communication with an external endpoint. This field is a relationship field. **Relationship Name** IntegrationDefinitionName

Field	Details
	Relationship Type Lookup **Refers To** IntegrationProviderDef
`IsSkipBranch`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the remaining steps in the fulfillment step group are skipped from execution when the Execute On Rule condition is set. This field is available in API version 62.0 and later. The default value is `false`.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The most recent date when a user referenced this record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The most recent date when a user viewed this record.
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The name of the fulfillment step definition.
`OmniscriptName`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** For internal use only.

Field	Details
`PointOfNoReturn`	**Type** multipicklist **Properties** Create, Filter, Nillable, Update **Description** The type of source change applied to the line item. This field is available in API version 62.0 and later. Valid value is: • `Changes Denied`
`ResumeOnConditionData`	**Type** textarea **Properties** Create, Nillable, Update **Description** The condition for resuming the paused fulfillment step. The condition is defined as a rule or a set of rules in JSON format. This field is available in API version 66.0 and later.
`ResumeOnRuleId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The expression set for the fulfillment step definition. The step is completed when the expression set is `true`. This field is a polymorphic relationship field. **Relationship Name** ResumeOnRule **Relationship Type** Lookup **Refers To** ExpressionSet
`RunAsUserId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The user context to run a fulfillment step when the fulfillment operator wants to override the default autoproc user. This field is a relationship field.

Field	Details
	Relationship Name RunAsUser **Relationship Type** Lookup **Refers To** User
Scope	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The scope of the fulfillment step definition. For example, `Bundle` or `Order`. Valid values are: • `Bundle` • `CrossPlan` • `LineItem` • `Plan` The default value is `Plan`.
StepDefinitionGroupId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** For internal use only. This field is a relationship field. **Relationship Name** StepDefinitionGroup **Relationship Type** Master-detail **Refers To** FulfillmentStepDefinitionGroup (the master object)
StepType	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** For internal use only.

Field	Details
	Valid values are: • `AutoTask` • `Callout` • `ManualTask` • `Milestone` • `Pause` • `StagedAssetize`
`TaskAllocationType`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The method of assigning the manual step. This field is available in API version 62.0 and later. Valid value is: • `RoundRobin` • `LeastLoaded` • `ContextBased`
`UsageType`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The business vertical that uses fulfillment orchestration. For example, Financial Services Cloud. Valid values are: • `IntegrationOrchestrator` • `OrderFulfillment`

FulfillmentStepDefinitionGroup

Represents a set of fulfillment step definitions. This object is available in API version 61.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The most recent date when a user referenced this record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The most recent date when a user viewed this record.
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The name of the fulfillment step definition group.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the user who created the record. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
UsageType	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The business that uses Fulfillment Orchestration.

Field	Details
	Possible values are: • `IntegrationOrchestrator` • `Generic` • `InsuranceRuleAction`

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

FulfillmentStepDefinitionGroupHistory on page 2873

History is available for tracked fields of the object.

FulfillmentStepDefinitionGroupShare on page 2877

Sharing is available for the object.

FulfillmentStepDependency

Represents a dependency between tasks by defining the order between a task and one that depends on it. This object is available in API version 61.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`DependencyDefinitionId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The name of the fulfillment step dependency definition. This field is a relationship field. **Relationship Name** DependencyDefinition **Refers To** FulfillmentStepDependencyDef

Field	Details
`DependentStepId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The name of a fulfillment step that depends on this step. This field is a relationship field. **Relationship Name** DependentStep **Relationship Type** Master-detail **Refers To** FulfillmentStep
`DependsOnStepId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The name of the fulfillment step that this step depends on. That is, the name of the step that must be executed before this one can run. This field is a relationship field. **Relationship Name** DependsOnStep **Refers To** FulfillmentStep
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** An automatically generated name for the fulfillment step dependency.

FulfillmentStepDependencyDef

Represents a dependency that must be created between two fulfillment step records. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`DependencyScope`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** The scope of the fulfillment step dependency definition. For example, Order or Order Item. Valid values are: • `Bundle` • `LineItem` • `Plan` • `CrossPlan` • `Custom` The default value is `Plan`.
`DependsOnStepDefinitionId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The fulfillment step definition that must be executed before this step. This field is a relationship field. **Relationship Name** DependsOnStepDefinition **Refers To** FulfillmentStepDefinition
`FulfillmentStepDefinitionId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The identifier of the fulfillment step definition. This field is a relationship field.

Field	Details
	Relationship Name FulfillmentStepDefinition **Relationship Type** Master-detail **Refers To** FulfillmentStepDefinition (the master object)
IsCompensateInReverse	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the order to insert the compensated group steps is reversed when a fulfillment step is canceled (`true`) or not (`false`). The default value is `false`. This field is available in API version 63.0 and later.
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The name of the fulfillment step dependency definition.
PropagateStateToDependentStep	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The state that's propagated to the dependent fulfillment step when the source fulfillment step is amended or canceled in the fulfillment plan. Valid values are: • `Amended` • `Both` • `Canceled` • `None` This field is available in API version 63.0 and later.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

FulfillmentStepDependencyDefChangeEvent on page 2864
Change events are available for the object.

FulfillmentStepDependencyDefFeed on page 2866
Feed tracking is available for the object.

FulfillmentStepDependencyDefHistory on page 2873
History is available for tracked fields of the object.

FulfillmentStepDependencyDefOwnerSharingRule on page 2875
Sharing rules are available for the object.

FulfillmentStepDependencyDefShare on page 2877
Sharing is available for the object.

FulfillmentStepJeopardyRule

Represents the duration and tolerance for the step in the fulfillment process to allow the overall tracking of rules and risks. This object is available in API version 61.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`EstimatedDuration`	**Type** int **Properties** Create, Filter, Group, Sort, Update **Description** The estimated time to complete the fulfillment step.
`EstimatedDurationUnit`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** The unit of measurement that applies to the estimated time to complete the fulfillment step. Valid values are: • `Days`

Field	Details
	• `Hours` • `Minutes` • `Seconds` The default value is `Minutes`.
`FlowDefinition`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The flow definition that's associated with the `AutoTask` type of the fulfillment step.
`IntegrationDefinitionId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The integration definition that's associated with the `Callout` type of the fulfillment step. This field is a relationship field. **Relationship Name** IntegrationDefinition **Relationship Type** Lookup **Refers To** IntegrationProviderDef
`JeopardyThreshold`	**Type** int **Properties** Create, Filter, Group, Sort, Update **Description** The value indicating the threshold after which the fulfillment step is in jeopardy.
`JeopardyThresholdUnit`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** The unit of measurement that applies to the jeopardy threshold value. Valid values are:

Field	Details
	• `Days` • `Hours` • `Minutes` • `Seconds` The default value is `Minutes`.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** For internal use only.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** For internal use only.
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** For internal use only.
`OwnerId`	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** For internal use only. This field is a polymorphic relationship field. **Relationship Name** Owner **Relationship Type** Lookup **Refers To** Group, User

Field	Details
StepType	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The type of fulfillment step that's affected by the jeopardy rule. Valid values are: • `AutoTask` • `Callout` • `ManualTask` • `Milestone` • `Pause`

FulfillmentStepSource

Represents a link between a fulfillment step and the corresponding order lines. This object is available in API version 61.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** An automatically generated name for the fulfillment step source.
SourceIdentifier	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update

Field	Details
	Description The identifier of the source order line item (order item or fulfillment order line).
`SourceLineItemId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** For internal use only. This field is a polymorphic relationship field. **Relationship Name** SourceLineItem **Refers To** FulfillmentOrderLineItem, OrderItem
`StepId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The identifier of the fulfillment step. This field is a relationship field. **Relationship Name** Step **Relationship Type** Master-detail **Refers To** FulfillmentStep (the master object)
`VersionGroupIdentifier`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the version group that's assigned to the fulfillment step source item. This field is available in API version 64.0 and later.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

FulfillmentStepSourceChangeEvent on page 2864
Change events are available for the object.

FulfillmentTaskAssignmentRule

Represents a set of actions that assign a task to a user or queue. This object is available in API version 63.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`ConditionData`	**Type** textarea **Properties** Create, Nillable, Update **Description** The condition for executing the fulfillment task assignment rule. The condition is defined as a rule or a set of rules in JSON format. This field is available in API version 66.0 and later.
`ConditionId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Condition ID that's used to determine the task assignment. This field is a polymorphic relationship field. **Relationship Name** Condition **Refers To** ExpressionSet
`DestinationId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** Destination ID of the task assignment such as, Queue or User. This field is a polymorphic relationship field.

Field	Details
	Relationship Name Destination **Refers To** Group, User
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The most recent date when a user referenced this record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The most recent date when a user viewed this record.
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The name of the object that specifies the condition used to determine the task assignment.
`OwnerId`	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the user who created the task assignment rule record. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
`Priority`	**Type** int

Field	Details
	Properties Create, Filter, Group, Sort, Update **Description** The priority of the rule for execution.
`SourceId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** Source ID of the task assignment such as Queue. This field is a relationship field. **Relationship Name** Source **Refers To** Group
`TaskAllocationType`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The method of assigning the manual step. Valid values are: • `ContextBased` • `LeastLoaded` • `RoundRobin`
`UsageType`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Name of the usage type. Possible values are: • `Fulfillment` • `Generic` • `InsuranceRuleAction`—Insurance Rule Action • `IntegrationOrchestrator`—Integration Orchestrator The default value is `Fulfillment`.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

FulfillmentTaskAssignmentRuleFeed
Feed tracking is available for the object.

FulfillmentTaskAssignmentRuleHistory
History is available for tracked fields of the object.

FulfillmentTaskAssignmentRuleShare
Sharing is available for the object.

FulfillmentWorkspace

Represents a visual designer for fulfillment plans that can have multiple step groups and their dependencies. This object is available in API version 61.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`Description`	**Type** textarea **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The description of the fulfillment workspace.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The most recent date when a user referenced this record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The most recent date when a user viewed this record.

Field	Details
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The name of the fulfillment workspace.
`OwnerId`	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the user who owns this record. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

FulfillmentWorkspaceHistory on page 2873
History is available for tracked fields of the object starting API version 65.0.

FulfillmentWorkspaceShare on page 2877
Sharing is available for the object.

FulfillmentWorkspaceItem

Represents information about the attributes that are used in the definition for a fulfillment step group. This object is available in API version 61.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
FulfillmentStepDefinitionGroupId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** For internal use only. This field is a relationship field. **Relationship Name** FulfillmentStepDefinitionGroup **Relationship Type** Master-detail **Refers To** FulfillmentStepDefinitionGroup (the detail object)
FulfillmentWorkspaceId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The ID of the parent fulfillment workspace that's related to this record. This field is a relationship field. **Relationship Name** FulfillmentWorkspace **Relationship Type** Master-detail **Refers To** FulfillmentWorkspace (the master object)
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The name of the fulfillment workspace item.
ShowOrder	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update

Field	Details
	Description The display sequence value of the fulfillment workspace item.

ProductDecompEnrichmentRule

Represents mappings between fields and attributes. Enrichment rules are part of a decomposition rule, and are used to propagate data to fulfillment order lines. This object is available in API version 61.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`CalculationDefinitionId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** An expression set or a decision matrix that calculates the destination value. This field is a polymorphic relationship field. **Relationship Name** CalculationDefinition **Refers To** DecisionMatrixDefinition, ExpressionSet
`CalculationMethod`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The type of enrichment rule. Valid values are: • `Ad-verbatim`—As Is • `Static-Lookup`—List Lookup

Field	Details
	• `Expression-Set`—Available in API version 64.0 and later
`DecompositionRuleId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The identifier of the decomposition rule. This field is a relationship field. **Relationship Name** DecompositionRule **Relationship Type** Parent-child **Refers To** ProductFulfillmentDecompRule (the master object)
`DestinationApiName`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The API name of the destination field or attribute code of an attribute.
`DestinationAttributeDefinitionId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** For internal use only. This field is a relationship field. **Relationship Name** DestinationAttributeDefinition **Refers To** AttributeDefinition
`DestinationAttributeIdentifier`	**Type** string **Properties** Create, Filter, Group, idLookup, Nillable, Sort, Update

Field	Details
	Description The destination entity attribute that is mapped from the source entity attribute. This field can store a Salesforce AttributeDefinition ID or an external identifier. This field is available in API version 65.0 and later.
DestinationContextTag	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The name of the destination context definition.
DestinationType	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The destination type for mapping. Valid values are: • `Attribute` • `Field`
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The most recent date when a user referenced this record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The most recent date when a user viewed this record
ListMappingGroupId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update

Field	Details
	Description For internal use only. This field is a relationship field. **Relationship Name** ListMappingGroup **Refers To** ValTfrmGrp
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The name of the decomposition rule.
`RuleEnforcement`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies whether the rule applies to all fulfillment requests or only to specific ones. Valid values are: • `AllFulfillmentRequests` • `InitialFulfillmentRequest` This field is available in API version 63.0 and later.
`SourceApiName`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The API name of the source field or attribute code of an attribute.
`SourceAttributeDefinitionId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** For internal use only. This field is a relationship field.

Field	Details
	Relationship Name SourceAttributeDefinition **Refers To** AttributeDefinition
SourceAttributeIdentifier	**Type** string **Properties** Create, Filter, Group, idLookup, Nillable, Sort, Update **Description** The source entity attribute that is mapped to the destination entity attribute. This field can store a Salesforce AttributeDefinition ID or an external identifier. This field is available in API version 65.0 and later.
SourceContextTag	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The source type for the context definition.
SourceType	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The source type for mapping. Valid values are: • `Attribute` • `Field`

ProdtDecompEnrchVarMap

Represents the mapping of a field context tag or an attribute to a variable within an expression set. This object is available in API version 64.0 and later.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`,

Fields

Field	Details
`AttributeDefinitionId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The attribute definition the expression set variable is mapped to when creating the enrichment rule. This field is a relationship field. **Relationship Name** AttributeDefinition **Refers To** AttributeDefinition
`ExpressionSetVarName`	**Type** string **Properties** Filter, Group, Sort **Description** The name of the variable that's mapped to the expression set defined in the enrichment rule.
`FieldContextTagName`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The name of the field context tag to which the expression set variable is mapped when creating the enrichment rule.
`ProductAttributeIdentifier`	**Type** string **Properties** Create, Filter, Group, idLookup, Nillable, Sort, Update **Description** The product attribute from an internal or external catalog, used by DRO to copy data during the enrichment process. This field is available in API version 65.0 and later.
`ProductDecompEnrichmentRuleId`	**Type** reference

Field	Details
	Properties Filter, Group, Sort **Description** The rule that contains the mappings between the fields and attributes for the decomposing product. This field is a relationship field. **Relationship Name** ProductDecompEnrichmentRule **Relationship Type** Master-detail **Refers To** ProductDecompEnrichmentRule (the master object)
`VariableType`	**Type** picklist **Properties** Defaulted on create, Filter, Group, Restricted picklist, Sort **Description** Specifies whether the expression set variable is an input or output variable. Valid values are: • `Input` • `Output` The default value is `Input`.

ProductFulfillmentDecompRule

Represents a rule that determines how an order is broken into sub-orders with specific technical details that help in order fulfillment. It can be applied to a commercial or a technical product. This object is available in API version 61.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`ConditionData`	**Type** textarea

Field	Details
	Properties Create, Nillable, Update **Description** The condition for executing the product fulfillment decomposition. The condition is defined as a rule or a set of rules in JSON format. This field is available in API version 66.0 and later.
`DestinationProductId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The destination product for the decomposition rule. This field is a relationship field. **Relationship Name** DestinationProduct **Relationship Type** Lookup **Refers To** Product2
`DestinationIdentifier`	**Type** string **Properties** Create, Filter, Group, idLookup, Nillable, Sort, Update **Description** The destination entity in the product fulfillment decomposition rule. This field can store a Salesforce product ID or an external identifier. This field is available in API version 65.0 and later.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The most recent date when a user referenced this record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort

Field	Details
	Description The most recent date when a user viewed this record.
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The name of the decomposition rule.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the user who created the record. This field is a polymorphic relationship field. **Relationship Name** Owner **Relationship Type** Lookup **Refers To** Group, User
Priority	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The priority of the decomposition rule. Decomposition rules are executed in order of priority.
SourceProductClassificationId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The classification of the source product that's used for decomposition. This field is available in API version 62.0 and later. This field is a relationship field. **Relationship Name** SourceProductClassification

Field	Details
	Refers To ProductClassification
SourceClassIdentifier	**Type** string **Properties** Create, Filter, Group, idLookup, Nillable, Sort, Update **Description** The source classification entity in the product fulfillment decomposition rule. This field can store a Salesforce product ID or an external identifier. This field is available in API version 65.0 and later.
SourceIdentifier	**Type** string **Properties** Create, Filter, Group, idLookup, Nillable, Sort, Update **Description** The source entity in the product fulfillment decomposition rule. This field can store a Salesforce product ID or an external identifier. This field is available in API version 65.0 and later.
SourceProductId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** Source product for the decomposition rule. This field is a relationship field. **Relationship Name** SourceProduct **Relationship Type** Lookup **Refers To** Product2

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

ProductFulfillmentDecompRuleShare on page 2877

Sharing is available for the object.

ProductFulfillmentScenario

Represents a link between a product and the corresponding group of fulfillment steps that's necessary to fulfill that product. This object is available in API version 61.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`Action`	**Type** multipicklist **Properties** Create, Filter, Nillable, Update **Description** For internal use only. Valid values are: • `Add` • `Amend` • `Cancel` • `NoChange` • `Renew`
`ConditionData`	**Type** textarea **Properties** Create, Nillable, Update **Description** The condition for executing the product fulfillment scenario. The condition is defined as a rule or a set of rules in JSON format. This field is available in API version 66.0 and later.
`FulfillmentStepDefnGroupId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The fulfillment step definition group associated with the product fulfillment scenario. This field is a relationship field.

Field	Details
	Relationship Name FulfillmentStepDefnGroup **Relationship Type** Lookup **Refers To** FulfillmentStepDefinitionGroup
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** For internal use only.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** For internal use only.
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The name of the product fulfillment scenario.
`OwnerId`	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** For internal use only. This field is a polymorphic relationship field. **Relationship Name** Owner **Relationship Type** Lookup **Refers To** Group, User

Field	Details
ProductClassificationId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The product classification associated with the product fulfillment scenario. This field is a relationship field. **Relationship Name** ProductClassification **Relationship Type** Lookup **Refers To** ProductClassification
ProductId	**Type** reference **Properties** Create, Filter, Group, Sort, Update, Nillable (Available in API version 64.0 and later) **Description** The product associated with the product fulfillment scenario. This field is a relationship field. **Relationship Name** Product **Relationship Type** Lookup **Refers To** Product2
SourceClassIdentifier	**Type** string **Properties** Create, Filter, Group, idLookup, Nillable, Sort, Update **Description** The source classification entity in the product fulfillment scenario. This field can store a Salesforce Product Class ID or an external identifier. This field is available in API version 65.0 and later.
SourceIdentifier	**Type** string **Properties** Create, Filter, Group, idLookup, Nillable, Sort, Update

Field	Details
	Description The source entity in the product fulfillment scenario. This field can store a Salesforce product ID or an external identifier. This field is available in API version 65.0 and later.
UsageType	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Name of the usage type. This field is available in API version 66.0 and later. Possible values are: • `Fulfillment` • `Generic` • `InsuranceRuleAction`—Insurance Rule Action • `IntegrationOrchestrator`—Integration Orchestrator • `StageManagement`—Stage Management The default value is `Fulfillment`.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

ProductFulfillmentScenarioShare on page 2877
Sharing is available for the object.

SalesTrxnDeleteEvent

Represents the platform event that triggers the deletion of sales transaction fulfillment request records when the corresponding reference records are deleted. This object is available in API version 64.0 and later.

Supported Calls

`create()`, `describeSObjects()`

Fields

Field	Details
ReferenceObjectIdentifier	**Type** string

Field	Details
	Properties Create, Nillable **Description** Object identifier for the sales transaction fulfillment request to be deleted.

SalesTransactionFulfillReq

Represents the statuses of all the sub-orders that belong to the selected commercial or technical product. This object is available in API version 62.0 and later.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`

Fields

Field	Details
`AssetizationStatus`	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** Specifies the status of the assetization. Valid values are: • `Completed` • `Failed` • `InProgress` • `NotStarted` • `Rejected` • `NotApplicable`- Available in API version 64.0 and later.
`DecompositionStatus`	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** Specifies the status of the decomposition. Valid values are: • `Completed`

Field	Details
	• `Failed` • `InProgress` • `NotStarted` • `Rejected` • `NotApplicable`- Available in API version 64.0 and later.
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The name of the sales transaction fulfillment request.
`OrchestrationGroupKey`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The identifier of the group of sales transactions that require synchronization before processing. This field is available in API version 63.0 and later.
`OwnerId`	**Type** reference **Properties** Filter, Group, Sort **Description** The ID of the user who created the request record. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
`PlanCompositionStatus`	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** For internal use only. Valid values are:

Field	Details
	• `Completed` • `Failed` • `InProgress` • `NotStarted` • `Rejected` • `NotApplicable` - Available in API version 64.0 and later.
`PlanExecutionStatus`	**Type** picklist **Properties** Filter, Group, Nillable, Sort **Description** Specifies the status of the plan execution. Valid values are: • `InProgress` • `Frozen` • `Freezing`
`PlanId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The identifier of the plan. This field is a relationship field. **Relationship Name** Plan **Refers To** FulfillmentPlan
`PreviousRequestId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The identifier of the previous fulfillment request. This field is a relationship field. **Relationship Name** PreviousRequest

Field	Details
	Refers To SalesTransactionFulfillReq
ReferenceObjectIdentifier	**Type** reference **Properties** Filter, Group, idLookup, Nillable, Sort **Description** The ID of the sales transaction record. This field is available in API version 64.0 and later.
SalesTransactionType	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** Specifies the type of sales transaction that's processed by the fulfillment request. Valid values are: • StandardOrder • GenericAdapter—Available in API version 64.0 and later.
Status	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** Specifies the overall status of the fulfillment. Valid values are: • Created • Freezing • Frozen • Fulfilled • Fulfilling • Rejected • Superseded

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

SalesTransactionFulfillReqShare on page 2877
Sharing is available for the object.

ValTfrm

Represents mappings between fields and attributes. Enrichment rules are part of a decomposition rule, and are used to propagate data to fulfillment order lines. This object is available in API version 61.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`InputDate`	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The date of value entry.
`InputDatetime`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The time of value entry.
`InputNumber`	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The value of input number.
`InputPicklistValueId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update

Field	Details
	Description The identifier of the input list of values. This field is a relationship field. **Relationship Name** InputPicklistValue **Refers To** AttributePicklistValue
`InputString`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The value of input text.
`IsInputBoolean`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates if a value was entered (`true`) or not (`false`). The default value is `false`.
`IsOutputBoolean`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates if there was an output value (`true`) or not (`false`). The default value is `false`.
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The name of the related transformation group.
`OutputDate`	**Type** date

Field	Details
	Properties Create, Filter, Group, Nillable, Sort, Update **Description** The date of matched output.
`OutputDatetime`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The time of matched output.
`OutputNumber`	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The value of output number.
`OutputPicklistValueId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The identifier of the output list of values. This field is a relationship field. **Relationship Name** OutputPicklistValue **Refers To** AttributePicklistValue
`OutputString`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The value of output text.
`ValueTransformGroupId`	**Type** reference

Field	Details
	Properties Create, Filter, Group, Sort **Description** The identifier of the related transformation group. This field is a relationship field. **Relationship Name** ValueTransformGroup **Relationship Type** Master-detail **Refers To** ValTfrmGrp (the master object)

ValTfrmGrp

Represents a rule that determines how an order is broken into sub-orders with specific technical details that help in order fulfillment. The rule can be applied to a commercial or a technical product. This object is available in API version 61.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`DestinationPrimitiveType`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The data type of the output value. Valid values are: • `Boolean` • `Currency` • `Date` • `Datetime` • `Number` • `Percent`—Picklist Value • `Text`

Field	Details
IsDestinationEnumerated	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the output is a list of values (`true`) or not (`false`). The default value is `false`.
IsSourceEnumerated	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the input is from a list of values (`true`) or not (`false`). The default value is `false`.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The most recent date when a user referenced this record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The most recent date when a user viewed this record.
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The name of the list mapping.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update

Field	Details
	Description The ID of the user who owns this record. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
SourcePrimitiveType	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The data type of the input value. Valid values are: • Boolean • Currency • Date • Datetime • Number • Percent—Picklist Value • Text
UsageType	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The data mapping feature that uses this value transformation group. Valid value is DFOListMapping

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

ValTfrmGrpShare on page 2877
Sharing is available for the object.

Dynamic Revenue Orchestrator Standard Invocable Actions

Use standard invocable actions to submit an order or a sales transaction to Dynamic Revenue Orchestrator (DRO) for fulfillment.

Decompose Sales Transaction Action

Decompose a sales transaction, such as a quote, order, or order summary.

Freeze Sales Transaction Action

Freeze a sales transaction to disable the modification of a line item.

Get Point Of No Return Action

Get details about the point of no return milestone for each line item in a sales transaction.

Orchestrate Sales Transaction Action

Initiate the orchestration process for a sales transaction. This action executes only the plan composition and orchestration phases, without performing decomposition.

Orchestrate Transaction Action

Orchestrate a transaction for any domain-specific object, such as a collection plan for Revenue billing, that requires the composition and execution of a fulfillment plan.

Submit Order Action

Submit an order to Dynamic Revenue Orchestrator (DRO) for fulfillment.

Submit Sales Transaction Action

Initiate the fulfillment process of any sales transaction, such as a quote, an order, or an order summary.

Unfreeze Sales Transaction Action

Unfreeze a sales transaction to enable the modification of a line item.

SEE ALSO:

Actions Developer Guide: Overview

REST API Developer Guide: Invocable Actions Standard

Decompose Sales Transaction Action

Decompose a sales transaction, such as a quote, order, or order summary.

Specify the ID of the sales transaction to decompose. The decomposition process includes these steps.

- Intake process
- Sales transaction decomposition

The intake process occurs synchronously or asynchronously, as specified with the `intakeRequestType` input parameter. You can also specify the decomposition priority using the `fulfillmentPriority` parameter.

This action executes only the decomposition process and stops before orchestration. You can execute custom logic in between the decomposition and orchestration processes.

This action is available in API version 67.0 and later.

Special Access Rules

The Decompose Sales Transaction action is available in Enterprise, Unlimited, and Developer Editions of Revenue Management. See the required permissions to access and call this invocable action.

Supported REST HTTP Methods

URI

`/services/data/v`***`66.0`***`/actions/standard/decomposeSalesTransaction`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer`***`token`***

Inputs

Input	Details
`allowOverrideOfPointOfNoReturn`	**Type** boolean **Description** Indicates whether to override the point of no return setting for the fulfillment step (`true`) or not (`false`). The default value is `false`.
`fulfillmentAdapter`	**Type** string **Description** Required. Type of fulfillment adapter. Valid values are: • `StandardOrder` • `GenericAdapter`
`fulfillmentPriority`	**Type** string **Description** Priority to fulfill the sales transaction. Valid values are: • `High` • `Bulk` • `Default`

Input	Details
`hydratedContextId`	**Type** string **Description** ID of the hydrated context.
`intakeRequestType`	**Type** string **Description** Type of request to process the intake. Valid values are: • `Synchronous` • `Asynchronous`
`priorityLimitAction`	**Type** string **Description** Type of action to perform when the priority limit is reached. Valid values are: • `Reject` • `Downgrade` This parameter is applicable only when you specify the `fulfillmentPriority` parameter.
`salesTransactionId`	**Type** id **Description** Required. ID of the sales transaction to submit.

Outputs

Output	Details
`errorCode`	**Type** string **Description** Code that corresponds to the type of error encountered.
`requested FulfillmentPriority`	**Type** string

Output	Details
	Description Priority to fulfill the sales transaction. Valid values are: • `High` • `Bulk` • `Default`
`requestId`	**Type** string **Description** Request ID of the invocation.
`resolved FulfillmentPriority`	**Type** string **Description** Resolved priority to fulfill the sales transaction. Valid values are: • `High` • `Bulk` • `Default`
`submitStatus`	**Type** string **Description** Submission status of the sales transaction for decomposition. Valid values are: • `Success` • `Error` • `Submitted` • `Rejected`
`usedContextId`	**Type** string **Description** ID of the used context that updates the decomposition process.

Example

POST

This sample request is for the Decompose Sales Transaction action.

```
{
  "inputs": [
    {
      "fulfillmentAdapter": "StandardOrder",
      "intakeRequestType": "Synchronous",
      "salesTransactionId": "801xx000003GYexAAG"
    }
  ]
}
```

This sample response is for the Decompose Sales Transaction action.

```
[
  {
    "actionName": "decomposeSalesTransaction",
    "errors": null,
    "invocationId": null,
    "isSuccess": true,
    "outcome": null,
    "outputValues": {
      "requestId": "ee3ded2e-fe43-401b-a54d-9124d48a0b72",
      "requestedFulfillmentPriority": "Default",
      "submitStatus": "SUCCESS",
     "usedContextId": "0000000s21to18g00091764127969531 80a8259def914e1abbd863dde076b71f",

      "resolvedFulfillmentPriority": "Default"
    },
    "sortOrder": -1,
    "version": 1
  }
]
```

Freeze Sales Transaction Action

Freeze a sales transaction to disable the modification of a line item.

A line item can reach a milestone in the fulfillment process, which is known as a point of no return, where the line item can't accept modifications. Get details about the point of no return for each line item of a sales transaction.

This action is available in API version 64.0 and later.

Special Access Rules

The Freeze Sales Transaction action is available in Enterprise, Unlimited, and Developer Editions of Revenue Cloud. See the required permissions to access and call this invocable action.

Supported REST HTTP Methods

URI
`/services/data/v`***`67.0`***`/actions/standard/freezeSalesTransaction`

Formats
JSON, XML

HTTP Methods
POST

Authentication
`Authorization: Bearer`***`token`***

Inputs

Input	Details
`salesTransactionId`	**Type** id **Description** Required. ID of the sales transaction that's submitted to the Dynamic Revenue Orchestrator.

Outputs

Output	Details
`errorCode`	**Type** string **Description** Code indicating the type of error.
`orchestrationPlanId`	**Type** id **Description** ID of the created orchestration plan.
`planState`	**Type** string **Description** Status of the created orchestration plan. Valid values are: • `FAILURE` • `NOTSTARTED` • `PENDING`

Output	Details
	• `COMPLETED` • `FROZEN` • `INPROGRESS`
`pointOfNoReturnDetailForLineItemsList`	**Type** string **Description** Collection of sales transaction line items, where each item includes a boolean value indicating whether it has reached the point of no return.
`requestId`	**Type** string **Description** ID of the request to get the point of no return details.
`salesTransactionId`	**Type** string **Description** ID of the sales transaction that's submitted to the Dynamic Revenue Orchestrator.

Example

POST

This sample request is for the Freeze Sales Transaction action.

```
{
    "inputs": [
        {
            "salesTransactionId": "801SG00000jQO1ZYAW"
        }
    ]
}
```

This sample response is for the Freeze Sales Transaction action.

```
[
  {
    "actionName": "freezeSalesTransaction",
    "errors": null,
    "invocationId": null,
    "isSuccess": true,
    "outcome": null,
    "outputValues": {
      "salesTransactionId": "801SG00000jQO1ZYAW",
      "orchestrationPlanId": "13VSG00000229Z72AI",
```

```
      "requestId": "452789a6-f2ab-4079-8aca-a11dbfef6a45",
      "pointOfNoReturnDetailForLineItemsList": "802SG000007D0B4YAK\": {\namendAllowed:
 false,\nanyChangesAllowed: true,\ncancelAllowed: false\n}, \n802SG000007D0B3YAK\":
{\namendAllowed: false,\nanyChangesAllowed: true,\ncancelAllowed: false\n}",
      "planState": "Frozen"
    },
    "sortOrder": -1,
    "version": 1
  }
]
```


Get Point Of No Return Action

Get details about the point of no return milestone for each line item in a sales transaction.

A line item can reach a milestone in the fulfillment process, which is known as a point of no return, where the line item can't accept modifications. Get details about the point of no return for each line item of a sales transaction.

This action is available in API version 64.0 and later.

Special Access Rules

The Get Point Of No Return action is available in Enterprise, Unlimited, and Developer Editions of Revenue Management. See the required permissions to access and call this invocable action.

Supported REST HTTP Methods

URI

`/services/data/v67.0/actions/standard/getPointOfNoReturn`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearertoken`

Inputs

Input	Details
`salesTransactionId`	**Type** id **Description** Required. ID of the sales transaction to get the point of no return details for.

Outputs

Output	Details
`errorCode`	**Type** string **Description** Code indicating the type of error.
`lineItemsPointOfNoReturnInfo`	**Type** string **Description** Line items with the point of no return details.
`planId`	**Type** id **Description** ID of the composed fulfillment plan.
`planState`	**Type** string **Description** State of the fulfillment plan.
`requestId`	**Type** string **Description** Request ID of the invocation.
`salesTransactionId`	**Type** string **Description** ID of the submitted sales transaction.

Example

POST

This sample request is for the Get Point Of No Return action.

```
{
  "inputs": [
    {
      "salesTransactionId": "801SG00000jQO1ZYAW"
```

```
    }
  ]
}
```

This sample response is for the Get Point Of No Return action.

```
[
  {
    "actionName": "getPointOfNoReturn",
    "errors": null,
    "invocationId": null,
    "isSuccess": true,
    "outcome": null,
    "outputValues": {
      "planId": "13VSG00000229Z72AI",
      "requestId": "35c9388b-d30d-4d68-aae5-c109c8bff7ef",
      "lineItemsPointOfNoReturnInfo": "802SG000007D0B4YAK\": {\namendAllowed:
false,\nanyChangesAllowed: true,\ncancelAllowed: false\n}, \n802SG000007D0B3YAK\":
{\namendAllowed: false,\nanyChangesAllowed: true,\ncancelAllowed: false\n}",
      "salesTransactionId": "801SG00000jQO1ZYAW",
      "planState": "InProgress"
    },
    "sortOrder": -1,
    "version": 1
  }
]
```

Orchestrate Sales Transaction Action

Initiate the orchestration process for a sales transaction. This action executes only the plan composition and orchestration phases, without performing decomposition.

Specify the ID of the sales transaction to orchestrate. The decomposition process includes these steps.

- Intake process
- Fulfillment orchestration

The intake process occurs synchronously or asynchronously, as specified with the `intakeRequestType` input parameter. You can also specify the orchestration priority using the `fulfillmentPriority` parameter.

Use this action when the sales transaction is decomposed either by a previous decomposition action or by custom logic before orchestrating the fulfillment plan.

This action is available in API version 67.0 and later.

Special Access Rules

The Orchestrate Sales Transaction action is available in Enterprise, Unlimited, and Developer Editions of Revenue Management. See the required permissions to access and call this invocable action.

Supported REST HTTP Methods

URI

/services/data/v***67.0***/actions/standard/orchestrateSalesTransaction

Formats
JSON, XML

HTTP Methods
POST

Authentication
`Authorization: Bearer`***`token`***

Inputs

Input	Details
`allowOverride OfPointOfNoReturn`	**Type** boolean **Description** Indicates whether to override the point-of-no-return setting for the fulfillment step (`true`) or not (`false`). The default value is `false`.
`fulfillmentAdapter`	**Type** string **Description** Required. Type of fulfillment adapter. Valid values are: • `StandardOrder` • `GenericAdapter`
`fulfillmentPriority`	**Type** string **Description** Priority to fulfill the sales transaction. Valid values are: • `High` • `Bulk` • `Default`
`hydratedContextId`	**Type** string **Description** ID of the hydrated context.
`intakeRequestType`	**Type** string

Input	Details
	Description Type of request to process the intake. Valid values are: • `Synchronous` • `Asynchronous`
`priorityLimitAction`	**Type** string **Description** Type of action to perform when the priority limit is reached. Valid values are: • `Reject` • `Downgrade` This parameter is applicable only when you specify the `fulfillmentPriority` parameter.
`salesTransactionId`	**Type** id **Description** Required. ID of the sales transaction to submit.

Outputs

Output	Details
`errorCode`	**Type** string **Description** Code indicating the type of error.
`fulfillmentPlanId`	**Type** id **Description** ID of the composed fulfillment plan.
`requestedFulfillmentPriority`	**Type** string **Description** Priority to fulfill the sales transaction. Valid values are: • `High` • `Bulk`

Output	Details
	• `Default`
`requestId`	**Type** string **Description** Request ID of the invocation.
`resolved FulfillmentPriority`	**Type** string **Description** Resolved priority to fulfill the sales transaction. Valid values are: • `High` • `Bulk` • `Default`
`submitStatus`	**Type** string **Description** Submission status of the sales transaction for orchestration. Valid values are: • `SUCCESS` • `ERROR` • `SUBMITTED` • `REJECTED`
`usedContextId`	**Type** string **Description** ID of the context that updates the orchestration process.

Example

POST

This sample request is for the Orchestrate Sales Transaction action.

```
{
    "inputs": [
        {
            "fulfillmentAdapter": "StandardOrder",
            "intakeRequestType": "Synchronous",
            "salesTransactionId": "801xx000003GYgZAAW"
        }
```

```
    ]
}
```

This sample response is for the Orchestrate Sales Transaction action.


```
[
  {
    "actionName": "orchestrateSalesTransaction",
    "errors": null,
    "invocationId": null,
    "isSuccess": true,
    "outcome": null,
    "outputValues": {
     "usedContextId": "0000000s21to18g00091764134566956e2100424de0d4af8869669df515e24cb",

      "requestId": "ac2a9d18-c702-43ee-bc08-1c03061b549c",
      "fulfillmentPlanId": "13Vxx0000004CFUEA2",
      "submitStatus": "SUCCESS",
      "resolvedFulfillmentPriority": "Default",
      "requestedFulfillmentPriority": "Default"
    },
    "sortOrder": -1,
    "version": 1
  }
]
```


Orchestrate Transaction Action

Orchestrate a transaction for any domain-specific object, such as a collection plan for Revenue billing, that requires the composition and execution of a fulfillment plan.

Specify the ID of the transaction to orchestrate and the orchestration type. The orchestration process includes:

- Composition of the orchestration plan
- Execution of the fulfillment plan

This action is available in API version 66.0 and later.

Special Access Rules

The Orchestrate Transaction action is available in Enterprise, Unlimited, and Developer Editions of Revenue Management. See the required permissions to access and call this invocable action.

Supported REST HTTP Methods

URI

/services/data/v***67.0***/actions/standard/orchestrateTransaction

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer`***`token`***

Inputs

Input	Details
`transactionId`	**Type** string **Description** Required. ID of the business object or domain object to be orchestrated, such as a Collection Plan ID.
`orchestrationType`	**Type** string **Description** Required. Type of orchestration that's used to orchestrate the transaction. Valid values are: • `Generic` • `Fulfillment` • `Billing`

Outputs

Output	Details
`requestId`	**Type** string **Description** Request ID of the invocation.
`errorCode`	**Type** string **Description** Code that corresponds to the type of encountered error.
`fulfillmentPlanId`	**Type** string **Description** ID of the composed fulfillment plan.

Output	Details
`submitStatus`	**Type** string **Description** Submission status of the transaction that's orchestrated. Valid values are: • `Success` • `Error`

Example

POST

This sample request is for the Orchestrate Transaction action.

```
{
  "inputs": [
    {
      "transactionId": "801xx000003GYexAAG",
      "orchestrationType": "Fulfillment"
    }
  ]
}
```

This sample response is for the Orchestrate Transaction action.

```
[
  {
    "actionName": "orchestrateTransaction",
    "errors": null,
    "invocationId": null,
    "isSuccess": true,
    "outcome": null,
    "outputValues": {
      "requestId": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
      "fulfillmentPlanId":
"0000000s21to18g000917641279695318 0a8259def914e1abbd863dde076b71f",
      "submitStatus": "SUCCESS"
    },
    "sortOrder": -1,
    "version": 1
  }
]
```

Submit Order Action

Submit an order to Dynamic Revenue Orchestrator (DRO) for fulfillment.

By using the Submit Order action, you can perform:

- Order decomposition

- Fulfillment orchestration that's driven through message queues
- Dynamic plan composition that's based on the incoming order

This action is available in API version 61.0 and later.

Special Access Rules

The Submit Order action is available in Enterprise, Unlimited, and Developer Editions of Revenue Cloud. See the required permissions to access and call this invocable action.

Supported REST HTTP Methods

URI

`/services/data/v67.0/actions/standard/submitOrder`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer token`

Inputs

Input	Details
`orderId`	**Type** string **Description** Required. ID of the order to submit to DRO.
`callType`	**Type** string **Description** Optional. Mode that the order intake must be processed in. Valid values are: • `Synchronous` • `Asynchronous` The default value is `Asynchronous`.
`contextId`	**Type** string

Input	Details
	Description Optional. ID of the hydrated context. See Context Service.

Outputs

Output	Details
`errorCode`	**Type** string **Description** Error code for the failed request, if any.
`fulfillmentPlanId`	**Type** string **Description** ID of the orchestrated fulfillment plan that's generated. Returned only if the `callType` value is `Synchronous`.
`requestId`	**Type** string **Description** Unique ID of the invocation request that helps identify a single request.
`submitStatus`	**Type** string **Description** Submit status of the invocation request. Valid values are: • `SUCCESS` • `ERROR` • `SUBMITTED` • `REJECTED`
`usedContextId`	**Type** string **Description** Hydrated context ID that's used in this request, which can be different from the `contextId` input.

Example

POST

This example shows a sample request.

```
{
    "inputs": [
        {
            "orderId": "801RM0000007yGaYAI",
            "callType": "Synchronous"
        }
    ]
}
```

This example shows a sample response when the call type is synchronous.

```
[
    {
   "actionName":"submitOrder"
   "errors":NULL
   "invocationId":NULL
   "isSuccess":true
   "outputValues":{
   "requestId":"a161cfda-868c-41d2-b589-7c7d7ff2d4c1"
   "submitStatus":"SUCCESS"
   "usedContextId":"e275e930923106ee7e39cbfa232e38252bd4d63f4ea2dd956b7301e243554134"
   "fulfillmentPlanId":"13VZM00000000062AA"
}
]
```

This example shows a sample response when the call type is asynchronous or isn't specified.

```
[
{
    "actionName": "submitOrder",
    "errors": null,
    "isSuccess": true,
    "outputValues": {
    "submitStatus": "SUBMITTED"
    "requestId": "a161cfda-868c-41d2-b589-7c7d7ff2d4c1"
}
}
]
```

This example shows a sample response when a validation error occurs.

```
[
    {
    "actionName": "submitOrder",
    "errors": [
    {
        "statusCode": "UNKNOWN_EXCEPTION",
        "message": "Missing required input parameter: orderId",
        "fields": []
     }
  ],
```

```
        "invocationId": null,
        "isSuccess": false,
        "outputValues": {
          "requestId": "4c7d8ebb-6b0b-4852-a8a0-b67e0d36a73e",
          "errorCode": "DRO_INTERNAL_ERROR"
        },
    }
]
```

Explainability Action Logs

To troubleshoot or debug errors, retrieve a list of explainability action logs. See Action Logs.

Get logs for order intake

This resource example includes sample query parameters to retrieve the action logs for an order intake.

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/decision-explainer/action-logs?applicationSubType=DroSubmit&applicationType=7&processType=DroSubmit&primaryFilter=801xx000003GYzAAG
```

This example shows the sample response. The `actionLog` property contains the action logs.

```
{
  "actionLogs": [
    {
      "actionContextCode": "801NA000000XKPUYA4",
      "actionLog": {
        "OrderIntakeStatus": "",
        "OrderIntakeStatusMessage": "",
        "OrderId": "801NA000000XKPUYA4",
        "SubmitMode": "Synchronous",
        "UniqueRequestId": "d00c2aa7-56b0-411a-9b51-83d9fe2e4440",
       "ContexId": "a0ca3c2296c82ad071a5efa83e8974793910f44ebddb21d37f007ce4889b65d7",

        "DesActionSpecDevName": "DroSubmitAction",
        "ContextDefinition": "SalesTransactionContext__stdctx"
      },
      "additionalFilter": "undef",
      "applicationLogDate": "Thu Mar 28 12:36:26 GMT 2024",
      "applicationSubtype": "DroSubmit",
      "applicationType": "7",
      "explainabilitySpecName": "DroSubmitAction",
      "isChunked": false,
      "name": "DroSubmitAction",
      "primaryFilter": "801NA000000XKPUYA4",
      "processType": "DroSubmit",
      "secondaryFilter": "undef",
      "uniqueIdentifier": "02c4bae9-d8b0-42f6-b031-e983d4247c76"
    }
  ],
  "queryMore": ""
}
```

Get logs for decomposition scope

This resource example includes sample query parameters to retrieve the decomposition-related action logs.

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/decision-explainer/action-logs?applicationSubType=DroDecomp&applicationType=7&processType=DroDecomp&primaryFilter=801xx000003GYzAAG
```

This example shows the sample response. The `actionLog` property contains the action logs.


```
{
  "actionLogs": [
    {
      "actionContextCode": "801NA000000XKPeYAO",
      "actionLog": {
        "CandidateDecompositionRules": [
          {
            "decompRuleId": "13UNA0000004C932AE",
            "SourceProductId": "01tNA0000007NP5YAM",
            "AssociatedOlis": [
              "802NA000002lWjOYAU"
            ],
            "DestinationProductId": "01tNA0000007NOvYAM",
            "DestinationProductScope": "Order",
            "ConfiguredContextRuleId": "null"
          },
          {
            "decompRuleId": "13UNA0000004ITY2A2",
            "SourceProductId": "01tNA0000007NP5YAM",
            "AssociatedOlis": [
              "802NA000002lWjOYAU"
            ],
            "DestinationProductId": "01tNA0000007NOqYAM",
            "DestinationProductScope": "Order",
            "ConfiguredContextRuleId": "null"
          },
          {
            "decompRuleId": "13UNA0000004C942AE",
            "SourceProductId": "01tNA0000007NPAYA2",
            "AssociatedOlis": [
              "802NA000002lWjKYAU"
            ],
            "DestinationProductId": "01tNA0000007NOwYAM",
            "DestinationProductScope": "Order",
            "ConfiguredContextRuleId": "null"
          },
          {
            "decompRuleId": "13UNA0000004ITd2AM",
            "SourceProductId": "01tNA0000007NPAYA2",
            "AssociatedOlis": [
              "802NA000002lWjKYAU"
            ],
            "DestinationProductId": "01tNA0000007NP0YAM",
            "DestinationProductScope": "Order",
            "ConfiguredContextRuleId": "null"
          },
          {
```

```
      "decompRuleId": "13UNA0000004C952AE",
      "SourceProductId": "01tNA0000007NPBYA2",
      "AssociatedOlis": [
        "802NA000002lWjLYAU"
      ],
      "DestinationProductId": "01tNA0000007NOxYAM",
      "DestinationProductScope": "Order",
      "ConfiguredContextRuleId": "9QwNA0000000CTu0AM"
    },
    {
      "decompRuleId": "13UNA0000004C982AE",
      "SourceProductId": "01tNA0000007NPCYA2",
      "AssociatedOlis": [
        "802NA000002lWjNYAU"
      ],
      "DestinationProductId": "01tNA0000007NOqYAM",
      "DestinationProductScope": "Order",
      "ConfiguredContextRuleId": "null"
    },
    {
      "decompRuleId": "13UNA0000004C992AE",
      "SourceProductId": "01tNA0000007NPCYA2",
      "AssociatedOlis": [
        "802NA000002lWjNYAU"
      ],
      "DestinationProductId": "01tNA0000007NOrYAM",
      "DestinationProductScope": "Order",
      "ConfiguredContextRuleId": "9QwNA0000000CTt0AM"
    }
  ],
  "SelectedDecompositionRules": [
    {
      "OliId": "802NA000002lWjKYAU",
      "DecompositionRuleIds": [
        "13UNA0000004C942AE",
        "13UNA0000004ITd2AM"
      ]
    },
    {
      "OliId": "802NA000002lWjOYAU",
      "DecompositionRuleIds": [
        "13UNA0000004C932AE",
        "13UNA0000004ITY2A2"
      ]
    },
    {
      "OliId": "802NA000002lWjNYAU",
      "DecompositionRuleIds": [
        "13UNA0000004C982AE"
      ]
    }
  ],
  "OliScopeDetails": [
    {
```

```
        "OliId": "802NA000002lWjLYAU",
        "ParentOliId": "802NA000002lWjOYAU",
        "BundleRootOli": "802NA000002lWjOYAU"
      },
      {
        "OliId": "802NA000002lWjMYAU",
        "ParentOliId": "802NA000002lWjOYAU",
        "BundleRootOli": "802NA000002lWjOYAU"
      },
      {
        "OliId": "802NA000002lWjKYAU",
        "ParentOliId": "802NA000002lWjOYAU",
        "BundleRootOli": "802NA000002lWjOYAU"
      },
      {
        "OliId": "802NA000002lWjNYAU",
        "ParentOliId": "802NA000002lWjOYAU",
        "BundleRootOli": "802NA000002lWjOYAU"
      }
    ],
    "FoliComputationDetails": [
      {
        "FoliId": "0a4NA000003HYbWYAW",
        "ComputedAction": "AMEND",
        "ComputedQuantity": "3.0"
      },
      {
        "FoliId": "0a4NA000003HYbXYAW",
        "ComputedAction": "AMEND",
        "ComputedQuantity": "6.0"
      },
      {
        "FoliId": "0a4NA000003HYbYYAW",
        "ComputedAction": "AMEND",
        "ComputedQuantity": "3.0"
      },
      {
        "FoliId": "0a4NA000003HYbZYAW",
        "ComputedAction": "AMEND",
        "ComputedQuantity": "3.0"
      }
    ],
    "FlsrDetails": [
      {
        "FlsrId": "16ANA000005idR82AI",
        "FlsrGuid": "WQoYClnAszG9axKxWxoe",
        "SourceId": "802NA000002lWjOYAU",
        "FoliId": "0a4NA000003HYbWYAW"
      },
      {
        "FlsrId": "16ANA000005idR92AI",
        "FlsrGuid": "80xWFpBK8sDmTFRV3Dn0",
        "SourceId": "802NA000002lWjOYAU",
        "FoliId": "0a4NA000003HYbXYAW"
```

```
          },
          {
            "FlsrId": "16ANA000005idRA2AY",
            "FlsrGuid": "nAql9VD3KiziZTFrfYXq",
            "SourceId": "802NA000002lWjNYAU",
            "FoliId": "0a4NA000003HYbXYAW"
          },
          {
            "FlsrId": "16ANA000005idRB2AY",
            "FlsrGuid": "wgCJT6NJ3ESolFXP5ajQ",
            "SourceId": "802NA000002lWjKYAU",
            "FoliId": "0a4NA000003HYbYYAW"
          },
          {
            "FlsrId": "16ANA000005idRC2AY",
            "FlsrGuid": "bvVhA34WvTBOYbG4q2Fj",
            "SourceId": "802NA000002lWjKYAU",
            "FoliId": "0a4NA000003HYbZYAW"
          }
        ],
        "OrderId": "801NA000000XKPeYAO",
        "SubmitMode": "Synchronous",
        "UniqueRequestId": "b9ac5971-0d71-45c6-a18e-f472976eaeae",
       "ContextId": "1422049ffa3cc42d8c060b9a7732be360bd62a346652cda9ab6e5fd54cd03eca",

        "DesActionSpecDevName": "DroDecompScopeAction",
        "ContextDefinition": "SalesTransactionContext__stdctx"
      },
      "additionalFilter": "undef",
      "applicationLogDate": "Fri Mar 29 10:02:26 GMT 2024",
      "applicationSubtype": "DroDcmp",
      "applicationType": "7",
      "explainabilitySpecName": "DroDecompScopeAction",
      "isChunked": false,
      "name": "DroDecompScopeAction",
      "primaryFilter": "801NA000000XKPeYAO",
      "processType": "DcmpScp",
      "secondaryFilter": "undef",
      "uniqueIdentifier": "c1e11bc0-53e5-4f62-9ae8-1096e79a62b6"
    }
  ],
  "queryMore": ""
}
```


Get logs for decomposition enrichment

This resource example includes sample query parameters to retrieve the logs for decomposition enrichment tasks. For example, conversion of order items to fulfillment order line items.

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/decision-explainer/action-logs?applicationSubType=DroDcmp&applicationType=7&processType=DcmpEnrich&primaryFilter=801xx000003YzAAG&secondaryFilter=9xQ2B4OOXkuG
```

The `secondaryFilter` property is optional. If this property is specified, the API returns the enrichment rule details for a decomposition rule. If this property is unspecified, the API returns all enrichment rule details for the order.

This example shows the sample response. The `actionLog` property contains the action logs.


```
{
  "actionLogs": [
    {
      "actionContextCode": "801NA000000XKPUYA4",
      "actionLog": {
        "FlsrUid": "1FxfIhMq53OIpE9Hzs5w",
        "EnrichmentRulesExecutionDetails": [
          {
            "EnrichmentRuleId": "13TNA00000000eG2AQ",
            "DecompRuleId": "13UNA0000004C942AE",
            "FlsrUid": "1FxfIhMq53OIpE9Hzs5w",
            "SourceType": "Attribute",
            "SourceAttributeId": "0tjNA0000004CygYAE",
            "TargetType": "Attribute",
            "TargetAttributeId": "0tjNA0000004CyDYAU",
            "CalculationMethod": "Ad-verbatim",
            "ExecutionStatus": "SUCCESS"
          },
          {
            "EnrichmentRuleId": "13TNA00000000eH2AQ",
            "DecompRuleId": "13UNA0000004C942AE",
            "FlsrUid": "1FxfIhMq53OIpE9Hzs5w",
            "SourceType": "Attribute",
            "SourceAttributeId": "0tjNA0000004CylYAE",
            "TargetType": "Attribute",
            "TargetAttributeId": "0tjNA0000004CyIYAU",
            "CalculationMethod": "Ad-verbatim",
            "ExecutionStatus": "SUCCESS"
          },
          {
            "EnrichmentRuleId": "13TNA00000000eI2AQ",
            "DecompRuleId": "13UNA0000004C942AE",
            "FlsrUid": "1FxfIhMq53OIpE9Hzs5w",
            "SourceType": "Attribute",
            "SourceAttributeId": "0tjNA0000004CyqYAE",
            "TargetType": "Attribute",
            "TargetAttributeId": "0tjNA0000004CyhYAE",
            "CalculationMethod": "Ad-verbatim",
            "ExecutionStatus": "SUCCESS"
          },
          {
            "EnrichmentRuleId": "13TNA00000000eJ2AQ",
            "DecompRuleId": "13UNA0000004C942AE",
            "FlsrUid": "1FxfIhMq53OIpE9Hzs5w",
            "SourceType": "Attribute",
            "SourceAttributeId": "0tjNA0000004CzPYAU",
            "TargetType": "Attribute",
            "TargetAttributeId": "0tjNA0000004CymYAE",
            "CalculationMethod": "Ad-verbatim",
            "ExecutionStatus": "SUCCESS"
          },
          {
            "EnrichmentRuleId": "13TNA00000000eK2AQ",
```

```
    "DecompRuleId": "13UNA0000004C942AE",
    "FlsrUid": "1FxfIhMq53OIpE9Hzs5w",
    "SourceType": "Attribute",
    "SourceAttributeId": "0tjNA0000004CzUYAU",
    "TargetType": "Attribute",
    "TargetAttributeId": "0tjNA0000004CyrYAE",
    "CalculationMethod": "Ad-verbatim",
    "ExecutionStatus": "SUCCESS"
  },
  {
    "EnrichmentRuleId": "13TNA00000000eL2AQ",
    "DecompRuleId": "13UNA0000004C942AE",
    "FlsrUid": "1FxfIhMq53OIpE9Hzs5w",
    "SourceType": "Attribute",
    "SourceAttributeId": "0tjNA0000004CzZYAU",
    "TargetType": "Attribute",
    "TargetAttributeId": "0tjNA0000004CywYAE",
    "CalculationMethod": "Ad-verbatim",
    "ExecutionStatus": "SUCCESS"
  },
  {
    "EnrichmentRuleId": "13TNA00000000eM2AQ",
    "DecompRuleId": "13UNA0000004C942AE",
    "FlsrUid": "1FxfIhMq53OIpE9Hzs5w",
    "SourceType": "Attribute",
    "SourceAttributeId": "0tjNA0000004D0dYAE",
    "TargetType": "Attribute",
    "TargetAttributeId": "0tjNA0000004Cz1YAE",
    "CalculationMethod": "Ad-verbatim",
    "ExecutionStatus": "ERROR_DATA_TYPE_MISMATCH"
  },
  {
    "EnrichmentRuleId": "13TNA00000000eN2AQ",
    "DecompRuleId": "13UNA0000004C942AE",
    "FlsrUid": "1FxfIhMq53OIpE9Hzs5w",
    "SourceType": "Attribute",
    "SourceAttributeId": "0tjNA0000004D0iYAE",
    "TargetType": "Attribute",
    "TargetAttributeId": "0tjNA0000004CzQYAU",
    "CalculationMethod": "Ad-verbatim",
    "ExecutionStatus": "ERROR_DATA_TYPE_MISMATCH"
  },
  {
    "EnrichmentRuleId": "13TNA00000000eO2AQ",
    "DecompRuleId": "13UNA0000004C942AE",
    "FlsrUid": "1FxfIhMq53OIpE9Hzs5w",
    "SourceType": "Attribute",
    "SourceAttributeId": "0tjNA0000004D18YAE",
    "TargetType": "Attribute",
    "TargetAttributeId": "0tjNA0000004CzVYAU",
    "CalculationMethod": "Ad-verbatim",
    "ExecutionStatus": "ERROR_DATA_TYPE_MISMATCH"
  },
  {
```

```
            "EnrichmentRuleId": "13TNA00000000eP2AQ",
            "DecompRuleId": "13UNA0000004C942AE",
            "FlsrUid": "1FxfIhMq53OIpE9Hzs5w",
            "SourceType": "Attribute",
            "SourceAttributeId": "0tjNA0000004D1QYAU",
            "TargetType": "Attribute",
            "TargetAttributeId": "0tjNA0000004CzaYAE",
            "CalculationMethod": "Ad-verbatim",
            "ExecutionStatus": "ERROR_DATA_TYPE_MISMATCH"
          }
        ],
        "OrderId": "801NA000000XKPUYA4",
        "SubmitMode": "Synchronous",
        "UniqueRequestId": "d00c2aa7-56b0-411a-9b51-83d9fe2e4440",
       "ContextId": "a0ca3c2296c82ad071a5efa83e8974793910f44ebddb21d37f007ce4889b65d7",

        "DesActionSpecDevName": "DroDecompEnrichmentAction",
        "ContextDefinition": "SalesTransactionContext__stdctx"
      },
      "additionalFilter": "undef",
      "applicationLogDate": "Thu Mar 28 12:36:25 GMT 2024",
      "applicationSubtype": "DroDcmp",
      "applicationType": "7",
      "explainabilitySpecName": "DroDecompEnrichmentAction",
      "isChunked": false,
      "name": "DroDecompEnrichmentAction",
      "primaryFilter": "801NA000000XKPUYA4",
      "processType": "DcmpEnrich",
      "secondaryFilter": "1FxfIhMq53OIpE9Hzs5w",
      "uniqueIdentifier": "0bd42942-12b2-4c2e-9f10-b77ee85fd20b"
    }
  ],
  "queryMore": ""
}
```

Get logs for plan composition

This resource example includes sample query parameters to retrieve the logs for plan composition.

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/decision-explainer/action-logs?applicationSubType=DroPcmp&applicationType=7&processType=PcmpSteps&primaryFilter=801xx000003GYzAAG
```

This example shows the sample response. The `actionLog` property contains the action logs.

```
{
  "actionLogs": [
    {
      "actionContextCode": "801NA000000XKPeYAO",
      "actionLog": {
        "PlanCompositionStatus": "",
        "PlanCompositionStatusMessage": "",
        "CandidateProductFulfillmentScenarios": [
          {
            "ProductFulfillmentScenarioId": "1axNA0000004C93YAE",
            "ProductId": "01tNA0000007NP5YAM",
            "FulfillmentStepDefinitionGroupId": "13oNA0000004CARYA2",
```

```
        "LineItemIds": [
          "802NA000002lWjOYAU"
        ],
        "ContextRulesetId": "9QwNA0000000CXq0AM"
      }
    ],
    "SelectedProductFulfillmentScenarios": [
      null
    ],
    "SelectedProductFulfillmentScenariosByOli": [
      {
        "OliId": "0a4NA000003HYbFYAW",
        "ProductFulfillmentScenarios": [
          null
        ]
      },
      {
        "OliId": "0a4NA000003HYbXYAW",
        "ProductFulfillmentScenarios": [
          null
        ]
      },
      {
        "OliId": "0a4NA000003HYbIYAW",
        "ProductFulfillmentScenarios": [
          null
        ]
      },
      {
        "OliId": "802NA000002lWjOYAU",
        "ProductFulfillmentScenarios": [
          null
        ]
      }
    ],
    "PlanId": "13VNA000000OQ9W2AW",
    "FulfillmentStepCreationStatus": "FulfillmentStepsCreated",
    "FulfillmentStepDependencyCreationStatus": "FulfilmentStepDepCreated",
    "OrderId": "801NA000000XKPeYAO",
    "SubmitMode": "Synchronous",
    "UniqueRequestId": "b9ac5971-0d71-45c6-a18e-f472976eaeae",
   "ContextId": "1422049ffa3cc42d8c060b9a7732be360bd62a346652cda9ab6e5fd54cd03eca",

    "DesActionSpecDevName": "DroPlanCompStepsAction",
    "ContextDefinition": "SalesTransactionContext__stdctx"
  },
  "additionalFilter": "undef",
  "applicationLogDate": "Fri Mar 29 10:02:27 GMT 2024",
  "applicationSubtype": "DroPcmp",
  "applicationType": "7",
  "explainabilitySpecName": "DroPlanCompStepsAction",
  "isChunked": false,
  "name": "DroPlanCompStepsAction",
  "primaryFilter": "801NA000000XKPeYAO",
```

```
      "processType": "PcmpSteps",
      "secondaryFilter": "undef",
      "uniqueIdentifier": "bb532c9b-a88f-4add-8cd8-bd385359b1ad"
    }
  ],
  "queryMore": ""
}
```

Capture logs for an order

This resource example includes sample query parameters to retrieve the logs for an order.

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/decision-explainer/action-logs?actionContextCode=801DU00000000cBJYA0&applicationType=7
```

This example shows the sample response. The `actionLog` property contains the action logs.

```
{
  "actionLogs": [
    {
      "actionContextCode": "801xx000003GZSxAAO",
      "actionLog":
[illegible]

      "additionalFilter": "undef",
      "applicationLogDate": "Tue May 07 13:05:28 GMT 2024",
      "applicationSubtype": "DroDcmp",
      "applicationType": "7",
      "explainabilitySpecName": "DroDecompEnrichmentAction",
      "isChunked": false,
      "name": "DroDecompEnrichmentAction",
      "primaryFilter": "801xx000003GZSxAAO",
      "processType": "DcmpEnrich",
      "secondaryFilter": "qrh0Z3xv6buew1BW0wdn",
      "uniqueIdentifier": "768f7d03-6770-4939-980f-a6f42ad07cde"
    },
    {
      "actionContextCode": "801xx000003GZSxAAO",
      "actionLog":
[illegible]

      "additionalFilter": "undef",
      "applicationLogDate": "Tue May 07 13:05:28 GMT 2024",
      "applicationSubtype": "DroDcmp",
      "applicationType": "7",
      "explainabilitySpecName": "DroDecompEnrichmentAction",
      "isChunked": false,
      "name": "DroDecompEnrichmentAction",
      "primaryFilter": "801xx000003GZSxAAO",
      "processType": "DcmpEnrich",
      "secondaryFilter": "hjow5L3YcntZsnNuEd5n",
      "uniqueIdentifier": "7ce652e3-81a0-4b2d-a072-88b012c83864"
    },
    {
      "actionContextCode": "801xx000003GZSxAAO",
      "actionLog":
[illegible]
```

```
      "additionalFilter": "undef",
      "applicationLogDate": "Tue May 07 13:05:32 GMT 2024",
      "applicationSubtype": "DroDcmp",
      "applicationType": "7",
      "explainabilitySpecName": "DroDecompScopeAction",
      "isChunked": false,
      "name": "DroDecompScopeAction",
      "primaryFilter": "801xx000003GZSxAAO",
      "processType": "DcmpScp",
      "secondaryFilter": "undef",
      "uniqueIdentifier": "9f633a63-67d0-4f90-b4dc-c368defe8b44"
    },
    {
      "actionContextCode": "801xx000003GZSxAAO",
      "actionLog":
[illegible]

      "additionalFilter": "undef",
      "applicationLogDate": "Tue May 07 13:05:33 GMT 2024",
      "applicationSubtype": "DroPcmp",
      "applicationType": "7",
      "explainabilitySpecName": "DroPlanCompStepsAction",
      "isChunked": false,
      "name": "DroPlanCompStepsAction",
      "primaryFilter": "801xx000003GZSxAAO",
      "processType": "PcmpSteps",
      "secondaryFilter": "undef",
      "uniqueIdentifier": "47a956a6-ae52-4e4a-843b-08b404beb52a"
    },
    {
      "actionContextCode": "801xx000003GZSxAAO",
      "actionLog":
[illegible]

      "additionalFilter": "undef",
      "applicationLogDate": "Tue May 07 13:06:30 GMT 2024",
      "applicationSubtype": "DroSubmit",
      "applicationType": "7",
      "explainabilitySpecName": "DroSubmitAction",
      "isChunked": false,
      "name": "DroSubmitAction",
      "primaryFilter": "801xx000003GZSxAAO",
      "processType": "DroSubmit",
      "secondaryFilter": "undef",
      "uniqueIdentifier": "278caec4-680c-4df8-a1b2-608e05e44baa"
    }
  ],
  "queryMore": ""
}
```

Submit Sales Transaction Action

Initiate the fulfillment process of any sales transaction, such as a quote, an order, or an order summary.

Specify the ID of the sales transaction to be fulfilled. The fulfillment process includes these steps.

- Intake process
- Fulfillment orchestration

The intake process happens synchronously or asynchronously, which is specified by using the `intakeRequestType` input parameter. You can also specify the priority for the execution of the fulfillment process, which is specified by using the `fulfillmentPriority` parameter.

This action is available in API version 63.0 and later.

Special Access Rules

The Submit Sales Transaction action is available in Enterprise, Unlimited, and Developer Editions of Revenue Management. See the required permissions to access and call this invocable action.

Supported REST HTTP Methods

URI
`/services/data/v67.0/actions/standard/submitSalesTransaction`

Formats
JSON, XML

HTTP Methods
POST

Authentication
`Authorization: Bearer token`

Inputs

Input	Details
`allowOverrideOfPointOfNoReturn`	**Type** boolean **Description** Indicates whether to override the point of no return setting for the fulfillment step (`true`) or not (`false`). The default value is `false`. Available in API version 64.0 and later.
`fulfillmentAdapter`	**Type** string **Description** Type of fulfillment adapter. Valid values are: • `StandardOrder` • `GenericAdapter`—Available in API version 64.0 and later.

Input	Details
`fulfillmentPriority`	**Type** string **Description** Priority to fulfill the sales transaction. Valid values are: • `High` • `Bulk` • `Default`
`hydratedContextId`	**Type** string **Description** ID of the hydrated context.
`intakeRequestType`	**Type** string **Description** Type of request to process the intake. Valid values are: • `Synchronous` • `Asynchronous`
`priorityLimitAction`	**Type** string **Description** Type of action to perform when the priority limit is reached. Valid values are: • `Reject` • `Downgrade` This parameter is applicable only when you specify the `fulfillmentPriority` parameter.
`salesTransactionId`	**Type** id **Description** Required. ID of the sales transaction to submit.

Outputs

Output	Details
errorCode	**Type** string **Description** Code indicating the type of error.
fulfillmentPlanId	**Type** id **Description** ID of the composed fulfillment plan.
requestedFulfillmentPriority	**Type** string **Description** Priority to fulfill the sales transaction. Valid values are: • High • Bulk • Default
requestId	**Type** string **Description** Request ID of the invocation.
resolvedFulfillmentPriority	**Type** string **Description** Resolved priority to fulfill the sales transaction.
submitStatus	**Type** string **Description** Submit status of the invocation.
usedContextId	**Type** string **Description** ID of the used context that updates the decomposition process.

Example

POST

This sample request is for the Submit Sales Transaction action.

```
{
  "inputs": [
    {
      "allowOverrideOfPointOfNoReturn": false,
      "salesTransactionId": "801DV000000CbIPYA0",
      "intakeRequestType": "Synchronous",
      "fulfillmentAdapter": "StandardOrder",
      "fulfillmentPriority": "Default",
      "priorityLimitAction": "Reject"
    }
  ]
}
```

This sample response is for the Submit Sales Transaction action.

```
[
  {
    "actionName": "submitSalesTransaction",
    "errors": null,
    "invocationId": null,
    "isSuccess": true,
    "outcome": null,
    "outputValues": {
      "submitStatus": "SUCCESS",
      "resolvedFulfillmentPriority": "Default",
      "requestedFulfillmentPriority": "Default",
     "usedContextId": "0abc8db32b30d09c5051e4561f0b39d938a3bd8db4ccb13e04d41019e427211d",

      "requestId": "927f72b7-85e0-4b5d-b92e-c265f41898f0",
      "fulfillmentPlanId": "13VDV00000008M92AI"
    },
    "sortOrder": -1,
    "version": 1
  }
]
```

Unfreeze Sales Transaction Action

Unfreeze a sales transaction to enable the modification of a line item.

A line item can reach a milestone in the fulfillment process, which is known as a point of no return, where the line item can't accept modifications. Get details about the point of no return for each line item of a sales transaction.

This action is available in API version 64.0 and later.

Special Access Rules

The Unfreeze Sales Transaction action is available in Enterprise, Unlimited, and Developer Editions of Revenue Management. See the required permissions to access and call this invocable action.

Supported REST HTTP Methods

URI
`/services/data/v67.0/actions/standard/unfreezeSalesTransaction`

Formats
JSON, XML

HTTP Methods
POST

Authentication
`Authorization: Bearer`***token***

Inputs

Input	Details
`salesTransactionId`	**Type** id **Description** Required. ID of the sales transaction that's submitted to the Dynamic Revenue Orchestrator.

Outputs

Output	Details
`errorCode`	**Type** string **Description** Code indicating the type of error.
`orchestrationPlanId`	**Type** id **Description** ID of the created orchestration plan.
`planState`	**Type** string **Description** Status of the created orchestration plan. Valid values are: • `FAILURE` • `NOTSTARTED` • `PENDING`

Output	Details
	• `COMPLETED` • `FROZEN` • `INPROGRESS`
`requestId`	**Type** string **Description** ID of the request to get the point of no return details.
`salesTransactionId`	**Type** string **Description** ID of the sales transaction that's submitted to the Dynamic Revenue Orchestrator.

Example

POST

This sample request is for the Unfreeze Sales Transaction action.

```
{
  "inputs": [
    {
      "salesTransactionId": "801SG00000jQO1ZYAW"
    }
  ]
}
```

This sample response is for the Unfreeze Sales Transaction action.

```
[
  {
    "actionName": "unfreezeSalesTransaction",
    "errors": null,
    "invocationId": null,
    "isSuccess": true,
    "outcome": null,
    "outputValues": {
      "orchestrationPlanId": "13VSG00000229Z72AI",
      "salesTransactionId": "801SG00000jQO1ZYAW",
      "planState": "InProgress",
      "requestId": "e9f2d961-b218-4911-8fee-8de31937850d"
    },
    "sortOrder": -1,
    "version": 1
  }
]
```

Dynamic Revenue Orchestrator Metadata API Types

Metadata API enables you to access some types and feature settings that you can customize in the user interface.

Flow for Dynamic Revenue Orchestrator
Represents the metadata associated with a flow. With Flow, you can create an application that navigates users through a series of screens to query and update records in the database. You can also execute logic and provide branching capability based on user input to build dynamic applications.

DynamicFulfillmentOrchestratorSettings
Represents the settings for Dynamic Revenue Orchestrator.

OrchestrationPlanCtxMapping
Represents the context mapping that Dynamic Revenue Orchestrator (DRO) uses to generate and orchestrate a plan for an object, such as a non-sales transaction for billing or another generic business process.

SEE ALSO:

Metadata API Developer Guide: Understanding Metadata API

Flow for Dynamic Revenue Orchestrator

Represents the metadata associated with a flow. With Flow, you can create an application that navigates users through a series of screens to query and update records in the database. You can also execute logic and provide branching capability based on user input to build dynamic applications.

FlowActionCall

Dynamic Revenue Orchestrator exposes additional actionType values for the FlowActionCall Metadata type.

Field Name	Field Type	Description
`actionType`	InvocableActionType (enumeration of type string)	Required. The action type. Additional valid values only for Dynamic Revenue Orchestrator include: • `decomposeSalesTransaction`—Decompose a sales transaction, such as a quote, order, or order summary. • `freezeSalesTransaction`—Freeze a sales transaction to disable the modification of a line item. • `getPointOfNoReturn`—Get details about the point of no return milestone for each line item in a sales transaction. • `orchestrateSalesTransaction`—Initiate the orchestration process for a sales transaction. • `orchestrateTransaction`—Orchestrate a transaction for any domain-specific object, such as a collection plan for Revenue billing, that requires the composition and execution of a fulfillment plan.

Field Name	Field Type	Description
		• `submitOrder`—Submit an order to Dynamic Revenue Orchestrator (DRO) for fulfillment. • `submitSalesTransaction`—Initiate the fulfillment process of any sales transaction, such as a quote, an order, or an order summary. • `unfreezeSalesTransaction`—Unfreeze a sales transaction to enable the modification of a line item.

DynamicFulfillmentOrchestratorSettings

Represents the settings for Dynamic Revenue Orchestrator.

Parent Type and Manifest Access

This type extends the Metadata metadata type and inherits its `fullName` field.

In the package manifest, all the settings metadata types for the org are accessed using the "Settings" name. See Settings for more details.

File Suffix and Directory Location

`DynamicFulfillmentOrchestratorSettings` values are stored in the `DynamicFulfillmentOrchestratorSettings.settings` file in the `settings` folder. The `.settings` files are different from other named components, because there is only one settings file for each settings component.

Version

DynamicFulfillmentOrchestratorSettings components are available in API version 61.0 and later.

Fields

Field Name	Description
`enableDFOFallout`	**Field Type** boolean **Description** Indicates whether to enable fallout management to handle fallouts and retry policies (`true`) or not (`false`). The default value is `false`. See Turn On Feature to Manage Fallout.
`enableDFOJeopardy`	**Field Type** boolean

Field Name	Description
	Description Indicates whether to enable management of Service Level Agreements (`true`) or not (`false`). The default value is `false`. See Turn On Feature to Manage Service Level Agreements.
`enableDFOPref`	**Field Type** boolean **Description** Indicates whether to enable features of Dynamic Revenue Orchestrator (`true`) or not (`false`). The default value is `false`. See Turn On Dynamic Revenue Orchestrator.
`enableDROFutureDatedTasks`	**Field Type** boolean **Description** Indicates whether to enable the Future Dated Steps feature (`true`) or not (`false`). The default value is `false`. See Enable Future Dated Steps. Available in API version 63.0 and later.
`enableDROInflightRequest`	**Field Type** boolean **Description** Indicates whether to allow changes to fulfillment requests that are in progress (`true`) or not (`false`). The default value is `false`. Available in API version 64.0 and later.
`enableDROTaskSource`	**Field Type** boolean **Description** Indicates whether to link a Salesforce manual task to a fulfillment step source (`true`) or not (`false`). The default value is `false`. See Enable the Linking of Task to Step Source. Available in API version 63.0 and later.

Declarative Metadata Sample Definition

The following is an example of a DynamicFulfillmentOrchestratorSettings component.

```
<DynamicFulfillmentOrchestratorSettings xmlns="http://soap.sforce.com/2006/04/metadata">
     <enableDFOPref>true</enableDFOPref>
     <enableDFOFallout>true</enableDFOFallout>
     <enableDFOJeopardy>true</enableDFOJeopardy>
     <enableDROFutureDatedTasks>true</enableDROFutureDatedTasks>
     <enableDROInflightRequest>true</enableDROInflightRequest>
     <enableDROTaskSource>true</enableDROTaskSource>
</DynamicFulfillmentOrchestratorSettings>
```

The following is an example `package.xml` that references the previous definition.

```
<?xml version="1.0" encoding="UTF-8"?>
<Package xmlns="http://soap.sforce.com/2006/04/metadata">
    <types>
        <members>DynamicFulfillmentOrchestrator</members>
        <name>Settings</name>
    </types>
    <version>67.0</version>
</Package>
```

Wildcard Support in the Manifest File

The wildcard character `*` (asterisk) in the `package.xml` manifest file doesn't apply to metadata types for feature settings. The wildcard applies only when retrieving all settings, not for an individual setting. For details, see Settings. For information about using the manifest file, see Deploying and Retrieving Metadata with the Zip File.

OrchestrationPlanCtxMapping

Represents the context mapping that Dynamic Revenue Orchestrator (DRO) uses to generate and orchestrate a plan for an object, such as a non-sales transaction for billing or another generic business process.

Parent Type

This type extends the Metadata metadata type and inherits its `fullName` field.

File Suffix and Directory Location

OrchestrationPlanCtxMapping components have the suffix `.orchestrationPlanCtxMapping` and are stored in the `orchestrationPlanCtxMappings` folder.

Version

OrchestrationPlanCtxMapping components are available in API version 67.0 and later.

Fields

Field Name	Description
`contextDefinition`	**Field Type** string **Description** Required. ContextDefinition that the orchestration use case is based on.
`contextItemNode`	**Field Type** string

Field Name	Description
	Description Name of the item-level node in the context definition. The item node represents the line items or child records of the root node, such as the items in an order.
`contextMapping`	**Field Type** string **Description** Required. Name of the context mapping that maps the source object fields to the nodes in the context definition.
`contextRootNode`	**Field Type** string **Description** Required. Name of the root node in the context definition. The root node represents the parent record that the orchestration plan is generated for, such as an order.
`label`	**Field Type** string **Description** Required. UI label for the orchestration plan context mapping.
`objectName`	**Field Type** string **Description** Required. API name of the object that the orchestration plan is generated for. The combination of `objectName` and `orchestrationType` must be unique.
`orchestrationType`	**Field Type** PlanUsageType (enumeration of type string) **Description** Required. Type of orchestration plan that this context mapping applies to. Valid values are: • `Billing` • `Fulfillment` • `Generic` • `IntegrationOrchestrator`

Field Name	Description
	• `InsuranceRuleAction` • `OrderFulfillment` • `StageManagement`

Declarative Metadata Sample Definition

This sample shows the definition of an OrchestrationPlanCtxMapping component for a fulfillment plan that orchestrates an order.

```
<?xml version="1.0" encoding="UTF-8"?>
<OrchestrationPlanCtxMapping xmlns="http://soap.sforce.com/2006/04/metadata">
    <label>Order Fulfillment Context Mapping</label>
    <orchestrationType>Fulfillment</orchestrationType>
    <contextDefinition>SalesTransactionContext__stdctx</contextDefinition>
    <objectName>Order</objectName>
    <contextMapping>OrderEntitiesMapping</contextMapping>
    <contextRootNode>SalesTransaction</contextRootNode>
    <contextItemNode>SalesTransactionItem</contextItemNode>
</OrchestrationPlanCtxMapping>
```

This sample `package.xml` references the previous definition.

```
<?xml version="1.0" encoding="UTF-8"?>
<Package xmlns="http://soap.sforce.com/2006/04/metadata">
    <types>
        <members>*</members>
        <name>OrchestrationPlanCtxMapping</name>
    </types>
    <version>67.0</version>
</Package>
```

Wildcard Support in the Manifest File

This metadata type supports the wildcard character `*` (asterisk) in the `package.xml` manifest file. For information about using the manifest file, see Deploying and Retrieving Metadata with the Zip File.

Dynamic Revenue Orchestrator Tooling API Objects

Tooling API exposes metadata used in developer tooling that you can access through REST or SOAP. Tooling API's SOQL capabilities for many metadata types allow you to retrieve smaller pieces of metadata.

CustomFulfillmentScopeCnfg
Represents a user-defined scope to define and customize scope-specific validation and orchestration for flexible fulfillment. This object is available in API version 64.0 and later.

OrchestrationPlanCtxMapping
Represents an orchestration plan context mapping entry in the org. This entry is used to connect the business data in an object to the orchestration logic within Dynamic Revenue Orchestrator (DRO) by using the orchestration type. This object is available in API version 66.0.0 and later.

SEE ALSO:
Tooling API Developer Guide: Introducing Tooling API

CustomFulfillmentScopeCnfg

Represents a user-defined scope to define and customize scope-specific validation and orchestration for flexible fulfillment. This object is available in API version 64.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported SOAP API Calls

`create()`, `delete()`, `describeSObjects()`, `query()`, `retrieve()`, `update()`, `upsert()`

Supported REST API Methods

`DELETE, GET, HEAD, PATCH, POST, Query`

Fields

Field	Details
`AssetContextTag`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Context tag that's used to derive custom scope value from assets. This field is available in API version 65.0 and later.
`DeveloperName`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** Required. Name of the custom fulfillment scope.

Field	Details
DoesParticipatingAssetImpact	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Required. Indicates whether the technical assets related to sales transactions impact the fulfillment line item actions (`true`) or not (`false`). If set to `true`, Dynamic Revenue Orchestrator reuses the existing technical assets with the same custom value. The default value is `false`.
FallbackScope	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** Scope that's used when custom scope can't be determined. Valid value is: • `LineItem` If an empty value is returned, then ID of the line item is used. The default value is `LineItem`.
IsAssetized	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the fulfillment line items grouped by the custom scope are assetized `true` or not (`false`). If set to `false`, then the scope can't be associated with assetizable products. The default value is `false`.
ItemContextTag	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** Required. Context tag that's used to derive custom scope from item context nodes. The supported value is of string type only.

Field	Details
`Language`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The language of the CustomFulfillmentScopeCnfg tooling API object.
`ManageableState`	**Type** ManageableState enumerated list **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** Indicates the manageable state of the specified component that is contained in a package: • `beta` • `deleted` • `deprecated` • `deprecatedEditable` • `installed` • `installedEditable` • `released` • `unmanaged`
`MasterLabel`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** Label of the custom fulfillment scope.
NamespacePrefix	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The namespace prefix associated with this object. Each Developer Edition organization that creates a managed package has a unique namespace prefix. Limit: 15 characters. You can refer to a component in a managed package by using the ***`namespacePrefix__componentName`*** notation. The namespace prefix can have one of the following values: • In Developer Edition organizations, the namespace prefix is set to the namespace prefix of the organization for all objects that support it. There is an exception if an object is in

Field	Details
	an installed managed package. In that case, the object has the namespace prefix of the installed managed package. This field's value is the namespace prefix of the Developer Edition organization of the package developer. • In organizations that are not Developer Edition organizations, `NamespacePrefix` is only set for objects that are part of an installed managed package. There is no namespace prefix for all other objects.

OrchestrationPlanCtxMapping

Represents an orchestration plan context mapping entry in the org. This entry is used to connect the business data in an object to the orchestration logic within Dynamic Revenue Orchestrator (DRO) by using the orchestration type. This object is available in API version 66.0.0 and later.

Supported Calls

`create()`, `delete()`, `describeSObjects()`, `query()`, `retrieve()`, `update()`, `upsert()`

Special Access Rules

Fields

Field	Details
`ContextDefinition`	**Type** reference **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. Name of the context definition to be used for the object.
`ContextItemNode`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Required. Name of the node that represents an item in the context definition.
`ContextMapping`	**Type** string **Properties** Create, Filter, Group, Sort, Update

Field	Details
	Description Required. Name of the context definition mapping to apply. This mapping must be related to the specified context definition.
`ContextRootNode`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** Required. Name of the node that represents the root in the context definition.
`DeveloperName`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** Required. The unique name of the object in the API. This name can contain only underscores and alphanumeric characters, and must be unique in your org. It must begin with a letter, not include spaces, not end with an underscore, and not contain two consecutive underscores. In managed packages, this field prevents naming conflicts on package installations. With this field, a developer can change the object's name in a managed package and the changes are reflected in a subscriber's organization. Label is **Record Type Name**.
`Language`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The language of the OrchestrationPlanCtxMapping tooling API object. Valid values are:
`ManageableState`	**Type** ManageableState enumerated list **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** Indicates the manageable state of the specified component that is contained in a package: • `beta` • `deleted` • `deprecated` • `deprecatedEditable`

Field	Details
	• `installed` • `installedEditable` • `released` • `unmanaged`
`MasterLabel`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** Label for the orchestration plan context mapping.
`NamespacePrefix`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The namespace prefix associated with this object. Each Developer Edition organization that creates a managed package has a unique namespace prefix. Limit: 15 characters. You can refer to a component in a managed package by using the ***`namespacePrefix__componentName`*** notation. The namespace prefix can have one of the following values: • In Developer Edition organizations, the namespace prefix is set to the namespace prefix of the organization for all objects that support it. There is an exception if an object is in an installed managed package. In that case, the object has the namespace prefix of the installed managed package. This field's value is the namespace prefix of the Developer Edition organization of the package developer. • In organizations that are not Developer Edition organizations, `NamespacePrefix` is only set for objects that are part of an installed managed package. There is no namespace prefix for all other objects.
`ObjectName`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** API name of the object for orchestration.
`OrchestrationType`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update

Field	Details
	Description Name of the type of the orchestration plan. Valid value is `Generic`.

Callouts in Dynamic Revenue Orchestrator

Use callout step types to make HTTP calls to an external system. Callout step types use the Data Consumption Framework and Process Type Integration for interface management and communication with external systems.

Use callout step types to implement these features.

- Endpoint configuration
- Authentication parameters
- Default sync and async implementation
- Extensions or Apex Interfaces
- Error handling and fallout management
- Request and response logging
- Timeouts

Dynamic Revenue Orchestrator supports these types of callouts.

Type	Description
Standard Fulfillment Provider	Use this callout provider to use a fixed payload structure to request an order. You can also define Omnistudio Integration Procedures for request transformation and response handling.
Apex Type Provider	Use this callout provider to implement custom integration scenarios via the `industriesintegrationfwk.ProcessIntegrationProvider` Apex interface.
External Services Defined Provider	Use this callout provider to implement custom integration scenarios via external services through the OpenAPI specification. You can also define Omnistudio Integration Procedures for request transformation and response handling. To use this callout, the Omnistudio Admin and Omnistudio Runtime permissions are required.

Callouts support these integration patterns.

Type	Description
Sync	The callout step is completed when the external system returns a response.

Type	Description
Async	The callout step is set to `In Progress` when the external system returns the acknowledgment response with 202 HTTP code. The system waits until the external system sends a callback response to complete the callout step. See Asynchronous Interaction Pattern on page 1945.

Configure Callout Settings

Before you set up a callout provider, configure the callout settings. The settings include the creation of a named credential and an external credential, the creation of an integration definition, and the configuration of a fulfillment step definition.

Standard Fulfillment Provider

The Standard Fulfillment Provider or `CalloutIntegrationProvider` is a provider for the order fulfillment usage type. A Fulfillment Designer can configure this provider.

Apex Type Provider

Implement custom integration logic via Apex by using the Apex Type Provider. This provider requires an Apex Integration Developer to implement a custom Apex adapter interface.

External Services Defined Provider

Generate interface contract and Apex types by using external services and Open API compatible schema.

Asynchronous Interaction Pattern

To specify an asynchronous request, you must add the callback URI to the integration definition for Standard Fulfillment Provider or Apex Type Provider as an optional attribute.

Input and Output Transformation Processors

Use input and output processors to process a standard fulfillment request before sending it to an external system.

SEE ALSO:

Salesforce Help: Data Consumption Framework

Configure Callout Settings

Before you set up a callout provider, configure the callout settings. The settings include the creation of a named credential and an external credential, the creation of an integration definition, and the configuration of a fulfillment step definition.

Meet these prerequisites before you configure the callout settings.

- Configure named and external credentials to define access to an external system. Specify a named credential as the callout endpoint and an external credential to configure the authentication protocol.

  See Create Named Credentials and External Credentials.

- Create integration definitions to connect Salesforce with an external system. Integration definitions use APIs to perform operations in both Salesforce and the external system. You can create Apex Defined, External Services Defined, or Standard integration definitions.

  See Create an Integration Definition.

EDITIONS

Available in: **Developer**, **Enterprise**, and **Unlimited** Editions

USER PERMISSIONS

To create authenticated callouts:

- External Credentials Principal Access Permission

- Define a fulfillment step with `Callout` as the step type. Additionally, set the integration definition name and integration user. See Callout Fulfillment Step.

Standard Fulfillment Provider

The Standard Fulfillment Provider or `CalloutIntegrationProvider` is a provider for the order fulfillment usage type. A Fulfillment Designer can configure this provider.

The Standard Fulfillment Provider includes these features.

- Predefined payload with sales transaction items and fulfillment items data, including attribute values
- Fixed integration parameters such as timeouts, credentials, encoding styles, and path. See Integration Definition for Standard Fulfillment Provider on page 1937 for details about the integration parameters.
- Modified requests and responses via Omnistudio Integration Procedures
- Asynchronous interaction pattern. See Asynchronous Interaction Pattern on page 1945.
- Request and response logging
- Error handling and retry policies. See Fallout Design and Management.
- Request transformation and response handling via integration procedures

To configure the callout settings for Standard Fulfillment Provider, see Configuration Steps.

Request Payload

This example shows a sample request payload structure with data for sales transaction items and fulfillment transaction items. This data is derived using context definition mappings that are defined through settings. See Context Definitions in Order Orchestration.

```
{
  "AccountId": "001xx000003GbsxAAC",
  "OrderId": "801xx000003GbsyAAC",
  "StepId": "802xx000001ndnyAAA",
  "PlanSourceId": "802xx000001ndnyAAA",
  "StepSourceId": "13Wxx0000004Cdh",
  "CorrelationId": "callout:13Wxx0000004Cdh:",
  "SalesTransactionItems": [
    {
      "ArePartialPeriodsAllowed": "false",
      "ParentReference": "801xx000003GbsyAAC",
      "IsAssetizable": "true",
      "ProductName": "iPhone18",
      "ProductCode": "iPhone18",
      "SalesTransactionItemParent": "801xx000003GbsyAAC",
      "Attributes": [
        {
          "AttributeKey": "0tjxx00000000ePAAQ",
          "ParentReference": "802xx000001ndnxAAA",
          "AttributeValue": "true",
          "AttributeName": "IsDeliveryNotificationNeeded",
          "AttributeDefinitionCode": "FDT5",
          "SalesTransactionItemAttrParent": "802xx000001ndnxAAA"
        },
        {
          "AttributeKey": "0tjxx00000000eNAAQ",
```

```
        "ParentReference": "802xx000001ndnxAAA",
        "AttributeValue": "10",
        "AttributeName": "FDTCharge",
        "AttributeDefinitionCode": "FDT3",
        "SalesTransactionItemAttrParent": "802xx000001ndnxAAA"
      }
    ],
    "Product": "01txx0000006igoAAA",
    "Quantity": "1.0",
    "ListPrice": "1.0",
    "ItemTotalAdjustmentAmount": "0.0",
    "SalesTransactionItemSource": "802xx000001ndnxAAA",
    "PricebookEntry": "01uxx0000008zT8AAI",
    "StartDate": "2024-10-15T00:00:00.000Z",
    "UnitPrice": "1.0",
    "RoundedLineAmount": "1.0",
    "EndQuantity": "1.0",
    "TotalTaxAmount": "0.0",
    "IsItemLocked": "false",
    "TotalPrice": "1.0",
    "ProductBasedOn": "11Bxx000002C9M2EAK",
    "SalesTransactionActionType": "Add",
    "SalesTransactionAction": "8OAxx0000004DGOGA2"
  },
  {
    "ArePartialPeriodsAllowed": "false",
    "ParentReference": "801xx000003GbsyAAC",
    "IsAssetizable": "true",
    "ProductName": "iPhone19",
    "ProductCode": "iPhone19",
    "SalesTransactionItemParent": "801xx000003GbsyAAC",
    "Attributes": [
      {
        "AttributeKey": "0tjxx00000000eQAAQ",
        "ParentReference": "802xx000001ndnyAAA",
        "AttributeValue": "false",
        "AttributeName": "IsDelivered",
        "AttributeDefinitionCode": "FDT6",
        "SalesTransactionItemAttrParent": "802xx000001ndnyAAA"
      }
    ],
    "Product": "01txx0000006igpAAA",
    "Quantity": "1.0",
    "ListPrice": "1.0",
    "ItemTotalAdjustmentAmount": "0.0",
    "SalesTransactionItemSource": "802xx000001ndnyAAA",
    "PricebookEntry": "01uxx0000008zTAAAY",
    "StartDate": "2024-10-20T00:00:00.000Z",
    "UnitPrice": "1.0",
    "RoundedLineAmount": "1.0",
    "EndQuantity": "1.0",
    "TotalTaxAmount": "0.0",
    "IsItemLocked": "false",
    "TotalPrice": "1.0",
```

```
      "ProductBasedOn": "11Bxx000002C9M2EAK",
      "SalesTransactionActionType": "Add",
      "SalesTransactionAction": "8OAxx0000004DGOGA2",
      "SalesTrxnItemDescription": "TrackingRef:TRK-HUBSWQT18-4238"
    }
  ],
  "FulfillmentTransactionItems": [
    {
      "ParentReference": "0a3xx00000000JOAAY",
      "OriginalQuantity": "2.0",
      "FulfillmentItemSource": "0a4xx00000000JOAAY",
      "Attributes": [
        {
          "FulfillmentOrderLineId": "0a4xx00000000JOAAY",
          "LineAttributeDefinitionCode": "FDT6",
          "ParentReference": "0a4xx00000000JOAAY",
          "LineAttributeName": "IsDelivered",
          "LineAttributeValue": "false",
          "LineAttributeKey": "0tjxx00000000eQAAQ"
        },
        {
          "FulfillmentOrderLineId": "0a4xx00000000JOAAY",
          "LineAttributeDefinitionCode": "FDT2",
          "ParentReference": "0a4xx00000000JOAAY",
          "LineAttributeName": "FDTColor",
          "LineAttributeKey": "0tjxx00000000eMAAQ"
        },
        {
          "FulfillmentOrderLineId": "0a4xx00000000JOAAY",
          "LineAttributeDefinitionCode": "FDT1",
          "ParentReference": "0a4xx00000000JOAAY",
          "LineAttributeName": "FDTSize",
          "LineAttributeKey": "0tjxx00000000eLAAQ"
        }
      ],
      "FulfillmentItemAction": "ADD",
      "FulfillmentItemProductCode": "iPhone-Pack",
      "FulfillmentOrderItemQuantity": "2.0",
      "FulfillmentItemProductName": "iPhone-Pack",
      "FulfillmentOrderId": "0a3xx00000000JOAAY",
      "FulfillmentItemTypeCode": "Product",
      "FulfillmentItemType": "Order Product",
      "FulfillmentOrderNumber": "FO-0002",
      "FulfillmentItemStartDate": "2024-10-20T00:00:00.000Z",
      "FulfillmentItemProductId": "01txx0000006igqAAA"
    },
    {
      "ParentReference": "0a3xx00000000JNAAY",
      "OriginalQuantity": "2.0",
      "FulfillmentItemSource": "0a4xx00000000JNAAY",
      "Attributes": [
        {
          "FulfillmentOrderLineId": "0a4xx00000000JNAAY",
          "LineAttributeDefinitionCode": "FDT2",
```

```
            "ParentReference": "0a4xx00000000JNAAY",
            "LineAttributeName": "FDTColor",
            "LineAttributeKey": "0tjxx00000000eMAAQ"
          },
          {
            "FulfillmentOrderLineId": "0a4xx00000000JNAAY",
            "LineAttributeDefinitionCode": "FDT5",
            "ParentReference": "0a4xx00000000JNAAY",
            "LineAttributeName": "IsDeliveryNotificationNeeded",
            "LineAttributeValue": "false",
            "LineAttributeKey": "0tjxx00000000ePAAQ"
          },
          {
            "FulfillmentOrderLineId": "0a4xx00000000JNAAY",
            "LineAttributeDefinitionCode": "FDT3",
            "ParentReference": "0a4xx00000000JNAAY",
            "LineAttributeName": "FDTCharge",
            "LineAttributeKey": "0tjxx00000000eNAAQ"
          }
        ],
        "FulfillmentItemAction": "ADD",
        "FulfillmentItemProductCode": "iPhone-Tech",
        "FulfillmentOrderItemQuantity": "2.0",
        "FulfillmentItemProductName": "iPhone-Tech",
        "FulfillmentOrderId": "0a3xx00000000JNAAY",
        "FulfillmentItemTypeCode": "Product",
        "FulfillmentItemType": "Order Product",
        "FulfillmentOrderNumber": "FO-0001",
        "FulfillmentItemStartDate": "2024-10-10T00:00:00.000Z",
        "FulfillmentItemProductId": "01txx0000006igrAAA"
      }
    ]
}
```


The step source ID of the fulfillment step, which is either a related order or fulfillment item, determines the payload composition with these considerations.

- If the step source ID is an order item, then all order items of the order are included in the payload under the SalesTransactionItems node.
- If the step source ID is a fulfillment item, then all fulfillment items of the fulfillment order are included in the payload under the FulfillmentTransactionItems node.
- If a product attribute doesn't have a defined code, then the attribute is excluded from the payload.

Considerations

Keep these considerations in mind for the request payload.

- The maximum request payload size limit is 12 MB.
- The default timeout period is five seconds, and the maximum timeout period is 120 seconds.
- The encoding style for fulfillment order line items, sales transaction items, and attributes can be configured through the Integration Definition for Standard Fulfillment Provider on page 1937.

Error Handling

To verify if the callout request was successful, check the `status` value in the payload. If the status is undefined, then these HTTP codes indicate a successful response.

- `200`
- `201`
- `202`
- `203`
- `204`
- `205`
- `206`
- `302`
- `304`

If the request isn't successful, then the fulfillment step state is marked as `Fatally Failed`.

Integration Definition Configurations

You can configure these additional features for the integration definition.

- Select the **Save the request and response as attachments to the record** checkbox for the integration definition to save request and response payloads as attachments to the Integration Provider Execution record. Content publish limits apply when saving request and response payloads as attachments. Use Shield Platform Encryption for secure storage of sensitive information.
- Define Input and Output Processors for the Integration Definition for the pre-processing of the standard fulfillment request before you send the request to an external system. See Omnistudio Integration Procedures.

See Create an Integration Definition.

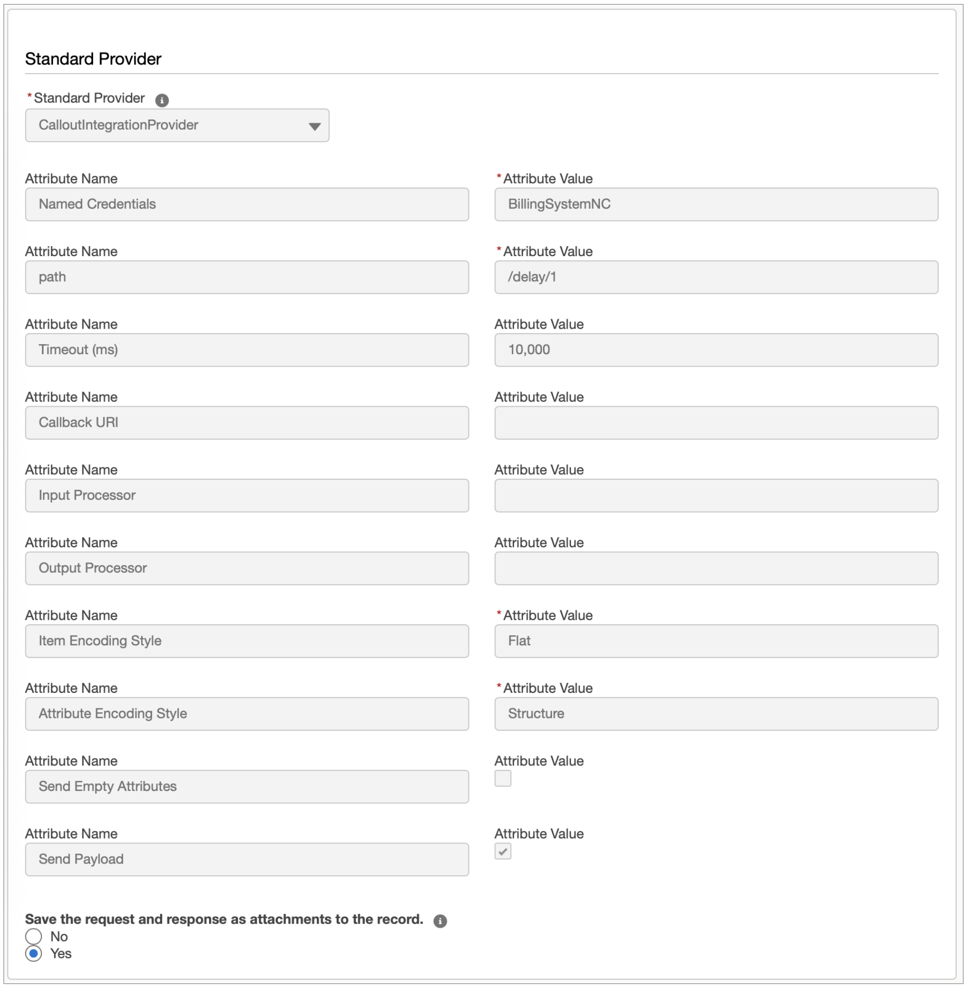
Integration Definition for Standard Fulfillment Provider

Use supported attribute values of an integration definition for a Standard Fulfillment Provider to implement features as per your requirement.

Integration Definition for Standard Fulfillment Provider

Use supported attribute values of an integration definition for a Standard Fulfillment Provider to implement features as per your requirement.

The Standard Fulfillment Provider supports these attribute values.



Attribute Name	Description
Named Credentials	API name of the associated named credentials.
Path	Additional string that's added to the URL specified in the named credential endpoint.
Timeout (ms)	Number of milliseconds to set as a timeout parameter for the HTTP connection.
Callback URI	Attribute used by a third party as the callback to process steps in case of a async callout. Additionally, this attribute also indicates these scenarios. • The configured callout is an async callout. • If callback URI is defined, then it's included as the `ResponseUri` parameter value in the default payload for Standard Fulfillment Provider. For example, a `ResponseUri` parameter value as `/services/apexrest/async/callout`.

Attribute Name	Description
	• The executor expects 202 response code from the remote destination. • The 202 response code keeps the Fulfillment Step state as Running whereas 200 response code completes it.
Input Processor	`Type_Subtype` or `Id` of OmniProcess to process the input payload.
Output Processor	`Type_Subtype` or `Id` of OmniProcess to process the output payload.
Item Encoding Style	Required. Encoding style for fulfillment transaction items and sales transaction items. Valid values are: • `Flat`—Includes a list of fulfillment transaction items and sales transaction items without any hierarchy details in the payload. • `Structure`—Includes a list of all child line items if the step's source line item has a hierarchy of line items associated with it. If item encoding style is configured as `Structure` on a fulfillment order line item or a sales transaction item, then the payload contains the details of the line item that the fulfillment order line item or sales transaction item is decomposed from. If the fulfillment order line item or sales transaction item is decomposed from the root of a bundle order line item or sales transaction item, then the payload contains the entire order line item or sales transaction item bundle structure. The default value is `Flat`.
Attribute Encoding Style	Required. Generates the payload for attributes based on the specified encoding style. Valid values are: • `Flat`—Includes specific details in key-value pair format that doesn't include granular details of attributes. • `Structure`—This payload structure includes granular details of attributes. The default value is `Structure`.
Send Empty Attributes	If this checkbox is selected, the request payload includes empty attribute values wherever applicable.
Send Payload	If this checkbox is selected, a default request payload is generated to optimize processing performance. If this checkbox isn't selected, a value for the input processor is required.

Apex Type Provider

Implement custom integration logic via Apex by using the Apex Type Provider. This provider requires an Apex Integration Developer to implement a custom Apex adapter interface.

Use a custom Apex adapter to:

- Include and use integration parameters in adapter implementations such as timeouts, credentials, and path.
- Generate and transform requests and responses via Omnistudio Integration Procedures or Extract, Transform, Load (ETL) libraries.
- Enable asynchronous interaction pattern.
- Enable request and response logging by implementing a specific Apex interface.
- Handle errors, and integrate with fallout rules.

To configure the callout settings for Apex Type Provider, see Configuration Steps.

Integration Adapter Implementation

Implement the `industriesintegrationfwk.ProcessIntegrationProvider` Apex interface. Use the `industriesintegrationfwk.ApexProviderAttr` class to define and access the attribute values in the integration provider definition.

```
global with sharing class DROSampleOrderAdapter implements
industriesintegrationfwk.ProcessIntegrationProvider {

    // Named credential attribute
    private static final String MOCK_CALLOUT_NAMED_CREDENTIAL =
'callout:DROOrderInterfaceNamedCred';
    private final static Integer TIMEOUT = 10000; // Request timeout in milliseconds, or
can be defined on Integration Definition as an Attribute


    private static final industriesintegrationfwk.ApexProviderAttr NAMED_CRED_ATTR = new
industriesintegrationfwk.ApexProviderAttr('Named Credential',
    'Named_Credential', 'DROOrderInterfaceNamedCred', true, 'String');
    private static final industriesintegrationfwk.ApexProviderAttr ENDPOINT_ATTR = new
industriesintegrationfwk.ApexProviderAttr('Endpoint URI',
    'Endpoint_URI', '/v1/orderitems/', true, 'String');


    /**
     * @param requestGuid                Request Globally Unique Identifier (GUID) provided
 by the client
     * @param inputRecordId              Input Record ID — value is taken from Fulfillment
 Step > Fulfillment Step Source > Source Line Item
     * @param payload                    Payload to be passed to the Provider Class (empty
 in DRO)
     * @param attributes                 Map of configuration attributes
     * @return IntegrationCalloutResponse Response sent to the client
    */
   global static industriesintegrationfwk.IntegrationCalloutResponse executeCallout(String
 requestGuid, String inputRecordId, String payload, Map<String, Object> attributes) {
        // create request
        String msgBody = '{\"message\":\"Hello\",'
            + '\n\"requestGuid\":\"' + requestGuid + '\",\n'
```

```
            + '\n\"inputRecordId\":\"' + inputRecordId + '\",\n'
            + '\n\"payload\":\"' + payload + '\"}'
            ;

        String endpointUri = (String) attributes.get('Endpoint_URI');

        // Make a call
        HttpResponse response = makeCallout(endpointUri, msgBody);
        // process response and pass details to the DRO Callout Step processor by using
IntegrationCalloutResponse(isSuccess, ResponseCode, ErrorMessage)

         industriesintegrationfwk.IntegrationCalloutResponse integrationCalloutResponse =
 handleResponse(response);

        return integrationCalloutResponse;
    }

    // define configurable attriobutes
    global static List<industriesintegrationfwk.ApexProviderAttr> getProviderAttributes()
 {
        List<industriesintegrationfwk.ApexProviderAttr> defaults = new
List<industriesintegrationfwk.ApexProviderAttr>();
        defaults.add(NAMED_CRED_ATTR);
        defaults.add(ENDPOINT_ATTR);
        // add any attributes such as endpoint, timeout, interface, and params
        // ...
        return defaults;
    }

    // Call Mock Service API.
    private static HttpResponse makeCallout(String endpointUri, String msgBody) {

        // Construct the request object
        String endPoint = MOCK_CALLOUT_NAMED_CREDENTIAL + endpointUri;
        HttpRequest request = new HttpRequest();
        request.setMethod('POST');
        request.setHeader('Content-Type', 'application/json');
        request.setHeader('Accept', 'application/json');
        request.setEndpoint(endPoint);
        request.setTimeout(TIMEOUT);
        request.setBody(msgBody);

        // Send request
        HttpResponse response = new Http().send(request);
        return response;
    }
}
```

Error Handling

This sample shows how to pass errors from response to orchestration via the `IntegrationResponse` interface in the Apex provider implementation. See ProcessIntegrationProvider Interface.

```
public static industriesintegrationfwk.IntegrationCalloutResponse handleResponse(HttpResponse
 response) {

        industriesintegrationfwk.IntegrationCalloutResponse integrationCalloutResponse;

        Map<String, Object> responseGroup = getResponseGroupAfterCallout(response);

        if(response.getStatusCode() == 200)
        {
            // SUCCESS
            integrationCalloutResponse = new
industriesintegrationfwk.IntegrationCalloutResponse(true);
                    integrationCalloutResponse.setResponseCode(response.getStatusCode());

              integrationCalloutResponse.setReturnValue(responseGroup);

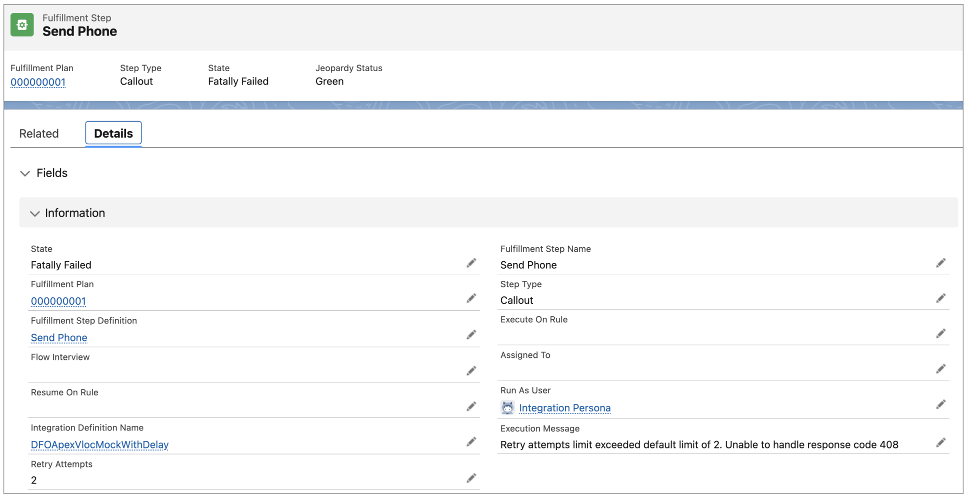
        } else {
            //FAILURE - pass error to DRO Fallout Handling to apply Retry Policies
            integrationCalloutResponse = new
industriesintegrationfwk.IntegrationCalloutResponse(false);
            integrationCalloutResponse.setResponseCode(response.getStatusCode());
            integrationCalloutResponse.setReturnValue(responseGroup);
            integrationCalloutResponse.setErrorMessage('Unable to process request.');
        }
        return integrationCalloutResponse;
        }

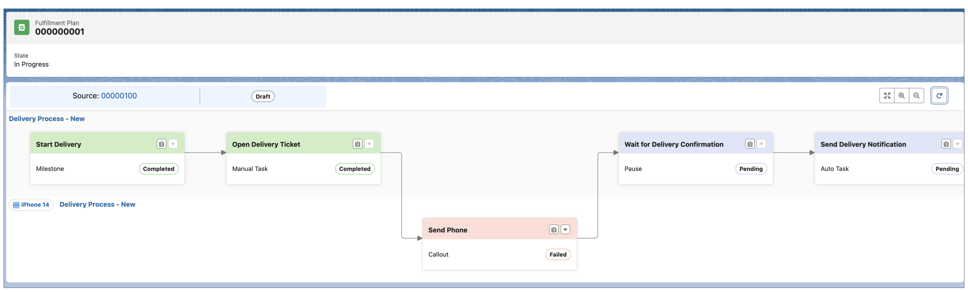
    // Process Response payLoad
   private static Map<String, Object> getResponseGroupAfterCallout(HttpResponse response)
 {
        Map<String, Object> responseGroup = new Map<String, Object>();
        if (response.getStatusCode() == 200) {
            responseGroup.put('isSuccess', true);
        } else {
            responseGroup.put('isSuccess', false);
        }
        responseGroup.put('response', getResponseMap(response.getBody()));
        return responseGroup;
    }

    // Convert response string into Map
    private static Map<String,Object> getResponseMap(String responseBody) {
        try {
            Map<String,Object> responseBodyMap = (Map<String,Object>)
JSON.deserializeUntyped(responseBody);
            return responseBodyMap;
        } catch (Exception e) {
            Map<String, Object> responseMap = new Map<String,Object>();
            responseMap.put('response', responseBody);
            return responseMap;
```

```
            }
        }
```

The fulfillment step state changes to `Fatally Failed`, and the error message is saved in the `Execution Message` field.





Apex Advanced Interface Implementation

To use advanced features such as logging, the Apex provider must implement the Apex Provider Advanced interface. You must use the HttpBaseProvider client for HTTP requests.

This example shows a sample of the `ProcessIntegrationProviderAdvanced` Apex interface implementation.

```
global with sharing class DFOApexVlocMockWithDelay
implements industriesintegrationfwk.ProcessIntegrationProviderAdvanced {
...
global static industriesintegrationfwk.IntegrationCalloutResponse executeCallout(String
requestGuid, String inputRecordId, String payload, Map<String, Object> attributes,
industriesintegrationfwk.HttpBaseProvider httpProvider) {
...
String endPoint = MOCK_CALLOUT_NAMED_CREDENTIAL + '/delay/5';
HttpRequest request = new HttpRequest();
request.setMethod('POST');
request.setHeader('Content-Type', 'application/json');
request.setHeader('Accept', 'application/json');
request.setEndpoint(endPoint);
request.setTimeout(TIMEOUT);
request.setBody(msgBody);
```

```
// Send request
//HttpResponse response = httpProvider.httpCallout(request);
HttpResponse response = new Http().send(request);
```

Integration Definition Configuration

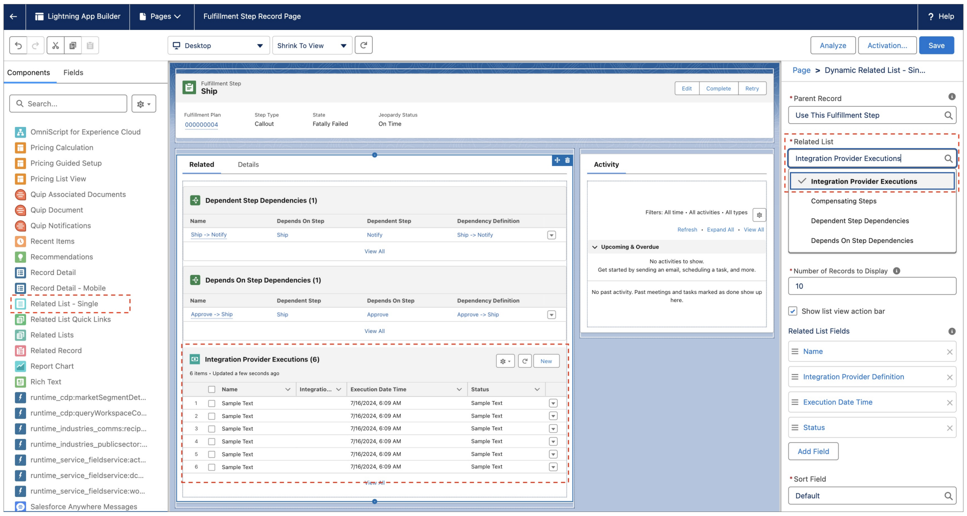
Select the **Save the request and response as attachments to the record** checkbox for the integration definition to save request and response payloads as attachments to the Integration Provider Execution record. Content publish limits apply when saving request and response payloads as attachments. Use Shield Platform Encryption for secure storage of sensitive information.

See Create an Integration Definition.

Log Records

Integration Provider execution records are created on every request and associated with the fulfillment step. The referenced object identifier is set to order item or fulfillment order line item ID. SeeIntegrationProviderExecution.

To view the Integration Provider execution records, configure the Fulfillment Step record page to include the Integration Provider Executions related list by using Lightning App Builder. See Lightning App Builder.



External Services Defined Provider

Generate interface contract and Apex types by using external services and Open API compatible schema.

To configure the callout settings for External Services Defined Provider, see Configuration Steps.

As an integration specialist user or admin user, perform these steps.

- Set up an external service and actions.
- In the external service definition, include integration parameters such as error codes, credentials, and path.

External Service

Use external services for outbound integrations from Salesforce by using low-code, process-based integrations to enhance your Apex integrations. See External Services.

Integration Definition Configurations

You can configure these additional features for the integration definition.

- Select the **Save the request and response as attachments to the record** checkbox for the integration definition to save request and response payloads as attachments to the Integration Provider Execution record. Content publish limits apply when saving request and response payloads as attachments. Use Shield Platform Encryption for secure storage of sensitive information.
- Define Input and Output Processors for the Integration Definition for the pre-processing of the standard fulfillment request before you send the request to an external system. See Omnistudio Integration Procedures.

See Create an Integration Definition.

Step Definition

Set the created integration definition on the Step Definition record with `Callout` as the step type.

Asynchronous Interaction Pattern

To specify an asynchronous request, you must add the callback URI to the integration definition for Standard Fulfillment Provider or Apex Type Provider as an optional attribute.

Standard Fulfillment Provider

- Provide the callback URI in the integration definition to accept the callback.
- Use these steps to set up the async callout.
  - Use the `ResponseUri` key to retrieve the Callback URI value from the payload.
  - Use the `StepId` key to retrieve the Fulfillment Step value from the payload. If order item ID is returned instead of Fulfillment Step ID, use `CorrelationId` key to extract the value of the step ID. For example, `callout:13Wxx0000004CAf:`.
  - Pass the StepId value to the callback URI endpoint. Return 202 response code to ensure the Fulfillment Step is in `Running` state. Otherwise, the 200 response code completes the step.

    Here's a sample payload.

    ```
    {
        "ResponseUri": "/services/apexrest/async/callout",
        "StepId": "802xx000001nb1MAAQ",
        "CorrelationId": "callout:13Wxx0000004CAf:",
        ...//Other values
    }
    ```
- Extract the step ID from the request. Find the Fulfillment Step ID in the org by using the extracted step ID and update the Fulfillment Step state to `Completed`.

Apex Type Provider

- Provide the callback URI in the integration definition to accept the callback.
- Set up the async callout by using Apex Type Provider.

- Create an Apex class that implements the `ProcessIntegrationProvider` interface.

  Here's a sample that shows the addition of a callback URI to the provider attribute for Apex Type Provider.

```
global with sharing class DFOAsyncApexSample implements
industriesintegrationfwk.ProcessIntegrationProvider {

    private static final industriesintegrationfwk.ApexProviderAttr NAMED_CRED_ATTR
  =
          new industriesintegrationfwk.ApexProviderAttr('Named Credential',
          'Named_Credential', 'DFOCalloutMockCreds', true, 'String');

    private static final industriesintegrationfwk.ApexProviderAttr CALLBACK_URL_ATTR
  =
          new industriesintegrationfwk.ApexProviderAttr('Callback URL',
          'Callback_URL', 'DFOCallbackUrl', true, 'String');

     global static List<industriesintegrationfwk.ApexProviderAttr>
getProviderAttributes() {
          List<industriesintegrationfwk.ApexProviderAttr> defaults = new
List<industriesintegrationfwk.ApexProviderAttr>();
          defaults.add(NAMED_CRED_ATTR);
          defaults.add(CALLBACK_URL_ATTR);

          return defaults;
     }
```

- Specify the Fulfillment Step ID as the value of the `requestGuid` parameter of the `executeCallout` method. You can also extract the Step ID from the `CorrelationId` key value. For example, `"callout:13Wxx0000004CAf:"`.

  Here's a sample payload that assigns the provider attribute values.

```
public interface ProcessIntegrationProvider {
    IntegrationCalloutResponse executeCallout(String requestGuid,
    String inputRecordId, String payload, Map<String, Object> attributes);

    List<ProviderAttr> getProviderAttributes();
}
```

- Set response code to 202 to indicate that the callout is asynchronously executed and in `Processing` state.

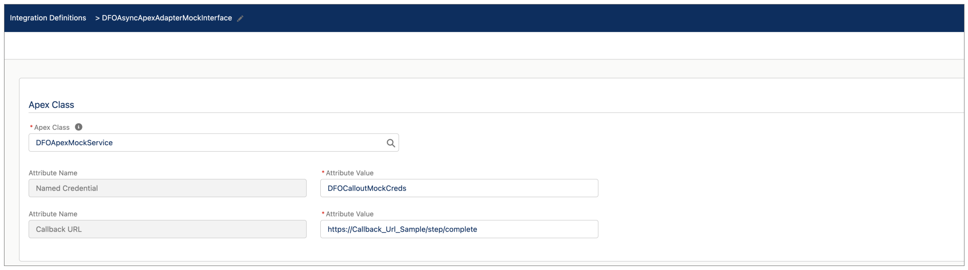
  Here's a sample payload that sets the response code.

```
global with sharing class DFOAsyncApexSample implements
industriesintegrationfwk.ProcessIntegrationProvider {
...
IntegrationCalloutResponse executeCallout(String requestGuid,
    String inputRecordId, String payload, Map<String, Object> attributes){
    // execute http request
    ...
    // set accepted if no failures
    IntegrationCalloutResponse acceptedResponse =
        new IntegrationCalloutResponse(true);
    acceptedResponse.setResponseCode(202);

    return new IntegrationCalloutResponse();
```

```
        }
}
```

- The callback URI is visible in the integration definition.



- Extract the step ID from the request. Find the Fulfillment Step ID in the org by using the extracted step ID and update the Fulfillment Step state to `Completed`.

```
FulfillmentStep step = new FulfillmentStep(id = 'stepId', State = 'Completed');
upsert step;
```

Input and Output Transformation Processors

Use input and output processors to process a standard fulfillment request before sending it to an external system.

Prerequisites

- Omnistudio license is required.
- Omnistudio Admin permission set license is assigned to Integration Configuration User (Fulfillment Designer).
- The input and output procedure attributes of an integration definition, which are available from Setup, are assigned with the Omnistudio Integration Procedure request and response. You can use `Type_Subtype` or `Id` of OmniProcess as the values for attributes.

When a callout step is executed, these steps are followed.

- The defined integration procedures are identified for request and response handling from an integration definition.
- The input processor generates the request by using `Fulfillment Step Source > SourceIdentifier` as the `InputRecordId` input parameter value. For example, the ID of an order item.
- The output processor handles the response by passing a map to the Integration Procedure service. The results from the Integration Procedure are used to identify any errors and details are passed to Dynamic Revenue Orchestrator.

Dynamic Revenue Orchestrator Platform Events

Salesforce publishes standard platform events in response to an action that occurred in the org or to report errors.

FulfillmentSourceChangeEvent

Notifies updates to the sources orchestrated by Dynamic Revenue Orchestrator like order product or fulfillment order product. This object is available in API version 66.0 and later.

SalesTrxnDecompositionEvent
Notifies when the decomposition process status changes. This object is available in API version 66.0 and later.

FulfillmentSourceChangeEvent

Notifies updates to the sources orchestrated by Dynamic Revenue Orchestrator like order product or fulfillment order product. This object is available in API version 66.0 and later.

Supported Calls

`describeSObjects()`

Supported Subscribers

Subscriber	Supported?
Apex Triggers	✓
Flows	✓
Processes	✗
Pub/Sub API	✓
Streaming API (CometD)	✓

Streaming API Subscription Channel

`/event/SalesTrxnDecompositionEvent`

Special Access Rules

This object is available in orgs with Revenue Management.

Event Delivery Allocation Enforced

Yes

Fields

Field	Details
`EventUuid`	**Type** string **Properties** Nillable

Field	Details
	Description A universally unique identifier (UUID) that identifies a platform event message.
`RecordIdentifier`	**Type** string **Properties** Create, Nillable **Description** The identifier of the source record that has been updated.
`ReplayId`	**Type** string **Properties** Nillable **Description** Represents an ID value that is populated by the system and refers to the position of the event in the event stream. Replay ID values aren't guaranteed to be contiguous for consecutive events. A subscriber can store a replay ID value and use it on resubscription to retrieve missed events that are within the retention window.

SalesTrxnDecompositionEvent

Notifies when the decomposition process status changes. This object is available in API version 66.0 and later.

Supported Calls

`describeSObjects()`

Supported Subscribers

Subscriber	Supported?
Apex Triggers	✓
Flows	✓
Processes	✗
Pub/Sub API	✓
Streaming API (CometD)	✓

Streaming API Subscription Channel

`/event/SalesTrxnDecompositionEvent`

Special Access Rules

This object is available in orgs with Revenue Management.

Event Delivery Allocation Enforced

Yes

Fields

Field	Details
`ErrorCode`	**Type** string **Properties** Create, Nillable **Description** The error code returned for a decomposition process failure.
`EventUuid`	**Type** string **Properties** Nillable **Description** A universally unique identifier (UUID) that identifies a platform event message.
`ReplayId`	**Type** string **Properties** Nillable **Description** Represents an ID value that is populated by the system and refers to the position of the event in the event stream. Replay ID values aren’t guaranteed to be contiguous for consecutive events. A subscriber can store a replay ID value and use it on resubscription to retrieve missed events that are within the retention window.
`SalesTransactionIdentifier`	**Type** string **Properties** Create, Nillable

Field	Details
	Description The sales transaction being decomposed.
`Status`	**Type** string **Properties** Create, Nillable **Description** The status of the decomposition process.

CHAPTER 11 Usage Management

In this chapter ...

Provide transparent, accurate, and efficient management of usage data and estimated usage amount to enhance revenue management capabilities.

EDITIONS

Available in: Lightning Experience

Available in: **Enterprise**, **Unlimited**, and **Developer** Editions

Usage Management Standard Objects

The Usage Management data model provides objects and fields to set up and manage consumption of usage-based products.

ProductUsageGrant

Represents the details of a grant associated with a resource, product, or service, such as the purchased quantity, renewal and rollover policy, and validity of the grant. This object is available in API version 62.0 and later.

ProductUsageResource

Represents the mapping of a product and its usage resources. This object is available in API version 64.0 and later.

ProductUsageResourcePolicy

Represents the policies applicable to the usage resource when it's associated with a sellable product. These policies are derived from the parent usage resource and can be overridden when setting up usage modeling.This object is available in API version 65 and later.

TransactionUsageEntitlement

Represents the details of each usage entitlement that's granted with the purchased sellable product, such as quantity and date when the entitlements were granted. This object is available in API version 63.0 and later.

TransactionJournal

Represents consumption details of a usage resource that are recorded for creating usage summaries. This object is available for usage management in API version 63.0 and later.

UnitOfMeasure

Defines the units and systems of units used to account for quantities of a usage resource. This object is available for usage management in API version 62.0 and later.

UnitOfMeasureClass

Represents a standard unit of measure dimension. This object is available in API version 63.0 and later.

UsageBillingPeriodItem

Represents the calculated overages for the usage entitlement and the amount that's charged for these overages. This object is available in API version 63.0 and later.

UsageCmtAssetRelatedObj

Represents the relation between an asset for the commitment-based usage product and an asset, account, contract, or custom object. This object is available in API version 64.0 and later.

UsageCommitmentPolicy

Represents the set of rules that determines how commitments are applied to a usage resource. This object is available in API version 65 and later.

UsageEntitlementAccount

Represents the entitlement account details related to the asset that holds the wallet with the granted units. This object is available in API version 63.0 and later.

UsageEntitlementBucket

Represents a usage entitlement that's granted with the sellable product. This object is available in API version 63.0 and later.

UsageEntitlementEntry

Represents the usage entitlement details, such as the usage consumption, rollovers, and details of expired units for each tenure. This object is available in API version 63.0 and later.

UsageGrantRenewalPolicy

Represents a policy about the rollover of a usage grant. This object is available in API version 62.0 and later.

UsageGrantRolloverPolicy

Represents a policy about the rollover of a usage grant.This object is available in API version 62.0 and later.

UsageOveragePolicy

Represents the set of rules that determine the management of usage resource's units consumed beyond the granted limit. This object is available in API version 65 and later.

UsagePrdGrantBindingPolicy

Represents the association of a usage resource's grants with a sellable product. This object is available in API version 63.0 and later.

UsageRatableSummary

Represents the aggregation of the usage summaries that are used to calculate the rate at which the overages are charged. This object is available in API version 63.0 and later.

UsageRatableSumCmtAssetRt

Represents the rate that's calculated and applicable for the usage resource associated with the commitment assets related to an anchor. This object is available in API version 65.0 and later.

UsageResource

Represents an entitlement granted to a user or party by a provider, such as data storage, computing power, bandwidth, or any other product or service. Additionally, this object is used to represent resources consumed over time. This object is available in API version 62.0 and later.

UsageResourcePolicy

Represents the policies applicable to the usage resource whether it's associated with a sellable product or not. This object is available in API version 65 and later.

UsageResourceBillingPolicy

Represents information about how the usage is accumulated before rating a usage resource.This object is available in API version 62.0 and later.

UsageSummary

Represents the aggregation of the entries in the transaction journal for a usage entitlement for a specified period. This object is available in API version 63.0 and later.

SEE ALSO:

Object Reference for the Salesforce Platform: Overview of Salesforce Objects and Fields

SOAP API Developer Guide: Introduction to SOAP API

ProductUsageGrant

Represents the details of a grant associated with a resource, product, or service, such as the purchased quantity, renewal and rollover policy, and validity of the grant. This object is available in API version 62.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

To create, update, and delete product usage grant records, you must have the Usage Management Design Time permission set license.

Fields

Field	Details
`DrawdownOrder`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The order that's used to debit entitlement consumption from the usage entitlement bucket. Valid values are: • `ExpiringFirst` • `GrantedFirst` • `GrantedLast` This field is deprecated and will be retired in a future version.
`EffectiveEndDate`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date and time until when the grant remains effective.
`EffectiveStartDate`	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** The date and time when the grant becomes effective.
`Label`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The identifying label for the product usage grant.

Field	Details
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when this record was last referenced.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed a record related to this record.
OverageChargeable	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies whether to charge for overages. Valid values are: • No • Yes This field is deprecated and will be retired in a future version.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the owner of the product usage grant. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
ProductOfferId	**Type** reference

Field	Details
	Properties Create, Filter, Group, Sort, Update **Description** The sellable product that grants the usage resource. This field is a relationship field. This field is deprecated and will be retired in a future version. **Relationship Name** ProductOffer **Refers To** Product2
`ProductSellingModelId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The product selling model associated with the product usage grant. This field is available in API version 65.0 and later. This field is a relationship field. **Relationship Name** ProductSellingModel **Refers To** ProductSellingModel
`ProductUsageGrantNum`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The number of each resource grant map that starts with one and is consecutive.
`ProductUsageResourceId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The product usage resource associated with the product usage grant. Available in API version 64.0 and later. This field is a relationship field. **Relationship Name** ProductUsageResource

Field	Details
	Refers To ProductUsageResource
`Quantity`	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The quantity of the granted resource.
`RenewalPolicyId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The grant renewal policy associated with the product usage grant. This field is a relationship field. **Relationship Name** RenewalPolicy **Refers To** UsageGrantRenewalPolicy
`RolloverPolicyId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The grant rollover policy associated with the product usage grant. This field is a relationship field. **Relationship Name** RolloverPolicy **Refers To** UsageGrantRolloverPolicy
`Status`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The status of the product usage grant. Valid values are:

Field	Details
	• `Active` • `Draft` • `Inactive`
`Type`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** The type of model that defines how the usage resource is consumed. Valid values are: • `Commit` • `Grant` The default value is `Grant`. Available in API version 64.0 and later.
`UnitOfMeasureClassId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The unit of measure class associated with the product usage grant. This field is a relationship field. **Relationship Name** UnitOfMeasureClass **Refers To** UnitOfMeasureClass
`UnitOfMeasureId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The unit for measure associated with the product usage grant. This value when specified, overrides the default unit of measure defined in the associated unit of measure class. This field is a relationship field. **Relationship Name** UnitOfMeasure **Refers To** UnitOfMeasure

Field	Details
`UsageDefinitionProductId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The sellable product associated with the usage resource that's used to retrieve tax policy, calculate rating during overages, and other invoicing actions. This field is a relationship field. **Relationship Name** UsageDefinitionProduct **Refers To** Product2
`UsageModelType`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The ID of the unit of measure associated with the product. The type of usage model for a product or service. `Anchor` is the main subscription product or service. `Pack` is the add-on product or service that grants additional usage resources for consumption. `Commit` is the product or service with a specific committed quantity of consumption. Valid values are: • `Anchor` • `Pack` • `Token Commitment` • `Quantity Commitment` • `Monetary Commitment` This field is available in API version 65.0 and later.
`UsageResourceId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The usage resource associated with the product usage grant. This field is a relationship field. This field is deprecated and will be retired in a future version. **Relationship Name** UsageResource

Field	Details
	Refers To UsageResource
`ValidityPeriodTerm`	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The period until when the resource grant is valid.
`ValidityPeriodUnit`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The length of a validity period for the resource grant, when used with the `ValidityPeriodTerm` field. Valid values are: • `Month` • `None` • `Year`

Associated Objects

This object has these associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

ProductUsageGrantHistory
History is available for tracked fields of the object.

ProductUsageGrantOwnerSharingRule
Sharing rules are available for the object.

ProductUsageGrantShare
Sharing is available for the object.

ProductUsageResource

Represents the mapping of a product and its usage resources. This object is available in API version 64.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

Fields

Field	Details
`EffectiveEndDate`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date and time when the relationship between the product and the usage resource stops being active, and any usage tracking or billing related to this relationship ends.
`EffectiveStartDate`	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** The date and time when the relationship between the product and the usage resource becomes active or effective, and any usage tracking or billing related to this relationship begins.
IsOptional	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the product usage resource is optional when the associated product is one of the commitment usage model types (`true`) or not (`false`). The default value is `false`. This field is available in API version 65.0 and later.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when this record was last referenced.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort

Field	Details
	Description The timestamp for when the current user last viewed a record related to this record.
`OwnerId`	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the owner of the product usage grant. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
`ProductId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The sellable product that grants the usage resource. This field is a relationship field. **Relationship Name** ProductOffer **Refers To** Product2
`ProductUsageResourceNum`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The number of each resource grant map that starts with one and is consecutive.
`Status`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The status of the product usage resource record. Valid values are:

Field	Details
	• `Active` • `Draft` • `Inactive`
TokenResourceId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The usage resource of category `Token` that's associated with the selected usage resource. This field is available in API version 65.0 and later. This field is a relationship field. **Relationship Name** TokenResource **Refers To** UsageResource
`UsageResourceId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The usage resource associated with the product usage grant. This field is a relationship field. **Relationship Name** UsageResource **Refers To** UsageResource

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

ProductUsageResourceChangeEvent
Change events are available for the object.

ProductUsageResourceFeed
Feed tracking is available for the object.

ProductUsageResourceHistory
History is available for tracked fields of the object.

ProductUsageResourceOwnerSharingRule
Sharing rules are available for the object.

ProductUsageResourceShare
Sharing is available for the object.

ProductUsageResourcePolicy

Represents the policies applicable to the usage resource when it's associated with a sellable product. These policies are derived from the parent usage resource and can be overridden when setting up usage modeling.This object is available in API version 65 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Date and time when the current user last viewed or modified this record, a record related to this record, or a list view.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Date and time when the current user last viewed or modified this record.
`ProductSellingModelId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The product selling model associated with this policy. This field is a relationship field.

Field	Details
	Relationship Name ProductSellingModel **Refers To** ProductSellingModel
`ProductUsageResourceId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The product usage resource associated with this policy. This field is a relationship field. **Relationship Name** ProductUsageResource **Relationship Type** Master-detail **Refers To** ProductUsageResource (the master object)
`RatingFrequencyPolicyId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The rating frequency policy associated with the usage resource. This field is a relationship field. **Relationship Name** RatingFrequencyPolicy **Refers To** RatingFrequencyPolicy
`UsageAggregationPolicyId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The usage aggregation policy associated with the usage resource. This field is a relationship field. **Relationship Name** UsageAggregationPolicy

Field	Details
	Refers To UsageResourceBillingPolicy
`UsageCommitmentPolicyId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The usage commitment policy associated with the usage resource. This field is a relationship field. **Relationship Name** UsageCommitmentPolicy **Refers To** UsageCommitmentPolicy
`UsageOveragePolicyId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The usage overage policy associated with the usage resource. This field is a relationship field. **Relationship Name** UsageOveragePolicy **Refers To** UsageOveragePolicy

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

ProductUsageResourcePolicyFeed
Feed tracking is available for the object.

ProductUsageResourcePolicyHistory
History is available for tracked fields of the object.

TransactionUsageEntitlement

Represents the details of each usage entitlement that's granted with the purchased sellable product, such as quantity and date when the entitlements were granted. This object is available in API version 63.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

To create, update, and delete transaction usage entitlement records, you must have the Usage Management Run Time permission set license.

Fields

Field	Details
`AccountId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The account that's associated with the usage entitlement. This field is a relationship field. **Relationship Name** Account **Refers To** Account
`ActionType`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The type of action that resulted in the transaction usage entitlement. Valid values are: • `Amend` • `Cancellation` • `New` • `Ramp` • `Renewal`
`AssetId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update

Field	Details
	Description The asset associated with the sellable product. This field is a relationship field. **Relationship Name** Asset **Refers To** Asset
`ChargeForOverage`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The action to be taken when the entitlements are used beyond their grant values. Valid values are: • `No`—Don't charge for overconsumption • `Yes`—Charge for overconsumption
`DrawdownOrder`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The order that's used to debit entitlement consumption from the usage entitlement bucket. Valid values are: • `ExpiringFirst` • `GrantedFirst` • `GrantedLast`
`EffectiveEndDateTime`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date and time when the active transaction usage entitlement ends.
`EffectiveStartDateTime`	**Type** dateTime **Properties** Create, Filter, Sort, Update

Field	Details
	Description The date and time when the transaction usage entitlement becomes active.
`EntitlementQuantity`	**Type** double **Properties** Create, Filter, Sort, Update **Description** The entitlement quantity for the usage resource.
`EntitlementProcessingStatus`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Indicates whether the transaction usage entitlement has been processed by entitlement service to be used in Billing. Available in API version 65.0 and later. Possible values are: • `PENDING`—Pending • `PROCESSED`—Processed The default value is `PENDING`.
`EntitlementUomClassId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The unit of measure class for the usage entitlement. This field is a relationship field. **Relationship Name** EntitlementUomClass **Refers To** UnitOfMeasureClass
`EntitlementUomId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The unit of measure to calculate the usage entitlement.

Field	Details
	This field is a relationship field. **Relationship Name** EntitlementUom **Refers To** UnitOfMeasure
ExternalOrderItem	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The custom or external order item that's associated with the entitlement.
GrantBindingTargetId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The target associated with the entitlements that are granted with the sellable product. This field is a relationship field. **Relationship Name** GrantBindingTarget **Refers To** Account, Asset, BindingObjectCustomExt, Contract
GrantType	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** Specifies the type of model that defines how the usage resource is consumed. Available in API version 65.0 and later. Possible values are: • `Commit` • `Grant` The default value is `Grant`.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort

Field	Details
	Description The date when this record was last referenced.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed a record related to this record.
Name	**Type** string **Properties** Auto number, Defaulted on create, Filter, idLookup, Sort **Description** Autogenerated identifier for the transaction usage entitlement record.
NetQuantity	**Type** double **Properties** Create, Filter, Sort, Update **Description** The total quantity that combines the amended quantity with the initial transaction quantity in the order item.
OriginalEndDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The end date of the transaction usage entitlement when the related order was assetized, amended, renewed, or canceled.
OrderItemId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The order item that's associated with the entitlement. This field is a polymorphic relationship field.

Field	Details
	Relationship Name OrderItem **Refers To** OrderItem, WorkOrderLineItem
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the owner of the transaction usage entitlement. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
PricebookEntryId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The price book entry that's associated with the sellable product. This field is a relationship field. **Relationship Name** PricebookEntry **Refers To** PricebookEntry
ProductId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The sellable product for which the entitlement is granted. This field is a relationship field. **Relationship Name** Product **Refers To** Product2

Field	Details
RatingFrequencyPolicyId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The sellable product for which the entitlement is granted. Available in API version 64.0 and later. This field is deprecated and will be retired in a future version. This field is a relationship field. **Relationship Name** RatingFrequencyPolicy **Refers To** RatingFrequencyPolicy
TokenResourceId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The usage resource of category Token associated with the usage resource related to the usage product added in the quote line item. Available in API version 65.0 and later. This field is a relationship field. **Relationship Name** TokenResource **Refers To** UsageResource
TransactionQuantity	**Type** double **Properties** Create, Filter, Sort, Update **Description** The transaction quantity in the order for the usage entitlement.
UsageAggregationPolicyId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The usage aggregation policy for this entitlement. This field is deprecated and will be retired in a future version. This field is a relationship field.

Field	Details
	Relationship Name UsageAggregationPolicy **Refers To** UsageResourceBillingPolicy
UsageGrantRefreshPolicyId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The usage grant refresh policy that's associated with the transaction usage entitlement. This field is a relationship field. **Relationship Name** UsageGrantRefreshPolicy **Refers To** UsageGrantRenewalPolicy
UsageGrantRolloverPolicyId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The usage grant rollover policy that's associated with the transaction usage entitlement. This field is a relationship field. **Relationship Name** UsageGrantRolloverPolicy **Refers To** UsageGrantRolloverPolicy
UsageModelType	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The type of usage model for a product or service. `Anchor` is the main subscription product or service. `Pack` is the add-on product or service that grants additional usage resources for consumption. `Commit` is the product or service with a specific committed quantity of consumption. Valid values are: • `Anchor` • `Monetary Commitment`

Field	Details
	• `Pack` • `Quantity Commitment` • `Token Commitment` Available in API version 64.0 and later.
`UsageResourceId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The usage resource record that's associated with the transaction usage entitlement. This field is a relationship field. **Relationship Name** UsageResource **Refers To** UsageResource
`ValidityPeriodTerm`	**Type** int **Properties** Create, Filter, Group, Sort, Update **Description** The duration for which the usage resource grant is valid, when used with the validity period units.
`ValidityPeriodUnit`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** The length of a validity period for the usage resource grant, when used with the validity period term. Valid values are: • `Month` • `None` • `Year`

Associated Objects

This object has these associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

TransactionUsageEntitlementFeed
Feed tracking is available for the object.

TransactionUsageEntitlementHistory
History is available for tracked fields of the object.

TransactionUsageEntitlementOwnerSharingRule
Sharing rules are available for the object.

TransactionUsageEntitlementShare
Sharing is available for the object.

TransactionJournal

Represents consumption details of a usage resource that are recorded for creating usage summaries. This object is available for usage management in API version 63.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You must have the Usage Management Run Time permission set license.

Fields

Field	Details
`AccountId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The account associated with the consumption. This field is a relationship field. **Relationship Name** Account **Refers To** Account
`ActivityDate`	**Type** dateTime

Field	Details
	Properties Create, Filter, Sort, Update **Description** Required. The date of the transaction.
CurrencyIsoCode	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** The three-letter ISO 4217 currency code of the monetary currency in which the transaction took place. Possible values are: • AED—UAE Dirham • EUR—Euro • GBP—British Pound • INR—Indian Rupee • JPY—Japanese Yen • USD—U.S. Dollar The default value is USD.
EndDate	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The end date associated with the transaction.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The most recent date on which a user referenced this record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort

Field	Details
	Description The most recent date on which a user viewed this record.
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The autogenerated number of the transaction journal.
`Quantity`	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The number of units consumed.
`QuantityUnitOfMeasureId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The unit of measure for the associated usage resource. This field is a relationship field. **Relationship Name** QuantityUnitOfMeasure **Refers To** UnitOfMeasure
`ReferenceRecordId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The asset associated with the transaction journal. This field is a polymorphic relationship field. **Relationship Name** ReferenceRecord **Refers To** Asset

Field	Details
StartDate	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The start date associated with the transaction.
Status	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. The status of the transaction. Possible values are: • Cancelled • Error • Pending • Processed
UniqueIdentifier	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The unique identifier of the transaction journal.
UsageResourceId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The usage resource that's associated with the transaction journal. This field is a relationship field. **Relationship Name** UsageResource **Refers To** UsageResource
UsageSummaryId	**Type** reference

Field	Details
	Properties Create, Filter, Group, Nillable, Sort, Update **Description** The usage summary that's associated with the transaction journal. Available in API version 64.0 and later. This field is a relationship field. **Relationship Name** UsageSummary **Refers To** UsageSummary
`UsageType`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort **Description** The type of usage. Valid value is `UsageManagement`.

UnitOfMeasure

Defines the units and systems of units used to account for quantities of a usage resource. This object is available for usage management in API version 62.0 and later.

Examples of units of measure include Liter (for volume), Kilogram (for weight), and single units (such as Can, sachet, and packet).

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`ConversionFactor`	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The factor or rate that's used to convert this unit of measurement to the base unit.

Field	Details
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when this record was last referenced.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed a record related to this record.
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** Autogenerated identifier for the unit of measure record.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the owner of the unit of measure record. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
Sequence	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The sequence number assigned to the unit of measure.

Field	Details
Status	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The status of the unit of measure. Valid values are: • Active • Draft • Inactive
Type	**Type** picklist **Properties** Create, Filter, Group, Sort, Update **Description** The type of the unit of measure. For example, weight, distance, period.
UnitCode	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** The code for the unit of measure.
UnitOfMeasureClassId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The unit of measure class associated with the unit of measure. This field is a relationship field. **Relationship Name** UnitOfMeasureClass **Refers To** UnitOfMeasureClass

Associated Objects

This object has these associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

UnitOfMeasureChangeEvent
Change events are available for the object.

UnitOfMeasureShare
Sharing is available for the object.

UnitOfMeasureClass

Represents a standard unit of measure dimension. This object is available in API version 63.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`BaseUnitOfMeasureId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The base unit of measure that's used to calculate and display quantities. This field is a relationship field. **Relationship Name** BaseUnitOfMeasure **Refers To** UnitOfMeasure
`Code`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** The unique user-defined string for the unit of measure class.
`DefaultUnitOfMeasureId`	**Type** reference

Field	Details
	Properties Create, Filter, Group, Nillable, Sort, Update **Description** The default unit of measure that's used to calculate and display quantities. This measure can be different from the base unit of measure and overridden by individual resources. This field is a relationship field. **Relationship Name** DefaultUnitOfMeasure **Refers To** UnitOfMeasure
`Description`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The description of the unit of measure class.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when this record was last referenced.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed a record related to this record.
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** Autogenerated identifier for the unit of measure class record.
`Status`	**Type** picklist

Field	Details
	Properties Create, Filter, Group, Restricted picklist, Sort, Update **Description** The status of the unit of measure class. Valid values are: • `Active` • `Draft` • `Inactive`
`Type`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** The type of the unit of measure class. Valid values are: • `Currency` • `Token` • `Usage` The default value is `Usage`. Available in API version 64.0 and later. This is a required field.

Associated Objects

This object has these associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

UnitOfMeasureClassHistory
History is available for tracked fields of the object.

UsageBillingPeriodItem

Represents the calculated overages for the usage entitlement and the amount that's charged for these overages. This object is available in API version 63.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

To create, update, and delete transaction usage entitlement records, you must have the Usage Management Run Time permission set license.

Fields

Field	Details
AccountId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The account that's associated with the billing period item. This field is a relationship field. **Relationship Name** Account **Refers To** Account
AssetId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The asset that's associated with the usage billing period item. This field is a relationship field. **Relationship Name** Asset **Refers To** Asset
EndDateTime	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** The end date and time of the billing period.
ErrorCode	**Type** picklist

Field	Details
	Properties Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The error code when the billing request fails. Valid value is: • `INTERNAL_ERROR`
`ErrorDescription`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The error description that corresponds to the error code.
`GrantBindingTargetId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The record that's associated with the usage entitlement account. This field is a relationship field. **Relationship Name** GrantBindingTarget **Refers To** Account, Asset, BindingObjectCustomExt, Contract
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when this record was last referenced.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed a record related to this record.

Field	Details
OverageAmount	**Type** currency **Properties** Create, Filter, Sort, Update **Description** The amount calculated for the overage.
OverageAmountDerived	**Type** double **Properties** Filter, Nillable, Sort **Description** The numeric value specified in the `OverageAmount` field to process Data processing Engine (DPE) jobs. This field is a calculated field.
OverageQuantity	**Type** double **Properties** Create, Filter, Sort, Update **Description** The quantity of the usage entitlement that was overused for the specified billing period.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the owner of the usage billing period item. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
StartDateTime	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** The start date and time of the billing period.

Field	Details
Status	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The status of the usage billing period item. Valid values are: • Invoiced • InvoicingFailed • LiableSummaryComplete • New • ReadyForInvoicing • Inactive—Available in API version 65.0 and later. • InProgress—Available in API version 66.0 and later.
TotalUsedQuantity	**Type** double **Properties** Create, Filter, Sort, Update **Description** The total quantity of usage entitlement that was used for the billing period. This includes granted and overused entitlements.
UoMId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The unit of measure for the overage quantity of usage entitlement. This field is a relationship field. **Relationship Name** UoM **Refers To** UnitOfMeasure
UsageBillingPeriodItemNum	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort

Field	Details
	Description Autogenerated identifier for the usage billing period item record.
UsageEntitlementAccountId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The usage entitlement account associated with the source usage entitlement bucket. This field is a relationship field. **Relationship Name** UsageEntitlementAccount **Refers To** UsageEntitlementAccount
UsageEntitlementBucketId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The usage entitlement bucket that's associated with the billing period item. This field is a relationship field. **Relationship Name** UsageEntitlementBucket **Refers To** UsageEntitlementBucket
UsageResourceId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The usage resource associated with the usage entitlement bucket. This field is a relationship field. **Relationship Name** UsageResource **Refers To** UsageResource

Associated Objects

This object has these associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

UsageBillingPeriodItemFeed
Feed tracking is available for the object.

UsageBillingPeriodItemHistory
History is available for tracked fields of the object.

UsageBillingPeriodItemOwnerSharingRule
Sharing rules are available for the object.

UsageBillingPeriodItemShare
Sharing is available for the object.

UsageCmtAssetRelatedObj

Represents the relation between an asset for the commitment-based usage product and an asset, account, contract, or custom object. This object is available in API version 64.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
AssetId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The asset associated with the commitment-based usage product. This field is a relationship field. **Relationship Name** Asset **Relationship Type** Master-detail **Refers To** Asset (the master object)

Field	Details
EffectiveEndDateTime	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date and time until when the commitment for the related asset remains effective.
EffectiveStartDateTime	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** The date and time when the commitment for the related asset becomes effective.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when this record was last referenced.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed a record related to this record.
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Autogenerated identifier for the usage commitment asset related object record.
RelatedObjectId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The asset, account, or contract associated with the asset.

Field	Details
	This field is a polymorphic relationship field. **Relationship Name** RelatedObject **Refers To** Account, Asset, Contract Only Asset is supported in API version 64.0.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

UsageCmtAssetRelatedObjFeed
Feed tracking is available for the object.

UsageCmtAssetRelatedObjHistory
History is available for tracked fields of the object.

UsageCommitmentPolicy

Represents the set of rules that determines how commitments are applied to a usage resource. This object is available in API version 65 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

Fields

Field	Details
`CommitmentRate`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** The rate that's applicable to the usage resource's units consumed post the commitment is utilized, but the commitment period is still active. Valid values are: • `Bounded Object Rate`

Field	Details
	• `Lowest Commitment Rate` The default value is `Lowest Commitment Rate`.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Date and time when the current user last viewed or modified this record, a record related to this record, or a list view.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Date and time when the current user last viewed or modified this record.
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The name of the usage commitment policy.

UsageEntitlementAccount

Represents the entitlement account details related to the asset that holds the wallet with the granted units. This object is available in API version 63.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
AccountId	**Type** reference

Field	Details
	Properties Create, Filter, Group, Sort, Update **Description** The account that's associated with the usage entitlement. This field is a relationship field. **Relationship Name** Account **Refers To** Account
BillDayOfMonth	**Type** int **Properties** Create, Filter, Group, Sort, Update **Description** The day of the month that the bill is generated on and that ends the billing period.
BillingPeriodEndDateTime	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** The date and time when the billing period of the usage entitlements ends.
BillingPeriodStartDateTime	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** The date and time when the billing period of the usage entitlements starts.
BillingPeriodTerm	**Type** int **Properties** Create, Filter, Group, Sort, Update **Description** The frequency at which the bill is generated.
BillingPeriodUnit	**Type** picklist

Field	Details
	Properties Create, Filter, Group, Restricted picklist, Sort, Update **Description** The unit to measure the billing frequency. Valid value is: • MONTH
EffectiveEndDateTime	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date and time when the usage entitlement account ends.
EffectiveStartDateTime	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** The date and time when the usage entitlement account starts.
GrantBindingTargetId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The target that's associated with the usage entitlement account. This field is a relationship field. **Relationship Name** GrantBindingTarget **Refers To** Account, Asset, BindingObjectCustomExt, Contract
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when this record was last referenced.

Field	Details
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed a record related to this record.
Name	**Type** string **Properties** Auto number, Defaulted on create, Filter, idLookup, Sort **Description** Autogenerated identifier for the usage entitlement account record.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the owner of the usage entitlement account. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
PricebookEntryId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The price book entry that's associated with the sellable product. This field is a relationship field. **Relationship Name** PricebookEntry **Refers To** PricebookEntry
ProductId	**Type** reference

Field	Details
	Properties Filter, Group, Nillable, Sort **Description** The sellable product for which the entitlements are granted. This field is a relationship field. **Relationship Name** Product **Refers To** Product2

Associated Objects

This object has these associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

UsageEntitlementAccountFeed
Feed tracking is available for the object.

UsageEntitlementAccountHistory
History is available for tracked fields of the object.

UsageEntitlementAccountOwnerSharingRule
Sharing rules are available for the object.

UsageEntitlementAccountShare
Sharing is available for the object.

UsageEntitlementBucket

Represents a usage entitlement that's granted with the sellable product. This object is available in API version 63.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`BucketBalance`	**Type** double **Properties** Create, Filter, Sort, Update

Field	Details
	Description The balance of the usage entitlement bucket after each transaction.
BucketBalanceUomId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The unit that's used to measure the usage entitlement bucket balance. This field is a relationship field. **Relationship Name** BucketBalanceUom **Refers To** UnitOfMeasure
CompletedRollovers	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The number of rollovers that were completed for the entitlements.
ConsumedEntitlement	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The entitlements that have been consumed from this bucket.
EffectiveEndDateTime	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date and time when the usage entitlement bucket period ends.
EffectiveStartDateTime	**Type** dateTime **Properties** Create, Filter, Sort, Update

Field	Details
	Description The date and time when the usage entitlement bucket becomes effective.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when this record was last referenced.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed a record related to this record.
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Autogenerated identifier for the usage entitlement bucket record.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the owner of the usage entitlement bucket. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
ParentId	**Type** reference **Properties** Create, Filter, Group, Sort, Update

Field	Details
	Description The parent record that's associated with the usage entitlement bucket. This field is a polymorphic relationship field. **Relationship Name** Parent **Refers To** UsageEntitlementAccount, UsageEntitlementBucket
ProvisionalBucketBalance	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The provisional entitlement balance available in the usage bucket. Available in API version 66.0 and later.
TotalAsOfBalance	**Type** double **Properties** Filter, Nillable, Sort **Description** The balance of the entitlements granted with the usage resource for a Usage entitlement account.
TotalConsumedEntitlement	**Type** double **Properties** Filter, Nillable, Sort **Description** The entitlements that have been consumed from this bucket.
TotalProvisionalBalance	**Type** double **Properties** Filter, Nillable, Sort **Description** The total provisional entitlement balance in the usage entitlement account as of the specified date. Available in API version 66.0 and later.
TransactionUsageEntitlementId	**Type** reference

Field	Details
	Properties Create, Filter, Group, Nillable, Sort, Update **Description** The transaction usage entitlement associated with this usage entitlement bucket. This field is a relationship field. **Relationship Name** TransactionUsageEntitlement **Refers To** TransactionUsageEntitlement
`UsageResourceId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The usage resource that's associated with the usage entitlement bucket. This field is a relationship field. **Relationship Name** UsageResource **Refers To** UsageResource

Associated Objects

This object has these associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

UsageEntitlementBucketFeed
Feed tracking is available for the object.

UsageEntitlementBucketHistory
History is available for tracked fields of the object.

UsageEntitlementBucketOwnerSharingRule
Sharing rules are available for the object.

UsageEntitlementBucketShare
Sharing is available for the object.

UsageEntitlementEntry

Represents the usage entitlement details, such as the usage consumption, rollovers, and details of expired units for each tenure. This object is available in API version 63.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`ClosingBalance`	**Type** double **Properties** Create, Filter, Sort, Update **Description** The quantity of the entitlement after adjusting the transaction quantity for the current entitlement validity period.
`EffectiveEndDateTime`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date and time when the usage entitlement entry ends.
`EffectiveStartDateTime`	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** The date and time when the usage entitlement entry starts.
`EntitlementBucketId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The usage entitlement bucket that's associated with the entitlement entry. This field is a relationship field. **Relationship Name** EntitlementBucket

Field	Details
	Relationship Type Master-detail **Refers To** UsageEntitlementBucket (the master object)
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when this record was last referenced.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed a record related to this record.
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Autogenerated identifier for the usage entitlement entry record.
OpeningBalance	**Type** double **Properties** Create, Filter, Sort, Update **Description** The starting quantity of the entitlement for the current entitlement validity period.
ParentEntitlementBucketId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The parent usage entitlement bucket record that's associated with the entitlement entry. This field is a relationship field.

Field	Details
	Relationship Name ParentEntitlementBucket **Refers To** UsageEntitlementBucket
TransactionDate	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** The date when the transaction usage entitlement entry was updated.
TransactionQuantity	**Type** double **Properties** Create, Filter, Sort, Update **Description** The quantity of the entitlement used during the current entitlement validity period.
TransactionReason	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The reason for the usage entitlement entry. Valid values are: • Amend • Cancel • Consumption • Original • Refresh • Renew • Rollover
TransactionType	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The type of transaction.

Field	Details
	Valid values are: • `Credit` • `Debit` • `Expired`
`TransactionUsageEntitlementId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The transaction usage entitlement that's associated with this usage entitlement entry. This field is a relationship field. **Relationship Name** TransactionUsageEntitlement **Refers To** TransactionUsageEntitlement
`TransactionalBucketId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The usage entitlement bucket that's used to debit the consumption of entitlements. This field is a relationship field. **Relationship Name** TransactionalBucket **Refers To** UsageEntitlementBucket
`UomId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The unit of measure that's associated with the usage entitlement. This field is a relationship field. **Relationship Name** Uom **Refers To** UnitOfMeasure

Field	Details
UsageSummaryId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The usage summary that's associated with the usage entitlement. Available in API version 64.0 and later. This field is a relationship field. **Relationship Name** UsageSummary **Refers To** UsageSummary

Associated Objects

This object has these associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

UsageEntitlementEntryFeed
Feed tracking is available for the object.

UsageEntitlementEntryHistory
History is available for tracked fields of the object.

UsageGrantRenewalPolicy

Represents a policy about the rollover of a usage grant. This object is available in API version 62.0 and later.

A usage grant renewal policy is used if you want to never renew a usage grant or renew on a specific frequency.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
Code	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update

Field	Details
	Description A unique user-defined string for the usage grant renewal policy.
`IsRenewalAllowed`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the policy renewal is allowed (`true`) or not (`false`). If `true`, then the policy can be renewed. The default value is `false`.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when this record was last referenced.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed a record related to this record.
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The name of the renewal policy record.
`RenewalFrequency`	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The frequency of the policy renewals, when used with the `RenewalFrequencyUnit` field.

Field	Details
RenewalFrequencyUnit	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The renewal duration for a policy. Valid values are: • Month • Quarter • Year
Status	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The status of the renewal policy. Valid values are: • Active • Draft • Inactive

Associated Objects

This object has these associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

UsageGrantRenewalPolicyFeed
Feed tracking is available for the object.

UsageGrantRenewalPolicyHistory
History is available for tracked fields of the object.

UsageGrantRolloverPolicy

Represents a policy about the rollover of a usage grant.This object is available in API version 62.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
Code	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** A unique user-defined string for the usage grant rollover policy.
IsRolloverAllowed	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the policy allows the rollover of the usage grant. The default value is `false`.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when this record was last referenced.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed a record related to this record.
MaximumRolloverCount	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The maximum number of times that the usage grant can roll over.
Name	**Type** string

Field	Details
	Properties Create, Filter, Group, idLookup, Sort, Update **Description** The name of the rollover policy record.
ShouldAllowRolloverExpiry	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the rollover for the associated usage grant is allowed to expire. The default value is `false`.
Status	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Specifies the status of the rollover policy. Possible values are: • `Active` • `Draft` • `Inactive`

Associated Objects

This object has these associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

UsageGrantRolloverPolicyFeed
Feed tracking is available for the object.

UsageGrantRolloverPolicyHistory
History is available for tracked fields of the object.

UsageOveragePolicy

Represents the set of rules that determine the management of usage resource's units consumed beyond the granted limit. This object is available in API version 65 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Date and time when the current user last viewed or modified this record, a record related to this record, or a list view.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Date and time when the current user last viewed or modified this record.
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The name of the usage overage policy.
`OverageChargeable`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Specifies whether the overconsumption beyond the granted quantity is to be charged. Valid values are: • `No` • `Yes`

UsagePrdGrantBindingPolicy

Represents the association of a usage resource's grants with a sellable product. This object is available in API version 63.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`GrantBindingTargetType`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The type of binding instance to which the grant is added. Valid values are: • `Custom` • `Product` • `Contract` • `Account`
`GrantBindingType`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The type of binding that indicates where the grant is added. Valid values are: • `Self` • `Target`
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when this record was last referenced.

Field	Details
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed a record related to this record.
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** Autogenerated identifier for the usage product grant binding policy.
Product2Id	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The sellable product associated with the usage product grant binding policy. This field is a relationship field. **Relationship Name** Product2 **Refers To** Product2

Associated Objects

This object has these associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

UsagePrdGrantBindingPolicyFeed
Feed tracking is available for the object.

UsagePrdGrantBindingPolicyHistory
History is available for tracked fields of the object.

UsageRatableSummary

Represents the aggregation of the usage summaries that are used to calculate the rate at which the overages are charged. This object is available in API version 63.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`AccountId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The account that's associated with the ratable summary. This field is a relationship field. **Relationship Name** Account **Refers To** Account
`AssetId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The asset that's associated with the ratable summary. This field is a relationship field. **Relationship Name** Asset **Refers To** Asset
`EndDateTime`	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** The end date and time of the ratable period.
`ErrorCode`	**Type** picklist

Field	Details
	Properties Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The code that represents the error encountered when the ratable summary is rated. Valid value is: • `INTERNAL_ERROR`
`ErrorDescription`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The error description that corresponds to the error code.
`GrantBindingTargetId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The record that's associated with the usage ratable summary. Available in API version 64.0 and later. This field is a relationship field. **Relationship Name** GrantBindingTarget **Refers To** Account, Asset, BindingObjectCustomExt, Contract
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when this record was last referenced.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed a record related to this record.

Field	Details
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Autogenerated identifier for the usage ratable summary record.
`NetUnitRate`	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The rate that's used to calculate the amount for each unit of usage entitlement overage.
`NetUnitRateUomId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The unit of measure for the net unit rate applied to the overage. This field is a relationship field. **Relationship Name** NetUnitRateUom **Refers To** UnitOfMeasure
`OverageQuantity`	**Type** double **Properties** Create, Filter, Sort, Update **Description** The quantity of the usage entitlements that were overused.
`OverageQuantityUomId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The unit of measure that's used for overage quantity. This field is a relationship field.

Field	Details
	Relationship Name OverageQuantityUom **Refers To** UnitOfMeasure
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the owner of the usage ratable summary. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
RatingDecisionDateTime	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** The date and time when the decision about rating consumption is done.
RatingExecutionIdentifier	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the rating service execution that's used to retrieve waterfall logs.
RatingRequestId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID that invokes the rating service that's used to rate the usage ratable summary. This field is a relationship field. **Relationship Name** RatingRequest

Field	Details
	Refers To RatingRequest
`SourceUsageResourceId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The usage resource that's rated in tokens when usage ratable summary is created. This field is available in API version 65.0 and later. This field is a relationship field. **Relationship Name** SourceUsageResource **Refers To** UsageResource
`StartDateTime`	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** The start date and time of the ratable period.
`Status`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The status of the ratable summary. Valid values are: • `New` • `RatingComplete` • `RatingFailed` • `SummaryCreated` • `Inactive`—Available in API version 65.0 and later. • `InProgress`—Available in API version 66.0 and later.
`TierQuantity`	**Type** double

Field	Details
	Properties Create, Filter, Sort, Update **Description** The quantity that impacts the net unit rate of a usage entitlement.
`TierQuantityUomId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The unit of measure for the tier quantity. This field is a relationship field. **Relationship Name** TierQuantityUom **Refers To** UnitOfMeasure
`TotalAmount`	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The total amount after calculating the overage based on the net unit rate.
`UsageEntitlementAccountId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The usage entitlement account that's associated with the source of the ratable summary. This field is a relationship field. **Relationship Name** UsageEntitlementAccount **Refers To** UsageEntitlementAccount
`UsageEntitlementBucketId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update

Field	Details
	Description The usage entitlement bucket that's associated with the source of the ratable summary. This field is a relationship field. **Relationship Name** UsageEntitlementBucket **Refers To** UsageEntitlementBucket
`UsageResourceId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The usage resource associated with the usage entitlement bucket. This field is a relationship field. **Relationship Name** UsageResource **Refers To** UsageResource

Associated Objects

This object has these associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

UsageRatableSummaryHistory
History is available for tracked fields of the object.

UsageRatableSummaryOwnerSharingRule
Sharing rules are available for the object.

UsageRatableSummaryShare
Sharing is available for the object.

UsageRatableSumCmtAssetRt

Represents the rate that's calculated and applicable for the usage resource associated with the commitment assets related to an anchor. This object is available in API version 65.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

Fields

Field	Details
`CommitmentAssetId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The asset ID associated with the Usage Ratable Summary. This field is a relationship field. **Relationship Name** CommitmentAsset **Refers To** Asset
`ErrorCode`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies the unique code generated when an error occurs. Valid value is `INTERNAL_ERROR`.
`ErrorDescription`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Specifies the description of the error that occurred.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort

Field	Details
	Description The date when this record was last referenced.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed this record.
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Name of the Usage Commitment Summary record.
NetUnitRate	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The calculated per-unit rate for usage after applying commitment-specific discounts during the rating process.
UsageRatableSummaryId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The usage ratable summary associated with the usage commitment summary record. This field is a relationship field. **Relationship Name** UsageRatableSummary **Relationship Type** Master-detail **Refers To** UsageRatableSummary (the master object)

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

UsageRatableSumCmtAssetRtHistory
History is available for tracked fields of the object.

UsageResource

Represents an entitlement granted to a user or party by a provider, such as data storage, computing power, bandwidth, or any other product or service. Additionally, this object is used to represent resources consumed over time. This object is available in API version 62.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`Category`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The category of the usage resource that's used to organize and understand the product grant maps. Valid values are: • `Currency`—Available in API version 65.0 and later. • `Usage` • `Token`—Available in API version 64.0 and later.
`Code`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The unique user-defined string for the usage resource.
`DefaultUnitOfMeasureId`	**Type** reference

Field	Details
	Properties Create, Filter, Group, Nillable, Sort, Update **Description** The default unit of measure for the given resource. The default value can be overridden with an alternate default unit of measure for a given resource. This field is a relationship field. **Relationship Name** DefaultUnitOfMeasure **Refers To** UnitOfMeasure
`Description`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The description of the usage resource.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when this record was last referenced.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed a record related to this record.
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The name of the usage resource record.
`OwnerId`	**Type** reference

Field	Details
	Properties Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the owner of the usage resource record. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
`Status`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The status of the usage resource. Valid values are: • `Active` • `Draft` • `Inactive`
`TokenResourceId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The usage resource of Category `Token` that's used to charge this usage resource. For example, you can select a usage resource Credits (Token category) to rate the usage resource Data (Usage category). This field is available in API version 65.0 and later. **Relationship Name** TokenResource **Refers To** UsageResource
`UnitOfMeasureClassId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The unit of measure class that's used with the resource to define the units in which this resource is measured.

Field	Details
	This field is a relationship field. **Relationship Name** UnitOfMeasureClass **Refers To** UnitOfMeasureClass
`UsageDefinitionProductId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The product associated with the usage resource to retrieve tax policy, calculate rating during overages, and other invoicing actions. This field is a relationship field. **Relationship Name** UsageDefinitionProduct **Refers To** Product2
`UsageResourceBillingPolicyId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The usage resource billing policy that defines how the usage resource can be aggregated before it's sent for rating. This field is a relationship field. This field is deprecated and will be retired in a future version. **Relationship Name** UsageResourceBillingPolicy **Refers To** UsageResourceBillingPolicy

Associated Objects

This object has these associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

UsageResourceFeed
Feed tracking is available for the object.

UsageResourceHistory
History is available for tracked fields of the object.

UsageResourceOwnerSharingRule

Sharing rules are available for the object.

UsageResourceShare

Sharing is available for the object.

UsageResourcePolicy

Represents the policies applicable to the usage resource whether it's associated with a sellable product or not. This object is available in API version 65 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

Fields

Field	Details
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Date and time when the current user last viewed or modified this record, a record related to this record, or a list view.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Date and time when the current user last viewed or modified this record.
`RatingFrequencyPolicyId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update

Field	Details
	Description The rating frequency policy associated with the usage resource. This field is a relationship field. **Relationship Name** RatingFrequencyPolicy **Refers To** RatingFrequencyPolicy
UsageAggregationPolicyId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The usage aggregation policy associated with the usage resource. This field is a relationship field. **Relationship Name** UsageAggregationPolicy **Refers To** UsageResourceBillingPolicy
UsageCommitmentPolicyId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The usage commitment policy associated with the usage resource. This field is a relationship field. **Relationship Name** UsageCommitmentPolicy **Refers To** UsageCommitmentPolicy
UsageOveragePolicyId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The usage overage policy associated with the usage resource. This field is a relationship field.

Field	Details
	Relationship Name UsageOveragePolicy **Refers To** UsageOveragePolicy
UsageResourceId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The usage resource associated with the usage resource policy. This field is a relationship field. **Relationship Name** UsageResource **Relationship Type** Master-detail **Refers To** UsageResource (the master object)

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

UsageResourcePolicyFeed
Feed tracking is available for the object.

UsageResourcePolicyHistory
History is available for tracked fields of the object.

UsageResourceBillingPolicy

Represents information about how the usage is accumulated before rating a usage resource.This object is available in API version 62.0 and later.

A usage resource billing policy object is used to configure the properties of usage resources related to how aggregation is performed on the usage records before rating. Usage resource billing policies are defined at the usage resource level and can be reused across multiple usage resources.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
Code	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** A unique user-defined string for the usage resource billing policy.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when this record was last referenced.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed a record related to this record.
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The name of the usage resource billing policy record.
Status	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The status of the policy. Valid values are: • Active • Draft • Inactive

Field	Details
UsageAccumulationMethod	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update, Defaulted On Create **Description** The method used to accumulate the usage. Valid values are: • `Peak` • `Sum`
UsageAccumulationPeriod	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The duration for which the usage is accumulated. Valid values are: • `Daily` • `Monthly`

Associated Objects

This object has these associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

UsageResourceBillingPolicyFeed
Feed tracking is available for the object.

UsageResourceBillingPolicyHistory
History is available for tracked fields of the object.

UsageSummary

Represents the aggregation of the entries in the transaction journal for a usage entitlement for a specified period. This object is available in API version 63.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
AccountId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The account that's associated with the sellable product that was purchased. This field is a relationship field. **Relationship Name** Account **Refers To** Account
AssetId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The asset that's associated with the entitlement that was summarized. This field is a relationship field. **Relationship Name** Asset **Refers To** Asset
ConsumptionUnits	**Type** double **Properties** Create, Filter, Sort, Update **Description** The quantity that was used by the usage entitlement bucket.
DebitedUnits	**Type** double **Properties** Create, Filter, Sort, Update **Description** The units that are debited from the associated usage entitlement bucket.

Field	Details
EndDateTime	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** The end date and time of the usage summary.
GrantBindingTargetId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The record that's associated with the usage summary. Available in API version 64.0 and later. This field is a relationship field. **Relationship Name** GrantBindingTarget **Refers To** Account, Asset, BindingObjectCustomExt, Contract
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when this record was last referenced.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp for when the current user last viewed a record related to this record.
LiableSummaryId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The usage billing period item that's associated with this usage summary. This field is a relationship field.

Field	Details
	Relationship Name LiableSummary **Refers To** UsageBillingPeriodItem
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Autogenerated identifier for the usage summary record.
OverageUnits	**Type** double **Properties** Create, Filter, Sort, Update **Description** The quantity that was overused by the usage entitlement bucket.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the owner of the usage summary. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
ParentUsageSummaryId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The parent usage summary that's associated with this usage summary. This field is available in API version 65.0 and later. This field is a relationship field. **Relationship Name** ParentUsageSummary

Field	Details
	Refers To UsageSummary
`RatableSummaryId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ratable summary that's associated with this usage summary. This field is a relationship field. **Relationship Name** RatableSummary **Refers To** UsageRatableSummary
`StartDateTime`	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** The start date and time of the usage summary.
`Status`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The status of the usage summary. Valid values are: • `DrawdownComplete` • `LiableSummaryComplete` • `New` • `RatableSummaryComplete` • `Rated` • `UsageSummaryComplete` • `UsageSummaryInProgress` • `Inactive`—Available in API version 65.0 and later.
`UomId`	**Type** reference

Field	Details
	Properties Create, Filter, Group, Sort, Update **Description** The unit of measure for the usage resource. This value overrides the default unit of measure defined in the associated unit of measure class. This field is a relationship field. **Relationship Name** Uom **Refers To** UnitOfMeasure
`UsageEntitlementAccountId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The usage entitlement account that's associated with the usage summary. This field is a relationship field. **Relationship Name** UsageEntitlementAccount **Refers To** UsageEntitlementAccount
`UsageEntitlementBucketId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The usage entitlement bucket that's associated with the usage summary. This field is a relationship field. **Relationship Name** UsageEntitlementBucket **Refers To** UsageEntitlementBucket
`UsageResourceId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The usage resource that's associated with the source of the usage entitlement bucket.

Field	Details
	This field is a relationship field. **Relationship Name** UsageResource **Refers To** UsageResource

Associated Objects

This object has these associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

UsageSummaryHistory
History is available for tracked fields of the object.

UsageSummaryOwnerSharingRule
Sharing rules are available for the object.

UsageSummaryShare
Sharing is available for the object.

Usage Management Fields on Standard Objects

Usage Management adds standard and custom fields to some standard Salesforce objects. These fields are available only in orgs where Usage Management is enabled. This object is available in API version 63.0 and later.

Usage Management Fields on Product2

Standard and custom fields extend the standard Product2 object for use in Usage Management to represent information about products.

Fields

Field	Details
`UsageModelType`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The type of usage model for a product or service. Valid values are:

Field	Details
	• `Anchor`—The main subscription product or service. Available in API version 62.0 and later. • `Pack`—The add-on product or service that grants additional usage resources for consumption. Available in API version 62.0 and later. • `Monetary Commitment`—An agreement by a customer to spend a minimum amount for a product or service in a defined period. Available in API version 65.0 and later. • `Quantity Commitment`—An agreement by a customer to use a minimum quantity of a product or service in a defined period. Available in API version 65.0 and later. • `Token Commitement`—An agreement by a customer to use a minimum quantity of tokens for a product or service in a defined period. Available in API version 65.0 and later.

SEE ALSO:

Product2

Usage Management Standard Invocable Actions

Learn more about the standard invocable actions available with Usage Management.

Invoke Summary Creation Action

Invoke the service that creates various summaries, such as usage, ratable, and liable summaries where the usage amount is zero. The service also checks and updates the billing period of the usage entitlement account if the billing period is expired.

Process Consumption Overages Action

Process consumption overages for the usage summary records with `SummaryComplete` status. This action uses the entitlement service to process the overages.

Refresh Usage Entitlement Bucket Action

Refresh entitlements by evaluating the usage entitlement bucket records and creating a new usage entitlement entry.

Retrigger Entitlement Creation Process Action

Retrigger entitlement creation process for failed or unprocessed assets.

SEE ALSO:

Actions Developer Guide: Overview

REST API Developer Guide: Invocable Actions Standard

Salesforce Help: Usage Management

Invoke Summary Creation Action

Invoke the service that creates various summaries, such as usage, ratable, and liable summaries where the usage amount is zero. The service also checks and updates the billing period of the usage entitlement account if the billing period is expired.

This action is available in API version 63.0 and later.

Special Access Rules

The Invoke Summary Creation action is available in Enterprise, Developer, and Unlimited Editions where Usage Management is enabled. To use this action, you need the Usage Management Run Time User permission.

Supported REST HTTP Methods

URI

`/services/data/v`***`67.0`***`/actions/standard/invokeSummaryCreationService`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer` ***`token`***

Inputs

Input	Details
usageEntitlementAccountId	**Type** string **Description** Required. ID of the usage entitlement account record that's used to create summaries.

Outputs

None.

Example

POST

This example shows a sample request for the Invoke Summary Creation action.

```
{
  "inputs": [
    {
      "usageEntitlementAccountId": "3ttDU00000000iZYAQ"
    }
  ]
}
```

This example shows a sample response for the Invoke Summary Creation action.


```
{
  "actionName": "invokeSummaryCreationService",
  "errors": null,
  "isSuccess": true
}
```


SEE ALSO:

Salesforce Help: Permission Set Licenses, Personas, and User Permissions for Usage Management

Process Consumption Overages Action

Process consumption overages for the usage summary records with `SummaryComplete` status. This action uses the entitlement service to process the overages.

This action is available in API version 63.0 and later.

Special Access Rules

The Process Consumption Overages action is available in Enterprise, Developer, and Unlimited Editions where Usage Management is enabled. To use this action, you need the Usage Management Run Time User permission.

Supported REST HTTP Methods

URI

`/services/data/v`***`67.0`***`/actions/standard/processConsumptionOverages`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer` ***`token`***

Inputs

Input	Details
`usageRatableSummaryId`	**Type** string **Description** Required. ID of the usage ratable summary record that contains the consumption details, and is used to calculate consumption overages and create usage entitlement entry records.

Outputs

None.

Example

POST

This example shows a sample request for the Process Consumption Overages action.

```
{
  "inputs": [
    {
      "usageRatableSummaryId": "3ttDU00000000iZYAQ"
    }
  ]
}
```

This example shows a sample response for the Process Consumption Overages action.

```
{
  "actionName": "processConsumptionOverages",
  "errors": null,
  "isSuccess": true
}
```

SEE ALSO:

Salesforce Help: Permission Set Licenses, Personas, and User Permissions for Usage Management

Refresh Usage Entitlement Bucket Action

Refresh entitlements by evaluating the usage entitlement bucket records and creating a new usage entitlement entry.

This action is available in API version 63.0 and later.

Special Access Rules

The Refresh Usage Entitlement Bucket action is available in Enterprise, Developer, and Unlimited Editions where Usage Management is enabled. To use this action, you need the Usage Management Run Time User permission.

Supported REST HTTP Methods

URI

`/services/data/v`***`67.0`***`/actions/standard/refreshUsageEntitlementBucket`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer` ***`token`***

Inputs

Input	Details
transactionUsageEntitlementId	**Type** string **Description** Required. ID of the transaction usage entitlement record that's associated with the usage entitlement buckets that you need to refresh.

Outputs

None.

Example

POST

This example shows a sample request for the Refresh Usage Entitlement Bucket action.

```
{
  "inputs": [
    {
      "transactionUsageEntitlementId": "3ttDU00000000iZYAQ"
    }
  ]
}
```

This example shows a sample response for the Refresh Usage Entitlement Bucket action.

```
{
  "actionName": "refreshUsageEntitlementBucket",
  "errors": null,
  "isSuccess": true
}
```

SEE ALSO:

Salesforce Help: Permission Set Licenses, Personas, and User Permissions for Usage Management

Retrigger Entitlement Creation Process Action

Retrigger entitlement creation process for failed or unprocessed assets.

Trigger the entitlement creation process again in these scenarios.

- Process failed assets in the asset to entitlement journey.
- Assetize or create wallets for assets without corresponding records in Usage Management.

This action is available in API version 65.0 and later.

Special Access Rules

The Retrigger Entitlement Creation Process action is available in Enterprise, Developer, and Unlimited Editions where Usage Management is enabled. To use this action, you need the Usage Management Run Time User permission.

Supported REST HTTP Methods

URI

`/services/data/v67.0/actions/standard/retriggerEntlCreaProc`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearertoken`

Inputs

Input	Details
`assetId`	**Type** string **Description** Required. ID of the asset for which you want to trigger the asset to entitlement process again.

Outputs

None.

Example

POST

Here's a sample request for the Retrigger Entitlement Creation Process action.

```
{
  "inputs": [
    {
      "assetId": "02iSB000000JzZFYA0"
    }
  ]
}
```

Here's a sample response for the Retrigger Entitlement Creation Process action.


```
[
  {
    "actionName": "retriggerEntlCreaProc",
    "errors": null,
    "invocationId": null,
    "isSuccess": true,
    "outcome": null,
    "outputValues": null,
    "sortOrder": -1,
    "version": 1
  }
]
```


Usage Management Business APIs

Use the Usage Management Business APIs to get details of a usage-based product that's associated with an asset, an order item, or a quote line item.

This table lists the available Usage Management resources.

Resource	Description
`/asset-management/assets/`***`assetId`***`/usage-details` (GET)	Get details of a usage-based product associated with an asset. This covers details of grants, resources, and configured rates for the product, including negotiated rates in case of a rate override.
`/commerce/sales-orders/line-items/`***`orderItemId`***`/usage-details` (GET)	Get details of a usage-based product associated with an order item.
`/commerce/quotes/line-items/`***`quoteLineItemId`***`/usage-details` (GET)	Get details of a usage-based product associated with a quote line item.
`/revenue/usage-management/binding-objects/`***`bindingObjectId`***`/actions/usage-details` (GET)	Get details of grants, resources, rates, and any configured policies for a specified binding object.
`/revenue/usage-management/consumption/actions/trace` (POST)	Get a comprehensive breakdown of overage charges and resource drawdown, enabling you to view information that's applicable to specific invoice lines.

Resource	Description
`/revenue/usage-management/usage-products/actions/validate` (POST)	Validate cross-object relationships and business rules for usage-based products.

Resources
Learn more about the available Usage Management API resources.

Request Bodies
Learn more about the available request bodies of Usage Management APIs.

Response Bodies
Learn more about the available response bodies of Usage Management APIs.

SEE ALSO:

Connect REST API Developer Guide: Introduction

Resources

Learn more about the available Usage Management API resources.

Asset Usage Details (GET)
Get details of a usage-based product associated with an asset. This covers details of grants, resources, and configured rates for the product, including negotiated rates in case of a rate override.

Binding Object Usage Details (GET)
Get details of grants, resources, rates, and any configured policies for a specified binding object.

Consumption Traceabilities (POST)
Get a comprehensive breakdown of overage charges and resource drawdown, enabling you to view information that's applicable to specific invoice lines.

Order Item Usage Details (GET)
Get details of a usage-based product associated with an order item.

Quote Line Item Usage Details (GET)
Get details of a usage-based product associated with a quote line item.

Usage Product Activation (POST)
Activate a usage product and its related records, such as usage resources, grants, policies, and rate card entries, in a single request.

Usage Product Validation (POST)
Validate cross-object relationships and business rules for usage-based products.

Asset Usage Details (GET)

Get details of a usage-based product associated with an asset. This covers details of grants, resources, and configured rates for the product, including negotiated rates in case of a rate override.

Here are the details that this API returns.

- Grants and resources for the product, if rates aren't configured.
- Grants, resources, and any configured rates for the product. The rates are returned by the Rate Plan (GET) API.
- Resources that include grants, if applicable, and any negotiated rates for the product in case of a rate override request.

This API doesn't return binding target rates. Use the Binding Object Usage Details API to retrieve binding target rates.

Resource

```
/asset-management/assets/assetId/usage-details
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/asset-management/assets/02iRM0000000tCdYAI/usage-details
```

Available version

63.0

HTTP methods

GET

Path parameter for GET

Parameter Name	Type	Description	Required or Optional	Available Version
`assetId`	String	ID of the asset.	Required	63.0

Query parameters for GET

Parameter Name	Type	Description	Required or Optional	Available Version
`effectiveDate`	String	Date that's used to search for the applicable rate card entries.	Required	63.0
`optionalFields`	String[]	Custom fields that you can use to query these objects. • AssetRateCardEntry • AssetRateAdjustment	Optional	63.0

Response body for GET

Usage Details

Binding Object Usage Details (GET)

Get details of grants, resources, rates, and any configured policies for a specified binding object.

Use this API to display the details for a binding object during the selling journey. Additionally, display the details after assetization on the selected binding objects.

The supported binding objects are Account, Contract, BindingObjectCustomExt, or Anchor Asset that's not bound to a target.

Resource

```
/revenue/usage-management/binding-objects/bindingObjectId/actions/usage-details
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/revenue/usage-management/binding-objects/1BRxx0000004C9EGAU/actions/usage-details?effectiveDate=2025-08-07
```

Available version

65.0

HTTP methods

GET

Path parameter for GET

Parameter Name	Type	Description	Required or Optional	Available Version
`bindingObjectId`	String	ID of the binding object.	Required	65.0

Query parameter for GET

Parameter Name	Type	Description	Required or Optional	Available Version
`effectiveDate`	String	Date filter that's used to retrieve the grants, rates, and applicable policies as of the specified date in `yyyy-MM-dd` format.	Required	65.0

Response body for GET

Binding Object Usage Detail

Consumption Traceabilities (POST)

Get a comprehensive breakdown of overage charges and resource drawdown, enabling you to view information that's applicable to specific invoice lines.

This API provides an automated, clear, and traceable breakdown of all charges with details of specific rates, tiers, and discounts.

Resource

```
/revenue/usage-management/consumption/actions/trace
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/revenue/usage-management/consumption/actions/trace
```

Available version

66.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "liableSummaryIds": [
    "1HG000000000001"
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
liableSummaryIds	String[]	List of liable summary IDs to trace the consumption.	Required	66.0

Response body for POST

Consumption Traceabilities

Order Item Usage Details (GET)

Get details of a usage-based product associated with an order item.

Here are the details that this API returns.

- Grants and resources for the product, if rates aren't configured.
- Grants, resources, and any configured rates for the product. The rates are returned by the Rate Plan (GET) API.
- Resources that include grants, if applicable, and any negotiated rates for the product in case of a rate override request.

This API doesn't return binding target rates. Use the Binding Object Usage Details API to retrieve binding target rates.

Resource

```
/commerce/sales-orders/line-items/orderItemId/usage-details
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/sales-orders/line-items/802SG000003vZ15YAE/usage-details
```

```
https://yourInstance.salesforce.com/services/data
/v67.0/commerce/sales-orders/line-items/802SG000003vZ15YAE/usage-details?optionalFields=OrderItemRateCardEntry.MyCustomDate__c,OrderItemRateCardEntry.MyCustomNumber__c,OrderItemRateAdjustment.MyCustomString__c
```

Available version

63.0

HTTP methods

GET

Path parameter for GET

Parameter Name	Type	Description	Required or Optional	Available Version
orderItemId	String	ID of the order item.	Required	63.0

Query parameters for GET

Parameter Name	Type	Description	Required or Optional	Available Version
`effectiveDate`	String	Date that's used to search for the applicable rate card entries.	Optional	63.0
`optionalFields`	String[]	Custom fields that you can use to query these objects. • OrderItemRateCardEntry • OrderItemRateAdjustment	Optional	63.0

Response body for GET

Usage Details

Quote Line Item Usage Details (GET)

Get details of a usage-based product associated with a quote line item.

Here are the details that this API returns.

- Grants and resources for the product, if rates aren't configured.
- Grants, resources, and any configured rates for the product. The rates are returned by the Rate Plan (GET) API.
- Resources that include grants, if applicable, and any negotiated rates for the product in case of a rate override request.

This API doesn't return binding target rates. Use the Binding Object Usage Details API to retrieve binding target rates.

Resource

```
/commerce/quotes/line-items/quoteLineItemId/usage-details
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/quotes/line-items/0QLXXXXXXXXXXXABC/usage-details
```

Available version

62.0

HTTP methods

GET

Request parameters for GET

Parameter Name	Type	Description	Required or Optional	Available Version
`effectiveDate`	String	Date that's used to search for the applicable rate card entries.	Optional	62.0
`optionalFields`	String[]	Custom fields that you can use to query these objects. • QuoteLineRateCardEntry	Optional	62.0

Parameter Name	Type	Description	Required or Optional	Available Version
		• QuoteLineRateAdjustment		

Response body for GET

Usage Details

Usage Product Activation (POST)

Activate a usage product and its related records, such as usage resources, grants, policies, and rate card entries, in a single request.

Keep these considerations in mind when you use this API.

- In version 67.0, each request activates one product. If a request includes more than one product, the API returns the `MAX_LIMIT_EXCEEDED` error.
- Each record in the request stays in Draft status until activation runs. If any record for the product fails to activate, the API rolls back all activations for that product.
- The records activate in the order required by their dependencies, so a child record never activates before its parent.
- Each request activates a maximum of 200 records, including the product and all of its associated child records. If the request exceeds 200 records, the API returns the `MAX_LIMIT_EXCEEDED` error.

Resource

```
/revenue/usage-management/usage-products/actions/activate
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/revenue/usage-management/usage-products/actions/activate
```

Available version

67.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "shouldValidateProductSetup": false,
  "activationRequests": [
    {
      "productId": "01txx0000006i2gAAA",
      "usageResourceIds": [
        "0hUxx000000001",
        "0hUxx000000002"
      ]
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
activationRequests	Usage Activation Request Input[]	List of activation requests. Each entry identifies a product and the usage resources to activate for that product. This list accepts only one entry. If you include more, the API returns the `MAX_LIMIT_EXCEEDED` error.	Required	67.0
shouldValidateProductSetup	Boolean	Indicates whether to run product setup validation before activation (`true`) or not (`false`). The default value is `true`.	Optional	67.0

Response body for POST

Usage Product Activation

Usage Product Validation (POST)

Validate cross-object relationships and business rules for usage-based products.

This API returns validation results with errors and warnings.

Resource

```
/revenue/usage-management/usage-products/actions/validate
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/revenue/usage-management/usage-products/actions/validate
```

Available version

66.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "productIds": [
    "01txx0000006i2gAAA",
    "01txx0000006j2gAAA"
  ],
  "startDate": "2024-01-01T00:00:00Z",
  "endDate": "2024-12-31T23:59:59Z"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`productIds`	String[]	List of product IDs to be validated. The maximum limit is `10` valid product IDs.	Required	66.0
`startDate`	String	Start date of the date range in which all active records are validated.	Optional	66.0
`endDate`	String	End date of the date range in which all active records are validated.	Optional	66.0

Response body for POST

Usage Product Validation

Request Bodies

Learn more about the available request bodies of Usage Management APIs.

Consumption Traceabilities Input

Input representation of the details of the liable summary IDs that are used to trace the consumption.

Usage Activation Request Input

Input representation for a single product entry in a usage product activation request. Each entry identifies one product and the usage resources to activate for that product.

Usage Product Activation Input

Input representation of a usage product activation request.

Usage Product Validation Input

Input representation of the usage product validation request.

Consumption Traceabilities Input

Input representation of the details of the liable summary IDs that are used to trace the consumption.

JSON example

```
{
  "liableSummaryIds": [
    "1HG000000000001"
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
liableSummaryIds	String[]	List of liable summary IDs to trace the consumption.	Required	66.0

Usage Activation Request Input

Input representation for a single product entry in a usage product activation request. Each entry identifies one product and the usage resources to activate for that product.

JSON example

```
{
  "productId": "01txx0000006i2gAAA",
  "usageResourceIds": [
    "0hUxx000000001",
    "0hUxx000000002"
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
productId	String	ID of the Product2 record whose associated usage records are activated.	Required	67.0
usageResourceIds	String[]	List of usage resource IDs to activate for the given product. The activation extends to all design-time records linked to these resources, such as product usage grants, usage policies, units of measure, units of measure classes, and rate card entries. If this property is empty or omitted, all usage records associated with the given product are activated.	Optional	67.0

Usage Product Activation Input

Input representation of a usage product activation request.

JSON example

```
{
  "shouldValidateProductSetup": false,
  "activationRequests": [
    {
```

```
        "productId": "01txx0000006i2gAAA",
        "usageResourceIds": [
          "0hUxx000000001",
          "0hUxx000000002"
        ]
      }
    ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
activation Requests	Usage Activation Request Input[]	List of activation requests. Each entry identifies a product and the usage resources to activate for that product. This list accepts only one entry. If you include more, the API returns the MAX_LIMIT_EXCEEDED error.	Required	67.0
shouldValidate ProductSetup	Boolean	Indicates whether to run product setup validation before activation (true) or not (false). The default value is true.	Optional	67.0

Usage Product Validation Input

Input representation of the usage product validation request.

JSON example

```
{
  "productIds": [
    "01txx0000006i2gAAA",
    "01txx0000006j2gAAA"
  ],
  "startDate": "2024-01-01T00:00:00Z",
  "endDate": "2024-12-31T23:59:59Z"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
productIds	String[]	List of product IDs to be validated. The maximum limit is 10 valid product IDs.	Required	66.0
startDate	String	Start date of the date range in which all active records are validated.	Optional	66.0
endDate	String	End date of the date range in which all active records are validated.	Optional	66.0

Response Bodies

Learn more about the available response bodies of Usage Management APIs.

Asset Detail

Output representation of the details of a specific asset.

Billing Period

Output representation of the details of a specific billing period.

Binding Object Detail

Output representation of the list of records with the binding target details.

Binding Object Grant Detail

Output representation of the details of usage resource grants for a specified binding object.

Binding Object Rate Adjustments

Output representation of the details of binding target rate adjustments.

Binding Object Rate

Output representation of the details of Binding Object Rates object or Asset Rates object.

Binding Object Resource Grant And Policy Detail

Output representation of the details of resource grants and binding policies.

Binding Object Resource Policy Detail

Output representation of the details of a usage resource policy.

Binding Object Usage Detail

Output representation of the usage details of a binding object.

Consumption Source Detail

Output representation of the details of a specific consumption source.

Consumption Traceabilities

Output representation of the overage and resource drawdown details.

Errors

Output representation of the group of error messages with the same error code.

Error Details

Output representation of the top-level error detail when validation fails.

Error Message

Output representation of the details of records that failed with this specific error.

Error Warning Details

Output representation of the individual warning or error message with associated record details.

Consumption Traceabilities Data

Output representation of the list of asset details.

Fields Response

Output representation of the details of the optional fields on the usage-based selling-related objects.

Grant Detail

Output representation of the details of a grant from the ProductUsageGrant, LineItemUsageResourceGrant, or TransactionUsageEntitlement objects.

[Generic Error Details](#)

Output representation of the error details encountered during the API request.

[Lookup Detail](#)

Output representation of the details of a usage resource record.

[Policy Details](#)

Output representation of the details of a policy.

[Product Activation Result](#)

Output representation of the activation outcome for a single product, including any errors that occurred while activating the product or its associated usage records.

[Product Validation Result](#)

Output representation of the validation result for a specific product.

[Rate Adjustments](#)

Output representation of the details of a rate adjustment.

[Rate and Consumption Source Detail](#)

Output representation of the details of a specific rate and its consumption source.

[Rate Card Entry](#)

Output representation of the details of a rate card entry.

[Record Details](#)

Output representation of the record details including ID and name.

[Resource Detail](#)

Output representation of the details of a specific usage resource.

[Resource Policy Detail](#)

Output representation of the details of a usage resource policy.

[Usage Activation Error](#)

Output representation of a single error encountered while activating a usage product or one of its related records.

[Usage Activation Result](#)

Output representation that groups the activation outcome and the validation outcome for the product in a usage product activation request.

[Usage Details](#)

Output representation of the usage details of a quote, an order, or an asset.

[Usage Details Error Response](#)

Output representation of the details of an error related to usage details.

[Usage Product Activation](#)

Output representation of a usage product activation response.

[Usage Product Validation](#)

Output representation of all the performed validations.

[Usage Resource Grant And Policy Detail](#)

Output representation of the details of a usage resource grant and policy.

[Validation Error](#)

Output representation of the validation errors grouped by rule name.

Validation Result

Output representation of the validation results grouped by rule name.

Validation Warning

Output representation of the validation warnings grouped by rule name.

Warnings

Output representation of a group of warning messages with the same warning code.

Warning Message

Output representation of the details of records that triggered this specific warning.

Asset Detail

Output representation of the details of a specific asset.

JSON example


```
{
  "assets": [
    {
      "assetId": "ASSET1",
      "usageEntitlementAccountId": "1EA000000000001",
      "grantBindingTargetId": "1GB000000000001",
      "billingPeriods": [
        {
          "startDate": "2025-01-01",
          "endDate": "2025-01-31",
          "resources": [
            {
              "liableSummaryId": "1HG000000000001",
              "usageResourceId": "1BX000000000004",
              "usageResourceName": "SF Credits",
              "usageResourceUomId": "1UM000000000001",
              "usageResourceUomUnitCode": "CREDIT",
              "resourceTotalOverageQuantity": 333.33,
              "resourceTotalOverageAmount": 333.33,
              "resourceTotalConsumption": 1500,
              "rateAndConsumptionSources": [
                {
                  "startDate": "2025-01-01",
                  "endDate": "2025-01-31",
                  "rateUomId": "USD",
                  "ratableSummaryId": "URS3",
                  "ratingExecutionId": "1RE000000000001",
                  "overageQuantity": 333.33,
                  "overageAmount ": 333.33,
                  "totalConsumption": 1500,
                  "netUnitRate": 1,
                  "consumptionSources": [
                    {
                      "consumptionSourceId": "1AE000000000001",
                      "consumptionUnit": 500
                    },
                    {
```

```
                    "consumptionSourceId": "1CO000000000001",
                    "consumptionUnit": 375,
                    "commitRate": 1.5,
                    "targetRate": 2,
                    "cmtAssetRatableSummaryId": "URSCARID1"
                  },
                  {
                    "consumptionSourceId": "1CO000000000002",
                    "consumptionUnit": 125,
                    "commitRate": 0.75,
                    "targetRate": 1,
                    "cmtAssetRatableSummaryId": "URSCARID2"
                  }
                ]
              }
            ]
          }
        ]
      }
    ]
  }
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
assetId	String	Unique identifier of the related asset.	Big, 66.0	66.0
billingPeriods	Billing Period[]	List of billing periods for the asset.	Big, 66.0	66.0
grantBinding TargetId	String	ID of the object the consumption is bound to.	Big, 66.0	66.0
usageEntitlement AccountId	String	ID of the account that holds the usage entitlement.	Big, 66.0	66.0

Billing Period

Output representation of the details of a specific billing period.

JSON example

```
{
  "billingPeriods": [
    {
      "startDate": "2025-01-01",
      "endDate": "2025-01-31",
      "resources": [
        {
          "liableSummaryId": "1HG000000000001",
          "usageResourceId": "1BX000000000004",
          "usageResourceName": "SF Credits",
          "usageResourceUomId": "1UM000000000001",
```

```
            "usageResourceUomUnitCode": "CREDIT",
            "resourceTotalOverageQuantity": 333.33,
            "resourceTotalOverageAmount": 333.33,
            "resourceTotalConsumption": 1500,
            "rateAndConsumptionSources": [
              {
                "startDate": "2025-01-01",
                "endDate": "2025-01-31",
                "rateUomId": "USD",
                "ratableSummaryId": "URS3",
                "ratingExecutionId": "1RE000000000001",
                "overageQuantity": 333.33,
                "overageAmount ": 333.33,
                "totalConsumption": 1500,
                "netUnitRate": 1,
                "consumptionSources": [
                  {
                    "consumptionSourceId": "1AE000000000001",
                    "consumptionUnit": 500
                  },
                  {
                    "consumptionSourceId": "1CO000000000001",
                    "consumptionUnit": 375,
                    "commitRate": 1.5,
                    "targetRate": 2,
                    "cmtAssetRatableSummaryId": "URSCARID1"
                  },
                  {
                    "consumptionSourceId": "1CO000000000002",
                    "consumptionUnit": 125,
                    "commitRate": 0.75,
                    "targetRate": 1,
                    "cmtAssetRatableSummaryId": "URSCARID2"
                  }
                ]
              }
            ]
          }
        ]
      }
    ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
endDate	String	End date of the billing period.	Big, 66.0	66.0
resources	Resource Detail[]	List of usage resources within the billing period.	Big, 66.0	66.0
startDate	String	Start date of the billing period.	Big, 66.0	66.0

Binding Object Detail

Output representation of the list of records with the binding target details.

JSON example

This example includes the details of a binding object.


```
{
  "records": [
    {
      "bindingObjectRate": {
        "id": "1QNSB0000001JyH4AU,1QNSB0000001JyI4AU",
        "negotiatedRate": null,
        "negotiatedRateAdjustments": [
          {
            "lowerBound": 101,
            "name": null,
            "rateAdjustmentId": "1DMSB000001N3C74AK",
            "rateAdjustmentType": "Amount",
            "rateAdjustmentValue": 10,
            "tierUnitOfMeasure": "USD",
            "upperBound": null
          },
          {
            "lowerBound": 1,
            "name": null,
            "rateAdjustmentId": "1DMSB000001N3C64AK",
            "rateAdjustmentType": "Percentage",
            "rateAdjustmentValue": 30,
            "tierUnitOfMeasure": "USD",
            "upperBound": 100
          }
        ],
        "rate": 100,
        "rateCardEntryId": "1CJSB000000207R4AQ,1CJSB000000207S4AQ",
        "rateUnitOfMeasureName": "USD"
      },
      "bindingObjectResourceGrantAndPolicyDetail": {
        "bindingObjectGrantDetail": [
          {
            "effectiveEndDate": "Sat Oct 04 23:59:59 GMT 2025",
            "effectiveStartDate": "Fri Sep 05 00:00:00 GMT 2025",
            "grantType": "Grant",
            "id": "1B0SB0000000Eiv0AE",
            "product": {
              "id": "01tSB000006XMtqYAG"
            },
            "quantity": 100,
            "record": {
              "id": "02iSB000000IoETYA0"
            },
            "unitOfMeasure": {
              "id": "0hESB0000003yfp2AA"
            },
            "usageRefreshPolicy": {
```

```
              "id": "1BYSB0000001lOH4AY",
              "negotiable": null
            },
            "usageRolloverPolicy": {
              "id": "1BVSB000000A1xJ4AS",
              "negotiable": null
            },
            "validityPeriodTerm": 1,
            "validityPeriodUnit": "Month"
          }
        ],
        "bindingObjectResourcePolicyDetail": {
          "drawdownOrder": "ExpiringFirst",
          "id": "1X2SB00000002WT0AY",
          "ratingFrequencyPolicy": {
            "id": "1HJSB0000000G3B4AU",
            "negotiable": null
          },
          "usageAggregationPolicy": {
            "id": "1cfSB0000001xHPYAY",
            "negotiable": null
          },
          "usageCommitmentPolicy": {
            "id": null,
            "negotiable": null
          },
          "usageOveragePolicy": {
            "id": "7UkSB00000002OP0AY",
            "negotiable": null
          }
        }
      },
      "usageResource": {
        "id": "1BRSB0000001x4h4AA"
      }
    }
  ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`bindingObjectRate`	Binding Object Rate	Details about the `Binding Object Rates` object or `Asset Rates` object.	Big, 65.0	65.0
`bindingObjectResourceGrantAndPolicyDetail`	Binding Object Resource Grant and Policy Detail	Details about the resource grants and binding policies.	Big, 65.0	65.0
`usageResource`	Lookup Detail	Details of the usage resource such as the ID of the record.	Big, 65.0	65.0

Binding Object Grant Detail

Output representation of the details of usage resource grants for a specified binding object.

JSON example

This example includes the details of usage resource grants for a specified binding object.


```
{
  "bindingObjectGrantDetail": [
    {
      "effectiveEndDate": "Sat Oct 04 23:59:59 GMT 2025",
      "effectiveStartDate": "Fri Sep 05 00:00:00 GMT 2025",
      "grantType": "Grant",
      "id": "1B0SB0000000Eiv0AE",
      "product": {
        "id": "01tSB000006XMtqYAG"
      },
      "quantity": 100,
      "record": {
        "id": "02iSB000000IoETYA0"
      },
      "unitOfMeasure": {
        "id": "0hESB0000003yfp2AA"
      },
      "usageRefreshPolicy": {
        "id": "1BYSB0000001lOH4AY",
        "negotiable": null
      },
      "usageRolloverPolicy": {
        "id": "1BVSB000000A1xJ4AS",
        "negotiable": null
      },
      "validityPeriodTerm": 1,
      "validityPeriodUnit": "Month"
    }
  ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`effectiveEndDate`	String	Effective end date for the asset lifecycle.	Big, 65.0	65.0
`effectiveStartDate`	String	Effective start date for the asset lifecycle.	Big, 65.0	65.0
`grantType`	String	Type of usage resource grant.	Big, 65.0	65.0
`id`	String	ID of the `Transaction Usage Entitlement` record responsible for the grant.	Big, 65.0	65.0
`product`	Lookup Detail	Details of the product.	Big, 65.0	65.0

Property Name	Type	Description	Filter Group and Version	Available Version
`quantity`	Double	Quantity of the binding object usage resource grant.	Big, 65.0	65.0
`record`	Lookup Detail	ID of the asset.	Big, 65.0	65.0
`unitOfMeasure`	Lookup Detail	unit of measure of the usage resource.	Big, 65.0	65.0
`usageRefreshPolicy`	Policy Detail	ID of the usage grant refresh policy.	Big, 65.0	65.0
`usageRolloverPolicy`	Policy Detail	ID of the usage grant rollover policy.	Big, 65.0	65.0
`validityPeriodTerm`	Double	Validity period term of the usage resource grant.	Big, 65.0	65.0
`validityPeriodUnit`	String	Validity period unit of the usage resource grant.	Big, 65.0	65.0

Binding Object Rate Adjustments

Output representation of the details of binding target rate adjustments.

JSON example

This example includes the details of binding target rate adjustments.

```
{
  "negotiatedRateAdjustments": [
    {
      "lowerBound": 101,
      "name": null,
      "rateAdjustmentId": "1DMSB000001N3C74AK",
      "rateAdjustmentType": "Amount",
      "rateAdjustmentValue": 10,
      "tierUnitOfMeasure": "USD",
      "upperBound": null
    },
    {
      "lowerBound": 1,
      "name": null,
      "rateAdjustmentId": "1DMSB000001N3C64AK",
      "rateAdjustmentType": "Percentage",
      "rateAdjustmentValue": 30,
      "tierUnitOfMeasure": "USD",
      "upperBound": 100
    }
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
lowerBound	Double	Minimum quantity for the adjustment to be applicable.	Small, 65.0	65.0
name	String	Name of the tier or binding object rate adjustment.	Small, 65.0	65.0
rateAdjustmentId	String	ID of the binding object rate adjustment.	Small, 65.0	65.0
rateAdjustmentType	String	Type of the binding object rate adjustment.	Small, 65.0	65.0
rateAdjustmentValue	Double	Value of the binding object rate adjustment.	Small, 65.0	65.0
tierUnitOfMeasure	String	Unit of measure that represents the tier or binding object rate adjustment.	Small, 65.0	65.0
upperBound	Double	Maximum quantity for the adjustment to be applicable.	Small, 65.0	65.0

Binding Object Rate

Output representation of the details of Binding Object Rates object or Asset Rates object.

JSON example

This example includes the details of a binding object rate.


```
{
  "bindingObjectRate": {
    "id": "1QNSB0000001JyH4AU,1QNSB0000001JyI4AU",
    "negotiatedRate": null,
    "negotiatedRateAdjustments": [
      {
        "lowerBound": 101,
        "name": null,
        "rateAdjustmentId": "1DMSB000001N3C74AK",
        "rateAdjustmentType": "Amount",
        "rateAdjustmentValue": 10,
        "tierUnitOfMeasure": "USD",
        "upperBound": null
      },
      {
        "lowerBound": 1,
        "name": null,
        "rateAdjustmentId": "1DMSB000001N3C64AK",
        "rateAdjustmentType": "Percentage",
        "rateAdjustmentValue": 30,
        "tierUnitOfMeasure": "USD",
        "upperBound": 100
      }
    ],
    "rate": 100,
    "rateCardEntryId": "1CJSB000000207R4AQ,1CJSB000000207S4AQ",
    "rateUnitOfMeasureName": "USD"
```

```
    }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`id`	String	ID of the `Binding Object Rate Card Entry` or `Asset Rate Card Entry` object.	Big, 65.0	65.0
`negotiated Rate`	Double	Negotiated rate available in the `Binding Object Rate Card Entry` or `Asset Rate Card Entry` object.	Big, 65.0	65.0
`negotiatedRate Adjustments`	Binding Object Rate Adjustments[]	List of rate adjustments available in the `Binding Object Rate Adjustment` or `Asset Rate Adjustment`.	Big, 65.0	65.0
`rate`	Double	Rate of the rate card entry associated to the `Binding Object Rate Card Entry` or `Asset Rate Card Entry` object.	Big, 65.0	65.0
`rateCard EntryId`	String	ID of the rate card entry associated to the `Binding Object Rate Card Entry` or `Asset Rate Card Entry`.	Big, 65.0	65.0
`rateUnitOf MeasureName`	String	Rate unit of measure of the rates.	Big, 65.0	65.0

Binding Object Resource Grant And Policy Detail

Output representation of the details of resource grants and binding policies.

JSON example

This example includes the details of resource grants and binding policies for a specified binding object.


```
{
  "bindingObjectResourceGrantAndPolicyDetail": {
    "bindingObjectGrantDetail": [
      {
        "effectiveEndDate": "Sat Oct 04 23:59:59 GMT 2025",
        "effectiveStartDate": "Fri Sep 05 00:00:00 GMT 2025",
        "grantType": "Grant",
        "id": "1B0SB0000000Eiv0AE",
        "product": {
          "id": "01tSB000006XMtqYAG"
        },
        "quantity": 100,
        "record": {
          "id": "02iSB000000IoETYA0"
```

```
        },
        "unitOfMeasure": {
          "id": "0hESB0000003yfp2AA"
        },
        "usageRefreshPolicy": {
          "id": "1BYSB0000001lOH4AY",
          "negotiable": null
        },
        "usageRolloverPolicy": {
          "id": "1BVSB000000A1xJ4AS",
          "negotiable": null
        },
        "validityPeriodTerm": 1,
        "validityPeriodUnit": "Month"
      }
    ],
    "bindingObjectResourcePolicyDetail": {
      "drawdownOrder": "ExpiringFirst",
      "id": "1X2SB00000002WT0AY",
      "ratingFrequencyPolicy": {
        "id": "1HJSB0000000G3B4AU",
        "negotiable": null
      },
      "usageAggregationPolicy": {
        "id": "1cfSB0000001xHPYAY",
        "negotiable": null
      },
      "usageCommitmentPolicy": {
        "id": null,
        "negotiable": null
      },
      "usageOveragePolicy": {
        "id": "7UkSB00000002OP0AY",
        "negotiable": null
      }
    }
  }
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
bindingObject GrantDetail	Binding Object Grant Detail[]	Details of the negotiated resource grants for the specified binding object.	Big, 65.0	65.0
bindingObject ResourcePolicy Detail	Binding Object Resource Policy Detail	Detail of the binding policies.	Big, 65.0	65.0

Binding Object Resource Policy Detail

Output representation of the details of a usage resource policy.

JSON example

This example includes the details of a usage resource policy.


```
{
  "bindingObjectResourcePolicyDetail": {
    "drawdownOrder": "ExpiringFirst",
    "id": "1X2SB00000002WT0AY",
    "ratingFrequencyPolicy": {
      "id": "1HJSB0000000G3B4AU",
      "negotiable": null
    },
    "usageAggregationPolicy": {
      "id": "1cfSB0000001xHPYAY",
      "negotiable": null
    },
    "usageCommitmentPolicy": {
      "id": null,
      "negotiable": null
    },
    "usageOveragePolicy": {
      "id": "7UkSB00000002OP0AY",
      "negotiable": null
    }
  }
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
drawdownOrder	String	Specifies the order or way to process the drawdown. See Usage Management Essentials to know more about a drawdown process.	Big, 65.0	65.0
id	String	ID of the Binding Object Usage Resource Policy record.	Big, 65.0	65.0
ratingFrequencyPolicy	Policy Detail	Details of the rating frequency policy.	Big, 65.0	65.0
usageAggregationPolicy	Policy Detail	Details of the usage aggregation policy.	Big, 65.0	65.0
usageCommitmentPolicy	Policy Detail	Details of the commitment policy of the usage resource.	Big, 65.0	65.0
usageOveragePolicy	Policy Detail	Details of the overage policy of the usage resource.	Big, 65.0	65.0

Binding Object Usage Detail

Output representation of the usage details of a binding object.

JSON example

This example shows a sample successful response.


```
{
  "records": [
    {
      "bindingObjectRate": {
        "id": "1QNSB0000001JyH4AU,1QNSB0000001JyI4AU",
        "negotiatedRate": null,
        "negotiatedRateAdjustments": [
          {
            "lowerBound": 101,
            "name": null,
            "rateAdjustmentId": "1DMSB000001N3C74AK",
            "rateAdjustmentType": "Amount",
            "rateAdjustmentValue": 10,
            "tierUnitOfMeasure": "USD",
            "upperBound": null
          },
          {
            "lowerBound": 1,
            "name": null,
            "rateAdjustmentId": "1DMSB000001N3C64AK",
            "rateAdjustmentType": "Percentage",
            "rateAdjustmentValue": 30,
            "tierUnitOfMeasure": "USD",
            "upperBound": 100
          }
        ],
        "rate": 100,
        "rateCardEntryId": "1CJSB000000207R4AQ,1CJSB000000207S4AQ",
        "rateUnitOfMeasureName": "USD"
      },
      "bindingObjectResourceGrantAndPolicyDetail": {
        "bindingObjectGrantDetail": [
          {
            "effectiveEndDate": "Sat Oct 04 23:59:59 GMT 2025",
            "effectiveStartDate": "Fri Sep 05 00:00:00 GMT 2025",
            "grantType": "Grant",
            "id": "1B0SB0000000Eiv0AE",
            "product": {
              "id": "01tSB000006XMtqYAG"
            },
            "quantity": 100,
            "record": {
              "id": "02iSB000000IoETYA0"
            },
            "unitOfMeasure": {
              "id": "0hESB0000003yfp2AA"
            },
            "usageRefreshPolicy": {
              "id": "1BYSB0000001lOH4AY",
              "negotiable": null
            },
```

```
            "usageRolloverPolicy": {
              "id": "1BVSB000000A1xJ4AS",
              "negotiable": null
            },
            "validityPeriodTerm": 1,
            "validityPeriodUnit": "Month"
          }
        ],
        "bindingObjectResourcePolicyDetail": {
          "drawdownOrder": "ExpiringFirst",
          "id": "1X2SB00000002WT0AY",
          "ratingFrequencyPolicy": {
            "id": "1HJSB0000000G3B4AU",
            "negotiable": null
          },
          "usageAggregationPolicy": {
            "id": "1cfSB0000001xHPYAY",
            "negotiable": null
          },
          "usageCommitmentPolicy": {
            "id": null,
            "negotiable": null
          },
          "usageOveragePolicy": {
            "id": "7UkSB00000002OP0AY",
            "negotiable": null
          }
        }
      },
      "usageResource": {
        "id": "1BRSB0000001x4h4AA"
      }
    }
  ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
errors	Usage Detail Error Response[]	List of errors encountered during the processing of the API request.	Small, 65.0	65.0
records	Binding Object Detail[]	List of records that contains the binding target details.	Small, 65.0	65.0

Consumption Source Detail

Output representation of the details of a specific consumption source.

JSON example


```
{
  "consumptionSources": [
    {
```

```
      "consumptionSourceId": "1AE000000000001",
      "consumptionUnit": 500
    },
    {
      "consumptionSourceId": "1CO000000000001",
      "consumptionUnit": 375,
      "commitRate": 1.5,
      "targetRate": 2,
      "cmtAssetRatableSummaryId": "URSCARID1"
    },
    {
      "consumptionSourceId": "1CO000000000002",
      "consumptionUnit": 125,
      "commitRate": 0.75,
      "targetRate": 1,
      "cmtAssetRatableSummaryId": "URSCARID2"
    }
  ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
cmtAssetRatableSummaryId	String	ID for querying rating waterfall.	Big, 66.0	66.0
commitRate	Double	Net unit rate at which drawdown is done for commitment products.	Big, 66.0	66.0
consumptionSourceId	String	Object on which the consumption was recorded.	Big, 66.0	66.0
consumptionUnit	Double	Recorded quantity of consumption.	Big, 66.0	66.0
targetRate	Double	Input unit rate which is used for commitment products.	Big, 66.0	66.0

Consumption Traceabilities

Output representation of the overage and resource drawdown details.

JSON example


```
{
  "success": true,
  "data": {
    "assets": [
      {
        "assetId": "ASSET1",
        "usageEntitlementAccountId": "1EA000000000001",
        "grantBindingTargetId": "1GB000000000001",
        "billingPeriods": [
          {
```

```
            "startDate": "2025-01-01",
            "endDate": "2025-01-31",
            "resources": [
              {
                "liableSummaryId": "1HG000000000001",
                "usageResourceId": "1BX000000000004",
                "usageResourceName": "SF Credits",
                "usageResourceUomId": "1UM000000000001",
                "usageResourceUomUnitCode": "CREDIT",
                "resourceTotalOverageQuantity": 333.33,
                "resourceTotalOverageAmount": 333.33,
                "resourceTotalConsumption": 1500,
                "rateAndConsumptionSources": [
                  {
                    "startDate": "2025-01-01",
                    "endDate": "2025-01-31",
                    "rateUomId": "USD",
                    "ratableSummaryId": "URS3",
                    "ratingExecutionId": "1RE000000000001",
                    "overageQuantity": 333.33,
                    "overageAmount ": 333.33,
                    "totalConsumption": 1500,
                    "netUnitRate": 1,
                    "consumptionSources": [
                      {
                        "consumptionSourceId": "1AE000000000001",
                        "consumptionUnit": 500
                      },
                      {
                        "consumptionSourceId": "1CO000000000001",
                        "consumptionUnit": 375,
                        "commitRate": 1.5,
                        "targetRate": 2,
                        "cmtAssetRatableSummaryId": "URSCARID1"
                      },
                      {
                        "consumptionSourceId": "1CO000000000002",
                        "consumptionUnit": 125,
                        "commitRate": 0.75,
                        "targetRate": 1,
                        "cmtAssetRatableSummaryId": "URSCARID2"
                      }
                    ]
                  }
                ]
              }
            ]
          }
        ]
      }
    ],
    "error": null
  }
}
```

zProperty Name	Type	Description	Filter Group and Version	Available Version
`data`	Consumption Traceabilities Data[]	Payload that contains the traceability details.	Big, 66.0	66.0
`error`	Generic Error Details[]	Error details if the request isn't successful.	Big, 66.0	66.0
`success`	Boolean	Indicates whether the API request is successful (`true`) or not (`false`).	Big, 66.0	66.0

Errors

Output representation of the group of error messages with the same error code.

JSON example

This example shows a group of error messages with the same error code.


```
{
  "errors": [
    {
      "errorCode": "EFFECTIVITY_MISMATCH",
      "errorMessages": [
        {
          "errorMessage": "PUR and RCE effective date ranges must have overlap for proper
 rating functionality",
          "errorDetails": []
        }
      ]
    }
  ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`errorCode`	String	Standardized error code. For example, `EFFECTIVITY_MISMATCH`.	Big, 66.0	66.0
`errorMessages`	Error Message[]	List of error messages for records that failed with this error code.	Big, 66.0	66.0

Error Details

Output representation of the top-level error detail when validation fails.

JSON example

This example shows a sample error response.


```
{
  "errors": [
    {
```

```
      "errorCode": "VALIDATION_FAILED",
      "message": "Product validation completed with cross-entity errors",
      "products": [
        {
          "productId": "01txx0000006i2gAAA",
          "validationResult": {
            "validationErrors": [],
            "validationWarnings": []
          }
        }
      ]
    }
  ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`errorCode`	String	Standardized error code. For example, `VALIDATION_FAILED`.	Big, 66.0	66.0
`message`	String	Human-readable error message that describes the overall validation failure.	Big, 66.0	66.0
`products`	Product Validation Result[]	List of product validation results.	Big, 66.0	66.0

Error Message

Output representation of the details of records that failed with this specific error.

JSON example

This example shows an error message with details.


```
{
  "errorMessages": [
    {
      "errorMessage": "PUR and RCE effective date ranges must have overlap for proper
rating functionality",
      "errorDetails": [
        {
          "relatedObjectAPIName": "ProductUsageRule",
          "records": [
            {
              "id": "a0bxx0000004CqZAAU",
              "name": "PUR-001"
            }
          ]
        }
      ]
    }
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
errorMessage	String	Human-readable error message that describes the validation failure.	Big, 66.0	66.0
errorDetails	Error Warning Details[]	Details of records that failed with this specific error.	Big, 66.0	66.0

Error Warning Details

Output representation of the individual warning or error message with associated record details.

JSON example

This example shows error details with record information.

```
{
  "errorDetails": [
    {
      "relatedObjectAPIName": "ProductUsageRule",
      "records": [
        {
          "id": "a0bxx0000004CqZAAU",
          "name": "PUR-001"
        }
      ]
    }
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
relatedObjectAPIName	String	API name of the related object.	Big, 66.0	66.0
records	Record Details[]	Details of records that triggered the specific warning or error.	Big, 66.0	66.0

Consumption Traceabilities Data

Output representation of the list of asset details.

JSON example

```
{
  "data": {
    "assets": [
      {
        "assetId": "ASSET1",
        "usageEntitlementAccountId": "1EA000000000001",
        "grantBindingTargetId": "1GB000000000001",
        "billingPeriods": [
          {
```

```
            "startDate": "2025-01-01",
            "endDate": "2025-01-31",
            "resources": [
              {
                "liableSummaryId": "1HG000000000001",
                "usageResourceId": "1BX000000000004",
                "usageResourceName": "SF Credits",
                "usageResourceUomId": "1UM000000000001",
                "usageResourceUomUnitCode": "CREDIT",
                "resourceTotalOverageQuantity": 333.33,
                "resourceTotalOverageAmount": 333.33,
                "resourceTotalConsumption": 1500,
                "rateAndConsumptionSources": [
                  {
                    "startDate": "2025-01-01",
                    "endDate": "2025-01-31",
                    "rateUomId": "USD",
                    "ratableSummaryId": "URS3",
                    "ratingExecutionId": "1RE000000000001",
                    "overageQuantity": 333.33,
                    "overageAmount ": 333.33,
                    "totalConsumption": 1500,
                    "netUnitRate": 1,
                    "consumptionSources": [
                      {
                        "consumptionSourceId": "1AE000000000001",
                        "consumptionUnit": 500
                      },
                      {
                        "consumptionSourceId": "1CO000000000001",
                        "consumptionUnit": 375,
                        "commitRate": 1.5,
                        "targetRate": 2,
                        "cmtAssetRatableSummaryId": "URSCARID1"
                      },
                      {
                        "consumptionSourceId": "1CO000000000002",
                        "consumptionUnit": 125,
                        "commitRate": 0.75,
                        "targetRate": 1,
                        "cmtAssetRatableSummaryId": "URSCARID2"
                      }
                    ]
                  }
                ]
              }
            ]
          }
        ]
      }
    ]
  }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
assets	Asset Detail[]	List of assets for the specified liable summaries.	Big, 66.0	66.0

Fields Response

Output representation of the details of the optional fields on the usage-based selling-related objects.

JSON Example

```
"fields": {
  "MyCustomDate__c": {
    "displayValue": "2024-09-24",
    "value": "2024-09-24T17:46:30.662Z"
  },
  "MyCustomNumber__c": {
    "displayValue": "20.0",
    "value": 20
  }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
displayValue	String	Display value of a field.	Big, 63.0	63.0
value	Object	Value of a field in its original data form.	Big, 63.0	63.0

Grant Detail

Output representation of the details of a grant from the ProductUsageGrant, LineItemUsageResourceGrant, or TransactionUsageEntitlement objects.

JSON Example

```
{
  "negotiatedGrantDetail": {
    "grantType": "Grant",
    "id": "1X6xx00000000OECAY",
    "quantity": 100,
    "usageGrantNegotiable": "Negotiable",
    "usageRefreshPolicy": {
      "displayName": null,
      "id": "1BYxx0000004C92GAE",
      "negotiable": "Non-Negotiable"
    },
    "usageRolloverPolicy": {
      "displayName": null,
      "id": "1BVxx0000004C92GAE",
      "negotiable": "Non-Negotiable"
    },
```

```
    "validityPeriodTerm": 12,
    "validityPeriodUnit": "Month"
  }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
grantType	String	Details about the grant type.	Big, 65.0	65.0
id	String	ID of the usage resource grant.	Big, 65.0	65.0
quantity	Double	Quantity of the negotiated usage resource grant.	Big, 65.0	65.0
usageGrantNegotiable	String	Specifies whether the grant is negotiable or not.	Big, 65.0	65.0
usageRefreshPolicy	Policy Detail	ID of the usage grant refresh policy.	Big, 65.0	65.0
usageRolloverPolicy	Policy Detail	ID of the usage grant rollover policy.	Big, 65.0	65.0
validityPeriodTerm	Double	Validity period term of the grant.	Big, 65.0	65.0
validityPeriodUnit	String	Validity period unit of the grant.	Big, 65.0	65.0

Generic Error Details

Output representation of the error details encountered during the API request.

JSON example

```
{
  "data": [],
  "error": {
    "errorCode": "INVALID_API_INPUT",
    "message": "Liable summary IDs cannot be null or empty."
  },
  "success": false
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
errorCode	String	Error code that represents the type of error.	Big, 67.0	67.0
message	String	Detailed error message that specifies the cause of failure.	Big, 67.0	67.0

Lookup Detail

Output representation of the details of a usage resource record.

JSON example

This example includes the details of a usage resource.


```
{
  "usageResource": {
    "id": "1BRxx0000004C9EGAU"
  }
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
id	String	ID of the usage resource record.	Big, 65.0	65.0

Policy Details

Output representation of the details of a policy.

JSON example

This example includes the details for different policy types.


```
{
  "bindingObjectGrantDetail": [
    {
      "effectiveEndDate": "Sat Oct 04 23:59:59 GMT 2025",
      "effectiveStartDate": "Fri Sep 05 00:00:00 GMT 2025",
      "grantType": "Grant",
      "id": "1B0SB0000000Eiv0AE",
      "product": {
        "id": "01tSB000006XMtqYAG"
      },
      "quantity": 100,
      "record": {
        "id": "02iSB000000IoETYA0"
      },
      "unitOfMeasure": {
        "id": "0hESB0000003yfp2AA"
      },
      "usageRefreshPolicy": {
        "id": "1BYSB0000001lOH4AY",
        "negotiable": null
      },
      "usageRolloverPolicy": {
        "id": "1BVSB000000A1xJ4AS",
        "negotiable": null
      },
      "validityPeriodTerm": 1,
      "validityPeriodUnit": "Month"
    }
```

```
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`id`	String	ID of the policy.	Big, 65.0	65.0
`negotiable`	String	Indicates whether the policy is negotiable. Valid values are: • `Negotiable` • `Non-Negotiable`	Big, 65.0	65.0

Product Activation Result

Output representation of the activation outcome for a single product, including any errors that occurred while activating the product or its associated usage records.

JSON example


```
{
  "productId": "01txx0000006i2hAAA",
  "errors": [
    {
      "productUsageResourceId": "0iUxx000000009",
      "usageResourceId": "0hUxx000000003",
      "message": "Usage Resource not found for product",
      "objectApiName": "ProductUsageResource",
      "fieldName": "UsageResourceId",
      "recordId": null,
      "recordName": null
    }
  ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`errors`	Usage Activation Error[]	List of errors that occurred while activating the product or its associated usage records.	Big, 67.0	67.0
`productId`	String	ID of the Product2 record that this activation outcome describes.	Big, 67.0	67.0

Product Validation Result

Output representation of the validation result for a specific product.

JSON example

This example shows the product validation result.

```
{
  "products": [
    {
      "productId": "01txx0000006i2gAAA",
      "validationResult": {
        "validationErrors": [],
        "validationWarnings": []
      }
    }
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`productId`	String	Unique product ID that's being validated.	Big, 67.0	67.0
`validationResult`	Validation Result	Validation results grouped by rule name.	Big, 67.0	67.0

Rate Adjustments

Output representation of the details of a rate adjustment.

JSON Example

```
      "rateAdjustments": [
        {
           "fields": {},
           "lowerBound": 0.0,
           "name": null,
           "negotiatedRateAdjustmentId": null,
           "rateAdjustmentId": "1ENxx0000004C9BGAU",
           "rateAdjustmentType": "Percentage",
           "rateAdjustmentValue": 10.0,
           "tierUnitOfMeasure": "USD",
           "upperBound": 50.0
        }
    ]
```

If the `negotiable` property value that’s associated with a rate card entry is blank, then the data is derived from Product Catalog Management. If it isn’t blank, then the data is derived from Rate Management.

Property Name	Type	Description	Filter Group and Version	Available Version
`fields`	Map<String, Fields Response>	List of optional fields and their values that belong to the rate adjustment object.	Big, 63.0	63.0
`lowerBound`	Double	Minimum quantity for the adjustment to be applicable.	Small, 63.0	63.0

Property Name	Type	Description	Filter Group and Version	Available Version
name	String	Name of the tier.	Small, 63.0	63.0
negotiatedRateAdjustmentId	String	ID of the negotiated rate adjustment.	Small, 63.0	63.0
rateAdjustmentId	String	ID of the rate adjustment.	Small, 63.0	63.0
rateAdjustmentType	String	Type of the rate adjustment.	Small, 63.0	63.0
rateAdjustmentValue	Double	Value of the rate adjustment.	Small, 63.0	63.0
tierUnitOfMeasure	String	Unit of measure representing the tier.	Small, 63.0	63.0
upperBound	Double	Maximum quantity for the adjustment to be applicable.	Small, 63.0	63.0

Rate and Consumption Source Detail

Output representation of the details of a specific rate and its consumption source.

JSON example


```
{
  "rateAndConsumptionSources": [
    {
      "startDate": "2025-01-01",
      "endDate": "2025-01-31",
      "rateUomId": "USD",
      "ratableSummaryId": "URS3",
      "ratingExecutionId": "1RE000000000001",
      "overageQuantity": 333.33,
      "overageAmount ": 333.33,
      "totalConsumption": 1500,
      "netUnitRate": 1,
      "consumptionSources": [
        {
          "consumptionSourceId": "1AE000000000001",
          "consumptionUnit": 500
        },
        {
          "consumptionSourceId": "1CO000000000001",
          "consumptionUnit": 375,
          "commitRate": 1.5,
          "targetRate": 2,
          "cmtAssetRatableSummaryId": "URSCARID1"
        },
        {
          "consumptionSourceId": "1CO000000000002",
```

```
            "consumptionUnit": 125,
            "commitRate": 0.75,
            "targetRate": 1,
            "cmtAssetRatableSummaryId": "URSCARID2"
          }
        ]
      }
    ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
consumptionSourceDetails	Consumption Source Detail[]	Details on the consumption source.	Big, 66.0	66.0
endDate	String	End date for the specific rate period.	Big, 66.0	66.0
netUnitRate	Double	Net unit rate.	Big, 66.0	66.0
overageAmt	Double	Amount for the overage quantity.	Big, 66.0	66.0
overageQty	Double	Quantity of consumption that was rated as overage.	Big, 66.0	66.0
ratableSummaryId	String	Unique ID for the summary used in rating.	Big, 66.0	66.0
ratableSummaryStatus	String	Status of ratable summary.	Big, 66.0	66.0
rateUomId	String	Unit of measure for the rate.	Big, 66.0	66.0
rateUomName	String	Name of the rate unit of measure	Big, 66.0	66.0
ratingExecutionId	String	Execution ID of the rating.	Big, 66.0	66.0
sourceUsageResourceId	String	ID of the source usage resource.	Big, 66.0	66.0
sourceUsageResourceName	String	Name of the source usage resource.	Big, 66.0	66.0
startDate	String	Start date for the specific rate period.	Big, 66.0	66.0
totalConsumption	Double	Total quantity consumed for this rate.	Big, 66.0	66.0

Rate Card Entry

Output representation of the details of a rate card entry.

JSON Example


```
{
  "records": [
    {
      "bindingInstanceTargetType": "Product",
      "bindingInstanceType": "Target",
      "chargeForOverages": "Yes",
      "fields": {},
      "isOptional": false,
      "name": "Paddle Board",
      "negotiable": "Negotiable,Non-Negotiable",
      "negotiatedRate": 20,
      "negotiatedRateCardEntryId": "1ELxx0000004C9JGAU,1ELxx0000004C9KGAU",
      "quantity": 15,
      "rate": 5,
      "rateAdjustments": [
        {
          "fields": {},
          "lowerBound": 0,
          "name": "Tier 1",
          "negotiatedRateAdjustmentId": "1ENxx0000004C9BGAU",
          "rateAdjustmentId": "1ENxx0000004C9BGAU",
          "rateAdjustmentType": "Percentage",
          "rateAdjustmentValue": 10,
          "tierUnitOfMeasure": "USD",
          "upperBound": 50
        }
      ],
      "rateCardEntryId": "1CJxx0000004C9IGAU,1CJxx0000004C9JGAU",
      "rateUnitOfMeasureName": "USD",
      "unitOfMeasure": "GB",
      "usageResourceGrantAndPolicyDetail": {
        "grantDetail": {
          "grantType": "Grant",
          "id": "1BXxx0000004C9lGAE",
          "quantity": 100,
          "usageGrantNegotiable": "Negotiable",
          "usageRefreshPolicy": {
            "id": "1BYxx0000004C92GAE",
            "negotiable": "Non-Negotiable"
          },
          "usageRolloverPolicy": {
            "id": "1BVxx0000004C93GAE",
            "negotiable": "Non-Negotiable"
          },
          "validityPeriodTerm": 1,
          "validityPeriodUnit": "Month"
        },
        "negotiatedGrantDetail": {
          "grantType": "Grant",
          "id": "1X6xx00000000OECAY",
          "quantity": 100,
          "usageGrantNegotiable": "Negotiable",
```

```
          "usageRefreshPolicy": {
            "id": "1BYxx0000004C92GAE",
            "negotiable": "Non-Negotiable"
          },
          "usageRolloverPolicy": {
            "id": "1BVxx0000004C93GAE",
            "negotiable": "Non-Negotiable"
          },
          "validityPeriodTerm": 1,
          "validityPeriodUnit": "Month"
        },
        "negotiatedResourcePolicyDetail": {
          "id": "1X5xx00000000OECAY",
          "ratingFrequencyPolicy": {
            "id": null,
            "negotiable": null
          },
          "usageAggregationPolicy": {
            "id": null,
            "negotiable": null
          },
          "usageCommitmentPolicy": {
            "id": "7Pexx0000004C92CAE",
            "negotiable": "Non-Negotiable"
          },
          "usageOveragePolicy": {
            "id": null,
            "negotiable": null
          }
        },
        "resourcePolicyDetail": {
          "id": "7Suxx0000004C9kCAE",
          "ratingFrequencyPolicy": {
            "id": null,
            "negotiable": null
          },
          "usageAggregationPolicy": {
            "id": null,
            "negotiable": null
          },
          "usageCommitmentPolicy": {
            "id": "7Pexx0000004C92CAE",
            "negotiable": "Non-Negotiable"
          },
          "usageOveragePolicy": {
            "id": null,
            "negotiable": null
          }
        }
      },
      "usageResourceId": "1BRxx0000004C9CGAU"
    }
  ]
}
```

If the `negotiable` property value is blank, then the data is derived from Product Catalog Management. If it isn't blank, then the data is derived from Rate Management.

Property Name	Type	Description	Filter Group and Version	Available Version
`binding Instance TargetType`	String	Type of the target object that's associated with this transaction.	Big, 63.0	63.0
`bindingInstance Type`	String	Type of the binding instance. Valid values are: • `Self` • `Target`	Big, 63.0	63.0
`chargeFor Overages`	String	Specifies whether overage is permitted. Valid values are: • `Yes` • `No` • `NA`	Big, 63.0	63.0
`fields`	Map<String, Fields Response>	List of optional fields and their values that's associated with the rate card entry object.	Big, 63.0	63.0
`isOptional`	Boolean	Indicates whether the product usage resource is optional when the associated product is one of the commitment usage model types (`true`) or not (`false`).	Big, 65.0	65.0
`name`	String	Name of the resource.	Big, 63.0	63.0
`negotiable`	String	Type of the base rate and the tier rate, if applicable. Valid values are: • `Negotiable` • `Non-Negotiable`	Big, 63.0	63.0
`negotiated Rate`	Double	User-overridden overage rate.	Big, 63.0	63.0
`negotiated RateCard EntryId`	String	ID of the negotiated rate card entry and the tier rate card entry, if applicable.	Big, 63.0	63.0
`quantity`	Double	Amount granted for the resource.	Big, 63.0	63.0
`rate`	Double	Base overage rate.	Big, 63.0	63.0
`rate Adjustments`	Rate Adjustments[]	List of tiers associated with the rate card entry, if applicable.	Big, 63.0	63.0

Property Name	Type	Description	Filter Group and Version	Available Version
rateCard EntryId	String	ID of the base rate card entry and the tier rate card entry, if applicable.	Big, 63.0	63.0
rateUnit OfMeasureName	String	Unit of measure for rates in the rate card entry.	Big, 64.0	64.0
unitOfMeasure	String	Unit of measure of the grant. For example, Unit.	Big, 63.0	63.0
usageResource Id	String	ID of the usage resource.	Big, 65.0	65.0
usageResource GrantAndPolicy Detail	Usage Resource Grant And Policy Detail	Details of a usage resource grant and policy.	Big, 65.0	65.0

Record Details

Output representation of the record details including ID and name.

JSON example

This example shows the record details with ID and name.

```
{
  "records": [
    {
      "id": "a0bxx0000004CqZAAU",
      "name": "PUR-001"
    }
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
id	String	Unique identifier of the object.	Big, 66.0	66.0
name	String	Display name of the object.	Big, 66.0	66.0

Resource Detail

Output representation of the details of a specific usage resource.

JSON example

```
{
  "resources": [
    {
      "liableSummaryId": "1HG000000000001",
      "usageResourceId": "1BX000000000004",
```

```
        "usageResourceName": "SF Credits",
        "usageResourceUomId": "1UM000000000001",
        "usageResourceUomUnitCode": "CREDIT",
        "resourceTotalOverageQuantity": 333.33,
        "resourceTotalOverageAmount": 333.33,
        "resourceTotalConsumption": 1500,
        "rateAndConsumptionSources": [
          {
            "startDate": "2025-01-01",
            "endDate": "2025-01-31",
            "rateUomId": "USD",
            "ratableSummaryId": "URS3",
            "ratingExecutionId": "1RE000000000001",
            "overageQuantity": 333.33,
            "overageAmount ": 333.33,
            "totalConsumption": 1500,
            "netUnitRate": 1,
            "consumptionSources": [
              {
                "consumptionSourceId": "1AE000000000001",
                "consumptionUnit": 500
              },
              {
                "consumptionSourceId": "1CO000000000001",
                "consumptionUnit": 375,
                "commitRate": 1.5,
                "targetRate": 2,
                "cmtAssetRatableSummaryId": "URSCARID1"
              },
              {
                "consumptionSourceId": "1CO000000000002",
                "consumptionUnit": 125,
                "commitRate": 0.75,
                "targetRate": 1,
                "cmtAssetRatableSummaryId": "URSCARID2"
              }
            ]
          }
        ]
      }
    ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`liableSummaryId`	String	Unique identifier for the consumption summary.	Big, 66.0	66.0
`liableSummary Status`	String	Status of liable summary.	Big, 66.0	66.0

Property Name	Type	Description	Filter Group and Version	Available Version
rateAndConsumptionSources	Rate And Consumption Source Detail[]	Details on rates and consumption sources.	Big, 66.0	66.0
resourceTotalConsumption	Double	Total quantity of consumption for the resource.	Big, 66.0	66.0
resourceTotalOverageAmount	Double	Total amount for the overage.	Big, 66.0	66.0
resourceTotalOverageQuantity	Double	Total overage quantity.	Big, 66.0	66.0
usageResourceId	String	ID of the specific usage resource.	Big, 66.0	66.0
usageResourceName	String	Name of the usage resource.	Big, 66.0	66.0
usageResourceUomId	String	Unit of measure ID for the usage resource.	Big, 66.0	66.0
usageResourceUomUnitCode	String	Unit of measure code for the usage resource.	Big, 66.0	66.0

Resource Policy Detail

Output representation of the details of a usage resource policy.

Property Name	Type	Description	Filter Group and Version	Available Version
id	String	ID of the usage resource policy.	Big, 65.0	65.0
ratingFrequencyPolicy	Policy Detail	Details of the rating frequency policy.	Big, 65.0	65.0
usageAggregationPolicy	Policy Detail	Details of the usage aggregation policy.	Big, 65.0	65.0
usageCommitmentPolicy	Policy Detail	Details of the usage commitment policy.	Big, 65.0	65.0
usageOveragePolicy	Policy Detail	Details of the usage overage policy.	Big, 65.0	65.0

Usage Activation Error

Output representation of a single error encountered while activating a usage product or one of its related records.

JSON example


```
{
  "productUsageResourceId": "0iUxx0000000678",
  "usageResourceId": "0hUxx000000004",
  "message": "Related Unit of measure records is inactive, activate it first.",
  "objectApiName": "ProductUsageGrant",
  "fieldName": "DefaultUnitofMeasure",
  "recordId": "1BXSM000000404f4AA",
  "recordName": "PUG-000000001"
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`fieldName`	String	API name of the field that caused the error.	Big, 67.0	67.0
`message`	String	Human-readable description of the error.	Big, 67.0	67.0
`objectApiName`	String	API name of the object on which the error occurred.	Big, 67.0	67.0
`productUsageResourceId`	String	ID of the ProductUsageResource record that links the product and the usage resource for which the error occurred.	Big, 67.0	67.0
`recordId`	String	ID of the record that failed activation.	Big, 67.0	67.0
`recordName`	String	Name of the record that failed activation.	Big, 67.0	67.0
`usageResourceId`	String	ID of the usage resource record from the input request for which the error occurred.	Big, 67.0	67.0

Usage Activation Result

Output representation that groups the activation outcome and the validation outcome for the product in a usage product activation request.

JSON example


```
{
  "productActivationResults": [
    {
      "productId": "01txx0000006i2hAAA",
      "errors": []
    }
  ],
  "productValidationResults": [
    {
      "productId": "01txx0000006i2gAAA",
      "validationResult": {
        "validationErrors": [],
        "validationWarnings": []
      }
    }
```

```
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`productActivationResults`	Product Activation Result[]	List of product activation results.	Big, 67.0	67.0
`productValidationResults`	Product Validation Result[]	List of product validation results.	Big, 67.0	67.0

Usage Details

Output representation of the usage details of a quote, an order, or an asset.

JSON example

This sample response shows resources that include grants, if applicable, and resources without rates when you retrieve the usage details of an order item.


```
{
  "records": [
    {
      "bindingInstanceTargetType": null,
      "bindingInstanceType": null,
      "chargeForOverages": "Yes",
      "isOptional": false,
      "fields": {},
      "name": "API Calls",
      "negotiable": ",",
      "negotiatedRate": null,
      "negotiatedRateCardEntryId": ",",
      "quantity": 1000,
      "rate": null,
      "rateAdjustments": [],
      "rateCardEntryId": ",",
      "rateUnitOfMeasureName": "USD",
      "unitOfMeasure": null,
      "usageResourceGrantAndPolicyDetail": {
        "grantDetail": {
          "grantType": "Grant",
          "id": "1BXxx0000004C91GAE",
          "quantity": 100,
          "usageGrantNegotiable": "Negotiable",
          "usageRefreshPolicy": {
            "id": "1BYxx0000004C92GAE",
            "negotiable": "Non-Negotiable"
          },
          "usageRolloverPolicy": {
            "id": "1BVxx0000004C93GAE",
            "negotiable": "Non-Negotiable"
          },
          "validityPeriodTerm": 1,
```

```
          "validityPeriodUnit": "Month"
        },
        "negotiatedGrantDetail": {
          "grantType": "Grant",
          "id": "1X6xx00000000OECAY",
          "quantity": 100,
          "usageGrantNegotiable": "Negotiable",
          "usageRefreshPolicy": {
            "id": "1BYxx0000004C92GAE",
            "negotiable": "Non-Negotiable"
          },
          "usageRolloverPolicy": {
            "id": "1BVxx0000004C93GAE",
            "negotiable": "Non-Negotiable"
          },
          "validityPeriodTerm": 1,
          "validityPeriodUnit": "Month"
        },
        "negotiatedResourcePolicyDetail": {
          "id": "1X5xx00000000OECAY",
          "ratingFrequencyPolicy": {
            "id": null,
            "negotiable": null
          },
          "usageAggregationPolicy": {
            "id": null,
            "negotiable": null
          },
          "usageCommitmentPolicy": {
            "id": "7Pexx0000004C92CAE",
            "negotiable": "Non-Negotiable"
          },
          "usageOveragePolicy": {
            "id": null,
            "negotiable": null
          }
        },
        "resourcePolicyDetail": {
          "id": "7Suxx0000004C9kCAE",
          "ratingFrequencyPolicy": {
            "id": null,
            "negotiable": null
          },
          "usageAggregationPolicy": {
            "id": null,
            "negotiable": null
          },
          "usageCommitmentPolicy": {
            "id": "7Pexx0000004C92CAE",
            "negotiable": "Non-Negotiable"
          },
          "usageOveragePolicy": {
            "id": null,
            "negotiable": null
```

```
        }
      }
    },
    "usageResourceId": "1BRxx0000004C9CGAU"
  }
 ]
}
```

This example shows a sample response without negotiated rates when you retrieve the usage details of an order item.

```
{
  "records": [
    {
      "bindingInstanceTargetType": "Product",
      "bindingInstanceType": "Target",
      "chargeForOverages": "Yes",
      "isOptional": false,
      "fields": {},
      "name": "Paddle Board",
      "negotiable": "Negotiable,Non-Negotiable",
      "negotiatedRate": null,
      "negotiatedRateCardEntryId": ",",
      "quantity": 15,
      "rate": 5,
      "rateAdjustments": [
        {
          "fields": {},
          "lowerBound": 0,
          "name": null,
          "negotiatedRateAdjustmentId": null,
          "rateAdjustmentId": "1ENxx0000004C9BGAU",
          "rateAdjustmentType": "Percentage",
          "rateAdjustmentValue": 10,
          "tierUnitOfMeasure": "USD",
          "upperBound": 50
        }
      ],
      "rateCardEntryId": "1CJxx0000004C9IGAU,1CJxx0000004C9JGAU",
      "rateUnitOfMeasureName": "USD",
      "unitOfMeasure": "GB",
      "usageResourceGrantAndPolicyDetail": {
        "grantDetail": {
          "grantType": "Grant",
          "id": "1BXxx0000004C91GAE",
          "quantity": 100,
          "usageGrantNegotiable": "Negotiable",
          "usageRefreshPolicy": {
            "id": "1BYxx0000004C92GAE",
            "negotiable": "Non-Negotiable"
          },
          "usageRolloverPolicy": {
            "id": "1BVxx0000004C93GAE",
            "negotiable": "Non-Negotiable"
          },
          "validityPeriodTerm": 1,
```

```
      "validityPeriodUnit": "Month"
    },
    "negotiatedGrantDetail": {
      "grantType": "Grant",
      "id": "1X6xx00000000OECAY",
      "quantity": 100,
      "usageGrantNegotiable": "Negotiable",
      "usageRefreshPolicy": {
        "id": "1BYxx0000004C92GAE",
        "negotiable": "Non-Negotiable"
      },
      "usageRolloverPolicy": {
        "id": "1BVxx0000004C93GAE",
        "negotiable": "Non-Negotiable"
      },
      "validityPeriodTerm": 1,
      "validityPeriodUnit": "Month"
    },
    "negotiatedResourcePolicyDetail": {
      "id": "1X5xx00000000OECAY",
      "ratingFrequencyPolicy": {
        "id": null,
        "negotiable": null
      },
      "usageAggregationPolicy": {
        "id": null,
        "negotiable": null
      },
      "usageCommitmentPolicy": {
        "id": "7Pexx0000004C92CAE",
        "negotiable": "Non-Negotiable"
      },
      "usageOveragePolicy": {
        "id": null,
        "negotiable": null
      }
    },
    "resourcePolicyDetail": {
      "id": "7Suxx0000004C9kCAE",
      "ratingFrequencyPolicy": {
        "id": null,
        "negotiable": null
      },
      "usageAggregationPolicy": {
        "id": null,
        "negotiable": null
      },
      "usageCommitmentPolicy": {
        "id": "7Pexx0000004C92CAE",
        "negotiable": "Non-Negotiable"
      },
      "usageOveragePolicy": {
        "id": null,
        "negotiable": null
```

```
            }
          }
        },
        "usageResourceId": "1BRxx0000004C9CGAU"
      }
    ]
}
```

This sample response shows negotiated rates when you retrieve the usage details of an order item. The negotiated rates are derived from these objects for assets, order items, or quote line items.

- AssetRateCardEntry
- AssetRateAdjustment
- OrderItemRateCardEntry
- OrderItemRateAdjustment
- QuoteLineRateCardEntry
- QuoteLineRateAdjustment

```
{
  "records": [
    {
      "bindingInstanceTargetType": "Product",
      "bindingInstanceType": "Target",
      "chargeForOverages": "Yes",
      "isOptional": false,
      "fields": {},
      "name": "Paddle Board",
      "negotiable": "Negotiable,Non-Negotiable",
      "negotiatedRate": 20,
      "negotiatedRateCardEntryId": "1ELxx0000004C9JGAU,1ELxx0000004C9KGAU",
      "quantity": 15,
      "rate": 5,
      "rateAdjustments": [
        {
          "fields": {},
          "lowerBound": 0,
          "name": "Tier 1",
          "negotiatedRateAdjustmentId": "1ENxx0000004C9BGAU",
          "rateAdjustmentId": "1ENxx0000004C9BGAU",
          "rateAdjustmentType": "Percentage",
          "rateAdjustmentValue": 10,
          "tierUnitOfMeasure": "USD",
          "upperBound": 50
        }
      ],
      "rateCardEntryId": "1CJxx0000004C9IGAU,1CJxx0000004C9JGAU",
      "rateUnitOfMeasureName": "USD",
      "unitOfMeasure": "GB",
      "usageResourceGrantAndPolicyDetail": {
        "grantDetail": {
          "grantType": "Grant",
          "id": "1BXxx0000004C91GAE",
          "quantity": 100,
```

```
    "usageGrantNegotiable": "Negotiable",
    "usageRefreshPolicy": {
      "id": "1BYxx0000004C92GAE",
      "negotiable": "Non-Negotiable"
    },
    "usageRolloverPolicy": {
      "id": "1BVxx0000004C93GAE",
      "negotiable": "Non-Negotiable"
    },
    "validityPeriodTerm": 1,
    "validityPeriodUnit": "Month"
  },
  "negotiatedGrantDetail": {
    "grantType": "Grant",
    "id": "1X6xx00000000OECAY",
    "quantity": 100,
    "usageGrantNegotiable": "Negotiable",
    "usageRefreshPolicy": {
      "id": "1BYxx0000004C92GAE",
      "negotiable": "Non-Negotiable"
    },
    "usageRolloverPolicy": {
      "id": "1BVxx0000004C93GAE",
      "negotiable": "Non-Negotiable"
    },
    "validityPeriodTerm": 1,
    "validityPeriodUnit": "Month"
  },
  "negotiatedResourcePolicyDetail": {
    "id": "1X5xx00000000OECAY",
    "ratingFrequencyPolicy": {
      "id": null,
      "negotiable": null
    },
    "usageAggregationPolicy": {
      "id": null,
      "negotiable": null
    },
    "usageCommitmentPolicy": {
      "id": "7Pexx0000004C92CAE",
      "negotiable": "Non-Negotiable"
    },
    "usageOveragePolicy": {
      "id": null,
      "negotiable": null
    }
  },
  "resourcePolicyDetail": {
    "id": "7Suxx0000004C9kCAE",
    "ratingFrequencyPolicy": {
      "id": null,
      "negotiable": null
    },
    "usageAggregationPolicy": {
```

```
            "id": null,
            "negotiable": null
          },
          "usageCommitmentPolicy": {
            "id": "7Pexx0000004C92CAE",
            "negotiable": "Non-Negotiable"
          },
          "usageOveragePolicy": {
            "id": null,
            "negotiable": null
          }
        }
      },
      "usageResourceId": "1BRxx0000004C9CGAU"
    }
  ]
}
```


This sample response shows details of the custom fields when you retrieve the usage details of an order item.


```
{
  "records": [
    {
      "fields": {
        "MyCustomDate__c": {
          "displayValue": "2024-09-24",
          "value": "2024-09-24T17:46:30.662Z"
        },
        "MyCustomNumber__c": {
          "displayValue": "20.0",
          "value": 20
        }
      },
      "isOptional": false,
      "bindingInstanceTargetType": "Product",
      "bindingInstanceType": "Target",
      "chargeForOverages": "Yes",
      "name": "Therapy",
      "negotiable": "Negotiable,Non-Negotiable",
      "negotiatedRate": 20,
      "negotiatedRateCardEntryId": "1ELxx0000004C9JGAU,1ELxx0000004C9KGAU",
      "quantity": 15,
      "rate": 5,
      "rateAdjustments": [
        {
          "fields": {
            "MyCustomString__c": {
              "displayValue": "My Custom String",
              "value": "MyCustomString"
            }
          },
          "lowerBound": 0,
          "name": "Tier 1",
          "negotiatedRateAdjustmentId": "1ENxx0000004C9BGAU",
          "rateAdjustmentId": "1ENxx0000004C9BGAU",
```

```
        "rateAdjustmentType": "Percentage",
        "rateAdjustmentValue": 10,
        "tierUnitOfMeasure": "USD",
        "upperBound": 50
      }
    ],
    "rateCardEntryId": "1CJxx0000004C9IGAU,1CJxx0000004C9JGAU",
    "rateUnitOfMeasureName": "USD",
    "unitOfMeasure": "GB",
    "usageResourceGrantAndPolicyDetail": {
      "grantDetail": {
        "grantType": "Grant",
        "id": "1BXxx0000004C9lGAE",
        "quantity": 100,
        "usageGrantNegotiable": "Negotiable",
        "usageRefreshPolicy": {
          "id": "1BYxx0000004C92GAE",
          "negotiable": "Non-Negotiable"
        },
        "usageRolloverPolicy": {
          "id": "1BVxx0000004C93GAE",
          "negotiable": "Non-Negotiable"
        },
        "validityPeriodTerm": 1,
        "validityPeriodUnit": "Month"
      },
      "negotiatedGrantDetail": {
        "grantType": "Grant",
        "id": "1X6xx00000000OECAY",
        "quantity": 100,
        "usageGrantNegotiable": "Negotiable",
        "usageRefreshPolicy": {
          "id": "1BYxx0000004C92GAE",
          "negotiable": "Non-Negotiable"
        },
        "usageRolloverPolicy": {
          "id": "1BVxx0000004C93GAE",
          "negotiable": "Non-Negotiable"
        },
        "validityPeriodTerm": 1,
        "validityPeriodUnit": "Month"
      },
      "negotiatedResourcePolicyDetail": {
        "id": "1X5xx00000000OECAY",
        "ratingFrequencyPolicy": {
          "id": null,
          "negotiable": null
        },
        "usageAggregationPolicy": {
          "id": null,
          "negotiable": null
        },
        "usageCommitmentPolicy": {
          "id": "7Pexx0000004C92CAE",
```

```
            "negotiable": "Non-Negotiable"
          },
          "usageOveragePolicy": {
            "id": null,
            "negotiable": null
          }
        },
        "resourcePolicyDetail": {
          "id": "7Suxx0000004C9kCAE",
          "ratingFrequencyPolicy": {
            "id": null,
            "negotiable": null
          },
          "usageAggregationPolicy": {
            "id": null,
            "negotiable": null
          },
          "usageCommitmentPolicy": {
            "id": "7Pexx0000004C92CAE",
            "negotiable": "Non-Negotiable"
          },
          "usageOveragePolicy": {
            "id": null,
            "negotiable": null
          }
        }
      },
      "usageResourceId": "1BRxx0000004C9CGAU"
    }
  ]
}
```

This sample response shows values for the type and target type of a binding instance if the `BindingInstanceTargetId` value is available in the QuoteLineItem, OrderItem, and AssetStatePeriod objects. Additionally, the rate, negotiated rates, and rate card entry values show the associated details of the binding object.

```
{
  "records": [
    {
      "bindingInstanceTargetType": "Account",
      "bindingInstanceType": "Target",
      "chargeForOverages": "Yes",
      "fields": {},
      "isOptional": false,
      "name": "Data Transfers",
      "negotiable": "Non-Negotiable,Non-Negotiable",
      "negotiatedRate": null,
      "negotiatedRateCardEntryId": "1QNxx0000004CQmGAM,1QNxx0000004CLwGAM",
      "quantity": 250,
      "rate": 0.098765,
      "rateAdjustments": [
        {
          "fields": {},
          "lowerBound": 0,
          "name": null,
```

```
        "negotiatedRateAdjustmentId": null,
        "rateAdjustmentId": "1Soxx0000004CAeCAM",
        "rateAdjustmentType": "Percentage",
        "rateAdjustmentValue": 0,
        "tierUnitOfMeasure": "USD",
        "upperBound": 121
      }
    ],
    "rateCardEntryId": "1CJxx0000004C9CGAU,1CJxx0000004C9GGAU",
    "rateUnitOfMeasureName": "USD",
    "unitOfMeasure": "GB",
    "usageResourceGrantAndPolicyDetail": {
      "grantDetail": {
        "grantType": "Grant",
        "id": "1BXxx0000004C9lGAE",
        "quantity": 100,
        "usageGrantNegotiable": "Negotiable",
        "usageRefreshPolicy": {
          "id": "1BYxx0000004C92GAE",
          "negotiable": "Non-Negotiable"
        },
        "usageRolloverPolicy": {
          "id": "1BVxx0000004C93GAE",
          "negotiable": "Non-Negotiable"
        },
        "validityPeriodTerm": 1,
        "validityPeriodUnit": "Month"
      },
      "negotiatedGrantDetail": {
        "grantType": "Grant",
        "id": "1X6xx00000000OECAY",
        "quantity": 100,
        "usageGrantNegotiable": "Negotiable",
        "usageRefreshPolicy": {
          "id": "1BYxx0000004C92GAE",
          "negotiable": "Non-Negotiable"
        },
        "usageRolloverPolicy": {
          "id": "1BVxx0000004C93GAE",
          "negotiable": "Non-Negotiable"
        },
        "validityPeriodTerm": 1,
        "validityPeriodUnit": "Month"
      },
      "negotiatedResourcePolicyDetail": {
        "id": "1X5xx00000000OECAY",
        "ratingFrequencyPolicy": {
          "id": null,
          "negotiable": null
        },
        "usageAggregationPolicy": {
          "id": null,
          "negotiable": null
        },
```

```
            "usageCommitmentPolicy": {
              "id": "7Pexx0000004C92CAE",
              "negotiable": "Non-Negotiable"
            },
            "usageOveragePolicy": {
              "id": null,
              "negotiable": null
            }
          },
          "resourcePolicyDetail": {
            "id": "7Suxx0000004C9kCAE",
            "ratingFrequencyPolicy": {
              "id": null,
              "negotiable": null
            },
            "usageAggregationPolicy": {
              "id": null,
              "negotiable": null
            },
            "usageCommitmentPolicy": {
              "id": "7Pexx0000004C92CAE",
              "negotiable": "Non-Negotiable"
            },
            "usageOveragePolicy": {
              "id": null,
              "negotiable": null
            }
          }
        },
        "usageResourceId": "1BRxx0000004C9CGAU"
      }
    ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
errors	Usage Detail Error Response[]	List of errors encountered during the processing of the API request.	Small, 63.0	63.0
records	Rate Card Entry[]	List of rate card entry records.	Small, 63.0	63.0

Usage Details Error Response

Output representation of the details of an error related to usage details.

JSON Example

```
{
   "errors": [
        {
            "referenceId": "MyOrderItem",
            "errorCode": "INVALID_API_INPUT",
            "message": "Something has failed"
```

```
        }
    ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
errorCode	String	Code associated with the error.	Small, 63.0	63.0
message	String	Message associated with the error.	Small, 63.0	63.0
referenceId	String	Unique ID that's associated with the specific error for tracking and reference purposes.	Small, 63.0	63.0

Usage Product Activation

Output representation of a usage product activation response.

JSON example


```
{
  "success": true,
  "errors": null,
  "activatedProducts": [
    "01txx0000006i2hAAB"
  ],
  "activationResults": {
    "productActivationResults": [
      {
        "productId": "01txx0000006i2hAAA",
        "errors": [
          {
            "productUsageResourceId": "0iUxx000000009",
            "usageResourceId": "0hUxx000000003",
            "message": "Usage Resource not found for product",
            "objectApiName": "ProductUsageResource",
            "fieldName": "UsageResourceId",
            "recordId": null,
            "recordName": null
          },
          {
            "productUsageResourceId": "0iUxx0000000678",
            "usageResourceId": "0hUxx000000004",
           "message": "Related Unit of measure records is inactive, activate it first.",

            "objectApiName": "ProductUsageGrant",
            "fieldName": "DefaultUnitofMeasure",
            "recordId": "1BXSM000000404f4AA",
            "recordName": "PUG-000000001"
          }
        ]
      }
    ],
    "productValidationResults": [
```

```
      {
        "productId": "01txx0000006i2gAAA",
        "validationResult": {
          "validationErrors": [
            {
              "ruleName": "Usage vs Rating Effectivity",
              "errors": [
                {
                  "errorCode": "EFFECTIVITY_MISMATCH",
                  "errorMessages": [
                    {
                      "errorMessage": "PUR and RCE effective date ranges must overlap
for proper rating functionality.",
                      "errorDetails": [
                        {
                          "relatedObjectAPIName": "ProductUsageResource",
                          "records": [
                            {
                              "id": "0Wnxx0000004CDEAA2",
                              "name": "API Calls PUR"
                            }
                          ]
                        },
                        {
                          "relatedObjectAPIName": "RateCardEntry",
                          "records": [
                            {
                              "id": "0Rcxx0000004FGHAA2",
                              "name": "API Calls Rate Entry"
                            }
                          ]
                        }
                      ]
                    }
                  ]
                }
              ]
            }
          ],
          "validationWarnings": []
        }
      }
    ]
  }
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`activated Products`	String[]	ID of the product whose associated usage records were activated. The array contains at most one entry.	Big, 67.0	67.0

Property Name	Type	Description	Filter Group and Version	Available Version
activation Results	Usage Activation Result	Activation and validation outcome for the product in the request.	Big, 67.0	67.0
errors	Generic Error[]	List of top-level API errors. For example, a guardrail violation or an invalid request.	Big, 67.0	67.0
success	Boolean	Indicates whether the API request was processed (`true`) or rejected at the top level (`false`).	Big, 67.0	67.0

Usage Product Validation

Output representation of all the performed validations.

JSON example

This example shows the validation output with errors and warnings.

```
{
  "errors": [
    {
      "errorCode": "VALIDATION_FAILED",
      "message": "Product validation completed with cross-entity errors",
      "products": [
        {
          "productId": "01txx0000006i2gAAA",
          "validationResult": {
            "validationErrors": [
              {
                "errors": [
                  {
                    "errorCode": "EFFECTIVITY_MISMATCH",
                    "errorMessages": [
                      {
                        "errorMessage": "PUR and RCE effective date ranges must have
overlap for proper rating functionality",
                        "errorDetails": [
                          {
                            "relatedObjectAPIName": "ProductUsageRule",
                            "records": [
                              {
                                "id": "a0bxx0000004CqZAAU",
                                "name": "PUR-001"
                              }
                            ]
                          }
                        ]
                      }
                    ]
                  }
                ],
                "ruleName": "Usage vs Rating Effectivity"
```

```
            }
          ],
          "validationWarnings": []
        }
      }
    ]
  }
],
"success": false
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`success`	Boolean	Indicates whether the product validation passed (`true`) or failed (`false`).	Big, 66.0	66.0
`errors`	Error Details[]	List of top-level error details when validation fails.	Big, 66.0	66.0

Usage Resource Grant And Policy Detail

Output representation of the details of a usage resource grant and policy.

JSON Example

```
{
  "usageResourceGrantAndPolicyDetail": {
    "grantDetail": {
      "grantType": "Grant",
      "id": "1BXxx0000004C91GAE",
      "quantity": 100,
      "usageGrantNegotiable": "Negotiable",
      "usageRefreshPolicy": {
        "id": "1BYxx0000004C92GAE",
        "negotiable": "Non-Negotiable"
      },
      "usageRolloverPolicy": {
        "id": "1BVxx0000004C93GAE",
        "negotiable": "Non-Negotiable"
      },
      "validityPeriodTerm": 1,
      "validityPeriodUnit": "Month"
    },
    "negotiatedGrantDetail": {
      "grantType": "Grant",
      "id": "1X6xx00000000OECAY",
      "quantity": 100,
      "usageGrantNegotiable": "Negotiable",
      "usageRefreshPolicy": {
        "id": "1BYxx0000004C92GAE",
        "negotiable": "Non-Negotiable"
      },
      "usageRolloverPolicy": {
```

```
          "id": "1BVxx0000004C93GAE",
          "negotiable": "Non-Negotiable"
        },
        "validityPeriodTerm": 1,
        "validityPeriodUnit": "Month"
      },
      "negotiatedResourcePolicyDetail": {
        "id": "1X5xx00000000OECAY",
        "ratingFrequencyPolicy": {
          "id": null,
          "negotiable": null
        },
        "usageAggregationPolicy": {
          "id": null,
          "negotiable": null
        },
        "usageCommitmentPolicy": {
          "id": "7Pexx0000004C92CAE",
          "negotiable": "Non-Negotiable"
        },
        "usageOveragePolicy": {
          "id": null,
          "negotiable": null
        }
      },
      "resourcePolicyDetail": {
        "id": "7Suxx0000004C9kCAE",
        "ratingFrequencyPolicy": {
          "id": null,
          "negotiable": null
        },
        "usageAggregationPolicy": {
          "id": null,
          "negotiable": null
        },
        "usageCommitmentPolicy": {
          "id": "7Pexx0000004C92CAE",
          "negotiable": "Non-Negotiable"
        },
        "usageOveragePolicy": {
          "id": null,
          "negotiable": null
        }
      }
    }
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
grantDetail	Grant Detail	Details about the grants from the ProductUsageGrant object.	Big, 65.0	65.0

Property Name	Type	Description	Filter Group and Version	Available Version
negotiated GrantDetail	Grant Detail	Details about the negotiated grants from the LineItemUsageResourceGrant or TransactionUsageEntitlement object.	Big, 65.0	65.0
negotiated Resource PolicyDetail	Resource Policy Detail	Details about the policy from the LineItemUsageResourcePolicy or BindingObjectUsageResourcePolicy object.	Big, 65.0	65.0
resourcePolicy Detail	Resource Policy Detail	Details about the policy from the ProductUsageResourcePolicy object.	Big, 65.0	65.0

Validation Error

Output representation of the validation errors grouped by rule name.

JSON example

This example shows a validation error grouped by rule name.

```
{
  "validationErrors": [
    {
      "ruleName": "Usage vs Rating Effectivity",
      "errors": [
        {
          "errorCode": "EFFECTIVITY_MISMATCH",
          "errorMessages": []
        }
      ]
    }
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
ruleName	String	Name of the validation rule. For example, `Usage vs Rating Effectivity`.	Big, 66.0	66.0
errors	Errors[]	List of error code groups for this validation.	Big, 66.0	66.0

Validation Result

Output representation of the validation results grouped by rule name.

JSON example

This example shows a validation result with errors and warnings.

```
{
  "validationResult": {
    "validationErrors": [
```

```
        {
          "ruleName": "Usage vs Rating Effectivity",
          "errors": []
        }
      ],
      "validationWarnings": [
        {
          "ruleName": "Performance Optimization",
          "warnings": []
        }
      ]
    }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
validationErrors	Validation Error[]	List of validation errors grouped by rule name.	Big, 66.0	66.0
validationWarnings	Validation Warning[]	List of validation warnings grouped by rule name.	Big, 66.0	66.0

Validation Warning

Output representation of the validation warnings grouped by rule name.

JSON example

This example shows a validation warning grouped by rule name.

```
{
  "validationWarnings": [
    {
      "ruleName": "Performance Optimization",
      "warnings": [
        {
          "warningCode": "PERFORMANCE_SUBOPTIMAL",
          "warningMessages": []
        }
      ]
    }
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
ruleName	String	Name of the validation rule. For example, Performance Optimization.	Big, 66.0	66.0
warnings	Warnings[]	List of warning code groups for this validation.	Big, 66.0	66.0

Warnings

Output representation of a group of warning messages with the same warning code.

JSON example

This example shows a group of warning messages with the same warning code.

```
{
  "warnings": [
    {
      "warningCode": "PERFORMANCE_SUBOPTIMAL",
      "warningMessages": [
        {
          "warningMessage": "PUR and RCE date ranges could be optimized for better
performance",
          "warningDetails": []
        }
      ]
    }
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
warningCode	String	Standardized warning code. For example, PERFORMANCE_SUBOPTIMAL.	Big, 66.0	66.0
warningMessages	Warning Message[]	List of warning messages for records that triggered with this warning.	Big, 66.0	66.0

Warning Message

Output representation of the details of records that triggered this specific warning.

JSON example

This example shows a warning message with details.

```
{
  "warningMessages": [
    {
      "warningMessage": "PUR and RCE date ranges could be optimized for better
performance",
      "warningDetails": [
        {
          "relatedObjectAPIName": "ProductUsageRule",
          "records": [
            {
              "id": "a0bxx0000004CqZAAU",
              "name": "PUR-001"
            }
          ]
        }
      ]
```

```
        }
    ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`warningMessage`	String	Human-readable warning message that describes the validation concern.	Big, 66.0	66.0
`warningDetails`	Error Warning Details[]	Details of records that triggered this specific warning.	Big, 66.0	66.0

Usage Management Metadata API Types

Metadata API enables you to access some types and feature settings that you can customize in the user interface.

Flow for Usage Management
Represents the metadata associated with a flow. With Flow, you can create an application that navigates users through a series of screens to query and update records in the database. You can also execute logic and provide branching capability based on user input to build dynamic applications.

IndustriesUsageSettings
Represents the settings for Usage Management.

SEE ALSO:

Metadata API Developer Guide: Understanding Metadata API

Flow for Usage Management

Represents the metadata associated with a flow. With Flow, you can create an application that navigates users through a series of screens to query and update records in the database. You can also execute logic and provide branching capability based on user input to build dynamic applications.

FlowActionCall

Usage Management exposes additional actionType values for the FlowActionCall Metadata type.

Field Name	Field Type	Description
`actionType`	InvocableActionType (enumeration of type string)	Required. The action type. Additional valid values only for Usage Management include: • `invokeSummaryCreationService`—Invoke the service that creates various summaries, such as usage, ratable, and liable summaries where the usage amount is zero. The service also checks

Field Name	Field Type	Description
		and updates the billing period of the usage entitlement account if the billing period is expired. • `processConsumptionOverages`—Process consumption overages for the usage summary records with `SummaryComplete` status. This action uses the entitlement service to process the overages. • `refreshUsageEntitlementBucket`—Refresh entitlements by evaluating the usage entitlement bucket records and creating a new usage entitlement entry. • `retriggerEntlCreaProc`—Retrigger entitlement creation process for failed or unprocessed assets.

IndustriesUsageSettings

Represents the settings for Usage Management.

Parent Type and Manifest Access

This type extends the Metadata metadata type and inherits its `fullName` field.

In the package manifest, all the settings metadata types for the org are accessed using the "Settings" name. See Settings for more details.

File Suffix and Directory Location

`IndustriesUsageSettings` values are stored in the `IndustriesUsage.settings` file in the `settings` folder. The `.settings` files are different from other named components, because there is only one settings file for each settings component.

Version

IndustriesUsageSettings components are available in API version 63.0 and later.

Fields

Field Name	Description
`enableUsage`	**Field Type** boolean **Description** Indicates whether to create and access Usage Management objects to manage product usage and determine rates based on usage (`true`) or not (`false`). The default value is `false`.

Declarative Metadata Sample Definition

The following is an example of an IndustriesUsageSettings component.

```
<?xml version="1.0" encoding="UTF-8"?>
<IndustriesUsageSettings xmlns="http://soap.sforce.com/2006/04/metadata">
    <enableUsage>true</enableUsage>
</IndustriesUsageSettings>
```

The following is an example `package.xml` that references the previous definition.

```
<?xml version="1.0" encoding="UTF-8"?>
<Package xmlns="http://soap.sforce.com/2006/04/metadata">
    <types>
        <members>IndustriesUsage</members>
        <name>Settings</name>
    </types>
    <version>[ftest]</version>
</Package>
```

Wildcard Support in the Manifest File

The wildcard character `*` (asterisk) in the `package.xml` manifest file doesn't apply to metadata types for feature settings. The wildcard applies only when retrieving all settings, not for an individual setting. For details, see Settings. For information about using the manifest file, see Deploying and Retrieving Metadata with the Zip File.

CHAPTER 12 Billing

In this chapter ...

Automate processes related to billing, credit application, and invoice generation. Generate billing schedules by using context service, and create invoices from a billing schedule.

SEE ALSO:

Salesforce Help: Assign Permissions to Access Billing Features

EDITIONS

Available in: Lightning Experience

Available in: **Enterprise**, **Unlimited**, and **Developer** Editions

Billing Standard Objects

The Billing data model provides objects and fields to manage billing and tax configurations, credit memos, and invoices, and accounting periods for legal entities.

AccountingPeriod

Represents information about a time period for which businesses prepare reports and analyze performance. Each billing transaction is associated with an accounting period. This object is available in API version 62.0 and later.

BillingArrangement

Represents the arrangement for invoicing a transaction's billing amount to one or more accounts. The arrangement specifies whether the total amount must be invoiced to the owning account or a different billing account, or whether the invoices must be split among multiple billing accounts. This object is available in API version 66.0 and later.

BillingArrangementLine

Represents the billing account, billing profile, and the percentage of billing amount to be invoiced. Each billing arrangement line results in a separate invoice addressed to the selected billing account. This object is available in API version 66.0 and later.

BillingBatchScheduler

Represents a scheduled processing job that triggers recurring invoice batch runs. This object is available in API version 62.0 and later.

BillingBatchFilterCriteria

Represents the filter that all eligible billing schedules must satisfy in order to be picked up by an invoice run. This object is available in API version 62.0 and later.

BillingMilestonePlan

Represents a structured approach to invoicing where invoices are scheduled based on predefined milestones. This object is available in API version 63.0 and later.

BillingMilestonePlanItem

Represents a specific billing milestone within the billing milestone plan that's used to manage and track billing based on the completion of certain deliverables or stages. This object is available in API version 63.0 and later.

BillingPeriodItem

Represents a payment period for an invoice. The billing period item is used to pass billing information to an invoice line. This object is available in API version 62.0 and later.

BillingPolicy

Represents information about a set of billing treatments that define the rules to invoice a customer for an order item. This object is available in API version 62.0 and later.

BillingSchedule

Represents information about the order item that's used in the invoicing process. This object is available in API version 62.0 and later.

BillingScheduleGroup

Represents a consolidated view of all the billing schedules related to the order items generated from one asset, including new orders and amendment orders. This object is available in API version 62.0 and later.

BillingTreatment

Represents information about the billing of an order item. This object is available in API version 62.0 and later.

BillingTreatmentItem

Represents information about allocation of the total amount of an order item to billing schedules throughout the order item's lifecycle. This object is available in API version 62.0 and later.

BsgRelationship

Represents a relationship between billing schedule groups to support bundles where one parent billing schedule group has multiple child billing schedule groups. This object is available in API version 62.0 and later.

CaseServiceProcessExtension

Represents additional information for a case record based on the type of service process for which the case was raised. This object is available in API version 67.0 and later.

CreditMemo

Represents a document that's used to reduce the amount that a buyer owes a seller under the terms of an earlier invoice. This object is available in API version 62.0 and later.

CreditMemoAddressGroup

Represents the storage of the buyer's address information, which is used to determine the tax credit amount for a buyer when a credit memo is issued. This object is available in API version 62.0 and later.

CreditMemoInvApplication

Represents information about the application of a credit memo to an invoice. This object is available in API version 62.0 and later.

CreditMemoLine

Represents the product, service, adjustment, or tax line items included in a credit memo. This object is available in API version 62.0 and later.

CreditMemoLineInvoiceLine

Represents a junction between a credit memo line and an invoice line. This object is available in API version 62.0 and later.

CreditMemoLineTax

Represents tax information of a credit memo line of type `Tax`. This object is available in API version 62.0 and later.

DebitMemo

Represents the document used to charge an additional amount to a buyer by a seller. An invoice is generated for the debit memo in the next invoice run. This object is available in API version 65.0 and later.

DebitMemoAddress

Represents the buyer's address information, which is used to determine the tax amount for a buyer when a debit memo is issued. This object is available in API version 65.0 and later.

DebitMemoLine

Represents the additional charge amount that the buyer must pay for the product, service, or debit memo line tax that's related to the debit memo. This object is available in API version 65.0 and later.

DebitMemoLineTax

Represents the tax information for a debit memo line. This object is available in API version 66.0 and later.

GeneralLedgerAccount

Represents information about the accounting codes, types, and names that are used to store and organize financial transactions. This object is available in API version 63.0 and later.

GeneralLedgerAcctAsgntRule

Represents information about the rule based on which general ledger accounts are assigned to transaction journals that are created for billing transactions. This object is available in API version 63.0 and later.

GeneralLdgrAcctPrdSummary

Represents a junction between a general ledger account and a legal entity accounting period. Stores information about the total credit amount, total debit amount, opening balance, and closing balance of a general ledger account for a specific legal entity accounting period. This object is available in API version 65.0 and later.

GeneralLedgerJrnlEntryRule

Represents information about the transaction journal entry rule, based on which transaction journals are created for the selected credit and debit general ledger accounts, transaction amount field, and percentage. This object is available in API version 65.0 and later.

InvBatchDraftToPostedRun

Represents information about the batch job that posts all invoices with the status as `Draft` that are generated by the invoice batch run associated with the billing schedule. This object is available in API version 62.0 and later.

Invoice

Represents information about a financial document describing the total amount a buyer must pay for provided products or services. This object is available in API version 62.0 and later.

InvoiceAddressGroup

Represents the storage of the buyer's address information. This object is available in API version 62.0 and later.

InvoiceBatchRun

Represents a batch processing job in Billing. During an invoice batch run, all billing schedules that meet the specified criteria are processed, resulting in the generation of invoices. This object is available in API version 62.0 and later.

InvoiceBatchRunCriteria

Represents a batch processing job and its required criteria in Billing. During an invoice batch run, all billing schedules that meet the specified criteria are processed, resulting in the generation of invoices. This object is available in API version 62.0 and later.

InvoiceBatchRunRecovery

Represents information about the recovery procedure of an invoice batch run. This object is available in API version 62.0 and later.

InvoiceDocument

Represents the PDF document generated for an invoice. This object is available in API version 63.0 and later.

InvoiceLine

Represents the amount that a buyer must pay for a product, service, or fee. Invoice lines are created based on the amount of an order line. This object is available in API version 62.0 and later.

InvoiceLineRelationship

Represents a relationship between invoice line items to support bundles where one parent invoice line has multiple child invoice lines. This object is available in API version 62.0 and later.

InvoiceLineTax

Represents tax information of an invoice line of type `Tax`. This object is available in API version 62.0 and later.

LegalEntity

Represents the way an organization is structured. An organization can be a single legal entity or it can comprise more than one legal entity. This object is available in API version 62.0 and later.

LegalEntyAccountingPeriod

Represents a junction between a legal entity and an accounting period. This object is available in API version 62.0 and later.

PaymentBatchRun

Represents a batch processing job that processes payments in Billing. During a payment batch run, all the payment schedules that meet the specified criteria are processed and the corresponding Payment records are created. These payments are then applied to invoices or invoice lines. This object is available in API version 64.0 and later.

PaymentLineInvoiceLine

Represents information about a payment line that's applied to or unapplied from an invoice line. This object is available in API version 64.0 and later.

PaymentRetryRule

Represents the specific payment retry rule for a failed payment schedule item. Each rule defines actionable parameters such as the maximum number of retries for the failed records and time intervals between subsequent retry attempts. This object is available in API version 66.0 and later.

PaymentRetryRuleSet

Represents the payment retry rule definition that defines how failed payments are retried based on the error codes across various retry categories. This object is available in API version 66.0 and later.

PaymentSchedule

Represents information about a set of payments that a customer wants to collect at different times for a certain record. A schedule contains one or more payment schedule items, where each item represents one payment to be processed. Each of a schedule's items can have different payment configuration fields, such as payment methods, payment dates, and payment accounts. When a payment scheduler launches a payment run, the run evaluates active payment schedule items, and picks them up for payment processing if they match the scheduler's payment criteria. This object is available in API version 64.0 and later.

PaymentSchedulePolicy

Represents information about the configuration for the payment schedule. This object is available in API version 64.0 and later.

PaymentScheduleTreatment

Represents information about the processing of payment schedules including the payment method and the payment amount for the payment schedule. This object is available in API version 64.0 and later.

PaymentScheduleTreatmentDtl

Represents information about the processing of payment schedules after the corresponding invoices are posted. This object is available in API version 64.0 and later.

PymtSchdDistributionMethod

Represents information about the partial payments that the total payment is divided into. This object is available in API version 64.0 and later.

PaymentScheduleItem

Represents information about a payment to be processed. Each schedule item can have different payment configuration fields, such as payment methods, payment dates, and payment accounts. When a payment scheduler launches a payment run, the run evaluates active payment schedule items, and picks them up for payment processing if they match the scheduler's payment criteria. This object is available in API version 64.0 and later.

PaymentTerm

Represents an agreement between a buyer and a seller about when payment is due for an invoice. This object is available in API version 62.0 and later.

PaymentTermItem

Represents configuration of a payment term. This object is available in API version 62.0 and later.

RevenueTransactionErrorLog

Represents the details of errors that occurred during the processing of a request. The error record persists until a new error with the same category, primary record, and, if necessary, related record occurs. This object is available in API version 62.0 and later.

SeqPolicySelectionCondition

Represents the condition used to determine which sequence policy is applied to a record. This object is available in API version 65.0 and later.

SequenceGapReconciliation

Represents a missing sequence value identified during reconciliation, which can be used later to ensure there are no gaps in the sequence policy numbers. This object is available in API version 65.0 and later.

SequencePolicy

Represents the configuration of rules and parameters for generating unique, sequential numbers for records. Stores settings such as numbering patterns, prefixes, suffixes, sequence start numbers, increment values, and filter criteria to ensure accurate and compliant numbering. This object is available in API version 65.0 and later.

TaxEngine

Represents information about an instance of a tax engine provider as well as the merchant credentials for that specific instance. This object is available in API version 62.0 and later.

TaxEngineInteractionLog

Represents a record of a communication with an external tax engine following a tax calculation request. This object is available in API version 62.0 and later.

TaxEngineProvider

Represents general information about a service that manages a tax engine. Tax engine providers have a one-to-many relationship with tax engines, where the tax engine record represents a specific configuration of a tax engine that can be assigned to multiple order items. This object is available in API version 62.0 and later.

TaxPolicy

Represents information about a group of tax treatments, where each treatment represents parameters to determine how a particular product is taxed for a transaction line item. Tax policies are related to products, which pass the policy on to the resulting order items and in turn the billing schedules. This object is available in API version 62.0 and later.

TaxTreatment

Represents information about tax calculation by external engines. Each product requires a tax policy to determine whether to apply tax. Each tax policy requires at least one tax treatment. The tax treatments determine how taxable products are taxed. This object is available in API version 62.0 and later.

Tax Treatment Item

Represents tax code information that's used to calculate tax for a product by a specific tax engine. This object is available in API version 66.0 and later.

SEE ALSO:

Object Reference for the Salesforce Platform: Overview of Salesforce Objects and Fields

SOAP API Developer Guide: Introduction to SOAP API

AccountingPeriod

Represents information about a time period for which businesses prepare reports and analyze performance. Each billing transaction is associated with an accounting period. This object is available in API version 62.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need the Accounts Receivables Admin permission set to access this object.

Fields

Field	Details
EndDate	**Type** date **Properties** Create, Filter, Group, Sort, Update **Description** Required. The end date of an accounting period.
FinancialYear	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** Required. The financial year in which an accounting period falls.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed an accounting period indirectly, for example, through a list view or related record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed an accounting period. If this value is null, it’s possible that the user only accessed the accounting period or a related list view (LastReferencedDate), but not viewed the accounting period itself.
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** Required. The auto-generated name of an accounting period. The name is a combination of the accounting period's start date, start month, end date, and end month.

Field	Details
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Required. The user who owns an Accounting Period record. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
StartDate	**Type** date **Properties** Create, Filter, Group, Sort, Update **Description** Required. The start date of an accounting period.
Status	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. The status of an accounting period. Valid values are: • Closed • Open
TotalAssetsAmount	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The sum of all the assets from legal entity accounting periods associated with the accounting period. This field is available in API version 65.0 and later.
TotalEquitiesAmount	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update

Field	Details
	Description The sum of all the equities from legal entity accounting periods associated with the accounting period. This field is available in API version 65.0 and later.
`TotalExpensesAmount`	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The sum of all the expenses from legal entity accounting periods associated with the accounting period. This field is available in API version 65.0 and later.
`TotalLiabilitiesAmount`	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The sum of all the liabilities from legal entity accounting periods associated with the accounting period. This field is available in API version 65.0 and later.
`TotalRevenueAmount`	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The sum of all the revenue from legal entity accounting periods associated with the accounting period. This field is available in API version 65.0 and later.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

AccountingPeriodShare **on page 2877**
Sharing is available for the object.

AccountingPeriodFeed **on page 2866**
Feed tracking is available for the object.

AccountingPeriodHistory **on page 2873**
History is available for tracked fields of the object.

BillingArrangement

Represents the arrangement for invoicing a transaction's billing amount to one or more accounts. The arrangement specifies whether the total amount must be invoiced to the owning account or a different billing account, or whether the invoices must be split among multiple billing accounts. This object is available in API version 66.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `update()`, `upsert()`

Fields

Field	Details
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the billing arrangement record was last referenced.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date the billing arrangement record was last viewed.
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** Required. A user-defined name for the billing arrangement.
`OwnerId`	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the owner of the billing arrangement record. This field is a polymorphic relationship field.

Field	Details
	Relationship Name Owner **Refers To** Group, User
RemainderBillingPercentage	**Type** double **Properties** Filter, Sort **Description** Required. The percentage of the billing amount that isn’t assigned to any of the billing accounts.
ShouldBillRemainderToAccount	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether to bill the remainder billing percentage to the owning account (`true`) or not (`false`).
Status	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. Specifies the status of the billing arrangement. Valid values are: • `Active` • `Draft` • `Inactive`
VersionNumber	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** A numerical value indicating the current version of the billing arrangement record. The version number increases when new billing arrangement line records are added or existing billing arrangement line records are updated.

BillingArrangementLine

Represents the billing account, billing profile, and the percentage of billing amount to be invoiced. Each billing arrangement line results in a separate invoice addressed to the selected billing account. This object is available in API version 66.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`AccountId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** Required. The ID of the account that's related to the billing arrangement line. This field is a relationship field. **Relationship Name** Account **Refers To** Account
`BillingAccountId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** Required. The ID of the billing account record that's related to the billing arrangement line. This field is a relationship field. **Relationship Name** BillingAccount **Refers To** BillingAccount
`BillingArrangementId`	**Type** reference

Field	Details
	Properties Create, Filter, Group, Sort **Description** Required. The ID of the billing arrangement record that's related to the billing arrangement line. This field is a relationship field. **Relationship Name** BillingArrangement **Relationship Type** Master-detail **Refers To** BillingArrangement (the master object)
`BillingArrangementLineName`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort
`BillingPercentage`	**Type** double **Properties** Create, Defaulted on create, Filter, Sort, Update **Description** Required. The percentage of the billing amount to be assigned to the billing account of the billing arrangement line.
`ShouldBillRemainder`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the remainder percentage must be billed to the billing account of the billing arrangement line (`true`) or not (`false`). The default value is `false`.
`VersionNumber`	**Type** int **Properties** Filter, Group, Nillable, Sort

Field	Details
	Description A numerical value indicating the current version of the billing arrangement record. The version number increases when new billing arrangement line records are added or existing billing arrangement line records are updated.

BillingBatchScheduler

Represents a scheduled processing job that triggers recurring invoice batch runs. This object is available in API version 62.0 and later.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`

Special Access Rules

You need the Billing Admin permission set to access this object.

Fields

Field	Details
`BillingSchedulerName`	**Type** string **Properties** Filter, Group, idLookup, Sort **Description** Required. The name of the scheduler.
`Comments`	**Type** textarea **Properties** Filter, Nillable, Sort **Description** Additional details about the billing batch scheduler.
`CronExpression`	**Type** string **Properties** Filter, Group, Sort **Description** Required. This field determines how often the scheduler recurs.

Field	Details
EndDate	**Type** date **Properties** Filter, Group, Nillable, Sort **Description** The date when the scheduler stops triggering batch processing jobs.
FrequencyCadence	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** Required. The frequency of the scheduler triggering the invoice batch run. Valid values are: • `Daily`— The scheduled job recurs every day. • `Monthly`— The scheduled job recurs every month. • `Once`— The scheduled job occurs one time and doesn't recur. • `Weekly`— The scheduled job recurs every week.
FrequencyOptions	**Type** textarea **Properties** Nillable **Description** This field is a derived field that stores the scheduler configuration.
JobType	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** Required. The type of batch processing job that the scheduler triggers. Valid value is `Invoice` for which the scheduler starts a batch invoice run.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed a billing batch scheduler record indirectly, for example, through a list view or related record.

Field	Details
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed a billing batch scheduler record. If this value is null, it's possible that the user only accessed the billing batch scheduler record or a related list view (`LastReferencedDate`), but not viewed the billing batch scheduler record itself.
NextRunTime	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date and timestamp of the next scheduled batch invoice run in the user's time zone.
OwnerId	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The ID of the user who created the scheduler. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
RecurringSubType	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The frequency at which the batch processing job recurs when the `FrequencyCadence` field value is set to `Monthly`. Valid values are: • `Every`— The processing job recurs at every instance of the frequency of the value. For example, if the `RecurringSubType` field value is `Every` and the `FrequencyCadence` field value is `Weekly`, then the batch processing job recurs every week.

Field	Details
	• `SpecificDate`— The scheduler triggers the batch processing job on the selected date. For example, if the selected date is 5, and the `FrequencyCadence` field value is `Monthly`, then the job recurs on the fifth day of each month.
`RecurringType`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The frequency at which the batch processing job is repeated when the `FrequencyCadence` field value is set to `Weekly`. Valid value is `Every`.
`RecursOn`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The interval at which the scheduler triggers a batch processing job. If the `FrequencyCadence` field value is `Monthly`, you must select either the specific date or the interval when the schedule triggers the job. Valid values are: • `First` • `Fourth` • `Last` • `Second` • `Third` **Example:** To configure the scheduler to trigger the job on the first Monday of the month, set the following fields: • `FrequencyCadence=Monthly` • `RecursOn=First` • `RecursOnDay= Monday`
`RecursOnDate`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The date on which the scheduler triggers a batch processing job.

Field	Details
	Example: To configure the scheduler to trigger the job on the fifth day of the month, set the following fields: • `FrequencyCadence=Monthly` • `RecursOnDate=` 5 **Example:** To configure the scheduler to trigger the job on the second to last day of the month, set the following fields: • `FrequencyCadence=Monthly` • `RecursOnDate=SecondToLast` If you select `Last`, `SecondToLast`, or `ThirdToLast`, the date of the batch processing job varies depending on the number of days in the month. For example, consider `SecondToLast` is selected. If the month has 30 days, such as June, then the batch processing job occurs on the 28th day. If the month has 31 days, such as July, then the batch processing job occurs on the 29th day.
`RecursOnDay`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The day on which the scheduler triggers a batch processing job. If the `FrequencyCadence` field value is set to `Weekly`, then you must select the day when the scheduler runs. The scheduler recurs every week on the selected day; for example, weekly on Monday. Valid values are: • `Sunday` • `Monday` • `Tuesday` • `Wednesday` • `Thursday` • `Friday` • `Saturday`
`RunCriteriaId`	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The ID of the filter criteria that's defined for the invoice batch run or the payment batch run. This field is a polymorphic relationship field.

Field	Details
	Relationship Name RunCriteria **Refers To** InvoiceBatchRunCriteria
`ShouldExcludeWkendAndHldy`	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Required. Indicates whether weekends and holidays are excluded from the billing schedule (`true`) or not (`false`). The default value is `false`.
`ShouldStartRunImmediately`	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Required. Indicates whether the invoice scheduler must start the run immediately (`true`) or not (`false`). The default value is `false`. Available in API version 63.0 and later.
`StartDate`	**Type** date **Properties** Filter, Group, Sort **Description** Required. The date when the scheduler triggers its first batch processing job.
`StartTime`	**Type** time **Properties** Filter, Sort **Description** Required. The time when the scheduler triggers the batch processing job.
`Status`	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort

Field	Details
	Description Required. The status of the scheduler. Only active schedulers can trigger batch processing jobs. Valid values are: • `Active` • `Canceled` • `Draft` • `Inactive`
`TimeZone`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The time zone is either the value selected when the run was configured, or it's the user's time zone. The time zone is shown in Greenwich Mean Time (GMT).

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

BillingBatchSchedulerShare on page 2877
Sharing is available for the object.

BillingBatchFilterCriteria

Represents the filter that all eligible billing schedules must satisfy in order to be picked up by an invoice run. This object is available in API version 62.0 and later.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`

Special Access Rules

You need the Billing Admin permission set to access this object.

Fields

Field	Details
`BatchCriteriaId`	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The ID of the batch criteria record. This field is a polymorphic relationship field. **Relationship Name** BatchCriteria **Refers To** InvoiceBatchRunCriteria
`BillingBatchFilterCriteriaNumber`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. The auto-generated reference number for the billing batch filter criteria.
`ColumnEnum`	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** Required. The column or field to which the filter criteria are applied.
`CriteriaFieldType`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The data type of the custom or standard criteria field. Valid values are: • `CustomBoolean` • `CustomCurrency` • `CustomDate` • `CustomLookup` • `CustomNumber`

Field	Details
	• `CustomPercent` • `CustomText` Available in API version 63.0 and later.
`CriteriaSequence`	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The order in which the filter criteria are applied on the billing batch.
`CustomCriteriaFieldName`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The API name of the custom field on the object specified in the `ObjectName` field. The filter criteria is applied on the custom field. This object is available in API version 65.0 and later.
`InvoiceRunMatchingValue`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** A value to match during an invoice run. This field is useful for filtering the invoices based on specific criteria during the billing process.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed this record indirectly, for example, through a list view or related record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort

Field	Details
	Description The timestamp when the current user last viewed this record or list view. If this value is null, and `LastReferenceDate` is not null, the user accessed this record or list view indirectly.
`ObjectName`	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** Required. The object on which the filter criteria are applied. Valid values are: • `AccountBillingAccount` — Available in API version 63.0 and later. • `BillingAccount` — Available in API version 63.0 and later. • `BillingSchedule` • `BillingScheduleGroup` • `CreditMemo` — Available in API version 63.0 and later. • `CreditMemoInvApplication` — Available in API version 63.0 and later. • `CreditMemoLine` — Available in API version 63.0 and later. • `CreditMemoLineInvoiceLine` — Available in API version 63.0 and later. • `CreditMemoLineTax` — Available in API version 63.0 and later. • `DebitMemoLine` — Available in API version 65.0 and later. • `Invoice` — Available in API version 63.0 and later. • `InvoiceLine` — Available in API version 63.0 and later. • `InvoiceLineTax` — Available in API version 63.0 and later. • `Payment` — Available in API version 64.0 and later. • `PaymentGateway` — Available in API version 64.0 and later. • `PaymentLineInvoice` — Available in API version 64.0 and later. • `PaymentLineInvoiceLine` — Available in API version 64.0 and later. • `PaymentSchedule` — Available in API version 64.0 and later. • `PaymentScheduleItem` — Available in API version 64.0 and later. • `Refund` — Available in API version 64.0 and later. • `RefundLinePayment` — Available in API version 64.0 and later.
`Operation`	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort

Field	Details
	Description Required. The type of comparison or logical operation to be performed on the specified column or field. Valid values are: • `Contains` — Available in API version 63.0 and later. • `Equals` • `GreaterThan` — Available in API version 63.0 and later. • `GreaterThanOrEqualTo` — Available in API version 63.0 and later. • `InList` • `LessThan` — Available in API version 63.0 and later. • `LessThanOrEqualTo` • `NotEquals` — Available in API version 63.0 and later. • `OfType` • `StartsWith` — Available in API version 63.0 and later.
`OwnerId`	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The ID of the user who created the billing batch filter criteria. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
`PaymentRunMatchingValue`	**Type** picklist **Properties** Filter, Group, Nillable, Sort **Description** A value to match during a payment run. This field is useful for filtering the payments based on specific criteria during the billing process. This field is visible but isn't used in API version 62.0.
`StandardCriteriaField`	**Type** string **Properties** Filter, Group, Nillable, Sort

Field	Details
	Description The name of the standard field for the object specified in the `ObjectName` field. The filter criteria is applied on the standard field. Available in API version 63.0 and later.
`Value`	**Type** string **Properties** Filter, Group, Sort **Description** Required. The value to be used in the filter criteria.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

BillingBatchFilterCriteriaHistory on page 2873
History is available for tracked fields of the object.

BillingMilestonePlan

Represents a structured approach to invoicing where invoices are scheduled based on predefined milestones. This object is available in API version 63.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `update()`, `upsert()`

Special Access Rules

You need Revenue Cloud Billing license and the Billing Admin permission set to access this object.

Fields

Field	Details
`BillingTreatmentId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** Required. The billing treatment associated with the billing milestone plan.

Field	Details
	This field is a relationship field. **Relationship Name** BillingTreatment **Refers To** BillingTreatment
Description	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Additional details about the billing milestone plan.
ExternalReference	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The external reference item that links a billing milestone plan item to the original transaction item. Available in API version 65.0 and later.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed a billing milestone plan record indirectly, for example, through a list view or related record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed a billing milestone plan record. If this value is null, it's possible that the user only accessed the billing milestone plan record or a related list view (LastReferencedDate), but not viewed the billing milestone plan record itself.
Name	**Type** string

Field	Details
	Properties Create, Filter, Group, idLookup, Sort, Update **Description** Required. The name of the billing milestone plan.
ReferenceItemAmount	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The total reference item amount to be billed.
ReferenceItemId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the reference item that links a billing milestone plan item to the original order item. This field is a polymorphic relationship field. **Relationship Name** ReferenceItem **Refers To** BillingSchedule, OrderItem
Status	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. The status of the billing milestone plan. Valid values are: • Active • Cancelled • Completely Billed • Draft

BillingMilestonePlanItem

Represents a specific billing milestone within the billing milestone plan that's used to manage and track billing based on the completion of certain deliverables or stages. This object is available in API version 63.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `update()`, `upsert()`

Special Access Rules

You need Revenue Cloud Billing license and the Billing Admin permission set to access this object.

Fields

Field	Details
`BillingMilestonePlanId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The billing milestone plan associated with the billing milestone plan item. This field is a relationship field. **Relationship Name** BillingMilestonePlan **Relationship Type** Master-detail **Refers To** BillingMilestonePlan (the master object)
`BillingScheduleGroupId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The billing schedule group that's related to the billing milestone plan item. Available in API version 64.0 and later. This field is a relationship field. **Relationship Name** BillingScheduleGroup **Refers To** BillingScheduleGroup

Field	Details
CommencementDate	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The action that triggers the start of the billing milestone plan for a date-based milestone. Valid value is: • OrderProductActivation
CommencementDateOffset	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The offset applied to the commencement date to determine the milestone achievement date for a date-based milestone.
CommencementDateOffsetUnit	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The unit of time for the commencement date offset. Valid values are: • Days • Months • Years
Description	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Additional details about the billing milestone plan item.
FlatAmount	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update

Field	Details
	Description The amount in terms of units of currency, such as $10 or $21.52, to invoice from the order item. Used only when `Type` field has a value of `FlatAmount`.
`IsMilestoneAccomplished`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Required. Indicates whether the billing treatment is for milestone billing (`true`) or not (`false`).
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed a billing milestone plan item record indirectly, for example, through a list view or related record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed a billing milestone plan item record. If this value is null, it's possible that the user only accessed the billing milestone plan item record or a related list view (`LastReferencedDate`), but not viewed the billing milestone plan item record itself.
`MilestoneAccomplishmentDate`	**Type** date **Properties** Filter, Group, Nillable, Sort **Description** The date the milestone is achieved for date-based milestones. For event-based milestones, this field indicates the date when the milestone is manually marked as completed.
`MilestoneAmount`	**Type** currency **Properties** Filter, Nillable, Sort

Field	Details
	Description The amount that's billed when the milestone is reached or completed.
MilestoneType	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The milestone type for the billing treatment item. Valid values are: • Date • Event
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** Required. The name of the billing milestone plan item.
Percentage	**Type** percent **Properties** Create, Filter, Nillable, Sort, Update **Description** The percentage,such as 10% or 12.5%, to invoice from the order item. Used only when Type field has a value of Percentage.
ServicePeriodEnd	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The end date of the service associated with the milestone. Valid value is: • Order Product End Date
ServicePeriodStart	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update

Field	Details
	Description The start date of the service associated with the milestone. Valid value is: • `Order Product Start Date`
`Status`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The status of the billing milestone plan item. Valid values are: • `Cancelled` • `Draft` • `Error` • `Invoiced` • `Ready for Invoicing` • `Waiting for Milestone Accomplishment`
`Type`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. Specifies whether billing schedules created from this billing treatment item are based on a flat amount or a percentage of the order item's total amount. Valid values are: • `FlatAmount` • `Percentage` • `Remainder`

BillingPeriodItem

Represents a payment period for an invoice. The billing period item is used to pass billing information to an invoice line. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`

Special Access Rules

You need the Billing Admin permission set to access this object.

Fields

Field	Details
`Amount`	**Type** currency **Properties** Filter, Sort **Description** Required. The price for the billing period item. This field is used to calculate the invoice line's amount.
`BillingPeriodEndDate`	**Type** date **Properties** Filter, Group, Nillable, Sort **Description** The end date of a billing period that's used to calculate the invoice line's end date.
`BillingPeriodItemNumber`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. An auto-generated number identifying the billing period item.
`BillingPeriodStartDate`	**Type** date **Properties** Filter, Group, Nillable, Sort **Description** The start date of a billing period item that's used to calculate the invoice line's start date.
`BillingScheduleGroupId`	**Type** reference

Field	Details
	Properties Filter, Group, Nillable, Sort **Description** The ID of the parent billing schedule group record that's related to the billing period item. Available in API version 64.0 and later. This field is a relationship field. **Relationship Name** BillingScheduleGroup **Refers To** BillingScheduleGroup
`BillingScheduleId`	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The ID of the parent billing schedule record of the billing period item. This field is a relationship field. **Relationship Name** BillingSchedule **Relationship Type** Master-detail **Refers To** BillingSchedule (the master object)
`InvoiceBatchRunId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the parent invoice batch run record of the billing period item. This field is a relationship field. **Relationship Name** InvoiceBatchRun **Refers To** InvoiceBatchRun
`InvoiceId`	**Type** reference **Properties** Filter, Group, Nillable, Sort

Field	Details
	Description The ID of the parent Invoice record of the Invoice Line record that's generated from the billing period item. This field is a relationship field. **Relationship Name** Invoice **Refers To** Invoice
`InvoiceLineId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the Invoice Line record that's generated from the billing period item. This field is populated only when a billing period item is generated by an invoice batch run. Otherwise, this field is empty. This field is a relationship field. **Relationship Name** InvoiceLine **Refers To** InvoiceLine
`InvoiceStatus`	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** Required. The status of the invoice that contains the invoice line created from the billing period item. Valid values are: • `Canceled`— The invoice for this billing period item was canceled. • `Draft`— The invoice has been created but hasn't been posted. • `DraftInProgress`— The invoice hasn't been created yet. When the invoice is created, the `InvoiceStatus` field value is changed to `Draft`. If the invoice generation process fails, the `InvoiceStatus` field value shows `DraftInProgress`. • `Error`— The invoice for this billing period item was generated in error. • `Pending`— The invoice for this billing period item is being generated. • `Posted`— An invoice line based on this billing period has been created and added successfully to the invoice.

Field	Details
	• `PostingInProgress`— An invoice line based on this billing period has been created and is in the process of being added to the invoice. • `Voided`— An invoice line based on this billing period was voided. • `VoidInProgress`— An invoice line based on this billing period is in the process of being voided.
`OverageQuantity`	**Type** double **Properties** Filter, Nillable, Sort **Description** The quantity of the usage overage that has exceeded the initial grant. This field is available in API version 63.0 and later with Revenue Cloud Billing license.
`RelatedMilestonePlanItemId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The billing milestone plan item that creates the billing period item. This field is available in API version 63.0 and later with Revenue Cloud Billing license. This field is a relationship field. **Relationship Name** RelatedMilestonePlanItem **Refers To** BillingMilestonePlanItem
`RelatedUsageBillPeriodItemId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The method for retrieving billing information for usage-based products. This field is available in API version 63.0 and later with Revenue Cloud Billing license. This field is a relationship field. **Relationship Name** RelatedUsageBillPeriodItem **Refers To** UsageBillingPeriodItem
`Status`	**Type** picklist

Field	Details
	Properties Filter, Group, Restricted picklist, Sort **Description** Required. The status of the billing period item. Draft billing period items aren't evaluated for invoice line creation. Valid values are: • `Canceled` • `Draft` • `Reviewed`
`TotalUsedQuantity`	**Type** double **Properties** Filter, Nillable, Sort **Description** The total quantity of the usage resource consumed during the billing period. This includes granted entitlements, excess consumption, or consumption below the granted entitlements. This field is available in API version 63.0 and later with Revenue Cloud Billing license.
`UnitOfMeasureId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The units used to measure the overage. For example, bytes or minutes. Available in API version 63.0 and later. This field is a relationship field. **Relationship Name** UnitOfMeasure **Refers To** UnitOfMeasure

BillingPolicy

Represents information about a set of billing treatments that define the rules to invoice a customer for an order item. This object is available in API version 62.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need the Billing Admin permission set to access this object.

Fields

Field	Details
`BillingTreatmentSelection`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. Defines how billing treatments are assigned to order items and assets that are related to the billing policy. Valid values are: • `Default` • `LegalEntity` • `Manual`
`DefaultBillingTreatmentId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** When `BillingTreatmentSelection` has a value of `Default`, the selected billing treatment is used for all the order items and assets that are related to the billing policy. This field is a relationship field. **Relationship Name** DefaultBillingTreatment **Refers To** BillingTreatment
`Description`	**Type** textarea **Properties** Create, Filter, Nillable, Sort, Update **Description** Additional details about the billing policy.
`LastReferencedDate`	**Type** dateTime

Field	Details
	Properties Filter, Nillable, Sort **Description** The timestamp when the current user last accessed a billing policy indirectly, for example, through a list view or related record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed a billing policy. If this value is null, it's possible that the user only accessed the billing policy or a related list view (`LastReferencedDate`), but not viewed the accounting period itself.
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** Required. The name of the billing policy.
`Status`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. The status of the billing policy. Valid values are: • `Active` • `Draft` • `Inactive`

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

BillingPolicyHistory on page 2873

History is available for tracked fields of the object.

BillingSchedule

Represents information about the order item that's used in the invoicing process. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `update()`

Special Access Rules

You need the Billing Admin permission set to access this object.

Fields

Field	Details
`BillDayOfMonth`	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The day of the month on which a recurring billing process is scheduled to occur.
`BilledAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The total amount (excluding tax) that has been invoiced from the billing schedule. This field is a calculated field.
`BillingAccountId`	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The ID of the account that's related to the billing schedule. This field is a relationship field. **Relationship Name** BillingAccount

Field	Details
	Refers To Account
BillingMethod	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The type of billing used for the source item. Valid values are: • Evergreen • OrderAmount • Usage—Available in API version 63.0 and later with Revenue Cloud Billing license.
BillingMilestonePlanId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The billing milestone plan associated with the billing schedule. This field is available in API version 63.0 and later with Revenue Cloud Billing license. This field is a relationship field. **Relationship Name** BillingMilestonePlan **Refers To** BillingMilestonePlan
BillingPeriodAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The amount to be invoiced each billing period. For example, if the billing period is monthly, this field shows the monthly amount that appears on the invoice line.
BillingScheduleEndDate	**Type** date **Properties** Filter, Group, Nillable, Sort

Field	Details
	Description The last date that the billing schedule is available for invoicing. This value is inherited from the `EndDate` field on the order item.
`BillingScheduleGroupId`	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The billing schedule group that contains the billing schedule. Billing schedules are grouped when they have the same source order item. The source order item is the original order item that a customer bought. Afterwards, if the customer amends, cancels, or renews the order item, a new billing schedule is created with the `BillingScheduleGroupId` for the original order item. This field is a relationship field. **Relationship Name** BillingScheduleGroup **Relationship Type** Master-detail **Refers To** BillingScheduleGroup (the master object)
`BillingScheduleNumber`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. The auto-generated reference number for the billing schedule.
`BillingScheduleStartDate`	**Type** date **Properties** Filter, Group, Nillable, Sort **Description** The date when the billing schedule is available for invoicing. This value is inherited from the `ServiceDate` field on the order item.
`BillingTerm`	**Type** int **Properties** Filter, Group, Nillable, Sort

Field	Details
	Description The duration for which the customer is invoiced.
BillingTermUnit	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** Required. The unit of measurement of the billing term. Valid values are: • BillingMilestonePlan—Available in API version 63.0 and later with Revenue Cloud Billing license. • Day • Month • OneTime • Quarter • Semi-Annual • Year
BillingTreatmentId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the billing treatment record associated with the billing schedule. Available in API version 67.0 and later. This field is a relationship field. **Relationship Name** BillingTreatment **Refers To** BillingTreatment
BillingTreatmentItemId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The billing treatment item that's used to configure invoiceable amounts on the billing schedule. This field is a relationship field.

Field	Details
	Relationship Name BillingTreatmentItem **Refers To** BillingTreatmentItem
BillingType	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The timing of invoicing for a product or service relative to its delivery to the customer. Valid values are: • Advance— Invoices a product or service before its delivery. • Arrears — Invoices a product or service after its delivery.
CancellationDate	**Type** date **Properties** Filter, Group, Nillable, Sort **Description** The date from when the user can no longer access the service. For example, if a service ends on August 31, the cancellation date is September 1 because that's the date from when the user can no longer use the service. Billing schedules past their cancellation date aren't invoiced.
Category	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** Required. The category of the billing schedule. Valid values are: • Amendment— A billing schedule for an order that was amended. Available in API version 67.0 and later. • AmendQuantity— A billing schedule for an order that changes the quantity. • Cancellation — A billing schedule for an order that was canceled. • Original — A billing schedule for the initial order. • Renewal — A billing schedule for an order that was renewed.
CustomInvoiceGroupKey	**Type** string

Field	Details
	Properties Filter, Group, Restricted picklist, Sort **Description** The group identifier for which an invoice must be generated when the invoice group type is `Custom`. This field is available in API version 63.0 and later with Revenue Cloud Billing license.
`InvBatchDraftToPostedRunId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The batch job that posts all invoices with a status as `Draft` that are generated by the invoice batch run associated with the billing schedule. This field is a relationship field. **Relationship Name** InvBatchDraftToPostedRun **Refers To** InvBatchDraftToPostedRun
`InvoiceBatchRunId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The invoice batch run that evaluated the billing schedule and its billing period items to generate an invoice. This field is a relationship field. **Relationship Name** InvoiceBatchRun **Refers To** InvoiceBatchRun
`InvoiceGroupType`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies whether to generate an invoice for a billing schedule, a custom group, or the default group of account, bill-to-contact, payment term, currency, and tax engine. Valid values are:

Field	Details
	• `Billing Schedule` • `Custom` • `Default` This field is available in API version 63.0 and later with Revenue Cloud Billing license.
`InvoiceRunMatchingValue`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The batch value used by the invoice run that evaluated the billing schedule. During an invoice run, billing schedules with the same batch value, including null, are grouped to the same invoice run.
`LineAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The amount associated with a specific line item in the billing schedule. Available in API version 63.0 and later.
`NetUnitPrice`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The net unit price of the order item for which the billing schedule was created. Available in API version 64.0 and later.
`NextBillingDate`	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The date that the next billing period starts for the invoice. This date is used to calculate which invoice lines are included on an invoice. When an invoice scheduler or API evaluates an order for invoicing, billing schedules with a next billing date on or before the invoice's target date are included in the invoice.
`NextChargeFromDate`	**Type** date

Field	Details
	Properties Create, Filter, Group, Nillable, Sort, Update **Description** The date that the billing schedule is invoiced in the upcoming billing period. For example, if you invoiced a customer for a billing period of 01/01/24 through 01/31/24, the billing schedule's `NextChargeFromDate` is 02/01/22.
`OriginalBillingScheduleId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** If the billing schedule is an amended or canceled billing schedule, then this field shows the original billing schedule. Otherwise, this field is null. This field is a relationship field. **Relationship Name** OriginalBillingSchedule **Refers To** BillingSchedule
`PendingAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The amount from the current billing term that hasn't been billed yet. For example, the unbilled amount for a month, quarter, or year, depending on this billing schedule's billing term.
PricingTermUnit	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** Required. The unit of time for which the pricing of the product applies. Valid values are: • `Months` • `Quarterly` • `Semi-Annual` • `Annual` Available in API version 66.0 and later.

Field	Details
ProcessingOrder	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** Specifies the order in which billing transactions are applied. Available in API version 67.0 and later. This field is an autogenerated field.
Quantity	**Type** double **Properties** Filter, Nillable, Sort **Description** The quantity of the order item that created the billing schedule.
Reference	**Type** reference **Properties** Filter, Group, idLookup, Sort **Description** The ID of the parent record of the reference item for which the billing schedule is created. Available in API version 61.0 and later.
ReferenceEntityId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The parent order of the order item that created the billing schedule. This field is a polymorphic relationship field. **Relationship Name** ReferenceEntity **Refers To** Order
ReferenceEntityItemId	**Type** reference **Properties** Filter, Group, Nillable, Sort

Field	Details
	Description The order item or asset that created the billing schedule. This field is a polymorphic relationship field. **Relationship Name** ReferenceEntityItem **Refers To** OrderItem, OrderItemAdjustmentLineItem
`ReferenceItem`	**Type** reference **Properties** Filter, Group, idLookup, Sort **Description** The ID of the transaction line item record for which the billing schedule is created. Available in API version 61.0 and later.
SubCategory	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** The subcategory of the billing schedule. Available in API version 67.0 and later. Valid values are: • `AsIsRenewal` • `BillingFrequencyAmendment` • `ChangeEndDate` • `DeltaPriceAmend` • `PriceAmend` • `QuantityAmendment` • `SimplifiedCancel` • `SimplifiedRenewal`
`Status`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. The status of the order item that the billing schedule represents. Valid values are:

Field	Details
	• `WaitingForMilestoneEventAccomplishment`—Available in API version 63.0 and later with Revenue Cloud Billing license. • `CompletelyBilled` • `Error` • `Processing` • `ReadyForInvoicing`
`TaxTreatmentId`	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The ID of the treatment that's used to calculate tax for the billing schedule. This value is defined based on the order item's tax policy. This field is a relationship field. **Relationship Name** TaxTreatment **Refers To** TaxTreatment
`TotalAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The total amount of the order item represented by the billing schedule.
`Type`	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** The type of the billing schedule. Valid values are: • `LineItemAdjustedCharge` • `LineItemAdjustment` • `LineItemCharge`
`UnitOfMeasureId`	**Type** reference

Field	Details
	Properties Filter, Group, Nillable, Sort **Description** The ID of the unit of measure for the billing schedule. Available in API version 63.0 and later. This field is a relationship field. **Relationship Name** UnitOfMeasure **Refers To** UnitOfMeasure
`UnitPrice`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The price for an individual unit of the billing schedule's parent order item, including charges, adjustments, and discounts. This value is inherited from the order item's `UnitPrice` field.
`UsageResourceId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The usage resource associated with the billing schedule. This field is available in API version 63.0 and later with Revenue Cloud Billing license. This field is a relationship field. **Relationship Name** UsageResource **Refers To** UsageResource

BillingScheduleGroup

Represents a consolidated view of all the billing schedules related to the order items generated from one asset, including new orders and amendment orders. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `update()`

Special Access Rules

You need the Billing Admin permission set to access this object.

Fields

Field	Details
`AnchorProdtBillingSchdGrpId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The billing schedule group for the anchor product in a usage-based product. The anchor product contains the grants for the usage-based product. This field is available in API version 63.0 and later with Revenue Cloud Billing license. This field is a relationship field. **Relationship Name** AnchorProdtBillingSchdGrp **Refers To** BillingScheduleGroup
`BillDayOfMonth`	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The day of the month on which a recurring billing process is scheduled to occur.
`BillToContactId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the contact related to the billing schedule group. This field can't be modified when related billing schedules are in processing. This field is a relationship field. **Relationship Name** BillToContact

Field	Details
	Refers To Contact
BillingAccountId	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The ID of the account that's related to the billing schedule group. This field is a relationship field. **Relationship Name** BillingAccount **Refers To** Account
BillingAddress	**Type** address **Properties** Filter, Nillable **Description** The compound form of the billing address. Read only. See Address Compound Fields for details on compound address fields.
BillingArrangementId	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The ID of the billing arrangement that's related to the billing schedule group. Available in API version 66.0 and later. This field is a relationship field. **Relationship Name** BillingArrangement **Refers To** BillingArrangement
BillingMethod	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort

Field	Details
	Description Required. The type of billing used for the source item. Valid values are: • `Evergreen` • `OrderAmount` • `Usage`
`BillingResumptionDate`	**Type** date **Properties** Filter, Group, Nillable, Sort **Description** The date when billing for the asset related to the billing schedule group is resumed. Available in API version 63.0 and later.
`BillingScheduleGroupNumber`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. The auto-generated reference number for the billing schedule group.
`BillingStartMonth`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** This is a read-only field used with annual billing. The field shows the numbers from 1 through 12, which indicate the month when billing begins for an annual subscription. For example, if billing starts in January, the value is 1; if billing starts in June, the value is 6.
`BillingSuspensionDate`	**Type** date **Properties** Filter, Group, Nillable, Sort **Description** The date when billing for the asset related to the billing schedule group is suspended. Available in API version 63.0 and later.
`BillingTerm`	**Type** int

Field	Details
	Properties Filter, Group, Nillable, Sort **Description** This field is used with `BillingTermUnit`field to define a billing cycle. For example, bill every 20 days or every two months. In this example, the `BillingTerm` field value is 20 and the `BillingTermUnit` field value is `Day`.
`BillingTermUnit`	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** Required. The unit of measurement of the billing term. Valid values are: • `BillingMilestonePlan`—Available in API version 63.0 and later with Revenue Cloud Billing license. • `Day` • `Month` • `OneTime` • `Quarter` • `Semi-Annual` • `Year`
`BillingTreatmentId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The billing treatment item that's used to configure invoiceable amounts on the billing schedule group. This field is a relationship field. **Relationship Name** BillingTreatment **Refers To** BillingTreatment
`BillingType`	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort

Field	Details
	Description Required. The timing of invoicing for a product or service relative to its delivery to the customer. Valid values are: • `Advance`— Invoices a product or service before its delivery. • `Arrears`— Invoices a product or service after its delivery.
`BindingInstanceId`	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The asset or custom object record associated with the billing schedule group. This field is available in API version 63.0 and later with Revenue Cloud Billing license. This field is a polymorphic relationship field. **Relationship Name** BindingInstance **Refers To** Asset
`CancellationDate`	**Type** date **Properties** Filter, Group, Nillable, Sort **Description** The date from when the user can no longer access the service. For example, if a service ends on August 31, the cancellation date is September 1 because that’s the date from when the user can no longer use the service. Billing schedules past their cancellation date aren't invoiced.
`Controller`	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** Required. During the invoicing process, this field determines which date is used when the billing schedule group and billing schedule have a related field with conflicting values. For example, when Controller has a value of `BillingScheduleGroup`, if the billing schedule's billing day of month is 5 while the billing schedule group's billing day of month is 10, the invoice is sent on the 10th day of the month. Valid values are:

Field	Details
	• `BillingScheduleGroup` • `None`
`EffectiveNextBillingDate`	**Type** date **Properties** Filter, Group, Nillable, Sort **Description** The earliest `NextBillingDate` from all billing schedules in the billing schedule group. This field is a reference field that isn't used for any features or calculations. This field is a calculated field.
`EndDate`	**Type** date **Properties** Filter, Group, Nillable, Sort **Description** The end date of the billing schedule group.
`ExternalBindingInstance`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The custom target associated with the entitlements that are granted with the sellable product. This field is available in API version 63.0 and later with Revenue Cloud Billing license.
`ExternalRefRecordIdentifier`	**Type** string **Properties** Filter, Group, Nillable, Sort, Update **Description** The identifier of the external record for which the billing schedule group was created. Available in API version 64.0 and later.
`LegalEntityId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the legal entity that's related to the billing schedule group. This field is a relationship field.

Field	Details
	Relationship Name LegalEntity **Refers To** LegalEntity
NextBillingDateOverride	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The next billing date for all the billing schedules related to the billing schedule group. If specified, this date overrides the next billing dates of the billing schedules. Available in API version 63.0 and later.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Required. The ID of the user who created the billing schedule group. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
PaymentTermId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the payment term used in this billing schedule group. This field can’t be modified when related billing schedules are in processing. This field is a relationship field. **Relationship Name** PaymentTerm **Refers To** PaymentTerm
PeriodBoundary	**Type** picklist

Field	Details
	Properties Filter, Group, Nillable, Restricted picklist, Sort **Description** This field is inherited from the order item's parent quote line item or sales transaction item. The period boundary determines the start and end date of the billing periods. Valid values are: • `AlignToCalendar` • `Anniversary` • `DayOfPeriod` • `LastDayOfPeriod`
`Product2Id`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The product that was charged or ordered to create the billing schedule group. This field is a relationship field. **Relationship Name** Product2 **Refers To** Product2
`ProductName`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The name of the product for the order item that's represented by each billing schedule in the billing schedule group.
`ProrationPolicyId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the proration policy that applies to this billing schedule group. The proration policy defines how time periods are calculated for orders. For example, whether partial periods are allowed. This field is inherited from the shared proration policy for each billing schedule in the billing schedule group. This field is a relationship field.

Field	Details
	Relationship Name ProrationPolicy **Refers To** ProrationPolicy
ReferenceEntityId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The asset used to create the billing schedules in the billing schedule group. This field is a relationship field. **Relationship Name** ReferenceEntity **Refers To** Asset
ReferenceRecordId	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The ID of the Salesforce record for which the billing schedule group was created. Available in API version 64.0 and later. This field is a polymorphic relationship field. **Relationship Name** BillingScheduleGroups
SavedPaymentMethodId	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The ID of the SavedPaymentMethod record that's used to collect payment for the billing schedule group. Available in API version 64.0 and later. This field is a relationship field. **Relationship Name** BillingScheduleGroups **Refers To** SavedPaymentMethod

Field	Details
ShipFromAddress	**Type** address **Properties** Filter, Nillable **Description** The address from which the product in the billing schedule group is shipped. Available in API version 64.0 and later.
ShippingAddress	**Type** address **Properties** Filter, Nillable **Description** The compound form of the shipping address. See Address Compound Fields for details on compound address fields. This field is a read-only field.
StartDate	**Type** date **Properties** Filter, Group, Nillable, Sort **Description** The earliest start date from all billing schedules in the billing schedule group.
TaxTreatmentId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the tax treatment record that's used to calculate tax for the billing schedule group. This value is defined based on the order item's tax policy. This field is a relationship field. **Relationship Name** TaxTreatment **Refers To** TaxTreatment
TotalBilledAmount	**Type** currency **Properties** Filter, Nillable, Sort

Field	Details
	Description The amount that has been invoiced for the billing schedules within the billing schedule group. This field is a calculated field.
`TotalPendingAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The amount that hasn't yet been invoiced for the billing schedules within the billing schedule group. This field is a calculated field.
UsageType	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The usage type for the billing schedule group. Valid values are: • `Revenue Cloud` • `Commerce Cloud` • `Insurance Cloud` Available in API version 66.0 and later.

BillingTreatment

Represents information about the billing of an order item. This object is available in API version 62.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need the Billing Admin permission set to access this object.

Fields

Field	Details
BillingPolicyId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The billing policy that's related to the billing treatment. This field is a relationship field. **Relationship Name** BillingPolicy **Refers To** BillingPolicy
CanChangeBillingFrequency	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Indicates whether the billing frequency can be changed for the billing treatment (`true`) or not (`false`). The default value is `false`.
Description	**Type** textarea **Properties** Create, Filter, Nillable, Sort, Update **Description** Additional details about the billing treatment.
ExcludeFromBilling	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. Specifies whether any order items assigned to the treatment are excluded from billing. Valid values are: • `No` • `Yes`

Field	Details
IsMilestoneBilling	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Required. Indicates whether the billing treatment is for milestone billing (`true`) or not (`false`). The default value is `false`. This field is available in API version 63.0 and later with Revenue Cloud Billing license.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed a billing treatment indirectly, for example, through a list view or related record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed a billing treatment. If this value is null, it’s possible that the user only accessed the billing treatment or a related list view (`LastReferencedDate`), but not viewed the billing treatment itself.
LegalEntityId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The legal entity that's used to assign the treatment to order items when the parent billing policy's `BillingTreatmentSelection` field value is `LegalEntity`. This field is a relationship field. **Relationship Name** LegalEntity **Refers To** LegalEntity
Name	**Type** string

Field	Details
	Properties Create, Filter, Group, idLookup, Sort, Update **Description** Required. The name of the billing treatment.
`Status`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. The status of the billing treatment. Valid values are: • `Active` • `Draft` • `Inactive` Draft or inactive billing treatments can't be assigned to order items.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

BillingTreatmentHistory on page 2873
History is available for tracked fields of the object.

BillingTreatmentItem

Represents information about allocation of the total amount of an order item to billing schedules throughout the order item's lifecycle. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need the Billing Admin permission set to access this object.

Fields

Field	Details
`BillingTreatmentId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** Required. The parent billing treatment for the billing treatment item. This field is a relationship field. **Relationship Name** BillingTreatment **Relationship Type** Master-detail **Refers To** BillingTreatment (the master object)
`BillingType`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. The timing of invoicing for a product or service relative to its delivery to the customer. Advance billing invoices a product or service before its delivery, whereas arrears billing invoices it after delivery. The billing system assesses the billing type to determine the next billing date for an order product. Valid values are: • `Advance`—If the order item is billed in advance, the order's billing day of month is evaluated to choose the nearest date on or before the order product's start date. For example, if a monthly order product's start date is January 1, and the order's billing day of month is 15, the next billing date is December 15. • `Arrears`—If the order item is billed in arrears, the order's billing day of month is evaluated to choose the nearest date after the order product's start date. For example, if a monthly order product's start date is January 1, and the order's billing day of month is 15, the order product's next billing date is January 15. • `None`
`Controller`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. During the invoicing process, this field determines which value is used when the billing schedule group and billing schedule have a shared field with different values. For

Field	Details
	example, when Controller has a value of `BillingScheduleGroup`, if the billing schedule's billing day of month is 5 while the billing schedule group's billing day of month is 10, the value of 10 is used. Valid values are: • `BillingScheduleGroup` • `None`
`Description`	**Type** textarea **Properties** Create, Filter, Nillable, Sort, Update **Description** Additional details about the billing treatment item.
`FlatAmount`	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The amount in terms of units of currency (such as $10 or $21.52) to invoice from the order item. Used only when Type has a value of `FlatAmount`.
`Handling0Amount`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. Specifies how invoices are generated for billing period items that have an amount of $0. Valid values are: • `CreateInvoice` • `None`
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed a billing treatment item indirectly, for example, through a list view or related record.

Field	Details
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed a billing treatment item. If this value is null, it's possible that the user only accessed the billing treatment item or a related list view (`LastReferencedDate`), but not viewed the billing treatment item itself.
MilestoneStartDate	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The occurrence whose date is used to calculate the milestone accomplishment date for the associated billing milestone plan item based on the provided offset details. Valid value is: • `OrderProductActivation` This field is available in API version 63.0 and later with Revenue Cloud Billing license.
MilestoneStartDateOffset	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The calculated value that determines the time difference from the start of a milestone. This value is calculated from the milestone start date record. This field is available in API version 63.0 and later with Revenue Cloud Billing license.
MilestoneStartDateOffsetUnit	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The unit type to use with the milestone start date offset, which defines the milestone date. Valid values are: • `Days` • `Months` • `Years` This field is available in API version 63.0 and later with Revenue Cloud Billing license.

Field	Details
MilestoneType	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The milestone type for the billing treatment item. Valid values are: • Date • Event This field is available in API version 63.0 and later with Revenue Cloud Billing license.
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** Required. The name of the billing treatment item.
Percentage	**Type** percent **Properties** Create, Filter, Nillable, Sort, Update **Description** The percentage (such as 10% or 12.5%) to invoice from the order item. Used only when Type field has a value of Percentage.
ProcessingOrder	**Type** int **Properties** Create, Filter, Group, Sort, Update **Description** Required. The order in which billing schedules are created based on each billing treatment item. Lower numbers are evaluated first. For example, if your billing treatment has a billing treatment item that invoices at 25 percentage and a ProcessingOrder of 1, and another item that invoices at 75 percentage and a ProcessingOrder of 2, your first billing schedule will be for 25% of the order item's total amount, and your second billing schedule will be for 75% of the order item's total amount.
Sequencing	**Type** picklist

Field	Details
	Properties Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. The number used to start invoice numbers on invoices generated from this billing treatment item. Valid values are: • `Manual` • `None`
`Status`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. The status of the billing treatment item. Valid values are: • `Active` • `Draft`– Draft billing treatment items aren't evaluated for creating billing schedules.
`Type`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. Specifies whether billing schedules created from this billing treatment item are based on a flat amount or a percentage of the order item's total amount. Valid values are: • `FlatAmount` • `Percentage` • `Remainder`

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

BillingTreatmentItemHistory on page 2873
History is available for tracked fields of the object.

BsgRelationship

Represents a relationship between billing schedule groups to support bundles where one parent billing schedule group has multiple child billing schedule groups. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`

Special Access Rules

You need the Billing Admin permission set to access this object.

Fields

Field	Details
`AssociatedBsgId`	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The ID of the related billing schedule group. In a bundle relationship, this billing schedule group is the child. This field is a relationship field. **Relationship Name** AssociatedBsg **Refers To** BillingScheduleGroup
`AssociatedBsgPricing`	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** Required. This field describes how the related billing schedule group is priced relative to the primary billing schedule group. Valid values are: • `IncludedInBundlePrice` • `NotIncludedInBundlePrice`
`AssociatedBsgRole`	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort

Field	Details
	Description Required. This field describes the role of the related billing schedule group in the relationship. Valid values are: • `AddOnComponent` • `BundleComponent` • `ClassificationComponent` • `SetComponent`
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed a billing schedule group record indirectly, for example, through a list view or related record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed a billing schedule group relationship record. If this value is null, it's possible that the user only accessed the billing schedule group relationship record or a related list view (`LastReferencedDate`), but not viewed the billing schedule group relationship record itself.
`MainBsgId`	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The ID of the primary billing schedule group. In a bundle relationship, this billing schedule group is the parent. This field is a relationship field. **Relationship Name** MainBsg **Relationship Type** Master-detail **Refers To** BillingScheduleGroup (the master object)

Field	Details
MainBsgRole	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** Required. This field describes the role of the primary billing schedule group in the relationship. Valid values are: • AddOn • Bundle • Set
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. The auto-generated reference number for the billing schedule relationship.
ProductRelationshipTypeId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the relationship type between the main and associated billing schedule group. This field is a relationship field. **Relationship Name** ProductRelationshipType **Refers To** ProductRelationshipType

CaseServiceProcessExtension

Represents additional information for a case record based on the type of service process for which the case was raised. This object is available in API version 67.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need the Billing Admin, Billing Operations User, or Billing Customer Service User permission set to access this object.

Fields

Field	Details
`CaseId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The case record related to the invoice. **Relationship Name** Case **Relationship Type** Master-detail **Refers To** Case
`CaseServiceProcessExtensionNumber`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The autogenerated number of the case.
`CustomServiceProcess`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** Specifies the name of the service process when the type of service process is `Custom`.
`IntakeMethod`	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort, Update

Field	Details
	Description Specifies whether the case was raised internally within the org or by using the self-service Billing portal. Valid values are: • `Internal` • `External`
`ReferenceRecordId`	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The record ID associated with the case record such as an invoice record. This field is a polymorphic relationship field. **Relationship Name** CaseServiceProcessExtensionsByReference
`ServiceProcess`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** Specifies the type of service process that created the case. Valid values are: • `Extend Invoice Due Date` • `Invoice Charge Correction` • `Suspend Billing` • `Update Bill-To Contact` • `Other Billing Inquiries` • `Custom`
`ReferenceRecordId`	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The record ID associated with the case record such as an invoice record. This field is a polymorphic relationship field. **Relationship Name** CaseServiceProcessExtensionsByReference

Field	Details
`UsageType`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** Specifies the name of the cloud or function that uses the CaseServiceProcessExtension object. Valid value is `Billing`.

Usage

This section covers a sample Apex class to support backward compatibility for cases created before the availability of the `CaseServiceProcessExtension` object. The class finds billing cases that don't yet have extension records, maps service process values from service catalog extended attributes for supported templates. This class also populates required extension fields such as service process, reference record ID, usage type, and intake method. For other billing queries where template entries aren't available in extended attributes, the class optionally accepts product IDs to identify and process those cases. The implementation runs as batch Apex for large data volumes and reports processing results in logs.

This example shows a sample Apex class. You can modify this class as per your requirements.

```
/**
 * Batchable Apex class to backfill CaseServiceProcessExtension records
 * for billing cases that were created in API version 66.0 and earlier.
 *
 * Identifies billing cases with no CaseServiceProcessExtension and populates the extension
 record.
 *
 * Additionally, accepts a set of Product IDs to identify billing cases and populates the
 extension record
 * for other billing inquiries.
 *
 * Usage:
 *   Set<Id> productIds = new Set<Id>{'01tXX0000000001', '01tXX0000000002'};
 *   Database.executeBatch(new BillingCaseServiceProcessExtensionBatch(productIds), 200);

 *
 */
public class BillingCaseServiceProcessExtensionBatch implements Database.Batchable<SObject>,
 Database.Stateful {

   private static final Map<String, String> TEXT_VALUE_TO_SERVICE_PROCESS = new Map<String,
 String>{
        'billing_svc_update_billToContact' => 'UpdateBillToContact',
        'billing_svc_due_date_extension' => 'ExtendInvoiceDueDate',
        'billing_svc_incorrect_invoice_charges' => 'InvoiceChargeDispute',
        'billing_svc_suspend_billing' => 'SuspendBilling'
    };

   private static final Map<String, String> PRODUCT_NAME_KEYWORD_TO_SERVICE_PROCESS = new
 Map<String, String>{
```

```
        'bill to contact' => 'UpdateBillToContact',
        'due date' => 'ExtendInvoiceDueDate',
        'invoice charge' => 'InvoiceChargeDispute',
        'suspend billing' => 'SuspendBilling',
        'other billing' => 'OtherBillingInquiry'
    };

    private final Set<Id> productIds;
    private Integer totalProcessed = 0;
    private Integer totalCreated = 0;
    private Integer totalErrors = 0;

    public BillingCaseServiceProcessExtensionBatch(Set<Id> productIds) {
        this.productIds = productIds != null ? productIds : new Set<Id>();
    }

    public Database.QueryLocator start(Database.BatchableContext bc) {
        return Database.getQueryLocator([
            SELECT Id, SourceId
            FROM Case
            WHERE SourceId != NULL
              AND Id NOT IN (SELECT CaseId FROM CaseServiceProcessExtension)
        ]);
    }

    public void execute(Database.BatchableContext bc, List<Case> scope) {
        totalProcessed += scope.size();

        // Build CaseId -> SourceId map and collect unique SourceIds for bulk query
        // Filter for ServiceCatalogRequest IDs (prefix '946')
        Map<Id, Id> caseIdToSourceId = new Map<Id, Id>();
        Set<Id> sourceIds = new Set<Id>();
        for (Case c : scope) {
            if (c.SourceId != null && String.valueOf(c.SourceId).startsWith('946')) {
                caseIdToSourceId.put(c.Id, c.SourceId);
                sourceIds.add(c.SourceId);
            }
        }

        // Single merged query: billing text values + 3tt reference records
        List<SvcCatalogReqExtdAttrVal> attrVals = [
            SELECT SvcCatalogRequestId, TextValue
            FROM SvcCatalogReqExtdAttrVal
            WHERE SvcCatalogRequestId IN :sourceIds
              AND (TextValue IN :TEXT_VALUE_TO_SERVICE_PROCESS.keySet()
                   OR TextValue LIKE '3tt%')
        ];

        // Step 1: Identify billing ServiceCatalogRequestIds and their ServiceProcess
        Map<Id, String> sourceIdToServiceProcess = new Map<Id, String>();
        for (SvcCatalogReqExtdAttrVal attr : attrVals) {
            if (TEXT_VALUE_TO_SERVICE_PROCESS.containsKey(attr.TextValue)) {
                sourceIdToServiceProcess.put(
                    attr.SvcCatalogRequestId,
```

```
                    TEXT_VALUE_TO_SERVICE_PROCESS.get(attr.TextValue)
                );
            }
        }

       // Step 2: Identify remaining billing cases via ProductId on ServiceCatalogRequest

        Map<Id, String> sourceIdToCustomServiceProcess = new Map<Id, String>();
        if (!this.productIds.isEmpty()) {
            Set<Id> unmatchedSourceIds = new Set<Id>();
            for (Id sid : sourceIds) {
                if (!sourceIdToServiceProcess.containsKey(sid)) {
                    unmatchedSourceIds.add(sid);
                }
            }

            if (!unmatchedSourceIds.isEmpty()) {
                List<SvcCatalogRequest> catalogRequests = [
                    SELECT Id, ProductId, Product.Name
                    FROM SvcCatalogRequest
                    WHERE Id IN :unmatchedSourceIds
                      AND ProductId IN :productIds
                ];

                for (SvcCatalogRequest scr : catalogRequests) {
                    String serviceProcess =
deriveServiceProcessFromProductName(scr.Product.Name);
                    sourceIdToServiceProcess.put(scr.Id, serviceProcess);
                    if (serviceProcess == 'Custom' && scr.Product.Name != null) {
                        sourceIdToCustomServiceProcess.put(scr.Id, scr.Product.Name);
                    }
                }
            }
        }

        // Step 3: Extract Invoice ReferenceRecord values for all confirmed billing cases

        Map<Id, Id> sourceIdToReferenceRecord = new Map<Id, Id>();
        for (SvcCatalogReqExtdAttrVal attr : attrVals) {
            if (attr.TextValue != null
                && attr.TextValue.startsWith('3tt')
                && sourceIdToServiceProcess.containsKey(attr.SvcCatalogRequestId)) {
              sourceIdToReferenceRecord.put(attr.SvcCatalogRequestId, (Id) attr.TextValue);

            }
        }

      // Step 4: For invoice charge dispute and other billing enquiries, check for dispute
 records
        // and prefer the dispute invoice ID over the text value-based invoice ID.
        Set<Id> disputeServiceProcessCaseIds = new Set<Id>();
        Map<Id, Id> disputeCaseIdToSourceId = new Map<Id, Id>();
        for (Case c : scope) {
            Id sourceId = caseIdToSourceId.get(c.Id);
```

```
            if (sourceId != null && sourceIdToServiceProcess.containsKey(sourceId)) {
                String serviceProcess = sourceIdToServiceProcess.get(sourceId);
                if (serviceProcess == 'InvoiceChargeDispute' || serviceProcess ==
'OtherBillingInquiry') {
                    disputeServiceProcessCaseIds.add(c.Id);
                    disputeCaseIdToSourceId.put(c.Id, sourceId);
                }
            }
        }

        if (!disputeServiceProcessCaseIds.isEmpty()) {
            List<Dispute> disputes = [
                SELECT Id, CaseId, InvoiceId
                FROM Dispute
                WHERE CaseId IN :disputeServiceProcessCaseIds
                  AND InvoiceId != NULL
            ];

            for (Dispute dispute : disputes) {
               // Override with Dispute's InvoiceId (this takes precedence over TextValue)

                Id sourceId = disputeCaseIdToSourceId.get(dispute.CaseId);
                if (sourceId != null) {
                    sourceIdToReferenceRecord.put(sourceId, dispute.InvoiceId);
                }
            }
        }

        // Build CaseServiceProcessExtension records
        List<CaseServiceProcessExtension> extensions = new
List<CaseServiceProcessExtension>();
        for (Case c : scope) {
            Id sourceId = caseIdToSourceId.get(c.Id);
            if (sourceIdToServiceProcess.containsKey(sourceId)) {
                CaseServiceProcessExtension ext = new CaseServiceProcessExtension();
                ext.CaseId = c.Id;
                ext.ServiceProcess = sourceIdToServiceProcess.get(sourceId);
                ext.UsageType = 'Billing';
                ext.IntakeMethod = 'External';
                if (sourceIdToReferenceRecord.containsKey(sourceId)) {
                    ext.ReferenceRecordId = sourceIdToReferenceRecord.get(sourceId);
                }
                if (sourceIdToCustomServiceProcess.containsKey(sourceId)) {
                 ext.CustomServiceProcess = sourceIdToCustomServiceProcess.get(sourceId);

                }
                extensions.add(ext);
            }
        }

        if (!extensions.isEmpty()) {
            List<Database.SaveResult> results = Database.insert(extensions, false);
            for (Database.SaveResult sr : results) {
                if (sr.isSuccess()) {
```

```
                    totalCreated++;
                } else {
                    totalErrors++;
                    for (Database.Error err : sr.getErrors()) {
                        System.debug(
                            LoggingLevel.ERROR,
                            'BillingCaseServiceProcessExtensionBatch insert error: ' +
err.getMessage()
                        );
                    }
                }
            }
        }
    }

    public void finish(Database.BatchableContext bc) {
        System.debug(
            LoggingLevel.INFO,
            'BillingCaseServiceProcessExtensionBatch complete. ' +
            'Cases processed: ' + totalProcessed +
            ', Extensions created: ' + totalCreated +
            ', Errors: ' + totalErrors
        );
    }

    private static String deriveServiceProcessFromProductName(String productName) {
        if (String.isBlank(productName)) {
            return 'Custom';
        }
        String lowerName = productName.toLowerCase();
        for (String keyword : PRODUCT_NAME_KEYWORD_TO_SERVICE_PROCESS.keySet()) {
            if (lowerName.contains(keyword)) {
                return PRODUCT_NAME_KEYWORD_TO_SERVICE_PROCESS.get(keyword);
            }
        }
        return 'Custom';
    }
}
```

CreditMemo

Represents a document that's used to reduce the amount that a buyer owes a seller under the terms of an earlier invoice. This object is available in API version 62.0 and later.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `update()`

Special Access Rules

You need the Credit Memo Operations User permission set to access this object.

Fields

Field	Details
`Balance`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The amount of the credit memo that's available for allocation.
`BillToContactId`	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** This field is inherited from the account's `Bill to Account` field. This field is a relationship field. **Relationship Name** BillToContact **Refers To** Contact
`BillingAccountId`	**Type** reference **Properties** Filter, Group, Sort, Update **Description** Required. The ID of the customer account associated with this credit memo. This field is a relationship field. **Relationship Name** BillingAccount **Refers To** Account
`Category`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** Specifies whether the credit memo is created due to writing off an invoice, converting negative invoice lines, or referencing an existing record. Valid values are:

Field	Details
	• `InvoiceWriteOff` • `Referenced` • `Standalone` This field is available in API version 64.0 and later.
`CorpCrcyCnvTotAmtWithTax`	**Type** double **Properties** Filter, Nillable, Sort **Description** The sum of the total amount with tax on the credit memo in corporate currency. Available in API version 63.0 and later.
`CorporateCurrencyCvsnDate`	**Type** date **Properties** Filter, Group, Nillable, Sort **Description** The date on which the credit memo amounts are converted to the corporate currency. Available in API version 63.0 and later.
`CorporateCurrencyCvsnRate`	**Type** double **Properties** Filter, Nillable, Sort **Description** The exchange rate that's used to convert credit memo amounts to the corporate currency. Available in API version 63.0 and later.
`CorporateCurrencyIsoCode`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The currency ISO code of the corporate currency. Available in API version 63.0 and later.
`CreationMode`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** Specifies whether the credit memo originated in Salesforce or an external system.

Field	Details
	Valid values are: • `External` • `Salesforce`
`CreditDate`	**Type** date **Properties** Filter, Group, Nillable, Sort **Description** The date when the credit memo was posted.
`CreditMemoNumber`	**Type** string **Properties** Filter, Group, Nillable, Sort, Update **Description** The number of the credit memo.
`Description`	**Type** string **Properties** Filter, Group, Nillable, Sort, Update **Description** Additional details about the credit memo.
`DocumentNumber`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. The auto-generated reference number for the credit memo.
`EffectiveDate`	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The effective date of the credit memo. If this field is empty, the credit date is used. For reporting purposes only; this field drives no other logic.
`ExternalReference`	**Type** string

Field	Details
	Properties Filter, Group, Nillable, Sort, Update **Description** An external system's ID for the credit memo.
ExternalReferenceDataSource	**Type** string **Properties** Filter, Group, Nillable, Sort, Update **Description** The name of the external system that also contains the credit memo.
FuncCrcyCnvTotAmtWithTax	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The total amount with tax value in functional currency. Available in API version 66.0 and later.
FunctionalCurrencyCvsnDate	**Type** date **Properties** Filter, Group, Nillable, Sort, Update **Description** The date on which the total amount with tax value is converted into functional currency. Available in API version 66.0 and later.
FunctionalCurrencyCvsnRate	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The exchange rate that's used to convert the total amount with tax value into functional currency. Available in API version 66.0 and later.
FunctionalCurrencyIsoCode	**Type** string **Properties** Filter, Group, Nillable, Sort, Update **Description** The ISO code of the functional currency. Available in API version 66.0 and later.

Field	Details
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed a credit memo record indirectly, for example, through a list view or related record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed a credit memo record. If this value is null, it's possible that the user only accessed the credit memo record or a related list view (LastReferencedDate), but not viewed the credit memo record itself.
LegalEntityAccountingPeriodId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The legal entity accounting period record that's related to the credit memo. This field is available in API version 65.0 and later. This field is a relationship field. **Relationship Name** LegalEntityAccountingPeriod **Refers To** LegalEntyAccountingPeriod
LegalEntityId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The legal entity record that's related to the credit memo. This field is available in API version 65.0 and later. This field is a relationship field. **Relationship Name** LegalEntity

Field	Details
	Refers To LegalEntity
`NetCreditsApplied`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The total difference between the credit applied to and credit unapplied from the invoice. This field is a calculated field.
`OwnerId`	**Type** reference **Properties** Filter, Group, Sort, Update **Description** Required. The ID of the user who owns a credit memo record. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
`ReasonCode`	**Type** picklist **Properties** Filter, Group, Nillable, Sort **Description** The reason code for the credit memo's category. For example, BD can be the reason code when the credit memo's category is `InvoiceWriteOff` and the reason for the invoice write-off is bad debt. This field is available in API version 64.0 and later.
`ReferenceEntityId`	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The ID of the record that this credit memo was generated from. For example, the order, order summary, or invoice. This field is a polymorphic relationship field.

Field	Details
	Relationship Name ReferenceEntity **Refers To** Invoice, Order
SettlementLevel	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** Specifies whether the credit memo amount was applied to an invoice or an invoice line.
SourceAction	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** Specifies which Salesforce API created the credit memo. Valid values are: • Invoice • NegativeInvoiceLineConversion • Standalone • VoidPostedInvoice • WriteOffPostedInvoice—Available in API version 64.0 and later.
Status	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort, Update **Description** Required. The status of the credit memo. Valid values are: • Canceled • Draft • Error • Pending • Posted • Void In Progress—Available in API version 64.0 and later. • Voided

Field	Details
TotalAdjustmentAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of the `TotalAmount` field values for the credit memo’s adjustment lines. This field is a calculated field.
TotalAdjustmentAmountWithTax	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of the credit memo’s adjustment line amounts, including tax. Available in API versions 62.0 and 63.0.
TotalAdjustmentTaxAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of the credit memo’s adjustment line tax. Adjustment line balances are excluded. Available in API versions 62.0 and 63.0.
TotalAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of the credit memo’s `TotalLineAmount` and `TotalAdjustmentAmount` field values. This field is a calculated field.
TotalAmountWithTax	**Type** currency **Properties** Filter, Nillable, Sort **Description** The total credit memo amount, with tax included. This field is a calculated field.

Field	Details
`TotalChargeAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of the `TotalAmount` field values for the credit memo’s charge lines. This field is a calculated field.
`TotalChargeAmountWithTax`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of the credit memo’s charge line amounts, including tax. Available in API versions 62.0 and 63.0.
`TotalChargeTaxAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of the credit memo’s charge tax amount. Available in API versions 62.0 and 63.0.
`TotalTaxAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of the `TotalAmount` field values for the credit memo’s tax lines. This field is a calculated field.
`TotalTaxesCapturedAtHeader`	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The total tax calculated at the transaction header level. Available in API version 66.0 and later.

CreditMemoAddressGroup

Represents the storage of the buyer's address information, which is used to determine the tax credit amount for a buyer when a credit memo is issued. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`

Special Access Rules

You need the Credit Memo Operations User permission set to access this object.

Fields

Field	Details
`Address`	**Type** address **Properties** Filter, Nillable **Description** The buyer's address.
`CreditMemoAddressGroupNumber`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. The auto-generated reference number for the credit memo address group.
`CreditMemoId`	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The ID of the credit memo associated with the address group. This field is a relationship field. **Relationship Name** CreditMemo

Field	Details
	Relationship Type Master-detail **Refers To** CreditMemo (the master object)
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed a credit memo address group record indirectly, for example, through a list view or related record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed a credit memo address group record. If this value is null, it's possible that the user only accessed the credit memo address group record or a related list view (`LastReferencedDate`), but not viewed the credit memo address group record itself.

CreditMemoInvApplication

Represents information about the application of a credit memo to an invoice. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `update()`

Special Access Rules

You need the Credit Memo Operations User permission set to access this object.

Fields

Field	Details
`Amount`	**Type** currency **Properties** Filter, Sort **Description** Required. The amount of the credit memo that was applied to or unapplied from the invoice.
`AppliedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when the credit memo was applied. If the credit memo invoice application's type is `Unapplied`, this value is inherited from the `Applied` date of the credit memo referenced in the `AssociatedLineId` field.
`AssociatedLineId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** ID of the original credit memo invoice application for a credit memo invoice application that represents an unapplied credit memo. This field is a relationship field. **Relationship Name** AssociatedLine **Refers To** CreditMemoInvApplication
`CreditMemoBalance`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The balance of a credit memo after it's applied or unapplied. This field is a snapshot of the credit memo's balance after the action. This field isn't updated after further changes to the credit memo balance.
`CreditMemoId`	**Type** reference

Field	Details
	Properties Filter, Group, Sort **Description** Required. The credit memo that was applied or unapplied. This field is a relationship field. **Relationship Name** CreditMemo **Refers To** CreditMemo
`CreditMemoInvoiceNumber`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. An auto-generated number identifying the credit memo invoice application.
`Date`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when the credit memo amount was applied to the invoice.
`Description`	**Type** string **Properties** Filter, Group, Nillable, Sort, Update **Description** Additional details about the credit memo applied to an invoice.
`EffectiveDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The effective date of the application or unapplication of credit. You can provide this value when applying or unapplying the credit memo. This field is optional and provided only for reporting purposes. It doesn't affect the credit memo invoice application's other fields.

Field	Details
HasBeenUnapplied	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** Specifies whether this credit memo application has been unapplied from the target invoice. Valid values are: • NA • No • Yes
ImpactAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The net adjustment to the invoice's balance after a credit memo is applied or unapplied. If a credit memo was applied, this value is the negative version of the credit memo invoice application's Amount field. If a credit memo was unapplied, this value is the positive version of the credit memo invoice application's Amount field. This field is a calculated field.
InvoiceBalance	**Type** currency **Properties** Filter, Nillable, Sort **Description** The balance of the credit memo after a credit memo is applied or unapplied. This field is a snapshot of the credit memo's balance after the action. This field isn't updated after further changes to the credit memo balance.
InvoiceId	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The invoice to which the credit memo is applied. This field is a relationship field. **Relationship Name** Invoice

Field	Details
	Relationship Type Master-detail **Refers To** Invoice (the master object)
LegalEntityAccountingPeriodId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The legal entity accounting period record that's related to the credit memo invoice application. This field is available in API version 67.0 and later. This field is a relationship field. **Relationship Name** LegalEntityAccountingPeriod **Refers To** LegalEntyAccountingPeriod
LegalEntityId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The legal entity record that's related to the credit memo invoice application. This field is available in API version 67.0 and later. This field is a relationship field. **Relationship Name** LegalEntity **Refers To** LegalEntity
Type	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** Required. Specifies whether the credit memo line application was generated because of an apply action (application) or an unapply action (unapplication). Valid values are: • Applied • Unapplied

Field	Details
`UnappliedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when this application was unapplied from the target invoice.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

CreditMemoInvApplicationFeed on page 2866
Feed tracking is available for the object.

CreditMemoInvApplicationHistory on page 2873
History is available for tracked fields of the object.

CreditMemoLine

Represents the product, service, adjustment, or tax line items included in a credit memo. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `update()`

Special Access Rules

You need the Credit Memo Operations User permission set to access this object.

Fields

Field	Details
`AdjustmentAmount`	**Type** currency **Properties** Filter, Nillable, Sort, Update **Description** The amount of this credit memo line item if its type is Adjustment.

Field	Details
AdjustmentAmountWithTax	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of the adjustment amount and the adjustment tax amount. Available in API versions 62.0 and 63.0.
AdjustmentTaxAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The amount of the tax related to the adjustment amount. Available in API versions 62.0 and 63.0.
Balance	**Type** currency **Properties** Filter, Nillable, Sort **Description** The amount of the credit memo line available for allocation.
BillingAddressId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the billing address related to this credit memo line. This field is a relationship field. **Relationship Name** BillingAddress **Refers To** CreditMemoAddressGroup
ChargeAmount	**Type** currency **Properties** Filter, Nillable, Sort, Update **Description** The amount of this credit memo line item if its type is Charge.

Field	Details
ChargeAmountWithTax	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of the charge amount and the charge tax amount. Available in API versions 62.0 and 63.0.
ChargeTaxAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The amount of the tax related to the charge amount. Available in API versions 62.0 and 63.0.
CorpCurrencyCnvChargeAmt	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The amount of the credit memo line item if its type is `Charge` in corporate currency. Available in API version 63.0 and later.
CorpCurrencyCnvTotalTaxAmt	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The sum of the total amount with the tax on the credit memo line in corporate currency. Available in API version 63.0 and later.
CorporateCurrencyCvsnDate	**Type** date **Properties** Filter, Group, Nillable, Sort, Update **Description** The date on which the credit memo line amounts are converted to the corporate currency. Available in API version 63.0 and later.
CorporateCurrencyCvsnRate	**Type** double

Field	Details
	Properties Filter, Nillable, Sort, Update **Description** The exchange rate that's used to convert the credit memo line amounts to the corporate currency. Available in API version 63.0 and later.
CorporateCurrencyIsoCode	**Type** string **Properties** Filter, Group, Nillable, Sort, Update **Description** The currency ISO code of the corporate currency. Available in API version 63.0 and later.
CreditMemoId	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The ID of the parent credit memo. This field is a relationship field. **Relationship Name** CreditMemo **Relationship Type** Master-detail **Refers To** CreditMemo (the master object)
Description	**Type** string **Properties** Filter, Group, Nillable, Sort, Update **Description** Additional details about the credit memo line.
EndDate	**Type** date **Properties** Filter, Group, Nillable, Sort, Update **Description** The description of the credit memo line.

Field	Details
FuncCrcyCnvTotalTaxAmount	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The total tax amount value in functional currency. Available in API version 66.0 and later.
FuncCurrencyCnvChargeAmt	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The charge amount value in functional currency. Available in API version 66.0 and later.
FunctionalCurrencyCvsnDate	**Type** date **Properties** Filter, Group, Nillable, Sort, Update **Description** The date on which the credit memo line amounts are converted into functional currency. Available in API version 66.0 and later.
FunctionalCurrencyCvsnRate	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The exchange rate that's used to convert credit memo line amounts into functional currency. Available in API version 66.0 and later.
FunctionalCurrencyIsoCode	**Type** string **Properties** Filter, Group, Nillable, Sort, Update **Description** The ISO code of the functional currency. Available in API version 66.0 and later.
LegalEntityAccountingPeriodId	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update

Field	Details
	Description The ID of the legal entity accounting period record of the credit memo line. This field is a relationship field. **Relationship Name** LegalEntityAccountingPeriod **Refers To** LegalEntyAccountingPeriod
LegalEntityId	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The ID of the legal entity record related to the credit memo line. This field is a relationship field. **Relationship Name** LegalEntity **Refers To** LegalEntity
LineAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The amount of the credit memo line. This field is a calculated field.
Name	**Type** string **Properties** Filter, Group, idLookup, Sort, Update **Description** Required. The name of the credit memo line.
NetCreditsApplied	**Type** currency **Properties** Filter, Nillable, Sort

Field	Details
	Description The total amount applied to invoice lines from the credit memo. This amount is calculated by subtracting the total unapplied credit amount from the total applied credit amount. This field is a calculated field.
`Product2Id`	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The ID of the product or service being credited in the credit memo line. This field is a relationship field. **Relationship Name** Product2 **Refers To** Product2
`ReferenceEntityItemId`	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The order product or invoice line corresponding to this credit memo line. This field is a polymorphic relationship field. **Relationship Name** ReferenceEntityItem **Refers To** InvoiceLine, OrderItem
`ReferenceEntityItemType`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The type of transaction that generated the credit memo line. Valid values are: • `Delivery Charge` • `Fee` • `Order Product`

Field	Details
ReferenceEntityItemTypeCode	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The type of object that generated the credit memo line. Valid values are: • `Charge` • `Product`
ShipFromAddressId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The address from which the product in the credit memo line is shipped. Available in API version 64.0 and later. This field is a relationship field. **Relationship Name** ShipFromAddress **Refers To** CreditMemoAddressGroup
ShippingAddressId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the shipping address. This field is a relationship field. **Relationship Name** ShippingAddress **Refers To** CreditMemoAddressGroup
StartDate	**Type** date **Properties** Filter, Group, Nillable, Sort, Update

Field	Details
	Description The first date of the billing for the service for credit memo lines generated from a time-based service.
`Status`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The status of the credit memo line. This field is inherited from the credit memo.
`TaxAmount`	**Type** currency **Properties** Filter, Nillable, Sort, Update **Description** The total tax amount for the credit memo.
`TaxTreatmentId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the tax treatment record for the credit memo line. This field is a relationship field. **Relationship Name** TaxTreatment **Refers To** TaxTreatment

CreditMemoLineInvoiceLine

Represents a junction between a credit memo line and an invoice line. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `update()`

Special Access Rules

You need the Credit Memo Operations User permission set to access this object.

Fields

Field	Details
Amount	**Type** currency **Properties** Filter, Sort **Description** Required. The amount that's been applied to or unapplied from the invoice line.
AppliedDateTime	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date and time when the credit memo line was applied to the invoice line.
CreditMemoLineBalance	**Type** currency **Properties** Filter, Nillable, Sort **Description** The balance of the credit memo line after it's applied to or unapplied from the invoice line.
CreditMemoLineId	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The ID of the credit memo line record that's applied to or unapplied from the invoice line. This field is a relationship field. **Relationship Name** CreditMemoLine **Refers To** CreditMemoLine
CreditMemoLineInvoiceLineNumber	**Type** string

Field	Details
	Properties Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. An auto-generated number identifying the credit memo line invoice line.
`Description`	**Type** textarea **Properties** Filter, Group, Nillable, Sort, Update **Description** Additional details about the credit memo line invoice line.
`EffectiveDateTime`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date and time when the credit memo line's application to or unapplication from the invoice line becomes effective.
`ImpactAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The credit memo line invoice line’s financial impact against the customer’s accounts receivable. If the credit memo line invoice line's type is `Applied`, the impact amount is the negative equivalent of the credit memo line invoice line's amount. Otherwise, the impact amount is equal to the credit memo line invoice line's amount. This field is a calculated field.
`InvoiceLineBalance`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The balance of the invoice line after the credit memo line was applied or unapplied.
`InvoiceLineId`	**Type** reference **Properties** Filter, Group, Sort

Field	Details
	Description Required. The ID of the invoice line record to which the credit memo line has been applied or unapplied. This field is a relationship field. **Relationship Name** InvoiceLine **Relationship Type** Master-detail **Refers To** InvoiceLine (the master object)
`LegalEntityAccountingPeriodId`	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The ID of the legal entity accounting period record related to the credit memo line invoice line. This field is a relationship field. **Relationship Name** LegalEntityAccountingPeriod **Refers To** LegalEntyAccountingPeriod
`LegalEntityId`	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The ID of the legal entity record related to the credit memo line invoice line. This field is a relationship field. **Relationship Name** LegalEntity **Refers To** LegalEntity
`RelatedCrMemoLineInvcLineId`	**Type** reference **Properties** Filter, Group, Nillable, Sort

Field	Details
	Description The ID of the related credit memo line invoice line record of the type `Applied` when the credit memo line invoice line's type is `Unapplied`. This field is a relationship field. **Relationship Name** RelatedCrMemoLineInvcLine **Refers To** CreditMemoLineInvoiceLine
`Type`	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** Required. Specifies whether the credit memo line has been applied to or unapplied from the invoice line. Valid values are: • `Applied` • `Unapplied`
`UnappliedDateTime`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date and time when the credit memo line was unapplied from the invoice line.
`UnappliedStatus`	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** Required. Specifies whether the credit memo line has been unapplied from the invoice line. Valid values are: • `NA` • `No` • `Yes`

CreditMemoLineTax

Represents tax information of a credit memo line of type `Tax`. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `update()`

Special Access Rules

You need the Tax Admin permission set to access this object.

Fields

Field	Details
`BillingAddressId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The billing address of the parent credit memo line. This field is a relationship field. **Relationship Name** BillingAddress **Refers To** CreditMemoAddressGroup
`CalculationStatus`	**Type** picklist **Properties** Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort **Description** The status of the tax calculation. Valid values are: • `Complete` • `Error` • `None` The default value is `None`.

Field	Details
CorpCrcyCnvTaxAmount	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The total tax amount of the credit memo line tax in corporate currency. Available in API version 63.0 and later.
CorporateCurrencyCvsnDate	**Type** date **Properties** Filter, Group, Nillable, Sort, Update **Description** The date on which the credit memo line tax amounts are converted to the corporate currency. Available in API version 63.0 and later.
CorporateCurrencyCvsnRate	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The exchange rate that's used to convert the credit memo line tax amounts to the corporate currency. Available in API version 63.0 and later.
CorporateCurrencyIsoCode	**Type** string **Properties** Filter, Group, Nillable, Sort, Update **Description** The currency ISO code of the corporate currency. Available in API version 63.0 and later.
CreditMemoLineId	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The ID of the parent charge or adjustment credit memo line record that a credit memo line tax is related to. This field is a relationship field. **Relationship Name** CreditMemoLine

Field	Details
	Relationship Type Master-detail **Refers To** CreditMemoLine (the master object)
`CreditMemoLineTaxNumber`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. An auto-generated number identifying a credit memo line tax.
`Description`	**Type** string **Properties** Filter, Group, Nillable, Sort, Update **Description** Additional details about the credit memo line tax.
`EndDate`	**Type** date **Properties** Filter, Group, Nillable, Sort **Description** The end date of a credit memo line tax.
`FuncCrcyCnvTaxAmount`	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The tax amount value in functional currency. Available in API version 66.0 and later.
`FunctionalCurrencyCvsnDate`	**Type** date **Properties** Filter, Group, Nillable, Sort, Update **Description** The date on which the tax amount value is converted into functional currency. Available in API version 66.0 and later.

Field	Details
FunctionalCurrencyCvsnRate	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The exchange rate that's used to convert the tax amount value into functional currency. Available in API version 66.0 and later.
FunctionalCurrencyIsoCode	**Type** string **Properties** Filter, Group, Nillable, Sort, Update **Description** The ISO code of the functional currency. Available in API version 66.0 and later.
LegalEntityAccountingPeriodId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the legal entity accounting period record that's related to a credit memo line tax. This field is a relationship field. **Relationship Name** LegalEntityAccountingPeriod **Refers To** LegalEntyAccountingPeriod
LegalEntityId	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The ID of the legal entity record that's related to a credit memo line tax. This field is a relationship field. **Relationship Name** LegalEntity **Refers To** LegalEntity
Product2Id	**Type** reference

Field	Details
	Properties Filter, Group, Nillable, Sort **Description** The product or service being credited in the parent credit memo line. Available in API version 62.0 only. This field is a relationship field. **Relationship Name** Product2 **Refers To** Product2
`ReferenceEntityItemId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The invoice line tax corresponding to a credit memo line tax. This field is a relationship field. **Relationship Name** ReferenceEntityItem **Refers To** InvoiceLineTax
`ShipFromAddressId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ship from address specified in the parent credit memo line. Available in API version 64.0 and later. This field is a relationship field. **Relationship Name** ShipFromAddress **Refers To** CreditMemoAddressGroup
`ShippingAddressId`	**Type** reference **Properties** Filter, Group, Nillable, Sort

Field	Details
	Description The shipping address of the parent credit memo line. This field is a relationship field. **Relationship Name** ShippingAddress **Refers To** CreditMemoAddressGroup
`StartDate`	**Type** date **Properties** Filter, Group, Nillable, Sort **Description** The start date of a credit memo line tax.
`TaxAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The total amount of a credit memo line tax.
`TaxCode`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The code that's used to calculate the tax rate for the parent credit memo line.
`TaxDocumentNumber`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The document number that's used to calculate the tax rate for the parent credit memo line.
`TaxEffectiveDate`	**Type** date **Properties** Filter, Group, Nillable, Sort

Field	Details
	Description The date used to calculate the tax amount.
TaxName	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The user-defined name for a credit memo line tax.
TaxRate	**Type** percent **Properties** Filter, Nillable, Sort **Description** The percentage that's used to calculate tax.
TaxTransactionNumber	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The number of the transaction in the external tax engine that calculated tax for the parent credit memo line.
TaxTreatmentId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the tax treatment record that's related to a credit memo line tax. This field is a relationship field. **Relationship Name** TaxTreatment **Refers To** TaxTreatment

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

CreditMemoLineTaxFeed on page 2866
Feed tracking is available for the object.

CreditMemoLineTaxHistory on page 2873
History is available for tracked fields of the object.

DebitMemo

Represents the document used to charge an additional amount to a buyer by a seller. An invoice is generated for the debit memo in the next invoice run. This object is available in API version 65.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `update()`, `upsert()`

Special Access Rules

You need Revenue Cloud Billing license and one of these permission sets to access this object.

- Billing Admin permission set
- Billing Operations User permission set
- Payments Admin permission set
- Payments Operation User permission set
- Credit Memo Operations User permission set

Fields

Field	Details
`AccountId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The account that's related to the debit memo record. This field is a relationship field. **Relationship Name** Account **Refers To** Account
`CorpCrcyCnvTotChargeAmt`	**Type** double **Properties** Filter, Nillable, Sort, Update

Field	Details
	Description The total charge amount of the debit memo in corporate currency.
CorporateCurrencyCvsnDate	**Type** date **Properties** Filter, Group, Nillable, Sort, Update **Description** The exchange date that's used to convert the debit memo amounts into corporate currency.
CorporateCurrencyCvsnRate	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The exchange rate that's used to convert the debit memo amounts into corporate currency.
CorporateCurrencyIsoCode	**Type** string **Properties** Filter, Group, Nillable, Sort, Update **Description** The currency ISO code of the corporate currency.
Description	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The description of the debit memo.
DocumentNumber	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** An autogenerated number assigned to the debit memo record.
FuncCrcyCnvTotChargeAmt	**Type** double **Properties** Filter, Nillable, Sort, Update

Field	Details
	Description The total charge amount value in functional currency. Available in API version 66.0 and later.
`FunctionalCurrencyCvsnDate`	**Type** date **Properties** Filter, Group, Nillable, Sort, Update **Description** The exchange date that's used to convert the total charge amount value into functional currency. Available in API version 66.0 and later.
`FunctionalCurrencyCvsnRate`	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The exchange rate that's used to convert the total charge amount value into functional currency. Available in API version 66.0 and later.
`FunctionalCurrencyIsoCode`	**Type** string **Properties** Filter, Group, Nillable, Sort, Update **Description** The ISO code of the functional currency. Available in API version 66.0 and later.
`InvoiceGenerationStatus`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The status of invoice generation for the debit memo. Valid values are: • `Draft Invoice Generated` • `None` • `Posted Invoice Generated` • `Processing` • `Ready for Invoice Generation` • `Not Applicable`—Available in API version 66.0 and later.

Field	Details
`InvoiceMatchingRefKeyId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Indicates the product or Asset ID. This key, combined with the account, currency, and legal entity, is used to find a matching invoice. The invoice lines converted from debit memo lines are then added to this matched invoice. This field is a polymorphic relationship field. **Relationship Name** InvoiceMatchingRefKey **Refers To** Asset, Product2
`InvoiceMatchingRefName`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Indicates the product's name. This text, combined with the account, currency, and legal entity, is used to find a matching invoice. The invoice lines converted from debit memo lines are then added to this matched invoice.
`IsManuallyProcessed`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the debit memo lines of the debit memo are converted to invoice lines manually (`true`) or not (`false`). The default value is `false`.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed a debit memo address group record indirectly, for example, through a list view or related record.
`LastViewedDate`	**Type** dateTime

Field	Details
	Properties Filter, Nillable, Sort **Description** The timestamp when the current user last viewed a debit memo address record. If this value is null, it's possible that the user only accessed the debit memo address record or a related list view (`LastReferencedDate`), but not viewed the debit memo address record itself.
`LegalEntityId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The legal entity record that's related to the debit memo. This field is a relationship field. **Relationship Name** LegalEntity **Refers To** LegalEntity
`NextBillingDate`	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The date indicates when the next billing period begins for the debit memo. This date is used to determine when invoices are created for the debit memos. When the invoice scheduler or API evaluates debit memos to generate invoices, those with a next billing date on or before the user-specified target date are included in the invoice batch run.
`OwnerId`	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the owner of this object. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User

Field	Details
PostedDate	**Type** date **Properties** Filter, Group, Nillable, Sort **Description** The date when the debit memo is posted.
ReasonCode	**Type** picklist **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Specifies a dynamic picklist of reasons for generating debit memos.
ReferenceRecord	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The record ID associated with the debit memo record. The record ID is a credit memo record. Available in API version 66.0 and later. This field is a polymorphic relationship field. **Relationship Name** InvoiceMatchingRefKey **Refers To** Credit Memo
Status	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The status of the debit memo. Valid values are: • Canceled • Draft • Posted • Void • Pending—Available in API version 66.0 and later. • Error—Available in API version 66.0 and later.

Field	Details
TotalAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of the debit memo's TotalLineAmount and TotalAdjustmentAmount field values. Available in API version 66.0 and later. This field is a calculated field.
TotalChargeAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of the charge amount field values of the debit memo lines. This field is a calculated field.
TotalTaxAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of the TotalAmount field values for the debit memo's tax lines. Available in API version 66.0 and later. This field is a calculated field.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

DebitMemoFeed on page 2866
Feed tracking is available for the object.

DebitMemoHistory on page 2873
History is available for tracked fields of the object.

DebitMemoOwnerSharingRule on page 2875
Sharing rules are available for the object.

DebitMemoShare on page 2877
Sharing is available for the object.

DebitMemoAddress

Represents the buyer's address information, which is used to determine the tax amount for a buyer when a debit memo is issued. This object is available in API version 65.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need Revenue Cloud Billing license and one of these permission sets to access this object.

- Billing Admin permission set
- Billing Operations User permission set
- Payments Admin permission set
- Payments Operation User permission set
- Credit Memo Operations User permission set

Fields

Field	Details
`Address`	**Type** address **Properties** Filter **Description** The billing or shipping address of the debit memo.
`DebitMemoAddressNumber`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** The auto-generated reference number for the debit memo address.
`DebitMemoId`	**Type** reference **Properties** Create, Filter, Group, Sort

Field	Details
	Description The ID of the debit memo associated with the address. This field is a relationship field. **Relationship Name** DebitMemo **Relationship Type** Master-detail **Refers To** DebitMemo (the master object)
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed a debit memo address record indirectly, for example, through a list view or related record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed a debit memo address record. If this value is null, it's possible that the user only accessed the debit memo address record or a related list view (`LastReferencedDate`), but not viewed the debit memo address record itself.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

DebitMemoAddressHistory on page 2873
History is available for tracked fields of the object.

DebitMemoLine

Represents the additional charge amount that the buyer must pay for the product, service, or debit memo line tax that's related to the debit memo. This object is available in API version 65.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `update()`, `upsert()`

Special Access Rules

You need Revenue Cloud Billing license and one of these permission sets to access this object.

- Billing Admin permission set
- Billing Operations User permission set
- Payments Admin permission set
- Payments Operation User permission set
- Credit Memo Operations User permission set

Fields

Field	Details
`BillingAddressId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The billing address record that's related to the debit memo line. This field is a relationship field. **Relationship Name** BillingAddress **Refers To** DebitMemoAddress
`ChargeAmount`	**Type** currency **Properties** Create, Filter, Sort, Update **Description** The amount of the debit memo line item.
`CorpCurrencyCnvChargeAmt`	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The charge amount of the debit memo line in corporate currency.

Field	Details
CorporateCurrencyCvsnDate	**Type** date **Properties** Filter, Group, Nillable, Sort, Update **Description** The date on which the debit memo line amounts are converted into corporate currency.
CorporateCurrencyCvsnRate	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The exchange rate that's used to convert debit memo line amounts into corporate currency.
CorporateCurrencyIsoCode	**Type** string **Properties** Filter, Group, Nillable, Sort, Update **Description** The currency ISO code of the corporate currency.
DebitMemoId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The parent debit memo record that's related to the debit memo line record. This field is a relationship field. **Relationship Name** DebitMemo **Relationship Type** Master-detail **Refers To** DebitMemo (the master object)
Description	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The description of the debit memo line.

Field	Details
EndDate	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The end date of the debit memo line that determines which billing cycle it must be associated with.
FuncCrcyCnvTotalTaxAmt	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The total tax amount value in functional currency. Available in API version 66.0 and later.
FuncCrcyCnvChargeAmount	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The charge amount value in functional currency. Available in API version 66.0 and later.
FunctionalCurrencyCvsnDate	**Type** date **Properties** Filter, Group, Nillable, Sort, Update **Description** The date on which the debit memo line amounts is converted into functional currency. Available in API version 66.0 and later.
FunctionalCurrencyCvsnRate	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The exchange rate that's used to convert the debit memo line amounts into functional currency. Available in API version 66.0 and later.
FunctionalCurrencyIsoCode	**Type** string **Properties** Filter, Group, Nillable, Sort, Update

Field	Details
	Description The ISO code of the functional currency. Available in API version 66.0 and later.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed a debit memo address record indirectly, for example, through a list view or related record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed a debit memo address record. If this value is null, it's possible that the user only accessed the debit memo address record or a related list view (LastReferencedDate), but not viewed the debit memo address record itself.
LegalEntityAccountingPeriodId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The legal entity accounting period record that's related to the debit memo line. This field is a relationship field. **Relationship Name** LegalEntityAccountingPeriod **Refers To** LegalEntyAccountingPeriod
LegalEntityId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The legal entity record that's related to the debit memo line. This field is a relationship field. **Relationship Name** LegalEntity

Field	Details
	Refers To LegalEntity
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** An auto-generated name assigned to the debit memo line.
`Product2Id`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the product or service being debited in the debit memo line. This field is a relationship field. **Relationship Name** Product2 **Refers To** Product2
`ReferenceRecordId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The record ID associated with the debit memo line record. The record ID is an asset, invoice line, contract, or refund record. This field is a polymorphic relationship field. **Relationship Name** ReferenceRecord **Refers To** Asset, Contract, InvoiceLine, Refund, Credit Memo Line
`ShippingAddressId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The shipping address record that’s related to the debit memo line.

Field	Details
	This field is a relationship field. **Relationship Name** ShippingAddress **Refers To** DebitMemoAddress
`StartDate`	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The start date of the debit memo line that determines which billing cycle it must be associated with.
`TaxTreatmentId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the tax treatment record that's related to the debit memo line. This field is a relationship field. **Relationship Name** TaxTreatment **Refers To** TaxTreatment

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

DebitMemoLineFeed on page 2866
Feed tracking is available for the object.

DebitMemoLineHistory on page 2873
History is available for tracked fields of the object.

DebitMemoLineTax

Represents the tax information for a debit memo line. This object is available in API version 66.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`

Special Access Rules

You need Revenue Cloud Billing license and the Tax Admin permission set to access this object.

Fields

Field	Details
`BillingAddressId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The billing address specified in the parent debit memo line. This field is a relationship field. **Relationship Name** BillingAddress **Refers To** DebitMemoAddress
`CurrencyIsoCode`	**Type** picklist **Properties** Defaulted on create, Filter, Group, Restricted picklist, Sort **Description** Required. The currency ISO code of the corporate currency.
`DebitMemoLineId`	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The parent debit memo line that the debit memo tax line is related to. This field is a relationship field. **Relationship Name** DebitMemoLine **Relationship Type** Master-detail **Refers To** DebitMemoLine (the master object)

Field	Details
DebitMemoLineTaxNumber	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. An auto-generated number that identifies a debit memo line tax.
Description	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The description of the debit memo line tax.
EndDate	**Type** date **Properties** Filter, Group, Nillable, Sort **Description** The end date of the debit memo line tax that determines which billing cycle it's associated with.
JurisdictionTaxCode	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The code of the jurisdiction that's used to calculate the tax rate for the parent debit memo line.
JurisdictionTaxName	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The user-defined name for the debit memo line tax.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort

Field	Details
	Description The timestamp when the current user last accessed a debit memo line tax record indirectly, for example, through a list view or related record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed a debit memo line tax record. If this value is null, it's possible that the user only accessed the debit memo line tax record or a related list view (`LastReferencedDate`), but not viewed the debit memo line tax record itself.
`LegalEntityAccountingPeriodId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The legal entity accounting period record associated with the debit memo line. This field is a relationship field. **Relationship Name** LegalEntityAccountingPeriod **Refers To** LegalEntyAccountingPeriod
`LegalEntityId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The legal entity record associated with the debit memo line. This field is a relationship field. **Relationship Name** LegalEntity **Refers To** LegalEntity
`ShipFromAddressId`	**Type** reference **Properties** Filter, Group, Nillable, Sort

Field	Details
	Description The shipping address specified in the parent debit memo line. This field is a relationship field. **Relationship Name** ShipFromAddress **Refers To** DebitMemoAddress
StartDate	**Type** date **Properties** Filter, Group, Nillable, Sort **Description** The start date of the debit memo line tax that determines which billing cycle it's associated with.
TaxAmount	**Type** currency **Properties** Filter, Sort **Description** Required. The total amount of the debit memo line tax.
TaxRate	**Type** percent **Properties** Filter, Nillable, Sort **Description** The percentage that's used to calculate tax amount based on the charge amount.
TaxTransactionNumber	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The transaction number in the external tax engine that calculated the tax for the parent debit memo line.

GeneralLedgerAccount

Represents information about the accounting codes, types, and names that are used to store and organize financial transactions. This object is available in API version 63.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need Revenue Cloud Billing license and the Accounts Receivables Admin permission set to access this object.

Fields

Field	Details
`AccountingCode`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The code that's used to organize information about the general ledger account.
`AccountingName`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** The user-specified name for the general ledger account.
`AccountingType`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The accounting type for the general ledger account.
`Description`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Additional details about the general ledger account.

Field	Details
FinancialStatement	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The financial statement that's created by using the information from the general ledger account.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed this record indirectly, for example, through a list view or related record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed this record or list view. If this value is null, and `LastReferenceDate` is not null, the user accessed this record or list view indirectly.
LegalEntityId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The legal entity that's related to the general ledger account. This field is a relationship field. **Relationship Name** LegalEntity **Refers To** LegalEntity
Name	**Type** string **Properties** Filter, Group, idLookup, Sort

Field	Details
	Description An auto-generated name identifying the general ledger account, which is a combination of the accounting code and the account name.
`OwnerId`	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the owner of this object or ID of the creator of this object. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
`Type`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Specifies the accounting type for the general ledger account. This field is available in API version 65.0 and later. Valid values are: • `Asset` • `Liability` • `Equity` • `Revenue` • `Expense` • `Others`

GeneralLedgerAcctAsgntRule

Represents information about the rule based on which general ledger accounts are assigned to transaction journals that are created for billing transactions. This object is available in API version 63.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need Revenue Cloud Billing license and the Accounts Receivables Admin permission set to access this object.

Fields

Field	Details
CreditGeneralLedgerAccountId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The general ledger account for a credit transaction that's related to the general ledger account assignment rule. This field is a relationship field. **Relationship Name** CreditGeneralLedgerAccount **Refers To** GeneralLedgerAccount
CurrencyIsoCode	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The currency code of the general ledger account assignment rule for Salesforce orgs with multicurrency enabled. Valid values are: • AUD—Australian Dollar • EUR—Euro • SGD—Singapore Dollar • USD—U.S. Dollar The default value is USD.
DebitGeneralLedgerAccountId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The general ledger account for a debit transaction that's related to the general ledger account assignment rule. This field is a relationship field.

Field	Details
	Relationship Name DebitGeneralLedgerAccount **Refers To** GeneralLedgerAccount
FilterCriteria	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The filter criteria for the general ledger account assignment rule. Valid values are: • All • Any • Custom
FilterLogic	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The filter logic for the general ledger account assignment rule when the filter criteria is Custom. Transaction journals are created for the transactions of the selected type that meet the defined filter logic.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed this record indirectly, for example, through a list view or related record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed this record or list view. If this value is null, and LastReferenceDate is not null, the user accessed this record or list view indirectly.

Field	Details
LegalEntityId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The legal entity that's related to the general ledger account assignment rule. This field is a relationship field. **Relationship Name** LegalEntity **Refers To** LegalEntity
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** A user-defined name for the general ledger account assignment rule.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** The ID of the owner of the general ledger account assignment rule record. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
Priority	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The priority of the general ledger account assignment rule when there are multiple general ledger account assignment rules defined for a transaction type.
Status	**Type** picklist

Field	Details
	Properties Create, Filter, Group, Restricted picklist, Sort, Update **Description** The status of the general ledger account assignment rule. Valid values are: • `Active` • `Inactive`
`TransactionAmountField`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The amount field on the transaction type that's used to record the credit or debit amount in the transaction journals that are generated for that specific transaction type. This field is available in API version 64.0 and later.
`TransactionType`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The transaction type that's related to the general ledger account assignment rule. Valid values are: • `CreditMemo` • `CreditMemoInvApplication` • `CreditMemoLine` • `CreditMemoLineInvoiceLine` • `CreditMemoLineTax` • `DebitMemoLine`—Available in API version 65.0 and later. • `Invoice` • `InvoiceLine` • `InvoiceLineTax` • `Payment`—Available in API version 64.0 and later. • `PaymentLineInvoice`—Available in API version 64.0 and later. • `PaymentLineInvoiceLine`—Available in API version 64.0 and later. • `Refund`—Available in API version 64.0 and later. • `RefundLinePayment`—Available in API version 64.0 and later.

GeneralLdgrAcctPrdSummary

Represents a junction between a general ledger account and a legal entity accounting period. Stores information about the total credit amount, total debit amount, opening balance, and closing balance of a general ledger account for a specific legal entity accounting period. This object is available in API version 65.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need Revenue Cloud Billing license and the Accounts Receivables Admin permission set to access this object.

Fields

Field	Details
`ClosingBalanceAmount`	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The closing balance amount for a general ledger accounting period summary is calculated based on the general ledger account's type. For asset and expense type general ledger accounts, it's the opening balance amount plus total debit amount minus total credit amount. For liability, equity, and revenue type general ledger accounts, it's the opening balance amount plus total credit amount minus total debit amount.
`GeneralLedgerAccountId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The general ledger account that's related to the general ledger accounting period summary. This field is a relationship field. **Relationship Name** GeneralLedgerAccount **Relationship Type** Master-detail **Refers To** GeneralLedgerAccount (the master object)

Field	Details
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed general ledger accounting period summary record indirectly, for example, through a list view or related record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed general ledger accounting period summary record or list view. If this value is null, and `LastReferenceDate` is not null, the user accessed this record or list view indirectly.
LegalEntityAccountingPeriodId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The legal entity accounting period that's related to the general ledger account. This field is a relationship field. **Relationship Name** LegalEntityAccountingPeriod **Refers To** LegalEntyAccountingPeriod
OpeningBalanceAmount	**Type** currency **Properties** Create, Filter, Sort, Update **Description** The opening balance is the same as the closing balance of the previous general ledger account period summary.
TotalCreditAmount	**Type** currency **Properties** Create, Filter, Sort, Update

Field	Details
	Description The sum of the credit fields from all transaction journals of the general ledger account for a specific legal entity accounting period.
`TotalDebitAmount`	**Type** currency **Properties** Create, Filter, Sort, Update **Description** The sum of the debit fields from all transaction journals of the general ledger account for a specific legal entity accounting period.

GeneralLedgerJrnlEntryRule

Represents information about the transaction journal entry rule, based on which transaction journals are created for the selected credit and debit general ledger accounts, transaction amount field, and percentage. This object is available in API version 65.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need Revenue Cloud Billing license and the Accounts Receivables Admin permission set to access this object.

Fields

Field	Details
`CreditGeneralLedgerAccountId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The general ledger account for a credit transaction. This field is a relationship field. **Relationship Name** CreditGeneralLedgerAccount

Field	Details
	Refers To GeneralLedgerAccount
DebitGeneralLedgerAccountId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** The general ledger account for a debit transaction. This field is a relationship field. **Relationship Name** DebitGeneralLedgerAccount **Refers To** GeneralLedgerAccount
GeneralLedgerAcctAsgntRuleId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The general ledger account assignment rule that's related to the general ledger journal entry rule. This field is a relationship field. **Relationship Name** GeneralLedgerAcctAsgntRule **Relationship Type** Master-detail **Refers To** GeneralLedgerAcctAsgntRule (the master object)
Percentage	**Type** percent **Properties** Create, Filter, Sort, Update **Description** The percentage of the amount field value that's used to record the credit or debit amount in the transaction journals generated for the general ledger journal entry rule.
TransactionAmountField	**Type** string

Field	Details
	Properties Create, Filter, Group, Sort, Update **Description** The amount field on the transaction type that's used to record the credit or debit amount in the transaction journals generated for that specific transaction type.

InvBatchDraftToPostedRun

Represents information about the batch job that posts all invoices with the status as `Draft` that are generated by the invoice batch run associated with the billing schedule. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `update()`

Special Access Rules

You need the Billing Operations User permission set to access this object.

Fields

Field	Details
`BatchJobId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The batch management job that's used to track the progress of the invoice batch run. This field is available in API version 67.0 and later. This field is a relationship field. **Relationship Name** BatchJob **Refers To** BatchJob
`Comments`	**Type** textarea

Field	Details
	Properties Filter, Nillable, Sort, Update **Description** Additional notes or comments for the invoice batch draft to posted run record.
CompletionDateTime	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The completion date and time of this invoice batch draft to posted run process.
CreditApplicationStatus	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The status of a credit application to the invoices that are posted by the invoice batch draft to posted run process. Valid values are: • Completed • Failed • NotApplicable
InvBatchDraftToPostedRunNumber	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. The auto-generated number of the invoice batch draft to posted run record.
InvoiceBatchRunId	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The ID of the invoice batch run record associated with the invoice batch draft to posted run record. This field is a relationship field. **Relationship Name** InvoiceBatchRun

Field	Details
	Relationship Type Master-detail **Refers To** InvoiceBatchRun (the master object)
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed an invoice batch draft to posted run record indirectly, for example, through a list view or related record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed an invoice batch draft to posted run record. If this value is null, it's possible that the user only accessed the invoice batch draft to posted run record or a related list view (`LastReferencedDate`), but not viewed the invoice batch draft to posted run record itself.
`StartDateTime`	**Type** dateTime **Properties** Filter, Sort **Description** Required. The start date and time of the invoice batch draft to posted run process.
`Status`	**Type** picklist **Properties** Defaulted on create, Filter, Group, Restricted picklist, Sort **Description** Required. The status of the invoice batch draft to posted run record. Valid values are: • `Completed` • `CompletedWithErrors` • `Failed` • `Started`

Field	Details
	The default value is `Started`.
`TotalDraftInvoicesSelected`	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The total number of invoices that are selected to be processed by the invoice batch draft to posted run process.
`TotalInvoiceAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The total amount of all the invoices posted by the invoice batch draft to posted run process.
`TotalInvoicesFailed`	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The total number of invoices that the invoice batch draft to posted run process failed to post.
`TotalPostedInvoices`	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The total number of invoices posted by the invoice batch draft to posted run process.

Invoice

Represents information about a financial document describing the total amount a buyer must pay for provided products or services. This object is available in API version 62.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `update()`, `upsert()`

Special Access Rules

You need the Billing Operations User permission set to access this object.

Fields

Field	Details
`Balance`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The outstanding balance of an invoice. The balance is equal to the invoice's total amount with tax, excluding payments and adjustments.
`BalanceValue`	**Type** double **Properties** Filter, Nillable, Sort **Description** The balance value is derived from an existing balance field in an invoice. Available in API version 65.0 and later. This field is a calculated field.
`BillToContactId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The bill to contact ID of the account that's related to an invoice. This field is a relationship field. **Relationship Name** BillToContact **Refers To** Contact
`BillingAccountId`	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** Required. The ID of the customer account related to an invoice.

Field	Details
	This field is a relationship field. **Relationship Name** BillingAccount **Refers To** Account
`BillingArrangementId`	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The ID of the billing arrangement record used to generate the invoice. Available in API version 66.0 and later. This field is a relationship field. **Relationship Name** BilllingArrangement **Refers To** BilllingArrangement
`BillingAccountId`	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The billing account for which the invoice is generated. Available in API version 66.0 and later. This field is a relationship field. **Relationship Name** BillingAccount **Refers To** BillingAccount
`BillingArrangementVerNumber`	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The version number of the billing arrangement record that's used to generate the invoice. Available in API version 66.0 and later. This field is an autogenerated field.

Field	Details
CorrelationIdentifier	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** A unique identifier that’s used to associate and track invoices that are generated based on the same billing arrangement. Available in API version 66.0 and later. This field is an autogenerated field.
CorpCrcyCnvTotAmtWithTax	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The sum of the total amount with tax on the related invoice lines in corporate currency. Available in API version 63.0 and later.
CorporateCurrencyCvsnDate	**Type** date **Properties** Filter, Group, Nillable, Sort, Update **Description** The date on which the invoice’s total amount with tax was converted to the corporate currency. Available in API version 63.0 and later.
CorporateCurrencyCvsnRate	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The exchange rate that's used to convert invoice’s total amount with tax to the corporate currency. Available in API version 63.0 and later.
CorporateCurrencyIsoCode	**Type** string **Properties** Filter, Group, Nillable, Sort, Update **Description** The currency ISO code of the corporate currency. Available in API version 63.0 and later.
DaysInvoiceOpen	**Type** int

Field	Details
	Properties Filter, Group, Nillable, Sort **Description** The number of days from the day an invoice was created to the day it was paid. This field is a calculated field.
DaysInvoiceOverdue	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The number of days since the date when the payment for an invoice was due. This field is a calculated field.
Description	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Additional details about the invoice.
DocumentNumber	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. An auto-generated number identifying an invoice.
DueDate	**Type** date **Properties** Create, Filter, Group, Sort, Update **Description** Required. The date by when the customer must pay an invoice.
FullSettlementDate	**Type** date **Properties** Filter, Group, Nillable, Sort **Description** The date when the customer has paid an invoice in full.

Field	Details
`FuncCrcyCnvTotAmtWithTax`	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The functional currency converted to total amount with tax based on the dated exchange rate. Available in API version 66.0 and later.
`FunctionalCurrencyCvsnDate`	**Type** date **Properties** Filter, Group, Nillable, Sort, Update **Description** The date at which the exchange rate is used for functional currency conversions. Available in API version 66.0 and later.
`FunctionalCurrencyCvsnRate`	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The exchange rate that's used for functional currency conversions. Available in API version 66.0 and later.
`FunctionalCurrencyIsoCode`	**Type** string **Properties** Filter, Group, Nillable, Sort, Update **Description** The ISO code of the functional currency. Available in API version 66.0 and later.
`InvBatchDraftToPostedRunId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the invoice batch to posted run record that generated a posted invoice from a draft invoice. This field is a relationship field. **Relationship Name** InvBatchDraftToPostedRun

Field	Details
	Refers To InvBatchDraftToPostedRun
`InvoiceBatchRunId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the invoice batch run record that generated an invoice. This field is a relationship field. **Relationship Name** InvoiceBatchRun **Refers To** InvoiceBatchRun
`InvoiceDate`	**Type** date **Properties** Create, Filter, Group, Sort, Update **Description** Required. The date when an invoice was posted. The payment term's net is added to this date to calculate the invoice's due date. For example, an invoice with an invoice date of 04/01 and the related payment term's period as 30 days has a due date of 05/01.
`InvoiceLockedDateTime`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date and time at which an invoice is locked.
`InvoiceNumber`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Required. The user-specified reference number for the invoice.
`IsBillingScheduleGroupSkipped`	**Type** boolean

Field	Details
	Properties Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the invoice is generated from a billing transaction directly and if the billing schedule group wasn't created for the transaction (`true`), or if the invoice is generated from the billing schedule group that was created for the transaction (`false`). The default value is `false`. Available in API version 63.0 and later.
`IsInvoiceLocked`	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort, Update **Description** Required. Indicates whether an invoice is locked for editing or not (`true`) or not (`false`). The default value is `false`.
`LastEmailDispatchStatus`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** Indicates the dispatch status of the last email that was sent for the invoice. Available in API version 65.0 and later. Valid values are: • `Failed` • `Sent With Invoice Document` • `Sent Without Invoice Document`
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed an invoice record indirectly, for example, through a list view or related record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort

Field	Details
	Description The timestamp when the current user last viewed an invoice record. If this value is null, it's possible that the user only accessed the invoice record or a related list view (`LastReferencedDate`), but not viewed the invoice record itself.
`LegalEntityAccountingPeriodId`	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The legal entity accounting period record that's related to the invoice. Available in API version 65.0 and later. This field is a relationship field. **Relationship Name** LegalEntityAccountingPeriod **Refers To** LegalEntyAccountingPeriod
`LegalEntityId`	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The legal entity used in this invoice. Available in API version 65.0 and later. This field is a relationship field. **Relationship Name** LegalEntity **Refers To** LegalEntity
`NetCreditsApplied`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The net credits applied to an invoice. This amount is calculated by subtracting the sum of all the unapplied lines from the sum of all the applied lines. This field is a calculated field.
`NetPaymentsApplied`	**Type** currency

Field	Details
	Properties Filter, Nillable, Sort **Description** The net payments applied to an invoice. This field is visible from API version 62.0 but can only be used from API version 64.0 and later with the Revenue Cloud Billing license. This field is a calculated field.
`OwnerId`	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Required. The ID of the user who owns an Invoice record. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
`PaymentExclusionReason`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The reason for skipping the creation of payment schedules and payment schedule items for the invoice. Available in API version 64.0 and later with the Revenue Cloud Billing license.
PaymentScheduleId	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The ID of the payment schedule record that's related to the invoice. Available in API version 67.0 and later. This field is a relationship field. **Relationship Name** PaymentSchedule **Refers To** PaymentSchedule

Field	Details
PaymentTermId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The payment term used for an invoice. This field is a relationship field. **Relationship Name** PaymentTerm **Refers To** PaymentTerm
PostedDate	**Type** date **Properties** Filter, Group, Nillable, Sort **Description** The date when an invoice was posted.
ReferenceEntityId	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The ID of the Debit Memo, Credit Memo, or Order record that created an invoice. This field is a polymorphic relationship field. **Relationship Name** ReferenceEntity **Refers To** DebitMemo, CreditMemo, Order
SavedPaymentMethodId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the Saved Payment Method record that's used for the invoice. Available in API version 64.0 and later with the Revenue Cloud Billing license. This field is a relationship field. **Relationship Name** SavedPaymentMethod

Field	Details
	Refers To SavedPaymentMethod
`SequencePolicyId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The sequence policy associated with the invoice. Available in API version 65.0 and later. This field is a relationship field. **Relationship Name** SequencePolicy **Refers To** SequencePolicy
`SettlementLevel`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The level at which the credit was applied. Valid values are: • `Invoice` • `InvoiceLine`
`SettlementStatus`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The status of an invoice's payment. Valid values are: • `Not Applicable` • `Not Settled` • `Partially Settled` • `Settled`
`ShouldExcludePayment`	**Type** boolean

Field	Details
	Properties Defaulted on create, Filter, Group, Sort **Description** Indicates whether to skip creating payment schedules and payment schedule items for the invoice (`true`). Available in API version 64.0 and later with the Revenue Cloud Billing license. The default value is `false`.
`Status`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. The status of an invoice. Valid values are: • `Canceled`— Indicates that the invoice was generated and later canceled. • `Draft`— Indicates that the invoice is a draft. • `Draft In Progress`— Indicates that the draft invoice is in progress. • `Error`— Indicates that an error occurred when processing the invoice. • `Pending`— Indicates that the invoice is being processed. • `Posted`— Indicates that the invoice has been generated and sent to the customer. • `Posting In Progress`—Indicates that the invoice posting is in progress. • `Void In Progress`—Indicates that the invoice is pending a status change. • `Voided`—The invoice's status after the API successfully voids the invoice.
`TotalAdjustmentAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of the invoice's adjustment line amounts. This field is a calculated field.
`TotalAdjustmentAmountWithTax`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of the invoice's adjustment line amounts, including tax.

Field	Details
TotalAdjustmentTaxAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The total amount of tax applied to the invoice line’s adjustment lines.
TotalAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of the total amount on the invoice’s lines. This field is a calculated field.
TotalAmountWithTax	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of the total amount with tax on the invoice’s lines. This field is a calculated field.
TotalChargeAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of the amount fields on the invoice’s charge-type invoice lines. This field is a calculated field.
TotalChargeAmountWithTax	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of the amount fields on the invoice’s charge-type invoice lines, including tax.
TotalChargeTaxAmount	**Type** currency

Field	Details
	Properties Filter, Nillable, Sort **Description** The total amount of tax applied to the invoice's charge lines.
`TotalConvertedNegAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of all negative invoice lines that were converted to a credit memo. For example, if one negative invoice line was for -$10 and one was for -$15, the total amount that's converted to a credit memo is -$25. This field is a calculated field.
`TotalTaxAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of the tax amount on the invoice lines. This field is a calculated field.
TotalTaxesCapturedAtHeader	**Type** currency **Properties** Filter, Nillable, Sort **Description** The total tax amount for the invoice, captured at the header level by the tax engine. Available in API version 66.0 and later.
`UniqueIdentifier`	**Type** string **Properties** Filter, Group, idLookup, Nillable, Sort **Description** The unique identifier of an externally ingested invoice that's used to ensure that duplicate invoices aren't inserted. Available in API version 63.0 and later.
`WriteOffStatus`	**Type** picklist

Field	Details
	Properties Filter, Group, Nillable, Restricted picklist, Sort **Description** The status of the invoice write-off request. This field is unique within your organization. Valid values are: • `Completed` • `Failed` • `InProgress` Available in API version 64.0 and later. This field is available in API version 64.0 and later with the Revenue Cloud Billing license.
`WriteOffTotalChargeAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The total charge amount of the invoice that's written off. This field is unique within your organization. This field is available in API version 64.0 and later with the Revenue Cloud Billing license.
`WriteOffTotalTaxAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The total tax amount of the invoice that's written off. This field is unique within your organization. This field is available in API version 64.0 and later with the Revenue Cloud Billing license.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

InvoiceFeed on page 2866
Feed tracking is available for the object.

InvoiceHistory on page 2873
History is available for tracked fields of the object.

InvoiceShare on page 2877
Sharing is available for the object.

InvoiceAddressGroup

Represents the storage of the buyer's address information. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`

Special Access Rules

You need the Billing Operations User permission set to access this object.

Fields

Field	Details
`Address`	**Type** address **Properties** Filter, Nillable **Description** The buyer's address. See Compound Field Considerations and Limitations for details on geolocation compound fields.
`InvoiceAddressGroupNumber`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. The auto-generated reference number for the invoice address group.
`InvoiceId`	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The ID of the invoice associated with the address group. This field is a relationship field. **Relationship Name** Invoice

Field	Details
	Relationship Type Master-detail **Refers To** Invoice (the master object)

InvoiceBatchRun

Represents a batch processing job in Billing. During an invoice batch run, all billing schedules that meet the specified criteria are processed, resulting in the generation of invoices. This object is available in API version 62.0 and later.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `update()`

Special Access Rules

You need the Billing Admin permission set to access this object.

Fields

Field	Details
`BatchJobId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The batch management job that's used to track the progress of the invoice batch run. This field is available in API version 67.0 and later. This field is a relationship field. **Relationship Name** BatchJob **Refers To** BatchJob
`BillingBatchSchedulerId`	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The ID of the related billing batch scheduler.

Field	Details
	This field is a relationship field. **Relationship Name** BillingBatchScheduler **Refers To** BillingBatchScheduler
`Comments`	**Type** textarea **Properties** Filter, Nillable, Sort, Update **Description** Additional notes or comments for the invoice batch run.
`CompletionTime`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the invoice batch run finished processing.
`CreditApplicationStatus`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The status of applying credit memos or credit memo lines to invoices or invoice lines during invoices batch runs. Valid values are: • `Completed` • `Failed` • `NotApplicable`
`DocGenCompletionTime`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The completion time of generating invoice documents by the invoice batch run. This field is available in API version 63.0 and later with Revenue Cloud Billing license.

Field	Details
DocGenStartTime	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The start time of generating invoice documents by the invoice batch run. This field is available in API version 63.0 and later with Revenue Cloud Billing license.
DocGenStatus	**Type** picklist **Properties** Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort **Description** The status of the invoice documents generated by the invoice batch run. Valid values are: • `Canceled` • `Completed` • `Failed` • `In Progress` • `Not Started` • `Paused` The default value is `Not Started`. This field is available in API version 63.0 and later with Revenue Cloud Billing license.
InvoiceBatchRunNumber	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. The auto-generated reference number for the invoice batch run.
InvoiceDocsGenFailed	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The number of invoice documents that the invoice batch run failed to generate. This field is available in API version 63.0 and later with Revenue Cloud Billing license.

Field	Details
InvoiceDocsGenerated	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The number of invoice documents that the invoice batch run generated. This field is available in API version 63.0 and later with Revenue Cloud Billing license.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed an invoice batch run record indirectly, for example, through a list view or related record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed an invoice batch run record. If this value is null, it's possible that the user only accessed the invoice batch run record or a related list view (LastReferencedDate), but not viewed the invoice batch run record itself.
OwnerId	**Type** reference **Properties** Filter, Group, Sort, Update **Description** Required. The ID of the user who created the related billing batch scheduler. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
RecoveryStatus	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort

Field	Details
	Description The status of the invoice batch run recovery process. Valid values are: • `CompletelyRecovered`—All billing schedules included in the recovery run were reset to `Ready for Invoicing`. These billing schedules are included in the next scheduled invoice batch run. • `PartiallyRecovered`—Some, but not all, billing schedules that were part of the recovery run were reset to `Ready for Invoicing`. The billing schedules that were recovered are included in the next scheduled invoice batch run. The billing schedules that weren't successfully recovered must be manually reset to `Ready for Invoicing` so they can be processed. • `RecoveryFailed`—The recovery job was unsuccessful. • `RecoveryStarted`—The recovery job is in progress.
`StartTime`	**Type** dateTime **Properties** Filter, Sort **Description** Required. The timestamp when the invoice batch run started processing.
`Status`	**Type** picklist **Properties** Defaulted on create, Filter, Group, Restricted picklist, Sort **Description** Required. The status of the invoice batch run. Valid values are: • `Canceled` • `Completed` • `Failed` • `Started` • `Stopped` The default value is `Started`.
`StatusSubtype`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort

Field	Details
	Description The status subtype of the invoice batch run. The subtype corresponds to the status of the invoice batch run. Valid values are: • `Billing Schedules Filtering Started`—The data processing engine (DPE) selection service has started. The selection service includes the Billing Schedule records that meet the conditions defined when scheduling the invoice run. • `Billing Schedules Filtering In Progress`—The invoice run is identifying the billing schedules to include for invoice generation. • `Billing Schedules Filtering Completed`—The invoice run has identified the billing schedules to include for invoice generation. • `Invoice Generation Started`—The invoice generation service has started. • `Invoice Generation In Progress`—The invoice run is generating invoices for the identified billing schedules. • `Invoice Generation Completed`—The invoice run has finished generating invoices for the identified billing schedules. • `Invoice Tax Calculation In Progress`—The tax calculation step has started. • `Billing Schedule and Billing Schedule Group Updates In Progress`—The post processing step has started. • `Invoice Generation Summarization In Progress`—The invoice generation summarization step has started. • `Completed`—The invoice batch run has completed successfully. • `Failed`—The invoice batch run was unsuccessful. Available in API version 63.0 and later.
TotBillSchdUpdtDurDrftToPost	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The total duration taken to update the billing schedule from `Draft` to `Posted` status during an invoice batch run.
TotalBillSchedRecovered	**Type** int **Properties** Filter, Group, Nillable, Sort

Field	Details
	Description The number of billing schedules that were a part of the recovery run that were reset to `Ready for Invoicing`. These billing schedules are included in the next scheduled invoice batch run.
`TotalBillSchedUnrecovered`	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The number of billing schedules that were a part of the recovery run that weren't reset to `Ready for Invoicing`. These billing schedules that weren't successfully recovered must be manually reset to `Ready for Invoicing` so they can be processed.
`TotalBillingSchedulesFailed`	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The total number of billing schedules that weren't successfully processed. When a billing schedule isn't successfully processed, then the system doesn't generate an invoice for it. For details about errors, check the Revenue Transaction Error Log.
`TotalBsSuccessfullyProcessed`	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The total number of billing schedules for which the system was able to generate and process invoices.
`TotalDraftInvoiceAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The total amount of all invoices in the `Draft` status.
`TotalDraftInvoices`	**Type** int **Properties** Filter, Group, Nillable, Sort

Field	Details
	Description The total number of invoices in the `Draft` status.
`TotalFilteredBillingSchedules`	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The total number of billing schedules that met the invoice run scheduler's matching criteria. The matching criteria specify which billing schedules are included in the invoice batch run. The field label is **Total Matching Billing Schedules**.
`TotalInvSuccessfullyProcessed`	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The total number of invoices that were successfully processed.
`TotalInvoicedAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The total amount of income including taxes that's represented by the successfully processed invoices.
`TotalInvoicesCanceled`	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The total number of invoices that weren't processed. To find out what went wrong, check the Revenue Transaction Error Log. Then fix the errors and run the invoice batch run recovery process.
`TotalInvoicesFailed`	**Type** int **Properties** Filter, Group, Nillable, Sort

Field	Details
	Description The total number of invoices that weren't processed successfully. To find out what went wrong, check the Revenue Transaction Error Log. Then fix the errors and run the invoice batch run recovery process.
`TotalInvoicesGenerated`	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The total number of invoices that were generated from the billing schedules processed by the invoice batch run.
`TotalPostedInvoices`	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The total number of invoices in `Posted` status that were generated from the billing schedules.

InvoiceBatchRunCriteria

Represents a batch processing job and its required criteria in Billing. During an invoice batch run, all billing schedules that meet the specified criteria are processed, resulting in the generation of invoices. This object is available in API version 62.0 and later.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`

Special Access Rules

You need the Billing Admin permission set to access this object.

Fields

Field	Details
`Comments`	**Type** textarea **Properties** Filter, Nillable, Sort

Field	Details
	Description Additional notes or comments for the invoice batch run criteria.
`CriteriaExpression`	**Type** textarea **Properties** Filter, Nillable, Sort **Description** The formula that specifies criteria for filtering the billing schedules. For example, you can filter billing schedules by the currency code.
`CriteriaMatchType`	**Type** picklist **Properties** Defaulted on create, Filter, Group, Restricted picklist, Sort **Description** Required. The type of matching criteria required for the batch. Valid value is `MatchAll`. The default value is `MatchAll`.
`ExpectedInvoiceStatus`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The type of invoice a batch run generates. Valid values are: • `Draft` • `Posted`
`InvoiceBatchRunCriteriaNumber`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. The auto-generated reference number for the invoice batch run criteria.
`InvoiceDate`	**Type** date **Properties** Filter, Group, Nillable, Sort

Field	Details
	Description The date displayed on the invoice. This date is also used for tax calculations.
`InvoiceDateOffset`	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The offset that's applied to the target date to calculate the invoice date.
`IsInvoiceDateFromRunDate`	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Required. Indicates whether the invoice date is derived from the run date (`true`) or not (`false`). The default value is `false`. Available in API version 63.0 and later.
`OwnerId`	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The ID of the user who created the invoice batch run criteria. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
`TargetDate`	**Type** date **Properties** Filter, Group, Nillable, Sort **Description** The target date for the invoice run. Billing schedules having the next billing date before this date are picked up for invoicing.
`TargetDateOffset`	**Type** int

Field	Details
	Properties Filter, Group, Nillable, Sort **Description** The offset that's applied to the next run date to calculate the target date.

InvoiceBatchRunRecovery

Represents information about the recovery procedure of an invoice batch run. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `update()`

Special Access Rules

You need the Billing Admin permission set to access this object.

Fields

Field	Details
`BatchJobId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The batch management job that's used to track the progress of the invoice batch run. This field is available in API version 67.0 and later. This field is a relationship field. **Relationship Name** BatchJob **Refers To** BatchJob
`Comments`	**Type** textarea **Properties** Filter, Nillable, Sort, Update

Field	Details
	Description Additional notes or comments for the invoice batch run recovery.
CompletionTime	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the invoice batch run recovery finished processing.
InvoiceBatchRunId	**Type** reference **Properties** Filter, Group, Sort **Description** Required. A unique identifier of the invoice batch run that's related to this recovery run. This field is a relationship field. **Relationship Name** InvoiceBatchRun **Relationship Type** Master-detail **Refers To** InvoiceBatchRun (the master object)
InvoiceBatchRunRecoveryNumber	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. A unique identifier of the recovery process for the invoice batch run.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed an invoice batch run recovery record indirectly, for example, through a list view or related record.
LastViewedDate	**Type** dateTime

Field	Details
	Properties Filter, Nillable, Sort **Description** The timestamp when the current user last viewed an invoice batch run recovery record. If this value is null, it's possible that the user only accessed the invoice batch run recovery record or a related list view (`LastReferencedDate`), but not viewed the invoice batch run recovery record itself.
`StartTime`	**Type** dateTime **Properties** Filter, Sort **Description** Required. The timestamp when the invoice batch run recovery started processing.
`Status`	**Type** picklist **Properties** Defaulted on create, Filter, Group, Restricted picklist, Sort **Description** Required. The final state of the recovery process for the invoice batch runs. Valid values are: • `Completed` • `CompletedWithErrors` • `Failed` • `Started` The default value is `Started`.

InvoiceDocument

Represents the PDF document generated for an invoice. This object is available in API version 63.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`

Special Access Rules

You need Revenue Cloud Billing license, and the Billing Admin permission set or the Billing Operations User permission set to access this object.

Fields

Field	Details
ContentDocumentId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the generated PDF document. This field is a relationship field. **Relationship Name** ContentDocument **Refers To** ContentDocument
DateGenerated	**Type** date **Properties** Filter, Group, Nillable, Sort **Description** The date on which the PDF is generated.
DocumentGenerationProcessId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the document generation process that contains the information used to create the PDF invoice. This field is a relationship field. **Relationship Name** DocumentGenerationProcess **Refers To** DocumentGenerationProcess
DocumentNumber	**Type** string

Field	Details
	Properties Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. A unique number assigned to the PDF invoice document.
`ErrorMessage`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** Any errors that occur during PDF generation.
`InvoiceId`	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The ID of the Invoice to which the Invoice Doc is attached. This field is a relationship field. **Relationship Name** Invoice **Relationship Type** Master-detail **Refers To** Invoice (the master object)
`Status`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The status of the PDF generation process. Valid values are: • `Blocked` • `Cancelled` • `Failure` • `Pending` • `Success`

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

InvoiceDocumentFeed on page 2866
Feed tracking is available for the object.

InvoiceDocumentHistory on page 2873
History is available for tracked fields of the object.

InvoiceLine

Represents the amount that a buyer must pay for a product, service, or fee. Invoice lines are created based on the amount of an order line. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `update()`

Special Access Rules

You need the Billing Admin permission set to access this object.

Fields

Field	Details
AdjustmentAmount	**Type** currency **Properties** Filter, Nillable, Sort, Update **Description** The sum of adjustments made to the invoice line.
AdjustmentAmountWithTax	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of adjustment amounts, including associated taxes related to the invoice line.
AdjustmentTaxAmount	**Type** currency

Field	Details
	Properties Filter, Nillable, Sort **Description** The sum of tax adjustments to the invoice line.
`Balance`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The outstanding balance for an invoice line. This amount is equal to the invoice’s total amount with tax after deducting the payments made.
`BillingAddressId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** This field is related to an `InvoiceAddressGroup` field containing the billing address for the invoice line. For example, one `InvoiceAddressGroup` field is related to the invoiceLine's `BillingAddressID` field, and another `InvoiceAddressGroup` field is related to the invoiceLine's `ShippingAddressId` field. This field is a relationship field. **Relationship Name** BillingAddress **Refers To** InvoiceAddressGroup
`BillingScheduleGroupId`	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The ID of the billing schedule group for the invoice line. This field is a relationship field. **Relationship Name** BillingScheduleGroup **Refers To** BillingScheduleGroup

Field	Details
BillingScheduleId	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The ID of the billing schedule for the invoice line. Edit access is enabled for this field. You must not modify this field when the invoice line is related to a posted invoice. This field is a relationship field. **Relationship Name** BillingSchedule **Refers To** BillingSchedule
ChargeAmount	**Type** currency **Properties** Filter, Nillable, Sort, Update **Description** The sum of charges made to the invoice line.
ChargeAmountWithTax	**Type** currency **Properties** Filter, Nillable, Sort **Description** The amount on a charge invoice line, including tax.
ChargeConvertedNegAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The amount on a charge invoice line that's converted to credit.
ChargeTaxAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The tax to be applied on a charge invoice line.

Field	Details
ConvertedNegAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The amount from an invoice line that’s converted to credit.
CorpCurrencyCnvChargeAmt	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The sum of charges made to the invoice line in corporate currency. Available in API version 63.0 and later.
CorpCurrencyCnvTotalTaxAmt	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The total tax amount of the invoice line in corporate currency. Available in API version 63.0 and later.
CorporateCurrencyCvsnDate	**Type** date **Properties** Filter, Group, Nillable, Sort, Update **Description** The date on which the invoice line amounts are converted to corporate currency. Available in API version 63.0 and later.
CorporateCurrencyCvsnRate	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The exchange rate that's used to convert the invoice line amounts to corporate currency. Available in API version 63.0 and later.
CorporateCurrencyIsoCode	**Type** string

Field	Details
	Properties Filter, Group, Nillable, Sort, Update **Description** The currency ISO code of the corporate currency. Available in API version 63.0 and later.
`Description`	**Type** string **Properties** Filter, Group, Nillable, Sort, Update **Description** The description of the invoice line.
`FuncCrcyCnvTotalTaxAmt`	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The total tax amount value in functional currency. Available in API version 66.0 and later.
`FuncCurrencyCnvChargeAmt`	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The charge amount value in functional currency. Available in API version 66.0 and later.
`FunctionalCurrencyCvsnDate`	**Type** date **Properties** Filter, Group, Nillable, Sort, Update **Description** The date on which the invoice line amount values are converted into functional currency. Available in API version 66.0 and later.
`FunctionalCurrencyCvsnRate`	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The exchange rate that's used to convert the invoice line amount values into functional currency. Available in API version 66.0 and later.

Field	Details
FunctionalCurrencyIsoCode	**Type** string **Properties** Filter, Group, Nillable, Sort, Update **Description** The ISO code of the functional currency. Available in API version 66.0 and later.
GroupReferenceEntityItemId	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The ID of the record that the invoice line corresponds to. This field is a polymorphic relationship field. **Relationship Name** GroupReferenceEntityItem **Refers To** DebitMemoLine, OrderItem, OrderItemAdjustmentLineItem, OrderItemDetail, QuoteLineDetail, QuoteLineItem
HasMultipleItems	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Required. Indicates whether this field merges items from the same billing period (`true`) or not (`false`). The default value is `false`.
InvoiceId	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The ID of the invoice record that contains this invoice line. This field is a relationship field. **Relationship Name** Invoice **Relationship Type** Master-detail

Field	Details
	Refers To Invoice (the master object)
InvoiceLineEndDate	**Type** date **Properties** Filter, Group, Sort, Update **Description** Required. The end date of the billing for the service for invoice lines made from a time-based service.
InvoiceLineStartDate	**Type** date **Properties** Filter, Group, Sort, Update **Description** Required. The first date of the billing for the service for invoice lines made from a time-based service.
InvoiceStatus	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The status of the invoice line. This field is inherited from the invoice's status.
IsUsageBasedInvoiceLine	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Required. Indicates whether the product is usage-based (`true`) or not (`false`). The default value is `false`. This field is available in API version 63.0 and later with Revenue Cloud Billing license.
LegalEntityAccountingPeriodId	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The ID of the legal entity accounting period record used in this invoice line.

Field	Details
	This field is a relationship field. **Relationship Name** LegalEntityAccountingPeriod **Refers To** LegalEntyAccountingPeriod
`LegalEntityId`	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The ID of the legal entity record used in this invoice line. This field is a relationship field. **Relationship Name** LegalEntity **Refers To** LegalEntity
`LineAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The amount of the invoice line. This field is a calculated field.
`Name`	**Type** string **Properties** Filter, Group, idLookup, Sort, Update **Description** Required. The name of the invoice line.
`NetCreditsApplied`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The total credit memo line amount applied to the invoice line. This amount is calculated by subtracting the unapplied credit memo line amount from the applied credit memo line amount.

Field	Details
	This field is a calculated field.
NetPaymentsApplied	**Type** currency **Properties** Filter, Nillable, Sort **Description** The total payment amount that's applied to the invoice line after unapplication of payments. This amount is calculated by subtracting the unapplied payment line invoice line amount from the applied payment line invoice line amount. Available in API version 61.0 and later. This field is a calculated field.
OverageUnitOfMeasure	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The unit that's used to measure the overage. For example, byte or minute. This field is available in API version 63.0 and later with Revenue Cloud Billing license.
ParentInvoiceLineId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the parent invoice line record. This field is a relationship field. **Relationship Name** ParentInvoiceLine **Refers To** InvoiceLine
Product2Id	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The ID of the product that was charged or ordered to create the invoice line. This field is a relationship field. **Relationship Name** Product2

Field	Details
	Refers To Product2
Quantity	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The number of units of the order product that created the invoice line.
ReferenceEntityItemId	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The order item or adjustment item that created the invoice line. This field is a polymorphic relationship field. **Relationship Name** ReferenceEntityItem **Refers To** OrderItem, OrderItemAdjustmentLineItem
ReferenceEntityItemType	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The type of transaction that created the invoice line. Valid values are: • Delivery Charge • Fee • Order Product
ReferenceEntityItemTypeCode	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The type of object that created the invoice line. Valid values are:

Field	Details
	• `Charge` • `Product`
`ShipFromAddressId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The shipping origin of the invoiced product. Available in API version 64.0 and later. This field is a relationship field. **Relationship Name** ShipFromAddress **Refers To** InvoiceAddressGroup
`ShippingAddressId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the shipping address record associated with the invoice line. This field is a relationship field. **Relationship Name** ShippingAddress **Refers To** InvoiceAddressGroup
`TaxAmount`	**Type** currency **Properties** Filter, Nillable, Sort, Update **Description** The total tax for the invoice line.
`TaxProcessingStatus`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The status of the tax processing of the invoice line. Valid values are:

Field	Details
	• `Estimated` • `Pending` • `Posted`
`TaxTreatmentId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The tax treatment used on this invoice line. This field is a relationship field. **Relationship Name** TaxTreatment **Refers To** TaxTreatment
`Type`	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** The type of the invoice line. Valid values are: • `Adjustment` • `Charge` • `Tax`
`UnitOfMeasureId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The unit of measure of the product associated with the invoice line. Available in API version 63.0 and later. This field is a relationship field. **Relationship Name** UnitOfMeasure **Refers To** UnitOfMeasure

Field	Details
UnitPrice	**Type** currency **Properties** Filter, Nillable, Sort, Update **Description** The price for one unit of the item on the invoice line.
UsageOverageQuantity	**Type** double **Properties** Filter, Nillable, Sort **Description** The quantity of the usage overage that’s being invoiced. This field is available in API version 63.0 and later with Revenue Cloud Billing license.
UsageProductBillSchdGrpId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the usage-based product billing schedule group associated with the invoice line. This field is available in API version 63.0 and later with Revenue Cloud Billing license. This field is a relationship field. **Relationship Name** UsageProductBillSchdGrp **Refers To** BillingScheduleGroup
UsageProductId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the usage-based product that was charged or ordered to create the invoice line. This field is available in API version 63.0 and later with Revenue Cloud Billing license. This field is a relationship field. **Relationship Name** UsageProduct **Refers To** Product2

InvoiceLineRelationship

Represents a relationship between invoice line items to support bundles where one parent invoice line has multiple child invoice lines. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`

Special Access Rules

You need the Billing Admin permission set to access this object.

Fields

Field	Details
`AssociatedInvoiceLineId`	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The ID of the related invoice line record. In a bundle relationship, this invoice line is the child. This field is a relationship field. **Relationship Name** AssociatedInvoiceLine **Refers To** InvoiceLine
`AssociatedInvoiceLinePricing`	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** Required. This field describes how the related invoice line is priced relative to the primary invoice line. Valid values are: • `IncludedInBundlePrice` • `NotIncludedInBundlePrice`

Field	Details
AssociatedInvoiceLineRole	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** Required. This field describes the role of the related invoice line in the relationship. The value is derived from the AssociatedProductRoleCat field of the ProductRelationshipType object. Valid values are: • AddOnComponent • BundleComponent • ClassificationComponent • SetComponent
InvoiceId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the related invoice record. This field is a relationship field. **Relationship Name** Invoice **Refers To** Invoice
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed an invoice line relationship record indirectly, for example, through a list view or related record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed an invoice line relationship record. If this value is null, it's possible that the user only accessed the invoice line relationship record or

Field	Details
	a related list view (`LastReferencedDate`), but not viewed the invoice line relationship record itself.
`MainInvoiceLineId`	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The ID of the primary invoice line record. In a bundle relationship, this invoice line is the parent. This field is a relationship field. **Relationship Name** MainInvoiceLine **Relationship Type** Master-detail **Refers To** InvoiceLine (the master object)
`MainInvoiceLineRole`	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** Required. This field describes the role of the primary invoice line in the relationship. The value is derived from the `MainProductRoleCat` field of the ProductRelationshipType object. Valid values are: • `AddOn` • `Bundle` • `Set`
`Name`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. An auto-generated number identifying the invoice line relationship.
`ProductRelationshipTypeId`	**Type** reference **Properties** Filter, Group, Nillable, Sort

Field	Details
	Description The ID of the product relationship type record between the main and associated invoice lines. This field is a relationship field. **Relationship Name** ProductRelationshipType **Refers To** ProductRelationshipType

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

InvoiceLineRelationshipFeed **on page 2866**
Feed tracking is available for the object.

InvoiceLineRelationshipHistory **on page 2873**
History is available for tracked fields of the object.

InvoiceLineTax

Represents tax information of an invoice line of type `Tax`. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `update()`

Special Access Rules

You need the Tax Admin permission set to access this object.

Fields

Field	Details
`BillingAddressId`	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update

Field	Details
	Description The billing address of the parent invoice line. Edit access is enabled for this field. You must not modify this field when the invoice line is related to a posted invoice. This field is a relationship field. **Relationship Name** BillingAddress **Refers To** InvoiceAddressGroup
`ConvertedNegAmount`	**Type** currency **Properties** Filter, Nillable, Sort, Update **Description** The amount from the parent invoice line that's converted to credit. Edit access is enabled for this field. You must not modify this field when the invoice line is related to a posted invoice.
`CorpCrcyCnvTaxAmount`	**Type** double **Properties** Filter, Group, Nillable, Sort **Description** The total tax amount of the invoice line tax in corporate currency. Available in API version 63.0 and later.
`CorporateCurrencyCvsnDate`	**Type** date **Properties** Filter, Group, Nillable, Sort **Description** The date on which the invoice line tax amounts are converted to the corporate currency. Available in API version 63.0 and later.
`CorporateCurrencyCvsnRate`	**Type** double **Properties** Filter, Nillable, Sort **Description** The exchange rate that's used to convert the invoice line tax amounts to the corporate currency. Available in API version 63.0 and later.

Field	Details
CorporateCurrencyIsoCode	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The currency ISO code of the corporate currency. Available in API version 63.0 and later.
Description	**Type** string **Properties** Filter, Group, Nillable, Sort, Update **Description** Additional details about the invoice line tax.
EndDate	**Type** date **Properties** Filter, Group, Sort, Update **Description** Required. The end date of an invoice line tax.
FuncCrcyCnvTaxAmount	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The tax amount value in functional currency. Available in API version 66.0 and later.
FunctionalCurrencyCvsnDate	**Type** date **Properties** Filter, Group, Nillable, Sort, Update **Description** The date on which the tax amount value is converted into functional currency. Available in API version 66.0 and later.
FunctionalCurrencyCvsnRate	**Type** double **Properties** Filter, Nillable, Sort, Update

Field	Details
	Description The exchange rate that's used to convert the tax amount value into functional currency. Available in API version 66.0 and later.
FunctionalCurrencyIsoCode	**Type** string **Properties** Filter, Group, Nillable, Sort, Update **Description** The ISO code of the functional currency. Available in API version 66.0 and later.
InvoiceLineId	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The ID of the parent charge or adjustment invoice line record that an invoice line tax is related to. This field is a relationship field. **Relationship Name** InvoiceLine **Relationship Type** Master-detail **Refers To** InvoiceLine (the master object)
InvoiceLineTaxNumber	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. An auto-generated number identifying an invoice line tax.
LegalEntityAccountingPeriodId	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The ID of the legal entity accounting period record that's related to an invoice line tax. This field is a relationship field.

Field	Details
	Relationship Name LegalEntityAccountingPeriod **Refers To** LegalEntyAccountingPeriod
LegalEntityId	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The ID of the legal entity that's related to an invoice line tax. This field is a relationship field. **Relationship Name** LegalEntity **Refers To** LegalEntity
Product2Id	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The product that was charged or ordered to create the parent invoice line. Available in API version 62.0 only. This field is a relationship field. **Relationship Name** Product2 **Refers To** Product2
Quantity	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The number of units of the order product that created the parent invoice line. Available in API version 62.0 only.
ShipFromAddressId	**Type** reference

Field	Details
	Properties Create, Filter, Group, Nillable, Sort, Update **Description** The ship from address specified in the parent invoice line. Edit access is enabled for this field. You must not modify this field when the invoice line is related to a posted invoice. Available in API version 64.0 and later. This field is a relationship field. **Relationship Name** ShipFromAddress **Refers To** InvoiceAddressGroup
ShippingAddressId	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The ID of the shipping address record of the parent invoice line. Edit access is enabled for this field. You must not modify this field when the invoice line is related to a posted invoice. This field is a relationship field. **Relationship Name** ShippingAddress **Refers To** InvoiceAddressGroup
StartDate	**Type** date **Properties** Filter, Group, Sort, Update **Description** Required. The start date of an invoice line tax.
TaxAmount	**Type** currency **Properties** Filter, Nillable, Sort, Update **Description** The total amount of an invoice line tax. Edit access is enabled for this field. You must not modify this field when the invoice line is related to a posted invoice. Available in API version 65.0 and later.

Field	Details
TaxCode	**Type** string **Properties** Filter, Group, Nillable, Sort, Update **Description** The code that's used to calculate the tax rate for the parent invoice line. Edit access is enabled for this field. You must not modify this field when the invoice line is related to a posted invoice.
TaxDocumentNumber	**Type** string **Properties** Filter, Group, Nillable, Sort, Update **Description** The number of the latest record in the external tax engine in which the parent invoice line is included. Edit access is enabled for this field. You must not modify this field when the invoice line is related to a posted invoice.
TaxEffectiveDate	**Type** date **Properties** Filter, Group, Nillable, Sort, Update **Description** The date used to calculate the tax amount. Edit access is enabled for this field. You must not modify this field when the invoice line is related to a posted invoice.
TaxExemptAmount	**Type** currency **Properties** Filter, Nillable, Sort, Update **Description** The amount that's exempted from tax. Edit access is enabled for this field. You must not modify this field when the invoice line is related to a posted invoice.
TaxName	**Type** string **Properties** Filter, Group, Nillable, Sort, Update **Description** The user-defined name for an invoice line tax. Edit access is enabled for this field. You must not modify this field when the invoice line is related to a posted invoice.

Field	Details
TaxProcessingStatus	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The processing status of the invoice line tax. Valid values are: • Estimated • Pending • Posted
TaxRate	**Type** percent **Properties** Filter, Nillable, Sort, Update **Description** The percentage of the order product price that's used to calculate tax. Edit access is enabled for this field. You must not modify this field when the invoice line is related to a posted invoice.
TaxTransactionNumber	**Type** string **Properties** Filter, Group, Nillable, Sort, Update **Description** The number of the transaction in the external tax engine that calculated tax for the parent invoice line.
TaxTreatmentId	**Type** reference **Properties** Filter, Group, Nillable, Sort, Update **Description** The ID of the tax treatment record that's related to an invoice line tax. Edit access is enabled for this field. You must not modify this field when the invoice line is related to a posted invoice. This field is a relationship field. **Relationship Name** TaxTreatment **Refers To** TaxTreatment

Field	Details
UnitPrice	**Type** currency **Properties** Filter, Nillable, Sort, Update **Description** The price per unit of the order product that's related to an invoice line tax. Available in API version 62.0 only.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

InvoiceLineTaxFeed on page 2866
Feed tracking is available for the object.

InvoiceLineTaxHistory on page 2873
History is available for tracked fields of the object.

LegalEntity

Represents the way an organization is structured. An organization can be a single legal entity or it can comprise more than one legal entity. This object is available in API version 62.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need the Accounts Receivables Admin permission set to access this object.

Fields

Field	Details
CompanyName	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The name of the company that this legal entity represents.

Field	Details
Description	**Type** textarea **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Additional details about the legal entity.
EmailTemplateId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** A template that's used to send emails for the legal entity. This field is available in API version 65.0 and later. This field is a relationship field. **Relationship Name** EmailTemplate **Refers To** EmailTemplate
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed a legal entity indirectly, for example, through a list view or related record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed a legal entity. If this value is null, it’s possible that the user only accessed the legal entity or a related list view (`LastReferencedDate`), but not viewed the legal entity itself.
LegalEntityAddress	**Type** address **Properties** Filter, Nillable

Field	Details
	Description The address of the company that this legal entity represents. See Address Compound Fields for details on compound address fields.
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** Required. The name of the legal entity.
`OwnerId`	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Required. The ID of the record owner. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
`ShouldAttachInvoiceDocToEmail`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether to attach the invoice PDF document to the email that's sent for the legal entity (`true`) or not (`false`). This field is available in API version 65.0 and later. The default value is `false`.
`Status`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The status of the legal entity. Valid values are: • `Active` • `Inactive`

LegalEntyAccountingPeriod

Represents a junction between a legal entity and an accounting period. This object is available in API version 62.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need the Accounts Receivables Admin permission set to access this object.

Fields

Field	Details
`AccountingPeriodId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the accounting period record that's related to a legal entity accounting period. This field is a relationship field. **Relationship Name** AccountingPeriod **Refers To** AccountingPeriod
`ClosureStage`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies the closure stage of the legal entity accounting period. This field is available in API version 65.0 and later. Valid values are: • `CloseLegalEntityAccountingPeriod` • `Completed` • `Open` • `CreateGeneralLedgerAccountingPeriodSummaries` • `CreateUnrealizedGainOrLossTransactionJournals`

Field	Details
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed a legal entity accounting period record indirectly, for example, through a list view or related record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed a legal entity accounting period record. If this value is null, it’s possible that the user only accessed the legal entity accounting period record or a related list view (`LastReferencedDate`), but not viewed the legal entity accounting period record itself.
LegalEntityId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** Required. The ID of the legal entity record that's related to a legal entity accounting period. This field is a relationship field. **Relationship Name** LegalEntity **Refers To** LegalEntity
Name	**Type** string **Properties** Filter, Group, idLookup, Sort **Description** Required. The auto-generated name of a legal entity accounting period. The name is a combination of the names of the legal entity and the accounting period.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update

Field	Details
	Description Required. The ID of the user who owns a legal entity accounting period record. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
Status	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort **Description** Required. The status of a legal entity accounting period record. Valid values are: • Closed • Error • Open • PendingClosure • PendingReopen • Reopened
TotalAssetsAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The total assets for a legal entity accounting period is a roll up summary of the closing balances of the general ledger accounting period summary records that are related to an asset type general ledger account. This field is available in API version 65.0 and later.
TotalEquitiesAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The total equities for a legal entity accounting period is a roll up summary of the closing balances of the general ledger accounting period summary records that are related to an equity type general ledger account. This field is available in API version 65.0 and later.

Field	Details
`TotalExpensesAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The total expenses for a legal entity accounting period is a roll up summary of the closing balances of the general ledger accounting period summary records that are related to an expense type general ledger account. This field is available in API version 65.0 and later.
`TotalLiabilitiesAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The total liabilities for a legal entity accounting period is a roll up summary of the closing balances of the general ledger accounting period summary records that are related to a liability type general ledger account. This field is available in API version 65.0 and later.
`TotalRevenueAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The total revenue for a legal entity accounting period is a roll up summary of the closing balances of the general ledger accounting period summary records that are related to a revenue type general ledger account. This field is available in API version 65.0 and later.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

LegalEntyAccountingPeriodShare on page 2877
Sharing is available for the object.

PaymentBatchRun

Represents a batch processing job that processes payments in Billing. During a payment batch run, all the payment schedules that meet the specified criteria are processed and the corresponding Payment records are created. These payments are then applied to invoices or invoice lines. This object is available in API version 64.0 and later.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `update()`

Special Access Rules

You need the Revenue Cloud Billing license, and the Payment Admin permission set or the Payment Operations User permission set to access this object.

Fields

Field	Details
`BillingBatchSchedulerId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** Required. The ID of the billing batch scheduler that's used to schedule the payment batch run. This field is a relationship field. **Relationship Name** BillingBatchScheduler **Refers To** BillingBatchScheduler
`Comments`	**Type** textarea **Properties** Filter, Nillable, Sort, Update **Description** Additional details about the payment batch run.
`CompletionTime`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date and time when the payment batch run completed processing payments.
`IsPaymentRetry`	**Type** boolean **Properties** Filter, Nillable, Sort **Description** Indicates whether the payment batch run is for retrying a failed payment (`true`) or not (`false`). The default value is `false`. Available in API version 66.0 and later.

Field	Details
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed this record indirectly, for example, through a list view or related record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed this record or list view. If this value is null, and `LastReferenceDate` is not null, the user accessed this record or list view indirectly.
OwnerId	**Type** reference **Properties** Filter, Group, Sort, Update **Description** Required. The ID of the owner of the Payment Batch Run record. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
PaymentBatchRunNumber	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. The auto-generated reference number for the payment batch run.
StartTime	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date and time when the payment batch run started processing payments.

Field	Details
Status	**Type** picklist **Properties** Defaulted on create, Filter, Group, Restricted picklist, Sort **Description** Required. The status of the payment batch run. Valid values are: • `Completed` • `Failed` • `New` • `Started` • `Stopped` The default value is `New`.
TotalFailedScheduleItems	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The total number of payment schedules that weren't processed by the payment batch run. When a payment schedule isn't processed, the system doesn't generate a Payment record for it. For details about errors, check the Revenue Transaction Error Log records for the payment batch run.
TotalFilteredScheduleItems	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The total number of payment schedule items that meet the payment run scheduler's matching criteria. The matching criteria identifies the payment schedule items that are included for processing by the payment batch run.
TotalProcessedScheduleItems	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The total number of payment schedule items that were processed by the payment batch run.

Field	Details
TotalScheduleItemsApplied	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The total number of payment schedule items that were processed and had corresponding payments also applied to invoices or invoice lines.
TotalScheduleItemsApplyFailed	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The total number of payment schedule items that were processed but had corresponding payments that weren't applied to invoices or invoice lines.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

PaymentBatchRunShare on page 2877
Sharing is available for the object.

PaymentLineInvoiceLine

Represents information about a payment line that's applied to or unapplied from an invoice line. This object is available in API version 64.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `update()`, `upsert()`

Special Access Rules

You need the Revenue Cloud Billing license, and the Payment Admin permission set or the Payment Operations User permission set to access this object.

Fields

Field	Details
AccountId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort **Description** The account of the customer who made a payment that's related to the payment line invoice line. This field is a relationship field. **Relationship Name** Account **Refers To** Account
Amount	**Type** currency **Properties** Create, Filter, Sort **Description** Required. The amount that's been applied or unapplied by a payment line.
AppliedDateTime	**Type** dateTime **Properties** Create, Filter, Nillable, Sort **Description** The date and time when a payment line was applied to an invoice line.
AppliedImpactAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** If the payment line invoice line's `Type` value is `Applied`, the applied impact amount is the same as the `ImpactAmount` value. The applied impact amount is 0 when the `Type` value is `Unapplied`. This field is a calculated field.
Description	**Type** textarea

Field	Details
	Properties Create, Filter, Nillable, Sort, Update **Description** Additional details about the payment line invoice line.
`EffectiveDateTime`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort **Description** The date and time when a payment line's application to or unapplication from an invoice line becomes effective.
`ImpactAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** If the payment line invoice line's `Type` value is `Applied`, the impact amount is the negative equivalent of the payment line invoice line's `Amount` value. Otherwise, it's equal to the payment line invoice line's `Amount` value. This field is a calculated field.
`InvoiceLineBalance`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The balance of the invoice line after a payment line was applied to it or unapplied from it.
`InvoiceLineId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** Required. The invoice line to which a payment line has been applied or unapplied. This field is a relationship field. **Relationship Name** InvoiceLine **Relationship Type** Master-detail

Field	Details
	Refers To InvoiceLine (the master object)
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed this record indirectly, for example, through a list view or related record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed this record or list view. If this value is null, and `LastReferenceDate` is not null, the user accessed this record or list view indirectly.
`LegalEntityAccountingPeriodId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The legal entity accounting period related to the payment line invoice line. This field is a relationship field. **Relationship Name** LegalEntityAccountingPeriod **Refers To** LegalEntyAccountingPeriod
`LegalEntityId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The legal entity related to the payment line invoice line. This field is a relationship field. **Relationship Name** LegalEntity

Field	Details
	Refers To LegalEntity
PaymentBalance	**Type** currency **Properties** Filter, Nillable, Sort **Description** The balance of the Payment record after it was applied to or unapplied from an invoice line.
PaymentId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** Required. The parent Payment record that's related to the payment line invoice line. This field is a relationship field. **Relationship Name** Payment **Refers To** Payment
PaymentLineInvoiceLineNumber	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. An auto-generated number identifying the payment line invoice line.
RelatedPaymentLineInvcLineId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort **Description** The related payment line invoice line with the `Type` value as `Applied` when the payment line invoice line's `Type` value is `Unapplied`. This field is a relationship field. **Relationship Name** RelatedPaymentLineInvcLine **Refers To** PaymentLineInvoiceLine

Field	Details
Type	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort **Description** Required. Specifies whether a payment line has been applied to or unapplied from an invoice line. Valid values are: • Applied • Unapplied
UnappliedDateTime	**Type** dateTime **Properties** Create, Filter, Nillable, Sort **Description** The date and time when a payment line was unapplied from an invoice line.
UnappliedStatus	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort **Description** Required. Specifies whether a payment line has been unapplied from an invoice line. Valid values are: • NA—Not Applicable • No • Yes

PaymentRetryRule

Represents the specific payment retry rule for a failed payment schedule item. Each rule defines actionable parameters such as the maximum number of retries for the failed records and time intervals between subsequent retry attempts. This object is available in API version 66.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`Description`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The description of the business details for the payment retry rule.
`IntervalUnit`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The unit of time after which a payment batch run must retry the failed payments. Valid values are: • `Days` • `Hours` • `Minutes`
`IntervalValue`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The single or comma-separated numeric values, after which the payment batch run must retry the failed payments.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed the payment retry rule record.
`LastViewedDate`	**Type** dateTime

Field	Details
	Properties Filter, Nillable, Sort **Description** The timestamp when the current user last viewed the payment retry rule record.
`MaximumRetryCount`	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The maximum number of retries on the failed payment records.
`PaymentGatewayErrorCategory`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The category of payment gateway errors for which the payment retry rule must be applied. Valid values are: • `CardLimit`—Card Limit Decline • `GatewayConnection`—Gateway Connection Error • `PaymentInformation`—Invalid Payment Details • `PaymentProcessing`—Payment Processing Error • `Security`—Security Failure • `Unknown`—Unknown Error • `ValidationFailure`—Internal Validation Error
`PaymentGatewayErrorCode`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The raw error code of the payment gateway response.
`PaymentGatewayId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The unique identifier of the payment gateway for which the payment retry rule must be applied.

Field	Details
	This field is a relationship field. **Relationship Name** PaymentGateway **Refers To** PaymentGateway
`PaymentRetryRuleSetId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** The payment retry rule set of the payment retry rule. This field is a relationship field. **Relationship Name** PaymentRetryRuleSet **Relationship Type** Master-detail **Refers To** PaymentRetryRuleSet (the master object)
`RetryIntervalType`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies whether the payment retry rule must be applied to the failed payment records at fixed or staggered intervals of time. Valid values are: • `Fixed` • `Staggered`

PaymentRetryRuleSet

Represents the payment retry rule definition that defines how failed payments are retried based on the error codes across various retry categories. This object is available in API version 66.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
CanUseAltrnPaymentMethod	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether an alternate payment method can be used on the last retry of the failed payment schedule item (`true`) or not (`false`). When set to `true`, the mostly recently added payment method is used to retry payments on the failed payment schedule items. The default value is `false`.
DefaultIntervalUnit	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The unit of time after which a payment batch run must retry the failed payments. The related payment retry rules inherit this value by default. Valid values are: • `Days` • `Hours` • `Minutes`
DefaultIntervalValue	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** The single or comma-separated numeric values, after which the payment batch run must retry the failed payments. The related payment retry rules inherit this value by default.
DefaultMaximumRetryCount	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The maximum number of retries on the failed payment records. The related payment retry rules inherit this value by default.

Field	Details
DefaultRetryIntervalType	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Specifies whether the payment retry rule set must be applied to the failed payment records at fixed or staggered intervals of time. The related payment retry rules inherit this value by default. Valid values are: • `Fixed` • `Staggered`
Description	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The description of the business details for the payment retry rule set.
IsDefaultRuleSet	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the payment retry rule must be set as default for the org (`true`) or not (`false`). The default value is `false`.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed the payment retry rule set record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed the payment retry rule set record.

Field	Details
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The user-specified name for the payment retry rule set.
Status	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Specifies the status of the payment retry rule set. Valid values are: • `Active` • `Draft`

PaymentSchedule

Represents information about a set of payments that a customer wants to collect at different times for a certain record. A schedule contains one or more payment schedule items, where each item represents one payment to be processed. Each of a schedule's items can have different payment configuration fields, such as payment methods, payment dates, and payment accounts. When a payment scheduler launches a payment run, the run evaluates active payment schedule items, and picks them up for payment processing if they match the scheduler's payment criteria. This object is available in API version 64.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need the Revenue Cloud Billing license, and the Payment Admin permission set or the Payment Operations User permission set to access this object.

Fields

Field	Details
AvailableRequestedAmount	**Type** currency

Field	Details
	Properties Filter, Nillable, Sort **Description** The payment schedule's remaining amount available for the creation of payment schedule items. This value is calculated by deducting the `TotalLineRequestedAmount` value from the `TotalRequestedAmount` value. This field is a calculated field.
`CollectionPlanItemId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The collection plan item for the payment schedule that's related to an invoice. This field is a relationship field. **Relationship Name** CollectionPlanItem **Refers To** CollectionPlanItem
`Comments`	**Type** textarea **Properties** Create, Filter, Nillable, Sort, Update **Description** Additional details about the payment schedule.
`DefaultPaymentAccountId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** When a payment run creates payments from a payment schedule item, it sets the payment's account to the item's `PaymentAccountId` value. When the payment schedule item is created, the item's `PaymentAccountId` inherits the schedule's `DefaultPaymentAccountId`. However, you can override the `PaymentAccountId` with a different account as needed. If you do, future payments made from the item use the new account. This field is a relationship field. **Relationship Name** DefaultPaymentAccount

Field	Details
	Refers To Account
DefaultPaymentMethodId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** When a payment run creates payments from a payment schedule item, it sets the payment's payment method to the item's `PaymentMethodId`. When the payment schedule item is created, the item's `PaymentMethodId` inherits the schedule's `DefaultPaymentMethodId`. However, you can override the `PaymentMethodId` with a different method as needed. If you do, future payments made from the item will use the new method. This field is a polymorphic relationship field. **Relationship Name** DefaultPaymentMethod **Refers To** CardPaymentMethod, SavedPaymentMethod
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed this record indirectly, for example, through a list view or related record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed this record or list view. If this value is null, and `LastReferenceDate` is not null, the user accessed this record or list view indirectly.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Required. The ID of the owner of the Payment Schedule record.

Field	Details
	This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
PaymentScheduleNumber	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. The auto-generated reference number for the payment schedule.
PaymentSource	**Type** picklist **Properties** Create, Filter, Group, Nillable, Sort, Update **Description**
ReferenceEntityId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The object record that receives payments as a result of the related payment schedule items that are processed. This field is a polymorphic relationship field. **Relationship Name** ReferenceEntity **Refers To** CollectionPlan, Contract, Invoice, Order, Quote
RemainingAmountToBeProcessed	**Type** currency **Properties** Filter, Nillable, Sort **Description** The total pending amount of payment schedule items that haven't yet been processed for payment. This value is calculated by deducting the `TotalProcessedAmount` value from the `TotalLineRequestedAmount` value. This field is a calculated field.

Field	Details
`Status`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. The status of the payment schedule. Valid values are: • `Canceled`—Payment runs can't evaluate payment schedules or use them to create payments. • `Completed`–All of the payment schedule's payment schedule items have been processed for payments. • `Draft`—The payment schedule can be edited and configured. Payment runs don't evaluate draft payment schedules. • `Open`—The payment schedule is available for payment run evaluation.
`TotalAppliedAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The amount of all the related payment schedule items that have been applied to payments. This field is a calculated field.
`TotalCanceledAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of all `RequestedAmount` values of all the related payment schedule items with a status of `Canceled`. This field is a calculated field.
`TotalPaymentScheduleItemAmount`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The total amount allocated from the payment schedule to its payment schedule items. This value is calculated by deducting the sum of each payment schedule item's `CanceledAmount` from the sum of each payment schedule item's `RequestedAmount`.

Field	Details
	This field is a calculated field.
TotalPaymentsReceived	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of all the total payment received for all the related payment schedule items. This field is a calculated field.
TotalProcessedAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of `ProcessedAmount` values of all the related payment schedule items with a status of `Processed`. This field is a calculated field.
TotalRequestedAmount	**Type** currency **Properties** Create, Filter, Sort, Update **Description** Required. The total amount available for a payment schedule to distribute to its payment schedule items. The sum of payment schedule items can't be greater than the `TotalLineRequestedAmount` value of the parent payment schedule.
UsageType	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The feature using the payment schedule. Valid value is: • `CollectionPlan`

PaymentSchedulePolicy

Represents information about the configuration for the payment schedule. This object is available in API version 64.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need the Revenue Cloud Billing license, and the Payment Admin permission set or the Payment Operations User permission set to access this object.

Fields

Field	Details
`DefaultTreatmentId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The default payment schedule treatment. This field is a relationship field. **Relationship Name** DefaultTreatment **Refers To** PaymentScheduleTreatment
`Description`	**Type** textarea **Properties** Create, Filter, Nillable, Sort, Update **Description** The user-entered description for the payment schedule policy.
`IsOrgDefault`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** If the payment schedule policy is the default policy for the org, this value is set to `true`. If not, this value is set to `false`. An org can have only one default payment policy. The default value is `false`.
`LastReferencedDate`	**Type** dateTime

Field	Details
	Properties Filter, Nillable, Sort **Description** The timestamp when the current user last accessed this record indirectly, for example, through a list view or related record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed this record or list view. If this value is null, and LastReferenceDate is not null, the user accessed this record or list view indirectly.
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** The user-entered name of the payment schedule policy.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Required. The ID of the owner of the PaymentSchedulePolicy record. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
Status	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The status of the payment schedule policy. Valid values are: • Active

Field	Details
	• `Canceled` • `Draft` • `Inactive`
`TreatmentSelection`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Specifies the payment schedule treatment. Valid value is: • `Default`—Uses the payment schedule treatment indicated by the `DefaultTreatmentId` field.

PaymentScheduleTreatment

Represents information about the processing of payment schedules including the payment method and the payment amount for the payment schedule. This object is available in API version 64.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need the Revenue Cloud Billing license, and the Payment Admin permission set or the Payment Operations User permission set to access this object.

Fields

Field	Details
`Description`	**Type** textarea **Properties** Create, Filter, Nillable, Sort, Update **Description** The user-entered description of the payment schedule treatment.
`DueDateWindow`	**Type** int

Field	Details
	Properties Filter, Group, Nillable, Sort **Description** The number that indicates the due date window of the payment schedule treatment. Available in API version 67.0 and later. This field is a calculated field.
`GroupingSource`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Specifies how the payment schedule treatments are grouped, such as by an invoice or account. Available in API version 67.0 and later. Valid values are: • `Invoice` • `Account`
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed this record indirectly, for example, through a list view or related record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed this record or list view. If this value is null, and `LastReferenceDate` is not null, the user accessed this record or list view indirectly.
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** Required. The user-entered name of the payment schedule treatment.

Field	Details
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Required. The ID of the owner of the PaymentScheduleTreatment record. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
PaymentSchedulePolicyId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the related payment schedule policy. This field is a relationship field. **Relationship Name** PaymentSchedulePolicy **Refers To** PaymentSchedulePolicy
Status	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. The status of the payment schedule treatment. Valid values are: • Active • Canceled • Draft • Inactive
TriggerSource	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update

Field	Details
	Description Required. The action that caused the payment schedule treatment to be created. Valid value is: • `InvoicePosted`

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

PaymentScheduleTreatmentShare on page 2877
Sharing is available for the object.

PaymentScheduleTreatmentDtl

Represents information about the processing of payment schedules after the corresponding invoices are posted. This object is available in API version 64.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need the Revenue Cloud Billing license, and the Payment Admin permission set or the Payment Operations User permission set to access this object.

Fields

Field	Details
`DateOffset`	**Type** int **Properties** Create, Filter, Group, Sort, Update **Description** Required. The date offset is subtracted from the `ProcessingDateReference` value to determine the processing date. It must be a number equal to or less than 0. For example, if the invoice due date is 10/17/2025 and the date offset is -7, the payment schedule item is processed by jobs that run on or before 10/10/2025.

Field	Details
`Description`	**Type** textarea **Properties** Create, Filter, Nillable, Sort, Update **Description** The user-entered description for the payment schedule treatment detail.
`InstallmentPaymentType`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. Indicates how the payment amount is divided into multiple payments. Valid value is: • `Percentage`
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed this record indirectly, for example, through a list view or related record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed this record or list view. If this value is null, and `LastReferenceDate` is not null, the user accessed this record or list view indirectly.
`PaymentMethodSelectionType`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. Indicates how the payment method is specified. Valid values are: • `Manual`—The user enters the payment method.

Field	Details
	• `MostRecentAutopay`—The payment method is the most recently used automatic payment method. • `DefaultSavedPaymentMethod`—The default payment method used for processing payments. Available in API version 65.0 and later.
`PaymentRunMatchingValue`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The value for the condition specified for the payment run criteria. When a payment batch run starts, all the payment schedule items that meet the specified criteria are processed.
`PaymentScheduleTreatmentDetailNumber`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. The system-generated unique identifier for the payment schedule treatment detail.
`PaymentScheduleTreatmentId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** Required. The ID of the related payment schedule treatment. This field is a relationship field. **Relationship Name** PaymentScheduleTreatment **Relationship Type** Master-detail **Refers To** PaymentScheduleTreatment (the master object)
`Percentage`	**Type** percent **Properties** Create, Filter, Nillable, Sort, Update **Description** The percentage of the source amount that's used to create the payment schedule.

Field	Details
ProcessingDateReference	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. The source of the reference date. Valid value is: • InvoiceDueDate
PymtSchdDistributionMethodId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** Required. The distribution method that contains the information on how to create the payment schedule items. This field is a relationship field. **Relationship Name** PymtSchdDistributionMethod **Refers To** PymtSchdDistributionMethod

PymtSchdDistributionMethod

Represents information about the partial payments that the total payment is divided into. This object is available in API version 64.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need the Revenue Cloud Billing license, and the Payment Admin permission set or the Payment Operations User permission set to access this object.

Fields

Field	Details
`Description`	**Type** textarea **Properties** Create, Filter, Nillable, Sort, Update **Description** The user-entered details of the payment schedule distribution method.
`DistributionCount`	**Type** int **Properties** Create, Filter, Group, Sort, Update **Description** Required. The number of payment schedule items for the payment schedule. The payment schedule items are used to distribute the payment schedule's total payment into partial payments.
`DistributionMethodType`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. The time interval of the payment distribution method. Valid value is: • `FullDistribution`—The full amount on the payment schedule is distributed to a single payment schedule item.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed this record indirectly, for example, through a list view or related record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort

Field	Details
	Description The timestamp when the current user last viewed this record or list view. If this value is null, and `LastReferenceDate` is not null, the user accessed this record or list view indirectly.
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** Required. The user-entered name for the payment schedule distribution method.
`OwnerId`	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Required. The ID of the owner of the PaymentScheduleDistributionMethod record. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

PymtSchdDistributionMethodOwnerSharingRule on page 2873
Sharing rules are available for the object.

PymtSchdDistributionMethodShare
Sharing is available for the object.

PaymentScheduleItem

Represents information about a payment to be processed. Each schedule item can have different payment configuration fields, such as payment methods, payment dates, and payment accounts. When a payment scheduler launches a payment run, the run evaluates active payment schedule items, and picks them up for payment processing if they match the scheduler's payment criteria. This object is available in API version 64.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need the Revenue Cloud Billing license, and the Payment Admin permission set or the Payment Operations User permission set to access this object.

Fields

Field	Details
AppliedAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The amount of a credit memo or payment that's applied to the parent reference entity and is excluded for payment collection.
Comments	**Type** textarea **Properties** Create, Filter, Nillable, Sort, Update **Description** Additional details about the payment schedule item.
LastPaymentGatewayLogId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The most recent payment gateway log created after a payment gateway request is sent to process a payment based on the payment schedule item. This field is a relationship field. **Relationship Name** LastPaymentGatewayLog **Refers To** PaymentGatewayLog
LastReferencedDate	**Type** dateTime

Field	Details
	Properties Filter, Nillable, Sort **Description** The timestamp when the current user last accessed this record indirectly, for example, through a list view or related record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed this record or list view. If this value is null, and `LastReferenceDate` is not null, the user accessed this record or list view indirectly.
`NextPaymentRetryTime`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort **Description** The slated time of the next payment retry for the payment schedule item. Available in API version 66.0 and later.
`NextPaymentRetryTime`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort **Description** The slated time of the next payment retry for the payment schedule item. Available in API version 66.0 and later.
`PaymentAccountId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The account assigned to payments made from the payment schedule item. When a payment schedule item is created, its `PaymentAccountId` inherits the payment schedule's `DefaultPaymentAccountId`. However, you can provide a new `PaymentAccountId` at any time. If you change the `PaymentAccountId`, only payments made after the change use the new account. This field is a relationship field.

Field	Details
	Relationship Name PaymentAccount **Refers To** Account
PaymentBatchRunId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The payment batch run that evaluated the payment schedule item for payment processing. This field is a relationship field. **Relationship Name** PaymentBatchRun **Refers To** PaymentBatchRun
PaymentGatewayErrorCategory	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The category of payment gateway errors for the payment schedule item. Valid values are: • CardLimit—Card Limit Decline • GatewayConnection—Gateway Connection Error • PaymentInformation—Invalid Payment Details • PaymentProcessing—Payment Processing Error • Security—Security Failure • Unknown—Unknown Error • ValidationFailure—Internal Validation Error Available in API version 66.0 and later.
PaymentGatewayErrorCode	**Type** string **Properties** Create, Filter, Group, Nillable, Sort **Description** The raw error code from the payment gateway for the payment schedule item. Available in API version 66.0 and later.

Field	Details
PaymentGatewayRespStatus	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The status of the payment gateway response when the payment schedule item is processed. Valid values are: • Decline • Indeterminate • PermanentFail • RequiresReview • Success • SystemError • ValidationError See SalesforceResultCode Enum for more information about the values.
PaymentId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The payment that a payment run created for the payment schedule item after picking up the parent payment schedule. This field is unique within your organization. This field is a relationship field. **Relationship Name** Payment **Refers To** Payment
PaymentMethodId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The payment method assigned to payments created from the payment schedule item. When a payment schedule item is created, its `PaymentMethodId` inherits the payment schedule's `DefaultPaymentMethodId`. However, you can provide a new `PaymentMethodId` at any time. If you change the `PaymentMethodId`, only payments made after the change use the new account. This field is a polymorphic relationship field.

Field	Details
	Relationship Name PaymentMethod **Refers To** CardPaymentMethod, SavedPaymentMethod
`PaymentProcessingMessage`	**Type** string **Properties** Filter, Nillable, Sort **Description** Information about whether the payment creation process has completed.
`PaymentRetryCount`	**Type** int **Properties** Create, Filter, Group, Nillable, Sort **Description** The number of payment retries that were attempted on the payment schedule item. Available in API version 66.0 and later.
`PaymentRunMatchingValue`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The value used to match a payment schedule item to a payment run based on the payment run's matching criteria.
`PaymentScheduleId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** Required. The parent payment schedule for the payment schedule item. This field is a relationship field. **Relationship Name** PaymentSchedule **Relationship Type** Master-detail **Refers To** PaymentSchedule (the master object)

Field	Details
PaymentScheduleItemNumber	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. The auto-generated reference number for the payment schedule item.
PaymentSource	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The process that created a payment based on the payment schedule item. Valid values are: • External Payment • Payment Run
PaymentsReceived	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The amount that's been received for the payment created for the payment schedule item.
ProcessedAmount	**Type** currency **Properties** Filter, Nillable, Sort **Description** The amount of the payment schedule item that has been processed for payment and converted to a payment record.
RequestedAmount	**Type** currency **Properties** Create, Filter, Sort, Update **Description** Required. The initial amount of the payment schedule item upon creation.
Status	**Type** picklist

Field	Details
	Properties Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. The status of the payment schedule item. Valid values are: • `Applied`—The payment schedule item has been successfully applied. • `Apply Failed`—The payment run encountered an error when attempting to process the payment schedule item for payment. For more information, review the payment run's revenue transaction error log. • `Canceled`—The payment schedule item can't be picked up by payment runs for processing. When a user or process changes the item's status to `Canceled`, the item's `CanceledAmount` becomes `RequestedAmount – ProcessedAmount`. • `Draft`—Payment schedule items are created with this status. • `Failed`—The payment run has failed to process the payment schedule item for payment. • `Processed`—The payment run has processed the payment schedule item for payment. • `Processing`–The payment run is processing the payment schedule item for payment. • `Ready for Processing`—The payment schedule item is ready to be processed by a payment run.
`TargetPaymentProcessingDate`	**Type** date **Properties** Create, Filter, Group, Sort, Update **Description** Required. The date when the payment run makes a payment request to the payment gateway for the payment schedule item.
`UsageType`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The feature using the payment schedule item. Valid value is: • `CollectionPlan`

PaymentTerm

Represents an agreement between a buyer and a seller about when payment is due for an invoice. This object is available in API version 62.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need the Billing Admin permission set to access this object.

Fields

Field	Details
`Description`	**Type** textarea **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Additional details about the payment term.
`IsDefault`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Required. Indicates whether the payment term is the default term for your Salesforce org (`true`) or not (`false`). The default value is `false`.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed this record indirectly, for example, through a list view or related record.
`LastViewedDate`	**Type** dateTime

Field	Details
	Properties Filter, Nillable, Sort **Description** The timestamp when the current user last viewed this record or list view. If this value is null, and `LastReferenceDate` field is not null, the user accessed this record or list view indirectly.
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** Required. The name of the payment term. This name appears on the invoice.
`Status`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. Specifies whether the payment term is available for use on invoices. Possible values are: • `Active` • `Draft` • `Inactive` The default value is `Draft`.

PaymentTermItem

Represents configuration of a payment term. This object is available in API version 62.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need the Billing Admin permission set to access this object.

Fields

Field	Details
Description	**Type** textarea **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Additional details about the payment term item.
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed this record indirectly, for example, through a list view or related record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed this record or list view. If this value is null, and `LastReferenceDate` field is not null, the user accessed this record or list view indirectly.
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. The autogenerated name of the payment term item.
PaymentTermId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** Required. The ID of the parent payment term record. This field is a relationship field. **Relationship Name** PaymentTerm

Field	Details
	Relationship Type Master-detail **Refers To** PaymentTerm (the master object)
`PaymentTimeframe`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. The period by when the payment is expected. Valid value is: • `Standard`— The payment must be made by the date specified in the payment term. If payment isn't received by the due date, the payment becomes overdue.
`Period`	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The number of units in the payment period. Used with the `PeriodUnit` field. For example, to define a payment term of Net 30, enter 30 as the value of the `Period` field and select `Days` as the value of the `PeriodUnit` field.
`PeriodUnit`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The unit of time for the payment period. For example, to define a payment term of Net 30, enter 30 as the value of the `Period` field and select `Days` as the value of the `PeriodUnit` field. The valid value is: • `Days`
`Type`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. Specifies how the payment term and invoice due date are derived.

Field	Details
	Valid values are: • `EOM-Plus-X`—Calculates the invoice due date by adding the `Period` field value to the last day of the month in which the invoice is posted. Available in API version 63.0 and later. • `Period-Based`—Calculates the invoice due date by adding the `Period` field value to the invoice date.

RevenueTransactionErrorLog

Represents the details of errors that occurred during the processing of a request. The error record persists until a new error with the same category, primary record, and, if necessary, related record occurs. This object is available in API version 62.0 and later.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`

Special Access Rules

You need the Billing Operations User permission set to access this object.

Fields

Field	Details
`AsyncOperationTrackerId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the async operation tracker record created by the request. Async operation tracker records contain information about the status of the asynchronous process initiated by the request. This field is a relationship field. **Relationship Name** AsyncOperationTracker **Refers To** AsyncOperationTracker
`BillingScheduleGroupId`	**Type** reference **Properties** Filter, Group, Nillable, Sort

Field	Details
	Description The ID of the group related to the billing schedule with errors. This field is available in API version 64.0 and later. This field is a relationship field. **Relationship Name** BillingScheduleGroup **Refers To** BillingScheduleGroup
`Category`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** Specifies the context of the source of error. Valid values are: • `ApplyAPI` • AutomatedNegativeInvoiceLineConversion • AutomaticRefunds • `ConvertNegativeInvoiceLineToCredit`—Available in API version 56.0 and later. • `Core Invoice Generation Failure` • `CreditInvoiceAPI` • `CreditMemoRecoveryApi`—Available in API version 66.0 and later. • `CreditTaxIntegrationAPI` • `BulkCurrencyConversion`—Available in API version 66.0 and later. • `InitiateAmendment`—Available in API version 56.0 and later. • `InitiateCancel` • `InitiateRenewal` • `InsufficientAccess`—Insufficient Access to start invoice run. • `InvoiceBatchRun` • `InvoiceBatchRunDebitConversion`—Available in API version 66.0 and later. • `InvoiceBatchRunInvoiceGeneration` • `InvoiceBatchRunPostProcessor` • `InvoiceBatchRunPreProcessor` • `InvoiceBatchRunRecovery` • `InvoiceBatchRunSelectionStep` • `InvoiceBatchRunSplitInvoiceGeneration`—Available in API version 66.0 and later.

Field	Details
	• `InvoiceBatchRunSummarizer` • `InvoiceBatchRunTaxProcessor` • `MaterialLineGeneration`—Available in API version 58.0 and later. • `Invalid Tax API Input` • `Invalid Tax Integration Input` • `LoadSalesRecipientData`—Available in API version 66.0 and later. • `OrderTaxCalculationFailure`—Available in API version 61.0 and later. • `OrderToAsset` • `OrderItemToAsset`—Available in API version 59.0 and later. • `OrderToBillingSchedule` • `PaymentSale` • `PaymentScheduleGeneration`—Available in API version 56.0 and later. • `ProcessAutoQuote`—Available in API version 66.0 and later. • `PstBaseStepFailure`—Available in API version 66.0 and later. • `QuotePriceCalculationFailure`—Available in API version 61.0 and later. • `QuoteTaxCalculationFailure`—Available in API version 61.0 and later. • `QuoteToOrder`—Available in API version 56.0 and later. • `Post Tax API Failure` • `Post-Credit Tax Failure` • `Pre-Credit Tax Failure` • `SetupEnergyAgreement`—Available in API version 66.0 and later. • `StandaloneCreditAPI` • `StatementOfAccountGeneration`—Available in API version 66.0 and later. • `Tax API Failure` • `TransactionToContract`—Available in API version 59.0 and later. • `Unknown Failure`—Available in API version 56.0 and later. • `VoidCreditMemo`—Available in API version 66.0 and later. • `VoidPostedInvoiceAPI`
`ConfiguratorErrorMessage`	**Type** textarea **Properties** Nillable **Description** The text of the error message. This field is available in API version 66.0 and later.
`ErrorCode`	**Type** string

Field	Details
	Properties Filter, Group, Nillable, Sort **Description** The error code shown during the request processing, such as INVALID_INPUT.
`ErrorLogNumber`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. An auto-generated unique ID that identifies the error.
`ErrorMessage`	**Type** textarea **Properties** Create **Description** This field contains information about the error and how to resolve it.
`OwnerId`	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The ID of the user who made the request that resulted in the creation of the error log. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
`PrimaryRecord2Id`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the record that's associated with this error. For example, if the error occurred while creating an invoice line from an order line, the primary2 ID is the ID of the order line. This field is available in API version 66.0 and later. This field is a polymorphic relationship field.

Field	Details
	Relationship Name PrimaryRecord2 **Refers To** Account, CreditMemoInvApplication, CreditMemoLine, CreditMemoLineInvoiceLine, CreditMemoLineTax, DebitMemo, DebitMemoLine, InvoiceLine, InvoiceLineTax, LegalEntyAccountingPeriod, Order, PaymentLineInvoice, PaymentLineInvoiceLine, RefundLinePayment
PrimaryRecordId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the record that's associated with this error. For example, if the error occurred while creating an invoice from an order, the primary ID is the ID of the order. This field is a polymorphic relationship field. **Relationship Name** PrimaryRecord **Refers To** Asset, AsyncOperationTracker, BillingBatchScheduler, BillingSchedule, CreditMemo, InvBatchDraftToPostedRun, Invoice, InvoiceBatchRun, InvoiceBatchRunRecovery, Order, OrderItem, RevenueAsyncOperation, RuleLibraryVersion
PrimaryTextRecord	**Type** string **Properties** Create, Filter, Group, Nillable, Sort **Description** The identifier of the primary record associated with the error log. For example, if the error occurred while creating an invoice from an order, the primary text Record ID is the ID of the order. There are two other fields in the same entity, PrimaryRecord and PrimaryRecord2, that are polymorphic fields, so they're limited to storing IDs from objects in their respective domains. Use this text field for all objects. This field is available in API version 63.0 and later.
RelatedRecord2Id	**Type** reference **Properties** Filter, Group, Nillable, Sort

Field	Details
	Description Optional. The ID of a record that can provide additional context about the error. For example, if `PrimaryRecord2Id` is the ID of an order, this field could be the ID of an invoice line. This field is available in API version 66.0 and later. This field is a polymorphic relationship field. **Relationship Name** RelatedRecord2 **Refers To** CreditMemoLine, CreditMemoLineInvoiceLine, CreditMemoLineTax, DebitMemo, GeneralLedgerAccount, GeneralLedgerAcctAsgntRule, InvoiceLine, InvoiceLineTax, Payment, PaymentLineInvoice, PaymentLineInvoiceLine, Refund, RefundLinePayment
`RelatedRecordId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of a record that can provide additional context about the error. For example, if `PrimaryRecordId` is the ID of an order, this field could be the ID of an order item. This field is a polymorphic relationship field. **Relationship Name** RelatedRecord **Refers To** Asset, BillingBatchScheduler, BillingSchedule, BillingScheduleGroup, CreditMemo, CreditMemoLine, InvBatchDraftToPostedRun, Invoice, InvoiceBatchRun, InvoiceLine, OrderItem
`Request`	**Type** textarea **Properties** Nillable **Description** Optional. A field providing additional information linking the error with the request. This field is available in API version 66.0 and later.
`RequestIdentifier`	**Type** string **Properties** Filter, Group, idLookup, Nillable, Sort **Description** A unique ID returned by the request. Use this ID to identify the revenue transaction error log records for a specific request.

Field	Details
`RevenueAsyncOperationId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the revenue async operation record created by the request. Revenue async operation records contain information about the status of the asynchronous process initiated by the request. This field is available in API version 57.0 and later. This field is a relationship field. **Relationship Name** RevenueAsyncOperation **Relationship Type** Lookup **Refers To** RevenueAsyncOperation
`Severity`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort **Description** The severity type for the error message. This field is available in API version 66.0 and later. Valid values are: • `Error` • `Warning`

SeqPolicySelectionCondition

Represents the condition used to determine which sequence policy is applied to a record. This object is available in API version 65.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need the Billing Admin permission set to access this object.

Fields

Field	Details
ConditionNumber	**Type** int **Properties** Create, Filter, Group, Sort, Update **Description** Required. A unique number that's assigned to a condition in a sequence policy.
FilterField	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** Required. The field used in the filter condition.
FilterFieldType	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The data type of the filter field. Valid values are: • Boolean • Currency • Date • DateTime • MultiPicklist • Number • Percent • Picklist • Reference • Text
FilterValue	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** Required. The value in the filter condition.

Field	Details
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed a sequence policy selection condition record indirectly, for example, through a list view or related record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed a sequence policy selection condition record. If this value is null, it's possible that the user only accessed the sequence policy selection condition record or a related list view (LastReferencedDate), but not viewed the sequence policy selection condition record itself.
Operator	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. The relational operator used to compare the filter field with the filter value. Valid values are: • Equals • Not Equals
SequencePolicyId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** Required. The parent sequence policy associated with the sequence policy selection condition. Deleting a sequencing policy automatically removes all its associated criteria. This field is a relationship field. **Relationship Name** SequencePolicy **Relationship Type** Master-detail

Field	Details
	Refers To SequencePolicy (the master object)
SequencePolicySelectionConditionName	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. The name of the sequence policy selection condition.

Usage

You can create, update, or delete sequence policy selection condition records by using sObject API from API version 67.0 and later.

Create a sequence policy selection condition record by making a POST request to this resource.

```
https://yourInstance.salesforce.com/services/data/v67.0/sobjects/SeqPolicySelectionCondition
```

This example shows a sample request.

```
{
  "SequencePolicyId": "1VdSG00000001fF0AQ",
  "FilterField": "Status",
  "FilterValue": "Draft",
  "Operator": "Equals",
  "ConditionNumber": 1
}
{
  "SequencePolicyId": "1VdSG00000001fF0AQ",
  "FilterField": "Description",
  "FilterValue": "Testing",
  "Operator": "Equals",
  "ConditionNumber": 2
}
```

Update a sequence policy selection condition record record by making a PATCH request to this resource.

```
https://yourInstance.salesforce.com/services/data/v67.0/sobjects/SeqPolicySelectionCondition/id
```

This example shows a sample request.

```
{
 "FilterValue": "654"
}
```

Delete a sequence policy selection condition record by making a DELETE request to this resource.

```
https://yourInstance.salesforce.com/services/data/v67.0/sobjects/SeqPolicySelectionCondition/id
```

SequenceGapReconciliation

Represents a missing sequence value identified during reconciliation, which can be used later to ensure there are no gaps in the sequence policy numbers. This object is available in API version 65.0 and later.

Supported Calls

`describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`

Special Access Rules

You need the Billing Admin permission set to access this object.

Fields

Field	Details
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed sequence gap reconciliation record indirectly, for example, through a list view or related record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed a sequence gap reconciliation record. If this value is null, it’s possible that the user only accessed the sequence gap reconciliation record or a related list view (`LastReferencedDate`), but not viewed the sequence gap reconciliation record itself.
`OwnerId`	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The ID of the user who owns a sequence gap reconciliation record. This field is a polymorphic relationship field. **Relationship Name** Owner

Field	Details
	Refers To Group, User
SequenceGapReconciliationNumber	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. The reconciled sequence pattern value.
SequencePolicyId	**Type** reference **Properties** Filter, Group, Sort **Description** Required. The parent sequence policy associated with the gap reconciliation. Deleting a sequencing policy automatically removes all its associated criteria. This field is a relationship field. **Relationship Name** SequencePolicy **Refers To** SequencePolicy
SequenceValue	**Type** long **Properties** Filter, Group, Sort **Description** Required. The number that was missed during sequence policy generation.
Status	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The current status of the missed number. Valid values are: • `Assigned` • `Blocked` • `Unassigned` • `Under Review`

SequencePolicy

Represents the configuration of rules and parameters for generating unique, sequential numbers for records. Stores settings such as numbering patterns, prefixes, suffixes, sequence start numbers, increment values, and filter criteria to ensure accurate and compliant numbering. This object is available in API version 65.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need the Billing Admin permission set to access this object.

Fields

Field	Details
`DateStampFormat`	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** Required. The format of the stamp date that's appended to the sequence number. Valid values are: • `MM-YYYY`—Month Year (MM-YYYY) • `MM-dd-yyyy`—Month Day Year (MM-DD-YYYY) • `None` • `YYYY`—Year (YYYY) • `YYYY-YY`—Org Fiscal Year (YYYY–YY)
`Description`	**Type** textarea **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Additional details about the sequencing policy.
`EffectiveFromDateTime`	**Type** dateTime **Properties** Create, Filter, Sort, Update

Field	Details
	Description Required. The date and time when the policy becomes effective.
`ExpirationDateTime`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** The date and time when the policy expires.
`IncrementByNumber`	**Type** long **Properties** Filter, Group, Sort **Description** Required. The sequence number increment value.
`IsActive`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Required. Indicates whether the policy is active (`true`) or not (`false`). The default value is `false`.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed a sequence policy record indirectly, for example, through a list view or related record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed a sequence policy record. If this value is null, it's possible that the user only accessed the sequence policy record or a related list view (`LastReferencedDate`), but not viewed the sequence policy record itself.

Field	Details
MaximumSequenceNumber	**Type** long **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The maximum number the sequence number can reach. The maximum width is determined by the maximum sequence number value.
MinimumSequenceNumberWidth	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The minimum number of digits a sequence number must have. For example, if the minimum width is set to 3, sequence numbers appear as 001, 002, and so on. If the maximum sequence number is 99999, the width would be 5 digits.
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** Required. The name of the sequence policy.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Required. The ID of the user who owns a sequence policy record. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
Prefix	**Type** string **Properties** Filter, Group, Nillable, Sort

Field	Details
	Description A string added to the start of the sequence number.
SelectionLogic	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The logic that determines the records to which the sequence policy applies.
SequenceMode	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. Specifies how sequence numbers are generated. Valid values are: • Basic—Gaps are allowed, such as when a record is canceled or rolled back. • Gapless—Numbers follow one after another with no gaps.
SequencePattern	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** Required. The structure of a sequence.
SequenceStartNumber	**Type** long **Properties** Create, Filter, Group, Sort, Update **Description** Required. The starting sequence number.
Suffix	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** A string added to the end of the sequence number.

Field	Details
`TargetObject`	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** Required. The object to which the policy is applied.
`TimeZone`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The time zone applicable to the date and time related fields of the policy. When not specified, it takes the user's time zone by default. The time zone is shown in Greenwich Mean Time (GMT).

Usage

You can create, update, or delete sequence policy records by using sObject API from API version 67.0 and later.

Create a sequence policy record by making a POST request to this resource.

```
https://yourInstance.salesforce.com/services/data/v67.0/sobjects/SequencePolicy
```

This example shows a sample request.

```
{
  "Name": "Invoice Sequence Policy Testing",
  "Description": "Sequence policy for invoice numbering",
  "TargetObject": "Invoice",
  "SequenceMode": "Gapless",
  "SequencePattern": "IN-{Date}-{SequenceValue}",
  "DateStampFormat": "MM-dd-yyyy",
  "TimeZone": "America/Los_Angeles",
  "IsActive": false,
  "SequenceStartNumber": 1,
  "IncrementByNumber": 1,
  "MaximumSequenceNumber": 99999,
  "MinimumSequenceNumberWidth": 4,
  "EffectiveFromDateTime": "2026-03-01T00:00:00.000Z",
  "ExpirationDateTime": "2027-03-01T00:00:00.000Z"
}
```

Update a sequence policy record by making a PATCH request to this resource.

```
https://yourInstance.salesforce.com/services/data/v67.0/sobjects/SequencePolicy/id
```

This example shows a sample request.

```
{
  "Description": "Sequence policy for invoice numbering testing",
  "IsActive": true,
  "MaximumSequenceNumber": 88888
}
{
  "SelectionLogic": "1 AND 2"
}
```

Delete a sequence policy record by making a DELETE request to this resource.

```
https://yourInstance.salesforce.com/services/data/v67.0/sobjects/SequencePolicy/id
```

TaxEngine

Represents information about an instance of a tax engine provider as well as the merchant credentials for that specific instance. This object is available in API version 62.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need the Tax Admin permission set to access this object.

Fields

Field	Details
Description	**Type** textarea **Properties** Create, Filter, Nillable, Sort, Update **Description** The description of an instance of the tax engine provider and merchant credential.
ExternalReference	**Type** string **Properties** Create, Filter, Group, idLookup, Nillable, Sort, Update **Description** Information about the external platform used for the tax engine.

Field	Details
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed a tax engine record indirectly, for example, through a list view or related record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed a tax engine record. If this value is null, it's possible that the user only accessed the tax engine record or a related list view (LastReferencedDate), but not viewed the tax engine record itself.
MerchantCredentialId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Required. The merchant credential setup object in Salesforce. The Tax Calculation API sends the merchant credential's information to the external tax engine that's used for tax calculation. This field is a relationship field. **Relationship Name** MerchantCredential **Refers To** NamedCredential
SellerCode	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Required. The seller code of the transaction for which the tax engine integration log was captured.
ShouldCaptureTaxesAtHeader	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Sort, Update

Field	Details
	Description Required. Indicates whether taxes are captured at the header level (`true`) or not (`false`). The default value is `false`. Available in API version 66.0 and later.
`Status`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. The status of the tax engine record. Valid values are: • `Active` • `Inactive`
`TaxEngineAddress`	**Type** address **Properties** Filter **Description** Required. The compound form of the tax engine address. See Address Compound Fields for details on compound address fields. This field is a read-only field.
`TaxEngineName`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** Required. The name of the tax engine record.
`TaxEngineProviderId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Required. The tax engine provider that's related to the tax engine record. This field is a relationship field. **Relationship Name** TaxEngineProvider

Field	Details
	Refers To TaxEngineProvider
TaxPrvdAccountIdentifier	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The unique identifier of the external tax provider's account. Available in API version 63.0 and later.
Type	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The type of the tax engine that's used to calculate tax. Valid values are: • CommerceTaxExtension • RevenueCloudTaxExtension • StandardTaxEngine • RevenueStandardTaxEngine—Available in API version 66.0 and later. • StripeNative Available in API versions 63.0 and later.

TaxEngineInteractionLog

Represents a record of a communication with an external tax engine following a tax calculation request. This object is available in API version 62.0 and later.

Supported Calls

delete(), describeLayout(), describeSObjects(), getDeleted(), getUpdated(), query(), retrieve(), undelete()

Special Access Rules

You need the Tax Admin permission set to access this object.

Fields

Field	Details
Description	**Type** textarea **Properties** Filter, Nillable, Sort **Description** Additional details about the tax engine interaction log.
DocumentCode	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** Document code of the transaction for which the tax engine integration log was captured.
EffectiveDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date that the tax engine request takes effect. This date is available for reference and bookkeeping only and doesn't have any impact on tax calculation.
InteractionHttpStatusCode	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The HHTP result code of the external callout made to a third-party tax engine provider. Refer to your third-party tax engine provider's documentation for details about the specific codes returned.
InteractionType	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** Indicates the type of request made to the tax engine. Valid value is: • `CalculateTax`

Field	Details
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed a tax engine interaction log record indirectly, for example, through a list view or related record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed a tax engine interaction log record. If this value is null, it's possible that the user only accessed the tax engine interaction log record or a related list view (`LastReferencedDate`), but not viewed the tax engine interaction log record itself.
ReferenceEntity	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The record on which tax was calculated.
RequestBody	**Type** base64 **Properties** Nillable **Description** The content of the tax calculation API request.
RequestContentType	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The type of data passed in the request. For example, `application/html` or `text/csv`.
RequestLength	**Type** int

Field	Details
	Properties Filter, Group, Nillable, Sort **Description** The character length of text within the request body.
`RequestName`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The name of the request.
`RequestTaxTransactionType`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The type of tax transaction request sent to the tax engine provider. Available in API version 65.0 and later. Valid values are: • `Credit` • `Debit` • `VoidOrCredit` • `VoidOrDebit`
`ResponseContentType`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The method used to deliver the tax calculation API response, such as `application/html` or `text/vnd.salesforce.quip-template`.
`ResponseLength`	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** The character length of text within the response body.

Field	Details
ResponseName	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The name of the response from the tax engine.
ResponseTaxTransactionType	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The type of tax transaction response received from the tax engine provider. Available in API version 65.0 and later. Valid values are: • Credit • Debit • Void
ResultCode	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The code describing the result of the request. Valid values are: • AdapterException • ReferenceDocumentCodeMissing • Success • TaxEngineError • ValidationError
TaxEngineId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the tax engine used in the tax calculation process. This field is a relationship field.

Field	Details
	Relationship Name TaxEngine **Refers To** TaxEngine
`TaxEngineInteractionLogNumber`	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** A system-generated number for a log entry.

TaxEngineProvider

Represents general information about a service that manages a tax engine. Tax engine providers have a one-to-many relationship with tax engines, where the tax engine record represents a specific configuration of a tax engine that can be assigned to multiple order items. This object is available in API version 62.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `query()`, `retrieve()`, `update()`, `upsert()`

Special Access Rules

You need the Tax Admin permission set to access this object.

Fields

Field	Details
`ApexAdapterId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the Apex adapter used by the tax provider. This field is unique within your organization. This field is a relationship field. **Relationship Name** ApexAdapter **Refers To** ApexClass

Field	Details
CustomMetadataTypeApiName	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The API name of the custom metadata type that defines field mappings for tax callout requests and responses. The custom metadata type name is referenced by the tax engine provider to identify the metadata configuration to use. Available in API version 65.0 and later.
Description	**Type** textarea **Properties** Create, Filter, Nillable, Sort, Update **Description** Additional details about the tax engine provider.
DeveloperName	**Type** string **Properties** Create, Filter, Group, Sort, Update **Description** Required. The API name for the tax engine provider record.
Language	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The language used by the tax engine provider. Values appear based on their language codes in Salesforce, such as `da` for Danish or `th` for Thai.
MasterLabel	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** Required. The label used for the tax engine's API in Salesforce.
NamespacePrefix	**Type** string **Properties** Filter, Group, Nillable, Sort

Field	Details
	Description The Apex namespace prefix of the API used for the tax engine.

TaxPolicy

Represents information about a group of tax treatments, where each treatment represents parameters to determine how a particular product is taxed for a transaction line item. Tax policies are related to products, which pass the policy on to the resulting order items and in turn the billing schedules. This object is available in API version 62.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need the Tax Admin permission set to access this object.

Fields

Field	Details
`DefaultTaxTreatmentId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the default tax treatment record. When you order a product, the order product, and in turn the billing schedule receives this tax treatment. This field is a relationship field. **Relationship Name** DefaultTaxTreatment **Refers To** TaxTreatment
`Description`	**Type** textarea **Properties** Create, Filter, Nillable, Sort, Update **Description** Additional details about the tax policy.

Field	Details
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed an invoice batch draft to posted run record indirectly, for example, through a list view or related record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed a tax policy record. If this value is null, it's possible that the user only accessed the tax policy record or a related list view (LastReferencedDate), but not viewed the tax policy record itself.
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** Required. The name of the tax policy.
Status	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. The status of the tax policy. To calculate tax for order products, products must have an active tax policy. Tax policies are created with a Draft status before being assigned to a product or order product. After activating a tax policy, you can't edit certain policy fields. Valid values are: • Active • Draft • Inactive
TreatmentSelection	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update

Field	Details
	Description Required. Specifies the selection of tax treatments to billing schedules that are related to the tax policy. Valid values are: • Default —Billing schedules receive the tax treatment defined in the tax policy's `DefaultTreatmentId` field. • `LegalEntity`—Billing schedules receive the tax treatment based on matching legal entities between itself and the tax treatment. • `Manual`—Billing schedules don't receive tax treatments based on the tax policy. You must specify the treatment.

TaxTreatment

Represents information about tax calculation by external engines. Each product requires a tax policy to determine whether to apply tax. Each tax policy requires at least one tax treatment. The tax treatments determine how taxable products are taxed. This object is available in API version 62.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need the Tax Admin permission set to access this object.

Fields

Field	Details
`Description`	**Type** textarea **Properties** Create, Filter, Nillable, Sort, Update **Description** Additional details about the tax treatment.
`IsTaxable`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update

Field	Details
	Description Required. Indicates whether to calculate tax for the order items covered by the tax treatment (`true`) or not (`false`). When this value is `true`, the CalculateTax API is called for the order item during the order item's creation. The default value is `false`.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed a tax treatment record indirectly, for example, through a list view or related record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed a tax treatment record. If this value is null, it's possible that the user only accessed the tax treatment record or a related list view (`LastReferencedDate`), but not viewed the tax treatment record itself.
`LegalEntityId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the legal entity record that's related to the tax treatment. This field is a relationship field. **Relationship Name** LegalEntity **Refers To** LegalEntity
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** Required. The name of the tax treatment.

Field	Details
`ProductCode`	**Type** string **Properties** Create, Filter, Group, idLookup, Nillable, Sort, Update **Description** The code of the product that the tax treatment applies to.
`ProductId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the product that the tax treatment applies to. This field is a relationship field. **Relationship Name** Product **Refers To** Product2
`ShouldUseTaxTreatmentItems`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Required. Indicates whether the tax codes and product codes of the related tax treatment items must be used when requesting tax calculations (`true`) or not (`false`). The default value is `false`. The tax code and product code of the tax treatment are ignored by default. Available in API version 66.0 and later.
`Status`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. The status of the tax treatment record. Valid values are: • `Active` • `Draft` • `Inactive`

Field	Details
`TaxCode`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The reference code that's used when an external tax engine calculates tax.
`TaxEngineId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the tax engine record that's related to the tax treatment. When tax is calculated for an order item, the tax engine that's related to order item's tax treatment is used. If the tax treatment's `IsTaxable` field value is `True`, the treatment requires a tax engine. This field is a relationship field. **Relationship Name** TaxEngine **Refers To** TaxEngine
`TaxPolicyId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the tax treatment's parent tax policy. This field is a relationship field. **Relationship Name** TaxPolicy **Refers To** TaxPolicy

Tax Treatment Item

Represents tax code information that's used to calculate tax for a product by a specific tax engine. This object is available in API version 66.0 and later.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need the Billing Admin permission set or Tax Admin permission set to access this object.

Fields

Field	Details
`Description`	**Type** textarea **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Additional details about the tax treatment item.
`LastReferencedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed a tax treatment item record indirectly, for example, through a list view or related record.
`LastViewedDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed a tax treatment item record. If this value is null, it's possible that the user only accessed the tax treatment item record or a related list view (`LastReferencedDate`), but not viewed the tax treatment item record itself.
`Name`	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** Required. The name of the tax treatment item.

Field	Details
ProductCode	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The code of the product that the tax treatment item applies to.
ProductId	**Type** reference **Properties** Create, Filter, Group, Sort, Update **Description** Required. The ID of the product that the tax treatment item applies to. This field is a relationship field. **Relationship Name** Product **Refers To** Product2
TaxCode	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The reference code that's used when an external tax engine calculates tax.
TaxTreatmentId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** Required. The parent tax treatment associated with the tax treatment item record. This field is a relationship field. **Relationship Name** TaxTreatment **Relationship Type** Master-detail **Refers To** TaxTreatment (the master object)

Billing Fields on Standard Objects

Billing adds standard fields to some standard Salesforce objects of other features to represent information specific to Billing. These fields are available only in orgs where Billing is enabled.

Billing Fields on AccountBillingAccount

Standard fields extend the AccountBillingAccount object for use in Billing to represent information about default billing accounts. This object is available in API version 63.0 and later.

Billing Fields on BillingAccount

Standard fields extend the BillingAccount object for use in Billing to represent information about the billing suspension date and the billing resumption date. This object is available in API version 63.0 and later.

Billing Fields on CollectionPlan

Standard fields extend the CollectionPlan object for use in Billing to represent information about the total invoice balance. This object is available in API version 64.0 and later.

Billing Fields on CollectionPlanItem

Standard fields extend the CollectionPlanItem object for use in Billing to represent information about the invoice balance. This object is available in API version 64.0 and later.

Billing Fields on ExpressionSet

Standard fields extend the ExpressionSet object for use in Billing. These fields represent information about an expression set that performs a series of calculations by using lookups and user-defined variables and constants to calculate taxes. This object is available in API version 66.0 and later.

Billing Fields on Dispute

Represents the details of a billing dispute that involves one invoice and one or more disputed invoice lines. The details include the disputed amount, the approved amount, and the dispute type, subtype and status. This object is available in API version 66.0 and later.

Billing Fields on DisputeItem

Represents a specific invoice line or charge that's being disputed. The details include the total transaction amount, transaction date, disputed amount, reason, and status of the dispute. This object is available in API version 66.0 and later.

Billing Fields on Payment

Standard fields extend the Payment object for use in Billing to represent information about corporate currency, transaction amounts in corporate currency, and accounting periods for legal entities. This object is available in API version 64.0 and later.

Billing Fields on PaymentLineInvoice

Standard fields extend the PaymentLineInvoice object for use in Billing to represent information about legal entities and legal entity accounting periods. This object is available in API version 64.0 and later.

Billing Fields on Refund

Standard fields extend the Refund object for use in Billing to represent information about corporate currency, transaction amounts in corporate currency, and accounting periods for legal entities. This object is available in API version 64.0 and later.

Billing Fields on RefundLinePayment

Standard fields extend the Refund Line Payment object for use in Billing to represent information about accounting periods for legal entities. This object is available in API version 64.0 and later.

Billing Fields on TaxRate

Standard fields extend the TaxRate object for use in Billing. These fields represent information about the tax rate for a transaction that's determined by the applicable tax code and country. This object is available in API version 66.0 and later.

[Billing Fields on TransactionJournal]
Standard fields extend the TransactionJournal object for use in Billing to represent information about the general ledger accounts for billing transactions. This object is available in API version 63.0 and later.

Billing Fields on AccountBillingAccount

Standard fields extend the AccountBillingAccount object for use in Billing to represent information about default billing accounts. This object is available in API version 63.0 and later.

Supported Calls

`describeLayout(), describeSObjects(), getDeleted(), getUpdated(), query(), retrieve(), search()`

Special Access Rules

You need the Billing Admin permission set, Billing Operations User permission set, or Billing Customer Service User permission set access to this object.

Fields

Field	Details
`IsDefaultBillingAccount`	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Indicates whether the account is the default billing account (`true`) or not (`false`). The default value is `false`.

SEE ALSO:

Energy and Utilities Cloud Developer Guide: AccountBillingAccount

Billing Fields on BillingAccount

Standard fields extend the BillingAccount object for use in Billing to represent information about the billing suspension date and the billing resumption date. This object is available in API version 63.0 and later.

Supported Calls

`describeLayout(), describeSObjects(), getDeleted(), getUpdated(), query(), retrieve(), search()`

Special Access Rules

You need the Billing Admin permission set, Billing Operations User permission set, or Billing Customer Service User permission set access to this object.

Fields

Field	Details
AccountID	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Specifies the account that's related to the billing account. Available in API version 65.0 and later. This field is a relationship field. **Relationship Name** Account **Refers To** Account
AddTaxIdentificationDetails	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The additional tax identification details, such as a Peppol ID. Available in API version 66.0 and later.
BillDayOfMonth	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** A number from 1 to 31 that indicates the billing date. Available in API version 64.0 and later.
BillToContactId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The contact associated with the billing account. Available in API version 64.0 and later. This field is a relationship field.

Field	Details
	Relationship Name BillToContact **Refers To** Contact
BillingResumptionDate	**Type** date **Properties** Filter, Group, Nillable, Sort **Description** The date when billing for the account is resumed.
BillingSuspensionDate	**Type** date **Properties** Filter, Group, Nillable, Sort **Description** The date when billing for the account is suspended.
DeliveryTerms	**Type** picklist **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The standard code in internal trades used to define shipping responsibilities. Valid values are: • CFR • CIF • CIO • CPT • CUS • DAF • DAP • DDP • DDU • DEQ • DES • DPU • EXW • FAS

Field	Details
	• FCA • FOB • SUP The default value is USD. Available in API version 66.0 and later.
DoesSkipAutomaticPayments	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether to skip the automatic creation of payment schedules and payment schedule items for posted invoices related to billing accounts (true) or not (false). This boolean value is considered when the automatic creation of payment schedules and payment schedule items is turned on. Available in API version 65.0 and later.
EmailTemplateId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The template used for sending billing account emails. Available in API version 65.0 and later. This field is a relationship field. **Relationship Name** EmailTemplate **Refers To** Email template
InvoiceDocumentTemplateId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The document template that's used to generate PDF documents for the invoices that are related to the billing account. Available in API version 66.0 and later. This field is a relationship field. **Relationship Name** InvoiceDocumentTemplate **Refers To** DocumentTemplate

Field	Details
`IsDefaultBillingProfile`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the billing account is the default billing profile (`true`) or not (`false`). Available in API version 65.0 and later.
`PaymentTermId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The payment term associated with the billing account. Available in API version 64.0 and later. This field is a relationship field. **Relationship Name** PaymentTerm **Refers To** PaymentTerm
`SavedPaymentMethodId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The saved payment method that's used to collect payments for the default billing profile of an account when the automatic creation of payment schedules and payment schedule items is turned on. Available in API version 65.0 and later. This field is a relationship field. **Relationship Name** SavedPaymentMethod **Refers To** SavedPaymentMethod
`ShippingAddress`	**Type** address **Properties** Filter, Nillable **Description** The shipping address of the billing account. Available in API version 64.0 and later.

Field	Details
ShouldAttachInvoiceDocToEmail	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether to attach the invoice PDF document to the email that's sent to the billing account. Available in API version 65.0 and later. The default value is false.
TaxExemptionExpirationDate	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The date until which the tax exemption is valid. Available in API version 66.0 and later.
TaxExemptionNumber	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** A unique identifier used to indicate that a transaction is tax exempt. Available in API version 66.0 and later.
TaxExemptionStatus	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The status of the tax exemption with values defined by the user. Available in API version 66.0 and later.
TaxIdentificationNumber	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The tax identification number such as a VAT ID. Available in API version 66.0 and later.

SEE ALSO:

Energy and Utilities Cloud Developer Guide: BillingAccount

Billing Fields on CollectionPlan

Standard fields extend the CollectionPlan object for use in Billing to represent information about the total invoice balance. This object is available in API version 64.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need the Revenue Cloud Billing license, and the Billing Collections and Recovery Specialist permission set to access this object.

Fields

Field	Details
`TotalInvoiceBalance`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The sum of all invoice balances for collection plan items associated with the collection plan.
OverdueRiskIndicator	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Indicates the overdue risk level for the collection plan. Available in API version 67.0 and later. Valid values are: • `Low` • `Medium` • `High`

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

CollectionPlanChangeEvent on page 2864
Change events are available for the object.

CollectionPlanFeed on page 2866
Feed tracking is available for the object.

CollectionPlanHistory on page 2873
History is available for tracked fields of the object.

CollectionPlanShare on page 2877
Sharing is available for the object.

SEE ALSO:

Industries Common Resources Developer Guide: CollectionPlan

Billing Fields on CollectionPlanItem

Standard fields extend the CollectionPlanItem object for use in Billing to represent information about the invoice balance. This object is available in API version 64.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need the Revenue Cloud Billing license, and the Billing Collections and Recovery Specialist permission set to access this object.

Fields

Field	Details
`InvoiceBalance`	**Type** currency **Properties** Filter, Nillable, Sort **Description** The balance amount of the invoice associated with the collection plan item.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

CollectionPlanItemChangeEvent on page 2864
Change events are available for the object.

CollectionPlanItemFeed on page 2866
Feed tracking is available for the object.

CollectionPlanItemHistory on page 2873
History is available for tracked fields of the object

SEE ALSO:
Industries Common Resources Developer Guide: CollectionPlanItem

Billing Fields on ExpressionSet

Standard fields extend the ExpressionSet object for use in Billing. These fields represent information about an expression set that performs a series of calculations by using lookups and user-defined variables and constants to calculate taxes. This object is available in API version 66.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
`UsageType`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Restricted picklist, Sort, Update **Description** Required. The Revenue Standard Tax Calculation value that's using the expression set to calculate taxes.

Billing Fields on Dispute

Represents the details of a billing dispute that involves one invoice and one or more disputed invoice lines. The details include the disputed amount, the approved amount, and the dispute type, subtype and status. This object is available in API version 66.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need the Revenue Cloud Billing license, and the Billing Operations User or the Billing Customer Service User permission set to access this object.

Fields

Field	Details
InvoiceId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The invoice ID associated with the dispute. This field is a relationship field. **Relationship Name** Invoice **Refers To** Invoice
RevisedDueDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The revised due date for the invoice.
BillingSuspensionDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when billing for the account is suspended.
BillingResumptionDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date when billing for the account is resumed.
RevisedBillToContact	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The revised billing contact for the invoice.

Field	Details
	This field is a relationship field. **Relationship Name** Contact **Refers To** Contact
`ResolutionAction`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The action taken to resolve the dispute. Valid values are: • `Resolve with No Action` • `Issue Credit Memo`
`ResolutionActionStatus`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The status of the resolution action. Valid values are: • `In Progress` • `Closed` • `Error`
`MaySetContactAsDefault`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the revised billing contact is set as default for future invoices (`true`) or not (`false`). The default value is `false`.
`UsageType`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update

Field	Details
	Description Specifies the name of the cloud or function that uses the Dispute object. Valid value is `Billing`.

SEE ALSO:

Financial Services Cloud Developer Guide: Dispute

Billing Fields on DisputeItem

Represents a specific invoice line or charge that's being disputed. The details include the total transaction amount, transaction date, disputed amount, reason, and status of the dispute. This object is available in API version 66.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need the Revenue Cloud Billing license, and the Billing Operations User or the Billing Customer Service User permission set to access this object.

Fields

Field	Details
`ApprovedAmount`	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The amount that's approved to be credited to the customer.
`InvoiceLineId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The invoice line ID associated with the dispute. This field is a relationship field.

Field	Details
	Relationship Name Invoice Line **Refers To** Invoice Line

SEE ALSO:

Financial Services Cloud Developer Guide: DisputeItem

Billing Fields on Payment

Standard fields extend the Payment object for use in Billing to represent information about corporate currency, transaction amounts in corporate currency, and accounting periods for legal entities. This object is available in API version 64.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need the Revenue Cloud Billing license, and the Payment Admin permission set or the Payment Operations User permission set to access this object.

Fields

Field	Details
`CorporateCurrencyCnvAmount`	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The payment amount in corporate currency.
`CorporateCurrencyCvsnDate`	**Type** date **Properties** Filter, Group, Nillable, Sort, Update **Description** The date on which the payment amount is converted into corporate currency.

Field	Details
CorporateCurrencyCvsnRate	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The exchange rate that's used to convert the payment amount into corporate currency.
CorporateCurrencyIsoCode	**Type** string **Properties** Filter, Group, Nillable, Sort, Update **Description** The currency ISO code of the corporate currency.
FunctionalCurrencyCnvAmount	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The amount value in functional currency. Available in API version 66.0 and later.
FunctionalCurrencyCvsnDate	**Type** date **Properties** Filter, Group, Nillable, Sort, Update **Description** The date on which the amount value is converted into functional currency. Available in API version 66.0 and later.
FunctionalCurrencyCvsnRate	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The exchange rate that's used to convert the amount value into functional currency. Available in API version 66.0 and later.
FunctionalCurrencyIsoCode	**Type** string **Properties** Filter, Group, Nillable, Sort, Update

Field	Details
	Description The ISO code of the functional currency. Available in API version 66.0 and later.
`LegalEntityAccountingPeriodId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The legal entity accounting period related to the payment. This field is a relationship field. **Relationship Name** LegalEntityAccountingPeriod **Refers To** LegalEntityAccountingPeriod
`LegalEntityId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The legal entity related to the payment. This field is a relationship field. **Relationship Name** LegalEntity **Refers To** LegalEntity

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

PaymentFeed on page 2866
Feed tracking is available for the object.

SEE ALSO:

Object Reference for the Salesforce Platform: Payment

Billing Fields on PaymentLineInvoice

Standard fields extend the PaymentLineInvoice object for use in Billing to represent information about legal entities and legal entity accounting periods. This object is available in API version 64.0 and later.

Supported Calls

`create()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `update()`, `upsert()`

Special Access Rules

You need the Revenue Cloud Billing license, and the Payment Admin permission set or the Payment Operations User permission set to access this object.

Fields

Field	Details
`LegalEntityAccountingPeriodId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The legal entity accounting period related to the payment line invoice. This field is a relationship field. **Relationship Name** LegalEntityAccountingPeriod **Refers To** LegalEntyAccountingPeriod
`LegalEntityId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The legal entity related to the payment line invoice. This field is a relationship field. **Relationship Name** LegalEntity **Refers To** LegalEntity

SEE ALSO:

Object Reference for the Salesforce Platform: PaymentLineInvoice

Billing Fields on Refund

Standard fields extend the Refund object for use in Billing to represent information about corporate currency, transaction amounts in corporate currency, and accounting periods for legal entities. This object is available in API version 64.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need the Revenue Cloud Billing license, and the Payment Admin permission set or the Payment Operations User permission set to access this object.

Fields

Field	Details
`CorporateCurrencyCnvAmount`	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The refund amount in corporate currency.
`CorporateCurrencyCvsnDate`	**Type** date **Properties** Filter, Group, Nillable, Sort, Update **Description** The date on which the refund amount is converted into corporate currency.
`CorporateCurrencyCvsnRate`	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The exchange rate that's used to convert the refund amount into corporate currency.
`CorporateCurrencyIsoCode`	**Type** string **Properties** Filter, Group, Nillable, Sort, Update

Field	Details
	Description The currency ISO code of the corporate currency.
FunctionalCurrencyCnvAmount	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The amount value in functional currency. Available in API version 66.0 and later.
FunctionalCurrencyCvsnDate	**Type** date **Properties** Filter, Group, Nillable, Sort, Update **Description** The date on which the amount value is converted into functional currency. Available in API version 66.0 and later.
FunctionalCurrencyCvsnRate	**Type** double **Properties** Filter, Nillable, Sort, Update **Description** The exchange rate that's used to convert the amount value into functional currency. Available in API version 66.0 and later.
FunctionalCurrencyIsoCode	**Type** string **Properties** Filter, Group, Nillable, Sort, Update **Description** The ISO code of the functional currency. Available in API version 66.0 and later.
LegalEntityAccountingPeriodId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The legal entity accounting period related to the refund. This field is a relationship field.

Field	Details
	Relationship Name LegalEntityAccountingPeriod **Refers To** LegalEntityAccountingPeriod
`LegalEntityId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The legal entity related to the refund. This field is a relationship field. **Relationship Name** LegalEntity **Refers To** LegalEntity

SEE ALSO:

Object Reference for the Salesforce Platform: Refund

Billing Fields on RefundLinePayment

Standard fields extend the Refund Line Payment object for use in Billing to represent information about accounting periods for legal entities. This object is available in API version 64.0 and later.

Special Access Rules

You need the Revenue Cloud Billing license, and the Payment Admin permission set or the Payment Operations User permission set to access this object.

Fields

Field	Details
`LegalEntityAccountingPeriodId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The legal entity accounting period related to the refund. This field is a relationship field.

Field	Details
	Relationship Name LegalEntityAccountingPeriod **Refers To** LegalEntityAccountingPeriod
`LegalEntityId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The legal entity related to the refund. This field is a relationship field. **Relationship Name** LegalEntity **Refers To** LegalEntity

Billing Fields on TaxRate

Standard fields extend the TaxRate object for use in Billing. These fields represent information about the tax rate for a transaction that's determined by the applicable tax code and country. This object is available in API version 66.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need the Billing Admin permission set or Tax Admin permission set access to this object.

Fields

Field	Details
`ApplicationBasis`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies whether the tax rate is applied on the net or gross amount.

Field	Details
	Valid values are: • `Gross` • `Net` The default value is `Gross`.
`City`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The city to which the tax rate applies.
`Country`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The country name that's derived from the `GeoCountry` field value. This field is a calculated field.
`CurrencyIsoCode`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The currency ISO code that's applicable to the tax rate.
`EndDate`	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The date until when the tax rate is valid.
`FlatTaxAmount`	**Type** double **Properties** Create, Filter, Nillable, Sort, Update **Description** The flat tax amount that's applied to the transaction.

Field	Details
LastReferencedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last accessed a tax rate record indirectly, for example, through a list view or related record.
LastViewedDate	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The timestamp when the current user last viewed a tax rate record. If this value is null, it’s possible that the user only accessed the debit memo line tax record or a related list view (LastReferencedDate), but not viewed the debit memo line tax record itself.
LegalEntityId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort **Description** The legal entity to which the tax rate applies. This field is a relationship field. **Relationship Name** LegalEntity **Refers To** LegalEntity
Name	**Type** string **Properties** Autonumber, Defaulted on create, Filter, idLookup, Sort **Description** Required. The name of the invoice line.
OwnerId	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update

Field	Details
	Description Required. The ID of the user who owns a TaxRate record. This field is a polymorphic relationship field. **Relationship Name** Owner **Refers To** Group, User
ProductCode	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The code of the product for which the tax rate applies.
RateUsageType	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort **Description** Specifies whether the tax rate is created for Commerce or Revenue Cloud. Valid values are: • Commerce • RevCloud The default value is Commerce.
StartDate	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The date from when the tax rate is valid.
State	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The state name that’s derived from the GeoState field value. This field is a calculated field.

Field	Details
ZipCode	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The postal or ZIP code to which the tax rate applies.

Billing Fields on TransactionJournal

Standard fields extend the TransactionJournal object for use in Billing to represent information about the general ledger accounts for billing transactions. This object is available in API version 63.0 and later.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `undelete()`, `update()`, `upsert()`

Special Access Rules

You need the Revenue Cloud Billing license and the Accounts Receivables Admin permission set to access this object.

Fields

Field	Details
ActivityDate	**Type** dateTime **Properties** Create, Filter, Sort, Update **Description** The date and time when a billing transaction record is posted or processed.
Credit	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The transaction record amount when a credit general ledger account is specified.
CreditGeneralLedgerAccountId	**Type** reference

Field	Details
	Properties Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the general ledger account for a debit transaction that's related to the transaction journal. This field is a relationship field. **Relationship Name** CreditGeneralLedgerAccount **Refers To** GeneralLedgerAccount
`Debit`	**Type** currency **Properties** Create, Filter, Nillable, Sort, Update **Description** The transaction record amount when a debit general ledger account is specified.
`DebitGeneralLedgerAccountId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The ID of the General Ledger (GL) Treatment based on which transaction journals must be generated for the billing transaction. This field is a relationship field. **Relationship Name** DebitGeneralLedgerAccount **Refers To** GeneralLedgerAccount
`ForeignExchangeGainOrLossType`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Specifies the type of foreign exchange gain or loss for the transaction journal. This field is available in API version 65.0 and later. Valid values are: • `Realized` • `Unrealized`

Field	Details
	• `UnrealizedReversal`
`GeneralLedgerAcctAsgntRuleId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The general ledger account assignment rule used to assign general ledger accounts to transaction journals that's created for billing transactions. This field is a relationship field. **Relationship Name** GeneralLedgerAcctAsgntRule **Refers To** GeneralLedgerAcctAsgntRule
`GenlLdgrJournalEntryRuleId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The general ledger journal entry rule used to assign general ledger accounts to transaction journals created for billing transactions. This field is available in API version 65.0 and later. This field is a relationship field. **Relationship Name** GenlLdgrJournalEntryRule **Refers To** GeneralLedgerJrnlEntryRule
`LegalEntityId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The legal entity that's related to the transaction journal. This field is a relationship field. **Relationship Name** LegalEntity **Refers To** LegalEntity

Field	Details
ReferenceTransactionRecordId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The transaction record for which the transaction journal is created. This field is a polymorphic relationship field. **Relationship Name** ReferenceTransactionRecord **Refers To** CreditMemo, CreditMemoInvApplication, CreditMemoLine, CreditMemoLineInvoiceLine, CreditMemoLineTax, Invoice, InvoiceLine, InvoiceLineTax, Payment, PaymentLineInvoice, PaymentLineInvoiceLine, Refund
TransactionType	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The transaction type that's related to the transaction journal. Valid values are: • CreditMemo • CreditMemoInvApplication • CreditMemoLine • CreditMemoLineTax • CreditMemoLineInvoiceLine • DebitMemoLine—Available in API version 65.0 and later. • Invoice • InvoiceLine • InvoiceLineTax • Payment—Available in API version 64.0 and later. • PaymentLineInvoice—Available in API version 64.0 and later. • PaymentLineInvoiceLine—Available in API version 64.0 and later. • Refund—Available in API version 64.0 and later. • RefundLine—Available in API version 64.0 and later.
UniqueIdentifier	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update

Field	Details
	Description An auto-generated identifier for the transaction journal when the usage type is Billing. The identifier is a combination of the reference transaction record ID, the general ledger account assignment rule ID, and other internal-only fields.
`UsageType`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort **Description** The type of usage. Valid value is: • `Billing`

SEE ALSO:

Loyalty Management Developer Guide: TransactionJournal

Salesforce Payments Objects in Billing

Billing provides access to some standard Salesforce Payments objects that you can use to authorize payments, connect to an external payment gateway, group all the payment transactions that are processed for a particular transaction, and store payment methods. You can access these objects with the Revenue Cloud Billing license.

For more information about these Salesforce Payments objects, refer to these resources.

- PaymentAuthAdjustment
- PaymentAuthorization
- PaymentGateway
- PaymentGatewayLog
- PaymentGatewayProvider
- PaymentGroup
- PaymentMethod

Billing Platform Events

Salesforce publishes standard platform events in response to an action that occurred in the org or to report errors. For example, the InvoiceProcessedEvent platform event sends notification to the customer when the billing invoice activity is complete. You can subscribe to a standard platform event by using the subscription mechanism that the event supports.

[BillingScheduleCreatedEvent]
Notifies subscribers when the `/commerce/invoicing/billing-schedules/actions/create` request is complete. This object is available in API version 63.0 and later.

[CreditInvoiceProcessedEvent]
Represents the notification to the customers after the process initiated by the `/commerce/invoicing/invoices/{invoiceId}/actions/credit` request is complete. This object is available in API version 62.0 and later.

[CreditMemoProcessedEvent]
Represents the notification to the customers after the process initiated by the `/commerce/invoicing/credit-memos` request is complete. This object is available in API version 62.0 and later.

[InvoiceProcessedEvent]
Represents the notification to the customers after the process started by the `/commerce/billing/invoices` request is complete. The process groups billing schedules by grouping keys and creates one invoice per grouping key. The `InvoiceProcessedEvent` platform event is a top-level object that contains a list of `InvoiceProcessedDetailEvents` platform events, where each detail event represents an attempt to create one invoice. This object is available in API version 62.0 and later.

[NegInvcLineProcessedEvent]
Represents the notification to the customers when a negative invoice line is converted to a credit memo This object is available in API version 62.0 and later.

[SequenceAssignedEvent]
Represents the notification to customers about the assignment of a sequence to a target record. This process is initiated by the `/sequences/actions/assign` request. This object is available in API version 65.0 and later.

[VoidInvoiceProcessedEvent]
Represents the notification to the customers after the process started by the `/commerce/invoicing/invoices/{invoiceId}/actions/void` request is complete. The request attempts to void an invoice by crediting an invoice and changing its status to `Voided`, which prevents further changes. This object is available in API version 62.0 and later.

BillingScheduleCreatedEvent

Notifies subscribers when the `/commerce/invoicing/billing-schedules/actions/create` request is complete. This object is available in API version 63.0 and later.

Supported Calls

`describeSObjects()`

Supported Subscribers

Subscriber	Supported?
Apex Triggers	✓
Flows	✓
Processes	✓

Subscriber	Supported?
Pub/Sub API	✓
Streaming API (CometD)	✓

Subscription Channel

`/event/billingschedulecreatedevent`

Event Delivery Allocation Enforced

No

Special Access Rules

This object is available when Billing is enabled.

Fields

Field	Details
`BillingScheduleCreatedEventDetail`	**Type** BillSchdCreatedEventDetail on page 2434 **Properties** Nillable **Description** One `BillingScheduleCreatedEventDetail` entry is created for each order item in the `BillingScheduleCreatedEvent` request. One `BillSchdCreatedEventDetail` entry is created for each error that occurred.
`CorrelationIdentifier`	**Type** string **Properties** Nillable **Description** Reserved for future use.
`EventUuid`	**Type** string **Properties** Nillable **Description** A universally unique identifier (UUID) that identifies a platform event message.

Field	Details
`ReplayId`	**Type** string **Properties** Nillable **Description** Represents an ID value that is populated by the system and refers to the position of the event in the event stream. Replay ID values aren’t guaranteed to be contiguous for consecutive events. A subscriber can store a replay ID value and use it on resubscription to retrieve missed events that are within the retention window.
`RequestIdentifier`	**Type** string **Properties** Nillable **Description** ID returned in the `CreateBillingScheduleFromOrderItem` response. Use this ID to identify the `BillingScheduleCreatedEvent` for a specific request.

BillSchdCreatedEventDetail
Contains details about each order item in the `/commerce/invoicing/billing-schedules/actions/create` request and any errors that occurred while processing the request. This object is included in an `BillingScheduleCreatedEvent` message. You can't subscribe to the `BillSchdCreatedEventDetail` platform event directly. This object is available in API version 63.0 and later.

BillSchdCreatedEventDetail

Contains details about each order item in the `/commerce/invoicing/billing-schedules/actions/create` request and any errors that occurred while processing the request. This object is included in an `BillingScheduleCreatedEvent` message. You can't subscribe to the `BillSchdCreatedEventDetail` platform event directly. This object is available in API version 63.0 and later.

Supported Calls

`describeSObjects()`

Special Access Rules

This object is available when Billing is enabled in your org.

Fields

Field	Details
`BillingScheduleId`	**Type** reference **Properties** Nillable **Description** If the request was successful, this field contains the ID of the billing schedule for the order item. This field is a relationship field. **Relationship Name** BillingSchedule **Refers To** BillingSchedule
`ErrorCode`	**Type** string **Properties** Nillable **Description** If the request wasn't successful, this field contains the error code.
`ErrorMessage`	**Type** string **Properties** Nillable **Description** If the request wasn't successful, this field contains the error message.
`EventUuid`	**Type** string **Properties** Nillable **Description** A universally unique identifier (UUID) that identifies a platform event message.
`IsSuccess`	**Type** boolean **Properties** Defaulted on create

Field	Details
	Description Indicates whether the request to create a billing schedule for the order item is successful (`true`) or not (`false`). The default value is `false`.
`OrderItemId`	**Type** reference **Properties** **Description** The ID of the order item used in the `/actions/standardCreateBillingScheduleFromOrderItem` REST request. This field is a relationship field. **Relationship Name** OrderItem **Refers To** OrderItem

CreditInvoiceProcessedEvent

Represents the notification to the customers after the process initiated by the `/commerce/invoicing/invoices/{invoiceId}/actions/credit` request is complete. This object is available in API version 62.0 and later.

Supported Calls

`describeSObjects()`

Supported Subscribers

Subscriber	Supported?
Apex Triggers	✓
Flows	✓
Processes	✓
Pub/Sub API	✓
Streaming API (CometD)	✓

Subscription Channel

`/event/CreditInvoiceProcessedEvent`

Event Delivery Allocation Enforced

No

Special Access Rules

This object is available when Billing is enabled.

Fields

Field	Details
CorrelationIdentifier	**Type** string **Properties** Nillable **Description** Reserved for future use.
CrMemoProcessErrDtlEvents	**Type** CrMemoProcessErrDtlEvent[] **Properties** Nillable **Description** A compilation of error messages and error codes for a failed request. See the `ErrorDetails` field for error messages and error codes.
CreditMemoId	**Type** reference **Properties** Nillable **Description** The credit memo created as the result of a successful request. This field is a relationship field. **Relationship Name** CreditMemo **Refers To** CreditMemo
ErrorDetails	**Type** string **Properties** Nillable

Field	Details
	Description If the request fails, this field shows error messages, error codes, and the ID of the record on which the errors occurred.
`EventUuid`	**Type** string **Properties** Nillable **Description** A universally unique identifier (UUID) that identifies a platform event message.
`InvoiceId`	**Type** reference **Properties** Nillable **Description** The invoice credited as the result of a successful request. This field is a relationship field. **Relationship Name** Invoice **Refers To** Invoice
`IsSuccess`	**Type** boolean **Properties** Defaulted on create **Description** Required. Indicates whether the request was successful (`true`) or not (`false`). The default value is `false`.
`ReplayId`	**Type** string **Properties** Nillable **Description** An ID value that's populated by the system and refers to the position of the event in the event stream. Replay ID values aren't guaranteed to be contiguous for consecutive events. A subscriber can store a replay ID value and use it on resubscription to retrieve missed events that are within the retention window.

Field	Details
RequestIdentifier	**Type** string **Properties** Nillable **Description** The unique ID returned in the response. Use this ID to identify the event for a specific request.

[CrMemoProcessErrDtlEvent](#)

Represents the information about errors that occurred while creating or applying a credit memo as part of a request. This object is included in a `CreditInvoiceProcessedEvent`, `CreditMemoProcessedEvent`, `NegInvcLineProcessedEvent`, or `VoidInvoiceProcessedEvent` platform event message. You can't subscribe to `CrMemoProcessErrDtlEvent` platform event directly. This object is available in API version 62.0 and later.

CrMemoProcessErrDtlEvent

Represents the information about errors that occurred while creating or applying a credit memo as part of a request. This object is included in a `CreditInvoiceProcessedEvent`, `CreditMemoProcessedEvent`, `NegInvcLineProcessedEvent`, or `VoidInvoiceProcessedEvent` platform event message. You can't subscribe to `CrMemoProcessErrDtlEvent` platform event directly. This object is available in API version 62.0 and later.

Supported Calls

`describeSObjects()`

Special Access Rules

This object is available when Billing is enabled.

Fields

Field	Details
ErrorCode	**Type** string **Properties** Nillable **Description** Reference code for the type of error that occurred.
ErrorMessage	**Type** string **Properties** Nillable

Field	Details
	Description Information about the error that occurred during processing.
`ErrorSourceId`	**Type** reference **Properties** Nillable **Description** The ID of the record on which the error occurred during the credit memo creation process and the application process. This field is a polymorphic relationship field. **Relationship Name** ErrorSource **Refers To** CreditMemo, CreditMemoLine, Invoice, InvoiceLine
`EventUuid`	**Type** string **Properties** Nillable **Description** A universally unique identifier (UUID) that identifies a platform event message.

CreditMemoProcessedEvent

Represents the notification to the customers after the process initiated by the `/commerce/invoicing/credit-memos` request is complete. This object is available in API version 62.0 and later.

Supported Calls

`describeSObjects()`

Supported Subscribers

Subscriber	Supported?
Apex Triggers	✓
Flows	✓
Processes	✓
Pub/Sub API	✓

Subscriber	Supported?
Streaming API (CometD)	✓

Subscription Channel

`/event/CreditMemoProcessedEvent`

Event Delivery Allocation Enforced

No

Special Access Rules

This object is available when Billing is enabled.

Fields

Field	Details
`CorrelationIdentifier`	**Type** string **Properties** Nillable **Description** Reserved for future use.
`CrMemoProcessErrDtlEvents`	**Type** CrMemoProcessErrDtlEvent[] **Properties** Nillable **Description** A compilation of error messages and error codes for a failed request. See the `ErrorDetails` field for error messages and error codes.
`CreditMemoId`	**Type** reference **Properties** Nillable **Description** The credit memo created as the result of a successful request. This field is a relationship field. **Relationship Name** CreditMemo

Field	Details
	Refers To CreditMemo
`ErrorDetails`	**Type** string **Properties** Nillable **Description** If the request fails, this field shows error messages, error codes, and the ID of the record on which the errors occurred.
`EventUuid`	**Type** string **Properties** Nillable **Description** A universally unique identifier (UUID) that identifies a platform event message.
`IsSuccess`	**Type** boolean **Properties** Defaulted on create **Description** Required. Indicates whether the request was successful (`true`) or not (`false`). The default value is `false`.
`ReplayId`	**Type** string **Properties** Nillable **Description** Represents an ID value that's populated by the system and refers to the position of the event in the event stream. Replay ID values aren't guaranteed to be contiguous for consecutive events. A subscriber can store a replay ID value and use it on resubscription to retrieve missed events that are within the retention window.
`RequestIdentifier`	**Type** string **Properties** Nillable **Description** The unique ID returned in the response. Use this ID to identify the event for a specific request.

InvoiceProcessedEvent

Represents the notification to the customers after the process started by the `/commerce/billing/invoices` request is complete. The process groups billing schedules by grouping keys and creates one invoice per grouping key. The `InvoiceProcessedEvent` platform event is a top-level object that contains a list of `InvoiceProcessedDetailEvents` platform events, where each detail event represents an attempt to create one invoice. This object is available in API version 62.0 and later.

Supported Calls

`describeSObjects()`

Supported Subscribers

Subscriber	Supported?
Apex Triggers	✓
Flows	✓
Processes	✓
Pub/Sub API	✓
Streaming API (CometD)	✓

Subscription Channel

`/event/InvoiceProcessedEvent`

Event Delivery Allocation Enforced

No

Special Access Rules

This object is available when Billing is enabled.

Fields

Field	Details
`CorrelationIdentifier`	**Type** string **Properties** Nillable **Description** Reserved for future use.

Field	Details
EventUuid	**Type** string **Properties** Nillable **Description** A universally unique identifier (UUID) that identifies a platform event message.
InvoiceErrorDetailEvent	**Type** InvoiceErrorDetailEvent[] **Properties** Nillable **Description** Information about errors that occurred during processing.
InvoiceProcessedDetailEvents	**Type** InvoiceProcessedDetailEvent[] **Properties** Nillable **Description** A list of `InvoiceProcessedDetailEvent` records. Each record contains information about an attempt to create an invoice from one or more billing schedules that share a grouping key.
IsSuccess	**Type** boolean **Properties** Defaulted on create **Description** Required. Indicates whether the Create Order from Invoice action was successful (`true`) or not (`false`). The default value is `false`.
ReplayId	**Type** string **Properties** Nillable **Description** An identification (ID) value that's populated by the system and refers to the position of the event in the event stream. Replay ID values aren't guaranteed to be contiguous for consecutive events. A user can store a replay ID value and use it on resubscription to retrieve missed events that are within the retention window.

Field	Details
`RequestIdentifier`	**Type** string **Properties** Nillable **Description** A unique ID returned in the `/commerce/billing/invoices` response. Use this ID to identify the event for a specific request.

InvoiceErrorDetailEvent

Represents information about the errors that occurred during the processing of a `/commerce/billing/invoices` request. This object is included in an `InvoiceProcessedEvent` platform event message. You can't subscribe to `InvoiceProcessedEvent` platform event directly. This object is available in API version 62.0 and later.

InvoiceProcessedDetailEvent

Represents the notification to customers regarding the results of an attempt to create an invoice from billing schedules as part of `/commerce/billing/invoices` request. The `InvoiceProcessedDetailEvent` platform event contains the results of an attempt to create an invoice from one or more billing schedules that share a grouping key. Each `InvoiceProcessedDetailEvent`platform event for an action is grouped within the parent `InvoiceProcessedDetailEvent` platform event. This object is available in API version 62.0 and later.

InvoiceErrorDetailEvent

Represents information about the errors that occurred during the processing of a `/commerce/billing/invoices` request. This object is included in an `InvoiceProcessedEvent` platform event message. You can't subscribe to `InvoiceProcessedEvent` platform event directly. This object is available in API version 62.0 and later.

Supported Calls

`describeSObjects()`

Special Access Rules

This object is available when Billing is enabled.

Fields

Field	Details
`ErrorCode`	**Type** string **Properties** None **Description** Reference code for the type of error that occurred.

Field	Details
`ErrorMessage`	**Type** string **Properties** None **Description** Information about the error that occurred during processing.
`ErrorSourceId`	**Type** reference **Properties** Nillable **Description** The ID of the record where the error occurred. This record can be an invoice or a billing schedule. This field is a polymorphic relationship field. **Relationship Name** ErrorSource **Refers To** BillingSchedule, Invoice
`EventUuid`	**Type** string **Properties** Nillable **Description** A universally unique identifier (UUID) that identifies a platform event message.

InvoiceProcessedDetailEvent

Represents the notification to customers regarding the results of an attempt to create an invoice from billing schedules as part of `/commerce/billing/invoices` request. The `InvoiceProcessedDetailEvent` platform event contains the results of an attempt to create an invoice from one or more billing schedules that share a grouping key. Each `InvoiceProcessedDetailEvent`platform event for an action is grouped within the parent `InvoiceProcessedDetailEvent` platform event. This object is available in API version 62.0 and later.

Supported Calls

`describeSObjects()`

Supported Subscribers

Subscriber	Supported?
Apex Triggers	✓
Flows	✓
Processes	✓
Pub/Sub API	✓
Streaming API (CometD)	✓

Subscription Channel

`/event/InvoiceProcessedEvent`

Event Delivery Allocation Enforced

No

Special Access Rules

This object is available when Billing is enabled.

Fields

Field	Details
`EventUuid`	**Type** string **Properties** Nillable **Description** A universally unique identifier (UUID) that identifies a platform event message.
`InvoiceErrorDetailEvents`	**Type** InvoiceErrorDetailEvent[] **Properties** Nillable **Description** A list of errors that occurred while attempting to create the invoice.
`InvoiceId`	**Type** reference **Properties** Nillable

Field	Details
	Description The ID of the new invoice. This field is a relationship field. **Relationship Name** Invoice **Refers To** Invoice
`InvoiceStatus`	**Type** string **Properties** Nillable **Description** The value of the `Status` field on the invoice.
`IsSuccess`	**Type** boolean **Properties** Defaulted on create **Description** Required. Indicates whether the invoice creation attempt was successful (`true`) or not (`false`). The default value is `false`.

NegInvcLineProcessedEvent

Represents the notification to the customers when a negative invoice line is converted to a credit memo This object is available in API version 62.0 and later.

Supported Calls

`describeSObjects()`

Supported Subscribers

Subscriber	Supported?
Apex Triggers	✓
Flows	✓
Processes	✓

Subscriber	Supported?
Pub/Sub API	✓
Streaming API (CometD)	✓

Subscription Channel

`/event/NegInvcLineProcessedEvent`

Event Delivery Allocation Enforced

No

Special Access Rules

This object is available when Billing is enabled.

Fields

Field	Details
`CorrelationIdentifier`	**Type** string **Properties** Nillable **Description** Reserved for future use.
`CrMemoProcessErrDtlEvents`	**Type** CrMemoProcessErrDtlEvent **Properties** Nillable **Description** A compilation of error messages and error codes for a failed request. See the `ErrorDetails` field for error messages and error codes.
`CreditMemoId`	**Type** reference **Properties** Nillable **Description** The ID of the credit memo created as a result of the successful conversion of a negative invoice line. This field is a relationship field.

Field	Details
	Relationship Name CreditMemo **Refers To** CreditMemo
`ErrorDetails`	**Type** string **Properties** Nillable **Description** If the request fails, this field shows error messages, error codes, and the ID of the record on which the errors occurred.
`EventUuid`	**Type** string **Properties** Nillable **Description** A universally unique identifier (UUID) that identifies a platform event message.
`InvoiceId`	**Type** reference **Properties** Nillable **Description** The ID of the invoice that this event is in reference to. This field is a relationship field. **Relationship Name** Invoice **Refers To** Invoice
`IsAutomatedNegativeInvoiceLineConversion`	**Type** boolean **Properties** Defaulted on create **Description** Indicates whether this event is generated either by an automated process to convert negative invoice lines to credit memos (`true`) or by a manual process (`false`). The default value is `false`.

Field	Details
IsSuccess	**Type** boolean **Properties** Defaulted on create **Description** Required. Indicates whether the negative invoice lines were converted successfully to credit memos (`true`) or not (`false`). The default value is `false`.
ReplayId	**Type** string **Properties** Nillable **Description** Represents an ID value that's populated by the system and refers to the position of the event in the event stream. Replay ID values aren't guaranteed to be contiguous for consecutive events. A subscriber can store a replay ID value and use it on resubscription to retrieve missed events that are within the retention window.
RequestIdentifier	**Type** string **Properties** Nillable **Description** The unique identifier of the request. This field is always empty.

SequenceAssignedEvent

Represents the notification to customers about the assignment of a sequence to a target record. This process is initiated by the `/sequences/actions/assign` request. This object is available in API version 65.0 and later.

Supported Calls

`describeSObjects()`

Supported Subscribers

Subscriber	Supported?
Apex Triggers	✓
Flows	✓

Subscriber	Supported?
Processes	
Pub/Sub API	
Streaming API (CometD)	

Subscription Channel

`/event/SequenceAssignedEvent`

Event Delivery Allocation Enforced

No

Special Access Rules

This object is available when Billing and Sequential Numbering is enabled.

Fields

Field	Details
`EventUuid`	**Type** string **Properties** Nillable **Description** A universally unique identifier (UUID) that identifies a platform event message.
`ReplayId`	**Type** string **Properties** Nillable **Description** Represents an ID value that's populated by the system and refers to the position of the event in the event stream. Replay ID values aren't guaranteed to be contiguous for consecutive events. A subscriber can store a replay ID value and use it on resubscription to retrieve missed events that are within the retention window.
`SequenceAssignmentDateTime`	**Type** dateTime **Properties** Nillable

Field	Details
	Description The date and time when the sequence pattern value was assigned to the target record.
`SequencePatternValue`	**Type** string **Properties** Nillable **Description** The complete sequence value that's assigned to the target record.
`SequencePolicyIdentifier`	**Type** string **Properties** Nillable **Description** The ID of the sequence policy that's related to the event.
`SequenceValue`	**Type** string **Properties** Nillable **Description** The sequence value that's assigned to the target record.
`TargetObjectName`	**Type** string **Properties** Nillable **Description** The name of the object to which the sequence policy is applicable.
`TargetRecordIdentifier`	**Type** string **Properties** Nillable **Description** The ID of the target record to which the sequence policy is applied.

VoidInvoiceProcessedEvent

Represents the notification to the customers after the process started by the `/commerce/invoicing/invoices/{invoiceId}/actions/void` request is complete. The request attempts to void an invoice by crediting an invoice and changing its status to `Voided`, which prevents further changes. This object is available in API version 62.0 and later.

Supported Calls

`describeSObjects()`

Supported Subscribers

Subscriber	Supported?
Apex Triggers	✓
Flows	✓
Processes	✓
Pub/Sub API	✓
Streaming API (CometD)	✓

Subscription Channel

`/event/VoidInvoiceProcessedEvent`

Event Delivery Allocation Enforced

No

Special Access Rules

This object is available when Billing is enabled.

Fields

Field	Details
`CorrelationIdentifier`	**Type** string **Properties** Nillable **Description** Reserved for future use.

Field	Details
CrMemoProcessErrDtlEvents	**Type** CrMemoProcessErrDtlEvent[] **Properties** Nillable **Description** A compilation of error messages and error codes for a failed request. See the `ErrorDetails` field for error messages and error codes.
CreditMemoId	**Type** reference **Properties** Nillable **Description** The ID of the credit memo created to void the invoice as the result of a successful request. This field is a relationship field. **Relationship Name** CreditMemo **Refers To** CreditMemo
ErrorDetails	**Type** string **Properties** Nillable **Description** If the request fails, this field shows error messages, error codes, and the ID of the record on which the errors occurred.
EventUuid	**Type** string **Properties** Nillable **Description** A universally unique identifier (UUID) that identifies a platform event message.
InvoiceId	**Type** reference **Properties** Nillable **Description** The invoice that was voided as the result of a successful request.

Field	Details
	This field is a relationship field. **Relationship Name** Invoice **Refers To** Invoice
`IsSuccess`	**Type** boolean **Properties** Defaulted on create **Description** Required. Indicates whether the request was successful (`true`) or not (`false`). The default value is `false`.
`ReplayId`	**Type** string **Properties** Nillable **Description** Represents an ID value that's populated by the system and refers to the position of the event in the event stream. Replay ID values aren't guaranteed to be contiguous for consecutive events. A subscriber can store a replay ID value and use it on resubscription to retrieve missed events that are within the retention window.
`RequestIdentifier`	**Type** string **Properties** Nillable **Description** The unique ID returned in the `/commerce/billing/invoices/{invoiceId}/actions/void` response. Use this ID to identify the event for a specific request.

Billing Standard Invocable Actions

Use standard invocable actions to automate processes such as credit application, billing schedules creation, and invoice management.

Commerce Payments Invocable Actions

This table provides a list of the invocable actions of Commerce Payments, which can be used for Billing.

Resource	Description
/services/data/v**67.0**/actions/standard/applyPayment (POST)	Apply a payment record to an invoice header by creating a PaymentLineInvoice record with Applied type.
/services/data/v**67.0**/actions/standard/paymentSale (POST)	Capture a payment without any prior authorization, and create a payment record.

Billing provides these invocable actions.

Apply Credit Action
Apply a credit memo or credit memo line to an invoice or invoice line, respectively.

Apply Payments and Credits by Rules Action
Apply payments and credits to posted invoices by adhering to the specified rules.

Automate Refund Action
Initiate refund orchestration for a credit memo generated from a subscription cancellation or negative amendment.

Create Billing Schedules From Billing Transaction Action
Create one or more billing schedules for a specified billing transaction ID.

Create Standalone Billing Schedules Action
Creates billing schedules for internal or external transaction records by calling the Create Standalone Billing Schedules API.

Extend Invoice Due Date Action
Update the due date on an invoice to accommodate payment extensions or resolve billing disputes.

Generate Account Statement
Generates a comprehensive account statement for a specified account with transaction history and balance information.

Generate Invoice Documents Action
Asynchronously generate PDF documents for the invoices associated with an invoice batch run record that are in the `Draft` or `Posted` status.

Issue Credit Memo Action
Issue credit memos for disputed invoices to resolve billing disputes.

Post Draft Credit Memo Action
Post a draft credit memo to a credit memo record for review and approval.

Post Draft Invoice Action
Update the status of an invoice from `Draft` to `Posted` for a credit memo application.

Post Draft Invoice Batch Run Action
Update the status of a batch of invoices from `Draft` to `Posted` for a credit memo application.

Recover Billing Schedules Action
Recover one or more billing schedules in the `Error` or `Processing` status.

Send Dunning Email Action

Run an orchestration that sends dunning process emails for collection plans to recover overdue revenue and notify customers about amounts still due.

Suspend Billing Action

Suspend or resume the billing of an account to handle billing disputes.

Update Bill To Contact Action

Update the Bill to Contact detail on an invoice to ensure accurate billing communication and routing.

Unapply Credit Action

Unapply a credit memo or credit memo line from an invoice or invoice line, respectively.

Unapply Payment Action

Unapplies a payment that's already been applied to an invoice or invoice line by crediting the amount back to the payment and the invoice or invoice line.

Void Posted Credit Memo Action

Invoke the Void Posted Credit Memo API by providing a credit memo ID.

Write Off Invoices Action

Write off partially paid or unpaid invoices to manage pending debts and to maintain accurate financial records. This action calls the Posted Invoice List Write-Off (POST) API.

SEE ALSO:

Actions Developer Guide: Overview

REST API Developer Guide: Invocable Actions Standard

Apply Credit Action

Apply a credit memo or credit memo line to an invoice or invoice line, respectively.

This action credits the amount of the credit memo or credit memo line to the corresponding invoice or invoice line, reducing both their balances.

This action is available in API version 62.0 and later.

Special Access Rules

The Apply Credit action is available in Enterprise, Developer, and Unlimited Editions where Billing is enabled. To use this action, you need the Credit Memo Operations User permission set.

Supported REST HTTP Methods

URI

`/services/data/v67.0/actions/standard/applyCredit`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer token`

Inputs

Input	Details
`appliedCreditAmount`	**Type** double **Description** Required. Credit amount applied to an invoice or invoice line.
`creditSourceRecordId`	**Type** string **Description** Required. ID of the credit memo or credit memo line that's applied to an invoice or invoice line.
`creditTargetRecordId`	**Type** string **Description** Required. ID of the invoice or invoice line record that the credit is applied to.
`description`	**Type** string **Description** Additional details about the credit memo or credit memo line to be applied to an invoice or invoice line.
`effectiveDate`	**Type** string **Description** Date to use for applying the credit memo to an invoice or invoice line.

Outputs

Output	Details
`recordId`	**Type** string **Description** ID of the credit memo invoice application or credit memo line invoice line record of type `Applied` that the action created.

Example

POST

This example shows a sample request for the Apply Credit action.

```
{
  "inputs": [
    {
      "appliedCreditAmount": 20,
      "creditTargetRecordId": "3ttDU00000000iZYAQ",
      "creditSourceRecordId": "50gDU000000007NYAQ",
      "description": "Applied credit memo to an invoice",
      "effectiveDate": "2024-08-27"
    }
  ]
}
```

This example shows a sample response for the Apply Credit action.

```
{
  "actionName": "applyCredit",
  "errors": null,
  "invocationId": null,
  "isSuccess": true,
  "outputValues": {
    "recordId": "4sFDU00000000652AA"
  },
  "sortOrder": -1,
  "version": 1
}
```

Apply Payments and Credits by Rules Action

Apply payments and credits to posted invoices by adhering to the specified rules.

This action uses predefined logic to allocate payments and credits, reducing any manual intervention and errors.

This action is available in API version 66.0 and later.

Supported REST HTTP Methods

URI

`/services/data/v67.0/actions/standard/applyPaymentsAndCreditsByRules`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer token`

Inputs

Input	Details
`accountId`	**Type** reference **Description** Required. The ID of the account to perform the settlement of payments and credits against invoices in adherence with the applied rules.
`targetDate`	**Type** date **Description** Optional. The date used to select invoices and invoice lines with a posted date equal to or later than the target date to apply payments and credits.

Outputs

Output	Details
`rulesApplicationResponse`	**Type** Apex-defined **Description** The Apex RulesApplicationResponse record that contains the results of rules application including the list of applied rules, application details, and any errors.

Example

POST

Here's a sample request for the Apply Payments and Credits by Rules action.


```
{
  "inputs": [
    {
      "accountId": "001AAC0001NmajhYBB",
      "targetDate": "2025-08-27"
    }
  ]
}
```


Here's a sample success response for the Apply Payments and Credits by Rules action.


```
[
  {
    "actionName": "applyPaymentsAndCreditsByRules",
    "errors": null,
    "invocationId": null,
    "isSuccess": true,
    "outcome": null,
    "outputValues": {
      "rulesApplicationResponse": {
        "rulesApplicationSummary": {
          "totalPaymentApplications": 1,
          "totalCreditMemoApplications": 1,
          "fetchedPaymentsCount": 1,
          "fetchedCreditMemosCount": 1,
          "areAllInvoicesConsidered": true
        },
        "isSuccess": true,
        "errors": null,
        "appliedRules": [
          "Match Balance",
          "Prioritize Highest Balance Invoices"
        ]
      }
    },
    "sortOrder": -1,
    "version": 1
  }
]
```


Here's a sample error response for the Apply Payments and Credits by Rules action.


```
[
  {
    "actionName": "applyPaymentsAndCreditsByRules",
    "errors": null,
    "invocationId": null,
    "isSuccess": true,
    "outcome": null,
    "outputValues": {
```

```
      "rulesApplicationResponse": {
        "rulesApplicationSummary": null,
        "isSuccess": false,
        "errors": [
          {
            "message": "We couldn't find eligible invoices to apply rules-based credits
 and payments. ",
            "errorCode": null
          }
        ],
        "appliedRules": null
      }
    },
    "sortOrder": -1,
    "version": 1
  }
]
```


Automate Refund Action

Initiate refund orchestration for a credit memo generated from a subscription cancellation or negative amendment.

When you cancel a subscription or make a change that lowers the price (a negative amendment), a credit memo is generated. A credit memo is a record that represents the amount to be returned. Use this action to process the refund for that credit memo.

During an invoice batch run, Revenue Cloud Billing uses these two org preferences to run the refund orchestration.

- Conversion of negative invoice lines to credit memo and credit memo lines—When turned on, this org preference creates a credit memo from the negative invoice lines of a cancellation or a negative amendment, and then emits a credit memo event.
- Issue refunds and settle balances — Listens to the credit memo event and starts the refund orchestration for the converted credit memo.

Turn off these org preferences to call this action directly and choose which credit memo receives a refund.

In both modes, the orchestration runs these steps in order.

- Remove payments from the original invoice. Partial removal isn't supported, so each selected payment is removed in full. For example, if an invoice has two payments P1 ($20) and P2 ($15) and the refund amount is $30, the orchestration removes both P1 and P2 in full for a total of $35.
- Apply back any extra amount to the invoice. In the example above, $35 was removed but only $30 is needed for the refund. The orchestration applies the extra $5 from P2 back to the original invoice.
- Apply the credit memo to the original invoice to record the credit.
- Send a refund request to the payment gateway. The payment gateway is the service that processes the customer's payment method. For example, a credit card processor. In the example above, the orchestration sends two separate refund requests, one for P1 ($20) and one for P2 ($10, the portion not applied back).
- Record each refund against its original payment.

The orchestration selects the fewest payments to unapply so that the total unapplied amount is at least equal to the refund amount.

This action is available in API version 67.0 and later.

For more information about automated refunds in Subscription Management, see Automate a refund (Subscription Management reference).

Special Access Rules

The Automate Refund action is available in Enterprise, Unlimited, and Developer Editions of Revenue Cloud.

Supported REST HTTP Methods

URI

`/services/data/v67.0/actions/standard/automateRefund`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearertoken`

Inputs

Input	Details
`creditMemoId`	**Type** ID **Description** Required. ID of the credit memo to generate a refund.
`invoiceId`	**Type** ID **Description** ID of the invoice that the credit memo converts from. This ID matches the value in the Reference Entity field of the credit memo object. When you omit this input, the action reads the invoice ID from the Reference Entity field. When you provide this input, the value must match the Reference Entity field. If the values don’t match, the action returns a validation error..

Outputs

Output	Details
`statusUrl`	**Type** string **Description** URL to poll for the `RefundAsyncOperation` status. Use this URL to access the tracking record for the credit memo and check the current orchestration run status.

Example

POST

This sample request is for the Automate Refund action.

```
{
  "inputs": [
    {
      "creditMemoId": "3ttxx0000000001AAA",
      "invoiceId": "3ttxx0000000002BBB"
    }
  ]
}
```

This sample response shows a successful invocation with a status URL.

```
[
  {
    "actionName": "automateRefund",
    "errors": null,
    "invocationId": null,
    "isSuccess": true,
    "outcome": null,
    "outputValues": {
      "statusUrl": "/services/data/v67.0/async-operations/00Dxx0000000000EAA/status"
    },
    "sortOrder": -1,
    "version": 1
  }
]
```

This sample shows an error response for an invalid credit memo ID.

```
[
  {
    "actionName": "automateRefund",
    "errors": [
      {
        "statusCode": "INVALID_INPUT",
        "message": "The value \"INVALID\" for the creditMemoId parameter is invalid.",
        "fields": []
      }
    ],
    "invocationId": null,
    "isSuccess": false,
    "outcome": null,
    "outputValues": null,
    "sortOrder": -1,
    "version": 1
  }
]
```

Create Billing Schedules From Billing Transaction Action

Create one or more billing schedules for a specified billing transaction ID.

This action calls the Create Billing Schedules for Orders (POST) API to retrieve the billing transaction items associated with the billing transaction ID. The API generates the corresponding billing schedules for each of the billing transaction items for operations such as transaction modifications, renewals, cancellations, and new sales.

This action is available in API version 62.0 and later.

Special Access Rules

The Create Billing Schedules From Billing Transaction action is available in Enterprise, Unlimited, and Developer Editions where Billing is enabled. To use this action, you need the Create Billing Schedules From Billing Transactions API permission set.

Supported REST HTTP Methods

URI

`/services/data/v67.0/actions/standard/createBillingSchedulesFromBillingTransaction`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer token`

Inputs

Input	Details
`billingTransactionId`	**Type** string **Description** Required. ID of the billing transaction record to create one or more billing schedules for.

Outputs

Output	Details
`requestId`	**Type** string **Description** Universally Unique Identifier (UUID) that's used to track the status of the asynchronous action.
`statusUrl`	**Type** string

Output	Details
	Description URL that's used to check the status of the API request.

Example

POST

This example shows a sample request for the Create Billing Schedules From Billing Transaction action.

```
{
  "inputs": [
    {
      "billingTransactionId": "801xx000003JztvAAC"
    }
  ]
}
```

This example shows a sample response for the Create Billing Schedules From Billing Transaction action.

```
{
  "actionName": "createBillingSchedulesFromBillingTransaction",
  "errors": null,
  "isSuccess": true,
  "outputValues": {
    "requestId": "4sFDU00000000652AA",
   "statusUrl": "/services/data/v62.0/sobjects/AsyncOperationTracker/16Pxx0000004NhAEAU"

  },
  "sortOrder": -1,
  "version": 1
}
```

SEE ALSO:

Revenue Cloud Developer Guide: Context-Aware Billing Schedule API

Create Standalone Billing Schedules Action

Creates billing schedules for internal or external transaction records by calling the Create Standalone Billing Schedules API.

This action is available in API version 64.0 and later.

See Create Standalone Billing Schedules (POST) API to know more about the mandatory and optional tags, sample transaction details, and sample payloads for various types of transactions.

Special Access Rules

This action is available in Enterprise, Unlimited, and Developer Editions with the Revenue Cloud Billing license. To use this action, you need the Billing Operations permission set.

Supported REST HTTP Methods

URI

`/services/data/v67.0/actions/standard/createBillingSchedulesFromTrxn`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer token`

Inputs

Input	Details
`contextDefinitionName`	**Type** string **Description** Required. Name of the context definition that contains the mappings for the transaction record and billing schedules.
`readContextMappingName`	**Type** string **Description** Required. Name of the context mapping with the mapping for the transaction.
`saveContextMappingName`	**Type** string **Description** Required. Name of the context mapping with the mapping for the billing schedule and billing schedule group. The save context mapping is used to save the billing schedule.
`transactionDetails`	**Type** string **Description** Required. A JSON string containing the ID of the transaction record that the billing schedule is created for and other additional transaction details.

Outputs

Output	Details
`requestId`	**Type** string **Description** Unique request identifier that you can use to poll the asynchronous request.
`statusUrl`	**Type** string **Description** Status URL that’s used to track the operation.

Example

POST

Here's a sample request for the Create Billing Schedules From Transaction action.

```
{
  "inputs": [
    {
      "transactionDetails":
[illegible]

      "contextDefinitionName": "StandaloneBillingContext__stdctx",
      "readContextMappingName": "TransactionMapping",
      "saveContextMappingName": "BSGEntitiesMapping"
    }
  ]
}
```

Here's a sample response for the Create Billing Schedules From Transaction action.

```
{
  "actionName": "createBillingSchedulesFromTrxn",
  "errors": null,
  "isSuccess": true,
  "outputValues": {
    "requestId": "16PZ6000000CnKRMA0",
   "statusUrl": "/services/data/v64.0/sobjects/AsyncOperationTracker/16PZ6000000CnKRMA0"

  },
  "sortOrder": -1,
  "version": 1
}
```

Extend Invoice Due Date Action

Update the due date on an invoice to accommodate payment extensions or resolve billing disputes.

Specify the invoice ID and the revised due date to adjust the payment timeline. Use this action to extend or change the due date on an invoice.

This action is available in API version 66.0 and later.

Special Access Rules

The Extend Invoice Due Date Action is available in Enterprise, Developer, and Unlimited Editions where Dispute Management is enabled in Billing.

Supported REST HTTP Methods

URI

`/services/data/v`***`67.0`***`/actions/standard/blngSvcExtendInvoiceDueDate`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer`***`token`***

Inputs

Input	Details
`invoiceId`	**Type** string **Description** ID of the invoice whose due date must be revised.
`revisedDueDateTime`	**Type** date **Description** Date and time that's to be updated as the new due date on the invoice.

Outputs

Output	Details
`additionalInformation`	**Type** string

Output	Details
	Description Any additional information to be included in the response.
`isSuccess`	**Type** boolean **Description** Indicates whether the due date is updated on the invoice record (`true`) or not (`false`).

Example

POST

This sample request is for the Extend Invoice Due Date action.

```
{
  "inputs": [
    {
      "invoiceID": "801xx000003GYexAAG",
      "revisedDueDate": "2025-03-31"
    }
  ]
}
```

This sample response is for the Extend Invoice Due Date action.

```
[
  {
    "actionName": "blngSvcExtendInvoiceDueDate",
    "errors": null,
    "invocationId": null,
    "isSuccess": true,
    "outcome": null,
    "outputValues": {
      "isSuccess": true,
      "additionalInformation": "{\"status\":\"Due date updated successfully\"}"
    },
    "sortOrder": -1,
    "version": 1
  }
]
```

Generate Account Statement

Generates a comprehensive account statement for a specified account with transaction history and balance information.

An account statement includes these details.

- Transaction History—List of all financial transactions, such as invoices, credit memos, debit memos, payments, and refunds, within a specified date range.
- Open Balances—Non-zero balances for invoices and other transactions.

- Account Summary—Aggregated totals including beginning balance, total charges, total payments, and ending balance.
- Multi-currency Support—Transactions across multiple currencies.

This action is available in API version 66.0 and later.

Supported REST HTTP Methods

URI

`/services/data/v67.0/actions/standard/generateAccountStatement`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer token`

Inputs

Input	Details
`accountId`	**Type** string **Description** Required. ID of the account for which to generate the statement of account.
`associatedAccountIds`	**Type** string **Description** List of associated account IDs from hierarchy to include in the statement. You can specify a maximum of 50 accounts.
`correlationId`	**Type** string **Description** Correlation ID that tracks messages related to the request. These messages are logged in Splunk by the different services involved in the request. If this ID isn't specified, a random Universally Unique Identifier (UUID) is created.
`customFields`	**Type** string **Description** JSON string specifying custom fields to include in the statement. As a best practice, we recommend that you limit the number of custom fields up to 30 fields. Valid objects to specify the associated fields are:

Input	Details
	• `Invoice` • `CreditMemo` • `Payment` • `DebitMemo` • `Refund` • `Account` Here's the expected format. `{"Invoice": ["field1", "field2"], "Account": {"base": ["field1"], "associated": ["field2"]}}` Available in API version 67.0 and later.
`documentTemplateId`	**Type** string **Description** Document template ID to use for PDF generation. If you don't specify a value, the default account statement template is used.
`shouldShowOpenBalancesOnly`	**Type** boolean **Description** Indicates whether to show only accounts with non-zero balances (`true`) or complete transaction history within date range (`false`).
`sortBy`	**Type** string **Description** Required. Criteria for sorting transactions. The default and valid value is `Date`.
`sortingOrder`	**Type** string **Description** Sort order for transactions. Valid values are: • `Ascending` • `Descending` The default value is `Desc`.

Input	Details
`startDate`	**Type** date **Description** Required. The date from when the transaction history is to be considered in YYYY MM DD format. This parameter value is required when the `shouldShowOpenBalancesOnly` parameter value is `false`. The date must be within the last 90 days and can't be in the future.
`transactionTypes`	**Type** string **Description** List of transaction types to include in the statement. If you don't specify a value or if it's empty, all transaction types are included. Valid values are: • `All` • `Invoice` • `CreditMemo` • `Payment` • `DebitMemo` • `Refund`

Outputs

Output	Details
`accountId`	**Type** string **Description** ID of the account for which to generate the account statement.
`requestIdentifier`	**Type** string **Description** Unique request identifier for tracking the async operation.
`statusUrl`	**Type** string **Description** URL that tracks the status of the request and retrieves the generated document.

Output	Details
templateId	**Type** string **Description** Document template ID that's used for PDF generation.

Example

POST

This example shows a sample request for the Generate Account Statement action.

```
{
  "inputs": [
    {
      "accountId": "001000000000001AAA",
      "shouldShowOpenBalancesOnly": false,
      "startDate": "2025-01-01",
      "transactionTypes": [
        "All"
      ],
      "sortBy": "Date",
      "sortingOrder": "Descending",
      "associatedAccountIds": [
        "001000000000002AAA",
        "001000000000003AAA"
      ],
      "documentTemplateId": "a0T000000000001AAA",
      "customFields": "{\"Invoice\": [\"TotalAmountWithTax\"], \"Account\": {\"base\":
 [\"Phone\", \"Fax\"], \"associated\": [\"Website\"]}}}"
    }
  ]
}
```

This example shows a sample response for the Generate Account Statement action.

```
{
  "accountId": "001AAC0001O5W9pYBF",
  "errors": [],
  "requestIdentifier": "3267407f-0c96-4b8a-a8e0-bb7a71c91a2f",
  "statusUrl": "16PAAC000008AIM",
  "success": true,
  "templateId": "2dtAAC000002oiIYBQ"
}
```

Generate Invoice Documents Action

Asynchronously generate PDF documents for the invoices associated with an invoice batch run record that are in the `Draft` or `Posted` status.

This action uses the ID of the invoice batch run record to find the draft or posted invoices from the batch and to generate the PDF documents for its invoices that are in the `Draft` or `Posted` status. This action creates a DocGenerationBatchProcess record that contains the Document Generation Process and Invoice Document records for each of the invoices. This action is available in API version 63.0 and later with Revenue Cloud Billing.

Special Access Rules

The Generate Invoice Documents action is available in Enterprise, Unlimited, and Developer Editions where Revenue Cloud Billing is enabled. The org must have Billing Docgen enabled and an active Invoice Document Template to generate PDF documents. To use this action, you need either the Billing Operations User or the Billing Admin permission set, along with the Docgen Designer and Docgen Designer Standard User permission sets.

Supported REST HTTP Methods

URI

`/services/data/v`***`67.0`***`/actions/standard/generateInvoiceDocuments`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer` ***`token`***

Inputs

Input	Details
`invoiceBatchRunId`	**Type** string **Description** Required. ID of the invoice batch run record that created the draft or posted invoices.

Outputs

Output	Details
`requestId`	**Type** string **Description** An alphanumeric string to track the status of the document generation request.
`requestStatus`	**Type** boolean

Output	Details
	Description Indicates whether the operation is successful (`true`) or not (`false`).

Example

POST

This example shows a sample request for the Generate Invoice Documents action.

```
{
  "inputs": [
    {
      "invoiceBatchRunId": "5IRSG000001Az014AC"
    }
  ]
}
```

This example shows a sample response for the Generate Invoice Documents action.

```
{
  "actionName": "generateInvoiceDocuments",
  "errors": null,
  "isSuccess": true,
  "outputValues": {
    "requestId": "4sFDU00000000652AA",
    "requestStatus": true
  }
}
```

Issue Credit Memo Action

Issue credit memos for disputed invoices to resolve billing disputes.

Specify the credit request list to automate the creation of credit memos during the dispute resolution process. The credit request list includes the invoice ID, the total credit amount, an optional description, and a detailed list of specific invoice lines with their corresponding credit amounts.

This action is available in API version 66.0 and later.

Special Access Rules

The Issue Credit Memo Action is available in Enterprise, Developer, and Unlimited Editions where Dispute Management is enabled in Billing.

Supported REST HTTP Methods

URI

`/services/data/v67.0/actions/standard/blngDsptIssueCreditMemo`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer`***`token`***

Inputs

Input	Details
`creditRequestList`	**Type** Apex-defined **Description** List of Apex-defined creditRequestInputRepresentations objects that contains the invoice ID, credit amount, description, and a list of creditLineRequestInputRepresentations objects. The creditLineRequestInputRepresentations objects include the invoice line ID, credit line amount, and a description. See CreditLineRequestInputRepresentations Apex class.

Outputs

Output	Details
`creditResponse`	**Type** Apex-defined **Description** An Apex-defined creditResponseOutputRepresentations object that contains the status and any additional information. See CreditResponseOutputRepresentations Apex class.

Example

POST

This sample request is for the Issue Credit Memo action.

```
{
  "inputs": [
    {
      "creditRequestList": [
        {
          "invoiceId": "801xx000003GYexAAG",
          "creditAmount": 100,
          "description": "Credit for disputed charges",
          "creditLineRequestInputRepresentations": [
```

```
              {
                "invoiceLineId": "801xx000003GYeyAAG",
                "creditLineAmount": 50,
                "description": "Line-level credit"
              }
            ]
          }
        ]
      }
    ]
}
```

This sample response is for the Issue Credit Memo action.

```
[
  {
    "actionName": "blngDsptIssueCreditMemo",
    "errors": null,
    "invocationId": null,
    "isSuccess": true,
    "outcome": null,
    "outputValues": {
      "creditResponse": {
        "success": true,
        "message": "Credit memo issued successfully"
      }
    },
    "sortOrder": -1,
    "version": 1
  }
]
```

Post Draft Credit Memo Action

Post a draft credit memo to a credit memo record for review and approval.

This action is available in API version 65.0 and later.

Supported REST HTTP Methods

URI

`/services/data/v`***`67.0`***`/actions/standard/postDraftCreditMemo`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer`***`token`***

Inputs

Input	Details
`correlationId`	**Type** string **Description** Splunk correlation ID to use to track messages that are related to the request and logged in Splunk by the different services involved in the request. If not specified, the service creates a random Universally Unique Identifier (UUID).
`creditMemoId`	**Type** string **Description** Required. ID of the credit memo record in `Draft` status to be posted.

Outputs

Output	Details
`requestIdentifier`	**Type** string **Description** The UUID that's used to track the status of the asynchronous action.
`statusUrl`	**Type** string **Description** The URL to use to check the status of the request.

Example

POST

This example shows a sample request that contains the ID of the credit memo to be posted.

```
{
  "inputs": [
    {
      "creditMemoId": "50gDU00000001MdYAI"
    }
  ]
}
```

This example shows a sample successful response.


```
[
  {
    "actionName": "postDraftCreditMemo",
    "errors": [],
    "invocationId": null,
    "isSuccess": true,
    "outcome": null,
    "outputValues": {
      "statusUrl":
"/services/data/v67.0/sobjects/AsyncOperationTracker/16PDU000000A4nw2AC",
      "requestIdentifier": "d3a9d9ce-2a83-4a08-bcf3-df0348a0008c"
    },
    "sortOrder": -1,
    "version": 1
  }
]
```


Post Draft Invoice Action

Update the status of an invoice from `Draft` to `Posted` for a credit memo application.

This action uses the ID of the draft invoice and triggers an asynchronous process to post the invoice. This action is available in API version 62.0 and later.

Special Access Rules

The Post Draft Invoice action is available in Enterprise, Unlimited, and Developer Editions where Billing is enabled. To use this action, you need the Billing Operations User permission set.

Supported REST HTTP Methods

URI
`/services/data/v67.0/actions/standard/postDraftInvoice`

Formats
JSON, XML

HTTP Methods
POST

Authentication
`Authorization: Bearer token`

Inputs

Input	Details
correlationId	**Type** string

Input	Details
	Description Splunk correlation ID to track the messages that are related to the request and are logged in Splunk by the different services involved in the request. If the ID isn't specified, the service creates a random Universally Unique Identifier (UUID).
`invoiceId`	**Type** string **Description** Required. ID of the `Draft` invoice to be posted.

Outputs

Output	Details
`requestIdentifier`	**Type** string **Description** UUID that's used to track the status of the asynchronous action.
`statusUrl`	**Type** string **Description** URL that's used to check the status of the API request.

Example

POST

This example shows a sample request for the Post Draft Invoice action.

```
{
  "inputs": [
    {
      "invoiceId": "3ttDU00000000iZYAQ"
    }
  ]
}
```

This example shows a sample response for the Post Draft Invoice action.

```
{
  "actionName": "postDraftInvoice",
  "errors": null,
```

```
  "isSuccess": true,
  "outputValues": {
    "requestIdentifier": "4sFDU00000000652AA",
   "statusUrl": "/services/data/v62.0/sobjects/AsyncOperationTracker/16Pxx0000004NhAEAU"

  }
}
```


Post Draft Invoice Batch Run Action

Update the status of a batch of invoices from `Draft` to `Posted` for a credit memo application.

This action uses the ID of the invoice batch run record to find draft invoices from the batch and to post the draft invoices to an invoice record. This action is available in API version 62.0 and later.

Special Access Rules

The Post Draft Invoice Batch Run action is available in Enterprise, Unlimited, and Developer Editions where Billing is enabled. To use this action, you need the Billing Operations User permission set.

Supported REST HTTP Methods

URI

`/services/data/v`***`67.0`***`/actions/standard/postDraftInvoiceBatchRun`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer` ***`token`***

Inputs

Input	Details
`invoiceBatchRunId`	**Type** string **Description** Required. ID of the invoice batch run record that created the draft invoices.

Outputs

Output	Details
`invBatchDraftToPostedRunId`	**Type** string **Description** ID of the record that's created to track the batch process of posting draft invoices. These draft invoices are associated with the parent invoice batch run record.

Example

POST

This example shows a sample request for the Post Draft Invoice Batch Run action.

```
{
  "inputs": [
    {
      "invoiceBatchRunId": "5IRSG000001Az014AC"
    }
  ]
}
```

This example shows a sample response for the Post Draft Invoice Batch Run action.

```
{
  "actionName": "postDraftInvoiceBatchRun",
  "errors": null,
  "isSuccess": true,
  "outputValues": {
    "invBatchDraftToPostedRunId": "4sFDU00000000652AA"
  }
}
```

Recover Billing Schedules Action

Recover one or more billing schedules in the `Error` or `Processing` status.

This action uses the ID of the billing schedule record in the `Error` or `Processing` status to retrieve the latest generated invoice. This action also retrieves any other billing schedules in the `Error` or `Processing` status associated with that invoice.

This action is available in API version 62.0 and later.

Special Access Rules

The Recover Billing Schedules action is available in Enterprise, Unlimited, and Developer Editions where Billing is enabled. To use this action, you need the Manage Errors Using Invoice Error Recovery API permission set.

Supported REST HTTP Methods

URI

`/services/data/v67.0/actions/standard/recoverBillingSchedules`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer token`

Inputs

Input	Details
`billingScheduleId`	**Type** string **Description** Required. ID of the billing schedule record in the `Error` or `Processing` status.

Outputs

Output	Details
`successBillingScheduleIds`	**Type** string **Description** Comma-separated list of IDs associated with the parent billing schedule record in the `Error` or `Processing` status.

Example

POST

This example shows a sample request for the Recover Billing Schedules action.

```
{
  "inputs": [
    {
      "billingScheduleId": "801xx000003JztvAAC"
    }
  ]
}
```

This example shows a sample response for the Recover Billing Schedules action.

```
{
  "actionName": "recoverBillingSchedules",
  "errors": null,
  "isSuccess": true,
  "outputValues": {
    "successBillingScheduleIds": ["4sFDU00000000652AA", 16Pxx0000004NhAEAU]
  },
  "sortOrder": -1,
  "version": 1
}
```

Send Dunning Email Action

Run an orchestration that sends dunning process emails for collection plans to recover overdue revenue and notify customers about amounts still due.

Specify the collection plan for the dunning emails. Optionally, specify the email template by name or by ID. The action triggers an orchestration that sends dunning process emails for a collection plan based on your configured timeline.

This action is available in API version 67.0 and later.

Special Access Rules

The Send Dunning Email action is available in Enterprise, Unlimited, and Developer Editions of Revenue Cloud.

Supported REST HTTP Methods

URI

/services/data/v***67.0***/actions/standard/blngSendDunningEmail

Formats

JSON, XML

HTTP Methods

POST

Authentication

Authorization: Bearer***token***

Inputs

Input	Details
collectionPlanId	**Type** string **Description** Required. ID of the collection plan record for dunning emails.

Input	Details
emailTemplateNameOrId	**Type** string **Description** Email template name or ID to use for the reminder email. If not specified, the default email template is used.

Outputs

Output	Details
isDunningEmailSent	**Type** boolean **Description** Indicates whether the dunning email related to the collection plan was sent (`true`) or not (`false`).
additionalInformation	**Type** string **Description** Additional information to be passed on with the response.

Example

POST

This sample request is for the Send Dunning Email action.

```
{
  "inputs": [
    {
      "collectionPlanId": "0PLxxxxxxxxxxxxxx",
      "emailTemplateNameOrId": "Dunning_Reminder_Template"
    }
  ]
}
```

This sample response is for the Send Dunning Email action.

```
[
  {
    "actionName": "blngSendDunningEmail",
    "errors": null,
    "invocationId": null,
    "isSuccess": true,
    "outcome": null,
```

```
    "outputValues": {
      "isDunningEmailSent": true,
      "additionalInformation": null
    },
    "sortOrder": -1,
    "version": 1
  }
]
```


Suspend Billing Action

Suspend or resume the billing of an account to handle billing disputes.

Specify the account ID and a date when billing must be suspended. Optionally, specify the date when billing must be resumed. The action suspends billing for the account from the suspension date and, if provided, resumes billing on the resumption date.

This action is available in API version 66.0 and later.

Special Access Rules

The Suspend Billing Action is available in Enterprise, Developer, and Unlimited Editions where Dispute Management is enabled in Billing.

Supported REST HTTP Methods

URI

`/services/data/v`***`67.0`***`/actions/standard/blngSvcSuspendBilling`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer`***`token`***

Inputs

Input	Details
`accountId`	**Type** string **Description** ID of the account for which billing must be suspended.
`resumptionDate`	**Type** date **Description** Date when the billing must be resumed.

Input	Details
suspensionDate	**Type** date **Description** Date when the billing must be suspended.

Outputs

Output	Details
additionalInformation	**Type** string **Description** Any additional information to be included in the response.
isSuccess	**Type** boolean **Description** Indicates whether the billing related to the specified account was suspended (`true`) or not (`false`). The default value is `false`.

Example

POST

This sample request is for the Suspend Billing action.

```
{
  "inputs": [
    {
      "accountId": "001xx000003GYexAAG",
      "suspensionDate": "2025-03-01",
      "resumptionDate": "2025-03-15"
    }
  ]
}
```

This sample response is for the Suspend Billing action.

```
[
  {
    "actionName": "blngSvcSuspendBilling",
    "errors": null,
    "invocationId": null,
    "isSuccess": true,
    "outcome": null,
```

```
    "outputValues": {
      "isSuccess": true,
      "additionalInformation": "{\"status\":\"Billing suspended successfully\"}"
    },
    "sortOrder": -1,
    "version": 1
  }
]
```


Update Bill To Contact Action

Update the Bill to Contact detail on an invoice to ensure accurate billing communication and routing.

Specify the invoice ID and the new Bill to Contact detail. Optionally, indicate whether the new contact must be set as the default for future invoices. Use this action to change the contact to be billed on an invoice, for example, when handling a billing dispute.

This action is available in API version 66.0 and later.

Special Access Rules

The Update Bill To Contact Action is available in Enterprise, Developer, and Unlimited Editions where Dispute Management is enabled in Billing.

Supported REST HTTP Methods

URI

`/services/data/v`***`67.0`***`/actions/standard/blngSvcUpdateBillToContact`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer`***`token`***

Inputs

Input	Details
`invoiceId`	**Type** string **Description** ID of the invoice whose Bill to Contact detail must be updated.
`newBillToContactId`	**Type** string **Description** ID of the Bill to Contact record to be updated on the invoice.

Input	Details
`setAsDefault`	**Type** boolean **Description** Indicates whether the new Bill to Contact detail must be set as default for future invoices (`true`) or not (`false`). The default value is `false`.

Outputs

Output	Details
`additionalInformation`	**Type** string **Description** Any additional information to be included in the response.
`isSuccess`	**Type** boolean **Description** Indicates whether the Bill to Contact detail is updated on the invoice record (`true`) or not (`false`).

Example

POST

This sample request is for the Update Bill To Contact action.

```
{
  "inputs": [
    {
      "invoiceID": "3ttxx0000000001AAA",
      "newBillToContactid": "003xx000004XYZPAA4",
      "setAsDefault": true
    }
  ]
}
```

This sample response is for the Update Bill To Contact action.

```
[
  {
    "actionName": "blngSvcUpdateBillToContact",
    "errors": null,
    "invocationId": null,
    "isSuccess": true,
```

```
    "outcome": null,
    "outputValues": {
      "isSuccess": true,
      "additionalInformation": "{\"status\":\"Success\",\"notes\":\"Default billing
contact updated.\"}"
    },
    "sortOrder": -1,
    "version": 1
  }
]
```


Unapply Credit Action

Unapply a credit memo or credit memo line from an invoice or invoice line, respectively.

This action unapplies the credit from an invoice or invoice line. This process involves crediting the applied amount of the credit memo invoice application or credit memo line invoice line record to the related credit memo or credit memo line and invoice or invoice line.

This action is available in API version 62.0 and later.

Special Access Rules

The Apply Credit action is available in Enterprise, Developer, and Unlimited Editions where Billing is enabled. To use this action, you need the Credit Memo Operations User permission set.

Supported REST HTTP Methods

URI
`/services/data/v`***`67.0`***`/actions/standard/unapplyCredit`

Formats
JSON, XML

HTTP Methods
POST

Authentication
`Authorization: Bearer` ***`token`***

Inputs

Input	Details
`effectiveDate`	**Type** string **Description** Date when the credit is unapplied from an invoice or invoice line.
`description`	**Type** string

Input	Details
	Description Additional details about the credit memo invoice application or credit memo line invoice line record of type `Applied` that's processed.
`recordId`	**Type** string **Description** Required. ID of the credit memo invoice application or credit memo line invoice line record of type `Applied` that's processed to unapply the credit.

Outputs

Output	Details
`recordId`	**Type** string **Description** ID of the credit memo invoice application or credit memo line invoice line record of type `Unapplied` that the action created.

Example

POST

This example shows a sample request for the Unapply Credit action.

```
{
  "inputs": [
    {
      "recordId": "4sFDU000000005g2AA",
      "description": "Unapplied credit memo from an invoice",
      "effectiveDate": "2024-08-27"
    }
  ]
}
```

This example shows a sample response for the Unapply Credit action.

```
  {
    "actionName": "unapplyCredit",
    "errors": null,
    "invocationId": null,
    "isSuccess": true,
    "outputValues": {
```

```
      "recordId": "4sFDU00000000602AA"
    },
    "sortOrder": -1,
    "version": 1
  }
```


Unapply Payment Action

Unapplies a payment that's already been applied to an invoice or invoice line by crediting the amount back to the payment and the invoice or invoice line.

If the **Apply Payments to Invoices** setting is enabled, payments can be applied to invoices, which creates the PaymentLineInvoice records of type `Applied`. If the **Apply Payments to Invoices feature** setting is disabled, payments can be applied to invoice lines, which creates the PaymentLineInvoiceLine records of type `Applied`.

When payments are unapplied, the PaymentLineInvoice or PaymentLineInvoiceLine records of type `Applied` has the associated PaymentLineInvoice or PaymentLineInvoiceLine records of type `Unapplied`.

This action is available in API version 64.0 and later.

Special Access Rules

This action is available in Enterprise, Unlimited, and Developer Editions where Billing is enabled. To use this action, you need the Payment Ops permission set.

Supported REST HTTP Methods

URI

`/services/data/v`***`67.0`***`/actions/standard/unapplyPayment`

Formats

JSON, XML

HTTP Methods

POST

Authentication

`Authorization: Bearer`***`token`***

Inputs

Input	Details
`description`	**Type** string **Description** Additional details about the payment line invoice or payment line invoice line that's processed to unapply the payment.

Input	Details
`effectiveDateTime`	**Type** datetime **Description** Date and time to use for unapplying the payment from an invoice or invoice line.
`recordId`	**Type** id **Description** Required ID of the payment line invoice or payment line invoice line record of type `Applied` that's processed to unapply the payment.

Outputs

Output	Details
`recordId`	**Type** id **Description** ID of the payment line invoice or payment line invoice line record of type `Unapplied` that the action created.
`unappliedDateTime`	**Type** datetime **Description** Date and time when the payment was unapplied.

Example

POST

This sample request is for the Unapply Payment action.

```
{
  "inputs": [
    {
      "description": "Unapply payment",
      "effectiveDateTime": "2024-08-11T07:53:15.000Z",
      "recordId": "1PLR000000000dDOAQ"
    }
  ]
}
```

This sample response is for the Unapply Payment action.


```
{
  "actionName": "unapplyPayment",
  "errors": null,
  "isSuccess": true,
  "outputValues": {
    "recordId": "1PLR000000000dDOAQ",
    "unappliedDateTime": "2024-08-11T08:09:01.000Z"
  },
  "sortOrder": -1,
  "version": 1
}
```


Void Posted Credit Memo Action

Invoke the Void Posted Credit Memo API by providing a credit memo ID.

This action is available in API version 66.0 and later.

Supported REST HTTP Methods

URI
`/services/data/v67.0/actions/standard/voidPostedCreditMemo`

Formats
JSON, XML

HTTP Methods
POST

Authentication
`Authorization: Bearertoken`

Inputs

Input	Details
`creditMemoId`	**Type** string **Description** Required. ID of the credit memo record in posted status to be voided.

Outputs

Output	Details
`debitMemoId`	**Type** string

Output	Details
	Description ID of the created debit memo record.
`statusUrl`	**Type** string **Description** URL to use to check the status of the request.

Example

POST

Here's a sample request for the Void Posted Credit Memo action.

```
{
  "inputs": [
    {
      "creditMemoId": "50gSG0000001y5NYAQ"
    }
  ]
}
```

Here's a sample response for the Void Posted Credit Memo action.

```
[
  {
    "actionName": "voidPostedCreditMemo",
    "errors": [],
    "invocationId": null,
    "isSuccess": true,
    "outcome": null,
    "outputValues": {
      "debitMemoId": "4DmSG000001YcIP0A0",
      "statusUrl":
"/services/data/v67.0/sobjects/AsyncOperationTracker/16PSG000001qlyL2AQ"
    },
    "sortOrder": -1,
    "version": 1
  }
]
```

Write Off Invoices Action

Write off partially paid or unpaid invoices to manage pending debts and to maintain accurate financial records. This action calls the Posted Invoice List Write-Off (POST) API.

This action is available in API version 64.0 and later.

Special Access Rules

The Write Off Invoices action is available in Enterprise, Developer, and Unlimited Editions where Billing is enabled. To use this action, you need the Billing Operations User and Credit Memo Operations User permission sets.

Supported REST HTTP Methods

URI

`/services/data/v67.0/actions/standard/writeOffInvoices`

Formats

JSON, XML

HTTP Methods

GET

Authentication

`Authorization: Bearertoken`

Notes

You can also call the associated Connect REST API endpoint or InvoiceWriteOff Apex methods. See Posted Invoice List Write-Off (POST) API or InvoiceWriteOff Namespace.

Inputs

Input	Details
`writeOffInvoiceInputList`	**Type** Apex-defined **Description** Required. A collection of Apex input records that contain details about the invoices to be written off. See InvoiceWriteOff Namespace for the list of input parameters.

Outputs

Output	Details
`writeOffInvoiceResponseList`	**Type** Apex-defined **Description** A collection of Apex output records that contain details about the invoices that were written off. See InvoiceWriteOff Namespace for the list of output parameters.

Example

GET

This sample request is for the Write Off Invoices action.

```
{
  "inputs": [
    {
      "apexClass": "InvoiceWriteOff__WriteOffInvoiceInputList",
      "bytelength": 0,
      "configuration": false,
      "defaultValue": null,
     "description": "A collection of Apex WriteOffInvoiceInputList records that contain
 details about the invoices to be written-off.",
      "label": "WriteOffInvoiceInputList",
      "maxOccurs": 1,
      "name": "writeOffInvoiceInputList",
      "picklistValues": null,
      "placeholderText": null,
      "required": true,
      "sObjectType": null,
      "setupReferenceType": null,
      "toolingType": null,
      "type": null
    }
  ]
}
```

This sample response is for the Write Off Invoices action.

```
{
  "outputs": [
    {
      "additionalAttributes": null,
      "apexClass": "InvoiceWriteOff__WriteOffInvoiceResponseList",
     "description": "A collection Apex WriteOffInvoiceResponseList records that contain
 details about the invoices that were written off.",
      "label": "WriteOffInvoiceResponseList",
      "maxOccurs": 1,
      "name": "writeOffInvoiceResponseList",
      "picklistValues": null,
      "sobjectType": null,
      "type": null
    }
  ]
}
```

Billing Business APIs

Use the Billing Business APIs to manage credit application and to handle billing scenarios.

These sections list the available resources.

Credits

Resource	Description
/commerce/invoicing/credit-memos/**creditMemoId**/actions/apply (POST)	Adjust or correct already issued invoices by applying an existing credit memo to an invoice.
/commerce/invoicing/credit-memo-inv-applications/**creditMemoInvApplicationId**/actions/unapply (POST)	Unapply a credit memo from an invoice and return the invoice and the credit memo to their pre-application states.
/commerce/invoicing/credit-memo-lines/**creditMemoLineId**/actions/apply (POST)	Adjust or correct already issued invoices by applying an existing credit memo line to an invoice line.
/commerce/invoicing/credit-memo-line-invoice-line/**creditMemoLineInvoiceLineId**/actions/unapply (POST)	Unapply a credit memo line from an invoice line and return the invoice line and the credit memo line to their pre-application states.
/commerce/invoicing/credit-memos/actions/generate (POST)	Create a credit memo without applying it to an invoice. You can credit the invoice at a later date.
/commerce/invoicing/invoices/**invoiceId**/actions/void (POST)	Void a posted invoice to rebill the customer, if necessary.
/commerce/invoicing/invoices/**invoiceId**/actions/convert-to-credit on page 2559 (POST)	Convert a list of invoice lines with a negative amount into a posted credit memo. This conversion is applicable for a single invoice at a time.
/commerce/invoicing/invoices/**invoiceId**/actions/credit on page 2530 (POST)	Create a credit memo and apply it to an invoice. The credit memo can fully or partially credit the invoice.
/commerce/invoicing/credit/collection/actions/post on page 2563 (POST)	Post a draft credit memo to a credit memo record for review and approval.
/commerce/billing/credit-memos/**creditMemoId**/actions/void on page 2583 (POST)	Void a credit memo in posted state.

Billing Schedules

Resource	Description
/commerce/invoicing/billing-schedules/actions/create (POST)	Generate billing schedules for orders by using context service.
/commerce/invoicing/standalone/billing-schedules/actions/create (POST)	Generate billing schedules from any internal or external transaction by using context service.
/commerce/invoicing/billing-schedules/collection/actions/recover (POST)	Recover the latest generated invoice associated with the billing schedules in the `Error` or `Processing` status.
/commerce/invoicing/actions/suspend-billing (POST)	Suspend billing for billing schedule groups or an account for a predefined period.
/commerce/invoicing/actions/resume-billing (POST)	Resume billing for billing schedule groups or an account that's currently on hold.

Invoices

Resource	Description
/commerce/invoicing/invoices/collection/actions/post (POST)	Update the status of the invoice from Draft to Posted.
/commerce/invoicing/invoice-batch-runs/***invoiceBatchRunId***/actions/draft-to-posted (POST)	Update a batch of invoices from `Draft` to `Posted` status for a credit memo application.
/commerce/invoicing/invoices/collection/actions/preview (POST)	Generate preview invoices, which includes the estimated tax amounts, for a billing transaction for the next two billing periods.
/commerce/invoicing/invoices/collection/actions/ingest (POST)	Ingest or generate an invoice from an internal or external billing transaction data.
/commerce/billing/invoices/invoice-batch-docgen/***invoiceBatchRunId***/actions/***actionName*** (POST)	Asynchronously generate PDF documents for the invoices that are in the `Draft` or `Posted` status and are associated with an invoice batch run record.

Resource	Description
/commerce/billing/invoices/invoice-batch-docgen/***invoiceBatchRunId***/actions/***actionName*** (POST)	Asynchronously regenerate PDF documents for the invoices that are in the `Draft` or `Posted` status and failed in an earlier invoice batch run.
`/commerce/invoicing/invoices/actions/write-off` (POST)	Create credit memos with the total charge amount on the invoice as the write-off amount and close the invoice.
`/commerce/invoicing/invoice-batch-runs/actions/send-email` (POST)	Send emails for the posted invoices of a specified invoice batch run ID.
`/commerce/invoicing/invoices/collection/actions/generate` (POST)	Create an invoice for an account, order, or a list of billing schedules.
`/revenue/billing/transactions/actions/apply` (POST)	Apply payments and credits to an account's invoices based on specified rules defined on the Billing Settings page.
`/revenue/billing/document/actions/generate` (POST)	Generate an invoice document for a record, and update any junction object record.

Invoice Scheduler

Resource	Description
`/commerce/invoicing/invoice-schedulers` (POST)	Create or update an invoice scheduler to automatically generate invoices. Use the criteria and filters of the invoice scheduler to set up the invoice run schedules based on your requirements.
/commerce/invoicing/invoice-batch-runs/***invoiceBatchRunId***/actions/recover (POST)	Recover records associated with a failed invoice run. Recovery is required only when billing schedules remain in the `Processing`, `Void In Progress`, or `Error` status.

Invoice Sequencing

Resource	Description
/connect/sequences/policy (POST)	Create a sequence policy to configure a unique, sequential number for posted invoices or credit memos.
/connect/sequences/policy/***sequencePolicyId*** (PATCH)	Update the settings of a sequence policy that defines how unique, sequential numbers are generated by using specific patterns, values, and filters.
/connect/sequences/actions/assign (POST)	Assign sequence pattern values to objects based on the configured sequence policy.
/connect/sequences/gap-reconciliation (POST)	Restore a missing sequence value identified by using this API in gapless-enabled sequences. This sequence value can be used later in the subsequent sequence policy numbering, ensuring there are no gaps.

Account Statement

Resource	Description
/revenue/billing/accounts/***accountId***/statement (POST)	Generate comprehensive account statement with transaction history and balance information.

Payments

Resource	Description
/commerce/billing/payments/***paymentId***/actions/apply (POST)	Allocate the balance of a payment to reduce the balance of an invoice. The response includes an ID of the payment line invoice or payment line invoice line that represents the

Resource	Description
	payment balance allocated against the invoice.
/commerce/billing/payments/***paymentId***/paymentlines/***paymentLineId***/actions/unapply (POST)	Revert the application of a payment line from an invoice, and return the payment and invoices to their preapplication state. Use this API to correct an input during the payment application process.
/commerce/billing/refunds/***refundId***/actions/apply (POST)	Make a refund transaction against a payment.

Tax Calculation

Resource	Description
/commerce/taxes/actions/calculate (POST)	Calculate tax for a transaction.
/commerce/invoicing/invoices/collection/actions/calculate-estimated-tax (POST)	Calculate estimated tax for invoices with invoice lines that have the `TaxProcessingStatus` as either `Pending` or `Estimated`.

Salesforce Commerce Payments API

Resource	Description
/commerce/payments/payment-methods (POST)	Tokenize a payment method.
/commerce/payments/sales (POST)	Make a payment sale.
/commerce/payments/payments/***paymentId***/refunds (POST)	Create a refund for a payment.
/commerce/payments/authorizations (POST)	Authorize a payment.
/commerce/payments/authorizations/***authorizationId***/reversals (POST)	Reverse an authorized payment.
/commerce/payments/authorizations/***authorizationId***/captures (POST)	Capture an authorized payment.

Billing Business API Limits

Learn about the default limits on the usage of the Billing business APIs.

Resources

Learn more about the available Billing API resources.

Request Bodies

Learn more about the available request bodies of Billing APIs.

Response Bodies

Learn more about the available response bodies of Billing APIs.

SEE ALSO:

Connect REST API Developer Guide: Introduction

Billing Business API Limits

Learn about the default limits on the usage of the Billing business APIs.

Refer to these default API limits and applicable recommendations to scale beyond these default limits.

Limit	Value	Scale Recommendations
The number of billing schedules that can be processed by the Create Invoices By Using Billing Schedules API.	200	Use Invoice Scheduler to scale up to 2000 invoice lines. See Invoice Scheduler or Batch Invoice Scheduler (POST) API.
The number of invoice lines that can be generated by the Create Invoices By Using Billing Schedules API.	200	Use Invoice Scheduler to scale up to 2000 invoice lines. See Invoice Scheduler or Batch Invoice Scheduler (POST) API.
The number of billing schedules that can be processed by the Recover Billing Schedule List API.	200	Use this API to follow the default API limits.
The number of invoices that can be processed by the Apply Credit Memos API.	300	Use this API recursively to apply left-over credits after 300 invoice lines.
The number of invoice lines that can be processed by the Apply Credit Memo Lines API.	300	Use this API iteratively for every 300 invoice lines to support more than 300 invoice lines.
The number of invoices that can be processed by the Create and Apply Credit Memos API.	300	Use this API to follow the default API limits. The limits include the count for both charge and tax lines.
The number of credit memo lines of type Charge that can be created by the Create a Standalone Credit Memo API.	300	Use this API to follow the default API limits. The limits include the count for both charge and tax lines.

EDITIONS

Available in: Lightning Experience

Available in: **Enterprise**, **Unlimited**, and **Developer** Editions

Limit	Value	Scale Recommendations
The number of invoice lines that can be processed by the Convert Negative Invoice Lines to Credits API, which excludes the number of associated invoice line taxes.	300	Use this API to follow the default API limits. The limits include the count for invoice charge lines only.
The number of billing transaction items that can be processed by the Create Billing Schedules for Orders API.	1000	This API supports creation of 1000 order lines as billing schedules.
The number of reference IDs that can be processed by the Suspend Billing API and the Resume Billing API.	200	Use this API to follow the default API limits.
The number of invoice lines that's supported by the Invoice Draft to Posted Status API.	200	Use this API to follow the default API limits.
The number of records that can be processed by the Invoice Ingestion API.	500	Use this API to follow the default API limits. The limits include the count for all invoices, invoice lines, taxes, and address group records.
The number of invoice lines or billing schedules that can be processed by the Invoice Preview API.	200	Use this API to follow the default API limits.
The number of invoice lines that can be processed by the Void a Posted Invoice API.	2000	Use this API to follow the default API limits.
The number of invoices that can be processed by the Posted Invoice List Write-Off API.	300	Use this API to follow the default API limits.
The number of invoice lines that can be generated on an invoice by the Batch Invoice Scheduler (POST) API. The number of records that can be recovered from a failed invoice run by the Batch Invoice Run Recovery (POST) API.	2000	Use this API to support the creation or recovery of up to 2,000 invoice lines on a single invoice.
The number of invoice lines that's supported by the Batch Invoices Draft to Posted Status (POST) API.	2000	Use this API to support 2000 invoice lines with no limit on the number of supported invoices.
The number of invoices that's supported by the Tax Calculation API.	1	Use this API to follow the default API limits.
The number of invoice lines that's supported by the Tax Calculation API.	500	The request can include 500 invoice lines. To avoid limit-related issues, test your TaxEngineAdapter Apex interface's implementation to make sure that it adheres to the Apex limit for total heap size.

Limit	Value	Scale Recommendations
The number of invoice lines that's supported by the Tax Calculation API interface when used with the Invoice Creation API.	200	The request can include 200 invoice lines. To avoid limit-related issues, test your TaxEngineAdapter Apex interface's implementation to make sure that it adheres to the Apex limit for total heap size.
The number of invoice lines that's supported by the Tax Calculation API interface when used with the Invoice Batch Run API.	2000	The request can include 2000 invoice lines. To avoid limit-related issues, test your TaxEngineAdapter Apex interface's implementation to make sure that it adheres to the Apex limit for total heap size.
The number of records that's supported by the Payment Line Apply API.	1	Payment supports application for either an Invoice or InvoiceLine record based on the settlement-level preferences.
The number of records that's supported by the Payment Line Unapply API.	1	Payment supports unapplication for either a PaymentLineInvoice or PaymentLineInvoiceLine record based on the settlement-level preferences.

Resources

Learn more about the available Billing API resources.

Sequence Gap Reconciliation (POST)
: Restore a missing sequence value identified by using this API in gapless-enabled sequences. This sequence value can be used later in the subsequent sequence policy numbering, ensuring there are no gaps.

Sequence Assignment (POST)
: Assign sequence pattern values to objects based on the configured sequence policy.

Batch Invoices Document Generation (POST)
: Asynchronously generate PDF documents for the invoices that are in the `Draft` or `Posted` status and are associated with an invoice batch run record.

Batch Invoices Document Generation Retry (POST)
: Asynchronously regenerate PDF documents for the invoices that are in the `Draft` or `Posted` status and failed in an earlier invoice batch run.

Batch Invoices Draft to Posted Status (POST)
: Update a batch of invoices from `Draft` to `Posted` status for a credit memo application.

Batch Invoice Scheduler (POST, PUT)
: Create or update an invoice scheduler to automatically generate invoices. Use the criteria and filters of the invoice scheduler to set up the invoice run schedules based on your requirements.

Batch Payment Scheduler (POST)
: Create a payment scheduler to automate and process payment runs on a recurring basis.

Billing Arrangement (GET)

Retrieve a billing arrangement and its associated billing arrangement lines.

Billing Schedule Recovery List (POST)

Recover the latest generated invoice associated with the billing schedules in the `Error` or `Processing` status.

Create Billing Schedules for Orders (POST)

Generate billing schedules for orders by using context service.

Create Sequence Policy (POST)

Create a sequence policy to configure a unique, sequential number for posted invoices or credit memos.

Create Standalone Billing Schedules (POST)

Generate billing schedules from any internal or external transaction by using context service.

Create and Apply Credit Memo (POST)

Create a credit memo and apply it to an invoice. The credit memo can fully or partially credit the invoice.

Apply Credit Memo (POST)

Adjust or correct already issued invoices by applying an existing credit memo to an invoice.

Unapply Credit Memo (POST)

Unapply a credit memo from an invoice and return the invoice and the credit memo to their pre-application states.

Apply Credit Memo Line (POST)

Adjust or correct already issued invoices by applying an existing credit memo line to an invoice line.

Unapply Credit Memo Line (POST)

Unapply a credit memo line from an invoice line and return the invoice line and the credit memo line to their pre-application states.

Generate On-Demand Invoice Document (POST)

Generate an invoice document for a record, and update any junction object record.

Generate Account Statement (POST)

Generate comprehensive account statement with transaction history and balance information.

Invoice Creation (POST)

Create an invoice for an account, order, or a list of billing schedules.

Invoice Draft to Posted Status (POST)

Update the status of the invoice from Draft to Posted.

Invoice Ingestion (POST)

Ingest or generate an invoice from an internal or external billing transaction data.

Invoice Estimated Tax Calculation (POST)

Calculate estimated tax for invoices with invoice lines that have the `TaxProcessingStatus` as either `Pending` or `Estimated`.

Invoice Preview (POST)

Generate preview invoices, which includes the estimated tax amounts, for a billing transaction for the next two billing periods.

Invoice Run Recovery (POST)

Recover records associated with a failed invoice run. Recovery is required only when billing schedules remain in the `Processing`, `Void In Progress`, or `Error` status.

Negative Invoice Lines to Credit Conversion (POST)

Convert a list of invoice lines with a negative amount into a posted credit memo. This conversion is applicable for a single invoice at a time.

Payment Line Apply (POST)

Allocate the balance of a payment to reduce the balance of an invoice. The response includes an ID of the payment line invoice or payment line invoice line that represents the payment balance allocated against the invoice.

Payment Line Unapply (POST)

Revert the application of a payment line from an invoice, and return the payment and invoices to their preapplication state. Use this API to correct an input during the payment application process.

Payment Scheduler Update (PATCH)

Activate or deactivate a payment scheduler. You can set the status of a payment scheduler to `Active`, `Canceled`, `Draft`, or `Inactive`.

Post a Draft Memo (POST)

Post a draft credit memo to a credit memo record for review and approval.

Rules Application (POST)

Apply payments and credits to an account's invoices based on specified rules defined on the Billing Settings page.

Posted Invoice List Write-Off (POST)

Create credit memos with the total charge amount on the invoice as the write-off amount and close the invoice.

Void a Posted Invoice (POST)

Void a posted invoice to rebill the customer, if necessary.

Refund Line Apply (POST)

Make a refund transaction against a payment.

Send Emails for Posted Invoices (POST)

Send emails for the posted invoices of a specified invoice batch run ID.

Resume Billing (POST)

Resume billing for billing schedule groups or an account that's currently on hold.

Suspend Billing (POST)

Suspend billing for billing schedule groups or an account for a predefined period.

Standalone Credit Memo (POST)

Create a credit memo without applying it to an invoice. You can credit the invoice at a later date.

Tax Calculation (POST)

Calculate tax for a transaction.

Update Sequence Policy (PATCH)

Update the settings of a sequence policy that defines how unique, sequential numbers are generated by using specific patterns, values, and filters.

Void Posted Credit Memo (POST)

Void a credit memo in posted state.

Sequence Gap Reconciliation (POST)

Restore a missing sequence value identified by using this API in gapless-enabled sequences. This sequence value can be used later in the subsequent sequence policy numbering, ensuring there are no gaps.

Special Access Rules

You need the Billing Admin permission set to use this API.

Resource

```
/connect/sequences/gap-reconciliation
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/sequences/gap-reconciliation
```

Available version

65.0

HTTP methods

POST

Request body for POST

JSON example

This example shows a sample request that specifies the list of sequence policies for gap reconciliation.

```
{
  "sequencePolicyIds": [
    "1vdxx0000000abc",
    "1vdxx0000000def"
  ]
}
```

This example shows a sample request that specifies the target invoice object for gap reconciliation.

```
{
  "targetObjects": [
    "Invoice"
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`sequence PolicyIds`	String[]	List of IDs of the sequence policies.	Required if the `targetObjects` property isn't specified. You must not specify both properties.	65.0
`target Objects`	String[]	List of objects to which the policies are applied. Valid values are: • `Invoice` • `CreditMemo`—Available in API version 66.0 and later.	Required if the `sequencePolicyIds` property isn't specified. You must not specify both properties.	65.0

Response body for POST

Sequence Gap Reconciliation

Sequence Assignment (POST)

Assign sequence pattern values to objects based on the configured sequence policy.

Special Access Rules

You need the Billing Admin permission set to use this API.

Resource

```
/connect/sequences/actions/assign
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/sequences/actions/assign
```

Available version

65.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "targetObjectIds": [
    "3ttxx00000005nhAAA",
    "3ttxx00000006bhAAA"
  ],
  "sequencePolicyId": "1Vdxx0000004CFU"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
sequence PolicyId	String	ID of the sequence policy.	Optional	65.0
shouldPublish Platform Event	Boolean	Indicates whether to publish a platform event when a sequence is assigned to a target record (`true`) or not (`false`).	Optional	65.0
target ObjectIds	String[]	List of records to which the sequence pattern values are assigned.	Required	65.0

Response body for POST

Sequences Assignment

Batch Invoices Document Generation (POST)

Asynchronously generate PDF documents for the invoices that are in the `Draft` or `Posted` status and are associated with an invoice batch run record.

Special Access Rules

This API is available with Revenue Cloud Billing. To use this API, enable Document Generation for Billing. Additionally, you need either the Billing Operations User permission set or the Billing Admin permission set, along with the Docgen Designer permission set and Docgen Designer Standard User permission set.

Resource

```
/commerce/billing/invoices/invoice-batch-docgen/invoiceBatchRunId/actions/actionName
```

- The *invoiceBatchRunId* parameter is the ID of the invoice batch run record that created the `Draft` or `Posted` invoices.
- The *actionName* parameter is the name of the action you want to perform on the specified invoice batch run record.

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/billing/invoices/invoice-batch-docgen/5KPxx0025063GSmSAX/actions/run
```

Available version

63.0

HTTP methods

POST

Response body for POST

Batch Invoice Document Generation

Batch Invoices Document Generation Retry (POST)

Asynchronously regenerate PDF documents for the invoices that are in the `Draft` or `Posted` status and failed in an earlier invoice batch run.

Special Access Rules

This API is available with Revenue Cloud Billing. To use this API, enable Document Generation for Billing. Additionally, you need either the Billing Operations User permission set or the Billing Admin permission set, along with the Docgen Designer permission set and Docgen Designer Standard User permission set.

Resource

```
/commerce/billing/invoices/invoice-batch-docgen/invoiceBatchRunId/actions/actionName
```

- The *invoiceBatchRunId* parameter is the ID of the invoice batch run record that created the `Draft` or `Posted` invoices.
- The *actionName* parameter is the name of the action you want to perform on the specified invoice batch run record. In this case, the action is to retry generating the `Draft` or `Posted` invoices that failed earlier.

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/billing/invoices/invoice-batch-docgen/5IRxx0000004GSmGAM/actions/retry
```

Available version

63.0

HTTP methods

POST

Response body for POST

Batch Invoice Document Generation

Batch Invoices Draft to Posted Status (POST)

Update a batch of invoices from `Draft` to `Posted` status for a credit memo application.

Special Access Rules

To use this API, you need the Billing Operations User permission set.

Resource

```
/commerce/invoicing/invoice-batch-runs/invoiceBatchRunId/actions/draft-to-posted
```

This API posts the draft invoices and changes the status of the invoices from `Draft` to `Posted`.

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/invoicing/invoice-batch-runs/5IRNZ0000000cA94AI/actions/draft-to-posted
```

Available version

62.0

HTTP methods

POST

Path parameter for POST

Name	Type	Description	Required or Optional	Available Version
`invoiceBatchRunId`	String	ID of the invoice batch run record that creates the draft invoices.	Required	62.0

Response body for POST

Invoice Batch Draft To Posted

Batch Invoice Scheduler (POST, PUT)

Create or update an invoice scheduler to automatically generate invoices. Use the criteria and filters of the invoice scheduler to set up the invoice run schedules based on your requirements.

Special Access Rules

You need the Billing Operations User and Data Pipelines Base User permission sets to use this API. See Data Pipelines.

Resource

```
/commerce/invoicing/invoice-schedulers
```

```
/commerce/invoicing/invoice-schedulers/billingBatchSchedulerId
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/invoicing/invoice-schedulers
```

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/invoicing/invoice-schedulers/5BSxx0000004TwGGAU
```

Available version

62.0

HTTP methods

POST, PUT

PUT is supported in version 63.0 and later for invoice schedulers with `Draft` or `Inactive` status.

Request body for POST

JSON example

This example shows a sample request to create an invoice scheduler that generates invoices once.


```
{
  "schedulerName": "InvoiceScheduler",
  "startDate": "2024-05-06",
  "invoiceStatus": "POSTED",
  "preferredTime": "00:45",
  "targetDate": "2024-05-22",
  "invoiceDate": "2024-05-22",
  "frequencyCadence": "Once",
  "frequencyCadenceOptions": {},
  "timezone": "Asia/Kolkata",
  "status": "Active",
  "filterCriteria": [
    {
      "operation": "InList",
      "value": "Batch 2,Batch 3,Batch 4",
      "criteriaSequence": 1,
      "objectName": "BillingSchedule",
      "fieldName": "InvoiceRunMatchingValue"
    },
    {
      "operation": "Equals",
      "value": "001xx000003GZG5AAO",
      "criteriaSequence": 2,
      "objectName": "BillingSchedule",
      "fieldName": "BillingAccount"
    },
    {
      "operation": "Equals",
      "value": "0fwxx0000000001AAA",
      "criteriaSequence": 3,
      "objectName": "BillingScheduleGroup",
      "fieldName": "LegalEntity"
    },
    {
      "operation": "InList",
      "value": "OneTime,Recurring",
      "criteriaSequence": 4,
      "objectName": "BillingSchedule",
      "fieldName": "BillingTermUnit"
    },
    {
      "operation": "Equals",
      "value": "USD",
      "criteriaSequence": 5,
      "objectName": "BillingSchedule",
      "fieldName": "Currency_Iso_code"
```

```
    }
  ]
}
```

This example shows a sample request to create an invoice scheduler that generates invoices daily.


```
{
  "schedulerName": "InvoiceScheduler",
  "startDate": "2024-05-06",
  "endDate": "2026-05-06",
  "invoiceStatus": "POSTED",
  "preferredTime": "00:45",
  "targetDateOffset": 0,
  "invoiceDateOffset": 0,
  "isInvoiceDateFromRunDate": true,
  "frequencyCadence": "Daily",
  "frequencyCadenceOptions": {},
  "timezone": "Asia/Kolkata",
  "status": "Active",
  "filterCriteria": [
    {
      "operation": "InList",
      "value": "Batch 2,Batch 3,Batch 4",
      "criteriaSequence": 1,
      "objectName": "BillingSchedule",
      "fieldName": "InvoiceRunMatchingValue"
    },
    {
      "operation": "Equals",
      "value": "001xx000003GZG5AAO",
      "criteriaSequence": 2,
      "objectName": "BillingSchedule",
      "fieldName": "BillingAccount"
    },
    {
      "operation": "Equals",
      "value": "0fwxx0000000001AAA",
      "criteriaSequence": 3,
      "objectName": "BillingScheduleGroup",
      "fieldName": "LegalEntity"
    },
    {
      "operation": "InList",
      "value": "OneTime,Recurring",
      "criteriaSequence": 4,
      "objectName": "BillingSchedule",
      "fieldName": "BillingTermUnit"
    },
    {
      "operation": "InList",
      "value": "USD",
      "criteriaSequence": 5,
      "objectName": "BillingSchedule",
      "fieldName": "Currency_Iso_code"
    }
```

```
  ]
}
```

This example shows a sample request to create an invoice scheduler that generates invoices weekly.

```
{
  "schedulerName": "InvoiceScheduler",
  "startDate": "2024-05-06",
  "endDate": "2026-05-06",
  "invoiceStatus": "POSTED",
  "preferredTime": "00:45",
  "targetDateOffset": 0,
  "invoiceDateOffset": 0,
  "isInvoiceDateFromRunDate": false,
  "frequencyCadence": "Weekly",
  "frequencyCadenceOptions": {
    "recursOnDay": "Sunday"
  },
  "timezone": "Asia/Kolkata",
  "status": "Active",
  "filterCriteria": [
    {
      "operation": "InList",
      "value": "Batch 2,Batch 3,Batch 4",
      "criteriaSequence": 1,
      "objectName": "BillingSchedule",
      "fieldName": "InvoiceRunMatchingValue"
    },
    {
      "operation": "Equals",
      "value": "001xx000003GZG5AAO",
      "criteriaSequence": 2,
      "objectName": "BillingSchedule",
      "fieldName": "BillingAccount"
    },
    {
      "operation": "Equals",
      "value": "0fwxx0000000001AAA",
      "criteriaSequence": 3,
      "objectName": "BillingScheduleGroup",
      "fieldName": "LegalEntity"
    },
    {
      "operation": "InList",
      "value": "OneTime,Recurring",
      "criteriaSequence": 4,
      "objectName": "BillingSchedule",
      "fieldName": "BillingTermUnit"
    },
    {
      "operation": "Equals",
      "value": "USD",
      "criteriaSequence": 5,
      "objectName": "BillingSchedule",
      "fieldName": "Currency_Iso_code"
```

```
    }
  ]
}
```

This example shows a sample request to create an invoice scheduler that generates invoices monthly on a specific date.

```
{
  "schedulerName": "InvoiceScheduler",
  "startDate": "2024-05-06",
  "endDate": "2026-05-06",
  "invoiceStatus": "POSTED",
  "preferredTime": "00:45",
  "targetDateOffset": 0,
  "invoiceDateOffset": 0,
  "isInvoiceDateFromRunDate": false,
  "frequencyCadence": "Monthly",
  "frequencyCadenceOptions": {
    "recurringSubType": "SpecificDate",
    "recursOnDate": "L-1",
    "shouldExcludeWkendAndHldy": true
  },
  "timezone": "Asia/Kolkata",
  "status": "Active",
  "filterCriteria": [
    {
      "operation": "InList",
      "value": "Batch 2,Batch 3,Batch 4",
      "criteriaSequence": 1,
      "objectName": "BillingSchedule",
      "fieldName": "InvoiceRunMatchingValue"
    },
    {
      "operation": "Equals",
      "value": "001xx000003GZG5AAO",
      "criteriaSequence": 2,
      "objectName": "BillingSchedule",
      "fieldName": "BillingAccount"
    },
    {
      "operation": "Equals",
      "value": "0fwxx0000000001AAA",
      "criteriaSequence": 3,
      "objectName": "BillingScheduleGroup",
      "fieldName": "LegalEntity"
    },
    {
      "operation": "InList",
      "value": "OneTime,Recurring",
      "criteriaSequence": 4,
      "objectName": "BillingSchedule",
      "fieldName": "BillingTermUnit"
    },
    {
      "operation": "Equals",
      "value": "USD",
```

```
      "criteriaSequence": 5,
      "objectName": "BillingSchedule",
      "fieldName": "Currency_Iso_code"
    }
  ]
}
```

This example shows a sample request to create an invoice scheduler that runs immediately.

```
{
    "schedulerName": "InvoiceScheduler",
    "status": "Draft",
    "invoiceStatus": "Posted",
    "frequencyCadenceOptions": {
        "shouldStartRunImmediately": true
    },
    "frequencyCadence": "Once",
    "targetDate": "2024-08-28",
    "invoiceDate": "2024-08-28",
    "filterCriteria": [
        {
            "operation": "InList",
            "value": "Batch 2,Batch 3,Batch 4",
            "criteriaSequence": 1,
            "objectName": "BillingSchedule",
            "fieldName": "InvoiceRunMatchingValue"
        },
        {
            "operation": "Equals",
            "value": "001xx000003GZG5AAO",
            "criteriaSequence": 2,
            "objectName": "BillingSchedule",
            "fieldName": "BillingAccount"
        },
        {
            "operation": "Equals",
            "value": "0fwxx0000000001AAA",
            "criteriaSequence": 3,
            "objectName": "BillingScheduleGroup",
            "fieldName": "LegalEntity"
        },
        {
            "operation": "InList",
            "value": "OneTime,Recurring",
            "criteriaSequence": 4,
            "objectName": "BillingSchedule",
            "fieldName": "BillingTermUnit"
        },
        {
            "operation": "Equals",
            "value": "USD",
            "criteriaSequence": 5,
            "objectName": "BillingSchedule",
            "fieldName": "Currency_Iso_code"
        }
```

```
    ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
endDate	String	End date of the invoice scheduler.	Optional	63.0
filter Criteria	Batch Invoice Filter Criteria Input[]	List of line items of the filter criteria.	Optional	62.0
frequency Cadence	String	Frequency to run the invoice scheduler. Valid values are: • Once • Daily • Weekly • Monthly	Required	62.0
frequency Cadence Options	Frequency Cadence Options	Frequency cadence options for the invoice scheduler.	Required	62.0
invoiceDate	String	Date shown on the invoice. This date is also used for tax calculations.	Required if the frequency Cadence property is set to Once.	62.0
invoiceDate Offset	Integer	Offset applied to the target date, which is the number of days added to or subtracted from the invoice date, to calculate the updated invoice date.	Required if the frequency Cadence property is set to Daily, Weekly, or Monthly.	62.0
invoice Status	String	Status of the invoice that specifies the expected invoice status from an invoice batch run. Valid values are: • Draft • Posted	Required	62.0
isInvoice DateFrom RunDate	Boolean	Indicates whether the invoice date is applicable from the date when the invoice scheduler is run (true) or not (false).	Optional	63.0
preferred Time	String	Preferred time for the invoice batch run.	Required	62.0

Name	Type	Description	Required or Optional	Available Version
scheduler Name	String	Name of the invoice scheduler, which must be unique in your org.	Required	62.0
startDate	String	Start date of the invoice scheduler.	Required	62.0
status	String	Status of the invoice scheduler. Valid values are: • Draft • Active • Inactive	Required	62.0
targetDate	String	Target date of the invoice batch run. Billing schedules having the next billing date before this date are picked up for invoicing. The target date must be less than or equal to the maximum allowed target date for the org.	Required if the frequency Cadence property is set to Once.	62.0
target DateOffset	Integer	Target date offset applied to the next invoice run date to calculate the target date. The offset is the number of days added to or subtracted from the next billing date.	Required if the frequency Cadence property is set to Daily, Weekly, or Monthly.	62.0
timezone	String	Time zone that's applicable for the invoice scheduler.	Optional	62.0

Response body for POST

Batch Invoice Scheduler

Batch Payment Scheduler (POST)

Create a payment scheduler to automate and process payment runs on a recurring basis.

Special Access Rules

You need the Payment Ops permission set to use this API.

Resource

```
/commerce/payments/payment-schedulers/
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/payments/payment-schedulers/
```

Available version

64.0

HTTP methods

POST

Request body for POST

JSON example

JSON example

```
{
  "schedulerName": "Payment Scheduler",
  "startDate": "2022-01-01",
  "endDate": "2022-12-31",
  "preferredTime": "02:05:00.000",
  "frequencyCadence": "Monthly",
  "recursEveryMonthOnDay": "28",
  "criteriaMatchType": "MatchAny",
  "status": "Active",
  "filterCriteria": [
    {
      "objectName": "PaymentScheduleItem",
      "fieldName": "PaymentRunMatchingValue",
      "operation": "Equals",
      "value": "1",
      "criteriaSequence": 1
    },
    {
      "objectName": "PaymentScheduleItem",
      "fieldName": "PaymentRunMatchingValue",
      "operation": "Equals",
      "value": "2",
      "criteriaSequence": 2
    },
    {
      "objectName": "PaymentScheduleItem",
      "fieldName": "PaymentRunMatchingValue",
      "operation": "Equals",
      "value": "3",
      "criteriaSequence": 3
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`criteria MatchType`	String	Match type for the criteria of the payment scheduler. Valid values are: • `Match Any` • `Match None`	Required if the `frequencyCadence` property is set to `Monthly`.	64.0

Name	Type	Description	Required or Optional	Available Version
endDate	String	End date of the payment scheduler.	Required if the frequencyCadence property is set to Monthly.	64.0
frequencyCadence	String	Frequency cadence of the payment scheduler. Valid values are: • Once • Daily • Weekly • Monthly	Required	64.0
filterCriteria	Payment Run Batch Filter Criteria Input	List of criteria that are used to filter the payment run details.	Required if the frequencyCadence property is set to Monthly.	64.0
preferredTime	String	Preferred time for the payment scheduler run.	Required	64.0
recursEveryMonthOnDay	String	Date when the payment scheduler recurs.	Required if the frequencyCadence property is set to Monthly.	64.0
schedulerName	String	Name of the payment scheduler.	Required	64.0
startDate	String	Start date of the payment scheduler.	Required	64.0
status	String	Status of the payment scheduler. Valid values are: • Active • Canceled • Draft • Inactive	Required	64.0

Response body for POST

Batch Payments Scheduler

Billing Arrangement (GET)

Retrieve a billing arrangement and its associated billing arrangement lines.

This API fetches billing information, including the split percentage allocation for accounts, the remainder percentage, and whether adjustments are applied to the owning account.

Resource

```
/revenue/billing/billing-arrangement/billingArrangementId
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/revenue/billing/billing-arrangement//revenue/billing/billing-arrangement/1bdxx000000004rAAA
```

Available version

66.0

HTTP methods

GET

Request parameter for GET

Parameter Name	Type	Description	Required or Optional	Available Version
billing ArrangementId	String	Unique ID of the billing arrangement to retrieve.	Required	66.0

Response body for GET

Billing Arrangement

Billing Schedule Recovery List (POST)

Recover the latest generated invoice associated with the billing schedules in the `Error` or `Processing` status.

Billing schedules include critical details such as the amount to be billed, next billing date, and status. An invoice can be associated with one or more billing schedules. When an invoice is generated or posted, the billing schedules are updated to reflect the accurate state of the invoice. The billing schedules associated with an invoice are marked in the `Error` status if any of the invoicing processes have errors. Use this API to recover the invoice associated with the billing schedules in the `Error` or `Processing` status.

Special Access Rules

You need the Manage Errors Using Invoice Error Recovery API permission set to use this API.

Resource

```
/commerce/invoicing/billing-schedules/collection/actions/recover
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/invoicing/billing-schedules/collection/actions/recover
```

Available version

62.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "billingScheduleIds": ["44bDU00000000XXYAY"]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
billing ScheduleIds	String[]	IDs of the billing schedules to recover the invoice for. You can recover one billing schedule per API request.	Required	62.0

Response body for POST

Billing Schedule Recovery List

Create Billing Schedules for Orders (POST)

Generate billing schedules for orders by using context service.

Special Access Rules

The org must have the standard billing context definition with the target mapping. This context definition is available in orgs with Billing enabled. Additionally, you need the Create Billing Schedules From Billing Transactions API permission set and the Context Service Runtime permission set to use this API.

See these requirements to learn more about the configuration prerequisites.

Resource

```
/commerce/invoicing/billing-schedules/actions/create
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/invoicing/billing-schedules/actions/create
```

Available version

62.0

HTTP methods

POST

Request body for POST

JSON example

```
{
    "billingTransactionIds": [ "801xx000003H1H9AAK"]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`billing Transaction Ids`	String[]	ID of the billing transaction. This property value is the ID of the order if the source of the billing request is for the Order object. If the order product associated with the specified order ID doesn't have an associated billing treatment ID, the API considers the default billing treatment ID. The generated billing schedule group has the default billing treatment ID. The API supports only one billing transaction ID in the input.	Required	62.0

Response body for POST

Context-Aware Billing Schedule

Create Sequence Policy (POST)

Create a sequence policy to configure a unique, sequential number for posted invoices or credit memos.

A sequence policy ensures unique identification and systematic tracking of financial transactions, supporting regulatory compliance and preventing fraud. You can create a sequence policy for invoices or credit memos only.

Special Access Rules

You need the Billing Admin permission set to use this API.

Resource

```
/connect/sequences/policy
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/sequences/policy
```

Available version

65.0

HTTP methods

POST

Request body for POST

JSON example

This example shows a sample request to create a sequence policy.

```
{
  "name": "Sample Sequence Policy",
  "description": "This is a sample sequence policy.",
  "effectiveFromDateTime": "2025-08-10",
  "expirationDateTime": "2025-09-20",
  "isActive": true,
  "sequenceMode": "Basic",
  "targetObject": "Invoice",
  "dateStampFormat": "Yyyy",
  "sequenceStartNumber": 1,
  "incrementNumber": 1,
  "maximumSequenceNumber": 1000,
  "minimumSequenceNumberWidth": 2,
  "filterCriteria": "Custom",
  "selectionLogic": "selectionLogic",
  "sequencePattern": "INV-{SequenceValue}-abc",
  "selectionCondition": [
    {
      "filterField": "AppType",
      "operator": "Equals",
      "filterValue": "RLM",
      "conditionNumber": 1
    },
    {
      "filterField": "Status",
      "operator": "Equals",
      "filterValue": "Posted",
      "conditionNumber": 2
    },
    {
      "filterField": "LegalEntity",
      "operator": "Equals",
      "filterValue": "US",
      "conditionNumber": 3
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`dateStampFormat`	String	Format of the stamp date that's appended to the sequence number. You can't edit this property by using the Update Sequence Policy API.	Required	65.0
`description`	String	Additional details about the sequence policy.	Optional	65.0

Name	Type	Description	Required or Optional	Available Version
effectiveFrom DateTime	String	Date and time when the policy becomes effective. The default value is the current date.	Required	65.0
expiration DateTime	String	Date and time when the policy expires.	Optional	65.0
filter Criteria	String	Criteria to filter the target objects.	Required	65.0
increment Number	Integer	Value by which the sequence number increases until it reaches the maximum number. This value must be greater than or equal to 1. The default value is 1. You can't edit this property by using the Update Sequence Policy API.	Required	65.0
isActive	Boolean	Indicates whether the policy is active (true) or not (false). The default value is false.	Required	65.0
maximumSequence Number	Integer	Maximum number the sequence number can reach.	Optional	65.0
minimumSequence NumberWidth	Integer	Minimum number of digits a sequence number must have. You can't edit this property by using the Update Sequence Policy API.	Optional	65.0
name	String	Name of the sequence policy.	Required	65.0
selection Condition	Selection Condition Input[]	Criteria to determine which sequence policy is applied to a record. This property includes conditions based on any standard or custom fields of the record.	Optional	65.0
selection Logic	String	Logic that determines the objects that the sequence policy applies to.	Optional	65.0
sequenceMode	String	Specifies how sequence numbers are generated. Valid values are: • Basic—Assigns sequential numbers without gap reconciliation. • Gapless—Assigns sequential numbers with gap reconciliation. The usage of this value ensures that the posted invoices or credit memos don't have any numbering gaps for audits and compliance.	Required	65.0

Name	Type	Description	Required or Optional	Available Version
		You can't edit this property by using the Update Sequence Policy API.		
`sequencePattern`	String	Pattern structure that's followed for the sequence.	Required	65.0
`sequenceStartNumber`	Integer	Starting value of the sequence number, which must be greater than or equal to `0`. The default value is `1`. You can't edit this property by using the Update Sequence Policy API.	Required	65.0
`targetObject`	String	Object that the policy is applied to. Valid values are: • `Invoice` • `CreditMemo`—Available in API version 66.0 and later. You can't edit this property by using the Update Sequence Policy API.	Required	65.0
`timeZone`	String	Time zone that's applicable for the sequence policy.	Optional	65.0

Response body for POST

Sequence Policy

Create Standalone Billing Schedules (POST)

Generate billing schedules from any internal or external transaction by using context service.

You can create a billing schedule from any sObject such as WorkOrder, Cart, Order, Opportunity, and insurance policy. Or, you can also generate billing schedules from order items.

The Create Standalone Billing Schedules (POST) API uses the StandaloneBillingContext context definition to hydrate the context of the transaction. The context definition includes these mappings.

- The TransactionMapping maps the fields of the transaction to the attributes of the Transaction node.
- The BSGEntitiesMapping maps the attributes of the Billing Schedule node, the Billing Schedule Group node, and Billing Schedule Group Relationship node to the fields of the corresponding Salesforce objects.

For the StandaloneBillingContext context definition to hydrate all the required data, transaction data for the mandatory context tags are required. Here are the topics that mention the mandatory and optional tags, sample transaction details, and sample payloads for various types of transactions.

- One-Time New Sale Transaction on page 2600
- Term-Defined New Sale Transaction on page 2603
- Evergreen New Sale Transaction on page 2607
- New Sale Transaction With Bundled Products on page 2611

- New Sale Transaction With Ramped Products on page 2617
- New Sale Transaction With Usage Products on page 2622
- Amended Transaction on page 2626
- Renewal Transaction on page 2636
- Early Renewal Transaction on page 2641
- Canceled Transaction on page 2647

Special Access Rules

This API is available with the Revenue Cloud Billing license.

Resource

```
/commerce/invoicing/standalone/billing-schedules/actions/create
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/invoicing/standalone/billing-schedules/actions/create
```

Available version

64.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "transactionDetails": "{\"nodeName\": [{\"id\":\"801Az00000aynKZIAY\",
\"businessSobjectType\": \"Order\"}]}",
  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext",
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "OrderTransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
transactionContextDetails	Standalone Billing Schedule Metadata Input[]	Details of the context definition and its mappings that are used to hydrate the transaction data and save it in the appropriate Billing fields.	Required	64.0
transactionDetails	String	Input JSON data that includes the ID of the transaction record for which the billing schedule must be created and other additional transaction details. The API request supports a single mapping ID. You can send separate	Required	64.0

Name	Type	Description	Required or Optional	Available Version
		requests for line items and line details by using their respective mapping IDs. However, this approach can result in duplicate billing schedules for the same line items and line details.		

Response body for POST

Context-Aware Billing Schedule

SEE ALSO:

Salesforce Help: Context Service

Industries Common Resources Developer Guide: Context Service

BillingScheduleGroup

Create and Apply Credit Memo (POST)

Create a credit memo and apply it to an invoice. The credit memo can fully or partially credit the invoice.

Use this API to adjust an outstanding invoice balance or rectify errors in an invoice. In the API request, pass a list of invoice lines to credit. Keep these considerations in mind when you use this API.

- The request must contain at least one invoice line. Each invoice line must have the invoice line's ID, the amount to credit, and any optional tax details. The invoice lines must be a part of the invoice passed in the resource.
- The amount to credit must not exceed the charge or adjustment amount of an individual invoice line.
- The request body's credit amount inclusive of taxes must not exceed the target invoice line's amount inclusive of taxes, except for taxes calculated through an external tax service.
- The request body's total credit amount inclusive of taxes calculated through an external tax service must not exceed the outstanding invoice balance, which is also inclusive of taxes.

This API creates and posts a credit memo. The credit memo has one credit memo line for each invoice line passed in the API request. The invoice balance reduces by a value equal to the credit memo's balance. This API modifies the balance of a posted invoice or invoice line based on the specified credit application level for your org. See Apply Credits to Posted Invoices or Invoice Lines.

Special Access Rules

You need the Credit Memo Operations User permission set to use this API.

Resource

```
/commerce/invoicing/invoices/invoiceId/actions/credit
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/invoicing/invoices/3ttR0000000008NEIAY/actions/credit
```

Available version

62.0

HTTP methods

POST

Path parameter for POST

Name	Type	Description	Required or Optional	Available Version
`invoiceId`	String	ID of the invoice to be credited partially or fully. The status of the invoice must be `Posted`.	Required	62.0

Request body for POST

JSON example

This example shows a sample request with the `Calculate` tax strategy.

```
{
  "type": "POSTED",
  "taxStrategy": "Calculate",
  "taxEffectiveDate": "2024-07-20",
  "effectiveDate": "2024-07-20",
  "description": "Credit Invoice",
  "invoiceLines": [
    {
      "invoiceLineId": "5TVR00000004SiqOBE",
      "amountToCredit": 5
    }
  ]
}
```

This example shows a sample request with the `CopyFromInvoiceLine` tax strategy.

```
{
  "type": "POSTED",
  "taxStrategy": "CopyFromInvoiceLine",
  "effectiveDate": "2020-05-22",
  "description": "Credit Invoice",
  "invoiceLines": [
    {
      "invoiceLineId": "5TVR00000004SiqOBE",
      "amountToCredit": "5",
      "taxStrategy": "CopyFromInvoiceLine"
    }
  ]
}
```

This example shows a sample request with the `ManualOverride` and `CopyFromInvoiceLine` tax strategies.

```
{
  "type": "POSTED",
  "taxStrategy": "ManualOverride",
  "taxEffectiveDate": "2021-08-01",
  "effectiveDate": "2021-08-01",
  "description": "Credit issued because product was malfunctioning.",
  "invoiceLines": [
    {
```

```
      "invoiceLineId": "5TVR00000004SiqOBE",
      "amountToCredit": 100,
      "taxStrategy": "ManualOverride",
      "taxEffectiveDate": "2021-08-01T21:22:41.000Z",
      "taxes": [
        {
          "taxAmount": 15,
          "taxName": "abc",
          "taxCode": "taxCode",
          "taxRate": 7
        }
      ],
      "addresses": {
        "billingAddress": {
          "street": "1 Market St #300",
          "city": "San Francisco",
          "state": "CA",
          "country": "US",
          "postalCode": "94105",
          "latitude": "37.789901",
          "longitude": "-122.396923"
        },
        "shippingAddress": {
          "street": "415 Mission St",
          "city": "San Francisco",
          "state": "CA",
          "country": "US",
          "postalCode": "94105",
          "latitude": "37.789901",
          "longitude": "-122.396923"
        }
      }
    },
    {
      "invoiceLineId": "5TVR00000004SiqOAE",
      "amountToCredit": 200,
      "taxStrategy": "CopyFromInvoiceLine"
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`description`	String	Description for the credit memo to be created.	Optional	62.0
`effective Date`	String	Date when the credit memo takes effect.	Optional	62.0
`invoiceLines`	Credit Invoice Line Input[]	List of the invoice lines to be credited. The invoice line IDs must be related to the invoice ID specified in the API request.	Required	62.0

Name	Type	Description	Required or Optional	Available Version
		If invoice lines aren't specified, the API request results in an error.		
taxEffective Date	String	Date when the tax takes effect to recalculate the taxes.	Optional	62.0
taxStrategy	String	Tax strategy to be applied across invoice lines. You can override the tax strategy at the individual invoice line level or at the tax line level. Valid values are: • Ignore—Specifies that the creation of tax lines must be ignored. • ManualOverride—Specifies that the provided tax values must be considered for taxes. • CopyFromInvoiceLine—Specifies that tax values must be copied from the invoice line. • Calculate—Specifies that tax must be calculated by using the API.	Required	62.0
type	String	Type of credit memo to be created. Valid values are Posted and Draft. Specify Draft as a value in your request to create draft credit memos.	Optional	62.0

Response body for POST

Revenue Async Line Level

Apply Credit Memo (POST)

Adjust or correct already issued invoices by applying an existing credit memo to an invoice.

Specify the credit memo ID and the amounts to be applied, with the total of all applied amounts not exceeding the credit memo's balance.

The credit amount for each invoice can't surpass the original charge or adjustment amount, and the overall credit amount must not exceed the invoice's outstanding balance. The exceptions include any taxes calculated by an external service.

For example, your organization sold 10 tablets at US$500 each, totaling $5000, to a vendor who later reported that 6 tablets were defective. Your accounts receivable team creates a $3000 credit memo by using the standalone Credit Memo API. Additionally, the team applies this credit to the original invoice by using the Apply Credit Memo API.

When the credit application level is Invoice Line, this API applies the credit memo amount to all invoice lines starting from the highest amount until the full credit is applied or all lines are settled.

Special Access Rules

You need the Credit Memo Operations User permission set to use this API.

Resource

```
/commerce/invoicing/credit-memos/creditMemoId/actions/apply
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/invoicing/credit-memos/50gSG0000000XATYA2/actions/apply
```

Available version

62.0

HTTP methods

POST

Path parameter for POST

Name	Type	Description	Required or Optional	Available Version
`creditMemoId`	String	ID of the credit memo record.	Required	62.0

Request body for POST

JSON example

```
{
  "applications": [
    {
      "appliedToId": "3ttxx000000003FAAQ",
      "amount": 10,
      "description": "Apply to invoice for refund",
      "effectiveDate": "2024-07-01"
    },
    {
      "appliedToId": "3ttxx0000000001AAA",
      "amount": 100
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`applications`	Credit Memo Apply Application Input[]	List of one or more applications to apply the credit memo for. Each application represents an invoice that's credited by using the balance of the specified credit memo.	Required	62.0

Response body for POST

Credit Memo Apply List

Unapply Credit Memo (POST)

Unapply a credit memo from an invoice and return the invoice and the credit memo to their pre-application states.

Use this resource if an error occurred when a credit is issued. For example, if an incorrect credit memo is applied to an invoice, or if a credit memo is created for an incorrect amount, use this resource to unapply the credit memo.

Special Access Rules

You need the Credit Memo Operations User permission set to use this API.

Resource

```
/commerce/invoicing/credit-memo-inv-applications/creditMemoInvApplicationId/actions/unapply
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/invoicing/credit-memo-inv-applications/4sESG000002nxPB2AY/actions/unapply
```

Available version

62.0

HTTP methods

POST

Path parameter for POST

Name	Type	Description	Required or Optional	Available Version
`creditMemoInvApplicationId`	String	ID of the credit memo invoice application.	Required	62.0

Request body for POST

JSON example

```
{
  "description": "Unapply credit memo from invoice to revert an error",
  "effectiveDate": "2024-07-01"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`description`	String	Explanation or reason for unapplying the credit memo.	Optional	62.0
`effectiveDate`	String	Effective date for the credit memo.	Optional	62.0

Response body for POST

Credit Memo Unapply

Apply Credit Memo Line (POST)

Adjust or correct already issued invoices by applying an existing credit memo line to an invoice line.

Special Access Rules

You need the Credit Memo Operations User permission set to use this API.

Resource

```
/commerce/invoicing/credit-memo-lines/creditMemoLineId/actions/apply
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/invoicing/credit-memo-lines/9yxSG0000000IUHYA2/actions/apply
```

Available version

62.0

HTTP methods

POST

Path parameter for POST

Name	Type	Description	Required or Optional	Available Version
`creditMemoLineId`	String	ID of the credit memo line record.	Required	62.0

Request body for POST

JSON example

```
{
 "applyCreditDetails": [
 {
 "invoiceLineId": "5TVSG0000002ZJR4A2",
 "appliedAmount": 5,
 "description": "Apply to invoice line 1",
 "effectiveDate": "2024-07-01"
 },
 {
 "invoiceLineId": "5TVSG0000002ZJS4A2",
 "appliedAmount": 10,
 "description": "Apply to invoice line 2",
 "effectiveDate": "2024-07-01"
 }
 ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
applyCreditDetails	Credit Memo Line Application Input[]	List of one or more applications to apply the credit memo line for. Each application represents an invoice line that's credited by using the balance of the specified credit memo line.	Required	62.0

Response body for POST

Credit Memo Line Applied

Unapply Credit Memo Line (POST)

Unapply a credit memo line from an invoice line and return the invoice line and the credit memo line to their pre-application states.

Special Access Rules

You need the Credit Memo Operations User permission set to use this API.

Resource

```
/commerce/invoicing/credit-memo-line-invoice-line/creditMemoLineInvoiceLineId/actions/unapply
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/invoicing/credit-memo-line-invoice-line/4sGSG000000024Q92AI/actions/unapply
```

Available version

62.0

HTTP methods

POST

Path parameter for POST

Name	Type	Description	Required or Optional	Available Version
creditMemoLineInvoiceLineId	String	ID of the credit memo line invoice line record.	Required	62.0

Request body for POST

JSON example

```
{
 "description": "Unapply a credit memo line from invoice line 1",
 "effectiveDate": "2024-07-01"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`description`	String	Explanation or reason for unapplying the credit memo line.	Optional	62.0
`effectiveDate`	String	Effective date for the credit memo line.	Optional	62.0

Response body for POST

Credit Memo Line Unapplied

Generate On-Demand Invoice Document (POST)

Generate an invoice document for a record, and update any junction object record.

Resource

```
/revenue/billing/document/actions/generate
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/revenue/billing/document/actions/generate
```

Available version

66.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "recordId": "3ttasd98776266777",
  "shouldForceRegenerate": true,
  "documentTemplateId": "0694x000000XyzABC",
  "documentTitle": "DOC-00000012",
  "tokenData":
[illegible]
 Market
St\",\"CompanyName\":\"260.1\",\"TotalAmt\":\"$7.99\",\"TotalTax\":\"$0.00\",\"InvoiceDueDate\":\"10/08/2025\",\"CompanyPostalCode\":\"94105\",\"InvoiceDate\":\"09/08/2025\",\"CompanyCity\":\"San
 Francisco\",\"CompanyCountry\":\"US\",\"AccountName\":\"Sonia BAT Account\"}"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`documentTemplateId`	String	Document template ID to use for PDF generation. The document template	Optional	66.0

Name	Type	Description	Required or Optional	Available Version
		must be active with UsageType as `INVOICE` If you don't specify this value, the system auto-resolves by using the default template.		
`documentTitle`	String	Custom title for the generated document. If you don't specify a value, the value is auto-generated.	Optional	66.0
`recordId`	String	Record ID of the object the document is generated for. You can specify invoice ID only.	Required	66.0
`shouldForceRegenerate`	Boolean	Indicates whether to regenerate the document (`true`) or not (`false`). If set to `true`, this API generates a document, replacing any existing ones. If set to `false`, this API skips the generation of document if invoice already has an associated document. The default value is `false`.	Optional	66.0
`tokenData`	String	Token data to generate the document.	Optional	66.0

Response body for POST

On-Demand Document Generation Response

Generate Account Statement (POST)

Generate comprehensive account statement with transaction history and balance information.

An account statement includes these details.

- Transaction History—List of all financial transactions, such as invoices, credit memos, debit memos, payments, and refunds, within a specified date range.
- Open Balances—Non-zero balances for invoices and other transactions.
- Account Summary—Aggregated totals including beginning balance, total charges, total payments, and ending balance.
- Multi-currency Support—Transactions across multiple currencies.

Resource

```
/revenue/billing/accounts/accountId/statement
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/revenue/billing/accounts/accountId/statement
```

Available version

66.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "shouldShowOpenBalancesOnly": false,
  "startDate": "2025-09-01",
  "transactionTypes": {
    "transactionTypes": [
      "Invoice"
    ]
  },
  "sortBy": "DueDate",
  "sortingOrder": "Ascending",
  "associatedAccountIds": {
    "associatedAccountIds": [
      "001xx000003DGb5AAG",
      "001xx000003DGb6AAG"
    ]
  },
  "documentTemplateId": "0TRxx000000002YGAQ",
  "correlationId": "monthly-statement-sept-2025",
  "customFields": "{\"Invoice\": [\"TotalAmountWithTax\"], \"Account\": {\"base\":
[\"Phone\", \"Fax\"], \"associated\": [\"Website\"]}}}"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
associatedAccountIds	String[]	List of associated account IDs from hierarchy to include in the statement. You can specify up to 50 accounts.	Optional	66.0
correlationId	String	Correlation ID for tracking the request across systems.	Optional	66.0
customFields	String	JSON string specifying custom fields to include in the statement. As a best practice, we recommend that you limit the number of custom fields up to 30 fields. Valid objects to specify the associated fields are:	Optional	67.0

Name	Type	Description	Required or Optional	Available Version
		• `Invoice` • `CreditMemo` • `Payment` • `DebitMemo` • `Refund` • `Account` Here's the expected format. `{"Invoice": ["field1", "field2"], "Account": {"base": ["field1"], "associated": ["field2"]}}`		
documentTemplateId	String	Document template ID to use for PDF generation. If you don't specify a value, the system auto-resolves by using the default template.	Optional	66.0
shouldShowOpenBalancesOnly	Boolean	Indicates whether to show open balances only (`true`) or not (`false`). If set to `true`, the API shows only accounts with non-zero balances. If set to `false`, this API shows complete transaction history within the date range from the start date.	Optional	66.0
sortBy	String	Criteria for sorting transactions. The default and valid value is `Date`.	Optional	66.0
sortingOrder	String	Sort order for transactions. Valid values are: • `Ascending` • `Descending` The default value is `Descending`.	Optional	66.0
startDate	String	Start date for the transaction history. The required format is `YYYY-MM-DD`. The system processes records up to 90 days from this date.	Required	66.0
transactionTypes	String[]	List of transaction types to include in the statement. If you don't specify a value or	Optional	66.0

Name	Type	Description	Required or Optional	Available Version
		the value is empty, all transaction types are included. Valid values are: • `All` • `CreditMemo` • `DebitMemo` • `Invoice` • `Payment` • `Refund`		

Response body for POST

Account Statement Response

Invoice Creation (POST)

Create an invoice for an account, order, or a list of billing schedules.

This API request creates billing period items for the matching billing schedules. The billing period items are created for the period between the next charge date of the billing schedule and the specified target date. Invoice lines are created by processing these billing period items. These invoice lines are then grouped into invoices based on the defined grouping criteria on the billing schedule.

This API also applies any available credits on an account to settle an invoice and to reduce it's balance. To apply available credits, ensure the **Apply Credits to Posted Invoices** setting from Billing Settings is turned on.

Special Access Rules

You need the Generate Invoices From Billing Schedule API, Billing Operations User, or Billing Customer Service permission set to use this API.

Resource

```
/commerce/invoicing/invoices/collection/actions/generate
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/invoicing/invoices/collection/actions/generate
```

Available version

62.0

HTTP methods

POST

Request body for POST

Note: The `billingTransactionId` property takes precedence over the `accountId` property followed by the `billingScheduleIds` property when values for these properties are specified in the input request.

JSON example

```
{
  "accountId": "001SG00000mYtRWYA0",
```

```
  "action": "Posted",
  "billingScheduleIds": [
    "44bSG000000CVeMYAW"
  ],
  "billingTransactionId": "801SG00000mYtaXYAS",
  "correlationId": null,
  "invoiceDate": "2024-01-12",
  "targetDate": "2024-01-12"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`accountId`	String	ID of the account record to create the invoices for.	Required if the `billing ScheduleIds` or `billing TransactionId` property isn't specified.	63.0
`action`	String	Type of invoice to be created. Valid values are: • `Draft` • `Posted`	Required	62.0
`billing ScheduleIds`	String[]	List of billing schedule IDs that's used to create the invoices. You can specify a maximum of 200 billing schedules.	Required if the `accountId` or `billing TransactionId` property isn't specified.	62.0
`billing TransactionId`	String	ID of the billing transaction record, which is the order ID, to create the invoices for.	Required if the `accountId` or `billing ScheduleIds` property isn't specified.	63.0
`correlation Id`	String	Property that's tagged against the published `InvoiceProcessedEvent` event, if specified.	Optional	62.0
`invoiceDate`	String	Stamping date of the invoice in ISO 8601 format.	Required	62.0

Name	Type	Description	Required or Optional	Available Version
`targetDate`	String	Date in ISO 8601 format used to decide the billing periods that are included to create invoices.	Required	62.0

Response body for POST

[Revenue Async Response](#)

Invoice Draft to Posted Status (POST)

Update the status of the invoice from Draft to Posted.

This API calls an external tax engine or provides information to your tax adapter implementation to calculate taxes for the draft invoice, post the invoice, and update the related billing schedules and billing periods.

Special Access Rules

You need the Billing Operations User permission set to use this API.

Resource

```
/commerce/invoicing/invoices/collection/actions/post
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/invoicing/invoices/collection/actions/post
```

Available version

62.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "invoiceIds": ["3ttxx0000004CIjAAM"]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`correlationId`	String	Splunk correlation ID to track the messages that are related to the request and are logged in Splunk by the different services involved in the request. If the ID isn't specified, the service creates a random Universally Unique Identifier (UUID).	Optional	62.0

Name	Type	Description	Required or Optional	Available Version
`invoiceIds`	String[]	IDs of the invoice records in `Draft` status to be posted. You can post one draft invoice per API request.	Required	62.0

Response body for POST

Revenue Async Response

Invoice Ingestion (POST)

Ingest or generate an invoice from an internal or external billing transaction data.

Create invoices from any external or internal transaction. For example, create invoices by using external orders, policies, quotes, or internal data, such as assets or quotes.

You can also perform these additional tasks.

- Ingest invoices from any external system into Billing to manage invoice operations such as issuing credits against an invoice or voiding an invoice.
- Generate invoices from debit memos from API version 66.0 and later.

Keep these considerations in mind when you use this API.

- If the Convert Negative Invoice Lines to Credit Memo Lines feature isn't enabled, use the Negative Invoice Lines to Credit Conversion API to create a credit memo against an ingested negative invoice line.
- Use the Apply Credit Action to apply a credit memo to an invoice.
- To ingest legacy invoices, create at least one invoice line.
- To generate invoices from debit memos, specify the debit memo ID as the `referenceEntityId` property value in the request body of this API. Make sure that the debit memo is in `Posted` status and is ready for invoice generation. The invoice is generated in `Draft` status.

Resource

```
/commerce/invoicing/invoices/collection/actions/ingest
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/invoicing/invoices/collection/actions/ingest
```

Available version

63.0

HTTP methods

POST

Request body for POST

JSON example to ingest a draft invoice with a tax callout

```
{
  "invoices": [
    {
      "shouldCalculateTax": true,
```

```
      "taxCalculationStatus": "Estimated",
      "graph": {
        "graphId": "CreateInvoice",
        "records": [
          {
            "referenceId": "refAccount",
            "record": {
              "attributes": {
                "type": "Account",
                "method": "GET",
                "id": "ExternalId__c/123"
              }
            }
          },
          {
            "referenceId": "refContact",
            "record": {
              "attributes": {
                "type": "Contact",
                "method": "GET",
                "id": "ExternalId__c/123"
              }
            }
          },
          {
            "referenceId": "refInvoice",
            "record": {
              "attributes": {
                "type": "Invoice",
                "method": "POST"
              },
              "billingAccountId": "@{refAccount.Id}",
              "billToContactId": "@{refContact.Id}",
              "paymentTermId": "2OXxx0000004CFUGA2",
              "referenceEntityId": "801xx000003GeQQAA0",
              "status": "Draft",
              "invoiceDate": "2024-12-19",
              "currencyIsoCode": "USD",
              "dueDate": "2024-12-19",
              "invoiceNumber": "DOC-10",
              "description": "Sample Invoice",
              "uniqueIdentifier": "5873af8f-f007-4aa0-9e3d-53a08c3f59de"
            }
          },
          {
            "referenceId": "refBillingAddress",
            "record": {
              "attributes": {
                "type": "InvoiceAddressGroup",
                "method": "POST"
              },
              "street": "123 Main St",
              "city": "NewYork",
              "postalCode": "10001",
```

```
            "state": "New York",
            "country": "US",
            "longitude": "123.456",
            "latitude": "78.910",
            "invoiceId": "@{refInvoice.id}"
          }
        },
        {
          "referenceId": "refShippingAddress",
          "record": {
            "attributes": {
              "type": "InvoiceAddressGroup",
              "method": "POST"
            },
            "street": "123 Main St",
            "city": "NewYork",
            "postalCode": "10001",
            "state": "New York",
            "country": "US",
            "longitude": "123.456",
            "latitude": "78.910",
            "invoiceId": "@{refInvoice.id}"
          }
        },
        {
          "referenceId": "refShipFromAddress",
          "record": {
            "attributes": {
              "type": "InvoiceAddressGroup",
              "method": "POST"
            },
            "street": "1 Market Street",
            "city": "San Francisco",
            "postalCode": "94105",
            "state": "CA",
            "country": "US",
            "longitude": "123.456",
            "latitude": "78.910",
            "invoiceId": "@{refInvoice.id}"
          }
        },
        {
          "referenceId": "refInvoiceLine1",
          "record": {
            "attributes": {
              "type": "InvoiceLine",
              "method": "POST"
            },
            "name": "productName1",
            "product2Id": "01txx0000006ic4AAA",
            "invoiceLineStartDate": "2024-11-10",
            "invoiceLineEndDate": "2024-11-13",
            "quantity": "10",
            "unitPrice": "10",
```

```
      "chargeAmount": "100",
      "invoiceId": "@{refInvoice.id}",
      "referenceEntityItemId": "802xx000001neB9AAI",
      "billingAddressId": "@{refBillingAddress.id}",
      "shippingAddressId": "@{refShippingAddress.id}",
      "shipFromAddressId": "@{refShipFromAddress.id}",
      "taxTreatmentId": "1ttxx0000000BOTAA2",
      "legalEntityId": "0fwxx0000000001AAA",
      "legalEntityAccountingPeriodId": "1HLxx0000004C92GAE"
    }
  },
  {
    "referenceId": "refInvoiceLine2",
    "record": {
      "attributes": {
        "type": "InvoiceLine",
        "method": "POST"
      },
      "name": "productName2",
      "product2Id": "01txx0000006ic4AAA",
      "invoiceLineStartDate": "2024-11-10",
      "invoiceLineEndDate": "2024-11-15",
      "quantity": "10",
      "unitPrice": "10",
      "chargeAmount": "100",
      "invoiceId": "@{refInvoice.id}",
      "referenceEntityItemId": "802xx000001neB9AAI",
      "billingAddressId": "@{refBillingAddress.id}",
      "shippingAddressId": "@{refShippingAddress.id}",
      "shipFromAddressId": "@{refShipFromAddress.id}",
      "taxTreatmentId": "1ttxx00000001DpAAI",
      "legalEntityId": "0fwxx0000000001AAA",
      "legalEntityAccountingPeriodId": "1HLxx0000004C92GAE"
    }
  },
  {
    "referenceId": "refInvoiceLineTax",
    "record": {
      "attributes": {
        "type": "InvoiceLineTax",
        "method": "POST"
      },
      "taxAmount": 7.25,
      "taxCode": "CA-94121",
      "taxName": "CALIFORNIA",
      "taxRate": 0.25,
      "taxEffectiveDate": "2024-11-10",
      "taxDocumentNumber": "123",
      "taxExemptAmount": 0,
      "taxTransactionNumber": "kl",
      "description": "Associated tax line.",
      "invoiceLineId": "@{refInvoiceLine1.id}"
    }
  },
```

```
          {
            "referenceId": "refInvoiceLineTax1",
            "record": {
              "attributes": {
                "type": "InvoiceLineTax",
                "method": "POST"
              },
              "taxAmount": "10",
              "taxCode": "CA-94121",
              "taxName": "CALIFORNIA",
              "taxRate": 0.25,
              "taxEffectiveDate": "2024-11-10",
              "taxDocumentNumber": "123",
              "taxExemptAmount": 0,
              "taxTransactionNumber": "125",
              "invoiceLineId": "@{refInvoiceLine2.id}",
              "description": "Associated tax line."
            }
          }
        ]
      }
    }
  ]
}
```


JSON example to ingest a draft invoice without a tax callout


```
{
  "invoices": [
    {
      "taxCalculationStatus": "Estimated",
      "graph": {
        "graphId": "123",
        "records": [
          {
            "referenceId": "refAccount",
            "record": {
              "attributes": {
                "type": "Account",
                "method": "GET",
                "id": "TestExternalId__c/123"
              }
            }
          },
          {
            "referenceId": "refInvoice",
            "record": {
              "attributes": {
                "type": "Invoice",
                "method": "POST"
              },
              "billingAccountId": "@{refAccount.Id}",
              "billToContactId": "003xx000004Wk8qAAC",
              "paymentTermId": "2OXxx0000004CFUGA2",
              "referenceEntityId": "801xx000003GeQQAA0",
```

```
          "status": "Draft",
          "invoiceDate": "2024-12-19",
          "currencyIsoCode": "USD",
          "dueDate": "2024-12-19",
          "invoiceNumber": "DOC-10",
          "description": "testInvoice",
          "uniqueIdentifier": "c76011a1-e113-49d9-9c54-3b5c68950ada"
        }
      },
      {
        "referenceId": "refBillingAddress",
        "record": {
          "attributes": {
            "type": "InvoiceAddressGroup",
            "method": "POST"
          },
          "street": "123 Main St",
          "city": "NewYork",
          "postalCode": "10001",
          "state": "New York",
          "country": "US",
          "longitude": "123.456",
          "latitude": "78.910",
          "invoiceId": "@{refInvoice.id}"
        }
      },
      {
        "referenceId": "refShippingAddress",
        "record": {
          "attributes": {
            "type": "InvoiceAddressGroup",
            "method": "POST"
          },
          "street": "123 Main St",
          "city": "NewYork",
          "postalCode": "10001",
          "state": "New York",
          "country": "US",
          "longitude": "123.456",
          "latitude": "78.910",
          "invoiceId": "@{refInvoice.id}"
        }
      },
      {
        "referenceId": "refShipFromAddress",
        "record": {
          "attributes": {
            "type": "InvoiceAddressGroup",
            "method": "POST"
          },
          "street": "1 Market Street",
          "city": "San Francisco",
          "postalCode": "94105",
          "state": "CA",
```

```
        "country": "US",
        "longitude": "123.456",
        "latitude": "78.910",
        "invoiceId": "@{refInvoice.id}"
      }
    },
    {
      "referenceId": "refInvoiceLine1",
      "record": {
        "attributes": {
          "type": "InvoiceLine",
          "method": "POST"
        },
        "name": "productName1",
        "product2Id": "01txx0000006ic4AAA",
        "invoiceLineStartDate": "2024-11-10",
        "invoiceLineEndDate": "2024-11-13",
        "quantity": "10",
        "unitPrice": "10",
        "chargeAmount": "100",
        "invoiceId": "@{refInvoice.id}",
        "referenceEntityItemId": "802xx000001neB9AAI",
        "billingAddressId": "@{refBillingAddress.id}",
        "shippingAddressId": "@{refShippingAddress.id}",
        "shipFromAddressId": "@{refShipFromAddress.id}",
        "taxTreatmentId": "1ttxx0000000BOTAA2",
        "legalEntityId": "0fwxx0000000001AAA",
        "legalEntityAccountingPeriodId": "1HLxx0000004C92GAE"
      }
    },
    {
      "referenceId": "refInvoiceLine2",
      "record": {
        "attributes": {
          "type": "InvoiceLine",
          "method": "POST"
        },
        "name": "productName2",
        "product2Id": "01txx0000006ic4AAA",
        "invoiceLineStartDate": "2024-11-10",
        "invoiceLineEndDate": "2024-11-15",
        "quantity": "10",
        "unitPrice": "10",
        "chargeAmount": "100",
        "invoiceId": "@{refInvoice.id}",
        "referenceEntityItemId": "802xx000001neB9AAI",
        "billingAddressId": "@{refBillingAddress.id}",
        "shippingAddressId": "@{refShippingAddress.id}",
        "shipFromAddressId": "@{refShipFromAddress.id}",
        "taxTreatmentId": "1ttxx00000001DpAAI",
        "legalEntityId": "0fwxx0000000001AAA",
        "legalEntityAccountingPeriodId": "1HLxx0000004C92GAE"
      }
    },
```

```
        {
          "referenceId": "refInvoiceLineTax",
          "record": {
            "attributes": {
              "type": "InvoiceLineTax",
              "method": "POST"
            },
            "taxAmount": 7.25,
            "taxCode": "CA-94121",
            "taxName": "CALIFORNIA",
            "taxRate": 0.25,
            "taxEffectiveDate": "2024-11-10",
            "taxDocumentNumber": "123",
            "taxExemptAmount": 0,
            "taxTransactionNumber": "kl",
            "description": "description for tax Line",
            "invoiceLineId": "@{refInvoiceLine1.id}"
          }
        },
        {
          "referenceId": "refInvoiceLineTax1",
          "record": {
            "attributes": {
              "type": "InvoiceLineTax",
              "method": "POST"
            },
            "taxAmount": "10",
            "taxCode": "CA-94121",
            "taxName": "CALIFORNIA",
            "taxRate": 0.25,
            "taxEffectiveDate": "2024-11-10",
            "taxDocumentNumber": "123",
            "taxExemptAmount": 0,
            "taxTransactionNumber": "125",
            "invoiceLineId": "@{refInvoiceLine2.id}",
            "description": "description for tax Line"
          }
        }
      ]
    }
  }
 ]
}
```

JSON example to ingest posted invoices

```
{
  "invoices": [
    {
      "taxCalculationStatus": "Posted",
      "graph": {
        "graphId": "123",
        "records": [
          {
            "referenceId": "refAccount",
```

```
      "record": {
        "attributes": {
          "type": "Account",
          "method": "GET",
          "id": "ExternalId__c/123"
        }
      }
    },
    {
      "referenceId": "refInvoice",
      "record": {
        "attributes": {
          "type": "Invoice",
          "method": "POST"
        },
        "billingAccountId": "001SG00000njpF3YAI",
        "billToContactId": "003xx000004Wk8qAAC",
        "paymentTermId": "2OXxx0000004CFUGA2",
        "referenceEntityId": "801xx000003GeQQAA0",
        "status": "Posted",
        "invoiceDate": "2024-12-19",
        "currencyIsoCode": "USD",
        "dueDate": "2024-12-19",
        "invoiceNumber": "DOC-10",
        "description": "Sample Invoice",
        "uniqueIdentifier": "9994b2c4-c0c3-47c3-806f-ae6e1f16bac3"
      }
    },
    {
      "referenceId": "refBillingAddress",
      "record": {
        "attributes": {
          "type": "InvoiceAddressGroup",
          "method": "POST"
        },
        "street": "123 Main St",
        "city": "NewYork",
        "postalCode": "10001",
        "state": "New York",
        "country": "US",
        "longitude": "123.456",
        "latitude": "78.910",
        "invoiceId": "@{refInvoice.id}"
      }
    },
    {
      "referenceId": "refShippingAddress",
      "record": {
        "attributes": {
          "type": "InvoiceAddressGroup",
          "method": "POST"
        },
        "street": "123 Main St",
        "city": "NewYork",
```

```
          "postalCode": "10001",
          "state": "New York",
          "country": "US",
          "longitude": "123.456",
          "latitude": "78.910",
          "invoiceId": "@{refInvoice.id}"
        }
      },
      {
        "referenceId": "refShipFromAddress",
        "record": {
          "attributes": {
            "type": "InvoiceAddressGroup",
            "method": "POST"
          },
          "street": "1 Market Street",
          "city": "San Francisco",
          "postalCode": "94105",
          "state": "CA",
          "country": "US",
          "longitude": "123.456",
          "latitude": "78.910",
          "invoiceId": "@{refInvoice.id}"
        }
      },
      {
        "referenceId": "refInvoiceLine1",
        "record": {
          "attributes": {
            "type": "InvoiceLine",
            "method": "POST"
          },
          "name": "productName1",
          "product2Id": "01txx0000006ic4AAA",
          "invoiceLineStartDate": "2024-11-10",
          "invoiceLineEndDate": "2024-11-13",
          "quantity": "10",
          "unitPrice": "10",
          "chargeAmount": "100",
          "invoiceId": "@{refInvoice.id}",
          "referenceEntityItemId": "802xx000001neB9AAI",
          "billingAddressId": "@{refBillingAddress.id}",
          "shippingAddressId": "@{refShippingAddress.id}",
          "shipFromAddressId": "@{refShipFromAddress.id}",
          "taxTreatmentId": "1ttxx0000000BOTAA2",
          "legalEntityId": "0fwxx0000000001AAA",
          "legalEntityAccountingPeriodId": "1HLxx0000004C92GAE"
        }
      },
      {
        "referenceId": "refInvoiceLine2",
        "record": {
          "attributes": {
            "type": "InvoiceLine",
```

```
                "method": "POST"
              },
              "name": "productName2",
              "product2Id": "01txx0000006ic4AAA",
              "invoiceLineStartDate": "2024-11-10",
              "invoiceLineEndDate": "2024-11-15",
              "quantity": "10",
              "unitPrice": "10",
              "chargeAmount": "100",
              "invoiceId": "@{refInvoice.id}",
              "referenceEntityItemId": "802xx000001neB9AAI",
              "billingAddressId": "@{refBillingAddress.id}",
              "shippingAddressId": "@{refShippingAddress.id}",
              "shipFromAddressId": "@{refShipFromAddress.id}",
              "taxTreatmentId": "1ttxx00000001DpAAI",
              "legalEntityId": "0fwxx0000000001AAA",
              "legalEntityAccountingPeriodId": "1HLxx0000004C92GAE"
            }
          },
          {
            "referenceId": "refInvoiceLineTax1",
            "record": {
              "attributes": {
                "type": "InvoiceLineTax",
                "method": "POST"
              },
              "taxAmount": "10",
              "taxCode": "CA-94121",
              "taxName": "CALIFORNIA",
              "taxRate": 0.25,
              "taxEffectiveDate": "2024-11-10",
              "taxDocumentNumber": "123",
              "taxExemptAmount": 0,
              "taxTransactionNumber": "125",
              "invoiceLineId": "@{refInvoiceLine1.id}",
              "description": "Associated tax line."
            }
          }
        ]
      }
    }
  ]
}
```

JSON example to generate invoices from debit memos

```
{
  "invoices": [
    {
      "shouldCalculateTax": true,
      "taxCalculationStatus": "Estimated",
      "graph": {
        "graphId": "CreateInvoice",
        "records": [
          {
```

```
            "referenceId": "refInvoice",
            "record": {
              "attributes": {
                "type": "Invoice",
                "method": "POST"
              },
              "accountId": "001SB00001V3RxNYAV",
              "billToContactId": "003xx000004WhUQAA0",
              "referenceEntityId": "4DmAAC000008UP70BM",
              "status": "Draft",
              "invoiceDate": "2024-12-19",
              "dueDate": "2024-12-19",
              "description": "Sample Invoice"
            }
          }
        ]
      }
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`invoices`	Invoice Ingestion Input[]	List of invoices to be generated or ingested, enabling the submission and processing of multiple invoices in a single request. This API supports one invoice per request. To send 25 requests at a time, see the Composite Batch request.	Required	63.0

Response body for POST

Invoice Ingestion

Invoice Estimated Tax Calculation (POST)

Calculate estimated tax for invoices with invoice lines that have the `TaxProcessingStatus` as either `Pending` or `Estimated`.

Resource

```
/commerce/invoicing/invoices/collection/actions/calculate-estimated-tax
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/invoicing/invoices/collection/actions/calculate-estimated-tax
```

Available version

63.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "invoiceIds": ["3ttxx0000004CIjAAM"]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
correlationId	String	Splunk correlation ID to track the messages that are related to the request and are logged in Splunk by the different services involved in the request. If the ID isn't specified, the service creates a random Universally Unique Identifier (UUID).	Optional	63.0
invoiceIds	String[]	IDs of the invoices for which the estimated tax must be calculated. You can specify one invoice per API request.	Required	63.0

Response body for POST

Revenue Async Response

Invoice Preview (POST)

Generate preview invoices, which includes the estimated tax amounts, for a billing transaction for the next two billing periods.

Invoice preview provides details of the upcoming invoices for orders or quotes within the system.

Special Access Rules

You need the Billing Operations User or Billing Customer Service permission set to use this API.

See these requirements to learn more about the configuration prerequisites.

Resource

```
/commerce/invoicing/invoices/collection/actions/preview
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/invoicing/invoices/collection/actions/preview
```

Available version

63.0

HTTP methods

POST

Request body for POST

JSON example

```
{
    "billingTransactionId": "801Z600000004LoIAI",
    "numberOfBillingPeriods": 2,
    "previewDate": "2024-12-04"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`billing TransactionId`	String	ID of the record to generate the preview invoices for.	Required	63.0
`numberOf BillingPeriods`	Integer	Number of billing periods that the invoice preview is generated for. If unspecified, the default value is `2`.	Optional	64.0
`previewDate`	String	The date on which the preview invoice is generated. For the first invoice, the preview date is the target date for generating the invoice. For the second invoice, the target date is calculated based on the preview date and the minimum billing frequency of the transactions. The default value is the current date.	Optional	63.0

Response body for POST

Invoice Preview

Invoice Run Recovery (POST)

Recover records associated with a failed invoice run. Recovery is required only when billing schedules remain in the `Processing`, `Void In Progress`, or `Error` status.

Special Access Rules

To use this API, you need the Invoice Scheduler API permission set.

Resource

```
/commerce/invoicing/invoice-batch-runs/invoiceBatchRunId/actions/recover
```

The *`invoiceBatchRunId`* parameter is the ID of the failed invoice batch run record whose details you want to retrieve.

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/invoicing/invoice-batch-runs/5IRxx0000004TwGGAU/actions/recover
```

Available version

62.0

HTTP methods

POST

Response body for POST

Invoice Batch Run Recovery on page 2739

Negative Invoice Lines to Credit Conversion (POST)

Convert a list of invoice lines with a negative amount into a posted credit memo. This conversion is applicable for a single invoice at a time.

Keep these considerations in mind when you use this API.

- All invoice lines must be related to the same invoice.
- The invoice line must have a negative amount.
- The invoice line must not be a previously converted credit memo.
- The invoice must have the `Posted` status.
- The invoice must not have any active settlements such as credit applications.

Special Access Rules

You need the Credit Memo Operations User permission set to use this API.

Resource

```
/commerce/invoicing/invoices/invoiceId/actions/convert-to-credit
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/invoicing/invoices/3ttxx00000000XtAAI/actions/convert-to-credit
```

Available version

62.0

HTTP methods

POST

Path parameter for POST

Name	Type	Description	Required or Optional	Available Version
`invoiceId`	String	ID of the invoice whose negative invoice lines must be converted into a posted credit memo.	Required	62.0

Request body for POST

JSON example

```
{
"invoiceLines": ["5TVxx0000004C92GAE", "5TVxx0000004C93GAE"],
"description": "Convert negative invoice lines into credit",
```

```
"effectiveDate":"2022-05-18"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`description`	String	Description stamped on the credit memo that's created after the negative invoice line conversion.	Optional	62.0
`effectiveDate`	String	Date stamped on the credit memo that's created after the negative invoice line conversion.	Required	62.0
`invoiceLines`	String[]	Complete list of the negative invoice lines along with the associated invoice line taxes. The specified negative invoice lines are converted into a posted credit memo.	Optional	62.0

Response body for POST

Convert Negative Invoice Lines

Payment Line Apply (POST)

Allocate the balance of a payment to reduce the balance of an invoice. The response includes an ID of the payment line invoice or payment line invoice line that represents the payment balance allocated against the invoice.

Use the Commerce Payments APIs to send your payment and refund details to external payment gateways for processing against a customer's bank. See Commerce Payments resources to check the APIs for payment gateways, payment captures, and payment authorizations.

Resource

```
/commerce/billing/payments/paymentId/actions/apply
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/billing/payments/1PLR000000000dDOAQ/actions/apply
```

Available version

64.0

HTTP methods

POST

Path parameter for POST

Name	Type	Description	Required or Optional	Available Version
`paymentId`	String	ID of the payment record.	Required	64.0

Request body for POST

JSON example

```
{
  "appliedToId": "3ttR000000001IkIAI",
  "amount": 10,
  "effectiveDate": "2020-08-11T07:53:15.000Z",
  "comments": "Apply payment",
  "associatedAccountId": "001R00000060AyuIAE"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
amount	Double	Amount that's applied. The amount must be less than the invoice line and payment balance.	Required	64.0
appliedToId	String	ID of the invoice line that this payment is applied to. Specify the IDs for these records. • Invoice • Invoice Line	Required	64.0
associatedAccountId	String	ID of the associated account.	Optional	64.0
comments	String	Comments that you can add to the payment line application.	Optional	64.0
effectiveDate	String	Date from which the payment line application takes effect.	Optional	64.0

Response body for POST

Payment Line Apply

Payment Line Unapply (POST)

Revert the application of a payment line from an invoice, and return the payment and invoices to their preapplication state. Use this API to correct an input during the payment application process.

For example, you can use this API to revert an incorrect payment that's applied to an invoice, or to rectify an incorrect amount.

Use the Commerce Payments APIs to send your payment and refund details to external payment gateways for processing against a customer's bank. See Commerce Payments resources to check the APIs for payment gateways, payment captures, and payment authorizations.

Resource

```
/commerce/billing/payments/paymentId/paymentlines/paymentLineId/actions/unapply
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/billing/payments/0aQ1j000000001O/paymentlines/1PLR000000000dDOAQ/actions/unapply
```

Available version

64.0

HTTP methods

POST

Path parameters for POST

Name	Type	Description	Required or Optional	Available Version
`paymentId`	String	ID of the payment record.	Required	64.0
`paymentLineId`	String	ID of the payment line record.	Required	64.0

Request body for POST

JSON example

```
{
  "effectiveDate": "2025-05-22T11:30:25.000Z",
  "comments": "Unapply payment"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`comments`	String	Comments that you can add when you revert a payment line application.	Optional	64.0
`effectiveDate`	String	Date from when the reversal of the payment line application is in effect.	Optional	64.0

Response body for POST

Payment Line Unapply

Payment Scheduler Update (PATCH)

Activate or deactivate a payment scheduler. You can set the status of a payment scheduler to `Active`, `Canceled`, `Draft`, or `Inactive`.

Special Access Rules

You need the Payment Ops permission set to use this API.

Resource

```
/commerce/payments/payment-schedulers/billingBatchSchedulerId
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/payments/payment-schedulers/5BSxx0000004TwGGAU
```

Available version

64.0

HTTP methods

PATCH

Path parameter for POST

Name	Type	Description	Required or Optional	Available Version
billingBatch SchedulerId	String	ID of the payment scheduler record.	Required	64.0

Request body for PATCH

JSON example

JSON example

```
{
  "status": "Active"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
status	String	Status that must be set to activate or deactivate a payment scheduler. Valid values are: • Active • Canceled • Draft • Inactive	Required	64.0

Response body for PATCH

Payments Scheduler Update

Post a Draft Memo (POST)

Post a draft credit memo to a credit memo record for review and approval.

Resource

```
/commerce/invoicing/credit/collection/actions/post
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v65.0/commerce/invoicing/credit/collection/actions/post
```

Available version

65.0

HTTP methods

POST

Request body for POST

JSON example

```
{
"creditMemoIds": ["50gDU00000001MnYAI"]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
correlationId	String	Splunk correlation ID to use to track messages that are related to the request and logged in Splunk by the different services involved in the request. If not specified, the service creates a random Universally Unique Identifier (UUID).	Optional	65.0
creditMemoIds	String[]	ID of the credit memo record in Draft status to be posted. You can post one draft credit memo per API request.	Required	65.0

Response body for POST

Credit Memo Post

Rules Application (POST)

Apply payments and credits to an account's invoices based on specified rules defined on the Billing Settings page.

This API uses predefined logic to allocate payments and credits, reducing any manual intervention and errors.

Resource

```
/revenue/billing/transactions/actions/apply
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/revenue/billing/transactions/actions/apply
```

Available version

66.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "accountId": "001xx000003DGbQAAW",
  "targetDate": "2024-01-15"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`accountId`	String	ID of the account to perform the settlement of payments and credits against invoices in adherence with the applied rules.	Required	66.0
`targetDate`	String	The date used to select invoices and invoice lines with a posted date equal to or later than the target date to apply payments and credits.	Optional	66.0

Response body for POST

Rules Application

Posted Invoice List Write-Off (POST)

Create credit memos with the total charge amount on the invoice as the write-off amount and close the invoice.

You can write off invoices to maintain accurate financial records and to prioritize invoices with a higher probability of payment, which is essential for compliance with accounting standards.

Special Access Rules

To use this API, you need the Billing Operations User or Credit Memo Operations User permission set.

Resource

```
/commerce/invoicing/invoices/actions/write-off
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/invoicing/invoices/actions/write-off
```

Available version

64.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "invoices": [
    {
      "invoiceId": "3ttxx00000000cjAAA",
      "reasonCode": "Bad Debt",
      "description": "Bad Debt"
    },
    {
      "invoiceId": "3ttxx00000000cjAAA",
      "reasonCode": "Concession",
      "description": "Concession"
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`invoices`	Posted Invoice Write-Off Input[]	Details of the invoices that you want to write off.	Required	64.0

Response body for POST

Posted Invoice List Write-Off

Void a Posted Invoice (POST)

Void a posted invoice to rebill the customer, if necessary.

This API request changes the invoice status from `Posted` to `Void In Progress`. The invoice remains in the `Void In Progress` status until the credit is applied and financial fields are recalculated on the invoice's related billing period items and billing schedule. The invoice status changes to `Voided` after all recalculations are completed.

Keep these considerations in mind when you use this API.

- The balance and total amount on the invoice must be equal. If these amounts aren't equal due to payments or credits, the API request fails.
- You can't call other APIs on an invoice with the `Void In Progress` status. You also can't update the invoice fields.
- You can void only the most recently posted invoice on a billing schedule.
- To void an invoice that has payments or credits, use the Credit Memo Unapply (POST) API.

Credit Memos

The void process creates a credit memo, which contains one credit memo line for each invoice line, including tax lines. For example, if the invoice line has a balance of US$20, the related credit memo line has a balance of $20. The credit memo's balance is then allocated to the invoice header's balance, reducing it to zero. A credit memo invoice application is created to record the details of the void process.

Negative Invoice Lines

If an invoice has negative invoice lines that aren't converted to a credit memo, you can use this endpoint to void the posted invoice.

Special Access Rules

You need the Void a Posted Invoice API permission set to use this API.

Resource

```
/commerce/invoicing/invoices/invoiceId/actions/void
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/invoicing/invoices/3ttxx00000000XtAAI/actions/void
```

Available version

62.0

HTTP methods

POST

Path parameter for POST

Properties

Name	Type	Description	Required or Optional	Available Version
`invoiceId`	String	ID of the posted invoice to be voided.	Required	62.0

Response body for POST

Revenue Async Response

Refund Line Apply (POST)

Make a refund transaction against a payment.

Resource

```
/commerce/billing/refunds/refundId/actions/apply
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/billing/refunds/0cbVc0000000G4nIAE/actions/apply
```

Available version

64.0

HTTP methods

POST

Path parameter for POST

Name	Type	Description	Required or Optional	Available Version
`refundId`	String	ID of the refund record.	Required	64.0

Request body for POST

JSON example

```
{
  "appliedToId": "0aQR00000004ZkKMAU",
  "amount": 10,
  "effectiveDate": "2020-08-11T07:53:15.000Z",
  "comments": "Payment application."
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`amount`	Double	Amount to refund.	Required	64.0
`appliedToId`	String	ID of a payment or credit memo record. The refund is applied to this object.	Required	64.0
`comments`	String	Additional details of the refund request.	Optional	64.0
`effectiveDate`	String	Date from when the refund is in effect.	Optional	64.0

Response body for POST

Refund Line Apply

Send Emails for Posted Invoices (POST)

Send emails for the posted invoices of a specified invoice batch run ID.

Each email includes a custom message template in the email body and an attachment of the Invoice Document PDF. The email template is based on the user preferences defined on the Billing Profile, Legal Entity, or Billing Settings page.

Special Access Rules

Enable the **Configure Email Delivery Settings** toggle from the Billing Settings page from Setup.

Resource

```
/commerce/invoicing/invoice-batch-runs/actions/send-email
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/invoicing/invoice-batch-runs/actions/send-email
```

Available version

65.0

HTTP methods

POST

Request body for POST

JSON example

```
{
  "invoiceBatchRunId": "5IRLT000001SIJB4A4"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`invoiceBatchRunId`	String	ID of the invoice batch run record to send emails for the posted invoices of an invoice batch run.	Required	65.0

Response body for POST

Send Email Response

Resume Billing (POST)

Resume billing for billing schedule groups or an account that's currently on hold.

Resource

```
/commerce/invoicing/actions/resume-billing
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/invoicing/actions/resume-billing
```

Available version

63.0

HTTP methods

POST

Request body for POST

JSON example

```
{
    "referenceIds":
    [
        {
          "referenceId": "001DU000001o2UwYAI",
          "resumeDate": "2024-11-27"
        }
    ]

}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`referenceIds`	Resume Billing Object Input[]	Input representation of the account or billing schedule group IDs to resume the billing operation for.	Required	63.0

Response body for POST

Suspend Resume Billing

Suspend Billing (POST)

Suspend billing for billing schedule groups or an account for a predefined period.

Resource

```
/commerce/invoicing/actions/suspend-billing
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/invoicing/actions/suspend-billing
```

Available version

63.0

HTTP methods

POST

Request body for POST

JSON example

```
{
    "referenceIds":
    [
        {
            "referenceId": "001DU000001o2UwYAI",
            "suspendDate": "2024-11-27",
            "resumeDate": "2024-12-27"
        }
    ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`referenceIds`	Suspend Billing Object Input[]	Input representation of the account or billing schedule group IDs to suspend the billing operation for.	Required	63.0

Response body for POST

Suspend Resume Billing

Standalone Credit Memo (POST)

Create a credit memo without applying it to an invoice. You can credit the invoice at a later date.

In the API request, specify the credit memo header information, charge parameters, adjustment parameters, and tax parameters. A credit memo requires at least one credit memo line. The credit memo line can be a charge or an adjustment.

Specify the credit memo lines that you want as lists of charges and adjustments. Each credit memo line must be related to a product.

Special Access Rules

You need the Credit Memo Operations User permission set to use this API.

Resource

```
/commerce/invoicing/credit-memos/actions/generate
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/invoicing/credit-memos/actions/generate
```

Available version

62.0

HTTP methods

POST

Request body for POST

JSON example

This example shows a sample request with the `Ignore` tax strategy.

```
{
  "billingAccountId": "001j000000WCFB800x",
  "type": "Posted",
  "description": "Standalone credit memo with ignored tax.",
  "currencyIsoCode": "USD",
  "taxStrategy": "Ignore",
  "charges": [
    {
      "chargeAmount": 100,
      "productId": "01tR0000000njDiIAI"
    }
  ]
}
```

This example shows a sample request with the `ManualOverride` tax strategy.

```
{
  "billingAccountId": "001DU000001nhoPYAQ",
  "description": "creditmemo-1",
  "currencyIsoCode": "USD",
  "taxStrategy": "ManualOverride",
  "charges": [
    {
      "productId": "01tDU000000EpKkYAK",
```

```
      "chargeAmount": 1000,
      "taxes": [
        {
          "taxAmount": 7.25,
          "taxCode": "CA-94121",
          "taxName": "CALIFORNIA",
          "taxRate": 0.25
        }
      ]
    }
  ]
}
```

This example shows a sample request with the `Calculate` tax strategy for a charge line.

```
{
  "billingAccountId": "001DU000001nhoPYAQ",
  "description": "Standalone Credit Memo",
  "type": "Posted",
  "currencyIsoCode": "USD",
  "taxStrategy": "Ignore",
  "charges": [
    {
      "productId": "01tR0000000njDiIAI",
      "chargeAmount": 10,
      "taxStrategy": "Calculate",
      "treatmentId": "1ttxx00000001VZAAY"
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`billToContactId`	String	Contact related to the credit memo.	Optional	62.0
`billingAccountId`	String	ID of the account that the credit is issued to.	Required	62.0
`charges`	Standalone Credit Memo Charge Input[]	Charge lines of the credit memo. Requires at least one charge line.	Required	62.0
`currencyIsoCode`	String	ISO code currency of the new credit that's issued.	Optional	62.0
`description`	String	Description for the new credit that's issued.	Optional	62.0
`effectiveDate`	String	Effective date of the credit memo. If the value isn't specified, then it's null.	Optional	62.0

Name	Type	Description	Required or Optional	Available Version
externalReference	String	ID of the external reference for the credit memo.	Optional	62.0
externalReferenceDataSource	String	Source of the external reference for the credit memo.	Optional	62.0
taxEffectiveDate	String	Effective date of the credit memo tax. If the value isn't specified, then it's null.	Optional	62.0
taxStrategy	String	Specifies how tax lines must be created for the standalone credit memos. Valid values are: • `Ignore`—Specifies that the creation of tax lines must be ignored. • `Manual Override`—Specifies that the provided tax values must be considered for taxes. • `Calculate`—Specifies that tax must be calculated by using the API.	Required	62.0
type	String	Type of credit memo to be created. Valid values are `Posted` and `Draft`. Specify `Draft` as a value in your request to create draft credit memos.	Optional	62.0

Response body for POST

Revenue Async Response

Tax Calculation (POST)

Calculate tax for a transaction.

Special Access Rules

To use this API, you need the CalculateTaxes API permission set.

Resource

```
/commerce/taxes/actions/calculate
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/taxes/actions/calculate
```

Available version

62.0

HTTP methods

POST

Request body for POST

JSON example

This example shows a tax request for a specified invoice as the reference entity.


```
{
  "addresses": {
    "billTo": {
      "street": "123 Main Street",
      "city": "Bainbridge Island",
      "state": "WA",
      "postalCode": "98110",
      "country": "US"
    },
    "soldTo": {
      "street": "123 Main Street",
      "city": "Bainbridge Island",
      "state": "WA",
      "postalCode": "98110",
      "country": "US"
    },
    "shipFrom": {
      "street": "123 Alaskan Way",
      "city": "Seattle",
      "state": "WA",
      "country": "US",
      "postalCode": "98101"
    },
    "shipTo": {
      "street": "123 Main street",
      "city": "Bainbridge Island",
      "state": "WA",
      "postalCode": "98110",
      "country": "US"
    }
  },
  "currencyIsoCode": "USD",
  "customerDetails": {
    "accountId": "001xx000003HYD5AAO"
  },
  "description": "Monthly invoice for account 001xx000003HYEhAAO",
  "documentCode": "3ttxx0000000C7d_Debit-4wAxx00000000ODEAY",
  "effectiveDate": "2022-03-09T10:30:41.000Z",
  "isCommit": true,
  "lineItems": [
    {
      "quantity": 1,
      "amount": 100,
      "taxCode": "TX0001",
      "productCode": "Y0001",
      "productSKU": "PRES-RX-12896745",
      "description": "New Product",
      "effectiveDate": "2022-03-09T10:30:41.000Z",
      "lineNumber": "5TVxx0000004C92GAE",
      "addresses": {
```

```
        "shipFrom": {
          "street": "123 Alaskan Way",
          "city": "Seattle",
          "state": "WA",
          "country": "US",
          "postalCode": "98101",
          "latitude": 45.12,
          "longitude": 45.12
        },
        "shipTo": {
          "street": "123 Auburn Blvd",
          "city": "Sacramento",
          "state": "CA",
          "country": "US",
          "postalCode": "95841",
          "latitude": 45.12,
          "longitude": 45.12
        },
        "soldTo": {
          "street": "123 Auburn Blvd",
          "city": "Sacramento",
          "state": "CA",
          "country": "US",
          "postalCode": "95841",
          "latitude": 45.12,
          "longitude": 45.12
        },
        "billTo": {
          "street": "123 Auburn Blvd",
          "city": "Sacramento",
          "state": "CA",
          "country": "US",
          "postalCode": "95841",
          "latitude": 45.12,
          "longitude": 45.12
        }
      }
    }
  ],
  "referenceDocumentCode" = null,
  "referenceEntityId": "3ttxx0000000C7d",
  "shouldVoidTax": false,
  "taxEngineId": "4wAxx0000000g8v",
  "taxType": "Actual",
  "taxTransactionType": "Debit",
  "transactionDate": "2022-03-09T10:30:41.000Z"
}
```

This example shows a tax request for a specified credit memo as the reference entity.

```
{
  "addresses": {
    "billTo": {
      "street": "123 Main Street",
      "city": "Bainbridge Island",
```

```
      "state": "WA",
      "postalCode": "98110",
      "country": "US"
    },
    "soldTo": {
      "street": "123 Main Street",
      "city": "Bainbridge Island",
      "state": "WA",
      "postalCode": "98110",
      "country": "US"
    },
    "shipFrom": {
      "street": "123 Alaskan Way",
      "city": "Seattle",
      "state": "WA",
      "country": "US",
      "postalCode": "98101"
    },
    "shipTo": {
      "street": "123 Main street",
      "city": "Bainbridge Island",
      "state": "WA",
      "postalCode": "98110",
      "country": "US"
    }
  },
  "currencyIsoCode": "USD",
  "customerDetails": {
    "accountId": "001xx000003HYD5AAO"
  },
  "description": "Monthly credit memo for account 001xx000003HYEhAAO",
  "documentCode": "50gxx000000g27KAAQ-4wAxx00000000ODEAY",
  "effectiveDate": "2022-03-09T10:30:41.000Z",
  "isCommit": true,
  "lineItems": [
    {
      "quantity": 1,
      "amount": 100,
      "taxCode": "TX0001",
      "productCode": "Y0001",
      "productSKU": "PRES-RX-12896745",
      "description": "New Product",
      "effectiveDate": "2022-03-09T10:30:41.000Z",
      "lineNumber": "5TVxx0000004C92GAE",
      "addresses": {
        "shipFrom": {
          "street": "123 Alaskan Way",
          "city": "Seattle",
          "state": "WA",
          "country": "US",
          "postalCode": "98101",
          "latitude": 45.12,
          "longitude": 45.12
        },
```

```
        "shipTo": {
          "street": "123 Auburn Blvd",
          "city": "Sacramento",
          "state": "CA",
          "country": "US",
          "postalCode": "95841",
          "latitude": 45.12,
          "longitude": 45.12
        },
        "soldTo": {
          "street": "123 Auburn Blvd",
          "city": "Sacramento",
          "state": "CA",
          "country": "US",
          "postalCode": "95841",
          "latitude": 45.12,
          "longitude": 45.12
        },
        "billTo": {
          "street": "123 Auburn Blvd",
          "city": "Sacramento",
          "state": "CA",
          "country": "US",
          "postalCode": "95841",
          "latitude": 45.12,
          "longitude": 45.12
        }
      }
    }
  ],
  "referenceDocumentCode" = null,
  "referenceEntityId": "50gxx000000g27K",
  "shouldVoidTax": false,
  "taxEngineId": "4wAxx0000000g8v",
  "taxType": "Actual",
  "taxTransactionType": "Debit",
  "transactionDate": "2022-03-09T10:30:41.000Z"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
addresses	Addresses	Address details for tax calculation.	Optional	62.0
currencyIsoCode	String	Currency ISO code that's used for tax calculation.	Optional	62.0
customerDetails	Customer Details	Customer details for determining the applicable tax.	Optional	62.0
description	String	Description of the tax transaction.	Optional	62.0

Name	Type	Description	Required or Optional	Available Version
documentCode	String	Unique identifier for the tax document. If the documentCode property isn't specified, the tax engine auto-generates it.	Optional	62.0
effectiveDate	String	Date when the tax rules are applied. If a tax rate changes on a specific date, the effectiveDate property ensures the correct rate is applied based on the transaction's timing. For credit-based tax callouts, specify the original invoice date.	Optional	62.0
isCommit	Boolean	Indicates whether to submit the transaction to the tax engine for reporting (true) or not (false). This property value is true for invoices and credit memos with the Posted status.	Optional	62.0
lineItems	Line Item Input []	Details of the line items for calculating the applicable tax.	Required	62.0
reference DocumentCode	String	Reference document code. For subsequent transactions such as a credit tax, this property value specifies the original document code. For credit-based tax callouts, specify the original invoice ID.	Optional	62.0
reference EntityId	String	ID of the related quote, invoice, and other transaction documents.	Optional	62.0
sellerDetails	Seller Details Input	Seller details for tax calculation.	Optional	62.0
shouldVoidTax	Boolean	Indicates whether to void the tax transaction associated with a document that's mentioned as the referenceDocumentCode property value with taxType property value as Actual and isCommit property value set to true. Keep these considerations in mind when you use this property.	Optional	65.0

Name	Type	Description	Required or Optional	Available Version
		• If the `shouldVoidTax` property value is set to `true`, then the operation returns a response with `documentCode` property value updated to `referenceDocumentCode` property value that was originally sent in the request payload. The response also includes the `taxTransactionType` property value as `Void`. This indicates that the document specified in the `referenceDocumentCode` property value is voided. • If document is locked or you can't void the tax transaction for any reason, then you can use the Tax Calculation request to perform another transaction such as a Credit Tax request. In this scenario, the response includes the `documentCode` property value that was sent in the request payload. • If the document that's mentioned in the `referenceDocumentCode` property value isn't available in the tax engine, then an error response occurs with ResultCode on page 1607 value as `ReferenceDocumentCodeMissing`.		
`taxEngineId`	String	ID of the tax engine that's used to calculate tax.	Required	62.0
`tax TransactionType`	String	Type of the tax transaction. Valid values are: • `Debit`—Increases tax liability for the seller, requiring the seller to pay tax on the transaction. • `Credit`—Decreases tax liability for the seller, resulting in a tax refund for the seller. • `Void`—Reserved for internal use. The default value is `Debit`.	Optional	62.0

Name	Type	Description	Required or Optional	Available Version
taxType	String	Type of the tax. Valid values are: • Actual—Exact tax amount that's calculated based on actual sales. • Estimated— Estimated tax amount, which is adjusted later to match actual tax calculations. For draft invoices and quote records, the tax type is marked as Estimated. After draft invoices are posted and the status changes to Posted, the tax type is updated to Actual. Similarly, when a quote record is finalized and converted into an actual order, the tax type is also updated to Actual.	Required	62.0
transactionDate	String	Date of the transaction that appears on the invoice, order, and other transaction documents.	Required	62.0

Response body for POST

Tax Calculation

Update Sequence Policy (PATCH)

Update the settings of a sequence policy that defines how unique, sequential numbers are generated by using specific patterns, values, and filters.

Special Access Rules

You need the Billing Admin permission set to use this API.

Resource

```
/connect/sequences/policy/sequencePolicyId
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/connect/sequences/policy/1Vdxx0000000GRNAA2
```

Available version

65.0

HTTP methods

PATCH

Request body for PATCH

JSON example

This example shows a sample request to create a sequence policy.

```
{
  "name": "Sample Sequence Policy",
  "description": "This is a sample sequence policy.",
  "effectiveFromDateTime": "2025-08-10",
  "expirationDateTime": "2025-09-20",
  "isActive": true,
  "sequenceMode": "Basic",
  "targetObject": "Invoice",
  "dateStampFormat": "Yyyy",
  "sequenceStartNumber": 1,
  "incrementNumber": 1,
  "maximumSequenceNumber": 1000,
  "minimumSequenceNumberWidth": 2,
  "filterCriteria": "Custom",
  "selectionLogic": "selectionLogic",
  "sequencePattern": "INV-{SequenceValue}-abc",
  "selectionCondition": [
    {
      "filterField": "AppType",
      "operator": "Equals",
      "filterValue": "RLM",
      "conditionNumber": 1
    },
    {
      "filterField": "Status",
      "operator": "Equals",
      "filterValue": "Posted",
      "conditionNumber": 2
    },
    {
      "filterField": "LegalEntity",
      "operator": "Equals",
      "filterValue": "US",
      "conditionNumber": 3
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
dateStamp Format	String	Format of the stamp date that's appended to the sequence number. You can't edit this property by using the Update Sequence Policy API.	Required	65.0
description	String	Additional details about the sequence policy.	Optional	65.0

Name	Type	Description	Required or Optional	Available Version
effectiveFrom DateTime	String	Date and time when the policy becomes effective. The default value is the current date.	Required	65.0
expiration DateTime	String	Date and time when the policy expires.	Optional	65.0
filter Criteria	String	Criteria to filter the target objects.	Required	65.0
increment Number	Integer	Value by which the sequence number increases until it reaches the maximum number. This value must be greater than or equal to 1. The default value is 1. You can't edit this property by using the Update Sequence Policy API.	Required	65.0
isActive	Boolean	Indicates whether the policy is active (true) or not (false). The default value is false.	Required	65.0
maximumSequence Number	Integer	Maximum number the sequence number can reach.	Optional	65.0
minimumSequence NumberWidth	Integer	Minimum number of digits a sequence number must have. You can't edit this property by using the Update Sequence Policy API.	Optional	65.0
name	String	Name of the sequence policy.	Required	65.0
selection Condition	Selection Condition Input[]	Criteria to determine which sequence policy is applied to a record. This property includes conditions based on any standard or custom fields of the record.	Optional	65.0
selection Logic	String	Logic that determines the objects that the sequence policy applies to.	Optional	65.0
sequenceMode	String	Specifies how sequence numbers are generated. Valid values are: • Basic—Assigns sequential numbers without gap reconciliation. • Gapless—Assigns sequential numbers with gap reconciliation. The usage of this value ensures that the posted invoices or credit memos don't have any numbering gaps for audits and compliance.	Required	65.0

Name	Type	Description	Required or Optional	Available Version
		You can't edit this property by using the Update Sequence Policy API.		
`sequencePattern`	String	Pattern structure that's followed for the sequence.	Required	65.0
`sequenceStartNumber`	Integer	Starting value of the sequence number, which must be greater than or equal to `0`. The default value is `1`. You can't edit this property by using the Update Sequence Policy API.	Required	65.0
`targetObject`	String	Object that the policy is applied to. Valid values are: • `Invoice` • `CreditMemo`—Available in API version 66.0 and later. You can't edit this property by using the Update Sequence Policy API.	Required	65.0
`timeZone`	String	Time zone that's applicable for the sequence policy.	Optional	65.0

Response body for PATCH

Sequence Policy

Void Posted Credit Memo (POST)

Void a credit memo in posted state.

The API response returns a unique identifier. In case of any asynchronous errors, a RevenueTransactionErrorLog record is created.

Resource

```
/commerce/billing/credit-memos/creditMemoId/actions/void
```

Resource example

```
https://yourInstance.salesforce.com/services/data/v67.0/commerce/billing/credit-memos/50gVB0000003khRYAQ/actions/void
```

Available version

66.0

HTTP methods

POST

Path parameter for POST

Properties

Name	Type	Description	Required or Optional	Available Version
`creditMemoId`	String	ID of the posted credit memo to be voided.	Required	66.0

Request body for POST

As this API accepts a credit memo ID as a path parameter in the URL, specify an empty request body.

Response body for POST

Void Posted Credit Memo

Request Bodies

Learn more about the available request bodies of Billing APIs.

Address Input

Input representation of the details of an address.

Addresses Input

Input representation of the details of the addresses for calculating tax.

Batch Invoice Filter Criteria Input

Input representation of the filter criteria for an invoice batch run.

Batch Invoice Scheduler Input

Input representation of the details of the request to create an invoice scheduler.

Billing Schedule Recovery Input

Input representation of the details of the billing schedules to recover the associated invoice.

Context-Aware Billing Schedule

Input representation of the billing transaction details.

Standalone Billing Schedule Input

Input representation of the request to create a billing schedule based on transaction details. This representation includes the transaction and context service details.

Standalone Billing Schedule Metadata Input

Input representation of the metadata details to create a billing schedule. This representation includes the name of the context definition and context mapping along with the mapping details of the transaction, billing schedule, and billing schedule group.

Convert Negative Invoice Lines Input

Input representation of the details of the request to convert a list of negative invoice lines into a credit.

Credit Invoice Input

Input representation of the details of the request to create a credit memo.

Credit Invoice Line Input

Input representation of the details of the invoice lines to be credited.

Credit Invoice Line Tax Input

Input representation of the details of the tax lines to be created manually for the invoice line.

Credit Memo Addresses Input

Input representation of the details of the billing and shipping addresses.

Credit Memo Apply Input

Input representation of the request to apply a credit memo to an invoice.

Credit Memo Apply Application Input

Input representation of the request to specify one or more applications to apply a credit memo for, with each application representing an invoice.

Credit Memo Draft to Posted Input

Input representation of the request to post a draft credit memo.

Credit Memo Unapply Input

Input representation of the request to unapply a credit memo from an invoice.

Credit Memo Line Apply Input

Input representation of the details of the request to apply a credit memo line to an invoice line.

Credit Memo Line Application Input

Input representation of the request to specify one or more applications to apply a credit memo line for, with each application representing an invoice line.

Credit Memo Line Unapply Input

Input representation of the details of the request to unapply a credit memo line from an invoice line.

Customer Details Input

Input representation of the customer details for tax calculation.

Frequency Cadence Options

Input representation of the frequency cadence options for an invoice scheduler.

Graph Record for Invoice Ingestion

A Graph record is an object that's a part of the graph structure, representing both the fields and relationships among different objects. Each record in the graph can contain attributes, which are fields of the object, and references to other related records.

Invoice Draft To Posted Input

Input representation of the details of the draft invoice that's posted.

Invoice Estimated Tax Calculation Input

Details of the invoice for which the estimated tadsfsefx must be calculated.

Invoice Ingestion Input

Input representation of the details of the invoice to be processed. The details include the tax processing status, user preferences for tax callouts, and associated object graph representation.

Invoice Input for Ingestion

Input representation of the details of the invoice that must be generated for or ingested into Billing.

Invoice Input

Input representation of the details of the billing schedule.

Invoice Preview Input

Input representation of the details of the billing transaction that the preview invoices are generated for.

Line Item Input

Input representation of the details of the line item for tax calculation.

On-Demand Document Generation Input

Input representation of the details to generate a comprehensive on-demand document for specified record types.

Payment Line Apply Input

Input representation of the payment line details. This representation covers details on allocation of a payment to a specific invoice line. It also provides additional context through optional fields such as associated account and effective date.

Payment Line Unapply Input

Input representation of the payment line details. This representation covers fields that you can specify to revert a payment line application to their preapplication state.

Payment Run Batch Filter Criteria Input

Input representation of the filter criteria for an invoice batch run. This representation covers the criteria and sequence for filtering payment run details. It specifies the field and object names, comparison operations, and values to be used for filtering.

Payment Batch Scheduler Input

Input representation of the details of the request to create a payment scheduler. This representation sets the rules and timing for a payment scheduler, including match types, dates, frequency, and filter criteria.

Payment Scheduler Update Input

Input representation of the details of the request to update the status of a payment scheduler. This representation defines the status of a payment scheduler, which can be set to Active, Canceled, Draft, or Inactive.

Posted Invoice List Write-Off Input

Input representation of the request to write off a list of posted invoices. This representation includes the details of invoices that you want to write off.

Posted Invoice Write-Off Input

Input representation of the details of the request to write off a posted invoice. This representation includes invoice details such as invoice ID and reason for writing off invoices.

Refund Line Apply Input

Input representation of the details of a transaction refund request. This representation outlines the properties of a refund, including the refund amount and ID of the payment or credit memo record that the refund is applied to.

Resume Billing Input

Input representation of the details of the request to resume the billing operation for an account or a billing schedule group.

Resume Billing Object Input

Input representation of the details such as the ID of the account or billing schedule group along with the effective date. These details are used to start the billing operation.

Rules Application Input

Input representation for applying payments and credits to invoices based on rules.

Selection Condition Input

Input representation of the criteria that's used to determine which sequencing policy is applied to a record. The criteria stores the conditions based on any standard or custom fields of the record.

Seller Details Input

Input representation of the seller details for tax calculation.

Send Email Input

Input representation of the request to send an email for an invoice batch run.

Sequence Policy Input

Input representation of the configured rules and properties to generate unique, sequential numbers for objects.

Sequence Gap Reconciliation Input

Input representation of the details that are used to identify and reconcile gaps in sequence values based on the sequence policy or target object.

Sequences Assignment Input

Input representation of the details of the target objects to which the sequence pattern values are assigned.

Standalone Credit Memo Charge Input

Input representation of the details of the charge lines of a credit memo.

Standalone Credit Memo Input

Input representation of the details required to create a standalone credit memo.

Standalone Credit Memo Tax Input

Input representation of the details of the tax request.

Account Statement Input

Input representation of the details required to generate a comprehensive account statement with transaction history and balance information.

Suspend Billing Input

Input representation of the details of the request to suspend the billing operation for an account or a billing schedule group.

Suspend Billing Object Input

Input representation of the details such as the ID of the account or billing schedule group along with the effective dates. These details are used to suspend the billing operation.

Tax Calculation Input

Input representation of the details of the request to calculate tax.

Void Posted Credit Memo Input

Input representation of the details of a credit memo to be voided.

Void Posted Invoice Input

Input representation of the details of the invoice to be voided.

Address Input

Input representation of the details of an address.

JSON example

JSON example

```
"billingAddress": {
  "street": "1 Market St #300",
  "city": "San Francisco",
  "state": "CA",
  "country": "US",
  "postalCode": "94105",
  "latitude": "37.789901",
  "longitude": "-122.396923"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
city	String	Address city.	Optional	62.0
country	String	Address country.	Optional	62.0
latitude	Double	Latitude for the address.	Optional	62.0
locationCode	String	Location code for the address.	Optional	62.0
longitude	Double	Longitude for the address.	Optional	62.0
postalCode	String	Postal code for the address.	Optional	62.0
state	String	Address state.	Optional	62.0
street	String	Address street.	Optional	62.0

Addresses Input

Input representation of the details of the addresses for calculating tax.

JSON example


```
{
  "addresses": {
    "billTo": {
      "street": "123 Main Street",
      "city": "Bainbridge Island",
      "state": "WA",
      "postalCode": "98110",
      "country": "US"
    },
    "soldTo": {
      "street": "123 Main Street",
      "city": "Bainbridge Island",
      "state": "WA",
      "postalCode": "98110",
      "country": "US"
    },
    "shipFrom": {
      "street": "123 Alaskan Way",
      "city": "Seattle",
      "state": "WA",
      "country": "US",
      "postalCode": "98101"
    },
    "shipTo": {
      "street": "123 Main street",
      "city": "Bainbridge Island",
      "state": "WA",
      "postalCode": "98110",
```

```
        "country": "US"
      }
    }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
billTo	Address Input	Billing address of the item.	Optional	62.0
shipFrom	Address Input	Address that the item is shipped from.	Optional	62.0
shipTo	Address Input	Address that the item is shipped to.	Required	62.0
soldTo	Address Input	Address of the entity that the item is sold to.	Optional	62.0

Batch Invoice Filter Criteria Input

Input representation of the filter criteria for an invoice batch run.

JSON example

```
    "filterCriteria": [
        {
            "operation": "InList",
            "value": "Batch 2,Batch 3,Batch 4",
            "criteriaSequence": 1,
            "objectName": "BillingSchedule",
            "fieldName": "InvoiceRunMatchingValue"
        },
        {
            "operation": "Equals",
            "value": "001xx000003GZG5AAO",
            "criteriaSequence": 2,
            "objectName": "BillingSchedule",
            "fieldName": "BillingAccount"
        },
        {
            "operation": "Equals",
            "value": "0fwxx0000000001AAA",
            "criteriaSequence": 3,
            "objectName": "BillingScheduleGroup",
            "fieldName": "LegalEntity"
        },
        {
            "operation": "InList",
            "value": "OneTime,Recurring",
            "criteriaSequence": 4,
            "objectName": "BillingSchedule",
            "fieldName": "BillingTermUnit"
        },
        {
```

```
            "operation": "Equals",
            "value": "USD",
            "criteriaSequence": 5,
            "objectName": "BillingSchedule",
            "fieldName": "Currency_Iso_code"
        }
    ]
```

Properties

Name	Type	Description	Required or Optional	Available Version
`fieldName`	String	Name of the field that this filter is applicable for. Valid values are: • `Currency_Iso_code`—If multiple currencies are enabled for the org, then this value is required. • `InvoiceRunMatchingValue` • `BillingAccount` • `LegalEntity` • `BillingTermUnit`	Required	62.0
`objectName`	Object	Name of the object that the filter is applicable for. Valid values are: • `BillingSchedule` • `BillingScheduleGroup`	Required	62.0
`operation`	String	Operation to be performed for comparison. Valid values are: • `Equals` • `InList`—This value supports only the `InvoiceRunMatchingValue` and `BillingTermUnit` fields with API version 62.0 and later. In addition, this value supports the `CurrencyIsoCode` field with API version 63.0 and later. • `NotEquals`	Required	62.0
`value`	String	Value for the filter criteria.	Required	62.0

Batch Invoice Scheduler Input

Input representation of the details of the request to create an invoice scheduler.

JSON example

This example shows a sample request to create an invoice scheduler that generates invoices once.


```
{
  "schedulerName": "InvoiceScheduler",
  "startDate": "2024-05-06",
  "invoiceStatus": "POSTED",
  "preferredTime": "00:45",
  "targetDate": "2024-05-22",
  "invoiceDate": "2024-05-22",
  "frequencyCadence": "Once",
  "frequencyCadenceOptions": {},
  "timezone": "Asia/Kolkata",
  "status": "Active",
  "filterCriteria": [
    {
      "operation": "InList",
      "value": "Batch 2,Batch 3,Batch 4",
      "criteriaSequence": 1,
      "objectName": "BillingSchedule",
      "fieldName": "InvoiceRunMatchingValue"
    },
    {
      "operation": "Equals",
      "value": "001xx000003GZG5AAO",
      "criteriaSequence": 2,
      "objectName": "BillingSchedule",
      "fieldName": "BillingAccount"
    },
    {
      "operation": "Equals",
      "value": "0fwxx0000000001AAA",
      "criteriaSequence": 3,
      "objectName": "BillingScheduleGroup",
      "fieldName": "LegalEntity"
    },
    {
      "operation": "InList",
      "value": "OneTime,Recurring",
      "criteriaSequence": 4,
      "objectName": "BillingSchedule",
      "fieldName": "BillingTermUnit"
    },
    {
      "operation": "Equals",
      "value": "USD",
      "criteriaSequence": 5,
      "objectName": "BillingSchedule",
      "fieldName": "Currency_Iso_code"
    }
  ]
}
```

This example shows a sample request to create an invoice scheduler that generates invoices daily.

```
{
  "schedulerName": "InvoiceScheduler",
  "startDate": "2024-05-06",
  "endDate": "2026-05-06",
  "invoiceStatus": "POSTED",
  "preferredTime": "00:45",
  "targetDateOffset": 0,
  "invoiceDateOffset": 0,
  "isInvoiceDateFromRunDate": true,
  "frequencyCadence": "Daily",
  "frequencyCadenceOptions": {},
  "timezone": "Asia/Kolkata",
  "status": "Active",
  "filterCriteria": [
    {
      "operation": "InList",
      "value": "Batch 2,Batch 3,Batch 4",
      "criteriaSequence": 1,
      "objectName": "BillingSchedule",
      "fieldName": "InvoiceRunMatchingValue"
    },
    {
      "operation": "Equals",
      "value": "001xx000003GZG5AAO",
      "criteriaSequence": 2,
      "objectName": "BillingSchedule",
      "fieldName": "BillingAccount"
    },
    {
      "operation": "Equals",
      "value": "0fwxx0000000001AAA",
      "criteriaSequence": 3,
      "objectName": "BillingScheduleGroup",
      "fieldName": "LegalEntity"
    },
    {
      "operation": "InList",
      "value": "OneTime,Recurring",
      "criteriaSequence": 4,
      "objectName": "BillingSchedule",
      "fieldName": "BillingTermUnit"
    },
    {
      "operation": "InList",
      "value": "USD",
      "criteriaSequence": 5,
      "objectName": "BillingSchedule",
      "fieldName": "Currency_Iso_code"
    }
  ]
}
```

This example shows a sample request to create an invoice scheduler that generates invoices weekly.


```
{
  "schedulerName": "InvoiceScheduler",
  "startDate": "2024-05-06",
  "endDate": "2026-05-06",
  "invoiceStatus": "POSTED",
  "preferredTime": "00:45",
  "targetDateOffset": 0,
  "invoiceDateOffset": 0,
  "isInvoiceDateFromRunDate": false,
  "frequencyCadence": "Weekly",
  "frequencyCadenceOptions": {
    "recursOnDay": "Sunday"
  },
  "timezone": "Asia/Kolkata",
  "status": "Active",
  "filterCriteria": [
    {
      "operation": "InList",
      "value": "Batch 2,Batch 3,Batch 4",
      "criteriaSequence": 1,
      "objectName": "BillingSchedule",
      "fieldName": "InvoiceRunMatchingValue"
    },
    {
      "operation": "Equals",
      "value": "001xx000003GZG5AAO",
      "criteriaSequence": 2,
      "objectName": "BillingSchedule",
      "fieldName": "BillingAccount"
    },
    {
      "operation": "Equals",
      "value": "0fwxx0000000001AAA",
      "criteriaSequence": 3,
      "objectName": "BillingScheduleGroup",
      "fieldName": "LegalEntity"
    },
    {
      "operation": "InList",
      "value": "OneTime,Recurring",
      "criteriaSequence": 4,
      "objectName": "BillingSchedule",
      "fieldName": "BillingTermUnit"
    },
    {
      "operation": "Equals",
      "value": "USD",
      "criteriaSequence": 5,
      "objectName": "BillingSchedule",
      "fieldName": "Currency_Iso_code"
    }
  ]
}
```

This example shows a sample request to create an invoice scheduler that generates invoices monthly on a specific date.

```
{
  "schedulerName": "InvoiceScheduler",
  "startDate": "2024-05-06",
  "endDate": "2026-05-06",
  "invoiceStatus": "POSTED",
  "preferredTime": "00:45",
  "targetDateOffset": 0,
  "invoiceDateOffset": 0,
  "isInvoiceDateFromRunDate": false,
  "frequencyCadence": "Monthly",
  "frequencyCadenceOptions": {
    "recurringSubType": "SpecificDate",
    "recursOnDate": "L-1",
    "shouldExcludeWkendAndHldy": true
  },
  "timezone": "Asia/Kolkata",
  "status": "Active",
  "filterCriteria": [
    {
      "operation": "InList",
      "value": "Batch 2,Batch 3,Batch 4",
      "criteriaSequence": 1,
      "objectName": "BillingSchedule",
      "fieldName": "InvoiceRunMatchingValue"
    },
    {
      "operation": "Equals",
      "value": "001xx000003GZG5AAO",
      "criteriaSequence": 2,
      "objectName": "BillingSchedule",
      "fieldName": "BillingAccount"
    },
    {
      "operation": "Equals",
      "value": "0fwxx0000000001AAA",
      "criteriaSequence": 3,
      "objectName": "BillingScheduleGroup",
      "fieldName": "LegalEntity"
    },
    {
      "operation": "InList",
      "value": "OneTime,Recurring",
      "criteriaSequence": 4,
      "objectName": "BillingSchedule",
      "fieldName": "BillingTermUnit"
    },
    {
      "operation": "Equals",
      "value": "USD",
      "criteriaSequence": 5,
      "objectName": "BillingSchedule",
      "fieldName": "Currency_Iso_code"
    }
```

```
  ]
}
```

This example shows a sample request to create an invoice scheduler that runs immediately.

```
{
    "schedulerName": "InvoiceScheduler",
    "status": "Draft",
    "invoiceStatus": "Posted",
    "frequencyCadenceOptions": {
        "shouldStartRunImmediately": true
    },
    "frequencyCadence": "Once",
    "targetDate": "2024-08-28",
    "invoiceDate": "2024-08-28",
    "filterCriteria": [
        {
            "operation": "InList",
            "value": "Batch 2,Batch 3,Batch 4",
            "criteriaSequence": 1,
            "objectName": "BillingSchedule",
            "fieldName": "InvoiceRunMatchingValue"
        },
        {
            "operation": "Equals",
            "value": "001xx000003GZG5AAO",
            "criteriaSequence": 2,
            "objectName": "BillingSchedule",
            "fieldName": "BillingAccount"
        },
        {
            "operation": "Equals",
            "value": "0fwxx0000000001AAA",
            "criteriaSequence": 3,
            "objectName": "BillingScheduleGroup",
            "fieldName": "LegalEntity"
        },
        {
            "operation": "InList",
            "value": "OneTime,Recurring",
            "criteriaSequence": 4,
            "objectName": "BillingSchedule",
            "fieldName": "BillingTermUnit"
        },
        {
            "operation": "Equals",
            "value": "USD",
            "criteriaSequence": 5,
            "objectName": "BillingSchedule",
            "fieldName": "Currency_Iso_code"
        }
    ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
endDate	String	End date of the invoice scheduler.	Optional	63.0
filterCriteria	Batch Invoice Filter Criteria Input[]	List of line items of the filter criteria.	Optional	62.0
frequencyCadence	String	Frequency to run the invoice scheduler. Valid values are: • Once • Daily • Weekly • Monthly	Required	62.0
frequencyCadenceOptions	Frequency Cadence Options	Frequency cadence options for the invoice scheduler.	Required	62.0
invoiceDate	String	Date shown on the invoice. This date is also used for tax calculations.	Required if the frequencyCadence property is set to Once.	62.0
invoiceDateOffset	Integer	Offset applied to the target date, which is the number of days added to or subtracted from the invoice date, to calculate the updated invoice date.	Required if the frequencyCadence property is set to Daily, Weekly, or Monthly.	62.0
invoiceStatus	String	Status of the invoice that specifies the expected invoice status from an invoice batch run. Valid values are: • Draft • Posted	Required	62.0
isInvoiceDateFromRunDate	Boolean	Indicates whether the invoice date is applicable from the date when the invoice scheduler is run (true) or not (false).	Optional	63.0
preferredTime	String	Preferred time for the invoice batch run.	Required	62.0
schedulerName	String	Name of the invoice scheduler, which must be unique in your org.	Required	62.0
startDate	String	Start date of the invoice scheduler.	Required	62.0

Name	Type	Description	Required or Optional	Available Version
`status`	String	Status of the invoice scheduler. Valid values are: • `Draft` • `Active` • `Inactive`	Required	62.0
`targetDate`	String	Target date of the invoice batch run. Billing schedules having the next billing date before this date are picked up for invoicing. The target date must be less than or equal to the maximum allowed target date for the org.	Required if the `frequency Cadence` property is set to `Once`.	62.0
`target DateOffset`	Integer	Target date offset applied to the next invoice run date to calculate the target date. The offset is the number of days added to or subtracted from the next billing date.	Required if the `frequency Cadence` property is set to `Daily`, `Weekly`, or `Monthly`.	62.0
`timezone`	String	Time zone that's applicable for the invoice scheduler.	Optional	62.0

Billing Schedule Recovery Input

Input representation of the details of the billing schedules to recover the associated invoice.

JSON example

```
{
  "billingScheduleIds": ["44bDU00000000XXYAY"]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`billing ScheduleIds`	String[]	IDs of the billing schedules to recover the invoice for. You can recover one billing schedule per API request.	Required	62.0

Context-Aware Billing Schedule

Input representation of the billing transaction details.

JSON example

```
{
    "billingTransactionIds": [ "801xx000003H1H9AAK"]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`billingTransactionIds`	String[]	ID of the billing transaction. This property value is the ID of the order if the source of the billing request is for the Order object. If the order product associated with the specified order ID doesn't have an associated billing treatment ID, the API considers the default billing treatment ID. The generated billing schedule group has the default billing treatment ID. The API supports only one billing transaction ID in the input.	Required	62.0

Standalone Billing Schedule Input

Input representation of the request to create a billing schedule based on transaction details. This representation includes the transaction and context service details.

The Create Standalone Billing Schedules (POST) API uses the StandaloneBillingContext context definition to hydrate the context of the transaction. The context definition includes these mappings.

- The TransactionMapping maps the fields of the transaction to the attributes of the Transaction node.
- The BSGEntitiesMapping maps the attributes of the Billing Schedule node, the Billing Schedule Group node, and Billing Schedule Group Relationship node to the fields of the corresponding Salesforce objects.

For the StandaloneBillingContext context definition to hydrate all the required data, transaction data for the mandatory context tags are required. Here are the topics that mention the mandatory and optional tags, sample transaction details, and sample payloads for various types of transactions.

- One-Time New Sale Transaction on page 2600
- Term-Defined New Sale Transaction on page 2603
- Evergreen New Sale Transaction on page 2607
- New Sale Transaction With Bundled Products on page 2611
- New Sale Transaction With Ramped Products on page 2617
- New Sale Transaction With Usage Products on page 2622
- Amended Transaction on page 2626
- Renewal Transaction on page 2636
- Early Renewal Transaction on page 2641
- Canceled Transaction on page 2647

JSON example

```
{
  "transactionDetails": "{\"nodeName\": [{\"id\":\"801Az00000aynKZIAY\",
\"businessSobjectType\": \"Order\"}]}",
  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext",
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "OrderTransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`transaction ContextDetails`	Standalone Billing Schedule Metadata Input[]	Details of the context definition and its mappings that are used to hydrate the transaction data and save it in the appropriate Billing fields.	Required	64.0
`transaction Details`	String	Input JSON data that includes the ID of the transaction record for which the billing schedule must be created and other additional transaction details. The API request supports a single mapping ID. You can send separate requests for line items and line details by using their respective mapping IDs. However, this approach can result in duplicate billing schedules for the same line items and line details.	Required	64.0

One-Time New Sale Transaction

Understand the required values and key considerations before you create a billing schedule for a new sale transaction with the OneTime selling model type.

Term-Defined New Sale Transaction

Understand the required values and key considerations before you create a billing schedule for a new sale transaction with the TermDefined selling model type.

Evergreen New Sale Transaction

Understand the required values and key considerations before you create a billing schedule for a new sale transaction with the Evergreen selling model type.

New Sale Transaction With Bundled Products

Understand the required values and key considerations before you create a billing schedule for a new sale transaction with bundled products.

New Sale Transaction With Ramped Products

Understand the required values and key considerations before you create a billing schedule for a new sale transaction with ramped products.

New Sale Transaction With Usage Products

Understand the required values and key considerations before you create a billing schedule for a new sale transaction with usage-based products.

Amended Transaction

Learn how amendments update quantity, price, and end date in billing schedules along with delta-only price transactions and billing-frequency changes at the billing schedule level.

Renewal Transaction

Create a renewal billing schedule for termed subscriptions that continues without a gap, starting the day after the current billing schedule group ends and running for the same term length.

Early Renewal Transaction

Early Renewal ends the current term at an effective date and starts a renewed term to a new end date, creating two billing schedules in one request. The request creates a negative segment for the remaining original term and a positive segment for the renewal term.

Canceled Transaction

Cancel an entire billing schedule from a specified date. Specify only the cancellation date and billing schedule group identifier. The API computes the logic to split the cancellation across overlapping billing schedules and creates one cancel transaction per segment.

One-Time New Sale Transaction

Understand the required values and key considerations before you create a billing schedule for a new sale transaction with the OneTime selling model type.

Considerations

Keep these important considerations in mind when you provide the details for new sale transactions with the OneTime selling model type.

- The `BillingActionType__std` value for any new sale transaction must be `Add`.
- When a new sale transaction is for the OneTime selling model type, the SellingModelType value must be `OneTime`.
- If you don't provide the `TaxTreatmentId__std` value, the `BillingTreatmentId__std` value, or the `LegalEntityId__std` value, the default `TaxTreatmentId` for the transaction, `BillingTreatmentId`, and `LegalEntityId` of your Salesforce org is considered. If your org doesn't have any default values, an error occurs.
- If your transaction is for an sObject record, only specify the record ID as the id value in the `transactionDetails` property value. All the other transaction details are automatically fetched from the sObject record if those details are mapped in the context definition.

JSON example

This example includes the transaction data for a new sale transaction with the OneTime selling model type.

```
{
  "Transaction": [
    {
      "id": "ot1",
      "SellingModelType__std": "OneTime",
      "ParentTransactionId__std": "sample",
      "TransactionId__std": "ot1",
```

```
      "ProductName__std": "OneTimeProduct",
      "StartDate__std": "2025-02-20",
      "UnitPrice__std": 10,
      "Quantity__std": 5,
      "TotalPrice__std": 50,
      "TaxTreatmentId__std": "1ttLT000000EV7DYAW",
      "BillingTreatmentId__std": "1BTLT00000008yT4AQ",
      "AccountId__std": "001LT00000dPVrlYAG",
      "BillingProfileId__std": "001LT00000dPVrlYAG",
      "BillingCity__std": "HYD",
      "ShippingCity__std": "VSKP",
      "BillingActionType__std": "Add",
      "CurrencyIsoCode__std": "USD"
    }
  ]
}
```

This example shows the request payload to create a billing schedule for a new sale transaction with the OneTime selling model type.

```
{
  "transactionDetails":
[illegible]

  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext__stdctx",
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "TransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```

When you're creating a billing schedule for a new sale transaction with the OneTime selling model type, make sure that you specify the required values in the `transactionDetails` property value.

Context Tag in the Transaction Node of the StandaloneBillingContext Context Definition	Description	Mapped Field	Required or Optional
`ParentTransactionId__std`	The ID of the parent transaction record. For example, if the transaction is at a level similar to that of an Order Item record, the parent transaction will be at a level similar to that of an Order record.	If the transaction is a child Order or Quote record, the ReferenceEntityId field on the BillingScheduleobject is populated. For other sObject records or external records, the Reference field on the BillingSchedule object is populated.	Optional

Context Tag in the Transaction Node of the StandaloneBillingContext Context Definition	Description	Mapped Field	Required or Optional
`AccountId__std`	The ID of the Account record that's related to the transaction.	BillingAccountId field on the BillingScheduleGroup object	Required
`BillingProfileId__std`	The ID of the billing profile that’s related to the transaction, which is used to resolve billing configurations if not specified at transaction level. See Billing Profile requirements.	Not applicable.	Optional
`BillingCity__std`	The city in the billing address of the transaction.	BillingCity field on the BillingScheduleGroup object	Any one of the billing address fields is required.
`ShippingCity__std`	The city in the shipping address of the transaction.	ShippingCity field on the BillingScheduleGroup object	Any one of the shipping address fields is required.
`SellingModelType__std`	The selling model type indicates whether the transaction is for a one-time product, a term-defined product, or an evergreen product. Specify `OneTime` as the SellingModelType for the transaction that's related to a product that's sold once.	BillingMethod on the BillingSchedule object is populated as OrderAmount.	Required
`TransactionId__std`	The ID of the transaction that you want to create a billing schedule for.	If the transaction is an OrderItem, OrderItemAdjustmentLineItem, OrderItemDetail, QuoteLineDetail, or QuoteLineItem record, the ReferenceEntityItemId field on the BillingScheduleobject is populated. For other sObject records or external records, the ReferenceItem field on the BillingSchedule object is populated.	Required
`ProductName__std`	The name of the product that's related to the transaction that you want to create a billing schedule for.	ProductName field on the BillingScheduleGroup object	Either the ProductName or the ProductId is required
`StartDate__std`	The start date of the transaction.	BillingScheduleStartDate field on the BillingSchedule object	Required

Context Tag in the Transaction Node of the StandaloneBillingContext Context Definition	Description	Mapped Field	Required or Optional
`Quantity__std`	The quantity of the transaction.	Quantity field on the BillingSchedule object	Required
`TaxTreatmentId__std`	The ID of the tax treatment that's used to calculate tax for the transaction. If you don't specify a TaxTreatmentId, the org-default TaxTreatmentId is considered.	TaxTreatmentId field on the BillingScheduleGroup object	Required
`BillingTreatmentId__std`	The ID of the billing treatment that's used to create the billing schedule for the transaction. If you don't specify a BillingTreatmentId, the org-default BillingTreatmentId is considered.	BillingTreatmentId field on the BillingScheduleGroup object	Required
`CurrencyIsoCode__std`	The currency code of the transaction amount if your Salesforce org has multi-currency enabled.	CurrencyIsoCode field on the BillingScheduleGroup object	Required for Salesforce orgs with multiple currencies enabled
`BillingActionType__std`	The action that you want to perform for the transaction. Valid value is `Add`.	Category field on the BillingSchedule object is populated as Original.	Required
`UnitPrice__std`	The unit price of the transaction.	UnitPrice field on the BillingSchedule object	Either the UnitPrice or the NetUnitPrice value is required
`TotalPrice__std`	The total price of the transaction.	TotalAmount field on the BillingSchedule object	Required

SEE ALSO:

BillingSchedule
BillingScheduleGroup
BsgRelationship

Term-Defined New Sale Transaction

Understand the required values and key considerations before you create a billing schedule for a new sale transaction with the TermDefined selling model type.

Considerations

Keep these important considerations in mind when you provide the details for new sale transactions with the TermDefined selling model type.

- The `BillingActionType__std` value for any new sale transaction must be `Add`.
- When a new sale transaction is for the TermDefined selling model type, the SellingModelType value must be `TermDefined`.

- When a new sale transaction is for the TermDefined selling model type, the `BillingTermUnit__std` and `PeriodBoundary__std` values are required.
- If you don't provide the `TaxTreatmentId__std` value, the `BillingTreatmentId__std` value, or the `LegalEntityId__std` value for the transaction, the default `TaxTreatmentId`, `BillingTreatmentId`, and `LegalEntityId` of your Salesforce org is considered. If your org doesn't have any default values, an error occurs.
- If your transaction is for an sObject record, only specify the record ID as the id value in the transactionDetails property value. All the other transaction details are automatically fetched from the sObject record if those details are mapped in the context definition.

JSON example

This example includes the transaction data for a new sale transaction with the TermDefined selling model type.

```
{
  "Transaction": [
    {
      "id": "ter1",
      "SellingModelType__std": "TermDefined",
      "ParentTransactionId__std": "sample",
      "TransactionId__std": "ter1",
      "ProductName__std": "TermedQuarterlyProduct",
      "StartDate__std": "2025-01-01",
      "EndDate__std": "2025-12-31",
      "PeriodBoundary__std": "DayOfPeriod",
      "BillingDayOfMonth__std": 1,
      "UnitPrice__std": 10,
      "Quantity__std": 5,
      "TotalPrice__std": 120,
      "TaxTreatmentId__std": "1ttLT000000EV7DYAW",
      "BillingTreatmentId__std": "1BTLT00000008yT4AQ",
      "AccountId__std": "001LT00000dPVrlYAG",
      "BillingProfileId__std": "001LT00000dPVrlYAG",
      "BillingCity__std": "HYD",
      "ShippingCity__std": "VSKP",
      "BillingTermUnit__std": "Month",
      "BillingActionType__std": "Add",
      "CurrencyIsoCode__std": "USD"
    }
  ]
}
```

This example shows the request payload to create a billing schedule for a new sale transaction with the TermDefined selling model type.

```
{
  "transactionDetails":
[illegible]

  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext__stdctx",
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "TransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```

When you're creating a billing schedule for a new sale transaction with the TermDefined selling model type, make sure that you specify the mandatory values in the `transactionDetails` property value.

Context Tag in the Transaction Node of the StandaloneBillingContext Context Definition	Description	Mapped Field	Required or Optional
`ParentTransactionId__std`	The ID of the parent transaction record. For example, if the transaction is at a level similar to that of an Order Item record, the parent transaction will be at a level similar to that of an Order record.	If the transaction is a child Order or Quote record, the ReferenceEntityId field on the BillingScheduleobject is populated. For other sObject records or external records, the Reference field on the BillingSchedule object is populated.	Required
`AccountId__std`	The ID of the Account record that's related to the transaction.	BillingAccountId field on the BillingScheduleGroup object	Required
`BillingProfileId__std`	The ID of the billing profile (Billing Account record) that's related to the transaction. See Billing Profile requirements.	BillingAccountId field on the BillingSchedule object	Optional
`BillingCity__std`	The city in the billing address of the transaction.	BillingCity field on the BillingScheduleGroup object	Any one of the billing address fields is required.
`ShippingCity__std`	The city in the shipping address of the transaction.	ShippingCity field on the BillingScheduleGroup object	Any one of the shipping address fields is required.
`SellingModelType__std`	The selling model type indicates whether the transaction is for a one-time product, a term-defined product, or an evergreen product. Specify `TermDefined` as the SellingModelType for the transaction that's related to a product that's sold for a specific term.	BillingMethod on the BillingSchedule object is populated as OrderAmount.	Required
`TransactionId__std`	The ID of the transaction that you want to create a billing schedule for.	If the transaction is an OrderItem, OrderItemAdjustmentLineItem, OrderItemDetail, QuoteLineDetail, or QuoteLineItem record, the ReferenceEntityItemId	Required

Context Tag in the Transaction Node of the StandaloneBillingContext Context Definition	Description	Mapped Field	Required or Optional
		field on the BillingScheduleobject is populated. For other sObject records or external records, the ReferenceItem field on the BillingSchedule object is populated.	
`ProductName__std`	The name of the product that's related to the transaction that you want to create a billing schedule for.	ProductName field on the BillingScheduleGroup object	Either the ProductName or the ProductId is required
`PeriodBoundary__std`	The period boundary determines the start and end date of the billing period. Valid values are: • `AlignToCalendar` • `Anniversary` • `DayOfPeriod` • `LastDayOfPeriod`	PeriodBoundary field on the BillingScheduleGroup object	Required
`BillingDayOfMonth__std`	The day of the month on which a recurring billing process is scheduled to occur for the transaction.	BillDayOfMonth field on the BillingScheduleGroup object	Required
`EndDate__std`	The end date of the transaction.	BillingScheduleEndDate field on the BillingSchedule object	Required
`StartDate__std`	The start date of the transaction.	BillingScheduleStartDate field on the BillingSchedule object	Required
`Quantity__std`	The quantity of the transaction.	Quantity field on the BillingSchedule object	Required
`TaxTreatmentId__std`	The ID of the tax treatment that's used to calculate tax for the transaction. If you don't specify a TaxTreatmentId, the org-default TaxTreatmentId is considered.	TaxTreatmentId field on the BillingScheduleGroup object	Required
`BillingTermUnit__std`	The unit of measurement of the billing term. Valid values are: • `Month` • `Quarter`	BillingTermUnit field on the BillingSchedule object	Required

Context Tag in the Transaction Node of the StandaloneBillingContext Context Definition	Description	Mapped Field	Required or Optional
	• Semi-Annual • Year		
BillingTreatmentId__std	The ID of the billing treatment that's used to create the billing schedule for the transaction. If you don't specify a BillingTreatmentId, the org-default BillingTreatmentId is considered.	BillingTreatmentId field on the BillingScheduleGroup object	Required
CurrencyIsoCode__std	The currency code of the transaction amount if your Salesforce org has multi-currency enabled.	CurrencyIsoCode field on the BillingScheduleGroup object	Required for Salesforce orgs with multiple currencies enabled
BillingActionType__std	The action that you want to perform for the transaction. Valid value is Add.	Category field on the BillingSchedule object is populated as Original.	Required
UnitPrice__std	The unit price of the transaction.	UnitPrice field on the BillingSchedule object	Either the UnitPrice or the NetUnitPrice value is required
TotalPrice__std	The total cost to the customer for the transaction, from the start date to the end date.	TotalAmount field on the BillingSchedule object	Required

SEE ALSO:

BillingSchedule

BillingScheduleGroup

BsgRelationship

Evergreen New Sale Transaction

Understand the required values and key considerations before you create a billing schedule for a new sale transaction with the Evergreen selling model type.

Considerations

Keep these important considerations in mind when you provide the details for new sale transactions with the Evergreen selling model type.

- The `BillingActionType__std` value for any new sale transaction must be `Add`.
- When a new sale transaction is for the Evergreen selling model type, the SellingModelType value must be `Evergreen`.
- When a new sale transaction is for the Evergreen selling model type, the BillingTermUnit and PeriodBoundary values are required.
- If you don't provide the `TaxTreatmentId__std` value, the `BillingTreatmentId__std` value, or the `LegalEntityId__std` value for the transaction, the default `TaxTreatmentId`, `BillingTreatmentId`, and `LegalEntityId` of your Salesforce org is considered. If your org doesn't have any default values, an error occurs.

- If your transaction is for an sObject record, only specify the record ID as the id value in the transactionDetails property value. All the other transaction details are automatically fetched from the sObject record if those details are mapped in the context definition.

JSON example

This example includes the transaction data for a new sale transaction with the Evergreen selling model type.

```
{
  "Transaction": [
    {
      "id": "evg1",
      "SellingModelType__std": "Evergreen",
      "ParentTransactionId__std": "sample",
      "TransactionId__std": "evg1",
      "ProductName__std": "EvergreenSemiAnnualProduct",
      "StartDate__std": "2025-01-01",
      "PeriodBoundary__std": "DayOfPeriod",
      "BillingDayOfMonth__std": 1,
      "BillingStartMonth__std": 3,
      "UnitPrice__std": 10,
      "Quantity__std": 1,
      "TotalPrice__std": 10,
      "TaxTreatmentId__std": "1ttLT000000EV7DYAW",
      "BillingTreatmentId__std": "1BTLT00000008yT4AQ",
      "AccountId__std": "001LT00000dPVrlYAG",
      "BillingProfileId__std": "001LT00000dPVrlYAG",
      "BillingCity__std": "HYD",
      "ShippingCity__std": "VSKP",
      "BillingTermUnit__std": "Semi-Annual",
      "BillingActionType__std": "Add",
      "CurrencyIsoCode__std": "USD"
    }
  ]
}
```

This example shows the request payload to create a billing schedule for a new sale transaction with the Evergreen selling model type.

```
{
  "transactionDetails":
[illegible]

  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext__stdctx",
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "TransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```

When you're creating a billing schedule for a new sale transaction with the Evergreen selling model type, make sure that you specify the mandatory values in the `transactionDetails` property value.

Context Tag in the Transaction Node of the StandaloneBillingContext Context Definition	Description	Mapped Field	Required or Optional
`ParentTransactionId__std`	The ID of the parent transaction record. For example, if the transaction is at a level similar to that of an Order Item record, the parent transaction will be at a level similar to that of an Order record.	If the transaction is a child Order or Quote record, the ReferenceEntityId field on the BillingScheduleobject is populated. For other sObject records or external records, the Reference field on the BillingSchedule object is populated.	Optional
`AccountId__std`	The ID of the Account record that's related to the transaction.	BillingAccountId field on the BillingScheduleGroup object	Required
`BillingProfileId__std`	The ID of the billing profile (Billing Account record) that's related to the transaction. See Billing Profile requirements.	BillingAccountId field on the BillingSchedule object	Optional
`BillingCity__std`	The city in the billing address of the transaction.	BillingCity field on the BillingScheduleGroup object	Any one of the billing address fields is required.
`ShippingCity__std`	The city in the shipping address of the transaction.	ShippingCity field on the BillingScheduleGroup object	Any one of the shipping address fields is required.
`SellingModelType__std`	The selling model type indicates whether the transaction is for a one-time product, a term-defined product, or an evergreen product. Specify `Evergreen` as the SellingModelType for the transaction that's related to a product that's sold for an indefinite term.	BillingMethod on the BillingSchedule object is populated as OrderAmount.	Required
`TransactionId__std`	The ID of the transaction that you want to create a billing schedule for.	If the transaction is an OrderItem, OrderItemAdjustmentLineItem, OrderItemDetail, QuoteLineDetail, or QuoteLineItem record, the ReferenceEntityItemId field on the BillingScheduleobject is populated. For other	Required

Context Tag in the Transaction Node of the StandaloneBillingContext Context Definition	Description	Mapped Field	Required or Optional
		sObject records or external records, the ReferenceItem field on the BillingSchedule object is populated.	
`ProductName__std`	The name of the product that's related to the transaction that you want to create a billing schedule for.	ProductName field on the BillingScheduleGroup object	Either the ProductName or the ProductId is required
`PeriodBoundary__std`	The period boundary determines the start and end date of the billing period. Valid values are: • `AlignToCalendar` • `Anniversary` • `DayOfPeriod` • `LastDayOfPeriod`	PeriodBoundary field on the BillingScheduleGroup object	Required
`BillingDayOfMonth__std`	The day of the month on which a recurring billing process is scheduled to occur for the transaction.	BillDayOfMonth field on the BillingScheduleGroup object	Required
`BillingStartMonth__std`	The month when billing begins for an annual subscription. This value can be any number from `1` through `12`. For example, if billing starts in January, the value is `1`. If billing starts in June, the value is `6`.	BillingStartMonth field on the BillingScheduleGroup object	Required
`StartDate__std`	The start date of the transaction.	BillingScheduleStartDate field on the BillingSchedule object	Required
`Quantity__std`	The quantity of the transaction.	Quantity field on the BillingSchedule object	Required
`TaxTreatmentId__std`	The ID of the tax treatment that's used to calculate tax for the transaction. If you don't specify a TaxTreatmentId, the org-default TaxTreatmentId is considered.	TaxTreatmentId field on the BillingScheduleGroup object	Required
`BillingTermUnit__std`	The unit of measurement of the billing term. Valid values are: • `Month` • `Quarter` • `Semi-Annual` • `Year`	BillingTermUnit field on the BillingSchedule object	Required

Context Tag in the Transaction Node of the StandaloneBillingContext Context Definition	Description	Mapped Field	Required or Optional
BillingTreatmentId__std	The ID of the billing treatment that's used to create the billing schedule for the transaction. If you don't specify a BillingTreatmentId, the org-default BillingTreatmentId is considered.	BillingTreatmentId field on the BillingScheduleGroup object	Required
CurrencyIsoCode__std	The currency code of the transaction amount if your Salesforce org has multi-currency enabled.	CurrencyIsoCode field on the BillingScheduleGroup object	Required for Salesforce orgs with multiple currencies enabled
BillingActionType__std	The action that you want to perform for the transaction. Valid value is `Add`.	Category field on the BillingSchedule object is populated as Original.	Required
UnitPrice__std	The unit price of the transaction.	UnitPrice field on the BillingSchedule object	Either the UnitPrice or the NetUnitPrice value is required
TotalPrice__std	The total cost to the customer for the transaction for each billing period, such as month, quarter, or year, based on the billing frequency.	TotalAmount field on the BillingSchedule object	Required

SEE ALSO:

BillingSchedule

BillingScheduleGroup

BsgRelationship

New Sale Transaction With Bundled Products

Understand the required values and key considerations before you create a billing schedule for a new sale transaction with bundled products.

Considerations

Keep these important considerations in mind when you provide the details for new sale transactions with bundled products.

- The `BillingActionType__std` value for any new sale transaction must be `Add`.
- For new sale transactions with bundled products, the `MainTransactionId__std` value is required for all the child transactions.
- If you don't provide the `TaxTreatmentId__std` value, the `BillingTreatmentId__std` value, or the `LegalEntityId__std` value for the transaction, the default `TaxTreatmentId`, `BillingTreatmentId`, and `LegalEntityId` of your Salesforce org is considered. If your org doesn't have any default values, an error occurs.
- If your transaction is for an sObject record, only specify the record ID as the id value in the `transactionDetails` property value. All the other transaction details are automatically fetched from the sObject record if those details are mapped in the context definition.
- To understand the requirements for the various selling model types, see these resources.

- One-Time New Sale Transaction on page 2600
- Term-Defined New Sale Transaction on page 2603
- Evergreen New Sale Transaction on page 2607

JSON example

This sample is for a new sale transaction for bundled products with two levels of nesting and with these selling model types.

- The main root product's selling model type is `OneTime`.
- The second root product's selling model type is `OneTime`.
- The child products' selling model types are `TermDefined` and `Evergreen`.

```
{
  "Transaction": [
    {
      "id": "sample1",
      "SellingModelType__std": "OneTime",
      "ParentTransactionId__std": "sample",
      "TransactionId__std": "sample1",
      "ProductName__std": "OneTimeProduct",
      "StartDate__std": "2025-02-20",
      "UnitPrice__std": 10,
      "Quantity__std": 1,
      "TotalPrice__std": 10,
      "TaxTreatmentId__std": "1ttLT000000EV7DYAW",
      "BillingTreatmentId__std": "1BTLT00000008yT4AQ",
      "AccountId__std": "001LT00000dPVrlYAG",
      "BillToContactId__std": "003LT0000038bALYAY",
      "BillingCity__std": "HYD",
      "ShippingCity__std": "VSKP",
      "BillingActionType__std": "Add",
      "CurrencyIsoCode__std": "USD"
    },
    {
      "id": "sample2",
      "SellingModelType__std": "OneTime",
      "ParentTransactionId__std": "abc",
      "TransactionId__std": "sample2",
      "ProductName__std": "OneTimeProduct",
      "StartDate__std": "2025-02-20",
      "UnitPrice__std": 10,
      "Quantity__std": 1,
      "TotalPrice__std": 10,
      "TaxTreatmentId__std": "1ttLT000000EV7DYAW",
      "BillingTreatmentId__std": "1BTLT00000008yT4AQ",
      "AccountId__std": "001LT00000dPVrlYAG",
      "BillToContactId__std": "003LT0000038bALYAY",
      "BillingActionType__std": "Add",
      "CurrencyIsoCode__std": "USD",
      "BillingCity__std": "HYD",
      "ShippingCity__std": "VSKP",
      "AssociatedTransactionPricing__std": "IncludedInBundlePrice",
      "MainTransactionRole__std": "Bundle",
      "AssociatedTransactionRole__std": "BundleComponent",
```

```
      "MainTransactionId__std": "sample1"
    },
    {
      "id": "sample3",
      "SellingModelType__std": "TermDefined",
      "ParentTransactionId__std": "sample",
      "TransactionId__std": "sample3",
      "ProductName__std": "TermedQuarterlyProduct",
      "StartDate__std": "2025-01-01",
      "EndDate__std": "2025-12-31",
      "PeriodBoundary__std": "DayOfPeriod",
      "BillingDayOfMonth__std": 1,
      "BillingStartMonth__std": 2,
      "UnitPrice__std": 10,
      "Quantity__std": 1,
      "TotalPrice__std": 120,
      "TaxTreatmentId__std": "1ttLT000000EV7DYAW",
      "BillingTreatmentId__std": "1BTLT00000008yT4AQ",
      "AccountId__std": "001LT00000dPVrlYAG",
      "BillToContactId__std": "003LT0000038bALYAY",
      "BillingTermUnit__std": "Quarter",
      "BillingActionType__std": "Add",
      "CurrencyIsoCode__std": "USD",
      "BillingCity__std": "HYD",
      "ShippingCity__std": "VSKP",
      "AssociatedTransactionPricing__std": "IncludedInBundlePrice",
      "MainTransactionRole__std": "Bundle",
      "AssociatedTransactionRole__std": "BundleComponent",
      "MainTransactionId__std": "sample2"
    },
    {
      "id": "sample4",
      "SellingModelType__std": "Evergreen",
      "ParentTransactionId__std": "sample",
      "TransactionId__std": "sample4",
      "ProductName__std": "EvergreenSemiAnnualProduct",
      "StartDate__std": "2025-01-01",
      "PeriodBoundary__std": "DayOfPeriod",
      "BillingDayOfMonth__std": 1,
      "BillingStartMonth__std": 3,
      "UnitPrice__std": 10,
      "Quantity__std": 1,
      "TotalPrice__std": 10,
      "TaxTreatmentId__std": "1ttLT000000EV7DYAW",
      "BillingTreatmentId__std": "1BTLT00000008yT4AQ",
      "AccountId__std": "001LT00000dPVrlYAG",
      "BillToContactId__std": "003LT0000038bALYAY",
      "BillingTermUnit__std": "Semi-Annual",
      "BillingActionType__std": "Add",
      "BillingPostalCode__std": "94105",
      "CurrencyIsoCode__std": "USD",
      "BillingCity__std": "HYD",
      "ShippingCity__std": "VSKP",
      "AssociatedTransactionPricing__std": "IncludedInBundlePrice",
```

```
      "MainTransactionRole__std": "Bundle",
      "AssociatedTransactionRole__std": "BundleComponent",
      "MainTransactionId__std": "sample2"
    }
  ]
}
```


This example shows the request payload to create a billing schedule for a new sale transaction for bundled products.


```
{
  "transactionDetails":
[redacted]

  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext__stdctx",
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "TransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```


When you're creating a billing schedule for a new sale transaction with bundled products, make sure that you specify the mandatory values in the `transactionDetails` property value.

Context Tag in the Transaction Node of the StandaloneBillingContext Context Definition	Description	Mapped Field	Required or Optional
`BillToContactId__std`	The ID of the Contact record that's related to the transaction.	BillToContactId field on the BillingScheduleGroup object	Required
`ParentTransactionId__std`	The ID of the parent transaction record. For example, if the transaction is at a level similar to that of an Order Item record, the parent transaction will be at a level similar to that of an Order record.	If the transaction is a child Order or Quote record, the ReferenceEntityId field on the BillingScheduleobject is populated. For other sObject records or external records, the Reference field on the BillingSchedule object is populated.	Optional
`AccountId__std`	The ID of the Account record that's related to the transaction.	BillingAccountId field on the BillingScheduleGroup object	Required
`BillingCity__std`	The city in the billing address of the transaction.	BillingCity field on the BillingScheduleGroup object	Any one of the billing address fields is required.

Context Tag in the Transaction Node of the StandaloneBillingContext Context Definition	Description	Mapped Field	Required or Optional
`BillingPostalCode__std`	The postal code in the billing address of the transaction.	BillingPostalCode field on the BillingScheduleGroup object	Any one of the billing address fields is required.
`ShippingCity__std`	The city in the shipping address of the transaction.	ShippingCity field on the BillingScheduleGroup object	Any one of the shipping address fields is required.
`SellingModelType__std`	The selling model type indicates whether the transaction is for a one-time product, a term-defined product, or an evergreen product. Specify `OneTime` as the SellingModelType for the transaction that's related to a product that's sold once.	BillingMethod on the BillingSchedule object is populated as OrderAmount.	Required
`TransactionId__std`	The ID of the transaction that you want to create a billing schedule for.	If the transaction is an OrderItem, OrderItemAdjustmentLineItem, OrderItemDetail, QuoteLineDetail, or QuoteLineItem record, the ReferenceEntityItemId field on the BillingScheduleobject is populated. For other sObject records or external records, the ReferenceItem field on the BillingSchedule object is populated.	Required
`ProductName__std`	The name of the product that's related to the transaction that you want to create a billing schedule for.	ProductName field on the BillingScheduleGroup object	Either the ProductName or the ProductId value is required
`PeriodBoundary__std`	The period boundary determines the start and end date of the billing period. Valid values are: • `AlignToCalendar` • `Anniversary` • `DayOfPeriod` • `LastDayOfPeriod`	PeriodBoundary field on the BillingScheduleGroup object	Required for child products with the `TermDefined` or `Evergreen` selling model types.
`BillingDayOfMonth__std`	The day of the month on which a recurring billing process is scheduled to occur for the transaction.	BillDayOfMonth field on the BillingScheduleGroup object	Required for child products with the `TermDefined` or

Context Tag in the Transaction Node of the StandaloneBillingContext Context Definition	Description	Mapped Field	Required or Optional
			`Evergreen` selling model types.
`BillingStartMonth__std`	The month when billing begins for an annual subscription. This value can be any number from `1` through `12`. For example, if billing starts in January, the value is `1`. If billing starts in June, the value is `6`.	BillingStartMonth field on the BillingScheduleGroup object	Required for child products with the `Evergreen` selling model type.
`EndDate__std`	The end date of the transaction.	BillingScheduleEndDate field on the BillingSchedule object	Required for child products with the `TermDefined` selling model type.
`StartDate__std`	The start date of the transaction.	BillingScheduleStartDate field on the BillingSchedule object	Required
`Quantity__std`	The quantity of the transaction.	Quantity field on the BillingSchedule object	Required
`TaxTreatmentId__std`	The ID of the tax treatment that's used to calculate tax for the transaction. If you don't specify a TaxTreatmentId, the org-default TaxTreatmentId is considered.	TaxTreatmentId field on the BillingScheduleGroup object	Required
`BillingTermUnit__std`	The unit of measurement of the billing term. Valid values are: • `Month` • `Quarter` • `Semi-Annual` • `Year`	BillingTermUnit field on the BillingSchedule object	Required for child products with the `TermDefined` or `Evergreen` selling model types.
`BillingTreatmentId__std`	The ID of the billing treatment that's used to create the billing schedule for the transaction. If you don't specify a BillingTreatmentId, the org-default BillingTreatmentId is considered.	BillingTreatmentId field on the BillingScheduleGroup object	Required
`CurrencyIsoCode__std`	The currency code of the transaction amount if your Salesforce org has multi-currency enabled.	CurrencyIsoCode field on the BillingScheduleGroup object	Required for Salesforce orgs with multiple currencies enabled.
`BillingActionType__std`	The action that you want to perform for the transaction. Valid value is `Add`.	Category field on the BillingSchedule object is populated as Original.	Required

Context Tag in the Transaction Node of the StandaloneBillingContext Context Definition	Description	Mapped Field	Required or Optional
`UnitPrice__std`	The unit price of the transaction.	UnitPrice field on the BillingSchedule object	Either the UnitPrice or the NetUnitPrice value is required
`TotalPrice__std`	The total price of the transaction.	TotalAmount field on the BillingSchedule object	Required
`MainTransactionId__std`	When your transaction is part of a bundle, this value is the ID of the main transaction record for all the child transactions.	MainBsgId field on the BsgRelationship object	Required for all the child products.
`MainTransactionRole__std`	When your transaction is part of a bundle, this value is the role of the primary transaction in the bundle relationship. Valid values are: • `AddOn` • `Bundle` • `Set`	MainBsgRole field on the BsgRelationship object	Required for all the child products.
`AssociatedTransactionRole__std`	When your transaction is part of a bundle, this value is the role of the child transaction in the bundle relationship. Valid values are: • • `BundleComponent` • `ClassificationComponent`	AssociatedBsgRole field on the BsgRelationship object	Required for all the child products.
`AssociatedTransactionPricing__std`	When your transaction is part of a bundle, this value describes how the child transaction is priced in relation to the primary transaction. Valid values are: • `IncludedInBundlePrice` • `NotIncludedInBundlePrice`	AssociatedBsgPricing field on the BsgRelationship object	Required for all the child products.

SEE ALSO:

BillingSchedule

BillingScheduleGroup

BsgRelationship

New Sale Transaction With Ramped Products

Understand the required values and key considerations before you create a billing schedule for a new sale transaction with ramped products.

Considerations

Keep these important considerations in mind when you provide the details for new sale transactions with ramped products.

- The `BillingActionType__std` value for any new sale transaction must be `Add`.
- For new sale transactions with a ramped product, the `RampIdentifier__std` value is required.
- If you don't provide the `TaxTreatmentId__std` value, the `BillingTreatmentId__std` value, or the `LegalEntityId__std` value for the transaction, the default `TaxTreatmentId`, `BillingTreatmentId`, and `LegalEntityId` of your Salesforce org is considered. If your org doesn't have any default values, an error occurs.
- The billing schedules for transactions with the same `RampIdentifier__std` value are created for the same billing schedule group.
- To understand the requirements for the various selling model types, see these resources.
  - One-Time New Sale Transaction on page 2600
  - Term-Defined New Sale Transaction on page 2603
  - Evergreen New Sale Transaction on page 2607

JSON example

This example includes the transaction data for a new sale transaction with ramped products and a `TermDefined` selling model.


```
{
  "Transaction": [
    {
      "id": "ter1",
      "SellingModelType__std": "TermDefined",
      "ParentTransactionId__std": "sample",
      "TransactionId__std": "ter1",
      "ProductName__std": "TermedQuarterlyProduct",
      "StartDate__std": "2025-01-01",
      "EndDate__std": "2026-12-31",
      "PeriodBoundary__std": "DayOfPeriod",
      "BillingDayOfMonth__std": 1,
      "UnitPrice__std": 10,
      "Quantity__std": 5,
      "TotalPrice__std": 120,
      "TaxTreatmentId__std": "1ttLT000000EV7DYAW",
      "BillingTreatmentId__std": "1BTLT00000008yT4AQ",
      "AccountId__std": "001LT00000dPVrlYAG",
      "BillingProfileId__std": "001LT00000dPVrlYAG",
      "BillingCity__std": "HYD",
      "ShippingCity__std": "VSKP",
      "BillingTermUnit__std": "Month",
      "BillingActionType__std": "Add",
      "CurrencyIsoCode__std": "USD",
      "RampIdentifier__std": "ramp1"
    },
    {
      "id": "ter2",
      "SellingModelType__std": "TermDefined",
      "ParentTransactionId__std": "sample",
      "TransactionId__std": "ter2",
      "ProductName__std": "TermedQuarterlyProduct",
      "StartDate__std": "2025-01-01",
```

```
        "EndDate__std": "2026-12-31",
        "PeriodBoundary__std": "DayOfPeriod",
        "BillingDayOfMonth__std": 1,
        "UnitPrice__std": 10,
        "Quantity__std": 5,
        "TotalPrice__std": 120,
        "TaxTreatmentId__std": "1ttLT000000EV7DYAW",
        "BillingTreatmentId__std": "1BTLT00000008yT4AQ",
        "AccountId__std": "001LT00000dPVrlYAG",
        "BillingProfileId__std": "001LT00000dPVrlYAG",
        "BillingCity__std": "HYD",
        "ShippingCity__std": "VSKP",
        "BillingTermUnit__std": "Month",
        "BillingActionType__std": "Add",
        "CurrencyIsoCode__std": "USD",
        "RampIdentifier__std": "ramp1"
      }
    ]
}
```

This example shows the request payload to create a billing schedule for a new sale transaction with ramped products and a `TermDefined` selling model.

```
{
  "transactionDetails":
[illegible]

  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext__stdctx",
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "TransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```

When you're creating a billing schedule for a new sale transaction with ramped products, make sure that you specify the mandatory values in the `transactionDetails` property value.

Name	Description	Mapped Field	Required or Optional
`ParentTransactionId__std`	The ID of the parent transaction record. For example, if the transaction is at a level similar to that of an Order Item record, the parent transaction will be at a level similar to that of an Order record.	If the transaction is a child Order or Quote record, the ReferenceEntityId field on the BillingScheduleobject is populated. For other sObject records or external records, the Reference field on the BillingSchedule object is populated.	Optional

Name	Description	Mapped Field	Required or Optional
`AccountId__std`	The ID of the Account record that's related to the transaction.	BillingAccountId field on the BillingScheduleGroup object	Required
`BillingProfileId__std`	The ID of the billing profile (Billing Account record) that's related to the transaction. See Billing Profile requirements.	BillingAccountId field on the BillingSchedule object	Optional
`BillingCity__std`	The city in the billing address of the transaction.	BillingCity field on the BillingScheduleGroup object	Any one of the billing address fields is required.
`ShippingCity__std`	The city in the shipping address of the transaction.	ShippingCity field on the BillingScheduleGroup object	Any one of the shipping address fields is required.
`SellingModelType__std`	The selling model type indicates whether the transaction is for a one-time product, a term-defined product, or an evergreen product. Specify `OneTime` as the SellingModelType for the transaction that's related to a product that's sold once.	BillingMethod on the BillingSchedule object is populated as OrderAmount.	Required
`TransactionId__std`	The ID of the transaction that you want to create a billing schedule for.	If the transaction is an OrderItem, OrderItemAdjustmentLineItem, OrderItemDetail, QuoteLineDetail, or QuoteLineItem record, the ReferenceEntityItemId field on the BillingScheduleobject is populated. For other sObject records or external records, the ReferenceItem field on the BillingSchedule object is populated.	Required
`ProductName__std`	The name of the product that's related to the transaction that you want to create a billing schedule for.	ProductName field on the BillingScheduleGroup object	Either the ProductName or the ProductId is required
`PeriodBoundary__std`	The period boundary determines the start and end date of the billing period. Valid values are: • `AlignToCalendar` • `Anniversary` • `DayOfPeriod` • `LastDayOfPeriod`	PeriodBoundary field on the BillingScheduleGroup object	Required for products with the `TermDefined` or `Evergreen` selling model types.

Name	Description	Mapped Field	Required or Optional
`BillingDayOfMonth__std`	The day of the month on which a recurring billing process is scheduled to occur for the transaction.	BillDayOfMonth field on the BillingScheduleGroup object	Required for products with the `TermDefined` or `Evergreen` selling model types.
`EndDate__std`	The end date of the transaction.	BillingScheduleEndDate field on the BillingSchedule object	Required for products with the `TermDefined` selling model type.
`StartDate__std`	The start date of the transaction.	BillingScheduleStartDate field on the BillingSchedule object	Required
`Quantity__std`	The quantity of the transaction.	Quantity field on the BillingSchedule object	Required
`TaxTreatmentId__std`	The ID of the tax treatment that's used to calculate tax for the transaction. If you don't specify a TaxTreatmentId, the org-default TaxTreatmentId is considered.	TaxTreatmentId field on the BillingScheduleGroup object	Required
`BillingTermUnit__std`	The unit of measurement of the billing term. Valid values are: • `Month` • `Quarter` • `Semi-Annual` • `Year`	BillingTermUnit field on the BillingSchedule object	Required for products with the `TermDefined` or `Evergreen` selling model types.
`BillingTreatmentId__std`	The ID of the billing treatment that's used to create the billing schedule for the transaction. If you don't specify a BillingTreatmentId, the org-default BillingTreatmentId is considered.	BillingTreatmentId field on the BillingScheduleGroup object	Required
`CurrencyIsoCode__std`	The currency code of the transaction amount if your Salesforce org has multi-currency enabled.	CurrencyIsoCode field on the BillingScheduleGroup object	Required for Salesforce orgs with multiple currencies enabled.
`RampIdentifier__std`	When the transaction is part of a ramp deal, the RampIdentifier value is a unique identifier for that ramp. All the transactions with the same RampIdentifier value are grouped to have billing schedules for the same billing schedule group.	This value isn't populated on any BillingSchedule or BillingScheduleGroup field.	Required
`BillingActionType__std`	The action that you want to perform for the transaction. Valid value is `Add`.	Category field on the BillingSchedule object is populated as Original.	Required

Name	Description	Mapped Field	Required or Optional
`UnitPrice__std`	The unit price of the transaction.	UnitPrice field on the BillingSchedule object	Either the UnitPrice or the NetUnitPrice value is required
`TotalPrice__std`	The total price of the transaction.	TotalAmount field on the BillingSchedule object	Required

New Sale Transaction With Usage Products

Understand the required values and key considerations before you create a billing schedule for a new sale transaction with usage-based products.

Considerations

Keep these important considerations in mind when you provide the details for new sale transactions with a usage-based product.

- The `BillingActionType__std` value for any new sale transaction must be `Add`.
- For new sale transactions with a usage-based product, the `ProductUsageModelType__std` and `BindingInstance__std` values are required.
- If you don't provide the `TaxTreatmentId__std` value, the `BillingTreatmentId__std` value, or the `LegalEntityId__std` value for the transaction, the default `TaxTreatmentId`, `BillingTreatmentId`, and `LegalEntityId` of your Salesforce org is considered. If your org doesn't have any default values, an error occurs.
- To understand the requirements for the various selling model types, see these resources.
  - One-Time New Sale Transaction on page 2600
  - Term-Defined New Sale Transaction on page 2603
  - Evergreen New Sale Transaction on page 2607

JSON example

This sample is for a new sale transaction for a usage-based product with a `TermDefined` selling model.

```
{
  "Transaction": [
    {
      "id": "termDefined",
      "SellingModelType__std": "TermDefined",
      "ParentTransactionId__std": "sample",
      "TransactionId__std": "onetimeUsage2",
      "StartDate__std": "2025-01-01",
      "EndDate__std": "2025-12-31",
      "PeriodBoundary__std": "DayOfPeriod",
      "BillingDayOfMonth__std": 1,
      "BillingTermUnit__std": "Month",
      "UnitPrice__std": 10,
      "Quantity__std": 1,
      "TotalPrice__std": 120,
      "AccountId__std": "001xx000003GgEJAA0",
      "BillingProfileId__std": "001LT00000dPVrlYAG",
      "BillingActionType__std": "Add",
      "CurrencyIsoCode__std": "USD",
```

```
        "BillingCity__std": "Hyderabad",
        "ShippingCity__std": "SFO",
        "ProductUsageModelType__std": "Anchor",
        "BindingInstance__std": "001xx000003GgChAAK",
        "ProductId__std": "01txx0000006i3DAAQ"
      }
    ]
}
```

This example shows the request payload to create a billing schedule for a new sale transaction for usage-based products with a `TermDefined` selling model.

```
{
  "transactionDetails":
[illegible]

  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext__stdctx",
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "TransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```

Before using the Create Standalone Billing Schedules API for usage-based products, Create a Usage Product Grant Binding Policy record with the grant binding type as `Target`.

When you're creating a billing schedule for a new sale transaction with a usage-based product, make sure that you specify the mandatory values in the `transactionDetails` property value.

Context Tag in the Transaction Node of the StandaloneBillingContext Context Definition	Description	Mapped Field	Required or Optional
`ParentTransactionId__std`	The ID of the parent transaction record. For example, if the transaction is at a level similar to that of an Order Item record, the parent transaction will be at a level similar to that of an Order record.	If the transaction is a child Order or Quote record, the ReferenceEntityId field on the BillingScheduleobject is populated. For other sObject records or external records, the Reference field on the BillingSchedule object is populated.	Optional
`AccountId__std`	The ID of the Account record that's related to the transaction.	BillingAccountId field on the BillingScheduleGroup object	Required

Context Tag in the Transaction Node of the StandaloneBillingContext Context Definition	Description	Mapped Field	Required or Optional
`BillingProfileId__std`	The ID of the billing profile (Billing Account record) that's related to the transaction. See Billing Profile requirements.	BillingAccountId field on the BillingSchedule object	Optional
`BillingCity__std`	The city in the billing address of the transaction.	BillingCity field on the BillingScheduleGroup object	Any one of the billing address fields is required.
`ShippingCity__std`	The city in the shipping address of the transaction.	ShippingCity field on the BillingScheduleGroup object	Any one of the shipping address fields is required.
`SellingModelType__std`	The selling model type indicates whether the transaction is for a one-time product, a term-defined product, or an evergreen product. Specify `OneTime` as the SellingModelType for the transaction that's related to a product that's sold once.	BillingMethod on the BillingSchedule object is populated as OrderAmount.	Required
`TransactionId__std`	The ID of the transaction that you want to create a billing schedule for.	If the transaction is an OrderItem, OrderItemAdjustmentLineItem, OrderItemDetail, QuoteLineDetail, or QuoteLineItem record, the ReferenceEntityItemId field on the BillingScheduleobject is populated. For other sObject records or external records, the ReferenceItem field on the BillingSchedule object is populated.	Required
`ProductId__std`	The ID of the product that's related to the transaction that you want to create a billing schedule for.	Product2Id field on the BillingScheduleGroup object	Either the ProductName or the ProductId is required
`PeriodBoundary__std`	The period boundary determines the start and end date of the billing period. Valid values are: • `AlignToCalendar` • `Anniversary` • `DayOfPeriod` • `LastDayOfPeriod`	PeriodBoundary field on the BillingScheduleGroup object	Required for products with the `TermDefined` or `Evergreen` selling model types.

Context Tag in the Transaction Node of the StandaloneBillingContext Context Definition	Description	Mapped Field	Required or Optional
`BillingDayOfMonth__std`	The day of the month on which a recurring billing process is scheduled to occur for the transaction.	BillDayOfMonth field on the BillingScheduleGroup object	Required for products with the `TermDefined` or `Evergreen` selling model types.
`EndDate__std`	The end date of the transaction.	BillingScheduleEndDate field on the BillingSchedule object	Required for products with the `TermDefined` selling model type.
`StartDate__std`	The start date of the transaction.	BillingScheduleStartDate field on the BillingSchedule object	Required
`Quantity__std`	The quantity of the transaction.	Quantity field on the BillingSchedule object	Required
`BillingTermUnit__std`	The unit of measurement of the billing term. Valid values are: • `Month` • `Quarter` • `Semi-Annual` • `Year`	BillingTermUnit field on the BillingSchedule object	Required for products with the `TermDefined` or `Evergreen` selling model types.
`CurrencyIsoCode__std`	The currency code of the transaction amount if your Salesforce org has multi-currency enabled.	CurrencyIsoCode field on the BillingScheduleGroup object	Required for Salesforce orgs with multiple currencies enabled.
`BillingActionType__std`	The action that you want to perform for the transaction. Valid value is `Add`.	Category field on the BillingSchedule object is populated as Original.	Required
`UnitPrice__std`	The unit price of the transaction.	UnitPrice field on the BillingSchedule object	Either the UnitPrice or the NetUnitPrice value is required
`TotalPrice__std`	The total price of the transaction.	TotalAmount field on the BillingSchedule object	Required
`ProductUsageModelType__std`	The type of usage model for the transaction product or service. Valid values are: • `Anchor`—Anchor is the main subscription product or service.	BillingMethod value field on the BillingSchedule object is populated as OrderAmount.	Required

Context Tag in the Transaction Node of the StandaloneBillingContext Context Definition	Description	Mapped Field	Required or Optional
	• `Pack`—Pack is the add-on product or service that grants additional usage resources for consumption.		
`BindingInstance__std`	The ID of the Asset record or custom object record that's related to the transaction.	BindingInstanceId on the BillingScheduleGroup object	Required

Amended Transaction

Learn how amendments update quantity, price, and end date in billing schedules along with delta-only price transactions and billing-frequency changes at the billing schedule level.

Considerations

Keep these important considerations in mind when providing the details for amended transactions.

- The `BillingActionType__std` value for any amended transaction must be `Amend`.
- When you provide a negative quantity, make sure that it's less than the total quantity of the related transaction.
- If your transaction is for an sObject record, only specify the record ID as the id value in the `transactionDetails` property value. All the other transaction details are automatically fetched from the sObject record if those details are mapped in the context definition.

Amendment Payload Types

The Create Standalone Billing Schedules API accepts two amendment payload types in the `transactionDetails` node. Use the type that matches the level of detail you want to provide in the request.

Simplified amendment payloads support these scenarios.

- Negative amendment to reduce quantity
- Positive amendment to add quantity
- Price amendment to change net or unit price from an effective date
- End date change to extend or shorten the subscription term
- Delta price change to pass a price delta as a single adjustment
- Billing frequency update at the billing schedule group level

Detailed amendment payloads cover the same business outcomes when you specify all transaction details and financial fields.

Simplified amendment payload

Set `BillingActionType__std` to `Amend`. Set `BillingSubActionType__std` to `QuantityAmend`, `PriceAmend`, `EndDateChange`, `DeltaPriceAmend`, or `BillingFrequencyChange` depending on the scenario. Include `TransactionId__std` and `StartDate__std`. Identify the source with either `RelatedTransactionId__std` (one billing schedule) or `BillingScheduleGroupId__std` (a billing schedule group). The examples in the scenario-specific sections in this topic use `RelatedTransactionId__std`. For amendments by using a billing schedule group as source, use `BillingScheduleGroupId__std` instead.

Negative amendment (QuantityAmend)

Reduce quantity from a start date by specifying the reduction quantity as a negative number along with the start date.

If you specify a related transaction, the reduction is applied against that transaction's billing schedules. If you specify a billing schedule group ID, the API applies reductions in Last In, First Out (LIFO) order within that billing schedule group.

The API determines which billing schedules to reduce, validates that sufficient net quantity exists, and checks whether price can be derived through proration or whether `TotalPrice` can be accepted when specified.

This sample includes a negative `Quantity__std` for the quantity to remove from the effective date.

```
{
  "Transaction": [
    {
      "TransactionId__std": "a2",
      "RelatedTransactionId__std": "a1",
      "StartDate__std": "2025-05-01",
      "Quantity__std": -2,
      "BillingActionType__std": "Amend",
      "BillingSubActionType__std": "QuantityAmend"
    }
  ]
}
```

This sample shows the request payload for the same negative quantity amendment.

```
{
  "transactionDetails":
"{\"Transaction\":[{\"TransactionId__std\":\"a2\",\"RelatedTransactionId__std\":\"a1\",\"StartDate__std\":\"2025-05-01\",\"Quantity__std\":-2,\"BillingActionType__std\":\"Amend\",\"BillingSubActionType__std\":\"QuantityAmend\"}]}",

  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext__stdctx",
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "TransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```

This sample shows the transaction data that references a billing schedule group.

```
{
  "Transaction": [
    {
      "TransactionId__std": "a2",
      "BillingScheduleGroupId__std": "9Ws00000AXB00AAI",
      "StartDate__std": "2025-05-01",
      "Quantity__std": -2,
      "BillingActionType__std": "Amend",
     "BillingSubActionType__std": "QuantityAmendment"
    }
  ]
}
```

This sample shows the request payload when billing schedule group is used as a reference.

```
{
  "transactionDetails":
"{\"Transaction\":[{\"TransactionId__std\":\"a2\",\"BillingScheduleGroupId__std\":\"9Ws00000AXB00AAI\",\"StartDate__std\":\"2025-05-01\",\"Quantity__std\":-2,\"BillingActionType__std\":\"Amend\",\"BillingSubActionType__std\":\"QuantityAmendment\"}]}",

  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext__stdctx",
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "TransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```

Positive amendment (QuantityAmend)

Add quantity from a start date. Specify minimal information in the payload. The API derives the billing schedule to amend and, if not specified, the unit price and total price. Use a positive `Quantity__std` for the quantity to add from the effective date.

```
{
  "Transaction": [
    {
      "TransactionId__std": "a2",
      "RelatedTransactionId__std": "a1",
      "StartDate__std": "2025-05-01",
      "Quantity__std": 2,
      "BillingActionType__std": "Amend",
      "BillingSubActionType__std": "QuantityAmend"
    }
  ]
}
```

This sample shows the request payload for the same positive quantity amendment.

```
{
  "transactionDetails":
"{\"Transaction\":[{\"TransactionId__std\":\"a2\",\"RelatedTransactionId__std\":\"a1\",\"StartDate__std\":\"2025-05-01\",\"Quantity__std\":2,\"BillingActionType__std\":\"Amend\",\"BillingSubActionType__std\":\"QuantityAmend\"}]}",

  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext__stdctx",
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "TransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```

This sample shows the transaction data that references a billing schedule group.

```
{
  "Transaction": [
    {
      "TransactionId__std": "a2",
      "BillingScheduleGroupId__std": "9Ws00000AXB00AAI",
      "StartDate__std": "2025-05-01",
      "Quantity__std": 2,
      "BillingActionType__std": "Amend",
      "BillingSubActionType__std": "QuantityAmendment"
```

```
    }
  ]
}
```

This sample shows the request payload when billing schedule group is used as a reference.

```
{
  "transactionDetails":
"{\"Transaction\":[{\"TransactionId__std\":\"ter2\",\"BillingScheduleGroupId__std\":\"9s600XX00A\",\"StartDate__std\":\"2025-05-01\",\"Quantity__std\":2,\"BillingActionType__std\":\"Amend\",\"BillingSubActionType__std\":\"QuantityAmend\"}]}",

  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext__stdctx",
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "TransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```

Price amendment (PriceAmend)

Change the price from a given start date. For example, apply a new net unit price or unit price. The API cancels the existing billing schedule(s) from that date and creates new positive amendment billing schedule with the updated price . Specify the new `NetUnitPrice__std` or `UnitPrice__std` effective on `StartDate__std`. This sample uses net unit price.

```
{
  "Transaction": [
    {
      "TransactionId__std": "ter3",
      "RelatedTransactionId__std": "ter1",
      "StartDate__std": "2026-04-01",
      "NetUnitPrice__std": 20,
      "BillingActionType__std": "Amend",
      "BillingSubActionType__std": "PriceAmend"
    }
  ]
}
```

This sample shows the request payload for the same price amendment.

```
{
  "transactionDetails":
"{\"Transaction\":[{\"TransactionId__std\":\"ter3\",\"RelatedTransactionId__std\":\"ter1\",\"StartDate__std\":\"2026-04-01\",\"NetUnitPrice__std\":20,\"BillingActionType__std\":\"Amend\",\"BillingSubActionType__std\":\"PriceAmend\"}]}",

  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext__stdctx",
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "TransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```

This sample shows the transaction data that references a billing schedule group with delta quantity to amend price on an entire billing schedule group.

```
{
  "Transaction": [
    {
```

```
      "TransactionId__std": "ter4",
      "BillingScheduleGroupId__std": "9Wsxx000000026fCAA",
      "StartDate__std": "2026-06-01",
      "NetUnitPrice__std": 30,
      "Quantity__std": 1,
      "BillingActionType__std": "Amend",
      "BillingSubActionType__std": "PriceAmend"
    }
  ]
}
```

This sample shows the request payload when billing schedule group is used as a reference.

```
{
  "transactionDetails":
"{\"Transaction\":[{\"TransactionId__std\":\"ter4\",\"BillingScheduleGroupId__std\":\"9Wsxx000000026fCAA\",\"StartDate__std\":\"2026-06-01\",\"NetUnitPrice__std\":30,\"Quantity__std\":1,\"BillingActionType__std\":\"Amend\",\"BillingSubActionType__std\":\"PriceAmend\"}]}",

  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext__stdctx",
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "TransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```

End date change (EndDateChange)

Change the end date of the subscription or of a specific billing schedule under a billing schedule group from a specified start date. The API cancels the existing billing schedule(s) from that start date and creates a new positive amendment billing schedule from the start date to the new end date.

Specify `StartDate__std` to specify when the change is effective and `EndDate__std` for the new end of the term.

```
{
  "Transaction": [
    {
      "TransactionId__std": "ter3",
      "RelatedTransactionId__std": "ter1",
      "StartDate__std": "2026-05-01",
      "EndDate__std": "2026-09-01",
      "BillingActionType__std": "Amend",
      "BillingSubActionType__std": "EndDateChange"
    }
  ]
}
```

This sample shows the request payload for the same end date change on one billing schedule by using related transaction ID.

```
{
  "transactionDetails":
"{\"Transaction\":[{\"TransactionId__std\":\"ter3\",\"RelatedTransactionId__std\":\"ter1\",\"StartDate__std\":\"2026-05-01\",\"EndDate__std\":\"2026-09-01\",\"BillingActionType__std\":\"Amend\",\"BillingSubActionType__std\":\"EndDateChange\"}]}",

  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext__stdctx",
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "TransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
```

```
  }
}
```

This sample shows the transaction data that references a billing schedule group for end date change on one billing schedule group.

```
{
  "Transaction": [
    {
      "TransactionId__std": "ter3",
      "BillingScheduleGroupId__std": "9Wsxx000000026fCAA",
      "StartDate__std": "2026-05-01",
      "EndDate__std": "2026-09-01",
      "BillingActionType__std": "Amend",
      "BillingSubActionType__std": "ChangeEndDate"
    }
  ]
}
```

This sample shows the request payload when billing schedule group is used as a reference.

```
{
  "transactionDetails":
"{\"Transaction\":[{\"TransactionId__std\":\"ter3\",\"BillingScheduleGroupId__std\":\"9Wsxx000000026fCAA\",\"StartDate__std\":\"2026-05-01\",\"EndDate__std\":\"2026-09-01\",\"BillingActionType__std\":\"Amend\",\"BillingSubActionType__std\":\"ChangeEndDate\"}]}",

  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext__stdctx",
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "TransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```

Delta price change (DeltaPriceAmend)

Change a subscription's price mid-term. For example, manage modifications such as add or remove features, or apply special discounts and price uplifts.

Send `Quantity__std` as `0` with `TotalPrice__std` set to the delta amount, or send a non-zero quantity with `NetUnitPrice__std` as the delta net unit price. This sample shows the zero-quantity delta total price pattern.

```
{
  "Transaction": [
    {
      "TransactionId__std": "amendmentSampleId1",
      "RelatedTransactionId__std": "ter1",
      "StartDate__std": "2025-07-01",
      "TotalPrice__std": 600,
      "Quantity__std": 0,
      "BillingActionType__std": "Amend",
      "BillingSubActionType__std": "DeltaPriceAmend"
    }
  ]
}
```

This sample shows the request payload for the same delta price change.

```
{
  "transactionDetails":
"{\"Transaction\":[{\"TransactionId__std\":\"amendDeltaSample1\",\"RelatedTransactionId__std\":\"ter1\",\"StartDate__std\":\"2025-07-01\",\"DeltaPrice__std\":60,\"Quantity__std\":1,\"BillingActionType__std\":\"Amend\",\"BillingSubActionType__std\":\"DeltaPriceAmend\"}]}",

  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext__stdctx",
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "TransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```

Billing frequency update at billing schedule group (BillingFrequencyChange)

Change how often the subscription bills for all billing schedules in a billing schedule group. Set `BillingSubActionType__std` to `BillingFrequencyChange`, identify the group with `BillingScheduleGroupId__std`, set `BillingTermUnit__std` to the new term unit (for example, `Quarter`), and set `StartDate__std` when the new frequency is effective. Include `TransactionId__std`. Optionally, include `BillingStartMonth__std` when the billing selling model requires it for that term unit.

```
{
  "Transaction": [
    {
      "TransactionId__std": "bfTer1",
      "BillingScheduleGroupId__std": "9Ws00000AXB00AAI",
      "StartDate__std": "2025-05-01",
      "BillingTermUnit__std": "Quarter",
      "BillingActionType__std": "Amend",
      "BillingSubActionType__std": "BillingFrequencyChange"
    }
  ]
}
```

This sample shows the request payload for the same billing frequency update.

```
{
  "transactionDetails":
"{\"Transaction\":[{\"TransactionId__std\":\"bfTer1\",\"BillingScheduleGroupId__std\":\"9Ws00000AXB00AAI\",\"StartDate__std\":\"2025-05-01\",\"BillingTermUnit__std\":\"Quarter\",\"BillingActionType__std\":\"Amend\",\"BillingSubActionType__std\":\"BillingFrequencyChange\"}]}",

  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext__stdctx",
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "TransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```

Detailed amendment payload

Send each amendment as its own transaction with `BillingActionType__std` set to `Amend`. Include identifiers, `RelatedTransactionId__std`, and the financial fields for each line. Use this payload type when you already computed quantities, unit prices, and total prices. This sample includes one positive amendment and two negative amendments for the same related transaction.

This sample includes the transaction data for an amendment.

```
{
  "Transaction": [
    {
      "id": "sampleA1",
      "ParentTransactionId__std": "sample",
      "TransactionId__std": "sampleA1",
      "RelatedTransactionId__std": "ter1",
      "StartDate__std": "2025-04-01",
      "UnitPrice__std": 10,
      "Quantity__std": 1,
      "TotalPrice__std": 90,
      "BillingActionType__std": "Amend"
    },
    {
      "id": "sampleA2",
      "ParentTransactionId__std": "sample",
      "TransactionId__std": "sampleA2",
      "RelatedTransactionId__std": "ter1",
      "StartDate__std": "2025-03-01",
      "UnitPrice__std": 10,
      "Quantity__std": -2,
      "TotalPrice__std": -200,
      "BillingActionType__std": "Amend"
    },
    {
      "id": "sampleA3",
      "ParentTransactionId__std": "sample",
      "TransactionId__std": "sampleA3",
      "RelatedTransactionId__std": "ter1",
      "StartDate__std": "2025-03-01",
      "UnitPrice__std": 10,
      "Quantity__std": -1,
      "TotalPrice__std": -100,
      "BillingActionType__std": "Amend"
    }
  ]
}
```

This sample shows the request payload to create a billing schedule for an amendment.

```
{
  "transactionDetails":
[illegible]

  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext__stdctx",
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "TransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```

When you're creating a billing schedule for an amended transaction, make sure that you specify the mandatory values in the transactionDetails property value.

Context Tag in the Transaction Node of the StandaloneBillingContext Context Definition	Description	Mapped Field	Required or Optional
`ParentTransactionId__std`	The ID of the parent transaction record. For example, if the transaction is at a level similar to that of an Order Item record, the parent transaction will be at a level similar to that of an Order record.	If the transaction is a child Order or Quote record, the ReferenceEntityId field on the BillingScheduleobject is populated. For other sObject records or external records, the Reference field on the BillingSchedule object is populated.	Optional
`TransactionId__std`	The unique identifier reference to apply to the billing schedule with amendment or updated billing frequency. The Reference and ReferenceItem field on a billing schedule has this value appended with a unique ID generated string.	If the transaction is an OrderItem, OrderItemAdjustmentLineItem, OrderItemDetail, QuoteLineDetail, or QuoteLineItem record, the ReferenceEntityItemId field on the BillingScheduleobject is populated. For other sObject records or external records, the ReferenceItem field on the BillingSchedule object is populated.	Required
`BillingStartMonth__std`	The month when billing begins for an annual subscription. This value can be any number from `1` through `12`. For example, if billing starts in January, the value is `1`. If billing starts in June, the value is `6`.	BillingStartMonth field on the BillingScheduleGroup object	Optional. Required if the billing term unit is Quarter, Semi-Annual, or Annual.
`EndDate__std`	The end date of the transaction.	BillingScheduleEndDate field on the BillingSchedule object	Optional. Required in case of an end date change of a subscription.
`StartDate__std`	The start date of the transaction.	BillingScheduleStartDate field on the BillingSchedule object	Required
`Quantity__std`	The quantity of the transaction.	Quantity field on the BillingSchedule object	Required

Context Tag in the Transaction Node of the StandaloneBillingContext Context Definition	Description	Mapped Field	Required or Optional
`BillingTermUnit__std`	The unit of measurement of the billing term. Valid values are: • `Month` • `Quarter` • `Semi-Annual` • `Year`	BillingTermUnit field on the BillingSchedule object	Optional
`RelatedTransactionId__std`	The ID of the related transaction that must be referenced for an amendment.	This value isn't populated on any BillingSchedule or BillingScheduleGroup field.	Required
`BillingScheduleGroupId__std`	ID of the billing schedule group that must be referenced for an amendment.	Not applicable.	Specify either `RelatedTransactionId__std` or `BillingScheduleGroupId__std` in your request if you're sending a simplified payload.
`BillingActionType__std`	The action that you want to perform for the transaction. Specify `Amend` as the BillingActionType for amended transactions.	Category field on the BillingSchedule object is populated as AmendQuantity.	Required
`BillingSubActionType__std`	The sub-action that you want to perform for the transaction. Specify `PriceAmend`, `QuantityAmend`, or `DeltaPriceAmend` as the BillingActionType to specify minimal details in the payload for a price, quantity, or delta price amendment respectively. Specify `EndDateChange` as the BillingActionType to change the end date of a subscription. Specify `BillingFrequencyChange` as the BillingActionType to update the billing frequency at the billing schedule group level. Available in API version 67.0 and later.	SubCategory field on the BillingSchedule object	Optional
`UnitPrice__std`	The unit price of the transaction.	UnitPrice field on the BillingSchedule object	Either the UnitPrice or the NetUnitPrice value is required

Context Tag in the Transaction Node of the StandaloneBillingContext Context Definition	Description	Mapped Field	Required or Optional
`TotalPrice__std`	The total price of the transaction.	TotalAmount field on the BillingSchedule object	Required
`NetUnitPrice__std`	The net unit price of the transaction.	NetUnitPrice field on the BillingSchedule object	Either the UnitPrice or the NetUnitPrice value is required

Renewal Transaction

Create a renewal billing schedule for termed subscriptions that continues without a gap, starting the day after the current billing schedule group ends and running for the same term length.

Considerations

Keep these important considerations in mind when you specify the details for renewal transactions.

- The `BillingActionType__std` value for renewal transactions must be `Renew`.
- The quantity for renewal transactions must be a nonzero positive number.
- The start date of the renewal transaction must be before the maximum end date of all the existing billing schedules associated with the billing schedule group.
- The end date of the renewal transaction must be after the maximum end date of all the existing transactions associated with the billing schedule group of the related transaction.
- Specify a `RelatedTransactionId__std` value with a positive quantity.
- For a particular RelatedTransactionId, you can create a billing schedule only for a single renewal transaction.
- For renewal transactions, the `TaxTreatmentId__std` value, the `BillingTreatmentId__std` value, and the `LegalEntityId__std` value of the related transaction are considered.

Renewal Payload Types

The Create Standalone Billing Schedules API accepts two renewal payload types in the `transactionDetails` node. Use the type that matches the level of detail you want to provide in the request.

Simplified renewal payloads support these scenarios.

- Renewal to start a new term after the related transaction or billing schedule group reaches its end
- As-is renewal to renew each existing billing schedule at its own quantity and net unit price

Detailed renewal payloads cover the same outcomes when you specify all transaction details and financial fields.

Simplified renewal payload

Set `BillingActionType__std` to `Renew`. Set `BillingSubActionType__std` to `SimplifiedRenewal` for a standard renewal, or to `AsIsRenewal` for an as-is renewal. Include `TransactionId__std`. Identify the source with either `RelatedTransactionId__std` (one billing schedule) or `BillingScheduleGroupId__std` (a billing schedule group). For a standard renewal, you can omit dates and pricing. The API can derive start date, end date, quantity, and price from the related transaction record or billing schedule group. You can add optional dates or pricing when you want to override derived values.

The examples in this topic use `RelatedTransactionId__std`. For renewals with billing schedule group as source, use `BillingScheduleGroupId__std` instead.

Renewal (SimplifiedRenewal)

Use this payload structure when the new term begins after the prior term ends. This sample shows the minimum fields. Keep these considerations in mind when you specify a simplified payload.

- If you specify a billing schedule group ID, start date is required. If you omit the start date and specify a related transaction ID, the API sets the start date to a day after the end date of the related billing schedule.
- If you specify a billing schedule group ID, end date is required. If you omit the end date and specify a related transaction ID, the API sets the end date by adding one full billing schedule term to that billing schedule's end date (plus one day).
- If you specify a billing schedule group ID, unit price and net unit price are required. If you omit the unit price and net unit price, and specify a related transaction ID, the API sets the unit price and net unit price values from the related billing schedule.
- If you specify a billing schedule group ID, quantity is required. If you omit the quantity and specify a related transaction ID, the API sets the renewal quantity from the related billing schedule.

This sample shows the transaction data for the simplified renewal with related transaction ID as a reference.

```
{
  "Transaction": [
    {
      "TransactionId__std": "ren1",
      "RelatedTransactionId__std": "ter1",
      "BillingActionType__std": "Renew",
      "BillingSubActionType__std": "SimplifiedRenewal"
    }
  ]
}
```

This sample shows the request payload for the simplified renewal with related transaction ID as a reference.

```
{
  "transactionDetails":
"{\"Transaction\":[{\"TransactionId__std\":\"ren1\",\"RelatedTransactionId__std\":\"ter1\",\"BillingActionType__std\":\"Renew\",\"BillingSubActionType__std\":\"SimplifiedRenewal\"}]}",

  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext__stdctx",
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "TransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```

This sample shows the transaction data for the simplified renewal with billing schedule group ID as a reference.

```
{
  "Transaction": [
    {
      "TransactionId__std": "ter3",
      "BillingScheduleGroupId__std": "9Ws00000AXB00AAI",
      "BillingActionType__std": "Renew",
      "BillingSubActionType__std": "SimplifiedRenewal",
      "Quantity__std": 5,
      "UnitPrice__std": 10,
      "StartDate__std": "2025-12-15",
```

```
      "EndDate__std": "2026-12-14"
    }
  ]
}
```

This sample shows the request payload for the simplified renewal with billing schedule group ID as a reference.

```
{
  "transactionDetails":
"{\"Transaction\":[{\"TransactionId__std\":\"ter3\",\"BillingScheduleGroupId__std\":\"9Ws00000AXB00AAI\",\"BillingActionType__std\":\"Renew\",\"BillingSubActionType__std\":\"SimplifiedRenewal\",\"Quantity__std\":\"5\",\"UnitPrice__std\":\"10\",\"StartDate__std\":\"2025-12-15\",\"EndDate__std\":\"2026-12-14\"}]}",

  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext__stdctx",
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "TransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```

As-is renewal (AsIsRenewal)

Use this payload structure when multiple billing schedules under the reference have different quantities or net unit prices and you want one renewal billing schedule per source schedule.

```
{
  "Transaction": [
    {
      "TransactionId__std": "asIsRen1",
      "RelatedTransactionId__std": "ter1",
      "BillingActionType__std": "Renew",
      "BillingSubActionType__std": "AsIsRenewal"
    }
  ]
}
```

This sample shows the request payload for the same as-is renewal.

```
{
  "transactionDetails":
"{\"Transaction\":[{\"TransactionId__std\":\"asIsRen1\",\"RelatedTransactionId__std\":\"ter1\",\"BillingActionType__std\":\"Renew\",\"BillingSubActionType__std\":\"AsIsRenewal\"}]}",

  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext__stdctx",
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "TransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```

This sample shows the request payload for the as-is renewal with billing schedule group ID as a reference.

```
{
  "Transaction": [
    {
      "TransactionId__std": "asIsRenew2",
      "BillingScheduleGroupId__std": "9Ws00000AXB00AAI",
      "BillingActionType__std": "Renew",
```

```
      "BillingSubActionType__std": "AsIsRenewal"
    }
  ]
}
```

This sample shows the request payload for the as-is renewal with billing schedule group ID as a reference.

```
{
  "transactionDetails":
"{\"Transaction\":[{\"TransactionId__std\":\"asIsRenew2\",\"BillingScheduleGroupId__std\":\"9s000000AX0AA1\",\"BillingActionType__std\":\"Renew\",\"BillingSubActionType__std\":\"AsIsRenewal\"}]}",

  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext__stdctx",
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "TransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```

Detailed renewal payload

Send each renewal row with `BillingActionType__std` set to `Renew` and include identifiers, related transaction, dates, quantity, unit price, and total price. Use this payload type when you already computed the renewal lines yourself.

This sample includes the transaction data for a standard renewal after the end date.

```
{
  "Transaction": [
    {
      "id": "temp1001",
      "TotalPrice__std": 10,
      "Quantity__std": 1,
      "UnitPrice__std": 10,
      "StartDate__std": "2026-03-01",
      "EndDate__std": "2027-12-31",
      "TransactionId__std": "temp1001",
      "BillingActionType__std": "Renew",
      "RelatedTransactionId__std": "temp1"
    }
  ]
}
```

This sample shows the request payload to create a billing schedule for a renewal.

```
{
  "transactionDetails":
"{\"Transaction\":[{\"id\":\"temp1001\",\"TotalPrice__std\":10,\"Quantity__std\":1,\"UnitPrice__std\":10,\"StartDate__std\":\"2026-03-01\",\"EndDate__std\":\"2027-12-31\",\"TransactionId__std\":\"temp1001\",\"BillingActionType__std\":\"Renew\",\"RelatedTransactionId__std\":\"temp1\"}]}",

  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext__stdctx",
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "TransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```

When you renew a new-sale transaction after its end date, those transactions are called renewal transactions. For example, a new-sale transaction starts on 01/01/2025 and ends on 12/31/2025, and you renew it to start on 03/01/2026 and end on 12/31/2026.

When you're creating a billing schedule for a renewal transaction, make sure that you provide the mandatory values in the `transactionDetails` property value.

Context Tag in the Transaction Node of the StandaloneBillingContext Context Definition	Description	Mapped Field	Required or Optional
`TransactionId__std`	The unique identifier reference to apply to the renewal billing schedule. The Reference and ReferenceItem field on a billing schedule has this value appended with a unique ID generated string.	If the transaction is an OrderItem, OrderItemAdjustmentLineItem, OrderItemDetail, QuoteLineDetail, or QuoteLineItem record, the ReferenceEntityItemId field on the BillingScheduleobject is populated. For other sObject records or external records, the ReferenceItem field on the BillingSchedule object is populated.	Required
`EndDate__std`	The end date of the transaction.	BillingScheduleEndDate field on the BillingSchedule object	Required
`StartDate__std`	The start date of the transaction.	BillingScheduleStartDate field on the BillingSchedule object	Required
`Quantity__std`	The quantity of the transaction.	Quantity field on the BillingSchedule object	Required
`RelatedTransactionId__std`	The ID of the original transaction that's related to the renewal transaction.	This value isn't populated on any BillingSchedule or BillingScheduleGroup field.	Required
`BillingScheduleGroupId__std`	ID of the billing schedule group that must be referenced to renew the billing schedule group.	Not applicable.	Specify either `RelatedTransactionId__std` or `BillingScheduleGroupId__std` in your request if you're sending a simplified payload.

Context Tag in the Transaction Node of the StandaloneBillingContext Context Definition	Description	Mapped Field	Required or Optional
`BillingActionType__std`	The action that you want to perform for the transaction. Specify `Renew` as the BillingActionType for renewal transactions.	Category field on the BillingSchedule object is populated as Renewal.	Required
`BillingSubActionType__std`	The sub-action that you want to perform for the transaction. Specify `AsIsRenewal` as the BillingActionType to specify minimal details in the payload for an as-is renewal. Specify `SimplifiedRenewal` as the BillingActionType to specify minimal details in the payload for a renewal or an early renewal. Available in API version 67.0 and later.	SubCategory field on the BillingSchedule object	Optional
`UnitPrice__std`	The unit price of the transaction.	UnitPrice field on the BillingSchedule object	Either the UnitPrice or the NetUnitPrice value is required
`TotalPrice__std`	The total price of the transaction.	TotalAmount field on the BillingSchedule object	Required

Early Renewal Transaction

Early Renewal ends the current term at an effective date and starts a renewed term to a new end date, creating two billing schedules in one request. The request creates a negative segment for the remaining original term and a positive segment for the renewal term.

Considerations

Keep these important considerations in mind when you specify the details for early renewal transactions.

- The `BillingActionType__std` value for early renewal transactions must be `Renew`.
- When you're creating billing schedules for early renewal transactions, your input payload must have details for two transactions.
  - A transaction to cancel the existing new-sale transaction. The details of this transaction must include a negative quantity to cancel billing for the remaining period. The quantity for this transaction must be the negative equivalent of the total quantity of all the existing billing schedules associated with the same billing schedule group.
  - Another transaction that specifies the early renewal details, which must have a positive quantity.
  - The `BillingActionType__std` value for both these transactions must be `Renew`.
- The start date of the early renewal transaction must be:
  - Before the end date of the related transaction.
  - Before the maximum end date of all the existing billing schedules associated with the billing schedule group of the related transaction.

- The same as the start date of all the existing billing schedules associated with the same billing schedule group or after it.

- The end date for the early renewal transaction must be after the maximum end date of all the existing transactions associated with the billing schedule group of the related transaction.
- For early renewal transactions, the `TaxTreatmentId__std` value, the `BillingTreatmentId__std` value, and the `LegalEntityId__std` value of the related transaction are considered.

Early Renewal Payload Types

The Create Standalone Billing Schedules API accepts two early renewal payload types in the `transactionDetails` node. Use the type that matches the level of detail you want to provide in the request.

Simplified early renewal payloads support these scenarios.

- Early renewal by related transaction by using `SimplifiedRenewal` sub-action type with an effective start date before the current term end and a new end date.
- Early renewal by billing schedule group by using `SimplifiedRenewal` sub-action type with the same date rules at the group level
- Early as-is renewal by using `AsIsRenewal` sub-action type when each underlying billing schedule should renew at its own quantity and net unit price from an effective date before the group end.

For renewal that begins after the prior term or billing schedule group ends, see Renewal Transaction on page 2636.

Detailed early renewal payloads use separate cancel and renewal rows when you supply full transaction and financial fields yourself.

Simplified early renewal payload

Set `BillingActionType__std` to `Renew`. Set `BillingSubActionType__std` to `SimplifiedRenewal` for a blended early renewal, or to `AsIsRenewal` for an early as-is renewal. Include `TransactionId__std`, `StartDate__std` (before the related transaction or group end), and `EndDate__std` for the new term. Identify the source with either `RelatedTransactionId__std` (one billing schedule) or `BillingScheduleGroupId__std` (a billing schedule group). The API can derive quantity and pricing when you omit those fields.

Early renewal by related transaction (SimplifiedRenewal)

```
{
  "Transaction": [
    {
      "TransactionId__std": "earlyRen1",
      "RelatedTransactionId__std": "ter1",
      "StartDate__std": "2025-12-15",
      "EndDate__std": "2026-12-14",
      "BillingActionType__std": "Renew",
      "BillingSubActionType__std": "SimplifiedRenewal"
    }
  ]
}
```

This sample shows the request payload for the same simplified early renewal by using related transaction ID.

```
{
  "transactionDetails":
"{\"Transaction\":[{\"TransactionId__std\":\"earlyRen1\",\"RelatedTransactionId__std\":\"ter1\",\"StartDate__std\":\"2025-12-15\",\"EndDate__std\":\"2026-12-14\",\"BillingActionType__std\":\"Renew\",\"BillingSubActionType__std\":\"SimplifiedRenewal\"}]}",

  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext__stdctx",
```

```
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "TransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```

Early renewal by billing schedule group (SimplifiedRenewal)

```
{
  "Transaction": [
    {
      "TransactionId__std": "earlyRen2",
      "BillingScheduleGroupId__std": "9Wsxx00000001ovCAA",
      "BillingActionType__std": "Renew",
      "BillingSubActionType__std": "SimplifiedRenewal",
      "Quantity__std": 5,
      "UnitPrice__std": 10,
      "StartDate__std": "2025-12-15",
      "EndDate__std": "2026-12-14"
    }
  ]
}
```

This sample shows the request payload for the same simplified early renewal by using billing schedule group.

```
{
  "transactionDetails":
"{\"Transaction\":[{\"TransactionId__std\":\"earlyRen2\",\"BillingScheduleGroupId__std\":\"9Wsxx00000001ovCAA\",\"BillingActionType__std\":\"Renew\",\"BillingSubActionType__std\":\"SimplifiedRenewal\",\"Quantity__std\":5,\"UnitPrice__std\":10,\"StartDate__std\":\"2025-12-15\",\"EndDate__std\":\"2026-12-14\"}]}",

  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext__stdctx",
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "TransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```

Early as-is renewal (AsIsRenewal)

Use this pattern when the billing schedule group has multiple billing schedules with different quantities or net unit prices and the renewal effective date is before the group end.

```
{
  "Transaction": [
    {
      "TransactionId__std": "asIsEarly1",
      "BillingScheduleGroupId__std": "9Wsxx00000001ovCAA",
      "StartDate__std": "2025-12-15",
      "EndDate__std": "2026-12-14",
      "BillingActionType__std": "Renew",
      "BillingSubActionType__std": "AsIsRenewal"
    }
  ]
}
```

This sample shows the request payload for the same early as-is renewal.

```
{
  "transactionDetails":
[illegible]

  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext__stdctx",
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "TransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```

Detailed early renewal payload

Send two transactions with `BillingActionType__std` set to `Renew` for each row. The first transaction ends billing for the remainder of the current term with a negative quantity. The second transaction starts the new term with a positive quantity. Include identifiers, related transaction, dates, unit price, and total price for each row.

This sample includes the transaction data for an early renewal.

```
{
  "Transaction": [
    {
      "id": "temp71",
      "TotalPrice__std": -10,
      "Quantity__std": -1,
      "UnitPrice__std": 10,
      "StartDate__std": "2026-12-31",
      "EndDate__std": "2027-12-31",
      "TransactionId__std": "temp71",
      "BillingActionType__std": "Renew",
      "RelatedTransactionId__std": "test21"
    },
    {
      "id": "temp72",
      "TotalPrice__std": 10,
      "Quantity__std": 1,
      "UnitPrice__std": 10,
      "StartDate__std": "2026-12-31",
      "EndDate__std": "2027-12-31",
      "TransactionId__std": "temp72",
      "BillingActionType__std": "Renew",
      "RelatedTransactionId__std": "test21"
    }
  ]
}
```

This sample shows the request payload to create a billing schedule for an early renewal.

```
{
  "transactionDetails":
[illegible]

  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext__stdctx",
```

```
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "TransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```

When you renew a new-sale transaction before its end date, those transactions are called early renewal transactions. For example, a new-sale transaction starts on 01/01/2025 and ends on 12/31/2025, and you renew it to start on 12/01/2025 and end on 12/31/2026.

When a billing schedule is created for an early renewal transaction, it results in these updates.

- The cancellation date on the original billing schedule is populated.
- A billing schedule with a negative quantity is created and can be used for a refund.
- A billing schedule with a positive quantity is created for the early renewal transaction.

When you're creating a billing schedule for an early renewal transaction, make sure that you provide the mandatory values in the `transactionDetails` property value.

Context Tag in the Transaction Node of the StandaloneBillingContext Context Definition	Description	Mapped Field	Required or Optional
`TransactionId__std`	The unique identifier reference to apply to the renewal billing schedule. The Reference and ReferenceItem field on a billing schedule has this value appended with a unique ID generated string.	If the transaction is an OrderItem, OrderItemAdjustmentLineItem, OrderItemDetail, QuoteLineDetail, or QuoteLineItem record, the ReferenceEntityItemId field on the BillingScheduleobject is populated. For other sObject records or external records, the ReferenceItem field on the BillingSchedule object is populated.	Required
`EndDate__std`	The end date of the transaction.	BillingScheduleEndDate field on the BillingSchedule object	Required for a positive transaction. Optional for a negative transaction.
`StartDate__std`	The start date of the transaction.	BillingScheduleStartDate field on the BillingSchedule object	Required
`Quantity__std`	The quantity of the transaction.	Quantity field on the BillingSchedule object	Required

Context Tag in the Transaction Node of the StandaloneBillingContext Context Definition	Description	Mapped Field	Required or Optional
`RelatedTransactionId__std`	The ID of the transaction against which you're doing an early renewal. This transaction must have a positive quantity.	This value isn't populated on any BillingSchedule or BillingScheduleGroup field.	Required
`BillingScheduleGroupId__std`	ID of the billing schedule group that must be referenced to renew each billing schedule in the billing schedule group early.	Not applicable.	Specify either `RelatedTransactionId__std` or `BillingScheduleGroupId__std` in your request if you're sending a simplified payload.
`BillingScheduleGroupId__std`	ID of the billing schedule group that must be referenced to renew each billing schedule in the billing schedule group early.	Not applicable.	Specify either `RelatedTransactionId__std` or `BillingScheduleGroupId__std` in your request if you're sending a simplified payload.
`BillingActionType__std`	The action that you want to perform for the transaction. Specify `Renew` as the BillingActionType for early renewal transactions.	Category field on the BillingSchedule object is populated as Renewal.	Required
`BillingSubActionType__std`	The sub-action that you want to perform for the transaction. Specify `AsIsRenewal` as the BillingActionType to specify minimal details in the payload for an as-is renewal. Specify `SimplifiedRenewal` as the BillingActionType to specify minimal details in the payload for a renewal or an early renewal. Available in API version 67.0 and later.	SubCategory field on the BillingSchedule object	Optional
`UnitPrice__std`	The unit price of the transaction.	UnitPrice field on the BillingSchedule object	Either the UnitPrice or the NetUnitPrice value is required
`TotalPrice__std`	The total price of the transaction.	TotalAmount field on the BillingSchedule object	Required

Canceled Transaction

Cancel an entire billing schedule from a specified date. Specify only the cancellation date and billing schedule group identifier. The API computes the logic to split the cancellation across overlapping billing schedules and creates one cancel transaction per segment.

Considerations

Keep these important considerations in mind when you provide the details for canceled transactions.

- The `BillingActionType__std` value for any amended transaction must be `Cancel`.
- If you provide a quantity, make sure that it matches the total quantity of the related transaction.
- If there are multiple canceled transactions associated with the same related transaction, specify the same start date for all of these transactions.
- Specify the canceled transaction details for all the billing schedules that are associated with the same billing schedule group.
- If your transaction is for an sObject record, only specify the record ID as the id value in the `transactionDetails` property value. All the other transaction details are automatically fetched from the sObject record if those details are mapped in the context definition.

Cancellation Payload Types

The Create Standalone Billing Schedules API accepts two cancellation payload types in `transactionDetails` node. Use the type that matches the level of details you want to provide in the request.

Simplified cancellation payload

Send a single transaction in the `Transaction` array. Set `BillingActionType__std` to `Cancel` and `BillingSubActionType__std` to `SimplifiedCancel`. Identify the source to cancel with `RelatedTransactionId__std` (one billing schedule) or `BillingScheduleGroupId__std` (a billing schedule group), and include `TransactionId__std` and `StartDate__std` for the effective cancellation date.

This sample shows the transaction data with `BillingScheduleGroupId__std` to cancel a billing schedule group.

```
{
  "Transaction": [
    {
      "TransactionId__std": "cTer5",
      "BillingScheduleGroupId__std": "9Wsxx000000009hCAA",
      "StartDate__std": "2026-08-01",
      "BillingActionType__std": "Cancel",
      "BillingSubActionType__std": "SimplifiedCancel"
    }
  ]
}
```

This sample shows a simplified cancellation request.

```
{
  "transactionDetails":
"{\"Transaction\":[{\"TransactionId__std\":\"cTer5\",\"BillingScheduleGroupId__std\":\"9Wsxx000000009hCAA\",\"StartDate__std\":\"2026-08-01\",\"BillingActionType__std\":\"Cancel\",\"BillingSubActionType__std\":\"SimplifiedCancel\"}]}",

  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext__stdctx",
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "TransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
```

```
  }
}
```

This sample shows the transaction data with `RelatedTransactionId__std` to cancel a billing schedule.


```
{
  "Transaction": [
    {
      "TransactionId__std": "ter4",
      "RelatedTransactionId__std": "evg1",
      "StartDate__std": "2026-02-01",
      "BillingActionType__std": "Cancel",
      "BillingSubActionType__std": "SimplifiedCancel"
    }
  ]
}
```


This sample shows a simplified cancellation request.


```
{
  "transactionDetails":
"{\"Transaction\":[{\"TransactionId__std\":\"ter4\",\"RelatedTransactionId__std\":
\"evg1\",\"StartDate__std\":\"2026-02-01\",\"BillingActionType__std\":
\"Cancel\",\"BillingSubActionType__std\":\"SimplifiedCancel\"}]}",
  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext__stdctx",
    "readContextMappingName": "TransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```


Detailed cancellation payload

Send one cancel transaction for each positive billing schedule that you want to end. Each transaction uses `BillingActionType__std` set to `Cancel`. You specify `RelatedTransactionId__std` and the financial fields, such as quantity, unit price, and total price, for each line. Use this sample structure when you already computed the cancel lines and amounts yourself.

This sample shows a detailed cancellation for a single billing schedule. It uses a compact set of fields and does not use `SimplifiedCancel` sub-action type. For a detailed cancellation with multiple transactions, see the following sample.

This sample includes the transaction data for a cancellation.


```
{
  "Transaction": [
    {
      "id": "termedCancel1",
      "ParentTransactionId__std": "sample",
      "TransactionId__std": "termedCancel1",
      "RelatedTransactionId__std": "ot1",
      "StartDate__std": "2025-03-01",
      "UnitPrice__std": 10,
      "TotalPrice__std": -50,
      "BillingActionType__std": "Cancel"
    }
  ]
}
```

This sample shows the request payload to create a billing schedule for a cancellation.

```
{
  "transactionDetails":
"{\"Transaction\":[{\"id\":\"createCancel1\",\"ParentTransactionId__std\":\"sample\",\"TransactionId__std\":\"createCancel1\",\"ReferenceTransactionId__std\":\"01\",\"StartDate__std\":\"2025-03-01\",\"UnitPrice__std\":\"10\",\"TotalPrice__std\":\"50\",\"BillingActionType__std\":\"Cancel\"}]}",

  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext__stdctx",
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "TransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```

When you're creating a billing schedule for a canceled transaction, make sure that you specify the mandatory values in the `transactionDetails` property value.

Context Tag in the Transaction Node of the StandaloneBillingContext Context Definition	Description	Mapped Field	Required or Optional
`ParentTransactionId__std`	The ID of the parent transaction record. For example, if the transaction is at a level similar to that of an Order Item record, the parent transaction will be at a level similar to that of an Order record.	If the transaction is a child Order or Quote record, the ReferenceEntityId field on the BillingScheduleobject is populated. For other sObject records or external records, the Reference field on the BillingSchedule object is populated.	Optional
`TransactionId__std`	The unique identifier reference to apply to the canceled billing schedule. The Reference and ReferenceItem field on a billing schedule has this value appended with a unique ID generated string.	If the transaction is an OrderItem, OrderItemAdjustmentLineItem, OrderItemDetail, QuoteLineDetail, or QuoteLineItem record, the ReferenceEntityItemId field on the BillingScheduleobject is populated. For other sObject records or external records, the ReferenceItem field on the BillingSchedule object is populated.	Optional if you're sending a simplified payload. If you don't specify this value, the API creates unique transaction IDs.

Context Tag in the Transaction Node of the StandaloneBillingContext Context Definition	Description	Mapped Field	Required or Optional
`StartDate__std`	The start date of the transaction.	BillingScheduleStartDate field on the BillingSchedule object	Required
`RelatedTransactionId__std`	The ID of the related transaction that must be referenced to cancel one billing schedule.	This value isn't populated on any BillingSchedule or BillingScheduleGroup field.	Required
`BillingScheduleGroupId__std`	ID of the billing schedule group that must be referenced to cancel the billing schedule group.	Not applicable.	Specify either `RelatedTransactionId__std` or `BillingScheduleGroupId__std` in your request if you're sending a simplified payload.
`BillingActionType__std`	The action that you want to perform for the transaction. Specify `Cancel` as the BillingActionType for canceled transactions.	Category field on the BillingSchedule object is populated as Cancellation.	Required
`BillingSubActionType__std`	The sub-action that you want to perform for the transaction. Specify `SimplifiedCancel` as the BillingActionType to specify minimal details in the payload for canceled transactions. Available in API version 67.0 and later.	SubCategory field on the BillingSchedule object	Optional
`UnitPrice__std`	The unit price of the transaction.	UnitPrice field on the BillingSchedule object	Either the UnitPrice or the NetUnitPrice value is required
`TotalPrice__std`	The total price of the transaction.	TotalAmount field on the BillingSchedule object	Required
`NetUnitPrice__std`	The net unit price of the transaction.	NetUnitPrice field on the BillingSchedule object	Either the UnitPrice or the NetUnitPrice value is required

Standalone Billing Schedule Metadata Input

Input representation of the metadata details to create a billing schedule. This representation includes the name of the context definition and context mapping along with the mapping details of the transaction, billing schedule, and billing schedule group.

JSON example

```
{
  "transactionDetails": "{\"nodeName\": [{\"id\":\"001SG000004FvlGYAS\",
\"businessSobjectType\": \"Account\", \"Quantity\":\"4\" , \"Name\": \"TestAccount\"}]}",

  "transactionContextDetails": {
    "contextDefinitionName": "StandaloneBillingContext",
    "intraContextCustomMappingName": "CustomContextMapping",
    "readContextMappingName": "OrderTransactionMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```

This example shows a sample request to generate billing schedules from order items by specifying the BillingContext standard context definition as the context definition name.

```
{
  "transactionDetails": "{\"nodeName\": [{\"id\":\"001SG000004FvlGYAS\",
\"businessSobjectType\": \"Account\", \"Quantity\":\"4\" , \"Name\": \"TestAccount\"}]}",

  "transactionContextDetails": {
    "contextDefinitionName": "BillingContext__stdctx",
    "readContextMappingName": "OrderEntitiesMapping",
    "saveContextMappingName": "BSGEntitiesMapping"
  }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`contextDefinitionName`	String	Name of the context definition that's used to hydrate the context to generate the billing schedule. To generate billing schedules from order items, specify the BillingContext standard context definition as the context definition name. Applicable from API version 66.0 and later.	Required	64.0
`intraContextCustomMappingName`	String	Name of the cross-context custom mapping that's used to map Billing fields to Transaction fields. Use this mapping to populate the Billing fields with the values stored in custom transaction fields.	Optional	65.0
`readContextMappingName`	String	Name of the context mapping with the mapping for the transaction.	Required	64.0

Name	Type	Description	Required or Optional	Available Version
saveContextMappingName	String	Name of the context mapping with the mapping for the billing schedule and billing schedule group.	Required	64.0

Convert Negative Invoice Lines Input

Input representation of the details of the request to convert a list of negative invoice lines into a credit.

JSON example

```
{
"invoiceLines": ["5TVxx0000004C92GAE", "5TVxx0000004C93GAE"],
"description": "Convert negative invoice lines into credit",
"effectiveDate":"2022-05-18"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
description	String	Description stamped on the credit memo that's created after the negative invoice line conversion.	Optional	62.0
effectiveDate	String	Date stamped on the credit memo that's created after the negative invoice line conversion.	Required	62.0
invoiceLines	String[]	Complete list of the negative invoice lines along with the associated invoice line taxes. The specified negative invoice lines are converted into a posted credit memo.	Optional	62.0

Credit Invoice Input

Input representation of the details of the request to create a credit memo.

JSON example

This example shows a sample request with the `Calculate` tax strategy.

```
{
  "type": "POSTED",
  "taxStrategy": "Calculate",
  "taxEffectiveDate": "2024-07-20",
  "effectiveDate": "2024-07-20",
  "description": "Credit Invoice",
  "invoiceLines": [
    {
      "invoiceLineId": "5TVR00000004SiqOBE",
```

```
      "amountToCredit": 5
    }
  ]
}
```

This example shows a sample request with the `CopyFromInvoiceLine` tax strategy.

```
{
  "type": "POSTED",
  "taxStrategy": "CopyFromInvoiceLine",
  "effectiveDate": "2020-05-22",
  "description": "Credit Invoice",
  "invoiceLines": [
    {
      "invoiceLineId": "5TVR00000004SiqOBE",
      "amountToCredit": "5",
      "taxStrategy": "CopyFromInvoiceLine"
    }
  ]
}
```

This example shows a sample request with the `ManualOverride` and `CopyFromInvoiceLine` tax strategies.

```
{
  "type": "POSTED",
  "taxStrategy": "ManualOverride",
  "taxEffectiveDate": "2021-08-01",
  "effectiveDate": "2021-08-01",
  "description": "Credit issued because product was malfunctioning.",
  "invoiceLines": [
    {
      "invoiceLineId": "5TVR00000004SiqOBE",
      "amountToCredit": 100,
      "taxStrategy": "ManualOverride",
      "taxEffectiveDate": "2021-08-01T21:22:41.000Z",
      "taxes": [
        {
          "taxAmount": 15,
          "taxName": "abc",
          "taxCode": "taxCode",
          "taxRate": 7
        }
      ],
      "addresses": {
        "billingAddress": {
          "street": "1 Market St #300",
          "city": "San Francisco",
          "state": "CA",
          "country": "US",
          "postalCode": "94105",
          "latitude": "37.789901",
          "longitude": "-122.396923"
        },
        "shippingAddress": {
          "street": "415 Mission St",
```

```
          "city": "San Francisco",
          "state": "CA",
          "country": "US",
          "postalCode": "94105",
          "latitude": "37.789901",
          "longitude": "-122.396923"
        }
      }
    },
    {
      "invoiceLineId": "5TVR00000004SiqOAE",
      "amountToCredit": 200,
      "taxStrategy": "CopyFromInvoiceLine"
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`description`	String	Description for the credit memo to be created.	Optional	62.0
`effectiveDate`	String	Date when the credit memo takes effect.	Optional	62.0
`invoiceLines`	Credit Invoice Line Input[]	List of the invoice lines to be credited. The invoice line IDs must be related to the invoice ID specified in the API request. If invoice lines aren't specified, the API request results in an error.	Required	62.0
`taxEffectiveDate`	String	Date when the tax takes effect to recalculate the taxes.	Optional	62.0
`taxStrategy`	String	Tax strategy to be applied across invoice lines. You can override the tax strategy at the individual invoice line level or at the tax line level. Valid values are: • `Ignore`—Specifies that the creation of tax lines must be ignored. • `ManualOverride`—Specifies that the provided tax values must be considered for taxes. • `CopyFromInvoiceLine`—Specifies that tax values must be copied from the invoice line. • `Calculate`—Specifies that tax must be calculated by using the API.	Required	62.0

Name	Type	Description	Required or Optional	Available Version
`type`	String	Type of credit memo to be created. Valid values are `Posted` and `Draft`. Specify `Draft` as a value in your request to create draft credit memos.	Optional	62.0

Credit Invoice Line Input

Input representation of the details of the invoice lines to be credited.

JSON example


```
"invoiceLines": [
  {
    "invoiceLineId": "5TVR00000004SiqOBE",
    "amountToCredit": 100,
    "taxStrategy": "ManualOverride",
    "taxEffectiveDate": "2021-08-01T21:22:41.000Z",
    "taxes": [
      {
        "taxAmount": 15,
        "taxName": "abc",
        "taxCode": "taxCode",
        "taxRate": 7
      }
    ],
    "addresses": {
      "billingAddress": {
        "street": "1 Market St #300",
        "city": "San Francisco",
        "state": "CA",
        "country": "US",
        "postalCode": "94105",
        "latitude": "37.789901",
        "longitude": "-122.396923"
      },
      "shippingAddress": {
        "street": "415 Mission St",
        "city": "San Francisco",
        "state": "CA",
        "country": "US",
        "postalCode": "94105",
        "latitude": "37.789901",
        "longitude": "-122.396923"
      }
    }
  },
  {
    "invoiceLineId": "5TVR00000004SiqOAE",
    "amountToCredit": 200,
    "taxStrategy": "CopyFromInvoiceLine"
```

```
    }
]
```

Properties

Name	Type	Description	Required or Optional	Available Version
`addresses`	Credit Memo Addresses Input[]	Addresses to be created manually for this invoice line and the overridden tax lines. These addresses are only applicable if this invoice line is using the `ManualOverride` tax strategy.	Optional	62.0
`amountToCredit`	Double	Amount to be credited from this invoice line.	Required	62.0
`invoiceLineId`	String	ID of the invoice line record to be credited. The invoice line ID must be related to the invoice ID specified in the API request.	Required	62.0
`isTaxOnlyCredit`	Boolean	Indicates whether the applicable tax amount is credited for the charge or adjustment amount (`true`), or the applicable tax amount is credited along with the charge or adjustment amount (`false`). The default value is `false`.	Optional	62.0
`taxEffectiveDate`	String	Date when the tax takes effect and the invoice line is credited.	Optional	62.0
`taxStrategy`	String	Tax strategy for crediting the invoice line. This tax strategy takes precedence over the `taxStrategy` property value specified in the Credit Invoice Input. Valid values are: • `Ignore`—Specifies that the creation of tax lines must be ignored. • `ManualOverride`—Specifies that the provided tax values must be considered for taxes. • `CopyFromInvoiceLine`—Specifies that tax values must be copied from the invoice line. • `Calculate`—Specifies that tax must be calculated by using the API.	Optional	62.0
`taxes`	Credit Invoice Line Tax Input[]	List of tax lines to be created manually for this invoice line.	Required if the `taxStrategy` property value is `ManualOverride`.	62.0

Credit Invoice Line Tax Input

Input representation of the details of the tax lines to be created manually for the invoice line.

JSON example

```
      "taxes": [
        {
          "taxAmount": 15,
          "taxName": "abc",
          "taxCode": "taxCode",
          "taxRate": 7
        }
      ]
```

Properties

Name	Type	Description	Required or Optional	Available Version
taxAmount	Double	Amount of tax to be applied related to this invoice line.	Required	62.0
taxCode	String	Tax code to be applied related to this invoice line to create the tax line.	Optional	62.0
taxName	String	Name of tax to be applied related to this invoice line.	Optional	62.0
taxRate	Double	Tax rate used to create the tax line.	Optional	62.0

Credit Memo Addresses Input

Input representation of the details of the billing and shipping addresses.

JSON example

```
{
  "billingAddress": {
    "street": "1 Market St #300",
    "city": "San Francisco",
    "state": "CA",
    "country": "US",
    "postalCode": "94105",
    "latitude": "37.789901",
    "longitude": "-122.396923"
  },
  "shippingAddress": {
    "street": "415 Mission St",
    "city": "San Francisco",
    "state": "CA",
    "country": "US",
    "postalCode": "94105",
    "latitude": "37.789901",
    "longitude": "-122.396923"
```

```
    }
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`billing Address`	Address Input[]	Billing address for charge or adjustment line.	Optional	62.0
`shipping Address`	Address Input[]	Shipping address for charge or adjustment line.	Optional	62.0

Credit Memo Apply Input

Input representation of the request to apply a credit memo to an invoice.

JSON example

```
{
  "applications": [
    {
      "appliedToId": "3ttxx000000003FAAQ",
      "amount": 10,
      "description": "Apply to invoice for refund",
      "effectiveDate": "2024-07-01"
    },
    {
      "appliedToId": "3ttxx0000000001AAA",
      "amount": 100
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`applications`	Credit Memo Apply Application Input[]	List of one or more applications to apply the credit memo for. Each application represents an invoice that's credited by using the balance of the specified credit memo.	Required	62.0

Credit Memo Apply Application Input

Input representation of the request to specify one or more applications to apply a credit memo for, with each application representing an invoice.

JSON example

```
"applications": [
  {
    "appliedToId": "3ttxx000000003FAAQ",
    "amount": 10,
    "description": "Apply to invoice for refund",
    "effectiveDate": "2024-07-01"
  },
  {
    "appliedToId": "3ttxx0000000001AAA",
    "amount": 100
  }
]
```

Properties

Name	Type	Description	Required or Optional	Available Version
`amount`	Double	Credit amount to be applied to the invoice.	Required	62.0
`appliedToId`	String	ID of the invoice record to apply the credit for.	Required	62.0
`description`	String	Explanation or reason for applying the credit memo.	Optional	62.0
`effectiveDate`	String	Effective date for the credit memo.	Optional	62.0

Credit Memo Draft to Posted Input

Input representation of the request to post a draft credit memo.

JSON example

```
{
"creditMemoIds": ["50gDU00000001MnYAI"]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`correlationId`	String	Splunk correlation ID to use to track messages that are related to the request and logged in Splunk by the different services involved in the request. If not specified, the service creates a random Universally Unique Identifier (UUID).	Optional	65.0

Name	Type	Description	Required or Optional	Available Version
creditMemoIds	String[]	ID of the credit memo record in Draft status to be posted. You can post one draft credit memo per API request.	Required	65.0

Credit Memo Unapply Input

Input representation of the request to unapply a credit memo from an invoice.

JSON example


```
{
  "description": "Unapply credit memo from invoice to revert an error",
  "effectiveDate": "2024-07-01"
}
```


Properties

Name	Type	Description	Required or Optional	Available Version
description	String	Explanation or reason for unapplying the credit memo.	Optional	62.0
effectiveDate	String	Effective date for the credit memo.	Optional	62.0

Credit Memo Line Apply Input

Input representation of the details of the request to apply a credit memo line to an invoice line.

JSON example


```
{
 "applyCreditDetails": [
 {
 "invoiceLineId": "5TVSG0000002ZJR4A2",
 "appliedAmount": 5,
 "description": "Apply to invoice line 1",
 "effectiveDate": "2024-07-01"
 },
 {
 "invoiceLineId": "5TVSG0000002ZJS4A2",
 "appliedAmount": 10,
 "description": "Apply to invoice line 2",
 "effectiveDate": "2024-07-01"
 }
 ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
applyCredit Details	Credit Memo Line Application Input[]	List of one or more applications to apply the credit memo line for. Each application represents an invoice line that's credited by using the balance of the specified credit memo line.	Required	62.0

Credit Memo Line Application Input

Input representation of the request to specify one or more applications to apply a credit memo line for, with each application representing an invoice line.

JSON example

```
"applyCreditDetails": [
{
"invoiceLineId": "5TVSG0000002ZJR4A2",
"appliedAmount": 5,
"description": "Apply to invoice line 1",
"effectiveDate": "2024-07-01"
},
{
"invoiceLineId": "5TVSG0000002ZJS4A2",
"appliedAmount": 10,
"description": "Apply to invoice line 2",
"effectiveDate": "2024-07-01"
}
]
```

Properties

Name	Type	Description	Required or Optional	Available Version
appliedAmount	Double	Credit amount to be applied to the invoice line.	Required	62.0
description	String	Explanation or reason for applying the credit memo line.	Optional	62.0
effectiveDate	String	Effective date for the credit memo line.	Optional	62.0
invoiceLineId	String	ID of the invoice line record to apply the credit for.	Required	62.0

Credit Memo Line Unapply Input

Input representation of the details of the request to unapply a credit memo line from an invoice line.

JSON example

```
{
 "description": "Unapply a credit memo line from invoice line 1",
 "effectiveDate": "2024-07-01"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
description	String	Explanation or reason for unapplying the credit memo line.	Optional	62.0
effectiveDate	String	Effective date for the credit memo line.	Optional	62.0

Customer Details Input

Input representation of the customer details for tax calculation.

JSON example

```
{
    "accountId": "001R000000000zSMAQ"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
accountId	String	Salesforce account ID of the customer.	Optional	62.0

Frequency Cadence Options

Input representation of the frequency cadence options for an invoice scheduler.

JSON example

```
  "frequencyCadenceOptions": {
        "recurringSubType" : "Every",
        "recursOn" : "First",
        "recursOnDay" : "Sunday",
        "shouldExcludeWkendAndHldy": true
    }
```

Properties

Name	Type	Description	Required or Optional	Available Version
`recurringSub Type`	String	Subtype of the recurring frequency for the invoice run. Valid values are: • `Every`—Specifies if the invoice scheduler must generate the invoices on a recurring frequency on a specific cadence. Use this value to generate invoices on a specific day of the month. For example, you can specify that the invoice scheduler must generate the invoices every first Monday of the month. • `SpecificDate`—Specifies if the invoice scheduler must generate the invoices on a recurring frequency on a specific date. Use this value to generate invoices on a monthly basis on a specific date.	Required if the `frequency Cadence` property is set to `Monthly`.	62.0
`recursOn`	String	Cadence that specifies when the invoice scheduler must generate the invoices on a recurring frequency. For example, you can specify that the invoice scheduler must generate the invoices every first Monday of the month. Valid values are: • `First` • `Second` • `Third` • `Fourth` • `Last`	Required if the `frequency Cadence` property is set to `Monthly`.	62.0
`recursOnDate`	String	Date when the invoice scheduler must generate the invoices on a specific date. The supported values are: • 1 through 28—Specify any date from 1 through 28. • `L`—Specifies that the invoice scheduler must generate the invoices on the last day of the month. • `L-1`—Specifies that the invoice scheduler must generate the invoices on the second to last day of the month.	Required if the `recurringSub Type` property is set to `SpecificDate`.	62.0

Name	Type	Description	Required or Optional	Available Version
		• L-2—Specifies that the invoice scheduler must generate the invoices on the third to last day of the month.		
recursOnDay	String	Day of the week when the invoice scheduler must generate the invoices on a recurring frequency. For example, you can specify that the invoice scheduler must generate the invoices every Monday or every first Monday of a month. Valid values are: • Sunday • Monday • Tuesday • Wednesday • Thursday • Friday • Saturday	Required if the frequency Cadence property is set to Weekly or Monthly.	62.0
shouldExclude WkendAndHldy	Boolean	Indicates whether to exclude weekends and holidays from the billing schedule (true) or not (false).	Required if the frequency Cadence property is set to Monthly.	62.0

Graph Record for Invoice Ingestion

A Graph record is an object that's a part of the graph structure, representing both the fields and relationships among different objects. Each record in the graph can contain attributes, which are fields of the object, and references to other related records.

Invoice ingestion supports a Graph record count of 500. The supported Graph attribute types for invoice ingestion are Account, Contact, Invoice, InvoiceLine, InvoiceLineTax, and InvoiceAddressGroup.

A Graph record has these properties and associated fields.

Field	Description
referenceId	Unique identifier of the record that's used to reference the record in the graph.
record	Object containing the actual data for the record.
record.attributes.type	Standard object of the referenced record. In this scenario, this field indicates an InvoiceAddressGroup attribute type.
record.attributes.method	Method that defines the operation on the record. For example, POST to create a record, PUT to update a record, or GET to get record data.

JSON example

```
{
  "referenceId": "refBillingAddress",
  "record": {
    "attributes": {
      "type": "InvoiceAddressGroup",
      "method": "POST"
    },
    //Contains the actual data
  }
}
```

Account Record

Keep these considerations in mind when you specify an Account record as the Graph attribute type.

- The supported Graph attribute method is GET only.
- The supported record count that you can specify for each account is one only.
- The id field on attributes is the value of the externalId field API Name in Account. Use this field to resolve the Salesforce Account ID from the custom external ID field. See Use an 'External ID' to set the values for audit fields.

JSON example

```
{
  "referenceId": "refAccount",
  "record": {
    "attributes": {
      "type": "Account",
      "method": "GET",
      "id": "AccountId__c/7661eaaf-f527-4dcf-beb3-301f3eddcd9e"
    }
  }
}
```

Contact Record

Keep these considerations in mind when you specify a Contact record as the Graph attribute type.

- The supported Graph attribute method is GET only.
- The supported record count that you can specify for Contact is one only.
- The id field on attributes is the value of the externalId field API Name in Contact. Use this field to resolve the Salesforce Contact ID from the custom external ID field. See Use an 'External ID' to set the values for audit fields.

JSON example

```
{
  "referenceId": "refContact",
  "record": {
    "attributes": {
      "type": "Contact",
      "method": "GET",
      "id": "ContactId__c/7661eaaf-f527-4dcf-beb3-301f3eddcd9e"
    }
```

```
  }
}
```

Invoice Record

Keep these considerations in mind when you specify an Invoice record as the Graph attribute type.

- The supported Graph attribute method is POST only.
- The supported record count that you can specify for Invoice is one only.
- An invoice with `Draft` status can't have a posted date.
- You can't use an invoice with `Draft` status when the `taxCalculationStatus` property value is `Posted`. The `taxCalculationStatus` property must be `Estimated` or `Pending`.
- You can't use an invoice with `Posted` status when the `shouldCalculateTax` property is `true`.
- An invoice with `Posted` status can't include the `Estimated` or `Pending` as the `taxCalculationStatus` property values. The tax processing status must be `Posted`.
- An invoice with `Posted` status with an invoice line marked as `taxable` must include an `invoiceLineTax` for that invoice line.
- An invoice with `Posted` status with an invoice line marked as `nontaxable` must not include an `invoiceLineTax` for that invoice line.
- The `invoice.CreationMode` field value is `External`. This field differentiates the invoices created from a billing schedule or an invoice ingestion. For invoices created from a billing schedule, the value is Salesforce. For invoices created from an invoice ingestion, the value is `External`.
- The due date of an invoice graph is in the future or with a valid payment term, or your Salesforce org has a default payment term set.

JSON example

```
{
  "referenceId": "refInvoice",
  "record": {
    "attributes": {
      "type": "Invoice",
      "method": "POST"
    },
    "invoiceDate": "2024-01-01",
    "billingAccountId": "001xx000003Dzo9AAC",
    "billToContactId": "003xx000004TzFoAAI",
    "paymentTermId": "2OXxx0000004CFUGA2",
    "referenceEntityId": "801xx000003GeQQAA0",
    "status": "Draft",
    "currencyIsoCode": "USD",
    "dueDate": "2024-02-01",
    "postedDate": "2024-01-02",
    "invoiceNumber": "INV-12345",
    "uniqueIdentifier": "1a9380c1-8042-422d-bcc5-70c3f51c2588",
    "description": "Consulting Services"
  }
}
```

Table 16: Properties

NAME	Standard Object Field	Description	Required or Optional
`invoiceDate`	Invoice.invoiceDate	Date when the invoice was created or issued.	Required
`billingAccountId`	Invoice.billingAccountId	Billing account associated with this invoice, which you can resolve with the `externalId` field by using the GET method.	Required
`billToContactId`	Invoice.billToContactId	Contact to whom the invoice is billed, which you can resolve with the `externalId` field by using the GET method.	Required
`paymentTermId`	Invoice.paymentTermId	Payment term associated with the invoice. If a value isn't specified, then the default payment term ID is used.	Optional
`referenceEntityId`	Invoice.referenceEntityId	Reference to a related object, if applicable. To generate invoices from debit memos, specify the debit memo ID as the property value. Make sure that the debit memo is in `Posted` status. Available in API version 66.0 and later. The invoice is generated in `Draft` status.	Optional
`status`	Invoice.status	Status of the invoice. Valid values are: • `Draft` • `Posted`. The default value is `Draft`.	Optional
`currencyIsoCode`	Invoice.currencyIsoCode	ISO code representing the currency of the invoice. This property must be specified if multi-currency is enabled in the organization.	Optional
`dueDate`	Invoice.dueDate	Due date for the invoice payment. If a value isn't specified, then this value is set based on the payment term.	Optional
`postedDate`	Invoice.postedDate	Date the invoice was posted to the system. Until API version 64.0, the default value is the current date irrespective of the value that's specified in the input payload. In API version 65.0 and later, the posted date specified in the input payload is considered.	Required in API version 65.0 and later.
`invoiceNumber`	Invoice.invoiceNumber	Unique identifier for the invoice.	Optional
`uniqueIdentifier`	Invoice.uniqueIdentifier	Unique identifier for the invoice. This property is used as the idempotency key to avoid duplicate invoice generation.	Optional

NAME	Standard Object Field	Description	Required or Optional
description	Invoice.description	Short description of the invoice or the billed items.	Optional

Invoice Line Record

Keep these considerations in mind when you specify an Invoice Line record as the Graph attribute type.

- The supported Graph attribute method is POST only.
- The Graph must include at least one Graph record with the InvoiceLine attribute type.
- The invoice line end date must be greater than or equal to the invoice line start date.

JSON example

```
{
  "referenceId": "refInvoiceLine",
  "record": {
    "attributes": {
      "type": "InvoiceLine",
      "method": "POST"
    },
    "name": "Parle G",
    "invoiceLineStartDate": "2024-01-01",
    "invoiceLineEndDate": "2024-01-31",
    "quantity": 10,
    "unitPrice": 25,
    "chargeAmount": 250,
    "invoiceId": "@{refInvoice.id}",
    "shippingAddressId": "@{refShippingAddress.id}",
    "billingAddressId": "@{refBillingAddress.id}",
    "referenceEntityItemId": "802xx000001neB9AAI",
    "taxTreatmentId": "1ttxx0000000BOTAA2",
    "legalEntityId": "0fwxx0000000001AAA",
    "legalEntityAccountingPeriodId": "1HLxx0000004C92GAE",
    "product2Id": "01txx0000006ic4AAA",
    "usageProductId": "01txx0000006ic5AAA",
    "isUsageBasedInvoiceLine": false,
    "usageOverageQuantity": 5,
    "unitOfMeasureId": "0hExx0000000001EAA",
    "description": "Sample product description"
  }
}
```

Table 17: Properties

NAME	Standard Object Field	Description	Required or Optional
name	InvoiceLine.name	Name or description of the invoice line item.	Required
invoiceLineStartDate	InvoiceLine.invoiceLineStartDate	Start date for the invoice line period. For example, the date when the product or service is provided.	Required

NAME	Standard Object Field	Description	Required or Optional
`invoiceLineEndDate`	InvoiceLine.invoiceLineEndDate	End date for the invoice line period. For example, the date when the product or service ends.	Required
`quantity`	InvoiceLine.quantity	Quantity of the product or billed service in this invoice line.	Required
`unitPrice`	InvoiceLine.unitPrice	Price per unit of the product or billed service.	Required
`chargeAmount`	InvoiceLine.chargeAmount	Total charge amount for this invoice line, which is calculated as quantity * unit price.	Required
`invoiceId`	InvoiceLine.invoiceId	Reference to the invoice that this invoice line is associated with.	Required
`shippingAddressId`	InvoiceLine.shippingAddressId	Shipping address associated with this invoice line.	Required
`billingAddressId`	InvoiceLine.billingAddressId	Billing address associated with this invoice line.	Required
`referenceEntityItemId`	InvoiceLine.referenceEntityItemId	Reference to a related object record for this invoice line.	Optional
`taxTreatmentId`	InvoiceLine.taxTreatmentId	Tax treatment applied to this invoice line such as taxable or nontaxable. If the `TaxTreatmentId` property isn't specified in the request, it's retrieved from the organization's default values. If the organization's defaults aren't set, an error is thrown.	Optional
`legalEntityId`	InvoiceLine.legalEntityId	Legal entity that this invoice line belongs to. If the `LegalEntityId` property isn't specified in the request, it's retrieved from the organization's default values. If the organization's defaults aren't set, an error is thrown.	Optional
`legalEntityAccountingPeriodId`	InvoiceLine.legalEntityAccountingPeriodId	Accounting period for the legal entity that this invoice line belongs to.	Optional
`product2Id`	InvoiceLine.product2Id	Identifier for the product that's billed in this invoice line.	Optional
`usageProductId`	InvoiceLine.usageProductId	Identifier for the usage-based product if this invoice line is for usage-based billing.	Optional
`isUsageBasedInvoiceLine`	InvoiceLine.isUsageBasedInvoiceLine	Boolean value that indicates whether this invoice line is for a usage-based product (`true`) or not (`false`).	Optional
`usageOverageQuantity`	InvoiceLine.usageOverageQuantity	Quantity of usage overage for this invoice line, if applicable.	Optional
`unitOfMeasureId`	InvoiceLine.unitOfMeasureId	Identifier for the unit of measure for the product or service, such as each, kg, or lb.	Optional

NAME	Standard Object Field	Description	Required or Optional
description	InvoiceLine.description	Description of the invoice line item with additional details.	Optional

Tax Record

Keep these considerations in mind when you specify a Tax record as the Graph attribute type.

- The supported Graph attribute method is POST only.
- The associated standard object is InvoiceLineTax.
- The invoiceLineTax graph record must not include the `taxCalculationStatus` property value as `Pending`.

JSON example

```
{
    "referenceId": "refInvoiceLineTax",
    "record": {
        "attributes": {
            "type": "InvoiceLineTax",
            "method": "POST"
        },
        "taxTransactionNumber": "TX123456789",
        "taxAmount": 15.00,
        "taxRate": 0.08,
        "taxName": "Sales Tax",
        "taxCode": "TX-SALES",
        "taxEffectiveDate": "2024-01-01",
        "invoiceLine": "@{refInvoiceLine.id}",
        "taxDocumentNumber": "TAXDOC987654",
        "taxExemptAmount": 0.00,
        "description": "Exempt from VAT"
    }
}
```

Table 18: Properties

NAME	Standard Object Field	Description	Required or Optional
taxTransactionNumber	InvoiceLineTax.taxTransactionNumber	Unique identifier of the tax transaction related to this invoice line tax.	Required
taxAmount	InvoiceLineTax.taxAmount	Total tax amount applied to the invoice line.	Required
taxRate	InvoiceLineTax.taxRate	Rate at which the tax is applied on the invoice line.	Required
taxName	InvoiceLineTax.taxName	Name of the tax applied on the invoice line, such as Sales Tax or Value Added Tax (VAT).	Required
taxCode	InvoiceLineTax.taxCode	Tax code that corresponds to the applied tax rate.	Required
taxEffectiveDate	InvoiceLineTax.taxEffectiveDate	Effective date from which the tax rate applies.	Required

NAME	Standard Object Field	Description	Required or Optional
invoiceLine	InvoiceLineTax.invoiceLine	Reference to the invoice line this tax record is associated with.	Required
taxDocumentNumber	InvoiceLineTax.taxDocumentNumber	Document number associated with the tax transaction, such as tax return document or government filing number.	Required
taxExemptAmount	InvoiceLine.billingAddressId	Billing address associated with this invoice line.	Required
description	InvoiceLineTax.description	Additional details about the tax.	Optional

Shipping Address Record

Keep these considerations in mind when you specify a Shipping Address record as the Graph attribute type.

- The supported Graph attribute method is POST only.
- The associated standard object is InvoiceAddressGroup.

JSON example

```
{
    "referenceId": "refShippingAddress",
    "record": {
        "attributes": {
            "type": "InvoiceAddressGroup",
            "method": "POST"
        },
        "street": "456 Elm St",
        "city": "Springfield",
        "postalCode": "62701",
        "state": "IL",
        "country": "USA",
        "longitude": "-89.6501",
        "latitude": "39.7817",
        "invoiceId": "@{refInvoice.id}"
    }
}
```

Table 19: Properties

NAME	Standard Object Field	Description	Required or Optional
invoiceId	InvoiceAddressGroup.invoice	Unique identifier of the invoice associated with this shipping address.	Required
street	InvoiceAddressGroup.street	Street address where the goods or services are shipped to.	Required
city	InvoiceAddressGroup.city	City where the shipment is delivered.	Required
postalCode	InvoiceAddressGroup.postalCode	Postal or ZIP code for the shipping address.	Required

NAME	Standard Object Field	Description	Required or Optional
state	InvoiceAddressGroup.state	State or province of the delivery location.	Required
country	InvoiceAddressGroup.country	Country where the shipment is delivered.	Required
longitude	InvoiceAddressGroup.longitude	Geographic longitude of the shipping address location for mapping purposes.	Optional
latitude	InvoiceAddressGroup.latitude	Geographic latitude of the shipping address location for mapping purposes.	Optional

Billing Address Record

Keep these considerations in mind when you specify a Billing Address record as the Graph attribute type.

- The supported Graph attribute method is POST only.
- The associated standard object is InvoiceAddressGroup.

JSON example

```
{
    "referenceId": "refBillingAddress",
    "record": {
        "attributes": {
            "type": "InvoiceAddressGroup",
            "method": "POST"
        },
        "street": "789 Oak St",
        "city": "Los Angeles",
        "postalCode": "90001",
        "state": "CA",
        "country": "USA",
        "longitude": "-118.2437",
        "latitude": "34.0522",
        "invoiceId": "@{refInvoice.id}"
    }
}
```

Table 20: Properties

NAME	Standard Object Field	Description	Required or Optional
invoiceId	InvoiceAddressGroup.invoice	Unique identifier of the invoice associated with this billing address.	Required
street	InvoiceAddressGroup.street	Street address for billing purposes.	Required
city	InvoiceAddressGroup.city	City for billing address.	Required
postalCode	InvoiceAddressGroup.postalCode	Postal or ZIP code for the billing address.	Required
state	InvoiceAddressGroup.state	State or province for the billing address.	Required
country	InvoiceAddressGroup.country	Country for the billing address.	Required

NAME	Standard Object Field	Description	Required or Optional
`longitude`	InvoiceAddressGroup.longitude	Geographic longitude of the billing address location for mapping purposes.	Optional
`latitude`	InvoiceAddressGroup.latitude	Geographic latitude of the billing address location for mapping purposes.	Optional

Invoice Draft To Posted Input

Input representation of the details of the draft invoice that's posted.

JSON example

```
{
  "invoiceIds": ["3ttxx0000004CIjAAM"]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`correlationId`	String	Splunk correlation ID to track the messages that are related to the request and are logged in Splunk by the different services involved in the request. If the ID isn't specified, the service creates a random Universally Unique Identifier (UUID).	Optional	62.0
`invoiceIds`	String[]	IDs of the invoice records in `Draft` status to be posted. You can post one draft invoice per API request.	Required	62.0

Invoice Estimated Tax Calculation Input

Details of the invoice for which the estimated tadsfsefx must be calculated.

JSON example

```
{
  "invoiceIds": ["3ttxx0000004CIjAAM"]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`correlationId`	String	Splunk correlation ID to track the messages that are related to the request and are	Optional	63.0

Name	Type	Description	Required or Optional	Available Version
		logged in Splunk by the different services involved in the request. If the ID isn't specified, the service creates a random Universally Unique Identifier (UUID).		
`invoiceIds`	String[]	IDs of the invoices for which the estimated tax must be calculated. You can specify one invoice per API request.	Required	63.0

Invoice Ingestion Input

Input representation of the details of the invoice to be processed. The details include the tax processing status, user preferences for tax callouts, and associated object graph representation.

JSON example to ingest a draft invoice with a tax callout

```
{
  "invoices": [
    {
      "shouldCalculateTax": true,
      "taxCalculationStatus": "Estimated",
      "graph": {
        "graphId": "CreateInvoice",
        "records": [
          {
            "referenceId": "refAccount",
            "record": {
              "attributes": {
                "type": "Account",
                "method": "GET",
                "id": "ExternalId__c/123"
              }
            }
          },
          {
            "referenceId": "refContact",
            "record": {
              "attributes": {
                "type": "Contact",
                "method": "GET",
                "id": "ExternalId__c/123"
              }
            }
          },
          {
            "referenceId": "refInvoice",
            "record": {
              "attributes": {
                "type": "Invoice",
                "method": "POST"
```

```
      },
      "billingAccountId": "@{refAccount.Id}",
      "billToContactId": "@{refContact.Id}",
      "paymentTermId": "2OXxx0000004CFUGA2",
      "referenceEntityId": "801xx000003GeQQAA0",
      "status": "Draft",
      "invoiceDate": "2024-12-19",
      "currencyIsoCode": "USD",
      "dueDate": "2024-12-19",
      "invoiceNumber": "DOC-10",
      "description": "Sample Invoice",
      "uniqueIdentifier": "5873af8f-f007-4aa0-9e3d-53a08c3f59de"
    }
  },
  {
    "referenceId": "refBillingAddress",
    "record": {
      "attributes": {
        "type": "InvoiceAddressGroup",
        "method": "POST"
      },
      "street": "123 Main St",
      "city": "NewYork",
      "postalCode": "10001",
      "state": "New York",
      "country": "US",
      "longitude": "123.456",
      "latitude": "78.910",
      "invoiceId": "@{refInvoice.id}"
    }
  },
  {
    "referenceId": "refShippingAddress",
    "record": {
      "attributes": {
        "type": "InvoiceAddressGroup",
        "method": "POST"
      },
      "street": "123 Main St",
      "city": "NewYork",
      "postalCode": "10001",
      "state": "New York",
      "country": "US",
      "longitude": "123.456",
      "latitude": "78.910",
      "invoiceId": "@{refInvoice.id}"
    }
  },
  {
    "referenceId": "refShipFromAddress",
    "record": {
      "attributes": {
        "type": "InvoiceAddressGroup",
        "method": "POST"
```

```
        },
        "street": "1 Market Street",
        "city": "San Francisco",
        "postalCode": "94105",
        "state": "CA",
        "country": "US",
        "longitude": "123.456",
        "latitude": "78.910",
        "invoiceId": "@{refInvoice.id}"
      }
    },
    {
      "referenceId": "refInvoiceLine1",
      "record": {
        "attributes": {
          "type": "InvoiceLine",
          "method": "POST"
        },
        "name": "productName1",
        "product2Id": "01txx0000006ic4AAA",
        "invoiceLineStartDate": "2024-11-10",
        "invoiceLineEndDate": "2024-11-13",
        "quantity": "10",
        "unitPrice": "10",
        "chargeAmount": "100",
        "invoiceId": "@{refInvoice.id}",
        "referenceEntityItemId": "802xx000001neB9AAI",
        "billingAddressId": "@{refBillingAddress.id}",
        "shippingAddressId": "@{refShippingAddress.id}",
        "shipFromAddressId": "@{refShipFromAddress.id}",
        "taxTreatmentId": "1ttxx0000000BOTAA2",
        "legalEntityId": "0fwxx0000000001AAA",
        "legalEntityAccountingPeriodId": "1HLxx0000004C92GAE"
      }
    },
    {
      "referenceId": "refInvoiceLine2",
      "record": {
        "attributes": {
          "type": "InvoiceLine",
          "method": "POST"
        },
        "name": "productName2",
        "product2Id": "01txx0000006ic4AAA",
        "invoiceLineStartDate": "2024-11-10",
        "invoiceLineEndDate": "2024-11-15",
        "quantity": "10",
        "unitPrice": "10",
        "chargeAmount": "100",
        "invoiceId": "@{refInvoice.id}",
        "referenceEntityItemId": "802xx000001neB9AAI",
        "billingAddressId": "@{refBillingAddress.id}",
        "shippingAddressId": "@{refShippingAddress.id}",
        "shipFromAddressId": "@{refShipFromAddress.id}",
```

```
              "taxTreatmentId": "1ttxx00000001DpAAI",
              "legalEntityId": "0fwxx0000000001AAA",
              "legalEntityAccountingPeriodId": "1HLxx0000004C92GAE"
            }
          },
          {
            "referenceId": "refInvoiceLineTax",
            "record": {
              "attributes": {
                "type": "InvoiceLineTax",
                "method": "POST"
              },
              "taxAmount": 7.25,
              "taxCode": "CA-94121",
              "taxName": "CALIFORNIA",
              "taxRate": 0.25,
              "taxEffectiveDate": "2024-11-10",
              "taxDocumentNumber": "123",
              "taxExemptAmount": 0,
              "taxTransactionNumber": "kl",
              "description": "Associated tax line.",
              "invoiceLineId": "@{refInvoiceLine1.id}"
            }
          },
          {
            "referenceId": "refInvoiceLineTax1",
            "record": {
              "attributes": {
                "type": "InvoiceLineTax",
                "method": "POST"
              },
              "taxAmount": "10",
              "taxCode": "CA-94121",
              "taxName": "CALIFORNIA",
              "taxRate": 0.25,
              "taxEffectiveDate": "2024-11-10",
              "taxDocumentNumber": "123",
              "taxExemptAmount": 0,
              "taxTransactionNumber": "125",
              "invoiceLineId": "@{refInvoiceLine2.id}",
              "description": "Associated tax line."
            }
          }
        ]
      }
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`correlationId`	String	Splunk correlation ID to track the messages that are related to the request and are logged in Splunk by the different services involved in the request. If the ID isn't specified, the service creates a random Universally Unique Identifier (UUID).	Optional	63.0
`graph`	Object Graph Input	Graph that represents the invoice structure for invoice ingestion or generation. The supported Graph attribute types for invoice ingestion are Account, Contact, Invoice, InvoiceLine, InvoiceLineTax, and InvoiceAddressGroup. See Graph Record for Invoice Ingestion.	Required	63.0
`shouldCalculateTax`	Boolean	Indicates whether the estimated tax must be calculated for the ingested invoice (`true`) or not (`false`). The default value is `false`.	Optional	63.0
`taxCalculationStatus`	String	Status of the tax calculation, which is saved on the invoice line. Valid values are: • `Estimated` • `Pending` • `Posted` The default value is `Pending`.	Optional	63.0

Invoice Input for Ingestion

Input representation of the details of the invoice that must be generated for or ingested into Billing.

JSON example to ingest a draft invoice with a tax callout

```
{
  "invoices": [
    {
      "shouldCalculateTax": true,
      "taxCalculationStatus": "Estimated",
      "graph": {
        "graphId": "CreateInvoice",
        "records": [
          {
            "referenceId": "refAccount",
            "record": {
```

```
        "attributes": {
          "type": "Account",
          "method": "GET",
          "id": "ExternalId__c/123"
        }
      }
    },
    {
      "referenceId": "refContact",
      "record": {
        "attributes": {
          "type": "Contact",
          "method": "GET",
          "id": "ExternalId__c/123"
        }
      }
    },
    {
      "referenceId": "refInvoice",
      "record": {
        "attributes": {
          "type": "Invoice",
          "method": "POST"
        },
        "billingAccountId": "@{refAccount.Id}",
        "billToContactId": "@{refContact.Id}",
        "paymentTermId": "2OXxx0000004CFUGA2",
        "referenceEntityId": "801xx000003GeQQAA0",
        "status": "Draft",
        "invoiceDate": "2024-12-19",
        "currencyIsoCode": "USD",
        "dueDate": "2024-12-19",
        "invoiceNumber": "DOC-10",
        "description": "Sample Invoice",
        "uniqueIdentifier": "5873af8f-f007-4aa0-9e3d-53a08c3f59de"
      }
    },
    {
      "referenceId": "refBillingAddress",
      "record": {
        "attributes": {
          "type": "InvoiceAddressGroup",
          "method": "POST"
        },
        "street": "123 Main St",
        "city": "NewYork",
        "postalCode": "10001",
        "state": "New York",
        "country": "US",
        "longitude": "123.456",
        "latitude": "78.910",
        "invoiceId": "@{refInvoice.id}"
      }
    },
```

```
{
  "referenceId": "refShippingAddress",
  "record": {
    "attributes": {
      "type": "InvoiceAddressGroup",
      "method": "POST"
    },
    "street": "123 Main St",
    "city": "NewYork",
    "postalCode": "10001",
    "state": "New York",
    "country": "US",
    "longitude": "123.456",
    "latitude": "78.910",
    "invoiceId": "@{refInvoice.id}"
  }
},
{
  "referenceId": "refShipFromAddress",
  "record": {
    "attributes": {
      "type": "InvoiceAddressGroup",
      "method": "POST"
    },
    "street": "1 Market Street",
    "city": "San Francisco",
    "postalCode": "94105",
    "state": "CA",
    "country": "US",
    "longitude": "123.456",
    "latitude": "78.910",
    "invoiceId": "@{refInvoice.id}"
  }
},
{
  "referenceId": "refInvoiceLine1",
  "record": {
    "attributes": {
      "type": "InvoiceLine",
      "method": "POST"
    },
    "name": "productName1",
    "product2Id": "01txx0000006ic4AAA",
    "invoiceLineStartDate": "2024-11-10",
    "invoiceLineEndDate": "2024-11-13",
    "quantity": "10",
    "unitPrice": "10",
    "chargeAmount": "100",
    "invoiceId": "@{refInvoice.id}",
    "referenceEntityItemId": "802xx000001neB9AAI",
    "billingAddressId": "@{refBillingAddress.id}",
    "shippingAddressId": "@{refShippingAddress.id}",
    "shipFromAddressId": "@{refShipFromAddress.id}",
    "taxTreatmentId": "1ttxx0000000BOTAA2",
```

```
        "legalEntityId": "0fwxx0000000001AAA",
        "legalEntityAccountingPeriodId": "1HLxx0000004C92GAE"
      }
    },
    {
      "referenceId": "refInvoiceLine2",
      "record": {
        "attributes": {
          "type": "InvoiceLine",
          "method": "POST"
        },
        "name": "productName2",
        "product2Id": "01txx0000006ic4AAA",
        "invoiceLineStartDate": "2024-11-10",
        "invoiceLineEndDate": "2024-11-15",
        "quantity": "10",
        "unitPrice": "10",
        "chargeAmount": "100",
        "invoiceId": "@{refInvoice.id}",
        "referenceEntityItemId": "802xx000001neB9AAI",
        "billingAddressId": "@{refBillingAddress.id}",
        "shippingAddressId": "@{refShippingAddress.id}",
        "shipFromAddressId": "@{refShipFromAddress.id}",
        "taxTreatmentId": "1ttxx00000001DpAAI",
        "legalEntityId": "0fwxx0000000001AAA",
        "legalEntityAccountingPeriodId": "1HLxx0000004C92GAE"
      }
    },
    {
      "referenceId": "refInvoiceLineTax",
      "record": {
        "attributes": {
          "type": "InvoiceLineTax",
          "method": "POST"
        },
        "taxAmount": 7.25,
        "taxCode": "CA-94121",
        "taxName": "CALIFORNIA",
        "taxRate": 0.25,
        "taxEffectiveDate": "2024-11-10",
        "taxDocumentNumber": "123",
        "taxExemptAmount": 0,
        "taxTransactionNumber": "kl",
        "description": "Associated tax line.",
        "invoiceLineId": "@{refInvoiceLine1.id}"
      }
    },
    {
      "referenceId": "refInvoiceLineTax1",
      "record": {
        "attributes": {
          "type": "InvoiceLineTax",
          "method": "POST"
        },
```

```
                "taxAmount": "10",
                "taxCode": "CA-94121",
                "taxName": "CALIFORNIA",
                "taxRate": 0.25,
                "taxEffectiveDate": "2024-11-10",
                "taxDocumentNumber": "123",
                "taxExemptAmount": 0,
                "taxTransactionNumber": "125",
                "invoiceLineId": "@{refInvoiceLine2.id}",
                "description": "Associated tax line."
              }
            }
          ]
        }
      }
  ]
}
```

JSON example to ingest a draft invoice without a tax callout

```
{
  "invoices": [
    {
      "taxCalculationStatus": "Estimated",
      "graph": {
        "graphId": "123",
        "records": [
          {
            "referenceId": "refAccount",
            "record": {
              "attributes": {
                "type": "Account",
                "method": "GET",
                "id": "TestExternalId__c/123"
              }
            }
          },
          {
            "referenceId": "refInvoice",
            "record": {
              "attributes": {
                "type": "Invoice",
                "method": "POST"
              },
              "billingAccountId": "@{refAccount.Id}",
              "billToContactId": "003xx000004Wk8qAAC",
              "paymentTermId": "2OXxx0000004CFUGA2",
              "referenceEntityId": "801xx000003GeQQAA0",
              "status": "Draft",
              "invoiceDate": "2024-12-19",
              "currencyIsoCode": "USD",
              "dueDate": "2024-12-19",
              "invoiceNumber": "DOC-10",
              "description": "testInvoice",
              "uniqueIdentifier": "c76011a1-e113-49d9-9c54-3b5c68950ada"
```

```
    }
  },
  {
    "referenceId": "refBillingAddress",
    "record": {
      "attributes": {
        "type": "InvoiceAddressGroup",
        "method": "POST"
      },
      "street": "123 Main St",
      "city": "NewYork",
      "postalCode": "10001",
      "state": "New York",
      "country": "US",
      "longitude": "123.456",
      "latitude": "78.910",
      "invoiceId": "@{refInvoice.id}"
    }
  },
  {
    "referenceId": "refShippingAddress",
    "record": {
      "attributes": {
        "type": "InvoiceAddressGroup",
        "method": "POST"
      },
      "street": "123 Main St",
      "city": "NewYork",
      "postalCode": "10001",
      "state": "New York",
      "country": "US",
      "longitude": "123.456",
      "latitude": "78.910",
      "invoiceId": "@{refInvoice.id}"
    }
  },
  {
    "referenceId": "refShipFromAddress",
    "record": {
      "attributes": {
        "type": "InvoiceAddressGroup",
        "method": "POST"
      },
      "street": "1 Market Street",
      "city": "San Francisco",
      "postalCode": "94105",
      "state": "CA",
      "country": "US",
      "longitude": "123.456",
      "latitude": "78.910",
      "invoiceId": "@{refInvoice.id}"
    }
  },
  {
```

```
    "referenceId": "refInvoiceLine1",
    "record": {
      "attributes": {
        "type": "InvoiceLine",
        "method": "POST"
      },
      "name": "productName1",
      "product2Id": "01txx0000006ic4AAA",
      "invoiceLineStartDate": "2024-11-10",
      "invoiceLineEndDate": "2024-11-13",
      "quantity": "10",
      "unitPrice": "10",
      "chargeAmount": "100",
      "invoiceId": "@{refInvoice.id}",
      "referenceEntityItemId": "802xx000001neB9AAI",
      "billingAddressId": "@{refBillingAddress.id}",
      "shippingAddressId": "@{refShippingAddress.id}",
      "shipFromAddressId": "@{refShipFromAddress.id}",
      "taxTreatmentId": "1ttxx0000000BOTAA2",
      "legalEntityId": "0fwxx0000000001AAA",
      "legalEntityAccountingPeriodId": "1HLxx0000004C92GAE"
    }
  },
  {
    "referenceId": "refInvoiceLine2",
    "record": {
      "attributes": {
        "type": "InvoiceLine",
        "method": "POST"
      },
      "name": "productName2",
      "product2Id": "01txx0000006ic4AAA",
      "invoiceLineStartDate": "2024-11-10",
      "invoiceLineEndDate": "2024-11-15",
      "quantity": "10",
      "unitPrice": "10",
      "chargeAmount": "100",
      "invoiceId": "@{refInvoice.id}",
      "referenceEntityItemId": "802xx000001neB9AAI",
      "billingAddressId": "@{refBillingAddress.id}",
      "shippingAddressId": "@{refShippingAddress.id}",
      "shipFromAddressId": "@{refShipFromAddress.id}",
      "taxTreatmentId": "1ttxx00000001DpAAI",
      "legalEntityId": "0fwxx0000000001AAA",
      "legalEntityAccountingPeriodId": "1HLxx0000004C92GAE"
    }
  },
  {
    "referenceId": "refInvoiceLineTax",
    "record": {
      "attributes": {
        "type": "InvoiceLineTax",
        "method": "POST"
      },
```

```
            "taxAmount": 7.25,
            "taxCode": "CA-94121",
            "taxName": "CALIFORNIA",
            "taxRate": 0.25,
            "taxEffectiveDate": "2024-11-10",
            "taxDocumentNumber": "123",
            "taxExemptAmount": 0,
            "taxTransactionNumber": "kl",
            "description": "description for tax Line",
            "invoiceLineId": "@{refInvoiceLine1.id}"
          }
        },
        {
          "referenceId": "refInvoiceLineTax1",
          "record": {
            "attributes": {
              "type": "InvoiceLineTax",
              "method": "POST"
            },
            "taxAmount": "10",
            "taxCode": "CA-94121",
            "taxName": "CALIFORNIA",
            "taxRate": 0.25,
            "taxEffectiveDate": "2024-11-10",
            "taxDocumentNumber": "123",
            "taxExemptAmount": 0,
            "taxTransactionNumber": "125",
            "invoiceLineId": "@{refInvoiceLine2.id}",
            "description": "description for tax Line"
          }
        }
      ]
    }
  }
 ]
}
```

JSON example to ingest posted invoices

```
{
  "invoices": [
    {
      "taxCalculationStatus": "Posted",
      "graph": {
        "graphId": "123",
        "records": [
          {
            "referenceId": "refAccount",
            "record": {
              "attributes": {
                "type": "Account",
                "method": "GET",
                "id": "ExternalId__c/123"
              }
            }
```

```
      },
      {
        "referenceId": "refInvoice",
        "record": {
          "attributes": {
            "type": "Invoice",
            "method": "POST"
          },
          "billingAccountId": "001SG00000njpF3YAI",
          "billToContactId": "003xx000004Wk8qAAC",
          "paymentTermId": "2OXxx0000004CFUGA2",
          "referenceEntityId": "801xx000003GeQQAA0",
          "status": "Posted",
          "invoiceDate": "2024-12-19",
          "currencyIsoCode": "USD",
          "dueDate": "2024-12-19",
          "invoiceNumber": "DOC-10",
          "description": "Sample Invoice",
          "uniqueIdentifier": "9994b2c4-c0c3-47c3-806f-ae6e1f16bac3"
        }
      },
      {
        "referenceId": "refBillingAddress",
        "record": {
          "attributes": {
            "type": "InvoiceAddressGroup",
            "method": "POST"
          },
          "street": "123 Main St",
          "city": "NewYork",
          "postalCode": "10001",
          "state": "New York",
          "country": "US",
          "longitude": "123.456",
          "latitude": "78.910",
          "invoiceId": "@{refInvoice.id}"
        }
      },
      {
        "referenceId": "refShippingAddress",
        "record": {
          "attributes": {
            "type": "InvoiceAddressGroup",
            "method": "POST"
          },
          "street": "123 Main St",
          "city": "NewYork",
          "postalCode": "10001",
          "state": "New York",
          "country": "US",
          "longitude": "123.456",
          "latitude": "78.910",
          "invoiceId": "@{refInvoice.id}"
        }
```

```
        },
        {
          "referenceId": "refShipFromAddress",
          "record": {
            "attributes": {
              "type": "InvoiceAddressGroup",
              "method": "POST"
            },
            "street": "1 Market Street",
            "city": "San Francisco",
            "postalCode": "94105",
            "state": "CA",
            "country": "US",
            "longitude": "123.456",
            "latitude": "78.910",
            "invoiceId": "@{refInvoice.id}"
          }
        },
        {
          "referenceId": "refInvoiceLine1",
          "record": {
            "attributes": {
              "type": "InvoiceLine",
              "method": "POST"
            },
            "name": "productName1",
            "product2Id": "01txx0000006ic4AAA",
            "invoiceLineStartDate": "2024-11-10",
            "invoiceLineEndDate": "2024-11-13",
            "quantity": "10",
            "unitPrice": "10",
            "chargeAmount": "100",
            "invoiceId": "@{refInvoice.id}",
            "referenceEntityItemId": "802xx000001neB9AAI",
            "billingAddressId": "@{refBillingAddress.id}",
            "shippingAddressId": "@{refShippingAddress.id}",
            "shipFromAddressId": "@{refShipFromAddress.id}",
            "taxTreatmentId": "1ttxx0000000BOTAA2",
            "legalEntityId": "0fwxx0000000001AAA",
            "legalEntityAccountingPeriodId": "1HLxx0000004C92GAE"
          }
        },
        {
          "referenceId": "refInvoiceLine2",
          "record": {
            "attributes": {
              "type": "InvoiceLine",
              "method": "POST"
            },
            "name": "productName2",
            "product2Id": "01txx0000006ic4AAA",
            "invoiceLineStartDate": "2024-11-10",
            "invoiceLineEndDate": "2024-11-15",
            "quantity": "10",
```

```
              "unitPrice": "10",
              "chargeAmount": "100",
              "invoiceId": "@{refInvoice.id}",
              "referenceEntityItemId": "802xx000001neB9AAI",
              "billingAddressId": "@{refBillingAddress.id}",
              "shippingAddressId": "@{refShippingAddress.id}",
              "shipFromAddressId": "@{refShipFromAddress.id}",
              "taxTreatmentId": "1ttxx00000001DpAAI",
              "legalEntityId": "0fwxx0000000001AAA",
              "legalEntityAccountingPeriodId": "1HLxx0000004C92GAE"
            }
          },
          {
            "referenceId": "refInvoiceLineTax1",
            "record": {
              "attributes": {
                "type": "InvoiceLineTax",
                "method": "POST"
              },
              "taxAmount": "10",
              "taxCode": "CA-94121",
              "taxName": "CALIFORNIA",
              "taxRate": 0.25,
              "taxEffectiveDate": "2024-11-10",
              "taxDocumentNumber": "123",
              "taxExemptAmount": 0,
              "taxTransactionNumber": "125",
              "invoiceLineId": "@{refInvoiceLine1.id}",
              "description": "Associated tax line."
            }
          }
        ]
      }
    }
  ]
}
```


JSON example to generate invoices from debit memos


```
{
  "invoices": [
    {
      "shouldCalculateTax": true,
      "taxCalculationStatus": "Estimated",
      "graph": {
        "graphId": "CreateInvoice",
        "records": [
          {
            "referenceId": "refInvoice",
            "record": {
              "attributes": {
                "type": "Invoice",
                "method": "POST"
              },
              "accountId": "001SB00001V3RxNYAV",
```

```
                "billToContactId": "003xx000004WhUQAA0",
                "referenceEntityId": "4DmAAC000008UP70BM",
                "status": "Draft",
                "invoiceDate": "2024-12-19",
                "dueDate": "2024-12-19",
                "description": "Sample Invoice"
              }
            }
          ]
        }
      }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
invoices	Invoice Ingestion Input[]	List of invoices to be generated or ingested, enabling the submission and processing of multiple invoices in a single request. This API supports one invoice per request. To send 25 requests at a time, see the Composite Batch request.	Required	63.0

Invoice Input

Input representation of the details of the billing schedule.

JSON example

```
{
  "accountId": "001SG00000mYtRWYA0",
  "action": "Posted",
  "billingScheduleIds": [
    "44bSG000000CVeMYAW"
  ],
  "billingTransactionId": "801SG00000mYtaXYAS",
  "correlationId": null,
  "invoiceDate": "2024-01-12",
  "targetDate": "2024-01-12"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
accountId	String	ID of the account record to create the invoices for.	Required if the billing ScheduleIds or billing	63.0

Name	Type	Description	Required or Optional	Available Version
			TransactionId property isn't specified.	
action	String	Type of invoice to be created. Valid values are: • Draft • Posted	Required	62.0
billing ScheduleIds	String[]	List of billing schedule IDs that's used to create the invoices. You can specify a maximum of 200 billing schedules.	Required if the accountId or billing TransactionId property isn't specified.	62.0
billing TransactionId	String	ID of the billing transaction record, which is the order ID, to create the invoices for.	Required if the accountId or billing ScheduleIds property isn't specified.	63.0
correlationId	String	Property that's tagged against the published InvoiceProcessedEvent event, if specified.	Optional	62.0
invoiceDate	String	Stamping date of the invoice in ISO 8601 format.	Required	62.0
targetDate	String	Date in ISO 8601 format used to decide the billing periods that are included to create invoices.	Required	62.0

Invoice Preview Input

Input representation of the details of the billing transaction that the preview invoices are generated for.

JSON example

```
{
    "billingTransactionId": "801Z600000004LoIAI",
    "numberOfBillingPeriods": 2,
    "previewDate": "2024-12-04"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
billing TransactionId	String	ID of the record to generate the preview invoices for.	Required	63.0
numberOf BillingPeriods	Integer	Number of billing periods that the invoice preview is generated for. If unspecified, the default value is 2.	Optional	64.0
previewDate	String	The date on which the preview invoice is generated. For the first invoice, the preview date is the target date for generating the invoice. For the second invoice, the target date is calculated based on the preview date and the minimum billing frequency of the transactions. The default value is the current date.	Optional	63.0

Line Item Input

Input representation of the details of the line item for tax calculation.

JSON example


```
{
  "lineItems": [
    {
      "quantity": 1,
      "amount": 100,
      "taxCode": "TX0001",
      "productCode": "Y0001",
      "productSKU": "PRES-RX-12896745",
      "description": "New Product",
      "effectiveDate": "2022-03-09T10:30:41.000Z",
      "lineNumber": "5TVxx0000004C92GAE",
      "addresses": {
        "shipFrom": {
          "street": "123 Alaskan Way",
          "city": "Seattle",
          "state": "WA",
          "country": "US",
          "postalCode": "98101",
          "latitude": 45.12,
          "longitude": 45.12
        },
        "shipTo": {
          "street": "123 Auburn Blvd",
          "city": "Sacramento",
          "state": "CA",
```

```
          "country": "US",
          "postalCode": "95841",
          "latitude": 45.12,
          "longitude": 45.12
        },
        "soldTo": {
          "street": "123 Auburn Blvd",
          "city": "Sacramento",
          "state": "CA",
          "country": "US",
          "postalCode": "95841",
          "latitude": 45.12,
          "longitude": 45.12
        },
        "billTo": {
          "street": "123 Auburn Blvd",
          "city": "Sacramento",
          "state": "CA",
          "country": "US",
          "postalCode": "95841",
          "latitude": 45.12,
          "longitude": 45.12
        }
      }
    }
  ]
}
```


Properties

Name	Type	Description	Required or Optional	Available Version
addresses	Addresses Input	Address types that specify different locations associated with the transaction of the line item.	Optional	62.0
amount	Double	Total amount for the line item.	Required	62.0
description	String	Description of the line item.	Optional	62.0
effectiveDate	String	Date that tax rules are applicable from. This property value overrides the global effective date.	Optional	62.0
legalEntity	String	Legal entity that's related to the tax treatment.	Optional	63.0
lineNumber	String	Unique identifier for the line item.	Required	62.0
productCode	String	Product code of the line item according to the tax treatment rules of the tax engine.	Optional	62.0
productId	String	ID of the product.	Optional	63.0

Name	Type	Description	Required or Optional	Available Version
productSKU	String	Unique identifier of a product that can be used to identify products that are exempted from tax.	Optional	64.0
quantity	Double	Quantity of the line item.	Required	62.0
taxCode	String	Tax code of the line item according to the tax treatment rules of the tax engine.	Optional	62.0
unitPrice	Double	Unit price of the product.	Optional	63.0

On-Demand Document Generation Input

Input representation of the details to generate a comprehensive on-demand document for specified record types.

JSON example

```
{
  "recordId": "3ttasd98776266777",
  "shouldForceRegenerate": true,
  "documentTemplateId": "0694x000000XyzABC",
  "documentTitle": "DOC-00000012",
  "tokenData":
[illegible]
 Market
St\",\"CompanyName\":\"260.1\",\"TotalAmt\":\"$7.99\",\"TotalTax\":\"$0.00\",\"InvoiceDueDate\":\"10/08/2025\",\"CompanyPostalCode\":\"94105\",\"InvoiceDate\":\"09/08/2025\",\"CompanyCity\":\"San
 Francisco\",\"CompanyCountry\":\"US\",\"AccountName\":\"Sonia BAT Account\"}"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
documentTemplateId	String	Document template ID to use for PDF generation. The document template must be active with UsageType as `INVOICE` If you don't specify this value, the system auto-resolves by using the default template.	Optional	66.0
documentTitle	String	Custom title for the generated document. If you don't specify a value, the value is auto-generated.	Optional	66.0
recordId	String	Record ID of the object the document is generated for. You can specify invoice ID only.	Required	66.0

Name	Type	Description	Required or Optional	Available Version
shouldForceRegenerate	Boolean	Indicates whether to regenerate the document (`true`) or not (`false`). If set to `true`, this API generates a document, replacing any existing ones. If set to `false`, this API skips the generation of document if invoice already has an associated document. The default value is `false`.	Optional	66.0
tokenData	String	Token data to generate the document.	Optional	66.0

Payment Line Apply Input

Input representation of the payment line details. This representation covers details on allocation of a payment to a specific invoice line. It also provides additional context through optional fields such as associated account and effective date.

JSON example

```
{
  "appliedToId": "3ttR000000001IkIAI",
  "amount": 10,
  "effectiveDate": "2020-08-11T07:53:15.000Z",
  "comments": "Apply payment",
  "associatedAccountId": "001R00000060AyuIAE"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
amount	Double	Amount that's applied. The amount must be less than the invoice line and payment balance.	Required	64.0
appliedToId	String	ID of the invoice line that this payment is applied to. Specify the IDs for these records. • Invoice • Invoice Line	Required	64.0
associated AccountId	String	ID of the associated account.	Optional	64.0
comments	String	Comments that you can add to the payment line application.	Optional	64.0
effectiveDate	String	Date from which the payment line application takes effect.	Optional	64.0

Payment Line Unapply Input

Input representation of the payment line details. This representation covers fields that you can specify to revert a payment line application to their preapplication state.

JSON example

```
{
  "effectiveDate": "2025-05-22T11:30:25.000Z",
  "comments": "Unapply payment"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
comments	String	Comments that you can add when you revert a payment line application.	Optional	64.0
effectiveDate	String	Date from when the reversal of the payment line application is in effect.	Optional	64.0

Payment Run Batch Filter Criteria Input

Input representation of the filter criteria for an invoice batch run. This representation covers the criteria and sequence for filtering payment run details. It specifies the field and object names, comparison operations, and values to be used for filtering.

JSON example

```
  "filterCriteria": [
    {
      "objectName": "PaymentScheduleItem",
      "fieldName": "PaymentRunMatchingValue",
      "operation": "Equals",
      "value": "1",
      "criteriaSequence": 1
    },
    {
      "objectName": "PaymentScheduleItem",
      "fieldName": "PaymentRunMatchingValue",
      "operation": "Equals",
      "value": "2",
      "criteriaSequence": 2
    },
    {
      "objectName": "PaymentScheduleItem",
      "fieldName": "PaymentRunMatchingValue",
      "operation": "Equals",
      "value": "3",
      "criteriaSequence": 3
    }
  ]
```

Properties

Name	Type	Description	Required or Optional	Available Version
criteriaSequence	Integer	Sequence that's used to filter the payment run details.	Required	64.0
fieldName	String	Name of the field that this filter is applicable for. Valid value is PaymentRunMatchingValue.	Required	64.0
objectName	Object	Name of the object that the filter is applicable for. Valid value is PaymentScheduleItem.	Required	64.0
operation	String	Operation that's used for comparison. Valid values are: • Equals • InList • NotEquals	Required	64.0
value	String	Value that's used for the filter criteria.	Required	64.0

Payment Batch Scheduler Input

Input representation of the details of the request to create a payment scheduler. This representation sets the rules and timing for a payment scheduler, including match types, dates, frequency, and filter criteria.

JSON example

JSON example

```
{
  "schedulerName": "Payment Scheduler",
  "startDate": "2022-01-01",
  "endDate": "2022-12-31",
  "preferredTime": "02:05:00.000",
  "frequencyCadence": "Monthly",
  "recursEveryMonthOnDay": "28",
  "criteriaMatchType": "MatchAny",
  "status": "Active",
  "filterCriteria": [
    {
      "objectName": "PaymentScheduleItem",
      "fieldName": "PaymentRunMatchingValue",
      "operation": "Equals",
      "value": "1",
      "criteriaSequence": 1
    },
    {
      "objectName": "PaymentScheduleItem",
```

```
        "fieldName": "PaymentRunMatchingValue",
        "operation": "Equals",
        "value": "2",
        "criteriaSequence": 2
      },
      {
        "objectName": "PaymentScheduleItem",
        "fieldName": "PaymentRunMatchingValue",
        "operation": "Equals",
        "value": "3",
        "criteriaSequence": 3
      }
    ]
  }
```

Properties

Name	Type	Description	Required or Optional	Available Version
criteria MatchType	String	Match type for the criteria of the payment scheduler. Valid values are: • Match Any • Match None	Required if the frequencyCadence property is set to Monthly.	64.0
endDate	String	End date of the payment scheduler.	Required if the frequencyCadence property is set to Monthly.	64.0
frequency Cadence	String	Frequency cadence of the payment scheduler. Valid values are: • Once • Daily • Weekly • Monthly	Required	64.0
filter Criteria	Payment Run Batch Filter Criteria Input	List of criteria that are used to filter the payment run details.	Required if the frequencyCadence property is set to Monthly.	64.0
preferredTime	String	Preferred time for the payment scheduler run.	Required	64.0
recursEvery MonthOnDay	String	Date when the payment scheduler recurs.	Required if the frequencyCadence property is set to Monthly.	64.0

Name	Type	Description	Required or Optional	Available Version
schedulerName	String	Name of the payment scheduler.	Required	64.0
startDate	String	Start date of the payment scheduler.	Required	64.0
status	String	Status of the payment scheduler. Valid values are: • Active • Canceled • Draft • Inactive	Required	64.0

Payment Scheduler Update Input

Input representation of the details of the request to update the status of a payment scheduler. This representation defines the status of a payment scheduler, which can be set to Active, Canceled, Draft, or Inactive.

JSON example

JSON example

```
{
  "status": "Active"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
status	String	Status that must be set to activate or deactivate a payment scheduler. Valid values are: • Active • Canceled • Draft • Inactive	Required	64.0

Posted Invoice List Write-Off Input

Input representation of the request to write off a list of posted invoices. This representation includes the details of invoices that you want to write off.

JSON example

```
{
  "invoices": [
    {
      "invoiceId": "3ttxx00000000cjAAA",
      "reasonCode": "Bad Debt",
      "description": "Bad Debt"
    },
    {
      "invoiceId": "3ttxx00000000cjAAA",
      "reasonCode": "Concession",
      "description": "Concession"
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`invoices`	Posted Invoice Write-Off Input[]	Details of the invoices that you want to write off.	Required	64.0

Posted Invoice Write-Off Input

Input representation of the details of the request to write off a posted invoice. This representation includes invoice details such as invoice ID and reason for writing off invoices.

JSON example

```
{
  "invoices": [
    {
      "invoiceId": "3ttxx00000000cjAAA",
      "reasonCode": "Bad Debt",
      "description": "Bad Debt"
    },
    {
      "invoiceId": "3ttxx00000000cjAAA",
      "reasonCode": "Concession",
      "description": "Concession"
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`invoiceId`	String	ID of the invoice record that you want to write off.	Required	64.0

Name	Type	Description	Required or Optional	Available Version
reason	String	Reason for writing off invoices.	Optional	64.0
reasonCode	String	Code that categorizes the write-off reason. For example, if the reason for the invoice write-off is a disputed amount, the reason code can be Disputed Amount (DA).	Required	64.0

Refund Line Apply Input

Input representation of the details of a transaction refund request. This representation outlines the properties of a refund, including the refund amount and ID of the payment or credit memo record that the refund is applied to.

JSON example

```
{
  "appliedToId": "0aQR00000004ZkKMAU",
  "amount": 10,
  "effectiveDate": "2020-08-11T07:53:15.000Z",
  "comments": "Payment application."
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
amount	Double	Amount to refund.	Required	64.0
appliedToId	String	ID of a payment or credit memo record. The refund is applied to this object.	Required	64.0
comments	String	Additional details of the refund request.	Optional	64.0
effectiveDate	String	Date from when the refund is in effect.	Optional	64.0

Resume Billing Input

Input representation of the details of the request to resume the billing operation for an account or a billing schedule group.

JSON example

```
{
    "referenceIds":
    [
        {
          "referenceId": "001DU000001o2UwYAI",
          "resumeDate": "2024-11-27"
        }
    ]
```

```
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
referenceIds	Resume Billing Object Input[]	Input representation of the account or billing schedule group IDs to resume the billing operation for.	Required	63.0

Resume Billing Object Input

Input representation of the details such as the ID of the account or billing schedule group along with the effective date. These details are used to start the billing operation.

JSON example

```
{
    "referenceIds":
    [
        {
          "referenceId": "001DU000001o2UwYAI",
          "resumeDate": "2024-11-27"
        }
    ]

}
```

Properties

Name	Type	Description	Required or Optional	Available Version
referenceId	String	ID of the account or billing schedule group to resume the billing operation for.	Required	63.0
resumeDate	String	Date when the billing operation is resumed. If a date isn't specified, the default value is today's date. The billing operation starts immediately and any future suspension dates aren't applicable.	Required	63.0

Rules Application Input

Input representation for applying payments and credits to invoices based on rules.

JSON example

```
{
  "accountId": "001xx000003DGbQAAW",
```

```
  "targetDate": "2024-01-15"
}
```


Properties

Name	Type	Description	Required or Optional	Available Version
`accountId`	String	ID of the account to perform the settlement of payments and credits against invoices in adherence with the applied rules.	Required	66.0
`targetDate`	String	The date used to select invoices and invoice lines with a posted date equal to or later than the target date to apply payments and credits.	Optional	66.0

Selection Condition Input

Input representation of the criteria that's used to determine which sequencing policy is applied to a record. The criteria stores the conditions based on any standard or custom fields of the record.

JSON example


```
{
  "selectionCondition": [
    {
      "filterField": "AppType",
      "operator": "Equals",
      "filterValue": "RLM",
      "conditionNumber": 1
    },
    {
      "filterField": "Status",
      "operator": "Equals",
      "filterValue": "Posted",
      "conditionNumber": 2
    },
    {
      "filterField": "LegalEntity",
      "operator": "Equals",
      "filterValue": "US",
      "conditionNumber": 3
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`conditionNumber`	Integer	Unique number that's assigned to a condition in a sequence policy.	Required	65.0
`filterField`	String	Field used in the filter condition.	Required	65.0
`filterValue`	String	Value in the filter condition.	Required	65.0
`operator`	String	Relational operator that's used to compare the filter field with the filter value. Valid values are: • `Equals` • `NotEquals`	Required	65.0

Seller Details Input

Input representation of the seller details for tax calculation.

JSON example

```
{
  "code": "ADIDAS"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`code`	String	Seller code as specified in the tax engine.	Required	62.0

Send Email Input

Input representation of the request to send an email for an invoice batch run.

JSON example

```
{
  "invoiceBatchRunId": "5IRLT000001SIJB4A4"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
invoiceBatchRunId	String	ID of the invoice batch run record to send emails for the posted invoices of an invoice batch run.	Required	65.0

Sequence Policy Input

Input representation of the configured rules and properties to generate unique, sequential numbers for objects.

JSON example

This example shows a sample request to create a sequence policy.


```
{
  "name": "Sample Sequence Policy",
  "description": "This is a sample sequence policy.",
  "effectiveFromDateTime": "2025-08-10",
  "expirationDateTime": "2025-09-20",
  "isActive": true,
  "sequenceMode": "Basic",
  "targetObject": "Invoice",
  "dateStampFormat": "Yyyy",
  "sequenceStartNumber": 1,
  "incrementNumber": 1,
  "maximumSequenceNumber": 1000,
  "minimumSequenceNumberWidth": 2,
  "filterCriteria": "Custom",
  "selectionLogic": "selectionLogic",
  "sequencePattern": "INV-{SequenceValue}-abc",
  "selectionCondition": [
    {
      "filterField": "AppType",
      "operator": "Equals",
      "filterValue": "RLM",
      "conditionNumber": 1
    },
    {
      "filterField": "Status",
      "operator": "Equals",
      "filterValue": "Posted",
      "conditionNumber": 2
    },
    {
      "filterField": "LegalEntity",
      "operator": "Equals",
      "filterValue": "US",
      "conditionNumber": 3
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
dateStamp Format	String	Format of the stamp date that's appended to the sequence number. You can't edit this property by using the Update Sequence Policy API.	Required	65.0
description	String	Additional details about the sequence policy.	Optional	65.0
effectiveFrom DateTime	String	Date and time when the policy becomes effective. The default value is the current date.	Required	65.0
expiration DateTime	String	Date and time when the policy expires.	Optional	65.0
filter Criteria	String	Criteria to filter the target objects.	Required	65.0
increment Number	Integer	Value by which the sequence number increases until it reaches the maximum number. This value must be greater than or equal to `1`. The default value is `1`. You can't edit this property by using the Update Sequence Policy API.	Required	65.0
isActive	Boolean	Indicates whether the policy is active (`true`) or not (`false`). The default value is `false`.	Required	65.0
maximumSequence Number	Integer	Maximum number the sequence number can reach.	Optional	65.0
minimumSequence NumberWidth	Integer	Minimum number of digits a sequence number must have. You can't edit this property by using the Update Sequence Policy API.	Optional	65.0
name	String	Name of the sequence policy.	Required	65.0
selection Condition	Selection Condition Input[]	Criteria to determine which sequence policy is applied to a record. This property includes conditions based on any standard or custom fields of the record.	Optional	65.0
selection Logic	String	Logic that determines the objects that the sequence policy applies to.	Optional	65.0
sequenceMode	String	Specifies how sequence numbers are generated. Valid values are:	Required	65.0

Name	Type	Description	Required or Optional	Available Version
		• `Basic`—Assigns sequential numbers without gap reconciliation. • `Gapless`—Assigns sequential numbers with gap reconciliation. The usage of this value ensures that the posted invoices or credit memos don't have any numbering gaps for audits and compliance. You can't edit this property by using the Update Sequence Policy API.		
`sequencePattern`	String	Pattern structure that's followed for the sequence.	Required	65.0
`sequenceStartNumber`	Integer	Starting value of the sequence number, which must be greater than or equal to `0`. The default value is `1`. You can't edit this property by using the Update Sequence Policy API.	Required	65.0
`targetObject`	String	Object that the policy is applied to. Valid values are: • `Invoice` • `CreditMemo`—Available in API version 66.0 and later. You can't edit this property by using the Update Sequence Policy API.	Required	65.0
`timeZone`	String	Time zone that's applicable for the sequence policy.	Optional	65.0

Sequence Gap Reconciliation Input

Input representation of the details that are used to identify and reconcile gaps in sequence values based on the sequence policy or target object.

JSON example

This example shows a sample request that specifies the list of sequence policies for gap reconciliation.

```
{
  "sequencePolicyIds": [
    "1vdxx0000000abc",
    "1vdxx0000000def"
  ]
}
```

This example shows a sample request that specifies the target invoice object for gap reconciliation.

```
{
  "targetObjects": [
    "Invoice"
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
sequence PolicyIds	String[]	List of IDs of the sequence policies.	Required if the targetObjects property isn't specified. You must not specify both properties.	65.0
targetObjects	String[]	List of objects to which the policies are applied. Valid values are: • Invoice • CreditMemo—Available in API version 66.0 and later.	Required if the sequencePolicyIds property isn't specified. You must not specify both properties.	65.0

Sequences Assignment Input

Input representation of the details of the target objects to which the sequence pattern values are assigned.

JSON example

```
{
  "targetObjectIds": [
    "3ttxx00000005nhAAA",
    "3ttxx00000006bhAAA"
  ],
  "sequencePolicyId": "1Vdxx0000004CFU"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
sequence PolicyId	String	ID of the sequence policy.	Optional	65.0
shouldPublish PlatformEvent	Boolean	Indicates whether to publish a platform event when a sequence is assigned to a target record (true) or not (false).	Optional	65.0

Name	Type	Description	Required or Optional	Available Version
`target ObjectIds`	String[]	List of records to which the sequence pattern values are assigned.	Required	65.0

Standalone Credit Memo Charge Input

Input representation of the details of the charge lines of a credit memo.

JSON example

```
"charges": [
  {
    "chargeAmount": 100,
    "productId": "01tR0000000njDiIAI"
  }
]
```

Properties

Name	Type	Description	Required or Optional	Available Version
`addresses`	Credit Memo Addresses Input	Details of the billing and shipping addresses.	Optional	62.0
`chargeAmount`	Double	Charge amount for the credit memo.	Required	62.0
`description`	String	Description of the created credit memo charge line.	Optional	62.0
`endDate`	String	End date of the credit memo charge line.	Optional	62.0
`isTax OnlyCredit`	Boolean	Indicates whether the credit is for tax only (`true`) or not (`false`).	Optional	62.0
`productId`	String	ID of the product record that the credit memo is issued on.	Optional	62.0
`productName`	String	Name of the product that the credit memo is issued on.	Optional	62.0
`startDate`	String	Start date of the credit memo charge line.	Optional	62.0
`taxEffective Date`	String	Date from when the tax is applicable.	Optional	62.0
`taxes`	Standalone Credit Memo Tax Input[]	List of credit memo tax lines.	Optional	62.0
`taxStrategy`	String	Tax strategy to be applied to this credit memo charge line, child treatment lines, and tax lines. You can override the tax	Optional	62.0

Name	Type	Description	Required or Optional	Available Version
		strategy at the individual credit memo lines or tax lines level. Valid values are: • `Ignore`—Specifies that the creation of tax lines must be ignored. • `Manual Override`—Specifies that the provided tax values must be considered for taxes. • `Calculate`—Specifies that tax must be calculated by using the API.		
`treatmentId`	String	ID of the tax treatment record that's used to calculate tax.	Optional	62.0

Standalone Credit Memo Input

Input representation of the details required to create a standalone credit memo.

JSON example

This example shows a sample request with the `Ignore` tax strategy.

```
{
  "billingAccountId": "001j000000WCFB800x",
  "type": "Posted",
  "description": "Standalone credit memo with ignored tax.",
  "currencyIsoCode": "USD",
  "taxStrategy": "Ignore",
  "charges": [
    {
      "chargeAmount": 100,
      "productId": "01tR0000000njDiIAI"
    }
  ]
}
```

This example shows a sample request with the `ManualOverride` tax strategy.

```
{
  "billingAccountId": "001DU000001nhoPYAQ",
  "description": "creditmemo-1",
  "currencyIsoCode": "USD",
  "taxStrategy": "ManualOverride",
  "charges": [
    {
      "productId": "01tDU000000EpKkYAK",
      "chargeAmount": 1000,
      "taxes": [
        {
          "taxAmount": 7.25,
          "taxCode": "CA-94121",
```

```
            "taxName": "CALIFORNIA",
            "taxRate": 0.25
          }
        ]
      }
    ]
}
```

This example shows a sample request with the `Calculate` tax strategy for a charge line.

```
{
  "billingAccountId": "001DU000001nhoPYAQ",
  "description": "Standalone Credit Memo",
  "type": "Posted",
  "currencyIsoCode": "USD",
  "taxStrategy": "Ignore",
  "charges": [
    {
      "productId": "01tR0000000njDiIAI",
      "chargeAmount": 10,
      "taxStrategy": "Calculate",
      "treatmentId": "1ttxx00000001VZAAY"
    }
  ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`billToContactId`	String	Contact related to the credit memo.	Optional	62.0
`billingAccountId`	String	ID of the account that the credit is issued to.	Required	62.0
`charges`	Standalone Credit Memo Charge Input[]	Charge lines of the credit memo. Requires at least one charge line.	Required	62.0
`currencyIsoCode`	String	ISO code currency of the new credit that's issued.	Optional	62.0
`description`	String	Description for the new credit that's issued.	Optional	62.0
`effectiveDate`	String	Effective date of the credit memo. If the value isn't specified, then it's null.	Optional	62.0
`externalReference`	String	ID of the external reference for the credit memo.	Optional	62.0
`externalReferenceDataSource`	String	Source of the external reference for the credit memo.	Optional	62.0

Name	Type	Description	Required or Optional	Available Version
taxEffectiveDate	String	Effective date of the credit memo tax. If the value isn't specified, then it's null.	Optional	62.0
taxStrategy	String	Specifies how tax lines must be created for the standalone credit memos. Valid values are: • Ignore—Specifies that the creation of tax lines must be ignored. • Manual Override—Specifies that the provided tax values must be considered for taxes. • Calculate—Specifies that tax must be calculated by using the API.	Required	62.0
type	String	Type of credit memo to be created. Valid values are Posted and Draft. Specify Draft as a value in your request to create draft credit memos.	Optional	62.0

Standalone Credit Memo Tax Input

Input representation of the details of the tax request.

JSON example

```
[
  {
    "taxAmount": 200,
    "taxName": "Federal Tax",
    "taxRate": 1
  },
  {
    "taxAmount": 500,
    "taxName": "State Tax",
    "taxRate": 1
  }
]
```

Properties

Name	Type	Description	Required or Optional	Available Version
taxAmount	Double	Amount of tax to be applied.	Required	62.0
taxCode	String	Tax code to be used to create the tax line.	Optional	62.0
taxName	String	Name of tax to be applied.	Optional	62.0

Name	Type	Description	Required or Optional	Available Version
taxRate	Double	Tax rate to be used to create the tax line.	Optional	62.0

Account Statement Input

Input representation of the details required to generate a comprehensive account statement with transaction history and balance information.

JSON example

```
{
  "shouldShowOpenBalancesOnly": false,
  "startDate": "2025-09-01",
  "transactionTypes": {
    "transactionTypes": [
      "Invoice"
    ]
  },
  "sortBy": "DueDate",
  "sortingOrder": "Ascending",
  "associatedAccountIds": {
    "associatedAccountIds": [
      "001xx000003DGb5AAG",
      "001xx000003DGb6AAG"
    ]
  },
  "documentTemplateId": "0TRxx000000002YGAQ",
  "correlationId": "monthly-statement-sept-2025",
  "customFields": "{\"Invoice\": [\"TotalAmountWithTax\"], \"Account\": {\"base\":
[\"Phone\", \"Fax\"], \"associated\": [\"Website\"]}}}"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
associatedAccountIds	String[]	List of associated account IDs from hierarchy to include in the statement. You can specify up to 50 accounts.	Optional	66.0
correlationId	String	Correlation ID for tracking the request across systems.	Optional	66.0
customFields	String	JSON string specifying custom fields to include in the statement. As a best practice, we recommend that you limit the number of custom fields up to 30 fields. Valid objects to specify the associated fields are:	Optional	67.0

Name	Type	Description	Required or Optional	Available Version
		• `Invoice` • `CreditMemo` • `Payment` • `DebitMemo` • `Refund` • `Account` Here's the expected format. `{"Invoice": ["field1", "field2"], "Account": {"base": ["field1"], "associated": ["field2"]}}`		
`documentTemplateId`	String	Document template ID to use for PDF generation. If you don't specify a value, the system auto-resolves by using the default template.	Optional	66.0
`shouldShowOpenBalancesOnly`	Boolean	Indicates whether to show open balances only (`true`) or not (`false`). If set to `true`, the API shows only accounts with non-zero balances. If set to `false`, this API shows complete transaction history within the date range from the start date.	Optional	66.0
`sortBy`	String	Criteria for sorting transactions. The default and valid value is `Date`.	Optional	66.0
`sortingOrder`	String	Sort order for transactions. Valid values are: • `Ascending` • `Descending` The default value is `Descending`.	Optional	66.0
`startDate`	String	Start date for the transaction history. The required format is `YYYY-MM-DD`. The system processes records up to 90 days from this date.	Required	66.0
`transactionTypes`	String[]	List of transaction types to include in the statement. If you don't specify a value or	Optional	66.0

Name	Type	Description	Required or Optional	Available Version
		the value is empty, all transaction types are included. Valid values are: • `All` • `CreditMemo` • `DebitMemo` • `Invoice` • `Payment` • `Refund`		

Suspend Billing Input

Input representation of the details of the request to suspend the billing operation for an account or a billing schedule group.

JSON example

```
{
    "referenceIds":
    [
        {
            "referenceId": "001DU000001o2UwYAI",
            "suspendDate": "2024-11-27",
            "resumeDate": "2024-12-27"
        }
    ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`referenceIds`	Suspend Billing Object Input[]	Input representation of the account or billing schedule group IDs to suspend the billing operation for.	Required	63.0

Suspend Billing Object Input

Input representation of the details such as the ID of the account or billing schedule group along with the effective dates. These details are used to suspend the billing operation.

JSON example

```
{
    "referenceIds":
    [
        {
            "referenceId": "001DU000001o2UwYAI",
```

```
            "suspendDate": "2024-11-27",
            "resumeDate": "2024-12-27"
        }
    ]
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
`referenceId`	String	ID of the account or billing schedule group to suspend the billing operation for.	Required	63.0
`resumeDate`	String	Date until when the account or billing schedule group is suspended for billing.	Required	63.0
`suspendDate`	String	Date when the account or billing schedule group is suspended for billing.	Required	63.0

Tax Calculation Input

Input representation of the details of the request to calculate tax.

JSON example

This example shows a tax request for a specified invoice as the reference entity.

```
{
  "addresses": {
    "billTo": {
      "street": "123 Main Street",
      "city": "Bainbridge Island",
      "state": "WA",
      "postalCode": "98110",
      "country": "US"
    },
    "soldTo": {
      "street": "123 Main Street",
      "city": "Bainbridge Island",
      "state": "WA",
      "postalCode": "98110",
      "country": "US"
    },
    "shipFrom": {
      "street": "123 Alaskan Way",
      "city": "Seattle",
      "state": "WA",
      "country": "US",
      "postalCode": "98101"
    },
    "shipTo": {
      "street": "123 Main street",
      "city": "Bainbridge Island",
      "state": "WA",
```

```
      "postalCode": "98110",
      "country": "US"
    }
  },
  "currencyIsoCode": "USD",
  "customerDetails": {
    "accountId": "001xx000003HYD5AAO"
  },
  "description": "Monthly invoice for account 001xx000003HYEhAAO",
  "documentCode": "3ttxx0000000C7d_Debit-4wAxx00000000ODEAY",
  "effectiveDate": "2022-03-09T10:30:41.000Z",
  "isCommit": true,
  "lineItems": [
    {
      "quantity": 1,
      "amount": 100,
      "taxCode": "TX0001",
      "productCode": "Y0001",
      "productSKU": "PRES-RX-12896745",
      "description": "New Product",
      "effectiveDate": "2022-03-09T10:30:41.000Z",
      "lineNumber": "5TVxx0000004C92GAE",
      "addresses": {
        "shipFrom": {
          "street": "123 Alaskan Way",
          "city": "Seattle",
          "state": "WA",
          "country": "US",
          "postalCode": "98101",
          "latitude": 45.12,
          "longitude": 45.12
        },
        "shipTo": {
          "street": "123 Auburn Blvd",
          "city": "Sacramento",
          "state": "CA",
          "country": "US",
          "postalCode": "95841",
          "latitude": 45.12,
          "longitude": 45.12
        },
        "soldTo": {
          "street": "123 Auburn Blvd",
          "city": "Sacramento",
          "state": "CA",
          "country": "US",
          "postalCode": "95841",
          "latitude": 45.12,
          "longitude": 45.12
        },
        "billTo": {
          "street": "123 Auburn Blvd",
          "city": "Sacramento",
          "state": "CA",
```

```
          "country": "US",
          "postalCode": "95841",
          "latitude": 45.12,
          "longitude": 45.12
        }
      }
    }
  ],
  "referenceDocumentCode" = null,
  "referenceEntityId": "3ttxx0000000C7d",
  "shouldVoidTax": false,
  "taxEngineId": "4wAxx0000000g8v",
  "taxType": "Actual",
  "taxTransactionType": "Debit",
  "transactionDate": "2022-03-09T10:30:41.000Z"
}
```

This example shows a tax request for a specified credit memo as the reference entity.

```
{
  "addresses": {
    "billTo": {
      "street": "123 Main Street",
      "city": "Bainbridge Island",
      "state": "WA",
      "postalCode": "98110",
      "country": "US"
    },
    "soldTo": {
      "street": "123 Main Street",
      "city": "Bainbridge Island",
      "state": "WA",
      "postalCode": "98110",
      "country": "US"
    },
    "shipFrom": {
      "street": "123 Alaskan Way",
      "city": "Seattle",
      "state": "WA",
      "country": "US",
      "postalCode": "98101"
    },
    "shipTo": {
      "street": "123 Main street",
      "city": "Bainbridge Island",
      "state": "WA",
      "postalCode": "98110",
      "country": "US"
    }
  },
  "currencyIsoCode": "USD",
  "customerDetails": {
    "accountId": "001xx000003HYD5AAO"
  },
  "description": "Monthly credit memo for account 001xx000003HYEhAAO",
```

```
"documentCode": "50gxx000000g27KAAQ-4wAxx00000000ODEAY",
"effectiveDate": "2022-03-09T10:30:41.000Z",
"isCommit": true,
"lineItems": [
  {
    "quantity": 1,
    "amount": 100,
    "taxCode": "TX0001",
    "productCode": "Y0001",
    "productSKU": "PRES-RX-12896745",
    "description": "New Product",
    "effectiveDate": "2022-03-09T10:30:41.000Z",
    "lineNumber": "5TVxx0000004C92GAE",
    "addresses": {
      "shipFrom": {
        "street": "123 Alaskan Way",
        "city": "Seattle",
        "state": "WA",
        "country": "US",
        "postalCode": "98101",
        "latitude": 45.12,
        "longitude": 45.12
      },
      "shipTo": {
        "street": "123 Auburn Blvd",
        "city": "Sacramento",
        "state": "CA",
        "country": "US",
        "postalCode": "95841",
        "latitude": 45.12,
        "longitude": 45.12
      },
      "soldTo": {
        "street": "123 Auburn Blvd",
        "city": "Sacramento",
        "state": "CA",
        "country": "US",
        "postalCode": "95841",
        "latitude": 45.12,
        "longitude": 45.12
      },
      "billTo": {
        "street": "123 Auburn Blvd",
        "city": "Sacramento",
        "state": "CA",
        "country": "US",
        "postalCode": "95841",
        "latitude": 45.12,
        "longitude": 45.12
      }
    }
  }
],
"referenceDocumentCode" = null,
```

```
  "referenceEntityId": "50gxx000000g27K",
  "shouldVoidTax": false,
  "taxEngineId": "4wAxx0000000g8v",
  "taxType": "Actual",
  "taxTransactionType": "Debit",
  "transactionDate": "2022-03-09T10:30:41.000Z"
}
```

Properties

Name	Type	Description	Required or Optional	Available Version
addresses	Addresses	Address details for tax calculation.	Optional	62.0
currencyIsoCode	String	Currency ISO code that's used for tax calculation.	Optional	62.0
customerDetails	Customer Details	Customer details for determining the applicable tax.	Optional	62.0
description	String	Description of the tax transaction.	Optional	62.0
documentCode	String	Unique identifier for the tax document. If the documentCode property isn't specified, the tax engine auto-generates it.	Optional	62.0
effectiveDate	String	Date when the tax rules are applied. If a tax rate changes on a specific date, the effectiveDate property ensures the correct rate is applied based on the transaction's timing. For credit-based tax callouts, specify the original invoice date.	Optional	62.0
isCommit	Boolean	Indicates whether to submit the transaction to the tax engine for reporting (true) or not (false). This property value is true for invoices and credit memos with the Posted status.	Optional	62.0
lineItems	Line Item Input []	Details of the line items for calculating the applicable tax.	Required	62.0
reference DocumentCode	String	Reference document code. For subsequent transactions such as a credit tax, this property value specifies the original document code.	Optional	62.0

Name	Type	Description	Required or Optional	Available Version
		For credit-based tax callouts, specify the original invoice ID.		
`referenceEntityId`	String	ID of the related quote, invoice, and other transaction documents.	Optional	62.0
`sellerDetails`	Seller Details Input	Seller details for tax calculation.	Optional	62.0
`shouldVoidTax`	Boolean	Indicates whether to void the tax transaction associated with a document that's mentioned as the `referenceDocumentCode` property value with `taxType` property value as `Actual` and `isCommit` property value set to `true`. Keep these considerations in mind when you use this property. • If the `shouldVoidTax` property value is set to `true`, then the operation returns a response with `documentCode` property value updated to `referenceDocumentCode` property value that was originally sent in the request payload. The response also includes the `taxTransactionType` property value as `Void`. This indicates that the document specified in the `referenceDocumentCode` property value is voided. • If document is locked or you can't void the tax transaction for any reason, then you can use the Tax Calculation request to perform another transaction such as a Credit Tax request. In this scenario, the response includes the `documentCode` property value that was sent in the request payload. • If the document that's mentioned in the `referenceDocumentCode` property value isn't available in the tax engine, then an error response occurs with ResultCode on page 1607 value as `ReferenceDocumentCodeMissing`.	Optional	65.0

Name	Type	Description	Required or Optional	Available Version
`taxEngineId`	String	ID of the tax engine that's used to calculate tax.	Required	62.0
`tax TransactionType`	String	Type of the tax transaction. Valid values are: • `Debit`—Increases tax liability for the seller, requiring the seller to pay tax on the transaction. • `Credit`—Decreases tax liability for the seller, resulting in a tax refund for the seller. • `Void`—Reserved for internal use. The default value is `Debit`.	Optional	62.0
`taxType`	String	Type of the tax. Valid values are: • `Actual`—Exact tax amount that's calculated based on actual sales. • `Estimated`— Estimated tax amount, which is adjusted later to match actual tax calculations. For draft invoices and quote records, the tax type is marked as `Estimated`. After draft invoices are posted and the status changes to `Posted`, the tax type is updated to `Actual`. Similarly, when a quote record is finalized and converted into an actual order, the tax type is also updated to `Actual`.	Required	62.0
`transactionDate`	String	Date of the transaction that appears on the invoice, order, and other transaction documents.	Required	62.0

Void Posted Credit Memo Input

Input representation of the details of a credit memo to be voided.

Properties

Name	Type	Description	Required or Optional	Available Version
`creditMemoId`	String	ID of the posted credit memo to be voided.	Required	66.0

Void Posted Invoice Input

Input representation of the details of the invoice to be voided.

Properties

Name	Type	Description	Required or Optional	Available Version
`invoiceId`	String	ID of the posted invoice to be voided.	Required	62.0

Response Bodies

Learn more about the available response bodies of Billing APIs.

Addresses
Output representation of the details of the addresses that are used for calculating tax.

Address
Output representation of the location code associated with an address.

Batch Invoice Document Generation
Output representation of the request to generate or regenerate the PDF documents for the invoices that are in the `Draft` or `Posted` status.

Error Response
Output representation of the error details encountered during the API request.

Error Response for Batch Invoice Document Generation
Output representation of the error details associated with the Batch Invoice Document Generation API.

Batch Invoice Scheduler
Output representation of the details of an invoice scheduler.

Batch Payment Scheduler
Output representation of the details of a payment scheduler.

Billing Arrangement
Output representation that contains the details of a billing arrangement, including its status, configuration settings, and associated lines.

Billing Arrangement Line
Output representation that contains the details of a specific line item within a billing arrangement, defining how charges are split for an account.

Billing Batch Scheduler
Output representation of the details of a created invoice or payment scheduler.

Billing Schedule Recovery List
Output representation of the recovered details of the billing schedules and associated invoice.

Billing Schedule Recovery
Output representation of the details of the recovered billing schedules.

Context-Aware Billing Schedule
Output representation of the context-aware billing schedule.

Context-Aware Billing Schedule Error

Output representation of the error response related to the generation of the billing schedule.

Convert Negative Invoice Lines

Output representation of the details of the created memo along with the status of the request.

Credit Memo Apply

Output representation of the list of applied credit memo results.

Credit Memo Apply List

Output representation of the list of applied credit memo results.

Credit Memo Post

Output representation of the request to post a credit memo.

Credit Memo Unapply

Output representation of the details of the credit memo invoice application record with the status of the request.

Credit Memo Line Applied

Output representation of the list of applied credit memo line results.

Credit Memo Line Applied Response

Output representation of the list of applied credit memo line results.

Credit Memo Line Unapplied

Output representation of the details of the credit memo line invoice line record with the status of the request.

Document Details

Output representation of the details of the generated document.

Invoice Batch Draft To Posted

Output representation of the batch update details of the invoices from `Draft` to `Posted` status.

Invoice Batch Run Recovery

Output representation of the details of the invoice batch run recovery record.

Invoice Ingestion

Output representation of the details of the generated invoices.

Invoice Ingestion Details

Output representation of the details of a generated invoice.

Invoice Ingestion Output Error

Output representation of the details of an invoice generation error.

Invoice Line Preview

Output representation of the invoice line preview result.

Invoice Preview

Output representation of the invoice preview result.

Invoice Preview Result

Output representation of the list of preview invoices that are generated for the billing transaction.

Invoice Recovery

Output representation of the details of the recovered invoice and billing schedules.

Line Item

Output representation of the details of the line item.

[On-Demand Document Generation Response]

Output representation of the details of the generated document along with error response.

[Payment Line Apply]

Output representation of the details of the applied payment line. The details include the ID of the payment record and date when the payment line was applied.

[Payment Line Unapply]

Output representation of the details of the reversed payment line application. The details include the ID of the payment line record and date when the payment line application was reversed.

[Payment Scheduler Update]

Output representation of the details of the updated payment scheduler. This representation covers the updated status value of the specified payment scheduler.

[Posted Invoice List Write-Off]

Output representation of the list of invoices that are written off.

[Posted Invoice Write-Off]

Output representation of the details of a posted invoice that's written off.

[Posted Invoice Write-Off Error]

Output representation of the error response that's associated with a request to write off a posted invoice.

[Reference Line Error]

Output representation of the details of the line level errors.

[Refund Line Applied Response]

Output representation of the details of an applied refund. This representation includes the properties of a refund line, such as the date when the refund is applied against a payment and ID of the refund line record.

[Revenue Async Line Level]

Output representation of the result of the API request for the async line level operations.

[Revenue Async Response]

Output representation of the result of the API request with the request identifier.

[Rules Application]

Output representation of the details of rules application.

[Rules Application Error]

Output representation of the error details for rules application failure.

[Rules Application Summary]

Output representation of the summary of the application rule. This includes the number of payments and credit memos for the account, the total number of payments and credit memos that's applied to invoices, and whether all invoices for an account are considered (true) or not (false).

[Send Email Error]

Output representation of the error response of the API request to send emails for posted invoices.

[Send Email Response]

Output representation of the API request to send emails for posted invoices.

[Sequence Error]

Output representation of the error response that's associated with a request to create or update a sequence policy, or assign sequences.

Sequence Gap Reconciliation Error

Output representation of the errors encountered during the processing of the API request.

Sequence Gap Reconciliation

Output representation of the details of the sequence gap reconciliation.

Sequence Policy

Output representation that shows the status of the assigned sequence pattern values.

Sequences Assignment

Output representation with the status of the assigned sequence pattern values.

Sequences Assignment Result

Output representation of the details of the assigned sequence values to target objects.

Statement of Account

Output representation of the details of the generated account statement with async tracking details.

Suspend Resume Billing

Output representation of the list of accounts and billing schedule groups, which are suspended or resumed for billing operations.

Suspend Resume Billing Object

Output representation of the details of accounts and billing schedule groups, which are suspended or resumed for billing operations, along with the status of the API request.

Tax Amount Details

Output representation of the details of the tax amount.

Tax Calculation

Output representation of the details of the calculated tax.

Tax Details

Output representation of the tax details for each line item.

Tax Engine Log

Output representation of the logs that the tax engine generates.

Tax Imposition

Output representation of the details of the imposed tax.

Tax Jurisdiction

Output representation of the details of the tax jurisdiction for the tax line.

Void Posted Credit Memo

Output representation of the request to void a posted credit memo.

Addresses

Output representation of the details of the addresses that are used for calculating tax.

JSON example

```
{
  "addresses": {
    "shipFrom": {
      "locationCode": "67890"
    },
    "shipTo": {
```

```
      "locationCode": "12345"
    },
    "soldTo": {
      "locationCode": "12345"
    }
  }
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
shipFrom	Address on page 2726	Address that the item is shipped from.	Big, 62.0	62.0
shipTo	Address on page 2726	Address that the item is shipped to.	Big, 62.0	62.0
soldTo	Address on page 2726	Address that the item is sold to.	Big, 62.0	62.0

Address

Output representation of the location code associated with an address.

JSON example


```
{
    "locationCode": "123456789"
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
locationCode	String	Unique location code of the address.	Big, 62.0	62.0

Batch Invoice Document Generation

Output representation of the request to generate or regenerate the PDF documents for the invoices that are in the `Draft` or `Posted` status.

JSON example


```
{
  "errors": [
    {
      "errorCode": "API_DISABLED_FOR_ORG",
      "errorMessage": "Document Generation is not enabled for this org!"
    },
    {
      "errorCode": "INVALID_API_INPUT",
      "errorMessage": "Invalid Invoice Batch Run Id"
    }
  ],
  "requestIdentifier": "5IRDU000000009i4AA",
  "success": false
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`errors`	Batch Invoice Document Generation Error []	Details of the error if the operation fails.	Big, 63.0	63.0
`requestIdentifier`	String	Unique ID that's associated with the specific error, and is used for tracking and referencing the request.	Big, 63.0	63.0
`success`	Boolean	Indicates whether the operation is successful (`true`) or not (`false`).	Big, 63.0	63.0

Error Response

Output representation of the error details encountered during the API request.

JSON example

This example shows a sample error response.

```
{
  "errors": [
    {
      "errorCode": "INVALID_STATUS",
      "message": "CreditMemo 50gxx00000000XtAAI is not in the Posted status."
    }
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`errorCode`	String	Code that indicates the type of error.	Big, 66.0	66.0
`message`	String	Message stating the reason for error, if any.	Big, 66.0	66.0

Error Response for Batch Invoice Document Generation

Output representation of the error details associated with the Batch Invoice Document Generation API.

JSON example

```
  "errors": {
    "errorCode": "API_DISABLED_FOR_ORG",
    "message": "Document Generation is not enabled for this org!"
  }
```

Property Name	Type	Description	Filter Group and Version	Available Version
`errorCode`	String	Code for the resultant error.	Big, 63.0	63.0

Property Name	Type	Description	Filter Group and Version	Available Version
`errorMessage`	String	Error message for the resultant error.	Big, 63.0	63.0

Batch Invoice Scheduler

Output representation of the details of an invoice scheduler.

JSON example


```
{
    "billingBatchScheduler": {
        "id": "5BSxx0000004TwGGAU"
    }
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`billingBatch Scheduler`	Billing Batch Scheduler[]	Details of the created invoice scheduler.	Big, 62.0	62.0

Batch Payment Scheduler

Output representation of the details of a payment scheduler.

JSON example


```
{
  "billingBatchScheduler": {
    "id": "5BSxx0000004TwGGAU"
  }
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`billingBatch Scheduler`	Billing Batch Scheduler	Details of the payment scheduler.	Big, 64.0	64.0

Billing Arrangement

Output representation that contains the details of a billing arrangement, including its status, configuration settings, and associated lines.

JSON example


```
{
  "billingArrangementId": "1bdxx000000004rAAA",
  "name": "Acme Shared Billing Definition",
  "isAdjustmentToOwnAccount": true,
  "remainderPercentage": 15,
```

```
  "versionNumber": 1,
  "numOfAssociatedBSGs": 5,
  "status": "Active",
  "billingArrangementLines": [
    {
      "billingArrangementLineId": "1blxx000000006TAAQ",
      "accountId": "accId1",
      "billingAccountId": "bAccId1",
      "isRemainderAdjustmentAccount": false,
      "percentage": 60
    },
    {
      "billingArrangementLineId": "1blxx000000001dAAA",
      "accountId": "accId2",
      "billingAccountId": "bAccId2",
      "isRemainderAdjustmentAccount": false,
      "percentage": 25
    }
  ],
  "isSuccess": true,
  "error": null
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`billing ArrangementId`	String	Unique ID of the billing arrangement.	Big, 66.0	66.0
`billing ArrangementLines`	Billing Arrangement Line[]	List of billing arrangement lines associated with the latest version of the billing arrangement.	Big, 66.0	66.0
`error`	Error Response Output	Error details if the request was unsuccessful.	Big, 66.0	66.0
`isAdjustment ToOwnAccount`	Boolean	Indicates whether the remainder of the bill is adjusted to the owning account (`true`) or not (`false`).	Big, 66.0	66.0
`isSuccess`	Boolean	Indicates whether the request was successfully processed (`true`) or not (`false`).	Big, 66.0	66.0
`name`	String	Name of the billing arrangement.	Big, 66.0	66.0
`numOf AssociatedBSGs`	Integer	Total number of billing schedule groups associated with the billing arrangement.	Big, 66.0	66.0
`remainderPercentage`	Double	Remaining percentage of the bill after all line allocations are calculated.	Big, 66.0	66.0

Property Name	Type	Description	Filter Group and Version	Available Version
`status`	String	Status of the billing arrangement. Valid values are: • `Draft` • `Active` • `Inactive`	Big, 66.0	66.0
`versionNumber`	Integer	Current version number of the billing arrangement.	Big, 66.0	66.0

Billing Arrangement Line

Output representation that contains the details of a specific line item within a billing arrangement, defining how charges are split for an account.

JSON example


```
{
  "billingArrangementLineId": "1blxx000000006TAAQ",
  "accountId": "accId1",
  "billingAccountId": "bAccId1",
  "isRemainderAdjustmentAccount": false,
  "percentage": 60.00
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`accountId`	String	ID of the account associated with the billing line.	Big, 66.0	66.0
`billing AccountId`	String	ID of the billing account responsible for this portion of the bill.	Big, 66.0	66.0
`billingArrangement LineId`	String	Unique ID of the billing arrangement line.	Big, 66.0	66.0
`isRemainder AdjustmentAccount`	Boolean	Indicates whether the specific line is designated as the adjustment account for any remainder (`true`) or not (`false`).	Big, 66.0	66.0
`percentage`	Double	Percentage of the charge allocated to this line.	Big, 66.0	66.0

Billing Batch Scheduler

Output representation of the details of a created invoice or payment scheduler.

JSON example

```
    "billingBatchScheduler": {
        "id": "5BSxx0000004TwGGAU"
    }
```

Property Name	Type	Description	Filter Group and Version	Available Version
id	String	ID of the invoice or payment scheduler.	Big, 62.0	62.0

Billing Schedule Recovery List

Output representation of the recovered details of the billing schedules and associated invoice.

JSON example

```
{
  "recoveryResults": [
    {
      "billingSchedules": [
        {
          "billingScheduleId": "44bDU00000000XX",
          "billingScheduleStatus": "ReadyForInvoicing"
        }
      ],
      "invoiceErrors": [],
      "invoiceId": null,
      "success": true
    }
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
recovery Results	Invoice Recovery[]	Details of the recovered invoice and billing schedules.	Big, 62.0	62.0

Billing Schedule Recovery

Output representation of the details of the recovered billing schedules.

JSON example

```
    "billingSchedules": [
        {
          "billingScheduleId": "44bDU00000000XX",
          "billingScheduleStatus": "ReadyForInvoicing"
        }
    ]
```

Property Name	Type	Description	Filter Group and Version	Available Version
billing ScheduleId	String	ID of the billing schedule.	Big, 62.0	62.0
billing Schedule Status	String	Flag that indicates the billing schedule status.	Big, 62.0	62.0

Context-Aware Billing Schedule

Output representation of the context-aware billing schedule.

JSON example

This request shows a sample success response.

```
{
    "errors": null,
    "requestIdentifier": "16Pxx0000004CaS",
   "statusURL": "/services/data/v67.0/sobjects/AsyncOperationTracker/16Pxx0000004CaSEAU",

    "success": true
}
```

This request shows a sample error response.

```
{
    "errors": [
        {
            "errorCode": "REQUIRED_FIELD_MISSING",
            "errorMessage": "Required fields are missing: billToContact",
            "referenceId": "802xx000001nmb5"
        }
    ],
    "requestIdentifier": "16Pxx0000004CYq",
   "statusURL": "/services/data/v67.0/sobjects/AsyncOperationTracker/16Pxx0000004CYqEAM",

    "success": false
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
errors	Context Aware Billing Schedule Error[]	Error response if the generation of the billing schedule fails.	Big, 62.0	62.0
request Identifier	String	Unique request identifier that you can use to poll the asynchronous request.	Big, 62.0	62.0
statusURL	String	Status URL to track the operation.	Big, 62.0	62.0

Property Name	Type	Description	Filter Group and Version	Available Version
`success`	Boolean	Indicates whether the processing of the billing schedule is successful (`true`) or not (`false`).	Big, 62.0	62.0

Context-Aware Billing Schedule Error

Output representation of the error response related to the generation of the billing schedule.

```
    "errors": [
        {
            "errorCode": "REQUIRED_FIELD_MISSING",
            "errorMessage": "Required fields are missing: billToContact",
            "referenceId": "802xx000001nmb5"
        }
    ]
```

Property Name	Type	Description	Filter Group and Version	Available Version
`errorCode`	String	Code for the resultant error.	Big, 62.0	62.0
`errorMessage`	String	Error message for the resultant error.	Big, 62.0	62.0
`referenceId`	String	Reference ID of the source item that resulted in the error.	Big, 62.0	62.0

Convert Negative Invoice Lines

Output representation of the details of the created memo along with the status of the request.

JSON example

```
{
"id": "50gxx000000g0WwAAI",
"success": true,
"errors": []
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`errors`	Error Response	List of errors encountered during the processing of the API request.	Big, 62.0	62.0
`id`	String	ID of the credit memo that's created after the conversion of the negative invoice lines.	Small, 62.0	62.0
`success`	Boolean	Indicates whether the API request is successful (`true`) or not (`false`).	Small, 62.0	62.0

Credit Memo Apply

Output representation of the list of applied credit memo results.

JSON example


```
  "applyCreditResults" : [ {
    "appliedToId" : "3ttxx000000003FAAQ",
    "errors" : null,
    "id" : "4sFxx00000002ppEAA",
    "success" : true
  }, {
    "appliedToId" : "3ttxx0000000001AAA",
    "errors" : null,
    "id" : "4sFxx00000002pqEAA",
    "success" : true
  } ]
```


Property Name	Type	Description	Filter Group and Version	Available Version
appliedToId	String	ID of the invoice record that the credit is applied to.	Big, 62.0	62.0
errors	Error Response	List of errors encountered during the processing of the API request.	Big, 62.0	62.0
id	String	ID of the credit memo invoice application.	Big, 62.0	62.0
success	Boolean	Indicates whether the credit memo is successfully applied (`true`) or not (`false`).	Big, 62.0	62.0

Credit Memo Apply List

Output representation of the list of applied credit memo results.

JSON example


```
{
  "applyCreditResults" : [ {
    "appliedToId" : "3ttxx000000003FAAQ",
    "errors" : null,
    "id" : "4sFxx00000002ppEAA",
    "success" : true
  }, {
    "appliedToId" : "3ttxx0000000001AAA",
    "errors" : null,
    "id" : "4sFxx00000002pqEAA",
    "success" : true
  } ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
applyCredit Results	Credit Memo Apply[]	Output list of the applied credit memo results.	Big, 62.0	62.0

Credit Memo Post

Output representation of the request to post a credit memo.

JSON example

This example shows a sample successful response.

```
{
  "errors": [],
  "requestIdentifier": "d3a9d9ce-2a83-4a08-bcf3-df0348a0008c",
 "statusURL": "/services/data/v67.0/sobjects/AsyncOperationTracker/16PDU000000A4oz2AC",

  "success": true
}
```

This example shows a sample error response.

```
{
  "errors": [
    {
      "errorCode": "BAD_REQUEST",
      "message": "You can post up to 1 credit memo at a time."
    }
  ],
  "requestIdentifier": "9065b043-dcc3-4dcf-b5a1-55fdf5a79f7b",
  "statusURL": "",
  "success": false
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
errors		List of errors encountered during the processing of the API request.	Big, 65.0	65.0
requestIdentifier	String	Unique request identifier for the request.	Big, 65.0	65.0
statusURL	String	Status URL for tracking this operation.	Big, 65.0	65.0
success	Boolean	Indicates whether the API request was successful (true) or not (false).	Big, 65.0	65.0

Credit Memo Unapply

Output representation of the details of the credit memo invoice application record with the status of the request.

JSON example


```
{
    "errors": [],
    "id": "4sFxx00000002ppEAA",
    "success": true
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
errors	Error Response	List of errors encountered during the processing of the API request.	Big, 62.0	62.0
id	String	ID of the credit memo invoice application record.	Small, 62.0	62.0
success	Boolean	Indicates whether the credit memo is successfully unapplied (true) or not (false).	Small, 62.0	62.0

Credit Memo Line Applied

Output representation of the list of applied credit memo line results.

JSON example


```
{
"appliedCreditResponses": [
 {
  "creditMemoLineInvoiceLineId": "4sGSG0000002pMb2AI",
  "errors": null,
  "invoiceLineId": "5TVSG0000003CuH4AU",
  "success": true
 }
]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
applied CreditResponses	Credit Memo Line Applied Response[]	Output list for the applied credit memo line results.	Big, 62.0	62.0

Credit Memo Line Applied Response

Output representation of the list of applied credit memo line results.

JSON example


```
"appliedCreditResponses": [
 {
  "creditMemoLineInvoiceLineId": "4sGSG0000002pMb2AI",
```

```
    "errors": null,
    "invoiceLineId": "5TVSG0000003CuH4AU",
    "success": true
  }
]
```


Property Name	Type	Description	Filter Group and Version	Available Version
`creditMemoLineInvoiceLineId`	String	ID of the credit memo line invoice line ID.	Big, 62.0	62.0
`errors`	Error Response	List of errors encountered during the processing of the API request.	Big, 62.0	62.0
`invoiceLineId`	String	ID of the invoice line record that the credit is applied to.	Big, 62.0	62.0
`success`	Boolean	Indicates whether the credit memo line is successfully applied (`true`) or not (`false`).	Big, 62.0	62.0

Credit Memo Line Unapplied

Output representation of the details of the credit memo line invoice line record with the status of the request.

JSON example


```
 {
 "creditMemoLineInvoiceLineId": "4sGSG0000002pOD2AY",
 "errors": [],
 "success": true
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`creditMemoLineInvoiceLineId`	String	ID of the credit memo line invoice line record.	Small, 62.0	62.0
`errors`	Error Response	List of errors encountered during the processing of the API request.	Big, 62.0	62.0
`success`	Boolean	Indicates whether the credit memo line is successfully unapplied (`true`) or not (`false`).	Small, 62.0	62.0

Document Details

Output representation of the details of the generated document.

JSON example

```
{
  "details": {
    "documentEntityId": "1BBSM00000003KT4AY",
    "documentGenerationProcessId": "0nnSM00000009jFYAQ",
    "documentTitle": "DOC-000000197",
    "existingDocumentId": "069SM000002PYQLYA4",
    "recordId": "3ttSM0000000o58YAA",
    "documentTemplateId": "0694x000000XyzABC"
  }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
documentEntityId	String	Identifier of the related object record that manages the relationship between the source record and the generated document. For example, InvoiceDocument for Invoice records.	Big, 66.0	66.0
documentGenerationProcessId	String	Identifier of the document generation process record that's created to track this document generation request.	Big, 66.0	66.0
documentTemplateId	String	Identifier of the document template that’s used to generate the PDF.	Big, 66.0	66.0
documentTitle	String	Document title or name of the generated document.	Big, 66.0	66.0
existingDocumentId	String	Identifier of the existing content document if a document is already generated and is available to view.	Big, 66.0	66.0
recordId	String	Identifier of the source record the document is generated for.	Big, 66.0	66.0

Invoice Batch Draft To Posted

Output representation of the batch update details of the invoices from `Draft` to `Posted` status.

JSON example

```
{
  "invoiceBatchDraftToPostedId": "4sFDU00000000652AA",
  "success": true
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
invoiceBatchDraftToPostedId	String	ID of the invoice batch draft to posted run record that's created to track the batch process of posting the draft invoices that are associated with the parent invoice batch run record.	Small, 62.0	62.0
success	Boolean	Indicates whether the API request is successful (`true`) or not (`false`).	Small, 62.0	62.0

Invoice Batch Run Recovery

Output representation of the details of the invoice batch run recovery record.

JSON example

```
{
  "invoiceBatchRunRecoveryId": ["0wBxx0000004TtwGAU"]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
invoiceBatchRunRecoveryId	String	ID of the invoice batch run recovery record for the specified invoice batch run. This record represents the background recovery process.	Small, 62.0	62.0

Invoice Ingestion

Output representation of the details of the generated invoices.

JSON example

```
{
  "invoices": [
    {
      "errors": null,
      "invoiceId": "3ttxx0000000GRNAA2",
      "requestIdentifier": "16e80fbc-e7b3-4462-9439-745647fcf0a8",
      "statusURL":
"/services/data/v63.0/sobjects/AsyncOperationTracker/16Pxx0000004Gz2EAE",
      "success": true
    }
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
invoices	Invoice Ingestion []	Result that contains the details for each generated invoice.	Big, 63.0	63.0

Invoice Ingestion Details

Output representation of the details of a generated invoice.

JSON example


```
{
  "invoices": [
    {
      "errors": null,
      "invoiceId": "3ttxx0000000GRNAA2",
      "requestIdentifier": "16e80fbc-e7b3-4462-9439-745647fcf0a8",
      "statusURL":
"/services/data/v63.0/sobjects/AsyncOperationTracker/16Pxx0000004Gz2EAE",
      "success": true
    }
  ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
errors	Invoice Ingestion Output Error[]	Details of the errors if the request was unsuccessful.	Small, 63.0	63.0
invoiceId	String	ID of the Invoice record that's created in Salesforce.	Small, 63.0	63.0
request Identifier	String	Unique request identifier that can be used to poll the async request.	Small, 63.0	63.0
statusURL	String	Status URL for tracking the estimated tax callout operation.	Big, 63.0	63.0
success	Boolean	Indicates whether the API request was successful (true) or not (false).	Small, 63.0	63.0

Invoice Ingestion Output Error

Output representation of the details of an invoice generation error.

Property Name	Type	Description	Filter Group and Version	Available Version
errorCode	String	Code that indicates the type of error.	Small, 63.0	63.0

Property Name	Type	Description	Filter Group and Version	Available Version
`message`	String	Message that states the reason for error, if any.	Small, 63.0	63.0
`referenceId`	String	Reference ID that maps to the subrequest's response and can be used to reference the response in later subrequests.	Small, 63.0	63.0

Invoice Line Preview

Output representation of the invoice line preview result.

JSON example


```
"invoiceLineDetailList": [
      {
        "billingFrequency": "OneTime",
        "chargeAmount": "1990.00",
        "endDate": "2024-07-18",
        "lineAmount": "2189.00",
        "productName": "Laptop",
        "quantity": "10.0",
        "startDate": "2024-07-18",
        "taxAmount": "199.0",
        "unitPrice": "199.0"
      }
    ]
```


Property Name	Type	Description	Filter Group and Version	Available Version
`billing Frequency`	String	The frequency at which the invoice line is created for the billing transaction.	Big, 63.0	63.0
`endDate`	String	The end date of the billing for the service for invoice lines made from a time-based service.	Big, 63.0	63.0
`lineAmount`	String	The amount of the invoice line.	Big, 63.0	63.0
`productName`	String	The name of the product that was charged or ordered to create the invoice line.	Big, 63.0	63.0
`quantity`	String	The number of units of the order product that created the invoice line.	Big, 63.0	63.0
`startDate`	String	The first date of the billing for the service for invoice lines made from a time-based service.	Big, 63.0	63.0
`taxAmount`	String	The tax amount for the invoice line.	Big, 63.0	63.0

Property Name	Type	Description	Filter Group and Version	Available Version
unitPrice	String	The price for one unit of the item on the invoice line.	Big, 63.0	63.0

Invoice Preview

Output representation of the invoice preview result.

JSON example


```
"invoiceDetailList": [
  {
    "accountId": "001Z6000005aQj8",
    "currencyIsoCode": "USD",
    "dueDate": "2025-01-03",
    "invoiceDate": "2024-12-04",
    "invoiceLineDetailList": [
      {
        "billingFrequency": "OneTime",
        "chargeAmount": "1990.00",
        "endDate": "2024-07-18",
        "lineAmount": "2189.00",
        "productName": "Laptop",
        "quantity": "10.0",
        "startDate": "2024-07-18",
        "taxAmount": "199.0",
        "unitPrice": "199.0"
      }
    ],
    "totalAmount": "1990.00",
    "totalAmountWithTax": "2189.00",
    "totalTaxAmount": "199.0"
  }
]
```


Property Name	Type	Description	Filter Group and Version	Available Version
accountId	String	The account ID of the billing transaction.	Big, 63.0	63.0
currencyCode	String	The currency code for the amount fields.	Big, 63.0	63.0
dueDate	String	The date by when the customer must pay an invoice that's generated for the billing transaction.	Big, 63.0	63.0
invoiceDate	String	The date when an invoice that's generated for the billing transaction is posted.	Big, 63.0	63.0
invoiceLines	Invoice Line Preview[]	The details of the invoice lines related to the invoices that are generated for the billing transaction.	Big, 63.0	63.0

Property Name	Type	Description	Filter Group and Version	Available Version
totalAmount	String	The sum of the total amount on the invoice lines of an invoice that's generated for the billing transaction.	Big, 63.0	63.0
totalAmount WithTax	String	The sum of the total amount including tax on the invoice lines of an invoice that's generated for the billing transaction.	Big, 63.0	63.0
totalTax Amount	String	The sum of the tax amount on the invoice lines of an invoice that's generated for the billing transaction.	Big, 63.0	63.0

Invoice Preview Result

Output representation of the list of preview invoices that are generated for the billing transaction.

JSON example


```
{
  "invoiceDetailList": [
    {
      "accountId": "001Z6000005aQj8",
      "currencyIsoCode": "USD",
      "dueDate": "2025-01-03",
      "invoiceDate": "2024-12-04",
      "invoiceLineDetailList": [
        {
          "billingFrequency": "OneTime",
          "chargeAmount": "1990.00",
          "endDate": "2024-07-18",
          "lineAmount": "2189.00",
          "productName": "Laptop",
          "quantity": "10.0",
          "startDate": "2024-07-18",
          "taxAmount": "199.0",
          "unitPrice": "199.0"
        }
      ],
      "totalAmount": "1990.00",
      "totalAmountWithTax": "2189.00",
      "totalTaxAmount": "199.0"
    }
  ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
invoiceDetail List	Invoice Preview[]	Details of the invoices that are generated for the billing transaction.	Big, 63.0	63.0

Invoice Recovery

Output representation of the details of the recovered invoice and billing schedules.

JSON example


```
{
  "recoveryResults": [
    {
      "billingSchedules": [
        {
          "billingScheduleId": "44bDU00000000XX",
          "billingScheduleStatus": "ReadyForInvoicing"
        }
      ],
      "invoiceErrors": [],
      "invoiceId": null,
      "success": true
    }
  ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`billing Schedules`	Billing Schedule Recovery[]	Billing schedules associated with this invoice.	Big, 62.0	62.0
`invoiceErrors`	Error Response	List of errors encountered during the invoice recovery.	Big, 62.0	62.0
`invoiceId`	String	ID of the recovered invoice.	Big, 62.0	62.0
`invoiceStatus`	String	Flag that indicates the invoice status.	Big, 62.0	62.0
`success`	Boolean	Indicates whether the overall transaction was successful or not (`true`) or not (`false`).	Big, 62.0	62.0

Line Item

Output representation of the details of the line item.

JSON example


```
{
  "lineItems": [
    {
      "addresses": {
        "shipFrom": {
          "locationCode": "67890"
        },
        "shipTo": {
          "locationCode": "12345"
        },
```

```
          "soldTo": {
            "locationCode": "12345"
          }
        },
        "amountDetails": {
          "exemptAmount": 0,
          "taxAmount": 12.5,
          "totalAmount": 100,
          "totalAmountWithTax": 112.5
        },
        "effectiveDate": "2022-03-09T10:55:38.416Z",
        "lineNumber": "001xx000003HYEiAAO",
        "productCode": "Y0001",
        "quantity": 1,
        "taxCode": "TX0001",
        "taxes": [
          {
            "exemptAmount": 0,
            "exemptReason": "NoExemption",
            "imposition": {
              "type": "General"
            },
            "jurisdiction": {
              "country": "US",
              "id": "63000",
              "level": "CIT",
              "name": "SEATTLE",
              "region": "WA",
              "stateAssignedNo": "1726"
            },
            "rate": 12.5,
            "tax": 12.5,
            "taxId": "11000378132466",
            "taxableAmount": 100
          }
        ]
      }
    ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`addresses`	Addresses	Different types of addresses that are used for the transaction of the line item.	Big, 62.0	62.0
`amountDetails`	Tax Amount Details	Details of the transaction amount and the taxes applicable for the line item.	Big, 62.0	62.0
`effectiveDate`	String	Date when the tax rules are applied on the line item.	Big, 62.0	62.0
`lineNumber`	String	Unique identifier of the line item.	Big, 62.0	62.0

Property Name	Type	Description	Filter Group and Version	Available Version
productCode	String	Product code of the line item as defined by the tax engine.	Big, 62.0	62.0
quantity	Double	Quantity of the line item.	Big, 62.0	62.0
taxCode	String	Tax code of the line item according to the tax engine.	Big, 62.0	62.0
taxes	Tax Details []	Tax details of the line item.	Big, 62.0	62.0

On-Demand Document Generation Response

Output representation of the details of the generated document along with error response.

JSON example


```
{
  "requestIdentifier": "aad8ee2d-d94d-4760-8759-a2c5d63ce6db",
  "isSuccess": true,
  "details": {
    "documentEntityId": "1BBSM00000003KT4AY",
    "documentGenerationProcessId": "0nnSM00000009jFYAQ",
    "documentTitle": "DOC-000000197",
    "existingDocumentId": "069SM000002PYQLYA4",
    "recordId": "3ttSM0000000o58YAA",
    "documentTemplateId": "0694x000000XyzABC"
  },
  "errors": []
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
details	Document Details[]	Details of the generated document.	Big, 66.0	66.0
errors	Error Response[]	Error details of any issues encountered during the enqueue operation.	Big, 66.0	66.0
isSuccess	Boolean	Status of the enqueuing PDF generation task.	Big, 66.0	66.0
requestIdentifier	String	Unique request identifier that's used to poll the asynchronous request.	Big, 66.0	66.0

Payment Line Apply

Output representation of the details of the applied payment line. The details include the ID of the payment record and date when the payment line was applied.

JSON example

```
{
  "appliedDate": "2020-08-11T08:09:01.000Z",
  "id": "1PLR000000000dDOAQ"
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
appliedDate	String	Date when the payment line was applied.	Big, 64.0	64.0
id	String	ID of the payment line record.	Big, 64.0	64.0

Payment Line Unapply

Output representation of the details of the reversed payment line application. The details include the ID of the payment line record and date when the payment line application was reversed.

JSON example

```
{
  "unappliedDate": "2020-08-11T08:09:01.000Z",
  "id": "1PLR000000000dDOAQ"
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
id	String	ID of the payment line record.	Big, 64.0	64.0
unappliedDate	String	Date when the payment line application was reversed.	Big, 64.0	64.0

Payment Scheduler Update

Output representation of the details of the updated payment scheduler. This representation covers the updated status value of the specified payment scheduler.

JSON example

```
{
  "status": "Active"
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
status	String	Updated status value of the specified payment scheduler. Valid values are: • Active	Big, 64.0	64.0

Property Name	Type	Description	Filter Group and Version	Available Version
		• `Canceled` • `Draft` • `Inactive`		

Posted Invoice List Write-Off

Output representation of the list of invoices that are written off.

JSON example


```
{
  "result": [
    {
      "requestIdentifier": null,
      "invoiceId": "3t00000000CwAGI",
      "success": false,
      "errors": {
        "errorcode": "INVALID_API_INPUT",
        "errorMessage": "Reason is missing."
      }
    },
    {
      "requestIdentifier": 37612787,
      "invoiceId": "3t00000000CwAAI",
      "success": true,
      "errors": {
        "errorcode": null,
        "errorMessage": null
      }
    }
  ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`result`	Posted Invoice Write-Off []	Details of the invoices for which the write-off process is initiated.	Big, 64.0	64.0

Posted Invoice Write-Off

Output representation of the details of a posted invoice that's written off.

JSON example


```
{
  "result": [
    {
      "requestIdentifier": null,
```

```
      "invoiceId": "3t00000000CwAGI",
      "success": false,
      "errors": {
        "errorcode": "INVALID_API_INPUT",
        "errorMessage": "Reason is missing"
      }
    },
    {
      "requestIdentifier": 37612787,
      "invoiceId": "3t00000000CwAAI",
      "success": true,
      "errors": {
        "errorcode": null,
        "errorMessage": null
      }
    }
  ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`errors`	Posted Invoice Write-Off Error[]	If the request fails, this property contains a list of errors.	Small, 64.0	64.0
`invoiceId`	String	ID of the invoice record that's written off.	Big, 64.0	64.0
`request Identifier`	String	If the request is successful, this property contains an asynchronous API request identifier for an invoice ID.	Big, 64.0	64.0
`success`	Boolean	Indicates whether the invoice write-off request was successful (`true`) or not (`false`).	Big, 64.0	64.0

Posted Invoice Write-Off Error

Output representation of the error response that's associated with a request to write off a posted invoice.

JSON example


```
{
  "errors": {
    "errorcode": "INVALID_API_INPUT",
    "errorMessage": "Reason is missing"
  }
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`errorCode`	String	Code that represents the error.	Small, 64.0	64.0
`errorMessage`	String	Message that describes the error.	Small, 64.0	64.0

Reference Line Error

Output representation of the details of the line level errors.

JSON example

If the API request fails, the `referenceLineErrorResults` property contains a list of errors grouped by the invoice line IDs.

```
[
  {
    "referenceLineId": "5TV9A000007x2gz",
    "errors": [
      {
        "errorCode": "INVALID_INPUT",
        "message": "Invalid invoice line id"
      }
    ]
  }
]
```

Property Name	Type	Description	Filter Group and Version	Available Version
`errors`	Error Response	List of errors with error code and error message for the specified invoice line ID.	Big, 62.0	62.0
`reference LineId`	String	ID of the invoice line specified in the API request that has an issue, causing the API request to fail.	Small, 62.0	62.0

Refund Line Applied Response

Output representation of the details of an applied refund. This representation includes the properties of a refund line, such as the date when the refund is applied against a payment and ID of the refund line record.

JSON example

```
{
  "appliedDate": "2020-08-11T08:09:01.000Z",
  "id": "0dRR000000000CsMAI"
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`appliedDate`	String	Date when the refund is applied against a payment.	Big, 64.0	64.0
`id`	String	ID of the refund line record.	Big, 64.0	64.0

Revenue Async Line Level

Output representation of the result of the API request for the async line level operations.

JSON example


```
{
"success": true,
"requestIdentifier": "237e9877-e79b-12d4-a765-321741963000",
"errors":[]
}
```


If the API request fails, the `referenceLineErrorResults` property contains a list of errors grouped by the invoice line IDs.


```
[
  {
    "referenceLineId": "5TV9A000007x2gz",
    "errors": [
      {
        "errorCode": "INVALID_INPUT",
        "message": "Invalid invoice line id"
      }
    ]
  }
]
```


Property Name	Type	Description	Filter Group and Version	Available Version
`errors`	Error Response	Details of errors, if any.	Big, 62.0	62.0
`referenceLineErrorResults`	Reference Line Error[]	List of errors grouped by the invoice line IDs if the API request fails.	Big, 62.0	62.0
`referenceLineType`	String	Reference type for the reference line entity in the `referenceLineErrorResults` property.	Big, 62.0	62.0
`requestIdentifier`	String	Unique identifier of the request.	Big, 62.0	62.0
`statusURL`	String	URL to track the status of the operation.	Big, 62.0	62.0
`success`	Boolean	Indicates whether the API request is successful (`true`) or not (`false`).	Big, 62.0	62.0

Revenue Async Response

Output representation of the result of the API request with the request identifier.

JSON example


```
{
  "errors": null,
  "requestIdentifier": "ae6f23bc-f056-44b7-aa4d-c7f6fc5e0cf4",
```

```
    "success": true
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`errors`	Error Response	Details of errors, if any.	Big, 62.0	62.0
`request Identifier`	String	Unique identifier of the request.	Big, 62.0	62.0
`success`	Boolean	Indicates whether the API request is successful (`true`) or not (`false`).	Big, 62.0	62.0

Rules Application

Output representation of the details of rules application.

JSON example

This example shows the rules application response.

```
{
  "isSuccess": true,
  "rulesApplicationSummary": {
    "fetchedPaymentsCount": 250,
    "fetchedCreditMemosCount": 250,
    "totalPaymentApplications": 250,
    "totalCreditMemoApplications": 200,
    "areAllInvoicesConsidered": true
  },
  "appliedRules": [
    "Match ID",
    "Match Balance"
  ],
  "errors": []
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`isSuccess`	Boolean	Indicates whether the rules are applied (`true`) or not (`false`).	Big, 66.0	66.0
`rulesApplication Summary`	Rules Application Summary	Summary of the application rule. This includes the number of payments and credit memos for the account, the total number of payments and credit memos that are applied to invoices, and whether all invoices for an account are considered (`true`) or not (`false`).	Big, 66.0	66.0
`appliedRules`	String[]	Rules that were applied. Valid values are:	Big, 66.0	66.0

Property Name	Type	Description	Filter Group and Version	Available Version
		• `Match ID` • `Match Balance` • `Prioritize Highest Balance Invoices` • `Prioritize Oldest Invoices`		
`errors`	Rules Application Error[]	Details of the errors if the API request was unsuccessful.	Big, 66.0	66.0

Rules Application Error

Output representation of the error details for rules application failure.

JSON example

This example shows the rules application error.

```
{
  "errors": [
    {
      "errorCode": "INVALID_ACCOUNT",
      "message": "The specified account does not exist or is not accessible."
    }
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`errorCode`	String	Code that indicates the type of error.	Big, 66.0	66.0
`message`	String	Message that states the reason for error, if any.	Big, 66.0	66.0

Rules Application Summary

Output representation of the summary of the application rule. This includes the number of payments and credit memos for the account, the total number of payments and credit memos that's applied to invoices, and whether all invoices for an account are considered (true) or not (false).

JSON example

This example shows the rules application summary.

```
{
  "rulesApplicationSummary": {
    "fetchedPaymentsCount": 5,
    "fetchedCreditMemosCount": 2,
    "totalPaymentApplications": 3,
```

```
    "totalCreditMemoApplications": 1,
    "areAllInvoicesConsidered": true
  }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
fetched PaymentsCount	Integer	Total number of payments related to the account.	Big, 66.0	66.0
fetchedCredit MemosCount	Integer	Total number of credit memos or credit memo lines related to the account.	Big, 66.0	66.0
totalPayment Applications	Integer	Total number of payments that are applied to the invoices or invoice lines.	Big, 66.0	66.0
totalCredit Memo Applications	Integer	Total number of credit memos that are applied to the invoice or invoice lines.	Big, 66.0	66.0
areAll Invoices Considered	Boolean	Indicates whether all invoices are considered (`true`) or not (`false`).	Big, 66.0	66.0

Send Email Error

Output representation of the error response of the API request to send emails for posted invoices.

JSON example

This example shows a sample response with an error scenario.

```
{
  "errors": [
    {
      "errorCode": "INVALID_API_INPUT",
      "errorMessage": "Specify a valid invoiceBatchRunId."
    }
  ],
  "requestIdentifier": "5IRxx0000004CKKGA2",
  "success": false
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
errorCode	String	Unique code for the error.	Big, 65.0	65.0
errorMessage	String	Descriptive message for the error.	Big, 65.0	65.0

Send Email Response

Output representation of the API request to send emails for posted invoices.

JSON example

This example shows a sample response when the request is successful.

```
{
  "errors": [],
  "requestIdentifier": "5IRLT000001SIJB4A4",
  "success": true
}
```

This example shows a sample response with an error scenario.

```
{
  "errors": [
    {
      "errorCode": "INVALID_API_INPUT",
      "errorMessage": "Specify a valid invoiceBatchRunId."
    }
  ],
  "requestIdentifier": "5IRxx0000004CKKGA2",
  "success": false
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`errors`	Send Email Error[]	List of errors that occurred while trying to enqueue the request. In case of errors while sending emails, you can check the Revenue Transaction Error Logs of the InvoiceBatchRun record and associated Invoice record. An error occurs for these scenarios. • The contact email or user isn't configured. • The configured email template isn't valid. • Email is sent through a contact email, and the contact doesn't have an associated user. In this scenario, the emails are counted against the general email limits. An error is thrown if you exceed the email limits.	Big, 65.0	65.0
`requestIdentifier`	String	Unique identifier for the request, which corresponds to the invoice batch run ID.	Big, 65.0	65.0

Property Name	Type	Description	Filter Group and Version	Available Version
`success`	Boolean	Indicates whether the request to start the email sending process was successfully enqueued (`true`) or not (`false`).	Big, 65.0	65.0

Sequence Error

Output representation of the error response that's associated with a request to create or update a sequence policy, or assign sequences.

JSON example

This example shows a sample error response.

```
{
  "error": {
    "errorCode": "INVALID_INPUT",
    "message": "Specify a valid selectionLogic."
  },
  "isSuccess": false,
  "sequencePolicyId": null
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`errorCode`	String	Code for the resultant error.	Big, 65.0	65.0
`message`	String	Error message for the resultant error.	Big, 65.0	65.0

Sequence Gap Reconciliation Error

Output representation of the errors encountered during the processing of the API request.

JSON example

This example shows a sample error response.

```
{
  "jobId": "",
  "sequencePolicyIds": [
    "1vdxx0000000abc",
    "1vdxx0000000def"
  ],
  "targetObjects": [
    "Invoice"
  ],
  "status": "NotSubmitted",
  "submittedAt": "",
  "error": {
    "errorCode": "INVALID_INPUT",
    "message": "Specify a value for either sequencePolicyIds or targetObjects."
  }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
errorCode	String	Code for the resultant error.	Big, 65.0	65.0
errorMessage	String	Error message for the resultant error.	Big, 65.0	65.0

Sequence Gap Reconciliation

Output representation of the details of the sequence gap reconciliation.

JSON example

This example shows a sample successful response.


```
{
  "jobId": "0B1xx0000006P12",
  "sequencePolicyIds": [
    "1vdxx0000000uyr",
    "1vdxx0000000lrf"
  ],
  "targetObjects": [
    "Invoice"
  ],
  "status": "Submitted",
  "submittedAt": "2025-06-05T09:12:28Z",
  "error": {}
}
```


This example shows a sample error response for an invalid input.


```
{
  "jobId": "",
  "sequencePolicyIds": [
    "1vdxx0000000abc",
    "1vdxx0000000def"
  ],
  "targetObjects": [
    "Invoice"
  ],
  "status": "NotSubmitted",
  "submittedAt": "",
  "error": {
    "errorCode": "INVALID_INPUT",
    "message": "Missing required field in the request."
  }
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
error	Sequence Gap Reconciliation Error[]	List of errors encountered during the processing of the API request.	Big, 65.0	65.0

Property Name	Type	Description	Filter Group and Version	Available Version
`jobId`	String	Unique identifier assigned to the sequence gap reconciliation asynchronous process.	Big, 65.0	65.0
`sequence PolicyIds`	String[]	List of IDs of the sequence policies.	Big, 65.0	65.0
`status`	String	Status of the sequence gap reconciliation API request. Valid values are: • `Submitted` • `NotSubmitted`	Big, 65.0	65.0
`submittedAt`	String	Date and time when the reconciliation request was submitted to the async job.	Big, 65.0	65.0
`targetObjects`	String[]	List of objects to which the policies are applied.	Big, 65.0	65.0

Sequence Policy

Output representation that shows the status of the assigned sequence pattern values.

JSON example

This example shows a sample successful response.

```
{
  "error": null,
  "isSuccess": true,
  "sequencePolicyId": "1Vdxx0000000GRNAA2"
}
```

This example shows a sample error response.

```
{
  "error": {
    "errorCode": "INVALID_INPUT",
    "message": "Specify a valid selectionLogic."
  },
  "isSuccess": false,
  "sequencePolicyId": null
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`error`	Sequence Error[]	Details of any error that encountered during the processing of the API request.	Big, 65.0	65.0
`isSuccess`	Boolean	Indicates whether the sequence policy is generated (`true`) or not (`false`).	Big, 65.0	65.0

Property Name	Type	Description	Filter Group and Version	Available Version
sequencePolicyId	String	ID of the sequence policy.	Big, 65.0	65.0

Sequences Assignment

Output representation with the status of the assigned sequence pattern values.

JSON example

This example shows a sample successful response.


```
{
  "errors": null,
  "sequencesAssignment": [
    {
      "errors": null,
      "isSuccess": true,
      "sequencePatternValue": "INV-1234-2025-04-12-001",
      "targetObjectId": "3ttxx000000085dAAA"
    }
  ],
  "status": "Success"
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
errors	Sequence Error[]	Error encountered during the processing of the API request.	Big, 65.0	65.0
sequencesAssignment	Sequences Assignment Result[]	Details of the sequence pattern values assignment.	Big, 65.0	65.0
status	String	Status of the sequence policy assignment. Valid values are: • PartialSuccess • Success • Failed	Big, 65.0	65.0

Sequences Assignment Result

Output representation of the details of the assigned sequence values to target objects.

JSON example

This example shows a sample successful response.


```
{
  "errors": null,
  "sequencesAssignment": [
    {
```

```
      "errors": null,
      "isSuccess": true,
      "sequencePatternValue": "INV-1234-2025-04-12-001",
      "targetObjectId": "3ttxx000000085dAAA"
    }
  ],
  "status": "Success"
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
`errors`	Sequence Error[]	Error encountered during the processing of the API request.	Big, 65.0	65.0
`isSuccess`	Boolean	Indicates whether the sequence pattern value is assigned (`true`) or not (`false`).	Big, 65.0	65.0
`sequencePatternValue`	String	Sequence pattern value assigned to the target object.	Big, 65.0	65.0
`sequencePolicyId`	String	ID of the sequence policy assigned to the target object.	Big, 65.0	65.0
`targetObjectId`	String	Record to which the sequence pattern value is assigned.	Big, 65.0	65.0

Statement of Account

Output representation of the details of the generated account statement with async tracking details.

JSON example

This example shows a sample successful response.


```
{
  "requestIdentifier": "a0Bxx000000001ZEAQ",
  "statusURL":
"/services/data/v60.0/commerce/billing/document-generation-process/a0Bxx000000001ZEAQ/status",

  "success": true,
  "accountId": "001xx000003DGb2AAG",
  "templateId": "0TRxx000000001XGAQ",
  "errors": null
}
```


This example shows a sample error response.


```
{
  "success": false,
  "errors": [
    {
      "code": "ACCOUNT_NOT_FOUND",
      "message": "Account with ID 001XX000004DKy5YAG not found",
      "field": "accountId"
```

```
        }
    ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`accountId`	String	Primary account ID the statement is generated for.	Big, 66.0	66.0
`errors`	Error Response[]	List of errors encountered during the processing of the API request.	Big, 66.0	66.0
`requestIdentifier`	String	Unique identifier for the request.	Big, 66.0	66.0
`statusURL`	String	Status URL to track the operation.	Big, 66.0	66.0
`success`	Boolean	Indicates whether the API request is successful (`true`) or not (`false`).	Big, 66.0	66.0
`templateId`	String	Document template ID that's used to generate the PDF.	Big, 66.0	66.0

Suspend Resume Billing

Output representation of the list of accounts and billing schedule groups, which are suspended or resumed for billing operations.

JSON example

```
{
    "result":{
        {
            "referenceId": "001DU000001o2UwYAI",
            "success": true,
            "errorcode": null,
            "errorMessage":null
        },
        {
            "referenceId": "001DU000001o2UuYAI",
            "success": false,
            "errorcode": "INVALID_API_INPUT",
            "errorMessage":"Billing is already suspended for 9Wsxx000000006TCAQ."
        }
    }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`result`	Suspend Resume Billing Object[]	Details of the accounts or billing schedule groups that the suspend or resume billing request is initiated for.	Big, 63.0	63.0

Suspend Resume Billing Object

Output representation of the details of accounts and billing schedule groups, which are suspended or resumed for billing operations, along with the status of the API request.

JSON example

```
{
    "result":{
        {
        "referenceId": "1",
        "isSuccess": true,
        "errorcode": null,
        "errorMessage":null
        }
    }
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`errorCode`	String	Code indicating the type of error.	Big, 63.0	63.0
`errorMessage`	String	Message stating the reason for the error, if any.	Big, 63.0	63.0
`isSuccess`	Boolean	Indicates whether the API request is successful (`true`) or not (`false`).	Big, 63.0	63.0
`referenceId`	String	ID of the account or billing schedule group that the suspend or resume billing request is initiated for.	Big, 63.0	63.0

Tax Amount Details

Output representation of the details of the tax amount.

JSON example

```
{
    "exemptAmount": 0.0,
    "taxAmount": 12.5,
    "totalAmount": 100.0,
    "totalAmountWithTax": 112.5
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`exemptAmount`	Double	Amount exempted from taxation.	Big, 62.0	62.0
`taxAmount`	Double	Tax amount applicable to the transaction.	Big, 62.0	62.0
`totalAmount`	Double	Total amount without tax.	Big, 62.0	62.0

Property Name	Type	Description	Filter Group and Version	Available Version
totalAmount WithTax	Double	Total amount with tax. The totalAmountWithTax property value is the sum of the taxAmount and totalAmount property values.	Big, 62.0	62.0

Tax Calculation

Output representation of the details of the calculated tax.

JSON example


```
{
    "referenceEntityId": "001xx000003HYEhAAO",
    "status": "Committed",
    "taxEngineLogs": [
        {
            "createdDate": "2022-03-09T10:55:38.000Z",
            "id": "3l1xx00000000PpAAI",
            "resultCode": "Success"
        }
    ],
    "taxTransactionType": "Debit",
    "taxType": "Actual",
    "transactionDate": "2022-03-09T10:30:41.000Z",
    "addresses": {
        "shipFrom": {
            "locationCode": "67890"
        },
        "shipTo": {
            "locationCode": "23456"
        },
        "soldTo": {
            "locationCode": "23456"
        }
    },
    "amountDetails": {
        "exemptAmount": 0.0,
        "taxAmount": 12.5,
        "totalAmount": 100.0,
        "totalAmountWithTax": 112.5
    },
    "currencyIsoCode": "USD",
    "description": "Monthly invoice for account 001xx000003HYEhAAO",
    "documentCode": "INV-003",
    "effectiveDate": "2022-03-09T10:55:38.410Z",
    "lineItems": [
        {
            "addresses": {
                "shipFrom": {
                    "locationCode": "67890"
```

```
            },
            "shipTo": {
                "locationCode": "12345"
            },
            "soldTo": {
                "locationCode": "12345"
            }
        },
        "amountDetails": {
            "exemptAmount": 0.0,
            "taxAmount": 12.5,
            "totalAmount": 100.0,
            "totalAmountWithTax": 112.5
        },
        "effectiveDate": "2022-03-09T10:55:38.416Z",
        "lineNumber": "001xx000003HYEiAAO",
        "productCode": "Y0001",
        "quantity": 1.0,
        "taxCode": "TX0001",
        "taxes": [
            {
                "exemptAmount": 0.0,
                "exemptReason": "NoExemption",
                "imposition": {
                    "type": "General"
                },
                "jurisdiction": {
                    "country": "US",
                    "id": "63000",
                    "level": "CIT",
                    "name": "SEATTLE",
                    "region": "WA",
                    "stateAssignedNo": "1726"
                },
                "rate": 12.5,
                "tax": 12.5,
                "taxId": "11000378132466",
                "taxableAmount": 100.0
            }
        ]
    }
  ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
adapterError	Error Response	Details of the adapter error.	Big, 62.0	62.0
addresses	Addresses	Addresses that are used for calculating tax.	Big, 62.0	62.0
amountDetails	Tax Amount Details	Details of the transaction amount and taxes.	Big, 62.0	62.0
currencyIsoCode	String	Currency ISO code that's used for tax calculation.	Big, 62.0	62.0

Property Name	Type	Description	Filter Group and Version	Available Version
description	String	Description of the transaction status.	Big, 62.0	62.0
documentCode	String	Unique identifier for the tax document. If the documentCode property isn't specified, the tax engine auto-generates it.	Big, 62.0	62.0
effectiveDate	String	Date when the tax rules are applied.	Big, 62.0	62.0
lineItems	Line Item []	Line items that the taxes are calculated for.	Big, 62.0	62.0
reference DocumentCode	String	Reference document code. For subsequent transactions such as a credit tax, the referenceDocumentCode property value refers to the original document code.	Big, 62.0	62.0
reference EntityId	String	ID of the related quote, invoice, or other transaction documents.	Big, 62.0	62.0
status	String	Commit status of the tax transaction. Valid values are: • Committed—Committed transaction, which is marked for tax reporting. • Uncommitted— Uncommitted transaction, which isn't marked for tax reporting.	Big, 62.0	62.0
taxEngineLogs	Tax Engine Log []	Records that the tax engine generates while calculating taxes for the transaction.	Big, 62.0	62.0
tax TransactionId	String	Unique ID for the tax transaction.	Big, 62.0	62.0
transactionDate	String	Date of the transaction that appears on the invoice, order, and other transaction documents.	Big, 62.0	62.0
taxTransaction Type	String	Tax transaction type. Valid values are: • Debit—Increases tax liability for the user, requiring the user to pay tax on the transaction. • Credit—Decreases tax liability for the user, resulting in a tax refund for the user. • Void—Specifies that the tax engine has voided the document that's mentioned as the referenceDocumentCode property value.	Big, 62.0	62.0

Property Name	Type	Description	Filter Group and Version	Available Version
taxType	String	Type of calculated tax. Valid values are: • Actual—Exact tax amount that's calculated based on the actual sales. • Estimated— Estimated tax amount, which is adjusted later to match the actual tax calculations.	Big, 62.0	62.0

Tax Details

Output representation of the tax details for each line item.

JSON example


```
{
  "taxes": [
    {
      "exemptAmount": 0,
      "exemptReason": "NoExemption",
      "imposition": {
        "type": "General"
      },
      "jurisdiction": {
        "country": "US",
        "id": "63000",
        "level": "CIT",
        "name": "SEATTLE",
        "region": "WA",
        "stateAssignedNo": "1726"
      },
      "rate": 12.5,
      "tax": 12.5,
      "taxId": "11000378132466",
      "taxableAmount": 100
    }
  ]
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
exemptAmount	Double	Amount exempted from taxation for the line item.	Big, 62.0	62.0
exemptReason	String	Reason for the tax exemption.	Big, 62.0	62.0
imposition	Tax Imposition	Tax imposition details applicable to the line item.	Big, 62.0	62.0
jurisdiction	Tax Jurisdiction	Details of the tax jurisdiction.	Big, 62.0	62.0

Property Name	Type	Description	Filter Group and Version	Available Version
`rate`	Double	Tax rate applied to the taxable amount to calculate the tax.	Big, 62.0	62.0
`tax`	Double	Actual amount of tax that's applicable to the line item.	Big, 62.0	62.0
`taxId`	String	Unique identifier such as a code or a number that's assigned to a specific tax. This value helps users identify which type of tax is applied.	Big, 62.0	62.0
`taxableAmount`	Double	Total amount of the line item that's eligible for taxation.	Big, 62.0	62.0

Tax Engine Log

Output representation of the logs that the tax engine generates.

JSON example

```
{
  "taxEngineLogs": [
    {
      "createdDate": "2022-03-09T10:55:38.000Z",
      "id": "3l1xx00000000PpAAI",
      "resultCode": "Success"
    }
  ]
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
`createdDate`	String	Date when the tax engine creates the log.	Big, 62.0	62.0
`id`	String	ID of the tax engine log record.	Big, 62.0	62.0
`resultCode`	String	Result code associated with the created log.	Big, 62.0	62.0

Tax Imposition

Output representation of the details of the imposed tax.

JSON example

```
{
  "type": "Parish",
  "name": "Burbank"
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
name	String	Name of the tax imposition.	Big, 62.0	62.0
type	String	Type of the tax imposition.	Big, 62.0	62.0

Tax Jurisdiction

Output representation of the details of the tax jurisdiction for the tax line.

JSON example


```
{
    "country": "US",
    "id": "63000",
    "level": "CIT",
    "name": "SEATTLE",
    "region": "WA",
    "stateAssignedNo": "1726"
}
```


Property Name	Type	Description	Filter Group and Version	Available Version
country	String	Country of the tax jurisdiction.	Big, 62.0	62.0
id	String	ID of the tax jurisdiction.	Big, 62.0	62.0
level	String	Level of the tax jurisdiction, for example, State and Federal.	Big, 62.0	62.0
name	String	Name of the tax jurisdiction authority.	Big, 62.0	62.0
region	String	Parent region of the tax jurisdiction.	Big, 62.0	62.0
stateAssignedNo	String	Number of the assigned state.	Big, 62.0	62.0

Void Posted Credit Memo

Output representation of the request to void a posted credit memo.

JSON example

This example shows a sample success response.


```
{
  "errors": null,
  "statusURL": "/services/data/v67.0/sobjects/AsyncOperationTracker/16Pxx0000004LdkEAE",

  "debitMemoId": "4Dmxx00000000XtAAK",
  "success": true
}
```

This example shows a sample error response.

```
{
  "errors": [
    {
      "errorCode": "INVALID_STATUS",
      "message": "CreditMemo 50gxx00000000XtAAI is not in the Posted status."
    }
  ],
  "debitMemoId": "",
  "success": false
}
```

Property Name	Type	Description	Filter Group and Version	Available Version
debitMemoId	String	ID of the created debit memo.	Big, 66.0	66.0
errors	Error Response[]	List of errors specific to this API request that were encountered during voiding the credit memo.	Big, 66.0	66.0
isSuccess	Boolean	Indicates whether the API request was successful (true) or not (false).	Big, 66.0	66.0
statusURL	String	Status URL for tracking this operation.	Big, 66.0	66.0

Billing Apex Reference

Billing provides the ConnectApi namespace (also called Connect in Apex) that contains classes for accessing the same capabilities that are available in the Billing Business APIs. Additionally, you can use other built-in Apex classes and interfaces grouped by namespace.

This table lists the available Billing Apex methods with the associated Connect REST API.

Apex Method	Connect REST API
convertNegativeInvoiceLines(ConvertNegativeInvoiceLinesInput, invoiceId)	Negative Invoice Lines to Credit Conversion (POST)
generateInvoices(inputRequest)	Invoices By Using Billing Schedules (POST)
recoverBillingSchedules(inputRequest)	Billing Schedule Recovery List (POST)
creditInvoice(CreditInvoiceInput, invoiceId)	Credit Memo Create and Apply (POST)
applyCreditMemos(CreditMemoApplyInput, creditMemoId)	Credit Memo Apply List(POST)
unapplyCreditMemos(CreditMemoUnapplyInput, creditMemoInvApplicationId)	Credit Memo Unapply (POST)

Apex Method	Connect REST API
applyCreditMemoLines(CreditMemoLineApplyInput, creditMemoLineId)	Credit Memo Line Apply (POST)
unapplyCreditMemoLines(CreditMemoLineUnapplyInput, creditMemoLineInvoiceLineId)	Credit Memo Line Unapply (POST)
triggerInvoiceBatchDraftToPosted(invoiceBatchRunId)	Invoices Batch Draft to Posted Status (POST)
postDraftInvoices(inputRequest)	Invoice Draft to Posted Status (POST)
createCreditMemos(CreditMemoInputRequest)	Credit Memo Create and Apply (POST)
voidPostedInvoice(invoiceId)	Posted Invoice Voidance (POST)
calculateTax(calculateTax)	Tax Calculation (POST)
applyPaymentLine(PaymentLineApplyInput, paymentId)	Payment Line Apply (POST)
unapplyPaymentLine(PaymentLineUnapplyInput, paymentId, paymentLineId)	Payment Line Unapply (POST)
applyRefundLine(RefundLineApplyInput, refundId)	Refund Line Apply (POST)
reconcileSequences(sequenceGapReconciliationInputRepresentation)	Sequence Gap Reconciliation (POST)
sequenceAssignment(sequencesAssignmentInputRepresentation)	Sequence Assignment (POST)
voidPostedCreditMemo(VoidPostedCreditMemoInput, creditMemoId)	Void Posted Credit Memo (POST)

For more information about Apex classes that are available for Commerce Payments, see CommercePayments Namespace.

ConnectApi Namespace
The `ConnectApi` namespace (also called Connect in Apex) provides classes to manage credit applications and billing scenarios.

InvoiceWriteOff Namespace
Create credit memos with the total charge amount on the invoice as the write-off amount and close the invoice.

IssueCreditMemo Namespace
Issue credit memos from disputed invoices. Use this namespace to create and apply credit memos against invoices or invoice lines based on dispute adjustments.

RulesAppln Namespace
Apply payments and credits to posted invoices by adhering to the specified rules.

TaxEngineAdapter Interface
Retrieves and evaluates the details from a tax engine to define tax details.

`ConnectApi` Namespace

The `ConnectApi` namespace (also called Connect in Apex) provides classes to manage credit applications and billing scenarios.

These classes are available in the `ConnectApi` namespace.

BatchInvoiceApplication Class

Update a batch of invoices from Draft to Posted status for a credit memo application. by using the BatchInvoiceApplication class.

Billing Class

Manage billing scenarios by using the Billing class. You can convert negative invoice lines, create and apply a credit memo to an invoice, generate invoices, and recover billing schedules.

CreditMemoApply Class

Manage credit memo applications by using the CreditMemoApply class.

CreditMemoLineApply Class

Manage credit memo line applications by using the CreditMemoLineApply class.

HarmonizeBilling Class

Update the status of the invoice from Draft to Posted by using the HarmonizeBilling class.

PaymentsBilling Class

Use the PaymentsBilling class to allocate the balance of a payment to reduce the balance of an invoice. Additionally, revert the application of a payment line from an invoice.

SequencingWithoutAura Class

Manage invoice sequencing processes by using the SequencingWithoutAura class.

ConnectApi Input Classes

Billing includes these Apex input classes.

ConnectApi Output Classes

Billing includes these Apex output classes.

ConnectApi Enums

Enums specific to the `ConnectApi` namespace.

BatchInvoiceApplication Class

Update a batch of invoices from Draft to Posted status for a credit memo application. by using the BatchInvoiceApplication class.

Namespace

ConnectApi

BatchInvoiceApplication Methods

These methods are for `BatchInvoiceApplication`. All methods are static.

triggerInvoiceBatchDraftToPosted(invoiceBatchRunId)

Update a batch of invoices from Draft to Posted status for a credit memo application.

`triggerInvoiceBatchDraftToPosted(invoiceBatchRunId)`

Update a batch of invoices from Draft to Posted status for a credit memo application.

API Version

62.0

Requires Chatter

No

Signature

```
public static ConnectApi.InvoiceBatchDraftToPostedResult
triggerInvoiceBatchDraftToPosted(String invoiceBatchRunId)
```

Parameters

`invoiceBatchRunId`

Type: String

ID of the invoice batch run record that creates the draft invoices.

Return Value

Type: `ConnectApi.InvoiceBatchDraftToPostedResult` on page 2805

Usage

You need the Billing Operations User permission set to use this method.

This method posts the draft invoices and changes the status of the invoices from `Draft` to `Posted`.

Billing Class

Manage billing scenarios by using the Billing class. You can convert negative invoice lines, create and apply a credit memo to an invoice, generate invoices, and recover billing schedules.

Namespace

ConnectApi

Billing Methods

These methods are for `Billing`. All methods are static.

convertNegativeInvoiceLines(ConvertNegativeInvoiceLinesInput, invoiceId)

Convert a list of invoice lines with a negative amount into a posted credit memo. This conversion is applicable for a single invoice at a time.

createCreditMemos(CreditMemoInputRequest)
: Create a credit memo without applying it to an invoice. You can credit the invoice at a later date.

creditInvoice(CreditInvoiceInput, invoiceId)
: Create a credit memo and apply it to an invoice. The credit memo can fully or partially credit the invoice.

generateInvoices(inputRequest)
: Create an invoice from a billing schedule.

recoverBillingSchedules(inputRequest)
: Recover the latest generated invoice associated with the billing schedules in the Error or Processing status.

voidPostedCreditMemo(VoidPostedCreditMemoInput, creditMemoId)
: Invoke the Void Posted Credit Memo API by providing a credit memo ID.

voidPostedInvoice(invoiceId)
: Void a posted invoice to rebill the customer, if necessary.

convertNegativeInvoiceLines(ConvertNegativeInvoiceLinesInput, invoiceId)

Convert a list of invoice lines with a negative amount into a posted credit memo. This conversion is applicable for a single invoice at a time.

API Version

62.0

Requires Chatter

No

Signature

```
public static ConnectApi.ConvertNegativeInvoiceLinesResult
convertNegativeInvoiceLines(ConnectApi.ConvertNegativeInvoiceLinesInputRequest
ConvertNegativeInvoiceLinesInput, String invoiceId)
```

```
ConnectApi.Billing, convertNegativeInvoiceLines,
[ConnectApi.ConvertNegativeInvoiceLinesInputRequest, String],
ConnectApi.ConvertNegativeInvoiceLinesResult
```

Parameters

ConvertNegativeInvoiceLinesInput
: Type: ConnectApi.ConvertNegativeInvoiceLinesInputRequest on page 2791

  Input parameters to convert a negative invoice line to a credit.

invoiceId
: Type: String

  ID of the invoice whose negative invoice lines must be converted.

Return Value

Type: ConnectApi.ConvertNegativeInvoiceLinesResult on page 2805

Usage

You need the Credit Memo Operations User permission set to use this method.

Keep these considerations in mind when you use this method.

- All invoice lines must be related to the same invoice.
- The invoice line must have a negative amount.
- The invoice line must not be a previously converted credit memo.
- The invoice must have the `Posted` status.
- The invoice must not have any active settlements such as credit applications.

`createCreditMemos(CreditMemoInputRequest)`

Create a credit memo without applying it to an invoice. You can credit the invoice at a later date.

API Version

62.0

Requires Chatter

No

Signature

```
public static ConnectApi.RevenueAsyncRepresentation
createCreditMemos(ConnectApi.StandaloneCreditMemoInputRequest CreditMemoInputRequest)
```

Parameters

`CreditMemoInputRequest`
Type: `ConnectApi.StandaloneCreditMemoInputRequest` on page 2800

Input representation of the details required to create a standalone credit memo.

Return Value

Type: `ConnectApi.RevenueAsyncRepresentation` on page 2807

Usage

You need the Credit Memo Operations User permission set to use this method.

Specify the credit memo header information, charge parameters, adjustment parameters, and tax parameters. A credit memo requires at least one credit memo line. The credit memo line can be a charge or an adjustment.

Specify the credit memo lines that you want as lists of charges and adjustments. Each credit memo line must be related to a product.

`creditInvoice(CreditInvoiceInput, invoiceId)`

Create a credit memo and apply it to an invoice. The credit memo can fully or partially credit the invoice.

API Version

62.0

Requires Chatter

No

Signature

```
public static ConnectApi.RevenueAsyncLineLevelOutputResponse
creditInvoice(ConnectApi.CreditInvoiceInputRequest CreditInvoiceInput, String invoiceId)
```

```
ConnectApi.Billing, creditInvoice, [ConnectApi.CreditInvoiceInputRequest, String],
ConnectApi.RevenueAsyncLineLevelOutputResponse
```

Parameters

`CreditInvoiceInput`

Type: `ConnectApi.CreditInvoiceInputRequest` on page 2792

Input representation of the details of the request to create a credit memo.

`invoiceId`

Type: String

ID of the invoice to be credited partially or fully. The status of the invoice must be `Posted`.

Return Value

Type: `ConnectApi.RevenueAsyncLineLevelOutputResponse` on page 2807

Usage

Use this method to adjust an outstanding invoice balance or rectify errors in an invoice. Pass a list of invoice lines to credit. Keep these considerations in mind when you use this API.

- The request must contain at least one invoice line. Each invoice line must have the invoice line's ID, the amount to credit, and any optional tax details. The invoice lines must be a part of the invoice passed in the resource.
- The amount to credit must not exceed the charge or adjustment amount of an individual invoice line.
- The request body's credit amount inclusive of taxes must not exceed the target invoice line's amount inclusive of taxes, except for taxes calculated through an external tax service.
- The request body's total credit amount inclusive of taxes calculated through an external tax service must not exceed the outstanding invoice balance, which is also inclusive of taxes.

This method creates and posts a credit memo. The credit memo has one credit memo line for each invoice line passed in the API request. The invoice's balance is then reduced by a value equal to the credit memo's balance. This API modifies the balance of a posted invoice or invoice line based on the specified credit application level for your org. See Apply Credits to Posted Invoices or Invoice Lines.

`generateInvoices(inputRequest)`

Create an invoice from a billing schedule.

API Version

62.0

Requires Chatter

No

Signature

`public static ConnectApi.RevenueAsyncRepresentation generateInvoices(ConnectApi.InvoiceInputRepresentation inputRequest)`

```
ConnectApi.Billing, generateInvoices, [ConnectApi.InvoiceInputRepresentation],
ConnectApi.RevenueAsyncRepresentation
```

Parameters

inputRequest

Type: ConnectApi.InvoiceInputRepresentation on page 2797

Input representation of the details of the billing schedule.

Return Value

Type: ConnectApi.RevenueAsyncRepresentation on page 2807

Usage

You need the Generate Invoices From Billing Schedule API permission set to use this method.

This method creates one billing period item for each unbilled period between the billing schedule's next billing date and the invoice's target date. Additionally, one invoice line is created for each billing period item. This method creates up to six billing periods per request.

`recoverBillingSchedules(inputRequest)`

Recover the latest generated invoice associated with the billing schedules in the Error or Processing status.

API Version

62.0

Requires Chatter

No

Signature

```
public static ConnectApi.BillingScheduleRecoveryResults
recoverBillingSchedules(ConnectApi.BillingScheduleRecoveryInputRequest inputRequest)
```

```
ConnectApi.Billing, recoverBillingSchedules,
[ConnectApi.BillingScheduleRecoveryInputRequest], ConnectApi.BillingScheduleRecoveryResults
```

Parameters

inputRequest

Type: ConnectApi.BillingScheduleRecoveryInputRequest on page 2791

Input representation of the details of the billing schedules to recover the associated invoice.

Return Value

Type: ConnectApi.BillingScheduleRecoveryResults on page 2804

Usage

You need the Manage Errors Using Invoice Error Recovery API permission set to use this method.

Billing schedules include critical details such as the amount to be billed, next billing date, and status. An invoice can be associated with one or more billing schedules. When an invoice is generated or posted, the billing schedules are updated to reflect the accurate state of the invoice. The billing schedules associated with an invoice are marked in the `Error` status if any of the invoicing processes have errors. Use this method to recover the invoice associated with the billing schedules in the `Error` or `Processing` status.

voidPostedCreditMemo(VoidPostedCreditMemoInput, creditMemoId)

Invoke the Void Posted Credit Memo API by providing a credit memo ID.

API Version

66.0

Requires Chatter

No

Signature

```
public static ConnectApi.VoidPostedCreditMemoOutputRepresentation
voidPostedCreditMemo(ConnectApi.VoidPostedCreditMemoInputRepresentation
VoidPostedCreditMemoInput, String creditMemoId)
```

```
ConnectApi.Billing, voidPostedCreditMemo,
[ConnectApi.VoidPostedCreditMemoInputRepresentation, String],
ConnectApi.VoidPostedCreditMemoOutputRepresentation
```

Parameters

VoidPostedCreditMemoInput

Type: `ConnectApi.VoidPostedCreditMemoInputRepresentation`

Input representation of the details of a credit memo to be voided.

creditMemoId

Type: String

Required. ID of the credit memo record in posted status to be voided.

Return Value

Type: `ConnectApi.VoidPostedCreditMemoOutputRepresentation`

voidPostedInvoice(invoiceId)

Void a posted invoice to rebill the customer, if necessary.

API Version

62.0

Requires Chatter

No

Signature

```
public static ConnectApi.RevenueAsyncRepresentation voidPostedInvoice(String invoiceId)
```

Parameters

invoiceId

Type: String

ID of the posted invoice to be voided.

Return Value

Type: `ConnectApi.RevenueAsyncRepresentation` on page 2807

Usage

You need the Void a Posted Invoice API permission set to use this method.

This method changes the invoice status from `Posted` to `Void In Progress`. The invoice remains in the `Void In Progress` status until the credit is applied and financial fields are recalculated on the invoice's related billing period items and billing schedule. The invoice status changes to `Voided` after all recalculations are completed.

Keep these considerations in mind when you use this method.

- The balance and total amount on the invoice must be equal. If these amounts aren't equal due to payments or credits, the API request fails.

- You can't call other APIs on an invoice with the `Void In Progress` status. You also can't update the invoice fields.
- You can void only the most recently posted invoice on a billing schedule.
- To void an invoice that has payments or credits, use the Credit Memo Unapply (POST) API.

Credit Memos

The void process creates a credit memo, which contains one credit memo line for each invoice line, including tax lines. For example, if the invoice line has a balance of US$20, the related credit memo line has a balance of $20. The credit memo's balance is then allocated to the invoice header's balance, reducing it to zero. A credit memo invoice application is created to record the details of the void process.

Negative Invoice Lines

If an invoice has negative invoice lines that aren't converted to a credit memo, you can use this endpoint to void the posted invoice.

CreditMemoApply Class

Manage credit memo applications by using the CreditMemoApply class.

Namespace

ConnectApi

CreditMemoApply Methods

These methods are for `CreditMemoApply`. All methods are static.

applyCreditMemos(CreditMemoApplyInput, creditMemoId)
Adjust or correct already issued invoices by applying an existing credit memo to an invoice.

unapplyCreditMemos(CreditMemoUnapplyInput, creditMemoInvApplicationId)
Unapply a credit memo from an invoice and return the invoice and the credit memo to their pre-application states.

`applyCreditMemos(CreditMemoApplyInput, creditMemoId)`

Adjust or correct already issued invoices by applying an existing credit memo to an invoice.

API Version

62.0

Requires Chatter

No

Signature

```
public static ConnectApi.ApplyCreditResults
applyCreditMemos(ConnectApi.CreditMemoApplyInputRequest CreditMemoApplyInput, String
creditMemoId)
```

Parameters

CreditMemoApplyInput

Type: ConnectApi.CreditMemoApplyInputRequest on page 2795

Input representation of the request to apply a credit memo to an invoice.

creditMemoId

Type: String

ID of the credit memo record.

Return Value

Type: ConnectApi.ApplyCreditResults on page 2803

Usage

Specify the credit memo ID and the amounts to be applied, with the total of all applied amounts not exceeding the credit memo's balance.

The credit amount for each invoice can't surpass the original charge or adjustment amount, and the overall credit amount must not exceed the invoice's outstanding balance. The exceptions include any taxes calculated by an external service.

For example, your organization sold 10 tablets at US$500 each, totaling $5000, to a vendor who later reported that 6 tablets were defective. Using this method, your accounts receivable team creates a $3000 credit memo and applies this credit to the original invoice.

unapplyCreditMemos(CreditMemoUnapplyInput, creditMemoInvApplicationId)

Unapply a credit memo from an invoice and return the invoice and the credit memo to their pre-application states.

API Version

62.0

Requires Chatter

No

Signature

```
public static ConnectApi.UnapplyCreditResult
unapplyCreditMemos(ConnectApi.CreditMemoUnapplyInputRequest CreditMemoUnapplyInput,
String creditMemoInvApplicationId)
```

Parameters

CreditMemoUnapplyInput

Type: ConnectApi.CreditMemoUnapplyInputRequest on page 2796

Input representation of the request to unapply a credit memo from an invoice.

creditMemoInvApplicationId

Type: String

ID of the credit memo invoice application.

Return Value

Type: ConnectApi.UnapplyCreditResult on page 2809

Usage

Use this method if an error occurred when a credit is issued. For example, if an incorrect credit memo is applied to an invoice, or if a credit memo is created for an incorrect amount, use this method to unapply the credit memo.

CreditMemoLineApply Class

Manage credit memo line applications by using the CreditMemoLineApply class.

Namespace

ConnectApi

CreditMemoLineApply Methods

These methods are for `CreditMemoLineApply`. All methods are static.

applyCreditMemoLines(CreditMemoLineApplyInput, creditMemoLineId)
: Adjust or correct already issued invoices by applying an existing credit memo line to an invoice line.

unapplyCreditMemoLines(CreditMemoLineUnapplyInput, creditMemoLineInvoiceLineId)
: Unapply a credit memo line from an invoice line and return the invoice line and the credit memo line to their pre-application states.

`applyCreditMemoLines(CreditMemoLineApplyInput, creditMemoLineId)`

Adjust or correct already issued invoices by applying an existing credit memo line to an invoice line.

API Version

62.0

Requires Chatter

No

Signature

```
public static ConnectApi.CreditMemoLineAppliedResponse
applyCreditMemoLines(ConnectApi.CreditMemoLineApplyInput CreditMemoLineApplyInput,
String creditMemoLineId)
```

Parameters

`CreditMemoLineApplyInput`
: Type: ConnectApi.CreditMemoLineApplyInput on page 2795

: Input representation of the details of the request to apply a credit memo line to an invoice line.

creditMemoLineId

Type: String

ID of the credit memo line record.

Return Value

Type: ConnectApi.CreditMemoLineAppliedResponse on page 2804

Usage

You need the Credit Memo Operations User permission set to use this method.

unapplyCreditMemoLines(CreditMemoLineUnapplyInput, creditMemoLineInvoiceLineId)

Unapply a credit memo line from an invoice line and return the invoice line and the credit memo line to their pre-application states.

API Version

62.0

Requires Chatter

No

Signature

```
public static ConnectApi.CreditMemoLineUnappliedResponse
unapplyCreditMemoLines(ConnectApi.CreditMemoLineUnapplyInput CreditMemoLineUnapplyInput,
String creditMemoLineInvoiceLineId)
```

Parameters

CreditMemoLineUnapplyInput

Type: ConnectApi.CreditMemoLineUnapplyInput on page 2796

Input representation of the details of the request to unapply a credit memo line from an invoice line.

creditMemoLineInvoiceLineId

Type: String

ID of the credit memo line invoice line record.

Return Value

Type: ConnectApi.CreditMemoLineUnappliedResponse on page 2804

Usage

You need the Credit Memo Operations User permission set to use this method.

HarmonizeBilling Class

Update the status of the invoice from Draft to Posted by using the HarmonizeBilling class.

Namespace

ConnectApi

HarmonizeBilling Methods

These methods are for `HarmonizeBilling`. All methods are static.

`postDraftInvoices(inputRequest)`

Update the status of the invoice from Draft to Posted.

API Version

62.0

Requires Chatter

No

Signature

```
public static ConnectApi.RevenueAsyncRepresentation
postDraftInvoices(ConnectApi.InvoiceDraftToPostedInputRequest inputRequest)
```

Parameters

inputRequest
Type: `ConnectApi.InvoiceDraftToPostedInputRequest` on page 2796

Input representation of the details of the draft invoice that's posted.

Return Value

Type: `ConnectApi.RevenueAsyncRepresentation` on page 2807

Usage

You need the Billing Operations User permission set to use this method.

This method calls an external tax engine to calculate taxes for the draft invoice, posts the invoice, and updates the related billing schedules and billing periods.

PaymentsBilling Class

Use the PaymentsBilling class to allocate the balance of a payment to reduce the balance of an invoice. Additionally, revert the application of a payment line from an invoice.

Namespace

ConnectApi

PaymentsBilling Methods

These methods are for `PaymentsBilling`. All methods are static.

`applyPaymentLine(PaymentLineApplyInput, paymentId)`

Allocate the balance of a payment to reduce the balance of an invoice. The response includes an ID of the payment line invoice that represents the payment balance allocated against the invoice.

API Version

64.0

Requires Chatter

No

Signature

```
public static ConnectApi.PaymentLineApplyResponse
applyPaymentLine(ConnectApi.PaymentLineApplyRequest PaymentLineApplyInput, String
paymentId)
```

Parameters

PaymentLineApplyInput
Type: `ConnectApi.PaymentLineApplyRequest` on page 2797

Input representation of the payment line details.

paymentId

Type: String

ID of the payment record.

Return Value

Type: `ConnectApi.PaymentLineApplyResponse` on page 2806

Usage

Use the Commerce Payments APIs to send your payment and refund details to external payment gateways for processing against a customer's bank. See Commerce Payments resources to check the APIs for payment gateways, payment captures, and payment authorizations.

applyRefundLine(RefundLineApplyInput, refundId)

Make a refund transaction against a payment.

API Version

64.0

Requires Chatter

No

Signature

```
public static ConnectApi.RefundLineApplyResponse
applyRefundLine(ConnectApi.RefundLineApplyRequest RefundLineApplyInput, String refundId)
```

Parameters

RefundLineApplyInput

Type: `ConnectApi.RefundLineApplyRequest` on page 2798

Input representation of the details of a transaction refund request. This representation outlines the properties of a refund, including the refund amount and ID of the payment or credit memo record that the refund is applied to.

refundId

Type: String

ID of the refund record.

Return Value

Type: `ConnectApi.RefundLineApplyResponse` on page 2806

`unapplyPaymentLine(PaymentLineUnapplyInput, paymentId, paymentLineId)`

Revert the application of a payment line from an invoice, and return the payment and invoices to their pre-application state. Use this method if you need to correct an input during the payment application process.

API Version

64.0

Requires Chatter

No

Signature

```
public static ConnectApi.PaymentLineUnapplyResponse
unapplyPaymentLine(ConnectApi.PaymentLineUnapplyRequest PaymentLineUnapplyInput, String
paymentId, String paymentLineId)
```

Parameters

`PaymentLineUnapplyInput`
: Type: `ConnectApi.PaymentLineUnapplyRequest` on page 2798

  Input representation of the payment line details.

`paymentId`
: Type: String

  ID of the payment record.

`paymentLineId`
: Type: String

  ID of the payment line record.

Return Value

Type: `ConnectApi.PaymentLineUnapplyResponse` on page 2806

Usage

Use the Commerce Payments APIs to send your payment and refund details to external payment gateways for processing against a customer's bank. See Commerce Payments resources to check the APIs for payment gateways, payment captures, and payment authorizations.

`unapplyPaymentLine(PaymentLineUnapplyInput, paymentLineId)`

Revert the application of a payment line from an invoice, and return the payment and invoices to their pre-application state. Use this method if you need to correct an input during the payment application process.

API Version

64.0

Requires Chatter

No

Signature

```
public static ConnectApi.PaymentLineUnapplyResponse
unapplyPaymentLine(ConnectApi.PaymentLineUnapplyRequest PaymentLineUnapplyInput, String
paymentLineId)
```

Parameters

PaymentLineUnapplyInput
: Type: `ConnectApi.PaymentLineUnapplyRequest` on page 2798

  Input representation of the payment line details.

paymentLineId
: Type: String

  ID of the payment line record.

Return Value

Type: `ConnectApi.PaymentLineUnapplyResponse` on page 2806

Usage

Use the Commerce Payments APIs to send your payment and refund details to external payment gateways for processing against a customer's bank. See Commerce Payments resources to check the APIs for payment gateways, payment captures, and payment authorizations.

SequencingWithoutAura Class

Manage invoice sequencing processes by using the SequencingWithoutAura class.

Namespace

ConnectApi

SequencingWithoutAura Methods

These methods are for `SequencingWithoutAura`. All methods are static.

reconcileSequences(sequenceGapReconciliationInputRepresentation)
: Restore a missing sequence value identified by using this API in gapless-enabled sequences. This sequence value can be used later in the subsequent sequence policy numbering, ensuring there are no gaps.

sequenceAssignment(sequencesAssignmentInputRepresentation)
: Assign sequence pattern values to objects based on the configured sequence policy.

`reconcileSequences(sequenceGapReconciliationInputRepresentation)`

Restore a missing sequence value identified by using this API in gapless-enabled sequences. This sequence value can be used later in the subsequent sequence policy numbering, ensuring there are no gaps.

API Version

65.0

Requires Chatter

No

Signature

```
public static ConnectApi.sequenceGapReconciliationOutputRepresentation
reconcileSequences(ConnectApi.sequenceGapReconciliationInputRepresentation
sequenceGapReconciliationInputRepresentation)
```

Parameters

`sequenceGapReconciliationInputRepresentation`

Type: `ConnectApi.SequenceGapReconciliationInputRepresentation` on page 2798

The details of the input used to identify and reconcile gaps in sequence values based on the sequence policy or target object.

Return Value

Type: `ConnectApi.SequenceGapReconciliationOutputRepresentation` on page 2808

Usage

You need the Billing Admin permission set to use this method.

`sequenceAssignment(sequencesAssignmentInputRepresentation)`

Assign sequence pattern values to objects based on the configured sequence policy.

API Version

65.0

Requires Chatter

No

Signature

```
public static ConnectApi.SequencesAssignmentOutputRepresentation
sequenceAssignment(ConnectApi.SequencesAssignmentInputRepresentation
sequencesAssignmentInputRepresentation)
```

Parameters

sequencesAssignmentInputRepresentation

Type: `ConnectApi.SequencesAssignmentInputRepresentation` on page 2799

The details of the target objects to which the sequence pattern values are assigned.

Return Value

Type: `ConnectApi.SequencesAssignmentOutputRepresentation` on page 2808

Usage

You need the Billing Admin permission set to use this method.

ConnectApi Input Classes

Billing includes these Apex input classes.

ConnectApi.ApplicationsRequest
Connect API representation of an application item input request for credit memo apply api

ConnectApi.BillingAddressRequest
Input representation of the details of an address.

ConnectApi.BillingScheduleRecoveryInputRepresentation
Input representation of the details of the billing schedules to recover the associated invoice.

ConnectApi.ConvertNegativeInvoiceLinesInputRequest
Input representation of the request details to convert a negative invoice line into a credit.

ConnectApi.CreditDetailsApplyInput
Input representation of the request to specify one or more applications to apply a credit memo line for, with each application representing an invoice line.

ConnectApi.CreditInvoiceInputRequest
Input representation of the details of the request to create a credit memo.

ConnectApi.CreditInvoiceInvoiceLine
Input representation of the details of the invoice lines to be credited.

ConnectApi.CreditInvoiceInvoiceLineTax
Input representation of the details of the tax lines to be created manually for the invoice line.

ConnectApi.CreditMemoAddressesInputRequest
Input representation of the details of the billing and shipping addresses.

ConnectApi.CreditMemoApplyInputRequest
Input representation of the request to apply a credit memo to an invoice.

ConnectApi.CreditMemoLineApplyInput
Input representation of the details of the request to apply a credit memo line to an invoice line.

ConnectApi.CreditMemoLineUnapplyInput
Input representation of the details of the request to unapply a credit memo line from an invoice line.

ConnectApi.CreditMemoUnapplyInputRequest

Input representation of the request to unapply a credit memo from an invoice.

ConnectApi.InvoiceDraftToPostedInputRequest

Input representation of the details of the draft invoice that's posted.

ConnectApi.InvoiceInputRepresentation

Input representation of the details of the billing schedule.

ConnectApi.PaymentLineApplyRequest

Input representation of the payment line details. This representation covers details on allocation of a payment to a specific invoice line. It also provides additional context through optional fields, such as associated account and effective date.

ConnectApi.PaymentLineUnapplyRequest

Input representation of the payment line details. This representation covers fields that you can specify to revert a payment line application.

ConnectApi.RefundLineApplyRequest

Input representation of the details of a transaction refund request. This representation outlines the properties of a refund, including the refund amount and ID of the payment or credit memo record that the refund is applied to.

ConnectApi.SequenceGapReconciliationInputRepresentation

The details of the input used to identify and reconcile gaps in sequence values based on the sequence policy or target object.

ConnectApi.SequencesAssignmentInputRepresentation

The details of the target objects to which the sequence pattern values are assigned.

ConnectApi.StandaloneCreditMemoChargeInputRequest

Input representation of the details of the charge lines of a credit memo.

ConnectApi.StandaloneCreditMemoInputRequest

Input representation of the details required to create a standalone credit memo.

ConnectApi.StandaloneCreditMemoTaxInputRequest

Connect API representation of Tax input request

ConnectApi.VoidPostedCreditMemoInputRepresentation

Input representation of the details of a credit memo to be voided.

ConnectApi.ApplicationsRequest

Connect API representation of an application item input request for credit memo apply api

Property	Type	Description	Required or Optional	Available Version
`amount`	Double	Credit amount to be applied to the invoice.	Required	62.0
`appliedToId`	String	ID of the invoice record to apply the credit for.	Required	62.0
`description`	String	Explanation or reason for applying the credit memo.	Optional	62.0
`effectiveDate`	String	Effective date for the credit memo.	Optional	62.0

ConnectApi.BillingAddressRequest

Input representation of the details of an address.

Property	Type	Description	Required or Optional	Available Version
city	String	Address city.	Optional	62.0
country	String	Address country.	Optional	62.0
latitude	Double	Latitude for the address.	Optional	62.0
locationCode	String	Location code for the address.	Optional	62.0
longitude	Double	Longitude for the address.	Optional	62.0
postalCode	String	Postal code for the address.	Optional	62.0
state	String	Address state.	Optional	62.0
street	String	Address street.	Optional	62.0

ConnectApi.BillingScheduleRecoveryInputRepresentation

Input representation of the details of the billing schedules to recover the associated invoice.

Property	Type	Description	Required or Optional	Available Version
billing ScheduleIds	List<String>	IDs of the billing schedules to recover the invoice for. You can recover one billing schedule per API request.	Required	62.0

ConnectApi.ConvertNegativeInvoiceLinesInputRequest

Input representation of the request details to convert a negative invoice line into a credit.

Property	Type	Description	Required or Optional	Available Version
description	String	Description stamped on the credit memo that's created after the negative invoice line conversion.	Optional	62.0
effectiveDate	String	Date stamped on the credit memo that's created after the negative invoice line conversion.	Required	62.0

Property	Type	Description	Required or Optional	Available Version
`invoiceId`	String	ID of the invoice whose negative invoice lines must be converted into a posted credit memo.	Required	62.0
`invoiceLines`	List<String>	Complete list of the negative invoice lines along with the associated invoice line taxes. The specified negative invoice lines are converted into a posted credit memo.	Required	62.0

ConnectApi.CreditDetailsApplyInput

Input representation of the request to specify one or more applications to apply a credit memo line for, with each application representing an invoice line.

Property	Type	Description	Required or Optional	Available Version
`appliedAmount`	Double	Credit amount to be applied to the invoice line.	Required	62.0
`description`	String	Explanation or reason for applying the credit memo line.	Optional	62.0
`effectiveDate`	String	Effective date for the credit memo line.	Optional	62.0
`invoiceLineId`	String	ID of the invoice line record to apply the credit for.	Required	62.0

ConnectApi.CreditInvoiceInputRequest

Input representation of the details of the request to create a credit memo.

Property	Type	Description	Required or Optional	Available Version
`description`	String	Description for the credit memo to be created.	Optional	62.0
`effectiveDate`	String	Date when the credit memo takes effect.	Optional	62.0
`invoiceLines`	List<ConnectApi.CreditInvoiceLine>	List of the invoice lines to be credited. The invoice line IDs must be related to the invoice ID specified in the API request. If invoice lines aren't specified, the API request results in an error.	Required	62.0

Property	Type	Description	Required or Optional	Available Version
`taxEffectiveDate`	String	Date when the tax takes effect to recalculate the taxes.	Optional	62.0
`taxStrategy`	ConnectApi.TaxStrategyEnum	Tax strategy to be applied across invoice lines. You can override the tax strategy at the individual invoice line level or at the tax line level. Valid values are: • `Ignore`—Specifies that the creation of tax lines must be ignored. • `ManualOverride`—Specifies that the provided tax values must be considered for taxes. • `CopyFromInvoiceLine`—Specifies that tax values must be copied from the invoice line. • `Calculate`—Specifies that tax must be calculated by using the API.	Required	62.0
`type`	ConnectApi.CreditMemoTypeEnum	Type of credit memo to be created. Valid values are `Posted` and `Draft`. Specify `Draft` as a value in your request to create draft credit memos.	Optional	62.0

ConnectApi.CreditInvoiceInvoiceLine

Input representation of the details of the invoice lines to be credited.

Property	Type	Description	Required or Optional	Available Version
`addresses`	ConnectApi.CreditMemoAddressInputRequest	Addresses to be created manually for this invoice line and the overridden tax lines. These addresses are only applicable if this invoice line is using the `ManualOverride` tax strategy.	Optional	62.0
`amountToCredit`	Double	Amount to be credited from this invoice line.	Required	62.0
`invoiceLineId`	String	ID of the invoice line record to be credited. The invoice line ID must be related to the invoice ID specified in the API request.	Required	62.0

Property	Type	Description	Required or Optional	Available Version
`isTaxOnlyCredit`	Boolean	Indicates whether the applicable tax amount is credited for the charge or adjustment amount (`true`), or the applicable tax amount is credited along with the charge or adjustment amount (`false`). The default value is `false`.	Optional	62.0
`taxEffectiveDate`	String	Date when the tax takes effect and the invoice line is credited.	Optional	62.0
`taxStrategy`	ConnectApi.TaxStrategyEnum	Tax strategy for crediting the invoice line. This tax strategy takes precedence over the `taxStrategy` property value specified in the Credit Invoice Input. Valid values are: • `Ignore`—Specifies that the creation of tax lines must be ignored. • `ManualOverride`—Specifies that the provided tax values must be considered for taxes. • `CopyFromInvoiceLine`—Specifies that tax values must be copied from the invoice line. • `Calculate`—Specifies that tax must be calculated by using the API.	Optional	62.0
`taxes`	List<ConnectApi.CreditInvoiceInvoiceLineTax>	List of tax lines to be created manually for this invoice line.	Required if the `taxStrategy` property value is `ManualOverride`.	62.0

ConnectApi.CreditInvoiceInvoiceLineTax

Input representation of the details of the tax lines to be created manually for the invoice line.

Property	Type	Description	Required or Optional	Available Version
`taxAmount`	Double	Amount of tax to be applied related to this invoice line.	Required	62.0
`taxCode`	String	Tax code to be applied related to this invoice line to create the tax line.	Optional	62.0

Property	Type	Description	Required or Optional	Available Version
taxName	String	Name of tax to be applied related to this invoice line.	Optional	62.0
taxRate	Double	Tax rate used to create the tax line.	Optional	62.0

ConnectApi.CreditMemoAddressesInputRequest

Input representation of the details of the billing and shipping addresses.

Property	Type	Description	Required or Optional	Available Version
billingAddress	ConnectApi.BillingAddressRequest	Billing address for charge or adjustment line.	Optional	62.0
shippingAddress	ConnectApi.BillingAddressRequest	Shipping address for charge or adjustment line.	Optional	62.0

ConnectApi.CreditMemoApplyInputRequest

Input representation of the request to apply a credit memo to an invoice.

Property	Type	Description	Required or Optional	Available Version
applications	List<ConnectApi.ApplicationsRequest>	List of one or more applications to apply the credit memo for. Each application represents an invoice that's credited by using the balance of the specified credit memo.	Required	62.0
creditMemoId	String	ID of the credit memo record.	Required	62.0

ConnectApi.CreditMemoLineApplyInput

Input representation of the details of the request to apply a credit memo line to an invoice line.

Property	Type	Description	Required or Optional	Available Version
applyCreditDetails	List<ConnectApi.CreditDetailsApplyInput>	List of one or more applications to apply the credit memo line for. Each application represents an invoice line that's credited by using the balance of the specified credit memo line.	Required	62.0

Property	Type	Description	Required or Optional	Available Version
creditMemoLineId	String	ID of the credit memo line record.	Required	62.0

ConnectApi.CreditMemoLineUnapplyInput

Input representation of the details of the request to unapply a credit memo line from an invoice line.

Property	Type	Description	Required or Optional	Available Version
creditMemoLineInvoiceLineId	String	ID of the credit memo line invoice line record.	Required	62.0
description	String	Explanation or reason for unapplying the credit memo line.	Optional	62.0
effectiveDate	String	Effective date for the credit memo line.	Optional	62.0

ConnectApi.CreditMemoUnapplyInputRequest

Input representation of the request to unapply a credit memo from an invoice.

Property	Type	Description	Required or Optional	Available Version
creditMemoInvApplicationId	String	ID of the credit memo invoice application.	Required	62.0
description	String	Explanation or reason for unapplying the credit memo.	Optional	62.0
effectiveDate	String	Effective date for the credit memo.	Optional	62.0

ConnectApi.InvoiceDraftToPostedInputRequest

Input representation of the details of the draft invoice that's posted.

Property	Type	Description	Required or Optional	Available Version
correlationId	String	Splunk correlation ID to track the messages that are related to the request and are logged in Splunk by the different services involved in the request. If the ID isn't specified, the service creates a random Universally Unique Identifier (UUID).	Optional	62.0
invoiceIds	List<String>	IDs of the invoice records in Draft status to be posted. You can post one draft invoice per API request.	Required	62.0

ConnectApi.InvoiceInputRepresentation

Input representation of the details of the billing schedule.

Property	Type	Description	Required or Optional	Available Version
action	ConnectApi.InvoiceAction	Type of invoice to be created. Valid values are: • Draft • Posted	Required	62.0
billing ScheduleIds	List<String>	List of billing schedule IDs that's used to create the invoices. You can specify a maximum of 200 billing schedules.	Required	62.0
correlationId	String	Property that's tagged against the published InvoiceProcessedEvent event, if specified.	Optional	62.0
invoiceDate	String	Stamping date of the invoice in ISO 8601 format.	Required	62.0
targetDate	String	Date in ISO 8601 format used to decide the billing periods that are included to create invoices.	Required	62.0

ConnectApi.PaymentLineApplyRequest

Input representation of the payment line details. This representation covers details on allocation of a payment to a specific invoice line. It also provides additional context through optional fields, such as associated account and effective date.

Property	Type	Description	Required or Optional	Available Version
amount	Double	Amount that's applied. The amount must be less than the invoice line and payment balance.	Required	64.0
appliedToId	String	ID of the invoice line that this payment is applied to. Specify the IDs for these records. • Invoice • Invoice Line	Required	64.0
associated AccountId	String	ID of the associated account.	Optional	64.0
comments	String	Comments that you can add to the payment line application.	Optional	64.0

Property	Type	Description	Required or Optional	Available Version
`effectiveDate`	Datetime	Date from which the payment line application takes effect.	Optional	64.0

ConnectApi.PaymentLineUnapplyRequest

Input representation of the payment line details. This representation covers fields that you can specify to revert a payment line application.

Property	Type	Description	Required or Optional	Available Version
`comments`	String	Comments that you can add when you revert a payment line application.	Optional	64.0
`effectiveDate`	Datetime	Date from when the reversal of the payment line application is in effect.	Optional	64.0

ConnectApi.RefundLineApplyRequest

Input representation of the details of a transaction refund request. This representation outlines the properties of a refund, including the refund amount and ID of the payment or credit memo record that the refund is applied to.

Property	Type	Description	Required or Optional	Available Version
`amount`	Double	Amount to refund.	Required	64.0
`appliedToId`	String	ID of a payment or credit memo record. The refund is applied to this object.	Required	64.0
`comments`	String	Additional details of the refund request.	Optional	64.0
`effectiveDate`	Datetime	Date from when the refund is in effect.	Optional	64.0

ConnectApi.SequenceGapReconciliationInputRepresentation

The details of the input used to identify and reconcile gaps in sequence values based on the sequence policy or target object.

Property	Type	Description	Required or Optional	Available Version
`sequencePolicyIds`	List<`String`>	List of IDs of the sequence policies.	Required if the `targetObjects` property isn't	65.0

Property	Type	Description	Required or Optional	Available Version
			specified. You must not specify both properties.	
targetObjects	List<String>	List of objects to which the policies are applied.	Required if the sequencePolicyIds property isn't specified. You must not specify both properties.	65.0

ConnectApi.SequencesAssignmentInputRepresentation

The details of the target objects to which the sequence pattern values are assigned.

Property	Type	Description	Required or Optional	Available Version
sequencePolicyId	String	ID of the sequence policy.	Optional	65.0
shouldPublishPlatformEvent	Boolean	Indicates whether to publish a platform event when a sequence is assigned to a target record (true) or not (false).	Optional	65.0
targetObjectIds	List<String>	List of records to which the sequence pattern values are assigned.	Required	65.0

ConnectApi.StandaloneCreditMemoChargeInputRequest

Input representation of the details of the charge lines of a credit memo.

Property	Type	Description	Required or Optional	Available Version
addresses	Credit Memo Addresses Input	Details of the billing and shipping addresses.	Optional	62.0
chargeAmount	Double	Charge amount for the credit memo.	Required	62.0
description	String	Description of the created credit memo charge line.	Optional	62.0
endDate	String	End date of the credit memo charge line.	Optional	62.0
isTaxOnlyCredit	Boolean	Indicates whether the credit is for tax only (true) or not (false).	Optional	62.0
productId	String	ID of the product record that the credit memo is issued on.	Optional	62.0

Property	Type	Description	Required or Optional	Available Version
productName	String	Name of the product that the credit memo is issued on.	Optional	62.0
startDate	String	Start date of the credit memo charge line.	Optional	62.0
taxEffectiveDate	String	Date from when the tax is applicable.	Optional	62.0
taxes	List<ConnectApi.StandaloneCreditMemoTaxInputRequest>	List of credit memo tax lines.	Optional	62.0
taxStrategy	ConnectApi.StandaloneTaxStrategyEnum	Tax strategy to be applied to this credit memo charge line, child treatment lines, and tax lines. You can override the tax strategy at the individual credit memo lines or tax lines level. Valid values are: • Ignore—Specifies that the creation of tax lines must be ignored. • Manual Override—Specifies that the provided tax values must be considered for taxes. • Calculate—Specifies that tax must be calculated by using the API.	Optional	62.0
treatmentId	String	ID of the tax treatment record that's used to calculate tax.	Optional	62.0

ConnectApi.StandaloneCreditMemoInputRequest

Input representation of the details required to create a standalone credit memo.

Property	Type	Description	Required or Optional	Available Version
billToContactId	String	Contact related to the credit memo.	Optional	62.0
billingAccountId	String	ID of the account that the credit is issued to.	Required	62.0
charges	List<ConnectApi.StandaloneCreditMemoChargeInputRequest>	Charge lines of the credit memo. Requires at least one charge line.	Required	62.0
currencyIsoCode	String	ISO code currency of the new credit that's issued.	Optional	62.0
description	String	Description for the new credit that's issued.	Optional	62.0
effectiveDate	String	Effective date of the credit memo. If the value isn't specified, then it's null.	Optional	62.0

Property	Type	Description	Required or Optional	Available Version
externalReference	String	ID of the external reference for the credit memo.	Optional	62.0
externalReferenceDataSource	String	Source of the external reference for the credit memo.	Optional	62.0
taxEffectiveDate	String	Effective date of the credit memo tax. If the value isn't specified, then it's null.	Optional	62.0
taxStrategy	ConnectApi.StandaloneTaxStrategyEnum	Specifies how tax lines must be created for the standalone credit memos. Valid values are: • Ignore—Specifies that the creation of tax lines must be ignored. • Manual Override—Specifies that the provided tax values must be considered for taxes. • Calculate—Specifies that tax must be calculated by using the API.	Required	62.0
type	ConnectApi.CreditMemoTypeEnum	Type of credit memo to be created. Valid values are Posted and Draft. Specify Draft as a value in your request to create draft credit memos.	Optional	62.0

ConnectApi.StandaloneCreditMemoTaxInputRequest

Connect API representation of Tax input request

Property	Type	Description	Required or Optional	Available Version
taxAmount	Double	Amount of tax to be applied.	Required	62.0
taxCode	String	Tax code to be used to create the tax line.	Optional	62.0
taxName	String	Name of tax to be applied.	Optional	62.0
taxRate	Double	Tax rate to be used to create the tax line.	Optional	62.0

ConnectApi.VoidPostedCreditMemoInputRepresentation

Input representation of the details of a credit memo to be voided.

Property	Type	Description	Required or Optional	Available Version
`creditMemoId`	String	ID of the posted credit memo to be voided.	Required	66.0

ConnectApi Output Classes

Billing includes these Apex output classes.

ConnectApi.AppliedCreditResponse
Output representation of the list of applied credit memo line results.

ConnectApi.ApplyCreditResults
Output representation of the list of applied credit memo results.

ConnectApi.ApplyCreditResult
Connect API representation of credit memo apply output result

ConnectApi.BillingScheduleRecoveryResult
Output representation of the details of the recovered billing schedules.

ConnectApi.BillingScheduleRecoveryResults
Output representation of the recovered details of the billing schedules and associated invoice.

ConnectApi.CreditMemoLineAppliedResponse
Output representation of the list of applied credit memo line results.

ConnectApi.CreditMemoLineUnappliedResponse
Output representation of the details of the credit memo line invoice line record with the status of the request.

ConnectApi.ConvertNegativeInvoiceLinesResult
Output representation of the details of the credit memo along with the status of the request.

ConnectApi.InvoiceBatchDraftToPostedResult
Output representation of the batch update details of the invoices from `Draft` to `Posted` status.

ConnectApi.InvoiceRecoveryResult
Output representation of the details of the recovered invoice and billing schedules.

ConnectApi.PaymentLineApplyResponse
Output representation of the details of the applied payment line. The details include the ID of the payment record and date when the payment line was applied.

ConnectApi.PaymentLineUnapplyResponse
Output representation of the details of the reversed payment line application. The details include the ID of the payment line record and date when the payment line application was reversed.

ConnectApi.ReferenceLineError
Output representation of the details of the line level errors.

ConnectApi.RefundLineApplyResponse
Output representation of the details of an applied refund. This representation includes the properties of a refund line, such as the date when the refund is applied against a payment and ID of the refund line record.

ConnectApi.RevenueAsyncLineLevelOutputResponse

Output representation of the result of the API request for the async line level operations.

ConnectApi.RevenueAsyncRepresentation

Output representation of the result of the API request with the request identifier.

ConnectApi.SequenceErrorOutputRepresentation

Output representation of the error response that's associated with a request to create or update a sequence policy, or assign sequences.

ConnectApi.SequenceGapReconciliationOutputRepresentation

Output representation of the details of the sequence gap reconciliation.

ConnectApi.SequenceGapReconciliationErrorOutputRepresentation

List of errors encountered during the processing of the API request.

ConnectApi.SequencesAssignmentOutputRepresentation

Output representation showing the status of the assigned sequence pattern values.

ConnectApi.SequencesAssignmentResultOutputRepresentation

Output representation of the details of the assigned sequence values to target objects.

ConnectApi.UnapplyCreditResult

Output representation of the details of the credit memo invoice application record with the status of the request.

ConnectApi.VoidPostedCreditMemoOutputRepresentation

Output representation of the request to void a posted credit memo.

ConnectApi.AppliedCreditResponse

Output representation of the list of applied credit memo line results.

Property Name	Type	Description	Available Version
creditMemoLineInvoiceLineId	String	ID of the credit memo line invoice line ID.	62.0
errors	List<ConnectApi.ErrorResponse>	List of errors encountered during the processing of the API request.	62.0
invoiceLineId	String	ID of the invoice line record that the credit is applied to.	62.0
success	Boolean	Indicates whether the credit memo line is successfully applied (`true`) or not (`false`).	62.0

ConnectApi.ApplyCreditResults

Output representation of the list of applied credit memo results.

Property Name	Type	Description	Available Version
applyCreditResults	List<ConnectApi.ApplyCreditResult>	Output list of the applied credit memo results.	62.0

ConnectApi.ApplyCreditResult

Connect API representation of credit memo apply output result

Property Name	Type	Description	Available Version
appliedToId	String	ID of the invoice record that the credit is applied to.	62.0
errors	List<ConnectApi.ErrorResponse>	List of errors encountered during the processing of the API request.	62.0
id	String	ID of the credit memo invoice application.	62.0
success	Boolean	Indicates whether the credit memo is successfully applied (true) or not (false).	62.0

ConnectApi.BillingScheduleRecoveryResult

Output representation of the details of the recovered billing schedules.

Property Name	Type	Description	Available Version
billing ScheduleId	String	ID of the billing schedule.	62.0
billingSchedule Status	String	Flag that indicates the billing schedule status.	62.0

ConnectApi.BillingScheduleRecoveryResults

Output representation of the recovered details of the billing schedules and associated invoice.

Property Name	Type	Description	Available Version
recoveryResults	List<ConnectApi.InvoiceRecoveryResult>	Details of the recovered invoice and billing schedules.	62.0

ConnectApi.CreditMemoLineAppliedResponse

Output representation of the list of applied credit memo line results.

Property Name	Type	Description	Available Version
appliedCreditResponses	List<ConnectApi.AppliedCreditResponse>	Output list for the applied credit memo line results.	62.0

ConnectApi.CreditMemoLineUnappliedResponse

Output representation of the details of the credit memo line invoice line record with the status of the request.

Property Name	Type	Description	Available Version
creditMemoLineInvoiceLineId	String	ID of the credit memo line invoice line record.	62.0
errors	List<ConnectApi.ErrorResponse>	List of errors encountered during the processing of the API request.	62.0

Property Name	Type	Description	Available Version
`success`	Boolean	Indicates whether the credit memo line is successfully unapplied (`true`) or not (`false`).	62.0

ConnectApi.ConvertNegativeInvoiceLinesResult

Output representation of the details of the credit memo along with the status of the request.

Property Name	Type	Description	Available Version
`errors`	List<ConnectApi.ErrorResponse>	List of errors encountered during the processing of the API request.	62.0
`id`	String	ID of the credit memo that's created after the conversion of the negative invoice lines.	62.0
`success`	Boolean	Indicates whether the API request is successful (`true`) or not (`false`).	62.0

ConnectApi.InvoiceBatchDraftToPostedResult

Output representation of the batch update details of the invoices from `Draft` to `Posted` status.

Property Name	Type	Description	Available Version
`invoiceBatchDraftToPostedId`	String	ID of the invoice batch draft to posted run record that's created to track the batch process of posting the draft invoices that are associated with the parent invoice batch run record.	62.0
`success`	Boolean	Indicates whether the API request is successful (`true`) or not (`false`).	62.0

ConnectApi.InvoiceRecoveryResult

Output representation of the details of the recovered invoice and billing schedules.

Property Name	Type	Description	Available Version
`billingSchedules`	List<ConnectApi.BillingScheduleRecoveryResult>	Billing schedules associated with this invoice.	62.0
`invoiceErrors`	List<ConnectApi.ErrorResponse>	List of errors encountered during the invoice recovery.	62.0
`invoiceId`	String	ID of the recovered invoice.	62.0
`invoiceStatus`	String	Flag that indicates the invoice status.	62.0

Property Name	Type	Description	Available Version
`success`	Boolean	Indicates whether the overall transaction was successful or not (`true`) or not (`false`).	62.0

ConnectApi.PaymentLineApplyResponse

Output representation of the details of the applied payment line. The details include the ID of the payment record and date when the payment line was applied.

Property Name	Type	Description	Available Version
`appliedDate`	Datetime	Date when the payment line was applied.	64.0
`id`	String	ID of the payment line record.	64.0

ConnectApi.PaymentLineUnapplyResponse

Output representation of the details of the reversed payment line application. The details include the ID of the payment line record and date when the payment line application was reversed.

Property Name	Type	Description	Available Version
`id`	String	ID of the payment line record.	64.0
`unappliedDate`	Datetime	Date when the payment line application was reversed.	64.0

ConnectApi.ReferenceLineError

Output representation of the details of the line level errors.

Property Name	Type	Description	Available Version
`errors`	List<ConnectApi.ErrorResponse>	List of errors with error code and error message for the specified invoice line ID.	62.0
`referenceLineId`	String	ID of the invoice line specified in the API request that has an issue, causing the API request to fail.	62.0

ConnectApi.RefundLineApplyResponse

Output representation of the details of an applied refund. This representation includes the properties of a refund line, such as the date when the refund is applied against a payment and ID of the refund line record.

Property Name	Type	Description	Available Version
appliedDate	Datetime	Date when the refund is applied against a payment.	64.0
id	String	ID of the refund line record.	64.0

ConnectApi.RevenueAsyncLineLevelOutputResponse

Output representation of the result of the API request for the async line level operations.

Property Name	Type	Description	Available Version
errors	List<ConnectApi.ErrorResponse>	Details of errors, if any.	62.0
referenceLineErrorResults	List<ConnectApi.ReferenceLineError>	List of errors grouped by the invoice line IDs if the API request fails.	62.0
referenceLineType	String	Reference type for the reference line entity in the referenceLineErrorResults property.	62.0
requestIdentifier	String	Unique identifier of the request.	62.0
statusURL	String	URL to track the status of the operation.	62.0
success	Boolean	Indicates whether the API request is successful (true) or not (false).	62.0

ConnectApi.RevenueAsyncRepresentation

Output representation of the result of the API request with the request identifier.

Property Name	Type	Description	Available Version
errors	List<ConnectApi.ErrorResponse>	Details of errors, if any.	62.0
request Identifier	String	Unique identifier of the request.	62.0
success	Boolean	Indicates whether the API request is successful (true) or not (false).	62.0

ConnectApi.SequenceErrorOutputRepresentation

Output representation of the error response that's associated with a request to create or update a sequence policy, or assign sequences.

Property Name	Type	Description	Available Version
`errorCode`	String	Code for the resultant error.	65.0
`message`	String	Error message for the resultant error.	65.0

ConnectApi.SequenceGapReconciliationOutputRepresentation

Output representation of the details of the sequence gap reconciliation.

Property Name	Type	Description	Available Version
`error`	ConnectApi.SequenceGapReconciliationErrorOutputRepresentation	List of errors encountered during the processing of the API request.	65.0
`jobId`	String	Unique identifier assigned to sequence gap reconciliation asynchronous process.	65.0
`sequencePolicyIds`	List<String>	List of IDs of the sequence policies.	65.0
`status`	`StatusEnum`	The status of sequence reconciliation API request. Valid values are: • `Submitted` • `NotSubmitted`	65.0
`submittedAt`	String	Date and time when the reconciliation request was submitted to the async job.	65.0
`targetObjects`	List<String>	List of objects to which the policies are applied.	65.0

ConnectApi.SequenceGapReconciliationErrorOutputRepresentation

List of errors encountered during the processing of the API request.

Property Name	Type	Description	Available Version
`errorCode`	String	Code for the resultant error.	65.0
`errorMessage`	String	Error message for the resultant error.	65.0

ConnectApi.SequencesAssignmentOutputRepresentation

Output representation showing the status of the assigned sequence pattern values.

Property Name	Type	Description	Available Version
`errors`	List<ConnectApi.SequenceErrorOutputRepresentation>	Error encountered during the processing of the API request.	65.0
`sequencesAssignment`	List<ConnectApi.SequencesAssignmentResultOutputRepresentation>	Details of the sequence pattern values assignment.	65.0

Property Name	Type	Description	Available Version
`status`	SequenceResponseStatusEnum	Status of the sequence policy assignment. Valid values are: • `PartialSuccess` • `Success` • `Failed`	65.0

ConnectApi.SequencesAssignmentResultOutputRepresentation

Output representation of the details of the assigned sequence values to target objects.

Property Name	Type	Description	Available Version
`errors`	List<ConnectApi.SequenceErrorOutputRepresentation>	Error encountered during the processing of the API request.	65.0
`isSuccess`	Boolean	Indicates whether the sequence pattern value was assigned (`true`) or not (`false`).	65.0
`sequencePatternValue`	String	Sequence pattern value assigned to the target object.	65.0
`sequencePolicyId`	String	ID of the sequence policy assigned to the target object.	65.0
`targetObjectId`	String	Record to which the sequence pattern value is assigned.	65.0

ConnectApi.UnapplyCreditResult

Output representation of the details of the credit memo invoice application record with the status of the request.

Property Name	Type	Description	Available Version
`errors`	List<ConnectApiErrorResponse>	List of errors encountered during the processing of the API request.	62.0
`id`	String	ID of the credit memo invoice application record.	62.0
`success`	Boolean	Indicates whether the credit memo is successfully unapplied (`true`) or not (`false`).	62.0

ConnectApi.VoidPostedCreditMemoOutputRepresentation

Output representation of the request to void a posted credit memo.

Property Name	Type	Description	Available Version
`debitMemoId`	String	ID of the created debit memo.	66.0

Property Name	Type	Description	Available Version
errors	List<ConnectApi.ErrorResponse>	List of errors specific to this API request that were encountered during voiding the credit memo.	66.0
isSuccess	Boolean	Indicates whether the API request was successful (true) or not (false).	66.0
statusURL	String	Status URL for tracking this operation.	66.0

ConnectApi Enums

Enums specific to the ConnectApi namespace.

ConnectApi enums inherit all properties and methods of Apex enums.

Enums are not versioned. Enum values are returned in all API versions.

Enum	Description
ConnectApi.InvoiceAction	Type of invoice to be created. Valid values are: • Draft • Posted
ConnectApi.TaxStrategyEnum	Tax strategy to be applied across invoice lines. You can override the tax strategy at the individual invoice line level or at the tax line level. Valid values are: • Ignore—Specifies that the creation of tax lines must be ignored. • ManualOverride—Specifies that the provided tax values must be considered for taxes. • CopyFromInvoiceLine—Specifies that tax values must be copied from the invoice line. • Calculate—Specifies that tax must be calculated by using the API.
ConnectApi.CreditMemoTypeEnum	Type of credit memo to be created. Valid values are Posted and Draft. Specify Draft as a value in your request to create draft credit memos.
ConnectApi.StandaloneTaxStrategyEnum	Specifies how tax lines must be created for the standalone credit memos. Valid values are: • Ignore—Specifies that the creation of tax lines must be ignored. • Manual Override—Specifies that the provided tax values must be considered for taxes. • Calculate—Specifies that tax must be calculated by using the API.
ConnectApi.SequenceResponseStatusEnum	Status of the sequence policy assignment. Valid values are: • PartialSuccess • Success • Failed

Enum	Description
`ConnectApi.StatusEnum`	The status of sequence reconciliation API request. Valid values are: • `Submitted` • `NotSubmitted`

InvoiceWriteOff Namespace

Create credit memos with the total charge amount on the invoice as the write-off amount and close the invoice.

The `InvoiceWriteOff` namespace includes these classes.

Usage

You need the Billing Operations User or Credit Memo Operations User permission set to access this namespace.

WriteOffInvoiceInput Class

Contains invoice details that are used for the request to write off an invoice.

Namespace

InvoiceWriteOff on page 2811

WriteOffInvoiceInput Constructors

Learn more about the constructors available with the WriteOffInvoiceInput class.

The `WriteOffInvoiceInput` class includes these constructors.

WriteOffInvoiceInput(invoiceId, reasonCode, reason)
Initializes the WriteOffInvoiceInput class that stores the invoice details and reason for writing off invoices.

WriteOffInvoiceInput()
Initializes the WriteOffInvoiceInput class.

`WriteOffInvoiceInput(invoiceId, reasonCode, reason)`

Initializes the WriteOffInvoiceInput class that stores the invoice details and reason for writing off invoices.

Signature

`public WriteOffInvoiceInput(String invoiceId, String reasonCode, String reason)`

Parameters

`invoiceId`
Type: String

ID of the invoice record that you want to write off.

`reasonCode`
Type: String

Code that categorizes the write-off reason. For example, if the reason for the invoice write-off is a disputed amount, the reason code can be Disputed Amount (DA).

`reason`
Type: String

Reason for writing off invoices.

`WriteOffInvoiceInput()`

Initializes the WriteOffInvoiceInput class.

Signature

`public WriteOffInvoiceInput()`

WriteOffInvoiceInput Properties

Learn more about the properties available with the WriteOffInvoiceInput class.

The `WriteOffInvoiceInput` class includes these properties.

invoiceId
Sets the ID of the invoice record that must be written off.

reason
Sets the reason for writing off invoices.

reasonCode
Sets the code that categorizes the write-off reason.

`invoiceId`

Sets the ID of the invoice record that must be written off.

Signature

`public String invoiceId {get; set;}`

Property Value

Type: String

`reason`

Sets the reason for writing off invoices.

Signature

`public String reason {get; set;}`

Property Value

Type: String

`reasonCode`

Sets the code that categorizes the write-off reason.

Signature

`public String reasonCode {get; set;}`

Property Value

Type: String

WriteOffInvoiceInputList Class

Contains invoice details to write off a list of posted invoices.

Namespace

InvoiceWriteOff on page 2811

WriteOffInvoiceInputList Constructors
Learn more about the constructors available with the WriteOffInvoiceInputList class.

WriteOffInvoiceInputList Properties
Learn more about the properties available with the WriteOffInvoiceInputList class.

WriteOffInvoiceInputList Constructors

Learn more about the constructors available with the WriteOffInvoiceInputList class.

The `WriteOffInvoiceInputList` class includes these constructors.

WriteOffInvoiceInputList(writeOffInvoiceInputList)
Initializes the WriteOffInvoiceInputList class that stores the details of invoices that you want to write off.

WriteOffInvoiceInputList()
Initializes the WriteOffInvoiceInputList class.

WriteOffInvoiceInputList(writeOffInvoiceInputList)

Initializes the WriteOffInvoiceInputList class that stores the details of invoices that you want to write off.

Signature

```
public WriteOffInvoiceInputList(List<InvoiceWriteOff.WriteOffInvoiceInput>
writeOffInvoiceInputList)
```

Parameters

writeOffInvoiceInputList
Type: List<InvoiceWriteOff.WriteOffInvoiceInput on page 2811 >

Input representation of the request to write off a list of posted invoices.

WriteOffInvoiceInputList()

Initializes the WriteOffInvoiceInputList class.

Signature

```
public WriteOffInvoiceInputList()
```

WriteOffInvoiceInputList Properties

Learn more about the properties available with the WriteOffInvoiceInputList class.

The `WriteOffInvoiceInputList` class includes these properties.

writeOffInvoiceInputList
Input representation of the request to write off a list of posted invoices.

writeOffInvoiceInputList

Input representation of the request to write off a list of posted invoices.

Signature

```
public List<InvoiceWriteOff.WriteOffInvoiceInput> writeOffInvoiceInputList {get; set;}
```

Property Value

Type: List<nvoiceWriteOff.WriteOffInvoiceInput on page 2811>

WriteOffInvoiceResponse Class

Contains properties to store the response details to the request to write off a posted invoice.

Namespace

InvoiceWriteOff on page 2811

WriteOffInvoiceResponse Constructors
Learn more about the constructors available with the WriteOffInvoiceResponse class.

WriteOffInvoiceResponse Properties
Learn more about the properties available with the WriteOffInvoiceResponse class.

WriteOffInvoiceResponse Constructors

Learn more about the constructors available with the WriteOffInvoiceResponse class.

The `WriteOffInvoiceResponse` class includes these constructors.

WriteOffInvoiceResponse(errors, invoiceId, requestIdentifier, success)
Initializes the WriteOffInvoiceResponse class that stores the response details to the request to write off a posted invoice.

`WriteOffInvoiceResponse(errors, invoiceId, requestIdentifier, success)`

Initializes the WriteOffInvoiceResponse class that stores the response details to the request to write off a posted invoice.

Signature

```
public WriteOffInvoiceResponse(InvoiceWriteOff.WriteOffInvoiceResponseError errors,
String invoiceId, String requestIdentifier, Boolean success)
```

Parameters

errors
Type: InvoiceWriteOff.WriteOffInvoiceResponseError on page 2817

If the request fails, this property contains a list of errors.

invoiceId
Type: String

ID of the invoice record that's written off.

requestIdentifier
Type: String

If the request is successful, this property contains an asynchronous API request identifier for an invoice ID.

success

Type: Boolean

Indicates whether the invoice write-off request was successful (`true`) or not (`false`).

WriteOffInvoiceResponse Properties

Learn more about the properties available with the WriteOffInvoiceResponse class.

The `WriteOffInvoiceResponse` class includes these properties.

errors

Get the list of errors if the request to write off posted invoices fails.

invoiceId

Get the ID of the invoice record that's written off.

requestIdentifier

Get the identifier of the asynchronous API request for an invoice ID if the request is successful.

success

Get the request status of the invoice write-off request.

errors

Get the list of errors if the request to write off posted invoices fails.

Signature

```
public InvoiceWriteOff.WriteOffInvoiceResponseError errors {get; set;}
```

Property Value

Type: InvoiceWriteOff.WriteOffInvoiceResponseError on page 2817

invoiceId

Get the ID of the invoice record that's written off.

Signature

```
public String invoiceId {get; set;}
```

Property Value

Type: String

requestIdentifier

Get the identifier of the asynchronous API request for an invoice ID if the request is successful.

Signature

```
public String requestIdentifier {get; set;}
```

Property Value

Type: String

success

Get the request status of the invoice write-off request.

Signature

```
public Boolean success {get; set;}
```

Property Value

Type: Boolean

WriteOffInvoiceResponseError Class

Contains properties to store the error response that's associated with a request to write off a posted invoice.

Namespace

InvoiceWriteOff on page 2811

WriteOffInvoiceResponseError Constructors
Learn more about the constructors available with the WriteOffInvoiceResponseError class.

WriteOffInvoiceResponseError Properties
Learn more about the properties available with the WriteOffInvoiceResponseError class.

WriteOffInvoiceResponseError Constructors

Learn more about the constructors available with the WriteOffInvoiceResponseError class.

The `WriteOffInvoiceResponseError` class includes these constructors.

WriteOffInvoiceResponseError(errorCode, errorMessage)
Initializes the WriteOffInvoiceResponseError class that stores the error response that's associated with a request to write off a posted invoice.

WriteOffInvoiceResponseError(errorCode, errorMessage)

Initializes the WriteOffInvoiceResponseError class that stores the error response that's associated with a request to write off a posted invoice.

Signature

```
public WriteOffInvoiceResponseError(String errorCode, String errorMessage)
```

Parameters

errorCode
Type: String

Code that represents the error.

errorMessage
Type: String

Message that describes the error.

WriteOffInvoiceResponseError Properties

Learn more about the properties available with the WriteOffInvoiceResponseError class.

The `WriteOffInvoiceResponseError` class includes these properties.

errorCode
Get the error code details.

errorMessage
Get the error message details.

errorCode

Get the error code details.

Signature

```
public String errorCode {get; set;}
```

Property Value

Type: String

errorMessage

Get the error message details.

Signature

```
public String errorMessage {get; set;}
```

Property Value

Type: String

WriteOffInvoiceResponseList Class

Contains properties to store the response details of the list of invoices that are written off.

Namespace

InvoiceWriteOff on page 2811

WriteOffInvoiceResponseList Constructors

Learn more about the constructors available with the WriteOffInvoiceResponseList class.

WriteOffInvoiceResponseList Properties

Learn more about the properties available with the WriteOffInvoiceResponseList class.

WriteOffInvoiceResponseList Constructors

Learn more about the constructors available with the WriteOffInvoiceResponseList class.

The `WriteOffInvoiceResponseList` class includes these constructors.

WriteOffInvoiceResponseList(writeOffInvoiceResponseList)

Initializes the WriteOffInvoiceResponseList class that stores the response details of the list of invoices that are written off.

WriteOffInvoiceResponseList()

Initializes the WriteOffInvoiceResponseList class.

`WriteOffInvoiceResponseList(writeOffInvoiceResponseList)`

Initializes the WriteOffInvoiceResponseList class that stores the response details of the list of invoices that are written off.

Signature

```
public WriteOffInvoiceResponseList(List<InvoiceWriteOff.WriteOffInvoiceResponse>
writeOffInvoiceResponseList)
```

Parameters

`writeOffInvoiceResponseList`

Type: List<InvoiceWriteOff.WriteOffInvoiceResponse on page 2815>

Details of the invoices for which the write-off process is initiated.

`WriteOffInvoiceResponseList()`

Initializes the WriteOffInvoiceResponseList class.

Signature

```
public WriteOffInvoiceResponseList()
```

WriteOffInvoiceResponseList Properties

Learn more about the properties available with the WriteOffInvoiceResponseList class.

The `WriteOffInvoiceResponseList` class includes these properties.

writeOffInvoiceResponseList

Get the details of the invoices for which the write-off posted invoice process is initiated.

Signature

```
public List<InvoiceWriteOff.WriteOffInvoiceResponse> writeOffInvoiceResponseList {get;
set;}
```

Property Value

Type: List<InvoiceWriteOff.WriteOffInvoiceResponse on page 2815>

IssueCreditMemo Namespace

Issue credit memos from disputed invoices. Use this namespace to create and apply credit memos against invoices or invoice lines based on dispute adjustments.

You can access this namespace if Dispute Management is enabled in Billing.

The `IssueCreditMemo` namespace includes these classes.

CreditLineRequestInputRepresentations Class

Represents a single line-level credit request. Specifies the invoice line to credit, the amount to apply, and an optional description.

Namespace

IssueCreditMemo on page 2820

CreditLineRequestInputRepresentations Constructors

The `CreditLineRequestInputRepresentations` class includes these constructors.

[CreditLineRequestInputRepresentations(invoiceLineId, creditLineAmount, description)](#)
Creates a credit line request for the specified invoice line, amount, and description.

[CreditLineRequestInputRepresentations()](#)
Creates an empty credit line request.

`CreditLineRequestInputRepresentations(invoiceLineId, creditLineAmount, description)`

Creates a credit line request for the specified invoice line, amount, and description.

Signature

`public CreditLineRequestInputRepresentations(String invoiceLineId, Double creditLineAmount, String description)`

Parameters

`invoiceLineId`
Type: String

The ID of the invoice line to which the credit applies.

`creditLineAmount`
Type: Double

The monetary amount to credit for this invoice line.

`description`
Type: String

Optional description or reason for the credit line.

`CreditLineRequestInputRepresentations()`

Creates an empty credit line request.

Signature

`public CreditLineRequestInputRepresentations()`

CreditLineRequestInputRepresentations Properties

The `CreditLineRequestInputRepresentations` class includes these properties.

[creditLineAmount](#)
The monetary amount to credit for this invoice line.

[description](#)
Optional description or reason for the credit line.

invoiceLineId
The ID of the invoice line to which the credit applies.

creditLineAmount

The monetary amount to credit for this invoice line.

Signature

```
public Double creditLineAmount {get; set;}
```

Property Value

Type: Double

description

Optional description or reason for the credit line.

Signature

```
public String description {get; set;}
```

Property Value

Type: String

invoiceLineId

The ID of the invoice line to which the credit applies.

Signature

```
public String invoiceLineId {get; set;}
```

Property Value

Type: String

CreditRequestInputRepresentations Class

Represents a credit request for an invoice. Contains invoice and dispute identifiers, total credit amount, category, and line-level credit details for issuing a credit memo.

Namespace

IssueCreditMemo on page 2820

CreditRequestInputRepresentations Constructors
CreditRequestInputRepresentations Properties

CreditRequestInputRepresentations Constructors

The `CreditRequestInputRepresentations` class includes these constructors.

CreditRequestInputRepresentations(invoiceId, creditAmount, description, disputeId, category, creditLineRequestInputRepresentations)
Creates a credit request with the given invoice, amount, description, dispute, category, and line-level credit details.

CreditRequestInputRepresentations(invoiceId, creditAmount, description, creditLineRequestInputRepresentations)
Creates a credit request with the given invoice, amount, description, and line-level credit details.

CreditRequestInputRepresentations()
Creates an empty credit request.

`CreditRequestInputRepresentations(invoiceId, creditAmount, description, disputeId, category, creditLineRequestInputRepresentations)`

Creates a credit request with the given invoice, amount, description, dispute, category, and line-level credit details.

Signature

```
public CreditRequestInputRepresentations(String invoiceId, Double creditAmount, String
description, String disputeId, String category,
List<IssueCreditMemo.CreditLineRequestInputRepresentations>
creditLineRequestInputRepresentations)
```

Parameters

invoiceId
Type: String

ID of the invoice to credit.

creditAmount
Type: Double

Total credit amount to apply to the invoice.

description
Type: String

Optional description for the credit request.

disputeId
Type: String

ID of the billing dispute associated with this credit request.

category
Type: String

Category of the credit memo.

creditLineRequestInputRepresentations
Type: List<IssueCreditMemo.CreditLineRequestInputRepresentations on page 2820 >

List of line-level credit requests for this invoice.

`CreditRequestInputRepresentations(invoiceId, creditAmount, description, creditLineRequestInputRepresentations)`

Creates a credit request with the given invoice, amount, description, and line-level credit details.

Signature

`public CreditRequestInputRepresentations(String invoiceId, Double creditAmount, String description, List<IssueCreditMemo.CreditLineRequestInputRepresentations> creditLineRequestInputRepresentations)`

Parameters

`invoiceId`
Type: String

ID of the invoice to credit.

`creditAmount`
Type: Double

Total credit amount to apply to the invoice.

`description`
Type: String

Optional description for the credit request.

`creditLineRequestInputRepresentations`
Type: List<IssueCreditMemo.CreditLineRequestInputRepresentations on page 2820 >

List of line-level credit requests for this invoice.

`CreditRequestInputRepresentations()`

Creates an empty credit request.

Signature

`public CreditRequestInputRepresentations()`

CreditRequestInputRepresentations Properties

The `CreditRequestInputRepresentations` class includes these properties.

category
The credit memo category.

creditAmount
The total credit amount to apply to the invoice.

creditLineRequestInputRepresentations
List of line-level credit requests for this invoice.

description
Optional description for the credit request.

disputeId
The ID of the billing dispute associated with this credit request.

invoiceId
The ID of the invoice to credit.

category

The credit memo category.

Signature

```
public String category {get; set;}
```

Property Value

Type: String

creditAmount

The total credit amount to apply to the invoice.

Signature

```
public Double creditAmount {get; set;}
```

Property Value

Type: Double

creditLineRequestInputRepresentations

List of line-level credit requests for this invoice.

Signature

```
public List<IssueCreditMemo.CreditLineRequestInputRepresentations>
creditLineRequestInputRepresentations {get; set;}
```

Property Value

Type: List<IssueCreditMemo.CreditLineRequestInputRepresentations on page 2820 >

description

Optional description for the credit request.

Signature

```
public String description {get; set;}
```

Property Value

Type: String

disputeId

The ID of the billing dispute associated with this credit request.

Signature

```
public String disputeId {get; set;}
```

Property Value

Type: String

invoiceId

The ID of the invoice to credit.

Signature

```
public String invoiceId {get; set;}
```

Property Value

Type: String

CreditResponseOutputRepresentations Class

Represents the result of a credit memo operation. Indicates success or failure and any additional information or message.

Namespace

IssueCreditMemo on page 2820

Usage

Example

- CreditResponseOutputRepresentations Constructors
- CreditResponseOutputRepresentations Properties

CreditResponseOutputRepresentations Constructors

The `CreditResponseOutputRepresentations` class includes these constructors.

[CreditResponseOutputRepresentations(success, additionalInformation)](#)
Creates a response with the given success flag and additional information.

`CreditResponseOutputRepresentations(success, additionalInformation)`

Creates a response with the given success flag and additional information.

Signature

`public CreditResponseOutputRepresentations(Boolean success, String additionalInformation)`

Parameters

`success`
Type: Boolean

Indicates whether the credit memo is issued successfully (`true`) or not (`false`).

`additionalInformation`
Type: String

Additional information or message, such as error details or confirmation.

CreditResponseOutputRepresentations Properties

The `CreditResponseOutputRepresentations` class includes these properties.

[additionalInformation](#)
Additional information or message, such as error details or confirmation.

[success](#)
Indicates whether the credit memo is issued successfully (true) or not (false).

`additionalInformation`

Additional information or message, such as error details or confirmation.

Signature

`public String additionalInformation {get; set;}`

Property Value

Type: String

`success`

Indicates whether the credit memo is issued successfully (true) or not (false).

Signature

`public Boolean success {get; set;}`

Property Value

Type: Boolean

RulesAppln Namespace

Apply payments and credits to posted invoices by adhering to the specified rules.

The `RulesAppln` namespace includes these classes.

RulesApplicationErrorResponse Class

Contains properties to store error details that occurred during the rules application.

RulesApplicationResponse Class

Contains properties to store the response details for the rules application request.

RulesApplicationSummaryResponse Class

Contains properties to store the summary details of the rules application, including payment and credit memo counts and application statistics.

RulesApplicationErrorResponse Class

Contains properties to store error details that occurred during the rules application.

Namespace

RulesAppln on page 2828

RulesApplicationErrorResponse Constructors

Learn more about the constructors available with the RulesApplicationErrorResponse class.

RulesApplicationErrorResponse Properties

Learn more about the properties available with the RulesApplicationErrorResponse class.

RulesApplicationErrorResponse Constructors

Learn more about the constructors available with the RulesApplicationErrorResponse class.

The `RulesApplicationErrorResponse` class includes these constructors.

RulesApplicationErrorResponse(errorCode, message)

Initializes the RulesApplicationErrorResponse class that stores error details that occurred during the rules application.

RulesApplicationErrorResponse()

Initializes an empty instance of the RulesApplicationErrorResponse class.

`RulesApplicationErrorResponse(errorCode, message)`

Initializes the RulesApplicationErrorResponse class that stores error details that occurred during the rules application.

Signature

`public RulesApplicationErrorResponse(String errorCode, String message)`

```
RulesAppln.RulesApplicationErrorResponse, newinstance, [String, String],
RulesAppln.RulesApplicationErrorResponse
```

Parameters

errorCode
Type: String

Error code that identifies the type of error that occurred during the rules application.

message
Type: String

Error message that describes the error that occurred during the rules application.

RulesApplicationErrorResponse()

Initializes an empty instance of the RulesApplicationErrorResponse class.

Signature

`public RulesApplicationErrorResponse()`

```
RulesAppln.RulesApplicationErrorResponse, newinstance, [],
RulesAppln.RulesApplicationErrorResponse
```

RulesApplicationErrorResponse Properties

Learn more about the properties available with the RulesApplicationErrorResponse class.

The `RulesApplicationErrorResponse` class includes these properties.

errorCode
Get the error code that identifies the type of error that occurred during the rules application.

message
Get the error message that describes the error that occurred during the rules application.

errorCode

Get the error code that identifies the type of error that occurred during the rules application.

Signature

`public String errorCode {get; set;}`

```
RulesAppln.RulesApplicationErrorResponse, errorCode
```

Property Value

Type: String

message

Get the error message that describes the error that occurred during the rules application.

Signature

```
public String message {get; set;}
```

```
RulesAppln.RulesApplicationErrorResponse, message
```

Property Value

Type: String

RulesApplicationResponse Class

Contains properties to store the response details for the rules application request.

Namespace

RulesAppln on page 2828

RulesApplicationResponse Constructors

Learn more about the constructors available with the RulesApplicationResponse class.

The `RulesApplicationResponse` class includes these constructors.

RulesApplicationResponse(isSuccess, appliedRules, rulesApplicationSummary, errors)

Initializes the RulesApplicationResponse class that stores the response details for the rules application request.

Signature

`public RulesApplicationResponse(Boolean isSuccess, List<String> appliedRules, RulesAppln.RulesApplicationSummaryResponse rulesApplicationSummary, List<RulesAppln.RulesApplicationErrorResponse> errors)`

```
RulesAppln.RulesApplicationResponse, newinstance, [Boolean, List<String>,
RulesAppln.RulesApplicationSummaryResponse, List<RulesAppln.RulesApplicationErrorResponse>],
 RulesAppln.RulesApplicationResponse
```

Parameters

isSuccess
Type: Boolean

Indicates whether the rules are applied (`true`) or not (`false`).

appliedRules
Type: List<String>

Comma-delimited list of applied rules.

rulesApplicationSummary
Type: RulesAppln.RulesApplicationSummaryResponse on page 2833

Details of the rules application that includes these details.

- Payment and credit memo count for the account
- Payment and credit memo count that's applied to invoices or invoice lines
- Details of rules

errors
Type: List<RulesAppln.RulesApplicationErrorResponse on page 2828 >

List of error responses that occurred during the rules application.

RulesApplicationResponse()

Initializes an empty instance of the RulesApplicationResponse class.

Signature

`public RulesApplicationResponse()`

```
RulesAppln.RulesApplicationResponse, newinstance, [], RulesAppln.RulesApplicationResponse
```

RulesApplicationResponse Properties

Learn more about the properties available with the RulesApplicationResponse class.

The `RulesApplicationResponse` class includes these properties.

appliedRules
Get the comma-delimited list of rules that were applied.

[errors](#)

Get the list of error responses that occurred during the rules application.

[isSuccess](#)

Get the boolean value that indicates whether the rules were applied (true) or not (false).

[rulesApplicationSummary](#)

Get the details of the rules application that includes payment and credit memo count for the account, payment and credit memo count that's applied to invoices or invoice lines, and whether the exact amount rule is applied.

appliedRules

Get the comma-delimited list of rules that were applied.

Signature

`public List<String> appliedRules {get; set;}`

```
RulesAppln.RulesApplicationResponse, appliedRules
```

Property Value

Type: List<String>

errors

Get the list of error responses that occurred during the rules application.

Signature

`public List<RulesAppln.RulesApplicationErrorResponse> errors {get; set;}`

```
RulesAppln.RulesApplicationResponse, errors
```

Property Value

Type: List<RulesAppln.RulesApplicationErrorResponse on page 2828 >

isSuccess

Get the boolean value that indicates whether the rules were applied (true) or not (false).

Signature

`public Boolean isSuccess {get; set;}`

```
RulesAppln.RulesApplicationResponse, isSuccess
```

Property Value

Type: Boolean

`rulesApplicationSummary`

Get the details of the rules application that includes payment and credit memo count for the account, payment and credit memo count that's applied to invoices or invoice lines, and whether the exact amount rule is applied.

Signature

```
public RulesAppln.RulesApplicationSummaryResponse rulesApplicationSummary {get; set;}
```

```
RulesAppln.RulesApplicationResponse, rulesApplicationSummary
```

Property Value

Type: RulesAppln.RulesApplicationSummaryResponse on page 2833

RulesApplicationSummaryResponse Class

Contains properties to store the summary details of the rules application, including payment and credit memo counts and application statistics.

Namespace

RulesAppln on page 2828

RulesApplicationSummaryResponse Constructors

Learn more about the constructors available with the RulesApplicationSummaryResponse class.

The `RulesApplicationSummaryResponse` class includes these constructors.

`RulesApplicationSummaryResponse(fetchedPaymentsCount, fetchedCreditMemosCount, totalPaymentApplications, totalCreditMemoApplications, areAllInvoicesConsidered)`

Initializes the RulesApplicationSummaryResponse class that stores the summary details of the rules application.

Signature

`public RulesApplicationSummaryResponse(Integer fetchedPaymentsCount, Integer fetchedCreditMemosCount, Integer totalPaymentApplications, Integer totalCreditMemoApplications, Boolean areAllInvoicesConsidered)`

```
RulesAppln.RulesApplicationSummaryResponse, newinstance, [Integer, Integer, Integer,
Integer, Boolean], RulesAppln.RulesApplicationSummaryResponse
```

Parameters

fetchedPaymentsCount
    Type: Integer

    Number of payment records retrieved for an account.

fetchedCreditMemosCount
    Type: Integer

    Number of credit memo or credit memo line records retrieved for an account.

totalPaymentApplications
    Type: Integer

    Number of payments that are applied to invoices and invoice lines.

totalCreditMemoApplications
    Type: Integer

    Number of credit memos that are applied to the invoice or invoice lines.

areAllInvoicesConsidered
    Type: Boolean

    Indicates whether all invoices are considered (`true`) or not (`false`).

RulesApplicationSummaryResponse()

Initializes an empty instance of the RulesApplicationSummaryResponse class.

Signature

`public RulesApplicationSummaryResponse()`

```
RulesAppln.RulesApplicationSummaryResponse, newinstance, [],
RulesAppln.RulesApplicationSummaryResponse
```

RulesApplicationSummaryResponse Properties

Learn more about the properties available with the RulesApplicationSummaryResponse class.

The `RulesApplicationSummaryResponse` class includes these properties.

areAllInvoicesConsidered
    Indicates whether all invoices are considered (true) or not (false).

fetchedCreditMemosCount
    Get the number of credit memo or credit memo line records retrieved for an account.

fetchedPaymentsCount

Get the number of payment records retrieved for an account.

totalCreditMemoApplications

Get the number of credit memos that are applied to the invoice or invoice lines.

totalPaymentApplications

Get the number of payments that are applied to invoices and invoice lines.

areAllInvoicesConsidered

Indicates whether all invoices are considered (true) or not (false).

Signature

`public Boolean areAllInvoicesConsidered {get; set;}`

```
RulesAppln.RulesApplicationSummaryResponse, areAllInvoicesConsidered
```

Property Value

Type: Boolean

fetchedCreditMemosCount

Get the number of credit memo or credit memo line records retrieved for an account.

Signature

`public Integer fetchedCreditMemosCount {get; set;}`

```
RulesAppln.RulesApplicationSummaryResponse, fetchedCreditMemosCount
```

Property Value

Type: Integer

fetchedPaymentsCount

Get the number of payment records retrieved for an account.

Signature

`public Integer fetchedPaymentsCount {get; set;}`

```
RulesAppln.RulesApplicationSummaryResponse, fetchedPaymentsCount
```

Property Value

Type: Integer

`totalCreditMemoApplications`

Get the number of credit memos that are applied to the invoice or invoice lines.

Signature

`public Integer totalCreditMemoApplications {get; set;}`

```
RulesAppln.RulesApplicationSummaryResponse, totalCreditMemoApplications
```

Property Value

Type: Integer

`totalPaymentApplications`

Get the number of payments that are applied to invoices and invoice lines.

Signature

`public Integer totalPaymentApplications {get; set;}`

```
RulesAppln.RulesApplicationSummaryResponse, totalPaymentApplications
```

Property Value

Type: Integer

TaxEngineAdapter Interface

Retrieves and evaluates the details from a tax engine to define tax details.

You can extend the TaxEngineAdapter interface to define a custom tax adapter based on your requirements. Use the custom tax adapter with Billing services to implement standard tax.

Create a custom object and associated fields to store tax details, such as tax rate for a country. For example, create a custom object named CountryTaxRate with Country_Code and Tax_Rate fields. Create records to define the details for these fields.

TaxEngineAdapter Methods
Learn more about the available methods with the `TaxEngineAdapter` class.

TaxEngineAdapter Example Implementation
Refer to the example implementation of the `TaxEngineAdapter` interface to accept information from a tax engine and evaluate the information to define tax details.

Tax Mappings for Invoices and Credits

You can extend and customize the existing tax interface by using custom metadata types and tax mappings. These customizations help you with unique business requirements such as the inclusion of specific data for accurate calculations and audits.

SEE ALSO:

Salesforce Help: Tax Calculation for Invoices

Billing Business APIs: Tax Calculation (POST)

Billing Standard Objects: TaxEngineProvider, TaxEngine, TaxPolicy, and TaxTreatment

Tax Engine Reference Gateway Adapter

TaxEngineAdapter Methods

Learn more about the available methods with the `TaxEngineAdapter` class.

The `TaxEngineAdapter` class includes these methods.

processRequest(requestType)

The `processRequest` method takes an instance of `TaxEngineContext` class and returns a response with the calculated tax details through the `TaxDetailsResponse` class or an error response through the `ErrorResponse` class.

`processRequest(requestType)`

The `processRequest` method takes an instance of `TaxEngineContext` class and returns a response with the calculated tax details through the `TaxDetailsResponse` class or an error response through the `ErrorResponse` class.

Signature

```
global commercetax.TaxEngineResponse processRequest(commercetax.TaxEngineContext var1)
```

Parameters

`var1`

Type: TaxEngineContext

Wrapper class that stores information about the type of a tax calculation request.

Return Value

Type: TaxEngineResponse

Generic interface representing a response from a tax engine.

TaxEngineAdapter Example Implementation

Refer to the example implementation of the `TaxEngineAdapter` interface to accept information from a tax engine and evaluate the information to define tax details.

Namespace

See commercetax namespace to view the list of available classes.

Usage

The `TaxEngineAdapter` interface accepts information from the tax engine through the `TaxEngineContext` class. The interface evaluates the information to define tax in the response with details, such as tax amount and addresses. The response is used to update and create entities in the Salesforce org.

Considerations

From Winter '26, the available state and country values from orgs that are configured with State and Country/Territory Picklists are also supported.

Example: Use these steps to build a sample tax adapter implementation. Each tax adapter implementation varies based on your implementation requirements. Customize this example to suit your business requirements.

See Tax Engine Reference Gateway Adapter for reference implementations of a tax engine adapter.

- Get the tax rates from the created custom object. See TaxEngineAdapter Interface on page 2836.
- The custom adapter class implements the `TaxEngineAdapter` interface. The `processRequest` method takes an instance of `TaxEngineContext` class and returns a response with the calculated tax details through the `TaxDetailsResponse` class or an error response through the `ErrorResponse` class.

```
global virtual class StandardTaxAdapter implements commercetax.TaxEngineAdapter {
    global commercetax.TaxEngineResponse processRequest(commercetax.TaxEngineContext
 taxEngineContext) {
        commercetax.RequestType requestType = taxEngineContext.getRequestType();

        // Map of tax field name to the value of the corresponding entity field
(EntityFieldName) at header level
        Map<String, Object> customTaxAttributeHeaderLevelMap = new Map<String,
Object>();
        // Map of tax field name to the value of the corresponding entity field
(EntityFieldName) at line level
        Map<String, Object> customTaxAttributeLineLevelMap = new Map<String,
Object>();
        // Map of tax field name to the value of the corresponding entity field
(EntityFieldName) at line tax level
        Map<String, Object> customTaxAttributeLineTaxLevelMap = new Map<String,
Object>();

       commercetax.CustomTaxAttributesResponse customTaxAttributeHeaderLevelResponse
 = new commercetax.CustomTaxAttributesResponse();
        commercetax.CustomTaxAttributesResponse customTaxAttributeLineLevelResponse
 = new commercetax.CustomTaxAttributesResponse();
       commercetax.CustomTaxAttributesResponse customTaxAttributeLineTaxLevelResponse
 = new commercetax.CustomTaxAttributesResponse();


customTaxAttributeHeaderLevelResponse.setData(customTaxAttributeHeaderLevelMap);
       customTaxAttributeLineLevelResponse.setData(customTaxAttributeLineLevelMap);
```

```
customTaxAttributeLineTaxLevelResponse.setData(customTaxAttributeLineTaxLevelMap);

        commercetax.CalculateTaxRequest request = (commercetax.CalculateTaxRequest)
 taxEngineContext.getRequest();
        if (request.documentCode == null) {
            return new
commercetax.ErrorResponse(commercetax.resultcode.TaxEngineError, '404', 'documentCode
 is mandatory');
        }
        if (request.documentCode == 'TaxEngineError') {
            return new
commercetax.ErrorResponse(commercetax.resultcode.TaxEngineError, '504', 'documentCode
 - not supported');
        }
        if (request.documentCode == 'simulateUnhandledExceptionInAdapter') {
            Integer foo = 5 / 0;
        }
        if (request.documentCode == 'simulateValidationFailureInAdapter') {
            return new commercetax.ErrorResponse(
                commercetax.resultcode.TaxEngineError,
                '400',
                'validations for documentCode failed in adapter'
            );
        }
        if (request.documentCode == 'simulateMalformedErrorInAdapter') {
            return new
commercetax.ErrorResponse(commercetax.resultcode.TaxEngineError, null, 'malformed
adapter error response');
        }
        if (request.documentCode == 'simulateTaxEngineProcessFailure') {
            return new
commercetax.ErrorResponse(commercetax.resultcode.TaxEngineError, '500', 'Tax Engine
 couldnt process your request');
        }
        if (request.documentCode == 'simulateReferenceDocumentCodeMissing') {
            return new
commercetax.ErrorResponse(commercetax.resultcode.ReferenceDocumentCodeMissing, '400',
 'Ref Document Code not found');
        }

        if (requestType == commercetax.RequestType.CalculateTax) {
            commercetax.calculatetaxtype type = request.taxtype;
            String docCode = '';
            if (request.DocumentCode == 'simulateEmptyDocumentCode')
                docCode = '';
            else if (request.DocumentCode != null)
                docCode = request.DocumentCode;
            else if (request.ReferenceEntityId != null)
                docCode = request.ReferenceEntityId;
            else
                docCode = String.valueOf(getRandomInteger(0, 2147483647));
            commercetax.CalculateTaxResponse response = new
commercetax.CalculateTaxResponse();
```

```
            response.setCustomTaxAttributes(customTaxAttributeHeaderLevelResponse);

            if (request.isCommit == true) {
                response.setStatus(commercetax.TaxTransactionStatus.Committed);
            } else {
                response.setStatus(commercetax.TaxTransactionStatus.Uncommitted);
            }

            if (request.shouldVoidTax) {
                if
(request.documentCode.startsWith('simulateCalculateWhenRefDocIsLocked')) {
                    response.setDocumentCode(docCode);
                 response.setReferenceDocumentCode(request.referenceDocumentCode);

                    if (request.taxTransactionType == null) {

response.setTaxTransactionType(commercetax.TaxTransactionType.Debit);
                    } else {
                        response.setTaxTransactionType(request.taxTransactionType);

                    }
                } else if
(request.documentCode.startsWith('simulateRandomRefDocumentCode')) {
                    response.setDocumentCode(docCode);

response.setReferenceDocumentCode('simulateRandomRefDocumentCode2');
                    response.setTaxTransactionType(request.taxTransactionType);
                } else if (request.documentCode.startsWith('simulateRandomCode'))
{
                    response.setDocumentCode('simulateRandomCode2');
                    response.setReferenceDocumentCode(null);

response.setTaxTransactionType(commercetax.TaxTransactionType.Void);
                } else {
                    response.setDocumentCode(request.referenceDocumentCode);
                    response.setReferenceDocumentCode(null);

response.setTaxTransactionType(commercetax.TaxTransactionType.Void);
                }
            } else {
                response.setDocumentCode(docCode);
                response.setReferenceDocumentCode(request.referenceDocumentCode);

                if (request.taxTransactionType == null) {

response.setTaxTransactionType(commercetax.TaxTransactionType.Debit);
                } else {
                    response.setTaxTransactionType(request.taxTransactionType);
                }
            }

            response.setTaxType(type);
            response.setStatusDescription('statusDescription');
            if (request.sellerDetails.code == 'testSellerCode') {
```

```
                response.setDescription('SellerCode fetched from TaxEngine entity');

            } else {
                response.setDescription('description');
            }
            response.setEffectiveDate(system.now());
            if (request.transactionDate == null) {
                response.setTransactionDate(system.now());
            } else {
                response.setTransactionDate(request.transactionDate);
            }

            if (request.currencyIsoCode == null || request.currencyIsoCode == '')
{
                response.setCurrencyIsoCode('USD');
            } else {
                response.setCurrencyIsoCode(request.currencyIsoCode);
            }
            response.setReferenceEntityId(request.ReferenceEntityId);
            Double totalTax = 0.0;
            Double totalAmount = 0.0;
            Map<String, Double> countryTaxRateMap = getTaxes(request.lineItems);
            List<commercetax.LineItemResponse> lineItemResponses = new
List<commercetax.LineItemResponse>();
            for (Commercetax.TaxLineItemRequest lineItem : request.lineItems) {
                String country = getCountryFromLineItem(lineItem);
                if (country == null) {
                    return new commercetax.ErrorResponse(
                        commercetax.resultcode.TaxEngineError,
                        '400',
                        'Country is mandatory for each line item'
                    );
                }

                Double taxRate = countryTaxRateMap.get(country);
                if (taxRate == null) {
                    return new commercetax.ErrorResponse(
                        commercetax.resultcode.TaxEngineError,
                        '404',
                        'No tax rate found for the specified country: ' + country
                    );
                }
                commercetax.AddressesResponse addressesRes = new
commercetax.AddressesResponse();
                if (request.DocumentCode == 'SetsNullForResponseWithoutException')
 {
                    addressesRes.setShipFrom(null);
                    addressesRes.setShipTO(null);
                    addressesRes.setSoldTo(null);
                } else {
                    setAddresses(addressesRes, lineItem);
                    //System.debug('Line item addresses: ' + addressesRes);
                }
                //System.debug('ProductSKU: ' + lineItem.productSKU);
```

```
                //System.debug('ReferenceDocumentCode: ' +
lineItem.referenceDocumentCode);
                commercetax.LineItemResponse lineItemResponse = new
commercetax.LineItemResponse();

lineItemResponse.setCustomTaxAttributes(customTaxAttributeLineLevelResponse);
                Double totalLineTax = 0;
                List<commercetax.TaxDetailsResponse> taxDetailsResponses = new
List<commercetax.TaxDetailsResponse>();
                for (integer i = 0; i < 1; i++) {
                    Integer rate = 1;
                    Double taxableAmount = lineItem.amount;
                    commercetax.TaxDetailsResponse taxDetailsResponse = new
commercetax.TaxDetailsResponse();
                    taxDetailsResponse.setRate(Double.valueOf(rate));
                    taxDetailsResponse.setTaxableAmount(taxableAmount);
                    Double tax = taxableAmount * rate;
                    totalLineTax += tax;
                    taxDetailsResponse.setTax(taxableAmount * rate);
                    taxDetailsResponse.setExemptAmount(0);
                    taxDetailsResponse.setExemptReason('exemptReason');
                    taxDetailsResponse.setTaxRegionId('taxRegionId');
                    taxDetailsResponse.setTaxId(String.valueOf(getRandomInteger(0,
 2323233)));
                    taxDetailsResponse.setSerCode('serCode');

taxDetailsResponse.setCustomTaxAttributes(customTaxAttributeLineTaxLevelResponse);
                   taxDetailsResponse.setTaxAuthorityTypeId('taxAuthorityTypeId');

                 if (request.DocumentCode == 'SetsNullForResponseWithoutException')
 {
                        taxDetailsResponse.setImposition(null);
                    } else {
                        commercetax.ImpositionResponse imposition = new
commercetax.ImpositionResponse();
                        imposition.setSubType('subtype');
                        imposition.setType('type');
                        taxDetailsResponse.setImposition(imposition);
                    }

                 if (request.DocumentCode == 'SetsNullForResponseWithoutException')
 {
                        taxDetailsResponse.setJurisdiction(null);
                    } else {
                        commercetax.JurisdictionResponse jurisdiction = new
commercetax.JurisdictionResponse();
                        jurisdiction.setCountry('country');
                        jurisdiction.setRegion('region');
                        jurisdiction.setName('name');
                        jurisdiction.setStateAssignedNumber('stateAssignedNo');
                        jurisdiction.setId('id');
                        jurisdiction.setLevel('level');
                        taxDetailsResponse.setJurisdiction(jurisdiction);
                    }
```

```
                    taxDetailsResponses.add(taxDetailsResponse);
                }
                lineItemResponse.setTaxes(taxDetailsResponses);
                totalTax += totalLineTax;
                totalAmount += lineItem.amount;

                if (request.DocumentCode == 'SetsNullForResponseWithException') {
                    lineItemResponse.setAmountDetails(null);
                } else {
                    commercetax.AmountDetailsResponse amountResponse = new
commercetax.AmountDetailsResponse();
                    amountResponse.setTotalAmountWithTax(totalTax + totalAmount);
                    amountResponse.setExemptAmount(0);
                    amountResponse.setTotalAmount(totalAmount);
                    amountResponse.setTaxAmount(totalTax);
                    lineItemResponse.setAmountDetails(amountResponse);
                }
                lineItemResponse.setEffectiveDate(system.now());
                lineItemResponse.setTaxCode(lineItem.taxCode);
                lineItemResponse.setProductCode(lineItem.ProductCode);
                lineItemResponse.setLineNumber(lineItem.linenumber);
                lineItemResponse.setIsTaxable(true);
                lineItemResponse.setQuantity(lineItem.quantity);
                if (request.DocumentCode == 'SetsNullForResponseWithoutException')
 {
                    lineItemResponse.setAddresses(null);
                } else {
                    lineItemResponse.setAddresses(addressesRes);
                }
                lineItemResponses.add(lineItemResponse);
            }
            if (request.DocumentCode == 'SetsNullForResponseWithException') {
                lineItemResponses.add(null);
            }
            if (request.documentCode == 'nolines') {
                // logic to skip adding lines to response
            } else {
                response.setLineItems(lineItemResponses);
            }
            if (request.DocumentCode == 'SetsNullForResponseWithException') {
                response.setAmountDetails(null);
            } else {
                commercetax.AmountDetailsResponse headerAmountResponse = new
commercetax.AmountDetailsResponse();
               headerAmountResponse.setTotalAmountWithTax(totalTax + totalAmount);

                headerAmountResponse.setExemptAmount(0);
                headerAmountResponse.setTotalAmount(totalAmount);
                headerAmountResponse.setTaxAmount(totalTax);
                response.setAmountDetails(headerAmountResponse);
            }
            commercetax.AddressesResponse addressesRes = new
commercetax.AddressesResponse();
```

```
            commercetax.AddressResponse addRes = new commercetax.AddressResponse();

             addRes.setLocationCode('street, city, state, country, postalCode');
             addressesRes.setShipFrom(addRes);
             addressesRes.setShipTO(addRes);
             addressesRes.setSoldTo(addRes);
             response.setAddresses(addressesRes);
             return response;
        } else
             return null;
    }

    public static Integer getRandomInteger(Integer min, Integer max) {
        return min + (Integer.valueOf(Math.random()) * (max - min));
    }

    // Method to get the tax rates for the line items based on the address country
 codes
    private Map<String, Double> getTaxes(List<commercetax.TaxLineItemRequest>
lineItems) {
        Set<String> countryCodes = new Set<String>();
        // Collecting all unique the country codes
        for (commercetax.TaxLineItemRequest lineItem : lineItems) {
            String country = getCountryFromLineItem(lineItem);
            if (country != null) {
                countryCodes.add(country);
            }
        }
        // Query tax rates for the unique country codes and store it in a map
        Map<String, Double> countryTaxRateMap = new Map<String, Double>();
        if (!countryCodes.isEmpty()) {
            // Query the tax rate from the custom object based on the country code
            List<CountryTaxRate__c> taxRates = [
                SELECT Country_Code__c, Tax_Rate__c
                FROM CountryTaxRate__c
                WHERE Country_Code__c IN :countryCodes
            ];
            for (CountryTaxRate__c taxRate : taxRates) {
              countryTaxRateMap.put(taxRate.Country_Code__c, taxRate.Tax_Rate__c);

            }
        }
        return countryTaxRateMap;
    }
    // Method to retrieve the country for the line item
    private String getCountryFromLineItem(commercetax.TaxLineItemRequest lineItem)
 {
        if (lineItem.addresses != null) {
           commercetax.TaxAddressRequest addressRequest = lineItem.addresses.shipTo;

            if (addressRequest != null && addressRequest.country != null) {
                return addressRequest.country;
            }
        }
```

```
        return null;
    }

    // Method to set addresses in the line item response
    private void setAddresses(commercetax.AddressesResponse addressesRes,
Commercetax.TaxLineItemRequest lineItem) {
        commercetax.LineTaxAddressesRequest addressesReq = lineItem.addresses;
        if (addressesReq == null) {
            return;
        }
        addressesRes.setShipFrom(getShipFromAddressRes(addressesReq));
        addressesRes.setShipTo(getShipToAddressRes(addressesReq));
        addressesRes.setSoldTo(getSoldToAddressRes(addressesReq));
    }

    private commercetax.AddressResponse
getShipFromAddressRes(commercetax.LineTaxAddressesRequest addressesReq) {
        return getAddressRes(addressesReq.shipFrom);
    }

    private commercetax.AddressResponse
getShipToAddressRes(commercetax.LineTaxAddressesRequest addressesReq) {
        return getAddressRes(addressesReq.shipTo);
    }

    private commercetax.AddressResponse
getSoldToAddressRes(commercetax.LineTaxAddressesRequest addressesReq) {
        if (addressesReq.soldTo != null) {
            return getAddressRes(addressesReq.soldTo);
        } else {
            return getAddressRes(addressesReq.billTo);
        }
    }

   private commercetax.AddressResponse getAddressRes(commercetax.TaxAddressRequest
 addressReq) {
        if (addressReq == null) {
            return null;
        }

       commercetax.AddressResponse addressRes = new commercetax.AddressResponse();


        List<String> addressEle = new List<String>();
        if (addressReq.street != null)
            addressEle.add(addressReq.street);
        if (addressReq.city != null)
            addressEle.add(addressReq.city);
        if (addressReq.state != null)
            addressEle.add(addressReq.state);
        if (addressReq.country != null)
            addressEle.add(addressReq.country);
        if (addressReq.postalCode != null)
            addressEle.add(addressReq.postalCode);
```

```
        addressRes.setLocationCode(String.join(addressEle, ', '));
        return addressRes;
    }
}
```

- Select `StandardTaxAdapter` as the provider when you create a tax engine record in your org.

Tax Mappings for Invoices and Credits

You can extend and customize the existing tax interface by using custom metadata types and tax mappings. These customizations help you with unique business requirements such as the inclusion of specific data for accurate calculations and audits.

Here are some prerequisites before you work with the tax mappings for invoices and credits.

- See custom metadata types to specify all your tax mapping definitions.
- See details on extension of your tax contract with custom fields.
- Tax callout extensions are supported for the Invoice, Invoice Line, Invoice Line Tax, Credit Memo, Credit Memo Line, and Credit Memo Line Tax objects.

Request Mappings for Header Attributes

This table defines the request mappings between the header attributes of a tax callout and fields of applicable objects.

Header Attributes	Invoice Mapping	Negative Invoice Mapping	Credit Mapping
currencyIsoCode	Invoice.CurrencyISOCode	Invoice.CurrencyISOCode	CreditMemo.CurrencyISOCode
isCommit	If status is `DRAFT` then this value is `False`. If status is `POSTED`, then this value is `True`.	If status is `DRAFT` then this value is `False`. If status is POSTED, then this value is `True`.	This value is `True` until Summer '25. From Winter '26, if status is `DRAFT`, then this value is `False`. If status is `POSTED`, then this value is `True`.
referenceEntityId	Invoice.ID	Invoice.ID	CreditMemo.ID
taxEngineId	TaxTreatment.TaxEngine.ID	TaxTreatment.TaxEngine.ID	TaxTreatment.TaxEngine.ID
transactionDate	SystemDate	SystemDate	SystemDate
sellerDetails			
code	TaxEngine.SellerCode	TaxEngine.SellerCode	TaxEngine.SellerCode
customerDetails			
accountId	Invoice.BillingAccount.ID	Invoice.BillingAccount.ID	CreditMemo.BillingAccount.ID
code	NULL	NULL	NULL
exemptionNo	Invoice.BillingAccount.TaxExemptionNumber. Available from Spring '26.	Invoice.BillingAccount.TaxExemptionNumber Available from Spring '26.	CreditMemo.BillingAccount.TaxExemptionNumber Available from Spring '26.
exemptionReason	NULL	NULL	NULL

Header Attributes	Invoice Mapping	Negative Invoice Mapping	Credit Mapping
exemptionStatus	Invoice.BillingAccount.TaxExemptionStatus Available from Spring '26.	Invoice.BillingAccount.TaxExemptionStatus Available from Spring '26.	CreditMemo.BillingAccount.TaxExemptionStatus Available from Spring '26.
exemptionExpirationDate	Invoice.BillingAccount.TaxExemptionExpirationDate Available from Spring '26.	Invoice.BillingAccount.TaxExemptionExpirationDate Available from Spring '26.	CreditMemo.BillingAccount.TaxExemptionExpirationDate Available from Spring '26.
deliveryTerms	Invoice.BillingAccount.DeliveryTerms Available from Spring '26.	Invoice.BillingAccount.DeliveryTerms Available from Spring '26.	CreditMemo.BillingAccount.DeliveryTerms Available from Spring '26.
billingProfileId	Invoice.BillingAccount.ID Available from Spring '26.	Invoice.BillingAccount.ID Available from Spring '26.	CreditMemo.BillingAccount.ID Available from Spring '26.
additionalTaxIdentificationDetails	Invoice.BillingAccount.AddTaxIdentificationDetails Available from Spring '26.	Invoice.BillingAccount.AddTaxIdentificationDetails Available from Spring '26.	CreditMemo.BillingAccount.AddTaxIdentificationDetails Available from Spring '26.
taxIdentificationNumber	Invoice.BillingAccount.TaxIdentificationNumber Available from Spring '26.	Invoice.BillingAccount.TaxIdentificationNumber Available from Spring '26.	CreditMemo.BillingAccount.TaxIdentificationNumber Available from Spring '26.
taxType	If status is `DRAFT`, then this value is `Estimated`. If status is `POSTED`, then this value is `Actual`.	If status is `DRAFT`, then this value is `Estimated`. If status is `POSTED`, then this value is `Actual`.	This value is `Actual` until Summer '25. From Winter '26, if status is `DRAFT`, then this value is `Estimated`. If status is POSTED, then this value is `Actual`.
taxTransactionType	Debit	Credit	Credit
effectiveDate	invoice.InvoiceDate	If you're creating from negative lines, then this value is the original `Invoice.InvoiceDate` value.	If you're using standalone credit memo, then this value is the `CreditMemo.CreditDate` value.
addresses			
billTo	NULL	NULL	NULL
shipTo	NULL	NULL	NULL
shipFrom	NULL	NULL	NULL
soldTo	NULL	NULL	NULL
taxEngineAddress	TaxEngine.Address	TaxEngine.Address	TaxEngine.Address
referenceDocumentCode	NULL	If you're converting from negative lines, then this value is the original document code `Invoice.ID & "Debit" & TaxEngine.ID`.	If you're using standalone credit memo, then this value is NULL. Otherwise, this value is the `documentCode` value of the original invoice.

Header Attributes	Invoice Mapping	Negative Invoice Mapping	Credit Mapping
description	Invoice.Description	Invoice.Description	Invoice.Description
documentCode	`Invoice.ID & "_Debit_" & TaxEngine.ID`	`Invoice.ID & "_Credit_" & TaxEngine.ID`	`CreditMemo.ID & "_Credit_" & TaxEngine.ID`
shouldVoid	FALSE	From Winter '26, if the request is to void a posted invoice, then this value is TRUE. Else, it is FALSE.	FALSE
lineItems	Refer to the next line attributes section.	Refer to the next line attributes section.	Refer to the next line attributes section.

Request Mappings for Line Attributes

This table defines the request mappings between the line attributes of a tax callout and fields of applicable objects.

Line Attributes	Invoice Mapping	Negative Invoices	Credit Mapping
taxCode	TaxTreatment.TaxCode	TaxTreatment.TaxCode	TaxTreatment.TaxCode
productCode	TaxTreatment.ProductCode	TaxTreatment.ProductCode	TaxTreatment.ProductCode
productId	InvoiceLine.ProductId	InvoiceLine.ProductId	ProductId
amount	InvoiceLine.ChargeAmount	InvoiceLine.ChargeAmount	CreditMemoLine.ChargeAmount
effectiveDate	InvoiceLine.Invoice.InvoiceDate	If you're creating from negative lines, then this value is the original `Invoice.InvoiceDate` value.	CreditMemoLine.CreditMemo.CreditDate
lineNumber	InvoiceLine.ID	InvoiceLine.ID	CreditMemoLine.ID
description	InvoiceLine.Description	InvoiceLine.Description	CreditMemoLine.Description
quantity	InvoiceLine.Quantity	InvoiceLine.Quantity	NULL
addresses			
billTo	InvoiceLine.BillingAddress.Address	InvoiceLine.BillingAddress.Address	CreditMemoLine.BillingAddress.Address
shipTo	InvoiceLine.ShippingAddress.Address	InvoiceLine.ShippingAddress.Address	CreditMemoLine.ShippingAddress.Address
shipFrom	InvoiceLine.ShippingFrom.Address	InvoiceLine.ShippingFrom.Address	CreditMemoLine.ShippingFrom.Address
soldTo	NULL	NULL	NULL
productsku	InvoiceLine.Product.productCode	InvoiceLine.Product.productCode	CreditMemoLine.Product.productCode
referenceDocumentCode	NULL	If you're converting from negative lines, then this value is	If you're using standalone credit memo, then this value is NULL.

Line Attributes	Invoice Mapping	Negative Invoices	Credit Mapping
		the original `Document Code - "Invoice.ID" & "Debit" & TaxEngine.ID`.	Otherwise, this value is the `documentCode` value of the original invoice.

Response Mappings for Header Attributes

This table defines the response mappings between the header attributes of a tax callout and fields of applicable objects. This response structure is used to create the InvoiceLineTax records.

Header Attributes	Invoice Mapping	Negative Invoice Mapping	Credit Mapping
currencyIsoCode	Invoice.CurrencyISOCode	Invoice.CurrencyISOCode	CreditMemo.CurrencyISOCode
isCommit	Not persisted.	Not persisted.	Not persisted.
referenceEntityId	Invoice.ID	Invoice.ID	CreditMemo.ID
taxEngineId	TaxTreatment.TaxEngine.ID	TaxTreatment.TaxEngine.ID	TaxTreatment.TaxEngine.ID
transactionDate	SystemDate	SystemDate	SystemDate
sellerDetails			
code	TaxEngine.SellerCode	TaxEngine.SellerCode	TaxEngine.SellerCode
customerDetails			
accountId	Invoice.BillingAccount.ID	Invoice.BillingAccount.ID	Invoice.BillingAccount.ID
code	Not persisted.	Not persisted.	Not persisted.
exemptionNo	Not persisted.	Not persisted.	Not persisted.
exemptionReason	Not persisted.	Not persisted.	Not persisted.
taxType	If status is `DRAFT`, then this value is `Estimated`. If status is `POSTED`, then this value is `Actual`.	If status is `DRAFT`, then this value is `Estimated`. If status is `POSTED`, then this value is `Actual`.	Actual
taxTransactionType	Debit	Credit	Credit
effectiveDate	invoice.InvoiceDate	If you're creating from negative lines, then this value is the original `Invoice.InvoiceDate` value.	If you're using standalone credit memo, then this value is the `CreditMemo.CreditDate` value. If you're creating from negative lines, then this value is the original `Invoice.InvoiceDate` value.
addresses			

Header Attributes	Invoice Mapping	Negative Invoice Mapping	Credit Mapping
billTo	Not persisted.	Not persisted.	Not persisted.
shipTo	Not persisted.	Not persisted.	Not persisted.
shipFrom	Not persisted.	Not persisted.	Not persisted.
soldTo	Not persisted.	Not persisted.	Not persisted.
taxEngineAddress	Not persisted.	Not persisted.	Not persisted.
referenceDocumentCode	Not persisted.	Not persisted.	Not persisted.
description	Not persisted.	Not persisted.	Not persisted.
documentCode	Invoice.ID & "_Debit_" & TaxEngine.ID	Invoice.ID & "_Credit_" & TaxEngine.ID	CreditMemo.ID & "_Credit_" & TaxEngine.ID
status	Committed		
taxEngineLogs			
resultCode	TaxEngineInteractionLog.ResultCode	TaxEngineInteractionLog.ResultCode	TaxEngineInteractionLog.ResultCode
createddate	Not persisted.	Not persisted.	Not persisted.
Id	Not persisted.	Not persisted.	Not persisted.
transactionDate	Not persisted.	Not persisted.	Not persisted.
amountDetails			
exemptAmount	Not persisted.	Not persisted.	Not persisted.
taxAmount	If header tax is enabled, then this value is `Invoice.TotalTaxesCapturedAtHeader`.	If header tax is enabled, then this value is `Invoice.TotalTaxesCapturedAtHeader`.	If header tax is enabled, then this value is `CreditMemo.TotalTaxesCapturedAtHeader`.
totalAmount	Not persisted.	Not persisted.	Not persisted.
totalAmountWithTax	Not persisted.	Not persisted.	Not persisted.
lineItems	Refer to the next line attributes section.	Refer to the next line attributes section.	Refer to the next line attributes section.

Response Mappings for Line Attributes

This table defines the response mappings between the line attributes of a tax callout and fields of applicable objects.

Line Attributes	Invoice Mapping	Negative Invoices	Credit Mapping
taxCode	InvoiceLineTax.taxCode	InvoiceLineTax.taxCode	InvoiceLineTax.taxCode
productCode	Not persisted.	Not persisted.	Not persisted.
productId	Not persisted.	Not persisted.	Not persisted.
amountDetails			

Line Attributes	Invoice Mapping	Negative Invoices	Credit Mapping
exemptAmount	Not persisted.	Not persisted.	Not persisted.
taxAmount	Not persisted.	Not persisted.	Not persisted.
totalAmount	Not persisted.	Not persisted.	Not persisted.
totalAmountWithTax	Not persisted.	Not persisted.	Not persisted.
effectiveDate	InvoiceLineTax.taxEffectiveDate	InvoiceLineTax.taxEffectiveDate	CreditMemoLineTax.taxEffectiveDate
lineNumber	InvoiceLineTax.InvoiceLine	InvoiceLineTax.InvoiceLine	CreditMemoLineTax.CreditMemoLine
description	Not persisted.	Not persisted.	Not persisted.
quantity	Not persisted.	Not persisted.	Not persisted.
addresses			
billTo	Not persisted.	Not persisted.	Not persisted.
shipTo	Not persisted.	Not persisted.	Not persisted.
shipFrom	Not persisted.	Not persisted.	Not persisted.
soldTo	Not persisted.	Not persisted.	Not persisted.
productsku	Not persisted.	Not persisted.	Not persisted.
referenceDocumentCode	Not persisted.	Not persisted.	Not persisted.
taxes	Refer to the next tax attributes section.	Refer to the next tax attributes section.	Refer to the next tax attributes section.

Response Mappings for Tax Attributes

This table defines the response mappings between the tax attributes of a tax callout and fields of applicable objects.

Line Attributes	Invoice Mapping	Negative Invoices	Credit Mapping
exemptAmount	InvoiceLineTax.taxExemptAmount	InvoiceLineTax.taxExemptAmount	Not persisted.
exemptReason	Not persisted.	Not persisted.	Not persisted.
imposition			
type	Not persisted.	Not persisted.	Not persisted.
Name	InvoiceLineTax.TaxName	InvoiceLineTax.TaxName	CreditMemoLineTax.TaxName
jurisdiction			
country	Not persisted.	Not persisted.	Not persisted.
id	Not persisted.	Not persisted.	Not persisted.
level	Not persisted.	Not persisted.	Not persisted.
name	Not persisted.	Not persisted.	Not persisted.

Line Attributes	Invoice Mapping	Negative Invoices	Credit Mapping
region	Not persisted.	Not persisted.	Not persisted.
stateAssignedNo	Not persisted.	Not persisted.	Not persisted.
rate	InvoiceLineTax.taxRate	InvoiceLineTax.taxRate	CreditMemoLineTax.TaxRate
tax	InvoiceLineTax.taxAmount	InvoiceLineTax.taxAmount	CreditMemoLineTax.TaxAmount
taxId	InvoiceLineTax.TaxTransactionNumber	InvoiceLineTax.TaxTransactionNumber	CreditMemoLineTax.TaxTransactionNumber
taxableAmount	Not persisted.	Not persisted.	Not persisted.

Billing Metadata API Types

Metadata API enables you to access some types and feature settings that you can customize in the user interface.

BillingSettings
Represents the settings for Salesforce Billing.

Flow for Billing
Represents the metadata associated with a flow. With Flow, you can create an application that navigates users through a series of screens to query and update records in the database. You can also execute logic and provide branching capability based on user input to build dynamic applications.

PaymentsSharingSettings
Represents the settings to enable account-based sharing to view details related to Revenue Cloud Billing on the objects for Payments and Refunds.

SEE ALSO:

Metadata API Developer Guide: Understanding Metadata API

BillingSettings

Represents the settings for Salesforce Billing.

Parent Type and Manifest Access

This type extends the Metadata metadata type and inherits its `fullName` field.

In the package manifest, all the settings metadata types for the org are accessed using the “Settings” name. See Settings for more details.

File Suffix and Directory Location

The `BillingSettings` values are stored in the `BillingSettings.settings` file in the `settings` folder. The `.settings` files are different from other named components, because there’s only one settings file for each settings component.

Version

BillingSettings components are available in API version 62.0 and later.

Special Access Rules

These settings are available when Billing is enabled.

Fields

Field Name	Description
acctRecGlAccount	**Field Type** string **Description** General ledger account to record the credit amount for unrealized or realized losses and the debit amount for unrealized or realized gains in transaction journals. Available in API version 64.0 and later.
billingContextDefinition	**Field Type** string **Description** Name of the context definition that the Create Billing Schedules for Orders API uses to understand your order data. Available in API version 64.0 and later.
billingContextSourceMapping	**Field Type** string **Description** Name of the context mapping that links Order fields to billing transaction context nodes. Available in API version 64.0 and later.
billingIntraCtxtSrcMapping	**Field Type** string **Description** Name of the custom context mapping that maps your custom or standard Order fields to billing transaction context nodes. Available in API version 64.0 and later.
defaultAPClosureDPEDefnName	**Field Type** string **Description** Org-wide default value to specify the Data Processing Engine (DPE) definition to close legal entity accounting periods. Available in API version 64.0 and later.
defaultApplyCreditMemoFlow	**Field Type** string

Field Name	Description
	Description Default flow that's used to apply the credit memo to invoices. Available in API version 64.0 and later.
`defaultBillingTreatment`	**Field Type** string **Description** Org-wide default value to specify the name of the billing treatment. Available in API version 64.0 and later.
`defaultEmailTemplate`	**Field Type** string **Description** Default email template to send the generated invoice PDFs. Available in API version 64.0 and later.
`defaultInvPreviewTemplate`	**Field Type** string **Description** Default template to generate PDFs of invoice previews. Available in API version 64.0 and later.
`defaultInvoiceDocTemplate`	**Field Type** string **Description** Default template to generate PDFs of invoices. Available in API version 64.0 and later.
`defaultLegalEntity`	**Field Type** string **Description** Org-wide default value to specify the name of the legal entity. Available in API version 64.0 and later.
`defaultTaxTreatment`	**Field Type** string **Description** Org-wide default value to specify the name of the tax treatment. Available in API version 64.0 and later.
`enableBillingDisputeManagement`	**Field Type** boolean

Field Name	Description
	Description Indicates whether to enable Dispute Management (`true`) or not (`false`). The default value is `false`. Available in API version 66.0 and later.
`enableBillingSetup`	**Field Type** boolean **Description** Indicates whether to enable Billing setting (`true`) or not (`false`). The default value is `false`.
`enableCreditMemoSequenceService`	**Field Type** boolean **Description** Indicates whether to mandate the application of sequence policy for credit memos (`true`) or not (`false`). The default value is `false`. Available in API version 66.0 and later with Revenue Cloud Billing.
`enableCrMemoApplicationToPostedInvoices`	**Field Type** boolean **Description** Indicates whether to enable Apply Credits to Posted Invoices setting (`true`) or not (`false`). The default value is `false`. This setting automates settlement of invoices through application of credits to posted invoices. The credit application level determines whether credits are automatically applied to invoices or invoice lines.
`enableFailedPaymentsRetry`	**Field Type** boolean **Description** Indicates whether to retry failed payment schedule items automatically based on the defined payment retry rules (`true`) or not (`false`). The default value is `false`. Available in API version 66.0 and later.
`enableForeignExchangeTrxnJrnlCreation`	**Field Type** boolean **Description** Indicates whether to create Transaction Journal records for invoices that hold balance amounts (partially settled and not fully settled posted invoices) to record foreign exchange unrealized gains or losses during the closure activity of a legal entity accounting period. The default value is `false`. Available in API version 65.0 and later with Revenue Cloud Billing.

Field Name	Description
enableInvoiceEmailDelivery	**Field Type** boolean **Description** Indicates whether to enable Configure Email Delivery Settings (`true`) or not (`false`). The default value is `false`. Available in API version 63.0 and later with Revenue Cloud Billing.
enableInvoicePdfGeneration	**Field Type** boolean **Description** Indicates whether to enable Document Generation setting (`true`) or not (`false`). The default value is `false`. Available in API version 63.0 and later with Revenue Cloud Billing.
enableInvoiceSequenceService	**Field Type** boolean **Description** Indicates whether to mandate the application of sequence policy for posted invoices (`true`) or not (`false`). The default value is `false`. If enabled, each posted invoice is assigned an invoice number. Available in API version 65.0 and later with Revenue Cloud Billing.
enableNegInvoiceLnConversionToCrMemoLn	**Field Type** boolean **Description** Indicates whether to enable Convert Negative Invoice Lines to Credit Memo Lines setting (`true`) or not (`false`). The default value is `false`.
enablePaymentSchedulesAndItemsCreation	**Field Type** boolean **Description** Indicates whether to create a default payment schedule policy and payment schedule treatment (`true`) or not (`false`). The default value is `false`. If enabled, payment schedules and payment schedule items are created during financial transactions such as posting of invoices. Available in API version 64.0 and later with Revenue Cloud Billing.
enableRefundIssuingAndBalanceSettlement	**Field Type** boolean **Description** Indicates whether to issue refunds and settle balances (`true`) or not (`false`). The default value is `false`.

Field Name	Description
	If enabled, refunds are issued and credit memos are applied to any remaining invoice balance when customers amend or cancel an order. Available in API version 67.0 and later.
`enableTransactionJournalCreation`	**Field Type** boolean **Description** Indicates whether to create Transaction Journal records based on the defined general ledger account assignment rules for the billing entities when billing transaction records are created or updated (`true`) or not (`false`). The default value is `false`. Available in API version 63.0 and later with Revenue Cloud Billing. Billing transaction records include these transaction types. • Invoice • Invoice Line • Invoice Line Tax • Credit Memo • Credit Memo Line • Credit Memo Line Tax • Payment • Refund • Payment Line Invoice • Payment Line Invoice Line • Credit Memo Inv Application • Credit Memo Line Invoice Line
`enableTransactionsApplicationToInvoices`	**Field Type** boolean **Description** Indicates whether to enable Credit Application Level setting (`true`) or not (`false`). The default value is `false`. **Revenue Cloud Advanced** This setting applies balances of credit memos to invoices or balances of credit memo lines to invoice lines. For the latter, amounts and balances on the invoices are rolled-up from the related invoice lines. **Revenue Cloud Billing** This setting applies balances of credit memos and payments to invoices or balances of credit memo lines and payments lines to invoice lines. For the latter, amounts and balances on the invoices are rolled-up from the related invoice lines.

Field Name	Description
enableTrxnAmountsStorageInCorpCurrency	**Field Type** boolean **Description** Indicates whether to allow conversion of amounts of the Invoice, Invoice Line, Credit Memo, and Credit Memo Line records to your corporate currency (`true`) or not (`false`). The default value is `false`. Available in API version 63.0 and later. Store the converted amounts in corporate currency-specific amount fields.
realisedGainGlAccount	**Field Type** string **Description** Name of the general ledger account to record realized gains in transaction journals. Available in API version 64.0 and later.
realisedLossGlAccount	**Field Type** string **Description** Name of the general ledger account to record realized losses in transaction journals. Available in API version 64.0 and later.
ruleBasedCrAndPymtAppln	**Field Type** string **Description** Automates the settlement of the posted invoices by applying payments and credits that meet the specified application rules. The rules application level determines whether payments or credits are applied first to the invoices. The ruleset displays a list of selectable rules. Available in API version 66.0 and later.
unrealisedGainGlAccount	**Field Type** string **Description** Name of the general ledger account to record unrealized gains in transaction journals. Available in API version 64.0 and later.
unrealisedLossGlAccount	**Field Type** string **Description** Name of the general ledger account to record unrealized losses in transaction journals. Available in API version 64.0 and later.

Declarative Metadata Sample Definition

The following is an example of a BillingSettings component.

```
<BillingSettings xmlns="http://soap.sforce.com/2006/04/metadata">
    <enableBillingSetup>true</enableBillingSetup>
    <enableForeignExchangeTrxnJrnlCreation>true</enableForeignExchangeTrxnJrnlCreation>
    <enableInvoicePdfGeneration>true</enableInvoicePdfGeneration>
   <enableTransactionsApplicationToInvoices>true</enableTransactionsApplicationToInvoices>

   <enableCrMemoApplicationToPostedInvoices>true</enableCrMemoApplicationToPostedInvoices>

    <enableInvoiceEmailDelivery>true</enableInvoiceEmailDelivery>
    <enableInvoiceSequenceService>true</enableInvoiceSequenceService>
    <enableTransactionJournalCreation>true</enableTransactionJournalCreation>
    <enableTrxnAmountsStorageInCorpCurrency>true</enableTrxnAmountsStorageInCorpCurrency>

    <enablePaymentSchedulesAndItemsCreation>true</enablePaymentSchedulesAndItemsCreation>
</BillingSettings>
```

The following is an example `package.xml` that references the previous definition.

```
<?xml version="1.0" encoding="UTF-8"?>
<Package xmlns="http://soap.sforce.com/2006/04/metadata">
    <types>
        <members>Billing</members>
        <name>Settings</name>
    </types>
    <version>67.0</version>
</Package>
```

Wildcard Support in the Manifest File

The wildcard character `*` (asterisk) in the `package.xml` manifest file doesn't apply to metadata types for feature settings. The wildcard applies only when retrieving all settings, not for an individual setting. For details, see Settings. For information about using the manifest file, see Deploying and Retrieving Metadata with the Zip File.

Flow for Billing

Represents the metadata associated with a flow. With Flow, you can create an application that navigates users through a series of screens to query and update records in the database. You can also execute logic and provide branching capability based on user input to build dynamic applications.

FlowActionCall

Billing exposes additional actionType values for the FlowActionCall Metadata type.

Field Name	Field Type	Description
`actionType`	InvocableActionType (enumeration of type string)	Required. The action type. Additional valid values only for Billing include:

Field Name	Field Type	Description
		• `applyCredit`—Apply a credit memo or credit memo line to an invoice or invoice line, respectively. • `unapplyCredit`—Unapply a credit memo or credit memo line from an invoice or invoice line, respectively. • `postDraftInvoice`—Update the status of an invoice from `Draft` to `Posted` for a credit memo application. • `postDraftInvoiceBatchRun`—Update the status of a batch of invoices from `Draft` to `Posted` for a credit memo application. • `createBillingSchedulesFromBillingTransaction`—Create one or more billing schedules for a specified billing transaction ID. • `recoverBillingSchedules`—Recover one or more billing schedules in the `Error` or `Processing` status. • `generateInvoiceDocuments`—Asynchronously generate PDF documents for the invoices associated with an invoice batch run record that are in the `Draft` or `Posted` status. • `createBillingSchedulesFromTrxn`—Creates billing schedules for internal or external transaction records by calling the Create Standalone Billing Schedules API. • `unapplyPayment`—Unapplies a payment that's already been applied to an invoice or invoice line by crediting the amount back to the payment and the invoice or invoice line. • `writeOffInvoices`—Write off partially paid or unpaid invoices to manage pending debts and to maintain accurate financial records. This action calls the Posted Invoice List Write-Off (POST) API. • `assignSequences`—Assigns sequence pattern values to target invoice records based on the specified sequence policy. • `postDraftCreditMemo`—Post a draft credit memo to a credit memo record for review and approval. • `generateAccountStatement`—Generates a comprehensive account statement for a specified account with transaction history and balance information. • `blngDsptIssueCreditMemo`—Issue credit memos for disputed invoices to resolve billing disputes. • `blngSvcExtendInvoiceDueDate`—Update the due date on an invoice to accommodate payment extensions or resolve billing disputes. • `blngSvcSuspendBilling`—Suspend or resume the billing of an account to handle billing disputes. • `blngSvcUpdateBillToContact`—Update the Bill to Contact detail on an invoice to ensure accurate billing communication and routing.

Field Name	Field Type	Description
		• `blngSendDunningEmail`—Run an orchestration that sends dunning process emails for collection plans to recover overdue revenue and notify customers about amounts still due. • `automateRefund`—Initiate refund orchestration for a credit memo generated from a subscription cancellation or negative amendment.

PaymentsSharingSettings

Represents the settings to enable account-based sharing to view details related to Revenue Cloud Billing on the objects for Payments and Refunds.

Use account-based sharing to view Revenue Cloud Billing details on these objects.

- Payment
- Payment Authorization
- Payment Authorization Adjustment
- Refund
- Saved Payment Method

Parent Type and Manifest Access

This type extends the Metadata metadata type and inherits its `fullName` field.

In the package manifest, all the settings metadata types for the org are accessed using the "Settings" name. See Settings for more details.

File Suffix and Directory Location

The `PaymentsSharingSettings` values are stored in the `PaymentsSharing.settings` file in the `settings` folder. The `.settings` files are different from other named components, because there's only one settings file for each settings component.

Version

PaymentsSharingSettings components are available in API version 64.0 and later.

Special Access Rules

If you don't have the View All Records permission to the Payments and Refunds objects, then you can:

- View the records only when the records have a value for the `Account` field and a shared Account ID.
- Use Payment Sale API and Payment Authorization API irrespective of whether the Account ID is null or shared.
- Use Payment Capture API, Authorization Reversal API, Create Payment Refund Billing API, and Apply Refunds to Payments API only when the corresponding Authorization or Payment record has a shared Account ID.

See Commerce Payments resources.

Fields

Field Name	Description
delegatePaymentSharingToAccount	**Field Type** boolean **Description** Indicates whether sharing for these objects must be delegated to the corresponding Account record (`true`) or not (`false`). • Payment • Payment Authorization • Payment Authorization Adjustment • Refund • Saved Payment Method The default value is `false`. If this field's value is set to `true`, you get access to these objects based on the access you have for the Account object. For example, if you have Read access to the Account object, you get Read access to the objects for Payments and Refunds. For saved payment method, the sharing is delegated to merchant account instead of account if the user has Manage Saved Payment Methods user permission.

Declarative Metadata Sample Definition

The following is an example of a PaymentsSharingSettings component.

```
<?xml version="1.0" encoding="UTF-8"?>
<PaymentsSharingSettings xmlns="http://soap.sforce.com/2006/04/metadata">
    <delegatePaymentSharingToAccount>true</delegatePaymentSharingToAccount>
</PaymentsSharingSettings>
```

The following is an example `package.xml` that references the previous definition.

```
<?xml version="1.0" encoding="UTF-8"?>
<Package xmlns="http://soap.sforce.com/2006/04/metadata">
    <types>
        <members>PaymentsSharing</members>
        <name>Settings</name>
    </types>
    <version>67.0</version>
</Package>
```

Wildcard Support in the Manifest File

The wildcard character * (asterisk) in the `package.xml` manifest file doesn't apply to metadata types for feature settings. The wildcard applies only when retrieving all settings, not for an individual setting. For details, see Settings. For information about using the manifest file, see Deploying and Retrieving Metadata with the Zip File.

CHAPTER 13 Revenue Management Associated Objects

In this chapter ...

This section provides a list of objects associated to standard objects of Revenue Management with their standard fields.

Some fields may not be listed for some objects. To see the system fields for each object, see System Fields in the *Object Reference for Salesforce and Lightning Platform*.

To verify the complete list of fields for an object, use a describe call from the API or inspect with an appropriate tool. For example, inspect the WSDL or use a schema viewer.

*StandardObjectName*ChangeEvent

A ChangeEvent object is available for each object that supports Change Data Capture. You can subscribe to a stream of change events using Change Data Capture to receive data tied to record changes in Salesforce. Changes include record creation, updates to an existing record, deletion of a record, and undeletion of a record. A change event isn't a Salesforce object—it doesn't support CRUD operations or queries. It's included in the object reference so you can discover which Salesforce objects support change events.

Supported Calls

`describeSObjects()`

Special Access Rules

- All objects may not be available in your org. Some objects require specific feature settings and permissions to be enabled.
- For more special access rules, if any, see the documentation for the standard object. For example, for AccountChangeEvent, see the special access rules for Account.

Change Event Support

Change events are available for all custom objects and a subset of standard objects. Change events that correspond to custom settings are partially supported. They aren't supported in Apex triggers but are supported in other types of subscribers. For more information about standard object support, see the Objects That Support Change Events section below.

Change Event Name

The name of a change event is based on the name of the corresponding object for which it captures the changes.

Standard Object Change Event Name

```
<Standard_Object_Name>ChangeEvent
```

Example: `AccountChangeEvent`

Custom Object Change Event Name

```
<Custom_Object_Name>__ChangeEvent
```

Example: `MyCustomObject__ChangeEvent`

Change Event Fields

The fields that a change event can include correspond to the fields on the associated parent Salesforce object, with a few exceptions. For example, AccountChangeEvent fields correspond to the fields on Account.

The fields that a change event doesn't include are:

- The `IsDeleted` system field.
- The `SystemModStamp` system field.
- Any field whose value isn't on the record and is derived from another record or from a formula, except roll-up summary fields, which are included. Examples are formula fields. Examples of fields with derived values include `LastActivityDate` and `PhotoUrl`.

Each change event also contains header fields. The header fields are included inside the `ChangeEventHeader` field. They contain information about the event, such as whether the change was an update or delete and the name of the object, like Account.

In addition to the event payload, the event schema ID is included in the `schema` field. Also included is the event-specific field, `replayId`, which is used for retrieving past events.

Event Message Example

This example is an event message in JSON format for a new account record creation.

```
{
  "schema": "IeRuaY6cbI_HsV8Rv1Mc5g",
  "payload": {
    "ChangeEventHeader": {
      "entityName": "Account",
      "recordIds": [
        "<record_ID>"
      ],
      "changeType": "CREATE",
      "changeOrigin": "com/salesforce/api/soap/51.0;client=SfdcInternalAPI/",
      "transactionKey": "0002343d-9d90-e395-ed20-cf416ba652ad",
      "sequenceNumber": 1,
      "commitTimestamp": 1612912679000,
      "commitNumber": 10716283339728,
      "commitUser": "<User_ID>"
    },
    "Name": "Acme",
    "Description": "Everyone is talking about the cloud. But what does it mean?",
    "OwnerId": "<Owner_ID>",
    "CreatedDate": "2021-02-09T23:17:59Z",
    "CreatedById": "<User_ID>",
    "LastModifiedDate": "2021-02-09T23:17:59Z",
    "LastModifiedById": "<User_ID>"
  },
  "event": {
    "replayId": 6
  }
}
```

API Version and Schema

When you subscribe to change events, the subscription uses the latest API version and the event messages received reflect the latest field definitions. For more information, see API Version and Event Schema in the *Change Data Capture Developer Guide*.

Usage

For more information about Change Data Capture, see Change Data Capture Developer Guide.

*StandardObjectName*Feed

*StandardObjectName*Feed is the model for all feed objects associated with standard objects. These objects represent the posts and feed-tracked changes of a standard object.

The object name is variable and uses *StandardObjectName*Feed syntax. For example, AccountFeed represents the posts and feed-tracked changes on an account record. We list the available associated feed objects at the end of this topic. For specific version information, see the documentation for the standard object.

Supported Calls

`delete()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`

Special Access Rules

In the internal org, users can delete all feed items they created. This rule varies in Experience Cloud sites where threaded discussions and delete-blocking are enabled. Site members can delete all feed items they created, provided the feed items don't have content nested under them—like a comment, answer, or reply. Where the feed item has nested content, only feed moderators and users with the Modify All Data permission can delete threads.

To delete feed items they didn't create, users must have one of these permissions:

- Modify All Data
- Modify All Records on the parent object, like Account for AccountFeed
- Moderate Chatter

  Note: Users with the Moderate Chatter permission can delete only the feed items and comments they can see.

  Only users with this permission can delete items in unlisted groups.

For more special access rules, if any, see the documentation for the standard object. For example, for AccountFeed, see the special access rules for Account.

Fields

Field	Details
BestCommentId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** The ID of the comment marked as best answer on a question post. This field is available in API version 44.0 and later.
Body	**Type** textarea

Field	Details
	Properties Nillable, Sort **Description** The body of the post. Required when `Type` is `TextPost`. Optional when `Type` is `ContentPost` or `LinkPost`.
`CommentCount`	**Type** int **Properties** Filter, Group, Sort **Description** The number of comments associated with this feed item. In a feed that supports pre-moderation, `CommentCount` isn't updated until a comment is published. For example, say that you comment on a post that already has one published comment and your comment triggers moderation. Now there are two comments on the post, but the count says there's only one. In a moderated feed, comments aren't counted until approved by an admin or someone with Can Approve Feed Post and Comment or Modify All Data. Feed moderation has implications on how you retrieve feed comments. In a moderated feed, rather than retrieving comments by looping through `CommentCount`, go through pagination until the end of comments is returned.
`ConnectionId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** When a PartnerNetworkConnection modifies a record that is tracked, the `CreatedBy` field contains the ID of the system administrator. The `ConnectionId` contains the ID of the PartnerNetworkConnection. Available if Salesforce to Salesforce is enabled for your organization.
`ContentData`	**Type** base64 **Properties** Nillable **Description** Available in API version 36.0 and earlier only. Required if `Type` is `ContentPost`. Encoded file data in any format, and can't be 0 bytes. Setting this field automatically sets `Type` to `ContentPost`.
`ContentDescription`	**Type** textarea

Field	Details
	Properties Nillable, Sort **Description** Available in API version 36.0 and earlier only. The description of the file specified in `ContentData`.
`ContentFileName`	**Type** string **Properties** Group, Nillable, Sort **Description** Available in API version 36.0 and earlier only. This field is required if `Type` is `ContentPost`.The name of the file uploaded to the feed. Setting `ContentFileName` automatically sets `Type` to `ContentPost`.
`ContentSize`	**Type** int **Properties** Group, Nillable, Sort **Description** Available in API version 36.0 and earlier only. The size of the file (in bytes) uploaded to the feed. This field is read-only and is automatically determined during insert.
`ContentType`	**Type** string **Properties** Group, Nillable, Sort **Description** Available in API version 36.0 and earlier only. The MIME type of the file uploaded to the feed. This field is read-only and is automatically determined during insert.
`FeedPostId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** This field was removed in API version 22.0, and is available in earlier versions for backward compatibility only. ID of the associated FeedPost. A FeedPost represents the following types of changes in a feed item: changes to tracked fields, text posts, link posts, and content posts.

Field	Details
`InsertedById`	**Type** reference **Properties** Group, Nillable, Sort **Description** ID of the user who added this item to the feed. For example, if an application migrates posts and comments from another application into a feed, the `InsertedBy` value is set to the ID of the context user.
`isRichText`	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Indicates whether the feed item `Body` contains rich text. If you post a rich text feed comment using SOAP API, set `IsRichText` to `true` and escape HTML entities from the body. Otherwise, the post is rendered as plain text. Rich text supports the following HTML tags: • `<p>`Though the ` ` tag isn't supported, you can use `<p> </p>` to create lines. • `<a>` • `<b>` • `<code>` • `<i>` • `<u>` • `<s>` • `<ul>` • `<ol>` • `<li>` • `<img>` The `<img>` tag is accessible only through the API and must reference files in Salesforce similar to this example: `<img src="sfdc://069B0000000omjh"></img>` In API version 35.0 and later, the system replaces special characters in rich text with escaped HTML. In API version 34.0 and prior, all rich text appears as a plain-text representation.
`LikeCount`	**Type** int **Properties** Filter, Group, Sort

Field	Details
	Description The number of likes associated with this feed item.
LinkUrl	**Type** url **Properties** Nillable, Sort **Description** The URL of a LinkPost.
NetworkScope	**Type** picklist **Properties** Group, Nillable, Restricted picklist, Sort **Description** Specifies whether this feed item is available in the default Experience Cloud site, a specific Experience Cloud site, or all sites. This field is available in API version 26.0 and later, if digital experiences is enabled for your org. NetworkScope can have the following values: • NetworkId—The ID of the Experience Cloud site in which the FeedItem is available. If left empty, the feed item is only available in the default Experience Cloud site. • AllNetworks—The feed item is available in all Experience Cloud sites. Note the following exceptions for NetworkScope: • Only feed items with a Group or User parent can set a NetworkId or a null value for NetworkScope. • For feed items with a record parent, users can set NetworkScope only to AllNetworks. • You can't filter a feed item on the NetworkScope field.
ParentId	**Type** reference **Properties** Filter, Group, Sort **Description** ID of the record that is tracked in the feed. The detail page for the record displays the feed.
RelatedRecordId	**Type** reference **Properties** Group, Nillable, Sort

Field	Details
	Description ID of the ContentVersion record associated with a `ContentPost`. This field is null for all posts except `ContentPost`.
`Title`	**Type** string **Properties** Group, Nillable, Sort **Description** The title of the feed item. When the `Type` is `LinkPost`, the `LinkUrl` is the URL and this field is the link name.
`Type`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** The type of feed item: • `ActivityEvent`—indirectly generated event when a user or the API adds a Task associated with a feed-enabled parent record (excluding email tasks on cases). Also occurs when a user or the API adds or updates a Task or Event associated with a case record (excluding email and call logging). For a recurring Task with CaseFeed disabled, one event is generated for the series only. For a recurring Task with CaseFeed enabled, events are generated for the series and each occurrence. • `AdvancedTextPost`—created when a user posts a group announcement and, in Lightning Experience as of API version 39.0 and later, when a user shares a post. • `AnnouncementPost`—Not used. • `ApprovalPost`—generated when a user submits an approval. • `BasicTemplateFeedItem`—Not used. • `CanvasPost`—a post made by a canvas app posted on a feed. • `CollaborationGroupCreated`—generated when a user creates a public group. • `CollaborationGroupUnarchived`—Not used. • `ContentPost`—a post with an attached file. • `CreatedRecordEvent`—generated when a user creates a record from the publisher. • `DashboardComponentAlert`—generated when a dashboard metric or gauge exceeds a user-defined threshold. • `DashboardComponentSnapshot`—created when a user posts a dashboard snapshot on a feed. • `LinkPost`—a post with an attached URL. • `PollPost`—a poll posted on a feed.

Field	Details
	• `ProfileSkillPost`—generated when a skill is added to a user's Chatter profile. • `QuestionPost`—generated when a user posts a question. • `ReplyPost`—generated when Chatter Answers posts a reply. • `RypplePost`—generated when a user creates a Thanks badge in WDC. • `TextPost`—a direct text entry on a feed. • `TrackedChange`—a change or group of changes to a tracked field. • `UserStatus`—automatically generated when a user adds a post. Deprecated. The following values appear in the `Type` picklist for all feed objects but apply only to CaseFeed: • `CaseCommentPost`—generated event when a user adds a case comment for a case object • `EmailMessageEvent`—generated event when an email related to a case object is sent or received • `CallLogPost`—generated event when a user logs a call for a case through the user interface. CTI calls also generate this event. • `ChangeStatusPost`—generated event when a user changes the status of a case • `AttachArticleEvent`—generated event when a user attaches an article to a case If you set `Type` to `ContentPost`, also specify `ContentData` and `ContentFileName`.
`Visibility`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** Specifies whether this feed item is available to all users or internal users only. This field is available in API version 26.0 and later, if digitial experiences is enabled for your organization. `Visibility` can have the following values: • `AllUsers`—The feed item is available to all users who have permission to see the feed item. • `InternalUsers`—The feed item is available to internal users only. Note the following exceptions for `Visibility`: • For record posts, `Visibility` is set to `InternalUsers` for all internal users by default. • External users can set `Visibility` only to `AllUsers`. • On user and group posts, only internal users can set `Visibility` to `InternalUsers`.

Usage

A feed for an object is automatically created when a user enables feed tracking for the object. Use feeds to track changes to records. For example, `AccountFeed` tracks changes to an account record. Use feed objects to retrieve the content of feed fields, such as type of feed or feed ID.

- `NewsFeed` and `UserProfileFeed` are available in API version 18.0 through API version 26.0. In API version 27.0 and later, `NewsFeed` and `UserProfileFeed` are no longer available in SOAP API. Use Connect REST API to access `NewsFeed` and `UserProfileFeed`.

  Use the `NewsFeed` object to query and retrieve lead feed items associated with a converted lead record.

- For `NewsFeed` and `UserProfileFeed`, users who don't have the View All Data permission have the following limitations when querying records: Must specify a `LIMIT` clause and the limit must be less than or equal to 1000. Can include a `WHERE` clause that references object fields, but can't include references to fields in related objects. For example, you can filter by `CreatedDate` or `ParentId`, but not by `Parent.Name`. Can include an `ORDER BY` clause that references object fields, but can't include references to fields in related objects. For example, `ORDER BY CreatedDate` or `ParentId`, but not by `Parent.Name`. To query for the most recent feed items, `ORDER BY CreatedDate DESC, Id DESC`.

  Note the following SOQL restrictions. No SOQL limit if logged-in user has View All Data permission. If not, specify a `LIMIT` clause of 1,000 records or fewer. SOQL `ORDER BY` on fields using relationships isn't available. Use `ORDER BY` on fields on the root object in the SOQL query.

- The name *`Article Type`*__Feed is variable, where *Article Type* is the object name for the article type associated with the article. For example, `Offer__Feed` represents a feed on an article of type *Offer*.
- Field Service must be enabled in your organization for `ServiceAppointmentFeed`, `ServiceCrewFeed`, `ServiceMemberFeed`, `ServiceResourceCapacityFeed`, `ServiceResourceFeed`, `ServiceResourceSkillFeed`, `ServiceTerritoryFeed`, `ServiceTerritoryMemberFeed`, and `SkillRequirementFeed`.
- For `WorkOrderFeed`, Work Orders or Field Service must be enabled in your organization.
- On `UserFeed`, if you use the `FeedComment` object to comment on a user record, the user can delete the comment. For example, if John Smith adds a comment to the feed on Sasha Jones' user record, Sasha can delete the comment.

*`StandardObjectName`*History

*`StandardObjectName`*History is the model for all history objects associated with standard objects. These objects represent the history of changes to the values in the fields of a standard object.

The object name is variable and uses *`StandardObjectName`*History syntax. For example, AccountHistory represents the history of changes to the values of an account record's fields. We list the available associated history objects at the end of this topic. For specific version information, see the documentation for the standard object.

Supported Calls

`describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`

You can also enable `delete()` in API version 42.0 and later. See Enable delete of Field History and Field History Archive.

Special Access Rules

For specific special access rules, if any, see the documentation for the standard object. For example, for AccountHistory, see the special access rules for Account.

Fields

Field Name	Details
*StandardObjectName*Id	**Type** reference **Properties** Filter, Group, Sort **Description** ID of the standard object.
DataType	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort **Description** Data type of the field that was changed.
Field	**Type** picklist **Properties** Filter, Group, Restricted picklist, Sort **Description** Name of the field that was changed.
NewValue	**Type** anyType **Properties** Nillable, Sort **Description** New value of the field that was changed.
OldValue	**Type** anyType **Properties** Nillable, Sort **Description** Old value of the field that was changed.

*StandardObjectName*OwnerSharingRule

*StandardObjectName*OwnerSharingRule is the model for all owner sharing rule objects associated with standard objects. These objects represent a rule for sharing a standard object with users other than the owner.

The object name is variable and uses *StandardObjectName*OwnerSharingRule syntax. For example, ChannelProgramOwnerSharingRule is a rule for sharing a channel program with users other than the channel program owner. The available associated owner sharing rule objects are listed at the end of this topic. For specific version information, see the standard object documentation.

Note: To enable access to this object, contact Salesforce customer support. But we recommend that you use Metadata API to programmatically update owner sharing rules instead because it triggers automatic sharing rule recalculation. The SharingRules Metadata API type is enabled for all orgs.

Supported Calls

`create()`, `delete()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `update()`, `upsert()`

Special Access Rules

For specific special access rules, if any, see the documentation for the standard object. For example, for ChannelProgramOwnerSharingRule, see the special access rules for ChannelProgram.

Fields

Field Name	Details
AccessLevel	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** Determines the level of access users have to records. Values are: • Read (read only) • Edit (read/write)
Description	**Type** textarea **Properties** Create, Filter, Nillable, Sort, Update **Description** Description of the sharing rule. Maximum length is 1,000 characters.

Field Name	Details
DeveloperName	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The unique name of the object in the API. This name can contain only underscores and alphanumeric characters and must be unique in your org. It must begin with a letter, not include spaces, not end with an underscore, and not contain two consecutive underscores. In managed packages, this field prevents naming conflicts on package installations. With this field, a developer can change the object's name in a managed package, and the changes are reflected in a subscriber's organization. When creating large sets of data, always specify a unique `DeveloperName` for each record. If no `DeveloperName` is specified, performance can slow while Salesforce generates one for each record.
GroupId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** ID of the source group. Records that are owned by users in the source group trigger the rule to give access.
Name	**Type** string **Properties** Create, Filter, Group, idLookup, Sort, Update **Description** Label of the sharing rule as it appears in the UI. Maximum length is 80 characters.
UserOrGroupId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** ID of the user or group that you're granting access to.

*StandardObjectName*Share

*StandardObjectName*Share is the model for all share objects associated with standard objects. These objects represent a sharing entry on the standard object.

The object name is variable and uses *StandardObjectName*Share syntax. For example, AccountBrandShare is a sharing entry on an account brand. For specific version information, see the standard object documentation.

You can only create, edit, and delete sharing entries for standard objects whose `RowCause` field is set to `Manual`. Sharing entries for standard objects with different `RowCause` values are created as a result of your Salesforce org's sharing configuration and are read-only. For some sharing mechanisms, such as sharing sets, sharing entries aren't stored at all.

Note: While Salesforce currently maintains read-only sharing entries for multiple sharing mechanisms, it's possible that we'll stop storing certain share records to improve performance. As a best practice, don't create customizations that rely on the availability of these sharing entries. Any changes to sharing behavior will be communicated before they occur.

Supported Calls

`create()`, `delete()`, `describeSObjects()`, `query()`, `retrieve()`, `update()`, `upsert()`

Special Access Rules

For specific special access rules, if any, see the documentation for the standard object. For example, for AccountBrandShare, see the special access rules for AccountBrand.

Fields

Field Name	Details
`AccessLevel`	**Type** picklist **Properties** Create, Filter, Group, Restricted picklist, Sort, Update **Description** The level of access allowed. Values are: • `All` (owner) • `Edit` (read/write) • `Read` (read only)
`ParentId`	**Type** reference **Properties** Create, Filter, Group, Sort **Description** ID of the parent record.

Field Name	Details
RowCause	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort **Description** Reason that the sharing entry exists.
UserOrGroupId	**Type** reference **Properties** Create, Filter, Group, Sort **Description** ID of the user or group that has been given access to the object.

Event

Represents an event in the calendar. In the user interface, event and task records are collectively referred to as activities.

Important: Where possible, we changed noninclusive terms to align with our company value of Equality. We maintained certain terms to avoid any effect on customer implementations.

Note:

- An EventRelation object can't be related to a child event, and child events don't include the invitee related list.
- `query()`, `delete()`, and `update()` aren't allowed with events related to more than one contact in API versions 25.0 and earlier.
- `create()` and `update()` aren't available for read-only fields on Lightning Experience event series.
- `upsert()` and `undelete()` aren't supported for syncing changes made to events through the API using the feature Lightning Sync.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Details
AcceptedEventInviteeIds	**Type** JunctionIdList

Field	Details
	Properties Create, Update **Description** A string array of contact or lead IDs who accepted this event. This `JunctionIdList` is linked to the `AcceptedEventRelation` child relationship. Warning: Adding a `JunctionIdList` field name to the `fieldsToNull` property deletes all related junction records. This action can't be undone.
`AccountId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** Represents the ID of the related account. The `AccountId` is determined as follows. If the value of `WhatId` is any of the following objects, then Salesforce uses that object's `AccountId`. • Account • Opportunity • Contract • Custom object that's a child of Account If the value of the `WhatId` field is any other object, and the value of the `WhoId` field is a contact object, then Salesforce uses that contact's `AccountId`. If your org uses Shared Activities, Salesforce uses the `AccountId` of the primary contact. Otherwise, Salesforce sets the value of the `AccountId` field to `null`. This is a relationship field. **Relationship Name** Account **Relationship Type** Lookup **Refers To** Account
`ActivityDate`	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Contains the event's due date if the `IsAllDayEvent` flag is set to `true`. This field is a date field with a timestamp that's always set to midnight in the Coordinated Universal Time

Field	Details
	(UTC) time zone. Don't attempt to alter the timestamp to account for time zone differences. Label is **Due Date Only**. This field is required in API versions 12.0 and earlier if the `IsAllDayEvent` flag is set to `true`. The value for this field and `StartDateTime` must match, or one of them must be `null`.
`ActivityDateTime`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** Contains the event's due date if the `IsAllDayEvent` flag is set to `false`. The time portion of this field is always transferred in the Coordinated Universal Time (UTC) time zone. Translate the time portion to or from a local time zone for the user or the application, as appropriate. Label is **Due Date Time**. This field is required in API versions 12.0 and earlier if the `IsAllDayEvent` flag is set to `false`. The value for this field and `StartDateTime` must match, or one of them must be `null`.
`ClientGuid`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** The client globally unique identifier identifies the external API client used to create the event. Label is **Client GUID**.
`CurrencyIsoCode`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Available only for orgs with the multicurrency feature enabled. Contains the ISO code for any currency allowed by the organization.
`DeclinedEventInviteeIds`	**Type** JunctionIdLIst **Properties** Create, Update **Description** A string array of contact, lead, or user IDs who declined this event. This `JunctionIdList` is linked to the `DeclinedEventRelation` child relationship.

Field	Details
	Warning: Adding a `JunctionIdList` field name to the `fieldsToNull` property deletes all related junction records. This action can't be undone.
`Description`	**Type** textarea **Properties** Create, Nillable, Update **Description** Contains a text description of the event. Limit: 32,000 characters.
`Division`	**Type** picklist **Properties** Defaulted on create, Filter, Group, Restricted picklist, Sort **Description** A logical segment of your organization's data. For example, if your company is organized into different business units, you could create a division for each business unit, such as "North America," "Healthcare," or "Consulting." Available only if the organization has the Division permission enabled.
`DurationInMinutes`	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Contains the event length, in minutes. Even though this field represents a temporal value, it's an integer type—not a Date/Time type. Required in API versions 12.0 and earlier if `IsAllDayEvent` is false. In API versions 13.0 and later, this field is optional, depending on the following: • If `IsAllDayEvent` is true, you can supply a value for either `DurationInMinutes` or `EndDateTime`. Supplying values in both fields is allowed if the values add up to the same amount of time. If both fields are `null`, the duration defaults to one day. • If `IsAllDayEvent` is false, a value must be supplied for either `DurationInMinutes` or `EndDateTime`. Supplying values in both fields is allowed if the values add up to the same amount of time. If the multiday event feature is enabled, then API versions 13.0 and later support values greater than 1440 for the `DurationInMinutes` field. API versions 12.0 and earlier can't access event objects whose `DurationInMinutes` is greater than 1440. For more information, see **Multiday Events**. Depending on your API version, errors with the `DurationInMinutes` and `EndDateTime` fields may appear in different places. • Versions 38.0 and before—Errors always appear in the `DurationInMinutes` field.

Field	Details
	• Versions 39.0 and later—If there's no value for the `DurationInMinutes` field, errors appear in the `EndDateTime` field. Otherwise, they appear in the `DurationInMinutes` field.
`EndDate`	**Type** date **Properties** Filter, Group, Nillable, Sort **Description** Read-only. Available in API versions 46.0 and later. This field supplies the date value that appears in the EndDateTime field. This field is a date field with a timestamp that is always set to midnight in the Coordinated Universal Time (UTC) time zone.
`EndDateTime`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** Available in API versions 13.0 and later. The time portion of this field is always transferred in the Coordinated Universal Time (UTC) time zone. Translate the time portion to or from a local time zone for the user or the application, as appropriate. This field is optional, depending on the following: • If `IsAllDayEvent` is true, you can supply a value for either `DurationInMinutes` or `EndDateTime`. Supplying values in both fields is allowed if the values add up to the same amount of time. If both fields are `null`, the duration defaults to one day. • If `IsAllDayEvent` is false, a value must be supplied for either `DurationInMinutes` or `EndDateTime`. Supplying values in both fields is allowed if the values add up to the same amount of time. Depending on your API version, errors with the `DurationInMinutes` and `EndDateTime` fields may appear in different places. • Versions 38.0 and before—Errors always appear in the `DurationInMinutes` field. • Versions 39.0 and later—If there's no value for the `DurationInMinutes` field, errors appear in the `EndDateTime` field. Otherwise, they appear in the `DurationInMinutes` field.
`EventSubtype`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort **Description** Provides standard subtypes to facilitate creating and searching for events. This field isn't updateable.

Field	Details
EventWhoIds	**Type** JunctionIdList **Properties** Create, Update **Description** A string array of contact or lead IDs used to create many-to-many relationships with a shared event. EventWhoIds is available when the shared activities setting is enabled. The first contact or lead ID in the list becomes the primary WhoId if you don't specify a primary WhoId. If you set the EventWhoIds field to null, all entries in the list are deleted and the value of WhoId is added as the first entry. Warning: Adding a JunctionIdList field name to the fieldsToNull property deletes all related junction records. This action can't be undone.
GroupEventType	**Type** picklist **Properties** Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort **Description** Read-only. Available in API versions 19.0 and later. The possible values are: • 0 (Non–group event)—An event with no invitees. • 1 (Group event)—An event with invitees. • 2 (Proposed event)—An event created when a user requests a meeting with a contact, lead, or person account using the Salesforce user interface. When the user confirms the meeting, the proposed event becomes a group event. You can't create, edit, or delete proposed events in the API. This value is no longer used in API version 41.0 and later. • 3 (IsRecurrence2 Series Pattern)—An event representing an event series recurrence pattern in Lightning Experience.
IsAllDayEvent	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the ActivityDate field (true) or the ActivityDateTime field (false) is used to define the date or time of the event. Label is **All-Day Event**. See also DurationInMinutes and EndDateTime.
IsArchived	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort

Field	Details
	Description Indicates whether the event has been archived.
IsChild	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Indicates whether the event is a child of another event (`true`) or not (`false`). For a child event, you can update `IsReminderSet` and `ReminderDateTime` only. You can query and delete a child event. If the objects related to the child event are different from those objects related to the parent event (this difference is possible if you use API version 25.0 or earlier) and one of the objects related to the child event is deleted, the objects related to the parent event are updated to ensure data integrity.
IsClientManaged	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Indicates whether the event is managed by an external client. If the value of this field is false, the event isn't owned or managed by an external client, and Salesforce can be used to update it. If the value is true, Salesforce can be used to change only noncritical fields on the event. Label is **Is Client Managed**.
IsGroupEvent	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Indicates whether the event is a group event—that is, whether it has invitees (`true`) or not (`false`).
IsPrivate	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether users other than the creator of the event can (`false`) or can't (`true`) see the event details when viewing the event user's calendar. However, users with the View All Data or Modify All Data permission can see private events in reports and searches, or when viewing other users' calendars. Private events can't be associated with opportunities, accounts, cases, campaigns, contracts, leads, or contacts. Label is **Private**.

Field	Details
IsRecurrence	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort **Description** Indicates whether a Salesforce Classic event is scheduled to repeat itself (`true`) or only occurs one time (`false`). This field is read-only when updating records, but not when creating them. If this field value is `true`, then `RecurrenceEndDateOnly`, `RecurrenceStartDateTime`, `RecurrenceType`, and any recurrence fields associated with the given recurrence type must be populated. Label is **Create recurring series of events**.
IsRecurrence2	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Read-only. This field is available in API version 44.0 and later. Indicates whether a Lightning Experience event is scheduled to repeat (`true`) or only occurs one time (`false)`. If this field value is true, then `Recurrence2PatternText` and `Recurrence2PatternVersion` must be populated. Label is **Repeat**.
IsRecurrence2Exception	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Read-only. This field is available in API version 44.0 and later. Indicates whether an individual event in a Lightning Experience event series has a recurrence pattern that’s different from the rest of the series, making it an exception.
IsRecurrence2Exclusion	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Read-only. This field is available in API version 44.0 and later. Indicates when updates to a Lightning Experience event series recurrence pattern have been made, but affect future event occurrences only. For past event occurrences, `IsRecurrence2Exclusion` is set to `true`, excluding past occurrences from the series recurrence pattern.
IsReminderSet	**Type** boolean

Field	Details
	Properties Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether the activity is a reminder (`true`) or not (`false`).
`IsVisibleInSelfService`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether an event associated with an object can be viewed in the Customer Portal (`true`) or not (`false`). If your org has enabled digital experiences, events marked `IsVisibleInSelfService` are visible to any external user in the Experience Cloud site, as long as the user has access to the record the event was created on. This field is available when • Customer Portal or partner portal is enabled OR • Digital experiences is enabled and you have Customer Portal or partner portal licenses
`Location`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Contains the location of the event.
`OwnerId`	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Contains the ID of the user or public calendar who owns the event. Label is **Assigned to ID**. This is a polymorphic relationship field. **Relationship Name** Owner **Relationship Type** Lookup **Refers To** Calendar, User

Field	Details
`Recurrence2PatternStartDate`	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** Read-only. This field is available in API version 44.0 and later. Indicates the date and time when the Lightning Experience event series begins. The time portion of this field is always transferred in the Coordinated Universal Time (UTC) time zone. Translate the time portion to or from a local time zone for the user or the application, as appropriate.
`Recurrence2PatternText`	**Type** textarea **Properties** Create, Nillable **Description** The RRULE that describes the recurrence pattern for Lightning Experience event series. Supports a subset of the RFC 5545 standard for internet calendaring and scheduling. See the Event Series section in this topic for usage examples. This field has a maximum length of 512 characters. This field is available in API version 44.0 and later, and has the `Create` property in API version 52.0 and later.
`Recurrence2PatternTimeZone`	**Type** string **Properties** Filter, Group, Nillable, Sort **Description** This field is available in API version 44.0 and later. Indicates the time zone in which the Lightning Experience event series was created or updated. This field uses standard Java TimeZone IDs. For example, America/Los_Angeles.
`Recurrence2PatternVersion`	**Type** picklist **Properties** Filter, Group, Nillable, Restricted picklist, Sort, **Description** Read-only. This field is available in API version 44.0 and later. Indicates the standard specifications for Lightning Experience event series recurrence patterns. The only possible value is `1` (RFC 5545 v4 RRULE)—RFC 5545 is a standard set of specifications for internet calendaring and scheduling that `IsRecurrence2` adheres to for series recurrence patterns. RFC 5545 specifications for series recurrence patterns are called RRULES. For examples of RRULE usage, see the Lightning Experience Event Series and Recurring Events section in this topic.

Field	Details
RecurrenceActivityId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** Read-only. Not required on create. Contains the ID of the main record of the Salesforce Classic recurring event. Subsequent occurrences have the same value in this field.
RecurrenceDayOfMonth	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Indicates the day of the month on which the event repeats.
RecurrenceDayOfWeekMask	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Indicates the day or days of the week on which the Salesforce Classic recurring event repeats. This field contains a bitmask. The values are as follows: • Sunday = 1 • Monday = 2 • Tuesday = 4 • Wednesday = 8 • Thursday = 16 • Friday = 32 • Saturday = 64 Multiple days are represented as the sum of their numerical values. For example, Tuesday and Thursday = 4 + 16 = 20.
RecurrenceEndDateOnly	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Indicates the last date on which the event repeats. For multiday Salesforce Classic recurring events, this date is the day on which the last occurrence starts. This field is a date field with a timestamp that is always set to midnight in the Coordinated Universal Time (UTC) time zone. Don't attempt to alter the timestamp to account for time zone differences.

Field	Details
RecurrenceInstance	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Indicates the frequency of the Salesforce Classic event's recurrence. For example, `2nd` or `3rd`.
RecurrenceInterval	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Indicates the interval between Salesforce Classic recurring events.
RecurrenceMonthOfYear	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Indicates the month in which the Salesforce Classic recurring event repeats.
RecurrenceStartDateTime	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** Indicates the date and time when the Salesforce Classic recurring event begins. The value must precede the `RecurrenceEndDateOnly`. The time portion of this field is always transferred in the Coordinated Universal Time (UTC) time zone. Translate the time portion to or from a local time zone for the user or the application, as appropriate.
RecurrenceTimeZoneSidKey	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Indicates the time zone associated with a Salesforce Classic recurring event. For example, "UTC-8:00" for Pacific Standard Time.
RecurrenceType	**Type** picklist

Field	Details
	Properties Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Indicates how often the Salesforce Classic event repeats. For example, daily, weekly, or every nth month (where "nth" is defined in `RecurrenceInstance`).
`ReminderDateTime`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** Represents the time when the reminder is scheduled to fire, if `IsReminderSet` is set to `true`. If `IsReminderSet` is set to `false`, then the user may have deselected the reminder checkbox in the Salesforce user interface, or the reminder has already fired at the time indicated by the value.
`ShowAs`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Indicates how this event appears when another user views the calendar: Busy, Out of Office, or Free. Label is **Show Time As**.
`StartDateTime`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** Indicates the start date and time of the event. Available in versions 13.0 and later. If the Event `IsAllDayEvent` flag is set to true (indicating that it's an all-day Event), then the event start date information is contained in the `StartDateTime` field. The time portion of this field is always transferred in the Coordinated Universal Time (UTC) time zone. Translate the time portion to or from a local time zone for the user or the application, as appropriate. If the Event `IsAllDayEvent` flag is set to false (indicating that it isn't an all-day event), then the event start date information is contained in the `StartDateTime` field. The time portion is always transferred in the Coordinated Universal Time (UTC) time zone. You need to translate the time portion to or from a local time zone for the user or the application, as appropriate. If this field has a value, then `ActivityDate` and `ActivityDateTime` must either be `null` or match the value of this field.

Field	Details
`Subject`	**Type** combobox **Properties** Create, Filter, Nillable, Sort, Update **Description** The subject line of the event, such as Call, Email, or Meeting. Limit: 255 characters.
`Type`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Indicates the event type, such as Call, Email, or Meeting.
`UndecidedEventInviteeIds`	**Type** JunctionIdList **Properties** Create, Update **Description** A string array of contact, lead, or user IDs who are undecided about this event. This `JunctionIdList` is linked to the `UndecidedEventRelation` child relationship. Warning: Adding a `JunctionIdList` field name to the `fieldsToNull` property deletes all related junction records. This action can't be undone.
`WhatCount`	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** Available if your organization has enabled Shared Activities. Represents the count of related EventRelations pertaining to the `WhatId`. The count of the `WhatId` must be *1* or less.
`WhatId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The `WhatId` represents nonhuman objects such as accounts, opportunities, campaigns, cases, or custom objects. `WhatIds` are polymorphic. Polymorphic means a `WhatId` is equivalent to the ID of a related object. The label is `Related To ID`. This is a polymorphic relationship field.

Field	Details
	Relationship Name What **Relationship Type** Lookup **Refers To** Account, Accreditation, AssessmentIndicatorDefinition, AssessmentTask, AssessmentTaskContentDocument, AssessmentTaskDefinition, AssessmentTaskOrder, Asset, AssetRelationship, AssignedResource, Award, BoardCertification, BusinessLicense, BusinessMilestone, BusinessProfile, Campaign, CareBarrier, CareBarrierDeterminant, CareBarrierType, CareDeterminant, CareDeterminantType, CareDiagnosis, CareInterventionType, CareMetricTarget, CareObservation, CareObservationComponent, CarePgmProvHealthcareProvider, CarePreauth, CarePreauthItem, CareProgram, CareProgramCampaign, CareProgramEligibilityRule, CareProgramEnrollee, CareProgramEnrolleeProduct, CareProgramEnrollmentCard, CareProgramGoal, CareProgramProduct, CareProgramProvider, CareProgramTeamMember, CareProviderAdverseAction, CareProviderFacilitySpecialty, CareProviderSearchableField, CareRegisteredDevice, CareRequest, CareRequestDrug, CareRequestExtension, CareRequestItem, CareSpecialty, CareSpecialtyTaxonomy, CareTaxonomy, Case, CommSubscriptionConsent, ContactEncounter, ContactEncounterParticipant, ContactRequest, Contract, CoverageBenefit, CoverageBenefitItem, CreditMemo, DelegatedAccount, DocumentChecklistItem, EnrollmentEligibilityCriteria, HealthcareFacility, HealthcareFacilityNetwork, HealthcarePayerNetwork, HealthcarePractitionerFacility, HealthcareProvider, HealthcareProviderNpi, HealthcareProviderSpecialty, HealthcareProviderTaxonomy, IdentityDocument, Image, IndividualApplication, Invoice, ListEmail, Location, MemberPlan, Opportunity, Order, OtherComponentTask, PartyConsent, PersonLifeEvent, PlanBenefit, PlanBenefitItem, ProcessException, Product2, ProductItem, ProductRequest, ProductRequestLineItem, ProductTransfer, PurchaserPlan, ReceivedDocument, ResourceAbsence, ReturnOrder, ReturnOrderLineItem, ServiceAppointment, ServiceResource, Shift, Shipment, ShipmentItem, Solution, Visit, VisitedParty, VolunteerProject, WorkOrder, WorkOrderLineItem
`WhoCount`	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** Available to organizations that have Shared Activities enabled. Represents the count of related EventRelations pertaining to the `WhoId`.
`WhoId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update

Field	Details
	Description The WhoId represents a human such as a lead or a contact. WhoIds are polymorphic. Polymorphic means a WhoId is equivalent to a contact's ID or a lead's ID. The label is `Name ID`. If Shared Activities is enabled, the value of this field is the ID of the related lead or primary contact. If you add, update, or remove the WhoId field, you might encounter problems with triggers, workflows, and data validation rules that are associated with the record. The label is `Name ID`. If the `JunctionIdList` field is used, all `WhoIds` are included in the relationship list. Beginning in API version 37.0, if the contact or lead ID in the `WhoId` field isn't in the `EventWhoIds` list, no error occurs and the ID is added to the `EventWhoIds` as the primary `WhoId`. If `WhoId` is set to null, an arbitrary ID from the existing `EventWhoIds` list is promoted to the primary position. This is a polymorphic relationship field. **Relationship Name** Who **Relationship Type** Lookup **Refers To** Contact, Lead

Usage

Use Event to manage calendar appointments.

Queries on events are denied before they time out if they involve amounts of data that are deemed too large. In such cases, the exception code `OPERATION_TOO_LARGE` is returned. If you receive `OPERATION_TOO_LARGE`, refactor your query to return or scan a smaller amount of data.

When querying for events with a specific due date, you must filter on both the `ActivityDateTimeand` and `ActivityDate` fields. For example to find all events with a due date of February 14, 2003, you need two filters:

- One filter with the `ActivityDate` field equal to the Coordinated Universal Time (UTC) time zone on February 14, 2003.
- One filter with the `ActivityDate` field greater than or equal to midnight on February 14, 2003 in the user's local time zone AND less than or equal to midnight on February 15, 2003 in the user's local time zone.

Alternatively, in API version 13.0 and later, you can find events with a specific due date by filtering on `StartDateTime`. For example, to find all events with a due date of February 14, 2003, filter with the `StartDateTime` greater than or equal to midnight on February 14, 2003 in the user's local time zone AND less than or equal to midnight on February 15, 2003 in the user's local time zone.

The `EventId` field of an EventRelation object always points to the master record. An invitee on a group event can query the EventRelation object to view the master record.

Multiday Events

- Multiday events are available in API version 13.0 and later. Also, in earlier versions SOQL queries don't return multiday events.

- Multiday events are enabled through the user interface from Setup by entering *Activity Settings* in the Quick Find box, then selecting **Activity Settings**.
- If the multiday event feature is enabled, then API versions 13.0 and later support values greater than 1440 for the DurationInMinutes field. API versions 12.0 and earlier can't access event objects whose DurationInMinutes is greater than 1440.
- Multiday events can't exceed 14 days.

Event Series and Recurring Events

In Lightning Experience, events with multiple occurrences are called event series, and are indicated when the IsRecurrence2 field is set to true. In Salesforce Classic, events with multiple occurrences are called recurring events, and are indicated when the IsRecurrence field is set to true. Both fields can't be set to true for the same event.

- Lightning Experience event series are available in API version 44.0 and later as read-only fields. Recurrence patterns, specified by the Recurrence2PatternText field, are creatable in API version 52.0 and later. Salesforce Classic recurring events are available in API version 7.0 and later. In earlier versions, SOQL queries don't return any Lightning Experience event series.
- After an event is created, you can't change the values of IsRecurrence2 or IsRecurrence from true to false or vice versa.
- You can't set fields associated with IsRecurrence2 for events where IsRecurrence is set to true, or vice versa.
- For Lightning Experience event series where IsRecurrence2 is true, if you'd like to delete a single or all remaining events, use the REST API call. For Salesforce Classic recurring events where IsRecurrence is true, all past and future events in the series are removed when you delete the recurring event series through the API. However, when you delete the recurring event series through the user interface, only future occurrences are removed.
- For Lightning Experience event series in API version 58.0 and later, when you change a future event, events in the entire series also change. When you change a past event, IsRecurrence2Exception is set to true and only that past event changes.
- When creating a Salesforce Classic recurring event series, the duration of the event must be 24 hours or less. When the Salesforce Classic recurring event series is created, you can extend the length of individual occurrences beyond 24 hours if Multiday events are enabled; see **Multiday Events**.
- For Salesforce Classic recurring events, RecurrenceStartDateTime, RecurrenceEndDateOnly, RecurrenceType, and any properties associated with the given recurrence type (see the Recurrence Field Usage for Salesforce Classic Recurring Events table) must be populated.
- When updating a Salesforce Classic recurring event series, it's not possible to update the EventRelation for the event series object and the EventRelation for the series object occurrences at the same time.
- Lightning Experience event series have no series ID, so it's not possible to locate other occurrences in the series. In Salesforce Classic recurring events, you can use RecurrenceActivityId to locate other occurrences.
- For both Lightning Experience event series and Salesforce Classic recurring events, when a series repeats every day, month, or year, you can only schedule occurrences one time per day, month, or year. The week option lets you schedule occurrences multiple days per week.

Limits for Lightning Experience event series and limits for Salesforce Classic recurring events also apply.

Lightning Experience Event Series and Recurring Events

Use the Recurrence2PatternText field to specify the recurrence pattern for Lightning Experience event series. These recurrence patterns, called reference rules or RRULES, support a subset of the RFC 5545 standards. This table includes common RRULE examples.

Recurrence Pattern	RRULE Example
Every day for five days	RRULE:FREQ=DAILY;INTERVAL=1;COUNT=5

Recurrence Pattern	RRULE Example
Every Monday, Tuesday, Wednesday, Thursday, and Friday with no end date	`RRULE:FREQ=WEEKLY;INTERVAL=1;BYDAY=MO,TU,WE,TH,FR`
Every two weeks on Monday and Friday for 10 occurrences	`RRULE:FREQ=WEEKLY;INTERVAL=2;BYDAY=MO,FR;COUNT=10`
Monthly on the first day of the month until January 1, 2020	`RRULE:FREQ=MONTHLY;INTERVAL=1;BYMONTHDAY=1;UNTIL=20200101T100000Z`
Yearly on July 4th for three years (in this example, specify the date using StartDateTime)	`RRULE:FREQ=YEARLY;INTERVAL=1;BYMONTH=7;BYMONTHDAY=4;COUNT=3`
Daily until January 1, 2022 with no end date	`RRULE:FREQ=DAILY;UNTIL=20220101T000000Z`
Every third Friday of the month with no end date	`RRULE:FREQ=MONTHLY;BYSETPOS=3;BYDAY=FR`

The RRULE defined by `Recurrence2PatternText` supports a subset of the RFC 5545 standard for internet calendaring and scheduling. Supported RRULE parts include FREQ, BYMONTH, BYMONTHDAY, BYDAY, WKST, BYSETPOS, INTERVAL, UNTIL, and COUNT.

When the event record is saved, the RRULE might be modified to follow the required format:

- The RRULE parts are placed in the following order: FREQ, BYMONTH, BYMONTHDAY, BYDAY, WKST, BYSETPOS, INTERVAL, UNTIL, and COUNT.
- Any missing default values are inserted. For example, if the RRULE doesn't include INTERVAL, then `INTERVAL=1` is added.
- The RRULE is prefaced with `RRULE:` if that preface is missing.

RRULE Part	Supported RFC 5545 Implementation
FREQ	Required. Indicates the type of recurrence rule. Allowed values are: • `DAILY`—supported parts include FREQ, INTERVAL, UNTIL, and COUNT. • `WEEKLY`—supported parts include INTERVAL, UNTIL, COUNT, BYDAY, and WKST. BYDAY is required, but can't be preceded by a number. For example, to indicate weekly on Tuesday and Thursday until September 1, 2023, use `RRULE:FREQ=WEEKLY;UNTIL=20230901T000000Z;BYDAY=TU,TH` • `MONTHLY`—supported patterns include: – BYMONTHDAY For example, to indicate monthly on the third day of the month use: `RRULE:FREQ=MONTHLY;BYMONTHDAY=3` – BYDAY and BYSETPOS For example, to indicate the last weekday of the month, use `RRULE:FREQ=MONTHLY;BYDAY=MO,TU,WE,TH,FR;BYSETPOS=-1` – BYDAY, where the BYDAY values are specified with a numeric value For example, to indicate monthly on the first Friday for 10 occurrences, use `RRULE:FREQ=MONTHLY;COUNT=10;BYDAY=1FR` • `YEARLY`—supported patterns include: – BYMONTH, BYDAY, and BYSETPOS

RRULE Part	Supported RFC 5545 Implementation
	For example, to indicate every year on the second Friday of January, use `RRULE:FREQ=YEARLY;BYMONTH=1;BYDAY=FR;BYSETPOS=2` – BYMONTH and BYMONTHDAY For example, to indicate every year on October 31, use `RRULE:FREQ=YEARLY;BYMONTH=10;BYMONTHDAY=31` For example, to create a maintenance pattern such as twice in May, and September on 7th and 15th; and one time in June/July/August on the 1st, use two RRULEs: `RRULE: FREQ=MONTHLY; BYMONTH=5,9; BYMONTHDAY=7, 15` `RRULE:FREQ=MONTHLY; BYMONTH=6,7,8; BYMONTHDAY=1`
BYMONTH	The month. Valid values are `1` to `12`.
BYMONTHDAY	The day of the month. Valid values are `1` to `31`. If BYMONTHDAY is `31` and the month has fewer than 31 days, the event is created on the last day of the month.
BYDAY	A comma-separated list of days of the week. Valid values are `SU`, `MO`, `TU`, `WE`, `TH`, `FR`, `SA`. For RRULES with yearly or monthly frequency, BYDAY must be one of: • a single day • weekend days • weekdays • every day of the week Each BYDAY value can be preceded by an integer that indicates the nth occurrence of a specific day within the monthly or yearly RRULE. Allowed values are `–1`, `1`, `2`, `3`, and `4`. You can't use different numbers in the BYDAY values. For example, this RRULE isn't supported: `RRULE:FREQ=MONTHLY;INTERVAL=2;COUNT=10;BYDAY=1SU,-1SU` If BYDAY values are prefaced with a number, the RRULE can't include BYSETPOS.
WKST	Specifies the day on which the workweek starts. Valid values are `MO`, `TU`, `WE`, `TH`, `FR`, `SA`, and `SU`. Default value is based on the user's locale.
BYSETPOS	A comma-separated list of values that correspond to the nth occurrence within the set of recurrence instances specified by the rule. Valid values are `–1`, `1`, `2`, `3`, or `4`. Default value is `1`. For example, to indicate the last weekday of the month, use: `RRULE:FREQ=MONTHLY;BYDAY=MO,TU,WE,TH,FR;BYSETPOS=-1`
INTERVAL	The repetition interval. Valid values are: • an integer between `1` and `999` if FREQ=DAILY • an integer between `1` and `99` if FREQ=WEEKLY or FREQ=MONTHLY • `1` if FREQ=YEARLY Default value is `1`.

RRULE Part	Supported RFC 5545 Implementation
UNTIL	Specifies the datetime in UTC format when the recurrence rule stops. The supported format is yyyyMMddTHHmmssZ, for example: 20210419T083000Z. An RRULE can't contain both UNTIL and COUNT. A recurring event without either UNTIL or COUNT repeats indefinitely.
COUNT	The number of occurrences. Allowed values are 1 to 999. An RRULE can't contain both UNTIL and COUNT. A recurring event without either UNTIL or COUNT repeats indefinitely.
BYWEEKNO	Specifies a comma-separated list of values that specify weeks of the year. Valid values are: • `1 to 53` • `-53 to -1` For example, to indicate specific weeks in a year, use: `RRULE: BYWEEKNO=20, -20`. This rule part can't be used when the FREQ rule part is set to anything other than YEARLY. For example, 3 represents the third week of the year. Note: Assuming a Monday week start, week 53 can only occur when Thursday is January 1 or if it's a leap year and Wednesday is January 1.
BYYEARDAY	Specifies a comma-separated list of values that specify days of the year. Valid values are: • `1 to 366` • `-366 to -1` For example, to indicate specific days in a year, use: `RRULE:BYYEARDAY=1,100,200`; or, `RRULE:BYYEARDAY=1,-2`.

Salesforce Classic Event Series and Recurring Events

This table describes the usage of recurrence fields for Salesforce Classic recurring events. Each recurrence type must have all of its properties set. All unused properties must be set to null.

RecurrenceType Value	Properties	Example Pattern
RecursDaily	RecurrenceInterval	Every second day
RecursEveryWeekday	RecurrenceDayOfWeekMask	Every weekday - can't be Saturday or Sunday
RecursMonthly	RecurrenceDayOfMonth RecurrenceInterval	Every second month, on the third day of the month
RecursMonthlyNth	RecurrenceInterval RecurrenceInstance RecurrenceDayOfWeekMask	Every second month, on the last Friday of the month
RecursWeekly	RecurrenceInterval RecurrenceDayOfWeekMask	Every three weeks on Wednesday and Friday
RecursYearly	RecurrenceDayOfMonth RecurrenceMonthOfYear	Every March on the 26th day of the month

RecurrenceType Value	Properties	Example Pattern
RecursYearlyNth	RecurrenceDayOfWeekMask RecurrenceInstanceRecurrenceMonthOfYear	The first Saturday in every October

Attendees, Invitees, and Resources

The field `GroupEventType` indicates that event participants are included on an event. You can add a resource to an event only when the resource is available. The only attendance status that can be assigned to resources is Accepted. Events can't be saved when resources you've added aren't available.

JunctionIdList

To create an event using `JunctionIdList`, IDs are pulled from the related contacts and both the event and the `EventRelation` records are created in one API call. If the `EventRelation` fails, the event is rolled back because it's all done in a single API call.

```
public void createEventNew(Contact[] contacts) {
 String[] contactIds = new String[contacts.size()];
 for (int i = 0; i < contacts.size(); i++) {
  contactIds[i] = contacts[i].getID();
 }
 Event event = new Event();
 event.setSubject("New Event");
 event.setEventWhoIds(contactIds);
 SaveResult[] results = null;
 try {
  results = connection.create(new Event[] {
   task
  });
 } catch (ConnectionException ce) {
  ce.printStackTrace();
 }
}
```

Syncing Events with Lightning Sync

Attendee statuses (Accepted or Maybe, Declined, or No Response) sync from Microsoft® Exchange or Google to Salesforce, but not from Salesforce to Exchange or Google. Be wary of creating API flows that update attendee status in Salesforce for users set up to sync both ways. Eventually the original Exchange or Google status overrides the update made in Salesforce.

Shared Field-Level Security for Event and Task Objects

Metadata deployments for the Event object must include the field-level security for the Task object. Shared field-level security prevents each object from changing the field-level security of the associated object.

Metadata deployments that include field-level security for only one of either the Event or Task objects can cause field-level security changes to the other object that aren't reflected in the metadata.

- If field-level security is enabled for one object, then field-level security is enabled for both objects.
- If field-level security is disabled for one object, then it's disabled for both objects.

Note: A missing entry in the metadata is treated as field-level security being disabled.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

EventChangeEvent (API version 44.0)
Change events are available for the object.

EventFeed (API version 20.0)
Feed tracking is available for the object.

Task

Represents a business activity such as making a phone call or other to-do items. In the user interface, Task and Event records are collectively referred to as activities.

Note: Task fields related to calls are exclusive to Salesforce CRM Call Center. Also, `query()`, `delete()`, and `update()` aren't allowed with tasks related to more than one contact in API versions 23.0 and earlier.

Supported Calls

`create()`, `delete()`, `describeLayout()`, `describeSObjects()`, `getDeleted()`, `getUpdated()`, `query()`, `retrieve()`, `search()`, `undelete()`, `update()`, `upsert()`

Fields

Field	Field Type
`AccountId`	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** Represents the ID of the related Account. The `AccountId` is determined as follows. If the value of `WhatId` is any of the following objects, then Salesforce uses that object's `AccountId`. • Account • Opportunity • Contract • Custom object that is a child of Account If the value of the `WhatId`field is any other object, and the value of the `WhoId` field is a Contact object, then Salesforce uses that contact's `AccountId`. (If your organization uses Shared Activities, then Salesforce uses the `AccountId` of the primary contact.) Otherwise, Salesforce sets the value of the `AccountId` field to `null`. For information on IDs, see ID Field Type.

Field	Field Type
	This is a relationship field. **Relationship Name** Account **Relationship Type** Lookup **Refers To** Account
`ActivityDate`	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Represents the due date of the task. This field has a timestamp that is always set to midnight in the Coordinated Universal Time (UTC) time zone. The timestamp is not relevant; do not attempt to alter it to accommodate time zone differences. Label is **Due Date**. This field can't be set or updated for a recurring task (`IsRecurrence` is `true`).
`CallDisposition`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Represents the result of a given call, for example, "we'll call back," or "call unsuccessful." Limit is 255 characters. Not subject to field-level security, available for any user in an organization with Salesforce CRM Call Center.
`CallDurationInSeconds`	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** Duration of the call in seconds. Not subject to field-level security, available for any user in an organization with Salesforce CRM Call Center.
`CallObject`	**Type** string **Properties** Create, Filter, Group, Nillable, Sort, Update

Field	Field Type
	Description Name of a call center. Limit is 255 characters. Not subject to field-level security, available for any user in an organization with Salesforce CRM Call Center.
CallType	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The type of call being answered: Inbound, Internal, or Outbound.
CompletedDateTime	**Type** dateTime **Properties** Filter, Nillable, Sort **Description** The date and time the task was saved with a Closed status. • For insert, if the task is saved with a Closed status the field is set. If the task is saved with an Open status the field is set to NULL. • For update, if the task is saved with a new Closed status, the field is reset. If the task is saved with a new non-closed status, the field is reset to NULL. If the task is saved with the same closed status (that is, unchanged) there is no change to the field. The status is a dynamic enum. If the Closed mapping is changed it won't cause an update of existing tasks. Only new insert/update operations are affected.
ConnectionReceivedId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** ID of the PartnerNetworkConnection that shared this record with your organization. This field is available if you enabled Salesforce to Salesforce.
ConnectionSentId	**Type** reference **Properties** Filter, Group, Nillable, Sort

Field	Field Type
	Description ID of the PartnerNetworkConnection that you shared this record with. This field is available if you enabled Salesforce to Salesforce. This field is supported using API versions earlier than 15.0. In all other API versions, this field's value is null. You can use the new PartnerNetworkRecordConnection object to forward records to connections.
`Description`	**Type** textarea **Properties** Create, Nillable, Update **Description** Contains a text description of the task.
`IsArchived`	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Indicates whether the event has been archived. The default value of this field is `false`.
`IsClosed`	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Indicates whether the task has been completed (`true`) or not (`false`). The default value of this field is `false`. Is only set indirectly via the `Status` picklist. Label is **Closed**.
`IsHighPriority`	**Type** boolean **Properties** Defaulted on create, Filter, Group, Sort **Description** Indicates a high-priority task. This field is derived from the `Priority` field. The default value of this field is `false`.
`IsRecurrence`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort

Field	Field Type
	Description Indicates whether the task is scheduled to repeat itself (`true`) or only occurs once (`false`). The default value of this field is `false`. This field is read-only on update, but not on create. If this field value is `true`, then `RecurrenceStartDateOnly`, `RecurrenceEndDateOnly`, `RecurrenceType`, and any recurrence fields associated with the given recurrence type must be populated. See Recurring Tasks.
`IsReminderSet`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether a popup reminder has been set for the task (`true`) or not (`false`). The default value of this field is `false`.
`IsVisibleInSelfService`	**Type** boolean **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Indicates whether a task associated with an object can be viewed in the Customer Portal (`true`) or not (`false`). If your organization has digital experiences enabled, tasks marked `IsVisibleInSelfService` are visible to any external user in the Experience Cloud site, as long as the user has access to the record the task was created on.
`OwnerId`	**Type** reference **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** ID of the User or Group who owns the record. Label is **Assigned To ID**. This field accepts Groups of type Queue only. In the user interface, Group IDs correspond with the queue's list view names. To create or update tasks assigned to Group, use v48.0 or later. This is a polymorphic relationship field. **Relationship Name** Owner **Relationship Type** Lookup **Refers To** Group, User

Field	Field Type
Priority	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Required. Indicates the importance or urgency of a task, such as high or low. The default value of this field is `Normal`.
RecurrenceActivityId	**Type** reference **Properties** Filter, Group, Nillable, Sort **Description** Read-only. Not required on create. ID of the main record of the recurring task. Subsequent occurrences have the same value in this field.
RecurrenceDayOfMonth	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The day of the month in which the task repeats.
RecurrenceDayOfWeekMask	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The day or days of the week on which the task repeats. This field contains a bitmask. The values are as follows: • Sunday = 1 • Monday = 2 • Tuesday = 4 • Wednesday = 8 • Thursday = 16 • Friday = 32 • Saturday = 64 Multiple days are represented as the sum of their numerical values. For example, Tuesday and Thursday = 4 + 16 = 20.

Field	Field Type
RecurrenceEndDateOnly	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The last date on which the task repeats. This field has a timestamp that is always set to midnight in the Coordinated Universal Time (UTC) time zone. The timestamp is not relevant; do not attempt to alter it to accommodate time zone differences.
RecurrenceInstance	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The frequency of the recurring task. Possible values are: • First—1st • Fourth—4th • Last—last • Second—2nd • Third—3rd
RecurrenceInterval	**Type** int **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The interval between recurring tasks.
RecurrenceMonthOfYear	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The month of the year in which the task repeats.
RecurrenceRegeneratedType	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update

Field	Field Type
	Description Represents what triggers a repeating task to repeat. Add this field to a page layout together with the `RecurrenceInterval` field, which determines the number of days between the triggering date (due date or close date) and the due date of the next repeating task in the series. Label is **Repeat This Task**. This field has the following picklist values: • **None**: The task doesn't repeat. • **After due date**: The next repeating task will be due the specified number of days after the current task's due date. • **After the task is closed**: The next repeating task will be due the specified number of days after the current task is closed. • **(Task closed)**: This task, now closed, was opened as part of a repeating series. Note: When tasks in a series are set to repeat after their due date, Salesforce doesn't create recurrences that would have been due in the past. Instead, Salesforce keeps adding the interval until a repeated task has a due date in the future. For example, suppose that someone sets a task to repeat three days after it's due. But, that person doesn't complete the task (mark it Closed) until five days after it's due. Instead of creating a task that's already overdue, Salesforce gives the new task a due date of tomorrow. This due date is equivalent to 6 days after the due date; two intervals of three days each. If that person completes the repeating task (marks it Closed) before the due date, the next task is still due three days after the due date.
RecurrenceStartDateOnly	**Type** date **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The date when the recurring task begins. Must be a date and time before `RecurrenceEndDateOnly`.
RecurrenceTimeZoneSidKey	**Type** picklist **Properties** Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** The time zone associated with the recurring task. For example, "UTC-8:00" for Pacific Standard Time.
RecurrenceType	**Type** picklist

Field	Field Type
	Properties Create, Filter, Group, Nillable, Restricted picklist, Sort, Update **Description** Indicates how often the task repeats. For example, daily, weekly, or every nth month (where "nth" is defined in `RecurrenceInstance`).
`ReminderDateTime`	**Type** dateTime **Properties** Create, Filter, Nillable, Sort, Update **Description** Represents the time when the reminder is scheduled to fire, if `IsReminderSet` is set to `true`. If `IsReminderSet` is set to `false`, then the user may have deselected the reminder checkbox in the Salesforce user interface, or the reminder has already fired at the time indicated by the value.
`Status`	**Type** picklist **Properties** Create, Defaulted on create, Filter, Group, Sort, Update **Description** Required. Indicates the status of the task. The default value of this field is `Not Started`. Each predefined `Status` field implies a value for the `IsClosed` flag. To obtain picklist values, query the TaskStatus object. Possible values are: • Completed • Deferred • In Progress • Not Started • Waiting on someone else Note: This field can't be updated for recurring tasks (`IsRecurrence` is `true`).
`Subject`	**Type** combobox **Properties** Create, Filter, Nillable, Sort, Update **Description** The subject line of the task, such as "Call" or "Send Quote." Limit: 255 characters.
`TaskSubtype`	**Type** picklist

Field	Field Type
	Properties Create, Filter, Group, Nillable, Restricted picklist, Sort **Description** Provides standard subtypes to facilitate creating and searching for specific task subtypes. This field isn't updateable. `TaskSubtype` values: • `Task` • `Email` • `LinkedIn` —Available in API version 56.0 and later. • `List Email` • `Cadence` • `Call` Note: The `Cadence` subtype is an internal value used by Sales Engagement, and can't be set manually.
`TaskWhoIds`	**Type** JunctionIdList **Properties** Create, Update **Description** A string array of contact or lead IDs related to this task. This `JunctionIdList` field is linked to the `TaskWhoRelations` child relationship. `TaskWhoIds` is only available when the shared activities setting is enabled. The first contact or lead ID in the list becomes the primary `WhoId` if you don't specify a primary `WhoId`. If you set the `EventWhoIds` field to null, all entries in the list are deleted and the value of `WhoId` is added as the first entry. Warning: Adding a `JunctionIdList` field name to the `fieldsToNull` property deletes all related junction records. This action can't be undone.
`Type`	**Type** picklist **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The type of task, such as Call or Meeting.
`WhatCount`	**Type** int **Properties** Filter, Group, Nillable, Sort

Field	Field Type
	Description Available to organizations that have Shared Activities enabled. Count of related TaskRelations pertaining to `WhatId`. Count of the `WhatId` must be *1* or less.
`WhatId`	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The `WhatId` represents nonhuman objects such as accounts, opportunities, campaigns, cases, or custom objects. `WhatId`s are polymorphic. Polymorphic means a `WhatId` is equivalent to the ID of a related object. The label is `Related To ID`. This is a polymorphic relationship field. **Relationship Name** What **Relationship Type** Lookup **Refers To** Account, Accreditation, AssessmentIndicatorDefinition, AssessmentTask, AssessmentTaskContentDocument, AssessmentTaskDefinition, AssessmentTaskOrder, Asset, AssetRelationship, AssignedResource, Award, BoardCertification, BusinessLicense, BusinessMilestone, BusinessProfile, Campaign, CareBarrier, CareBarrierDeterminant, CareBarrierType, CareDeterminant, CareDeterminantType, CareDiagnosis, CareInterventionType, CareMetricTarget, CareObservation, CareObservationComponent, CarePgmProvHealthcareProvider, CarePreauth, CarePreauthItem, CareProgram, CareProgramCampaign, CareProgramEligibilityRule, CareProgramEnrollee, CareProgramEnrolleeProduct, CareProgramEnrollmentCard, CareProgramGoal, CareProgramProduct, CareProgramProvider, CareProgramTeamMember, CareProviderAdverseAction, CareProviderFacilitySpecialty, CareProviderSearchableField, CareRegisteredDevice, CareRequest, CareRequestDrug, CareRequestExtension, CareRequestItem, CareSpecialty, CareSpecialtyTaxonomy, CareTaxonomy, Case, CommSubscriptionConsent, ContactEncounter, ContactEncounterParticipant, ContactRequest, Contract, CoverageBenefit, CoverageBenefitItem, CreditMemo, DelegatedAccount, DocumentChecklistItem, EnrollmentEligibilityCriteria, HealthcareFacility, HealthcareFacilityNetwork, HealthcarePayerNetwork, HealthcarePractitionerFacility, HealthcareProvider, HealthcareProviderNpi, HealthcareProviderSpecialty, HealthcareProviderTaxonomy, IdentityDocument, Image, IndividualApplication, Invoice, ListEmail, Location, MemberPlan, Opportunity, Order, OtherComponentTask, PartyConsent, PersonLifeEvent, PlanBenefit, PlanBenefitItem, ProcessException, Product2, ProductItem, ProductRequest, ProductRequestLineItem, ProductTransfer, PurchaserPlan, ReceivedDocument, ResourceAbsence, ReturnOrder, ReturnOrderLineItem, ServiceAppointment, ServiceResource, Shift, Shipment, ShipmentItem, Solution, Visit, VisitedParty, VolunteerProject, WorkOrder, WorkOrderLineItem

Field	Field Type
WhoCount	**Type** int **Properties** Filter, Group, Nillable, Sort **Description** Available to organizations that have Shared Activities enabled. Count of related TaskRelations pertaining to `WhoId`.
WhoId	**Type** reference **Properties** Create, Filter, Group, Nillable, Sort, Update **Description** The WhoId represents a human such as a lead or a contact. WhoIds are polymorphic. Polymorphic means a WhoId is equivalent to a contact's ID or a lead's ID. The label is `Name ID`. If Shared Activities is enabled, the value of this field is the ID of the related lead or primary contact. If you add, update, or remove the WhoId field, you might encounter problems with triggers, workflows, and data validation rules that are associated with the record. The label is `Name ID`. Beginning in API version 37.0, if the contact or lead ID in the `WhoId` field is not in the `TaskWhoIds` list, no error occurs and the ID is added to the `TaskWhoIds` as the primary `WhoId`. If `WhoId` is set to null, an arbitrary ID from the existing `TaskWhoIds` list is promoted to the primary position. This is a polymorphic relationship field. **Relationship Name** Who **Relationship Type** Lookup **Refers To** Contact, Lead

Usage

Recurring Tasks

- Recurring tasks are available in API version 16.0 and later.
- After a task is created, it can't be changed from recurring to nonrecurring or vice versa.
- When a user creates a series of recurring tasks, Salesforce creates a main record and subsequent occurrences. For the main record, `IsRecurrence` is set to `true` and other fields that define the recurrence pattern are populated. The ID of the main record of the recurring task is saved in the subsequent occurrences, in the `RecurrenceActivityId` field.

- When you delete a recurring task series through the API, all open and closed task occurrences in the series are removed. However, when you delete a recurring task series through the user interface, only open tasks occurrences (`IsClosed` is `false`) in the series are removed.
- If `IsRecurrence` is `true`, then `RecurrenceStartDateOnly`, `RecurrenceEndDateOnly`, `RecurrenceType`, and any properties associated with the given recurrence type (see the following table) must be populated.
- When you change the `RecurrenceStartDateOnly` field or the recurrence pattern, all open tasks occurrences in the series are deleted and new open task occurrences are created based on the new recurrence pattern. The following fields determine the recurrence pattern: `RecurrenceType`, `RecurrenceTimeZoneSidKey`, `RecurrenceInterval`, `RecurrenceDayOfWeekMask`, `RecurrenceDayOfMonth`, `RecurrenceInstance`, and `RecurrenceMonthOfYear`.
- When you change the value of `RecurrenceEndDateOnly` to an earlier date (for example, from January 20 to January 10), all open task occurrences in the series with the `ActivityDate` value greater than the new end date value are deleted. Other open and closed task occurrences in the series are not affected.
- When you change the value of `RecurrenceEndDateOnly` to a later date (for example, from January 10 to January 20), new task occurrences are created up to the new end date. Existing open and closed tasks in the series are not affected.

This table describes the usage of recurrence fields for Salesforce Classic recurring events. Each recurrence type must have all of its properties set. All unused properties must be set to null.

RecurrenceType Value	Properties	Example Pattern
RecursDaily	RecurrenceInterval	Every second day
RecursEveryWeekday	RecurrenceDayOfWeekMask	Every weekday - can't be Saturday or Sunday
RecursMonthly	RecurrenceDayOfMonth RecurrenceInterval	Every second month, on the third day of the month
RecursMonthlyNth	RecurrenceInterval RecurrenceInstance RecurrenceDayOfWeekMask	Every second month, on the last Friday of the month
RecursWeekly	RecurrenceInterval RecurrenceDayOfWeekMask	Every three weeks on Wednesday and Friday
RecursYearly	RecurrenceDayOfMonth RecurrenceMonthOfYear	Every March on the 26th day of the month
RecursYearlyNth	RecurrenceDayOfWeekMask RecurrenceInstanceRecurrenceMonthOfYear	The first Saturday in every October

JunctionIdList

The `JunctionIdList` field is now implemented in the Event and Task objects. With a single API call, it's easy to create many-to-many relationships between the Event or Task object with contacts, leads, or users.

To create a Task with related Contacts without `JunctionIdList`, you first have to create the task, then use the returned task ID to create the `TaskRelation` records. If the `TaskRelation` save call fails, error handling is your responsibility because the task has already been committed to the database.

```
public void createTasksOld(Contact[] contacts) {
 Task task = new Task();
 task.setSubject("New Task");
 SaveResult[] results = null;
```

```
  try {
   results = connection.create(new Task[] {
    task
   });
   if (results[0].isSuccess()) {
    TaskRelation[] relations = new TaskRelation[contacts.size()];
    for (int i = 0; i < contacts.length; i++) {
     relations[i] = new TaskRelation();
     relations[i].setTaskId(results[0].getID());
     relations[i].setRelationId(contacts[i].getID());
    }
    results = connection.create(relations);
   }
  } catch (ConnectionException ce) {
   ce.printStackTrace();
  }
}
```

To create a task using `JuncionIdList`, IDs are pulled from the related contacts and both the task and the `TaskRelation` records are created in one API call. If the `TaskRelation` fails, the task is rolled back because it's all done in a single API call.

```
public void createTaskNew(Contact[] contacts) {
  String[] contactIds = new String[contacts.size()];
  for (int i = 0; i < contacts.size(); i++) {
   contactIds[i] = contacts[i].getID();
  }
  Task task = new Task();
  task.setSubject("New Task");
  task.setTaskWhoIds(contactIds);
  SaveResult[] results = null;
  try {
   results = connection.create(new Task[] {
    task
   });
  } catch (ConnectionException ce) {
   ce.printStackTrace();
  }
}
```

Shared Field-Level Security for Event and Task Objects

Metadata deployments for the Task object should always include the field-level security for the Event object. Shared field-level security prevents each object from changing the field-level security of the associated object.

Metadata deployments that include field-level security for only one of either the Event or Task objects can cause field-level security changes to the other object that aren't reflected in the metadata.

- If field-level security is enabled for one object, then field-level security is enabled for both objects.
- If field-level security is disabled for one object, then it's disabled for both objects.

Note: A missing entry in the metadata is treated as field-level security being disabled.

Associated Objects

This object has the following associated objects. If the API version isn't specified, they're available in the same API versions as this object. Otherwise, they're available in the specified API version and later.

TaskChangeEvent (API version 44.0)
Change events are available for the object.

TaskFeed (API version 20.0)
Feed tracking is available for the object.